

Core Flight Executive Users Guide

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1 Core Flight Executive Documentation	1
1.1 Background	2
1.1.1 Core Flight Executive (cFE) Goals	3
1.2 Applicable Documents	3
1.3 Version Numbers	4
1.3.1 Version Number Semantics	4
1.3.2 How and Where Defined	4
1.3.3 Identifying Development Builds	5
1.3.4 Templates for the short and long version string	5
1.4 Dependencies	5
1.5 Acronyms	5
1.6 cFE Executive Services Overview	7
1.6.1 Terminology	8
1.6.2 Software Reset	10
1.6.3 Reset Types and Subtypes	10
1.6.4 Exception and Reset (ER) Log	10
1.6.5 Application and Child Task Management	11
1.6.6 Starting an Application	11
1.6.7 Stopping an Application	11
1.6.8 Restarting an Application	12
1.6.9 Reloading an Application	12
1.6.10 Listing Current Applications	12
1.6.11 Listing Current Tasks	13
1.6.12 Loading Common Libraries	13
1.6.13 Basic File System	13
1.6.14 Performance Data Collection	14
1.6.15 Critical Data Store	15
1.6.16 Memory Pool	16
1.6.17 System Log	18
1.6.18 Version Identification	18
1.6.19 Frequently Asked Questions about Executive Services	18
1.7 cFE Executive Services Commands	18
1.8 cFE Executive Services Telemetry	20
1.9 cFE Executive Services Configuration Parameters	20
1.10 cFE Event Services Overview	25
1.10.1 Event Message Format	26
1.10.2 Local Event Log	27
1.10.3 Event Message Control	27
1.10.4 Event Message Filtering	29

1.10.5 EVS Registry	29
1.10.6 EVS Counters	30
1.10.7 Resetting EVS Counters	30
1.10.8 Effects of a Processor Reset on EVS	31
1.10.9 EVS squelching of misbehaving apps	31
1.10.10 Frequently Asked Questions about Event Services	32
1.11 cFE Event Services Commands	33
1.12 cFE Event Services Telemetry	34
1.13 cFE Event Services Configuration Parameters	35
1.14 cFE Software Bus Overview	36
1.14.1 Software Bus Terminology	36
1.14.2 Autonomous Actions	38
1.14.3 Operation of the SB Software	39
1.14.4 Frequently Asked Questions about Software Bus	43
1.15 cFE Software Bus Commands	44
1.16 cFE Software Bus Telemetry	45
1.17 cFE Software Bus Configuration Parameters	45
1.18 cFE Table Services Overview	47
1.18.1 Managing Tables	47
1.18.2 cFE Table Types and Table Options	48
1.18.3 Table Registry	50
1.18.4 Table Services Telemetry	51
1.18.5 Effects of Processor Reset on Tables	51
1.18.6 Frequently Asked Questions about Table Services	52
1.19 cFE Table Services Commands	53
1.20 cFE Table Services Telemetry	54
1.21 cFE Table Services Configuration Parameters	54
1.22 cFE Time Services Overview	56
1.22.1 Time Components	57
1.22.2 Time Structure	58
1.22.3 Time Formats	58
1.22.4 Time Configuration	59
1.22.5 Time Services Paradigm(s)	66
1.22.6 Flywheeling	67
1.22.7 Time State	68
1.22.8 Initialization	68
1.22.9 Normal Operation	69
1.22.10 Frequently Asked Questions about Time Services	74
1.23 cFE Time Services Commands	74

1.24 cFE Time Services Telemetry	75
1.25 cFE Time Services Configuration Parameters	75
1.26 cFE Event Message Cross Reference	76
1.27 cFE Command Mnemonic Cross Reference	77
1.28 cFE Telemetry Mnemonic Cross Reference	81
2 Glossary of Terms	91
3 cFE Application Programmer's Interface (API) Reference	92
3.1 Executive Services API	92
3.2 Events Services API	94
3.3 File Services API	94
3.4 Message API	95
3.5 Resource ID API	96
3.6 Software Bus Services API	96
3.7 Table Services API	97
3.8 Time Services API	97
4 Osal API Documentation	98
4.1 OSAL Introduction	100
4.2 File System Overview	100
4.3 File Descriptors In Osal	101
4.4 Timer Overview	101
5 cFE Mission Configuration Parameters	101
6 Topic Index	102
6.1 Topics	102
7 Data Structure Index	105
7.1 Data Structures	105
8 File Index	115
8.1 File List	115
9 Topic Documentation	121
9.1 cFE Return Code Defines	121
9.1.1 Detailed Description	126
9.1.2 Macro Definition Documentation	126
9.2 cFE Resource ID APIs	145
9.2.1 Detailed Description	145
9.2.2 Function Documentation	145

9.3 cFE Entry/Exit APIs	147
9.3.1 Detailed Description	148
9.3.2 Function Documentation	148
9.4 cFE Application Control APIs	149
9.4.1 Detailed Description	149
9.4.2 Function Documentation	149
9.5 cFE Application Behavior APIs	151
9.5.1 Detailed Description	152
9.5.2 Function Documentation	152
9.6 cFE Information APIs	155
9.6.1 Detailed Description	156
9.6.2 Function Documentation	156
9.7 cFE Child Task APIs	164
9.7.1 Detailed Description	164
9.7.2 Function Documentation	164
9.8 cFE Miscellaneous APIs	168
9.8.1 Detailed Description	168
9.8.2 Function Documentation	168
9.9 cFE Critical Data Store APIs	170
9.9.1 Detailed Description	170
9.9.2 Function Documentation	170
9.10 cFE Memory Manager APIs	174
9.10.1 Detailed Description	175
9.10.2 Function Documentation	175
9.11 cFE Performance Monitor APIs	182
9.11.1 Detailed Description	183
9.11.2 Macro Definition Documentation	183
9.11.3 Function Documentation	184
9.12 cFE Generic Counter APIs	184
9.12.1 Detailed Description	185
9.12.2 Function Documentation	185
9.13 cFE Registration APIs	190
9.13.1 Detailed Description	190
9.13.2 Function Documentation	190
9.14 cFE Send Event APIs	191
9.14.1 Detailed Description	191
9.14.2 Function Documentation	191
9.15 cFE Reset Event Filter APIs	195
9.15.1 Detailed Description	195

9.15.2 Function Documentation	195
9.16 cFE File Header Management APIs	196
9.16.1 Detailed Description	196
9.16.2 Function Documentation	196
9.17 cFE File Utility APIs	200
9.17.1 Detailed Description	200
9.17.2 Function Documentation	200
9.18 cFE Generic Message APIs	204
9.18.1 Detailed Description	205
9.18.2 Function Documentation	205
9.19 cFE Message Primary Header APIs	205
9.19.1 Detailed Description	206
9.19.2 Function Documentation	206
9.20 cFE Message Extended Header APIs	214
9.20.1 Detailed Description	214
9.20.2 Function Documentation	214
9.21 cFE Message Secondary Header APIs	219
9.21.1 Detailed Description	220
9.21.2 Function Documentation	220
9.22 cFE Message Id APIs	225
9.22.1 Detailed Description	225
9.22.2 Function Documentation	225
9.23 cFE Message Integrity APIs	227
9.23.1 Detailed Description	227
9.23.2 Function Documentation	227
9.24 cFE Pipe Management APIs	228
9.24.1 Detailed Description	229
9.24.2 Function Documentation	229
9.25 cFE Message Subscription Control APIs	233
9.25.1 Detailed Description	234
9.25.2 Function Documentation	234
9.26 cFE Send/Receive Message APIs	239
9.26.1 Detailed Description	239
9.26.2 Function Documentation	239
9.27 cFE Zero Copy APIs	241
9.27.1 Detailed Description	241
9.27.2 Function Documentation	241
9.28 cFE Message Characteristics APIs	243
9.28.1 Detailed Description	244

9.28.2 Function Documentation	244
9.29 cFE Message ID APIs	247
9.29.1 Detailed Description	248
9.29.2 Function Documentation	248
9.30 cFE SB Pipe options	252
9.30.1 Detailed Description	252
9.30.2 Macro Definition Documentation	252
9.31 cFE Registration APIs	253
9.31.1 Detailed Description	253
9.31.2 Function Documentation	253
9.32 cFE Manage Table Content APIs	257
9.32.1 Detailed Description	257
9.32.2 Function Documentation	257
9.33 cFE Access Table Content APIs	262
9.33.1 Detailed Description	263
9.33.2 Function Documentation	263
9.34 cFE Get Table Information APIs	267
9.34.1 Detailed Description	267
9.34.2 Function Documentation	267
9.35 cFE Table Type Defines	270
9.35.1 Detailed Description	270
9.35.2 Macro Definition Documentation	270
9.36 cFE Get Current Time APIs	272
9.36.1 Detailed Description	272
9.36.2 Function Documentation	272
9.37 cFE Get Time Information APIs	275
9.37.1 Detailed Description	275
9.37.2 Function Documentation	275
9.38 cFE Time Arithmetic APIs	277
9.38.1 Detailed Description	277
9.38.2 Function Documentation	277
9.39 cFE Time Conversion APIs	279
9.39.1 Detailed Description	280
9.39.2 Function Documentation	280
9.40 cFE External Time Source APIs	281
9.40.1 Detailed Description	282
9.40.2 Function Documentation	282
9.41 cFE Miscellaneous Time APIs	285
9.41.1 Detailed Description	286

9.41.2 Function Documentation	286
9.42 cFE Resource ID base values	287
9.42.1 Detailed Description	288
9.42.2 Enumeration Type Documentation	288
9.43 cFE Clock State Flag Defines	289
9.43.1 Detailed Description	289
9.43.2 Macro Definition Documentation	289
9.44 OSAL Semaphore State Defines	291
9.44.1 Detailed Description	291
9.44.2 Macro Definition Documentation	291
9.45 OSAL Binary Semaphore APIs	291
9.45.1 Detailed Description	292
9.45.2 Function Documentation	292
9.46 OSAL BSP low level access APIs	296
9.46.1 Detailed Description	296
9.46.2 Function Documentation	296
9.47 OSAL Real Time Clock APIs	297
9.47.1 Detailed Description	298
9.47.2 Function Documentation	298
9.48 OSAL Core Operation APIs	312
9.48.1 Detailed Description	313
9.48.2 Function Documentation	313
9.49 OSAL Condition Variable APIs	315
9.49.1 Detailed Description	316
9.49.2 Function Documentation	316
9.50 OSAL Counting Semaphore APIs	322
9.50.1 Detailed Description	322
9.50.2 Function Documentation	322
9.51 OSAL Directory APIs	326
9.51.1 Detailed Description	326
9.51.2 Function Documentation	326
9.52 OSAL Return Code Defines	329
9.52.1 Detailed Description	331
9.52.2 Macro Definition Documentation	331
9.53 OSAL Error Info APIs	336
9.53.1 Detailed Description	336
9.53.2 Function Documentation	336
9.54 OSAL File Access Option Defines	339
9.54.1 Detailed Description	339

9.54.2 Macro Definition Documentation	339
9.55 OSAL Reference Point For Seek Offset Defines	339
9.55.1 Detailed Description	339
9.55.2 Macro Definition Documentation	339
9.56 OSAL Standard File APIs	340
9.56.1 Detailed Description	341
9.56.2 Function Documentation	341
9.57 OSAL File System Level APIs	353
9.57.1 Detailed Description	354
9.57.2 Function Documentation	354
9.58 OSAL Heap APIs	360
9.58.1 Detailed Description	361
9.58.2 Function Documentation	361
9.59 OSAL Object Type Defines	361
9.59.1 Detailed Description	362
9.59.2 Macro Definition Documentation	362
9.60 OSAL Object ID Utility APIs	364
9.60.1 Detailed Description	364
9.60.2 Function Documentation	364
9.61 OSAL Dynamic Loader and Symbol APIs	369
9.61.1 Detailed Description	369
9.61.2 Function Documentation	369
9.62 OSAL Mutex APIs	372
9.62.1 Detailed Description	373
9.62.2 Function Documentation	373
9.63 OSAL Network ID APIs	376
9.63.1 Detailed Description	376
9.63.2 Function Documentation	376
9.64 OSAL Printf APIs	377
9.64.1 Detailed Description	377
9.64.2 Function Documentation	377
9.65 OSAL Message Queue APIs	378
9.65.1 Detailed Description	378
9.65.2 Function Documentation	378
9.66 OSAL RwLock APIs	382
9.66.1 Detailed Description	382
9.66.2 Function Documentation	382
9.67 OSAL Select APIs	386
9.67.1 Detailed Description	387

9.67.2 Function Documentation	387
9.68 OSAL Shell APIs	392
9.68.1 Detailed Description	393
9.68.2 Function Documentation	393
9.69 OSAL Socket Address APIs	393
9.69.1 Detailed Description	393
9.69.2 Function Documentation	394
9.70 OSAL Socket Management APIs	396
9.70.1 Detailed Description	397
9.70.2 Function Documentation	397
9.71 OSAL Task APIs	407
9.71.1 Detailed Description	408
9.71.2 Function Documentation	408
9.72 OSAL Time Base APIs	413
9.72.1 Detailed Description	413
9.72.2 Function Documentation	413
9.73 OSAL Timer APIs	417
9.73.1 Detailed Description	417
9.73.2 Function Documentation	417
10 Data Structure Documentation	422
10.1 CCSDS_ExtendedHeader Struct Reference	422
10.1.1 Detailed Description	423
10.1.2 Field Documentation	423
10.2 CCSDS_PrimaryHeader Struct Reference	423
10.2.1 Detailed Description	423
10.2.2 Field Documentation	423
10.3 CFE_Config_ArrayValue Struct Reference	424
10.3.1 Detailed Description	424
10.3.2 Field Documentation	424
10.4 CFE_Config_IdNameEntry Struct Reference	424
10.4.1 Detailed Description	424
10.4.2 Field Documentation	425
10.5 CFE_Config_ValueBuffer Union Reference	425
10.5.1 Detailed Description	425
10.5.2 Field Documentation	425
10.6 CFE_Config_ValueEntry Struct Reference	425
10.6.1 Detailed Description	425
10.6.2 Field Documentation	425

10.7 CFE_ES_AppInfo Struct Reference	426
10.7.1 Detailed Description	427
10.7.2 Field Documentation	427
10.8 CFE_ES_AppNameCmd_Payload Struct Reference	430
10.8.1 Detailed Description	430
10.8.2 Field Documentation	431
10.9 CFE_ES_AppReloadCmd_Payload Struct Reference	431
10.9.1 Detailed Description	431
10.9.2 Field Documentation	431
10.10 CFE_ES_BlockStats Struct Reference	431
10.10.1 Detailed Description	432
10.10.2 Field Documentation	432
10.11 CFE_ES_CDSRegDumpRec Struct Reference	432
10.11.1 Detailed Description	433
10.11.2 Field Documentation	433
10.12 CFE_ES_ClearERLogCmd Struct Reference	433
10.12.1 Detailed Description	434
10.12.2 Field Documentation	434
10.13 CFE_ES_ClearSysLogCmd Struct Reference	434
10.13.1 Detailed Description	434
10.13.2 Field Documentation	434
10.14 CFE_ES_DeleteCDSCmd Struct Reference	434
10.14.1 Detailed Description	434
10.14.2 Field Documentation	435
10.15 CFE_ES_DeleteCDSCmd_Payload Struct Reference	435
10.15.1 Detailed Description	435
10.15.2 Field Documentation	435
10.16 CFE_ES_DumpCDSRegistryCmd Struct Reference	435
10.16.1 Detailed Description	436
10.16.2 Field Documentation	436
10.17 CFE_ES_DumpCDSRegistryCmd_Payload Struct Reference	436
10.17.1 Detailed Description	436
10.17.2 Field Documentation	436
10.18 CFE_ES_FileNameCmd Struct Reference	436
10.18.1 Detailed Description	437
10.18.2 Field Documentation	437
10.19 CFE_ES_FileNameCmd_Payload Struct Reference	437
10.19.1 Detailed Description	437
10.19.2 Field Documentation	437

10.20 CFE_ES_HousekeepingTlm Struct Reference	438
10.20.1 Detailed Description	438
10.20.2 Field Documentation	438
10.21 CFE_ES_HousekeepingTlm_Payload Struct Reference	438
10.21.1 Detailed Description	440
10.21.2 Field Documentation	440
10.22 CFE_ES_MemAddress_t Struct Reference	446
10.22.1 Detailed Description	447
10.22.2 Field Documentation	447
10.23 CFE_ES_MemOffset_t Struct Reference	447
10.23.1 Detailed Description	447
10.23.2 Field Documentation	447
10.24 CFE_ES_MemPoolStats Struct Reference	448
10.24.1 Detailed Description	448
10.24.2 Field Documentation	448
10.25 CFE_ES_MemStatsTlm Struct Reference	449
10.25.1 Detailed Description	449
10.25.2 Field Documentation	449
10.26 CFE_ES_NoopCmd Struct Reference	449
10.26.1 Detailed Description	450
10.26.2 Field Documentation	450
10.27 CFE_ES_OneAppTlm Struct Reference	450
10.27.1 Detailed Description	450
10.27.2 Field Documentation	450
10.28 CFE_ES_OneAppTlm_Payload Struct Reference	450
10.28.1 Detailed Description	451
10.28.2 Field Documentation	451
10.29 CFE_ES_OverWriteSysLogCmd Struct Reference	451
10.29.1 Detailed Description	451
10.29.2 Field Documentation	451
10.30 CFE_ES_OverWriteSysLogCmd_Payload Struct Reference	452
10.30.1 Detailed Description	452
10.30.2 Field Documentation	452
10.31 CFE_ES_PoolAlign Union Reference	452
10.31.1 Detailed Description	452
10.31.2 Field Documentation	453
10.32 CFE_ES_PoolStatsTlm_Payload Struct Reference	453
10.32.1 Detailed Description	453
10.32.2 Field Documentation	453

10.33 CFE_ES_QueryAllCmd Struct Reference	454
10.33.1 Detailed Description	454
10.33.2 Field Documentation	454
10.34 CFE_ES_QueryAllTasksCmd Struct Reference	454
10.34.1 Detailed Description	454
10.34.2 Field Documentation	455
10.35 CFE_ES_QueryOneCmd Struct Reference	455
10.35.1 Detailed Description	455
10.35.2 Field Documentation	455
10.36 CFE_ES_ReloadAppCmd Struct Reference	455
10.36.1 Detailed Description	456
10.36.2 Field Documentation	456
10.37 CFE_ES_ResetCountersCmd Struct Reference	456
10.37.1 Detailed Description	456
10.37.2 Field Documentation	456
10.38 CFE_ES_ResetPRCountCmd Struct Reference	456
10.38.1 Detailed Description	457
10.38.2 Field Documentation	457
10.39 CFE_ES_RestartAppCmd Struct Reference	457
10.39.1 Detailed Description	457
10.39.2 Field Documentation	457
10.40 CFE_ES_RestartCmd Struct Reference	457
10.40.1 Detailed Description	458
10.40.2 Field Documentation	458
10.41 CFE_ES_RestartCmd_Payload Struct Reference	458
10.41.1 Detailed Description	458
10.41.2 Field Documentation	458
10.42 CFE_ES_SendHkCmd Struct Reference	459
10.42.1 Detailed Description	459
10.42.2 Field Documentation	459
10.43 CFE_ES_SendMemPoolStatsCmd Struct Reference	459
10.43.1 Detailed Description	459
10.43.2 Field Documentation	459
10.44 CFE_ES_SendMemPoolStatsCmd_Payload Struct Reference	460
10.44.1 Detailed Description	460
10.44.2 Field Documentation	460
10.45 CFE_ES_SetMaxPRCountCmd Struct Reference	460
10.45.1 Detailed Description	460
10.45.2 Field Documentation	461

10.46 CFE_ES_SetMaxPRCountCmd_Payload Struct Reference	461
10.46.1 Detailed Description	461
10.46.2 Field Documentation	461
10.47 CFE_ES_SetPerfFilterMaskCmd Struct Reference	461
10.47.1 Detailed Description	462
10.47.2 Field Documentation	462
10.48 CFE_ES_SetPerfFilterMaskCmd_Payload Struct Reference	462
10.48.1 Detailed Description	462
10.48.2 Field Documentation	462
10.49 CFE_ES_SetPerfTriggerMaskCmd Struct Reference	463
10.49.1 Detailed Description	463
10.49.2 Field Documentation	463
10.50 CFE_ES_SetPerfTrigMaskCmd_Payload Struct Reference	463
10.50.1 Detailed Description	463
10.50.2 Field Documentation	464
10.51 CFE_ES_StartApp Struct Reference	464
10.51.1 Detailed Description	464
10.51.2 Field Documentation	464
10.52 CFE_ES_StartAppCmd_Payload Struct Reference	464
10.52.1 Detailed Description	465
10.52.2 Field Documentation	465
10.53 CFE_ES_StartPerfCmd_Payload Struct Reference	466
10.53.1 Detailed Description	466
10.53.2 Field Documentation	466
10.54 CFE_ES_StartPerfDataCmd Struct Reference	466
10.54.1 Detailed Description	467
10.54.2 Field Documentation	467
10.55 CFE_ES_StopAppCmd Struct Reference	467
10.55.1 Detailed Description	467
10.55.2 Field Documentation	467
10.56 CFE_ES_StopPerfCmd_Payload Struct Reference	468
10.56.1 Detailed Description	468
10.56.2 Field Documentation	468
10.57 CFE_ES_StopPerfDataCmd Struct Reference	468
10.57.1 Detailed Description	468
10.57.2 Field Documentation	469
10.58 CFE_ES_TaskInfo Struct Reference	469
10.58.1 Detailed Description	469
10.58.2 Field Documentation	470

10.59 CFE_ES_WriteERLogCmd Struct Reference	. 470
10.59.1 Detailed Description	. 471
10.59.2 Field Documentation	. 471
10.60 CFE_ES_WriteSysLogCmd Struct Reference	. 471
10.60.1 Detailed Description	. 471
10.60.2 Field Documentation	. 471
10.61 CFE_EVS_AddEventFilterCmd Struct Reference	. 472
10.61.1 Detailed Description	. 472
10.61.2 Field Documentation	. 472
10.62 CFE_EVS_AppDataCmd_Payload Struct Reference	. 472
10.62.1 Detailed Description	. 472
10.62.2 Field Documentation	. 473
10.63 CFE_EVS_AppNameBitMaskCmd_Payload Struct Reference	. 473
10.63.1 Detailed Description	. 473
10.63.2 Field Documentation	. 473
10.64 CFE_EVS_AppNameCmd_Payload Struct Reference	. 474
10.64.1 Detailed Description	. 474
10.64.2 Field Documentation	. 474
10.65 CFE_EVS_AppNameEventIDCmd_Payload Struct Reference	. 474
10.65.1 Detailed Description	. 474
10.65.2 Field Documentation	. 474
10.66 CFE_EVS_AppNameEventIDMaskCmd_Payload Struct Reference	. 475
10.66.1 Detailed Description	. 475
10.66.2 Field Documentation	. 475
10.67 CFE_EVS_AppTlmData Struct Reference	. 475
10.67.1 Detailed Description	. 476
10.67.2 Field Documentation	. 476
10.68 CFE_EVS_BinFilter Struct Reference	. 477
10.68.1 Detailed Description	. 477
10.68.2 Field Documentation	. 477
10.69 CFE_EVS_BitMaskCmd_Payload Struct Reference	. 477
10.69.1 Detailed Description	. 477
10.69.2 Field Documentation	. 478
10.70 CFE_EVS_ClearLogCmd Struct Reference	. 478
10.70.1 Detailed Description	. 478
10.70.2 Field Documentation	. 478
10.71 CFE_EVS_DeleteEventFilterCmd Struct Reference	. 478
10.71.1 Detailed Description	. 478
10.71.2 Field Documentation	. 479

10.72 CFE_EVS_DisableAppEventsCmd Struct Reference	479
10.72.1 Detailed Description	479
10.72.2 Field Documentation	479
10.73 CFE_EVS_DisableAppEventTypeCmd Struct Reference	479
10.73.1 Detailed Description	480
10.73.2 Field Documentation	480
10.74 CFE_EVS_DisableEventTypeCmd Struct Reference	480
10.74.1 Detailed Description	480
10.74.2 Field Documentation	480
10.75 CFE_EVS_DisablePortsCmd Struct Reference	481
10.75.1 Detailed Description	481
10.75.2 Field Documentation	481
10.76 CFE_EVS_EnableAppEventsCmd Struct Reference	481
10.76.1 Detailed Description	481
10.76.2 Field Documentation	482
10.77 CFE_EVS_EnableAppEventTypeCmd Struct Reference	482
10.77.1 Detailed Description	482
10.77.2 Field Documentation	482
10.78 CFE_EVS_EnableEventTypeCmd Struct Reference	482
10.78.1 Detailed Description	483
10.78.2 Field Documentation	483
10.79 CFE_EVS_EnablePortsCmd Struct Reference	483
10.79.1 Detailed Description	483
10.79.2 Field Documentation	483
10.80 CFE_EVS_HousekeepingTlm Struct Reference	484
10.80.1 Detailed Description	484
10.80.2 Field Documentation	484
10.81 CFE_EVS_HousekeepingTlm_Payload Struct Reference	484
10.81.1 Detailed Description	485
10.81.2 Field Documentation	485
10.82 CFE_EVS_LogFileCmd_Payload Struct Reference	487
10.82.1 Detailed Description	487
10.82.2 Field Documentation	487
10.83 CFE_EVS_LongEventTlm Struct Reference	487
10.83.1 Detailed Description	488
10.83.2 Field Documentation	488
10.84 CFE_EVS_LongEventTlm_Payload Struct Reference	488
10.84.1 Detailed Description	488
10.84.2 Field Documentation	488

10.85 CFE_EVS_NoopCmd Struct Reference	489
10.85.1 Detailed Description	489
10.85.2 Field Documentation	489
10.86 CFE_EVS_PacketID Struct Reference	489
10.86.1 Detailed Description	490
10.86.2 Field Documentation	490
10.87 CFE_EVS_ResetAllFiltersCmd Struct Reference	490
10.87.1 Detailed Description	491
10.87.2 Field Documentation	491
10.88 CFE_EVS_ResetAppCounterCmd Struct Reference	491
10.88.1 Detailed Description	491
10.88.2 Field Documentation	491
10.89 CFE_EVS_ResetCountersCmd Struct Reference	492
10.89.1 Detailed Description	492
10.89.2 Field Documentation	492
10.90 CFE_EVS_ResetFilterCmd Struct Reference	492
10.90.1 Detailed Description	492
10.90.2 Field Documentation	492
10.91 CFE_EVS_SendHkCmd Struct Reference	493
10.91.1 Detailed Description	493
10.91.2 Field Documentation	493
10.92 CFE_EVS_SetEventFormatCode_Payload Struct Reference	493
10.92.1 Detailed Description	493
10.92.2 Field Documentation	493
10.93 CFE_EVS_SetEventFormatModeCmd Struct Reference	494
10.93.1 Detailed Description	494
10.93.2 Field Documentation	494
10.94 CFE_EVS_SetFilterCmd Struct Reference	494
10.94.1 Detailed Description	495
10.94.2 Field Documentation	495
10.95 CFE_EVS_SetLogMode_Payload Struct Reference	495
10.95.1 Detailed Description	495
10.95.2 Field Documentation	495
10.96 CFE_EVS_SetLogModeCmd Struct Reference	496
10.96.1 Detailed Description	496
10.96.2 Field Documentation	496
10.97 CFE_EVS_ShortEventTlm Struct Reference	496
10.97.1 Detailed Description	496
10.97.2 Field Documentation	496

10.98 CFE_EVS_ShortEventTlm_Payload Struct Reference	497
10.98.1 Detailed Description	497
10.98.2 Field Documentation	497
10.99 CFE_EVS_WriteAppDataFileCmd Struct Reference	497
10.99.1 Detailed Description	497
10.99.2 Field Documentation	498
10.100 CFE_EVS_WriteLogDataFileCmd Struct Reference	498
10.100.1 Detailed Description	498
10.100.2 Field Documentation	498
10.101 CFE_FS_FileWriteMetaData Struct Reference	498
10.101.1 Detailed Description	499
10.101.2 Field Documentation	499
10.102 CFE_FS_Header Struct Reference	500
10.102.1 Detailed Description	500
10.102.2 Field Documentation	500
10.103 CFE_SB_AllSubscriptionsTlm Struct Reference	501
10.103.1 Detailed Description	501
10.103.2 Field Documentation	502
10.104 CFE_SB_AllSubscriptionsTlm_Payload Struct Reference	502
10.104.1 Detailed Description	502
10.104.2 Field Documentation	502
10.105 CFE_SB_DisableRouteCmd Struct Reference	503
10.105.1 Detailed Description	503
10.105.2 Field Documentation	503
10.106 CFE_SB_DisableSubReportingCmd Struct Reference	503
10.106.1 Detailed Description	504
10.106.2 Field Documentation	504
10.107 CFE_SB_EnableRouteCmd Struct Reference	504
10.107.1 Detailed Description	504
10.107.2 Field Documentation	504
10.108 CFE_SB_EnableSubReportingCmd Struct Reference	504
10.108.1 Detailed Description	504
10.108.2 Field Documentation	505
10.109 CFE_SB_HousekeepingTlm Struct Reference	505
10.109.1 Detailed Description	505
10.109.2 Field Documentation	505
10.110 CFE_SB_HousekeepingTlm_Payload Struct Reference	505
10.110.1 Detailed Description	506
10.110.2 Field Documentation	506

10.111 CFE_SB_Msg Union Reference	509
10.111.1 Detailed Description	509
10.111.2 Field Documentation	509
10.112 CFE_SB_MsgId_t Struct Reference	510
10.112.1 Detailed Description	510
10.112.2 Field Documentation	510
10.113 CFE_SB_MsgMapFileEntry Struct Reference	510
10.113.1 Detailed Description	510
10.113.2 Field Documentation	510
10.114 CFE_SB_NoopCmd Struct Reference	511
10.114.1 Detailed Description	511
10.114.2 Field Documentation	511
10.115 CFE_SB_PipeDepthStats Struct Reference	511
10.115.1 Detailed Description	511
10.115.2 Field Documentation	512
10.116 CFE_SB_PipeInfoEntry Struct Reference	512
10.116.1 Detailed Description	513
10.116.2 Field Documentation	513
10.117 CFE_SB_Qos_t Struct Reference	514
10.117.1 Detailed Description	514
10.117.2 Field Documentation	515
10.118 CFE_SB_ResetCountersCmd Struct Reference	515
10.118.1 Detailed Description	515
10.118.2 Field Documentation	515
10.119 CFE_SB_RouteCmd_Payload Struct Reference	515
10.119.1 Detailed Description	516
10.119.2 Field Documentation	516
10.120 CFE_SB_RoutingFileEntry Struct Reference	516
10.120.1 Detailed Description	516
10.120.2 Field Documentation	517
10.121 CFE_SB_SendHkCmd Struct Reference	517
10.121.1 Detailed Description	517
10.121.2 Field Documentation	518
10.122 CFE_SB_SendPrevSubsCmd Struct Reference	518
10.122.1 Detailed Description	518
10.122.2 Field Documentation	518
10.123 CFE_SB_SendSbStatsCmd Struct Reference	518
10.123.1 Detailed Description	518
10.123.2 Field Documentation	518

10.124 CFE_SB_SingleSubscriptionTlm Struct Reference	518
10.124.1 Detailed Description	519
10.124.2 Field Documentation	519
10.125 CFE_SB_SingleSubscriptionTlm_Payload Struct Reference	519
10.125.1 Detailed Description	519
10.125.2 Field Documentation	520
10.126 CFE_SB_StatsTlm Struct Reference	520
10.126.1 Detailed Description	520
10.126.2 Field Documentation	520
10.127 CFE_SB_StatsTlm_Payload Struct Reference	521
10.127.1 Detailed Description	522
10.127.2 Field Documentation	522
10.128 CFE_SB_SubEntries Struct Reference	524
10.128.1 Detailed Description	524
10.128.2 Field Documentation	524
10.129 CFE_SB_WriteFileInfoCmd_Payload Struct Reference	525
10.129.1 Detailed Description	525
10.129.2 Field Documentation	525
10.130 CFE_SB_WriteMapInfoCmd Struct Reference	525
10.130.1 Detailed Description	525
10.130.2 Field Documentation	526
10.131 CFE_SB_WritePipeInfoCmd Struct Reference	526
10.131.1 Detailed Description	526
10.131.2 Field Documentation	526
10.132 CFE_SB_WriteRoutingInfoCmd Struct Reference	526
10.132.1 Detailed Description	527
10.132.2 Field Documentation	527
10.133 CFE_TBL_AbortLoadCmd Struct Reference	527
10.133.1 Detailed Description	527
10.133.2 Field Documentation	527
10.134 CFE_TBL_AbortLoadCmd_Payload Struct Reference	528
10.134.1 Detailed Description	528
10.134.2 Field Documentation	528
10.135 CFE_TBL_ActivateCmd Struct Reference	528
10.135.1 Detailed Description	528
10.135.2 Field Documentation	529
10.136 CFE_TBL_ActivateCmd_Payload Struct Reference	529
10.136.1 Detailed Description	529
10.136.2 Field Documentation	529

10.137 CFE_TBL_CombinedFileHdr Struct Reference	529
10.137.1 Detailed Description	530
10.137.2 Field Documentation	530
10.138 CFE_TBL_DelCDSCmd_Payload Struct Reference	530
10.138.1 Detailed Description	530
10.138.2 Field Documentation	530
10.139 CFE_TBL_DeleteCDSCmd Struct Reference	530
10.139.1 Detailed Description	531
10.139.2 Field Documentation	531
10.140 CFE_TBL_DumpCmd Struct Reference	531
10.140.1 Detailed Description	531
10.140.2 Field Documentation	531
10.141 CFE_TBL_DumpCmd_Payload Struct Reference	532
10.141.1 Detailed Description	532
10.141.2 Field Documentation	532
10.142 CFE_TBL_DumpRegistryCmd Struct Reference	533
10.142.1 Detailed Description	533
10.142.2 Field Documentation	533
10.143 CFE_TBL_DumpRegistryCmd_Payload Struct Reference	533
10.143.1 Detailed Description	533
10.143.2 Field Documentation	533
10.144 CFE_TBL_File_Hdr Struct Reference	534
10.144.1 Detailed Description	534
10.144.2 Field Documentation	534
10.145 CFE_TBL_FileDef Struct Reference	535
10.145.1 Detailed Description	535
10.145.2 Field Documentation	535
10.146 CFE_TBL_HousekeepingTlm Struct Reference	536
10.146.1 Detailed Description	536
10.146.2 Field Documentation	536
10.147 CFE_TBL_HousekeepingTlm_Payload Struct Reference	537
10.147.1 Detailed Description	538
10.147.2 Field Documentation	538
10.148 CFE_TBL_Info Struct Reference	541
10.148.1 Detailed Description	541
10.148.2 Field Documentation	541
10.149 CFE_TBL_LoadCmd Struct Reference	543
10.149.1 Detailed Description	543
10.149.2 Field Documentation	543

10.150 CFE_TBL_LoadCmd_Payload Struct Reference	543
10.150.1 Detailed Description	544
10.150.2 Field Documentation	544
10.151 CFE_TBL_NoopCmd Struct Reference	544
10.151.1 Detailed Description	544
10.151.2 Field Documentation	544
10.152 CFE_TBL_NotifyCmd Struct Reference	544
10.152.1 Detailed Description	544
10.152.2 Field Documentation	545
10.153 CFE_TBL_NotifyCmd_Payload Struct Reference	545
10.153.1 Detailed Description	545
10.153.2 Field Documentation	545
10.154 CFE_TBL_ResetCountersCmd Struct Reference	545
10.154.1 Detailed Description	546
10.154.2 Field Documentation	546
10.155 CFE_TBL_SendHkCmd Struct Reference	546
10.155.1 Detailed Description	546
10.155.2 Field Documentation	546
10.156 CFE_TBL_SendRegistryCmd Struct Reference	546
10.156.1 Detailed Description	546
10.156.2 Field Documentation	547
10.157 CFE_TBL_SendRegistryCmd_Payload Struct Reference	547
10.157.1 Detailed Description	547
10.157.2 Field Documentation	547
10.158 CFE_TBL_TableRegistryTlm Struct Reference	547
10.158.1 Detailed Description	548
10.158.2 Field Documentation	548
10.159 CFE_TBL_TblRegPacket_Payload Struct Reference	548
10.159.1 Detailed Description	549
10.159.2 Field Documentation	549
10.160 CFE_TBL_ValidateCmd Struct Reference	551
10.160.1 Detailed Description	552
10.160.2 Field Documentation	552
10.161 CFE_TBL_ValidateCmd_Payload Struct Reference	552
10.161.1 Detailed Description	552
10.161.2 Field Documentation	552
10.162 CFE_TIME_AddAdjustCmd Struct Reference	553
10.162.1 Detailed Description	553
10.162.2 Field Documentation	553

10.163 CFE_TIME_AddDelayCmd Struct Reference	553
10.163.1 Detailed Description	553
10.163.2 Field Documentation	553
10.164 CFE_TIME_AddOneHzAdjustmentCmd Struct Reference	554
10.164.1 Detailed Description	554
10.164.2 Field Documentation	554
10.165 CFE_TIME_DiagnosticTlm Struct Reference	554
10.165.1 Detailed Description	554
10.165.2 Field Documentation	555
10.166 CFE_TIME_DiagnosticTlm_Payload Struct Reference	555
10.166.1 Detailed Description	557
10.166.2 Field Documentation	557
10.167 CFE_TIME_FakeToneCmd Struct Reference	563
10.167.1 Detailed Description	564
10.167.2 Field Documentation	564
10.168 CFE_TIME_HousekeepingTlm Struct Reference	564
10.168.1 Detailed Description	564
10.168.2 Field Documentation	564
10.169 CFE_TIME_HousekeepingTlm_Payload Struct Reference	564
10.169.1 Detailed Description	565
10.169.2 Field Documentation	565
10.170 CFE_TIME_LeapsCmd_Payload Struct Reference	566
10.170.1 Detailed Description	567
10.170.2 Field Documentation	567
10.171 CFE_TIME_NoopCmd Struct Reference	567
10.171.1 Detailed Description	567
10.171.2 Field Documentation	567
10.172 CFE_TIME_OneHzAdjustmentCmd_Payload Struct Reference	567
10.172.1 Detailed Description	567
10.172.2 Field Documentation	568
10.173 CFE_TIME_OneHzCmd Struct Reference	568
10.173.1 Detailed Description	568
10.173.2 Field Documentation	568
10.174 CFE_TIME_ResetCountersCmd Struct Reference	568
10.174.1 Detailed Description	568
10.174.2 Field Documentation	569
10.175 CFE_TIME_SendDiagnosticCmd Struct Reference	569
10.175.1 Detailed Description	569
10.175.2 Field Documentation	569

10.176 CFE_TIME_SendHkCmd Struct Reference	569
10.176.1 Detailed Description	569
10.176.2 Field Documentation	569
10.177 CFE_TIME_SetLeapSecondsCmd Struct Reference	570
10.177.1 Detailed Description	570
10.177.2 Field Documentation	570
10.178 CFE_TIME_SetMETCmd Struct Reference	570
10.178.1 Detailed Description	570
10.178.2 Field Documentation	570
10.179 CFE_TIME_SetSignalCmd Struct Reference	571
10.179.1 Detailed Description	571
10.179.2 Field Documentation	571
10.180 CFE_TIME_SetSourceCmd Struct Reference	571
10.180.1 Detailed Description	572
10.180.2 Field Documentation	572
10.181 CFE_TIME_SetStateCmd Struct Reference	572
10.181.1 Detailed Description	572
10.181.2 Field Documentation	572
10.182 CFE_TIME_SetSTCFCmd Struct Reference	573
10.182.1 Detailed Description	573
10.182.2 Field Documentation	573
10.183 CFE_TIME_SetTimeCmd Struct Reference	573
10.183.1 Detailed Description	573
10.183.2 Field Documentation	573
10.184 CFE_TIME_SignalCmd_Payload Struct Reference	574
10.184.1 Detailed Description	574
10.184.2 Field Documentation	574
10.185 CFE_TIME_SourceCmd_Payload Struct Reference	574
10.185.1 Detailed Description	574
10.185.2 Field Documentation	575
10.186 CFE_TIME_StateCmd_Payload Struct Reference	575
10.186.1 Detailed Description	575
10.186.2 Field Documentation	575
10.187 CFE_TIME_SubAdjustCmd Struct Reference	575
10.187.1 Detailed Description	576
10.187.2 Field Documentation	576
10.188 CFE_TIME_SubDelayCmd Struct Reference	576
10.188.1 Detailed Description	576
10.188.2 Field Documentation	576

10.189 CFE_TIME_SubOneHzAdjustmentCmd Struct Reference	576
10.189.1 Detailed Description	577
10.189.2 Field Documentation	577
10.190 CFE_TIME_SysTime Struct Reference	577
10.190.1 Detailed Description	577
10.190.2 Field Documentation	578
10.191 CFE_TIME_TimeCmd_Payload Struct Reference	578
10.191.1 Detailed Description	578
10.191.2 Field Documentation	578
10.192 CFE_TIME_ToneDataCmd Struct Reference	578
10.192.1 Detailed Description	579
10.192.2 Field Documentation	579
10.193 CFE_TIME_ToneDataCmd_Payload Struct Reference	579
10.193.1 Detailed Description	579
10.193.2 Field Documentation	579
10.194 CFE_TIME_ToneSignalCmd Struct Reference	580
10.194.1 Detailed Description	580
10.194.2 Field Documentation	580
10.195 OS_bin_sem_prop_t Struct Reference	580
10.195.1 Detailed Description	580
10.195.2 Field Documentation	581
10.196 OS_condvar_prop_t Struct Reference	581
10.196.1 Detailed Description	581
10.196.2 Field Documentation	581
10.197 OS_count_sem_prop_t Struct Reference	581
10.197.1 Detailed Description	582
10.197.2 Field Documentation	582
10.198 os_dirent_t Struct Reference	582
10.198.1 Detailed Description	582
10.198.2 Field Documentation	582
10.199 OS_FdSet Struct Reference	583
10.199.1 Detailed Description	583
10.199.2 Field Documentation	583
10.200 OS_file_prop_t Struct Reference	583
10.200.1 Detailed Description	583
10.200.2 Field Documentation	583
10.201 os_fsinfo_t Struct Reference	584
10.201.1 Detailed Description	584
10.201.2 Field Documentation	584

10.202 <code>os_fstat_t</code> Struct Reference	585
10.202.1 Detailed Description	585
10.202.2 Field Documentation	585
10.203 <code>OS_heap_prop_t</code> Struct Reference	585
10.203.1 Detailed Description	586
10.203.2 Field Documentation	586
10.204 <code>OS_module_address_t</code> Struct Reference	586
10.204.1 Detailed Description	586
10.204.2 Field Documentation	587
10.205 <code>OS_module_prop_t</code> Struct Reference	587
10.205.1 Detailed Description	588
10.205.2 Field Documentation	588
10.206 <code>OS_mut_sem_prop_t</code> Struct Reference	588
10.206.1 Detailed Description	588
10.206.2 Field Documentation	588
10.207 <code>OS_queue_prop_t</code> Struct Reference	589
10.207.1 Detailed Description	589
10.207.2 Field Documentation	589
10.208 <code>OS_rwlock_prop_t</code> Struct Reference	589
10.208.1 Detailed Description	589
10.208.2 Field Documentation	590
10.209 <code>OS_SockAddr_t</code> Struct Reference	590
10.209.1 Detailed Description	590
10.209.2 Field Documentation	590
10.210 <code>OS_SockAddrData_t</code> Union Reference	590
10.210.1 Detailed Description	591
10.210.2 Field Documentation	591
10.211 <code>OS_socket_optval</code> Union Reference	591
10.211.1 Detailed Description	591
10.211.2 Field Documentation	592
10.212 <code>OS_socket_prop_t</code> Struct Reference	592
10.212.1 Detailed Description	592
10.212.2 Field Documentation	592
10.213 <code>OS_static_symbol_record_t</code> Struct Reference	592
10.213.1 Detailed Description	593
10.213.2 Field Documentation	593
10.214 <code>OS_statvfs_t</code> Struct Reference	593
10.214.1 Detailed Description	593
10.214.2 Field Documentation	593

10.215 OS_task_prop_t Struct Reference	594
10.215.1 Detailed Description	594
10.215.2 Field Documentation	594
10.216 OS_time_t Struct Reference	594
10.216.1 Detailed Description	595
10.216.2 Field Documentation	595
10.217 OS_timebase_prop_t Struct Reference	595
10.217.1 Detailed Description	595
10.217.2 Field Documentation	595
10.218 OS_timer_prop_t Struct Reference	596
10.218.1 Detailed Description	596
10.218.2 Field Documentation	596
11 File Documentation	597
11.1 build/osal_public_api/inc/osconfig.h File Reference	597
11.1.1 Macro Definition Documentation	598
11.2 osconfig.h	604
11.3 cfe_perfids.h File Reference	605
11.3.1 Detailed Description	605
11.3.2 Macro Definition Documentation	606
11.4 cfe_perfids.h	607
11.5 example_2x32bit_cfe_es_memaddress.h File Reference	608
11.5.1 Detailed Description	608
11.5.2 Macro Definition Documentation	608
11.5.3 Function Documentation	609
11.6 example_2x32bit_cfe_es_memaddress.h	610
11.7 example_32bit_cfe_es_memaddress.h File Reference	611
11.7.1 Detailed Description	611
11.7.2 Macro Definition Documentation	611
11.7.3 Typedef Documentation	612
11.8 example_32bit_cfe_es_memaddress.h	612
11.9 example_64bit_cfe_es_memaddress.h File Reference	613
11.9.1 Detailed Description	613
11.9.2 Macro Definition Documentation	614
11.9.3 Typedef Documentation	614
11.10 example_64bit_cfe_es_memaddress.h	614
11.11 example_mission_cfg.h File Reference	615
11.11.1 Detailed Description	616
11.11.2 Macro Definition Documentation	616

11.12 example_mission_cfg.h	626
11.13 example_platform_cfg.h File Reference	628
11.13.1 Detailed Description	632
11.13.2 Macro Definition Documentation	632
11.14 example_platform_cfg.h	672
11.15 cfe/docs/src/cfe_api.dox File Reference	676
11.16 cfe/docs/src/cfe_es.dox File Reference	676
11.17 cfe/docs/src/cfe_evs.dox File Reference	676
11.18 cfe/docs/src/cfe_frontpage.dox File Reference	676
11.19 cfe/docs/src/cfe_glossary.dox File Reference	676
11.20 cfe/docs/src/cfe_sb.dox File Reference	676
11.21 cfe/docs/src/cfe_tbl.dox File Reference	676
11.22 cfe/docs/src/cfe_time.dox File Reference	676
11.23 cfe/docs/src/cfe_xref.dox File Reference	676
11.24 cfe/docs/src/cfs_versions.dox File Reference	676
11.25 cfe/modules/config/fsw/inc/cfe_config_external.h File Reference	676
11.25.1 Detailed Description	676
11.25.2 Function Documentation	677
11.26 cfe_config_external.h	677
11.27 cfe/modules/config/fsw/inc/cfe_config_init.h File Reference	677
11.27.1 Detailed Description	677
11.27.2 Function Documentation	677
11.28 cfe_config_init.h	678
11.29 cfe/modules/config/fsw/inc/cfe_config_lookup.h File Reference	678
11.29.1 Detailed Description	678
11.29.2 Function Documentation	678
11.30 cfe_config_lookup.h	678
11.31 cfe/modules/config/fsw/inc/cfe_config_nametable.h File Reference	679
11.31.1 Detailed Description	679
11.31.2 Typedef Documentation	679
11.31.3 Variable Documentation	679
11.32 cfe_config_nametable.h	680
11.33 cfe/modules/config/fsw/inc/cfe_config_set.h File Reference	680
11.33.1 Detailed Description	680
11.33.2 Function Documentation	680
11.34 cfe_config_set.h	681
11.35 cfe/modules/config/fsw/inc/cfe_config_table.h File Reference	681
11.35.1 Detailed Description	682
11.35.2 Typedef Documentation	682

11.35.3 Enumeration Type Documentation	682
11.36 cfe_config_table.h	682
11.37 cfe/modules/core_api/config/default_cfe_core_api_base_msgid_values.h File Reference	683
11.37.1 Detailed Description	683
11.37.2 Macro Definition Documentation	683
11.38 default_cfe_core_api_base_msgid_values.h	684
11.39 cfe/modules/core_api/config/default_cfe_core_api_interface_cfg_values.h File Reference	684
11.39.1 Detailed Description	684
11.39.2 Macro Definition Documentation	684
11.40 default_cfe_core_api_interface_cfg_values.h	684
11.41 cfe/modules/core_api/config/default_cfe_core_api_msgid_mapping.h File Reference	685
11.41.1 Detailed Description	685
11.41.2 Macro Definition Documentation	685
11.42 default_cfe_core_api_msgid_mapping.h	687
11.43 cfe/modules/core_api/config/default_cfe_mission_cfg.h File Reference	687
11.43.1 Detailed Description	687
11.44 default_cfe_mission_cfg.h	687
11.45 cfe/modules/core_api/config/default_cfe_msgids.h File Reference	688
11.45.1 Detailed Description	688
11.46 default_cfe_msgids.h	688
11.47 cfe/modules/core_api/fsw/inc/cfe.h File Reference	689
11.47.1 Detailed Description	689
11.48 cfe.h	689
11.49 cfe/modules/core_api/fsw/inc/cfe_config.h File Reference	690
11.49.1 Detailed Description	690
11.49.2 Function Documentation	690
11.50 cfe_config.h	694
11.51 cfe/modules/core_api/fsw/inc/cfe_config_api_typedefs.h File Reference	694
11.51.1 Detailed Description	695
11.51.2 Macro Definition Documentation	695
11.51.3 Typedef Documentation	695
11.52 cfe_config_api_typedefs.h	695
11.53 cfe/modules/core_api/fsw/inc/cfe_core_api_base_msgids.h File Reference	696
11.53.1 Detailed Description	696
11.54 cfe_core_api_base_msgids.h	696
11.55 cfe/modules/core_api/fsw/inc/cfe_core_api_interface_cfg.h File Reference	697
11.55.1 Detailed Description	697
11.55.2 Macro Definition Documentation	697
11.56 cfe_core_api_interface_cfg.h	699

11.57 cfe/modules/core_api/fsw/inc/cfe_endian.h File Reference	700
11.57.1 Detailed Description	700
11.57.2 Macro Definition Documentation	700
11.58 cfe_endian.h	700
11.59 cfe/modules/core_api/fsw/inc/cfe_error.h File Reference	701
11.59.1 Detailed Description	708
11.59.2 Macro Definition Documentation	708
11.59.3 Typedef Documentation	709
11.59.4 Function Documentation	710
11.60 cfe_error.h	710
11.61 cfe/modules/core_api/fsw/inc/cfe_es.h File Reference	715
11.61.1 Detailed Description	718
11.61.2 Macro Definition Documentation	718
11.62 cfe_es.h	718
11.63 cfe/modules/core_api/fsw/inc/cfe_es_api_typedefs.h File Reference	721
11.63.1 Detailed Description	722
11.63.2 Macro Definition Documentation	722
11.63.3 Typedef Documentation	724
11.63.4 Enumeration Type Documentation	725
11.64 cfe_es_api_typedefs.h	726
11.65 cfe/modules/core_api/fsw/inc/cfe_evs.h File Reference	728
11.65.1 Detailed Description	728
11.65.2 Macro Definition Documentation	729
11.66 cfe_evs.h	729
11.67 cfe/modules/core_api/fsw/inc/cfe_evs_api_typedefs.h File Reference	730
11.67.1 Detailed Description	731
11.67.2 Macro Definition Documentation	731
11.67.3 Typedef Documentation	732
11.68 cfe_evs_api_typedefs.h	732
11.69 cfe/modules/core_api/fsw/inc/cfe_fs.h File Reference	733
11.69.1 Detailed Description	734
11.70 cfe_fs.h	734
11.71 cfe/modules/core_api/fsw/inc/cfe_fs_api_typedefs.h File Reference	735
11.71.1 Detailed Description	735
11.71.2 Typedef Documentation	736
11.71.3 Enumeration Type Documentation	737
11.72 cfe_fs_api_typedefs.h	737
11.73 cfe/modules/core_api/fsw/inc/cfe_msg.h File Reference	739
11.73.1 Detailed Description	741

11.74	cfe_msg.h	741
11.75	cfe/modules/core_api/fsw/inc/cfe_msg_api_typedefs.h File Reference	743
11.75.1	Detailed Description	744
11.75.2	Macro Definition Documentation	744
11.75.3	Typedef Documentation	744
11.75.4	Enumeration Type Documentation	746
11.76	cfe_msg_api_typedefs.h	747
11.77	cfe/modules/core_api/fsw/inc/cfe_resourceid.h File Reference	748
11.77.1	Detailed Description	749
11.77.2	Macro Definition Documentation	750
11.77.3	Typedef Documentation	750
11.77.4	Function Documentation	751
11.78	cfe_resourceid.h	755
11.79	cfe/modules/core_api/fsw/inc/cfe_resourceid_api_typedefs.h File Reference	756
11.79.1	Detailed Description	757
11.79.2	Macro Definition Documentation	757
11.80	cfe_resourceid_api_typedefs.h	757
11.81	cfe/modules/core_api/fsw/inc/cfe_sb.h File Reference	758
11.81.1	Detailed Description	760
11.81.2	Macro Definition Documentation	760
11.82	cfe_sb.h	760
11.83	cfe/modules/core_api/fsw/inc/cfe_sb_api_typedefs.h File Reference	762
11.83.1	Detailed Description	763
11.83.2	Macro Definition Documentation	763
11.83.3	Typedef Documentation	765
11.84	cfe_sb_api_typedefs.h	765
11.85	cfe/modules/core_api/fsw/inc/cfe_tbl.h File Reference	766
11.85.1	Detailed Description	768
11.85.2	Function Documentation	768
11.86	cfe_tbl.h	768
11.87	cfe/modules/core_api/fsw/inc/cfe_tbl_api_typedefs.h File Reference	770
11.87.1	Detailed Description	771
11.87.2	Macro Definition Documentation	771
11.87.3	Typedef Documentation	772
11.87.4	Enumeration Type Documentation	773
11.88	cfe_tbl_api_typedefs.h	773
11.89	cfe/modules/core_api/fsw/inc/cfe_tbl_filedef.h File Reference	775
11.89.1	Detailed Description	775
11.89.2	Macro Definition Documentation	775

11.89.3 Typedef Documentation	776
11.90 cfe_tbl_filedef.h	776
11.91 cfe/modules/core_api/fsw/inc/cfe_time.h File Reference	777
11.91.1 Detailed Description	778
11.91.2 Macro Definition Documentation	778
11.92 cfe_time.h	779
11.93 cfe/modules/core_api/fsw/inc/cfe_time_api_typedefs.h File Reference	780
11.93.1 Detailed Description	781
11.93.2 Macro Definition Documentation	781
11.93.3 Typedef Documentation	781
11.93.4 Enumeration Type Documentation	782
11.94 cfe_time_api_typedefs.h	782
11.95 cfe/modules/core_api/fsw/inc/cfe_version.h File Reference	783
11.95.1 Detailed Description	783
11.95.2 Macro Definition Documentation	783
11.96 cfe_version.h	785
11.97 cfe/modules/es/config/default_cfe_es_extern_typedefs.h File Reference	786
11.97.1 Detailed Description	787
11.97.2 Typedef Documentation	788
11.97.3 Enumeration Type Documentation	791
11.98 default_cfe_es_extern_typedefs.h	793
11.99 cfe/modules/es/config/default_cfe_es_fcncode_values.h File Reference	796
11.99.1 Detailed Description	796
11.99.2 Macro Definition Documentation	796
11.99.3 Enumeration Type Documentation	796
11.100 default_cfe_es_fcncode_values.h	797
11.101 cfe/modules/es/config/default_cfe_es_interface_cfg_values.h File Reference	798
11.101.1 Detailed Description	798
11.101.2 Macro Definition Documentation	798
11.102 default_cfe_es_interface_cfg_values.h	798
11.103 cfe/modules/es/config/default_cfe_es_internal_cfg_values.h File Reference	799
11.103.1 Detailed Description	799
11.103.2 Macro Definition Documentation	799
11.104 default_cfe_es_internal_cfg_values.h	799
11.105 cfe/modules/es/config/default_cfe_es_memaddress.h File Reference	800
11.105.1 Detailed Description	800
11.105.2 Macro Definition Documentation	800
11.105.3 Typedef Documentation	801
11.106 default_cfe_es_memaddress.h	801

11.107 cfe/modules/es/config/default_cfe_es_mission_cfg.h File Reference	802
11.107.1 Detailed Description	802
11.108 default_cfe_es_mission_cfg.h	802
11.109 cfe/modules/es/config/default_cfe_es_msg.h File Reference	802
11.109.1 Detailed Description	803
11.110 default_cfe_es_msg.h	803
11.111 cfe/modules/es/config/default_cfe_es_msgdefs.h File Reference	803
11.111.1 Detailed Description	805
11.111.2 Typedef Documentation	805
11.111.3 Enumeration Type Documentation	807
11.112 default_cfe_es_msgdefs.h	807
11.113 cfe/modules/es/config/default_cfe_es_msgid_values.h File Reference	810
11.113.1 Detailed Description	810
11.113.2 Macro Definition Documentation	810
11.114 default_cfe_es_msgid_values.h	810
11.115 cfe/modules/es/config/default_cfe_es_msgids.h File Reference	811
11.115.1 Detailed Description	811
11.115.2 Macro Definition Documentation	811
11.116 default_cfe_es_msgids.h	812
11.117 cfe/modules/es/config/default_cfe_es_msgstruct.h File Reference	812
11.117.1 Detailed Description	814
11.117.2 Typedef Documentation	814
11.118 default_cfe_es_msgstruct.h	817
11.119 cfe/modules/es/config/default_cfe_es_platform_cfg.h File Reference	819
11.119.1 Detailed Description	819
11.120 default_cfe_es_platform_cfg.h	820
11.121 cfe/modules/es/config/default_cfe_es_topicid_values.h File Reference	820
11.121.1 Detailed Description	820
11.121.2 Macro Definition Documentation	820
11.122 default_cfe_es_topicid_values.h	820
11.123 cfe/modules/es/fsw/inc/cfe_es_eventids.h File Reference	821
11.123.1 Detailed Description	824
11.123.2 Macro Definition Documentation	824
11.124 cfe_es_eventids.h	846
11.125 cfe/modules/es/fsw/inc/cfe_es_fcncodes.h File Reference	849
11.125.1 Detailed Description	849
11.125.2 Macro Definition Documentation	850
11.126 cfe_es_fcncodes.h	871
11.127 cfe/modules/es/fsw/inc/cfe_es_interface_cfg.h File Reference	872

11.127.1 Detailed Description	872
11.127.2 Macro Definition Documentation	872
11.128 cfe_es_interface_cfg.h	875
11.129 cfe/modules/es/fsw/inc/cfe_es_internal_cfg.h File Reference	876
11.129.1 Detailed Description	881
11.129.2 Macro Definition Documentation	881
11.130 cfe_es_internal_cfg.h	906
11.131 cfe/modules/es/fsw/inc/cfe_es_topicids.h File Reference	909
11.131.1 Detailed Description	909
11.131.2 Macro Definition Documentation	910
11.132 cfe_es_topicids.h	911
11.133 cfe/modules/evs/config/default_cfe_evs_extern_typedefs.h File Reference	911
11.133.1 Detailed Description	912
11.133.2 Typedef Documentation	912
11.133.3 Enumeration Type Documentation	913
11.134 default_cfe_evs_extern_typedefs.h	914
11.135 cfe/modules/evs/config/default_cfe_evs_fcncode_values.h File Reference	915
11.135.1 Detailed Description	916
11.135.2 Macro Definition Documentation	916
11.135.3 Enumeration Type Documentation	916
11.136 default_cfe_evs_fcncode_values.h	917
11.137 cfe/modules/evs/config/default_cfe_evs_interface_cfg_values.h File Reference	917
11.137.1 Detailed Description	918
11.137.2 Macro Definition Documentation	918
11.138 default_cfe_evs_interface_cfg_values.h	918
11.139 cfe/modules/evs/config/default_cfe_evs_internal_cfg_values.h File Reference	918
11.139.1 Detailed Description	918
11.139.2 Macro Definition Documentation	919
11.140 default_cfe_evs_internal_cfg_values.h	919
11.141 cfe/modules/evs/config/default_cfe_evs_mission_cfg.h File Reference	919
11.141.1 Detailed Description	919
11.142 default_cfe_evs_mission_cfg.h	919
11.143 cfe/modules/evs/config/default_cfe_evs_msg.h File Reference	920
11.143.1 Detailed Description	920
11.144 default_cfe_evs_msg.h	920
11.145 cfe/modules/evs/config/default_cfe_evs_msgdefs.h File Reference	921
11.145.1 Detailed Description	922
11.145.2 Macro Definition Documentation	922
11.145.3 Typedef Documentation	923

11.146	default_cfe_evs_msgdefs.h	925
11.147	cfe/modules/evs/config/default_cfe_evs_msgid_values.h File Reference	926
11.147.1	Detailed Description	927
11.147.2	Macro Definition Documentation	927
11.148	default_cfe_evs_msgid_values.h	927
11.149	cfe/modules/evs/config/default_cfe_evs_msgids.h File Reference	927
11.149.1	Detailed Description	928
11.149.2	Macro Definition Documentation	928
11.150	default_cfe_evs_msgids.h	928
11.151	cfe/modules/evs/config/default_cfe_evs_msgstruct.h File Reference	929
11.151.1	Detailed Description	930
11.151.2	Typedef Documentation	930
11.152	default_cfe_evs_msgstruct.h	932
11.153	cfe/modules/evs/config/default_cfe_evs_platform_cfg.h File Reference	935
11.153.1	Detailed Description	935
11.154	default_cfe_evs_platform_cfg.h	935
11.155	cfe/modules/evs/config/default_cfe_evs_topicid_values.h File Reference	936
11.155.1	Detailed Description	936
11.155.2	Macro Definition Documentation	936
11.156	default_cfe_evs_topicid_values.h	936
11.157	cfe/modules/evs/fsw/inc/cfe_evs_eventids.h File Reference	936
11.157.1	Detailed Description	938
11.157.2	Macro Definition Documentation	938
11.158	cfe_evs_eventids.h	948
11.159	cfe/modules/evs/fsw/inc/cfe_evs_fcncodes.h File Reference	950
11.159.1	Detailed Description	950
11.159.2	Macro Definition Documentation	950
11.160	cfe_evs_fcncodes.h	968
11.161	cfe/modules/evs/fsw/inc/cfe_evs_interface_cfg.h File Reference	969
11.161.1	Detailed Description	969
11.161.2	Macro Definition Documentation	969
11.162	cfe_evs_interface_cfg.h	970
11.163	cfe/modules/evs/fsw/inc/cfe_evs_internal_cfg.h File Reference	970
11.163.1	Detailed Description	971
11.163.2	Macro Definition Documentation	971
11.164	cfe_evs_internal_cfg.h	976
11.165	cfe/modules/evs/fsw/inc/cfe_evs_topicids.h File Reference	977
11.165.1	Detailed Description	977
11.165.2	Macro Definition Documentation	977

11.166	cfe_evs_topicids.h	978
11.167	cfe/modules/fs/config/default_cfe_fs_extern_typedefs.h File Reference	979
11.167.1	Detailed Description	979
11.168	default_cfe_fs_extern_typedefs.h	979
11.169	cfe/modules/fs/config/default_cfe_fs_filedef.h File Reference	980
11.169.1	Detailed Description	980
11.169.2	Typedef Documentation	980
11.169.3	Enumeration Type Documentation	981
11.170	default_cfe_fs_filedef.h	982
11.171	cfe/modules/fs/config/default_cfe_fs_interface_cfg_values.h File Reference	982
11.171.1	Detailed Description	983
11.171.2	Macro Definition Documentation	983
11.172	default_cfe_fs_interface_cfg_values.h	983
11.173	cfe/modules/fs/config/default_cfe_fs_mission_cfg.h File Reference	983
11.173.1	Detailed Description	983
11.174	default_cfe_fs_mission_cfg.h	984
11.175	cfe/modules/fs/fsw/inc/cfe_fs_interface_cfg.h File Reference	984
11.175.1	Detailed Description	984
11.175.2	Macro Definition Documentation	984
11.176	cfe_fs_interface_cfg.h	985
11.177	cfe/modules/msg/fsw/inc/ccsds_hdr.h File Reference	985
11.177.1	Detailed Description	986
11.177.2	Typedef Documentation	986
11.178	ccsds_hdr.h	986
11.179	cfe/modules/resourceid/fsw/inc/cfe_core_resourceid_basevalues.h File Reference	987
11.179.1	Detailed Description	988
11.180	cfe_core_resourceid_basevalues.h	988
11.181	cfe/modules/resourceid/fsw/inc/cfe_resourceid_basevalue.h File Reference	989
11.181.1	Detailed Description	989
11.181.2	Macro Definition Documentation	990
11.182	cfe_resourceid_basevalue.h	990
11.183	cfe/modules/sb/config/default_cfe_sb_extern_typedefs.h File Reference	991
11.183.1	Detailed Description	991
11.183.2	Typedef Documentation	992
11.183.3	Enumeration Type Documentation	992
11.184	default_cfe_sb_extern_typedefs.h	993
11.185	cfe/modules/sb/config/default_cfe_sb_fcncode_values.h File Reference	994
11.185.1	Detailed Description	994
11.185.2	Macro Definition Documentation	994

11.185.3 Enumeration Type Documentation	994
11.186 default_cfe_sb_fcncode_values.h	995
11.187 cfe/modules/sb/config/default_cfe_sb_interface_cfg_values.h File Reference	995
11.187.1 Detailed Description	995
11.187.2 Macro Definition Documentation	996
11.188 default_cfe_sb_interface_cfg_values.h	996
11.189 cfe/modules/sb/config/default_cfe_sb_internal_cfg_values.h File Reference	996
11.189.1 Detailed Description	996
11.189.2 Macro Definition Documentation	996
11.190 default_cfe_sb_internal_cfg_values.h	997
11.191 cfe/modules/sb/config/default_cfe_sb_mission_cfg.h File Reference	997
11.191.1 Detailed Description	997
11.192 default_cfe_sb_mission_cfg.h	997
11.193 cfe/modules/sb/config/default_cfe_sb_msg.h File Reference	998
11.193.1 Detailed Description	998
11.194 default_cfe_sb_msg.h	998
11.195 cfe/modules/sb/config/default_cfe_sb_msgdefs.h File Reference	998
11.195.1 Detailed Description	999
11.195.2 Typedef Documentation	1000
11.196 default_cfe_sb_msgdefs.h	1001
11.197 cfe/modules/sb/config/default_cfe_sb_msgid_values.h File Reference	1003
11.197.1 Detailed Description	1003
11.197.2 Macro Definition Documentation	1003
11.198 default_cfe_sb_msgid_values.h	1004
11.199 cfe/modules/sb/config/default_cfe_sb_msgids.h File Reference	1004
11.199.1 Detailed Description	1004
11.199.2 Macro Definition Documentation	1005
11.200 default_cfe_sb_msgids.h	1005
11.201 cfe/modules/sb/config/default_cfe_sb_msgstruct.h File Reference	1006
11.201.1 Detailed Description	1007
11.201.2 Typedef Documentation	1007
11.202 default_cfe_sb_msgstruct.h	1008
11.203 cfe/modules/sb/config/default_cfe_sb_platform_cfg.h File Reference	1010
11.203.1 Detailed Description	1010
11.204 default_cfe_sb_platform_cfg.h	1010
11.205 cfe/modules/sb/config/default_cfe_sb_topicid_values.h File Reference	1010
11.205.1 Detailed Description	1010
11.205.2 Macro Definition Documentation	1011
11.206 default_cfe_sb_topicid_values.h	1011

11.207 cfe/modules/sb/fsw/inc/cfe_sb_eventids.h File Reference	1011
11.207.1 Detailed Description	1014
11.207.2 Macro Definition Documentation	1014
11.208 cfe_sb_eventids.h	1031
11.209 cfe/modules/sb/fsw/inc/cfe_sb_fcncodes.h File Reference	1033
11.209.1 Detailed Description	1033
11.209.2 Macro Definition Documentation	1033
11.210 cfe_sb_fcncodes.h	1042
11.211 cfe/modules/sb/fsw/inc/cfe_sb_interface_cfg.h File Reference	1043
11.211.1 Detailed Description	1043
11.211.2 Macro Definition Documentation	1043
11.212 cfe_sb_interface_cfg.h	1044
11.213 cfe/modules/sb/fsw/inc/cfe_sb_internal_cfg.h File Reference	1045
11.213.1 Detailed Description	1047
11.213.2 Macro Definition Documentation	1048
11.214 cfe_sb_internal_cfg.h	1059
11.215 cfe/modules/sb/fsw/inc/cfe_sb_topicids.h File Reference	1061
11.215.1 Detailed Description	1061
11.215.2 Macro Definition Documentation	1061
11.216 cfe_sb_topicids.h	1063
11.217 cfe/modules/tbl/config/default_cfe_tbl_extern_typedefs.h File Reference	1063
11.217.1 Detailed Description	1064
11.217.2 Typedef Documentation	1064
11.217.3 Enumeration Type Documentation	1065
11.218 default_cfe_tbl_extern_typedefs.h	1065
11.219 cfe/modules/tbl/config/default_cfe_tbl_fcncode_values.h File Reference	1066
11.219.1 Detailed Description	1066
11.219.2 Macro Definition Documentation	1066
11.219.3 Enumeration Type Documentation	1067
11.220 default_cfe_tbl_fcncode_values.h	1067
11.221 cfe/modules/tbl/config/default_cfe_tbl_interface_cfg_values.h File Reference	1068
11.221.1 Detailed Description	1068
11.221.2 Macro Definition Documentation	1068
11.222 default_cfe_tbl_interface_cfg_values.h	1068
11.223 cfe/modules/tbl/config/default_cfe_tbl_internal_cfg_values.h File Reference	1068
11.223.1 Detailed Description	1068
11.223.2 Macro Definition Documentation	1069
11.224 default_cfe_tbl_internal_cfg_values.h	1069
11.225 cfe/modules/tbl/config/default_cfe_tbl_mission_cfg.h File Reference	1069

11.225.1 Detailed Description	1069
11.225.2 Macro Definition Documentation	1070
11.226 default_cfe_tbl_mission_cfg.h	1071
11.227 cfe/modules/tbl/config/default_cfe_tbl_msg.h File Reference	1071
11.227.1 Detailed Description	1071
11.228 default_cfe_tbl_msg.h	1071
11.229 cfe/modules/tbl/config/default_cfe_tbl_msgdefs.h File Reference	1072
11.229.1 Detailed Description	1073
11.229.2 Typedef Documentation	1073
11.230 default_cfe_tbl_msgdefs.h	1074
11.231 cfe/modules/tbl/config/default_cfe_tbl_msgid_values.h File Reference	1076
11.231.1 Detailed Description	1077
11.231.2 Macro Definition Documentation	1077
11.232 default_cfe_tbl_msgid_values.h	1077
11.233 cfe/modules/tbl/config/default_cfe_tbl_msgids.h File Reference	1077
11.233.1 Detailed Description	1078
11.233.2 Macro Definition Documentation	1078
11.234 default_cfe_tbl_msgids.h	1078
11.235 cfe/modules/tbl/config/default_cfe_tbl_msgstruct.h File Reference	1079
11.235.1 Detailed Description	1080
11.235.2 Typedef Documentation	1080
11.236 default_cfe_tbl_msgstruct.h	1081
11.237 cfe/modules/tbl/config/default_cfe_tbl_platform_cfg.h File Reference	1082
11.237.1 Detailed Description	1084
11.237.2 Macro Definition Documentation	1084
11.238 default_cfe_tbl_platform_cfg.h	1091
11.239 cfe/modules/tbl/config/default_cfe_tbl_topicid_values.h File Reference	1092
11.239.1 Detailed Description	1092
11.239.2 Macro Definition Documentation	1092
11.240 default_cfe_tbl_topicid_values.h	1092
11.241 cfe/modules/tbl/fsw/inc/cfe_tbl_eventids.h File Reference	1093
11.241.1 Detailed Description	1095
11.241.2 Macro Definition Documentation	1096
11.242 cfe_tbl_eventids.h	1113
11.243 cfe/modules/tbl/fsw/inc/cfe_tbl_fcncodes.h File Reference	1115
11.243.1 Detailed Description	1115
11.243.2 Macro Definition Documentation	1116
11.244 cfe_tbl_fcncodes.h	1125
11.245 cfe/modules/tbl/fsw/inc/cfe_tbl_interface_cfg.h File Reference	1125

11.245.1 Detailed Description	1125
11.245.2 Macro Definition Documentation	1126
11.246 cfe_tbl_interface_cfg.h	1127
11.247 cfe/modules/tbl/fsw/inc/cfe_tbl_internal_cfg.h File Reference	1127
11.247.1 Detailed Description	1128
11.247.2 Macro Definition Documentation	1129
11.248 cfe_tbl_internal_cfg.h	1136
11.249 cfe/modules/tbl/fsw/inc/cfe_tbl_topicsids.h File Reference	1137
11.249.1 Detailed Description	1137
11.249.2 Macro Definition Documentation	1137
11.250 cfe_tbl_topicsids.h	1138
11.251 cfe/modules/time/config/default_cfe_time_extern_typedefs.h File Reference	1139
11.251.1 Detailed Description	1140
11.251.2 Typedef Documentation	1140
11.251.3 Enumeration Type Documentation	1142
11.252 default_cfe_time_extern_typedefs.h	1144
11.253 cfe/modules/time/config/default_cfe_time_fcncode_values.h File Reference	1145
11.253.1 Detailed Description	1146
11.253.2 Macro Definition Documentation	1146
11.253.3 Enumeration Type Documentation	1146
11.254 default_cfe_time_fcncode_values.h	1146
11.255 cfe/modules/time/config/default_cfe_time_interface_cfg_values.h File Reference	1147
11.255.1 Detailed Description	1147
11.255.2 Macro Definition Documentation	1147
11.256 default_cfe_time_interface_cfg_values.h	1148
11.257 cfe/modules/time/config/default_cfe_time_internal_cfg_values.h File Reference	1148
11.257.1 Detailed Description	1148
11.257.2 Macro Definition Documentation	1148
11.258 default_cfe_time_internal_cfg_values.h	1148
11.259 cfe/modules/time/config/default_cfe_time_mission_cfg.h File Reference	1149
11.259.1 Detailed Description	1149
11.260 default_cfe_time_mission_cfg.h	1149
11.261 cfe/modules/time/config/default_cfe_time_msg.h File Reference	1150
11.261.1 Detailed Description	1150
11.262 default_cfe_time_msg.h	1150
11.263 cfe/modules/time/config/default_cfe_time_msgdefs.h File Reference	1150
11.263.1 Detailed Description	1152
11.263.2 Typedef Documentation	1152
11.264 default_cfe_time_msgdefs.h	1153

11.265 cfe/modules/time/config/default_cfe_time_msgid_values.h File Reference	1155
11.265.1 Detailed Description	1156
11.265.2 Macro Definition Documentation	1156
11.266 default_cfe_time_msgid_values.h	1156
11.267 cfe/modules/time/config/default_cfe_time_msgids.h File Reference	1157
11.267.1 Detailed Description	1157
11.267.2 Macro Definition Documentation	1157
11.268 default_cfe_time_msgids.h	1158
11.269 cfe/modules/time/config/default_cfe_time_msgstruct.h File Reference	1159
11.269.1 Detailed Description	1160
11.269.2 Typedef Documentation	1160
11.270 default_cfe_time_msgstruct.h	1162
11.271 cfe/modules/time/config/default_cfe_time_platform_cfg.h File Reference	1164
11.271.1 Detailed Description	1164
11.272 default_cfe_time_platform_cfg.h	1164
11.273 cfe/modules/time/config/default_cfe_time_topicid_values.h File Reference	1165
11.273.1 Detailed Description	1165
11.273.2 Macro Definition Documentation	1165
11.274 default_cfe_time_topicid_values.h	1165
11.275 cfe/modules/time/fsw/inc/cfe_time_eventids.h File Reference	1165
11.275.1 Detailed Description	1167
11.275.2 Macro Definition Documentation	1167
11.276 cfe_time_eventids.h	1176
11.277 cfe/modules/time/fsw/inc/cfe_time_fcncodes.h File Reference	1177
11.277.1 Detailed Description	1178
11.277.2 Macro Definition Documentation	1178
11.278 cfe_time_fcncodes.h	1192
11.279 cfe/modules/time/fsw/inc/cfe_time_interface_cfg.h File Reference	1193
11.279.1 Detailed Description	1194
11.279.2 Macro Definition Documentation	1195
11.280 cfe_time_interface_cfg.h	1201
11.281 cfe/modules/time/fsw/inc/cfe_time_internal_cfg.h File Reference	1202
11.281.1 Detailed Description	1203
11.281.2 Macro Definition Documentation	1203
11.282 cfe_time_internal_cfg.h	1210
11.283 cfe/modules/time/fsw/inc/cfe_time_topicsids.h File Reference	1211
11.283.1 Detailed Description	1211
11.283.2 Macro Definition Documentation	1211
11.284 cfe_time_topicsids.h	1213

11.285	osal/docs/src/osal_frontpage.dox File Reference	1214
11.286	osal/docs/src/osal_fs.dox File Reference	1214
11.287	osal/docs/src/osal_timer.dox File Reference	1214
11.288	osal/src/os/inc/common_types.h File Reference	1214
11.288.1	Detailed Description	1215
11.288.2	Macro Definition Documentation	1215
11.288.3	Typedef Documentation	1216
11.288.4	Function Documentation	1218
11.289	common_types.h	1219
11.290	osal/src/os/inc/osapi-binsem.h File Reference	1221
11.290.1	Detailed Description	1221
11.291	osapi-binsem.h	1222
11.292	osal/src/os/inc/osapi-bsp.h File Reference	1222
11.292.1	Detailed Description	1223
11.293	osapi-bsp.h	1223
11.294	osal/src/os/inc/osapi-clock.h File Reference	1224
11.294.1	Detailed Description	1226
11.294.2	Macro Definition Documentation	1226
11.294.3	Enumeration Type Documentation	1226
11.295	osapi-clock.h	1226
11.296	osal/src/os/inc/osapi-common.h File Reference	1229
11.296.1	Detailed Description	1230
11.296.2	Typedef Documentation	1230
11.296.3	Enumeration Type Documentation	1231
11.297	osapi-common.h	1231
11.298	osal/src/os/inc/osapi-condvar.h File Reference	1232
11.298.1	Detailed Description	1233
11.299	osapi-condvar.h	1233
11.300	osal/src/os/inc/osapi-constants.h File Reference	1234
11.300.1	Detailed Description	1234
11.300.2	Macro Definition Documentation	1234
11.301	osapi-constants.h	1235
11.302	osal/src/os/inc/osapi-countsem.h File Reference	1235
11.302.1	Detailed Description	1236
11.303	osapi-countsem.h	1236
11.304	osal/src/os/inc/osapi-dir.h File Reference	1237
11.304.1	Detailed Description	1237
11.304.2	Macro Definition Documentation	1238
11.305	osapi-dir.h	1238

11.306 osal/src/os/inc/osapi-error.h File Reference	1238
11.306.1 Detailed Description	1241
11.306.2 Macro Definition Documentation	1241
11.306.3 Typedef Documentation	1241
11.307 osapi-error.h	1242
11.308 osal/src/os/inc/osapi-file.h File Reference	1243
11.308.1 Detailed Description	1245
11.308.2 Macro Definition Documentation	1245
11.308.3 Enumeration Type Documentation	1246
11.309 osapi-file.h	1246
11.310 osal/src/os/inc/osapi-filesys.h File Reference	1248
11.310.1 Detailed Description	1249
11.310.2 Macro Definition Documentation	1249
11.311 osapi-filesys.h	1249
11.312 osal/src/os/inc/osapi-heap.h File Reference	1250
11.312.1 Detailed Description	1251
11.313 osapi-heap.h	1251
11.314 osal/src/os/inc/osapi-idmap.h File Reference	1251
11.314.1 Detailed Description	1253
11.314.2 Macro Definition Documentation	1253
11.315 osapi-idmap.h	1253
11.316 osal/src/os/inc/osapi-macros.h File Reference	1254
11.316.1 Detailed Description	1255
11.316.2 Macro Definition Documentation	1255
11.317 osapi-macros.h	1256
11.318 osal/src/os/inc/osapi-module.h File Reference	1257
11.318.1 Detailed Description	1258
11.318.2 Macro Definition Documentation	1258
11.319 osapi-module.h	1259
11.320 osal/src/os/inc/osapi-mutex.h File Reference	1260
11.320.1 Detailed Description	1260
11.321 osapi-mutex.h	1261
11.322 osal/src/os/inc/osapi-network.h File Reference	1261
11.322.1 Detailed Description	1261
11.323 osapi-network.h	1262
11.324 osal/src/os/inc/osapi-printf.h File Reference	1262
11.324.1 Detailed Description	1262
11.325 osapi-printf.h	1262
11.326 osal/src/os/inc/osapi-queue.h File Reference	1263

11.326.1 Detailed Description	1263
11.327 osapi-queue.h	1264
11.328 osal/src/os/inc/osapi-rwlock.h File Reference	1264
11.328.1 Detailed Description	1265
11.329 osapi-rwlock.h	1265
11.330 osal/src/os/inc/osapi-select.h File Reference	1266
11.330.1 Detailed Description	1266
11.330.2 Enumeration Type Documentation	1267
11.331 osapi-select.h	1267
11.332 osal/src/os/inc/osapi-shell.h File Reference	1268
11.332.1 Detailed Description	1268
11.333 osapi-shell.h	1268
11.334 osal/src/os/inc/osapi-sockets.h File Reference	1269
11.334.1 Detailed Description	1270
11.334.2 Macro Definition Documentation	1271
11.334.3 Typedef Documentation	1271
11.334.4 Enumeration Type Documentation	1271
11.335 osapi-sockets.h	1272
11.336 osal/src/os/inc/osapi-task.h File Reference	1274
11.336.1 Detailed Description	1275
11.336.2 Macro Definition Documentation	1276
11.336.3 Typedef Documentation	1276
11.336.4 Function Documentation	1277
11.337 osapi-task.h	1277
11.338 osal/src/os/inc/osapi-timebase.h File Reference	1278
11.338.1 Detailed Description	1278
11.338.2 Typedef Documentation	1279
11.339 osapi-timebase.h	1279
11.340 osal/src/os/inc/osapi-timer.h File Reference	1279
11.340.1 Detailed Description	1280
11.340.2 Typedef Documentation	1280
11.341 osapi-timer.h	1280
11.342 osal/src/os/inc/osapi-version.h File Reference	1281
11.342.1 Detailed Description	1282
11.342.2 Macro Definition Documentation	1282
11.342.3 Function Documentation	1284
11.343 osapi-version.h	1285
11.344 osal/src/os/inc/osapi.h File Reference	1286
11.344.1 Detailed Description	1287

11.345 osapi.h	1287
11.346 psp/fsw/inc/cfe_psp.h File Reference	1288
11.346.1 Macro Definition Documentation	1288
11.346.2 Function Documentation	1289
11.347 cfe_psp.h	1289
11.348 psp/fsw/inc/cfe_psp_cache_api.h File Reference	1290
11.348.1 Function Documentation	1290
11.349 cfe_psp_cache_api.h	1291
11.350 psp/fsw/inc/cfe_psp_cds_api.h File Reference	1291
11.350.1 Function Documentation	1292
11.351 cfe_psp_cds_api.h	1293
11.352 psp/fsw/inc/cfe_psp_eepromaccess_api.h File Reference	1294
11.352.1 Function Documentation	1294
11.353 cfe_psp_eepromaccess_api.h	1297
11.354 psp/fsw/inc/cfe_psp_endian.h File Reference	1297
11.354.1 Detailed Description	1298
11.354.2 Function Documentation	1298
11.355 cfe_psp_endian.h	1299
11.356 psp/fsw/inc/cfe_psp_error.h File Reference	1300
11.356.1 Detailed Description	1301
11.356.2 Macro Definition Documentation	1301
11.356.3 Typedef Documentation	1302
11.356.4 Function Documentation	1303
11.357 cfe_psp_error.h	1303
11.358 psp/fsw/inc/cfe_psp_exception_api.h File Reference	1304
11.358.1 Function Documentation	1304
11.359 cfe_psp_exception_api.h	1306
11.360 psp/fsw/inc/cfe_psp_id_api.h File Reference	1306
11.360.1 Function Documentation	1307
11.361 cfe_psp_id_api.h	1307
11.362 psp/fsw/inc/cfe_psp_memaccess_api.h File Reference	1308
11.362.1 Function Documentation	1309
11.363 cfe_psp_memaccess_api.h	1312
11.364 psp/fsw/inc/cfe_psp_memrange_api.h File Reference	1313
11.364.1 Macro Definition Documentation	1314
11.364.2 Function Documentation	1315
11.365 cfe_psp_memrange_api.h	1319
11.366 psp/fsw/inc/cfe_psp_port_api.h File Reference	1320
11.366.1 Function Documentation	1321

11.367 cfe_psp_port_api.h	1323
11.368 psp/fsw/inc/cfe_psp_ssr_api.h File Reference	1324
11.368.1 Function Documentation	1324
11.369 cfe_psp_ssr_api.h	1325
11.370 psp/fsw/inc/cfe_psp_timertick_api.h File Reference	1325
11.370.1 Macro Definition Documentation	1326
11.370.2 Function Documentation	1326
11.371 cfe_psp_timertick_api.h	1327
11.372 psp/fsw/inc/cfe_psp_version_api.h File Reference	1328
11.372.1 Function Documentation	1329
11.373 cfe_psp_version_api.h	1330
11.374 psp/fsw/inc/cfe_psp_watchdog_api.h File Reference	1330
11.374.1 Macro Definition Documentation	1332
11.374.2 Function Documentation	1334
11.375 cfe_psp_watchdog_api.h	1336
Index	1339

1 Core Flight Executive Documentation

- General Information and Concepts
 - [Background](#)
 - [Applicable Documents](#)
 - [Version Numbers](#)
 - [Dependencies](#)
 - [Acronyms](#)
 - [Glossary of Terms](#)
- Executive Services (ES)
 - [cFE Executive Services Overview](#)
 - [cFE Executive Services Commands](#)
 - [cFE Executive Services Telemetry](#)
 - [ES Event Message Reference](#)
 - [cFE Executive Services Configuration Parameters](#)
- Events Services (EVS)
 - [cFE Event Services Overview](#)
 - [cFE Event Services Commands](#)
 - [cFE Event Services Telemetry](#)
 - [EVS Event Message Reference](#)
 - [cFE Event Services Configuration Parameters](#)

- [Software Bus Services \(SB\)](#)
 - [cFE Software Bus Overview](#)
 - [cFE Software Bus Commands](#)
 - [cFE Software Bus Telemetry](#)
 - [SB Event Message Reference](#)
 - [cFE Software Bus Configuration Parameters](#)
- [Table Services \(TBL\)](#)
 - [cFE Table Services Overview](#)
 - [cFE Table Services Commands](#)
 - [cFE Table Services Telemetry](#)
 - [TBL Event Message Reference](#)
 - [cFE Table Services Configuration Parameters](#)
- [Time Services \(TIME\)](#)
 - [cFE Time Services Overview](#)
 - [cFE Time Services Commands](#)
 - [cFE Time Services Telemetry](#)
 - [TIME Event Message Reference](#)
 - [cFE Time Services Configuration Parameters](#)
- [cFE Event Message Cross Reference](#)
- [cFE Command Mnemonic Cross Reference](#)
- [cFE Telemetry Mnemonic Cross Reference](#)
- [cFE Application Programmer's Interface \(API\) Reference](#)

1.1 Background

The Core Flight Executive (cFE) is an application development and run-time environment. The cFE provides a set of core services including Software Bus (messaging), Time, Event (Alerts), Executive (startup and runtime), and Table services. The cFE defines an application programming interface (API) for each service which serves as the basis for application development.

The cFE Software Bus service provides a publish and subscribe messaging system that allows applications to easily plug and play into the system. Applications subscribe to cFE services at runtime, making system modifications easy. Facilitating rapid prototyping, new applications can be compiled, linked, loaded, and started without requiring the entire system to be rebuilt.

Each service comes complete with a built in application that allows users to interface with each service. To support reuse and project independence, the cFE contains a configurable set of requirements and code. The configurable parameters allow the cFE to be tailored for each environment including desk-top and closed loop simulation environments. This provides the ability to run and test software applications on a developer's desktop and then deploy that same software without changes to the embedded system. In addition the cFE includes the following software development tools:

- Unit Test Framework (UTF) for unit testing applications developed via the cFE
- Software Timing Analyzer that provides visibility into the real-time performance of embedded systems software
- Table Builder
- Command and Telemetry utilities

The cFE is one of the components of the Core Flight System (cFS), a platform and project independent reusable software framework and set of reusable software applications. There are three key aspects to the cFS architecture: a dynamic run-time environment, layered software, and a component based design. The combination of these key aspects along with an implementation targeted to the embedded software domain makes it suitable for reuse on any number of NASA flight projects and/or embedded software systems.

The pivotal design feature, abstracting the software architecture from the hardware and forming the basis of reuse, is component layering. Each layer of the architecture "hides" its implementation and technology details from the other layers by defining and using standard Application Programming Interfaces (APIs). The internals of a layer can be changed without affecting other layers' internals and components.

The layers include an OS Abstraction Layer (OSAL), Platform Support Package (PSP) layer, core Flight Executive (cFE) layer, and an Application layer. The cFE layer runs on top of the PSP and OSAL layers. The cFE comes complete with a build environment, deployment guide, API reference guide, and provides a sample PSP. The OSAL is available open source and once integrated into the cFE build environment, developers will be ready to build and run the system and start developing their mission/project specific applications that easily plug and play into the system.

1.1.1 Core Flight Executive (cFE) Goals

The main long term goal of the cFE is to form the basis for a platform and project independent reusable software framework. The cFE with the OSAL allow the development of portable embedded system software that is independent of a particular Real Time Operating System and hardware platform. A secondary long term goal is to create a standardized, product-line approach for development of embedded aerospace flight software.

1.1.1.1 Functional and Community Goals

The cFE allows embedded system software to be developed and tested on desktop workstations and ported to the target platform without changing a single line of code, providing a shorter development and debug time. The cFE is an enabler of software collaboration amongst all users promoting the growth of the application and library layers where new applications, libraries, tools, and lessons learned can be contributed and shared.

It is important for application developers to realize the long term and functional goals of the cFE. With a standard set of services providing a standard API, all applications developed with the cFE have an opportunity to become useful on future missions through code reuse. In order to achieve this goal, applications must be written with care to ensure that their code does not have dependencies on specific hardware, software or compilers. The cFE and the underlying generic operating system API (OS API) have been designed to insulate the cFE Application developer from hardware and software dependencies. The developer, however, must make the effort to identify the proper methods through the cFE and OS API to satisfy their software requirements and not be tempted to take a "short-cut" and accomplish their goal with a direct hardware or operating system software interface.

1.2 Applicable Documents

Document Title	Link
cFE System (L4) Requirements Document	cfe/docs/'cfe requirements.docx'
cFE Functional (L5) Requirements Document	cfe/docs/cFE_FunctionalRequirements.csv
cFE Application Developers Guide	cfe/docs/'cFE Application Developers Guide.md'
cFE User's Guide (includes API)	Autogenerated from code, provided with releases in cFE repository
OS Abstraction Layer (OSAL) API	Autogenerated from code, provided with releases in OSAL repository

1.3 Version Numbers

1.3.1 Version Number Semantics

The version number is a sequence of four numbers, generally separated by dots when written. These are, in order, the Major number, the Minor number, the Revision number, and the Mission Revision number.

It is important to note that version numbers are only updated upon official releases of tagged versions, **NOT** on development builds. We aim to follow the Semantic Versioning v2.0 specification with our versioning.

The MAJOR number is incremented on release to indicate when there is a change to an API that may cause existing, correctly-written cFS components to stop working. It may also be incremented for a release that contains changes deemed to be of similar impact, even if there are no actual changes to the API.

The MINOR number is incremented on release to indicate the addition of features to the API which do not break the existing code. It may also be incremented for a release that contains changes deemed to be of similar impact, even if there are no actual updates to the API.

The REVISION number shall be incremented on changes that benefit from unique identification such as bug fixes or major documentation updates. The Revision number may also be updated if there are other changes contained within a release that make it desirable for applications to distinguish one release from another. **WARNING:** The revision number is set to the number 99 in development builds. To distinguish between development builds refer to the BUILD_NUMBER and BUILD_BASELINE detailed in the section "Identifying Development Builds".

The Mission Rev Version number is set to zero in all official releases, and is reserved for the mission use.

1.3.2 How and Where Defined

The version numbers are provided as simple macros defined in the [cfe_version.h](#) header file as part of the API definition; these macros must expand to simple integer values, so that they can be used in simple if directives by the macro preprocessor.

Note the Mission Rev number is provided for missions to be able to identify unique changes they have made to the released software (via clone and own). Specifically, the values 1-254 are reserved for mission use to denote patches/customizations while 0 and 0xFF are reserved for cFS open-source development use (pending resolution of nasa/cFS#440).

1.3.3 Identifying Development Builds

In order to distinguish between development versions, we also provide a BUILD_NUMBER.

The BUILD_NUMBER reflects the number of commits since the BUILD_BASELINE, a baseline git tag, for each particular component. The BUILD_NUMBER integer monotonically increases for a given baseline. The BUILD_BASELINE identifies the current development cycle and is a git tag with format vMAJOR.MINOR.REVISION. The Codename used in the version string also refers to the current development cycle. When a new baseline tag and codename are created, the BUILD_NUMBER resets to zero and begins increasing from a new baseline.

1.3.4 Templates for the short and long version string

See [cfe_version.h](#) for the standard layout and definition of version information. The apps and repositories follow the same pattern by replacing the CFE_ prefix with the appropriate name; for example, osal uses OS_, psp uses CFE_↵PSP_IMPL, and so on.

Suggested pattern for development:

- CFSCOMPONENT_SRC_VERSION: REFERENCE_GIT_TAG"+dev"BUILD_NUMBER
 - Example: "v6.8.0-rc1+dev123"
- CFSCOMPONENT_VERSION_STRING: "CFSCOMPONENT DEVELOPMENT BUILD "CFSCOMPONENT_↵SRC_VERSION" (Codename: CFSCONSTELLATION), Last Official Release: MAJOR.MINOR.REVISION"
 - Example: "cFE DEVELOPMENT BUILD v6.8.0-rc1+dev123 (Codename: Bootes), Last Official Release: cfe v6.7.0"

Suggested pattern for official releases:

- CFSCOMPONENT_SRC_VERSION: OFFICIAL_GIT_TAG
 - Example: "v7.0.0"
- COMPONENT_VERSION_STRING: "CFSCOMPONENT OFFICIAL RELEASE "CFSCOMPONENT_SRC_↵VERSION" (Codename: CFSCONSTELLATION)"
 - Example: "cFE OFFICIAL RELEASE v7.0.0 (Codename: Caelum)"

1.4 Dependencies

The Core Flight Executive (cFE) is required to be built with the Operating System Abstraction Layer (OSAL) and Platform Support Package (PSP) components of the Core Flight System (cFS). It is always recommended to build with the latest versions of each of the components as backward compatibility may not be supported.

Several internal data structures within the cFE use the "char" data type. This data type is typically 1 byte in storage size with a value range -128 to 127 or 0 to 255. The size of the "char" data type and whether or not the type is signed or unsigned can change across platforms. The cFE assumes use of the "char" data type as an **8-bit type**.

1.5 Acronyms

Acronym	Description
AC	Attitude Control
ACE	Attitude Control Electronics
ACS	Attitude Control System
API	Application Programming Interface
APID	CCSDS Application ID
App	Application
CCSDS	Consultative Committee for Space Data Systems
CDH, C&DH	Command and Data Handling
cFE	core Flight Executive
cFS	core Flight System
CM	Configuration Management
CMD	Command
CPU	Central Processing Unit
EDAC	Error Detection and Correction
EEPROM	Electrically Erasable Programmable Read-Only Memory
ES	Executive Services
EVS	Event Services
FC	Function Code
FDC	Failure Detection and Correction
FSW	Flight Software
HW, H/W	Hardware
ICD	Interface Control Document
MET	Mission Elapsed Time
MID	Message ID
OS	Operating System
OSAL	Operating System Abstraction Layer
PID	Pipeline ID
PKT	Packet
PSP	Platform Support Package
RAM	Random-Access Memory
SB	Software Bus
SDO	Solar Dynamics Observatory
ST5	Space Technology Five
STCF	Spacecraft Time Correlation Factor
SW, S/W	Software
TAI	International Atomic Time
TBD	To Be Determined
TBL	Table Services
TID	Task ID
TIME	Time Services
TLM	Telemetry
UTC	Coordinated Universal Time

1.6 cFE Executive Services Overview

Executive Services (ES) is one of the five core Flight Executive components. ES is the primary interface to the underlying Operating System, providing a high level interface to system control facilities. The ES component is responsible for starting up and restarting the cFE, starting up, shutting down, and restarting cFE Applications, logging errors and performance data, and providing a persistent memory store for cFE Applications.

The interfaces to the ES task include the Ground Interface (commands and telemetry) and the Application Programmer Interfaces (APIs). The ES task interfaces to the OS through the OS Abstraction Layer (OSAL) and platform through the Platform Support Package (PSP).

The functionality provided by the ES task include Software Reset, Application and Child Task Management, Basic File System, Performance Data Collection, Critical Data Store, Memory Pool, System Log, Shell Command.

For additional detail on Executive Services, see the following sections:

- [Terminology](#)
- [Software Reset](#)
 - [Reset Types and Subtypes](#)
 - [Exception and Reset \(ER\) Log](#)
- [Application and Child Task Management](#)
 - [Starting an Application](#)
 - [Stopping an Application](#)
 - [Restarting an Application](#)
 - [Reloading an Application](#)
 - [Listing Current Applications](#)
 - [Listing Current Tasks](#)
 - [Loading Common Libraries](#)
- [Basic File System](#)
- [Performance Data Collection](#)
- [Critical Data Store](#)

- [Memory Pool](#)
- [System Log](#)
- [Version Identification](#)
- [Frequently Asked Questions about Executive Services](#)

1.6.1 Terminology

The following sections describe terminology that is very relevant to understanding the Executive Services:

- ["Application" and "cFE Application"](#)
- ["Task"](#)
- ["Startup Script"](#)

1.6.1.1 "Application" and "cFE Application"

Application

The term 'Application' as defined in the [Glossary of Terms](#) is *a set of data and functions that is treated as a single entity by the cFE. cFE resources are allocated on a per-Application basis. Applications are made up of a Main Task and zero or more Child Tasks.*

cFE Application

A 'cFE Application' is an application that is external to the cFE and designed to interface to the cFE through the APIs. It is created through an entry in the ["Startup Script"](#) (with the 'Object Type' field set to CFE_APP) or by way of the [CFE_ES_START_APP_CC](#) ground command.

When referring to one of the five applications internal to the cFE (ES, EVS, SB, TIME or TBL), the term 'Service' or 'Core Application' is typically used.

A listing of cFE applications can be acquired by using the [CFE_ES_QUERY_ALL_CC](#) ground command. This listing will include the cFE internal applications as well as cFE applications that are loaded and running.

1.6.1.2 "Task"

A Task is a thread of execution in the operating system, often associated with a cFE Application. Each cFE Application has a Main task providing its CPU context, stack and other OS resources. In addition, each cFE Application can create multiple Child Tasks which are closely associated with the Parent Task and cFE Application.

In a traditional Real Time Operating System such as vxWorks, the cFE Application Main task and child tasks end up being mapped to these OS tasks in the same shared memory space. For example, a Stored Command cFE Application that consists of a cFE Main Task and 10 Relative Time Sequence Child Tasks would have 11 tasks on a vxWorks system. The only association between these tasks exists in the cFE.

In a memory protected process oriented Operating System, the intention is to have a cFE Application implemented as a memory protected process with its own virtual address space. In this Process Model, each cFE Child Task would be a thread in the parent Process, much like a Unix process with multiple threads. In this model, the Stored Command example with a cFE Main Task and 10 Relative Time Sequence Child Tasks would consist of a Unix Process and 10 pthreads, all under the same virtual address space.

1.6.1.3 "Startup Script"

The startup script is a text file, written by the user that contains a list of entries (one entry for each application) and is used by the ES application for automating the startup of applications. For a processor reset, ES checks for the `CFE_PLATFORM_ES_VOLATILE_STARTUP_FILE` first, and if it doesn't exist or for a power on reset ES uses the file passed in to `CFE_ES_Main` (typically `CFE_PLATFORM_ES_NONVOL_STARTUP_FILE` but dependent on the PSP).

The fields in a single entry include:

Object Type	<code>CFE_APP</code> for an Application, or <code>CFE_LIB</code> for a library.
Path/Filename	This is a cFE Virtual filename, not a vxWorks device/pathname
Entry Point	This is the name of the "main" function for App.
CFE Name	The cFE name for the APP or Library
Priority	This is the Priority of the App, not used for a Library
Stack Size	This is the Stack size for the App, not used for a Library
Load Address	This is the Optional Load Address for the App or Library. It is currently not implemented so it should always be 0x0.
Exception Action	This is the Action the cFE should take if the Application has an exception. • 0 = Do a cFE Processor Reset • Non-Zero = Just restart the Application

Immediately after the cFE completes its initialization, the ES Application first looks for the volatile startup script. The location in the file system is defined by the cFE platform configuration parameter named `CFE_PLATFORM_ES_VOLATILE_STARTUP_FILE`. This configuration parameter contains a path as well as a file-name. If the file is found, ES begins to startup the applications that are listed in the file. If ES does not find the file, it attempts to open the `CFE_PLATFORM_ES_NONVOL_STARTUP_FILE`.

If ES finds the volatile startup script, the attempt to open the nonvolatile startup script is bypassed.

Any errors encountered in the startup script processing are written to the [System Log](#). The [System Log](#) may also contain positive acknowledge messages regarding the startup script processing.

The startup script delivered with the cFE (`cfe_es_startup.scr`) also has some detailed information about the fields and the settings.

1.6.2 Software Reset

The ES Software Reset provides a command to [reset the cFE](#) as well as [resetting individual applications](#). Because applications are dependent on the cFE services, it is not possible to reset the cFE without affecting the applications. Therefore, a command to reset the cFE will also reset every application that is running at the time the command is received.

Also include is the Exception and Reset (ER) Log, which has a command for [dumping](#) or [clearing](#) the log and telemetry to show the number of entries in the log. In addition to the ER log, the user may find information about the most recent reset in the ES task housekeeping telemetry.

The ES Software Reset also provides a command to [set the maximum number of processor resets](#) before ES issues a power-on reset. There is a corresponding 'processor resets' counter in ES housekeeping telemetry that may be [reset through another ES command](#).

1.6.3 Reset Types and Subtypes

The Reset Type is sent to the ground in the ES housekeeping packet and tells how the current running version of the cFE was invoked. The possible Reset Types expected in the telemetry field are [CFE_PSP_RST_TYPE_POWERON](#) and [CFE_PSP_RST_TYPE_PROCESSOR](#). There is a third Reset Type defined in the ES code as [CFE_ES_APP_RESTART](#) which applies only to restarting an individual application and is covered in more detail in the section titled Application and Child Task.

The Reset Subtype is also sent in the ES housekeeping packet and gives more detail about the type of reset that started the execution of the current running version of the cFE. The possible Reset Subtypes are [CFE_PSP_RST_SUBTYPE_POWER_CYCLE](#), [CFE_PSP_RST_SUBTYPE_PUSH_BUTTON](#), [CFE_PSP_RST_SUBTYPE_HW_SPECIAL_BUTTON](#), [CFE_PSP_RST_SUBTYPE_HW_WATCHDOG](#), [CFE_PSP_RST_SUBTYPE_RESET_COMMAND](#), [CFE_PSP_RST_SUBTYPE_EXCEPTION](#), [CFE_PSP_RST_SUBTYPE_UNDEFINED_RESET](#), [CFE_PSP_RST_SUBTYPE_HWDEBUG_RESET](#), [CFE_PSP_RST_SUBTYPE_BAN](#)

1.6.4 Exception and Reset (ER) Log

The Exception and Reset Log contains detailed information about past resets and exceptions. To view the information the [CFE_ES_WRITE_ER_LOG_CC](#) command must be sent. This command will write the log to a binary file. The path and filename may be specified in the command. If the filename command field contains an empty string, the configuration parameter [CFE_PLATFORM_ES_DEFAULT_ER_LOG_FILE](#) is used to specify the path and filename. Use the ground system to get the file and display the contents. There is also a command to clear the ER log, [CFE_ES_CLEAR_ER_LOG_CC](#).

The size of the ER log is defined by the platform configuration parameter [CFE_PLATFORM_ES_ER_LOG_ENTRIES](#). This log is preserved after a processor reset and held in the ES reset area.

A count of the number of entries in the log is present in the ES housekeeping telemetry. This count can be used with the configuration parameter [CFE_PLATFORM_ES_ER_LOG_ENTRIES](#) to calculate the fullness of the log.

The information contained in a single log entry is defined by the structure `CFE_ES_ERLog_t`.

1.6.5 Application and Child Task Management

The ES Application and Child Task Management provides the user with full control over starting and stopping applications as well as querying information regarding applications, tasks and library routines.

There is no command to start or stop a child task. Child tasks can be controlled (started, stopped or deleted) only by the parent application through an API call.

This provides a way for the user to load a set of library routines, (via the startup script) without starting a corresponding task. See the section related to library routines for more detail.

The ES task maintains a counter for the number of registered applications, number of registered child tasks and the number of registered libraries in the ES housekeeping data.

1.6.6 Starting an Application

There are two ways to start an application, through the ground command [CFE_ES_START_APP_CC](#) or through the startup script. In either case, the object file must be loaded on board before the command is sent or before the startup script is executed. The startup script contains a list of applications and library routines to load and start immediately after the cFE finishes its startup sequence. The parameters in the command, match the elements of an entry in the startup script.

The format of the Start Application command, is defined in the structure [CFE_ES_StartAppCmd_t](#). The members of the structure include, application name, entry point, filename, stack size, load address, exception action and priority.

If the command fails for any reason, an error event will be sent stating the reason for the failure. There may be additional information in the system log that can be viewed by sending the ES command to dump the system log.

After starting an application, the ES task sends an informational event message displaying the application name, file-name of the object and the application ID. The new application will then show up in the query list downloaded in response to the [CFE_ES_QUERY_ALL_CC](#) command.

1.6.7 Stopping an Application

Stopping an application can be done through the ground command [CFE_ES_STOP_APP_CC](#). This command will terminate the application execution and all child tasks created by the application, free the system resources that it allocated and delete the corresponding object file.

The process of stopping an application is done in a controlled manner when the application is properly using the return code from the call to the [CFE_ES_RunLoop](#). When the application properly uses this function, the ES task starts a timer and (via the return code) tells the application to exit at its own convenience. This gives the application time to free its own resources and do any cleanup that may be required before terminating itself by calling [CFE_ES_ExitApp](#). If the timer expires and the application still exists, then ES must 'kill' the application. When the application is killed, ES attempts to cleanup the applications resources as best it could. In this case there is no guarantee that all the system resources are properly released.

The format of the Stop Application command, is defined in the structure [CFE_ES_StopAppCmd_t](#). The only parameter in the command is an application name.

If the command fails for any reason, an error event will be sent stating the reason for the failure. There may be additional information in the system log that can be viewed by sending the ES command to dump the system log.

After stopping an application, the ES task sends a debug message stating the name of the application. After executing the command, the application (or any resources it allocated) should no longer be listed in any cFE tables or files.

1.6.8 Restarting an Application

The [CFE_ES_RESTART_APP_CC](#) command is used to restart an application using the same file name as the last start.

This command checks for file existence, the application is running, and the application is not a core app. If valid, the application restart is requested.

When requested, ES stops the application, unloads the object file, loads the object file using the previous file name, and restarts an application using the parameters defined when the application was previously started, either through the startup script or by way of the [CFE_ES_START_APP_CC](#) command.

1.6.9 Reloading an Application

The [CFE_ES_RELOAD_APP_CC](#) command is used to reload an application using a new file name.

This command performs the same actions as [CFE_ES_RESTART_APP_CC](#) only using the new file.

1.6.10 Listing Current Applications

There are two options for receiving information about applications, the [CFE_ES_QUERY_ONE_CC](#) command can be used to get details about a single application. This command takes an application name as its only parameter and the application information is sent as a software bus packet that can be telemetered to the ground.

Or the [CFE_ES_QUERY_ALL_CC](#) command can be used to get information about all the applications that are currently registered with ES. This command writes the application data to a file and has a one parameter which specifies the path and filename of the output file.

For either command, the following Application information is made available:

- **Application ID** - The Application ID assigned by the cFE to the Application
- **Type Identifier** - Identifies whether the Application is a CORE App or an EXTERNAL App
- **Name** - The Application Name
- **Entry Point** - The symbolic name for the entry point into the Application
- **Filename** - The name of the file the Application was loaded from
- **Stack Size** - The number of bytes allocated for the Application's stack
- **Load Address** - The starting address of memory where the Application was loaded
- **Load Size** - The size, in bytes, of the Application when loaded into memory

- **Start Address** - The physical address that maps to the Entry Point
- **Exception Action** - A flag that identifies whether the Processor should undergo a Restart or whether just the Application should restart upon an exception condition within the Application
- **Priority** - The assigned priority for the Application
- **Main Task ID** - The Task ID assigned to the main task associated with the Application
- **Main Task Name** - The name of the main task associated with the Application
- **Number of Child Tasks** - The number of child tasks spawned by the main task

For a description of the format in which this data is dumped, see [CFE_ES_ApplInfo_t](#).

1.6.11 Listing Current Tasks

The [CFE_ES_QUERY_ALL_TASKS_CC](#) command is used to get a list of child tasks that are currently registered with ES. The following information is provided for each registered task:

- **Task ID** - The Task ID associated with the specified task
- **Task Name** - The name of the Task
- **Application ID** - The ID for the Application the Task is associated with
- **Application Name** - The name of the Application the Task is associated with

1.6.12 Loading Common Libraries

Library routines may be loaded only through the startup script. There is an option that allows a library routine initialization function to be executed after the library is loaded. Refer to the cFE Application Developers Guide for more information regarding Library Routines and startup scripts. The startup script delivered with the cFE (`cfe_es_startup.scr`) also has some detailed information about library routines.

1.6.13 Basic File System

ES provides minimal functionality to initialize, read, and write cfe File headers.

1.6.14 Performance Data Collection

The Performance Data Collection provides precise timing information for each software application similar to how a logic analyzer can trigger and filter data.

API calls are inserted by the development team at key points in the code. The basic operation is to start the data collection, wait some amount of time, then send the command to stop the data collection. When the stop command is received, the ES task writes all the data from the buffer to a file. The file can then be imported to analysis tools for viewing. The size of the buffer is configurable through the [CFE_PLATFORM_ES_PERF_DATA_BUFFER_SIZE](#) platform configuration parameter.

Additional information follows:

- [Performance Data Collection Trigger Masks](#)
- [Starting to Collect Performance Data](#)
- [Stopping the Collection of Performance Data](#)
- [Viewing the Collection of Performance Data](#)

1.6.14.1 Performance Data Collection Trigger Masks

The trigger mask is used to control precisely when to start collecting the data. There is a bit in the trigger mask for every marker used in the code. After a start command is received, the trigger mask is read and dictates when to begin storing data in the buffer.

If the trigger mask is set to all zeros, then the collection will begin immediately after the start command and continue until a stop command is received. In this case the buffer behaves in a 'circular' manner.

1.6.14.2 Starting to Collect Performance Data

The [CFE_ES_START_PERF_DATA_CC](#) command is used to start the data collection process. The ES task sends a debug event when the command is received. It is not possible to start a collection if the buffer-to-file write is in process from an earlier collection. There is an ES telemetry point that can be used to ensure there is not a buffer-to-file write in progress. This ES telemetry point is called 'Perf Data to Write' and begins counting down from 'Data Count' to zero. If this counter is zero, it is ok to send the start command. If any errors are encountered when the start command is received, the details will be displayed in an error event message.

1.6.14.3 Stopping the Collection of Performance Data

The [CFE_ES_STOP_PERF_DATA_CC](#) command is used to stop the data collection process and write the buffer data to a file. The path and filename may be specified in the command. If the filename command field contains an empty string, the configuration parameter [CFE_PLATFORM_ES_DEFAULT_PERF_DUMP_FILENAME](#) is used to specify the path and filename. The number of entries written to the file is determined by the 'data count' variable, which is sent in the ES housekeeping telemetry packet. To ensure cpu hogging does not occur during the write process, ES creates a low priority child task to perform the file write operation. This child task will write a number of entries, then sleep for a short time to give tasks of lower priority a chance to run. The number of entries between delays, and the delay time is displayed in the debug event at the time the stop command is received.

1.6.14.4 Viewing the Collection of Performance Data

To view the performance data, the file created as a result of the stop command must be transferred to the ground and imported into a viewing tool. See <https://github.com/nasa/perfutils-java> as an example.

1.6.15 Critical Data Store

Some missions are required, for health, safety and mission success criteria, to survive Processor Resets. These mission requirements frequently flow down to Attitude Control and/or Command and Data Handling requirements that force an Application developer to design a mechanism for retaining software state information through a Processor Reset. The cFE provides the Critical Data Store to assist the developer in meeting these requirements.

The Critical Data Store is an area of memory that is not cleared during a Processor Reset. In addition, the contents of memory are validated when accessed with a Data Integrity Value that helps to ensure the contents have not been corrupted. Each processor platform, through the design of its Board Support Package, can implement this area of memory in a number of ways to ensure the contents survive a Processor Reset. Applications can allocate a section of this memory for their use in a way similar to the [cFE Table Services Overview](#).

When an Application registers a Critical Data Store (CDS), the Executive Services allocates a section of the Critical Data Store memory for the application's use and assigns the Application specified name to the memory area. The operator can find and learn the characteristics of these Critical Data Stores by using the [Dump CDS Registry Command](#). This command will dump the contents of the CDS Registry maintained by the Executive Services into a file that can be downlinked and examined by the operator.

The CDS Registry dump will identify the following information for each registered CDS:

- **Handle** - the numeric identifier used by an Application to access the contents of the CDS
- **Size** - the number of bytes allocated to the specified CDS
- **Table Flag** - a flag that indicates whether the CDS is associated with a [Critical Tables](#) (when non-zero) or not (when equal to zero).
- **Name** - a processor specific name that uniquely identifies the CDS. The name comes in two parts, "AppName . ↔ CDSName". AppName identifies which Application registered the CDS. CDSName is the name the Application assigned to the CDS.

The format of the CDS Registry Dump File is a cFE Standard File header (see [CFE_FS_Header_t](#)) followed by one or more CDS Registry Dump File Records (see [CFE_ES_CDSRegDumpRec_t](#)).

1.6.16 Memory Pool

Refer to the cFE Application Developers Guide for additional information.

Applications that are designed for generic missions, frequently have to wait until run-time before allocating memory for buffers, data records, etc.

The cFE provides a memory allocation algorithm that may be used by an application to manage its block of memory. The user provides a pointer to its memory block and a list of block sizes and the cFE provides 'get' and 'put' API's to the user for managing its memory pool.

Run-time memory allocation in an embedded system can be risky because of the potential problem of memory fragmentation. Memory fragmentation is also referred to as External Fragmentation and is defined in the wikipedia as:

External fragmentation is the phenomenon in which free storage becomes divided into many small pieces over time. It is a weakness of certain storage allocation algorithms, occurring when an application allocates and deallocates ("frees") regions of storage of varying sizes, and the allocation algorithm responds by leaving the allocated and deallocated regions interspersed. The result is that, although free storage is available, it is effectively unusable because it is divided into pieces that are too small to satisfy the demands of the application. The term "external" refers to the fact that the unusable storage is outside the allocated regions.

To help prevent this from happening, the cFE has integrated a memory allocation algorithm that is designed to create blocks at run-time, based on the size of the blocks requested. After a reset, there are no blocks created, the memory pool is said to be unconfigured. As requests for memory blocks are made, the memory pool first tries to use blocks that have been created but are no longer in use. If it cannot find an available block, it will create a new one. The created blocks remain until a reset occurs.

This algorithm is recommended when the size of the requests and the peak rate of requests can be pre-determined. It is highly recommended that adequate margin is designed into the pool size. The memory pool should never get close to being fully configured (i.e. not enough memory to create a new block). If the memory does become fully configured, requests for new size blocks will fail, regardless of whether the created blocks are in-use or not. The margin on the memory pool can be monitored by viewing the 'free bytes' member of the memory pool statistics. The memory pool statistics are dumped only when commanded by way of the ES command [CFE_ES_SEND_MEM_POOL_STATS_CC](#).

A user of the ES memory pool begins by tailoring the memory pool for the particular use, by defining a list of block sizes and allocating a block of memory. These block size definitions simply give the memory pool a set of sizes to choose from. They do not configure the memory pool in any way and they do not affect the size of the pool. The cFE defines a default set of block sizes in the `cfe_platform_cfg.h` file.

If the default block sizes are used, the application will create the pool using the simpler [CFE_ES_PoolCreate](#) API. This API takes a pointer to the first byte of the memory pool (allocated by the application) and a size parameter. The API returns a handle to be used for the get and put requests.

If the defaults are not sufficient, the user must define the block sizes and use the [CFE_ES_PoolCreateEx](#) API.

After receiving a positive response from the PoolCreate API, the memory pool is ready to accept requests, but at this point it is completely unconfigured (meaning there are no blocks created). The first valid request (via [CFE_ES_GetPoolBuf](#) API) after creating the pool will always cause the memory pool to create a block and return a pointer to the new block. The size of the block depends on the size definitions mentioned earlier. If there is not an exact match between the requested and defined sizes, then the memory pool will create and return the smallest block that meets the following criteria: is a defined size and large enough to hold the request.

If another request for that size comes in before the first block was released through the [CFE_ES_PutPoolBuf](#) API, then the memory pool will create a second block of that size and return a pointer to the second block. If both blocks were then released through the [CFE_ES_PutPoolBuf](#) API and the memory pool statistics were dumped via the [CFE_ES_SEND_MEM_POOL_STATS_CC](#) command, the number of blocks created would be two. The number of 'free bytes' in the pool would be the size of the pool minus the sum of the following items:

- the size of the two blocks created (even though they are not 'in-use').
- a buffer descriptor for each of the two blocks created (2 * 12 bytes)
- a 168 byte pool descriptor Refer to the cFE Applications Developers Guide for more details.

This allocation algorithm does have its limits. There are certain conditions that can place the memory pool in an undesired state. For instance, if a burst of get requests were received for the same block size, the memory pool may create a large number of blocks of that size. If this is a one-time burst, the memory pool would be configured with this large number of blocks that may no longer be needed. This scenario would use up the 'free bytes' margin in an undesired way. It should be noted that once the blocks are created, they cannot be deleted by any means other than a processor or power-on reset. It is highly recommended that the memory pool statistics be carefully monitored to ensure that the 'free-bytes' margin is sufficient (which is typically dictated by mission requirements).

An operator can obtain information about an Application's Memory Pool by using the [Telemeter Memory Pool Statistics Command](#).

This command will cause Executive Services to extract pertinent statistics from the data used to manage the Memory Pool and telemeter them to the ground in the [Memory Pool Statistics Telemetry Packet](#).

In order to obtain the statistics associated with a memory pool, the operator **MUST** have the correct Memory Handle as reported by the Application who owns the Memory Pool. **It should be noted that an inappropriate Memory Pool Handle can (and likely will) cause the system software to crash!** Within the cFE itself, there are three cFE Core Applications that make use of the Executive Services Memory Pool API. These are Software Bus (SB), Event Services (EVS) and Table Services (TBL). Each of these cFE Core Applications report their memory pool handles in telemetry.

The [Memory Pool Statistics Telemetry Packet](#) contains the following information:

- **Memory Pool Handle** - the handle, as provided by the operator in the [Telemeter Memory Pool Statistics Command](#). This repeating of the handle in telemetry ensures the operator knows which Memory Pool Statistics are being viewed
- **Pool Size** - The total size of the memory pool (in bytes)
- **Number Blocks Requested** - The total number of memory blocks requested for allocation
- **Number of Errors** - The total number of errors encountered when a block was released
- **Number of Free Bytes** - The total number of bytes in the Memory Pool that have never been allocated to a Memory Block
- **Block Statistics** - For each specified size of memory block (of which there are [CFE_MISSION_ES_POOL_MAX_BUCKETS](#)), the following statistics are kept
 - **Block Size** - The size, in bytes, of all blocks of this type
 - **Number of Blocks Allocated** - The number of this sized block which are currently allocated and in use
 - **Number of Blocks Free** - The number of this size block which have been in use previously but are no longer being used

1.6.17 System Log

The System Log is an array of bytes that contains back-to-back printf type messages from applications. The cFE internal applications use this log when errors are encountered during initialization before the Event Manager is fully initialized. To view the information the [CFE_ES_WRITE_SYS_LOG_CC](#) command must be sent. This command will write the log to a binary file. The path and filename may be specified in the command. If the filename command field contains an empty string, the configuration parameter [CFE_PLATFORM_ES_DEFAULT_SYSLOG_FILE](#) is used to specify the path and filename. Use the ground system to get the file and display the contents. The [CFE_ES_CLEAR_SYS_LOG_CC](#) is used to clear the System log.

The size of the System log is defined by the platform configuration parameter [CFE_PLATFORM_ES_SYSTEM_LOG_SIZE](#). This log is preserved after a processor reset and held in the ES reset area.

A count of the number of entries in the log is present in the ES housekeeping telemetry.

1.6.18 Version Identification

Version information is reported at startup, and upon receipt of a No-op command

1.6.19 Frequently Asked Questions about Executive Services

None submitted

1.7 cFE Executive Services Commands

Upon receipt of any command, the Executive Services application will confirm that the message length embedded within the header (from [CFE_MSG_GetSize\(\)](#)) matches the expected length of that message, based on the size of the C structure defining that command. If there is any discrepancy between the expected and actual message size, ES will generate the [CFE_ES_LEN_ERR_EID](#) event, increment the command error counter (`$sc_$cpu_ES_CMDEC`), and the command will *not* be accepted for processing.

The following is a list of commands that are processed by the cFE Executive Services Task.

Global [CFE_ES_CLEAR_ER_LOG_CC](#)

Clears the contents of the Exception and Reset Log

Global [CFE_ES_CLEAR_SYS_LOG_CC](#)

Clear Executive Services System Log

Global [CFE_ES_DELETE_CDS_CC](#)

Delete Critical Data Store

Global [CFE_ES_DUMP_CDS_REGISTRY_CC](#)

Dump Critical Data Store Registry to a File

Global [CFE_ES_NOOP_CC](#)

Executive Services No-Op

Global CFE_ES_OVER_WRITE_SYS_LOG_CC

Set Executive Services System Log Mode to Discard/Overwrite

Global CFE_ES_QUERY_ALL_CC

Writes all Executive Services Information on all loaded modules to a File

Global CFE_ES_QUERY_ALL_TASKS_CC

Writes a list of All Executive Services Tasks to a File

Global CFE_ES_QUERY_ONE_CC

Request Executive Services Information on a specified module

Global CFE_ES_RELOAD_APP_CC

Stops, Unloads, Loads from the command specified File and Restarts an Application

Global CFE_ES_RESET_COUNTERS_CC

Executive Services Reset Counters

Global CFE_ES_RESET_PR_COUNT_CC

Resets the Processor Reset Counter to Zero

Global CFE_ES_RESTART_APP_CC

Stops, Unloads, Loads using the previous File name, and Restarts an Application

Global CFE_ES_RESTART_CC

Executive Services Processor / Power-On Reset

Global CFE_ES_SEND_MEM_POOL_STATS_CC

Telemeter Memory Pool Statistics

Global CFE_ES_SET_MAX_PR_COUNT_CC

Configure the Maximum Number of Processor Resets before a Power-On Reset

Global CFE_ES_SET_PERF_FILTER_MASK_CC

Set Performance Analyzer's Filter Masks

Global CFE_ES_SET_PERF_TRIGGER_MASK_CC

Set Performance Analyzer's Trigger Masks

Global CFE_ES_START_APP_CC

Load and Start an Application

Global CFE_ES_START_PERF_DATA_CC

Start Performance Analyzer

Global CFE_ES_STOP_APP_CC

Stop and Unload Application

Global CFE_ES_STOP_PERF_DATA_CC

Stop Performance Analyzer and write data file

Global CFE_ES_WRITE_ER_LOG_CC

Writes Exception and Reset Log to a File

Global CFE_ES_WRITE_SYS_LOG_CC

Writes contents of Executive Services System Log to a File

1.8 cFE Executive Services Telemetry

The following are telemetry packets generated by the cFE Executive Services Task.

Global [CFE_ES_HousekeepingTlm_Payload_t](#)

Executive Services Housekeeping Packet

Global [CFE_ES_HousekeepingTlm_t](#)

Executive Services Housekeeping Packet

Global [CFE_ES_MemStatsTlm_t](#)

Memory Pool Statistics Packet

Global [CFE_ES_OneAppTlm_Payload_t](#)

Single Application Information Packet

Global [CFE_ES_OneAppTlm_t](#)

Single Application Information Packet

Global [CFE_ES_PoolStatsTlm_Payload_t](#)

Memory Pool Statistics Packet

1.9 cFE Executive Services Configuration Parameters

The following are configuration parameters used to configure the cFE Executive Services either for each platform or for a mission as a whole.

Global [CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN](#)

Maximum Length of Full CDS Name in messages

Maximum Length of Full CDS Name in messages

Global [CFE_MISSION_ES_CDS_MAX_NAME_LENGTH](#)

Maximum Length of CDS Name

Maximum Length of CDS Name

Global [CFE_MISSION_ES_DEFAULT_CRC](#)

Mission Default CRC algorithm

Mission Default CRC algorithm

Global [CFE_MISSION_ES_MAX_APPLICATIONS](#)

Mission Max Apps in a message

Mission Max Apps in a message

Global [CFE_MISSION_ES_PERF_MAX_IDS](#)

Define Max Number of Performance IDs for messages

Define Max Number of Performance IDs for messages

Global [CFE_MISSION_ES_POOL_MAX_BUCKETS](#)

Maximum number of block sizes in pool structures

Maximum number of block sizes in pool structures

Global CFE_PLATFORM_CORE_MAX_STARTUP_MSEC

CFE core application startup timeout

Global CFE_PLATFORM_ES_APP_KILL_TIMEOUT

Define ES Application Kill Timeout

Define ES Application Kill Timeout

Global CFE_PLATFORM_ES_APP_SCAN_RATE

Define ES Application Control Scan Rate

Define ES Application Control Scan Rate

Global CFE_PLATFORM_ES_CDS_MAX_NUM_ENTRIES

Define Maximum Number of Registered CDS Blocks

Define Maximum Number of Registered CDS Blocks

Global CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_01

Define ES Critical Data Store Memory Pool Block Sizes

Define ES Critical Data Store Memory Pool Block Sizes

Global CFE_PLATFORM_ES_CDS_SIZE

Define Critical Data Store Size

Define Critical Data Store Size

Global CFE_PLATFORM_ES_DEFAULT_APP_LOG_FILE

Default Application Information Filename

Default Application Information Filename

Global CFE_PLATFORM_ES_DEFAULT_CDS_REG_DUMP_FILE

Default Critical Data Store Registry Filename

Default Critical Data Store Registry Filename

Global CFE_PLATFORM_ES_DEFAULT_ER_LOG_FILE

Default Exception and Reset (ER) Log Filename

Default Exception and Reset (ER) Log Filename

Global CFE_PLATFORM_ES_DEFAULT_PERF_DUMP_FILENAME

Default Performance Data Filename

Default Performance Data Filename

Global CFE_PLATFORM_ES_DEFAULT_POR_SYSLOG_MODE

Define Default System Log Mode following Power On Reset

Define Default System Log Mode following Power On Reset

Global CFE_PLATFORM_ES_DEFAULT_PR_SYSLOG_MODE

Define Default System Log Mode following Processor Reset

Define Default System Log Mode following Processor Reset

Global CFE_PLATFORM_ES_DEFAULT_STACK_SIZE

Define Default Stack Size for an Application

Define Default Stack Size for an Application

Global CFE_PLATFORM_ES_DEFAULT_SYSLOG_FILE

Default System Log Filename

Default System Log Filename

Global CFE_PLATFORM_ES_DEFAULT_TASK_LOG_FILE

Default Application Information Filename

Default Application Information Filename

Global CFE_PLATFORM_ES_ER_LOG_ENTRIES

Define Max Number of ER (Exception and Reset) log entries

Define Max Number of ER (Exception and Reset) log entries

Global CFE_PLATFORM_ES_ER_LOG_MAX_CONTEXT_SIZE

Maximum size of CPU Context in ES Error Log

Maximum size of CPU Context in ES Error Log

Global CFE_PLATFORM_ES_MAX_APPLICATIONS

Define Max Number of Applications

Define Max Number of Applications

Global CFE_PLATFORM_ES_MAX_GEN_COUNTERS

Define Max Number of Generic Counters

Define Max Number of Generic Counters

Global CFE_PLATFORM_ES_MAX_LIBRARIES

Define Max Number of Shared libraries

Define Max Number of Shared libraries

Global CFE_PLATFORM_ES_MAX_MEMORY_POOLS

Maximum number of memory pools

Maximum number of memory pools

Global CFE_PLATFORM_ES_MAX_PROCESSOR_RESETS

Define Number of Processor Resets Before a Power On Reset

Define Number of Processor Resets Before a Power On Reset

Global CFE_PLATFORM_ES_MEM_BLOCK_SIZE_01

Define Default ES Memory Pool Block Sizes

Define Default ES Memory Pool Block Sizes

Global CFE_PLATFORM_ES_MEMPOOL_ALIGN_SIZE_MIN

Define Memory Pool Alignment Size

Define Memory Pool Alignment Size

Global CFE_PLATFORM_ES_NONVOL_DISK_MOUNT_STRING

Default virtual path for persistent storage

Default virtual path for persistent storage

Global CFE_PLATFORM_ES_NONVOL_STARTUP_FILE

ES Nonvolatile Startup Filename

ES Nonvolatile Startup Filename

Global CFE_PLATFORM_ES_PERF_CHILD_MS_DELAY

Define Performance Analyzer Child Task Delay

Define Performance Analyzer Child Task Delay

Global CFE_PLATFORM_ES_PERF_CHILD_PRIORITY

Define Performance Analyzer Child Task Priority

Define Performance Analyzer Child Task Priority

Global CFE_PLATFORM_ES_PERF_CHILD_STACK_SIZE

Define Performance Analyzer Child Task Stack Size

Define Performance Analyzer Child Task Stack Size

Global CFE_PLATFORM_ES_PERF_DATA_BUFFER_SIZE

Define Max Size of Performance Data Buffer

Define Max Size of Performance Data Buffer

Global CFE_PLATFORM_ES_PERF_ENTRIES_BTWN_DLYS

Define Performance Analyzer Child Task Number of Entries Between Delay

Define Performance Analyzer Child Task Number of Entries Between Delay

Global CFE_PLATFORM_ES_PERF_FILTMASK_ALL

Define Filter Mask Setting for Enabling All Performance Entries

Define Filter Mask Setting for Enabling All Performance Entries

Global CFE_PLATFORM_ES_PERF_FILTMASK_INIT

Define Default Filter Mask Setting for Performance Data Buffer

Define Default Filter Mask Setting for Performance Data Buffer

Global CFE_PLATFORM_ES_PERF_FILTMASK_NONE

Define Filter Mask Setting for Disabling All Performance Entries

Define Filter Mask Setting for Disabling All Performance Entries

Global CFE_PLATFORM_ES_PERF_TRIGMASK_ALL

Define Filter Trigger Setting for Enabling All Performance Entries

Define Filter Trigger Setting for Enabling All Performance Entries

Global CFE_PLATFORM_ES_PERF_TRIGMASK_INIT

Define Default Filter Trigger Setting for Performance Data Buffer

Define Default Filter Trigger Setting for Performance Data Buffer

Global CFE_PLATFORM_ES_PERF_TRIGMASK_NONE

Define Default Filter Trigger Setting for Disabling All Performance Entries

Define Default Filter Trigger Setting for Disabling All Performance Entries

Global CFE_PLATFORM_ES_POOL_MAX_BUCKETS

Maximum number of block sizes in pool structures

Maximum number of block sizes in pool structures

Global CFE_PLATFORM_ES_RAM_DISK_MOUNT_STRING

Default virtual path for volatile storage

Default virtual path for volatile storage

Global CFE_PLATFORM_ES_RAM_DISK_NUM_SECTORS

ES Ram Disk Number of Sectors

ES Ram Disk Number of Sectors

Global CFE_PLATFORM_ES_RAM_DISK_PERCENT_RESERVED

Percentage of Ram Disk Reserved for Decompressing Apps

Percentage of Ram Disk Reserved for Decompressing Apps

Global CFE_PLATFORM_ES_RAM_DISK_SECTOR_SIZE

ES Ram Disk Sector Size

ES Ram Disk Sector Size

Global [CFE_PLATFORM_ES_START_TASK_PRIORITY](#)

Define ES Task Priority

Define ES Task Priority

Global [CFE_PLATFORM_ES_START_TASK_STACK_SIZE](#)

Define ES Task Stack Size

Define ES Task Stack Size

Global [CFE_PLATFORM_ES_STARTUP_SCRIPT_TIMEOUT_MSEC](#)

Startup script timeout

Startup script timeout

Global [CFE_PLATFORM_ES_STARTUP_SYNC_POLL_MSEC](#)

Poll timer for startup sync delay

Poll timer for startup sync delay

Global [CFE_PLATFORM_ES_SYSTEM_LOG_SIZE](#)

Define Size of the cFE System Log.

Define Size of the cFE System Log.

Global [CFE_PLATFORM_ES_USER_RESERVED_SIZE](#)

Define User Reserved Memory Size

Define User Reserved Memory Size

Global [CFE_PLATFORM_ES_VOLATILE_STARTUP_FILE](#)

ES Volatile Startup Filename

ES Volatile Startup Filename

Global [CFE_PLATFORM_EVS_START_TASK_PRIORITY](#)

Define EVS Task Priority

Define EVS Task Priority

Global [CFE_PLATFORM_EVS_START_TASK_STACK_SIZE](#)

Define EVS Task Stack Size

Define EVS Task Stack Size

Global [CFE_PLATFORM_SB_MEM_BLOCK_SIZE_01](#)

Define SB Memory Pool Block Sizes

Define SB Memory Pool Block Sizes

Global [CFE_PLATFORM_SB_POOL_MAX_BUCKETS](#)

Number of block sizes in SB memory pool structure

Global [CFE_PLATFORM_SB_START_TASK_PRIORITY](#)

Define SB Task Priority

Define SB Task Priority

Global [CFE_PLATFORM_SB_START_TASK_STACK_SIZE](#)

Define SB Task Stack Size

Define SB Task Stack Size

Global [CFE_PLATFORM_TBL_START_TASK_PRIORITY](#)

Define TBL Task Priority

Define TBL Task Priority

Define TBL Task Priority

Global [CFE_PLATFORM_TBL_START_TASK_STACK_SIZE](#)

Define TBL Task Stack Size

Define TBL Task Stack Size

Define TBL Task Stack Size

1.10 cFE Event Services Overview

Event Services (EVS) provides centralized control for the processing of event messages originating from the EVS task itself, other cFE core applications (ES, SB, TIME, and TBL), and from cFE applications. Event messages are asynchronous messages that are used to inform the operator of a significant event from within the context of a registered application or core service. EVS provides various ways to filter event messages in order to manage event message generation.

Note for messages outside the context of a registered application (for example early in app initialization or if registration fails) [CFE_ES_WriteToSysLog](#) can be used for reporting.

For more information on cFE Event Services, see the following sections:

- [Event Message Format](#)
- [Local Event Log](#)
- [Event Message Control](#)
- [Event Message Filtering](#)
- [EVS Registry](#)
- [EVS Counters](#)
- [Resetting EVS Counters](#)
- [Effects of a Processor Reset on EVS](#)
- [EVS squelching of misbehaving apps](#)
- [Frequently Asked Questions about Event Services](#)

1.10.1 Event Message Format

Event messages are software bus messages that contain the following fields:

- Timestamp
- Event Type
- Spacecraft ID
- Processor ID
- Application Name
- Event ID
- Message

The *Timestamp* corresponds to when the event was generated, in spacecraft time. The *Event Type* is one of the following: DEBUG, INFO, ERROR or CRITICAL. The *Spacecraft ID* and *Processor ID* identify the spacecraft and processor from which the event was generated. Note that the *Spacecraft ID* is defined in the `cfe_mission_cfg.h` file; The *Processor ID* is defined in the appropriate `cfe_platform_cfg.h` file. The *Application Name* refers to the Application that issued the event message as specified on application startup (either startup script or app start command). The *Event ID* is an Application unique number that identifies the event. The *Message* is an ASCII text string describing the event. Event messages may have parameters associated with the event message. EVS formats the parameters such that they are part of the ASCII text string that make up the event message.

In order to accommodate missions that have limited telemetry bandwidth, EVS can be configured such that the ASCII text string part of the event message is omitted, thus reducing the size of each event message. This is referred to as *Short Format*; Event messages including the ASCII text string are referred to as *Long Format*. The default setting is specified in the `cfe_platform_cfg.h` file. EVS also provides commands in order to set the mode (short or long).

Since the design of the cFE's Software Bus is based on run-time registration, no predetermined message routing is defined, hence it is not truly correct to say that events are generated as telemetry. Technically, EVS generates events in the form of software bus messages. Applications such as Telemetry Output and Data Storage can then subscribe to these messages making them telemetry. For the purposes of this document, any references to telemetry assumes that a telemetry application subscribes to the EVS event software bus message and routes it to the ground as telemetry. Note that short format event messages on the Software Bus have different message lengths than long form messages and do not include any part of the long format message string.

The EVS can be configured via ground command to send event messages out one or more message ports. These message ports may include ports such as debug, console, and UART. Messages sent out of the message ports will be in ASCII text format. This is generally used for lab purposes. Note that the event mode (short or long) does affect the event message content sent out these message ports.

1.10.2 Local Event Log

In addition to generating a software bus message, EVS logs the event message to a Local Event Log. Note that this is an optional feature that must be enabled via the `cfe_platform_cfg.h` file. The Local Event Log resides on the same processor as the EVS which is used to store events without relying on an external bus. In multi-processor cFE configurations the Local Event Buffer preserves event messages during non-deterministic processor initialization sequences and during failure scenarios. In order to obtain the contents of the Local Event Log, a command must be sent to write the contents of the buffer to a file which can then be sent to the ground via a file transfer mechanism. Note that event messages stored in the EVS Local Event Log are always long format messages and are not affected by the event mode (short or long).

EVS provides a command in order to [clear the Local Event Log](#) .

1.10.2.1 Local Event Log Mode

EVS can be configured to control the Local Event Log to either discard or overwrite the contents of the log when it becomes full. If the mode is set to overwrite, the log is treated like a circular buffer, overwriting the oldest event message contained in the log first. This control is configured by default in the `cfe_platform_cfg.h` file but can be modified by [a command](#) .

1.10.3 Event Message Control

In order for an application to be serviced by EVS, it must be registered with EVS. EVS provides various commands in order to control the event messages that are generated as software bus messages.

1.10.3.1 Event Message Control - By Type

The highest level of event message control that EVS provides is the ability to enable and disable event message types. As mentioned above, there are four event types. They are:

1. DEBUG
2. INFORMATION
3. ERROR
4. CRITICAL

When commands are sent to [enable](#) or [disable](#) a particular type of event message, ALL event messages of the specified type are affected. Typically, event messages of type DEBUG are disabled on-orbit. Note that EVS provides the capability to affect multiple types within one command using a bit mask. Note also that the configuration parameter [CFE_PLATFORM_EVS_DEFAULT_TYPE_FLAG](#) in the `cfe_platform_cfg.h` file specifies which event message types are enabled/disabled by default.

1.10.3.2 Event Message Control - By Application

Commands are available to [enable](#) and [disable](#) the generation of event messages for a particular application. The result is that ALL event messages for the specified Application are affected (i.e. enabled or disabled).

1.10.3.3 Event Message Control - By Event Type for an Application

EVS also provides the capability to [enable](#) / [disable](#) an event type for a particular application. Note that EVS provides the capability to affect multiple event types within one command using a bit mask.

1.10.3.4 Event Message Control - Individual Events

There are two ways to control the generation of individual events depending on whether the application's event message has been registered with EVS or not.

1.10.3.4.1 Modifying a registered event message filter

When an application registers with EVS, the application has the option of specifying the events that it wants to register for filtering along with the [Event Message Filtering](#) (only the Binary Filtering Scheme exists currently). Note that applications are limited in the number of events that they can register for filtering (see [CFE_PLATFORM_EVS_MAX_EVENT_FILTERS](#) in `cfe_platform_cfg.h` for the mission defined limit). The filtering method uses a mask to determine if the message is forwarded to the software bus, making it available in telemetry (see [Event Message Filtering](#) for a description on filtering). Commands are available to [modify the filter mask](#) for any registered event.

An on-orbit mission, for example, might be experiencing a problem resulting in an application's event message being repeatedly issued, flooding the downlink. If the event message for the application is registered with EVS, then a command can be issued to set the event message filter to the specified value in order to prevent flooding of the downlink.

1.10.3.4.2 Adding/Removing an event message for filtering

Commands are also available to add filtering for those events that are not registered for filtering. Once an event is [registered for filtering](#) , the filter can be modified (see above) or [removed](#) .

An on-orbit mission, for example, might be experiencing a problem resulting in an event message being repeatedly issued, flooding the downlink. If the event message was not registered with EVS for filtering then the ground can add (i.e. register) the offending application's event for filtering (much like an application registers the event during initialization).

EVS also supports the ability to [remove](#) (i.e. unregister) an application's event message. Once it is removed, the event will no longer be filtered. Note that commands issued to disable events by event type, by application or by event type for an application are still valid and could affect this particular event.

1.10.4 Event Message Filtering

EVS uses a hexadecimal bit mask that controls how often a message is filtered. An event's filter mask is bit-wise ANDed with the event's event counter. There is one event counter for each event ID. If the result of the ANDing is zero then the message is sent.

Filter masks can be set so that one out of 1, 2, 4, 8 events are sent. Some examples of masks that use this pattern are: (0x0000, Every one), (0x0001, One of every 2), (0x0003, One of every 4), and (0x0007, One of every 8).

Filter masks can also be set so that only the first *n* events are sent. For example, the mask 0xFFFF generates one event message and then stops. Note that when the filter counter is reset to zero by command, this will restart the counting and enable *n* more events to be sent.

Event messages will be filtered until CFE_EVS_MAX_FILTER_COUNT events of the filtered event ID from the application have been received. After this, the filtering will become locked (no more of that event will be received by the ground) until the filter is either reset or deleted by ground command. This is to prevent the counter from rolling over, which would cause some filters to behave improperly. An event message will be sent when this maximum count is reached.

The following shows an example of how filtering works using a filter mask of x'0001', resulting in sending every other event:

	packet x	packet X+1	packet X+2	packet X+3	packet X+4	...
Event ID counter	x'0000'	x'0001'	x'0002'	x'0003'	x'0004'	
Event Filter mask	x'0001'	x'0001'	x'0001'	x'0001'	x'0001'	
Bitwise AND results	x'0000'	x'0001'	x'0000'	x'0001'	x'0000'	
Send event?	Yes	No	Yes	No	Yes	

In this example, the ground uses a filter mask of x'FFFE' resulting in the first two events being sent and then no more.

	packet x	packet X+1	packet X+2	packet X+3	packet X+4	...
Event ID counter	x'0000'	x'0001'	x'0002'	x'0003'	x'0004'	
Event Filter mask	x'FFFE'	x'FFFE'	x'FFFE'	x'FFFE'	x'FFFE'	
Bitwise AND results	x'0000'	x'0000'	x'0002'	x'0002'	x'0004'	
Send event?	Yes	Yes	No	No	No	

See [cfe_evs.h](#) for predefined macro values which can be used for masks.

1.10.5 EVS Registry

EVS maintains information on each registered application and all events registered for an application.

The registry contains the following information for each Registered Application:

- Active Flag - If equal to FALSE (0), all events from this Application are Filtered

- Event Count - Total number of events issued by this Application. Note that this value stop incrementing at 65535.

The following information for each Filtered Event (up to [CFE_PLATFORM_EVS_MAX_EVENT_FILTERS](#))↔
:

- Event ID - Event ID for event whose filter has been defined
- Mask - Binary Filter mask value (see [Event Message Filtering](#) for an explanation)
- Count - Current number of times this Event ID has been issued by this Application

1.10.6 EVS Counters

There are 2 types of counters in EVS housekeeping telemetry:

- Total events sent counter
- Number of events sent for each Application

The difference is that the first one is the sum of all of the event messages sent. Both of these represent events that are actually sent (by EVS to the software bus). If an event message is filtered or disabled, neither counter is incremented.

There are other counters available that show how many event messages were generated by an App, however, these are only available for those events that are registered for filtering hence if you have a message that is not registered for filtering and the message type (e.g. DEBUG) is disabled then you won't know if the event was ever issued by an application. These counters are available by sending a command to [write the EVS Application Data](#) and transferring the file to the ground.

1.10.7 Resetting EVS Counters

As far as reset commands, there are 4 commands available:

1. [Reset the total events sent counter](#)
2. [Reset the events sent counter for a particular Application](#) - e.g. reset the LC application events counter
3. [Reset all of the event counters for a particular registered event for a particular Application](#) - e.g. Reset event counter for Event ID 5 for the LC Application.
4. [Reset all of the event counters for ALL registered events for a particular App](#) - e.g. Reset all registered event counters for LC.

Note that there is currently no way to reset ALL of the events sent counters for all of the Apps with one command.

1.10.8 Effects of a Processor Reset on EVS

On a processor reset, the EVS Registry is cleared such that applications must re-register with EVS in order to use EVS services. All counters are also cleared with the exceptions of those listed below.

On a processor reset, the following EVS data is preserved (if the cFE is configured to include an [Local Event Log](#)):

- Local Event Log if the Local Event Log Mode is configured to Discard (1). If the Local Event Log Mode is configured to Overwrite (0), the contents of the log may be overwritten depending on the size and contents of the log prior to the reset.
- Local Event Log Full Flag
- Local Event Log overflow counter

The Local Event Log Mode (overwrite/discard) is set to the configured value specified in the `cfe_platform_cfg.h` file. The default value is Discard (1). Discard mode will guarantee the contents of the event log are preserved over a processor restart.

This provides the ground with the capability to write the Local Event Log to a file and transfer it to the ground in order to help debug a reset.

1.10.9 EVS squelching of misbehaving apps

Event squelching is an optional feature for suppressing excessive events from misbehaving apps. It is enabled by setting `CFE_PLATFORM_EVS_MAX_APP_EVENT_BURST` to a nonzero positive value, and `CFE_PLATFORM_EVS_APP_EVENTS_PER_SEC` equal to or less than that value.

`CFE_PLATFORM_EVS_MAX_APP_EVENT_BURST` controls the maximum events that can be sent at a given moment, and `CFE_PLATFORM_EVS_APP_EVENTS_PER_SEC` is the sustained event throughput per second.

The suppression mechanism initializes with $CFE_PLATFORM_EVS_MAX_APP_EVENT_BURST * 1000$ credits. Each event costs 1000 credits. Credits are restored at a rate of $CFE_PLATFORM_EVS_APP_EVENTS_PER_SEC * 1000$ up to a maximum balance of $CFE_PLATFORM_EVS_MAX_APP_EVENT_BURST * 1000$, and the maximum "debt" is $-CFE_PLATFORM_EVS_MAX_APP_EVENT_BURST * 1000$. When the credit count crosses from positive to negative, a squelched event message is emitted and events are suppressed, until the credit count becomes positive again.

Figure EVS-1 is a notional state diagram of the event squelching mechanism.

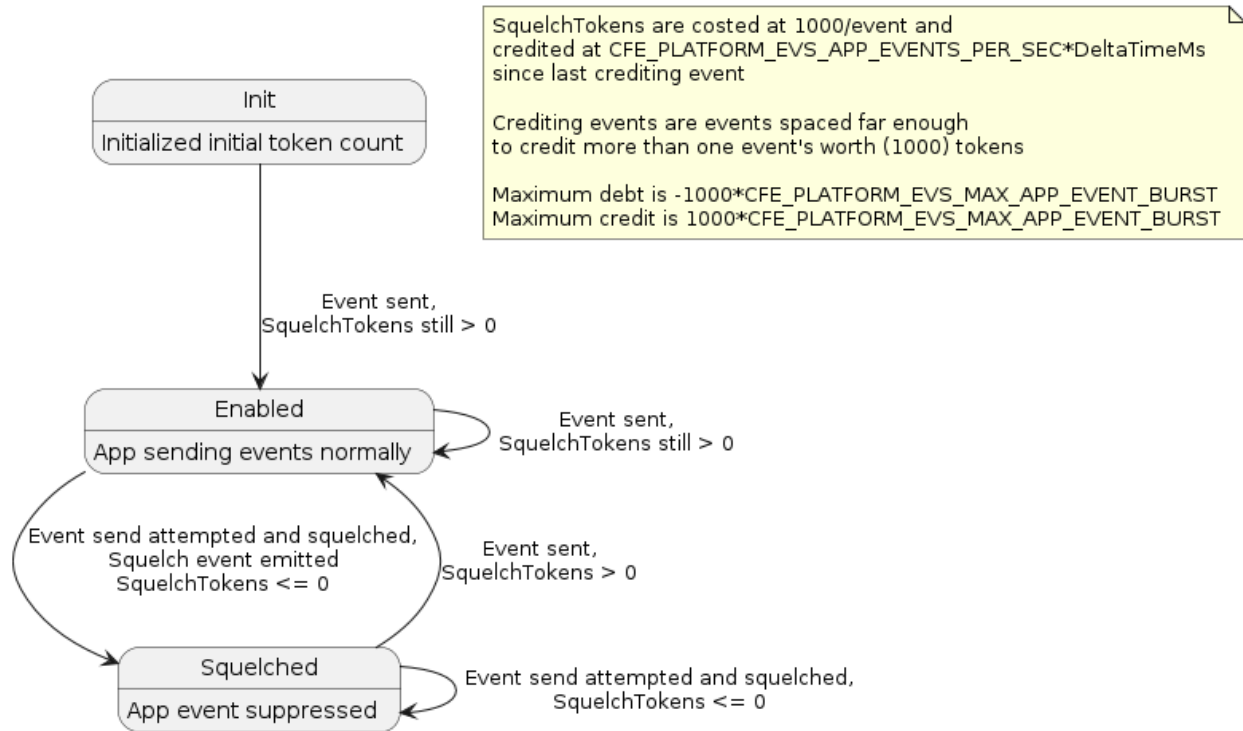


Figure 1 Figure EVS-1: EVS Squelching State Diagram

1.10.10 Frequently Asked Questions about Event Services

(Q) My telemetry stream is being flooded with the same event message. How do I make it stop?

The most direct way to stop an event message from flooding your downlink stream is to send a command to EVS to filter the offending event (see [Event Message Control](#) or [\\$sc_\\$cpu_EVS_SetBinFiltrMask](#)). In order to stop the event message from being sent, a bit mask of '0xFFFF' should be used. If the event is not currently registered for filtering, the event message must be added using the command [\\$sc_\\$cpu_EVS_AddEvtFiltr](#).

(Q) I filtered an event message and would now like to see it again. What do I do in order to see those events again?

If the event message that you are interested is registered with EVS for filtering, then you have 2 options:

1. You can use the [\\$sc_\\$cpu_EVS_SetBinFiltrMask](#) command using a bit mask of '0x0000' which will result in getting all of the events for that Event Id
2. You can remove the registration of that event with EVS (see [\\$sc_\\$cpu_EVS_DelEvtFiltr](#)).

Note that option (1) is the preferred method.

(Q) What is the purpose of DEBUG event messages?

Event message of type "DEBUG" are primarily used during flight software development in order to provide information that is most likely not needed on orbit. Some commands send debug event messages as verification that a

command request was received. When writing the EVS local event log to a file, for example, an event message of type *DEBUG* is issued. On orbit, this event message is probably not needed. Instead, the command counter is used for command verification.

(Q) How do I find out which events are registered for filtering?

EVS provides a command (`$sc_$cpu_EVS_WriteAppData2File`) which generates a file containing all of the applications that have registered with EVS and all of the filters that are registered for each application. Note that EVS merely generates the file. The file must be transferred to the ground in order to view it.

(Q) Why do I see event messages in my console window?

By default, the events are configured to transmit out a "port" that shows event messages in the console

(Q) What is the difference between event services and the ES System Log

Events are within the context of an App or cFE Service (requires registration with ES). The system log can be written to outside of the Application or cFE Service context, for example during application startup to report errors before registration.

1.11 cFE Event Services Commands

Upon receipt of any command, the Event Services application will confirm that the message length embedded within the header (from `CFE_MSG_GetSize()`) matches the expected length of that message, based on the size of the C structure defining that command. If there is any discrepancy between the expected and actual message size, EVS will generate the `CFE_EVS_LEN_ERR_EID` event, increment the command error counter (`$sc_$cpu_EVS_CMDEC`), and the command will *not* be accepted for processing.

The following is a list of commands that are processed by the cFE Event Services Task.

Global `CFE_EVS_ADD_EVENT_FILTER_CC`

Add Application Event Filter

Global `CFE_EVS_CLEAR_LOG_CC`

Clear Event Log

Global `CFE_EVS_DELETE_EVENT_FILTER_CC`

Delete Application Event Filter

Global `CFE_EVS_DISABLE_APP_EVENT_TYPE_CC`

Disable Application Event Type

Global `CFE_EVS_DISABLE_APP_EVENTS_CC`

Disable Event Services for an Application

Global `CFE_EVS_DISABLE_EVENT_TYPE_CC`

Disable Event Type

Global `CFE_EVS_DISABLE_PORTS_CC`

Disable Event Services Output Ports

Global `CFE_EVS_ENABLE_APP_EVENT_TYPE_CC`

Enable Application Event Type

Global [CFE_EVS_ENABLE_APP_EVENTS_CC](#)

Enable Event Services for an Application

Global [CFE_EVS_ENABLE_EVENT_TYPE_CC](#)

Enable Event Type

Global [CFE_EVS_ENABLE_PORTS_CC](#)

Enable Event Services Output Ports

Global [CFE_EVS_NOOP_CC](#)

Event Services No-Op

Global [CFE_EVS_RESET_ALL_FILTERS_CC](#)

Reset All Event Filters for an Application

Global [CFE_EVS_RESET_APP_COUNTER_CC](#)

Reset Application Event Counters

Global [CFE_EVS_RESET_COUNTERS_CC](#)

Event Services Reset Counters

Global [CFE_EVS_RESET_FILTER_CC](#)

Reset an Event Filter for an Application

Global [CFE_EVS_SET_EVENT_FORMAT_MODE_CC](#)

Set Event Format Mode

Global [CFE_EVS_SET_FILTER_CC](#)

Set Application Event Filter

Global [CFE_EVS_SET_LOG_MODE_CC](#)

Set Logging Mode

Global [CFE_EVS_WRITE_APP_DATA_FILE_CC](#)

Write Event Services Application Information to File

Global [CFE_EVS_WRITE_LOG_DATA_FILE_CC](#)

Write Event Log to File

1.12 cFE Event Services Telemetry

The following are telemetry packets generated by the cFE Event Services Task.

Global [CFE_EVS_HousekeepingTlm_Payload_t](#)

Event Services Housekeeping Telemetry Packet

Global [CFE_EVS_LongEventTlm_Payload_t](#)

Event Message Telemetry Packet (Long format)

Global [CFE_EVS_ShortEventTlm_Payload_t](#)

Event Message Telemetry Packet (Short format)

1.13 cFE Event Services Configuration Parameters

The following are configuration parameters used to configure the cFE Event Services either for each platform or for a mission as a whole.

Global **CFE_MISSION_EVS_MAX_MESSAGE_LENGTH**

Maximum Event Message Length

Maximum Event Message Length

Global **CFE_PLATFORM_EVS_APP_EVENTS_PER_SEC**

Sustained number of event messages per second per app before squelching

Sustained number of event messages per second per app before squelching

Global **CFE_PLATFORM_EVS_DEFAULT_APP_DATA_FILE**

Default EVS Application Data Filename

Default EVS Application Data Filename

Global **CFE_PLATFORM_EVS_DEFAULT_LOG_FILE**

Default Event Log Filename

Default Event Log Filename

Global **CFE_PLATFORM_EVS_DEFAULT_LOG_MODE**

Default EVS Local Event Log Mode

Default EVS Local Event Log Mode

Global **CFE_PLATFORM_EVS_DEFAULT_MSG_FORMAT_MODE**

Default EVS Message Format Mode

Default EVS Message Format Mode

Global **CFE_PLATFORM_EVS_DEFAULT_TYPE_FLAG**

Default EVS Event Type Filter Mask

Default EVS Event Type Filter Mask

Global **CFE_PLATFORM_EVS_LOG_MAX**

Maximum Number of Events in EVS Local Event Log

Maximum Number of Events in EVS Local Event Log

Global **CFE_PLATFORM_EVS_MAX_APP_EVENT_BURST**

Maximum number of event before squelching

Maximum number of event before squelching

Global **CFE_PLATFORM_EVS_MAX_EVENT_FILTERS**

Define Maximum Number of Event Filters per Application

Define Maximum Number of Event Filters per Application

Global **CFE_PLATFORM_EVS_PORT_DEFAULT**

Default EVS Output Port State

Default EVS Output Port State

1.14 cFE Software Bus Overview

The Software Bus (SB) handles communication between software tasks on a processor. All tasks communicate with each other, with hardware devices, and with the ground by sending command and telemetry messages. The software bus provides an application programming interface (API) to other tasks for sending and receiving messages. This API is independent of the underlying operating system so that tasks can use the same interface regardless of which processor they reside on. Refer to the [cFE Application Programmer's Interface \(API\) Reference](#) for detailed information about the API functions.

The software bus is used internally by the flight software, and normally does not require attention from the ground. However, because of the scalability and the dynamic nature of the software bus, it is strongly recommended that each project carefully review the SB statistics and SB memory pool to be sure adequate margin is met on the configurable items.

The cFE software bus uses a dynamic protocol and builds its routing table at run-time through the SB subscribe API's. Also the cFE software bus pipes are created at run-time through the [CFE_SB_CreatePipe](#) API. Because the routing is established, and pipes are created at run-time, it is necessary to have a clear view of the routing details on command. The cFE software bus allows the user to dump the routing table, the pipe table, the message map and the statistics packet. Each of these items are described in detail in the corresponding section of this document.

- [Software Bus Terminology](#)
- [Autonomous Actions](#)
- [Operation of the SB Software](#)
- [Frequently Asked Questions about Software Bus](#)

1.14.1 Software Bus Terminology

In order to fully understand the Software Bus, it is imperative that the basic terms used to describe its features are also understood. Below are the critical terms that help identify what the Software Bus accomplishes for each Application:

- [Messages](#)
- [Pipes](#)
- [Subscriptions](#)
- [Memory](#)

1.14.1.1 Messages

The sole purpose of the software bus is to provide applications a way to send messages to each other. The term message and the term packet are used interchangeably throughout this document. A message is a combined set of bytes with a predefined format that is used as the basis of communication on a spacecraft. All commands, telemetry, and other data that are passed between the ground and the spacecraft, and between subsystems of the spacecraft, are considered to be messages. The most common message format is CCSDS (Consultative Committee for Space Data Systems) in [CCSDS Space Packet Protocol](#), but can be customized by replacing the message module.

There are two general types of messages - commands (or command packets) and telemetry (or telemetry packets). Command packets are sent to a particular software task from the ground (or another task). Telemetry packets are sent from a particular software task to the ground (or other tasks).

The concept of a message identifier is utilized to provide abstraction from header implementation, often abbreviated as message ID, MsgId, or MID. Header and message identifier values should not be accessed directly to avoid implementation specific dependencies.

Telemetry packets typically contain a timestamp that indicates when the packet was produced. Command packets typically contain a command code that identifies the particular type of command.

The message module provides APIs for 'setting' and 'getting' the fields in the header of the message. The message module was separated from software bus to enable users to customize message headers without requiring clone and own of the entire cfe repository. To customize, remove the built in msg module from the build and replace with custom implementation. See sample target definitions folder for examples.

Following the header is the user defined message data.

1.14.1.2 Pipes

The destinations to which messages are sent are called pipes. These are queues that can hold messages until they are read out and processed by a task. Each pipe is created at run-time through the [CFE_SB_CreatePipe](#) API. The pipe name and the pipe depth are given as arguments in the API. The pipe identifier (or PipeId) is given back to the caller after the API is executed. Each pipe can be read by only one task, but a task may read more than one pipe. Only the pipe owner is allowed to subscribe to messages on the pipe.

The Pipe IDs are specific to a particular processor (that is, the same ID number may refer to a different pipe on each processor). The pipe information for all pipes that have been created, may be requested at anytime by sending the ['Write Pipe Info' SB command](#). The software bus also provides a set of figures regarding capacity, current utilization and high water marks relevant to pipes. This information may be requested by sending the command to [dump the SB statistics packet](#).

1.14.1.3 Subscriptions

A subscription is a run-time request for a particular message to be sent to a particular pipe. If the caller of the subscribe API is not the owner of the pipe, the request is rejected and an error event is sent. The application that creates the pipe is considered the owner of the pipe. The pipe specified in the subscription is sometimes referred to as the destination of the message. There are a maximum number of destinations for a particular message. This value is specified by the platform configuration parameter [CFE_PLATFORM_SB_MAX_DEST_PER_PKT](#).

As subscriptions are received, the destinations are added to the head of a linked list. During the sending of a message, the list is traversed beginning at the head of the list. Therefore the message will first be sent to the last subscriber. If an application has timing constraints and needs to receive a message in the shortest possible time, the developer may consider holding off its subscription until other applications have subscribed to the message.

The message limit specifies the maximum number of messages (with the specified Message ID) that are allowed on the specified pipe at any time. This limit is specified by the application at the time of the subscription. If the application uses the [CFE_SB_Subscribe](#) API, a message limit default value of four is used. If this default value is not sufficient, the caller would use the [CFE_SB_SubscribeEx](#) API that allows the message limit to be specified.

The software bus also provides the user with an option to unsubscribe to a message. The [unsubscribe API](#) takes two parameters, Message ID and Pipe ID. Only the owner of a pipe may unsubscribe to messages on that pipe.

1.14.1.4 Memory

The software bus statically allocates a block of memory for message buffers and subscription blocks. The size of this memory block is defined by the platform configuration parameter [CFE_PLATFORM_SB_BUF_MEMORY_BYTES](#). The memory is managed by the cFE ES memory pool and is used only by the software bus. The ES memory pool allows an application to define the block sizes for the pool at compile time. These sizes are defined by the platform configuration parameters prefixed with [CFE_SB_MEM_BLOCK_SIZE](#) (for example, [CFE_PLATFORM_SB_MEM_BLOCK_SIZE_01](#)). It is recommended that a project tailor these values for the mission, based on the software bus packet sizes.

At the time a message is sent, two buffers are allocated from the pool. One for a buffer descriptor ([CFE_SB_BufferD_t](#)) and one for the size of the packet. Both buffers are returned to the pool when the message has been received by all recipients. More precisely, if there is one recipient for a message, the message buffers will be released on the following call to [CFE_SB_ReceiveBuffer](#) for the pipe that received the buffer.

Also when subscriptions are received through the subscribe API's, the software bus allocates a subscription block ([CFE_SB_DestinationD_t](#)) from the pool. The subscription blocks are returned to the pool if and when the subscription is nullified through a [CFE_SB_Unsubscribe](#) call.

The software bus provides a set of figures regarding memory capacity, current memory utilization and high water marks relevant to the SB memory pool. This information may be requested by sending the command to dump the SB statistics packet. In addition, the current memory utilization value and the 'unmarked memory' value ([CFE_PLATFORM_SB_BUF_MEMORY_BYTES](#) minus peak memory in use) are sent in software bus housekeeping telemetry. The unmarked memory value should be monitored regularly to ensure that the value (in bytes) does not continue to decline or approach zero. If this value were to approach zero, there is a possibility that memory requests would fail which may inhibit the sending of a message. The current memory utilization value should also be monitored to ensure the system contains no memory leaks. The value (in bytes) should remain stable under nominal conditions. Refer to the ES users guide for more information regarding the ES Memory Pool.

1.14.2 Autonomous Actions

The software bus is primarily a set of library routines that are called by other software tasks to send and receive packets. The software bus does not perform any operations autonomously, except for sending event messages if errors are detected during the transfer of packets.

As do other tasks, the SB task sends out housekeeping telemetry when requested through the 'Send Housekeeping Data' command.

1.14.3 Operation of the SB Software

- [Initialization](#)
- [All Resets](#)
- [Message Routing](#)
- [Packet Sequence Values](#)
- [Message Limit Error](#)
- [Pipe Overflow Error](#)
- [SB Event Filtering](#)
- [Diagnostic Data](#)
- [Control of Packet Routing](#)
- [Quality of Service](#)
- [Known Problem](#)

1.14.3.1 Initialization

No action is required by the ground to initialize the software bus. The software bus initializes internal data structures and tables the same way regardless of the type of reset.

1.14.3.2 All Resets

The software bus does not preserve any information across a reset of any kind. The software bus initializes internal data structures and tables the same way regardless of the type of reset. The routing is reestablished as the system initializes. It is normal procedure for each task of the system to create the pipe or pipes it needs and do all of its subscriptions during task initialization.

After any reset the following statements are true:

- The routing table is cleared and does not contain any routes.

- All subscriptions are lost and must be regenerated.
- The pipe table contains no data, all pipes must be recreated.
- Any packets in transit at the time of the reset are lost.
- The sequence counters for telemetry packets will begin again with a value of one.

1.14.3.3 Message Routing

In the software bus, all messages are processed in a similar way. The software bus uses the Message ID and the packet length fields (contained in the header) for routing the message to the destination pipe. If either of these two fields do not pass validation, the software bus generates an error event and aborts the delivery process. The software bus performs some validation checks by simply checking message header values against mission or platform configuration parameters. Messages originating from various tasks or instruments are routed to one or more pipes, where they wait until read by a task. The routing configuration for each message is established when applications call one of the SB subscribe APIs. The subscribe APIs take a Message ID and a Pipe ID as parameters. The routing for each packet is stored in SB memory and may be requested at any time by sending the 'Send Routing Info' command. The software bus also provides a set of figures regarding capacity, current utilization and high water marks relevant to the routing. This information may be requested by sending the command to dump the SB statistics packet.

1.14.3.4 Packet Sequence Values

The sequence count behavior depends on if the message is a command type or telemetry type.

The sequence counter for command messages is not altered by the software bus.

For a telemetry message, the behavior is controlled via API input parameters when sending. When enabled, the software bus will populate the packet sequence counter using an internal counter that gets initialized upon the first subscription to the message (first message will have a packet sequence counter value of 1). From that point on each send request will increment the counter by one, regardless of the number of destinations or if there is an active subscription.

After a rollover condition the sequence counter will be a value of zero for one instance. The sequence counter is incremented after all the checks have passed prior to the actual sending of the message. This includes the parameter checks and the memory allocation check.

When disabled, the original message will not be altered. This method of message delivery is recommended for situations where the sender did not generate the packet, such as a network interface application passing a packet from a remote system to the local software bus.

1.14.3.5 Message Limit Error

Before placing a message on a pipe, the software bus checks the message limit to ensure the maximum number of packets in transit to the destination is not exceeded. If placing the message on the pipe would exceed the message limit, then the action of sending to that pipe is aborted and the 'Message Limit Error' event is sent. This condition will typically occur when an application that receives the packets does not respond quickly enough, or if the sender of the packets produces them too quickly.

This condition occurs often during development and during integration, for example when a remote processor gets reset or a 1553 cable becomes disconnected. Because of the common occurrences, the event may have filtering associated with it. Any filtering for this event would be performed by the cFE Event Services (EVS). Filtering for SB events may be specified in the cFE platform configuration file or may be commanded after the system initializes.

If this error occurs during nominal conditions, it could be an indication that the 'message limit' is not set correctly. The message limit is given at the time of the subscription and given as a parameter in the subscribe API. With the [CFE_SB_Subscribe](#) API, the SB uses a default message limit value specified by [CFE_PLATFORM_SB_DEFAULT_MSG_LIMIT](#). This constant is currently set to a value of four. If the default value is insufficient, the message limit value can be specified in the [CFE_SB_SubscribeEx](#) API.

A related failure is the pipe overflow condition, which can occur if the total number of packets (of all kinds) sent to a particular pipe is too large.

1.14.3.6 Pipe Overflow Error

Another common error that occurs during the send process is the pipe overflow error. This condition occurs if the total number of packets (of all kinds) sent to a particular pipe is too large. If this error occurs too frequently, it may be an indication that the pipe depth is not set correctly. The pipe depth is given at the time the pipe is created as a parameter in the [CFE_SB_CreatePipe](#) API.

1.14.3.7 SB Event Filtering

Most filtering for SB events is performed by the cFE Event Services (EVS). Filtering for SB events may be specified in the cFE platform configuration file or may be commanded after the system initializes. There is no SB event log that limits the number of events based on the capacity of the log, as in the heritage software bus.

There is one case in which events are filtered by the software bus instead of event services. This occurs when the software bus needs to suppress events so that a fatal recursive event condition does not transpire. Because error cases encountered when sending a message generate an event, and events cause a message to be sent a calling sequence could cause a stack overflow if the recursion is not properly terminated. The cFE software bus detects this condition and properly terminates the recursion. This is done by using a set of flags (one flag per event in the Send API) which determine whether an API has relinquished its stack. If the software bus needs to send an event that may cause recursion, the flag is set and the event is sent. If sending the event would cause the same event again, the event call will be bypassed, terminating the recursion. The result is that the user will see only one event instead of the many events that would normally occur without the protection. The heritage software bus did not have this condition because it stored events in the software bus event log and another thread would read them out at a later time.

1.14.3.8 Diagnostic Data

The cFE software bus provides a set of commands to dump SB diagnostic data to help troubleshoot problems or check configuration settings. These commands allow the user to view the routing table, the pipe table or the message map. The message map is a lookup table used during a send operation to give fast access to the routing table index that corresponds to the message being sent.

The software bus also provides a statistics packet that can be used to tune the configuration parameters. This information is sent to the ground in the form of an SB packet when the corresponding command is received. The cFE limits the number of system pipes, unique Message IDs, buffer memory, messages on a pipe and subscriptions per Message ID. These limits are configurable through cFE platform and mission configuration parameters. The statistics packet was designed to let the project verify that these user settings provide the necessary margin to meet requirements.

The SB statistics information shows 'Currently In Use' figures, 'High Water Mark' figures and 'Max Allowed' figures for the following: buffer memory, messages on each pipe (pipe depth stats), System Pipes, Unique Message IDs and total subscriptions.

Depending on the task-scheduling implementation details of the operating system, it is possible to see the peak messages on a pipe occasionally exceed the depth of the pipe. The "Peak Messages In Use" parameter is included in the SB statistics packet under the pipe depth stats.

1.14.3.9 Control of Packet Routing

The software bus allows the ground to disable and enable the sending of packets of a specified Message ID to a specified pipe. All destinations that are needed for normal operation are enabled by default. Modifying the routing of packets may be required for the following reasons:

- In flight, one can enable diagnostic packets to see them on the ground.
- During testing, one can disable a destination to simulate an anomaly.

1.14.3.10 Quality of Service

The software bus has a parameter in the [CFE_SB_SubscribeEx](#) API named Quality, which means Quality of Service (QOS) for off-board routing and is of the type [CFE_SB_Qos_t](#). This structure has two members named priority and reliability. The Quality parameter is currently unused by the software bus. It is a placeholder to be used with the future software bus capability of inter-processor communication. Although currently the software bus does not implement quality of service.

A default quality of services is provided via the [CFE_SB_DEFAULT_QOS](#) macro.

1.14.3.11 Known Problem

The software bus may perform unexpectedly under an unlikely corner-case scenario. This scenario was revealed in a stress test. The stress test was designed to deplete the Software Bus memory pool by having a high priority application continuously send 1000 byte packets to a lower priority application until the memory pool code returned an error code and sent the following event. "CFE_ES:getPoolBuf err:Request won't fit in remaining memory" At this point the higher priority sending application would stop executing. This would allow the lower priority receiving application to begin receiving the 1000 byte packets. After the receiving app processed all of the packets, the memory was restored to the memory pool as expected. The SB memory-in-use telemetry was zero because there were no software bus packets in transit. At this point any attempt to send a new-sized packet on the software bus was be rejected. The ES memory pool stated that the "... Request won't fit in remaining memory" even though there was currently no memory in use.

The simplest way to prevent this behavior is to ensure that there is margin when sizing the SB memory pool. To check the margin, monitor the "Peak Memory in Use" vs. the configuration parameter [CFE_PLATFORM_SB_BUF_MEMORY_BYTES](#) which indicates the amount allocated.

1.14.4 Frequently Asked Questions about Software Bus

(Q) How is the memory pool handle (sent in SB housekeeping telemetry) intended to be used?

The memory pool handle is used to analyze the SB memory pool statistics. The cFE ES command ([CFE_ES_SEND_MEM_POOL_STATS_CC](#)) to dump the memory pool statistics takes the pool handle as a parameter. These statistics tell how the SB memory pool is configured and gives details on margin. An improperly configured SB memory pool may inhibit communication. This may occur if there is not enough margin to create a block of the size needed for a transfer. Refer to the ES memory pool users guide for more details. [Memory Pool](#)

(Q) When sending a message, what message header fields are critical for routing the message?

To route the message properly, the software bus uses only the Message ID and packet length fields from the header of the message. If the packet length field is incorrect, then the buffer allocation for the message will also be incorrect. This may appear to the receiver as a truncated message or a message with unknown data added to the end of the message.

(Q) How many copies of the message are performed in a typical message delivery?

There is a single copy of the message performed when sending a message (from the callers memory space) using [CFE_SB_TransmitMsg](#). When transmitting the message, the software bus copies the message from the callers memory space into a buffer in the software bus memory space. There is also the option to request a buffer from SB, write directly to the buffer and send via [CFE_SB_TransmitBuffer](#). This is equivalent to the previous zero copy implementation. The [CFE_SB_ReceiveBuffer](#) API gives the user back a pointer to the buffer. When working with the buffers, the additional complexity to be aware of is the buffer is only available to the app from the request to send (on the sending side), or from the receive until the next receive on the same pipe on the receiving side. If the data is required outside that scope, the app needs a local copy.

(Q) When does the software bus free the buffer during a typical message delivery process? Or how long is the message, and the pointer to the buffer in the [CFE_SB_ReceiveBuffer](#) valid?

After receiving a buffer by calling [CFE_SB_ReceiveBuffer](#), the buffer received is valid until the next call to [CFE_SB_ReceiveBuffer](#) with the same Pipe Id. If the caller needs the message longer than the next call to [CFE_SB_ReceiveBuffer](#), the caller must copy the message to its memory space.

(Q) The first parameter in the [CFE_SB_ReceiveBuffer](#) API is a pointer to a pointer which can get confusing. How can I be sure that the correct address is given for this parameter.

Typically a caller declares a ptr of type `CFE_SB_Buffer_t` (i.e. `CFE_SB_Buffer_t *Ptr`) then gives the address of that pointer (`&Ptr`) as this parameter. After a successful call to [CFE_SB_ReceiveBuffer](#), `Ptr` will point to the first byte of the software bus buffer. This should be used as a read-only pointer. In systems with an MMU, writes to this pointer may cause a memory protection fault.

(Q) Why am I not seeing expected Message Limit error events or Pipe Overflow events?

It is possible the events are being filtered by cFE Event Services. The filtering for this event may be specified in the platform configuration file or it may have been commanded after the system initializes.

There is a corresponding counter for each of these conditions. First verify that the condition is happening by viewing the counter in SB HK telemetry. If the condition is happening, you can view the SB filter information through the EVS App Data Main page by clicking the 'go to' button for SB. The event Id for these events can be learned through a previous event or from the [cfe_sb_eventids.h](#) file.

(Q) Why does the SB provide event filtering through the platform configuration file?

To give the user the ability to filter events before an EVS command can be sent. During system initialization, there are many conditions occurring that can cause a flood of SB events such as No Subscribers, Pipe Overflow and MsgId to Pipe errors. This gives the user a way to limit these events.

(Q) Why does SB have so many debug event messages?

The SB debug messages are positive acknowledgments that an action (like receiving a cmd, creating a pipe or subscribing to a message) has occurred. They are intended to help isolate system problems. For instance, if an expected response to a command is not happening, it may be possible to repeat the scenario with the debug event turned on to verify that the command was successfully received.

(Q) How is the QOS parameter in the [CFE_SB_SubscribeEx](#) used by the software bus?

The QOS parameter is currently unused by the software bus. It is a placeholder to be used with the future software bus capability of inter-processor communication. Setting the QOS as [CFE_SB_DEFAULT_QOS](#) will ensure seamless integration when the software bus is expanded to support inter-processor communication.

(Q) Can I confirm my software bus buffer was delivered?

There is no built in mechanism for confirming delivery (it could span systems). This could be accomplished by generating a response message from the receiver.

1.15 cFE Software Bus Commands

Upon receipt of any command, the Software Bus application will confirm that the message length embedded within the header (from [CFE_MSG_GetSize\(\)](#)) matches the expected length of that message, based on the size of the C structure defining that command. If there is any discrepancy between the expected and actual message size, SB will generate the [CFE_SB_LEN_ERR_EID](#) event, increment the command error counter (`$sc_$cpu_SB_CMDEC`), and the command will *not* be accepted for processing.

The following is a list of commands that are processed by the cFE Software Bus Task.

Global [CFE_SB_DISABLE_ROUTE_CC](#)

Disable Software Bus Route

Global CFE_SB_DISABLE_SUB_REPORTING_CC

Disable Subscription Reporting Command

Global CFE_SB_ENABLE_ROUTE_CC

Enable Software Bus Route

Global CFE_SB_ENABLE_SUB_REPORTING_CC

Enable Subscription Reporting Command

Global CFE_SB_NOOP_CC

Software Bus No-Op

Global CFE_SB_RESET_COUNTERS_CC

Software Bus Reset Counters

Global CFE_SB_SEND_PREV_SUBS_CC

Send Previous Subscriptions Command

Global CFE_SB_SEND_SB_STATS_CC

Send Software Bus Statistics

Global CFE_SB_WRITE_MAP_INFO_CC

Write Map Info to a File

Global CFE_SB_WRITE_PIPE_INFO_CC

Write Pipe Info to a File

Global CFE_SB_WRITE_ROUTING_INFO_CC

Write Software Bus Routing Info to a File

1.16 cFE Software Bus Telemetry

The following are telemetry packets generated by the cFE Software Bus Task.

Global CFE_SB_AllSubscriptionsTlm_Payload_t

SB Previous Subscriptions Packet

Global CFE_SB_HousekeepingTlm_Payload_t

Software Bus task housekeeping Packet

Global CFE_SB_SingleSubscriptionTlm_Payload_t

SB Subscription Report Packet

Global CFE_SB_StatsTlm_Payload_t

SB Statistics Telemetry Packet

1.17 cFE Software Bus Configuration Parameters

The following are configuration parameters used to configure the cFE Software Bus either for each platform or for a mission as a whole.

Global CFE_MISSION_SB_MAX_PIPES

Maximum Number of pipes that SB command/telemetry messages may hold

Maximum Number of pipes that SB command/telemetry messages may hold

Global CFE_MISSION_SB_MAX_MSG_SIZE

Maximum SB Message Size

Maximum SB Message Size

Global CFE_MISSION_SB_SUB_ENTRIES_PER_PKT

Maximum Number of subscription entries per subscription report packet

Global CFE_PLATFORM_ENDIAN

Platform Endian Indicator

Global CFE_PLATFORM_SB_BUF_MEMORY_BYTES

Size of the SB buffer memory pool

Size of the SB buffer memory pool

Global CFE_PLATFORM_SB_DEFAULT_MAP_FILENAME

Default Message Map Filename

Default Message Map Filename

Global CFE_PLATFORM_SB_DEFAULT_MSG_LIMIT

Default Subscription Message Limit

Default Subscription Message Limit

Global CFE_PLATFORM_SB_DEFAULT_PIPE_FILENAME

Default Pipe Information Filename

Default Pipe Information Filename

Global CFE_PLATFORM_SB_DEFAULT_ROUTING_FILENAME

Default Routing Information Filename

Default Routing Information Filename

Global CFE_PLATFORM_SB_FILTERED_EVENT1

SB Event Filtering

SB Event Filtering

Global CFE_PLATFORM_SB_HIGHEST_VALID_MSGID

Highest Valid Message Id

Highest Valid Message Id

Global CFE_PLATFORM_SB_MAX_DEST_PER_PKT

Maximum Number of unique local destinations a single MsgId can have

Maximum Number of unique local destinations a single MsgId can have

Global CFE_PLATFORM_SB_MAX_MSG_IDS

Maximum Number of Unique Message IDs SB Routing Table can hold

Maximum Number of Unique Message IDs SB Routing Table can hold

Global CFE_PLATFORM_SB_MAX_PIPES

Maximum Number of Unique Pipes SB Routing Table can hold

Maximum Number of Unique Pipes SB Routing Table can hold

1.18 cFE Table Services Overview

Applications often organize sets of their parameters into logical units called tables. These are typically constant parameters that can change the behavior of a flight software algorithm and are only intended to be modified by operations personnel. Examples of this would be attitude control gains, sensor scalefactors, telemetry filter settings, etc.

Table Services (TBL) provides a centralized control of flight software tables. Operations personnel would interact with TBL in order to dump the contents of current tables, load new table images, verify the contents of a table image and manage Critical tables.

None of the cFE core applications (EVS, SB, ES, TIME, or TBL) use tables, and it is possible to build cFE without Table Services if not needed or an alternative parameter management mechanism is to be utilized.

For additional detail on Tables and how to manage them, see the following sections:

- [Managing Tables](#)
- [cFE Table Types and Table Options](#)
- [Table Registry](#)
- [Table Services Telemetry](#)
- [Effects of Processor Reset on Tables](#)
- [Frequently Asked Questions about Table Services](#)

1.18.1 Managing Tables

In order to effectively manage tables, an operator needs to understand how cFE Applications manage tables from their end. There are a number of methods that cFE Applications typically use to manage their tables. Each method is appropriate based upon the nature of the contents of the table.

cFE Applications are required to periodically check to see if their table is to be validated, updated (or in the case of dump-only tables, dumped). Most Applications perform this periodic management at the same time as housekeeping requests are processed. This table management is performed by the cFE Application that "owns" a table (ie - the cFE Application that registered the table with cFE Table Services). It is possible for cFE Applications to "share" a table with other cFE Applications. An Application that shares a table does not typically perform any of the management duties associated with that table.

A table can have one of two different types and a number of different options. These are discussed further in later sections. An operator should understand the chosen type and selected options for a particular table before attempting to modify a table's contents.

To understand the methods of maintaining a table, it is important that the terminology be clear. A table has two images: "Active" and "Inactive". The Active table is the one that a cFE Application is currently accessing when it executes. The

Inactive table is a copy of the Active table that an operator (or on-board process such as a stored command processor) can manipulate and change to have a newly desired set of data.

To create an Inactive table image on board, the operator would be required to perform a "Load" to the table. Loads are table images stored in on-board files. The Load can contain either a complete table image or just a part of a table image. If the Load contains just a portion, the Inactive image is first initialized with the contents of the Active image and then the portion identified in the Load file is written on top of the Active image. After the initial Load, an operator can continue to manipulate the Inactive table image with additional partial table load images. This allows the operator to reconfigure the contents of multiple portions of the table before deciding to "Validate" and/or "Activate" it.

Some cFE Applications provide special functions that will examine a table image to determine if the contents are logically sound. This function is referred to as the "Validation Function." When a cFE Application assigns a Validation Function to a table during the table registration process, it is then requiring that a Validation be performed before the table can be Activated. When an operator requests a Validation of a table image, they are sending a request to the owning Application to execute the associated Validation Function on that image. The results of this function are then reported in telemetry. If the Validation is successful, the operator is free to perform a table Activation. If the Validation fails, the operator would be required to make additional changes to the Inactive table image and attempt another Validation before commanding an Activation.

To change an Inactive table image into the Active table image, an operator must Activate a table. When an operator sends the table Activation command, they are notifying the table's owning Application that a new table image is available. It is then up to the Application to determine when is the best time to perform the "Update" of the table. When an Application performs an Update, the contents of the Inactive table image become the Active table image.

1.18.2 cFE Table Types and Table Options

A cFE Application Developer has several choices when creating a cFE Application. There are two basic types of tables: single buffered and double buffered. In addition to these two basic types there are a small variety of options possible with each table. These options control special characteristics of the table such as whether it is dump-only, critical or whether it has an application defined location in memory.

Each choice has its advantages and disadvantages. The developer chooses the appropriate type based upon the requirements of the application. Anyone operating a particular cFE Application must understand the nature of the type and options selected for a particular table before they can successfully understand how to perform updates, validations, etc.

For more information on the different types of tables available, see the following sections:

- Table Types
 - [Single Buffered Tables](#)
 - [Double Buffered Tables](#)
- Table Options
 - [Tables with Validation Functions](#)
 - [Critical Tables](#)
 - [User Defined Address Tables](#)
 - [Dump Only Tables](#)

1.18.2.1 Single Buffered Tables

The default table type for a cFE Application to use is a single buffered table. The principle advantage of a single buffered table is that it can share one of several shared table buffers for uploaded and pending table images. Since many cFE Applications have relatively small tables that are not changed at time critical moments or are not changed very often during a mission, single buffered tables represent the most memory resource efficient method of being managed.

The number of single buffered tables that can have inactive table images being manipulated at one time is specified by a TBL Services configuration parameter ([CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS](#)) found in the `cfe_platform_cfg.h` file associated with the processor in question. This parameter identifies the number of shared table buffers that are available.

Since inactive single buffered table images share a common resource, it may not be prudent for an operator to load an image and then delay on the image's activation for an extended period of time.

Single buffered tables are allowed to be critical (see [Critical Tables](#)), dump-only (see [Dump Only Tables](#)) and/or have a user-defined address (see [User Defined Address Tables](#)).

1.18.2.2 Double Buffered Tables

Under certain conditions, a cFE Application Developer may choose to use a double buffered table type within their application. Double buffered tables retain a dedicated inactive image of the table data. With a dedicated inactive table image available, double buffered tables are then capable of efficiently swapping table contents and/or delaying the activation of a table's contents for an indeterminate amount of time.

Some cFE Applications prefer to delay the Activation of a table until a specified time (e.g. - a Spacecraft Ephemeris). These tables are typically defined as double buffered tables so that the Inactive image can be left sitting untouched for an extended period of time without interfering with shared resources for other tables. Then the Application can perform the Update when the time is right.

Applications which have unusually large tables may decide to conserve memory resources by making them double buffered. This is because the shared buffers used by single buffered tables must be sized to match the largest table. If there is one table that is unusually large, there is little reason to allocate up to [CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS](#) number of buffers that size. A double buffered table will only allocate ONE extra buffer of that size.

Performance minded Applications that are required to perform processing with tight timing deadlines may choose to use double buffered tables because the Update for a double buffered table is deterministic and quick.

1.18.2.3 Tables with Validation Functions

Applications that associate Validation Functions with their tables when the tables are registered are effectively requiring that the contents of a table be logically Validated before it is Activated. The cFE will refuse to let a table with an associated Validation Function be Activated until a successful Validation on the Inactive table image has occurred.

Tables that are NOT assigned a Validation Function are assumed to be valid regardless of the contents of the table image. These tables do not require a Validation Command prior to Activation.

1.18.2.4 Critical Tables

Applications that must be able to recover quickly from a Processor Reset may select the "Critical" table option when registering their table. Table Services automatically creates a Critical Data Store for the table and ensures that the contents of the Critical Data Store are updated whenever a Table Activation occurs.

If a Processor Reset happens, when the Application attempts to Register the table again, Table Services automatically locates the associated Critical Data Store and initializes the Table with the saved contents.

1.18.2.5 User Defined Address Tables

In order to provide a mechanism for Flight Software Maintenance teams to quickly create a table image for dumping contents of memory that isn't normally loaded by the ground, there is an option to create User-Defined Address tables. These tables, when they are first registered, provide a memory address where the Active image of the table is to be maintained. Normally, the address is specified by Table Services from its memory pool.

By specifying the address, the Flight Software Maintenance team can create a Dump-Only table that contains the contents of a data structure that is not normally accessible via telemetry or table dumps. Then, on command, the Flight Software Maintenance team can periodically dump the data structure's contents to an on-board file(s) that can then be transferred to the ground for later analysis.

1.18.2.6 Dump Only Tables

On occasion, cFE Applications require a segment of memory in which the Application writes data. The typical cFE Table is not normally modified directly by an Application but only via Load and Activate commands from either the Ground or Stored Command Processor. However, for those situations where an Application wishes to modify the contents of a data structure and the Application is limited in its telemetry bandwidth so that the modified data cannot be telemetered, the Application can create a Dump-Only table.

Dump-Only tables are not allowed to be modified via the Load/Validate/Activate process most other tables are. They are only supposed to be modified by onboard Applications. The Operator can still command a Dump which will be processed by the table's owning Application when it manages its tables. By letting the Application perform the dump, the Operator can feel confident that the table contents are a complete snapshot in time and not corrupted by taking a snapshot while the Application was in the process of modifying its contents.

1.18.3 Table Registry

When Applications register tables, Table Services retains pertinent information on the table in the Table Registry. The following information (along with other information that is less important for an operator) is kept for each table:

- The Application ID of the Application that Registered the table
- The full name of the table
- The size, in bytes, of the table
- Pointers to the start addresses of the Table's image buffers, Active and Inactive (if appropriate)
- A pointer to the start address of a Validation Function

- A flag indicating whether a table image has been loaded into an Inactive buffer
- A flag indicating whether the table is Critical and its associated CDS Handle if it is
- A flag indicating whether the table has ever been loaded (initialized)
- A flag indicating whether the table is Dump Only
- A flag indicating whether the table has an Update Pending
- A flag indicating whether the table is double buffered or not
- The System Time when the Table was last Updated
- The filename of the last file loaded into the table
- The File Creation Time for the last file used to load the contents of the table

This information can be obtained by either sending the Dump Registry command which will put all of the information from the Table Registry into an onboard file for later downlink or the operator can send a command to Telemeter the Registry Entry for a single table. This will cause the pertinent registry entry for a single table to be sent via a telemetry packet.

The API function [CFE_TBL_Register\(\)](#) returns either CFE_SUCCESS or CFE_TBL_INFO_RECOVERED_TBL to indicate that the table was successfully registered. The difference is whether the table data was recovered from CDS as part of the registration. There are several error return values that describe why the function failed to register the table but nothing related to why the restoration from CDS might have failed. There is, however, a message written to the System Error Log by Table Services that can be dumped by the ground to get this information. Note that failure to restore a table from CDS is not an expected error and requires some sort of data corruption to occur.

1.18.4 Table Services Telemetry

Table Services produces two different telemetry packets. The first packet, referred to as the Table Services Housekeeping Packet, is routinely produced by Table Services upon receipt of the Housekeeping Request message that is typically sent to all Applications by an on board scheduler. The contents and format of this packet are described in detail at [CFE_TBL_HousekeepingTlm_t](#).

1.18.5 Effects of Processor Reset on Tables

When a processor resets, the Table Registry is re-initialized. All Applications must, therefore, re-register and re-initialize their tables. The one exception, however, is if the Application has previously tagged a table as "Critical" during Table Registration, then Table Services will attempt to locate a table image for that table stored in the Critical Data Store. Table Services also attempts to locate the Critical Table Registry which is also maintained in the Critical Data Store.

If Table Services is able to find a valid table image for a Critical table in the Critical Data Store, the contents of the table are automatically loaded into the table and the Application is notified that the table does not require additional initialization.

1.18.6 Frequently Asked Questions about Table Services

(Q) Is it an error to load a table image that is smaller than the registered size?

Table images that are smaller than the declared size of a table fall into one of two categories.

If the starting offset of the table image (as specified in the Table Image secondary file header) is not equal to zero, then the table image is considered to be a "partial" table load. Partial loads are valid as long as a table has been previously loaded with a non-"partial" table image.

If the starting offset of the table image is zero and the size is less than the declared size of the table, the image is considered "short" but valid. This feature allows application developers to use variable length tables.

(Q) I tried to validate a table and received the following event message that said the event failed:

MyApp validation failed for Inactive 'MyApp.MyTable', Status=0x####

What happened?

The event message indicates the application who owns the table has discovered a problem with the contents of the image. The code number following the 'Status' keyword is defined by the Application. The documentation for the specified Application should be referred to in order to identify the exact nature of the problem.

(Q) What commands do I use to load a table with a new image?

There are a number of steps required to load a table.

- 1. The operator needs to create a cFE Table Services compatible table image file with the desired data contained in it. This can be accomplished by creating a 'C' source file, compiling it with the appropriate cross compiler for the onboard platform and then running the `elf2cfetbl` utility on the resultant object file.*
- 2. The file needs to be loaded into the onboard processor's filesystem using whichever file transfer protocol is used for that mission.*
- 3. The [Load Command](#) is sent next to tell Table Services to load the table image file into the Inactive Table Image Buffer for the table identified in the file.*
- 4. The [Validate Command](#) is then sent to validate the contents of the inactive table image. This will ensure the file was not corrupted or improperly defined. The results of the validation are reported in Table Services Housekeeping Telemetry. If a table does not have a validation function associated with it, the operator may wish to compare the computed CRC to verify the table contents match what was intended.*
- 5. Upon successful validation, the operator then sends the [Activate Command](#). The application owning the table should, within a reasonable amount of time, perform a table update and send an event message.*

(Q) What causes cFE Table Services to generate the following sys log message:

CFE_TBL: GetAddressInternal-App(%d) attempt to access unowned Tbl Handle=%d

When an application sharing its table(s) with one or more applications is reloaded, the reloaded application's table handle(s) are released. cFE Table Services sees that the table(s) are shared and keeps a 'shadow' version of the table in the Table Services registry. The registry will show the released, shared tables with no name. When the applications sharing the table attempt to access the table via the 'old', released handle, Table Services will return an error code to

the applications and generate the sys log message. The applications may then unregister the 'old' handle(s) in order to remove the released, shared table(s) from the Table Services registry and share the newly loaded application table(s).

(Q) When does the Table Services Abort Table Load command need to be issued?

The Abort command should be used whenever a table image has been loaded but the application has not yet activated it and the operator no longer wants the table to be loaded.

The purpose of the Abort command is to free a previously allocated table buffer. It should be noted, however, that multiple table loads to the SAME table without an intervening activation or abort, will simply OVERWRITE the previous table load using the SAME buffer.

Therefore, the most likely scenarios that would lead to a needed abort are as follows:

1. Operator loads a table and realizes immediately that the load is not wanted.
2. Operator loads a table and performs a validation on it. Regardless of whether the table passes or fails the validation, if the operator no longer wants to activate the table, the abort command should be issued.
It should be noted that a table image that fails activation is retained in the inactive buffer for diagnosis, if necessary. It is NOT released until it is aborted or overwritten and successfully validated and activated.
3. A table image was loaded; the image was successfully validated; the command for activation was sent; but the application fails to perform the activation.

The Abort command will free the table buffer and clear the activation request.

This situation can occur when either the application is improperly designed and fails to adequately manage its tables (sometimes seen in the lab during development) or the application is "hung" and not performing as it should.

1.19 cFE Table Services Commands

Upon receipt of any command, the Table Services application will confirm that the message length embedded within the header (from `CFE_MSG_GetSize()`) matches the expected length of that message, based on the size of the C structure defining that command. If there is any discrepancy between the expected and actual message size, TBL will generate the `CFE_TBL_LEN_ERR_EID` event, increment the command error counter (`$sc$_cpu_TBL_CMDEC`), and the command will *not* be accepted for processing.

The following is a list of commands that are processed by the cFE Table Services Task.

Global `CFE_TBL_ABORT_LOAD_CC`

Abort Table Load

Global `CFE_TBL_ACTIVATE_CC`

Activate Table

Global `CFE_TBL_DELETE_CDS_CC`

Delete Critical Table from Critical Data Store

Global `CFE_TBL_DUMP_CC`

Dump Table

Global `CFE_TBL_DUMP_REGISTRY_CC`

Dump Table Registry

Global CFE_TBL_LOAD_CC

Load Table

Global CFE_TBL_NOOP_CC

Table No-Op

Global CFE_TBL_RESET_COUNTERS_CC

Table Reset Counters

Global CFE_TBL_SEND_REGISTRY_CC

Telemeter One Table Registry Entry

Global CFE_TBL_VALIDATE_CC

Validate Table

1.20 cFE Table Services Telemetry

The following are telemetry packets generated by the cFE Table Services Task.

Global CFE_TBL_HousekeepingTlm_Payload_t

Table Services Housekeeping Packet

Global CFE_TBL_TblRegPacket_Payload_t

Table Registry Info Packet

1.21 cFE Table Services Configuration Parameters

The following are configuration parameters used to configure the cFE Table Services either for each platform or for a mission as a whole.

Global CFE_MISSION_TBL_MAX_FULL_NAME_LEN

Maximum Length of Full Table Name in messages

Maximum Length of Full Table Name in messages

Maximum Length of Full Table Name in messages

Global CFE_MISSION_TBL_MAX_NAME_LENGTH

Maximum Table Name Length

Maximum Table Name Length

Maximum Table Name Length

Global CFE_PLATFORM_TBL_BUF_MEMORY_BYTES

Size of Table Services Table Memory Pool

Size of Table Services Table Memory Pool

Size of Table Services Table Memory Pool

Global CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE

Default Filename for a Table Registry Dump

Default Filename for a Table Registry Dump

Default Filename for a Table Registry Dump

Global CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES

Maximum Number of Critical Tables that can be Registered
Maximum Number of Critical Tables that can be Registered
Maximum Number of Critical Tables that can be Registered

Global CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE

Maximum Size Allowed for a Double Buffered Table
Maximum Size Allowed for a Double Buffered Table
Maximum Size Allowed for a Double Buffered Table

Global CFE_PLATFORM_TBL_MAX_NUM_HANDLES

Maximum Number of Table Handles
Maximum Number of Table Handles
Maximum Number of Table Handles

Global CFE_PLATFORM_TBL_MAX_NUM_TABLES

Maximum Number of Tables Allowed to be Registered
Maximum Number of Tables Allowed to be Registered
Maximum Number of Tables Allowed to be Registered

Global CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS

Maximum Number of Simultaneous Table Validations
Maximum Number of Simultaneous Table Validations
Maximum Number of Simultaneous Table Validations

Global CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS

Maximum Number of Simultaneous Loads to Support
Maximum Number of Simultaneous Loads to Support
Maximum Number of Simultaneous Loads to Support

Global CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE

Maximum Size Allowed for a Single Buffered Table
Maximum Size Allowed for a Single Buffered Table
Maximum Size Allowed for a Single Buffered Table

Global CFE_PLATFORM_TBL_VALID_PRID_1

Processor ID values used for table load validation
Processor ID values used for table load validation
Processor ID values used for table load validation

Global CFE_PLATFORM_TBL_VALID_PRID_COUNT

Number of Processor ID's specified for validation
Number of Processor ID's specified for validation
Number of Processor ID's specified for validation

Global CFE_PLATFORM_TBL_VALID_SCID_1

Spacecraft ID values used for table load validation
Spacecraft ID values used for table load validation
Spacecraft ID values used for table load validation

Global CFE_PLATFORM_TBL_VALID_SCID_COUNT

Number of Spacecraft ID's specified for validation
Number of Spacecraft ID's specified for validation
Number of Spacecraft ID's specified for validation

1.22 cFE Time Services Overview

The cFE Time Service (TIME) is one of the cFE core services. TIME provides time correlation, distribution and synchronization services. TIME exists in two varieties: a Time Server responsible for maintaining the master time reference for all remote systems, and a Time Client responsible for synchronizing to that master time reference.

Since TIME is a generic implementation aimed to meet the needs of a variety of mission configurations, there are numerous configuration parameters, which dictate the behavior of TIME (see `cfe_mission_cfg.h` and `cfe_platform_cfg.h` for the specific mission configuration).

With the exception of those sections specific to Time Clients and Servers, this document assumes the most common physical environment - one instantiation of cFE installed on a single processor. Therefore, TIME represents cFE Time Services configured as a Time Server.

For additional detail on Time Services and how to manage it, see the following sections:

- [Time Components](#)
- [Time Structure](#)
- [Time Formats](#)
- [Time Configuration](#)
 - [Time Format Selection](#)
 - [Enabling Fake Tone Signal](#)
 - [Selecting Tone and Data Ordering](#)
 - [Specifying Tone and Data Window](#)
 - [Specifying Time Server/Client](#)
 - [Specifying Time Tone Byte Order](#)
 - [Virtual MET](#)
 - [Specifying Time Source](#)
 - [Specifying Time Signal](#)
- [Time Services Paradigm\(s\)](#)
- [Flywheeling](#)

- [Time State](#)
- [Initialization](#)
 - [Power-On Reset](#)
 - [Processor Reset](#)
- [Initialization](#)
 - [Power-On Reset](#)
 - [Processor Reset](#)
- [Normal Operation](#)
 - [Client](#)
 - [Server](#)
 - * [Setting Time](#)
 - * [Adjusting Time](#)
 - * [Setting MET](#)
- [Frequently Asked Questions about Time Services](#)

1.22.1 Time Components

Time knowledge is stored in several pieces, so that the time information can more easily be manipulated and utilized. These components include:

The **Ground Epoch** is an arbitrary date and time that establishes the zero point for spacecraft time calculations. The selection of the epoch is mission specific, although in the past, it was common to select the same epoch as defined for the Operating System used by the computers hosting the ground system software. Recent mission epoch selections have also included using zero seconds after midnight, Jan 1, 2001.

Spacecraft Time is the number of seconds (and fraction of a second) since the ground epoch. Spacecraft time is the sum of **Mission Elapsed Time** (MET) and the **Spacecraft Time Correlation Factor** (STCF). By definition, MET is a measure of time since launch or separation. However, for most missions the MET actually represents the amount of time since powering on the hardware containing the MET timer. The STCF correlates the MET to the ground epoch.

The **Tone** is the signal that MET seconds have incremented. In most hardware configurations, the tone is synonymous with the **1 PPS** signal. The tone signal may be generated by a local hardware timer, or by an external event (GPS

receiver, spacewire time tick, 1553 bus signal, etc). TIME may also be configured to simulate the tone for lab environments that do not have the necessary hardware to provide a tone signal. Note that MET sub-seconds will be zero at the instant of the tone.

Time at the Tone is the spacecraft time at the most recent "valid" tone.

Time since the Tone is the amount of time since the tone (usually less than one second). This value is often measured using the local processor clock. Upon detecting the tone signal, TIME stores the contents of the local processor clock to facilitate this measurement.

Thus, **Current Spacecraft Time** is the sum of "time at the tone" and "time since the tone".

Leap Seconds occur to keep clocks correlated to astronomical observations. The modern definition of a second (9,192,631,770 oscillations of a cesium-133 atom) is constant while the earth's rotation has been slow by a small fraction of a second per day. The **International Earth Rotation and Reference System Service** (IERS) maintains the count of leap seconds as a signed whole number that is subject to update twice a year. Although it is possible to have a negative leap second count if the earth rotates too fast, it is highly unlikely. The initial count of leap seconds (10) was established in January of 1972 and the first leap second was added to the initial count in June of 1972. The most recent leap seconds are announced by the International Earth Rotation Service (IERS): <https://www.iers.org> in IERS Bulletin C (leap second announcements). Search the IERS site for "Bulletin C" to obtain the latest issue/announcement.

1.22.2 Time Structure

The cFE implementation of the **System Time Structure** is a modified version of the CCSDS Unsegmented Time Code (CUC) which includes 4 bytes of seconds, and 4 bytes of subseconds, where a subsecond is equivalent to $1/(2^{32})$ seconds. The system time structure is used by TIME to store current time, time at the tone, time since the tone, the MET, the STCF and command arguments for time adjustments. Note that typically the 32 bits of seconds and the upper 16 bits of subseconds are used for time stamping Software bus messages, but this is dependent on the underlying definition.

The system time structure is defined as follows:

```
typedef struct {
    uint32    Seconds;        /* Number of seconds */
    uint32    Subseconds;     /* Number of 2^(-32) subseconds */
} CFE_TIME_SysTime_t;
```

1.22.3 Time Formats

International Atomic Time (TAI) is one of two time formats supported by cFE TIME. TAI is the number of seconds and sub-seconds elapsed since the ground epoch as measured with the atomic clock previously described. TAI has no reference to leap seconds and is calculated using the following equation:

$$\text{TAI} = \text{MET} + \text{STCF}$$

It should be noted that TAI is only "true" TAI when the selected ground epoch is the same as the TAI epoch (zero seconds after midnight, January 1, 1958). However, nothing precludes configuring cFE TIME to calculate time in the TAI format and setting the STCF to correlate to any other epoch definition.

Coordinated Universal Time (UTC) is the other time format supported by cFE TIME. UTC differs from TAI in the fact that UTC includes a leap seconds adjustment. TIME computes UTC using the following equation:

$$\text{UTC} = \text{TAI} - \text{Leap Seconds}.$$

The preceding UTC equation might seem to imply that TAI includes leap seconds and UTC does not - which is not the case. In fact, the UTC calculation includes a leap seconds adjustment that subtracts leap seconds from the same time components used to create TAI. Alternatively, it might be less confusing to express the UTC equation as follows:

$$\text{UTC} = \text{MET} + \text{STCF} - \text{Leap Seconds}$$

1.22.4 Time Configuration

All configurations of TIME require a local processor source for a 1Hz interrupt and access to a local clock with a resolution fine enough that it can be used to measure short periods of elapsed time. The local interrupt is used to wake-up TIME at a regular interval for the purpose of verifying that the tone is being received. The local clock is used to measure time since the tone and to provide coarse verification that the tone is occurring at approximately one second intervals. The presumption is that the tone is the most accurate timer in the system and, within reason, is to be trusted. Note that nothing precludes the use of the MET as the local clock, assuming the MET is both local and provides sub-second data. However, the tone must not be used as the source for the local 1Hz interrupt.

Consider the following brief description of three hypothetical hardware configurations. These sample systems may be used as reference examples to help clarify the descriptions of the various TIME configuration selections.

In the first system, there is no MET timer and therefore no tone signal. The MET is a count of the number of "fake" tones generated by TIME software. There is no validation performed regarding the quality of time data. This hardware configuration is a common lab environment using COTS equipment.

In the second system, the MET timer is a hardware register that is directly accessible by TIME. When MET seconds increment, a processor interrupt signals the tone. Upon detecting the tone, TIME can read the MET to establish the time at the tone. To verify that the tone is valid, TIME need only validate that this tone signal occurred approximately one second after the previous tone signal (as measured with the local clock).

In the third system, the MET is located on hardware connected via spacewire. When MET seconds increment, a spacewire time tick triggers a local processor interrupt to signal the tone. Shortly after announcing the tone, the hardware containing the MET also generates a spacewire data packet containing the MET value corresponding to the tone. TIME must wait until both the tone and data packet have been received before validating the tone. The tone must have occurred approximately one second after the previous tone signal and the data packet must have been received within a specified window in time following the tone.

The hardware design choice for how the tone signal is distributed is not material to TIME configuration. The software detecting the tone need only call the cFE API function announcing the arrival of the tone. This function is designed to be called from interrupt handlers.

For detail on each of the individual configuration settings for cFE Time Services, see the following sections:

- [Time Format Selection](#)
- [Enabling Fake Tone Signal](#)
- [Selecting Tone and Data Ordering](#)
- [Specifying Tone and Data Window](#)
- [Specifying Time Server/Client](#)
- [Specifying Time Tone Byte Order](#)
- [Virtual MET](#)
- [Specifying Time Source](#)
- [Specifying Time Signal](#)

1.22.4.1 Time Format Selection

Time format is defined in the mission configuration header file.

This selection defines the default time format as TAI or UTC. The API functions to get time in either specific format are still enabled, but the API function to get time in the default format will follow this selection. Enable one, and **only one**, of the following time format definitions:

```
#define CFE_MISSION_TIME_CFG_DEFAULT_TAI  TRUE
#define CFE_MISSION_TIME_CFG_DEFAULT_UTC  FALSE
```

or

```
#define CFE_MISSION_TIME_CFG_DEFAULT_TAI  FALSE
#define CFE_MISSION_TIME_CFG_DEFAULT_UTC  TRUE
```

The choice of time format is a mission specific decision and is not directly affected by the hardware configuration.

See also

[CFE_MISSION_TIME_CFG_DEFAULT_TAI](#), [CFE_MISSION_TIME_CFG_DEFAULT_UTC](#)

1.22.4.2 Enabling Fake Tone Signal

The fake tone is defined in the mission configuration header file.

If this selection is set to TRUE, TIME will generate a "fake" tone signal by calling the same API function as would be called upon detection of the "real" tone signal. Enable the fake tone only for hardware configurations that do not provide a tone signal.

```
#define CFE_MISSION_TIME_CFG_FAKE_TONE  TRUE
```

Hypothetical hardware configuration number one (described above) would enable the fake tone signal.

See also

[CFE_MISSION_TIME_CFG_FAKE_TONE](#)

1.22.4.3 Selecting Tone and Data Ordering

Tone and data order is defined in the mission configuration header file.

This selection defines which comes first - the tone or the time at the tone data. Does the time data describe the tone that already occurred, or the tone that has not yet occurred? This decision may be driven by the hardware design but can also be arbitrary. Enable one, and only one, of the following:

```
#define CFE_MISSION_TIME_AT_TONE_WAS
#define CFE_MISSION_TIME_AT_TONE_WILL_BE
```

Hypothetical hardware configuration number three (described [Time Configuration](#) above) would enable "time at the tone was".

See also

[CFE_MISSION_TIME_AT_TONE_WAS](#), [CFE_MISSION_TIME_AT_TONE_WILL_BE](#)

1.22.4.4 Specifying Tone and Data Window

The tone and data window is defined in the mission configuration header file.

In concert with the definition of tone and data order, this selection defines the valid window in time for the second of the pair to follow the first. Both must be defined, units are micro-seconds.

```
#define CFE_MISSION_TIME_MIN_ELAPSED 0
#define CFE_MISSION_TIME_MAX_ELAPSED 100000
```

Hypothetical hardware configuration number three (described above) might use these values which describe a window that begins immediately after the tone and lasts for one tenth of a second.

See also

[CFE_MISSION_TIME_MIN_ELAPSED](#), [CFE_MISSION_TIME_MAX_ELAPSED](#)

1.22.4.5 Specifying Time Server/Client

Configure TIME as a client only when the target system has multiple processors running separate instantiations of the cFE. One instantiation must be configured as the server and the remainder configured as clients. If the target system has only one processor running the cFE, then TIME must be configured as a server.

Enable one, and only one, of the following definitions in the platform configuration header file:

```
#define CFE_PLATFORM_TIME_CFG_SERVER TRUE
#define CFE_PLATFORM_TIME_CFG_CLIENT FALSE
```

or

```
#define CFE_PLATFORM_TIME_CFG_SERVER FALSE
#define CFE_PLATFORM_TIME_CFG_CLIENT TRUE
```

See also

[CFE_PLATFORM_TIME_CFG_SERVER](#), [CFE_PLATFORM_TIME_CFG_CLIENT](#)

1.22.4.6 Specifying Time Tone Byte Order

By default, the CFE time tone message is a payload of integers in platform-endian order (containing the tone's timestamp, the leap seconds, and state information.) In some configurations, it may be better to have the payload produced in big-endian order—particularly in mixed-endian environments.

In order to force the tone message to be in big-endian order, you must define the following:

```
#define CFE_PLATFORM_TIME_CFG_BIGENDIAN
```

1.22.4.7 Virtual MET

This configuration option refers to whether the MET is local to this instantiation of TIME. If the MET is not local then TIME must be configured as using a virtual MET.

Therefore, all TIME clients must be configured as using a virtual MET. If the MET was local to any TIME client, then that instantiation of TIME would have to be the server.

TIME servers must be configured as using a virtual MET

1.22.4.8 Specifying Time Source

TIME configuration provides the ability to specify where the source for time data is originating - either internal or external. In hypothetical system one, the MET is internal. In system two, TIME cannot directly read the MET, therefore time data must be received from an external source.

This selection also enables a command interface to switch between internal and external input. When commanded to use internal time data, TIME will ignore the external data. However, TIME will continue to use the API function as the trigger to generate a "time at the tone" command packet regardless of the internal/external command selection.

Set the following definition to TRUE only for TIME servers using an external time data source.

```
#define CFE_PLATFORM_TIME_CFG_SOURCE  TRUE
```

The remainder of this section pertains only to TIME servers configured to accept external time data.

When configured to accept external time data, TIME requires an additional definition for the type of external data (GPS, MET, spacecraft time, etc.). This selection will enable an API function specific to the selected data type. Regardless of how the time data is received, the receiver need only pass the data to the appropriate API function.

TIME servers using an external time data source must set one, and only one, of the following to TRUE, for example:

```
#define CFE_PLATFORM_TIME_CFG_SRC_MET    TRUE
#define CFE_PLATFORM_TIME_CFG_SRC_GPS    FALSE
#define CFE_PLATFORM_TIME_CFG_SRC_TIME   FALSE
```

configuration definitions for the particular source.

If the `cfe_platform_cfg.h` file contains `"#define CFE_PLATFORM_TIME_CFG_SOURCE TRUE"` then time is configured to allow switching between internal and external time sources (see [CFE_TIME_SET_SOURCE_CC](#)). If this configuration parameter is set to FALSE then the command to set the source will be rejected.

If this configuration parameter is set to TRUE then ONE and ONLY ONE of the following configuration parameters must also be set TRUE in order to specify the external time source, for example:

```
#define CFE_PLATFORM_TIME_CFG_SRC_MET    TRUE
#define CFE_PLATFORM_TIME_CFG_SRC_GPS    FALSE
#define CFE_PLATFORM_TIME_CFG_SRC_TIME   FALSE
```

Note that Internal MET source depends on available hardware. It may be the local count of tone signals, the contents of a hardware register or an OS specific time function.

Note also that when configured to use an external time source, commands to set the time will be overwritten.

See also

[CFE_PLATFORM_TIME_CFG_SRC_MET](#), [CFE_PLATFORM_TIME_CFG_SRC_GPS](#), [CFE_PLATFORM_TIME_CFG_SRC_TIME](#)

1.22.4.9 Specifying Time Signal

Some hardware configurations support a primary and redundant tone signal selection. Setting the following configuration definition to TRUE will result in enabling a TIME command to select the active tone signal.

```
#define CFE_PLATFORM_TIME_CFG_SIGNAL TRUE
```

Note: this feature requires additional custom software to make the physical signal switch.

See also

[CFE_PLATFORM_TIME_CFG_SIGNAL](#)

1.22.4.10 Time Format Selection

Time format is defined in the mission configuration header file.

This selection defines the default time format as TAI or UTC. The API functions to get time in either specific format are still enabled, but the API function to get time in the default format will follow this selection. Enable one, and **only one**, of the following time format definitions:

```
#define CFE_MISSION_TIME_CFG_DEFAULT_TAI TRUE
#define CFE_MISSION_TIME_CFG_DEFAULT_UTC FALSE
```

or

```
#define CFE_MISSION_TIME_CFG_DEFAULT_TAI FALSE
#define CFE_MISSION_TIME_CFG_DEFAULT_UTC TRUE
```

The choice of time format is a mission specific decision and is not directly affected by the hardware configuration.

See also

[CFE_MISSION_TIME_CFG_DEFAULT_TAI](#), [CFE_MISSION_TIME_CFG_DEFAULT_UTC](#)

1.22.4.11 Enabling Fake Tone Signal

The fake tone is defined in the mission configuration header file.

If this selection is set to TRUE, TIME will generate a "fake" tone signal by calling the same API function as would be called upon detection of the "real" tone signal. Enable the fake tone only for hardware configurations that do not provide a tone signal.

```
#define CFE_MISSION_TIME_CFG_FAKE_TONE TRUE
```

Hypothetical hardware configuration number one (described above) would enable the fake tone signal.

See also

[CFE_MISSION_TIME_CFG_FAKE_TONE](#)

1.22.4.12 Selecting Tone and Data Ordering

Tone and data order is defined in the mission configuration header file.

This selection defines which comes first - the tone or the time at the tone data. Does the time data describe the tone that already occurred, or the tone that has not yet occurred? This decision may be driven by the hardware design but can also be arbitrary. Enable one, and only one, of the following:

```
#define CFE_MISSION_TIME_AT_TONE_WAS
#define CFE_MISSION_TIME_AT_TONE_WILL_BE
```

Hypothetical hardware configuration number three (described [Time Configuration](#) above) would enable "time at the tone was".

See also

[CFE_MISSION_TIME_AT_TONE_WAS](#), [CFE_MISSION_TIME_AT_TONE_WILL_BE](#)

1.22.4.13 Specifying Tone and Data Window

The tone and data window is defined in the mission configuration header file.

In concert with the definition of tone and data order, this selection defines the valid window in time for the second of the pair to follow the first. Both must be defined, units are micro-seconds.

```
#define CFE_MISSION_TIME_MIN_ELAPSED 0
#define CFE_MISSION_TIME_MAX_ELAPSED 100000
```

Hypothetical hardware configuration number three (described above) might use these values which describe a window that begins immediately after the tone and lasts for one tenth of a second.

See also

[CFE_MISSION_TIME_MIN_ELAPSED](#), [CFE_MISSION_TIME_MAX_ELAPSED](#)

1.22.4.14 Specifying Time Server/Client

Configure TIME as a client only when the target system has multiple processors running separate instantiations of the cFE. One instantiation must be configured as the server and the remainder configured as clients. If the target system has only one processor running the cFE, then TIME must be configured as a server.

Enable one, and only one, of the following definitions in the platform configuration header file:

```
#define CFE_PLATFORM_TIME_CFG_SERVER TRUE
#define CFE_PLATFORM_TIME_CFG_CLIENT FALSE
```

or

```
#define CFE_PLATFORM_TIME_CFG_SERVER FALSE
#define CFE_PLATFORM_TIME_CFG_CLIENT TRUE
```

See also

[CFE_PLATFORM_TIME_CFG_SERVER](#), [CFE_PLATFORM_TIME_CFG_CLIENT](#)

1.22.4.15 Specifying Time Tone Byte Order

By default, the CFE time tone message is a payload of integers in platform-endian order (containing the tone's timestamp, the leap seconds, and state information.) In some configurations, it may be better to have the payload produced in big-endian order—particularly in mixed-endian environments.

In order to force the tone message to be in big-endian order, you must define the following:

```
#define CFE_PLATFORM_TIME_CFG_BIGENDIAN
```

1.22.4.16 Virtual MET

This configuration option refers to whether the MET is local to this instantiation of TIME. If the MET is not local then TIME must be configured as using a virtual MET.

Therefore, all TIME clients must be configured as using a virtual MET. If the MET was local to any TIME client, then that instantiation of TIME would have to be the server.

TIME servers must be configured as using a virtual MET

1.22.4.17 Specifying Time Source

TIME configuration provides the ability to specify where the source for time data is originating - either internal or external. In hypothetical system one, the MET is internal. In system two, TIME cannot directly read the MET, therefore time data must be received from an external source.

This selection also enables a command interface to switch between internal and external input. When commanded to use internal time data, TIME will ignore the external data. However, TIME will continue to use the API function as the trigger to generate a "time at the tone" command packet regardless of the internal/external command selection.

Set the following definition to TRUE only for TIME servers using an external time data source.

```
#define CFE_PLATFORM_TIME_CFG_SOURCE TRUE
```

The remainder of this section pertains only to TIME servers configured to accept external time data.

When configured to accept external time data, TIME requires an additional definition for the type of external data (GPS, MET, spacecraft time, etc.). This selection will enable an API function specific to the selected data type. Regardless of how the time data is received, the receiver need only pass the data to the appropriate API function.

TIME servers using an external time data source must set one, and only one, of the following to TRUE, for example:

```
#define CFE_PLATFORM_TIME_CFG_SRC_MET    TRUE
#define CFE_PLATFORM_TIME_CFG_SRC_GPS    FALSE
#define CFE_PLATFORM_TIME_CFG_SRC_TIME   FALSE
```

configuration definitions for the particular source.

If the `cfe_platform_cfg.h` file contains `"#define CFE_PLATFORM_TIME_CFG_SOURCE TRUE"` then time is configured to allow switching between internal and external time sources (see [CFE_TIME_SET_SOURCE_CC](#)). If this configuration parameter is set to `FALSE` then the command to set the source will be rejected.

If this configuration parameter is set to `TRUE` then ONE and ONLY ONE of the following configuration parameters must also be set `TRUE` in order to specify the external time source, for example:

```
#define CFE_PLATFORM_TIME_CFG_SRC_MET    TRUE
#define CFE_PLATFORM_TIME_CFG_SRC_GPS    FALSE
#define CFE_PLATFORM_TIME_CFG_SRC_TIME   FALSE
```

Note that Internal MET source depends on available hardware. It may be the local count of tone signals, the contents of a hardware register or an OS specific time function.

Note also that when configured to use an external time source, commands to set the time will be overwritten.

See also

[CFE_PLATFORM_TIME_CFG_SRC_MET](#), [CFE_PLATFORM_TIME_CFG_SRC_GPS](#), [CFE_PLATFORM_TIME_CFG_SRC_TIME](#)

1.22.4.18 Specifying Time Signal

Some hardware configurations support a primary and redundant tone signal selection. Setting the following configuration definition to `TRUE` will result in enabling a `TIME` command to select the active tone signal.

```
#define CFE_PLATFORM_TIME_CFG_SIGNAL    TRUE
```

Note: this feature requires additional custom software to make the physical signal switch.

See also

[CFE_PLATFORM_TIME_CFG_SIGNAL](#)

1.22.5 Time Services Paradigm(s)

In order for the cFE Time Services to work for a particular mission, the methods of obtaining time, distributing time and translating time must follow some standard paradigms used in previous missions. The following describes this expected context:

Mission dependent hardware provides the Tone. When this Tone message is received, `TIME` latches the local time based on the local clock. Note that in lab environments, a simulated Tone capability exists which uses an SB message. Mission dependent hardware also provides the "time at the tone" message based on the hardware latched time and the reference times stored by `TIME` Server. The `TIME` Client then updates its local reference time based on the local hardware latched time at the Tone and the provided Time-at-Tone message packet when certain checks (such as the Validity bit being set) pass.

When used in an environment that includes multiple processors, each running a separate instantiation of cFE software, the presumption is that TIME will be distributed in a client/server relationship. In this model, one processor will have TIME configured as the server and the other processors as clients. The TIME server will maintain the various time components and publish a "time at the tone" message to provide synchronized time to the TIME clients. Environments that have only a single instance of TIME must be configured as a TIME server.

In all configurations, the final step in calculating the time "right now" for any instantiation of TIME is to use a local processor clock to measure the "time since the tone".

The specific MET hardware properties will determine whether the MET value can be modified. However, the cFE design is such that there should never be a need to purposefully change or reset the MET.

Regardless of the physical hardware implementation for the MET (elapsed seconds, elapsed ticks, etc.), cFE TIME will convert the hardware MET value into a System Time Format structure for time calculations and will report the converted value in telemetry. cFE TIME will also maintain and report the STCF in a System Time Format structure.

cFE TIME has no knowledge of the current epoch; it is up to the user to keep time on the spacecraft correlated to an epoch. An exception might appear to be the epoch definition required in the cFE mission configuration definition file. However, this definition is for use only by the API functions that convert spacecraft time and file system time, and the API function that prints spacecraft time as a date and time text string. The cFE "get time" functions are independent of the ground epoch.

The mission configuration parameters, [CFE_MISSION_TIME_CFG_DEFAULT_TAI](#) and [CFE_MISSION_TIME_CFG_DEFAULT_UTC](#) specify the default time format. Applications are encouraged to use the [CFE_TIME_GetTime](#) API, which returns time in the format specified by this configuration parameter.

1.22.6 Flywheeling

Flywheeling occurs when TIME is not getting a valid tone signal or external "time at the tone" message. While this has minimal impact on internal operations, it can result in the drifting apart of times being stored by different spacecraft systems.

Flywheeling occurs when at least one of the following conditions is true:

- loss of tone signal
- loss of "time at the tone" data packet
- signal and packet not within valid window
- commanded into fly-wheel mode

If the TIME server is in Flywheel mode then the TIME client is also in flywheel mode.

1.22.7 Time State

Clock state is a combination of factors, most significantly whether the spacecraft time has been accurately set and whether Time Service is operating in FLYWHEEL mode. A ground command is provided to set the state to reflect when the ground has determined the spacecraft time is now correct, or that time is no longer correct. This information will be distributed to Time Clients, and in turn, to any interested sub-systems. If time has not been set then TIME services reports the state of time as invalid, regardless of whether time is flywheeling or not. Also, this command may be used to force a Time Server or Time Client into FLYWHEEL mode. Use of FLYWHEEL mode is mainly for debug purposes although, in extreme circumstances, it may be of value to force Time Service not to rely on normal time updates. Note that when commanded into FLYWHEEL mode, the Time Service will remain so until receipt of another "set state" command setting the state into a mode other than FLYWHEEL. Note also that setting the clock state to VALID or INVALID on a Time Client that is currently getting time updates from the Time Server will have very limited effect. As soon as the Time Client receives the next time update, the VALID/INVALID selection will be set to that of the Time Server. However, setting a Time Client to FLYWHEEL cannot be overridden by the Time Server since the Time Client will ignore time updates from the Time Server while in FLYWHEEL mode.

1.22.8 Initialization

No action is required by the ground to initialize the TIME software; however, time variables in the TIME Server must be set by command to allow correct time to propagate.

For a description of what happens during each type of reset, see below:

- [Power-On Reset](#)
- [Processor Reset](#)

1.22.8.1 Power-On Reset

TIME initializes all counters in housekeeping telemetry, sets the Validity state to Invalid, and initializes the STCF, Leap Seconds, and 1 Hz Adjustment to zero.

1.22.8.2 Processor Reset

In the event of a processor reset, the following time values are preserved:

- MET
- STCF
- Leap Seconds

- Clock Signal Selection
- Current Time Client Delay (if applicable)

Note that since it is virtually impossible for TIME services to validate the actual data that is saved across a processor reset, a signature pattern is written to the preserved area. On a processor reset, TIME queries that signature to make sure that it matches what is expected. If the signature does not match, then TIME is initialized as if a cFE power-on reset occurred.

1.22.8.3 Power-On Reset

TIME initializes all counters in housekeeping telemetry, sets the Validity state to Invalid, and initializes the STCF, Leap Seconds, and 1 Hz Adjustment to zero.

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- Leap Seconds
- Clock Signal Selection
- Current Time Client Delay (if applicable)

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1.22.9 Normal Operation

The following sections describe the operator's responsibilities for maintaining time under nominal conditions:

- [Client](#)
- [Server](#)

1.22.9.1 Client

Under normal operation, TIME Client systems do not require any attention from the ground, however TIME clients do provide commands to set the persistent latency between the server and client. Latency can be either added or subtracted to the current TIME client time calculation to account for the latency.

1.22.9.2 Server

TIME Servers require maintenance by the operations team to ensure the spacecraft is maintaining a time that can be successfully correlated to other entities. The following sections describe the commands that the operations team can use to help maintain a proper time reference:

- [Setting Time](#)
- [Adjusting Time](#)
- [Setting MET](#)

1.22.9.2.1 Setting Time

The Time Server provides commands to set time. The new time value represents the desired offset from mission-defined time epoch and takes effect immediately upon execution of this command. Time Service will calculate a new STCF value based on the current MET and the desired new time using one of the following:

If Time Service is configured to compute current time as TAI:

```
STCF = new time - current MET
current time = current MET + STCF
```

If Time Service is configured to compute current time as UTC:

```
STCF = ((new time) - (current MET)) + Leap Seconds
current time = ((current MET) + STCF) - Leap Seconds
```

See also

[CFE_TIME_SET_TIME_CC](#)

1.22.9.2.2 Adjusting Time

The TIME Server includes commands to set the STCF, Leap Seconds, and Validity state. The STCF should be set implicitly using the [CFE_TIME_SET_TIME_CC](#) or explicitly using [CFE_TIME_SET_STCF_CC](#). TIME provides the ability to command a one time adjustment ([CFE_TIME_ADD_ADJUST_CC](#) and [CFE_TIME_SUB_ADJUST_CC](#)) to the current STCF. In addition there is a 1Hz adjustment ([CFE_TIME_ADD_ONE_HZ_ADJUSTMENT_CC](#) and [CFE_TIME_SUB_ONE_HZ_ADJUSTMENT_CC](#)) that can be made to the STCF to compensate for oscillator drift. Mission specific ground correlation should be used to assist in determining the proper values to use. The Leap Seconds should be set to the current TAI-UTC. Note that the International Earth Rotation and Reference Systems Service Bulletin C, which defines the current difference, reports it as UTC-TAI, and thus that value must be negated. **The Leap Seconds value will always be a positive number.** The Validity state does not have to be set to invalid to change the STCF or Leap Seconds, and should be set to valid at any time that the TIME Server time reference should be synchronized to by the other systems.

See also

[CFE_TIME_ADD_ADJUST_CC](#), [CFE_TIME_SUB_ADJUST_CC](#), [CFE_TIME_SET_STCF_CC](#), [CFE_TIME_ADD_ONE_HZ_ADJUSTMENT_CC](#), [CFE_TIME_SUB_ONE_HZ_ADJUSTMENT_CC](#), [CFE_TIME_SET_LEAP_SECONDS_CC](#)

1.22.9.2.3 Setting MET

The TIME Server provides the capability to set the MET. Note that the MET (as implemented for cFE Time Service) is a logical representation and not a physical timer. Thus, setting the MET is not dependent on whether the hardware supports a MET register that can be written to. Note also that Time Service "assumes" that during normal operation, the MET is synchronized to the tone signal. Therefore, unless operating in FLYWHEEL mode, the sub-seconds portion of the MET will be set to zero at the next tone signal interrupt. The new MET takes effect immediately upon execution of this command.

See also

[CFE_TIME_SET_MET_CC](#)

1.22.9.3 Client

Under normal operation, TIME Client systems do not require any attention from the ground, however TIME clients do provide commands to set the persistent latency between the server and client. Latency can be either added or subtracted to the current TIME client time calculation to account for the latency.

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- [Setting Time](#)
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The Time Server provides commands to set time. The new time value represents the desired offset from mission-defined time epoch and takes effect immediately upon execution of this command. Time Service will calculate a new STCF value based on the current MET and the desired new time using one of the following:

If Time Service is configured to compute current time as TAI:

```
STCF = new time - current MET
current time = current MET + STCF
```

If Time Service is configured to compute current time as UTC:

```
STCF = ((new time) - (current MET)) + Leap Seconds
current time = ((current MET) + STCF) - Leap Seconds
```

See also

[CFE_TIME_SET_TIME_CC](#)

1.22.9.4.2 Adjusting Time

The TIME Server includes commands to set the STCF, Leap Seconds, and Validity state. The STCF should be set implicitly using the [CFE_TIME_SET_TIME_CC](#) or explicitly using [CFE_TIME_SET_STCF_CC](#). TIME provides the ability to command a one time adjustment ([CFE_TIME_ADD_ADJUST_CC](#) and [CFE_TIME_SUB_ADJUST_CC](#)) to the current STCF. In addition there is a 1Hz adjustment ([CFE_TIME_ADD_ONE_HZ_ADJUSTMENT_CC](#) and [CFE_TIME_SUB_ONE_HZ_ADJUSTMENT_CC](#)) that can be made to the STCF to compensate for oscillator drift. Mission specific ground correlation should be used to assist in determining the proper values to use. The Leap Seconds should be set to the current TAI-UTC. Note that the International Earth Rotation and Reference Systems Service Bulletin C, which defines the current difference, reports it as UTC-TAI, and thus that value must be negated. **The Leap Seconds value will always be a positive number.** The Validity state does not have to be set to invalid to change the STCF or Leap Seconds, and should be set to valid at any time that the TIME Server time reference should be synchronized to by the other systems.

See also

[CFE_TIME_ADD_ADJUST_CC](#), [CFE_TIME_SUB_ADJUST_CC](#), [CFE_TIME_SET_STCF_CC](#), [CFE_TIME_ADD_ONE_HZ_ADJUSTMENT_CC](#), [CFE_TIME_SUB_ONE_HZ_ADJUSTMENT_CC](#), [CFE_TIME_SET_LEAP_SECONDS_CC](#)

1.22.9.4.3 Setting MET

The TIME Server provides the capability to set the MET. Note that the MET (as implemented for cFE Time Service) is a logical representation and not a physical timer. Thus, setting the MET is not dependent on whether the hardware supports a MET register that can be written to. Note also that Time Service "assumes" that during normal operation, the MET is synchronized to the tone signal. Therefore, unless operating in FLYWHEEL mode, the sub-seconds portion of the MET will be set to zero at the next tone signal interrupt. The new MET takes effect immediately upon execution of this command.

See also

[CFE_TIME_SET_MET_CC](#)

1.22.9.4.4 Setting Time

The Time Server provides commands to set time. The new time value represents the desired offset from mission-defined time epoch and takes effect immediately upon execution of this command. Time Service will calculate a new STCF value based on the current MET and the desired new time using one of the following:

If Time Service is configured to compute current time as TAI:

```
STCF = new time - current MET
current time = current MET + STCF
```

If Time Service is configured to compute current time as UTC:

```
STCF = ((new time) - (current MET)) + Leap Seconds
current time = ((current MET) + STCF) - Leap Seconds
```

See also

[CFE_TIME_SET_TIME_CC](#)

1.22.9.4.5 Adjusting Time

The TIME Server includes commands to set the STCF, Leap Seconds, and Validity state. The STCF should be set implicitly using the [CFE_TIME_SET_TIME_CC](#) or explicitly using [CFE_TIME_SET_STCF_CC](#). TIME provides the ability to command a one time adjustment ([CFE_TIME_ADD_ADJUST_CC](#) and [CFE_TIME_SUB_ADJUST_CC](#)) to the current STCF. In addition there is a 1Hz adjustment ([CFE_TIME_ADD_ONE_HZ_ADJUSTMENT_CC](#) and [CFE_TIME_SUB_ONE_HZ_ADJUSTMENT_CC](#)) that can be made to the STCF to compensate for oscillator drift. Mission specific ground correlation should be used to assist in determining the proper values to use. The Leap Seconds should be set to the current TAI-UTC. Note that the International Earth Rotation and Reference Systems Service Bulletin C, which defines the current difference, reports it as UTC-TAI, and thus that value must be negated. **The Leap Seconds value will always be a positive number.** The Validity state does not have to be set to invalid to change the STCF or Leap Seconds, and should be set to valid at any time that the TIME Server time reference should be synchronized to by the other systems.

See also

[CFE_TIME_ADD_ADJUST_CC](#), [CFE_TIME_SUB_ADJUST_CC](#), [CFE_TIME_SET_STCF_CC](#), [CFE_TIME_ADD_ONE_HZ_ADJUSTMENT_CC](#), [CFE_TIME_SUB_ONE_HZ_ADJUSTMENT_CC](#), [CFE_TIME_SET_LEAP_SECONDS_CC](#)

1.22.9.4.6 Setting MET

The TIME Server provides the capability to set the MET. Note that the MET (as implemented for cFE Time Service) is a logical representation and not a physical timer. Thus, setting the MET is not dependent on whether the hardware supports a MET register that can be written to. Note also that Time Service "assumes" that during normal operation, the MET is synchronized to the tone signal. Therefore, unless operating in FLYWHEEL mode, the sub-seconds portion of the MET will be set to zero at the next tone signal interrupt. The new MET takes effect immediately upon execution of this command.

See also

[CFE_TIME_SET_MET_CC](#)

1.22.10 Frequently Asked Questions about Time Services

None submitted

1.23 cFE Time Services Commands

Upon receipt of any command, the Time Services application will confirm that the message length embedded within the header (from `CFE_MSG_GetSize()`) matches the expected length of that message, based on the size of the C structure defining that command. If there is any discrepancy between the expected and actual message size, TIME will generate the `CFE_TIME_LEN_ERR_EID` event, increment the command error counter (`$sc_$cpu_TIME_CMDEC`), and the command will *not* be accepted for processing.

The following is a list of commands that are processed by the cFE Time Services Task.

Global `CFE_TIME_ADD_ADJUST_CC`

Add Delta to Spacecraft Time Correlation Factor

Global `CFE_TIME_ADD_DELAY_CC`

Add Time to Tone Time Delay

Global `CFE_TIME_ADD_ONE_HZ_ADJUSTMENT_CC`

Add Delta to Spacecraft Time Correlation Factor each 1Hz

Global `CFE_TIME_NOOP_CC`

Time No-Op

Global `CFE_TIME_RESET_COUNTERS_CC`

Time Reset Counters

Global `CFE_TIME_SEND_DIAGNOSTIC_CC`

Request TIME Diagnostic Telemetry

Global `CFE_TIME_SET_LEAP_SECONDS_CC`

Set Leap Seconds

Global `CFE_TIME_SET_MET_CC`

Set Mission Elapsed Time

Global `CFE_TIME_SET_SIGNAL_CC`

Set Tone Signal Source

Global `CFE_TIME_SET_SOURCE_CC`

Set Time Source

Global `CFE_TIME_SET_STATE_CC`

Set Time State

Global `CFE_TIME_SET_STCF_CC`

Set Spacecraft Time Correlation Factor

Global `CFE_TIME_SET_TIME_CC`

Set Spacecraft Time

Global `CFE_TIME_SUB_ADJUST_CC`

Subtract Delta from Spacecraft Time Correlation Factor

Global `CFE_TIME_SUB_DELAY_CC`

Subtract Time from Tone Time Delay

Global `CFE_TIME_SUB_ONE_HZ_ADJUSTMENT_CC`

Subtract Delta from Spacecraft Time Correlation Factor each 1Hz

1.24 cFE Time Services Telemetry

The following are telemetry packets generated by the cFE Time Services Task.

Global `CFE_TIME_DiagnosticTlm_Payload_t`

Time Services Diagnostics Packet

Global `CFE_TIME_HousekeepingTlm_Payload_t`

Time Services Housekeeping Packet

1.25 cFE Time Services Configuration Parameters

The following are configuration parameters used to configure the cFE Time Services either for each platform or for a mission as a whole.

Global `CFE_MISSION_TIME_AT_TONE_WAS`

Default Time and Tone Order

Default Time and Tone Order

Global `CFE_MISSION_TIME_CFG_DEFAULT_TAI`

Default Time Format

Default Time Format

Global `CFE_MISSION_TIME_CFG_FAKE_TONE`

Default Time Format

Default Time Format

Global `CFE_MISSION_TIME_DEF_MET_SECS`

Default Time Values

Default Time Values

Global `CFE_MISSION_TIME_EPOCH_YEAR`

Default EPOCH Values

Default EPOCH Values

Global `CFE_MISSION_TIME_FS_FACTOR`

Time File System Factor

Time File System Factor

Global `CFE_MISSION_TIME_MIN_ELAPSED`

Min and Max Time Elapsed

Min and Max Time Elapsed

Global `CFE_PLATFORM_TIME_CFG_LATCH_FLY`

Define Periodic Time to Update Local Clock Tone Latch

Define Periodic Time to Update Local Clock Tone Latch

Global `CFE_PLATFORM_TIME_CFG_SERVER`

Time Server or Time Client Selection

Time Server or Time Client Selection

Global [CFE_PLATFORM_TIME_CFG_SIGNAL](#)

Include or Exclude the Primary/Redundant Tone Selection Cmd

Include or Exclude the Primary/Redundant Tone Selection Cmd

Global [CFE_PLATFORM_TIME_CFG_SOURCE](#)

Include or Exclude the Internal/External Time Source Selection Cmd

Include or Exclude the Internal/External Time Source Selection Cmd

Global [CFE_PLATFORM_TIME_CFG_SRC_MET](#)

Choose the External Time Source for Server only

Choose the External Time Source for Server only

Global [CFE_PLATFORM_TIME_CFG_START_FLY](#)

Define Time to Start Flywheel Since Last Tone

Define Time to Start Flywheel Since Last Tone

Global [CFE_PLATFORM_TIME_CFG_TONE_LIMIT](#)

Define Timing Limits From One Tone To The Next

Define Timing Limits From One Tone To The Next

Global [CFE_PLATFORM_TIME_CFG_VIRTUAL](#)

Local MET or Virtual MET Selection for Time Servers

Time Tone In Big-Endian Order

Local MET or Virtual MET Selection for Time Servers

Global [CFE_PLATFORM_TIME_MAX_DELTA_SECS](#)

Define the Max Delta Limits for Time Servers using an Ext Time Source

Define the Max Delta Limits for Time Servers using an Ext Time Source

Global [CFE_PLATFORM_TIME_MAX_LOCAL_SECS](#)

Define the Local Clock Rollover Value in seconds and subseconds

Define the Local Clock Rollover Value in seconds and subseconds

Global [CFE_PLATFORM_TIME_START_TASK_PRIORITY](#)

Define TIME Task Priorities

Define TIME Task Priorities

Global [CFE_PLATFORM_TIME_START_TASK_STACK_SIZE](#)

Define TIME Task Stack Sizes

Define TIME Task Stack Sizes

1.26 cFE Event Message Cross Reference

The following cross reference maps the text associated with each cFE Event Message to its Event Message Identifier. A user can search this page for the text of the message they wish to learn more about and then click on the associated Event Message Identifier to obtain more information.

1.27 cFE Command Mnemonic Cross Reference

The following cross reference maps the cFE command codes to Command Mnemonics. To learn about the details of a particular command, click on its associated command code.

Global [CFE_ES_CLEAR_ER_LOG_CC](#)

\$sc_\$cpu_ES_ClearERLog

Global [CFE_ES_CLEAR_SYS_LOG_CC](#)

\$sc_\$cpu_ES_ClearSysLog

Global [CFE_ES_DELETE_CDS_CC](#)

\$sc_\$cpu_ES_DeleteCDS

Global [CFE_ES_DUMP_CDS_REGISTRY_CC](#)

\$sc_\$cpu_ES_WriteCDS2File

Global [CFE_ES_NOOP_CC](#)

\$sc_\$cpu_ES_NOOP

Global [CFE_ES_OVER_WRITE_SYS_LOG_CC](#)

\$sc_\$cpu_ES_OverwriteSysLogMode

Global [CFE_ES_QUERY_ALL_CC](#)

\$sc_\$cpu_ES_WriteAppInfo2File

Global [CFE_ES_QUERY_ALL_TASKS_CC](#)

\$sc_\$cpu_ES_WriteTaskInfo2File

Global [CFE_ES_QUERY_ONE_CC](#)

\$sc_\$cpu_ES_QueryApp

Global [CFE_ES_RELOAD_APP_CC](#)

\$sc_\$cpu_ES_ReloadApp

Global [CFE_ES_RESET_COUNTERS_CC](#)

\$sc_\$cpu_ES_ResetCtrs

Global [CFE_ES_RESET_PR_COUNT_CC](#)

\$sc_\$cpu_ES_ResetPRCnt

Global [CFE_ES_RESTART_APP_CC](#)

\$sc_\$cpu_ES_ResetApp

Global [CFE_ES_RESTART_CC](#)

\$sc_\$cpu_ES_ProcessorReset, \$sc_\$cpu_ES_PowerOnReset

Global [CFE_ES_SEND_MEM_POOL_STATS_CC](#)

\$sc_\$cpu_ES_PoolStats

Global [CFE_ES_SET_MAX_PR_COUNT_CC](#)

\$sc_\$cpu_ES_SetMaxPRCnt

Global [CFE_ES_SET_PERF_FILTER_MASK_CC](#)

\$sc_\$cpu_ES_LAFilterMask

Global [CFE_ES_SET_PERF_TRIGGER_MASK_CC](#)

\$sc_\$cpu_ES_LATriggerMask

Global [CFE_ES_START_APP_CC](#)

\$sc_\$cpu_ES_StartApp

Global CFE_ES_START_PERF_DATA_CC`$sc_$cpu_ES_StartLAData`**Global CFE_ES_STOP_APP_CC**`$sc_$cpu_ES_StopApp`**Global CFE_ES_STOP_PERF_DATA_CC**`$sc_$cpu_ES_StopLAData`**Global CFE_ES_WRITE_ER_LOG_CC**`$sc_$cpu_ES_WriteERLog2File`**Global CFE_ES_WRITE_SYS_LOG_CC**`$sc_$cpu_ES_WriteSysLog2File`**Global CFE_EVS_ADD_EVENT_FILTER_CC**`$sc_$cpu_EVS_AddEvtFiltr`**Global CFE_EVS_CLEAR_LOG_CC**`$sc_$cpu_EVS_ClrLog`**Global CFE_EVS_DELETE_EVENT_FILTER_CC**`$sc_$cpu_EVS_DelEvtFiltr`**Global CFE_EVS_DISABLE_APP_EVENT_TYPE_CC**`$sc_$cpu_EVS_DisAppEvtType, $sc_$cpu_EVS_DisAppEvtTypeMask`**Global CFE_EVS_DISABLE_APP_EVENTS_CC**`$sc_$cpu_EVS_DisAppEvGen`**Global CFE_EVS_DISABLE_EVENT_TYPE_CC**`$sc_$cpu_EVS_DisEventType, $sc_$cpu_EVS_DisEventTypeMask`**Global CFE_EVS_DISABLE_PORTS_CC**`$sc_$cpu_EVS_DisPort, $sc_$cpu_EVS_DisPortMask`**Global CFE_EVS_ENABLE_APP_EVENT_TYPE_CC**`$sc_$cpu_EVS_EnaAppEvtType, $sc_$cpu_EVS_EnaAppEvtTypeMask`**Global CFE_EVS_ENABLE_APP_EVENTS_CC**`$sc_$cpu_EVS_EnaAppEvGen`**Global CFE_EVS_ENABLE_EVENT_TYPE_CC**`$sc_$cpu_EVS_EnaEventType, $sc_$cpu_EVS_EnaEventTypeMask`**Global CFE_EVS_ENABLE_PORTS_CC**`$sc_$cpu_EVS_EnaPort, $sc_$cpu_EVS_EnaPortMask`**Global CFE_EVS_NOOP_CC**`$sc_$cpu_EVS_NOOP`**Global CFE_EVS_RESET_ALL_FILTERS_CC**`$sc_$cpu_EVS_RstAllFiltrs`**Global CFE_EVS_RESET_APP_COUNTER_CC**`$sc_$cpu_EVS_RstAppCtrs`**Global CFE_EVS_RESET_COUNTERS_CC**`$sc_$cpu_EVS_ResetCtrs`**Global CFE_EVS_RESET_FILTER_CC**`$sc_$cpu_EVS_RstBinFiltrCtr`

Global CFE_EVS_SET_EVENT_FORMAT_MODE_CC`$sc_$cpu_EVS_SetEvtFmt`**Global CFE_EVS_SET_FILTER_CC**`$sc_$cpu_EVS_SetBinFiltrMask`**Global CFE_EVS_SET_LOG_MODE_CC**`$sc_$cpu_EVS_SetLogMode`**Global CFE_EVS_WRITE_APP_DATA_FILE_CC**`$sc_$cpu_EVS_WriteAppData2File`**Global CFE_EVS_WRITE_LOG_DATA_FILE_CC**`$sc_$cpu_EVS_WriteLog2File`**Global CFE_SB_DISABLE_ROUTE_CC**`$sc_$cpu_SB_DisRoute`**Global CFE_SB_DISABLE_SUB_REPORTING_CC**`$sc_$cpu_SB_DisSubRptg`**Global CFE_SB_ENABLE_ROUTE_CC**`$sc_$cpu_SB_EnaRoute`**Global CFE_SB_ENABLE_SUB_REPORTING_CC**`$sc_$cpu_SB_EnaSubRptg`**Global CFE_SB_NOOP_CC**`$sc_$cpu_SB_NOOP`**Global CFE_SB_RESET_COUNTERS_CC**`$sc_$cpu_SB_ResetCtrs`**Global CFE_SB_SEND_PREV_SUBS_CC**`$sc_$cpu_SB_SendPrevSubs`**Global CFE_SB_SEND_SB_STATS_CC**`$sc_$cpu_SB_DumpStats`**Global CFE_SB_WRITE_MAP_INFO_CC**`$sc_$cpu_SB_WriteMap2File`**Global CFE_SB_WRITE_PIPE_INFO_CC**`$sc_$cpu_SB_WritePipe2File`**Global CFE_SB_WRITE_ROUTING_INFO_CC**`$sc_$cpu_SB_WriteRouting2File`**Global CFE_TBL_ABORT_LOAD_CC**`$sc_$cpu_TBL_LOADABORT`**Global CFE_TBL_ACTIVATE_CC**`$sc_$cpu_TBL_ACTIVATE`**Global CFE_TBL_DELETE_CDS_CC**`$sc_$cpu_TBL_DeleteCDS`**Global CFE_TBL_DUMP_CC**`$sc_$cpu_TBL_DUMP`**Global CFE_TBL_DUMP_REGISTRY_CC**`$sc_$cpu_TBL_WriteReg2File`

Global CFE_TBL_LOAD_CC

\$sc_\$cpu_TBL_Load

Global CFE_TBL_NOOP_CC

\$sc_\$cpu_TBL_NOOP

Global CFE_TBL_RESET_COUNTERS_CC

\$sc_\$cpu_TBL_ResetCtrs

Global CFE_TBL_SEND_REGISTRY_CC

\$sc_\$cpu_TBL_TLMReg

Global CFE_TBL_VALIDATE_CC

\$sc_\$cpu_TBL_VALIDATE

Global CFE_TIME_ADD_ADJUST_CC

\$sc_\$cpu_TIME_AddSTCFAdj

Global CFE_TIME_ADD_DELAY_CC

\$sc_\$cpu_TIME_AddClockLat

Global CFE_TIME_ADD_ONE_HZ_ADJUSTMENT_CC

\$sc_\$cpu_TIME_Add1HzSTCF

Global CFE_TIME_NOOP_CC

\$sc_\$cpu_TIME_NOOP

Global CFE_TIME_RESET_COUNTERS_CC

\$sc_\$cpu_TIME_ResetCtrs

Global CFE_TIME_SEND_DIAGNOSTIC_CC

\$sc_\$cpu_TIME_RequestDiag

Global CFE_TIME_SET_LEAP_SECONDS_CC

\$sc_\$cpu_TIME_SetClockLeap

Global CFE_TIME_SET_MET_CC

\$sc_\$cpu_TIME_SetClockMET

Global CFE_TIME_SET_SIGNAL_CC

\$sc_\$cpu_TIME_SetSignal

Global CFE_TIME_SET_SOURCE_CC

\$sc_\$cpu_TIME_SetSource

Global CFE_TIME_SET_STATE_CC

\$sc_\$cpu_TIME_SetState

Global CFE_TIME_SET_STCF_CC

\$sc_\$cpu_TIME_SetClockSTCF

Global CFE_TIME_SET_TIME_CC

\$sc_\$cpu_TIME_SetClock

Global CFE_TIME_SUB_ADJUST_CC

\$sc_\$cpu_TIME_SubSTCFAdj

Global CFE_TIME_SUB_DELAY_CC

\$sc_\$cpu_TIME_SubClockLat

Global CFE_TIME_SUB_ONE_HZ_ADJUSTMENT_CC

\$sc_\$cpu_TIME_Sub1HzSTCF

1.28 cFE Telemetry Mnemonic Cross Reference

The following cross reference maps the cFE telemetry packet members to their associated ground system telemetry mnemonics.

Global CFE_ES_AppInfo::AddressesAreValid

\$sc_\$cpu_ES_AddrsValid

Global CFE_ES_AppInfo::BSSAddress

\$sc_\$cpu_ES_BSSAddress

Global CFE_ES_AppInfo::BSSSize

\$sc_\$cpu_ES_BSSSize

Global CFE_ES_AppInfo::CodeAddress

\$sc_\$cpu_ES_CodeAddress

Global CFE_ES_AppInfo::CodeSize

\$sc_\$cpu_ES_CodeSize

Global CFE_ES_AppInfo::DataAddress

\$sc_\$cpu_ES_DataAddress

Global CFE_ES_AppInfo::DataSize

\$sc_\$cpu_ES_DataSize

Global CFE_ES_AppInfo::EntryPoint [CFE_MISSION_MAX_API_LEN]

\$sc_\$cpu_ES_AppEntryPt[OS_MAX_API_NAME]

Global CFE_ES_AppInfo::ExceptionAction

\$sc_\$cpu_ES_ExceptnActn

Global CFE_ES_AppInfo::ExecutionCounter

\$sc_\$cpu_ES_ExecutionCtr

Global CFE_ES_AppInfo::FileName [CFE_MISSION_MAX_PATH_LEN]

\$sc_\$cpu_ES_AppFilename[OS_MAX_PATH_LEN]

Global CFE_ES_AppInfo::MainTaskId

\$sc_\$cpu_ES_MainTaskId

Global CFE_ES_AppInfo::MainTaskName [CFE_MISSION_MAX_API_LEN]

\$sc_\$cpu_ES_MainTaskName[OS_MAX_API_NAME]

Global CFE_ES_AppInfo::Name [CFE_MISSION_MAX_API_LEN]

\$sc_\$cpu_ES_AppName[OS_MAX_API_NAME]

Global CFE_ES_AppInfo::NumOfChildTasks

\$sc_\$cpu_ES_ChildTasks

Global CFE_ES_AppInfo::Priority

\$sc_\$cpu_ES_Priority

Global CFE_ES_AppInfo::ResourceId

\$sc_\$cpu_ES_AppID

Global CFE_ES_AppInfo::StackSize

\$sc_\$cpu_ES_StackSize

Global CFE_ES_AppInfo::StartAddress

\$sc_\$cpu_ES_StartAddr

Global CFE_ES_AppInfo::Type

\$sc_\$cpu_ES_AppType

Global CFE_ES_HousekeepingTlm_Payload::BootSource

\$sc_\$cpu_ES_BootSource

Global CFE_ES_HousekeepingTlm_Payload::CFECoreChecksum

\$sc_\$cpu_ES_CKSUM

Global CFE_ES_HousekeepingTlm_Payload::CFEMajorVersion

\$sc_\$cpu_ES_CFEMAJORVER

Global CFE_ES_HousekeepingTlm_Payload::CFEMinorVersion

\$sc_\$cpu_ES_CFEMINORVER

Global CFE_ES_HousekeepingTlm_Payload::CFEMissionRevision

\$sc_\$cpu_ES_CFEMISSIONREV

Global CFE_ES_HousekeepingTlm_Payload::CFERevision

\$sc_\$cpu_ES_CFEREVISION

Global CFE_ES_HousekeepingTlm_Payload::CommandCounter

\$sc_\$cpu_ES_CMDPC

Global CFE_ES_HousekeepingTlm_Payload::CommandErrorCounter

\$sc_\$cpu_ES_CMDEC

Global CFE_ES_HousekeepingTlm_Payload::ERLogEntries

\$sc_\$cpu_ES_ERLOGENTRIES

Global CFE_ES_HousekeepingTlm_Payload::ERLogIndex

\$sc_\$cpu_ES_ERLOGINDEX

Global CFE_ES_HousekeepingTlm_Payload::HeapBlocksFree

\$sc_\$cpu_ES_HeapBlocksFree

Global CFE_ES_HousekeepingTlm_Payload::HeapBytesFree

\$sc_\$cpu_ES_HeapBytesFree

Global CFE_ES_HousekeepingTlm_Payload::HeapMaxBlockSize

\$sc_\$cpu_ES_HeapMaxBlkSize

Global CFE_ES_HousekeepingTlm_Payload::MaxProcessorResets

\$sc_\$cpu_ES_MaxProcResets

Global CFE_ES_HousekeepingTlm_Payload::OSALMajorVersion

\$sc_\$cpu_ES_OSMAJORVER

Global CFE_ES_HousekeepingTlm_Payload::OSALMinorVersion

\$sc_\$cpu_ES_OSMINORVER

Global CFE_ES_HousekeepingTlm_Payload::OSALMissionRevision

\$sc_\$cpu_ES_OSMISSIONREV

Global CFE_ES_HousekeepingTlm_Payload::OSALRevision

\$sc_\$cpu_ES_OSREVISION

Global CFE_ES_HousekeepingTlm_Payload::PerfDataCount

\$sc_\$cpu_ES_PerfDataCnt

Global CFE_ES_HousekeepingTlm_Payload::PerfDataEnd

\$sc_\$cpu_ES_PerfDataEnd

Global CFE_ES_HousekeepingTlm_Payload::PerfDataStart
\$sc_\$cpu_ES_PerfDataStart

Global CFE_ES_HousekeepingTlm_Payload::PerfDataToWrite
\$sc_\$cpu_ES_PerfData2Write

Global CFE_ES_HousekeepingTlm_Payload::PerfFilterMask [CFE_MISSION_ES_PERF_MAX_IDS/32]
\$sc_\$cpu_ES_PerfFiltrMask[MaskCnt]

Global CFE_ES_HousekeepingTlm_Payload::PerfMode
\$sc_\$cpu_ES_PerfMode

Global CFE_ES_HousekeepingTlm_Payload::PerfState
\$sc_\$cpu_ES_PerfState

Global CFE_ES_HousekeepingTlm_Payload::PerfTriggerCount
\$sc_\$cpu_ES_PerfTrigCnt

Global CFE_ES_HousekeepingTlm_Payload::PerfTriggerMask [CFE_MISSION_ES_PERF_MAX_IDS/32]
\$sc_\$cpu_ES_PerfTrigMask[MaskCnt]

Global CFE_ES_HousekeepingTlm_Payload::ProcessorResets
\$sc_\$cpu_ES_ProcResetCnt

Global CFE_ES_HousekeepingTlm_Payload::PSPMajorVersion
\$sc_\$cpu_ES_PSPMAJORVER

Global CFE_ES_HousekeepingTlm_Payload::PSPMinorVersion
\$sc_\$cpu_ES_PSPMINORVER

Global CFE_ES_HousekeepingTlm_Payload::PSPMissionRevision
\$sc_\$cpu_ES_PSPMISSIONREV

Global CFE_ES_HousekeepingTlm_Payload::PSPRevision
\$sc_\$cpu_ES_PSPREVISION

Global CFE_ES_HousekeepingTlm_Payload::RegisteredCoreApps
\$sc_\$cpu_ES_RegCoreApps

Global CFE_ES_HousekeepingTlm_Payload::RegisteredExternalApps
\$sc_\$cpu_ES_RegExtApps

Global CFE_ES_HousekeepingTlm_Payload::RegisteredLibs
\$sc_\$cpu_ES_RegLibs

Global CFE_ES_HousekeepingTlm_Payload::RegisteredTasks
\$sc_\$cpu_ES_RegTasks

Global CFE_ES_HousekeepingTlm_Payload::ResetSubtype
\$sc_\$cpu_ES_ResetSubtype

Global CFE_ES_HousekeepingTlm_Payload::ResetType
\$sc_\$cpu_ES_ResetType

Global CFE_ES_HousekeepingTlm_Payload::SysLogBytesUsed
\$sc_\$cpu_ES_SYSLOGBYTEUSED

Global CFE_ES_HousekeepingTlm_Payload::SysLogEntries
\$sc_\$cpu_ES_SYSLOGENTRIES

Global CFE_ES_HousekeepingTlm_Payload::SysLogMode
\$sc_\$cpu_ES_SYSLOGMODE

Global CFE_ES_HousekeepingTlm_Payload::SysLogSize

\$sc_\$cpu_ES_SYSLOGSIZE

Global CFE_ES_MemPoolStats::BlockStats [CFE_MISSION_ES_POOL_MAX_BUCKETS]

\$sc_\$cpu_ES_BlkStats[BLK_SIZES]

Global CFE_ES_MemPoolStats::CheckErrCtr

\$sc_\$cpu_ES_BlkErrCTR

Global CFE_ES_MemPoolStats::NumBlocksRequested

\$sc_\$cpu_ES_BlksREQ

Global CFE_ES_MemPoolStats::NumFreeBytes

\$sc_\$cpu_ES_FreeBytes

Global CFE_ES_MemPoolStats::PoolSize

\$sc_\$cpu_ES_PoolSize

Global CFE_ES_PoolStatsTlm_Payload::PoolHandle

\$sc_\$cpu_ES_PoolHandle

Global CFE_EVS_AppTlmData::AppEnableStatus

\$sc_\$cpu_EVS_APP[CFE_PLATFORM_ES_MAX_APPLICATIONS].APPENASTAT

Global CFE_EVS_AppTlmData::AppID

\$sc_\$cpu_EVS_APP[CFE_PLATFORM_ES_MAX_APPLICATIONS].APPID

Global CFE_EVS_AppTlmData::AppMessageSentCounter

\$sc_\$cpu_EVS_APP[CFE_PLATFORM_ES_MAX_APPLICATIONS].APPMMSGSENTC

Global CFE_EVS_AppTlmData::AppMessageSquelchedCounter

\$sc_\$cpu_EVS_APP[CFE_PLATFORM_ES_MAX_APPLICATIONS].SQUELCHEDC

Global CFE_EVS_HousekeepingTlm_Payload::AppData [CFE_MISSION_ES_MAX_APPLICATIONS]

\$sc_\$cpu_EVS_APP[CFE_PLATFORM_ES_MAX_APPLICATIONS]

Global CFE_EVS_HousekeepingTlm_Payload::CommandCounter

\$sc_\$cpu_EVS_CMDPC

Global CFE_EVS_HousekeepingTlm_Payload::CommandErrorCounter

\$sc_\$cpu_EVS_CMDEC

Global CFE_EVS_HousekeepingTlm_Payload::LogEnabled

\$sc_\$cpu_EVS_LOGENABLED

Global CFE_EVS_HousekeepingTlm_Payload::LogFullFlag

\$sc_\$cpu_EVS_LOGFULL

Global CFE_EVS_HousekeepingTlm_Payload::LogMode

\$sc_\$cpu_EVS_LOGMODE

Global CFE_EVS_HousekeepingTlm_Payload::LogOverflowCounter

\$sc_\$cpu_EVS_LOGOVERFLOWC

Global CFE_EVS_HousekeepingTlm_Payload::MessageFormatMode

\$sc_\$cpu_EVS_MSGFMTMODE

Global CFE_EVS_HousekeepingTlm_Payload::MessageSendCounter

\$sc_\$cpu_EVS_MSGSENTC

Global CFE_EVS_HousekeepingTlm_Payload::MessageTruncCounter

\$sc_\$cpu_EVS_MSGTRUNC

Global CFE_EVS_HousekeepingTlm_Payload::OutputPort
\$sc_\$cpu_EVS_OUTPUTPORT

Global CFE_EVS_HousekeepingTlm_Payload::UnregisteredAppCounter
\$sc_\$cpu_EVS_UNREGAPPC

Global CFE_EVS_LongEventTlm_Payload::Message [CFE_MISSION_EVS_MAX_MESSAGE_LENGTH]
\$sc_\$cpu_EVS_EVENT[CFE_MISSION_EVS_MAX_MESSAGE_LENGTH]

Global CFE_EVS_PacketID::AppName [CFE_MISSION_MAX_API_LEN]
\$sc_\$cpu_EVS_APPNAME[OS_MAX_API_NAME]

Global CFE_EVS_PacketID::EventID
\$sc_\$cpu_EVS_EVENTID

Global CFE_EVS_PacketID::EventType
\$sc_\$cpu_EVS_EVENTTYPE

Global CFE_EVS_PacketID::ProcessorID
\$sc_\$cpu_EVS_PROCESSORID

Global CFE_EVS_PacketID::SpacecraftID
\$sc_\$cpu_EVS_SCID

Global CFE_SB_HousekeepingTlm_Payload::CommandCounter
\$sc_\$cpu_SB_CMDPC

Global CFE_SB_HousekeepingTlm_Payload::CommandErrorCounter
\$sc_\$cpu_SB_CMDEC

Global CFE_SB_HousekeepingTlm_Payload::CreatePipeErrorCounter
\$sc_\$cpu_SB_NewPipeEC

Global CFE_SB_HousekeepingTlm_Payload::DuplicateSubscriptionsCounter
\$sc_\$cpu_SB_DupSubCnt

Global CFE_SB_HousekeepingTlm_Payload::GetPipeIDByNameErrorCounter
\$sc_\$cpu_SB_GetPipeIDByNameEC

Global CFE_SB_HousekeepingTlm_Payload::InternalErrorCounter
\$sc_\$cpu_SB_InternalEC

Global CFE_SB_HousekeepingTlm_Payload::MemInUse
\$sc_\$cpu_SB_MemInUse

Global CFE_SB_HousekeepingTlm_Payload::MemPoolHandle
\$sc_\$cpu_SB_MemPoolHdl

Global CFE_SB_HousekeepingTlm_Payload::MsgLimitErrorCounter
\$sc_\$cpu_SB_MsgLimEC

Global CFE_SB_HousekeepingTlm_Payload::MsgReceiveErrorCounter
\$sc_\$cpu_SB_MsgRecEC

Global CFE_SB_HousekeepingTlm_Payload::MsgSendErrorCounter
\$sc_\$cpu_SB_MsgSndEC

Global CFE_SB_HousekeepingTlm_Payload::NoSubscribersCounter
\$sc_\$cpu_SB_NoSubEC

Global CFE_SB_HousekeepingTlm_Payload::PipeOptsErrorCounter
\$sc_\$cpu_SB_PipeOptsEC

Global CFE_SB_HousekeepingTlm_Payload::PipeOverflowErrorCounter

\$sc_\$cpu_SB_PipeOvrEC

Global CFE_SB_HousekeepingTlm_Payload::SubscribeErrorCounter

\$sc_\$cpu_SB_SubscrEC

Global CFE_SB_HousekeepingTlm_Payload::UnmarkedMem

\$sc_\$cpu_SB_UnMarkedMem

Global CFE_SB_PipeDepthStats::CurrentQueueDepth

\$sc_\$cpu_SB_Stat.SB_SMPDS[CFE_PLATFORM_SB_MAX_PIPES].SB_PDINUSE

Global CFE_SB_PipeDepthStats::MaxQueueDepth

\$sc_\$cpu_SB_Stat.SB_SMPDS[CFE_PLATFORM_SB_MAX_PIPES].SB_PDDEPTH

Global CFE_SB_PipeDepthStats::PeakQueueDepth

\$sc_\$cpu_SB_Stat.SB_SMPDS[CFE_PLATFORM_SB_MAX_PIPES].SB_PDPKINUSE

Global CFE_SB_PipeDepthStats::PipeId

\$sc_\$cpu_SB_Stat.SB_SMPDS[CFE_PLATFORM_SB_MAX_PIPES].SB_PDPIPEID

Global CFE_SB_PipeDepthStats::Spare

\$sc_\$cpu_SB_Stat.SB_SMPDS[CFE_PLATFORM_SB_MAX_PIPES].SB_PDSPARE

Global CFE_SB_StatsTlm_Payload::MaxMemAllowed

\$sc_\$cpu_SB_Stat.SB_SMMBMALW

Global CFE_SB_StatsTlm_Payload::MaxMsgIdsAllowed

\$sc_\$cpu_SB_Stat.SB_SMMMIDALW

Global CFE_SB_StatsTlm_Payload::MaxPipeDepthAllowed

\$sc_\$cpu_SB_Stat.SB_SMMPDALW

Global CFE_SB_StatsTlm_Payload::MaxPipesAllowed

\$sc_\$cpu_SB_Stat.SB_SMMPALW

Global CFE_SB_StatsTlm_Payload::MaxSubscriptionsAllowed

\$sc_\$cpu_SB_Stat.SB_SMMSALW

Global CFE_SB_StatsTlm_Payload::MemInUse

\$sc_\$cpu_SB_Stat.SB_SMBMIU

Global CFE_SB_StatsTlm_Payload::MsgIdsInUse

\$sc_\$cpu_SB_Stat.SB_SMMIDIU

Global CFE_SB_StatsTlm_Payload::PeakMemInUse

\$sc_\$cpu_SB_Stat.SB_SMPBMIU

Global CFE_SB_StatsTlm_Payload::PeakMsgIdsInUse

\$sc_\$cpu_SB_Stat.SB_SMPMIDIU

Global CFE_SB_StatsTlm_Payload::PeakPipesInUse

\$sc_\$cpu_SB_Stat.SB_SMPPIU

Global CFE_SB_StatsTlm_Payload::PeakSBBuffersInUse

\$sc_\$cpu_SB_Stat.SB_SMPSBBIU

Global CFE_SB_StatsTlm_Payload::PeakSubscriptionsInUse

\$sc_\$cpu_SB_Stat.SB_SMPSIU

Global CFE_SB_StatsTlm_Payload::PipeDepthStats [CFE_MISSION_SB_MAX_PIPES]

\$sc_\$cpu_SB_Stat.SB_SMPDS[CFE_PLATFORM_SB_MAX_PIPES]

Global CFE_SB_StatsTlm_Payload::PipesInUse

\$sc_\$cpu_SB_Stat.SB_SMPIU

Global CFE_SB_StatsTlm_Payload::SBBuffersInUse

\$sc_\$cpu_SB_Stat.SB_SMSBBIU

Global CFE_SB_StatsTlm_Payload::SubscriptionsInUse

\$sc_\$cpu_SB_Stat.SB_SMSIU

Global CFE_TBL_HousekeepingTlm_Payload::ActiveBuffer

\$sc_\$cpu_TBL_LastValBuf

Global CFE_TBL_HousekeepingTlm_Payload::ByteAlignPad1

\$sc_\$cpu_TBL_ByteAlignPad1

Global CFE_TBL_HousekeepingTlm_Payload::CommandCounter

\$sc_\$cpu_TBL_CMDPC

Global CFE_TBL_HousekeepingTlm_Payload::CommandErrorCounter

\$sc_\$cpu_TBL_CMDEC

Global CFE_TBL_HousekeepingTlm_Payload::FailedValCounter

\$sc_\$cpu_TBL_ValFailedCtr

Global CFE_TBL_HousekeepingTlm_Payload::LastFileDumped [CFE_MISSION_MAX_PATH_LEN]

\$sc_\$cpu_TBL_LastFileDumped[OS_MAX_PATH_LEN]

Global CFE_TBL_HousekeepingTlm_Payload::LastFileLoaded [CFE_MISSION_MAX_PATH_LEN]

\$sc_\$cpu_TBL_LastFileLoaded[OS_MAX_PATH_LEN]

Global CFE_TBL_HousekeepingTlm_Payload::LastTableLoaded [CFE_MISSION_TBL_MAX_FULL_NAME_LEN]

\$sc_\$cpu_TBL_LastTableLoaded[CFE_TBL_MAX_FULL_NAME_LEN]

Global CFE_TBL_HousekeepingTlm_Payload::LastUpdatedTable [CFE_MISSION_TBL_MAX_FULL_NAME_LEN]

\$sc_\$cpu_TBL_LastUpdTblName[CFE_TB_MAX_FULL_NAME_LEN]

Global CFE_TBL_HousekeepingTlm_Payload::LastUpdateTime

\$sc_\$cpu_TBL_LastUpdTime, \$sc_\$cpu_TBL_SECONDS, \$sc_\$cpu_TBL_SUBSECONDS

Global CFE_TBL_HousekeepingTlm_Payload::LastValCrc

\$sc_\$cpu_TBL_LastValCRC

Global CFE_TBL_HousekeepingTlm_Payload::LastValStatus

\$sc_\$cpu_TBL_LastValS

Global CFE_TBL_HousekeepingTlm_Payload::LastValTableName [CFE_MISSION_TBL_MAX_FULL_NAME_LEN]

\$sc_\$cpu_TBL_LastValTblName[CFE_TB_MAX_FULL_NAME_LEN]

Global CFE_TBL_HousekeepingTlm_Payload::MemPoolHandle

\$sc_\$cpu_TBL_MemPoolHandle

Global CFE_TBL_HousekeepingTlm_Payload::NumFreeSharedBufs

\$sc_\$cpu_TBL_NumFreeShrBuf

Global CFE_TBL_HousekeepingTlm_Payload::NumLoadPending

\$sc_\$cpu_TBL_NumUpdatesPend

Global CFE_TBL_HousekeepingTlm_Payload::NumTables

\$sc_\$cpu_TBL_NumTables

Global CFE_TBL_HousekeepingTlm_Payload::NumValRequests

\$sc_\$cpu_TBL_ValReqCtr

Global CFE_TBL_HousekeepingTlm_Payload::SuccessValCounter

\$sc_\$cpu_TBL_ValSuccessCtr

Global CFE_TBL_HousekeepingTlm_Payload::ValidationCounter

\$sc_\$cpu_TBL_ValCompltdCtr

Global CFE_TBL_TblRegPacket_Payload::ActiveBufferAddr

\$sc_\$cpu_TBL_ActBufAdd

Global CFE_TBL_TblRegPacket_Payload::Crc

\$sc_\$cpu_TBL_CRC

Global CFE_TBL_TblRegPacket_Payload::Critical

\$sc_\$cpu_TBL_Spare3

Global CFE_TBL_TblRegPacket_Payload::DoubleBuffered

\$sc_\$cpu_TBL_DblBuffered

Global CFE_TBL_TblRegPacket_Payload::DumpOnly

\$sc_\$cpu_TBL_DumpOnly

Global CFE_TBL_TblRegPacket_Payload::FileTime

\$sc_\$cpu_TBL_FILECTIME

Global CFE_TBL_TblRegPacket_Payload::InactiveBufferAddr

\$sc_\$cpu_TBL_IActBufAdd

Global CFE_TBL_TblRegPacket_Payload::LastFileLoaded [CFE_MISSION_MAX_PATH_LEN]

\$sc_\$cpu_TBL_LastFileUpd[OS_MAX_PATH_LEN]

Global CFE_TBL_TblRegPacket_Payload::LoadPending

\$sc_\$cpu_TBL_UpdatePndng

Global CFE_TBL_TblRegPacket_Payload::Name [CFE_MISSION_TBL_MAX_FULL_NAME_LEN]

\$sc_\$cpu_TBL_Name[CFE_TB_MAX_FULL_NAME_LEN]

Global CFE_TBL_TblRegPacket_Payload::OwnerAppName [CFE_MISSION_MAX_API_LEN]

\$sc_\$cpu_TBL_OwnerApp[OS_MAX_API_NAME]

Global CFE_TBL_TblRegPacket_Payload::Size

\$sc_\$cpu_TBL_SIZE

Global CFE_TBL_TblRegPacket_Payload::TableLoadedOnce

\$sc_\$cpu_TBL_LoadedOnce

Global CFE_TBL_TblRegPacket_Payload::TimeOfLastUpdate

\$sc_\$cpu_TBL_TimeLastUpd, \$sc_\$cpu_TBL_TLUSECONDS, \$sc_\$cpu_TBL_TLUSUBSECONDS

Global CFE_TBL_TblRegPacket_Payload::ValidationFuncPtr

\$sc_\$cpu_TBL_ValFuncPtr

Global CFE_TIME_DiagnosticTlm_Payload::AtToneDelay

\$sc_\$cpu_TIME_DLatentS, \$sc_\$cpu_TIME_DLatentSs

Global CFE_TIME_DiagnosticTlm_Payload::AtToneLatch

\$sc_\$cpu_TIME_DTValidS, \$sc_\$cpu_TIME_DTValidSs

Global CFE_TIME_DiagnosticTlm_Payload::AtToneLeapSeconds

\$sc_\$cpu_TIME_DLeapS

Global CFE_TIME_DiagnosticTlm_Payload::AtToneMET`$sc_$cpu_TIME_DTMETS, $sc_$cpu_TIME_DTMETSs`**Global CFE_TIME_DiagnosticTlm_Payload::AtToneSTCF**`$sc_$cpu_TIME_DSTCFS, $sc_$cpu_TIME_DSTCFSS`**Global CFE_TIME_DiagnosticTlm_Payload::ClockFlyState**`$sc_$cpu_TIME_DFlywheel`**Global CFE_TIME_DiagnosticTlm_Payload::ClockSetState**`$sc_$cpu_TIME_DValid`**Global CFE_TIME_DiagnosticTlm_Payload::ClockSignal**`$sc_$cpu_TIME_DSignal`**Global CFE_TIME_DiagnosticTlm_Payload::ClockSource**`$sc_$cpu_TIME_DSource`**Global CFE_TIME_DiagnosticTlm_Payload::ClockStateAPI**`$sc_$cpu_TIME_DAPIState`**Global CFE_TIME_DiagnosticTlm_Payload::ClockStateFlags**`$sc_$cpu_TIME_DStateFlags, $sc_$cpu_TIME_DFlagSet, $sc_$cpu_TIME_DFlagFly, $sc_$cpu_TIME_DFlagSrc,
$sc_$cpu_TIME_DFlagPri, $sc_$cpu_TIME_DFlagSfly, $sc_$cpu_TIME_DFlagCfly, $sc_$cpu_TIME_DFlagAdj,
$sc_$cpu_TIME_DFlag1Hzd, $sc_$cpu_TIME_DFlagClat, $sc_$cpu_TIME_DFlagSorC, $sc_$cpu_TIME_DFlag↔
NIU`**Global CFE\_TIME\_DiagnosticTlm\_Payload::CurrentLatch**`$sc_$cpu_TIME_DLocalS, $sc_$cpu_TIME_DLocalSs`**Global CFE\_TIME\_DiagnosticTlm\_Payload::CurrentMET**`$sc_$cpu_TIME_DMETS, $sc_$cpu_TIME_DMETSs`**Global CFE\_TIME\_DiagnosticTlm\_Payload::CurrentTAI**`$sc_$cpu_TIME_DTAIS, $sc_$cpu_TIME_DTAISS`**Global CFE\_TIME\_DiagnosticTlm\_Payload::CurrentUTC**`$sc_$cpu_TIME_DUTCs, $sc_$cpu_TIME_DUTCSS`**Global CFE\_TIME\_DiagnosticTlm\_Payload::DataStoreStatus**`$sc_$cpu_TIME_DataStStat`**Global CFE\_TIME\_DiagnosticTlm\_Payload::DelayDirection**`$sc_$cpu_TIME_DLatentDir`**Global CFE\_TIME\_DiagnosticTlm\_Payload::Forced2Fly**`$sc_$cpu_TIME_DCMD2Fly`**Global CFE\_TIME\_DiagnosticTlm\_Payload::LocalIntCounter**`$sc_$cpu_TIME_D1HzISRCNT`**Global CFE\_TIME\_DiagnosticTlm\_Payload::LocalTaskCounter**`$sc_$cpu_TIME_D1HzTaskCNT`**Global CFE\_TIME\_DiagnosticTlm\_Payload::MaxElapsed**`$sc_$cpu_TIME_DMaxWindow`**Global CFE\_TIME\_DiagnosticTlm\_Payload::MaxLocalClock**`$sc_$cpu_TIME_DWrapS, $sc_$cpu_TIME_DWrapSs`

Global CFE_TIME_DiagnosticTlm_Payload::MinElapsed

\$sc_\$cpu_TIME_DMinWindow

Global CFE_TIME_DiagnosticTlm_Payload::OneHzAdjust

\$sc_\$cpu_TIME_D1HzAdjS, \$sc_\$cpu_TIME_D1HzAdjSs

Global CFE_TIME_DiagnosticTlm_Payload::OneHzDirection

\$sc_\$cpu_TIME_D1HzAdjDir

Global CFE_TIME_DiagnosticTlm_Payload::OneTimeAdjust

\$sc_\$cpu_TIME_DAdjustS, \$sc_\$cpu_TIME_DAdjustSs

Global CFE_TIME_DiagnosticTlm_Payload::OneTimeDirection

\$sc_\$cpu_TIME_DAdjustDir

Global CFE_TIME_DiagnosticTlm_Payload::ServerFlyState

\$sc_\$cpu_TIME_DSrvFly

Global CFE_TIME_DiagnosticTlm_Payload::TimeSinceTone

\$sc_\$cpu_TIME_DElapsedS, \$sc_\$cpu_TIME_DElapsedSs

Global CFE_TIME_DiagnosticTlm_Payload::ToneDataCounter

\$sc_\$cpu_TIME_DTatTCNT

Global CFE_TIME_DiagnosticTlm_Payload::ToneDataLatch

\$sc_\$cpu_TIME_DTDS, \$sc_\$cpu_TIME_DTDSs

Global CFE_TIME_DiagnosticTlm_Payload::ToneIntCounter

\$sc_\$cpu_TIME_DTsISRCNT

Global CFE_TIME_DiagnosticTlm_Payload::ToneIntErrorCounter

\$sc_\$cpu_TIME_DTsISRERR

Global CFE_TIME_DiagnosticTlm_Payload::ToneMatchCounter

\$sc_\$cpu_TIME_DVerifyCNT

Global CFE_TIME_DiagnosticTlm_Payload::ToneMatchErrorCounter

\$sc_\$cpu_TIME_DVerifyER

Global CFE_TIME_DiagnosticTlm_Payload::ToneOverLimit

\$sc_\$cpu_TIME_DMaxSs

Global CFE_TIME_DiagnosticTlm_Payload::ToneSignalCounter

\$sc_\$cpu_TIME_DTSDetCNT

Global CFE_TIME_DiagnosticTlm_Payload::ToneSignalLatch

\$sc_\$cpu_TIME_DTTS, \$sc_\$cpu_TIME_DTTSs

Global CFE_TIME_DiagnosticTlm_Payload::ToneTaskCounter

\$sc_\$cpu_TIME_DTsTaskCNT

Global CFE_TIME_DiagnosticTlm_Payload::ToneUnderLimit

\$sc_\$cpu_TIME_DMinSs

Global CFE_TIME_DiagnosticTlm_Payload::VersionCounter

\$sc_\$cpu_TIME_DVersionCNT

Global CFE_TIME_DiagnosticTlm_Payload::VirtualMET

\$sc_\$cpu_TIME_DLogicalMET

Global CFE_TIME_HousekeepingTlm_Payload::AdjustmentFactor

\$sc_\$cpu_TIME_ADJ

Global CFE_TIME_HousekeepingTlm_Payload::ClockStateAPI

\$sc_\$cpu_TIME_DAPIState

Global CFE_TIME_HousekeepingTlm_Payload::ClockStateFlags

\$sc_\$cpu_TIME_StateFlg, \$sc_\$cpu_TIME_FlagSet, \$sc_\$cpu_TIME_FlagFly, \$sc_\$cpu_TIME_FlagSrc, \$sc_\$cpu_TIME_FlagPri, \$sc_\$cpu_TIME_FlagSfly, \$sc_\$cpu_TIME_FlagCfly, \$sc_\$cpu_TIME_FlagAdj, \$sc_\$cpu_TIME_Flag1Hzd, \$sc_\$cpu_TIME_FlagClat, \$sc_\$cpu_TIME_FlagSorC, \$sc_\$cpu_TIME_FlagNIU

Global CFE_TIME_HousekeepingTlm_Payload::CommandCounter

\$sc_\$cpu_TIME_CMDPC

Global CFE_TIME_HousekeepingTlm_Payload::CommandErrorCounter

\$sc_\$cpu_TIME_CMDEC

Global CFE_TIME_HousekeepingTlm_Payload::LeapSeconds

\$sc_\$cpu_TIME_LeapSecs

Global CFE_TIME_HousekeepingTlm_Payload::MET

\$sc_\$cpu_TIME_MET

Global CFE_TIME_HousekeepingTlm_Payload::STCF

\$sc_\$cpu_TIME_STCF

2 Glossary of Terms

Term	Definition
Application (or App)	A set of data and functions that is treated as a single entity by the cFE. cFE resources are allocated on a per-Application basis. Applications are made up of a Main Task and zero or more Child Tasks.
Application ID	A processor unique reference to an Application. NOTE: This is different from a CCSDS Application ID which is referred to as an "APID."
Application Programmer's Interface (API)	A set of routines, protocols, and tools for building software applications
Platform Support Package (PSP)	A collection of user-provided facilities that interface an OS and the cFE with a specific hardware platform. The PSP is responsible for hardware initialization.
Child Task	A separate thread of execution that is spawned by an Application's Main Task.
Command	A Software Bus Message defined by the receiving Application. Commands can originate from other onboard Applications or from the ground.
Core Flight Executive (cFE)	A runtime environment and a set of services for hosting FSW Applications
Critical Data Store (CDS)	A collection of data that is not modified by the OS or cFE following a Processor Reset.
Cyclic Redundancy Check	A polynomial based method for checking that a data set has remained unchanged from one time period to another.
Developer	Anyone who is coding a cFE Application.
Event Data	Data describing an Event that is supplied to the cFE Event Service. The cFE includes this data in an Event Message .
Event Filter	A numeric value (bit mask) used to determine how frequently to output an application Event Message defined by its Event ID .

Term	Definition
Event Format Mode	Defines the Event Message Format downlink option: short or long. The short format is used when there is limited telemetry bandwidth and is binary. The long format is in ASCII and is used for logging to a Local Event Log and to an Event Message Port.
Event ID	A numeric literal used to uniquely name an Application event.
Event Type	A numeric literal used to identify the type of an Application event. An event type may be CFE_EVS_EventType_DEBUG , CFE_EVS_EventType_INFORMATION , CFE_EVS_EventType_ERROR , or CFE_EVS_EventType_CRITICAL .
Event Message	A data item used to notify the user and/or an external Application of a significant event. Event Messages include a time-stamp of when the message was generated, a processor unique identifier, an Application ID , the Event Type (DEBUG,INFO,ERROR or CRITICAL), and Event Data . An Event Message can either be real-time or playback from a Local Event Log.

3 cFE Application Programmer's Interface (API) Reference

3.1 Executive Services API

- [cFE Entry/Exit APIs](#)
 - [CFE_ES_Main](#) - cFE Main Entry Point used by Board Support Package to start cFE
 - [CFE_ES_ResetCFE](#) - Reset the cFE Core and all cFE Applications.
- [cFE Application Control APIs](#)
 - [CFE_ES_RestartApp](#) - Restart a single cFE Application.
 - [CFE_ES_ReloadApp](#) - Reload a single cFE Application.
 - [CFE_ES_DeleteApp](#) - Delete a cFE Application.
- [cFE Application Behavior APIs](#)
 - [CFE_ES_RunLoop](#) - Check for Exit, Restart, or Reload commands.
 - [CFE_ES_WaitForStartupSync](#) - Allow an Application to Wait for the "OPERATIONAL" global system state.
 - [CFE_ES_WaitForSystemState](#) - Allow an Application to Wait for a minimum global system state.
 - [CFE_ES_IncrementTaskCounter](#) - Increments the execution counter for the calling task.
 - [CFE_ES_ExitApp](#) - Exit a cFE Application.
- [cFE Information APIs](#)
 - [CFE_ES_GetResetType](#) - Return the most recent Reset Type.
 - [CFE_ES_GetAppID](#) - Get an Application ID for the calling Application.
 - [CFE_ES_GetTaskID](#) - Get the task ID of the calling context.
 - [CFE_ES_GetAppIDByName](#) - Get an Application ID associated with a specified Application name.
 - [CFE_ES_GetLibIDByName](#) - Get a Library ID associated with a specified Library name.
 - [CFE_ES_GetAppName](#) - Get an Application name for a specified Application ID.
 - [CFE_ES_GetLibName](#) - Get a Library name for a specified Library ID.
 - [CFE_ES_GetAppInfo](#) - Get Application Information given a specified App ID.
 - [CFE_ES_GetTaskInfo](#) - Get Task Information given a specified Task ID.

- [CFE_ES_GetLibInfo](#) - Get Library Information given a specified Resource ID.
- [CFE_ES_GetModuleInfo](#) - Get Information given a specified Resource ID.
- [cFE Child Task APIs](#)
 - [CFE_ES_CreateChildTask](#) - Creates a new task under an existing Application.
 - [CFE_ES_GetTaskIDByName](#) - Get a Task ID associated with a specified Task name.
 - [CFE_ES_GetTaskName](#) - Get a Task name for a specified Task ID.
 - [CFE_ES_DeleteChildTask](#) - Deletes a task under an existing Application.
 - [CFE_ES_ExitChildTask](#) - Exits a child task.
- [cFE Critical Data Store APIs](#)
 - [CFE_ES_RegisterCDS](#) - Reserve space (or re-obtain previously reserved space) in the Critical Data Store (CDS)
 - [CFE_ES_GetCDSBlockIDByName](#) - Get a CDS Block ID associated with a specified CDS Block name.
 - [CFE_ES_GetCDSBlockName](#) - Get a Block name for a specified Block ID.
 - [CFE_ES_CopyToCDS](#) - Save a block of data in the Critical Data Store (CDS)
 - [CFE_ES_RestoreFromCDS](#) - Recover a block of data from the Critical Data Store (CDS)
- [cFE Memory Manager APIs](#)
 - [CFE_ES_PoolCreate](#) - Initializes a memory pool created by an application while using a semaphore during processing.
 - [CFE_ES_PoolCreateEx](#) - Initializes a memory pool created by an application with application specified block sizes.
 - [CFE_ES_PoolCreateNoSem](#) - Initializes a memory pool created by an application without using a semaphore during processing.
 - [CFE_ES_PoolDelete](#) - Deletes a memory pool that was previously created.
 - [CFE_ES_GetPoolBuf](#) - Gets a buffer from the memory pool created by [CFE_ES_PoolCreate](#) or [CFE_ES_PoolCreateNoSem](#).
 - [CFE_ES_PutPoolBuf](#) - Releases a buffer from the memory pool that was previously allocated via [CFE_ES_GetPoolBuf](#).
 - [CFE_ES_GetMemPoolStats](#) - Extracts the statistics maintained by the memory pool software.
 - [CFE_ES_GetPoolBufInfo](#) - Gets info on a buffer previously allocated via [CFE_ES_GetPoolBuf](#).
- [cFE Performance Monitor APIs](#)
 - [CFE_ES_PerfLogEntry](#) - Entry marker for use with Software Performance Analysis Tool.
 - [CFE_ES_PerfLogExit](#) - Exit marker for use with Software Performance Analysis Tool.
 - [CFE_ES_PerfLogAdd](#) - Adds a new entry to the data buffer.
- [cFE Generic Counter APIs](#)
 - [CFE_ES_RegisterGenCounter](#) - Register a generic counter.
 - [CFE_ES_DeleteGenCounter](#) - Delete a generic counter.
 - [CFE_ES_IncrementGenCounter](#) - Increments the specified generic counter.
 - [CFE_ES_SetGenCount](#) - Set the specified generic counter.
 - [CFE_ES_GetGenCount](#) - Get the specified generic counter count.
 - [CFE_ES_GetGenCounterIDByName](#) - Get the Id associated with a generic counter name.

- [CFE_ES_GetGenCounterName](#) - Get a Counter name for a specified Counter ID.
- [cFE Miscellaneous APIs](#)
 - [CFE_ES_BackgroundWakeup](#) - Wakes up the CFE background task.
 - [CFE_ES_CalculateCRC](#) - Calculate a CRC on a block of memory.
 - [CFE_ES_WriteToSysLog](#) - Write a string to the cFE System Log.
 - [CFE_ES_ProcessAsyncEvent](#) - Notification that an asynchronous event was detected by the underlying OS/PSP.
 - [CFE_ES_StatusToString](#) - Convert status to a string.
- [cFE Resource ID APIs](#)
 - [CFE_ES_AppID_ToIndex](#) - Obtain an index value correlating to an ES Application ID.
 - [CFE_ES_LibID_ToIndex](#) - Obtain an index value correlating to an ES Library ID.
 - [CFE_ES_TaskID_ToIndex](#) - Obtain an index value correlating to an ES Task ID.
 - [CFE_ES_CounterID_ToIndex](#) - Obtain an index value correlating to an ES Counter ID.

3.2 Events Services API

- [cFE Registration APIs](#)
 - [CFE_EVS_Register](#) - Register an application for receiving event services.
- [cFE Send Event APIs](#)
 - [CFE_EVS_SendEvent](#) - Generate a software event.
 - [CFE_EVS_SendEventWithAppID](#) - Generate a software event given the specified Application ID.
 - [CFE_EVS_SendTimedEvent](#) - Generate a software event with a specific time tag.
- [cFE Reset Event Filter APIs](#)
 - [CFE_EVS_ResetFilter](#) - Resets the calling application's event filter for a single event ID.
 - [CFE_EVS_ResetAllFilters](#) - Resets all of the calling application's event filters.

3.3 File Services API

- [cFE File Header Management APIs](#)
 - [CFE_FS_ReadHeader](#) - Read the contents of the Standard cFE File Header.
 - [CFE_FS_InitHeader](#) - Initializes the contents of the Standard cFE File Header.
 - [CFE_FS_WriteHeader](#) - Write the specified Standard cFE File Header to the specified file.
 - [CFE_FS_SetTimestamp](#) - Modifies the Time Stamp field in the Standard cFE File Header for the specified file.
- [cFE File Utility APIs](#)
 - [CFE_FS_GetDefaultMountPoint](#) - Get the default virtual mount point for a file category.
 - [CFE_FS_GetDefaultExtension](#) - Get the default filename extension for a file category.
 - [CFE_FS_ParseInputFileNameEx](#) - Parse a filename input from an input buffer into a local buffer.
 - [CFE_FS_ParseInputFileName](#) - Parse a filename string from the user into a local buffer.
 - [CFE_FS_ExtractFilenameFromPath](#) - Extracts the filename from a unix style path and filename string.
 - [CFE_FS_BackgroundFileDumpRequest](#) - Register a background file dump request.
 - [CFE_FS_BackgroundFileDumpsPending](#) - Query if a background file write request is currently pending.

3.4 Message API

- [cFE Generic Message APIs](#)
 - [CFE_MSG_Init](#) - Initialize a message.
- [cFE Message Primary Header APIs](#)
 - [CFE_MSG_GetSize](#) - Gets the total size of a message.
 - [CFE_MSG_SetSize](#) - Sets the total size of a message.
 - [CFE_MSG_GetType](#) - Gets the message type.
 - [CFE_MSG_SetType](#) - Sets the message type.
 - [CFE_MSG_GetHeaderVersion](#) - Gets the message header version.
 - [CFE_MSG_SetHeaderVersion](#) - Sets the message header version.
 - [CFE_MSG_GetHasSecondaryHeader](#) - Gets the message secondary header boolean.
 - [CFE_MSG_SetHasSecondaryHeader](#) - Sets the message secondary header boolean.
 - [CFE_MSG_GetApId](#) - Gets the message application ID.
 - [CFE_MSG_SetApId](#) - Sets the message application ID.
 - [CFE_MSG_GetSegmentationFlag](#) - Gets the message segmentation flag.
 - [CFE_MSG_SetSegmentationFlag](#) - Sets the message segmentation flag.
 - [CFE_MSG_GetSequenceCount](#) - Gets the message sequence count.
 - [CFE_MSG_SetSequenceCount](#) - Sets the message sequence count.
 - [CFE_MSG_GetNextSequenceCount](#) - Gets the next sequence count value (rolls over if appropriate)
- [cFE Message Extended Header APIs](#)
 - [CFE_MSG_GetEDSVersion](#) - Gets the message EDS version.
 - [CFE_MSG_SetEDSVersion](#) - Sets the message EDS version.
 - [CFE_MSG_GetEndian](#) - Gets the message endian.
 - [CFE_MSG_SetEndian](#) - Sets the message endian.
 - [CFE_MSG_GetPlaybackFlag](#) - Gets the message playback flag.
 - [CFE_MSG_SetPlaybackFlag](#) - Sets the message playback flag.
 - [CFE_MSG_GetSubsystem](#) - Gets the message subsystem.
 - [CFE_MSG_SetSubsystem](#) - Sets the message subsystem.
 - [CFE_MSG_GetSystem](#) - Gets the message system.
 - [CFE_MSG_SetSystem](#) - Sets the message system.
- [cFE Message Secondary Header APIs](#)
 - [CFE_MSG_GenerateChecksum](#) - Calculates and sets the checksum of a message.
 - [CFE_MSG_ValidateChecksum](#) - Validates the checksum of a message.
 - [CFE_MSG_SetFcnCode](#) - Sets the function code field in a message.
 - [CFE_MSG_GetFcnCode](#) - Gets the function code field from a message.
 - [CFE_MSG_GetMsgTime](#) - Gets the time field from a message.
 - [CFE_MSG_SetMsgTime](#) - Sets the time field in a message.
- [cFE Message Id APIs](#)
 - [CFE_MSG_GetMsgId](#) - Gets the message id from a message.
 - [CFE_MSG_SetMsgId](#) - Sets the message id bits in a message.
 - [CFE_MSG_GetTypeFromMsgId](#) - Gets message type using message ID.

3.5 Resource ID API

- cFE Resource Misc APIs
 - [CFE_ResourceId_ToInteger](#) - Convert a resource ID to an integer.
 - [CFE_ResourceId_FromInteger](#) - Convert an integer to a resource ID.
 - [CFE_ResourceId_Equal](#) - Compare two Resource ID values for equality.
 - [CFE_ResourceId_IsDefined](#) - Check if a resource ID value is defined.
 - [CFE_ResourceId_GetBase](#) - Get the Base value (type/category) from a resource ID value.
 - [CFE_ResourceId_GetSerial](#) - Get the Serial Number (sequential ID) from a resource ID value.
 - [CFE_ResourceId_FindNext](#) - Locate the next resource ID that maps to an available table entry.
 - [CFE_ResourceId_ToIndex](#) - Internal routine to aid in converting an ES resource ID to an array index.

3.6 Software Bus Services API

- [cFE Pipe Management APIs](#)
 - [CFE_SB_CreatePipe](#) - Creates a new software bus pipe.
 - [CFE_SB_DeletePipe](#) - Delete a software bus pipe.
 - [CFE_SB_PipeId_ToIndex](#) - Obtain an index value correlating to an SB Pipe ID.
 - [CFE_SB_SetPipeOpts](#) - Set options on a pipe.
 - [CFE_SB_GetPipeOpts](#) - Get options on a pipe.
 - [CFE_SB_GetPipeName](#) - Get the pipe name for a given id.
 - [CFE_SB_GetPipeIdByName](#) - Get pipe id by pipe name.
- [cFE Message Subscription Control APIs](#)
 - [CFE_SB_Subscribe](#) - Subscribe to a message on the software bus with default parameters.
 - [CFE_SB_SubscribeEx](#) - Subscribe to a message on the software bus.
 - [CFE_SB_SubscribeLocal](#) - Subscribe to a message while keeping the request local to a cpu.
 - [CFE_SB_Unsubscribe](#) - Remove a subscription to a message on the software bus.
 - [CFE_SB_UnsubscribeLocal](#) - Remove a subscription to a message on the software bus on the current CPU.
- [cFE Send/Receive Message APIs](#)
 - [CFE_SB_TransmitMsg](#) - Transmit a message.
 - [CFE_SB_ReceiveBuffer](#) - Receive a message from a software bus pipe.
- [cFE Zero Copy APIs](#)
 - [CFE_SB_AllocateMessageBuffer](#) - Get a buffer pointer to use for "zero copy" SB sends.
 - [CFE_SB_ReleaseMessageBuffer](#) - Release an unused "zero copy" buffer pointer.
 - [CFE_SB_TransmitBuffer](#) - Transmit a buffer.
- [cFE Message Characteristics APIs](#)
 - [CFE_SB_SetUserDataLength](#) - Sets the length of user data in a software bus message.
 - [CFE_SB_TimeStampMsg](#) - Sets the time field in a software bus message with the current spacecraft time.
 - [CFE_SB_MessageStringSet](#) - Copies a string into a software bus message.
 - [CFE_SB_GetUserData](#) - Get a pointer to the user data portion of a software bus message.
 - [CFE_SB_GetUserDataLength](#) - Gets the length of user data in a software bus message.

- [CFE_SB_MessageStringGet](#) - Copies a string out of a software bus message.
- [cFE Message ID APIs](#)
 - [CFE_SB_IsValidMsgId](#) - Identifies whether a given [CFE_SB_MsgId_t](#) is valid.
 - [CFE_SB_MsgId_Equal](#) - Identifies whether two [CFE_SB_MsgId_t](#) values are equal.
 - [CFE_SB_MsgIdToValue](#) - Converts a [CFE_SB_MsgId_t](#) to a normal integer.
 - [CFE_SB_ValueToMsgId](#) - Converts a normal integer into a [CFE_SB_MsgId_t](#).

3.7 Table Services API

- [cFE Registration APIs](#)
 - [CFE_TBL_Register](#) - Register a table with cFE to obtain Table Management Services.
 - [CFE_TBL_Share](#) - Obtain handle of table registered by another application.
 - [CFE_TBL_Unregister](#) - Unregister a table.
- [cFE Manage Table Content APIs](#)
 - [CFE_TBL_Load](#) - Load a specified table with data from specified source.
 - [CFE_TBL_Update](#) - Update contents of a specified table, if an update is pending.
 - [CFE_TBL_Validate](#) - Perform steps to validate the contents of a table image.
 - [CFE_TBL_Manage](#) - Perform standard operations to maintain a table.
 - [CFE_TBL_DumpToBuffer](#) - Copies the contents of a Dump Only Table to a shared buffer.
 - [CFE_TBL_Modified](#) - Notify cFE Table Services that table contents have been modified by the Application.
- [cFE Access Table Content APIs](#)
 - [CFE_TBL_GetAddress](#) - Obtain the current address of the contents of the specified table.
 - [CFE_TBL_GetAddresses](#) - Obtain the current addresses of an array of specified tables.
 - [CFE_TBL_ReleaseAddress](#) - Release previously obtained pointer to the contents of the specified table.
 - [CFE_TBL_ReleaseAddresses](#) - Release the addresses of an array of specified tables.
- [cFE Get Table Information APIs](#)
 - [CFE_TBL_GetStatus](#) - Obtain current status of pending actions for a table.
 - [CFE_TBL_GetInfo](#) - Obtain characteristics/information of/about a specified table.
 - [CFE_TBL_NotifyByMessage](#) - Instruct cFE Table Services to notify Application via message when table requires management.

3.8 Time Services API

- [cFE Get Current Time APIs](#)
 - [CFE_TIME_GetTime](#) - Get the current spacecraft time.
 - [CFE_TIME_GetTAI](#) - Get the current TAI (MET + SCTF) time.
 - [CFE_TIME_GetUTC](#) - Get the current UTC (MET + SCTF - Leap Seconds) time.
 - [CFE_TIME_GetMET](#) - Get the current value of the Mission Elapsed Time (MET).
 - [CFE_TIME_GetMETseconds](#) - Get the current seconds count of the mission-elapsed time.
 - [CFE_TIME_GetMETsubsecs](#) - Get the current sub-seconds count of the mission-elapsed time.
- [cFE Get Time Information APIs](#)

- [CFE_TIME_GetSTCF](#) - Get the current value of the spacecraft time correction factor (STCF).
 - [CFE_TIME_GetLeapSeconds](#) - Get the current value of the leap seconds counter.
 - [CFE_TIME_GetClockState](#) - Get the current state of the spacecraft clock.
 - [CFE_TIME_GetClockInfo](#) - Provides information about the spacecraft clock.
- [cFE Time Arithmetic APIs](#)
 - [CFE_TIME_Add](#) - Adds two time values.
 - [CFE_TIME_Subtract](#) - Subtracts two time values.
 - [CFE_TIME_Compare](#) - Enumerated types identifying the relative relationships of two times.
- [cFE Time Conversion APIs](#)
 - [CFE_TIME_MET2SCTime](#) - Convert specified MET into Spacecraft Time.
 - [CFE_TIME_Sub2MicroSecs](#) - Converts a sub-seconds count to an equivalent number of microseconds.
 - [CFE_TIME_Micro2SubSecs](#) - Converts a number of microseconds to an equivalent sub-seconds count.
- [cFE External Time Source APIs](#)
 - [CFE_TIME_ExternalTone](#) - Provides the 1 Hz signal from an external source.
 - [CFE_TIME_ExternalMET](#) - Provides the Mission Elapsed Time from an external source.
 - [CFE_TIME_ExternalGPS](#) - Provide the time from an external source that has data common to GPS receivers.
 - [CFE_TIME_ExternalTime](#) - Provide the time from an external source that measures time relative to a known epoch.
 - [CFE_TIME_RegisterSynchCallback](#) - Registers a callback function that is called whenever time synchronization occurs.
 - [CFE_TIME_UnregisterSynchCallback](#) - Unregisters a callback function that is called whenever time synchronization occurs.
- [cFE Miscellaneous Time APIs](#)
 - [CFE_TIME_Print](#) - Print a time value as a string.
 - [CFE_TIME_Local1HzISR](#) - This function is called via a timer callback set up at initialization of the TIME service.

4 Osal API Documentation

- General Information and Concepts
 - [OSAL Introduction](#)
- Core
 - [OSAL Return Code Defines](#)
 - [OSAL Object Type Defines](#)
 - APIs
 - * [OSAL Core Operation APIs](#)
 - * [OSAL Object ID Utility APIs](#)
 - * [OSAL Task APIs](#)
 - * [OSAL Message Queue APIs](#)
 - * [OSAL Heap APIs](#)

- * [OSAL Error Info APIs](#)
 - * [OSAL Select APIs](#)
 - * [OSAL Printf APIs](#)
 - * [OSAL BSP low level access APIs](#)
 - * [OSAL Real Time Clock APIs](#)
 - * [OSAL Shell APIs](#)
 - [Common Reference](#)
 - [Return Code Reference](#)
 - [Id Map Reference](#)
 - [Clock Reference](#)
 - [Task Reference](#)
 - [Message Queue Reference](#)
 - [Heap Reference](#)
 - [Select Reference](#)
 - [Printf Reference](#)
 - [BSP Reference](#)
 - [Shell Reference](#)
- [File System](#)
 - [File System Overview](#)
 - [File Descriptors In Osal](#)
 - [OSAL File Access Option Defines](#)
 - [OSAL Reference Point For Seek Offset Defines](#)
 - [APIs](#)
 - * [OSAL Standard File APIs](#)
 - * [OSAL Directory APIs](#)
 - * [OSAL File System Level APIs](#)
 - [File System Reference](#)
 - [File Reference](#)
 - [Directory Reference](#)
- [Object File Loader](#)
 - [APIs](#)
 - * [OSAL Dynamic Loader and Symbol APIs](#)
 - [File Loader Reference](#)
- [Network](#)
 - [APIs](#)
 - * [OSAL Network ID APIs](#)
 - * [OSAL Socket Address APIs](#)
 - * [OSAL Socket Management APIs](#)
 - [Network Reference](#)
 - [Socket Reference](#)
- [Timer](#)

- [Timer Overview](#)
- [APIs](#)
 - * [OSAL Time Base APIs](#)
 - * [OSAL Timer APIs](#)
- [Timer Reference](#)
- [Time Base Reference](#)
- Semaphore and Mutex
 - [OSAL Semaphore State Defines](#)
 - [APIs](#)
 - * [OSAL Binary Semaphore APIs](#)
 - * [OSAL Counting Semaphore APIs](#)
 - * [OSAL Mutex APIs](#)
 - [Binary Semaphore Reference](#)
 - [Counting Semaphore Reference](#)
 - [Mutex Reference](#)

4.1 OSAL Introduction

The goal of this library is to promote the creation of portable and reusable real time embedded system software. Given the necessary OS abstraction layer implementations, the same embedded software should compile and run on a number of platforms ranging from spacecraft computer systems to desktop PCs.

The OS Application Program Interfaces (APIs) are broken up into core, file system, loader, network, and timer APIs. See the related document sections for full descriptions.

Note

The majority of these APIs should be called from a task running in the context of an OSAL application and in general should not be called from an ISR. There are a few exceptions, such as the ability to give a binary semaphore from an ISR.

4.2 File System Overview

The File System API is a thin wrapper around a selection of POSIX file APIs. In addition the File System API presents a common directory structure and volume view regardless of the underlying system type. For example, vxWorks uses MS-DOS style volume names and directories where a vxWorks RAM disk might have the volume “RAM:0”. With this File System API, volumes are represented as Unix-style paths where each volume is mounted on the root file system:

- RAM:0/file1.dat becomes /mnt/ram/file1.dat
- FL:0/file2.dat becomes /mnt/fl/file2.dat

This abstraction allows the applications to use the same paths regardless of the implementation and it also allows file systems to be simulated on a desktop system for testing. On a desktop Linux system, the file system abstraction can be set up to map virtual devices to a regular directory. This is accomplished through the `OS_mkfs` call, `OS_mount` call, and a BSP specific volume table that maps the virtual devices to real devices or underlying file systems.

In order to make this file system volume abstraction work, a “Volume Table” needs to be provided in the Board Support Package of the application. The table has the following fields:

- Device Name: This is the name of the virtual device that the Application uses. Common names are “ramdisk1”, “flash1”, or “volatile1” etc. But the name can be any unique string.

- **Physical Device Name:** This is an implementation specific field. For vxWorks it is not needed and can be left blank. For a File system based implementation, it is the “mount point” on the root file system where all of the volume will be mounted. A common place for this on Linux could be a user’s home directory, “/tmp”, or even the current working directory “.”. In the example of “/tmp” all of the directories created for the volumes would be under “/tmp” on the Linux file system. For a real disk device in Linux, such as a RAM disk, this field is the device name “/dev/ram0”.
- **Volume Type:** This field defines the type of volume. The types are: FS_BASED which uses the existing file system, RAM_DISK which uses a RAM_DISK device in vxWorks, RTEMS, or Linux, FLASH_DISK_FORMAT which uses a flash disk that is to be formatted before use, FLASH_DISK_INIT which uses a flash disk with an existing format that is just to be initialized before it’s use, EEPROM which is for an EEPROM or PROM based system.
- **Volatile Flag:** This flag indicates that the volume or disk is a volatile disk (RAM disk) or a non-volatile disk, that retains its contents when the system is rebooted. This should be set to TRUE or FALSE.
- **Free Flag:** This is an internal flag that should be set to FALSE or zero.
- **Is Mounted Flag:** This is an internal flag that should be set to FALSE or zero. Note that a “pre-mounted” FS_↵ BASED path can be set up by setting this flag to one.
- **Volume Name:** This is an internal field and should be set to a space character “ ”.
- **Mount Point Field:** This is an internal field and should be set to a space character “ ”.
- **Block Size Field:** This is used to record the block size of the device and does not need to be set by the user.

4.3 File Descriptors In Osal

The OSAL uses abstracted file descriptors. This means that the file descriptors passed back from the OS_open and OS_creat calls will only work with other OSAL OS_* calls. The reasoning for this is as follows:

Because the OSAL now keeps track of all file descriptors, OSAL specific information can be associated with a specific file descriptor in an OS independent way. For instance, the path of the file that the file descriptor points to can be easily retrieved. Also, the OSAL task ID of the task that opened the file can also be retrieved easily. Both of these pieces of information are very useful when trying to determine statistics for a task, or the entire system. This information can all be retrieved with a single API, OS_FDGetInfo.

All of the possible file system calls are not implemented. "Special" files requiring OS specific control/operations are by nature not portable. Abstraction in this case is not possible, so the raw OS calls should be used (including open/close/etc). Mixing with OSAL calls is not supported for such cases. [OS_TranslatePath](#) is available to support using open directly by an app and maintain abstraction on the file system.

There are some small drawbacks with the OSAL file descriptors. Because the related information is kept in a table, there is a define called OS_MAX_NUM_OPEN_FILES that defines the maximum number of file descriptors available. This is a configuration parameter, and can be changed to fit your needs.

Also, if you open or create a file not using the OSAL calls (OS_open or OS_creat) then none of the other OS_* calls that accept a file descriptor as a parameter will work (the results of doing so are undefined). Therefore, if you open a file with the underlying OS's open call, you must continue to use the OS's calls until you close the file descriptor. Be aware that by doing this your software may no longer be OS agnostic.

4.4 Timer Overview

The timer API is a generic interface to the OS timer facilities. It is implemented using the POSIX timers on Linux and vxWorks and the native timer API on RTEMS. The number of timers supported is controlled by the configuration parameter OS_MAX_TIMERS.

5 cFE Mission Configuration Parameters

Global [CFE_MISSION_ES_CMD_TOPICID](#)

cFE Portable Message Numbers for Commands

Global [CFE_MISSION_ES_HK_TLM_TOPICID](#)

cFE Portable Message Numbers for Telemetry

Global [CFE_MISSION_EVS_CMD_TOPICID](#)

cFE Portable Message Numbers for Commands

Global [CFE_MISSION_EVS_HK_TLM_TOPICID](#)

cFE Portable Message Numbers for Telemetry

Global [CFE_MISSION_MAX_API_LEN](#)

cFE Maximum length for API names within data exchange structures

cFE Maximum length for API names within data exchange structures

Global [CFE_MISSION_MAX_FILE_LEN](#)

cFE Maximum length for filenames within data exchange structures

cFE Maximum length for filenames within data exchange structures

Global [CFE_MISSION_MAX_NUM_FILES](#)

cFE Maximum number of files in a message/data exchange

cFE Maximum number of files in a message/data exchange

Global [CFE_MISSION_MAX_PATH_LEN](#)

cFE Maximum length for pathnames within data exchange structures

cFE Maximum length for pathnames within data exchange structures

Global [CFE_MISSION_SB_CMD_TOPICID](#)

cFE Portable Message Numbers for Commands

Global [CFE_MISSION_SB_HK_TLM_TOPICID](#)

cFE Portable Message Numbers for Telemetry

Global [CFE_MISSION_TBL_CMD_TOPICID](#)

cFE Portable Message Numbers for Commands

Global [CFE_MISSION_TBL_HK_TLM_TOPICID](#)

cFE Portable Message Numbers for Telemetry

Global [CFE_MISSION_TIME_CMD_TOPICID](#)

cFE Portable Message Numbers for Commands

Global [CFE_MISSION_TIME_DATA_CMD_TOPICID](#)

cFE Portable Message Numbers for Global Messages

Global [CFE_MISSION_TIME_HK_TLM_TOPICID](#)

cFE Portable Message Numbers for Telemetry

6 Topic Index

6.1 Topics

Here is a list of all topics with brief descriptions:

cFE Return Code Defines	121
cFE Resource ID APIs	145
cFE Entry/Exit APIs	147

cFE Application Control APIs	149
cFE Application Behavior APIs	151
cFE Information APIs	155
cFE Child Task APIs	164
cFE Miscellaneous APIs	168
cFE Critical Data Store APIs	170
cFE Memory Manager APIs	174
cFE Performance Monitor APIs	182
cFE Generic Counter APIs	184
cFE Registration APIs	190
cFE Send Event APIs	191
cFE Reset Event Filter APIs	195
cFE File Header Management APIs	196
cFE File Utility APIs	200
cFE Generic Message APIs	204
cFE Message Primary Header APIs	205
cFE Message Extended Header APIs	214
cFE Message Secondary Header APIs	219
cFE Message Id APIs	225
cFE Message Integrity APIs	227
cFE Pipe Management APIs	228
cFE Message Subscription Control APIs	233
cFE Send/Receive Message APIs	239
cFE Zero Copy APIs	241
cFE Message Characteristics APIs	243
cFE Message ID APIs	247
cFE SB Pipe options	252
cFE Registration APIs	253
cFE Manage Table Content APIs	257
cFE Access Table Content APIs	262

cFE Get Table Information APIs	267
cFE Table Type Defines	270
cFE Get Current Time APIs	272
cFE Get Time Information APIs	275
cFE Time Arithmetic APIs	277
cFE Time Conversion APIs	279
cFE External Time Source APIs	281
cFE Miscellaneous Time APIs	285
cFE Resource ID base values	287
cFE Clock State Flag Defines	289
OSAL Semaphore State Defines	291
OSAL Binary Semaphore APIs	291
OSAL BSP low level access APIs	296
OSAL Real Time Clock APIs	297
OSAL Core Operation APIs	312
OSAL Condition Variable APIs	315
OSAL Counting Semaphore APIs	322
OSAL Directory APIs	326
OSAL Return Code Defines	329
OSAL Error Info APIs	336
OSAL File Access Option Defines	339
OSAL Reference Point For Seek Offset Defines	339
OSAL Standard File APIs	340
OSAL File System Level APIs	353
OSAL Heap APIs	360
OSAL Object Type Defines	361
OSAL Object ID Utility APIs	364
OSAL Dynamic Loader and Symbol APIs	369
OSAL Mutex APIs	372
OSAL Network ID APIs	376

OSAL Printf APIs	377
OSAL Message Queue APIs	378
OSAL RwLock APIs	382
OSAL Select APIs	386
OSAL Shell APIs	392
OSAL Socket Address APIs	393
OSAL Socket Management APIs	396
OSAL Task APIs	407
OSAL Time Base APIs	413
OSAL Timer APIs	417

7 Data Structure Index

7.1 Data Structures

Here are the data structures with brief descriptions:

CCSDS_ExtendedHeader	
CCSDS packet extended header	422
CCSDS_PrimaryHeader	
CCSDS packet primary header	423
CFE_Config_ArrayValue	
Wrapper type for array configuration	424
CFE_Config_IdNameEntry	424
CFE_Config_ValueBuffer	425
CFE_Config_ValueEntry	425
CFE_ES_ApplInfo	
Application Information	426
CFE_ES_AppNameCmd_Payload	
Generic application name command payload	430
CFE_ES_AppReloadCmd_Payload	
Reload Application Command Payload	431
CFE_ES_BlockStats	
Block statistics	431
CFE_ES_CDSRegDumpRec	
CDS Register Dump Record	432
CFE_ES_ClearERLogCmd	433

CFE_ES_ClearSysLogCmd	434
CFE_ES_DeleteCDSCmd Delete Critical Data Store Command	434
CFE_ES_DeleteCDSCmd_Payload Delete Critical Data Store Command Payload	435
CFE_ES_DumpCDSRegistryCmd Dump CDS Registry Command	435
CFE_ES_DumpCDSRegistryCmd_Payload Dump CDS Registry Command Payload	436
CFE_ES_FileNameCmd Generic file name command	436
CFE_ES_FileNameCmd_Payload Generic file name command payload	437
CFE_ES_HousekeepingTlm	438
CFE_ES_HousekeepingTlm_Payload	438
CFE_ES_MemAddress_t Type used for memory addresses in command and telemetry messages	446
CFE_ES_MemOffset_t Type used for memory sizes and offsets in commands and telemetry	447
CFE_ES_MemPoolStats Memory Pool Statistics	448
CFE_ES_MemStatsTlm	449
CFE_ES_NoopCmd	449
CFE_ES_OneAppTlm	450
CFE_ES_OneAppTlm_Payload	450
CFE_ES_OverWriteSysLogCmd Overwrite/Discard System Log Configuration Command Payload	451
CFE_ES_OverWriteSysLogCmd_Payload Overwrite/Discard System Log Configuration Command Payload	452
CFE_ES_PoolAlign Pool Alignment	452
CFE_ES_PoolStatsTlm_Payload	453
CFE_ES_QueryAllCmd	454
CFE_ES_QueryAllTasksCmd	454
CFE_ES_QueryOneCmd	455

CFE_ES_ReloadAppCmd	
Reload Application Command	455
CFE_ES_ResetCountersCmd	456
CFE_ES_ResetPRCountCmd	456
CFE_ES_RestartAppCmd	457
CFE_ES_RestartCmd	
Restart cFE Command	457
CFE_ES_RestartCmd_Payload	
Restart cFE Command Payload	458
CFE_ES_SendHkCmd	459
CFE_ES_SendMemPoolStatsCmd	
Send Memory Pool Statistics Command	459
CFE_ES_SendMemPoolStatsCmd_Payload	
Send Memory Pool Statistics Command Payload	460
CFE_ES_SetMaxPRCountCmd	
Set Maximum Processor Reset Count Command	460
CFE_ES_SetMaxPRCountCmd_Payload	
Set Maximum Processor Reset Count Command Payload	461
CFE_ES_SetPerfFilterMaskCmd	
Set Performance Analyzer Filter Mask Command	461
CFE_ES_SetPerfFilterMaskCmd_Payload	
Set Performance Analyzer Filter Mask Command Payload	462
CFE_ES_SetPerfTriggerMaskCmd	
Set Performance Analyzer Trigger Mask Command	463
CFE_ES_SetPerfTrigMaskCmd_Payload	
Set Performance Analyzer Trigger Mask Command Payload	463
CFE_ES_StartApp	
Start Application Command	464
CFE_ES_StartAppCmd_Payload	
Start Application Command Payload	464
CFE_ES_StartPerfCmd_Payload	
Start Performance Analyzer Command Payload	466
CFE_ES_StartPerfDataCmd	
Start Performance Analyzer Command	466
CFE_ES_StopAppCmd	467
CFE_ES_StopPerfCmd_Payload	
Stop Performance Analyzer Command Payload	468

CFE_ES_StopPerfDataCmd	
Stop Performance Analyzer Command	468
CFE_ES_TaskInfo	
Task Information	469
CFE_ES_WriteERLogCmd	470
CFE_ES_WriteSysLogCmd	471
CFE_EVS_AddEventFilterCmd	472
CFE_EVS_AppDataCmd_Payload	
Write Event Services Application Information to File Command Payload	472
CFE_EVS_AppNameBitMaskCmd_Payload	
Generic App Name and Bitmask Command Payload	473
CFE_EVS_AppNameCmd_Payload	
Generic App Name Command Payload	474
CFE_EVS_AppNameEventIDCmd_Payload	
Generic App Name and Event ID Command Payload	474
CFE_EVS_AppNameEventIDMaskCmd_Payload	
Generic App Name, Event ID, Mask Command Payload	475
CFE_EVS_AppTlmData	475
CFE_EVS_BinFilter	
Event message filter definition structure	477
CFE_EVS_BitMaskCmd_Payload	
Generic Bitmask Command Payload	477
CFE_EVS_ClearLogCmd	478
CFE_EVS_DeleteEventFilterCmd	478
CFE_EVS_DisableAppEventsCmd	479
CFE_EVS_DisableAppEventTypeCmd	479
CFE_EVS_DisableEventTypeCmd	480
CFE_EVS_DisablePortsCmd	481
CFE_EVS_EnableAppEventsCmd	481
CFE_EVS_EnableAppEventTypeCmd	482
CFE_EVS_EnableEventTypeCmd	482
CFE_EVS_EnablePortsCmd	483
CFE_EVS_HousekeepingTlm	484
CFE_EVS_HousekeepingTlm_Payload	484

CFE_EVS_LogFileCmd_Payload	
Write Event Log to File Command Payload	487
CFE_EVS_LongEventTlm	487
CFE_EVS_LongEventTlm_Payload	488
CFE_EVS_NoopCmd	489
CFE_EVS_PacketID	489
CFE_EVS_ResetAllFiltersCmd	490
CFE_EVS_ResetAppCounterCmd	491
CFE_EVS_ResetCountersCmd	492
CFE_EVS_ResetFilterCmd	492
CFE_EVS_SendHkCmd	493
CFE_EVS_SetEventFormatCode_Payload	
Set Event Format Mode Command Payload	493
CFE_EVS_SetEventFormatModeCmd	
Set Event Format Mode Command	494
CFE_EVS_SetFilterCmd	494
CFE_EVS_SetLogMode_Payload	
Set Log Mode Command Payload	495
CFE_EVS_SetLogModeCmd	
Set Log Mode Command	496
CFE_EVS_ShortEventTlm	496
CFE_EVS_ShortEventTlm_Payload	497
CFE_EVS_WriteAppDataFileCmd	
Write Event Services Application Information to File Command	497
CFE_EVS_WriteLogDataFileCmd	
Write Event Log to File Command	498
CFE_FS_FileWriteMetaData	
External Metadata/State object associated with background file writes	498
CFE_FS_Header	
Standard cFE File header structure definition	500
CFE_SB_AllSubscriptionsTlm	501
CFE_SB_AllSubscriptionsTlm_Payload	502
CFE_SB_DisableRouteCmd	503
CFE_SB_DisableSubReportingCmd	503

CFE_SB_EnableRouteCmd	504
CFE_SB_EnableSubReportingCmd	504
CFE_SB_HousekeepingTlm	505
CFE_SB_HousekeepingTlm_Payload	505
CFE_SB_Msg	
Software Bus generic message	509
CFE_SB_MsgId_t	
CFE_SB_MsgId_t type definition	510
CFE_SB_MsgMapFileEntry	
SB Map File Entry	510
CFE_SB_NoopCmd	511
CFE_SB_PipeDepthStats	
SB Pipe Depth Statistics	511
CFE_SB_PipeInfoEntry	
SB Pipe Information File Entry	512
CFE_SB_Qos_t	
Quality Of Service Type Definition	514
CFE_SB_ResetCountersCmd	515
CFE_SB_RouteCmd_Payload	
Enable/Disable Route Command Payload	515
CFE_SB_RoutingFileEntry	
SB Routing File Entry	516
CFE_SB_SendHkCmd	517
CFE_SB_SendPrevSubsCmd	518
CFE_SB_SendSbStatsCmd	518
CFE_SB_SingleSubscriptionTlm	518
CFE_SB_SingleSubscriptionTlm_Payload	519
CFE_SB_StatsTlm	520
CFE_SB_StatsTlm_Payload	521
CFE_SB_SubEntries	
SB Previous Subscriptions Entry	524
CFE_SB_WriteFileInfoCmd_Payload	
Write File Info Command Payload	525
CFE_SB_WriteMapInfoCmd	525

CFE_SB_WritePipeInfoCmd	526
CFE_SB_WriteRoutingInfoCmd	526
CFE_TBL_AbortLoadCmd Abort Load Command	527
CFE_TBL_AbortLoadCmd_Payload Abort Load Command Payload	528
CFE_TBL_ActivateCmd Activate Table Command	528
CFE_TBL_ActivateCmd_Payload Activate Table Command Payload	529
CFE_TBL_CombinedFileHdr Complete header for CFE table files	529
CFE_TBL_DelCDSCmd_Payload Delete Critical Table CDS Command Payload	530
CFE_TBL_DeleteCDSCmd Delete Critical Table CDS Command	530
CFE_TBL_DumpCmd	531
CFE_TBL_DumpCmd_Payload Dump Table Command Payload	532
CFE_TBL_DumpRegistryCmd Dump Registry Command	533
CFE_TBL_DumpRegistryCmd_Payload Dump Registry Command Payload	533
CFE_TBL_File_Hdr The definition of the header fields that are included in CFE Table Data files	534
CFE_TBL_FileDef Table File summary object	535
CFE_TBL_HousekeepingTlm	536
CFE_TBL_HousekeepingTlm_Payload	537
CFE_TBL_Info Table Info	541
CFE_TBL_LoadCmd Load Table Command	543
CFE_TBL_LoadCmd_Payload Load Table Command Payload	543
CFE_TBL_NoopCmd	544

CFE_TBL_NotifyCmd	544
CFE_TBL_NotifyCmd_Payload	
Table Management Notification Command Payload	545
CFE_TBL_ResetCountersCmd	545
CFE_TBL_SendHkCmd	546
CFE_TBL_SendRegistryCmd	
Send Table Registry Command	546
CFE_TBL_SendRegistryCmd_Payload	
Send Table Registry Command Payload	547
CFE_TBL_TableRegistryTlm	547
CFE_TBL_TblRegPacket_Payload	548
CFE_TBL_ValidateCmd	
Validate Table Command	551
CFE_TBL_ValidateCmd_Payload	
Validate Table Command Payload	552
CFE_TIME_AddAdjustCmd	553
CFE_TIME_AddDelayCmd	553
CFE_TIME_AddOneHzAdjustmentCmd	554
CFE_TIME_DiagnosticTlm	554
CFE_TIME_DiagnosticTlm_Payload	555
CFE_TIME_FakeToneCmd	563
CFE_TIME_HousekeepingTlm	564
CFE_TIME_HousekeepingTlm_Payload	564
CFE_TIME_LeapsCmd_Payload	
Set leap seconds command payload	566
CFE_TIME_NoopCmd	567
CFE_TIME_OneHzAdjustmentCmd_Payload	
Generic seconds, subseconds command payload	567
CFE_TIME_OneHzCmd	568
CFE_TIME_ResetCountersCmd	568
CFE_TIME_SendDiagnosticCmd	569
CFE_TIME_SendHkCmd	569

CFE_TIME_SetLeapSecondsCmd	
Set leap seconds command	570
CFE_TIME_SetMETCmd	570
CFE_TIME_SetSignalCmd	
Set tone signal source command	571
CFE_TIME_SetSourceCmd	
Set time data source command	571
CFE_TIME_SetStateCmd	
Set clock state command	572
CFE_TIME_SetSTCFCmd	573
CFE_TIME_SetTimeCmd	573
CFE_TIME_SignalCmd_Payload	
Set tone signal source command payload	574
CFE_TIME_SourceCmd_Payload	
Set time data source command payload	574
CFE_TIME_StateCmd_Payload	
Set clock state command payload	575
CFE_TIME_SubAdjustCmd	575
CFE_TIME_SubDelayCmd	576
CFE_TIME_SubOneHzAdjustmentCmd	576
CFE_TIME_SysTime	
Data structure used to hold system time values	577
CFE_TIME_TimeCmd_Payload	
Generic seconds, microseconds command payload	578
CFE_TIME_ToneDataCmd	
Time at tone data command	578
CFE_TIME_ToneDataCmd_Payload	
Time at tone data command payload	579
CFE_TIME_ToneSignalCmd	580
OS_bin_sem_prop_t	
OSAL binary semaphore properties	580
OS_condvar_prop_t	
OSAL condition variable properties	581
OS_count_sem_prop_t	
OSAL counting semaphore properties	581
os_dirent_t	
Directory entry	582

OS_FdSet	An abstract structure capable of holding several OSAL IDs	583
OS_file_prop_t	OSAL file properties	583
os_fsinfo_t	OSAL file system info	584
os_fstat_t	File system status	585
OS_heap_prop_t	OSAL heap properties	585
OS_module_address_t	OSAL module address properties	586
OS_module_prop_t	OSAL module properties	587
OS_mut_sem_prop_t	OSAL mutex properties	588
OS_queue_prop_t	OSAL queue properties	589
OS_rwlock_prop_t	OSAL rwlock properties	589
OS_SockAddr_t	Encapsulates a generic network address	590
OS_SockAddrData_t	Storage buffer for generic network address	590
OS_socket_optval	Socket option value	591
OS_socket_prop_t	Encapsulates socket properties	592
OS_static_symbol_record_t	Associates a single symbol name with a memory address	592
OS_statvfs_t		593
OS_task_prop_t	OSAL task properties	594
OS_time_t	OSAL time interval structure	594
OS_timebase_prop_t	Time base properties	595

OS_timer_prop_t	
Timer properties	596

8 File Index

8.1 File List

Here is a list of all files with brief descriptions:

build/osal_public_api/inc/osconfig.h	597
cfe_perfids.h	605
example_2x32bit_cfe_es_memaddress.h	608
example_32bit_cfe_es_memaddress.h	611
example_64bit_cfe_es_memaddress.h	613
example_mission_cfg.h	615
example_platform_cfg.h	628
cfe/modules/config/fsw/inc/cfe_config_external.h	676
cfe/modules/config/fsw/inc/cfe_config_init.h	677
cfe/modules/config/fsw/inc/cfe_config_lookup.h	678
cfe/modules/config/fsw/inc/cfe_config_nametable.h	679
cfe/modules/config/fsw/inc/cfe_config_set.h	680
cfe/modules/config/fsw/inc/cfe_config_table.h	681
cfe/modules/core_api/config/default_cfe_core_api_base_msgid_values.h	683
cfe/modules/core_api/config/default_cfe_core_api_interface_cfg_values.h	684
cfe/modules/core_api/config/default_cfe_core_api_msgid_mapping.h	685
cfe/modules/core_api/config/default_cfe_mission_cfg.h	687
cfe/modules/core_api/config/default_cfe_msgids.h	688
cfe/modules/core_api/fsw/inc/cfe.h	689
cfe/modules/core_api/fsw/inc/cfe_config.h	690
cfe/modules/core_api/fsw/inc/cfe_config_api_typedefs.h	694
cfe/modules/core_api/fsw/inc/cfe_core_api_base_msgids.h	696
cfe/modules/core_api/fsw/inc/cfe_core_api_interface_cfg.h	697
cfe/modules/core_api/fsw/inc/cfe_endian.h	700
cfe/modules/core_api/fsw/inc/cfe_error.h	701

cfe/modules/core_api/fsw/inc/cfe_es.h	715
cfe/modules/core_api/fsw/inc/cfe_es_api_typedefs.h	721
cfe/modules/core_api/fsw/inc/cfe_evs.h	728
cfe/modules/core_api/fsw/inc/cfe_evs_api_typedefs.h	730
cfe/modules/core_api/fsw/inc/cfe_fs.h	733
cfe/modules/core_api/fsw/inc/cfe_fs_api_typedefs.h	735
cfe/modules/core_api/fsw/inc/cfe_msg.h	739
cfe/modules/core_api/fsw/inc/cfe_msg_api_typedefs.h	743
cfe/modules/core_api/fsw/inc/cfe_resourceid.h	748
cfe/modules/core_api/fsw/inc/cfe_resourceid_api_typedefs.h	756
cfe/modules/core_api/fsw/inc/cfe_sb.h	758
cfe/modules/core_api/fsw/inc/cfe_sb_api_typedefs.h	762
cfe/modules/core_api/fsw/inc/cfe_tbl.h	766
cfe/modules/core_api/fsw/inc/cfe_tbl_api_typedefs.h	770
cfe/modules/core_api/fsw/inc/cfe_tbl_filedef.h	775
cfe/modules/core_api/fsw/inc/cfe_time.h	777
cfe/modules/core_api/fsw/inc/cfe_time_api_typedefs.h	780
cfe/modules/core_api/fsw/inc/cfe_version.h	783
cfe/modules/es/config/default_cfe_es_extern_typedefs.h	786
cfe/modules/es/config/default_cfe_es_fcncode_values.h	796
cfe/modules/es/config/default_cfe_es_interface_cfg_values.h	798
cfe/modules/es/config/default_cfe_es_internal_cfg_values.h	799
cfe/modules/es/config/default_cfe_es_memaddress.h	800
cfe/modules/es/config/default_cfe_es_mission_cfg.h	802
cfe/modules/es/config/default_cfe_es_msg.h	802
cfe/modules/es/config/default_cfe_es_msgdefs.h	803
cfe/modules/es/config/default_cfe_es_msgid_values.h	810
cfe/modules/es/config/default_cfe_es_msgids.h	811
cfe/modules/es/config/default_cfe_es_msgstruct.h	812
cfe/modules/es/config/default_cfe_es_platform_cfg.h	819

cfe/modules/es/config/default_cfe_es_topicid_values.h	820
cfe/modules/es/fsw/inc/cfe_es_eventids.h	821
cfe/modules/es/fsw/inc/cfe_es_fcncodes.h	849
cfe/modules/es/fsw/inc/cfe_es_interface_cfg.h	872
cfe/modules/es/fsw/inc/cfe_es_internal_cfg.h	876
cfe/modules/es/fsw/inc/cfe_es_topicids.h	909
cfe/modules/evs/config/default_cfe_evs_extern_typedefs.h	911
cfe/modules/evs/config/default_cfe_evs_fcncode_values.h	915
cfe/modules/evs/config/default_cfe_evs_interface_cfg_values.h	917
cfe/modules/evs/config/default_cfe_evs_internal_cfg_values.h	918
cfe/modules/evs/config/default_cfe_evs_mission_cfg.h	919
cfe/modules/evs/config/default_cfe_evs_msg.h	920
cfe/modules/evs/config/default_cfe_evs_msgdefs.h	921
cfe/modules/evs/config/default_cfe_evs_msgid_values.h	926
cfe/modules/evs/config/default_cfe_evs_msgids.h	927
cfe/modules/evs/config/default_cfe_evs_msgstruct.h	929
cfe/modules/evs/config/default_cfe_evs_platform_cfg.h	935
cfe/modules/evs/config/default_cfe_evs_topicid_values.h	936
cfe/modules/evs/fsw/inc/cfe_evs_eventids.h	936
cfe/modules/evs/fsw/inc/cfe_evs_fcncodes.h	950
cfe/modules/evs/fsw/inc/cfe_evs_interface_cfg.h	969
cfe/modules/evs/fsw/inc/cfe_evs_internal_cfg.h	970
cfe/modules/evs/fsw/inc/cfe_evs_topicids.h	977
cfe/modules/fs/config/default_cfe_fs_extern_typedefs.h	979
cfe/modules/fs/config/default_cfe_fs_filedef.h	980
cfe/modules/fs/config/default_cfe_fs_interface_cfg_values.h	982
cfe/modules/fs/config/default_cfe_fs_mission_cfg.h	983
cfe/modules/fs/fsw/inc/cfe_fs_interface_cfg.h	984
cfe/modules/msg/fsw/inc/ccsds_hdr.h	985
cfe/modules/resourceid/fsw/inc/cfe_core_resourceid_basevalues.h	987

cfe/modules/resourceid/fsw/inc/cfe_resourceid_basevalue.h	989
cfe/modules/sb/config/default_cfe_sb_extern_typedefs.h	991
cfe/modules/sb/config/default_cfe_sb_fcncode_values.h	994
cfe/modules/sb/config/default_cfe_sb_interface_cfg_values.h	995
cfe/modules/sb/config/default_cfe_sb_internal_cfg_values.h	996
cfe/modules/sb/config/default_cfe_sb_mission_cfg.h	997
cfe/modules/sb/config/default_cfe_sb_msg.h	998
cfe/modules/sb/config/default_cfe_sb_msgdefs.h	998
cfe/modules/sb/config/default_cfe_sb_msgid_values.h	1003
cfe/modules/sb/config/default_cfe_sb_msgids.h	1004
cfe/modules/sb/config/default_cfe_sb_msgstruct.h	1006
cfe/modules/sb/config/default_cfe_sb_platform_cfg.h	1010
cfe/modules/sb/config/default_cfe_sb_topicid_values.h	1010
cfe/modules/sb/fsw/inc/cfe_sb_eventids.h	1011
cfe/modules/sb/fsw/inc/cfe_sb_fcncodes.h	1033
cfe/modules/sb/fsw/inc/cfe_sb_interface_cfg.h	1043
cfe/modules/sb/fsw/inc/cfe_sb_internal_cfg.h	1045
cfe/modules/sb/fsw/inc/cfe_sb_topicids.h	1061
cfe/modules/tbl/config/default_cfe_tbl_extern_typedefs.h	1063
cfe/modules/tbl/config/default_cfe_tbl_fcncode_values.h	1066
cfe/modules/tbl/config/default_cfe_tbl_interface_cfg_values.h	1068
cfe/modules/tbl/config/default_cfe_tbl_internal_cfg_values.h	1068
cfe/modules/tbl/config/default_cfe_tbl_mission_cfg.h	1069
cfe/modules/tbl/config/default_cfe_tbl_msg.h	1071
cfe/modules/tbl/config/default_cfe_tbl_msgdefs.h	1072
cfe/modules/tbl/config/default_cfe_tbl_msgid_values.h	1076
cfe/modules/tbl/config/default_cfe_tbl_msgids.h	1077
cfe/modules/tbl/config/default_cfe_tbl_msgstruct.h	1079
cfe/modules/tbl/config/default_cfe_tbl_platform_cfg.h	1082
cfe/modules/tbl/config/default_cfe_tbl_topicid_values.h	1092

cfe/modules/tbl/fsw/inc/cfe_tbl_eventids.h	1093
cfe/modules/tbl/fsw/inc/cfe_tbl_fcncodes.h	1115
cfe/modules/tbl/fsw/inc/cfe_tbl_interface_cfg.h	1125
cfe/modules/tbl/fsw/inc/cfe_tbl_internal_cfg.h	1127
cfe/modules/tbl/fsw/inc/cfe_tbl_topicids.h	1137
cfe/modules/time/config/default_cfe_time_extern_typedefs.h	1139
cfe/modules/time/config/default_cfe_time_fcncode_values.h	1145
cfe/modules/time/config/default_cfe_time_interface_cfg_values.h	1147
cfe/modules/time/config/default_cfe_time_internal_cfg_values.h	1148
cfe/modules/time/config/default_cfe_time_mission_cfg.h	1149
cfe/modules/time/config/default_cfe_time_msg.h	1150
cfe/modules/time/config/default_cfe_time_msgdefs.h	1150
cfe/modules/time/config/default_cfe_time_msgid_values.h	1155
cfe/modules/time/config/default_cfe_time_msgids.h	1157
cfe/modules/time/config/default_cfe_time_msgstruct.h	1159
cfe/modules/time/config/default_cfe_time_platform_cfg.h	1164
cfe/modules/time/config/default_cfe_time_topicid_values.h	1165
cfe/modules/time/fsw/inc/cfe_time_eventids.h	1165
cfe/modules/time/fsw/inc/cfe_time_fcncodes.h	1177
cfe/modules/time/fsw/inc/cfe_time_interface_cfg.h	1193
cfe/modules/time/fsw/inc/cfe_time_internal_cfg.h	1202
cfe/modules/time/fsw/inc/cfe_time_topicids.h	1211
osal/src/os/inc/common_types.h	1214
osal/src/os/inc/osapi-binsem.h	1221
osal/src/os/inc/osapi-bsp.h	1222
osal/src/os/inc/osapi-clock.h	1224
osal/src/os/inc/osapi-common.h	1229
osal/src/os/inc/osapi-condvar.h	1232
osal/src/os/inc/osapi-constants.h	1234
osal/src/os/inc/osapi-countsem.h	1235

osal/src/os/inc/osapi-dir.h	1237
osal/src/os/inc/osapi-error.h	1238
osal/src/os/inc/osapi-file.h	1243
osal/src/os/inc/osapi-filesys.h	1248
osal/src/os/inc/osapi-heap.h	1250
osal/src/os/inc/osapi-idmap.h	1251
osal/src/os/inc/osapi-macros.h	1254
osal/src/os/inc/osapi-module.h	1257
osal/src/os/inc/osapi-mutex.h	1260
osal/src/os/inc/osapi-network.h	1261
osal/src/os/inc/osapi-printf.h	1262
osal/src/os/inc/osapi-queue.h	1263
osal/src/os/inc/osapi-rwlock.h	1264
osal/src/os/inc/osapi-select.h	1266
osal/src/os/inc/osapi-shell.h	1268
osal/src/os/inc/osapi-sockets.h	1269
osal/src/os/inc/osapi-task.h	1274
osal/src/os/inc/osapi-timebase.h	1278
osal/src/os/inc/osapi-timer.h	1279
osal/src/os/inc/osapi-version.h	1281
osal/src/os/inc/osapi.h	1286
psp/fsw/inc/cfe_psp.h	1288
psp/fsw/inc/cfe_psp_cache_api.h	1290
psp/fsw/inc/cfe_psp_cds_api.h	1291
psp/fsw/inc/cfe_psp_eepromaccess_api.h	1294
psp/fsw/inc/cfe_psp_endian.h	1297
psp/fsw/inc/cfe_psp_error.h	
CFE PSP Error header	1300
psp/fsw/inc/cfe_psp_exception_api.h	1304
psp/fsw/inc/cfe_psp_id_api.h	1306

psp/fsw/inc/cfe_psp_memaccess_api.h	1308
psp/fsw/inc/cfe_psp_memrange_api.h	1313
psp/fsw/inc/cfe_psp_port_api.h	1320
psp/fsw/inc/cfe_psp_ssr_api.h	1324
psp/fsw/inc/cfe_psp_timertick_api.h	1325
psp/fsw/inc/cfe_psp_version_api.h	1328
psp/fsw/inc/cfe_psp_watchdog_api.h	1330

9 Topic Documentation

9.1 cFE Return Code Defines

Macros

- #define [CFE_SUCCESS](#) ((CFE_Status_t)0)
Successful execution.
- #define [CFE_STATUS_NO_COUNTER_INCREMENT](#) ((CFE_Status_t)0x48000001)
No Counter Increment.
- #define [CFE_STATUS_WRONG_MSG_LENGTH](#) ((CFE_Status_t)0xc8000002)
Wrong Message Length.
- #define [CFE_STATUS_UNKNOWN_MSG_ID](#) ((CFE_Status_t)0xc8000003)
Unknown Message ID.
- #define [CFE_STATUS_BAD_COMMAND_CODE](#) ((CFE_Status_t)0xc8000004)
Bad Command Code.
- #define [CFE_STATUS_EXTERNAL_RESOURCE_FAIL](#) ((CFE_Status_t)0xc8000005)
External failure.
- #define [CFE_STATUS_REQUEST_ALREADY_PENDING](#) ((int32)0xc8000006)
Request already pending.
- #define [CFE_STATUS_VALIDATION_FAILURE](#) ((int32)0xc8000007)
Request or input value failed basic structural validation.
- #define [CFE_STATUS_RANGE_ERROR](#) ((int32)0xc8000008)
Request or input value is out of range.
- #define [CFE_STATUS_INCORRECT_STATE](#) ((int32)0xc8000009)
Cannot process request at this time.
- #define [CFE_STATUS_NOT_IMPLEMENTED](#) ((CFE_Status_t)0xc800ffff)
Not Implemented.
- #define [CFE_EVS_UNKNOWN_FILTER](#) ((CFE_Status_t)0xc2000001)
Unknown Filter.
- #define [CFE_EVS_APP_NOT_REGISTERED](#) ((CFE_Status_t)0xc2000002)
Application Not Registered.
- #define [CFE_EVS_APP_ILLEGAL_APP_ID](#) ((CFE_Status_t)0xc2000003)
Illegal Application ID.
- #define [CFE_EVS_APP_FILTER_OVERLOAD](#) ((CFE_Status_t)0xc2000004)
Application Filter Overload.

- `#define CFE_EVS_RESET_AREA_POINTER ((CFE_Status_t)0xc2000005)`
Reset Area Pointer Failure.
- `#define CFE_EVS_EVT_NOT_REGISTERED ((CFE_Status_t)0xc2000006)`
Event Not Registered.
- `#define CFE_EVS_FILE_WRITE_ERROR ((CFE_Status_t)0xc2000007)`
File Write Error.
- `#define CFE_EVS_INVALID_PARAMETER ((CFE_Status_t)0xc2000008)`
Invalid Pointer.
- `#define CFE_EVS_APP_SQUELCHED ((CFE_Status_t)0xc2000009)`
Event squelched.
- `#define CFE_EVS_NOT_IMPLEMENTED ((CFE_Status_t)0xc200ffff)`
Not Implemented.
- `#define CFE_ES_ERR_RESOURCEID_NOT_VALID ((CFE_Status_t)0xc4000001)`
Resource ID is not valid.
- `#define CFE_ES_ERR_NAME_NOT_FOUND ((CFE_Status_t)0xc4000002)`
Resource Name Error.
- `#define CFE_ES_ERR_APP_CREATE ((CFE_Status_t)0xc4000004)`
Application Create Error.
- `#define CFE_ES_ERR_CHILD_TASK_CREATE ((CFE_Status_t)0xc4000005)`
Child Task Create Error.
- `#define CFE_ES_ERR_SYS_LOG_FULL ((CFE_Status_t)0xc4000006)`
System Log Full.
- `#define CFE_ES_ERR_MEM_BLOCK_SIZE ((CFE_Status_t)0xc4000008)`
Memory Block Size Error.
- `#define CFE_ES_ERR_LOAD_LIB ((CFE_Status_t)0xc4000009)`
Load Library Error.
- `#define CFE_ES_BAD_ARGUMENT ((CFE_Status_t)0xc400000a)`
Bad Argument.
- `#define CFE_ES_ERR_CHILD_TASK_REGISTER ((CFE_Status_t)0xc400000b)`
Child Task Register Error.
- `#define CFE_ES_CDS_ALREADY_EXISTS ((CFE_Status_t)0x4400000d)`
CDS Already Exists.
- `#define CFE_ES_CDS_INSUFFICIENT_MEMORY ((CFE_Status_t)0xc400000e)`
CDS Insufficient Memory.
- `#define CFE_ES_CDS_INVALID_NAME ((CFE_Status_t)0xc400000f)`
CDS Invalid Name.
- `#define CFE_ES_CDS_INVALID_SIZE ((CFE_Status_t)0xc4000010)`
CDS Invalid Size.
- `#define CFE_ES_CDS_INVALID ((CFE_Status_t)0xc4000012)`
CDS Invalid.
- `#define CFE_ES_CDS_ACCESS_ERROR ((CFE_Status_t)0xc4000013)`
CDS Access Error.
- `#define CFE_ES_FILE_IO_ERR ((CFE_Status_t)0xc4000014)`
File IO Error.
- `#define CFE_ES_RST_ACCESS_ERR ((CFE_Status_t)0xc4000015)`
Reset Area Access Error.
- `#define CFE_ES_ERR_APP_REGISTER ((CFE_Status_t)0xc4000017)`

- *Application Register Error.*
• #define CFE_ES_ERR_CHILD_TASK_DELETE ((CFE_Status_t)0xc4000018)
- *Child Task Delete Error.*
• #define CFE_ES_ERR_CHILD_TASK_DELETE_MAIN_TASK ((CFE_Status_t)0xc4000019)
- *Child Task Delete Passed Main Task.*
• #define CFE_ES_CDS_BLOCK_CRC_ERR ((CFE_Status_t)0xc400001A)
- *CDS Block CRC Error.*
• #define CFE_ES_MUT_SEM_DELETE_ERR ((CFE_Status_t)0xc400001B)
- *Mutex Semaphore Delete Error.*
• #define CFE_ES_BIN_SEM_DELETE_ERR ((CFE_Status_t)0xc400001C)
- *Binary Semaphore Delete Error.*
• #define CFE_ES_COUNT_SEM_DELETE_ERR ((CFE_Status_t)0xc400001D)
- *Counting Semaphore Delete Error.*
• #define CFE_ES_QUEUE_DELETE_ERR ((CFE_Status_t)0xc400001E)
- *Queue Delete Error.*
• #define CFE_ES_FILE_CLOSE_ERR ((CFE_Status_t)0xc400001F)
- *File Close Error.*
• #define CFE_ES_CDS_WRONG_TYPE_ERR ((CFE_Status_t)0xc4000020)
- *CDS Wrong Type Error.*
• #define CFE_ES_CDS_OWNER_ACTIVE_ERR ((CFE_Status_t)0xc4000022)
- *CDS Owner Active Error.*
• #define CFE_ES_APP_CLEANUP_ERR ((CFE_Status_t)0xc4000023)
- *Application Cleanup Error.*
• #define CFE_ES_TIMER_DELETE_ERR ((CFE_Status_t)0xc4000024)
- *Timer Delete Error.*
• #define CFE_ES_BUFFER_NOT_IN_POOL ((CFE_Status_t)0xc4000025)
- *Buffer Not In Pool.*
• #define CFE_ES_TASK_DELETE_ERR ((CFE_Status_t)0xc4000026)
- *Task Delete Error.*
• #define CFE_ES_OPERATION_TIMED_OUT ((CFE_Status_t)0xc4000027)
- *Operation Timed Out.*
• #define CFE_ES_LIB_ALREADY_LOADED ((CFE_Status_t)0x44000028)
- *Library Already Loaded.*
• #define CFE_ES_ERR_SYS_LOG_TRUNCATED ((CFE_Status_t)0x44000029)
- *System Log Message Truncated.*
• #define CFE_ES_NO_RESOURCE_IDS_AVAILABLE ((CFE_Status_t)0xc400002B)
- *Resource ID is not available.*
• #define CFE_ES_POOL_BLOCK_INVALID ((CFE_Status_t)0xc400002C)
- *Invalid pool block.*
• #define CFE_ES_ERR_DUPLICATE_NAME ((CFE_Status_t)0xc400002E)
- *Duplicate Name Error.*
• #define CFE_ES_NOT_IMPLEMENTED ((CFE_Status_t)0xc400ffff)
- *Not Implemented.*
• #define CFE_FS_BAD_ARGUMENT ((CFE_Status_t)0xc6000001)
- *Bad Argument.*
• #define CFE_FS_INVALID_PATH ((CFE_Status_t)0xc6000002)
- *Invalid Path.*

- #define CFE_FS_FNAME_TOO_LONG ((CFE_Status_t)0xc6000003)
Filename Too Long.
- #define CFE_FS_NOT_IMPLEMENTED ((CFE_Status_t)0xc600ffff)
Not Implemented.
- #define CFE_SB_TIME_OUT ((CFE_Status_t)0xca000001)
Time Out.
- #define CFE_SB_NO_MESSAGE ((CFE_Status_t)0xca000002)
No Message.
- #define CFE_SB_BAD_ARGUMENT ((CFE_Status_t)0xca000003)
Bad Argument.
- #define CFE_SB_MAX_PIPES_MET ((CFE_Status_t)0xca000004)
Max Pipes Met.
- #define CFE_SB_PIPE_CR_ERR ((CFE_Status_t)0xca000005)
Pipe Create Error.
- #define CFE_SB_PIPE_RD_ERR ((CFE_Status_t)0xca000006)
Pipe Read Error.
- #define CFE_SB_MSG_TOO_BIG ((CFE_Status_t)0xca000007)
Message Too Big.
- #define CFE_SB_BUF_ALOC_ERR ((CFE_Status_t)0xca000008)
Buffer Allocation Error.
- #define CFE_SB_MAX_MSGS_MET ((CFE_Status_t)0xca000009)
Max Messages Met.
- #define CFE_SB_MAX_DESTS_MET ((CFE_Status_t)0xca00000a)
Max Destinations Met.
- #define CFE_SB_INTERNAL_ERR ((CFE_Status_t)0xca00000c)
Internal Error.
- #define CFE_SB_WRONG_MSG_TYPE ((CFE_Status_t)0xca00000d)
Wrong Message Type.
- #define CFE_SB_BUFFER_INVALID ((CFE_Status_t)0xca00000e)
Buffer Invalid.
- #define CFE_SB_NOT_IMPLEMENTED ((CFE_Status_t)0xca00ffff)
Not Implemented.
- #define CFE_TBL_ERR_INVALID_HANDLE ((CFE_Status_t)0xcc000001)
Invalid Handle.
- #define CFE_TBL_ERR_INVALID_NAME ((CFE_Status_t)0xcc000002)
Invalid Name.
- #define CFE_TBL_ERR_INVALID_SIZE ((CFE_Status_t)0xcc000003)
Invalid Size.
- #define CFE_TBL_INFO_UPDATE_PENDING ((CFE_Status_t)0x4c000004)
Update Pending.
- #define CFE_TBL_ERR_NEVER_LOADED ((CFE_Status_t)0xcc000005)
Never Loaded.
- #define CFE_TBL_ERR_REGISTRY_FULL ((CFE_Status_t)0xcc000006)
Registry Full.
- #define CFE_TBL_WARN_DUPLICATE ((CFE_Status_t)0x4c000007)
Duplicate Warning.
- #define CFE_TBL_ERR_NO_ACCESS ((CFE_Status_t)0xcc000008)

- *No Access.*
- #define CFE_TBL_ERR_UNREGISTERED ((CFE_Status_t)0xcc000009)
- *Unregistered.*
- #define CFE_TBL_ERR_HANDLES_FULL ((CFE_Status_t)0xcc00000B)
- *Handles Full.*
- #define CFE_TBL_ERR_DUPLICATE_DIFF_SIZE ((CFE_Status_t)0xcc00000C)
- *Duplicate Table With Different Size.*
- #define CFE_TBL_ERR_DUPLICATE_NOT_OWNED ((CFE_Status_t)0xcc00000D)
- *Duplicate Table And Not Owned.*
- #define CFE_TBL_INFO_UPDATED ((CFE_Status_t)0x4c00000E)
- *Updated.*
- #define CFE_TBL_ERR_NO_BUFFER_AVAIL ((CFE_Status_t)0xcc00000F)
- *No Buffer Available.*
- #define CFE_TBL_ERR_DUMP_ONLY ((CFE_Status_t)0xcc000010)
- *Dump Only Error.*
- #define CFE_TBL_ERR_ILLEGAL_SRC_TYPE ((CFE_Status_t)0xcc000011)
- *Illegal Source Type.*
- #define CFE_TBL_ERR_LOAD_IN_PROGRESS ((CFE_Status_t)0xcc000012)
- *Load In Progress.*
- #define CFE_TBL_ERR_FILE_TOO_LARGE ((CFE_Status_t)0xcc000014)
- *File Too Large.*
- #define CFE_TBL_WARN_SHORT_FILE ((CFE_Status_t)0x4c000015)
- *Short File Warning.*
- #define CFE_TBL_ERR_BAD_CONTENT_ID ((CFE_Status_t)0xcc000016)
- *Bad Content ID.*
- #define CFE_TBL_INFO_NO_UPDATE_PENDING ((CFE_Status_t)0x4c000017)
- *No Update Pending.*
- #define CFE_TBL_INFO_TABLE_LOCKED ((CFE_Status_t)0x4c000018)
- *Table Locked.*
- #define CFE_TBL_INFO_VALIDATION_PENDING ((CFE_Status_t)0x4c000019)
- #define CFE_TBL_INFO_NO_VALIDATION_PENDING ((CFE_Status_t)0x4c00001A)
- #define CFE_TBL_ERR_BAD_SUBTYPE_ID ((CFE_Status_t)0xcc00001B)
- *Bad Subtype ID.*
- #define CFE_TBL_ERR_FILE_SIZE_INCONSISTENT ((CFE_Status_t)0xcc00001C)
- *File Size Inconsistent.*
- #define CFE_TBL_ERR_NO_STD_HEADER ((CFE_Status_t)0xcc00001D)
- *No Standard Header.*
- #define CFE_TBL_ERR_NO_TBL_HEADER ((CFE_Status_t)0xcc00001E)
- *No Table Header.*
- #define CFE_TBL_ERR_FILENAME_TOO_LONG ((CFE_Status_t)0xcc00001F)
- *Filename Too Long.*
- #define CFE_TBL_ERR_FILE_FOR_WRONG_TABLE ((CFE_Status_t)0xcc000020)
- *File For Wrong Table.*
- #define CFE_TBL_ERR_LOAD_INCOMPLETE ((CFE_Status_t)0xcc000021)
- *Load Incomplete.*
- #define CFE_TBL_WARN_PARTIAL_LOAD ((CFE_Status_t)0x4c000022)
- *Partial Load Warning.*

- #define CFE_TBL_ERR_PARTIAL_LOAD ((CFE_Status_t)0xcc000023)
Partial Load Error.
- #define CFE_TBL_INFO_DUMP_PENDING ((CFE_Status_t)0x4c000024)
Dump Pending.
- #define CFE_TBL_ERR_INVALID_OPTIONS ((CFE_Status_t)0xcc000025)
Invalid Options.
- #define CFE_TBL_WARN_NOT_CRITICAL ((CFE_Status_t)0x4c000026)
Not Critical Warning.
- #define CFE_TBL_INFO_RECOVERED_TBL ((CFE_Status_t)0x4c000027)
Recovered Table.
- #define CFE_TBL_ERR_BAD_SPACECRAFT_ID ((CFE_Status_t)0xcc000028)
Bad Spacecraft ID.
- #define CFE_TBL_ERR_BAD_PROCESSOR_ID ((CFE_Status_t)0xcc000029)
Bad Processor ID.
- #define CFE_TBL_MESSAGE_ERROR ((CFE_Status_t)0xcc00002a)
Message Error.
- #define CFE_TBL_ERR_SHORT_FILE ((CFE_Status_t)0xcc00002b)
- #define CFE_TBL_ERR_ACCESS ((CFE_Status_t)0xcc00002c)
- #define CFE_TBL_BAD_ARGUMENT ((CFE_Status_t)0xcc00002d)
Bad Argument.
- #define CFE_TBL_INFO_NO_DUMP_PENDING ((CFE_Status_t)0x4c00002e)
No Dump Pending.
- #define CFE_TBL_NOT_IMPLEMENTED ((CFE_Status_t)0xcc00ffff)
Not Implemented.
- #define CFE_TIME_NOT_IMPLEMENTED ((CFE_Status_t)0xce00ffff)
Not Implemented.
- #define CFE_TIME_INTERNAL_ONLY ((CFE_Status_t)0xce000001)
Internal Only.
- #define CFE_TIME_OUT_OF_RANGE ((CFE_Status_t)0xce000002)
Out Of Range.
- #define CFE_TIME_TOO_MANY_SYNCH_CALLBACKS ((CFE_Status_t)0xce000003)
Too Many Sync Callbacks.
- #define CFE_TIME_CALLBACK_NOT_REGISTERED ((CFE_Status_t)0xce000004)
Callback Not Registered.
- #define CFE_TIME_BAD_ARGUMENT ((CFE_Status_t)0xce000005)
Bad Argument.

9.1.1 Detailed Description

9.1.2 Macro Definition Documentation

CFE_ES_APP_CLEANUP_ERR

```
#define CFE_ES_APP_CLEANUP_ERR ((CFE_Status_t)0xc4000023)
```

Application Cleanup Error.

Occurs when an attempt was made to Clean Up an application which involves calling Table, EVS, and SB cleanup functions, then deleting all ES resources, child tasks, and unloading the object module. The approach here is to keep going even though one of these steps had an error. There will be syslog messages detailing each problem.

Definition at line 588 of file [cfe_error.h](#).

CFE_ES_BAD_ARGUMENT

```
#define CFE_ES_BAD_ARGUMENT ((CFE_Status_t)0xc400000a)
```

Bad Argument.

Bad parameter passed into an ES API.

Definition at line 399 of file [cfe_error.h](#).

CFE_ES_BIN_SEM_DELETE_ERR

```
#define CFE_ES_BIN_SEM_DELETE_ERR ((CFE_Status_t)0xc400001c)
```

Binary Semaphore Delete Error.

Occurs when trying to delete a Binary Semaphore that belongs to a task that ES is cleaning up.

Definition at line 527 of file [cfe_error.h](#).

CFE_ES_BUFFER_NOT_IN_POOL

```
#define CFE_ES_BUFFER_NOT_IN_POOL ((CFE_Status_t)0xc4000025)
```

Buffer Not In Pool.

The specified address is not in the memory pool.

Definition at line 605 of file [cfe_error.h](#).

CFE_ES_CDS_ACCESS_ERROR

```
#define CFE_ES_CDS_ACCESS_ERROR ((CFE_Status_t)0xc4000013)
```

CDS Access Error.

The CDS was inaccessible

Definition at line 458 of file [cfe_error.h](#).

CFE_ES_CDS_ALREADY_EXISTS

```
#define CFE_ES_CDS_ALREADY_EXISTS ((CFE_Status_t)0x4400000d)
```

CDS Already Exists.

The Application is receiving the pointer to a CDS that was already present.

Definition at line 415 of file [cfe_error.h](#).

CFE_ES_CDS_BLOCK_CRC_ERR

```
#define CFE_ES_CDS_BLOCK_CRC_ERR ((CFE_Status_t)0xc400001a)
```

CDS Block CRC Error.

Occurs when trying to read a CDS Data block and the CRC of the current data does not match the stored CRC for the data. Either the contents of the CDS Data Block are corrupted or the CDS Control Block is corrupted.

Definition at line 509 of file [cfe_error.h](#).

CFE_ES_CDS_INSUFFICIENT_MEMORY

```
#define CFE_ES_CDS_INSUFFICIENT_MEMORY ((CFE_Status_t)0xc400000e)
```

CDS Insufficient Memory.

The Application is requesting a CDS Block that is larger than the remaining CDS memory.

Definition at line 424 of file [cfe_error.h](#).

CFE_ES_CDS_INVALID

```
#define CFE_ES_CDS_INVALID ((CFE_Status_t)0xc4000012)
```

CDS Invalid.

The CDS contents are invalid.

Definition at line 450 of file [cfe_error.h](#).

CFE_ES_CDS_INVALID_NAME

```
#define CFE_ES_CDS_INVALID_NAME ((CFE_Status_t)0xc400000f)
```

CDS Invalid Name.

The Application is requesting a CDS Block with an invalid ASCII string name. Either the name is too long (> [CFE_MISSION_ES_CDS_MAX_NAME_LENGTH](#)) or was an empty string.

Definition at line [433](#) of file [cfe_error.h](#).

CFE_ES_CDS_INVALID_SIZE

```
#define CFE_ES_CDS_INVALID_SIZE ((CFE_Status_t)0xc4000010)
```

CDS Invalid Size.

The Application is requesting a CDS Block or Pool with a size beyond the applicable limits, either too large or too small/zero.

Definition at line [442](#) of file [cfe_error.h](#).

CFE_ES_CDS_OWNER_ACTIVE_ERR

```
#define CFE_ES_CDS_OWNER_ACTIVE_ERR ((CFE_Status_t)0xc4000022)
```

CDS Owner Active Error.

Occurs when an attempt was made to delete a CDS when an application with the same name associated with the CDS is still present. CDSs can ONLY be deleted when Applications that created them are not present in the system.

Definition at line [575](#) of file [cfe_error.h](#).

CFE_ES_CDS_WRONG_TYPE_ERR

```
#define CFE_ES_CDS_WRONG_TYPE_ERR ((CFE_Status_t)0xc4000020)
```

CDS Wrong Type Error.

Occurs when Table Services is trying to delete a Critical Data Store that is not a Critical Table Image or when Executive Services is trying to delete a Critical Table Image.

Definition at line [564](#) of file [cfe_error.h](#).

CFE_ES_COUNT_SEM_DELETE_ERR

```
#define CFE_ES_COUNT_SEM_DELETE_ERR ((CFE_Status_t)0xc400001D)
```

Counting Semaphore Delete Error.

Occurs when trying to delete a Counting Semaphore that belongs to a task that ES is cleaning up.

Definition at line [536](#) of file [cfe_error.h](#).

CFE_ES_ERR_APP_CREATE

```
#define CFE_ES_ERR_APP_CREATE ((CFE_Status_t)0xc4000004)
```

Application Create Error.

There was an error loading or creating the App.

Definition at line [358](#) of file [cfe_error.h](#).

CFE_ES_ERR_APP_REGISTER

```
#define CFE_ES_ERR_APP_REGISTER ((CFE_Status_t)0xc4000017)
```

Application Register Error.

Occurs when a task cannot be registered in ES global tables

Definition at line [482](#) of file [cfe_error.h](#).

CFE_ES_ERR_CHILD_TASK_CREATE

```
#define CFE_ES_ERR_CHILD_TASK_CREATE ((CFE_Status_t)0xc4000005)
```

Child Task Create Error.
There was an error creating a child task.
Definition at line 366 of file [cfe_error.h](#).

CFE_ES_ERR_CHILD_TASK_DELETE

```
#define CFE_ES_ERR_CHILD_TASK_DELETE ((CFE_Status_t)0xc4000018)
```

Child Task Delete Error.
There was an error deleting a child task.
Definition at line 490 of file [cfe_error.h](#).

CFE_ES_ERR_CHILD_TASK_DELETE_MAIN_TASK

```
#define CFE_ES_ERR_CHILD_TASK_DELETE_MAIN_TASK ((CFE_Status_t)0xc4000019)
```

Child Task Delete Passed Main Task.
There was an attempt to delete a cFE App Main Task with the [CFE_ES_DeleteChildTask](#) API.
Definition at line 499 of file [cfe_error.h](#).

CFE_ES_ERR_CHILD_TASK_REGISTER

```
#define CFE_ES_ERR_CHILD_TASK_REGISTER ((CFE_Status_t)0xc400000b)
```

Child Task Register Error.
Errors occurred when trying to register a child task.
Definition at line 407 of file [cfe_error.h](#).

CFE_ES_ERR_DUPLICATE_NAME

```
#define CFE_ES_ERR_DUPLICATE_NAME ((CFE_Status_t)0xc400002E)
```

Duplicate Name Error.
Resource creation failed due to the name already existing in the system.
Definition at line 668 of file [cfe_error.h](#).

CFE_ES_ERR_LOAD_LIB

```
#define CFE_ES_ERR_LOAD_LIB ((CFE_Status_t)0xc4000009)
```

Load Library Error.
Could not load the shared library.
Definition at line 391 of file [cfe_error.h](#).

CFE_ES_ERR_MEM_BLOCK_SIZE

```
#define CFE_ES_ERR_MEM_BLOCK_SIZE ((CFE_Status_t)0xc4000008)
```

Memory Block Size Error.
The block size requested is invalid.
Definition at line 383 of file [cfe_error.h](#).

CFE_ES_ERR_NAME_NOT_FOUND

```
#define CFE_ES_ERR_NAME_NOT_FOUND ((CFE_Status_t)0xc4000002)
```

Resource Name Error.
There is no match in the system for the given name.
Definition at line 350 of file [cfe_error.h](#).

CFE_ES_ERR_RESOURCEID_NOT_VALID

```
#define CFE_ES_ERR_RESOURCEID_NOT_VALID ((CFE_Status_t)0xc4000001)
```

Resource ID is not valid.

This error indicates that the passed in resource identifier (App ID, Lib ID, Counter ID, etc) did not validate.

Definition at line 342 of file [cfe_error.h](#).

CFE_ES_ERR_SYS_LOG_FULL

```
#define CFE_ES_ERR_SYS_LOG_FULL ((CFE_Status_t)0xc4000006)
```

System Log Full.

The cFE system Log is full. This error means the message was not logged at all

Definition at line 375 of file [cfe_error.h](#).

CFE_ES_ERR_SYS_LOG_TRUNCATED

```
#define CFE_ES_ERR_SYS_LOG_TRUNCATED ((CFE_Status_t)0x44000029)
```

System Log Message Truncated.

This information code means the last syslog message was truncated due to insufficient space in the log buffer.

Definition at line 640 of file [cfe_error.h](#).

CFE_ES_FILE_CLOSE_ERR

```
#define CFE_ES_FILE_CLOSE_ERR ((CFE_Status_t)0xc400001F)
```

File Close Error.

Occurs when trying to close a file that belongs to a task that ES is cleaning up.

Definition at line 554 of file [cfe_error.h](#).

CFE_ES_FILE_IO_ERR

```
#define CFE_ES_FILE_IO_ERR ((CFE_Status_t)0xc4000014)
```

File IO Error.

Occurs when a file operation fails

Definition at line 466 of file [cfe_error.h](#).

CFE_ES_LIB_ALREADY_LOADED

```
#define CFE_ES_LIB_ALREADY_LOADED ((CFE_Status_t)0x44000028)
```

Library Already Loaded.

Occurs if CFE_ES_LoadLibrary detects that the requested library name is already loaded.

Definition at line 631 of file [cfe_error.h](#).

CFE_ES_MUT_SEM_DELETE_ERR

```
#define CFE_ES_MUT_SEM_DELETE_ERR ((CFE_Status_t)0xc400001B)
```

Mutex Semaphore Delete Error.

Occurs when trying to delete a Mutex that belongs to a task that ES is cleaning up.

Definition at line 518 of file [cfe_error.h](#).

CFE_ES_NO_RESOURCE_IDS_AVAILABLE

```
#define CFE_ES_NO_RESOURCE_IDS_AVAILABLE ((CFE_Status_t)0xc400002B)
```

Resource ID is not available.

This error indicates that the maximum resource identifiers (App ID, Lib ID, Counter ID, etc) has already been reached and a new ID cannot be allocated.

Definition at line 650 of file [cfe_error.h](#).

CFE_ES_NOT_IMPLEMENTED

```
#define CFE_ES_NOT_IMPLEMENTED ((CFE_Status_t)0xc400ffff)
```

Not Implemented.

Current version of cFE does not have the function or the feature of the function implemented. This could be due to either an early build of the cFE for this platform or the platform does not support the specified feature.

Definition at line 679 of file [cfe_error.h](#).

CFE_ES_OPERATION_TIMED_OUT

```
#define CFE_ES_OPERATION_TIMED_OUT ((CFE_Status_t)0xc4000027)
```

Operation Timed Out.

Occurs if the timeout for a given operation was exceeded

Definition at line 622 of file [cfe_error.h](#).

CFE_ES_POOL_BLOCK_INVALID

```
#define CFE_ES_POOL_BLOCK_INVALID ((CFE_Status_t)0xc400002C)
```

Invalid pool block.

Software attempted to "put" a block back into a pool which does not appear to belong to that pool. This may mean the pool has become unusable due to memory corruption.

Definition at line 660 of file [cfe_error.h](#).

CFE_ES_QUEUE_DELETE_ERR

```
#define CFE_ES_QUEUE_DELETE_ERR ((CFE_Status_t)0xc400001E)
```

Queue Delete Error.

Occurs when trying to delete a Queue that belongs to a task that ES is cleaning up.

Definition at line 545 of file [cfe_error.h](#).

CFE_ES_RST_ACCESS_ERR

```
#define CFE_ES_RST_ACCESS_ERR ((CFE_Status_t)0xc4000015)
```

Reset Area Access Error.

Occurs when the BSP is not successful in returning the reset area address.

Definition at line 474 of file [cfe_error.h](#).

CFE_ES_TASK_DELETE_ERR

```
#define CFE_ES_TASK_DELETE_ERR ((CFE_Status_t)0xc4000026)
```

Task Delete Error.

Occurs when trying to delete a task that ES is cleaning up.

Definition at line 614 of file [cfe_error.h](#).

CFE_ES_TIMER_DELETE_ERR

```
#define CFE_ES_TIMER_DELETE_ERR ((CFE_Status_t)0xc4000024)
```

Timer Delete Error.

Occurs when trying to delete a Timer that belongs to a task that ES is cleaning up.

Definition at line 597 of file [cfe_error.h](#).

CFE_EVS_APP_FILTER_OVERLOAD

```
#define CFE_EVS_APP_FILTER_OVERLOAD ((CFE_Status_t)0xc2000004)
```

Application Filter Overload.

Number of Application event filters input upon registration is greater than [CFE_PLATFORM_EVS_MAX_EVENT_FILTERS](#)

Definition at line 276 of file [cfe_error.h](#).

CFE_EVS_APP_ILLEGAL_APP_ID

```
#define CFE_EVS_APP_ILLEGAL_APP_ID ((CFE_Status_t)0xc2000003)
```

Illegal Application ID.

Application ID returned by [CFE_ES_GetAppIDByName](#) is greater than [CFE_PLATFORM_ES_MAX_APPLICATIONS](#)

Definition at line 267 of file [cfe_error.h](#).

CFE_EVS_APP_NOT_REGISTERED

```
#define CFE_EVS_APP_NOT_REGISTERED ((CFE_Status_t)0xc2000002)
```

Application Not Registered.

Calling application never previously called [CFE_EVS_Register](#)

Definition at line 258 of file [cfe_error.h](#).

CFE_EVS_APP_SQUELCHED

```
#define CFE_EVS_APP_SQUELCHED ((CFE_Status_t)0xc2000009)
```

Event squelched.

Event squelched due to being sent at too high a rate

Definition at line 318 of file [cfe_error.h](#).

CFE_EVS_EVT_NOT_REGISTERED

```
#define CFE_EVS_EVT_NOT_REGISTERED ((CFE_Status_t)0xc2000006)
```

Event Not Registered.

[CFE_EVS_ResetFilter](#) EventID argument was not found in any event filter registered by the calling application.

Definition at line 294 of file [cfe_error.h](#).

CFE_EVS_FILE_WRITE_ERROR

```
#define CFE_EVS_FILE_WRITE_ERROR ((CFE_Status_t)0xc2000007)
```

File Write Error.

A file write error occurred while processing an EVS command

Definition at line 302 of file [cfe_error.h](#).

CFE_EVS_INVALID_PARAMETER

```
#define CFE_EVS_INVALID_PARAMETER ((CFE_Status_t)0xc2000008)
```

Invalid Pointer.

Invalid parameter supplied to EVS command

Definition at line 310 of file [cfe_error.h](#).

CFE_EVS_NOT_IMPLEMENTED

```
#define CFE_EVS_NOT_IMPLEMENTED ((CFE_Status_t)0xc200ffff)
```

Not Implemented.

Current version of cFE does not have the function or the feature of the function implemented. This could be due to either an early build of the cFE for this platform or the platform does not support the specified feature.

Definition at line 329 of file [cfe_error.h](#).

CFE_EVS_RESET_AREA_POINTER

```
#define CFE_EVS_RESET_AREA_POINTER ((CFE_Status_t)0xc2000005)
```

Reset Area Pointer Failure.

Could not get pointer to the ES Reset area, so we could not get the pointer to the EVS Log.

Definition at line 285 of file [cfe_error.h](#).

CFE_EVS_UNKNOWN_FILTER

```
#define CFE_EVS_UNKNOWN_FILTER ((CFE_Status_t)0xc2000001)
```

Unknown Filter.

[CFE_EVS_Register](#) FilterScheme parameter was illegal

Definition at line 250 of file [cfe_error.h](#).

CFE_FS_BAD_ARGUMENT

```
#define CFE_FS_BAD_ARGUMENT ((CFE_Status_t)0xc6000001)
```

Bad Argument.

A parameter given by a caller to a File Services API did not pass validation checks.

Definition at line 692 of file [cfe_error.h](#).

CFE_FS_FNAME_TOO_LONG

```
#define CFE_FS_FNAME_TOO_LONG ((CFE_Status_t)0xc6000003)
```

Filename Too Long.

FS filename string is too long

Definition at line 708 of file [cfe_error.h](#).

CFE_FS_INVALID_PATH

```
#define CFE_FS_INVALID_PATH ((CFE_Status_t)0xc6000002)
```

Invalid Path.

FS was unable to extract a filename from a path string

Definition at line 700 of file [cfe_error.h](#).

CFE_FS_NOT_IMPLEMENTED

```
#define CFE_FS_NOT_IMPLEMENTED ((CFE_Status_t)0xc600ffff)
```

Not Implemented.

Current version of cFE does not have the function or the feature of the function implemented. This could be due to either an early build of the cFE for this platform or the platform does not support the specified feature.

Definition at line 719 of file [cfe_error.h](#).

CFE_SB_BAD_ARGUMENT

```
#define CFE_SB_BAD_ARGUMENT ((CFE_Status_t)0xca000003)
```

Bad Argument.

A parameter given by a caller to a Software Bus API did not pass validation checks.

Definition at line 750 of file [cfe_error.h](#).

CFE_SB_BUF_ALOC_ERR

```
#define CFE_SB_BUF_ALOC_ERR ((CFE_Status_t)0xca000008)
```

Buffer Allocation Error.

Returned when the memory in the SB message buffer pool has been depleted. The amount of memory in the pool is dictated by the configuration parameter [CFE_PLATFORM_SB_BUF_MEMORY_BYTES](#) specified in the `cfe_platform_cfg.h` file. Also the memory statistics, including current utilization figures and high water marks for the SB Buffer memory pool can be monitored by sending a Software Bus command to send the SB statistics packet.
Definition at line 808 of file [cfe_error.h](#).

CFE_SB_BUFFER_INVALID

```
#define CFE_SB_BUFFER_INVALID ((CFE_Status_t)0xca00000e)
```

Buffer Invalid.

This error code will be returned when a request to release or send a zero copy buffer is invalid, such as if the handle or buffer is not correct or the buffer was previously released.

Definition at line 859 of file [cfe_error.h](#).

CFE_SB_INTERNAL_ERR

```
#define CFE_SB_INTERNAL_ERR ((CFE_Status_t)0xca00000c)
```

Internal Error.

This error code will be returned by the [CFE_SB_Subscribe](#) API if the code detects an internal index is out of range. The most likely cause would be a Single Event Upset.

Definition at line 840 of file [cfe_error.h](#).

CFE_SB_MAX_DESTS_MET

```
#define CFE_SB_MAX_DESTS_MET ((CFE_Status_t)0xca00000a)
```

Max Destinations Met.

Will be returned when calling one of the SB subscription API's if the SB routing table cannot accommodate another destination for a particular the given message ID. This occurs when the number of destinations in use meets the platform configuration parameter [CFE_PLATFORM_SB_MAX_DEST_PER_PKT](#).

Definition at line 830 of file [cfe_error.h](#).

CFE_SB_MAX_MSGS_MET

```
#define CFE_SB_MAX_MSGS_MET ((CFE_Status_t)0xca000009)
```

Max Messages Met.

Will be returned when calling one of the SB subscription API's if the SB routing table cannot accommodate another unique message ID because the platform configuration parameter [CFE_PLATFORM_SB_MAX_MSG_IDS](#) has been met.

Definition at line 818 of file [cfe_error.h](#).

CFE_SB_MAX_PIPES_MET

```
#define CFE_SB_MAX_PIPES_MET ((CFE_Status_t)0xca000004)
```

Max Pipes Met.

This error code will be returned from [CFE_SB_CreatePipe](#) when the SB cannot accommodate the request to create a pipe because the maximum number of pipes ([CFE_PLATFORM_SB_MAX_PIPES](#)) are in use. This configuration parameter is defined in the `cfe_platform_cfg.h` file.

Definition at line 761 of file [cfe_error.h](#).

CFE_SB_MSG_TOO_BIG

```
#define CFE_SB_MSG_TOO_BIG ((CFE_Status_t)0xca000007)
```

Message Too Big.

The size field in the message header indicates the message exceeds the max Software Bus message size. The max size is defined by configuration parameter [CFE_MISSION_SB_MAX_SB_MSG_SIZE](#) in `cfe_mission_cfg.h`.

Definition at line 795 of file [cfe_error.h](#).

CFE_SB_NO_MESSAGE

```
#define CFE_SB_NO_MESSAGE ((CFE_Status_t)0xca000002)
```

No Message.

When "Polling" a pipe for a message in [CFE_SB_ReceiveBuffer](#), this return value indicates that there was not a message on the pipe.

Definition at line 741 of file [cfe_error.h](#).

CFE_SB_NOT_IMPLEMENTED

```
#define CFE_SB_NOT_IMPLEMENTED ((CFE_Status_t)0xca00ffff)
```

Not Implemented.

Current version of cFE does not have the function or the feature of the function implemented. This could be due to either an early build of the cFE for this platform or the platform does not support the specified feature.

Definition at line 870 of file [cfe_error.h](#).

CFE_SB_PIPE_CR_ERR

```
#define CFE_SB_PIPE_CR_ERR ((CFE_Status_t)0xca000005)
```

Pipe Create Error.

The maximum number of queues([OS_MAX_QUEUES](#)) are in use. Or possibly a lower level problem with creating the underlying queue has occurred such as a lack of memory. If the latter is the problem, the status code displayed in the event must be tracked.

Definition at line 772 of file [cfe_error.h](#).

CFE_SB_PIPE_RD_ERR

```
#define CFE_SB_PIPE_RD_ERR ((CFE_Status_t)0xca000006)
```

Pipe Read Error.

This return value indicates an error at the Queue read level. This error typically cannot be corrected by the caller. Some possible causes are: queue was not properly initialized or created, the number of bytes read from the queue was not the number of bytes requested in the read. The queue id is invalid. Similar errors regarding the pipe will be caught by higher level code in the Software Bus.

Definition at line 785 of file [cfe_error.h](#).

CFE_SB_TIME_OUT

```
#define CFE_SB_TIME_OUT ((CFE_Status_t)0xca000001)
```

Time Out.

In [CFE_SB_ReceiveBuffer](#), this return value indicates that a packet has not been received in the time given in the "timeout" parameter.

Definition at line 732 of file [cfe_error.h](#).

CFE_SB_WRONG_MSG_TYPE

```
#define CFE_SB_WRONG_MSG_TYPE ((CFE_Status_t)0xca00000d)
```

Wrong Message Type.

This error code will be returned when a request such as [CFE_MSG_SetMsgTime](#) is made on a packet that does not include a field for msg time.

Definition at line 849 of file [cfe_error.h](#).

CFE_STATUS_BAD_COMMAND_CODE

```
#define CFE_STATUS_BAD_COMMAND_CODE ((CFE_Status_t)0xc8000004)
```

Bad Command Code.

This error code will be returned when a message identification process determined that the command code is does not correspond to any known value

Definition at line 182 of file [cfe_error.h](#).

CFE_STATUS_EXTERNAL_RESOURCE_FAIL

```
#define CFE_STATUS_EXTERNAL_RESOURCE_FAIL ((CFE_Status_t)0xc8000005)
```

External failure.

This error indicates that the operation failed for some reason outside the scope of CFE. The real failure may have been in OSAL, PSP, or another dependent library.

Details of the original failure should be written to syslog and/or a system event before returning this error.

Definition at line 194 of file [cfe_error.h](#).

CFE_STATUS_INCORRECT_STATE

```
#define CFE_STATUS_INCORRECT_STATE ((int32)0xc8000009)
```

Cannot process request at this time.

The system is not currently in the correct state to accept the request at this time.

Definition at line 227 of file [cfe_error.h](#).

CFE_STATUS_NO_COUNTER_INCREMENT

```
#define CFE_STATUS_NO_COUNTER_INCREMENT ((CFE_Status_t)0x48000001)
```

No Counter Increment.

Informational code indicating that a command was processed successfully but that the command counter should *not* be incremented.

Definition at line 155 of file [cfe_error.h](#).

CFE_STATUS_NOT_IMPLEMENTED

```
#define CFE_STATUS_NOT_IMPLEMENTED ((CFE_Status_t)0xc800ffff)
```

Not Implemented.

Current version does not have the function or the feature of the function implemented. This could be due to either an early build for this platform or the platform does not support the specified feature.

Definition at line 238 of file [cfe_error.h](#).

CFE_STATUS_RANGE_ERROR

```
#define CFE_STATUS_RANGE_ERROR ((int32)0xc8000008)
```

Request or input value is out of range.

A message, table, or function call input contained a value that was outside the acceptable range, and the request was rejected.

Definition at line 219 of file [cfe_error.h](#).

CFE_STATUS_REQUEST_ALREADY_PENDING

```
#define CFE_STATUS_REQUEST_ALREADY_PENDING ((int32)0xc8000006)
```

Request already pending.

Commands or requests are already pending or the pending request limit has been reached. No more requests can be made until the current request(s) complete.

Definition at line 203 of file [cfe_error.h](#).

CFE_STATUS_UNKNOWN_MSG_ID

```
#define CFE_STATUS_UNKNOWN_MSG_ID ((CFE_Status_t)0xc8000003)
```

Unknown Message ID.

This error code will be returned when a message identification process determined that the message ID does not correspond to a known value

Definition at line 173 of file [cfe_error.h](#).

CFE_STATUS_VALIDATION_FAILURE

```
#define CFE_STATUS_VALIDATION_FAILURE ((int32)0xc8000007)
```

Request or input value failed basic structural validation.

A message or table input was not in the proper format to be understood and processed by an application, and was rejected.

Definition at line 211 of file [cfe_error.h](#).

CFE_STATUS_WRONG_MSG_LENGTH

```
#define CFE_STATUS_WRONG_MSG_LENGTH ((CFE_Status_t)0xc8000002)
```

Wrong Message Length.

This error code will be returned when a message validation process determined that the message length is incorrect

Definition at line 164 of file [cfe_error.h](#).

CFE_SUCCESS

```
#define CFE_SUCCESS ((CFE_Status_t)0)
```

Successful execution.

Operation was performed successfully

Definition at line 147 of file [cfe_error.h](#).

CFE_TBL_BAD_ARGUMENT

```
#define CFE_TBL_BAD_ARGUMENT ((CFE_Status_t)0xcc00002d)
```

Bad Argument.

A parameter given by a caller to a Table API did not pass validation checks.

Definition at line 1281 of file [cfe_error.h](#).

CFE_TBL_ERR_ACCESS

```
#define CFE_TBL_ERR_ACCESS ((CFE_Status_t)0xcc00002c)
```

Error code indicating that the TBL file could not be opened by the OS.

Definition at line 1272 of file [cfe_error.h](#).

CFE_TBL_ERR_BAD_CONTENT_ID

```
#define CFE_TBL_ERR_BAD_CONTENT_ID ((CFE_Status_t)0xcc000016)
```

Bad Content ID.

The calling Application called [CFE_TBL_Load](#) with a filename that specified a file whose content ID was not that of a table image.

Definition at line 1064 of file [cfe_error.h](#).

CFE_TBL_ERR_BAD_PROCESSOR_ID

```
#define CFE_TBL_ERR_BAD_PROCESSOR_ID ((CFE_Status_t)0xcc000029)
```

Bad Processor ID.

The selected table file failed validation for Processor ID. The platform configuration file has verification of table files enabled for Processor ID and an attempt was made to load a table with an invalid Processor ID in the table file header. Definition at line 1252 of file [cfe_error.h](#).

CFE_TBL_ERR_BAD_SPACECRAFT_ID

```
#define CFE_TBL_ERR_BAD_SPACECRAFT_ID ((CFE_Status_t)0xcc000028)
```

Bad Spacecraft ID.

The selected table file failed validation for Spacecraft ID. The platform configuration file has verification of table files enabled for Spacecraft ID and an attempt was made to load a table with an invalid Spacecraft ID in the table file header. Definition at line 1241 of file [cfe_error.h](#).

CFE_TBL_ERR_BAD_SUBTYPE_ID

```
#define CFE_TBL_ERR_BAD_SUBTYPE_ID ((CFE_Status_t)0xcc00001B)
```

Bad Subtype ID.

The calling Application tried to access a table file whose Subtype identifier indicated it was not a table image file. Definition at line 1105 of file [cfe_error.h](#).

CFE_TBL_ERR_DUMP_ONLY

```
#define CFE_TBL_ERR_DUMP_ONLY ((CFE_Status_t)0xcc000010)
```

Dump Only Error.

The calling Application has attempted to perform a load on a table that was created with "Dump Only" attributes. Definition at line 1016 of file [cfe_error.h](#).

CFE_TBL_ERR_DUPLICATE_DIFF_SIZE

```
#define CFE_TBL_ERR_DUPLICATE_DIFF_SIZE ((CFE_Status_t)0xcc00000C)
```

Duplicate Table With Different Size.

An application attempted to register a table with the same name as a table that is already in the registry. The size of the new table is different from the size already in the registry.

Definition at line 977 of file [cfe_error.h](#).

CFE_TBL_ERR_DUPLICATE_NOT_OWNED

```
#define CFE_TBL_ERR_DUPLICATE_NOT_OWNED ((CFE_Status_t)0xcc00000D)
```

Duplicate Table And Not Owned.

An application attempted to register a table with the same name as a table that is already in the registry. The previously registered table is owned by a different application.

Definition at line 987 of file [cfe_error.h](#).

CFE_TBL_ERR_FILE_FOR_WRONG_TABLE

```
#define CFE_TBL_ERR_FILE_FOR_WRONG_TABLE ((CFE_Status_t)0xcc000020)
```

File For Wrong Table.

The calling Application tried to load a table using a file whose header indicated that it was for a different table.

Definition at line 1149 of file [cfe_error.h](#).

CFE_TBL_ERR_FILE_SIZE_INCONSISTENT

```
#define CFE_TBL_ERR_FILE_SIZE_INCONSISTENT ((CFE_Status_t)0xcc00001C)
```

File Size Inconsistent.

The calling Application tried to access a table file whose Subtype identifier indicated it was not a table image file.

Definition at line 1114 of file [cfe_error.h](#).

CFE_TBL_ERR_FILE_TOO_LARGE

```
#define CFE_TBL_ERR_FILE_TOO_LARGE ((CFE_Status_t)0xcc000014)
```

File Too Large.

The calling Application called [CFE_TBL_Load](#) with a filename that specified a file that contained more data than the size of the table OR which contained more data than specified in the table header.

Definition at line [1044](#) of file [cfe_error.h](#).

CFE_TBL_ERR_FILENAME_TOO_LONG

```
#define CFE_TBL_ERR_FILENAME_TOO_LONG ((CFE_Status_t)0xcc00001F)
```

Filename Too Long.

The calling Application tried to load a table using a filename that was too long.

Definition at line [1140](#) of file [cfe_error.h](#).

CFE_TBL_ERR_HANDLES_FULL

```
#define CFE_TBL_ERR_HANDLES_FULL ((CFE_Status_t)0xcc00000B)
```

Handles Full.

An application attempted to create a table and the Table Handle Array already used all CFE_PLATFORM_TBL_MAX←_NUM_HANDLES in it.

Definition at line [967](#) of file [cfe_error.h](#).

CFE_TBL_ERR_ILLEGAL_SRC_TYPE

```
#define CFE_TBL_ERR_ILLEGAL_SRC_TYPE ((CFE_Status_t)0xcc000011)
```

Illegal Source Type.

The calling Application called [CFE_TBL_Load](#) with an illegal value for the second parameter.

Definition at line [1025](#) of file [cfe_error.h](#).

CFE_TBL_ERR_INVALID_HANDLE

```
#define CFE_TBL_ERR_INVALID_HANDLE ((CFE_Status_t)0xcc000001)
```

Invalid Handle.

The calling Application attempted to pass a Table handle that represented too large an index or identified a Table Access Descriptor that was not used.

Definition at line [884](#) of file [cfe_error.h](#).

CFE_TBL_ERR_INVALID_NAME

```
#define CFE_TBL_ERR_INVALID_NAME ((CFE_Status_t)0xcc000002)
```

Invalid Name.

The calling Application attempted to register a table whose name length exceeded the platform configuration value of [CFE_MISSION_TBL_MAX_NAME_LENGTH](#) or was zero characters long.

Definition at line [894](#) of file [cfe_error.h](#).

CFE_TBL_ERR_INVALID_OPTIONS

```
#define CFE_TBL_ERR_INVALID_OPTIONS ((CFE_Status_t)0xcc000025)
```

Invalid Options.

The calling Application has used an illegal combination of table options. A summary of the illegal combinations are as follows:

#CFE_TBL_OPT_USR_DEF_ADDR cannot be combined with any of the following:

1. [CFE_TBL_OPT_DBL_BUFFER](#)
2. [CFE_TBL_OPT_LOAD_DUMP](#)
3. [CFE_TBL_OPT_CRITICAL](#)

#CFE_TBL_OPT_DBL_BUFFER cannot be combined with the following:

1. [CFE_TBL_OPT_USR_DEF_ADDR](#)
2. [CFE_TBL_OPT_DUMP_ONLY](#)

Definition at line 1206 of file [cfe_error.h](#).

CFE_TBL_ERR_INVALID_SIZE

```
#define CFE_TBL_ERR_INVALID_SIZE ((CFE_Status_t)0xcc000003)
```

Invalid Size.

The calling Application attempted to register a table: a) that was a double buffered table with size greater than [CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE](#) b) that was a single buffered table with size greater than [CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE](#) c) that had a size of zero

Definition at line 905 of file [cfe_error.h](#).

CFE_TBL_ERR_LOAD_IN_PROGRESS

```
#define CFE_TBL_ERR_LOAD_IN_PROGRESS ((CFE_Status_t)0xcc000012)
```

Load In Progress.

The calling Application called [CFE_TBL_Load](#) when another Application was trying to load the table.

Definition at line 1034 of file [cfe_error.h](#).

CFE_TBL_ERR_LOAD_INCOMPLETE

```
#define CFE_TBL_ERR_LOAD_INCOMPLETE ((CFE_Status_t)0xcc000021)
```

Load Incomplete.

The calling Application tried to load a table file whose header claimed the load was larger than what was actually read from the file.

Definition at line 1158 of file [cfe_error.h](#).

CFE_TBL_ERR_NEVER_LOADED

```
#define CFE_TBL_ERR_NEVER_LOADED ((CFE_Status_t)0xcc000005)
```

Never Loaded.

Table has not been loaded with data.

Definition at line 921 of file [cfe_error.h](#).

CFE_TBL_ERR_NO_ACCESS

```
#define CFE_TBL_ERR_NO_ACCESS ((CFE_Status_t)0xcc000008)
```

No Access.

The calling application either failed when calling [CFE_TBL_Register](#), failed when calling [CFE_TBL_Share](#) or forgot to call either one.

Definition at line 949 of file [cfe_error.h](#).

CFE_TBL_ERR_NO_BUFFER_AVAIL

```
#define CFE_TBL_ERR_NO_BUFFER_AVAIL ((CFE_Status_t)0xcc00000F)
```

No Buffer Available.

The calling Application has tried to allocate a working buffer but none were available.

Definition at line 1007 of file [cfe_error.h](#).

CFE_TBL_ERR_NO_STD_HEADER

```
#define CFE_TBL_ERR_NO_STD_HEADER ((CFE_Status_t)0xcc00001D)
```

No Standard Header.

The calling Application tried to access a table file whose standard cFE File Header was the wrong size, etc.

Definition at line 1122 of file [cfe_error.h](#).

CFE_TBL_ERR_NO_TBL_HEADER

```
#define CFE_TBL_ERR_NO_TBL_HEADER ((CFE_Status_t)0xcc00001E)
```

No Table Header.

The calling Application tried to access a table file whose standard cFE Table File Header was the wrong size, etc.

Definition at line 1131 of file [cfe_error.h](#).

CFE_TBL_ERR_PARTIAL_LOAD

```
#define CFE_TBL_ERR_PARTIAL_LOAD ((CFE_Status_t)0xcc000023)
```

Partial Load Error.

The calling Application tried to load a table file whose header claimed the load did not start with the first byte and the table image had NEVER been loaded before. Partial loads are not allowed on uninitialized tables. It should be noted that [CFE_TBL_WARN_SHORT_FILE](#) also indicates a partial load.

Definition at line 1180 of file [cfe_error.h](#).

CFE_TBL_ERR_REGISTRY_FULL

```
#define CFE_TBL_ERR_REGISTRY_FULL ((CFE_Status_t)0xcc000006)
```

Registry Full.

An application attempted to create a table and the Table registry already contained [CFE_PLATFORM_TBL_MAX_NUM_TABLES](#) in it.

Definition at line 930 of file [cfe_error.h](#).

CFE_TBL_ERR_SHORT_FILE

```
#define CFE_TBL_ERR_SHORT_FILE ((CFE_Status_t)0xcc00002b)
```

Error code indicating that the TBL file is shorter than indicated in the file header.

Definition at line 1266 of file [cfe_error.h](#).

CFE_TBL_ERR_UNREGISTERED

```
#define CFE_TBL_ERR_UNREGISTERED ((CFE_Status_t)0xcc000009)
```

Unregistered.

The calling application is trying to access a table that has been unregistered.

Definition at line 958 of file [cfe_error.h](#).

CFE_TBL_INFO_DUMP_PENDING

```
#define CFE_TBL_INFO_DUMP_PENDING ((CFE_Status_t)0x4c000024)
```

Dump Pending.

The calling Application should call [CFE_TBL_Manage](#) for the specified table. The ground has requested a dump of the Dump-Only table and needs to synchronize with the owning application.
Definition at line 1190 of file [cfe_error.h](#).

CFE_TBL_INFO_NO_DUMP_PENDING

```
#define CFE_TBL_INFO_NO_DUMP_PENDING ((CFE_Status_t)0x4c00002e)
```

No Dump Pending.

The calling Application invoked [CFE_TBL_DumpToBuffer](#) on a table that did not have a pending dump request
Definition at line 1290 of file [cfe_error.h](#).

CFE_TBL_INFO_NO_UPDATE_PENDING

```
#define CFE_TBL_INFO_NO_UPDATE_PENDING ((CFE_Status_t)0x4c000017)
```

No Update Pending.

The calling Application has attempted to update a table without a pending load.
Definition at line 1072 of file [cfe_error.h](#).

CFE_TBL_INFO_NO_VALIDATION_PENDING

```
#define CFE_TBL_INFO_NO_VALIDATION_PENDING ((CFE_Status_t)0x4c00001A)
```

No Validation Pending

The calling Application tried to validate a table that did not have a validation request pending.
Definition at line 1096 of file [cfe_error.h](#).

CFE_TBL_INFO_RECOVERED_TBL

```
#define CFE_TBL_INFO_RECOVERED_TBL ((CFE_Status_t)0x4c000027)
```

Recovered Table.

The calling Application registered a critical table whose previous contents were discovered in the Critical Data Store. The discovered contents were copied back into the newly registered table as the table's initial contents.

NOTE: In this situation, the contents of the table are **NOT** validated using the table's validation function.

Definition at line 1230 of file [cfe_error.h](#).

CFE_TBL_INFO_TABLE_LOCKED

```
#define CFE_TBL_INFO_TABLE_LOCKED ((CFE_Status_t)0x4c000018)
```

Table Locked.

The calling Application tried to update a table that is locked by another user.
Definition at line 1080 of file [cfe_error.h](#).

CFE_TBL_INFO_UPDATE_PENDING

```
#define CFE_TBL_INFO_UPDATE_PENDING ((CFE_Status_t)0x4c000004)
```

Update Pending.

The calling Application has identified a table that has a load pending.
Definition at line 913 of file [cfe_error.h](#).

CFE_TBL_INFO_UPDATED

```
#define CFE_TBL_INFO_UPDATED ((CFE_Status_t)0x4c00000E)
```

Updated.

The calling Application has identified a table that has been updated.

NOTE: This is a nominal return code informing the calling application that the table identified in the call has had its contents updated since the last time the application obtained its address or status.

Definition at line 998 of file [cfe_error.h](#).

CFE_TBL_INFO_VALIDATION_PENDING

```
#define CFE_TBL_INFO_VALIDATION_PENDING ((CFE_Status_t)0x4c000019)
```

Validation Pending

The calling Application should call [CFE_TBL_Validate](#) for the specified table.

Definition at line 1088 of file [cfe_error.h](#).

CFE_TBL_MESSAGE_ERROR

```
#define CFE_TBL_MESSAGE_ERROR ((CFE_Status_t)0xcc00002a)
```

Message Error.

Error code indicating that the TBL command was not processed successfully and that the error counter should be incremented.

Definition at line 1260 of file [cfe_error.h](#).

CFE_TBL_NOT_IMPLEMENTED

```
#define CFE_TBL_NOT_IMPLEMENTED ((CFE_Status_t)0xcc00ffff)
```

Not Implemented.

Current version of cFE does not have the function or the feature of the function implemented. This could be due to either an early build of the cFE for this platform or the platform does not support the specified feature.

Definition at line 1301 of file [cfe_error.h](#).

CFE_TBL_WARN_DUPLICATE

```
#define CFE_TBL_WARN_DUPLICATE ((CFE_Status_t)0x4c000007)
```

Duplicate Warning.

This is an error that the registration is trying to replace an existing table with the same name. The previous table stays in place and the new table is rejected.

Definition at line 940 of file [cfe_error.h](#).

CFE_TBL_WARN_NOT_CRITICAL

```
#define CFE_TBL_WARN_NOT_CRITICAL ((CFE_Status_t)0x4c000026)
```

Not Critical Warning.

The calling Application attempted to register a table as "Critical". Table Services failed to create an appropriate Critical Data Store (See System Log for reason) to save the table contents. The table will be treated as a normal table from now on.

Definition at line 1217 of file [cfe_error.h](#).

CFE_TBL_WARN_PARTIAL_LOAD

```
#define CFE_TBL_WARN_PARTIAL_LOAD ((CFE_Status_t)0x4c000022)
```

Partial Load Warning.

The calling Application tried to load a table file whose header claimed the load did not start with the first byte. It should be noted that [CFE_TBL_WARN_SHORT_FILE](#) also indicates a partial load.

Definition at line 1168 of file [cfe_error.h](#).

CFE_TBL_WARN_SHORT_FILE

```
#define CFE_TBL_WARN_SHORT_FILE ((CFE_Status_t)0x4c000015)
```

Short File Warning.

The calling Application called [CFE_TBL_Load](#) with a filename that specified a file that started with the first byte of the table but contained less data than the size of the table. It should be noted that [CFE_TBL_WARN_PARTIAL_LOAD](#) also indicates a partial load (one that starts at a non-zero offset).

Definition at line 1055 of file [cfe_error.h](#).

CFE_TIME_BAD_ARGUMENT

```
#define CFE_TIME_BAD_ARGUMENT ((CFE_Status_t)0xce000005)
```

Bad Argument.

A parameter given by a caller to a TIME Services API did not pass validation checks.

Definition at line 1373 of file [cfe_error.h](#).

CFE_TIME_CALLBACK_NOT_REGISTERED

```
#define CFE_TIME_CALLBACK_NOT_REGISTERED ((CFE_Status_t)0xce000004)
```

Callback Not Registered.

An attempt to unregister a cFE Time Services Synchronization callback has failed because the specified callback function was not located in the Synchronization Callback Registry.

Definition at line 1364 of file [cfe_error.h](#).

CFE_TIME_INTERNAL_ONLY

```
#define CFE_TIME_INTERNAL_ONLY ((CFE_Status_t)0xce000001)
```

Internal Only.

One of the TIME Services API functions to set the time with data from an external time source has been called, but TIME Services has been commanded to not accept external time data. However, the command is still a signal for the Time Server to generate a "time at the tone" command packet using internal data.

Definition at line 1328 of file [cfe_error.h](#).

CFE_TIME_NOT_IMPLEMENTED

```
#define CFE_TIME_NOT_IMPLEMENTED ((CFE_Status_t)0xce00ffff)
```

Not Implemented.

Current version of cFE does not have the function or the feature of the function implemented. This could be due to either an early build of the cFE for this platform or the platform does not support the specified feature.

Definition at line 1316 of file [cfe_error.h](#).

CFE_TIME_OUT_OF_RANGE

```
#define CFE_TIME_OUT_OF_RANGE ((CFE_Status_t)0xce000002)
```

Out Of Range.

One of the TIME Services API functions to set the time with data from an external time source has been called, but TIME Services has determined that the new time data is invalid. However, the command is still a signal for the Time Server to generate a "time at the tone" command packet using internal data.

Note that the test for invalid time update data only occurs if TIME Services has previously been commanded to set the clock state to "valid".

Definition at line 1343 of file [cfe_error.h](#).

CFE_TIME_TOO_MANY_SYNCH_CALLBACKS

```
#define CFE_TIME_TOO_MANY_SYNCH_CALLBACKS ((CFE_Status_t)0xce000003)
```

Too Many Sync Callbacks.

An attempt to register too many cFE Time Services Synchronization callbacks has been made. Only one callback function is allowed per application. It is expected that the application itself will distribute the single callback to child threads as needed.

Definition at line 1354 of file [cfe_error.h](#).

9.2 cFE Resource ID APIs

Functions

- [CFE_Status_t CFE_ES_AppID_ToIndex](#) ([CFE_ES_AppId_t](#) AppID, [uint32](#) *Idx)
Obtain an index value correlating to an ES Application ID.
- [int32 CFE_ES_LibID_ToIndex](#) ([CFE_ES_LibId_t](#) LibId, [uint32](#) *Idx)
Obtain an index value correlating to an ES Library ID.
- [CFE_Status_t CFE_ES_TaskID_ToIndex](#) ([CFE_ES_TaskId_t](#) TaskID, [uint32](#) *Idx)
Obtain an index value correlating to an ES Task ID.
- [CFE_Status_t CFE_ES_CounterID_ToIndex](#) ([CFE_ES_CounterId_t](#) CounterId, [uint32](#) *Idx)
Obtain an index value correlating to an ES Counter ID.

9.2.1 Detailed Description

9.2.2 Function Documentation

CFE_ES_AppID_ToIndex()

```
CFE_Status_t CFE_ES_AppID_ToIndex (
    CFE_ES_AppId_t AppID,
    uint32 * Idx )
```

Obtain an index value correlating to an ES Application ID.

This calculates a zero based integer value that may be used for indexing into a local resource table/array.

Index values are only guaranteed to be unique for resources of the same type. For instance, the indices corresponding to two [valid] application IDs will never overlap, but the index of an application and a library ID may be the same. Furthermore, indices may be reused if a resource is deleted and re-created.

Note

There is no inverse of this function - indices cannot be converted back to the original AppID value. The caller should retain the original ID for future use.

Parameters

in	<i>AppID</i>	Application ID to convert
out	<i>Idx</i>	Buffer where the calculated index will be stored (must not be null)

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.

CFE_ES_CounterID_ToIndex()

```
CFE_Status_t CFE_ES_CounterID_ToIndex (
    CFE_ES_CounterId_t CounterId,
    uint32 * Idx )
```

Obtain an index value correlating to an ES Counter ID.

This calculates a zero based integer value that may be used for indexing into a local resource table/array.

Index values are only guaranteed to be unique for resources of the same type. For instance, the indices corresponding to two [valid] Counter IDs will never overlap, but the index of a Counter and a library ID may be the same. Furthermore, indices may be reused if a resource is deleted and re-created.

Note

There is no inverse of this function - indices cannot be converted back to the original CounterID value. The caller should retain the original ID for future use.

Parameters

in	<i>Counter↔ Id</i>	Counter ID to convert
out	<i>Idx</i>	Buffer where the calculated index will be stored (must not be null)

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.

CFE_ES_LibID_ToIndex()

```
int32 CFE_ES_LibID_ToIndex (
    CFE_ES_LibId_t LibId,
    uint32 * Idx )
```

Obtain an index value correlating to an ES Library ID.

This calculates a zero based integer value that may be used for indexing into a local resource table/array.

Index values are only guaranteed to be unique for resources of the same type. For instance, the indices corresponding to two [valid] Library IDs will never overlap, but the index of an Library and a library ID may be the same. Furthermore, indices may be reused if a resource is deleted and re-created.

Note

There is no inverse of this function - indices cannot be converted back to the original LibID value. The caller should retain the original ID for future use.

Parameters

in	<i>Lib↔ Id</i>	Library ID to convert
out	<i>Idx</i>	Buffer where the calculated index will be stored (must not be null)

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

<i>CFE_SUCCESS</i>	Successful execution.
<i>CFE_ES_ERR_RESOURCEID_NOT_VALID</i>	Resource ID is not valid.

CFE_ES_TaskID_ToIndex()

```
CFE_Status_t CFE_ES_TaskID_ToIndex (
    CFE_ES_TaskId_t TaskID,
    uint32 * Idx )
```

Obtain an index value correlating to an ES Task ID.

This calculates a zero based integer value that may be used for indexing into a local resource table/array.

Index values are only guaranteed to be unique for resources of the same type. For instance, the indices corresponding to two [valid] Task IDs will never overlap, but the index of a Task and a library ID may be the same. Furthermore, indices may be reused if a resource is deleted and re-created.

Note

There is no inverse of this function - indices cannot be converted back to the original TaskID value. The caller should retain the original ID for future use.

Parameters

in	<i>TaskID</i>	Task ID to convert
out	<i>Idx</i>	Buffer where the calculated index will be stored (must not be null)

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

<i>CFE_SUCCESS</i>	Successful execution.
<i>CFE_ES_ERR_RESOURCEID_NOT_VALID</i>	Resource ID is not valid.
<i>CFE_ES_BAD_ARGUMENT</i>	Bad Argument.

9.3 cFE Entry/Exit APIs**Functions**

- void [CFE_ES_Main](#) (uint32 StartType, uint32 StartSubtype, uint32 ModelId, const char *StartFilePath)
cFE Main Entry Point used by Board Support Package to start cFE
- [CFE_Status_t CFE_ES_ResetCFE](#) (uint32 ResetType)
Reset the cFE Core and all cFE Applications.

9.3.1 Detailed Description

9.3.2 Function Documentation

CFE_ES_Main()

```
void CFE_ES_Main (
    uint32 StartType,
    uint32 StartSubtype,
    uint32 ModeId,
    const char * StartFilePath )
```

cFE Main Entry Point used by Board Support Package to start cFE

Description

cFE main entry point. This is the entry point into the cFE software. It is called only by the Board Support Package software.

Assumptions, External Events, and Notes:

None

Parameters

in	<i>StartType</i>	Identifies whether this was a CFE_PSP_RST_TYPE_POWERON or CFE_PSP_RST_TYPE_PROCESSOR .
in	<i>StartSubtype</i>	Specifies, in more detail, what caused the <i>StartType</i> identified above. See CFE_PSP_RST_SUBTYPE_POWER_CYCLE for possible examples.
in	<i>ModeId</i>	Identifies the source of the Boot as determined by the BSP.
in	<i>StartFilePath</i>	Identifies the startup file to use to initialize the cFE apps.

See also

[CFE_ES_ResetCFE](#)

CFE_ES_ResetCFE()

```
CFE_Status_t CFE_ES_ResetCFE (
    uint32 ResetType )
```

Reset the cFE Core and all cFE Applications.

Description

This API causes an immediate reset of the cFE Kernel and all cFE Applications. The caller can specify whether the reset should clear all memory ([CFE_PSP_RST_TYPE_POWERON](#)) or try to retain volatile memory areas ([CFE_PSP_RST_TYPE_PROCESSOR](#)).

Assumptions, External Events, and Notes:

None

Parameters

in	<i>ResetType</i>	Identifies the type of reset desired. Allowable settings are: • CFE_PSP_RST_TYPE_POWERON - Causes all memory to be cleared • CFE_PSP_RST_TYPE_PROCESSOR - Attempts to retain volatile disk, critical data store and user reserved memory.
----	------------------	--

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_BAD_ARGUMENT	Bad Argument.
CFE_ES_NOT_IMPLEMENTED	Not Implemented.

See also

[CFE_ES_Main](#)

9.4 cFE Application Control APIs

Functions

- [CFE_Status_t CFE_ES_RestartApp](#) ([CFE_ES_AppId_t](#) AppID)
Restart a single cFE Application.
- [CFE_Status_t CFE_ES_ReloadApp](#) ([CFE_ES_AppId_t](#) AppID, const char *AppFileName)
Reload a single cFE Application.
- [CFE_Status_t CFE_ES_DeleteApp](#) ([CFE_ES_AppId_t](#) AppID)
Delete a cFE Application.

9.4.1 Detailed Description

9.4.2 Function Documentation

CFE_ES_DeleteApp()

```
CFE_Status_t CFE_ES_DeleteApp (
    CFE_ES_AppId_t AppID )
```

Delete a cFE Application.

Description

This API causes a cFE Application to be stopped deleted.

Assumptions, External Events, and Notes:

None

Parameters

in	<i>AppID</i>	Identifies the application to be reset.
----	--------------	---

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_SUCCESS	Successful execution.

See also

[CFE_ES_RestartApp](#), [CFE_ES_ReloadApp](#)

CFE_ES_ReloadApp()

```
CFE_Status_t CFE_ES_ReloadApp (
    CFE_ES_AppId_t AppID,
    const char * AppFileName )
```

Reload a single cFE Application.

Description

This API causes a cFE Application to be stopped and restarted from the specified file.

Assumptions, External Events, and Notes:

The filename is checked for existence prior to load. A missing file will be reported and the reload operation will be aborted prior to unloading the app.

Goes through the standard CFE_ES_CleanUpApp which unloads, then attempts a load using the specified file name. In the event that an application cannot be reloaded due to a corrupt file, the application may no longer be reloaded when given a valid load file (it has been deleted and no longer exists). To recover, the application may be started by loading the application via the ES_STARTAPP command ([CFE_ES_START_APP_CC](#)).

Parameters

in	<i>AppID</i>	Identifies the application to be reset.
in	<i>AppFileName</i>	Identifies the new file to start (must not be null)

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_SUCCESS	Successful execution.
CFE_ES_FILE_IO_ERR	File IO Error.

See also

[CFE_ES_RestartApp](#), [CFE_ES_DeleteApp](#), [CFE_ES_START_APP_CC](#)

CFE_ES_RestartApp()

```
CFE_Status_t CFE_ES_RestartApp (
    CFE_ES_AppId_t AppID )
```

Restart a single cFE Application.

Description

This API causes a cFE Application to be unloaded and restarted from the same file name as the last start.

Assumptions, External Events, and Notes:

The filename is checked for existence prior to load. A missing file will be reported and the reload operation will be aborted prior to unloading the app.

Goes through the standard [CFE_ES_CleanUpApp](#) which unloads, then attempts a load using the original file name. In the event that an application cannot be reloaded due to a missing file or any other load issue, the application may no longer be restarted or reloaded when given a valid load file (the app has been deleted and no longer exists). To recover, the application may be started by loading the application via the [ES_STARTAPP](#) command ([CFE_ES_START_APP_CC](#)).

Parameters

in	<i>AppID</i>	Identifies the application to be reset.
----	--------------	---

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_ES_FILE_IO_ERR	File IO Error.
CFE_SUCCESS	Successful execution.

See also

[CFE_ES_ReloadApp](#), [CFE_ES_DeleteApp](#)

9.5 cFE Application Behavior APIs

Functions

- void [CFE_ES_ExitApp](#) (uint32 ExitStatus)
Exit a cFE Application.
- bool [CFE_ES_RunLoop](#) (uint32 *RunStatus)
Check for Exit, Restart, or Reload commands.
- [CFE_Status_t](#) [CFE_ES_WaitForSystemState](#) (uint32 MinSystemState, uint32 TimeOutMilliseconds)
Allow an Application to Wait for a minimum global system state.

- void [CFE_ES_WaitForStartupSync](#) (uint32 TimeOutMilliseconds)
Allow an Application to Wait for the "OPERATIONAL" global system state.
- void [CFE_ES_IncrementTaskCounter](#) (void)
Increments the execution counter for the calling task.

9.5.1 Detailed Description

9.5.2 Function Documentation

CFE_ES_ExitApp()

```
void CFE_ES_ExitApp (
    uint32 ExitStatus )
```

Exit a cFE Application.

Description

This API is the "Exit Point" for the cFE application

Assumptions, External Events, and Notes:

None

Parameters

in	ExitStatus	Acceptable values are: • CFE_ES_RunStatus_APP_EXIT - Indicates that the Application wants to exit normally. • CFE_ES_RunStatus_APP_ERROR - Indicates that the Application is quitting with an error. • CFE_ES_RunStatus_CORE_APP_INIT_ERROR - Indicates that the Core Application could not Init. • CFE_ES_RunStatus_CORE_APP_RUNTIME_ERROR - Indicates that the Core Application had a runtime failure.
----	------------	---

See also

[CFE_ES_RunLoop](#)

CFE_ES_IncrementTaskCounter()

```
void CFE_ES_IncrementTaskCounter (
    void )
```

Increments the execution counter for the calling task.

Description

This routine increments the execution counter that is stored for the calling task. It can be called from cFE Application main tasks, child tasks, or cFE Core application main tasks. Normally, the call is not necessary from a cFE Application, since the [CFE_ES_RunLoop](#) call increments the counter for the Application.

Assumptions, External Events, and Notes:

NOTE: This API is not needed for Applications that call the CFE_ES_RunLoop call.

See also

[CFE_ES_RunLoop](#)

CFE_ES_RunLoop()

```
bool CFE_ES_RunLoop (
    uint32 * RunStatus )
```

Check for Exit, Restart, or Reload commands.

Description

This is the API that allows an app to check for exit requests from the system, or request shutdown from the system.

Assumptions, External Events, and Notes:

This API updates the internal task counter tracked by ES for the calling task. For ES to report application counters correctly this API should be called from the main app task as part of it's main processing loop.

In the event of a externally initiated app shutdown request (such as the APP_STOP, APP_RELOAD, and APP_RESTART commands) or if a system error occurs requiring the app to be shut down administratively, this function returns "false" and optionally sets the "RunStatus" output to further indicate the specific application state.

If "RunStatus" is passed as non-NULL, it should point to a local status variable containing the requested status to ES. Normally, this should be initialized to [CFE_ES_RunStatus_APP_RUN](#) during application start up, and should remain as this value during normal operation.

If "RunStatus" is set to [CFE_ES_RunStatus_APP_EXIT](#) or [CFE_ES_RunStatus_APP_ERROR](#) on input, this acts as a shutdown request - [CFE_ES_RunLoop\(\)](#) function will return "false", and a shutdown will be initiated similar to if ES had been externally commanded to shut down the app.

If "RunStatus" is not used, it should be passed as NULL. In this mode, only the boolean return value is relevant, which will indicate if an externally-initiated shutdown request is pending.

Parameters

in, out	<i>RunStatus</i>	Optional pointer to a variable containing the desired run status
---------	------------------	--

Returns

Boolean indicating application should continue running

Return values

<i>true</i>	Application should continue running
<i>false</i>	Application should not continue running

See also

[CFE_ES_ExitApp](#)

CFE_ES_WaitForStartupSync()

```
void CFE_ES_WaitForStartupSync (
    uint32 TimeOutMilliseconds )
```

Allow an Application to Wait for the "OPERATIONAL" global system state.

Description

This is the API that allows an app to wait for the rest of the apps to complete their entire initialization before continuing. It is most useful for applications such as Health and Safety or the Scheduler that need to wait until applications exist and are running before sending out packets to them.

This is a specialized wrapper for [CFE_ES_WaitForSystemState](#) for compatibility with applications using this API.

Assumptions, External Events, and Notes:

This API should only be called as the last item of an Apps initialization. In addition, this API should only be called by an App that is started from the ES Startup file. It should not be used by an App that is started after the system is running. (Although it will cause no harm)

Parameters

in	<i>TimeOutMilliseconds</i>	The timeout value in Milliseconds. This parameter must be at least 1000. Lower values will be rounded up. There is not an option to wait indefinitely to avoid hanging a critical application because a non-critical app did not start.
----	----------------------------	---

See also

[CFE_ES_RunLoop](#)

CFE_ES_WaitForSystemState()

```
CFE_Status_t CFE_ES_WaitForSystemState (
    uint32 MinSystemState,
    uint32 TimeOutMilliseconds )
```

Allow an Application to Wait for a minimum global system state.

Description

This is the API that allows an app to wait for the rest of the apps to complete a given stage of initialization before continuing.

This gives finer grained control than [CFE_ES_WaitForStartupSync](#)

Assumptions, External Events, and Notes:

This API assumes that the caller has also been initialized sufficiently to satisfy the global system state it is waiting for, and the apps own state will be updated accordingly.

Parameters

in	<i>MinSystemState</i>	Determine the state of the App
in	<i>TimeoutMilliseconds</i>	The timeout value in Milliseconds. There is not an option to wait indefinitely to avoid hanging a critical application because a non-critical app did not start.

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	State successfully achieved
CFE_ES_OPERATION_TIMED_OUT	(return value only verified in coverage test) Timeout was reached

See also

[CFE_ES_RunLoop](#)

9.6 cFE Information APIs

Functions

- [int32 CFE_ES_GetResetType](#) (uint32 *ResetSubtypePtr)
Return the most recent Reset Type.
- [CFE_Status_t CFE_ES_GetAppID](#) (CFE_ES_AppId_t *AppIdPtr)
Get an Application ID for the calling Application.
- [CFE_Status_t CFE_ES_GetTaskID](#) (CFE_ES_TaskId_t *TaskIdPtr)
Get the task ID of the calling context.
- [CFE_Status_t CFE_ES_GetAppIDByName](#) (CFE_ES_AppId_t *AppIdPtr, const char *AppName)
Get an Application ID associated with a specified Application name.
- [CFE_Status_t CFE_ES_GetLibIDByName](#) (CFE_ES_LibId_t *LibIdPtr, const char *LibName)
Get a Library ID associated with a specified Library name.
- [CFE_Status_t CFE_ES_GetAppName](#) (char *AppName, CFE_ES_AppId_t AppId, size_t BufferLength)
Get an Application name for a specified Application ID.
- [CFE_Status_t CFE_ES_GetLibName](#) (char *LibName, CFE_ES_LibId_t LibId, size_t BufferLength)
Get a Library name for a specified Library ID.
- [CFE_Status_t CFE_ES_GetAppInfo](#) (CFE_ES_AppInfo_t *AppInfo, CFE_ES_AppId_t AppId)
Get Application Information given a specified App ID.
- [CFE_Status_t CFE_ES_GetTaskInfo](#) (CFE_ES_TaskInfo_t *TaskInfo, CFE_ES_TaskId_t TaskId)
Get Task Information given a specified Task ID.
- [int32 CFE_ES_GetLibInfo](#) (CFE_ES_AppInfo_t *LibInfo, CFE_ES_LibId_t LibId)
Get Library Information given a specified Resource ID.
- [int32 CFE_ES_GetModuleInfo](#) (CFE_ES_AppInfo_t *ModuleInfo, CFE_ResourceId_t ResourceId)
Get Information given a specified Resource ID.

9.6.1 Detailed Description

9.6.2 Function Documentation

CFE_ES_GetAppID()

```
CFE_Status_t CFE_ES_GetAppID (
    CFE_ES_AppId_t * AppIdPtr )
```

Get an Application ID for the calling Application.

Description

This routine retrieves the cFE Application ID for the calling Application.

Assumptions, External Events, and Notes:

NOTE: **All** tasks associated with the Application would return the same Application ID.

Parameters

out	<i>AppIdPtr</i>	Pointer to variable that is to receive the Application's ID (must not be null). *AppIdPtr will be set to the application ID of the calling Application.
-----	-----------------	---

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also

[CFE_ES_GetResetType](#), [CFE_ES_GetAppIDByName](#), [CFE_ES_GetAppName](#), [CFE_ES_GetTaskInfo](#)

CFE_ES_GetAppIDByName()

```
CFE_Status_t CFE_ES_GetAppIDByName (
    CFE_ES_AppId_t * AppIdPtr,
    const char * AppName )
```

Get an Application ID associated with a specified Application name.

Description

This routine retrieves the cFE Application ID associated with a specified Application name.

Assumptions, External Events, and Notes:

None

Parameters

out	<i>AppIdPtr</i>	Pointer to variable that is to receive the Application's ID (must not be null).
in	<i>AppName</i>	Pointer to null terminated character string containing an Application name (must not be null).

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_ERR_NAME_NOT_FOUND	Resource Name Error.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also

[CFE_ES_GetAppID](#), [CFE_ES_GetAppName](#), [CFE_ES_GetAppInfo](#)

CFE_ES_GetAppInfo()

```
CFE_Status_t CFE_ES_GetAppInfo (
    CFE_ES_AppInfo_t * AppInfo,
    CFE_ES_AppId_t AppId )
```

Get Application Information given a specified App ID.

Description

This routine retrieves the information about an App associated with a specified App ID. The information includes all of the information ES maintains for an application (documented in the `CFE_ES_AppInfo_t` type)

Assumptions, External Events, and Notes:

None

Parameters

out	<i>AppInfo</i>	Pointer to a structure (must not be null) that will be filled with resource name and memory addresses information.
in	<i>AppId</i>	ID of application to obtain information about

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also

[CFE_ES_GetAppID](#), [CFE_ES_GetAppIDByName](#), [CFE_ES_GetAppName](#)

CFE_ES_GetAppName()

```
CFE_Status_t CFE_ES_GetAppName (
    char * AppName,
    CFE_ES_AppId_t AppId,
    size_t BufferLength )
```

Get an Application name for a specified Application ID.

Description

This routine retrieves the cFE Application name associated with a specified Application ID.

Assumptions, External Events, and Notes:

In the case of a failure ([CFE_ES_ERR_RESOURCEID_NOT_VALID](#)), an empty string is returned.

Parameters

out	<i>AppName</i>	Pointer to a character array (must not be null) of at least <i>BufferLength</i> in size that will be filled with the appropriate Application name.
in	<i>AppId</i>	Application ID of Application whose name is being requested.
in	<i>BufferLength</i>	The maximum number of characters, including the null terminator, that can be put into the <i>AppName</i> buffer. This routine will truncate the name to this length, if necessary.

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also

[CFE_ES_GetAppID](#), [CFE_ES_GetAppIDByName](#), [CFE_ES_GetAppInfo](#)

CFE_ES_GetLibIDByName()

```
CFE_Status_t CFE_ES_GetLibIDByName (
    CFE_ES_LibId_t * LibIdPtr,
    const char * LibName )
```

Get a Library ID associated with a specified Library name.

Description

This routine retrieves the cFE Library ID associated with a specified Library name.

Assumptions, External Events, and Notes:

None

Parameters

out	<i>LibIdPtr</i>	Pointer to variable that is to receive the Library's ID (must not be null).
in	<i>LibName</i>	Pointer to null terminated character string containing a Library name (must not be null).

ReturnsExecution status, see [cFE Return Code Defines](#)**Return values**

CFE_SUCCESS	Successful execution.
CFE_ES_ERR_NAME_NOT_FOUND	Resource Name Error.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also[CFE_ES_GetLibName](#)**CFE_ES_GetLibInfo()**

```
int32 CFE_ES_GetLibInfo (
    CFE_ES_AppInfo_t * LibInfo,
    CFE_ES_LibId_t LibId )
```

Get Library Information given a specified Resource ID.

Description

This routine retrieves the information about a Library associated with a specified ID. The information includes all of the information ES maintains for this resource type (documented in the `CFE_ES_AppInfo_t` type).

This shares the same output structure as `CFE_ES_GetAppInfo`, such that informational commands can be executed against either applications or libraries. When applied to a library, the task information in the structure will be omitted, as libraries do not have tasks associated.

Assumptions, External Events, and Notes:

None

Parameters

out	<i>LibInfo</i>	Pointer to a structure (must not be null) that will be filled with resource name and memory addresses information.
in	<i>LibId</i>	ID of application to obtain information about

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also

[CFE_ES_GetLibIDByName](#), [CFE_ES_GetLibName](#)

CFE_ES_GetLibName()

```
CFE_Status_t CFE_ES_GetLibName (
    char * LibName,
    CFE_ES_LibId_t LibId,
    size_t BufferLength )
```

Get a Library name for a specified Library ID.

Description

This routine retrieves the cFE Library name associated with a specified Library ID.

Assumptions, External Events, and Notes:

In the case of a failure ([CFE_ES_ERR_RESOURCEID_NOT_VALID](#)), an empty string is returned.

Parameters

out	<i>LibName</i>	Pointer to a character array (must not be null) of at least <i>BufferLength</i> in size that will be filled with the Library name.
in	<i>LibId</i>	Library ID of Library whose name is being requested.
in	<i>BufferLength</i>	The maximum number of characters (must not be zero), including the null terminator, that can be put into the <i>LibName</i> buffer. This routine will truncate the name to this length, if necessary.

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also

[CFE_ES_GetLibIDByName](#)

CFE_ES_GetModuleInfo()

```
int32 CFE_ES_GetModuleInfo (
    CFE_ES_AppInfo_t * ModuleInfo,
    CFE_ResourceId_t ResourceId )
```

Get Information given a specified Resource ID.

Description

This routine retrieves the information about an Application or Library associated with a specified ID.

This is a wrapper API that in turn calls either CFE_ES_GetAppInfo or CFE_ES_GetLibInfo if passed an AppId or LibId, respectively.

This allows commands originally targeted to operate on AppIDs to be easily ported to operate on either Libraries or Applications, where relevant.

Assumptions, External Events, and Notes:

None

Parameters

out	<i>ModuleInfo</i>	Pointer to a structure (must not be null) that will be filled with resource name and memory addresses information.
in	<i>ResourceId</i>	ID of application or library to obtain information about

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also

[CFE_ES_GetLibInfo](#), [CFE_ES_GetAppInfo](#)

CFE_ES_GetResetType()

```
int32 CFE_ES_GetResetType (
    uint32 * ResetSubtypePtr )
```

Return the most recent Reset Type.

Description

Provides the caller with codes that identifies the type of Reset the processor most recently underwent. The caller can also obtain information on what caused the reset by supplying a pointer to a variable that will be filled with the Reset Sub-Type.

Assumptions, External Events, and Notes:

None

Parameters

<i>in, out</i>	<i>ResetSubtypePtr</i>	Pointer to <code>uint32</code> type variable in which the Reset Sub-Type will be stored. The caller can set this pointer to <code>NULL</code> if the Sub-Type is of no interest. ResetSubtypePtr If the provided pointer was not <code>NULL</code> , the Reset Sub-Type is stored at the given address. For a list of possible Sub-Type values, see "Reset Sub-Types" .
----------------	------------------------	--

Returns

Processor reset type

Return values

CFE_PSP_RST_TYPE_POWERON	
CFE_PSP_RST_TYPE_PROCESSOR	

See also

[CFE_ES_GetAppID](#), [CFE_ES_GetAppIDByName](#), [CFE_ES_GetAppName](#), [CFE_ES_GetTaskInfo](#)

CFE_ES_GetTaskID()

```
CFE_Status_t CFE_ES_GetTaskID (
    CFE_ES_TaskId_t * TaskIdPtr )
```

Get the task ID of the calling context.

Description

This retrieves the current task context from OSAL

Assumptions, External Events, and Notes:

Applications which desire to call other CFE ES services such as [CFE_ES_TaskGetInfo\(\)](#) should use this API rather than getting the ID from OSAL directly via [OS_TaskGetId\(\)](#).

Parameters

<i>out</i>	<i>TaskIdPtr</i>	Pointer to variable that is to receive the ID (must not be null). Will be set to the ID of the calling task.
------------	------------------	--

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_ES_BAD_ARGUMENT	Bad Argument.

CFE_ES_GetTaskInfo()

```
CFE_Status_t CFE_ES_GetTaskInfo (
    CFE_ES_TaskInfo_t * TaskInfo,
    CFE_ES_TaskId_t TaskId )
```

Get Task Information given a specified Task ID.

Description

This routine retrieves the information about a Task associated with a specified Task ID. The information includes Task Name, and Parent/Creator Application ID.

Assumptions, External Events, and Notes:

None

Parameters

out	<i>TaskInfo</i>	Pointer to a CFE_ES_TaskInfo_t structure (must not be null) that holds the specific task information. *TaskInfo is the filled out CFE_ES_TaskInfo_t structure containing the Task Name, Parent App Name, Parent App ID among other fields.
in	<i>TaskId</i>	Application ID of Application whose name is being requested.

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also

[CFE_ES_GetTaskID](#), [CFE_ES_GetTaskIDByName](#), [CFE_ES_GetTaskName](#)

9.7 cFE Child Task APIs

Functions

- [CFE_Status_t CFE_ES_CreateChildTask](#) ([CFE_ES_TaskId_t](#) *TaskIdPtr, const char *TaskName, [CFE_ES_ChildTaskMainFuncPtr_t](#) FunctionPtr, [CFE_ES_StackPointer_t](#) StackPtr, [size_t](#) StackSize, [CFE_ES_TaskPriority_Atom_t](#) Priority, [uint32](#) Flags)
Creates a new task under an existing Application.
- [CFE_Status_t CFE_ES_GetTaskIDByName](#) ([CFE_ES_TaskId_t](#) *TaskIdPtr, const char *TaskName)
Get a Task ID associated with a specified Task name.
- [CFE_Status_t CFE_ES_GetTaskName](#) (char *TaskName, [CFE_ES_TaskId_t](#) TaskId, [size_t](#) BufferLength)
Get a Task name for a specified Task ID.
- [CFE_Status_t CFE_ES_DeleteChildTask](#) ([CFE_ES_TaskId_t](#) TaskId)
Deletes a task under an existing Application.
- void [CFE_ES_ExitChildTask](#) (void)
Exits a child task.

9.7.1 Detailed Description

9.7.2 Function Documentation

CFE_ES_CreateChildTask()

```
CFE_Status_t CFE_ES_CreateChildTask (
    CFE_ES_TaskId_t * TaskIdPtr,
    const char * TaskName,
    CFE_ES_ChildTaskMainFuncPtr_t FunctionPtr,
    CFE_ES_StackPointer_t StackPtr,
    size_t StackSize,
    CFE_ES_TaskPriority_Atom_t Priority,
    uint32 Flags )
```

Creates a new task under an existing Application.

Description

This routine creates a new task (a separate execution thread) owned by the calling Application.

Assumptions, External Events, and Notes:

None

Parameters

out	<i>TaskIdPtr</i>	A pointer to a variable that will be filled in with the new task's ID (must not be null). TaskIdPtr is the Task ID of the newly created child task.
in	<i>TaskName</i>	A pointer to a string containing the desired name of the new task (must not be null). This can be up to OS_MAX_API_NAME characters, including the trailing null.
in	<i>FunctionPtr</i>	A pointer to the function that will be spawned as a new task (must not be null).
in	<i>StackPtr</i>	A pointer to the location where the child task's stack pointer should start. NOTE: Not all underlying operating systems support this parameter. The CFE_ES_TASK_STACK_ALLOCATE constant may be passed to indicate that the stack should be dynamically allocated.

Parameters

in	<i>StackSize</i>	The number of bytes to allocate for the new task's stack (must not be zero).
in	<i>Priority</i>	The priority for the new task. Lower numbers are higher priority, with 0 being the highest priority.
in	<i>Flags</i>	Reserved for future expansion.

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_ERR_CHILD_TASK_CREATE	Child Task Create Error.
CFE_ES_BAD_ARGUMENT	Bad Argument.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.

See also

[CFE_ES_DeleteChildTask](#), [CFE_ES_ExitChildTask](#)

CFE_ES_DeleteChildTask()

```
CFE_Status_t CFE_ES_DeleteChildTask (
    CFE_ES_TaskId_t TaskId )
```

Deletes a task under an existing Application.

Description

This routine deletes a task under an Application specified by the `TaskId` obtained when the child task was created using the [CFE_ES_CreateChildTask](#) API.

Assumptions, External Events, and Notes:

None

Parameters

in	<i>TaskId</i>	The task ID previously obtained when the Child Task was created with the CFE_ES_CreateChildTask API.
----	---------------	--

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_ERR_CHILD_TASK_DELETE	(return value only verified in coverage test) Child Task Delete Error.

Return values

CFE_ES_ERR_CHILD_TASK_DELETE_MAIN_TASK	Child Task Delete Passed Main Task.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.

See also

[CFE_ES_CreateChildTask](#), [CFE_ES_ExitChildTask](#)

CFE_ES_ExitChildTask()

```
void CFE_ES_ExitChildTask (
    void )
```

Exits a child task.

Description

This routine allows the current executing child task to exit and be deleted by ES.

Assumptions, External Events, and Notes:

This function cannot be called from an Application's Main Task.

Note

This function does not return a value, but if it does return at all, it is assumed that the Task was either unregistered or this function was called from a cFE Application's main task.

See also

[CFE_ES_CreateChildTask](#), [CFE_ES_DeleteChildTask](#)

CFE_ES_GetTaskIDByName()

```
CFE_Status_t CFE_ES_GetTaskIDByName (
    CFE_ES_TaskId_t * TaskIdPtr,
    const char * TaskName )
```

Get a Task ID associated with a specified Task name.

Description

This routine retrieves the cFE Task ID associated with a specified Task name.

Assumptions, External Events, and Notes:

None

Parameters

out	<i>TaskIdPtr</i>	Pointer to variable that is to receive the Task's ID (must not be null).
in	<i>TaskName</i>	Pointer to null terminated character string containing a Task name (must not be null).

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_ERR_NAME_NOT_FOUND	Resource Name Error.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also

[CFE_ES_GetTaskName](#)

CFE_ES_GetTaskName()

```
CFE_Status_t CFE_ES_GetTaskName (
    char * TaskName,
    CFE_ES_TaskId_t TaskId,
    size_t BufferLength )
```

Get a Task name for a specified Task ID.

Description

This routine retrieves the cFE Task name associated with a specified Task ID.

Assumptions, External Events, and Notes:

In the case of a failure ([CFE_ES_ERR_RESOURCEID_NOT_VALID](#)), an empty string is returned.

Parameters

out	<i>TaskName</i>	Pointer to a character array (must not be null) of at least <i>BufferLength</i> in size that will be filled with the Task name.
in	<i>TaskId</i>	Task ID of Task whose name is being requested.
in	<i>BufferLength</i>	The maximum number of characters, including the null terminator, that can be put into the <i>TaskName</i> buffer. This routine will truncate the name to this length, if necessary.

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also

[CFE_ES_GetTaskIDByName](#)

9.8 cFE Miscellaneous APIs

Functions

- void [CFE_ES_BackgroundWakeup](#) (void)
Wakes up the CFE background task.
- [CFE_Status_t](#) [CFE_ES_WriteToSysLog](#) (const char *SpecStringPtr,...) [OS_PRINTF](#)(1)
Write a string to the cFE System Log.
- [CFE_Status_t](#) [uint32](#) [CFE_ES_CalculateCRC](#) (const void *DataPtr, size_t DataLength, [uint32](#) InputCRC, [CFE_ES_CrcType_Enum_t](#) TypeCRC)
Calculate a CRC on a block of memory.
- void [CFE_ES_ProcessAsyncEvent](#) (void)
Notification that an asynchronous event was detected by the underlying OS/PSP.

9.8.1 Detailed Description

9.8.2 Function Documentation

CFE_ES_BackgroundWakeup()

```
void CFE_ES_BackgroundWakeup (  
    void )
```

Wakes up the CFE background task.

Description

Normally the ES background task wakes up at a periodic interval. Whenever new background work is added, this can be used to wake the task early, which may reduce the delay between adding the job and the job getting processed.

Assumptions, External Events, and Notes:

Note the amount of work that the background task will perform is pro-rated based on the amount of time elapsed since the last wakeup. Waking the task early will not cause the background task to do more work than it otherwise would - it just reduces the delay before work starts initially.

CFE_ES_CalculateCRC()

```
CFE\_Status\_t uint32 CFE_ES_CalculateCRC (  
    const void * DataPtr,  
    size_t DataLength,  
    uint32 InputCRC,  
    CFE\_ES\_CrcType\_Enum\_t TypeCRC )
```

Calculate a CRC on a block of memory.

Description

This routine calculates a cyclic redundancy check (CRC) on a block of memory. The CRC algorithm used is determined by the last parameter.

Assumptions, External Events, and Notes:

None

Parameters

in	<i>DataPtr</i>	Pointer to the base of the memory block.
in	<i>DataLength</i>	The number of bytes in the memory block.
in	<i>InputCRC</i>	A starting value for use in the CRC calculation. This parameter allows the user to calculate the CRC of non-contiguous blocks as a single value. Nominally, the user should set this value to zero.
in	<i>TypeCRC</i>	One of the following CRC algorithm selections defined in CFE_ES_CrcType_Enum_t

Returns

The result of the CRC calculation on the specified memory block. If the TypeCRC is unimplemented will return 0. If DataPtr is null or DataLength is 0, will return InputCRC

CFE_ES_ProcessAsyncEvent()

```
void CFE_ES_ProcessAsyncEvent (
    void )
```

Notification that an asynchronous event was detected by the underlying OS/PSP.

Description

This hook routine is called from the PSP when an exception or other asynchronous system event occurs

Assumptions, External Events, and Notes:

The PSP must guarantee that this function is only invoked from a context which may use OSAL primitives. In general this means that it shouldn't be *directly* invoked from an ISR/signal context.

CFE_ES_WriteToSysLog()

```
CFE_Status_t CFE_ES_WriteToSysLog (
    const char * SpecStringPtr,
    ... )
```

Write a string to the cFE System Log.

Description

This routine writes a formatted string to the cFE system log. This can be used to record very low-level errors that can't be reported using the Event Services. This function is used in place of printf for flight software. It should be used for significant startup events, critical errors, and conditionally compiled debug software.

Assumptions, External Events, and Notes:

None

Parameters

in	<i>SpecStringPtr</i>	The format string for the log message (must not be null). This is similar to the format string for a printf() call.
----	----------------------	---

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_ERR_SYS_LOG_FULL	System Log Full.
CFE_ES_BAD_ARGUMENT	Bad Argument.

9.9 cFE Critical Data Store APIs

Functions

- [CFE_Status_t CFE_ES_RegisterCDS](#) ([CFE_ES_CDSHandle_t](#) *CDSHandlePtr, [size_t](#) BlockSize, [const char](#) *Name)
Reserve space (or re-obtain previously reserved space) in the Critical Data Store (CDS)
- [CFE_Status_t CFE_ES_GetCDSBlockIDByName](#) ([CFE_ES_CDSHandle_t](#) *BlockIdPtr, [const char](#) *BlockName)
Get a CDS Block ID associated with a specified CDS Block name.
- [CFE_Status_t CFE_ES_GetCDSBlockName](#) ([char](#) *BlockName, [CFE_ES_CDSHandle_t](#) BlockId, [size_t](#) BufferLength)
Get a Block name for a specified Block ID.
- [CFE_Status_t CFE_ES_CopyToCDS](#) ([CFE_ES_CDSHandle_t](#) Handle, [const void](#) *DataToCopy)
Save a block of data in the Critical Data Store (CDS)
- [CFE_Status_t CFE_ES_RestoreFromCDS](#) ([void](#) *RestoreToMemory, [CFE_ES_CDSHandle_t](#) Handle)
Recover a block of data from the Critical Data Store (CDS)

9.9.1 Detailed Description

9.9.2 Function Documentation

CFE_ES_CopyToCDS()

```
CFE_Status_t CFE_ES_CopyToCDS (
    CFE_ES_CDSHandle_t Handle,
    const void * DataToCopy )
```

Save a block of data in the Critical Data Store (CDS)

Description

This routine copies a specified block of memory into the Critical Data Store that had been previously registered via [CFE_ES_RegisterCDS](#). The block of memory to be copied must be at least as big as the size specified when registering the CDS.

Assumptions, External Events, and Notes:

None

Parameters

in	<i>Handle</i>	The handle of the CDS block that was previously obtained from CFE_ES_RegisterCDS .
in	<i>DataToCopy</i>	A Pointer to the block of memory to be copied into the CDS (must not be null).

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also

[CFE_ES_RegisterCDS](#), [CFE_ES_RestoreFromCDS](#)

CFE_ES_GetCDSBlockIDByName()

```
CFE_Status_t CFE_ES_GetCDSBlockIDByName (
    CFE_ES_CDSHandle_t * BlockIdPtr,
    const char * BlockName )
```

Get a CDS Block ID associated with a specified CDS Block name.

Description

This routine retrieves the CDS Block ID associated with a specified CDS Block name.

Assumptions, External Events, and Notes:

None

Parameters

out	<i>BlockIdPtr</i>	Pointer to variable that is to receive the CDS Block ID (must not be null).
in	<i>BlockName</i>	Pointer to null terminated character string containing a CDS Block name (must not be null).

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_ERR_NAME_NOT_FOUND	Resource Name Error.
CFE_ES_BAD_ARGUMENT	Bad Argument.
CFE_ES_NOT_IMPLEMENTED	The processor does not support a Critical Data Store.

See also

[CFE_ES_GetCDSBlockName](#)

CFE_ES_GetCDSBlockName()

```
CFE_Status_t CFE_ES_GetCDSBlockName (
    char * BlockName,
    CFE_ES_CDSHandle_t BlockId,
    size_t BufferLength )
```

Get a Block name for a specified Block ID.

Description

This routine retrieves the cFE Block name associated with a specified Block ID.

Assumptions, External Events, and Notes:

In the case of a failure ([CFE_ES_ERR_RESOURCEID_NOT_VALID](#)), an empty string is returned.

Parameters

out	<i>BlockName</i>	Pointer to a character array (must not be null) of at least <i>BufferLength</i> in size that will be filled with the CDS Block name.
in	<i>BlockId</i>	Block ID/Handle of CDS registry entry whose name is being requested.
in	<i>BufferLength</i>	The maximum number of characters, including the null terminator, that can be put into the <i>BlockName</i> buffer. This routine will truncate the name to this length, if necessary.

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_ES_BAD_ARGUMENT	Bad Argument.
CFE_ES_NOT_IMPLEMENTED	The processor does not support a Critical Data Store.

See also

[CFE_ES_GetCDSBlockIDByName](#)

CFE_ES_RegisterCDS()

```
CFE_Status_t CFE_ES_RegisterCDS (
    CFE_ES_CDSHandle_t * CDSHandlePtr,
    size_t BlockSize,
    const char * Name )
```

Reserve space (or re-obtain previously reserved space) in the Critical Data Store (CDS)

Description

This routine allocates a block of memory in the Critical Data Store and associates it with the calling Application. The memory can survive an Application restart as well as a Processor Reset.

Assumptions, External Events, and Notes:

This function does *not* clear or otherwise initialize/modify the data within the CDS block. If this function returns [CFE_ES_CDS_ALREADY_EXISTS](#) the block may already have valid data in it.

If a new CDS block is reserved (either because the name did not exist, or existed as a different size) it is the responsibility of the calling application to fill the CDS block with valid data. This is indicated by a [CFE_SUCCESS](#) return code, and in this case the calling application should ensure that it also calls [CFE_ES_CopyToCDS\(\)](#) to fill the block with valid data.

Parameters

out	<i>CDSHandlePtr</i>	Pointer Application's variable that will contain the CDS Memory Block Handle (must not be null). HandlePtr is the handle of the CDS block that can be used in CFE_ES_CopyToCDS and CFE_ES_RestoreFromCDS .
in	<i>BlockSize</i>	The number of bytes needed in the CDS (must not be zero).
in	<i>Name</i>	A pointer to a character string (must not be null) containing an application unique name of CFE_MISSION_ES_CDS_MAX_NAME_LENGTH characters or less.

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	The memory block was successfully created in the CDS.
CFE_ES_NOT_IMPLEMENTED	The processor does not support a Critical Data Store.
CFE_ES_CDS_ALREADY_EXISTS	CDS Already Exists.
CFE_ES_CDS_INVALID_SIZE	CDS Invalid Size.
CFE_ES_CDS_INVALID_NAME	CDS Invalid Name.
CFE_ES_BAD_ARGUMENT	Bad Argument.
CFE_ES_CDS_INVALID	(return value only verified in coverage test) CDS Invalid.

See also

[CFE_ES_CopyToCDS](#), [CFE_ES_RestoreFromCDS](#)

CFE_ES_RestoreFromCDS()

```
CFE_Status_t CFE_ES_RestoreFromCDS (
    void * RestoreToMemory,
    CFE_ES_CDSHandle_t Handle )
```

Recover a block of data from the Critical Data Store (CDS)

Description

This routine copies data from the Critical Data Store identified with the `Handle` into the area of memory pointed to by the `RestoreToMemory` pointer. The area of memory to be copied into must be at least as big as the size specified when registering the CDS. The recovery will indicate an error if the data integrity check maintained by the CDS indicates the contents of the CDS have changed. However, the contents will still be copied into the specified area of memory.

Assumptions, External Events, and Notes:

None

Parameters

in	<i>Handle</i>	The handle of the CDS block that was previously obtained from CFE_ES_RegisterCDS .
out	<i>RestoreToMemory</i>	A Pointer to the block of memory (must not be null) that is to be restored with the contents of the CDS. *RestoreToMemory is the contents of the specified CDS.

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_ES_CDS_BLOCK_CRC_ERR	(return value only verified in coverage test) CDS Block CRC Error.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also

[CFE_ES_RegisterCDS](#), [CFE_ES_CopyToCDS](#)

9.10 cFE Memory Manager APIs

Functions

- [CFE_Status_t CFE_ES_PoolCreateNoSem](#) ([CFE_ES_MemHandle_t](#) *PoolID, void *MemPtr, size_t Size)
Initializes a memory pool created by an application without using a semaphore during processing.
- [CFE_Status_t CFE_ES_PoolCreate](#) ([CFE_ES_MemHandle_t](#) *PoolID, void *MemPtr, size_t Size)
Initializes a memory pool created by an application while using a semaphore during processing.
- [CFE_Status_t CFE_ES_PoolCreateEx](#) ([CFE_ES_MemHandle_t](#) *PoolID, void *MemPtr, size_t Size, [uint16](#) NumBlockSizes, const size_t *BlockSizes, bool UseMutex)
Initializes a memory pool created by an application with application specified block sizes.
- [CFE_Status_t CFE_ES_PoolCreateEx_WithAlignment](#) ([CFE_ES_MemHandle_t](#) *PoolID, void *MemPtr, size_t Size, [uint16](#) NumBlockSizes, const size_t *BlockSizes, bool UseMutex, size_t Alignment)
Implements CFE_ES_PoolCreateEx with added param for alignment added for coverage purposes)
- [int32 CFE_ES_PoolDelete](#) ([CFE_ES_MemHandle_t](#) PoolID)
Deletes a memory pool that was previously created.
- [int32 CFE_ES_GetPoolBuf](#) ([CFE_ES_MemPoolBuf_t](#) *BufPtr, [CFE_ES_MemHandle_t](#) Handle, size_t Size)

Gets a buffer from the memory pool created by [CFE_ES_PoolCreate](#) or [CFE_ES_PoolCreateNoSem](#).

- [CFE_Status_t](#) [CFE_ES_GetPoolBufInfo](#) ([CFE_ES_MemHandle_t](#) Handle, [CFE_ES_MemPoolBuf_t](#) BufPtr)

Gets info on a buffer previously allocated via [CFE_ES_GetPoolBuf](#).

- [int32](#) [CFE_ES_PutPoolBuf](#) ([CFE_ES_MemHandle_t](#) Handle, [CFE_ES_MemPoolBuf_t](#) BufPtr)

Releases a buffer from the memory pool that was previously allocated via [CFE_ES_GetPoolBuf](#).

- [CFE_Status_t](#) [CFE_ES_GetMemPoolStats](#) ([CFE_ES_MemPoolStats_t](#) *BufPtr, [CFE_ES_MemHandle_t](#) Handle)

Extracts the statistics maintained by the memory pool software.

9.10.1 Detailed Description

9.10.2 Function Documentation

CFE_ES_GetMemPoolStats()

```
CFE_Status_t CFE_ES_GetMemPoolStats (
    CFE_ES_MemPoolStats_t * BufPtr,
    CFE_ES_MemHandle_t Handle )
```

Extracts the statistics maintained by the memory pool software.

Description

This routine fills the [CFE_ES_MemPoolStats_t](#) data structure with the statistics maintained by the memory pool software. These statistics can then be telemetered by the calling Application.

Assumptions, External Events, and Notes:

None

Parameters

out	<i>BufPtr</i>	Pointer to CFE_ES_MemPoolStats_t data structure (must not be null) to be filled with memory statistics. *BufPtr is the Memory Pool Statistics stored in given data structure.
in	<i>Handle</i>	The handle to the memory pool whose statistics are desired.

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also

[CFE_ES_PoolCreate](#), [CFE_ES_PoolCreateNoSem](#), [CFE_ES_PoolCreateEx](#), [CFE_ES_GetPoolBuf](#), [CFE_ES_PutPoolBuf](#)

CFE_ES_GetPoolBuf()

```
int32 CFE_ES_GetPoolBuf (
    CFE_ES_MemPoolBuf_t * BufPtr,
```

```
CFE_ES_MemHandle_t Handle,
size_t Size )
```

Gets a buffer from the memory pool created by [CFE_ES_PoolCreate](#) or [CFE_ES_PoolCreateNoSem](#).

Description

This routine obtains a block of memory from the memory pool supplied by the calling application.

Assumptions, External Events, and Notes:

1. The size allocated from the memory pool is, at a minimum, 12 bytes more than requested.

Parameters

out	<i>BufPtr</i>	A pointer to the Application's pointer (must not be null) in which will be stored the address of the allocated memory buffer. *BufPtr is the address of the requested buffer.
in	<i>Handle</i>	The handle to the memory pool as returned by CFE_ES_PoolCreate or CFE_ES_PoolCreateNoSem .
in	<i>Size</i>	The size of the buffer requested. NOTE: The size allocated may be larger.

Returns

Bytes Allocated, or error code [cFE Return Code Defines](#)

Return values

CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_ES_ERR_MEM_BLOCK_SIZE	Memory Block Size Error.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also

[CFE_ES_PoolCreate](#), [CFE_ES_PoolCreateNoSem](#), [CFE_ES_PoolCreateEx](#), [CFE_ES_PutPoolBuf](#), [CFE_ES_GetMemPoolStats](#), [CFE_ES_GetPoolBufInfo](#)

CFE_ES_GetPoolBufInfo()

```
CFE_Status_t CFE_ES_GetPoolBufInfo (
    CFE_ES_MemHandle_t Handle,
    CFE_ES_MemPoolBuf_t BufPtr )
```

Gets info on a buffer previously allocated via [CFE_ES_GetPoolBuf](#).

Description

This routine gets info on a buffer in the memory pool.

Assumptions, External Events, and Notes:

None

Parameters

in	<i>Handle</i>	The handle to the memory pool as returned by CFE_ES_PoolCreate or CFE_ES_PoolCreateNoSem .
in	<i>BufPtr</i>	A pointer to the memory buffer to provide status for (must not be null).

Returns

Size of the buffer if successful, or status code if not successful, see [cFE Return Code Defines](#)

Return values

CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_ES_BUFFER_NOT_IN_POOL	Buffer Not In Pool.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also

[CFE_ES_PoolCreate](#), [CFE_ES_PoolCreateNoSem](#), [CFE_ES_PoolCreateEx](#), [CFE_ES_GetPoolBuf](#), [CFE_ES_GetMemPoolStats](#), [CFE_ES_PutPoolBuf](#)

CFE_ES_PoolCreate()

```
CFE_Status_t CFE_ES_PoolCreate (
    CFE_ES_MemHandle_t * PoolID,
    void * MemPtr,
    size_t Size )
```

Initializes a memory pool created by an application while using a semaphore during processing.

Description

This routine initializes a pool of memory supplied by the calling application. When a memory pool created by this routine is processed, mutex handling will be performed.

Assumptions, External Events, and Notes:

1. The size of the pool must be an integral number of 32-bit words
2. The start address of the pool must be 32-bit aligned
3. 168 bytes are used for internal bookkeeping, therefore, they will not be available for allocation.

Parameters

out	<i>PoolID</i>	A pointer to the variable the caller wishes to have the memory pool handle kept in (must not be null). PoolID is the memory pool handle.
in	<i>MemPtr</i>	A Pointer to the pool of memory created by the calling application (must not be null). This address must be aligned suitably for the processor architecture. The CFE_ES_STATIC_POOL_TYPE macro may be used to assist in creating properly aligned memory pools.
in	<i>Size</i>	The size of the pool of memory (must not be zero). Note that this must be an integral multiple of the memory alignment of the processor architecture.

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also

[CFE_ES_PoolCreateNoSem](#), [CFE_ES_PoolCreateEx](#), [CFE_ES_GetPoolBuf](#), [CFE_ES_PutPoolBuf](#), [CFE_ES_GetMemPoolStats](#)

CFE_ES_PoolCreateEx()

```
CFE_Status_t CFE_ES_PoolCreateEx (
    CFE_ES_MemHandle_t * PoolID,
    void * MemPtr,
    size_t Size,
    uint16 NumBlockSizes,
    const size_t * BlockSizes,
    bool UseMutex )
```

Initializes a memory pool created by an application with application specified block sizes.

Description

This routine initializes a pool of memory supplied by the calling application.

Assumptions, External Events, and Notes:

1. The size of the pool must be an integral number of 32-bit words
2. The start address of the pool must be 32-bit aligned
3. 168 bytes are used for internal bookkeeping, therefore, they will not be available for allocation.

Parameters

out	<i>PoolID</i>	A pointer to the variable the caller wishes to have the memory pool handle kept in (must not be null). PoolID is the memory pool handle.
in	<i>MemPtr</i>	A Pointer to the pool of memory created by the calling application (must not be null). This address must be aligned suitably for the processor architecture. The CFE_ES_STATIC_POOL_TYPE macro may be used to assist in creating properly aligned memory pools.
in	<i>Size</i>	The size of the pool of memory (must not be zero). Note that this must be an integral multiple of the memory alignment of the processor architecture.
in	<i>NumBlockSizes</i>	The number of different block sizes specified in the <code>BlockSizes</code> array. If set larger than CFE_PLATFORM_ES_POOL_MAX_BUCKETS , CFE_ES_BAD_ARGUMENT will be returned. If <code>BlockSizes</code> is null and <code>NumBlockSizes</code> is 0, <code>NubBlockSizes</code> will be set to CFE_PLATFORM_ES_POOL_MAX_BUCKETS .
in	<i>BlockSizes</i>	Pointer to an array of sizes to be used instead of the default block sizes specified by CFE_PLATFORM_ES_MEM_BLOCK_SIZE_01 through CFE_PLATFORM_ES_MAX_BLOCK_SIZE . If the pointer is equal to NULL, the default block sizes are used.
in	<i>UseMutex</i>	Flag indicating whether the new memory pool will be processing with mutex handling or not. Valid parameter values are CFE_ES_USE_MUTEX and CFE_ES_NO_MUTEX .

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_BAD_ARGUMENT	Bad Argument.
CFE_ES_NO_RESOURCE_IDS_AVAILABLE	Resource ID is not available.
CFE_STATUS_EXTERNAL_RESOURCE_FAIL	(return value only verified in coverage test) External failure.

See also

[CFE_ES_PoolCreate](#), [CFE_ES_PoolCreateNoSem](#), [CFE_ES_GetPoolBuf](#), [CFE_ES_PutPoolBuf](#), [CFE_ES_GetMemPoolStats](#)

CFE_ES_PoolCreateEx_WithAlignment()

```
CFE_Status_t CFE_ES_PoolCreateEx_WithAlignment (
    CFE_ES_MemHandle_t * PoolID,
    void * MemPtr,
    size_t Size,
    uint16 NumBlockSizes,
    const size_t * BlockSizes,
    bool UseMutex,
    size_t Alignment )
```

Implements CFE_ES_PoolCreateEx with added param for alignment added for coverage purposes)

Description

This routine initializes a pool of memory supplied by the calling application.

Assumptions, External Events, and Notes:

1. The size of the pool must be an integral number of 32-bit words
2. The start address of the pool must be 32-bit aligned
3. 168 bytes are used for internal bookkeeping, therefore, they will not be available for allocation.

Parameters

out	<i>PoolID</i>	A pointer to the variable the caller wishes to have the memory pool handle kept in (must not be null). PoolID is the memory pool handle.
in	<i>MemPtr</i>	A Pointer to the pool of memory created by the calling application (must not be null). This address must be aligned suitably for the processor architecture. The CFE_ES_STATIC_POOL_TYPE macro may be used to assist in creating properly aligned memory pools.
in	<i>Size</i>	The size of the pool of memory (must not be zero). Note that this must be an integral multiple of the memory alignment of the processor architecture.
in	<i>NumBlockSizes</i>	The number of different block sizes specified in the <code>BlockSizes</code> array. If set larger than CFE_PLATFORM_ES_POOL_MAX_BUCKETS , CFE_ES_BAD_ARGUMENT will be returned. If <code>BlockSizes</code> is null and <code>NumBlockSizes</code> is 0, <code>NumBlockSizes</code> will be set to CFE_PLATFORM_ES_POOL_MAX_BUCKETS .

Parameters

in	<i>BlockSizes</i>	Pointer to an array of sizes to be used instead of the default block sizes specified by CFE_PLATFORM_ES_MEM_BLOCK_SIZE_01 through CFE_PLATFORM_ES_MAX_BLOCK_SIZE . If the pointer is equal to NULL, the default block sizes are used.
in	<i>UseMutex</i>	Flag indicating whether the new memory pool will be processing with mutex handling or not. Valid parameter values are CFE_ES_USE_MUTEX and CFE_ES_NO_MUTEX
out	<i>Alignment</i>	Size of struct alignment

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_BAD_ARGUMENT	Bad Argument.
CFE_ES_NO_RESOURCE_IDS_AVAILABLE	Resource ID is not available.
CFE_STATUS_EXTERNAL_RESOURCE_FAIL	(return value only verified in coverage test) External failure.

See also

[CFE_ES_PoolCreate](#), [CFE_ES_PoolCreateNoSem](#), [CFE_ES_GetPoolBuf](#), [CFE_ES_PutPoolBuf](#), [CFE_ES_GetMemPoolStats](#)

CFE_ES_PoolCreateNoSem()

```
CFE_Status_t CFE_ES_PoolCreateNoSem (
    CFE_ES_MemHandle_t * PoolID,
    void * MemPtr,
    size_t Size )
```

Initializes a memory pool created by an application without using a semaphore during processing.

Description

This routine initializes a pool of memory supplied by the calling application. When a memory pool created by this routine is processed, no mutex handling is performed.

Assumptions, External Events, and Notes:

1. The size of the pool must be an integral number of 32-bit words
2. The start address of the pool must be 32-bit aligned
3. 168 bytes are used for internal bookkeeping, therefore, they will not be available for allocation.

Parameters

out	<i>PoolID</i>	A pointer to the variable the caller wishes to have the memory pool handle kept in (must not be null). PoolID is the memory pool handle.
in	<i>MemPtr</i>	A Pointer to the pool of memory created by the calling application (must not be null). This address must be aligned suitably for the processor architecture. The CFE_ES_STATIC_POOL_TYPE macro may be used to assist in creating properly aligned memory pools.

Parameters

in	Size	The size of the pool of memory (must not be zero). Note that this must be an integral multiple of the memory alignment of the processor architecture.
----	------	---

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also

[CFE_ES_PoolCreate](#), [CFE_ES_PoolCreateEx](#), [CFE_ES_GetPoolBuf](#), [CFE_ES_PutPoolBuf](#), [CFE_ES_GetMemPoolStats](#)

CFE_ES_PoolDelete()

```
int32 CFE_ES_PoolDelete (
    CFE_ES_MemHandle_t PoolID )
```

Deletes a memory pool that was previously created.

Description

This routine removes the pool ID and frees the global table entry for future re-use.

Assumptions, External Events, and Notes:

All buffers associated with the pool become invalid after this call. The application should ensure that buffers/references to the pool are returned before deleting the pool.

Parameters

in	PoolID	The ID of the pool to delete
----	--------	------------------------------

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.

See also

[CFE_ES_PoolCreate](#), [CFE_ES_PoolCreateNoSem](#), [CFE_ES_GetPoolBuf](#), [CFE_ES_PutPoolBuf](#), [CFE_ES_GetMemPoolStats](#)

CFE_ES_PutPoolBuf()

```
int32 CFE_ES_PutPoolBuf (
    CFE_ES_MemHandle_t Handle,
    CFE_ES_MemPoolBuf_t BufPtr )
```

Releases a buffer from the memory pool that was previously allocated via [CFE_ES_GetPoolBuf](#).

Description

This routine releases a buffer back into the memory pool.

Assumptions, External Events, and Notes:

None

Parameters

in	<i>Handle</i>	The handle to the memory pool as returned by CFE_ES_PoolCreate or CFE_ES_PoolCreateNoSem .
in	<i>BufPtr</i>	A pointer to the memory buffer to be released (must not be null).

Returns

Bytes released, or error code [cFE Return Code Defines](#)

Return values

CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_ES_BAD_ARGUMENT	Bad Argument.
CFE_ES_BUFFER_NOT_IN_POOL	Buffer Not In Pool.
CFE_ES_POOL_BLOCK_INVALID	Invalid pool block.

See also

[CFE_ES_PoolCreate](#), [CFE_ES_PoolCreateNoSem](#), [CFE_ES_PoolCreateEx](#), [CFE_ES_GetPoolBuf](#), [CFE_ES_GetMemPoolStats](#), [CFE_ES_GetPoolBufInfo](#)

9.11 cFE Performance Monitor APIs**Macros**

- #define [CFE_ES_PerfLogEntry](#)(id) ([CFE_ES_PerfLogAdd](#)(id, 0))
Entry marker for use with Software Performance Analysis Tool.
- #define [CFE_ES_PerfLogExit](#)(id) ([CFE_ES_PerfLogAdd](#)(id, 1))
Exit marker for use with Software Performance Analysis Tool.

Functions

- void [CFE_ES_PerfLogAdd](#) (uint32 Marker, uint32 EntryExit)
Adds a new entry to the data buffer.

9.11.1 Detailed Description

9.11.2 Macro Definition Documentation

CFE_ES_PerfLogEntry

```
#define CFE_ES_PerfLogEntry(  
    id ) (CFE_ES_PerfLogAdd(id, 0))
```

Entry marker for use with Software Performance Analysis Tool.

Description

This macro logs the entry or start event/marker for the specified entry *id*. This macro, in conjunction with the [CFE_ES_PerfLogExit](#), is used by the Software Performance Analysis tool.

Assumptions, External Events, and Notes:

None

Parameters

<i>in</i>	<i>id</i>	Identifier of the specific event or marker.
-----------	-----------	---

See also

[CFE_ES_PerfLogExit](#), [CFE_ES_PerfLogAdd](#)

Definition at line 1520 of file [cfe_es.h](#).

CFE_ES_PerfLogExit

```
#define CFE_ES_PerfLogExit(  
    id ) (CFE_ES_PerfLogAdd(id, 1))
```

Exit marker for use with Software Performance Analysis Tool.

Description

This macro logs the exit or end event/marker for the specified entry *id*. This macro, in conjunction with the [CFE_ES_PerfLogEntry](#), is used by the Software Performance Analysis tool.

Assumptions, External Events, and Notes:

None

Parameters

<i>in</i>	<i>id</i>	Identifier of the specific event or marker.
-----------	-----------	---

See also

[CFE_ES_PerfLogEntry](#), [CFE_ES_PerfLogAdd](#)

Definition at line 1539 of file [cfe_es.h](#).

9.11.3 Function Documentation

CFE_ES_PerfLogAdd()

```
void CFE_ES_PerfLogAdd (
    uint32 Marker,
    uint32 EntryExit )
```

Adds a new entry to the data buffer.

Function called by [CFE_ES_PerfLogEntry](#) and [CFE_ES_PerfLogExit](#) macros

Description

This function logs the entry and exit marker for the specified `id`. This function is used by the Software Performance Analysis tool.

Assumptions, External Events, and Notes:

`Marker` limited to the range of 0 to [CFE_MISSION_ES_PERF_MAX_IDS](#) - 1. Any performance ids outside of this range will be ignored and will be flagged as an error.

This function implements a circular buffer using an array. `DataStart` points to first stored entry `DataEnd` points to next available entry if `DataStart == DataEnd` then the buffer is either empty or full depending on the value of the `DataCount`. Time is stored as 2 32 bit integers, (`TimerLower32`, `TimerUpper32`): `TimerLower32` is the current value of the hardware timer register. `TimerUpper32` is the number of times the timer has rolled over.

Parameters

in	<i>Marker</i>	Identifier of the specific event or marker.
in	<i>EntryExit</i>	Used to specify Entry(0) or Exit(1)

See also

[CFE_ES_PerfLogEntry](#), [CFE_ES_PerfLogExit](#)

9.12 cFE Generic Counter APIs

Functions

- [CFE_Status_t CFE_ES_RegisterGenCounter](#) ([CFE_ES_CounterId_t](#) *CounterIdPtr, const char *CounterName)
Register a generic counter.
- [CFE_Status_t CFE_ES_DeleteGenCounter](#) ([CFE_ES_CounterId_t](#) CounterId)
Delete a generic counter.
- [CFE_Status_t CFE_ES_IncrementGenCounter](#) ([CFE_ES_CounterId_t](#) CounterId)
Increments the specified generic counter.
- [CFE_Status_t CFE_ES_SetGenCount](#) ([CFE_ES_CounterId_t](#) CounterId, [uint32](#) Count)
Set the specified generic counter.
- [CFE_Status_t CFE_ES_GetGenCount](#) ([CFE_ES_CounterId_t](#) CounterId, [uint32](#) *Count)
Get the specified generic counter count.

- [CFE_Status_t CFE_ES_GetGenCounterIDByName](#) ([CFE_ES_CounterId_t](#) *CounterIdPtr, const char *CounterName)

Get the Id associated with a generic counter name.

- [CFE_Status_t CFE_ES_GetGenCounterName](#) (char *CounterName, [CFE_ES_CounterId_t](#) CounterId, size_t BufferLength)

Get a Counter name for a specified Counter ID.

9.12.1 Detailed Description

9.12.2 Function Documentation

CFE_ES_DeleteGenCounter()

```
CFE_Status_t CFE_ES_DeleteGenCounter (
    CFE\_ES\_CounterId\_t CounterId )
```

Delete a generic counter.

Description

This routine deletes a previously registered generic counter.

Assumptions, External Events, and Notes:

None.

Parameters

in	<i>CounterId</i>	The Counter Id of the newly created counter.
----	------------------	--

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also

[CFE_ES_IncrementGenCounter](#), [CFE_ES_RegisterGenCounter](#), [CFE_ES_SetGenCount](#), [CFE_ES_GetGenCount](#), [CFE_ES_GetGenCounterIDByName](#)

CFE_ES_GetGenCount()

```
CFE_Status_t CFE_ES_GetGenCount (
    CFE\_ES\_CounterId\_t CounterId,
    uint32 * Count )
```

Get the specified generic counter count.

Description

This routine gets the value of a generic counter.

Assumptions, External Events, and Notes:

None.

Parameters

in	<i>CounterId</i>	The Counter to get the value from.
out	<i>Count</i>	Buffer to store value of the Counter (must not be null).

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also

[CFE_ES_RegisterGenCounter](#), [CFE_ES_DeleteGenCounter](#), [CFE_ES_SetGenCount](#), [CFE_ES_IncrementGenCounter](#),
[CFE_ES_GetGenCounterIDByName](#)

CFE_ES_GetGenCounterIDByName()

```
CFE_Status_t CFE_ES_GetGenCounterIDByName (
    CFE_ES_CounterId_t * CounterIdPtr,
    const char * CounterName )
```

Get the Id associated with a generic counter name.

Description

This routine gets the Counter Id for a generic counter specified by name.

Assumptions, External Events, and Notes:

None.

Parameters

out	<i>CounterIdPtr</i>	Pointer to variable that is to receive the Counter's ID (must not be null).
in	<i>CounterName</i>	Pointer to null terminated character string containing a Counter name (must not be null).

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_ERR_NAME_NOT_FOUND	Resource Name Error.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also

[CFE_ES_GetGenCounterName](#)

CFE_ES_GetGenCounterName()

```
CFE_Status_t CFE_ES_GetGenCounterName (
    char * CounterName,
    CFE_ES_CounterId_t CounterId,
    size_t BufferLength )
```

Get a Counter name for a specified Counter ID.

Description

This routine retrieves the cFE Counter name associated with a specified Counter ID.

Assumptions, External Events, and Notes:

In the case of a failure ([CFE_ES_ERR_RESOURCEID_NOT_VALID](#)), an empty string is returned.

Parameters

out	<i>CounterName</i>	Pointer to a character array (must not be null) of at least <code>BufferLength</code> in size that will be filled with the Counter name.
in	<i>CounterId</i>	ID of Counter whose name is being requested.
in	<i>BufferLength</i>	The maximum number of characters, including the null terminator (must not be zero), that can be put into the <code>CounterName</code> buffer. This routine will truncate the name to this length, if necessary.

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also

[CFE_ES_GetGenCounterIDByName](#)

CFE_ES_IncrementGenCounter()

```
CFE_Status_t CFE_ES_IncrementGenCounter (
    CFE_ES_CounterId_t CounterId )
```

Increments the specified generic counter.

Description

This routine increments the specified generic counter.

Assumptions, External Events, and Notes:

None.

Parameters

in	<i>CounterId</i>	The Counter to be incremented.
----	------------------	--------------------------------

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also

[CFE_ES_RegisterGenCounter](#), [CFE_ES_DeleteGenCounter](#), [CFE_ES_SetGenCount](#), [CFE_ES_GetGenCount](#),
[CFE_ES_GetGenCounterIDByName](#)

CFE_ES_RegisterGenCounter()

```
CFE_Status_t CFE_ES_RegisterGenCounter (
    CFE_ES_CounterId_t * CounterIdPtr,
    const char * CounterName )
```

Register a generic counter.

Description

This routine registers a generic thread-safe counter which can be used for inter-task management.

Assumptions, External Events, and Notes:

The initial value of all newly registered counters is 0.

Parameters

out	<i>CounterIdPtr</i>	Buffer to store the Counter Id of the newly created counter (must not be null).
in	<i>CounterName</i>	The Name of the generic counter (must not be null).

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_BAD_ARGUMENT	Bad Argument.
CFE_ES_ERR_DUPLICATE_NAME	Duplicate Name Error.
CFE_ES_NO_RESOURCE_IDS_AVAILABLE	Resource ID is not available.

See also

[CFE_ES_IncrementGenCounter](#), [CFE_ES_DeleteGenCounter](#), [CFE_ES_SetGenCount](#), [CFE_ES_GetGenCount](#), [CFE_ES_GetGenCounterIDByName](#)

CFE_ES_SetGenCount()

```
CFE_Status_t CFE_ES_SetGenCount (
    CFE_ES_CounterId_t CounterId,
    uint32 Count )
```

Set the specified generic counter.

Description

This routine sets the specified generic counter to the specified value.

Assumptions, External Events, and Notes:

None.

Parameters

in	<i>CounterId</i>	The Counter to be set.
in	<i>Count</i>	The new value of the Counter.

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_BAD_ARGUMENT	Bad Argument.

See also

[CFE_ES_RegisterGenCounter](#), [CFE_ES_DeleteGenCounter](#), [CFE_ES_IncrementGenCounter](#), [CFE_ES_GetGenCount](#), [CFE_ES_GetGenCounterIDByName](#)

9.13 cFE Registration APIs

Functions

- [CFE_Status_t CFE_EVS_Register](#) (const void *Filters, uint16 NumEventFilters, uint16 FilterScheme)
Register an application for receiving event services.

9.13.1 Detailed Description

9.13.2 Function Documentation

CFE_EVS_Register()

```
CFE_Status_t CFE_EVS_Register (
    const void * Filters,
    uint16 NumEventFilters,
    uint16 FilterScheme )
```

Register an application for receiving event services.

Description

This routine registers an application with event services and allocates/initializes the internal data structures used to support this application's events. An application may not send events unless it has called this routine. The routine also accepts a filter array structure for applications requiring event filtering. In the current implementation of the EVS, only the binary filtering scheme is supported. See section TBD of the cFE Application Programmer's Guide for a description of the behavior of binary filters. Applications may call [CFE_EVS_Register](#) more than once, but each call will wipe out all filters registered by previous calls (filter registration is NOT cumulative).

Assumptions, External Events, and Notes:

Note: Event filters can be added, deleted or modified by ground commands. All filtering schemes include a default setting that results in no filtering (such as [CFE_EVS_NO_FILTER](#) for binary filters).

Filter Scheme: Binary

Code: CFE_EVS_EventFilter_BINARY

Filter Structure:

```
typedef struct CFE_EVS_BinFilter {
    uint16 EventID,
    uint16 Mask ;
} CFE_EVS_BinFilter_t;
```

Parameters

in	<i>Filters</i>	Pointer to an array of event message filters, or NULL if no filtering is desired. The structure of an event message filter depends on the FilterScheme selected. (see Filter Schemes mentioned above)
in	<i>NumEventFilters</i>	The number of event message filters included in this call. This must be less than or equal to the maximum number of events allowed per application (CFE_PLATFORM_EVS_MAX_EVENT_FILTERS).
in	<i>FilterScheme</i>	The event filtering scheme that this application will use. For the first implementation of the event services, only filter type CFE_EVS_EventFilter_BINARY will be supported.

Returns

Execution status below or from [CFE_ES_GetAppID](#), see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_EVS_APP_FILTER_OVERLOAD	Application Filter Overload.
CFE_EVS_UNKNOWN_FILTER	Unknown Filter.
CFE_EVS_APP_ILLEGAL_APP_ID	Illegal Application ID.
CFE_ES_BAD_ARGUMENT	Bad Argument.

9.14 cFE Send Event APIs

Functions

- [CFE_Status_t](#) [CFE_EVS_SendEvent](#) ([uint16](#) EventID, [CFE_EVS_EventType_Enum_t](#) EventType, const char *Spec,...) [OS_PRINTF](#)(3)
Generate a software event.
- [CFE_Status_t](#) [CFE_Status_t](#) [CFE_EVS_SendEventWithAppID](#) ([uint16](#) EventID, [CFE_EVS_EventType_Enum_t](#) EventType, [CFE_ES_AppID_t](#) AppID, const char *Spec,...) [OS_PRINTF](#)(4)
Generate a software event given the specified Application ID.
- [CFE_Status_t](#) [CFE_Status_t](#) [CFE_Status_t](#) [CFE_EVS_SendTimedEvent](#) ([CFE_TIME_SysTime_t](#) Time, [uint16](#) EventID, [CFE_EVS_EventType_Enum_t](#) EventType, const char *Spec,...) [OS_PRINTF](#)(4)
Generate a software event with a specific time tag.

9.14.1 Detailed Description

9.14.2 Function Documentation

CFE_EVS_SendEvent()

```
CFE_Status_t CFE_EVS_SendEvent (
    uint16 EventID,
    CFE_EVS_EventType_Enum_t EventType,
    const char * Spec,
    ... )
```

Generate a software event.

Description

This routine generates a software event message. If the EventID is not filtered, the event will be sent as a software bus message, optionally logged in the local event log, and optionally sent as an ASCII text string out the enabled output port(s).

Assumptions, External Events, and Notes:

This API only works within the context of a registered application or core service. For messages outside the context of a registered application (for example early in app initialization or if registration fails) [CFE_ES_WriteToSysLog](#) can be used for reporting.

Parameters

in	<i>EventID</i>	A numeric literal used to uniquely identify an application event. The <code>EventID</code> is defined and supplied by the application sending the event.
in	<i>EventType</i>	A numeric literal used to classify an event, one of: • CFE_EVS_EventType_DEBUG • CFE_EVS_EventType_INFORMATION • CFE_EVS_EventType_ERROR • CFE_EVS_EventType_CRITICAL
in	<i>Spec</i>	A pointer to a null terminated text string (must not be null) describing the output format for the event. This is the same type of format string used for the ANSI <code>printf</code> function. Nominally the post-conversion string is limited to 80 characters, but this limit is configurable through the parameter CFE_MISSION_EVS_MAX_MESSAGE_LENGTH . Characters beyond the limit will be truncated. Do not use floating point conversions (f, e, E, g, and G) in the format string unless your application will be running in a system that supports floating point arithmetic. Do not use non-printable characters (\t, \n, etc.) in the format string; they will mess up the formatting when the events are displayed on the ground system.

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_EVS_APP_NOT_REGISTERED	Application Not Registered.
CFE_EVS_APP_ILLEGAL_APP_ID	Illegal Application ID.
CFE_EVS_INVALID_PARAMETER	Invalid Pointer.

See also

[CFE_EVS_SendEventWithAppID](#), [CFE_EVS_SendTimedEvent](#)

CFE_EVS_SendEventWithAppID()

```
CFE_Status_t CFE_Status_t CFE_EVS_SendEventWithAppID (
    uint16 EventID,
    CFE_EVS_EventType_Enum_t EventType,
    CFE_ES_AppId_t AppID,
    const char * Spec,
    ... )
```

Generate a software event given the specified Application ID.

Description

This routine generates a software event message. If the `EventID` is not filtered, the event will be sent as a software bus message, optionally logged in the local event log, and optionally sent as an ASCII text string out the enabled output port(s). Note that this function should really only be used from within an API in order to preserve the context of an Application's event. In general, [CFE_EVS_SendEvent](#) should be used.

Assumptions, External Events, and Notes:

The Application ID must correspond to a registered application or core service. For messages outside the context of a registered application (for example early in app initialization or if registration fails) [CFE_ES_WriteToSysLog](#) can be used for reporting.

Parameters

in	<i>EventID</i>	A numeric literal used to uniquely identify an application event. The <code>EventID</code> is defined and supplied by the application sending the event.
in	<i>EventType</i>	A numeric literal used to classify an event, one of: • CFE_EVS_EventType_DEBUG • CFE_EVS_EventType_INFORMATION • CFE_EVS_EventType_ERROR • CFE_EVS_EventType_CRITICAL
in	<i>AppID</i>	The Application ID from which the event message should appear.
in	<i>Spec</i>	A pointer to a null terminated text string (must not be null) describing the output format for the event. This is the same type of format string used for the ANSI <code>printf</code> function. Nominally the post-conversion string is limited to 80 characters, but this limit is configurable through the parameter CFE_MISSION_EVS_MAX_MESSAGE_LENGTH . Characters beyond the limit will be truncated. Do not use floating point conversions (f, e, E, g, and G) in the format string unless your application will be running in a system that supports floating point arithmetic. Do not use non-printable characters (\t, \n, etc.) in the format string; they will mess up the formatting when the events are displayed on the ground system.

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_EVS_APP_NOT_REGISTERED	Application Not Registered.
CFE_EVS_APP_ILLEGAL_APP_ID	Illegal Application ID.
CFE_EVS_INVALID_PARAMETER	Invalid Pointer.

See also

[CFE_EVS_SendEvent](#), [CFE_EVS_SendTimedEvent](#)

CFE_EVS_SendTimedEvent()

```
CFE_Status_t CFE_Status_t CFE_Status_t CFE_EVS_SendTimedEvent (
    CFE_TIME_SysTime_t Time,
    uint16 EventID,
    CFE_EVS_EventType_Enum_t EventType,
    const char * Spec,
    ... )
```

Generate a software event with a specific time tag.

Description

This routine is the same as [CFE_EVS_SendEvent](#) except that the caller specifies the event time instead of having the EVS use the current spacecraft time. This routine should be used in situations where an error condition is detected at one time, but the event message is reported at a later time.

Assumptions, External Events, and Notes:

This API only works within the context of a registered application or core service. For messages outside the context of a registered application (for example early in app initialization or if registration fails) [CFE_ES_WriteToSysLog](#) can be used for reporting.

Parameters

in	<i>Time</i>	The time to include in the event. This will usually be a time returned by the function CFE_TIME_GetTime .
in	<i>EventID</i>	A numeric literal used to uniquely identify an application event. The <code>EventID</code> is defined and supplied by the application sending the event.
in	<i>EventType</i>	A numeric literal used to classify an event, one of: • CFE_EVS_EventType_DEBUG • CFE_EVS_EventType_INFORMATION • CFE_EVS_EventType_ERROR • CFE_EVS_EventType_CRITICAL
in	<i>Spec</i>	A pointer to a null terminated text string (must not be null) describing the output format for the event. This is the same type of format string used for the ANSI <code>printf</code> function. Nominally the post-conversion string is limited to 80 characters, but this limit is configurable through the parameter CFE_MISSION_EVS_MAX_MESSAGE_LENGTH . Characters beyond the limit will be truncated. Do not use floating point conversions (f, e, E, g, and G) in the format string unless your application will be running in a system that supports floating point arithmetic. Do not use non-printable characters (\t, \n, etc.) in the format string; they will mess up the formatting when the events are displayed on the ground system.

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_EVS_APP_NOT_REGISTERED	Application Not Registered.
CFE_EVS_APP_ILLEGAL_APP_ID	Illegal Application ID.
CFE_EVS_INVALID_PARAMETER	Invalid Pointer.

See also

[CFE_EVS_SendEvent](#), [CFE_EVS_SendEventWithAppID](#)

9.15 cFE Reset Event Filter APIs

Functions

- [CFE_Status_t CFE_EVS_ResetFilter](#) (uint16 EventID)
Resets the calling application's event filter for a single event ID.
- [CFE_Status_t CFE_EVS_ResetAllFilters](#) (void)
Resets all of the calling application's event filters.

9.15.1 Detailed Description

9.15.2 Function Documentation

CFE_EVS_ResetAllFilters()

```
CFE_Status_t CFE_EVS_ResetAllFilters (
    void )
```

Resets all of the calling application's event filters.

Description

This routine resets all the calling application's event filter counters to zero, providing a quick and convenient method for resetting event filters.

Assumptions, External Events, and Notes:

None

Returns

Execution status below or from [CFE_ES_GetAppID](#), see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_EVS_APP_NOT_REGISTERED	Application Not Registered.
CFE_EVS_APP_ILLEGAL_APP_ID	Illegal Application ID.

See also

[CFE_EVS_ResetFilter](#)

CFE_EVS_ResetFilter()

```
CFE_Status_t CFE_EVS_ResetFilter (
    uint16 EventID )
```

Resets the calling application's event filter for a single event ID.

Description

Resets the filter such that the next event is treated like the first. For example, if the filter was set to only send the first event, the next event following the reset would be sent.

Assumptions, External Events, and Notes:

None

Parameters

in	<i>EventID</i>	A numeric literal used to uniquely identify an application event. The <code>EventID</code> is defined and supplied by the application sending the event.
----	----------------	--

Returns

Execution status below or from [CFE_ES_GetAppID](#), see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_EVS_APP_NOT_REGISTERED	Application Not Registered.
CFE_EVS_APP_ILLEGAL_APP_ID	Illegal Application ID.
CFE_EVS_EVT_NOT_REGISTERED	Event Not Registered.

See also

[CFE_EVS_ResetAllFilters](#)

9.16 cFE File Header Management APIs

Functions

- [CFE_Status_t CFE_FS_ReadHeader](#) ([CFE_FS_Header_t](#) *Hdr, [osal_id_t](#) FileDes)
Read the contents of the Standard cFE File Header.
- void [CFE_FS_InitHeader](#) ([CFE_FS_Header_t](#) *Hdr, const char *Description, [uint32](#) SubType)
Initializes the contents of the Standard cFE File Header.
- [CFE_Status_t CFE_FS_WriteHeader](#) ([osal_id_t](#) FileDes, [CFE_FS_Header_t](#) *Hdr)
Write the specified Standard cFE File Header to the specified file.
- [CFE_Status_t CFE_FS_SetTimestamp](#) ([osal_id_t](#) FileDes, [CFE_TIME_SysTime_t](#) NewTimestamp)
Modifies the Time Stamp field in the Standard cFE File Header for the specified file.

9.16.1 Detailed Description

9.16.2 Function Documentation

CFE_FS_InitHeader()

```
void CFE_FS_InitHeader (
    CFE\_FS\_Header\_t * Hdr,
    const char * Description,
    uint32 SubType )
```

Initializes the contents of the Standard cFE File Header.

Description

This API will clear the specified [CFE_FS_Header_t](#) variable and initialize the description field with the specified value

Parameters

in	<i>Hdr</i>	Pointer to a variable of type CFE_FS_Header_t that will be cleared and initialized
in	<i>Description</i>	Initializes Header's Description (must not be null)
in	<i>SubType</i>	Initializes Header's SubType

See also

[CFE_FS_WriteHeader](#)

CFE_FS_ReadHeader()

```
CFE_Status_t CFE_FS_ReadHeader (
    CFE_FS_Header_t * Hdr,
    osal_id_t FileDes )
```

Read the contents of the Standard cFE File Header.

Description

This API will fill the specified [CFE_FS_Header_t](#) variable with the contents of the Standard cFE File Header of the file identified by the given File Descriptor.

Assumptions, External Events, and Notes:

1. The File has already been successfully opened using [OS_OpenCreate](#) and the caller has a legitimate File Descriptor.
2. File offset behavior: Agnostic on entry since it will move the offset to the start of the file, on success the offset will be at the end of the header, undefined offset behavior for error cases.

Parameters

out	<i>Hdr</i>	Pointer to a variable of type CFE_FS_Header_t (must not be null) that will be filled with the contents of the Standard cFE File Header. *Hdr is the contents of the Standard cFE File Header for the specified file.
in	<i>FileDes</i>	File Descriptor obtained from a previous call to OS_OpenCreate that is associated with the file whose header is to be read.

Returns

Bytes read or error status from OSAL

Return values

CFE_FS_BAD_ARGUMENT	Bad Argument.
-------------------------------------	---------------

Note

This function invokes OSAL API routines and the current implementation may return OSAL error codes to the caller if failure occurs. In a future version of CFE, the status codes will be converted to a value in [cFE Return Code Defines](#).

See also

[CFE_FS_WriteHeader](#)

CFE_FS_SetTimestamp()

```
CFE_Status_t CFE_FS_SetTimestamp (
    osal_id_t FileDes,
    CFE_TIME_SysTime_t NewTimestamp )
```

Modifies the Time Stamp field in the Standard cFE File Header for the specified file.

Description

This API will modify the [timestamp](#) found in the Standard cFE File Header of the specified file. The timestamp will be replaced with the time specified by the caller.

Assumptions, External Events, and Notes:

1. The File has already been successfully opened using [OS_OpenCreate](#) and the caller has a legitimate File Descriptor.
2. The `NewTimestamp` field has been filled appropriately by the Application.
3. File offset behavior: Agnostic on entry since it will move the offset, on success the offset will be at the end of the time stamp, undefined offset behavior for error cases.

Parameters

in	<i>FileDes</i>	File Descriptor obtained from a previous call to OS_OpenCreate that is associated with the file whose header is to be read.
in	<i>NewTimestamp</i>	A CFE_TIME_SysTime_t data structure containing the desired time to be put into the file's Standard cFE File Header.

Returns

Execution status, see [cFE Return Code Defines](#), or OSAL status

Return values

CFE_STATUS_EXTERNAL_RESOURCE_FAIL	(return value only verified in coverage test) External failure.
CFE_SUCCESS	Successful execution.

Note

This function invokes OSAL API routines and the current implementation may return OSAL error codes to the caller if failure occurs. In a future version of CFE, the status codes will be converted to a value in [cFE Return Code Defines](#).

CFE_FS_WriteHeader()

```
CFE_Status_t CFE_FS_WriteHeader (
    osal_id_t FileDes,
    CFE_FS_Header_t * Hdr )
```

Write the specified Standard cFE File Header to the specified file.

Description

This API will output the specified [CFE_FS_Header_t](#) variable, with some fields automatically updated, to the specified file as the Standard cFE File Header. This API will automatically populate the following fields in the specified [CFE_FS_Header_t](#):

1. [ContentType](#) - Filled with 0x63464531 ('cFE1')
2. [Length](#) - Filled with the sizeof([CFE_FS_Header_t](#))
3. [SpacecraftID](#) - Filled with the Spacecraft ID
4. [ProcessorID](#) - Filled with the Processor ID
5. [ApplicationID](#) - Filled with the Application ID
6. [TimeSeconds](#) - Filled with the Time, in seconds, as obtained by [CFE_TIME_GetTime](#)
7. [TimeSubSeconds](#) - Filled with the Time, subseconds, as obtained by [CFE_TIME_GetTime](#)

Assumptions, External Events, and Notes:

1. The File has already been successfully opened using [OS_OpenCreate](#) and the caller has a legitimate File Descriptor.
2. The [SubType](#) field has been filled appropriately by the Application.
3. The [Description](#) field has been filled appropriately by the Application.
4. File offset behavior: Agnostic on entry since it will move the offset to the start of the file, on success the offset will be at the end of the header, undefined offset behavior for error cases.

Parameters

in	<i>FileDes</i>	File Descriptor obtained from a previous call to OS_OpenCreate that is associated with the file whose header is to be read.
out	<i>Hdr</i>	Pointer to a variable of type CFE_FS_Header_t (must not be null) that will be filled with the contents of the Standard cFE File Header. *Hdr is the contents of the Standard cFE File Header for the specified file.

Returns

Bytes read or error status from OSAL

Return values

CFE_FS_BAD_ARGUMENT	Bad Argument.
-------------------------------------	---------------

Note

This function invokes OSAL API routines and the current implementation may return OSAL error codes to the caller if failure occurs. In a future version of CFE, the status codes will be converted to a value in [cFE Return Code Defines](#).

See also

[CFE_FS_ReadHeader](#)

9.17 cFE File Utility APIs**Functions**

- `const char * CFE\_FS\_GetDefaultMountPoint (CFE\_FS\_FileCategory\_t FileCategory)`
Get the default virtual mount point for a file category.
- `const char * CFE\_FS\_GetDefaultExtension (CFE\_FS\_FileCategory\_t FileCategory)`
Get the default filename extension for a file category.
- `int32 CFE\_FS\_ParseInputFileNameEx (char *OutputBuffer, const char *InputBuffer, size_t OutputBufSize, size_t InputBufSize, const char *DefaultInput, const char *DefaultPath, const char *DefaultExtension)`
Parse a filename input from an input buffer into a local buffer.
- `int32 CFE\_FS\_ParseInputFileName (char *OutputBuffer, const char *InputName, size_t OutputBufSize, CFE\_FS\_FileCategory\_t FileCategory)`
Parse a filename string from the user into a local buffer.
- `CFE\_Status\_t CFE\_FS\_ExtractFilenameFromPath (const char *OriginalPath, char *FileNameOnly)`
Extracts the filename from a unix style path and filename string.
- `int32 CFE\_FS\_BackgroundFileDumpRequest (CFE\_FS\_FileWriteMetaData\_t *Meta)`
Register a background file dump request.
- `bool CFE\_FS\_BackgroundFileDumpsPending (const CFE\_FS\_FileWriteMetaData\_t *Meta)`
Query if a background file write request is currently pending.

9.17.1 Detailed Description**9.17.2 Function Documentation****CFE_FS_BackgroundFileDumpsPending()**

```
bool CFE_FS_BackgroundFileDumpIsPending (
    const CFE\_FS\_FileWriteMetaData\_t * Meta )
```

Query if a background file write request is currently pending.

Description

This returns "true" while the request is on the background work queue This returns "false" once the request is complete and removed from the queue.

Assumptions, External Events, and Notes:

None

Parameters

<code>in, out</code>	<i>Meta</i>	The background file write persistent state object (must not be null)
----------------------	-------------	--

Returns

boolean value indicating if request is already pending

Return values

<i>true</i>	if request is pending
<i>false</i>	if request is not pending

CFE_FS_BackgroundFileDumpRequest()

```
int32 CFE_FS_BackgroundFileDumpRequest (
    CFE_FS_FileWriteMetaData_t * Meta )
```

Register a background file dump request.

Description

Puts the previously-initialized metadata into the pending request queue

Assumptions, External Events, and Notes:

Metadata structure should be stored in a persistent memory area (not on stack) as it must remain accessible by the file writer task throughout the asynchronous job operation.

Parameters

<i>in, out</i>	<i>Meta</i>	The background file write persistent state object (must not be null)
----------------	-------------	--

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

<i>CFE_FS_BAD_ARGUMENT</i>	Bad Argument.
<i>CFE_FS_INVALID_PATH</i>	Invalid Path.
<i>CFE_STATUS_REQUEST_ALREADY_PENDING</i>	Request already pending.
<i>CFE_SUCCESS</i>	Successful execution.

CFE_FS_ExtractFilenameFromPath()

```
CFE_Status_t CFE_FS_ExtractFilenameFromPath (
    const char * OriginalPath,
    char * FileNameOnly )
```

Extracts the filename from a unix style path and filename string.

Description

This API will take the original unix path/filename combination and extract the base filename. Example: Given the path/filename : "/cf/apps/myapp.o.gz" this function will return the filename: "myapp.o.gz".

Assumptions, External Events, and Notes:

1. The paths and filenames used here are the standard unix style filenames separated by "/" characters.
2. The extracted filename (including terminator) is no longer than [OS_MAX_PATH_LEN](#)

Parameters

in	<i>OriginalPath</i>	The original path (must not be null)
out	<i>FileNameOnly</i>	The filename that is extracted from the path (must not be null)

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_FS_BAD_ARGUMENT	Bad Argument.
CFE_FS_FNAME_TOO_LONG	Filename Too Long.
CFE_FS_INVALID_PATH	Invalid Path.
CFE_SUCCESS	Successful execution.

CFE_FS_GetDefaultExtension()

```
const char * CFE_FS_GetDefaultExtension (
    CFE\_FS\_FileCategory\_t FileCategory )
```

Get the default filename extension for a file category.

Certain file types may have an extension that varies from system to system. This is primarily an issue for application modules which are ".so" on Linux systems, ".dll" on Windows, ".o" on VxWorks, ".obj" on RTEMS, and so on.

This uses a combination of compile-time configuration and hints from the build environment to get the default/expected extension for a given file category.

Returns

String containing the extension

Return values

<i>NULL</i>	if no default extension is known for the given file category
-------------	--

CFE_FS_GetDefaultMountPoint()

```
const char * CFE_FS_GetDefaultMountPoint (
    CFE\_FS\_FileCategory\_t FileCategory )
```

Get the default virtual mount point for a file category.

Certain classes of files generally reside in a common directory, mainly either the persistent storage (/cf typically) or ram disk (/ram typically).

Ephemeral status files are generally in the ram disk while application modules and scripts are generally in the persistent storage.

This returns the expected directory for a given class of files in the form of a virtual OSAL mount point string.

Returns

String containing the mount point

Return values

<i>NULL</i>	if no mount point is known for the given file category
-------------	--

CFE_FS_ParseInputFileName()

```
int32 CFE_FS_ParseInputFileName (
    char * OutputBuffer,
    const char * InputName,
    size_t OutputBufSize,
    CFE_FS_FileCategory_t FileCategory )
```

Parse a filename string from the user into a local buffer.

Description

Simplified API for [CFE_FS_ParseInputFileNameEx\(\)](#) where input is always known to be a non-empty, null terminated string and the fixed-length input buffer not needed. For instance this may be used where the input is a fixed string from `cfe_platform_cfg.h` or similar.

Assumptions, External Events, and Notes:

The parameters are organized such that this is basically like `strncpy()` with an extra argument, and existing file name accesses which use a direct copy can easily change to use this instead.

See also

[CFE_FS_ParseInputFileNameEx\(\)](#)

Parameters

out	<i>OutputBuffer</i>	Buffer to store result (must not be null).
in	<i>InputName</i>	A null terminated input string (must not be null).
in	<i>OutputBufSize</i>	Maximum Size of output buffer (must not be zero).
in	<i>FileCategory</i>	The generalized category of file (implies default path/extension)

Returns

Execution status, see [cFE Return Code Defines](#)

CFE_FS_ParseInputFileNameEx()

```
int32 CFE_FS_ParseInputFileNameEx (
    char * OutputBuffer,
    const char * InputBuffer,
    size_t OutputBufSize,
    size_t InputBufSize,
    const char * DefaultInput,
```



```
const char * DefaultPath,
const char * DefaultExtension )
```

Parse a filename input from an input buffer into a local buffer.

Description

This provides a more user friendly way to specify file names, using default values for the path and extension, which can vary from system to system.

If InputBuffer is null or its length is zero, then DefaultInput is used as if it was the content of the input buffer.

If either the pathname or extension is missing from the input, it will be added from defaults, with the complete fully-qualified filename stored in the output buffer.

Assumptions, External Events, and Notes:

1. The paths and filenames used here are the standard unix style filenames separated by "/" (path) and "." (extension) characters.
2. Input Buffer has a fixed max length. Parsing will not exceed InputBufSize, and does not need to be null terminated. However parsing will stop at the first null char, when the input is shorter than the maximum.

Parameters

out	<i>OutputBuffer</i>	Buffer to store result (must not be null).
in	<i>InputBuffer</i>	A input buffer that may contain a file name (e.g. from command) (must not be null).
in	<i>OutputBufSize</i>	Maximum Size of output buffer (must not be zero).
in	<i>InputBufSize</i>	Maximum Size of input buffer.
in	<i>DefaultInput</i>	Default value to use for input if InputBffer is empty
in	<i>DefaultPath</i>	Default value to use for pathname if omitted from input
in	<i>DefaultExtension</i>	Default value to use for extension if omitted from input

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_FS_BAD_ARGUMENT	Bad Argument.
CFE_FS_FNAME_TOO_LONG	Filename Too Long.
CFE_FS_INVALID_PATH	Invalid Path.
CFE_SUCCESS	Successful execution.

9.18 cFE Generic Message APIs

Functions

- [CFE_Status_t](#) [CFE_MSG_Init](#) ([CFE_MSG_Message_t](#) *MsgPtr, [CFE_SB_MsgId_t](#) MsgId, [CFE_MSG_Size_t](#) Size)

Initialize a message.

9.18.1 Detailed Description

9.18.2 Function Documentation

CFE_MSG_Init()

```
CFE_Status_t CFE_MSG_Init (
    CFE_MSG_Message_t * MsgPtr,
    CFE_SB_MsgId_t MsgId,
    CFE_MSG_Size_t Size )
```

Initialize a message.

Description

This routine initialize a message. The entire message is set to zero (based on size), defaults are set, then the size and bits from MsgId are set.

Parameters

out	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null).
in	<i>MsgId</i>	MsgId that corresponds to message
in	<i>Size</i>	Total size of the message (used to set length field)

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

9.19 cFE Message Primary Header APIs

Functions

- [CFE_Status_t CFE_MSG_GetSize](#) (const [CFE_MSG_Message_t](#) *MsgPtr, [CFE_MSG_Size_t](#) *Size)
Gets the total size of a message.
- [CFE_Status_t CFE_MSG_SetSize](#) ([CFE_MSG_Message_t](#) *MsgPtr, [CFE_MSG_Size_t](#) Size)
Sets the total size of a message.
- [CFE_Status_t CFE_MSG_GetType](#) (const [CFE_MSG_Message_t](#) *MsgPtr, [CFE_MSG_Type_t](#) *Type)
Gets the message type.
- [CFE_Status_t CFE_MSG_SetType](#) ([CFE_MSG_Message_t](#) *MsgPtr, [CFE_MSG_Type_t](#) Type)
Sets the message type.
- [CFE_Status_t CFE_MSG_GetHeaderVersion](#) (const [CFE_MSG_Message_t](#) *MsgPtr, [CFE_MSG_HeaderVersion_t](#) *Version)
Gets the message header version.
- [CFE_Status_t CFE_MSG_SetHeaderVersion](#) ([CFE_MSG_Message_t](#) *MsgPtr, [CFE_MSG_HeaderVersion_t](#) Version)
Sets the message header version.

- `CFE_Status_t CFE_MSG_GetHasSecondaryHeader` (const `CFE_MSG_Message_t` *MsgPtr, bool *HasSecondary)
Gets the message secondary header boolean.
- `CFE_Status_t CFE_MSG_SetHasSecondaryHeader` (`CFE_MSG_Message_t` *MsgPtr, bool HasSecondary)
Sets the message secondary header boolean.
- `CFE_Status_t CFE_MSG_GetApId` (const `CFE_MSG_Message_t` *MsgPtr, `CFE_MSG_ApId_t` *ApId)
Gets the message application ID.
- `CFE_Status_t CFE_MSG_SetApId` (`CFE_MSG_Message_t` *MsgPtr, `CFE_MSG_ApId_t` ApId)
Sets the message application ID.
- `CFE_Status_t CFE_MSG_GetSegmentationFlag` (const `CFE_MSG_Message_t` *MsgPtr, `CFE_MSG_SegmentationFlag_t` *SegFlag)
Gets the message segmentation flag.
- `CFE_Status_t CFE_MSG_SetSegmentationFlag` (`CFE_MSG_Message_t` *MsgPtr, `CFE_MSG_SegmentationFlag_t` SegFlag)
Sets the message segmentation flag.
- `CFE_Status_t CFE_MSG_GetSequenceCount` (const `CFE_MSG_Message_t` *MsgPtr, `CFE_MSG_SequenceCount_t` *SeqCnt)
Gets the message sequence count.
- `CFE_Status_t CFE_MSG_SetSequenceCount` (`CFE_MSG_Message_t` *MsgPtr, `CFE_MSG_SequenceCount_t` SeqCnt)
Sets the message sequence count.
- `CFE_MSG_SequenceCount_t CFE_MSG_GetNextSequenceCount` (`CFE_MSG_SequenceCount_t` SeqCnt)
Gets the next sequence count value (rolls over if appropriate)

9.19.1 Detailed Description

9.19.2 Function Documentation

CFE_MSG_GetApId()

```
CFE_Status_t CFE_MSG_GetApId (
    const CFE_MSG_Message_t * MsgPtr,
    CFE_MSG_ApId_t * ApId )
```

Gets the message application ID.

Description

This routine gets the message application ID.

Parameters

in	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null).
out	<i>ApId</i>	Application ID (must not be null)

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

<code>CFE_SUCCESS</code>	Successful execution.
<code>CFE_MSG_BAD_ARGUMENT</code>	Error - bad argument.

CFE_MSG_GetHasSecondaryHeader()

```
CFE_Status_t CFE_MSG_GetHasSecondaryHeader (
    const CFE_MSG_Message_t * MsgPtr,
    bool * HasSecondary )
```

Gets the message secondary header boolean.

Description

This routine gets the message secondary header boolean.

Parameters

in	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null).
out	<i>HasSecondary</i>	Has secondary header flag (must not be null)

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

CFE_MSG_GetHeaderVersion()

```
CFE_Status_t CFE_MSG_GetHeaderVersion (
    const CFE_MSG_Message_t * MsgPtr,
    CFE_MSG_HeaderVersion_t * Version )
```

Gets the message header version.

Description

This routine gets the message header version.

Parameters

in	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null).
out	<i>Version</i>	Header version (must not be null)

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

CFE_MSG_GetNextSequenceCount()

```
CFE_MSG_SequenceCount_t CFE_MSG_GetNextSequenceCount (
    CFE_MSG_SequenceCount_t SeqCnt )
```

Gets the next sequence count value (rolls over if appropriate)

Description

Abstract method to get the next valid sequence count value. Will roll over to zero for any input value greater than or equal to the maximum possible sequence count value given the field in the header.

Parameters

in	<i>SeqCnt</i>	Sequence count
----	---------------	----------------

Returns

The next valid sequence count value

CFE_MSG_GetSegmentationFlag()

```
CFE_Status_t CFE_MSG_GetSegmentationFlag (
    const CFE_MSG_Message_t * MsgPtr,
    CFE_MSG_SegmentationFlag_t * SegFlag )
```

Gets the message segmentation flag.

Description

This routine gets the message segmentation flag

Parameters

in	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null).
out	<i>SegFlag</i>	Segmentation flag (must not be null)

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

<i>CFE_SUCCESS</i>	Successful execution.
<i>CFE_MSG_BAD_ARGUMENT</i>	Error - bad argument.

CFE_MSG_GetSequenceCount()

```
CFE_Status_t CFE_MSG_GetSequenceCount (
    const CFE_MSG_Message_t * MsgPtr,
    CFE_MSG_SequenceCount_t * SeqCnt )
```

Gets the message sequence count.

Description

This routine gets the message sequence count.

Parameters

in	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null).
out	<i>SeqCnt</i>	Sequence count (must not be null)

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

CFE_MSG_GetSize()

```
CFE_Status_t CFE_MSG_GetSize (
    const CFE_MSG_Message_t * MsgPtr,
    CFE_MSG_Size_t * Size )
```

Gets the total size of a message.

Description

This routine gets the total size of the message.

Parameters

in	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null).
out	<i>Size</i>	Total message size (must not be null)

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

CFE_MSG_GetType()

```
CFE_Status_t CFE_MSG_GetType (
    const CFE_MSG_Message_t * MsgPtr,
    CFE_MSG_Type_t * Type )
```

Gets the message type.

Description

This routine gets the message type.

Parameters

in	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null).
out	<i>Type</i>	Message type (must not be null)

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

CFE_MSG_SetApId()

```
CFE_Status_t CFE_MSG_SetApId (
    CFE_MSG_Message_t * MsgPtr,
    CFE_MSG_ApId_t ApId )
```

Sets the message application ID.

Description

This routine sets the message application ID. Typically set at initialization using the MsgId, but API available to set bits that may not be included in MsgId.

Parameters

in, out	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null).
in	<i>ApId</i>	Application ID

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

CFE_MSG_SetHasSecondaryHeader()

```
CFE_Status_t CFE_MSG_SetHasSecondaryHeader (
    CFE_MSG_Message_t * MsgPtr,
    bool HasSecondary )
```

Sets the message secondary header boolean.

Description

This routine sets the message secondary header boolean. Typically only set within message initialization and not used by APPs.

Parameters

in, out	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null).
in	<i>HasSecondary</i>	Has secondary header flag

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

CFE_MSG_SetHeaderVersion()

```
CFE_Status_t CFE_MSG_SetHeaderVersion (
    CFE_MSG_Message_t * MsgPtr,
    CFE_MSG_HeaderVersion_t Version )
```

Sets the message header version.

Description

This routine sets the message header version. Typically only set within message initialization and not used by APPs.

Parameters

in, out	<i>MsgPtr</i>	A pointer to the buffer that contains the message.
in	<i>Version</i>	Header version

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

CFE_MSG_SetSegmentationFlag()

```
CFE_Status_t CFE_MSG_SetSegmentationFlag (
    CFE_MSG_Message_t * MsgPtr,
    CFE_MSG_SegmentationFlag_t SegFlag )
```

Sets the message segmentation flag.

Description

This routine sets the message segmentation flag.

Parameters

in, out	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null).
in	<i>SegFlag</i>	Segmentation flag

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

CFE_MSG_SetSequenceCount()

```
CFE_Status_t CFE_MSG_SetSequenceCount (
    CFE_MSG_Message_t * MsgPtr,
    CFE_MSG_SequenceCount_t SeqCnt )
```

Sets the message sequence count.

Description

This routine sets the message sequence count.

Parameters

in, out	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null).
in	<i>SeqCnt</i>	Sequence count

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

CFE_MSG_SetSize()

```
CFE_Status_t CFE_MSG_SetSize (
    CFE_MSG_Message_t * MsgPtr,
    CFE_MSG_Size_t Size )
```

Sets the total size of a message.

Description

This routine sets the total size of the message.

Parameters

in, out	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null).
in	<i>Size</i>	Total message size

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

CFE_MSG_SetType()

```
CFE_Status_t CFE_MSG_SetType (
    CFE_MSG_Message_t * MsgPtr,
    CFE_MSG_Type_t Type )
```

Sets the message type.

Description

This routine sets the message type.

Parameters

in, out	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null).
in	<i>Type</i>	Message type

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

9.20 cFE Message Extended Header APIs

Functions

- `CFE_Status_t CFE_MSG_GetEDSVersion` (const `CFE_MSG_Message_t` *MsgPtr, `CFE_MSG_EDSVersion_t` *Version)
Gets the message EDS version.
- `CFE_Status_t CFE_MSG_SetEDSVersion` (`CFE_MSG_Message_t` *MsgPtr, `CFE_MSG_EDSVersion_t` Version)
Sets the message EDS version.
- `CFE_Status_t CFE_MSG_GetEndian` (const `CFE_MSG_Message_t` *MsgPtr, `CFE_MSG_Endian_t` *Endian)
Gets the message endian.
- `CFE_Status_t CFE_MSG_SetEndian` (`CFE_MSG_Message_t` *MsgPtr, `CFE_MSG_Endian_t` Endian)
Sets the message endian.
- `CFE_Status_t CFE_MSG_GetPlaybackFlag` (const `CFE_MSG_Message_t` *MsgPtr, `CFE_MSG_PlaybackFlag_t` *PlayFlag)
Gets the message playback flag.
- `CFE_Status_t CFE_MSG_SetPlaybackFlag` (`CFE_MSG_Message_t` *MsgPtr, `CFE_MSG_PlaybackFlag_t` PlayFlag)
Sets the message playback flag.
- `CFE_Status_t CFE_MSG_GetSubsystem` (const `CFE_MSG_Message_t` *MsgPtr, `CFE_MSG_Subsystem_t` *Subsystem)
Gets the message subsystem.
- `CFE_Status_t CFE_MSG_SetSubsystem` (`CFE_MSG_Message_t` *MsgPtr, `CFE_MSG_Subsystem_t` Subsystem)
Sets the message subsystem.
- `CFE_Status_t CFE_MSG_GetSystem` (const `CFE_MSG_Message_t` *MsgPtr, `CFE_MSG_System_t` *System)
Gets the message system.
- `CFE_Status_t CFE_MSG_SetSystem` (`CFE_MSG_Message_t` *MsgPtr, `CFE_MSG_System_t` System)
Sets the message system.

9.20.1 Detailed Description

9.20.2 Function Documentation

CFE_MSG_GetEDSVersion()

```
CFE_Status_t CFE_MSG_GetEDSVersion (
    const CFE_MSG_Message_t * MsgPtr,
    CFE_MSG_EDSVersion_t * Version )
```

Gets the message EDS version.

Description

This routine gets the message EDS version.

Parameters

in	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null).
out	<i>Version</i>	EDS Version (must not be null)

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

CFE_MSG_GetEndian()

```
CFE_Status_t CFE_MSG_GetEndian (
    const CFE_MSG_Message_t * MsgPtr,
    CFE_MSG_Endian_t * Endian )
```

Gets the message endian.

Description

This routine gets the message endian.

Parameters

in	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null).
out	<i>Endian</i>	Endian (must not be null)

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

CFE_MSG_GetPlaybackFlag()

```
CFE_Status_t CFE_MSG_GetPlaybackFlag (
    const CFE_MSG_Message_t * MsgPtr,
    CFE_MSG_PlaybackFlag_t * PlayFlag )
```

Gets the message playback flag.

Description

This routine gets the message playback flag.

Parameters

in	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null).
out	<i>PlayFlag</i>	Playback Flag (must not be null)

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

CFE_MSG_GetSubsystem()

```
CFE_Status_t CFE_MSG_GetSubsystem (
    const CFE_MSG_Message_t * MsgPtr,
    CFE_MSG_Subsystem_t * Subsystem )
```

Gets the message subsystem.

Description

This routine gets the message subsystem

Parameters

in	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null).
out	<i>Subsystem</i>	Subsystem (must not be null)

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

CFE_MSG_GetSystem()

```
CFE_Status_t CFE_MSG_GetSystem (
    const CFE_MSG_Message_t * MsgPtr,
    CFE_MSG_System_t * System )
```

Gets the message system.

Description

This routine gets the message system id

Parameters

in	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null).
out	<i>System</i>	System (must not be null)

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

CFE_MSG_SetEDSVersion()

```
CFE_Status_t CFE_MSG_SetEDSVersion (
    CFE_MSG_Message_t * MsgPtr,
    CFE_MSG_EDSVersion_t Version )
```

Sets the message EDS version.

Description

This routine sets the message EDS version.

Parameters

in, out	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null).
in	<i>Version</i>	EDS Version

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

CFE_MSG_SetEndian()

```
CFE_Status_t CFE_MSG_SetEndian (
    CFE_MSG_Message_t * MsgPtr,
    CFE_MSG_Endian_t Endian )
```

Sets the message endian.

Description

This routine sets the message endian. Invalid endian selection will set big endian.

Parameters

in, out	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null).
in	<i>Endian</i>	Endian

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

CFE_MSG_SetPlaybackFlag()

```
CFE_Status_t CFE_MSG_SetPlaybackFlag (
    CFE_MSG_Message_t * MsgPtr,
    CFE_MSG_PlaybackFlag_t PlayFlag )
```

Sets the message playback flag.

Description

This routine sets the message playback flag.

Parameters

in, out	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null).
in	<i>PlayFlag</i>	Playback Flag

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

CFE_MSG_SetSubsystem()

```
CFE_Status_t CFE_MSG_SetSubsystem (
    CFE_MSG_Message_t * MsgPtr,
    CFE_MSG_Subsystem_t Subsystem )
```

Sets the message subsystem.

Description

This routine sets the message subsystem. Some bits may be set at initialization using the MsgId, but API available to set bits that may not be included in MsgId.

Parameters

in, out	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null).
in	<i>Subsystem</i>	Subsystem

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

CFE_MSG_SetSystem()

```
CFE_Status_t CFE_MSG_SetSystem (
    CFE_MSG_Message_t * MsgPtr,
    CFE_MSG_System_t System )
```

Sets the message system.

Description

This routine sets the message system id. Some bits may be set at initialization using the MsgId, but API available to set bits that may not be included in MsgId.

Parameters

in, out	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null).
in	<i>System</i>	System

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

9.21 cFE Message Secondary Header APIs**Functions**

- [CFE_Status_t CFE_MSG_GenerateChecksum](#) ([CFE_MSG_Message_t](#) *MsgPtr)
Calculates and sets the checksum of a message.
- [CFE_Status_t CFE_MSG_ValidateChecksum](#) (const [CFE_MSG_Message_t](#) *MsgPtr, bool *IsValid)
Validates the checksum of a message.
- [CFE_Status_t CFE_MSG_SetFcnCode](#) ([CFE_MSG_Message_t](#) *MsgPtr, [CFE_MSG_FcnCode_t](#) FcnCode)
Sets the function code field in a message.
- [CFE_Status_t CFE_MSG_GetFcnCode](#) (const [CFE_MSG_Message_t](#) *MsgPtr, [CFE_MSG_FcnCode_t](#) *FcnCode)
Gets the function code field from a message.
- [CFE_Status_t CFE_MSG_GetMsgTime](#) (const [CFE_MSG_Message_t](#) *MsgPtr, [CFE_TIME_SysTime_t](#) *Time)

Gets the time field from a message.

- [CFE_Status_t](#) [CFE_MSG_SetMsgTime](#) ([CFE_MSG_Message_t](#) *MsgPtr, [CFE_TIME_SysTime_t](#) NewTime)

Sets the time field in a message.

9.21.1 Detailed Description

9.21.2 Function Documentation

CFE_MSG_GenerateChecksum()

```
CFE\_Status\_t CFE_MSG_GenerateChecksum (
    CFE\_MSG\_Message\_t * MsgPtr )
```

Calculates and sets the checksum of a message.

Description

This routine calculates the checksum of a message according to an implementation-defined algorithm. Then, it sets the checksum field in the message with the calculated value. The contents and location of this field will depend on the underlying implementation of messages. It may be a checksum, a CRC, or some other algorithm.

Assumptions, External Events, and Notes:

- If the underlying implementation of messages does not include a checksum field, then this routine will return [CFE_MSG_WRONG_MSG_TYPE](#)

Parameters

<i>in, out</i>	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null).
----------------	---------------	---

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.
CFE_MSG_WRONG_MSG_TYPE	Error - wrong type.

CFE_MSG_GetFcnCode()

```
CFE\_Status\_t CFE_MSG_GetFcnCode (
    const CFE\_MSG\_Message\_t * MsgPtr,
    CFE\_MSG\_FcnCode\_t * FcnCode )
```

Gets the function code field from a message.

Description

This routine gets the function code from a message.

Assumptions, External Events, and Notes:

- If the underlying implementation of messages does not include a function code field, then this routine will set FcnCode to zero and return [CFE_MSG_WRONG_MSG_TYPE](#)

Parameters

in	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null).
out	<i>FcnCode</i>	The function code from the message (must not be null)

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.
CFE_MSG_WRONG_MSG_TYPE	Error - wrong type.

CFE_MSG_GetMsgTime()

```
CFE_Status_t CFE_MSG_GetMsgTime (
    const CFE_MSG_Message_t * MsgPtr,
    CFE_TIME_SysTime_t * Time )
```

Gets the time field from a message.

Description

This routine gets the time from a message.

Assumptions, External Events, and Notes:

- If the underlying implementation of messages does not include a time field, then this routine will set Time to zero and return [CFE_MSG_WRONG_MSG_TYPE](#)
- Note default implementation of command messages do not have a time field.

Parameters

in	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null).
out	<i>Time</i>	Time from the message (must not be null)

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.
CFE_MSG_WRONG_MSG_TYPE	Error - wrong type.

CFE_MSG_SetFcnCode()

```
CFE_Status_t CFE_MSG_SetFcnCode (
    CFE_MSG_Message_t * MsgPtr,
    CFE_MSG_FcnCode_t FcnCode )
```

Sets the function code field in a message.

Description

This routine sets the function code of a message.

Assumptions, External Events, and Notes:

- If the underlying implementation of messages does not include a function code field, then this routine will do nothing to the message contents and will return [CFE_MSG_WRONG_MSG_TYPE](#).

Parameters

in, out	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null).
in	<i>FcnCode</i>	The function code to include in the message.

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.
CFE_MSG_WRONG_MSG_TYPE	Error - wrong type.

CFE_MSG_SetMsgTime()

```
CFE_Status_t CFE_MSG_SetMsgTime (
    CFE_MSG_Message_t * MsgPtr,
    CFE_TIME_SysTime_t NewTime )
```

Sets the time field in a message.

Description

This routine sets the time of a message. Most applications will want to use [CFE_SB_TimeStampMsg](#) instead of this function. But, when needed, this API can be used to set multiple messages with identical time stamps.

Assumptions, External Events, and Notes:

- If the underlying implementation of messages does not include a time field, then this routine will do nothing to the message contents and will return [CFE_MSG_WRONG_MSG_TYPE](#).
- Note default implementation of command messages do not have a time field.

Parameters

in, out	<i>MsgPtr</i>	A pointer to the message (must not be null).
in	<i>NewTime</i>	The time to include in the message. This will usually be a time from CFE_TIME_GetTime .

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.
CFE_MSG_WRONG_MSG_TYPE	Error - wrong type.

CFE_MSG_ValidateChecksum()

```
CFE_Status_t CFE_MSG_ValidateChecksum (
    const CFE_MSG_Message_t * MsgPtr,
    bool * IsValid )
```

Validates the checksum of a message.

Description

This routine validates the checksum of a message according to an implementation-defined algorithm.

Assumptions, External Events, and Notes:

- If the underlying implementation of messages does not include a checksum field, then this routine will return [CFE_MSG_WRONG_MSG_TYPE](#) and set the *IsValid* parameter false.

Parameters

in	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null). This must point to the first byte of the message header.
out	<i>IsValid</i>	Checksum validation result (must not be null) • true - valid • false - invalid or not supported/implemented

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.
CFE_MSG_WRONG_MSG_TYPE	Error - wrong type.

9.22 cFE Message Id APIs

Functions

- [CFE_Status_t CFE_MSG_GetMsgId](#) (const [CFE_MSG_Message_t](#) *MsgPtr, [CFE_SB_MsgId_t](#) *MsgId)
Gets the message id from a message.
- [CFE_Status_t CFE_MSG_SetMsgId](#) ([CFE_MSG_Message_t](#) *MsgPtr, [CFE_SB_MsgId_t](#) MsgId)
Sets the message id bits in a message.
- [CFE_Status_t CFE_MSG_GetTypeFromMsgId](#) ([CFE_SB_MsgId_t](#) MsgId, [CFE_MSG_Type_t](#) *Type)
Gets message type using message ID.

9.22.1 Detailed Description

9.22.2 Function Documentation

CFE_MSG_GetMsgId()

```
CFE_Status_t CFE_MSG_GetMsgId (
    const CFE_MSG_Message_t * MsgPtr,
    CFE_SB_MsgId_t * MsgId )
```

Gets the message id from a message.

Description

This routine gets the message id from a message. The message id is a hash of bits in the message header, used by the software bus for routing. Message id needs to be unique for each endpoint in the system.

Parameters

in	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null).
out	<i>MsgId</i>	Message id (must not be null)

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

CFE_MSG_GetTypeFromMsgId()

```
CFE_Status_t CFE_MSG_GetTypeFromMsgId (
    CFE_SB_MsgId_t MsgId,
    CFE_MSG_Type_t * Type )
```

Gets message type using message ID.

Description

This routine gets the message type using the message ID

Parameters

in	<i>MsgId</i>	Message id
out	<i>Type</i>	Message type (must not be null)

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

CFE_MSG_SetMsgId()

```
CFE_Status_t CFE_MSG_SetMsgId (
    CFE_MSG_Message_t * MsgPtr,
    CFE_SB_MsgId_t MsgId )
```

Sets the message id bits in a message.

Description

This routine sets the message id bits in a message. The message id is a hash of bits in the message header, used by the software bus for routing. Message id needs to be unique for each endpoint in the system.

Note

This API only sets the bits in the header that make up the message ID. No other values in the header are modified.

The user should ensure that this function is only called with a valid MsgId parameter value. If called with an invalid value, the results are implementation-defined. The implementation may or may not return the error code [CFE_MSG_BAD_ARGUMENT](#) in this case.

Parameters

in, out	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null).
in	<i>MsgId</i>	Message id

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

9.23 cFE Message Integrity APIs

Functions

- [CFE_Status_t CFE_MSG_OriginationAction](#) ([CFE_MSG_Message_t](#) *MsgPtr, [size_t](#) BufferSize, [bool](#) *IsAcceptable)
Perform any necessary actions on a newly-created message, prior to sending.
- [CFE_Status_t CFE_MSG_VerificationAction](#) (const [CFE_MSG_Message_t](#) *MsgPtr, [size_t](#) BufferSize, [bool](#) *IsAcceptable)
Checks message integrity/acceptability.

9.23.1 Detailed Description

9.23.2 Function Documentation

CFE_MSG_OriginationAction()

```
CFE_Status_t CFE_MSG_OriginationAction (
    CFE_MSG_Message_t * MsgPtr,
    size_t BufferSize,
    bool * IsAcceptable )
```

Perform any necessary actions on a newly-created message, prior to sending.

Description

This routine updates and/or appends any necessary fields on a message, is invoked via SB just prior to broadcasting the message. The actions include updating any values that should be computed/updated per message, including:

- setting the sequence number
- updating the timestamp, if present
- computing any error control or checksum fields, if present

The MSG module implementation determines which header fields meet this criteria and how they should be computed. The BufferSize parameter indicates the allocation size message of the buffer that holds the message (i.e. the message envelope size). In some implementations, the allocated buffer may include extra space in order to append a CRC or digital signature.

See also

[CFE_MSG_VerificationAction](#)

Parameters

in, out	<i>MsgPtr</i>	A pointer to the buffer that contains the message (must not be null).
in	<i>BufferSize</i>	The size of the buffer encapsulating the message
out	<i>IsAcceptable</i>	Output variable to be set, indicates message acceptability (must not be null)

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

CFE_MSG_VerificationAction()

```
CFE_Status_t CFE_MSG_VerificationAction (
    const CFE_MSG_Message_t * MsgPtr,
    size_t BufferSize,
    bool * IsAcceptable )
```

Checks message integrity/acceptability.

Description

This routine validates that any error-control field(s) in the message header matches the expected value.

The specific function of this API is entirely dependent on the header fields and may be a no-op if no error checking is implemented. In that case, it will always output "true".

Note

Due to the fact that software bus uses a multicast architecture, this function must not modify the message, as the buffer may be shared among multiple receivers. This should generally be the inverse of [CFE_MSG_OriginationAction\(\)](#), but on the origination side it may update header fields and/or modify the message, on the verification/receive side it must only check those fields, not modify them.

See also

[CFE_MSG_OriginationAction](#)

Parameters

in	<i>MsgPtr</i>	Message Pointer (must not be null)
in	<i>BufferSize</i>	The size of the buffer encapsulating the message
out	<i>IsAcceptable</i>	Output variable to be set, indicates message acceptability (must not be null)

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_MSG_BAD_ARGUMENT	Error - bad argument.

9.24 cFE Pipe Management APIs**Functions**

- [CFE_Status_t CFE_SB_CreatePipe](#) ([CFE_SB_PipeId_t](#) *PipeIdPtr, [uint16](#) Depth, const char *PipeName)
Creates a new software bus pipe.
- [CFE_Status_t CFE_SB_DeletePipe](#) ([CFE_SB_PipeId_t](#) PipeId)
Delete a software bus pipe.
- [CFE_Status_t CFE_SB_PipeId_ToIndex](#) ([CFE_SB_PipeId_t](#) PipeId, [uint32](#) *Idx)
Obtain an index value correlating to an SB Pipe ID.
- [CFE_Status_t CFE_SB_SetPipeOpts](#) ([CFE_SB_PipeId_t](#) PipeId, [uint8](#) Opts)

Set options on a pipe.

- [CFE_Status_t CFE_SB_GetPipeOpts](#) ([CFE_SB_PipeId_t](#) PipeId, [uint8](#) *OptsPtr)

Get options on a pipe.

- [CFE_Status_t CFE_SB_GetPipeName](#) ([char](#) *PipeNameBuf, [size_t](#) PipeNameSize, [CFE_SB_PipeId_t](#) PipeId)

Get the pipe name for a given id.

- [CFE_Status_t CFE_SB_GetPipeIdByName](#) ([CFE_SB_PipeId_t](#) *PipeIdPtr, [const char](#) *PipeName)

Get pipe id by pipe name.

9.24.1 Detailed Description

9.24.2 Function Documentation

CFE_SB_CreatePipe()

```
CFE_Status_t CFE_SB_CreatePipe (
    CFE_SB_PipeId_t * PipeIdPtr,
    uint16 Depth,
    const char * PipeName )
```

Creates a new software bus pipe.

Description

This routine creates and initializes an input pipe that the calling application can use to receive software bus messages. By default, no messages are routed to the new pipe. So, the application must use [CFE_SB_Subscribe](#) to specify which messages it wants to receive on this pipe.

Assumptions, External Events, and Notes:

None

Parameters

out	<i>PipeIdPtr</i>	A pointer to a variable of type CFE_SB_PipeId_t (must not be null), which will be filled in with the pipe ID information by the CFE_SB_CreatePipe routine. *PipeIdPtr is the identifier for the created pipe.
in	<i>Depth</i>	The maximum number of messages that will be allowed on this pipe at one time.
in	<i>PipeName</i>	A string (must not be null) to be used to identify this pipe in error messages and routing information telemetry. The string must be no longer than OS_MAX_API_NAME (including terminator). Longer strings will be truncated.

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_SB_BAD_ARGUMENT	Bad Argument.
CFE_SB_MAX_PIPES_MET	Max Pipes Met.
CFE_SB_PIPE_CR_ERR	Pipe Create Error.

See also

[CFE_SB_DeletePipe](#) [CFE_SB_GetPipeOpts](#) [CFE_SB_SetPipeOpts](#) [CFE_SB_GetPipeIdByName](#)

CFE_SB_DeletePipe()

```
CFE_Status_t CFE_SB_DeletePipe (
    CFE_SB_PipeId_t PipeId )
```

Delete a software bus pipe.

Description

This routine deletes an input pipe and cleans up all data structures associated with the pipe. All subscriptions made for this pipe by calls to [CFE_SB_Subscribe](#) will be automatically removed from the SB routing tables. Any messages in the pipe will be discarded.

Applications should not call this routine for all of their SB pipes as part of their orderly shutdown process, as the pipe will be deleted by the support framework at the appropriate time.

Assumptions, External Events, and Notes:

None

Parameters

in	<i>PipeId</i>	The pipe ID (obtained previously from CFE_SB_CreatePipe) of the pipe to be deleted.
----	---------------	--

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_SB_BAD_ARGUMENT	Bad Argument.

See also

[CFE_SB_CreatePipe](#) [CFE_SB_GetPipeOpts](#) [CFE_SB_SetPipeOpts](#) [CFE_SB_GetPipeIdByName](#)

CFE_SB_GetPipeIdByName()

```
CFE_Status_t CFE_SB_GetPipeIdByName (
    CFE_SB_PipeId_t * PipeIdPtr,
    const char * PipeName )
```

Get pipe id by pipe name.

Description

This routine finds the pipe id for a pipe name.

Parameters

in	<i>PipeName</i>	The name of the pipe (must not be null).
out	<i>PipeldPtr</i>	The Pipeld for that name (must not be null).

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_SB_BAD_ARGUMENT	Bad Argument.

See also

[CFE_SB_CreatePipe](#) [CFE_SB_DeletePipe](#) [CFE_SB_SetPipeOpts](#) [CFE_SB_PIPEOPTS_IGNOREMINE](#)

CFE_SB_GetPipeName()

```
CFE_Status_t CFE_SB_GetPipeName (
    char * PipeNameBuf,
    size_t PipeNameSize,
    CFE_SB_PipeId_t PipeId )
```

Get the pipe name for a given id.

Description

This routine finds the pipe name for a pipe id.

Parameters

out	<i>PipeNameBuf</i>	The buffer to receive the pipe name (must not be null).
in	<i>PipeNameSize</i>	The size (in chars) of the PipeName buffer (must not be zero).
in	<i>Pipeld</i>	The Pipeld for that name.

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_SB_BAD_ARGUMENT	Bad Argument.

See also

[CFE_SB_CreatePipe](#) [CFE_SB_DeletePipe](#) [CFE_SB_SetPipeOpts](#) [CFE_SB_GetPipeldByName](#)

CFE_SB_GetPipeOpts()

```
CFE_Status_t CFE_SB_GetPipeOpts (
    CFE_SB_PipeId_t PipeId,
    uint8 * OptsPtr )
```

Get options on a pipe.

Description

This routine gets the current options on a pipe.

Parameters

in	<i>PipeId</i>	The pipe ID of the pipe to get options from.
out	<i>OptsPtr</i>	A bit field of options: cFE SB Pipe options (must not be null)

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_SB_BAD_ARGUMENT	Bad Argument.

See also

[CFE_SB_CreatePipe](#) [CFE_SB_DeletePipe](#) [CFE_SB_SetPipeOpts](#) [CFE_SB_GetPipeIdByName](#) [CFE_SB_PIPEOPTS_IGNOREMIN](#)

CFE_SB_PipeId_ToIndex()

```
CFE_Status_t CFE_SB_PipeId_ToIndex (
    CFE_SB_PipeId_t PipeID,
    uint32 * Idx )
```

Obtain an index value correlating to an SB Pipe ID.

This calculates a zero based integer value that may be used for indexing into a local resource table/array.

Index values are only guaranteed to be unique for resources of the same type. For instance, the indices corresponding to two [valid] application IDs will never overlap, but the index of a pipe ID and an app ID may be the same. Furthermore, indices may be reused if a resource is deleted and re-created.

Note

There is no inverse of this function - indices cannot be converted back to the original PipeID value. The caller should retain the original ID for future use.

Parameters

in	<i>PipeID</i>	Pipe ID to convert
out	<i>Idx</i>	Buffer where the calculated index will be stored (must not be null)

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.

CFE_SB_SetPipeOpts()

```
CFE_Status_t CFE_SB_SetPipeOpts (
    CFE_SB_PipeId_t PipeId,
    uint8 Opts )
```

Set options on a pipe.

Description

This routine sets (or clears) options to alter the pipe's behavior. Options are (re)set every call to this routine.

Parameters

in	<i>PipeId</i>	The pipe ID of the pipe to set options on.
in	<i>Opts</i>	A bit field of options: cFE SB Pipe options

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_SB_BAD_ARGUMENT	Bad Argument.

See also

[CFE_SB_CreatePipe](#) [CFE_SB_DeletePipe](#) [CFE_SB_GetPipeOpts](#) [CFE_SB_GetPipeIdByName](#) [CFE_SB_PIPEOPTS_IGNOREMIS](#)

9.25 cFE Message Subscription Control APIs**Functions**

- [CFE_Status_t CFE_SB_SubscribeEx](#) ([CFE_SB_MsgId_t](#) MsgId, [CFE_SB_PipeId_t](#) PipeId, [CFE_SB_Qos_t](#) Quality, [uint16](#) MsgLim)
Subscribe to a message on the software bus.
- [CFE_Status_t CFE_SB_Subscribe](#) ([CFE_SB_MsgId_t](#) MsgId, [CFE_SB_PipeId_t](#) PipeId)
Subscribe to a message on the software bus with default parameters.
- [CFE_Status_t CFE_SB_SubscribeLocal](#) ([CFE_SB_MsgId_t](#) MsgId, [CFE_SB_PipeId_t](#) PipeId, [uint16](#) MsgLim)
Subscribe to a message while keeping the request local to a cpu.
- [CFE_Status_t CFE_SB_Unsubscribe](#) ([CFE_SB_MsgId_t](#) MsgId, [CFE_SB_PipeId_t](#) PipeId)

Remove a subscription to a message on the software bus.

- [CFE_Status_t](#) [CFE_SB_UnsubscribeLocal](#) ([CFE_SB_MsgId_t](#) MsgId, [CFE_SB_PipeId_t](#) PipeId)

Remove a subscription to a message on the software bus on the current CPU.

9.25.1 Detailed Description

9.25.2 Function Documentation

CFE_SB_Subscribe()

```
CFE_Status_t CFE_SB_Subscribe (
    CFE_SB_MsgId_t MsgId,
    CFE_SB_PipeId_t PipeId )
```

Subscribe to a message on the software bus with default parameters.

Description

This routine adds the specified pipe to the destination list for the specified message ID. This is the same as [CFE_SB_SubscribeEx](#) with the Quality field set to [CFE_SB_DEFAULT_QOS](#) and MsgLim set to [CFE_PLATFORM_SB_DEFAULT_MSG_LIMIT](#) (4).

Assumptions, External Events, and Notes:

Note: As subscriptions are received, the destinations are added to the head of a linked list. During the sending of a message, the list is traversed beginning at the head of the list. Therefore the message will first be sent to the last subscriber. If an application has timing constraints and needs to receive a message in the shortest possible time, the developer may consider holding off its subscription until other applications have subscribed to the message.

Parameters

in	MsgId	The message ID of the message to be subscribed to.
in	PipeId	The pipe ID of the pipe the subscribed message should be sent to.

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_SB_MAX_MSGS_MET	(return value only verified in coverage test) Max Messages Met.
CFE_SB_MAX_DESTS_MET	Max Destinations Met.
CFE_SB_BAD_ARGUMENT	Bad Argument.
CFE_SB_BUF_ALLOC_ERR	(return value only verified in coverage test) Buffer Allocation Error.

See also

[CFE_SB_SubscribeEx](#), [CFE_SB_SubscribeLocal](#), [CFE_SB_Unsubscribe](#), [CFE_SB_UnsubscribeLocal](#)

CFE_SB_SubscribeEx()

```
CFE_Status_t CFE_SB_SubscribeEx (
    CFE_SB_MsgId_t MsgId,
    CFE_SB_PipeId_t PipeId,
    CFE_SB_Qos_t Quality,
    uint16 MsgLim )
```

Subscribe to a message on the software bus.

Description

This routine adds the specified pipe to the destination list associated with the specified message ID.

Assumptions, External Events, and Notes:

Note: As subscriptions are received, the destinations are added to the head of a linked list. During the sending of a message, the list is traversed beginning at the head of the list. Therefore the message will first be sent to the last subscriber. If an application has timing constraints and needs to receive a message in the shortest possible time, the developer may consider holding off its subscription until other applications have subscribed to the message.

Parameters

in	<i>MsgId</i>	The message ID of the message to be subscribed to.
in	<i>PipeId</i>	The pipe ID of the pipe the subscribed message should be sent to.
in	<i>Quality</i>	The requested Quality of Service (QoS) required of the messages. Most callers will use CFE_SB_DEFAULT_QOS for this parameter.
in	<i>MsgLim</i>	The maximum number of messages with this Message ID to allow in this pipe at the same time.

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_SB_MAX_MSGS_MET	(return value only verified in coverage test) Max Messages Met.
CFE_SB_MAX_DESTS_MET	Max Destinations Met.
CFE_SB_BAD_ARGUMENT	Bad Argument.
CFE_SB_BUF_ALLOC_ERR	(return value only verified in coverage test) Buffer Allocation Error.

See also

[CFE_SB_Subscribe](#), [CFE_SB_SubscribeLocal](#), [CFE_SB_Unsubscribe](#), [CFE_SB_UnsubscribeLocal](#)

CFE_SB_SubscribeLocal()

```
CFE_Status_t CFE_SB_SubscribeLocal (
    CFE_SB_MsgId_t MsgId,
```



```
CFE_SB_PipeId_t PipeId,
uint16 MsgLim )
```

Subscribe to a message while keeping the request local to a cpu.

Description

This routine adds the specified pipe to the destination list for the specified message ID. This is similar to [CFE_SB_SubscribeEx](#) with the Quality field set to [CFE_SB_DEFAULT_QOS](#) and MsgLim set to [CFE_PLATFORM_SB_DEFAULT_MSG_LIMIT](#), but will not report the subscription.

Software Bus Network (SBN) application is an example use case, where local subscriptions should not be reported to peers.

Assumptions, External Events, and Notes:

- This API is typically only used by Software Bus Network (SBN) Application

Parameters

in	<i>MsgId</i>	The message ID of the message to be subscribed to.
in	<i>PipeId</i>	The pipe ID of the pipe the subscribed message should be sent to.
in	<i>MsgLim</i>	The maximum number of messages with this Message ID to allow in this pipe at the same time.

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_SB_MAX_MSGS_MET	(return value only verified in coverage test) Max Messages Met.
CFE_SB_MAX_DESTS_MET	Max Destinations Met.
CFE_SB_BAD_ARGUMENT	Bad Argument.
CFE_SB_BUF_ALOC_ERR	(return value only verified in coverage test) Buffer Allocation Error.

See also

[CFE_SB_Subscribe](#), [CFE_SB_SubscribeEx](#), [CFE_SB_Unsubscribe](#), [CFE_SB_UnsubscribeLocal](#)

CFE_SB_Unsubscribe()

```
CFE_Status_t CFE_SB_Unsubscribe (
    CFE_SB_MsgId_t MsgId,
    CFE_SB_PipeId_t PipeId )
```

Remove a subscription to a message on the software bus.

Description

This routine removes the specified pipe from the destination list for the specified message ID.

Assumptions, External Events, and Notes:

If the Pipe is not subscribed to MsgId, the CFE_SB_UNSUB_NO_SUBS_EID event will be generated and CFE_SUCCESS will be returned

Parameters

in	<i>MsgId</i>	The message ID of the message to be unsubscribed.
in	<i>PipeId</i>	The pipe ID of the pipe the subscribed message should no longer be sent to.

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_SB_BAD_ARGUMENT	Bad Argument.

See also

[CFE_SB_Subscribe](#), [CFE_SB_SubscribeEx](#), [CFE_SB_SubscribeLocal](#), [CFE_SB_UnsubscribeLocal](#)

CFE_SB_UnsubscribeLocal()

```
CFE_Status_t CFE_SB_UnsubscribeLocal (
    CFE_SB_MsgId_t MsgId,
    CFE_SB_PipeId_t PipeId )
```

Remove a subscription to a message on the software bus on the current CPU.

Description

This routine removes the specified pipe from the destination list for the specified message ID on the current CPU.

Assumptions, External Events, and Notes:

This API is typically only used by Software Bus Network (SBN) Application. If the Pipe is not subscribed to MsgId, the CFE_SB_UNSUB_NO_SUBS_EID event will be generated and [CFE_SUCCESS](#) will be returned

Parameters

in	<i>MsgId</i>	The message ID of the message to be unsubscribed.
in	<i>PipeId</i>	The pipe ID of the pipe the subscribed message should no longer be sent to.

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_SB_BAD_ARGUMENT	Bad Argument.

See also

[CFE_SB_Subscribe](#), [CFE_SB_SubscribeEx](#), [CFE_SB_SubscribeLocal](#), [CFE_SB_Unsubscribe](#)

9.26 cFE Send/Receive Message APIs

Functions

- [CFE_Status_t CFE_SB_TransmitMsg](#) (const [CFE_MSG_Message_t](#) *MsgPtr, bool IsOrigination)
Transmit a message.
- [CFE_Status_t CFE_SB_ReceiveBuffer](#) ([CFE_SB_Buffer_t](#) **BufPtr, [CFE_SB_PipeId_t](#) PipeId, [int32](#) TimeOut)
Receive a message from a software bus pipe.

9.26.1 Detailed Description

9.26.2 Function Documentation

CFE_SB_ReceiveBuffer()

```
CFE_Status_t CFE_SB_ReceiveBuffer (
    CFE_SB_Buffer_t ** BufPtr,
    CFE_SB_PipeId_t PipeId,
    int32 TimeOut )
```

Receive a message from a software bus pipe.

Description

This routine retrieves the next message from the specified pipe. If the pipe is empty, this routine will block until either a new message comes in or the timeout value is reached.

Assumptions, External Events, and Notes:

Note - If an error occurs in this API, the *BufPtr value may be NULL or random. Therefore, it is recommended that the return code be tested for CFE_SUCCESS before processing the message.

Parameters

in, out	<i>BufPtr</i>	A pointer to the software bus buffer to receive to (must not be null). Typically a caller declares a ptr of type CFE_SB_Buffer_t (i.e. CFE_SB_Buffer_t *Ptr) then gives the address of that pointer (&Ptr) as this parameter. After a successful receipt of a message, *BufPtr will point to the first byte of the software bus buffer. This should be used as a read-only pointer (in systems with an MMU, writes to this pointer may cause a memory protection fault). The *BufPtr is valid only until the next call to CFE_SB_ReceiveBuffer for the same pipe.
in	<i>PipeId</i>	The pipe ID of the pipe containing the message to be obtained.
in	<i>TimeOut</i>	The number of milliseconds to wait for a new message if the pipe is empty at the time of the call. This can also be set to CFE_SB_POLL for a non-blocking receive or CFE_SB_PEND_FOREVER to wait forever for a message to arrive.

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

<i>CFE_SUCCESS</i>	Successful execution.
<i>CFE_SB_BAD_ARGUMENT</i>	Bad Argument.
<i>CFE_SB_TIME_OUT</i>	Time Out.
<i>CFE_SB_PIPE_RD_ERR</i>	(return value only verified in coverage test) Pipe Read Error.
<i>CFE_SB_NO_MESSAGE</i>	No Message.

CFE_SB_TransmitMsg()

```
CFE_Status_t CFE_SB_TransmitMsg (
    const CFE_MSG_Message_t * MsgPtr,
    bool IsOrigination )
```

Transmit a message.

Description

This routine copies the specified message into a software bus buffer which is then transmitted to all subscribers. The software bus will read the message ID from the message header to determine which pipes should receive the message.

The IsOrigination parameter should be passed as "true" if the message was newly constructed by the sender and is being sent for the first time. This enables the message origination actions as determined by the CFE MSG module, which may include (but not limited to):

- Updating sequence number
- Updating timestamp
- Calculating a CRC, checksum, or other message error control field

Conversely, when forwarding a message that originated from an external entity (e.g. messages passing through CI or SBN), the parameter should be passed as "false" to not overwrite existing data.

Assumptions, External Events, and Notes:

- This routine will not normally wait for the receiver tasks to process the message before returning control to the caller's task.
- However, if a higher priority task is pending and subscribed to this message, that task may get to run before returning control to the caller.
- In previous versions of CFE, the boolean parameter referred to the sequence number header of telemetry messages only. This has been extended to apply more generically to any headers, as determined by the CFE MSG implementation.

Parameters

in	<i>MsgPtr</i>	A pointer to the message to be sent (must not be null). This must point to the first byte of the message header.
in	<i>IsOrigination</i>	Update the headers of the message

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_SB_BAD_ARGUMENT	Bad Argument.
CFE_SB_MSG_TOO_BIG	Message Too Big.
CFE_SB_BUF_ALLOC_ERR	(return value only verified in coverage test) Buffer Allocation Error.

9.27 cFE Zero Copy APIs

Functions

- [CFE_SB_Buffer_t * CFE_SB_AllocateMessageBuffer](#) (size_t MsgSize)
Get a buffer pointer to use for "zero copy" SB sends.
- [CFE_Status_t CFE_SB_ReleaseMessageBuffer](#) ([CFE_SB_Buffer_t](#) *BufPtr)
Release an unused "zero copy" buffer pointer.
- [CFE_Status_t CFE_SB_TransmitBuffer](#) ([CFE_SB_Buffer_t](#) *BufPtr, bool IsOrigination)
Transmit a buffer.

9.27.1 Detailed Description

9.27.2 Function Documentation

CFE_SB_AllocateMessageBuffer()

```
CFE_SB_Buffer_t * CFE_SB_AllocateMessageBuffer (
    size_t MsgSize )
```

Get a buffer pointer to use for "zero copy" SB sends.

Description

This routine can be used to get a pointer to one of the software bus' internal memory buffers that are used for sending messages. The caller can use this memory buffer to build an SB message, then send it using the [CFE_SB_TransmitBuffer\(\)](#) function. This interface avoids an extra copy of the message from the user's memory buffer to the software bus internal buffer.

Assumptions, External Events, and Notes:

1. The pointer returned by [CFE_SB_AllocateMessageBuffer\(\)](#) is only good for one call to [CFE_SB_TransmitBuffer\(\)](#).
2. Once a buffer has been successfully transmitted (as indicated by a successful return from [CFE_SB_TransmitBuffer\(\)](#)) the buffer becomes owned by the SB application. It will automatically be freed by SB once all recipients have finished reading it.
3. Applications must not de-reference the message pointer (for reading or writing) after the call to [CFE_SB_TransmitBuffer\(\)](#).
4. If [CFE_SB_ReleaseMessageBuffer](#) should be used only if a message is not transmitted

Parameters

in	<i>MsgSize</i>	The size of the SB message buffer the caller wants (including the SB message header).
----	----------------	---

Returns

A pointer to a memory buffer that message data can be written to for use with [CFE_SB_TransmitBuffer\(\)](#).

CFE_SB_ReleaseMessageBuffer()

```
CFE_Status_t CFE_SB_ReleaseMessageBuffer (
    CFE_SB_Buffer_t * BufPtr )
```

Release an unused "zero copy" buffer pointer.

Description

This routine can be used to release a pointer to one of the software bus' internal memory buffers.

Assumptions, External Events, and Notes:

1. This function is not needed for normal "zero copy" transfers. It is needed only for cleanup when an application gets a pointer using [CFE_SB_AllocateMessageBuffer\(\)](#), but (due to some error condition) never uses that pointer in a call to [CFE_SB_TransmitBuffer\(\)](#).

Parameters

in	BufPtr	A pointer to the SB internal buffer (must not be null). This must be a pointer returned by a call to CFE_SB_AllocateMessageBuffer() , but never used in a call to CFE_SB_TransmitBuffer() .
----	--------	---

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_SB_BUFFER_INVALID	Buffer Invalid.

CFE_SB_TransmitBuffer()

```
CFE_Status_t CFE_SB_TransmitBuffer (
    CFE_SB_Buffer_t * BufPtr,
    bool IsOrigination )
```

Transmit a buffer.

Description

This routine sends a message that has been created directly in an internal SB message buffer by an application (after a call to [CFE_SB_AllocateMessageBuffer\(\)](#)). This interface is more complicated than the normal [CFE_SB_TransmitMsg](#) interface, but it avoids an extra copy of the message from the user's memory buffer to the software bus internal buffer. The "zero copy" interface can be used to improve performance in high-rate, high-volume software bus traffic.

The IsOrigination parameter should be passed as "true" if the message was newly constructed by the sender and is being sent for the first time. This enables the message origination actions as determined by the CFE MSG module, which may include (but not limited to):

- Updating sequence number

- Updating timestamp
- Calculating a CRC, checksum, or other message error control field

Conversely, when forwarding a message that originated from an external entity (e.g. messages passing through CI or SBN), the parameter should be passed as "false" to not overwrite existing data.

Assumptions, External Events, and Notes:

1. A handle returned by [CFE_SB_AllocateMessageBuffer](#) is "consumed" by a *successful* call to [CFE_SB_TransmitBuffer](#).
2. If this function returns CFE_SUCCESS, this indicates the zero copy handle is now owned by software bus, and is no longer owned by the calling application, and should not be re-used.
3. However if this function fails (returns any error status) it does not change the state of the buffer at all, meaning the calling application still owns it. (a failure means the buffer is left in the same state it was before the call).
4. Applications should be written as if [CFE_SB_AllocateMessageBuffer](#) is equivalent to a `malloc()` and a successful call to [CFE_SB_TransmitBuffer](#) is equivalent to a `free()`.
5. Applications must not de-reference the message pointer (for reading or writing) after a successful call to [CFE_SB_TransmitBuffer](#).
6. This function will increment and apply the internally tracked sequence counter if set to do so.

Parameters

in	<i>BufPtr</i>	A pointer to the buffer to be sent (must not be null).
in	<i>IsOrigination</i>	Update applicable header field(s) of a newly constructed message

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_SB_BAD_ARGUMENT	Bad Argument.
CFE_SB_MSG_TOO_BIG	Message Too Big.

9.28 cFE Message Characteristics APIs

Functions

- void [CFE_SB_SetUserDataLength](#) ([CFE_MSG_Message_t](#) *MsgPtr, [size_t](#) DataLength)
Sets the length of user data in a software bus message.
- void [CFE_SB_TimeStampMsg](#) ([CFE_MSG_Message_t](#) *MsgPtr)
Sets the time field in a software bus message with the current spacecraft time.
- [int32](#) [CFE_SB_MessageStringSet](#) (char *DestStringPtr, const char *SourceStringPtr, [size_t](#) DestMaxSize, [size_t](#) SourceMaxSize)
Copies a string into a software bus message.
- void * [CFE_SB_GetUserData](#) ([CFE_MSG_Message_t](#) *MsgPtr)
Get a pointer to the user data portion of a software bus message.

- `size_t CFE_SB_GetUserDataLength` (const `CFE_MSG_Message_t` *MsgPtr)
Gets the length of user data in a software bus message.
- `int32 CFE_SB_MessageStringGet` (char *DestStringPtr, const char *SourceStringPtr, const char *DefaultString, size_t DestMaxSize, size_t SourceMaxSize)
Copies a string out of a software bus message.

9.28.1 Detailed Description

9.28.2 Function Documentation

CFE_SB_GetUserData()

```
void * CFE_SB_GetUserData (
    CFE_MSG_Message_t * MsgPtr )
```

Get a pointer to the user data portion of a software bus message.

Description

This routine returns a pointer to the user data portion of a software bus message. SB message header formats can be different for each deployment of the cFE. So, applications should use this function and avoid hard coding offsets into their SB message buffers.

Assumptions, External Events, and Notes:

None

Parameters

in	<i>MsgPtr</i>	A pointer to the buffer that contains the software bus message (must not be null).
----	---------------	--

Returns

A pointer to the first byte of user data within the software bus message.

CFE_SB_GetUserDataLength()

```
size_t CFE_SB_GetUserDataLength (
    const CFE_MSG_Message_t * MsgPtr )
```

Gets the length of user data in a software bus message.

Description

This routine returns the size of the user data in a software bus message.

Assumptions, External Events, and Notes:

None

Parameters

in	<i>MsgPtr</i>	A pointer to the buffer that contains the software bus message (must not be null). This must point to the first byte of the message header.
----	---------------	---

Returns

The size (in bytes) of the user data in the software bus message.

Return values

0	if an error occurs, such as if the MsgPtr argument is not valid.
---	--

CFE_SB_MessageStringGet()

```
int32 CFE_SB_MessageStringGet (
    char * DestStringPtr,
    const char * SourceStringPtr,
    const char * DefaultString,
    size_t DestMaxSize,
    size_t SourceMaxSize )
```

Copies a string out of a software bus message.

Description

Strings within software bus messages have a defined/fixed maximum length, and may not necessarily be null terminated within the message. This presents a possible issue when using the C library functions to copy strings out of a message.

This function should replace use of C library functions such as strcpy/strncpy when copying strings out of software bus messages to local storage buffers.

Up to [SourceMaxSize] or [DestMaxSize-1] (whichever is smaller) characters will be copied from the source buffer to the destination buffer, and a NUL termination character will be written to the destination buffer as the last character.

If the DefaultString pointer is non-NULL, it will be used in place of the source string if the source is an empty string. This is typically a string constant that comes from the platform configuration, allowing default values to be assumed for fields that are unspecified.

IMPORTANT - the default string, if specified, must be null terminated. This will be the case if a string literal is passed in (the typical/expected use case).

If the default is NULL, then only the source string will be copied, and the result will be an empty string if the source was empty.

If the destination buffer is too small to store the entire string, it will be truncated, but it will still be null terminated.

Parameters

out	<i>DestStringPtr</i>	Pointer to destination buffer (must not be null)
in	<i>SourceStringPtr</i>	Pointer to source buffer (component of SB message definition)
in	<i>DefaultString</i>	Default string to use if source is empty
in	<i>DestMaxSize</i>	Size of destination storage buffer (must not be zero)
in	<i>SourceMaxSize</i>	Size of source buffer as defined by the message definition

Returns

Number of characters copied or error code, see [cFE Return Code Defines](#)

Return values

CFE_SB_BAD_ARGUMENT	Bad Argument.
-------------------------------------	---------------

CFE_SB_MessageStringSet()

```
int32 CFE_SB_MessageStringSet (
    char * DestStringPtr,
    const char * SourceStringPtr,
    size_t DestMaxSize,
    size_t SourceMaxSize )
```

Copies a string into a software bus message.

Description

Strings within software bus messages have a defined/fixed maximum length, and may not necessarily be null terminated within the message. This presents a possible issue when using the C library functions to copy strings out of a message.

This performs a very similar function to "strncpy()" except that the sizes of *both* buffers are passed in. Neither buffer is required to be null-terminated, but copying will stop after the first termination character is encountered.

If the destination buffer is not completely filled by the source data (such as if the supplied string was shorter than the allotted length) the destination buffer will be padded with NUL characters up to the size of the buffer, similar to what strncpy() does. This ensures that the entire destination buffer is set.

Note

If the source string buffer is already guaranteed to be null terminated, then there is no difference between the C library "strncpy()" function and this implementation. It is only necessary to use this when termination of the source buffer is not guaranteed.

Parameters

out	<i>DestStringPtr</i>	Pointer to destination buffer (component of SB message definition) (must not be null)
in	<i>SourceStringPtr</i>	Pointer to source buffer (must not be null)
in	<i>DestMaxSize</i>	Size of destination buffer as defined by the message definition
in	<i>SourceMaxSize</i>	Size of source buffer

Returns

Number of characters copied or error code, see [cFE Return Code Defines](#)

Return values

CFE_SB_BAD_ARGUMENT	Bad Argument.
-------------------------------------	---------------

CFE_SB_SetUserDataLength()

```
void CFE_SB_SetUserDataLength (
    CFE_MSG_Message_t * MsgPtr,
    size_t DataLength )
```

Sets the length of user data in a software bus message.

Description

This routine sets the field in the SB message header that determines the size of the user data in a software bus message. SB message header formats can be different for each deployment of the cFE. So, applications should use this function rather than trying to poke a length value directly into their SB message buffers.

Assumptions, External Events, and Notes:

- You must set a valid message ID in the SB message header before calling this function.

Parameters

in	<i>MsgPtr</i>	A pointer to the buffer that contains the software bus message (must not be null). This must point to the first byte of the message header.
in	<i>DataLength</i>	The length to set (size of the user data, in bytes).

CFE_SB_TimeStampMsg()

```
void CFE_SB_TimeStampMsg (
    CFE_MSG_Message_t * MsgPtr )
```

Sets the time field in a software bus message with the current spacecraft time.

Description

This routine sets the time of a software bus message with the current spacecraft time. This will be the same time that is returned by the function [CFE_TIME_GetTime](#).

Assumptions, External Events, and Notes:

- If the underlying implementation of software bus messages does not include a time field, then this routine will do nothing.

Parameters

in	<i>MsgPtr</i>	A pointer to the buffer that contains the software bus message (must not be null). This must point to the first byte of the message header.
----	---------------	---

9.29 cFE Message ID APIs**Functions**

- bool [CFE_SB_IsValidMsgId](#) (CFE_SB_MsgId_t MsgId)
Identifies whether a given CFE_SB_MsgId_t is valid.
- static bool [CFE_SB_MsgId_Equal](#) (CFE_SB_MsgId_t MsgId1, CFE_SB_MsgId_t MsgId2)
Identifies whether two CFE_SB_MsgId_t values are equal.
- static CFE_SB_MsgId_Atom_t [CFE_SB_MsgIdToValue](#) (CFE_SB_MsgId_t MsgId)
Converts a CFE_SB_MsgId_t to a normal integer.
- static CFE_SB_MsgId_t [CFE_SB_ValueToMsgId](#) (CFE_SB_MsgId_Atom_t MsgIdValue)
Converts a normal integer into a CFE_SB_MsgId_t.
- CFE_SB_MsgId_Atom_t [CFE_SB_CmdTopicIdToMsgId](#) (uint16 TopicId, uint16 InstanceNum)
Converts a topic ID and instance number combination into a MsgID value integer.
- CFE_SB_MsgId_Atom_t [CFE_SB_TlmTopicIdToMsgId](#) (uint16 TopicId, uint16 InstanceNum)
Converts a topic ID and instance number combination into a MsgID value integer.
- CFE_SB_MsgId_Atom_t [CFE_SB_GlobalCmdTopicIdToMsgId](#) (uint16 TopicId)
Converts a topic ID to a MsgID value integer for Global commands.

- [CFE_SB_MsgId_Atom_t CFE_SB_GlobalTlmTopicIdToMsgId \(uint16 TopicId\)](#)
Converts a topic ID to a MsgID value integer for Global telemetry.
- [CFE_SB_MsgId_Atom_t CFE_SB_LocalCmdTopicIdToMsgId \(uint16 TopicId\)](#)
Converts a topic ID to a MsgID value integer for local commands.
- [CFE_SB_MsgId_Atom_t CFE_SB_LocalTlmTopicIdToMsgId \(uint16 TopicId\)](#)
Converts a topic ID to a MsgID value integer for local telemetry.

9.29.1 Detailed Description

9.29.2 Function Documentation

CFE_SB_CmdTopicIdToMsgId()

```
CFE_SB_MsgId_Atom_t CFE_SB_CmdTopicIdToMsgId (
    uint16 TopicId,
    uint16 InstanceNum )
```

Converts a topic ID and instance number combination into a MsgID value integer.

Description

This function accepts a data pair of topic ID + instance number and returns the corresponding MsgID Value (integer) for commands.

Assumptions and Notes:

A topic ID identifies a certain data stream from an application, for example the CFE Software bus ground commands (CFE_MISSION_SB_CMD_TOPICID). In contrast to MsgID, the topic ID is consistent across all CPUs in a system, whereas each CPU instance will have a unique MsgID.

CPU instance numbers are 1-based. The instance number of 0 is reserved for "global" MsgID values that are the same on all CPUs. The PSP function may be used to obtain the current CPU number for the host processor.

See also

[CFE_SB_TlmTopicIdToMsgId\(\)](#), [CFE_PSP_GetProcessorId\(\)](#)

Returns

Integer representation of the [CFE_SB_MsgId_t](#)

CFE_SB_GlobalCmdTopicIdToMsgId()

```
CFE_SB_MsgId_Atom_t CFE_SB_GlobalCmdTopicIdToMsgId (
    uint16 TopicId )
```

Converts a topic ID to a MsgID value integer for Global commands.

Description

This is a wrapper around [CFE_SB_CmdTopicIdToMsgId\(\)](#) for topic IDs which are the same on all CPUs within a system (i.e. not specific to a certain processor)

Assumptions and Notes:

Global MsgIDs may be used when only a single instance of a service exists within the system. The CFE framework does not use this feature for commands, but is defined for future use.

See also

[CFE_SB_CmdTopicIdToMsgId\(\)](#), [CFE_SB_LocalCmdTopicIdToMsgId\(\)](#)

Returns

Integer representation of the [CFE_SB_MsgId_t](#)

CFE_SB_GlobalTlmTopicIdToMsgId()

```
CFE_SB_MsgId_t CFE_SB_GlobalTlmTopicIdToMsgId (
    uint16 TopicId )
```

Converts a topic ID to a MsgID value integer for Global telemetry.

Description

This is a wrapper around [CFE_SB_TlmTopicIdToMsgId\(\)](#) for topic IDs which are the same on all CPUs within a system (i.e. not specific to a certain processor)

Assumptions and Notes:

Global MsgIDs may be used when only a single instance of a service exists within the system. An example for such telemetry is the time synchronization service published by CFE_TIME.

See also

[CFE_SB_TlmTopicIdToMsgId\(\)](#), [CFE_SB_LocalTlmTopicIdToMsgId\(\)](#)

Returns

Integer representation of the [CFE_SB_MsgId_t](#)

CFE_SB_IsValidMsgId()

```
bool CFE_SB_IsValidMsgId (
    CFE_SB_MsgId_t MsgId )
```

Identifies whether a given [CFE_SB_MsgId_t](#) is valid.

Description

Implements a basic sanity check on the value provided

Returns

Boolean message ID validity indicator

Return values

<i>true</i>	Message ID is within the valid range
<i>false</i>	Message ID is not within the valid range

CFE_SB_LocalCmdTopicIdToMsgId()

```
CFE_SB_MsgId_Atom_t CFE_SB_LocalCmdTopicIdToMsgId (
    uint16 TopicId )
```

Converts a topic ID to a MsgID value integer for local commands.

Description

This is a wrapper around [CFE_SB_CmdTopicIdToMsgId\(\)](#) for topic IDs which are unique on all CPUs within a system (i.e. specific to a certain processor)

Assumptions and Notes:

This assumes the caller is referring to a service running on the same processor instance as itself.

See also

[CFE_SB_CmdTopicIdToMsgId\(\)](#), [CFE_SB_LocalTlmTopicIdToMsgId\(\)](#)

Returns

Integer representation of the [CFE_SB_MsgId_t](#)

CFE_SB_LocalTlmTopicIdToMsgId()

```
CFE_SB_MsgId_Atom_t CFE_SB_LocalTlmTopicIdToMsgId (
    uint16 TopicId )
```

Converts a topic ID to a MsgID value integer for local telemetry.

Description

This is a wrapper around [CFE_SB_TlmTopicIdToMsgId\(\)](#) for topic IDs which are unique on all CPUs within a system (i.e. specific to a certain processor)

Assumptions and Notes:

This assumes the caller is referring to a service running on the same processor instance as itself.

See also

[CFE_SB_TlmTopicIdToMsgId\(\)](#), [CFE_SB_LocalCmdTopicIdToMsgId\(\)](#)

Returns

Integer representation of the [CFE_SB_MsgId_t](#)

CFE_SB_MsgId_Equal()

```
static bool CFE_SB_MsgId_Equal (
    CFE_SB_MsgId_t MsgId1,
    CFE_SB_MsgId_t MsgId2 ) [inline], [static]
```

Identifies whether two [CFE_SB_MsgId_t](#) values are equal.

Description

In cases where the `CFE_SB_MsgId_t` type is not a simple integer type, it may not be possible to do a direct equality check. This inline function provides an abstraction for the equality check between two `CFE_SB_MsgId_t` values.

Applications should transition to using this function to compare MsgId values for equality to remain compatible with future versions of cFE.

Returns

Boolean message ID equality indicator

Return values

<i>true</i>	Message IDs are Equal
<i>false</i>	Message IDs are not Equal

Definition at line 796 of file `cfe_sb.h`.

References `CFE_SB_MSGID_UNWRAP_VALUE`.

CFE_SB_MsgIdToValue()

```
static CFE_SB_MsgId_Atom_t CFE_SB_MsgIdToValue (
    CFE_SB_MsgId_t MsgId ) [inline], [static]
```

Converts a `CFE_SB_MsgId_t` to a normal integer.

Description

In cases where the `CFE_SB_MsgId_t` type is not a simple integer type, it is not possible to directly display the value in a printf-style statement, use it in a switch() statement, or other similar use cases.

This inline function provides the ability to map a `CFE_SB_MsgId_t` type back into a simple integer value.

Applications should transition to using this function wherever a `CFE_SB_MsgId_t` type needs to be used as an integer.

Assumptions and Notes:

This negates the type safety that was gained by using a non- integer type for the `CFE_SB_MsgId_t` value. This should only be used in specific cases such as UI display (printf, events, etc) where the value is being sent externally. Any internal API calls should be updated to use the `CFE_SB_MsgId_t` type directly, rather than an integer type.

Returns

Integer representation of the `CFE_SB_MsgId_t`

Definition at line 827 of file `cfe_sb.h`.

References `CFE_SB_MSGID_UNWRAP_VALUE`.

CFE_SB_TlmTopicIdToMsgId()

```
CFE_SB_MsgId_Atom_t CFE_SB_TlmTopicIdToMsgId (
    uint16 TopicId,
    uint16 InstanceNum )
```

Converts a topic ID and instance number combination into a MsgID value integer.

Description

This function accepts a data pair of topic ID + instance number and returns the corresponding MsgID Value (integer) for telemetry.

Assumptions and Notes:

A topic ID identifies a certain data stream from an application, for example the CFE Software bus housekeeping telemetry (CFE_MISSION_SB_HK_TLM_TOPICID). In contrast to MsgID, the topic ID is consistent across all CPUs in a system, whereas each CPU instance will have a unique MsgID.

CPU instance numbers are 1-based. The instance number of 0 is reserved for "global" MsgID values that are the same on all CPUs. The PSP function may be used to obtain the current CPU number for the host processor.

See also

[CFE_SB_CmdTopicIdToMsgId\(\)](#), [CFE_PSP_GetProcessorId\(\)](#)

Returns

Integer representation of the [CFE_SB_MsgId_t](#)

CFE_SB_ValueToMsgId()

```
static CFE_SB_MsgId_t CFE_SB_ValueToMsgId (
    CFE_SB_MsgId_Atom_t MsgIdValue ) [inline], [static]
```

Converts a normal integer into a [CFE_SB_MsgId_t](#).

Description

In cases where the [CFE_SB_MsgId_t](#) type is not a simple integer type, it is not possible to directly use an integer value supplied via a define or similar method.

This inline function provides the ability to map an integer value into a corresponding [CFE_SB_MsgId_t](#) value. Applications should transition to using this function wherever an integer needs to be used for a [CFE_SB_MsgId_t](#).

Assumptions and Notes:

This negates the type safety that was gained by using a non- integer type for the [CFE_SB_MsgId_t](#) value. This should only be used in specific cases where the value is coming from an external source. Any internal API calls should be updated to return the [CFE_SB_MsgId_t](#) type directly, rather than an integer type.

Returns

[CFE_SB_MsgId_t](#) representation of the integer

Definition at line 856 of file [cfe_sb.h](#).

References [CFE_SB_MSGID_WRAP_VALUE](#).

9.30 cFE SB Pipe options

Macros

- `#define CFE_SB_PIPEOPTS_IGNOREMINE 0x00000001`
Messages sent by the app that owns this pipe will not be sent to this pipe.

9.30.1 Detailed Description

9.30.2 Macro Definition Documentation

CFE_SB_PIPEOPTS_IGNOREMINE

```
#define CFE_SB_PIPEOPTS_IGNOREMINE 0x00000001
```

Messages sent by the app that owns this pipe will not be sent to this pipe.

Definition at line 129 of file [cfe_sb_api_typedefs.h](#).

9.31 cFE Registration APIs

Functions

- [CFE_Status_t CFE_TBL_Register](#) ([CFE_TBL_Handle_t](#) *TblHandlePtr, const char *Name, size_t Size, [uint16](#) TblOptionFlags, [CFE_TBL_CallbackFuncPtr_t](#) TblValidationFuncPtr)
Register a table with cFE to obtain Table Management Services.
- [CFE_Status_t CFE_TBL_Share](#) ([CFE_TBL_Handle_t](#) *TblHandlePtr, const char *TblName)
Obtain handle of table registered by another application.
- [CFE_Status_t CFE_TBL_Unregister](#) ([CFE_TBL_Handle_t](#) TblHandle)
Unregister a table.

9.31.1 Detailed Description

9.31.2 Function Documentation

CFE_TBL_Register()

```
CFE_Status_t CFE_TBL_Register (
    CFE_TBL_Handle_t * TblHandlePtr,
    const char * Name,
    size_t Size,
    uint16 TblOptionFlags,
    CFE_TBL_CallbackFuncPtr_t TblValidationFuncPtr )
```

Register a table with cFE to obtain Table Management Services.

Description

When an application is created and initialized, it is responsible for creating its table images via the TBL API. The application must inform the Table Service of the table name, table size and selection of optional table features.

Assumptions, External Events, and Notes:

Note: This function call can block. Therefore, interrupt service routines should NOT create their own tables. An application should create any table(s) and provide the handle(s) to the interrupt service routine.

Parameters

out	<i>TblHandlePtr</i>	a pointer to a CFE_TBL_Handle_t type variable (must not be null) that will be assigned the table's handle. The table handle is required for other API calls when accessing the data contained in the table. *TblHandlePtr is the handle used to identify table to cFE when performing Table operations. This value is returned at address specified by TblHandlePtr.
in	<i>Name</i>	The raw table name. This name will be combined with the name of the application to produce a name of the form "AppName.RawTableName". This application specific name will be used in commands for modifying or viewing the contents of the table.
in	<i>Size</i>	The size, in bytes, of the table to be created (must not be zero). This is the size that will be allocated as a shared memory resource between the Table Management Service and the calling application.

Parameters

in	<i>TblOptionFlags</i>	Flag bits indicating selected options for table. A bitwise OR of the following option flags: • CFE_TBL_OPT_DEFAULT - The default setting for table options is a combination of CFE_TBL_OPT_SNGL_BUFFER and CFE_TBL_OPT_LOAD_DUMP. See below for a description of these two options. This option is mutually exclusive with the CFE_TBL_OPT_DBL_BUFFER, CFE_TBL_OPT_DUMP_ONLY and CFE_TBL_OPT_USR_DEF_ADDR options. • CFE_TBL_OPT_SNGL_BUFFER - When this option is selected, the table will use a shared session table for performing table modifications and a memory copy from the session table to the "active" table buffer will occur when the table is updated. This is the preferred option since it will minimize memory usage. This option is mutually exclusive with the CFE_TBL_OPT_DBL_BUFFER option • CFE_TBL_OPT_DBL_BUFFER - When this option is selected, two instances of the table are created. One is considered the "active" table and the other the "inactive" table. Whenever table modifications occur, they do not require the use of a common session table. Modifications occur in the "inactive" buffer. Then, when it is time to update the table, the pointer to the "active" table is changed to point to the "inactive" buffer thus making it the new "active" buffer. This feature is most useful for time critical applications (ie - interrupt service routines, etc). This option is mutually exclusive with the CFE_TBL_OPT_SNGL_BUFFER and CFE_TBL_OPT_DEFAULT option. • CFE_TBL_OPT_LOAD_DUMP - When this option is selected, the Table Service is allowed to perform all operations on the specified table. This option is mutually exclusive with the CFE_TBL_OPT_DUMP_ONLY option. • CFE_TBL_OPT_DUMP_ONLY - When this option is selected, the Table Service will not perform table loads to this table. This does not prevent, however, a task from writing to the table via an address obtained with the CFE_TBL_GetAddress API function. This option is mutually exclusive with the CFE_TBL_OPT_LOAD_DUMP and CFE_TBL_OPT_DEFAULT options. If the Application wishes to specify their own block of memory as the Dump Only table, they need to also include the CFE_TBL_OPT_USR_DEF_ADDR option explained below. • CFE_TBL_OPT_NOT_USR_DEF - When this option is selected, Table Services allocates memory for the table and, in the case of a double buffered table, it allocates the same amount of memory again for the second buffer. This option is mutually exclusive with the CFE_TBL_OPT_USR_DEF_ADDR option. • CFE_TBL_OPT_USR_DEF_ADDR - When this option is selected, the Table Service will not allocate memory for the table. Table Services will require the Application to identify the location of the active table buffer via the CFE_TBL_Load function. This option implies the CFE_TBL_OPT_DUMP_ONLY and the CFE_TBL_OPT_SNGL_BUFFER options and is mutually exclusive of the CFE_TBL_OPT_DBL_BUFFER option. • CFE_TBL_OPT_CRITICAL - When this option is selected, the Table Service will automatically allocate space in the Critical Data Store (CDS) for the table and ensure that the contents in the CDS are the same as the contents of the currently active buffer for the table. This option is mutually exclusive of the CFE_TBL_OPT_USR_DEF_ADDR and CFE_TBL_OPT_DUMP_ONLY options. It should also be noted that the use of this option with double buffered tables will prevent the update of the double buffered table from being quick and it could be blocked. Therefore, critical tables should not be
		of the currently active buffer for the table. This option is mutually exclusive of the CFE_TBL_OPT_USR_DEF_ADDR and CFE_TBL_OPT_DUMP_ONLY options. It should also be noted that the use of this option with double buffered tables will prevent the update of the double buffered table from being quick and it could be blocked. Therefore, critical tables should not be

Parameters

in	<i>TblValidationFuncPtr</i>	is a pointer to a function that will be executed in the context of the Table Management Service when the contents of a table need to be validated. If set to NULL, then the Table Management Service will assume any data is valid. If the value is not NULL, it must be a pointer to a function with the following prototype: int32 CallbackFunc(void *TblPtr); where TblPtr will be a pointer to the table data that is to be verified. When the function returns CFE_SUCCESS, the data is considered valid and ready for a commit. When the function returns a negative value, the data is considered invalid and an Event Message will be issued containing the returned value. If the function should return a positive number, the table is considered invalid and the return code is considered invalid. Validation functions must return either CFE_SUCCESS or a negative number (whose value is at the developer's discretion). The validation function will be executed in the Application's context so that Event Messages describing the validation failure are possible from within the function.
----	-----------------------------	---

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_TBL_INFO_RECOVERED_TBL	Recovered Table.
CFE_TBL_ERR_DUPLICATE_DIFF_SIZE	Duplicate Table With Different Size.
CFE_TBL_ERR_DUPLICATE_NOT_OWNED	Duplicate Table And Not Owned.
CFE_TBL_ERR_REGISTRY_FULL	Registry Full.
CFE_TBL_ERR_HANDLES_FULL	Handles Full.
CFE_TBL_ERR_INVALID_SIZE	Invalid Size.
CFE_TBL_ERR_INVALID_NAME	Invalid Name.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_TBL_BAD_ARGUMENT	Bad Argument.
CFE_TBL_ERR_INVALID_OPTIONS	Invalid Options.
CFE_TBL_WARN_DUPLICATE	Duplicate Warning.
CFE_TBL_WARN_NOT_CRITICAL	Not Critical Warning.

See also

[CFE_TBL_Unregister](#), [CFE_TBL_Share](#)

CFE_TBL_Share()

```
CFE_Status_t CFE_TBL_Share (
    CFE_TBL_Handle_t * TblHandlePtr,
    const char * TblName )
```

Obtain handle of table registered by another application.

Description

After a table has been created, other applications can gain access to that table via the table handle. In order for two or more applications to share a table, the applications that do not create the table must obtain the handle using this function.

Assumptions, External Events, and Notes:

None

Parameters

out	<i>TblHandlePtr</i>	A pointer to a CFE_TBL_Handle_t type variable (must not be null) that will be assigned the table's handle. The table handle is required for other API calls when accessing the data contained in the table. *TblHandlePtr is the handle used to identify table to cFE when performing Table operations. This value is returned at the address specified by TblHandlePtr.
in	<i>TblName</i>	The application specific name of the table of the form "AppName.RawTableName", where RawTableName is the name specified in the CFE_TBL_Register API call. Example: "ACS.TamParams" for a table called "TamParams" that was registered by the application called "ACS".

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_TBL_ERR_HANDLES_FULL	Handles Full.
CFE_TBL_ERR_INVALID_NAME	Invalid Name.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_TBL_BAD_ARGUMENT	Bad Argument.

See also

[CFE_TBL_Unregister](#), [CFE_TBL_Register](#)

CFE_TBL_Unregister()

```
CFE_Status_t CFE_TBL_Unregister (
    CFE_TBL_Handle_t TblHandle )
```

Unregister a table.

Description

When an application is being removed from the system, ES will clean up/free all the application related resources including tables so apps are not required to call this function.

A valid use-case for this API is to unregister a shared table if access is no longer needed or the owning application was removed from the system (CS app is an example).

Typically apps should only register tables during initialization and registration/unregistration by the owning application during operation should be avoided. If unavoidable, special care needs to be taken (especially for shared tables) to avoid race conditions due to competing requests from multiple tasks.

Note the table will not be removed from memory until all table access links have been removed (registration and all shared access).

Assumptions, External Events, and Notes:

None

Parameters

in	<i>TblHandle</i>	Handle, previously obtained from CFE_TBL_Register or CFE_TBL_Share , that identifies the Table to be unregistered.
----	------------------	--

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_TBL_ERR_NO_ACCESS	No Access.
CFE_TBL_ERR_INVALID_HANDLE	Invalid Handle.

See also

[CFE_TBL_Share](#), [CFE_TBL_Register](#)

9.32 cFE Manage Table Content APIs

Functions

- [CFE_Status_t CFE_TBL_Load](#) ([CFE_TBL_Handle_t](#) TblHandle, [CFE_TBL_SrcEnum_t](#) SrcType, const void *SrcDataPtr)
Load a specified table with data from specified source.
- [CFE_Status_t CFE_TBL_Update](#) ([CFE_TBL_Handle_t](#) TblHandle)
Update contents of a specified table, if an update is pending.
- [CFE_Status_t CFE_TBL_Validate](#) ([CFE_TBL_Handle_t](#) TblHandle)
Perform steps to validate the contents of a table image.
- [CFE_Status_t CFE_TBL_Manage](#) ([CFE_TBL_Handle_t](#) TblHandle)
Perform standard operations to maintain a table.
- [CFE_Status_t CFE_TBL_DumpToBuffer](#) ([CFE_TBL_Handle_t](#) TblHandle)
Copies the contents of a Dump Only Table to a shared buffer.
- [CFE_Status_t CFE_TBL_Modified](#) ([CFE_TBL_Handle_t](#) TblHandle)
Notify cFE Table Services that table contents have been modified by the Application.

9.32.1 Detailed Description

9.32.2 Function Documentation

CFE_TBL_DumpToBuffer()

```
CFE_Status_t CFE_TBL_DumpToBuffer (
    CFE_TBL_Handle_t TblHandle )
```

Copies the contents of a Dump Only Table to a shared buffer.

Description

Typically, apps should just call [CFE_TBL_Manage](#) as part of routine processing which will perform validation, update, or dump if pending. This API is provided for the case where just a dump should be performed.

Assumptions, External Events, and Notes:

If the table does not have a dump pending status, nothing will occur (no error, no dump)

Parameters

in	<i>TblHandle</i>	Handle of Table to be dumped.
----	------------------	-------------------------------

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_TBL_ERR_NO_ACCESS	No Access.
CFE_TBL_ERR_INVALID_HANDLE	Invalid Handle.
CFE_TBL_INFO_DUMP_PENDING	Dump Pending.

See also

[CFE_TBL_Manage](#)

CFE_TBL_Load()

```
CFE_Status_t CFE_TBL_Load (
    CFE_TBL_Handle_t TblHandle,
    CFE_TBL_SrcEnum_t SrcType,
    const void * SrcDataPtr )
```

Load a specified table with data from specified source.

Description

Once an application has created a table ([CFE_TBL_Register](#)), it must provide the values that initialize the contents of that table. The application accomplishes this with one of two different TBL API calls. This function call initializes the table with values that are held in a data structure.

Assumptions, External Events, and Notes:

This function call can block. Therefore, interrupt service routines should NOT initialize their own tables. An application should initialize any table(s) prior to providing the handle(s) to the interrupt service routine.

Parameters

in	<i>TblHandle</i>	Handle, previously obtained from CFE_TBL_Register or CFE_TBL_Share , that identifies the Table to be loaded.
in	<i>SrcType</i>	Flag indicating the nature of the given <i>SrcDataPtr</i> below. This value can be any one of the following: • CFE_TBL_SRC_FILE - File source When this option is selected, the <i>SrcDataPtr</i> will be interpreted as a pointer to a null terminated character string. The string should specify the full path and filename of the file containing the initial data contents of the table. • CFE_TBL_SRC_ADDRESS - Address source When this option is selected, the <i>SrcDataPtr</i> will be interpreted as a pointer to a memory location that is the beginning of the initialization data for loading the table OR, in the case of a "user defined" dump only table, the address of the active table itself. The block of memory is assumed to be of the same size specified in the CFE_TBL_Register function Size parameter.
in	<i>SrcDataPtr</i>	Pointer (must not be null) to either a character string specifying a filename or a memory address of a block of binary data to be loaded into a table or, if the table was registered with the CFE_TBL_OPT_USR_DEF_ADDR option, the address of the active table buffer.

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_TBL_ERR_NO_ACCESS	No Access.
CFE_TBL_ERR_INVALID_HANDLE	Invalid Handle.
CFE_TBL_ERR_DUMP_ONLY	Dump Only Error.
CFE_TBL_ERR_ILLEGAL_SRC_TYPE	Illegal Source Type.
CFE_TBL_ERR_LOAD_IN_PROGRESS	Load In Progress.
CFE_TBL_ERR_LOAD_INCOMPLETE	Load Incomplete.
CFE_TBL_ERR_NO_BUFFER_AVAIL	No Buffer Available.
CFE_TBL_ERR_ACCESS	
CFE_TBL_ERR_FILE_TOO_LARGE	File Too Large.
CFE_TBL_ERR_BAD_CONTENT_ID	Bad Content ID.
CFE_TBL_ERR_BAD_SUBTYPE_ID	Bad Subtype ID.
CFE_TBL_ERR_NO_STD_HEADER	No Standard Header.
CFE_TBL_ERR_NO_TBL_HEADER	No Table Header.
CFE_TBL_ERR_PARTIAL_LOAD	Partial Load Error.
CFE_TBL_BAD_ARGUMENT	Bad Argument.

See also

[CFE_TBL_Update](#), [CFE_TBL_Validate](#), [CFE_TBL_Manage](#)

CFE_TBL_Manage()

```
CFE_Status_t CFE_TBL_Manage (
    CFE_TBL_Handle_t TblHandle )
```

Perform standard operations to maintain a table.

Description

Applications should call this API periodically to process pending requests for update, validation, or dump to buffer. Typically, the application that created the table would call this function at the start or conclusion of any routine processing cycle.

Assumptions, External Events, and Notes:

None

Parameters

in	<i>TblHandle</i>	Handle, previously obtained from CFE_TBL_Register or CFE_TBL_Share , that identifies the Table to be managed.
----	------------------	---

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_TBL_INFO_UPDATED	Updated.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_TBL_ERR_NO_ACCESS	No Access.
CFE_TBL_ERR_INVALID_HANDLE	Invalid Handle.
CFE_TBL_INFO_DUMP_PENDING	Dump Pending.
CFE_TBL_INFO_UPDATE_PENDING	Update Pending.
CFE_TBL_INFO_VALIDATION_PENDING	

See also

[CFE_TBL_Update](#), [CFE_TBL_Validate](#), [CFE_TBL_Load](#), [CFE_TBL_DumpToBuffer](#)

CFE_TBL_Modified()

```
CFE_Status_t CFE_TBL_Modified (
    CFE_TBL_Handle_t TblHandle )
```

Notify cFE Table Services that table contents have been modified by the Application.

Description

This API notifies Table Services that the contents of the specified table has been modified by the Application. This notification is important when a table has been registered as "Critical" because Table Services can then update the contents of the table kept in the Critical Data Store.

Assumptions, External Events, and Notes:

None

Parameters

in	<i>TblHandle</i>	Handle of Table that was modified.
----	------------------	------------------------------------

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_TBL_ERR_NO_ACCESS	No Access.
CFE_TBL_ERR_INVALID_HANDLE	Invalid Handle.

See also

[CFE_TBL_Manage](#)

CFE_TBL_Update()

```
CFE_Status_t CFE_TBL_Update (
    CFE_TBL_Handle_t TblHandle )
```

Update contents of a specified table, if an update is pending.

Description

Typically, apps should just call [CFE_TBL_Manage](#) as part of routine processing which will perform validation, update, or dump if pending. This API is provided for the case where just an update should be performed.

Assumptions, External Events, and Notes:

None

Parameters

in	<i>TblHandle</i>	Handle, previously obtained from CFE_TBL_Register or CFE_TBL_Share , that identifies the Table to be updated.
----	------------------	---

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_TBL_INFO_NO_UPDATE_PENDING	No Update Pending.

Return values

<code>CFE_ES_ERR_RESOURCEID_NOT_VALID</code>	Resource ID is not valid.
<code>CFE_TBL_ERR_NO_ACCESS</code>	No Access.
<code>CFE_TBL_ERR_INVALID_HANDLE</code>	Invalid Handle.

See also

[CFE_TBL_Load](#), [CFE_TBL_Validate](#), [CFE_TBL_Manage](#)

CFE_TBL_Validate()

```
CFE_Status_t CFE_TBL_Validate (
    CFE_TBL_Handle_t TblHandle )
```

Perform steps to validate the contents of a table image.

Description

Typically, apps should just call [CFE_TBL_Manage](#) as part of routine processing which will perform validation, update, or dump if pending. This API is provided for the case where just a validation should be performed.

Assumptions, External Events, and Notes:

None

Parameters

in	<i>TblHandle</i>	Handle, previously obtained from CFE_TBL_Register or CFE_TBL_Share , that identifies the Table to be managed.
----	------------------	---

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

<code>CFE_SUCCESS</code>	Successful execution.
<code>CFE_TBL_INFO_NO_VALIDATION_PENDING</code>	
<code>CFE_ES_ERR_RESOURCEID_NOT_VALID</code>	Resource ID is not valid.
<code>CFE_TBL_ERR_NO_ACCESS</code>	No Access.
<code>CFE_TBL_ERR_INVALID_HANDLE</code>	Invalid Handle.

See also

[CFE_TBL_Update](#), [CFE_TBL_Manage](#), [CFE_TBL_Load](#)

9.33 cFE Access Table Content APIs**Functions**

- [`CFE\_Status\_t CFE\_TBL\_GetAddress`](#) (void **TblPtr, [CFE_TBL_Handle_t](#) TblHandle)

Obtain the current address of the contents of the specified table.

- [CFE_Status_t CFE_TBL_ReleaseAddress](#) ([CFE_TBL_Handle_t](#) TblHandle)

Release previously obtained pointer to the contents of the specified table.

- [CFE_Status_t CFE_TBL_GetAddresses](#) (void **TblPtrs[], [uint16](#) NumTables, const [CFE_TBL_Handle_t](#) TblHandles[])

Obtain the current addresses of an array of specified tables.

- [CFE_Status_t CFE_TBL_ReleaseAddresses](#) ([uint16](#) NumTables, const [CFE_TBL_Handle_t](#) TblHandles[])

Release the addresses of an array of specified tables.

9.33.1 Detailed Description

9.33.2 Function Documentation

CFE_TBL_GetAddress()

```
CFE_Status_t CFE_TBL_GetAddress (
    void ** TblPtr,
    CFE_TBL_Handle_t TblHandle )
```

Obtain the current address of the contents of the specified table.

Description

When a table has been created and initialized, it is available to any application that can identify it with its unique handle. In order to view the data contained in the table, an application must call this function or [CFE_TBL_GetAddresses](#).

Assumptions, External Events, and Notes:

1. This call can be a blocking call when the table is not double buffered and is shared with another application of lower priority that just happens to be in the middle of a table update of the specific table. If this occurs, the application performing the table update will automatically have its priority elevated in order to release the resource as soon as possible.
2. An application must always release the returned table address using the [CFE_TBL_ReleaseAddress](#) or [CFE_TBL_ReleaseAddresses](#) function prior to either a [CFE_TBL_Update](#) call or any blocking call (e.g. - pending on software bus message, etc). Table updates cannot occur while table addresses have not been released.
3. [CFE_TBL_ERR_NEVER_LOADED](#) will be returned if the table has never been loaded (either from file or from a block of memory), but the function will still return a valid table pointer to a table with all zero content. This pointer must be released with the [CFE_TBL_ReleaseAddress](#) API before the table can be loaded with data.

Parameters

out	<i>TblPtr</i>	The address of a pointer (must not be null) that will be loaded with the address of the first byte of the table. This pointer can then be typecast by the calling application to the appropriate table data structure. *TblPtr is the address of the first byte of data associated with the specified table.
in	<i>TblHandle</i>	Handle, previously obtained from CFE_TBL_Register or CFE_TBL_Share , that identifies the Table whose address is to be returned.

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_TBL_INFO_UPDATED	Updated.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_TBL_ERR_NO_ACCESS	No Access.
CFE_TBL_ERR_INVALID_HANDLE	Invalid Handle.
CFE_TBL_ERR_UNREGISTERED	Unregistered.
CFE_TBL_ERR_NEVER_LOADED	Never Loaded.
CFE_TBL_BAD_ARGUMENT	Bad Argument.

See also

[CFE_TBL_ReleaseAddress](#), [CFE_TBL_GetAddresses](#), [CFE_TBL_ReleaseAddresses](#)

CFE_TBL_GetAddresses()

```
CFE_Status_t CFE_TBL_GetAddresses (
    void ** TblPtrs[],
    uint16 NumTables,
    const CFE_TBL_Handle_t TblHandles[] )
```

Obtain the current addresses of an array of specified tables.

Description

When a table has been created and initialized, it is available to any application that can identify it with its unique handle. In order to view the data contained in the table, an application must call this function or [CFE_TBL_GetAddress](#).

Assumptions, External Events, and Notes:

1. This call can be a blocking call when the table is not double buffered and is shared with another application of lower priority that just happens to be in the middle of a table update of the specific table. If this occurs, the application performing the table update will automatically have its priority elevated in order to release the resource as soon as possible.
2. An application must always release the returned table address using the [CFE_TBL_ReleaseAddress](#) or [CFE_TBL_ReleaseAddresses](#) function prior to either a [CFE_TBL_Update](#) call or any blocking call (e.g. - pending on software bus message, etc). Table updates cannot occur while table addresses have not been released.
3. [CFE_TBL_ERR_NEVER_LOADED](#) will be returned if the table has never been loaded (either from file or from a block of memory), but the function will still return a valid table pointer to a table with all zero content. This pointer must be released with the [CFE_TBL_ReleaseAddress](#) API before the table can be loaded with data.

Parameters

out	<i>TblPtrs</i>	Array of Pointers (must not be null) to variables that calling Application wishes to hold the start addresses of the Tables. *TblPtrs is an array of addresses of the first byte of data associated with the specified tables.
in	<i>NumTables</i>	Size of TblPtrs and TblHandles arrays.
in	<i>TblHandles</i>	Array of Table Handles, previously obtained from CFE_TBL_Register or CFE_TBL_Share , of those tables whose start addresses are to be obtained.

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_TBL_INFO_UPDATED	Updated.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_TBL_ERR_NO_ACCESS	No Access.
CFE_TBL_ERR_INVALID_HANDLE	Invalid Handle.
CFE_TBL_ERR_UNREGISTERED	Unregistered.
CFE_TBL_ERR_NEVER_LOADED	Never Loaded.
CFE_TBL_BAD_ARGUMENT	Bad Argument.

See also

[CFE_TBL_GetAddress](#), [CFE_TBL_ReleaseAddress](#), [CFE_TBL_ReleaseAddresses](#)

CFE_TBL_ReleaseAddress()

```
CFE_Status_t CFE_TBL_ReleaseAddress (
    CFE_TBL_Handle_t TblHandle )
```

Release previously obtained pointer to the contents of the specified table.

Description

Each application is **required** to release a table address obtained through the [CFE_TBL_GetAddress](#) function.

Assumptions, External Events, and Notes:

An application must always release the returned table address using the [CFE_TBL_ReleaseAddress](#) function prior to either a [CFE_TBL_Update](#) call or any blocking call (e.g. - pending on software bus message, etc). Table updates cannot occur while table addresses have not been released.

Parameters

in	<i>TblHandle</i>	Handle, previously obtained from CFE_TBL_Register or CFE_TBL_Share , that identifies the Table whose address is to be released.
----	------------------	---

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_TBL_INFO_UPDATED	Updated.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_TBL_ERR_NO_ACCESS	No Access.
CFE_TBL_ERR_INVALID_HANDLE	Invalid Handle.
CFE_TBL_ERR_NEVER_LOADED	Never Loaded.

See also

[CFE_TBL_GetAddress](#), [CFE_TBL_GetAddresses](#), [CFE_TBL_ReleaseAddresses](#)

CFE_TBL_ReleaseAddresses()

```
CFE_Status_t CFE_TBL_ReleaseAddresses (
    uint16 NumTables,
    const CFE_TBL_Handle_t TblHandles[] )
```

Release the addresses of an array of specified tables.

Description

Each application is **required** to release a table address obtained through the [CFE_TBL_GetAddress](#) function.

Assumptions, External Events, and Notes:

An application must always release the returned table address using the [CFE_TBL_ReleaseAddress](#) function prior to either a [CFE_TBL_Update](#) call or any blocking call (e.g. - pending on software bus message, etc). Table updates cannot occur while table addresses have not been released.

Parameters

in	<i>NumTables</i>	Size of TblHandles array.
in	<i>TblHandles</i>	Array of Table Handles (must not be null), previously obtained from CFE_TBL_Register or CFE_TBL_Share , of those tables whose start addresses are to be released.

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_TBL_INFO_UPDATED	Updated.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_TBL_ERR_NO_ACCESS	No Access.
CFE_TBL_ERR_INVALID_HANDLE	Invalid Handle.

Return values

CFE_TBL_ERR_NEVER_LOADED	Never Loaded.
CFE_TBL_BAD_ARGUMENT	Bad Argument.

See also

[CFE_TBL_GetAddress](#), [CFE_TBL_ReleaseAddress](#), [CFE_TBL_GetAddresses](#)

9.34 cFE Get Table Information APIs

Functions

- [CFE_Status_t CFE_TBL_GetStatus](#) ([CFE_TBL_Handle_t](#) TblHandle)
Obtain current status of pending actions for a table.
- [CFE_Status_t CFE_TBL_GetInfo](#) ([CFE_TBL_Info_t](#) *TblInfoPtr, const char *TblName)
Obtain characteristics/information of/about a specified table.
- [CFE_Status_t CFE_TBL_NotifyByMessage](#) ([CFE_TBL_Handle_t](#) TblHandle, [CFE_SB_MsgId_t](#) MsgId, [CFE_MSG_FcnCode_t](#) CommandCode, [uint32](#) Parameter)
Instruct cFE Table Services to notify Application via message when table requires management.

9.34.1 Detailed Description

9.34.2 Function Documentation

CFE_TBL_GetInfo()

```
CFE_Status_t CFE_TBL_GetInfo (
    CFE_TBL_Info_t * TblInfoPtr,
    const char * TblName )
```

Obtain characteristics/information of/about a specified table.

Description

This API provides the registry information associated with the specified table. The function fills the given data structure with the data found in the Table Registry.

Assumptions, External Events, and Notes:

None

Parameters

out	<i>TblInfoPtr</i>	A pointer to a CFE_TBL_Info_t data structure (must not be null) that is to be populated with table characteristics and information. *TblInfoPtr is the description of the tables characteristics and registry information stored in the CFE_TBL_Info_t data structure format.
in	<i>TblName</i>	The application specific name (must not be null) of the table of the form "AppName.RawTableName", where RawTableName is the name specified in the CFE_TBL_Register API call. Example: "ACS.TamParams" for a table called "TamParams" that was registered by the application called "ACS".

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_TBL_ERR_INVALID_NAME	Invalid Name.
CFE_TBL_BAD_ARGUMENT	Bad Argument.

See also

[CFE_TBL_GetStatus](#)

CFE_TBL_GetStatus()

```
CFE_Status_t CFE_TBL_GetStatus (
    CFE_TBL_Handle_t TblHandle )
```

Obtain current status of pending actions for a table.

Description

An application is **required** to perform a periodic check for an update or a validation request for all the tables that it creates. Typically, the application that created the table would call this function at the start or conclusion of any routine processing cycle. If a table update or validation request is pending, the Application should follow up with a call to [CFE_TBL_Update](#) or [CFE_TBL_Validate](#) respectively.

Assumptions, External Events, and Notes:

None

Parameters

in	<i>TblHandle</i>	Handle, previously obtained from CFE_TBL_Register or CFE_TBL_Share , that identifies the Table to be managed.
----	------------------	---

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_TBL_INFO_UPDATE_PENDING	Update Pending.
CFE_TBL_INFO_VALIDATION_PENDING	
CFE_TBL_INFO_DUMP_PENDING	Dump Pending.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_TBL_ERR_NO_ACCESS	No Access.
CFE_TBL_ERR_INVALID_HANDLE	Invalid Handle.

Note

Some status return codes are "success" while being non-zero. This behavior will change in the future.

See also

[CFE_TBL_Manage](#), [CFE_TBL_Update](#), [CFE_TBL_Validate](#), [CFE_TBL_GetInfo](#)

CFE_TBL_NotifyByMessage()

```
CFE_Status_t CFE_TBL_NotifyByMessage (
    CFE_TBL_Handle_t TblHandle,
    CFE_SB_MsgId_t MsgId,
    CFE_MSG_FcnCode_t CommandCode,
    uint32 Parameter )
```

Instruct cFE Table Services to notify Application via message when table requires management.

Description

This API instructs Table Services to send a message to the calling Application whenever the specified table requires management by the application. This feature allows applications to avoid polling table services via the [CFE_TBL_Manage](#) call to determine whether a table requires updates, validation, etc. This API should be called following the [CFE_TBL_Register](#) API whenever the owning application requires this feature.

Assumptions, External Events, and Notes:

- Only the application that owns the table is allowed to register a notification message
- Recommend **NOT** using the ground command MID which typically impacts command counters. The typical approach is to use a unique MID for inter-task communications similar to how schedulers typically trigger application housekeeping messages.

Parameters

in	<i>TblHandle</i>	Handle of Table with which the message should be associated.
in	<i>MsgId</i>	Message ID to be used in notification message sent by Table Services.
in	<i>CommandCode</i>	Command Code value to be placed in secondary header of message sent by Table Services.
in	<i>Parameter</i>	Application defined value to be passed as a parameter in the message sent by Table Services. Suggested use includes an application's table index that allows the same MsgId and Command Code to be used for all table management notifications.

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_ES_ERR_RESOURCEID_NOT_VALID	Resource ID is not valid.
CFE_TBL_ERR_NO_ACCESS	No Access.
CFE_TBL_ERR_INVALID_HANDLE	Invalid Handle.

See also

[CFE_TBL_Register](#)

9.35 cFE Table Type Defines

Macros

- #define [CFE_TBL_OPT_BUFFER_MSK](#) (0x0001)
Table buffer mask.
- #define [CFE_TBL_OPT_SNGL_BUFFER](#) (0x0000)
Single buffer table.
- #define [CFE_TBL_OPT_DBL_BUFFER](#) (0x0001)
Double buffer table.
- #define [CFE_TBL_OPT_LD_DMP_MSK](#) (0x0002)
Table load/dump mask.
- #define [CFE_TBL_OPT_LOAD_DUMP](#) (0x0000)
Load/Dump table.
- #define [CFE_TBL_OPT_DUMP_ONLY](#) (0x0002)
Dump only table.
- #define [CFE_TBL_OPT_USR_DEF_MSK](#) (0x0004)
Table user defined mask.
- #define [CFE_TBL_OPT_NOT_USR_DEF](#) (0x0000)
Not user defined table.
- #define [CFE_TBL_OPT_USR_DEF_ADDR](#) (0x0006)
User Defined table,.
- #define [CFE_TBL_OPT_CRITICAL_MSK](#) (0x0008)
Table critical mask.
- #define [CFE_TBL_OPT_NOT_CRITICAL](#) (0x0000)
Not critical table.
- #define [CFE_TBL_OPT_CRITICAL](#) (0x0008)
Critical table.
- #define [CFE_TBL_OPT_DEFAULT](#) ([CFE_TBL_OPT_SNGL_BUFFER](#) | [CFE_TBL_OPT_LOAD_DUMP](#))
Default table options.

9.35.1 Detailed Description

9.35.2 Macro Definition Documentation

CFE_TBL_OPT_BUFFER_MSK

```
#define CFE_TBL_OPT_BUFFER_MSK (0x0001)
```

Table buffer mask.
Definition at line 49 of file [cfe_tbl_api_typedefs.h](#).

CFE_TBL_OPT_CRITICAL

```
#define CFE_TBL_OPT_CRITICAL (0x0008)
```

Critical table.
Definition at line 64 of file [cfe_tbl_api_typedefs.h](#).

CFE_TBL_OPT_CRITICAL_MSK

```
#define CFE_TBL_OPT_CRITICAL_MSK (0x0008)
```

Table critical mask.
Definition at line 62 of file [cfe_tbl_api_typedefs.h](#).

CFE_TBL_OPT_DBL_BUFFER

```
#define CFE_TBL_OPT_DBL_BUFFER (0x0001)
```

Double buffer table.
Definition at line 51 of file [cfe_tbl_api_typedefs.h](#).

CFE_TBL_OPT_DEFAULT

```
#define CFE_TBL_OPT_DEFAULT (CFE_TBL_OPT_SNGL_BUFFER | CFE_TBL_OPT_LOAD_DUMP)
```

Default table options.
Definition at line 67 of file [cfe_tbl_api_typedefs.h](#).

CFE_TBL_OPT_DUMP_ONLY

```
#define CFE_TBL_OPT_DUMP_ONLY (0x0002)
```

Dump only table.
Definition at line 55 of file [cfe_tbl_api_typedefs.h](#).

CFE_TBL_OPT_LD_DMP_MSK

```
#define CFE_TBL_OPT_LD_DMP_MSK (0x0002)
```

Table load/dump mask.
Definition at line 53 of file [cfe_tbl_api_typedefs.h](#).

CFE_TBL_OPT_LOAD_DUMP

```
#define CFE_TBL_OPT_LOAD_DUMP (0x0000)
```

Load/Dump table.
Definition at line 54 of file [cfe_tbl_api_typedefs.h](#).

CFE_TBL_OPT_NOT_CRITICAL

```
#define CFE_TBL_OPT_NOT_CRITICAL (0x0000)
```

Not critical table.
Definition at line 63 of file [cfe_tbl_api_typedefs.h](#).

CFE_TBL_OPT_NOT_USR_DEF

```
#define CFE_TBL_OPT_NOT_USR_DEF (0x0000)
```

Not user defined table.
Definition at line 58 of file [cfe_tbl_api_typedefs.h](#).

CFE_TBL_OPT_SNGL_BUFFER

```
#define CFE_TBL_OPT_SNGL_BUFFER (0x0000)
```

Single buffer table.
Definition at line 50 of file [cfe_tbl_api_typedefs.h](#).

CFE_TBL_OPT_USR_DEF_ADDR

```
#define CFE_TBL_OPT_USR_DEF_ADDR (0x0006)  
User Defined table,.
```

Note

Automatically includes [CFE_TBL_OPT_DUMP_ONLY](#) option

Definition at line 60 of file [cfe_tbl_api_typedefs.h](#).

CFE_TBL_OPT_USR_DEF_MSK

```
#define CFE_TBL_OPT_USR_DEF_MSK (0x0004)  
Table user defined mask.  
Definition at line 57 of file cfe\_tbl\_api\_typedefs.h.
```

9.36 cFE Get Current Time APIs

Functions

- [CFE_TIME_SysTime_t CFE_TIME_GetTime](#) (void)
Get the current spacecraft time.
- [CFE_TIME_SysTime_t CFE_TIME_GetTAI](#) (void)
Get the current TAI (MET + SCTF) time.
- [CFE_TIME_SysTime_t CFE_TIME_GetUTC](#) (void)
Get the current UTC (MET + SCTF - Leap Seconds) time.
- [CFE_TIME_SysTime_t CFE_TIME_GetMET](#) (void)
Get the current value of the Mission Elapsed Time (MET).
- [uint32 CFE_TIME_GetMETseconds](#) (void)
Get the current seconds count of the mission-elapsed time.
- [uint32 CFE_TIME_GetMETsubsecs](#) (void)
Get the current sub-seconds count of the mission-elapsed time.

9.36.1 Detailed Description

9.36.2 Function Documentation

CFE_TIME_GetMET()

```
CFE_TIME_SysTime_t CFE_TIME_GetMET (  
    void )
```

Get the current value of the Mission Elapsed Time (MET).

Description

This routine returns the current mission-elapsed time (MET). MET is usually derived from a hardware-based clock that is not adjusted during normal operations. Callers of this routine should not assume that the MET return value has any specific relationship to any ground-based time standard.

Assumptions, External Events, and Notes:

None

Returns

The current MET

See also

[CFE_TIME_GetTime](#), [CFE_TIME_GetTAI](#), [CFE_TIME_GetUTC](#), [CFE_TIME_GetMETseconds](#), [CFE_TIME_GetMETsubsecs](#), [CFE_TIME_MET2SCTime](#)

CFE_TIME_GetMETseconds()

```
uint32 CFE_TIME_GetMETseconds (
    void )
```

Get the current seconds count of the mission-elapsed time.

Description

This routine is the same as [CFE_TIME_GetMET](#), except that it returns only the integer seconds portion of the MET time.

Assumptions, External Events, and Notes:

None

Returns

The current MET seconds

See also

[CFE_TIME_GetTime](#), [CFE_TIME_GetTAI](#), [CFE_TIME_GetUTC](#), [CFE_TIME_GetMET](#), [CFE_TIME_GetMETsubsecs](#), [CFE_TIME_MET2SCTime](#)

CFE_TIME_GetMETsubsecs()

```
uint32 CFE_TIME_GetMETsubsecs (
    void )
```

Get the current sub-seconds count of the mission-elapsed time.

Description

This routine is the same as [CFE_TIME_GetMET](#), except that it returns only the integer sub-seconds portion of the MET time. Each count is equal to $2^{(-32)}$ seconds.

Assumptions, External Events, and Notes:

None

Returns

The current MET sub-seconds

See also

[CFE_TIME_GetTime](#), [CFE_TIME_GetTAI](#), [CFE_TIME_GetUTC](#), [CFE_TIME_GetMET](#), [CFE_TIME_GetMETseconds](#), [CFE_TIME_MET2SCTime](#)

CFE_TIME_GetTAI()

```
CFE_TIME_SysTime_t CFE_TIME_GetTAI (  
    void )
```

Get the current TAI (MET + SCTF) time.

Description

This routine returns the current TAI time to the caller. TAI is an international time standard that does not include leap seconds. This routine should only be used in situations where TAI is absolutely required. Applications that call [CFE_TIME_GetTAI](#) may not be portable to all missions. Maintenance of correct TAI in flight is not guaranteed under all mission operations scenarios. To maintain re-usability across missions, most applications should be using [CFE_TIME_GetTime](#), rather than the specific routines for getting UTC/TAI directly.

Assumptions, External Events, and Notes:

1. The "TAI" time returned is referenced to the mission-defined time epoch, which may or may not be the same as the standard TAI epoch.
2. Even though TAI does not include leap seconds, the time returned by this function can still jump forward or backward without warning when the spacecraft clock is set or adjusted by operators. Applications using this function must be able to handle these time discontinuities gracefully.

Returns

The current spacecraft time in TAI

See also

[CFE_TIME_GetTime](#), [CFE_TIME_GetUTC](#), [CFE_TIME_GetMET](#), [CFE_TIME_GetMETseconds](#), [CFE_TIME_GetMETsubsecs](#)

CFE_TIME_GetTime()

```
CFE_TIME_SysTime_t CFE_TIME_GetTime (  
    void )
```

Get the current spacecraft time.

Description

This routine returns the current spacecraft time, which is the amount of time elapsed since the epoch as set in mission configuration. The time returned is either TAI (no leap seconds) or UTC (including leap seconds). This choice is made in the mission configuration file by defining either [CFE_MISSION_TIME_CFG_DEFAULT_TAI](#) or [CFE_MISSION_TIME_CFG_DEFAULT_UTC](#) as true at compile time. To maintain re-usability across missions, most applications should be using this function rather than the specific routines for getting UTC/TAI directly.

Assumptions, External Events, and Notes:

None

Returns

The current spacecraft time in default format

See also

[CFE_TIME_GetTAI](#), [CFE_TIME_GetUTC](#), [CFE_TIME_GetMET](#), [CFE_TIME_GetMETseconds](#), [CFE_TIME_GetMETsubsecs](#)

CFE_TIME_GetUTC()

```
CFE_TIME_SysTime_t CFE_TIME_GetUTC (
    void )
```

Get the current UTC (MET + SCTF - Leap Seconds) time.

Description

This routine returns the current UTC time to the caller. This routine should only be used in situations where UTC is absolutely required. Applications that call [CFE_TIME_GetUTC](#) may not be portable to all missions. Maintenance of correct UTC in flight is not guaranteed under all mission operations scenarios. If UTC is maintained in flight, it will jump backwards occasionally due to leap second adjustments. To maintain re-usability across missions, most applications should be using [CFE_TIME_GetTime](#), rather than the specific routines for getting UTC/TAI directly.

Assumptions, External Events, and Notes:

Note: The "UTC" time returned is referenced to the mission-defined time epoch, which may or may not be the same as the standard UTC epoch.

Returns

The current spacecraft time in UTC

See also

[CFE_TIME_GetTime](#), [CFE_TIME_GetTAI](#), [CFE_TIME_GetMET](#), [CFE_TIME_GetMETseconds](#), [CFE_TIME_GetMETsubsecs](#)

9.37 cFE Get Time Information APIs**Functions**

- [CFE_TIME_SysTime_t CFE_TIME_GetSTCF](#) (void)
Get the current value of the spacecraft time correction factor (STCF).
- [int16 CFE_TIME_GetLeapSeconds](#) (void)
Get the current value of the leap seconds counter.
- [CFE_TIME_ClockState_Enum_t CFE_TIME_GetClockState](#) (void)
Get the current state of the spacecraft clock.
- [uint16 CFE_TIME_GetClockInfo](#) (void)
Provides information about the spacecraft clock.

9.37.1 Detailed Description**9.37.2 Function Documentation****CFE_TIME_GetClockInfo()**

```
uint16 CFE_TIME_GetClockInfo (
    void )
```

Provides information about the spacecraft clock.

Description

This routine returns information on the spacecraft clock in a bit mask.

Assumptions, External Events, and Notes:

None

Returns

Spacecraft clock information, [cFE Clock State Flag Defines](#). To extract the information from the returned value, the flags can be used as in the following:

if ((ReturnValue & CFE_TIME_FLAG_XXXXXX) == CFE_TIME_FLAG_XXXXXX) then the following definition of the CFE_TIME_FLAG_XXXXXX is true.

See also

[CFE_TIME_GetSTCF](#), [CFE_TIME_GetLeapSeconds](#), [CFE_TIME_GetClockState](#)

CFE_TIME_GetClockState()

```
CFE_TIME_ClockState_Enum_t CFE_TIME_GetClockState (
    void )
```

Get the current state of the spacecraft clock.

Description

This routine returns the spacecraft clock state. Applications that are highly dependent on valid time may want to call this routine before taking actions based on the times returned by the various clock routines

Assumptions, External Events, and Notes:

None

Returns

The current spacecraft clock state

See also

[CFE_TIME_GetSTCF](#), [CFE_TIME_GetLeapSeconds](#), [CFE_TIME_GetClockInfo](#)

CFE_TIME_GetLeapSeconds()

```
int16 CFE_TIME_GetLeapSeconds (
    void )
```

Get the current value of the leap seconds counter.

Description

This routine returns the current value of the leap seconds counter. This is the delta seconds between international atomic time (TAI) and universal coordinated time (UTC). There is no API provided to set or adjust leap seconds or SCTF, those actions should be done by command only. This API is provided for applications to be able to include leap seconds in their data products to aid in time correlation during downstream science data processing. Note that some mission operations teams do not maintain the leap seconds count, preferring to adjust the STCF instead. Users of this function should check with their mission ops team to see how they are planning to handle leap seconds.

Assumptions, External Events, and Notes:

None

Returns

The current spacecraft leap seconds.

See also

[CFE_TIME_GetSTCF](#), [CFE_TIME_GetClockState](#), [CFE_TIME_GetClockInfo](#)

CFE_TIME_GetSTCF()

```
CFE_TIME_SysTime_t CFE_TIME_GetSTCF (
    void )
```

Get the current value of the spacecraft time correction factor (STCF).

Description

This routine returns the current value of the spacecraft time correction factor. This is the delta time between the MET and the TAI time. There is no API provided to set or adjust leap seconds or SCTF, those actions should be done by command only. This API is provided for applications to be able to include STCF in their data products to aid in time correlation during downstream science data processing.

Assumptions, External Events, and Notes:

Does not include leap seconds

Returns

The current SCTF

See also

[CFE_TIME_GetLeapSeconds](#), [CFE_TIME_GetClockState](#), [CFE_TIME_GetClockInfo](#)

9.38 cFE Time Arithmetic APIs**Functions**

- [CFE_TIME_SysTime_t CFE_TIME_Add \(CFE_TIME_SysTime_t Time1, CFE_TIME_SysTime_t Time2\)](#)
Adds two time values.
- [CFE_TIME_SysTime_t CFE_TIME_Subtract \(CFE_TIME_SysTime_t Time1, CFE_TIME_SysTime_t Time2\)](#)
Subtracts two time values.
- [CFE_TIME_Compare_t CFE_TIME_Compare \(CFE_TIME_SysTime_t TimeA, CFE_TIME_SysTime_t TimeB\)](#)
Compares two time values.

9.38.1 Detailed Description**9.38.2 Function Documentation****CFE_TIME_Add()**

```
CFE_TIME_SysTime_t CFE_TIME_Add (
    CFE_TIME_SysTime_t Time1,
    CFE_TIME_SysTime_t Time2 )
```

Adds two time values.

Description

This routine adds the two specified times and returns the result. Normally, at least one of the input times should be a value representing a delta time. Adding two absolute times together will not cause an error, but the result will probably be meaningless.

Assumptions, External Events, and Notes:

None

Parameters

in	<i>Time1</i>	The first time to be added.
in	<i>Time2</i>	The second time to be added.

Returns

The sum of the two times. If the sum is greater than the maximum value that can be stored in a [CFE_TIME_SysTime_t](#), the result will roll over (this is not considered an error).

See also

[CFE_TIME_Subtract](#), [CFE_TIME_Compare](#)

CFE_TIME_Compare()

```
CFE_TIME_Compare_t CFE_TIME_Compare (
    CFE_TIME_SysTime_t TimeA,
    CFE_TIME_SysTime_t TimeB )
```

Compares two time values.

Description

This routine compares two time values to see which is "greater". It is important that applications use this function rather than trying to directly compare the component pieces of times. This function will handle roll-over cases seamlessly, which may not be intuitively obvious. The cFE's internal representation of time "rolls over" when the 32 bit seconds count reaches 0xFFFFFFFF. Also, subtracting a delta time from an absolute time close to the epoch could result in "roll under". The strange cases that result from these situations can be handled by defining the comparison function for times as follows: Plot the two times on the circumference of a circle where 0 is at the top and 0x80000000 is at the bottom. If the shortest arc from time A to time B runs clockwise around the circle, then time A is less than time B. If the shortest arc from A to B runs counter-clockwise, then time A is greater than time B.

Assumptions, External Events, and Notes:

None

Parameters

in	<i>TimeA</i>	The first time to compare.
in	<i>TimeB</i>	The second time to compare.

Returns

The result of comparing the two times.

Return values

CFE_TIME_EQUAL	The two specified times are considered to be equal.
CFE_TIME_A_GT_B	The first specified time is considered to be after the second specified time.
CFE_TIME_A_LT_B	The first specified time is considered to be before the second specified time.

See also

[CFE_TIME_Add](#), [CFE_TIME_Subtract](#)

CFE_TIME_Subtract()

```
CFE_TIME_SysTime_t CFE_TIME_Subtract (
    CFE_TIME_SysTime_t Time1,
    CFE_TIME_SysTime_t Time2 )
```

Subtracts two time values.

Description

This routine subtracts time2 from time1 and returns the result. The time values can represent either absolute or delta times, but not all combinations make sense.

- AbsTime - AbsTime = DeltaTime
- AbsTime - DeltaTime = AbsTime
- DeltaTime - DeltaTime = DeltaTime
- DeltaTime - AbsTime = garbage

Assumptions, External Events, and Notes:

None

Parameters

in	<i>Time1</i>	The base time.
in	<i>Time2</i>	The time to be subtracted from the base time.

Returns

The result of subtracting the two times. If the subtraction results in an underflow, the result will roll over (this is not considered an error).

See also

[CFE_TIME_Add](#), [CFE_TIME_Compare](#)

9.39 cFE Time Conversion APIs

Functions

- [CFE_TIME_SysTime_t CFE_TIME_MET2SCTime](#) ([CFE_TIME_SysTime_t](#) METTime)
Convert specified MET into Spacecraft Time.
- [uint32 CFE_TIME_Sub2MicroSecs](#) ([uint32](#) SubSeconds)
Converts a sub-seconds count to an equivalent number of microseconds.
- [uint32 CFE_TIME_Micro2SubSecs](#) ([uint32](#) MicroSeconds)
Converts a number of microseconds to an equivalent sub-seconds count.

9.39.1 Detailed Description

9.39.2 Function Documentation

CFE_TIME_MET2SCTime()

```
CFE_TIME_SysTime_t CFE_TIME_MET2SCTime (
    CFE_TIME_SysTime_t METTime )
```

Convert specified MET into Spacecraft Time.

Description

This function returns Spacecraft Time given MET. Note that Spacecraft Time is returned as either UTC or TAI depending on whether the mission configuration parameter [CFE_MISSION_TIME_CFG_DEFAULT_UTC](#) or [CFE_MISSION_TIME_CFG_DEFAULT_TAI](#) was set to true at compile time.

Assumptions, External Events, and Notes:

None

Parameters

in	<i>METTime</i>	The MET to be converted.
----	----------------	--------------------------

Returns

Spacecraft Time (UTC or TAI) corresponding to the specified MET

See also

[CFE_TIME_GetMET](#), [CFE_TIME_GetMETseconds](#), [CFE_TIME_GetMETsubsecs](#), [CFE_TIME_Sub2MicroSecs](#), [CFE_TIME_Micro2SubSecs](#)

CFE_TIME_Micro2SubSecs()

```
uint32 CFE_TIME_Micro2SubSecs (
    uint32 MicroSeconds )
```

Converts a number of microseconds to an equivalent sub-seconds count.

Description

This routine converts from microseconds (each tick is 1e-06 seconds) to a subseconds count (each tick is $1 / 2^{32}$ seconds).

Assumptions, External Events, and Notes:

None

Parameters

in	<i>MicroSeconds</i>	The sub-seconds count to convert.
----	---------------------	-----------------------------------

Returns

The equivalent number of subseconds. If the number of microseconds passed in is greater than one second, (i.e. > 999,999), the return value is equal to 0xffffffff.

See also

[CFE_TIME_MET2SCTime](#), [CFE_TIME_Sub2MicroSecs](#),

CFE_TIME_Sub2MicroSecs()

```
uint32 CFE_TIME_Sub2MicroSecs (
    uint32 SubSeconds )
```

Converts a sub-seconds count to an equivalent number of microseconds.

Description

This routine converts from a sub-seconds count (each tick is 1 / 2³² seconds) to microseconds (each tick is 1e-06 seconds).

Assumptions, External Events, and Notes:

None

Parameters

in	<i>SubSeconds</i>	The sub-seconds count to convert.
----	-------------------	-----------------------------------

Returns

The equivalent number of microseconds.

See also

[CFE_TIME_MET2SCTime](#), [CFE_TIME_Micro2SubSecs](#),

9.40 cFE External Time Source APIs**Functions**

- void [CFE_TIME_ExternalTone](#) (void)
Provides the 1 Hz signal from an external source.
- void [CFE_TIME_ExternalMET](#) (CFE_TIME_SysTime_t NewMET)
Provides the Mission Elapsed Time from an external source.
- void [CFE_TIME_ExternalGPS](#) (CFE_TIME_SysTime_t NewTime, int16 NewLeaps)
Provide the time from an external source that has data common to GPS receivers.
- void [CFE_TIME_ExternalTime](#) (CFE_TIME_SysTime_t NewTime)
Provide the time from an external source that measures time relative to a known epoch.
- CFE_Status_t [CFE_TIME_RegisterSynchCallback](#) (CFE_TIME_SynchCallbackPtr_t CallbackFuncPtr)
Registers a callback function that is called whenever time synchronization occurs.
- CFE_Status_t [CFE_TIME_UnregisterSynchCallback](#) (CFE_TIME_SynchCallbackPtr_t CallbackFuncPtr)
Unregisters a callback function that is called whenever time synchronization occurs.

9.40.1 Detailed Description

9.40.2 Function Documentation

CFE_TIME_ExternalGPS()

```
void CFE_TIME_ExternalGPS (
    CFE_TIME_SysTime_t NewTime,
    int16 NewLeaps )
```

Provide the time from an external source that has data common to GPS receivers.

Description

This routine provides a method to provide cFE TIME with current time data acquired from an external source. There is a presumption that this function will be called at the appropriate time (relative to the tone) such that this call may be used by cFE TIME as the signal to generate the "time at the tone" data command. The "time at the tone" data command must arrive within the configuration parameter specified window for tone signal and data packet verification.

Internally, cFE TIME will calculate a new STCF as the difference between this new time value and the spacecraft MET value at the tone. This allows cFE TIME to always calculate time as the sum of MET and STCF. The value of STCF will change only as much as the drift factor between spacecraft MET and the external time source.

Assumptions, External Events, and Notes:

- This routine is included in the API only when 3 specific configuration parameters are set to true. The first is [CFE_PLATFORM_TIME_CFG_SERVER](#) which defines this instantiation of cFE TIME as a time server (not a client). The second required configuration parameter is [CFE_PLATFORM_TIME_CFG_SOURCE](#) which enables time source selection commands to the cFE TIME task, and further enables configuration definitions for the selected type of external time data. The third configuration parameter required for this routine is [CFE_PLATFORM_TIME_CFG_SRC_GPS](#), which indicates that the external time data consists of a time value relative to a known epoch, plus a leap seconds value.

Parameters

in	<i>NewTime</i>	The MET value at the next (or previous) 1 Hz tone signal.
in	<i>NewLeaps</i>	The Leap Seconds value used to calculate time as UTC.

See also

[CFE_TIME_ExternalTone](#), [CFE_TIME_ExternalMET](#), [CFE_TIME_ExternalTime](#)

CFE_TIME_ExternalMET()

```
void CFE_TIME_ExternalMET (
    CFE_TIME_SysTime_t NewMET )
```

Provides the Mission Elapsed Time from an external source.

Description

This routine provides a method to provide cFE TIME with MET acquired from an external source. There is a presumption that this function will be called at the appropriate time (relative to the tone) such that this call may be used by cFE TIME as the signal to generate the "time at the tone" data command. The "time at the tone" data command must arrive within the configuration parameter specified window for tone signal and data packet verification.

The MET value at the tone "should" have zero subseconds. Although the interface accepts non-zero values for sub-seconds, it may be harmful to other applications that expect zero subseconds at the moment of the tone. Any decision to use non-zero subseconds should be carefully considered.

Assumptions, External Events, and Notes:

- This routine is included in the API only when 3 specific configuration parameters are set to true. The first is [CFE_PLATFORM_TIME_CFG_SERVER](#) which defines this instantiation of cFE TIME as a time server (not a client). The second required configuration parameter is [CFE_PLATFORM_TIME_CFG_SOURCE](#) which enables time source selection commands to the cFE TIME task, and further enables configuration definitions for the selected type of external time data. The third configuration parameter required for this routine is [CFE_PLATFORM_TIME_CFG_SRC_MET](#), which indicates that the external time data consists of MET.

Parameters

in	<i>NewMET</i>	The MET value at the next (or previous) 1 Hz tone signal.
----	---------------	---

See also

[CFE_TIME_ExternalTone](#), [CFE_TIME_ExternalGPS](#), [CFE_TIME_ExternalTime](#)

CFE_TIME_ExternalTime()

```
void CFE_TIME_ExternalTime (
    CFE_TIME_SysTime_t NewTime )
```

Provide the time from an external source that measures time relative to a known epoch.

Description

This routine provides a method to provide cFE TIME with current time data acquired from an external source. There is a presumption that this function will be called at the appropriate time (relative to the tone) such that this call may be used by cFE TIME as the signal to generate the "time at the tone" data command. The "time at the tone" data command must arrive within the configuration specified window for tone signal and data packet verification.

Internally, cFE TIME will calculate a new STCF as the difference between this new time value and the spacecraft MET value at the tone. This allows cFE TIME to always calculate time as the sum of MET and STCF. The value of STCF will change only as much as the drift factor between spacecraft MET and the external time source.

Assumptions, External Events, and Notes:

- This routine is included in the API only when 3 specific configuration parameters are set to true. The first is [CFE_PLATFORM_TIME_CFG_SERVER](#) which defines this instantiation of cFE TIME as a time server (not a client). The second required configuration parameter is [CFE_PLATFORM_TIME_CFG_SOURCE](#) which enables time source selection commands to the cFE TIME task, and further enables configuration definitions for the selected type of external time data. The third configuration parameter required for this routine is [CFE_PLATFORM_TIME_CFG_SRC_TIME](#), which indicates that the external time data consists of a time value relative to a known epoch.

Parameters

in	<i>NewTime</i>	The MET value at the next (or previous) 1 Hz tone signal.
----	----------------	---

See also

[CFE_TIME_ExternalTone](#), [CFE_TIME_ExternalMET](#), [CFE_TIME_ExternalGPS](#)

CFE_TIME_ExternalTone()

```
void CFE_TIME_ExternalTone (
    void )
```

Provides the 1 Hz signal from an external source.

Description

This routine provides a method for cFE TIME software to be notified of the occurrence of the 1Hz tone signal without knowledge of the specific hardware design. Regardless of the source of the tone, this routine should be called as soon as possible after detection to allow cFE TIME software the opportunity to latch the local clock as close as possible to the instant of the tone.

Assumptions, External Events, and Notes:

- This routine may be called directly from within the context of an interrupt handler.

See also

[CFE_TIME_ExternalMET](#), [CFE_TIME_ExternalGPS](#), [CFE_TIME_ExternalTime](#)

CFE_TIME_RegisterSynchCallback()

```
CFE_Status_t CFE_TIME_RegisterSynchCallback (
    CFE_TIME_SynchCallbackPtr_t CallbackFuncPtr )
```

Registers a callback function that is called whenever time synchronization occurs.

Description

This routine passes a callback function pointer for an Application that wishes to be notified whenever a legitimate time synchronization signal (typically a 1 Hz) is received.

Assumptions, External Events, and Notes:

Only a single callback per application is supported, and this function should only be called from a single thread within each application (typically the apps main thread). If an application requires triggering multiple child tasks at 1Hz, it should distribute the timing signal internally, rather than registering for multiple callbacks.

Parameters

in	<i>CallbackFuncPtr</i>	Function to call at synchronization interval (must not be null)
----	------------------------	---

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
-----------------------------	-----------------------

Return values

CFE_TIME_TOO_MANY_SYNC_CALLBACKS	Too Many Sync Callbacks.
CFE_TIME_BAD_ARGUMENT	Bad Argument.

See also

[CFE_TIME_UnregisterSynchCallback](#)

CFE_TIME_UnregisterSynchCallback()

```
CFE_Status_t CFE_TIME_UnregisterSynchCallback (
    CFE_TIME_SynchCallbackPtr_t CallbackFuncPtr )
```

Unregisters a callback function that is called whenever time synchronization occurs.

Description

This routine removes the specified callback function pointer from the list of Callback functions that are called whenever a time synchronization (typically the 1Hz signal) is received.

Assumptions, External Events, and Notes:

Only a single callback per application is supported, and this function should only be called from a single thread within each application (typically the apps main thread).

Parameters

in	CallbackFuncPtr	Function to remove from synchronization call list (must not be null)
----	---------------------------------	--

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_TIME_CALLBACK_NOT_REGISTERED	Callback Not Registered.
CFE_TIME_BAD_ARGUMENT	Bad Argument.

See also

[CFE_TIME_RegisterSynchCallback](#)

9.41 cFE Miscellaneous Time APIs**Functions**

- [CFE_Status_t CFE_TIME_Print](#) (char *PrintBuffer, [CFE_TIME_SysTime_t](#) TimeToPrint)
Print a time value as a string.
- void [CFE_TIME_Local1HzISR](#) (void)
This function is called via a timer callback set up at initialization of the TIME service.

9.41.1 Detailed Description

9.41.2 Function Documentation

CFE_TIME_Local1HzISR()

```
void CFE_TIME_Local1HzISR (
    void )
```

This function is called via a timer callback set up at initialization of the TIME service.

Description

Drives the time processing logic from the system PSP layer. This must be called once per second based on a hardware interrupt or OS kernel signal.

Assumptions, External Events, and Notes:

This will update the global data structures accordingly, incrementing each by the 1Hz amount.

CFE_TIME_Print()

```
CFE_Status_t CFE_TIME_Print (
    char * PrintBuffer,
    CFE_TIME_SysTime_t TimeToPrint )
```

Print a time value as a string.

Description

This routine prints the specified time to the specified string buffer in the following format:

yyyy-ddd-hh:mm:ss.xxxxx\0

where:

- `YYYY` = year
- `ddd` = Julian day of the year
- `hh` = hour of the day (0 to 23)
- `mm` = minute (0 to 59)
- `ss` = second (0 to 59)
- `xxxxx` = subsecond formatted as a decimal fraction (1/4 second = 0.25000)
- `\0` = trailing null

Assumptions, External Events, and Notes:

- The value of the time argument is simply added to the configuration definitions for the ground epoch and converted into a fixed length string in the buffer provided by the caller.
- A loss of data during the string conversion will occur if the computed year exceeds 9999. However, a year that large would require an unrealistic definition for the ground epoch since the maximum amount of time represented by a `CFE_TIME_SysTime` structure is approximately 136 years.

Parameters

out	<i>PrintBuffer</i>	Pointer to a character array (must not be null) of at least CFE_TIME_PRINTED_STRING_SIZE characters in length. *PrintBuffer is the time as a character string as described above.
in	<i>TimeToPrint</i>	The time to print into the character array.

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

CFE_SUCCESS	Successful execution.
CFE_TIME_BAD_ARGUMENT	Bad Argument.

9.42 cFE Resource ID base values

Enumerations

- enum {
[CFE_RESOURCEID_ES_TASKID_BASE_OFFSET](#) = OS_OBJECT_TYPE_OS_TASK , [CFE_RESOURCEID_ES_APPID_BASE_C](#)
= OS_OBJECT_TYPE_USER + 1 , [CFE_RESOURCEID_ES_LIBID_BASE_OFFSET](#) = OS_OBJECT_TYPE_↵
USER + 2 , [CFE_RESOURCEID_ES_COUNTID_BASE_OFFSET](#) = OS_OBJECT_TYPE_USER + 3 ,
[CFE_RESOURCEID_ES_POOLID_BASE_OFFSET](#) = OS_OBJECT_TYPE_USER + 4 , [CFE_RESOURCEID_ES_CDSBLOCKID_](#)
= OS_OBJECT_TYPE_USER + 5 , [CFE_RESOURCEID_SB_PIPEID_RESOURCE_BASE_OFFSET](#) = OS_↵
OBJECT_TYPE_USER + 6 , [CFE_RESOURCEID_CONFIGID_BASE_OFFSET](#) = OS_OBJECT_TYPE_USER
+ 7 ,
[CFE_RESOURCEID_TBL_VALRESULTID_BASE_OFFSET](#) = OS_OBJECT_TYPE_USER + 8 , [CFE_RESOURCEID_TBL_DUMP](#)
= OS_OBJECT_TYPE_USER + 9 , [CFE_RESOURCEID_TBL_LOADBUFFID_BASE_OFFSET](#) = OS_↵
OBJECT_TYPE_USER + 10 , [CFE_RESOURCEID_TBL_ACCESSID_BASE_OFFSET](#) = OS_OBJECT_↵
TYPE_USER + 11 ,
[CFE_RESOURCEID_TBL_REGID_BASE_OFFSET](#) = OS_OBJECT_TYPE_USER + 12 }
- enum {
[CFE_ES_TASKID_BASE](#) = CFE_RESOURCEID_MAKE_BASE(CFE_RESOURCEID_ES_TASKID_BASE_↵
OFFSET) , [CFE_ES_APPID_BASE](#) = CFE_RESOURCEID_MAKE_BASE(CFE_RESOURCEID_ES_APPID_↵
_BASE_OFFSET) , [CFE_ES_LIBID_BASE](#) = CFE_RESOURCEID_MAKE_BASE(CFE_RESOURCEID_↵
_ES_LIBID_BASE_OFFSET) , [CFE_ES_COUNTID_BASE](#) = CFE_RESOURCEID_MAKE_BASE(CFE_↵
RESOURCEID_ES_COUNTID_BASE_OFFSET) ,
[CFE_ES_POOLID_BASE](#) = CFE_RESOURCEID_MAKE_BASE(CFE_RESOURCEID_ES_POOLID_BASE_↵
_OFFSET) , [CFE_ES_CDSBLOCKID_BASE](#) = CFE_RESOURCEID_MAKE_BASE(CFE_RESOURCEID_↵
ES_CDSBLOCKID_BASE_OFFSET) , [CFE_SB_PIPEID_BASE](#) = CFE_RESOURCEID_MAKE_BASE(CFE_↵
RESOURCEID_SB_PIPEID_RESOURCE_BASE_OFFSET) , [CFE_CONFIGID_BASE](#) = CFE_RESOURCEID_↵
_MAKE_BASE(CFE_RESOURCEID_CONFIGID_BASE_OFFSET) ,
[CFE_TBL_VALRESULTID_BASE](#) = CFE_RESOURCEID_MAKE_BASE(CFE_RESOURCEID_TBL_VALRESULTID_↵
_BASE_OFFSET) , [CFE_TBL_DUMPCTRLID_BASE](#) = CFE_RESOURCEID_MAKE_BASE(CFE_RESOURCEID_↵
_TBL_DUMPCTRLID_BASE_OFFSET) , [CFE_TBL_LOADBUFFID_BASE](#) = CFE_RESOURCEID_MAKE_↵
_BASE(CFE_RESOURCEID_TBL_LOADBUFFID_BASE_OFFSET) , [CFE_TBL_HANDLE_BASE](#) = CFE_↵
RESOURCEID_MAKE_BASE(CFE_RESOURCEID_TBL_ACCESSID_BASE_OFFSET) ,
[CFE_TBL_REGID_BASE](#) = CFE_RESOURCEID_MAKE_BASE(CFE_RESOURCEID_TBL_REGID_BASE_↵
OFFSET) }

9.42.1 Detailed Description

9.42.2 Enumeration Type Documentation

anonymous enum

anonymous enum

Enumerator

CFE_RESOURCEID_ES_TASKID_BASE_OFFSET	
CFE_RESOURCEID_ES_APPID_BASE_OFFSET	
CFE_RESOURCEID_ES_LIBID_BASE_OFFSET	
CFE_RESOURCEID_ES_COUNTID_BASE_OFFSET	
CFE_RESOURCEID_ES_POOLID_BASE_OFFSET	
CFE_RESOURCEID_ES_CDSBLOCKID_BASE_OFFSET	
CFE_RESOURCEID_SB_PIPEID_RESOURCE_BASE_OFFSET	
CFE_RESOURCEID_CONFIGID_BASE_OFFSET	
CFE_RESOURCEID_TBL_VALRESULTID_BASE_OFFSET	
CFE_RESOURCEID_TBL_DUMPCTRLID_BASE_OFFSET	
CFE_RESOURCEID_TBL_LOADBUFFID_BASE_OFFSET	
CFE_RESOURCEID_TBL_ACCESSID_BASE_OFFSET	
CFE_RESOURCEID_TBL_REGID_BASE_OFFSET	

Definition at line 48 of file [cfe_core_resourceid_basevalues.h](#).

anonymous enum

anonymous enum

Enumerator

CFE_ES_TASKID_BASE	
CFE_ES_APPID_BASE	
CFE_ES_LIBID_BASE	
CFE_ES_COUNTID_BASE	
CFE_ES_POOLID_BASE	
CFE_ES_CDSBLOCKID_BASE	
CFE_SB_PIPEID_BASE	
CFE_CONFIGID_BASE	
CFE_TBL_VALRESULTID_BASE	
CFE_TBL_DUMPCTRLID_BASE	
CFE_TBL_LOADBUFFID_BASE	
CFE_TBL_HANDLE_BASE	
CFE_TBL_REGID_BASE	

Definition at line 87 of file [cfe_core_resourceid_basevalues.h](#).

9.43 cFE Clock State Flag Defines

Macros

- #define `CFE_TIME_FLAG_CLKSET` 0x8000
The spacecraft time has been set.
- #define `CFE_TIME_FLAG_FLYING` 0x4000
This instance of Time Services is flywheeling.
- #define `CFE_TIME_FLAG_SRCINT` 0x2000
The clock source is set to "internal".
- #define `CFE_TIME_FLAG_SIGPRI` 0x1000
The clock signal is set to "primary".
- #define `CFE_TIME_FLAG_SRVFLY` 0x0800
The Time Server is in flywheel mode.
- #define `CFE_TIME_FLAG_CMDFLY` 0x0400
This instance of Time Services was commanded into flywheel mode.
- #define `CFE_TIME_FLAG_ADDADJ` 0x0200
One time STCF Adjustment is to be done in positive direction.
- #define `CFE_TIME_FLAG_ADD1HZ` 0x0100
1 Hz STCF Adjustment is to be done in a positive direction
- #define `CFE_TIME_FLAG_ADDTCL` 0x0080
Time Client Latency is applied in a positive direction.
- #define `CFE_TIME_FLAG_SERVER` 0x0040
This instance of Time Services is a Time Server.
- #define `CFE_TIME_FLAG_GDTONE` 0x0020
The tone received is good compared to the last tone received.
- #define `CFE_TIME_FLAG_REFERR` 0x0010
GetReference read error, will be set if unable to get a consistent ref value.
- #define `CFE_TIME_FLAG_UNUSED` 0x000F
Reserved flags - should be zero.

9.43.1 Detailed Description

9.43.2 Macro Definition Documentation

`CFE_TIME_FLAG_ADD1HZ`

#define `CFE_TIME_FLAG_ADD1HZ` 0x0100
1 Hz STCF Adjustment is to be done in a positive direction
Definition at line 42 of file [default_cfe_time_msgdefs.h](#).

`CFE_TIME_FLAG_ADDADJ`

#define `CFE_TIME_FLAG_ADDADJ` 0x0200
One time STCF Adjustment is to be done in positive direction.
Definition at line 41 of file [default_cfe_time_msgdefs.h](#).

`CFE_TIME_FLAG_ADDTCL`

#define `CFE_TIME_FLAG_ADDTCL` 0x0080
Time Client Latency is applied in a positive direction.
Definition at line 43 of file [default_cfe_time_msgdefs.h](#).

CFE_TIME_FLAG_CLKSET

```
#define CFE_TIME_FLAG_CLKSET 0x8000
```

The spacecraft time has been set.

Definition at line 35 of file [default_cfe_time_msgdefs.h](#).

CFE_TIME_FLAG_CMDFLY

```
#define CFE_TIME_FLAG_CMDFLY 0x0400
```

This instance of Time Services was commanded into flywheel mode.

Definition at line 40 of file [default_cfe_time_msgdefs.h](#).

CFE_TIME_FLAG_FLYING

```
#define CFE_TIME_FLAG_FLYING 0x4000
```

This instance of Time Services is flywheeling.

Definition at line 36 of file [default_cfe_time_msgdefs.h](#).

CFE_TIME_FLAG_GDTONE

```
#define CFE_TIME_FLAG_GDTONE 0x0020
```

The tone received is good compared to the last tone received.

Definition at line 45 of file [default_cfe_time_msgdefs.h](#).

CFE_TIME_FLAG_REFERR

```
#define CFE_TIME_FLAG_REFERR 0x0010
```

GetReference read error, will be set if unable to get a consistent ref value.

Definition at line 47 of file [default_cfe_time_msgdefs.h](#).

CFE_TIME_FLAG_SERVER

```
#define CFE_TIME_FLAG_SERVER 0x0040
```

This instance of Time Services is a Time Server.

Definition at line 44 of file [default_cfe_time_msgdefs.h](#).

CFE_TIME_FLAG_SIGPRI

```
#define CFE_TIME_FLAG_SIGPRI 0x1000
```

The clock signal is set to "primary".

Definition at line 38 of file [default_cfe_time_msgdefs.h](#).

CFE_TIME_FLAG_SRCINT

```
#define CFE_TIME_FLAG_SRCINT 0x2000
```

The clock source is set to "internal".

Definition at line 37 of file [default_cfe_time_msgdefs.h](#).

CFE_TIME_FLAG_SRVFLY

```
#define CFE_TIME_FLAG_SRVFLY 0x0800
```

The Time Server is in flywheel mode.

Definition at line 39 of file [default_cfe_time_msgdefs.h](#).

CFE_TIME_FLAG_UNUSED

```
#define CFE_TIME_FLAG_UNUSED 0x000F
```

Reserved flags - should be zero.

Definition at line 48 of file [default_cfe_time_msgdefs.h](#).

9.44 OSAL Semaphore State Defines**Macros**

- `#define OS_SEM_FULL 1`
Semaphore full state.
- `#define OS_SEM_EMPTY 0`
Semaphore empty state.

9.44.1 Detailed Description**9.44.2 Macro Definition Documentation****OS_SEM_EMPTY**

```
#define OS_SEM_EMPTY 0
```

Semaphore empty state.

Definition at line 35 of file [osapi-binsem.h](#).

OS_SEM_FULL

```
#define OS_SEM_FULL 1
```

Semaphore full state.

Definition at line 34 of file [osapi-binsem.h](#).

9.45 OSAL Binary Semaphore APIs**Functions**

- `int32 OS_BinSemCreate (osal_id_t *sem_id, const char *sem_name, uint32 sem_initial_value, uint32 options)`
Creates a binary semaphore.
- `int32 OS_BinSemFlush (osal_id_t sem_id)`
Unblock all tasks pending on the specified semaphore.
- `int32 OS_BinSemGive (osal_id_t sem_id)`
Increment the semaphore value.
- `int32 OS_BinSemTake (osal_id_t sem_id)`
Decrement the semaphore value.
- `int32 OS_BinSemTimedWait (osal_id_t sem_id, uint32 msec)`
Decrement the semaphore value with a timeout.
- `int32 OS_BinSemDelete (osal_id_t sem_id)`
Deletes the specified Binary Semaphore.
- `int32 OS_BinSemGetIdByName (osal_id_t *sem_id, const char *sem_name)`
Find an existing semaphore ID by name.
- `int32 OS_BinSemGetInfo (osal_id_t sem_id, OS_bin_sem_prop_t *bin_prop)`
Fill a property object buffer with details regarding the resource.

9.45.1 Detailed Description

9.45.2 Function Documentation

OS_BinSemCreate()

```
int32 OS_BinSemCreate (
    osal_id_t * sem_id,
    const char * sem_name,
    uint32 sem_initial_value,
    uint32 options )
```

Creates a binary semaphore.

Creates a binary semaphore with initial value specified by `sem_initial_value` and name specified by `sem_name`. `sem_id` will be returned to the caller

Parameters

out	<i>sem_id</i>	will be set to the non-zero ID of the newly-created resource (must not be null)
in	<i>sem_name</i>	the name of the new resource to create (must not be null)
in	<i>sem_initial_value</i>	the initial value of the binary semaphore
in	<i>options</i>	Reserved for future use, should be passed as 0.

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if <code>sem_name</code> or <code>sem_id</code> are NULL
OS_ERR_NAME_TOO_LONG	name length including null terminator greater than OS_MAX_API_NAME
OS_ERR_NO_FREE_IDS	if all of the semaphore ids are taken
OS_ERR_NAME_TAKEN	if this is already the name of a binary semaphore
OS_SEM_FAILURE	if the OS call failed (return value only verified in coverage test)

OS_BinSemDelete()

```
int32 OS_BinSemDelete (
    osal_id_t sem_id )
```

Deletes the specified Binary Semaphore.

This is the function used to delete a binary semaphore in the operating system. This also frees the respective `sem_id` to be used again when another semaphore is created.

Parameters

in	<i>sem_id</i>	The object ID to delete
----	---------------	-------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<i>OS_SUCCESS</i>	Successful execution.
<i>OS_ERR_INVALID_ID</i>	if the id passed in is not a valid binary semaphore
<i>OS_SEM_FAILURE</i>	if an unspecified failure occurs (return value only verified in coverage test)

OS_BinSemFlush()

```
int32 OS_BinSemFlush (
    osal_id_t sem_id )
```

Unblock all tasks pending on the specified semaphore.

The function unblocks all tasks pending on the specified semaphore. However, this function does not change the state of the semaphore.

Parameters

in	<i>sem_id</i>	The object ID to operate on
----	---------------	-----------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<i>OS_SUCCESS</i>	Successful execution.
<i>OS_ERR_INVALID_ID</i>	if the id passed in is not a binary semaphore
<i>OS_SEM_FAILURE</i>	if an unspecified failure occurs (return value only verified in coverage test)

OS_BinSemGetIdByName()

```
int32 OS_BinSemGetIdByName (
    osal_id_t * sem_id,
    const char * sem_name )
```

Find an existing semaphore ID by name.

This function tries to find a binary sem Id given the name of a bin_sem The id is returned through sem_id

Parameters

out	<i>sem_id</i>	will be set to the ID of the existing resource
in	<i>sem_name</i>	the name of the existing resource to find (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<i>OS_SUCCESS</i>	Successful execution.
-----------------------------------	-----------------------

Return values

<i>OS_INVALID_POINTER</i>	is semid or sem_name are NULL pointers
<i>OS_ERR_NAME_TOO_LONG</i>	name length including null terminator greater than <i>OS_MAX_API_NAME</i>
<i>OS_ERR_NAME_NOT_FOUND</i>	if the name was not found in the table

OS_BinSemGetInfo()

```
int32 OS_BinSemGetInfo (
    osal_id_t sem_id,
    OS_bin_sem_prop_t * bin_prop )
```

Fill a property object buffer with details regarding the resource.

This function will pass back a pointer to structure that contains all of the relevant info(name and creator) about the specified binary semaphore.

Parameters

in	<i>sem_id</i>	The object ID to operate on
out	<i>bin_prop</i>	The property object buffer to fill (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<i>OS_SUCCESS</i>	Successful execution.
<i>OS_ERR_INVALID_ID</i>	if the id passed in is not a valid semaphore
<i>OS_INVALID_POINTER</i>	if the bin_prop pointer is null
<i>OS_ERR_NOT_IMPLEMENTED</i>	Not implemented.

OS_BinSemGive()

```
int32 OS_BinSemGive (
    osal_id_t sem_id )
```

Increment the semaphore value.

The function unlocks the semaphore referenced by sem_id by performing a semaphore unlock operation on that semaphore. If the semaphore value resulting from this operation is positive, then no threads were blocked waiting for the semaphore to become unlocked; the semaphore value is simply incremented for this semaphore.

Parameters

in	<i>sem_id</i>	The object ID to operate on
----	---------------	-----------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<code>OS_SUCCESS</code>	Successful execution.
<code>OS_ERR_INVALID_ID</code>	if the id passed in is not a binary semaphore
<code>OS_SEM_FAILURE</code>	if an unspecified failure occurs (return value only verified in coverage test)

OS_BinSemTake()

```
int32 OS_BinSemTake (
    osal_id_t sem_id )
```

Decrement the semaphore value.

The locks the semaphore referenced by `sem_id` by performing a semaphore lock operation on that semaphore. If the semaphore value is currently zero, then the calling thread shall not return from the call until it either locks the semaphore or the call is interrupted.

Parameters

in	<i>sem_id</i>	The object ID to operate on
----	---------------	-----------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<code>OS_SUCCESS</code>	Successful execution.
<code>OS_ERR_INVALID_ID</code>	the Id passed in is not a valid binary semaphore
<code>OS_SEM_FAILURE</code>	if an unspecified failure occurs (return value only verified in coverage test)

OS_BinSemTimedWait()

```
int32 OS_BinSemTimedWait (
    osal_id_t sem_id,
    uint32 msec )
```

Decrement the semaphore value with a timeout.

The function locks the semaphore referenced by `sem_id`. However, if the semaphore cannot be locked without waiting for another process or thread to unlock the semaphore, this wait shall be terminated when the specified timeout, msec, expires.

Parameters

in	<i>sem_id</i>	The object ID to operate on
in	<i>msec</i>	The maximum amount of time to block, in milliseconds

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<code>OS_SUCCESS</code>	Successful execution.
<code>OS_SEM_TIMEOUT</code>	if semaphore was not relinquished in time
<code>OS_ERR_INVALID_ID</code>	if the ID passed in is not a valid semaphore ID
<code>OS_SEM_FAILURE</code>	if an unspecified failure occurs (return value only verified in coverage test)

9.46 OSAL BSP low level access APIs

Functions

- void [`OS\_BSP\_SetResourceTypeConfig`](#) ([`uint32`](#) ResourceType, [`uint32`](#) ConfigOptionValue)
- [`uint32`](#) [`OS\_BSP\_GetResourceTypeConfig`](#) ([`uint32`](#) ResourceType)
- [`uint32`](#) [`OS\_BSP\_GetArgC`](#) (void)
- char *const * [`OS\_BSP\_GetArgV`](#) (void)
- void [`OS\_BSP\_SetExitCode`](#) ([`int32`](#) code)

9.46.1 Detailed Description

These are for OSAL internal BSP information access to pass any BSP-specific boot/command line/startup arguments through to the application, and return a status code back to the OS after exit.

Not intended for user application use

9.46.2 Function Documentation

`OS_BSP_GetArgC()`

```
uint32 OS_BSP_GetArgC (
    void )
```

`OS_BSP_GetArgV()`

```
char *const * OS_BSP_GetArgV (
    void )
```

`OS_BSP_GetResourceTypeConfig()`

```
uint32 OS_BSP_GetResourceTypeConfig (
    uint32 ResourceType )
```

`OS_BSP_SetExitCode()`

```
void OS_BSP_SetExitCode (
    int32 code )
```

`OS_BSP_SetResourceTypeConfig()`

```
void OS_BSP_SetResourceTypeConfig (
    uint32 ResourceType,
    uint32 ConfigOptionValue )
```

9.47 OSAL Real Time Clock APIs

Functions

- [int32 OS_GetMonotonicTime](#) ([OS_time_t](#) *time_struct)
Get the monotonic time.
- [int32 OS_GetLocalTime](#) ([OS_time_t](#) *time_struct)
Get the local time.
- [int32 OS_SetLocalTime](#) (const [OS_time_t](#) *time_struct)
Set the local time.
- [OS_time_t OS_TimeFromRelativeMilliseconds](#) ([int32](#) relative_msec)
Gets an absolute time value relative to the current time.
- [int32 OS_TimeToRelativeMilliseconds](#) ([OS_time_t](#) time)
Gets a relative time value from an absolute time.
- static [int64 OS_TimeGetTotalSeconds](#) ([OS_time_t](#) tm)
Get interval from an [OS_time_t](#) object normalized to whole number of seconds.
- static [OS_time_t OS_TimeFromTotalSeconds](#) ([int64](#) tm)
Get an [OS_time_t](#) interval object from an integer number of seconds.
- static [int64 OS_TimeGetTotalMilliseconds](#) ([OS_time_t](#) tm)
Get interval from an [OS_time_t](#) object normalized to millisecond units.
- static [OS_time_t OS_TimeFromTotalMilliseconds](#) ([int64](#) tm)
Get an [OS_time_t](#) interval object from a integer number of milliseconds.
- static [int64 OS_TimeGetTotalMicroseconds](#) ([OS_time_t](#) tm)
Get interval from an [OS_time_t](#) object normalized to microsecond units.
- static [OS_time_t OS_TimeFromTotalMicroseconds](#) ([int64](#) tm)
Get an [OS_time_t](#) interval object from a integer number of microseconds.
- static [int64 OS_TimeGetTotalNanoseconds](#) ([OS_time_t](#) tm)
Get interval from an [OS_time_t](#) object normalized to nanosecond units.
- static [OS_time_t OS_TimeFromTotalNanoseconds](#) ([int64](#) tm)
Get an [OS_time_t](#) interval object from a integer number of nanoseconds.
- static [int64 OS_TimeGetFractionalPart](#) ([OS_time_t](#) tm)
Get subseconds portion (fractional part only) from an [OS_time_t](#) object.
- static [uint32 OS_TimeGetSubsecondsPart](#) ([OS_time_t](#) tm)
Get 32-bit normalized subseconds (fractional part only) from an [OS_time_t](#) object.
- static [uint32 OS_TimeGetMillisecondsPart](#) ([OS_time_t](#) tm)
Get milliseconds portion (fractional part only) from an [OS_time_t](#) object.
- static [uint32 OS_TimeGetMicrosecondsPart](#) ([OS_time_t](#) tm)
Get microseconds portion (fractional part only) from an [OS_time_t](#) object.
- static [uint32 OS_TimeGetNanosecondsPart](#) ([OS_time_t](#) tm)
Get nanoseconds portion (fractional part only) from an [OS_time_t](#) object.
- static [OS_time_t OS_TimeAssembleFromNanoseconds](#) ([int64](#) seconds, [uint32](#) nanoseconds)
Assemble/Convert a number of seconds + nanoseconds into an [OS_time_t](#) interval.
- static [OS_time_t OS_TimeAssembleFromMicroseconds](#) ([int64](#) seconds, [uint32](#) microseconds)
Assemble/Convert a number of seconds + microseconds into an [OS_time_t](#) interval.
- static [OS_time_t OS_TimeAssembleFromMilliseconds](#) ([int64](#) seconds, [uint32](#) milliseconds)
Assemble/Convert a number of seconds + milliseconds into an [OS_time_t](#) interval.
- static [OS_time_t OS_TimeAssembleFromSubseconds](#) ([int64](#) seconds, [uint32](#) subseconds)
Assemble/Convert a number of seconds + subseconds into an [OS_time_t](#) interval.

- static [OS_time_t](#) [OS_TimeAdd](#) ([OS_time_t](#) time1, [OS_time_t](#) time2)
Computes the sum of two time intervals.
- static [OS_time_t](#) [OS_TimeSubtract](#) ([OS_time_t](#) time1, [OS_time_t](#) time2)
Computes the difference between two time intervals.
- static bool [OS_TimeEqual](#) ([OS_time_t](#) time1, [OS_time_t](#) time2)
Checks if two time values are equal.
- static [int8_t](#) [OS_TimeGetSign](#) ([OS_time_t](#) time)
Checks the sign of the time value.
- static [int8_t](#) [OS_TimeCompare](#) ([OS_time_t](#) time1, [OS_time_t](#) time2)
Compares two time values.

9.47.1 Detailed Description

9.47.2 Function Documentation

OS_GetLocalTime()

```
int32 OS_GetLocalTime (
    OS_time_t * time_struct )
```

Get the local time.

This function gets the local time from the underlying OS.

Note

Mission time management typically uses the cFE Time Service

Parameters

out	<i>time_struct</i>	An OS_time_t that will be set to the current time (must not be null)
-----	--------------------	--

Returns

Get local time status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if time_struct is null

OS_GetMonotonicTime()

```
int32 OS_GetMonotonicTime (
    OS_time_t * time_struct )
```

Get the monotonic time.

This function gets the monotonic time from the underlying OS.

Monotonic time differs from local or real (wall-clock) time in that it cannot be set. As a result, the time is always guaranteed to be increasing, and never moves backwards or has discontinuities (with some exceptions - see POSIX documentation).

The Epoch is undefined- In many implementations, the epoch is the system boot time, and the clock increases indefinitely so long as the system remains powered on.

Note

Sometimes a monotonic clock is implemented in the operating system and sometimes it is implemented in platform-specific hardware.

If the operating system does not provide a monotonic clock, then this function may return [OS_ERR_NOT_IMPLEMENTED](#). However, a platform-specific monotonic clock may still exist, in the form of a CPU register or external oscillator. This API only accesses a monotonic clock if one is provided by the operating system. To read a platform-specific monotonic clock in CFE, see [CFE_PSP_GetTime\(\)](#).

Parameters

out	<i>time_struct</i>	An OS_time_t that will be set to the monotonic time (must not be null)
-----	--------------------	--

Returns

Get monotonic time status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if <i>time_struct</i> is null
OS_ERR_NOT_IMPLEMENTED	if operating system does not implement a monotonic clock

OS_SetLocalTime()

```
int32 OS_SetLocalTime (
    const OS_time_t * time_struct )
```

Set the local time.

This function sets the local time on the underlying OS.

Note

Mission time management typically uses the cFE Time Services

Parameters

in	<i>time_struct</i>	An OS_time_t containing the current time (must not be null)
----	--------------------	---

Returns

Set local time status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if <i>time_struct</i> is null

OS_TimeAdd()

```
static OS_time_t OS_TimeAdd (
    OS_time_t time1,
    OS_time_t time2 ) [inline], [static]
```

Computes the sum of two time intervals.

Parameters

in	<i>time1</i>	The first interval
in	<i>time2</i>	The second interval

Returns

The sum of the two intervals (time1 + time2)

Definition at line 562 of file [osapi-clock.h](#).

References [OS_time_t::ticks](#).

OS_TimeAssembleFromMicroseconds()

```
static OS_time_t OS_TimeAssembleFromMicroseconds (
    int64 seconds,
    uint32 microseconds ) [inline], [static]
```

Assemble/Convert a number of seconds + microseconds into an [OS_time_t](#) interval.

This creates an [OS_time_t](#) value using a whole number of seconds and a fractional part in units of microseconds. This is the inverse of [OS_TimeGetTotalSeconds\(\)](#) and [OS_TimeGetMicrosecondsPart\(\)](#), and should recreate the original [OS_time_t](#) value from these separate values (aside from any potential conversion losses due to limited resolution of the data types/units).

See also

[OS_TimeGetTotalSeconds\(\)](#), [OS_TimeGetMicrosecondsPart\(\)](#)

Parameters

in	<i>seconds</i>	Whole number of seconds
in	<i>microseconds</i>	Number of microseconds (fractional part only)

Returns

The input arguments represented as an [OS_time_t](#) interval

Definition at line 497 of file [osapi-clock.h](#).

References [OS_TIME_TICKS_PER_SECOND](#), [OS_TIME_TICKS_PER_USEC](#), and [OS_time_t::ticks](#).

OS_TimeAssembleFromMilliseconds()

```
static OS_time_t OS_TimeAssembleFromMilliseconds (
    int64 seconds,
    uint32 milliseconds ) [inline], [static]
```

Assemble/Convert a number of seconds + milliseconds into an [OS_time_t](#) interval.

This creates an [OS_time_t](#) value using a whole number of seconds and a fractional part in units of milliseconds. This is the inverse of [OS_TimeGetTotalSeconds\(\)](#) and [OS_TimeGetMillisecondsPart\(\)](#), and should recreate the original [OS_time_t](#) value from these separate values (aside from any potential conversion losses due to limited resolution of the data types/units).

See also

[OS_TimeGetTotalSeconds\(\)](#), [OS_TimeGetMillisecondsPart\(\)](#)

Parameters

in	<i>seconds</i>	Whole number of seconds
in	<i>milliseconds</i>	Number of milliseconds (fractional part only)

Returns

The input arguments represented as an [OS_time_t](#) interval

Definition at line 521 of file [osapi-clock.h](#).

References [OS_TIME_TICKS_PER_MSEC](#), [OS_TIME_TICKS_PER_SECOND](#), and [OS_time_t::ticks](#).

OS_TimeAssembleFromNanoseconds()

```
static OS_time_t OS_TimeAssembleFromNanoseconds (
    int64 seconds,
    uint32 nanoseconds ) [inline], [static]
```

Assemble/Convert a number of seconds + nanoseconds into an [OS_time_t](#) interval.

This creates an [OS_time_t](#) value using a whole number of seconds and a fractional part in units of nanoseconds. This is the inverse of [OS_TimeGetTotalSeconds\(\)](#) and [OS_TimeGetNanosecondsPart\(\)](#), and should recreate the original [OS_time_t](#) value from these separate values (aside from any potential conversion losses due to limited resolution of the data types/units).

See also

[OS_TimeGetTotalSeconds\(\)](#), [OS_TimeGetNanosecondsPart\(\)](#)

Parameters

in	<i>seconds</i>	Whole number of seconds
in	<i>nanoseconds</i>	Number of nanoseconds (fractional part only)

Returns

The input arguments represented as an [OS_time_t](#) interval

Definition at line 473 of file [osapi-clock.h](#).

References [OS_TIME_TICK_RESOLUTION_NS](#), [OS_TIME_TICKS_PER_SECOND](#), and [OS_time_t::ticks](#).

OS_TimeAssembleFromSubseconds()

```
static OS_time_t OS_TimeAssembleFromSubseconds (
    int64 seconds,
    uint32 subseconds ) [inline], [static]
```

Assemble/Convert a number of seconds + subseconds into an [OS_time_t](#) interval.

This creates an [OS_time_t](#) value using a whole number of seconds and a fractional part in units of sub-seconds ($1/2^{32}$). This is the inverse of [OS_TimeGetTotalSeconds\(\)](#) and [OS_TimeGetSubsecondsPart\(\)](#), and should recreate the original [OS_time_t](#) value from these separate values (aside from any potential conversion losses due to limited resolution of the data types/units).

See also

[OS_TimeGetTotalSeconds\(\)](#), [OS_TimeGetNanosecondsPart\(\)](#)

Parameters

in	<i>seconds</i>	Whole number of seconds
in	<i>subseconds</i>	Number of subseconds (32 bit fixed point fractional part)

Returns

The input arguments represented as an [OS_time_t](#) interval

Definition at line [544](#) of file [osapi-clock.h](#).

References [OS_TIME_TICKS_PER_SECOND](#), and [OS_time_t::ticks](#).

OS_TimeCompare()

```
static int8_t OS_TimeCompare (
    OS_time_t time1,
    OS_time_t time2 ) [inline], [static]
```

Compares two time values.

Parameters

in	<i>time1</i>	The first time
in	<i>time2</i>	The second time

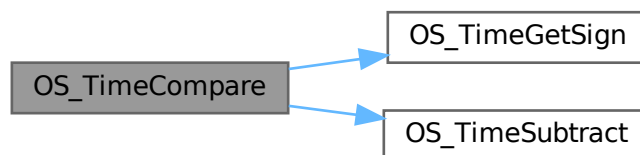
Return values

-1	if the time1 < time2
0	if the times are equal
1	if the time1 > time2

Definition at line [624](#) of file [osapi-clock.h](#).

References [OS_TimeGetSign\(\)](#), and [OS_TimeSubtract\(\)](#).

Here is the call graph for this function:



OS_TimeEqual()

```
static bool OS_TimeEqual (
    OS_time_t time1,
    OS_time_t time2 ) [inline], [static]
```

Checks if two time values are equal.

Parameters

in	<i>time1</i>	The first time value
in	<i>time2</i>	The second time value

Return values

<i>true</i>	if the two values are equal
<i>false</i>	if the two values are not equal

Definition at line 593 of file [osapi-clock.h](#).

References [OS_time_t::ticks](#).

OS_TimeFromRelativeMilliseconds()

```
OS_time_t OS_TimeFromRelativeMilliseconds (
    int32 relative_msec )
```

Gets an absolute time value relative to the current time.

This function adds the given interval, expressed in milliseconds, to the current clock and returns the result.

Note

This is intended to ease transitioning from a relative timeout value to an absolute timeout value. The result can be passed to any function that accepts an absolute timeout, to mimic the behavior of a relative timeout.

Parameters

in	<i>relative_msec</i>	A relative time interval, in milliseconds
----	----------------------	---

Returns

Absolute time value after adding interval

OS_TimeFromTotalMicroseconds()

```
static OS_time_t OS_TimeFromTotalMicroseconds (
    int64 tm ) [inline], [static]
```

Get an [OS_time_t](#) interval object from a integer number of microseconds.

This is the inverse operation of [OS_TimeGetTotalMicroseconds\(\)](#), converting the total number of microseconds into an [OS_time_t](#) value.

See also

[OS_TimeGetTotalMicroseconds\(\)](#)

Parameters

<code>in</code>	<code>tm</code>	Time interval value, in microseconds
-----------------	-----------------	--------------------------------------

Returns

[OS_time_t](#) value representing the interval

Definition at line 310 of file [osapi-clock.h](#).

References [OS_TIME_TICKS_PER_USEC](#).

OS_TimeFromTotalMilliseconds()

```
static OS\_time\_t OS_TimeFromTotalMilliseconds (  
    int64 tm ) [inline], [static]
```

Get an [OS_time_t](#) interval object from a integer number of milliseconds.

This is the inverse operation of [OS_TimeGetTotalMilliseconds\(\)](#), converting the total number of milliseconds into an [OS_time_t](#) value.

See also

[OS_TimeGetTotalMilliseconds\(\)](#)

Parameters

<code>in</code>	<code>tm</code>	Time interval value, in milliseconds
-----------------	-----------------	--------------------------------------

Returns

[OS_time_t](#) value representing the interval

Definition at line 276 of file [osapi-clock.h](#).

References [OS_TIME_TICKS_PER_MSEC](#).

OS_TimeFromTotalNanoseconds()

```
static OS\_time\_t OS_TimeFromTotalNanoseconds (  
    int64 tm ) [inline], [static]
```

Get an [OS_time_t](#) interval object from a integer number of nanoseconds.

This is the inverse operation of [OS_TimeGetTotalNanoseconds\(\)](#), converting the total number of nanoseconds into an [OS_time_t](#) value.

See also

[OS_TimeGetTotalNanoseconds\(\)](#)

Parameters

<code>in</code>	<code>tm</code>	Time interval value, in nanoseconds
-----------------	-----------------	-------------------------------------

Returns

[OS_time_t](#) value representing the interval

Definition at line 349 of file [osapi-clock.h](#).

References [OS_TIME_TICK_RESOLUTION_NS](#).

OS_TimeFromTotalSeconds()

```
static OS_time_t OS_TimeFromTotalSeconds (
    int64 tm ) [inline], [static]
```

Get an [OS_time_t](#) interval object from an integer number of seconds.

This is the inverse operation of [OS_TimeGetTotalSeconds\(\)](#), converting the total number of seconds into an [OS_time_t](#) value.

See also

[OS_TimeGetTotalSeconds\(\)](#)

Parameters

in	tm	Time interval value, in seconds
----	----	---------------------------------

Returns

[OS_time_t](#) value representing the interval

Definition at line 242 of file [osapi-clock.h](#).

References [OS_TIME_TICKS_PER_SECOND](#).

OS_TimeGetFractionalPart()

```
static int64 OS_TimeGetFractionalPart (
    OS_time_t tm ) [inline], [static]
```

Get subseconds portion (fractional part only) from an [OS_time_t](#) object.

Extracts the fractional part from a given [OS_time_t](#) object. Units returned are in ticks, not normalized to any standard time unit.

Parameters

in	tm	Time interval value
----	----	---------------------

Returns

Fractional/subsecond portion of time interval in ticks

Definition at line 365 of file [osapi-clock.h](#).

References [OS_TIME_TICKS_PER_SECOND](#), and [OS_time_t::ticks](#).

Referenced by [OS_TimeGetMicrosecondsPart\(\)](#), [OS_TimeGetMillisecondsPart\(\)](#), [OS_TimeGetNanosecondsPart\(\)](#), and [OS_TimeGetSubsecondsPart\(\)](#).

OS_TimeGetMicrosecondsPart()

```
static uint32 OS_TimeGetMicrosecondsPart (
    OS_time_t tm ) [inline], [static]
```

Get microseconds portion (fractional part only) from an [OS_time_t](#) object.

Extracts the fractional part from a given [OS_time_t](#) object normalized to units of microseconds.

This function may be used to adapt applications initially implemented using an older OSAL version where [OS_time_t](#) was a structure containing a "seconds" and "microsecs" field.

This function will obtain a value that is compatible with the "microsecs" field of [OS_time_t](#) as it was defined in previous versions of OSAL, as well as the "tv_usec" field of POSIX-style "struct timeval" values.

See also

[OS_TimeGetTotalSeconds\(\)](#)

Parameters

<i>in</i>	<i>tm</i>	Time interval value
-----------	-----------	---------------------

Returns

Number of microseconds in time interval

Definition at line 433 of file [osapi-clock.h](#).

References [OS_TIME_TICKS_PER_USEC](#), and [OS_TimeGetFractionalPart\(\)](#).

Here is the call graph for this function:



OS_TimeGetMillisecondsPart()

```
static uint32 OS_TimeGetMillisecondsPart (  
    OS_time_t tm ) [inline], [static]
```

Get milliseconds portion (fractional part only) from an [OS_time_t](#) object.

Extracts the fractional part from a given [OS_time_t](#) object normalized to units of milliseconds.

See also

[OS_TimeGetTotalSeconds\(\)](#)

Parameters

<i>in</i>	<i>tm</i>	Time interval value
-----------	-----------	---------------------

Returns

Number of milliseconds in time interval

Definition at line 408 of file [osapi-clock.h](#).

References [OS_TIME_TICKS_PER_MSEC](#), and [OS_TimeGetFractionalPart\(\)](#).

Here is the call graph for this function:

**OS_TimeGetNanosecondsPart()**

```
static uint32 OS_TimeGetNanosecondsPart (
    OS_time_t tm ) [inline], [static]
```

Get nanoseconds portion (fractional part only) from an [OS_time_t](#) object.

Extracts the only number of nanoseconds from a given [OS_time_t](#) object.

This function will obtain a value that is compatible with the "tv_nsec" field of POSIX-style "struct timespec" values.

See also

[OS_TimeGetTotalSeconds\(\)](#)

Parameters

in	tm	Time interval value
----	----	---------------------

Returns

Number of nanoseconds in time interval

Definition at line 452 of file [osapi-clock.h](#).

References [OS_TIME_TICK_RESOLUTION_NS](#), and [OS_TimeGetFractionalPart\(\)](#).

Here is the call graph for this function:

**OS_TimeGetSign()**

```
static int8_t OS_TimeGetSign (
    OS_time_t time ) [inline], [static]
```


Checks the sign of the time value.

Parameters

<i>in</i>	<i>time</i>	The time to check
-----------	-------------	-------------------

Return values

-1	if the time value is negative / below 0
0	if the time value is 0
1	if the time value is positive / above 0

Definition at line 608 of file [osapi-clock.h](#).

References [OS_time_t::ticks](#).

Referenced by [OS_TimeCompare\(\)](#).

OS_TimeGetSubsecondsPart()

```
static uint32 OS_TimeGetSubsecondsPart (
    OS_time_t tm ) [inline], [static]
```

Get 32-bit normalized subseconds (fractional part only) from an [OS_time_t](#) object.

Extracts the fractional part from a given [OS_time_t](#) object in maximum precision, with units of $2^{(-32)}$ sec. This is a base-2 fixed-point fractional value with the point left-justified in the 32-bit value (i.e. left of MSB).

This is (mostly) compatible with the CFE "subseconds" value, where 0x80000000 represents exactly one half second, and 0 represents a full second.

Parameters

<i>in</i>	<i>tm</i>	Time interval value
-----------	-----------	---------------------

Returns

Fractional/subsecond portion of time interval as 32-bit fixed point value

Definition at line 384 of file [osapi-clock.h](#).

References [OS_TIME_TICKS_PER_SECOND](#), and [OS_TimeGetFractionalPart\(\)](#).

Here is the call graph for this function:

**OS_TimeGetTotalMicroseconds()**

```
static int64 OS_TimeGetTotalMicroseconds (
    OS_time_t tm ) [inline], [static]
```

Get interval from an [OS_time_t](#) object normalized to microsecond units.

Note this refers to the complete interval, not just the fractional part.

See also

[OS_TimeFromTotalMicroseconds\(\)](#)

Parameters

<code>in</code>	<code>tm</code>	Time interval value
-----------------	-----------------	---------------------

Returns

Whole number of microseconds in time interval

Definition at line 293 of file [osapi-clock.h](#).

References [OS_TIME_TICKS_PER_USEC](#), and [OS_time_t::ticks](#).

OS_TimeGetTotalMilliseconds()

```
static int64 OS_TimeGetTotalMilliseconds (  
    OS_time_t tm ) [inline], [static]
```

Get interval from an [OS_time_t](#) object normalized to millisecond units.

Note this refers to the complete interval, not just the fractional part.

See also

[OS_TimeFromTotalMilliseconds\(\)](#)

Parameters

<code>in</code>	<code>tm</code>	Time interval value
-----------------	-----------------	---------------------

Returns

Whole number of milliseconds in time interval

Definition at line 259 of file [osapi-clock.h](#).

References [OS_TIME_TICKS_PER_MSEC](#), and [OS_time_t::ticks](#).

OS_TimeGetTotalNanoseconds()

```
static int64 OS_TimeGetTotalNanoseconds (  
    OS_time_t tm ) [inline], [static]
```

Get interval from an [OS_time_t](#) object normalized to nanosecond units.

Note this refers to the complete interval, not just the fractional part.

Note

There is no protection against overflow of the 64-bit return value. Applications must use caution to ensure that the interval does not exceed the representable range of a signed 64 bit integer - approximately 140 years.

See also

[OS_TimeFromTotalNanoseconds](#)

Parameters

in	<i>tm</i>	Time interval value
----	-----------	---------------------

Returns

Whole number of microseconds in time interval

Definition at line 332 of file [osapi-clock.h](#).

References [OS_TIME_TICK_RESOLUTION_NS](#), and [OS_time_t::ticks](#).

OS_TimeGetTotalSeconds()

```
static int64 OS_TimeGetTotalSeconds (
    OS_time_t tm ) [inline], [static]
```

Get interval from an [OS_time_t](#) object normalized to whole number of seconds.

Extracts the number of whole seconds from a given [OS_time_t](#) object, discarding any fractional component.

This may also replace a direct read of the "seconds" field from the [OS_time_t](#) object from previous versions of OSAL, where the structure was defined with separate seconds/microseconds fields.

See also

[OS_TimeGetMicrosecondsPart\(\)](#)

[OS_TimeFromTotalSeconds\(\)](#)

Parameters

in	<i>tm</i>	Time interval value
----	-----------	---------------------

Returns

Whole number of seconds in time interval

Definition at line 225 of file [osapi-clock.h](#).

References [OS_TIME_TICKS_PER_SECOND](#), and [OS_time_t::ticks](#).

OS_TimeSubtract()

```
static OS_time_t OS_TimeSubtract (
    OS_time_t time1,
    OS_time_t time2 ) [inline], [static]
```

Computes the difference between two time intervals.

Parameters

in	<i>time1</i>	The first interval
in	<i>time2</i>	The second interval

Returns

The difference of the two intervals (time1 - time2)

Definition at line 577 of file [osapi-clock.h](#).

References [OS_time_t::ticks](#).

Referenced by [OS_TimeCompare\(\)](#).

OS_TimeToRelativeMilliseconds()

```
int32 OS_TimeToRelativeMilliseconds (
    OS_time_t time )
```

Gets a relative time value from an absolute time.

This function computes the number of milliseconds until the given absolute time value is reached in the system clock.

Note

This is intended to ease transitioning from a relative timeout value to an absolute timeout value. The result can be passed to any function that accepts a relative timeout, to mimic the behavior of an absolute timeout.

The return value of this function is intended to be compatible with the relative timeout parameter of various OSAL APIs e.g. [OS_TimedRead\(\)](#) / [OS_TimedWrite\(\)](#)

Parameters

in	<i>time</i>	An absolute time value
----	-------------	------------------------

Returns

Milliseconds until time value will be reached

Return values

<i>OS_CHECK</i>	(0) if time is the current time or is in the past
<i>OS_PEND</i>	(-1) if time is far in the future (not expressable as an int32)

9.48 OSAL Core Operation APIs**Functions**

- void [OS_Application_Startup](#) (void)
Application startup.
- void [OS_Application_Run](#) (void)
Application run.
- int32 [OS_API_Init](#) (void)
Initialization of API.
- void [OS_API_Teardown](#) (void)
Teardown/de-initialization of OSAL API.
- void [OS_IdleLoop](#) (void)
Background thread implementation - waits forever for events to occur.
- void [OS_DeleteAllObjects](#) (void)
delete all resources created in OSAL.
- void [OS_ApplicationShutdown](#) (uint8 flag)
Initiate orderly shutdown.
- void [OS_ApplicationExit](#) (int32 Status)
Exit/Abort the application.
- int32 [OS_RegisterEventHandler](#) (OS_EventHandler_t handler)
Callback routine registration.
- size_t [OS_strlen](#) (const char *s, size_t maxlen)
get string length

9.48.1 Detailed Description

These are for OSAL core operations for startup/initialization, running, and shutdown. Typically only used in bsps, unit tests, psp, etc.

Not intended for user application use

9.48.2 Function Documentation

OS_API_Init()

```
int32 OS_API_Init (
    void )
```

Initialization of API.

This function returns initializes the internal data structures of the OS Abstraction Layer. It must be called in the application startup code before calling any other OS routines.

Returns

Execution status, see [OSAL Return Code Defines](#). Any error code (negative) means the OSAL can not be initialized. Typical platform specific response is to abort since additional OSAL calls will have undefined behavior.

Return values

OS_SUCCESS	Successful execution.
OS_ERROR	Failed execution. (return value only verified in coverage test)

OS_API_Teardown()

```
void OS_API_Teardown (
    void )
```

Teardown/de-initialization of OSAL API.

This is the inverse of [OS_API_Init\(\)](#). It will release all OS resources and return the system to a state similar to what it was prior to invoking [OS_API_Init\(\)](#) initially.

Normally for embedded applications, the OSAL is initialized after boot and will remain initialized in memory until the processor is rebooted. However for testing and development purposes, it is potentially useful to reset back to initial conditions.

For testing purposes, this API is designed/intended to be compatible with the [UtTest_AddTeardown\(\)](#) routine provided by the UT-Assert subsystem.

Note

This is a "best-effort" routine and it may not always be possible/guaranteed to recover all resources, particularly in the case of off-nominal conditions, or if a resource is used outside of OSAL.

For example, while this will attempt to unload all dynamically-loaded modules, doing so may not be possible and/or may induce undefined behavior if resources are in use by tasks/functions outside of OSAL.

OS_Application_Run()

```
void OS_Application_Run (
    void )
```

Application run.

Run abstraction such that the same BSP can be used for operations and testing.

OS_Application_Startup()

```
void OS_Application_Startup (
    void )
```

Application startup.

Startup abstraction such that the same BSP can be used for operations and testing.

OS_ApplicationExit()

```
void OS_ApplicationExit (
    int32 Status )
```

Exit/Abort the application.

Indicates that the OSAL application should exit and return control to the OS This is intended for e.g. scripted unit testing where the test needs to end without user intervention.

This function does not return. Production code typically should not ever call this.

Note

This exits the entire process including tasks that have been created.

OS_ApplicationShutdown()

```
void OS_ApplicationShutdown (
    uint8 flag )
```

Initiate orderly shutdown.

Indicates that the OSAL application should perform an orderly shutdown of ALL tasks, clean up all resources, and exit the application.

This allows the task currently blocked in [OS_IdleLoop\(\)](#) to wake up, and for that function to return to its caller.

This is preferred over e.g. [OS_ApplicationExit\(\)](#) which exits immediately and does not provide for any means to clean up first.

Parameters

<i>in</i>	<i>flag</i>	set to true to initiate shutdown, false to cancel
-----------	-------------	---

OS_DeleteAllObjects()

```
void OS_DeleteAllObjects (
    void )
```

delete all resources created in OSAL.

provides a means to clean up all resources allocated by this instance of OSAL. It would typically be used during an orderly shutdown but may also be helpful for testing purposes.

OS_IdleLoop()

```
void OS_IdleLoop (
    void )
```

Background thread implementation - waits forever for events to occur.

This should be called from the BSP main routine or initial thread after all other board and application initialization has taken place and all other tasks are running.

Typically just waits forever until "OS_shutdown" flag becomes true.

OS_RegisterEventHandler()

```
int32 OS_RegisterEventHandler (
    OS_EventHandler_t handler )
```

Callback routine registration.

This hook enables the application code to perform extra platform-specific operations on various system events such as resource creation/deletion.

Note

Some events are invoked while the resource is "locked" and therefore application-defined handlers for these events should not block or attempt to access other OSAL resources.

Parameters

in	<i>handler</i>	The application-provided event handler (must not be null)
----	----------------	---

Returns

Execution status, see [OSAL Return Code Defines](#).

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if handler is NULL

OS_strnlen()

```
size_t OS_strnlen (
    const char * s,
    size_t maxlen )
```

get string length

Provides an OSAL routine to get the functionality of the (non-C99) "strnlen()" function, via the C89/C99 standard "memchr()" function instead.

Parameters

in	<i>s</i>	The input string
in	<i>maxlen</i>	Maximum length to check

Return values

<i>Length</i>	of the string or maxlen, whichever is smaller.
---------------	--

9.49 OSAL Condition Variable APIs**Functions**

- [int32 OS_CondVarCreate](#) ([osal_id_t](#) *var_id, const char *var_name, [uint32](#) options)
Creates a condition variable resource.

- `int32 OS_CondVarLock (osal_id_t var_id)`
Locks/Acquires the underlying mutex associated with a condition variable.
- `int32 OS_CondVarUnlock (osal_id_t var_id)`
Unlocks/Releases the underlying mutex associated with a condition variable.
- `int32 OS_CondVarSignal (osal_id_t var_id)`
Signals the condition variable resource referenced by var_id.
- `int32 OS_CondVarBroadcast (osal_id_t var_id)`
Broadcasts the condition variable resource referenced by var_id.
- `int32 OS_CondVarWait (osal_id_t var_id)`
Waits on the condition variable object referenced by var_id.
- `int32 OS_CondVarTimedWait (osal_id_t var_id, const OS_time_t *abs_wakeup_time)`
Time-limited wait on the condition variable object referenced by var_id.
- `int32 OS_CondVarDelete (osal_id_t var_id)`
Deletes the specified condition variable.
- `int32 OS_CondVarGetIdByName (osal_id_t *var_id, const char *var_name)`
Find an existing condition variable ID by name.
- `int32 OS_CondVarGetInfo (osal_id_t var_id, OS_condvar_prop_t *condvar_prop)`
Fill a property object buffer with details regarding the resource.

9.49.1 Detailed Description

9.49.2 Function Documentation

OS_CondVarBroadcast()

```
int32 OS_CondVarBroadcast (
    osal_id_t var_id )
```

Broadcasts the condition variable resource referenced by var_id.

This function may be used to indicate when the state of a data object has been changed.

If there are threads blocked on the condition variable object referenced by var_id when this function is called, all threads will be unblocked.

Note that although all threads are unblocked, because the mutex is re-acquired before the wait function returns, only a single task will be testing the condition at a given time. The order with which each blocked task runs is determined by the scheduling policy.

Parameters

in	<i>var</i> ↔ <i>_id</i>	The object ID to operate on
----	----------------------------	-----------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<i>OS_SUCCESS</i>	Successful execution.
<i>OS_ERR_INVALID_ID</i>	if the id passed in is not a valid condition variable

OS_CondVarCreate()

```
int32 OS_CondVarCreate (
    osal_id_t * var_id,
    const char * var_name,
    uint32 options )
```

Creates a condition variable resource.

A condition variable adds a more sophisticated synchronization option for mutexes, such that it can operate on arbitrary user-defined conditions rather than simply a counter or boolean (as in the case of simple semaphores).

Creating a condition variable resource in OSAL will in turn create both a basic mutex as well as a synchronization overlay. The underlying mutex is similar to the mutex functionality provided by the OSAL mutex subsystem, and can be locked and unlocked normally.

This mutex is intended to protect access to any arbitrary user-defined data object that serves as the condition being tested.

A task that needs a particular state of the object should follow this general flow:

- Lock the underlying mutex
- Test for the condition being waited for (a user-defined check on user-defined data)
- If condition IS NOT met, then call [OS_CondVarWait\(\)](#) to wait, then repeat test
- If condition IS met, then unlock the underlying mutex and continue

A task that changes the state of the object should follow this general flow:

- Lock the underlying mutex
- Change the state as necessary
- Call either [OS_CondVarSignal\(\)](#) or [OS_CondVarBroadcast\(\)](#)
- Unlock the underlying mutex

Parameters

out	<i>var_id</i>	will be set to the non-zero ID of the newly-created resource (must not be null)
in	<i>var_name</i>	the name of the new resource to create (must not be null)
in	<i>options</i>	reserved for future use. Should be passed as 0.

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if <i>var_id</i> or <i>var_name</i> are NULL
OS_ERR_NAME_TOO_LONG	name length including null terminator greater than OS_MAX_API_NAME
OS_ERR_NO_FREE_IDS	if there are no more free condition variable Ids
OS_ERR_NAME_TAKEN	if there is already a condition variable with the same name

OS_CondVarDelete()

```
int32 OS_CondVarDelete (
    osal_id_t var_id )
```

Deletes the specified condition variable.

Delete the condition variable and releases any related system resources.

Parameters

in	<i>var_id</i>	The object ID to delete
----	---------------	-------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the id passed in is not a valid condvar

OS_CondVarGetIdByName()

```
int32 OS_CondVarGetIdByName (
    osal_id_t * var_id,
    const char * var_name )
```

Find an existing condition variable ID by name.

This function tries to find an existing condition variable ID given the name. The id is returned through var_id.

Parameters

out	<i>var_id</i>	will be set to the ID of the existing resource
in	<i>var_name</i>	the name of the existing resource to find (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	is var_id or var_name are NULL pointers
OS_ERR_NAME_TOO_LONG	name length including null terminator greater than OS_MAX_API_NAME
OS_ERR_NAME_NOT_FOUND	if the name was not found in the table

OS_CondVarGetInfo()

```
int32 OS_CondVarGetInfo (
    osal_id_t var_id,
```

```
OS_condvar_prop_t * condvar_prop )
```

Fill a property object buffer with details regarding the resource.

This function will fill a structure to contain the information (name and creator) about the specified condition variable.

Parameters

in	<i>var_id</i>	The object ID to operate on
out	<i>condvar_prop</i>	The property object buffer to fill (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the id passed in is not a valid semaphore
OS_INVALID_POINTER	if the mut_prop pointer is null

OS_CondVarLock()

```
int32 OS_CondVarLock (
    osal_id_t var_id )
```

Locks/Acquires the underlying mutex associated with a condition variable.

The mutex should always be locked by a task before reading or modifying the data object associated with a condition variable.

Note

This lock must be acquired by a task before invoking [OS_CondVarWait\(\)](#) or [OS_CondVarTimedWait\(\)](#) on the same condition variable.

Parameters

in	<i>var↔ _id</i>	The object ID to operate on
----	---------------------	-----------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the id passed in is not a valid condition variable

OS_CondVarSignal()

```
int32 OS_CondVarSignal (
    osal_id_t var_id )
```

Signals the condition variable resource referenced by `var_id`.

This function may be used to indicate when the state of a data object has been changed.

If there are threads blocked on the condition variable object referenced by `var_id` when this function is called, one of those threads will be unblocked, as determined by the scheduling policy.

Parameters

in	<i>var_id</i>	The object ID to operate on
----	---------------	-----------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the id passed in is not a valid condition variable

OS_CondVarTimedWait()

```
int32 OS_CondVarTimedWait (
    osal_id_t var_id,
    const OS_time_t * abs_wakeup_time )
```

Time-limited wait on the condition variable object referenced by `var_id`.

Identical in operation to [OS_CondVarWait\(\)](#), except that the maximum amount of time that the task will be blocked is limited.

The `abs_wakeup_time` refers to the absolute time of the system clock at which the task should be unblocked to run, regardless of the state of the condition variable. This refers to the same system clock that is the subject of the [OS_GetLocalTime\(\)](#) API.

Parameters

in	<i>var_id</i>	The object ID to operate on
in	<i>abs_wakeup_time</i>	The system time at which the task should be unblocked (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	the id passed in is not a valid condvar

OS_CondVarUnlock()

```
int32 OS_CondVarUnlock (
    osal_id_t var_id )
```

Unlocks/Releases the underlying mutex associated with a condition variable.

The mutex should be unlocked by a task once reading or modifying the data object associated with a condition variable is complete.

Parameters

in	<i>var</i> ↔ <i>_id</i>	The object ID to operate on
----	----------------------------	-----------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the id passed in is not a valid condition variable

OS_CondVarWait()

```
int32 OS_CondVarWait (
    osal_id_t var_id )
```

Waits on the condition variable object referenced by var_id.

The calling task will be blocked until another task calls the function [OS_CondVarSignal\(\)](#) or [OS_CondVarBroadcast\(\)](#) on the same condition variable.

The underlying mutex associated with the condition variable must be locked and owned by the calling task at the time this function is invoked. As part of this call, the mutex will be unlocked as the task blocks. This is done in such a way that there is no possibility that another task could acquire the mutex before the calling task has actually blocked.

This atomicity with respect to blocking the task and unlocking the mutex is a critical difference between condition variables and other synchronization primitives. It avoids a window of opportunity where inherent in the simpler synchronization resource types where the state of the data could change between the time that the calling task tested the state and the time that the task actually blocks on the sync resource.

Parameters

in	<i>var</i> ↔ <i>_id</i>	The object ID to operate on
----	----------------------------	-----------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	the id passed in is not a valid condvar

9.50 OSAL Counting Semaphore APIs

Functions

- [int32 OS_CountSemCreate](#) ([osal_id_t](#) *sem_id, const char *sem_name, [uint32](#) sem_initial_value, [uint32](#) options)
Creates a counting semaphore.
- [int32 OS_CountSemGive](#) ([osal_id_t](#) sem_id)
Increment the semaphore value.
- [int32 OS_CountSemTake](#) ([osal_id_t](#) sem_id)
Decrement the semaphore value.
- [int32 OS_CountSemTimedWait](#) ([osal_id_t](#) sem_id, [uint32](#) msecs)
Decrement the semaphore value with timeout.
- [int32 OS_CountSemDelete](#) ([osal_id_t](#) sem_id)
Deletes the specified counting Semaphore.
- [int32 OS_CountSemGetIdByName](#) ([osal_id_t](#) *sem_id, const char *sem_name)
Find an existing semaphore ID by name.
- [int32 OS_CountSemGetInfo](#) ([osal_id_t](#) sem_id, [OS_count_sem_prop_t](#) *count_prop)
Fill a property object buffer with details regarding the resource.

9.50.1 Detailed Description

9.50.2 Function Documentation

OS_CountSemCreate()

```
int32 OS_CountSemCreate (
    osal_id_t * sem_id,
    const char * sem_name,
    uint32 sem_initial_value,
    uint32 options )
```

Creates a counting semaphore.

Creates a counting semaphore with initial value specified by sem_initial_value and name specified by sem_name. sem_id will be returned to the caller.

Note

Underlying RTOS implementations may or may not impose a specific upper limit to the value of a counting semaphore. If the OS has a specific limit and the sem_initial_value exceeds this limit, then [OS_INVALID_SEM_VALUE](#) is returned. On other implementations, any 32-bit integer value may be acceptable. For maximum portability, it is recommended to keep counting semaphore values within the range of a "short int" (i.e. between 0 and 32767). Many platforms do accept larger values, but may not be guaranteed.

Parameters

out	<i>sem_id</i>	will be set to the non-zero ID of the newly-created resource (must not be null)
in	<i>sem_name</i>	the name of the new resource to create (must not be null)
in	<i>sem_initial_value</i>	the initial value of the counting semaphore
in	<i>options</i>	Reserved for future use, should be passed as 0.

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<i>OS_SUCCESS</i>	Successful execution.
<i>OS_INVALID_POINTER</i>	if sen name or sem_id are NULL
<i>OS_ERR_NAME_TOO_LONG</i>	name length including null terminator greater than <i>OS_MAX_API_NAME</i>
<i>OS_ERR_NO_FREE_IDS</i>	if all of the semaphore ids are taken
<i>OS_ERR_NAME_TAKEN</i>	if this is already the name of a counting semaphore
<i>OS_INVALID_SEM_VALUE</i>	if the semaphore value is too high (return value only verified in coverage test)
<i>OS_SEM_FAILURE</i>	if an unspecified implementation error occurs (return value only verified in coverage test)

OS_CountSemDelete()

```
int32 OS_CountSemDelete (
    osal_id_t sem_id )
```

Deletes the specified counting Semaphore.

Parameters

in	<i>sem_id</i>	The object ID to delete
----	---------------	-------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<i>OS_SUCCESS</i>	Successful execution.
<i>OS_ERR_INVALID_ID</i>	if the id passed in is not a valid counting semaphore
<i>OS_SEM_FAILURE</i>	if an unspecified implementation error occurs (return value only verified in coverage test)

OS_CountSemGetIdByName()

```
int32 OS_CountSemGetIdByName (
    osal_id_t * sem_id,
    const char * sem_name )
```

Find an existing semaphore ID by name.

This function tries to find a counting sem Id given the name of a count_sem The id is returned through sem_id

Parameters

out	<i>sem_id</i>	will be set to the ID of the existing resource
in	<i>sem_name</i>	the name of the existing resource to find (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<i>OS_SUCCESS</i>	Successful execution.
<i>OS_INVALID_POINTER</i>	is semid or sem_name are NULL pointers
<i>OS_ERR_NAME_TOO_LONG</i>	name length including null terminator greater than <i>OS_MAX_API_NAME</i>
<i>OS_ERR_NAME_NOT_FOUND</i>	if the name was not found in the table

OS_CountSemGetInfo()

```
int32 OS_CountSemGetInfo (
    osal_id_t sem_id,
    OS_count_sem_prop_t * count_prop )
```

Fill a property object buffer with details regarding the resource.

This function will pass back a pointer to structure that contains all of the relevant info(name and creator) about the specified counting semaphore.

Parameters

in	<i>sem_id</i>	The object ID to operate on
out	<i>count_prop</i>	The property object buffer to fill (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<i>OS_SUCCESS</i>	Successful execution.
<i>OS_ERR_INVALID_ID</i>	if the id passed in is not a valid semaphore
<i>OS_INVALID_POINTER</i>	if the count_prop pointer is null
<i>OS_ERR_NOT_IMPLEMENTED</i>	Not implemented.

OS_CountSemGive()

```
int32 OS_CountSemGive (
    osal_id_t sem_id )
```

Increment the semaphore value.

The function unlocks the semaphore referenced by sem_id by performing a semaphore unlock operation on that semaphore. If the semaphore value resulting from this operation is positive, then no threads were blocked waiting for the semaphore to become unlocked; the semaphore value is simply incremented for this semaphore.

Parameters

in	<i>sem_id</i>	The object ID to operate on
----	---------------	-----------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the id passed in is not a counting semaphore
OS_SEM_FAILURE	if an unspecified implementation error occurs (return value only verified in coverage test)

OS_CountSemTake()

```
int32 OS_CountSemTake (
    osal_id_t sem_id )
```

Decrement the semaphore value.

The locks the semaphore referenced by sem_id by performing a semaphore lock operation on that semaphore. If the semaphore value is currently zero, then the calling thread shall not return from the call until it either locks the semaphore or the call is interrupted.

Parameters

in	<i>sem_id</i>	The object ID to operate on
----	---------------	-----------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	the Id passed in is not a valid counting semaphore
OS_SEM_FAILURE	if an unspecified implementation error occurs (return value only verified in coverage test)

OS_CountSemTimedWait()

```
int32 OS_CountSemTimedWait (
    osal_id_t sem_id,
    uint32 msec )
```

Decrement the semaphore value with timeout.

The function locks the semaphore referenced by sem_id. However, if the semaphore cannot be locked without waiting for another process or thread to unlock the semaphore, this wait shall be terminated when the specified timeout, msec, expires.

Parameters

in	<i>sem_id</i>	The object ID to operate on
in	<i>msec</i>	The maximum amount of time to block, in milliseconds

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_SEM_TIMEOUT	if semaphore was not relinquished in time
OS_ERR_INVALID_ID	if the ID passed in is not a valid semaphore ID
OS_SEM_FAILURE	if an unspecified implementation error occurs (return value only verified in coverage test)

9.51 OSAL Directory APIs**Functions**

- [int32 OS_DirectoryOpen](#) ([osal_id_t](#) *dir_id, const char *path)
Opens a directory.
- [int32 OS_DirectoryClose](#) ([osal_id_t](#) dir_id)
Closes an open directory.
- [int32 OS_DirectoryRewind](#) ([osal_id_t](#) dir_id)
Rewinds an open directory.
- [int32 OS_DirectoryRead](#) ([osal_id_t](#) dir_id, [os_dirent_t](#) *dirent)
Reads the next name in the directory.
- [int32 OS_mkdir](#) (const char *path, [uint32](#) access)
Makes a new directory.
- [int32 OS_rmdir](#) (const char *path)
Removes a directory from the file system.

9.51.1 Detailed Description**9.51.2 Function Documentation****OS_DirectoryClose()**

```
int32 OS_DirectoryClose (
    osal_id_t dir_id )
```

Closes an open directory.

The directory referred to by dir_id will be closed

Parameters

in	<i>dir_id</i>	The handle ID of the directory
----	---------------	--------------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the directory handle is invalid

OS_DirectoryOpen()

```
int32 OS_DirectoryOpen (
    osal_id_t * dir_id,
    const char * path )
```

Opens a directory.

Prepares for reading the files within a directory

Parameters

out	<i>dir_id</i>	Location to store handle ID of the directory (must not be null)
in	<i>path</i>	The directory to open (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if dir_id or path is NULL
OS_FS_ERR_PATH_TOO_LONG	if the path argument exceeds the maximum length
OS_FS_ERR_PATH_INVALID	if the path argument is not valid
OS_ERROR	if the directory could not be opened

OS_DirectoryRead()

```
int32 OS_DirectoryRead (
    osal_id_t dir_id,
    os_dirent_t * dirent )
```

Reads the next name in the directory.

Obtains directory entry data for the next file from an open directory

Parameters

in	<i>dir_id</i>	The handle ID of the directory
out	<i>dirent</i>	Buffer to store directory entry information (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if dirent argument is NULL
OS_ERR_INVALID_ID	if the directory handle is invalid
OS_ERROR	at the end of the directory or if the OS call otherwise fails

OS_DirectoryRewind()

```
int32 OS_DirectoryRewind (
    osal_id_t dir_id )
```

Rewinds an open directory.

Resets a directory read handle back to the first file.

Parameters

in	<i>dir_id</i>	The handle ID of the directory
----	---------------	--------------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the directory handle is invalid

OS_mkdir()

```
int32 OS_mkdir (
    const char * path,
    uint32 access )
```

Makes a new directory.

Makes a directory specified by path.

Parameters

in	<i>path</i>	The new directory name (must not be null)
in	<i>access</i>	The permissions for the directory (reserved for future use)

Note

Current implementations do not utilize the "access" parameter. Applications should still pass the intended value ([OS_READ_WRITE](#) or [OS_READ_ONLY](#)) to be compatible with future implementations.

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if path is NULL
OS_FS_ERR_PATH_TOO_LONG	if the path is too long to be stored locally
OS_FS_ERR_PATH_INVALID	if path cannot be parsed
OS_ERROR	if the OS call fails (return value only verified in coverage test)

OS_rmdir()

```
int32 OS_rmdir (
    const char * path )
```

Removes a directory from the file system.

Removes a directory from the structure. The directory must be empty prior to this operation.

Parameters

in	<i>path</i>	The directory to remove
----	-------------	-------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if path is NULL
OS_FS_ERR_PATH_INVALID	if path cannot be parsed
OS_FS_ERR_PATH_TOO_LONG	
OS_ERROR	if the directory remove operation failed (return value only verified in coverage test)

9.52 OSAL Return Code Defines**Macros**

- `#define OS_SUCCESS (0)`
Successful execution.
- `#define OS_ERROR (-1)`
Failed execution.
- `#define OS_INVALID_POINTER (-2)`
Invalid pointer.
- `#define OS_ERROR_ADDRESS_MISALIGNED (-3)`
Address misalignment.
- `#define OS_ERROR_TIMEOUT (-4)`
Error timeout.
- `#define OS_INVALID_INT_NUM (-5)`
Invalid Interrupt number.
- `#define OS_SEM_FAILURE (-6)`
Semaphore failure.
- `#define OS_SEM_TIMEOUT (-7)`
Semaphore timeout.
- `#define OS_QUEUE_EMPTY (-8)`
Queue empty.
- `#define OS_QUEUE_FULL (-9)`
Queue full.
- `#define OS_QUEUE_TIMEOUT (-10)`
Queue timeout.

- `#define OS_QUEUE_INVALID_SIZE (-11)`
Queue invalid size.
- `#define OS_QUEUE_ID_ERROR (-12)`
Queue ID error.
- `#define OS_ERR_NAME_TOO_LONG (-13)`
name length including null terminator greater than `OS_MAX_API_NAME`
- `#define OS_ERR_NO_FREE_IDS (-14)`
No free IDs.
- `#define OS_ERR_NAME_TAKEN (-15)`
Name taken.
- `#define OS_ERR_INVALID_ID (-16)`
Invalid ID.
- `#define OS_ERR_NAME_NOT_FOUND (-17)`
Name not found.
- `#define OS_ERR_SEM_NOT_FULL (-18)`
Semaphore not full.
- `#define OS_ERR_INVALID_PRIORITY (-19)`
Invalid priority.
- `#define OS_INVALID_SEM_VALUE (-20)`
Invalid semaphore value.
- `#define OS_ERR_FILE (-27)`
File error.
- `#define OS_ERR_NOT_IMPLEMENTED (-28)`
Not implemented.
- `#define OS_TIMER_ERR_INVALID_ARGS (-29)`
Timer invalid arguments.
- `#define OS_TIMER_ERR_TIMER_ID (-30)`
Timer ID error.
- `#define OS_TIMER_ERR_UNAVAILABLE (-31)`
Timer unavailable.
- `#define OS_TIMER_ERR_INTERNAL (-32)`
Timer internal error.
- `#define OS_ERR_OBJECT_IN_USE (-33)`
Object in use.
- `#define OS_ERR_BAD_ADDRESS (-34)`
Bad address.
- `#define OS_ERR_INCORRECT_OBJ_STATE (-35)`
Incorrect object state.
- `#define OS_ERR_INCORRECT_OBJ_TYPE (-36)`
Incorrect object type.
- `#define OS_ERR_STREAM_DISCONNECTED (-37)`
Stream disconnected.
- `#define OS_ERR_OPERATION_NOT_SUPPORTED (-38)`
Requested operation not support on supplied object(s)
- `#define OS_ERR_INVALID_SIZE (-40)`
Invalid Size.
- `#define OS_ERR_OUTPUT_TOO_LARGE (-41)`

Size of output exceeds limit

- `#define OS_ERR_INVALID_ARGUMENT (-42)`
Invalid argument value (other than ID or size)
- `#define OS_ERR_TRY_AGAIN (-43)`
Failure is temporary in nature, call may be repeated.
- `#define OS_ERR_EMPTY_SET (-44)`
Address or name lookup returned no results.
- `#define OS_FS_ERR_PATH_TOO_LONG (-103)`
FS path too long.
- `#define OS_FS_ERR_NAME_TOO_LONG (-104)`
FS name too long.
- `#define OS_FS_ERR_DRIVE_NOT_CREATED (-106)`
FS drive not created.
- `#define OS_FS_ERR_DEVICE_NOT_FREE (-107)`
FS device not free.
- `#define OS_FS_ERR_PATH_INVALID (-108)`
FS path invalid.

9.52.1 Detailed Description

The specific status/return code definitions listed in this section may be extended or refined in future versions of OSAL.

Note

Application developers should assume that any OSAL API may return any status value listed here. While the documentation of each OSAL API function indicates the return/status values that function may directly generate, functions may also pass through other status codes from related functions, so that list should not be considered absolute/exhaustive.

The `int32` data type should be used to store an OSAL status code. Negative values will always represent errors, while non-negative values indicate success. Most APIs specifically return `OS_SUCCESS` (0) upon successful execution, but some return a nonzero value, such as data size.

Ideally, in order to more easily adapt to future OSAL versions and status code extensions/refinements, applications should typically check for errors as follows:

```
int32 status;
status = OS_TaskCreate(...); (or any other API)
if (status < OS_SUCCESS)
{
    handle or report error....
    may also check for specific codes here.
}
else
{
    handle normal/successful status...
}
```

9.52.2 Macro Definition Documentation

OS_ERR_BAD_ADDRESS

`#define OS_ERR_BAD_ADDRESS (-34)`

Bad address.

Definition at line 124 of file [osapi-error.h](#).

OS_ERR_EMPTY_SET

#define OS_ERR_EMPTY_SET (-44)
Address or name lookup returned no results.
Definition at line 133 of file [osapi-error.h](#).

OS_ERR_FILE

#define OS_ERR_FILE (-27)
File error.
Definition at line 117 of file [osapi-error.h](#).

OS_ERR_INCORRECT_OBJ_STATE

#define OS_ERR_INCORRECT_OBJ_STATE (-35)
Incorrect object state.
Definition at line 125 of file [osapi-error.h](#).

OS_ERR_INCORRECT_OBJ_TYPE

#define OS_ERR_INCORRECT_OBJ_TYPE (-36)
Incorrect object type.
Definition at line 126 of file [osapi-error.h](#).

OS_ERR_INVALID_ARGUMENT

#define OS_ERR_INVALID_ARGUMENT (-42)
Invalid argument value (other than ID or size)
Definition at line 131 of file [osapi-error.h](#).

OS_ERR_INVALID_ID

#define OS_ERR_INVALID_ID (-16)
Invalid ID.
Definition at line 112 of file [osapi-error.h](#).

OS_ERR_INVALID_PRIORITY

#define OS_ERR_INVALID_PRIORITY (-19)
Invalid priority.
Definition at line 115 of file [osapi-error.h](#).

OS_ERR_INVALID_SIZE

#define OS_ERR_INVALID_SIZE (-40)
Invalid Size.
Definition at line 129 of file [osapi-error.h](#).

OS_ERR_NAME_NOT_FOUND

#define OS_ERR_NAME_NOT_FOUND (-17)
Name not found.
Definition at line 113 of file [osapi-error.h](#).

OS_ERR_NAME_TAKEN

#define OS_ERR_NAME_TAKEN (-15)
Name taken.
Definition at line 111 of file [osapi-error.h](#).

OS_ERR_NAME_TOO_LONG

#define OS_ERR_NAME_TOO_LONG (-13)
name length including null terminator greater than [OS_MAX_API_NAME](#)
Definition at line 109 of file [osapi-error.h](#).

OS_ERR_NO_FREE_IDS

#define OS_ERR_NO_FREE_IDS (-14)
No free IDs.
Definition at line 110 of file [osapi-error.h](#).

OS_ERR_NOT_IMPLEMENTED

#define OS_ERR_NOT_IMPLEMENTED (-28)
Not implemented.
Definition at line 118 of file [osapi-error.h](#).

OS_ERR_OBJECT_IN_USE

#define OS_ERR_OBJECT_IN_USE (-33)
Object in use.
Definition at line 123 of file [osapi-error.h](#).

OS_ERR_OPERATION_NOT_SUPPORTED

#define OS_ERR_OPERATION_NOT_SUPPORTED (-38)
Requested operation not support on supplied object(s)
Definition at line 128 of file [osapi-error.h](#).

OS_ERR_OUTPUT_TOO_LARGE

#define OS_ERR_OUTPUT_TOO_LARGE (-41)
Size of output exceeds limit

Definition at line 130 of file [osapi-error.h](#).

OS_ERR_SEM_NOT_FULL

#define OS_ERR_SEM_NOT_FULL (-18)
Semaphore not full.
Definition at line 114 of file [osapi-error.h](#).

OS_ERR_STREAM_DISCONNECTED

#define OS_ERR_STREAM_DISCONNECTED (-37)
Stream disconnected.
Definition at line 127 of file [osapi-error.h](#).

OS_ERR_TRY_AGAIN

```
#define OS_ERR_TRY_AGAIN (-43)
```

Failure is temporary in nature, call may be repeated.

Definition at line 132 of file [osapi-error.h](#).

OS_ERROR

```
#define OS_ERROR (-1)
```

Failed execution.

Definition at line 97 of file [osapi-error.h](#).

OS_ERROR_ADDRESS_MISALIGNED

```
#define OS_ERROR_ADDRESS_MISALIGNED (-3)
```

Address misalignment.

Definition at line 99 of file [osapi-error.h](#).

OS_ERROR_TIMEOUT

```
#define OS_ERROR_TIMEOUT (-4)
```

Error timeout.

Definition at line 100 of file [osapi-error.h](#).

OS_FS_ERR_DEVICE_NOT_FREE

```
#define OS_FS_ERR_DEVICE_NOT_FREE (-107)
```

FS device not free.

Definition at line 146 of file [osapi-error.h](#).

OS_FS_ERR_DRIVE_NOT_CREATED

```
#define OS_FS_ERR_DRIVE_NOT_CREATED (-106)
```

FS drive not created.

Definition at line 145 of file [osapi-error.h](#).

OS_FS_ERR_NAME_TOO_LONG

```
#define OS_FS_ERR_NAME_TOO_LONG (-104)
```

FS name too long.

Definition at line 144 of file [osapi-error.h](#).

OS_FS_ERR_PATH_INVALID

```
#define OS_FS_ERR_PATH_INVALID (-108)
```

FS path invalid.

Definition at line 147 of file [osapi-error.h](#).

OS_FS_ERR_PATH_TOO_LONG

```
#define OS_FS_ERR_PATH_TOO_LONG (-103)
```

FS path too long.

Definition at line 143 of file [osapi-error.h](#).

OS_INVALID_INT_NUM

#define OS_INVALID_INT_NUM (-5)
Invalid Interrupt number.
Definition at line 101 of file [osapi-error.h](#).

OS_INVALID_POINTER

#define OS_INVALID_POINTER (-2)
Invalid pointer.
Definition at line 98 of file [osapi-error.h](#).

OS_INVALID_SEM_VALUE

#define OS_INVALID_SEM_VALUE (-20)
Invalid semaphore value.
Definition at line 116 of file [osapi-error.h](#).

OS_QUEUE_EMPTY

#define OS_QUEUE_EMPTY (-8)
Queue empty.
Definition at line 104 of file [osapi-error.h](#).

OS_QUEUE_FULL

#define OS_QUEUE_FULL (-9)
Queue full.
Definition at line 105 of file [osapi-error.h](#).

OS_QUEUE_ID_ERROR

#define OS_QUEUE_ID_ERROR (-12)
Queue ID error.
Definition at line 108 of file [osapi-error.h](#).

OS_QUEUE_INVALID_SIZE

#define OS_QUEUE_INVALID_SIZE (-11)
Queue invalid size.
Definition at line 107 of file [osapi-error.h](#).

OS_QUEUE_TIMEOUT

#define OS_QUEUE_TIMEOUT (-10)
Queue timeout.
Definition at line 106 of file [osapi-error.h](#).

OS_SEM_FAILURE

#define OS_SEM_FAILURE (-6)
Semaphore failure.
Definition at line 102 of file [osapi-error.h](#).

OS_SEM_TIMEOUT

`#define OS_SEM_TIMEOUT (-7)`
Semaphore timeout.
Definition at line 103 of file [osapi-error.h](#).

OS_SUCCESS

`#define OS_SUCCESS (0)`
Successful execution.
Definition at line 96 of file [osapi-error.h](#).

OS_TIMER_ERR_INTERNAL

`#define OS_TIMER_ERR_INTERNAL (-32)`
Timer internal error.
Definition at line 122 of file [osapi-error.h](#).

OS_TIMER_ERR_INVALID_ARGS

`#define OS_TIMER_ERR_INVALID_ARGS (-29)`
Timer invalid arguments.
Definition at line 119 of file [osapi-error.h](#).

OS_TIMER_ERR_TIMER_ID

`#define OS_TIMER_ERR_TIMER_ID (-30)`
Timer ID error.
Definition at line 120 of file [osapi-error.h](#).

OS_TIMER_ERR_UNAVAILABLE

`#define OS_TIMER_ERR_UNAVAILABLE (-31)`
Timer unavailable.
Definition at line 121 of file [osapi-error.h](#).

9.53 OSAL Error Info APIs**Functions**

- static long [OS_StatusToInteger](#) ([osal_status_t](#) Status)
Convert a status code to a native "long" type.
- [int32 OS_GetErrorName](#) ([int32](#) error_num, [os_err_name_t](#) *err_name)
Convert an error number to a string.
- char * [OS_StatusToString](#) ([osal_status_t](#) status, [os_status_string_t](#) *status_string)
Convert status to a string.

9.53.1 Detailed Description**9.53.2 Function Documentation****OS_GetErrorName()**

```
int32 OS_GetErrorName (  
    int32 error_num,  
    os\_err\_name\_t * err_name )
```

Convert an error number to a string.

Parameters

in	<i>error_num</i>	Error number to convert
out	<i>err_name</i>	Buffer to store error string

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	if successfully converted to a string
OS_INVALID_POINTER	if <i>err_name</i> is NULL
OS_ERROR	if error could not be converted

OS_StatusToInteger()

```
static long OS_StatusToInteger (
    osal\_status\_t Status ) [inline], [static]
```

Convert a status code to a native "long" type.

For printing or logging purposes, this converts the given status code to a "long" (signed integer) value. It should be used in conjunction with the "%ld" conversion specifier in printf-style statements.

Parameters

in	<i>Status</i>	Execution status, see OSAL Return Code Defines
----	---------------	--

Returns

Same status value converted to the "long" data type

Definition at line [166](#) of file [osapi-error.h](#).

OS_StatusToString()

```
char * OS_StatusToString (
    osal\_status\_t status,
    os\_status\_string\_t * status_string )
```

Convert status to a string.

Parameters

in	<i>status</i>	Status value to convert
out	<i>status_string</i>	Buffer to store status converted to string

Returns

Passed in string pointer

9.54 OSAL File Access Option Defines**Macros**

- `#define OS_READ_ONLY 0`
- `#define OS_WRITE_ONLY 1`
- `#define OS_READ_WRITE 2`

9.54.1 Detailed Description**9.54.2 Macro Definition Documentation****OS_READ_ONLY**

```
#define OS_READ_ONLY 0
```

Read only file access
Definition at line 35 of file [osapi-file.h](#).

OS_READ_WRITE

```
#define OS_READ_WRITE 2
```

Read write file access
Definition at line 37 of file [osapi-file.h](#).

OS_WRITE_ONLY

```
#define OS_WRITE_ONLY 1
```

Write only file access
Definition at line 36 of file [osapi-file.h](#).

9.55 OSAL Reference Point For Seek Offset Defines**Macros**

- `#define OS_SEEK_SET 0`
- `#define OS_SEEK_CUR 1`
- `#define OS_SEEK_END 2`

9.55.1 Detailed Description**9.55.2 Macro Definition Documentation****OS_SEEK_CUR**

```
#define OS_SEEK_CUR 1
```

Seek offset current
Definition at line 44 of file [osapi-file.h](#).

OS_SEEK_END

```
#define OS_SEEK_END 2
```

Seek offset end
Definition at line 45 of file [osapi-file.h](#).

OS_SEEK_SET

```
#define OS_SEEK_SET 0
```

Seek offset set

Definition at line 43 of file [osapi-file.h](#).

9.56 OSAL Standard File APIs

Functions

- [int32 OS_OpenCreate](#) ([osal_id_t](#) *filedes, const char *path, [int32](#) flags, [int32](#) access_mode)
Open or create a file.
- [int32 OS_close](#) ([osal_id_t](#) filedес)
Closes an open file handle.
- [int32 OS_read](#) ([osal_id_t](#) filedес, void *buffer, [size_t](#) nbytes)
Read from a file handle.
- [int32 OS_write](#) ([osal_id_t](#) filedес, const void *buffer, [size_t](#) nbytes)
Write to a file handle.
- [int32 OS_TimedReadAbs](#) ([osal_id_t](#) filedес, void *buffer, [size_t](#) nbytes, [OS_time_t](#) abstime)
File/Stream input read with a timeout.
- [int32 OS_TimedRead](#) ([osal_id_t](#) filedес, void *buffer, [size_t](#) nbytes, [int32](#) timeout)
File/Stream input read with a timeout.
- [int32 OS_TimedWriteAbs](#) ([osal_id_t](#) filedес, const void *buffer, [size_t](#) nbytes, [OS_time_t](#) abstime)
File/Stream output write with a timeout.
- [int32 OS_TimedWrite](#) ([osal_id_t](#) filedес, const void *buffer, [size_t](#) nbytes, [int32](#) timeout)
File/Stream output write with a timeout.
- [int32 OS_FileAllocate](#) ([osal_id_t](#) filedес, [osal_offset_t](#) offset, [osal_offset_t](#) len)
Pre-allocates space at the given file location.
- [int32 OS_FileTruncate](#) ([osal_id_t](#) filedес, [osal_offset_t](#) len)
Changes the size of the file.
- [int32 OS_chmod](#) (const char *path, [uint32](#) access_mode)
Changes the permissions of a file.
- [int32 OS_stat](#) (const char *path, [os_fstat_t](#) *filestats)
Obtain information about a file or directory.
- [int32 OS_lseek](#) ([osal_id_t](#) filedес, [osal_offset_t](#) offset, [uint32](#) whence)
Seeks to the specified position of an open file.
- [int32 OS_remove](#) (const char *path)
Removes a file from the file system.
- [int32 OS_rename](#) (const char *old_filename, const char *new_filename)
Renames a file.
- [int32 OS_cp](#) (const char *src, const char *dest)
Copies a single file from src to dest.
- [int32 OS_mv](#) (const char *src, const char *dest)
Move a single file from src to dest.
- [int32 OS_FDGetInfo](#) ([osal_id_t](#) filedес, [OS_file_prop_t](#) *fd_prop)
Obtain information about an open file.
- [int32 OS_FileOpenCheck](#) (const char *Filename)
Checks to see if a file is open.
- [int32 OS_CloseAllFiles](#) (void)

Close all open files.

- `int32 OS_CloseFileByName` (const char *Filename)

Close a file by filename.

9.56.1 Detailed Description

9.56.2 Function Documentation

OS_chmod()

```
int32 OS_chmod (
    const char * path,
    uint32 access_mode )
```

Changes the permissions of a file.

Parameters

in	<i>path</i>	File to change (must not be null)
in	<i>access_mode</i>	Desired access mode - see OSAL File Access Option Defines

Note

Some file systems do not implement permissions. If the underlying OS does not support this operation, then `OS_ERR_NOT_IMPLEMENTED` is returned.

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<code>OS_SUCCESS</code>	Successful execution. (return value only verified in coverage test)
<code>OS_ERR_NOT_IMPLEMENTED</code>	if the filesystem does not support this call
<code>OS_INVALID_POINTER</code>	if the path argument is NULL

OS_close()

```
int32 OS_close (
    osal_id_t filedes )
```

Closes an open file handle.

This closes regular file handles and any other file-like resource, such as network streams or pipes.

Parameters

in	<i>filedes</i>	The handle ID to operate on
----	----------------	-----------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<i>OS_SUCCESS</i>	Successful execution.
<i>OS_ERR_INVALID_ID</i>	if the file descriptor passed in is invalid
<i>OS_ERROR</i>	if an unexpected/unhandled error occurs (return value only verified in coverage test)

OS_CloseAllFiles()

```
int32 OS_CloseAllFiles (
    void )
```

Close all open files.

Closes All open files that were opened through the OSAL

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<i>OS_SUCCESS</i>	Successful execution.
<i>OS_ERROR</i>	if one or more file close returned an error (return value only verified in coverage test)

OS_CloseFileByName()

```
int32 OS_CloseFileByName (
    const char * Filename )
```

Close a file by filename.

Allows a file to be closed by name. This will only work if the name passed in is the same name used to open the file.

Parameters

in	<i>Filename</i>	The file to close (must not be null)
----	-----------------	--------------------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<i>OS_SUCCESS</i>	Successful execution.
<i>OS_FS_ERR_PATH_INVALID</i>	if the file is not found
<i>OS_ERROR</i>	if the file close returned an error (return value only verified in coverage test)
<i>OS_INVALID_POINTER</i>	if the filename argument is NULL

OS_cp()

```
int32 OS_cp (
    const char * src,
    const char * dest )
```

Copies a single file from src to dest.

Note

The behavior of this API on an open file is not defined at the OSAL level due to dependencies on the underlying OS which may or may not allow the related operation based on a variety of potential configurations. For portability, it is recommended that applications ensure the file is closed prior to removal.

Parameters

in	<i>src</i>	The source file to operate on (must not be null)
in	<i>dest</i>	The destination file (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERROR	if the file could not be accessed
OS_INVALID_POINTER	if src or dest are NULL
OS_FS_ERR_PATH_INVALID	if path cannot be parsed
OS_FS_ERR_PATH_TOO_LONG	if the paths given are too long to be stored locally
OS_FS_ERR_NAME_TOO_LONG	if the dest name is too long to be stored locally

OS_FDGetInfo()

```
int32 OS_FDGetInfo (
    osal_id_t filedes,
    OS_file_prop_t * fd_prop )
```

Obtain information about an open file.

Copies the information of the given file descriptor into a structure passed in

Parameters

in	<i>filedes</i>	The handle ID to operate on
out	<i>fd_prop</i>	Storage buffer for file information (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the file descriptor passed in is invalid
OS_INVALID_POINTER	if the fd_prop argument is NULL

OS_FileAllocate()

```
int32 OS_FileAllocate (
    osal_id_t filedes,
    osal_offset_t offset,
    osal_offset_t len )
```

Pre-allocates space at the given file location.

Instructs the underlying OS/Filesystem to pre-allocate filesystem blocks for the given file location and length. After this, future writes into the same file area will not fail due to lack of space.

Parameters

in	<i>filedes</i>	The handle ID to operate on
in	<i>offset</i>	The offset within the file
in	<i>len</i>	The length of space to pre-allocate

Note

Some file systems do not implement this capability

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution. (return value only verified in coverage test)
OS_ERR_OUTPUT_TOO_LARGE	if this would case the file to be too large
OS_ERR_OPERATION_NOT_SUPPORTED	if the filesystem does not support this

OS_FileOpenCheck()

```
int32 OS_FileOpenCheck (
    const char * Filename )
```

Checks to see if a file is open.

This function takes a filename and determines if the file is open. The function will return success if the file is open.

Parameters

in	<i>Filename</i>	The file to operate on (must not be null)
----	-----------------	---

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	if the file is open
OS_ERROR	if the file is not open
OS_INVALID_POINTER	if the filename argument is NULL

OS_FileTruncate()

```
int32 OS_FileTruncate (
    osal_id_t filedes,
    osal_offset_t len )
```

Changes the size of the file.

This changes the size of the file on disk to the specified value. It may either extend or truncate the file.

Note

No data is written to the extended file area when a file is grown using this API call. Depending on the file system in use, data blocks may not be allocated at the same time (a sparse file). Data blocks are allocated at the time file data is written into the extended area.

Parameters

in	<i>filedes</i>	The handle ID to operate on
in	<i>len</i>	The desired length of the file

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution. (return value only verified in coverage test)
OS_ERR_OUTPUT_TOO_LARGE	if this would case the file to be too large

OS_lseek()

```
int32 OS_lseek (
    osal_id_t filedes,
    osal_offset_t offset,
    uint32 whence )
```

Seeks to the specified position of an open file.

Sets the read/write pointer to a specific offset in a specific file.

Parameters

in	<i>filedes</i>	The handle ID to operate on
in	<i>offset</i>	The file offset to seek to
in	<i>whence</i>	The reference point for offset, see OSAL Reference Point For Seek Offset Defines

Returns

Byte offset from the beginning of the file or appropriate error code, see [OSAL Return Code Defines](#)

Return values

OS_ERR_INVALID_ID	if the file descriptor passed in is invalid
OS_ERROR	if OS call failed (return value only verified in coverage test)

OS_mv()

```
int32 OS_mv (
    const char * src,
    const char * dest )
```

Move a single file from src to dest.

This first attempts to rename the file, which is faster if the source and destination reside on the same file system.

If this fails, it falls back to copying the file and removing the original.

Note

The behavior of this API on an open file is not defined at the OSAL level due to dependencies on the underlying OS which may or may not allow the related operation based on a variety of potential configurations. For portability, it is recommended that applications ensure the file is closed prior to removal.

Parameters

in	<i>src</i>	The source file to operate on (must not be null)
in	<i>dest</i>	The destination file (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERROR	if the file could not be renamed.
OS_INVALID_POINTER	if src or dest are NULL
OS_FS_ERR_PATH_INVALID	if path cannot be parsed
OS_FS_ERR_PATH_TOO_LONG	if the paths given are too long to be stored locally
OS_FS_ERR_NAME_TOO_LONG	if the dest name is too long to be stored locally

OS_OpenCreate()

```
int32 OS_OpenCreate (
    osal_id_t * filedes,
    const char * path,
    int32 flags,
    int32 access_mode )
```

Open or create a file.

Implements the same as OS_open/OS_creat but follows the OSAL paradigm of outputting the ID/descriptor separately from the return value, rather than relying on the user to convert it back.

Parameters

out	<i>filedes</i>	The handle ID (OS_OBJECT_ID_UNDEFINED on failure) (must not be null)
in	<i>path</i>	File name to create or open (must not be null)
in	<i>flags</i>	The file permissions - see OS_file_flag_t
in	<i>access_mode</i>	Intended access mode - see OSAL File Access Option Defines

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERROR	if the command was not executed properly
OS_INVALID_POINTER	if pointer argument was NULL
OS_ERR_NO_FREE_IDS	if all available file handles are in use
OS_FS_ERR_NAME_TOO_LONG	if the filename portion of the path exceeds OS_MAX_FILE_NAME
OS_FS_ERR_PATH_INVALID	if the path argument is not valid
OS_FS_ERR_PATH_TOO_LONG	if the path argument exceeds OS_MAX_PATH_LEN

OS_read()

```
int32 OS_read (
    osal_id_t filedes,
    void * buffer,
    size_t nbytes )
```

Read from a file handle.

Reads up to nbytes from a file, and puts them into buffer.

If the file position is at the end of file (or beyond, if the OS allows) then this function will return 0.

Parameters

in	<i>filedes</i>	The handle ID to operate on
out	<i>buffer</i>	Storage location for file data (must not be null)
in	<i>nbytes</i>	Maximum number of bytes to read (must not be zero)

Note

All OSAL error codes are negative int32 values. Failure of this call can be checked by testing if the result is less than 0.

Returns

A non-negative byte count or appropriate error code, see [OSAL Return Code Defines](#)

Return values

OS_INVALID_POINTER	if buffer is a null pointer
OS_ERR_INVALID_SIZE	if the passed-in size is not valid
OS_ERROR	if OS call failed (return value only verified in coverage test)
OS_ERR_INVALID_ID	if the file descriptor passed in is invalid
0	if at end of file/stream data

OS_remove()

```
int32 OS_remove (
    const char * path )
```

Removes a file from the file system.

Removes a given filename from the drive

Note

The behavior of this API on an open file is not defined at the OSAL level due to dependencies on the underlying OS which may or may not allow the related operation based on a variety of potential configurations. For portability, it is recommended that applications ensure the file is closed prior to removal.

Parameters

in	<i>path</i>	The file to operate on (must not be null)
----	-------------	---

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERROR	if there is no device or the driver returns error
OS_INVALID_POINTER	if path is NULL
OS_FS_ERR_PATH_TOO_LONG	if path is too long to be stored locally
OS_FS_ERR_PATH_INVALID	if path cannot be parsed
OS_FS_ERR_NAME_TOO_LONG	if the name of the file to remove is too long

OS_rename()

```
int32 OS_rename (
    const char * old_filename,
    const char * new_filename )
```

Renames a file.

Changes the name of a file, where the source and destination reside on the same file system.

Note

The behavior of this API on an open file is not defined at the OSAL level due to dependencies on the underlying OS which may or may not allow the related operation based on a variety of potential configurations. For portability, it is recommended that applications ensure the file is closed prior to removal.

Parameters

in	<i>old_filename</i>	The original filename (must not be null)
in	<i>new_filename</i>	The desired filename (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERROR	if the file could not be opened or renamed.
OS_INVALID_POINTER	if old_filename or new_filename are NULL
OS_FS_ERR_PATH_INVALID	if path cannot be parsed
OS_FS_ERR_PATH_TOO_LONG	if the paths given are too long to be stored locally
OS_FS_ERR_NAME_TOO_LONG	if the new name is too long to be stored locally

OS_stat()

```
int32 OS_stat (
    const char * path,
    os_fstat_t * filestats )
```

Obtain information about a file or directory.

Returns information about a file or directory in an [os_fstat_t](#) structure

Parameters

in	<i>path</i>	The file to operate on (must not be null)
out	<i>filestats</i>	Buffer to store file information (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if path or filestats is NULL
OS_FS_ERR_PATH_TOO_LONG	if the path is too long to be stored locally
OS_FS_ERR_NAME_TOO_LONG	if the name of the file is too long to be stored
OS_FS_ERR_PATH_INVALID	if path cannot be parsed
OS_ERROR	if the OS call failed

OS_TimedRead()

```
int32 OS_TimedRead (
    osal_id_t filedes,
    void * buffer,
    size_t nbytes,
    int32 timeout )
```

File/Stream input read with a timeout.

This implements a time-limited read and is primarily intended for use with sockets but may also work with any other stream-like resource that the underlying OS supports, such as pipes or special devices.

If data is immediately available on the file/socket, this will return that data along with the actual number of bytes that were immediately available. It will not block.

If the file position is at the end of file or end of stream data (e.g. if the remote end has closed the connection), then this function will immediately return 0 without blocking for the timeout period.

If no data is immediately available, but the underlying resource/stream is still connected to a peer, this will wait up to the given timeout for additional data to appear. If no data appears within the timeout period, then this returns the [OS_ERROR_TIMEOUT](#) status code. This allows the caller to differentiate an open (but idle) socket connection from a connection which has been closed by the remote peer.

In all cases this will return successfully as soon as at least 1 byte of actual data is available. It will not attempt to read the entire input buffer.

If an EOF condition occurs prior to timeout, this function returns zero.

Parameters

in	<i>filedes</i>	The handle ID to operate on
out	<i>buffer</i>	Storage location for file data (must not be null)
in	<i>nbytes</i>	Maximum number of bytes to read (must not be zero)
in	<i>timeout</i>	Maximum time to wait, in milliseconds, relative to current time (OS_PEND = forever)

Returns

Byte count on success or appropriate error code, see [OSAL Return Code Defines](#)

Return values

OS_ERROR_TIMEOUT	if no data became available during timeout period
OS_ERR_INVALID_ID	if the file descriptor passed in is invalid
OS_ERR_INVALID_SIZE	if the passed-in size is not valid
OS_INVALID_POINTER	if the passed-in buffer is not valid
0	if at end of file/stream data

OS_TimedReadAbs()

```
int32 OS_TimedReadAbs (
    osal_id_t filedes,
    void * buffer,
    size_t nbytes,
    OS_time_t abstime )
```

File/Stream input read with a timeout.

This implements a time-limited read and is primarily intended for use with sockets but may also work with any other stream-like resource that the underlying OS supports, such as pipes or special devices.

If data is immediately available on the file/socket, this will return that data along with the actual number of bytes that were immediately available. It will not block.

If the file position is at the end of file or end of stream data (e.g. if the remote end has closed the connection), then this function will immediately return 0 without blocking for the timeout period.

If no data is immediately available, but the underlying resource/stream is still connected to a peer, this will wait up to the given timeout for additional data to appear. If no data appears within the timeout period, then this returns the [OS_ERROR_TIMEOUT](#) status code. This allows the caller to differentiate an open (but idle) socket connection from a connection which has been closed by the remote peer.

In all cases this will return successfully as soon as at least 1 byte of actual data is available. It will not attempt to read the entire input buffer.

If an EOF condition occurs prior to timeout, this function returns zero.

Parameters

in	<i>filedes</i>	The handle ID to operate on
out	<i>buffer</i>	Storage location for file data (must not be null)
in	<i>nbytes</i>	Maximum number of bytes to read (must not be zero)
in	<i>abstime</i>	Absolute time at which this function should return, if no data is readable

Returns

Byte count on success or appropriate error code, see [OSAL Return Code Defines](#)

Return values

OS_ERROR_TIMEOUT	if no data became available during timeout period
OS_ERR_INVALID_ID	if the file descriptor passed in is invalid
OS_ERR_INVALID_SIZE	if the passed-in size is not valid
OS_INVALID_POINTER	if the passed-in buffer is not valid
0	if at end of file/stream data

OS_TimedWrite()

```
int32 OS_TimedWrite (
    osal_id_t filedes,
    const void * buffer,
    size_t nbytes,
    int32 timeout )
```

File/Stream output write with a timeout.

This implements a time-limited write and is primarily intended for use with sockets but may also work with any other stream-like resource that the underlying OS supports.

If output buffer space is immediately available on the file/socket, this will place data into the buffer and return the actual number of bytes that were queued for output. It will not block.

If no output buffer space is immediately available, this will wait up to the given timeout for space to become available. If no space becomes available within the timeout period, then this returns an error code (not zero).

In all cases this will return successfully as soon as at least 1 byte of actual data is output. It will *not* attempt to write the entire output buffer.

If an EOF condition occurs prior to timeout, this function returns zero.

Parameters

in	<i>filedes</i>	The handle ID to operate on
in	<i>buffer</i>	Source location for file data (must not be null)
in	<i>nbytes</i>	Maximum number of bytes to read (must not be zero)
in	<i>timeout</i>	Maximum time to wait, in milliseconds, relative to current time (OS_PEND = forever)

Returns

A non-negative byte count or appropriate error code, see [OSAL Return Code Defines](#)

Return values

OS_ERROR_TIMEOUT	if no data became available during timeout period
OS_ERR_INVALID_ID	if the file descriptor passed in is invalid
OS_ERR_INVALID_SIZE	if the passed-in size is not valid
OS_INVALID_POINTER	if the passed-in buffer is not valid
0	if file/stream cannot accept any more data

OS_TimedWriteAbs()

```
int32 OS_TimedWriteAbs (
    osal_id_t filedes,
    const void * buffer,
    size_t nbytes,
    OS_time_t abstime )
```

File/Stream output write with a timeout.

This implements a time-limited write and is primarily intended for use with sockets but may also work with any other stream-like resource that the underlying OS supports.

If output buffer space is immediately available on the file/socket, this will place data into the buffer and return the actual number of bytes that were queued for output. It will not block.

If no output buffer space is immediately available, this will wait up to the given timeout for space to become available. If no space becomes available within the timeout period, then this returns an error code (not zero).

In all cases this will return successfully as soon as at least 1 byte of actual data is output. It will *not* attempt to write the entire output buffer.

If an EOF condition occurs prior to timeout, this function returns zero.

Parameters

in	<i>filedes</i>	The handle ID to operate on
in	<i>buffer</i>	Source location for file data (must not be null)
in	<i>nbytes</i>	Maximum number of bytes to read (must not be zero)
in	<i>abstime</i>	Absolute time at which this function should return, if no data is readable

Returns

A non-negative byte count or appropriate error code, see [OSAL Return Code Defines](#)

Return values

OS_ERROR_TIMEOUT	if no data became available during timeout period
OS_ERR_INVALID_ID	if the file descriptor passed in is invalid
OS_ERR_INVALID_SIZE	if the passed-in size is not valid
OS_INVALID_POINTER	if the passed-in buffer is not valid
0	if file/stream cannot accept any more data

OS_write()

```
int32 OS_write (
```

```
osal_id_t filedes,
const void * buffer,
size_t nbytes )
```

Write to a file handle.

Writes to a file. copies up to a maximum of nbytes of buffer to the file described in filedes

Parameters

in	<i>filedes</i>	The handle ID to operate on
in	<i>buffer</i>	Source location for file data (must not be null)
in	<i>nbytes</i>	Maximum number of bytes to read (must not be zero)

Note

All OSAL error codes are negative int32 values. Failure of this call can be checked by testing if the result is less than 0.

Returns

A non-negative byte count or appropriate error code, see [OSAL Return Code Defines](#)

Return values

OS_INVALID_POINTER	if buffer is NULL
OS_ERR_INVALID_SIZE	if the passed-in size is not valid
OS_ERROR	if OS call failed (return value only verified in coverage test)
OS_ERR_INVALID_ID	if the file descriptor passed in is invalid
0	if file/stream cannot accept any more data

9.57 OSAL File System Level APIs

Functions

- [int32 OS_FileSysAddFixedMap](#) ([osal_id_t](#) *filesys_id, const char *phys_path, const char *virt_path)
Create a fixed mapping between an existing directory and a virtual OSAL mount point.
- [int32 OS_mkfs](#) (char *address, const char *devname, const char *volname, size_t blocksize, [osal_blockcount_t](#) numblocks)
Makes a file system on the target.
- [int32 OS_mount](#) (const char *devname, const char *mountpoint)
Mounts a file system.
- [int32 OS_initfs](#) (char *address, const char *devname, const char *volname, size_t blocksize, [osal_blockcount_t](#) numblocks)
Initializes an existing file system.
- [int32 OS_rmfs](#) (const char *devname)
Removes a file system.
- [int32 OS_unmount](#) (const char *mountpoint)
Unmounts a mounted file system.
- [int32 OS_FileSysStatVolume](#) (const char *name, [OS_statvfs_t](#) *statbuf)
Obtains information about size and free space in a volume.

- [int32 OS_chkfs](#) (const char *name, bool repair)
Checks the health of a file system and repairs it if necessary.
- [int32 OS_FS_GetPhysDriveName](#) (char *PhysDriveName, const char *MountPoint)
Obtains the physical drive name associated with a mount point.
- [int32 OS_TranslatePath](#) (const char *VirtualPath, char *LocalPath)
Translates an OSAL Virtual file system path to a host Local path.
- [int32 OS_GetFsInfo](#) (os_fsinfo_t *filesys_info)
Returns information about the file system.

9.57.1 Detailed Description

9.57.2 Function Documentation

OS_chkfs()

```
int32 OS_chkfs (
    const char * name,
    bool repair )
```

Checks the health of a file system and repairs it if necessary.
Checks the drives for inconsistencies and optionally also repairs it

Note

not all operating systems implement this function. If the underlying OS does not provide a facility to check the volume, then OS_ERR_NOT_IMPLEMENTED will be returned.

Parameters

in	<i>name</i>	The device/path to operate on (must not be null)
in	<i>repair</i>	Whether to also repair inconsistencies

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution. (return value only verified in coverage test)
OS_INVALID_POINTER	Name is NULL
OS_ERR_NOT_IMPLEMENTED	Not implemented.
OS_FS_ERR_PATH_TOO_LONG	if the name is too long
OS_ERROR	Failed execution. (return value only verified in coverage test)

OS_FileSysAddFixedMap()

```
int32 OS_FileSysAddFixedMap (
    osal_id_t * filesys_id,
    const char * phys_path,
    const char * virt_path )
```

Create a fixed mapping between an existing directory and a virtual OSAL mount point.

This mimics the behavior of a "FS_BASED" entry in the VolumeTable but is registered at runtime. It is intended to be called by the PSP/BSP prior to starting the application.

Note

OSAL virtual mount points are required to be a single, non-empty top-level directory name. Virtual path names always follow the form /<virt_mount_point>/<relative_path>/<file>. Only the relative path may be omitted/empty (i.e. /<virt_mount_point>/<file>) but the virtual mount point must be present and not an empty string. In particular this means it is not possible to directly refer to files in the "root" of the native file system from OSAL. However it is possible to create a virtual map to the root, such as by calling:

```
OS_FileSysAddFixedMap(&fs_id, "/", "/root");
```

Parameters

out	<i>filesys_id</i>	A buffer to store the ID of the file system mapping (must not be null)
in	<i>phys_path</i>	The native system directory (an existing mount point) (must not be null)
in	<i>virt_path</i>	The virtual mount point of this filesystem (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_FS_ERR_PATH_TOO_LONG	if the overall <i>phys_path</i> is too long
OS_ERR_NAME_TOO_LONG	if the <i>phys_path</i> basename (filesystem name) is too long
OS_INVALID_POINTER	if any argument is NULL

OS_FileSysStatVolume()

```
int32 OS_FileSysStatVolume (
    const char * name,
    OS_statvfs_t * statbuf )
```

Obtains information about size and free space in a volume.

Populates the supplied [OS_statvfs_t](#) structure, which includes the block size and total/free blocks in a file system volume.

This replaces two older OSAL calls:

`OS_fsBlocksFree()` is determined by reading the `blocks_free` output struct member `OS_fsBytesFree()` is determined by multiplying `blocks_free` by the `block_size` member

Parameters

in	<i>name</i>	The device/path to operate on (must not be null)
out	<i>statbuf</i>	Output structure to populate (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<i>OS_SUCCESS</i>	Successful execution.
<i>OS_INVALID_POINTER</i>	if name or statbuf is NULL
<i>OS_FS_ERR_PATH_TOO_LONG</i>	if the name is too long
<i>OS_ERROR</i>	if an unexpected/unhandled OS error occurs (return value only verified in coverage test)

OS_FS_GetPhysDriveName()

```
int32 OS_FS_GetPhysDriveName (
    char * PhysDriveName,
    const char * MountPoint )
```

Obtains the physical drive name associated with a mount point.

Returns the name of the physical volume associated with the drive, when given the OSAL mount point of the drive

Parameters

out	<i>PhysDriveName</i>	Buffer to store physical drive name (must not be null)
in	<i>MountPoint</i>	OSAL mount point (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<i>OS_SUCCESS</i>	Successful execution.
<i>OS_INVALID_POINTER</i>	if either parameter is NULL
<i>OS_ERR_NAME_NOT_FOUND</i>	if the MountPoint is not mounted in OSAL
<i>OS_FS_ERR_PATH_TOO_LONG</i>	if the MountPoint is too long

OS_GetFsInfo()

```
int32 OS_GetFsInfo (
    os_fsinfo_t * filesys_info )
```

Returns information about the file system.

Returns information about the file system in an [os_fsinfo_t](#). This includes the number of open files and file systems

Parameters

out	<i>filesys_info</i>	Buffer to store filesystem information (must not be null)
-----	---------------------	---

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<i>OS_SUCCESS</i>	Successful execution.
<i>OS_INVALID_POINTER</i>	if <code>filesystem_info</code> is NULL

OS_initfs()

```
int32 OS_initfs (
    char * address,
    const char * devname,
    const char * volname,
    size_t blocksize,
    osal_blockcount_t numblocks )
```

Initializes an existing file system.

Initializes a file system on the target.

Note

The "volname" parameter of RAM disks should always begin with the string "RAM", e.g. "RAMDISK" or "RAM0", "↔ RAM1", etc if multiple devices are created. The underlying implementation uses this to select the correct filesystem type/format, and this may also be used to differentiate between RAM disks and real physical disks.

Parameters

in	<i>address</i>	The address at which to start the new disk. If <code>address == 0</code> , then space will be allocated by the OS
in	<i>devname</i>	The underlying kernel device to use, if applicable. (must not be null)
in	<i>volname</i>	The name of the volume (see note) (must not be null)
in	<i>blocksize</i>	The size of a single block on the drive
in	<i>numblocks</i>	The number of blocks to allocate for the drive

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<i>OS_SUCCESS</i>	Successful execution.
<i>OS_INVALID_POINTER</i>	if <code>devname</code> or <code>volname</code> are NULL
<i>OS_FS_ERR_PATH_TOO_LONG</i>	if the name is too long
<i>OS_FS_ERR_DEVICE_NOT_FREE</i>	if the volume table is full
<i>OS_FS_ERR_DRIVE_NOT_CREATED</i>	if an unexpected/unhandled OS error occurs (return value only verified in coverage test)

OS_mkfs()

```
int32 OS_mkfs (
    char * address,
    const char * devname,
```

```

    const char * volname,
    size_t blocksize,
    osal_blockcount_t numblocks )

```

Makes a file system on the target.

Makes a file system on the target. Highly dependent on underlying OS and dependent on OS volume table definition.

Note

The "volname" parameter of RAM disks should always begin with the string "RAM", e.g. "RAMDISK" or "RAM0", "←RAM1", etc if multiple devices are created. The underlying implementation uses this to select the correct filesystem type/format, and this may also be used to differentiate between RAM disks and real physical disks.

Parameters

in	<i>address</i>	The address at which to start the new disk. If address == 0 space will be allocated by the OS.
in	<i>devname</i>	The underlying kernel device to use, if applicable. (must not be null)
in	<i>volname</i>	The name of the volume (see note) (must not be null)
in	<i>blocksize</i>	The size of a single block on the drive
in	<i>numblocks</i>	The number of blocks to allocate for the drive

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if devname or volname is NULL
OS_FS_ERR_PATH_TOO_LONG	if the overall devname or volname is too long
OS_FS_ERR_DEVICE_NOT_FREE	if the volume table is full
OS_FS_ERR_DRIVE_NOT_CREATED	if an unexpected/unhandled OS error occurs (return value only verified in coverage test)

OS_mount()

```

int32 OS_mount (
    const char * devname,
    const char * mountpoint )

```

Mounts a file system.

Mounts a file system / block device at the given mount point.

Parameters

in	<i>devname</i>	The name of the drive to mount. devname is the same from OS_mkfs (must not be null)
in	<i>mountpoint</i>	The name to call this disk from now on (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<i>OS_SUCCESS</i>	Successful execution.
<i>OS_ERR_NAME_NOT_FOUND</i>	if the device name does not exist in OSAL
<i>OS_FS_ERR_PATH_TOO_LONG</i>	if the mount point string is too long
<i>OS_INVALID_POINTER</i>	if any argument is NULL
<i>OS_ERROR</i>	if an unexpected/unhandled OS error occurs (return value only verified in coverage test)

OS_rmfs()

```
int32 OS_rmfs (
    const char * devname )
```

Removes a file system.

This function will remove or un-map the target file system. Note that this is not the same as un-mounting the file system.

Parameters

in	<i>devname</i>	The name of the "generic" drive (must not be null)
----	----------------	--

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<i>OS_SUCCESS</i>	Successful execution.
<i>OS_INVALID_POINTER</i>	if devname is NULL
<i>OS_FS_ERR_PATH_TOO_LONG</i>	if the devname is too long
<i>OS_ERR_NAME_NOT_FOUND</i>	if the devname does not exist in OSAL
<i>OS_ERROR</i>	if an unexpected/unhandled OS error occurs (return value only verified in coverage test)

OS_TranslatePath()

```
int32 OS_TranslatePath (
    const char * VirtualPath,
    char * LocalPath )
```

Translates an OSAL Virtual file system path to a host Local path.

Translates a virtual path to an actual system path name

Note

The buffer provided in the LocalPath argument is required to be at least OS_MAX_PATH_LEN characters in length.

Parameters

in	<i>VirtualPath</i>	OSAL virtual path name (must not be null)
out	<i>LocalPath</i>	Buffer to store native/translated path name (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if either parameter is NULL
OS_FS_ERR_NAME_TOO_LONG	if the filename component is too long
OS_FS_ERR_PATH_INVALID	if either parameter cannot be interpreted as a path
OS_FS_ERR_PATH_TOO_LONG	if either input or output pathnames are too long

OS_unmount()

```
int32 OS_unmount (
    const char * mountpoint )
```

Unmounts a mounted file system.

This function will unmount a drive from the file system and make all open file descriptors useless.

Note

Any open file descriptors referencing this file system should be closed prior to unmounting a drive

Parameters

in	<i>mountpoint</i>	The mount point to remove from OS_mount (must not be null)
----	-------------------	--

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if name is NULL
OS_FS_ERR_PATH_TOO_LONG	if the absolute path given is too long
OS_ERR_NAME_NOT_FOUND	if the mountpoint is not mounted in OSAL
OS_ERROR	if an unexpected/unhandled OS error occurs (return value only verified in coverage test)

9.58 OSAL Heap APIs**Functions**

- [int32 OS_HeapGetInfo](#) ([OS_heap_prop_t](#) *heap_prop)

Return current info on the heap.

9.58.1 Detailed Description

9.58.2 Function Documentation

OS_HeapGetInfo()

```
int32 OS_HeapGetInfo (
    OS_heap_prop_t * heap_prop )
```

Return current info on the heap.

Parameters

out	heap_prop	Storage buffer for heap info
-----	-----------	------------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if the heap_prop argument is NULL

9.59 OSAL Object Type Defines

Macros

- #define [OS_OBJECT_TYPE_UNDEFINED](#) 0x00
Object type undefined.
- #define [OS_OBJECT_TYPE_OS_TASK](#) 0x01
Object task type.
- #define [OS_OBJECT_TYPE_OS_QUEUE](#) 0x02
Object queue type.
- #define [OS_OBJECT_TYPE_OS_COUNTSEM](#) 0x03
Object counting semaphore type.
- #define [OS_OBJECT_TYPE_OS_BINSEM](#) 0x04
Object binary semaphore type.
- #define [OS_OBJECT_TYPE_OS_MUTEX](#) 0x05
Object mutex type.
- #define [OS_OBJECT_TYPE_OS_STREAM](#) 0x06
Object stream type.
- #define [OS_OBJECT_TYPE_OS_DIR](#) 0x07
Object directory type.
- #define [OS_OBJECT_TYPE_OS_TIMEBASE](#) 0x08
Object timebase type.
- #define [OS_OBJECT_TYPE_OS_TIMECB](#) 0x09
Object timer callback type.
- #define [OS_OBJECT_TYPE_OS_MODULE](#) 0x0A
Object module type.
- #define [OS_OBJECT_TYPE_OS_FILESYS](#) 0x0B

Object file system type.

- `#define OS_OBJECT_TYPE_OS_CONSOLE 0x0C`

Object console type.

- `#define OS_OBJECT_TYPE_OS_CONDVAR 0x0D`

Object condition variable type.

- `#define OS_OBJECT_TYPE_OS_RWLOCK 0x0E`

Object readers-writer lock type.

- `#define OS_OBJECT_TYPE_USER 0x10`

Object user type.

9.59.1 Detailed Description

9.59.2 Macro Definition Documentation

OS_OBJECT_TYPE_OS_BINSEM

```
#define OS_OBJECT_TYPE_OS_BINSEM 0x04
```

Object binary semaphore type.

Definition at line 42 of file [osapi-idmap.h](#).

OS_OBJECT_TYPE_OS_CONDVAR

```
#define OS_OBJECT_TYPE_OS_CONDVAR 0x0D
```

Object condition variable type.

Definition at line 51 of file [osapi-idmap.h](#).

OS_OBJECT_TYPE_OS_CONSOLE

```
#define OS_OBJECT_TYPE_OS_CONSOLE 0x0C
```

Object console type.

Definition at line 50 of file [osapi-idmap.h](#).

OS_OBJECT_TYPE_OS_COUNTSEM

```
#define OS_OBJECT_TYPE_OS_COUNTSEM 0x03
```

Object counting semaphore type.

Definition at line 41 of file [osapi-idmap.h](#).

OS_OBJECT_TYPE_OS_DIR

```
#define OS_OBJECT_TYPE_OS_DIR 0x07
```

Object directory type.

Definition at line 45 of file [osapi-idmap.h](#).

OS_OBJECT_TYPE_OS_FILESYS

```
#define OS_OBJECT_TYPE_OS_FILESYS 0x0B
```

Object file system type.

Definition at line 49 of file [osapi-idmap.h](#).

OS_OBJECT_TYPE_OS_MODULE

```
#define OS_OBJECT_TYPE_OS_MODULE 0x0A
```

Object module type.

Definition at line 48 of file [osapi-idmap.h](#).

OS_OBJECT_TYPE_OS_MUTEX

```
#define OS_OBJECT_TYPE_OS_MUTEX 0x05
```

Object mutex type.
Definition at line 43 of file [osapi-idmap.h](#).

OS_OBJECT_TYPE_OS_QUEUE

```
#define OS_OBJECT_TYPE_OS_QUEUE 0x02
```

Object queue type.
Definition at line 40 of file [osapi-idmap.h](#).

OS_OBJECT_TYPE_OS_RWLOCK

```
#define OS_OBJECT_TYPE_OS_RWLOCK 0x0E
```

Object readers-writer lock type.
Definition at line 52 of file [osapi-idmap.h](#).

OS_OBJECT_TYPE_OS_STREAM

```
#define OS_OBJECT_TYPE_OS_STREAM 0x06
```

Object stream type.
Definition at line 44 of file [osapi-idmap.h](#).

OS_OBJECT_TYPE_OS_TASK

```
#define OS_OBJECT_TYPE_OS_TASK 0x01
```

Object task type.
Definition at line 39 of file [osapi-idmap.h](#).

OS_OBJECT_TYPE_OS_TIMEBASE

```
#define OS_OBJECT_TYPE_OS_TIMEBASE 0x08
```

Object timebase type.
Definition at line 46 of file [osapi-idmap.h](#).

OS_OBJECT_TYPE_OS_TIMECB

```
#define OS_OBJECT_TYPE_OS_TIMECB 0x09
```

Object timer callback type.
Definition at line 47 of file [osapi-idmap.h](#).

OS_OBJECT_TYPE_UNDEFINED

```
#define OS_OBJECT_TYPE_UNDEFINED 0x00
```

Object type undefined.
Definition at line 38 of file [osapi-idmap.h](#).

OS_OBJECT_TYPE_USER

```
#define OS_OBJECT_TYPE_USER 0x10
```

Object user type.
Definition at line 53 of file [osapi-idmap.h](#).

9.60 OSAL Object ID Utility APIs

Functions

- static unsigned long [OS_ObjectIdToInteger](#) ([osal_id_t](#) object_id)
Obtain an integer value corresponding to an object ID.
- static [osal_id_t](#) [OS_ObjectIdFromInteger](#) (unsigned long value)
Obtain an osal ID corresponding to an integer value.
- static bool [OS_ObjectIdEqual](#) ([osal_id_t](#) object_id1, [osal_id_t](#) object_id2)
Check two OSAL object ID values for equality.
- static bool [OS_ObjectIdDefined](#) ([osal_id_t](#) object_id)
Check if an object ID is defined.
- [int32](#) [OS_GetResourceName](#) ([osal_id_t](#) object_id, char *buffer, [size_t](#) buffer_size)
Obtain the name of an object given an arbitrary object ID.
- [osal_objtype_t](#) [OS_IdentifyObject](#) ([osal_id_t](#) object_id)
Obtain the type of an object given an arbitrary object ID.
- [int32](#) [OS_ConvertToArrayIndex](#) ([osal_id_t](#) object_id, [osal_index_t](#) *ArrayIndex)
Converts an abstract ID into a number suitable for use as an array index.
- [int32](#) [OS_ObjectIdToArrayIndex](#) ([osal_objtype_t](#) idtype, [osal_id_t](#) object_id, [osal_index_t](#) *ArrayIndex)
Converts an abstract ID into a number suitable for use as an array index.
- void [OS_ForEachObject](#) ([osal_id_t](#) creator_id, [OS_ArgCallback_t](#) callback_ptr, void *callback_arg)
call the supplied callback function for all valid object IDs
- void [OS_ForEachObjectOfType](#) ([osal_objtype_t](#) objtype, [osal_id_t](#) creator_id, [OS_ArgCallback_t](#) callback_ptr, void *callback_arg)
call the supplied callback function for valid object IDs of a specific type

9.60.1 Detailed Description

9.60.2 Function Documentation

OS_ConvertToArrayIndex()

```
int32 OS_ConvertToArrayIndex (
    osal\_id\_t object_id,
    osal\_index\_t * ArrayIndex )
```

Converts an abstract ID into a number suitable for use as an array index.

This will return a unique zero-based integer number in the range of [0,MAX) for any valid object ID. This may be used by application code as an array index for indexing into local tables.

Note

This does NOT verify the validity of the ID, that is left to the caller. This is only the conversion logic.

This routine accepts any object type, and returns a value based on the maximum number of objects for that type. This is equivalent to invoking [OS_ObjectIdToArrayIndex\(\)](#) with the idtype set to OS_OBJECT_TYPE_UNDEFINED.

See also

[OS_ObjectIdToArrayIndex](#)

Parameters

in	<i>object_id</i>	The object ID to operate on
out	<i>*ArrayIndex</i>	The Index to return (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the object_id argument is not valid
OS_INVALID_POINTER	if the ArrayIndex is NULL

OS_ForEachObject()

```
void OS_ForEachObject (
    osal_id_t creator_id,
    OS_ArgCallback_t callback_ptr,
    void * callback_arg )
```

call the supplied callback function for all valid object IDs

Loops through all defined OSAL objects of all types and calls callback_ptr on each one If creator_id is nonzero then only objects with matching creator id are processed.

Parameters

in	<i>creator_id</i>	Filter objects to those created by a specific task This may be passed as OS_OBJECT_CREATOR_ANY to return all objects
in	<i>callback_ptr</i>	Function to invoke for each matching object ID
in	<i>callback_arg</i>	Opaque Argument to pass to callback function (may be NULL)

OS_ForEachObjectOfType()

```
void OS_ForEachObjectOfType (
    osal_objtype_t objtype,
    osal_id_t creator_id,
    OS_ArgCallback_t callback_ptr,
    void * callback_arg )
```

call the supplied callback function for valid object IDs of a specific type

Loops through all defined OSAL objects of a specific type and calls callback_ptr on each one If creator_id is nonzero then only objects with matching creator id are processed.

Parameters

in	<i>objtype</i>	The type of objects to iterate
in	<i>creator_id</i>	Filter objects to those created by a specific task This may be passed as OS_OBJECT_CREATOR_ANY to return all objects
in	<i>callback_ptr</i>	Function to invoke for each matching object ID
in	<i>callback_arg</i>	Opaque Argument to pass to callback function (may be NULL)

OS_GetResourceName()

```
int32 OS_GetResourceName (
```

```
osal_id_t object_id,
char * buffer,
size_t buffer_size )
```

Obtain the name of an object given an arbitrary object ID.

All OSAL resources generally have a name associated with them. This allows application code to retrieve the name of any valid OSAL object ID.

Parameters

in	<i>object_id</i>	The object ID to operate on
out	<i>buffer</i>	Buffer in which to store the name (must not be null)
in	<i>buffer_size</i>	Size of the output storage buffer (must not be zero)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the passed-in ID is not a valid OSAL ID
OS_INVALID_POINTER	if the passed-in buffer is invalid
OS_ERR_NAME_TOO_LONG	if the name will not fit in the buffer provided

OS_IdentifyObject()

```
osal_objtype_t OS_IdentifyObject (
    osal_id_t object_id )
```

Obtain the type of an object given an arbitrary object ID.

Given an arbitrary object ID, get the type of the object

Parameters

in	<i>object_id</i>	The object ID to operate on
----	------------------	-----------------------------

Returns

The object type portion of the *object_id*, see [OSAL Object Type Defines](#) for expected values

OS_ObjectIdDefined()

```
static bool OS_ObjectIdDefined (
    osal_id_t object_id ) [inline], [static]
```

Check if an object ID is defined.

The OSAL ID values should be treated as abstract values by applications, and not directly manipulated using standard C operators.

This returns false if the ID is NOT a defined resource (i.e. free/empty/invalid).

Note

OS_ObjectIdDefined(OS_OBJECT_ID_UNDEFINED) is always guaranteed to be false.

Parameters

in	<i>object_id</i>	The first object ID
----	------------------	---------------------

Definition at line 151 of file [osapi-idmap.h](#).

References [OS_ObjectIdToInteger\(\)](#).

OS_ObjectIdEqual()

```
static bool OS_ObjectIdEqual (
    osal_id_t object_id1,
    osal_id_t object_id2 ) [inline], [static]
```

Check two OSAL object ID values for equality.

The OSAL ID values should be treated as abstract values by applications, and not directly manipulated using standard C operators.

This checks two values for equality, replacing the "==" operator.

Parameters

in	<i>object_id1</i>	The first object ID
in	<i>object_id2</i>	The second object ID

Returns

true if the object IDs are equal

Definition at line 130 of file [osapi-idmap.h](#).

References [OS_ObjectIdToInteger\(\)](#).

OS_ObjectIdFromInteger()

```
static osal_id_t OS_ObjectIdFromInteger (
    unsigned long value ) [inline], [static]
```

Obtain an osal ID corresponding to an integer value.

Provides the inverse of [OS_ObjectIdToInteger\(\)](#). Reconstitutes the original osal_id_t type from an integer representation.

Parameters

in	<i>value</i>	The integer representation of an OSAL ID
----	--------------	--

Returns

The ID value converted to an osal_id_t

Definition at line 103 of file [osapi-idmap.h](#).

OS_ObjectIdToArrayIndex()

```
int32 OS_ObjectIdToArrayIndex (
    osal_objtype_t idtype,
```

```
osal_id_t object_id,
osal_index_t * ArrayIndex )
```

Converts an abstract ID into a number suitable for use as an array index.

This will return a unique zero-based integer number in the range of [0,MAX) for any valid object ID. This may be used by application code as an array index for indexing into local tables.

This routine operates on a specific object type, and returns a value based on the maximum number of objects for that type.

If the idtype is passed as [OS_OBJECT_TYPE_UNDEFINED](#), then object type verification is skipped and any object ID will be accepted and converted to an index. In this mode, the range of the output depends on the actual passed-in object type.

If the idtype is passed as any other value, the passed-in ID value is first confirmed to be the correct type. This check will guarantee that the output is within an expected range; for instance, if the type is passed as [OS_OBJECT_TYPE_OS_TASK](#), then the output index is guaranteed to be between 0 and [OS_MAX_TASKS](#)-1 after successful conversion.

Parameters

in	<i>idtype</i>	The object type to convert
in	<i>object_id</i>	The object ID to operate on
out	<i>*ArrayIndex</i>	The Index to return (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the object_id argument is not valid
OS_INVALID_POINTER	if the ArrayIndex is NULL

OS_ObjectIdToInteger()

```
static unsigned long OS_ObjectIdToInteger (
    osal_id_t object_id ) [inline], [static]
```

Obtain an integer value corresponding to an object ID.

Obtains an integer representation of an object id, generally for the purpose of printing to the console or system logs.

The returned value is of the type "unsigned long" for direct use with printf-style functions. It is recommended to use the "%lx" conversion specifier as the hexadecimal encoding clearly delineates the internal fields.

Note

This provides the raw integer value and is *not* suitable for use as an array index, as the result is not zero-based. See the [OS_ConvertToArrayIndex\(\)](#) to obtain a zero-based index value.

Parameters

in	<i>object↔ _id</i>	The object ID
----	------------------------	---------------

Returns

integer value representation of object ID

Definition at line 81 of file [osapi-idmap.h](#).

Referenced by [OS_ObjectIdDefined\(\)](#), and [OS_ObjectIdEqual\(\)](#).

9.61 OSAL Dynamic Loader and Symbol APIs**Functions**

- [int32 OS_SymbolLookup](#) ([cpuaddr](#) *symbol_address, const char *symbol_name)
Find the Address of a Symbol.
- [int32 OS_ModuleSymbolLookup](#) ([osal_id_t](#) module_id, [cpuaddr](#) *symbol_address, const char *symbol_name)
Find the Address of a Symbol within a module.
- [int32 OS_SymbolTableDump](#) (const char *filename, size_t size_limit)
Dumps the system symbol table to a file.
- [int32 OS_ModuleLoad](#) ([osal_id_t](#) *module_id, const char *module_name, const char *filename, [uint32](#) flags)
Loads an object file.
- [int32 OS_ModuleUnload](#) ([osal_id_t](#) module_id)
Unloads the module file.
- [int32 OS_ModuleInfo](#) ([osal_id_t](#) module_id, [OS_module_prop_t](#) *module_info)
Obtain information about a module.

9.61.1 Detailed Description**9.61.2 Function Documentation****OS_ModuleInfo()**

```
int32 OS_ModuleInfo (
    osal_id_t module_id,
    OS_module_prop_t * module_info )
```

Obtain information about a module.

Returns information about the loadable module

Parameters

in	<i>module_id</i>	OSAL ID of the previously the loaded module
out	<i>module_info</i>	Buffer to store module information (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the module id invalid
OS_INVALID_POINTER	if the pointer to the ModuleInfo structure is invalid
OS_ERROR	if an other/unspecified error occurs (return value only verified in coverage test)

OS_ModuleLoad()

```
int32 OS_ModuleLoad (
    osal_id_t * module_id,
    const char * module_name,
    const char * filename,
    uint32 flags )
```

Loads an object file.

Loads an object file into the running operating system

The "flags" parameter may influence how the loaded module symbols are made available for use in the application. See [OS_MODULE_FLAG_LOCAL_SYMBOLS](#) and [OS_MODULE_FLAG_GLOBAL_SYMBOLS](#) for descriptions.

Parameters

out	<i>module_id</i>	Non-zero OSAL ID corresponding to the loaded module
in	<i>module_name</i>	Name of module (must not be null)
in	<i>filename</i>	File containing the object code to load (must not be null)
in	<i>flags</i>	Options for the loaded module

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if one of the parameters is NULL
OS_ERR_NO_FREE_IDS	if the module table is full
OS_ERR_NAME_TAKEN	if the name is in use
OS_ERR_NAME_TOO_LONG	if the module_name is too long
OS_FS_ERR_PATH_INVALID	if the filename argument is not valid
OS_ERROR	if an other/unspecified error occurs (return value only verified in coverage test)

OS_ModuleSymbolLookup()

```
int32 OS_ModuleSymbolLookup (
    osal_id_t module_id,
    cpuaddr * symbol_address,
    const char * symbol_name )
```

Find the Address of a Symbol within a module.

This is similar to [OS_SymbolLookup\(\)](#) but for a specific module ID. This should be used to look up a symbol in a module that has been loaded with the [OS_MODULE_FLAG_LOCAL_SYMBOLS](#) flag.

Parameters

in	<i>module_id</i>	Module ID that should contain the symbol
out	<i>symbol_address</i>	Set to the address of the symbol (must not be null)
in	<i>symbol_name</i>	Name of the symbol to look up (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERROR	if the symbol could not be found
OS_INVALID_POINTER	if one of the pointers passed in are NULL

OS_ModuleUnload()

```
int32 OS_ModuleUnload (
    osal_id_t module_id )
```

Unloads the module file.

Unloads the module file from the running operating system

Parameters

in	<i>module_id</i>	OSAL ID of the previously the loaded module
----	------------------	---

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the module id invalid
OS_ERROR	if an other/unspecified error occurs (return value only verified in coverage test)

OS_SymbolLookup()

```
int32 OS_SymbolLookup (
    cpuaddr * symbol_address,
    const char * symbol_name )
```

Find the Address of a Symbol.

This calls to the OS dynamic symbol lookup implementation, and/or checks a static symbol table for a matching symbol name.

The static table is intended to support embedded targets that do not have module loading capability or have it disabled.

Parameters

out	<i>symbol_address</i>	Set to the address of the symbol (must not be null)
in	<i>symbol_name</i>	Name of the symbol to look up (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERROR	if the symbol could not be found
OS_INVALID_POINTER	if one of the pointers passed in are NULL

OS_SymbolTableDump()

```
int32 OS_SymbolTableDump (
    const char * filename,
    size_t size_limit )
```

Dumps the system symbol table to a file.

Dumps the system symbol table to the specified filename

Note

Not all RTOS implementations support this API. If the underlying module subsystem does not provide a facility to iterate through the symbol table, then the [OS_ERR_NOT_IMPLEMENTED](#) status code is returned.

Parameters

in	<i>filename</i>	File to write to (must not be null)
in	<i>size_limit</i>	Maximum number of bytes to write

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_NOT_IMPLEMENTED	Not implemented.
OS_INVALID_POINTER	if the filename argument is NULL
OS_FS_ERR_PATH_INVALID	if the filename argument is not valid
OS_ERR_OUTPUT_TOO_LARGE	if the size_limit was reached before completing all symbols (return value only verified in coverage test)
OS_ERROR	if an other/unspecified error occurs (return value only verified in coverage test)

9.62 OSAL Mutex APIs**Functions**

- [int32 OS_MutSemCreate](#) ([osal_id_t](#) *sem_id, const char *sem_name, [uint32](#) options)

Creates a mutex semaphore.

- `int32 OS_MutSemGive (osal_id_t sem_id)`
Releases the mutex object referenced by `sem_id`.
- `int32 OS_MutSemTake (osal_id_t sem_id)`
Acquire the mutex object referenced by `sem_id`.
- `int32 OS_MutSemDelete (osal_id_t sem_id)`
Deletes the specified Mutex Semaphore.
- `int32 OS_MutSemGetIdByName (osal_id_t *sem_id, const char *sem_name)`
Find an existing mutex ID by name.
- `int32 OS_MutSemGetInfo (osal_id_t sem_id, OS_mut_sem_prop_t *mut_prop)`
Fill a property object buffer with details regarding the resource.

9.62.1 Detailed Description

9.62.2 Function Documentation

OS_MutSemCreate()

```
int32 OS_MutSemCreate (
    osal_id_t * sem_id,
    const char * sem_name,
    uint32 options )
```

Creates a mutex semaphore.

Mutex semaphores are always created in the unlocked (full) state.

Parameters

out	<code>sem_id</code>	will be set to the non-zero ID of the newly-created resource (must not be null)
in	<code>sem_name</code>	the name of the new resource to create (must not be null)
in	<code>options</code>	reserved for future use. Should be passed as 0.

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<code>OS_SUCCESS</code>	Successful execution.
<code>OS_INVALID_POINTER</code>	if <code>sem_id</code> or <code>sem_name</code> are NULL
<code>OS_ERR_NAME_TOO_LONG</code>	name length including null terminator greater than <code>OS_MAX_API_NAME</code>
<code>OS_ERR_NO_FREE_IDS</code>	if there are no more free mutex Ids
<code>OS_ERR_NAME_TAKEN</code>	if there is already a mutex with the same name
<code>OS_SEM_FAILURE</code>	if the OS call failed (return value only verified in coverage test)

OS_MutSemDelete()

```
int32 OS_MutSemDelete (
    osal_id_t sem_id )
```

Deletes the specified Mutex Semaphore.

Delete the semaphore. This also frees the respective `sem_id` such that it can be used again when another is created.

Parameters

in	<i>sem_id</i>	The object ID to delete
----	---------------	-------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the id passed in is not a valid mutex
OS_SEM_FAILURE	if an unspecified error occurs (return value only verified in coverage test)

OS_MutSemGetIdByName()

```
int32 OS_MutSemGetIdByName (
    osal_id_t * sem_id,
    const char * sem_name )
```

Find an existing mutex ID by name.

This function tries to find a mutex sem Id given the name of a mut_sem. The id is returned through sem_id

Parameters

out	<i>sem_id</i>	will be set to the ID of the existing resource
in	<i>sem_name</i>	the name of the existing resource to find (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	is semid or sem_name are NULL pointers
OS_ERR_NAME_TOO_LONG	name length including null terminator greater than OS_MAX_API_NAME
OS_ERR_NAME_NOT_FOUND	if the name was not found in the table

OS_MutSemGetInfo()

```
int32 OS_MutSemGetInfo (
    osal_id_t sem_id,
    OS_mut_sem_prop_t * mut_prop )
```

Fill a property object buffer with details regarding the resource.

This function will pass back a pointer to structure that contains all of the relevant info(name and creator) about the specified mutex semaphore.

Parameters

in	<i>sem_id</i>	The object ID to operate on
out	<i>mut_prop</i>	The property object buffer to fill (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the id passed in is not a valid semaphore
OS_INVALID_POINTER	if the mut_prop pointer is null

OS_MutSemGive()

```
int32 OS_MutSemGive (
    osal_id_t sem_id )
```

Releases the mutex object referenced by *sem_id*.

If there are threads blocked on the mutex object referenced by *mutex* when this function is called, resulting in the mutex becoming available, the scheduling policy shall determine which thread shall acquire the mutex.

Parameters

in	<i>sem_id</i>	The object ID to operate on
----	---------------	-----------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the id passed in is not a valid mutex
OS_SEM_FAILURE	if an unspecified error occurs (return value only verified in coverage test)

OS_MutSemTake()

```
int32 OS_MutSemTake (
    osal_id_t sem_id )
```

Acquire the mutex object referenced by *sem_id*.

If the mutex is already locked, the calling thread shall block until the mutex becomes available. This operation shall return with the mutex object referenced by *mutex* in the locked state with the calling thread as its owner.

Parameters

Parameters

in	<i>sem</i> ↔ <i>_id</i>	The object ID to operate on
----	----------------------------	-----------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	the id passed in is not a valid mutex
OS_SEM_FAILURE	if an unspecified error occurs (return value only verified in coverage test)

9.63 OSAL Network ID APIs

Functions

- [int32 OS_NetworkGetID](#) (void)
Gets the network ID of the local machine.
- [int32 OS_NetworkGetHostName](#) (char *host_name, size_t name_len)
Gets the local machine network host name.

9.63.1 Detailed Description

Provides some basic methods to query a network host name and ID

9.63.2 Function Documentation

OS_NetworkGetHostName()

```
int32 OS_NetworkGetHostName (
    char * host_name,
    size_t name_len )
```

Gets the local machine network host name.

If configured in the underlying network stack, this function retrieves the local hostname of the system.

Parameters

out	<i>host_name</i>	Buffer to hold name information (must not be null)
in	<i>name_len</i>	Maximum length of host name buffer (must not be zero)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_SIZE	if the name_len is zero
OS_INVALID_POINTER	if the host_name is NULL

OS_NetworkGetID()

```
int32 OS_NetworkGetID (
    void )
```

Gets the network ID of the local machine.

The ID is an implementation-defined value and may not be consistent in meaning across different platform types.

Note

This API may be removed in a future version of OSAL due to inconsistencies between platforms.

Returns

The ID or fixed value of -1 if the host id could not be found. Note it is not possible to differentiate between error codes and valid network IDs here. It is assumed, however, that -1 is never a valid ID.

9.64 OSAL Printf APIs**Functions**

- void [OS_printf](#) (const char *string,...) [OS_PRINTF](#)(1)
Abstraction for the system printf() call.
- void void [OS_printf_disable](#) (void)
This function disables the output from OS_printf.
- void [OS_printf_enable](#) (void)
This function enables the output from OS_printf.

9.64.1 Detailed Description**9.64.2 Function Documentation****OS_printf()**

```
void OS_printf (
    const char * string,
    ... )
```

Abstraction for the system printf() call.

This function abstracts out the printf type statements. This is useful for using OS- specific that will allow non-poll print statements for the real time systems.

Operates in a manner similar to the printf() call defined by the standard C library and takes all the parameters and formatting options of printf. This abstraction may implement additional buffering, if necessary, to improve the real-time performance of the call.

Strings (including terminator) longer than [OS_BUFFER_SIZE](#) will be truncated.

The output of this routine also may be dynamically enabled or disabled by the [OS_printf_enable\(\)](#) and [OS_printf_disable\(\)](#) calls, respectively.

Parameters

in	string	Format string, followed by additional arguments
----	--------	---

OS_printf_disable()

```
void OS_printf_disable (
    void )
```

This function disables the output from OS_printf.

OS_printf_enable()

```
void OS_printf_enable (
    void )
```

This function enables the output from OS_printf.

9.65 OSAL Message Queue APIs**Functions**

- [int32 OS_QueueCreate](#) ([osal_id_t](#) *queue_id, const char *queue_name, [osal_blockcount_t](#) queue_depth, [size_t](#) data_size, [uint32](#) flags)
Create a message queue.
- [int32 OS_QueueDelete](#) ([osal_id_t](#) queue_id)
Deletes the specified message queue.
- [int32 OS_QueueGet](#) ([osal_id_t](#) queue_id, void *data, [size_t](#) size, [size_t](#) *size_copied, [int32](#) timeout)
Receive a message on a message queue.
- [int32 OS_QueuePut](#) ([osal_id_t](#) queue_id, const void *data, [size_t](#) size, [uint32](#) flags)
Put a message on a message queue.
- [int32 OS_QueueGetIdByName](#) ([osal_id_t](#) *queue_id, const char *queue_name)
Find an existing queue ID by name.
- [int32 OS_QueueGetInfo](#) ([osal_id_t](#) queue_id, [OS_queue_prop_t](#) *queue_prop)
Fill a property object buffer with details regarding the resource.

9.65.1 Detailed Description**9.65.2 Function Documentation****OS_QueueCreate()**

```
int32 OS_QueueCreate (
    osal_id_t * queue_id,
    const char * queue_name,
    osal_blockcount_t queue_depth,
    size_t data_size,
    uint32 flags )
```

Create a message queue.

This is the function used to create a queue in the operating system. Depending on the underlying operating system, the memory for the queue will be allocated automatically or allocated by the code that sets up the queue. Queue names must be unique; if the name already exists this function fails. Names cannot be NULL.

Parameters

out	<i>queue_id</i>	will be set to the non-zero ID of the newly-created resource (must not be null)
in	<i>queue_name</i>	the name of the new resource to create (must not be null)
in	<i>queue_depth</i>	the maximum depth of the queue
in	<i>data_size</i>	the size of each entry in the queue (must not be zero)
in	<i>flags</i>	options for the queue (reserved for future use, pass as 0)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if a pointer passed in is NULL
OS_ERR_NAME_TOO_LONG	name length including null terminator greater than OS_MAX_API_NAME
OS_ERR_NO_FREE_IDS	if there are already the max queues created
OS_ERR_NAME_TAKEN	if the name is already being used on another queue
OS_ERR_INVALID_SIZE	if data_size is 0
OS_QUEUE_INVALID_SIZE	if the queue depth exceeds the limit
OS_ERROR	if the OS create call fails

OS_QueueDelete()

```
int32 OS_QueueDelete (
    osal_id_t queue_id )
```

Deletes the specified message queue.

This is the function used to delete a queue in the operating system. This also frees the respective queue_id to be used again when another queue is created.

Note

If There are messages on the queue, they will be lost and any subsequent calls to QueueGet or QueuePut to this queue will result in errors

Parameters

in	<i>queue↔ _id</i>	The object ID to delete
----	-----------------------	-------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the id passed in does not exist
OS_ERROR	if the OS call returns an unexpected error (return value only verified in coverage test)

OS_QueueGet()

```
int32 OS_QueueGet (
    osal_id_t queue_id,
    void * data,
    size_t size,
```



```

    size_t * size_copied,
    int32 timeout )

```

Receive a message on a message queue.

If a message is pending, it is returned immediately. Otherwise the calling task will block until a message arrives or the timeout expires.

Parameters

in	<i>queue_id</i>	The object ID to operate on
out	<i>data</i>	The buffer to store the received message (must not be null)
in	<i>size</i>	The size of the data buffer (must not be zero)
out	<i>size_copied</i>	Set to the actual size of the message (must not be null)
in	<i>timeout</i>	The maximum amount of time to block, or OS_PEND to wait forever

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the given ID does not exist
OS_INVALID_POINTER	if a pointer passed in is NULL
OS_QUEUE_EMPTY	if the Queue has no messages on it to be received
OS_QUEUE_TIMEOUT	if the timeout was OS_PEND and the time expired
OS_QUEUE_INVALID_SIZE	if the size copied from the queue was not correct
OS_ERROR	if the OS call returns an unexpected error (return value only verified in coverage test)

OS_QueueGetIdByName()

```

int32 OS_QueueGetIdByName (
    osal_id_t * queue_id,
    const char * queue_name )

```

Find an existing queue ID by name.

This function tries to find a queue Id given the name of the queue. The id of the queue is passed back in queue_id.

Parameters

out	<i>queue_id</i>	will be set to the ID of the existing resource
in	<i>queue_name</i>	the name of the existing resource to find (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if the name or id pointers are NULL

Return values

<code>OS_ERR_NAME_TOO_LONG</code>	name length including null terminator greater than <code>OS_MAX_API_NAME</code>
<code>OS_ERR_NAME_NOT_FOUND</code>	the name was not found in the table

OS_QueueGetInfo()

```
int32 OS_QueueGetInfo (
    osal_id_t queue_id,
    OS_queue_prop_t * queue_prop )
```

Fill a property object buffer with details regarding the resource.

This function will pass back a pointer to structure that contains all of the relevant info (name and creator) about the specified queue.

Parameters

in	<i>queue_id</i>	The object ID to operate on
out	<i>queue_prop</i>	The property object buffer to fill (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<code>OS_SUCCESS</code>	Successful execution.
<code>OS_INVALID_POINTER</code>	if <i>queue_prop</i> is NULL
<code>OS_ERR_INVALID_ID</code>	if the ID given is not a valid queue

OS_QueuePut()

```
int32 OS_QueuePut (
    osal_id_t queue_id,
    const void * data,
    size_t size,
    uint32 flags )
```

Put a message on a message queue.

Parameters

in	<i>queue_id</i>	The object ID to operate on
in	<i>data</i>	The buffer containing the message to put (must not be null)
in	<i>size</i>	The size of the data buffer (must not be zero)
in	<i>flags</i>	Currently reserved/unused, should be passed as 0

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the queue id passed in is not a valid queue
OS_INVALID_POINTER	if the data pointer is NULL
OS_QUEUE_INVALID_SIZE	if the data message is too large for the queue
OS_QUEUE_FULL	if the queue cannot accept another message
OS_ERROR	if the OS call returns an unexpected error (return value only verified in coverage test)

9.66 OSAL RwLock APIs**Functions**

- [int32 OS_RwLockCreate](#) ([osal_id_t](#) *rw_id, const char *rw_name, [uint32](#) options)
Creates an readers-writer lock (rwlock)
- [int32 OS_RwLockReadGive](#) ([osal_id_t](#) rw_id)
Releases the reader lock on the rwlock object referenced by rw_id.
- [int32 OS_RwLockWriteGive](#) ([osal_id_t](#) rw_id)
Releases the rwlock object referenced by rw_id.
- [int32 OS_RwLockReadTake](#) ([osal_id_t](#) rw_id)
Acquire the rwlock object as a read lock as referenced by rw_id.
- [int32 OS_RwLockWriteTake](#) ([osal_id_t](#) rw_id)
Acquire the rwlock object as a write lock as referenced by rw_id.
- [int32 OS_RwLockDelete](#) ([osal_id_t](#) rw_id)
Deletes the specified RwLock.
- [int32 OS_RwLockGetIdByName](#) ([osal_id_t](#) *rw_id, const char *rw_name)
Find an existing rwlock ID by name.
- [int32 OS_RwLockGetInfo](#) ([osal_id_t](#) rw_id, [OS_rwlock_prop_t](#) *rw_prop)
Fill a property object buffer with details regarding the resource.

9.66.1 Detailed Description**9.66.2 Function Documentation****OS_RwLockCreate()**

```
int32 OS_RwLockCreate (
    osal_id_t * rw_id,
    const char * rw_name,
    uint32 options )
```

Creates an readers-writer lock (rwlock)
RwLocks are always created in the unlocked (full) state.

Parameters

out	<i>rw_id</i>	will be set to the non-zero ID of the newly-created resource (must not be null)
in	<i>rw_name</i>	the name of the new resource to create (must not be null)
in	<i>options</i>	reserved for future use. Should be passed as 0.

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if sem_id or sem_name are NULL
OS_ERR_NAME_TOO_LONG	name length including null terminator greater than OS_MAX_API_NAME
OS_ERR_NO_FREE_IDS	if there are no more free rwlock Ids
OS_ERR_NAME_TAKEN	if there is already a rwlock with the same name
OS_ERROR	if the OS call failed (return value only verified in coverage test)

OS_RwLockDelete()

```
int32 OS_RwLockDelete (
    osal_id_t rw_id )
```

Deletes the specified RwLock.

Delete the rwlock. This also frees the respective rw_id such that it can be used again when another is created.

Parameters

in	<i>rw_id</i>	The object ID to delete
----	--------------	-------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the id passed in is not a valid rwlock
OS_ERROR	if an unspecified error occurs (return value only verified in coverage test)

OS_RwLockGetIdByName()

```
int32 OS_RwLockGetIdByName (
    osal_id_t * rw_id,
    const char * rw_name )
```

Find an existing rwlock ID by name.

This function tries to find a rwlock Id given the name of a rwlock. The id is returned through rw_id

Parameters

out	<i>rw_id</i>	will be set to the ID of the existing resource
in	<i>rw_name</i>	the name of the existing resource to find (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	is <i>rw_id</i> or <i>rw_name</i> are NULL pointers
OS_ERR_NAME_TOO_LONG	name length including null terminator greater than OS_MAX_API_NAME
OS_ERR_NAME_NOT_FOUND	if the name was not found in the table

OS_RwLockGetInfo()

```
int32 OS_RwLockGetInfo (
    osal_id_t rw_id,
    OS_rwlock_prop_t * rw_prop )
```

Fill a property object buffer with details regarding the resource.

This function will pass back a pointer to structure that contains all of the relevant info(name and creator) about the specified rwlock.

Parameters

in	<i>rw_id</i>	The object ID to operate on
out	<i>rw_prop</i>	The property object buffer to fill (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the id passed in is not a valid rwlock
OS_INVALID_POINTER	if the <i>rw_prop</i> pointer is null

OS_RwLockReadGive()

```
int32 OS_RwLockReadGive (
    osal_id_t rw_id )
```

Releases the reader lock on the rwlock object referenced by *rw_id*.

If this function is called by a thread which holds a read lock on this rwlock object, and there are other threads currently holding read locks, the rwlock will remain in the read locked state. If the function is called by the last thread which held a read lock, the rwlock becomes unlocked.

If this function is called by a thread which holds a write lock on this rwlock object, the behavior is undefined.

If there are threads blocked on the rwlock object referenced by *rw_id* when this function is called, resulting in the rwlock becoming available, the scheduling policy shall determine which thread shall acquire the rwlock.

Parameters

in	<i>rw↔ _id</i>	The object ID to operate on
----	--------------------	-----------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the id passed in is not a valid rwlock
OS_ERROR	if an unspecified error occurs (return value only verified in coverage test)

OS_RwLockReadTake()

```
int32 OS_RwLockReadTake (
    osal_id_t rw_id )
```

Acquire the rwlock object as a read lock as referenced by *rw_id*.

If the rwlock is currently write locked, as in a thread holds a write lock on this rwlock object, the current thread will block until the rwlock becomes available to readers.

If the rwlock is currently read locked and a thread intending to obtain a write lock is waiting, whether or not the current thread is allowed to obtain a read lock depends on the underlying OS rwlock implementation.

If the rwlock is currently read locked and there are no threads waiting to obtain a write lock, the current thread is allowed to obtain a read lock.

If the rwlock is unlocked, the current thread will obtain a read lock on this rwlock object.

Parameters

in	<i>rw↔ _id</i>	The object ID to operate on
----	--------------------	-----------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	the id passed in is not a valid rwlock
OS_ERROR	if an unspecified error occurs (return value only verified in coverage test)

OS_RwLockWriteGive()

```
int32 OS_RwLockWriteGive (
    osal_id_t rw_id )
```

Releases the rwlock object referenced by *rw_id*.

If this function is called by a thread which holds a write lock on this rwlock object, the rwlock becomes unlocked.

If this function is called by a thread which holds a read lock on this rwlock object, the behavior is undefined. If there are threads blocked on the rwlock object referenced by `rwlock` when this function is called, resulting in the rwlock becoming available, the scheduling policy shall determine which thread shall acquire the rwlock.

Parameters

in	<code>rw↔ _id</code>	The object ID to operate on
----	--------------------------	-----------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the id passed in is not a valid rwlock
OS_ERROR	if an unspecified error occurs (return value only verified in coverage test)

OS_RwLockWriteTake()

```
int32 OS_RwLockWriteTake (
    osal_id_t rw_id )
```

Acquire the rwlock object as a write lock as referenced by `rw_id`.

If the rwlock is currently write or read locked, the current thread will block until the rwlock becomes available. If the rwlock is open, the current thread will acquire a write lock on this rwlock object.

Parameters

in	<code>rw↔ _id</code>	The object ID to operate on
----	--------------------------	-----------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	the id passed in is not a valid rwlock
OS_ERROR	if an unspecified error occurs (return value only verified in coverage test)

9.67 OSAL Select APIs

Functions

- [int32 OS_SelectMultipleAbs](#) ([OS_FdSet](#) *ReadSet, [OS_FdSet](#) *WriteSet, [OS_time_t](#) abs_timeout)
Wait for events across multiple file handles.
- [int32 OS_SelectMultiple](#) ([OS_FdSet](#) *ReadSet, [OS_FdSet](#) *WriteSet, [int32](#) msec)

Wait for events across multiple file handles.

- `int32 OS_SelectSingleAbs (osal_id_t objid, uint32 *StateFlags, OS_time_t abs_timeout)`

Wait for events on a single file handle.

- `int32 OS_SelectSingle (osal_id_t objid, uint32 *StateFlags, int32 msecs)`

Wait for events on a single file handle.

- `int32 OS_SelectFdZero (OS_FdSet *Set)`

Clear a FdSet structure.

- `int32 OS_SelectFdAdd (OS_FdSet *Set, osal_id_t objid)`

Add an ID to an FdSet structure.

- `int32 OS_SelectFdClear (OS_FdSet *Set, osal_id_t objid)`

Clear an ID from an FdSet structure.

- `bool OS_SelectFdsSet (const OS_FdSet *Set, osal_id_t objid)`

Check if an FdSet structure contains a given ID.

9.67.1 Detailed Description

9.67.2 Function Documentation

OS_SelectFdAdd()

```
int32 OS_SelectFdAdd (
    OS_FdSet * Set,
    osal_id_t objid )
```

Add an ID to an FdSet structure.

After this call the set will contain the given OSAL ID

Parameters

in, out	Set	Pointer to OS_FdSet object to operate on (must not be null)
in	objid	The handle ID to add to the set

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if argument is NULL
OS_ERR_INVALID_ID	if the objid is not a valid handle

OS_SelectFdClear()

```
int32 OS_SelectFdClear (
    OS_FdSet * Set,
    osal_id_t objid )
```

Clear an ID from an FdSet structure.

After this call the set will no longer contain the given OSAL ID

Parameters

<i>in, out</i>	<i>Set</i>	Pointer to OS_FdSet object to operate on (must not be null)
<i>in</i>	<i>objid</i>	The handle ID to remove from the set

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if argument is NULL
OS_ERR_INVALID_ID	if the objid is not a valid handle

OS_SelectFdsSet()

```
bool OS_SelectFdsSet (  
    const OS\_FdSet * Set,  
    osal\_id\_t objid )
```

Check if an FdSet structure contains a given ID.

Parameters

<i>in</i>	<i>Set</i>	Pointer to OS_FdSet object to operate on (must not be null)
<i>in</i>	<i>objid</i>	The handle ID to check for in the set

Returns

Boolean set status

Return values

<i>true</i>	FdSet structure contains ID
<i>false</i>	FDSet structure does not contain ID

OS_SelectFdZero()

```
int32 OS_SelectFdZero (  
    OS\_FdSet * Set )
```

Clear a FdSet structure.

After this call the set will contain no OSAL IDs

Parameters

<i>out</i>	<i>Set</i>	Pointer to OS_FdSet object to clear (must not be null)
------------	------------	--

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if argument is NULL

OS_SelectMultiple()

```
int32 OS_SelectMultiple (
    OS_FdSet * ReadSet,
    OS_FdSet * WriteSet,
    int32 msec )
```

Wait for events across multiple file handles.

Wait for any of the given sets of IDs to become readable or writable

This function will block until any of the following occurs:

- At least one OSAL ID in the ReadSet is readable
- At least one OSAL ID in the WriteSet is writable
- The timeout has elapsed

The sets are input/output parameters. On entry, these indicate the file handle(s) to wait for. On exit, these are set to the actual file handle(s) that have activity.

If the timeout occurs this returns an error code and all output sets should be empty.

The timeout is expressed in milliseconds, relative to the time that the API was invoked. Use [OS_SelectMultipleAbs\(\)](#) for higher timing precision.

Note

This does not lock or otherwise protect the file handles in the given sets. If a filehandle supplied via one of the FdSet arguments is closed or modified by another while this function is in progress, the results are undefined. Because of this limitation, it is recommended to use [OS_SelectSingle\(\)](#) whenever possible.

Parameters

in, out	<i>ReadSet</i>	Set of handles to check/wait to become readable
in, out	<i>WriteSet</i>	Set of handles to check/wait to become writable
in	<i>msec</i>	Indicates the timeout. Positive values will wait up to that many milliseconds. Zero will not wait (poll). Negative values will wait forever (pend)

See also

[OS_SelectMultipleAbs\(\)](#)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<i>OS_SUCCESS</i>	If any handle in the ReadSet or WriteSet is readable or writable, respectively
<i>OS_ERROR_TIMEOUT</i>	If no handles in the ReadSet or WriteSet became readable or writable within the timeout
<i>OS_ERR_OPERATION_NOT_SUPPORTED</i>	if a specified handle does not support select
<i>OS_ERR_INVALID_ID</i>	if no valid handles were contained in the ReadSet/WriteSet

OS_SelectMultipleAbs()

```
int32 OS_SelectMultipleAbs (
    OS_FdSet * ReadSet,
    OS_FdSet * WriteSet,
    OS_time_t abs_timeout )
```

Wait for events across multiple file handles.

Wait for any of the given sets of IDs to become readable or writable

This function will block until any of the following occurs:

- At least one OSAL ID in the ReadSet is readable
- At least one OSAL ID in the WriteSet is writable
- The timeout has elapsed

The sets are input/output parameters. On entry, these indicate the file handle(s) to wait for. On exit, these are set to the actual file handle(s) that have activity.

If the timeout occurs this returns an error code and all output sets should be empty.

This API is identical to [OS_SelectMultiple\(\)](#) except for the timeout parameter. In this call, timeout is expressed as an absolute value of the OS clock, in the same time domain as obtained via [OS_GetLocalTime\(\)](#). This allows for a more precise timeout than what is possible via the normal [OS_SelectMultiple\(\)](#).

Note

This does not lock or otherwise protect the file handles in the given sets. If a filehandle supplied via one of the FdSet arguments is closed or modified by another while this function is in progress, the results are undefined. Because of this limitation, it is recommended to use [OS_SelectSingle\(\)](#) whenever possible.

Parameters

in, out	<i>ReadSet</i>	Set of handles to check/wait to become readable
in, out	<i>WriteSet</i>	Set of handles to check/wait to become writable
in	<i>abs_timeout</i>	The absolute time that the call may block until

See also

[OS_SelectMultiple\(\)](#)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<i>OS_SUCCESS</i>	If any handle in the ReadSet or WriteSet is readable or writable, respectively
<i>OS_ERROR_TIMEOUT</i>	If no handles in the ReadSet or WriteSet became readable or writable within the timeout
<i>OS_ERR_OPERATION_NOT_SUPPORTED</i>	if a specified handle does not support select
<i>OS_ERR_INVALID_ID</i>	if no valid handles were contained in the ReadSet/WriteSet

OS_SelectSingle()

```
int32 OS_SelectSingle (
    osal_id_t objid,
    uint32 * StateFlags,
    int32 msec )
```

Wait for events on a single file handle.

Wait for a single OSAL filehandle to change state

This function can be used to wait for a single OSAL stream ID to become readable or writable. On entry, the "StateFlags" parameter should be set to the desired state (OS_STREAM_STATE_READABLE and/or OS_STREAM_STATE_WRITEABLE) and upon return the flags will be set to the state actually detected.

As this operates on a single ID, the filehandle is protected during this call, such that another thread accessing the same handle will return an error. However, it is important to note that once the call returns then other threads may then also read/write and affect the state before the current thread can service it.

To mitigate this risk the application may prefer to use the OS_TimedRead/OS_TimedWrite calls.

The timeout is expressed in milliseconds, relative to the time that the API was invoked. Use [*OS_SelectSingleAbs\(\)*](#) for higher timing precision.

Parameters

in	<i>objid</i>	The handle ID to select on
in, out	<i>StateFlags</i>	State flag(s) (readable or writable) (must not be null)
in	<i>msec</i>	Indicates the timeout. Positive values will wait up to that many milliseconds. Zero will not wait (poll). Negative values will wait forever (pend)

See also

[*OS_SelectSingleAbs\(\)*](#)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<i>OS_SUCCESS</i>	If the handle is readable and/or writable, as requested
<i>OS_ERROR_TIMEOUT</i>	If the handle did not become readable or writable within the timeout
<i>OS_INVALID_POINTER</i>	if argument is NULL
<i>OS_ERR_INVALID_ID</i>	if the objid is not a valid handle

OS_SelectSingleAbs()

```
int32 OS_SelectSingleAbs (
    osal_id_t objid,
    uint32 * StateFlags,
    OS_time_t abs_timeout )
```

Wait for events on a single file handle.

Wait for a single OSAL filehandle to change state

This function can be used to wait for a single OSAL stream ID to become readable or writable. On entry, the "StateFlags" parameter should be set to the desired state (OS_STREAM_STATE_READABLE and/or OS_STREAM_STATE_WRITEABLE) and upon return the flags will be set to the state actually detected.

As this operates on a single ID, the filehandle is protected during this call, such that another thread accessing the same handle will return an error. However, it is important to note that once the call returns then other threads may then also read/write and affect the state before the current thread can service it.

To mitigate this risk the application may prefer to use the OS_TimedRead/OS_TimedWrite calls.

This API is identical to [OS_SelectSingle\(\)](#) except for the timeout parameter. In this call, timeout is expressed as an absolute value of the OS clock, in the same time domain as obtained via [OS_GetLocalTime\(\)](#). This allows for a more precise timeout than what is possible via the normal [OS_SelectSingle\(\)](#).

Parameters

in	<i>objid</i>	The handle ID to select on
in, out	<i>StateFlags</i>	State flag(s) (readable or writable) (must not be null)
in	<i>abs_timeout</i>	The absolute time that the call may block until

See also

[OS_SelectSingle\(\)](#)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	If the handle is readable and/or writable, as requested
OS_ERROR_TIMEOUT	If the handle did not become readable or writable within the timeout
OS_INVALID_POINTER	if argument is NULL
OS_ERR_INVALID_ID	if the objid is not a valid handle

9.68 OSAL Shell APIs

Functions

- [int32 OS_ShellOutputToFile](#) (const char *Cmd, [osal_id_t](#) filedes)

Executes the command and sends output to a file.

9.68.1 Detailed Description

9.68.2 Function Documentation

OS_ShellOutputToFile()

```
int32 OS_ShellOutputToFile (
    const char * Cmd,
    osal_id_t filedes )
```

Executes the command and sends output to a file.

Takes a shell command in and writes the output of that command to the specified file. The output file must be opened previously with write access (OS_WRITE_ONLY or OS_READ_WRITE).

Parameters

in	<i>Cmd</i>	Command to pass to shell (must not be null)
in	<i>filedes</i>	File to send output to.

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERROR	if the command was not executed properly
OS_INVALID_POINTER	if Cmd argument is NULL
OS_ERR_INVALID_ID	if the file descriptor passed in is invalid

9.69 OSAL Socket Address APIs

Functions

- [int32 OS_SocketAddrInit](#) ([OS_SockAddr_t](#) *Addr, [OS_SocketDomain_t](#) Domain)
Initialize a socket address structure to hold an address of the given family.
- [int32 OS_SocketAddrToString](#) (char *buffer, size_t buflen, const [OS_SockAddr_t](#) *Addr)
Get a string representation of a network host address.
- [int32 OS_SocketAddrFromString](#) ([OS_SockAddr_t](#) *Addr, const char *string)
Set a network host address from a string representation.
- [int32 OS_SocketAddrGetPort](#) (uint16 *PortNum, const [OS_SockAddr_t](#) *Addr)
Get the port number of a network address.
- [int32 OS_SocketAddrSetPort](#) ([OS_SockAddr_t](#) *Addr, uint16 PortNum)
Set the port number of a network address.

9.69.1 Detailed Description

These functions provide a means to manipulate network addresses in a manner that is (mostly) agnostic to the actual network address type.

Every network address should be representable as a string (i.e. dotted decimal IP, etc). This can serve as the "common denominator" to all address types.

9.69.2 Function Documentation

OS_SocketAddrFromString()

```
int32 OS_SocketAddrFromString (
    OS_SockAddr_t * Addr,
    const char * string )
```

Set a network host address from a string representation.

The specific format of the output string depends on the address family.

The address structure should have been previously initialized using [OS_SocketAddrInit\(\)](#) to set the address family type.

Note

For IPv4, this would typically be the dotted-decimal format (X.X.X.X). It is up to the discretion of the underlying implementation whether to accept hostnames, as this depends on the availability of DNS services. Since many embedded deployments do not have name services, this should not be relied upon.

Parameters

out	<i>Addr</i>	The address buffer to initialize (must not be null)
in	<i>string</i>	The string to initialize the address from (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if argument is NULL
OS_ERROR	if the string cannot be converted to an address

OS_SocketAddrGetPort()

```
int32 OS_SocketAddrGetPort (
    uint16 * PortNum,
    const OS_SockAddr_t * Addr )
```

Get the port number of a network address.

For network protocols that have the concept of a port number (such as TCP/IP and UDP/IP) this function gets the port number from the address structure.

Parameters

out	<i>PortNum</i>	Buffer to store the port number (must not be null)
in	<i>Addr</i>	The network address buffer (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<i>OS_SUCCESS</i>	Successful execution.
<i>OS_INVALID_POINTER</i>	if argument is NULL
<i>OS_ERR_BAD_ADDRESS</i>	if the address domain is not compatible

OS_SocketAddrInit()

```
int32 OS_SocketAddrInit (
    OS_SockAddr_t * Addr,
    OS_SocketDomain_t Domain )
```

Initialize a socket address structure to hold an address of the given family. The address is set to a suitable default value for the family.

Parameters

out	<i>Addr</i>	The address buffer to initialize (must not be null)
in	<i>Domain</i>	The address family

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<i>OS_SUCCESS</i>	Successful execution.
<i>OS_INVALID_POINTER</i>	if Addr argument is NULL
<i>OS_ERR_NOT_IMPLEMENTED</i>	if the system does not implement the requested domain

OS_SocketAddrSetPort()

```
int32 OS_SocketAddrSetPort (
    OS_SockAddr_t * Addr,
    uint16 PortNum )
```

Set the port number of a network address.

For network protocols that have the concept of a port number (such as TCP/IP and UDP/IP) this function sets the port number from the address structure.

Parameters

out	<i>Addr</i>	The network address buffer (must not be null)
in	<i>PortNum</i>	The port number to set

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<i>OS_SUCCESS</i>	Successful execution.
<i>OS_INVALID_POINTER</i>	if argument is NULL
<i>OS_ERR_BAD_ADDRESS</i>	if the address domain is not compatible

OS_SocketAddrToString()

```
int32 OS_SocketAddrToString (
    char * buffer,
    size_t buflen,
    const OS_SockAddr_t * Addr )
```

Get a string representation of a network host address.

The specific format of the output string depends on the address family.

This string should be suitable to pass back into [OS_SocketAddrFromString\(\)](#) which should recreate the same network address, and it should also be meaningful to a user of printed or logged as a C string.

Note

For IPv4, this would typically be the dotted-decimal format (X.X.X.X).

Parameters

out	<i>buffer</i>	Buffer to hold the output string (must not be null)
in	<i>buflen</i>	Maximum length of the output string (must not be zero)
in	<i>Addr</i>	The network address buffer to convert (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<i>OS_SUCCESS</i>	Successful execution.
<i>OS_INVALID_POINTER</i>	if argument is NULL
<i>OS_ERR_INVALID_SIZE</i>	if passed-in buflen is not valid
<i>OS_ERROR</i>	if the address cannot be converted to string, or string buffer too small

9.70 OSAL Socket Management APIs**Functions**

- [int32 OS_SocketOpen](#) ([osal_id_t](#) *sock_id, [OS_SocketDomain_t](#) Domain, [OS_SocketType_t](#) Type)
Opens a socket.
- [int32 OS_SocketBind](#) ([osal_id_t](#) sock_id, const [OS_SockAddr_t](#) *Addr)
Binds a socket to a given local address and enter listening (server) mode.
- [int32 OS_SocketListen](#) ([osal_id_t](#) sock_id)
Places the specified socket into a listening state.
- [int32 OS_SocketBindAddress](#) ([osal_id_t](#) sock_id, const [OS_SockAddr_t](#) *Addr)

Binds a socket to a given local address.

- `int32 OS_SocketConnectAbs (osal_id_t sock_id, const OS_SockAddr_t *Addr, OS_time_t abs_timeout)`

Connects a socket to a given remote address.

- `int32 OS_SocketConnect (osal_id_t sock_id, const OS_SockAddr_t *Addr, int32 timeout)`

Connects a socket to a given remote address.

- `int32 OS_SocketShutdown (osal_id_t sock_id, OS_SocketShutdownMode_t Mode)`

Implement graceful shutdown of a stream socket.

- `int32 OS_SocketAcceptAbs (osal_id_t sock_id, osal_id_t *connsock_id, OS_SockAddr_t *Addr, OS_time_t abs_timeout)`

Waits for and accept the next incoming connection on the given socket.

- `int32 OS_SocketAccept (osal_id_t sock_id, osal_id_t *connsock_id, OS_SockAddr_t *Addr, int32 timeout)`

Waits for and accept the next incoming connection on the given socket.

- `int32 OS_SocketRecvFromAbs (osal_id_t sock_id, void *buffer, size_t buflen, OS_SockAddr_t *RemoteAddr, OS_time_t abs_timeout)`

Reads data from a message-oriented (datagram) socket.

- `int32 OS_SocketRecvFrom (osal_id_t sock_id, void *buffer, size_t buflen, OS_SockAddr_t *RemoteAddr, int32 timeout)`

Reads data from a message-oriented (datagram) socket.

- `int32 OS_SocketSendTo (osal_id_t sock_id, const void *buffer, size_t buflen, const OS_SockAddr_t *RemoteAddr)`

Sends data to a message-oriented (datagram) socket.

- `int32 OS_SocketGetIdByName (osal_id_t *sock_id, const char *sock_name)`

Gets an OSAL ID from a given name.

- `int32 OS_SocketGetInfo (osal_id_t sock_id, OS_socket_prop_t *sock_prop)`

Gets information about an OSAL Socket ID.

- `int32 OS_SocketGetOption (osal_id_t sock_id, OS_socket_option_t opt_id, OS_socket_optval_t *optval)`

Gets the value of a socket option.

- `int32 OS_SocketSetOption (osal_id_t sock_id, OS_socket_option_t opt_id, const OS_socket_optval_t *optval)`

Sets the value of a socket option.

9.70.1 Detailed Description

These functions are loosely related to the BSD Sockets API but made to be more consistent with other OSAL API functions. That is, they operate on OSAL IDs (32-bit opaque number values) and return an OSAL error code.

OSAL Socket IDs are very closely related to File IDs and share the same ID number space. Additionally, the file `OS_read()` / `OS_write()` / `OS_close()` calls also work on sockets.

Note that all of functions may return `OS_ERR_NOT_IMPLEMENTED` if network support is not configured at compile time.

9.70.2 Function Documentation

OS_SocketAccept()

```
int32 OS_SocketAccept (
    osal_id_t sock_id,
    osal_id_t * connsock_id,
    OS_SockAddr_t * Addr,
    int32 timeout )
```

Waits for and accept the next incoming connection on the given socket.

This is used for sockets operating in a "server" role. The socket must be a stream type (connection-oriented) and previously bound to a local address using `OS_SocketBind()`. This will block the caller up to the given timeout or until an incoming connection request occurs, whichever happens first.

The new stream connection is then returned to the caller and the original server socket ID can be reused for the next connection.

The timeout is expressed in milliseconds, relative to the time that the API was invoked. Use [OS_SocketAcceptAbs\(\)](#) for higher timing precision.

Parameters

in	<i>sock_id</i>	The server socket ID, previously bound using OS_SocketBind()
out	<i>connsock↔ _id</i>	The connection socket, a new ID that can be read/written (must not be null)
in	<i>Addr</i>	The remote address of the incoming connection (must not be null)
in	<i>timeout</i>	The maximum amount of time to wait, or OS_PEND to wait forever

See also

[OS_SocketAcceptAbs\(\)](#)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if argument is NULL
OS_ERR_INVALID_ID	if the sock_id parameter is not valid
OS_ERR_INCORRECT_OBJ_TYPE	if the handle is not a socket
OS_ERR_INCORRECT_OBJ_STATE	if the socket is not bound or already connected

OS_SocketAcceptAbs()

```
int32 OS_SocketAcceptAbs (
    osal_id_t sock_id,
    osal_id_t * connsock_id,
    OS_SockAddr_t * Addr,
    OS_time_t abs_timeout )
```

Waits for and accept the next incoming connection on the given socket.

This is used for sockets operating in a "server" role. The socket must be a stream type (connection-oriented) and previously bound to a local address using [OS_SocketBind\(\)](#). This will block the caller up to the given timeout or until an incoming connection request occurs, whichever happens first.

The new stream connection is then returned to the caller and the original server socket ID can be reused for the next connection.

This API is identical to [OS_SocketAccept\(\)](#) except for the timeout parameter. In this call, timeout is expressed as an absolute value of the OS clock, in the same time domain as obtained via [OS_GetLocalTime\(\)](#). This allows for a more precise timeout than what is possible via the normal [OS_SocketAccept\(\)](#).

Parameters

in	<i>sock_id</i>	The server socket ID, previously bound using OS_SocketBind()
out	<i>connsock↔ _id</i>	The connection socket, a new ID that can be read/written (must not be null)

Parameters

in	<i>Addr</i>	The remote address of the incoming connection (must not be null)
in	<i>abs_timeout</i>	The absolute time that the call may block until

See also

[OS_SocketAccept\(\)](#)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if argument is NULL
OS_ERR_INVALID_ID	if the sock_id parameter is not valid
OS_ERR_INCORRECT_OBJ_TYPE	if the handle is not a socket
OS_ERR_INCORRECT_OBJ_STATE	if the socket is not bound or already connected

OS_SocketBind()

```
int32 OS_SocketBind (
    osal_id_t sock_id,
    const OS_SockAddr_t * Addr )
```

Binds a socket to a given local address and enter listening (server) mode.

This is a convenience/compatibility routine to perform both [OS_SocketBindAddress\(\)](#) and [OS_SocketListen\(\)](#) operations in a single call, intended to simplify the setup for a server role.

If the socket is connectionless, then it only binds to the local address.

Parameters

in	<i>sock_id</i>	The socket ID
in	<i>Addr</i>	The local address to bind to (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the sock_id parameter is not valid
OS_INVALID_POINTER	if argument is NULL
OS_ERR_INCORRECT_OBJ_STATE	if the socket is already bound
OS_ERR_INCORRECT_OBJ_TYPE	if the handle is not a socket

OS_SocketBindAddress()

```
int32 OS_SocketBindAddress (
    osal_id_t sock_id,
    const OS_SockAddr_t * Addr )
```

Binds a socket to a given local address.

The specified socket will be bound to the local address and port, if available. This controls the source address reflected in network traffic transmitted via this socket.

After binding to the address, a stream socket may be followed by a call to either [OS_SocketListen\(\)](#) for a server role or to [OS_SocketConnect\(\)](#) for a client role.

Parameters

in	<i>sock_id</i>	The socket ID
in	<i>Addr</i>	The local address to bind to (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the sock_id parameter is not valid
OS_INVALID_POINTER	if argument is NULL
OS_ERR_INCORRECT_OBJ_STATE	if the socket is already bound
OS_ERR_INCORRECT_OBJ_TYPE	if the handle is not a socket

OS_SocketConnect()

```
int32 OS_SocketConnect (
    osal_id_t sock_id,
    const OS_SockAddr_t * Addr,
    int32 timeout )
```

Connects a socket to a given remote address.

The socket will be connected to the remote address and port, if available. This only applies to stream-oriented sockets. Calling this on a datagram socket will return an error (these sockets should use SendTo/RecvFrom).

The timeout is expressed in milliseconds, relative to the time that the API was invoked. Use [OS_SocketConnectAbs\(\)](#) for higher timing precision.

Parameters

in	<i>sock_id</i>	The socket ID
in	<i>Addr</i>	The remote address to connect to (must not be null)
in	<i>timeout</i>	The maximum amount of time to wait, or OS_PEND to wait forever

See also

[OS_SocketConnectAbs\(\)](#)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INCORRECT_OBJ_STATE	if the socket is already connected
OS_ERR_INVALID_ID	if the sock_id parameter is not valid
OS_ERR_INCORRECT_OBJ_TYPE	if the handle is not a socket
OS_INVALID_POINTER	if Addr argument is NULL

OS_SocketConnectAbs()

```
int32 OS_SocketConnectAbs (
    osal_id_t sock_id,
    const OS_SockAddr_t * Addr,
    OS_time_t abs_timeout )
```

Connects a socket to a given remote address.

The socket will be connected to the remote address and port, if available. This only applies to stream-oriented sockets. Calling this on a datagram socket will return an error (these sockets should use SendTo/RecvFrom).

This API is identical to [OS_SocketConnect\(\)](#) except for the timeout parameter. In this call, timeout is expressed as an absolute value of the OS clock, in the same time domain as obtained via [OS_GetLocalTime\(\)](#). This allows for a more precise timeout than what is possible via the normal [OS_SocketConnect\(\)](#).

Parameters

in	<i>sock_id</i>	The socket ID
in	<i>Addr</i>	The remote address to connect to (must not be null)
in	<i>abs_timeout</i>	The absolute time that the call may block until

See also

[OS_SocketConnect\(\)](#)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INCORRECT_OBJ_STATE	if the socket is already connected
OS_ERR_INVALID_ID	if the sock_id parameter is not valid
OS_ERR_INCORRECT_OBJ_TYPE	if the handle is not a socket
OS_INVALID_POINTER	if Addr argument is NULL

OS_SocketGetIdByName()

```
int32 OS_SocketGetIdByName (
    osal_id_t * sock_id,
    const char * sock_name )
```

Gets an OSAL ID from a given name.

Note

OSAL Sockets use generated names according to the address and type.

See also

[OS_SocketGetInfo\(\)](#)

Parameters

out	<i>sock_id</i>	Buffer to hold result (must not be null)
in	<i>sock_name</i>	Name of socket to find (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	is id or name are NULL pointers
OS_ERR_NAME_TOO_LONG	name length including null terminator greater than OS_MAX_API_NAME
OS_ERR_NAME_NOT_FOUND	if the name was not found in the table

OS_SocketGetInfo()

```
int32 OS_SocketGetInfo (
    osal_id_t sock_id,
    OS_socket_prop_t * sock_prop )
```

Gets information about an OSAL Socket ID.

OSAL Sockets use generated names according to the address and type. This allows applications to find the name of a given socket.

Parameters

in	<i>sock_id</i>	The socket ID
out	<i>sock_prop</i>	Buffer to hold socket information (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<i>OS_SUCCESS</i>	Successful execution.
<i>OS_ERR_INVALID_ID</i>	if the id passed in is not a valid socket
<i>OS_INVALID_POINTER</i>	if the sock_prop pointer is null

OS_SocketGetOption()

```
int32 OS_SocketGetOption (
    osal_id_t sock_id,
    OS_socket_option_t opt_id,
    OS_socket_optval_t * optval )
```

Gets the value of a socket option.

Gets the state of a per-socket option / configurable item

Parameters

in	<i>sock↔ _id</i>	The socket ID
in	<i>opt_id</i>	The socket option ID
out	<i>optval</i>	Buffer to hold socket option value (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<i>OS_SUCCESS</i>	Successful execution.
<i>OS_ERR_INVALID_ID</i>	if the id passed in is not a valid socket
<i>OS_INVALID_POINTER</i>	if the optval pointer is null

OS_SocketListen()

```
int32 OS_SocketListen (
    osal_id_t sock_id )
```

Places the specified socket into a listening state.

This function only applies to connection-oriented (stream) sockets that are intended to be used in a server-side role.

This places the socket into a state where it can accept incoming connections from clients.

Parameters

in	<i>sock↔ _id</i>	The socket ID
----	----------------------	---------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<code>OS_SUCCESS</code>	Successful execution.
<code>OS_ERR_INVALID_ID</code>	if the <code>sock_id</code> parameter is not valid
<code>OS_ERR_INCORRECT_OBJ_STATE</code>	if the socket is already listening
<code>OS_ERR_INCORRECT_OBJ_TYPE</code>	if the handle is not a stream socket

OS_SocketOpen()

```
int32 OS_SocketOpen (
    osal_id_t * sock_id,
    OS_SocketDomain_t Domain,
    OS_SocketType_t Type )
```

Opens a socket.

A new, unconnected and unbound socket is allocated of the given domain and type.

Parameters

out	<code>sock_id</code>	Buffer to hold the non-zero OSAL ID (must not be null)
in	<code>Domain</code>	The domain / address family of the socket (INET or INET6, etc)
in	<code>Type</code>	The type of the socket (STREAM or DATAGRAM)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<code>OS_SUCCESS</code>	Successful execution.
<code>OS_INVALID_POINTER</code>	if argument is NULL
<code>OS_ERR_NOT_IMPLEMENTED</code>	if the system does not implement the requested socket/address domain

OS_SocketRecvFrom()

```
int32 OS_SocketRecvFrom (
    osal_id_t sock_id,
    void * buffer,
    size_t buflen,
    OS_SockAddr_t * RemoteAddr,
    int32 timeout )
```

Reads data from a message-oriented (datagram) socket.

If a message is already available on the socket, this should immediately return that data without blocking. Otherwise, it may block up to the given timeout.

The timeout is expressed in milliseconds, relative to the time that the API was invoked. Use [OS_SocketRecvFromAbs\(\)](#) for higher timing precision.

Parameters

in	<code>sock_id</code>	The socket ID, previously bound using OS_SocketBind()
----	----------------------	---

Parameters

out	<i>buffer</i>	Pointer to message data receive buffer (must not be null)
in	<i>buflen</i>	The maximum length of the message data to receive (must not be zero)
out	<i>RemoteAddr</i>	Buffer to store the remote network address (may be NULL)
in	<i>timeout</i>	The maximum amount of time to wait or OS_PEND to wait forever

See also

[OS_SocketRecvFromAbs\(\)](#)

Returns

Count of actual bytes received or error status, see [OSAL Return Code Defines](#)

Return values

OS_INVALID_POINTER	if argument is NULL
OS_ERR_INVALID_SIZE	if passed-in buflen is not valid
OS_ERR_INVALID_ID	if the sock_id parameter is not valid
OS_ERR_INCORRECT_OBJ_TYPE	if the handle is not a socket

OS_SocketRecvFromAbs()

```
int32 OS_SocketRecvFromAbs (
    osal_id_t sock_id,
    void * buffer,
    size_t buflen,
    OS_SockAddr_t * RemoteAddr,
    OS_time_t abs_timeout )
```

Reads data from a message-oriented (datagram) socket.

If a message is already available on the socket, this should immediately return that data without blocking. Otherwise, it may block up to the given timeout.

This API is identical to [OS_SocketRecvFrom\(\)](#) except for the timeout parameter. In this call, timeout is expressed as an absolute value of the OS clock, in the same time domain as obtained via [OS_GetLocalTime\(\)](#). This allows for a more precise timeout than what is possible via the normal [OS_SocketRecvFrom\(\)](#).

Parameters

in	<i>sock_id</i>	The socket ID, previously bound using OS_SocketBind()
out	<i>buffer</i>	Pointer to message data receive buffer (must not be null)
in	<i>buflen</i>	The maximum length of the message data to receive (must not be zero)
out	<i>RemoteAddr</i>	Buffer to store the remote network address (may be NULL)
in	<i>abs_timeout</i>	The absolute time at which the call should return if nothing received

Returns

Count of actual bytes received or error status, see [OSAL Return Code Defines](#)

Return values

<i>OS_INVALID_POINTER</i>	if argument is NULL
<i>OS_ERR_INVALID_SIZE</i>	if passed-in buflen is not valid
<i>OS_ERR_INVALID_ID</i>	if the sock_id parameter is not valid
<i>OS_ERR_INCORRECT_OBJ_TYPE</i>	if the handle is not a socket

OS_SocketSendTo()

```
int32 OS_SocketSendTo (
    osal_id_t sock_id,
    const void * buffer,
    size_t buflen,
    const OS_SockAddr_t * RemoteAddr )
```

Sends data to a message-oriented (datagram) socket.

This sends data in a non-blocking mode. If the socket is not currently able to queue the message, such as if its outbound buffer is full, then this returns an error code.

Parameters

in	<i>sock_id</i>	The socket ID, which must be of the datagram type
in	<i>buffer</i>	Pointer to message data to send (must not be null)
in	<i>buflen</i>	The length of the message data to send (must not be zero)
in	<i>RemoteAddr</i>	Buffer containing the remote network address to send to

Returns

Count of actual bytes sent or error status, see [OSAL Return Code Defines](#)

Return values

<i>OS_INVALID_POINTER</i>	if argument is NULL
<i>OS_ERR_INVALID_SIZE</i>	if passed-in buflen is not valid
<i>OS_ERR_INVALID_ID</i>	if the sock_id parameter is not valid
<i>OS_ERR_INCORRECT_OBJ_TYPE</i>	if the handle is not a socket

OS_SocketSetOption()

```
int32 OS_SocketSetOption (
    osal_id_t sock_id,
    OS_socket_option_t opt_id,
    const OS_socket_optval_t * optval )
```

Sets the value of a socket option.

Sets the state of a per-socket option / configurable item

Parameters

in	<i>sock</i> ↔ <i>_id</i>	The socket ID
----	-----------------------------	---------------

Parameters

in	<i>opt_id</i>	The socket option ID
in	<i>optval</i>	Socket option value (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the id passed in is not a valid socket
OS_INVALID_POINTER	if the optval pointer is null

OS_SocketShutdown()

```
int32 OS_SocketShutdown (
    osal_id_t sock_id,
    OS_SocketShutdownMode_t Mode )
```

Implement graceful shutdown of a stream socket.

This can be utilized to indicate the end of data stream without immediately closing the socket, giving the remote side an indication that the data transfer is complete.

Parameters

in	<i>sock↔ _id</i>	The socket ID
in	<i>Mode</i>	Whether to shutdown reading, writing, or both.

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the sock_id parameter is not valid
OS_ERR_INVALID_ARGUMENT	if the Mode argument is not one of the valid options
OS_ERR_INCORRECT_OBJ_TYPE	if the handle is not a socket
OS_ERR_INCORRECT_OBJ_STATE	if the socket is not connected

9.71 OSAL Task APIs**Functions**

- [int32 OS_TaskCreate](#) ([osal_id_t](#) *task_id, const char *task_name, [osal_task_entry](#) function_pointer, [osal_stackptr_t](#) stack_pointer, [size_t](#) stack_size, [osal_priority_t](#) priority, [uint32](#) flags)

Creates a task and starts running it.

- `int32 OS_TaskDelete (osal_id_t task_id)`

Deletes the specified Task.

- `void OS_TaskExit (void)`

Exits the calling task.

- `int32 OS_TaskInstallDeleteHandler (osal_task_entry function_pointer)`

Installs a handler for when the task is deleted.

- `int32 OS_TaskDelay (uint32 millisecond)`

Delay a task for specified amount of milliseconds.

- `int32 OS_TaskSetPriority (osal_id_t task_id, osal_priority_t new_priority)`

Sets the given task to a new priority.

- `osal_id_t OS_TaskGetId (void)`

Obtain the task id of the calling task.

- `int32 OS_TaskGetIdByName (osal_id_t *task_id, const char *task_name)`

Find an existing task ID by name.

- `int32 OS_TaskGetInfo (osal_id_t task_id, OS_task_prop_t *task_prop)`

Fill a property object buffer with details regarding the resource.

- `int32 OS_TaskFindIdBySystemData (osal_id_t *task_id, const void *sysdata, size_t sysdata_size)`

Reverse-lookup the OSAL task ID from an operating system ID.

9.71.1 Detailed Description

9.71.2 Function Documentation

OS_TaskCreate()

```
int32 OS_TaskCreate (
    osal_id_t * task_id,
    const char * task_name,
    osal_task_entry function_pointer,
    osal_stackptr_t stack_pointer,
    size_t stack_size,
    osal_priority_t priority,
    uint32 flags )
```

Creates a task and starts running it.

Creates a task and passes back the id of the task created. Task names must be unique; if the name already exists this function fails. Names cannot be NULL.

Portable applications should always specify the actual stack size in the `stack_size` parameter, not 0. This size value is not enforced/checked by OSAL, but is simply passed through to the RTOS for stack creation. Some RTOS implementations may assume 0 means a default stack size while others may actually create a task with no stack.

Unlike `stack_size`, the `stack_pointer` is optional and can be specified as NULL. In that case, a stack of the requested size will be dynamically allocated from the system heap.

Parameters

out	<i>task_id</i>	will be set to the non-zero ID of the newly-created resource (must not be null)
in	<i>task_name</i>	the name of the new resource to create (must not be null)
in	<i>function_pointer</i>	the entry point of the new task (must not be null)
in	<i>stack_pointer</i>	pointer to the stack for the task, or NULL to allocate a stack from the system memory heap
in	<i>stack_size</i>	the size of the stack (must not be zero)
in	<i>priority</i>	initial priority of the new task
in	<i>flags</i>	initial options for the new task

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if any of the necessary pointers are NULL
OS_ERR_INVALID_SIZE	if the stack_size argument is zero
OS_ERR_NAME_TOO_LONG	name length including null terminator greater than OS_MAX_API_NAME
OS_ERR_INVALID_PRIORITY	if the priority is bad (return value only verified in coverage test)
OS_ERR_NO_FREE_IDS	if there can be no more tasks created
OS_ERR_NAME_TAKEN	if the name specified is already used by a task
OS_ERROR	if an unspecified/other error occurs (return value only verified in coverage test)

OS_TaskDelay()

```
int32 OS_TaskDelay (
    uint32 millisecond )
```

Delay a task for specified amount of milliseconds.

Causes the current thread to be suspended from execution for the period of millisecond. This is a scheduled wait (clock_nanosleep/rtems_task_wake_after/taskDelay), not a "busy" wait.

Parameters

in	<i>millisecond</i>	Amount of time to delay
----	--------------------	-------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERROR	if an unspecified/other error occurs (return value only verified in coverage test)

OS_TaskDelete()

```
int32 OS_TaskDelete (
    osal_id_t task_id )
```

Deletes the specified Task.

The task will be removed from the local tables. and the OS will be configured to stop executing the task at the next opportunity.

Parameters

in	<i>task_id</i>	The object ID to operate on
----	----------------	-----------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the ID given to it is invalid
OS_ERROR	if the OS delete call fails (return value only verified in coverage test)

OS_TaskExit()

```
void OS_TaskExit (
    void )
```

Exits the calling task.

The calling thread is terminated. This function does not return.

OS_TaskFindIdBySystemData()

```
int32 OS_TaskFindIdBySystemData (
    osal_id_t * task_id,
    const void * sysdata,
    size_t sysdata_size )
```

Reverse-lookup the OSAL task ID from an operating system ID.

This provides a method by which an external entity may find the OSAL task ID corresponding to a system-defined identifier (e.g. TASK_ID, pthread_t, rtems_id, etc).

Normally OSAL does not expose the underlying OS-specific values to the application, but in some circumstances, such as exception handling, the OS may provide this information directly to a BSP handler outside of the normal OSAL API.

Parameters

out	<i>task_id</i>	The buffer where the task id output is stored (must not be null)
in	<i>sysdata</i>	Pointer to the system-provided identification data
in	<i>sysdata_size</i>	Size of the system-provided identification data

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution. (return value only verified in coverage test)
OS_INVALID_POINTER	if a pointer argument is NULL

OS_TaskGetId()

```
osal_id_t OS_TaskGetId (
    void )
```

Obtain the task id of the calling task.

This function returns the task id of the calling task

Returns

Task ID, or zero if the operation failed (zero is never a valid task ID)

OS_TaskGetIdByName()

```
int32 OS_TaskGetIdByName (
    osal_id_t * task_id,
    const char * task_name )
```

Find an existing task ID by name.

This function tries to find a task Id given the name of a task

Parameters

out	<i>task_id</i>	will be set to the ID of the existing resource
in	<i>task_name</i>	the name of the existing resource to find (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if the pointers passed in are NULL
OS_ERR_NAME_TOO_LONG	name length including null terminator greater than OS_MAX_API_NAME
OS_ERR_NAME_NOT_FOUND	if the name wasn't found in the table

OS_TaskGetInfo()

```
int32 OS_TaskGetInfo (
    osal_id_t task_id,
    OS_task_prop_t * task_prop )
```

Fill a property object buffer with details regarding the resource.

This function will pass back a pointer to structure that contains all of the relevant info (creator, stack size, priority, name) about the specified task.

Parameters

in	<i>task_id</i>	The object ID to operate on
out	<i>task_prop</i>	The property object buffer to fill (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the ID passed to it is invalid
OS_INVALID_POINTER	if the task_prop pointer is NULL

OS_TaskInstallDeleteHandler()

```
int32 OS_TaskInstallDeleteHandler (
    osal_task_entry function_pointer )
```

Installs a handler for when the task is deleted.

This function is used to install a callback that is called when the task is deleted. The callback is called when OS_TaskDelete is called with the task ID. A task delete handler is useful for cleaning up resources that a task creates, before the task is removed from the system.

Parameters

in	<i>function_pointer</i>	function to be called when task exits
----	-------------------------	---------------------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_ERR_INVALID_ID	if the calling context is not an OSAL task
-----------------------------------	--

OS_TaskSetPriority()

```
int32 OS_TaskSetPriority (
    osal_id_t task_id,
    osal_priority_t new_priority )
```

Sets the given task to a new priority.

Parameters

in	<i>task_id</i>	The object ID to operate on
in	<i>new_priority</i>	Set the new priority

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the ID passed to it is invalid
OS_ERR_INVALID_PRIORITY	if the priority is greater than the max allowed (return value only verified in coverage test)
OS_ERROR	if an unspecified/other error occurs (return value only verified in coverage test)

9.72 OSAL Time Base APIs

Functions

- `int32 OS_TimeBaseCreate (osal_id_t *timebase_id, const char *timebase_name, OS_TimerSync_t external_sync)`
Create an abstract Time Base resource.
- `int32 OS_TimeBaseSet (osal_id_t timebase_id, uint32 start_time, uint32 interval_time)`
Sets the tick period for simulated time base objects.
- `int32 OS_TimeBaseDelete (osal_id_t timebase_id)`
Deletes a time base object.
- `int32 OS_TimeBaseGetIdByName (osal_id_t *timebase_id, const char *timebase_name)`
Find the ID of an existing time base resource.
- `int32 OS_TimeBaseGetInfo (osal_id_t timebase_id, OS_timebase_prop_t *timebase_prop)`
Obtain information about a timebase resource.
- `int32 OS_TimeBaseGetFreeRun (osal_id_t timebase_id, uint32 *freerun_val)`
Read the value of the timebase free run counter.

9.72.1 Detailed Description

9.72.2 Function Documentation

OS_TimeBaseCreate()

```
int32 OS_TimeBaseCreate (
    osal_id_t * timebase_id,
    const char * timebase_name,
    OS_TimerSync_t external_sync )
```

Create an abstract Time Base resource.

An OSAL time base is an abstraction of a "timer tick" that can, in turn, be used for measurement of elapsed time between events.

Time bases can be simulated by the operating system using the OS kernel-provided timing facilities, or based on a hardware timing source if provided by the BSP.

A time base object has a servicing task associated with it, that runs at elevated priority and will thereby interrupt user-level tasks when timing ticks occur.

If the `external_sync` function is passed as NULL, the operating system kernel timing resources will be utilized for a simulated timer tick.

If the `external_sync` function is not NULL, this should point to a BSP-provided function that will block the calling task until the next tick occurs. This can be used for synchronizing with hardware events.

Note

When provisioning a tunable RTOS kernel, such as RTEMS, the kernel should be configured to support at least `(OS_MAX_TASKS + OS_MAX_TIMEBASES)` threads, to account for the helper threads associated with time base objects.

This configuration API must not be used from the context of a timer callback. Timers should only be configured from the context of normal OSAL tasks.

Parameters

out	<code>timebase_id</code>	will be set to the non-zero ID of the newly-created resource (must not be null)
in	<code>timebase_name</code>	The name of the time base (must not be null)
in	<code>external_sync</code>	A synchronization function for BSP hardware-based timer ticks

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_NAME_TAKEN	if the name specified is already used
OS_ERR_NO_FREE_IDS	if there can be no more timebase resources created
OS_ERR_INCORRECT_OBJ_STATE	if called from timer/timebase context
OS_ERR_NAME_TOO_LONG	if the timebase_name is too long
OS_INVALID_POINTER	if a pointer argument is NULL

OS_TimeBaseDelete()

```
int32 OS_TimeBaseDelete (
    osal_id_t timebase_id )
```

Deletes a time base object.

The helper task and any other resources associated with the time base abstraction will be freed.

Note

This configuration API must not be used from the context of a timer callback. Timers should only be configured from the context of normal OSAL tasks.

Parameters

in	<i>timebase</i> ↔ <i>_id</i>	The timebase resource to delete
----	---------------------------------	---------------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the id passed in is not a valid timebase
OS_ERR_INCORRECT_OBJ_STATE	if called from timer/timebase context

OS_TimeBaseGetFreeRun()

```
int32 OS_TimeBaseGetFreeRun (
    osal_id_t timebase_id,
    uint32 * freerun_val )
```

Read the value of the timebase free run counter.

Poll the timer free-running time counter in a lightweight fashion.

The free run count is a monotonically increasing value reflecting the total time elapsed since the timebase inception.

Units are the same as the timebase itself, usually microseconds.

Applications may quickly and efficiently calculate relative time differences by polling this value and subtracting the previous counter value.

The absolute value of this counter is not relevant, because it will "roll over" after 2^{32} units of time. For a timebase with microsecond units, this occurs approximately every 4294 seconds, or about 1.2 hours.

Note

To ensure consistency of results, the application should sample the value at a minimum of two times the roll over frequency, and calculate the difference between the consecutive samples.

Parameters

in	<i>timebase_id</i>	The timebase to operate on
out	<i>freerun_val</i>	Buffer to store the free run counter (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the id passed in is not a valid timebase
OS_INVALID_POINTER	if pointer argument is NULL

OS_TimeBaseGetIdByName()

```
int32 OS_TimeBaseGetIdByName (
    osal_id_t * timebase_id,
    const char * timebase_name )
```

Find the ID of an existing time base resource.

Given a time base name, find and output the ID associated with it.

Note

This configuration API must not be used from the context of a timer callback. Timers should only be configured from the context of normal OSAL tasks.

Parameters

out	<i>timebase_id</i>	will be set to the non-zero ID of the matching resource (must not be null)
in	<i>timebase_name</i>	The name of the timebase resource to find (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
----------------------------	-----------------------

Return values

<i>OS_INVALID_POINTER</i>	if <code>timebase_id</code> or <code>timebase_name</code> are NULL pointers
<i>OS_ERR_NAME_TOO_LONG</i>	name length including null terminator greater than <i>OS_MAX_API_NAME</i>
<i>OS_ERR_NAME_NOT_FOUND</i>	if the name was not found in the table
<i>OS_ERR_INCORRECT_OBJ_STATE</i>	if called from timer/timebase context

OS_TimeBaseGetInfo()

```
int32 OS_TimeBaseGetInfo (
    osal_id_t timebase_id,
    OS_timebase_prop_t * timebase_prop )
```

Obtain information about a timebase resource.

Fills the buffer referred to by the `timebase_prop` parameter with relevant information about the time base resource.

This function will pass back a pointer to structure that contains all of the relevant info(name and creator) about the specified timebase.

Note

This configuration API must not be used from the context of a timer callback. Timers should only be configured from the context of normal OSAL tasks.

Parameters

in	<i>timebase_id</i>	The timebase resource ID
out	<i>timebase_prop</i>	Buffer to store timebase properties (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<i>OS_SUCCESS</i>	Successful execution.
<i>OS_ERR_INVALID_ID</i>	if the id passed in is not a valid timebase
<i>OS_INVALID_POINTER</i>	if the <code>timebase_prop</code> pointer is null
<i>OS_ERR_INCORRECT_OBJ_STATE</i>	if called from timer/timebase context

OS_TimeBaseSet()

```
int32 OS_TimeBaseSet (
    osal_id_t timebase_id,
    uint32 start_time,
    uint32 interval_time )
```

Sets the tick period for simulated time base objects.

This sets the actual tick period for timing ticks that are simulated by the RTOS kernel (i.e. the "external_sync" parameter on the call to [OS_TimeBaseCreate\(\)](#) is NULL).

The RTOS will be configured to wake up the helper thread at the requested interval.

This function has no effect for time bases that are using a BSP-provided `external_sync` function.

Note

This configuration API must not be used from the context of a timer callback. Timers should only be configured from the context of normal OSAL tasks.

Parameters

in	<i>timebase_id</i>	The timebase resource to configure
in	<i>start_time</i>	The amount of delay for the first tick, in microseconds.
in	<i>interval_time</i>	The amount of delay between ticks, in microseconds.

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the id passed in is not a valid timebase
OS_ERR_INCORRECT_OBJ_STATE	if called from timer/timebase context
OS_TIMER_ERR_INVALID_ARGS	if start_time or interval_time are out of range

9.73 OSAL Timer APIs**Functions**

- [int32 OS_TimerCreate](#) ([osal_id_t](#) *timer_id, const char *timer_name, [uint32](#) *clock_accuracy, [OS_TimerCallback_t](#) callback_ptr)
Create a timer object.
- [int32 OS_TimerAdd](#) ([osal_id_t](#) *timer_id, const char *timer_name, [osal_id_t](#) timebase_id, [OS_ArgCallback_t](#) callback_ptr, void *callback_arg)
Add a timer object based on an existing TimeBase resource.
- [int32 OS_TimerSet](#) ([osal_id_t](#) timer_id, [uint32](#) start_time, [uint32](#) interval_time)
Configures a periodic or one shot timer.
- [int32 OS_TimerDelete](#) ([osal_id_t](#) timer_id)
Deletes a timer resource.
- [int32 OS_TimerGetIdByName](#) ([osal_id_t](#) *timer_id, const char *timer_name)
Locate an existing timer resource by name.
- [int32 OS_TimerGetInfo](#) ([osal_id_t](#) timer_id, [OS_timer_prop_t](#) *timer_prop)
Gets information about an existing timer.

9.73.1 Detailed Description**9.73.2 Function Documentation****OS_TimerAdd()**

```
int32 OS_TimerAdd (
    osal_id_t * timer_id,
    const char * timer_name,
```

```
osal_id_t timebase_id,
OS_ArgCallback_t callback_ptr,
void * callback_arg )
```

Add a timer object based on an existing TimeBase resource.

A timer object is a resource that invokes the specified application-provided function upon timer expiration. Timers may be one-shot or periodic in nature.

This function uses an existing time base object to service this timer, which must exist prior to adding the timer. The precision of the timer is the same as that of the underlying time base object. Multiple timer objects can be created referring to a single time base object.

This routine also uses a different callback function prototype from [OS_TimerCreate\(\)](#), allowing a single opaque argument to be passed to the callback routine. The OSAL implementation does not use this parameter, and may be set NULL.

The callback function for this method should be declared according to the [OS_ArgCallback_t](#) function pointer type. The timer_id is passed in to the function by the OSAL, and the arg parameter is passed through from the callback_arg argument on this call.

Note

This configuration API must not be used from the context of a timer callback. Timers should only be configured from the context of normal OSAL tasks.

See also

[OS_ArgCallback_t](#)

Parameters

out	<i>timer_id</i>	Will be set to the non-zero resource ID of the timer object (must not be null)
in	<i>timer_name</i>	Name of the timer object (must not be null)
in	<i>timebase_id</i>	The time base resource to use as a reference
in	<i>callback_ptr</i>	Application-provided function to invoke (must not be null)
in	<i>callback_arg</i>	Opaque argument to pass to callback function, may be NULL

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if any parameters are NULL
OS_ERR_INVALID_ID	if the timebase_id parameter is not valid
OS_ERR_NAME_TOO_LONG	name length including null terminator greater than OS_MAX_API_NAME
OS_ERR_NAME_TAKEN	if the name is already in use by another timer.
OS_ERR_NO_FREE_IDS	if all of the timers are already allocated.
OS_ERR_INCORRECT_OBJ_STATE	if invoked from a timer context
OS_TIMER_ERR_INTERNAL	if there was an error programming the OS timer (return value only verified in coverage test)

OS_TimerCreate()

```
int32 OS_TimerCreate (
    osal_id_t * timer_id,
    const char * timer_name,
    uint32 * clock_accuracy,
    OS_TimerCallback_t callback_ptr )
```

Create a timer object.

A timer object is a resource that invokes the specified application-provided function upon timer expiration. Timers may be one-shot or periodic in nature.

This function creates a dedicated (hidden) time base object to service this timer, which is created and deleted with the timer object itself. The internal time base is configured for an OS simulated timer tick at the same interval as the timer.

The callback function should be declared according to the `OS_TimerCallback_t` function pointer type. The `timer_id` value is passed to the callback function.

Note

`clock_accuracy` comes from the underlying OS tick value. The nearest integer microsecond value is returned, so may not be exact.

This configuration API must not be used from the context of a timer callback. Timers should only be configured from the context of normal OSAL tasks.

See also

[OS_TimerCallback_t](#)

Parameters

out	<i>timer_id</i>	Will be set to the non-zero resource ID of the timer object (must not be null)
in	<i>timer_name</i>	Name of the timer object (must not be null)
out	<i>clock_accuracy</i>	Expected precision of the timer, in microseconds. This is the underlying tick value rounded to the nearest microsecond integer. (must not be null)
in	<i>callback_ptr</i>	The function pointer of the timer callback (must not be null).

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_INVALID_POINTER	if any parameters are NULL
OS_ERR_NAME_TOO_LONG	name length including null terminator greater than OS_MAX_API_NAME
OS_ERR_NAME_TAKEN	if the name is already in use by another timer.
OS_ERR_NO_FREE_IDS	if all of the timers are already allocated.
OS_ERR_INCORRECT_OBJ_STATE	if invoked from a timer context
OS_TIMER_ERR_INTERNAL	if there was an error programming the OS timer (return value only verified in coverage test)

OS_TimerDelete()

```
int32 OS_TimerDelete (
    osal_id_t timer_id )
```

Deletes a timer resource.

The application callback associated with the timer will be stopped, and the resources freed for future use.

Note

This configuration API must not be used from the context of a timer callback. Timers should only be configured from the context of normal OSAL tasks.

Parameters

in	<i>timer_id</i>	The timer ID to operate on
----	-----------------	----------------------------

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the timer_id is invalid.
OS_TIMER_ERR_INTERNAL	if there was a problem deleting the timer in the host OS (return value only verified in coverage test)
OS_ERR_INCORRECT_OBJ_STATE	if called from timer/timebase context

OS_TimerGetIdByName()

```
int32 OS_TimerGetIdByName (
    osal_id_t * timer_id,
    const char * timer_name )
```

Locate an existing timer resource by name.

Outputs the ID associated with the given timer, if it exists.

Note

This configuration API must not be used from the context of a timer callback. Timers should only be configured from the context of normal OSAL tasks.

Parameters

out	<i>timer_id</i>	Will be set to the timer ID corresponding to the name (must not be null)
in	<i>timer_name</i>	The timer name to find (must not be null)

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<code>OS_SUCCESS</code>	Successful execution.
<code>OS_INVALID_POINTER</code>	if <code>timer_id</code> or <code>timer_name</code> are NULL pointers
<code>OS_ERR_NAME_TOO_LONG</code>	name length including null terminator greater than <code>OS_MAX_API_NAME</code>
<code>OS_ERR_NAME_NOT_FOUND</code>	if the name was not found in the table
<code>OS_ERR_INCORRECT_OBJ_STATE</code>	if called from timer/timebase context

OS_TimerGetInfo()

```
int32 OS_TimerGetInfo (
    osal_id_t timer_id,
    OS_timer_prop_t * timer_prop )
```

Gets information about an existing timer.

This function takes `timer_id`, and looks it up in the OS table. It puts all of the information known about that timer into a structure pointer to by `timer_prop`.

Note

This configuration API must not be used from the context of a timer callback. Timers should only be configured from the context of normal OSAL tasks.

Parameters

in	<i>timer_id</i>	The timer ID to operate on
out	<i>timer_prop</i>	Buffer containing timer properties (must not be null) • creator: the OS task ID of the task that created this timer • name: the string name of the timer • start_time: the start time in microseconds, if any • interval_time: the interval time in microseconds, if any • accuracy: the accuracy of the timer in microseconds

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

<code>OS_SUCCESS</code>	Successful execution.
<code>OS_ERR_INVALID_ID</code>	if the id passed in is not a valid timer
<code>OS_INVALID_POINTER</code>	if the <code>timer_prop</code> pointer is null
<code>OS_ERR_INCORRECT_OBJ_STATE</code>	if called from timer/timebase context

OS_TimerSet()

```
int32 OS_TimerSet (
```

```
osal_id_t timer_id,
uint32 start_time,
uint32 interval_time )
```

Configures a periodic or one shot timer.

This function programs the timer with a start time and an optional interval time. The start time is the time in microseconds when the user callback function will be called. If the interval time is non-zero, the timer will be reprogrammed with that interval in microseconds to call the user callback function periodically. If the start time and interval time are zero, the function will return an error.

For a "one-shot" timer, the start_time configures the expiration time, and the interval_time should be passed as zero to indicate the timer is not to be automatically reset.

Note

The resolution of the times specified is limited to the clock accuracy returned in the OS_TimerCreate call. If the times specified in the start_msec or interval_msec parameters are less than the accuracy, they will be rounded up to the accuracy of the timer.

This configuration API must not be used from the context of a timer callback. Timers should only be configured from the context of normal OSAL tasks.

Parameters

in	<i>timer_id</i>	The timer ID to operate on
in	<i>start_time</i>	Time in microseconds to the first expiration
in	<i>interval_time</i>	Time in microseconds between subsequent intervals, value of zero will only call the user callback function once after the start_msec time.

Returns

Execution status, see [OSAL Return Code Defines](#)

Return values

OS_SUCCESS	Successful execution.
OS_ERR_INVALID_ID	if the timer_id is not valid.
OS_TIMER_ERR_INTERNAL	if there was an error programming the OS timer (return value only verified in coverage test)
OS_ERR_INCORRECT_OBJ_STATE	if called from timer/timebase context
OS_TIMER_ERR_INVALID_ARGS	if the start_time or interval_time is out of range, or both 0

10 Data Structure Documentation

10.1 CCSDS_ExtendedHeader Struct Reference

CCSDS packet extended header.

```
#include <ccsds_hdr.h>
```

Data Fields

- [uint8 Subsystem](#) [2]
subsystem qualifier

- [uint8 SystemId](#) [2]
system qualifier

10.1.1 Detailed Description

CCSDS packet extended header.
Definition at line 73 of file [ccsds_hdr.h](#).

10.1.2 Field Documentation

Subsystem

[uint8](#) CCSDS_ExtendedHeader::Subsystem[2]
subsystem qualifier
Definition at line 75 of file [ccsds_hdr.h](#).

SystemId

[uint8](#) CCSDS_ExtendedHeader::SystemId[2]
system qualifier
Definition at line 82 of file [ccsds_hdr.h](#).
The documentation for this struct was generated from the following file:

- [cfe/modules/msg/fsw/inc/ccsds_hdr.h](#)

10.2 CCSDS_PrimaryHeader Struct Reference

CCSDS packet primary header.
`#include <ccsds_hdr.h>`

Data Fields

- [uint8 StreamId](#) [2]
packet identifier word (stream ID)
- [uint8 Sequence](#) [2]
packet sequence word
- [uint8 Length](#) [2]
packet length word

10.2.1 Detailed Description

CCSDS packet primary header.
Definition at line 51 of file [ccsds_hdr.h](#).

10.2.2 Field Documentation

Length

[uint8](#) CCSDS_PrimaryHeader::Length[2]
packet length word
Definition at line 65 of file [ccsds_hdr.h](#).

Sequence

`uint8 CCSDS_PrimaryHeader::Sequence[2]`
packet sequence word
Definition at line 60 of file [ccsds_hdr.h](#).

StreamId

`uint8 CCSDS_PrimaryHeader::StreamId[2]`
packet identifier word (stream ID)
Definition at line 53 of file [ccsds_hdr.h](#).
The documentation for this struct was generated from the following file:

- [cfe/modules/msg/fsw/inc/ccsds_hdr.h](#)

10.3 CFE_Config_ArrayValue Struct Reference

Wrapper type for array configuration.
`#include <cfe_config_api_typedefs.h>`

Data Fields

- `size_t` [NumElements](#)
- `const void *` [ElementPtr](#)

10.3.1 Detailed Description

Wrapper type for array configuration.
This is a pair containing a size and pointer that is get/set via a single config table entry
Definition at line 59 of file [cfe_config_api_typedefs.h](#).

10.3.2 Field Documentation

ElementPtr

`const void* CFE_Config_ArrayValue::ElementPtr`
Definition at line 62 of file [cfe_config_api_typedefs.h](#).

NumElements

`size_t CFE_Config_ArrayValue::NumElements`
Definition at line 61 of file [cfe_config_api_typedefs.h](#).
The documentation for this struct was generated from the following file:

- [cfe/modules/core_api/fsw/inc/cfe_config_api_typedefs.h](#)

10.4 CFE_Config_IdNameEntry Struct Reference

`#include <cfe_config_nametable.h>`

Data Fields

- `const char *` [Name](#)

10.4.1 Detailed Description

Definition at line 33 of file [cfe_config_nametable.h](#).

10.4.2 Field Documentation

Name

`const char* CFE_Config_IdNameEntry::Name`

Definition at line 35 of file [cfe_config_nametable.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/config/fsw/inc/cfe_config_nametable.h](#)

10.5 CFE_Config_ValueBuffer Union Reference

```
#include <cfe_config_table.h>
```

Data Fields

- [uint32 AsInteger](#)
- `const void *` [AsPointer](#)

10.5.1 Detailed Description

Definition at line 43 of file [cfe_config_table.h](#).

10.5.2 Field Documentation

AsInteger

`uint32 CFE_Config_ValueBuffer::AsInteger`

Definition at line 45 of file [cfe_config_table.h](#).

AsPointer

`const void* CFE_Config_ValueBuffer::AsPointer`

Definition at line 46 of file [cfe_config_table.h](#).

The documentation for this union was generated from the following file:

- [cfe/modules/config/fsw/inc/cfe_config_table.h](#)

10.6 CFE_Config_ValueEntry Struct Reference

```
#include <cfe_config_table.h>
```

Data Fields

- [CFE_ConfigType_t ActualType](#)
- [CFE_Config_ValueBuffer_t Datum](#)

10.6.1 Detailed Description

Definition at line 49 of file [cfe_config_table.h](#).

10.6.2 Field Documentation

ActualType

`CFE_ConfigType_t CFE_Config_ValueEntry::ActualType`

Definition at line 51 of file [cfe_config_table.h](#).

Datum

`CFE_Config_ValueBuffer_t CFE_Config_ValueEntry::Datum`

Definition at line 52 of file `cfe_config_table.h`.

The documentation for this struct was generated from the following file:

- `cfe/modules/config/fsw/inc/cfe_config_table.h`

10.7 CFE_ES_AppInfo Struct Reference

Application Information.

```
#include <default_cfe_es_extern_typedefs.h>
```

Data Fields

- `CFE_ResourceId_t ResourceId`
Application or Library ID for this resource.
- `uint32 Type`
The type of App: CORE or EXTERNAL.
- `char Name [CFE_MISSION_MAX_API_LEN]`
The Registered Name of the Application.
- `char EntryPoint [CFE_MISSION_MAX_API_LEN]`
The Entry Point label for the Application.
- `char FileName [CFE_MISSION_MAX_PATH_LEN]`
The Filename of the file containing the Application.
- `CFE_ES_MemOffset_t StackSize`
The Stack Size of the Application.
- `uint32 AddressesAreValid`
Indicates that the Code, Data, and BSS addresses/sizes are valid.
- `uint8 Padding1 [4]`
- `CFE_ES_MemAddress_t CodeAddress`
The Address of the Application Code Segment.
- `CFE_ES_MemOffset_t CodeSize`
The Code Size of the Application.
- `CFE_ES_MemAddress_t DataAddress`
The Address of the Application Data Segment.
- `CFE_ES_MemOffset_t DataSize`
The Data Size of the Application.
- `CFE_ES_MemAddress_t BSSAddress`
The Address of the Application BSS Segment.
- `CFE_ES_MemOffset_t BSSSize`
The BSS Size of the Application.
- `CFE_ES_MemAddress_t StartAddress`
The Start Address of the Application.
- `CFE_ES_ExceptionAction_Enum_t ExceptionAction`
What should occur if Application has an exception (Restart Application OR Restart Processor)
- `uint8 Padding2`
- `CFE_ES_TaskPriority_Atom_t Priority`
The Priority of the Application.
- `CFE_ES_TaskId_t MainTaskId`

- The Application's Main Task ID.*
- [uint32 ExecutionCounter](#)
The Application's Main Task Execution Counter.
- [char MainTaskName \[CFE_MISSION_MAX_API_LEN\]](#)
The Application's Main Task ID.
- [uint32 NumOfChildTasks](#)
Number of Child tasks for an App.
- [uint8 Padding3 \[4\]](#)

10.7.1 Detailed Description

Application Information.

Structure that is used to provide information about an app. It is primarily used for the QueryOne and QueryAll Commands.

While this structure is primarily intended for Application info, it can also represent Library information where only a subset of the information applies.

Definition at line 376 of file [default_cfe_es_extern_typedefs.h](#).

10.7.2 Field Documentation

AddressesAreValid

[uint32](#) CFE_ES_AppInfo::AddressesAreValid

Indicates that the Code, Data, and BSS addresses/sizes are valid.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_AddrsValid

Definition at line 392 of file [default_cfe_es_extern_typedefs.h](#).

BSSAddress

[CFE_ES_MemAddress_t](#) CFE_ES_AppInfo::BSSAddress

The Address of the Application BSS Segment.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_BSSAddress

Definition at line 403 of file [default_cfe_es_extern_typedefs.h](#).

BSSSize

[CFE_ES_MemOffset_t](#) CFE_ES_AppInfo::BSSSize

The BSS Size of the Application.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_BSSSize

Definition at line 405 of file [default_cfe_es_extern_typedefs.h](#).

CodeAddress

[CFE_ES_MemAddress_t](#) CFE_ES_AppInfo::CodeAddress

The Address of the Application Code Segment.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_CodeAddress

Definition at line 395 of file [default_cfe_es_extern_typedefs.h](#).

CodeSize

`CFE_ES_MemOffset_t CFE_ES_AppInfo::CodeSize`

The Code Size of the Application.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_CodeSize

Definition at line 397 of file [default_cfe_es_extern_typedefs.h](#).

DataAddress

`CFE_ES_MemAddress_t CFE_ES_AppInfo::DataAddress`

The Address of the Application Data Segment.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_DataAddress

Definition at line 399 of file [default_cfe_es_extern_typedefs.h](#).

DataSize

`CFE_ES_MemOffset_t CFE_ES_AppInfo::DataSize`

The Data Size of the Application.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_DataSize

Definition at line 401 of file [default_cfe_es_extern_typedefs.h](#).

EntryPoint

`char CFE_ES_AppInfo::EntryPoint[CFE_MISSION_MAX_API_LEN]`

The Entry Point label for the Application.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_AppEntryPt[OS_MAX_API_NAME]

Definition at line 385 of file [default_cfe_es_extern_typedefs.h](#).

ExceptionAction

`CFE_ES_ExceptionAction_Enum_t CFE_ES_AppInfo::ExceptionAction`

What should occur if Application has an exception (Restart Application OR Restart Processor)

Telemetry Mnemonic(s) \$sc_\$cpu_ES_ExceptnActn

Definition at line 409 of file [default_cfe_es_extern_typedefs.h](#).

ExecutionCounter

`uint32 CFE_ES_AppInfo::ExecutionCounter`

The Application's Main Task Execution Counter.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_ExecutionCtr

Definition at line 417 of file [default_cfe_es_extern_typedefs.h](#).

FileName

`char CFE_ES_AppInfo::FileName[CFE_MISSION_MAX_PATH_LEN]`

The Filename of the file containing the Application.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_AppFilename[OS_MAX_PATH_LEN]

Definition at line 387 of file [default_cfe_es_extern_typedefs.h](#).

MainTaskId

`CFE_ES_TaskId_t CFE_ES_AppInfo::MainTaskId`

The Application's Main Task ID.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_MainTaskId

Definition at line 415 of file [default_cfe_es_extern_typedefs.h](#).

MainTaskName

`char CFE_ES_AppInfo::MainTaskName[CFE_MISSION_MAX_API_LEN]`

The Application's Main Task ID.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_MainTaskName[OS_MAX_API_NAME]

Definition at line 419 of file [default_cfe_es_extern_typedefs.h](#).

Name

`char CFE_ES_AppInfo::Name[CFE_MISSION_MAX_API_LEN]`

The Registered Name of the Application.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_AppName[OS_MAX_API_NAME]

Definition at line 383 of file [default_cfe_es_extern_typedefs.h](#).

NumOfChildTasks

`uint32 CFE_ES_AppInfo::NumOfChildTasks`

Number of Child tasks for an App.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_ChildTasks

Definition at line 421 of file [default_cfe_es_extern_typedefs.h](#).

Padding1

`uint8 CFE_ES_AppInfo::Padding1[4]`

Definition at line 394 of file [default_cfe_es_extern_typedefs.h](#).

Padding2

`uint8 CFE_ES_AppInfo::Padding2`

Definition at line 412 of file [default_cfe_es_extern_typedefs.h](#).

Padding3

`uint8 CFE_ES_AppInfo::Padding3[4]`

Definition at line 423 of file [default_cfe_es_extern_typedefs.h](#).

Priority

`CFE_ES_TaskPriority_Atom_t CFE_ES_AppInfo::Priority`

The Priority of the Application.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_Priority

Definition at line 413 of file [default_cfe_es_extern_typedefs.h](#).

ResourceId

`CFE_ResourceId_t CFE_ES_AppInfo::ResourceId`
Application or Library ID for this resource.

Telemetry Mnemonic(s) `$sc_$cpu_ES_AppID`

Definition at line 378 of file [default_cfe_es_extern_typedefs.h](#).

StackSize

`CFE_ES_MemOffset_t CFE_ES_AppInfo::StackSize`
The Stack Size of the Application.

Telemetry Mnemonic(s) `$sc_$cpu_ES_StackSize`

Definition at line 390 of file [default_cfe_es_extern_typedefs.h](#).

StartAddress

`CFE_ES_MemAddress_t CFE_ES_AppInfo::StartAddress`
The Start Address of the Application.

Telemetry Mnemonic(s) `$sc_$cpu_ES_StartAddr`

Definition at line 407 of file [default_cfe_es_extern_typedefs.h](#).

Type

`uint32 CFE_ES_AppInfo::Type`
The type of App: CORE or EXTERNAL.

Telemetry Mnemonic(s) `$sc_$cpu_ES_AppType`

Definition at line 380 of file [default_cfe_es_extern_typedefs.h](#).
The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_extern_typedefs.h](#)

10.8 CFE_ES_AppNameCmd_Payload Struct Reference

Generic application name command payload.
`#include <default_cfe_es_msgdefs.h>`

Data Fields

- char [Application](#) [[CFE_MISSION_MAX_API_LEN](#)]
ASCII text string containing Application or Library Name.

10.8.1 Detailed Description

Generic application name command payload.
For command details, see [CFE_ES_STOP_APP_CC](#), [CFE_ES_RESTART_APP_CC](#), [CFE_ES_QUERY_ONE_CC](#)
Definition at line 107 of file [default_cfe_es_msgdefs.h](#).

10.8.2 Field Documentation

Application

char CFE_ES_AppNameCmd_Payload::Application[CFE_MISSION_MAX_API_LEN]

ASCII text string containing Application or Library Name.

Definition at line 109 of file [default_cfe_es_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgdefs.h](#)

10.9 CFE_ES_AppReloadCmd_Payload Struct Reference

Reload Application Command Payload.

#include <default_cfe_es_msgdefs.h>

Data Fields

- char [Application](#) [CFE_MISSION_MAX_API_LEN]
ASCII text string containing Application Name.
- char [AppFileName](#) [CFE_MISSION_MAX_PATH_LEN]
Full path and filename of Application's executable image.

10.9.1 Detailed Description

Reload Application Command Payload.

For command details, see [CFE_ES_RELOAD_APP_CC](#)

Definition at line 118 of file [default_cfe_es_msgdefs.h](#).

10.9.2 Field Documentation

AppFileName

char CFE_ES_AppReloadCmd_Payload::AppFileName[CFE_MISSION_MAX_PATH_LEN]

Full path and filename of Application's executable image.

Definition at line 121 of file [default_cfe_es_msgdefs.h](#).

Application

char CFE_ES_AppReloadCmd_Payload::Application[CFE_MISSION_MAX_API_LEN]

ASCII text string containing Application Name.

Definition at line 120 of file [default_cfe_es_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgdefs.h](#)

10.10 CFE_ES_BlockStats Struct Reference

Block statistics.

#include <default_cfe_es_extern_typedefs.h>

Data Fields

- [CFE_ES_MemOffset_t](#) BlockSize
Number of bytes in each of these blocks.
- [uint32](#) NumCreated

Number of Memory Blocks of this size created.

- [uint32 NumFree](#)

Number of Memory Blocks of this size that are free.

10.10.1 Detailed Description

Block statistics.

Sub-Structure that is used to provide information about a specific block size/bucket within a memory pool.

Definition at line 476 of file [default_cfe_es_extern_typedefs.h](#).

10.10.2 Field Documentation

BlockSize

[CFE_ES_MemOffset_t](#) [CFE_ES_BlockStats::BlockSize](#)

Number of bytes in each of these blocks.

Definition at line 478 of file [default_cfe_es_extern_typedefs.h](#).

NumCreated

[uint32](#) [CFE_ES_BlockStats::NumCreated](#)

Number of Memory Blocks of this size created.

Definition at line 479 of file [default_cfe_es_extern_typedefs.h](#).

NumFree

[uint32](#) [CFE_ES_BlockStats::NumFree](#)

Number of Memory Blocks of this size that are free.

Definition at line 480 of file [default_cfe_es_extern_typedefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_extern_typedefs.h](#)

10.11 CFE_ES_CDSRegDumpRec Struct Reference

CDS Register Dump Record.

```
#include <default_cfe_es_extern_typedefs.h>
```

Data Fields

- [CFE_ES_CDSHandle_t](#) [Handle](#)
Handle of CDS.
- [CFE_ES_MemOffset_t](#) [Size](#)
Size, in bytes, of the CDS memory block.
- [bool](#) [Table](#)
Flag that indicates whether CDS contains a Critical Table.
- [char](#) [Name](#) [[CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN](#)]
Processor Unique Name of CDS.
- [uint8](#) [ByteAlignSpare](#) [3]
Spare bytes to ensure structure size is multiple of 4 bytes.

10.11.1 Detailed Description

CDS Register Dump Record.

Structure that is used to provide information about a critical data store. It is primarily used for the Dump CDS registry ([CFE_ES_DUMP_CDS_REGISTRY_CC](#)) command.

Note

There is not currently a telemetry message directly containing this data structure, but it does define the format of the data file generated by the Dump CDS registry command. Therefore it should be considered part of the overall telemetry interface.

Definition at line 461 of file [default_cfe_es_extern_typedefs.h](#).

10.11.2 Field Documentation

ByteAlignSpare

```
uint8 CFE_ES_CDSRegDumpRec::ByteAlignSpare[3]
```

Spare bytes to ensure structure size is multiple of 4 bytes.

Definition at line 467 of file [default_cfe_es_extern_typedefs.h](#).

Handle

```
CFE_ES_CDSHandle_t CFE_ES_CDSRegDumpRec::Handle
```

Handle of CDS.

Definition at line 463 of file [default_cfe_es_extern_typedefs.h](#).

Name

```
char CFE_ES_CDSRegDumpRec::Name[CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN]
```

Processor Unique Name of CDS.

Definition at line 466 of file [default_cfe_es_extern_typedefs.h](#).

Size

```
CFE_ES_MemOffset_t CFE_ES_CDSRegDumpRec::Size
```

Size, in bytes, of the CDS memory block.

Definition at line 464 of file [default_cfe_es_extern_typedefs.h](#).

Table

```
bool CFE_ES_CDSRegDumpRec::Table
```

Flag that indicates whether CDS contains a Critical Table.

Definition at line 465 of file [default_cfe_es_extern_typedefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_extern_typedefs.h](#)

10.12 CFE_ES_ClearERLogCmd Struct Reference

```
#include <default_cfe_es_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader

Command header.

10.12.1 Detailed Description

Definition at line 61 of file [default_cfe_es_msgstruct.h](#).

10.12.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_ES_ClearERLogCmd::CommandHeader

Command header.

Definition at line 63 of file [default_cfe_es_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgstruct.h](#)

10.13 CFE_ES_ClearSysLogCmd Struct Reference

```
#include <default_cfe_es_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.

10.13.1 Detailed Description

Definition at line 56 of file [default_cfe_es_msgstruct.h](#).

10.13.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_ES_ClearSysLogCmd::CommandHeader

Command header.

Definition at line 58 of file [default_cfe_es_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgstruct.h](#)

10.14 CFE_ES_DeleteCDSCmd Struct Reference

Delete Critical Data Store Command.

```
#include <default_cfe_es_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.
- [CFE_ES_DeleteCDSCmd_Payload_t](#) Payload
Command payload.

10.14.1 Detailed Description

Delete Critical Data Store Command.

Definition at line 185 of file [default_cfe_es_msgstruct.h](#).

10.14.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) [CFE_ES_DeleteCDSCmd::CommandHeader](#)

Command header.

Definition at line 187 of file [default_cfe_es_msgstruct.h](#).

Payload

[CFE_ES_DeleteCDSCmd_Payload_t](#) [CFE_ES_DeleteCDSCmd::Payload](#)

Command payload.

Definition at line 188 of file [default_cfe_es_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgstruct.h](#)

10.15 CFE_ES_DeleteCDSCmd_Payload Struct Reference

Delete Critical Data Store Command Payload.

```
#include <default_cfe_es_msgdefs.h>
```

Data Fields

- char [CdsName](#) [[CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN](#)]
ASCII text string containing name of CDS to delete.

10.15.1 Detailed Description

Delete Critical Data Store Command Payload.

For command details, see [CFE_ES_DELETE_CDS_CC](#)

Definition at line 143 of file [default_cfe_es_msgdefs.h](#).

10.15.2 Field Documentation

CdsName

char [CFE_ES_DeleteCDSCmd_Payload::CdsName](#) [[CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN](#)]

ASCII text string containing name of CDS to delete.

Definition at line 146 of file [default_cfe_es_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgdefs.h](#)

10.16 CFE_ES_DumpCDSRegistryCmd Struct Reference

Dump CDS Registry Command.

```
#include <default_cfe_es_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) [CommandHeader](#)
Command header.
- [CFE_ES_DumpCDSRegistryCmd_Payload_t](#) [Payload](#)
Command payload.

10.16.1 Detailed Description

Dump CDS Registry Command.

Definition at line 239 of file [default_cfe_es_msgstruct.h](#).

10.16.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_ES_DumpCDSRegistryCmd::CommandHeader

Command header.

Definition at line 241 of file [default_cfe_es_msgstruct.h](#).

Payload

[CFE_ES_DumpCDSRegistryCmd_Payload_t](#) CFE_ES_DumpCDSRegistryCmd::Payload

Command payload.

Definition at line 242 of file [default_cfe_es_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgstruct.h](#)

10.17 CFE_ES_DumpCDSRegistryCmd_Payload Struct Reference

Dump CDS Registry Command Payload.

```
#include <default_cfe_es_msgdefs.h>
```

Data Fields

- char [DumpFilename](#) [CFE_MISSION_MAX_PATH_LEN]
ASCII text string of full path and filename of file CDS Registry is to be written.

10.17.1 Detailed Description

Dump CDS Registry Command Payload.

For command details, see [CFE_ES_DUMP_CDS_REGISTRY_CC](#)

Definition at line 228 of file [default_cfe_es_msgdefs.h](#).

10.17.2 Field Documentation

DumpFilename

char CFE_ES_DumpCDSRegistryCmd_Payload::DumpFilename [CFE_MISSION_MAX_PATH_LEN]

ASCII text string of full path and filename of file CDS Registry is to be written.

Definition at line 230 of file [default_cfe_es_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgdefs.h](#)

10.18 CFE_ES_FileNameCmd Struct Reference

Generic file name command.

```
#include <default_cfe_es_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) `CommandHeader`
Command header.
- [CFE_ES_FileNameCmd_Payload_t](#) `Payload`
Command payload.

10.18.1 Detailed Description

Generic file name command.

Definition at line 89 of file [default_cfe_es_msgstruct.h](#).

10.18.2 Field Documentation**CommandHeader**

[CFE_MSG_CommandHeader_t](#) `CFE_ES_FileNameCmd::CommandHeader`
Command header.

Definition at line 91 of file [default_cfe_es_msgstruct.h](#).

Payload

[CFE_ES_FileNameCmd_Payload_t](#) `CFE_ES_FileNameCmd::Payload`
Command payload.

Definition at line 92 of file [default_cfe_es_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgstruct.h](#)

10.19 CFE_ES_FileNameCmd_Payload Struct Reference

Generic file name command payload.

`#include <default_cfe_es_msgdefs.h>`

Data Fields

- char [FileName](#) [[CFE_MISSION_MAX_PATH_LEN](#)]
ASCII text string containing full path and filename of file in which Application data is to be dumped.

10.19.1 Detailed Description

Generic file name command payload.

This format is shared by several executive services commands. For command details, see [CFE_ES_QUERY_ALL_CC](#), [CFE_ES_QUERY_ALL_TASKS_CC](#), [CFE_ES_WRITE_SYS_LOG_CC](#), and [CFE_ES_WRITE_ER_LOG_CC](#)

Definition at line 58 of file [default_cfe_es_msgdefs.h](#).

10.19.2 Field Documentation**FileName**

char `CFE_ES_FileNameCmd_Payload::FileName` [[CFE_MISSION_MAX_PATH_LEN](#)]

ASCII text string containing full path and filename of file in which Application data is to be dumped.

Definition at line 60 of file [default_cfe_es_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgdefs.h](#)

10.20 CFE_ES_HousekeepingTlm Struct Reference

```
#include <default_cfe_es_msgstruct.h>
```

Data Fields

- [CFE_MSG_TelemetryHeader_t](#) TelemetryHeader
Telemetry header.
- [CFE_ES_HousekeepingTlm_Payload_t](#) Payload
Telemetry payload.

10.20.1 Detailed Description

Name Executive Services Housekeeping Packet

Definition at line 272 of file [default_cfe_es_msgstruct.h](#).

10.20.2 Field Documentation

Payload

[CFE_ES_HousekeepingTlm_Payload_t](#) CFE_ES_HousekeepingTlm::Payload
Telemetry payload.

Definition at line 275 of file [default_cfe_es_msgstruct.h](#).

TelemetryHeader

[CFE_MSG_TelemetryHeader_t](#) CFE_ES_HousekeepingTlm::TelemetryHeader
Telemetry header.

Definition at line 274 of file [default_cfe_es_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgstruct.h](#)

10.21 CFE_ES_HousekeepingTlm_Payload Struct Reference

```
#include <default_cfe_es_msgdefs.h>
```

Data Fields

- [uint8](#) CommandCounter
The ES Application Command Counter.
- [uint8](#) CommandErrorCounter
The ES Application Command Error Counter.
- [uint16](#) CFECoreChecksum
Checksum of cFE Core Code.
- [uint8](#) CFEMajorVersion
Major Version Number of cFE.
- [uint8](#) CFEMinorVersion
Minor Version Number of cFE.
- [uint8](#) CFERevision
Sub-Minor Version Number of cFE.
- [uint8](#) CFEMissionRevision
Mission Version Number of cFE.

- [uint8 OSALMajorVersion](#)
OS Abstraction Layer Major Version Number.
- [uint8 OSALMinorVersion](#)
OS Abstraction Layer Minor Version Number.
- [uint8 OSALRevision](#)
OS Abstraction Layer Revision Number.
- [uint8 OSALMissionRevision](#)
OS Abstraction Layer MissionRevision Number.
- [uint8 PSPMajorVersion](#)
Platform Support Package Major Version Number.
- [uint8 PSPMinorVersion](#)
Platform Support Package Minor Version Number.
- [uint8 PSPRevision](#)
Platform Support Package Revision Number.
- [uint8 PSPMissionRevision](#)
Platform Support Package MissionRevision Number.
- [CFE_ES_MemOffset_t SysLogBytesUsed](#)
Total number of bytes used in system log.
- [CFE_ES_MemOffset_t SysLogSize](#)
Total size of the system log.
- [uint32 SysLogEntries](#)
Number of entries in the system log.
- [uint32 SysLogMode](#)
Write/Overwrite Mode.
- [uint32 ERLogIndex](#)
Current index of the ER Log (wraps around)
- [uint32 ERLogEntries](#)
Number of entries made in the ER Log since the power on.
- [uint32 RegisteredCoreApps](#)
Number of Applications registered with ES.
- [uint32 RegisteredExternalApps](#)
Number of Applications registered with ES.
- [uint32 RegisteredTasks](#)
Number of Tasks (main AND child tasks) registered with ES.
- [uint32 RegisteredLibs](#)
Number of Libraries registered with ES.
- [uint32 ResetType](#)
Reset type (PROCESSOR or POWERON)
- [uint32 ResetSubtype](#)
Reset Sub Type.
- [uint32 ProcessorResets](#)
Number of processor resets since last power on.
- [uint32 MaxProcessorResets](#)
Max processor resets before a power on is done.
- [uint32 BootSource](#)
Boot source (as provided from BSP)
- [uint32 PerfState](#)

Current state of Performance Analyzer.

- [uint32 PerfMode](#)

Current mode of Performance Analyzer.

- [uint32 PerfTriggerCount](#)

Number of Times Performance Analyzer has Triggered.

- [uint32 PerfFilterMask](#) [CFE_MISSION_ES_PERF_MAX_IDS/32]

Current Setting of Performance Analyzer Filter Masks.

- [uint32 PerfTriggerMask](#) [CFE_MISSION_ES_PERF_MAX_IDS/32]

Current Setting of Performance Analyzer Trigger Masks.

- [uint32 PerfDataStart](#)

Identifies First Stored Entry in Performance Analyzer Log.

- [uint32 PerfDataEnd](#)

Identifies Last Stored Entry in Performance Analyzer Log.

- [uint32 PerfDataCount](#)

Number of Entries Put Into the Performance Analyzer Log.

- [uint32 PerfDataToWrite](#)

Number of Performance Analyzer Log Entries Left to be Written to Log Dump File.

- [CFE_ES_MemOffset_t HeapBytesFree](#)

Number of free bytes remaining in the OS heap.

- [CFE_ES_MemOffset_t HeapBlocksFree](#)

Number of free blocks remaining in the OS heap.

- [CFE_ES_MemOffset_t HeapMaxBlockSize](#)

Number of bytes in the largest free block.

10.21.1 Detailed Description

Name Executive Services Housekeeping Packet

Definition at line 264 of file [default_cfe_es_msgdefs.h](#).

10.21.2 Field Documentation

BootSource

[uint32](#) CFE_ES_HousekeepingTlm_Payload::BootSource

Boot source (as provided from BSP)

Telemetry Mnemonic(s) \$sc_\$cpu_ES_BootSource

Definition at line 330 of file [default_cfe_es_msgdefs.h](#).

CFECoreChecksum

[uint16](#) CFE_ES_HousekeepingTlm_Payload::CFECoreChecksum

Checksum of cFE Core Code.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_CKSUM

Definition at line 271 of file [default_cfe_es_msgdefs.h](#).

CFEMajorVersion

`uint8 CFE_ES_HousekeepingTlm_Payload::CFEMajorVersion`
Major Version Number of cFE.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_CFEMAJORVER

Definition at line 273 of file [default_cfe_es_msgdefs.h](#).

CFEMinorVersion

`uint8 CFE_ES_HousekeepingTlm_Payload::CFEMinorVersion`
Minor Version Number of cFE.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_CFEMINORVER

Definition at line 275 of file [default_cfe_es_msgdefs.h](#).

CFEMissionRevision

`uint8 CFE_ES_HousekeepingTlm_Payload::CFEMissionRevision`
Mission Version Number of cFE.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_CFEMISSIONREV

Definition at line 279 of file [default_cfe_es_msgdefs.h](#).

CFERevision

`uint8 CFE_ES_HousekeepingTlm_Payload::CFERevision`
Sub-Minor Version Number of cFE.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_CFEREVISION

Definition at line 277 of file [default_cfe_es_msgdefs.h](#).

CommandCounter

`uint8 CFE_ES_HousekeepingTlm_Payload::CommandCounter`
The ES Application Command Counter.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_CMDPC

Definition at line 266 of file [default_cfe_es_msgdefs.h](#).

CommandErrorCounter

`uint8 CFE_ES_HousekeepingTlm_Payload::CommandErrorCounter`
The ES Application Command Error Counter.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_CMDEC

Definition at line 268 of file [default_cfe_es_msgdefs.h](#).

ERLogEntries

`uint32 CFE_ES_HousekeepingTlm_Payload::ERLogEntries`
Number of entries made in the ER Log since the power on.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_ERLOGENTRIES

Definition at line 310 of file [default_cfe_es_msgdefs.h](#).

ERLogIndex

`uint32 CFE_ES_HousekeepingTlm_Payload::ERLogIndex`
Current index of the ER Log (wraps around)

Telemetry Mnemonic(s) \$sc_\$cpu_ES_ERLOGINDEX

Definition at line 308 of file [default_cfe_es_msgdefs.h](#).

HeapBlocksFree

`CFE_ES_MemOffset_t CFE_ES_HousekeepingTlm_Payload::HeapBlocksFree`
Number of free blocks remaining in the OS heap.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_HeapBlocksFree

Definition at line 355 of file [default_cfe_es_msgdefs.h](#).

HeapBytesFree

`CFE_ES_MemOffset_t CFE_ES_HousekeepingTlm_Payload::HeapBytesFree`
Number of free bytes remaining in the OS heap.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_HeapBytesFree

Definition at line 353 of file [default_cfe_es_msgdefs.h](#).

HeapMaxBlockSize

`CFE_ES_MemOffset_t CFE_ES_HousekeepingTlm_Payload::HeapMaxBlockSize`
Number of bytes in the largest free block.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_HeapMaxBlkSize

Definition at line 357 of file [default_cfe_es_msgdefs.h](#).

MaxProcessorResets

`uint32 CFE_ES_HousekeepingTlm_Payload::MaxProcessorResets`
Max processor resets before a power on is done.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_MaxProcResets

Definition at line 328 of file [default_cfe_es_msgdefs.h](#).

OSALMajorVersion

`uint8 CFE_ES_HousekeepingTlm_Payload::OSALMajorVersion`
OS Abstraction Layer Major Version Number.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_OSMAJORVER

Definition at line 281 of file [default_cfe_es_msgdefs.h](#).

OSALMinorVersion

`uint8 CFE_ES_HousekeepingTlm_Payload::OSALMinorVersion`
OS Abstraction Layer Minor Version Number.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_OSMINORVER

Definition at line 283 of file [default_cfe_es_msgdefs.h](#).

OSALMissionRevision

`uint8 CFE_ES_HousekeepingTlm_Payload::OSALMissionRevision`
OS Abstraction Layer MissionRevision Number.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_OSMISSIONREV

Definition at line 287 of file [default_cfe_es_msgdefs.h](#).

OSALRevision

`uint8 CFE_ES_HousekeepingTlm_Payload::OSALRevision`
OS Abstraction Layer Revision Number.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_OSREVISION

Definition at line 285 of file [default_cfe_es_msgdefs.h](#).

PerfDataCount

`uint32 CFE_ES_HousekeepingTlm_Payload::PerfDataCount`
Number of Entries Put Into the Performance Analyzer Log.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_PerfDataCnt

Definition at line 348 of file [default_cfe_es_msgdefs.h](#).

PerfDataEnd

`uint32 CFE_ES_HousekeepingTlm_Payload::PerfDataEnd`
Identifies Last Stored Entry in Performance Analyzer Log.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_PerfDataEnd

Definition at line 346 of file [default_cfe_es_msgdefs.h](#).

PerfDataStart

`uint32 CFE_ES_HousekeepingTlm_Payload::PerfDataStart`
Identifies First Stored Entry in Performance Analyzer Log.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_PerfDataStart

Definition at line 344 of file [default_cfe_es_msgdefs.h](#).

PerfDataToWrite

`uint32 CFE_ES_HousekeepingTlm_Payload::PerfDataToWrite`
Number of Performance Analyzer Log Entries Left to be Written to Log Dump File.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_PerfData2Write

Definition at line 351 of file [default_cfe_es_msgdefs.h](#).

PerfFilterMask

`uint32 CFE_ES_HousekeepingTlm_Payload::PerfFilterMask[CFE_MISSION_ES_PERF_MAX_IDS/32]`
Current Setting of Performance Analyzer Filter Masks.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_PerfFtrMask[MaskCnt]

Definition at line 339 of file [default_cfe_es_msgdefs.h](#).

PerfMode

`uint32 CFE_ES_HousekeepingTlm_Payload::PerfMode`
Current mode of Performance Analyzer.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_PerfMode

Definition at line 335 of file [default_cfe_es_msgdefs.h](#).

PerfState

`uint32 CFE_ES_HousekeepingTlm_Payload::PerfState`
Current state of Performance Analyzer.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_PerfState

Definition at line 333 of file [default_cfe_es_msgdefs.h](#).

PerfTriggerCount

`uint32 CFE_ES_HousekeepingTlm_Payload::PerfTriggerCount`
Number of Times Performance Analyzer has Triggered.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_PerfTrigCnt

Definition at line 337 of file [default_cfe_es_msgdefs.h](#).

PerfTriggerMask

`uint32 CFE_ES_HousekeepingTlm_Payload::PerfTriggerMask[CFE_MISSION_ES_PERF_MAX_IDS/32]`
Current Setting of Performance Analyzer Trigger Masks.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_PerfTrigMask[MaskCnt]

Definition at line 342 of file [default_cfe_es_msgdefs.h](#).

ProcessorResets

`uint32 CFE_ES_HousekeepingTlm_Payload::ProcessorResets`
Number of processor resets since last power on.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_ProcResetCnt

Definition at line 326 of file [default_cfe_es_msgdefs.h](#).

PSPMajorVersion

`uint8 CFE_ES_HousekeepingTlm_Payload::PSPMajorVersion`
Platform Support Package Major Version Number.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_PSPMAJORVER

Definition at line 290 of file [default_cfe_es_msgdefs.h](#).

PSPMinorVersion

`uint8 CFE_ES_HousekeepingTlm_Payload::PSPMinorVersion`
Platform Support Package Minor Version Number.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_PSPMINORVER

Definition at line 292 of file [default_cfe_es_msgdefs.h](#).

PSPMissionRevision

`uint8 CFE_ES_HousekeepingTlm_Payload::PSPMissionRevision`
Platform Support Package MissionRevision Number.

Telemetry Mnemonic(s) `$sc_$cpu_ES_PSPMISSIONREV`

Definition at line 296 of file `default_cfe_es_msgdefs.h`.

PSPRevision

`uint8 CFE_ES_HousekeepingTlm_Payload::PSPRevision`
Platform Support Package Revision Number.

Telemetry Mnemonic(s) `$sc_$cpu_ES_PSPREVISION`

Definition at line 294 of file `default_cfe_es_msgdefs.h`.

RegisteredCoreApps

`uint32 CFE_ES_HousekeepingTlm_Payload::RegisteredCoreApps`
Number of Applications registered with ES.

Telemetry Mnemonic(s) `$sc_$cpu_ES_RegCoreApps`

Definition at line 313 of file `default_cfe_es_msgdefs.h`.

RegisteredExternalApps

`uint32 CFE_ES_HousekeepingTlm_Payload::RegisteredExternalApps`
Number of Applications registered with ES.

Telemetry Mnemonic(s) `$sc_$cpu_ES_RegExtApps`

Definition at line 315 of file `default_cfe_es_msgdefs.h`.

RegisteredLibs

`uint32 CFE_ES_HousekeepingTlm_Payload::RegisteredLibs`
Number of Libraries registered with ES.

Telemetry Mnemonic(s) `$sc_$cpu_ES_RegLibs`

Definition at line 319 of file `default_cfe_es_msgdefs.h`.

RegisteredTasks

`uint32 CFE_ES_HousekeepingTlm_Payload::RegisteredTasks`
Number of Tasks (main AND child tasks) registered with ES.

Telemetry Mnemonic(s) `$sc_$cpu_ES_RegTasks`

Definition at line 317 of file `default_cfe_es_msgdefs.h`.

ResetSubtype

`uint32 CFE_ES_HousekeepingTlm_Payload::ResetSubtype`
Reset Sub Type.

Telemetry Mnemonic(s) `$sc_$cpu_ES_ResetSubtype`

Definition at line 324 of file `default_cfe_es_msgdefs.h`.

ResetType

`uint32 CFE_ES_HousekeepingTlm_Payload::ResetType`
Reset type (PROCESSOR or POWERON)

Telemetry Mnemonic(s) \$sc_\$cpu_ES_ResetType

Definition at line 322 of file [default_cfe_es_msgdefs.h](#).

SysLogBytesUsed

`CFE_ES_MemOffset_t CFE_ES_HousekeepingTlm_Payload::SysLogBytesUsed`
Total number of bytes used in system log.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_SYSLOGBYTEUSED

Definition at line 299 of file [default_cfe_es_msgdefs.h](#).

SysLogEntries

`uint32 CFE_ES_HousekeepingTlm_Payload::SysLogEntries`
Number of entries in the system log.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_SYSLOGENTRIES

Definition at line 303 of file [default_cfe_es_msgdefs.h](#).

SysLogMode

`uint32 CFE_ES_HousekeepingTlm_Payload::SysLogMode`
Write/Overwrite Mode.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_SYSLOGMODE

Definition at line 305 of file [default_cfe_es_msgdefs.h](#).

SysLogSize

`CFE_ES_MemOffset_t CFE_ES_HousekeepingTlm_Payload::SysLogSize`
Total size of the system log.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_SYSLOGSIZE

Definition at line 301 of file [default_cfe_es_msgdefs.h](#).
The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgdefs.h](#)

10.22 CFE_ES_MemAddress_t Struct Reference

Type used for memory addresses in command and telemetry messages.
`#include <example_2x32bit_cfe_es_memaddress.h>`

Data Fields

- `uint32 bits` [2]

10.22.1 Detailed Description

Type used for memory addresses in command and telemetry messages.

For backward compatibility with existing CFE code this should be uint32, but if running on a 64-bit platform, addresses in telemetry will be truncated to 32 bits and therefore will not be valid.

On 64-bit platforms this can be a 64-bit address which will allow the full memory address in commands and telemetry, but this will break compatibility with existing control systems, and may also change the alignment/padding of messages. In either case this must be an unsigned type.

FSW code should access this value via the macros provided, which converts to the native "cpuaddr" type provided by OSAL. This macro provides independence between the message representation and local representation of a memory address.

Definition at line 117 of file [example_2x32bit_cfe_es_memaddress.h](#).

10.22.2 Field Documentation

bits

`uint32 CFE_ES_MemAddress_t::bits[2]`

Definition at line 119 of file [example_2x32bit_cfe_es_memaddress.h](#).

Referenced by [CFE_ES_MemAddress_FromNative\(\)](#), and [CFE_ES_MemAddress_ToNative\(\)](#).

The documentation for this struct was generated from the following file:

- [example_2x32bit_cfe_es_memaddress.h](#)

10.23 CFE_ES_MemOffset_t Struct Reference

Type used for memory sizes and offsets in commands and telemetry.

`#include <example_2x32bit_cfe_es_memaddress.h>`

Data Fields

- `uint32 bits [2]`

10.23.1 Detailed Description

Type used for memory sizes and offsets in commands and telemetry.

For backward compatibility with existing CFE code this should be uint32, but all telemetry information will be limited to 4GB in size as a result.

On 64-bit platforms this can be a 64-bit value which will allow larger memory objects, but this will break compatibility with existing control systems, and may also change the alignment/padding of messages.

In either case this must be an unsigned type.

Note

It is defined as two uint32s rather than a uint64 in case it is not aligned in the parent structure.

Definition at line 60 of file [example_2x32bit_cfe_es_memaddress.h](#).

10.23.2 Field Documentation

bits

`uint32 CFE_ES_MemOffset_t::bits[2]`

Definition at line 62 of file [example_2x32bit_cfe_es_memaddress.h](#).

Referenced by [CFE_ES_MemOffset_FromNative\(\)](#), and [CFE_ES_MemOffset_ToNative\(\)](#).

The documentation for this struct was generated from the following file:

- [example_2x32bit_cfe_es_memaddress.h](#)

10.24 CFE_ES_MemPoolStats Struct Reference

Memory Pool Statistics.

```
#include <default_cfe_es_extern_typedefs.h>
```

Data Fields

- [CFE_ES_MemOffset_t PoolSize](#)
Size of Memory Pool (in bytes)
- [uint32 NumBlocksRequested](#)
Number of times a memory block has been allocated.
- [uint32 CheckErrCtr](#)
Number of errors detected when freeing a memory block.
- [CFE_ES_MemOffset_t NumFreeBytes](#)
Number of bytes never allocated to a block.
- [CFE_ES_BlockStats_t BlockStats \[CFE_MISSION_ES_POOL_MAX_BUCKETS\]](#)
Contains stats on each block size.

10.24.1 Detailed Description

Memory Pool Statistics.

Structure that is used to provide information about a memory pool. Used by the Memory Pool Stats telemetry message.

See also

[CFE_ES_SEND_MEM_POOL_STATS_CC](#)

Definition at line 491 of file [default_cfe_es_extern_typedefs.h](#).

10.24.2 Field Documentation

BlockStats

```
CFE_ES_BlockStats_t CFE_ES_MemPoolStats::BlockStats[CFE_MISSION_ES_POOL_MAX_BUCKETS]
```

Contains stats on each block size.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_BlkStats[BLK_SIZES]

Definition at line 501 of file [default_cfe_es_extern_typedefs.h](#).

CheckErrCtr

```
uint32 CFE_ES_MemPoolStats::CheckErrCtr
```

Number of errors detected when freeing a memory block.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_BlkErrCTR

Definition at line 497 of file [default_cfe_es_extern_typedefs.h](#).

NumBlocksRequested

```
uint32 CFE_ES_MemPoolStats::NumBlocksRequested
```

Number of times a memory block has been allocated.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_BlksREQ

Definition at line 495 of file [default_cfe_es_extern_typedefs.h](#).

NumFreeBytes

`CFE_ES_MemOffset_t CFE_ES_MemPoolStats::NumFreeBytes`

Number of bytes never allocated to a block.

Telemetry Mnemonic(s) \$sc_\$cpu_ES_FreeBytes

Definition at line 499 of file [default_cfe_es_extern_typedefs.h](#).

PoolSize

`CFE_ES_MemOffset_t CFE_ES_MemPoolStats::PoolSize`

Size of Memory Pool (in bytes)

Telemetry Mnemonic(s) \$sc_\$cpu_ES_PoolSize

Definition at line 493 of file [default_cfe_es_extern_typedefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_extern_typedefs.h](#)

10.25 CFE_ES_MemStatsTlm Struct Reference

```
#include <default_cfe_es_msgstruct.h>
```

Data Fields

- [CFE_MSG_TelemetryHeader_t](#) TelemetryHeader
Telemetry header.
- [CFE_ES_PoolStatsTlm_Payload_t](#) Payload
Telemetry payload.

10.25.1 Detailed Description

Name Memory Pool Statistics Packet

Definition at line 263 of file [default_cfe_es_msgstruct.h](#).

10.25.2 Field Documentation**Payload**

`CFE_ES_PoolStatsTlm_Payload_t CFE_ES_MemStatsTlm::Payload`

Telemetry payload.

Definition at line 266 of file [default_cfe_es_msgstruct.h](#).

TelemetryHeader

`CFE_MSG_TelemetryHeader_t CFE_ES_MemStatsTlm::TelemetryHeader`

Telemetry header.

Definition at line 265 of file [default_cfe_es_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgstruct.h](#)

10.26 CFE_ES_NoopCmd Struct Reference

```
#include <default_cfe_es_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) `CommandHeader`
Command header.

10.26.1 Detailed Description

Definition at line 46 of file [default_cfe_es_msgstruct.h](#).

10.26.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) `CFE_ES_NoopCmd::CommandHeader`

Command header.

Definition at line 48 of file [default_cfe_es_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgstruct.h](#)

10.27 CFE_ES_OneAppTlm Struct Reference

```
#include <default_cfe_es_msgstruct.h>
```

Data Fields

- [CFE_MSG_TelemetryHeader_t](#) `TelemetryHeader`
Telemetry header.
- [CFE_ES_OneAppTlm_Payload_t](#) `Payload`
Telemetry payload.

10.27.1 Detailed Description

Name Single Application Information Packet

Definition at line 254 of file [default_cfe_es_msgstruct.h](#).

10.27.2 Field Documentation

Payload

[CFE_ES_OneAppTlm_Payload_t](#) `CFE_ES_OneAppTlm::Payload`

Telemetry payload.

Definition at line 257 of file [default_cfe_es_msgstruct.h](#).

TelemetryHeader

[CFE_MSG_TelemetryHeader_t](#) `CFE_ES_OneAppTlm::TelemetryHeader`

Telemetry header.

Definition at line 256 of file [default_cfe_es_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgstruct.h](#)

10.28 CFE_ES_OneAppTlm_Payload Struct Reference

```
#include <default_cfe_es_msgdefs.h>
```

Data Fields

- [CFE_ES_AppInfo_t AppInfo](#)

For more information, see [CFE_ES_AppInfo_t](#).

10.28.1 Detailed Description

Name Single Application Information Packet

Definition at line 243 of file [default_cfe_es_msgdefs.h](#).

10.28.2 Field Documentation**AppInfo**

[CFE_ES_AppInfo_t](#) CFE_ES_OneAppTlm_Payload::AppInfo

For more information, see [CFE_ES_AppInfo_t](#).

Definition at line 245 of file [default_cfe_es_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgdefs.h](#)

10.29 CFE_ES_OverWriteSysLogCmd Struct Reference

Overwrite/Discard System Log Configuration Command Payload.

#include <[default_cfe_es_msgstruct.h](#)>

Data Fields

- [CFE_MSG_CommandHeader_t CommandHeader](#)
Command header.
- [CFE_ES_OverWriteSysLogCmd_Payload_t Payload](#)
Command payload.

10.29.1 Detailed Description

Overwrite/Discard System Log Configuration Command Payload.

Definition at line 126 of file [default_cfe_es_msgstruct.h](#).

10.29.2 Field Documentation**CommandHeader**

[CFE_MSG_CommandHeader_t](#) CFE_ES_OverWriteSysLogCmd::CommandHeader

Command header.

Definition at line 128 of file [default_cfe_es_msgstruct.h](#).

Payload

[CFE_ES_OverWriteSysLogCmd_Payload_t](#) CFE_ES_OverWriteSysLogCmd::Payload

Command payload.

Definition at line 129 of file [default_cfe_es_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgstruct.h](#)

10.30 CFE_ES_OverWriteSysLogCmd_Payload Struct Reference

Overwrite/Discard System Log Configuration Command Payload.

```
#include <default_cfe_es_msgdefs.h>
```

Data Fields

- [uint32 Mode](#)

[CFE_ES_LogMode_DISCARD](#)=Throw away most recent messages, [CFE_ES_LogMode_OVERWRITE](#)=Overwrite oldest with most recent

10.30.1 Detailed Description

Overwrite/Discard System Log Configuration Command Payload.

For command details, see [CFE_ES_OVER_WRITE_SYS_LOG_CC](#)

Definition at line 70 of file [default_cfe_es_msgdefs.h](#).

10.30.2 Field Documentation

Mode

`uint32 CFE_ES_OverWriteSysLogCmd_Payload::Mode`

[CFE_ES_LogMode_DISCARD](#)=Throw away most recent messages, [CFE_ES_LogMode_OVERWRITE](#)=Overwrite oldest with most recent

Definition at line 72 of file [default_cfe_es_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgdefs.h](#)

10.31 CFE_ES_PoolAlign Union Reference

Pool Alignment.

```
#include <cfe_es_api_typedefs.h>
```

Data Fields

- `void * Ptr`
Aligned pointer.
- `long long int LongInt`
Aligned Long Integer.
- `long double LongDouble`
Aligned Long Double.

10.31.1 Detailed Description

Pool Alignment.

Union that can be used for minimum memory alignment of ES memory pools on the target. It contains the longest native data types such that the alignment of this structure should reflect the largest possible alignment requirements for any data on this processor.

Definition at line 145 of file [cfe_es_api_typedefs.h](#).

10.31.2 Field Documentation

LongDouble

`long double CFE_ES_PoolAlign::LongDouble`
 Aligned Long Double.
 Definition at line 150 of file [cfe_es_api_typedefs.h](#).

LongInt

`long long int CFE_ES_PoolAlign::LongInt`
 Aligned Long Integer.
 Definition at line 149 of file [cfe_es_api_typedefs.h](#).

Ptr

`void* CFE_ES_PoolAlign::Ptr`
 Aligned pointer.
 Definition at line 147 of file [cfe_es_api_typedefs.h](#).
 The documentation for this union was generated from the following file:

- [cfe/modules/core_api/fsw/inc/cfe_es_api_typedefs.h](#)

10.32 CFE_ES_PoolStatsTlm_Payload Struct Reference

`#include <default_cfe_es_msgdefs.h>`

Data Fields

- [CFE_ES_MemHandle_t PoolHandle](#)
Handle of memory pool whose stats are being telemetered.
- [uint8 Padding](#) [4]
- [CFE_ES_MemPoolStats_t PoolStats](#)
For more info, see [CFE_ES_MemPoolStats_t](#).

10.32.1 Detailed Description

Name Memory Pool Statistics Packet

Definition at line 251 of file [default_cfe_es_msgdefs.h](#).

10.32.2 Field Documentation

Padding

`uint8 CFE_ES_PoolStatsTlm_Payload::Padding` [4]
 Definition at line 255 of file [default_cfe_es_msgdefs.h](#).

PoolHandle

[CFE_ES_MemHandle_t](#) `CFE_ES_PoolStatsTlm_Payload::PoolHandle`
 Handle of memory pool whose stats are being telemetered.

Telemetry Mnemonic(s) `$sc_$cpu_ES_PoolHandle`

Definition at line 253 of file [default_cfe_es_msgdefs.h](#).

PoolStats

[CFE_ES_MemPoolStats_t](#) CFE_ES_PoolStatsTlm_Payload::PoolStats

For more info, see [CFE_ES_MemPoolStats_t](#).

Definition at line 256 of file [default_cfe_es_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgdefs.h](#)

10.33 CFE_ES_QueryAllCmd Struct Reference

```
#include <default_cfe_es_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.
- [CFE_ES_FileNameCmd_Payload_t](#) Payload
Command payload.

10.33.1 Detailed Description

Definition at line 99 of file [default_cfe_es_msgstruct.h](#).

10.33.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_ES_QueryAllCmd::CommandHeader

Command header.

Definition at line 101 of file [default_cfe_es_msgstruct.h](#).

Payload

[CFE_ES_FileNameCmd_Payload_t](#) CFE_ES_QueryAllCmd::Payload

Command payload.

Definition at line 102 of file [default_cfe_es_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgstruct.h](#)

10.34 CFE_ES_QueryAllTasksCmd Struct Reference

```
#include <default_cfe_es_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.
- [CFE_ES_FileNameCmd_Payload_t](#) Payload
Command payload.

10.34.1 Detailed Description

Definition at line 105 of file [default_cfe_es_msgstruct.h](#).

10.34.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_ES_QueryAllTasksCmd::CommandHeader

Command header.

Definition at line 107 of file [default_cfe_es_msgstruct.h](#).

Payload

[CFE_ES_FileNameCmd_Payload_t](#) CFE_ES_QueryAllTasksCmd::Payload

Command payload.

Definition at line 108 of file [default_cfe_es_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgstruct.h](#)

10.35 CFE_ES_QueryOneCmd Struct Reference

```
#include <default_cfe_es_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.
- [CFE_ES_AppNameCmd_Payload_t](#) Payload
Command payload.

10.35.1 Detailed Description

Definition at line 158 of file [default_cfe_es_msgstruct.h](#).

10.35.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_ES_QueryOneCmd::CommandHeader

Command header.

Definition at line 160 of file [default_cfe_es_msgstruct.h](#).

Payload

[CFE_ES_AppNameCmd_Payload_t](#) CFE_ES_QueryOneCmd::Payload

Command payload.

Definition at line 161 of file [default_cfe_es_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgstruct.h](#)

10.36 CFE_ES_ReloadAppCmd Struct Reference

Reload Application Command.

```
#include <default_cfe_es_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) `CommandHeader`
Command header.
- [CFE_ES_AppReloadCmd_Payload_t](#) `Payload`
Command payload.

10.36.1 Detailed Description

Reload Application Command.

Definition at line 167 of file [default_cfe_es_msgstruct.h](#).

10.36.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) `CFE_ES_ReloadAppCmd::CommandHeader`
Command header.

Definition at line 169 of file [default_cfe_es_msgstruct.h](#).

Payload

[CFE_ES_AppReloadCmd_Payload_t](#) `CFE_ES_ReloadAppCmd::Payload`
Command payload.

Definition at line 170 of file [default_cfe_es_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgstruct.h](#)

10.37 CFE_ES_ResetCountersCmd Struct Reference

```
#include <default_cfe_es_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) `CommandHeader`
Command header.

10.37.1 Detailed Description

Definition at line 51 of file [default_cfe_es_msgstruct.h](#).

10.37.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) `CFE_ES_ResetCountersCmd::CommandHeader`
Command header.

Definition at line 53 of file [default_cfe_es_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgstruct.h](#)

10.38 CFE_ES_ResetPRCountCmd Struct Reference

```
#include <default_cfe_es_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) `CommandHeader`
Command header.

10.38.1 Detailed Description

Definition at line 66 of file [default_cfe_es_msgstruct.h](#).

10.38.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) `CFE_ES_ResetPRCountCmd::CommandHeader`

Command header.

Definition at line 68 of file [default_cfe_es_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgstruct.h](#)

10.39 CFE_ES_RestartAppCmd Struct Reference

```
#include <default_cfe_es_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) `CommandHeader`
Command header.
- [CFE_ES_AppNameCmd_Payload_t](#) `Payload`
Command payload.

10.39.1 Detailed Description

Definition at line 152 of file [default_cfe_es_msgstruct.h](#).

10.39.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) `CFE_ES_RestartAppCmd::CommandHeader`

Command header.

Definition at line 154 of file [default_cfe_es_msgstruct.h](#).

Payload

[CFE_ES_AppNameCmd_Payload_t](#) `CFE_ES_RestartAppCmd::Payload`

Command payload.

Definition at line 155 of file [default_cfe_es_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgstruct.h](#)

10.40 CFE_ES_RestartCmd Struct Reference

Restart cFE Command.

```
#include <default_cfe_es_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) `CommandHeader`
Command header.
- [CFE_ES_RestartCmd_Payload_t](#) `Payload`
Command payload.

10.40.1 Detailed Description

Restart cFE Command.

Definition at line 79 of file [default_cfe_es_msgstruct.h](#).

10.40.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) `CFE_ES_RestartCmd::CommandHeader`

Command header.

Definition at line 81 of file [default_cfe_es_msgstruct.h](#).

Payload

[CFE_ES_RestartCmd_Payload_t](#) `CFE_ES_RestartCmd::Payload`

Command payload.

Definition at line 82 of file [default_cfe_es_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgstruct.h](#)

10.41 CFE_ES_RestartCmd_Payload Struct Reference

Restart cFE Command Payload.

```
#include <default_cfe_es_msgdefs.h>
```

Data Fields

- [uint16](#) `RestartType`
[CFE_PSP_RST_TYPE_PROCESSOR](#)=Processor Reset or [CFE_PSP_RST_TYPE_POWERON](#)=Power-On Reset

10.41.1 Detailed Description

Restart cFE Command Payload.

For command details, see [CFE_ES_RESTART_CC](#)

Definition at line 44 of file [default_cfe_es_msgdefs.h](#).

10.41.2 Field Documentation

RestartType

[uint16](#) `CFE_ES_RestartCmd_Payload::RestartType`

[CFE_PSP_RST_TYPE_PROCESSOR](#)=Processor Reset or [CFE_PSP_RST_TYPE_POWERON](#)=Power-On Reset

Definition at line 46 of file [default_cfe_es_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgdefs.h](#)

10.42 CFE_ES_SendHkCmd Struct Reference

```
#include <default_cfe_es_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) `CommandHeader`
Command header.

10.42.1 Detailed Description

Definition at line 71 of file [default_cfe_es_msgstruct.h](#).

10.42.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) `CFE_ES_SendHkCmd::CommandHeader`

Command header.

Definition at line 73 of file [default_cfe_es_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgstruct.h](#)

10.43 CFE_ES_SendMemPoolStatsCmd Struct Reference

Send Memory Pool Statistics Command.

```
#include <default_cfe_es_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) `CommandHeader`
Command header.
- [CFE_ES_SendMemPoolStatsCmd_Payload_t](#) `Payload`
Command payload.

10.43.1 Detailed Description

Send Memory Pool Statistics Command.

Definition at line 230 of file [default_cfe_es_msgstruct.h](#).

10.43.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) `CFE_ES_SendMemPoolStatsCmd::CommandHeader`

Command header.

Definition at line 232 of file [default_cfe_es_msgstruct.h](#).

Payload

[CFE_ES_SendMemPoolStatsCmd_Payload_t](#) `CFE_ES_SendMemPoolStatsCmd::Payload`

Command payload.

Definition at line 233 of file [default_cfe_es_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgstruct.h](#)

10.44 CFE_ES_SendMemPoolStatsCmd_Payload Struct Reference

Send Memory Pool Statistics Command Payload.

```
#include <default_cfe_es_msgdefs.h>
```

Data Fields

- char [Application](#) [CFE_MISSION_MAX_API_LEN]
 - *RESERVED* - should be all zeroes
- [CFE_ES_MemHandle_t](#) [PoolHandle](#)
Handle of Pool whose statistics are to be telemetered.

10.44.1 Detailed Description

Send Memory Pool Statistics Command Payload.

For command details, see [CFE_ES_SEND_MEM_POOL_STATS_CC](#)

Definition at line 216 of file [default_cfe_es_msgdefs.h](#).

10.44.2 Field Documentation

Application

```
char CFE_ES_SendMemPoolStatsCmd_Payload::Application[CFE_MISSION_MAX_API_LEN]
```

- *RESERVED* - should be all zeroes

Definition at line 218 of file [default_cfe_es_msgdefs.h](#).

PoolHandle

```
CFE\_ES\_MemHandle\_t CFE_ES_SendMemPoolStatsCmd_Payload::PoolHandle
```

Handle of Pool whose statistics are to be telemetered.

Definition at line 219 of file [default_cfe_es_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgdefs.h](#)

10.45 CFE_ES_SetMaxPRCountCmd Struct Reference

Set Maximum Processor Reset Count Command.

```
#include <default_cfe_es_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) [CommandHeader](#)
Command header.
- [CFE_ES_SetMaxPRCountCmd_Payload_t](#) [Payload](#)
Command payload.

10.45.1 Detailed Description

Set Maximum Processor Reset Count Command.

Definition at line 176 of file [default_cfe_es_msgstruct.h](#).

10.45.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_ES_SetMaxPRCountCmd::CommandHeader

Command header.

Definition at line 178 of file [default_cfe_es_msgstruct.h](#).

Payload

[CFE_ES_SetMaxPRCountCmd_Payload_t](#) CFE_ES_SetMaxPRCountCmd::Payload

Command payload.

Definition at line 179 of file [default_cfe_es_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgstruct.h](#)

10.46 CFE_ES_SetMaxPRCountCmd_Payload Struct Reference

Set Maximum Processor Reset Count Command Payload.

#include <[default_cfe_es_msgdefs.h](#)>

Data Fields

- [uint16 MaxPRCount](#)

New maximum number of Processor Resets before an automatic Power-On Reset is performed.

10.46.1 Detailed Description

Set Maximum Processor Reset Count Command Payload.

For command details, see [CFE_ES_SET_MAX_PR_COUNT_CC](#)

Definition at line 131 of file [default_cfe_es_msgdefs.h](#).

10.46.2 Field Documentation

MaxPRCount

[uint16](#) CFE_ES_SetMaxPRCountCmd_Payload::MaxPRCount

New maximum number of Processor Resets before an automatic Power-On Reset is performed.

Definition at line 133 of file [default_cfe_es_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgdefs.h](#)

10.47 CFE_ES_SetPerfFilterMaskCmd Struct Reference

Set Performance Analyzer Filter Mask Command.

#include <[default_cfe_es_msgstruct.h](#)>

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.
- [CFE_ES_SetPerfFilterMaskCmd_Payload_t](#) Payload
Command payload.

10.47.1 Detailed Description

Set Performance Analyzer Filter Mask Command.
Definition at line 212 of file [default_cfe_es_msgstruct.h](#).

10.47.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_ES_SetPerfFilterMaskCmd::CommandHeader
Command header.
Definition at line 214 of file [default_cfe_es_msgstruct.h](#).

Payload

[CFE_ES_SetPerfFilterMaskCmd_Payload_t](#) CFE_ES_SetPerfFilterMaskCmd::Payload
Command payload.
Definition at line 215 of file [default_cfe_es_msgstruct.h](#).
The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgstruct.h](#)

10.48 CFE_ES_SetPerfFilterMaskCmd_Payload Struct Reference

Set Performance Analyzer Filter Mask Command Payload.
`#include <default_cfe_es_msgdefs.h>`

Data Fields

- [uint32 FilterMaskNum](#)
Index into array of Filter Masks.
- [uint32 FilterMask](#)
New Mask for specified entry in array of Filter Masks.

10.48.1 Detailed Description

Set Performance Analyzer Filter Mask Command Payload.
For command details, see [CFE_ES_SET_PERF_FILTER_MASK_CC](#)
Definition at line 192 of file [default_cfe_es_msgdefs.h](#).

10.48.2 Field Documentation

FilterMask

[uint32](#) CFE_ES_SetPerfFilterMaskCmd_Payload::FilterMask
New Mask for specified entry in array of Filter Masks.
Definition at line 195 of file [default_cfe_es_msgdefs.h](#).

FilterMaskNum

[uint32](#) CFE_ES_SetPerfFilterMaskCmd_Payload::FilterMaskNum
Index into array of Filter Masks.
Definition at line 194 of file [default_cfe_es_msgdefs.h](#).
The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgdefs.h](#)

10.49 CFE_ES_SetPerfTriggerMaskCmd Struct Reference

Set Performance Analyzer Trigger Mask Command.

```
#include <default_cfe_es_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) *CommandHeader*
Command header.
- [CFE_ES_SetPerfTrigMaskCmd_Payload_t](#) *Payload*
Command payload.

10.49.1 Detailed Description

Set Performance Analyzer Trigger Mask Command.

Definition at line 221 of file [default_cfe_es_msgstruct.h](#).

10.49.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_ES_SetPerfTriggerMaskCmd::CommandHeader

Command header.

Definition at line 223 of file [default_cfe_es_msgstruct.h](#).

Payload

[CFE_ES_SetPerfTrigMaskCmd_Payload_t](#) CFE_ES_SetPerfTriggerMaskCmd::Payload

Command payload.

Definition at line 224 of file [default_cfe_es_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgstruct.h](#)

10.50 CFE_ES_SetPerfTrigMaskCmd_Payload Struct Reference

Set Performance Analyzer Trigger Mask Command Payload.

```
#include <default_cfe_es_msgdefs.h>
```

Data Fields

- [uint32](#) *TriggerMaskNum*
Index into array of Trigger Masks.
- [uint32](#) *TriggerMask*
New Mask for specified entry in array of Trigger Masks.

10.50.1 Detailed Description

Set Performance Analyzer Trigger Mask Command Payload.

For command details, see [CFE_ES_SET_PERF_TRIGGER_MASK_CC](#)

Definition at line 204 of file [default_cfe_es_msgdefs.h](#).

10.50.2 Field Documentation

TriggerMask

`uint32 CFE_ES_SetPerfTrigMaskCmd_Payload::TriggerMask`

New Mask for specified entry in array of Trigger Masks.

Definition at line 207 of file [default_cfe_es_msgdefs.h](#).

TriggerMaskNum

`uint32 CFE_ES_SetPerfTrigMaskCmd_Payload::TriggerMaskNum`

Index into array of Trigger Masks.

Definition at line 206 of file [default_cfe_es_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgdefs.h](#)

10.51 CFE_ES_StartApp Struct Reference

Start Application Command.

`#include <default_cfe_es_msgstruct.h>`

Data Fields

- [CFE_MSG_CommandHeader_t](#) `CommandHeader`
Command header.
- [CFE_ES_StartAppCmd_Payload_t](#) `Payload`
Command payload.

10.51.1 Detailed Description

Start Application Command.

Definition at line 135 of file [default_cfe_es_msgstruct.h](#).

10.51.2 Field Documentation

CommandHeader

`CFE_MSG_CommandHeader_t CFE_ES_StartApp::CommandHeader`

Command header.

Definition at line 137 of file [default_cfe_es_msgstruct.h](#).

Payload

`CFE_ES_StartAppCmd_Payload_t CFE_ES_StartApp::Payload`

Command payload.

Definition at line 138 of file [default_cfe_es_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgstruct.h](#)

10.52 CFE_ES_StartAppCmd_Payload Struct Reference

Start Application Command Payload.

`#include <default_cfe_es_msgdefs.h>`

Data Fields

- char [Application](#) [[CFE_MISSION_MAX_API_LEN](#)]
Name of Application to be started.
- char [AppEntryPoint](#) [[CFE_MISSION_MAX_API_LEN](#)]
Symbolic name of Application's entry point.
- char [AppFileName](#) [[CFE_MISSION_MAX_PATH_LEN](#)]
Full path and filename of Application's executable image.
- [CFE_ES_MemOffset_t](#) [StackSize](#)
Desired stack size for the new application.
- [CFE_ES_ExceptionAction_Enum_t](#) [ExceptionAction](#)
[CFE_ES_ExceptionAction_RESTART_APP](#)=On exception, restart Application, [CFE_ES_ExceptionAction_PROC_RESTART](#)=On exception, perform a Processor Reset
- [uint8](#) [Padding1](#)
- [CFE_ES_TaskPriority_Atom_t](#) [Priority](#)
Padding for alignment.
- [uint8](#) [Padding2](#) [4]

10.52.1 Detailed Description

Start Application Command Payload.

For command details, see [CFE_ES_START_APP_CC](#)

Definition at line 82 of file [default_cfe_es_msgdefs.h](#).

10.52.2 Field Documentation

AppEntryPoint

```
char CFE_ES_StartAppCmd_Payload::AppEntryPoint[CFE_MISSION_MAX_API_LEN]
```

Symbolic name of Application's entry point.

Definition at line 85 of file [default_cfe_es_msgdefs.h](#).

AppFileName

```
char CFE_ES_StartAppCmd_Payload::AppFileName[CFE_MISSION_MAX_PATH_LEN]
```

Full path and filename of Application's executable image.

Definition at line 87 of file [default_cfe_es_msgdefs.h](#).

Application

```
char CFE_ES_StartAppCmd_Payload::Application[CFE_MISSION_MAX_API_LEN]
```

Name of Application to be started.

Definition at line 84 of file [default_cfe_es_msgdefs.h](#).

ExceptionAction

```
CFE_ES_ExceptionAction_Enum_t CFE_ES_StartAppCmd_Payload::ExceptionAction
```

[CFE_ES_ExceptionAction_RESTART_APP](#)=On exception, restart Application, [CFE_ES_ExceptionAction_PROC_RESTART](#)=On exception, perform a Processor Reset

Definition at line 91 of file [default_cfe_es_msgdefs.h](#).

Padding1

```
uint8 CFE_ES_StartAppCmd_Payload::Padding1
```

Definition at line 96 of file [default_cfe_es_msgdefs.h](#).

Padding2

`uint8 CFE_ES_StartAppCmd_Payload::Padding2[4]`
Definition at line 98 of file [default_cfe_es_msgdefs.h](#).

Priority

`CFE_ES_TaskPriority_Atom_t CFE_ES_StartAppCmd_Payload::Priority`
Padding for alignment.
The new Applications runtime priority.
Definition at line 97 of file [default_cfe_es_msgdefs.h](#).

StackSize

`CFE_ES_MemOffset_t CFE_ES_StartAppCmd_Payload::StackSize`
Desired stack size for the new application.
Definition at line 90 of file [default_cfe_es_msgdefs.h](#).
The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgdefs.h](#)

10.53 CFE_ES_StartPerfCmd_Payload Struct Reference

Start Performance Analyzer Command Payload.
`#include <default_cfe_es_msgdefs.h>`

Data Fields

- [CFE_ES_PerfMode_Enum_t TriggerMode](#)
Desired trigger position (Start, Center, End). Values defined by [CFE_ES_PerfMode](#).

10.53.1 Detailed Description

Start Performance Analyzer Command Payload.
For command details, see [CFE_ES_START_PERF_DATA_CC](#)
Definition at line 168 of file [default_cfe_es_msgdefs.h](#).

10.53.2 Field Documentation

TriggerMode

`CFE_ES_PerfMode_Enum_t CFE_ES_StartPerfCmd_Payload::TriggerMode`
Desired trigger position (Start, Center, End). Values defined by [CFE_ES_PerfMode](#).
Definition at line 171 of file [default_cfe_es_msgdefs.h](#).
The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgdefs.h](#)

10.54 CFE_ES_StartPerfDataCmd Struct Reference

Start Performance Analyzer Command.
`#include <default_cfe_es_msgstruct.h>`

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.
- [CFE_ES_StartPerfCmd_Payload_t](#) Payload
Command payload.

10.54.1 Detailed Description

Start Performance Analyzer Command.

Definition at line 194 of file [default_cfe_es_msgstruct.h](#).

10.54.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_ES_StartPerfDataCmd::CommandHeader
Command header.

Definition at line 196 of file [default_cfe_es_msgstruct.h](#).

Payload

[CFE_ES_StartPerfCmd_Payload_t](#) CFE_ES_StartPerfDataCmd::Payload
Command payload.

Definition at line 197 of file [default_cfe_es_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgstruct.h](#)

10.55 CFE_ES_StopAppCmd Struct Reference

```
#include <default_cfe_es_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.
- [CFE_ES_AppNameCmd_Payload_t](#) Payload
Command payload.

10.55.1 Detailed Description

Definition at line 146 of file [default_cfe_es_msgstruct.h](#).

10.55.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_ES_StopAppCmd::CommandHeader
Command header.

Definition at line 148 of file [default_cfe_es_msgstruct.h](#).

Payload

[CFE_ES_AppNameCmd_Payload_t](#) CFE_ES_StopAppCmd::Payload

Command payload.

Definition at line 149 of file [default_cfe_es_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgstruct.h](#)

10.56 CFE_ES_StopPerfCmd_Payload Struct Reference

Stop Performance Analyzer Command Payload.

`#include <default_cfe_es_msgdefs.h>`

Data Fields

- [char DataFileName \[CFE_MISSION_MAX_PATH_LEN\]](#)
ASCII text string of full path and filename of file Performance Analyzer data is to be written.

10.56.1 Detailed Description

Stop Performance Analyzer Command Payload.

For command details, see [CFE_ES_STOP_PERF_DATA_CC](#)

Definition at line 180 of file [default_cfe_es_msgdefs.h](#).

10.56.2 Field Documentation

DataFileName

`char CFE_ES_StopPerfCmd_Payload::DataFileName[CFE_MISSION_MAX_PATH_LEN]`

ASCII text string of full path and filename of file Performance Analyzer data is to be written.

Definition at line 182 of file [default_cfe_es_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgdefs.h](#)

10.57 CFE_ES_StopPerfDataCmd Struct Reference

Stop Performance Analyzer Command.

`#include <default_cfe_es_msgstruct.h>`

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.
- [CFE_ES_StopPerfCmd_Payload_t](#) Payload
Command payload.

10.57.1 Detailed Description

Stop Performance Analyzer Command.

Definition at line 203 of file [default_cfe_es_msgstruct.h](#).

10.57.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_ES_StopPerfDataCmd::CommandHeader

Command header.

Definition at line 205 of file [default_cfe_es_msgstruct.h](#).

Payload

[CFE_ES_StopPerfCmd_Payload_t](#) CFE_ES_StopPerfDataCmd::Payload

Command payload.

Definition at line 206 of file [default_cfe_es_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgstruct.h](#)

10.58 CFE_ES_TaskInfo Struct Reference

Task Information.

#include <[default_cfe_es_extern_typedefs.h](#)>

Data Fields

- [CFE_ES_TaskId_t](#) TaskId
Task Id.
- [uint32](#) ExecutionCounter
Task Execution Counter.
- char [TaskName](#) [[CFE_MISSION_MAX_API_LEN](#)]
Task Name.
- [CFE_ES_AppId_t](#) AppId
Parent Application ID.
- char [AppName](#) [[CFE_MISSION_MAX_API_LEN](#)]
Parent Application Name.
- [CFE_ES_MemOffset_t](#) StackSize
- [CFE_ES_TaskPriority_Atom_t](#) Priority
- [uint8](#) Spare [2]

10.58.1 Detailed Description

Task Information.

Structure that is used to provide information about a task. It is primarily used for the Query All Tasks ([CFE_ES_QUERY_ALL_TASKS_CC](#)) command.

Note

There is not currently a telemetry message directly containing this data structure, but it does define the format of the data file generated by the Query All Tasks command. Therefore it should be considered part of the overall telemetry interface.

Definition at line 437 of file [default_cfe_es_extern_typedefs.h](#).

10.58.2 Field Documentation

AppId

`CFE_ES_AppId_t CFE_ES_TaskInfo::AppId`

Parent Application ID.

Definition at line 442 of file [default_cfe_es_extern_typedefs.h](#).

AppName

`char CFE_ES_TaskInfo::AppName[CFE_MISSION_MAX_API_LEN]`

Parent Application Name.

Definition at line 443 of file [default_cfe_es_extern_typedefs.h](#).

ExecutionCounter

`uint32 CFE_ES_TaskInfo::ExecutionCounter`

Task Execution Counter.

Definition at line 440 of file [default_cfe_es_extern_typedefs.h](#).

Priority

`CFE_ES_TaskPriority_Atom_t CFE_ES_TaskInfo::Priority`

Priority of task

Definition at line 445 of file [default_cfe_es_extern_typedefs.h](#).

Spare

`uint8 CFE_ES_TaskInfo::Spare[2]`

Spare bytes for alignment

Definition at line 446 of file [default_cfe_es_extern_typedefs.h](#).

StackSize

`CFE_ES_MemOffset_t CFE_ES_TaskInfo::StackSize`

Size of task stack

Definition at line 444 of file [default_cfe_es_extern_typedefs.h](#).

TaskId

`CFE_ES_TaskId_t CFE_ES_TaskInfo::TaskId`

Task Id.

Definition at line 439 of file [default_cfe_es_extern_typedefs.h](#).

TaskName

`char CFE_ES_TaskInfo::TaskName[CFE_MISSION_MAX_API_LEN]`

Task Name.

Definition at line 441 of file [default_cfe_es_extern_typedefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_extern_typedefs.h](#)

10.59 CFE_ES_WriteERLogCmd Struct Reference

```
#include <default_cfe_es_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) *CommandHeader*
Command header.
- [CFE_ES_FileNameCmd_Payload_t](#) *Payload*
Command payload.

10.59.1 Detailed Description

Definition at line 117 of file [default_cfe_es_msgstruct.h](#).

10.59.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_ES_WriteERLogCmd::CommandHeader
Command header.
Definition at line 119 of file [default_cfe_es_msgstruct.h](#).

Payload

[CFE_ES_FileNameCmd_Payload_t](#) CFE_ES_WriteERLogCmd::Payload
Command payload.
Definition at line 120 of file [default_cfe_es_msgstruct.h](#).
The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgstruct.h](#)

10.60 CFE_ES_WriteSysLogCmd Struct Reference

```
#include <default_cfe_es_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) *CommandHeader*
Command header.
- [CFE_ES_FileNameCmd_Payload_t](#) *Payload*
Command payload.

10.60.1 Detailed Description

Definition at line 111 of file [default_cfe_es_msgstruct.h](#).

10.60.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_ES_WriteSysLogCmd::CommandHeader
Command header.
Definition at line 113 of file [default_cfe_es_msgstruct.h](#).

Payload

[CFE_ES_FileNameCmd_Payload_t](#) CFE_ES_WriteSysLogCmd::Payload
Command payload.

Definition at line 114 of file [default_cfe_es_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/es/config/default_cfe_es_msgstruct.h](#)

10.61 CFE_EVS_AddEventFilterCmd Struct Reference

```
#include <default_cfe_evs_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.
- [CFE_EVS_AppNameEventIDMaskCmd_Payload_t](#) Payload
Command payload.

10.61.1 Detailed Description

Definition at line 195 of file [default_cfe_evs_msgstruct.h](#).

10.61.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_EVS_AddEventFilterCmd::CommandHeader
Command header.

Definition at line 197 of file [default_cfe_evs_msgstruct.h](#).

Payload

[CFE_EVS_AppNameEventIDMaskCmd_Payload_t](#) CFE_EVS_AddEventFilterCmd::Payload
Command payload.

Definition at line 198 of file [default_cfe_evs_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgstruct.h](#)

10.62 CFE_EVS_AppDataCmd_Payload Struct Reference

Write Event Services Application Information to File Command Payload.

```
#include <default_cfe_evs_msgdefs.h>
```

Data Fields

- char [AppDataFilename](#) [[CFE_MISSION_MAX_PATH_LEN](#)]
Filename where application data is to be written.

10.62.1 Detailed Description

Write Event Services Application Information to File Command Payload.

For command details, see [CFE_EVS_WRITE_APP_DATA_FILE_CC](#)

Definition at line 66 of file [default_cfe_evs_msgdefs.h](#).

10.62.2 Field Documentation

AppDataFilename

char CFE_EVS_AppDataCmd_Payload::AppDataFilename[CFE_MISSION_MAX_PATH_LEN]

Filename where application data is to be written.

Definition at line 68 of file [default_cfe_evs_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgdefs.h](#)

10.63 CFE_EVS_AppNameBitMaskCmd_Payload Struct Reference

Generic App Name and Bitmask Command Payload.

#include <default_cfe_evs_msgdefs.h>

Data Fields

- char [AppName](#) [CFE_MISSION_MAX_API_LEN]
Application name to use in the command.
- [uint8 BitMask](#)
BitMask to use in the command.
- [uint8 Spare](#)
Pad to even byte.

10.63.1 Detailed Description

Generic App Name and Bitmask Command Payload.

For command details, see [CFE_EVS_ENABLE_APP_EVENT_TYPE_CC](#) and/or [CFE_EVS_DISABLE_APP_EVENT_TYPE_CC](#)

Definition at line 138 of file [default_cfe_evs_msgdefs.h](#).

10.63.2 Field Documentation

AppName

char CFE_EVS_AppNameBitMaskCmd_Payload::AppName[CFE_MISSION_MAX_API_LEN]

Application name to use in the command.

Definition at line 140 of file [default_cfe_evs_msgdefs.h](#).

BitMask

[uint8](#) CFE_EVS_AppNameBitMaskCmd_Payload::BitMask

BitMask to use in the command.

Definition at line 141 of file [default_cfe_evs_msgdefs.h](#).

Spare

[uint8](#) CFE_EVS_AppNameBitMaskCmd_Payload::Spare

Pad to even byte.

Definition at line 142 of file [default_cfe_evs_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgdefs.h](#)

10.64 CFE_EVS_AppNameCmd_Payload Struct Reference

Generic App Name Command Payload.

```
#include <default_cfe_evs_msgdefs.h>
```

Data Fields

- char [AppName](#) [CFE_MISSION_MAX_API_LEN]
Application name to use in the command.

10.64.1 Detailed Description

Generic App Name Command Payload.

For command details, see [CFE_EVS_ENABLE_APP_EVENTS_CC](#), [CFE_EVS_DISABLE_APP_EVENTS_CC](#), [CFE_EVS_RESET_APP_COUNTER_CC](#) and/or [CFE_EVS_RESET_ALL_FILTERS_CC](#)

Definition at line 115 of file [default_cfe_evs_msgdefs.h](#).

10.64.2 Field Documentation

AppName

```
char CFE_EVS_AppNameCmd_Payload::AppName [CFE_MISSION_MAX_API_LEN]
```

Application name to use in the command.

Definition at line 117 of file [default_cfe_evs_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgdefs.h](#)

10.65 CFE_EVS_AppNameEventIDCmd_Payload Struct Reference

Generic App Name and Event ID Command Payload.

```
#include <default_cfe_evs_msgdefs.h>
```

Data Fields

- char [AppName](#) [CFE_MISSION_MAX_API_LEN]
Application name to use in the command.
- [uint16 EventID](#)
Event ID to use in the command.

10.65.1 Detailed Description

Generic App Name and Event ID Command Payload.

For command details, see [CFE_EVS_RESET_FILTER_CC](#) and [CFE_EVS_DELETE_EVENT_FILTER_CC](#)

Definition at line 126 of file [default_cfe_evs_msgdefs.h](#).

10.65.2 Field Documentation

AppName

```
char CFE_EVS_AppNameEventIDCmd_Payload::AppName [CFE_MISSION_MAX_API_LEN]
```

Application name to use in the command.

Definition at line 128 of file [default_cfe_evs_msgdefs.h](#).

EventID

`uint16 CFE_EVS_AppNameEventIDMaskCmd_Payload::EventID`

Event ID to use in the command.

Definition at line 129 of file [default_cfe_evs_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgdefs.h](#)

10.66 CFE_EVS_AppNameEventIDMaskCmd_Payload Struct Reference

Generic App Name, Event ID, Mask Command Payload.

```
#include <default_cfe_evs_msgdefs.h>
```

Data Fields

- `char` [AppName](#) [[CFE_MISSION_MAX_API_LEN](#)]
Application name to use in the command.
- `uint16` [EventID](#)
Event ID to use in the command.
- `uint16` [Mask](#)
Mask to use in the command.

10.66.1 Detailed Description

Generic App Name, Event ID, Mask Command Payload.

For command details, see [CFE_EVS_SET_FILTER_CC](#), [CFE_EVS_ADD_EVENT_FILTER_CC](#) and/or [CFE_EVS_DELETE_EVENT_FILTER_CC](#).

Definition at line 152 of file [default_cfe_evs_msgdefs.h](#).

10.66.2 Field Documentation**AppName**

`char CFE_EVS_AppNameEventIDMaskCmd_Payload::AppName` [[CFE_MISSION_MAX_API_LEN](#)]

Application name to use in the command.

Definition at line 154 of file [default_cfe_evs_msgdefs.h](#).

EventID

`uint16 CFE_EVS_AppNameEventIDMaskCmd_Payload::EventID`

Event ID to use in the command.

Definition at line 155 of file [default_cfe_evs_msgdefs.h](#).

Mask

`uint16 CFE_EVS_AppNameEventIDMaskCmd_Payload::Mask`

Mask to use in the command.

Definition at line 156 of file [default_cfe_evs_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgdefs.h](#)

10.67 CFE_EVS_AppTimData Struct Reference

```
#include <default_cfe_evs_msgdefs.h>
```


Data Fields

- [CFE_ES_AppId_t AppID](#)
Numerical application identifier.
- [uint16 AppMessageSentCounter](#)
Application message sent counter.
- [uint8 AppEnableStatus](#)
Application event service enable status.
- [uint8 AppMessageSquelchedCounter](#)
Number of events squelched.

10.67.1 Detailed Description

Definition at line 165 of file [default_cfe_evs_msgdefs.h](#).

10.67.2 Field Documentation

AppEnableStatus

[uint8](#) CFE_EVS_AppTlmData::AppEnableStatus

Application event service enable status.

Telemetry Mnemonic(s) \$sc_\$cpu_EVS_APP[CFE_PLATFORM_ES_MAX_APPLICATIONS].APPENASTAT

Definition at line 171 of file [default_cfe_evs_msgdefs.h](#).

AppID

[CFE_ES_AppId_t](#) CFE_EVS_AppTlmData::AppID

Numerical application identifier.

Telemetry Mnemonic(s) \$sc_\$cpu_EVS_APP[CFE_PLATFORM_ES_MAX_APPLICATIONS].APPID

Definition at line 167 of file [default_cfe_evs_msgdefs.h](#).

AppMessageSentCounter

[uint16](#) CFE_EVS_AppTlmData::AppMessageSentCounter

Application message sent counter.

Telemetry Mnemonic(s) \$sc_\$cpu_EVS_APP[CFE_PLATFORM_ES_MAX_APPLICATIONS].APPMSGSENTC

Definition at line 169 of file [default_cfe_evs_msgdefs.h](#).

AppMessageSquelchedCounter

[uint8](#) CFE_EVS_AppTlmData::AppMessageSquelchedCounter

Number of events squelched.

Telemetry Mnemonic(s) \$sc_\$cpu_EVS_APP[CFE_PLATFORM_ES_MAX_APPLICATIONS].SQUELCHEDC

Definition at line 173 of file [default_cfe_evs_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgdefs.h](#)

10.68 CFE_EVS_BinFilter Struct Reference

Event message filter definition structure.

```
#include <cf_evs_api_typedefs.h>
```

Data Fields

- [uint16 EventID](#)
Numerical event identifier.
- [uint16 Mask](#)
Binary filter mask value.

10.68.1 Detailed Description

Event message filter definition structure.

Definition at line 60 of file [cf_evs_api_typedefs.h](#).

10.68.2 Field Documentation

EventID

```
uint16 CFE_EVS_BinFilter::EventID
```

Numerical event identifier.

Definition at line 62 of file [cf_evs_api_typedefs.h](#).

Mask

```
uint16 CFE_EVS_BinFilter::Mask
```

Binary filter mask value.

Definition at line 63 of file [cf_evs_api_typedefs.h](#).

The documentation for this struct was generated from the following file:

- [cf/modules/core_api/fsw/inc/cf_evs_api_typedefs.h](#)

10.69 CFE_EVS_BitMaskCmd_Payload Struct Reference

Generic Bitmask Command Payload.

```
#include <default_cfe_evs_msgdefs.h>
```

Data Fields

- [uint8 BitMask](#)
BitMask to use in the command.
- [uint8 Spare](#)
Pad to even byte.

10.69.1 Detailed Description

Generic Bitmask Command Payload.

For command details, see [CFE_EVS_ENABLE_EVENT_TYPE_CC](#), [CFE_EVS_DISABLE_EVENT_TYPE_CC](#), [CFE_EVS_ENABLE_PORTS_CC](#) and/or [CFE_EVS_DISABLE_PORTS_CC](#)

Definition at line 102 of file [default_cfe_evs_msgdefs.h](#).

10.69.2 Field Documentation

BitMask

`uint8 CFE_EVS_BitMaskCmd_Payload::BitMask`

BitMask to use in the command.

Definition at line 104 of file [default_cfe_evs_msgdefs.h](#).

Spare

`uint8 CFE_EVS_BitMaskCmd_Payload::Spare`

Pad to even byte.

Definition at line 105 of file [default_cfe_evs_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgdefs.h](#)

10.70 CFE_EVS_ClearLogCmd Struct Reference

```
#include <default_cfe_evs_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.

10.70.1 Detailed Description

Definition at line 52 of file [default_cfe_evs_msgstruct.h](#).

10.70.2 Field Documentation

CommandHeader

`CFE_MSG_CommandHeader_t CFE_EVS_ClearLogCmd::CommandHeader`

Command header.

Definition at line 54 of file [default_cfe_evs_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgstruct.h](#)

10.71 CFE_EVS_DeleteEventFilterCmd Struct Reference

```
#include <default_cfe_evs_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.
- [CFE_EVS_AppNameEventIDCmd_Payload_t](#) Payload
Command payload.

10.71.1 Detailed Description

Definition at line 167 of file [default_cfe_evs_msgstruct.h](#).

10.71.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_EVS_DeleteEventFilterCmd::CommandHeader

Command header.

Definition at line 169 of file [default_cfe_evs_msgstruct.h](#).

Payload

[CFE_EVS_AppNameEventIDCmd_Payload_t](#) CFE_EVS_DeleteEventFilterCmd::Payload

Command payload.

Definition at line 170 of file [default_cfe_evs_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgstruct.h](#)

10.72 CFE_EVS_DisableAppEventsCmd Struct Reference

```
#include <default_cfe_evs_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.
- [CFE_EVS_AppNameCmd_Payload_t](#) Payload
Command payload.

10.72.1 Detailed Description

Definition at line 138 of file [default_cfe_evs_msgstruct.h](#).

10.72.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_EVS_DisableAppEventsCmd::CommandHeader

Command header.

Definition at line 140 of file [default_cfe_evs_msgstruct.h](#).

Payload

[CFE_EVS_AppNameCmd_Payload_t](#) CFE_EVS_DisableAppEventsCmd::Payload

Command payload.

Definition at line 141 of file [default_cfe_evs_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgstruct.h](#)

10.73 CFE_EVS_DisableAppEventTypeCmd Struct Reference

```
#include <default_cfe_evs_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) `CommandHeader`
Command header.
- [CFE_EVS_AppNameBitMaskCmd_Payload_t](#) `Payload`
Command payload.

10.73.1 Detailed Description

Definition at line 184 of file [default_cfe_evs_msgstruct.h](#).

10.73.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) `CFE_EVS_DisableAppEventTypeCmd::CommandHeader`
Command header.

Definition at line 186 of file [default_cfe_evs_msgstruct.h](#).

Payload

[CFE_EVS_AppNameBitMaskCmd_Payload_t](#) `CFE_EVS_DisableAppEventTypeCmd::Payload`
Command payload.

Definition at line 187 of file [default_cfe_evs_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgstruct.h](#)

10.74 CFE_EVS_DisableEventTypeCmd Struct Reference

```
#include <default_cfe_evs_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) `CommandHeader`
Command header.
- [CFE_EVS_BitMaskCmd_Payload_t](#) `Payload`
Command payload.

10.74.1 Detailed Description

Definition at line 121 of file [default_cfe_evs_msgstruct.h](#).

10.74.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) `CFE_EVS_DisableEventTypeCmd::CommandHeader`
Command header.

Definition at line 123 of file [default_cfe_evs_msgstruct.h](#).

Payload

[CFE_EVS_BitMaskCmd_Payload_t](#) CFE_EVS_DisableEventTypeCmd::Payload

Command payload.

Definition at line 124 of file [default_cfe_evs_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgstruct.h](#)

10.75 CFE_EVS_DisablePortsCmd Struct Reference

```
#include <default_cfe_evs_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.
- [CFE_EVS_BitMaskCmd_Payload_t](#) Payload
Command payload.

10.75.1 Detailed Description

Definition at line 109 of file [default_cfe_evs_msgstruct.h](#).

10.75.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_EVS_DisablePortsCmd::CommandHeader

Command header.

Definition at line 111 of file [default_cfe_evs_msgstruct.h](#).

Payload

[CFE_EVS_BitMaskCmd_Payload_t](#) CFE_EVS_DisablePortsCmd::Payload

Command payload.

Definition at line 112 of file [default_cfe_evs_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgstruct.h](#)

10.76 CFE_EVS_EnableAppEventsCmd Struct Reference

```
#include <default_cfe_evs_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.
- [CFE_EVS_AppNameCmd_Payload_t](#) Payload
Command payload.

10.76.1 Detailed Description

Definition at line 132 of file [default_cfe_evs_msgstruct.h](#).

10.76.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_EVS_EnableAppEventsCmd::CommandHeader

Command header.

Definition at line 134 of file [default_cfe_evs_msgstruct.h](#).

Payload

[CFE_EVS_AppNameCmd_Payload_t](#) CFE_EVS_EnableAppEventsCmd::Payload

Command payload.

Definition at line 135 of file [default_cfe_evs_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgstruct.h](#)

10.77 CFE_EVS_EnableAppEventTypeCmd Struct Reference

```
#include <default_cfe_evs_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.
- [CFE_EVS_AppNameBitMaskCmd_Payload_t](#) Payload
Command payload.

10.77.1 Detailed Description

Definition at line 178 of file [default_cfe_evs_msgstruct.h](#).

10.77.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_EVS_EnableAppEventTypeCmd::CommandHeader

Command header.

Definition at line 180 of file [default_cfe_evs_msgstruct.h](#).

Payload

[CFE_EVS_AppNameBitMaskCmd_Payload_t](#) CFE_EVS_EnableAppEventTypeCmd::Payload

Command payload.

Definition at line 181 of file [default_cfe_evs_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgstruct.h](#)

10.78 CFE_EVS_EnableEventTypeCmd Struct Reference

```
#include <default_cfe_evs_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) `CommandHeader`
Command header.
- [CFE_EVS_BitMaskCmd_Payload_t](#) `Payload`
Command payload.

10.78.1 Detailed Description

Definition at line 115 of file [default_cfe_evs_msgstruct.h](#).

10.78.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) `CFE_EVS_EnableEventTypeCmd::CommandHeader`
Command header.

Definition at line 117 of file [default_cfe_evs_msgstruct.h](#).

Payload

[CFE_EVS_BitMaskCmd_Payload_t](#) `CFE_EVS_EnableEventTypeCmd::Payload`
Command payload.

Definition at line 118 of file [default_cfe_evs_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgstruct.h](#)

10.79 CFE_EVS_EnablePortsCmd Struct Reference

```
#include <default_cfe_evs_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) `CommandHeader`
Command header.
- [CFE_EVS_BitMaskCmd_Payload_t](#) `Payload`
Command payload.

10.79.1 Detailed Description

Definition at line 103 of file [default_cfe_evs_msgstruct.h](#).

10.79.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) `CFE_EVS_EnablePortsCmd::CommandHeader`
Command header.

Definition at line 105 of file [default_cfe_evs_msgstruct.h](#).

Payload

[CFE_EVS_BitMaskCmd_Payload_t](#) CFE_EVS_EnablePortsCmd::Payload
Command payload.

Definition at line 106 of file [default_cfe_evs_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgstruct.h](#)

10.80 CFE_EVS_HousekeepingTlm Struct Reference

```
#include <default_cfe_evs_msgstruct.h>
```

Data Fields

- [CFE_MSG_TelemetryHeader_t](#) TelemetryHeader
Telemetry header.
- [CFE_EVS_HousekeepingTlm_Payload_t](#) Payload
Telemetry payload.

10.80.1 Detailed Description

Definition at line 213 of file [default_cfe_evs_msgstruct.h](#).

10.80.2 Field Documentation

Payload

[CFE_EVS_HousekeepingTlm_Payload_t](#) CFE_EVS_HousekeepingTlm::Payload
Telemetry payload.

Definition at line 216 of file [default_cfe_evs_msgstruct.h](#).

TelemetryHeader

[CFE_MSG_TelemetryHeader_t](#) CFE_EVS_HousekeepingTlm::TelemetryHeader
Telemetry header.

Definition at line 215 of file [default_cfe_evs_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgstruct.h](#)

10.81 CFE_EVS_HousekeepingTlm_Payload Struct Reference

```
#include <default_cfe_evs_msgdefs.h>
```

Data Fields

- [uint8 CommandCounter](#)
EVS Command Counter.
- [uint8 CommandErrorCounter](#)
EVS Command Error Counter.
- [uint8 MessageFormatMode](#)
Event message format mode (short/long)
- [uint8 MessageTruncCounter](#)
Event message truncation counter.

- [uint8 UnregisteredAppCounter](#)
Unregistered application message send counter.
- [uint8 OutputPort](#)
Output port mask.
- [uint8 LogFullFlag](#)
Local event log full flag.
- [uint8 LogMode](#)
Local event logging mode (overwrite/discard)
- [uint16 MessageSendCounter](#)
Event message send counter.
- [uint16 LogOverflowCounter](#)
Local event log overflow counter.
- [uint8 LogEnabled](#)
Current event log enable/disable state.
- [uint8 Padding](#) [3]
- [CFE_EVS_AppTlmData_t AppData](#) [CFE_MISSION_ES_MAX_APPLICATIONS]
Padding for 32 bit boundary.

10.81.1 Detailed Description

Name Event Services Housekeeping Telemetry Packet

Definition at line 180 of file [default_cfe_evs_msgdefs.h](#).

10.81.2 Field Documentation

AppData

[CFE_EVS_AppTlmData_t](#) CFE_EVS_HousekeepingTlm_Payload::AppData[CFE_MISSION_ES_MAX_APPLICATIONS]
Padding for 32 bit boundary.

Telemetry Mnemonic(s) \$sc_\$cpu_EVS_APP[CFE_PLATFORM_ES_MAX_APPLICATIONS]

Array of registered application table data

Definition at line 209 of file [default_cfe_evs_msgdefs.h](#).

CommandCounter

[uint8](#) CFE_EVS_HousekeepingTlm_Payload::CommandCounter
EVS Command Counter.

Telemetry Mnemonic(s) \$sc_\$cpu_EVS_CMDPC

Definition at line 182 of file [default_cfe_evs_msgdefs.h](#).

CommandErrorCounter

[uint8](#) CFE_EVS_HousekeepingTlm_Payload::CommandErrorCounter
EVS Command Error Counter.

Telemetry Mnemonic(s) \$sc_\$cpu_EVS_CMDEC

Definition at line 184 of file [default_cfe_evs_msgdefs.h](#).

LogEnabled

`uint8 CFE_EVS_HousekeepingTlm_Payload::LogEnabled`
Current event log enable/disable state.

Telemetry Mnemonic(s) `$sc_$cpu_EVS_LOGENABLED`

Definition at line 205 of file `default_cfe_evs_msgdefs.h`.

LogFullFlag

`uint8 CFE_EVS_HousekeepingTlm_Payload::LogFullFlag`
Local event log full flag.

Telemetry Mnemonic(s) `$sc_$cpu_EVS_LOGFULL`

Definition at line 195 of file `default_cfe_evs_msgdefs.h`.

LogMode

`uint8 CFE_EVS_HousekeepingTlm_Payload::LogMode`
Local event logging mode (overwrite/discard)

Telemetry Mnemonic(s) `$sc_$cpu_EVS_LOGMODE`

Definition at line 197 of file `default_cfe_evs_msgdefs.h`.

LogOverflowCounter

`uint16 CFE_EVS_HousekeepingTlm_Payload::LogOverflowCounter`
Local event log overflow counter.

Telemetry Mnemonic(s) `$sc_$cpu_EVS_LOGOVERFLOWC`

Definition at line 202 of file `default_cfe_evs_msgdefs.h`.

MessageFormatMode

`uint8 CFE_EVS_HousekeepingTlm_Payload::MessageFormatMode`
Event message format mode (short/long)

Telemetry Mnemonic(s) `$sc_$cpu_EVS_MSGFMTMODE`

Definition at line 186 of file `default_cfe_evs_msgdefs.h`.

MessageSendCounter

`uint16 CFE_EVS_HousekeepingTlm_Payload::MessageSendCounter`
Event message send counter.

Telemetry Mnemonic(s) `$sc_$cpu_EVS_MSGSENTC`

Definition at line 200 of file `default_cfe_evs_msgdefs.h`.

MessageTruncCounter

`uint8 CFE_EVS_HousekeepingTlm_Payload::MessageTruncCounter`
Event message truncation counter.

Telemetry Mnemonic(s) `$sc_$cpu_EVS_MSGTRUNC`

Definition at line 188 of file `default_cfe_evs_msgdefs.h`.

OutputPort

`uint8 CFE_EVS_HousekeepingTlm_Payload::OutputPort`
Output port mask.

Telemetry Mnemonic(s) `$sc_$cpu_EVS_OUTPUTPORT`

Definition at line 193 of file [default_cfe_evs_msgdefs.h](#).

Padding

`uint8 CFE_EVS_HousekeepingTlm_Payload::Padding[3]`
Definition at line 207 of file [default_cfe_evs_msgdefs.h](#).

UnregisteredAppCounter

`uint8 CFE_EVS_HousekeepingTlm_Payload::UnregisteredAppCounter`
Unregistered application message send counter.

Telemetry Mnemonic(s) `$sc_$cpu_EVS_UNREGAPPC`

Definition at line 191 of file [default_cfe_evs_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgdefs.h](#)

10.82 CFE_EVS_LogFileCmd_Payload Struct Reference

Write Event Log to File Command Payload.

`#include <default_cfe_evs_msgdefs.h>`

Data Fields

- `char LogFilename [CFE_MISSION_MAX_PATH_LEN]`
Filename where log data is to be written.

10.82.1 Detailed Description

Write Event Log to File Command Payload.

For command details, see [CFE_EVS_WRITE_LOG_DATA_FILE_CC](#)

Definition at line 55 of file [default_cfe_evs_msgdefs.h](#).

10.82.2 Field Documentation

LogFilename

`char CFE_EVS_LogFileCmd_Payload::LogFilename [CFE_MISSION_MAX_PATH_LEN]`

Filename where log data is to be written.

Definition at line 57 of file [default_cfe_evs_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgdefs.h](#)

10.83 CFE_EVS_LongEventTlm Struct Reference

`#include <default_cfe_evs_msgstruct.h>`

Data Fields

- [CFE_MSG_TelemetryHeader_t](#) TelemetryHeader
Telemetry header.
- [CFE_EVS_LongEventTlm_Payload_t](#) Payload
Telemetry payload.

10.83.1 Detailed Description

Definition at line 219 of file [default_cfe_evs_msgstruct.h](#).

10.83.2 Field Documentation

Payload

[CFE_EVS_LongEventTlm_Payload_t](#) CFE_EVS_LongEventTlm::Payload
Telemetry payload.
Definition at line 222 of file [default_cfe_evs_msgstruct.h](#).

TelemetryHeader

[CFE_MSG_TelemetryHeader_t](#) CFE_EVS_LongEventTlm::TelemetryHeader
Telemetry header.
Definition at line 221 of file [default_cfe_evs_msgstruct.h](#).
The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgstruct.h](#)

10.84 CFE_EVS_LongEventTlm_Payload Struct Reference

```
#include <default_cfe_evs_msgdefs.h>
```

Data Fields

- [CFE_EVS_PacketID_t](#) PacketID
Event packet information.
- char [Message](#) [[CFE_MISSION_EVS_MAX_MESSAGE_LENGTH](#)]
Event message string.
- [uint8](#) [Padding](#) [2]

10.84.1 Detailed Description

Name Event Message Telemetry Packet (Long format)

Definition at line 232 of file [default_cfe_evs_msgdefs.h](#).

10.84.2 Field Documentation

Message

char CFE_EVS_LongEventTlm_Payload::Message [[CFE_MISSION_EVS_MAX_MESSAGE_LENGTH](#)]
Event message string.

Telemetry Mnemonic(s) \$sc_\$cpu_EVS_EVENT[CFE_MISSION_EVS_MAX_MESSAGE_LENGTH]

Definition at line 235 of file [default_cfe_evs_msgdefs.h](#).

PacketID

[CFE_EVS_PacketID_t](#) CFE_EVS_LongEventTlm_Payload::PacketID
 Event packet information.
 Definition at line 234 of file [default_cfe_evs_msgdefs.h](#).

Padding

[uint8](#) CFE_EVS_LongEventTlm_Payload::Padding[2]
 Definition at line 237 of file [default_cfe_evs_msgdefs.h](#).
 The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgdefs.h](#)

10.85 CFE_EVS_NoopCmd Struct Reference

```
#include <default_cfe_evs_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.

10.85.1 Detailed Description

Definition at line 42 of file [default_cfe_evs_msgstruct.h](#).

10.85.2 Field Documentation**CommandHeader**

[CFE_MSG_CommandHeader_t](#) CFE_EVS_NoopCmd::CommandHeader
 Command header.
 Definition at line 44 of file [default_cfe_evs_msgstruct.h](#).
 The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgstruct.h](#)

10.86 CFE_EVS_PacketID Struct Reference

```
#include <default_cfe_evs_msgdefs.h>
```

Data Fields

- char [AppName](#) [CFE_MISSION_MAX_API_LEN]
Application name.
- [uint16](#) EventID
Numerical event identifier.
- [CFE_EVS_EventType_Enum_t](#) EventType
Numerical event type identifier.
- [uint32](#) SpacecraftID
Spacecraft identifier.
- [uint32](#) ProcessorID
Numerical processor identifier.

10.86.1 Detailed Description

Telemetry packet structures

Definition at line 215 of file [default_cfe_evs_msgdefs.h](#).

10.86.2 Field Documentation

AppName

```
char CFE_EVS_PacketID::AppName[CFE_MISSION_MAX_API_LEN]
```

Application name.

Telemetry Mnemonic(s) \$sc_\$cpu_EVS_APPNAME[OS_MAX_API_NAME]

Definition at line 217 of file [default_cfe_evs_msgdefs.h](#).

EventID

```
uint16 CFE_EVS_PacketID::EventID
```

Numerical event identifier.

Telemetry Mnemonic(s) \$sc_\$cpu_EVS_EVENTID

Definition at line 219 of file [default_cfe_evs_msgdefs.h](#).

EventType

```
CFE_EVS_EventType_Enum_t CFE_EVS_PacketID::EventType
```

Numerical event type identifier.

Telemetry Mnemonic(s) \$sc_\$cpu_EVS_EVENTTYPE

Definition at line 221 of file [default_cfe_evs_msgdefs.h](#).

ProcessorID

```
uint32 CFE_EVS_PacketID::ProcessorID
```

Numerical processor identifier.

Telemetry Mnemonic(s) \$sc_\$cpu_EVS_PROCESSORID

Definition at line 225 of file [default_cfe_evs_msgdefs.h](#).

SpacecraftID

```
uint32 CFE_EVS_PacketID::SpacecraftID
```

Spacecraft identifier.

Telemetry Mnemonic(s) \$sc_\$cpu_EVS_SCID

Definition at line 223 of file [default_cfe_evs_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgdefs.h](#)

10.87 CFE_EVS_ResetAllFiltersCmd Struct Reference

```
#include <default_cfe_evs_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) *CommandHeader*
Command header.
- [CFE_EVS_AppNameCmd_Payload_t](#) *Payload*
Command payload.

10.87.1 Detailed Description

Definition at line 150 of file [default_cfe_evs_msgstruct.h](#).

10.87.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) *CFE_EVS_ResetAllFiltersCmd::CommandHeader*
Command header.

Definition at line 152 of file [default_cfe_evs_msgstruct.h](#).

Payload

[CFE_EVS_AppNameCmd_Payload_t](#) *CFE_EVS_ResetAllFiltersCmd::Payload*
Command payload.

Definition at line 153 of file [default_cfe_evs_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgstruct.h](#)

10.88 CFE_EVS_ResetAppCounterCmd Struct Reference

```
#include <default_cfe_evs_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) *CommandHeader*
Command header.
- [CFE_EVS_AppNameCmd_Payload_t](#) *Payload*
Command payload.

10.88.1 Detailed Description

Definition at line 144 of file [default_cfe_evs_msgstruct.h](#).

10.88.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) *CFE_EVS_ResetAppCounterCmd::CommandHeader*
Command header.

Definition at line 146 of file [default_cfe_evs_msgstruct.h](#).

Payload

[CFE_EVS_AppNameCmd_Payload_t](#) CFE_EVS_ResetAppCounterCmd::Payload
Command payload.

Definition at line 147 of file [default_cfe_evs_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgstruct.h](#)

10.89 CFE_EVS_ResetCountersCmd Struct Reference

```
#include <default_cfe_evs_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.

10.89.1 Detailed Description

Definition at line 47 of file [default_cfe_evs_msgstruct.h](#).

10.89.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_EVS_ResetCountersCmd::CommandHeader
Command header.

Definition at line 49 of file [default_cfe_evs_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgstruct.h](#)

10.90 CFE_EVS_ResetFilterCmd Struct Reference

```
#include <default_cfe_evs_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.
- [CFE_EVS_AppNameEventIDCmd_Payload_t](#) Payload
Command payload.

10.90.1 Detailed Description

Definition at line 161 of file [default_cfe_evs_msgstruct.h](#).

10.90.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_EVS_ResetFilterCmd::CommandHeader
Command header.

Definition at line 163 of file [default_cfe_evs_msgstruct.h](#).

Payload

[CFE_EVS_AppNameEventIDCmd_Payload_t](#) CFE_EVS_ResetFilterCmd::Payload

Command payload.

Definition at line 164 of file [default_cfe_evs_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgstruct.h](#)

10.91 CFE_EVS_SendHkCmd Struct Reference

```
#include <default_cfe_evs_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.

10.91.1 Detailed Description

Definition at line 57 of file [default_cfe_evs_msgstruct.h](#).

10.91.2 Field Documentation**CommandHeader**

[CFE_MSG_CommandHeader_t](#) CFE_EVS_SendHkCmd::CommandHeader

Command header.

Definition at line 59 of file [default_cfe_evs_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgstruct.h](#)

10.92 CFE_EVS_SetEventFormatCode_Payload Struct Reference

Set Event Format Mode Command Payload.

```
#include <default_cfe_evs_msgdefs.h>
```

Data Fields

- [CFE_EVS_MsgFormat_Enum_t](#) MsgFormat
Mode to use in the command.
- [uint8](#) Spare
Pad to even byte.

10.92.1 Detailed Description

Set Event Format Mode Command Payload.

For command details, see [CFE_EVS_SET_EVENT_FORMAT_MODE_CC](#)

Definition at line 89 of file [default_cfe_evs_msgdefs.h](#).

10.92.2 Field Documentation**MsgFormat**

[CFE_EVS_MsgFormat_Enum_t](#) CFE_EVS_SetEventFormatCode_Payload::MsgFormat

Mode to use in the command.

Definition at line 91 of file [default_cfe_evs_msgdefs.h](#).

Spare

`uint8 CFE_EVS_SetEventFormatCode_Payload::Spare`

Pad to even byte.

Definition at line 92 of file [default_cfe_evs_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgdefs.h](#)

10.93 CFE_EVS_SetEventFormatModeCmd Struct Reference

Set Event Format Mode Command.

`#include <default_cfe_evs_msgstruct.h>`

Data Fields

- [CFE_MSG_CommandHeader_t](#) `CommandHeader`
Command header.
- [CFE_EVS_SetEventFormatMode_Payload_t](#) `Payload`
Command payload.

10.93.1 Detailed Description

Set Event Format Mode Command.

Definition at line 92 of file [default_cfe_evs_msgstruct.h](#).

10.93.2 Field Documentation

CommandHeader

`CFE_MSG_CommandHeader_t CFE_EVS_SetEventFormatModeCmd::CommandHeader`

Command header.

Definition at line 94 of file [default_cfe_evs_msgstruct.h](#).

Payload

`CFE_EVS_SetEventFormatMode_Payload_t CFE_EVS_SetEventFormatModeCmd::Payload`

Command payload.

Definition at line 95 of file [default_cfe_evs_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgstruct.h](#)

10.94 CFE_EVS_SetFilterCmd Struct Reference

`#include <default_cfe_evs_msgstruct.h>`

Data Fields

- [CFE_MSG_CommandHeader_t](#) `CommandHeader`
Command header.
- [CFE_EVS_AppNameEventIDMaskCmd_Payload_t](#) `Payload`
Command payload.

10.94.1 Detailed Description

Definition at line 201 of file [default_cfe_evs_msgstruct.h](#).

10.94.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_EVS_SetFilterCmd::CommandHeader

Command header.

Definition at line 203 of file [default_cfe_evs_msgstruct.h](#).

Payload

[CFE_EVS_AppNameEventIDMaskCmd_Payload_t](#) CFE_EVS_SetFilterCmd::Payload

Command payload.

Definition at line 204 of file [default_cfe_evs_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgstruct.h](#)

10.95 CFE_EVS_SetLogMode_Payload Struct Reference

Set Log Mode Command Payload.

#include <[default_cfe_evs_msgdefs.h](#)>

Data Fields

- [CFE_EVS_LogMode_Enum_t](#) LogMode
Mode to use in the command.
- [uint8](#) Spare
Pad to even byte.

10.95.1 Detailed Description

Set Log Mode Command Payload.

For command details, see [CFE_EVS_SET_LOG_MODE_CC](#)

Definition at line 77 of file [default_cfe_evs_msgdefs.h](#).

10.95.2 Field Documentation

LogMode

[CFE_EVS_LogMode_Enum_t](#) CFE_EVS_SetLogMode_Payload::LogMode

Mode to use in the command.

Definition at line 79 of file [default_cfe_evs_msgdefs.h](#).

Spare

[uint8](#) CFE_EVS_SetLogMode_Payload::Spare

Pad to even byte.

Definition at line 80 of file [default_cfe_evs_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgdefs.h](#)

10.96 CFE_EVS_SetLogModeCmd Struct Reference

Set Log Mode Command.

```
#include <default_cfe_evs_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) *CommandHeader*
Command header.
- [CFE_EVS_SetLogMode_Payload_t](#) *Payload*
Command payload.

10.96.1 Detailed Description

Set Log Mode Command.

Definition at line 83 of file [default_cfe_evs_msgstruct.h](#).

10.96.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_EVS_SetLogModeCmd::CommandHeader

Command header.

Definition at line 85 of file [default_cfe_evs_msgstruct.h](#).

Payload

[CFE_EVS_SetLogMode_Payload_t](#) CFE_EVS_SetLogModeCmd::Payload

Command payload.

Definition at line 86 of file [default_cfe_evs_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- cfe/modules/evs/config/[default_cfe_evs_msgstruct.h](#)

10.97 CFE_EVS_ShortEventTlm Struct Reference

```
#include <default_cfe_evs_msgstruct.h>
```

Data Fields

- [CFE_MSG_TelemetryHeader_t](#) *TelemetryHeader*
Telemetry header.
- [CFE_EVS_ShortEventTlm_Payload_t](#) *Payload*
Telemetry payload.

10.97.1 Detailed Description

Definition at line 225 of file [default_cfe_evs_msgstruct.h](#).

10.97.2 Field Documentation

Payload

[CFE_EVS_ShortEventTlm_Payload_t](#) CFE_EVS_ShortEventTlm::Payload

Telemetry payload.

Definition at line 228 of file [default_cfe_evs_msgstruct.h](#).

TelemetryHeader

[CFE_MSG_TelemetryHeader_t](#) CFE_EVS_ShortEventTlm::TelemetryHeader

Telemetry header.

Definition at line 227 of file [default_cfe_evs_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgstruct.h](#)

10.98 CFE_EVS_ShortEventTlm_Payload Struct Reference

```
#include <default_cfe_evs_msgdefs.h>
```

Data Fields

- [CFE_EVS_PacketID_t](#) PacketID
Event packet information.

10.98.1 Detailed Description

Name Event Message Telemetry Packet (Short format)

Definition at line 243 of file [default_cfe_evs_msgdefs.h](#).

10.98.2 Field Documentation

PacketID

[CFE_EVS_PacketID_t](#) CFE_EVS_ShortEventTlm_Payload::PacketID

Event packet information.

Definition at line 245 of file [default_cfe_evs_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgdefs.h](#)

10.99 CFE_EVS_WriteAppDataFileCmd Struct Reference

Write Event Services Application Information to File Command.

```
#include <default_cfe_evs_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.
- [CFE_EVS_AppDataCmd_Payload_t](#) Payload
Command payload.

10.99.1 Detailed Description

Write Event Services Application Information to File Command.

Definition at line 74 of file [default_cfe_evs_msgstruct.h](#).

10.99.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_EVS_WriteAppDataFileCmd::CommandHeader

Command header.

Definition at line 76 of file [default_cfe_evs_msgstruct.h](#).

Payload

[CFE_EVS_AppDataCmd_Payload_t](#) CFE_EVS_WriteAppDataFileCmd::Payload

Command payload.

Definition at line 77 of file [default_cfe_evs_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgstruct.h](#)

10.100 CFE_EVS_WriteLogDataFileCmd Struct Reference

Write Event Log to File Command.

```
#include <default_cfe_evs_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.
- [CFE_EVS_LogFileCmd_Payload_t](#) Payload
Command payload.

10.100.1 Detailed Description

Write Event Log to File Command.

Definition at line 65 of file [default_cfe_evs_msgstruct.h](#).

10.100.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_EVS_WriteLogDataFileCmd::CommandHeader

Command header.

Definition at line 67 of file [default_cfe_evs_msgstruct.h](#).

Payload

[CFE_EVS_LogFileCmd_Payload_t](#) CFE_EVS_WriteLogDataFileCmd::Payload

Command payload.

Definition at line 68 of file [default_cfe_evs_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/evs/config/default_cfe_evs_msgstruct.h](#)

10.101 CFE_FS_FileWriteMetaData Struct Reference

External Metadata/State object associated with background file writes.

```
#include <cfe_fs_api_typedefs.h>
```

Data Fields

- volatile bool [IsPending](#)
- char [FileName](#) [CFE_MISSION_MAX_PATH_LEN]
- uint32 [FileSubType](#)
- char [Description](#) [CFE_FS_HDR_DESC_MAX_LEN]
- [CFE_FS_FileWriteGetData_t](#) [GetData](#)
- [CFE_FS_FileWriteOnEvent_t](#) [OnEvent](#)

10.101.1 Detailed Description

External Metadata/State object associated with background file writes.

Applications intending to schedule background file write jobs should instantiate this object in static/global data memory.

This keeps track of the state of the file write request(s).

Definition at line 128 of file [cfe_fs_api_typedefs.h](#).

10.101.2 Field Documentation

Description

```
char CFE_FS_FileWriteMetaData::Description[CFE_FS_HDR_DESC_MAX_LEN]
```

Description of file (for FS header)

Definition at line 136 of file [cfe_fs_api_typedefs.h](#).

FileName

```
char CFE_FS_FileWriteMetaData::FileName[CFE_MISSION_MAX_PATH_LEN]
```

Name of file to write

Definition at line 132 of file [cfe_fs_api_typedefs.h](#).

FileSubType

```
uint32 CFE_FS_FileWriteMetaData::FileSubType
```

Type of file to write (for FS header)

Definition at line 135 of file [cfe_fs_api_typedefs.h](#).

GetData

```
CFE_FS_FileWriteGetData_t CFE_FS_FileWriteMetaData::GetData
```

Application callback to get a data record

Definition at line 138 of file [cfe_fs_api_typedefs.h](#).

IsPending

```
volatile bool CFE_FS_FileWriteMetaData::IsPending
```

Whether request is pending (volatile as it may be checked outside lock)

Definition at line 130 of file [cfe_fs_api_typedefs.h](#).

OnEvent

```
CFE_FS_FileWriteOnEvent_t CFE_FS_FileWriteMetaData::OnEvent
```

Application callback for abstract event processing

Definition at line 139 of file [cfe_fs_api_typedefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/core_api/fsw/inc/cfe_fs_api_typedefs.h](#)

10.102 CFE_FS_Header Struct Reference

Standard cFE File header structure definition.

```
#include <default_cfe_fs_filedef.h>
```

Data Fields

- [uint32 ContentType](#)
Identifies the content type (=cFE1'=0x63464531)
- [uint32 SubType](#)
Type of ContentType, if necessary.
- [uint32 Length](#)
Length of this header to support external processing.
- [uint32 SpacecraftID](#)
Spacecraft that generated the file.
- [uint32 ProcessorID](#)
Processor that generated the file.
- [uint32 ApplicationID](#)
Application that generated the file.
- [uint32 TimeSeconds](#)
File creation timestamp (seconds)
- [uint32 TimeSubSeconds](#)
File creation timestamp (sub-seconds)
- [char Description \[CFE_FS_HDR_DESC_MAX_LEN\]](#)
File description.

10.102.1 Detailed Description

Standard cFE File header structure definition.

Definition at line 181 of file [default_cfe_fs_filedef.h](#).

10.102.2 Field Documentation

ApplicationID

```
uint32 CFE_FS_Header::ApplicationID
```

Application that generated the file.

Definition at line 190 of file [default_cfe_fs_filedef.h](#).

ContentType

```
uint32 CFE_FS_Header::ContentType
```

Identifies the content type (=cFE1'=0x63464531)

Definition at line 183 of file [default_cfe_fs_filedef.h](#).

Description

```
char CFE_FS_Header::Description[CFE_FS_HDR_DESC_MAX_LEN]
```

File description.

Definition at line 195 of file [default_cfe_fs_filedef.h](#).

Length

```
uint32 CFE_FS_Header::Length
```

Length of this header to support external processing.

Definition at line 187 of file [default_cfe_fs_filedef.h](#).

ProcessorID

```
uint32 CFE_FS_Header::ProcessorID
```

Processor that generated the file.

Definition at line 189 of file [default_cfe_fs_filedef.h](#).

SpacecraftID

```
uint32 CFE_FS_Header::SpacecraftID
```

Spacecraft that generated the file.

Definition at line 188 of file [default_cfe_fs_filedef.h](#).

SubType

```
uint32 CFE_FS_Header::SubType
```

Type of ContentType, if necessary.

Standard SubType definitions can be found [here](#)

Definition at line 184 of file [default_cfe_fs_filedef.h](#).

TimeSeconds

```
uint32 CFE_FS_Header::TimeSeconds
```

File creation timestamp (seconds)

Definition at line 192 of file [default_cfe_fs_filedef.h](#).

TimeSubSeconds

```
uint32 CFE_FS_Header::TimeSubSeconds
```

File creation timestamp (sub-seconds)

Definition at line 193 of file [default_cfe_fs_filedef.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/fs/config/default_cfe_fs_filedef.h](#)

10.103 CFE_SB_AllSubscriptionsTIm Struct Reference

```
#include <default_cfe_sb_msgstruct.h>
```

Data Fields

- [CFE_MSG_TelemetryHeader_t](#) TelemetryHeader
Telemetry header.
- [CFE_SB_AllSubscriptionsTIm_Payload_t](#) Payload
Telemetry payload.

10.103.1 Detailed Description

Definition at line 138 of file [default_cfe_sb_msgstruct.h](#).

10.103.2 Field Documentation

Payload

`CFE_SB_AllSubscriptionsTlm_Payload_t` `CFE_SB_AllSubscriptionsTlm::Payload`

Telemetry payload.

Definition at line 141 of file [default_cfe_sb_msgstruct.h](#).

TelemetryHeader

`CFE_MSG_TelemetryHeader_t` `CFE_SB_AllSubscriptionsTlm::TelemetryHeader`

Telemetry header.

Definition at line 140 of file [default_cfe_sb_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/sb/config/default_cfe_sb_msgstruct.h](#)

10.104 CFE_SB_AllSubscriptionsTlm_Payload Struct Reference

```
#include <default_cfe_sb_msgdefs.h>
```

Data Fields

- [uint32 PktSegment](#)
Pkt number(starts at 1) in the series.
- [uint32 TotalSegments](#)
Total number of pkts needed to complete the request.
- [uint32 Entries](#)
Number of entries in the pkt.
- [CFE_SB_SubEntries_t Entry](#) [`CFE_MISSION_SB_SUB_ENTRIES_PER_PKT`]
Array of [CFE_SB_SubEntries_t](#) entries.

10.104.1 Detailed Description

Name SB Previous Subscriptions Packet

This structure defines the pkt(s) sent by SB that contains a list of all current subscriptions. This pkt is generated on cmd and intended to be used primarily by the Software Bus Networking Application (SBN). Typically, when the cmd is received there are more subscriptions than can fit in one pkt. The complete list of subscriptions is sent via a series of segmented pkts.

Definition at line 273 of file [default_cfe_sb_msgdefs.h](#).

10.104.2 Field Documentation

Entries

`uint32` `CFE_SB_AllSubscriptionsTlm_Payload::Entries`

Number of entries in the pkt.

Definition at line 277 of file [default_cfe_sb_msgdefs.h](#).

Entry

`CFE_SB_SubEntries_t` `CFE_SB_AllSubscriptionsTlm_Payload::Entry` [`CFE_MISSION_SB_SUB_ENTRIES_PER_PKT`]

Array of [CFE_SB_SubEntries_t](#) entries.

Definition at line 278 of file [default_cfe_sb_msgdefs.h](#).

PktSegment

`uint32 CFE_SB_AllSubscriptionsTlm_Payload::PktSegment`

Pkt number(starts at 1) in the series.

Definition at line 275 of file [default_cfe_sb_msgdefs.h](#).

TotalSegments

`uint32 CFE_SB_AllSubscriptionsTlm_Payload::TotalSegments`

Total number of pkts needed to complete the request.

Definition at line 276 of file [default_cfe_sb_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/sb/config/default_cfe_sb_msgdefs.h](#)

10.105 CFE_SB_DisableRouteCmd Struct Reference

```
#include <default_cfe_sb_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) *CommandHeader*
Command header.
- [CFE_SB_RouteCmd_Payload_t](#) *Payload*
Command payload.

10.105.1 Detailed Description

Definition at line 110 of file [default_cfe_sb_msgstruct.h](#).

10.105.2 Field Documentation

CommandHeader

`CFE_MSG_CommandHeader_t CFE_SB_DisableRouteCmd::CommandHeader`

Command header.

Definition at line 112 of file [default_cfe_sb_msgstruct.h](#).

Payload

`CFE_SB_RouteCmd_Payload_t CFE_SB_DisableRouteCmd::Payload`

Command payload.

Definition at line 113 of file [default_cfe_sb_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/sb/config/default_cfe_sb_msgstruct.h](#)

10.106 CFE_SB_DisableSubReportingCmd Struct Reference

```
#include <default_cfe_sb_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) *CommandHeader*

10.106.1 Detailed Description

Definition at line 60 of file [default_cfe_sb_msgstruct.h](#).

10.106.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_SB_DisableSubReportingCmd::CommandHeader

Definition at line 62 of file [default_cfe_sb_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/sb/config/default_cfe_sb_msgstruct.h](#)

10.107 CFE_SB_EnableRouteCmd Struct Reference

```
#include <default_cfe_sb_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.
- [CFE_SB_RouteCmd_Payload_t](#) Payload
Command payload.

10.107.1 Detailed Description

Definition at line 104 of file [default_cfe_sb_msgstruct.h](#).

10.107.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_SB_EnableRouteCmd::CommandHeader

Command header.

Definition at line 106 of file [default_cfe_sb_msgstruct.h](#).

Payload

[CFE_SB_RouteCmd_Payload_t](#) CFE_SB_EnableRouteCmd::Payload

Command payload.

Definition at line 107 of file [default_cfe_sb_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/sb/config/default_cfe_sb_msgstruct.h](#)

10.108 CFE_SB_EnableSubReportingCmd Struct Reference

```
#include <default_cfe_sb_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader

10.108.1 Detailed Description

Definition at line 55 of file [default_cfe_sb_msgstruct.h](#).

10.108.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_SB_EnableSubReportingCmd::CommandHeader

Definition at line 57 of file [default_cfe_sb_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/sb/config/default_cfe_sb_msgstruct.h](#)

10.109 CFE_SB_HousekeepingTlm Struct Reference

```
#include <default_cfe_sb_msgstruct.h>
```

Data Fields

- [CFE_MSG_TelemetryHeader_t](#) TelemetryHeader
Telemetry header.
- [CFE_SB_HousekeepingTlm_Payload_t](#) Payload
Telemetry payload.

10.109.1 Detailed Description

Definition at line 120 of file [default_cfe_sb_msgstruct.h](#).

10.109.2 Field Documentation

Payload

[CFE_SB_HousekeepingTlm_Payload_t](#) CFE_SB_HousekeepingTlm::Payload

Telemetry payload.

Definition at line 123 of file [default_cfe_sb_msgstruct.h](#).

TelemetryHeader

[CFE_MSG_TelemetryHeader_t](#) CFE_SB_HousekeepingTlm::TelemetryHeader

Telemetry header.

Definition at line 122 of file [default_cfe_sb_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/sb/config/default_cfe_sb_msgstruct.h](#)

10.110 CFE_SB_HousekeepingTlm_Payload Struct Reference

```
#include <default_cfe_sb_msgdefs.h>
```

Data Fields

- [uint8](#) CommandCounter
Count of valid commands received.
- [uint8](#) CommandErrorCounter
Count of invalid commands received.
- [uint8](#) NoSubscribersCounter
Count pkts sent with no subscribers.
- [uint8](#) MsgSendErrorCounter

- Count of message send errors.*
 - [uint8 MsgReceiveErrorCounter](#)
Count of message receive errors.
 - [uint8 InternalErrorCounter](#)
Count of queue read or write errors.
 - [uint8 CreatePipeErrorCounter](#)
Count of errors in create pipe API.
 - [uint8 SubscribeErrorCounter](#)
Count of errors in subscribe API.
 - [uint8 PipeOptsErrorCounter](#)
Count of errors in set/get pipe options API.
 - [uint8 DuplicateSubscriptionsCounter](#)
Count of duplicate subscriptions.
 - [uint8 GetPipeIdByNameErrorCounter](#)
Count of errors in get pipe id by name API.
 - [uint8 Padding](#)
 - [uint16 PipeOverflowErrorCounter](#)
Spare bytes to ensure alignment.
 - [uint16 MsgLimitErrorCounter](#)
Count of msg id to pipe errors.
 - [CFE_ES_MemHandle_t MemPoolHandle](#)
Handle to SB's Memory Pool.
 - [uint32 MemInUse](#)
Memory in use.
 - [uint32 UnmarkedMem](#)
cfg param CFE_PLATFORM_SB_BUF_MEMORY_BYTES minus Peak Memory in use

10.110.1 Detailed Description

Name Software Bus task housekeeping Packet

Definition at line 67 of file [default_cfe_sb_msgdefs.h](#).

10.110.2 Field Documentation

CommandCounter

[uint8](#) CFE_SB_HousekeepingTlm_Payload::CommandCounter
Count of valid commands received.

Telemetry Mnemonic(s) \$sc_\$cpu_SB_CMDPC

Definition at line 69 of file [default_cfe_sb_msgdefs.h](#).

CommandErrorCounter

[uint8](#) CFE_SB_HousekeepingTlm_Payload::CommandErrorCounter
Count of invalid commands received.

Telemetry Mnemonic(s) \$sc_\$cpu_SB_CMDEC

Definition at line 71 of file [default_cfe_sb_msgdefs.h](#).

CreatePipeErrorCounter

`uint8 CFE_SB_HousekeepingTlm_Payload::CreatePipeErrorCounter`
Count of errors in create pipe API.

Telemetry Mnemonic(s) `$sc_$cpu_SB_NewPipeEC`

Definition at line 82 of file [default_cfe_sb_msgdefs.h](#).

DuplicateSubscriptionsCounter

`uint8 CFE_SB_HousekeepingTlm_Payload::DuplicateSubscriptionsCounter`
Count of duplicate subscriptions.

Telemetry Mnemonic(s) `$sc_$cpu_SB_DupSubCnt`

Definition at line 88 of file [default_cfe_sb_msgdefs.h](#).

GetPipeIdByNameErrorCounter

`uint8 CFE_SB_HousekeepingTlm_Payload::GetPipeIdByNameErrorCounter`
Count of errors in get pipe id by name API.

Telemetry Mnemonic(s) `$sc_$cpu_SB_GetPipeIDByNameEC`

Definition at line 90 of file [default_cfe_sb_msgdefs.h](#).

InternalErrorCounter

`uint8 CFE_SB_HousekeepingTlm_Payload::InternalErrorCounter`
Count of queue read or write errors.

Telemetry Mnemonic(s) `$sc_$cpu_SB_InternalEC`

Definition at line 80 of file [default_cfe_sb_msgdefs.h](#).

MemInUse

`uint32 CFE_SB_HousekeepingTlm_Payload::MemInUse`
Memory in use.

Telemetry Mnemonic(s) `$sc_$cpu_SB_MemInUse`

Definition at line 102 of file [default_cfe_sb_msgdefs.h](#).

MemPoolHandle

`CFE_ES_MemHandle_t CFE_SB_HousekeepingTlm_Payload::MemPoolHandle`
Handle to SB's Memory Pool.

Telemetry Mnemonic(s) `$sc_$cpu_SB_MemPoolHdl`

Definition at line 99 of file [default_cfe_sb_msgdefs.h](#).

MsgLimitErrorCounter

`uint16 CFE_SB_HousekeepingTlm_Payload::MsgLimitErrorCounter`
Count of msg id to pipe errors.

Telemetry Mnemonic(s) `$sc_$cpu_SB_MsgLimEC`

Definition at line 96 of file [default_cfe_sb_msgdefs.h](#).

MsgReceiveErrorCounter

`uint8 CFE_SB_HousekeepingTlm_Payload::MsgReceiveErrorCounter`
Count of message receive errors.

Telemetry Mnemonic(s) `$sc_$cpu_SB_MsgRecEC`

Definition at line 78 of file [default_cfe_sb_msgdefs.h](#).

MsgSendErrorCounter

`uint8 CFE_SB_HousekeepingTlm_Payload::MsgSendErrorCounter`
Count of message send errors.

Telemetry Mnemonic(s) `$sc_$cpu_SB_MsgSndEC`

Definition at line 75 of file [default_cfe_sb_msgdefs.h](#).

NoSubscribersCounter

`uint8 CFE_SB_HousekeepingTlm_Payload::NoSubscribersCounter`
Count pkts sent with no subscribers.

Telemetry Mnemonic(s) `$sc_$cpu_SB_NoSubEC`

Definition at line 73 of file [default_cfe_sb_msgdefs.h](#).

Padding

`uint8 CFE_SB_HousekeepingTlm_Payload::Padding`
Definition at line 92 of file [default_cfe_sb_msgdefs.h](#).

PipeOptsErrorCounter

`uint8 CFE_SB_HousekeepingTlm_Payload::PipeOptsErrorCounter`
Count of errors in set/get pipe options API.

Telemetry Mnemonic(s) `$sc_$cpu_SB_PipeOptsEC`

Definition at line 86 of file [default_cfe_sb_msgdefs.h](#).

PipeOverflowErrorCounter

`uint16 CFE_SB_HousekeepingTlm_Payload::PipeOverflowErrorCounter`
Spare bytes to ensure alignment.

Telemetry Mnemonic(s) `$sc_$cpu_SB_PipeOvrEC`

Count of pipe overflow errors

Definition at line 94 of file [default_cfe_sb_msgdefs.h](#).

SubscribeErrorCounter

`uint8 CFE_SB_HousekeepingTlm_Payload::SubscribeErrorCounter`
Count of errors in subscribe API.

Telemetry Mnemonic(s) `$sc_$cpu_SB_SubscrEC`

Definition at line 84 of file [default_cfe_sb_msgdefs.h](#).

UnmarkedMem

`uint32 CFE_SB_HousekeepingTlm_Payload::UnmarkedMem`
 cfg param CFE_PLATFORM_SB_BUF_MEMORY_BYTES minus Peak Memory in use

Telemetry Mnemonic(s) \$sc_\$cpu_SB_UnMarkedMem

Definition at line 105 of file [default_cfe_sb_msgdefs.h](#).
 The documentation for this struct was generated from the following file:

- [cfe/modules/sb/config/default_cfe_sb_msgdefs.h](#)

10.111 CFE_SB_Msg Union Reference

Software Bus generic message.
`#include <cfe_sb_api_typedefs.h>`

Data Fields

- [CFE_MSG_Message_t](#) **Msg**
Base message type without enforced alignment.
- long long int [LongInt](#)
Align to support Long Integer.
- long double [LongDouble](#)
Align to support Long Double.

10.111.1 Detailed Description

Software Bus generic message.
 Definition at line 139 of file [cfe_sb_api_typedefs.h](#).

10.111.2 Field Documentation**LongDouble**

`long double CFE_SB_Msg::LongDouble`
 Align to support Long Double.
 Definition at line 143 of file [cfe_sb_api_typedefs.h](#).

LongInt

`long long int CFE_SB_Msg::LongInt`
 Align to support Long Integer.
 Definition at line 142 of file [cfe_sb_api_typedefs.h](#).

Msg

`CFE_MSG_Message_t CFE_SB_Msg::Msg`
 Base message type without enforced alignment.
 Definition at line 141 of file [cfe_sb_api_typedefs.h](#).
 The documentation for this union was generated from the following file:

- [cfe/modules/core_api/fsw/inc/cfe_sb_api_typedefs.h](#)

10.112 CFE_SB_MsgId_t Struct Reference

[CFE_SB_MsgId_t](#) type definition.

```
#include <default_cfe_sb_extern_typedefs.h>
```

Data Fields

- [CFE_SB_MsgId_Atom_t](#) Value

10.112.1 Detailed Description

[CFE_SB_MsgId_t](#) type definition.

Software Bus message identifier used in many SB APIs

Currently this is directly mapped to the underlying holding type (not wrapped) for compatibility with existing usage semantics in apps (mainly switch/case statements)

Note

In a future version it could become a type-safe wrapper similar to the route index, to avoid message IDs getting mixed between other integer values.

Definition at line 102 of file [default_cfe_sb_extern_typedefs.h](#).

10.112.2 Field Documentation

Value

[CFE_SB_MsgId_Atom_t](#) [CFE_SB_MsgId_t::Value](#)

Definition at line 104 of file [default_cfe_sb_extern_typedefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/sb/config/default_cfe_sb_extern_typedefs.h](#)

10.113 CFE_SB_MsgMapFileEntry Struct Reference

SB Map File Entry.

```
#include <default_cfe_sb_msgdefs.h>
```

Data Fields

- [CFE_SB_MsgId_t](#) MsgId
Message Id which has been subscribed to.
- [CFE_SB_RouteId_Atom_t](#) Index
Routing raw index value (0 based, not Route ID)

10.113.1 Detailed Description

SB Map File Entry.

Structure of one element of the map information in response to [CFE_SB_WRITE_MAP_INFO_CC](#)

Definition at line 225 of file [default_cfe_sb_msgdefs.h](#).

10.113.2 Field Documentation

Index

[CFE_SB_RouteId_Atom_t](#) [CFE_SB_MsgMapFileEntry::Index](#)

Routing raw index value (0 based, not Route ID)

Definition at line 228 of file [default_cfe_sb_msgdefs.h](#).

MsgId

[CFE_SB_MsgId_t](#) CFE_SB_MsgMapFileEntry::MsgId

Message Id which has been subscribed to.

Definition at line 227 of file [default_cfe_sb_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/sb/config/default_cfe_sb_msgdefs.h](#)

10.114 CFE_SB_NoopCmd Struct Reference

```
#include <default_cfe_sb_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader

10.114.1 Detailed Description

Definition at line 45 of file [default_cfe_sb_msgstruct.h](#).

10.114.2 Field Documentation**CommandHeader**

[CFE_MSG_CommandHeader_t](#) CFE_SB_NoopCmd::CommandHeader

Definition at line 47 of file [default_cfe_sb_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/sb/config/default_cfe_sb_msgstruct.h](#)

10.115 CFE_SB_PipeDepthStats Struct Reference

SB Pipe Depth Statistics.

```
#include <default_cfe_sb_msgdefs.h>
```

Data Fields

- [CFE_SB_Pipeld_t](#) Pipeld
Pipe Id associated with the stats below.
- [uint16](#) MaxQueueDepth
Number of messages the pipe can hold.
- [uint16](#) CurrentQueueDepth
Number of messages currently on the pipe.
- [uint16](#) PeakQueueDepth
Peak number of messages that have been on the pipe.
- [uint16](#) Spare
Spare word to ensure alignment.

10.115.1 Detailed Description

SB Pipe Depth Statistics.

Used in SB Statistics Telemetry Packet [CFE_SB_StatsTlm_t](#)

Definition at line 114 of file [default_cfe_sb_msgdefs.h](#).

10.115.2 Field Documentation

CurrentQueueDepth

`uint16 CFE_SB_PipeDepthStats::CurrentQueueDepth`

Number of messages currently on the pipe.

Telemetry Mnemonic(s) `$sc_$cpu_SB_Stat.SB_SMPDS[CFE_PLATFORM_SB_MAX_PIPES].SB_PDINUSE`

Definition at line 120 of file [default_cfe_sb_msgdefs.h](#).

MaxQueueDepth

`uint16 CFE_SB_PipeDepthStats::MaxQueueDepth`

Number of messages the pipe can hold.

Telemetry Mnemonic(s) `$sc_$cpu_SB_Stat.SB_SMPDS[CFE_PLATFORM_SB_MAX_PIPES].SB_PDDEPTH`

Definition at line 118 of file [default_cfe_sb_msgdefs.h](#).

PeakQueueDepth

`uint16 CFE_SB_PipeDepthStats::PeakQueueDepth`

Peak number of messages that have been on the pipe.

Telemetry Mnemonic(s) `$sc_$cpu_SB_Stat.SB_SMPDS[CFE_PLATFORM_SB_MAX_PIPES].SB_PDPKINUSE`

Definition at line 122 of file [default_cfe_sb_msgdefs.h](#).

PipeId

`CFE_SB_PipeId_t CFE_SB_PipeDepthStats::PipeId`

Pipe Id associated with the stats below.

Telemetry Mnemonic(s) `$sc_$cpu_SB_Stat.SB_SMPDS[CFE_PLATFORM_SB_MAX_PIPES].SB_PDPIPEID`

Definition at line 116 of file [default_cfe_sb_msgdefs.h](#).

Spare

`uint16 CFE_SB_PipeDepthStats::Spare`

Spare word to ensure alignment.

Telemetry Mnemonic(s) `$sc_$cpu_SB_Stat.SB_SMPDS[CFE_PLATFORM_SB_MAX_PIPES].SB_PDSPARE`

Definition at line 124 of file [default_cfe_sb_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/sb/config/default_cfe_sb_msgdefs.h](#)

10.116 CFE_SB_PipeInfoEntry Struct Reference

SB Pipe Information File Entry.

```
#include <default_cfe_sb_msgdefs.h>
```

Data Fields

- [CFE_SB_PipeId_t](#) PipeId
- [CFE_ES_AppId_t](#) AppId
- char [PipeName](#) [CFE_MISSION_MAX_API_LEN]
- char [AppName](#) [CFE_MISSION_MAX_API_LEN]
- [uint16](#) MaxQueueDepth
- [uint16](#) CurrentQueueDepth
- [uint16](#) PeakQueueDepth
- [uint16](#) SendErrors
- [uint8](#) Opts
- [uint8](#) Spare [3]

10.116.1 Detailed Description

SB Pipe Information File Entry.

This statistics structure is output as part of the CFE SB "Send Pipe Info" command (CFE_SB_SEND_PIPE_INFO_CC). Previous versions of CFE simply wrote the internal CFE_SB_PipeD_t object to the file, but this also contains information such as pointers which are not relevant outside the running CFE process.

By defining the pipe info structure separately, it also provides some independence, such that the internal CFE_SB_PipeD_t definition can evolve without changing the binary format of the information file.

Definition at line 143 of file [default_cfe_sb_msgdefs.h](#).

10.116.2 Field Documentation

AppId

[CFE_ES_AppId_t](#) CFE_SB_PipeInfoEntry::AppId

The runtime ID of the application that owns the pipe

Definition at line 146 of file [default_cfe_sb_msgdefs.h](#).

AppName

char CFE_SB_PipeInfoEntry::AppName[CFE_MISSION_MAX_API_LEN]

The Name of the application that owns the pipe

Definition at line 148 of file [default_cfe_sb_msgdefs.h](#).

CurrentQueueDepth

[uint16](#) CFE_SB_PipeInfoEntry::CurrentQueueDepth

The current depth of the pipe

Definition at line 150 of file [default_cfe_sb_msgdefs.h](#).

MaxQueueDepth

[uint16](#) CFE_SB_PipeInfoEntry::MaxQueueDepth

The allocated depth of the pipe (max capacity)

Definition at line 149 of file [default_cfe_sb_msgdefs.h](#).

Opts

[uint8](#) CFE_SB_PipeInfoEntry::Opts

Pipe options set (bitmask)

Definition at line 153 of file [default_cfe_sb_msgdefs.h](#).

PeakQueueDepth

`uint16 CFE_SB_PipeInfoEntry::PeakQueueDepth`
The peak depth of the pipe (high watermark)
Definition at line 151 of file [default_cfe_sb_msgdefs.h](#).

PipeId

`CFE_SB_PipeId_t CFE_SB_PipeInfoEntry::PipeId`
The runtime ID of the pipe
Definition at line 145 of file [default_cfe_sb_msgdefs.h](#).

PipeName

`char CFE_SB_PipeInfoEntry::PipeName[CFE_MISSION_MAX_API_LEN]`
The Name of the pipe
Definition at line 147 of file [default_cfe_sb_msgdefs.h](#).

SendErrors

`uint16 CFE_SB_PipeInfoEntry::SendErrors`
Number of errors when writing to this pipe
Definition at line 152 of file [default_cfe_sb_msgdefs.h](#).

Spare

`uint8 CFE_SB_PipeInfoEntry::Spare[3]`
Padding to make this structure a multiple of 4 bytes
Definition at line 154 of file [default_cfe_sb_msgdefs.h](#).
The documentation for this struct was generated from the following file:

- [cfe/modules/sb/config/default_cfe_sb_msgdefs.h](#)

10.117 CFE_SB_Qos_t Struct Reference

Quality Of Service Type Definition.
`#include <default_cfe_sb_extern_typedefs.h>`

Data Fields

- [uint8 Priority](#)
Specify high(1) or low(0) message priority for off-board routing, currently unused.
- [uint8 Reliability](#)
Specify high(1) or low(0) message transfer reliability for off-board routing, currently unused.

10.117.1 Detailed Description

Quality Of Service Type Definition.
Currently an unused parameter in [CFE_SB_SubscribeEx](#) Intended to be used for interprocessor communication only
Definition at line 119 of file [default_cfe_sb_extern_typedefs.h](#).

10.117.2 Field Documentation

Priority

`uint8 CFE_SB_Qos_t::Priority`

Specify high(1) or low(0) message priority for off-board routing, currently unused.

Definition at line 121 of file [default_cfe_sb_extern_typedefs.h](#).

Reliability

`uint8 CFE_SB_Qos_t::Reliability`

Specify high(1) or low(0) message transfer reliability for off-board routing, currently unused.

Definition at line 122 of file [default_cfe_sb_extern_typedefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/sb/config/default_cfe_sb_extern_typedefs.h](#)

10.118 CFE_SB_ResetCountersCmd Struct Reference

```
#include <default_cfe_sb_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader

10.118.1 Detailed Description

Definition at line 50 of file [default_cfe_sb_msgstruct.h](#).

10.118.2 Field Documentation

CommandHeader

`CFE_MSG_CommandHeader_t CFE_SB_ResetCountersCmd::CommandHeader`

Definition at line 52 of file [default_cfe_sb_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/sb/config/default_cfe_sb_msgstruct.h](#)

10.119 CFE_SB_RouteCmd_Payload Struct Reference

Enable/Disable Route Command Payload.

```
#include <default_cfe_sb_msgdefs.h>
```

Data Fields

- [CFE_SB_MsgId_t](#) MsgId
Message ID of route to be enabled or disabled [CFE_SB_MsgId_t](#).
- [CFE_SB_PipeId_t](#) Pipe
Pipe ID of route to be enabled or disabled [CFE_SB_PipeId_t](#).
- `uint8 Spare`
Spare byte to make command even number of bytes.

10.119.1 Detailed Description

Enable/Disable Route Command Payload.

This structure contains a definition used by two SB commands, 'Enable Route' [CFE_SB_ENABLE_ROUTE_CC](#) and 'Disable Route' [CFE_SB_DISABLE_ROUTE_CC](#). A route is the destination pipe for a particular message and is therefore defined as a MsgId and PipeId combination.

Definition at line 53 of file [default_cfe_sb_msgdefs.h](#).

10.119.2 Field Documentation

MsgId

[CFE_SB_MsgId_t](#) CFE_SB_RouteCmd_Payload::MsgId

Message ID of route to be enabled or disabled [CFE_SB_MsgId_t](#).

Definition at line 55 of file [default_cfe_sb_msgdefs.h](#).

Pipe

[CFE_SB_PipeId_t](#) CFE_SB_RouteCmd_Payload::Pipe

Pipe ID of route to be enabled or disabled [CFE_SB_PipeId_t](#).

Definition at line 56 of file [default_cfe_sb_msgdefs.h](#).

Spare

[uint8](#) CFE_SB_RouteCmd_Payload::Spare

Spare byte to make command even number of bytes.

Definition at line 57 of file [default_cfe_sb_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/sb/config/default_cfe_sb_msgdefs.h](#)

10.120 CFE_SB_RoutingFileEntry Struct Reference

SB Routing File Entry.

```
#include <default_cfe_sb_msgdefs.h>
```

Data Fields

- [CFE_SB_MsgId_t](#) MsgId
Message Id portion of the route.
- [CFE_SB_PipeId_t](#) PipeId
Pipe Id portion of the route.
- [uint8](#) State
Route Enabled or Disabled.
- [uint16](#) MsgCnt
Number of msgs with this MsgId sent to this PipeId.
- char [AppName](#) [[CFE_MISSION_MAX_API_LEN](#)]
Pipe Depth Statistics.
- char [PipeName](#) [[CFE_MISSION_MAX_API_LEN](#)]
Pipe Depth Statistics.

10.120.1 Detailed Description

SB Routing File Entry.

Structure of one element of the routing information in response to [CFE_SB_WRITE_ROUTING_INFO_CC](#)

Definition at line 210 of file [default_cfe_sb_msgdefs.h](#).

10.120.2 Field Documentation

AppName

`char CFE_SB_RoutingFileEntry::AppName[CFE_MISSION_MAX_API_LEN]`

Pipe Depth Statistics.

Definition at line 216 of file [default_cfe_sb_msgdefs.h](#).

MsgCnt

`uint16 CFE_SB_RoutingFileEntry::MsgCnt`

Number of msgs with this MsgId sent to this PipeId.

Definition at line 215 of file [default_cfe_sb_msgdefs.h](#).

MsgId

`CFE_SB_MsgId_t CFE_SB_RoutingFileEntry::MsgId`

Message Id portion of the route.

Definition at line 212 of file [default_cfe_sb_msgdefs.h](#).

PipeId

`CFE_SB_PipeId_t CFE_SB_RoutingFileEntry::PipeId`

Pipe Id portion of the route.

Definition at line 213 of file [default_cfe_sb_msgdefs.h](#).

PipeName

`char CFE_SB_RoutingFileEntry::PipeName[CFE_MISSION_MAX_API_LEN]`

Pipe Depth Statistics.

Definition at line 217 of file [default_cfe_sb_msgdefs.h](#).

State

`uint8 CFE_SB_RoutingFileEntry::State`

Route Enabled or Disabled.

Definition at line 214 of file [default_cfe_sb_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/sb/config/default_cfe_sb_msgdefs.h](#)

10.121 CFE_SB_SendHkCmd Struct Reference

```
#include <default_cfe_sb_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader

10.121.1 Detailed Description

Definition at line 75 of file [default_cfe_sb_msgstruct.h](#).

10.121.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_SB_SendHkCmd::CommandHeader

Definition at line 77 of file [default_cfe_sb_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/sb/config/default_cfe_sb_msgstruct.h](#)

10.122 CFE_SB_SendPrevSubsCmd Struct Reference

```
#include <default_cfe_sb_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader

10.122.1 Detailed Description

Definition at line 70 of file [default_cfe_sb_msgstruct.h](#).

10.122.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_SB_SendPrevSubsCmd::CommandHeader

Definition at line 72 of file [default_cfe_sb_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/sb/config/default_cfe_sb_msgstruct.h](#)

10.123 CFE_SB_SendSbStatsCmd Struct Reference

```
#include <default_cfe_sb_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader

10.123.1 Detailed Description

Definition at line 65 of file [default_cfe_sb_msgstruct.h](#).

10.123.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_SB_SendSbStatsCmd::CommandHeader

Definition at line 67 of file [default_cfe_sb_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/sb/config/default_cfe_sb_msgstruct.h](#)

10.124 CFE_SB_SingleSubscriptionTlm Struct Reference

```
#include <default_cfe_sb_msgstruct.h>
```

Data Fields

- [CFE_MSG_TelemetryHeader_t](#) TelemetryHeader
Telemetry header.
- [CFE_SB_SingleSubscriptionTlm_Payload_t](#) Payload
Telemetry payload.

10.124.1 Detailed Description

Definition at line 132 of file [default_cfe_sb_msgstruct.h](#).

10.124.2 Field Documentation**Payload**

[CFE_SB_SingleSubscriptionTlm_Payload_t](#) CFE_SB_SingleSubscriptionTlm::Payload

Telemetry payload.

Definition at line 135 of file [default_cfe_sb_msgstruct.h](#).

TelemetryHeader

[CFE_MSG_TelemetryHeader_t](#) CFE_SB_SingleSubscriptionTlm::TelemetryHeader

Telemetry header.

Definition at line 134 of file [default_cfe_sb_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/sb/config/default_cfe_sb_msgstruct.h](#)

10.125 CFE_SB_SingleSubscriptionTlm_Payload Struct Reference

```
#include <default_cfe_sb_msgdefs.h>
```

Data Fields

- [uint8](#) SubType
Subscription or Unsubscription.
- [CFE_SB_MsgId_t](#) MsgId
MsgId subscribed or unsubscribe to.
- [CFE_SB_Qos_t](#) Qos
Quality of Service, used only for interprocessor communication.
- [CFE_SB_Pipeld_t](#) Pipe
Destination pipe id to send above msg id

10.125.1 Detailed Description

Name SB Subscription Report Packet

This structure defines the pkt sent by SB when a subscription or a request to unsubscribe is received while subscription reporting is enabled. By default subscription reporting is disabled. This feature is intended to be used primarily by Software Bus Networking Application (SBN)

See also

[CFE_SB_ENABLE_SUB_REPORTING_CC](#), [CFE_SB_DISABLE_SUB_REPORTING_CC](#)

Definition at line 241 of file [default_cfe_sb_msgdefs.h](#).

10.125.2 Field Documentation

MsgId

[CFE_SB_MsgId_t](#) CFE_SB_SingleSubscriptionTlm_Payload::MsgId

MsgId subscribed or unsubscribe to.

Definition at line 244 of file [default_cfe_sb_msgdefs.h](#).

Pipe

[CFE_SB_PipeId_t](#) CFE_SB_SingleSubscriptionTlm_Payload::Pipe

Destination pipe id to send above msg id

Definition at line 246 of file [default_cfe_sb_msgdefs.h](#).

Qos

[CFE_SB_Qos_t](#) CFE_SB_SingleSubscriptionTlm_Payload::Qos

Quality of Service, used only for interprocessor communication.

Definition at line 245 of file [default_cfe_sb_msgdefs.h](#).

SubType

[uint8](#) CFE_SB_SingleSubscriptionTlm_Payload::SubType

Subscription or Unsubscription.

Definition at line 243 of file [default_cfe_sb_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/sb/config/default_cfe_sb_msgdefs.h](#)

10.126 CFE_SB_StatsTlm Struct Reference

```
#include <default_cfe_sb_msgstruct.h>
```

Data Fields

- [CFE_MSG_TelemetryHeader_t](#) TelemetryHeader
Telemetry header.
- [CFE_SB_StatsTlm_Payload_t](#) Payload
Telemetry payload.

10.126.1 Detailed Description

Definition at line 126 of file [default_cfe_sb_msgstruct.h](#).

10.126.2 Field Documentation

Payload

[CFE_SB_StatsTlm_Payload_t](#) CFE_SB_StatsTlm::Payload

Telemetry payload.

Definition at line 129 of file [default_cfe_sb_msgstruct.h](#).

TelemetryHeader

[CFE_MSG_TelemetryHeader_t](#) CFE_SB_StatsTlm::TelemetryHeader

Telemetry header.

Definition at line 128 of file [default_cfe_sb_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/sb/config/default_cfe_sb_msgstruct.h](#)

10.127 CFE_SB_StatsTlm_Payload Struct Reference

```
#include <default_cfe_sb_msgdefs.h>
```

Data Fields

- [uint32 MsgIdsInUse](#)
Current number of MsgIds with a destination.
- [uint32 PeakMsgIdsInUse](#)
Peak number of MsgIds with a destination.
- [uint32 MaxMsgIdsAllowed](#)
cFE Cfg Param [CFE_PLATFORM_SB_MAX_MSG_IDS](#)
- [uint32 PipesInUse](#)
Number of pipes currently in use.
- [uint32 PeakPipesInUse](#)
Peak number of pipes since last reboot.
- [uint32 MaxPipesAllowed](#)
cFE Cfg Param [CFE_PLATFORM_SB_MAX_PIPES](#)
- [uint32 MemInUse](#)
Memory bytes currently in use for SB msg transfers.
- [uint32 PeakMemInUse](#)
Peak memory bytes in use for SB msg transfers.
- [uint32 MaxMemAllowed](#)
cFE Cfg Param [CFE_PLATFORM_SB_BUF_MEMORY_BYTES](#)
- [uint32 SubscriptionsInUse](#)
Number of current subscriptions.
- [uint32 PeakSubscriptionsInUse](#)
Peak number of subscriptions.
- [uint32 MaxSubscriptionsAllowed](#)
product of [CFE_PLATFORM_SB_MAX_MSG_IDS](#) and [CFE_PLATFORM_SB_MAX_DEST_PER_PKT](#)
- [uint32 SBBuffersInUse](#)
Number of SB message buffers currently in use.
- [uint32 PeakSBBuffersInUse](#)
Max number of SB message buffers in use.
- [uint32 MaxPipeDepthAllowed](#)
Maximum allowed pipe depth.
- [CFE_SB_PipeDepthStats_t PipeDepthStats](#) [[CFE_MISSION_SB_MAX_PIPES](#)]
Pipe Depth Statistics [CFE_SB_PipeDepthStats_t](#).

10.127.1 Detailed Description

Name SB Statistics Telemetry Packet

SB Statistics packet sent in response to [CFE_SB_SEND_SB_STATS_CC](#)
Definition at line 162 of file [default_cfe_sb_msgdefs.h](#).

10.127.2 Field Documentation

MaxMemAllowed

[uint32](#) CFE_SB_StatsTlm_Payload::MaxMemAllowed
cFE Cfg Param [CFE_PLATFORM_SB_BUF_MEMORY_BYTES](#)

Telemetry Mnemonic(s) \$sc_\$cpu_SB_Stat.SB_SMMBMALW

Definition at line 182 of file [default_cfe_sb_msgdefs.h](#).

MaxMsgIdsAllowed

[uint32](#) CFE_SB_StatsTlm_Payload::MaxMsgIdsAllowed
cFE Cfg Param [CFE_PLATFORM_SB_MAX_MSG_IDS](#)

Telemetry Mnemonic(s) \$sc_\$cpu_SB_Stat.SB_SMMMIDALW

Definition at line 168 of file [default_cfe_sb_msgdefs.h](#).

MaxPipeDepthAllowed

[uint32](#) CFE_SB_StatsTlm_Payload::MaxPipeDepthAllowed
Maximum allowed pipe depth.

Telemetry Mnemonic(s) \$sc_\$cpu_SB_Stat.SB_SMMPDALW

Definition at line 198 of file [default_cfe_sb_msgdefs.h](#).

MaxPipesAllowed

[uint32](#) CFE_SB_StatsTlm_Payload::MaxPipesAllowed
cFE Cfg Param [CFE_PLATFORM_SB_MAX_PIPES](#)

Telemetry Mnemonic(s) \$sc_\$cpu_SB_Stat.SB_SMMPALW

Definition at line 175 of file [default_cfe_sb_msgdefs.h](#).

MaxSubscriptionsAllowed

[uint32](#) CFE_SB_StatsTlm_Payload::MaxSubscriptionsAllowed
product of [CFE_PLATFORM_SB_MAX_MSG_IDS](#) and [CFE_PLATFORM_SB_MAX_DEST_PER_PKT](#)

Telemetry Mnemonic(s) \$sc_\$cpu_SB_Stat.SB_SMMSALW

Definition at line 189 of file [default_cfe_sb_msgdefs.h](#).

MemInUse

[uint32](#) CFE_SB_StatsTlm_Payload::MemInUse
Memory bytes currently in use for SB msg transfers.

Telemetry Mnemonic(s) \$sc_\$cpu_SB_Stat.SB_SMBMIU

Definition at line 178 of file [default_cfe_sb_msgdefs.h](#).

MsgIdsInUse

`uint32 CFE_SB_StatsTlm_Payload::MsgIdsInUse`

Current number of MsgIds with a destination.

Telemetry Mnemonic(s) \$sc_\$cpu_SB_Stat.SB_SMMIDIU

Definition at line 164 of file [default_cfe_sb_msgdefs.h](#).

PeakMemInUse

`uint32 CFE_SB_StatsTlm_Payload::PeakMemInUse`

Peak memory bytes in use for SB msg transfers.

Telemetry Mnemonic(s) \$sc_\$cpu_SB_Stat.SB_SMPBMIU

Definition at line 180 of file [default_cfe_sb_msgdefs.h](#).

PeakMsgIdsInUse

`uint32 CFE_SB_StatsTlm_Payload::PeakMsgIdsInUse`

Peak number of MsgIds with a destination.

Telemetry Mnemonic(s) \$sc_\$cpu_SB_Stat.SB_SMPMIDIU

Definition at line 166 of file [default_cfe_sb_msgdefs.h](#).

PeakPipesInUse

`uint32 CFE_SB_StatsTlm_Payload::PeakPipesInUse`

Peak number of pipes since last reboot.

Telemetry Mnemonic(s) \$sc_\$cpu_SB_Stat.SB_SMPPIU

Definition at line 173 of file [default_cfe_sb_msgdefs.h](#).

PeakSBBuffersInUse

`uint32 CFE_SB_StatsTlm_Payload::PeakSBBuffersInUse`

Max number of SB message buffers in use.

Telemetry Mnemonic(s) \$sc_\$cpu_SB_Stat.SB_SMPSBBIU

Definition at line 195 of file [default_cfe_sb_msgdefs.h](#).

PeakSubscriptionsInUse

`uint32 CFE_SB_StatsTlm_Payload::PeakSubscriptionsInUse`

Peak number of subscriptions.

Telemetry Mnemonic(s) \$sc_\$cpu_SB_Stat.SB_SMPSIU

Definition at line 187 of file [default_cfe_sb_msgdefs.h](#).

PipeDepthStats

`CFE_SB_PipeDepthStats_t CFE_SB_StatsTlm_Payload::PipeDepthStats[CFE_MISSION_SB_MAX_PIPES]`

Pipe Depth Statistics [CFE_SB_PipeDepthStats_t](#).

Telemetry Mnemonic(s) \$sc_\$cpu_SB_Stat.SB_SMPDS[CFE_PLATFORM_SB_MAX_PIPES]

Definition at line 201 of file [default_cfe_sb_msgdefs.h](#).

PipesInUse

`uint32 CFE_SB_StatsTlm_Payload::PipesInUse`

Number of pipes currently in use.

Telemetry Mnemonic(s) `$sc_$cpu_SB_Stat.SB_SMPIU`

Definition at line 171 of file [default_cfe_sb_msgdefs.h](#).

SBBuffersInUse

`uint32 CFE_SB_StatsTlm_Payload::SBBuffersInUse`

Number of SB message buffers currently in use.

Telemetry Mnemonic(s) `$sc_$cpu_SB_Stat.SB_SMSBBIU`

Definition at line 193 of file [default_cfe_sb_msgdefs.h](#).

SubscriptionsInUse

`uint32 CFE_SB_StatsTlm_Payload::SubscriptionsInUse`

Number of current subscriptions.

Telemetry Mnemonic(s) `$sc_$cpu_SB_Stat.SB_SMSIU`

Definition at line 185 of file [default_cfe_sb_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/sb/config/default_cfe_sb_msgdefs.h](#)

10.128 CFE_SB_SubEntries Struct Reference

SB Previous Subscriptions Entry.

```
#include <default_cfe_sb_msgdefs.h>
```

Data Fields

- [CFE_SB_MsgId_t](#) `MsgId`
MsgId portion of the subscription.
- [CFE_SB_Qos_t](#) `Qos`
Qos portion of the subscription.
- [CFE_SB_Pipeld_t](#) `Pipe`
Pipeld portion of the subscription.

10.128.1 Detailed Description

SB Previous Subscriptions Entry.

This structure defines an entry used in the `CFE_SB_PrevSubsPkt_t` Intended to be used primarily by Software Bus Networking Application (SBN)

Used in structure definition [CFE_SB_AllSubscriptionsTlm_t](#)

Definition at line 257 of file [default_cfe_sb_msgdefs.h](#).

10.128.2 Field Documentation

MsgId

[CFE_SB_MsgId_t](#) `CFE_SB_SubEntries::MsgId`

MsgId portion of the subscription.

Definition at line 259 of file [default_cfe_sb_msgdefs.h](#).

Pipe

[CFE_SB_PipeId_t](#) CFE_SB_SubEntries::Pipe

PipeId portion of the subscription.

Definition at line 261 of file [default_cfe_sb_msgdefs.h](#).

Qos

[CFE_SB_Qos_t](#) CFE_SB_SubEntries::Qos

Qos portion of the subscription.

Definition at line 260 of file [default_cfe_sb_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/sb/config/default_cfe_sb_msgdefs.h](#)

10.129 CFE_SB_WriteFileInfoCmd_Payload Struct Reference

Write File Info Command Payload.

`#include <default_cfe_sb_msgdefs.h>`

Data Fields

- char [Filename](#) [[CFE_MISSION_MAX_PATH_LEN](#)]

Path and Filename of data to be loaded.

10.129.1 Detailed Description

Write File Info Command Payload.

This structure contains a generic definition used by SB commands that write to a file

Definition at line 40 of file [default_cfe_sb_msgdefs.h](#).

10.129.2 Field Documentation

Filename

char CFE_SB_WriteFileInfoCmd_Payload::Filename [[CFE_MISSION_MAX_PATH_LEN](#)]

Path and Filename of data to be loaded.

Definition at line 42 of file [default_cfe_sb_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/sb/config/default_cfe_sb_msgdefs.h](#)

10.130 CFE_SB_WriteMapInfoCmd Struct Reference

`#include <default_cfe_sb_msgstruct.h>`

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader

Command header.

- [CFE_SB_WriteFileInfoCmd_Payload_t](#) Payload

Command payload.

10.130.1 Detailed Description

Definition at line 95 of file [default_cfe_sb_msgstruct.h](#).

10.130.2 Field Documentation

CommandHeader

`CFE_MSG_CommandHeader_t CFE_SB_WriteMapInfoCmd::CommandHeader`

Command header.

Definition at line 97 of file [default_cfe_sb_msgstruct.h](#).

Payload

`CFE_SB_WriteFileInfoCmd_Payload_t CFE_SB_WriteMapInfoCmd::Payload`

Command payload.

Definition at line 98 of file [default_cfe_sb_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/sb/config/default_cfe_sb_msgstruct.h](#)

10.131 CFE_SB_WritePipeInfoCmd Struct Reference

```
#include <default_cfe_sb_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) `CommandHeader`
Command header.
- [CFE_SB_WriteFileInfoCmd_Payload_t](#) `Payload`
Command payload.

10.131.1 Detailed Description

Definition at line 89 of file [default_cfe_sb_msgstruct.h](#).

10.131.2 Field Documentation

CommandHeader

`CFE_MSG_CommandHeader_t CFE_SB_WritePipeInfoCmd::CommandHeader`

Command header.

Definition at line 91 of file [default_cfe_sb_msgstruct.h](#).

Payload

`CFE_SB_WriteFileInfoCmd_Payload_t CFE_SB_WritePipeInfoCmd::Payload`

Command payload.

Definition at line 92 of file [default_cfe_sb_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/sb/config/default_cfe_sb_msgstruct.h](#)

10.132 CFE_SB_WriteRoutingInfoCmd Struct Reference

```
#include <default_cfe_sb_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) `CommandHeader`
Command header.
- [CFE_SB_WriteFileInfoCmd_Payload_t](#) `Payload`
Command payload.

10.132.1 Detailed Description

Definition at line 83 of file [default_cfe_sb_msgstruct.h](#).

10.132.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) `CFE_SB_WriteRoutingInfoCmd::CommandHeader`
Command header.
Definition at line 85 of file [default_cfe_sb_msgstruct.h](#).

Payload

[CFE_SB_WriteFileInfoCmd_Payload_t](#) `CFE_SB_WriteRoutingInfoCmd::Payload`
Command payload.
Definition at line 86 of file [default_cfe_sb_msgstruct.h](#).
The documentation for this struct was generated from the following file:

- [cfe/modules/sb/config/default_cfe_sb_msgstruct.h](#)

10.133 CFE_TBL_AbortLoadCmd Struct Reference

Abort Load Command.
`#include <default_cfe_tbl_msgstruct.h>`

Data Fields

- [CFE_MSG_CommandHeader_t](#) `CommandHeader`
Command header.
- [CFE_TBL_AbortLoadCmd_Payload_t](#) `Payload`
Command payload.

10.133.1 Detailed Description

Abort Load Command.
Definition at line 124 of file [default_cfe_tbl_msgstruct.h](#).

10.133.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) `CFE_TBL_AbortLoadCmd::CommandHeader`
Command header.
Definition at line 126 of file [default_cfe_tbl_msgstruct.h](#).

Payload

[CFE_TBL_AbortLoadCmd_Payload_t](#) CFE_TBL_AbortLoadCmd::Payload

Command payload.

Definition at line 127 of file [default_cfe_tbl_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/tbl/config/default_cfe_tbl_msgstruct.h](#)

10.134 CFE_TBL_AbortLoadCmd_Payload Struct Reference

Abort Load Command Payload.

```
#include <default_cfe_tbl_msgdefs.h>
```

Data Fields

- char [TableName](#) [[CFE_MISSION_TBL_MAX_FULL_NAME_LEN](#)]
Full Name of Table whose load is to be aborted.

10.134.1 Detailed Description

Abort Load Command Payload.

For command details, see [CFE_TBL_ABORT_LOAD_CC](#)

Definition at line 146 of file [default_cfe_tbl_msgdefs.h](#).

10.134.2 Field Documentation

TableName

```
char CFE_TBL_AbortLoadCmd_Payload::TableName[CFE_MISSION_TBL_MAX_FULL_NAME_LEN]
```

Full Name of Table whose load is to be aborted.

ASCII string containing full table name identifier of a table whose load is to be aborted

Definition at line 148 of file [default_cfe_tbl_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/tbl/config/default_cfe_tbl_msgdefs.h](#)

10.135 CFE_TBL_ActivateCmd Struct Reference

Activate Table Command.

```
#include <default_cfe_tbl_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) [CommandHeader](#)
Command header.
- [CFE_TBL_ActivateCmd_Payload_t](#) [Payload](#)
Command payload.

10.135.1 Detailed Description

Activate Table Command.

Definition at line 88 of file [default_cfe_tbl_msgstruct.h](#).

10.135.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) [CFE_TBL_ActivateCmd::CommandHeader](#)

Command header.

Definition at line 90 of file [default_cfe_tbl_msgstruct.h](#).

Payload

[CFE_TBL_ActivateCmd_Payload_t](#) [CFE_TBL_ActivateCmd::Payload](#)

Command payload.

Definition at line 91 of file [default_cfe_tbl_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/tbl/config/default_cfe_tbl_msgstruct.h](#)

10.136 CFE_TBL_ActivateCmd_Payload Struct Reference

Activate Table Command Payload.

```
#include <default_cfe_tbl_msgdefs.h>
```

Data Fields

- char [TableName](#) [[CFE_MISSION_TBL_MAX_FULL_NAME_LEN](#)]
Full Name of Table to be activated.

10.136.1 Detailed Description

Activate Table Command Payload.

For command details, see [CFE_TBL_ACTIVATE_CC](#)

Definition at line 93 of file [default_cfe_tbl_msgdefs.h](#).

10.136.2 Field Documentation

TableName

char [CFE_TBL_ActivateCmd_Payload::TableName](#) [[CFE_MISSION_TBL_MAX_FULL_NAME_LEN](#)]

Full Name of Table to be activated.

ASCII string containing full table name identifier of table to be activated

Definition at line 95 of file [default_cfe_tbl_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/tbl/config/default_cfe_tbl_msgdefs.h](#)

10.137 CFE_TBL_CombinedFileHdr Struct Reference

Complete header for CFE table files.

```
#include <default_cfe_tbl_extern_typedefs.h>
```

Data Fields

- [CFE_FS_Header_t](#) Std
- [CFE_TBL_File_Hdr_t](#) Tbl

10.137.1 Detailed Description

Complete header for CFE table files.

Table files always have a combination of the standard file header and the table-specific file header. This struct just combines the two and makes for an easier item to pass around, simplifying APIs

Definition at line 80 of file [default_cfe_tbl_extern_typedefs.h](#).

10.137.2 Field Documentation

Std

[CFE_FS_Header_t](#) CFE_TBL_CombinedFileHdr::Std

Definition at line 82 of file [default_cfe_tbl_extern_typedefs.h](#).

Tbl

[CFE_TBL_File_Hdr_t](#) CFE_TBL_CombinedFileHdr::Tbl

Definition at line 83 of file [default_cfe_tbl_extern_typedefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/tbl/config/default_cfe_tbl_extern_typedefs.h](#)

10.138 CFE_TBL_DelCDSCmd_Payload Struct Reference

Delete Critical Table CDS Command Payload.

```
#include <default_cfe_tbl_msgdefs.h>
```

Data Fields

- char [TableName](#) [[CFE_MISSION_TBL_MAX_FULL_NAME_LEN](#)]
Full Name of Table whose CDS is to be deleted.

10.138.1 Detailed Description

Delete Critical Table CDS Command Payload.

For command details, see [CFE_TBL_DELETE_CDS_CC](#)

Definition at line 132 of file [default_cfe_tbl_msgdefs.h](#).

10.138.2 Field Documentation

TableName

char CFE_TBL_DelCDSCmd_Payload::TableName [[CFE_MISSION_TBL_MAX_FULL_NAME_LEN](#)]

Full Name of Table whose CDS is to be deleted.

ASCII string containing full table name identifier of a critical table whose CDS is to be deleted

Definition at line 134 of file [default_cfe_tbl_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/tbl/config/default_cfe_tbl_msgdefs.h](#)

10.139 CFE_TBL_DeleteCDSCmd Struct Reference

Delete Critical Table CDS Command.

```
#include <default_cfe_tbl_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) *CommandHeader*
Command header.
- [CFE_TBL_DelCDSCmd_Payload_t](#) *Payload*
Command payload.

10.139.1 Detailed Description

Delete Critical Table CDS Command.

Definition at line 115 of file [default_cfe_tbl_msgstruct.h](#).

10.139.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) *CFE_TBL_DeleteCDSCmd::CommandHeader*
Command header.

Definition at line 117 of file [default_cfe_tbl_msgstruct.h](#).

Payload

[CFE_TBL_DelCDSCmd_Payload_t](#) *CFE_TBL_DeleteCDSCmd::Payload*
Command payload.

Definition at line 118 of file [default_cfe_tbl_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/tbl/config/default_cfe_tbl_msgstruct.h](#)

10.140 CFE_TBL_DumpCmd Struct Reference

```
#include <default_cfe_tbl_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) *CommandHeader*
Command header.
- [CFE_TBL_DumpCmd_Payload_t](#) *Payload*
Command payload.

10.140.1 Detailed Description

/brief Dump Table Command

Definition at line 70 of file [default_cfe_tbl_msgstruct.h](#).

10.140.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) *CFE_TBL_DumpCmd::CommandHeader*
Command header.

Definition at line 72 of file [default_cfe_tbl_msgstruct.h](#).

Payload

[CFE_TBL_DumpCmd_Payload_t](#) CFE_TBL_DumpCmd::Payload

Command payload.

Definition at line 73 of file [default_cfe_tbl_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/tbl/config/default_cfe_tbl_msgstruct.h](#)

10.141 CFE_TBL_DumpCmd_Payload Struct Reference

Dump Table Command Payload.

```
#include <default_cfe_tbl_msgdefs.h>
```

Data Fields

- [CFE_TBL_BufferSelect_Enum_t](#) ActiveTableFlag
CFE_TBL_BufferSelect_INACTIVE=Inactive Table, CFE_TBL_BufferSelect_ACTIVE=Active Table
- char [TableName](#) [[CFE_MISSION_TBL_MAX_FULL_NAME_LEN](#)]
Full name of table to be dumped.
- char [DumpFilename](#) [[CFE_MISSION_MAX_PATH_LEN](#)]
Full Filename where data is to be written.

10.141.1 Detailed Description

Dump Table Command Payload.

For command details, see [CFE_TBL_DUMP_CC](#)

Definition at line 54 of file [default_cfe_tbl_msgdefs.h](#).

10.141.2 Field Documentation

ActiveTableFlag

[CFE_TBL_BufferSelect_Enum_t](#) CFE_TBL_DumpCmd_Payload::ActiveTableFlag

[CFE_TBL_BufferSelect_INACTIVE](#)=Inactive Table, [CFE_TBL_BufferSelect_ACTIVE](#)=Active Table

Selects either the "Inactive" ([CFE_TBL_BufferSelect_INACTIVE](#)) buffer or the "Active" ([CFE_TBL_BufferSelect_ACTIVE](#)) buffer to be dumped

Definition at line 56 of file [default_cfe_tbl_msgdefs.h](#).

DumpFilename

char CFE_TBL_DumpCmd_Payload::DumpFilename [[CFE_MISSION_MAX_PATH_LEN](#)]

Full Filename where data is to be written.

ASCII string containing full path of filename where data is to be dumped

Definition at line 65 of file [default_cfe_tbl_msgdefs.h](#).

TableName

char CFE_TBL_DumpCmd_Payload::TableName [[CFE_MISSION_TBL_MAX_FULL_NAME_LEN](#)]

Full name of table to be dumped.

ASCII string containing full table name identifier of table to be dumped

Definition at line 62 of file [default_cfe_tbl_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/tbl/config/default_cfe_tbl_msgdefs.h](#)

10.142 CFE_TBL_DumpRegistryCmd Struct Reference

Dump Registry Command.

```
#include <default_cfe_tbl_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) **CommandHeader**
Command header.
- [CFE_TBL_DumpRegistryCmd_Payload_t](#) **Payload**
Command payload.

10.142.1 Detailed Description

Dump Registry Command.

Definition at line 97 of file [default_cfe_tbl_msgstruct.h](#).

10.142.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_TBL_DumpRegistryCmd::CommandHeader

Command header.

Definition at line 99 of file [default_cfe_tbl_msgstruct.h](#).

Payload

[CFE_TBL_DumpRegistryCmd_Payload_t](#) CFE_TBL_DumpRegistryCmd::Payload

Command payload.

Definition at line 100 of file [default_cfe_tbl_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/tbl/config/default_cfe_tbl_msgstruct.h](#)

10.143 CFE_TBL_DumpRegistryCmd_Payload Struct Reference

Dump Registry Command Payload.

```
#include <default_cfe_tbl_msgdefs.h>
```

Data Fields

- char [DumpFilename](#) [[CFE_MISSION_MAX_PATH_LEN](#)]
Full Filename where dumped data is to be written.

10.143.1 Detailed Description

Dump Registry Command Payload.

For command details, see [CFE_TBL_DUMP_REGISTRY_CC](#)

Definition at line 105 of file [default_cfe_tbl_msgdefs.h](#).

10.143.2 Field Documentation

DumpFilename

char CFE_TBL_DumpRegistryCmd_Payload::DumpFilename [[CFE_MISSION_MAX_PATH_LEN](#)]

Full Filename where dumped data is to be written.

ASCII string containing full path of filename where registry is to be dumped

Definition at line 107 of file [default_cfe_tbl_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/tbl/config/default_cfe_tbl_msgdefs.h](#)

10.144 CFE_TBL_File_Hdr Struct Reference

The definition of the header fields that are included in CFE Table Data files.

```
#include <default_cfe_tbl_extern_typedefs.h>
```

Data Fields

- [uint32 Reserved](#)
- [uint32 Offset](#)
- [uint32 NumBytes](#)
- [char TableName](#) [[CFE_MISSION_TBL_MAX_FULL_NAME_LEN](#)]

10.144.1 Detailed Description

The definition of the header fields that are included in CFE Table Data files.

This header follows the CFE_FS header and precedes the actual table data.

Note

The Offset and NumBytes fields in the table header are to 32 bits for backward compatibility with existing CFE versions. This means that even on 64-bit CPUs, individual table files will be limited to 4GiB in size.

Definition at line 65 of file [default_cfe_tbl_extern_typedefs.h](#).

10.144.2 Field Documentation

NumBytes

```
uint32 CFE_TBL_File_Hdr::NumBytes
```

Number of bytes to load into table

Definition at line 69 of file [default_cfe_tbl_extern_typedefs.h](#).

Offset

```
uint32 CFE_TBL_File_Hdr::Offset
```

Byte Offset at which load should commence

Definition at line 68 of file [default_cfe_tbl_extern_typedefs.h](#).

Reserved

```
uint32 CFE_TBL_File_Hdr::Reserved
```

Future Use: NumTblSegments in File?

Definition at line 67 of file [default_cfe_tbl_extern_typedefs.h](#).

TableName

```
char CFE_TBL_File_Hdr::TableName[CFE_MISSION_TBL_MAX_FULL_NAME_LEN]
```

Fully qualified name of table to load

Definition at line 70 of file [default_cfe_tbl_extern_typedefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/tbl/config/default_cfe_tbl_extern_typedefs.h](#)

10.145 CFE_TBL_FileDef Struct Reference

Table File summary object.

```
#include <cfe_tbl_filedef.h>
```

Data Fields

- char [ObjectName](#) [64]
Name of instantiated variable that contains desired table image.
- char [TableName](#) [CFE_MISSION_TBL_MAX_FULL_NAME_LEN]
Name of Table as defined onboard.
- char [Description](#) [CFE_FS_HDR_DESC_MAX_LEN]
Description of table image that is included in cFE File Header.
- char [TgtFilename](#) [CFE_MISSION_MAX_FILE_LEN]
Default filename to be used for output of elf2cfetbl utility.
- [uint32](#) [ObjectSize](#)
Size, in bytes, of instantiated object.

10.145.1 Detailed Description

Table File summary object.

The definition of the file definition metadata that can be used by external tools (e.g. elf2cfetbl) to generate CFE table data files.

Definition at line 58 of file [cfe_tbl_filedef.h](#).

10.145.2 Field Documentation

Description

```
char CFE_TBL_FileDef::Description[CFE_FS_HDR_DESC_MAX_LEN]
```

Description of table image that is included in cFE File Header.

This is a free-form text string that can be any meaningful value

Definition at line 94 of file [cfe_tbl_filedef.h](#).

ObjectName

```
char CFE_TBL_FileDef::ObjectName[64]
```

Name of instantiated variable that contains desired table image.

Note

For consistency and future compatibility with auto-generated table files and table definitions, the "ObjectName" should match the table struct typedef name without the "_t" suffix. For example, the limit checker action table (ADT) is defined by a type called "LC_ADT_t", the ObjectName should be "LC_ADT".

This naming convention allows the type name to be inferred from the ObjectName (and vice-versa) without having to directly specify both the type name and object name here.

Although the traditional elf2cfetbl tool does not currently do any type checking, future tool versions may add more robust type verification and therefore need to know the type name as well as the object name.

Definition at line 76 of file [cfe_tbl_filedef.h](#).

ObjectSize

`uint32 CFE_TBL_FileDef::ObjectSize`

Size, in bytes, of instantiated object.

This may be used by tools to check for consistency between the actual defined table size and the expected table size.

This is set automatically via the `CFE_TBL_FILEDEF` macro.

Definition at line 112 of file `cfe_tbl_filedef.h`.

TableName

`char CFE_TBL_FileDef::TableName[CFE_MISSION_TBL_MAX_FULL_NAME_LEN]`

Name of Table as defined onboard.

This should be in the form of "APP_NAME.TABLE_NAME" where APP_NAME matches what the app is named at runtime (the 4th column of `cfe_es_startup.scr`) and TABLE_NAME matches the 2nd parameter of the call to `CFE_TBL_Register()`. Preferably the TABLE_NAME should also match the ObjectName here in this structure, although this is not strictly required, it helps keep things consistent.

Definition at line 87 of file `cfe_tbl_filedef.h`.

TgtFilename

`char CFE_TBL_FileDef::TgtFilename[CFE_MISSION_MAX_FILE_LEN]`

Default filename to be used for output of `elf2cfetbl` utility.

This must match the expected table file name, which is the name of the source file but the ".c" extension replaced with ".tbl". This is the filename only - do not include a directory/path name here, it can be copied to any runtime directory on the target by external scripts, but should not be renamed.

Definition at line 104 of file `cfe_tbl_filedef.h`.

The documentation for this struct was generated from the following file:

- `cfe/modules/core_api/fsw/inc/cfe_tbl_filedef.h`

10.146 CFE_TBL_HousekeepingTlm Struct Reference

```
#include <default_cfe_tbl_msgstruct.h>
```

Data Fields

- `CFE_MSG_TelemetryHeader_t` TelemetryHeader
Telemetry header.
- `CFE_TBL_HousekeepingTlm_Payload_t` Payload
Telemetry payload.

10.146.1 Detailed Description

Definition at line 147 of file `default_cfe_tbl_msgstruct.h`.

10.146.2 Field Documentation

Payload

`CFE_TBL_HousekeepingTlm_Payload_t` CFE_TBL_HousekeepingTlm::Payload

Telemetry payload.

Definition at line 150 of file `default_cfe_tbl_msgstruct.h`.

TelemetryHeader

`CFE_MSG_TelemetryHeader_t` CFE_TBL_HousekeepingTlm::TelemetryHeader

Telemetry header.

Definition at line 149 of file `default_cfe_tbl_msgstruct.h`.

The documentation for this struct was generated from the following file:

- `cfe/modules/tbl/config/default_cfe_tbl_msgstruct.h`

10.147 CFE_TBL_HousekeepingTlm_Payload Struct Reference

```
#include <default_cfe_tbl_msgdefs.h>
```

Data Fields

- `uint8 CommandCounter`
Count of valid commands received.
- `uint8 CommandErrorCounter`
Count of invalid commands received.
- `uint16 NumTables`
Number of Tables Registered.
- `uint16 NumLoadPending`
Number of Tables pending on Applications for their update.
- `uint16 ValidationCounter`
Number of completed table validations.
- `uint32 LastValCrc`
Data Integrity Value computed for last table validated.
- `int32 LastValStatus`
Returned status from validation function for last table validated.
- `bool ActiveBuffer`
Indicator of whether table buffer validated was 0=Inactive, 1=Active.
- `char LastValTableName [CFE_MISSION_TBL_MAX_FULL_NAME_LEN]`
Name of last table validated.
- `uint8 SuccessValCounter`
Total number of successful table validations.
- `uint8 FailedValCounter`
Total number of unsuccessful table validations.
- `uint8 NumValRequests`
Number of times Table Services has requested validations from Apps.
- `uint8 NumFreeSharedBufs`
Number of free Shared Working Buffers.
- `uint8 ByteAlignPad1`
Spare byte to ensure longword alignment.
- `CFE_ES_MemHandle_t MemPoolHandle`
Handle to TBL's memory pool.
- `CFE_TIME_SysTime_t LastUpdateTime`
Time of last table update.
- `char LastUpdatedTable [CFE_MISSION_TBL_MAX_FULL_NAME_LEN]`
Name of the last table updated.
- `char LastFileLoaded [CFE_MISSION_MAX_PATH_LEN]`

Path and Name of last table image file loaded.

- char [LastFileDumped](#) [CFE_MISSION_MAX_PATH_LEN]

Path and Name of last file dumped to.

- char [LastTableLoaded](#) [CFE_MISSION_TBL_MAX_FULL_NAME_LEN]

Name of the last table loaded.

10.147.1 Detailed Description

Name Table Services Housekeeping Packet

Definition at line 177 of file [default_cfe_tbl_msgdefs.h](#).

10.147.2 Field Documentation

ActiveBuffer

`bool CFE_TBL_HousekeepingTlm_Payload::ActiveBuffer`

Indicator of whether table buffer validated was 0=Inactive, 1=Active.

Telemetry Mnemonic(s) \$sc_\$cpu_TBL_LastValBuf

Definition at line 204 of file [default_cfe_tbl_msgdefs.h](#).

ByteAlignPad1

`uint8 CFE_TBL_HousekeepingTlm_Payload::ByteAlignPad1`

Spare byte to ensure longword alignment.

Telemetry Mnemonic(s) \$sc_\$cpu_TBL_ByteAlignPad1

Definition at line 220 of file [default_cfe_tbl_msgdefs.h](#).

CommandCounter

`uint8 CFE_TBL_HousekeepingTlm_Payload::CommandCounter`

Count of valid commands received.

Telemetry Mnemonic(s) \$sc_\$cpu_TBL_CMDPC

Definition at line 182 of file [default_cfe_tbl_msgdefs.h](#).

CommandErrorCounter

`uint8 CFE_TBL_HousekeepingTlm_Payload::CommandErrorCounter`

Count of invalid commands received.

Telemetry Mnemonic(s) \$sc_\$cpu_TBL_CMDEC

Definition at line 184 of file [default_cfe_tbl_msgdefs.h](#).

FailedValCounter

`uint8 CFE_TBL_HousekeepingTlm_Payload::FailedValCounter`

Total number of unsuccessful table validations.

Telemetry Mnemonic(s) \$sc_\$cpu_TBL_ValFailedCtr

Definition at line 210 of file [default_cfe_tbl_msgdefs.h](#).

LastFileDumped

char CFE_TBL_HousekeepingTlm_Payload::LastFileDumped[CFE_MISSION_MAX_PATH_LEN]
 Path and Name of last file dumped to.

Telemetry Mnemonic(s) \$sc_\$cpu_TBL_LastFileDumped[OS_MAX_PATH_LEN]

Definition at line 230 of file [default_cfe_tbl_msgdefs.h](#).

LastFileLoaded

char CFE_TBL_HousekeepingTlm_Payload::LastFileLoaded[CFE_MISSION_MAX_PATH_LEN]
 Path and Name of last table image file loaded.

Telemetry Mnemonic(s) \$sc_\$cpu_TBL_LastFileLoaded[OS_MAX_PATH_LEN]

Definition at line 228 of file [default_cfe_tbl_msgdefs.h](#).

LastTableLoaded

char CFE_TBL_HousekeepingTlm_Payload::LastTableLoaded[CFE_MISSION_TBL_MAX_FULL_NAME_LEN]
 Name of the last table loaded.

Telemetry Mnemonic(s) \$sc_\$cpu_TBL_LastTableLoaded[CFE_TBL_MAX_FULL_NAME_LEN]

Definition at line 232 of file [default_cfe_tbl_msgdefs.h](#).

LastUpdatedTable

char CFE_TBL_HousekeepingTlm_Payload::LastUpdatedTable[CFE_MISSION_TBL_MAX_FULL_NAME_LEN]
 Name of the last table updated.

Telemetry Mnemonic(s) \$sc_\$cpu_TBL_LastUpdTblName[CFE_TB_MAX_FULL_NAME_LEN]

Definition at line 226 of file [default_cfe_tbl_msgdefs.h](#).

LastUpdateTime

CFE_TIME_SysTime_t CFE_TBL_HousekeepingTlm_Payload::LastUpdateTime
 Time of last table update.

Telemetry Mnemonic(s) \$sc_\$cpu_TBL_LastUpdTime, \$sc_\$cpu_TBL_SECONDS, \$sc_\$cpu_TBL_SUBSECONDS

Definition at line 224 of file [default_cfe_tbl_msgdefs.h](#).

LastValCrc

uint32 CFE_TBL_HousekeepingTlm_Payload::LastValCrc
 Data Integrity Value computed for last table validated.

Telemetry Mnemonic(s) \$sc_\$cpu_TBL_LastValCRC

Definition at line 200 of file [default_cfe_tbl_msgdefs.h](#).

LastValStatus

int32 CFE_TBL_HousekeepingTlm_Payload::LastValStatus
 Returned status from validation function for last table validated.

Telemetry Mnemonic(s) \$sc_\$cpu_TBI_LastValS

Definition at line 202 of file [default_cfe_tbl_msgdefs.h](#).

LastValTableName

`char CFE_TBL_HousekeepingTlm_Payload::LastValTableName[CFE_MISSION_TBL_MAX_FULL_NAME_LEN]`
Name of last table validated.

Telemetry Mnemonic(s) \$sc_\$cpu_TBL_LastValTblName[CFE_TB_MAX_FULL_NAME_LEN]

Definition at line 206 of file [default_cfe_tbl_msgdefs.h](#).

MemPoolHandle

`CFE_ES_MemHandle_t CFE_TBL_HousekeepingTlm_Payload::MemPoolHandle`
Handle to TBL's memory pool.

Telemetry Mnemonic(s) \$sc_\$cpu_TBL_MemPoolHandle

Definition at line 222 of file [default_cfe_tbl_msgdefs.h](#).

NumFreeSharedBufs

`uint8 CFE_TBL_HousekeepingTlm_Payload::NumFreeSharedBufs`
Number of free Shared Working Buffers.

Telemetry Mnemonic(s) \$sc_\$cpu_TBL_NumFreeShrBuf

Definition at line 218 of file [default_cfe_tbl_msgdefs.h](#).

NumLoadPending

`uint16 CFE_TBL_HousekeepingTlm_Payload::NumLoadPending`
Number of Tables pending on Applications for their update.

Telemetry Mnemonic(s) \$sc_\$cpu_TBL_NumUpdatesPend

Definition at line 192 of file [default_cfe_tbl_msgdefs.h](#).

NumTables

`uint16 CFE_TBL_HousekeepingTlm_Payload::NumTables`
Number of Tables Registered.

Telemetry Mnemonic(s) \$sc_\$cpu_TBL_NumTables

Definition at line 190 of file [default_cfe_tbl_msgdefs.h](#).

NumValRequests

`uint8 CFE_TBL_HousekeepingTlm_Payload::NumValRequests`
Number of times Table Services has requested validations from Apps.

Telemetry Mnemonic(s) \$sc_\$cpu_TBL_ValReqCtr

Definition at line 212 of file [default_cfe_tbl_msgdefs.h](#).

SuccessValCounter

`uint8 CFE_TBL_HousekeepingTlm_Payload::SuccessValCounter`
Total number of successful table validations.

Telemetry Mnemonic(s) \$sc_\$cpu_TBL_ValSuccessCtr

Definition at line 208 of file [default_cfe_tbl_msgdefs.h](#).

ValidationCounter

`uint16` CFE_TBL_HousekeepingTlm_Payload::ValidationCounter
 Number of completed table validations.

Telemetry Mnemonic(s) \$sc_\$cpu_TBL_ValCompltdCtr

Definition at line 198 of file `default_cfe_tbl_msgdefs.h`.
 The documentation for this struct was generated from the following file:

- `cfe/modules/tbl/config/default_cfe_tbl_msgdefs.h`

10.148 CFE_TBL_Info Struct Reference

Table Info.

```
#include <cfe_tbl_api_typedefs.h>
```

Data Fields

- `size_t` [Size](#)
Size, in bytes, of Table.
- `uint32` [NumUsers](#)
Number of Apps with access to the table.
- `CFE_TIME_SysTime_t` [FileTime](#)
File creation time from last file loaded into table.
- `uint32` [Crc](#)
Most recently calculated CRC by TBL services on table contents.
- `CFE_TIME_SysTime_t` [TimeOfLastUpdate](#)
Time when Table was last updated.
- `bool` [TableLoadedOnce](#)
Flag indicating whether table has been loaded once or not.
- `bool` [DumpOnly](#)
Flag indicating Table is NOT to be loaded.
- `bool` [DoubleBuffered](#)
Flag indicating Table has a dedicated inactive buffer.
- `bool` [UserDefAddr](#)
Flag indicating Table address was defined by Owner Application.
- `bool` [Critical](#)
Flag indicating Table contents are maintained in a CDS.
- `char` [LastFileLoaded](#) [[CFE_MISSION_MAX_PATH_LEN](#)]
Filename of last file loaded into table.

10.148.1 Detailed Description

Table Info.

Definition at line 104 of file `cfe_tbl_api_typedefs.h`.

10.148.2 Field Documentation**Crc**

`uint32` CFE_TBL_Info::Crc
 Most recently calculated CRC by TBL services on table contents.
 Definition at line 109 of file `cfe_tbl_api_typedefs.h`.

Critical

```
bool CFE_TBL_Info::Critical
```

Flag indicating Table contents are maintained in a CDS.

Definition at line 115 of file [cfe_tbl_api_typedefs.h](#).

DoubleBuffered

```
bool CFE_TBL_Info::DoubleBuffered
```

Flag indicating Table has a dedicated inactive buffer.

Definition at line 113 of file [cfe_tbl_api_typedefs.h](#).

DumpOnly

```
bool CFE_TBL_Info::DumpOnly
```

Flag indicating Table is NOT to be loaded.

Definition at line 112 of file [cfe_tbl_api_typedefs.h](#).

FileTime

```
CFE_TIME_SysTime_t CFE_TBL_Info::FileTime
```

File creation time from last file loaded into table.

Definition at line 108 of file [cfe_tbl_api_typedefs.h](#).

LastFileLoaded

```
char CFE_TBL_Info::LastFileLoaded[CFE_MISSION_MAX_PATH_LEN]
```

Filename of last file loaded into table.

Definition at line 116 of file [cfe_tbl_api_typedefs.h](#).

NumUsers

```
uint32 CFE_TBL_Info::NumUsers
```

Number of Apps with access to the table.

Definition at line 107 of file [cfe_tbl_api_typedefs.h](#).

Size

```
size_t CFE_TBL_Info::Size
```

Size, in bytes, of Table.

Definition at line 106 of file [cfe_tbl_api_typedefs.h](#).

TableLoadedOnce

```
bool CFE_TBL_Info::TableLoadedOnce
```

Flag indicating whether table has been loaded once or not.

Definition at line 111 of file [cfe_tbl_api_typedefs.h](#).

TimeOfLastUpdate

```
CFE_TIME_SysTime_t CFE_TBL_Info::TimeOfLastUpdate
```

Time when Table was last updated.

Definition at line 110 of file [cfe_tbl_api_typedefs.h](#).

UserDefAddr

```
bool CFE_TBL_Info::UserDefAddr
```

Flag indicating Table address was defined by Owner Application.

Definition at line 114 of file [cfe_tbl_api_typedefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/core_api/fsw/inc/cfe_tbl_api_typedefs.h](#)

10.149 CFE_TBL_LoadCmd Struct Reference

Load Table Command.

```
#include <default_cfe_tbl_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.
- [CFE_TBL_LoadCmd_Payload_t](#) Payload
Command payload.

10.149.1 Detailed Description

Load Table Command.

Definition at line 61 of file [default_cfe_tbl_msgstruct.h](#).

10.149.2 Field Documentation**CommandHeader**

```
CFE_MSG_CommandHeader_t CFE_TBL_LoadCmd::CommandHeader
```

Command header.

Definition at line 63 of file [default_cfe_tbl_msgstruct.h](#).

Payload

```
CFE_TBL_LoadCmd_Payload_t CFE_TBL_LoadCmd::Payload
```

Command payload.

Definition at line 64 of file [default_cfe_tbl_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/tbl/config/default_cfe_tbl_msgstruct.h](#)

10.150 CFE_TBL_LoadCmd_Payload Struct Reference

Load Table Command Payload.

```
#include <default_cfe_tbl_msgdefs.h>
```

Data Fields

- char [LoadFilename](#) [[CFE_MISSION_MAX_PATH_LEN](#)]
Filename (and path) of data to be loaded.

10.150.1 Detailed Description

Load Table Command Payload.

For command details, see [CFE_TBL_LOAD_CC](#)

Definition at line 44 of file [default_cfe_tbl_msgdefs.h](#).

10.150.2 Field Documentation

LoadFilename

```
char CFE_TBL_LoadCmd_Payload::LoadFilename[CFE_MISSION_MAX_PATH_LEN]
```

Filename (and path) of data to be loaded.

Definition at line 46 of file [default_cfe_tbl_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/tbl/config/default_cfe_tbl_msgdefs.h](#)

10.151 CFE_TBL_NoopCmd Struct Reference

```
#include <default_cfe_tbl_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) *CommandHeader*
Command header.

10.151.1 Detailed Description

Definition at line 43 of file [default_cfe_tbl_msgstruct.h](#).

10.151.2 Field Documentation

CommandHeader

```
CFE_MSG_CommandHeader_t CFE_TBL_NoopCmd::CommandHeader
```

Command header.

Definition at line 45 of file [default_cfe_tbl_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/tbl/config/default_cfe_tbl_msgstruct.h](#)

10.152 CFE_TBL_NotifyCmd Struct Reference

```
#include <default_cfe_tbl_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) *CommandHeader*
Command header.
- [CFE_TBL_NotifyCmd_Payload_t](#) *Payload*
Command payload.

10.152.1 Detailed Description

/brief Table Management Notification Command

Definition at line 135 of file [default_cfe_tbl_msgstruct.h](#).

10.152.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) [CFE_TBL_NotifyCmd::CommandHeader](#)

Command header.

Definition at line 137 of file [default_cfe_tbl_msgstruct.h](#).

Payload

[CFE_TBL_NotifyCmd_Payload_t](#) [CFE_TBL_NotifyCmd::Payload](#)

Command payload.

Definition at line 138 of file [default_cfe_tbl_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/tbl/config/default_cfe_tbl_msgstruct.h](#)

10.153 CFE_TBL_NotifyCmd_Payload Struct Reference

Table Management Notification Command Payload.

```
#include <default_cfe_tbl_msgdefs.h>
```

Data Fields

- [uint32](#) [Parameter](#)

Application specified command parameter.

10.153.1 Detailed Description

Table Management Notification Command Payload.

Description

Whenever an application that owns a table calls the [CFE_TBL_NotifyByMessage](#) API following the table registration, Table services will generate the following command message with the application specified message ID, command code and parameter whenever the table requires management (e.g. - loads and validations).

Definition at line 164 of file [default_cfe_tbl_msgdefs.h](#).

10.153.2 Field Documentation

Parameter

[uint32](#) [CFE_TBL_NotifyCmd_Payload::Parameter](#)

Application specified command parameter.

Definition at line 166 of file [default_cfe_tbl_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/tbl/config/default_cfe_tbl_msgdefs.h](#)

10.154 CFE_TBL_ResetCountersCmd Struct Reference

```
#include <default_cfe_tbl_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) [CommandHeader](#)

Command header.

10.154.1 Detailed Description

Definition at line 48 of file [default_cfe_tbl_msgstruct.h](#).

10.154.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_TBL_ResetCountersCmd::CommandHeader

Command header.

Definition at line 50 of file [default_cfe_tbl_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/tbl/config/default_cfe_tbl_msgstruct.h](#)

10.155 CFE_TBL_SendHkCmd Struct Reference

```
#include <default_cfe_tbl_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.

10.155.1 Detailed Description

Definition at line 53 of file [default_cfe_tbl_msgstruct.h](#).

10.155.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_TBL_SendHkCmd::CommandHeader

Command header.

Definition at line 55 of file [default_cfe_tbl_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/tbl/config/default_cfe_tbl_msgstruct.h](#)

10.156 CFE_TBL_SendRegistryCmd Struct Reference

Send Table Registry Command.

```
#include <default_cfe_tbl_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.
- [CFE_TBL_SendRegistryCmd_Payload_t](#) Payload
Command payload.

10.156.1 Detailed Description

Send Table Registry Command.

Definition at line 106 of file [default_cfe_tbl_msgstruct.h](#).

10.156.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_TBL_SendRegistryCmd::CommandHeader

Command header.

Definition at line 108 of file [default_cfe_tbl_msgstruct.h](#).

Payload

[CFE_TBL_SendRegistryCmd_Payload_t](#) CFE_TBL_SendRegistryCmd::Payload

Command payload.

Definition at line 109 of file [default_cfe_tbl_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/tbl/config/default_cfe_tbl_msgstruct.h](#)

10.157 CFE_TBL_SendRegistryCmd_Payload Struct Reference

Send Table Registry Command Payload.

#include <[default_cfe_tbl_msgdefs.h](#)>

Data Fields

- char [TableName](#) [[CFE_MISSION_TBL_MAX_FULL_NAME_LEN](#)]
Full Name of Table whose registry entry is to be telemetered.

10.157.1 Detailed Description

Send Table Registry Command Payload.

For command details, see [CFE_TBL_SEND_REGISTRY_CC](#)

Definition at line 118 of file [default_cfe_tbl_msgdefs.h](#).

10.157.2 Field Documentation

TableName

char CFE_TBL_SendRegistryCmd_Payload::TableName [[CFE_MISSION_TBL_MAX_FULL_NAME_LEN](#)]

Full Name of Table whose registry entry is to be telemetered.

ASCII string containing full table name identifier of table whose registry entry is to be telemetered via [CFE_TBL_TableRegistryTlm_t](#)

Definition at line 120 of file [default_cfe_tbl_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/tbl/config/default_cfe_tbl_msgdefs.h](#)

10.158 CFE_TBL_TableRegistryTlm Struct Reference

#include <[default_cfe_tbl_msgstruct.h](#)>

Data Fields

- [CFE_MSG_TelemetryHeader_t](#) TelemetryHeader
Telemetry header.
- [CFE_TBL_TblRegPacket_Payload_t](#) Payload
Telemetry payload.

10.158.1 Detailed Description

Definition at line 153 of file [default_cfe_tbl_msgstruct.h](#).

10.158.2 Field Documentation

Payload

[CFE_TBL_TblRegPacket_Payload_t](#) CFE_TBL_TableRegistryTlm::Payload

Telemetry payload.

Definition at line 156 of file [default_cfe_tbl_msgstruct.h](#).

TelemetryHeader

[CFE_MSG_TelemetryHeader_t](#) CFE_TBL_TableRegistryTlm::TelemetryHeader

Telemetry header.

Definition at line 155 of file [default_cfe_tbl_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/tbl/config/default_cfe_tbl_msgstruct.h](#)

10.159 CFE_TBL_TblRegPacket_Payload Struct Reference

```
#include <default_cfe_tbl_msgdefs.h>
```

Data Fields

- [CFE_ES_MemOffset_t](#) Size
Size, in bytes, of Table.
- [uint32](#) Crc
Most recently calculated CRC of Table.
- [uint8](#) Padding1 [4]
- [CFE_ES_MemAddress_t](#) ActiveBufferAddr
Address of Active Buffer.
- [CFE_ES_MemAddress_t](#) InactiveBufferAddr
Address of Inactive Buffer.
- [CFE_ES_MemAddress_t](#) ValidationFuncPtr
Ptr to Owner App's function that validates tbl contents.
- [CFE_TIME_SysTime_t](#) TimeOfLastUpdate
Time when Table was last updated.
- [CFE_TIME_SysTime_t](#) FileTime
File creation time from last file loaded into table.
- bool [TableLoadedOnce](#)
Flag indicating whether table has been loaded once or not.
- bool [LoadPending](#)
Flag indicating an inactive buffer is ready to be copied.
- bool [DumpOnly](#)
Flag indicating Table is NOT to be loaded.
- bool [DoubleBuffered](#)
Flag indicating Table has a dedicated inactive buffer.
- char [Name](#) [[CFE_MISSION_TBL_MAX_FULL_NAME_LEN](#)]
Processor specific table name.

- char [LastFileLoaded](#) [CFE_MISSION_MAX_PATH_LEN]
Filename of last file loaded into table.
- char [OwnerAppName](#) [CFE_MISSION_MAX_API_LEN]
Name of owning application.
- bool [Critical](#)
Indicates whether table is Critical or not.
- uint8 [Padding2](#) [7]

10.159.1 Detailed Description

Name Table Registry Info Packet

Definition at line 239 of file [default_cfe_tbl_msgdefs.h](#).

10.159.2 Field Documentation

ActiveBufferAddr

[CFE_ES_MemAddress_t](#) CFE_TBL_TblRegPacket_Payload::ActiveBufferAddr
Address of Active Buffer.

Telemetry Mnemonic(s) \$sc_\$cpu_TBL_ActBufAdd

Definition at line 246 of file [default_cfe_tbl_msgdefs.h](#).

Crc

[uint32](#) CFE_TBL_TblRegPacket_Payload::Crc
Most recently calculated CRC of Table.

Telemetry Mnemonic(s) \$sc_\$cpu_TBL_CRC

Definition at line 243 of file [default_cfe_tbl_msgdefs.h](#).

Critical

[bool](#) CFE_TBL_TblRegPacket_Payload::Critical
Indicates whether table is Critical or not.

Telemetry Mnemonic(s) \$sc_\$cpu_TBL_Spare3

Definition at line 270 of file [default_cfe_tbl_msgdefs.h](#).

DoubleBuffered

[bool](#) CFE_TBL_TblRegPacket_Payload::DoubleBuffered
Flag indicating Table has a dedicated inactive buffer.

Telemetry Mnemonic(s) \$sc_\$cpu_TBL_DblBuffered

Definition at line 262 of file [default_cfe_tbl_msgdefs.h](#).

DumpOnly

[bool](#) CFE_TBL_TblRegPacket_Payload::DumpOnly
Flag indicating Table is NOT to be loaded.

Telemetry Mnemonic(s) \$sc_\$cpu_TBL_DumpOnly

Definition at line 260 of file [default_cfe_tbl_msgdefs.h](#).

FileTime

`CFE_TIME_SysTime_t CFE_TBL_TblRegPacket_Payload::FileTime`

File creation time from last file loaded into table.

Telemetry Mnemonic(s) \$sc_\$cpu_TBL_FILECTIME

Definition at line 254 of file [default_cfe_tbl_msgdefs.h](#).

InactiveBufferAddr

`CFE_ES_MemAddress_t CFE_TBL_TblRegPacket_Payload::InactiveBufferAddr`

Address of Inactive Buffer.

Telemetry Mnemonic(s) \$sc_\$cpu_TBL_IActBufAdd

Definition at line 248 of file [default_cfe_tbl_msgdefs.h](#).

LastFileLoaded

`char CFE_TBL_TblRegPacket_Payload::LastFileLoaded[CFE_MISSION_MAX_PATH_LEN]`

Filename of last file loaded into table.

Telemetry Mnemonic(s) \$sc_\$cpu_TBL_LastFileUpd[OS_MAX_PATH_LEN]

Definition at line 266 of file [default_cfe_tbl_msgdefs.h](#).

LoadPending

`bool CFE_TBL_TblRegPacket_Payload::LoadPending`

Flag indicating an inactive buffer is ready to be copied.

Telemetry Mnemonic(s) \$sc_\$cpu_TBL_UpdatePndng

Definition at line 258 of file [default_cfe_tbl_msgdefs.h](#).

Name

`char CFE_TBL_TblRegPacket_Payload::Name[CFE_MISSION_TBL_MAX_FULL_NAME_LEN]`

Processor specific table name.

Telemetry Mnemonic(s) \$sc_\$cpu_TBL_Name[CFE_TB_MAX_FULL_NAME_LEN]

Definition at line 264 of file [default_cfe_tbl_msgdefs.h](#).

OwnerAppName

`char CFE_TBL_TblRegPacket_Payload::OwnerAppName[CFE_MISSION_MAX_API_LEN]`

Name of owning application.

Telemetry Mnemonic(s) \$sc_\$cpu_TBL_OwnerApp[OS_MAX_API_NAME]

Definition at line 268 of file [default_cfe_tbl_msgdefs.h](#).

Padding1

`uint8 CFE_TBL_TblRegPacket_Payload::Padding1[4]`

Definition at line 245 of file [default_cfe_tbl_msgdefs.h](#).

Padding2

`uint8 CFE_TBL_TblRegPacket_Payload::Padding2[7]`
 Definition at line 272 of file [default_cfe_tbl_msgdefs.h](#).

Size

`CFE_ES_MemOffset_t CFE_TBL_TblRegPacket_Payload::Size`
 Size, in bytes, of Table.

Telemetry Mnemonic(s) `$sc_$cpu_TBL_SIZE`

Definition at line 241 of file [default_cfe_tbl_msgdefs.h](#).

TableLoadedOnce

`bool CFE_TBL_TblRegPacket_Payload::TableLoadedOnce`
 Flag indicating whether table has been loaded once or not.

Telemetry Mnemonic(s) `$sc_$cpu_TBL_LoadedOnce`

Definition at line 256 of file [default_cfe_tbl_msgdefs.h](#).

TimeOfLastUpdate

`CFE_TIME_SysTime_t CFE_TBL_TblRegPacket_Payload::TimeOfLastUpdate`
 Time when Table was last updated.

Telemetry Mnemonic(s) `$sc_$cpu_TBL_TimeLastUpd, $sc_$cpu_TBL_TLUSECONDS, $sc_$cpu_TBL_TLUSUBSECONDS`

Definition at line 252 of file [default_cfe_tbl_msgdefs.h](#).

ValidationFuncPtr

`CFE_ES_MemAddress_t CFE_TBL_TblRegPacket_Payload::ValidationFuncPtr`
 Ptr to Owner App's function that validates tbl contents.

Telemetry Mnemonic(s) `$sc_$cpu_TBL_ValFuncPtr`

Definition at line 250 of file [default_cfe_tbl_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/tbl/config/default_cfe_tbl_msgdefs.h](#)

10.160 CFE_TBL_ValidateCmd Struct Reference

Validate Table Command.

```
#include <default_cfe_tbl_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) *CommandHeader*
Command header.
- [CFE_TBL_ValidateCmd_Payload_t](#) *Payload*
Command payload.

10.160.1 Detailed Description

Validate Table Command.

Definition at line 79 of file [default_cfe_tbl_msgstruct.h](#).

10.160.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_TBL_ValidateCmd::CommandHeader

Command header.

Definition at line 81 of file [default_cfe_tbl_msgstruct.h](#).

Payload

[CFE_TBL_ValidateCmd_Payload_t](#) CFE_TBL_ValidateCmd::Payload

Command payload.

Definition at line 82 of file [default_cfe_tbl_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/tbl/config/default_cfe_tbl_msgstruct.h](#)

10.161 CFE_TBL_ValidateCmd_Payload Struct Reference

Validate Table Command Payload.

```
#include <default_cfe_tbl_msgdefs.h>
```

Data Fields

- [CFE_TBL_BufferSelect_Enum_t](#) ActiveTableFlag
CFE_TBL_BufferSelect_INACTIVE=Inactive Table, CFE_TBL_BufferSelect_ACTIVE=Active Table
- char [TableName](#) [[CFE_MISSION_TBL_MAX_FULL_NAME_LEN](#)]
Full Name of Table to be validated.

10.161.1 Detailed Description

Validate Table Command Payload.

For command details, see [CFE_TBL_VALIDATE_CC](#)

Definition at line 75 of file [default_cfe_tbl_msgdefs.h](#).

10.161.2 Field Documentation

ActiveTableFlag

[CFE_TBL_BufferSelect_Enum_t](#) CFE_TBL_ValidateCmd_Payload::ActiveTableFlag

[CFE_TBL_BufferSelect_INACTIVE](#)=Inactive Table, [CFE_TBL_BufferSelect_ACTIVE](#)=Active Table

Selects either the "Inactive" ([CFE_TBL_BufferSelect_INACTIVE](#)) buffer or the "Active" ([CFE_TBL_BufferSelect_ACTIVE](#)) buffer to be validated

Definition at line 77 of file [default_cfe_tbl_msgdefs.h](#).

TableName

char CFE_TBL_ValidateCmd_Payload::TableName [[CFE_MISSION_TBL_MAX_FULL_NAME_LEN](#)]

Full Name of Table to be validated.

ASCII string containing full table name identifier of table to be validated

Definition at line 83 of file [default_cfe_tbl_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/tbl/config/default_cfe_tbl_msgdefs.h](#)

10.162 CFE_TIME_AddAdjustCmd Struct Reference

```
#include <default_cfe_time_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) *CommandHeader*
Command header.
- [CFE_TIME_TimeCmd_Payload_t](#) *Payload*
Command payload.

10.162.1 Detailed Description

Definition at line 137 of file [default_cfe_time_msgstruct.h](#).

10.162.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_TIME_AddAdjustCmd::CommandHeader
Command header.
Definition at line 139 of file [default_cfe_time_msgstruct.h](#).

Payload

[CFE_TIME_TimeCmd_Payload_t](#) CFE_TIME_AddAdjustCmd::Payload
Command payload.
Definition at line 140 of file [default_cfe_time_msgstruct.h](#).
The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgstruct.h](#)

10.163 CFE_TIME_AddDelayCmd Struct Reference

```
#include <default_cfe_time_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) *CommandHeader*
Command header.
- [CFE_TIME_TimeCmd_Payload_t](#) *Payload*
Command payload.

10.163.1 Detailed Description

Definition at line 113 of file [default_cfe_time_msgstruct.h](#).

10.163.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_TIME_AddDelayCmd::CommandHeader
Command header.
Definition at line 115 of file [default_cfe_time_msgstruct.h](#).

Payload

[CFE_TIME_TimeCmd_Payload_t](#) CFE_TIME_AddDelayCmd::Payload

Command payload.

Definition at line 116 of file [default_cfe_time_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgstruct.h](#)

10.164 CFE_TIME_AddOneHzAdjustmentCmd Struct Reference

```
#include <default_cfe_time_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.
- [CFE_TIME_OneHzAdjustmentCmd_Payload_t](#) Payload
Command payload.

10.164.1 Detailed Description

Definition at line 160 of file [default_cfe_time_msgstruct.h](#).

10.164.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_TIME_AddOneHzAdjustmentCmd::CommandHeader

Command header.

Definition at line 162 of file [default_cfe_time_msgstruct.h](#).

Payload

[CFE_TIME_OneHzAdjustmentCmd_Payload_t](#) CFE_TIME_AddOneHzAdjustmentCmd::Payload

Command payload.

Definition at line 163 of file [default_cfe_time_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgstruct.h](#)

10.165 CFE_TIME_DiagnosticTlm Struct Reference

```
#include <default_cfe_time_msgstruct.h>
```

Data Fields

- [CFE_MSG_TelemetryHeader_t](#) TelemetryHeader
Telemetry header.
- [CFE_TIME_DiagnosticTlm_Payload_t](#) Payload
Telemetry payload.

10.165.1 Detailed Description

Definition at line 189 of file [default_cfe_time_msgstruct.h](#).

10.165.2 Field Documentation

Payload

[CFE_TIME_DiagnosticTlm_Payload_t](#) CFE_TIME_DiagnosticTlm::Payload

Telemetry payload.

Definition at line 192 of file [default_cfe_time_msgstruct.h](#).

TelemetryHeader

[CFE_MSG_TelemetryHeader_t](#) CFE_TIME_DiagnosticTlm::TelemetryHeader

Telemetry header.

Definition at line 191 of file [default_cfe_time_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgstruct.h](#)

10.166 CFE_TIME_DiagnosticTlm_Payload Struct Reference

```
#include <default_cfe_time_msgdefs.h>
```

Data Fields

- [CFE_TIME_SysTime_t](#) AtToneMET
MET at time of tone.
- [CFE_TIME_SysTime_t](#) AtToneSTCF
STCF at time of tone.
- [CFE_TIME_SysTime_t](#) AtToneDelay
Adjustment for slow tone detection.
- [CFE_TIME_SysTime_t](#) AtToneLatch
Local clock latched at time of tone.
- [int16](#) AtToneLeapSeconds
Leap Seconds at time of tone.
- [CFE_TIME_ClockState_Enum_t](#) ClockStateAPI
Clock state as per API.
- [CFE_TIME_SysTime_t](#) TimeSinceTone
Time elapsed since the tone.
- [CFE_TIME_SysTime_t](#) CurrentLatch
Local clock latched just "now".
- [CFE_TIME_SysTime_t](#) CurrentMET
MET at this instant.
- [CFE_TIME_SysTime_t](#) CurrentTAI
TAI at this instant.
- [CFE_TIME_SysTime_t](#) CurrentUTC
UTC at this instant.
- [int16](#) ClockSetState
Time has been "set".
- [int16](#) ClockFlyState
Current fly-wheel state.
- [int16](#) ClockSource
Internal vs external, etc.

- [int16 ClockSignal](#)
Primary vs redundant, etc.
- [int16 ServerFlyState](#)
Used by clients only.
- [int16 Forced2Fly](#)
Commanded into fly-wheel.
- [uint16 ClockStateFlags](#)
Clock State Flags.
- [int16 OneTimeDirection](#)
One time STCF adjustment direction (Add = 1, Sub = 2)
- [int16 OneHzDirection](#)
1Hz STCF adjustment direction
- [int16 DelayDirection](#)
Client latency adjustment direction.
- [CFE_TIME_SysTime_t OneTimeAdjust](#)
Previous one-time STCF adjustment.
- [CFE_TIME_SysTime_t OneHzAdjust](#)
Current 1Hz STCF adjustment.
- [CFE_TIME_SysTime_t ToneSignalLatch](#)
Local Clock latched at most recent tone signal.
- [CFE_TIME_SysTime_t ToneDataLatch](#)
Local Clock latched at arrival of tone data.
- [uint32 ToneMatchCounter](#)
Tone signal / data verification count.
- [uint32 ToneMatchErrorCounter](#)
Tone signal / data verification error count.
- [uint32 ToneSignalCounter](#)
Tone signal detected SB message count.
- [uint32 ToneDataCounter](#)
Time at the tone data SB message count.
- [uint32 ToneIntCounter](#)
Tone signal ISR execution count.
- [uint32 ToneIntErrorCounter](#)
Tone signal ISR error count.
- [uint32 ToneTaskCounter](#)
Tone task execution count.
- [uint32 VersionCounter](#)
Count of mods to time at tone reference data (version)
- [uint32 LocalIntCounter](#)
Local 1Hz ISR execution count.
- [uint32 LocalTaskCounter](#)
Local 1Hz task execution count.
- [uint32 VirtualMET](#)
Software MET.
- [uint32 MinElapsed](#)
Min tone signal / data pkt arrival window (Sub-seconds)
- [uint32 MaxElapsed](#)

Max tone signal / data pkt arrival window (Sub-seconds)

- [CFE_TIME_SysTime_t MaxLocalClock](#)

Max local clock value before rollover.

- [uint32 ToneOverLimit](#)

Max between tone signal interrupts.

- [uint32 ToneUnderLimit](#)

Min between tone signal interrupts.

- [uint32 DataStoreStatus](#)

Data Store status (preserved across processor reset)

10.166.1 Detailed Description

Name Time Services Diagnostics Packet

Definition at line 175 of file [default_cfe_time_msgdefs.h](#).

10.166.2 Field Documentation

AtToneDelay

[CFE_TIME_SysTime_t](#) CFE_TIME_DiagnosticTlm_Payload::AtToneDelay

Adjustment for slow tone detection.

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_DLatentS, \$sc_\$cpu_TIME_DLatentSs

Definition at line 184 of file [default_cfe_time_msgdefs.h](#).

AtToneLatch

[CFE_TIME_SysTime_t](#) CFE_TIME_DiagnosticTlm_Payload::AtToneLatch

Local clock latched at time of tone.

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_DTValidS, \$sc_\$cpu_TIME_DTValidSs

Definition at line 186 of file [default_cfe_time_msgdefs.h](#).

AtToneLeapSeconds

[int16](#) CFE_TIME_DiagnosticTlm_Payload::AtToneLeapSeconds

Leap Seconds at time of tone.

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_DLeapS

Definition at line 189 of file [default_cfe_time_msgdefs.h](#).

AtToneMET

[CFE_TIME_SysTime_t](#) CFE_TIME_DiagnosticTlm_Payload::AtToneMET

MET at time of tone.

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_DTMETS, \$sc_\$cpu_TIME_DTMETSs

Definition at line 180 of file [default_cfe_time_msgdefs.h](#).

AtToneSTCF

`CFE_TIME_SysTime_t` `CFE_TIME_DiagnosticTlm_Payload::AtToneSTCF`
STCF at time of tone.

Telemetry Mnemonic(s) `$sc_$cpu_TIME_DSTCFS`, `$sc_$cpu_TIME_DSTCFSS`

Definition at line 182 of file `default_cfe_time_msgdefs.h`.

ClockFlyState

`int16` `CFE_TIME_DiagnosticTlm_Payload::ClockFlyState`
Current fly-wheel state.

Telemetry Mnemonic(s) `$sc_$cpu_TIME_DFlywheel`

Definition at line 213 of file `default_cfe_time_msgdefs.h`.

ClockSetState

`int16` `CFE_TIME_DiagnosticTlm_Payload::ClockSetState`
Time has been "set".

Telemetry Mnemonic(s) `$sc_$cpu_TIME_DValid`

Definition at line 211 of file `default_cfe_time_msgdefs.h`.

ClockSignal

`int16` `CFE_TIME_DiagnosticTlm_Payload::ClockSignal`
Primary vs redundant, etc.

Telemetry Mnemonic(s) `$sc_$cpu_TIME_DSignal`

Definition at line 217 of file `default_cfe_time_msgdefs.h`.

ClockSource

`int16` `CFE_TIME_DiagnosticTlm_Payload::ClockSource`
Internal vs external, etc.

Telemetry Mnemonic(s) `$sc_$cpu_TIME_DSource`

Definition at line 215 of file `default_cfe_time_msgdefs.h`.

ClockStateAPI

`CFE_TIME_ClockState_Enum_t` `CFE_TIME_DiagnosticTlm_Payload::ClockStateAPI`
Clock state as per API.

Telemetry Mnemonic(s) `$sc_$cpu_TIME_DAPIState`

Definition at line 191 of file `default_cfe_time_msgdefs.h`.

ClockStateFlags

`uint16 CFE_TIME_DiagnosticTlm_Payload::ClockStateFlags`

Clock State Flags.

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_DStateFlags, \$sc_\$cpu_TIME_DFlagSet, \$sc_\$cpu_TIME_DFlagFly, \$sc_\$cpu_TIME_DFlagSrc, \$sc_\$cpu_TIME_DFlagPri, \$sc_\$cpu_TIME_DFlagSfly, \$sc_\$cpu_TIME_DFlagCfly, \$sc_\$cpu_TIME_DFlagAdj, \$sc_\$cpu_TIME_DFlag1Hzd, \$sc_\$cpu_TIME_DFlagClat, \$sc_\$cpu_TIME_DFlagSorC, \$sc_\$cpu_TIME_DFlagNIU

Definition at line 227 of file [default_cfe_time_msgdefs.h](#).

CurrentLatch

`CFE_TIME_SysTime_t CFE_TIME_DiagnosticTlm_Payload::CurrentLatch`

Local clock latched just "now".

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_DLocalS, \$sc_\$cpu_TIME_DLocalSs

Definition at line 199 of file [default_cfe_time_msgdefs.h](#).

CurrentMET

`CFE_TIME_SysTime_t CFE_TIME_DiagnosticTlm_Payload::CurrentMET`

MET at this instant.

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_DMETS, \$sc_\$cpu_TIME_DMETSs

Definition at line 201 of file [default_cfe_time_msgdefs.h](#).

CurrentTAI

`CFE_TIME_SysTime_t CFE_TIME_DiagnosticTlm_Payload::CurrentTAI`

TAI at this instant.

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_DTAIS, \$sc_\$cpu_TIME_DTAISS

Definition at line 203 of file [default_cfe_time_msgdefs.h](#).

CurrentUTC

`CFE_TIME_SysTime_t CFE_TIME_DiagnosticTlm_Payload::CurrentUTC`

UTC at this instant.

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_DUTCS, \$sc_\$cpu_TIME_DUTCSS

Definition at line 205 of file [default_cfe_time_msgdefs.h](#).

DataStoreStatus

`uint32 CFE_TIME_DiagnosticTlm_Payload::DataStoreStatus`

Data Store status (preserved across processor reset)

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_DataStStat

Definition at line 317 of file [default_cfe_time_msgdefs.h](#).

DelayDirection

`int16 CFE_TIME_DiagnosticTlm_Payload::DelayDirection`
Client latency adjustment direction.

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_DLatentDir

Definition at line 237 of file [default_cfe_time_msgdefs.h](#).

Forced2Fly

`int16 CFE_TIME_DiagnosticTlm_Payload::Forced2Fly`
Commanded into fly-wheel.

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_DCMD2Fly

Definition at line 221 of file [default_cfe_time_msgdefs.h](#).

LocalIntCounter

`uint32 CFE_TIME_DiagnosticTlm_Payload::LocalIntCounter`
Local 1Hz ISR execution count.

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_D1HzISRCNT

Definition at line 275 of file [default_cfe_time_msgdefs.h](#).

LocalTaskCounter

`uint32 CFE_TIME_DiagnosticTlm_Payload::LocalTaskCounter`
Local 1Hz task execution count.

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_D1HzTaskCNT

Definition at line 277 of file [default_cfe_time_msgdefs.h](#).

MaxElapsed

`uint32 CFE_TIME_DiagnosticTlm_Payload::MaxElapsed`
Max tone signal / data pkt arrival window (Sub-seconds)

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_DMaxWindow

Definition at line 297 of file [default_cfe_time_msgdefs.h](#).

MaxLocalClock

`CFE_TIME_SysTime_t CFE_TIME_DiagnosticTlm_Payload::MaxLocalClock`
Max local clock value before rollover.

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_DWrapS, \$sc_\$cpu_TIME_DWrapSs

Definition at line 303 of file [default_cfe_time_msgdefs.h](#).

MinElapsed

`uint32 CFE_TIME_DiagnosticTlm_Payload::MinElapsed`
Min tone signal / data pkt arrival window (Sub-seconds)

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_DMinWindow

Definition at line 295 of file [default_cfe_time_msgdefs.h](#).

OneHzAdjust

`CFE_TIME_SysTime_t CFE_TIME_DiagnosticTlm_Payload::OneHzAdjust`
Current 1Hz STCF adjustment.

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_D1HzAdjS, \$sc_\$cpu_TIME_D1HzAdjSs

Definition at line 245 of file [default_cfe_time_msgdefs.h](#).

OneHzDirection

`int16 CFE_TIME_DiagnosticTlm_Payload::OneHzDirection`
1Hz STCF adjustment direction

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_D1HzAdjDir

Definition at line 235 of file [default_cfe_time_msgdefs.h](#).

OneTimeAdjust

`CFE_TIME_SysTime_t CFE_TIME_DiagnosticTlm_Payload::OneTimeAdjust`
Previous one-time STCF adjustment.

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_DAdjustS, \$sc_\$cpu_TIME_DAdjustSs

Definition at line 243 of file [default_cfe_time_msgdefs.h](#).

OneTimeDirection

`int16 CFE_TIME_DiagnosticTlm_Payload::OneTimeDirection`
One time STCF adjustment direction (Add = 1, Sub = 2)

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_DAdjustDir

Definition at line 233 of file [default_cfe_time_msgdefs.h](#).

ServerFlyState

`int16 CFE_TIME_DiagnosticTlm_Payload::ServerFlyState`
Used by clients only.

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_DSrvFly

Definition at line 219 of file [default_cfe_time_msgdefs.h](#).

TimeSinceTone

`CFE_TIME_SysTime_t CFE_TIME_DiagnosticTlm_Payload::TimeSinceTone`
Time elapsed since the tone.

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_DElapsedS, \$sc_\$cpu_TIME_DElapsedSs

Definition at line 197 of file [default_cfe_time_msgdefs.h](#).

ToneDataCounter

`uint32 CFE_TIME_DiagnosticTlm_Payload::ToneDataCounter`
Time at the tone data SB message count.

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_DTatTCNT

Definition at line 265 of file [default_cfe_time_msgdefs.h](#).

ToneDataLatch

`CFE_TIME_SysTime_t CFE_TIME_DiagnosticTlm_Payload::ToneDataLatch`
Local Clock latched at arrival of tone data.

Telemetry Mnemonic(s) `$sc_$cpu_TIME_DTDS, $sc_$cpu_TIME_DTDSs`

Definition at line 253 of file [default_cfe_time_msgdefs.h](#).

ToneIntCounter

`uint32 CFE_TIME_DiagnosticTlm_Payload::ToneIntCounter`
Tone signal ISR execution count.

Telemetry Mnemonic(s) `$sc_$cpu_TIME_DTslSRCNT`

Definition at line 267 of file [default_cfe_time_msgdefs.h](#).

ToneIntErrorCounter

`uint32 CFE_TIME_DiagnosticTlm_Payload::ToneIntErrorCounter`
Tone signal ISR error count.

Telemetry Mnemonic(s) `$sc_$cpu_TIME_DTslSRERR`

Definition at line 269 of file [default_cfe_time_msgdefs.h](#).

ToneMatchCounter

`uint32 CFE_TIME_DiagnosticTlm_Payload::ToneMatchCounter`
Tone signal / data verification count.

Telemetry Mnemonic(s) `$sc_$cpu_TIME_DVerifyCNT`

Definition at line 259 of file [default_cfe_time_msgdefs.h](#).

ToneMatchErrorCounter

`uint32 CFE_TIME_DiagnosticTlm_Payload::ToneMatchErrorCounter`
Tone signal / data verification error count.

Telemetry Mnemonic(s) `$sc_$cpu_TIME_DVerifyER`

Definition at line 261 of file [default_cfe_time_msgdefs.h](#).

ToneOverLimit

`uint32 CFE_TIME_DiagnosticTlm_Payload::ToneOverLimit`
Max between tone signal interrupts.

Telemetry Mnemonic(s) `$sc_$cpu_TIME_DMaxSs`

Definition at line 309 of file [default_cfe_time_msgdefs.h](#).

ToneSignalCounter

`uint32 CFE_TIME_DiagnosticTlm_Payload::ToneSignalCounter`
Tone signal detected SB message count.

Telemetry Mnemonic(s) `$sc_$cpu_TIME_DTSDetCNT`

Definition at line 263 of file [default_cfe_time_msgdefs.h](#).

ToneSignalLatch

`CFE_TIME_SysTime_t CFE_TIME_DiagnosticTlm_Payload::ToneSignalLatch`
 Local Clock latched at most recent tone signal.

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_DTTS, \$sc_\$cpu_TIME_DTTSS

Definition at line 251 of file [default_cfe_time_msgdefs.h](#).

ToneTaskCounter

`uint32 CFE_TIME_DiagnosticTlm_Payload::ToneTaskCounter`
 Tone task execution count.

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_DTSTaskCNT

Definition at line 271 of file [default_cfe_time_msgdefs.h](#).

ToneUnderLimit

`uint32 CFE_TIME_DiagnosticTlm_Payload::ToneUnderLimit`
 Min between tone signal interrupts.

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_DMinSs

Definition at line 311 of file [default_cfe_time_msgdefs.h](#).

VersionCounter

`uint32 CFE_TIME_DiagnosticTlm_Payload::VersionCounter`
 Count of mods to time at tone reference data (version)

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_DVersionCNT

Definition at line 273 of file [default_cfe_time_msgdefs.h](#).

VirtualMET

`uint32 CFE_TIME_DiagnosticTlm_Payload::VirtualMET`
 Software MET.

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_DLogicalMET

Definition at line 283 of file [default_cfe_time_msgdefs.h](#).
 The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgdefs.h](#)

10.167 CFE_TIME_FakeToneCmd Struct Reference

```
#include <default_cfe_time_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) `CommandHeader`
Command header.

10.167.1 Detailed Description

Definition at line 62 of file [default_cfe_time_msgstruct.h](#).

10.167.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_TIME_FakeToneCmd::CommandHeader

Command header.

Definition at line 64 of file [default_cfe_time_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgstruct.h](#)

10.168 CFE_TIME_HousekeepingTlm Struct Reference

```
#include <default_cfe_time_msgstruct.h>
```

Data Fields

- [CFE_MSG_TelemetryHeader_t](#) TelemetryHeader
Telemetry header.
- [CFE_TIME_HousekeepingTlm_Payload_t](#) Payload
Telemetry payload.

10.168.1 Detailed Description

Definition at line 183 of file [default_cfe_time_msgstruct.h](#).

10.168.2 Field Documentation

Payload

[CFE_TIME_HousekeepingTlm_Payload_t](#) CFE_TIME_HousekeepingTlm::Payload

Telemetry payload.

Definition at line 186 of file [default_cfe_time_msgstruct.h](#).

TelemetryHeader

[CFE_MSG_TelemetryHeader_t](#) CFE_TIME_HousekeepingTlm::TelemetryHeader

Telemetry header.

Definition at line 185 of file [default_cfe_time_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgstruct.h](#)

10.169 CFE_TIME_HousekeepingTlm_Payload Struct Reference

```
#include <default_cfe_time_msgdefs.h>
```

Data Fields

- [uint8](#) CommandCounter
Time Command Execution Counter.
- [uint8](#) CommandErrorCounter

Time Command Error Counter.

- [uint16 ClockStateFlags](#)

State Flags.

- [CFE_TIME_ClockState_Enum_t ClockStateAPI](#)

API State.

- [int16 LeapSeconds](#)

Current Leaps Seconds.

- [CFE_TIME_SysTime_t MET](#)

Current MET (seconds+subseconds)

- [CFE_TIME_SysTime_t STCF](#)

Current STCF (seconds+subseconds)

- [CFE_TIME_SysTime_t AdjustmentFactor](#)

Current Adjustment Factor (seconds+subseconds)

10.169.1 Detailed Description

Name Time Services Housekeeping Packet

Definition at line 125 of file [default_cfe_time_msgdefs.h](#).

10.169.2 Field Documentation

AdjustmentFactor

[CFE_TIME_SysTime_t](#) CFE_TIME_HousekeepingTlm_Payload::AdjustmentFactor

Current Adjustment Factor (seconds+subseconds)

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_ADJ

- In time server mode, the adjustment factor is to compensate for drift over time This is known as the "1Hz STCF adjustment value" and is applied continuously during server operation
- In time client mode, the adjustment factor is a compensation for transit delay from the server This is known as the "Time at tone delay" and is applied to every time at tone update

Definition at line 166 of file [default_cfe_time_msgdefs.h](#).

ClockStateAPI

[CFE_TIME_ClockState_Enum_t](#) CFE_TIME_HousekeepingTlm_Payload::ClockStateAPI

API State.

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_DAPIState

Definition at line 140 of file [default_cfe_time_msgdefs.h](#).

ClockStateFlags

[uint16](#) CFE_TIME_HousekeepingTlm_Payload::ClockStateFlags

State Flags.

Telemetry Mnemonic(s) \$sc_\$cpu_TIME_StateFlg, \$sc_\$cpu_TIME_FlagSet, \$sc_\$cpu_TIME_FlagFly, \$sc_\$cpu_TIME_FlagSrc, \$sc_\$cpu_TIME_FlagPri, \$sc_\$cpu_TIME_FlagSfly, \$sc_\$cpu_TIME_FlagCfly, \$sc_\$cpu_TIME_FlagAdj, \$sc_\$cpu_TIME_Flag1Hzd, \$sc_\$cpu_TIME_FlagClat, \$sc_\$cpu_TIME_FlagSorC, \$sc_\$cpu_TIME_FlagNIU

Definition at line 138 of file [default_cfe_time_msgdefs.h](#).

CommandCounter

`uint8 CFE_TIME_HousekeepingTlm_Payload::CommandCounter`
Time Command Execution Counter.

Telemetry Mnemonic(s) `$sc_$cpu_TIME_CMDPC`

Definition at line 130 of file [default_cfe_time_msgdefs.h](#).

CommandErrorCounter

`uint8 CFE_TIME_HousekeepingTlm_Payload::CommandErrorCounter`
Time Command Error Counter.

Telemetry Mnemonic(s) `$sc_$cpu_TIME_CMDEC`

Definition at line 132 of file [default_cfe_time_msgdefs.h](#).

LeapSeconds

`int16 CFE_TIME_HousekeepingTlm_Payload::LeapSeconds`
Current Leaps Seconds.

Telemetry Mnemonic(s) `$sc_$cpu_TIME_LeapSecs`

Definition at line 146 of file [default_cfe_time_msgdefs.h](#).

MET

`CFE_TIME_SysTime_t CFE_TIME_HousekeepingTlm_Payload::MET`
Current MET (seconds+subseconds)

Telemetry Mnemonic(s) `$sc_$cpu_TIME_MET`

Definition at line 152 of file [default_cfe_time_msgdefs.h](#).

STCF

`CFE_TIME_SysTime_t CFE_TIME_HousekeepingTlm_Payload::STCF`
Current STCF (seconds+subseconds)

Telemetry Mnemonic(s) `$sc_$cpu_TIME_STCF`

Definition at line 153 of file [default_cfe_time_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgdefs.h](#)

10.170 CFE_TIME_LeapsCmd_Payload Struct Reference

Set leap seconds command payload.

```
#include <default_cfe_time_msgdefs.h>
```

Data Fields

- [int16 LeapSeconds](#)

10.170.1 Detailed Description

Set leap seconds command payload.

Definition at line 54 of file [default_cfe_time_msgdefs.h](#).

10.170.2 Field Documentation

LeapSeconds

`int16 CFE_TIME_LeapsCmd_Payload::LeapSeconds`

Definition at line 56 of file [default_cfe_time_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgdefs.h](#)

10.171 CFE_TIME_NoopCmd Struct Reference

```
#include <default_cfe_time_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) `CommandHeader`
Command header.

10.171.1 Detailed Description

Definition at line 37 of file [default_cfe_time_msgstruct.h](#).

10.171.2 Field Documentation

CommandHeader

`CFE_MSG_CommandHeader_t CFE_TIME_NoopCmd::CommandHeader`

Command header.

Definition at line 39 of file [default_cfe_time_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgstruct.h](#)

10.172 CFE_TIME_OneHzAdjustmentCmd_Payload Struct Reference

Generic seconds, subseconds command payload.

```
#include <default_cfe_time_msgdefs.h>
```

Data Fields

- [uint32](#) `Seconds`
- [uint32](#) `Subseconds`

10.172.1 Detailed Description

Generic seconds, subseconds command payload.

Definition at line 103 of file [default_cfe_time_msgdefs.h](#).

10.172.2 Field Documentation

Seconds

`uint32 CFE_TIME_OneHzAdjustmentCmd_Payload::Seconds`

Definition at line 105 of file [default_cfe_time_msgdefs.h](#).

Subseconds

`uint32 CFE_TIME_OneHzAdjustmentCmd_Payload::Subseconds`

Definition at line 106 of file [default_cfe_time_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgdefs.h](#)

10.173 CFE_TIME_OneHzCmd Struct Reference

```
#include <default_cfe_time_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) `CommandHeader`
Command header.

10.173.1 Detailed Description

Definition at line 52 of file [default_cfe_time_msgstruct.h](#).

10.173.2 Field Documentation

CommandHeader

`CFE_MSG_CommandHeader_t CFE_TIME_OneHzCmd::CommandHeader`

Command header.

Definition at line 54 of file [default_cfe_time_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgstruct.h](#)

10.174 CFE_TIME_ResetCountersCmd Struct Reference

```
#include <default_cfe_time_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) `CommandHeader`
Command header.

10.174.1 Detailed Description

Definition at line 42 of file [default_cfe_time_msgstruct.h](#).

10.174.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_TIME_ResetCountersCmd::CommandHeader

Command header.

Definition at line 44 of file [default_cfe_time_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgstruct.h](#)

10.175 CFE_TIME_SendDiagnosticCmd Struct Reference

```
#include <default_cfe_time_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.

10.175.1 Detailed Description

Definition at line 47 of file [default_cfe_time_msgstruct.h](#).

10.175.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_TIME_SendDiagnosticCmd::CommandHeader

Command header.

Definition at line 49 of file [default_cfe_time_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgstruct.h](#)

10.176 CFE_TIME_SendHkCmd Struct Reference

```
#include <default_cfe_time_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.

10.176.1 Detailed Description

Definition at line 67 of file [default_cfe_time_msgstruct.h](#).

10.176.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_TIME_SendHkCmd::CommandHeader

Command header.

Definition at line 69 of file [default_cfe_time_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgstruct.h](#)

10.177 CFE_TIME_SetLeapSecondsCmd Struct Reference

Set leap seconds command.

```
#include <default_cfe_time_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) *CommandHeader*
Command header.
- [CFE_TIME_LeapsCmd_Payload_t](#) *Payload*
Command payload.

10.177.1 Detailed Description

Set leap seconds command.

Definition at line 75 of file [default_cfe_time_msgstruct.h](#).

10.177.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_TIME_SetLeapSecondsCmd::CommandHeader

Command header.

Definition at line 77 of file [default_cfe_time_msgstruct.h](#).

Payload

[CFE_TIME_LeapsCmd_Payload_t](#) CFE_TIME_SetLeapSecondsCmd::Payload

Command payload.

Definition at line 78 of file [default_cfe_time_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgstruct.h](#)

10.178 CFE_TIME_SetMETCmd Struct Reference

```
#include <default_cfe_time_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) *CommandHeader*
Command header.
- [CFE_TIME_TimeCmd_Payload_t](#) *Payload*
Command payload.

10.178.1 Detailed Description

Definition at line 125 of file [default_cfe_time_msgstruct.h](#).

10.178.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_TIME_SetMETCmd::CommandHeader

Command header.

Definition at line 127 of file [default_cfe_time_msgstruct.h](#).

Payload

[CFE_TIME_TimeCmd_Payload_t](#) CFE_TIME_SetMETCmd::Payload

Command payload.

Definition at line 128 of file [default_cfe_time_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgstruct.h](#)

10.179 CFE_TIME_SetSignalCmd Struct Reference

Set tone signal source command.

```
#include <default_cfe_time_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.
- [CFE_TIME_SignalCmd_Payload_t](#) Payload
Command payload.

10.179.1 Detailed Description

Set tone signal source command.

Definition at line 102 of file [default_cfe_time_msgstruct.h](#).

10.179.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_TIME_SetSignalCmd::CommandHeader

Command header.

Definition at line 104 of file [default_cfe_time_msgstruct.h](#).

Payload

[CFE_TIME_SignalCmd_Payload_t](#) CFE_TIME_SetSignalCmd::Payload

Command payload.

Definition at line 105 of file [default_cfe_time_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgstruct.h](#)

10.180 CFE_TIME_SetSourceCmd Struct Reference

Set time data source command.

```
#include <default_cfe_time_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.
- [CFE_TIME_SourceCmd_Payload_t](#) Payload
Command payload.

10.180.1 Detailed Description

Set time data source command.

Definition at line 93 of file [default_cfe_time_msgstruct.h](#).

10.180.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_TIME_SetSourceCmd::CommandHeader

Command header.

Definition at line 95 of file [default_cfe_time_msgstruct.h](#).

Payload

[CFE_TIME_SourceCmd_Payload_t](#) CFE_TIME_SetSourceCmd::Payload

Command payload.

Definition at line 96 of file [default_cfe_time_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgstruct.h](#)

10.181 CFE_TIME_SetStateCmd Struct Reference

Set clock state command.

```
#include <default_cfe_time_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.
- [CFE_TIME_StateCmd_Payload_t](#) Payload
Command payload.

10.181.1 Detailed Description

Set clock state command.

Definition at line 84 of file [default_cfe_time_msgstruct.h](#).

10.181.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_TIME_SetStateCmd::CommandHeader

Command header.

Definition at line 86 of file [default_cfe_time_msgstruct.h](#).

Payload

[CFE_TIME_StateCmd_Payload_t](#) CFE_TIME_SetStateCmd::Payload

Command payload.

Definition at line 87 of file [default_cfe_time_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgstruct.h](#)

10.182 CFE_TIME_SetSTCFCmd Struct Reference

```
#include <default_cfe_time_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) *CommandHeader*
Command header.
- [CFE_TIME_TimeCmd_Payload_t](#) *Payload*
Command payload.

10.182.1 Detailed Description

Definition at line 131 of file [default_cfe_time_msgstruct.h](#).

10.182.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_TIME_SetSTCFCmd::CommandHeader

Command header.

Definition at line 133 of file [default_cfe_time_msgstruct.h](#).

Payload

[CFE_TIME_TimeCmd_Payload_t](#) CFE_TIME_SetSTCFCmd::Payload

Command payload.

Definition at line 134 of file [default_cfe_time_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgstruct.h](#)

10.183 CFE_TIME_SetTimeCmd Struct Reference

```
#include <default_cfe_time_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) *CommandHeader*
Command header.
- [CFE_TIME_TimeCmd_Payload_t](#) *Payload*
Command payload.

10.183.1 Detailed Description

Definition at line 149 of file [default_cfe_time_msgstruct.h](#).

10.183.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_TIME_SetTimeCmd::CommandHeader

Command header.

Definition at line 151 of file [default_cfe_time_msgstruct.h](#).

Payload

`CFE_TIME_TimeCmd_Payload_t` `CFE_TIME_SetTimeCmd::Payload`

Command payload.

Definition at line 152 of file [default_cfe_time_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgstruct.h](#)

10.184 CFE_TIME_SignalCmd_Payload Struct Reference

Set tone signal source command payload.

`#include <default_cfe_time_msgdefs.h>`

Data Fields

- [int16 ToneSource](#)

[CFE_TIME_ToneSignalSelect_PRIMARY](#)=Primary Source, [CFE_TIME_ToneSignalSelect_REDUNDANT](#)=Redundant Source

10.184.1 Detailed Description

Set tone signal source command payload.

Definition at line 84 of file [default_cfe_time_msgdefs.h](#).

10.184.2 Field Documentation

ToneSource

[int16](#) `CFE_TIME_SignalCmd_Payload::ToneSource`

[CFE_TIME_ToneSignalSelect_PRIMARY](#)=Primary Source, [CFE_TIME_ToneSignalSelect_REDUNDANT](#)=Redundant Source

Selects either the "Primary" or "Redundant" tone signal source

Definition at line 86 of file [default_cfe_time_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgdefs.h](#)

10.185 CFE_TIME_SourceCmd_Payload Struct Reference

Set time data source command payload.

`#include <default_cfe_time_msgdefs.h>`

Data Fields

- [int16 TimeSource](#)

[CFE_TIME_SourceSelect_INTERNAL](#)=Internal Source, [CFE_TIME_SourceSelect_EXTERNAL](#)=External Source

10.185.1 Detailed Description

Set time data source command payload.

Definition at line 74 of file [default_cfe_time_msgdefs.h](#).

10.185.2 Field Documentation

TimeSource

`int16 CFE_TIME_SourceCmd_Payload::TimeSource`

`CFE_TIME_SourceSelect_INTERNAL`=Internal Source, `CFE_TIME_SourceSelect_EXTERNAL`=External Source

Selects either the "Internal" and "External" clock source

Definition at line 76 of file [default_cfe_time_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgdefs.h](#)

10.186 CFE_TIME_StateCmd_Payload Struct Reference

Set clock state command payload.

`#include <default_cfe_time_msgdefs.h>`

Data Fields

- [CFE_TIME_ClockState_Enum_t](#) ClockState

`CFE_TIME_ClockState_INVALID`=Spacecraft time has not been accurately set, `CFE_TIME_ClockState_VALID`=Spacecraft clock has been accurately set, `CFE_TIME_ClockState_FLYWHEEL`=Force into FLYWHEEL mode

10.186.1 Detailed Description

Set clock state command payload.

Definition at line 62 of file [default_cfe_time_msgdefs.h](#).

10.186.2 Field Documentation

ClockState

`CFE_TIME_ClockState_Enum_t CFE_TIME_StateCmd_Payload::ClockState`

`CFE_TIME_ClockState_INVALID`=Spacecraft time has not been accurately set, `CFE_TIME_ClockState_VALID`=Spacecraft clock has been accurately set, `CFE_TIME_ClockState_FLYWHEEL`=Force into FLYWHEEL mode

Selects the current clock state

Definition at line 64 of file [default_cfe_time_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgdefs.h](#)

10.187 CFE_TIME_SubAdjustCmd Struct Reference

`#include <default_cfe_time_msgstruct.h>`

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader

Command header.

- [CFE_TIME_TimeCmd_Payload_t](#) Payload

Command payload.

10.187.1 Detailed Description

Definition at line 143 of file [default_cfe_time_msgstruct.h](#).

10.187.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_TIME_SubAdjustCmd::CommandHeader

Command header.

Definition at line 145 of file [default_cfe_time_msgstruct.h](#).

Payload

[CFE_TIME_TimeCmd_Payload_t](#) CFE_TIME_SubAdjustCmd::Payload

Command payload.

Definition at line 146 of file [default_cfe_time_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgstruct.h](#)

10.188 CFE_TIME_SubDelayCmd Struct Reference

```
#include <default_cfe_time_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) CommandHeader
Command header.
- [CFE_TIME_TimeCmd_Payload_t](#) Payload
Command payload.

10.188.1 Detailed Description

Definition at line 119 of file [default_cfe_time_msgstruct.h](#).

10.188.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_TIME_SubDelayCmd::CommandHeader

Command header.

Definition at line 121 of file [default_cfe_time_msgstruct.h](#).

Payload

[CFE_TIME_TimeCmd_Payload_t](#) CFE_TIME_SubDelayCmd::Payload

Command payload.

Definition at line 122 of file [default_cfe_time_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgstruct.h](#)

10.189 CFE_TIME_SubOneHzAdjustmentCmd Struct Reference

```
#include <default_cfe_time_msgstruct.h>
```

Data Fields

- [CFE_MSG_CommandHeader_t](#) `CommandHeader`
Command header.
- [CFE_TIME_OneHzAdjustmentCmd_Payload_t](#) `Payload`
Command payload.

10.189.1 Detailed Description

Definition at line 166 of file [default_cfe_time_msgstruct.h](#).

10.189.2 Field Documentation**CommandHeader**

[CFE_MSG_CommandHeader_t](#) `CFE_TIME_SubOneHzAdjustmentCmd::CommandHeader`

Command header.

Definition at line 168 of file [default_cfe_time_msgstruct.h](#).

Payload

[CFE_TIME_OneHzAdjustmentCmd_Payload_t](#) `CFE_TIME_SubOneHzAdjustmentCmd::Payload`

Command payload.

Definition at line 169 of file [default_cfe_time_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgstruct.h](#)

10.190 CFE_TIME_SysTime Struct Reference

Data structure used to hold system time values.

`#include <default_cfe_time_extern_typedefs.h>`

Data Fields

- [uint32](#) `Seconds`
Number of seconds since epoch.
- [uint32](#) `Subseconds`
Number of subseconds since epoch ($LSB = 2^{(-32)}$ seconds)

10.190.1 Detailed Description

Data structure used to hold system time values.

Description

The [CFE_TIME_SysTime_t](#) data structure is used to hold time values. Time is referred to as the elapsed time (in seconds and subseconds) since a specified epoch time. The subseconds field contains the number of $2^{(-32)}$ second intervals that have elapsed since the epoch.

Definition at line 41 of file [default_cfe_time_extern_typedefs.h](#).

10.190.2 Field Documentation

Seconds

`uint32 CFE_TIME_SysTime::Seconds`

Number of seconds since epoch.

Definition at line 43 of file [default_cfe_time_extern_typedefs.h](#).

Subseconds

`uint32 CFE_TIME_SysTime::Subseconds`

Number of subseconds since epoch (LSB = $2^{(-32)}$ seconds)

Definition at line 44 of file [default_cfe_time_extern_typedefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_extern_typedefs.h](#)

10.191 CFE_TIME_TimeCmd_Payload Struct Reference

Generic seconds, microseconds command payload.

`#include <default_cfe_time_msgdefs.h>`

Data Fields

- [uint32 Seconds](#)
- [uint32 MicroSeconds](#)

10.191.1 Detailed Description

Generic seconds, microseconds command payload.

Definition at line 94 of file [default_cfe_time_msgdefs.h](#).

10.191.2 Field Documentation

MicroSeconds

`uint32 CFE_TIME_TimeCmd_Payload::MicroSeconds`

Definition at line 97 of file [default_cfe_time_msgdefs.h](#).

Seconds

`uint32 CFE_TIME_TimeCmd_Payload::Seconds`

Definition at line 96 of file [default_cfe_time_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgdefs.h](#)

10.192 CFE_TIME_ToneDataCmd Struct Reference

Time at tone data command.

`#include <default_cfe_time_msgstruct.h>`

Data Fields

- [CFE_MSG_CommandHeader_t CommandHeader](#)
Command header.
- [CFE_TIME_ToneDataCmd_Payload_t Payload](#)
Command payload.

10.192.1 Detailed Description

Time at tone data command.

Definition at line 175 of file [default_cfe_time_msgstruct.h](#).

10.192.2 Field Documentation

CommandHeader

[CFE_MSG_CommandHeader_t](#) CFE_TIME_ToneDataCmd::CommandHeader

Command header.

Definition at line 177 of file [default_cfe_time_msgstruct.h](#).

Payload

[CFE_TIME_ToneDataCmd_Payload_t](#) CFE_TIME_ToneDataCmd::Payload

Command payload.

Definition at line 178 of file [default_cfe_time_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgstruct.h](#)

10.193 CFE_TIME_ToneDataCmd_Payload Struct Reference

Time at tone data command payload.

```
#include <default_cfe_time_msgdefs.h>
```

Data Fields

- [CFE_TIME_SysTime_t](#) AtToneMET
MET at time of tone.
- [CFE_TIME_SysTime_t](#) AtToneSTCF
STCF at time of tone.
- [int16](#) AtToneLeapSeconds
Leap Seconds at time of tone.
- [CFE_TIME_ClockState_Enum_t](#) AtToneState
Clock state at time of tone.

10.193.1 Detailed Description

Time at tone data command payload.

Definition at line 112 of file [default_cfe_time_msgdefs.h](#).

10.193.2 Field Documentation

AtToneLeapSeconds

[int16](#) CFE_TIME_ToneDataCmd_Payload::AtToneLeapSeconds

Leap Seconds at time of tone.

Definition at line 116 of file [default_cfe_time_msgdefs.h](#).

AtToneMET

[CFE_TIME_SysTime_t](#) CFE_TIME_ToneDataCmd_Payload::AtToneMET

MET at time of tone.

Definition at line 114 of file [default_cfe_time_msgdefs.h](#).

AtToneState

`CFE_TIME_ClockState_Enum_t CFE_TIME_ToneDataCmd_Payload::AtToneState`

Clock state at time of tone.

Definition at line 117 of file [default_cfe_time_msgdefs.h](#).

AtToneSTCF

`CFE_TIME_SysTime_t CFE_TIME_ToneDataCmd_Payload::AtToneSTCF`

STCF at time of tone.

Definition at line 115 of file [default_cfe_time_msgdefs.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgdefs.h](#)

10.194 CFE_TIME_ToneSignalCmd Struct Reference

`#include <default_cfe_time_msgstruct.h>`

Data Fields

- [CFE_MSG_CommandHeader_t](#) `CommandHeader`
Command header.

10.194.1 Detailed Description

Definition at line 57 of file [default_cfe_time_msgstruct.h](#).

10.194.2 Field Documentation

CommandHeader

`CFE_MSG_CommandHeader_t CFE_TIME_ToneSignalCmd::CommandHeader`

Command header.

Definition at line 59 of file [default_cfe_time_msgstruct.h](#).

The documentation for this struct was generated from the following file:

- [cfe/modules/time/config/default_cfe_time_msgstruct.h](#)

10.195 OS_bin_sem_prop_t Struct Reference

OSAL binary semaphore properties.

`#include <osapi-binsem.h>`

Data Fields

- `char name` [[OS_MAX_API_NAME](#)]
- [osal_id_t](#) `creator`
- [int32](#) `value`

10.195.1 Detailed Description

OSAL binary semaphore properties.

Definition at line 39 of file [osapi-binsem.h](#).

10.195.2 Field Documentation

creator

[osal_id_t](#) OS_bin_sem_prop_t::creator

Definition at line 42 of file [osapi-binsem.h](#).

name

char OS_bin_sem_prop_t::name[OS_MAX_API_NAME]

Definition at line 41 of file [osapi-binsem.h](#).

value

[int32](#) OS_bin_sem_prop_t::value

Definition at line 43 of file [osapi-binsem.h](#).

The documentation for this struct was generated from the following file:

- [osal/src/os/inc/osapi-binsem.h](#)

10.196 OS_condvar_prop_t Struct Reference

OSAL condition variable properties.

```
#include <osapi-condvar.h>
```

Data Fields

- char [name](#) [OS_MAX_API_NAME]
- [osal_id_t](#) creator

10.196.1 Detailed Description

OSAL condition variable properties.

Definition at line 34 of file [osapi-condvar.h](#).

10.196.2 Field Documentation

creator

[osal_id_t](#) OS_condvar_prop_t::creator

Definition at line 37 of file [osapi-condvar.h](#).

name

char OS_condvar_prop_t::name[OS_MAX_API_NAME]

Definition at line 36 of file [osapi-condvar.h](#).

The documentation for this struct was generated from the following file:

- [osal/src/os/inc/osapi-condvar.h](#)

10.197 OS_count_sem_prop_t Struct Reference

OSAL counting semaphore properties.

```
#include <osapi-countsem.h>
```

Data Fields

- char [name](#) [[OS_MAX_API_NAME](#)]
- [osal_id_t](#) creator
- [int32](#) value

10.197.1 Detailed Description

OSAL counting semaphore properties.

Definition at line [32](#) of file [osapi-countsem.h](#).

10.197.2 Field Documentation

creator

[osal_id_t](#) OS_count_sem_prop_t::creator

Definition at line [35](#) of file [osapi-countsem.h](#).

name

char OS_count_sem_prop_t::name [[OS_MAX_API_NAME](#)]

Definition at line [34](#) of file [osapi-countsem.h](#).

value

[int32](#) OS_count_sem_prop_t::value

Definition at line [36](#) of file [osapi-countsem.h](#).

The documentation for this struct was generated from the following file:

- [osal/src/os/inc/osapi-countsem.h](#)

10.198 os_dirent_t Struct Reference

Directory entry.

```
#include <osapi-dir.h>
```

Data Fields

- char [FileName](#) [[OS_MAX_FILE_NAME](#)]

10.198.1 Detailed Description

Directory entry.

Definition at line [32](#) of file [osapi-dir.h](#).

10.198.2 Field Documentation

FileName

char os_dirent_t::FileName [[OS_MAX_FILE_NAME](#)]

Definition at line [34](#) of file [osapi-dir.h](#).

The documentation for this struct was generated from the following file:

- [osal/src/os/inc/osapi-dir.h](#)

10.199 OS_FdSet Struct Reference

An abstract structure capable of holding several OSAL IDs.

```
#include <osapi-select.h>
```

Data Fields

- [uint8 object_ids](#) [(OS_MAX_NUM_OPEN_FILES+7)/8]

10.199.1 Detailed Description

An abstract structure capable of holding several OSAL IDs.

This is part of the select API and is manipulated using the related API calls. It should not be modified directly by applications.

Note: Math is to determine uint8 array size needed to represent single bit OS_MAX_NUM_OPEN_FILES objects, + 7 rounds up and 8 is the size of uint8.

See also

[OS_SelectFdZero\(\)](#), [OS_SelectFdAdd\(\)](#), [OS_SelectFdClear\(\)](#), [OS_SelectFdsSet\(\)](#)

Definition at line 44 of file [osapi-select.h](#).

10.199.2 Field Documentation

object_ids

```
uint8 OS_FdSet::object_ids[(OS_MAX_NUM_OPEN_FILES+7)/8]
```

Definition at line 46 of file [osapi-select.h](#).

The documentation for this struct was generated from the following file:

- [osal/src/os/inc/osapi-select.h](#)

10.200 OS_file_prop_t Struct Reference

OSAL file properties.

```
#include <osapi-file.h>
```

Data Fields

- [char Path](#) [OS_MAX_PATH_LEN]
- [osal_id_t User](#)
- [uint8 IsValid](#)

10.200.1 Detailed Description

OSAL file properties.

Definition at line 49 of file [osapi-file.h](#).

10.200.2 Field Documentation

IsValid

```
uint8 OS_file_prop_t::IsValid
```

Definition at line 53 of file [osapi-file.h](#).

Path

`char OS_file_prop_t::Path[OS_MAX_PATH_LEN]`
Definition at line 51 of file [osapi-file.h](#).

User

`osal_id_t OS_file_prop_t::User`
Definition at line 52 of file [osapi-file.h](#).

The documentation for this struct was generated from the following file:

- [osal/src/os/inc/osapi-file.h](#)

10.201 os_fsinfo_t Struct Reference

OSAL file system info.

```
#include <osapi-filesys.h>
```

Data Fields

- [uint32 MaxFds](#)
Total number of file descriptors.
- [uint32 FreeFds](#)
Total number that are free.
- [uint32 MaxVolumes](#)
Maximum number of volumes.
- [uint32 FreeVolumes](#)
Total number of volumes free.

10.201.1 Detailed Description

OSAL file system info.

Definition at line 35 of file [osapi-filesys.h](#).

10.201.2 Field Documentation

FreeFds

`uint32 os_fsinfo_t::FreeFds`
Total number that are free.
Definition at line 38 of file [osapi-filesys.h](#).

FreeVolumes

`uint32 os_fsinfo_t::FreeVolumes`
Total number of volumes free.
Definition at line 40 of file [osapi-filesys.h](#).

MaxFds

`uint32 os_fsinfo_t::MaxFds`
Total number of file descriptors.
Definition at line 37 of file [osapi-filesys.h](#).

MaxVolumes

`uint32 os_fsinfo_t::MaxVolumes`

Maximum number of volumes.

Definition at line 39 of file [osapi-filesys.h](#).

The documentation for this struct was generated from the following file:

- [osal/src/os/inc/osapi-filesys.h](#)

10.202 os_fstat_t Struct Reference

File system status.

```
#include <osapi-file.h>
```

Data Fields

- `uint32 FileModeBits`
- `OS_time_t FileTime`
- `size_t FileSize`

10.202.1 Detailed Description

File system status.

Note

This used to be directly typedef'ed to the "struct stat" from the C library

Some C libraries (glibc in particular) actually define member names to reference into sub-structures, so attempting to reuse a name like "st_mtime" might not work.

Definition at line 64 of file [osapi-file.h](#).

10.202.2 Field Documentation

FileModeBits

`uint32 os_fstat_t::FileModeBits`

Definition at line 66 of file [osapi-file.h](#).

FileSize

`size_t os_fstat_t::FileSize`

Definition at line 68 of file [osapi-file.h](#).

FileTime

`OS_time_t os_fstat_t::FileTime`

Definition at line 67 of file [osapi-file.h](#).

The documentation for this struct was generated from the following file:

- [osal/src/os/inc/osapi-file.h](#)

10.203 OS_heap_prop_t Struct Reference

OSAL heap properties.

```
#include <osapi-heap.h>
```

Data Fields

- [size_t free_bytes](#)
- [osal_blockcount_t free_blocks](#)
- [size_t largest_free_block](#)

10.203.1 Detailed Description

OSAL heap properties.

See also

[OS_HeapGetInfo\(\)](#)

Definition at line 36 of file [osapi-heap.h](#).

10.203.2 Field Documentation

free_blocks

[osal_blockcount_t](#) OS_heap_prop_t::free_blocks

Definition at line 39 of file [osapi-heap.h](#).

free_bytes

[size_t](#) OS_heap_prop_t::free_bytes

Definition at line 38 of file [osapi-heap.h](#).

largest_free_block

[size_t](#) OS_heap_prop_t::largest_free_block

Definition at line 40 of file [osapi-heap.h](#).

The documentation for this struct was generated from the following file:

- [osal/src/os/inc/osapi-heap.h](#)

10.204 OS_module_address_t Struct Reference

OSAL module address properties.

```
#include <osapi-module.h>
```

Data Fields

- [uint32 valid](#)
- [uint32 flags](#)
- [cpuaddr code_address](#)
- [cpuaddr code_size](#)
- [cpuaddr data_address](#)
- [cpuaddr data_size](#)
- [cpuaddr bss_address](#)
- [cpuaddr bss_size](#)

10.204.1 Detailed Description

OSAL module address properties.

Definition at line 78 of file [osapi-module.h](#).

10.204.2 Field Documentation

bss_address

`cpuaddr OS_module_address_t::bss_address`

Definition at line 86 of file [osapi-module.h](#).

bss_size

`cpuaddr OS_module_address_t::bss_size`

Definition at line 87 of file [osapi-module.h](#).

code_address

`cpuaddr OS_module_address_t::code_address`

Definition at line 82 of file [osapi-module.h](#).

code_size

`cpuaddr OS_module_address_t::code_size`

Definition at line 83 of file [osapi-module.h](#).

data_address

`cpuaddr OS_module_address_t::data_address`

Definition at line 84 of file [osapi-module.h](#).

data_size

`cpuaddr OS_module_address_t::data_size`

Definition at line 85 of file [osapi-module.h](#).

flags

`uint32 OS_module_address_t::flags`

Definition at line 81 of file [osapi-module.h](#).

valid

`uint32 OS_module_address_t::valid`

Definition at line 80 of file [osapi-module.h](#).

The documentation for this struct was generated from the following file:

- [osal/src/os/inc/osapi-module.h](#)

10.205 OS_module_prop_t Struct Reference

OSAL module properties.

```
#include <osapi-module.h>
```

Data Fields

- `cpuaddr entry_point`
- `cpuaddr host_module_id`
- `char filename [OS_MAX_PATH_LEN]`
- `char name [OS_MAX_API_NAME]`
- `OS_module_address_t addr`

10.205.1 Detailed Description

OSAL module properties.

Definition at line 91 of file [osapi-module.h](#).

10.205.2 Field Documentation

addr

[OS_module_address_t](#) OS_module_prop_t::addr

Definition at line 97 of file [osapi-module.h](#).

entry_point

[cpuaddr](#) OS_module_prop_t::entry_point

Definition at line 93 of file [osapi-module.h](#).

filename

char OS_module_prop_t::filename[OS_MAX_PATH_LEN]

Definition at line 95 of file [osapi-module.h](#).

host_module_id

[cpuaddr](#) OS_module_prop_t::host_module_id

Definition at line 94 of file [osapi-module.h](#).

name

char OS_module_prop_t::name[OS_MAX_API_NAME]

Definition at line 96 of file [osapi-module.h](#).

The documentation for this struct was generated from the following file:

- [osal/src/os/inc/osapi-module.h](#)

10.206 OS_mut_sem_prop_t Struct Reference

OSAL mutex properties.

```
#include <osapi-mutex.h>
```

Data Fields

- char [name](#) [OS_MAX_API_NAME]
- [osal_id_t](#) creator

10.206.1 Detailed Description

OSAL mutex properties.

Definition at line 32 of file [osapi-mutex.h](#).

10.206.2 Field Documentation

creator

[osal_id_t](#) OS_mut_sem_prop_t::creator

Definition at line 35 of file [osapi-mutex.h](#).

name

char OS_mut_sem_prop_t::name[OS_MAX_API_NAME]

Definition at line 34 of file [osapi-mutex.h](#).

The documentation for this struct was generated from the following file:

- [osal/src/os/inc/osapi-mutex.h](#)

10.207 OS_queue_prop_t Struct Reference

OSAL queue properties.

```
#include <osapi-queue.h>
```

Data Fields

- char [name](#) [OS_MAX_API_NAME]
- [osal_id_t](#) creator

10.207.1 Detailed Description

OSAL queue properties.

Definition at line 32 of file [osapi-queue.h](#).

10.207.2 Field Documentation

creator

[osal_id_t](#) OS_queue_prop_t::creator

Definition at line 35 of file [osapi-queue.h](#).

name

char OS_queue_prop_t::name[OS_MAX_API_NAME]

Definition at line 34 of file [osapi-queue.h](#).

The documentation for this struct was generated from the following file:

- [osal/src/os/inc/osapi-queue.h](#)

10.208 OS_rwlock_prop_t Struct Reference

OSAL rwlock properties.

```
#include <osapi-rwlock.h>
```

Data Fields

- char [name](#) [OS_MAX_API_NAME]
- [osal_id_t](#) creator

10.208.1 Detailed Description

OSAL rwlock properties.

Definition at line 32 of file [osapi-rwlock.h](#).

10.208.2 Field Documentation

creator

[osal_id_t](#) OS_rwlock_prop_t::creator

Definition at line 35 of file [osapi-rwlock.h](#).

name

char OS_rwlock_prop_t::name[[OS_MAX_API_NAME](#)]

Definition at line 34 of file [osapi-rwlock.h](#).

The documentation for this struct was generated from the following file:

- [osal/src/os/inc/osapi-rwlock.h](#)

10.209 OS_SockAddr_t Struct Reference

Encapsulates a generic network address.

```
#include <osapi-sockets.h>
```

Data Fields

- [size_t](#) [ActualLength](#)
Length of the actual address data.
- [OS_SockAddrData_t](#) [AddrData](#)
Abstract Address data.

10.209.1 Detailed Description

Encapsulates a generic network address.

This is just an abstract buffer type that holds a network address. It is allocated for the worst-case size defined by OS_SOCKADDR_MAX_LEN, and the real size is stored within.

Definition at line 110 of file [osapi-sockets.h](#).

10.209.2 Field Documentation

ActualLength

[size_t](#) OS_SockAddr_t::ActualLength

Length of the actual address data.

Definition at line 112 of file [osapi-sockets.h](#).

AddrData

[OS_SockAddrData_t](#) OS_SockAddr_t::AddrData

Abstract Address data.

Definition at line 113 of file [osapi-sockets.h](#).

The documentation for this struct was generated from the following file:

- [osal/src/os/inc/osapi-sockets.h](#)

10.210 OS_SockAddrData_t Union Reference

Storage buffer for generic network address.

```
#include <osapi-sockets.h>
```

Data Fields

- [uint8 Buffer](#) [[OS_SOCKADDR_MAX_LEN](#)]
Ensures length of at least OS_SOCKADDR_MAX_LEN.
- [uint32 AlignU32](#)
Ensures uint32 alignment.
- `void *` [AlignPtr](#)
Ensures pointer alignment.

10.210.1 Detailed Description

Storage buffer for generic network address.

This is a union type that helps to ensure a minimum alignment value for the data storage, such that it can be cast to the system-specific type without increasing alignment requirements.

Definition at line 96 of file [osapi-sockets.h](#).

10.210.2 Field Documentation

AlignPtr

```
void* OS_SockAddrData_t::AlignPtr
```

Ensures pointer alignment.

Definition at line 100 of file [osapi-sockets.h](#).

AlignU32

```
uint32 OS_SockAddrData_t::AlignU32
```

Ensures uint32 alignment.

Definition at line 99 of file [osapi-sockets.h](#).

Buffer

```
uint8 OS_SockAddrData_t::Buffer[OS_SOCKADDR_MAX_LEN]
```

Ensures length of at least OS_SOCKADDR_MAX_LEN.

Definition at line 98 of file [osapi-sockets.h](#).

The documentation for this union was generated from the following file:

- `osal/src/os/inc/osapi-sockets.h`

10.211 OS_socket_optval Union Reference

Socket option value.

```
#include <osapi-sockets.h>
```

Data Fields

- [int32 IntVal](#)

10.211.1 Detailed Description

Socket option value.

This is used with [OS_SocketGetOption\(\)](#) and [OS_SocketSetOption\(\)](#) to store the option value that is get or set, respectively. Currently only integers values are relevant but defining as a union will allow other types to be transparently added in the future if needed.

Definition at line 150 of file [osapi-sockets.h](#).

10.211.2 Field Documentation

IntVal

`int32 OS_socket_optval::IntVal`

Definition at line 152 of file [osapi-sockets.h](#).

The documentation for this union was generated from the following file:

- [osal/src/os/inc/osapi-sockets.h](#)

10.212 OS_socket_prop_t Struct Reference

Encapsulates socket properties.

```
#include <osapi-sockets.h>
```

Data Fields

- `char name [OS_MAX_API_NAME]`
Name of the socket.
- `osal_id_t creator`
OSAL TaskID which opened the socket.

10.212.1 Detailed Description

Encapsulates socket properties.

This is for consistency with other OSAL resource types. Currently no extra properties are exposed here but this could change in a future revision of OSAL as needed.

Definition at line 123 of file [osapi-sockets.h](#).

10.212.2 Field Documentation

creator

`osal_id_t OS_socket_prop_t::creator`

OSAL TaskID which opened the socket.

Definition at line 126 of file [osapi-sockets.h](#).

name

`char OS_socket_prop_t::name [OS_MAX_API_NAME]`

Name of the socket.

Definition at line 125 of file [osapi-sockets.h](#).

The documentation for this struct was generated from the following file:

- [osal/src/os/inc/osapi-sockets.h](#)

10.213 OS_static_symbol_record_t Struct Reference

Associates a single symbol name with a memory address.

```
#include <osapi-module.h>
```

Data Fields

- `const char * Name`
- `void(* Address )(void)`
- `const char * Module`

10.213.1 Detailed Description

Associates a single symbol name with a memory address.

If the OS_STATIC_SYMBOL_TABLE feature is enabled, then an array of these structures should be provided by the application. When the application needs to find a symbol address, the static table will be checked in addition to (or instead of) the OS/library-provided lookup function.

This static symbol allows systems that do not implement dynamic module loading to maintain the same semantics as dynamically loaded modules.

Definition at line 113 of file [osapi-module.h](#).

10.213.2 Field Documentation

Address

```
void(* OS_static_symbol_record_t::Address) (void)
```

Definition at line 116 of file [osapi-module.h](#).

Module

```
const char* OS_static_symbol_record_t::Module
```

Definition at line 117 of file [osapi-module.h](#).

Name

```
const char* OS_static_symbol_record_t::Name
```

Definition at line 115 of file [osapi-module.h](#).

The documentation for this struct was generated from the following file:

- [osal/src/os/inc/osapi-module.h](#)

10.214 OS_statvfs_t Struct Reference

```
#include <osapi-filesys.h>
```

Data Fields

- [size_t](#) [block_size](#)
- [osal_blockcount_t](#) [total_blocks](#)
- [osal_blockcount_t](#) [blocks_free](#)

10.214.1 Detailed Description

Definition at line 49 of file [osapi-filesys.h](#).

10.214.2 Field Documentation

block_size

```
size_t OS_statvfs_t::block_size
```

Block size of underlying FS

Definition at line 51 of file [osapi-filesys.h](#).

blocks_free

```
osal\_blockcount\_t OS_statvfs_t::blocks_free
```

Available blocks in underlying FS

Definition at line 53 of file [osapi-filesys.h](#).

total_blocks

`osal_blockcount_t OS_statvfs_t::total_blocks`

Total blocks in underlying FS

Definition at line 52 of file [osapi-filesys.h](#).

The documentation for this struct was generated from the following file:

- [osal/src/os/inc/osapi-filesys.h](#)

10.215 OS_task_prop_t Struct Reference

OSAL task properties.

```
#include <osapi-task.h>
```

Data Fields

- char [name](#) [[OS_MAX_API_NAME](#)]
- [osal_id_t](#) [creator](#)
- [size_t](#) [stack_size](#)
- [osal_priority_t](#) [priority](#)

10.215.1 Detailed Description

OSAL task properties.

Definition at line 57 of file [osapi-task.h](#).

10.215.2 Field Documentation**creator**

`osal_id_t OS_task_prop_t::creator`

Definition at line 60 of file [osapi-task.h](#).

name

`char OS_task_prop_t::name[OS_MAX_API_NAME]`

Definition at line 59 of file [osapi-task.h](#).

priority

`osal_priority_t OS_task_prop_t::priority`

Definition at line 62 of file [osapi-task.h](#).

stack_size

`size_t OS_task_prop_t::stack_size`

Definition at line 61 of file [osapi-task.h](#).

The documentation for this struct was generated from the following file:

- [osal/src/os/inc/osapi-task.h](#)

10.216 OS_time_t Struct Reference

OSAL time interval structure.

```
#include <osapi-clock.h>
```

Data Fields

- [int64 ticks](#)

10.216.1 Detailed Description

OSAL time interval structure.

This is used to represent a basic time interval.

When used with OS_GetLocalTime/OS_SetLocalTime, this represents the interval from the OS's epoch point, typically 01 Jan 1970 00:00:00 UTC on systems that have a persistent real time clock (RTC), or the system boot time if there is no RTC available.

Applications should not directly access fields within this structure, as the definition may change in future versions of OSAL. Instead, applications should use the accessor/conversion methods defined below.

Definition at line 45 of file [osapi-clock.h](#).

10.216.2 Field Documentation**ticks**

```
int64 OS_time_t::ticks
```

Ticks elapsed since reference point

Definition at line 47 of file [osapi-clock.h](#).

Referenced by [OS_TimeAdd\(\)](#), [OS_TimeAssembleFromMicroseconds\(\)](#), [OS_TimeAssembleFromMilliseconds\(\)](#), [OS_TimeAssembleFromNanoseconds\(\)](#), [OS_TimeAssembleFromSubseconds\(\)](#), [OS_TimeEqual\(\)](#), [OS_TimeGetFractionalPart\(\)](#), [OS_TimeGetSign\(\)](#), [OS_TimeGetTotalMicroseconds\(\)](#), [OS_TimeGetTotalMilliseconds\(\)](#), [OS_TimeGetTotalNanoseconds\(\)](#), [OS_TimeGetTotalSeconds\(\)](#), and [OS_TimeSubtract\(\)](#).

The documentation for this struct was generated from the following file:

- [osal/src/os/inc/osapi-clock.h](#)

10.217 OS_timebase_prop_t Struct Reference

Time base properties.

```
#include <osapi-timebase.h>
```

Data Fields

- char [name](#) [OS_MAX_API_NAME]
- [osal_id_t](#) creator
- [uint32](#) nominal_interval_time
- [uint32](#) freerun_time
- [uint32](#) accuracy

10.217.1 Detailed Description

Time base properties.

Definition at line 37 of file [osapi-timebase.h](#).

10.217.2 Field Documentation**accuracy**

```
uint32 OS_timebase_prop_t::accuracy
```

Definition at line 43 of file [osapi-timebase.h](#).

creator

`osal_id_t OS_timebase_prop_t::creator`
Definition at line 40 of file [osapi-timebase.h](#).

freerun_time

`uint32 OS_timebase_prop_t::freerun_time`
Definition at line 42 of file [osapi-timebase.h](#).

name

`char OS_timebase_prop_t::name[OS_MAX_API_NAME]`
Definition at line 39 of file [osapi-timebase.h](#).

nominal_interval_time

`uint32 OS_timebase_prop_t::nominal_interval_time`
Definition at line 41 of file [osapi-timebase.h](#).
The documentation for this struct was generated from the following file:

- [osal/src/os/inc/osapi-timebase.h](#)

10.218 OS_timer_prop_t Struct Reference

Timer properties.

```
#include <osapi-timer.h>
```

Data Fields

- `char name[OS_MAX_API_NAME]`
- `osal_id_t creator`
- `uint32 start_time`
- `uint32 interval_time`
- `uint32 accuracy`

10.218.1 Detailed Description

Timer properties.

Definition at line 37 of file [osapi-timer.h](#).

10.218.2 Field Documentation

accuracy

`uint32 OS_timer_prop_t::accuracy`
Definition at line 43 of file [osapi-timer.h](#).

creator

`osal_id_t OS_timer_prop_t::creator`
Definition at line 40 of file [osapi-timer.h](#).

interval_time

`uint32 OS_timer_prop_t::interval_time`
Definition at line 42 of file [osapi-timer.h](#).

name

```
char OS_timer_prop_t::name[OS_MAX_API_NAME]
```

Definition at line 39 of file [osapi-timer.h](#).

start_time

```
uint32 OS_timer_prop_t::start_time
```

Definition at line 41 of file [osapi-timer.h](#).

The documentation for this struct was generated from the following file:

- [osal/src/os/inc/osapi-timer.h](#)

11 File Documentation

11.1 build/osal_public_api/inc/osconfig.h File Reference

Macros

- `#define OSAL_CONFIG_INCLUDE_DYNAMIC_LOADER`
Configuration file Operating System Abstraction Layer.
- `#define OSAL_CONFIG_INCLUDE_NETWORK`
- `#define OSAL_CONFIG_INCLUDE_STATIC_LOADER`
- `#define OSAL_CONFIG_CONSOLE_ASYNC`
- `#define OS_MAX_TASKS 64`
The maximum number of to support.
- `#define OS_MAX_QUEUES 64`
The maximum number of queues to support.
- `#define OS_MAX_COUNT_SEMAPHORES 20`
The maximum number of counting semaphores to support.
- `#define OS_MAX_BIN_SEMAPHORES 20`
The maximum number of binary semaphores to support.
- `#define OS_MAX_MUTEXES 20`
The maximum number of mutexes to support.
- `#define OS_MAX_RWLOCKS 20`
The maximum number of rwlocks to support.
- `#define OS_MAX_CONDVARS 4`
The maximum number of condition variables to support.
- `#define OS_MAX_MODULES 20`
The maximum number of modules to support.
- `#define OS_MAX_TIMEBASES 5`
The maximum number of timebases to support.
- `#define OS_MAX_TIMERS 10`
The maximum number of timer callbacks to support.
- `#define OS_MAX_NUM_OPEN_FILES 50`
The maximum number of concurrently open files to support.
- `#define OS_MAX_NUM_OPEN_DIRS 4`
The maximum number of concurrently open directories to support.
- `#define OS_MAX_FILE_SYSTEMS 14`
The maximum number of file systems to support.

- `#define OS_MAX_SYM_LEN 64`
The maximum length of symbols.
- `#define OS_MAX_FILE_NAME 20`
The maximum length of OSAL file names.
- `#define OS_MAX_PATH_LEN 64`
The maximum length of OSAL path names.
- `#define OS_MAX_API_NAME 20`
The maximum length of OSAL resource names.
- `#define OS_SOCKADDR_MAX_LEN 28`
The maximum size of the socket address structure.
- `#define OS_BUFFER_SIZE 172`
The maximum size of output produced by a single `OS_printf()`
- `#define OS_BUFFER_MSG_DEPTH 100`
The maximum number of `OS_printf()` output strings to buffer.
- `#define OS_UTILITYTASK_PRIORITY 245`
Priority level of the background utility task.
- `#define OS_UTILITYTASK_STACK_SIZE 2048`
The stack size of the background utility task.
- `#define OS_MAX_CMD_LEN 1000`
The maximum size of a shell command.
- `#define OS_QUEUE_MAX_DEPTH 50`
The maximum depth of OSAL queues.
- `#define OS_SHELL_CMD_INPUT_FILE_NAME ""`
The name of the temporary file used to store shell commands.
- `#define OS_PRINTF_CONSOLE_NAME ""`
The name of the primary console device.
- `#define OS_ADD_TASK_FLAGS 0`
Flags added to all tasks on creation.
- `#define OS_MAX_CONSOLES 1`
The maximum number of console devices to support.
- `#define OS_MODULE_FILE_EXTENSION ".so"`
The system-specific file extension used on loadable module files.
- `#define OS_FS_DEV_NAME_LEN 32`
- `#define OS_FS_PHYS_NAME_LEN 64`
- `#define OS_FS_VOL_NAME_LEN 32`

11.1.1 Macro Definition Documentation

OS_ADD_TASK_FLAGS

```
#define OS_ADD_TASK_FLAGS 0
```

Flags added to all tasks on creation.

Added to the task flags on creation

Supports adding floating point support for all tasks when the OS requires it

Definition at line 262 of file `osconfig.h`.

OS_BUFFER_MSG_DEPTH

```
#define OS_BUFFER_MSG_DEPTH 100
```

The maximum number of [OS_printf\(\)](#) output strings to buffer.

Based on the OSAL_CONFIG_PRINTF_BUFFER_DEPTH configuration option

Definition at line 195 of file [osconfig.h](#).

OS_BUFFER_SIZE

```
#define OS_BUFFER_SIZE 172
```

The maximum size of output produced by a single [OS_printf\(\)](#)

Based on the OSAL_CONFIG_PRINTF_BUFFER_SIZE configuration option

Definition at line 188 of file [osconfig.h](#).

OS_FS_DEV_NAME_LEN

```
#define OS_FS_DEV_NAME_LEN 32
```

Device name length

Definition at line 289 of file [osconfig.h](#).

OS_FS_PHYS_NAME_LEN

```
#define OS_FS_PHYS_NAME_LEN 64
```

Physical drive name length

Definition at line 290 of file [osconfig.h](#).

OS_FS_VOL_NAME_LEN

```
#define OS_FS_VOL_NAME_LEN 32
```

Volume name length

Definition at line 291 of file [osconfig.h](#).

OS_MAX_API_NAME

```
#define OS_MAX_API_NAME 20
```

The maximum length of OSAL resource names.

Based on the OSAL_CONFIG_MAX_API_NAME configuration option

Note

This value must include a terminating NUL character

Definition at line 171 of file [osconfig.h](#).

OS_MAX_BIN_SEMAPHORES

```
#define OS_MAX_BIN_SEMAPHORES 20
```

The maximum number of binary semaphores to support.

Based on the OSAL_CONFIG_MAX_BIN_SEMAPHORES configuration option

Definition at line 66 of file [osconfig.h](#).

OS_MAX_CMD_LEN

```
#define OS_MAX_CMD_LEN 1000
```

The maximum size of a shell command.

This limit is only applicable if shell support is enabled.

Based on the OSAL_CONFIG_MAX_CMD_LEN configuration option

Note

This value must include a terminating NUL character

Definition at line 226 of file [osconfig.h](#).

OS_MAX_CONDVARS

```
#define OS_MAX_CONDVARS 4
```

The maximum number of condition variables to support.

Based on the OSAL_CONFIG_MAX_CONDVARS configuration option

Definition at line 87 of file [osconfig.h](#).

OS_MAX_CONSOLES

```
#define OS_MAX_CONSOLES 1
```

The maximum number of console devices to support.

Fixed value based on current OSAL implementation, not user configurable.

Definition at line 277 of file [osconfig.h](#).

OS_MAX_COUNT_SEMAPHORES

```
#define OS_MAX_COUNT_SEMAPHORES 20
```

The maximum number of counting semaphores to support.

Based on the OSAL_CONFIG_MAX_COUNT_SEMAPHORES configuration option

Definition at line 59 of file [osconfig.h](#).

OS_MAX_FILE_NAME

```
#define OS_MAX_FILE_NAME 20
```

The maximum length of OSAL file names.

This limit applies specifically to the file name portion, not the directory portion, of a path name.

Based on the OSAL_CONFIG_MAX_FILE_NAME configuration option

Note

This value must include a terminating NUL character

Definition at line 150 of file [osconfig.h](#).

OS_MAX_FILE_SYSTEMS

```
#define OS_MAX_FILE_SYSTEMS 14
```

The maximum number of file systems to support.

Based on the OSAL_CONFIG_MAX_FILE_SYSTEMS configuration option

Definition at line 129 of file [osconfig.h](#).

OS_MAX_MODULES

```
#define OS_MAX_MODULES 20
```

The maximum number of modules to support.

Based on the OSAL_CONFIG_MAX_MODULES configuration option

Definition at line 94 of file [osconfig.h](#).

OS_MAX_MUTEXES

```
#define OS_MAX_MUTEXES 20
```

The maximum number of mutexes to support.

Based on the OSAL_CONFIG_MAX_MUTEXES configuration option

Definition at line 73 of file [osconfig.h](#).

OS_MAX_NUM_OPEN_DIRS

```
#define OS_MAX_NUM_OPEN_DIRS 4
```

The maximum number of concurrently open directories to support.

Based on the OSAL_CONFIG_MAX_NUM_OPEN_DIRS configuration option

Definition at line 122 of file [osconfig.h](#).

OS_MAX_NUM_OPEN_FILES

```
#define OS_MAX_NUM_OPEN_FILES 50
```

The maximum number of concurrently open files to support.

Based on the OSAL_CONFIG_MAX_NUM_OPEN_FILES configuration option

Definition at line 115 of file [osconfig.h](#).

OS_MAX_PATH_LEN

```
#define OS_MAX_PATH_LEN 64
```

The maximum length of OSAL path names.

This limit applies to the overall length of a path name, including the file name and directory portions.

Based on the OSAL_CONFIG_MAX_PATH_LEN configuration option

Note

This value must include a terminating NUL character

Definition at line 162 of file [osconfig.h](#).

OS_MAX_QUEUES

```
#define OS_MAX_QUEUES 64
```

The maximum number of queues to support.

Based on the OSAL_CONFIG_MAX_QUEUES configuration option

Definition at line 52 of file [osconfig.h](#).

OS_MAX_RWLOCKS

```
#define OS_MAX_RWLOCKS 20
```

The maximum number of rwlocks to support.

Based on the OSAL_CONFIG_MAX_RWLOCKS configuration option

Definition at line 80 of file [osconfig.h](#).

OS_MAX_SYM_LEN

```
#define OS_MAX_SYM_LEN 64
```

The maximum length of symbols.

Based on the OSAL_CONFIG_MAX_SYM_LEN configuration option

Note

This value must include a terminating NUL character

Definition at line 138 of file [osconfig.h](#).

OS_MAX_TASKS

```
#define OS_MAX_TASKS 64
```

The maximum number of to support.

Based on the OSAL_CONFIG_MAX_TASKS configuration option

Definition at line 45 of file [osconfig.h](#).

OS_MAX_TIMEBASES

```
#define OS_MAX_TIMEBASES 5
```

The maximum number of timebases to support.

Based on the OSAL_CONFIG_MAX_TIMEBASES configuration option

Definition at line 101 of file [osconfig.h](#).

OS_MAX_TIMERS

```
#define OS_MAX_TIMERS 10
```

The maximum number of timer callbacks to support.

Based on the OSAL_CONFIG_MAX_TIMERS configuration option

Definition at line 108 of file [osconfig.h](#).

OS_MODULE_FILE_EXTENSION

```
#define OS_MODULE_FILE_EXTENSION ".so"
```

The system-specific file extension used on loadable module files.

Fixed value based on system selection, not user configurable.

Definition at line 284 of file [osconfig.h](#).

OS_PRINTF_CONSOLE_NAME

```
#define OS_PRINTF_CONSOLE_NAME ""
```

The name of the primary console device.

This is the device to which [OS_printf\(\)](#) output is written. The output may be configured to tag each line with this prefix for identification.

Based on the OSAL_CONFIG_PRINTF_CONSOLE_NAME configuration option

Definition at line 253 of file [osconfig.h](#).

OS_QUEUE_MAX_DEPTH

```
#define OS_QUEUE_MAX_DEPTH 50
```

The maximum depth of OSAL queues.

Based on the OSAL_CONFIG_QUEUE_MAX_DEPTH configuration option

Definition at line 233 of file [osconfig.h](#).

OS_SHELL_CMD_INPUT_FILE_NAME

```
#define OS_SHELL_CMD_INPUT_FILE_NAME ""
```

The name of the temporary file used to store shell commands.

This configuration is only applicable if shell support is enabled, and only necessary/relevant on some OS implementations.

Based on the OSAL_CONFIG_SHELL_CMD_INPUT_FILE_NAME configuration option

Definition at line 243 of file [osconfig.h](#).

OS_SOCKADDR_MAX_LEN

```
#define OS_SOCKADDR_MAX_LEN 28
```

The maximum size of the socket address structure.

This is part of the Socket API, and should be set large enough to hold the largest address type in use on the target system.

Based on the OSAL_CONFIG_SOCKADDR_MAX_LEN configuration option

Definition at line 181 of file [osconfig.h](#).

OS_UTILITYTASK_PRIORITY

```
#define OS_UTILITYTASK_PRIORITY 245
```

Priority level of the background utility task.

This task is responsible for writing buffered output of OS_printf to the actual console device, and any other future maintenance task.

Based on the OSAL_CONFIG_UTILITYTASK_PRIORITY configuration option

Definition at line 205 of file [osconfig.h](#).

OS_UTILITYTASK_STACK_SIZE

```
#define OS_UTILITYTASK_STACK_SIZE 2048
```

The stack size of the background utility task.

This task is responsible for writing buffered output of OS_printf to the actual console device, and any other future maintenance task.

Based on the OSAL_CONFIG_UTILITYTASK_STACK_SIZE configuration option

Definition at line 215 of file [osconfig.h](#).

OSAL_CONFIG_CONSOLE_ASYNC

```
#define OSAL_CONFIG_CONSOLE_ASYNC
```

Definition at line 28 of file [osconfig.h](#).

OSAL_CONFIG_INCLUDE_DYNAMIC_LOADER

```
#define OSAL_CONFIG_INCLUDE_DYNAMIC_LOADER
```

Configuration file Operating System Abstraction Layer.

The specific definitions in this file may only be modified by setting the respective OSAL configuration options in the CMake build.

Any direct modifications to the generated copy will be overwritten each time CMake executes.

Note

This file was automatically generated by CMake from /tmp/vdd_cfe_docs_azgfvx93/cFS/osal/default_config.cmake

Definition at line 21 of file [osconfig.h](#).

OSAL_CONFIG_INCLUDE_NETWORK

```
#define OSAL_CONFIG_INCLUDE_NETWORK
```

Definition at line 22 of file [osconfig.h](#).

OSAL_CONFIG_INCLUDE_STATIC_LOADER

```
#define OSAL_CONFIG_INCLUDE_STATIC_LOADER
```

Definition at line 24 of file [osconfig.h](#).

11.2 osconfig.h

[Go to the documentation of this file.](#)

```

00001
00015 #ifndef OSCONFIG_H
00016 #define OSCONFIG_H
00017
00018 /*
00019  * OSAL feature selection options from build config
00020  */
00021 #define OSAL_CONFIG_INCLUDE_DYNAMIC_LOADER
00022 #define OSAL_CONFIG_INCLUDE_NETWORK
00023 /* #undef OSAL_CONFIG_INCLUDE_DNS */
00024 #define OSAL_CONFIG_INCLUDE_STATIC_LOADER
00025 /* #undef OSAL_CONFIG_INCLUDE_SHELL */
00026 /* #undef OSAL_CONFIG_DEBUG_PRINTF */
00027 /* #undef OSAL_CONFIG_DEBUG_PERMISSIVE_MODE */
00028 #define OSAL_CONFIG_CONSOLE_ASYNC
00029
00030 /* #undef OSAL_CONFIG_BUGCHECK_DISABLE */
00031 /* #undef OSAL_CONFIG_BUGCHECK_STRICT */
00032
00033 /*
00034  * OSAL resource limits from build config
00035  *
00036  * (These are prefixed with OS_ for compatibility
00037  * with existing code referencing these symbols)
00038  */
00039
00045 #define OS_MAX_TASKS 64
00046
00052 #define OS_MAX_QUEUES 64
00053
00059 #define OS_MAX_COUNT_SEMAPHORES 20
00060
00066 #define OS_MAX_BIN_SEMAPHORES 20
00067
00073 #define OS_MAX_MUTEXES 20
00074
00080 #define OS_MAX_RWLOCKS 20
00081
00087 #define OS_MAX_CONDVARS 4
00088
00094 #define OS_MAX_MODULES 20
00095
00101 #define OS_MAX_TIMEBASES 5
00102
00108 #define OS_MAX_TIMERS 10
00109
00115 #define OS_MAX_NUM_OPEN_FILES 50
00116
00122 #define OS_MAX_NUM_OPEN_DIRS 4
00123
00129 #define OS_MAX_FILE_SYSTEMS 14
00130
00138 #define OS_MAX_SYM_LEN 64
00139
00150 #define OS_MAX_FILE_NAME 20
00151
00162 #define OS_MAX_PATH_LEN 64
00163
00171 #define OS_MAX_API_NAME 20
00172
00181 #define OS_SOCKADDR_MAX_LEN 28
00182
00188 #define OS_BUFFER_SIZE 172
00189
00195 #define OS_BUFFER_MSG_DEPTH 100
00196
00205 #define OS_UTILITYTASK_PRIORITY 245
00206
00215 #define OS_UTILITYTASK_STACK_SIZE 2048
00216
00226 #define OS_MAX_CMD_LEN 1000
00227
00233 #define OS_QUEUE_MAX_DEPTH 50
00234
00243 #define OS_SHELL_CMD_INPUT_FILE_NAME ""
00244
00253 #define OS_PRINTF_CONSOLE_NAME ""

```

```

00254
00262 #define OS_ADD_TASK_FLAGS                0
00263
00264 /*
00265  * OSAL fixed resource limits
00266  *
00267  * The resource limits here are not user-configurable, but
00268  * may be changed in a future revision of OSAL, so it is
00269  * still present in osconfig.h along with the others.
00270  */
00271
00277 #define OS_MAX_CONSOLES                    1
00278
00284 #define OS_MODULE_FILE_EXTENSION          ".so"
00285
00286 /*
00287  ** Length of a Device and Volume name
00288  */
00289 #define OS_FS_DEV_NAME_LEN                32
00290 #define OS_FS_PHYS_NAME_LEN               64
00291 #define OS_FS_VOL_NAME_LEN                32
00294 #endif /* OSCONFIG_H */

```

11.3 cfe_perfids.h File Reference

Macros

- `#define CFE_MISSION_ES_PERF_EXIT_BIT 31`
bit (31) is reserved by the perf utilities

cFE Performance Monitor IDs (Reserved IDs 0-31)

- `#define CFE_MISSION_ES_MAIN_PERF_ID 1`
Performance ID for Executive Services Task.
- `#define CFE_MISSION_EVS_MAIN_PERF_ID 2`
Performance ID for Events Services Task.
- `#define CFE_MISSION_TBL_MAIN_PERF_ID 3`
Performance ID for Table Services Task.
- `#define CFE_MISSION_SB_MAIN_PERF_ID 4`
Performance ID for Software Bus Services Task.
- `#define CFE_MISSION_SB_MSG_LIM_PERF_ID 5`
Performance ID for Software Bus Msg Limit Errors.
- `#define CFE_MISSION_SB_PIPE_OFLOW_PERF_ID 27`
Performance ID for Software Bus Pipe Overflow Errors.
- `#define CFE_MISSION_TIME_MAIN_PERF_ID 6`
Performance ID for Time Services Task.
- `#define CFE_MISSION_TIME_TONE1HZISR_PERF_ID 7`
Performance ID for 1 Hz Tone ISR.
- `#define CFE_MISSION_TIME_LOCAL1HZISR_PERF_ID 8`
Performance ID for 1 Hz Local ISR.
- `#define CFE_MISSION_TIME_SENDMET_PERF_ID 9`
Performance ID for Time ToneSendMET.
- `#define CFE_MISSION_TIME_LOCAL1HZTASK_PERF_ID 10`
Performance ID for 1 Hz Local Task.
- `#define CFE_MISSION_TIME_TONE1HZTASK_PERF_ID 11`
Performance ID for 1 Hz Tone Task.

11.3.1 Detailed Description

Purpose: This file contains the cFE performance IDs

Design Notes: Each performance id is used to identify something that needs to be measured. Performance ids are limited to the range of 0 to CFE_MISSION_ES_PERF_MAX_IDS - 1. Any performance ids outside of this range will be ignored and will be flagged as an error. Note that performance ids 0-31 are reserved for the cFE Core.

References:

Definition in file [cfe_perfids.h](#).

11.3.2 Macro Definition Documentation

CFE_MISSION_ES_MAIN_PERF_ID

```
#define CFE_MISSION_ES_MAIN_PERF_ID 1
```

Performance ID for Executive Services Task.
Definition at line 42 of file [cfe_perfids.h](#).

CFE_MISSION_ES_PERF_EXIT_BIT

```
#define CFE_MISSION_ES_PERF_EXIT_BIT 31
```

bit (31) is reserved by the perf utilities
Definition at line 38 of file [cfe_perfids.h](#).

CFE_MISSION_EVS_MAIN_PERF_ID

```
#define CFE_MISSION_EVS_MAIN_PERF_ID 2
```

Performance ID for Events Services Task.
Definition at line 43 of file [cfe_perfids.h](#).

CFE_MISSION_SB_MAIN_PERF_ID

```
#define CFE_MISSION_SB_MAIN_PERF_ID 4
```

Performance ID for Software Bus Services Task.
Definition at line 45 of file [cfe_perfids.h](#).

CFE_MISSION_SB_MSG_LIM_PERF_ID

```
#define CFE_MISSION_SB_MSG_LIM_PERF_ID 5
```

Performance ID for Software Bus Msg Limit Errors.
Definition at line 46 of file [cfe_perfids.h](#).

CFE_MISSION_SB_PIPE_OFLOW_PERF_ID

```
#define CFE_MISSION_SB_PIPE_OFLOW_PERF_ID 27
```

Performance ID for Software Bus Pipe Overflow Errors.
Definition at line 47 of file [cfe_perfids.h](#).

CFE_MISSION_TBL_MAIN_PERF_ID

```
#define CFE_MISSION_TBL_MAIN_PERF_ID 3
```

Performance ID for Table Services Task.
Definition at line 44 of file [cfe_perfids.h](#).

CFE_MISSION_TIME_LOCAL1HZISR_PERF_ID

```
#define CFE_MISSION_TIME_LOCAL1HZISR_PERF_ID 8
```

Performance ID for 1 Hz Local ISR.
Definition at line 51 of file [cfe_perfids.h](#).

CFE_MISSION_TIME_LOCAL1HZTASK_PERF_ID

#define CFE_MISSION_TIME_LOCAL1HZTASK_PERF_ID 10
 Performance ID for 1 Hz Local Task.
 Definition at line 54 of file [cfe_perfids.h](#).

CFE_MISSION_TIME_MAIN_PERF_ID

#define CFE_MISSION_TIME_MAIN_PERF_ID 6
 Performance ID for Time Services Task.
 Definition at line 49 of file [cfe_perfids.h](#).

CFE_MISSION_TIME_SENDMET_PERF_ID

#define CFE_MISSION_TIME_SENDMET_PERF_ID 9
 Performance ID for Time ToneSendMET.
 Definition at line 53 of file [cfe_perfids.h](#).

CFE_MISSION_TIME_TONE1HZISR_PERF_ID

#define CFE_MISSION_TIME_TONE1HZISR_PERF_ID 7
 Performance ID for 1 Hz Tone ISR.
 Definition at line 50 of file [cfe_perfids.h](#).

CFE_MISSION_TIME_TONE1HZTASK_PERF_ID

#define CFE_MISSION_TIME_TONE1HZTASK_PERF_ID 11
 Performance ID for 1 Hz Tone Task.
 Definition at line 55 of file [cfe_perfids.h](#).

11.4 cfe_perfids.h

[Go to the documentation of this file.](#)

```

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00016  * limitations under the License.
00017  *****/
00018
00035 #ifndef SAMPLE_PERFIDS_H
00036 #define SAMPLE_PERFIDS_H
00037
00038 #define CFE_MISSION_ES_PERF_EXIT_BIT 31
00042 #define CFE_MISSION_ES_MAIN_PERF_ID      1
00043 #define CFE_MISSION_EVS_MAIN_PERF_ID     2
00044 #define CFE_MISSION_TBL_MAIN_PERF_ID     3
00045 #define CFE_MISSION_SB_MAIN_PERF_ID     4
00046 #define CFE_MISSION_SB_MSG_LIM_PERF_ID  5
00047 #define CFE_MISSION_SB_PIPE_OFLOW_PERF_ID 27
00049 #define CFE_MISSION_TIME_MAIN_PERF_ID    6
00050 #define CFE_MISSION_TIME_TONE1HZISR_PERF_ID 7
00051 #define CFE_MISSION_TIME_LOCAL1HZISR_PERF_ID 8
00053 #define CFE_MISSION_TIME_SENDMET_PERF_ID 9
00054 #define CFE_MISSION_TIME_LOCAL1HZTASK_PERF_ID 10

```

```
00055 #define CFE_MISSION_TIME_TONE1HZTASK_PERF_ID 11
00059 #endif /* SAMPLE_PERFIDS_H */
```

11.5 example_2x32bit_cfe_es_memaddress.h File Reference

```
#include "common_types.h"
```

Data Structures

- struct [CFE_ES_MemOffset_t](#)
Type used for memory sizes and offsets in commands and telemetry.
- struct [CFE_ES_MemAddress_t](#)
Type used for memory addresses in command and telemetry messages.

Macros

- #define [CFE_ES_MEMOFFSET_C\(x\)](#) [CFE_ES_MemOffset_FromNative\(x\)](#)
Memory Offset initializer wrapper.
- #define [CFE_ES_MEMOFFSET_TO_SIZET\(x\)](#) [CFE_ES_MemOffset_ToNative\(&\(x\)\)](#)
Memory Offset to integer value (size_t) wrapper.
- #define [CFE_ES_MEMADDRESS_C\(x\)](#) [CFE_ES_MemAddress_FromNative\(\(cpuaddr\)\(x\)\)](#)
Memory Address initializer wrapper.
- #define [CFE_ES_MEMADDRESS_TO_PTR\(x\)](#) [\(\(void *\)CFE_ES_MemAddress_ToNative\(&\(x\)\)\)](#)
Memory Address to pointer wrapper.

Functions

- static [size_t](#) [CFE_ES_MemOffset_ToNative](#) (const [CFE_ES_MemOffset_t](#) *val)
- static [CFE_ES_MemOffset_t](#) [CFE_ES_MemOffset_FromNative](#) ([size_t](#) val)
- static [cpuaddr](#) [CFE_ES_MemAddress_ToNative](#) (const [CFE_ES_MemAddress_t](#) *val)
- static [CFE_ES_MemAddress_t](#) [CFE_ES_MemAddress_FromNative](#) ([cpuaddr](#) val)

11.5.1 Detailed Description

Defines memory addresses and offsets to be 64 bit integer values

This expands the traditional 32 bit memory addresses in commands and telemetry out to 64 bits for compatibility with modern CPUs, but does so as a pair of 32 bit values rather than a single 64 bit value. This is done to avoid the introduction of implicit padding if addresses or sizes are not aligned at 64 bit offsets within the respective parent structure. This does necessitate that all access to these values is done via the provided conversion macros. Attempts to directly assign these values to an integer type will fail to compile.

Definition in file [example_2x32bit_cfe_es_memaddress.h](#).

11.5.2 Macro Definition Documentation

CFE_ES_MEMADDRESS_C

```
#define CFE_ES_MEMADDRESS_C(  
    x ) CFE\_ES\_MemAddress\_FromNative((cpuaddr) (x))
```

Memory Address initializer wrapper.

A converter macro to use when initializing a [CFE_ES_MemAddress_t](#) from a pointer value of a different type.

Definition at line 144 of file [example_2x32bit_cfe_es_memaddress.h](#).

CFE_ES_MEMADDRESS_TO_PTR

```
#define CFE_ES_MEMADDRESS_TO_PTR(
    x ) ((void *)CFE_ES_MemAddress_ToNative(&(x)))
```

Memory Address to pointer wrapper.

A converter macro to use when interpreting a [CFE_ES_MemAddress_t](#) as a pointer value.

Definition at line 152 of file [example_2x32bit_cfe_es_memaddress.h](#).

CFE_ES_MEMOFFSET_C

```
#define CFE_ES_MEMOFFSET_C(
    x ) CFE_ES_MemOffset_FromNative(x)
```

Memory Offset initializer wrapper.

A converter macro to use when initializing a [CFE_ES_MemOffset_t](#) from an integer value of a different type.

Definition at line 88 of file [example_2x32bit_cfe_es_memaddress.h](#).

CFE_ES_MEMOFFSET_TO_SIZE_T

```
#define CFE_ES_MEMOFFSET_TO_SIZE_T(
    x ) CFE_ES_MemOffset_ToNative(&(x))
```

Memory Offset to integer value (size_t) wrapper.

A converter macro to use when interpreting a [CFE_ES_MemOffset_t](#) value as a "size_t" type

Definition at line 96 of file [example_2x32bit_cfe_es_memaddress.h](#).

11.5.3 Function Documentation**CFE_ES_MemAddress_FromNative()**

```
static CFE_ES_MemAddress_t CFE_ES_MemAddress_FromNative (
    cpuaddr val ) [inline], [static]
```

Definition at line 130 of file [example_2x32bit_cfe_es_memaddress.h](#).

References [CFE_ES_MemAddress_t::bits](#).

CFE_ES_MemAddress_ToNative()

```
static cpuaddr CFE_ES_MemAddress_ToNative (
    const CFE_ES_MemAddress_t * val ) [inline], [static]
```

Definition at line 122 of file [example_2x32bit_cfe_es_memaddress.h](#).

References [CFE_ES_MemAddress_t::bits](#).

CFE_ES_MemOffset_FromNative()

```
static CFE_ES_MemOffset_t CFE_ES_MemOffset_FromNative (
    size_t val ) [inline], [static]
```

Definition at line 73 of file [example_2x32bit_cfe_es_memaddress.h](#).

References [CFE_ES_MemOffset_t::bits](#).

CFE_ES_MemOffset_ToNative()

```
static size_t CFE_ES_MemOffset_ToNative (
    const CFE_ES_MemOffset_t * val ) [inline], [static]
```

Definition at line 65 of file [example_2x32bit_cfe_es_memaddress.h](#).

References [CFE_ES_MemOffset_t::bits](#).

11.6 example_2x32bit_cfe_es_memaddress.h

[Go to the documentation of this file.](#)

```

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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00035 #ifndef EXAMPLE_2X32BIT_CFE_ES_MEMADDRESS_H
00036 #define EXAMPLE_2X32BIT_CFE_ES_MEMADDRESS_H
00037
00038 #include "common_types.h"
00039
00040 /*
00041  * NOTE: On a typical Linux/GCC build environment, this produces
00042  * sizeof(CFE_ES_AppInfo_t) == 216 based on the current definition
00043  */
00044
00060 typedef struct
00061 {
00062     uint32 bits[2];
00063 } CFE_ES_MemOffset_t;
00064
00065 static inline size_t CFE_ES_MemOffset_ToNative(const CFE_ES_MemOffset_t *val)
00066 {
00067     /* this is done as two separate shifts, in case size_t is 32 bits */
00068     /* if that is the case then bits[1] should be entirely shifted out */
00069     size_t size = (size_t)val->bits[1] << 16;
00070     return ((size << 16) | (size_t)val->bits[0]);
00071 }
00072
00073 static inline CFE_ES_MemOffset_t CFE_ES_MemOffset_FromNative(size_t val)
00074 {
00075     CFE_ES_MemOffset_t offs;
00076     offs.bits[0] = (uint32)val;
00077     val >>= 16;
00078     offs.bits[1] = (val >> 16);
00079     return offs;
00080 }
00081
00088 #define CFE_ES_MEMOFFSET_C(x) CFE_ES_MemOffset_FromNative(x)
00089
00096 #define CFE_ES_MEMOFFSET_TO_SIZE_T(x) CFE_ES_MemOffset_ToNative(&(x))
00097
00117 typedef struct
00118 {
00119     uint32 bits[2];
00120 } CFE_ES_MemAddress_t;
00121
00122 static inline cpuaddr CFE_ES_MemAddress_ToNative(const CFE_ES_MemAddress_t *val)
00123 {
00124     /* this is done as two separate shifts, in case size_t is 32 bits */
00125     /* if that is the case then bits[1] should be entirely shifted out */
00126     cpuaddr size = (cpuaddr)val->bits[1] << 16;
00127     return ((size << 16) | (cpuaddr)val->bits[0]);
00128 }
00129
00130 static inline CFE_ES_MemAddress_t CFE_ES_MemAddress_FromNative(cpuaddr val)
00131 {
00132     CFE_ES_MemAddress_t offs;
00133     offs.bits[0] = (uint32)val;
00134     val >>= 16;
00135     offs.bits[1] = (val >> 16);
00136     return offs;
00137 }
00144 #define CFE_ES_MEMADDRESS_C(x) CFE_ES_MemAddress_FromNative((cpuaddr)(x))
00145

```

```

00152 #define CFE_ES_MEMADDRESS_TO_PTR(x) ((void *)CFE_ES_MemAddress_ToNative(&(x)))
00153
00154 #endif /* CFE_ES_EXTERN_TYPEDEFS_H */

```

11.7 example_32bit_cfe_es_memaddress.h File Reference

```
#include "common_types.h"
```

Macros

- #define [CFE_ES_MEMOFFSET_C\(x\)](#) ((CFE_ES_MemOffset_t)(x))
Memory Offset initializer wrapper.
- #define [CFE_ES_MEMOFFSET_TO_SIZET\(x\)](#) ((size_t)(x))
Memory Offset to integer value (size_t) wrapper.
- #define [CFE_ES_MEMADDRESS_C\(x\)](#) ((CFE_ES_MemAddress_t)((cpuaddr)(x) & 0xFFFFFFFF))
Memory Address initializer wrapper.
- #define [CFE_ES_MEMADDRESS_TO_PTR\(x\)](#) ((void *) (cpuaddr)(x))
Memory Address to pointer wrapper.

Typedefs

- typedef [uint32 CFE_ES_MemOffset_t](#)
Type used for memory sizes and offsets in commands and telemetry.
- typedef [uint32 CFE_ES_MemAddress_t](#)
Type used for memory addresses in command and telemetry messages.

11.7.1 Detailed Description

Example header file override that defines memory addresses and offsets to be a 32 bit integer values. This is backward compatible with prior CFS versions where all memory references were assumed to be 32 bits in size. It is still possible to execute this on a 64-bit CPU, but all address will be truncated to the lower 32 bits. One will also not be able to send commands that require a memory address, as the address will be incomplete. To use this implementation, clone this file as "cfe_es_memaddress.h" in your local defs dir. Definition in file [example_32bit_cfe_es_memaddress.h](#).

11.7.2 Macro Definition Documentation

CFE_ES_MEMADDRESS_C

```
#define CFE_ES_MEMADDRESS_C(
    x ) ((CFE_ES_MemAddress_t)((cpuaddr)(x) & 0xFFFFFFFF))
```

Memory Address initializer wrapper.

A converter macro to use when initializing a [CFE_ES_MemAddress_t](#) from a pointer value of a different type.

Definition at line 102 of file [example_32bit_cfe_es_memaddress.h](#).

CFE_ES_MEMADDRESS_TO_PTR

```
#define CFE_ES_MEMADDRESS_TO_PTR(
    x ) ((void *) (cpuaddr)(x))
```

Memory Address to pointer wrapper.

A converter macro to use when interpreting a [CFE_ES_MemAddress_t](#) as a pointer value.

Definition at line 110 of file [example_32bit_cfe_es_memaddress.h](#).

CFE_ES_MEMOFFSET_C

```
#define CFE_ES_MEMOFFSET_C(
    x ) ((CFE_ES_MemOffset_t)(x))
```

Memory Offset initializer wrapper.

A converter macro to use when initializing a [CFE_ES_MemOffset_t](#) from an integer value of a different type.

Definition at line 65 of file [example_32bit_cfe_es_memaddress.h](#).

CFE_ES_MEMOFFSET_TO_SIZE_T

```
#define CFE_ES_MEMOFFSET_TO_SIZE_T(
    x ) ((size_t)(x))
```

Memory Offset to integer value (size_t) wrapper.

A converter macro to use when interpreting a [CFE_ES_MemOffset_t](#) value as a "size_t" type

Definition at line 73 of file [example_32bit_cfe_es_memaddress.h](#).

11.7.3 Typedef Documentation

CFE_ES_MemAddress_t

```
typedef uint32 CFE_ES_MemAddress_t
```

Type used for memory addresses in command and telemetry messages.

For backward compatibility with existing CFE code this should be uint32, but if running on a 64-bit platform, addresses in telemetry will be truncated to 32 bits and therefore will not be valid.

On 64-bit platforms this can be a 64-bit address which will allow the full memory address in commands and telemetry, but this will break compatibility with existing control systems, and may also change the alignment/padding of messages. In either case this must be an unsigned type.

FSW code should access this value via the macros provided, which converts to the native "cpuaddr" type provided by OSAL. This macro provides independence between the message representation and local representation of a memory address.

Definition at line 94 of file [example_32bit_cfe_es_memaddress.h](#).

CFE_ES_MemOffset_t

```
typedef uint32 CFE_ES_MemOffset_t
```

Type used for memory sizes and offsets in commands and telemetry.

For backward compatibility with existing CFE code this should be uint32, but all telemetry information will be limited to 4GB in size as a result.

On 64-bit platforms this can be a 64-bit value which will allow larger memory objects, but this will break compatibility with existing control systems, and may also change the alignment/padding of messages.

In either case this must be an unsigned type.

Definition at line 57 of file [example_32bit_cfe_es_memaddress.h](#).

11.8 example_32bit_cfe_es_memaddress.h

[Go to the documentation of this file.](#)

```
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```

```

00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00035 #ifndef EXAMPLE_32BIT_CFE_ES_MEMADDRESS_H
00036 #define EXAMPLE_32BIT_CFE_ES_MEMADDRESS_H
00037
00038 #include "common_types.h"
00039
00040 /*
00041  * NOTE: On a typical Linux/GCC build environment, this produces
00042  * sizeof(CFE_ES_AppInfo_t) == 184 based on the current definition
00043  */
00044
00057 typedef uint32 CFE_ES_MemOffset_t;
00058
00065 #define CFE_ES_MEMOFFSET_C(x) ((CFE_ES_MemOffset_t)(x))
00066
00073 #define CFE_ES_MEMOFFSET_TO_SIZE_T(x) ((size_t)(x))
00074
00094 typedef uint32 CFE_ES_MemAddress_t;
00095
00102 #define CFE_ES_MEMADDRESS_C(x) ((CFE_ES_MemAddress_t)((cpuaddr)(x) & 0xFFFFFFFF))
00103
00110 #define CFE_ES_MEMADDRESS_TO_PTR(x) ((void *) (cpuaddr)(x))
00111
00112 #endif /* CFE_ES_EXTERN_TYPEDEFS_H */

```

11.9 example_64bit_cfe_es_memaddress.h File Reference

```
#include "common_types.h"
```

Macros

- #define [CFE_ES_MEMOFFSET_C\(x\)](#) ((CFE_ES_MemOffset_t)(x))
Memory Offset initializer wrapper.
- #define [CFE_ES_MEMOFFSET_TO_SIZE_T\(x\)](#) ((size_t)(x))
Memory Offset to integer value (size_t) wrapper.
- #define [CFE_ES_MEMADDRESS_C\(x\)](#) ((CFE_ES_MemAddress_t)(cpuaddr)(x))
Memory Address initializer wrapper.
- #define [CFE_ES_MEMADDRESS_TO_PTR\(x\)](#) ((void *) (cpuaddr)(x))
Memory Address to pointer wrapper.

Typedefs

- typedef [uint64 CFE_ES_MemOffset_t](#)
Type used for memory sizes and offsets in commands and telemetry.
- typedef [uint64 CFE_ES_MemAddress_t](#)
Type used for memory addresses in command and telemetry messages.

11.9.1 Detailed Description

Example header file override that defines memory addresses and offsets to be a full 64 bit integer value.

This is the simplest and most efficient approach to use 64 bit addressing, but may introduce unexpected padding in certain cases. On most systems a uint64 value needs to be aligned on a 64-bit boundary, and CFS has some TLM and CMD structures where this is not the case. These will be padded by the compiler and may cause interoperability issues. To use this implementation, clone this file as "cfe_es_memaddress.h" in your local defs dir.

Definition in file [example_64bit_cfe_es_memaddress.h](#).

11.9.2 Macro Definition Documentation

CFE_ES_MEMADDRESS_C

```
#define CFE_ES_MEMADDRESS_C(  
    x ) ((CFE_ES_MemAddress_t) (cpuaddr) (x))
```

Memory Address initializer wrapper.

A converter macro to use when initializing a [CFE_ES_MemAddress_t](#) from a pointer value of a different type.

Definition at line 81 of file [example_64bit_cfe_es_memaddress.h](#).

CFE_ES_MEMADDRESS_TO_PTR

```
#define CFE_ES_MEMADDRESS_TO_PTR(  
    x ) ((void *) (cpuaddr) (x))
```

Memory Address to pointer wrapper.

A converter macro to use when interpreting a [CFE_ES_MemAddress_t](#) as a pointer value.

Definition at line 89 of file [example_64bit_cfe_es_memaddress.h](#).

CFE_ES_MEMOFFSET_C

```
#define CFE_ES_MEMOFFSET_C(  
    x ) ((CFE_ES_MemOffset_t) (x))
```

Memory Offset initializer wrapper.

A converter macro to use when initializing a [CFE_ES_MemOffset_t](#) from an integer value of a different type.

Definition at line 58 of file [example_64bit_cfe_es_memaddress.h](#).

CFE_ES_MEMOFFSET_TO_SIZET

```
#define CFE_ES_MEMOFFSET_TO_SIZET(  
    x ) ((size_t) (x))
```

Memory Offset to integer value (size_t) wrapper.

A converter macro to use when interpreting a [CFE_ES_MemOffset_t](#) value as a "size_t" type

Definition at line 66 of file [example_64bit_cfe_es_memaddress.h](#).

11.9.3 Typedef Documentation

CFE_ES_MemAddress_t

```
typedef uint64 CFE_ES_MemAddress_t
```

Type used for memory addresses in command and telemetry messages.

Use a full 64-bit integer value for memory offsets

Definition at line 73 of file [example_64bit_cfe_es_memaddress.h](#).

CFE_ES_MemOffset_t

```
typedef uint64 CFE_ES_MemOffset_t
```

Type used for memory sizes and offsets in commands and telemetry.

Use a full 64-bit integer value for memory offsets

Definition at line 50 of file [example_64bit_cfe_es_memaddress.h](#).

11.10 example_64bit_cfe_es_memaddress.h

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```
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00005  * Administrator of the National Aeronautics and Space Administration.
```

```

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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00035 #ifndef EXAMPLE_64BIT_CFE_ES_MEMADDRESS_H
00036 #define EXAMPLE_64BIT_CFE_ES_MEMADDRESS_H
00037
00038 #include "common_types.h"
00039
00040 /*
00041  * NOTE: On a typical Linux/GCC build environment, this produces
00042  * sizeof(CFE_ES_AppInfo_t) == 224 based on the current definition
00043  */
00044
00050 typedef uint64 CFE_ES_MemOffset_t;
00051
00058 #define CFE_ES_MEMOFFSET_C(x) ((CFE_ES_MemOffset_t)(x))
00059
00066 #define CFE_ES_MEMOFFSET_TO_SIZE_T(x) ((size_t)(x))
00067
00073 typedef uint64 CFE_ES_MemAddress_t;
00074
00081 #define CFE_ES_MEMADDRESS_C(x) ((CFE_ES_MemAddress_t)(cpuaddr)(x))
00082
00089 #define CFE_ES_MEMADDRESS_TO_PTR(x) ((void *) (cpuaddr)(x))
00090
00091 #endif /* CFE_ES_EXTERN_TYPEDEFS_H */

```

11.11 example_mission_cfg.h File Reference

Macros

- #define CFE_MISSION_MAX_PATH_LEN 64
- #define CFE_MISSION_MAX_FILE_LEN 20
- #define CFE_MISSION_MAX_API_LEN 20
- #define CFE_MISSION_MAX_NUM_FILES 50
- #define CFE_MISSION_ES_MAX_APPLICATIONS 16
- #define CFE_MISSION_ES_PERF_MAX_IDS 128
- #define CFE_MISSION_ES_POOL_MAX_BUCKETS 17
- #define CFE_MISSION_ES_CDS_MAX_NAME_LENGTH 16
- #define CFE_MISSION_ES_DEFAULT_CRC CFE_ES_CrcType_CRC_16
- #define CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN (CFE_MISSION_ES_CDS_MAX_NAME_LENGTH + CFE_MISSION_MAX_API_LEN + 4)

Checksum/CRC algorithm identifiers

- #define CFE_MISSION_ES_CRC_8 CFE_ES_CrcType_CRC_8 /* 1 */
- #define CFE_MISSION_ES_CRC_16 CFE_ES_CrcType_CRC_16 /* 2 */
- #define CFE_MISSION_ES_CRC_32 CFE_ES_CrcType_CRC_32 /* 3 */
- #define CFE_MISSION_EVS_MAX_MESSAGE_LENGTH 122
- #define CFE_FS_HDR_DESC_MAX_LEN 32
- *Max length of description field in a standard cFE File Header.*
- #define CFE_FS_FILE_CONTENT_ID 0x63464531
- *Magic Number for cFE compliant files (= 'cFE1')*
- #define CFE_MISSION_SB_MAX_SB_MSG_SIZE 32768
- #define CFE_MISSION_SB_MAX_PIPES 32
- #define CFE_MISSION_TBL_MAX_NAME_LENGTH 16

- `#define CFE_MISSION_TBL_MAX_FULL_NAME_LEN (CFE_MISSION_TBL_MAX_NAME_LENGTH + CFE_MISSION_MAX_API_LEN + 4)`
- `#define CFE_MISSION_TIME_CFG_DEFAULT_TAI true`
- `#define CFE_MISSION_TIME_CFG_DEFAULT_UTC false`
- `#define CFE_MISSION_TIME_CFG_FAKE_TONE true`
- `#define CFE_MISSION_TIME_AT_TONE_WAS true`
- `#define CFE_MISSION_TIME_AT_TONE_WILL_BE false`
- `#define CFE_MISSION_TIME_MIN_ELAPSED 0`
- `#define CFE_MISSION_TIME_MAX_ELAPSED 200000`
- `#define CFE_MISSION_TIME_DEF_MET_SECS 1000`
- `#define CFE_MISSION_TIME_DEF_MET_SUBS 0`
- `#define CFE_MISSION_TIME_DEF_STCF_SECS 1000000`
- `#define CFE_MISSION_TIME_DEF_STCF_SUBS 0`
- `#define CFE_MISSION_TIME_DEF_LEAPS 37`
- `#define CFE_MISSION_TIME_DEF_DELAY_SECS 0`
- `#define CFE_MISSION_TIME_DEF_DELAY_SUBS 1000`
- `#define CFE_MISSION_TIME_EPOCH_YEAR 1980`
- `#define CFE_MISSION_TIME_EPOCH_DAY 1`
- `#define CFE_MISSION_TIME_EPOCH_HOUR 0`
- `#define CFE_MISSION_TIME_EPOCH_MINUTE 0`
- `#define CFE_MISSION_TIME_EPOCH_SECOND 0`
- `#define CFE_MISSION_TIME_EPOCH_MICROS 0`
- `#define CFE_MISSION_TIME_FS_FACTOR 789004800`

11.11.1 Detailed Description

This header file contains the mission configuration parameters and typedefs with mission scope.

This provides values for configurable items that affect the interface(s) of this module. This includes the CMD/TLM message interface, tables definitions, and any other data products that serve to exchange information with other entities.

Note

It is no longer necessary to provide this file directly in the defs directory, but if present, this file is still supported/usable for backward compatibility. To use this file, it should be called "cfe_mission_cfg.h".

Going forward, more fine-grained (module/purposes-specific) header files are included with each submodule. These may be overridden as necessary, but only if a definition within that file needs to be changed from the default. This approach will reduce the amount of duplicate/cloned definitions and better support alternative build configurations in the future.

Note that if this file is present, the fine-grained header files noted above will *not* be used.

Definition in file [example_mission_cfg.h](#).

11.11.2 Macro Definition Documentation

CFE_FS_FILE_CONTENT_ID

```
#define CFE_FS_FILE_CONTENT_ID 0x63464531
```

Magic Number for cFE compliant files (= 'cFE1')

Definition at line 313 of file [example_mission_cfg.h](#).

CFE_FS_HDR_DESC_MAX_LEN

```
#define CFE_FS_HDR_DESC_MAX_LEN 32
```

Max length of description field in a standard cFE File Header.

Definition at line 311 of file [example_mission_cfg.h](#).

CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN

```
#define CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN (CFE_MISSION_ES_CDS_MAX_NAME_LENGTH + CFE_MISSION_MAX_API_LEN + 4)
```

Purpose Maximum Length of Full CDS Name in messages

Description:

Indicates the maximum length (in characters) of the entire CDS name of the following form: "ApplicationName.CDSName"

This affects the layout of command/telemetry messages but does not affect run time behavior or internal allocation.

Limits

All CPUs within the same SB domain (mission) must share the same definition Note this affects the size of messages, so it must not cause any message to exceed the max length.

This value should be kept as a multiple of 4, to maintain alignment of any possible neighboring fields without implicit padding.

Definition at line 262 of file [example_mission_cfg.h](#).

CFE_MISSION_ES_CDS_MAX_NAME_LENGTH

```
#define CFE_MISSION_ES_CDS_MAX_NAME_LENGTH 16
```

Purpose Maximum Length of CDS Name

Description:

Indicates the maximum length (in characters) of the CDS name ('CDSName') portion of a Full CDS Name of the following form: "ApplicationName.CDSName"

This length does not need to include an extra character for NULL termination.

Limits

This value should be kept as a multiple of 4, to maintain alignment of any possible neighboring fields without implicit padding.

Definition at line 228 of file [example_mission_cfg.h](#).

CFE_MISSION_ES_CRC_16

```
#define CFE_MISSION_ES_CRC_16 CFE_ES_CrcType_CRC_16 /* 2 */
```

Definition at line 270 of file [example_mission_cfg.h](#).

CFE_MISSION_ES_CRC_32

```
#define CFE_MISSION_ES_CRC_32 CFE_ES_CrcType_CRC_32 /* 3 */
```

Definition at line 271 of file [example_mission_cfg.h](#).

CFE_MISSION_ES_CRC_8

```
#define CFE_MISSION_ES_CRC_8 CFE_ES_CrcType_CRC_8 /* 1 */
```

Definition at line 269 of file [example_mission_cfg.h](#).

CFE_MISSION_ES_DEFAULT_CRC

```
#define CFE_MISSION_ES_DEFAULT_CRC CFE_ES_CrcType_CRC_16
```

Purpose Mission Default CRC algorithm

Description:

Indicates the which CRC algorithm should be used as the default for verifying the contents of Critical Data Stores and when calculating Table Image data integrity values.

Limits

Currently only CFE_ES_CrcType_CRC_16 is supported (see brief in CFE_ES_CrcType_Enum definition in [cfe_es_api_typedefs.h](#))

Definition at line 242 of file [example_mission_cfg.h](#).

CFE_MISSION_ES_MAX_APPLICATIONS

```
#define CFE_MISSION_ES_MAX_APPLICATIONS 16
```

Purpose Mission Max Apps in a message

Description:

Indicates the maximum number of apps in a telemetry housekeeping message

This affects the layout of command/telemetry messages but does not affect run time behavior or internal allocation.

Limits

All CPUs within the same SB domain (mission) must share the same definition Note this affects the size of messages, so it must not cause any message to exceed the max length.

Definition at line 173 of file [example_mission_cfg.h](#).

CFE_MISSION_ES_PERF_MAX_IDS

```
#define CFE_MISSION_ES_PERF_MAX_IDS 128
```

Purpose Define Max Number of Performance IDs for messages

Description:

Defines the maximum number of perf ids allowed in command/telemetry messages

This affects the layout of command/telemetry messages but does not affect run time behavior or internal allocation.

Limits

All CPUs within the same SB domain (mission) must share the same definition Note this affects the size of messages, so it must not cause any message to exceed the max length.

Definition at line 190 of file [example_mission_cfg.h](#).

CFE_MISSION_ES_POOL_MAX_BUCKETS

```
#define CFE_MISSION_ES_POOL_MAX_BUCKETS 17
```

Purpose Maximum number of block sizes in pool structures

Description:

The upper limit for the number of block sizes supported in the generic pool implementation, which in turn implements the memory pools and CDS. This definition is used as the array size with the pool stats structure, and therefore should be consistent across all CPUs in a mission, as well as with the ground station.

There is also a platform-specific limit which may be fewer than this value.

Limits:

Must be at least one. No specific upper limit, but the number is anticipated to be reasonably small (i.e. tens, not hundreds). Large values have not been tested.

Definition at line 211 of file [example_mission_cfg.h](#).

CFE_MISSION_EVS_MAX_MESSAGE_LENGTH

```
#define CFE_MISSION_EVS_MAX_MESSAGE_LENGTH 122
```

Purpose Maximum Event Message Length

Description:

Indicates the maximum length (in characters) of the formatted text string portion of an event message

This length does not need to include an extra character for NULL termination.

Limits

Not Applicable

Definition at line 297 of file [example_mission_cfg.h](#).

CFE_MISSION_MAX_API_LEN

```
#define CFE_MISSION_MAX_API_LEN 20
```

Purpose cFE Maximum length for API names within data exchange structures

Description:

The value of this constant dictates the size of filenames within all structures used for external data exchange, such as Software bus messages and table definitions. This is typically the same as OS_MAX_API_LEN but that is OSAL dependent – and as such it definable on a per-processor/OS basis and hence may be different across multiple processors. It is recommended to set this to the value of the largest OS_MAX_API_LEN in use on any CPU on the mission.

This affects only the layout of command/telemetry messages and table definitions; internal allocation may use the platform-specific OS_MAX_API_LEN value.

This length must include an extra character for NULL termination.

Limits

All CPUs within the same SB domain (mission) must share the same definition Note this affects the size of messages, so it must not cause any message to exceed the max length.

This value should be kept as a multiple of 4, to maintain alignment of any possible neighboring fields without implicit padding.

Definition at line 125 of file [example_mission_cfg.h](#).

CFE_MISSION_MAX_FILE_LEN

```
#define CFE_MISSION_MAX_FILE_LEN 20
```

Purpose cFE Maximum length for filenames within data exchange structures

Description:

The value of this constant dictates the size of filenames within all structures used for external data exchange, such as Software bus messages and table definitions. This is typically the same as OS_MAX_FILE_LEN but that is OSAL dependent – and as such it definable on a per-processor/OS basis and hence may be different across multiple processors. It is recommended to set this to the value of the largest OS_MAX_FILE_LEN in use on any CPU on the mission.

This affects only the layout of command/telemetry messages and table definitions; internal allocation may use the platform-specific OS_MAX_FILE_LEN value.

This length must include an extra character for NULL termination.

Limits

All CPUs within the same SB domain (mission) and ground tools must share the same definition. Note this affects the size of messages, so it must not cause any message to exceed the max length.

This value should be kept as a multiple of 4, to maintain alignment of any possible neighboring fields without implicit padding.

Definition at line 99 of file [example_mission_cfg.h](#).

CFE_MISSION_MAX_NUM_FILES

```
#define CFE_MISSION_MAX_NUM_FILES 50
```

Purpose cFE Maximum number of files in a message/data exchange

Description:

The value of this constant dictates the maximum number of files within all structures used for external data exchange, such as Software bus messages and table definitions. This is typically the same as OS_MAX_NUM_↵_OPEN_FILES but that is OSAL dependent – and as such it definable on a per-processor/OS basis and hence may be different across multiple processors. It is recommended to set this to the value of the largest OS_MAX_NUM_↵_OPEN_FILES in use on any CPU on the mission.

This affects only the layout of command/telemetry messages and table definitions; internal allocation may use the platform-specific OS_MAX_NUM_OPEN_FILES value.

Limits

All CPUs within the same SB domain (mission) must share the same definition Note this affects the size of messages, so it must not cause any message to exceed the max length.

Definition at line 147 of file [example_mission_cfg.h](#).

CFE_MISSION_MAX_PATH_LEN

```
#define CFE_MISSION_MAX_PATH_LEN 64
```

Purpose cFE Maximum length for pathnames within data exchange structures

Description:

The value of this constant dictates the size of pathnames within all structures used for external data exchange, such as Software bus messages and table definitions. This is typically the same as OS_MAX_PATH_LEN but that is OSAL dependent – and as such it definable on a per-processor/OS basis and hence may be different across multiple processors. It is recommended to set this to the value of the largest OS_MAX_PATH_LEN in use on any CPU on the mission.

This affects only the layout of command/telemetry messages and table definitions; internal allocation may use the platform-specific OS_MAX_PATH_LEN value.

This length must include an extra character for NULL termination.

Limits

All CPUs within the same SB domain (mission) and ground tools must share the same definition. Note this affects the size of messages, so it must not cause any message to exceed the max length.

This value should be kept as a multiple of 4, to maintain alignment of any possible neighboring fields without implicit padding.

Definition at line 72 of file [example_mission_cfg.h](#).

CFE_MISSION_SB_MAX_PIPES

```
#define CFE_MISSION_SB_MAX_PIPES 32
```

Purpose Maximum Number of pipes that SB command/telemetry messages may hold

Description:

Dictates the maximum number of unique Pipes the SB message definitions will hold.

This affects the layout of command/telemetry messages but does not affect run time behavior or internal allocation.

Limits

All CPUs within the same SB domain (mission) must share the same definition Note this affects the size of messages, so it must not cause any message to exceed the max length.

Definition at line 356 of file [example_mission_cfg.h](#).

CFE_MISSION_SB_MAX_SB_MSG_SIZE

```
#define CFE_MISSION_SB_MAX_SB_MSG_SIZE 32768
```

Purpose Maximum SB Message Size

Description:

The following definition dictates the maximum message size allowed on the software bus. SB checks the pkt length field in the header of all messages sent. If the pkt length field indicates the message is larger than this define, SB sends an event and rejects the send.

Limits

This parameter has a lower limit of 6 (CCSDS primary header size). There are no restrictions on the upper limit however, the maximum message size is system dependent and should be verified. Total message size values that are checked against this configuration are defined by a 16 bit data word.

Definition at line 339 of file [example_mission_cfg.h](#).

CFE_MISSION_TBL_MAX_FULL_NAME_LEN

```
#define CFE_MISSION_TBL_MAX_FULL_NAME_LEN (CFE_MISSION_TBL_MAX_NAME_LENGTH + CFE_MISSION_MAX_API_LEN + 4)
```

Purpose Maximum Length of Full Table Name in messages

Description:

Indicates the maximum length (in characters) of the entire table name within software bus messages, in "App←Name.TableName" notation.

This affects the layout of command/telemetry messages but does not affect run time behavior or internal allocation.

Limits

All CPUs within the same SB domain (mission) must share the same definition Note this affects the size of messages, so it must not cause any message to exceed the max length.

This value should be kept as a multiple of 4, to maintain alignment of any possible neighboring fields without implicit padding.

Definition at line 401 of file [example_mission_cfg.h](#).

CFE_MISSION_TBL_MAX_NAME_LENGTH

```
#define CFE_MISSION_TBL_MAX_NAME_LENGTH 16
```

Purpose Maximum Table Name Length

Description:

Indicates the maximum length (in characters) of the table name ('TblName') portion of a Full Table Name of the following form: "ApplicationName.TblName"

This length does not need to include an extra character for NULL termination.

Limits

This value should be kept as a multiple of 4, to maintain alignment of any possible neighboring fields without implicit padding.

Definition at line 381 of file [example_mission_cfg.h](#).

CFE_MISSION_TIME_AT_TONE_WAS

```
#define CFE_MISSION_TIME_AT_TONE_WAS true
```

Purpose Default Time and Tone Order

Description:

Time Services may be configured to expect the time at the tone data packet to either precede or follow the tone signal. If the time at the tone data packet follows the tone signal, then the data within the packet describes what the time "was" at the tone. If the time at the tone data packet precedes the tone signal, then the data within the packet describes what the time "will be" at the tone. One, and only one, of the following symbols must be set to true:

- CFE_MISSION_TIME_AT_TONE_WAS
- CFE_MISSION_TIME_AT_TONE_WILL_BE Note: If Time Services is defined as using a simulated tone signal (see [CFE_MISSION_TIME_CFG_FAKE_TONE](#) above), then the tone data packet must follow the tone signal.

Limits

Either CFE_MISSION_TIME_AT_TONE_WAS or CFE_MISSION_TIME_AT_TONE_WILL_BE must be set to true. They may not both be true and they may not both be false.

Definition at line 466 of file [example_mission_cfg.h](#).

CFE_MISSION_TIME_AT_TONE_WILL_BE

```
#define CFE_MISSION_TIME_AT_TONE_WILL_BE false
```

Definition at line 467 of file [example_mission_cfg.h](#).

CFE_MISSION_TIME_CFG_DEFAULT_TAI

```
#define CFE_MISSION_TIME_CFG_DEFAULT_TAI true
```

Purpose Default Time Format

Description:

The following definitions select either UTC or TAI as the default (mission specific) time format. Although it is possible for an application to request time in a specific format, most callers should use [CFE_TIME_GetTime\(\)](#), which returns time in the default format. This avoids having to modify each individual caller when the default choice is changed.

Limits

if CFE_MISSION_TIME_CFG_DEFAULT_TAI is defined as true then CFE_MISSION_TIME_CFG_DEFAULT_UTC must be defined as false. if CFE_MISSION_TIME_CFG_DEFAULT_TAI is defined as false then CFE_MISSION_←
_TIME_CFG_DEFAULT_UTC must be defined as true.

Definition at line 430 of file [example_mission_cfg.h](#).

CFE_MISSION_TIME_CFG_DEFAULT_UTC

```
#define CFE_MISSION_TIME_CFG_DEFAULT_UTC false
```

Definition at line 431 of file [example_mission_cfg.h](#).

CFE_MISSION_TIME_CFG_FAKE_TONE

```
#define CFE_MISSION_TIME_CFG_FAKE_TONE true
```

Purpose Default Time Format

Description:

The following definition enables the use of a simulated time at the tone signal using a software bus message.

Limits

Not Applicable

Definition at line 443 of file [example_mission_cfg.h](#).

CFE_MISSION_TIME_DEF_DELAY_SECS

```
#define CFE_MISSION_TIME_DEF_DELAY_SECS 0
```

Definition at line 525 of file [example_mission_cfg.h](#).

CFE_MISSION_TIME_DEF_DELAY_SUBS

```
#define CFE_MISSION_TIME_DEF_DELAY_SUBS 1000
```

Definition at line 526 of file [example_mission_cfg.h](#).

CFE_MISSION_TIME_DEF_LEAPS

```
#define CFE_MISSION_TIME_DEF_LEAPS 37
```

Definition at line 523 of file [example_mission_cfg.h](#).

CFE_MISSION_TIME_DEF_MET_SECS

```
#define CFE_MISSION_TIME_DEF_MET_SECS 1000
```

Purpose Default Time Values

Description:

Default time values are provided to avoid problems due to time calculations performed after startup but before commands can be processed. For example, if the default time format is UTC then it is important that the sum of MET and STCF always exceed the value of Leap Seconds to prevent the UTC time calculation ($\text{time} = \text{MET} + \text{STCF} - \text{Leap Seconds}$) from resulting in a negative (very large) number.

Some past missions have also created known (albeit wrong) default timestamps. For example, assume the epoch is defined as Jan 1, 1970 and further assume the default time values are set to create a timestamp of Jan 1, 2000. Even though the year 2000 timestamps are wrong, it may be of value to keep the time within some sort of bounds acceptable to the software.

Note: Sub-second units are in micro-seconds (0 to 999,999) and all values must be defined

Limits

Not Applicable

Definition at line 517 of file [example_mission_cfg.h](#).

CFE_MISSION_TIME_DEF_MET_SUBS

```
#define CFE_MISSION_TIME_DEF_MET_SUBS 0
```

Definition at line 518 of file [example_mission_cfg.h](#).

CFE_MISSION_TIME_DEF_STCF_SECS

```
#define CFE_MISSION_TIME_DEF_STCF_SECS 1000000
```

Definition at line 520 of file [example_mission_cfg.h](#).

CFE_MISSION_TIME_DEF_STCF_SUBS

```
#define CFE_MISSION_TIME_DEF_STCF_SUBS 0
```

Definition at line 521 of file [example_mission_cfg.h](#).

CFE_MISSION_TIME_EPOCH_DAY

```
#define CFE_MISSION_TIME_EPOCH_DAY 1
```

Definition at line 544 of file [example_mission_cfg.h](#).

CFE_MISSION_TIME_EPOCH_HOUR

```
#define CFE_MISSION_TIME_EPOCH_HOUR 0
```

Definition at line 545 of file [example_mission_cfg.h](#).

CFE_MISSION_TIME_EPOCH_MICROS

```
#define CFE_MISSION_TIME_EPOCH_MICROS 0
```

Definition at line 548 of file [example_mission_cfg.h](#).

CFE_MISSION_TIME_EPOCH_MINUTE

```
#define CFE_MISSION_TIME_EPOCH_MINUTE 0
```

Definition at line 546 of file [example_mission_cfg.h](#).

CFE_MISSION_TIME_EPOCH_SECOND

```
#define CFE_MISSION_TIME_EPOCH_SECOND 0
```

Definition at line 547 of file [example_mission_cfg.h](#).

CFE_MISSION_TIME_EPOCH_YEAR

```
#define CFE_MISSION_TIME_EPOCH_YEAR 1980
```

Purpose Default EPOCH Values

Description:

Default ground time epoch values Note: these values are used only by the [CFE_TIME_Print\(\)](#) API function

Limits

Year - must be within 136 years Day - Jan 1 = 1, Feb 1 = 32, etc. Hour - 0 to 23 Minute - 0 to 59 Second - 0 to 59
Micros - 0 to 999999

Definition at line 543 of file [example_mission_cfg.h](#).

CFE_MISSION_TIME_FS_FACTOR

```
#define CFE_MISSION_TIME_FS_FACTOR 789004800
```

Purpose Time File System Factor

Description:

Define the s/c vs file system time conversion constant...

Note: this value is intended for use only by CFE TIME API functions to convert time values based on the ground system epoch (s/c time) to and from time values based on the file system epoch (fs time).

FS time = S/C time + factor S/C time = FS time - factor

Worksheet:

S/C epoch = Jan 1, 2005 (LRO ground system epoch) FS epoch = Jan 1, 1980 (vxWorks DOS file system epoch)

Delta = 25 years, 0 days, 0 hours, 0 minutes, 0 seconds

Leap years = 1980, 1984, 1988, 1992, 1996, 2000, 2004 (divisible by 4 – except if by 100 – unless also by 400)

1 year = 31,536,000 seconds 1 day = 86,400 seconds 1 hour = 3,600 seconds 1 minute = 60 seconds

25 years = 788,400,000 seconds 7 extra leap days = 604,800 seconds

total delta = 789,004,800 seconds

Limits

Not Applicable

Definition at line 586 of file [example_mission_cfg.h](#).

CFE_MISSION_TIME_MAX_ELAPSED

```
#define CFE_MISSION_TIME_MAX_ELAPSED 200000
```

Definition at line 492 of file [example_mission_cfg.h](#).

CFE_MISSION_TIME_MIN_ELAPSED

```
#define CFE_MISSION_TIME_MIN_ELAPSED 0
```

Purpose Min and Max Time Elapsed

Description:

Based on the definition of Time and Tone Order (CFE_MISSION_TIME_AT_TONE_WAS/WILL_BE) either the "time at the tone" signal or data packet will follow the other. This definition sets the valid window of time for the second of the pair to lag behind the first. Time Services will invalidate both the tone and packet if the second does not arrive within this window following the first.

For example, if the data packet follows the tone, it might be valid for the data packet to arrive between zero and 100,000 micro-seconds after the tone. But, if the tone follows the packet, it might be valid only if the packet arrived between 200,000 and 700,000 micro-seconds before the tone.

Note: units are in micro-seconds

Limits

0 to 999,999 decimal

Definition at line 491 of file [example_mission_cfg.h](#).

11.12 example_mission_cfg.h

[Go to the documentation of this file.](#)

```
00001 /*****
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"
00003  *
00004  * Copyright (c) 2023 United States Government as represented by the
00005  * Administrator of the National Aeronautics and Space Administration.
00006  * All Rights Reserved.
00007  *
00008  * Licensed under the Apache License, Version 2.0 (the "License"); you may
00009  * not use this file except in compliance with the License. You may obtain
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00011  *
00012  * Unless required by applicable law or agreed to in writing, software
00013  * distributed under the License is distributed on an "AS IS" BASIS,
00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00044 #ifndef EXAMPLE_MISSION_CFG_H
00045 #define EXAMPLE_MISSION_CFG_H
00046
00072 #define CFE_MISSION_MAX_PATH_LEN 64
00073
00099 #define CFE_MISSION_MAX_FILE_LEN 20
00100
00125 #define CFE_MISSION_MAX_API_LEN 20
00126
00147 #define CFE_MISSION_MAX_NUM_FILES 50
00148
```

```

00149 /*****
00150 *   CFE Executive Services (CFE_ES) Application Public Definitions
00151 *
00152 * This provides default values for configurable items that affect
00153 * the interface(s) of this module. This includes the CMD/TLM message
00154 * interface, tables definitions, and any other data products that
00155 * serve to exchange information with other entities.
00156 *
00157 */
00158
00173 #define CFE_MISSION_ES_MAX_APPLICATIONS 16
00174
00190 #define CFE_MISSION_ES_PERF_MAX_IDS 128
00191
00211 #define CFE_MISSION_ES_POOL_MAX_BUCKETS 17
00212
00228 #define CFE_MISSION_ES_CDS_MAX_NAME_LENGTH 16
00229
00242 #define CFE_MISSION_ES_DEFAULT_CRC CFE_ES_CrcType_CRC_16
00243
00262 #define CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN (CFE_MISSION_ES_CDS_MAX_NAME_LENGTH + CFE_MISSION_MAX_API_LEN
+ 4)
00263
00264 #ifndef CFE_OMIT_DEPRECATED_6_8
00265 /* These names have been converted to an enum in cfe_es_api_typedefs.h */
00266
00269 #define CFE_MISSION_ES_CRC_8 CFE_ES_CrcType_CRC_8 /* 1 */
00270 #define CFE_MISSION_ES_CRC_16 CFE_ES_CrcType_CRC_16 /* 2 */
00271 #define CFE_MISSION_ES_CRC_32 CFE_ES_CrcType_CRC_32 /* 3 */
00272
00273 #endif
00274
00275 /*****
00276 *   CFE Event Services (CFE_EVS) Application Public Definitions
00277 *
00278 * This provides default values for configurable items that affect
00279 * the interface(s) of this module. This includes the CMD/TLM message
00280 * interface, tables definitions, and any other data products that
00281 * serve to exchange information with other entities.
00282 *
00283 */
00284
00297 #define CFE_MISSION_EVS_MAX_MESSAGE_LENGTH 122
00298
00299 /*****
00300 *   CFE File Services (CFE_FS) Public Definitions
00301 *
00302 */
00303
00304 /*
00305 * NOTE: the value of CFE_FS_HDR_DESC_MAX_LEN, if modified, should
00306 * be constrained to multiples of 4, as it is used within a structure that
00307 * also contains uint32 types. This ensures that the entire structure
00308 * remains 32-bit aligned without the need for implicit padding bytes.
00309 */
00310
00311 #define CFE_FS_HDR_DESC_MAX_LEN 32
00313 #define CFE_FS_FILE_CONTENT_ID 0x63464531
00315 /*****
00316 *   CFE Software Bus (CFE_SB) Application Public Definitions
00317 *
00318 * This provides default values for configurable items that affect
00319 * the interface(s) of this module. This includes the CMD/TLM message
00320 * interface, tables definitions, and any other data products that
00321 * serve to exchange information with other entities.
00322 */
00323
00339 #define CFE_MISSION_SB_MAX_SB_MSG_SIZE 32768
00340
00356 #define CFE_MISSION_SB_MAX_PIPES 32
00357
00358 /*****
00359 *   CFE Table Services (CFE_TBL) Application Public Definitions
00360 *
00361 * This provides default values for configurable items that affect
00362 * the interface(s) of this module. This includes the CMD/TLM message
00363 * interface, tables definitions, and any other data products that
00364 * serve to exchange information with other entities.
00365 */
00366
00381 #define CFE_MISSION_TBL_MAX_NAME_LENGTH 16

```



```

00382
00401 #define CFE_MISSION_TBL_MAX_FULL_NAME_LEN (CFE_MISSION_TBL_MAX_NAME_LENGTH + CFE_MISSION_MAX_API_LEN + 4)
00402
00403 /*****
00404  *   CFE Time Services (CFE_TIME) Application Public Definitions
00405  *
00406  *   This provides default values for configurable items that affect
00407  *   the interface(s) of this module. This includes the CMD/TLM message
00408  *   interface, tables definitions, and any other data products that
00409  *   serve to exchange information with other entities.
00410  *
00411  */
00412
00430 #define CFE_MISSION_TIME_CFG_DEFAULT_TAI true
00431 #define CFE_MISSION_TIME_CFG_DEFAULT_UTC false
00432
00443 #define CFE_MISSION_TIME_CFG_FAKE_TONE true
00444
00466 #define CFE_MISSION_TIME_AT_TONE_WAS true
00467 #define CFE_MISSION_TIME_AT_TONE_WILL_BE false
00468
00491 #define CFE_MISSION_TIME_MIN_ELAPSED 0
00492 #define CFE_MISSION_TIME_MAX_ELAPSED 200000
00493
00517 #define CFE_MISSION_TIME_DEF_MET_SECS 1000
00518 #define CFE_MISSION_TIME_DEF_MET_SUBS 0
00519
00520 #define CFE_MISSION_TIME_DEF_STCF_SECS 1000000
00521 #define CFE_MISSION_TIME_DEF_STCF_SUBS 0
00522
00523 #define CFE_MISSION_TIME_DEF_LEAPS 37
00524
00525 #define CFE_MISSION_TIME_DEF_DELAY_SECS 0
00526 #define CFE_MISSION_TIME_DEF_DELAY_SUBS 1000
00527
00543 #define CFE_MISSION_TIME_EPOCH_YEAR 1980
00544 #define CFE_MISSION_TIME_EPOCH_DAY 1
00545 #define CFE_MISSION_TIME_EPOCH_HOUR 0
00546 #define CFE_MISSION_TIME_EPOCH_MINUTE 0
00547 #define CFE_MISSION_TIME_EPOCH_SECOND 0
00548 #define CFE_MISSION_TIME_EPOCH_MICROS 0
00549
00586 #define CFE_MISSION_TIME_FS_FACTOR 789004800
00587
00588 #endif /* EXAMPLE_MISSION_CFG_H */

```

11.13 example_platform_cfg.h File Reference

Macros

- #define CFE_PLATFORM_ENDIAN CCSDS_LITTLE_ENDIAN
- #define CFE_PLATFORM_CORE_MAX_STARTUP_MSEC 30000
- #define CFE_PLATFORM_ES_START_TASK_PRIORITY 68
- #define CFE_PLATFORM_ES_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
- #define CFE_PLATFORM_ES_NONVOL_DISK_MOUNT_STRING "/cf"
- #define CFE_PLATFORM_ES_RAM_DISK_MOUNT_STRING "/ram"
- #define CFE_PLATFORM_ES_MAX_APPLICATIONS 32
- #define CFE_PLATFORM_ES_MAX_LIBRARIES 10
- #define CFE_PLATFORM_ES_ER_LOG_ENTRIES 20
- #define CFE_PLATFORM_ES_ER_LOG_MAX_CONTEXT_SIZE 256
- #define CFE_PLATFORM_ES_SYSTEM_LOG_SIZE 3072
- #define CFE_PLATFORM_ES_MAX_GEN_COUNTERS 8
- #define CFE_PLATFORM_ES_APP_SCAN_RATE 1000
- #define CFE_PLATFORM_ES_APP_KILL_TIMEOUT 5
- #define CFE_PLATFORM_ES_RAM_DISK_SECTOR_SIZE 512
- #define CFE_PLATFORM_ES_RAM_DISK_NUM_SECTORS 4096
- #define CFE_PLATFORM_ES_RAM_DISK_PERCENT_RESERVED 30
- #define CFE_PLATFORM_ES_CDS_SIZE (128 * 1024)

- `#define CFE_PLATFORM_ES_USER_RESERVED_SIZE (1024 * 1024)`
- `#define CFE_PLATFORM_ES_MEMPOOL_ALIGN_SIZE_MIN 4`
- `#define CFE_PLATFORM_ES_NONVOL_STARTUP_FILE "/cf/cfe_es_startup.scr"`
- `#define CFE_PLATFORM_ES_VOLATILE_STARTUP_FILE "/ram/cfe_es_startup.scr"`
- `#define CFE_PLATFORM_ES_DEFAULT_APP_LOG_FILE "/ram/cfe_es_app_info.log"`
- `#define CFE_PLATFORM_ES_DEFAULT_TASK_LOG_FILE "/ram/cfe_es_taskinfo.log"`
- `#define CFE_PLATFORM_ES_DEFAULT_SYSLOG_FILE "/ram/cfe_es_syslog.log"`
- `#define CFE_PLATFORM_ES_DEFAULT_ER_LOG_FILE "/ram/cfe_erlog.log"`
- `#define CFE_PLATFORM_ES_DEFAULT_PERF_DUMP_FILENAME "/ram/cfe_es_perf.dat"`
- `#define CFE_PLATFORM_ES_DEFAULT_CDS_REG_DUMP_FILE "/ram/cfe_cds_reg.log"`
- `#define CFE_PLATFORM_ES_DEFAULT_POR_SYSLOG_MODE 0`
- `#define CFE_PLATFORM_ES_DEFAULT_PR_SYSLOG_MODE 1`
- `#define CFE_PLATFORM_ES_PERF_DATA_BUFFER_SIZE 10000`
- `#define CFE_PLATFORM_ES_PERF_FILTMASK_NONE 0`
- `#define CFE_PLATFORM_ES_PERF_FILTMASK_ALL ~CFE_PLATFORM_ES_PERF_FILTMASK_NONE`
- `#define CFE_PLATFORM_ES_PERF_FILTMASK_INIT CFE_PLATFORM_ES_PERF_FILTMASK_ALL`
- `#define CFE_PLATFORM_ES_PERF_TRIGMASK_NONE 0`
- `#define CFE_PLATFORM_ES_PERF_TRIGMASK_ALL ~CFE_PLATFORM_ES_PERF_TRIGMASK_NONE`
- `#define CFE_PLATFORM_ES_PERF_TRIGMASK_INIT CFE_PLATFORM_ES_PERF_TRIGMASK_NONE`
- `#define CFE_PLATFORM_ES_PERF_CHILD_PRIORITY 200`
- `#define CFE_PLATFORM_ES_PERF_CHILD_STACK_SIZE 4096`
- `#define CFE_PLATFORM_ES_PERF_CHILD_MS_DELAY 20`
- `#define CFE_PLATFORM_ES_PERF_ENTRIES_BTWN_DLYS 50`
- `#define CFE_PLATFORM_ES_DEFAULT_STACK_SIZE 8192`
- `#define CFE_PLATFORM_ES_CDS_MAX_NUM_ENTRIES 512`
- `#define CFE_PLATFORM_ES_MAX_PROCESSOR_RESETS 2`
- `#define CFE_PLATFORM_ES_POOL_MAX_BUCKETS 17`
- `#define CFE_PLATFORM_ES_MAX_MEMORY_POOLS 10`
- `#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_01 8`
- `#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_02 16`
- `#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_03 32`
- `#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_04 48`
- `#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_05 64`
- `#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_06 96`
- `#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_07 128`
- `#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_08 160`
- `#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_09 256`
- `#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_10 512`
- `#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_11 1024`
- `#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_12 2048`
- `#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_13 4096`
- `#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_14 8192`
- `#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_15 16384`
- `#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_16 32768`
- `#define CFE_PLATFORM_ES_MAX_BLOCK_SIZE 80000`
- `#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_01 8`
- `#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_02 16`
- `#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_03 32`
- `#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_04 48`
- `#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_05 64`
- `#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_06 96`

- `#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_07` 128
- `#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_08` 160
- `#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_09` 256
- `#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_10` 512
- `#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_11` 1024
- `#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_12` 2048
- `#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_13` 4096
- `#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_14` 8192
- `#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_15` 16384
- `#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_16` 32768
- `#define CFE_PLATFORM_ES_CDS_MAX_BLOCK_SIZE` 80000
- `#define CFE_PLATFORM_ES_STARTUP_SYNC_POLL_MSEC` 50
- `#define CFE_PLATFORM_ES_STARTUP_SCRIPT_TIMEOUT_MSEC` 1000
- `#define CFE_PLATFORM_EVS_START_TASK_PRIORITY` 61
- `#define CFE_PLATFORM_EVS_START_TASK_STACK_SIZE` `CFE_PLATFORM_ES_DEFAULT_STACK_SIZE`
- `#define CFE_PLATFORM_EVS_MAX_EVENT_FILTERS` 8
- `#define CFE_PLATFORM_EVS_MAX_APP_EVENT_BURST` 32
- `#define CFE_PLATFORM_EVS_APP_EVENTS_PER_SEC` 15
- `#define CFE_PLATFORM_EVS_DEFAULT_LOG_FILE` `"/ram/cfe_evs.log"`
- `#define CFE_PLATFORM_EVS_LOG_MAX` 20
- `#define CFE_PLATFORM_EVS_DEFAULT_APP_DATA_FILE` `"/ram/cfe_evs_app.dat"`
- `#define CFE_PLATFORM_EVS_PORT_DEFAULT` 0x0001
- `#define CFE_PLATFORM_EVS_DEFAULT_TYPE_FLAG` 0xE
- `#define CFE_PLATFORM_EVS_DEFAULT_LOG_MODE` 1
- `#define CFE_PLATFORM_EVS_DEFAULT_MSG_FORMAT_MODE` `CFE_EVS_MsgFormat_LONG`
- `#define CFE_PLATFORM_SB_MAX_MSG_IDS` 256
- `#define CFE_PLATFORM_SB_MAX_PIPES` 64
- `#define CFE_PLATFORM_SB_MAX_DEST_PER_PKT` 16
- `#define CFE_PLATFORM_SB_DEFAULT_MSG_LIMIT` 4
- `#define CFE_PLATFORM_SB_BUF_MEMORY_BYTES` 524288
- `#define CFE_PLATFORM_SB_HIGHEST_VALID_MSGID` 0x1FFF
- `#define CFE_PLATFORM_SB_DEFAULT_ROUTING_FILENAME` `"/ram/cfe_sb_route.dat"`
- `#define CFE_PLATFORM_SB_DEFAULT_PIPE_FILENAME` `"/ram/cfe_sb_pipe.dat"`
- `#define CFE_PLATFORM_SB_DEFAULT_MAP_FILENAME` `"/ram/cfe_sb_msgmap.dat"`
- `#define CFE_PLATFORM_SB_FILTERED_EVENT1` `CFE_SB_SEND_NO_SUBS_EID`
- `#define CFE_PLATFORM_SB_FILTER_MASK1` `CFE_EVS_FIRST_4_STOP`
- `#define CFE_PLATFORM_SB_FILTERED_EVENT2` `CFE_SB_DUP_SUBSCRIP_EID`
- `#define CFE_PLATFORM_SB_FILTER_MASK2` `CFE_EVS_FIRST_4_STOP`
- `#define CFE_PLATFORM_SB_FILTERED_EVENT3` `CFE_SB_MSGID_LIM_ERR_EID`
- `#define CFE_PLATFORM_SB_FILTER_MASK3` `CFE_EVS_FIRST_16_STOP`
- `#define CFE_PLATFORM_SB_FILTERED_EVENT4` `CFE_SB_Q_FULL_ERR_EID`
- `#define CFE_PLATFORM_SB_FILTER_MASK4` `CFE_EVS_FIRST_16_STOP`
- `#define CFE_PLATFORM_SB_FILTERED_EVENT5` 0
- `#define CFE_PLATFORM_SB_FILTER_MASK5` `CFE_EVS_NO_FILTER`
- `#define CFE_PLATFORM_SB_FILTERED_EVENT6` 0
- `#define CFE_PLATFORM_SB_FILTER_MASK6` `CFE_EVS_NO_FILTER`
- `#define CFE_PLATFORM_SB_FILTERED_EVENT7` 0
- `#define CFE_PLATFORM_SB_FILTER_MASK7` `CFE_EVS_NO_FILTER`
- `#define CFE_PLATFORM_SB_FILTERED_EVENT8` 0
- `#define CFE_PLATFORM_SB_FILTER_MASK8` `CFE_EVS_NO_FILTER`
- `#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_01` 8

- `#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_02` 16
- `#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_03` 20
- `#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_04` 36
- `#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_05` 64
- `#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_06` 96
- `#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_07` 128
- `#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_08` 160
- `#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_09` 256
- `#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_10` 512
- `#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_11` 1024
- `#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_12` 2048
- `#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_13` 4096
- `#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_14` 8192
- `#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_15` 16384
- `#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_16` 32768
- `#define CFE_PLATFORM_SB_MAX_BLOCK_SIZE` (CFE_MISSION_SB_MAX_SB_MSG_SIZE + 128)
- `#define CFE_PLATFORM_SB_START_TASK_PRIORITY` 64
- `#define CFE_PLATFORM_SB_START_TASK_STACK_SIZE` CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
- `#define CFE_PLATFORM_TBL_START_TASK_PRIORITY` 70
- `#define CFE_PLATFORM_TBL_START_TASK_STACK_SIZE` CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
- `#define CFE_PLATFORM_TBL_BUF_MEMORY_BYTES` 524288
- `#define CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE` 16384
- `#define CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE` 16384
- `#define CFE_PLATFORM_TBL_MAX_NUM_TABLES` 128
- `#define CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES` 32
- `#define CFE_PLATFORM_TBL_MAX_NUM_HANDLES` 256
- `#define CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS` 4
- `#define CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS` 10
- `#define CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE` "/ram/cfe_tbl_reg.log"
- `#define CFE_PLATFORM_TBL_VALID_SCID_COUNT` 0
- `#define CFE_PLATFORM_TBL_U32FROM4CHARS(_C1, _C2, _C3, _C4)` (((uint32)(_C1) << 24 | (uint32)(_C2) << 16 | (uint32)(_C3) << 8 | (uint32)(_C4))
- `#define CFE_PLATFORM_TBL_VALID_SCID_1` (0x42)
- `#define CFE_PLATFORM_TBL_VALID_SCID_2` (CFE_PLATFORM_TBL_U32FROM4CHARS('a', 'b', 'c', 'd'))
- `#define CFE_PLATFORM_TBL_VALID_PRID_COUNT` 0
- `#define CFE_PLATFORM_TBL_VALID_PRID_1` (1)
- `#define CFE_PLATFORM_TBL_VALID_PRID_2` (CFE_PLATFORM_TBL_U32FROM4CHARS('a', 'b', 'c', 'd'))
- `#define CFE_PLATFORM_TBL_VALID_PRID_3` 0
- `#define CFE_PLATFORM_TBL_VALID_PRID_4` 0
- `#define CFE_PLATFORM_TIME_CFG_SERVER` true
- `#define CFE_PLATFORM_TIME_CFG_CLIENT` false
- `#define CFE_PLATFORM_TIME_CFG_VIRTUAL` true
- `#define CFE_PLATFORM_TIME_CFG_SIGNAL` false
- `#define CFE_PLATFORM_TIME_CFG_SOURCE` false
- `#define CFE_PLATFORM_TIME_CFG_SRC_MET` false
- `#define CFE_PLATFORM_TIME_CFG_SRC_GPS` false
- `#define CFE_PLATFORM_TIME_CFG_SRC_TIME` false
- `#define CFE_PLATFORM_TIME_MAX_DELTA_SECS` 0
- `#define CFE_PLATFORM_TIME_MAX_DELTA_SUBS` 500000
- `#define CFE_PLATFORM_TIME_MAX_LOCAL_SECS` 27
- `#define CFE_PLATFORM_TIME_MAX_LOCAL_SUBS` 0

- `#define CFE_PLATFORM_TIME_CFG_TONE_LIMIT 20000`
- `#define CFE_PLATFORM_TIME_CFG_START_FLY 2`
- `#define CFE_PLATFORM_TIME_CFG_LATCH_FLY 8`
- `#define CFE_PLATFORM_TIME_START_TASK_PRIORITY 60`
- `#define CFE_PLATFORM_TIME_TONE_TASK_PRIORITY 25`
- `#define CFE_PLATFORM_TIME_ONEHZ_TASK_PRIORITY 25`
- `#define CFE_PLATFORM_TIME_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE`
- `#define CFE_PLATFORM_TIME_TONE_TASK_STACK_SIZE 4096`
- `#define CFE_PLATFORM_TIME_ONEHZ_TASK_STACK_SIZE 8192`

11.13.1 Detailed Description

This header file contains the internal configuration parameters and typedefs with platform scope.

This provides default values for configurable items that do NOT affect the interface(s) of this module. This includes internal parameters, path names, and limit value(s) that are relevant for a specific platform.

Note

It is no longer necessary to provide this file directly in the defs directory, but if present, this file is still supported/usable for backward compatibility. To use this file, it should be called "cfe_platform_cfg.h".

Going forward, more fine-grained (module/purposes-specific) header files are included with each submodule. These may be overridden as necessary, but only if a definition within that file needs to be changed from the default. This approach will reduce the amount of duplicate/cloned definitions and better support alternative build configurations in the future.

Note that if this file is present, the fine-grained header files noted above will *not* be used.

Definition in file [example_platform_cfg.h](#).

11.13.2 Macro Definition Documentation

CFE_PLATFORM_CORE_MAX_STARTUP_MSEC

```
#define CFE_PLATFORM_CORE_MAX_STARTUP_MSEC 30000
```

Purpose CFE core application startup timeout

Description:

The upper limit for the amount of time that the cFE core applications (ES, SB, EVS, TIME, TBL) are each allotted to reach their respective "ready" states.

The CFE "main" thread starts individual tasks for each of the core applications (except FS). Each of these must perform some initialization work before the next core application can be started, so the main thread waits to ensure that the application has reached the "ready" state before starting the next application.

If any core application fails to start, then it indicates a major problem with the system and startup is aborted.

Units are in milliseconds

Limits:

Must be defined as an integer value that is greater than or equal to zero.

Definition at line 84 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ENDIAN

```
#define CFE_PLATFORM_ENDIAN CCSDS_LITTLE_ENDIAN
```

Purpose Platform Endian Indicator

Description:

The value of this constant indicates the endianness of the target system

Limits

This parameter has a lower limit of 0 and an upper limit of 1.

Definition at line 60 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_APP_KILL_TIMEOUT

```
#define CFE_PLATFORM_ES_APP_KILL_TIMEOUT 5
```

Purpose Define ES Application Kill Timeout

Description:

ES Application Kill Timeout. This parameter controls the number of "scan periods" that ES will wait for an application to Exit after getting the signal Delete, Reload or Restart. The sequence works as follows:

1. ES will set the control request for an App to Delete/Restart/Reload and set this kill timer to the value in this parameter.
2. If the App is responding and Calls it's RunLoop function, it will drop out of it's main loop and call CFE_ES↵_ExitApp. Once it calls Exit App, then ES can delete, restart, or reload the app the next time it scans the app table.
3. If the App is not responding, the ES App will decrement this Kill Timeout value each time it runs. If the timeout value reaches zero, ES will kill the app.

The Kill timeout value depends on the [CFE_PLATFORM_ES_APP_SCAN_RATE](#). If the Scan Rate is 1000, or 1 second, and this [CFE_PLATFORM_ES_APP_KILL_TIMEOUT](#) is set to 5, then it will take 5 seconds to kill a non-responding App. If the Scan Rate is 250, or 1/4 second, and the [CFE_PLATFORM_ES_APP_KILL_TIMEOUT](#) is set to 2, then it will take 1/2 second to time out.

Limits

There is a lower limit of 1 and an upper limit of 100 on this configuration parameter. Units are number of [CFE_PLATFORM_ES_APP_SCAN_RATE](#) cycles.

Definition at line 274 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_APP_SCAN_RATE

```
#define CFE_PLATFORM_ES_APP_SCAN_RATE 1000
```

Purpose Define ES Application Control Scan Rate

Description:

ES Application Control Scan Rate. This parameter controls the speed that ES scans the Application Table looking for App Delete/Restart/Reload requests. All Applications are deleted, restarted, or reloaded by the ES Application. ES will periodically scan for control requests to process. The scan rate is controlled by this parameter, which is given in milliseconds. A value of 1000 means that ES will scan the Application Table once per second. Be careful not to set the value of this too low, because ES will use more CPU cycles scanning the table.

Limits

There is a lower limit of 100 and an upper limit of 20000 on this configuration parameter. millisecond units.

Definition at line 245 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_CDS_MAX_BLOCK_SIZE

```
#define CFE_PLATFORM_ES_CDS_MAX_BLOCK_SIZE 80000
```

Definition at line 816 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_CDS_MAX_NUM_ENTRIES

```
#define CFE_PLATFORM_ES_CDS_MAX_NUM_ENTRIES 512
```

Purpose Define Maximum Number of Registered CDS Blocks

Description:

Maximum number of registered CDS Blocks

Limits

There is a lower limit of 8. There are no restrictions on the upper limit however, the maximum number of CDS entries is system dependent and should be verified.

Definition at line 706 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_01

```
#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_01 8
```

Purpose Define ES Critical Data Store Memory Pool Block Sizes

Description:

Intermediate ES Critical Data Store Memory Pool Block Sizes

Limits

These sizes MUST be increasing and MUST be an integral multiple of 4.

Definition at line 800 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_02

```
#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_02 16
```

Definition at line 801 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_03

#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_03 32
Definition at line 802 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_04

#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_04 48
Definition at line 803 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_05

#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_05 64
Definition at line 804 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_06

#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_06 96
Definition at line 805 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_07

#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_07 128
Definition at line 806 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_08

#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_08 160
Definition at line 807 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_09

#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_09 256
Definition at line 808 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_10

#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_10 512
Definition at line 809 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_11

#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_11 1024
Definition at line 810 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_12

#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_12 2048
Definition at line 811 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_13

#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_13 4096
Definition at line 812 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_14

```
#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_14 8192
```

Definition at line 813 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_15

```
#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_15 16384
```

Definition at line 814 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_16

```
#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_16 32768
```

Definition at line 815 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_CDS_SIZE

```
#define CFE_PLATFORM_ES_CDS_SIZE (128 * 1024)
```

Purpose Define Critical Data Store Size

Description:

Defines the Critical Data Store (CDS) area size in bytes size. The CDS is one of four memory areas that are preserved during a processor reset. NOTE: Changing this value changes memory allocation, and may require changes to platform specific values (in CFE_PSP) such as USER_RESERVED_MEM in VxWorks depending on the memory areas being used for preserved data and on OS specific behavior.

Limits

There is a lower limit of 8192 and an upper limit of UINT_MAX (4 Gigabytes) on this configuration parameter.

Definition at line 351 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_DEFAULT_APP_LOG_FILE

```
#define CFE_PLATFORM_ES_DEFAULT_APP_LOG_FILE "/ram/cfe_es_app_info.log"
```

Purpose Default Application Information Filename

Description:

The value of this constant defines the filename used to store information pertaining to all of the Applications that are registered with Executive Services. This filename is used only when no filename is specified in the the command to query all system apps.

Limits

The length of each string, including the NULL terminator cannot exceed the [OS_MAX_PATH_LEN](#) value.

Definition at line 433 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_DEFAULT_CDS_REG_DUMP_FILE

```
#define CFE_PLATFORM_ES_DEFAULT_CDS_REG_DUMP_FILE "/ram/cfe_cds_reg.log"
```

Purpose Default Critical Data Store Registry Filename

Description:

The value of this constant defines the filename used to store the Critical Data Store Registry. This filename is used only when no filename is specified in the command to stop performance data collecting.

Limits

The length of each string, including the NULL terminator cannot exceed the [OS_MAX_PATH_LEN](#) value.

Definition at line 507 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_DEFAULT_ER_LOG_FILE

```
#define CFE_PLATFORM_ES_DEFAULT_ER_LOG_FILE "/ram/cfe_erlog.log"
```

Purpose Default Exception and Reset (ER) Log Filename

Description:

The value of this constant defines the filename used to store the Exception and Reset (ER) Log. This filename is used only when no filename is specified in the command to dump the ER log. No file specified in the cmd means the first character in the cmd filename is a NULL terminator (zero).

Limits

The length of each string, including the NULL terminator cannot exceed the [OS_MAX_PATH_LEN](#) value.

Definition at line 479 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_DEFAULT_PERF_DUMP_FILENAME

```
#define CFE_PLATFORM_ES_DEFAULT_PERF_DUMP_FILENAME "/ram/cfe_es_perf.dat"
```

Purpose Default Performance Data Filename

Description:

The value of this constant defines the filename used to store the Performance Data. This filename is used only when no filename is specified in the command to stop performance data collecting.

Limits

The length of each string, including the NULL terminator cannot exceed the [OS_MAX_PATH_LEN](#) value.

Definition at line 493 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_DEFAULT_POR_SYSLOG_MODE

```
#define CFE_PLATFORM_ES_DEFAULT_POR_SYSLOG_MODE 0
```

Purpose Define Default System Log Mode following Power On Reset

Description:

Defines the default mode for the operation of the ES System log following a power on reset. The log may operate in either Overwrite mode = 0, where once the log becomes full the oldest message in the log will be overwritten, or Discard mode = 1, where once the log becomes full the contents of the log are preserved and the new event is discarded. This constant may hold a value of either 0 or 1 depending on the desired default. Overwrite Mode = 0, Discard Mode = 1.

Limits

There is a lower limit of 0 and an upper limit of 1 on this configuration parameter.

Definition at line 525 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_DEFAULT_PR_SYSLOG_MODE

```
#define CFE_PLATFORM_ES_DEFAULT_PR_SYSLOG_MODE 1
```

Purpose Define Default System Log Mode following Processor Reset

Description:

Defines the default mode for the operation of the ES System log following a processor reset. The log may operate in either Overwrite mode = 0, where once the log becomes full the oldest message in the log will be overwritten, or Discard mode = 1, where once the log becomes full the contents of the log are preserved and the new event is discarded. This constant may hold a value of either 0 or 1 depending on the desired default. Overwrite Mode = 0, Discard Mode = 1.

Limits

There is a lower limit of 0 and an upper limit of 1 on this configuration parameter.

Definition at line 543 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_DEFAULT_STACK_SIZE

```
#define CFE_PLATFORM_ES_DEFAULT_STACK_SIZE 8192
```

Purpose Define Default Stack Size for an Application

Description:

This parameter defines a default stack size. This parameter is used by the cFE Core Applications.

Limits

There is a lower limit of 2048. There are no restrictions on the upper limit however, the maximum stack size is system dependent and should be verified. Most operating systems provide tools for measuring the amount of stack used by a task during operation. It is always a good idea to verify that no more than 1/2 of the stack is used.

Definition at line 693 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_DEFAULT_SYSLOG_FILE

```
#define CFE_PLATFORM_ES_DEFAULT_SYSLOG_FILE "/ram/cfe_es_syslog.log"
```

Purpose Default System Log Filename

Description:

The value of this constant defines the filename used to store important information (as ASCII text strings) that might not be able to be sent in an Event Message. This filename is used only when no filename is specified in the command to dump the system log. No file specified in the cmd means the first character in the cmd filename is a NULL terminator (zero).

Limits

The length of each string, including the NULL terminator cannot exceed the [OS_MAX_PATH_LEN](#) value.

Definition at line 464 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_DEFAULT_TASK_LOG_FILE

```
#define CFE_PLATFORM_ES_DEFAULT_TASK_LOG_FILE "/ram/cfe_es_taskinfo.log"
```

Purpose Default Application Information Filename

Description:

The value of this constant defines the filename used to store information pertaining to all of the Applications that are registered with Executive Services. This filename is used only when no filename is specified in the the command to query all system tasks.

Limits

The length of each string, including the NULL terminator cannot exceed the [OS_MAX_PATH_LEN](#) value.

Definition at line 448 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_ER_LOG_ENTRIES

```
#define CFE_PLATFORM_ES_ER_LOG_ENTRIES 20
```

Purpose Define Max Number of ER (Exception and Reset) log entries

Description:

Defines the maximum number of ER (Exception and Reset) log entries

Limits

There is a lower limit of 1. There are no restrictions on the upper limit however, the maximum number of log entries is system dependent and should be verified.

Definition at line 186 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_ER_LOG_MAX_CONTEXT_SIZE

```
#define CFE_PLATFORM_ES_ER_LOG_MAX_CONTEXT_SIZE 256
```

Purpose Maximum size of CPU Context in ES Error Log

Description:

This should be large enough to accommodate the CPU context information supplied by the PSP on the given platform.

Limits:

Must be greater than zero and a multiple of `sizeof(uint32)`. Limited only by the available memory and the number of entries in the error log. Any context information beyond this size will be truncated.

Definition at line 200 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_MAX_APPLICATIONS

```
#define CFE_PLATFORM_ES_MAX_APPLICATIONS 32
```

Purpose Define Max Number of Applications

Description:

Defines the maximum number of applications that can be loaded into the system. This number does not include child tasks.

Limits

There is a lower limit of 6. The lower limit corresponds to the cFE internal applications. There are no restrictions on the upper limit however, the maximum number of applications is system dependent and should be verified. AppIDs that are checked against this configuration are defined by a 32 bit data word.

Definition at line 159 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_MAX_BLOCK_SIZE

```
#define CFE_PLATFORM_ES_MAX_BLOCK_SIZE 80000
```

Definition at line 789 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_MAX_GEN_COUNTERS

```
#define CFE_PLATFORM_ES_MAX_GEN_COUNTERS 8
```

Purpose Define Max Number of Generic Counters

Description:

Defines the maximum number of Generic Counters that can be registered.

Limits

This parameter has a lower limit of 1 and an upper limit of 65535.

Definition at line 226 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_MAX_LIBRARIES

```
#define CFE_PLATFORM_ES_MAX_LIBRARIES 10
```

Purpose Define Max Number of Shared libraries

Description:

Defines the maximum number of cFE Shared libraries that can be loaded into the system.

Limits

There is a lower limit of 1. There are no restrictions on the upper limit however, the maximum number of libraries is system dependent and should be verified.

Definition at line 173 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_MAX_MEMORY_POOLS

```
#define CFE_PLATFORM_ES_MAX_MEMORY_POOLS 10
```

Purpose Maximum number of memory pools

Description:

The upper limit for the number of memory pools that can concurrently exist within the system.

The CFE_SB and CFE_TBL core subsystems each define a memory pool. Individual applications may also create memory pools, so this value should be set sufficiently high enough to support the applications being used on this platform.

Limits:

Must be at least 2 to support CFE core - SB and TBL pools. No specific upper limit.

Definition at line 754 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_MAX_PROCESSOR_RESETS

```
#define CFE_PLATFORM_ES_MAX_PROCESSOR_RESETS 2
```

Purpose Define Number of Processor Resets Before a Power On Reset

Description:

Number of Processor Resets before a Power On Reset is called. If set to 2, then 2 processor resets will occur, and the 3rd processor reset will be a power on reset instead.

Limits

There is a lower limit of 0. There are no restrictions on the upper limit however, the maximum number of processor resets may be system dependent and should be verified.

Definition at line 721 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_01

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_01 8
```

Purpose Define Default ES Memory Pool Block Sizes

Description:

Default Intermediate ES Memory Pool Block Sizes. If an application is using the CFE_ES Memory Pool APIs ([CFE_ES_PoolCreate](#), [CFE_ES_PoolCreateNoSem](#), [CFE_ES_GetPoolBuf](#) and [CFE_ES_PutPoolBuf](#)) but finds these sizes inappropriate for their use, they may wish to use the [CFE_ES_PoolCreateEx](#) API to specify their own intermediate block sizes

Limits

These sizes MUST be increasing and MUST be an integral multiple of 4. Also, CFE_PLATFORM_ES_MAX_BLOCK_SIZE must be larger than CFE_MISSION_SB_MAX_SB_MSG_SIZE and both CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE and CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE. Note that if Table Services have been removed from the CFE, the table size limits are still enforced although the table size definitions may be reduced.

Definition at line 773 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_02

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_02 16
```

Definition at line 774 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_03

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_03 32
```

Definition at line 775 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_04

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_04 48
```

Definition at line 776 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_05

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_05 64
```

Definition at line 777 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_06

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_06 96
```

Definition at line 778 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_07

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_07 128
```

Definition at line 779 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_08

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_08 160
```

Definition at line 780 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_09

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_09 256
```

Definition at line 781 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_10

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_10 512
```

Definition at line 782 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_11

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_11 1024
```

Definition at line 783 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_12

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_12 2048
```

Definition at line 784 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_13

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_13 4096
```

Definition at line 785 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_14

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_14 8192
```

Definition at line 786 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_15

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_15 16384
```

Definition at line 787 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_16

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_16 32768
```

Definition at line 788 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_MEMPOOL_ALIGN_SIZE_MIN

```
#define CFE_PLATFORM_ES_MEMPOOL_ALIGN_SIZE_MIN 4
```

Purpose Define Memory Pool Alignment Size

Description:

Ensures that buffers obtained from a memory pool are aligned to a certain minimum block size. Note the allocator will always align to the minimum required by the CPU architecture. This may be set greater than the CPU requirement as desired for optimal performance.

For some architectures/applications it may be beneficial to set this to the cache line size of the target CPU, or to use special SIMD instructions that require a more stringent memory alignment.

Limits

This must always be a power of 2, as it is used as a binary address mask.

Definition at line 390 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_NONVOL_DISK_MOUNT_STRING

```
#define CFE_PLATFORM_ES_NONVOL_DISK_MOUNT_STRING "/cf"
```

Purpose Default virtual path for persistent storage

Description:

This configures the default location in the virtual file system for persistent/non-volatile storage. Files such as the startup script, app/library dynamic modules, and configuration tables are expected to be stored in this directory.

Definition at line 127 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_NONVOL_STARTUP_FILE

```
#define CFE_PLATFORM_ES_NONVOL_STARTUP_FILE "/cf/cfe_es_startup.scr"
```

Purpose ES Nonvolatile Startup Filename

Description:

The value of this constant defines the path and name of the file that contains a list of modules that will be loaded and started by the cFE after the cFE finishes its startup sequence.

Limits

The length of each string, including the NULL terminator cannot exceed the [OS_MAX_PATH_LEN](#) value.

Definition at line 404 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_PERF_CHILD_MS_DELAY

```
#define CFE_PLATFORM_ES_PERF_CHILD_MS_DELAY 20
```

Purpose Define Performance Analyzer Child Task Delay

Description:

This parameter defines the delay time (in milliseconds) between performance data file writes performed by the Executive Services Performance Analyzer Child Task.

Limits

It is recommended this parameter be greater than or equal to 20ms. This parameter is limited by the maximum value allowed by the data type. In this case, the data type is an unsigned 32-bit integer, so the valid range is 0 to 0xFFFFFFFF.

Definition at line 667 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_PERF_CHILD_PRIORITY

```
#define CFE_PLATFORM_ES_PERF_CHILD_PRIORITY 200
```

Purpose Define Performance Analyzer Child Task Priority

Description:

This parameter defines the priority of the child task spawned by the Executive Services to write performance data to a file. Lower numbers are higher priority, with 1 being the highest priority in the case of a child task.

Limits

Valid range for a child task is 1 to 255 however, the priority cannot be higher (lower number) than the ES parent application priority.

Definition at line 638 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_PERF_CHILD_STACK_SIZE

```
#define CFE_PLATFORM_ES_PERF_CHILD_STACK_SIZE 4096
```

Purpose Define Performance Analyzer Child Task Stack Size

Description:

This parameter defines the stack size of the child task spawned by the Executive Services to write performance data to a file.

Limits

It is recommended this parameter be greater than or equal to 4KB. This parameter is limited by the maximum value allowed by the data type. In this case, the data type is an unsigned 32-bit integer, so the valid range is 0 to 0xFFFFFFFF.

Definition at line 652 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_PERF_DATA_BUFFER_SIZE

```
#define CFE_PLATFORM_ES_PERF_DATA_BUFFER_SIZE 10000
```

Purpose Define Max Size of Performance Data Buffer

Description:

Defines the maximum size of the performance data buffer. Units are number of performance data entries. An entry is defined by a 32 bit data word followed by a 64 bit time stamp.

Limits

There is a lower limit of 1025. There are no restrictions on the upper limit however, the maximum buffer size is system dependent and should be verified. The units are number of entries. An entry is defined by a 32 bit data word followed by a 64 bit time stamp.

Definition at line 559 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_PERF_ENTRIES_BTWN_DLYS

```
#define CFE_PLATFORM_ES_PERF_ENTRIES_BTWN_DLYS 50
```

Purpose Define Performance Analyzer Child Task Number of Entries Between Delay

Description:

This parameter defines the number of performance analyzer entries the Performance Analyzer Child Task will write to the file between delays.

Definition at line 677 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_PERF_FILTMASK_ALL

```
#define CFE_PLATFORM_ES_PERF_FILTMASK_ALL ~CFE_PLATFORM_ES_PERF_FILTMASK_NONE
```

Purpose Define Filter Mask Setting for Enabling All Performance Entries

Description:

Defines the filter mask for enabling all performance entries. The value is a bit mask. For each bit, 0 means the corresponding entry is disabled and 1 means it is enabled.

Definition at line 579 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_PERF_FILTMASK_INIT

```
#define CFE_PLATFORM_ES_PERF_FILTMASK_INIT CFE_PLATFORM_ES_PERF_FILTMASK_ALL
```

Purpose Define Default Filter Mask Setting for Performance Data Buffer

Description:

Defines the default filter mask for the performance data buffer. The value is a bit mask. For each bit, 0 means the corresponding entry is disabled and 1 means it is enabled.

Definition at line 590 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_PERF_FILTMASK_NONE

```
#define CFE_PLATFORM_ES_PERF_FILTMASK_NONE 0
```

Purpose Define Filter Mask Setting for Disabling All Performance Entries

Description:

Defines the filter mask for disabling all performance entries. The value is a bit mask. For each bit, 0 means the corresponding entry is disabled and 1 means it is enabled.

Definition at line 569 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_PERF_TRIGMASK_ALL

```
#define CFE_PLATFORM_ES_PERF_TRIGMASK_ALL ~CFE_PLATFORM_ES_PERF_TRIGMASK_NONE
```

Purpose Define Filter Trigger Setting for Enabling All Performance Entries

Description:

Defines the trigger mask for enabling all performance data entries. The value is a bit mask. For each bit, 0 means the trigger for the corresponding entry is disabled and 1 means it is enabled.

Definition at line 612 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_PERF_TRIGMASK_INIT

```
#define CFE_PLATFORM_ES_PERF_TRIGMASK_INIT CFE_PLATFORM_ES_PERF_TRIGMASK_NONE
```

Purpose Define Default Filter Trigger Setting for Performance Data Buffer

Description:

Defines the default trigger mask for the performance data buffer. The value is a 32-bit mask. For each bit, 0 means the trigger for the corresponding entry is disabled and 1 means it is enabled.

Definition at line 623 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_PERF_TRIGMASK_NONE

```
#define CFE_PLATFORM_ES_PERF_TRIGMASK_NONE 0
```

Purpose Define Default Filter Trigger Setting for Disabling All Performance Entries

Description:

Defines the default trigger mask for disabling all performance data entries. The value is a bit mask. For each bit, 0 means the trigger for the corresponding entry is disabled and 1 means it is enabled.

Definition at line 601 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_POOL_MAX_BUCKETS

```
#define CFE_PLATFORM_ES_POOL_MAX_BUCKETS 17
```

Purpose Maximum number of block sizes in pool structures

Description:

The upper limit for the number of block sizes supported in the generic pool implementation, which in turn implements the memory pools and CDS.

Limits:

Must be at least one. No specific upper limit, but the number is anticipated to be reasonably small (i.e. tens, not hundreds). Large values have not been tested.

The ES and CDS block size lists must correlate with this value

Definition at line 736 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_RAM_DISK_MOUNT_STRING

```
#define CFE_PLATFORM_ES_RAM_DISK_MOUNT_STRING "/ram"
```

Purpose Default virtual path for volatile storage

Description:

The [CFE_PLATFORM_ES_RAM_DISK_MOUNT_STRING](#) parameter is used to set the cFE mount path for the CFE RAM disk. This is a parameter for missions that do not want to use the default value of "/ram", or for missions that need to have a different value for different CPUs or Spacecraft. Note that the vxWorks OSAL cannot currently handle names that have more than one path separator in it. The names "/ram", "/ramdisk", "/disk123" will all work, but "/disks/ram" will not. Multiple separators can be used with the posix or RTEMS ports.

Definition at line 143 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_RAM_DISK_NUM_SECTORS

```
#define CFE_PLATFORM_ES_RAM_DISK_NUM_SECTORS 4096
```

Purpose ES Ram Disk Number of Sectors

Description:

Defines the ram disk number of sectors. The ram disk is one of four memory areas that are preserved on a processor reset. NOTE: Changing this value changes memory allocation, and may require changes to platform specific values (in CFE_PSP) such as USER_RESERVED_MEM in VxWorks depending on the memory areas being used for preserved data and on OS specific behavior.

Limits

There is a lower limit of 128. There are no restrictions on the upper limit however, the maximum number of RAM sectors is system dependent and should be verified.

Definition at line 310 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_RAM_DISK_PERCENT_RESERVED

```
#define CFE_PLATFORM_ES_RAM_DISK_PERCENT_RESERVED 30
```

Purpose Percentage of Ram Disk Reserved for Decompressing Apps

Description:

The [CFE_PLATFORM_ES_RAM_DISK_PERCENT_RESERVED](#) parameter is used to make sure that the Volatile (RAM) Disk has a defined amount of free space during a processor reset. The cFE uses the Volatile disk to decompress cFE applications during system startup. If this Volatile disk happens to get filled with logs and misc files, then a processor reset may not work, because there will be no room to decompress cFE apps. To solve that problem, this parameter sets the "Low Water Mark" for disk space on a Processor reset. It should be set to allow the largest cFE Application to be decompressed. During a Processor reset, if there is not sufficient space left on the disk, it will be re-formatted in order to clear up some space.

This feature can be turned OFF by setting the parameter to 0.

Limits

There is a lower limit of 0 and an upper limit of 75 on this configuration parameter. Units are percentage. A setting of zero will turn this feature off.

Definition at line 334 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_RAM_DISK_SECTOR_SIZE

```
#define CFE_PLATFORM_ES_RAM_DISK_SECTOR_SIZE 512
```

Purpose ES Ram Disk Sector Size

Description:

Defines the ram disk sector size. The ram disk is 1 of 4 memory areas that are preserved on a processor reset. NOTE: Changing this value changes memory allocation, and may require changes to platform specific values (in CFE_PSP) such as USER_RESERVED_MEM in VxWorks depending on the memory areas being used for preserved data and on OS specific behavior.

Limits

There is a lower limit of 128. There are no restrictions on the upper limit however, the maximum RAM disk sector size is system dependent and should be verified.

Definition at line 292 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_START_TASK_PRIORITY

```
#define CFE_PLATFORM_ES_START_TASK_PRIORITY 68
```

Purpose Define ES Task Priority

Description:

Defines the cFE_ES Task priority.

Limits

Not Applicable

Definition at line 100 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_START_TASK_STACK_SIZE

```
#define CFE_PLATFORM_ES_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
```

Purpose Define ES Task Stack Size

Description:

Defines the cFE_ES Task Stack Size

Limits

There is a lower limit of 2048 on this configuration parameter. There are no restrictions on the upper limit however, the maximum stack size is system dependent and should be verified. Most operating systems provide tools for measuring the amount of stack used by a task during operation. It is always a good idea to verify that no more than 1/2 of the stack is used.

Definition at line 115 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_STARTUP_SCRIPT_TIMEOUT_MSEC

```
#define CFE_PLATFORM_ES_STARTUP_SCRIPT_TIMEOUT_MSEC 1000
```

Purpose Startup script timeout

Description:

The upper limit for the total amount of time that all apps listed in the CFE ES startup script may take to all become ready.

Unlike the "core" app timeout, this is a soft limit; if the allotted time is exceeded, it probably indicates an issue with one of the apps, but does not cause CFE ES to take any additional action other than logging the event to the syslog. Units are in milliseconds

Limits:

Must be defined as an integer value that is greater than or equal to zero.

Definition at line 856 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_STARTUP_SYNC_POLL_MSEC

```
#define CFE_PLATFORM_ES_STARTUP_SYNC_POLL_MSEC 50
```

Purpose Poll timer for startup sync delay

Description:

During startup, some tasks may need to synchronize their own initialization with the initialization of other applications in the system.

CFE ES implements an API to accomplish this, that performs a task delay (sleep) while polling the overall system state until other tasks are ready.

This value controls the amount of time that the CFE_ES_ApplicationSyncDelay will sleep between each check of the system state. This should be large enough to allow other tasks to run, but not so large as to noticeably delay the startup completion.

Units are in milliseconds

Limits:

Must be defined as an integer value that is greater than or equal to zero.

Definition at line 838 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_SYSTEM_LOG_SIZE

```
#define CFE_PLATFORM_ES_SYSTEM_LOG_SIZE 3072
```

Purpose Define Size of the cFE System Log.

Description:

Defines the size in bytes of the cFE system log. The system log holds variable length strings that are terminated by a linefeed and null character.

Limits

There is a lower limit of 512. There are no restrictions on the upper limit however, the maximum system log size is system dependent and should be verified.

Definition at line 215 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_USER_RESERVED_SIZE

```
#define CFE_PLATFORM_ES_USER_RESERVED_SIZE (1024 * 1024)
```

Purpose Define User Reserved Memory Size

Description:

User Reserved Memory Size. This is the size in bytes of the cFE User reserved Memory area. This is a block of memory that is available for cFE application use. The address is obtained by calling [CFE_PSP_GetUserReservedArea](#). The User Reserved Memory is one of four memory areas that are preserved during a processor reset. NOTE: Changing this value changes memory allocation, and may require changes to platform specific values (in CFE_PSP) such as USER_RESERVED_MEM in VxWorks depending on the memory areas being used for preserved data and on OS specific behavior.

Limits

There is a lower limit of 1024 and an upper limit of UINT_MAX (4 Gigabytes) on this configuration parameter.

Definition at line 371 of file [example_platform_cfg.h](#).

CFE_PLATFORM_ES_VOLATILE_STARTUP_FILE

```
#define CFE_PLATFORM_ES_VOLATILE_STARTUP_FILE "/ram/cfe_es_startup.scr"
```

Purpose ES Volatile Startup Filename

Description:

The value of this constant defines the path and name of the file that contains a list of modules that will be loaded and started by the cFE after the cFE finishes its startup sequence.

Limits

The length of each string, including the NULL terminator cannot exceed the [OS_MAX_PATH_LEN](#) value.

Definition at line 418 of file [example_platform_cfg.h](#).

CFE_PLATFORM_EVS_APP_EVENTS_PER_SEC

```
#define CFE_PLATFORM_EVS_APP_EVENTS_PER_SEC 15
```

Purpose Sustained number of event messages per second per app before squelching

Description:

Sustained number of events that may be emitted per app per second.

Limits

This number must be less than or equal to [CFE_PLATFORM_EVS_MAX_APP_EVENT_BURST](#). Values lower than 8 may cause functional and unit test failures.

Definition at line 924 of file [example_platform_cfg.h](#).

CFE_PLATFORM_EVS_DEFAULT_APP_DATA_FILE

```
#define CFE_PLATFORM_EVS_DEFAULT_APP_DATA_FILE "/ram/cfe_evs_app.dat"
```

Purpose Default EVS Application Data Filename

Description:

The value of this constant defines the filename used to store the EVS Application Data(event counts/filtering information). This filename is used only when no filename is specified in the command to dump the event log.

Limits

The length of each string, including the NULL terminator cannot exceed the [OS_MAX_PATH_LEN](#) value.

Definition at line 965 of file [example_platform_cfg.h](#).

CFE_PLATFORM_EVS_DEFAULT_LOG_FILE

```
#define CFE_PLATFORM_EVS_DEFAULT_LOG_FILE "/ram/cfe_evs.log"
```

Purpose Default Event Log Filename

Description:

The value of this constant defines the filename used to store the Event Services local event log. This filename is used only when no filename is specified in the command to dump the event log.

Limits

The length of each string, including the NULL terminator cannot exceed the [OS_MAX_PATH_LEN](#) value.

Definition at line 938 of file [example_platform_cfg.h](#).

CFE_PLATFORM_EVS_DEFAULT_LOG_MODE

```
#define CFE_PLATFORM_EVS_DEFAULT_LOG_MODE 1
```

Purpose Default EVS Local Event Log Mode

Description:

Defines a state of overwrite(0) or discard(1) for the operation of the EVS local event log. The log may operate in either Overwrite mode = 0, where once the log becomes full the oldest event in the log will be overwritten, or Discard mode = 1, where once the log becomes full the contents of the log are preserved and the new event is discarded. Overwrite Mode = 0, Discard Mode = 1.

Limits

The valid settings are 0 or 1

Definition at line 1012 of file [example_platform_cfg.h](#).

CFE_PLATFORM_EVS_DEFAULT_MSG_FORMAT_MODE

```
#define CFE_PLATFORM_EVS_DEFAULT_MSG_FORMAT_MODE CFE_EVS_MsgFormat_LONG
```

Purpose Default EVS Message Format Mode

Description:

Defines the default message format (long or short) for event messages being sent to the ground. Choose between [CFE_EVS_MsgFormat_LONG](#) or [CFE_EVS_MsgFormat_SHORT](#).

Limits

The valid settings are [CFE_EVS_MsgFormat_LONG](#) or [CFE_EVS_MsgFormat_SHORT](#)

Definition at line 1025 of file [example_platform_cfg.h](#).

CFE_PLATFORM_EVS_DEFAULT_TYPE_FLAG

```
#define CFE_PLATFORM_EVS_DEFAULT_TYPE_FLAG 0xE
```

Purpose Default EVS Event Type Filter Mask

Description:

Defines a state of on or off for all four event types. The term event 'type' refers to the criticality level and may be Debug, Informational, Error or Critical. Each event type has a bit position. (bit 0 = Debug, bit 1 = Info, bit 2 = Error, bit 3 = Critical). This is a global setting, meaning it applies to all applications. To filter an event type, set its bit to zero. For example, 0xE means Debug = OFF, Info = ON, Error = ON, Critical = ON

Limits

The valid settings are 0x0 to 0xF.

Definition at line 996 of file [example_platform_cfg.h](#).

CFE_PLATFORM_EVS_LOG_MAX

```
#define CFE_PLATFORM_EVS_LOG_MAX 20
```

Purpose Maximum Number of Events in EVS Local Event Log

Description:

Dictates the EVS local event log capacity. Units are the number of events.

Limits

There are no restrictions on the lower and upper limits however, the maximum log size is system dependent and should be verified.

Definition at line 950 of file [example_platform_cfg.h](#).

CFE_PLATFORM_EVS_MAX_APP_EVENT_BURST

```
#define CFE_PLATFORM_EVS_MAX_APP_EVENT_BURST 32
```

Purpose Maximum number of event before squelching

Description:

Maximum number of events that may be emitted per app per second. Setting this to 0 will cause events to be unrestricted.

Limits

This number must be less than or equal to INT_MAX/1000

Definition at line 912 of file [example_platform_cfg.h](#).

CFE_PLATFORM_EVS_MAX_EVENT_FILTERS

```
#define CFE_PLATFORM_EVS_MAX_EVENT_FILTERS 8
```

Purpose Define Maximum Number of Event Filters per Application

Description:

Maximum number of events that may be filtered per application.

Limits

There are no restrictions on the lower and upper limits however, the maximum number of event filters is system dependent and should be verified.

Definition at line 900 of file [example_platform_cfg.h](#).

CFE_PLATFORM_EVS_PORT_DEFAULT

```
#define CFE_PLATFORM_EVS_PORT_DEFAULT 0x0001
```

Purpose Default EVS Output Port State

Description:

Defines the default port state (enabled or disabled) for the four output ports defined within the Event Service. Port 1 is usually the uart output terminal. To enable a port, set the proper bit to a 1. Bit 0 is port 1, bit 1 is port2 etc.

Limits

The valid settings are 0x0 to 0xF.

Definition at line 979 of file [example_platform_cfg.h](#).

CFE_PLATFORM_EVS_START_TASK_PRIORITY

```
#define CFE_PLATFORM_EVS_START_TASK_PRIORITY 61
```

Purpose Define EVS Task Priority

Description:

Defines the cFE_EVS Task priority.

Limits

Not Applicable

Definition at line 872 of file [example_platform_cfg.h](#).

CFE_PLATFORM_EVS_START_TASK_STACK_SIZE

```
#define CFE_PLATFORM_EVS_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
```

Purpose Define EVS Task Stack Size

Description:

Defines the cFE_EVS Task Stack Size

Limits

There is a lower limit of 2048 on this configuration parameter. There are no restrictions on the upper limit however, the maximum stack size is system dependent and should be verified. Most operating systems provide tools for measuring the amount of stack used by a task during operation. It is always a good idea to verify that no more than 1/2 of the stack is used.

Definition at line 887 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_BUF_MEMORY_BYTES

```
#define CFE_PLATFORM_SB_BUF_MEMORY_BYTES 524288
```

Purpose Size of the SB buffer memory pool

Description:

Dictates the size of the SB memory pool. For each message the SB sends, the SB dynamically allocates from this memory pool, the memory needed to process the message. The memory needed to process each message is msg size + msg descriptor(CFE_SB_BufferD_t). This memory pool is also used to allocate destination descriptors (CFE_SB_DestinationD_t) during the subscription process. To see the run-time, high-water mark and the current utilization figures regarding this parameter, send an SB command to 'Send Statistics Pkt'. Some memory statistics have been added to the SB housekeeping packet. NOTE: It is important to monitor these statistics to ensure the desired memory margin is met.

Limits

This parameter has a lower limit of 512 and an upper limit of UINT_MAX (4 Gigabytes).

Definition at line 1120 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_DEFAULT_MAP_FILENAME

```
#define CFE_PLATFORM_SB_DEFAULT_MAP_FILENAME "/ram/cfe_sb_msgmap.dat"
```

Purpose Default Message Map Filename

Description:

The value of this constant defines the filename used to store the software bus message map information. This filename is used only when no filename is specified in the command. The message map is a lookup table (array of 16bit words) that has an element for each possible MsgId value and holds the routing table index for that MsgId. The Msg Map provides fast access to the destinations of a message.

Limits

The length of each string, including the NULL terminator cannot exceed the [OS_MAX_PATH_LEN](#) value.

Definition at line 1190 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_DEFAULT_MSG_LIMIT

```
#define CFE_PLATFORM_SB_DEFAULT_MSG_LIMIT 4
```

Purpose Default Subscription Message Limit

Description:

Dictates the default Message Limit when using the [CFE_SB_Subscribe](#) API. This will limit the number of messages with a specific message ID that can be received through a subscription. This only changes the default; other message limits can be set on a per subscription basis using [CFE_SB_SubscribeEx](#) .

Limits

This parameter has a lower limit of 4 and an upper limit of 65535.

Definition at line 1098 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_DEFAULT_PIPE_FILENAME

```
#define CFE_PLATFORM_SB_DEFAULT_PIPE_FILENAME "/ram/cfe_sb_pipe.dat"
```

Purpose Default Pipe Information Filename

Description:

The value of this constant defines the filename used to store the software bus pipe information. This filename is used only when no filename is specified in the command.

Limits

The length of each string, including the NULL terminator cannot exceed the [OS_MAX_PATH_LEN](#) value.

Definition at line 1173 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_DEFAULT_ROUTING_FILENAME

```
#define CFE_PLATFORM_SB_DEFAULT_ROUTING_FILENAME "/ram/cfe_sb_route.dat"
```

Purpose Default Routing Information Filename

Description:

The value of this constant defines the filename used to store the software bus routing information. This filename is used only when no filename is specified in the command.

Limits

The length of each string, including the NULL terminator cannot exceed the [OS_MAX_PATH_LEN](#) value.

Definition at line 1159 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_FILTER_MASK1

```
#define CFE_PLATFORM_SB_FILTER_MASK1 CFE_EVS_FIRST_4_STOP
```

Definition at line 1208 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_FILTER_MASK2

```
#define CFE_PLATFORM_SB_FILTER_MASK2 CFE_EVS_FIRST_4_STOP
```

Definition at line 1211 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_FILTER_MASK3

```
#define CFE_PLATFORM_SB_FILTER_MASK3 CFE_EVS_FIRST_16_STOP
```

Definition at line 1214 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_FILTER_MASK4

```
#define CFE_PLATFORM_SB_FILTER_MASK4 CFE_EVS_FIRST_16_STOP
```

Definition at line 1217 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_FILTER_MASK5

```
#define CFE_PLATFORM_SB_FILTER_MASK5 CFE_EVS_NO_FILTER
```

Definition at line 1220 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_FILTER_MASK6

```
#define CFE_PLATFORM_SB_FILTER_MASK6 CFE_EVS_NO_FILTER
```

Definition at line 1223 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_FILTER_MASK7

```
#define CFE_PLATFORM_SB_FILTER_MASK7 CFE_EVS_NO_FILTER
```

Definition at line 1226 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_FILTER_MASK8

```
#define CFE_PLATFORM_SB_FILTER_MASK8 CFE_EVS_NO_FILTER
```

Definition at line 1229 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_FILTERED_EVENT1

```
#define CFE_PLATFORM_SB_FILTERED_EVENT1 CFE_SB_SEND_NO_SUBS_EID
```

Purpose SB Event Filtering

Description:

This group of configuration parameters dictates what SB events will be filtered through SB. The filtering will begin after the SB task initializes and stay in effect until a cmd to SB changes it. This allows the operator to set limits on the number of event messages that are sent during system initialization. NOTE: Set all unused event values and mask values to zero

Limits

This filtering applies only to SB events. These parameters have a lower limit of 0 and an upper limit of 65535.

Definition at line 1207 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_FILTERED_EVENT2

```
#define CFE_PLATFORM_SB_FILTERED_EVENT2 CFE_SB_DUP_SUBSCRIP_EID
```

Definition at line 1210 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_FILTERED_EVENT3

#define CFE_PLATFORM_SB_FILTERED_EVENT3 CFE_SB_MSGID_LIM_ERR_EID
Definition at line 1213 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_FILTERED_EVENT4

#define CFE_PLATFORM_SB_FILTERED_EVENT4 CFE_SB_Q_FULL_ERR_EID
Definition at line 1216 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_FILTERED_EVENT5

#define CFE_PLATFORM_SB_FILTERED_EVENT5 0
Definition at line 1219 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_FILTERED_EVENT6

#define CFE_PLATFORM_SB_FILTERED_EVENT6 0
Definition at line 1222 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_FILTERED_EVENT7

#define CFE_PLATFORM_SB_FILTERED_EVENT7 0
Definition at line 1225 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_FILTERED_EVENT8

#define CFE_PLATFORM_SB_FILTERED_EVENT8 0
Definition at line 1228 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_HIGHEST_VALID_MSGID

#define CFE_PLATFORM_SB_HIGHEST_VALID_MSGID 0x1FFF

Purpose Highest Valid Message Id

Description:

The value of this constant dictates the range of valid message ID's, from 0 to CFE_PLATFORM_SB_HIGHEST_VALID_MSGID (inclusive).

Although this can be defined differently across platforms, each platform can only publish/subscribe to message ids within their allowable range. Typically this value is set the same across all mission platforms to avoid this complexity.

Limits

This parameter has a lower limit is 1, and an upper limit of 0xFFFFFFFF.

When using the direct message map implementation for software bus routing, this value is used to size the map where a value of 0x1FFF results in a 16 KBytes map and 0xFFFF is 128 KBytes.

When using the hash implementation for software bus routing, a multiple of the CFE_PLATFORM_SB_MAX_MSG_IDS is used to size the message map. In that case the range selected here does not impact message map memory use, so it's reasonable to use up to the full range supported by the message ID implementation.

Definition at line 1145 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_MAX_BLOCK_SIZE

#define CFE_PLATFORM_SB_MAX_BLOCK_SIZE (CFE_MISSION_SB_MAX_SB_MSG_SIZE + 128)
Definition at line 1258 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_MAX_DEST_PER_PKT

```
#define CFE_PLATFORM_SB_MAX_DEST_PER_PKT 16
```

Purpose Maximum Number of unique local destinations a single MsgId can have

Description:

Dictates the maximum number of unique local destinations a single MsgId can have.

Limits

This parameter has a lower limit of 1. There are no restrictions on the upper limit however, the maximum number of destinations per packet is system dependent and should be verified. Destination number values that are checked against this configuration are defined by a 16 bit data word.

Definition at line 1083 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_MAX_MSG_IDS

```
#define CFE_PLATFORM_SB_MAX_MSG_IDS 256
```

Purpose Maximum Number of Unique Message IDs SB Routing Table can hold

Description:

Dictates the maximum number of unique MsgIds the SB routing table will hold. This constant has a direct effect on the size of SB's tables and arrays. Keeping this count as low as possible will save memory. To see the run-time, high-water mark and the current utilization figures regarding this parameter, send an SB command to 'Send Statistics Pkt'.

Limits

This must be a power of two if software bus message routing hash implementation is being used. Lower than 64 will cause unit test failures, and telemetry reporting is impacted below 32. There is no hard upper limit, but impacts memory footprint. For software bus message routing search implementation the number of msg ids subscribed to impacts performance.

Definition at line 1050 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_MAX_PIPES

```
#define CFE_PLATFORM_SB_MAX_PIPES 64
```

Purpose Maximum Number of Unique Pipes SB Routing Table can hold

Description:

Dictates the maximum number of unique Pipes the SB routing table will hold. This constant has a direct effect on the size of SB's tables and arrays. Keeping this count as low as possible will save memory. To see the run-time, high-water mark and the current utilization figures regarding this parameter, send an SB command to 'Send Statistics Pkt'.

Limits

This parameter has a lower limit of 1. This parameter must also be less than or equal to OS_MAX_QUEUES.

Definition at line 1067 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_01

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_01 8
```

Purpose Define SB Memory Pool Block Sizes

Description:

Software Bus Memory Pool Block Sizes

Limits

These sizes MUST be increasing and MUST be an integral multiple of 4. The number of block sizes defined cannot exceed [CFE_PLATFORM_ES_POOL_MAX_BUCKETS](#)

Definition at line [1242](#) of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_02

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_02 16
```

Definition at line [1243](#) of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_03

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_03 20
```

Definition at line [1244](#) of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_04

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_04 36
```

Definition at line [1245](#) of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_05

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_05 64
```

Definition at line [1246](#) of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_06

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_06 96
```

Definition at line [1247](#) of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_07

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_07 128
```

Definition at line [1248](#) of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_08

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_08 160
```

Definition at line [1249](#) of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_09

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_09 256
```

Definition at line [1250](#) of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_10

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_10 512
```

Definition at line 1251 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_11

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_11 1024
```

Definition at line 1252 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_12

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_12 2048
```

Definition at line 1253 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_13

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_13 4096
```

Definition at line 1254 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_14

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_14 8192
```

Definition at line 1255 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_15

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_15 16384
```

Definition at line 1256 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_16

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_16 32768
```

Definition at line 1257 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_START_TASK_PRIORITY

```
#define CFE_PLATFORM_SB_START_TASK_PRIORITY 64
```

Purpose Define SB Task Priority

Description:

Defines the cFE_SB Task priority.

Limits

Not Applicable

Definition at line 1269 of file [example_platform_cfg.h](#).

CFE_PLATFORM_SB_START_TASK_STACK_SIZE

```
#define CFE_PLATFORM_SB_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
```

Purpose Define SB Task Stack Size

Description:

Defines the cFE_SB Task Stack Size

Limits

There is a lower limit of 2048 on this configuration parameter. There are no restrictions on the upper limit however, the maximum stack size is system dependent and should be verified. Most operating systems provide tools for measuring the amount of stack used by a task during operation. It is always a good idea to verify that no more than 1/2 of the stack is used.

Definition at line 1284 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TBL_BUF_MEMORY_BYTES

```
#define CFE_PLATFORM_TBL_BUF_MEMORY_BYTES 524288
```

Purpose Size of Table Services Table Memory Pool

Description:

Defines the TOTAL size of the memory pool that cFE Table Services allocates from the system. The size must be large enough to provide memory for each registered table, the inactive buffers for double buffered tables and for the shared inactive buffers for single buffered tables.

Limits

The cFE does not place a limit on the size of this parameter.

Definition at line 1331 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE

```
#define CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE "/ram/cfe_tbl_reg.log"
```

Purpose Default Filename for a Table Registry Dump

Description:

Defines the file name used to store the table registry when no filename is specified in the dump registry command.

Limits

The length of each string, including the NULL terminator cannot exceed the [OS_MAX_PATH_LEN](#) value.

Definition at line 1445 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES

```
#define CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES 32
```

Purpose Maximum Number of Critical Tables that can be Registered

Description:

Defines the maximum number of critical tables supported by this processor's Table Services.

Limits

This number must be less than 32767. It should be recognized that this parameter determines the size of the Critical Table Registry which is maintained in the Critical Data Store. An excessively high number will waste Critical Data Store memory. Therefore, this number must not exceed the value defined in `CFE_ES_CDS_MAX_CRITICAL_TABLES`.

Definition at line 1386 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE

```
#define CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE 16384
```

Purpose Maximum Size Allowed for a Double Buffered Table

Description:

Defines the maximum allowed size (in bytes) of a double buffered table.

Limits

The cFE does not place a limit on the size of this parameter but it must be less than half of [CFE_PLATFORM_TBL_BUF_MEMORY_B](#)

Definition at line 1343 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TBL_MAX_NUM_HANDLES

```
#define CFE_PLATFORM_TBL_MAX_NUM_HANDLES 256
```

Purpose Maximum Number of Table Handles

Description:

Defines the maximum number of Table Handles.

Limits

This number must be less than 32767. This number must be at least as big as the number of tables ([CFE_PLATFORM_TBL_MAX_NUM_TABLES](#)) and should be set higher if tables are shared between applications.

Definition at line 1399 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TBL_MAX_NUM_TABLES

```
#define CFE_PLATFORM_TBL_MAX_NUM_TABLES 128
```

Purpose Maximum Number of Tables Allowed to be Registered

Description:

Defines the maximum number of tables supported by this processor's Table Services.

Limits

This number must be less than 32767. It should be recognized that this parameter determines the size of the Table Registry. An excessively high number will waste memory.

Definition at line 1372 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS

```
#define CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS 10
```

Purpose Maximum Number of Simultaneous Table Validations

Description:

Defines the maximum number of pending validations that the Table Services can handle at any one time. When a table has a validation function, a validation request is made of the application to perform that validation. This number determines how many of those requests can be outstanding at any one time.

Limits

This number must be less than 32767. An excessively high number will degrade system performance and waste memory. A number less than 20 is suggested but not required.

Definition at line 1432 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS

```
#define CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS 4
```

Purpose Maximum Number of Simultaneous Loads to Support

Description:

Defines the maximum number of single buffered tables that can be loaded simultaneously. This number is used to determine the number of shared buffers to allocate.

Limits

This number must be less than 32767. An excessively high number will degrade system performance and waste memory. A number less than 5 is suggested but not required.

Definition at line 1414 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE

```
#define CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE 16384
```

Purpose Maximum Size Allowed for a Single Buffered Table

Description:

Defines the maximum allowed size (in bytes) of a single buffered table. **NOTE:** This size determines the size of all shared table buffers. Therefore, this size will be multiplied by [CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS](#) below when allocating memory for shared tables.

Limits

The cFE does not place a limit on the size of this parameter but it must be small enough to allow for [CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS](#) number of tables to fit into [CFE_PLATFORM_TBL_BUF_MEMORY_BYTES](#).

Definition at line 1359 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TBL_START_TASK_PRIORITY

```
#define CFE_PLATFORM_TBL_START_TASK_PRIORITY 70
```

Purpose Define TBL Task Priority

Description:

Defines the cFE_TBL Task priority.

Limits

Not Applicable

Definition at line 1300 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TBL_START_TASK_STACK_SIZE

```
#define CFE_PLATFORM_TBL_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
```

Purpose Define TBL Task Stack Size

Description:

Defines the cFE_TBL Task Stack Size

Limits

There is a lower limit of 2048 on this configuration parameter. There are no restrictions on the upper limit however, the maximum stack size is system dependent and should be verified. Most operating systems provide tools for measuring the amount of stack used by a task during operation. It is always a good idea to verify that no more than 1/2 of the stack is used.

Definition at line 1315 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TBL_U32FROM4CHARS

```
#define CFE_PLATFORM_TBL_U32FROM4CHARS (
    _C1,
    _C2,
    _C3,
    _C4 ) ((uint32) (_C1) << 24 | (uint32) (_C2) << 16 | (uint32) (_C3) << 8 | (uint32) (_C4))
```

Definition at line 1467 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TBL_VALID_PRID_1

```
#define CFE_PLATFORM_TBL_VALID_PRID_1 (1)
```

Purpose Processor ID values used for table load validation

Description:

Defines the processor ID values used for validating the processor ID field in the table file header. To be valid, the spacecraft ID specified in the table file header must match one of the values defined here.

Limits

This value can be any 32 bit unsigned integer.

Definition at line 1516 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TBL_VALID_PRID_2

```
#define CFE_PLATFORM_TBL_VALID_PRID_2 (CFE_PLATFORM_TBL_U32FROM4CHARS('a', 'b', 'c', 'd'))
```

Definition at line 1517 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TBL_VALID_PRID_3

```
#define CFE_PLATFORM_TBL_VALID_PRID_3 0
```

Definition at line 1518 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TBL_VALID_PRID_4

```
#define CFE_PLATFORM_TBL_VALID_PRID_4 0
```

Definition at line 1519 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TBL_VALID_PRID_COUNT

```
#define CFE_PLATFORM_TBL_VALID_PRID_COUNT 0
```

Purpose Number of Processor ID's specified for validation

Description:

Defines the number of specified processor ID values that are verified during table loads. If the number is zero then no validation of the processor ID field in the table file header is performed when tables are loaded. Non-zero values indicate how many values from the list of processor ID's defined below are compared to the processor ID field in the table file header. The ELF2CFETBL tool may be used to create table files with specified processor ID values.

Limits

This number must be greater than or equal to zero and less than or equal to 4.

Definition at line 1502 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TBL_VALID_SCID_1

```
#define CFE_PLATFORM_TBL_VALID_SCID_1 (0x42)
```

Purpose Spacecraft ID values used for table load validation

Description:

Defines the spacecraft ID values used for validating the spacecraft ID field in the table file header. To be valid, the spacecraft ID specified in the table file header must match one of the values defined here.

Limits

This value can be any 32 bit unsigned integer.

Definition at line 1482 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TBL_VALID_SCID_2

```
#define CFE_PLATFORM_TBL_VALID_SCID_2 (CFE_PLATFORM_TBL_U32FROM4CHARS('a', 'b', 'c', 'd'))
```

Definition at line 1483 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TBL_VALID_SCID_COUNT

```
#define CFE_PLATFORM_TBL_VALID_SCID_COUNT 0
```

Purpose Number of Spacecraft ID's specified for validation

Description:

Defines the number of specified spacecraft ID values that are verified during table loads. If the number is zero then no validation of the spacecraft ID field in the table file header is performed when tables are loaded. Non-zero values indicate how many values from the list of spacecraft ID's defined below are compared to the spacecraft ID field in the table file header. The ELF2CFETBL tool may be used to create table files with specified spacecraft ID values.

Limits

This number must be greater than or equal to zero and less than or equal to 2.

Definition at line 1464 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TIME_CFG_CLIENT

```
#define CFE_PLATFORM_TIME_CFG_CLIENT false
```

Definition at line 1539 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TIME_CFG_LATCH_FLY

```
#define CFE_PLATFORM_TIME_CFG_LATCH_FLY 8
```

Purpose Define Periodic Time to Update Local Clock Tone Latch

Description:

Define Periodic Time to Update Local Clock Tone Latch. Applies only when in flywheel mode. This define dictates the period at which the simulated 'last tone' time is updated. Units are seconds.

Limits

Not Applicable

Definition at line 1696 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TIME_CFG_SERVER

```
#define CFE_PLATFORM_TIME_CFG_SERVER true
```

Purpose Time Server or Time Client Selection

Description:

This configuration parameter selects whether the Time task functions as a time "server" or "client". A time server generates the "time at the tone" packet which is received by time clients.

Limits

Enable one, and only one by defining either CFE_PLATFORM_TIME_CFG_SERVER or CFE_PLATFORM_TIME_CFG_CLIENT AS true. The other must be defined as false.

Definition at line 1538 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TIME_CFG_SIGNAL

```
#define CFE_PLATFORM_TIME_CFG_SIGNAL false
```

Purpose Include or Exclude the Primary/Redundant Tone Selection Cmd

Description:

Depending on the specific hardware system configuration, it may be possible to switch between a primary and redundant tone signal. If supported by hardware, this definition will enable command interfaces to select the active tone signal. Both Time Clients and Time Servers support this feature. Note: Set the CFE_PLATFORM_TIME_CFG_SIGNAL define to true to enable tone signal commands.

Limits

Not Applicable

Definition at line 1586 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TIME_CFG_SOURCE

```
#define CFE_PLATFORM_TIME_CFG_SOURCE false
```

Purpose Include or Exclude the Internal/External Time Source Selection Cmd

Description:

By default, Time Servers maintain time using an internal MET which may be a h/w register or software counter, depending on available hardware. The following definition enables command interfaces to switch between an internal MET, or external time data received from one of several supported external time sources. Only a Time Server may be configured to use external time data. Note: Set the CFE_PLATFORM_TIME_CFG_SOURCE define to true to include the Time Source Selection Command (command allows selection between the internal or external time source). Then choose the external source with the CFE_TIME_CFG_SRC_??? define.

Limits

Only applies if [CFE_PLATFORM_TIME_CFG_SERVER](#) is set to true.

Definition at line 1606 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TIME_CFG_SRC_GPS

```
#define CFE_PLATFORM_TIME_CFG_SRC_GPS false
```

Definition at line 1623 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TIME_CFG_SRC_MET

```
#define CFE_PLATFORM_TIME_CFG_SRC_MET false
```

Purpose Choose the External Time Source for Server only

Description:

If [CFE_PLATFORM_TIME_CFG_SOURCE](#) is set to true, then one of the following external time source types must also be set to true. Do not set any of the external time source types to true unless [CFE_PLATFORM_TIME_CFG_SOURCE](#) is set to true.

Limits

1. If `CFE_PLATFORM_TIME_CFG_SOURCE` is set to true then one and only one of the following three external time sources can and must be set true: `CFE_PLATFORM_TIME_CFG_SRC_MET`, `CFE_PLATFORM_TIME_CFG_SRC_GPS`, `CFE_PLATFORM_TIME_CFG_SRC_TIME`
2. Only applies if `CFE_PLATFORM_TIME_CFG_SERVER` is set to true.

Definition at line 1622 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TIME_CFG_SRC_TIME

```
#define CFE_PLATFORM_TIME_CFG_SRC_TIME false
```

Definition at line 1624 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TIME_CFG_START_FLY

```
#define CFE_PLATFORM_TIME_CFG_START_FLY 2
```

Purpose Define Time to Start Flywheel Since Last Tone

Description:

Define time to enter flywheel mode (in seconds since last tone data update) Units are microseconds as measured with the local clock.

Limits

Not Applicable

Definition at line 1683 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TIME_CFG_TONE_LIMIT

```
#define CFE_PLATFORM_TIME_CFG_TONE_LIMIT 20000
```

Purpose Define Timing Limits From One Tone To The Next

Description:

Defines limits to the timing of the 1Hz tone signal. A tone signal is valid only if it arrives within one second (plus or minus the tone limit) from the previous tone signal. Units are microseconds as measured with the local clock.

Limits

Not Applicable

Definition at line 1671 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TIME_CFG_VIRTUAL

```
#define CFE_PLATFORM_TIME_CFG_VIRTUAL true
```

Purpose Time Tone In Big-Endian Order

Description:

If this configuration parameter is defined, the CFE time server will publish time tones with payloads in big-endian order, and time clients will expect the tones to be in big-endian order. This is useful for mixed-endian environments. This will become obsolete once EDS is available and the CFE time tone message is defined.

Purpose Local MET or Virtual MET Selection for Time Servers

Description:

Depending on the specific hardware system configuration, it may be possible for Time Servers to read the "local" MET from a h/w register rather than having to track the MET as the count of tone signal interrupts (virtual MET)

Time Clients must be defined as using a virtual MET. Also, a Time Server cannot be defined as having both a h/w MET and an external time source (they both cannot synchronize to the same tone).

Note: "disable" this define (set to false) only for Time Servers with local hardware that supports a h/w MET that is synchronized to the tone signal !!!

Limits

Only applies if [CFE_PLATFORM_TIME_CFG_SERVER](#) is set to true.

Definition at line 1571 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TIME_MAX_DELTA_SECS

```
#define CFE_PLATFORM_TIME_MAX_DELTA_SECS 0
```

Purpose Define the Max Delta Limits for Time Servers using an Ext Time Source

Description:

If [CFE_PLATFORM_TIME_CFG_SOURCE](#) is set to true and one of the external time sources is also set to true, then the delta time limits for range checking is used.

When a new time value is received from an external source, the value is compared against the "expected" time value. If the delta exceeds the following defined amount, then the new time data will be ignored. This range checking is only performed after the clock state has been commanded to "valid". Until then, external time data is accepted unconditionally.

Limits

Applies only if both [CFE_PLATFORM_TIME_CFG_SERVER](#) and [CFE_PLATFORM_TIME_CFG_SOURCE](#) are set to true.

Definition at line 1643 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TIME_MAX_DELTA_SUBS

```
#define CFE_PLATFORM_TIME_MAX_DELTA_SUBS 500000
```

Definition at line 1644 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TIME_MAX_LOCAL_SECS

```
#define CFE_PLATFORM_TIME_MAX_LOCAL_SECS 27
```

Purpose Define the Local Clock Rollover Value in seconds and subseconds

Description:

Specifies the capability of the local clock. Indicates the time at which the local clock rolls over.

Limits

Not Applicable

Definition at line 1656 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TIME_MAX_LOCAL_SUBS

```
#define CFE_PLATFORM_TIME_MAX_LOCAL_SUBS 0
```

Definition at line 1657 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TIME_ONEHZ_TASK_PRIORITY

```
#define CFE_PLATFORM_TIME_ONEHZ_TASK_PRIORITY 25
```

Definition at line 1713 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TIME_ONEHZ_TASK_STACK_SIZE

```
#define CFE_PLATFORM_TIME_ONEHZ_TASK_STACK_SIZE 8192
```

Definition at line 1732 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TIME_START_TASK_PRIORITY

```
#define CFE_PLATFORM_TIME_START_TASK_PRIORITY 60
```

Purpose Define TIME Task Priorities

Description:

Defines the cFE_TIME Task priority. Defines the cFE_TIME Tone Task priority. Defines the cFE_TIME 1HZ Task priority.

Limits

There is a lower limit of zero and an upper limit of 255 on these configuration parameters. Remember that the meaning of each task priority is inverted – a "lower" number has a "higher" priority.

Definition at line 1711 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TIME_START_TASK_STACK_SIZE

```
#define CFE_PLATFORM_TIME_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
```

Purpose Define TIME Task Stack Sizes

Description:

Defines the cFE_TIME Main Task Stack Size Defines the cFE_TIME Tone Task Stack Size Defines the cFE_TIME 1HZ Task Stack Size

Limits

There is a lower limit of 2048 on these configuration parameters. There are no restrictions on the upper limit however, the maximum stack size is system dependent and should be verified. Most operating systems provide tools for measuring the amount of stack used by a task during operation. It is always a good idea to verify that no more than 1/2 of the stack is used.

Definition at line 1730 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TIME_TONE_TASK_PRIORITY

```
#define CFE_PLATFORM_TIME_TONE_TASK_PRIORITY 25
```

Definition at line 1712 of file [example_platform_cfg.h](#).

CFE_PLATFORM_TIME_TONE_TASK_STACK_SIZE

#define CFE_PLATFORM_TIME_TONE_TASK_STACK_SIZE 4096

Definition at line 1731 of file [example_platform_cfg.h](#).

11.14 example_platform_cfg.h

[Go to the documentation of this file.](#)

```

00001 /*****
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"
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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
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00016  * limitations under the License.
00017  *****/
00018
00043 #ifndef EXAMPLE_PLATFORM_CFG_H
00044 #define EXAMPLE_PLATFORM_CFG_H
00045
00046 /*****
00047  *
00048  *   CFE Core Private Config Definitions
00049  */
00050
00060 #define CFE_PLATFORM_ENDIAN CCSDS_LITTLE_ENDIAN
00061
00084 #define CFE_PLATFORM_CORE_MAX_STARTUP_MSEC 30000
00085
00086 /*****
00087  *
00088  *   CFE Executive Services (CFE_ES) Application Private Config Definitions
00089  */
00090
00100 #define CFE_PLATFORM_ES_START_TASK_PRIORITY 68
00101
00115 #define CFE_PLATFORM_ES_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
00116
00127 #define CFE_PLATFORM_ES_NONVOL_DISK_MOUNT_STRING "/cf"
00128
00143 #define CFE_PLATFORM_ES_RAM_DISK_MOUNT_STRING "/ram"
00144
00159 #define CFE_PLATFORM_ES_MAX_APPLICATIONS 32
00160
00173 #define CFE_PLATFORM_ES_MAX_LIBRARIES 10
00174
00186 #define CFE_PLATFORM_ES_ER_LOG_ENTRIES 20
00187
00200 #define CFE_PLATFORM_ES_ER_LOG_MAX_CONTEXT_SIZE 256
00201
00215 #define CFE_PLATFORM_ES_SYSTEM_LOG_SIZE 3072
00216
00226 #define CFE_PLATFORM_ES_MAX_GEN_COUNTERS 8
00227
00245 #define CFE_PLATFORM_ES_APP_SCAN_RATE 1000
00246
00274 #define CFE_PLATFORM_ES_APP_KILL_TIMEOUT 5
00275
00292 #define CFE_PLATFORM_ES_RAM_DISK_SECTOR_SIZE 512
00293
00310 #define CFE_PLATFORM_ES_RAM_DISK_NUM_SECTORS 4096
00311
00334 #define CFE_PLATFORM_ES_RAM_DISK_PERCENT_RESERVED 30
00335
00351 #define CFE_PLATFORM_ES_CDS_SIZE (128 * 1024)
00352
00371 #define CFE_PLATFORM_ES_USER_RESERVED_SIZE (1024 * 1024)
00372
00390 #define CFE_PLATFORM_ES_MEMPOOL_ALIGN_SIZE_MIN 4

```

```
00391
00404 #define CFE_PLATFORM_ES_NONVOL_STARTUP_FILE "/cf/cfe_es_startup.scr"
00405
00418 #define CFE_PLATFORM_ES_VOLATILE_STARTUP_FILE "/ram/cfe_es_startup.scr"
00419
00433 #define CFE_PLATFORM_ES_DEFAULT_APP_LOG_FILE "/ram/cfe_es_app_info.log"
00434
00448 #define CFE_PLATFORM_ES_DEFAULT_TASK_LOG_FILE "/ram/cfe_es_taskinfo.log"
00449
00464 #define CFE_PLATFORM_ES_DEFAULT_SYSLOG_FILE "/ram/cfe_es_syslog.log"
00465
00479 #define CFE_PLATFORM_ES_DEFAULT_ER_LOG_FILE "/ram/cfe_erlog.log"
00480
00493 #define CFE_PLATFORM_ES_DEFAULT_PERF_DUMP_FILENAME "/ram/cfe_es_perf.dat"
00494
00507 #define CFE_PLATFORM_ES_DEFAULT_CDS_REG_DUMP_FILE "/ram/cfe_cds_reg.log"
00508
00525 #define CFE_PLATFORM_ES_DEFAULT_POR_SYSLOG_MODE 0
00526
00543 #define CFE_PLATFORM_ES_DEFAULT_PR_SYSLOG_MODE 1
00544
00559 #define CFE_PLATFORM_ES_PERF_DATA_BUFFER_SIZE 10000
00560
00569 #define CFE_PLATFORM_ES_PERF_FILTMASK_NONE 0
00570
00579 #define CFE_PLATFORM_ES_PERF_FILTMASK_ALL ~CFE_PLATFORM_ES_PERF_FILTMASK_NONE
00580
00590 #define CFE_PLATFORM_ES_PERF_FILTMASK_INIT CFE_PLATFORM_ES_PERF_FILTMASK_ALL
00591
00601 #define CFE_PLATFORM_ES_PERF_TRIGMASK_NONE 0
00602
00612 #define CFE_PLATFORM_ES_PERF_TRIGMASK_ALL ~CFE_PLATFORM_ES_PERF_TRIGMASK_NONE
00613
00623 #define CFE_PLATFORM_ES_PERF_TRIGMASK_INIT CFE_PLATFORM_ES_PERF_TRIGMASK_NONE
00624
00638 #define CFE_PLATFORM_ES_PERF_CHILD_PRIORITY 200
00639
00652 #define CFE_PLATFORM_ES_PERF_CHILD_STACK_SIZE 4096
00653
00667 #define CFE_PLATFORM_ES_PERF_CHILD_MS_DELAY 20
00668
00677 #define CFE_PLATFORM_ES_PERF_ENTRIES_BTWN_DLYS 50
00678
00693 #define CFE_PLATFORM_ES_DEFAULT_STACK_SIZE 8192
00694
00706 #define CFE_PLATFORM_ES_CDS_MAX_NUM_ENTRIES 512
00707
00721 #define CFE_PLATFORM_ES_MAX_PROCESSOR_RESETS 2
00722
00736 #define CFE_PLATFORM_ES_POOL_MAX_BUCKETS 17
00737
00754 #define CFE_PLATFORM_ES_MAX_MEMORY_POOLS 10
00755
00773 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_01 8
00774 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_02 16
00775 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_03 32
00776 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_04 48
00777 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_05 64
00778 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_06 96
00779 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_07 128
00780 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_08 160
00781 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_09 256
00782 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_10 512
00783 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_11 1024
00784 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_12 2048
00785 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_13 4096
00786 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_14 8192
00787 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_15 16384
00788 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_16 32768
00789 #define CFE_PLATFORM_ES_MAX_BLOCK_SIZE 80000
00790
00800 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_01 8
00801 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_02 16
00802 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_03 32
00803 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_04 48
00804 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_05 64
00805 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_06 96
00806 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_07 128
00807 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_08 160
00808 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_09 256
00809 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_10 512
```

```
00810 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_11 1024
00811 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_12 2048
00812 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_13 4096
00813 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_14 8192
00814 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_15 16384
00815 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_16 32768
00816 #define CFE_PLATFORM_ES_CDS_MAX_BLOCK_SIZE 80000
00817
00838 #define CFE_PLATFORM_ES_STARTUP_SYNC_POLL_MSEC 50
00839
00856 #define CFE_PLATFORM_ES_STARTUP_SCRIPT_TIMEOUT_MSEC 1000
00857
00858 /*****
00859 */
00860 *   CFE Event Services (CFE_EVS) Application Private Config Definitions
00861 */
00862
00872 #define CFE_PLATFORM_EVS_START_TASK_PRIORITY 61
00873
00887 #define CFE_PLATFORM_EVS_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
00888
00900 #define CFE_PLATFORM_EVS_MAX_EVENT_FILTERS 8
00901
00912 #define CFE_PLATFORM_EVS_MAX_APP_EVENT_BURST 32
00913
00924 #define CFE_PLATFORM_EVS_APP_EVENTS_PER_SEC 15
00925
00938 #define CFE_PLATFORM_EVS_DEFAULT_LOG_FILE "/ram/cfe_evs.log"
00939
00950 #define CFE_PLATFORM_EVS_LOG_MAX 20
00951
00965 #define CFE_PLATFORM_EVS_DEFAULT_APP_DATA_FILE "/ram/cfe_evs_app.dat"
00966
00979 #define CFE_PLATFORM_EVS_PORT_DEFAULT 0x0001
00980
00996 #define CFE_PLATFORM_EVS_DEFAULT_TYPE_FLAG 0xE
00997
01012 #define CFE_PLATFORM_EVS_DEFAULT_LOG_MODE 1
01013
01025 #define CFE_PLATFORM_EVS_DEFAULT_MSG_FORMAT_MODE CFE_EVS_MsgFormat_LONG
01026
01027 /*****
01028 */
01029 *   CFE Software Bus (CFE_SB) Application Private Config Definitions
01030 */
01031
01050 #define CFE_PLATFORM_SB_MAX_MSG_IDS 256
01051
01067 #define CFE_PLATFORM_SB_MAX_PIPES 64
01068
01083 #define CFE_PLATFORM_SB_MAX_DEST_PER_PKT 16
01084
01098 #define CFE_PLATFORM_SB_DEFAULT_MSG_LIMIT 4
01099
01120 #define CFE_PLATFORM_SB_BUF_MEMORY_BYTES 524288
01121
01145 #define CFE_PLATFORM_SB_HIGHEST_VALID_MSGID 0x1FFF
01146
01159 #define CFE_PLATFORM_SB_DEFAULT_ROUTING_FILENAME "/ram/cfe_sb_route.dat"
01160
01173 #define CFE_PLATFORM_SB_DEFAULT_PIPE_FILENAME "/ram/cfe_sb_pipe.dat"
01174
01190 #define CFE_PLATFORM_SB_DEFAULT_MAP_FILENAME "/ram/cfe_sb_msgmap.dat"
01191
01207 #define CFE_PLATFORM_SB_FILTERED_EVENT1 CFE_SB_SEND_NO_SUBS_EID
01208 #define CFE_PLATFORM_SB_FILTER_MASK1 CFE_EVS_FIRST_4_STOP
01209
01210 #define CFE_PLATFORM_SB_FILTERED_EVENT2 CFE_SB_DUP_SUBSCRIP_EID
01211 #define CFE_PLATFORM_SB_FILTER_MASK2 CFE_EVS_FIRST_4_STOP
01212
01213 #define CFE_PLATFORM_SB_FILTERED_EVENT3 CFE_SB_MSGID_LIM_ERR_EID
01214 #define CFE_PLATFORM_SB_FILTER_MASK3 CFE_EVS_FIRST_16_STOP
01215
01216 #define CFE_PLATFORM_SB_FILTERED_EVENT4 CFE_SB_O_FULL_ERR_EID
01217 #define CFE_PLATFORM_SB_FILTER_MASK4 CFE_EVS_FIRST_16_STOP
01218
01219 #define CFE_PLATFORM_SB_FILTERED_EVENT5 0
01220 #define CFE_PLATFORM_SB_FILTER_MASK5 CFE_EVS_NO_FILTER
01221
01222 #define CFE_PLATFORM_SB_FILTERED_EVENT6 0
01223 #define CFE_PLATFORM_SB_FILTER_MASK6 CFE_EVS_NO_FILTER
```

```

01224
01225 #define CFE_PLATFORM_SB_FILTERED_EVENT7 0
01226 #define CFE_PLATFORM_SB_FILTER_MASK7 CFE_EVS_NO_FILTER
01227
01228 #define CFE_PLATFORM_SB_FILTERED_EVENT8 0
01229 #define CFE_PLATFORM_SB_FILTER_MASK8 CFE_EVS_NO_FILTER
01230
01242 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_01 8
01243 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_02 16
01244 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_03 20
01245 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_04 36
01246 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_05 64
01247 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_06 96
01248 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_07 128
01249 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_08 160
01250 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_09 256
01251 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_10 512
01252 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_11 1024
01253 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_12 2048
01254 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_13 4096
01255 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_14 8192
01256 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_15 16384
01257 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_16 32768
01258 #define CFE_PLATFORM_SB_MAX_BLOCK_SIZE (CFE_MISSION_SB_MAX_SB_MSG_SIZE + 128)
01259
01269 #define CFE_PLATFORM_SB_START_TASK_PRIORITY 64
01270
01284 #define CFE_PLATFORM_SB_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
01285
01286 /*****
01287 */
01288 * CFE Table Services (CFE_TBL) Application Private Config Definitions
01289 */
01290
01300 #define CFE_PLATFORM_TBL_START_TASK_PRIORITY 70
01301
01315 #define CFE_PLATFORM_TBL_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
01316
01317 /* Platform Configuration Parameters for Table Service (TBL) */
01318
01331 #define CFE_PLATFORM_TBL_BUF_MEMORY_BYTES 524288
01332
01343 #define CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE 16384
01344
01359 #define CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE 16384
01360
01372 #define CFE_PLATFORM_TBL_MAX_NUM_TABLES 128
01373
01386 #define CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES 32
01387
01399 #define CFE_PLATFORM_TBL_MAX_NUM_HANDLES 256
01400
01414 #define CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS 4
01415
01432 #define CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS 10
01433
01445 #define CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE "/ram/cfe_tbl_reg.log"
01446
01464 #define CFE_PLATFORM_TBL_VALID_SCID_COUNT 0
01465
01466 /* macro to construct 32 bit value from 4 chars */
01467 #define CFE_PLATFORM_TBL_U32FROM4CHARS(_C1, _C2, _C3, _C4) \
01468 ((uint32)(_C1) << 24 | (uint32)(_C2) << 16 | (uint32)(_C3) << 8 | (uint32)(_C4))
01469
01482 #define CFE_PLATFORM_TBL_VALID_SCID_1 (0x42)
01483 #define CFE_PLATFORM_TBL_VALID_SCID_2 (CFE_PLATFORM_TBL_U32FROM4CHARS('a', 'b', 'c', 'd'))
01484
01502 #define CFE_PLATFORM_TBL_VALID_PRID_COUNT 0
01503
01516 #define CFE_PLATFORM_TBL_VALID_PRID_1 (1)
01517 #define CFE_PLATFORM_TBL_VALID_PRID_2 (CFE_PLATFORM_TBL_U32FROM4CHARS('a', 'b', 'c', 'd'))
01518 #define CFE_PLATFORM_TBL_VALID_PRID_3 0
01519 #define CFE_PLATFORM_TBL_VALID_PRID_4 0
01520
01521 /*****
01522 */
01523 * CFE Time Service (CFE_TIME) Application Private Config Definitions
01524 */
01525
01538 #define CFE_PLATFORM_TIME_CFG_SERVER true
01539 #define CFE_PLATFORM_TIME_CFG_CLIENT false

```



```
01540
01551 #undef CFE_PLATFORM_TIME_CFG_BIGENDIAN
01552
01571 #define CFE_PLATFORM_TIME_CFG_VIRTUAL true
01572
01586 #define CFE_PLATFORM_TIME_CFG_SIGNAL false
01587
01606 #define CFE_PLATFORM_TIME_CFG_SOURCE false
01607
01622 #define CFE_PLATFORM_TIME_CFG_SRC_MET false
01623 #define CFE_PLATFORM_TIME_CFG_SRC_GPS false
01624 #define CFE_PLATFORM_TIME_CFG_SRC_TIME false
01625
01643 #define CFE_PLATFORM_TIME_MAX_DELTA_SECS 0
01644 #define CFE_PLATFORM_TIME_MAX_DELTA_SUBS 500000
01645
01656 #define CFE_PLATFORM_TIME_MAX_LOCAL_SECS 27
01657 #define CFE_PLATFORM_TIME_MAX_LOCAL_SUBS 0
01658
01671 #define CFE_PLATFORM_TIME_CFG_TONE_LIMIT 20000
01672
01683 #define CFE_PLATFORM_TIME_CFG_START_FLY 2
01684
01696 #define CFE_PLATFORM_TIME_CFG_LATCH_FLY 8
01697
01711 #define CFE_PLATFORM_TIME_START_TASK_PRIORITY 60
01712 #define CFE_PLATFORM_TIME_TONE_TASK_PRIORITY 25
01713 #define CFE_PLATFORM_TIME_ONEHZ_TASK_PRIORITY 25
01714
01730 #define CFE_PLATFORM_TIME_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
01731 #define CFE_PLATFORM_TIME_TONE_TASK_STACK_SIZE 4096
01732 #define CFE_PLATFORM_TIME_ONEHZ_TASK_STACK_SIZE 8192
01733
01734 #endif /* EXAMPLE_PLATFORM_CFG_H */
```

11.15 [cfe/docs/src/cfe_api.dox](#) File Reference

11.16 [cfe/docs/src/cfe_es.dox](#) File Reference

11.17 [cfe/docs/src/cfe_evs.dox](#) File Reference

11.18 [cfe/docs/src/cfe_frontpage.dox](#) File Reference

11.19 [cfe/docs/src/cfe_glossary.dox](#) File Reference

11.20 [cfe/docs/src/cfe_sb.dox](#) File Reference

11.21 [cfe/docs/src/cfe_tbl.dox](#) File Reference

11.22 [cfe/docs/src/cfe_time.dox](#) File Reference

11.23 [cfe/docs/src/cfe_xref.dox](#) File Reference

11.24 [cfe/docs/src/cfs_versions.dox](#) File Reference

11.25 [cfe/modules/config/fsw/inc/cfe_config_external.h](#) File Reference

```
#include "cfe_config_api_typedefs.h"
```

Functions

- void [CFE_Config_SetupPlatformConfigInfo](#) (void)

11.25.1 Detailed Description

Initialization for CFE configuration registry, external items

These are generated from the build system, they are not directly set

Definition in file [cfe_config_external.h](#).

11.25.2 Function Documentation

CFE_Config_SetupPlatformConfigInfo()

```
void CFE_Config_SetupPlatformConfigInfo (
    void )
```

11.26 cfe_config_external.h

[Go to the documentation of this file.](#)

```
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00016  * limitations under the License.
00017  *****/
00018
00027 #ifndef CFE_CONFIG_EXTERNAL_H
00028 #define CFE_CONFIG_EXTERNAL_H
00029
00030 #include "cfe_config_api_typedefs.h"
00031
00032 void CFE_Config_SetupPlatformConfigInfo(void);
00033
00034 #endif /* CFE_CONFIG_EXTERNAL_H */
```

11.27 cfe/modules/config/fsw/inc/cfe_config_init.h File Reference

```
#include "cfe_config_api_typedefs.h"
#include "cfe_config_ids.h"
```

Functions

- void [CFE_Config_SetupBasicBuildInfo](#) (void)
- [int32 CFE_Config_Init](#) (void)

11.27.1 Detailed Description

Function declarations for items implemented in init.c

Definition in file [cfe_config_init.h](#).

11.27.2 Function Documentation

CFE_Config_Init()

```
int32 CFE_Config_Init (
    void )
```

CFE_Config_SetupBasicBuildInfo()

```
void CFE_Config_SetupBasicBuildInfo (
    void )
```

11.28 cfe_config_init.h

[Go to the documentation of this file.](#)

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00017  *****/
00018
00025 #ifndef CFE_CONFIG_INIT_H
00026 #define CFE_CONFIG_INIT_H
00027
00028 #include "cfe_config_api_typedefs.h"
00029
00030 #include "cfe_config_ids.h"
00031
00032 void CFE_Config_SetupBasicBuildInfo(void);
00033 int32 CFE_Config_Init(void);
00034
00035 #endif /* CFE_CONFIG_INIT_H */
```

11.29 cfe/modules/config/fsw/inc/cfe_config_lookup.h File Reference

```
#include "cfe_config_api_typedefs.h"
#include "cfe_config_table.h"
```

Functions

- [CFE_Config_ValueEntry_t * CFE_Config_LocateConfigRecordById \(CFE_ConfigId_t ConfigId\)](#)
Gets the value record associated with a config ID.

11.29.1 Detailed Description

Function declarations for items implemented in lookup.c
Definition in file [cfe_config_lookup.h](#).

11.29.2 Function Documentation

CFE_Config_LocateConfigRecordById()

```
CFE_Config_ValueEntry_t * CFE_Config_LocateConfigRecordById (
    CFE_ConfigId_t ConfigId )
```

Gets the value record associated with a config ID.

11.30 cfe_config_lookup.h

[Go to the documentation of this file.](#)

```

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00017  *****/
00018
00025 #ifndef CFE_CONFIG_LOOKUP_H
00026 #define CFE_CONFIG_LOOKUP_H
00027
00028 /*
00029 ** Includes
00030 */
00031 #include "cfe_config_api_typedefs.h"
00032 #include "cfe_config_table.h"
00033
00037 CFE_Config_ValueEntry_t *CFE_Config_LocateConfigRecordByID(CFE_ConfigId_t ConfigId);
00038
00039 #endif /* CFE_CONFIG_LOOKUP_H */

```

11.31 cfe/modules/config/fsw/inc/cfe_config_nametable.h File Reference

```
#include "cfe_configid_offset.h"
```

Data Structures

- struct [CFE_Config_IdNameEntry](#)

Typedefs

- typedef struct [CFE_Config_IdNameEntry](#) [CFE_Config_IdNameEntry_t](#)

Variables

- const [CFE_Config_IdNameEntry_t](#) [CFE_CONFIGID_NAMETABLE](#) []

11.31.1 Detailed Description

This file contains the CFE configuration registry global data definitions.
Definition in file [cfe_config_nametable.h](#).

11.31.2 Typedef Documentation

CFE_Config_IdNameEntry_t

```
typedef struct CFE\_Config\_IdNameEntry CFE\_Config\_IdNameEntry\_t
```

11.31.3 Variable Documentation

CFE_CONFIGID_NAMETABLE

```
const CFE\_Config\_IdNameEntry\_t CFE\_CONFIGID\_NAMETABLE[] [extern]
```

11.32 cfe_config_nametable.h

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```

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00017  *****/
00018
00025 #ifndef CFE_CONFIG_NAMETABLE_H
00026 #define CFE_CONFIG_NAMETABLE_H
00027
00028 /*
00029 ** Includes
00030 */
00031 #include "cfe_configid_offset.h"
00032
00033 typedef struct CFE_Config_IdNameEntry
00034 {
00035     const char *Name;
00036 } CFE_Config_IdNameEntry_t;
00037
00038 extern const CFE_Config_IdNameEntry_t CFE_CONFIGID_NAMETABLE[];
00039
00040 #endif /* CFE_CONFIG_MAP_H */

```

11.33 cfe/modules/config/fsw/inc/cfe_config_set.h File Reference

```
#include "cfe_config_api_typedefs.h"
```

Functions

- void [CFE_Config_SetValue](#) (CFE_ConfigId_t ConfigId, uint32 Value)
- void [CFE_Config_SetObjPointer](#) (CFE_ConfigId_t ConfigId, const void *Ptr)
- void [CFE_Config_SetString](#) (CFE_ConfigId_t ConfigId, const char *Ptr)
- void [CFE_Config_SetArrayValue](#) (CFE_ConfigId_t ConfigId, const [CFE_Config_ArrayValue_t](#) *ArrayPtr)

11.33.1 Detailed Description

Function declarations for items implemented in set.c

Definition in file [cfe_config_set.h](#).

11.33.2 Function Documentation

CFE_Config_SetArrayValue()

```

void CFE_Config_SetArrayValue (
    CFE_ConfigId_t ConfigId,
    const CFE_Config_ArrayValue_t * ArrayPtr )

```

CFE_Config_SetObjPointer()

```

void CFE_Config_SetObjPointer (
    CFE_ConfigId_t ConfigId,

```

```
const void * Ptr )
```

CFE_Config_SetString()

```
void CFE_Config_SetString (
    CFE_ConfigId_t ConfigId,
    const char * Ptr )
```

CFE_Config_SetValue()

```
void CFE_Config_SetValue (
    CFE_ConfigId_t ConfigId,
    uint32 Value )
```

11.34 cfe_config_set.h

[Go to the documentation of this file.](#)

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00017  *****/
00018
00025 #ifndef CFE_CONFIG_SET_H
00026 #define CFE_CONFIG_SET_H
00027
00028 /*
00029 ** Includes
00030 */
00031 #include "cfe_config_api_typedefs.h"
00032
00033 void CFE_Config_SetValue(CFE_ConfigId_t ConfigId, uint32 Value);
00034 void CFE_Config_SetObjPointer(CFE_ConfigId_t ConfigId, const void *Ptr);
00035 void CFE_Config_SetString(CFE_ConfigId_t ConfigId, const char *Ptr);
00036 void CFE_Config_SetArrayValue(CFE_ConfigId_t ConfigId, const CFE_Config_ArrayValue_t *ArrayPtr);
00037
00038 #endif /* CFE_CONFIG_SET_H */
```

11.35 cfe/modules/config/fsw/inc/cfe_config_table.h File Reference

```
#include "common_types.h"
#include "cfe_config_ids.h"
```

Data Structures

- union [CFE_Config_ValueBuffer](#)
- struct [CFE_Config_ValueEntry](#)

Typedefs

- typedef enum [CFE_ConfigType](#) [CFE_ConfigType_t](#)
- typedef union [CFE_Config_ValueBuffer](#) [CFE_Config_ValueBuffer_t](#)
- typedef struct [CFE_Config_ValueEntry](#) [CFE_Config_ValueEntry_t](#)

Enumerations

- enum `CFE_ConfigType` {
`CFE_ConfigType_UNDEFINED`, `CFE_ConfigType_VALUE`, `CFE_ConfigType_STRING`, `CFE_ConfigType_POINTER`
, `CFE_ConfigType_ARRAY` }

11.35.1 Detailed Description

This file contains the CFE configuration registry global data definitions.
Definition in file [cfe_config_table.h](#).

11.35.2 Typedef Documentation

CFE_Config_ValueBuffer_t

```
typedef union CFE_Config_ValueBuffer CFE_Config_ValueBuffer_t
```

CFE_Config_ValueEntry_t

```
typedef struct CFE_Config_ValueEntry CFE_Config_ValueEntry_t
```

CFE_ConfigType_t

```
typedef enum CFE_ConfigType CFE_ConfigType_t
```

11.35.3 Enumeration Type Documentation

CFE_ConfigType

```
enum CFE_ConfigType
```

Enumerator

<code>CFE_ConfigType_UNDEFINED</code>	
<code>CFE_ConfigType_VALUE</code>	Value is an unsigned int
<code>CFE_ConfigType_STRING</code>	Value is a string pointer
<code>CFE_ConfigType_POINTER</code>	Value is a non-string object pointer
<code>CFE_ConfigType_ARRAY</code>	Value is a combination of length and pointer

Definition at line 34 of file [cfe_config_table.h](#).

11.36 cfe_config_table.h

[Go to the documentation of this file.](#)

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```

```

00016  * limitations under the License.
00017  *****/
00018
00025 #ifndef CFE_CONFIG_TABLE_H
00026 #define CFE_CONFIG_TABLE_H
00027
00028 /*
00029 ** Includes
00030 */
00031 #include "common_types.h"
00032 #include "cfe_config_ids.h"
00033
00034 typedef enum CFE_ConfigType
00035 {
00036     CFE_ConfigType_UNDEFINED,
00037     CFE_ConfigType_VALUE,
00038     CFE_ConfigType_STRING,
00039     CFE_ConfigType_POINTER,
00040     CFE_ConfigType_ARRAY,
00041 } CFE_ConfigType_t;
00042
00043 typedef union CFE_Config_ValueBuffer
00044 {
00045     uint32      AsInteger;
00046     const void *AsPointer;
00047 } CFE_Config_ValueBuffer_t;
00048
00049 typedef struct CFE_Config_ValueEntry
00050 {
00051     CFE_ConfigType_t      ActualType;
00052     CFE_Config_ValueBuffer_t Datum;
00053 } CFE_Config_ValueEntry_t;
00054
00055 #endif /* CFE_CONFIG_TABLE_H */

```

11.37 cfe/modules/core_api/config/default_cfe_core_api_base_msgid_values.h File Reference

Macros

- #define [CFE_PLATFORM_BASE_MIDVAL\(x\)](#) DEFAULT_CFE_PLATFORM_###_MID_BASE
Platform-specific MSGID base value.
- #define [CFE_GLOBAL_BASE_MIDVAL\(x\)](#) DEFAULT_GLOBAL_###_MID_BASE

11.37.1 Detailed Description

This header file contains the platform-specific base msg ID values and logic to convert a topic ID to a message ID value. This file is just a stand-in to permit CFE to build initially. It is expected that the user will provide the preferred values in a custom header.

Conventionally, CFS binaries are compiled for a specific CPU instance, with the MSGIDs for that CPU compiled directly into the binaries. This example file implements that paradigm. There should be a separate definition of this file for every CFS instance in the mission.

The intent of this design is to permit use of a separate, single header that defines all of the msgid values for all cpus, for example:

```

#define MY_CPU1_CMD_MID_BASE 0x1A00
#define MY_CPU2_CMD_MID_BASE 0x1B00
#define MY_CPU3_CMD_MID_BASE 0x1C00
...

```

Then this file can include that combined base msgid header, and define [CFE_PLATFORM_BASE_MIDVAL\(\)](#) to cherry-pick one of the values within that set. For example on CPU2 this can be defined as:

```

#define CFE_PLATFORM_BASE_MIDVAL(x) MY_CPU2_###_MID_BASE

```

Definition in file [default_cfe_core_api_base_msgid_values.h](#).

11.37.2 Macro Definition Documentation

CFE_GLOBAL_BASE_MIDVAL

```

#define CFE_GLOBAL_BASE_MIDVAL(

```



```
x ) DEFAULT_GLOBAL_##x##_MID_BASE
```

Definition at line 62 of file [default_cfe_core_api_base_msgid_values.h](#).

CFE_PLATFORM_BASE_MIDVAL

```
#define CFE_PLATFORM_BASE_MIDVAL(
```

```
x ) DEFAULT_CFE_PLATFORM_##x##_MID_BASE
```

Platform-specific MSGID base value.

Definition at line 61 of file [default_cfe_core_api_base_msgid_values.h](#).

11.38 default_cfe_core_api_base_msgid_values.h

[Go to the documentation of this file.](#)

```
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00017  *****/
00018
00053 #ifndef DEFAULT_CFE_CORE_API_BASE_MSGID_VALUES_H
00054 #define DEFAULT_CFE_CORE_API_BASE_MSGID_VALUES_H
00055
00061 #define CFE_PLATFORM_BASE_MIDVAL(x) DEFAULT_CFE_PLATFORM_##x##_MID_BASE
00062 #define CFE_GLOBAL_BASE_MIDVAL(x) DEFAULT_GLOBAL_##x##_MID_BASE
00063
00064 #endif /* DEFAULT_CFE_CORE_API_BASE_MSGID_VALUES_H */
```

11.39 cfe/modules/core_api/config/default_cfe_core_api_interface_cfg_values.h File Reference

Macros

- `#define CFE_MISSION_CORE_API_CFGVAL(x) DEFAULT_CFE_MISSION_CORE_API_##x`

11.39.1 Detailed Description

Purpose: This header file contains the mission configuration parameters and typedefs with mission scope.

Definition in file [default_cfe_core_api_interface_cfg_values.h](#).

11.39.2 Macro Definition Documentation

CFE_MISSION_CORE_API_CFGVAL

```
#define CFE_MISSION_CORE_API_CFGVAL(
```

```
x ) DEFAULT_CFE_MISSION_CORE_API_##x
```

Definition at line 31 of file [default_cfe_core_api_interface_cfg_values.h](#).

11.40 default_cfe_core_api_interface_cfg_values.h

[Go to the documentation of this file.](#)

```
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00016  * limitations under the License.
00017  *****/
00018
00027 #ifndef DEFAULT_CFE_CORE_API_INTERFACE_CFG_VALUES_H
00028 #define DEFAULT_CFE_CORE_API_INTERFACE_CFG_VALUES_H
00029
00030 /* Use the default configuration value for all */
00031 #define CFE_MISSION_CORE_API_CFGVAL(x) DEFAULT_CFE_MISSION_CORE_API_##x
00032
00033 #endif /* DEFAULT_CFE_CORE_API_INTERFACE_CFG_VALUES_H */

```

11.41 cfe/modules/core_api/config/default_cfe_core_api_msgid_mapping.h File Reference

```
#include "cfe_core_api_base_msgid_values.h"
```

Macros

- `#define CFE_PLATFORM_CMD_TOPICID_TO_MIDV(topic) (CFE_PLATFORM_BASE_MIDVAL(CMD) | (topic))`
Convert a command topic ID to a MsgID value.
- `#define DEFAULT_CFE_PLATFORM_CMD_MID_BASE 0x1800`
- `#define CFE_PLATFORM_TLM_TOPICID_TO_MIDV(topic) (CFE_PLATFORM_BASE_MIDVAL(TLM) | (topic))`
Convert a telemetry topic ID to a MsgID value.
- `#define DEFAULT_CFE_PLATFORM_TLM_MID_BASE 0x0800`
- `#define CFE_GLOBAL_CMD_TOPICID_TO_MIDV(topic) (CFE_GLOBAL_BASE_MIDVAL(CMD) | (topic))`
Convert a "global" command topic ID to a MsgID value.
- `#define DEFAULT_GLOBAL_CMD_MID_BASE 0x1860`
- `#define CFE_GLOBAL_TLM_TOPICID_TO_MIDV(topic) (CFE_GLOBAL_BASE_MIDVAL(TLM) | (topic))`
Convert a "global" telemetry topic ID to a MsgID value.
- `#define DEFAULT_GLOBAL_TLM_MID_BASE 0x0860`

11.41.1 Detailed Description

This header file contains the logic to convert a topic ID to a message ID value.

In an conventional deployment, the macro simply combines the base MID with the topic ID to produce the MID value. It is simple bitwise "OR" operation.

This logic is intended to be customizable. By overriding this file, these macros can be replaced with whatever logic is desired. However, the same logic must be used across all instances. In order to simply tune the base value used for a given instance, only the `cfs_base_msgids.h` file needs to be overridden.

Definition in file [default_cfe_core_api_msgid_mapping.h](#).

11.41.2 Macro Definition Documentation

CFE_GLOBAL_CMD_TOPICID_TO_MIDV

```
#define CFE_GLOBAL_CMD_TOPICID_TO_MIDV(  
    topic ) (CFE_GLOBAL_BASE_MIDVAL(CMD) | (topic))
```

Convert a "global" command topic ID to a MsgID value.

A global command is one that is not specific to an individual instance of CFE, but rather intended to be broadcast to all CFE instances at the same time.

This is otherwise identical to [CFE_PLATFORM_CMD_TOPICID_TO_MIDV](#)

Definition at line 74 of file [default_cfe_core_api_msgid_mapping.h](#).

CFE_GLOBAL_TLM_TOPICID_TO_MIDV

```
#define CFE_GLOBAL_TLM_TOPICID_TO_MIDV(  
    topic ) (CFE_GLOBAL_BASE_MIDVAL(TLM) | (topic))
```

Convert a "global" telemetry topic ID to a MsgID value.

A global telemetry is one that is not specific to an individual instance of CFE, but rather intended to be broadcast to all CFE instances at the same time.

This is otherwise identical to [CFE_PLATFORM_TLM_TOPICID_TO_MIDV](#)

Definition at line 85 of file [default_cfe_core_api_msgid_mapping.h](#).

CFE_PLATFORM_CMD_TOPICID_TO_MIDV

```
#define CFE_PLATFORM_CMD_TOPICID_TO_MIDV(  
    topic ) (CFE_PLATFORM_BASE_MIDVAL(CMD) | (topic))
```

Convert a command topic ID to a MsgID value.

This defines the logic to convert a topic ID value into a message ID value. This operates on integer values and should resolve at compile time such that it can be used in e.g. switch/case statements.

Note

The result of this conversion is a simple integer, thus also needs to go through [CFE_SB_ValueToMsgId\(\)](#) to obtain a properly-typed [CFE_SB_MsgId_t](#) for interacting with SB APIs.

Definition at line 49 of file [default_cfe_core_api_msgid_mapping.h](#).

CFE_PLATFORM_TLM_TOPICID_TO_MIDV

```
#define CFE_PLATFORM_TLM_TOPICID_TO_MIDV(  
    topic ) (CFE_PLATFORM_BASE_MIDVAL(TLM) | (topic))
```

Convert a telemetry topic ID to a MsgID value.

This defines the logic to convert a topic ID value into a message ID value. This operates on integer values and should resolve at compile time such that it can be used in e.g. switch/case statements.

Note

The result of this conversion is a simple integer, thus also needs to go through [CFE_SB_ValueToMsgId\(\)](#) to obtain a properly-typed [CFE_SB_MsgId_t](#) for interacting with SB APIs.

Definition at line 63 of file [default_cfe_core_api_msgid_mapping.h](#).

DEFAULT_CFE_PLATFORM_CMD_MID_BASE

```
#define DEFAULT_CFE_PLATFORM_CMD_MID_BASE 0x1800
```

Definition at line 50 of file [default_cfe_core_api_msgid_mapping.h](#).

DEFAULT_CFE_PLATFORM_TLM_MID_BASE

```
#define DEFAULT_CFE_PLATFORM_TLM_MID_BASE 0x0800
```

Definition at line 64 of file [default_cfe_core_api_msgid_mapping.h](#).

DEFAULT_GLOBAL_CMD_MID_BASE

```
#define DEFAULT_GLOBAL_CMD_MID_BASE 0x1860
```

Definition at line 75 of file [default_cfe_core_api_msgid_mapping.h](#).

DEFAULT_GLOBAL_TLM_MID_BASE

```
#define DEFAULT_GLOBAL_TLM_MID_BASE 0x0860
```

Definition at line 86 of file [default_cfe_core_api_msgid_mapping.h](#).

11.42 default_cfe_core_api_msgid_mapping.h

[Go to the documentation of this file.](#)

```
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00017  *****/
00018
00032 #ifndef DEFAULT_CFE_CORE_API_MSGID_MAPPING_H
00033 #define DEFAULT_CFE_CORE_API_MSGID_MAPPING_H
00034
00035 /* This header must provide CFE_PLATFORM_BASE_MIDVAL and CFE_PLATFORM_BASE_MIDVAL */
00036 #include "cfe_core_api_base_msgid_values.h"
00037
00049 #define CFE_PLATFORM_CMD_TOPICID_TO_MIDV(topic) (CFE_PLATFORM_BASE_MIDVAL(CMD) | (topic))
00050 #define DEFAULT_CFE_PLATFORM_CMD_MID_BASE 0x1800
00051
00063 #define CFE_PLATFORM_TLM_TOPICID_TO_MIDV(topic) (CFE_PLATFORM_BASE_MIDVAL(TLM) | (topic))
00064 #define DEFAULT_CFE_PLATFORM_TLM_MID_BASE 0x0800
00065
00074 #define CFE_GLOBAL_CMD_TOPICID_TO_MIDV(topic) (CFE_GLOBAL_BASE_MIDVAL(CMD) | (topic))
00075 #define DEFAULT_GLOBAL_CMD_MID_BASE 0x1860
00076
00085 #define CFE_GLOBAL_TLM_TOPICID_TO_MIDV(topic) (CFE_GLOBAL_BASE_MIDVAL(TLM) | (topic))
00086 #define DEFAULT_GLOBAL_TLM_MID_BASE 0x0860
00087
00088 #endif /* CFE_CORE_BASE_MSGIDS_H */
```

11.43 cfe/modules/core_api/config/default_cfe_mission_cfg.h File Reference

```
#include "cfe_core_api_interface_cfg.h"
#include "cfe_es_mission_cfg.h"
#include "cfe_evs_mission_cfg.h"
#include "cfe_sb_mission_cfg.h"
#include "cfe_tbl_mission_cfg.h"
#include "cfe_time_mission_cfg.h"
#include "cfe_fs_mission_cfg.h"
```

11.43.1 Detailed Description

Purpose: This header file contains the mission configuration parameters and typedefs with mission scope.

Definition in file [default_cfe_mission_cfg.h](#).

11.44 default_cfe_mission_cfg.h

[Go to the documentation of this file.](#)

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00017 *
00018
00028 #ifndef DEFAULT_CFE_MISSION_CFG_H
00029 #define DEFAULT_CFE_MISSION_CFG_H
00030
00031 #include "cfe_core_api_interface_cfg.h"
00032 #include "cfe_es_mission_cfg.h"
00033 #include "cfe_evs_mission_cfg.h"
00034 #include "cfe_sb_mission_cfg.h"
00035 #include "cfe_tbl_mission_cfg.h"
00036 #include "cfe_time_mission_cfg.h"
00037 #include "cfe_fs_mission_cfg.h"
00038
00039 #endif /* CFE_MISSION_CFG_H */

```

11.45 cfe/modules/core_api/config/default_cfe_msgids.h File Reference

```

#include "cfe_es_msgids.h"
#include "cfe_evs_msgids.h"
#include "cfe_sb_msgids.h"
#include "cfe_tbl_msgids.h"
#include "cfe_time_msgids.h"

```

11.45.1 Detailed Description

Purpose: This header file contains the Message Id's for messages used by the cFE core.
Definition in file [default_cfe_msgids.h](#).

11.46 default_cfe_msgids.h

[Go to the documentation of this file.](#)

```

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00017 *
00018
00028 #ifndef DEFAULT_CFE_MSGIDS_H
00029 #define DEFAULT_CFE_MSGIDS_H
00030
00031 /*
00032 ** Includes
00033 */
00034
00035 #include "cfe_es_msgids.h"
00036 #include "cfe_evs_msgids.h"

```

```

00037 #include "cfe_sb_msgids.h"
00038 #include "cfe_tbl_msgids.h"
00039 #include "cfe_time_msgids.h"
00040
00041 #endif /* CFE_MSGIDS_H */

```

11.47 cfe/modules/core_api/fsw/inc/cfe.h File Reference

```

#include "common_types.h"
#include "osapi.h"
#include "cfe_mission_cfg.h"
#include "cfe_error.h"
#include "cfe_es.h"
#include "cfe_evs.h"
#include "cfe_fs.h"
#include "cfe_sb.h"
#include "cfe_time.h"
#include "cfe_tbl.h"
#include "cfe_msg.h"
#include "cfe_resourceid.h"
#include "cfe_psp.h"

```

11.47.1 Detailed Description

Purpose: cFE header file

Author: David Kobe, the Hammers Company, Inc.

Notes: This header file centralizes the includes for all cFE Applications. It includes all header files necessary to completely define the cFE interface.

Definition in file [cfe.h](#).

11.48 cfe.h

[Go to the documentation of this file.](#)

```

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00016  * limitations under the License.
00017  *****/
00018
00032 #ifndef CFE_H
00033 #define CFE_H
00034
00035 #include "common_types.h" /* Define basic data types */
00036
00037 #include "osapi.h" /* Define OS API function prototypes */
00038
00039 #include "cfe_mission_cfg.h" /* Define mission configuration parameters */
00040
00041 #include "cfe_error.h" /* Define common cFE error codes */
00042
00043 #include "cfe_es.h" /* Define Executive Service API */
00044 #include "cfe_evs.h" /* Define Event Service API */
00045 #include "cfe_fs.h" /* Define File Service API */
00046 #include "cfe_sb.h" /* Define Software Bus API */

```

```

00047 #include "cfe_time.h" /* Define Time Service API */
00048 #include "cfe_tbl.h" /* Define Table Service API */
00049
00050 #include "cfe_msg.h" /* Define Message API */
00051 #include "cfe_resourceid.h" /* Define ResourceID API */
00052
00053 #include "cfe_psp.h" /* Define Platform Support Package API */
00054
00055 #endif /* CFE_H */

```

11.49 cfe/modules/core_api/fsw/inc/cfe_config.h File Reference

```

#include "common_types.h"
#include "cfe_config_api_typedefs.h"
#include "cfe_config_ids.h"

```

Functions

- [uint32 CFE_Config_GetValue \(CFE_ConfigId_t ConfigId\)](#)
Obtain an integer value correlating to an CFE configuration ID.
- [const void * CFE_Config_GetObjPointer \(CFE_ConfigId_t ConfigId\)](#)
Obtain a pointer value correlating to an CFE configuration ID.
- [CFE_Config_ArrayValue_t CFE_Config_GetArrayValue \(CFE_ConfigId_t ConfigId\)](#)
Obtain an array correlating to an CFE configuration ID.
- [const char * CFE_Config_GetString \(CFE_ConfigId_t ConfigId\)](#)
Obtain a string value correlating to an CFE configuration ID.
- [const char * CFE_Config_GetName \(CFE_ConfigId_t ConfigId\)](#)
Obtain the name of a CFE configuration ID.
- [CFE_ConfigId_t CFE_Config_GetIdByName \(const char *Name\)](#)
Obtain the ID value associated with a configuration name.
- [void CFE_Config_IterateAll \(void *Arg, CFE_Config_Callback_t Callback\)](#)
Iterate all known name/ID value pairs.
- [void CFE_Config_GetVersionString \(char *Buf, size_t Size, const char *Component, const char *SrcVersion, const char *CodeName, const char *LastOffcRel\)](#)
Obtain the version string for a cFS component or app.

11.49.1 Detailed Description

Title: cFE Status Code Definition Header File

Purpose: Common source of cFE API return status codes.

Design Notes:

References: Flight Software Branch C Coding Standard Version 1.0a

Definition in file [cfe_config.h](#).

11.49.2 Function Documentation

CFE_Config_GetArrayValue()

```

CFE_Config_ArrayValue_t CFE_Config_GetArrayValue (
    CFE_ConfigId_t ConfigId )

```

Obtain an array correlating to an CFE configuration ID.

Retrieves the CFE_Config_ArrayValue_t value associated with the specified key. This combines an array length (number of elements) and a pointer to the first element.

If no value has been set, or the key is not valid, this returns 0 / NULL.

Parameters

in	<i>Config↔ Id</i>	Configuration ID/Key to look up
----	-----------------------	---------------------------------

Returns

Value associated with key

Return values

<i>NULL</i>	if key is not defined or not set
-------------	----------------------------------

CFE_Config_GetIdByName()

```
CFE_ConfigId_t CFE_Config_GetIdByName (
    const char * Name )
```

Obtain the ID value associated with a configuration name.

Parameters

in	<i>Name</i>	The name of the ID to look up
----	-------------	-------------------------------

Returns

ID associated with name

Return values

<i>CFE_CONFIGID_UNDEFINED</i>	if the name did not correspond to a key
-------------------------------	---

CFE_Config_GetName()

```
const char * CFE_Config_GetName (
    CFE_ConfigId_t ConfigId )
```

Obtain the name of a CFE configuration ID.

Retrieves the printable name associated with the specified key.

Note

This function does not return NULL.

If the ID is not valid/known, then the implementation returns the special string '[unknown]' rather than NULL, so this function may be more easily used in printf() style calls.

Parameters

in	<i>Config↔ Id</i>	Configuration ID/Key to look up
----	-----------------------	---------------------------------

Returns

Name associated with key

CFE_Config_GetObjPointer()

```
const void * CFE_Config_GetObjPointer (
    CFE_ConfigId_t ConfigId )
```

Obtain a pointer value correlating to an CFE configuration ID.
Retrieves the pointer value associated with the specified key.
If no value has been set, or the key is not valid, this returns NULL.

Parameters

in	<i>ConfigId</i>	Configuration ID/Key to look up
----	-----------------	---------------------------------

Returns

Value associated with key

Return values

NULL	if key is not defined or not set
------	----------------------------------

CFE_Config_GetString()

```
const char * CFE_Config_GetString (
    CFE_ConfigId_t ConfigId )
```

Obtain a string value correlating to an CFE configuration ID.
Retrieves the string value associated with the specified key.
If no value has been set, or the key is not valid, this returns the special string "UNDEFINED"

Note

This function does not return NULL, so it can be used directly in printf-style calls.

Parameters

in	<i>ConfigId</i>	Configuration ID/Key to look up
----	-----------------	---------------------------------

Returns

String value associated with key

CFE_Config_GetValue()

```
uint32 CFE_Config_GetValue (
    CFE_ConfigId_t ConfigId )
```

Obtain an integer value correlating to an CFE configuration ID.
Retrieves the integer value associated with the specified key.

If no value has been set, or the key is not valid, this returns 0.

Parameters

in	<i>ConfigId</i>	Configuration ID/Key to look up
----	-----------------	---------------------------------

Returns

Value associated with key

Return values

0	if key is not defined or not set
---	----------------------------------

CFE_Config_GetVersionString()

```
void CFE_Config_GetVersionString (
    char * Buf,
    size_t Size,
    const char * Component,
    const char * SrcVersion,
    const char * CodeName,
    const char * LastOffcRel )
```

Obtain the version string for a cFS component or app.

Assembles a standardized version string associated with the specified component/app.

Parameters

in	<i>Buf</i>	Buffer to place version string in. Will be populated with standard version string containing the provided parameters (i.e.: "cFE DEVELOPMENT BUILD equuleus-rc1+dev0 (Codename equuleus), Last Official Release: cFE 6.7.0"
in	<i>Size</i>	Size of the provided buffer
in	<i>Component</i>	Component for which to get version string (i.e. "cFE")
in	<i>SrcVersion</i>	Source version identifier (i.e. "equuleus-rc1+dev0")
in	<i>CodeName</i>	Code name for the build (i.e. "equuleus")
in	<i>LastOffcRel</i>	Last official release (i.e. "6.7.0")

CFE_Config_IterateAll()

```
void CFE_Config_IterateAll (
    void * Arg,
    CFE_Config_Callback_t Callback )
```

Iterate all known name/ID value pairs.

Parameters

in	<i>Arg</i>	User-supplied opaque argument to pass to callback
in	<i>Callback</i>	User-supplied callback function to invoke for each ID

11.50 cfe_config.h

[Go to the documentation of this file.](#)

```

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00017  *****/
00018
00034 #ifndef CFE_CONFIG_H
00035 #define CFE_CONFIG_H
00036
00037 /* Include Files */
00038 #include "common_types.h"
00039 #include "cfe_config_api_typedefs.h"
00040 #include "cfe_config_ids.h"
00041
00054 uint32 CFE_Config_GetValue(CFE_ConfigId_t ConfigId);
00055
00068 const void *CFE_Config_GetObjPointer(CFE_ConfigId_t ConfigId);
00069
00083 CFE_Config_ArrayValue_t CFE_Config_GetArrayValue(CFE_ConfigId_t ConfigId);
00084
00100 const char *CFE_Config_GetString(CFE_ConfigId_t ConfigId);
00101
00117 const char *CFE_Config_GetName(CFE_ConfigId_t ConfigId);
00118
00127 CFE_ConfigId_t CFE_Config_GetIdByName(const char *Name);
00128
00135 void CFE_Config_IterateAll(void *Arg, CFE_Config_Callback_t Callback);
00136
00155 void CFE_Config_GetVersionString(char *Buf,
00156                                 size_t Size,
00157                                 const char *Component,
00158                                 const char *SrcVersion,
00159                                 const char *CodeName,
00160                                 const char *LastOffcRel);
00161
00162 #endif /* CFE_CONFIG_H */

```

11.51 cfe/modules/core_api/fsw/inc/cfe_config_api_typedefs.h File Reference

```

#include "common_types.h"
#include "cfe_resourceid_api_typedefs.h"

```

Data Structures

- struct [CFE_Config_ArrayValue](#)
Wrapper type for array configuration.

Macros

- #define [CFE_CONFIGID_C\(val\)](#) ((CFE_ConfigId_t)CFE_RESOURCEID_WRAP(val))
- #define [CFE_CONFIGID_UNDEFINED](#) CFE_CONFIGID_C(CFE_RESOURCEID_UNDEFINED)

Typedefs

- typedef CFE_RESOURCEID_BASE_TYPE [CFE_ConfigId_t](#)

A type for Configuration IDs.

- typedef void(* CFE_Config_Callback_t) (void *Arg, CFE_ConfigId_t Id, const char *Name)
- typedef struct CFE_Config_ArrayValue CFE_Config_ArrayValue_t

Wrapper type for array configuration.

11.51.1 Detailed Description

Title: cFE Status Code Definition Header File

Purpose: Common source of cFE API return status codes.

Design Notes:

References: Flight Software Branch C Coding Standard Version 1.0a

Definition in file [cfe_config_api_typedefs.h](#).

11.51.2 Macro Definition Documentation

CFE_CONFIGID_C

```
#define CFE_CONFIGID_C(  
    val ) ((CFE_ConfigId_t)CFE_RESOURCEID_WRAP(val))
```

Definition at line 48 of file [cfe_config_api_typedefs.h](#).

CFE_CONFIGID_UNDEFINED

```
#define CFE_CONFIGID_UNDEFINED CFE_CONFIGID_C(CFE_RESOURCEID_UNDEFINED)
```

Definition at line 49 of file [cfe_config_api_typedefs.h](#).

11.51.3 Typedef Documentation

CFE_Config_ArrayValue_t

```
typedef struct CFE_Config_ArrayValue CFE_Config_ArrayValue_t
```

Wrapper type for array configuration.

This is a pair containing a size and pointer that is get/set via a single config table entry

CFE_Config_Callback_t

```
typedef void(* CFE_Config_Callback_t) (void *Arg, CFE_ConfigId_t Id, const char *Name)
```

Definition at line 51 of file [cfe_config_api_typedefs.h](#).

CFE_ConfigId_t

```
typedef CFE_RESOURCEID_BASE_TYPE CFE_ConfigId_t
```

A type for Configuration IDs.

This is the type that is used for any API accepting or returning a configuration key ID

Definition at line 46 of file [cfe_config_api_typedefs.h](#).

11.52 cfe_config_api_typedefs.h

[Go to the documentation of this file.](#)

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00017  *****/
00018
00034 #ifndef CFE_CONFIG_API_TYPEDEFS_H
00035 #define CFE_CONFIG_API_TYPEDEFS_H
00036
00037 /* Include Files */
00038 #include "common_types.h"
00039 #include "cfe_resourceid_api_typedefs.h"
00040
00046 typedef CFE_RESOURCEID_BASE_TYPE CFE_ConfigId_t;
00047
00048 #define CFE_CONFIGID_C(val) ((CFE_ConfigId_t)CFE_RESOURCEID_WRAP(val))
00049 #define CFE_CONFIGID_UNDEFINED CFE_CONFIGID_C(CFE_RESOURCEID_UNDEFINED)
00050
00051 typedef void (*CFE_Config_Callback_t)(void *Arg, CFE_ConfigId_t Id, const char *Name);
00052
00059 typedef struct CFE_Config_ArrayValue
00060 {
00061     size_t      NumElements;
00062     const void *ElementPtr;
00063 } CFE_Config_ArrayValue_t;
00064
00065 #endif /* CFE_CONFIG_API_TYPEDEFS_H */

```

11.53 cfe/modules/core_api/fsw/inc/cfe_core_api_base_msgids.h File Reference

```
#include "cfe_core_api_msgid_mapping.h"
```

11.53.1 Detailed Description

This header file contains the logic to convert a topic ID to a message ID value.

Note

Traditionally in CFS the MsgID values were translated using an offset added to a base value that was specific to that CFS instance. The name of this file reflects that heritage and is preserved for backward compatibility with existing apps.

Although this file is called "base_msgids.h" the translation can take on any form, even a runtime lookup. The file name of "cfe_core_api_msgid_mapping.h" better conveys the intent that this is not limited to a base+offset paradigm.

This file is now just an alias/wrapper around the msgid_mapping header file to allow apps and libraries to continue building.

To customize the mapping, override "cfe_core_api_msgid_mapping.h"

Definition in file [cfe_core_api_base_msgids.h](#).

11.54 cfe_core_api_base_msgids.h

[Go to the documentation of this file.](#)

```

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```

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00017  *****/
00018
00042 #ifndef CFE_CORE_API_BASE_MSGIDS_H
00043 #define CFE_CORE_API_BASE_MSGIDS_H
00044
00045 #include "cfe_core_api_msgid_mapping.h"
00046
00047 #endif /* CFE_CORE_API_BASE_MSGIDS_H */

```

11.55 cfe/modules/core_api/fsw/inc/cfe_core_api_interface_cfg.h File Reference

```
#include "cfe_core_api_interface_cfg_values.h"
```

Macros

- #define [CFE_MISSION_MAX_PATH_LEN](#) [CFE_MISSION_CORE_API_CFGVAL](#)(MAX_PATH_LEN)
- #define [DEFAULT_CFE_MISSION_CORE_API_MAX_PATH_LEN](#) 64
- #define [CFE_MISSION_MAX_FILE_LEN](#) [CFE_MISSION_CORE_API_CFGVAL](#)(MAX_FILE_LEN)
- #define [DEFAULT_CFE_MISSION_CORE_API_MAX_FILE_LEN](#) 20
- #define [CFE_MISSION_MAX_API_LEN](#) [CFE_MISSION_CORE_API_CFGVAL](#)(MAX_API_LEN)
- #define [DEFAULT_CFE_MISSION_CORE_API_MAX_API_LEN](#) 20
- #define [CFE_MISSION_MAX_NUM_FILES](#) [CFE_MISSION_CORE_API_CFGVAL](#)(MAX_NUM_FILES)
- #define [DEFAULT_CFE_MISSION_CORE_API_MAX_NUM_FILES](#) 50

11.55.1 Detailed Description

Purpose: This header file contains the mission configuration parameters and typedefs with mission scope.
Definition in file [cfe_core_api_interface_cfg.h](#).

11.55.2 Macro Definition Documentation

CFE_MISSION_MAX_API_LEN

```
#define CFE_MISSION_MAX_API_LEN CFE\_MISSION\_CORE\_API\_CFGVAL (MAX_API_LEN)
```

Purpose cFE Maximum length for API names within data exchange structures

Description:

The value of this constant dictates the size of filenames within all structures used for external data exchange, such as Software bus messages and table definitions. This is typically the same as `OS_MAX_API_LEN` but that is OSAL dependent – and as such it definable on a per-processor/OS basis and hence may be different across multiple processors. It is recommended to set this to the value of the largest `OS_MAX_API_LEN` in use on any CPU on the mission.

This affects only the layout of command/telemetry messages and table definitions; internal allocation may use the platform-specific `OS_MAX_API_LEN` value.

This length must include an extra character for NULL termination.

Limits

All CPUs within the same SB domain (mission) must share the same definition Note this affects the size of messages, so it must not cause any message to exceed the max length.

This value should be kept as a multiple of 4, to maintain alignment of any possible neighboring fields without implicit padding.

Definition at line 112 of file [cfe_core_api_interface_cfg.h](#).

CFE_MISSION_MAX_FILE_LEN

```
#define CFE_MISSION_MAX_FILE_LEN CFE_MISSION_CORE_API_CFGVAL(MAX_FILE_LEN)
```

Purpose cFE Maximum length for filenames within data exchange structures

Description:

The value of this constant dictates the size of filenames within all structures used for external data exchange, such as Software bus messages and table definitions. This is typically the same as OS_MAX_FILE_LEN but that is OSAL dependent – and as such it definable on a per-processor/OS basis and hence may be different across multiple processors. It is recommended to set this to the value of the largest OS_MAX_FILE_LEN in use on any CPU on the mission.

This affects only the layout of command/telemetry messages and table definitions; internal allocation may use the platform-specific OS_MAX_FILE_LEN value.

This length must include an extra character for NULL termination.

Limits

All CPUs within the same SB domain (mission) and ground tools must share the same definition. Note this affects the size of messages, so it must not cause any message to exceed the max length.

This value should be kept as a multiple of 4, to maintain alignment of any possible neighboring fields without implicit padding.

Definition at line 85 of file [cfe_core_api_interface_cfg.h](#).

CFE_MISSION_MAX_NUM_FILES

```
#define CFE_MISSION_MAX_NUM_FILES CFE_MISSION_CORE_API_CFGVAL(MAX_NUM_FILES)
```

Purpose cFE Maximum number of files in a message/data exchange

Description:

The value of this constant dictates the maximum number of files within all structures used for external data exchange, such as Software bus messages and table definitions. This is typically the same as OS_MAX_NUM_↵_OPEN_FILES but that is OSAL dependent – and as such it definable on a per-processor/OS basis and hence may be different across multiple processors. It is recommended to set this to the value of the largest OS_MAX_NUM_↵_OPEN_FILES in use on any CPU on the mission.

This affects only the layout of command/telemetry messages and table definitions; internal allocation may use the platform-specific OS_MAX_NUM_OPEN_FILES value.

Limits

All CPUs within the same SB domain (mission) must share the same definition Note this affects the size of messages, so it must not cause any message to exceed the max length.

Definition at line 135 of file [cfe_core_api_interface_cfg.h](#).

CFE_MISSION_MAX_PATH_LEN

```
#define CFE_MISSION_MAX_PATH_LEN CFE_MISSION_CORE_API_CFGVAL(MAX_PATH_LEN)
```

Purpose cFE Maximum length for pathnames within data exchange structures

Description:

The value of this constant dictates the size of pathnames within all structures used for external data exchange, such as Software bus messages and table definitions. This is typically the same as OS_MAX_PATH_LEN but that is OSAL dependent – and as such it definable on a per-processor/OS basis and hence may be different across multiple processors. It is recommended to set this to the value of the largest OS_MAX_PATH_LEN in use on any CPU on the mission.

This affects only the layout of command/telemetry messages and table definitions; internal allocation may use the platform-specific OS_MAX_PATH_LEN value.

This length must include an extra character for NULL termination.

Limits

All CPUs within the same SB domain (mission) and ground tools must share the same definition. Note this affects the size of messages, so it must not cause any message to exceed the max length.

This value should be kept as a multiple of 4, to maintain alignment of any possible neighboring fields without implicit padding.

Definition at line 57 of file [cfe_core_api_interface_cfg.h](#).

DEFAULT_CFE_MISSION_CORE_API_MAX_API_LEN

```
#define DEFAULT_CFE_MISSION_CORE_API_MAX_API_LEN 20
```

Definition at line 113 of file [cfe_core_api_interface_cfg.h](#).

DEFAULT_CFE_MISSION_CORE_API_MAX_FILE_LEN

```
#define DEFAULT_CFE_MISSION_CORE_API_MAX_FILE_LEN 20
```

Definition at line 86 of file [cfe_core_api_interface_cfg.h](#).

DEFAULT_CFE_MISSION_CORE_API_MAX_NUM_FILES

```
#define DEFAULT_CFE_MISSION_CORE_API_MAX_NUM_FILES 50
```

Definition at line 136 of file [cfe_core_api_interface_cfg.h](#).

DEFAULT_CFE_MISSION_CORE_API_MAX_PATH_LEN

```
#define DEFAULT_CFE_MISSION_CORE_API_MAX_PATH_LEN 64
```

Definition at line 58 of file [cfe_core_api_interface_cfg.h](#).

11.56 cfe_core_api_interface_cfg.h

[Go to the documentation of this file.](#)

```
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00017  *****/
00018
00027 #ifndef CFE_CORE_API_INTERFACE_CFG_H
```



```

00028 #define CFE_CORE_API_INTERFACE_CFG_H
00029
00030 #include "cfe_core_api_interface_cfg_values.h"
00031
00057 #define CFE_MISSION_MAX_PATH_LEN CFE_MISSION_CORE_API_CFGVAL(MAX_PATH_LEN)
00058 #define DEFAULT_CFE_MISSION_CORE_API_MAX_PATH_LEN 64
00059
00085 #define CFE_MISSION_MAX_FILE_LEN CFE_MISSION_CORE_API_CFGVAL(MAX_FILE_LEN)
00086 #define DEFAULT_CFE_MISSION_CORE_API_MAX_FILE_LEN 20
00087
00112 #define CFE_MISSION_MAX_API_LEN CFE_MISSION_CORE_API_CFGVAL(MAX_API_LEN)
00113 #define DEFAULT_CFE_MISSION_CORE_API_MAX_API_LEN 20
00114
00135 #define CFE_MISSION_MAX_NUM_FILES CFE_MISSION_CORE_API_CFGVAL(MAX_NUM_FILES)
00136 #define DEFAULT_CFE_MISSION_CORE_API_MAX_NUM_FILES 50
00137
00138 #endif /* SAMPLE_MISSION_CFG_H */

```

11.57 cfe/modules/core_api/fsw/inc/cfe_endian.h File Reference

```
#include "common_types.h"
```

Macros

- #define [CFE_MAKE_BIG16](#)(n) (((n) & 0x00FF) << 8) | (((n) & 0xFF00) >> 8))
- #define [CFE_MAKE_BIG32](#)(n) (((n) & 0x000000FF) << 24) | (((n) & 0x0000FF00) << 8) | (((n) & 0x00FF0000) >> 8) | (((n) & 0xFF000000) >> 24))

11.57.1 Detailed Description

Purpose: Define macros to enforce big-endian/network byte order for 16 and 32 bit integers

Definition in file [cfe_endian.h](#).

11.57.2 Macro Definition Documentation

CFE_MAKE_BIG16

```
#define CFE_MAKE_BIG16(
    n ) (((n) & 0x00FF) << 8) | (((n) & 0xFF00) >> 8))
```

Definition at line 64 of file [cfe_endian.h](#).

CFE_MAKE_BIG32

```
#define CFE_MAKE_BIG32(
    n ) (((n) & 0x000000FF) << 24) | (((n) & 0x0000FF00) << 8) | (((n) & 0x00FF0000)
>> 8) | (((n) & 0xFF000000) >> 24))
```

Definition at line 65 of file [cfe_endian.h](#).

11.58 cfe_endian.h

[Go to the documentation of this file.](#)

```

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00017  *****/
00018
00027 #ifndef CFE_ENDIAN_H
00028 #define CFE_ENDIAN_H
00029
00030 /*
00031 ** Include Files
00032 */
00033 #include "common_types.h"
00034
00035 /*
00036 * SOFTWARE_BIG/LITTLE_BIT_ORDER COMPATIBILITY MACRO -
00037 *
00038 * This is provided only for backward compatibility. Do not write any new code that
00039 * uses this macro.
00040 */
00041 #if !defined(SOFTWARE_BIG_BIT_ORDER) && !defined(SOFTWARE_LITTLE_BIT_ORDER)
00042
00043 #if defined(__BYTE_ORDER) && __BYTE_ORDER == __BIG_ENDIAN || defined(__BIG_ENDIAN__) || defined(__ARMEB__)
00044 \
00045 || defined(__THUMBEB__) || defined(__AARCH64EB__) || defined(__MIBSEB__) || defined(__MIBSEB__) ||
00046 defined(__MIBSEB__)
00047 /* It is a big-endian target architecture */
00048 #define SOFTWARE_BIG_BIT_ORDER
00049 #elif defined(__BYTE_ORDER) && __BYTE_ORDER == __LITTLE_ENDIAN || defined(__LITTLE_ENDIAN__) ||
00050 defined(__ARMEL__) \
00051 || defined(__THUMBEL__) || defined(__AARCH64EL__) || defined(__MIPSEL__) || defined(__MIPSEL__) ||
00052 defined(__MIPSEL__) \
00053 || defined(__i386__) || defined(__i386__) || defined(__i686__) || defined(__i686__) || defined(__x86_64__)
00054 \
00055 || defined(__x86_64__)
00056 /* It is a little-endian target architecture */
00057 #define SOFTWARE_LITTLE_BIT_ORDER
00058 #else
00059 #error Unknown byte order on this platform
00060 #endif
00061 #endif /* !defined(SOFTWARE_BIG_BIT_ORDER) && !defined(SOFTWARE_LITTLE_BIT_ORDER) */
00062
00063 /* Macro to convert 16/32 bit types from platform "endianness" to Big Endian */
00064 #ifndef SOFTWARE_BIG_BIT_ORDER
00065 #define CFE_MAKE_BIG16(n) (n)
00066 #define CFE_MAKE_BIG32(n) (n)
00067 #else
00068 #define CFE_MAKE_BIG16(n) (((n) & 0x00FF) << 8) | ((n) & 0xFF00) >> 8)
00069 #define CFE_MAKE_BIG32(n) \
00070 (((n) & 0x000000FF) << 24) | (((n) & 0x0000FF00) << 8) | (((n) & 0x00FF0000) >> 8) | (((n) & 0xFF000000) >> 24))
00071 #endif
00072
00073 #endif /* CFE_ENDIAN_H */

```

11.59 cfe/modules/core_api/fsw/inc/cfe_error.h File Reference

```
#include "osapi.h"
```

Macros

- #define [CFE_STATUS_C\(X\)](#) ((CFE_Status_t)(X))
cFE Status macro for literal
- #define [CFE_STATUS_STRING_LENGTH](#) 11
cFE Status converted to string length limit
- #define [CFE_SEVERITY_BITMASK](#) ((CFE_Status_t)0xc0000000)
Error Severity Bitmask.
- #define [CFE_SEVERITY_SUCCESS](#) ((CFE_Status_t)0x00000000)
Severity Success.
- #define [CFE_SEVERITY_INFO](#) ((CFE_Status_t)0x40000000)

- *Severity Info.*
• #define CFE_SEVERITY_ERROR ((CFE_Status_t)0xc0000000)
- *Severity Error.*
• #define CFE_SERVICE_BITMASK ((CFE_Status_t)0x0e000000)
- *Error Service Bitmask.*
• #define CFE_EVENTS_SERVICE ((CFE_Status_t)0x02000000)
- *Event Service.*
• #define CFE_EXECUTIVE_SERVICE ((CFE_Status_t)0x04000000)
- *Executive Service.*
• #define CFE_FILE_SERVICE ((CFE_Status_t)0x06000000)
- *File Service.*
• #define CFE_GENERIC_SERVICE ((CFE_Status_t)0x08000000)
- *Generic Service.*
• #define CFE_SOFTWARE_BUS_SERVICE ((CFE_Status_t)0x0a000000)
- *Software Bus Service.*
• #define CFE_TABLE_SERVICE ((CFE_Status_t)0x0c000000)
- *Table Service.*
• #define CFE_TIME_SERVICE ((CFE_Status_t)0x0e000000)
- *Time Service.*
• #define CFE_SUCCESS ((CFE_Status_t)0)
- *Successful execution.*
• #define CFE_STATUS_NO_COUNTER_INCREMENT ((CFE_Status_t)0x48000001)
- *No Counter Increment.*
• #define CFE_STATUS_WRONG_MSG_LENGTH ((CFE_Status_t)0xc8000002)
- *Wrong Message Length.*
• #define CFE_STATUS_UNKNOWN_MSG_ID ((CFE_Status_t)0xc8000003)
- *Unknown Message ID.*
• #define CFE_STATUS_BAD_COMMAND_CODE ((CFE_Status_t)0xc8000004)
- *Bad Command Code.*
• #define CFE_STATUS_EXTERNAL_RESOURCE_FAIL ((CFE_Status_t)0xc8000005)
- *External failure.*
• #define CFE_STATUS_REQUEST_ALREADY_PENDING ((int32)0xc8000006)
- *Request already pending.*
• #define CFE_STATUS_VALIDATION_FAILURE ((int32)0xc8000007)
- *Request or input value failed basic structural validation.*
• #define CFE_STATUS_RANGE_ERROR ((int32)0xc8000008)
- *Request or input value is out of range.*
• #define CFE_STATUS_INCORRECT_STATE ((int32)0xc8000009)
- *Cannot process request at this time.*
• #define CFE_STATUS_NOT_IMPLEMENTED ((CFE_Status_t)0xc800ffff)
- *Not Implemented.*
• #define CFE_EVS_UNKNOWN_FILTER ((CFE_Status_t)0xc2000001)
- *Unknown Filter.*
• #define CFE_EVS_APP_NOT_REGISTERED ((CFE_Status_t)0xc2000002)
- *Application Not Registered.*
• #define CFE_EVS_APP_ILLEGAL_APP_ID ((CFE_Status_t)0xc2000003)
- *Illegal Application ID.*

- #define CFE_EVS_APP_FILTER_OVERLOAD ((CFE_Status_t)0xc2000004)
Application Filter Overload.
- #define CFE_EVS_RESET_AREA_POINTER ((CFE_Status_t)0xc2000005)
Reset Area Pointer Failure.
- #define CFE_EVS_EVT_NOT_REGISTERED ((CFE_Status_t)0xc2000006)
Event Not Registered.
- #define CFE_EVS_FILE_WRITE_ERROR ((CFE_Status_t)0xc2000007)
File Write Error.
- #define CFE_EVS_INVALID_PARAMETER ((CFE_Status_t)0xc2000008)
Invalid Pointer.
- #define CFE_EVS_APP_SQUELCHED ((CFE_Status_t)0xc2000009)
Event squelched.
- #define CFE_EVS_NOT_IMPLEMENTED ((CFE_Status_t)0xc200ffff)
Not Implemented.
- #define CFE_ES_ERR_RESOURCEID_NOT_VALID ((CFE_Status_t)0xc4000001)
Resource ID is not valid.
- #define CFE_ES_ERR_NAME_NOT_FOUND ((CFE_Status_t)0xc4000002)
Resource Name Error.
- #define CFE_ES_ERR_APP_CREATE ((CFE_Status_t)0xc4000004)
Application Create Error.
- #define CFE_ES_ERR_CHILD_TASK_CREATE ((CFE_Status_t)0xc4000005)
Child Task Create Error.
- #define CFE_ES_ERR_SYS_LOG_FULL ((CFE_Status_t)0xc4000006)
System Log Full.
- #define CFE_ES_ERR_MEM_BLOCK_SIZE ((CFE_Status_t)0xc4000008)
Memory Block Size Error.
- #define CFE_ES_ERR_LOAD_LIB ((CFE_Status_t)0xc4000009)
Load Library Error.
- #define CFE_ES_BAD_ARGUMENT ((CFE_Status_t)0xc400000a)
Bad Argument.
- #define CFE_ES_ERR_CHILD_TASK_REGISTER ((CFE_Status_t)0xc400000b)
Child Task Register Error.
- #define CFE_ES_CDS_ALREADY_EXISTS ((CFE_Status_t)0x4400000d)
CDS Already Exists.
- #define CFE_ES_CDS_INSUFFICIENT_MEMORY ((CFE_Status_t)0xc400000e)
CDS Insufficient Memory.
- #define CFE_ES_CDS_INVALID_NAME ((CFE_Status_t)0xc400000f)
CDS Invalid Name.
- #define CFE_ES_CDS_INVALID_SIZE ((CFE_Status_t)0xc4000010)
CDS Invalid Size.
- #define CFE_ES_CDS_INVALID ((CFE_Status_t)0xc4000012)
CDS Invalid.
- #define CFE_ES_CDS_ACCESS_ERROR ((CFE_Status_t)0xc4000013)
CDS Access Error.
- #define CFE_ES_FILE_IO_ERR ((CFE_Status_t)0xc4000014)
File IO Error.
- #define CFE_ES_RST_ACCESS_ERR ((CFE_Status_t)0xc4000015)

- *Reset Area Access Error.*
- #define CFE_ES_ERR_APP_REGISTER ((CFE_Status_t)0xc4000017)
- *Application Register Error.*
- #define CFE_ES_ERR_CHILD_TASK_DELETE ((CFE_Status_t)0xc4000018)
- *Child Task Delete Error.*
- #define CFE_ES_ERR_CHILD_TASK_DELETE_MAIN_TASK ((CFE_Status_t)0xc4000019)
- *Child Task Delete Passed Main Task.*
- #define CFE_ES_CDS_BLOCK_CRC_ERR ((CFE_Status_t)0xc400001A)
- *CDS Block CRC Error.*
- #define CFE_ES_MUT_SEM_DELETE_ERR ((CFE_Status_t)0xc400001B)
- *Mutex Semaphore Delete Error.*
- #define CFE_ES_BIN_SEM_DELETE_ERR ((CFE_Status_t)0xc400001C)
- *Binary Semaphore Delete Error.*
- #define CFE_ES_COUNT_SEM_DELETE_ERR ((CFE_Status_t)0xc400001D)
- *Counting Semaphore Delete Error.*
- #define CFE_ES_QUEUE_DELETE_ERR ((CFE_Status_t)0xc400001E)
- *Queue Delete Error.*
- #define CFE_ES_FILE_CLOSE_ERR ((CFE_Status_t)0xc400001F)
- *File Close Error.*
- #define CFE_ES_CDS_WRONG_TYPE_ERR ((CFE_Status_t)0xc4000020)
- *CDS Wrong Type Error.*
- #define CFE_ES_CDS_OWNER_ACTIVE_ERR ((CFE_Status_t)0xc4000022)
- *CDS Owner Active Error.*
- #define CFE_ES_APP_CLEANUP_ERR ((CFE_Status_t)0xc4000023)
- *Application Cleanup Error.*
- #define CFE_ES_TIMER_DELETE_ERR ((CFE_Status_t)0xc4000024)
- *Timer Delete Error.*
- #define CFE_ES_BUFFER_NOT_IN_POOL ((CFE_Status_t)0xc4000025)
- *Buffer Not In Pool.*
- #define CFE_ES_TASK_DELETE_ERR ((CFE_Status_t)0xc4000026)
- *Task Delete Error.*
- #define CFE_ES_OPERATION_TIMED_OUT ((CFE_Status_t)0xc4000027)
- *Operation Timed Out.*
- #define CFE_ES_LIB_ALREADY_LOADED ((CFE_Status_t)0x44000028)
- *Library Already Loaded.*
- #define CFE_ES_ERR_SYS_LOG_TRUNCATED ((CFE_Status_t)0x44000029)
- *System Log Message Truncated.*
- #define CFE_ES_NO_RESOURCE_IDS_AVAILABLE ((CFE_Status_t)0xc400002B)
- *Resource ID is not available.*
- #define CFE_ES_POOL_BLOCK_INVALID ((CFE_Status_t)0xc400002C)
- *Invalid pool block.*
- #define CFE_ES_ERR_DUPLICATE_NAME ((CFE_Status_t)0xc400002E)
- *Duplicate Name Error.*
- #define CFE_ES_NOT_IMPLEMENTED ((CFE_Status_t)0xc400ffff)
- *Not Implemented.*
- #define CFE_FS_BAD_ARGUMENT ((CFE_Status_t)0xc6000001)
- *Bad Argument.*

- #define CFE_FS_INVALID_PATH ((CFE_Status_t)0xc6000002)
Invalid Path.
- #define CFE_FS_FNAME_TOO_LONG ((CFE_Status_t)0xc6000003)
Filename Too Long.
- #define CFE_FS_NOT_IMPLEMENTED ((CFE_Status_t)0xc600ffff)
Not Implemented.
- #define CFE_SB_TIME_OUT ((CFE_Status_t)0xca000001)
Time Out.
- #define CFE_SB_NO_MESSAGE ((CFE_Status_t)0xca000002)
No Message.
- #define CFE_SB_BAD_ARGUMENT ((CFE_Status_t)0xca000003)
Bad Argument.
- #define CFE_SB_MAX_PIPES_MET ((CFE_Status_t)0xca000004)
Max Pipes Met.
- #define CFE_SB_PIPE_CR_ERR ((CFE_Status_t)0xca000005)
Pipe Create Error.
- #define CFE_SB_PIPE_RD_ERR ((CFE_Status_t)0xca000006)
Pipe Read Error.
- #define CFE_SB_MSG_TOO_BIG ((CFE_Status_t)0xca000007)
Message Too Big.
- #define CFE_SB_BUF_ALOC_ERR ((CFE_Status_t)0xca000008)
Buffer Allocation Error.
- #define CFE_SB_MAX_MSGS_MET ((CFE_Status_t)0xca000009)
Max Messages Met.
- #define CFE_SB_MAX_DESTS_MET ((CFE_Status_t)0xca00000a)
Max Destinations Met.
- #define CFE_SB_INTERNAL_ERR ((CFE_Status_t)0xca00000c)
Internal Error.
- #define CFE_SB_WRONG_MSG_TYPE ((CFE_Status_t)0xca00000d)
Wrong Message Type.
- #define CFE_SB_BUFFER_INVALID ((CFE_Status_t)0xca00000e)
Buffer Invalid.
- #define CFE_SB_NOT_IMPLEMENTED ((CFE_Status_t)0xca00ffff)
Not Implemented.
- #define CFE_TBL_ERR_INVALID_HANDLE ((CFE_Status_t)0xcc000001)
Invalid Handle.
- #define CFE_TBL_ERR_INVALID_NAME ((CFE_Status_t)0xcc000002)
Invalid Name.
- #define CFE_TBL_ERR_INVALID_SIZE ((CFE_Status_t)0xcc000003)
Invalid Size.
- #define CFE_TBL_INFO_UPDATE_PENDING ((CFE_Status_t)0x4c000004)
Update Pending.
- #define CFE_TBL_ERR_NEVER_LOADED ((CFE_Status_t)0xcc000005)
Never Loaded.
- #define CFE_TBL_ERR_REGISTRY_FULL ((CFE_Status_t)0xcc000006)
Registry Full.
- #define CFE_TBL_WARN_DUPLICATE ((CFE_Status_t)0x4c000007)

- Duplicate Warning.
 - #define CFE_TBL_ERR_NO_ACCESS ((CFE_Status_t)0xcc000008)
- No Access.
 - #define CFE_TBL_ERR_UNREGISTERED ((CFE_Status_t)0xcc000009)
- Unregistered.
 - #define CFE_TBL_ERR_HANDLES_FULL ((CFE_Status_t)0xcc00000B)
- Handles Full.
 - #define CFE_TBL_ERR_DUPLICATE_DIFF_SIZE ((CFE_Status_t)0xcc00000C)
- Duplicate Table With Different Size.
 - #define CFE_TBL_ERR_DUPLICATE_NOT_OWNED ((CFE_Status_t)0xcc00000D)
- Duplicate Table And Not Owned.
 - #define CFE_TBL_INFO_UPDATED ((CFE_Status_t)0x4c00000E)
- Updated.
 - #define CFE_TBL_ERR_NO_BUFFER_AVAIL ((CFE_Status_t)0xcc00000F)
- No Buffer Available.
 - #define CFE_TBL_ERR_DUMP_ONLY ((CFE_Status_t)0xcc000010)
- Dump Only Error.
 - #define CFE_TBL_ERR_ILLEGAL_SRC_TYPE ((CFE_Status_t)0xcc000011)
- Illegal Source Type.
 - #define CFE_TBL_ERR_LOAD_IN_PROGRESS ((CFE_Status_t)0xcc000012)
- Load In Progress.
 - #define CFE_TBL_ERR_FILE_TOO_LARGE ((CFE_Status_t)0xcc000014)
- File Too Large.
 - #define CFE_TBL_WARN_SHORT_FILE ((CFE_Status_t)0x4c000015)
- Short File Warning.
 - #define CFE_TBL_ERR_BAD_CONTENT_ID ((CFE_Status_t)0xcc000016)
- Bad Content ID.
 - #define CFE_TBL_INFO_NO_UPDATE_PENDING ((CFE_Status_t)0x4c000017)
- No Update Pending.
 - #define CFE_TBL_INFO_TABLE_LOCKED ((CFE_Status_t)0x4c000018)
- Table Locked.
 - #define CFE_TBL_INFO_VALIDATION_PENDING ((CFE_Status_t)0x4c000019)
- #define CFE_TBL_INFO_NO_VALIDATION_PENDING ((CFE_Status_t)0x4c00001A)
- #define CFE_TBL_ERR_BAD_SUBTYPE_ID ((CFE_Status_t)0xcc00001B)
- Bad Subtype ID.
 - #define CFE_TBL_ERR_FILE_SIZE_INCONSISTENT ((CFE_Status_t)0xcc00001C)
- File Size Inconsistent.
 - #define CFE_TBL_ERR_NO_STD_HEADER ((CFE_Status_t)0xcc00001D)
- No Standard Header.
 - #define CFE_TBL_ERR_NO_TBL_HEADER ((CFE_Status_t)0xcc00001E)
- No Table Header.
 - #define CFE_TBL_ERR_FILENAME_TOO_LONG ((CFE_Status_t)0xcc00001F)
- Filename Too Long.
 - #define CFE_TBL_ERR_FILE_FOR_WRONG_TABLE ((CFE_Status_t)0xcc000020)
- File For Wrong Table.
 - #define CFE_TBL_ERR_LOAD_INCOMPLETE ((CFE_Status_t)0xcc000021)
- Load Incomplete.

- #define `CFE_TBL_WARN_PARTIAL_LOAD` ((`CFE_Status_t`)0x4c000022)
Partial Load Warning.
- #define `CFE_TBL_ERR_PARTIAL_LOAD` ((`CFE_Status_t`)0xcc000023)
Partial Load Error.
- #define `CFE_TBL_INFO_DUMP_PENDING` ((`CFE_Status_t`)0x4c000024)
Dump Pending.
- #define `CFE_TBL_ERR_INVALID_OPTIONS` ((`CFE_Status_t`)0xcc000025)
Invalid Options.
- #define `CFE_TBL_WARN_NOT_CRITICAL` ((`CFE_Status_t`)0x4c000026)
Not Critical Warning.
- #define `CFE_TBL_INFO_RECOVERED_TBL` ((`CFE_Status_t`)0x4c000027)
Recovered Table.
- #define `CFE_TBL_ERR_BAD_SPACECRAFT_ID` ((`CFE_Status_t`)0xcc000028)
Bad Spacecraft ID.
- #define `CFE_TBL_ERR_BAD_PROCESSOR_ID` ((`CFE_Status_t`)0xcc000029)
Bad Processor ID.
- #define `CFE_TBL_MESSAGE_ERROR` ((`CFE_Status_t`)0xcc00002a)
Message Error.
- #define `CFE_TBL_ERR_SHORT_FILE` ((`CFE_Status_t`)0xcc00002b)
- #define `CFE_TBL_ERR_ACCESS` ((`CFE_Status_t`)0xcc00002c)
- #define `CFE_TBL_BAD_ARGUMENT` ((`CFE_Status_t`)0xcc00002d)
Bad Argument.
- #define `CFE_TBL_INFO_NO_DUMP_PENDING` ((`CFE_Status_t`)0x4c00002e)
No Dump Pending.
- #define `CFE_TBL_NOT_IMPLEMENTED` ((`CFE_Status_t`)0xcc00ffff)
Not Implemented.
- #define `CFE_TIME_NOT_IMPLEMENTED` ((`CFE_Status_t`)0xce00ffff)
Not Implemented.
- #define `CFE_TIME_INTERNAL_ONLY` ((`CFE_Status_t`)0xce000001)
Internal Only.
- #define `CFE_TIME_OUT_OF_RANGE` ((`CFE_Status_t`)0xce000002)
Out Of Range.
- #define `CFE_TIME_TOO_MANY_SYNC_CALLBACKS` ((`CFE_Status_t`)0xce000003)
Too Many Sync Callbacks.
- #define `CFE_TIME_CALLBACK_NOT_REGISTERED` ((`CFE_Status_t`)0xce000004)
Callback Not Registered.
- #define `CFE_TIME_BAD_ARGUMENT` ((`CFE_Status_t`)0xce000005)
Bad Argument.

Typedefs

- typedef `int32 CFE_Status_t`
cFE Status type for readability and eventually type safety
- typedef `char CFE_StatusString_t[CFE_STATUS_STRING_LENGTH]`
For the `CFE_ES_StatusToString()` function, to ensure everyone is making an array of the same length.

Functions

- char * [CFE_ES_StatusToString](#) (CFE_Status_t status, CFE_StatusString_t *status_string)
Convert status to a string.

11.59.1 Detailed Description

Title: cFE Status Code Definition Header File

Purpose: Common source of cFE API return status codes.

Design Notes:

References: Flight Software Branch C Coding Standard Version 1.0a

Definition in file [cfe_error.h](#).

11.59.2 Macro Definition Documentation

CFE_EVENTS_SERVICE

```
#define CFE_EVENTS_SERVICE ((CFE_Status_t)0x02000000)
```

Event Service.

Definition at line 126 of file [cfe_error.h](#).

CFE_EXECUTIVE_SERVICE

```
#define CFE_EXECUTIVE_SERVICE ((CFE_Status_t)0x04000000)
```

Executive Service.

Definition at line 127 of file [cfe_error.h](#).

CFE_FILE_SERVICE

```
#define CFE_FILE_SERVICE ((CFE_Status_t)0x06000000)
```

File Service.

Definition at line 128 of file [cfe_error.h](#).

CFE_GENERIC_SERVICE

```
#define CFE_GENERIC_SERVICE ((CFE_Status_t)0x08000000)
```

Generic Service.

Definition at line 129 of file [cfe_error.h](#).

CFE_SERVICE_BITMASK

```
#define CFE_SERVICE_BITMASK ((CFE_Status_t)0x0e000000)
```

Error Service Bitmask.

Definition at line 124 of file [cfe_error.h](#).

CFE_SEVERITY_BITMASK

```
#define CFE_SEVERITY_BITMASK ((CFE_Status_t)0xc0000000)
```

Error Severity Bitmask.

Definition at line 115 of file [cfe_error.h](#).

CFE_SEVERITY_ERROR

```
#define CFE_SEVERITY_ERROR ((CFE_Status_t)0xc0000000)
```

Severity Error.

Definition at line 119 of file [cfe_error.h](#).

CFE_SEVERITY_INFO

```
#define CFE_SEVERITY_INFO ((CFE_Status_t)0x40000000)
```

Severity Info.
Definition at line 118 of file [cfe_error.h](#).

CFE_SEVERITY_SUCCESS

```
#define CFE_SEVERITY_SUCCESS ((CFE_Status_t)0x00000000)
```

Severity Success.
Definition at line 117 of file [cfe_error.h](#).

CFE_SOFTWARE_BUS_SERVICE

```
#define CFE_SOFTWARE_BUS_SERVICE ((CFE_Status_t)0x0a000000)
```

Software Bus Service.
Definition at line 130 of file [cfe_error.h](#).

CFE_STATUS_C

```
#define CFE_STATUS_C(  
    X ) ((CFE_Status_t)(X))
```

cFE Status macro for literal
Definition at line 48 of file [cfe_error.h](#).

CFE_STATUS_STRING_LENGTH

```
#define CFE_STATUS_STRING_LENGTH 11
```

cFE Status converted to string length limit
Used for sizing CFE_StatusString_t intended for use in printing CFE_Status_t values Sized for 0x%08x and NULL
Definition at line 56 of file [cfe_error.h](#).

CFE_TABLE_SERVICE

```
#define CFE_TABLE_SERVICE ((CFE_Status_t)0x0c000000)
```

Table Service.
Definition at line 131 of file [cfe_error.h](#).

CFE_TIME_SERVICE

```
#define CFE_TIME_SERVICE ((CFE_Status_t)0x0e000000)
```

Time Service.
Definition at line 132 of file [cfe_error.h](#).

11.59.3 Typedef Documentation

CFE_Status_t

```
typedef int32 CFE_Status_t
```

cFE Status type for readability and eventually type safety
Definition at line 43 of file [cfe_error.h](#).

CFE_StatusString_t

```
typedef char CFE_StatusString_t[CFE_STATUS_STRING_LENGTH]
```

For the [CFE_ES_StatusToString\(\)](#) function, to ensure everyone is making an array of the same length.
Definition at line 62 of file [cfe_error.h](#).

11.59.4 Function Documentation

CFE_ES_StatusToString()

```
char * CFE_ES_StatusToString (
    CFE_Status_t status,
    CFE_StatusString_t * status_string )
```

Convert status to a string.

Parameters

in	<i>status</i>	Status value to convert
out	<i>status_string</i>	Buffer to store status converted to string

Returns

Passed in string pointer

11.60 cfe_error.h

[Go to the documentation of this file.](#)

```
00001 /*****
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"
00003  *
00004  * Copyright (c) 2023 United States Government as represented by the
00005  * Administrator of the National Aeronautics and Space Administration.
00006  * All Rights Reserved.
00007  *
00008  * Licensed under the Apache License, Version 2.0 (the "License"); you may
00009  * not use this file except in compliance with the License. You may obtain
00010  * a copy of the License at http://www.apache.org/licenses/LICENSE-2.0
00011  *
00012  * Unless required by applicable law or agreed to in writing, software
00013  * distributed under the License is distributed on an "AS IS" BASIS,
00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00034 #ifndef CFE_ERROR_H
00035 #define CFE_ERROR_H
00036
00037 /* Include Files */
00038 #include "osapi.h"
00039
00043 typedef int32 CFE_Status_t;
00044
00048 #define CFE_STATUS_C(X) ((CFE_Status_t)(X))
00049
00056 #define CFE_STATUS_STRING_LENGTH 11
00057
00062 typedef char CFE_StatusString_t[CFE_STATUS_STRING_LENGTH];
00063
00072 char *CFE_ES_StatusToString(CFE_Status_t status, CFE_StatusString_t *status_string);
00073
00074 /*
00075 ** Status Codes are 32 bit values formatted as follows:
00076 **
00077 **   3 3 2 2 2 2 2 2 2 2 2 1 1 1 1 1 1 1 1
00078 **   1 0 9 8 7 6 5 4 3 2 1 0 9 8 7 6 5 4 3 2 1 0
00079 **   +---+---+---+---+---+---+---+---+---+---+
00080 **   |Sev| R | Srv | Mission Defined |           Code           |
00081 **   +---+---+---+---+---+---+---+---+---+---+
00082 **
00083 **   where
00084 **
00085 **       Sev - is the severity code
00086 **
00087 **       00 - Success
00088 **       01 - Informational
00089 **       11 - Error
00090 **
```

```

00091 **      R - are reserved bits
00092 **
00093 **      Srv - is the cFE Service Identifier
00094 **
00095 **          000 - Not a cFE Service
00096 **          001 - Events Services
00097 **          010 - Executive Services
00098 **          011 - File Services
00099 **          100 - Generic code for all services
00100 **          101 - Software Bus Services
00101 **          110 - Tables Services
00102 **          111 - Time Services
00103 **
00104 **      Mission Defined - These bits are available for Mission
00105 **                        specific coding standards. They can
00106 **                        be used to classify error codes related
00107 **                        to mission specific library function calls, etc.
00108 **
00109 **      Code - is the status code
00110 */
00111
00112 /*
00113 ** Error Severity
00114 */
00115 #define CFE_SEVERITY_BITMASK ((CFE_Status_t)0xc0000000)
00117 #define CFE_SEVERITY_SUCCESS ((CFE_Status_t)0x00000000)
00118 #define CFE_SEVERITY_INFO    ((CFE_Status_t)0x40000000)
00119 #define CFE_SEVERITY_ERROR   ((CFE_Status_t)0xc0000000)
00121 /*
00122 ** cFE Service Identifiers
00123 */
00124 #define CFE_SERVICE_BITMASK ((CFE_Status_t)0x0e000000)
00126 #define CFE_EVENTS_SERVICE   ((CFE_Status_t)0x02000000)
00127 #define CFE_EXECUTIVE_SERVICE ((CFE_Status_t)0x04000000)
00128 #define CFE_FILE_SERVICE     ((CFE_Status_t)0x06000000)
00129 #define CFE_GENERIC_SERVICE  ((CFE_Status_t)0x08000000)
00130 #define CFE_SOFTWARE_BUS_SERVICE ((CFE_Status_t)0x0a000000)
00131 #define CFE_TABLE_SERVICE    ((CFE_Status_t)0x0c000000)
00132 #define CFE_TIME_SERVICE     ((CFE_Status_t)0x0e000000)
00134 /*
00135 ***** COMMON STATUS CODES *****
00136 */
00137
00147 #define CFE_SUCCESS ((CFE_Status_t)0)
00148
00155 #define CFE_STATUS_NO_COUNTER_INCREMENT ((CFE_Status_t)0x48000001)
00156
00164 #define CFE_STATUS_WRONG_MSG_LENGTH ((CFE_Status_t)0xc8000002)
00165
00173 #define CFE_STATUS_UNKNOWN_MSG_ID ((CFE_Status_t)0xc8000003)
00174
00182 #define CFE_STATUS_BAD_COMMAND_CODE ((CFE_Status_t)0xc8000004)
00183
00194 #define CFE_STATUS_EXTERNAL_RESOURCE_FAIL ((CFE_Status_t)0xc8000005)
00195
00203 #define CFE_STATUS_REQUEST_ALREADY_PENDING ((int32)0xc8000006)
00204
00211 #define CFE_STATUS_VALIDATION_FAILURE ((int32)0xc8000007)
00212
00219 #define CFE_STATUS_RANGE_ERROR ((int32)0xc8000008)
00220
00227 #define CFE_STATUS_INCORRECT_STATE ((int32)0xc8000009)
00228
00238 #define CFE_STATUS_NOT_IMPLEMENTED ((CFE_Status_t)0xc800ffff)
00239
00240 /*
00241 ***** EVENTS SERVICES STATUS CODES *****
00242 */
00243
00250 #define CFE_EVS_UNKNOWN_FILTER ((CFE_Status_t)0xc2000001)
00251
00258 #define CFE_EVS_APP_NOT_REGISTERED ((CFE_Status_t)0xc2000002)
00259
00267 #define CFE_EVS_APP_ILLEGAL_APP_ID ((CFE_Status_t)0xc2000003)
00268
00276 #define CFE_EVS_APP_FILTER_OVERLOAD ((CFE_Status_t)0xc2000004)
00277
00285 #define CFE_EVS_RESET_AREA_POINTER ((CFE_Status_t)0xc2000005)
00286
00294 #define CFE_EVS_EVT_NOT_REGISTERED ((CFE_Status_t)0xc2000006)
00295

```

```
00302 #define CFE_EVS_FILE_WRITE_ERROR ((CFE_Status_t)0xc2000007)
00303
00310 #define CFE_EVS_INVALID_PARAMETER ((CFE_Status_t)0xc2000008)
00311
00318 #define CFE_EVS_APP_SQUELCHED ((CFE_Status_t)0xc2000009)
00319
00329 #define CFE_EVS_NOT_IMPLEMENTED ((CFE_Status_t)0xc200ffff)
00330
00331 /*
00332 ***** EXECUTIVE SERVICES STATUS CODES *****
00333 */
00334
00342 #define CFE_ES_ERR_RESOURCEID_NOT_VALID ((CFE_Status_t)0xc4000001)
00343
00350 #define CFE_ES_ERR_NAME_NOT_FOUND ((CFE_Status_t)0xc4000002)
00351
00358 #define CFE_ES_ERR_APP_CREATE ((CFE_Status_t)0xc4000004)
00359
00366 #define CFE_ES_ERR_CHILD_TASK_CREATE ((CFE_Status_t)0xc4000005)
00367
00375 #define CFE_ES_ERR_SYS_LOG_FULL ((CFE_Status_t)0xc4000006)
00376
00383 #define CFE_ES_ERR_MEM_BLOCK_SIZE ((CFE_Status_t)0xc4000008)
00384
00391 #define CFE_ES_ERR_LOAD_LIB ((CFE_Status_t)0xc4000009)
00392
00399 #define CFE_ES_BAD_ARGUMENT ((CFE_Status_t)0xc400000a)
00400
00407 #define CFE_ES_ERR_CHILD_TASK_REGISTER ((CFE_Status_t)0xc400000b)
00408
00415 #define CFE_ES_CDS_ALREADY_EXISTS ((CFE_Status_t)0x4400000d)
00416
00424 #define CFE_ES_CDS_INSUFFICIENT_MEMORY ((CFE_Status_t)0xc400000e)
00425
00433 #define CFE_ES_CDS_INVALID_NAME ((CFE_Status_t)0xc400000f)
00434
00442 #define CFE_ES_CDS_INVALID_SIZE ((CFE_Status_t)0xc4000010)
00443
00450 #define CFE_ES_CDS_INVALID ((CFE_Status_t)0xc4000012)
00451
00458 #define CFE_ES_CDS_ACCESS_ERROR ((CFE_Status_t)0xc4000013)
00459
00466 #define CFE_ES_FILE_IO_ERR ((CFE_Status_t)0xc4000014)
00467
00474 #define CFE_ES_RST_ACCESS_ERR ((CFE_Status_t)0xc4000015)
00475
00482 #define CFE_ES_ERR_APP_REGISTER ((CFE_Status_t)0xc4000017)
00483
00490 #define CFE_ES_ERR_CHILD_TASK_DELETE ((CFE_Status_t)0xc4000018)
00491
00499 #define CFE_ES_ERR_CHILD_TASK_DELETE_MAIN_TASK ((CFE_Status_t)0xc4000019)
00500
00509 #define CFE_ES_CDS_BLOCK_CRC_ERR ((CFE_Status_t)0xc400001a)
00510
00518 #define CFE_ES_MUT_SEM_DELETE_ERR ((CFE_Status_t)0xc400001b)
00519
00527 #define CFE_ES_BIN_SEM_DELETE_ERR ((CFE_Status_t)0xc400001c)
00528
00536 #define CFE_ES_COUNT_SEM_DELETE_ERR ((CFE_Status_t)0xc400001d)
00537
00545 #define CFE_ES_QUEUE_DELETE_ERR ((CFE_Status_t)0xc400001e)
00546
00554 #define CFE_ES_FILE_CLOSE_ERR ((CFE_Status_t)0xc400001f)
00555
00564 #define CFE_ES_CDS_WRONG_TYPE_ERR ((CFE_Status_t)0xc4000020)
00565
00575 #define CFE_ES_CDS_OWNER_ACTIVE_ERR ((CFE_Status_t)0xc4000022)
00576
00588 #define CFE_ES_APP_CLEANUP_ERR ((CFE_Status_t)0xc4000023)
00589
00597 #define CFE_ES_TIMER_DELETE_ERR ((CFE_Status_t)0xc4000024)
00598
00605 #define CFE_ES_BUFFER_NOT_IN_POOL ((CFE_Status_t)0xc4000025)
00606
00614 #define CFE_ES_TASK_DELETE_ERR ((CFE_Status_t)0xc4000026)
00615
00622 #define CFE_ES_OPERATION_TIMED_OUT ((CFE_Status_t)0xc4000027)
00623
00631 #define CFE_ES_LIB_ALREADY_LOADED ((CFE_Status_t)0x44000028)
00632
00640 #define CFE_ES_ERR_SYS_LOG_TRUNCATED ((CFE_Status_t)0x44000029)
```

```
00641
00650 #define CFE_ES_NO_RESOURCE_IDS_AVAILABLE ((CFE_Status_t)0xc400002B)
00651
00660 #define CFE_ES_POOL_BLOCK_INVALID ((CFE_Status_t)0xc400002C)
00661
00668 #define CFE_ES_ERR_DUPLICATE_NAME ((CFE_Status_t)0xc400002E)
00669
00679 #define CFE_ES_NOT_IMPLEMENTED ((CFE_Status_t)0xc400ffff)
00680
00681 /*
00682 ***** FILE SERVICES STATUS CODES *****
00683 */
00684
00692 #define CFE_FS_BAD_ARGUMENT ((CFE_Status_t)0xc6000001)
00693
00700 #define CFE_FS_INVALID_PATH ((CFE_Status_t)0xc6000002)
00701
00708 #define CFE_FS_FNAME_TOO_LONG ((CFE_Status_t)0xc6000003)
00709
00719 #define CFE_FS_NOT_IMPLEMENTED ((CFE_Status_t)0xc600ffff)
00720
00721 /*
00722 ***** SOFTWARE BUS SERVICES STATUS CODES *****
00723 */
00724
00732 #define CFE_SB_TIME_OUT ((CFE_Status_t)0xca000001)
00733
00741 #define CFE_SB_NO_MESSAGE ((CFE_Status_t)0xca000002)
00742
00750 #define CFE_SB_BAD_ARGUMENT ((CFE_Status_t)0xca000003)
00751
00761 #define CFE_SB_MAX_PIPES_MET ((CFE_Status_t)0xca000004)
00762
00772 #define CFE_SB_PIPE_CR_ERR ((CFE_Status_t)0xca000005)
00773
00785 #define CFE_SB_PIPE_RD_ERR ((CFE_Status_t)0xca000006)
00786
00795 #define CFE_SB_MSG_TOO_BIG ((CFE_Status_t)0xca000007)
00796
00808 #define CFE_SB_BUF_ALOC_ERR ((CFE_Status_t)0xca000008)
00809
00818 #define CFE_SB_MAX_MSGS_MET ((CFE_Status_t)0xca000009)
00819
00830 #define CFE_SB_MAX_DESTS_MET ((CFE_Status_t)0xca00000a)
00831
00840 #define CFE_SB_INTERNAL_ERR ((CFE_Status_t)0xca00000c)
00841
00849 #define CFE_SB_WRONG_MSG_TYPE ((CFE_Status_t)0xca00000d)
00850
00859 #define CFE_SB_BUFFER_INVALID ((CFE_Status_t)0xca00000e)
00860
00870 #define CFE_SB_NOT_IMPLEMENTED ((CFE_Status_t)0xca00ffff)
00871
00872 /*
00873 ***** TABLE SERVICES STATUS CODES *****
00874 */
00875
00884 #define CFE_TBL_ERR_INVALID_HANDLE ((CFE_Status_t)0xcc000001)
00885
00894 #define CFE_TBL_ERR_INVALID_NAME ((CFE_Status_t)0xcc000002)
00895
00905 #define CFE_TBL_ERR_INVALID_SIZE ((CFE_Status_t)0xcc000003)
00906
00913 #define CFE_TBL_INFO_UPDATE_PENDING ((CFE_Status_t)0x4c000004)
00914
00921 #define CFE_TBL_ERR_NEVER_LOADED ((CFE_Status_t)0xcc000005)
00922
00930 #define CFE_TBL_ERR_REGISTRY_FULL ((CFE_Status_t)0xcc000006)
00931
00940 #define CFE_TBL_WARN_DUPLICATE ((CFE_Status_t)0x4c000007)
00941
00949 #define CFE_TBL_ERR_NO_ACCESS ((CFE_Status_t)0xcc000008)
00950
00958 #define CFE_TBL_ERR_UNREGISTERED ((CFE_Status_t)0xcc000009)
00959
00967 #define CFE_TBL_ERR_HANDLES_FULL ((CFE_Status_t)0xcc00000B)
00968
00977 #define CFE_TBL_ERR_DUPLICATE_DIFF_SIZE ((CFE_Status_t)0xcc00000C)
00978
00987 #define CFE_TBL_ERR_DUPLICATE_NOT_OWNED ((CFE_Status_t)0xcc00000D)
00988
```

```
00998 #define CFE_TBL_INFO_UPDATED ((CFE_Status_t)0x4c00000E)
00999
01007 #define CFE_TBL_ERR_NO_BUFFER_AVAIL ((CFE_Status_t)0xcc00000F)
01008
01016 #define CFE_TBL_ERR_DUMP_ONLY ((CFE_Status_t)0xcc000010)
01017
01025 #define CFE_TBL_ERR_ILLEGAL_SRC_TYPE ((CFE_Status_t)0xcc000011)
01026
01034 #define CFE_TBL_ERR_LOAD_IN_PROGRESS ((CFE_Status_t)0xcc000012)
01035
01044 #define CFE_TBL_ERR_FILE_TOO_LARGE ((CFE_Status_t)0xcc000014)
01045
01055 #define CFE_TBL_WARN_SHORT_FILE ((CFE_Status_t)0x4c000015)
01056
01064 #define CFE_TBL_ERR_BAD_CONTENT_ID ((CFE_Status_t)0xcc000016)
01065
01072 #define CFE_TBL_INFO_NO_UPDATE_PENDING ((CFE_Status_t)0x4c000017)
01073
01080 #define CFE_TBL_INFO_TABLE_LOCKED ((CFE_Status_t)0x4c000018)
01081
01088 #define CFE_TBL_INFO_VALIDATION_PENDING ((CFE_Status_t)0x4c000019)
01089
01096 #define CFE_TBL_INFO_NO_VALIDATION_PENDING ((CFE_Status_t)0x4c00001A)
01097
01105 #define CFE_TBL_ERR_BAD_SUBTYPE_ID ((CFE_Status_t)0xcc00001B)
01106
01114 #define CFE_TBL_ERR_FILE_SIZE_INCONSISTENT ((CFE_Status_t)0xcc00001C)
01115
01122 #define CFE_TBL_ERR_NO_STD_HEADER ((CFE_Status_t)0xcc00001D)
01123
01131 #define CFE_TBL_ERR_NO_TBL_HEADER ((CFE_Status_t)0xcc00001E)
01132
01140 #define CFE_TBL_ERR_FILENAME_TOO_LONG ((CFE_Status_t)0xcc00001F)
01141
01149 #define CFE_TBL_ERR_FILE_FOR_WRONG_TABLE ((CFE_Status_t)0xcc000020)
01150
01158 #define CFE_TBL_ERR_LOAD_INCOMPLETE ((CFE_Status_t)0xcc000021)
01159
01168 #define CFE_TBL_WARN_PARTIAL_LOAD ((CFE_Status_t)0x4c000022)
01169
01180 #define CFE_TBL_ERR_PARTIAL_LOAD ((CFE_Status_t)0xcc000023)
01181
01190 #define CFE_TBL_INFO_DUMP_PENDING ((CFE_Status_t)0x4c000024)
01191
01206 #define CFE_TBL_ERR_INVALID_OPTIONS ((CFE_Status_t)0xcc000025)
01207
01217 #define CFE_TBL_WARN_NOT_CRITICAL ((CFE_Status_t)0x4c000026)
01218
01230 #define CFE_TBL_INFO_RECOVERED_TBL ((CFE_Status_t)0x4c000027)
01231
01241 #define CFE_TBL_ERR_BAD_SPACECRAFT_ID ((CFE_Status_t)0xcc000028)
01242
01252 #define CFE_TBL_ERR_BAD_PROCESSOR_ID ((CFE_Status_t)0xcc000029)
01253
01260 #define CFE_TBL_MESSAGE_ERROR ((CFE_Status_t)0xcc00002a)
01261
01266 #define CFE_TBL_ERR_SHORT_FILE ((CFE_Status_t)0xcc00002b)
01267
01272 #define CFE_TBL_ERR_ACCESS ((CFE_Status_t)0xcc00002c)
01273
01281 #define CFE_TBL_BAD_ARGUMENT ((CFE_Status_t)0xcc00002d)
01282
01290 #define CFE_TBL_INFO_NO_DUMP_PENDING ((CFE_Status_t)0x4c00002e)
01291
01301 #define CFE_TBL_NOT_IMPLEMENTED ((CFE_Status_t)0xcc00ffff)
01302
01303 /*
01304 ***** TIME SERVICES STATUS CODES *****
01305 */
01306
01316 #define CFE_TIME_NOT_IMPLEMENTED ((CFE_Status_t)0xce00ffff)
01317
01328 #define CFE_TIME_INTERNAL_ONLY ((CFE_Status_t)0xce000001)
01329
01343 #define CFE_TIME_OUT_OF_RANGE ((CFE_Status_t)0xce000002)
01344
01354 #define CFE_TIME_TOO_MANY_SYNCH_CALLBACKS ((CFE_Status_t)0xce000003)
01355
01364 #define CFE_TIME_CALLBACK_NOT_REGISTERED ((CFE_Status_t)0xce000004)
01365
01373 #define CFE_TIME_BAD_ARGUMENT ((CFE_Status_t)0xce000005)
```

```
01376 #endif /* CFE_ERROR_H */
```

11.61 cfe/modules/core_api/fsw/inc/cfe_es.h File Reference

```
#include "common_types.h"
#include "cfe_error.h"
#include "cfe_es_api_typedefs.h"
#include "cfe_resourceid_api_typedefs.h"
```

Macros

- `#define OS_PRINTF(m, n)`
- `#define CFE_ES_PerfLogEntry(id) (CFE_ES_PerfLogAdd(id, 0))`
Entry marker for use with Software Performance Analysis Tool.
- `#define CFE_ES_PerfLogExit(id) (CFE_ES_PerfLogAdd(id, 1))`
Exit marker for use with Software Performance Analysis Tool.

Functions

- `CFE_Status_t CFE_ES_AppID_ToIndex (CFE_ES_AppId_t AppID, uint32 *Idx)`
Obtain an index value correlating to an ES Application ID.
- `int32 CFE_ES_LibID_ToIndex (CFE_ES_LibId_t LibId, uint32 *Idx)`
Obtain an index value correlating to an ES Library ID.
- `CFE_Status_t CFE_ES_TaskID_ToIndex (CFE_ES_TaskId_t TaskID, uint32 *Idx)`
Obtain an index value correlating to an ES Task ID.
- `CFE_Status_t CFE_ES_CounterID_ToIndex (CFE_ES_CounterId_t CounterId, uint32 *Idx)`
Obtain an index value correlating to an ES Counter ID.
- `void CFE_ES_Main (uint32 StartType, uint32 StartSubtype, uint32 ModelId, const char *StartFilePath)`
cFE Main Entry Point used by Board Support Package to start cFE
- `CFE_Status_t CFE_ES_ResetCFE (uint32 ResetType)`
Reset the cFE Core and all cFE Applications.
- `CFE_Status_t CFE_ES_RestartApp (CFE_ES_AppId_t AppID)`
Restart a single cFE Application.
- `CFE_Status_t CFE_ES_ReloadApp (CFE_ES_AppId_t AppID, const char *AppFileName)`
Reload a single cFE Application.
- `CFE_Status_t CFE_ES_DeleteApp (CFE_ES_AppId_t AppID)`
Delete a cFE Application.
- `void CFE_ES_ExitApp (uint32 ExitStatus)`
Exit a cFE Application.
- `bool CFE_ES_RunLoop (uint32 *RunStatus)`
Check for Exit, Restart, or Reload commands.
- `CFE_Status_t CFE_ES_WaitForSystemState (uint32 MinSystemState, uint32 TimeOutMilliseconds)`
Allow an Application to Wait for a minimum global system state.
- `void CFE_ES_WaitForStartupSync (uint32 TimeOutMilliseconds)`
Allow an Application to Wait for the "OPERATIONAL" global system state.
- `void CFE_ES_IncrementTaskCounter (void)`
Increments the execution counter for the calling task.
- `int32 CFE_ES_GetResetType (uint32 *ResetSubtypePtr)`

- Return the most recent Reset Type.*

 - [CFE_Status_t CFE_ES_GetAppID](#) ([CFE_ES_AppId_t](#) *AppIdPtr)

Get an Application ID for the calling Application.
- [CFE_Status_t CFE_ES_GetTaskID](#) ([CFE_ES_TaskId_t](#) *TaskIdPtr)

Get the task ID of the calling context.
- [CFE_Status_t CFE_ES_GetAppIDByName](#) ([CFE_ES_AppId_t](#) *AppIdPtr, const char *AppName)

Get an Application ID associated with a specified Application name.
- [CFE_Status_t CFE_ES_GetLibIDByName](#) ([CFE_ES_LibId_t](#) *LibIdPtr, const char *LibName)

Get a Library ID associated with a specified Library name.
- [CFE_Status_t CFE_ES_GetAppName](#) (char *AppName, [CFE_ES_AppId_t](#) AppId, size_t BufferLength)

Get an Application name for a specified Application ID.
- [CFE_Status_t CFE_ES_GetLibName](#) (char *LibName, [CFE_ES_LibId_t](#) LibId, size_t BufferLength)

Get a Library name for a specified Library ID.
- [CFE_Status_t CFE_ES_GetAppInfo](#) ([CFE_ES_AppInfo_t](#) *AppInfo, [CFE_ES_AppId_t](#) AppId)

Get Application Information given a specified App ID.
- [CFE_Status_t CFE_ES_GetTaskInfo](#) ([CFE_ES_TaskInfo_t](#) *TaskInfo, [CFE_ES_TaskId_t](#) TaskId)

Get Task Information given a specified Task ID.
- [int32 CFE_ES_GetLibInfo](#) ([CFE_ES_AppInfo_t](#) *LibInfo, [CFE_ES_LibId_t](#) LibId)

Get Library Information given a specified Resource ID.
- [int32 CFE_ES_GetModuleInfo](#) ([CFE_ES_AppInfo_t](#) *ModuleInfo, [CFE_ResourceId_t](#) ResourceId)

Get Information given a specified Resource ID.
- [CFE_Status_t CFE_ES_CreateChildTask](#) ([CFE_ES_TaskId_t](#) *TaskIdPtr, const char *TaskName, [CFE_ES_ChildTaskMainFuncPtr](#) FunctionPtr, [CFE_ES_StackPointer_t](#) StackPtr, size_t StackSize, [CFE_ES_TaskPriority_Atom_t](#) Priority, uint32 Flags)

Creates a new task under an existing Application.
- [CFE_Status_t CFE_ES_GetTaskIDByName](#) ([CFE_ES_TaskId_t](#) *TaskIdPtr, const char *TaskName)

Get a Task ID associated with a specified Task name.
- [CFE_Status_t CFE_ES_GetTaskName](#) (char *TaskName, [CFE_ES_TaskId_t](#) TaskId, size_t BufferLength)

Get a Task name for a specified Task ID.
- [CFE_Status_t CFE_ES_DeleteChildTask](#) ([CFE_ES_TaskId_t](#) TaskId)

Deletes a task under an existing Application.
- void [CFE_ES_ExitChildTask](#) (void)

Exits a child task.
- void [CFE_ES_BackgroundWakeup](#) (void)

Wakes up the CFE background task.
- [CFE_Status_t CFE_ES_WriteToSysLog](#) (const char *SpecStringPtr,...) [OS_PRINTF](#)(1)

Write a string to the cFE System Log.
- [CFE_Status_t uint32 CFE_ES_CalculateCRC](#) (const void *DataPtr, size_t DataLength, [uint32](#) InputCRC, [CFE_ES_CrcType_Enum_t](#) TypeCRC)

Calculate a CRC on a block of memory.
- void [CFE_ES_ProcessAsyncEvent](#) (void)

Notification that an asynchronous event was detected by the underlying OS/PSP.
- [CFE_Status_t CFE_ES_RegisterCDS](#) ([CFE_ES_CDSHandle_t](#) *CDSHandlePtr, size_t BlockSize, const char *Name)

Reserve space (or re-obtain previously reserved space) in the Critical Data Store (CDS)
- [CFE_Status_t CFE_ES_GetCDSBlockIDByName](#) ([CFE_ES_CDSHandle_t](#) *BlockIdPtr, const char *BlockName)

Get a CDS Block ID associated with a specified CDS Block name.

- [CFE_Status_t CFE_ES_GetCDSBlockName](#) (char *BlockName, [CFE_ES_CDSHandle_t](#) BlockId, size_t BufferLength)

Get a Block name for a specified Block ID.
- [CFE_Status_t CFE_ES_CopyToCDS](#) ([CFE_ES_CDSHandle_t](#) Handle, const void *DataToCopy)

Save a block of data in the Critical Data Store (CDS)
- [CFE_Status_t CFE_ES_RestoreFromCDS](#) (void *RestoreToMemory, [CFE_ES_CDSHandle_t](#) Handle)

Recover a block of data from the Critical Data Store (CDS)
- [CFE_Status_t CFE_ES_PoolCreateNoSem](#) ([CFE_ES_MemHandle_t](#) *PoolID, void *MemPtr, size_t Size)

Initializes a memory pool created by an application without using a semaphore during processing.
- [CFE_Status_t CFE_ES_PoolCreate](#) ([CFE_ES_MemHandle_t](#) *PoolID, void *MemPtr, size_t Size)

Initializes a memory pool created by an application while using a semaphore during processing.
- [CFE_Status_t CFE_ES_PoolCreateEx](#) ([CFE_ES_MemHandle_t](#) *PoolID, void *MemPtr, size_t Size, [uint16](#) NumBlockSizes, const size_t *BlockSizes, bool UseMutex)

Initializes a memory pool created by an application with application specified block sizes.
- [CFE_Status_t CFE_ES_PoolCreateEx_WithAlignment](#) ([CFE_ES_MemHandle_t](#) *PoolID, void *MemPtr, size_t Size, [uint16](#) NumBlockSizes, const size_t *BlockSizes, bool UseMutex, size_t Alignment)

Implements CFE_ES_PoolCreateEx with added param for alignment added for coverage purposes)
- [int32 CFE_ES_PoolDelete](#) ([CFE_ES_MemHandle_t](#) PoolID)

Deletes a memory pool that was previously created.
- [int32 CFE_ES_GetPoolBuf](#) ([CFE_ES_MemPoolBuf_t](#) *BufPtr, [CFE_ES_MemHandle_t](#) Handle, size_t Size)

Gets a buffer from the memory pool created by [CFE_ES_PoolCreate](#) or [CFE_ES_PoolCreateNoSem](#).
- [CFE_Status_t CFE_ES_GetPoolBufInfo](#) ([CFE_ES_MemHandle_t](#) Handle, [CFE_ES_MemPoolBuf_t](#) BufPtr)

Gets info on a buffer previously allocated via [CFE_ES_GetPoolBuf](#).
- [int32 CFE_ES_PutPoolBuf](#) ([CFE_ES_MemHandle_t](#) Handle, [CFE_ES_MemPoolBuf_t](#) BufPtr)

Releases a buffer from the memory pool that was previously allocated via [CFE_ES_GetPoolBuf](#).
- [CFE_Status_t CFE_ES_GetMemPoolStats](#) ([CFE_ES_MemPoolStats_t](#) *BufPtr, [CFE_ES_MemHandle_t](#) Handle)

Extracts the statistics maintained by the memory pool software.
- void [CFE_ES_PerfLogAdd](#) ([uint32](#) Marker, [uint32](#) EntryExit)

Adds a new entry to the data buffer.
- [CFE_Status_t CFE_ES_RegisterGenCounter](#) ([CFE_ES_CounterId_t](#) *CounterIdPtr, const char *CounterName)

Register a generic counter.
- [CFE_Status_t CFE_ES_DeleteGenCounter](#) ([CFE_ES_CounterId_t](#) CounterId)

Delete a generic counter.
- [CFE_Status_t CFE_ES_IncrementGenCounter](#) ([CFE_ES_CounterId_t](#) CounterId)

Increments the specified generic counter.
- [CFE_Status_t CFE_ES_SetGenCount](#) ([CFE_ES_CounterId_t](#) CounterId, [uint32](#) Count)

Set the specified generic counter.
- [CFE_Status_t CFE_ES_GetGenCount](#) ([CFE_ES_CounterId_t](#) CounterId, [uint32](#) *Count)

Get the specified generic counter count.
- [CFE_Status_t CFE_ES_GetGenCounterIDByName](#) ([CFE_ES_CounterId_t](#) *CounterIdPtr, const char *CounterName)

Get the Id associated with a generic counter name.
- [CFE_Status_t CFE_ES_GetGenCounterName](#) (char *CounterName, [CFE_ES_CounterId_t](#) CounterId, size_t BufferLength)

Get a Counter name for a specified Counter ID.

11.61.1 Detailed Description

Purpose: Unit specification for Executive Services library functions and macros.

References: Flight Software Branch C Coding Standard Version 1.0a cFE Flight Software Application Developers Guide

Notes:

Definition in file [cfe_es.h](#).

11.61.2 Macro Definition Documentation

OS_PRINTF

```
#define OS_PRINTF(
    m,
    n )
```

Definition at line 50 of file [cfe_es.h](#).

11.62 cfe_es.h

[Go to the documentation of this file.](#)

```
00001 /*****
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"
00003  *
00004  * Copyright (c) 2023 United States Government as represented by the
00005  * Administrator of the National Aeronautics and Space Administration.
00006  * All Rights Reserved.
00007  *
00008  * Licensed under the Apache License, Version 2.0 (the "License"); you may
00009  * not use this file except in compliance with the License. You may obtain
00010  * a copy of the License at http://www.apache.org/licenses/LICENSE-2.0
00011  *
00012  * Unless required by applicable law or agreed to in writing, software
00013  * distributed under the License is distributed on an "AS IS" BASIS,
00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00033 #ifndef CFE_ES_H
00034 #define CFE_ES_H
00035
00036 /*
00037 ** Includes
00038 */
00039 #include "common_types.h"
00040 #include "cfe_error.h"
00041 #include "cfe_es_api_typedefs.h"
00042 #include "cfe_resourceid_api_typedefs.h"
00043
00044 /*
00045 ** The OS_PRINTF macro may be defined by OSAL to enable
00046 ** printf-style argument checking. If using a version of OSAL
00047 ** that does not define this then define it as a no-op.
00048 */
00049 #ifndef OS_PRINTF
00050 #define OS_PRINTF(m, n)
00051 #endif
00052
00053 /*****
00054  *
00055  ** Exported Functions
00056  */
00057
00058 /*****
00059
00087 CFE_Status_t CFE_ES_AppID_ToIndex(CFE_ES_AppId_t AppID, uint32 *Idx);
00088
00112 int32 CFE_ES_LibID_ToIndex(CFE_ES_LibId_t LibId, uint32 *Idx);
00113
00138 CFE_Status_t CFE_ES_TaskID_ToIndex(CFE_ES_TaskId_t TaskID, uint32 *Idx);
00139
00163 CFE_Status_t CFE_ES_CounterID_ToIndex(CFE_ES_CounterId_t CounterId, uint32 *Idx);
00164
00167 *****/
```

```

00168
00173 /*****/
00196 void CFE_ES_Main(uint32 StartType, uint32 StartSubtype, uint32 ModeId, const char *StartFilePath);
00197
00198 /*****/
00223 CFE_Status_t CFE_ES_ResetCFE(uint32 ResetType);
00230 /*****/
00263 CFE_Status_t CFE_ES_RestartApp(CFE_ES_AppId_t AppID);
00264
00265 /*****/
00299 CFE_Status_t CFE_ES_ReloadApp(CFE_ES_AppId_t AppID, const char *AppFileName);
00300
00301 /*****/
00320 CFE_Status_t CFE_ES_DeleteApp(CFE_ES_AppId_t AppID);
00327 /*****/
00346 void CFE_ES_ExitApp(uint32 ExitStatus);
00347
00348 /*****/
00388 bool CFE_ES_RunLoop(uint32 *RunStatus);
00389
00390 /*****/
00418 CFE_Status_t CFE_ES_WaitForSystemState(uint32 MinSystemState, uint32 TimeOutMilliseconds);
00419
00420 /*****/
00449 void CFE_ES_WaitForStartupSync(uint32 TimeOutMilliseconds);
00450
00451 /*****/
00469 void CFE_ES_IncrementTaskCounter(void);
00476 /*****/
00502 int32 CFE_ES_GetResetType(uint32 *ResetSubtypePtr);
00503
00504 /*****/
00525 CFE_Status_t CFE_ES_GetAppID(CFE_ES_AppId_t *AppIdPtr);
00526
00527 /*****/
00548 CFE_Status_t CFE_ES_GetTaskID(CFE_ES_TaskId_t *TaskIdPtr);
00549
00550 /*****/
00572 CFE_Status_t CFE_ES_GetAppIDByName(CFE_ES_AppId_t *AppIdPtr, const char *AppName);
00573
00574 /*****/
00596 CFE_Status_t CFE_ES_GetLibIDByName(CFE_ES_LibId_t *LibIdPtr, const char *LibName);
00597
00598 /*****/
00626 CFE_Status_t CFE_ES_GetAppName(char *AppName, CFE_ES_AppId_t AppId, size_t BufferLength);
00627
00628 /*****/
00656 CFE_Status_t CFE_ES_GetLibName(char *LibName, CFE_ES_LibId_t LibId, size_t BufferLength);
00657
00658 /*****/
00682 CFE_Status_t CFE_ES_GetAppInfo(CFE_ES_AppInfo_t *AppInfo, CFE_ES_AppId_t AppId);
00683
00684 /*****/
00710 CFE_Status_t CFE_ES_GetTaskInfo(CFE_ES_TaskInfo_t *TaskInfo, CFE_ES_TaskId_t TaskId);
00711
00712 /*****/
00742 int32 CFE_ES_GetLibInfo(CFE_ES_AppInfo_t *LibInfo, CFE_ES_LibId_t LibId);
00743
00744 /*****/
00774 int32 CFE_ES_GetModuleInfo(CFE_ES_AppInfo_t *ModuleInfo, CFE_ResourceId_t ResourceId);
00775
00782 /*****/
00822 CFE_Status_t CFE_ES_CreateChildTask(CFE_ES_TaskId_t *TaskIdPtr,
00823                                     const char *TaskName,
00824                                     CFE_ES_ChildTaskMainFuncPtr_t FunctionPtr,
00825                                     CFE_ES_StackPointer_t StackPtr,
00826                                     size_t StackSize,
00827                                     CFE_ES_TaskPriority_Atom_t Priority,
00828                                     uint32 Flags);
00829
00830 /*****/
00852 CFE_Status_t CFE_ES_GetTaskIDByName(CFE_ES_TaskId_t *TaskIdPtr, const char *TaskName);
00853
00854 /*****/
00882 CFE_Status_t CFE_ES_GetTaskName(char *TaskName, CFE_ES_TaskId_t TaskId, size_t BufferLength);
00883
00884 /*****/
00907 CFE_Status_t CFE_ES_DeleteChildTask(CFE_ES_TaskId_t TaskId);
00908
00909 /*****/
00927 void CFE_ES_ExitChildTask(void);

```

```

00934 /*****/
00950 void CFE_ES_BackgroundWakeup(void);
00951
00952 /*****/
00975 CFE_Status_t CFE_ES_WriteToSysLog(const char *SpecStringPtr, ...) OS_PRINTF(1, 2);
00976
00977 /*****/
01003 uint32 CFE_ES_CalculateCRC(const void *DataPtr, size_t DataLength, uint32 InputCRC, CFE_ES_CrcType_Enum_t
TypeCRC);
01004
01005 /*****/
01020 void CFE_ES_ProcessAsyncEvent(void);
01021
01028 /*****/
01066 CFE_Status_t CFE_ES_RegisterCDS(CFE_ES_CDSHandle_t *CDSHandlePtr, size_t BlockSize, const char *Name);
01067
01068 /*****/
01091 CFE_Status_t CFE_ES_GetCDSBlockIDByName(CFE_ES_CDSHandle_t *BlockIdPtr, const char *BlockName);
01092
01093 /*****/
01122 CFE_Status_t CFE_ES_GetCDSBlockName(char *BlockName, CFE_ES_CDSHandle_t BlockId, size_t BufferLength);
01123
01124 /*****/
01148 CFE_Status_t CFE_ES_CopyToCDS(CFE_ES_CDSHandle_t Handle, const void *DataToCopy);
01149
01150 /*****/
01180 CFE_Status_t CFE_ES_RestoreFromCDS(void *RestoreToMemory, CFE_ES_CDSHandle_t Handle);
01187 /*****/
01220 CFE_Status_t CFE_ES_PoolCreateNoSem(CFE_ES_MemHandle_t *PoolID, void *MemPtr, size_t Size);
01221
01222 /*****/
01255 CFE_Status_t CFE_ES_PoolCreate(CFE_ES_MemHandle_t *PoolID, void *MemPtr, size_t Size);
01256
01257 /*****/
01303 CFE_Status_t CFE_ES_PoolCreateEx(CFE_ES_MemHandle_t *PoolID,
void *MemPtr,
size_t Size,
uint16 NumBlockSizes,
const size_t *BlockSizes,
bool UseMutex);
01309
01310 /*****/
01359 CFE_Status_t CFE_ES_PoolCreateEx_WithAlignment(CFE_ES_MemHandle_t *PoolID,
void *MemPtr,
size_t Size,
uint16 NumBlockSizes,
const size_t *BlockSizes,
bool UseMutex,
size_t Alignment);
01366
01367 /*****/
01389 int32 CFE_ES_PoolDelete(CFE_ES_MemHandle_t PoolID);
01390
01391 /*****/
01417 int32 CFE_ES_GetPoolBuf(CFE_ES_MemPoolBuf_t *BufPtr, CFE_ES_MemHandle_t Handle, size_t Size);
01418
01419 /*****/
01442 CFE_Status_t CFE_ES_GetPoolBufInfo(CFE_ES_MemHandle_t Handle, CFE_ES_MemPoolBuf_t BufPtr);
01443
01444 /*****/
01468 int32 CFE_ES_PutPoolBuf(CFE_ES_MemHandle_t Handle, CFE_ES_MemPoolBuf_t BufPtr);
01469
01470 /*****/
01496 CFE_Status_t CFE_ES_GetMemPoolStats(CFE_ES_MemPoolStats_t *BufPtr, CFE_ES_MemHandle_t Handle);
01503 /*****/
01520 #define CFE_ES_PerfLogEntry(id) (CFE_ES_PerfLogAdd(id, 0))
01521
01522 /*****/
01539 #define CFE_ES_PerfLogExit(id) (CFE_ES_PerfLogAdd(id, 1))
01540
01541 /*****/
01574 void CFE_ES_PerfLogAdd(uint32 Marker, uint32 EntryExit);
01581 /*****/
01606 CFE_Status_t CFE_ES_RegisterGenCounter(CFE_ES_CounterId_t *CounterIdPtr, const char *CounterName);
01607
01608 /*****/
01628 CFE_Status_t CFE_ES_DeleteGenCounter(CFE_ES_CounterId_t CounterId);
01629
01630 /*****/
01650 CFE_Status_t CFE_ES_IncrementGenCounter(CFE_ES_CounterId_t CounterId);
01651

```

```

01652 /*****
01674 CFE_Status_t CFE_ES_SetGenCount(CFE_ES_CounterId_t CounterId, uint32 Count);
01675
01676 /*****
01698 CFE_Status_t CFE_ES_GetGenCount(CFE_ES_CounterId_t CounterId, uint32 *Count);
01699
01700 /*****
01721 CFE_Status_t CFE_ES_GetGenCounterIDByName(CFE_ES_CounterId_t *CounterIdPtr, const char *CounterName);
01722
01723 /*****
01751 CFE_Status_t CFE_ES_GetGenCounterName(char *CounterName, CFE_ES_CounterId_t CounterId, size_t
    BufferLength);
01752
01755 #endif /* CFE_ES_H */

```

11.63 cfe/modules/core_api/fsw/inc/cfe_es_api_typedefs.h File Reference

```

#include "common_types.h"
#include "cfe_es_extern_typedefs.h"

```

Data Structures

- union [CFE_ES_PoolAlign](#)
Pool Alignment.

Macros

- #define [CFE_ES_STATIC_POOL_TYPE](#)(size)
Static Pool Type.
- #define [CFE_ES_MEMPOOLBUF_C](#)(x) ((CFE_ES_MemPoolBuf_t)(x))
Conversion macro to create buffer pointer from another type.
- #define [CFE_ES_NO_MUTEX](#) false
Indicates that the memory pool selection will not use a semaphore.
- #define [CFE_ES_USE_MUTEX](#) true
Indicates that the memory pool selection will use a semaphore.

Reset Type extensions

- #define [CFE_ES_APP_RESTART](#) [CFE_PSP_RST_TYPE_MAX](#)

Conversions for ES resource IDs

- #define [CFE_ES_APPID_C](#)(val) ((CFE_ES_AppId_t)CFE_RESOURCEID_WRAP(val))
- #define [CFE_ES_TASKID_C](#)(val) ((CFE_ES_TaskId_t)CFE_RESOURCEID_WRAP(val))
- #define [CFE_ES_LIBID_C](#)(val) ((CFE_ES_LibId_t)CFE_RESOURCEID_WRAP(val))
- #define [CFE_ES_COUNTERID_C](#)(val) ((CFE_ES_CounterId_t)CFE_RESOURCEID_WRAP(val))
- #define [CFE_ES_MEMHANDLE_C](#)(val) ((CFE_ES_MemHandle_t)CFE_RESOURCEID_WRAP(val))
- #define [CFE_ES_CDSHANDLE_C](#)(val) ((CFE_ES_CDSHandle_t)CFE_RESOURCEID_WRAP(val))

Type-specific initializers for "undefined" resource IDs

- #define [CFE_ES_APPID_UNDEFINED](#) [CFE_ES_APPID_C](#)(CFE_RESOURCEID_UNDEFINED)
- #define [CFE_ES_TASKID_UNDEFINED](#) [CFE_ES_TASKID_C](#)(CFE_RESOURCEID_UNDEFINED)
- #define [CFE_ES_LIBID_UNDEFINED](#) [CFE_ES_LIBID_C](#)(CFE_RESOURCEID_UNDEFINED)
- #define [CFE_ES_COUNTERID_UNDEFINED](#) [CFE_ES_COUNTERID_C](#)(CFE_RESOURCEID_UNDEFINED)
- #define [CFE_ES_MEMHANDLE_UNDEFINED](#) [CFE_ES_MEMHANDLE_C](#)(CFE_RESOURCEID_UNDEFINED)
- #define [CFE_ES_CDS_BAD_HANDLE](#) [CFE_ES_CDSHANDLE_C](#)(CFE_RESOURCEID_UNDEFINED)

Task Stack Constants

- #define [CFE_ES_TASK_STACK_ALLOCATE](#) NULL /* aka OS_TASK_STACK_ALLOCATE in proposed OSAL change */
Indicates that the stack for the child task should be dynamically allocated.

Typedefs

- typedef void(* [CFE_ES_TaskEntryFuncPtr_t](#)) (void)
Required Prototype of Task Main Functions.
- typedef int32(* [CFE_ES_LibraryEntryFuncPtr_t](#)) ([CFE_ES_LibId_t](#) LibId)
Required Prototype of Library Initialization Functions.
- typedef [CFE_ES_TaskEntryFuncPtr_t](#) [CFE_ES_ChildTaskMainFuncPtr_t](#)
Compatible typedef for ES child task entry point.
- typedef void * [CFE_ES_StackPointer_t](#)
Type for the stack pointer of tasks.
- typedef enum [CFE_ES_CrcType_Enum](#) [CFE_ES_CrcType_Enum_t](#)
Checksum/CRC algorithm identifiers.
- typedef union [CFE_ES_PoolAlign](#) [CFE_ES_PoolAlign_t](#)
Pool Alignment.
- typedef void * [CFE_ES_MemPoolBuf_t](#)
Pointer type used for memory pool API.

Enumerations

- enum [CFE_ES_CrcType_Enum](#) {
[CFE_ES_CrcType_NONE](#) = 0 , [CFE_ES_CrcType_16_ARC](#) = 1 , [CFE_ES_CrcType_MAX](#) = 2 , [CFE_ES_CrcType_CRC_16](#)
= [CFE_ES_CrcType_16_ARC](#) ,
[CFE_ES_CrcType_CRC_8](#) = [CFE_ES_CrcType_MAX](#) + 1 , [CFE_ES_CrcType_CRC_32](#) = [CFE_ES_CrcType_MAX](#) + 2 }
Checksum/CRC algorithm identifiers.

11.63.1 Detailed Description

Purpose: Unit specification for Executive Services library functions and macros.

References: Flight Software Branch C Coding Standard Version 1.0a cFE Flight Software Application Developers Guide

Notes:

Definition in file [cfe_es_api_typedefs.h](#).

11.63.2 Macro Definition Documentation

CFE_ES_APP_RESTART

```
#define CFE_ES_APP_RESTART CFE\_PSP\_RST\_TYPE\_MAX
```

Application only was reset (extend the PSP enumeration here)

Definition at line 57 of file [cfe_es_api_typedefs.h](#).

CFE_ES_APPID_C

```
#define CFE_ES_APPID_C(  
    val ) ((CFE\_ES\_AppId\_t) CFE_RESOURCEID_WRAP (val))
```

Definition at line 208 of file [cfe_es_api_typedefs.h](#).

CFE_ES_APPID_UNDEFINED

```
#define CFE_ES_APPID_UNDEFINED CFE\_ES\_APPID\_C(CFE_RESOURCEID_UNDEFINED)
```

Definition at line 220 of file [cfe_es_api_typedefs.h](#).

CFE_ES_CDS_BAD_HANDLE

```
#define CFE_ES_CDS_BAD_HANDLE CFE_ES_CDSHANDLE_C(CFE_RESOURCEID_UNDEFINED)
```

Definition at line 225 of file [cfe_es_api_typedefs.h](#).

CFE_ES_CDSHANDLE_C

```
#define CFE_ES_CDSHANDLE_C(  
    val ) ((CFE_ES_CDSHandle_t)CFE_RESOURCEID_WRAP(val))
```

Definition at line 213 of file [cfe_es_api_typedefs.h](#).

CFE_ES_COUNTERID_C

```
#define CFE_ES_COUNTERID_C(  
    val ) ((CFE_ES_CounterId_t)CFE_RESOURCEID_WRAP(val))
```

Definition at line 211 of file [cfe_es_api_typedefs.h](#).

CFE_ES_COUNTERID_UNDEFINED

```
#define CFE_ES_COUNTERID_UNDEFINED CFE_ES_COUNTERID_C(CFE_RESOURCEID_UNDEFINED)
```

Definition at line 223 of file [cfe_es_api_typedefs.h](#).

CFE_ES_LIBID_C

```
#define CFE_ES_LIBID_C(  
    val ) ((CFE_ES_LibId_t)CFE_RESOURCEID_WRAP(val))
```

Definition at line 210 of file [cfe_es_api_typedefs.h](#).

CFE_ES_LIBID_UNDEFINED

```
#define CFE_ES_LIBID_UNDEFINED CFE_ES_LIBID_C(CFE_RESOURCEID_UNDEFINED)
```

Definition at line 222 of file [cfe_es_api_typedefs.h](#).

CFE_ES_MEMHANDLE_C

```
#define CFE_ES_MEMHANDLE_C(  
    val ) ((CFE_ES_MemHandle_t)CFE_RESOURCEID_WRAP(val))
```

Definition at line 212 of file [cfe_es_api_typedefs.h](#).

CFE_ES_MEMHANDLE_UNDEFINED

```
#define CFE_ES_MEMHANDLE_UNDEFINED CFE_ES_MEMHANDLE_C(CFE_RESOURCEID_UNDEFINED)
```

Definition at line 224 of file [cfe_es_api_typedefs.h](#).

CFE_ES_MEMPOOLBUF_C

```
#define CFE_ES_MEMPOOLBUF_C(  
    x ) ((CFE_ES_MemPoolBuf_t)(x))
```

Conversion macro to create buffer pointer from another type.

In cases where the actual buffer pointer is computed, this macro aids in converting the computed address (typically an OSAL "cpuaddr" type) into a buffer pointer.

Note

Any address calculation needs to take machine alignment requirements into account.

Definition at line 193 of file [cfe_es_api_typedefs.h](#).

CFE_ES_NO_MUTEX

```
#define CFE_ES_NO_MUTEX false
```

Indicates that the memory pool selection will not use a semaphore.

Definition at line 240 of file [cfe_es_api_typedefs.h](#).

CFE_ES_STATIC_POOL_TYPE

```
#define CFE_ES_STATIC_POOL_TYPE(  
    size )
```

Value:

```
union  
{  
    CFE_ES_PoolAlign_t Align;  
    uint8 Data[size];  
}
```

Static Pool Type.

A macro to help instantiate static memory pools that are correctly aligned. This resolves to a union type that contains a member called "Data" that will be correctly aligned to be a memory pool and sized according to the argument.

Definition at line 160 of file [cfe_es_api_typedefs.h](#).

CFE_ES_TASK_STACK_ALLOCATE

```
#define CFE_ES_TASK_STACK_ALLOCATE NULL /* aka OS_TASK_STACK_ALLOCATE in proposed OSAL change */
```

Indicates that the stack for the child task should be dynamically allocated.

This value may be supplied as the Stack Pointer argument to `CFE_ES_ChildTaskCreate()` to indicate that the stack should be dynamically allocated.

Definition at line 237 of file [cfe_es_api_typedefs.h](#).

CFE_ES_TASKID_C

```
#define CFE_ES_TASKID_C(  
    val ) ((CFE_ES_TaskId_t)CFE_RESOURCEID_WRAP(val))
```

Definition at line 209 of file [cfe_es_api_typedefs.h](#).

CFE_ES_TASKID_UNDEFINED

```
#define CFE_ES_TASKID_UNDEFINED CFE_ES_TASKID_C(CFE_RESOURCEID_UNDEFINED)
```

Definition at line 221 of file [cfe_es_api_typedefs.h](#).

CFE_ES_USE_MUTEX

```
#define CFE_ES_USE_MUTEX true
```

Indicates that the memory pool selection will use a semaphore.

Definition at line 241 of file [cfe_es_api_typedefs.h](#).

11.63.3 Typedef Documentation

CFE_ES_ChildTaskMainFuncPtr_t

```
typedef CFE_ES_TaskEntryFuncPtr_t CFE_ES_ChildTaskMainFuncPtr_t
```

Compatible typedef for ES child task entry point.

All ES task functions (main + child) use the same entry point type.

Definition at line 77 of file [cfe_es_api_typedefs.h](#).

CFE_ES_CrcType_Enum_t

```
typedef enum CFE_ES_CrcType_Enum CFE_ES_CrcType_Enum_t
```

Checksum/CRC algorithm identifiers.

Currently only CFE_ES_CrcType_16_ARC is supported.

All defined CRC algorithms should include a check value, which is the accepted result of computing the CRC across the fixed string "123456789"

CFE_ES_LibraryEntryFuncPtr_t

```
typedef int32(* CFE_ES_LibraryEntryFuncPtr_t) (CFE_ES_LibId_t LibId)
```

Required Prototype of Library Initialization Functions.

Definition at line 69 of file [cfe_es_api_typedefs.h](#).

CFE_ES_MemPoolBuf_t

```
typedef void* CFE_ES_MemPoolBuf_t
```

Pointer type used for memory pool API.

This is used in the Get/Put API calls to refer to a pool buffer.

This pointer is expected to be type cast to the real object type after getting a new buffer. Using void* allows this type conversion to occur easily.

Note

Older versions of CFE implemented the API using a uint32*, which required explicit type casting everywhere it was called. Although the API type is now void* to make usage easier, the pool buffers are aligned to machine requirements - typically 64 bits.

Definition at line 181 of file [cfe_es_api_typedefs.h](#).

CFE_ES_PoolAlign_t

```
typedef union CFE_ES_PoolAlign CFE_ES_PoolAlign_t
```

Pool Alignment.

Union that can be used for minimum memory alignment of ES memory pools on the target. It contains the longest native data types such that the alignment of this structure should reflect the largest possible alignment requirements for any data on this processor.

CFE_ES_StackPointer_t

```
typedef void* CFE_ES_StackPointer_t
```

Type for the stack pointer of tasks.

This type is used in the CFE ES task API.

Definition at line 84 of file [cfe_es_api_typedefs.h](#).

CFE_ES_TaskEntryFuncPtr_t

```
typedef void(* CFE_ES_TaskEntryFuncPtr_t) (void)
```

Required Prototype of Task Main Functions.

Definition at line 68 of file [cfe_es_api_typedefs.h](#).

11.63.4 Enumeration Type Documentation

CFE_ES_CrcType_Enum

```
enum CFE_ES_CrcType_Enum
```

Checksum/CRC algorithm identifiers.

Currently only CFE_ES_CrcType_16_ARC is supported.

All defined CRC algorithms should include a check value, which is the accepted result of computing the CRC across the fixed string "123456789"

Enumerator

CFE_ES_CrcType_NONE	Reserved placeholder value. No computation is performed, always returns 0 as a result.
CFE_ES_CrcType_16_ARC	Implementation of CRC-16/ARC. Polynomial: 0x8005 Initialization: 0x0000 Reflect Input/Output: true Check value: 0xbb3d XorOut: 0x0000
CFE_ES_CrcType_MAX	Placeholder for end of normal enumeration list This should reflect the number of algorithms defined.
CFE_ES_CrcType_CRC_16	CRC-16 historically implied CRC-16/ARC CRC-8 historically defined but not implemented, value must not be 0
CFE_ES_CrcType_CRC_8	CRC-32 historically defined but not implemented, value must not be 0
CFE_ES_CrcType_CRC_32	

Definition at line 94 of file [cfe_es_api_typedefs.h](#).

11.64 cfe_es_api_typedefs.h

[Go to the documentation of this file.](#)

```

00001 /*****
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"
00003  *
00004  * Copyright (c) 2023 United States Government as represented by the
00005  * Administrator of the National Aeronautics and Space Administration.
00006  * All Rights Reserved.
00007  *
00008  * Licensed under the Apache License, Version 2.0 (the "License"); you may
00009  * not use this file except in compliance with the License. You may obtain
00010  * a copy of the license at http://www.apache.org/licenses/LICENSE-2.0
00011  *
00012  * Unless required by applicable law or agreed to in writing, software
00013  * distributed under the License is distributed on an "AS IS" BASIS,
00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00033 #ifndef CFE_ES_API_TYPEDEFS_H
00034 #define CFE_ES_API_TYPEDEFS_H
00035
00036 /*
00037  ** Includes
00038  */
00039 #include "common_types.h"
00040 #include "cfe_es_extern_typedefs.h"
00041
00042 /*
00043  ** Note about reset type and subtypes:
00044  **
00045  ** These values come from the PSP so the actual definition of these enumerations
00046  ** was moved to the PSP header file <cfe_psp.h>.
00047  **
00048  ** In the future the Electronic Data sheets (EDS) for PSP/ES
00049  ** will define the exact values to use in telemetry messages.
00050  */

```

```

00051
00052 /*
00053 ** Reset types
00054 */
00057 #define CFE_ES_APP_RESTART CFE_PSP_RST_TYPE_MAX
00060 /*****
00061 */
00062 ** Type Definitions
00063 */
00064
00065 /*
00066 ** Entry Function Prototypes
00067 */
00068 typedef void (*CFE_ES_TaskEntryFuncPtr_t)(void);
00069 typedef int32 (*CFE_ES_LibraryEntryFuncPtr_t)(
00070     CFE_ES_LibId_t LibId);
00077 typedef CFE_ES_TaskEntryFuncPtr_t CFE_ES_ChildTaskMainFuncPtr_t;
00078
00084 typedef void *CFE_ES_StackPointer_t; /* aka osal_stackptr_t in proposed OSAL change */
00085
00094 typedef enum CFE_ES_CrcType_Enum
00095 {
00101     CFE_ES_CrcType_NONE = 0,
00102
00113     CFE_ES_CrcType_16_ARC = 1,
00114
00119     CFE_ES_CrcType_MAX = 2,
00120
00121     /*
00122      * Backward compatibility values.
00123      * For compatibility with apps, these simplified symbols need to be defined,
00124      * and they also must map to nonzero values.
00125      */
00126
00128     CFE_ES_CrcType_CRC_16 = CFE_ES_CrcType_16_ARC,
00129
00131     CFE_ES_CrcType_CRC_8 = CFE_ES_CrcType_MAX + 1,
00132
00134     CFE_ES_CrcType_CRC_32 = CFE_ES_CrcType_MAX + 2,
00135
00136 } CFE_ES_CrcType_Enum_t;
00137
00145 typedef union CFE_ES_PoolAlign
00146 {
00147     void *Ptr;
00148     /* note -- native types (int/double) are intentional here */
00149     long long int LongInt;
00150     long double LongDouble;
00151 } CFE_ES_PoolAlign_t;
00152
00160 #define CFE_ES_STATIC_POOL_TYPE(size) \
00161     union \
00162     { \
00163         CFE_ES_PoolAlign_t Align; \
00164         uint8 Data[size]; \
00165     } \
00166
00181 typedef void *CFE_ES_MemPoolBuf_t;
00182
00193 #define CFE_ES_MEMPOOLBUF_C(x) ((CFE_ES_MemPoolBuf_t)(x))
00194
00198 /*
00199 * Conversion macros for each ES resource ID subtype
00200 *
00201 * These accept a generic/non-specific CFE_ResourceId_t value
00202 * and convert it to the corresponding resource-specific type.
00203 *
00204 * These should only be used when with the resource ID constants,
00205 * or where the code has confirmed or is determining the generic
00206 * identifier does correspond to a resource of that type.
00207 */
00208 #define CFE_ES_APPID_C(val) ((CFE_ES_AppId_t)CFE_RESOURCEID_WRAP(val))
00209 #define CFE_ES_TASKID_C(val) ((CFE_ES_TaskId_t)CFE_RESOURCEID_WRAP(val))
00210 #define CFE_ES_LIBID_C(val) ((CFE_ES_LibId_t)CFE_RESOURCEID_WRAP(val))
00211 #define CFE_ES_COUNTERID_C(val) ((CFE_ES_CounterId_t)CFE_RESOURCEID_WRAP(val))
00212 #define CFE_ES_MEMHANDLE_C(val) ((CFE_ES_MemHandle_t)CFE_RESOURCEID_WRAP(val))
00213 #define CFE_ES_CDSHANDLE_C(val) ((CFE_ES_CDSHandle_t)CFE_RESOURCEID_WRAP(val))
00214
00220 #define CFE_ES_APPID_UNDEFINED CFE_ES_APPID_C(CFE_RESOURCEID_UNDEFINED)
00221 #define CFE_ES_TASKID_UNDEFINED CFE_ES_TASKID_C(CFE_RESOURCEID_UNDEFINED)
00222 #define CFE_ES_LIBID_UNDEFINED CFE_ES_LIBID_C(CFE_RESOURCEID_UNDEFINED)

```

```

00223 #define CFE_ES_COUNTERID_UNDEFINED CFE_ES_COUNTERID_C(CFE_RESOURCEID_UNDEFINED)
00224 #define CFE_ES_MEMHANDLE_UNDEFINED CFE_ES_MEMHANDLE_C(CFE_RESOURCEID_UNDEFINED)
00225 #define CFE_ES_CDS_BAD_HANDLE CFE_ES_CDSHANDLE_C(CFE_RESOURCEID_UNDEFINED)
00237 #define CFE_ES_TASK_STACK_ALLOCATE NULL /* aka OS_TASK_STACK_ALLOCATE in proposed OSAL change */
00240 #define CFE_ES_NO_MUTEX false
00241 #define CFE_ES_USE_MUTEX true
00243 #endif /* CFE_ES_API_TYPEDEFS_H */

```

11.65 cfe/modules/core_api/fsw/inc/cfe_evs.h File Reference

```

#include "common_types.h"
#include "cfe_error.h"
#include "cfe_evs_api_typedefs.h"
#include "cfe_es_api_typedefs.h"
#include "cfe_time_api_typedefs.h"

```

Macros

- #define [CFE_EVS_Send](#)(E, T, ...) [CFE_EVS_SendEvent](#)((E), CFE_EVS_EventType_##T, __VA_ARGS__)
- #define [CFE_EVS_SendDbg](#)(E, ...) [CFE_EVS_Send](#)(E, DEBUG, __VA_ARGS__)
- #define [CFE_EVS_SendInfo](#)(E, ...) [CFE_EVS_Send](#)(E, INFORMATION, __VA_ARGS__)
- #define [CFE_EVS_SendErr](#)(E, ...) [CFE_EVS_Send](#)(E, ERROR, __VA_ARGS__)
- #define [CFE_EVS_SendCrit](#)(E, ...) [CFE_EVS_Send](#)(E, CRITICAL, __VA_ARGS__)

Functions

- [CFE_Status_t CFE_EVS_Register](#) (const void *Filters, [uint16](#) NumEventFilters, [uint16](#) FilterScheme)
Register an application for receiving event services.
- [CFE_Status_t CFE_EVS_SendEvent](#) ([uint16](#) EventID, [CFE_EVS_EventType_Enum_t](#) EventType, const char *Spec,...) [OS_PRINTF](#)(3)
Generate a software event.
- [CFE_Status_t CFE_Status_t CFE_EVS_SendEventWithAppID](#) ([uint16](#) EventID, [CFE_EVS_EventType_Enum_t](#) EventType, [CFE_ES_AppID_t](#) AppID, const char *Spec,...) [OS_PRINTF](#)(4)
Generate a software event given the specified Application ID.
- [CFE_Status_t CFE_Status_t CFE_Status_t CFE_EVS_SendTimedEvent](#) ([CFE_TIME_SysTime_t](#) Time, [uint16](#) EventID, [CFE_EVS_EventType_Enum_t](#) EventType, const char *Spec,...) [OS_PRINTF](#)(4)
Generate a software event with a specific time tag.
- [CFE_Status_t CFE_EVS_ResetFilter](#) ([uint16](#) EventID)
Resets the calling application's event filter for a single event ID.
- [CFE_Status_t CFE_EVS_ResetAllFilters](#) (void)
Resets all of the calling application's event filters.

11.65.1 Detailed Description

Title: Event Services API Application Library Header File

Purpose: Unit specification for Event services library functions and macros.

Design Notes:

References: Flight Software Branch C Coding Standard Version 1.0a

Definition in file [cfe_evs.h](#).

11.65.2 Macro Definition Documentation

CFE_EVS_Send

```
#define CFE_EVS_Send(
    E,
    T,
    ... ) CFE_EVS_SendEvent((E), CFE_EVS_EventType_#T, __VA_ARGS__)
```

Definition at line 46 of file [cfe_evs.h](#).

CFE_EVS_SendCrit

```
#define CFE_EVS_SendCrit(
    E,
    ... ) CFE_EVS_Send(E, CRITICAL, __VA_ARGS__)
```

Definition at line 50 of file [cfe_evs.h](#).

CFE_EVS_SendDbg

```
#define CFE_EVS_SendDbg(
    E,
    ... ) CFE_EVS_Send(E, DEBUG, __VA_ARGS__)
```

Definition at line 47 of file [cfe_evs.h](#).

CFE_EVS_SendErr

```
#define CFE_EVS_SendErr(
    E,
    ... ) CFE_EVS_Send(E, ERROR, __VA_ARGS__)
```

Definition at line 49 of file [cfe_evs.h](#).

CFE_EVS_SendInfo

```
#define CFE_EVS_SendInfo(
    E,
    ... ) CFE_EVS_Send(E, INFORMATION, __VA_ARGS__)
```

Definition at line 48 of file [cfe_evs.h](#).

11.66 cfe_evs.h

[Go to the documentation of this file.](#)

```
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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
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00016  * limitations under the License.
00017  *****/
00018
00033 #ifndef CFE_EVS_H
00034 #define CFE_EVS_H
00035
00036 /***** Include Files *****/
```

```

00037 #include "common_types.h" /* Basic data types */
00038 #include "cfe_error.h"
00039 #include "cfe_evs_api_typedefs.h"
00040 #include "cfe_es_api_typedefs.h"
00041 #include "cfe_time_api_typedefs.h"
00042
00043 /*
00044 ** Utility macros to make for simpler/more compact/readable code.
00045 */
00046 #define CFE_EVS_Send(E, T, ...) CFE_EVS_SendEvent((E), CFE_EVS_EventType_##T, __VA_ARGS__)
00047 #define CFE_EVS_SendDbg(E, ...) CFE_EVS_Send(E, DEBUG, __VA_ARGS__)
00048 #define CFE_EVS_SendInfo(E, ...) CFE_EVS_Send(E, INFORMATION, __VA_ARGS__)
00049 #define CFE_EVS_SendErr(E, ...) CFE_EVS_Send(E, ERROR, __VA_ARGS__)
00050 #define CFE_EVS_SendCrit(E, ...) CFE_EVS_Send(E, CRITICAL, __VA_ARGS__)
00051
00052 /***** Function Prototypes *****/
00053
00105 CFE_Status_t CFE_EVS_Register(const void *Filters, uint16 NumEventFilters, uint16 FilterScheme);
00155 CFE_Status_t CFE_EVS_SendEvent(uint16 EventID, CFE_EVS_EventType_Enum_t EventType, const char *Spec, ...)
00156     OS_PRINTF(3, 4);
00157
00205 CFE_Status_t CFE_EVS_SendEventWithAppID(uint16 EventID,
00206     CFE_EVS_EventType_Enum_t EventType,
00207     CFE_ES_AppId_t AppID,
00208     const char *Spec,
00209     ...) OS_PRINTF(4, 5);
00210
00258 CFE_Status_t CFE_EVS_SendTimedEvent(CFE_TIME_SysTime_t Time,
00259     uint16 EventID,
00260     CFE_EVS_EventType_Enum_t EventType,
00261     const char *Spec,
00262     ...) OS_PRINTF(4, 5);
00292 CFE_Status_t CFE_EVS_ResetFilter(uint16 EventID);
00293
00312 CFE_Status_t CFE_EVS_ResetAllFilters(void);
00315 #endif /* CFE_EVS_H */

```

11.67 cfe/modules/core_api/fsw/inc/cfe_evs_api_typedefs.h File Reference

```

#include "common_types.h"
#include "cfe_evs_extern_typedefs.h"

```

Data Structures

- struct [CFE_EVS_BinFilter](#)
Event message filter definition structure.

Macros

Common Event Filter Mask Values

Message is sent if (previous event count) & MASK == 0

- #define [CFE_EVS_NO_FILTER](#) 0x0000
Stops any filtering. All messages are sent.
- #define [CFE_EVS_FIRST_ONE_STOP](#) 0xFFFF
Sends the first event. All remaining messages are filtered.
- #define [CFE_EVS_FIRST_TWO_STOP](#) 0xFFFE
Sends the first 2 events. All remaining messages are filtered.
- #define [CFE_EVS_FIRST_4_STOP](#) 0xFFFC
Sends the first 4 events. All remaining messages are filtered.
- #define [CFE_EVS_FIRST_8_STOP](#) 0xFF8
Sends the first 8 events. All remaining messages are filtered.
- #define [CFE_EVS_FIRST_16_STOP](#) 0xFF0
Sends the first 16 events. All remaining messages are filtered.
- #define [CFE_EVS_FIRST_32_STOP](#) 0xFFE0

- Sends the first 32 events. All remaining messages are filtered.*
 - `#define CFE_EVS_FIRST_64_STOP 0xFFC0`
Sends the first 64 events. All remaining messages are filtered.
 - `#define CFE_EVS_EVERY_OTHER_ONE 0x0001`
Sends every other event.
 - `#define CFE_EVS_EVERY_OTHER_TWO 0x0002`
Sends two, filters one, sends two, filters one, etc.
 - `#define CFE_EVS_EVERY_FOURTH_ONE 0x0003`
Sends every fourth event message. All others are filtered.

Typedefs

- typedef struct `CFE_EVS_BinFilter` `CFE_EVS_BinFilter_t`
Event message filter definition structure.

11.67.1 Detailed Description

Title: Event Services API Application Library Header File

Purpose: Unit specification for Event services library functions and macros.

Design Notes:

References: Flight Software Branch C Coding Standard Version 1.0a

Definition in file [cfe_evs_api_typedefs.h](#).

11.67.2 Macro Definition Documentation

CFE_EVS_EVERY_FOURTH_ONE

```
#define CFE_EVS_EVERY_FOURTH_ONE 0x0003
```

Sends every fourth event message. All others are filtered.

Definition at line 54 of file [cfe_evs_api_typedefs.h](#).

CFE_EVS_EVERY_OTHER_ONE

```
#define CFE_EVS_EVERY_OTHER_ONE 0x0001
```

Sends every other event.

Definition at line 52 of file [cfe_evs_api_typedefs.h](#).

CFE_EVS_EVERY_OTHER_TWO

```
#define CFE_EVS_EVERY_OTHER_TWO 0x0002
```

Sends two, filters one, sends two, filters one, etc.

Definition at line 53 of file [cfe_evs_api_typedefs.h](#).

CFE_EVS_FIRST_16_STOP

```
#define CFE_EVS_FIRST_16_STOP 0xFFFF
```

Sends the first 16 events. All remaining messages are filtered.

Definition at line 49 of file [cfe_evs_api_typedefs.h](#).

CFE_EVS_FIRST_32_STOP

```
#define CFE_EVS_FIRST_32_STOP 0xFFE0
```

Sends the first 32 events. All remaining messages are filtered.

Definition at line 50 of file [cfe_evs_api_typedefs.h](#).

CFE_EVS_FIRST_4_STOP

```
#define CFE_EVS_FIRST_4_STOP 0xFFFC
```

Sends the first 4 events. All remaining messages are filtered.

Definition at line 47 of file [cfe_evs_api_typedefs.h](#).

CFE_EVS_FIRST_64_STOP

```
#define CFE_EVS_FIRST_64_STOP 0xFFC0
```

Sends the first 64 events. All remaining messages are filtered.

Definition at line 51 of file [cfe_evs_api_typedefs.h](#).

CFE_EVS_FIRST_8_STOP

```
#define CFE_EVS_FIRST_8_STOP 0xFFFF8
```

Sends the first 8 events. All remaining messages are filtered.

Definition at line 48 of file [cfe_evs_api_typedefs.h](#).

CFE_EVS_FIRST_ONE_STOP

```
#define CFE_EVS_FIRST_ONE_STOP 0xFFFFF
```

Sends the first event. All remaining messages are filtered.

Definition at line 45 of file [cfe_evs_api_typedefs.h](#).

CFE_EVS_FIRST_TWO_STOP

```
#define CFE_EVS_FIRST_TWO_STOP 0xFFFFE
```

Sends the first 2 events. All remaining messages are filtered.

Definition at line 46 of file [cfe_evs_api_typedefs.h](#).

CFE_EVS_NO_FILTER

```
#define CFE_EVS_NO_FILTER 0x0000
```

Stops any filtering. All messages are sent.

Definition at line 44 of file [cfe_evs_api_typedefs.h](#).

11.67.3 Typedef Documentation**CFE_EVS_BinFilter_t**

```
typedef struct CFE_EVS_BinFilter CFE_EVS_BinFilter_t
```

Event message filter definition structure.

11.68 cfe_evs_api_typedefs.h

[Go to the documentation of this file.](#)

```
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```

```

00016  * limitations under the License.
00017  *****/
00018
00033 #ifndef CFE_EVS_API_TYPEDEFS_H
00034 #define CFE_EVS_API_TYPEDEFS_H
00035
00036 /***** Include Files *****/
00037 #include "common_types.h" /* Basic data types */
00038 #include "cfe_evs_extern_typedefs.h"
00039
00044 #define CFE_EVS_NO_FILTER      0x0000
00045 #define CFE_EVS_FIRST_ONE_STOP 0xFFFF
00046 #define CFE_EVS_FIRST_TWO_STOP 0xFFFE
00047 #define CFE_EVS_FIRST_4_STOP   0xFFFC
00048 #define CFE_EVS_FIRST_8_STOP   0xFF8
00049 #define CFE_EVS_FIRST_16_STOP  0xFF0
00050 #define CFE_EVS_FIRST_32_STOP  0xFF0
00051 #define CFE_EVS_FIRST_64_STOP  0xFF0
00052 #define CFE_EVS_EVERY_OTHER_ONE 0x0001
00053 #define CFE_EVS_EVERY_OTHER_TWO 0x0002
00054 #define CFE_EVS_EVERY_FOURTH_ONE 0x0003
00057 /***** Structure Definitions *****/
00058
00060 typedef struct CFE_EVS_BinFilter
00061 {
00062     uint16 EventID;
00063     uint16 Mask;
00064 } CFE_EVS_BinFilter_t;
00065
00066 #endif /* CFE_EVS_API_TYPEDEFS_H */

```

11.69 cfe/modules/core_api/fsw/inc/cfe_fs.h File Reference

```

#include "common_types.h"
#include "osconfig.h"
#include "cfe_platform_cfg.h"
#include "cfe_error.h"
#include "cfe_fs_api_typedefs.h"
#include "cfe_fs_extern_typedefs.h"
#include "cfe_time_api_typedefs.h"

```

Functions

- [CFE_Status_t CFE_FS_ReadHeader](#) ([CFE_FS_Header_t](#) *Hdr, [osal_id_t](#) FileDes)
Read the contents of the Standard cFE File Header.
- void [CFE_FS_InitHeader](#) ([CFE_FS_Header_t](#) *Hdr, const char *Description, [uint32](#) SubType)
Initializes the contents of the Standard cFE File Header.
- [CFE_Status_t CFE_FS_WriteHeader](#) ([osal_id_t](#) FileDes, [CFE_FS_Header_t](#) *Hdr)
Write the specified Standard cFE File Header to the specified file.
- [CFE_Status_t CFE_FS_SetTimestamp](#) ([osal_id_t](#) FileDes, [CFE_TIME_SysTime_t](#) NewTimestamp)
Modifies the Time Stamp field in the Standard cFE File Header for the specified file.
- const char * [CFE_FS_GetDefaultMountPoint](#) ([CFE_FS_FileCategory_t](#) FileCategory)
Get the default virtual mount point for a file category.
- const char * [CFE_FS_GetDefaultExtension](#) ([CFE_FS_FileCategory_t](#) FileCategory)
Get the default filename extension for a file category.
- [int32 CFE_FS_ParseInputFileNameEx](#) (char *OutputBuffer, const char *InputBuffer, [size_t](#) OutputBufSize, [size_t](#) InputBufSize, const char *DefaultInput, const char *DefaultPath, const char *DefaultExtension)
Parse a filename input from an input buffer into a local buffer.
- [int32 CFE_FS_ParseInputFileName](#) (char *OutputBuffer, const char *InputName, [size_t](#) OutputBufSize, [CFE_FS_FileCategory_t](#) FileCategory)
Parse a filename string from the user into a local buffer.

- [CFE_Status_t CFE_FS_ExtractFilenameFromPath](#) (const char *OriginalPath, char *FileNameOnly)
Extracts the filename from a unix style path and filename string.
- [int32 CFE_FS_BackgroundFileDumpRequest](#) (CFE_FS_FileWriteMetaData_t *Meta)
Register a background file dump request.
- [bool CFE_FS_BackgroundFileDumpsPending](#) (const CFE_FS_FileWriteMetaData_t *Meta)
Query if a background file write request is currently pending.

11.69.1 Detailed Description

Purpose: cFE File Services (FS) library API header file

Author: S.Walling/Microtel

Definition in file [cfe_fs.h](#).

11.70 cfe_fs.h

[Go to the documentation of this file.](#)

```

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00016  * limitations under the License.
00017  *****/
00018
00028 #ifndef CFE_FS_H
00029 #define CFE_FS_H
00030
00031 /*
00032 ** Required header files...
00033 */
00034 #include "common_types.h"
00035 #include "osconfig.h"
00036 #include "cfe_platform_cfg.h"
00037 #include "cfe_error.h"
00038 #include "cfe_fs_api_typedefs.h"
00039 #include "cfe_fs_extern_typedefs.h"
00040 #include "cfe_time_api_typedefs.h"
00041
00046 /*****
00077 CFE_Status_t CFE_FS_ReadHeader(CFE_FS_Header_t *Hdr, osal_id_t FileDes);
00078
00079 /*****
00097 void CFE_FS_InitHeader(CFE_FS_Header_t *Hdr, const char *Description, uint32 SubType);
00098
00099 /*****
00144 CFE_Status_t CFE_FS_WriteHeader(osal_id_t FileDes, CFE_FS_Header_t *Hdr);
00145
00146 /*****
00177 CFE_Status_t CFE_FS_SetTimestamp(osal_id_t FileDes, CFE_TIME_SysTime_t NewTimestamp);
00184 /*****
00200 const char *CFE_FS_GetDefaultMountPoint(CFE_FS_FileCategory_t FileCategory);
00201
00202 /*****
00217 const char *CFE_FS_GetDefaultExtension(CFE_FS_FileCategory_t FileCategory);
00218
00219 /*****
00257 int32 CFE_FS_ParseInputFileNameEx(char *OutputBuffer,
00258                                   const char *InputBuffer,
00259                                   size_t OutputBufSize,
00260                                   size_t InputBufSize,
00261                                   const char *DefaultInput,
00262                                   const char *DefaultPath,

```

```

00263                                     const char *DefaultExtension);
00264
00265 /*****
00291 int32 CFE_FS_ParseInputFileName(char          *OutputBuffer,
00292                                     const char   *InputName,
00293                                     size_t       OutputBufSize,
00294                                     CFE_FS_FileCategory_t FileCategory);
00295
00296 /*****
00320 CFE_Status_t CFE_FS_ExtractFilenameFromPath(const char *OriginalPath, char *FileNameOnly);
00321
00322 /*****
00343 int32 CFE_FS_BackgroundFileDumpRequest(CFE_FS_FileWriteMetaData_t *Meta);
00344
00345 /*****
00363 bool CFE_FS_BackgroundFileDumpIsPending(const CFE_FS_FileWriteMetaData_t *Meta);
00364
00367 #endif /* CFE_FS_H */

```

11.71 cfe/modules/core_api/fsw/inc/cfe_fs_api_typedefs.h File Reference

```

#include "common_types.h"
#include "osconfig.h"
#include "cfe_mission_cfg.h"
#include "cfe_fs_extern_typedefs.h"

```

Data Structures

- struct [CFE_FS_FileWriteMetaData](#)
External Metadata/State object associated with background file writes.

Typedefs

- typedef bool(* [CFE_FS_FileWriteGetData_t](#)) (void *Meta, uint32 RecordNum, void **Buffer, size_t *BufSize)
- typedef void(* [CFE_FS_FileWriteOnEvent_t](#)) (void *Meta, [CFE_FS_FileWriteEvent_t](#) Event, int32 Status, uint32 RecordNum, size_t BlockSize, size_t Position)
- typedef struct [CFE_FS_FileWriteMetaData](#) [CFE_FS_FileWriteMetaData_t](#)
External Metadata/State object associated with background file writes.

Enumerations

- enum [CFE_FS_FileCategory_t](#) {
 [CFE_FS_FileCategory_UNKNOWN](#) , [CFE_FS_FileCategory_DYNAMIC_MODULE](#) , [CFE_FS_FileCategory_BINARY_DATA_DUMP](#) ,
 [CFE_FS_FileCategory_TEXT_LOG](#) ,
 [CFE_FS_FileCategory_SCRIPT](#) , [CFE_FS_FileCategory_TEMP](#) , [CFE_FS_FileCategory_MAX](#) }
Generalized file types/categories known to FS.
- enum [CFE_FS_FileWriteEvent_t](#) {
 [CFE_FS_FileWriteEvent_UNDEFINED](#) , [CFE_FS_FileWriteEvent_COMPLETE](#) , [CFE_FS_FileWriteEvent_CREATE_ERROR](#) ,
 [CFE_FS_FileWriteEvent_HEADER_WRITE_ERROR](#) ,
 [CFE_FS_FileWriteEvent_RECORD_WRITE_ERROR](#) , [CFE_FS_FileWriteEvent_MAX](#) }

11.71.1 Detailed Description

Purpose: cFE File Services (FS) library API header file

Author: S.Walling/Microtel

Definition in file [cfe_fs_api_typedefs.h](#).

11.71.2 Typedef Documentation

CFE_FS_FileWriteGetData_t

```
typedef bool(* CFE_FS_FileWriteGetData_t) (void *Meta, uint32 RecordNum, void **Buffer, size_t *BufSize)
```

Data Getter routine provided by requester

Outputs a data block. Should return true if the file is complete (last record/EOF), otherwise return false.

Parameters

in, out	<i>Meta</i>	Pointer to the metadata object
in	<i>RecordNum</i>	Incrementing record number counter
out	<i>Buffer</i>	Pointer to buffer data block, should be set by implementation
out	<i>BufSize</i>	Pointer to buffer data size, should be set by implementation

Returns

End of file status

Return values

<i>true</i>	if at last data record, and output file should be closed
<i>false</i>	if not at last record, more data records to write

Note

The implementation of this function must always set the "Buffer" and "BufSize" outputs. If no data is available, they may be set to NULL and 0, respectively.

Definition at line 98 of file [cfe_fs_api_typedefs.h](#).

CFE_FS_FileWriteMetaData_t

```
typedef struct CFE_FS_FileWriteMetaData CFE_FS_FileWriteMetaData_t
```

External Metadata/State object associated with background file writes.

Applications intending to schedule background file write jobs should instantiate this object in static/global data memory. This keeps track of the state of the file write request(s).

CFE_FS_FileWriteOnEvent_t

```
typedef void(* CFE_FS_FileWriteOnEvent_t) (void *Meta, CFE_FS_FileWriteEvent_t Event, int32 Status, uint32 RecordNum, size_t BlockSize, size_t Position)
```

Event generator routine provided by requester

Invoked from certain points in the file write process. Implementation may invoke [CFE_EVS_SendEvent\(\)](#) appropriately to inform of progress.

Parameters

in, out	<i>Meta</i>	Pointer to the metadata object
in	<i>Event</i>	Generalized type of event to report (not actual event ID)
in	<i>Status</i>	Generalized status code (may be from OSAL or CFE)
in	<i>RecordNum</i>	Record number counter at which event occurred

Parameters

in	<i>BlockSize</i>	Size of record being processed when event occurred (if applicable)
in	<i>Position</i>	File position/size when event occurred

Definition at line 114 of file [cfe_fs_api_typedefs.h](#).

11.71.3 Enumeration Type Documentation

CFE_FS_FileCategory_t

enum [CFE_FS_FileCategory_t](#)

Generalized file types/categories known to FS.

This defines different categories of files, where they may reside in different default locations of the virtualized file system. This is different from, and should not be confused with, the "SubType" field in the FS header. This value is only used at runtime for FS APIs and should not actually appear in any output file or message.

Enumerator

CFE_FS_FileCategory_UNKNOWN	Placeholder, unknown file category
CFE_FS_FileCategory_DYNAMIC_MODULE	Dynamically loadable apps/libraries (e.g. .so, .o, .dll, etc)
CFE_FS_FileCategory_BINARY_DATA_DUMP	Binary log file generated by various data dump commands
CFE_FS_FileCategory_TEXT_LOG	Text-based log file generated by various commands
CFE_FS_FileCategory_SCRIPT	Text-based Script files (e.g. ES startup script)
CFE_FS_FileCategory_TEMP	Temporary/Ephemeral files
CFE_FS_FileCategory_MAX	Placeholder, keep last

Definition at line 49 of file [cfe_fs_api_typedefs.h](#).

CFE_FS_FileWriteEvent_t

enum [CFE_FS_FileWriteEvent_t](#)

Enumerator

CFE_FS_FileWriteEvent_UNDEFINED	
CFE_FS_FileWriteEvent_COMPLETE	File is completed successfully
CFE_FS_FileWriteEvent_CREATE_ERROR	Unable to create/open file
CFE_FS_FileWriteEvent_HEADER_WRITE_ERROR	Unable to write FS header
CFE_FS_FileWriteEvent_RECORD_WRITE_ERROR	Unable to write data record
CFE_FS_FileWriteEvent_MAX	

Definition at line 69 of file [cfe_fs_api_typedefs.h](#).

11.72 cfe_fs_api_typedefs.h

[Go to the documentation of this file.](#)

```

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```

```

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00016  * limitations under the License.
00017  *****/
00018
00028 #ifndef CFE_FS_API_TYPEDEFS_H
00029 #define CFE_FS_API_TYPEDEFS_H
00030
00031 /*
00032 ** Required header files...
00033 */
00034 #include "common_types.h"
00035 #include "osconfig.h"
00036 #include "cfe_mission_cfg.h"
00037 #include "cfe_fs_extern_typedefs.h"
00038
00049 typedef enum
00050 {
00051     CFE_FS_FileCategory_UNKNOWN,
00052     CFE_FS_FileCategory_DYNAMIC_MODULE,
00053     CFE_FS_FileCategory_BINARY_DATA_DUMP,
00054     CFE_FS_FileCategory_TEXT_LOG,
00055     CFE_FS_FileCategory_SCRIPT,
00056     CFE_FS_FileCategory_TEMP,
00057     CFE_FS_FileCategory_MAX
00058 } CFE_FS_FileCategory_t;
00059
00060 /*
00061  * Because FS is a library not an app, it does not have its own context or
00062  * event IDs. The file writer runs in the context of the ES background task
00063  * on behalf of whatever App requested the file write.
00064  *
00065  * This is a list of abstract events associated with background file write jobs.
00066  * An app requesting the file write must supply a callback function to translate
00067  * these into its own event IDs for feedback (i.e. file complete, error conditions, etc).
00068  */
00069 typedef enum
00070 {
00071     CFE_FS_FileWriteEvent_UNDEFINED, /* placeholder, no-op, keep as 0 */
00072
00073     CFE_FS_FileWriteEvent_COMPLETE,
00074     CFE_FS_FileWriteEvent_CREATE_ERROR,
00075     CFE_FS_FileWriteEvent_HEADER_WRITE_ERROR,
00076     CFE_FS_FileWriteEvent_RECORD_WRITE_ERROR,
00077     CFE_FS_FileWriteEvent_MAX /* placeholder, no-op, keep last */
00078 } CFE_FS_FileWriteEvent_t;
00080
00098 typedef bool (*CFE_FS_FileWriteGetData_t)(void *Meta, uint32 RecordNum, void **Buffer, size_t *BufSize);
00099
00114 typedef void (*CFE_FS_FileWriteOnEvent_t)(void *Meta,
00115                                           CFE_FS_FileWriteEvent_t Event,
00116                                           int32 Status,
00117                                           uint32 RecordNum,
00118                                           size_t BlockSize,
00119                                           size_t Position);
00120
00128 typedef struct CFE_FS_FileWriteMetaData
00129 {
00130     volatile bool IsPending;
00132     char FileName[CFE_MISSION_MAX_PATH_LEN];
00134     /* Data for FS header */
00135     uint32 FileSubType;
00136     char Description[CFE_FS_HDR_DESC_MAX_LEN];
00138     CFE_FS_FileWriteGetData_t GetData;
00139     CFE_FS_FileWriteOnEvent_t OnEvent;
00140 } CFE_FS_FileWriteMetaData_t;
00141
00142 #endif /* CFE_FS_API_TYPEDEFS_H */

```

11.73 cfe/modules/core_api/fsw/inc/cfe_msg.h File Reference

```
#include "common_types.h"
#include "cfe_error.h"
#include "cfe_msg_hdr.h"
#include "cfe_msg_api_typedefs.h"
#include "cfe_es_api_typedefs.h"
#include "cfe_sb_api_typedefs.h"
#include "cfe_time_api_typedefs.h"
```

Functions

- [CFE_Status_t CFE_MSG_Init](#) ([CFE_MSG_Message_t](#) *MsgPtr, [CFE_SB_MsgId_t](#) MsgId, [CFE_MSG_Size_t](#) Size)
Initialize a message.
- [CFE_Status_t CFE_MSG_GetSize](#) (const [CFE_MSG_Message_t](#) *MsgPtr, [CFE_MSG_Size_t](#) *Size)
Gets the total size of a message.
- [CFE_Status_t CFE_MSG_SetSize](#) ([CFE_MSG_Message_t](#) *MsgPtr, [CFE_MSG_Size_t](#) Size)
Sets the total size of a message.
- [CFE_Status_t CFE_MSG_GetType](#) (const [CFE_MSG_Message_t](#) *MsgPtr, [CFE_MSG_Type_t](#) *Type)
Gets the message type.
- [CFE_Status_t CFE_MSG_SetType](#) ([CFE_MSG_Message_t](#) *MsgPtr, [CFE_MSG_Type_t](#) Type)
Sets the message type.
- [CFE_Status_t CFE_MSG_GetHeaderVersion](#) (const [CFE_MSG_Message_t](#) *MsgPtr, [CFE_MSG_HeaderVersion_t](#) *Version)
Gets the message header version.
- [CFE_Status_t CFE_MSG_SetHeaderVersion](#) ([CFE_MSG_Message_t](#) *MsgPtr, [CFE_MSG_HeaderVersion_t](#) Version)
Sets the message header version.
- [CFE_Status_t CFE_MSG_GetHasSecondaryHeader](#) (const [CFE_MSG_Message_t](#) *MsgPtr, bool *HasSecondary)
Gets the message secondary header boolean.
- [CFE_Status_t CFE_MSG_SetHasSecondaryHeader](#) ([CFE_MSG_Message_t](#) *MsgPtr, bool HasSecondary)
Sets the message secondary header boolean.
- [CFE_Status_t CFE_MSG_GetApId](#) (const [CFE_MSG_Message_t](#) *MsgPtr, [CFE_MSG_ApId_t](#) *ApId)
Gets the message application ID.
- [CFE_Status_t CFE_MSG_SetApId](#) ([CFE_MSG_Message_t](#) *MsgPtr, [CFE_MSG_ApId_t](#) ApId)
Sets the message application ID.
- [CFE_Status_t CFE_MSG_GetSegmentationFlag](#) (const [CFE_MSG_Message_t](#) *MsgPtr, [CFE_MSG_SegmentationFlag_t](#) *SegFlag)
Gets the message segmentation flag.
- [CFE_Status_t CFE_MSG_SetSegmentationFlag](#) ([CFE_MSG_Message_t](#) *MsgPtr, [CFE_MSG_SegmentationFlag_t](#) SegFlag)
Sets the message segmentation flag.
- [CFE_Status_t CFE_MSG_GetSequenceCount](#) (const [CFE_MSG_Message_t](#) *MsgPtr, [CFE_MSG_SequenceCount_t](#) *SeqCnt)
Gets the message sequence count.
- [CFE_Status_t CFE_MSG_SetSequenceCount](#) ([CFE_MSG_Message_t](#) *MsgPtr, [CFE_MSG_SequenceCount_t](#) SeqCnt)

- Sets the message sequence count.*

 - `CFE_MSG_SequenceCount_t CFE_MSG_GetNextSequenceCount (CFE_MSG_SequenceCount_t SeqCnt)`

Gets the next sequence count value (rolls over if appropriate)
- `CFE_Status_t CFE_MSG_GetEDSVersion (const CFE_MSG_Message_t *MsgPtr, CFE_MSG_EDSVersion_t *Version)`

Gets the message EDS version.
- `CFE_Status_t CFE_MSG_SetEDSVersion (CFE_MSG_Message_t *MsgPtr, CFE_MSG_EDSVersion_t Version)`

Sets the message EDS version.
- `CFE_Status_t CFE_MSG_GetEndian (const CFE_MSG_Message_t *MsgPtr, CFE_MSG_Endian_t *Endian)`

Gets the message endian.
- `CFE_Status_t CFE_MSG_SetEndian (CFE_MSG_Message_t *MsgPtr, CFE_MSG_Endian_t Endian)`

Sets the message endian.
- `CFE_Status_t CFE_MSG_GetPlaybackFlag (const CFE_MSG_Message_t *MsgPtr, CFE_MSG_PlaybackFlag_t *PlayFlag)`

Gets the message playback flag.
- `CFE_Status_t CFE_MSG_SetPlaybackFlag (CFE_MSG_Message_t *MsgPtr, CFE_MSG_PlaybackFlag_t PlayFlag)`

Sets the message playback flag.
- `CFE_Status_t CFE_MSG_GetSubsystem (const CFE_MSG_Message_t *MsgPtr, CFE_MSG_Subsystem_t *Subsystem)`

Gets the message subsystem.
- `CFE_Status_t CFE_MSG_SetSubsystem (CFE_MSG_Message_t *MsgPtr, CFE_MSG_Subsystem_t Subsystem)`

Sets the message subsystem.
- `CFE_Status_t CFE_MSG_GetSystem (const CFE_MSG_Message_t *MsgPtr, CFE_MSG_System_t *System)`

Gets the message system.
- `CFE_Status_t CFE_MSG_SetSystem (CFE_MSG_Message_t *MsgPtr, CFE_MSG_System_t System)`

Sets the message system.
- `CFE_Status_t CFE_MSG_GenerateChecksum (CFE_MSG_Message_t *MsgPtr)`

Calculates and sets the checksum of a message.
- `CFE_Status_t CFE_MSG_ValidateChecksum (const CFE_MSG_Message_t *MsgPtr, bool *IsValid)`

Validates the checksum of a message.
- `CFE_Status_t CFE_MSG_SetFcnCode (CFE_MSG_Message_t *MsgPtr, CFE_MSG_FcnCode_t FcnCode)`

Sets the function code field in a message.
- `CFE_Status_t CFE_MSG_GetFcnCode (const CFE_MSG_Message_t *MsgPtr, CFE_MSG_FcnCode_t *FcnCode)`

Gets the function code field from a message.
- `CFE_Status_t CFE_MSG_GetMsgTime (const CFE_MSG_Message_t *MsgPtr, CFE_TIME_SysTime_t *Time)`

Gets the time field from a message.
- `CFE_Status_t CFE_MSG_SetMsgTime (CFE_MSG_Message_t *MsgPtr, CFE_TIME_SysTime_t NewTime)`

Sets the time field in a message.
- `CFE_Status_t CFE_MSG_GetMsgId (const CFE_MSG_Message_t *MsgPtr, CFE_SB_MsgId_t *MsgId)`

Gets the message id from a message.
- `CFE_Status_t CFE_MSG_SetMsgId (CFE_MSG_Message_t *MsgPtr, CFE_SB_MsgId_t MsgId)`

Sets the message id bits in a message.
- `CFE_Status_t CFE_MSG_GetTypeFromMsgId (CFE_SB_MsgId_t MsgId, CFE_MSG_Type_t *Type)`

Gets message type using message ID.

- `CFE_Status_t CFE_MSG_OriginationAction` (`CFE_MSG_Message_t *MsgPtr`, `size_t BufferSize`, `bool *IsAcceptable`)
Perform any necessary actions on a newly-created message, prior to sending.
- `CFE_Status_t CFE_MSG_VerificationAction` (`const CFE_MSG_Message_t *MsgPtr`, `size_t BufferSize`, `bool *IsAcceptable`)
Checks message integrity/acceptability.

11.73.1 Detailed Description

Message access APIs

Definition in file [cfe_msg.h](#).

11.74 cfe_msg.h

[Go to the documentation of this file.](#)

```

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00016  * limitations under the License.
00017  *****/
00018
00025 #ifndef CFE_MSG_H
00026 #define CFE_MSG_H
00027
00028 /*
00029  * Includes
00030  */
00031 #include "common_types.h"
00032 #include "cfe_error.h"
00033 #include "cfe_msg_hdr.h"
00034 #include "cfe_msg_api_typedefs.h"
00035
00036 #include "cfe_es_api_typedefs.h"
00037 #include "cfe_sb_api_typedefs.h"
00038 #include "cfe_time_api_typedefs.h"
00039
00044 /*****/
00061 CFE_Status_t CFE_MSG_Init(CFE_MSG_Message_t *MsgPtr, CFE_SB_MsgId_t MsgId, CFE_MSG_Size_t Size);
00062
00069 /*****/
00083 CFE_Status_t CFE_MSG_GetSize(const CFE_MSG_Message_t *MsgPtr, CFE_MSG_Size_t *Size);
00084
00085 /*****/
00099 CFE_Status_t CFE_MSG_SetSize(CFE_MSG_Message_t *MsgPtr, CFE_MSG_Size_t Size);
00100
00101 /*****/
00115 CFE_Status_t CFE_MSG_GetType(const CFE_MSG_Message_t *MsgPtr, CFE_MSG_Type_t *Type);
00116
00117 /*****/
00131 CFE_Status_t CFE_MSG_SetType(CFE_MSG_Message_t *MsgPtr, CFE_MSG_Type_t Type);
00132
00133 /*****/
00147 CFE_Status_t CFE_MSG_GetHeaderVersion(const CFE_MSG_Message_t *MsgPtr, CFE_MSG_HeaderVersion_t *Version);
00148
00149 /*****/
00164 CFE_Status_t CFE_MSG_SetHeaderVersion(CFE_MSG_Message_t *MsgPtr, CFE_MSG_HeaderVersion_t Version);
00165
00166 /*****/
00180 CFE_Status_t CFE_MSG_GetHasSecondaryHeader(const CFE_MSG_Message_t *MsgPtr, bool *HasSecondary);
00181
00182 /*****/

```

```
00197 CFE_Status_t CFE_MSG_SetHasSecondaryHeader(CFE_MSG_Message_t *MsgPtr, bool HasSecondary);
00198
00199 /*****/
00213 CFE_Status_t CFE_MSG_GetApId(const CFE_MSG_Message_t *MsgPtr, CFE_MSG_ApId_t *ApId);
00214
00215 /*****/
00231 CFE_Status_t CFE_MSG_SetApId(CFE_MSG_Message_t *MsgPtr, CFE_MSG_ApId_t ApId);
00232
00233 /*****/
00247 CFE_Status_t CFE_MSG_GetSegmentationFlag(const CFE_MSG_Message_t *MsgPtr, CFE_MSG_SegmentationFlag_t
*SegFlag);
00248
00249 /*****/
00263 CFE_Status_t CFE_MSG_SetSegmentationFlag(CFE_MSG_Message_t *MsgPtr, CFE_MSG_SegmentationFlag_t SegFlag);
00264
00265 /*****/
00279 CFE_Status_t CFE_MSG_GetSequenceCount(const CFE_MSG_Message_t *MsgPtr, CFE_MSG_SequenceCount_t *SeqCnt);
00280
00281 /*****/
00295 CFE_Status_t CFE_MSG_SetSequenceCount(CFE_MSG_Message_t *MsgPtr, CFE_MSG_SequenceCount_t SeqCnt);
00296
00297 /*****/
00311 CFE_MSG_SequenceCount_t CFE_MSG_GetNextSequenceCount(CFE_MSG_SequenceCount_t SeqCnt);
00318 /*****/
00332 CFE_Status_t CFE_MSG_GetEDSVersion(const CFE_MSG_Message_t *MsgPtr, CFE_MSG_EDSVersion_t *Version);
00333
00334 /*****/
00348 CFE_Status_t CFE_MSG_SetEDSVersion(CFE_MSG_Message_t *MsgPtr, CFE_MSG_EDSVersion_t Version);
00349
00350 /*****/
00364 CFE_Status_t CFE_MSG_GetEndian(const CFE_MSG_Message_t *MsgPtr, CFE_MSG_Endian_t *Endian);
00365
00366 /*****/
00381 CFE_Status_t CFE_MSG_SetEndian(CFE_MSG_Message_t *MsgPtr, CFE_MSG_Endian_t Endian);
00382
00383 /*****/
00397 CFE_Status_t CFE_MSG_GetPlaybackFlag(const CFE_MSG_Message_t *MsgPtr, CFE_MSG_PlaybackFlag_t *PlayFlag);
00398
00399 /*****/
00413 CFE_Status_t CFE_MSG_SetPlaybackFlag(CFE_MSG_Message_t *MsgPtr, CFE_MSG_PlaybackFlag_t PlayFlag);
00414
00415 /*****/
00429 CFE_Status_t CFE_MSG_GetSubsystem(const CFE_MSG_Message_t *MsgPtr, CFE_MSG_Subsystem_t *Subsystem);
00430
00431 /*****/
00447 CFE_Status_t CFE_MSG_SetSubsystem(CFE_MSG_Message_t *MsgPtr, CFE_MSG_Subsystem_t Subsystem);
00448
00449 /*****/
00463 CFE_Status_t CFE_MSG_GetSystem(const CFE_MSG_Message_t *MsgPtr, CFE_MSG_System_t *System);
00464
00465 /*****/
00481 CFE_Status_t CFE_MSG_SetSystem(CFE_MSG_Message_t *MsgPtr, CFE_MSG_System_t System);
00488 /*****/
00511 CFE_Status_t CFE_MSG_GenerateChecksum(CFE_MSG_Message_t *MsgPtr);
00512
00513 /*****/
00537 CFE_Status_t CFE_MSG_ValidateChecksum(const CFE_MSG_Message_t *MsgPtr, bool *IsValid);
00538
00539 /*****/
00560 CFE_Status_t CFE_MSG_SetFcnCode(CFE_MSG_Message_t *MsgPtr, CFE_MSG_FcnCode_t FcnCode);
00561
00562 /*****/
00582 CFE_Status_t CFE_MSG_GetFcnCode(const CFE_MSG_Message_t *MsgPtr, CFE_MSG_FcnCode_t *FcnCode);
00583
00584 /*****/
00605 CFE_Status_t CFE_MSG_GetMsgTime(const CFE_MSG_Message_t *MsgPtr, CFE_TIME_SysTime_t *Time);
00606
00607 /*****/
00632 CFE_Status_t CFE_MSG_SetMsgTime(CFE_MSG_Message_t *MsgPtr, CFE_TIME_SysTime_t NewTime);
00633
00640 /*****/
00657 CFE_Status_t CFE_MSG_GetMsgId(const CFE_MSG_Message_t *MsgPtr, CFE_SB_MsgId_t *MsgId);
00658
00659 /*****/
00684 CFE_Status_t CFE_MSG_SetMsgId(CFE_MSG_Message_t *MsgPtr, CFE_SB_MsgId_t MsgId);
00685
00686 /*****/
00700 CFE_Status_t CFE_MSG_GetTypeFromMsgId(CFE_SB_MsgId_t MsgId, CFE_MSG_Type_t *Type);
00701
00708 /*****/
00737 CFE_Status_t CFE_MSG_OriginationAction(CFE_MSG_Message_t *MsgPtr, size_t BufferSize, bool *IsAcceptable);
```

```

00738
00739 /*****
00767 CFE_Status_t CFE_MSG_VerificationAction(const CFE_MSG_Message_t *MsgPtr, size_t BufferSize, bool
      *IsAcceptable);
00768
00771 #endif /* CFE_MSG_H */

```

11.75 cfe/modules/core_api/fsw/inc/cfe_msg_api_typedefs.h File Reference

```

#include "common_types.h"
#include "cfe_error.h"

```

Macros

- `#define CFE_MSG_BAD_ARGUMENT CFE_SB_BAD_ARGUMENT`
Error - bad argument.
- `#define CFE_MSG_NOT_IMPLEMENTED CFE_SB_NOT_IMPLEMENTED`
Error - not implemented.
- `#define CFE_MSG_WRONG_MSG_TYPE CFE_SB_WRONG_MSG_TYPE`
Error - wrong type.

Typedefs

- `typedef size_t CFE_MSG_Size_t`
Message size, note CCSDS maximum is UINT16_MAX+7.
- `typedef uint32 CFE_MSG_Checksum_t`
Message checksum (Oversized to avoid redefine)
- `typedef uint16 CFE_MSG_FcnCode_t`
Message function code.
- `typedef uint16 CFE_MSG_HeaderVersion_t`
Message header version.
- `typedef uint16 CFE_MSG_ApId_t`
Message application ID.
- `typedef uint16 CFE_MSG_SequenceCount_t`
Message sequence count.
- `typedef uint16 CFE_MSG_EDSVersion_t`
Message EDS version.
- `typedef uint16 CFE_MSG_Subsystem_t`
Message subsystem.
- `typedef uint16 CFE_MSG_System_t`
Message system.
- `typedef enum CFE_MSG_Type CFE_MSG_Type_t`
Message type.
- `typedef enum CFE_MSG_SegmentationFlag CFE_MSG_SegmentationFlag_t`
Segmentation flags.
- `typedef enum CFE_MSG_Endian CFE_MSG_Endian_t`
Endian flag.
- `typedef enum CFE_MSG_PlaybackFlag CFE_MSG_PlaybackFlag_t`
Playback flag.
- `typedef struct CFE_MSG_Message CFE_MSG_Message_t`

cFS generic base message

- typedef struct CFE_MSG_CommandHeader [CFE_MSG_CommandHeader_t](#)

cFS command header

- typedef struct CFE_MSG_TelemetryHeader [CFE_MSG_TelemetryHeader_t](#)

cFS telemetry header

Enumerations

- enum [CFE_MSG_Type](#) { [CFE_MSG_Type_Invalid](#) , [CFE_MSG_Type_Cmd](#) , [CFE_MSG_Type_Tlm](#) }
Message type.
- enum [CFE_MSG_SegmentationFlag](#) {
 [CFE_MSG_SegFlag_Invalid](#) , [CFE_MSG_SegFlag_Continue](#) , [CFE_MSG_SegFlag_First](#) , [CFE_MSG_SegFlag_Last](#)
 ,
 [CFE_MSG_SegFlag_Unsegmented](#) }
Segmentation flags.
- enum [CFE_MSG_Endian](#) { [CFE_MSG_Endian_Invalid](#) , [CFE_MSG_Endian_Big](#) , [CFE_MSG_Endian_Little](#) }
Endian flag.
- enum [CFE_MSG_PlaybackFlag](#) { [CFE_MSG_PlayFlag_Invalid](#) , [CFE_MSG_PlayFlag_Original](#) , [CFE_MSG_PlayFlag_Playback](#) }
Playback flag.

11.75.1 Detailed Description

Typedefs for Message API

- Separate from API so these can be adjusted for custom implementations

Definition in file [cfe_msg_api_typedefs.h](#).

11.75.2 Macro Definition Documentation

CFE_MSG_BAD_ARGUMENT

```
#define CFE_MSG_BAD_ARGUMENT CFE_SB_BAD_ARGUMENT
```

Error - bad argument.

Definition at line 39 of file [cfe_msg_api_typedefs.h](#).

CFE_MSG_NOT_IMPLEMENTED

```
#define CFE_MSG_NOT_IMPLEMENTED CFE_SB_NOT_IMPLEMENTED
```

Error - not implemented.

Definition at line 40 of file [cfe_msg_api_typedefs.h](#).

CFE_MSG_WRONG_MSG_TYPE

```
#define CFE_MSG_WRONG_MSG_TYPE CFE_SB_WRONG_MSG_TYPE
```

Error - wrong type.

Definition at line 41 of file [cfe_msg_api_typedefs.h](#).

11.75.3 Typedef Documentation

CFE_MSG_ApId_t

```
typedef uint16 CFE_MSG_ApId_t
```

Message application ID.

Definition at line 50 of file [cfe_msg_api_typedefs.h](#).

CFE_MSG_Checksum_t

```
typedef uint32 CFE_MSG_Checksum_t
```

Message checksum (Oversized to avoid redefine)

Definition at line 47 of file [cfe_msg_api_typedefs.h](#).

CFE_MSG_CommandHeader_t

```
typedef struct CFE_MSG_CommandHeader CFE_MSG_CommandHeader_t
```

cFS command header

Definition at line 107 of file [cfe_msg_api_typedefs.h](#).

CFE_MSG_EDSVersion_t

```
typedef uint16 CFE_MSG_EDSVersion_t
```

Message EDS version.

Definition at line 52 of file [cfe_msg_api_typedefs.h](#).

CFE_MSG_Endian_t

```
typedef enum CFE_MSG_Endian CFE_MSG_Endian_t
```

Endian flag.

CFE_MSG_FcnCode_t

```
typedef uint16 CFE_MSG_FcnCode_t
```

Message function code.

Definition at line 48 of file [cfe_msg_api_typedefs.h](#).

CFE_MSG_HeaderVersion_t

```
typedef uint16 CFE_MSG_HeaderVersion_t
```

Message header version.

Definition at line 49 of file [cfe_msg_api_typedefs.h](#).

CFE_MSG_Message_t

```
typedef struct CFE_MSG_Message CFE_MSG_Message_t
```

cFS generic base message

Definition at line 102 of file [cfe_msg_api_typedefs.h](#).

CFE_MSG_PlaybackFlag_t

```
typedef enum CFE_MSG_PlaybackFlag CFE_MSG_PlaybackFlag_t
```

Playback flag.

CFE_MSG_SegmentationFlag_t

```
typedef enum CFE_MSG_SegmentationFlag CFE_MSG_SegmentationFlag_t
```

Segmentation flags.

CFE_MSG_SequenceCount_t

```
typedef uint16 CFE_MSG_SequenceCount_t
```

Message sequence count.

Definition at line 51 of file [cfe_msg_api_typedefs.h](#).

CFE_MSG_Size_t

```
typedef size_t CFE_MSG_Size_t
```

Message size, note CCSDS maximum is UINT16_MAX+7.

Definition at line 46 of file [cfe_msg_api_typedefs.h](#).

CFE_MSG_Subsystem_t

```
typedef uint16 CFE_MSG_Subsystem_t
```

Message subsystem.

Definition at line 53 of file [cfe_msg_api_typedefs.h](#).

CFE_MSG_System_t

```
typedef uint16 CFE_MSG_System_t
```

Message system.

Definition at line 54 of file [cfe_msg_api_typedefs.h](#).

CFE_MSG_TelemetryHeader_t

```
typedef struct CFE_MSG_TelemetryHeader CFE_MSG_TelemetryHeader_t
```

cFS telemetry header

Definition at line 112 of file [cfe_msg_api_typedefs.h](#).

CFE_MSG_Type_t

```
typedef enum CFE_MSG_Type CFE_MSG_Type_t
```

Message type.

11.75.4 Enumeration Type Documentation**CFE_MSG_Endian**

```
enum CFE_MSG_Endian
```

Endian flag.

Enumerator

CFE_MSG_Endian_Invalid	Invalid endian setting.
CFE_MSG_Endian_Big	Big endian.
CFE_MSG_Endian_Little	Little endian.

Definition at line 75 of file [cfe_msg_api_typedefs.h](#).

CFE_MSG_PlaybackFlag

```
enum CFE_MSG_PlaybackFlag
```

Playback flag.

Enumerator

CFE_MSG_PlayFlag_Invalid	Invalid playback setting.
CFE_MSG_PlayFlag_Original	Original.
CFE_MSG_PlayFlag_Playback	Playback.

Definition at line 83 of file [cfe_msg_api_typedefs.h](#).

CFE_MSG_SegmentationFlag

enum [CFE_MSG_SegmentationFlag](#)

Segmentation flags.

Enumerator

CFE_MSG_SegFlag_Invalid	Invalid segmentation flag.
CFE_MSG_SegFlag_Continue	Continuation segment of User Data.
CFE_MSG_SegFlag_First	First segment of User Data.
CFE_MSG_SegFlag_Last	Last segment of User Data.
CFE_MSG_SegFlag_Unsegmented	Unsegmented data.

Definition at line 65 of file [cfe_msg_api_typedefs.h](#).

CFE_MSG_Type

enum [CFE_MSG_Type](#)

Message type.

Enumerator

CFE_MSG_Type_Invalid	Message type invalid, undefined, not implemented.
CFE_MSG_Type_Cmd	Command message type.
CFE_MSG_Type_Tlm	Telemetry message type.

Definition at line 57 of file [cfe_msg_api_typedefs.h](#).

11.76 cfe_msg_api_typedefs.h

[Go to the documentation of this file.](#)

```

00001 /*****
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"
00003  *
00004  * Copyright (c) 2023 United States Government as represented by the
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00008  * Licensed under the Apache License, Version 2.0 (the "License"); you may
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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00026 #ifndef CFE_MSG_API_TYPEDEFS_H
00027 #define CFE_MSG_API_TYPEDEFS_H
00028
00029 /*
00030  * Includes
00031  */
00032 #include "common_types.h"
00033 #include "cfe_error.h"
00034
00035 /*
00036  * Defines

```



```

00037  */
00038  /* Error bits won't fit a new module so reuse SB error codes */
00039  #define CFE_MSG_BAD_ARGUMENT      CFE_SB_BAD_ARGUMENT
00040  #define CFE_MSG_NOT_IMPLEMENTED   CFE_SB_NOT_IMPLEMENTED
00041  #define CFE_MSG_WRONG_MSG_TYPE    CFE_SB_WRONG_MSG_TYPE
00043  /*
00044   * Types
00045   */
00046  typedef size_t CFE_MSG_Size_t;
00047  typedef uint32 CFE_MSG_Checksum_t;
00048  typedef uint16 CFE_MSG_FcnCode_t;
00049  typedef uint16 CFE_MSG_HeaderVersion_t;
00050  typedef uint16 CFE_MSG_ApId_t;
00051  typedef uint16 CFE_MSG_SequenceCount_t;
00052  typedef uint16 CFE_MSG_EDSVersion_t;
00053  typedef uint16 CFE_MSG_Subsystem_t;
00054  typedef uint16 CFE_MSG_System_t;
00057  typedef enum CFE_MSG_Type
00058  {
00059      CFE_MSG_Type_Invalid,
00060      CFE_MSG_Type_Cmd,
00061      CFE_MSG_Type_Tlm
00062  } CFE_MSG_Type_t;
00063
00065  typedef enum CFE_MSG_SegmentationFlag
00066  {
00067      CFE_MSG_SegFlag_Invalid,
00068      CFE_MSG_SegFlag_Continue,
00069      CFE_MSG_SegFlag_First,
00070      CFE_MSG_SegFlag_Last,
00071      CFE_MSG_SegFlag_Unsegmented
00072  } CFE_MSG_SegmentationFlag_t;
00073
00075  typedef enum CFE_MSG_Endian
00076  {
00077      CFE_MSG_Endian_Invalid,
00078      CFE_MSG_Endian_Big,
00079      CFE_MSG_Endian_Little
00080  } CFE_MSG_Endian_t;
00081
00083  typedef enum CFE_MSG_PlaybackFlag
00084  {
00085      CFE_MSG_PlayFlag_Invalid,
00086      CFE_MSG_PlayFlag_Original,
00087      CFE_MSG_PlayFlag_Playback
00088  } CFE_MSG_PlaybackFlag_t;
00089
00090  /*
00091   * Abstract Message Base Types
00092   *
00093   * The concrete definition of these is provided by "cfe_msg_hdr.h" which is
00094   * user-selectable depending on the actual desired message definition. These
00095   * abstract types are used in the API definition; the API is defined based
00096   * on these abstract types, independent of the actual message definition.
00097   */
00098
00102  typedef struct CFE_MSG_Message CFE_MSG_Message_t;
00103
00107  typedef struct CFE_MSG_CommandHeader CFE_MSG_CommandHeader_t;
00108
00112  typedef struct CFE_MSG_TelemetryHeader CFE_MSG_TelemetryHeader_t;
00113
00114  #endif /* CFE_MSG_API_TYPEDEFS_H */

```

11.77 cfe/modules/core_api/fsw/inc/cfe_resourceid.h File Reference

```
#include "cfe_resourceid_api_typedefs.h"
```

Typedefs

- typedef CFE_ResourceId_t(* CFE_ResourceId_IncrementFunc_t) (CFE_ResourceId_t, void *)
Serial number increment function.
- typedef bool(* CFE_ResourceId_CheckFunc_t) (CFE_ResourceId_t)
Serial number availability check function.

Functions

- [uint32 CFE_ResourceId_GetBase](#) (CFE_ResourceId_t ResourceId)
Get the Base value (type/category) from a resource ID value.
- [uint32 CFE_ResourceId_GetSerial](#) (CFE_ResourceId_t ResourceId)
Get the Serial Number (sequential ID) from a resource ID value.
- CFE_ResourceId_t [CFE_ResourceId_FindNext](#) (CFE_ResourceId_t StartId, [uint32](#) TableSize, [CFE_ResourceId_CheckFunc_t](#) CheckFunc)
Locate the next resource ID that maps to an available table entry.
- CFE_ResourceId_t [CFE_ResourceId_FindNextEx](#) (CFE_ResourceId_t StartId, [CFE_ResourceId_IncrementFunc_t](#) IncrFunc, void *IncrArg, [CFE_ResourceId_CheckFunc_t](#) CheckFunc)
Locate the next resource ID that maps to an available table entry.
- [int32 CFE_ResourceId_ToIndex](#) (CFE_ResourceId_t Id, [uint32](#) BaseValue, [uint32](#) TableSize, [uint32](#) *Idx)
Internal routine to aid in converting an ES resource ID to an array index.

Resource ID test/conversion macros and inline functions

- [#define CFE_RESOURCEID_TO_ULONG\(id\) CFE_ResourceId_ToInteger\(CFE_RESOURCEID_UNWRAP\(id\)\)](#)
Convert a derived (app-specific) ID directly into an "unsigned long".
- [#define CFE_RESOURCEID_TEST_DEFINED\(id\) CFE_ResourceId_IsDefined\(CFE_RESOURCEID_UNWRAP\(id\)\)](#)
Determine if a derived (app-specific) ID is defined or not.
- [#define CFE_RESOURCEID_TEST_EQUAL\(id1, id2\) CFE_ResourceId_Equal\(CFE_RESOURCEID_UNWRAP\(id1\), CFE_RESOURCEID_UNWRAP\(id2\)\)](#)
Determine if two derived (app-specific) IDs are equal.
- static unsigned long [CFE_ResourceId_ToInteger](#) (CFE_ResourceId_t id)
Convert a resource ID to an integer.
- static CFE_ResourceId_t [CFE_ResourceId_FromInteger](#) (unsigned long Value)
Convert an integer to a resource ID.
- static bool [CFE_ResourceId_Equal](#) (CFE_ResourceId_t id1, CFE_ResourceId_t id2)
Compare two Resource ID values for equality.
- static bool [CFE_ResourceId_IsDefined](#) (CFE_ResourceId_t id)
Check if a resource ID value is defined.

11.77.1 Detailed Description

Contains global prototypes and definitions related to resource management and related CFE resource IDs.

A CFE ES Resource ID is a common way to identify CFE-managed resources such as apps, tasks, counters, memory pools, CDS blocks, and other entities.

Simple operations are provided as inline functions, which should alleviate the need to do direct manipulation of resource IDs:

- Check for undefined ID value
- Check for equality of two ID values
- Convert ID to simple integer (typically for printing/logging)
- Convert simple integer to ID (inverse of above)

Definition in file [cfe_resourceid.h](#).

11.77.2 Macro Definition Documentation

CFE_RESOURCEID_TEST_DEFINED

```
#define CFE_RESOURCEID_TEST_DEFINED(  
    id ) CFE_ResourceId_IsDefined(CFE_RESOURCEID_UNWRAP(id))
```

Determine if a derived (app-specific) ID is defined or not.

This generic routine is implemented as a macro so it is agnostic to the actual argument type, and it will evaluate correctly so long as the argument type is based on the CFE_RESOURCEID_BASE_TYPE.

Definition at line 97 of file [cfe_resourceid.h](#).

CFE_RESOURCEID_TEST_EQUAL

```
#define CFE_RESOURCEID_TEST_EQUAL(  
    id1,  
    id2 ) CFE_ResourceId_Equal(CFE_RESOURCEID_UNWRAP(id1), CFE_RESOURCEID_UNWRAP(id2))
```

Determine if two derived (app-specific) IDs are equal.

This generic routine is implemented as a macro so it is agnostic to the actual argument type, and it will evaluate correctly so long as the argument type is based on the CFE_RESOURCEID_BASE_TYPE.

Definition at line 105 of file [cfe_resourceid.h](#).

CFE_RESOURCEID_TO_ULONG

```
#define CFE_RESOURCEID_TO_ULONG(  
    id ) CFE_ResourceId_ToInteger(CFE_RESOURCEID_UNWRAP(id))
```

Convert a derived (app-specific) ID directly into an "unsigned long".

This generic routine is implemented as a macro so it is agnostic to the actual argument type, and it will evaluate correctly so long as the argument type is based on the CFE_RESOURCEID_BASE_TYPE.

There is no inverse of this macro, as it depends on the actual derived type desired. Applications needing to recreate an ID from an integer should use [CFE_ResourceId_FromInteger\(\)](#) combined with a cast/conversion to the correct/intended derived type, as needed.

Note

This evaluates as an "unsigned long" such that it can be used in printf()-style functions with the "%lx" modifier without extra casting, as this is the most typical use-case for representing an ID as an integer.

Definition at line 89 of file [cfe_resourceid.h](#).

11.77.3 Typedef Documentation

CFE_ResourceId_CheckFunc_t

```
typedef bool(* CFE_ResourceId_CheckFunc_t) (CFE_ResourceId_t)
```

Serial number availability check function.

Checks if the slot associated with a pending serial number is in use or not. Used with [CFE_ResourceId_FindNext\(\)](#) to find the next available serial number.

Return values

<i>true</i>	if the slot is already in use (unavailable)
<i>false</i>	if the slot is not in use (available)

Definition at line 70 of file [cfe_resourceid.h](#).

CFE_ResourceId_IncrementFunc_t

```
typedef CFE_ResourceId_t (* CFE_ResourceId_IncrementFunc_t) (CFE_ResourceId_t, void *)
```

Serial number increment function.

A helper function responsible for incrementing the serial number when iterating over all available resource slots. The default implementation of this function will treat all slots as equal, and simply increment to the next serial number.

An alternative function can be used with [CFE_ResourceId_FindNextEx\(\)](#) if there are special requirements for slot assignments/relationships.

Returns

Next serial number to check/test

Return values

<code>CFE_RESOURCEID_UNDEFINED</code>	if no more IDs are available to test
---------------------------------------	--------------------------------------

Definition at line 58 of file [cfe_resourceid.h](#).

11.77.4 Function Documentation**CFE_ResourceId_Equal()**

```
static bool CFE_ResourceId_Equal (
    CFE_ResourceId_t id1,
    CFE_ResourceId_t id2 ) [inline], [static]
```

Compare two Resource ID values for equality.

Parameters

in	<i>id1</i>	Resource ID to check
in	<i>id2</i>	Resource ID to check

Returns

true if id1 and id2 are equal, false otherwise.

Definition at line 160 of file [cfe_resourceid.h](#).

Referenced by [CFE_ResourceId_IsDefined\(\)](#).

CFE_ResourceId_FindNext()

```
CFE_ResourceId_t CFE_ResourceId_FindNext (
    CFE_ResourceId_t StartId,
    uint32 TableSize,
    CFE_ResourceId_CheckFunc_t CheckFunc )
```

Locate the next resource ID that maps to an available table entry.

This begins searching from StartId which should be the most recently issued ID for the resource category. This will then search for the next ID that maps to a table entry that is available for use. That is, it does not alias any valid/in-use ID when converted to an array index.

This is the simple form of the API that iterates over all slots equally in a round-robin fashion, and works for most use cases.

returns an undefined ID value if no open slots are available

Parameters

in	<i>StartId</i>	the last issued ID for the resource category (app, lib, etc).
in	<i>TableSize</i>	the maximum size of the target table
in	<i>CheckFunc</i>	a function to check if the given ID is available

Returns

Next ID value which does not map to a valid entry

Return values

<i>CFE_RESOURCEID_UNDEFINED</i>	if no open slots or bad arguments.
---	------------------------------------

CFE_ResourceId_FindNextEx()

```
CFE_ResourceId_t CFE_ResourceId_FindNextEx (
    CFE_ResourceId_t StartId,
    CFE_ResourceId_IncrementFunc_t IncrFunc,
    void * IncrArg,
    CFE_ResourceId_CheckFunc_t CheckFunc )
```

Locate the next resource ID that maps to an available table entry.

An extended form of [CFE_ResourceId_FindNext\(\)](#) that allows more control over the slots that are checked. This can be used if slots are not all equivalent and thus the simple round-robin approach is insufficient. The additional increment function should return the next ID to test/check, given the previous ID.

returns an undefined ID value if no open slots are available

Parameters

in	<i>StartId</i>	the last issued ID for the resource category (app, lib, etc).
in	<i>IncrFunc</i>	function to increment the id to the next value
in	<i>IncrArg</i>	opaque argument that is passed through to the increment function
in	<i>CheckFunc</i>	a function to check if the given ID is available

Returns

Next ID value which does not map to a valid entry

Return values

<i>CFE_RESOURCEID_UNDEFINED</i>	if no open slots or bad arguments.
---	------------------------------------

CFE_ResourceId_FromInteger()

```
static CFE_ResourceId_t CFE_ResourceId_FromInteger (
    unsigned long Value ) [inline], [static]
```

Convert an integer to a resource ID.

This is the inverse of [CFE_ResourceId_ToInteger\(\)](#), and reconstitutes the original CFE_ResourceId_t value from the integer representation.

This may be used, for instance, where an ID value is parsed from a text file or message using C library APIs such as `scanf()` or `strtoul()`.

See also

[CFE_ResourceId_ToInteger\(\)](#)

Parameters

in	<i>Value</i>	Integer value to convert
----	--------------	--------------------------

Returns

ID value corresponding to integer

Definition at line 148 of file [cfe_resourceid.h](#).

CFE_ResourceId_GetBase()

```
uint32 CFE_ResourceId_GetBase (
    CFE_ResourceId_t ResourceId )
```

Get the Base value (type/category) from a resource ID value.

This masks out the ID serial number to obtain the base value, which is different for each resource type.

Note

The value is NOT shifted or otherwise adjusted.

Parameters

in	<i>ResourceId</i>	the resource ID to decode
----	-------------------	---------------------------

Returns

The base value associated with that ID

CFE_ResourceId_GetSerial()

```
uint32 CFE_ResourceId_GetSerial (
    CFE_ResourceId_t ResourceId )
```

Get the Serial Number (sequential ID) from a resource ID value.

This masks out the ID base value to obtain the serial number, which is different for each entity created.

Parameters

in	<i>ResourceId</i>	the resource ID to decode
----	-------------------	---------------------------

Returns

The serial number associated with that ID

CFE_ResourceId_IsDefined()

```
static bool CFE_ResourceId_IsDefined (
    CFE_ResourceId_t id ) [inline], [static]
```

Check if a resource ID value is defined.

The constant [CFE_RESOURCEID_UNDEFINED](#) represents an undefined ID value, such that the expression:

```
CFE_ResourceId_IsDefined(CFE_RESOURCEID_UNDEFINED)
```

Always returns false.

Parameters

in	<i>id</i>	Resource ID to check
----	-----------	----------------------

Returns

True if the ID may refer to a defined entity, false if invalid/undefined.

Definition at line 178 of file [cfe_resourceid.h](#).

References [CFE_ResourceId_Equal\(\)](#), and [CFE_RESOURCEID_UNDEFINED](#).

Here is the call graph for this function:

**CFE_ResourceId_ToIndex()**

```
int32 CFE_ResourceId_ToIndex (
    CFE_ResourceId_t Id,
    uint32 BaseValue,
    uint32 TableSize,
    uint32 * Idx )
```

Internal routine to aid in converting an ES resource ID to an array index.

Parameters

in	<i>Id</i>	The resource ID
in	<i>BaseValue</i>	The respective ID base value corresponding to the ID type
in	<i>TableSize</i>	The actual size of the internal table (MAX index value + 1)
out	<i>Idx</i>	The output index

Returns

Execution status, see [cFE Return Code Defines](#)

Return values

<i>CFE_SUCCESS</i>	Successful execution.
<i>CFE_ES_BAD_ARGUMENT</i>	Bad Argument.
<i>CFE_ES_ERR_RESOURCEID_NOT_VALID</i>	Resource ID is not valid.

CFE_ResourceId_ToInteger()

```
static unsigned long CFE_ResourceId_ToInteger (
    CFE_ResourceId_t id ) [inline], [static]
```

Convert a resource ID to an integer.

This is primarily intended for logging purposes, such as writing to debug console, event messages, or log files, using printf-like APIs.

For compatibility with C library APIs, this returns an "unsigned long" type and should be used with the "%lx" format specifier in a printf format string.

Note

No assumptions should be made about the actual integer value, such as its base/range. It may be printed, but should not be modified or tested/compared using other arithmetic ops, and should never be used as the index to an array or table. See the related function [*CFE_ResourceId_ToIndex\(\)*](#) for cases where a zero-based array/table index is needed.

See also

[*CFE_ResourceId_FromInteger\(\)*](#)

Parameters

in	<i>id</i>	Resource ID to convert
----	-----------	------------------------

Returns

Integer value corresponding to ID

Definition at line 129 of file [cfe_resourceid.h](#).

11.78 cfe_resourceid.h

[Go to the documentation of this file.](#)

```
00001 /*****
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"
00003  *
00004  * Copyright (c) 2023 United States Government as represented by the
00005  * Administrator of the National Aeronautics and Space Administration.
00006  * All Rights Reserved.
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00008  * Licensed under the Apache License, Version 2.0 (the "License"); you may
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00010  * a copy of the License at http://www.apache.org/licenses/LICENSE-2.0
00011  *
00012  * Unless required by applicable law or agreed to in writing, software
00013  * distributed under the License is distributed on an "AS IS" BASIS,
00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00037 #ifndef CFE_RESOURCEID_H
00038 #define CFE_RESOURCEID_H
```



```

00039
00040 /*
00041  * The basic resource ID API definitions
00042  */
00043 #include "cfe_resourceid_api_typedefs.h"
00044
00058 typedef CFE_ResourceId_t (*CFE_ResourceId_IncrementFunc_t) (CFE_ResourceId_t, void *);
00059
00070 typedef bool (*CFE_ResourceId_CheckFunc_t) (CFE_ResourceId_t);
00071
00089 #define CFE_RESOURCEID_TO_ULONG(id) CFE_ResourceId_ToInteger(CFE_RESOURCEID_UNWRAP(id))
00090
00097 #define CFE_RESOURCEID_TEST_DEFINED(id) CFE_ResourceId_IsDefined(CFE_RESOURCEID_UNWRAP(id))
00098
00105 #define CFE_RESOURCEID_TEST_EQUAL(id1, id2) CFE_ResourceId_Equal(CFE_RESOURCEID_UNWRAP(id1),
    CFE_RESOURCEID_UNWRAP(id2))
00106
00129 static inline unsigned long CFE_ResourceId_ToInteger(CFE_ResourceId_t id)
00130 {
00131     return (unsigned long)CFE_RESOURCEID_UNWRAP(id);
00132 }
00133
00148 static inline CFE_ResourceId_t CFE_ResourceId_FromInteger(unsigned long Value)
00149 {
00150     return (CFE_ResourceId_t)CFE_RESOURCEID_WRAP(Value);
00151 }
00152
00160 static inline bool CFE_ResourceId_Equal(CFE_ResourceId_t id1, CFE_ResourceId_t id2)
00161 {
00162     return (CFE_RESOURCEID_UNWRAP(id1) == CFE_RESOURCEID_UNWRAP(id2));
00163 }
00164
00178 static inline bool CFE_ResourceId_IsDefined(CFE_ResourceId_t id)
00179 {
00180     return (!CFE_ResourceId_Equal(id, CFE_RESOURCEID_UNDEFINED));
00181 }
00182
00185 /*
00186  * Non-inline API functions provided by the Resource ID module
00187  */
00188
00200 uint32 CFE_ResourceId_GetBase(CFE_ResourceId_t ResourceId);
00201
00211 uint32 CFE_ResourceId_GetSerial(CFE_ResourceId_t ResourceId);
00212
00233 CFE_ResourceId_t
00234 CFE_ResourceId_FindNext(CFE_ResourceId_t StartId, uint32 TableSize, CFE_ResourceId_CheckFunc_t CheckFunc);
00235
00254 CFE_ResourceId_t CFE_ResourceId_FindNextEx(CFE_ResourceId_t      StartId,
00255                                             CFE_ResourceId_IncrementFunc_t IncrFunc,
00256                                             void *IncrArg,
00257                                             CFE_ResourceId_CheckFunc_t CheckFunc);
00258
00272 int32 CFE_ResourceId_ToIndex(CFE_ResourceId_t Id, uint32 BaseValue, uint32 TableSize, uint32 *Idx);
00273
00274 #endif /* CFE_RESOURCEID_H */

```

11.79 cfe/modules/core_api/fsw/inc/cfe_resourceid_api_typedefs.h File Reference

```
#include "cfe_resourceid_typedef.h"
```

Macros

Resource ID predefined values

- #define **CFE_RESOURCEID_UNDEFINED** ((CFE_ResourceId_t)CFE_RESOURCEID_WRAP(0))
A resource ID value that represents an undefined/unused resource.
- #define **CFE_RESOURCEID_RESERVED** ((CFE_ResourceId_t)CFE_RESOURCEID_WRAP(0xFFFFFFFF))
A resource ID value that represents a reserved entry.

11.79.1 Detailed Description

Contains global prototypes and definitions related to resource management and related CFE resource IDs.

A CFE ES Resource ID is a common way to identify CFE-managed resources such as apps, tasks, counters, memory pools, CDS blocks, and other entities.

Simple operations are provided as inline functions, which should alleviate the need to do direct manipulation of resource IDs:

- Check for undefined ID value
- Check for equality of two ID values
- Convert ID to simple integer (typically for printing/logging)
- Convert simple integer to ID (inverse of above)

Definition in file [cfe_resourceid_api_typedefs.h](#).

11.79.2 Macro Definition Documentation

CFE_RESOURCEID_RESERVED

```
#define CFE_RESOURCEID_RESERVED ((CFE_ResourceId_t)CFE_RESOURCEID_WRAP(0xFFFFFFFF))
```

A resource ID value that represents a reserved entry.

This is not a valid value for any resource type, but is used to mark table entries that are not available for use. For instance, this may be used while setting up an entry initially.

Definition at line 74 of file [cfe_resourceid_api_typedefs.h](#).

CFE_RESOURCEID_UNDEFINED

```
#define CFE_RESOURCEID_UNDEFINED ((CFE_ResourceId_t)CFE_RESOURCEID_WRAP(0))
```

A resource ID value that represents an undefined/unused resource.

This constant may be used to initialize local variables of the CFE_ResourceId_t type to a safe value that will not alias a valid ID.

By design, this value is also the result of zeroing a CFE_ResourceId_t type via standard functions like memset(), such that objects initialized using this method will also be set to safe values.

Definition at line 65 of file [cfe_resourceid_api_typedefs.h](#).

11.80 cfe_resourceid_api_typedefs.h

[Go to the documentation of this file.](#)

```
00001 /*****
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"
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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00037 #ifndef CFE_RESOURCEID_API_TYPEDEFS_H
00038 #define CFE_RESOURCEID_API_TYPEDEFS_H
00039
00040 /*
00041  * The basic resource ID typedef
00042  */
```

```

00043  * This is provided via the external resourceid library
00044  * and may be customized by the user/mission preferences.
00045  */
00046 #include "cfe_resourceid_typedef.h"
00047
00048 /*
00049 ** Defines
00050 */
00051
00065 #define CFE_RESOURCEID_UNDEFINED ((CFE_ResourceId_t)CFE_RESOURCEID_WRAP(0))
00066
00074 #define CFE_RESOURCEID_RESERVED ((CFE_ResourceId_t)CFE_RESOURCEID_WRAP(0xFFFFFFFF))
00075
00078 #endif /* CFE_RESOURCEID_API_TYPEDEFS_H */

```

11.81 cfe/modules/core_api/fsw/inc/cfe_sb.h File Reference

```

#include "common_types.h"
#include "cfe_error.h"
#include "cfe_sb_api_typedefs.h"
#include "cfe_es_api_typedefs.h"

```

Macros

- `#define CFE_BIT(x) (1 << (x))`
Places a one at bit positions 0 - 31.
- `#define CFE_SET(i, x) ((i) |= CFE_BIT(x))`
Sets bit x of i.
- `#define CFE_CLR(i, x) ((i) &= ~CFE_BIT(x))`
Clears bit x of i.
- `#define CFE_TST(i, x) (((i) & CFE_BIT(x)) != 0)`
true(non zero) if bit x of i is set

Functions

- `CFE_Status_t CFE_SB_CreatePipe (CFE_SB_Pipeld_t *PipeIdPtr, uint16 Depth, const char *PipeName)`
Creates a new software bus pipe.
- `CFE_Status_t CFE_SB_DeletePipe (CFE_SB_Pipeld_t PipeId)`
Delete a software bus pipe.
- `CFE_Status_t CFE_SB_Pipeld_ToIndex (CFE_SB_Pipeld_t PipeID, uint32 *Idx)`
Obtain an index value correlating to an SB Pipe ID.
- `CFE_Status_t CFE_SB_SetPipeOpts (CFE_SB_Pipeld_t PipeId, uint8 Opts)`
Set options on a pipe.
- `CFE_Status_t CFE_SB_GetPipeOpts (CFE_SB_Pipeld_t PipeId, uint8 *OptsPtr)`
Get options on a pipe.
- `CFE_Status_t CFE_SB_GetPipeName (char *PipeNameBuf, size_t PipeNameSize, CFE_SB_Pipeld_t PipeId)`
Get the pipe name for a given id.
- `CFE_Status_t CFE_SB_GetPipeldByName (CFE_SB_Pipeld_t *PipeIdPtr, const char *PipeName)`
Get pipe id by pipe name.
- `CFE_Status_t CFE_SB_SubscribeEx (CFE_SB_MsgId_t MsgId, CFE_SB_Pipeld_t PipeId, CFE_SB_Qos_t Quality, uint16 MsgLim)`
Subscribe to a message on the software bus.
- `CFE_Status_t CFE_SB_Subscribe (CFE_SB_MsgId_t MsgId, CFE_SB_Pipeld_t PipeId)`
Subscribe to a message on the software bus with default parameters.

- [CFE_Status_t CFE_SB_SubscribeLocal](#) ([CFE_SB_MsgId_t](#) MsgId, [CFE_SB_Pipeld_t](#) Pipeld, [uint16](#) MsgLim)
Subscribe to a message while keeping the request local to a cpu.
- [CFE_Status_t CFE_SB_Unsubscribe](#) ([CFE_SB_MsgId_t](#) MsgId, [CFE_SB_Pipeld_t](#) Pipeld)
Remove a subscription to a message on the software bus.
- [CFE_Status_t CFE_SB_UnsubscribeLocal](#) ([CFE_SB_MsgId_t](#) MsgId, [CFE_SB_Pipeld_t](#) Pipeld)
Remove a subscription to a message on the software bus on the current CPU.
- [CFE_Status_t CFE_SB_TransmitMsg](#) (const [CFE_MSG_Message_t](#) *MsgPtr, bool IsOrigination)
Transmit a message.
- [CFE_Status_t CFE_SB_ReceiveBuffer](#) ([CFE_SB_Buffer_t](#) **BufPtr, [CFE_SB_Pipeld_t](#) Pipeld, [int32](#) TimeOut)
Receive a message from a software bus pipe.
- [CFE_SB_Buffer_t](#) * [CFE_SB_AllocateMessageBuffer](#) ([size_t](#) MsgSize)
Get a buffer pointer to use for "zero copy" SB sends.
- [CFE_Status_t CFE_SB_ReleaseMessageBuffer](#) ([CFE_SB_Buffer_t](#) *BufPtr)
Release an unused "zero copy" buffer pointer.
- [CFE_Status_t CFE_SB_TransmitBuffer](#) ([CFE_SB_Buffer_t](#) *BufPtr, bool IsOrigination)
Transmit a buffer.
- void [CFE_SB_SetUserDataLength](#) ([CFE_MSG_Message_t](#) *MsgPtr, [size_t](#) DataLength)
Sets the length of user data in a software bus message.
- void [CFE_SB_TimeStampMsg](#) ([CFE_MSG_Message_t](#) *MsgPtr)
Sets the time field in a software bus message with the current spacecraft time.
- [int32 CFE_SB_MessageStringSet](#) (char *DestStringPtr, const char *SourceStringPtr, [size_t](#) DestMaxSize, [size_t](#) SourceMaxSize)
Copies a string into a software bus message.
- void * [CFE_SB_GetUserData](#) ([CFE_MSG_Message_t](#) *MsgPtr)
Get a pointer to the user data portion of a software bus message.
- [size_t CFE_SB_GetUserDataLength](#) (const [CFE_MSG_Message_t](#) *MsgPtr)
Gets the length of user data in a software bus message.
- [int32 CFE_SB_MessageStringGet](#) (char *DestStringPtr, const char *SourceStringPtr, const char *DefaultString, [size_t](#) DestMaxSize, [size_t](#) SourceMaxSize)
Copies a string out of a software bus message.
- bool [CFE_SB_IsValidMsgId](#) ([CFE_SB_MsgId_t](#) MsgId)
Identifies whether a given CFE_SB_MsgId_t is valid.
- static bool [CFE_SB_MsgId_Equal](#) ([CFE_SB_MsgId_t](#) MsgId1, [CFE_SB_MsgId_t](#) MsgId2)
Identifies whether two CFE_SB_MsgId_t values are equal.
- static [CFE_SB_MsgId_Atom_t CFE_SB_MsgIdToValue](#) ([CFE_SB_MsgId_t](#) MsgId)
Converts a CFE_SB_MsgId_t to a normal integer.
- static [CFE_SB_MsgId_t CFE_SB_ValueToMsgId](#) ([CFE_SB_MsgId_Atom_t](#) MsgIdValue)
Converts a normal integer into a CFE_SB_MsgId_t.
- [CFE_SB_MsgId_Atom_t CFE_SB_CmdTopicIdToMsgId](#) ([uint16](#) TopicId, [uint16](#) InstanceNum)
Converts a topic ID and instance number combination into a MsgID value integer.
- [CFE_SB_MsgId_Atom_t CFE_SB_TlmTopicIdToMsgId](#) ([uint16](#) TopicId, [uint16](#) InstanceNum)
Converts a topic ID and instance number combination into a MsgID value integer.
- [CFE_SB_MsgId_Atom_t CFE_SB_GlobalCmdTopicIdToMsgId](#) ([uint16](#) TopicId)
Converts a topic ID to a MsgID value integer for Global commands.
- [CFE_SB_MsgId_Atom_t CFE_SB_GlobalTlmTopicIdToMsgId](#) ([uint16](#) TopicId)
Converts a topic ID to a MsgID value integer for Global telemetry.
- [CFE_SB_MsgId_Atom_t CFE_SB_LocalCmdTopicIdToMsgId](#) ([uint16](#) TopicId)
Converts a topic ID to a MsgID value integer for local commands.
- [CFE_SB_MsgId_Atom_t CFE_SB_LocalTlmTopicIdToMsgId](#) ([uint16](#) TopicId)
Converts a topic ID to a MsgID value integer for local telemetry.

11.81.1 Detailed Description

Purpose: This header file contains all definitions for the cFE Software Bus Application Programmer's Interface.

Author: R.McGraw/SSI

Definition in file [cfe_sb.h](#).

11.81.2 Macro Definition Documentation

CFE_BIT

```
#define CFE_BIT(
    x ) (1 << (x))
```

Places a one at bit positions 0 - 31.

Definition at line 44 of file [cfe_sb.h](#).

CFE_CLR

```
#define CFE_CLR(
    i,
    x ) ((i) &= ~CFE_BIT(x))
```

Clears bit x of i.

Definition at line 46 of file [cfe_sb.h](#).

CFE_SET

```
#define CFE_SET(
    i,
    x ) ((i) |= CFE_BIT(x))
```

Sets bit x of i.

Definition at line 45 of file [cfe_sb.h](#).

CFE_TST

```
#define CFE_TST(
    i,
    x ) (((i) & CFE_BIT(x)) != 0)
```

true(non zero) if bit x of i is set

Definition at line 47 of file [cfe_sb.h](#).

11.82 cfe_sb.h

[Go to the documentation of this file.](#)

```
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00016  * limitations under the License.
00017  *****/
00018
00030 #ifndef CFE_SB_H
00031 #define CFE_SB_H
```

```

00032
00033 /*
00034 ** Includes
00035 */
00036 #include "common_types.h"
00037 #include "cfe_error.h"
00038 #include "cfe_sb_api_typedefs.h"
00039 #include "cfe_es_api_typedefs.h"
00040
00041 /*
00042 ** Macro Definitions
00043 */
00044 #define CFE_BIT(x) (1 << (x))
00045 #define CFE_SET(i, x) ((i) |= CFE_BIT(x))
00046 #define CFE_CLR(i, x) ((i) &= ~CFE_BIT(x))
00047 #define CFE_TST(i, x) ((i) & CFE_BIT(x)) != 0)
00049 /***** Function Prototypes *****/
00050
00055 /*****/
00089 CFE_Status_t CFE_SB_CreatePipe(CFE_SB_PipeId_t *PipeIdPtr, uint16 Depth, const char *PipeName);
00090
00091 /*****/
00118 CFE_Status_t CFE_SB_DeletePipe(CFE_SB_PipeId_t PipeId);
00119
00143 CFE_Status_t CFE_SB_PipeId_ToIndex(CFE_SB_PipeId_t PipeId, uint32 *Idx);
00144
00145 /*****/
00163 CFE_Status_t CFE_SB_SetPipeOpts(CFE_SB_PipeId_t PipeId, uint8 Opts);
00164
00165 /*****/
00182 CFE_Status_t CFE_SB_GetPipeOpts(CFE_SB_PipeId_t PipeId, uint8 *OptsPtr);
00183
00184 /*****/
00203 CFE_Status_t CFE_SB_GetPipeName(char *PipeNameBuf, size_t PipeNameSize, CFE_SB_PipeId_t PipeId);
00204
00205 /*****/
00222 CFE_Status_t CFE_SB_GetPipeIdByName(CFE_SB_PipeId_t *PipeIdPtr, const char *PipeName);
00229 /*****/
00267 CFE_Status_t CFE_SB_SubscribeEx(CFE_SB_MsgId_t MsgId, CFE_SB_PipeId_t PipeId, CFE_SB_Qos_t Quality, uint16
    MsgLim);
00268
00269 /*****/
00302 CFE_Status_t CFE_SB_Subscribe(CFE_SB_MsgId_t MsgId, CFE_SB_PipeId_t PipeId);
00303
00304 /*****/
00337 CFE_Status_t CFE_SB_SubscribeLocal(CFE_SB_MsgId_t MsgId, CFE_SB_PipeId_t PipeId, uint16 MsgLim);
00338
00339 /*****/
00362 CFE_Status_t CFE_SB_Unsubscribe(CFE_SB_MsgId_t MsgId, CFE_SB_PipeId_t PipeId);
00363
00364 /*****/
00388 CFE_Status_t CFE_SB_UnsubscribeLocal(CFE_SB_MsgId_t MsgId, CFE_SB_PipeId_t PipeId);
00395 /*****/
00438 CFE_Status_t CFE_SB_TransmitMsg(const CFE_MSG_Message_t *MsgPtr, bool IsOrigination);
00439
00440 /*****/
00479 CFE_Status_t CFE_SB_ReceiveBuffer(CFE_SB_Buffer_t **BufPtr, CFE_SB_PipeId_t PipeId, int32 Timeout);
00480
00487 /*****/
00515 CFE_SB_Buffer_t *CFE_SB_AllocateMessageBuffer(size_t MsgSize);
00516
00517 /*****/
00539 CFE_Status_t CFE_SB_ReleaseMessageBuffer(CFE_SB_Buffer_t *BufPtr);
00540
00541 /*****/
00591 CFE_Status_t CFE_SB_TransmitBuffer(CFE_SB_Buffer_t *BufPtr, bool IsOrigination);
00592
00599 /*****/
00619 void CFE_SB_SetUserDataLength(CFE_MSG_Message_t *MsgPtr, size_t DataLength);
00620
00621 /*****/
00637 void CFE_SB_TimeStampMsg(CFE_MSG_Message_t *MsgPtr);
00638
00639 /*****/
00672 int32 CFE_SB_MessageStringSet(char *DestStringPtr,
    const char *SourceStringPtr,
    size_t DestMaxSize,
    size_t SourceMaxSize);
00674
00675
00676
00677 /*****/
00694 void *CFE_SB_GetUserData(CFE_MSG_Message_t *MsgPtr);

```

```

00695
00696 /*****
00712 size_t CFE_SB_GetUserDataLength(const CFE_MSG_Message_t *MsgPtr);
00713
00714 /*****
00754 int32 CFE_SB_MessageStringGet(char *DestStringPtr,
00755                                const char *SourceStringPtr,
00756                                const char *DefaultString,
00757                                size_t DestMaxSize,
00758                                size_t SourceMaxSize);
00765 /*****
00776 bool CFE_SB_IsValidMsgId(CFE_SB_MsgId_t MsgId);
00777
00778 /*****
00796 static inline bool CFE_SB_MsgId_Equal(CFE_SB_MsgId_t MsgId1, CFE_SB_MsgId_t MsgId2)
00797 {
00798     return CFE_SB_MSGID_UNWRAP_VALUE(MsgId1) == CFE_SB_MSGID_UNWRAP_VALUE(MsgId2);
00799 }
00800
00801 /*****
00827 static inline CFE_SB_MsgId_Atom_t CFE_SB_MsgIdToValue(CFE_SB_MsgId_t MsgId)
00828 {
00829     return CFE_SB_MSGID_UNWRAP_VALUE(MsgId);
00830 }
00831
00832 /*****
00856 static inline CFE_SB_MsgId_t CFE_SB_ValueToMsgId(CFE_SB_MsgId_Atom_t MsgIdValue)
00857 {
00858     CFE_SB_MsgId_t Result = CFE_SB_MSGID_WRAP_VALUE(MsgIdValue);
00859     return Result;
00860 }
00861
00862 /*****
00884 CFE_SB_MsgId_Atom_t CFE_SB_CmdTopicIdToMsgId(uint16 TopicId, uint16 InstanceNum);
00885
00886 /*****
00908 CFE_SB_MsgId_Atom_t CFE_SB_TlmTopicIdToMsgId(uint16 TopicId, uint16 InstanceNum);
00909
00910 /*****
00928 CFE_SB_MsgId_Atom_t CFE_SB_GlobalCmdTopicIdToMsgId(uint16 TopicId);
00929
00930 /*****
00948 CFE_SB_MsgId_Atom_t CFE_SB_GlobalTlmTopicIdToMsgId(uint16 TopicId);
00949
00950 /*****
00967 CFE_SB_MsgId_Atom_t CFE_SB_LocalCmdTopicIdToMsgId(uint16 TopicId);
00968
00969 /*****
00986 CFE_SB_MsgId_Atom_t CFE_SB_LocalTlmTopicIdToMsgId(uint16 TopicId);
00987
00990 #endif /* CFE_SB_H */

```

11.83 cfe/modules/core_api/fsw/inc/cfe_sb_api_typedefs.h File Reference

```

#include "common_types.h"
#include "cfe_sb_extern_typedefs.h"
#include "cfe_msg_api_typedefs.h"
#include "cfe_resourceid_api_typedefs.h"
#include "cfe_msg_hdr.h"

```

Data Structures

- union [CFE_SB_Msg](#)
Software Bus generic message.

Macros

- #define [CFE_SB_POLL](#) 0
Option used with [CFE_SB_ReceiveBuffer](#) to request immediate pipe status.
- #define [CFE_SB_PEND_FOREVER](#) -1

- Option used with [CFE_SB_ReceiveBuffer](#) to force a wait for next message.
- `#define CFE_SB_SUBSCRIPTION 0`
Subtype specifier used in [CFE_SB_SingleSubscriptionTlm_t](#) by SBN App.
- `#define CFE_SB_UNSUBSCRIPTION 1`
Subtype specified used in [CFE_SB_SingleSubscriptionTlm_t](#) by SBN App.
- `#define CFE_SB_MSGID_WRAP_VALUE(val) { val }`
Translation macro to convert from MsgId integer values to opaque/abstract API values.
- `#define CFE_SB_MSGID_C(val) ((CFE_SB_MsgId_t)CFE_SB_MSGID_WRAP_VALUE(val))`
Translation macro to convert to MsgId integer values from a literal.
- `#define CFE_SB_MSGID_UNWRAP_VALUE(mid) ((mid).Value)`
Translation macro to convert to MsgId integer values from opaque/abstract API values.
- `#define CFE_SB_MSGID_RESERVED CFE_SB_MSGID_WRAP_VALUE(0)`
Reserved value for [CFE_SB_MsgId_t](#) that will not match any valid MsgId.
- `#define CFE_SB_INVALID_MSG_ID CFE_SB_MSGID_C(0)`
A literal of the [CFE_SB_MsgId_t](#) type representing an invalid ID.
- `#define CFE_SB_PIPEID_C(val) ((CFE_SB_Pipeld_t)CFE_RESOURCEID_WRAP(val))`
Cast/Convert a generic [CFE_ResourceId_t](#) to a [CFE_SB_Pipeld_t](#).
- `#define CFE_SB_INVALID_PIPE CFE_SB_PIPEID_C(CFE_RESOURCEID_UNDEFINED)`
A [CFE_SB_Pipeld_t](#) value which is always invalid.
- `#define CFE_SB_PIPEOPTS_IGNOREMINE 0x00000001`
Messages sent by the app that owns this pipe will not be sent to this pipe.
- `#define CFE_SB_DEFAULT_QOS ((CFE_SB_Qos_t) { 0 })`
Default Qos macro.

Typedefs

- `typedef union CFE_SB_Msg CFE_SB_Buffer_t`
Software Bus generic message.

11.83.1 Detailed Description

Purpose: This header file contains all definitions for the cFE Software Bus Application Programmer's Interface.

Author: R.McGraw/SSI

Definition in file [cfe_sb_api_typedefs.h](#).

11.83.2 Macro Definition Documentation

CFE_SB_DEFAULT_QOS

```
#define CFE_SB_DEFAULT_QOS ((CFE_SB_Qos_t) { 0 })
```

Default Qos macro.

Definition at line 132 of file [cfe_sb_api_typedefs.h](#).

CFE_SB_INVALID_MSG_ID

```
#define CFE_SB_INVALID_MSG_ID CFE_SB_MSGID_C(0)
```

A literal of the [CFE_SB_MsgId_t](#) type representing an invalid ID.

This value should be used for runtime initialization of [CFE_SB_MsgId_t](#) values.

Note

This may be a compound literal in a future revision. Per C99, compound literals are lvalues, not rvalues, so this value should not be used in static/compile-time data initialization. For static data initialization purposes (rvalue), [CFE_SB_MSGID_RESERVED](#) should be used instead. However, in the current implementation, they are equivalent.

Definition at line 110 of file [cfe_sb_api_typedefs.h](#).

CFE_SB_INVALID_PIPE

```
#define CFE_SB_INVALID_PIPE CFE_SB_PIPEID_C(CFE_RESOURCEID_UNDEFINED)
```

A [CFE_SB_PipeId_t](#) value which is always invalid.

This may be used as a safe initializer for [CFE_SB_PipeId_t](#) values

Definition at line 122 of file [cfe_sb_api_typedefs.h](#).

CFE_SB_MSGID_C

```
#define CFE_SB_MSGID_C(  
    val ) ((CFE_SB_MsgId_t)CFE_SB_MSGID_WRAP_VALUE(val))
```

Translation macro to convert to [MsgId](#) integer values from a literal.

This ensures that the literal is interpreted as the [CFE_SB_MsgId_t](#) type, rather than the default type associated with that literal (e.g. int/unsigned int).

Note

Due to constraints in C99 this style of initializer can only be used at runtime, not for static/compile-time initializers.

See also

[CFE_SB_ValueToMsgId\(\)](#)

Definition at line 77 of file [cfe_sb_api_typedefs.h](#).

CFE_SB_MSGID_RESERVED

```
#define CFE_SB_MSGID_RESERVED CFE_SB_MSGID_WRAP_VALUE(0)
```

Reserved value for [CFE_SB_MsgId_t](#) that will not match any valid [MsgId](#).

This rvalue macro can be used for static/compile-time data initialization to ensure that the initialized value does not alias to a valid [MsgId](#) object.

Definition at line 97 of file [cfe_sb_api_typedefs.h](#).

CFE_SB_MSGID_UNWRAP_VALUE

```
#define CFE_SB_MSGID_UNWRAP_VALUE(  
    mid ) ((mid).Value)
```

Translation macro to convert to [MsgId](#) integer values from opaque/abstract API values.

This conversion exists in macro form to allow compile-time evaluation for constants, and should not be used directly in application code.

For applications, use the [CFE_SB_MsgIdToValue\(\)](#) inline function instead.

See also

[CFE_SB_MsgIdToValue\(\)](#)

Definition at line 89 of file [cfe_sb_api_typedefs.h](#).

CFE_SB_MSGID_WRAP_VALUE

```
#define CFE_SB_MSGID_WRAP_VALUE(
    val ) { val }
```

Translation macro to convert from MsgId integer values to opaque/abstract API values.

This conversion exists in macro form to allow compile-time evaluation for constants, and should not be used directly in application code.

For applications, use the [CFE_SB_ValueToMsgId\(\)](#) inline function instead.

See also

[CFE_SB_ValueToMsgId\(\)](#)

Definition at line 64 of file [cfe_sb_api_typedefs.h](#).

CFE_SB_PEND_FOREVER

```
#define CFE_SB_PEND_FOREVER -1
```

Option used with [CFE_SB_ReceiveBuffer](#) to force a wait for next message.

Definition at line 46 of file [cfe_sb_api_typedefs.h](#).

CFE_SB_PIPEID_C

```
#define CFE_SB_PIPEID_C(
    val ) ((CFE_SB_PipeId_t)CFE_RESOURCEID_WRAP(val))
```

Cast/Convert a generic CFE_ResourceId_t to a CFE_SB_PipeId_t.

Definition at line 115 of file [cfe_sb_api_typedefs.h](#).

CFE_SB_POLL

```
#define CFE_SB_POLL 0
```

Option used with [CFE_SB_ReceiveBuffer](#) to request immediate pipe status.

Definition at line 45 of file [cfe_sb_api_typedefs.h](#).

CFE_SB_SUBSCRIPTION

```
#define CFE_SB_SUBSCRIPTION 0
```

Subtype specifier used in [CFE_SB_SingleSubscriptionTlm_t](#) by SBN App.

Definition at line 47 of file [cfe_sb_api_typedefs.h](#).

CFE_SB_UNSUBSCRIPTION

```
#define CFE_SB_UNSUBSCRIPTION 1
```

Subtype specified used in [CFE_SB_SingleSubscriptionTlm_t](#) by SBN App.

Definition at line 48 of file [cfe_sb_api_typedefs.h](#).

11.83.3 Typedef Documentation**CFE_SB_Buffer_t**

```
typedef union CFE_SB_Msg CFE_SB_Buffer_t
```

Software Bus generic message.

11.84 cfe_sb_api_typedefs.h

[Go to the documentation of this file.](#)

```
00001 /*****
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"
00003  */
```

```

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00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00030 #ifndef CFE_SB_API_TYPEDEFS_H
00031 #define CFE_SB_API_TYPEDEFS_H
00032
00033 /*
00034 ** Includes
00035 */
00036 #include "common_types.h"
00037 #include "cfe_sb_extern_typedefs.h"
00038 #include "cfe_msg_api_typedefs.h"
00039 #include "cfe_resourceid_api_typedefs.h"
00040 #include "cfe_msg_hdr.h"
00041
00042 /*
00043 ** Defines
00044 */
00045 #define CFE_SB_POLL 0
00046 #define CFE_SB PEND_FOREVER -1
00047 #define CFE_SB_SUBSCRIPTION 0
00048 #define CFE_SB_UNSUBSCRIPTION 1
00050 /* ----- */
00051 /* Macro Constants for use with the CFE_SB_MsgId_t type */
00052 /* ----- */
00053
00064 #define CFE_SB_MSGID_WRAP_VALUE(val) { val }
00065
00077 #define CFE_SB_MSGID_C(val) ((CFE_SB_MsgId_t)CFE_SB_MSGID_WRAP_VALUE(val))
00078
00089 #define CFE_SB_MSGID_UNWRAP_VALUE(mid) ((mid).Value)
00090
00097 #define CFE_SB_MSGID_RESERVED CFE_SB_MSGID_WRAP_VALUE(0)
00098
00110 #define CFE_SB_INVALID_MSG_ID CFE_SB_MSGID_C(0)
00111
00115 #define CFE_SB_PIPEID_C(val) ((CFE_SB_PipeId_t)CFE_RESOURCEID_WRAP(val))
00116
00122 #define CFE_SB_INVALID_PIPE CFE_SB_PIPEID_C(CFE_RESOURCEID_UNDEFINED)
00123
00128 #define CFE_SB_PIPEOPTS_IGNOREMINE \
00129     0x00000001
00132 #define CFE_SB_DEFAULT_QOS ((CFE_SB_Qos_t) { 0 })
00134 /*
00135 ** Type Definitions
00136 */
00137
00139 typedef union CFE_SB_Msg
00140 {
00141     CFE_MSG_Message_t Msg;
00142     long long int LongInt;
00143     long double LongDouble;
00144 } CFE_SB_Buffer_t;
00145
00146 #endif /* CFE_SB_API_TYPEDEFS_H */

```

11.85 cfe/modules/core_api/fsw/inc/cfe_tbl.h File Reference

```

#include "common_types.h"
#include "cfe_error.h"
#include "cfe_tbl_extern_typedefs.h"
#include "cfe_tbl_api_typedefs.h"
#include "cfe_sb_api_typedefs.h"

```

```
#include "cfe_resourceid.h"
```

Functions

- [CFE_Status_t CFE_TBL_Register](#) ([CFE_TBL_Handle_t](#) *TblHandlePtr, const char *Name, [size_t](#) Size, [uint16](#) TblOptionFlags, [CFE_TBL_CallbackFuncPtr_t](#) TblValidationFuncPtr)
Register a table with cFE to obtain Table Management Services.
- [CFE_Status_t CFE_TBL_Share](#) ([CFE_TBL_Handle_t](#) *TblHandlePtr, const char *TblName)
Obtain handle of table registered by another application.
- [CFE_Status_t CFE_TBL_Unregister](#) ([CFE_TBL_Handle_t](#) TblHandle)
Unregister a table.
- [CFE_Status_t CFE_TBL_Load](#) ([CFE_TBL_Handle_t](#) TblHandle, [CFE_TBL_SrcEnum_t](#) SrcType, const void *SrcDataPtr)
Load a specified table with data from specified source.
- [CFE_Status_t CFE_TBL_Update](#) ([CFE_TBL_Handle_t](#) TblHandle)
Update contents of a specified table, if an update is pending.
- [CFE_Status_t CFE_TBL_Validate](#) ([CFE_TBL_Handle_t](#) TblHandle)
Perform steps to validate the contents of a table image.
- [CFE_Status_t CFE_TBL_Manage](#) ([CFE_TBL_Handle_t](#) TblHandle)
Perform standard operations to maintain a table.
- [CFE_Status_t CFE_TBL_DumpToBuffer](#) ([CFE_TBL_Handle_t](#) TblHandle)
Copies the contents of a Dump Only Table to a shared buffer.
- [CFE_Status_t CFE_TBL_Modified](#) ([CFE_TBL_Handle_t](#) TblHandle)
Notify cFE Table Services that table contents have been modified by the Application.
- [CFE_Status_t CFE_TBL_GetAddress](#) (void **TblPtr, [CFE_TBL_Handle_t](#) TblHandle)
Obtain the current address of the contents of the specified table.
- [CFE_Status_t CFE_TBL_ReleaseAddress](#) ([CFE_TBL_Handle_t](#) TblHandle)
Release previously obtained pointer to the contents of the specified table.
- [CFE_Status_t CFE_TBL_GetAddresses](#) (void **TblPtrs[], [uint16](#) NumTables, const [CFE_TBL_Handle_t](#) TblHandles[])
Obtain the current addresses of an array of specified tables.
- [CFE_Status_t CFE_TBL_ReleaseAddresses](#) ([uint16](#) NumTables, const [CFE_TBL_Handle_t](#) TblHandles[])
Release the addresses of an array of specified tables.
- [CFE_Status_t CFE_TBL_GetStatus](#) ([CFE_TBL_Handle_t](#) TblHandle)
Obtain current status of pending actions for a table.
- [CFE_Status_t CFE_TBL_GetInfo](#) ([CFE_TBL_Info_t](#) *TblInfoPtr, const char *TblName)
Obtain characteristics/information of/about a specified table.
- [CFE_Status_t CFE_TBL_NotifyByMessage](#) ([CFE_TBL_Handle_t](#) TblHandle, [CFE_SB_MsgId_t](#) MsgId, [CFE_MSG_FcnCode_t](#) CommandCode, [uint32](#) Parameter)
Instruct cFE Table Services to notify Application via message when table requires management.

Conversions for TBL resource IDs

These inline functions enforce the type of the argument(s), such that comparing a table handle to some other type of resource ID will result in a compiler error.

- static bool [CFE_TBL_HandleID_IsEqual](#) ([CFE_TBL_HandleId_t](#) x, [CFE_TBL_HandleId_t](#) y)
- static unsigned long [CFE_TBL_HandleID_AsInt](#) ([CFE_TBL_HandleId_t](#) x)
- static bool [CFE_TBL_HandleID_IsDefined](#) ([CFE_TBL_HandleId_t](#) x)
- [CFE_TBL_HandleId_t CFE_TBL_HandleToID](#) ([CFE_TBL_Handle_t](#) TblHandle)
- [CFE_TBL_Handle_t CFE_TBL_HandleFromID](#) ([CFE_TBL_HandleId_t](#) TblId)

11.85.1 Detailed Description

Title: Table Services API Application Library Header File

Purpose: Unit specification for Table services library functions and macros.

Design Notes:

References: Flight Software Branch C Coding Standard Version 1.0a

Notes:

Definition in file [cfe_tbl.h](#).

11.85.2 Function Documentation

CFE_TBL_HandleFromID()

```
CFE_TBL_Handle_t CFE_TBL_HandleFromID (
    CFE_TBL_HandleId_t TblId )
```

Convert an ID to an API handle

CFE_TBL_HandleID_AsInt()

```
static unsigned long CFE_TBL_HandleID_AsInt (
    CFE_TBL_HandleId_t x ) [inline], [static]
```

Get the integer representation of the ID

Definition at line 65 of file [cfe_tbl.h](#).

References [CFE_RESOURCEID_TO_ULONG](#).

CFE_TBL_HandleID_IsDefined()

```
static bool CFE_TBL_HandleID_IsDefined (
    CFE_TBL_HandleId_t x ) [inline], [static]
```

Check if the ID is valid

Definition at line 73 of file [cfe_tbl.h](#).

References [CFE_RESOURCEID_TEST_DEFINED](#).

CFE_TBL_HandleID_IsEqual()

```
static bool CFE_TBL_HandleID_IsEqual (
    CFE_TBL_HandleId_t x,
    CFE_TBL_HandleId_t y ) [inline], [static]
```

Check for equality between two IDs

Definition at line 57 of file [cfe_tbl.h](#).

References [CFE_RESOURCEID_TEST_EQUAL](#).

CFE_TBL_HandleToID()

```
CFE_TBL_HandleId_t CFE_TBL_HandleToID (
    CFE_TBL_Handle_t TblHandle )
```

Convert an API handle to an ID

11.86 cfe_tbl.h

[Go to the documentation of this file.](#)

```
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00007  */
```

```

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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00036 #ifndef CFE_TBL_H
00037 #define CFE_TBL_H
00038
00039 /***** Include Files *****/
00040 #include "common_types.h" /* Basic Data Types */
00041 #include "cfe_error.h"
00042 #include "cfe_tbl_extern_typedefs.h"
00043 #include "cfe_tbl_api_typedefs.h"
00044 #include "cfe_sb_api_typedefs.h"
00045 #include "cfe_resourceid.h"
00046
00057 static inline bool CFE_TBL_HandleID_IsEqual(CFE_TBL_HandleId_t x, CFE_TBL_HandleId_t y)
00058 {
00059     return CFE_RESOURCEID_TEST_EQUAL(x, y);
00060 }
00061
00065 static inline unsigned long CFE_TBL_HandleID_AsInt(CFE_TBL_HandleId_t x)
00066 {
00067     return CFE_RESOURCEID_TO_ULONG(x);
00068 }
00069
00073 static inline bool CFE_TBL_HandleID_IsDefined(CFE_TBL_HandleId_t x)
00074 {
00075     return CFE_RESOURCEID_TEST_DEFINED(x);
00076 }
00077
00081 CFE_TBL_HandleId_t CFE_TBL_HandleToID(CFE_TBL_Handle_t TblHandle);
00082
00086 CFE_TBL_Handle_t CFE_TBL_HandleFromID(CFE_TBL_HandleId_t TblId);
00087
00090 /***** Function Prototypes *****/
00091
00096 /*****
00234 CFE_Status_t CFE_TBL_Register(CFE_TBL_Handle_t          *TblHandlePtr,
00235                               const char                *Name,
00236                               size_t                    Size,
00237                               uint16                    TblOptionFlags,
00238                               CFE_TBL_CallbackFuncPtr_t TblValidationFuncPtr);
00239
00240 /*****
00275 CFE_Status_t CFE_TBL_Share(CFE_TBL_Handle_t *TblHandlePtr, const char *TblName);
00276
00277 /*****
00314 CFE_Status_t CFE_TBL_Unregister(CFE_TBL_Handle_t TblHandle);
00321 /*****
00372 CFE_Status_t CFE_TBL_Load(CFE_TBL_Handle_t TblHandle, CFE_TBL_SrcEnum_t SrcType, const void *SrcDataPtr);
00373
00374 /*****
00400 CFE_Status_t CFE_TBL_Update(CFE_TBL_Handle_t TblHandle);
00401
00402 /*****
00428 CFE_Status_t CFE_TBL_Validate(CFE_TBL_Handle_t TblHandle);
00429
00430 /*****
00459 CFE_Status_t CFE_TBL_Manage(CFE_TBL_Handle_t TblHandle);
00460
00461 /*****
00486 CFE_Status_t CFE_TBL_DumpToBuffer(CFE_TBL_Handle_t TblHandle);
00487
00488 /*****
00512 CFE_Status_t CFE_TBL_Modified(CFE_TBL_Handle_t TblHandle);
00519 /*****
00567 CFE_Status_t CFE_TBL_GetAddress(void **TblPtr, CFE_TBL_Handle_t TblHandle);
00568
00569 /*****
00597 CFE_Status_t CFE_TBL_ReleaseAddress(CFE_TBL_Handle_t TblHandle);
00598
00599 /*****
00648 CFE_Status_t CFE_TBL_GetAddresses(void **TblPtrs[], uint16 NumTables, const CFE_TBL_Handle_t
    TblHandles[]);

```

```

00649
00650 /*****
00681 CFE_Status_t CFE_TBL_ReleaseAddresses(uint16 NumTables, const CFE_TBL_Handle_t TblHandles[]);
00688 *****/
00721 CFE_Status_t CFE_TBL_GetStatus(CFE_TBL_Handle_t TblHandle);
00722
00723 /*****
00752 CFE_Status_t CFE_TBL_GetInfo(CFE_TBL_Info_t *TblInfoPtr, const char *TblName);
00753 *****/
00754 /*****
00793 CFE_Status_t CFE_TBL_NotifyByMessage(CFE_TBL_Handle_t TblHandle,
00794                                     CFE_SB_MsgId_t      MsgId,
00795                                     CFE_MSG_FcnCode_t    CommandCode,
00796                                     uint32                Parameter);
00799 #endif /* CFE_TBL_H */

```

11.87 cfe/modules/core_api/fsw/inc/cfe_tbl_api_typedefs.h File Reference

```

#include "common_types.h"
#include "cfe_mission_cfg.h"
#include "cfe_tbl_extern_typedefs.h"
#include "cfe_time_extern_typedefs.h"
#include "cfe_resourceid_api_typedefs.h"

```

Data Structures

- struct [CFE_TBL_Info](#)
Table Info.

Macros

- #define [CFE_TBL_OPT_BUFFER_MSK](#) (0x0001)
Table buffer mask.
- #define [CFE_TBL_OPT_SNGL_BUFFER](#) (0x0000)
Single buffer table.
- #define [CFE_TBL_OPT_DBL_BUFFER](#) (0x0001)
Double buffer table.
- #define [CFE_TBL_OPT_LD_DMP_MSK](#) (0x0002)
Table load/dump mask.
- #define [CFE_TBL_OPT_LOAD_DUMP](#) (0x0000)
Load/Dump table.
- #define [CFE_TBL_OPT_DUMP_ONLY](#) (0x0002)
Dump only table.
- #define [CFE_TBL_OPT_USR_DEF_MSK](#) (0x0004)
Table user defined mask.
- #define [CFE_TBL_OPT_NOT_USR_DEF](#) (0x0000)
Not user defined table.
- #define [CFE_TBL_OPT_USR_DEF_ADDR](#) (0x0006)
User Defined table,.
- #define [CFE_TBL_OPT_CRITICAL_MSK](#) (0x0008)
Table critical mask.
- #define [CFE_TBL_OPT_NOT_CRITICAL](#) (0x0000)
Not critical table.
- #define [CFE_TBL_OPT_CRITICAL](#) (0x0008)

Critical table.

- #define [CFE_TBL_OPT_DEFAULT](#) ([CFE_TBL_OPT_SNGL_BUFFER](#) | [CFE_TBL_OPT_LOAD_DUMP](#))

Default table options.

- #define [CFE_TBL_MAX_FULL_NAME_LEN](#) ([CFE_MISSION_TBL_MAX_FULL_NAME_LEN](#))

Table maximum full name length.

- #define [CFE_TBL_BAD_TABLE_HANDLE](#) (([CFE_TBL_Handle_t](#))(-1))

Bad table handle.

- #define [CFE_TBL_HANDLE_EQ](#)(x, y) ((x) == (y))
- #define [CFE_TBL_HANDLE_INT](#)(x) ((unsigned long)(x))
- #define [CFE_TBL_HANDLE_IS_VALID](#)(x) ((x) != [CFE_TBL_BAD_TABLE_HANDLE](#))

Constants for table registry IDs, using the [CFE_TBL_RegId_t](#) type

- #define [CFE_TBL_REGID_C](#)(x) (([CFE_TBL_RegId_t](#))[CFE_RESOURCEID_WRAP](#)(x))
- #define [CFE_TBL_REGID_UNDEFINED](#) [CFE_TBL_REGID_C](#)([CFE_RESOURCEID_UNDEFINED](#))

Constants for table handle IDs, using the [CFE_TBL_HandleId_t](#) type

- #define [CFE_TBL_HANDLEID_C](#)(x) (([CFE_TBL_HandleId_t](#))[CFE_RESOURCEID_WRAP](#)(x))
- #define [CFE_TBL_HANDLEID_UNDEFINED](#) [CFE_TBL_HANDLEID_C](#)([CFE_RESOURCEID_UNDEFINED](#))

Typedefs

- typedef [int32](#)(* [CFE_TBL_CallbackFuncPtr_t](#)) (void *TblPtr)

Table Callback Function.

- typedef enum [CFE_TBL_SrcEnum](#) [CFE_TBL_SrcEnum_t](#)

Table Source.

- typedef struct [CFE_TBL_Info](#) [CFE_TBL_Info_t](#)

Table Info.

- typedef [int16](#) [CFE_TBL_Handle_t](#)

Enumerations

- enum [CFE_TBL_SrcEnum](#) { [CFE_TBL_SRC_FILE](#) = 0 , [CFE_TBL_SRC_ADDRESS](#) }

Table Source.

11.87.1 Detailed Description

Title: Table Services API Application Library Header File

Purpose: Unit specification for Table services library functions and macros.

Design Notes:

References: Flight Software Branch C Coding Standard Version 1.0a

Notes:

Definition in file [cfe_tbl_api_typedefs.h](#).

11.87.2 Macro Definition Documentation

[CFE_TBL_BAD_TABLE_HANDLE](#)

```
#define CFE_TBL_BAD_TABLE_HANDLE ((CFE_TBL_Handle_t)(-1))
```

Bad table handle.

Definition at line 173 of file [cfe_tbl_api_typedefs.h](#).

CFE_TBL_HANDLE_EQ

```
#define CFE_TBL_HANDLE_EQ(  
    x,  
    y ) ((x) == (y))
```

Definition at line 175 of file [cfe_tbl_api_typedefs.h](#).

CFE_TBL_HANDLE_INT

```
#define CFE_TBL_HANDLE_INT(  
    x ) ((unsigned long)(x))
```

Definition at line 176 of file [cfe_tbl_api_typedefs.h](#).

CFE_TBL_HANDLE_IS_VALID

```
#define CFE_TBL_HANDLE_IS_VALID(  
    x ) ((x) != CFE_TBL_BAD_TABLE_HANDLE)
```

Definition at line 177 of file [cfe_tbl_api_typedefs.h](#).

CFE_TBL_HANDLEID_C

```
#define CFE_TBL_HANDLEID_C(  
    x ) ((CFE_TBL_HandleId_t)CFE_RESOURCEID_WRAP(x))
```

Definition at line 136 of file [cfe_tbl_api_typedefs.h](#).

CFE_TBL_HANDLEID_UNDEFINED

```
#define CFE_TBL_HANDLEID_UNDEFINED CFE_TBL_HANDLEID_C(CFE_RESOURCEID_UNDEFINED)
```

Definition at line 137 of file [cfe_tbl_api_typedefs.h](#).

CFE_TBL_MAX_FULL_NAME_LEN

```
#define CFE_TBL_MAX_FULL_NAME_LEN (CFE_MISSION_TBL_MAX_FULL_NAME_LEN)
```

Table maximum full name length.

The full length of table names is defined at the mission scope. This is defined here to support applications that depend on [cfe_tbl.h](#) providing this value.

Definition at line 77 of file [cfe_tbl_api_typedefs.h](#).

CFE_TBL_REGID_C

```
#define CFE_TBL_REGID_C(  
    x ) ((CFE_TBL_RegId_t)CFE_RESOURCEID_WRAP(x))
```

Definition at line 128 of file [cfe_tbl_api_typedefs.h](#).

CFE_TBL_REGID_UNDEFINED

```
#define CFE_TBL_REGID_UNDEFINED CFE_TBL_REGID_C(CFE_RESOURCEID_UNDEFINED)
```

Definition at line 129 of file [cfe_tbl_api_typedefs.h](#).

11.87.3 Typedef Documentation**CFE_TBL_CallbackFuncPtr_t**

```
typedef int32(* CFE_TBL_CallbackFuncPtr_t) (void *TblPtr)
```

Table Callback Function.

Definition at line 82 of file [cfe_tbl_api_typedefs.h](#).

CFE_TBL_Handle_t

```
typedef int16 CFE_TBL_Handle_t
```

Definition at line 170 of file [cfe_tbl_api_typedefs.h](#).

CFE_TBL_Info_t

```
typedef struct CFE_TBL_Info CFE_TBL_Info_t
```

Table Info.

CFE_TBL_SrcEnum_t

```
typedef enum CFE_TBL_SrcEnum CFE_TBL_SrcEnum_t
```

Table Source.

11.87.4 Enumeration Type Documentation**CFE_TBL_SrcEnum**

```
enum CFE_TBL_SrcEnum
```

Table Source.

Enumerator

CFE_TBL_SRC_FILE	File source When this option is selected, the <code>SrcDataPtr</code> will be interpreted as a pointer to a null terminated character string. The string should specify the full path and filename of the file containing the initial data contents of the table.
CFE_TBL_SRC_ADDRESS	Address source When this option is selected, the <code>SrcDataPtr</code> will be interpreted as a pointer to a memory location that is the beginning of the initialization data for loading the table OR, in the case of a "user defined" dump only table, the address of the active table itself. The block of memory is assumed to be of the same size specified in the CFE_TBL_Register function Size parameter.

Definition at line 85 of file [cfe_tbl_api_typedefs.h](#).

11.88 cfe_tbl_api_typedefs.h

[Go to the documentation of this file.](#)

```

00001 /*****
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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00036 #ifndef CFE_TBL_API_TYPEDEFS_H
00037 #define CFE_TBL_API_TYPEDEFS_H
00038
00039 /***** Include Files *****/
00040 #include "common_types.h" /* Basic Data Types */
00041 #include "cfe_mission_cfg.h"
00042 #include "cfe_tbl_extern_typedefs.h"
00043 #include "cfe_time_extern_typedefs.h"

```

```

00044 #include "cfe_resourceid_api_typedefs.h"
00045
00049 #define CFE_TBL_OPT_BUFFER_MSK    (0x0001)
00050 #define CFE_TBL_OPT_SNGL_BUFFER    (0x0000)
00051 #define CFE_TBL_OPT_DBL_BUFFER    (0x0001)
00053 #define CFE_TBL_OPT_LD_DUMP_MSK    (0x0002)
00054 #define CFE_TBL_OPT_LOAD_DUMP    (0x0000)
00055 #define CFE_TBL_OPT_DUMP_ONLY    (0x0002)
00057 #define CFE_TBL_OPT_USR_DEF_MSK    (0x0004)
00058 #define CFE_TBL_OPT_NOT_USR_DEF    (0x0000)
00059 #define CFE_TBL_OPT_USR_DEF_ADDR \
00060     (0x0006)
00062 #define CFE_TBL_OPT_CRITICAL_MSK    (0x0008)
00063 #define CFE_TBL_OPT_NOT_CRITICAL    (0x0000)
00064 #define CFE_TBL_OPT_CRITICAL    (0x0008)
00067 #define CFE_TBL_OPT_DEFAULT    (CFE_TBL_OPT_SNGL_BUFFER | CFE_TBL_OPT_LOAD_DUMP)
00077 #define CFE_TBL_MAX_FULL_NAME_LEN    (CFE_MISSION_TBL_MAX_FULL_NAME_LEN)
00078
00079 /***** Data Type Definitions *****/
00080
00082 typedef int32 (*CFE_TBL_CallbackFuncPtr_t)(void *TblPtr);
00083
00085 typedef enum CFE_TBL_SrcEnum
00086 {
00087     CFE_TBL_SRC_FILE = 0,
00093     CFE_TBL_SRC_ADDRESS
00101 } CFE_TBL_SrcEnum_t;
00102
00104 typedef struct CFE_TBL_Info
00105 {
00106     size_t          Size;
00107     uint32          NumUsers;
00108     CFE_TIME_SysTime_t  FileTime;
00109     uint32          Crc;
00110     CFE_TIME_SysTime_t  TimeOfLastUpdate;
00111     bool            TableLoadedOnce;
00112     bool            DumpOnly;
00113     bool            DoubleBuffered;
00114     bool            UserDefAddr;
00115     bool            Critical;
00116     char            LastFileLoaded[CFE_MISSION_MAX_PATH_LEN];
00117 } CFE_TBL_Info_t;
00118
00119 /*
00120  * NOTE: CFE_TBL_HandleId_t and CFE_TBL_RegId_t types are now defined in
00121  * the cfe_tbl_extern_typedefs.h file which is included above. This file
00122  * provides the constants and macros to work with those types.
00123  */
00124
00128 #define CFE_TBL_REGID_C(x)          ((CFE_TBL_RegId_t)CFE_RESOURCEID_WRAP(x))
00129 #define CFE_TBL_REGID_UNDEFINED    CFE_TBL_REGID_C(CFE_RESOURCEID_UNDEFINED)
00130
00136 #define CFE_TBL_HANDLEID_C(x)        ((CFE_TBL_HandleId_t)CFE_RESOURCEID_WRAP(x))
00137 #define CFE_TBL_HANDLEID_UNDEFINED    CFE_TBL_HANDLEID_C(CFE_RESOURCEID_UNDEFINED)
00138
00141 /*
00142  * The historical name of the external table handle type was "CFE_TBL_Handle_t" and it was a simple int16
00143  * To facilitate migration of apps to use the macros, provide the old type behind the OMIT_DEPRECATED
00144  * switch
00145  */
00146 #ifndef CFE_OMIT_DEPRECATED_6_8
00147 /*
00148  * The preferred way is to use handle IDs which are type-safe.
00149  *
00150  * In this mode CFE_TBL_Handle_t and CFE_TBL_HandleId_t are the same thing
00151  * (a future version can remove the duplicate name)
00152  */
00153 typedef CFE_TBL_HandleId_t CFE_TBL_Handle_t;
00154
00156 #define CFE_TBL_BAD_TABLE_HANDLE    CFE_TBL_HANDLEID_UNDEFINED
00157
00158 #define CFE_TBL_HANDLE_EQ(x, y)      CFE_TBL_HandleID_IsEqual(x, y)
00159 #define CFE_TBL_HANDLE_INT(x)        CFE_TBL_HandleID_AsInt(x)
00160 #define CFE_TBL_HANDLE_IS_VALID(x)    CFE_TBL_HandleID_IsDefined(x)
00161
00162 #else
00163
00164 /*
00165  * The historical way is to use handle IDs which are simply an integer.
00166  */

```

```

00167  * In this mode CFE_TBL_Handle_t is a truncated form of CFE_TBL_HandleId_t that only has
00168  * the lower 16 bits. It is not type safe.
00169  */
00170 typedef int16 CFE_TBL_Handle_t;
00171
00173 #define CFE_TBL_BAD_TABLE_HANDLE ((CFE_TBL_Handle_t) (-1))
00174
00175 #define CFE_TBL_HANDLE_EQ(x, y) ((x) == (y))
00176 #define CFE_TBL_HANDLE_INT(x) ((unsigned long)(x))
00177 #define CFE_TBL_HANDLE_IS_VALID(x) ((x) != CFE_TBL_BAD_TABLE_HANDLE)
00178
00179 #endif
00180
00181 #endif /* CFE_TBL_API_TYPEDEFS_H */

```

11.89 cfe/modules/core_api/fsw/inc/cfe_tbl_filedef.h File Reference

```

#include "cfe_mission_cfg.h"
#include "common_types.h"
#include "cfe_tbl_extern_typedefs.h"
#include "cfe_fs_extern_typedefs.h"

```

Data Structures

- struct [CFE_TBL_FileDef](#)
Table File summary object.

Macros

- #define [CFE_TBL_FILEDEF](#)(ObjName, TblName, Desc, Filename) [CFE_TBL_FileDef_t](#) CFE_TBL_FileDef = {
#ObjName "\0", #TblName "\0", #Desc "\0", #Filename "\0", sizeof(ObjName) };
Macro to assist in with table definition object declaration.

Typedefs

- typedef struct [CFE_TBL_FileDef](#) [CFE_TBL_FileDef_t](#)
Table File summary object.

11.89.1 Detailed Description

Title: ELF2CFETBL Utility Header File for Table Images

Purpose: This header file provides a data structure definition and macro definition required in source code that is intended to be compiled into a cFE compatible Table Image file.

Design Notes:

Typically, a user would include this file in a ".c" file that contains nothing but a desired instantiation of values for a table image along with the macro defined below. After compilation, the resultant elf file can be processed using the 'elf2cfetbl' utility to generate a file that can be loaded onto a cFE flight system and successfully loaded into a table using the cFE Table Services.

References: Flight Software Branch C Coding Standard Version 1.0a

Notes:

Definition in file [cfe_tbl_filedef.h](#).

11.89.2 Macro Definition Documentation

CFE_TBL_FILEDEF

```

#define CFE_TBL_FILEDEF(
    ObjName,

```

```

    TblName,
    Desc,
    Filename ) CFE_TBL_FileDef_t CFE_TBL_FileDef = { #ObjName "\0", #TblName "\0",
#Desc "\0", #Filename "\0", sizeof(ObjName) };

```

Macro to assist in with table definition object declaration.

See notes in the [CFE_TBL_FileDef_t](#) structure type about naming conventions and recommended practices for the various fields.

The CFE_TBL_FILEDEF macro can be used to simplify the declaration of a table image when using the elf2cfetbl utility. Note that the macro adds a NULL at the end to ensure that it is null-terminated. (C allows a struct to be statically initialized with a string exactly the length of the array, which loses the null terminator.) This means the actual length limit of the fields are the above LEN - 1.

An example of the source code and how this macro would be used is as follows:

```

#include "cfe_tbl_filedef.h"

typedef struct MyTblStruct
{
    int    Int1;
    int    Int2;
    int    Int3;
    char   Char1;
} MyTblStruct_t;

MyTblStruct_t MyTblStruct = { 0x01020304, 0x05060708, 0x090A0B0C, 0x0D };

CFE_TBL_FILEDEF(MyTblStruct, MyApp.TableName, Table Utility Test Table, MyTblDefault.bin )

```

Definition at line 149 of file [cfe_tbl_filedef.h](#).

11.89.3 Typedef Documentation

CFE_TBL_FileDef_t

```
typedef struct CFE_TBL_FileDef CFE_TBL_FileDef_t
```

Table File summary object.

The definition of the file definition metadata that can be used by external tools (e.g. elf2cfetbl) to generate CFE table data files.

11.90 cfe_tbl_filedef.h

[Go to the documentation of this file.](#)

```

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00016  * limitations under the License.
00017  *****/
00018
00044 #ifndef CFE_TBL_FILEDEF_H
00045 #define CFE_TBL_FILEDEF_H
00046
00047 #include "cfe_mission_cfg.h"
00048 #include "common_types.h"
00049 #include "cfe_tbl_extern_typedefs.h" /* for "CFE_TBL_FileHdr_t" definition */
00050 #include "cfe_fs_extern_typedefs.h" /* for "CFE_FS_HDR_DESC_MAX_LEN" definition */
00051
00058 typedef struct CFE_TBL_FileDef
00059 {
00076     char ObjectName[64];
00077

```

```

00087     char TableName[CFE_MISSION_TBL_MAX_FULL_NAME_LEN];
00088
00094     char Description[CFE_FS_HDR_DESC_MAX_LEN];
00095
00104     char TgtFilename[CFE_MISSION_MAX_FILE_LEN];
00105
00112     uint32 ObjectSize;
00113 } CFE_TBL_FileDef_t;
00114
00149 #define CFE_TBL_FILEDEF(ObjName, TblName, Desc, Filename) \
00150     CFE_TBL_FileDef_t CFE_TBL_FileDef = { #ObjName "\0", #TblName "\0", #Desc "\0", #Filename "\0",
      sizeof(ObjName) };
00151
00152 /*****
00153
00154 #endif /* CFE_TBL_FILEDEF_H */

```

11.91 cfe/modules/core_api/fsw/inc/cfe_time.h File Reference

```

#include "common_types.h"
#include "cfe_error.h"
#include "cfe_time_api_typedefs.h"
#include "cfe_es_api_typedefs.h"

```

Macros

- `#define CFE_TIME_Copy(m, t)`
Time Copy.

Functions

- `CFE_TIME_SysTime_t CFE_TIME_GetTime` (void)
Get the current spacecraft time.
- `CFE_TIME_SysTime_t CFE_TIME_GetTAI` (void)
Get the current TAI (MET + SCTF) time.
- `CFE_TIME_SysTime_t CFE_TIME_GetUTC` (void)
Get the current UTC (MET + SCTF - Leap Seconds) time.
- `CFE_TIME_SysTime_t CFE_TIME_GetMET` (void)
Get the current value of the Mission Elapsed Time (MET).
- `uint32 CFE_TIME_GetMETseconds` (void)
Get the current seconds count of the mission-elapsed time.
- `uint32 CFE_TIME_GetMETsubsecs` (void)
Get the current sub-seconds count of the mission-elapsed time.
- `CFE_TIME_SysTime_t CFE_TIME_GetSTCF` (void)
Get the current value of the spacecraft time correction factor (STCF).
- `int16 CFE_TIME_GetLeapSeconds` (void)
Get the current value of the leap seconds counter.
- `CFE_TIME_ClockState_Enum_t CFE_TIME_GetClockState` (void)
Get the current state of the spacecraft clock.
- `uint16 CFE_TIME_GetClockInfo` (void)
Provides information about the spacecraft clock.
- `CFE_TIME_SysTime_t CFE_TIME_Add` (CFE_TIME_SysTime_t Time1, CFE_TIME_SysTime_t Time2)
Adds two time values.
- `CFE_TIME_SysTime_t CFE_TIME_Subtract` (CFE_TIME_SysTime_t Time1, CFE_TIME_SysTime_t Time2)
Subtracts two time values.

- [CFE_TIME_Compare_t CFE_TIME_Compare](#) ([CFE_TIME_SysTime_t](#) TimeA, [CFE_TIME_SysTime_t](#) TimeB)
Compares two time values.
- [CFE_TIME_SysTime_t CFE_TIME_MET2SCTime](#) ([CFE_TIME_SysTime_t](#) METTime)
Convert specified MET into Spacecraft Time.
- [uint32 CFE_TIME_Sub2MicroSecs](#) ([uint32](#) SubSeconds)
Converts a sub-seconds count to an equivalent number of microseconds.
- [uint32 CFE_TIME_Micro2SubSecs](#) ([uint32](#) MicroSeconds)
Converts a number of microseconds to an equivalent sub-seconds count.
- [void CFE_TIME_ExternalTone](#) ([void](#))
Provides the 1 Hz signal from an external source.
- [void CFE_TIME_ExternalMET](#) ([CFE_TIME_SysTime_t](#) NewMET)
Provides the Mission Elapsed Time from an external source.
- [void CFE_TIME_ExternalGPS](#) ([CFE_TIME_SysTime_t](#) NewTime, [int16](#) NewLeaps)
Provide the time from an external source that has data common to GPS receivers.
- [void CFE_TIME_ExternalTime](#) ([CFE_TIME_SysTime_t](#) NewTime)
Provide the time from an external source that measures time relative to a known epoch.
- [CFE_Status_t CFE_TIME_RegisterSynchCallback](#) ([CFE_TIME_SynchCallbackPtr_t](#) CallbackFuncPtr)
Registers a callback function that is called whenever time synchronization occurs.
- [CFE_Status_t CFE_TIME_UnregisterSynchCallback](#) ([CFE_TIME_SynchCallbackPtr_t](#) CallbackFuncPtr)
Unregisters a callback function that is called whenever time synchronization occurs.
- [CFE_Status_t CFE_TIME_Print](#) ([char *PrintBuffer](#), [CFE_TIME_SysTime_t](#) TimeToPrint)
Print a time value as a string.
- [void CFE_TIME_Local1HzISR](#) ([void](#))
This function is called via a timer callback set up at initialization of the TIME service.

11.91.1 Detailed Description

Purpose: cFE Time Services (TIME) library API header file

Author: S.Walling/Microtel

Notes:

Definition in file [cfe_time.h](#).

11.91.2 Macro Definition Documentation

CFE_TIME_Copy

```
#define CFE_TIME_Copy(
    m,
    t )
```

Value:

```
{
    (m)->Seconds      = (t)->Seconds; \
    (m)->Subseconds   = (t)->Subseconds; \
}
```

Time Copy.

Macro to copy systime into another systime. Preferred to use this macro as it does not require the two arguments to be exactly the same type, it will work with any two structures that define "Seconds" and "Subseconds" members.

Definition at line 48 of file [cfe_time.h](#).

11.92 cfe_time.h

[Go to the documentation of this file.](#)

```

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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00030 #ifndef CFE_TIME_H
00031 #define CFE_TIME_H
00032
00033 /*
00034 ** Includes
00035 */
00036 #include "common_types.h"
00037 #include "cfe_error.h"
00038 #include "cfe_time_api_typedefs.h"
00039 #include "cfe_es_api_typedefs.h"
00040
00048 #define CFE_TIME_Copy(m, t)
00049 {
00050     (m)->Seconds = (t)->Seconds;
00051     (m)->Subseconds = (t)->Subseconds;
00052 }
00053
00054 /*****
00055 */
00056 ** Exported Functions
00057 */
00058
00063 /*****
00085 CFE_TIME_SysTime_t CFE_TIME_GetTime(void);
00086
00087 /*****
00115 CFE_TIME_SysTime_t CFE_TIME_GetTAI(void);
00116
00117 /*****
00142 CFE_TIME_SysTime_t CFE_TIME_GetUTC(void);
00143
00144 /*****
00164 CFE_TIME_SysTime_t CFE_TIME_GetMET(void);
00165
00166 /*****
00183 uint32 CFE_TIME_GetMETseconds(void);
00184
00185 /*****
00203 uint32 CFE_TIME_GetMETsubsecs(void);
00210 /*****
00230 CFE_TIME_SysTime_t CFE_TIME_GetSTCF(void);
00231
00232 /*****
00257 int16 CFE_TIME_GetLeapSeconds(void);
00258
00259 /*****
00277 CFE_TIME_ClockState_Enum_t CFE_TIME_GetClockState(void);
00278
00279 /*****
00298 uint16 CFE_TIME_GetClockInfo(void);
00305 /*****
00329 CFE_TIME_SysTime_t CFE_TIME_Add(CFE_TIME_SysTime_t Time1, CFE_TIME_SysTime_t Time2);
00330
00331 /*****
00358 CFE_TIME_SysTime_t CFE_TIME_Subtract(CFE_TIME_SysTime_t Time1, CFE_TIME_SysTime_t Time2);
00359
00360 /*****
00395 CFE_TIME_Compare_t CFE_TIME_Compare(CFE_TIME_SysTime_t TimeA, CFE_TIME_SysTime_t TimeB);
00402 /*****/

```



```

00423 CFE_TIME_SysTime_t CFE_TIME_MET2SCTime(CFE_TIME_SysTime_t METTime);
00424
00425 /*****/
00443 uint32 CFE_TIME_Sub2MicroSecs(uint32 SubSeconds);
00444
00445 /*****/
00465 uint32 CFE_TIME_Micro2SubSecs(uint32 MicroSeconds);
00466
00473 /*****/
00492 void CFE_TIME_ExternalTone(void);
00493
00494 /*
00495 ** Function prototypes (external time source)...
00496 **
00497 ** If a Time Server has been configured to accept external time
00498 ** data, then one of the following functions will be enabled.
00499 **
00500 ** If the Time Server has also been commanded to use the external
00501 ** time data (as opposed to getting time data from a local MET)
00502 ** then the Time Server will use the external data in computing
00503 ** time.
00504 **
00505 ** However, regardless whether the external time data is accepted
00506 ** by the Time Server, these functions also act as the signal to
00507 ** create and distribute the "time at the tone" command packet.
00508 */
00509
00510 /*****/
00543 void CFE_TIME_ExternalMET(CFE_TIME_SysTime_t NewMET);
00544
00545 /*****/
00581 void CFE_TIME_ExternalGPS(CFE_TIME_SysTime_t NewTime, int16 NewLeaps);
00582
00583 /*****/
00617 void CFE_TIME_ExternalTime(CFE_TIME_SysTime_t NewTime);
00618
00619 /*****/
00644 CFE_Status_t CFE_TIME_RegisterSynchCallback(CFE_TIME_SynchCallbackPtr_t CallbackFuncPtr);
00645
00646 /*****/
00669 CFE_Status_t CFE_TIME_UnregisterSynchCallback(CFE_TIME_SynchCallbackPtr_t CallbackFuncPtr);
00670 /*****/
00714 CFE_Status_t CFE_TIME_Print(char *PrintBuffer, CFE_TIME_SysTime_t TimeToPrint);
00715
00716 /*****/
00730 void CFE_TIME_Local1HzISR(void);
00733 #endif /* CFE_TIME_H */

```

11.93 cfe/modules/core_api/fsw/inc/cfe_time_api_typedefs.h File Reference

```

#include "common_types.h"
#include "cfe_time_extern_typedefs.h"

```

Macros

- #define `CFE_TIME_PRINTED_STRING_SIZE` 24
Required size of buffer to be passed into `CFE_TIME_Print` (includes null terminator)
- #define `CFE_TIME_ZERO_VALUE` ((CFE_TIME_SysTime_t) { 0, 0 })

Typedefs

- typedef enum `CFE_TIME_Compare` `CFE_TIME_Compare_t`
Enumerated types identifying the relative relationships of two times.
- typedef `int32`(* `CFE_TIME_SynchCallbackPtr_t`) (void)
Time Synchronization Callback Function Ptr Type.

Enumerations

- enum [CFE_TIME_Compare](#) { [CFE_TIME_A_LT_B](#) = -1 , [CFE_TIME_EQUAL](#) = 0 , [CFE_TIME_A_GT_B](#) = 1 }
- Enumerated types identifying the relative relationships of two times.*

11.93.1 Detailed Description

Purpose: cFE Time Services (TIME) library API header file

Author: S.Walling/Microtel

Notes:

Definition in file [cfe_time_api_typedefs.h](#).

11.93.2 Macro Definition Documentation

CFE_TIME_PRINTED_STRING_SIZE

```
#define CFE_TIME_PRINTED_STRING_SIZE 24
```

Required size of buffer to be passed into [CFE_TIME_Print](#) (includes null terminator)

Definition at line 45 of file [cfe_time_api_typedefs.h](#).

CFE_TIME_ZERO_VALUE

```
#define CFE_TIME_ZERO_VALUE ((CFE_TIME_SysTime_t) { 0, 0 })
```

A general-purpose initializer for CFE_TIME_SysTime_t values.

Represents "time zero" in the CFE_TIME_SysTime_t domain. This can be used as a general purpose initializer for instantiations of the CFE_TIME_SysTime_t type.

Definition at line 54 of file [cfe_time_api_typedefs.h](#).

11.93.3 Typedef Documentation

CFE_TIME_Compare_t

```
typedef enum CFE\_TIME\_Compare CFE_TIME_Compare_t
```

Enumerated types identifying the relative relationships of two times.

Description

Since time fields contain numbers that are relative to an epoch time, then it is possible for a time value to be "negative". This can lead to some confusion about what relationship exists between two time values. To resolve this confusion, the cFE provides the API [CFE_TIME_Compare](#) which returns these enumerated values.

CFE_TIME_SynchCallbackPtr_t

```
typedef int32(* CFE_TIME_SynchCallbackPtr_t) (void)
```

Time Synchronization Callback Function Ptr Type.

Description

Applications that wish to get direct notification of the receipt of the cFE Time Synchronization signal (typically a 1 Hz signal), must register a callback function with the following prototype via the [CFE_TIME_RegisterSynchCallback](#) API.

Definition at line 84 of file [cfe_time_api_typedefs.h](#).

11.93.4 Enumeration Type Documentation

CFE_TIME_Compare

enum [CFE_TIME_Compare](#)

Enumerated types identifying the relative relationships of two times.

Description

Since time fields contain numbers that are relative to an epoch time, then it is possible for a time value to be "negative". This can lead to some confusion about what relationship exists between two time values. To resolve this confusion, the cFE provides the API [CFE_TIME_Compare](#) which returns these enumerated values.

Enumerator

CFE_TIME_A_LT_B	The first specified time is considered to be before the second specified time.
CFE_TIME_EQUAL	The two specified times are considered to be equal.
CFE_TIME_A_GT_↔ _B	The first specified time is considered to be after the second specified time.

Definition at line 69 of file [cfe_time_api_typedefs.h](#).

11.94 cfe_time_api_typedefs.h

[Go to the documentation of this file.](#)

```

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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00030 #ifndef CFE_TIME_API_TYPEDEFS_H
00031 #define CFE_TIME_API_TYPEDEFS_H
00032
00033 /*
00034 ** Includes
00035 */
00036 #include "common_types.h"
00037 #include "cfe_time_extern_typedefs.h"
00038
00039 /*****
00040 */
00041 ** Macro Definitions
00042 */
00043
00044 #define CFE_TIME_PRINTED_STRING_SIZE \
00045     24
00054 #define CFE_TIME_ZERO_VALUE ((CFE_TIME_SysTime_t) { 0, 0 })
00055
00056 /*****
00057 */
00058 ** Type Definitions
00059 */
00060
00069 typedef enum CFE_TIME_Compare
00070 {
00071     CFE_TIME_A_LT_B = -1,

```

```

00072     CFE_TIME_EQUAL    = 0,
00073     CFE_TIME_A_GT_B    = 1
00074 } CFE_TIME_Compare_t;
00075
00084 typedef int32 (*CFE_TIME_SynchCallbackPtr_t) (void);
00085
00086 #endif /* CFE_TIME_API_TYPEDEFS_H */

```

11.95 cfe/modules/core_api/fsw/inc/cfe_version.h File Reference

Macros

- `#define CFE_BUILD_NUMBER 0`
Development: Number of development git commits since CFE_BUILD_BASELINE.
- `#define CFE_BUILD_BASELINE "v7.0.1"`
Development: Reference git tag for build number.
- `#define CFE_BUILD_DEV_CYCLE "v7.0.1"`
Development: Release name for current development cycle.
- `#define CFE_BUILD_CODENAME "Draco"`
: Development: Code name for the current build
- `#define CFE_MAJOR_VERSION 7`
Major version number.
- `#define CFE_MINOR_VERSION 0`
Minor version number.
- `#define CFE_REVISION 1`
Revision version number.
- `#define CFE_LAST_OFFICIAL "v7.0.1"`
Last official release.
- `#define CFE_MISSION_REV 0x0`
Mission revision.
- `#define CFE_STR_HELPER(x) #x`
Convert argument to string.
- `#define CFE_STR(x) CFE_STR_HELPER(x)`
Expand macro before conversion.
- `#define CFE_SRC_VERSION CFE_BUILD_BASELINE "+dev" CFE_STR(CFE_BUILD_NUMBER)`
Short Build Version String.
- `#define CFE_CFG_MAX_VERSION_STR_LEN 256`
Max Version String length.

11.95.1 Detailed Description

Provide version identifiers for the cFE core. See [Version Numbers](#) for further details.
 Definition in file [cfe_version.h](#).

11.95.2 Macro Definition Documentation

CFE_BUILD_BASELINE

```
#define CFE_BUILD_BASELINE "v7.0.1"
```

Development: Reference git tag for build number.
 Definition at line 30 of file [cfe_version.h](#).

CFE_BUILD_CODENAME

```
#define CFE_BUILD_CODENAME "Draco"
```

: Development: Code name for the current build
Definition at line 32 of file [cfe_version.h](#).

CFE_BUILD_DEV_CYCLE

```
#define CFE_BUILD_DEV_CYCLE "v7.0.1"
```

Development: Release name for current development cycle.
Definition at line 31 of file [cfe_version.h](#).

CFE_BUILD_NUMBER

```
#define CFE_BUILD_NUMBER 0
```

Development: Number of development git commits since CFE_BUILD_BASELINE.
Definition at line 29 of file [cfe_version.h](#).

CFE_CFG_MAX_VERSION_STR_LEN

```
#define CFE_CFG_MAX_VERSION_STR_LEN 256
```

Max Version String length.
Maximum length that a cFE version string can be.
Definition at line 69 of file [cfe_version.h](#).

CFE_LAST_OFFICIAL

```
#define CFE_LAST_OFFICIAL "v7.0.1"
```

Last official release.
Definition at line 42 of file [cfe_version.h](#).

CFE_MAJOR_VERSION

```
#define CFE_MAJOR_VERSION 7
```

Major version number.
Definition at line 35 of file [cfe_version.h](#).

CFE_MINOR_VERSION

```
#define CFE_MINOR_VERSION 0
```

Minor version number.
Definition at line 36 of file [cfe_version.h](#).

CFE_MISSION_REV

```
#define CFE_MISSION_REV 0x0
```

Mission revision.
Values 1-254 are reserved for mission use to denote patches/customizations as needed. Value of 0 is reserved for official releases only Value of 255 (0xFF) is reserved for development and testing versions
Definition at line 51 of file [cfe_version.h](#).

CFE_REVISION

```
#define CFE_REVISION 1
```

Revision version number.
Definition at line 37 of file [cfe_version.h](#).

CFE_SRC_VERSION

```
#define CFE_SRC_VERSION CFE_BUILD_BASELINE "+dev" CFE_STR(CFE_BUILD_NUMBER)
```

Short Build Version String.

Short string identifying the build, see [Version Numbers](#) for suggested format for development and official releases.

Definition at line 62 of file [cfe_version.h](#).

CFE_STR

```
#define CFE_STR(  
    x ) CFE_STR_HELPER(x)
```

Expand macro before conversion.

Definition at line 54 of file [cfe_version.h](#).

CFE_STR_HELPER

```
#define CFE_STR_HELPER(  
    x ) #x
```

Convert argument to string.

Definition at line 53 of file [cfe_version.h](#).

11.96 cfe_version.h

[Go to the documentation of this file.](#)

```
00001 /*****  
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"  
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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.  
00015  * See the License for the specific language governing permissions and  
00016  * limitations under the License.  
00017  *****/  
00018  
00025 #ifndef CFE_VERSION_H  
00026 #define CFE_VERSION_H  
00027  
00028 /* Development Build Macro Definitions */  
00029 #define CFE_BUILD_NUMBER 0  
00030 #define CFE_BUILD_BASELINE "v7.0.1"  
00031 #define CFE_BUILD_DEV_CYCLE "v7.0.1"  
00032 #define CFE_BUILD_CODENAME "Draco"  
00034 /* See \ref cfsversions for definitions */  
00035 #define CFE_MAJOR_VERSION 7  
00036 #define CFE_MINOR_VERSION 0  
00037 #define CFE_REVISION 1  
00042 #define CFE_LAST_OFFICIAL "v7.0.1"  
00043  
00051 #define CFE_MISSION_REV 0x0  
00052  
00053 #define CFE_STR_HELPER(x) #x  
00054 #define CFE_STR(x) CFE_STR_HELPER(x)  
00062 #define CFE_SRC_VERSION CFE_BUILD_BASELINE "+dev" CFE_STR(CFE_BUILD_NUMBER)  
00063  
00069 #define CFE_CFG_MAX_VERSION_STR_LEN 256  
00070  
00071 #endif /* CFE_VERSION_H */
```

11.97 cfe/modules/es/config/default_cfe_es_extern_typedefs.h File Reference

```
#include "common_types.h"
#include "cfe_resourceid_typedef.h"
#include "cfe_mission_cfg.h"
#include "cfe_es_memaddress.h"
```

Data Structures

- struct [CFE_ES_AppInfo](#)
Application Information.
- struct [CFE_ES_TaskInfo](#)
Task Information.
- struct [CFE_ES_CDSRegDumpRec](#)
CDS Register Dump Record.
- struct [CFE_ES_BlockStats](#)
Block statistics.
- struct [CFE_ES_MemPoolStats](#)
Memory Pool Statistics.

Typedefs

- typedef [uint8 CFE_ES_LogMode_Enum_t](#)
Identifies handling of log messages after storage is filled.
- typedef [uint8 CFE_ES_ExceptionAction_Enum_t](#)
Identifies action to take if exception occurs.
- typedef [uint8 CFE_ES_AppType_Enum_t](#)
Identifies type of CFE application.
- typedef [uint32 CFE_ES_RunStatus_Enum_t](#)
Run Status and Exit Status identifiers.
- typedef [uint32 CFE_ES_SystemState_Enum_t](#)
The overall cFE System State.
- typedef [uint8 CFE_ES_LogEntryType_Enum_t](#)
Type of entry in the Error and Reset (ER) Log.
- typedef [uint32 CFE_ES_AppState_Enum_t](#)
Application Run State.
- typedef CFE_RESOURCEID_BASE_TYPE [CFE_ES_AppId_t](#)
A type for Application IDs.
- typedef CFE_RESOURCEID_BASE_TYPE [CFE_ES_TaskId_t](#)
A type for Task IDs.
- typedef CFE_RESOURCEID_BASE_TYPE [CFE_ES_LibId_t](#)
A type for Library IDs.
- typedef CFE_RESOURCEID_BASE_TYPE [CFE_ES_CounterId_t](#)
A type for Counter IDs.
- typedef CFE_RESOURCEID_BASE_TYPE [CFE_ES_MemHandle_t](#)
Memory Handle type.
- typedef CFE_RESOURCEID_BASE_TYPE [CFE_ES_CDSHandle_t](#)
CDS Handle type.

- typedef [uint16 CFE_ES_TaskPriority_Atom_t](#)
Type used for task priority in CFE ES as including the commands/telemetry messages.
- typedef struct [CFE_ES_AppInfo CFE_ES_AppInfo_t](#)
Application Information.
- typedef struct [CFE_ES_TaskInfo CFE_ES_TaskInfo_t](#)
Task Information.
- typedef struct [CFE_ES_CDSRegDumpRec CFE_ES_CDSRegDumpRec_t](#)
CDS Register Dump Record.
- typedef struct [CFE_ES_BlockStats CFE_ES_BlockStats_t](#)
Block statistics.
- typedef struct [CFE_ES_MemPoolStats CFE_ES_MemPoolStats_t](#)
Memory Pool Statistics.

Enumerations

- enum [CFE_ES_LogMode](#) { [CFE_ES_LogMode_OVERWRITE](#) = 0 , [CFE_ES_LogMode_DISCARD](#) = 1 }
- Label definitions associated with CFE_ES_LogMode_Enum_t.*
- enum [CFE_ES_ExceptionAction](#) { [CFE_ES_ExceptionAction_RESTART_APP](#) = 0 , [CFE_ES_ExceptionAction_PROC_RESTART](#) = 1 }
- Label definitions associated with CFE_ES_ExceptionAction_Enum_t.*
- enum [CFE_ES_AppType](#) { [CFE_ES_AppType_CORE](#) = 1 , [CFE_ES_AppType_EXTERNAL](#) = 2 , [CFE_ES_AppType_LIBRARY](#) = 3 }
- Label definitions associated with CFE_ES_AppType_Enum_t.*
- enum [CFE_ES_RunStatus](#) {
[CFE_ES_RunStatus_UNDEFINED](#) = 0 , [CFE_ES_RunStatus_APP_RUN](#) = 1 , [CFE_ES_RunStatus_APP_EXIT](#) = 2 , [CFE_ES_RunStatus_APP_ERROR](#) = 3 ,
[CFE_ES_RunStatus_SYS_EXCEPTION](#) = 4 , [CFE_ES_RunStatus_SYS_RESTART](#) = 5 , [CFE_ES_RunStatus_SYS_RELOAD](#) = 6 , [CFE_ES_RunStatus_SYS_DELETE](#) = 7 ,
[CFE_ES_RunStatus_CORE_APP_INIT_ERROR](#) = 8 , [CFE_ES_RunStatus_CORE_APP_RUNTIME_ERROR](#) = 9 , [CFE_ES_RunStatus_MAX](#) }
- Label definitions associated with CFE_ES_RunStatus_Enum_t.*
- enum [CFE_ES_SystemState](#) {
[CFE_ES_SystemState_UNDEFINED](#) = 0 , [CFE_ES_SystemState_EARLY_INIT](#) = 1 , [CFE_ES_SystemState_CORE_STARTUP](#) = 2 , [CFE_ES_SystemState_CORE_READY](#) = 3 ,
[CFE_ES_SystemState_APPS_INIT](#) = 4 , [CFE_ES_SystemState_OPERATIONAL](#) = 5 , [CFE_ES_SystemState_SHUTDOWN](#) = 6 , [CFE_ES_SystemState_MAX](#) }
- Label definitions associated with CFE_ES_SystemState_Enum_t.*
- enum [CFE_ES_LogEntryType](#) { [CFE_ES_LogEntryType_CORE](#) = 1 , [CFE_ES_LogEntryType_APPLICATION](#) = 2 }
- Label definitions associated with CFE_ES_LogEntryType_Enum_t.*
- enum [CFE_ES_AppState](#) {
[CFE_ES_AppState_UNDEFINED](#) = 0 , [CFE_ES_AppState_EARLY_INIT](#) = 1 , [CFE_ES_AppState_LATE_INIT](#) = 2 , [CFE_ES_AppState_RUNNING](#) = 3 ,
[CFE_ES_AppState_WAITING](#) = 4 , [CFE_ES_AppState_STOPPED](#) = 5 , [CFE_ES_AppState_MAX](#) }
- Label definitions associated with CFE_ES_AppState_Enum_t.*

11.97.1 Detailed Description

Declarations and prototypes for cfe_es_extern_typedefs module
Definition in file [default_cfe_es_extern_typedefs.h](#).

11.97.2 Typedef Documentation

CFE_ES_AppId_t

```
typedef CFE_RESOURCEID_BASE_TYPE CFE_ES_AppId_t
```

A type for Application IDs.

This is the type that is used for any API accepting or returning an App ID

Definition at line 314 of file [default_cfe_es_extern_typedefs.h](#).

CFE_ES_AppInfo_t

```
typedef struct CFE_ES_AppInfo CFE_ES_AppInfo_t
```

Application Information.

Structure that is used to provide information about an app. It is primarily used for the QueryOne and QueryAll Commands.

While this structure is primarily intended for Application info, it can also represent Library information where only a subset of the information applies.

CFE_ES_AppState_Enum_t

```
typedef uint32 CFE_ES_AppState_Enum_t
```

Application Run State.

The normal progression of APP states: UNDEFINED -> EARLY_INIT -> LATE_INIT -> RUNNING -> WAITING -> STOPPED

Note

These are defined in order so that relational comparisons e.g. if (STATEA < STATEB) are possible

See also

enum [CFE_ES_AppState](#)

Definition at line 307 of file [default_cfe_es_extern_typedefs.h](#).

CFE_ES_AppType_Enum_t

```
typedef uint8 CFE_ES_AppType_Enum_t
```

Identifies type of CFE application.

See also

enum [CFE_ES_AppType](#)

Definition at line 106 of file [default_cfe_es_extern_typedefs.h](#).

CFE_ES_BlockStats_t

```
typedef struct CFE_ES_BlockStats CFE_ES_BlockStats_t
```

Block statistics.

Sub-Structure that is used to provide information about a specific block size/bucket within a memory pool.

CFE_ES_CDSHandle_t

```
typedef CFE_RESOURCEID_BASE_TYPE CFE_ES_CDSHandle_t
```

CDS Handle type.

Data type used to hold Handles of Critical Data Stores. See [CFE_ES_RegisterCDS](#)

Definition at line 350 of file [default_cfe_es_extern_typedefs.h](#).

CFE_ES_CDSRegDumpRec_t

```
typedef struct CFE_ES_CDSRegDumpRec CFE_ES_CDSRegDumpRec_t
```

CDS Register Dump Record.

Structure that is used to provide information about a critical data store. It is primarily used for the Dump CDS registry ([CFE_ES_DUMP_CDS_REGISTRY_CC](#)) command.

Note

There is not currently a telemetry message directly containing this data structure, but it does define the format of the data file generated by the Dump CDS registry command. Therefore it should be considered part of the overall telemetry interface.

CFE_ES_CounterId_t

```
typedef CFE_RESOURCEID_BASE_TYPE CFE_ES_CounterId_t
```

A type for Counter IDs.

This is the type that is used for any API accepting or returning a Counter ID

Definition at line 335 of file [default_cfe_es_extern_typedefs.h](#).

CFE_ES_ExceptionAction_Enum_t

```
typedef uint8 CFE_ES_ExceptionAction_Enum_t
```

Identifies action to take if exception occurs.

See also

enum [CFE_ES_ExceptionAction](#)

Definition at line 78 of file [default_cfe_es_extern_typedefs.h](#).

CFE_ES_LibId_t

```
typedef CFE_RESOURCEID_BASE_TYPE CFE_ES_LibId_t
```

A type for Library IDs.

This is the type that is used for any API accepting or returning a Lib ID

Definition at line 328 of file [default_cfe_es_extern_typedefs.h](#).

CFE_ES_LogEntryType_Enum_t

```
typedef uint8 CFE_ES_LogEntryType_Enum_t
```

Type of entry in the Error and Reset (ER) Log.

See also

enum [CFE_ES_LogEntryType](#)

Definition at line 254 of file [default_cfe_es_extern_typedefs.h](#).

CFE_ES_LogMode_Enum_t

```
typedef uint8 CFE_ES_LogMode_Enum_t
```

Identifies handling of log messages after storage is filled.

See also

enum [CFE_ES_LogMode](#)

Definition at line 55 of file [default_cfe_es_extern_typedefs.h](#).

CFE_ES_MemHandle_t

```
typedef CFE_RESOURCEID_BASE_TYPE CFE_ES_MemHandle_t
```

Memory Handle type.

Data type used to hold Handles of Memory Pools created via CFE_ES_PoolCreate and CFE_ES_PoolCreateNoSem

Definition at line 343 of file [default_cfe_es_extern_typedefs.h](#).

CFE_ES_MemPoolStats_t

```
typedef struct CFE_ES_MemPoolStats CFE_ES_MemPoolStats_t
```

Memory Pool Statistics.

Structure that is used to provide information about a memory pool. Used by the Memory Pool Stats telemetry message.

See also

[CFE_ES_SEND_MEM_POOL_STATS_CC](#)

CFE_ES_RunStatus_Enum_t

```
typedef uint32 CFE_ES_RunStatus_Enum_t
```

Run Status and Exit Status identifiers.

See also

enum [CFE_ES_RunStatus](#)

Definition at line 174 of file [default_cfe_es_extern_typedefs.h](#).

CFE_ES_SystemState_Enum_t

```
typedef uint32 CFE_ES_SystemState_Enum_t
```

The overall cFE System State.

These values are used with the [CFE_ES_WaitForSystemState](#) API call to synchronize application startup.

Note

These are defined in order so that relational comparisons e.g. if (STATEA < STATEB) are possible

See also

enum [CFE_ES_SystemState](#)

Definition at line 231 of file [default_cfe_es_extern_typedefs.h](#).

CFE_ES_TaskId_t

```
typedef CFE_RESOURCEID_BASE_TYPE CFE_ES_TaskId_t
```

A type for Task IDs.

This is the type that is used for any API accepting or returning a Task ID

Definition at line 321 of file [default_cfe_es_extern_typedefs.h](#).

CFE_ES_TaskInfo_t

```
typedef struct CFE_ES_TaskInfo CFE_ES_TaskInfo_t
```

Task Information.

Structure that is used to provide information about a task. It is primarily used for the Query All Tasks ([CFE_ES_QUERY_ALL_TASKS_CC](#)) command.

Note

There is not currently a telemetry message directly containing this data structure, but it does define the format of the data file generated by the Query All Tasks command. Therefore it should be considered part of the overall telemetry interface.

CFE_ES_TaskPriority_Atom_t

```
typedef uint16 CFE_ES_TaskPriority_Atom_t
```

Type used for task priority in CFE ES as including the commands/telemetry messages.

Note

the valid range is only 0-255 (same as OSAL) but a wider type is used for backward compatibility in binary formats of messages.

Definition at line 360 of file [default_cfe_es_extern_typedefs.h](#).

11.97.3 Enumeration Type Documentation**CFE_ES_AppState**

```
enum CFE_ES_AppState
```

Label definitions associated with CFE_ES_AppState_Enum_t.

Enumerator

CFE_ES_AppState_UNDEFINED	Initial state before app thread is started.
CFE_ES_AppState_EARLY_INIT	App thread has started, app performing early initialization of its own data.
CFE_ES_AppState_LATE_INIT	Early/Local initialization is complete. First sync point.
CFE_ES_AppState_RUNNING	All initialization is complete. Second sync point.
CFE_ES_AppState_WAITING	Application is waiting on a Restart/Reload/Delete request.
CFE_ES_AppState_STOPPED	Application is stopped.
CFE_ES_AppState_MAX	Reserved entry, marker for the maximum state.

Definition at line 259 of file [default_cfe_es_extern_typedefs.h](#).

CFE_ES_AppType

```
enum CFE_ES_AppType
```

Label definitions associated with CFE_ES_AppType_Enum_t.

Enumerator

CFE_ES_AppType_CORE	CFE core application.
CFE_ES_AppType_EXTERNAL	CFE external application.
CFE_ES_AppType_LIBRARY	CFE library.

Definition at line 83 of file [default_cfe_es_extern_typedefs.h](#).

CFE_ES_ExceptionAction

```
enum CFE_ES_ExceptionAction
```

Label definitions associated with CFE_ES_ExceptionAction_Enum_t.

Enumerator

CFE_ES_ExceptionAction_RESTART_APP	Restart application if exception occurs.
CFE_ES_ExceptionAction_PROC_RESTART	Restart processor if exception occurs.

Definition at line 60 of file [default_cfe_es_extern_typedefs.h](#).

CFE_ES_LogEntryType

enum [CFE_ES_LogEntryType](#)

Label definitions associated with CFE_ES_LogEntryType_Enum_t.

Enumerator

CFE_ES_LogEntryType_CORE	Log entry from a core subsystem.
CFE_ES_LogEntryType_APPLICATION	Log entry from an application.

Definition at line 236 of file [default_cfe_es_extern_typedefs.h](#).

CFE_ES_LogMode

enum [CFE_ES_LogMode](#)

Label definitions associated with CFE_ES_LogMode_Enum_t.

Enumerator

CFE_ES_LogMode_OVERWRITE	Overwrite Log Mode.
CFE_ES_LogMode_DISCARD	Discard Log Mode.

Definition at line 37 of file [default_cfe_es_extern_typedefs.h](#).

CFE_ES_RunStatus

enum [CFE_ES_RunStatus](#)

Label definitions associated with CFE_ES_RunStatus_Enum_t.

Enumerator

CFE_ES_RunStatus_UNDEFINED	Reserved value, should not be used.
CFE_ES_RunStatus_APP_RUN	Indicates that the Application should continue to run.
CFE_ES_RunStatus_APP_EXIT	Indicates that the Application wants to exit normally.
CFE_ES_RunStatus_APP_ERROR	Indicates that the Application is quitting with an error.
CFE_ES_RunStatus_SYS_EXCEPTION	The cFE App caused an exception.
CFE_ES_RunStatus_SYS_RESTART	The system is requesting a restart of the cFE App.
CFE_ES_RunStatus_SYS_RELOAD	The system is requesting a reload of the cFE App.
CFE_ES_RunStatus_SYS_DELETE	The system is requesting that the cFE App is stopped.
CFE_ES_RunStatus_CORE_APP_INIT_ERROR	Indicates that the Core Application could not Init.
CFE_ES_RunStatus_CORE_APP_RUNTIME_ERROR	Indicates that the Core Application had a runtime failure.
CFE_ES_RunStatus_MAX	Reserved value, marker for the maximum state.

Definition at line 111 of file [default_cfe_es_extern_typedefs.h](#).

CFE_ES_SystemStateenum [CFE_ES_SystemState](#)Label definitions associated with [CFE_ES_SystemState_Enum_t](#).**Enumerator**

CFE_ES_SystemState_UNDEFINED	reserved
CFE_ES_SystemState_EARLY_INIT	single threaded mode while setting up CFE itself
CFE_ES_SystemState_CORE_STARTUP	core apps (CFE_ES_ObjectTable) are starting (multi-threaded)
CFE_ES_SystemState_CORE_READY	core is ready, starting other external apps/libraries (if any)
CFE_ES_SystemState_APPS_INIT	startup apps have all completed their early init, but not necessarily operational yet
CFE_ES_SystemState_OPERATIONAL	normal operation mode; all apps are RUNNING
CFE_ES_SystemState_SHUTDOWN	reserved for future use, all apps would be STOPPED
CFE_ES_SystemState_MAX	Reserved value, marker for the maximum state.

Definition at line 179 of file [default_cfe_es_extern_typedefs.h](#).**11.98 default_cfe_es_extern_typedefs.h**[Go to the documentation of this file.](#)

```

00001 /*****
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"
00003  *
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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00025 #ifndef DEFAULT_CFE_ES_EXTERN_TYPEDEFS_H
00026 #define DEFAULT_CFE_ES_EXTERN_TYPEDEFS_H
00027
00028 #include "common_types.h"
00029 #include "cfe_resourceid_typedef.h"
00030 #include "cfe_mission_cfg.h"
00031
00032 #include "cfe_es_memaddress.h"
00033
00037 enum CFE_ES_LogMode
00038 {
00042     CFE_ES_LogMode_OVERWRITE = 0,
00043
00047     CFE_ES_LogMode_DISCARD = 1
00048 };
00049
00055 typedef uint8 CFE_ES_LogMode_Enum_t;
00056
00060 enum CFE_ES_ExceptionAction
00061 {
00065     CFE_ES_ExceptionAction_RESTART_APP = 0,
00066
00070     CFE_ES_ExceptionAction_PROC_RESTART = 1
00071 };
00072
00078 typedef uint8 CFE_ES_ExceptionAction_Enum_t;
00079
00083 enum CFE_ES_AppType

```

```
00084 {
00088     CFE_ES_AppType_CORE = 1,
00089
00093     CFE_ES_AppType_EXTERNAL = 2,
00094
00098     CFE_ES_AppType_LIBRARY = 3
00099 };
00100
00106 typedef uint8 CFE_ES_AppType_Enum_t;
00107
00111 enum CFE_ES_RunStatus
00112 {
00116     CFE_ES_RunStatus_UNDEFINED = 0,
00117
00121     CFE_ES_RunStatus_APP_RUN = 1,
00122
00126     CFE_ES_RunStatus_APP_EXIT = 2,
00127
00131     CFE_ES_RunStatus_APP_ERROR = 3,
00132
00136     CFE_ES_RunStatus_SYS_EXCEPTION = 4,
00137
00141     CFE_ES_RunStatus_SYS_RESTART = 5,
00142
00146     CFE_ES_RunStatus_SYS_RELOAD = 6,
00147
00151     CFE_ES_RunStatus_SYS_DELETE = 7,
00152
00156     CFE_ES_RunStatus_CORE_APP_INIT_ERROR = 8,
00157
00161     CFE_ES_RunStatus_CORE_APP_RUNTIME_ERROR = 9,
00162
00166     CFE_ES_RunStatus_MAX
00167 };
00168
00174 typedef uint32 CFE_ES_RunStatus_Enum_t;
00175
00179 enum CFE_ES_SystemState
00180 {
00184     CFE_ES_SystemState_UNDEFINED = 0,
00185
00189     CFE_ES_SystemState_EARLY_INIT = 1,
00190
00194     CFE_ES_SystemState_CORE_STARTUP = 2,
00195
00199     CFE_ES_SystemState_CORE_READY = 3,
00200
00204     CFE_ES_SystemState_APPS_INIT = 4,
00205
00209     CFE_ES_SystemState_OPERATIONAL = 5,
00210
00214     CFE_ES_SystemState_SHUTDOWN = 6,
00215
00219     CFE_ES_SystemState_MAX
00220 };
00221
00231 typedef uint32 CFE_ES_SystemState_Enum_t;
00232
00236 enum CFE_ES_LogEntryType
00237 {
00241     CFE_ES_LogEntryType_CORE = 1,
00242
00246     CFE_ES_LogEntryType_APPLICATION = 2
00247 };
00248
00254 typedef uint8 CFE_ES_LogEntryType_Enum_t;
00255
00259 enum CFE_ES_AppState
00260 {
00264     CFE_ES_AppState_UNDEFINED = 0,
00265
00269     CFE_ES_AppState_EARLY_INIT = 1,
00270
00274     CFE_ES_AppState_LATE_INIT = 2,
00275
00279     CFE_ES_AppState_RUNNING = 3,
00280
00284     CFE_ES_AppState_WAITING = 4,
00285
00289     CFE_ES_AppState_STOPPED = 5,
00290
```

```

00294     CFE_ES_AppState_MAX
00295 };
00296
00307 typedef uint32 CFE_ES_AppState_Enum_t;
00308
00314 typedef CFE_RESOURCEID_BASE_TYPE CFE_ES_AppId_t;
00315
00321 typedef CFE_RESOURCEID_BASE_TYPE CFE_ES_TaskId_t;
00322
00328 typedef CFE_RESOURCEID_BASE_TYPE CFE_ES_LibId_t;
00329
00335 typedef CFE_RESOURCEID_BASE_TYPE CFE_ES_CounterId_t;
00336
00343 typedef CFE_RESOURCEID_BASE_TYPE CFE_ES_MemHandle_t;
00344
00350 typedef CFE_RESOURCEID_BASE_TYPE CFE_ES_CDSHandle_t;
00351
00360 typedef uint16 CFE_ES_TaskPriority_Atom_t;
00361
00362 /*
00363  * Data Structures shared between API and Message (CMD/TLM) interfaces
00364  */
00365
00376 typedef struct CFE_ES_AppInfo
00377 {
00378     CFE_ResourceId_t ResourceId;
00380     uint32          Type;
00383     char Name[CFE_MISSION_MAX_API_LEN];
00385     char EntryPoint[CFE_MISSION_MAX_API_LEN];
00387     char FileName[CFE_MISSION_MAX_PATH_LEN];
00390     CFE_ES_MemOffset_t StackSize;
00392     uint32              AddressesAreValid;
00394     uint8               Padding1[4];
00395     CFE_ES_MemAddress_t CodeAddress;
00397     CFE_ES_MemOffset_t CodeSize;
00399     CFE_ES_MemAddress_t DataAddress;
00401     CFE_ES_MemOffset_t DataSize;
00403     CFE_ES_MemAddress_t BSSAddress;
00405     CFE_ES_MemOffset_t BSSSize;
00407     CFE_ES_MemAddress_t StartAddress;
00409     CFE_ES_ExceptionAction_Enum_t ExceptionAction;
00412     uint8               Padding2;
00413     CFE_ES_TaskPriority_Atom_t Priority;
00415     CFE_ES_TaskId_t      MainTaskId;
00417     uint32               ExecutionCounter;
00419     char                 MainTaskName[CFE_MISSION_MAX_API_LEN];
00421     uint32               NumOfChildTasks;
00423     uint8               Padding3[4];
00424 } CFE_ES_AppInfo_t;
00425
00437 typedef struct CFE_ES_TaskInfo
00438 {
00439     CFE_ES_TaskId_t      TaskId;
00440     uint32               ExecutionCounter;
00441     char                 TaskName[CFE_MISSION_MAX_API_LEN];
00442     CFE_ES_AppId_t       AppId;
00443     char                 AppName[CFE_MISSION_MAX_API_LEN];
00444     CFE_ES_MemOffset_t   StackSize;
00445     CFE_ES_TaskPriority_Atom_t Priority;
00446     uint8               Spare[2];
00447 } CFE_ES_TaskInfo_t;
00448
00461 typedef struct CFE_ES_CDSRegDumpRec
00462 {
00463     CFE_ES_CDSHandle_t Handle;
00464     CFE_ES_MemOffset_t Size;
00465     bool               Table;
00466     char               Name[CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN];
00467     uint8              ByteAlignSpare[3];
00468 } CFE_ES_CDSRegDumpRec_t;
00469
00476 typedef struct CFE_ES_BlockStats
00477 {
00478     CFE_ES_MemOffset_t BlockSize;
00479     uint32              NumCreated;
00480     uint32              NumFree;
00481 } CFE_ES_BlockStats_t;
00482
00491 typedef struct CFE_ES_MemPoolStats
00492 {
00493     CFE_ES_MemOffset_t PoolSize;

```



```

00495     uint32             NumBlocksRequested;
00497     uint32             CheckErrCtr;
00499     CFE_ES_MemOffset_t  NumFreeBytes;
00501     CFE_ES_BlockStats_t BlockStats[CFE_MISSION_ES_POOL_MAX_BUCKETS];
00503 } CFE_ES_MemPoolStats_t;
00504
00505 #endif /* CFE_ES_EXTERN_TYPEDEFS_H */

```

11.99 cfe/modules/es/config/default_cfe_es_fcncode_values.h File Reference

Macros

- #define `CFE_ES_CCVAL(x) CFE_ES_FunctionCode_##x`

Enumerations

- enum `CFE_ES_FunctionCode_` {
`CFE_ES_FunctionCode_NOOP` = 0 , `CFE_ES_FunctionCode_RESET_COUNTERS` = 1 , `CFE_ES_FunctionCode_RESTART`
= 2 , `CFE_ES_FunctionCode_START_APP` = 4 ,
`CFE_ES_FunctionCode_STOP_APP` = 5 , `CFE_ES_FunctionCode_RESTART_APP` = 6 , `CFE_ES_FunctionCode_RELOAD_APP`
= 7 , `CFE_ES_FunctionCode_QUERY_ONE` = 8 ,
`CFE_ES_FunctionCode_QUERY_ALL` = 9 , `CFE_ES_FunctionCode_CLEAR_SYS_LOG` = 10 , `CFE_ES_FunctionCode_WRITE_S`
= 11 , `CFE_ES_FunctionCode_CLEAR_ER_LOG` = 12 ,
`CFE_ES_FunctionCode_WRITE_ER_LOG` = 13 , `CFE_ES_FunctionCode_START_PERF_DATA` = 14 ,
`CFE_ES_FunctionCode_STOP_PERF_DATA` = 15 , `CFE_ES_FunctionCode_SET_PERF_FILTER_MASK` =
16 ,
`CFE_ES_FunctionCode_SET_PERF_TRIGGER_MASK` = 17 , `CFE_ES_FunctionCode_OVER_WRITE_SYS_LOG`
= 18 , `CFE_ES_FunctionCode_RESET_PR_COUNT` = 19 , `CFE_ES_FunctionCode_SET_MAX_PR_COUNT` =
20 ,
`CFE_ES_FunctionCode_DELETE_CDS` = 21 , `CFE_ES_FunctionCode_SEND_MEM_POOL_STATS` = 22 ,
`CFE_ES_FunctionCode_DUMP_CDS_REGISTRY` = 23 , `CFE_ES_FunctionCode_QUERY_ALL_TASKS` = 24 }

11.99.1 Detailed Description

Specification for the CFE Executive Services (CFE_ES) command function codes

Note

This file should be strictly limited to the command/function code (CC) macro definitions. Other definitions such as enums, typedefs, or other macros should be placed in the msgdefs.h or msg.h files.

Definition in file [default_cfe_es_fcncode_values.h](#).

11.99.2 Macro Definition Documentation

CFE_ES_CCVAL

```

#define CFE_ES_CCVAL(
    x ) CFE_ES_FunctionCode_##x

```

Definition at line 35 of file [default_cfe_es_fcncode_values.h](#).

11.99.3 Enumeration Type Documentation

CFE_ES_FunctionCode_

```
enum CFE_ES_FunctionCode_
```

Enumerator

CFE_ES_FunctionCode_NOOP	
--------------------------	--

Enumerator

CFE_ES_FunctionCode_RESET_COUNTERS	
CFE_ES_FunctionCode_RESTART	
CFE_ES_FunctionCode_START_APP	
CFE_ES_FunctionCode_STOP_APP	
CFE_ES_FunctionCode_RESTART_APP	
CFE_ES_FunctionCode_RELOAD_APP	
CFE_ES_FunctionCode_QUERY_ONE	
CFE_ES_FunctionCode_QUERY_ALL	
CFE_ES_FunctionCode_CLEAR_SYS_LOG	
CFE_ES_FunctionCode_WRITE_SYS_LOG	
CFE_ES_FunctionCode_CLEAR_ER_LOG	
CFE_ES_FunctionCode_WRITE_ER_LOG	
CFE_ES_FunctionCode_START_PERF_DATA	
CFE_ES_FunctionCode_STOP_PERF_DATA	
CFE_ES_FunctionCode_SET_PERF_FILTER_MASK	
CFE_ES_FunctionCode_SET_PERF_TRIGGER_MASK	
CFE_ES_FunctionCode_OVER_WRITE_SYS_LOG	
CFE_ES_FunctionCode_RESET_PR_COUNT	
CFE_ES_FunctionCode_SET_MAX_PR_COUNT	
CFE_ES_FunctionCode_DELETE_CDS	
CFE_ES_FunctionCode_SEND_MEM_POOL_STATS	
CFE_ES_FunctionCode_DUMP_CDS_REGISTRY	
CFE_ES_FunctionCode_QUERY_ALL_TASKS	

Definition at line 37 of file [default_cfe_es_fcncode_values.h](#).

11.100 default_cfe_es_fcncode_values.h

[Go to the documentation of this file.](#)

```

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00017  *****/
00018
00028 #ifndef DEFAULT_CFE_ES_FCNCODE_VALUES_H
00029 #define DEFAULT_CFE_ES_FCNCODE_VALUES_H
00030
00031 /*****
00032  * Macro Definitions
00033  *****/
00034
00035 #define CFE_ES_CCVAL(x) CFE_ES_FunctionCode_##x
00036
00037 enum CFE_ES_FunctionCode_
00038 {
00039     CFE_ES_FunctionCode_NOOP = 0,

```

```

00040     CFE_ES_FunctionCode_RESET_COUNTERS      = 1,
00041     CFE_ES_FunctionCode_RESTART              = 2,
00042     CFE_ES_FunctionCode_START_APP            = 4,
00043     CFE_ES_FunctionCode_STOP_APP             = 5,
00044     CFE_ES_FunctionCode_RESTART_APP          = 6,
00045     CFE_ES_FunctionCode_RELOAD_APP           = 7,
00046     CFE_ES_FunctionCode_QUERY_ONE            = 8,
00047     CFE_ES_FunctionCode_QUERY_ALL            = 9,
00048     CFE_ES_FunctionCode_CLEAR_SYS_LOG        = 10,
00049     CFE_ES_FunctionCode_WRITE_SYS_LOG        = 11,
00050     CFE_ES_FunctionCode_CLEAR_ER_LOG         = 12,
00051     CFE_ES_FunctionCode_WRITE_ER_LOG         = 13,
00052     CFE_ES_FunctionCode_START_PERF_DATA      = 14,
00053     CFE_ES_FunctionCode_STOP_PERF_DATA       = 15,
00054     CFE_ES_FunctionCode_SET_PERF_FILTER_MASK = 16,
00055     CFE_ES_FunctionCode_SET_PERF_TRIGGER_MASK = 17,
00056     CFE_ES_FunctionCode_OVER_WRITE_SYS_LOG   = 18,
00057     CFE_ES_FunctionCode_RESET_PR_COUNT       = 19,
00058     CFE_ES_FunctionCode_SET_MAX_PR_COUNT     = 20,
00059     CFE_ES_FunctionCode_DELETE_CDS           = 21,
00060     CFE_ES_FunctionCode_SEND_MEM_POOL_STATS  = 22,
00061     CFE_ES_FunctionCode_DUMP_CDS_REGISTRY    = 23,
00062     CFE_ES_FunctionCode_QUERY_ALL_TASKS      = 24,
00063 };
00064
00065 #endif

```

11.101 cfe/modules/es/config/default_cfe_es_interface_cfg_values.h File Reference

Macros

- `#define CFE_MISSION_ES_CFGVAL(x) DEFAULT_CFE_MISSION_ES_##x`

11.101.1 Detailed Description

CFE Executive Services (CFE_ES) Application Public Definitions

This provides default values for configurable items that affect the interface(s) of this module. This includes the CMD/TLM message interface, tables definitions, and any other data products that serve to exchange information with other entities.

Note

This file may be overridden/supersceded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [default_cfe_es_interface_cfg_values.h](#).

11.101.2 Macro Definition Documentation

CFE_MISSION_ES_CFGVAL

```

#define CFE_MISSION_ES_CFGVAL(
    x ) DEFAULT_CFE_MISSION_ES_##x

```

Definition at line 36 of file [default_cfe_es_interface_cfg_values.h](#).

11.102 default_cfe_es_interface_cfg_values.h

[Go to the documentation of this file.](#)

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00017  *****/
00018
00032 #ifndef DEFAULT_CFE_ES_INTERFACE_CFG_VALUES_H
00033 #define DEFAULT_CFE_ES_INTERFACE_CFG_VALUES_H
00034
00035 /* Use the default configuration value for all */
00036 #define CFE_MISSION_ES_CFGVAL(x) DEFAULT_CFE_MISSION_ES_##x
00037
00038 #endif

```

11.103 cfe/modules/es/config/default_cfe_es_internal_cfg_values.h File Reference

Macros

- #define [CFE_PLATFORM_ES_CFGVAL\(x\)](#) DEFAULT_CFE_PLATFORM_ES_##x

11.103.1 Detailed Description

CFE Executive Services (CFE_ES) Application Private Config Definitions

This provides default values for configurable items that are internal to this module and do NOT affect the interface(s) of this module. Changes to items in this file only affect the local module and will be transparent to external entities that are using the public interface(s).

Note

This file may be overridden/superseded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [default_cfe_es_internal_cfg_values.h](#).

11.103.2 Macro Definition Documentation

CFE_PLATFORM_ES_CFGVAL

```
#define CFE_PLATFORM_ES_CFGVAL(
    x ) DEFAULT_CFE_PLATFORM_ES_##x
```

Definition at line 36 of file [default_cfe_es_internal_cfg_values.h](#).

11.104 default_cfe_es_internal_cfg_values.h

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```

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00017  *****/
00018
00032 #ifndef DEFAULT_CFE_ES_INTERNAL_CFG_VALUES_H
00033 #define DEFAULT_CFE_ES_INTERNAL_CFG_VALUES_H
00034
00035 /* Use the default configuration value for all */
00036 #define CFE_PLATFORM_ES_CFGVAL(x) DEFAULT_CFE_PLATFORM_ES_##x
00037
00038 #endif

```

11.105 cfe/modules/es/config/default_cfe_es_memaddress.h File Reference

```
#include "common_types.h"
```

Macros

- `#define CFE_ES_MEMOFFSET_C(x) ((CFE_ES_MemOffset_t)(x))`
Memory Offset initializer wrapper.
- `#define CFE_ES_MEMOFFSET_TO_SIZE_T(x) ((size_t)(x))`
Memory Offset to integer value (size_t) wrapper.
- `#define CFE_ES_MEMADDRESS_C(x) ((CFE_ES_MemAddress_t)(cpuaddr)(x))`
Memory Address initializer wrapper.
- `#define CFE_ES_MEMADDRESS_TO_PTR(x) ((void *) (cpuaddr)(x))`
Memory Address to pointer wrapper.

Typedefs

- `typedef uint64 CFE_ES_MemOffset_t`
Type used for memory sizes and offsets in commands and telemetry.
- `typedef uint64 CFE_ES_MemAddress_t`
Type used for memory addresses in command and telemetry messages.

11.105.1 Detailed Description

Example header file override that defines memory addresses and offsets to be a full 64 bit integer value.

This is the simplest and most efficient approach to use 64 bit addressing, but may introduce unexpected padding in certain cases. On most systems a uint64 value needs to be aligned on a 64-bit boundary, and CFS has some TLM and CMD structures where this is not the case. These will be padded by the compiler and may cause interoperability issues. To use this implementation, clone this file as "cfe_es_memaddress.h" in your local defs dir.

Definition in file [default_cfe_es_memaddress.h](#).

11.105.2 Macro Definition Documentation

CFE_ES_MEMADDRESS_C

```
#define CFE_ES_MEMADDRESS_C(  
    x ) ((CFE_ES_MemAddress_t) (cpuaddr) (x))
```

Memory Address initializer wrapper.

A converter macro to use when initializing a [CFE_ES_MemAddress_t](#) from a pointer value of a different type.

Definition at line 81 of file [default_cfe_es_memaddress.h](#).

CFE_ES_MEMADDRESS_TO_PTR

```
#define CFE_ES_MEMADDRESS_TO_PTR(  
    x ) ((void *) (cpuaddr) (x))
```

Memory Address to pointer wrapper.

A converter macro to use when interpreting a [CFE_ES_MemAddress_t](#) as a pointer value.

Definition at line 89 of file [default_cfe_es_memaddress.h](#).

CFE_ES_MEMOFFSET_C

```
#define CFE_ES_MEMOFFSET_C(
    x ) ((CFE_ES_MemOffset_t)(x))
```

Memory Offset initializer wrapper.

A converter macro to use when initializing a [CFE_ES_MemOffset_t](#) from an integer value of a different type.

Definition at line 58 of file [default_cfe_es_memaddress.h](#).

CFE_ES_MEMOFFSET_TO_SIZE_T

```
#define CFE_ES_MEMOFFSET_TO_SIZE_T(
    x ) ((size_t)(x))
```

Memory Offset to integer value (size_t) wrapper.

A converter macro to use when interpreting a [CFE_ES_MemOffset_t](#) value as a "size_t" type

Definition at line 66 of file [default_cfe_es_memaddress.h](#).

11.105.3 Typedef Documentation**CFE_ES_MemAddress_t**

```
typedef uint64 CFE_ES_MemAddress_t
```

Type used for memory addresses in command and telemetry messages.

Use a full 64-bit integer value for memory offsets

Definition at line 73 of file [default_cfe_es_memaddress.h](#).

CFE_ES_MemOffset_t

```
typedef uint64 CFE_ES_MemOffset_t
```

Type used for memory sizes and offsets in commands and telemetry.

Use a full 64-bit integer value for memory offsets

Definition at line 50 of file [default_cfe_es_memaddress.h](#).

11.106 default_cfe_es_memaddress.h

[Go to the documentation of this file.](#)

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00017  *****/
00018
00035 #ifndef DEFAULT_CFE_ES_MEMADDRESS_H
00036 #define DEFAULT_CFE_ES_MEMADDRESS_H
00037
00038 #include "common_types.h"
00039
00040 /*
00041  * NOTE: On a typical Linux/GCC build environment, this produces
00042  * sizeof(CFE_ES_AppInfo_t) == 224 based on the current definition
00043  */
00044
00050 typedef uint64 CFE_ES_MemOffset_t;
00051
00058 #define CFE_ES_MEMOFFSET_C(x) ((CFE_ES_MemOffset_t)(x))
```

```

00059
00066 #define CFE_ES_MEMOFFSET_TO_SIZE_T(x) ((size_t)(x))
00067
00073 typedef uint64 CFE_ES_MemAddress_t;
00074
00081 #define CFE_ES_MEMADDRESS_C(x) ((CFE_ES_MemAddress_t)(cpuaddr)(x))
00082
00089 #define CFE_ES_MEMADDRESS_TO_PTR(x) ((void *) (cpuaddr)(x))
00090
00091 #endif /* CFE_ES_EXTERN_TYPEDEFS_H */

```

11.107 cfe/modules/es/config/default_cfe_es_mission_cfg.h File Reference

```
#include "cfe_es_interface_cfg.h"
```

11.107.1 Detailed Description

CFE Executive Services (CFE_ES) Application Mission Configuration Header File

This is a compatibility header for the "mission_cfg.h" file that has traditionally provided public config definitions for each CFS app.

Note

This file may be overridden/superseded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [default_cfe_es_mission_cfg.h](#).

11.108 default_cfe_es_mission_cfg.h

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00016  * limitations under the License.
00017  *****/
00018
00031 #ifndef DEFAULT_CFE_ES_MISSION_CFG_H
00032 #define DEFAULT_CFE_ES_MISSION_CFG_H
00033
00034 #include "cfe_es_interface_cfg.h"
00035
00036 #endif

```

11.109 cfe/modules/es/config/default_cfe_es_msg.h File Reference

```

#include "cfe_mission_cfg.h"
#include "cfe_es_fcncodes.h"
#include "cfe_es_msgdefs.h"
#include "cfe_es_msgstruct.h"

```

11.109.1 Detailed Description

Specification for the CFE Executive Services (CFE_ES) command and telemetry message data types.

This is a compatibility header for the "cfe_es_msg.h" file that has traditionally provided the message definitions for cFS apps.

Note

This file may be overridden/superceded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [default_cfe_es_msg.h](#).

11.110 default_cfe_es_msg.h

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00017  *****/
00018
00031 #ifndef DEFAULT_CFE_ES_MSG_H
00032 #define DEFAULT_CFE_ES_MSG_H
00033
00034 #include "cfe_mission_cfg.h"
00035 #include "cfe_es_fcncodes.h"
00036 #include "cfe_es_msgdefs.h"
00037 #include "cfe_es_msgstruct.h"
00038
00039 #endif
```

11.111 cfe/modules/es/config/default_cfe_es_msgdefs.h File Reference

```
#include "common_types.h"
#include "cfe_mission_cfg.h"
#include "cfe_es_extern_typedefs.h"
#include "cfe_es_fcncodes.h"
```

Data Structures

- struct [CFE_ES_RestartCmd_Payload](#)
Restart cFE Command Payload.
- struct [CFE_ES_FileNameCmd_Payload](#)
Generic file name command payload.
- struct [CFE_ES_OverWriteSysLogCmd_Payload](#)
Overwrite/Discard System Log Configuration Command Payload.
- struct [CFE_ES_StartAppCmd_Payload](#)
Start Application Command Payload.
- struct [CFE_ES_AppNameCmd_Payload](#)

- *Generic application name command payload.*
- struct [CFE_ES_AppReloadCmd_Payload](#)
Reload Application Command Payload.
- struct [CFE_ES_SetMaxPRCountCmd_Payload](#)
Set Maximum Processor Reset Count Command Payload.
- struct [CFE_ES_DeleteCDSCmd_Payload](#)
Delete Critical Data Store Command Payload.
- struct [CFE_ES_StartPerfCmd_Payload](#)
Start Performance Analyzer Command Payload.
- struct [CFE_ES_StopPerfCmd_Payload](#)
Stop Performance Analyzer Command Payload.
- struct [CFE_ES_SetPerfFilterMaskCmd_Payload](#)
Set Performance Analyzer Filter Mask Command Payload.
- struct [CFE_ES_SetPerfTrigMaskCmd_Payload](#)
Set Performance Analyzer Trigger Mask Command Payload.
- struct [CFE_ES_SendMemPoolStatsCmd_Payload](#)
Send Memory Pool Statistics Command Payload.
- struct [CFE_ES_DumpCDSRegistryCmd_Payload](#)
Dump CDS Registry Command Payload.
- struct [CFE_ES_OneAppTlm_Payload](#)
- struct [CFE_ES_PoolStatsTlm_Payload](#)
- struct [CFE_ES_HousekeepingTlm_Payload](#)

Typedefs

- typedef struct [CFE_ES_RestartCmd_Payload](#) [CFE_ES_RestartCmd_Payload_t](#)
Restart cFE Command Payload.
- typedef struct [CFE_ES_FileNameCmd_Payload](#) [CFE_ES_FileNameCmd_Payload_t](#)
Generic file name command payload.
- typedef struct [CFE_ES_OverWriteSysLogCmd_Payload](#) [CFE_ES_OverWriteSysLogCmd_Payload_t](#)
Overwrite/Discard System Log Configuration Command Payload.
- typedef struct [CFE_ES_StartAppCmd_Payload](#) [CFE_ES_StartAppCmd_Payload_t](#)
Start Application Command Payload.
- typedef struct [CFE_ES_AppNameCmd_Payload](#) [CFE_ES_AppNameCmd_Payload_t](#)
Generic application name command payload.
- typedef struct [CFE_ES_AppReloadCmd_Payload](#) [CFE_ES_AppReloadCmd_Payload_t](#)
Reload Application Command Payload.
- typedef struct [CFE_ES_SetMaxPRCountCmd_Payload](#) [CFE_ES_SetMaxPRCountCmd_Payload_t](#)
Set Maximum Processor Reset Count Command Payload.
- typedef struct [CFE_ES_DeleteCDSCmd_Payload](#) [CFE_ES_DeleteCDSCmd_Payload_t](#)
Delete Critical Data Store Command Payload.
- typedef [uint32](#) [CFE_ES_PerfMode_Enum_t](#)
- typedef struct [CFE_ES_StartPerfCmd_Payload](#) [CFE_ES_StartPerfCmd_Payload_t](#)
Start Performance Analyzer Command Payload.
- typedef struct [CFE_ES_StopPerfCmd_Payload](#) [CFE_ES_StopPerfCmd_Payload_t](#)
Stop Performance Analyzer Command Payload.
- typedef struct [CFE_ES_SetPerfFilterMaskCmd_Payload](#) [CFE_ES_SetPerfFilterMaskCmd_Payload_t](#)
Set Performance Analyzer Filter Mask Command Payload.

- typedef struct [CFE_ES_SetPerfTrigMaskCmd_Payload](#) [CFE_ES_SetPerfTrigMaskCmd_Payload_t](#)
Set Performance Analyzer Trigger Mask Command Payload.
- typedef struct [CFE_ES_SendMemPoolStatsCmd_Payload](#) [CFE_ES_SendMemPoolStatsCmd_Payload_t](#)
Send Memory Pool Statistics Command Payload.
- typedef struct [CFE_ES_DumpCDSRegistryCmd_Payload](#) [CFE_ES_DumpCDSRegistryCmd_Payload_t](#)
Dump CDS Registry Command Payload.
- typedef struct [CFE_ES_OneAppTlm_Payload](#) [CFE_ES_OneAppTlm_Payload_t](#)
- typedef struct [CFE_ES_PoolStatsTlm_Payload](#) [CFE_ES_PoolStatsTlm_Payload_t](#)
- typedef struct [CFE_ES_HousekeepingTlm_Payload](#) [CFE_ES_HousekeepingTlm_Payload_t](#)

Enumerations

- enum [CFE_ES_PerfMode](#) { [CFE_ES_PerfTrigger_START](#) = 0 , [CFE_ES_PerfTrigger_CENTER](#) , [CFE_ES_PerfTrigger_END](#) }
- Labels for values to use in [CFE_ES_StartPerfCmd_Payload.TriggerMode](#).*

11.111.1 Detailed Description

Specification for the CFE Executive Services (CFE_ES) command and telemetry message constant definitions. For CFE_ES this is only the function/command code definitions
Definition in file [default_cfe_es_msgdefs.h](#).

11.111.2 Typedef Documentation

CFE_ES_AppNameCmd_Payload_t

typedef struct [CFE_ES_AppNameCmd_Payload](#) [CFE_ES_AppNameCmd_Payload_t](#)
Generic application name command payload.
For command details, see [CFE_ES_STOP_APP_CC](#), [CFE_ES_RESTART_APP_CC](#), [CFE_ES_QUERY_ONE_CC](#)

CFE_ES_AppReloadCmd_Payload_t

typedef struct [CFE_ES_AppReloadCmd_Payload](#) [CFE_ES_AppReloadCmd_Payload_t](#)
Reload Application Command Payload.
For command details, see [CFE_ES_RELOAD_APP_CC](#)

CFE_ES_DeleteCDSCmd_Payload_t

typedef struct [CFE_ES_DeleteCDSCmd_Payload](#) [CFE_ES_DeleteCDSCmd_Payload_t](#)
Delete Critical Data Store Command Payload.
For command details, see [CFE_ES_DELETE_CDS_CC](#)

CFE_ES_DumpCDSRegistryCmd_Payload_t

typedef struct [CFE_ES_DumpCDSRegistryCmd_Payload](#) [CFE_ES_DumpCDSRegistryCmd_Payload_t](#)
Dump CDS Registry Command Payload.
For command details, see [CFE_ES_DUMP_CDS_REGISTRY_CC](#)

CFE_ES_FileNameCmd_Payload_t

typedef struct [CFE_ES_FileNameCmd_Payload](#) [CFE_ES_FileNameCmd_Payload_t](#)
Generic file name command payload.
This format is shared by several executive services commands. For command details, see [CFE_ES_QUERY_ALL_CC](#), [CFE_ES_QUERY_ALL_TASKS_CC](#), [CFE_ES_WRITE_SYS_LOG_CC](#), and [CFE_ES_WRITE_ER_LOG_CC](#)

CFE_ES_HousekeepingTlm_Payload_t

```
typedef struct CFE_ES_HousekeepingTlm_Payload CFE_ES_HousekeepingTlm_Payload_t
```

Name Executive Services Housekeeping Packet

CFE_ES_OneAppTlm_Payload_t

```
typedef struct CFE_ES_OneAppTlm_Payload CFE_ES_OneAppTlm_Payload_t
```

Name Single Application Information Packet

CFE_ES_OverWriteSysLogCmd_Payload_t

```
typedef struct CFE_ES_OverWriteSysLogCmd_Payload CFE_ES_OverWriteSysLogCmd_Payload_t
```

Overwrite/Discard System Log Configuration Command Payload.

For command details, see [CFE_ES_OVER_WRITE_SYS_LOG_CC](#)

CFE_ES_PerfMode_Enum_t

```
typedef uint32 CFE_ES_PerfMode_Enum_t
```

Definition at line 160 of file [default_cfe_es_msgdefs.h](#).

CFE_ES_PoolStatsTlm_Payload_t

```
typedef struct CFE_ES_PoolStatsTlm_Payload CFE_ES_PoolStatsTlm_Payload_t
```

Name Memory Pool Statistics Packet

CFE_ES_RestartCmd_Payload_t

```
typedef struct CFE_ES_RestartCmd_Payload CFE_ES_RestartCmd_Payload_t
```

Restart cFE Command Payload.

For command details, see [CFE_ES_RESTART_CC](#)

CFE_ES_SendMemPoolStatsCmd_Payload_t

```
typedef struct CFE_ES_SendMemPoolStatsCmd_Payload CFE_ES_SendMemPoolStatsCmd_Payload_t
```

Send Memory Pool Statistics Command Payload.

For command details, see [CFE_ES_SEND_MEM_POOL_STATS_CC](#)

CFE_ES_SetMaxPRCountCmd_Payload_t

```
typedef struct CFE_ES_SetMaxPRCountCmd_Payload CFE_ES_SetMaxPRCountCmd_Payload_t
```

Set Maximum Processor Reset Count Command Payload.

For command details, see [CFE_ES_SET_MAX_PR_COUNT_CC](#)

CFE_ES_SetPerfFilterMaskCmd_Payload_t

```
typedef struct CFE_ES_SetPerfFilterMaskCmd_Payload CFE_ES_SetPerfFilterMaskCmd_Payload_t
```

Set Performance Analyzer Filter Mask Command Payload.

For command details, see [CFE_ES_SET_PERF_FILTER_MASK_CC](#)

CFE_ES_SetPerfTrigMaskCmd_Payload_t

typedef struct CFE_ES_SetPerfTrigMaskCmd_Payload CFE_ES_SetPerfTrigMaskCmd_Payload_t
 Set Performance Analyzer Trigger Mask Command Payload.
 For command details, see [CFE_ES_SET_PERF_TRIGGER_MASK_CC](#)

CFE_ES_StartAppCmd_Payload_t

typedef struct CFE_ES_StartAppCmd_Payload CFE_ES_StartAppCmd_Payload_t
 Start Application Command Payload.
 For command details, see [CFE_ES_START_APP_CC](#)

CFE_ES_StartPerfCmd_Payload_t

typedef struct CFE_ES_StartPerfCmd_Payload CFE_ES_StartPerfCmd_Payload_t
 Start Performance Analyzer Command Payload.
 For command details, see [CFE_ES_START_PERF_DATA_CC](#)

CFE_ES_StopPerfCmd_Payload_t

typedef struct CFE_ES_StopPerfCmd_Payload CFE_ES_StopPerfCmd_Payload_t
 Stop Performance Analyzer Command Payload.
 For command details, see [CFE_ES_STOP_PERF_DATA_CC](#)

11.111.3 Enumeration Type Documentation**CFE_ES_PerfMode**

enum CFE_ES_PerfMode
 Labels for values to use in [CFE_ES_StartPerfCmd_Payload.TriggerMode](#).

See also

[CFE_ES_StartPerfCmd_Payload](#)

Enumerator

CFE_ES_PerfTrigger_START	
CFE_ES_PerfTrigger_CENTER	
CFE_ES_PerfTrigger_END	

Definition at line 153 of file [default_cfe_es_msgdefs.h](#).

11.112 default_cfe_es_msgdefs.h

[Go to the documentation of this file.](#)

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00017  *****/
00018
00026 #ifndef DEFAULT_CFE_ES_MSGDEFS_H
00027 #define DEFAULT_CFE_ES_MSGDEFS_H
00028
00029 #include "common_types.h"
00030 #include "cfe_mission_cfg.h"
00031 #include "cfe_es_extern_typedefs.h"
00032 #include "cfe_es_fcncodes.h"
00033
00034 /*****/
00035 /* Command Message Data Formats */
00036 /*****/
00037
00044 typedef struct CFE_ES_RestartCmd_Payload
00045 {
00046     uint16 RestartType;
00048 } CFE_ES_RestartCmd_Payload_t;
00049
00058 typedef struct CFE_ES_FileNameCmd_Payload
00059 {
00060     char FileName[CFE_MISSION_MAX_PATH_LEN];
00062 } CFE_ES_FileNameCmd_Payload_t;
00063
00070 typedef struct CFE_ES_OverWriteSysLogCmd_Payload
00071 {
00072     uint32 Mode;
00074 } CFE_ES_OverWriteSysLogCmd_Payload_t;
00075
00082 typedef struct CFE_ES_StartAppCmd_Payload
00083 {
00084     char Application[CFE_MISSION_MAX_API_LEN];
00085     char AppEntryPoint[CFE_MISSION_MAX_API_LEN];
00087     char AppFileName[CFE_MISSION_MAX_PATH_LEN];
00090     CFE_ES_MemOffset_t StackSize;
00091     CFE_ES_ExceptionAction_Enum_t ExceptionAction;
00096     uint8 Padding1;
00097     CFE_ES_TaskPriority_Atom_t Priority;
00098     uint8 Padding2[4];
00099 } CFE_ES_StartAppCmd_Payload_t;
00100
00107 typedef struct CFE_ES_AppNameCmd_Payload
00108 {
00109     char Application[CFE_MISSION_MAX_API_LEN];
00110 } CFE_ES_AppNameCmd_Payload_t;
00111
00118 typedef struct CFE_ES_AppReloadCmd_Payload
00119 {
00120     char Application[CFE_MISSION_MAX_API_LEN];
00121     char AppFileName[CFE_MISSION_MAX_PATH_LEN];
00123 } CFE_ES_AppReloadCmd_Payload_t;
00124
00131 typedef struct CFE_ES_SetMaxPRCountCmd_Payload
00132 {
00133     uint16 MaxPRCount;
00135 } CFE_ES_SetMaxPRCountCmd_Payload_t;
00136
00143 typedef struct CFE_ES_DeleteCDSCmd_Payload
00144 {
00145     char
00146         CdsName[CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN];
00147 } CFE_ES_DeleteCDSCmd_Payload_t;
00148
00153 enum CFE_ES_PerfMode
00154 {
00155     CFE_ES_PerfTrigger_START = 0,
00156     CFE_ES_PerfTrigger_CENTER,
00157     CFE_ES_PerfTrigger_END
00158 };
00159
00160 typedef uint32 CFE_ES_PerfMode_Enum_t;
00161
00168 typedef struct CFE_ES_StartPerfCmd_Payload
00169 {
00170     CFE_ES_PerfMode_Enum_t
00171         TriggerMode;
00172 } CFE_ES_StartPerfCmd_Payload_t;
00173

```

```

00180 typedef struct CFE_ES_StopPerfCmd_Payload
00181 {
00182     char DataFileName[CFE_MISSION_MAX_PATH_LEN];
00184 } CFE_ES_StopPerfCmd_Payload_t;
00185
00192 typedef struct CFE_ES_SetPerfFilterMaskCmd_Payload
00193 {
00194     uint32 FilterMaskNum;
00195     uint32 FilterMask;
00196 } CFE_ES_SetPerfFilterMaskCmd_Payload_t;
00197
00204 typedef struct CFE_ES_SetPerfTrigMaskCmd_Payload
00205 {
00206     uint32 TriggerMaskNum;
00207     uint32 TriggerMask;
00208 } CFE_ES_SetPerfTrigMaskCmd_Payload_t;
00209
00216 typedef struct CFE_ES_SendMemPoolStatsCmd_Payload
00217 {
00218     char Application[CFE_MISSION_MAX_API_LEN];
00219     CFE_ES_MemHandle_t PoolHandle;
00220 } CFE_ES_SendMemPoolStatsCmd_Payload_t;
00221
00228 typedef struct CFE_ES_DumpCDSRegistryCmd_Payload
00229 {
00230     char DumpFilename[CFE_MISSION_MAX_PATH_LEN];
00232 } CFE_ES_DumpCDSRegistryCmd_Payload_t;
00233
00234 /*****
00235
00236 /*****
00237 /* Telemetry Interface Data Formats */
00238 /*****
00239
00243 typedef struct CFE_ES_OneAppTlm_Payload
00244 {
00245     CFE_ES_AppInfo_t AppInfo;
00246 } CFE_ES_OneAppTlm_Payload_t;
00247
00251 typedef struct CFE_ES_PoolStatsTlm_Payload
00252 {
00253     CFE_ES_MemHandle_t PoolHandle;
00255     uint8 Padding[4];
00256     CFE_ES_MemPoolStats_t PoolStats;
00257 } CFE_ES_PoolStatsTlm_Payload_t;
00258
00259 /*****
00260
00264 typedef struct CFE_ES_HousekeepingTlm_Payload
00265 {
00266     uint8 CommandCounter;
00268     uint8 CommandErrorCounter;
00271     uint16 CFECoreChecksum;
00273     uint8 CFEMajorVersion;
00275     uint8 CFEMinorVersion;
00277     uint8 CFERevision;
00279     uint8 CFEMissionRevision;
00281     uint8 OSALMajorVersion;
00283     uint8 OSALMinorVersion;
00285     uint8 OSALRevision;
00287     uint8 OSALMissionRevision;
00290     uint8 PSPMajorVersion;
00292     uint8 PSPMinorVersion;
00294     uint8 PSPRevision;
00296     uint8 PSPMissionRevision;
00299     CFE_ES_MemOffset_t SysLogBytesUsed;
00301     CFE_ES_MemOffset_t SysLogSize;
00303     uint32 SysLogEntries;
00305     uint32 SysLogMode;
00308     uint32 ERLogIndex;
00310     uint32 ERLogEntries;
00313     uint32 RegisteredCoreApps;
00315     uint32 RegisteredExternalApps;
00317     uint32 RegisteredTasks;
00319     uint32 RegisteredLibs;
00322     uint32 ResetType;
00324     uint32 ResetSubtype;
00326     uint32 ProcessorResets;
00328     uint32 MaxProcessorResets;
00330     uint32 BootSource;
00333     uint32 PerfState;

```

```

00335     uint32 PerfMode;
00337     uint32 PerfTriggerCount;
00339     uint32 PerfFilterMask[CFE_MISSION_ES_PERF_MAX_IDS / 32];
00341     uint32
00342         PerfTriggerMask[CFE_MISSION_ES_PERF_MAX_IDS / 32];
00344     uint32 PerfDataStart;
00346     uint32 PerfDataEnd;
00348     uint32 PerfDataCount;
00350     uint32
00351         PerfDataToWrite;
00353     CFE_ES_MemOffset_t HeapBytesFree;
00355     CFE_ES_MemOffset_t HeapBlocksFree;
00357     CFE_ES_MemOffset_t HeapMaxBlockSize;
00359 } CFE_ES_HousekeepingTlm_Payload_t;
00360
00361 #endif

```

11.113 cfe/modules/es/config/default_cfe_es_msgid_values.h File Reference

```

#include "cfe_core_api_base_msgids.h"
#include "cfe_es_topicids.h"

```

Macros

- #define `CFE_PLATFORM_ES_CMD_MIDVAL(x)` `CFE_PLATFORM_CMD_TOPICID_TO_MIDV(CFE_MISSION_ES_##x##_TOPICID)`
- #define `CFE_PLATFORM_ES_TLM_MIDVAL(x)` `CFE_PLATFORM_TLM_TOPICID_TO_MIDV(CFE_MISSION_ES_##x##_TOPICID)`

11.113.1 Detailed Description

CFE Executive Services (CFE_ES) Application Message IDs
Definition in file [default_cfe_es_msgid_values.h](#).

11.113.2 Macro Definition Documentation

CFE_PLATFORM_ES_CMD_MIDVAL

```

#define CFE_PLATFORM_ES_CMD_MIDVAL(
    x ) CFE_PLATFORM_CMD_TOPICID_TO_MIDV(CFE_MISSION_ES_##x##_TOPICID)

```

Definition at line 29 of file [default_cfe_es_msgid_values.h](#).

CFE_PLATFORM_ES_TLM_MIDVAL

```

#define CFE_PLATFORM_ES_TLM_MIDVAL(
    x ) CFE_PLATFORM_TLM_TOPICID_TO_MIDV(CFE_MISSION_ES_##x##_TOPICID)

```

Definition at line 30 of file [default_cfe_es_msgid_values.h](#).

11.114 default_cfe_es_msgid_values.h

[Go to the documentation of this file.](#)

```

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00016  * limitations under the License.
00017  *****/
00018
00023 #ifndef DEFAULT_CFE_ES_MSGID_VALUES_H
00024 #define DEFAULT_CFE_ES_MSGID_VALUES_H
00025
00026 #include "cfe_core_api_base_msgids.h"
00027 #include "cfe_es_topicids.h"
00028
00029 #define CFE_PLATFORM_ES_CMD_MIDVAL(x) CFE_PLATFORM_CMD_TOPICID_TO_MIDV(CFE_MISSION_ES_##x##_TOPICID)
00030 #define CFE_PLATFORM_ES_TLM_MIDVAL(x) CFE_PLATFORM_TLM_TOPICID_TO_MIDV(CFE_MISSION_ES_##x##_TOPICID)
00031
00032 #endif

```

11.115 cfe/modules/es/config/default_cfe_es_msgids.h File Reference

```
#include "cfe_es_msgid_values.h"
```

Macros

- `#define CFE_ES_CMD_MID CFE_PLATFORM_ES_CMD_MIDVAL(CMD)`
- `#define CFE_ES_SEND_HK_MID CFE_PLATFORM_ES_CMD_MIDVAL(SEND_HK)`
- `#define CFE_ES_HK_TLM_MID CFE_PLATFORM_ES_TLM_MIDVAL(HK_TLM)`
- `#define CFE_ES_APP_TLM_MID CFE_PLATFORM_ES_TLM_MIDVAL(APP_TLM)`
- `#define CFE_ES_MEMSTATS_TLM_MID CFE_PLATFORM_ES_TLM_MIDVAL(MEMSTATS_TLM)`

11.115.1 Detailed Description

CFE Executive Services (CFE_ES) Application Message IDs
Definition in file [default_cfe_es_msgids.h](#).

11.115.2 Macro Definition Documentation

CFE_ES_APP_TLM_MID

```
#define CFE_ES_APP_TLM_MID CFE_PLATFORM_ES_TLM_MIDVAL(APP_TLM)
```

Definition at line 38 of file [default_cfe_es_msgids.h](#).

CFE_ES_CMD_MID

```
#define CFE_ES_CMD_MID CFE_PLATFORM_ES_CMD_MIDVAL(CMD)
```

Definition at line 31 of file [default_cfe_es_msgids.h](#).

CFE_ES_HK_TLM_MID

```
#define CFE_ES_HK_TLM_MID CFE_PLATFORM_ES_TLM_MIDVAL(HK_TLM)
```

Definition at line 37 of file [default_cfe_es_msgids.h](#).

CFE_ES_MEMSTATS_TLM_MID

```
#define CFE_ES_MEMSTATS_TLM_MID CFE_PLATFORM_ES_TLM_MIDVAL(MEMSTATS_TLM)
```

Definition at line 39 of file [default_cfe_es_msgids.h](#).

CFE_ES_SEND_HK_MID

```
#define CFE_ES_SEND_HK_MID CFE_PLATFORM_ES_CMD_MIDVAL(SEND_HK)
```

Definition at line 32 of file [default_cfe_es_msgids.h](#).

11.116 default_cfe_es_msgids.h

[Go to the documentation of this file.](#)

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00017  *****/
00018
00023 #ifndef DEFAULT_CFE_ES_MSGIDS_H
00024 #define DEFAULT_CFE_ES_MSGIDS_H
00025
00026 #include "cfe_es_msgid_values.h"
00027
00028 /*
00029 ** cFE ES Command Message Id's
00030 */
00031 #define CFE_ES_CMD_MID          CFE_PLATFORM_ES_CMD_MIDVAL(CMD)
00032 #define CFE_ES_SEND_HK_MID      CFE_PLATFORM_ES_CMD_MIDVAL(SEND_HK)
00033
00034 /*
00035 ** CFE ES Telemetry Message Id's
00036 */
00037 #define CFE_ES_HK_TLM_MID       CFE_PLATFORM_ES_TLM_MIDVAL(HK_TLM)
00038 #define CFE_ES_APP_TLM_MID      CFE_PLATFORM_ES_TLM_MIDVAL(APP_TLM)
00039 #define CFE_ES_MEMSTATS_TLM_MID CFE_PLATFORM_ES_TLM_MIDVAL(MEMSTATS_TLM)
00040
00041 #endif

```

11.117 cfe/modules/es/config/default_cfe_es_msgstruct.h File Reference

```

#include "cfe_es_msgdefs.h"
#include "cfe_msg_hdr.h"
#include "cfe_mission_cfg.h"

```

Data Structures

- struct [CFE_ES_NoopCmd](#)
- struct [CFE_ES_ResetCountersCmd](#)
- struct [CFE_ES_ClearSysLogCmd](#)
- struct [CFE_ES_ClearERLogCmd](#)
- struct [CFE_ES_ResetPRCountCmd](#)
- struct [CFE_ES_SendHkCmd](#)
- struct [CFE_ES_RestartCmd](#)
 - Restart cFE Command.*
- struct [CFE_ES_FileNameCmd](#)
 - Generic file name command.*
- struct [CFE_ES_QueryAllCmd](#)
- struct [CFE_ES_QueryAllTasksCmd](#)
- struct [CFE_ES_WriteSysLogCmd](#)
- struct [CFE_ES_WriteERLogCmd](#)
- struct [CFE_ES_OverWriteSysLogCmd](#)
 - Overwrite/Discard System Log Configuration Command Payload.*

- struct [CFE_ES_StartApp](#)
Start Application Command.
- struct [CFE_ES_StopAppCmd](#)
- struct [CFE_ES_RestartAppCmd](#)
- struct [CFE_ES_QueryOneCmd](#)
- struct [CFE_ES_ReloadAppCmd](#)
Reload Application Command.
- struct [CFE_ES_SetMaxPRCountCmd](#)
Set Maximum Processor Reset Count Command.
- struct [CFE_ES_DeleteCDSCmd](#)
Delete Critical Data Store Command.
- struct [CFE_ES_StartPerfDataCmd](#)
Start Performance Analyzer Command.
- struct [CFE_ES_StopPerfDataCmd](#)
Stop Performance Analyzer Command.
- struct [CFE_ES_SetPerfFilterMaskCmd](#)
Set Performance Analyzer Filter Mask Command.
- struct [CFE_ES_SetPerfTriggerMaskCmd](#)
Set Performance Analyzer Trigger Mask Command.
- struct [CFE_ES_SendMemPoolStatsCmd](#)
Send Memory Pool Statistics Command.
- struct [CFE_ES_DumpCDSRegistryCmd](#)
Dump CDS Registry Command.
- struct [CFE_ES_OneAppTlm](#)
- struct [CFE_ES_MemStatsTlm](#)
- struct [CFE_ES_HousekeepingTlm](#)

Typedefs

- typedef struct [CFE_ES_NoopCmd](#) [CFE_ES_NoopCmd_t](#)
- typedef struct [CFE_ES_ResetCountersCmd](#) [CFE_ES_ResetCountersCmd_t](#)
- typedef struct [CFE_ES_ClearSysLogCmd](#) [CFE_ES_ClearSysLogCmd_t](#)
- typedef struct [CFE_ES_ClearERLogCmd](#) [CFE_ES_ClearERLogCmd_t](#)
- typedef struct [CFE_ES_ResetPRCountCmd](#) [CFE_ES_ResetPRCountCmd_t](#)
- typedef struct [CFE_ES_SendHkCmd](#) [CFE_ES_SendHkCmd_t](#)
- typedef struct [CFE_ES_RestartCmd](#) [CFE_ES_RestartCmd_t](#)
Restart cFE Command.
- typedef struct [CFE_ES_FileNameCmd](#) [CFE_ES_FileNameCmd_t](#)
Generic file name command.
- typedef struct [CFE_ES_QueryAllCmd](#) [CFE_ES_QueryAllCmd_t](#)
- typedef struct [CFE_ES_QueryAllTasksCmd](#) [CFE_ES_QueryAllTasksCmd_t](#)
- typedef struct [CFE_ES_WriteSysLogCmd](#) [CFE_ES_WriteSysLogCmd_t](#)
- typedef struct [CFE_ES_WriteERLogCmd](#) [CFE_ES_WriteERLogCmd_t](#)
- typedef struct [CFE_ES_OverWriteSysLogCmd](#) [CFE_ES_OverWriteSysLogCmd_t](#)
Overwrite/Discard System Log Configuration Command Payload.
- typedef struct [CFE_ES_StartApp](#) [CFE_ES_StartAppCmd_t](#)
Start Application Command.
- typedef struct [CFE_ES_StopAppCmd](#) [CFE_ES_StopAppCmd_t](#)
- typedef struct [CFE_ES_RestartAppCmd](#) [CFE_ES_RestartAppCmd_t](#)

- typedef struct [CFE_ES_QueryOneCmd](#) [CFE_ES_QueryOneCmd_t](#)
- typedef struct [CFE_ES_ReloadAppCmd](#) [CFE_ES_ReloadAppCmd_t](#)
Reload Application Command.
- typedef struct [CFE_ES_SetMaxPRCountCmd](#) [CFE_ES_SetMaxPRCountCmd_t](#)
Set Maximum Processor Reset Count Command.
- typedef struct [CFE_ES_DeleteCDSCmd](#) [CFE_ES_DeleteCDSCmd_t](#)
Delete Critical Data Store Command.
- typedef struct [CFE_ES_StartPerfDataCmd](#) [CFE_ES_StartPerfDataCmd_t](#)
Start Performance Analyzer Command.
- typedef struct [CFE_ES_StopPerfDataCmd](#) [CFE_ES_StopPerfDataCmd_t](#)
Stop Performance Analyzer Command.
- typedef struct [CFE_ES_SetPerfFilterMaskCmd](#) [CFE_ES_SetPerfFilterMaskCmd_t](#)
Set Performance Analyzer Filter Mask Command.
- typedef struct [CFE_ES_SetPerfTriggerMaskCmd](#) [CFE_ES_SetPerfTriggerMaskCmd_t](#)
Set Performance Analyzer Trigger Mask Command.
- typedef struct [CFE_ES_SendMemPoolStatsCmd](#) [CFE_ES_SendMemPoolStatsCmd_t](#)
Send Memory Pool Statistics Command.
- typedef struct [CFE_ES_DumpCDSRegistryCmd](#) [CFE_ES_DumpCDSRegistryCmd_t](#)
Dump CDS Registry Command.
- typedef struct [CFE_ES_OneAppTlm](#) [CFE_ES_OneAppTlm_t](#)
- typedef struct [CFE_ES_MemStatsTlm](#) [CFE_ES_MemStatsTlm_t](#)
- typedef struct [CFE_ES_HousekeepingTlm](#) [CFE_ES_HousekeepingTlm_t](#)

11.117.1 Detailed Description

Purpose: cFE Executive Services (ES) Command and Telemetry packet definition file.
Definition in file [default_cfe_es_msgstruct.h](#).

11.117.2 Typedef Documentation

CFE_ES_ClearERLogCmd_t

```
typedef struct CFE\_ES\_ClearERLogCmd CFE\_ES\_ClearERLogCmd\_t
```

CFE_ES_ClearSysLogCmd_t

```
typedef struct CFE\_ES\_ClearSysLogCmd CFE\_ES\_ClearSysLogCmd\_t
```

CFE_ES_DeleteCDSCmd_t

```
typedef struct CFE\_ES\_DeleteCDSCmd CFE\_ES\_DeleteCDSCmd\_t
```

Delete Critical Data Store Command.

CFE_ES_DumpCDSRegistryCmd_t

```
typedef struct CFE\_ES\_DumpCDSRegistryCmd CFE\_ES\_DumpCDSRegistryCmd\_t
```

Dump CDS Registry Command.

CFE_ES_FileNameCmd_t

```
typedef struct CFE\_ES\_FileNameCmd CFE\_ES\_FileNameCmd\_t
```

Generic file name command.

CFE_ES_HousekeepingTlm_t

```
typedef struct CFE_ES_HousekeepingTlm CFE_ES_HousekeepingTlm_t
```

Name Executive Services Housekeeping Packet

CFE_ES_MemStatsTlm_t

```
typedef struct CFE_ES_MemStatsTlm CFE_ES_MemStatsTlm_t
```

Name Memory Pool Statistics Packet

CFE_ES_NoopCmd_t

```
typedef struct CFE_ES_NoopCmd CFE_ES_NoopCmd_t
```

CFE_ES_OneAppTlm_t

```
typedef struct CFE_ES_OneAppTlm CFE_ES_OneAppTlm_t
```

Name Single Application Information Packet

CFE_ES_OverWriteSysLogCmd_t

```
typedef struct CFE_ES_OverWriteSysLogCmd CFE_ES_OverWriteSysLogCmd_t
```

Overwrite/Discard System Log Configuration Command Payload.

CFE_ES_QueryAllCmd_t

```
typedef struct CFE_ES_QueryAllCmd CFE_ES_QueryAllCmd_t
```

CFE_ES_QueryAllTasksCmd_t

```
typedef struct CFE_ES_QueryAllTasksCmd CFE_ES_QueryAllTasksCmd_t
```

CFE_ES_QueryOneCmd_t

```
typedef struct CFE_ES_QueryOneCmd CFE_ES_QueryOneCmd_t
```

CFE_ES_ReloadAppCmd_t

```
typedef struct CFE_ES_ReloadAppCmd CFE_ES_ReloadAppCmd_t
```

Reload Application Command.

CFE_ES_ResetCountersCmd_t

```
typedef struct CFE_ES_ResetCountersCmd CFE_ES_ResetCountersCmd_t
```

CFE_ES_ResetPRCountCmd_t

```
typedef struct CFE_ES_ResetPRCountCmd CFE_ES_ResetPRCountCmd_t
```

CFE_ES_RestartAppCmd_t

```
typedef struct CFE_ES_RestartAppCmd CFE_ES_RestartAppCmd_t
```

CFE_ES_RestartCmd_t

```
typedef struct CFE_ES_RestartCmd CFE_ES_RestartCmd_t
```

Restart cFE Command.

CFE_ES_SendHkCmd_t

```
typedef struct CFE_ES_SendHkCmd CFE_ES_SendHkCmd_t
```

CFE_ES_SendMemPoolStatsCmd_t

```
typedef struct CFE_ES_SendMemPoolStatsCmd CFE_ES_SendMemPoolStatsCmd_t
```

Send Memory Pool Statistics Command.

CFE_ES_SetMaxPRCountCmd_t

```
typedef struct CFE_ES_SetMaxPRCountCmd CFE_ES_SetMaxPRCountCmd_t
```

Set Maximum Processor Reset Count Command.

CFE_ES_SetPerfFilterMaskCmd_t

```
typedef struct CFE_ES_SetPerfFilterMaskCmd CFE_ES_SetPerfFilterMaskCmd_t
```

Set Performance Analyzer Filter Mask Command.

CFE_ES_SetPerfTriggerMaskCmd_t

```
typedef struct CFE_ES_SetPerfTriggerMaskCmd CFE_ES_SetPerfTriggerMaskCmd_t
```

Set Performance Analyzer Trigger Mask Command.

CFE_ES_StartAppCmd_t

```
typedef struct CFE_ES_StartApp CFE_ES_StartAppCmd_t
```

Start Application Command.

CFE_ES_StartPerfDataCmd_t

```
typedef struct CFE_ES_StartPerfDataCmd CFE_ES_StartPerfDataCmd_t
```

Start Performance Analyzer Command.

CFE_ES_StopAppCmd_t

```
typedef struct CFE_ES_StopAppCmd CFE_ES_StopAppCmd_t
```

CFE_ES_StopPerfDataCmd_t

```
typedef struct CFE_ES_StopPerfDataCmd CFE_ES_StopPerfDataCmd_t
```

Stop Performance Analyzer Command.

CFE_ES_WriteERLogCmd_t

```
typedef struct CFE_ES_WriteERLogCmd CFE_ES_WriteERLogCmd_t
```

CFE_ES_WriteSysLogCmd_t

```
typedef struct CFE_ES_WriteSysLogCmd CFE_ES_WriteSysLogCmd_t
```

11.118 default_cfe_es_msgstruct.h

Go to the documentation of this file.

```

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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00025 #ifndef DEFAULT_CFE_ES_MSGSTRUCT_H
00026 #define DEFAULT_CFE_ES_MSGSTRUCT_H
00027
00028 /*****
00029  * Includes
00030  *****/
00031 #include "cfe_es_msgdefs.h"
00032 #include "cfe_msg_hdr.h"
00033 #include "cfe_mission_cfg.h"
00034
00035 /*****
00036  * Command Message Data Formats *
00037  *****/
00038
00039 /*
00040  * The following commands all share the "NoArgs" format
00041  *
00042  * They are each given their own type name matching the command name, which
00043  * allows them to change independently in the future without changing the prototype
00044  * of the handler function.
00045  */
00046 typedef struct CFE_ES_NoopCmd
00047 {
00048     CFE_MSG_CommandHeader_t CommandHeader;
00049 } CFE_ES_NoopCmd_t;
00050
00051 typedef struct CFE_ES_ResetCountersCmd
00052 {
00053     CFE_MSG_CommandHeader_t CommandHeader;
00054 } CFE_ES_ResetCountersCmd_t;
00055
00056 typedef struct CFE_ES_ClearSysLogCmd
00057 {
00058     CFE_MSG_CommandHeader_t CommandHeader;
00059 } CFE_ES_ClearSysLogCmd_t;
00060
00061 typedef struct CFE_ES_ClearERLogCmd
00062 {
00063     CFE_MSG_CommandHeader_t CommandHeader;
00064 } CFE_ES_ClearERLogCmd_t;
00065
00066 typedef struct CFE_ES_ResetPRCountCmd
00067 {
00068     CFE_MSG_CommandHeader_t CommandHeader;
00069 } CFE_ES_ResetPRCountCmd_t;
00070
00071 typedef struct CFE_ES_SendHkCmd
00072 {
00073     CFE_MSG_CommandHeader_t CommandHeader;
00074 } CFE_ES_SendHkCmd_t;
00075
00079 typedef struct CFE_ES_RestartCmd
00080 {
00081     CFE_MSG_CommandHeader_t CommandHeader;
00082     CFE_ES_RestartCmd_Payload_t Payload;
00083 } CFE_ES_RestartCmd_t;
00084
00085 #ifndef jphfix
00089 typedef struct CFE_ES_FileNameCmd

```

```

00090 {
00091     CFE_MSG_CommandHeader_t    CommandHeader;
00092     CFE_ES_FileNameCmd_Payload_t Payload;
00093 } CFE_ES_FileNameCmd_t;
00094 #endif
00095 /*
00096  * Unique typedefs for each of the commands that utilize the FileNameCmd
00097  * packet format
00098  */
00099 typedef struct CFE_ES_QueryAllCmd
00100 {
00101     CFE_MSG_CommandHeader_t    CommandHeader;
00102     CFE_ES_FileNameCmd_Payload_t Payload;
00103 } CFE_ES_QueryAllCmd_t;
00104
00105 typedef struct CFE_ES_QueryAllTasksCmd
00106 {
00107     CFE_MSG_CommandHeader_t    CommandHeader;
00108     CFE_ES_FileNameCmd_Payload_t Payload;
00109 } CFE_ES_QueryAllTasksCmd_t;
00110
00111 typedef struct CFE_ES_WriteSysLogCmd
00112 {
00113     CFE_MSG_CommandHeader_t    CommandHeader;
00114     CFE_ES_FileNameCmd_Payload_t Payload;
00115 } CFE_ES_WriteSysLogCmd_t;
00116
00117 typedef struct CFE_ES_WriteERLogCmd
00118 {
00119     CFE_MSG_CommandHeader_t    CommandHeader;
00120     CFE_ES_FileNameCmd_Payload_t Payload;
00121 } CFE_ES_WriteERLogCmd_t;
00122
00126 typedef struct CFE_ES_OverWriteSysLogCmd
00127 {
00128     CFE_MSG_CommandHeader_t    CommandHeader;
00129     CFE_ES_OverWriteSysLogCmd_Payload_t Payload;
00130 } CFE_ES_OverWriteSysLogCmd_t;
00131
00135 typedef struct CFE_ES_StartApp
00136 {
00137     CFE_MSG_CommandHeader_t    CommandHeader;
00138     CFE_ES_StartAppCmd_Payload_t Payload;
00139 } CFE_ES_StartAppCmd_t;
00140
00141 /*
00142  * Like NoArgsCmd, this message definition is shared by multiple commands.
00143  * Create a separate typedef for each one so they can all evolve independently
00144  * without affecting the prototype.
00145  */
00146 typedef struct CFE_ES_StopAppCmd
00147 {
00148     CFE_MSG_CommandHeader_t    CommandHeader;
00149     CFE_ES_AppNameCmd_Payload_t Payload;
00150 } CFE_ES_StopAppCmd_t;
00151
00152 typedef struct CFE_ES_RestartAppCmd
00153 {
00154     CFE_MSG_CommandHeader_t    CommandHeader;
00155     CFE_ES_AppNameCmd_Payload_t Payload;
00156 } CFE_ES_RestartAppCmd_t;
00157
00158 typedef struct CFE_ES_QueryOneCmd
00159 {
00160     CFE_MSG_CommandHeader_t    CommandHeader;
00161     CFE_ES_AppNameCmd_Payload_t Payload;
00162 } CFE_ES_QueryOneCmd_t;
00163
00167 typedef struct CFE_ES_ReloadAppCmd
00168 {
00169     CFE_MSG_CommandHeader_t    CommandHeader;
00170     CFE_ES_AppReloadCmd_Payload_t Payload;
00171 } CFE_ES_ReloadAppCmd_t;
00172
00176 typedef struct CFE_ES_SetMaxPRCountCmd
00177 {
00178     CFE_MSG_CommandHeader_t    CommandHeader;
00179     CFE_ES_SetMaxPRCountCmd_Payload_t Payload;
00180 } CFE_ES_SetMaxPRCountCmd_t;
00181
00185 typedef struct CFE_ES_DeleteCDSCommand

```

```

00186 {
00187     CFE_MSG_CommandHeader_t    CommandHeader;
00188     CFE_ES_DeleteCDSCmd_Payload_t Payload;
00189 } CFE_ES_DeleteCDSCmd_t;
00190
00194 typedef struct CFE_ES_StartPerfDataCmd
00195 {
00196     CFE_MSG_CommandHeader_t    CommandHeader;
00197     CFE_ES_StartPerfCmd_Payload_t Payload;
00198 } CFE_ES_StartPerfDataCmd_t;
00199
00203 typedef struct CFE_ES_StopPerfDataCmd
00204 {
00205     CFE_MSG_CommandHeader_t    CommandHeader;
00206     CFE_ES_StopPerfCmd_Payload_t Payload;
00207 } CFE_ES_StopPerfDataCmd_t;
00208
00212 typedef struct CFE_ES_SetPerfFilterMaskCmd
00213 {
00214     CFE_MSG_CommandHeader_t    CommandHeader;
00215     CFE_ES_SetPerfFilterMaskCmd_Payload_t Payload;
00216 } CFE_ES_SetPerfFilterMaskCmd_t;
00217
00221 typedef struct CFE_ES_SetPerfTriggerMaskCmd
00222 {
00223     CFE_MSG_CommandHeader_t    CommandHeader;
00224     CFE_ES_SetPerfTrigMaskCmd_Payload_t Payload;
00225 } CFE_ES_SetPerfTriggerMaskCmd_t;
00226
00230 typedef struct CFE_ES_SendMemPoolStatsCmd
00231 {
00232     CFE_MSG_CommandHeader_t    CommandHeader;
00233     CFE_ES_SendMemPoolStatsCmd_Payload_t Payload;
00234 } CFE_ES_SendMemPoolStatsCmd_t;
00235
00239 typedef struct CFE_ES_DumpCDSRegistryCmd
00240 {
00241     CFE_MSG_CommandHeader_t    CommandHeader;
00242     CFE_ES_DumpCDSRegistryCmd_Payload_t Payload;
00243 } CFE_ES_DumpCDSRegistryCmd_t;
00244
00245 /******
00246
00247 /*****
00248 /* Telemetry Message Data Formats */
00249 /*****
00250
00254 typedef struct CFE_ES_OneAppTlm
00255 {
00256     CFE_MSG_TelemetryHeader_t    TelemetryHeader;
00257     CFE_ES_OneAppTlm_Payload_t Payload;
00258 } CFE_ES_OneAppTlm_t;
00259
00263 typedef struct CFE_ES_MemStatsTlm
00264 {
00265     CFE_MSG_TelemetryHeader_t    TelemetryHeader;
00266     CFE_ES_PoolStatsTlm_Payload_t Payload;
00267 } CFE_ES_MemStatsTlm_t;
00268
00272 typedef struct CFE_ES_HousekeepingTlm
00273 {
00274     CFE_MSG_TelemetryHeader_t    TelemetryHeader;
00275     CFE_ES_HousekeepingTlm_Payload_t Payload;
00276 } CFE_ES_HousekeepingTlm_t;
00277
00278 #endif

```

11.119 cfe/modules/es/config/default_cfe_es_platform_cfg.h File Reference

```

#include "cfe_es_mission_cfg.h"
#include "cfe_es_internal_cfg.h"

```

11.119.1 Detailed Description

CFE Executive Services (CFE_ES) Application Platform Configuration Header File

This is a compatibility header for the "platform_cfg.h" file that has traditionally provided both public and private config definitions for each CFS app.

These definitions are now provided in two separate files, one for the public/mission scope and one for internal scope.

Note

This file may be overridden/superseded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [default_cfe_es_platform_cfg.h](#).

11.120 default_cfe_es_platform_cfg.h

[Go to the documentation of this file.](#)

```
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00013  * distributed under the License is distributed on an "AS IS" BASIS,
00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00035 #ifndef DEFAULT_CFE_ES_PLATFORM_CFG_H
00036 #define DEFAULT_CFE_ES_PLATFORM_CFG_H
00037
00038 #include "cfe_es_mission_cfg.h"
00039 #include "cfe_es_internal_cfg.h"
00040
00041 #endif
```

11.121 cfe/modules/es/config/default_cfe_es_topicid_values.h File Reference

Macros

- #define [CFE_MISSION_ES_TIDVAL](#)(x) DEFAULT_CFE_MISSION_ES_###_TOPICID

11.121.1 Detailed Description

CFE Executive Services (CFE_ES) Application Topic IDs

Definition in file [default_cfe_es_topicid_values.h](#).

11.121.2 Macro Definition Documentation

CFE_MISSION_ES_TIDVAL

```
#define CFE_MISSION_ES_TIDVAL(
    x ) DEFAULT_CFE_MISSION_ES_###_TOPICID
```

Definition at line 26 of file [default_cfe_es_topicid_values.h](#).

11.122 default_cfe_es_topicid_values.h

[Go to the documentation of this file.](#)

```
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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00023 #ifndef DEFAULT_CFE_ES_TOPICID_VALUES_H
00024 #define DEFAULT_CFE_ES_TOPICID_VALUES_H
00025
00026 #define CFE_MISSION_ES_TIDVAL(x) DEFAULT_CFE_MISSION_ES_##x##_TOPICID
00027
00028 #endif

```

11.123 cfe/modules/es/fsw/inc/cfe_es_eventids.h File Reference

Macros

ES event IDs

- #define [CFE_ES_INIT_INF_EID](#) 1
ES Initialization Event ID.
- #define [CFE_ES_INITSTATS_INF_EID](#) 2
ES Initialization Statistics Information Event ID.
- #define [CFE_ES_NOOP_INF_EID](#) 3
ES No-op Command Success Event ID.
- #define [CFE_ES_RESET_INF_EID](#) 4
ES Reset Counters Command Success Event ID.
- #define [CFE_ES_START_INF_EID](#) 6
ES Start Application Command Success Event ID.
- #define [CFE_ES_STOP_DBG_EID](#) 7
ES Stop Application Command Request Success Event ID.
- #define [CFE_ES_STOP_INF_EID](#) 8
ES Stop Application Completed Event ID.
- #define [CFE_ES_RESTART_APP_DBG_EID](#) 9
ES Restart Application Command Request Success Event ID.
- #define [CFE_ES_RESTART_APP_INF_EID](#) 10
ES Restart Application Completed Event ID.
- #define [CFE_ES_RELOAD_APP_DBG_EID](#) 11
ES Reload Application Command Request Success Event ID.
- #define [CFE_ES_RELOAD_APP_INF_EID](#) 12
ES Reload Application Complete Event ID.
- #define [CFE_ES_EXIT_APP_INF_EID](#) 13
ES Nominal Exit Application Complete Event ID.
- #define [CFE_ES_ERREXIT_APP_INF_EID](#) 14
ES Error Exit Application Complete Event ID.
- #define [CFE_ES_ONE_APP_EID](#) 15
ES Query One Application Command Success Event ID.
- #define [CFE_ES_ALL_APPS_EID](#) 16
ES Query All Applications Command Success Event ID.
- #define [CFE_ES_SYSLOG1_INF_EID](#) 17
ES Clear System Log Command Success Event ID.
- #define [CFE_ES_SYSLOG2_EID](#) 18
ES Write System Log Command Success Event ID.

- #define CFE_ES_ERLOG1_INF_EID 19
ES Clear Exception Reset Log Command Success Event ID.
- #define CFE_ES_ERLOG2_EID 20
ES Write Exception Reset Log Complete Event ID.
- #define CFE_ES_MID_ERR_EID 21
ES Invalid Message ID Received Event ID.
- #define CFE_ES_CC1_ERR_EID 22
ES Invalid Command Code Received Event ID.
- #define CFE_ES_LEN_ERR_EID 23
ES Invalid Command Length Event ID.
- #define CFE_ES_BOOT_ERR_EID 24
ES Restart Command Invalid Restart Type Event ID.
- #define CFE_ES_START_ERR_EID 26
ES Start Application Command Application Creation Failed Event ID.
- #define CFE_ES_START_INVALID_FILENAME_ERR_EID 27
ES Start Application Command Invalid Filename Event ID.
- #define CFE_ES_START_INVALID_ENTRY_POINT_ERR_EID 28
ES Start Application Command Entry Point NULL Event ID.
- #define CFE_ES_START_NULL_APP_NAME_ERR_EID 29
ES Start Application Command App Name NULL Event ID.
- #define CFE_ES_START_PRIORITY_ERR_EID 31
ES Start Application Command Priority Too Large Event ID.
- #define CFE_ES_START_EXC_ACTION_ERR_EID 32
ES Start Application Command Exception Action Invalid Event ID.
- #define CFE_ES_ERREXIT_APP_ERR_EID 33
ES Error Exit Application Cleanup Failed Event ID.
- #define CFE_ES_STOP_ERR1_EID 35
ES Stop Application Command Request Failed Event ID.
- #define CFE_ES_STOP_ERR2_EID 36
ES Stop Application Command Get AppID By Name Failed Event ID.
- #define CFE_ES_STOP_ERR3_EID 37
ES Stop Application Cleanup Failed Event ID.
- #define CFE_ES_RESTART_APP_ERR1_EID 38
ES Restart Application Command Request Failed Event ID.
- #define CFE_ES_RESTART_APP_ERR2_EID 39
ES Restart Application Command Get AppID By Name Failed Event ID.
- #define CFE_ES_RESTART_APP_ERR3_EID 40
ES Restart Application Startup Failed Event ID.
- #define CFE_ES_RESTART_APP_ERR4_EID 41
ES Restart Application Cleanup Failed Event ID.
- #define CFE_ES_RELOAD_APP_ERR1_EID 42
ES Reload Application Command Request Failed Event ID.
- #define CFE_ES_RELOAD_APP_ERR2_EID 43
ES Reload Application Command Get AppID By Name Failed Event ID.
- #define CFE_ES_RELOAD_APP_ERR3_EID 44
ES Reload Application Startup Failed Event ID.
- #define CFE_ES_RELOAD_APP_ERR4_EID 45
ES Reload Application Cleanup Failed Event ID.
- #define CFE_ES_EXIT_APP_ERR_EID 46
ES Exit Application Cleanup Failed Event ID.
- #define CFE_ES_PCR_ERR1_EID 47
ES Process Control Invalid Exception State Event ID.
- #define CFE_ES_PCR_ERR2_EID 48
ES Process Control Unknown State Event ID.
- #define CFE_ES_ONE_ERR_EID 49

- ES Query One Application Data Command Transmit Message Failed Event ID.*
 - #define [CFE_ES_ONE_APPID_ERR_EID](#) 50
- ES Query One Application Data Command Get AppID By Name Failed Event ID.*
 - #define [CFE_ES_OSCREATE_ERR_EID](#) 51
- ES Query All Application Data Command File Creation Failed Event ID.*
 - #define [CFE_ES_WRHDR_ERR_EID](#) 52
- ES Query All Application Data Command File Write Header Failed Event ID.*
 - #define [CFE_ES_TASKWR_ERR_EID](#) 53
- ES Query All Application Data Command File Write App Data Failed Event ID.*
 - #define [CFE_ES_SYSLOG2_ERR_EID](#) 55
- ES Write System Log Command Filename Parse or File Creation Failed Event ID.*
 - #define [CFE_ES_ERLOG2_ERR_EID](#) 56
- ES Write Exception Reset Log Command Request or File Creation Failed Event ID.*
 - #define [CFE_ES_PERF_STARTCMD_EID](#) 57
- ES Start Performance Analyzer Data Collection Command Success Event ID.*
 - #define [CFE_ES_PERF_STARTCMD_ERR_EID](#) 58
- ES Start Performance Analyzer Data Collection Command Idle Check Failed Event ID.*
 - #define [CFE_ES_PERF_STARTCMD_TRIG_ERR_EID](#) 59
- ES Start Performance Analyzer Data Collection Command Invalid Trigger Event ID.*
 - #define [CFE_ES_PERF_STOPCMD_EID](#) 60
- ES Stop Performance Analyzer Data Collection Command Request Success Event ID.*
 - #define [CFE_ES_PERF_STOPCMD_ERR2_EID](#) 62
- ES Stop Performance Analyzer Data Collection Command Request Idle Check Failed Event ID.*
 - #define [CFE_ES_PERF_FILTMSKCMD_EID](#) 63
- ES Set Performance Analyzer Filter Mask Command Success Event ID.*
 - #define [CFE_ES_PERF_FILTMSKERR_EID](#) 64
- ES Set Performance Analyzer Filter Mask Command Invalid Index Event ID.*
 - #define [CFE_ES_PERF_TRIGMSKCMD_EID](#) 65
- ES Set Performance Analyzer Trigger Mask Command Success Event ID.*
 - #define [CFE_ES_PERF_TRIGMSKERR_EID](#) 66
- ES Set Performance Analyzer Trigger Mask Command Invalid Mask Event ID.*
 - #define [CFE_ES_PERF_LOG_ERR_EID](#) 67
- ES Stop Performance Analyzer Data Collection Command Filename Parse or File Create Failed Event ID.*
 - #define [CFE_ES_PERF_DATAWRITTEN_EID](#) 68
- Performance Log Write Success Event ID.*
 - #define [CFE_ES_CDS_REGISTER_ERR_EID](#) 69
- ES Register CDS API Failed Event ID.*
 - #define [CFE_ES_SYSLOGMODE_EID](#) 70
- ES Set System Log Overwrite Mode Command Success Event ID.*
 - #define [CFE_ES_ERR_SYSLOGMODE_EID](#) 71
- ES Set System Log Overwrite Mode Command Failed Event ID.*
 - #define [CFE_ES_RESET_PR_COUNT_EID](#) 72
- ES Set Processor Reset Counter to Zero Command Success Event ID.*
 - #define [CFE_ES_SET_MAX_PR_COUNT_EID](#) 73
- ES Set Maximum Processor Reset Limit Command Success Event ID.*
 - #define [CFE_ES_FILEWRITE_ERR_EID](#) 74
- ES File Write Failed Event ID.*
 - #define [CFE_ES_CDS_DELETE_ERR_EID](#) 76
- ES Delete CDS Command Delete Failed Event ID.*
 - #define [CFE_ES_CDS_NAME_ERR_EID](#) 77
- ES Delete CDS Command Lookup CDS Failed Event ID.*
 - #define [CFE_ES_CDS_DELETED_INFO_EID](#) 78
- ES Delete CDS Command Success Event ID.*
 - #define [CFE_ES_CDS_DELETE_TBL_ERR_EID](#) 79
- ES Delete CDS Command For Critical Table Event ID.*

- `#define CFE_ES_CDS_OWNER_ACTIVE_EID` 80
ES Delete CDS Command With Active Owner Event ID.
- `#define CFE_ES_TLM_POOL_STATS_INFO_EID` 81
ES Telemetry Memory Statistics Command Success Event ID.
- `#define CFE_ES_INVALID_POOL_HANDLE_ERR_EID` 82
ES Telemetry Memory Statistics Command Invalid Handle Event ID.
- `#define CFE_ES_CDS_REG_DUMP_INF_EID` 83
ES Write Critical Data Store Registry Command Success Event ID.
- `#define CFE_ES_CDS_DUMP_ERR_EID` 84
ES Write Critical Data Store Registry Command Record Write Failed Event ID.
- `#define CFE_ES_WRITE_CFE_HDR_ERR_EID` 85
ES Write Critical Data Store Registry Command Header Write Failed Event ID.
- `#define CFE_ES_CREATING_CDS_DUMP_ERR_EID` 86
ES Write Critical Data Store Registry Command Filename Parse or File Create Failed Event ID.
- `#define CFE_ES_TASKINFO_EID` 87
ES Write All Task Data Command Success Event ID.
- `#define CFE_ES_TASKINFO_OSCREATE_ERR_EID` 88
ES Write All Task Data Command Filename Parse or File Create Failed Event ID.
- `#define CFE_ES_TASKINFO_WRHDR_ERR_EID` 89
ES Write All Task Data Command Write Header Failed Event ID.
- `#define CFE_ES_TASKINFO_WR_ERR_EID` 90
ES Write All Task Data Command Write Data Failed Event ID.
- `#define CFE_ES_VERSION_INF_EID` 91
cFS Version Information Event ID
- `#define CFE_ES_BUILD_INF_EID` 92
cFS Build Information Event ID
- `#define CFE_ES_ERLOG_PENDING_ERR_EID` 93
ES Write Exception Reset Log Command Already In Progress Event ID.

11.123.1 Detailed Description

cFE Executive Services Event IDs
Definition in file [cfe_es_eventids.h](#).

11.123.2 Macro Definition Documentation

CFE_ES_ALL_APPS_EID

```
#define CFE_ES_ALL_APPS_EID 16
```

ES Query All Applications Command Success Event ID.

Type: DEBUG

Cause:

[ES Query All Applications Command](#) success.
Definition at line 206 of file [cfe_es_eventids.h](#).

CFE_ES_BOOT_ERR_EID

```
#define CFE_ES_BOOT_ERR_EID 24
```

ES Restart Command Invalid Restart Type Event ID.

Type: ERROR

Cause:

ES cFE Restart Command failure due to invalid restart type.

Definition at line 294 of file [cfe_es_eventids.h](#).

CFE_ES_BUILD_INF_EID

```
#define CFE_ES_BUILD_INF_EID 92
```

cFS Build Information Event ID

Type: INFORMATION

Cause:

ES Initialization complete and response to [ES NO-OP Command](#) .

The `Build` field identifies the build date, time, hostname and user identifier of the build host machine for the current running binary. The first string is the build date/time, and the second string is formatted as "user@hostname"

This additionally reports the configuration name that was selected by the user, which may affect various platform/mission limits.

By default, if not specified/overridden, the default values of these variables will be: `BUILDDATE` ==> the output of "date +%Y%m%d%H%M" `HOSTNAME` ==> the output of "hostname" `USER` ==> the output of "whoami"

The values can be overridden by setting an environment variable with the names above to the value desired for the field when running "make".

Definition at line 1047 of file [cfe_es_eventids.h](#).

CFE_ES_CC1_ERR_EID

```
#define CFE_ES_CC1_ERR_EID 22
```

ES Invalid Command Code Received Event ID.

Type: ERROR

Cause:

Invalid command code for message ID [CFE_ES_CMD_MID](#) received on the ES message pipe.

Definition at line 272 of file [cfe_es_eventids.h](#).

CFE_ES_CDS_DELETE_ERR_EID

#define CFE_ES_CDS_DELETE_ERR_EID 76
ES Delete CDS Command Delete Failed Event ID.
Type: ERROR

Cause:

[ES Delete CDS Command](#) failed while deleting, see reported status code or system log for details.
Definition at line 834 of file [cfe_es_eventids.h](#).

CFE_ES_CDS_DELETE_TBL_ERR_EID

#define CFE_ES_CDS_DELETE_TBL_ERR_EID 79
ES Delete CDS Command For Critical Table Event ID.
Type: ERROR

Cause:

[Delete CDS Command](#) failure due to the specified CDS name being a critical table. Critical Table images can only be deleted via a Table Services command, [CFE_TBL_DELETE_CDS_CC](#).
Definition at line 871 of file [cfe_es_eventids.h](#).

CFE_ES_CDS_DELETED_INFO_EID

#define CFE_ES_CDS_DELETED_INFO_EID 78
ES Delete CDS Command Success Event ID.
Type: INFORMATION

Cause:

[ES Delete CDS Command](#) success.
Definition at line 857 of file [cfe_es_eventids.h](#).

CFE_ES_CDS_DUMP_ERR_EID

#define CFE_ES_CDS_DUMP_ERR_EID 84
ES Write Critical Data Store Registry Command Record Write Failed Event ID.
Type: ERROR

Cause:

[ES Write Critical Data Store Registry Command](#) failed to write CDS record.
Definition at line 929 of file [cfe_es_eventids.h](#).

CFE_ES_CDS_NAME_ERR_EID

```
#define CFE_ES_CDS_NAME_ERR_EID 77
```

ES Delete CDS Command Lookup CDS Failed Event ID.

Type: ERROR

Cause:

[ES Delete CDS Command](#) failed due to the specified CDS name not found in the CDS Registry.
Definition at line 846 of file [cfe_es_eventids.h](#).

CFE_ES_CDS_OWNER_ACTIVE_EID

```
#define CFE_ES_CDS_OWNER_ACTIVE_EID 80
```

ES Delete CDS Command With Active Owner Event ID.

Type: ERROR

Cause:

[ES Delete CDS Command](#) failure due to the specifies CDS name is registered to an active application.
Definition at line 883 of file [cfe_es_eventids.h](#).

CFE_ES_CDS_REG_DUMP_INF_EID

```
#define CFE_ES_CDS_REG_DUMP_INF_EID 83
```

ES Write Critical Data Store Registry Command Success Event ID.

Type: DEBUG

Cause:

[ES Write Critical Data Store Registry Command](#) success.
Definition at line 917 of file [cfe_es_eventids.h](#).

CFE_ES_CDS_REGISTER_ERR_EID

```
#define CFE_ES_CDS_REGISTER_ERR_EID 69
```

ES Register CDS API Failed Event ID.

Type: ERROR

Cause:

[CFE_ES_RegisterCDS](#) API failure, see reported status code or system log for details.
Definition at line 766 of file [cfe_es_eventids.h](#).

CFE_ES_CREATING_CDS_DUMP_ERR_EID

```
#define CFE_ES_CREATING_CDS_DUMP_ERR_EID 86
```

ES Write Critical Data Store Registry Command Filename Parse or File Create Failed Event ID.

Type: ERROR

Cause:

[ES Write Critical Data Store Registry Command](#) failed to parse filename or open/create the file. OVERLOADED
Definition at line 953 of file [cfe_es_eventids.h](#).

CFE_ES_ERLOG1_INF_EID

```
#define CFE_ES_ERLOG1_INF_EID 19
```

ES Clear Exception Reset Log Command Success Event ID.

Type: INFORMATION

Cause:

[ES Clear Exception Reset Log Command](#) success.
Definition at line 239 of file [cfe_es_eventids.h](#).

CFE_ES_ERLOG2_EID

```
#define CFE_ES_ERLOG2_EID 20
```

ES Write Exception Reset Log Complete Event ID.

Type: DEBUG

Cause:

Request to write the Exception Reset log successfully completed.
Definition at line 250 of file [cfe_es_eventids.h](#).

CFE_ES_ERLOG2_ERR_EID

```
#define CFE_ES_ERLOG2_ERR_EID 56
```

ES Write Exception Reset Log Command Request or File Creation Failed Event ID.

Type: ERROR

Cause:

[ES Write Exception Reset Log Command](#) request failed or file creation failed. OVERLOADED
Definition at line 626 of file [cfe_es_eventids.h](#).

CFE_ES_ERLOG_PENDING_ERR_EID

```
#define CFE_ES_ERLOG_PENDING_ERR_EID 93
```

ES Write Exception Reset Log Command Already In Progress Event ID.

Type: ERROR

Cause:

[ES Write Exception Reset Log Command](#) failure due to a write already being in progress.

Definition at line 1059 of file [cfe_es_eventids.h](#).

CFE_ES_ERR_SYSLOGMODE_EID

```
#define CFE_ES_ERR_SYSLOGMODE_EID 71
```

ES Set System Log Overwrite Mode Command Failed Event ID.

Type: ERROR

Cause:

[ES Set System Log Overwrite Mode Command](#) failed due to invalid mode requested.

Definition at line 789 of file [cfe_es_eventids.h](#).

CFE_ES_ERREXIT_APP_ERR_EID

```
#define CFE_ES_ERREXIT_APP_ERR_EID 33
```

ES Error Exit Application Cleanup Failed Event ID.

Type: ERROR

Cause:

Error request to exit an application failed during application cleanup. Application and related resources will be in undefined state.

Definition at line 379 of file [cfe_es_eventids.h](#).

CFE_ES_ERREXIT_APP_INF_EID

```
#define CFE_ES_ERREXIT_APP_INF_EID 14
```

ES Error Exit Application Complete Event ID.

Type: INFORMATION

Cause:

Error request to exit an application successfully completed. This event indicates the Application exited due to an error condition. The details of the error that occurred should be given by the Application through an event message, System Log entry, or both.

Definition at line 184 of file [cfe_es_eventids.h](#).

CFE_ES_EXIT_APP_ERR_EID

```
#define CFE_ES_EXIT_APP_ERR_EID 46
```

ES Exit Application Cleanup Failed Event ID.
Type: ERROR

Cause:

Nominal request to exit an application failed during application cleanup. Application and related resources will be in undefined state.

Definition at line [522](#) of file [cfe_es_eventids.h](#).

CFE_ES_EXIT_APP_INF_EID

```
#define CFE_ES_EXIT_APP_INF_EID 13
```

ES Nominal Exit Application Complete Event ID.
Type: INFORMATION

Cause:

Nominal request to exit an application successfully completed. This event indicates the Application exited due to a nominal exit condition.

Definition at line [170](#) of file [cfe_es_eventids.h](#).

CFE_ES_FILEWRITE_ERR_EID

```
#define CFE_ES_FILEWRITE_ERR_EID 74
```

ES File Write Failed Event ID.
Type: ERROR

Cause:

ES File Write failure writing data to file. OVERLOADED

Definition at line [822](#) of file [cfe_es_eventids.h](#).

CFE_ES_INIT_INF_EID

```
#define CFE_ES_INIT_INF_EID 1
```

ES Initialization Event ID.
Type: INFORMATION

Cause:

Executive Services Task initialization complete.

Definition at line [42](#) of file [cfe_es_eventids.h](#).

CFE_ES_INITSTATS_INF_EID

```
#define CFE_ES_INITSTATS_INF_EID 2
```

ES Initialization Statistics Information Event ID.

Type: INFORMATION

Cause:

Executive Services Task initialization complete.

Definition at line 53 of file [cfe_es_eventids.h](#).

CFE_ES_INVALID_POOL_HANDLE_ERR_EID

```
#define CFE_ES_INVALID_POOL_HANDLE_ERR_EID 82
```

ES Telemeter Memory Statistics Command Invalid Handle Event ID.

Type: ERROR

Cause:

[ES Telemeter Memory Statistics Command](#) failure due to an invalid memory handle.

Definition at line 906 of file [cfe_es_eventids.h](#).

CFE_ES_LEN_ERR_EID

```
#define CFE_ES_LEN_ERR_EID 23
```

ES Invalid Command Length Event ID.

Type: ERROR

Cause:

Invalid length for the command code in message ID [CFE_ES_CMD_MID](#) received on the ES message pipe.

Definition at line 283 of file [cfe_es_eventids.h](#).

CFE_ES_MID_ERR_EID

```
#define CFE_ES_MID_ERR_EID 21
```

ES Invalid Message ID Received Event ID.

Type: ERROR

Cause:

Invalid message ID received on the ES message pipe.

Definition at line 261 of file [cfe_es_eventids.h](#).

CFE_ES_NOOP_INF_EID

#define CFE_ES_NOOP_INF_EID 3
ES No-op Command Success Event ID.

Type: INFORMATION

Cause:

[ES No-op Command](#) success.
Definition at line 64 of file [cfe_es_eventids.h](#).

CFE_ES_ONE_APP_EID

#define CFE_ES_ONE_APP_EID 15
ES Query One Application Command Success Event ID.

Type: DEBUG

Cause:

[ES Query One Application Command](#) success.
Definition at line 195 of file [cfe_es_eventids.h](#).

CFE_ES_ONE_APPID_ERR_EID

#define CFE_ES_ONE_APPID_ERR_EID 50
ES Query One Application Data Command Get AppID By Name Failed Event ID.

Type: ERROR

Cause:

[ES Query One Application Data Command](#) failed to get application ID from application name. Message will not be sent.
Definition at line 569 of file [cfe_es_eventids.h](#).

CFE_ES_ONE_ERR_EID

#define CFE_ES_ONE_ERR_EID 49
ES Query One Application Data Command Transmit Message Failed Event ID.

Type: ERROR

Cause:

[ES Query One Application Data Command](#) failed during message transmission.
Definition at line 557 of file [cfe_es_eventids.h](#).

CFE_ES_OSCREATE_ERR_EID

```
#define CFE_ES_OSCREATE_ERR_EID 51
```

ES Query All Application Data Command File Creation Failed Event ID.

Type: ERROR

Cause:

[ES Query All Application Data Command](#) failed to create file.

Definition at line [580](#) of file [cfe_es_eventids.h](#).

CFE_ES_PCR_ERR1_EID

```
#define CFE_ES_PCR_ERR1_EID 47
```

ES Process Control Invalid Exception State Event ID.

Type: ERROR

Cause:

Invalid Exception state encountered when processing requests for application state changes. Exceptions are processed immediately, so this state should never occur during routine processing.

Definition at line [534](#) of file [cfe_es_eventids.h](#).

CFE_ES_PCR_ERR2_EID

```
#define CFE_ES_PCR_ERR2_EID 48
```

ES Process Control Unknown State Event ID.

Type: ERROR

Cause:

Unknown state encountered when processing requests for application state changes.

Definition at line [545](#) of file [cfe_es_eventids.h](#).

CFE_ES_PERF_DATAWRITTEN_EID

```
#define CFE_ES_PERF_DATAWRITTEN_EID 68
```

Performance Log Write Success Event ID.

Type: DEBUG

Cause:

Request to write the performance log successfully completed.

Definition at line [755](#) of file [cfe_es_eventids.h](#).

CFE_ES_PERF_FILTMSKCMD_EID

```
#define CFE_ES_PERF_FILTMSKCMD_EID 63
```

ES Set Performance Analyzer Filter Mask Command Success Event ID.

Type: DEBUG

Cause:

[ES Set Performance Analyzer Filter Mask Command](#) success.

Definition at line 697 of file [cfe_es_eventids.h](#).

CFE_ES_PERF_FILTMSKERR_EID

```
#define CFE_ES_PERF_FILTMSKERR_EID 64
```

ES Set Performance Analyzer Filter Mask Command Invalid Index Event ID.

Type: ERROR

Cause:

[ES Set Performance Analyzer Filter Mask Command](#) failed filter index range check.

Definition at line 709 of file [cfe_es_eventids.h](#).

CFE_ES_PERF_LOG_ERR_EID

```
#define CFE_ES_PERF_LOG_ERR_EID 67
```

ES Stop Performance Analyzer Data Collection Command Filename Parse or File Create Failed Event ID.

Type: ERROR

Cause:

[ES Stop Performance Analyzer Data Collection Command](#) failed either parsing the file name or during open/creation of the file. OVERLOADED

Definition at line 744 of file [cfe_es_eventids.h](#).

CFE_ES_PERF_STARTCMD_EID

```
#define CFE_ES_PERF_STARTCMD_EID 57
```

ES Start Performance Analyzer Data Collection Command Success Event ID.

Type: DEBUG

Cause:

[ES Start Performance Analyzer Data Collection Command](#) success.

Definition at line 637 of file [cfe_es_eventids.h](#).

CFE_ES_PERF_STARTCMD_ERR_EID

```
#define CFE_ES_PERF_STARTCMD_ERR_EID 58
```

ES Start Performance Analyzer Data Collection Command Idle Check Failed Event ID.

Type: ERROR

Cause:

[ES Start Performance Analyzer Data Collection Command](#) failed due to already being started.
Definition at line 649 of file [cfe_es_eventids.h](#).

CFE_ES_PERF_STARTCMD_TRIG_ERR_EID

```
#define CFE_ES_PERF_STARTCMD_TRIG_ERR_EID 59
```

ES Start Performance Analyzer Data Collection Command Invalid Trigger Event ID.

Type: ERROR

Cause:

[ES Start Performance Analyzer Data Collection Command](#) failed due to invalid trigger mode.
Definition at line 661 of file [cfe_es_eventids.h](#).

CFE_ES_PERF_STOPCMD_EID

```
#define CFE_ES_PERF_STOPCMD_EID 60
```

ES Stop Performance Analyzer Data Collection Command Request Success Event ID.

Type: DEBUG

Cause:

[ES Stop Performance Analyzer Data Collection Command](#) success. Note this event signifies the request to stop and write the performance data has been successfully submitted. The successful completion will generate a [CFE_ES_PERF_DATAWRITTEN_EID](#) event.
Definition at line 674 of file [cfe_es_eventids.h](#).

CFE_ES_PERF_STOPCMD_ERR2_EID

```
#define CFE_ES_PERF_STOPCMD_ERR2_EID 62
```

ES Stop Performance Analyzer Data Collection Command Request Idle Check Failed Event ID.

Type: ERROR

Cause:

[ES Stop Performance Analyzer Data Collection Command](#) failed due to a write already in progress.
Definition at line 686 of file [cfe_es_eventids.h](#).

CFE_ES_PERF_TRIGMSKCMD_EID

```
#define CFE_ES_PERF_TRIGMSKCMD_EID 65
```

ES Set Performance Analyzer Trigger Mask Command Success Event ID.

Type: DEBUG

Cause:

[ES Set Performance Analyzer Trigger Mask Command](#) success.

Definition at line 720 of file [cfe_es_eventids.h](#).

CFE_ES_PERF_TRIGMSKERR_EID

```
#define CFE_ES_PERF_TRIGMSKERR_EID 66
```

ES Set Performance Analyzer Trigger Mask Command Invalid Mask Event ID.

Type: ERROR

Cause:

[ES Set Performance Analyzer Trigger Mask Command](#) failed the mask range check.

Definition at line 732 of file [cfe_es_eventids.h](#).

CFE_ES_RELOAD_APP_DBG_EID

```
#define CFE_ES_RELOAD_APP_DBG_EID 11
```

ES Reload Application Command Request Success Event ID.

Type: DEBUG

Cause:

[ES Reload Application Command](#) success. Note this event signifies the request to reload the application has been successfully submitted. The successful completion will generate a [CFE_ES_RELOAD_APP_INF_EID](#) event.

Definition at line 147 of file [cfe_es_eventids.h](#).

CFE_ES_RELOAD_APP_ERR1_EID

```
#define CFE_ES_RELOAD_APP_ERR1_EID 42
```

ES Reload Application Command Request Failed Event ID.

Type: ERROR

Cause:

[ES Reload Application Command](#) request failed.

Definition at line 473 of file [cfe_es_eventids.h](#).

CFE_ES_RELOAD_APP_ERR2_EID

```
#define CFE_ES_RELOAD_APP_ERR2_EID 43
```

ES Reload Application Command Get AppID By Name Failed Event ID.

Type: ERROR

Cause:

[ES Reload Application Command](#) failed to get application ID from application name. The application will not be reloaded.

Definition at line [485](#) of file [cfe_es_eventids.h](#).

CFE_ES_RELOAD_APP_ERR3_EID

```
#define CFE_ES_RELOAD_APP_ERR3_EID 44
```

ES Reload Application Startup Failed Event ID.

Type: ERROR

Cause:

Request to reload an application failed during application startup. The application will not be reloaded.

Definition at line [497](#) of file [cfe_es_eventids.h](#).

CFE_ES_RELOAD_APP_ERR4_EID

```
#define CFE_ES_RELOAD_APP_ERR4_EID 45
```

ES Reload Application Cleanup Failed Event ID.

Type: ERROR

Cause:

Request to reload an application failed during application cleanup. The application will not be reloaded and will be in an undefined state along with it's associated resources.

Definition at line [510](#) of file [cfe_es_eventids.h](#).

CFE_ES_RELOAD_APP_INF_EID

```
#define CFE_ES_RELOAD_APP_INF_EID 12
```

ES Reload Application Complete Event ID.

Type: INFORMATION

Cause:

Request to reload an application successfully completed.

Definition at line [158](#) of file [cfe_es_eventids.h](#).

CFE_ES_RESET_INF_EID

#define CFE_ES_RESET_INF_EID 4
ES Reset Counters Command Success Event ID.
Type: INFORMATION

Cause:

[ES Reset Counters Command](#) success.
Definition at line 75 of file [cfe_es_eventids.h](#).

CFE_ES_RESET_PR_COUNT_EID

#define CFE_ES_RESET_PR_COUNT_EID 72
ES Set Processor Reset Counter to Zero Command Success Event ID.
Type: INFORMATION

Cause:

[ES Set Processor Reset Counter to Zero Command](#) success.
Definition at line 800 of file [cfe_es_eventids.h](#).

CFE_ES_RESTART_APP_DBG_EID

#define CFE_ES_RESTART_APP_DBG_EID 9
ES Restart Application Command Request Success Event ID.
Type: DEBUG

Cause:

[ES Restart Application Command](#) success. Note this event signifies the request to restart the application has been successfully submitted. The successful completion will generate a [CFE_ES_RESTART_APP_INF_EID](#) event.
Definition at line 123 of file [cfe_es_eventids.h](#).

CFE_ES_RESTART_APP_ERR1_EID

#define CFE_ES_RESTART_APP_ERR1_EID 38
ES Restart Application Command Request Failed Event ID.
Type: ERROR

Cause:

[ES Restart Application Command](#) request failed.
Definition at line 425 of file [cfe_es_eventids.h](#).

CFE_ES_RESTART_APP_ERR2_EID

```
#define CFE_ES_RESTART_APP_ERR2_EID 39
```

ES Restart Application Command Get AppID By Name Failed Event ID.

Type: ERROR

Cause:

[ES Restart Application Command](#) failed to get application ID from application name. The application will not be restarted.

Definition at line [437](#) of file [cfe_es_eventids.h](#).

CFE_ES_RESTART_APP_ERR3_EID

```
#define CFE_ES_RESTART_APP_ERR3_EID 40
```

ES Restart Application Startup Failed Event ID.

Type: ERROR

Cause:

Request to restart an application failed during application startup. The application will not be restarted.

Definition at line [449](#) of file [cfe_es_eventids.h](#).

CFE_ES_RESTART_APP_ERR4_EID

```
#define CFE_ES_RESTART_APP_ERR4_EID 41
```

ES Restart Application Cleanup Failed Event ID.

Type: ERROR

Cause:

Request to restart an application failed during application cleanup. The application will not be restarted and will be in an undefined state along with it's associated resources.

Definition at line [462](#) of file [cfe_es_eventids.h](#).

CFE_ES_RESTART_APP_INF_EID

```
#define CFE_ES_RESTART_APP_INF_EID 10
```

ES Restart Application Completed Event ID.

Type: INFORMATION

Cause:

Request to restart an application successfully completed.

Definition at line [134](#) of file [cfe_es_eventids.h](#).

CFE_ES_SET_MAX_PR_COUNT_EID

```
#define CFE_ES_SET_MAX_PR_COUNT_EID 73
```

ES Set Maximum Processor Reset Limit Command Success Event ID.

Type: INFORMATION

Cause:

[ES Set Maximum Processor Reset Limit Command](#) success.

Definition at line 811 of file [cfe_es_eventids.h](#).

CFE_ES_START_ERR_EID

```
#define CFE_ES_START_ERR_EID 26
```

ES Start Application Command Application Creation Failed Event ID.

Type: ERROR

Cause:

[ES Start Application Command](#) failure during application creation after successful parameter validation.

Definition at line 306 of file [cfe_es_eventids.h](#).

CFE_ES_START_EXC_ACTION_ERR_EID

```
#define CFE_ES_START_EXC_ACTION_ERR_EID 32
```

ES Start Application Command Exception Action Invalid Event ID.

Type: ERROR

Cause:

[ES Start Application Command](#) failure due to invalid application exception action.

Definition at line 367 of file [cfe_es_eventids.h](#).

CFE_ES_START_INF_EID

```
#define CFE_ES_START_INF_EID 6
```

ES Start Application Command Success Event ID.

Type: INFORMATION

Cause:

[ES Start Application Command](#) success.

Definition at line 86 of file [cfe_es_eventids.h](#).

CFE_ES_START_INVALID_ENTRY_POINT_ERR_EID

```
#define CFE_ES_START_INVALID_ENTRY_POINT_ERR_EID 28
```

ES Start Application Command Entry Point NULL Event ID.

Type: ERROR

Cause:

[ES Start Application Command](#) failure due to a NULL Application Entry Point.
Definition at line 330 of file [cfe_es_eventids.h](#).

CFE_ES_START_INVALID_FILENAME_ERR_EID

```
#define CFE_ES_START_INVALID_FILENAME_ERR_EID 27
```

ES Start Application Command Invalid Filename Event ID.

Type: ERROR

Cause:

[ES Start Application Command](#) failure due to invalid filename.
Definition at line 318 of file [cfe_es_eventids.h](#).

CFE_ES_START_NULL_APP_NAME_ERR_EID

```
#define CFE_ES_START_NULL_APP_NAME_ERR_EID 29
```

ES Start Application Command App Name NULL Event ID.

Type: ERROR

Cause:

[ES Start Application Command](#) failure due to NULL Application Name.
Definition at line 342 of file [cfe_es_eventids.h](#).

CFE_ES_START_PRIORITY_ERR_EID

```
#define CFE_ES_START_PRIORITY_ERR_EID 31
```

ES Start Application Command Priority Too Large Event ID.

Type: ERROR

Cause:

[ES Start Application Command](#) failure due to a requested application priority greater than the maximum priority allowed for tasks as defined by the OS Abstraction Layer (OS_MAX_PRIORITY).
Definition at line 355 of file [cfe_es_eventids.h](#).

CFE_ES_STOP_DBG_EID

```
#define CFE_ES_STOP_DBG_EID 7
```

ES Stop Application Command Request Success Event ID.

Type: DEBUG

Cause:

[ES Stop Application Command](#) success. Note this event signifies the request to delete the application has been successfully submitted. The successful completion will generate a [CFE_ES_STOP_INF_EID](#) event.

Definition at line 99 of file [cfe_es_eventids.h](#).

CFE_ES_STOP_ERR1_EID

```
#define CFE_ES_STOP_ERR1_EID 35
```

ES Stop Application Command Request Failed Event ID.

Type: ERROR

Cause:

[ES Stop Application Command](#) request failed.

Definition at line 390 of file [cfe_es_eventids.h](#).

CFE_ES_STOP_ERR2_EID

```
#define CFE_ES_STOP_ERR2_EID 36
```

ES Stop Application Command Get AppID By Name Failed Event ID.

Type: ERROR

Cause:

[ES Stop Application Command](#) failed to get application ID from application name. The application will not be deleted.

Definition at line 402 of file [cfe_es_eventids.h](#).

CFE_ES_STOP_ERR3_EID

```
#define CFE_ES_STOP_ERR3_EID 37
```

ES Stop Application Cleanup Failed Event ID.

Type: ERROR

Cause:

Request to delete an application failed during application cleanup. Application and related resources will be in undefined state.

Definition at line 414 of file [cfe_es_eventids.h](#).

CFE_ES_STOP_INF_EID

```
#define CFE_ES_STOP_INF_EID 8
```

ES Stop Application Completed Event ID.

Type: INFORMATION

Cause:

Request to delete an application successfully completed.
Definition at line 110 of file [cfe_es_eventids.h](#).

CFE_ES_SYSLOG1_INF_EID

```
#define CFE_ES_SYSLOG1_INF_EID 17
```

ES Clear System Log Command Success Event ID.

Type: INFORMATION

Cause:

[ES Clear System Log Command](#) success.
Definition at line 217 of file [cfe_es_eventids.h](#).

CFE_ES_SYSLOG2_EID

```
#define CFE_ES_SYSLOG2_EID 18
```

ES Write System Log Command Success Event ID.

Type: DEBUG

Cause:

[ES Write System Log Command](#) success.
Definition at line 228 of file [cfe_es_eventids.h](#).

CFE_ES_SYSLOG2_ERR_EID

```
#define CFE_ES_SYSLOG2_ERR_EID 55
```

ES Write System Log Command Filename Parse or File Creation Failed Event ID.

Type: ERROR

Cause:

[ES Write System Log Command](#) failed parsing file name or creating the file. OVERLOADED
Definition at line 614 of file [cfe_es_eventids.h](#).

CFE_ES_SYSLOGMODE_EID

```
#define CFE_ES_SYSLOGMODE_EID 70
```

ES Set System Log Overwrite Mode Command Success Event ID.

Type: DEBUG

Cause:

[ES Set System Log Overwrite Mode Command](#) success.

Definition at line 777 of file [cfe_es_eventids.h](#).

CFE_ES_TASKINFO_EID

```
#define CFE_ES_TASKINFO_EID 87
```

ES Write All Task Data Command Success Event ID.

Type: DEBUG

Cause:

[ES Write All Task Data Command](#) success.

Definition at line 964 of file [cfe_es_eventids.h](#).

CFE_ES_TASKINFO_OSCREATE_ERR_EID

```
#define CFE_ES_TASKINFO_OSCREATE_ERR_EID 88
```

ES Write All Task Data Command Filename Parse or File Create Failed Event ID.

Type: ERROR

Cause:

[ES Write All Task Data Command](#) failed to parse the filename or open/create the file.

Definition at line 976 of file [cfe_es_eventids.h](#).

CFE_ES_TASKINFO_WR_ERR_EID

```
#define CFE_ES_TASKINFO_WR_ERR_EID 90
```

ES Write All Task Data Command Write Data Failed Event ID.

Type: ERROR

Cause:

[ES Write All Task Data Command](#) failed to write task data to file.

Definition at line 1000 of file [cfe_es_eventids.h](#).

CFE_ES_TASKINFO_WRHDR_ERR_EID

```
#define CFE_ES_TASKINFO_WRHDR_ERR_EID 89
```

ES Write All Task Data Command Write Header Failed Event ID.

Type: ERROR

Cause:

[ES Write All Task Data Command](#) failed to write file header.

Definition at line 988 of file [cfe_es_eventids.h](#).

CFE_ES_TASKWR_ERR_EID

```
#define CFE_ES_TASKWR_ERR_EID 53
```

ES Query All Application Data Command File Write App Data Failed Event ID.

Type: ERROR

Cause:

[ES Query All Application Data Command](#) failed to write file application data.

Definition at line 602 of file [cfe_es_eventids.h](#).

CFE_ES_TLM_POOL_STATS_INFO_EID

```
#define CFE_ES_TLM_POOL_STATS_INFO_EID 81
```

ES Telemeter Memory Statistics Command Success Event ID.

Type: DEBUG

Cause:

[ES Telemeter Memory Statistics Command](#) success.

Definition at line 894 of file [cfe_es_eventids.h](#).

CFE_ES_VERSION_INF_EID

```
#define CFE_ES_VERSION_INF_EID 91
```

cFS Version Information Event ID

Type: INFORMATION

Cause:

ES Initialization complete and response to [ES NO-OP Command](#).

A separate version info event will be generated for every module which is statically linked into the CFE core executable (e.g. OSAL, PSP, MSG, SBR, etc).

The version information reported in this event is derived from the source revision control system at build time, as opposed to manually-assigned semantic version numbers. It is intended to uniquely identify the actual source code that is currently running, to the extent this is possible.

The `Mission` version information also identifies the build configuration name, if available.

Definition at line 1021 of file [cfe_es_eventids.h](#).

CFE_ES_WRHDR_ERR_EID

```
#define CFE_ES_WRHDR_ERR_EID 52
```

ES Query All Application Data Command File Write Header Failed Event ID.

Type: ERROR

Cause:

[ES Query All Application Data Command](#) failed to write file header.

Definition at line 591 of file [cfe_es_eventids.h](#).

CFE_ES_WRITE_CFE_HDR_ERR_EID

```
#define CFE_ES_WRITE_CFE_HDR_ERR_EID 85
```

ES Write Critical Data Store Registry Command Header Write Failed Event ID.

Type: ERROR

Cause:

[ES Write Critical Data Store Registry Command](#) failed to write header.

Definition at line 941 of file [cfe_es_eventids.h](#).

11.124 cfe_es_eventids.h

[Go to the documentation of this file.](#)

```
00001 /*****
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"
00003  *
00004  * Copyright (c) 2023 United States Government as represented by the
00005  * Administrator of the National Aeronautics and Space Administration.
00006  * All Rights Reserved.
00007  *
00008  * Licensed under the Apache License, Version 2.0 (the "License"); you may
00009  * not use this file except in compliance with the License. You may obtain
00010  * a copy of the license at http://www.apache.org/licenses/LICENSE-2.0
00011  *
00012  * Unless required by applicable law or agreed to in writing, software
00013  * distributed under the License is distributed on an "AS IS" BASIS,
00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
```

```
00016  * limitations under the License.
00017  *****/
00018
00025 #ifndef CFE_ES_EVENTS_H
00026 #define CFE_ES_EVENTS_H
00027
00042 #define CFE_ES_INIT_INF_EID 1
00043
00053 #define CFE_ES_INITSTATS_INF_EID 2
00054
00064 #define CFE_ES_NOOP_INF_EID 3
00065
00075 #define CFE_ES_RESET_INF_EID 4
00076
00086 #define CFE_ES_START_INF_EID 6
00087
00099 #define CFE_ES_STOP_DBG_EID 7
00100
00110 #define CFE_ES_STOP_INF_EID 8
00111
00123 #define CFE_ES_RESTART_APP_DBG_EID 9
00124
00134 #define CFE_ES_RESTART_APP_INF_EID 10
00135
00147 #define CFE_ES_RELOAD_APP_DBG_EID 11
00148
00158 #define CFE_ES_RELOAD_APP_INF_EID 12
00159
00170 #define CFE_ES_EXIT_APP_INF_EID 13
00171
00184 #define CFE_ES_ERREXIT_APP_INF_EID 14
00185
00195 #define CFE_ES_ONE_APP_EID 15
00196
00206 #define CFE_ES_ALL_APPS_EID 16
00207
00217 #define CFE_ES_SYSLOG1_INF_EID 17
00218
00228 #define CFE_ES_SYSLOG2_EID 18
00229
00239 #define CFE_ES_ERLOG1_INF_EID 19
00240
00250 #define CFE_ES_ERLOG2_EID 20
00251
00261 #define CFE_ES_MID_ERR_EID 21
00262
00272 #define CFE_ES_CC1_ERR_EID 22
00273
00283 #define CFE_ES_LEN_ERR_EID 23
00284
00294 #define CFE_ES_BOOT_ERR_EID 24
00295
00306 #define CFE_ES_START_ERR_EID 26
00307
00318 #define CFE_ES_START_INVALID_FILENAME_ERR_EID 27
00319
00330 #define CFE_ES_START_INVALID_ENTRY_POINT_ERR_EID 28
00331
00342 #define CFE_ES_START_NULL_APP_NAME_ERR_EID 29
00343
00355 #define CFE_ES_START_PRIORITY_ERR_EID 31
00356
00367 #define CFE_ES_START_EXC_ACTION_ERR_EID 32
00368
00379 #define CFE_ES_ERREXIT_APP_ERR_EID 33
00380
00390 #define CFE_ES_STOP_ERR1_EID 35
00391
00402 #define CFE_ES_STOP_ERR2_EID 36
00403
00414 #define CFE_ES_STOP_ERR3_EID 37
00415
00425 #define CFE_ES_RESTART_APP_ERR1_EID 38
00426
00437 #define CFE_ES_RESTART_APP_ERR2_EID 39
00438
00449 #define CFE_ES_RESTART_APP_ERR3_EID 40
00450
00462 #define CFE_ES_RESTART_APP_ERR4_EID 41
00463
00473 #define CFE_ES_RELOAD_APP_ERR1_EID 42
```

```
00474
00485 #define CFE_ES_RELOAD_APP_ERR2_EID 43
00486
00497 #define CFE_ES_RELOAD_APP_ERR3_EID 44
00498
00510 #define CFE_ES_RELOAD_APP_ERR4_EID 45
00511
00522 #define CFE_ES_EXIT_APP_ERR_EID 46
00523
00534 #define CFE_ES_PCR_ERR1_EID 47
00535
00545 #define CFE_ES_PCR_ERR2_EID 48
00546
00557 #define CFE_ES_ONE_ERR_EID 49
00558
00569 #define CFE_ES_ONE_APPID_ERR_EID 50
00570
00580 #define CFE_ES_OSCREATE_ERR_EID 51
00581
00591 #define CFE_ES_WRHDR_ERR_EID 52
00592
00602 #define CFE_ES_TASKWR_ERR_EID 53
00603
00614 #define CFE_ES_SYSLOG2_ERR_EID 55
00615
00626 #define CFE_ES_ERLOG2_ERR_EID 56
00627
00637 #define CFE_ES_PERF_STARTCMD_EID 57
00638
00649 #define CFE_ES_PERF_STARTCMD_ERR_EID 58
00650
00661 #define CFE_ES_PERF_STARTCMD_TRIG_ERR_EID 59
00662
00674 #define CFE_ES_PERF_STOPCMD_EID 60
00675
00686 #define CFE_ES_PERF_STOPCMD_ERR2_EID 62
00687
00697 #define CFE_ES_PERF_FILTMSKCMD_EID 63
00698
00709 #define CFE_ES_PERF_FILTMSKERR_EID 64
00710
00720 #define CFE_ES_PERF_TRIGMSKCMD_EID 65
00721
00732 #define CFE_ES_PERF_TRIGMSKERR_EID 66
00733
00744 #define CFE_ES_PERF_LOG_ERR_EID 67
00745
00755 #define CFE_ES_PERF_DATAWRITTEN_EID 68
00756
00766 #define CFE_ES_CDS_REGISTER_ERR_EID 69
00767
00777 #define CFE_ES_SYSLOGMODE_EID 70
00778
00789 #define CFE_ES_ERR_SYSLOGMODE_EID 71
00790
00800 #define CFE_ES_RESET_PR_COUNT_EID 72
00801
00811 #define CFE_ES_SET_MAX_PR_COUNT_EID 73
00812
00822 #define CFE_ES_FILEWRITE_ERR_EID 74
00823
00834 #define CFE_ES_CDS_DELETE_ERR_EID 76
00835
00846 #define CFE_ES_CDS_NAME_ERR_EID 77
00847
00857 #define CFE_ES_CDS_DELETED_INFO_EID 78
00858
00871 #define CFE_ES_CDS_DELETE_TBL_ERR_EID 79
00872
00883 #define CFE_ES_CDS_OWNER_ACTIVE_EID 80
00884
00894 #define CFE_ES_TLM_POOL_STATS_INFO_EID 81
00895
00906 #define CFE_ES_INVALID_POOL_HANDLE_ERR_EID 82
00907
00917 #define CFE_ES_CDS_REG_DUMP_INF_EID 83
00918
00929 #define CFE_ES_CDS_DUMP_ERR_EID 84
00930
00941 #define CFE_ES_WRITE_CFE_HDR_ERR_EID 85
00942
```

```

00953 #define CFE_ES_CREATING_CDS_DUMP_ERR_EID 86
00954
00964 #define CFE_ES_TASKINFO_EID 87
00965
00976 #define CFE_ES_TASKINFO_OSCREATE_ERR_EID 88
00977
00988 #define CFE_ES_TASKINFO_WRHDR_ERR_EID 89
00989
01000 #define CFE_ES_TASKINFO_WR_ERR_EID 90
01001
01021 #define CFE_ES_VERSION_INF_EID 91
01022
01047 #define CFE_ES_BUILD_INF_EID 92
01048
01059 #define CFE_ES_ERLOG_PENDING_ERR_EID 93
01062 #endif /* CFE_ES_EVENTS_H */

```

11.125 cfe/modules/es/fsw/inc/cfe_es_fcncodes.h File Reference

```
#include "cfe_es_fcncode_values.h"
```

Macros

Executive Services Command Codes

- #define [CFE_ES_NOOP_CC](#) [CFE_ES_CCVAL](#)(NOOP)
- #define [CFE_ES_RESET_COUNTERS_CC](#) [CFE_ES_CCVAL](#)(RESET_COUNTERS)
- #define [CFE_ES_RESTART_CC](#) [CFE_ES_CCVAL](#)(RESTART)
- #define [CFE_ES_START_APP_CC](#) [CFE_ES_CCVAL](#)(START_APP)
- #define [CFE_ES_STOP_APP_CC](#) [CFE_ES_CCVAL](#)(STOP_APP)
- #define [CFE_ES_RESTART_APP_CC](#) [CFE_ES_CCVAL](#)(RESTART_APP)
- #define [CFE_ES_RELOAD_APP_CC](#) [CFE_ES_CCVAL](#)(RELOAD_APP)
- #define [CFE_ES_QUERY_ONE_CC](#) [CFE_ES_CCVAL](#)(QUERY_ONE)
- #define [CFE_ES_QUERY_ALL_CC](#) [CFE_ES_CCVAL](#)(QUERY_ALL)
- #define [CFE_ES_CLEAR_SYS_LOG_CC](#) [CFE_ES_CCVAL](#)(CLEAR_SYS_LOG)
- #define [CFE_ES_WRITE_SYS_LOG_CC](#) [CFE_ES_CCVAL](#)(WRITE_SYS_LOG)
- #define [CFE_ES_CLEAR_ER_LOG_CC](#) [CFE_ES_CCVAL](#)(CLEAR_ER_LOG)
- #define [CFE_ES_WRITE_ER_LOG_CC](#) [CFE_ES_CCVAL](#)(WRITE_ER_LOG)
- #define [CFE_ES_START_PERF_DATA_CC](#) [CFE_ES_CCVAL](#)(START_PERF_DATA)
- #define [CFE_ES_STOP_PERF_DATA_CC](#) [CFE_ES_CCVAL](#)(STOP_PERF_DATA)
- #define [CFE_ES_SET_PERF_FILTER_MASK_CC](#) [CFE_ES_CCVAL](#)(SET_PERF_FILTER_MASK)
- #define [CFE_ES_SET_PERF_TRIGGER_MASK_CC](#) [CFE_ES_CCVAL](#)(SET_PERF_TRIGGER_MASK)
- #define [CFE_ES_OVER_WRITE_SYS_LOG_CC](#) [CFE_ES_CCVAL](#)(OVER_WRITE_SYS_LOG)
- #define [CFE_ES_RESET_PR_COUNT_CC](#) [CFE_ES_CCVAL](#)(RESET_PR_COUNT)
- #define [CFE_ES_SET_MAX_PR_COUNT_CC](#) [CFE_ES_CCVAL](#)(SET_MAX_PR_COUNT)
- #define [CFE_ES_DELETE_CDS_CC](#) [CFE_ES_CCVAL](#)(DELETE_CDS)
- #define [CFE_ES_SEND_MEM_POOL_STATS_CC](#) [CFE_ES_CCVAL](#)(SEND_MEM_POOL_STATS)
- #define [CFE_ES_DUMP_CDS_REGISTRY_CC](#) [CFE_ES_CCVAL](#)(DUMP_CDS_REGISTRY)
- #define [CFE_ES_QUERY_ALL_TASKS_CC](#) [CFE_ES_CCVAL](#)(QUERY_ALL_TASKS)

11.125.1 Detailed Description

Specification for the CFE Executive Services (CFE_ES) command function codes

Note

This file should be strictly limited to the command/function code (CC) macro definitions. Other definitions such as enums, typedefs, or other macros should be placed in the msgdefs.h or msg.h files.

Definition in file [cfe_es_fcncodes.h](#).

11.125.2 Macro Definition Documentation

CFE_ES_CLEAR_ER_LOG_CC

```
#define CFE_ES_CLEAR_ER_LOG_CC CFE_ES_CCVAL(CLEAR_ER_LOG)
```

Name Clears the contents of the Exception and Reset Log

Description

This command causes the contents of the Executive Services Exception and Reset Log to be cleared.

Command Mnemonic(s) \$sc_\$cpu_ES_ClearERLog

Command Structure

[CFE_ES_ClearERLogCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **\$sc_\$cpu_ES_CMDPC** - command execution counter will increment
- The [CFE_ES_ERLOG1_INF_EID](#) informational event message will be generated.
- **\$sc_\$cpu_ES_ERLOGINDEX** - Index into Exception Reset Log goes to zero

Error Conditions

There are no error conditions for this command. If the Executive Services receives the command, the event is sent (although it may be filtered by EVS) and the counter is incremented unconditionally.

Criticality

This command is not dangerous. However, any previously logged data will be lost.

See also

[CFE_ES_CLEAR_SYS_LOG_CC](#), [CFE_ES_WRITE_SYS_LOG_CC](#), [CFE_ES_WRITE_ER_LOG_CC](#)

Definition at line 542 of file [cfe_es_fcncodes.h](#).

CFE_ES_CLEAR_SYS_LOG_CC

```
#define CFE_ES_CLEAR_SYS_LOG_CC CFE_ES_CCVAL(CLEAR_SYS_LOG)
```

Name Clear Executive Services System Log

Description

This command clears the contents of the Executive Services System Log.

Command Mnemonic(s) \$sc_\$cpu_ES_ClearSysLog

Command Structure

[CFE_ES_ClearSysLogCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_ES_CMDPC` - command execution counter will increment
- The [CFE_ES_SYSLOG1_INF_EID](#) informational event message will be generated.
- `$sc_$cpu_ES_SYSLOGBYTEUSED` - System Log Bytes Used will go to zero
- `$sc_$cpu_ES_SYSLOGENTRIES` - Number of System Log Entries will go to zero

Error Conditions

There are no error conditions for this command. If the Executive Services receives the command, the event is sent (although it may be filtered by EVS) and the counter is incremented unconditionally.

Criticality

This command is not dangerous. However, any previously logged data will be lost.

See also

[CFE_ES_WRITE_SYS_LOG_CC](#), [CFE_ES_CLEAR_ER_LOG_CC](#), [CFE_ES_WRITE_ER_LOG_CC](#), [CFE_ES_OVER_WRITE_SY](#)

Definition at line 465 of file [cfe_es_fcncodes.h](#).

CFE_ES_DELETE_CDS_CC

```
#define CFE_ES_DELETE_CDS_CC CFE_ES_CCVAL(DELETE_CDS)
```

Name Delete Critical Data Store

Description

This command allows the user to delete a Critical Data Store that was created by an Application that is now no longer executing.

Command Mnemonic(s) `$sc_$cpu_ES_DeleteCDS`

Command Structure

[CFE_ES_DeleteCDSCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_ES_CMDPC` - command execution counter will increment
- The [CFE_ES_CDS_DELETED_INFO_EID](#) informational event message will be generated.
- The specified CDS should no longer appear in a CDS Registry dump generated upon receipt of the [CFE_ES_DUMP_CDS_REGISTRY_CC](#) command

Error Conditions

This command may fail for the following reason(s):

- The specified CDS is the CDS portion of a Critical Table
- The specified CDS is not found in the CDS Registry
- The specified CDS is associated with an Application that is still active
- An error occurred while accessing the CDS memory (see the System Log for more details)

Evidence of failure may be found in the following telemetry:

- `$sc_$cpu_ES_CMDEC` - command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

This command is not critical because it is not possible to delete a CDS that is associated with an active application. However, deleting a CDS does eliminate any "history" that an application may be wishing to keep.

See also

[CFE_ES_DUMP_CDS_REGISTRY_CC](#), [CFE_TBL_DELETE_CDS_CC](#)

Definition at line 911 of file [cfe_es_fcncodes.h](#).

CFE_ES_DUMP_CDS_REGISTRY_CC

```
#define CFE_ES_DUMP_CDS_REGISTRY_CC CFE_ES_CCVAL(DUMP_CDS_REGISTRY)
```

Name Dump Critical Data Store Registry to a File

Description

This command allows the user to dump the Critical Data Store Registry to an onboard file.

Command Mnemonic(s) `$sc_$cpu_ES_WriteCDS2File`

Command Structure

[CFE_ES_DumpCDSRegistryCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_ES_CMDPC` - command execution counter will increment
- The [CFE_ES_CDS_REG_DUMP_INF_EID](#) debug event message will be generated.
- The file specified in the command (or the default specified by the [CFE_PLATFORM_ES_DEFAULT_CDS_REG_DUMP_FILE](#) configuration parameter) will be updated with the latest information.

Error Conditions

This command may fail for the following reason(s):

- The file name specified could not be parsed
- Error occurred while creating or writing to the dump file

Evidence of failure may be found in the following telemetry:

- `$sc_$cpu_ES_CMDEC` - command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

This command is not inherently dangerous. It will create a new file in the file system (or overwrite an existing one) and could, if performed repeatedly without sufficient file management by the operator, fill the file system.

See also

[CFE_ES_DELETE_CDS_CC](#), [CFE_TBL_DELETE_CDS_CC](#)

Definition at line 992 of file [cfe_es_fcncodes.h](#).

CFE_ES_NOOP_CC

```
#define CFE_ES_NOOP_CC CFE_ES_CCVAL(NOOP)
```

Name Executive Services No-Op

Description

This command performs no other function than to increment the command execution counter. The command may be used to verify general aliveness of the Executive Services task.

Command Mnemonic(s) `$sc_$cpu_ES_NOOP`

Command Structure

[CFE_ES_NoopCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_ES_CMDPC` - command execution counter will increment
- The [CFE_ES_BUILD_INF_EID](#) informational event message will be generated
- The [CFE_ES_NOOP_INF_EID](#) informational event message will be generated

Error Conditions

This command may fail for the following reason(s):

- The command packet length is incorrect

Evidence of failure may be found in the following telemetry:

- `$sc_$cpu_ES_CMDEC` - command error counter will increment
- the [CFE_ES_LEN_ERR_EID](#) error event message will be generated

Criticality

None

See also

Definition at line 75 of file [cfe_es_fcncodes.h](#).

CFE_ES_OVER_WRITE_SYS_LOG_CC

```
#define CFE_ES_OVER_WRITE_SYS_LOG_CC CFE_ES_CCVAL(OVER_WRITE_SYS_LOG)
```

Name Set Executive Services System Log Mode to Discard/Overwrite

Description

This command allows the user to configure the Executive Services to either discard new System Log messages when it is full or to overwrite the oldest messages.

Command Mnemonic(s) `$sc_$cpu_ES_OverwriteSysLogMode`

Command Structure

[CFE_ES_OverWriteSysLogCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_ES_CMDPC` - command execution counter will increment
- `$sc_$cpu_ES_SYSLOGMODE` - Current System Log Mode should reflect the commanded value
- The [CFE_ES_SYSLOGMODE_EID](#) debug event message will be generated.

Error Conditions

This command may fail for the following reason(s):

- The desired mode is neither [CFE_ES_LogMode_OVERWRITE](#) or [CFE_ES_LogMode_DISCARD](#)

Evidence of failure may be found in the following telemetry:

- `$sc_$cpu_ES_CMDEC` - command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

None. (It should be noted that "Overwrite" mode would allow a message identifying the cause of a problem to be lost by a subsequent flood of additional messages).

See also

[CFE_ES_CLEAR_SYS_LOG_CC](#), [CFE_ES_WRITE_SYS_LOG_CC](#)

Definition at line 794 of file [cfe_es_fcncodes.h](#).

CFE_ES_QUERY_ALL_CC

```
#define CFE_ES_QUERY_ALL_CC CFE_ES_CCVAL(QUERY_ALL)
```

Name Writes all Executive Services Information on all loaded modules to a File

Description

This command takes the information kept by Executive Services on all of the registered applications and libraries and writes it to the specified file.

Command Mnemonic(s) \$sc_\$cpu_ES_WriteAppInfo2File

Command Structure

CFE_ES_QueryAllCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- \$sc_\$cpu_ES_CMDPC - command execution counter will increment
- The CFE_ES_ALL_APPS_EID debug event message will be generated.
- The file specified in the command (or the default specified by the CFE_PLATFORM_ES_DEFAULT_APP_LOG_FILE configuration parameter) will be updated with the latest information.

Error Conditions

This command may fail for the following reason(s):

- The specified FileName cannot be parsed
- An Error occurs while trying to write to the file

Evidence of failure may be found in the following telemetry:

- \$sc_\$cpu_ES_CMDEC - command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

This command is not inherently dangerous. It will create a new file in the file system (or overwrite an existing one) and could, if performed repeatedly without sufficient file management by the operator, fill the file system.

See also

CFE_ES_QUERY_ONE_CC, CFE_ES_QUERY_ALL_TASKS_CC

Definition at line 430 of file cfe_es_fcncodes.h.

CFE_ES_QUERY_ALL_TASKS_CC

```
#define CFE_ES_QUERY_ALL_TASKS_CC CFE_ES_CCVAL(QUERY_ALL_TASKS)
```

Name Writes a list of All Executive Services Tasks to a File

Description

This command takes the information kept by Executive Services on all of the registered tasks and writes it to the specified file.

Command Mnemonic(s) `$sc_$cpu_ES_WriteTaskInfo2File`

Command Structure

`CFE_ES_QueryAllTasksCmd_t`

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_ES_CMDPC` - command execution counter will increment
- The `CFE_ES_TASKINFO_EID` debug event message will be generated.
- The file specified in the command (or the default specified by the `CFE_PLATFORM_ES_DEFAULT_TASK_LOG_FILE` configuration parameter) will be updated with the latest information.

Error Conditions

This command may fail for the following reason(s):

- The file name specified could not be parsed
- An Error occurs while trying to write to the file

Evidence of failure may be found in the following telemetry:

- `$sc_$cpu_ES_CMDEC` - command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

This command is not inherently dangerous. It will create a new file in the file system (or overwrite an existing one) and could, if performed repeatedly without sufficient file management by the operator, fill the file system.

See also

`CFE_ES_QUERY_ALL_CC`, `CFE_ES_QUERY_ONE_CC`

Definition at line 1034 of file `cfe_es_fcncodes.h`.

`CFE_ES_QUERY_ONE_CC`

```
#define CFE_ES_QUERY_ONE_CC CFE_ES_CCVAL(QUERY_ONE)
```

Name Request Executive Services Information on a specified module

Description

This command takes the information kept by Executive Services on the specified application or library and telemeters it to the ground.

Command Mnemonic(s) `$sc_$cpu_ES_QueryApp`

Command Structure

[CFE_ES_QueryOneCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_ES_CMDPC` - command execution counter will increment
- The [CFE_ES_ONE_APP_EID](#) debug event message will be generated.
- Receipt of the [CFE_ES_OneAppTlm_t](#) telemetry packet

Error Conditions

This command may fail for the following reason(s):

- The specified name is not recognized as an active application or library

Evidence of failure may be found in the following telemetry:

- `$sc_$cpu_ES_CMDEC` - command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

None

See also

[CFE_ES_QUERY_ALL_CC](#), [CFE_ES_QUERY_ALL_TASKS_CC](#)

Definition at line 388 of file [cfe_es_fcncodes.h](#).

CFE_ES_RELOAD_APP_CC

```
#define CFE_ES_RELOAD_APP_CC CFE_ES_CCVAL(RELOAD_APP)
```

Name Stops, Unloads, Loads from the command specified File and Restarts an Application

Description

This command halts and removes the specified Application from the system. Then it immediately loads the Application from the command specified file and restarts it. This command is especially useful for restarting a Command Ingest Application since once it has been stopped, no further commands can come in to restart it.

Command Mnemonic(s) `$sc_$cpu_ES_ReloadApp`

Command Structure

[CFE_ES_ReloadAppCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_ES_CMDPC` - command execution counter will increment
- The [CFE_ES_RELOAD_APP_DBG_EID](#) debug event message will be generated. NOTE: This event message only identifies that the reload process has been initiated, not that it has completed.

Error Conditions

This command may fail for the following reason(s):

- The specified application filename string cannot be parsed
- The specified application name is not recognized as an active application
- The specified application is one of the cFE's Core applications (ES, EVS, SB, TBL, TIME)

Evidence of failure may be found in the following telemetry:

- `$sc_$cpu_ES_CMDEC` - command error counter will increment
- A command specific error event message is issued for all error cases
- Additional information on the reason for command failure may be found in the System Log

Criticality

This command is not inherently dangerous, however the restarting of certain applications (e.g. - Spacecraft Attitude and Control) may have a detrimental effect on the spacecraft.

See also

[CFE_ES_START_APP_CC](#), [CFE_ES_STOP_APP_CC](#), [CFE_ES_RESTART_APP_CC](#)

Definition at line 352 of file [cfe_es_fcncodes.h](#).

CFE_ES_RESET_COUNTERS_CC

```
#define CFE_ES_RESET_COUNTERS_CC CFE\_ES\_CCVAL(RESET_COUNTERS)
```

Name Executive Services Reset Counters

Description

This command resets the following counters within the Executive Services housekeeping telemetry:

- Command Execution Counter
- Command Error Counter

Command Mnemonic(s) `$sc_$cpu_ES_ResetCtrs`

Command Structure

[CFE_ES_ResetCountersCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_ES_CMDPC` - command execution counter and error counter will be reset to zero
- The [CFE_ES_RESET_INF_EID](#) informational event message will be generated

Error Conditions

There are no error conditions for this command. If the Executive Services receives the command, the event is sent (although it may be filtered by EVS) and the counter is incremented unconditionally.

Criticality

This command is not inherently dangerous. However, it is possible for ground systems and on-board safing procedures to be designed such that they react to changes in the counter values that are reset by this command.

See also

[CFE_ES_RESET_PR_COUNT_CC](#)

Definition at line 112 of file [cfe_es_fcncodes.h](#).

CFE_ES_RESET_PR_COUNT_CC

```
#define CFE_ES_RESET_PR_COUNT_CC CFE_ES_CCVAL(RESET_PR_COUNT)
```

Name Resets the Processor Reset Counter to Zero

Description

This command allows the user to reset the Processor Reset Counter to zero. The Processor Reset Counter counts the number of Processor Resets that have occurred so as to identify when a Processor Reset should automatically be upgraded to a full Power-On Reset.

Command Mnemonic(s) `$sc_$cpu_ES_ResetPRCnt`

Command Structure

[CFE_ES_ResetPRCountCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_ES_CMDPC` - command execution counter will increment
- `$sc_$cpu_ES_ProcResetCnt` - Current number of processor resets will go to zero
- The [CFE_ES_RESET_PR_COUNT_EID](#) informational event message will be generated.

Error Conditions

There are no error conditions for this command. If the Executive Services receives the command, the event is sent (although it may be filtered by EVS) and the counter is incremented unconditionally.

Criticality

This command is not critical. The only impact would be that the system would have to have more processor resets before an automatic power-on reset occurred.

See also

[CFE_ES_SET_MAX_PR_COUNT_CC](#), [CFE_ES_RESET_COUNTERS_CC](#)

Definition at line 831 of file [cfe_es_fcncodes.h](#).

CFE_ES_RESTART_APP_CC

```
#define CFE_ES_RESTART_APP_CC CFE_ES_CCVAL (RESTART_APP)
```

Name Stops, Unloads, Loads using the previous File name, and Restarts an Application

Description

This command halts and removes the specified Application from the system. Then it immediately loads the Application from the same filename last used to start. This command is especially useful for restarting a Command Ingest Application since once it has been stopped, no further commands can come in to restart it.

Command Mnemonic(s) \$sc_\$cpu_ES_ResetApp

Command Structure

[CFE_ES_RestartAppCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **\$sc_\$cpu_ES_CMDPC** - command execution counter will increment
- The [CFE_ES_RESTART_APP_DBG_EID](#) debug event message will be generated. NOTE: This event message only identifies that the restart process has been initiated, not that it has completed.

Error Conditions

This command may fail for the following reason(s):

- The original file is missing
- The specified application name is not recognized as an active application
- The specified application is one of the cFE's Core applications (ES, EVS, SB, TBL, TIME)

Evidence of failure may be found in the following telemetry:

- **\$sc_\$cpu_ES_CMDEC** - command error counter will increment
- A command specific error event message is issued for all error cases
- Additional information on the reason for command failure may be found in the System Log

Criticality

This command is not inherently dangerous, however the restarting of certain applications (e.g. - Spacecraft Attitude and Control) may have a detrimental effect on the spacecraft.

See also

[CFE_ES_START_APP_CC](#), [CFE_ES_STOP_APP_CC](#), [CFE_ES_RELOAD_APP_CC](#)

Definition at line 306 of file [cfe_es_fcncodes.h](#).

CFE_ES_RESTART_CC

```
#define CFE_ES_RESTART_CC CFE_ES_CCVAL(RESTART)
```

Name Executive Services Processor / Power-On Reset

Description

This command restarts the cFE in one of two modes. The Power-On Reset will cause the cFE to restart as though the power were first applied to the processor. The Processor Reset will attempt to retain the contents of the volatile disk and the contents of the Critical Data Store. NOTE: If a requested Processor Reset should cause the Processor Reset Counter (`$sc_$cpu_ES_ProcResetCnt`) to exceed OR EQUAL the limit `CFE_PLATFORM_ES_MAX_PROCESSOR_RESETS` (which is reported in housekeeping telemetry as `$sc_↵$cpu_ES_MaxProcResets`), the command is **AUTOMATICALLY** upgraded to a Power-On Reset.

Command Mnemonic(s) `$sc_$cpu_ES_ProcessorReset`, `$sc_$cpu_ES_PowerOnReset`

Command Structure

`CFE_ES_RestartCmd_t`

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_ES_ProcResetCnt` - processor reset counter will increment (processor reset) or reset to zero (power-on reset)
 - `$sc_$cpu_ES_ResetType` - processor reset type will be updated
 - `$sc_$cpu_ES_ResetSubtype` - processor reset subtype will be updated
 - New entries in the Exception Reset Log and System Log can be found
- NOTE: Verification of a Power-On Reset is shown through the loss of data nominally retained through a Processor Reset
- NOTE: Since the reset of the processor resets the command execution counter (`$sc_$cpu_ES_CMDPC`), this counter **CANNOT** be used to verify command execution.

Error Conditions

This command may fail for the following reason(s):

- The `Restart Type` was not a recognized value.

Evidence of failure may be found in the following telemetry:

- `$sc_$cpu_ES_CMDEC` - command error counter will increment
- the `CFE_ES_BOOT_ERR_EID` error event message will be generated

Criticality

This command is, by definition, dangerous. Significant loss of data will occur. All processes and the cFE itself will be stopped and restarted. With the Power-On reset option, all data on the volatile disk and the contents of the Critical Data Store will be lost.

See also

`CFE_ES_RESET_PR_COUNT_CC`, `CFE_ES_SET_MAX_PR_COUNT_CC`

Definition at line 164 of file `cfe_es_fcncodes.h`.

CFE_ES_SEND_MEM_POOL_STATS_CC

```
#define CFE_ES_SEND_MEM_POOL_STATS_CC CFE_ES_CCVAL(SEND_MEM_POOL_STATS)
```

Name Telemeter Memory Pool Statistics

Description

This command allows the user to obtain a snapshot of the statistics maintained for a specified memory pool.

Command Mnemonic(s) \$sc_\$cpu_ES_PoolStats

Command Structure

[CFE_ES_SendMemPoolStatsCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- [\\$sc_\\$cpu_ES_CMDPC](#) - command execution counter will increment
- The [CFE_ES_TLM_POOL_STATS_INFO_EID](#) debug event message will be generated.
- The [Memory Pool Statistics Telemetry Packet](#) is produced

Error Conditions

This command may fail for the following reason(s):

- The specified handle is not associated with a known memory pool

Evidence of failure may be found in the following telemetry:

- [\\$sc_\\$cpu_ES_CMDEC](#) - command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

An incorrect Memory Pool Handle value can cause a system crash. Extreme care should be taken to ensure the memory handle value used in the command is correct.

See also

Definition at line [950](#) of file [cfe_es_fcncodes.h](#).

CFE_ES_SET_MAX_PR_COUNT_CC

```
#define CFE_ES_SET_MAX_PR_COUNT_CC CFE_ES_CCVAL(SET_MAX_PR_COUNT)
```

Name Configure the Maximum Number of Processor Resets before a Power-On Reset

Description

This command allows the user to specify the number of Processor Resets that are allowed before the next Processor Reset is upgraded to a Power-On Reset.

Command Mnemonic(s) \$sc_\$cpu_ES_SetMaxPRCnt

Command Structure

[CFE_ES_SetMaxPRCountCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_ES_CMDPC` - command execution counter will increment
- `$sc_$cpu_ES_MaxProcResets` - Current maximum number of processor resets before an automatic power-on reset will go to the command specified value.
- The [CFE_ES_SET_MAX_PR_COUNT_EID](#) informational event message will be generated.

Error Conditions

There are no error conditions for this command. If the Executive Services receives the command, the event is sent (although it may be filtered by EVS) and the counter is incremented unconditionally.

Criticality

If the operator were to set the Maximum Processor Reset Count to too high a value, the processor would require an inordinate number of consecutive processor resets before an automatic power-on reset would occur. This could potentially leave the spacecraft without any control for a significant amount of time if a processor reset fails to clear a problem.

See also

[CFE_ES_RESET_PR_COUNT_CC](#)

Definition at line 869 of file [cfe_es_fcncodes.h](#).

CFE_ES_SET_PERF_FILTER_MASK_CC

```
#define CFE_ES_SET_PERF_FILTER_MASK_CC CFE_ES_CCVAL(SET_PERF_FILTER_MASK)
```

Name Set Performance Analyzer's Filter Masks

Description

This command sets the Performance Analyzer's Filter Masks.

Command Mnemonic(s) `$sc_$cpu_ES_LAFilterMask`

Command Structure

[CFE_ES_SetPerfFilterMaskCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_ES_CMDPC` - command execution counter will increment
- `$sc_$cpu_ES_PerfFltrMask[MaskCnt]` - the current performance filter mask value(s) should reflect the commanded value
- The [CFE_ES_PERF_FILTMSKCMD_EID](#) debug event message will be generated.

Error Conditions

This command may fail for the following reason(s):

- The Filter Mask ID number is out of range

Evidence of failure may be found in the following telemetry:

- **\$sc_\$cpu_ES_CMDEC** - command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

Changing the filter masks may cause a small change in the Performance Analyzer's CPU utilization.

See also

[CFE_ES_START_PERF_DATA_CC](#), [CFE_ES_STOP_PERF_DATA_CC](#), [CFE_ES_SET_PERF_TRIGGER_MASK_CC](#)

Definition at line 717 of file [cfe_es_fcncodes.h](#).

CFE_ES_SET_PERF_TRIGGER_MASK_CC

```
#define CFE_ES_SET_PERF_TRIGGER_MASK_CC CFE_ES_CCVAL (SET_PERF_TRIGGER_MASK)
```

Name Set Performance Analyzer's Trigger Masks

Description

This command sets the Performance Analyzer's Trigger Masks.

Command Mnemonic(s) `$sc_$cpu_ES_LATriggerMask`

Command Structure

[CFE_ES_SetPerfTriggerMaskCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **\$sc_\$cpu_ES_CMDPC** - command execution counter will increment
- **\$sc_\$cpu_ES_PerfTrigMask[MaskCnt]** - the current performance trigger mask value(s) should reflect the commanded value
- The [CFE_ES_PERF_TRIGMSKCMD_EID](#) debug event message will be generated.

Error Conditions

This command may fail for the following reason(s):

- The Trigger Mask ID number is out of range

Evidence of failure may be found in the following telemetry:

- **\$sc_\$cpu_ES_CMDEC** - command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

Changing the trigger masks may cause a small change in the Performance Analyzer's CPU utilization.

See also

[CFE_ES_START_PERF_DATA_CC](#), [CFE_ES_STOP_PERF_DATA_CC](#), [CFE_ES_SET_PERF_FILTER_MASK_CC](#)

Definition at line 754 of file [cfe_es_fcncodes.h](#).

CFE_ES_START_APP_CC

```
#define CFE_ES_START_APP_CC CFE_ES_CCVAL(START_APP)
```

Name Load and Start an Application

Description

This command starts the specified application with the specified start address, stack size, etc options.

Command Mnemonic(s) \$sc_\$cpu_ES_StartApp

Command Structure

CFE_ES_StartAppCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- \$sc_\$cpu_ES_CMDPC - command execution counter will increment
- The CFE_ES_START_INF_EID informational event message will be generated

Error Conditions

This command may fail for the following reason(s):

- The specified application filename string cannot be parsed
- The specified application entry point is an empty string
- The specified application name is an empty string
- The specified priority is greater than 255
- The specified exception action is neither CFE_ES_ExceptionAction_RESTART_APP (0) or CFE_ES_ExceptionAction_PROC_I (1)
- The Operating System was unable to load the specified application file

Evidence of failure may be found in the following telemetry:

- \$sc_\$cpu_ES_CMDEC - command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

This command is not inherently dangerous although system resources could be taxed beyond their limits with the starting of erroneous or invalid applications.

See also

CFE_ES_STOP_APP_CC, CFE_ES_RESTART_APP_CC, CFE_ES_RELOAD_APP_CC

Definition at line 207 of file cfe_es_fcncodes.h.

CFE_ES_START_PERF_DATA_CC

```
#define CFE_ES_START_PERF_DATA_CC CFE_ES_CCVAL(START_PERF_DATA)
```

Name Start Performance Analyzer

Description

This command causes the Performance Analyzer to begin collecting data using the specified trigger mode.

Command Mnemonic(s) \$sc_\$cpu_ES_StartLAData

Command Structure

[CFE_ES_StartPerfDataCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **\$sc_\$cpu_ES_CMDPC** - command execution counter will increment
- **\$sc_\$cpu_ES_PerfState** - Current performance analyzer state will change to either WAITING FOR TRIGGER or, if conditions are appropriate fast enough, TRIGGERED.
- **\$sc_\$cpu_ES_PerfMode** - Performance Analyzer Mode will change to the commanded trigger mode (TRIGGER START, TRIGGER CENTER, or TRIGGER END).
- **\$sc_\$cpu_ES_PerfTrigCnt** - Performance Trigger Count will go to zero
- **\$sc_\$cpu_ES_PerfDataStart** - Data Start Index will go to zero
- **\$sc_\$cpu_ES_PerfDataEnd** - Data End Index will go to zero
- **\$sc_\$cpu_ES_PerfDataCnt** - Performance Data Counter will go to zero
- The [CFE_ES_PERF_STARTCMD_EID](#) debug event message will be generated.

Error Conditions

This command may fail for the following reason(s):

- A previous [CFE_ES_STOP_PERF_DATA_CC](#) command has not completely finished.
- An invalid trigger mode is requested.

Evidence of failure may be found in the following telemetry:

- **\$sc_\$cpu_ES_CMDEC** - command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

This command is not inherently dangerous but may cause a small increase in CPU utilization as the performance analyzer data is collected.

See also

[CFE_ES_STOP_PERF_DATA_CC](#), [CFE_ES_SET_PERF_FILTER_MASK_CC](#), [CFE_ES_SET_PERF_TRIGGER_MASK_CC](#)

Definition at line 630 of file [cfe_es_fcncodes.h](#).

CFE_ES_STOP_APP_CC

```
#define CFE_ES_STOP_APP_CC CFE_ES_CCVAL(STOP_APP)
```

Name Stop and Unload Application

Description

This command halts and removes the specified Application from the system. **NOTE:** This command should never be used on the Command Ingest application. This would prevent further commands from entering the system. If Command Ingest needs to be stopped and restarted, use [CFE_ES_RESTART_APP_CC](#) or [CFE_ES_RELOAD_APP_CC](#).

Command Mnemonic(s) \$sc_\$cpu_ES_StopApp

Command Structure

[CFE_ES_StopAppCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **\$sc_\$cpu_ES_CMDPC** - command execution counter will increment
- The [CFE_ES_STOP_DBG_EID](#) debug event message will be generated. NOTE: This event message only identifies that the stop request has been initiated, not that it has completed.
- Once the stop has successfully completed, the list of Applications and Tasks created in response to the **\$sc_\$cpu_ES_WriteAppInfo2File**, **\$sc_\$cpu_ES_WriteTaskInfo2File** **should** no longer contain the specified application.
- **\$sc_\$cpu_ES_RegTasks** - number of tasks will decrease after tasks associated with app (main task and any child tasks) are stopped
- **\$sc_\$cpu_ES_RegExtApps** - external application counter will decrement after app is cleaned up

Error Conditions

This command may fail for the following reason(s):

- The specified application name is not recognized as an active application
- The specified application is one of the cFE's Core applications (ES, EVS, SB, TBL, TIME)

Evidence of failure may be found in the following telemetry:

- **\$sc_\$cpu_ES_CMDEC** - command error counter will increment
- A command specific error event message is issued for all error cases
- Additional information on the reason for command failure may be found in the System Log

Criticality

This command is not inherently dangerous, however the removal of certain applications (e.g. - Spacecraft Attitude and Control) may have a detrimental effect on the spacecraft.

See also

[CFE_ES_START_APP_CC](#), [CFE_ES_RESTART_APP_CC](#), [CFE_ES_RELOAD_APP_CC](#)

Definition at line 260 of file [cfe_es_fcncodes.h](#).

CFE_ES_STOP_PERF_DATA_CC

```
#define CFE_ES_STOP_PERF_DATA_CC CFE_ES_CCVAL(STOP_PERF_DATA)
```

Name Stop Performance Analyzer and write data file

Description

This command stops the Performance Analyzer from collecting any more data, and writes all previously collected performance data to a log file.

Command Mnemonic(s) `$sc_$cpu_ES_StopLAData`

Command Structure

`CFE_ES_StopPerfDataCmd_t`

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_ES_CMDPC` - command execution counter will increment
- `$sc_$cpu_ES_PerfState` - Current performance analyzer state will change to IDLE.
- The `CFE_ES_PERF_STOPCMD_EID` debug event message will be generated to indicate that data collection has been stopped. NOTE: Performance log data is written to the file as a background job. This event indicates that the file write process is initiated, not that it has completed.
- The file specified in the command (or the default specified by the `CFE_PLATFORM_ES_DEFAULT_PERF_DUMP_FILENAME` configuration parameter) will be updated with the latest information.

Error Conditions

This command may fail for the following reason(s):

- The file name specified could not be parsed
- Log data from a previous Stop Performance Analyzer command is still being written to a file.

Evidence of failure may be found in the following telemetry:

- `$sc_$cpu_ES_CMDEC` - command error counter will increment
- A command specific error event message is issued for all error cases

NOTE: The performance analyzer data collection will still be stopped in the event of an error parsing the log file name or writing the log file.

Criticality

This command is not inherently dangerous. However, depending on configuration, performance data log files may be large in size and thus may fill the available storage.

See also

[CFE_ES_START_PERF_DATA_CC](#), [CFE_ES_SET_PERF_FILTER_MASK_CC](#), [CFE_ES_SET_PERF_TRIGGER_MASK_CC](#)

Definition at line 680 of file [cfe_es_fcncodes.h](#).

CFE_ES_WRITE_ER_LOG_CC

```
#define CFE_ES_WRITE_ER_LOG_CC CFE_ES_CCVAL(WRITE_ER_LOG)
```

Name Writes Exception and Reset Log to a File

Description

This command causes the contents of the Executive Services Exception and Reset Log to be written to the specified file.

Command Mnemonic(s) \$sc_\$cpu_ES_WriteERLog2File

Command Structure

[CFE_ES_WriteERLogCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **\$sc_\$cpu_ES_CMDPC** - command execution counter will increment
- The [CFE_ES_ERLOG2_EID](#) debug event message will be generated.
- The file specified in the command (or the default specified by the [CFE_PLATFORM_ES_DEFAULT_ER_LOG_FILE](#) configuration parameter) will be updated with the latest information.

Error Conditions

This command may fail for the following reason(s):

- A previous request to write the ER log has not yet completed
- The specified FileName cannot be parsed
- An Error occurs while trying to write to the file

Evidence of failure may be found in the following telemetry:

- **\$sc_\$cpu_ES_CMDEC** - command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

This command is not inherently dangerous. It will create a new file in the file system (or overwrite an existing one) and could, if performed repeatedly without sufficient file management by the operator, fill the file system.

See also

[CFE_ES_CLEAR_SYS_LOG_CC](#), [CFE_ES_WRITE_SYS_LOG_CC](#), [CFE_ES_CLEAR_ER_LOG_CC](#)

Definition at line 585 of file [cfe_es_fcncodes.h](#).

CFE_ES_WRITE_SYS_LOG_CC

```
#define CFE_ES_WRITE_SYS_LOG_CC CFE_ES_CCVAL(WRITE_SYS_LOG)
```

Name Writes contents of Executive Services System Log to a File

Description

This command causes the contents of the Executive Services System Log to be written to a log file.

Command Mnemonic(s) \$sc_\$cpu_ES_WriteSysLog2File

Command Structure

[CFE_ES_WriteSysLogCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **\$sc_\$cpu_ES_CMDPC** - command execution counter will increment
- The [CFE_ES_SYSLOG2_EID](#) debug event message will be generated.
- The file specified in the command (or the default specified by the [CFE_PLATFORM_ES_DEFAULT_SYSLOG_FILE](#) configuration parameter) will be updated with the latest information.

Error Conditions

This command may fail for the following reason(s):

- The specified FileName cannot be parsed
- An Error occurs while trying to write to the file

Evidence of failure may be found in the following telemetry:

- **\$sc_\$cpu_ES_CMDEC** - command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

This command is not inherently dangerous. It will create a new file in the file system (or overwrite an existing one) and could, if performed repeatedly without sufficient file management by the operator, fill the file system.

See also

[CFE_ES_CLEAR_SYS_LOG_CC](#), [CFE_ES_CLEAR_ER_LOG_CC](#), [CFE_ES_WRITE_ER_LOG_CC](#), [CFE_ES_OVER_WRITE_SYS_LOG_CC](#)

Definition at line 508 of file [cfe_es_fcncodes.h](#).

11.126 cfe_es_fcncodes.h

[Go to the documentation of this file.](#)

```

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00016  * limitations under the License.
00017  *****/
00018
00028 #ifndef CFE_ES_FCNCODES_H
00029 #define CFE_ES_FCNCODES_H
00030
00031 #include "cfe_es_fcncode_values.h"
00032
00033 /*****
00034  * Macro Definitions
00035  *****/
00036
00075 #define CFE_ES_NOOP_CC CFE_ES_CCVAL(NOOP)
00076
00112 #define CFE_ES_RESET_COUNTERS_CC CFE_ES_CCVAL(RESET_COUNTERS)
00113
00164 #define CFE_ES_RESTART_CC CFE_ES_CCVAL(RESTART)
00165
00207 #define CFE_ES_START_APP_CC CFE_ES_CCVAL(START_APP)
00208
00260 #define CFE_ES_STOP_APP_CC CFE_ES_CCVAL(STOP_APP)
00261
00306 #define CFE_ES_RESTART_APP_CC CFE_ES_CCVAL(RESTART_APP)
00307
00352 #define CFE_ES_RELOAD_APP_CC CFE_ES_CCVAL(RELOAD_APP)
00353
00388 #define CFE_ES_QUERY_ONE_CC CFE_ES_CCVAL(QUERY_ONE)
00389
00430 #define CFE_ES_QUERY_ALL_CC CFE_ES_CCVAL(QUERY_ALL)
00431
00465 #define CFE_ES_CLEAR_SYS_LOG_CC CFE_ES_CCVAL(CLEAR_SYS_LOG)
00466
00508 #define CFE_ES_WRITE_SYS_LOG_CC CFE_ES_CCVAL(WRITE_SYS_LOG)
00509
00542 #define CFE_ES_CLEAR_ER_LOG_CC CFE_ES_CCVAL(CLEAR_ER_LOG)
00543
00585 #define CFE_ES_WRITE_ER_LOG_CC CFE_ES_CCVAL(WRITE_ER_LOG)
00586
00630 #define CFE_ES_START_PERF_DATA_CC CFE_ES_CCVAL(START_PERF_DATA)
00631
00680 #define CFE_ES_STOP_PERF_DATA_CC CFE_ES_CCVAL(STOP_PERF_DATA)
00681
00717 #define CFE_ES_SET_PERF_FILTER_MASK_CC CFE_ES_CCVAL(SET_PERF_FILTER_MASK)
00718
00754 #define CFE_ES_SET_PERF_TRIGGER_MASK_CC CFE_ES_CCVAL(SET_PERF_TRIGGER_MASK)
00755
00794 #define CFE_ES_OVER_WRITE_SYS_LOG_CC CFE_ES_CCVAL(OVER_WRITE_SYS_LOG)
00795
00831 #define CFE_ES_RESET_PR_COUNT_CC CFE_ES_CCVAL(RESET_PR_COUNT)
00832
00869 #define CFE_ES_SET_MAX_PR_COUNT_CC CFE_ES_CCVAL(SET_MAX_PR_COUNT)
00870
00911 #define CFE_ES_DELETE_CDS_CC CFE_ES_CCVAL(DELETE_CDS)
00912
00950 #define CFE_ES_SEND_MEM_POOL_STATS_CC CFE_ES_CCVAL(SEND_MEM_POOL_STATS)
00951
00992 #define CFE_ES_DUMP_CDS_REGISTRY_CC CFE_ES_CCVAL(DUMP_CDS_REGISTRY)
00993
01034 #define CFE_ES_QUERY_ALL_TASKS_CC CFE_ES_CCVAL(QUERY_ALL_TASKS)
01035
01038 #endif

```

11.127 cfe/modules/es/fsw/inc/cfe_es_interface_cfg.h File Reference

```
#include "cfe_es_interface_cfg_values.h"
```

Macros

- #define CFE_MISSION_ES_MAX_APPLICATIONS CFE_MISSION_ES_CFGVAL(MAX_APPLICATIONS)
- #define DEFAULT_CFE_MISSION_ES_MAX_APPLICATIONS 16
- #define CFE_MISSION_ES_PERF_MAX_IDS CFE_MISSION_ES_CFGVAL(PERF_MAX_IDS)
- #define DEFAULT_CFE_MISSION_ES_PERF_MAX_IDS 128
- #define CFE_MISSION_ES_POOL_MAX_BUCKETS CFE_MISSION_ES_CFGVAL(POOL_MAX_BUCKETS)
- #define DEFAULT_CFE_MISSION_ES_POOL_MAX_BUCKETS 17
- #define CFE_MISSION_ES_CDS_MAX_NAME_LENGTH CFE_MISSION_ES_CFGVAL(CDS_MAX_NAME_LENGTH)
- #define DEFAULT_CFE_MISSION_ES_CDS_MAX_NAME_LENGTH 16
- #define CFE_MISSION_ES_DEFAULT_CRC CFE_MISSION_ES_CFGVAL(DEFAULT_CRC)
- #define DEFAULT_CFE_MISSION_ES_DEFAULT_CRC CFE_ES_CrcType_16_ARC
- #define CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN CFE_MISSION_ES_CFGVAL(CDS_MAX_FULL_NAME_LEN)
- #define DEFAULT_CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN 40

Checksum/CRC algorithm identifiers

- #define CFE_MISSION_ES_CRC_8 CFE_MISSION_ES_CFGVAL(CRC_8)
- #define DEFAULT_CFE_MISSION_ES_CRC_8 CFE_ES_CrcType_CRC_8
- #define CFE_MISSION_ES_CRC_16 CFE_MISSION_ES_CFGVAL(CRC_16)
- #define DEFAULT_CFE_MISSION_ES_CRC_16 CFE_ES_CrcType_CRC_16
- #define CFE_MISSION_ES_CRC_32 CFE_MISSION_ES_CFGVAL(CRC_32)
- #define DEFAULT_CFE_MISSION_ES_CRC_32 CFE_ES_CrcType_CRC_32

11.127.1 Detailed Description

CFE Executive Services (CFE_ES) Application Mission Configuration Header File

Note

This file may be overridden/superseded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [cfe_es_interface_cfg.h](#).

11.127.2 Macro Definition Documentation

CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN

```
#define CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN CFE_MISSION_ES_CFGVAL(CDS_MAX_FULL_NAME_LEN)
```

Purpose Maximum Length of Full CDS Name in messages

Description:

Indicates the maximum length (in characters) of the entire CDS name of the following form: "ApplicationName.CDSName"

This affects the layout of command/telemetry messages but does not affect run time behavior or internal allocation.

Limits

All CPUs within the same SB domain (mission) must share the same definition. Note this affects the size of messages, so it must not cause any message to exceed the max length.

This value should be kept as a multiple of 4, to maintain alignment of any possible neighboring fields without implicit padding.

Definition at line 146 of file [cfe_es_interface_cfg.h](#).

CFE_MISSION_ES_CDS_MAX_NAME_LENGTH

```
#define CFE_MISSION_ES_CDS_MAX_NAME_LENGTH CFE_MISSION_ES_CFGVAL(CDS_MAX_NAME_LENGTH)
```

Purpose Maximum Length of CDS Name

Description:

Indicates the maximum length (in characters) of the CDS name ('CDSName') portion of a Full CDS Name of the following form: "ApplicationName.CDSName"

This length does not need to include an extra character for NULL termination.

Limits

This value should be kept as a multiple of 4, to maintain alignment of any possible neighboring fields without implicit padding.

Definition at line 110 of file [cfe_es_interface_cfg.h](#).

CFE_MISSION_ES_CRC_16

```
#define CFE_MISSION_ES_CRC_16 CFE_MISSION_ES_CFGVAL(CRC_16)
```

Definition at line 158 of file [cfe_es_interface_cfg.h](#).

CFE_MISSION_ES_CRC_32

```
#define CFE_MISSION_ES_CRC_32 CFE_MISSION_ES_CFGVAL(CRC_32)
```

Definition at line 160 of file [cfe_es_interface_cfg.h](#).

CFE_MISSION_ES_CRC_8

```
#define CFE_MISSION_ES_CRC_8 CFE_MISSION_ES_CFGVAL(CRC_8)
```

Definition at line 156 of file [cfe_es_interface_cfg.h](#).

CFE_MISSION_ES_DEFAULT_CRC

```
#define CFE_MISSION_ES_DEFAULT_CRC CFE_MISSION_ES_CFGVAL(DEFAULT_CRC)
```

Purpose Mission Default CRC algorithm

Description:

Indicates the which CRC algorithm should be used as the default for verifying the contents of Critical Data Stores and when calculating Table Image data integrity values.

Limits

Currently only CFE_ES_CrcType_16_ARC is supported (see brief in CFE_ES_CrcType_Enum definition in [cfe_es_api_typedefs.h](#))

Definition at line 125 of file [cfe_es_interface_cfg.h](#).

CFE_MISSION_ES_MAX_APPLICATIONS

```
#define CFE_MISSION_ES_MAX_APPLICATIONS CFE_MISSION_ES_CFGVAL(MAX_APPLICATIONS)
```

Purpose Mission Max Apps in a message

Description:

Indicates the maximum number of apps in a telemetry housekeeping message

This affects the layout of command/telemetry messages but does not affect run time behavior or internal allocation.

Limits

All CPUs within the same SB domain (mission) must share the same definition Note this affects the size of messages, so it must not cause any message to exceed the max length.

Definition at line 47 of file [cfe_es_interface_cfg.h](#).

CFE_MISSION_ES_PERF_MAX_IDS

```
#define CFE_MISSION_ES_PERF_MAX_IDS CFE_MISSION_ES_CFGVAL(PERF_MAX_IDS)
```

Purpose Define Max Number of Performance IDs for messages

Description:

Defines the maximum number of perf ids allowed.

Each performance id is used to identify something that needs to be measured. Performance ids are limited to the range of 0 to [CFE_MISSION_ES_PERF_MAX_IDS](#) - 1. Any performance ids outside of this range will be ignored and will be flagged as an error.

This affects the layout of telemetry messages but does not affect run time behavior or internal allocation.

Limits

All CPUs within the same SB domain (mission) must share the same definition Note this affects the size of messages, so it must not cause any message to exceed the max length.

Definition at line 70 of file [cfe_es_interface_cfg.h](#).

CFE_MISSION_ES_POOL_MAX_BUCKETS

```
#define CFE_MISSION_ES_POOL_MAX_BUCKETS CFE_MISSION_ES_CFGVAL(POOL_MAX_BUCKETS)
```

Purpose Maximum number of block sizes in pool structures

Description:

The upper limit for the number of block sizes supported in the generic pool implementation, which in turn implements the memory pools and CDS. This definition is used as the array size with the pool stats structure, and therefore should be consistent across all CPUs in a mission, as well as with the ground station.

There is also a platform-specific limit which may be fewer than this value.

Limits:

Must be at least one. No specific upper limit, but the number is anticipated to be reasonably small (i.e. tens, not hundreds). Large values have not been tested.

Definition at line 92 of file [cfe_es_interface_cfg.h](#).

DEFAULT_CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN

```
#define DEFAULT_CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN 40
```

Definition at line 149 of file [cfe_es_interface_cfg.h](#).

DEFAULT_CFE_MISSION_ES_CDS_MAX_NAME_LENGTH

```
#define DEFAULT_CFE_MISSION_ES_CDS_MAX_NAME_LENGTH 16
```

Definition at line 111 of file [cfe_es_interface_cfg.h](#).

DEFAULT_CFE_MISSION_ES_CRC_16

```
#define DEFAULT_CFE_MISSION_ES_CRC_16 CFE_ES_CrcType_CRC_16
```

Definition at line 159 of file [cfe_es_interface_cfg.h](#).

DEFAULT_CFE_MISSION_ES_CRC_32

```
#define DEFAULT_CFE_MISSION_ES_CRC_32 CFE_ES_CrcType_CRC_32
```

Definition at line 161 of file [cfe_es_interface_cfg.h](#).

DEFAULT_CFE_MISSION_ES_CRC_8

```
#define DEFAULT_CFE_MISSION_ES_CRC_8 CFE_ES_CrcType_CRC_8
```

Definition at line 157 of file [cfe_es_interface_cfg.h](#).

DEFAULT_CFE_MISSION_ES_DEFAULT_CRC

```
#define DEFAULT_CFE_MISSION_ES_DEFAULT_CRC CFE_ES_CrcType_16_ARC
```

Definition at line 126 of file [cfe_es_interface_cfg.h](#).

DEFAULT_CFE_MISSION_ES_MAX_APPLICATIONS

```
#define DEFAULT_CFE_MISSION_ES_MAX_APPLICATIONS 16
```

Definition at line 48 of file [cfe_es_interface_cfg.h](#).

DEFAULT_CFE_MISSION_ES_PERF_MAX_IDS

```
#define DEFAULT_CFE_MISSION_ES_PERF_MAX_IDS 128
```

Definition at line 71 of file [cfe_es_interface_cfg.h](#).

DEFAULT_CFE_MISSION_ES_POOL_MAX_BUCKETS

```
#define DEFAULT_CFE_MISSION_ES_POOL_MAX_BUCKETS 17
```

Definition at line 93 of file [cfe_es_interface_cfg.h](#).

11.128 cfe_es_interface_cfg.h

[Go to the documentation of this file.](#)

```
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00017  *****/
00018
00028 #ifndef CFE_ES_INTERFACE_CFG_H
00029 #define CFE_ES_INTERFACE_CFG_H
00030
00031 #include "cfe_es_interface_cfg_values.h"
00032
00047 #define CFE_MISSION_ES_MAX_APPLICATIONS      CFE_MISSION_ES_CFGVAL(MAX_APPLICATIONS)
00048 #define DEFAULT_CFE_MISSION_ES_MAX_APPLICATIONS 16
00049
00070 #define CFE_MISSION_ES_PERF_MAX_IDS          CFE_MISSION_ES_CFGVAL(PERF_MAX_IDS)
00071 #define DEFAULT_CFE_MISSION_ES_PERF_MAX_IDS 128
00072
00092 #define CFE_MISSION_ES_POOL_MAX_BUCKETS      CFE_MISSION_ES_CFGVAL(POOL_MAX_BUCKETS)
00093 #define DEFAULT_CFE_MISSION_ES_POOL_MAX_BUCKETS 17
00094
00110 #define CFE_MISSION_ES_CDS_MAX_NAME_LENGTH   CFE_MISSION_ES_CFGVAL(CDS_MAX_NAME_LENGTH)
00111 #define DEFAULT_CFE_MISSION_ES_CDS_MAX_NAME_LENGTH 16
00112
00125 #define CFE_MISSION_ES_DEFAULT_CRC           CFE_MISSION_ES_CFGVAL(DEFAULT_CRC)
00126 #define DEFAULT_CFE_MISSION_ES_DEFAULT_CRC CFE_ES_CrcType_16_ARC
00127
00146 #define CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN CFE_MISSION_ES_CFGVAL(CDS_MAX_FULL_NAME_LEN)
00147
00148 /* Default value is calculated as: (CFE_MISSION_ES_CDS_MAX_NAME_LENGTH + CFE_MISSION_MAX_API_LEN + 4) */
00149 #define DEFAULT_CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN 40
00150
00151 #ifndef CFE_OMIT_DEPRECATED_6_8
00152 /* These names have been converted to an enum in cfe_es_api_typedefs.h */
00153
00156 #define CFE_MISSION_ES_CRC_8      CFE_MISSION_ES_CFGVAL(CRC_8)
00157 #define DEFAULT_CFE_MISSION_ES_CRC_8 CFE_ES_CrcType_CRC_8
00158 #define CFE_MISSION_ES_CRC_16     CFE_MISSION_ES_CFGVAL(CRC_16)
00159 #define DEFAULT_CFE_MISSION_ES_CRC_16 CFE_ES_CrcType_CRC_16
00160 #define CFE_MISSION_ES_CRC_32     CFE_MISSION_ES_CFGVAL(CRC_32)
00161 #define DEFAULT_CFE_MISSION_ES_CRC_32 CFE_ES_CrcType_CRC_32
00162
00163 #endif
00164
00165 #endif

```

11.129 cfe/modules/es/fsw/inc/cfe_es_internal_cfg.h File Reference

```

#include "cfe_es_mission_cfg.h"
#include "cfe_es_internal_cfg_values.h"

```

Macros

- #define CFE_PLATFORM_ES_START_TASK_PRIORITY CFE_PLATFORM_ES_CFGVAL(START_TASK_PRIORITY)
- #define DEFAULT_CFE_PLATFORM_ES_START_TASK_PRIORITY 68
- #define CFE_PLATFORM_ES_START_TASK_STACK_SIZE CFE_PLATFORM_ES_CFGVAL(START_TASK_STACK_SIZE)
- #define DEFAULT_CFE_PLATFORM_ES_START_TASK_STACK_SIZE 8192
- #define CFE_PLATFORM_ES_NONVOL_DISK_MOUNT_STRING CFE_PLATFORM_ES_CFGVAL(NONVOL_DISK_MOUNT_STRING)
- #define DEFAULT_CFE_PLATFORM_ES_NONVOL_DISK_MOUNT_STRING "/cf"
- #define CFE_PLATFORM_ES_RAM_DISK_MOUNT_STRING CFE_PLATFORM_ES_CFGVAL(RAM_DISK_MOUNT_STRING)
- #define DEFAULT_CFE_PLATFORM_ES_RAM_DISK_MOUNT_STRING "/ram"
- #define CFE_PLATFORM_ES_MAX_APPLICATIONS CFE_PLATFORM_ES_CFGVAL(MAX_APPLICATIONS)
- #define DEFAULT_CFE_PLATFORM_ES_MAX_APPLICATIONS 32
- #define CFE_PLATFORM_ES_MAX_LIBRARIES CFE_PLATFORM_ES_CFGVAL(MAX_LIBRARIES)

- #define `DEFAULT_CFE_PLATFORM_ES_MAX_LIBRARIES` 10
- #define `CFE_PLATFORM_ES_ER_LOG_ENTRIES` `CFE_PLATFORM_ES_CFGVAL(ER_LOG_ENTRIES)`
- #define `DEFAULT_CFE_PLATFORM_ES_ER_LOG_ENTRIES` 20
- #define `CFE_PLATFORM_ES_ER_LOG_MAX_CONTEXT_SIZE` `CFE_PLATFORM_ES_CFGVAL(ER_LOG_↵
MAX_CONTEXT_SIZE)`
- #define `DEFAULT_CFE_PLATFORM_ES_ER_LOG_MAX_CONTEXT_SIZE` 256
- #define `CFE_PLATFORM_ES_SYSTEM_LOG_SIZE` `CFE_PLATFORM_ES_CFGVAL(SYSTEM_LOG_SIZE)`
- #define `DEFAULT_CFE_PLATFORM_ES_SYSTEM_LOG_SIZE` 3072
- #define `CFE_PLATFORM_ES_MAX_GEN_COUNTERS` `CFE_PLATFORM_ES_CFGVAL(MAX_GEN_↵
COUNTERS)`
- #define `DEFAULT_CFE_PLATFORM_ES_MAX_GEN_COUNTERS` 8
- #define `CFE_PLATFORM_ES_APP_SCAN_RATE` `CFE_PLATFORM_ES_CFGVAL(APP_SCAN_RATE)`
- #define `DEFAULT_CFE_PLATFORM_ES_APP_SCAN_RATE` 1000
- #define `CFE_PLATFORM_ES_APP_KILL_TIMEOUT` `CFE_PLATFORM_ES_CFGVAL(APP_KILL_TIMEOUT)`
- #define `DEFAULT_CFE_PLATFORM_ES_APP_KILL_TIMEOUT` 5
- #define `CFE_PLATFORM_ES_RAM_DISK_SECTOR_SIZE` `CFE_PLATFORM_ES_CFGVAL(RAM_DISK_↵
SECTOR_SIZE)`
- #define `DEFAULT_CFE_PLATFORM_ES_RAM_DISK_SECTOR_SIZE` 512
- #define `CFE_PLATFORM_ES_RAM_DISK_NUM_SECTORS` `CFE_PLATFORM_ES_CFGVAL(RAM_DISK_↵
NUM_SECTORS)`
- #define `DEFAULT_CFE_PLATFORM_ES_RAM_DISK_NUM_SECTORS` 4096
- #define `CFE_PLATFORM_ES_RAM_DISK_PERCENT_RESERVED` `CFE_PLATFORM_ES_CFGVAL(RAM_↵
DISK_PERCENT_RESERVED)`
- #define `DEFAULT_CFE_PLATFORM_ES_RAM_DISK_PERCENT_RESERVED` 30
- #define `CFE_PLATFORM_ES_CDS_SIZE` `CFE_PLATFORM_ES_CFGVAL(CDS_SIZE)`
- #define `DEFAULT_CFE_PLATFORM_ES_CDS_SIZE` (128 * 1024)
- #define `CFE_PLATFORM_ES_USER_RESERVED_SIZE` `CFE_PLATFORM_ES_CFGVAL(USER_RESERVED↵
_SIZE)`
- #define `DEFAULT_CFE_PLATFORM_ES_USER_RESERVED_SIZE` (1024 * 1024)
- #define `CFE_PLATFORM_ES_MEMPOOL_ALIGN_SIZE_MIN` `CFE_PLATFORM_ES_CFGVAL(MEMPOOL_↵
ALIGN_SIZE_MIN)`
- #define `DEFAULT_CFE_PLATFORM_ES_MEMPOOL_ALIGN_SIZE_MIN` 4
- #define `CFE_PLATFORM_ES_NONVOL_STARTUP_FILE` `CFE_PLATFORM_ES_CFGVAL(NONVOL_↵
STARTUP_FILE)`
- #define `DEFAULT_CFE_PLATFORM_ES_NONVOL_STARTUP_FILE` "cf/cfe_es_startup.scr"
- #define `CFE_PLATFORM_ES_VOLATILE_STARTUP_FILE` `CFE_PLATFORM_ES_CFGVAL(VOLATILE_↵
STARTUP_FILE)`
- #define `DEFAULT_CFE_PLATFORM_ES_VOLATILE_STARTUP_FILE` "/ram/cfe_es_startup.scr"
- #define `CFE_PLATFORM_ES_DEFAULT_APP_LOG_FILE` `CFE_PLATFORM_ES_CFGVAL(DEFAULT_APP_↵
LOG_FILE)`
- #define `DEFAULT_CFE_PLATFORM_ES_DEFAULT_APP_LOG_FILE` "/ram/cfe_es_app_info.log"
- #define `CFE_PLATFORM_ES_DEFAULT_TASK_LOG_FILE` `CFE_PLATFORM_ES_CFGVAL(DEFAULT_↵
TASK_LOG_FILE)`
- #define `DEFAULT_CFE_PLATFORM_ES_DEFAULT_TASK_LOG_FILE` "/ram/cfe_es_taskinfo.log"
- #define `CFE_PLATFORM_ES_DEFAULT_SYSLOG_FILE` `CFE_PLATFORM_ES_CFGVAL(DEFAULT_↵
SYSLOG_FILE)`
- #define `DEFAULT_CFE_PLATFORM_ES_DEFAULT_SYSLOG_FILE` "/ram/cfe_es_syslog.log"
- #define `CFE_PLATFORM_ES_DEFAULT_ER_LOG_FILE` `CFE_PLATFORM_ES_CFGVAL(DEFAULT_ER_↵
LOG_FILE)`
- #define `DEFAULT_CFE_PLATFORM_ES_DEFAULT_ER_LOG_FILE` "/ram/cfe_erlog.log"
- #define `CFE_PLATFORM_ES_DEFAULT_PERF_DUMP_FILENAME` `CFE_PLATFORM_ES_CFGVAL(DEFAULT↵
_PERF_DUMP_FILENAME)`

```

• #define DEFAULT_CFE_PLATFORM_ES_DEFAULT_PERF_DUMP_FILENAME "/ram/cfe_es_perf.dat"
• #define CFE_PLATFORM_ES_DEFAULT_CDS_REG_DUMP_FILE CFE_PLATFORM_ES_CFGVAL(DEFAULT_CDS_REG_DUMP_FILE)
• #define DEFAULT_CFE_PLATFORM_ES_DEFAULT_CDS_REG_DUMP_FILE "/ram/cfe_cds_reg.log"
• #define CFE_PLATFORM_ES_DEFAULT_POR_SYSLOG_MODE CFE_PLATFORM_ES_CFGVAL(DEFAULT_POR_SYSLOG_MODE)
• #define DEFAULT_CFE_PLATFORM_ES_DEFAULT_POR_SYSLOG_MODE 0
• #define CFE_PLATFORM_ES_DEFAULT_PR_SYSLOG_MODE CFE_PLATFORM_ES_CFGVAL(DEFAULT_PR_SYSLOG_MODE)
• #define DEFAULT_CFE_PLATFORM_ES_DEFAULT_PR_SYSLOG_MODE 1
• #define CFE_PLATFORM_ES_PERF_DATA_BUFFER_SIZE CFE_PLATFORM_ES_CFGVAL(PERF_DATA_BUFFER_SIZE)
• #define DEFAULT_CFE_PLATFORM_ES_PERF_DATA_BUFFER_SIZE 10000
• #define CFE_PLATFORM_ES_PERF_FILTMASK_NONE CFE_PLATFORM_ES_CFGVAL(PERF_FILTMASK_NONE)
• #define DEFAULT_CFE_PLATFORM_ES_PERF_FILTMASK_NONE (0)
• #define CFE_PLATFORM_ES_PERF_FILTMASK_ALL CFE_PLATFORM_ES_CFGVAL(PERF_FILTMASK_ALL)
• #define DEFAULT_CFE_PLATFORM_ES_PERF_FILTMASK_ALL (~0)
• #define CFE_PLATFORM_ES_PERF_FILTMASK_INIT CFE_PLATFORM_ES_CFGVAL(PERF_FILTMASK_INIT)
• #define DEFAULT_CFE_PLATFORM_ES_PERF_FILTMASK_INIT (~0)
• #define CFE_PLATFORM_ES_PERF_TRIGMASK_NONE CFE_PLATFORM_ES_CFGVAL(PERF_TRIGMASK_NONE)
• #define DEFAULT_CFE_PLATFORM_ES_PERF_TRIGMASK_NONE 0
• #define CFE_PLATFORM_ES_PERF_TRIGMASK_ALL CFE_PLATFORM_ES_CFGVAL(PERF_TRIGMASK_ALL)
• #define DEFAULT_CFE_PLATFORM_ES_PERF_TRIGMASK_ALL ~0
• #define CFE_PLATFORM_ES_PERF_TRIGMASK_INIT CFE_PLATFORM_ES_CFGVAL(PERF_TRIGMASK_INIT)
• #define DEFAULT_CFE_PLATFORM_ES_PERF_TRIGMASK_INIT 0
• #define CFE_PLATFORM_ES_PERF_CHILD_PRIORITY CFE_PLATFORM_ES_CFGVAL(PERF_CHILD_PRIORITY)
• #define DEFAULT_CFE_PLATFORM_ES_PERF_CHILD_PRIORITY 200
• #define CFE_PLATFORM_ES_PERF_CHILD_STACK_SIZE CFE_PLATFORM_ES_CFGVAL(PERF_CHILD_STACK_SIZE)
• #define DEFAULT_CFE_PLATFORM_ES_PERF_CHILD_STACK_SIZE 4096
• #define CFE_PLATFORM_ES_PERF_CHILD_MS_DELAY CFE_PLATFORM_ES_CFGVAL(PERF_CHILD_MS_DELAY)
• #define DEFAULT_CFE_PLATFORM_ES_PERF_CHILD_MS_DELAY 20
• #define CFE_PLATFORM_ES_PERF_ENTRIES_BTWN_DLYS CFE_PLATFORM_ES_CFGVAL(PERF_ENTRIES_BTWN_DLYS)
• #define DEFAULT_CFE_PLATFORM_ES_PERF_ENTRIES_BTWN_DLYS 50
• #define CFE_PLATFORM_ES_DEFAULT_STACK_SIZE CFE_PLATFORM_ES_CFGVAL(DEFAULT_STACK_SIZE)
• #define DEFAULT_CFE_PLATFORM_ES_DEFAULT_STACK_SIZE 8192
• #define CFE_PLATFORM_ES_CDS_MAX_NUM_ENTRIES CFE_PLATFORM_ES_CFGVAL(CDS_MAX_NUM_ENTRIES)
• #define DEFAULT_CFE_PLATFORM_ES_CDS_MAX_NUM_ENTRIES 512
• #define CFE_PLATFORM_ES_MAX_PROCESSOR_RESETS CFE_PLATFORM_ES_CFGVAL(MAX_PROCESSOR_RESETS)
• #define DEFAULT_CFE_PLATFORM_ES_MAX_PROCESSOR_RESETS 2

```

- `#define CFE_PLATFORM_ES_POOL_MAX_BUCKETS CFE_PLATFORM_ES_CFGVAL(PPOOL_MAX_↵
BUCKETS)`
- `#define DEFAULT_CFE_PLATFORM_ES_POOL_MAX_BUCKETS 17`
- `#define CFE_PLATFORM_ES_MAX_MEMORY_POOLS CFE_PLATFORM_ES_CFGVAL(MAX_MEMORY_↵
POOLS)`
- `#define DEFAULT_CFE_PLATFORM_ES_MAX_MEMORY_POOLS 10`
- `#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_01 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_↵
_01)`
- `#define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_01 8`
- `#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_02 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_↵
_02)`
- `#define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_02 16`
- `#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_03 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_↵
_03)`
- `#define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_03 32`
- `#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_04 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_↵
_04)`
- `#define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_04 48`
- `#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_05 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_↵
_05)`
- `#define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_05 64`
- `#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_06 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_↵
_06)`
- `#define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_06 96`
- `#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_07 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_↵
_07)`
- `#define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_07 128`
- `#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_08 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_↵
_08)`
- `#define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_08 160`
- `#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_09 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_↵
_09)`
- `#define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_09 256`
- `#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_10 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_↵
_10)`
- `#define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_10 512`
- `#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_11 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_↵
_11)`
- `#define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_11 1024`
- `#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_12 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_↵
_12)`
- `#define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_12 2048`
- `#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_13 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_↵
_13)`
- `#define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_13 4096`
- `#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_14 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_↵
_14)`
- `#define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_14 8192`
- `#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_15 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_↵
_15)`
- `#define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_15 16384`

```

• #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_16 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_16)
• #define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_16 32768
• #define CFE_PLATFORM_ES_MAX_BLOCK_SIZE CFE_PLATFORM_ES_CFGVAL(MAX_BLOCK_SIZE)
• #define DEFAULT_CFE_PLATFORM_ES_MAX_BLOCK_SIZE 80000
• #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_01 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_01)
• #define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_01 8
• #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_02 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_02)
• #define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_02 16
• #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_03 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_03)
• #define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_03 32
• #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_04 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_04)
• #define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_04 48
• #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_05 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_05)
• #define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_05 64
• #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_06 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_06)
• #define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_06 96
• #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_07 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_07)
• #define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_07 128
• #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_08 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_08)
• #define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_08 160
• #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_09 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_09)
• #define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_09 256
• #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_10 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_10)
• #define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_10 512
• #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_11 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_11)
• #define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_11 1024
• #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_12 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_12)
• #define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_12 2048
• #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_13 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_13)
• #define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_13 4096
• #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_14 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_14)
• #define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_14 8192
• #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_15 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_15)
• #define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_15 16384
• #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_16 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_16)

```

- `#define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_16 32768`
- `#define CFE_PLATFORM_ES_CDS_MAX_BLOCK_SIZE CFE_PLATFORM_ES_CFGVAL(CDS_MAX_BLOCK_SIZE)`
- `#define DEFAULT_CFE_PLATFORM_ES_CDS_MAX_BLOCK_SIZE 80000`
- `#define CFE_PLATFORM_ES_STARTUP_SYNC_POLL_MSEC CFE_PLATFORM_ES_CFGVAL(STARTUP_SYNC_POLL_MSEC)`
- `#define DEFAULT_CFE_PLATFORM_ES_STARTUP_SYNC_POLL_MSEC 50`
- `#define CFE_PLATFORM_ES_STARTUP_SCRIPT_TIMEOUT_MSEC CFE_PLATFORM_ES_CFGVAL(STARTUP_SCRIPT_TIMEOUT_MSEC)`
- `#define DEFAULT_CFE_PLATFORM_ES_STARTUP_SCRIPT_TIMEOUT_MSEC 1000`

11.129.1 Detailed Description

CFE Executive Services (CFE_ES) Application Platform Configuration Header File

Note

This file may be overridden/superseded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [cfe_es_internal_cfg.h](#).

11.129.2 Macro Definition Documentation

CFE_PLATFORM_ES_APP_KILL_TIMEOUT

```
#define CFE_PLATFORM_ES_APP_KILL_TIMEOUT CFE_PLATFORM_ES_CFGVAL(APP_KILL_TIMEOUT)
```

Purpose Define ES Application Kill Timeout

Description:

ES Application Kill Timeout. This parameter controls the number of "scan periods" that ES will wait for an application to Exit after getting the signal Delete, Reload or Restart. The sequence works as follows:

1. ES will set the control request for an App to Delete/Restart/Reload and set this kill timer to the value in this parameter.
2. If the App is responding and Calls it's RunLoop function, it will drop out of it's main loop and call `CFE_ES_ExitApp`. Once it calls Exit App, then ES can delete, restart, or reload the app the next time it scans the app table.
3. If the App is not responding, the ES App will decrement this Kill Timeout value each time it runs. If the timeout value reaches zero, ES will kill the app.

The Kill timeout value depends on the [CFE_PLATFORM_ES_APP_SCAN_RATE](#). If the Scan Rate is 1000, or 1 second, and this [CFE_PLATFORM_ES_APP_KILL_TIMEOUT](#) is set to 5, then it will take 5 seconds to kill a non-responding App. If the Scan Rate is 250, or 1/4 second, and the [CFE_PLATFORM_ES_APP_KILL_TIMEOUT](#) is set to 2, then it will take 1/2 second to time out.

Limits

There is a lower limit of 1 and an upper limit of 100 on this configuration parameter. Units are number of [CFE_PLATFORM_ES_APP_SCAN_RATE](#) cycles.

Definition at line 228 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_APP_SCAN_RATE

```
#define CFE_PLATFORM_ES_APP_SCAN_RATE CFE_PLATFORM_ES_CFGVAL(APP_SCAN_RATE)
```

Purpose Define ES Application Control Scan Rate

Description:

ES Application Control Scan Rate. This parameter controls the speed that ES scans the Application Table looking for App Delete/Restart/Reload requests. All Applications are deleted, restarted, or reloaded by the ES Application. ES will periodically scan for control requests to process. The scan rate is controlled by this parameter, which is given in milliseconds. A value of 1000 means that ES will scan the Application Table once per second. Be careful not to set the value of this too low, because ES will use more CPU cycles scanning the table.

Limits

There is a lower limit of 100 and an upper limit of 20000 on this configuration parameter. millisecond units.

Definition at line 198 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_CDS_MAX_BLOCK_SIZE

```
#define CFE_PLATFORM_ES_CDS_MAX_BLOCK_SIZE CFE_PLATFORM_ES_CFGVAL(CDS_MAX_BLOCK_SIZE)
```

Definition at line 836 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_CDS_MAX_NUM_ENTRIES

```
#define CFE_PLATFORM_ES_CDS_MAX_NUM_ENTRIES CFE_PLATFORM_ES_CFGVAL(CDS_MAX_NUM_ENTRIES)
```

Purpose Define Maximum Number of Registered CDS Blocks

Description:

Maximum number of registered CDS Blocks

Limits

There is a lower limit of 8. There are no restrictions on the upper limit however, the maximum number of CDS entries is system dependent and should be verified.

Definition at line 689 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_01

```
#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_01 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_01)
```

Purpose Define ES Critical Data Store Memory Pool Block Sizes

Description:

Intermediate ES Critical Data Store Memory Pool Block Sizes

Limits

These sizes MUST be increasing and MUST be an integral multiple of 4.

Definition at line 804 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_02

#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_02 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_02)
Definition at line 806 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_03

#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_03 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_03)
Definition at line 808 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_04

#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_04 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_04)
Definition at line 810 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_05

#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_05 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_05)
Definition at line 812 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_06

#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_06 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_06)
Definition at line 814 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_07

#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_07 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_07)
Definition at line 816 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_08

#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_08 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_08)
Definition at line 818 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_09

#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_09 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_09)
Definition at line 820 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_10

#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_10 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_10)
Definition at line 822 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_11

#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_11 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_11)
Definition at line 824 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_12

#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_12 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_12)
Definition at line 826 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_13

```
#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_13 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_13)
```

Definition at line 828 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_14

```
#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_14 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_14)
```

Definition at line 830 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_15

```
#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_15 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_15)
```

Definition at line 832 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_16

```
#define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_16 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_16)
```

Definition at line 834 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_CDS_SIZE

```
#define CFE_PLATFORM_ES_CDS_SIZE CFE_PLATFORM_ES_CFGVAL(CDS_SIZE)
```

Purpose Define Critical Data Store Size

Description:

Defines the Critical Data Store (CDS) area size in bytes size. The CDS is one of four memory areas that are preserved during a processor reset. NOTE: Changing this value changes memory allocation, and may require changes to platform specific values (in CFE_PSP) such as USER_RESERVED_MEM in VxWorks depending on the memory areas being used for preserved data and on OS specific behavior.

Limits

There is a lower limit of 8192 and an upper limit of UINT_MAX (4 Gigabytes) on this configuration parameter.

Definition at line 309 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_DEFAULT_APP_LOG_FILE

```
#define CFE_PLATFORM_ES_DEFAULT_APP_LOG_FILE CFE_PLATFORM_ES_CFGVAL(DEFAULT_APP_LOG_FILE)
```

Purpose Default Application Information Filename

Description:

The value of this constant defines the filename used to store information pertaining to all of the Applications that are registered with Executive Services. This filename is used only when no filename is specified in the the command to query all system apps.

Limits

The length of each string, including the NULL terminator cannot exceed the [OS_MAX_PATH_LEN](#) value.

Definition at line 396 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_DEFAULT_CDS_REG_DUMP_FILE

```
#define CFE_PLATFORM_ES_DEFAULT_CDS_REG_DUMP_FILE CFE_PLATFORM_ES_CFGVAL(DEFAULT_CDS_REG_DUMP_↵  
FILE)
```

Purpose Default Critical Data Store Registry Filename

Description:

The value of this constant defines the filename used to store the Critical Data Store Registry. This filename is used only when no filename is specified in the command to stop performance data collecting.

Limits

The length of each string, including the NULL terminator cannot exceed the [OS_MAX_PATH_LEN](#) value.

Definition at line 475 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_DEFAULT_ER_LOG_FILE

```
#define CFE_PLATFORM_ES_DEFAULT_ER_LOG_FILE CFE_PLATFORM_ES_CFGVAL(DEFAULT_ER_LOG_FILE)
```

Purpose Default Exception and Reset (ER) Log Filename

Description:

The value of this constant defines the filename used to store the Exception and Reset (ER) Log. This filename is used only when no filename is specified in the command to dump the ER log. No file specified in the cmd means the first character in the cmd filename is a NULL terminator (zero).

Limits

The length of each string, including the NULL terminator cannot exceed the [OS_MAX_PATH_LEN](#) value.

Definition at line 445 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_DEFAULT_PERF_DUMP_FILENAME

```
#define CFE_PLATFORM_ES_DEFAULT_PERF_DUMP_FILENAME CFE_PLATFORM_ES_CFGVAL(DEFAULT_PERF_DUMP_↵  
FILENAME)
```

Purpose Default Performance Data Filename

Description:

The value of this constant defines the filename used to store the Performance Data. This filename is used only when no filename is specified in the command to stop performance data collecting.

Limits

The length of each string, including the NULL terminator cannot exceed the [OS_MAX_PATH_LEN](#) value.

Definition at line 460 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_DEFAULT_POR_SYSLOG_MODE

```
#define CFE_PLATFORM_ES_DEFAULT_POR_SYSLOG_MODE CFE_PLATFORM_ES_CFGVAL(DEFAULT_POR_SYSLOG_MODE)
```

Purpose Define Default System Log Mode following Power On Reset

Description:

Defines the default mode for the operation of the ES System log following a power on reset. The log may operate in either Overwrite mode = 0, where once the log becomes full the oldest message in the log will be overwritten, or Discard mode = 1, where once the log becomes full the contents of the log are preserved and the new event is discarded. This constant may hold a value of either 0 or 1 depending on the desired default. Overwrite Mode = 0, Discard Mode = 1.

Limits

There is a lower limit of 0 and an upper limit of 1 on this configuration parameter.

Definition at line 494 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_DEFAULT_PR_SYSLOG_MODE

```
#define CFE_PLATFORM_ES_DEFAULT_PR_SYSLOG_MODE CFE_PLATFORM_ES_CFGVAL(DEFAULT_PR_SYSLOG_MODE)
```

Purpose Define Default System Log Mode following Processor Reset

Description:

Defines the default mode for the operation of the ES System log following a processor reset. The log may operate in either Overwrite mode = 0, where once the log becomes full the oldest message in the log will be overwritten, or Discard mode = 1, where once the log becomes full the contents of the log are preserved and the new event is discarded. This constant may hold a value of either 0 or 1 depending on the desired default. Overwrite Mode = 0, Discard Mode = 1.

Limits

There is a lower limit of 0 and an upper limit of 1 on this configuration parameter.

Definition at line 513 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_DEFAULT_STACK_SIZE

```
#define CFE_PLATFORM_ES_DEFAULT_STACK_SIZE CFE_PLATFORM_ES_CFGVAL(DEFAULT_STACK_SIZE)
```

Purpose Define Default Stack Size for an Application

Description:

This parameter defines a default stack size. This parameter is used by the cFE Core Applications.

Limits

There is a lower limit of 2048. There are no restrictions on the upper limit however, the maximum stack size is system dependent and should be verified. Most operating systems provide tools for measuring the amount of stack used by a task during operation. It is always a good idea to verify that no more than 1/2 of the stack is used.

Definition at line 675 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_DEFAULT_SYSLOG_FILE

```
#define CFE_PLATFORM_ES_DEFAULT_SYSLOG_FILE CFE_PLATFORM_ES_CFGVAL(DEFAULT_SYSLOG_FILE)
```

Purpose Default System Log Filename

Description:

The value of this constant defines the filename used to store important information (as ASCII text strings) that might not be able to be sent in an Event Message. This filename is used only when no filename is specified in the command to dump the system log. No file specified in the cmd means the first character in the cmd filename is a NULL terminator (zero).

Limits

The length of each string, including the NULL terminator cannot exceed the [OS_MAX_PATH_LEN](#) value.

Definition at line 429 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_DEFAULT_TASK_LOG_FILE

```
#define CFE_PLATFORM_ES_DEFAULT_TASK_LOG_FILE CFE_PLATFORM_ES_CFGVAL(DEFAULT_TASK_LOG_FILE)
```

Purpose Default Application Information Filename

Description:

The value of this constant defines the filename used to store information pertaining to all of the Applications that are registered with Executive Services. This filename is used only when no filename is specified in the the command to query all system tasks.

Limits

The length of each string, including the NULL terminator cannot exceed the [OS_MAX_PATH_LEN](#) value.

Definition at line 412 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_ER_LOG_ENTRIES

```
#define CFE_PLATFORM_ES_ER_LOG_ENTRIES CFE_PLATFORM_ES_CFGVAL(ER_LOG_ENTRIES)
```

Purpose Define Max Number of ER (Exception and Reset) log entries

Description:

Defines the maximum number of ER (Exception and Reset) log entries

Limits

There is a lower limit of 1. There are no restrictions on the upper limit however, the maximum number of log entries is system dependent and should be verified.

Definition at line 135 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_ER_LOG_MAX_CONTEXT_SIZE

```
#define CFE_PLATFORM_ES_ER_LOG_MAX_CONTEXT_SIZE CFE_PLATFORM_ES_CFGVAL(ER_LOG_MAX_CONTEXT_SIZE)
```

Purpose Maximum size of CPU Context in ES Error Log

Description:

This should be large enough to accommodate the CPU context information supplied by the PSP on the given platform.

Limits:

Must be greater than zero and a multiple of sizeof(uint32). Limited only by the available memory and the number of entries in the error log. Any context information beyond this size will be truncated.

Definition at line 150 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_MAX_APPLICATIONS

```
#define CFE_PLATFORM_ES_MAX_APPLICATIONS CFE_PLATFORM_ES_CFGVAL(MAX_APPLICATIONS)
```

Purpose Define Max Number of Applications

Description:

Defines the maximum number of applications that can be loaded into the system. This number does not include child tasks.

Limits

There is a lower limit of 6. The lower limit corresponds to the cFE internal applications. There are no restrictions on the upper limit however, the maximum number of applications is system dependent and should be verified. ApplDs that are checked against this configuration are defined by a 32 bit data word.

Definition at line 106 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_MAX_BLOCK_SIZE

```
#define CFE_PLATFORM_ES_MAX_BLOCK_SIZE CFE_PLATFORM_ES_CFGVAL(MAX_BLOCK_SIZE)
```

Definition at line 792 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_MAX_GEN_COUNTERS

```
#define CFE_PLATFORM_ES_MAX_GEN_COUNTERS CFE_PLATFORM_ES_CFGVAL(MAX_GEN_COUNTERS)
```

Purpose Define Max Number of Generic Counters

Description:

Defines the maximum number of Generic Counters that can be registered.

Limits

This parameter has a lower limit of 1 and an upper limit of 65535.

Definition at line 178 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_MAX_LIBRARIES

```
#define CFE_PLATFORM_ES_MAX_LIBRARIES CFE_PLATFORM_ES_CFGVAL (MAX_LIBRARIES)
```

Purpose Define Max Number of Shared libraries

Description:

Defines the maximum number of cFE Shared libraries that can be loaded into the system.

Limits

There is a lower limit of 1. There are no restrictions on the upper limit however, the maximum number of libraries is system dependent and should be verified.

Definition at line 121 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_MAX_MEMORY_POOLS

```
#define CFE_PLATFORM_ES_MAX_MEMORY_POOLS CFE_PLATFORM_ES_CFGVAL (MAX_MEMORY_POOLS)
```

Purpose Maximum number of memory pools

Description:

The upper limit for the number of memory pools that can concurrently exist within the system.

The CFE_SB and CFE_TBL core subsystems each define a memory pool. Individual applications may also create memory pools, so this value should be set sufficiently high enough to support the applications being used on this platform.

Limits:

Must be at least 2 to support CFE core - SB and TBL pools. No specific upper limit.

Definition at line 740 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_MAX_PROCESSOR_RESETS

```
#define CFE_PLATFORM_ES_MAX_PROCESSOR_RESETS CFE_PLATFORM_ES_CFGVAL (MAX_PROCESSOR_RESETS)
```

Purpose Define Number of Processor Resets Before a Power On Reset

Description:

Number of Processor Resets before a Power On Reset is called. If set to 2, then 2 processor resets will occur, and the 3rd processor reset will be a power on reset instead.

Limits

There is a lower limit of 0. There are no restrictions on the upper limit however, the maximum number of processor resets may be system dependent and should be verified.

Definition at line 705 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_01

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_01 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_01)
```

Purpose Define Default ES Memory Pool Block Sizes

Description:

Default Intermediate ES Memory Pool Block Sizes. If an application is using the CFE_ES Memory Pool APIs ([CFE_ES_PoolCreate](#), [CFE_ES_PoolCreateNoSem](#), [CFE_ES_GetPoolBuf](#) and [CFE_ES_PutPoolBuf](#)) but finds these sizes inappropriate for their use, they may wish to use the [CFE_ES_PoolCreateEx](#) API to specify their own intermediate block sizes

Limits

These sizes MUST be increasing and MUST be an integral multiple of 4. Also, CFE_PLATFORM_ES_MAX_BLOCK_SIZE must be larger than CFE_MISSION_SB_MAX_SB_MSG_SIZE and both CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE and CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE. Note that if Table Services have been removed from the CFE, the table size limits are still enforced although the table size definitions may be reduced.

Definition at line 760 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_02

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_02 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_02)
```

Definition at line 762 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_03

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_03 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_03)
```

Definition at line 764 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_04

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_04 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_04)
```

Definition at line 766 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_05

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_05 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_05)
```

Definition at line 768 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_06

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_06 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_06)
```

Definition at line 770 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_07

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_07 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_07)
```

Definition at line 772 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_08

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_08 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_08)
```

Definition at line 774 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_09

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_09 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_09)
```

Definition at line 776 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_10

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_10 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_10)
```

Definition at line 778 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_11

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_11 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_11)
```

Definition at line 780 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_12

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_12 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_12)
```

Definition at line 782 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_13

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_13 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_13)
```

Definition at line 784 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_14

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_14 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_14)
```

Definition at line 786 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_15

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_15 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_15)
```

Definition at line 788 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_MEM_BLOCK_SIZE_16

```
#define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_16 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_16)
```

Definition at line 790 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_MEMPOOL_ALIGN_SIZE_MIN

```
#define CFE_PLATFORM_ES_MEMPOOL_ALIGN_SIZE_MIN CFE_PLATFORM_ES_CFGVAL(MEMPOOL_ALIGN_SIZE_MIN)
```

Purpose Define Memory Pool Alignment Size

Description:

Ensures that buffers obtained from a memory pool are aligned to a certain minimum block size. Note the allocator will always align to the minimum required by the CPU architecture. This may be set greater than the CPU requirement as desired for optimal performance.

For some architectures/applications it may be beneficial to set this to the cache line size of the target CPU, or to use special SIMD instructions that require a more stringent memory alignment.

Limits

This must always be a power of 2, as it is used as a binary address mask.

Definition at line 350 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_NONVOL_DISK_MOUNT_STRING

```
#define CFE_PLATFORM_ES_NONVOL_DISK_MOUNT_STRING CFE_PLATFORM_ES_CFGVAL(NONVOL_DISK_MOUNT_STRING)
```

Purpose Default virtual path for persistent storage

Description:

This configures the default location in the virtual file system for persistent/non-volatile storage. Files such as the startup script, app/library dynamic modules, and configuration tables are expected to be stored in this directory.

Definition at line 72 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_NONVOL_STARTUP_FILE

```
#define CFE_PLATFORM_ES_NONVOL_STARTUP_FILE CFE_PLATFORM_ES_CFGVAL(NONVOL_STARTUP_FILE)
```

Purpose ES Nonvolatile Startup Filename

Description:

The value of this constant defines the path and name of the file that contains a list of modules that will be loaded and started by the cFE after the cFE finishes its startup sequence.

Limits

The length of each string, including the NULL terminator cannot exceed the [OS_MAX_PATH_LEN](#) value.

Definition at line 365 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_PERF_CHILD_MS_DELAY

```
#define CFE_PLATFORM_ES_PERF_CHILD_MS_DELAY CFE_PLATFORM_ES_CFGVAL(PERF_CHILD_MS_DELAY)
```

Purpose Define Performance Analyzer Child Task Delay

Description:

This parameter defines the delay time (in milliseconds) between performance data file writes performed by the Executive Services Performance Analyzer Child Task.

Limits

It is recommended this parameter be greater than or equal to 20ms. This parameter is limited by the maximum value allowed by the data type. In this case, the data type is an unsigned 32-bit integer, so the valid range is 0 to 0xFFFFFFFF.

Definition at line 647 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_PERF_CHILD_PRIORITY

```
#define CFE_PLATFORM_ES_PERF_CHILD_PRIORITY CFE_PLATFORM_ES_CFGVAL(PERF_CHILD_PRIORITY)
```

Purpose Define Performance Analyzer Child Task Priority

Description:

This parameter defines the priority of the child task spawned by the Executive Services to write performance data to a file. Lower numbers are higher priority, with 1 being the highest priority in the case of a child task.

Limits

Valid range for a child task is 1 to 255 however, the priority cannot be higher (lower number) than the ES parent application priority.

Definition at line 616 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_PERF_CHILD_STACK_SIZE

```
#define CFE_PLATFORM_ES_PERF_CHILD_STACK_SIZE CFE_PLATFORM_ES_CFGVAL(PERF_CHILD_STACK_SIZE)
```

Purpose Define Performance Analyzer Child Task Stack Size

Description:

This parameter defines the stack size of the child task spawned by the Executive Services to write performance data to a file.

Limits

It is recommended this parameter be greater than or equal to 4KB. This parameter is limited by the maximum value allowed by the data type. In this case, the data type is an unsigned 32-bit integer, so the valid range is 0 to 0xFFFFFFFF.

Definition at line 631 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_PERF_DATA_BUFFER_SIZE

```
#define CFE_PLATFORM_ES_PERF_DATA_BUFFER_SIZE CFE_PLATFORM_ES_CFGVAL(PERF_DATA_BUFFER_SIZE)
```

Purpose Define Max Size of Performance Data Buffer

Description:

Defines the maximum size of the performance data buffer. Units are number of performance data entries. An entry is defined by a 32 bit data word followed by a 64 bit time stamp.

Limits

There is a lower limit of 1025. There are no restrictions on the upper limit however, the maximum buffer size is system dependent and should be verified. The units are number of entries. An entry is defined by a 32 bit data word followed by a 64 bit time stamp.

Definition at line 530 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_PERF_ENTRIES_BTWN_DLYS

```
#define CFE_PLATFORM_ES_PERF_ENTRIES_BTWN_DLYS CFE_PLATFORM_ES_CFGVAL(PERF_ENTRIES_BTWN_DLYS)
```

Purpose Define Performance Analyzer Child Task Number of Entries Between Delay

Description:

This parameter defines the number of performance analyzer entries the Performance Analyzer Child Task will write to the file between delays.

Definition at line 658 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_PERF_FILTMASK_ALL

```
#define CFE_PLATFORM_ES_PERF_FILTMASK_ALL CFE_PLATFORM_ES_CFGVAL(PERF_FILTMASK_ALL)
```

Purpose Define Filter Mask Setting for Enabling All Performance Entries

Description:

Defines the filter mask for enabling all performance entries. The value is a bit mask. For each bit, 0 means the corresponding entry is disabled and 1 means it is enabled.

Definition at line 552 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_PERF_FILTMASK_INIT

```
#define CFE_PLATFORM_ES_PERF_FILTMASK_INIT CFE_PLATFORM_ES_CFGVAL(PERF_FILTMASK_INIT)
```

Purpose Define Default Filter Mask Setting for Performance Data Buffer

Description:

Defines the default filter mask for the performance data buffer. The value is a bit mask. For each bit, 0 means the corresponding entry is disabled and 1 means it is enabled.

Definition at line 564 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_PERF_FILTMASK_NONE

```
#define CFE_PLATFORM_ES_PERF_FILTMASK_NONE CFE_PLATFORM_ES_CFGVAL(PERF_FILTMASK_NONE)
```

Purpose Define Filter Mask Setting for Disabling All Performance Entries

Description:

Defines the filter mask for disabling all performance entries. The value is a bit mask. For each bit, 0 means the corresponding entry is disabled and 1 means it is enabled.

Definition at line 541 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_PERF_TRIGMASK_ALL

```
#define CFE_PLATFORM_ES_PERF_TRIGMASK_ALL CFE_PLATFORM_ES_CFGVAL(PERF_TRIGMASK_ALL)
```

Purpose Define Filter Trigger Setting for Enabling All Performance Entries

Description:

Defines the trigger mask for enabling all performance data entries. The value is a bit mask. For each bit, 0 means the trigger for the corresponding entry is disabled and 1 means it is enabled.

Definition at line 588 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_PERF_TRIGMASK_INIT

```
#define CFE_PLATFORM_ES_PERF_TRIGMASK_INIT CFE_PLATFORM_ES_CFGVAL(PERF_TRIGMASK_INIT)
```

Purpose Define Default Filter Trigger Setting for Performance Data Buffer

Description:

Defines the default trigger mask for the performance data buffer. The value is a 32-bit mask. For each bit, 0 means the trigger for the corresponding entry is disabled and 1 means it is enabled.

Definition at line 600 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_PERF_TRIGMASK_NONE

```
#define CFE_PLATFORM_ES_PERF_TRIGMASK_NONE CFE_PLATFORM_ES_CFGVAL (PERF_TRIGMASK_NONE)
```

Purpose Define Default Filter Trigger Setting for Disabling All Performance Entries

Description:

Defines the default trigger mask for disabling all performance data entries. The value is a bit mask. For each bit, 0 means the trigger for the corresponding entry is disabled and 1 means it is enabled.

Definition at line 576 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_POOL_MAX_BUCKETS

```
#define CFE_PLATFORM_ES_POOL_MAX_BUCKETS CFE_PLATFORM_ES_CFGVAL (POOL_MAX_BUCKETS)
```

Purpose Maximum number of block sizes in pool structures

Description:

The upper limit for the number of block sizes supported in the generic pool implementation, which in turn implements the memory pools and CDS.

Limits:

Must be at least one. No specific upper limit, but the number is anticipated to be reasonably small (i.e. tens, not hundreds). Large values have not been tested.

The ES and CDS block size lists must correlate with this value

Definition at line 721 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_RAM_DISK_MOUNT_STRING

```
#define CFE_PLATFORM_ES_RAM_DISK_MOUNT_STRING CFE_PLATFORM_ES_CFGVAL (RAM_DISK_MOUNT_STRING)
```

Purpose Default virtual path for volatile storage

Description:

The [CFE_PLATFORM_ES_RAM_DISK_MOUNT_STRING](#) parameter is used to set the cFE mount path for the CFE RAM disk. This is a parameter for missions that do not want to use the default value of "/ram", or for missions that need to have a different value for different CPUs or Spacecraft. Note that the vxWorks OSAL cannot currently handle names that have more than one path separator in it. The names "/ram", "/ramdisk", "/disk123" will all work, but "/disks/ram" will not. Multiple separators can be used with the posix or RTEMS ports.

Definition at line 89 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_RAM_DISK_NUM_SECTORS

```
#define CFE_PLATFORM_ES_RAM_DISK_NUM_SECTORS CFE_PLATFORM_ES_CFGVAL (RAM_DISK_NUM_SECTORS)
```

Purpose ES Ram Disk Number of Sectors

Description:

Defines the ram disk number of sectors. The ram disk is one of four memory areas that are preserved on a processor reset. NOTE: Changing this value changes memory allocation, and may require changes to platform specific values (in CFE_PSP) such as USER_RESERVED_MEM in VxWorks depending on the memory areas being used for preserved data and on OS specific behavior.

Limits

There is a lower limit of 128. There are no restrictions on the upper limit however, the maximum number of RAM sectors is system dependent and should be verified.

Definition at line 266 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_RAM_DISK_PERCENT_RESERVED

```
#define CFE_PLATFORM_ES_RAM_DISK_PERCENT_RESERVED CFE_PLATFORM_ES_CFGVAL(RAM_DISK_PERCENT_RESERVED)
```

Purpose Percentage of Ram Disk Reserved for Decompressing Apps

Description:

The [CFE_PLATFORM_ES_RAM_DISK_PERCENT_RESERVED](#) parameter is used to make sure that the Volatile (RAM) Disk has a defined amount of free space during a processor reset. The cFE uses the Volatile disk to decompress cFE applications during system startup. If this Volatile disk happens to get filled with logs and misc files, then a processor reset may not work, because there will be no room to decompress cFE apps. To solve that problem, this parameter sets the "Low Water Mark" for disk space on a Processor reset. It should be set to allow the largest cFE Application to be decompressed. During a Processor reset, if there is not sufficient space left on the disk, it will be re-formatted in order to clear up some space.

This feature can be turned OFF by setting the parameter to 0.

Limits

There is a lower limit of 0 and an upper limit of 75 on this configuration parameter. Units are percentage. A setting of zero will turn this feature off.

Definition at line 291 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_RAM_DISK_SECTOR_SIZE

```
#define CFE_PLATFORM_ES_RAM_DISK_SECTOR_SIZE CFE_PLATFORM_ES_CFGVAL(RAM_DISK_SECTOR_SIZE)
```

Purpose ES Ram Disk Sector Size

Description:

Defines the ram disk sector size. The ram disk is 1 of 4 memory areas that are preserved on a processor reset. NOTE: Changing this value changes memory allocation, and may require changes to platform specific values (in CFE_PSP) such as USER_RESERVED_MEM in VxWorks depending on the memory areas being used for preserved data and on OS specific behavior.

Limits

There is a lower limit of 128. There are no restrictions on the upper limit however, the maximum RAM disk sector size is system dependent and should be verified.

Definition at line 247 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_START_TASK_PRIORITY

```
#define CFE_PLATFORM_ES_START_TASK_PRIORITY CFE_PLATFORM_ES_CFGVAL (START_TASK_PRIORITY)
```

Purpose Define ES Task Priority

Description:

Defines the cFE_ES Task priority.

Limits

Not Applicable

Definition at line 43 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_START_TASK_STACK_SIZE

```
#define CFE_PLATFORM_ES_START_TASK_STACK_SIZE CFE_PLATFORM_ES_CFGVAL (START_TASK_STACK_SIZE)
```

Purpose Define ES Task Stack Size

Description:

Defines the cFE_ES Task Stack Size

Limits

There is a lower limit of 2048 on this configuration parameter. There are no restrictions on the upper limit however, the maximum stack size is system dependent and should be verified. Most operating systems provide tools for measuring the amount of stack used by a task during operation. It is always a good idea to verify that no more than 1/2 of the stack is used.

Definition at line 59 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_STARTUP_SCRIPT_TIMEOUT_MSEC

```
#define CFE_PLATFORM_ES_STARTUP_SCRIPT_TIMEOUT_MSEC CFE_PLATFORM_ES_CFGVAL (STARTUP_SCRIPT_TIMEOUT↵_MSEC)
```

Purpose Startup script timeout

Description:

The upper limit for the total amount of time that all apps listed in the CFE ES startup script may take to all become ready.

Unlike the "core" app timeout, this is a soft limit; if the allotted time is exceeded, it probably indicates an issue with one of the apps, but does not cause CFE ES to take any additional action other than logging the event to the syslog. Units are in milliseconds

Limits:

Must be defined as an integer value that is greater than or equal to zero.

Definition at line 878 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_STARTUP_SYNC_POLL_MSEC

```
#define CFE_PLATFORM_ES_STARTUP_SYNC_POLL_MSEC CFE_PLATFORM_ES_CFGVAL (STARTUP_SYNC_POLL_MSEC)
```

Purpose Poll timer for startup sync delay

Description:

During startup, some tasks may need to synchronize their own initialization with the initialization of other applications in the system.

CFE ES implements an API to accomplish this, that performs a task delay (sleep) while polling the overall system state until other tasks are ready.

This value controls the amount of time that the CFE_ES_ApplicationSyncDelay will sleep between each check of the system state. This should be large enough to allow other tasks to run, but not so large as to noticeably delay the startup completion.

Units are in milliseconds

Limits:

Must be defined as an integer value that is greater than or equal to zero.

Definition at line 859 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_SYSTEM_LOG_SIZE

```
#define CFE_PLATFORM_ES_SYSTEM_LOG_SIZE CFE_PLATFORM_ES_CFGVAL (SYSTEM_LOG_SIZE)
```

Purpose Define Size of the cFE System Log.

Description:

Defines the size in bytes of the cFE system log. The system log holds variable length strings that are terminated by a linefeed and null character.

Limits

There is a lower limit of 512. There are no restrictions on the upper limit however, the maximum system log size is system dependent and should be verified.

Definition at line 166 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_USER_RESERVED_SIZE

```
#define CFE_PLATFORM_ES_USER_RESERVED_SIZE CFE_PLATFORM_ES_CFGVAL (USER_RESERVED_SIZE)
```

Purpose Define User Reserved Memory Size

Description:

User Reserved Memory Size. This is the size in bytes of the cFE User reserved Memory area. This is a block of memory that is available for cFE application use. The address is obtained by calling [CFE_PSP_GetUserReservedArea](#). The User Reserved Memory is one of four memory areas that are preserved during a processor reset. NOTE: Changing this value changes memory allocation, and may require changes to platform specific values (in CFE_PSP) such as USER_RESERVED_MEM in VxWorks depending on the memory areas being used for preserved data and on OS specific behavior.

Limits

There is a lower limit of 1024 and an upper limit of UINT_MAX (4 Gigabytes) on this configuration parameter.

Definition at line 330 of file [cfe_es_internal_cfg.h](#).

CFE_PLATFORM_ES_VOLATILE_STARTUP_FILE

```
#define CFE_PLATFORM_ES_VOLATILE_STARTUP_FILE CFE_PLATFORM_ES_CFGVAL(VOLATILE_STARTUP_FILE)
```

Purpose ES Volatile Startup Filename

Description:

The value of this constant defines the path and name of the file that contains a list of modules that will be loaded and started by the cFE after the cFE finishes its startup sequence.

Limits

The length of each string, including the NULL terminator cannot exceed the [OS_MAX_PATH_LEN](#) value.

Definition at line 380 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_APP_KILL_TIMEOUT

```
#define DEFAULT_CFE_PLATFORM_ES_APP_KILL_TIMEOUT 5
```

Definition at line 229 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_APP_SCAN_RATE

```
#define DEFAULT_CFE_PLATFORM_ES_APP_SCAN_RATE 1000
```

Definition at line 199 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_CDS_MAX_BLOCK_SIZE

```
#define DEFAULT_CFE_PLATFORM_ES_CDS_MAX_BLOCK_SIZE 80000
```

Definition at line 837 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_CDS_MAX_NUM_ENTRIES

```
#define DEFAULT_CFE_PLATFORM_ES_CDS_MAX_NUM_ENTRIES 512
```

Definition at line 690 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_01

```
#define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_01 8
```

Definition at line 805 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_02

```
#define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_02 16
```

Definition at line 807 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_03

```
#define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_03 32
```

Definition at line 809 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_04

```
#define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_04 48
```

Definition at line 811 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_05

#define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_05 64
Definition at line 813 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_06

#define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_06 96
Definition at line 815 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_07

#define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_07 128
Definition at line 817 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_08

#define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_08 160
Definition at line 819 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_09

#define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_09 256
Definition at line 821 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_10

#define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_10 512
Definition at line 823 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_11

#define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_11 1024
Definition at line 825 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_12

#define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_12 2048
Definition at line 827 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_13

#define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_13 4096
Definition at line 829 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_14

#define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_14 8192
Definition at line 831 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_15

#define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_15 16384
Definition at line 833 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_16

#define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_16 32768
Definition at line 835 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_CDS_SIZE

#define DEFAULT_CFE_PLATFORM_ES_CDS_SIZE (128 * 1024)
Definition at line 310 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_DEFAULT_APP_LOG_FILE

#define DEFAULT_CFE_PLATFORM_ES_DEFAULT_APP_LOG_FILE "/ram/cfe_es_app_info.log"
Definition at line 397 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_DEFAULT_CDS_REG_DUMP_FILE

#define DEFAULT_CFE_PLATFORM_ES_DEFAULT_CDS_REG_DUMP_FILE "/ram/cfe_cds_reg.log"
Definition at line 476 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_DEFAULT_ER_LOG_FILE

#define DEFAULT_CFE_PLATFORM_ES_DEFAULT_ER_LOG_FILE "/ram/cfe_erlog.log"
Definition at line 446 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_DEFAULT_PERF_DUMP_FILENAME

#define DEFAULT_CFE_PLATFORM_ES_DEFAULT_PERF_DUMP_FILENAME "/ram/cfe_es_perf.dat"
Definition at line 461 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_DEFAULT_POR_SYSLOG_MODE

#define DEFAULT_CFE_PLATFORM_ES_DEFAULT_POR_SYSLOG_MODE 0
Definition at line 495 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_DEFAULT_PR_SYSLOG_MODE

#define DEFAULT_CFE_PLATFORM_ES_DEFAULT_PR_SYSLOG_MODE 1
Definition at line 514 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_DEFAULT_STACK_SIZE

#define DEFAULT_CFE_PLATFORM_ES_DEFAULT_STACK_SIZE 8192
Definition at line 676 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_DEFAULT_SYSLOG_FILE

#define DEFAULT_CFE_PLATFORM_ES_DEFAULT_SYSLOG_FILE "/ram/cfe_es_syslog.log"
Definition at line 430 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_DEFAULT_TASK_LOG_FILE

#define DEFAULT_CFE_PLATFORM_ES_DEFAULT_TASK_LOG_FILE "/ram/cfe_es_taskinfo.log"
Definition at line 413 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_ER_LOG_ENTRIES

#define DEFAULT_CFE_PLATFORM_ES_ER_LOG_ENTRIES 20
Definition at line 136 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_ER_LOG_MAX_CONTEXT_SIZE

#define DEFAULT_CFE_PLATFORM_ES_ER_LOG_MAX_CONTEXT_SIZE 256
Definition at line 151 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_MAX_APPLICATIONS

#define DEFAULT_CFE_PLATFORM_ES_MAX_APPLICATIONS 32
Definition at line 107 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_MAX_BLOCK_SIZE

#define DEFAULT_CFE_PLATFORM_ES_MAX_BLOCK_SIZE 80000
Definition at line 793 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_MAX_GEN_COUNTERS

#define DEFAULT_CFE_PLATFORM_ES_MAX_GEN_COUNTERS 8
Definition at line 179 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_MAX_LIBRARIES

#define DEFAULT_CFE_PLATFORM_ES_MAX_LIBRARIES 10
Definition at line 122 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_MAX_MEMORY_POOLS

#define DEFAULT_CFE_PLATFORM_ES_MAX_MEMORY_POOLS 10
Definition at line 741 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_MAX_PROCESSOR_RESETS

#define DEFAULT_CFE_PLATFORM_ES_MAX_PROCESSOR_RESETS 2
Definition at line 706 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_01

#define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_01 8
Definition at line 761 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_02

#define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_02 16
Definition at line 763 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_03

#define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_03 32
Definition at line 765 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_04

#define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_04 48
Definition at line 767 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_05

#define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_05 64
Definition at line 769 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_06

#define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_06 96
Definition at line 771 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_07

#define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_07 128
Definition at line 773 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_08

#define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_08 160
Definition at line 775 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_09

#define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_09 256
Definition at line 777 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_10

#define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_10 512
Definition at line 779 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_11

#define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_11 1024
Definition at line 781 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_12

#define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_12 2048
Definition at line 783 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_13

#define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_13 4096
Definition at line 785 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_14

#define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_14 8192
Definition at line 787 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_15

#define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_15 16384
Definition at line 789 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_16

#define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_16 32768
Definition at line 791 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_MEMPOOL_ALIGN_SIZE_MIN

#define DEFAULT_CFE_PLATFORM_ES_MEMPOOL_ALIGN_SIZE_MIN 4
Definition at line 351 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_NONVOL_DISK_MOUNT_STRING

#define DEFAULT_CFE_PLATFORM_ES_NONVOL_DISK_MOUNT_STRING "/cf"
Definition at line 73 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_NONVOL_STARTUP_FILE

#define DEFAULT_CFE_PLATFORM_ES_NONVOL_STARTUP_FILE "/cf/cfe_es_startup.scr"
Definition at line 366 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_PERF_CHILD_MS_DELAY

#define DEFAULT_CFE_PLATFORM_ES_PERF_CHILD_MS_DELAY 20
Definition at line 648 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_PERF_CHILD_PRIORITY

#define DEFAULT_CFE_PLATFORM_ES_PERF_CHILD_PRIORITY 200
Definition at line 617 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_PERF_CHILD_STACK_SIZE

#define DEFAULT_CFE_PLATFORM_ES_PERF_CHILD_STACK_SIZE 4096
Definition at line 632 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_PERF_DATA_BUFFER_SIZE

#define DEFAULT_CFE_PLATFORM_ES_PERF_DATA_BUFFER_SIZE 10000
Definition at line 531 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_PERF_ENTRIES_BTWN_DLYS

#define DEFAULT_CFE_PLATFORM_ES_PERF_ENTRIES_BTWN_DLYS 50
Definition at line 659 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_PERF_FILTMASK_ALL

#define DEFAULT_CFE_PLATFORM_ES_PERF_FILTMASK_ALL (~0)
Definition at line 553 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_PERF_FILTMASK_INIT

```
#define DEFAULT_CFE_PLATFORM_ES_PERF_FILTMASK_INIT (~0)
```

Definition at line 565 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_PERF_FILTMASK_NONE

```
#define DEFAULT_CFE_PLATFORM_ES_PERF_FILTMASK_NONE (0)
```

Definition at line 542 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_PERF_TRIGMASK_ALL

```
#define DEFAULT_CFE_PLATFORM_ES_PERF_TRIGMASK_ALL ~0
```

Definition at line 589 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_PERF_TRIGMASK_INIT

```
#define DEFAULT_CFE_PLATFORM_ES_PERF_TRIGMASK_INIT 0
```

Definition at line 601 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_PERF_TRIGMASK_NONE

```
#define DEFAULT_CFE_PLATFORM_ES_PERF_TRIGMASK_NONE 0
```

Definition at line 577 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_POOL_MAX_BUCKETS

```
#define DEFAULT_CFE_PLATFORM_ES_POOL_MAX_BUCKETS 17
```

Definition at line 722 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_RAM_DISK_MOUNT_STRING

```
#define DEFAULT_CFE_PLATFORM_ES_RAM_DISK_MOUNT_STRING "/ram"
```

Definition at line 90 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_RAM_DISK_NUM_SECTORS

```
#define DEFAULT_CFE_PLATFORM_ES_RAM_DISK_NUM_SECTORS 4096
```

Definition at line 267 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_RAM_DISK_PERCENT_RESERVED

```
#define DEFAULT_CFE_PLATFORM_ES_RAM_DISK_PERCENT_RESERVED 30
```

Definition at line 292 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_RAM_DISK_SECTOR_SIZE

```
#define DEFAULT_CFE_PLATFORM_ES_RAM_DISK_SECTOR_SIZE 512
```

Definition at line 248 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_START_TASK_PRIORITY

```
#define DEFAULT_CFE_PLATFORM_ES_START_TASK_PRIORITY 68
```

Definition at line 44 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_START_TASK_STACK_SIZE

```
#define DEFAULT_CFE_PLATFORM_ES_START_TASK_STACK_SIZE 8192
```

Definition at line 60 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_STARTUP_SCRIPT_TIMEOUT_MSEC

```
#define DEFAULT_CFE_PLATFORM_ES_STARTUP_SCRIPT_TIMEOUT_MSEC 1000
```

Definition at line 879 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_STARTUP_SYNC_POLL_MSEC

```
#define DEFAULT_CFE_PLATFORM_ES_STARTUP_SYNC_POLL_MSEC 50
```

Definition at line 860 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_SYSTEM_LOG_SIZE

```
#define DEFAULT_CFE_PLATFORM_ES_SYSTEM_LOG_SIZE 3072
```

Definition at line 167 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_USER_RESERVED_SIZE

```
#define DEFAULT_CFE_PLATFORM_ES_USER_RESERVED_SIZE (1024 * 1024)
```

Definition at line 331 of file [cfe_es_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_ES_VOLATILE_STARTUP_FILE

```
#define DEFAULT_CFE_PLATFORM_ES_VOLATILE_STARTUP_FILE "/ram/cfe_es_startup.scr"
```

Definition at line 381 of file [cfe_es_internal_cfg.h](#).

11.130 cfe_es_internal_cfg.h

[Go to the documentation of this file.](#)

```
00001 /*****
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"
00003  *
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00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00028 #ifndef CFE_ES_INTERNAL_CFG_H
00029 #define CFE_ES_INTERNAL_CFG_H
00030
00031 #include "cfe_es_mission_cfg.h"
00032 #include "cfe_es_internal_cfg_values.h"
00033
00043 #define CFE_PLATFORM_ES_START_TASK_PRIORITY          CFE_PLATFORM_ES_CFGVAL(START_TASK_PRIORITY)
00044 #define DEFAULT_CFE_PLATFORM_ES_START_TASK_PRIORITY 68
00045
00059 #define CFE_PLATFORM_ES_START_TASK_STACK_SIZE        CFE_PLATFORM_ES_CFGVAL(START_TASK_STACK_SIZE)
00060 #define DEFAULT_CFE_PLATFORM_ES_START_TASK_STACK_SIZE 8192
00061
00072 #define CFE_PLATFORM_ES_NONVOL_DISK_MOUNT_STRING      CFE_PLATFORM_ES_CFGVAL(NONVOL_DISK_MOUNT_STRING)
00073 #define DEFAULT_CFE_PLATFORM_ES_NONVOL_DISK_MOUNT_STRING "/cf"
00074
00089 #define CFE_PLATFORM_ES_RAM_DISK_MOUNT_STRING          CFE_PLATFORM_ES_CFGVAL(RAM_DISK_MOUNT_STRING)
```

```
00090 #define DEFAULT_CFE_PLATFORM_ES_RAM_DISK_MOUNT_STRING "/ram"
00091
00106 #define CFE_PLATFORM_ES_MAX_APPLICATIONS CFE_PLATFORM_ES_CFGVAL(MAX_APPLICATIONS)
00107 #define DEFAULT_CFE_PLATFORM_ES_MAX_APPLICATIONS 32
00108
00121 #define CFE_PLATFORM_ES_MAX_LIBRARIES CFE_PLATFORM_ES_CFGVAL(MAX_LIBRARIES)
00122 #define DEFAULT_CFE_PLATFORM_ES_MAX_LIBRARIES 10
00123
00135 #define CFE_PLATFORM_ES_ER_LOG_ENTRIES CFE_PLATFORM_ES_CFGVAL(ER_LOG_ENTRIES)
00136 #define DEFAULT_CFE_PLATFORM_ES_ER_LOG_ENTRIES 20
00137
00150 #define CFE_PLATFORM_ES_ER_LOG_MAX_CONTEXT_SIZE CFE_PLATFORM_ES_CFGVAL(ER_LOG_MAX_CONTEXT_SIZE)
00151 #define DEFAULT_CFE_PLATFORM_ES_ER_LOG_MAX_CONTEXT_SIZE 256
00152
00166 #define CFE_PLATFORM_ES_SYSTEM_LOG_SIZE CFE_PLATFORM_ES_CFGVAL(SYSTEM_LOG_SIZE)
00167 #define DEFAULT_CFE_PLATFORM_ES_SYSTEM_LOG_SIZE 3072
00168
00178 #define CFE_PLATFORM_ES_MAX_GEN_COUNTERS CFE_PLATFORM_ES_CFGVAL(MAX_GEN_COUNTERS)
00179 #define DEFAULT_CFE_PLATFORM_ES_MAX_GEN_COUNTERS 8
00180
00198 #define CFE_PLATFORM_ES_APP_SCAN_RATE CFE_PLATFORM_ES_CFGVAL(APP_SCAN_RATE)
00199 #define DEFAULT_CFE_PLATFORM_ES_APP_SCAN_RATE 1000
00200
00228 #define CFE_PLATFORM_ES_APP_KILL_TIMEOUT CFE_PLATFORM_ES_CFGVAL(APP_KILL_TIMEOUT)
00229 #define DEFAULT_CFE_PLATFORM_ES_APP_KILL_TIMEOUT 5
00230
00247 #define CFE_PLATFORM_ES_RAM_DISK_SECTOR_SIZE CFE_PLATFORM_ES_CFGVAL(RAM_DISK_SECTOR_SIZE)
00248 #define DEFAULT_CFE_PLATFORM_ES_RAM_DISK_SECTOR_SIZE 512
00249
00266 #define CFE_PLATFORM_ES_RAM_DISK_NUM_SECTORS CFE_PLATFORM_ES_CFGVAL(RAM_DISK_NUM_SECTORS)
00267 #define DEFAULT_CFE_PLATFORM_ES_RAM_DISK_NUM_SECTORS 4096
00268
00291 #define CFE_PLATFORM_ES_RAM_DISK_PERCENT_RESERVED
CFE_PLATFORM_ES_CFGVAL(RAM_DISK_PERCENT_RESERVED)
00292 #define DEFAULT_CFE_PLATFORM_ES_RAM_DISK_PERCENT_RESERVED 30
00293
00309 #define CFE_PLATFORM_ES_CDS_SIZE CFE_PLATFORM_ES_CFGVAL(CDS_SIZE)
00310 #define DEFAULT_CFE_PLATFORM_ES_CDS_SIZE (128 * 1024)
00311
00330 #define CFE_PLATFORM_ES_USER_RESERVED_SIZE CFE_PLATFORM_ES_CFGVAL(USER_RESERVED_SIZE)
00331 #define DEFAULT_CFE_PLATFORM_ES_USER_RESERVED_SIZE (1024 * 1024)
00332
00350 #define CFE_PLATFORM_ES_MEMPOOL_ALIGN_SIZE_MIN CFE_PLATFORM_ES_CFGVAL(MEMPOOL_ALIGN_SIZE_MIN)
00351 #define DEFAULT_CFE_PLATFORM_ES_MEMPOOL_ALIGN_SIZE_MIN 4
00352
00365 #define CFE_PLATFORM_ES_NONVOL_STARTUP_FILE CFE_PLATFORM_ES_CFGVAL(NONVOL_STARTUP_FILE)
00366 #define DEFAULT_CFE_PLATFORM_ES_NONVOL_STARTUP_FILE "/cf/cfe_es_startup.scr"
00367
00380 #define CFE_PLATFORM_ES_VOLATILE_STARTUP_FILE CFE_PLATFORM_ES_CFGVAL(VOLATILE_STARTUP_FILE)
00381 #define DEFAULT_CFE_PLATFORM_ES_VOLATILE_STARTUP_FILE "/ram/cfe_es_startup.scr"
00382
00396 #define CFE_PLATFORM_ES_DEFAULT_APP_LOG_FILE CFE_PLATFORM_ES_CFGVAL(DEFAULT_APP_LOG_FILE)
00397 #define DEFAULT_CFE_PLATFORM_ES_DEFAULT_APP_LOG_FILE "/ram/cfe_es_app_info.log"
00398
00412 #define CFE_PLATFORM_ES_DEFAULT_TASK_LOG_FILE CFE_PLATFORM_ES_CFGVAL(DEFAULT_TASK_LOG_FILE)
00413 #define DEFAULT_CFE_PLATFORM_ES_DEFAULT_TASK_LOG_FILE "/ram/cfe_es_taskinfo.log"
00414
00429 #define CFE_PLATFORM_ES_DEFAULT_SYSLOG_FILE CFE_PLATFORM_ES_CFGVAL(DEFAULT_SYSLOG_FILE)
00430 #define DEFAULT_CFE_PLATFORM_ES_DEFAULT_SYSLOG_FILE "/ram/cfe_es_syslog.log"
00431
00445 #define CFE_PLATFORM_ES_DEFAULT_ER_LOG_FILE CFE_PLATFORM_ES_CFGVAL(DEFAULT_ER_LOG_FILE)
00446 #define DEFAULT_CFE_PLATFORM_ES_DEFAULT_ER_LOG_FILE "/ram/cfe_erlog.log"
00447
00460 #define CFE_PLATFORM_ES_DEFAULT_PERF_DUMP_FILENAME
CFE_PLATFORM_ES_CFGVAL(DEFAULT_PERF_DUMP_FILENAME)
00461 #define DEFAULT_CFE_PLATFORM_ES_DEFAULT_PERF_DUMP_FILENAME "/ram/cfe_es_perf.dat"
00462
00475 #define CFE_PLATFORM_ES_DEFAULT_CDS_REG_DUMP_FILE
CFE_PLATFORM_ES_CFGVAL(DEFAULT_CDS_REG_DUMP_FILE)
00476 #define DEFAULT_CFE_PLATFORM_ES_DEFAULT_CDS_REG_DUMP_FILE "/ram/cfe_cds_reg.log"
00477
00494 #define CFE_PLATFORM_ES_DEFAULT_POR_SYSLOG_MODE CFE_PLATFORM_ES_CFGVAL(DEFAULT_POR_SYSLOG_MODE)
00495 #define DEFAULT_CFE_PLATFORM_ES_DEFAULT_POR_SYSLOG_MODE 0
00496
00513 #define CFE_PLATFORM_ES_DEFAULT_PR_SYSLOG_MODE CFE_PLATFORM_ES_CFGVAL(DEFAULT_PR_SYSLOG_MODE)
00514 #define DEFAULT_CFE_PLATFORM_ES_DEFAULT_PR_SYSLOG_MODE 1
00515
00530 #define CFE_PLATFORM_ES_PERF_DATA_BUFFER_SIZE CFE_PLATFORM_ES_CFGVAL(PERF_DATA_BUFFER_SIZE)
00531 #define DEFAULT_CFE_PLATFORM_ES_PERF_DATA_BUFFER_SIZE 10000
00532
00541 #define CFE_PLATFORM_ES_PERF_FILTMASK_NONE CFE_PLATFORM_ES_CFGVAL(PERF_FILTMASK_NONE)
```



```
00542 #define DEFAULT_CFE_PLATFORM_ES_PERF_FILTMASK_NONE (0)
00543
00552 #define CFE_PLATFORM_ES_PERF_FILTMASK_ALL CFE_PLATFORM_ES_CFGVAL(PERF_FILTMASK_ALL)
00553 #define DEFAULT_CFE_PLATFORM_ES_PERF_FILTMASK_ALL (~0)
00554
00564 #define CFE_PLATFORM_ES_PERF_FILTMASK_INIT CFE_PLATFORM_ES_CFGVAL(PERF_FILTMASK_INIT)
00565 #define DEFAULT_CFE_PLATFORM_ES_PERF_FILTMASK_INIT (~0)
00566
00576 #define CFE_PLATFORM_ES_PERF_TRIGMASK_NONE CFE_PLATFORM_ES_CFGVAL(PERF_TRIGMASK_NONE)
00577 #define DEFAULT_CFE_PLATFORM_ES_PERF_TRIGMASK_NONE 0
00578
00588 #define CFE_PLATFORM_ES_PERF_TRIGMASK_ALL CFE_PLATFORM_ES_CFGVAL(PERF_TRIGMASK_ALL)
00589 #define DEFAULT_CFE_PLATFORM_ES_PERF_TRIGMASK_ALL ~0
00590
00600 #define CFE_PLATFORM_ES_PERF_TRIGMASK_INIT CFE_PLATFORM_ES_CFGVAL(PERF_TRIGMASK_INIT)
00601 #define DEFAULT_CFE_PLATFORM_ES_PERF_TRIGMASK_INIT 0
00602
00616 #define CFE_PLATFORM_ES_PERF_CHILD_PRIORITY CFE_PLATFORM_ES_CFGVAL(PERF_CHILD_PRIORITY)
00617 #define DEFAULT_CFE_PLATFORM_ES_PERF_CHILD_PRIORITY 200
00618
00631 #define CFE_PLATFORM_ES_PERF_CHILD_STACK_SIZE CFE_PLATFORM_ES_CFGVAL(PERF_CHILD_STACK_SIZE)
00632 #define DEFAULT_CFE_PLATFORM_ES_PERF_CHILD_STACK_SIZE 4096
00633
00647 #define CFE_PLATFORM_ES_PERF_CHILD_MS_DELAY CFE_PLATFORM_ES_CFGVAL(PERF_CHILD_MS_DELAY)
00648 #define DEFAULT_CFE_PLATFORM_ES_PERF_CHILD_MS_DELAY 20
00649
00658 #define CFE_PLATFORM_ES_PERF_ENTRIES_BTWN_DLYS CFE_PLATFORM_ES_CFGVAL(PERF_ENTRIES_BTWN_DLYS)
00659 #define DEFAULT_CFE_PLATFORM_ES_PERF_ENTRIES_BTWN_DLYS 50
00660
00675 #define CFE_PLATFORM_ES_DEFAULT_STACK_SIZE CFE_PLATFORM_ES_CFGVAL(DEFAULT_STACK_SIZE)
00676 #define DEFAULT_CFE_PLATFORM_ES_DEFAULT_STACK_SIZE 8192
00677
00689 #define CFE_PLATFORM_ES_CDS_MAX_NUM_ENTRIES CFE_PLATFORM_ES_CFGVAL(CDS_MAX_NUM_ENTRIES)
00690 #define DEFAULT_CFE_PLATFORM_ES_CDS_MAX_NUM_ENTRIES 512
00691
00705 #define CFE_PLATFORM_ES_MAX_PROCESSOR_RESETS CFE_PLATFORM_ES_CFGVAL(MAX_PROCESSOR_RESETS)
00706 #define DEFAULT_CFE_PLATFORM_ES_MAX_PROCESSOR_RESETS 2
00707
00721 #define CFE_PLATFORM_ES_POOL_MAX_BUCKETS CFE_PLATFORM_ES_CFGVAL(POOL_MAX_BUCKETS)
00722 #define DEFAULT_CFE_PLATFORM_ES_POOL_MAX_BUCKETS 17
00723
00740 #define CFE_PLATFORM_ES_MAX_MEMORY_POOLS CFE_PLATFORM_ES_CFGVAL(MAX_MEMORY_POOLS)
00741 #define DEFAULT_CFE_PLATFORM_ES_MAX_MEMORY_POOLS 10
00742
00760 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_01 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_01)
00761 #define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_01 8
00762 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_02 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_02)
00763 #define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_02 16
00764 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_03 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_03)
00765 #define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_03 32
00766 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_04 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_04)
00767 #define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_04 48
00768 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_05 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_05)
00769 #define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_05 64
00770 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_06 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_06)
00771 #define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_06 96
00772 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_07 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_07)
00773 #define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_07 128
00774 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_08 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_08)
00775 #define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_08 160
00776 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_09 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_09)
00777 #define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_09 256
00778 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_10 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_10)
00779 #define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_10 512
00780 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_11 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_11)
00781 #define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_11 1024
00782 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_12 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_12)
00783 #define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_12 2048
00784 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_13 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_13)
00785 #define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_13 4096
00786 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_14 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_14)
00787 #define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_14 8192
00788 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_15 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_15)
00789 #define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_15 16384
00790 #define CFE_PLATFORM_ES_MEM_BLOCK_SIZE_16 CFE_PLATFORM_ES_CFGVAL(MEM_BLOCK_SIZE_16)
00791 #define DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_16 32768
00792 #define CFE_PLATFORM_ES_MAX_BLOCK_SIZE CFE_PLATFORM_ES_CFGVAL(MAX_BLOCK_SIZE)
00793 #define DEFAULT_CFE_PLATFORM_ES_MAX_BLOCK_SIZE 80000
00794
00804 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_01 CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_01)
00805 #define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_01 8
```

```

00806 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_02      CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_02)
00807 #define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_02 16
00808 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_03      CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_03)
00809 #define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_03 32
00810 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_04      CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_04)
00811 #define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_04 48
00812 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_05      CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_05)
00813 #define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_05 64
00814 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_06      CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_06)
00815 #define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_06 96
00816 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_07      CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_07)
00817 #define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_07 128
00818 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_08      CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_08)
00819 #define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_08 160
00820 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_09      CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_09)
00821 #define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_09 256
00822 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_10      CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_10)
00823 #define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_10 512
00824 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_11      CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_11)
00825 #define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_11 1024
00826 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_12      CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_12)
00827 #define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_12 2048
00828 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_13      CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_13)
00829 #define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_13 4096
00830 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_14      CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_14)
00831 #define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_14 8192
00832 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_15      CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_15)
00833 #define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_15 16384
00834 #define CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_16      CFE_PLATFORM_ES_CFGVAL(CDS_MEM_BLOCK_SIZE_16)
00835 #define DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_16 32768
00836 #define CFE_PLATFORM_ES_CDS_MAX_BLOCK_SIZE          CFE_PLATFORM_ES_CFGVAL(CDS_MAX_BLOCK_SIZE)
00837 #define DEFAULT_CFE_PLATFORM_ES_CDS_MAX_BLOCK_SIZE 80000
00838
00859 #define CFE_PLATFORM_ES_STARTUP_SYNC_POLL_MSEC      CFE_PLATFORM_ES_CFGVAL(STARTUP_SYNC_POLL_MSEC)
00860 #define DEFAULT_CFE_PLATFORM_ES_STARTUP_SYNC_POLL_MSEC 50
00861
00878 #define CFE_PLATFORM_ES_STARTUP_SCRIPT_TIMEOUT_MSEC
      CFE_PLATFORM_ES_CFGVAL(STARTUP_SCRIPT_TIMEOUT_MSEC)
00879 #define DEFAULT_CFE_PLATFORM_ES_STARTUP_SCRIPT_TIMEOUT_MSEC 1000
00880
00881 #endif

```

11.131 cfe/modules/es/fsw/inc/cfe_es_topicids.h File Reference

```
#include "cfe_es_topicid_values.h"
```

Macros

- #define CFE_MISSION_ES_CMD_TOPICID CFE_MISSION_ES_TIDVAL(CMD)
- #define DEFAULT_CFE_MISSION_ES_CMD_TOPICID 6
- #define CFE_MISSION_ES_SEND_HK_TOPICID CFE_MISSION_ES_TIDVAL(SEND_HK)
- #define DEFAULT_CFE_MISSION_ES_SEND_HK_TOPICID 8
- #define CFE_MISSION_ES_HK_TLM_TOPICID CFE_MISSION_ES_TIDVAL(HK_TLM)
- #define DEFAULT_CFE_MISSION_ES_HK_TLM_TOPICID 0
- #define CFE_MISSION_ES_APP_TLM_TOPICID CFE_MISSION_ES_TIDVAL(APP_TLM)
- #define DEFAULT_CFE_MISSION_ES_APP_TLM_TOPICID 11
- #define CFE_MISSION_ES_MEMSTATS_TLM_TOPICID CFE_MISSION_ES_TIDVAL(MEMSTATS_TLM)
- #define DEFAULT_CFE_MISSION_ES_MEMSTATS_TLM_TOPICID 16

11.131.1 Detailed Description

CFE Executive Services (CFE_ES) Application Topic IDs

Definition in file [cfe_es_topicids.h](#).

11.131.2 Macro Definition Documentation

CFE_MISSION_ES_APP_TLM_TOPICID

```
#define CFE_MISSION_ES_APP_TLM_TOPICID CFE_MISSION_ES_TIDVAL (APP_TLM)
```

Definition at line 53 of file [cfe_es_topicsids.h](#).

CFE_MISSION_ES_CMD_TOPICID

```
#define CFE_MISSION_ES_CMD_TOPICID CFE_MISSION_ES_TIDVAL (CMD)
```

Purpose cFE Portable Message Numbers for Commands

Description:

Portable message numbers for the cFE ES command messages

Limits

Not Applicable

Definition at line 37 of file [cfe_es_topicsids.h](#).

CFE_MISSION_ES_HK_TLM_TOPICID

```
#define CFE_MISSION_ES_HK_TLM_TOPICID CFE_MISSION_ES_TIDVAL (HK_TLM)
```

Purpose cFE Portable Message Numbers for Telemetry

Description:

Portable message numbers for the cFE ES telemetry messages

Limits

Not Applicable

Definition at line 51 of file [cfe_es_topicsids.h](#).

CFE_MISSION_ES_MEMSTATS_TLM_TOPICID

```
#define CFE_MISSION_ES_MEMSTATS_TLM_TOPICID CFE_MISSION_ES_TIDVAL (MEMSTATS_TLM)
```

Definition at line 55 of file [cfe_es_topicsids.h](#).

CFE_MISSION_ES_SEND_HK_TOPICID

```
#define CFE_MISSION_ES_SEND_HK_TOPICID CFE_MISSION_ES_TIDVAL (SEND_HK)
```

Definition at line 39 of file [cfe_es_topicsids.h](#).

DEFAULT_CFE_MISSION_ES_APP_TLM_TOPICID

```
#define DEFAULT_CFE_MISSION_ES_APP_TLM_TOPICID 11
```

Definition at line 54 of file [cfe_es_topicsids.h](#).

DEFAULT_CFE_MISSION_ES_CMD_TOPICID

```
#define DEFAULT_CFE_MISSION_ES_CMD_TOPICID 6
```

Definition at line 38 of file [cfe_es_topicsids.h](#).

DEFAULT_CFE_MISSION_ES_HK_TLM_TOPICID

```
#define DEFAULT_CFE_MISSION_ES_HK_TLM_TOPICID 0
```

Definition at line 52 of file [cfe_es_topics.h](#).

DEFAULT_CFE_MISSION_ES_MEMSTATS_TLM_TOPICID

```
#define DEFAULT_CFE_MISSION_ES_MEMSTATS_TLM_TOPICID 16
```

Definition at line 56 of file [cfe_es_topics.h](#).

DEFAULT_CFE_MISSION_ES_SEND_HK_TOPICID

```
#define DEFAULT_CFE_MISSION_ES_SEND_HK_TOPICID 8
```

Definition at line 40 of file [cfe_es_topics.h](#).

11.132 cfe_es_topics.h

[Go to the documentation of this file.](#)

```
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00017  *****/
00018
00023 #ifndef CFE_ES_TOPICS_H
00024 #define CFE_ES_TOPICS_H
00025
00026 #include "cfe_es_topicid_values.h"
00027
00037 #define CFE_MISSION_ES_CMD_TOPICID          CFE_MISSION_ES_TIDVAL(CMD)
00038 #define DEFAULT_CFE_MISSION_ES_CMD_TOPICID  6
00039 #define CFE_MISSION_ES_SEND_HK_TOPICID      CFE_MISSION_ES_TIDVAL(SEND_HK)
00040 #define DEFAULT_CFE_MISSION_ES_SEND_HK_TOPICID 8
00041
00051 #define CFE_MISSION_ES_HK_TLM_TOPICID          CFE_MISSION_ES_TIDVAL(HK_TLM)
00052 #define DEFAULT_CFE_MISSION_ES_HK_TLM_TOPICID 0
00053 #define CFE_MISSION_ES_APP_TLM_TOPICID         CFE_MISSION_ES_TIDVAL(APP_TLM)
00054 #define DEFAULT_CFE_MISSION_ES_APP_TLM_TOPICID 11
00055 #define CFE_MISSION_ES_MEMSTATS_TLM_TOPICID    CFE_MISSION_ES_TIDVAL(MEMSTATS_TLM)
00056 #define DEFAULT_CFE_MISSION_ES_MEMSTATS_TLM_TOPICID 16
00057
00058 #endif
```

11.133 cfe/modules/evs/config/default_cfe_evs_extern_typedefs.h File Reference

```
#include "common_types.h"
```

Typedefs

- typedef [uint8 CFE_EVS_MsgFormat_Enum_t](#)
Identifies format of log messages.
- typedef [uint8 CFE_EVS_LogMode_Enum_t](#)
Identifies handling of log messages after storage is filled.

- typedef `uint16 CFE_EVS_EventType_Enum_t`
Identifies type of event message.
- typedef `uint8 CFE_EVS_EventFilter_Enum_t`
Identifies event filter schemes.
- typedef `uint8 CFE_EVS_EventOutput_Enum_t`
Identifies event output port.

Enumerations

- enum `CFE_EVS_MsgFormat` { `CFE_EVS_MsgFormat_SHORT` = 0 , `CFE_EVS_MsgFormat_LONG` = 1 }
Label definitions associated with CFE_EVS_MsgFormat_Enum_t.
- enum `CFE_EVS_LogMode` { `CFE_EVS_LogMode_OVERWRITE` = 0 , `CFE_EVS_LogMode_DISCARD` = 1 }
Label definitions associated with CFE_EVS_LogMode_Enum_t.
- enum `CFE_EVS_EventType` { `CFE_EVS_EventType_DEBUG` = 1 , `CFE_EVS_EventType_INFORMATION` = 2 , `CFE_EVS_EventType_ERROR` = 3 , `CFE_EVS_EventType_CRITICAL` = 4 }
Label definitions associated with CFE_EVS_EventType_Enum_t.
- enum `CFE_EVS_EventFilter` { `CFE_EVS_EventFilter_BINARY` = 0 }
Label definitions associated with CFE_EVS_EventFilter_Enum_t.
- enum `CFE_EVS_EventOutput` { `CFE_EVS_EventOutput_PORT1` = 1 , `CFE_EVS_EventOutput_PORT2` = 2 , `CFE_EVS_EventOutput_PORT3` = 3 , `CFE_EVS_EventOutput_PORT4` = 4 }
Label definitions associated with CFE_EVS_EventOutput_Enum_t.

11.133.1 Detailed Description

Declarations and prototypes for `cfe_evs_extern_typedefs` module
Definition in file [default_cfe_evs_extern_typedefs.h](#).

11.133.2 Typedef Documentation

CFE_EVS_EventFilter_Enum_t

typedef `uint8 CFE_EVS_EventFilter_Enum_t`
Identifies event filter schemes.

See also

enum [CFE_EVS_EventFilter](#)

Definition at line 125 of file [default_cfe_evs_extern_typedefs.h](#).

CFE_EVS_EventOutput_Enum_t

typedef `uint8 CFE_EVS_EventOutput_Enum_t`
Identifies event output port.

See also

enum [CFE_EVS_EventOutput](#)

Definition at line 158 of file [default_cfe_evs_extern_typedefs.h](#).

CFE_EVS_EventType_Enum_t

```
typedef uint16 CFE_EVS_EventType_Enum_t
```

Identifies type of event message.

See also

enum [CFE_EVS_EventType](#)

Definition at line 107 of file [default_cfe_evs_extern_typedefs.h](#).

CFE_EVS_LogMode_Enum_t

```
typedef uint8 CFE_EVS_LogMode_Enum_t
```

Identifies handling of log messages after storage is filled.

See also

enum [CFE_EVS_LogMode](#)

Definition at line 74 of file [default_cfe_evs_extern_typedefs.h](#).

CFE_EVS_MsgFormat_Enum_t

```
typedef uint8 CFE_EVS_MsgFormat_Enum_t
```

Identifies format of log messages.

See also

enum [CFE_EVS_MsgFormat](#)

Definition at line 51 of file [default_cfe_evs_extern_typedefs.h](#).

11.133.3 Enumeration Type Documentation**CFE_EVS_EventFilter**

```
enum CFE_EVS_EventFilter
```

Label definitions associated with CFE_EVS_EventFilter_Enum_t.

Enumerator

CFE_EVS_EventFilter_BINARY	Binary event filter.
----------------------------	----------------------

Definition at line 112 of file [default_cfe_evs_extern_typedefs.h](#).

CFE_EVS_EventOutput

```
enum CFE_EVS_EventOutput
```

Label definitions associated with CFE_EVS_EventOutput_Enum_t.

Enumerator

CFE_EVS_EventOutput_PORT1	Output Port 1.
CFE_EVS_EventOutput_PORT2	Output Port 2.
CFE_EVS_EventOutput_PORT3	Output Port 3.
CFE_EVS_EventOutput_PORT4	Output Port 4.

Definition at line 130 of file [default_cfe_evs_extern_typedefs.h](#).

CFE_EVS_EventType

enum [CFE_EVS_EventType](#)

Label definitions associated with CFE_EVS_EventType_Enum_t.

Enumerator

CFE_EVS_EventType_DEBUG	Events that are intended only for debugging, not nominal operations.
CFE_EVS_EventType_INFORMATION	Events that identify a state change or action that is not an error.
CFE_EVS_EventType_ERROR	Events that identify an error but are not catastrophic (e.g. - bad command).
CFE_EVS_EventType_CRITICAL	Events that identify errors that are unrecoverable autonomously.

Definition at line 79 of file [default_cfe_evs_extern_typedefs.h](#).

CFE_EVS_LogMode

enum [CFE_EVS_LogMode](#)

Label definitions associated with CFE_EVS_LogMode_Enum_t.

Enumerator

CFE_EVS_LogMode_OVERWRITE	Overwrite Log Mode.
CFE_EVS_LogMode_DISCARD	Discard Log Mode.

Definition at line 56 of file [default_cfe_evs_extern_typedefs.h](#).

CFE_EVS_MsgFormat

enum [CFE_EVS_MsgFormat](#)

Label definitions associated with CFE_EVS_MsgFormat_Enum_t.

Enumerator

CFE_EVS_MsgFormat_SHORT	Short Format Messages.
CFE_EVS_MsgFormat_LONG	Long Format Messages.

Definition at line 33 of file [default_cfe_evs_extern_typedefs.h](#).

11.134 default_cfe_evs_extern_typedefs.h

[Go to the documentation of this file.](#)

```

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```

```

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00017  *****/
00018
00025 #ifndef DEFAULT_CFE_EVS_EXTERN_TYPEDEFS_H
00026 #define DEFAULT_CFE_EVS_EXTERN_TYPEDEFS_H
00027
00028 #include "common_types.h"
00029
00033 enum CFE_EVS_MsgFormat
00034 {
00038     CFE_EVS_MsgFormat_SHORT = 0,
00039
00043     CFE_EVS_MsgFormat_LONG = 1
00044 };
00045
00051 typedef uint8 CFE_EVS_MsgFormat_Enum_t;
00052
00056 enum CFE_EVS_LogMode
00057 {
00061     CFE_EVS_LogMode_OVERWRITE = 0,
00062
00066     CFE_EVS_LogMode_DISCARD = 1
00067 };
00068
00074 typedef uint8 CFE_EVS_LogMode_Enum_t;
00075
00079 enum CFE_EVS_EventType
00080 {
00084     CFE_EVS_EventType_DEBUG = 1,
00085
00089     CFE_EVS_EventType_INFORMATION = 2,
00090
00094     CFE_EVS_EventType_ERROR = 3,
00095
00099     CFE_EVS_EventType_CRITICAL = 4
00100 };
00101
00107 typedef uint16 CFE_EVS_EventType_Enum_t;
00108
00112 enum CFE_EVS_EventFilter
00113 {
00117     CFE_EVS_EventFilter_BINARY = 0
00118 };
00119
00125 typedef uint8 CFE_EVS_EventFilter_Enum_t;
00126
00130 enum CFE_EVS_EventOutput
00131 {
00135     CFE_EVS_EventOutput_PORT1 = 1,
00136
00140     CFE_EVS_EventOutput_PORT2 = 2,
00141
00145     CFE_EVS_EventOutput_PORT3 = 3,
00146
00150     CFE_EVS_EventOutput_PORT4 = 4
00151 };
00152
00158 typedef uint8 CFE_EVS_EventOutput_Enum_t;
00159
00160 #endif /* CFE_EVS_EXTERN_TYPEDEFS_H */

```

11.135 cfe/modules/evs/config/default_cfe_evs_fcncode_values.h File Reference

Macros

- #define CFE_EVS_CCVAL(x) CFE_EVS_FunctionCode_##x

Enumerations

- enum CFE_EVS_FunctionCode_ {
CFE_EVS_FunctionCode_NOOP = 0, CFE_EVS_FunctionCode_RESET_COUNTERS = 1, CFE_EVS_FunctionCode_ENABLE_EVENT_TYPE = 2, CFE_EVS_FunctionCode_DISABLE_EVENT_TYPE = 3, CFE_EVS_FunctionCode_SET_EVENT_FORMAT_MODE = 4, CFE_EVS_FunctionCode_ENABLE_APP_EVENT_TYPE = 5, CFE_EVS_FunctionCode_DISABLE_APP_EVENT_TYPE = 6, CFE_EVS_FunctionCode_ENABLE_APP_EVENTS


```

= 7 ,
CFE_EVS_FunctionCode_DISABLE_APP_EVENTS = 8 , CFE_EVS_FunctionCode_RESET_APP_COUNTER
= 9 , CFE_EVS_FunctionCode_SET_FILTER = 10 , CFE_EVS_FunctionCode_ENABLE_PORTS = 11 ,
CFE_EVS_FunctionCode_DISABLE_PORTS = 12 , CFE_EVS_FunctionCode_RESET_FILTER = 13 ,
CFE_EVS_FunctionCode_RESET_ALL_FILTERS = 14 , CFE_EVS_FunctionCode_ADD_EVENT_FILTER =
15 ,
CFE_EVS_FunctionCode_DELETE_EVENT_FILTER = 16 , CFE_EVS_FunctionCode_WRITE_APP_DATA_FILE
= 17 , CFE_EVS_FunctionCode_WRITE_LOG_DATA_FILE = 18 , CFE_EVS_FunctionCode_SET_LOG_MODE
= 19 ,
CFE_EVS_FunctionCode_CLEAR_LOG = 20 }

```

11.135.1 Detailed Description

Specification for the CFE Event Services (CFE_EVS) command function codes

Note

This file should be strictly limited to the command/function code (CC) macro definitions. Other definitions such as enums, typedefs, or other macros should be placed in the msgdefs.h or msg.h files.

Definition in file [default_cfe_evs_fcncode_values.h](#).

11.135.2 Macro Definition Documentation

CFE_EVS_CCVAL

```

#define CFE_EVS_CCVAL(
    x ) CFE_EVS_FunctionCode_##x

```

Definition at line 35 of file [default_cfe_evs_fcncode_values.h](#).

11.135.3 Enumeration Type Documentation

CFE_EVS_FunctionCode_

```
enum CFE_EVS_FunctionCode_
```

Enumerator

CFE_EVS_FunctionCode_NOOP	
CFE_EVS_FunctionCode_RESET_COUNTERS	
CFE_EVS_FunctionCode_ENABLE_EVENT_TYPE	
CFE_EVS_FunctionCode_DISABLE_EVENT_TYPE	
CFE_EVS_FunctionCode_SET_EVENT_FORMAT_MODE	
CFE_EVS_FunctionCode_ENABLE_APP_EVENT_TYPE	
CFE_EVS_FunctionCode_DISABLE_APP_EVENT_TYPE	
CFE_EVS_FunctionCode_ENABLE_APP_EVENTS	
CFE_EVS_FunctionCode_DISABLE_APP_EVENTS	
CFE_EVS_FunctionCode_RESET_APP_COUNTER	
CFE_EVS_FunctionCode_SET_FILTER	
CFE_EVS_FunctionCode_ENABLE_PORTS	
CFE_EVS_FunctionCode_DISABLE_PORTS	
CFE_EVS_FunctionCode_RESET_FILTER	
CFE_EVS_FunctionCode_RESET_ALL_FILTERS	
CFE_EVS_FunctionCode_ADD_EVENT_FILTER	
CFE_EVS_FunctionCode_DELETE_EVENT_FILTER	

Enumerator

CFE_EVS_FunctionCode_WRITE_APP_DATA_FILE	
CFE_EVS_FunctionCode_WRITE_LOG_DATA_FILE	
CFE_EVS_FunctionCode_SET_LOG_MODE	
CFE_EVS_FunctionCode_CLEAR_LOG	

Definition at line 37 of file [default_cfe_evs_fcncode_values.h](#).

11.136 default_cfe_evs_fcncode_values.h

[Go to the documentation of this file.](#)

```

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00016  * limitations under the License.
00017  *****/
00018
00028 #ifndef DEFAULT_CFE_EVS_FCNCODE_VALUES_H
00029 #define DEFAULT_CFE_EVS_FCNCODE_VALUES_H
00030
00031 /*****
00032  * Macro Definitions
00033  *****/
00034
00035 #define CFE_EVS_CCVAL(x) CFE_EVS_FunctionCode_##x
00036
00037 enum CFE_EVS_FunctionCode_
00038 {
00039     CFE_EVS_FunctionCode_NOOP = 0,
00040     CFE_EVS_FunctionCode_RESET_COUNTERS = 1,
00041     CFE_EVS_FunctionCode_ENABLE_EVENT_TYPE = 2,
00042     CFE_EVS_FunctionCode_DISABLE_EVENT_TYPE = 3,
00043     CFE_EVS_FunctionCode_SET_EVENT_FORMAT_MODE = 4,
00044     CFE_EVS_FunctionCode_ENABLE_APP_EVENT_TYPE = 5,
00045     CFE_EVS_FunctionCode_DISABLE_APP_EVENT_TYPE = 6,
00046     CFE_EVS_FunctionCode_ENABLE_APP_EVENTS = 7,
00047     CFE_EVS_FunctionCode_DISABLE_APP_EVENTS = 8,
00048     CFE_EVS_FunctionCode_RESET_APP_COUNTER = 9,
00049     CFE_EVS_FunctionCode_SET_FILTER = 10,
00050     CFE_EVS_FunctionCode_ENABLE_PORTS = 11,
00051     CFE_EVS_FunctionCode_DISABLE_PORTS = 12,
00052     CFE_EVS_FunctionCode_RESET_FILTER = 13,
00053     CFE_EVS_FunctionCode_RESET_ALL_FILTERS = 14,
00054     CFE_EVS_FunctionCode_ADD_EVENT_FILTER = 15,
00055     CFE_EVS_FunctionCode_DELETE_EVENT_FILTER = 16,
00056     CFE_EVS_FunctionCode_WRITE_APP_DATA_FILE = 17,
00057     CFE_EVS_FunctionCode_WRITE_LOG_DATA_FILE = 18,
00058     CFE_EVS_FunctionCode_SET_LOG_MODE = 19,
00059     CFE_EVS_FunctionCode_CLEAR_LOG = 20,
00060 };
00061
00062 #endif

```

11.137 cfe/modules/evs/config/default_cfe_evs_interface_cfg_values.h File Reference

Macros

- #define [CFE_MISSION_EVS_CFGVAL\(x\)](#) DEFAULT_CFE_MISSION_EVS_##x

11.137.1 Detailed Description

CFE Event Services (CFE_EVS) Application Public Definitions

This provides default values for configurable items that affect the interface(s) of this module. This includes the CMD/TLM message interface, tables definitions, and any other data products that serve to exchange information with other entities.

Note

This file may be overridden/superceded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [default_cfe_evs_interface_cfg_values.h](#).

11.137.2 Macro Definition Documentation

CFE_MISSION_EVS_CFGVAL

```
#define CFE_MISSION_EVS_CFGVAL(  
    x ) DEFAULT_CFE_MISSION_EVS_##x
```

Definition at line 36 of file [default_cfe_evs_interface_cfg_values.h](#).

11.138 default_cfe_evs_interface_cfg_values.h

[Go to the documentation of this file.](#)

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00016  * limitations under the License.  
00017  *****/  
00018  
00032 #ifndef DEFAULT_CFE_EVS_INTERFACE_CFG_VALUES_H  
00033 #define DEFAULT_CFE_EVS_INTERFACE_CFG_VALUES_H  
00034  
00035 /* Use the default configuration value for all */  
00036 #define CFE_MISSION_EVS_CFGVAL(x) DEFAULT_CFE_MISSION_EVS_##x  
00037  
00038 #endif
```

11.139 cfe/modules/evs/config/default_cfe_evs_internal_cfg_values.h File Reference

Macros

- #define [CFE_PLATFORM_EVS_CFGVAL](#)(x) DEFAULT_CFE_PLATFORM_EVS_##x

11.139.1 Detailed Description

CFE Event Services (CFE_EVS) Application Private Config Definitions

This provides default values for configurable items that are internal to this module and do NOT affect the interface(s) of this module. Changes to items in this file only affect the local module and will be transparent to external entities that are using the public interface(s).

Note

This file may be overridden/superseded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [default_cfe_evs_internal_cfg_values.h](#).

11.139.2 Macro Definition Documentation**CFE_PLATFORM_EVS_CFGVAL**

```
#define CFE_PLATFORM_EVS_CFGVAL(  
    x ) DEFAULT_CFE_PLATFORM_EVS_##x
```

Definition at line 36 of file [default_cfe_evs_internal_cfg_values.h](#).

11.140 default_cfe_evs_internal_cfg_values.h

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00017  *****/  
00018  
00032 #ifndef DEFAULT_CFE_EVS_INTERNAL_CFG_VALUES_H  
00033 #define DEFAULT_CFE_EVS_INTERNAL_CFG_VALUES_H  
00034  
00035 /* Use the default configuration value for all */  
00036 #define CFE_PLATFORM_EVS_CFGVAL(x) DEFAULT_CFE_PLATFORM_EVS_##x  
00037  
00038 #endif
```

11.141 cfe/modules/evs/config/default_cfe_evs_mission_cfg.h File Reference

```
#include "cfe_evs_interface_cfg.h"
```

11.141.1 Detailed Description

CFE Event Services (CFE_EVS) Application Mission Configuration Header File

This is a compatibility header for the "mission_cfg.h" file that has traditionally provided public config definitions for each CFS app.

Note

This file may be overridden/superseded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [default_cfe_evs_mission_cfg.h](#).

11.142 default_cfe_evs_mission_cfg.h

[Go to the documentation of this file.](#)

```
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```

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00017  *****/
00018
00031 #ifndef DEFAULT_CFE_EVS_MISSION_CFG_H
00032 #define DEFAULT_CFE_EVS_MISSION_CFG_H
00033
00034 #include "cfe_evs_interface_cfg.h"
00035
00036 #endif

```

11.143 cfe/modules/evs/config/default_cfe_evs_msg.h File Reference

```

#include "cfe_mission_cfg.h"
#include "cfe_evs_fcncodes.h"
#include "cfe_evs_msgdefs.h"
#include "cfe_evs_msgstruct.h"

```

11.143.1 Detailed Description

Specification for the CFE Event Services (CFE_EVS) command and telemetry message data types.

This is a compatibility header for the "cfe_evs_msg.h" file that has traditionally provided the message definitions for cFS apps.

Note

This file may be overridden/superceded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [default_cfe_evs_msg.h](#).

11.144 default_cfe_evs_msg.h

[Go to the documentation of this file.](#)

```

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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
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00016  * limitations under the License.
00017  *****/
00018
00031 #ifndef DEFAULT_CFE_EVS_MSG_H
00032 #define DEFAULT_CFE_EVS_MSG_H
00033
00034 #include "cfe_mission_cfg.h"

```

```

00035 #include "cfe_evs_fcncodes.h"
00036 #include "cfe_evs_msgdefs.h"
00037 #include "cfe_evs_msgstruct.h"
00038
00039 #endif

```

11.145 cfe/modules/evs/config/default_cfe_evs_msgdefs.h File Reference

```

#include "common_types.h"
#include "cfe_mission_cfg.h"
#include "cfe_es_extern_typedefs.h"
#include "cfe_evs_extern_typedefs.h"
#include "cfe_evs_fcncodes.h"

```

Data Structures

- struct [CFE_EVS_LogFileCmd_Payload](#)
Write Event Log to File Command Payload.
- struct [CFE_EVS_AppDataCmd_Payload](#)
Write Event Services Application Information to File Command Payload.
- struct [CFE_EVS_SetLogMode_Payload](#)
Set Log Mode Command Payload.
- struct [CFE_EVS_SetEventFormatCode_Payload](#)
Set Event Format Mode Command Payload.
- struct [CFE_EVS_BitMaskCmd_Payload](#)
Generic Bitmask Command Payload.
- struct [CFE_EVS_AppNameCmd_Payload](#)
Generic App Name Command Payload.
- struct [CFE_EVS_AppNameEventIDCmd_Payload](#)
Generic App Name and Event ID Command Payload.
- struct [CFE_EVS_AppNameBitMaskCmd_Payload](#)
Generic App Name and Bitmask Command Payload.
- struct [CFE_EVS_AppNameEventIDMaskCmd_Payload](#)
Generic App Name, Event ID, Mask Command Payload.
- struct [CFE_EVS_AppTlmData](#)
- struct [CFE_EVS_HousekeepingTlm_Payload](#)
- struct [CFE_EVS_PacketID](#)
- struct [CFE_EVS_LongEventTlm_Payload](#)
- struct [CFE_EVS_ShortEventTlm_Payload](#)

Macros

- #define [CFE_EVS_DEBUG_BIT](#) (1 << (CFE_EVS_EventType_DEBUG - 1))
- #define [CFE_EVS_INFORMATION_BIT](#) (1 << (CFE_EVS_EventType_INFORMATION - 1))
- #define [CFE_EVS_ERROR_BIT](#) (1 << (CFE_EVS_EventType_ERROR - 1))
- #define [CFE_EVS_CRITICAL_BIT](#) (1 << (CFE_EVS_EventType_CRITICAL - 1))
- #define [CFE_EVS_PORT1_BIT](#) (1 << (CFE_EVS_EventOutput_PORT1 - 1))
- #define [CFE_EVS_PORT2_BIT](#) (1 << (CFE_EVS_EventOutput_PORT2 - 1))
- #define [CFE_EVS_PORT3_BIT](#) (1 << (CFE_EVS_EventOutput_PORT3 - 1))
- #define [CFE_EVS_PORT4_BIT](#) (1 << (CFE_EVS_EventOutput_PORT4 - 1))

Typedefs

- typedef struct [CFE_EVS_LogFileCmd_Payload](#) [CFE_EVS_LogFileCmd_Payload_t](#)
Write Event Log to File Command Payload.
- typedef struct [CFE_EVS_AppDataCmd_Payload](#) [CFE_EVS_AppDataCmd_Payload_t](#)
Write Event Services Application Information to File Command Payload.
- typedef struct [CFE_EVS_SetLogMode_Payload](#) [CFE_EVS_SetLogMode_Payload_t](#)
Set Log Mode Command Payload.
- typedef struct [CFE_EVS_SetEventFormatCode_Payload](#) [CFE_EVS_SetEventFormatMode_Payload_t](#)
Set Event Format Mode Command Payload.
- typedef struct [CFE_EVS_BitMaskCmd_Payload](#) [CFE_EVS_BitMaskCmd_Payload_t](#)
Generic Bitmask Command Payload.
- typedef struct [CFE_EVS_AppNameCmd_Payload](#) [CFE_EVS_AppNameCmd_Payload_t](#)
Generic App Name Command Payload.
- typedef struct [CFE_EVS_AppNameEventIDCmd_Payload](#) [CFE_EVS_AppNameEventIDCmd_Payload_t](#)
Generic App Name and Event ID Command Payload.
- typedef struct [CFE_EVS_AppNameBitMaskCmd_Payload](#) [CFE_EVS_AppNameBitMaskCmd_Payload_t](#)
Generic App Name and Bitmask Command Payload.
- typedef struct [CFE_EVS_AppNameEventIDMaskCmd_Payload](#) [CFE_EVS_AppNameEventIDMaskCmd_Payload_t](#)
Generic App Name, Event ID, Mask Command Payload.
- typedef struct [CFE_EVS_AppTlmData](#) [CFE_EVS_AppTlmData_t](#)
- typedef struct [CFE_EVS_HousekeepingTlm_Payload](#) [CFE_EVS_HousekeepingTlm_Payload_t](#)
- typedef struct [CFE_EVS_PacketID](#) [CFE_EVS_PacketID_t](#)
- typedef struct [CFE_EVS_LongEventTlm_Payload](#) [CFE_EVS_LongEventTlm_Payload_t](#)
- typedef struct [CFE_EVS_ShortEventTlm_Payload](#) [CFE_EVS_ShortEventTlm_Payload_t](#)

11.145.1 Detailed Description

Specification for the CFE Event Services (CFE_EVS) command and telemetry message payload structures and constant definitions.

Definition in file [default_cfe_evs_msgdefs.h](#).

11.145.2 Macro Definition Documentation

CFE_EVS_CRITICAL_BIT

```
#define CFE_EVS_CRITICAL_BIT (1 << (CFE_EVS_EventType_CRITICAL - 1))
```

Definition at line 37 of file [default_cfe_evs_msgdefs.h](#).

CFE_EVS_DEBUG_BIT

```
#define CFE_EVS_DEBUG_BIT (1 << (CFE_EVS_EventType_DEBUG - 1))
```

Definition at line 34 of file [default_cfe_evs_msgdefs.h](#).

CFE_EVS_ERROR_BIT

```
#define CFE_EVS_ERROR_BIT (1 << (CFE_EVS_EventType_ERROR - 1))
```

Definition at line 36 of file [default_cfe_evs_msgdefs.h](#).

CFE_EVS_INFORMATION_BIT

```
#define CFE_EVS_INFORMATION_BIT (1 << (CFE_EVS_EventType_INFORMATION - 1))
```

Definition at line 35 of file [default_cfe_evs_msgdefs.h](#).

CFE_EVS_PORT1_BIT

```
#define CFE_EVS_PORT1_BIT (1 << (CFE_EVS_EventOutput_PORT1 - 1))
```

Definition at line 40 of file [default_cfe_evs_msgdefs.h](#).

CFE_EVS_PORT2_BIT

```
#define CFE_EVS_PORT2_BIT (1 << (CFE_EVS_EventOutput_PORT2 - 1))
```

Definition at line 41 of file [default_cfe_evs_msgdefs.h](#).

CFE_EVS_PORT3_BIT

```
#define CFE_EVS_PORT3_BIT (1 << (CFE_EVS_EventOutput_PORT3 - 1))
```

Definition at line 42 of file [default_cfe_evs_msgdefs.h](#).

CFE_EVS_PORT4_BIT

```
#define CFE_EVS_PORT4_BIT (1 << (CFE_EVS_EventOutput_PORT4 - 1))
```

Definition at line 43 of file [default_cfe_evs_msgdefs.h](#).

11.145.3 Typedef Documentation**CFE_EVS_AppDataCmd_Payload_t**

```
typedef struct CFE_EVS_AppDataCmd_Payload CFE_EVS_AppDataCmd_Payload_t
```

Write Event Services Application Information to File Command Payload.
For command details, see [CFE_EVS_WRITE_APP_DATA_FILE_CC](#)

CFE_EVS_AppNameBitMaskCmd_Payload_t

```
typedef struct CFE_EVS_AppNameBitMaskCmd_Payload CFE_EVS_AppNameBitMaskCmd_Payload_t
```

Generic App Name and Bitmask Command Payload.
For command details, see [CFE_EVS_ENABLE_APP_EVENT_TYPE_CC](#) and/or [CFE_EVS_DISABLE_APP_EVENT_TYPE_CC](#)

CFE_EVS_AppNameCmd_Payload_t

```
typedef struct CFE_EVS_AppNameCmd_Payload CFE_EVS_AppNameCmd_Payload_t
```

Generic App Name Command Payload.
For command details, see [CFE_EVS_ENABLE_APP_EVENTS_CC](#), [CFE_EVS_DISABLE_APP_EVENTS_CC](#), [CFE_EVS_RESET_APP_COUNTER_CC](#) and/or [CFE_EVS_RESET_ALL_FILTERS_CC](#)

CFE_EVS_AppNameEventIDCmd_Payload_t

```
typedef struct CFE_EVS_AppNameEventIDCmd_Payload CFE_EVS_AppNameEventIDCmd_Payload_t
```

Generic App Name and Event ID Command Payload.
For command details, see [CFE_EVS_RESET_FILTER_CC](#) and [CFE_EVS_DELETE_EVENT_FILTER_CC](#)

CFE_EVS_AppNameEventIDMaskCmd_Payload_t

```
typedef struct CFE_EVS_AppNameEventIDMaskCmd_Payload CFE_EVS_AppNameEventIDMaskCmd_Payload_t
```

Generic App Name, Event ID, Mask Command Payload.
For command details, see [CFE_EVS_SET_FILTER_CC](#), [CFE_EVS_ADD_EVENT_FILTER_CC](#) and/or [CFE_EVS_DELETE_EVENT_FILTER_CC](#)

CFE_EVS_AppTlmData_t

```
typedef struct CFE_EVS_AppTlmData CFE_EVS_AppTlmData_t
```

CFE_EVS_BitMaskCmd_Payload_t

```
typedef struct CFE_EVS_BitMaskCmd_Payload CFE_EVS_BitMaskCmd_Payload_t
```

Generic Bitmask Command Payload.

For command details, see [CFE_EVS_ENABLE_EVENT_TYPE_CC](#), [CFE_EVS_DISABLE_EVENT_TYPE_CC](#), [CFE_EVS_ENABLE_PORTS_CC](#) and/or [CFE_EVS_DISABLE_PORTS_CC](#)

CFE_EVS_HousekeepingTlm_Payload_t

```
typedef struct CFE_EVS_HousekeepingTlm_Payload CFE_EVS_HousekeepingTlm_Payload_t
```

Name Event Services Housekeeping Telemetry Packet

CFE_EVS_LogFileCmd_Payload_t

```
typedef struct CFE_EVS_LogFileCmd_Payload CFE_EVS_LogFileCmd_Payload_t
```

Write Event Log to File Command Payload.

For command details, see [CFE_EVS_WRITE_LOG_DATA_FILE_CC](#)

CFE_EVS_LongEventTlm_Payload_t

```
typedef struct CFE_EVS_LongEventTlm_Payload CFE_EVS_LongEventTlm_Payload_t
```

Name Event Message Telemetry Packet (Long format)

CFE_EVS_PacketID_t

```
typedef struct CFE_EVS_PacketID CFE_EVS_PacketID_t
```

Telemetry packet structures

CFE_EVS_SetEventFormatMode_Payload_t

```
typedef struct CFE_EVS_SetEventFormatCode_Payload CFE_EVS_SetEventFormatMode_Payload_t
```

Set Event Format Mode Command Payload.

For command details, see [CFE_EVS_SET_EVENT_FORMAT_MODE_CC](#)

CFE_EVS_SetLogMode_Payload_t

```
typedef struct CFE_EVS_SetLogMode_Payload CFE_EVS_SetLogMode_Payload_t
```

Set Log Mode Command Payload.

For command details, see [CFE_EVS_SET_LOG_MODE_CC](#)

CFE_EVS_ShortEventTlm_Payload_t

```
typedef struct CFE_EVS_ShortEventTlm_Payload CFE_EVS_ShortEventTlm_Payload_t
```

Name Event Message Telemetry Packet (Short format)

11.146 default_cfe_evs_msgdefs.h

[Go to the documentation of this file.](#)

```

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00016  * limitations under the License.
00017  *****/
00018
00024 #ifndef DEFAULT_CFE_EVS_MSGDEFS_H
00025 #define DEFAULT_CFE_EVS_MSGDEFS_H
00026
00027 #include "common_types.h"
00028 #include "cfe_mission_cfg.h"
00029 #include "cfe_es_extern_typedefs.h"
00030 #include "cfe_evs_extern_typedefs.h"
00031 #include "cfe_evs_fcncodes.h"
00032
00033 /* Event Type bit masks */
00034 #define CFE_EVS_DEBUG_BIT (1 < (CFE_EVS_EventType_DEBUG - 1)) // 0x0001
00035 #define CFE_EVS_INFORMATION_BIT (1 < (CFE_EVS_EventType_INFORMATION - 1)) // 0x0002
00036 #define CFE_EVS_ERROR_BIT (1 < (CFE_EVS_EventType_ERROR - 1)) // 0x0004
00037 #define CFE_EVS_CRITICAL_BIT (1 < (CFE_EVS_EventType_CRITICAL - 1)) // 0x0008
00038
00039 /* Output Port bit masks */
00040 #define CFE_EVS_PORT1_BIT (1 < (CFE_EVS_EventOutput_PORT1 - 1)) // 0x0001
00041 #define CFE_EVS_PORT2_BIT (1 < (CFE_EVS_EventOutput_PORT2 - 1)) // 0x0002
00042 #define CFE_EVS_PORT3_BIT (1 < (CFE_EVS_EventOutput_PORT3 - 1)) // 0x0004
00043 #define CFE_EVS_PORT4_BIT (1 < (CFE_EVS_EventOutput_PORT4 - 1)) // 0x0008
00044
00045 /*****
00046  * Command Message Data Payloads */
00047 /*****
00048
00055 typedef struct CFE_EVS_LogFileCmd_Payload
00056 {
00057     char LogFilename[CFE_MISSION_MAX_PATH_LEN];
00058 } CFE_EVS_LogFileCmd_Payload_t;
00059
00066 typedef struct CFE_EVS_AppDataCmd_Payload
00067 {
00068     char AppDataFilename[CFE_MISSION_MAX_PATH_LEN];
00069 } CFE_EVS_AppDataCmd_Payload_t;
00070
00077 typedef struct CFE_EVS_SetLogMode_Payload
00078 {
00079     CFE_EVS_LogMode_Enum_t LogMode;
00080     uint8 Spare;
00081 } CFE_EVS_SetLogMode_Payload_t;
00082
00089 typedef struct CFE_EVS_SetEventFormatCode_Payload
00090 {
00091     CFE_EVS_MsgFormat_Enum_t MsgFormat;
00092     uint8 Spare;
00093 } CFE_EVS_SetEventFormatMode_Payload_t;
00094
00102 typedef struct CFE_EVS_BitMaskCmd_Payload
00103 {
00104     uint8 BitMask;
00105     uint8 Spare;
00106 } CFE_EVS_BitMaskCmd_Payload_t;
00107
00115 typedef struct CFE_EVS_AppNameCmd_Payload
00116 {
00117     char AppName[CFE_MISSION_MAX_API_LEN];
00118 } CFE_EVS_AppNameCmd_Payload_t;
00119
00126 typedef struct CFE_EVS_AppNameEventIDCmd_Payload

```

```

00127 {
00128     char    AppName[CFE_MISSION_MAX_API_LEN];
00129     uint16  EventID;
00130 } CFE_EVS_AppNameEventIDCmd_Payload_t;
00131
00132 typedef struct CFE_EVS_AppNameBitMaskCmd_Payload
00133 {
00134     char    AppName[CFE_MISSION_MAX_API_LEN];
00135     uint8   BitMask;
00136     uint8   Spare;
00137 } CFE_EVS_AppNameBitMaskCmd_Payload_t;
00138
00139 typedef struct CFE_EVS_AppNameEventIDMaskCmd_Payload
00140 {
00141     char    AppName[CFE_MISSION_MAX_API_LEN];
00142     uint16  EventID;
00143     uint16  Mask;
00144 } CFE_EVS_AppNameEventIDMaskCmd_Payload_t;
00145
00146 /*****
00147 /*****
00148 /* Telemetry Message Data Payloads */
00149 /*****
00150
00151 typedef struct CFE_EVS_AppTlmData
00152 {
00153     CFE_ES_AppId_t AppID;
00154     uint16         AppMessageSentCounter;
00155     uint8          AppEnableStatus;
00156     uint8          AppMessageSquelchedCounter;
00157 } CFE_EVS_AppTlmData_t;
00158
00159 typedef struct CFE_EVS_HousekeepingTlm_Payload
00160 {
00161     uint8 CommandCounter;
00162     uint8 CommandErrorCounter;
00163     uint8 MessageFormatMode;
00164     uint8 MessageTruncCounter;
00165     uint8 UnregisteredAppCounter;
00166     uint8 OutputPort;
00167     uint8 LogFullFlag;
00168     uint8 LogMode;
00169     uint16 MessageSendCounter;
00170     uint16 LogOverflowCounter;
00171     uint8 LogEnabled;
00172     uint8 Padding[3];
00173     CFE_EVS_AppTlmData_t AppData[CFE_MISSION_ES_MAX_APPLICATIONS];
00174 } CFE_EVS_HousekeepingTlm_Payload_t;
00175
00176 typedef struct CFE_EVS_PacketID
00177 {
00178     char    AppName[CFE_MISSION_MAX_API_LEN];
00179     uint16  EventID;
00180     CFE_EVS_EventType_Enum_t EventType;
00181     uint32  SpacecraftID;
00182     uint32  ProcessorID;
00183 } CFE_EVS_PacketID_t;
00184
00185 typedef struct CFE_EVS_LongEventTlm_Payload
00186 {
00187     CFE_EVS_PacketID_t PacketID;
00188     char    Message[CFE_MISSION_EVS_MAX_MESSAGE_LENGTH];
00189     uint8   Padding[2];
00190 } CFE_EVS_LongEventTlm_Payload_t;
00191
00192 typedef struct CFE_EVS_ShortEventTlm_Payload
00193 {
00194     CFE_EVS_PacketID_t PacketID;
00195 } CFE_EVS_ShortEventTlm_Payload_t;
00196
00197 #endif

```

11.147 cfe/modules/evs/config/default_cfe_evs_msgid_values.h File Reference

```

#include "cfe_core_api_base_msgids.h"
#include "cfe_evs_topicids.h"

```

Macros

- #define CFE_PLATFORM_EVS_CMD_MIDVAL(x) CFE_PLATFORM_CMD_TOPICID_TO_MIDV(CFE_MISSION_EVS_##x##_TOPICID)
- #define CFE_PLATFORM_EVS_TLM_MIDVAL(x) CFE_PLATFORM_TLM_TOPICID_TO_MIDV(CFE_MISSION_EVS_##x##_TOPICID)

11.147.1 Detailed Description

CFE Event Services (CFE_EVS) Application Message IDs
 Definition in file [default_cfe_evs_msgid_values.h](#).

11.147.2 Macro Definition Documentation**CFE_PLATFORM_EVS_CMD_MIDVAL**

```
#define CFE_PLATFORM_EVS_CMD_MIDVAL(
    x ) CFE_PLATFORM_CMD_TOPICID_TO_MIDV(CFE_MISSION_EVS_##x##_TOPICID)
```

Definition at line 29 of file [default_cfe_evs_msgid_values.h](#).

CFE_PLATFORM_EVS_TLM_MIDVAL

```
#define CFE_PLATFORM_EVS_TLM_MIDVAL(
    x ) CFE_PLATFORM_TLM_TOPICID_TO_MIDV(CFE_MISSION_EVS_##x##_TOPICID)
```

Definition at line 30 of file [default_cfe_evs_msgid_values.h](#).

11.148 default_cfe_evs_msgid_values.h

[Go to the documentation of this file.](#)

```
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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00023 #ifndef DEFAULT_CFE_EVS_MSGID_VALUES_H
00024 #define DEFAULT_CFE_EVS_MSGID_VALUES_H
00025
00026 #include "cfe_core_api_base_msgids.h"
00027 #include "cfe_evs_topicids.h"
00028
00029 #define CFE_PLATFORM_EVS_CMD_MIDVAL(x) CFE_PLATFORM_CMD_TOPICID_TO_MIDV(CFE_MISSION_EVS_##x##_TOPICID)
00030 #define CFE_PLATFORM_EVS_TLM_MIDVAL(x) CFE_PLATFORM_TLM_TOPICID_TO_MIDV(CFE_MISSION_EVS_##x##_TOPICID)
00031
00032 #endif
```

11.149 cfe/modules/evs/config/default_cfe_evs_msgids.h File Reference

```
#include "cfe_evs_msgid_values.h"
```

Macros

- `#define CFE_EVS_CMD_MID CFE_PLATFORM_EVS_CMD_MIDVAL(CMD)`
- `#define CFE_EVS_SEND_HK_MID CFE_PLATFORM_EVS_CMD_MIDVAL(SEND_HK)`
- `#define CFE_EVS_HK_TLM_MID CFE_PLATFORM_EVS_TLM_MIDVAL(HK_TLM)`
- `#define CFE_EVS_LONG_EVENT_MSG_MID CFE_PLATFORM_EVS_TLM_MIDVAL(LONG_EVENT_MSG)`
- `#define CFE_EVS_SHORT_EVENT_MSG_MID CFE_PLATFORM_EVS_TLM_MIDVAL(SHORT_EVENT_MSG)`

11.149.1 Detailed Description

CFE Event Services (CFE_EVS) Application Message IDs

Definition in file [default_cfe_evs_msgids.h](#).

11.149.2 Macro Definition Documentation

CFE_EVS_CMD_MID

```
#define CFE_EVS_CMD_MID CFE_PLATFORM_EVS_CMD_MIDVAL(CMD)
```

Definition at line 32 of file [default_cfe_evs_msgids.h](#).

CFE_EVS_HK_TLM_MID

```
#define CFE_EVS_HK_TLM_MID CFE_PLATFORM_EVS_TLM_MIDVAL(HK_TLM)
```

Definition at line 38 of file [default_cfe_evs_msgids.h](#).

CFE_EVS_LONG_EVENT_MSG_MID

```
#define CFE_EVS_LONG_EVENT_MSG_MID CFE_PLATFORM_EVS_TLM_MIDVAL(LONG_EVENT_MSG)
```

Definition at line 39 of file [default_cfe_evs_msgids.h](#).

CFE_EVS_SEND_HK_MID

```
#define CFE_EVS_SEND_HK_MID CFE_PLATFORM_EVS_CMD_MIDVAL(SEND_HK)
```

Definition at line 33 of file [default_cfe_evs_msgids.h](#).

CFE_EVS_SHORT_EVENT_MSG_MID

```
#define CFE_EVS_SHORT_EVENT_MSG_MID CFE_PLATFORM_EVS_TLM_MIDVAL(SHORT_EVENT_MSG)
```

Definition at line 40 of file [default_cfe_evs_msgids.h](#).

11.150 default_cfe_evs_msgids.h

[Go to the documentation of this file.](#)

```
00001 /*****
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"
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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00023 #ifndef DEFAULT_CFE_EVS_MSGIDS_H
```

```

00024 #define DEFAULT_CFE_EVS_MSGIDS_H
00025
00026 #include "cfe_evs_msgid_values.h"
00027
00028 /*
00029 ** cFE Command Message Id's
00030 */
00031
00032 #define CFE_EVS_CMD_MID          CFE_PLATFORM_EVS_CMD_MIDVAL(CMD)
00033 #define CFE_EVS_SEND_HK_MID     CFE_PLATFORM_EVS_CMD_MIDVAL(SEND_HK)
00034
00035 /*
00036 ** CFE Telemetry Message Id's
00037 */
00038 #define CFE_EVS_HK_TLM_MID       CFE_PLATFORM_EVS_TLM_MIDVAL(HK_TLM)
00039 #define CFE_EVS_LONG_EVENT_MSG_MID CFE_PLATFORM_EVS_TLM_MIDVAL(LONG_EVENT_MSG)
00040 #define CFE_EVS_SHORT_EVENT_MSG_MID CFE_PLATFORM_EVS_TLM_MIDVAL(SHORT_EVENT_MSG)
00041
00042 #endif

```

11.151 cfe/modules/evs/config/default_cfe_evs_msgstruct.h File Reference

```

#include "common_types.h"
#include "cfe_evs_msgdefs.h"
#include "cfe_evs_extern_typedefs.h"
#include "cfe_msg_hdr.h"

```

Data Structures

- struct [CFE_EVS_NoopCmd](#)
- struct [CFE_EVS_ResetCountersCmd](#)
- struct [CFE_EVS_ClearLogCmd](#)
- struct [CFE_EVS_SendHkCmd](#)
- struct [CFE_EVS_WriteLogDataFileCmd](#)
Write Event Log to File Command.
- struct [CFE_EVS_WriteAppDataFileCmd](#)
Write Event Services Application Information to File Command.
- struct [CFE_EVS_SetLogModeCmd](#)
Set Log Mode Command.
- struct [CFE_EVS_SetEventFormatModeCmd](#)
Set Event Format Mode Command.
- struct [CFE_EVS_EnablePortsCmd](#)
- struct [CFE_EVS_DisablePortsCmd](#)
- struct [CFE_EVS_EnableEventTypeCmd](#)
- struct [CFE_EVS_DisableEventTypeCmd](#)
- struct [CFE_EVS_EnableAppEventsCmd](#)
- struct [CFE_EVS_DisableAppEventsCmd](#)
- struct [CFE_EVS_ResetAppCounterCmd](#)
- struct [CFE_EVS_ResetAllFiltersCmd](#)
- struct [CFE_EVS_ResetFilterCmd](#)
- struct [CFE_EVS_DeleteEventFilterCmd](#)
- struct [CFE_EVS_EnableAppEventTypeCmd](#)
- struct [CFE_EVS_DisableAppEventTypeCmd](#)
- struct [CFE_EVS_AddEventFilterCmd](#)
- struct [CFE_EVS_SetFilterCmd](#)
- struct [CFE_EVS_HousekeepingTlm](#)
- struct [CFE_EVS_LongEventTlm](#)
- struct [CFE_EVS_ShortEventTlm](#)

Typedefs

- typedef struct [CFE_EVS_NoopCmd](#) [CFE_EVS_NoopCmd_t](#)
- typedef struct [CFE_EVS_ResetCountersCmd](#) [CFE_EVS_ResetCountersCmd_t](#)
- typedef struct [CFE_EVS_ClearLogCmd](#) [CFE_EVS_ClearLogCmd_t](#)
- typedef struct [CFE_EVS_SendHkCmd](#) [CFE_EVS_SendHkCmd_t](#)
- typedef struct [CFE_EVS_WriteLogDataFileCmd](#) [CFE_EVS_WriteLogDataFileCmd_t](#)
Write Event Log to File Command.
- typedef struct [CFE_EVS_WriteAppDataFileCmd](#) [CFE_EVS_WriteAppDataFileCmd_t](#)
Write Event Services Application Information to File Command.
- typedef struct [CFE_EVS_SetLogModeCmd](#) [CFE_EVS_SetLogModeCmd_t](#)
Set Log Mode Command.
- typedef struct [CFE_EVS_SetEventFormatModeCmd](#) [CFE_EVS_SetEventFormatModeCmd_t](#)
Set Event Format Mode Command.
- typedef struct [CFE_EVS_EnablePortsCmd](#) [CFE_EVS_EnablePortsCmd_t](#)
- typedef struct [CFE_EVS_DisablePortsCmd](#) [CFE_EVS_DisablePortsCmd_t](#)
- typedef struct [CFE_EVS_EnableEventTypeCmd](#) [CFE_EVS_EnableEventTypeCmd_t](#)
- typedef struct [CFE_EVS_DisableEventTypeCmd](#) [CFE_EVS_DisableEventTypeCmd_t](#)
- typedef struct [CFE_EVS_EnableAppEventsCmd](#) [CFE_EVS_EnableAppEventsCmd_t](#)
- typedef struct [CFE_EVS_DisableAppEventsCmd](#) [CFE_EVS_DisableAppEventsCmd_t](#)
- typedef struct [CFE_EVS_ResetAppCounterCmd](#) [CFE_EVS_ResetAppCounterCmd_t](#)
- typedef struct [CFE_EVS_ResetAllFiltersCmd](#) [CFE_EVS_ResetAllFiltersCmd_t](#)
- typedef struct [CFE_EVS_ResetFilterCmd](#) [CFE_EVS_ResetFilterCmd_t](#)
- typedef struct [CFE_EVS_DeleteEventFilterCmd](#) [CFE_EVS_DeleteEventFilterCmd_t](#)
- typedef struct [CFE_EVS_EnableAppEventTypeCmd](#) [CFE_EVS_EnableAppEventTypeCmd_t](#)
- typedef struct [CFE_EVS_DisableAppEventTypeCmd](#) [CFE_EVS_DisableAppEventTypeCmd_t](#)
- typedef struct [CFE_EVS_AddEventFilterCmd](#) [CFE_EVS_AddEventFilterCmd_t](#)
- typedef struct [CFE_EVS_SetFilterCmd](#) [CFE_EVS_SetFilterCmd_t](#)
- typedef struct [CFE_EVS_HousekeepingTlm](#) [CFE_EVS_HousekeepingTlm_t](#)
- typedef struct [CFE_EVS_LongEventTlm](#) [CFE_EVS_LongEventTlm_t](#)
- typedef struct [CFE_EVS_ShortEventTlm](#) [CFE_EVS_ShortEventTlm_t](#)

11.151.1 Detailed Description

Specification for the CFE Event Services (CFE_EVS) command and telemetry message frame structures. Definition in file [default_cfe_evs_msgstruct.h](#).

11.151.2 Typedef Documentation

CFE_EVS_AddEventFilterCmd_t

```
typedef struct CFE\_EVS\_AddEventFilterCmd CFE\_EVS\_AddEventFilterCmd\_t
```

CFE_EVS_ClearLogCmd_t

```
typedef struct CFE\_EVS\_ClearLogCmd CFE\_EVS\_ClearLogCmd\_t
```

CFE_EVS_DeleteEventFilterCmd_t

```
typedef struct CFE\_EVS\_DeleteEventFilterCmd CFE\_EVS\_DeleteEventFilterCmd\_t
```

CFE_EVS_DisableAppEventsCmd_t

```
typedef struct CFE\_EVS\_DisableAppEventsCmd CFE\_EVS\_DisableAppEventsCmd\_t
```

CFE_EVS_DisableAppEventTypeCmd_t

```
typedef struct CFE_EVS_DisableAppEventTypeCmd CFE_EVS_DisableAppEventTypeCmd_t
```

CFE_EVS_DisableEventTypeCmd_t

```
typedef struct CFE_EVS_DisableEventTypeCmd CFE_EVS_DisableEventTypeCmd_t
```

CFE_EVS_DisablePortsCmd_t

```
typedef struct CFE_EVS_DisablePortsCmd CFE_EVS_DisablePortsCmd_t
```

CFE_EVS_EnableAppEventsCmd_t

```
typedef struct CFE_EVS_EnableAppEventsCmd CFE_EVS_EnableAppEventsCmd_t
```

CFE_EVS_EnableAppEventTypeCmd_t

```
typedef struct CFE_EVS_EnableAppEventTypeCmd CFE_EVS_EnableAppEventTypeCmd_t
```

CFE_EVS_EnableEventTypeCmd_t

```
typedef struct CFE_EVS_EnableEventTypeCmd CFE_EVS_EnableEventTypeCmd_t
```

CFE_EVS_EnablePortsCmd_t

```
typedef struct CFE_EVS_EnablePortsCmd CFE_EVS_EnablePortsCmd_t
```

CFE_EVS_HousekeepingTlm_t

```
typedef struct CFE_EVS_HousekeepingTlm CFE_EVS_HousekeepingTlm_t
```

CFE_EVS_LongEventTlm_t

```
typedef struct CFE_EVS_LongEventTlm CFE_EVS_LongEventTlm_t
```

CFE_EVS_NoopCmd_t

```
typedef struct CFE_EVS_NoopCmd CFE_EVS_NoopCmd_t
```

CFE_EVS_ResetAllFiltersCmd_t

```
typedef struct CFE_EVS_ResetAllFiltersCmd CFE_EVS_ResetAllFiltersCmd_t
```

CFE_EVS_ResetAppCounterCmd_t

```
typedef struct CFE_EVS_ResetAppCounterCmd CFE_EVS_ResetAppCounterCmd_t
```

CFE_EVS_ResetCountersCmd_t

```
typedef struct CFE_EVS_ResetCountersCmd CFE_EVS_ResetCountersCmd_t
```

CFE_EVS_ResetFilterCmd_t

```
typedef struct CFE_EVS_ResetFilterCmd CFE_EVS_ResetFilterCmd_t
```


CFE_EVS_SendHkCmd_t

```
typedef struct CFE_EVS_SendHkCmd CFE_EVS_SendHkCmd_t
```

CFE_EVS_SetEventFormatModeCmd_t

```
typedef struct CFE_EVS_SetEventFormatModeCmd CFE_EVS_SetEventFormatModeCmd_t
```

Set Event Format Mode Command.

CFE_EVS_SetFilterCmd_t

```
typedef struct CFE_EVS_SetFilterCmd CFE_EVS_SetFilterCmd_t
```

CFE_EVS_SetLogModeCmd_t

```
typedef struct CFE_EVS_SetLogModeCmd CFE_EVS_SetLogModeCmd_t
```

Set Log Mode Command.

CFE_EVS_ShortEventTlm_t

```
typedef struct CFE_EVS_ShortEventTlm CFE_EVS_ShortEventTlm_t
```

CFE_EVS_WriteAppDataFileCmd_t

```
typedef struct CFE_EVS_WriteAppDataFileCmd CFE_EVS_WriteAppDataFileCmd_t
```

Write Event Services Application Information to File Command.

CFE_EVS_WriteLogDataFileCmd_t

```
typedef struct CFE_EVS_WriteLogDataFileCmd CFE_EVS_WriteLogDataFileCmd_t
```

Write Event Log to File Command.

11.152 default_cfe_evs_msgstruct.h

[Go to the documentation of this file.](#)

```
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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00024 #ifndef DEFAULT_CFE_EVS_MSGSTRUCT_H
00025 #define DEFAULT_CFE_EVS_MSGSTRUCT_H
00026
00027 /*****
00028  * Includes
00029  *****/
00030 #include "common_types.h"
00031 #include "cfe_evs_msgdefs.h"
00032 #include "cfe_evs_extern_typedefs.h"
00033 #include "cfe_msg_hdr.h"
00034
00035 /***** Structure Definitions *****/
00036
```

```

00037 /*
00038  * The NoArgsCmd is shared by several EVS command handlers.
00039  * Create a unique type for each one so the prototypes will follow the naming pattern,
00040  * allowing each command to evolve independently.
00041  */
00042 typedef struct CFE_EVS_NoopCmd
00043 {
00044     CFE_MSG_CommandHeader_t CommandHeader;
00045 } CFE_EVS_NoopCmd_t;
00046
00047 typedef struct CFE_EVS_ResetCountersCmd
00048 {
00049     CFE_MSG_CommandHeader_t CommandHeader;
00050 } CFE_EVS_ResetCountersCmd_t;
00051
00052 typedef struct CFE_EVS_ClearLogCmd
00053 {
00054     CFE_MSG_CommandHeader_t CommandHeader;
00055 } CFE_EVS_ClearLogCmd_t;
00056
00057 typedef struct CFE_EVS_SendHkCmd
00058 {
00059     CFE_MSG_CommandHeader_t CommandHeader;
00060 } CFE_EVS_SendHkCmd_t;
00061
00062 typedef struct CFE_EVS_WriteLogDataFileCmd
00063 {
00064     CFE_MSG_CommandHeader_t CommandHeader;
00065     CFE_EVS_LogFileCmd_Payload_t Payload;
00066 } CFE_EVS_WriteLogDataFileCmd_t;
00067
00068 typedef struct CFE_EVS_WriteAppDataFileCmd
00069 {
00070     CFE_MSG_CommandHeader_t CommandHeader;
00071     CFE_EVS_AppDataCmd_Payload_t Payload;
00072 } CFE_EVS_WriteAppDataFileCmd_t;
00073
00074 typedef struct CFE_EVS_SetLogModeCmd
00075 {
00076     CFE_MSG_CommandHeader_t CommandHeader;
00077     CFE_EVS_SetLogMode_Payload_t Payload;
00078 } CFE_EVS_SetLogModeCmd_t;
00079
00080 typedef struct CFE_EVS_SetEventFormatModeCmd
00081 {
00082     CFE_MSG_CommandHeader_t CommandHeader;
00083     CFE_EVS_SetEventFormatMode_Payload_t Payload;
00084 } CFE_EVS_SetEventFormatModeCmd_t;
00085
00086 typedef struct CFE_EVS_EnablePortsCmd
00087 {
00088     CFE_MSG_CommandHeader_t CommandHeader;
00089     CFE_EVS_BitMaskCmd_Payload_t Payload;
00090 } CFE_EVS_EnablePortsCmd_t;
00091
00092 typedef struct CFE_EVS_DisablePortsCmd
00093 {
00094     CFE_MSG_CommandHeader_t CommandHeader;
00095     CFE_EVS_BitMaskCmd_Payload_t Payload;
00096 } CFE_EVS_DisablePortsCmd_t;
00097
00098 typedef struct CFE_EVS_EnableEventTypeCmd
00099 {
00100     CFE_MSG_CommandHeader_t CommandHeader;
00101     CFE_EVS_BitMaskCmd_Payload_t Payload;
00102 } CFE_EVS_EnableEventTypeCmd_t;
00103
00104 typedef struct CFE_EVS_DisableEventTypeCmd
00105 {
00106     CFE_MSG_CommandHeader_t CommandHeader;
00107     CFE_EVS_BitMaskCmd_Payload_t Payload;
00108 } CFE_EVS_DisableEventTypeCmd_t;
00109
00110 /*
00111  * The CFE_EVS_AppNameCmd_t is shared by several EVS command handlers.
00112  * Create a unique type for each one so the prototypes will follow the naming pattern,
00113  * allowing each command to evolve independently.
00114  */
00115 typedef struct CFE_EVS_AppNameCmd_t
00116 {
00117     CFE_MSG_CommandHeader_t CommandHeader;
00118     CFE_EVS_AppNameCmd_Payload_t Payload;
00119 } CFE_EVS_AppNameCmd_t;

```

```

00130  * allowing each command to evolve independently.
00131  */
00132  typedef struct CFE_EVS_EnableAppEventsCmd
00133  {
00134      CFE_MSG_CommandHeader_t    CommandHeader;
00135      CFE_EVS_AppNameCmd_Payload_t Payload;
00136  } CFE_EVS_EnableAppEventsCmd_t;
00137
00138  typedef struct CFE_EVS_DisableAppEventsCmd
00139  {
00140      CFE_MSG_CommandHeader_t    CommandHeader;
00141      CFE_EVS_AppNameCmd_Payload_t Payload;
00142  } CFE_EVS_DisableAppEventsCmd_t;
00143
00144  typedef struct CFE_EVS_ResetAppCounterCmd
00145  {
00146      CFE_MSG_CommandHeader_t    CommandHeader;
00147      CFE_EVS_AppNameCmd_Payload_t Payload;
00148  } CFE_EVS_ResetAppCounterCmd_t;
00149
00150  typedef struct CFE_EVS_ResetAllFiltersCmd
00151  {
00152      CFE_MSG_CommandHeader_t    CommandHeader;
00153      CFE_EVS_AppNameCmd_Payload_t Payload;
00154  } CFE_EVS_ResetAllFiltersCmd_t;
00155
00156  /*
00157  * The CFE_EVS_AppNameEventIDCmd_t is shared by several EVS command handlers.
00158  * Create a unique type for each one so the prototypes will follow the naming pattern,
00159  * allowing each command to evolve independently.
00160  */
00161  typedef struct CFE_EVS_ResetFilterCmd
00162  {
00163      CFE_MSG_CommandHeader_t    CommandHeader;
00164      CFE_EVS_AppNameEventIDCmd_Payload_t Payload;
00165  } CFE_EVS_ResetFilterCmd_t;
00166
00167  typedef struct CFE_EVS_DeleteEventFilterCmd
00168  {
00169      CFE_MSG_CommandHeader_t    CommandHeader;
00170      CFE_EVS_AppNameEventIDCmd_Payload_t Payload;
00171  } CFE_EVS_DeleteEventFilterCmd_t;
00172
00173  /*
00174  * The CFE_EVS_AppNameBitMaskCmd_t is shared by several EVS command handlers.
00175  * Create a unique type for each one so the prototypes will follow the naming pattern,
00176  * allowing each command to evolve independently.
00177  */
00178  typedef struct CFE_EVS_EnableAppEventTypeCmd
00179  {
00180      CFE_MSG_CommandHeader_t    CommandHeader;
00181      CFE_EVS_AppNameBitMaskCmd_Payload_t Payload;
00182  } CFE_EVS_EnableAppEventTypeCmd_t;
00183
00184  typedef struct CFE_EVS_DisableAppEventTypeCmd
00185  {
00186      CFE_MSG_CommandHeader_t    CommandHeader;
00187      CFE_EVS_AppNameBitMaskCmd_Payload_t Payload;
00188  } CFE_EVS_DisableAppEventTypeCmd_t;
00189
00190  /*
00191  * The CFE_EVS_AppNameEventIDMaskCmd_t is shared by several EVS command handlers.
00192  * Create a unique type for each one so the prototypes will follow the naming pattern,
00193  * allowing each command to evolve independently.
00194  */
00195  typedef struct CFE_EVS_AddEventFilterCmd
00196  {
00197      CFE_MSG_CommandHeader_t    CommandHeader;
00198      CFE_EVS_AppNameEventIDMaskCmd_Payload_t Payload;
00199  } CFE_EVS_AddEventFilterCmd_t;
00200
00201  typedef struct CFE_EVS_SetFilterCmd
00202  {
00203      CFE_MSG_CommandHeader_t    CommandHeader;
00204      CFE_EVS_AppNameEventIDMaskCmd_Payload_t Payload;
00205  } CFE_EVS_SetFilterCmd_t;
00206
00207  /*
00208  *
00209  *
00210  */

```

```

00211 /*****/
00212
00213 typedef struct CFE_EVS_HousekeepingTlm
00214 {
00215     CFE_MSG_TelemetryHeader_t    TelemetryHeader;
00216     CFE_EVS_HousekeepingTlm_Payload_t Payload;
00217 } CFE_EVS_HousekeepingTlm_t;
00218
00219 typedef struct CFE_EVS_LongEventTlm
00220 {
00221     CFE_MSG_TelemetryHeader_t    TelemetryHeader;
00222     CFE_EVS_LongEventTlm_Payload_t Payload;
00223 } CFE_EVS_LongEventTlm_t;
00224
00225 typedef struct CFE_EVS_ShortEventTlm
00226 {
00227     CFE_MSG_TelemetryHeader_t    TelemetryHeader;
00228     CFE_EVS_ShortEventTlm_Payload_t Payload;
00229 } CFE_EVS_ShortEventTlm_t;
00230
00231 #endif

```

11.153 cfe/modules/evs/config/default_cfe_evs_platform_cfg.h File Reference

```

#include "cfe_evs_mission_cfg.h"
#include "cfe_evs_internal_cfg.h"

```

11.153.1 Detailed Description

CFE Event Services (CFE_EVS) Application Platform Configuration Header File

This is a compatibility header for the "platform_cfg.h" file that has traditionally provided both public and private config definitions for each CFS app.

These definitions are now provided in two separate files, one for the public/mission scope and one for internal scope.

Note

This file may be overridden/superseded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [default_cfe_evs_platform_cfg.h](#).

11.154 default_cfe_evs_platform_cfg.h

[Go to the documentation of this file.](#)

```

00001 /*****/
00002 * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"
00003 *
00004 * Copyright (c) 2023 United States Government as represented by the
00005 * Administrator of the National Aeronautics and Space Administration.
00006 * All Rights Reserved.
00007 *
00008 * Licensed under the Apache License, Version 2.0 (the "License"); you may
00009 * not use this file except in compliance with the License. You may obtain
00010 * a copy of the License at http://www.apache.org/licenses/LICENSE-2.0
00011 *
00012 * Unless required by applicable law or agreed to in writing, software
00013 * distributed under the License is distributed on an "AS IS" BASIS,
00014 * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015 * See the License for the specific language governing permissions and
00016 * limitations under the License.
00017 *****/
00018
00035 #ifndef DEFAULT_CFE_EVS_PLATFORM_CFG_H
00036 #define DEFAULT_CFE_EVS_PLATFORM_CFG_H
00037
00038 #include "cfe_evs_mission_cfg.h"
00039 #include "cfe_evs_internal_cfg.h"
00040
00041 #endif

```

11.155 cfe/modules/evs/config/default_cfe_evs_topicid_values.h File Reference

Macros

- #define [CFE_MISSION_EVS_TIDVAL](#)(x) DEFAULT_CFE_MISSION_EVS_##x##_TOPICID

11.155.1 Detailed Description

CFE Event Services (CFE_EVS) Application Topic IDs

Definition in file [default_cfe_evs_topicid_values.h](#).

11.155.2 Macro Definition Documentation

CFE_MISSION_EVS_TIDVAL

```
#define CFE_MISSION_EVS_TIDVAL(  
    x ) DEFAULT_CFE_MISSION_EVS_##x##_TOPICID
```

Definition at line 26 of file [default_cfe_evs_topicid_values.h](#).

11.156 default_cfe_evs_topicid_values.h

[Go to the documentation of this file.](#)

```
00001 /*****  
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"  
00003  *  
00004  * Copyright (c) 2023 United States Government as represented by the  
00005  * Administrator of the National Aeronautics and Space Administration.  
00006  * All Rights Reserved.  
00007  *  
00008  * Licensed under the Apache License, Version 2.0 (the "License"); you may  
00009  * not use this file except in compliance with the License. You may obtain  
00010  * a copy of the License at http://www.apache.org/licenses/LICENSE-2.0  
00011  *  
00012  * Unless required by applicable law or agreed to in writing, software  
00013  * distributed under the License is distributed on an "AS IS" BASIS,  
00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.  
00015  * See the License for the specific language governing permissions and  
00016  * limitations under the License.  
00017  *****/  
00018  
00023 #ifndef DEFAULT_CFE_EVS_TOPICID_VALUES_H  
00024 #define DEFAULT_CFE_EVS_TOPICID_VALUES_H  
00025  
00026 #define CFE_MISSION_EVS_TIDVAL(x) DEFAULT_CFE_MISSION_EVS_##x##_TOPICID  
00027  
00028 #endif
```

11.157 cfe/modules/evs/fsw/inc/cfe_evs_eventids.h File Reference

Macros

EVS event IDs

- #define [CFE_EVS_NOOP_EID](#) 0
EVS No-op Command Success Event ID.
- #define [CFE_EVS_STARTUP_EID](#) 1
EVS Initialization Event ID.
- #define [CFE_EVS_ERR_WRLOGFILE_EID](#) 2
EVS Write Event Log Command File Write Entry Failed Event ID.
- #define [CFE_EVS_ERR_CRLOGFILE_EID](#) 3
EVS Write Event Log Command Filename Parse or File Create Failed Event ID.
- #define [CFE_EVS_ERR_MSGID_EID](#) 5
EVS Invalid Message ID Received Event ID.
- #define [CFE_EVS_ERR_EVTIDNOREGS_EID](#) 6

- *EVS Command Event Not Registered For Filtering Event ID.*
• #define [CFE_EVS_ERR_APPNOREGS_EID](#) 7
- *EVS Command Application Not Registered With EVS Event ID.*
• #define [CFE_EVS_ERR_ILLAPPIDRANGE_EID](#) 8
- *EVS Command Get Application Data Failure Event ID.*
• #define [CFE_EVS_ERR_NOAPPIDFOUND_EID](#) 9
- *EVS Command Get Application ID Failure Event ID.*
• #define [CFE_EVS_ERR_ILLEGALFMTMOD_EID](#) 10
- *EVS Set Event Format Command Invalid Format Event ID.*
• #define [CFE_EVS_ERR_MAXREGSFILTER_EID](#) 11
- *EVS Add Filter Command Max Filters Exceeded Event ID.*
• #define [CFE_EVS_ERR_WRDATFILE_EID](#) 12
- *EVS Write Application Data Command Write Data Failure Event ID.*
• #define [CFE_EVS_ERR_CRDATFILE_EID](#) 13
- *EVS Write Application Data Command Filename Parse or File Create Failed Event ID.*
• #define [CFE_EVS_WRITE_HEADER_ERR_EID](#) 14
- *EVS Write File Header to Log File Failure Event ID.*
• #define [CFE_EVS_ERR_CC_EID](#) 15
- *EVS Invalid Command Code Received Event ID.*
• #define [CFE_EVS_RSTCNT_EID](#) 16
- *EVS Reset Counters Command Success Event ID.*
• #define [CFE_EVS_SETFILTERMSK_EID](#) 17
- *EVS Set Filter Command Success Event ID.*
• #define [CFE_EVS_ENAPORT_EID](#) 18
- *EVS Enable Ports Command Success Event ID.*
• #define [CFE_EVS_DISPORT_EID](#) 19
- *EVS Disable Ports Command Success Event ID.*
• #define [CFE_EVS_ENAEVTTYPE_EID](#) 20
- *EVS Enable Event Type Command Success Event ID.*
• #define [CFE_EVS_DISEVTTYPE_EID](#) 21
- *EVS Disable Event Type Command Success Event ID.*
• #define [CFE_EVS_SETEVTFMTMOD_EID](#) 22
- *EVS Set Event Format Mode Command Success Event ID.*
• #define [CFE_EVS_ENAAPPEVTTYPE_EID](#) 23
- *EVS Enable App Event Type Command Success Event ID.*
• #define [CFE_EVS_DISAPPENTTYPE_EID](#) 24
- *EVS Disable App Event Type Command Success Event ID.*
• #define [CFE_EVS_ENAAPPEVT_EID](#) 25
- *EVS Enable App Events Command Success Event ID.*
• #define [CFE_EVS_DISAPPEVT_EID](#) 26
- *EVS Disable App Events Command Success Event ID.*
• #define [CFE_EVS_RSTEVCNT_EID](#) 27
- *EVS Reset App Event Counter Command Success Event ID.*
• #define [CFE_EVS_RSTFILTER_EID](#) 28
- *EVS Reset App Event Filter Command Success Event ID.*
• #define [CFE_EVS_RSTALLFILTER_EID](#) 29
- *EVS Reset All Filters Command Success Event ID.*
• #define [CFE_EVS_ADDFILTER_EID](#) 30
- *EVS Add Event Filter Command Success Event ID.*
• #define [CFE_EVS_DELFILTER_EID](#) 31
- *EVS Delete Event Filter Command Success Event ID.*
• #define [CFE_EVS_WRDAT_EID](#) 32
- *EVS Write Application Data Command Success Event ID.*
• #define [CFE_EVS_WRLOG_EID](#) 33
- *EVS Write Event Log Command Success Event ID.*

- `#define CFE_EVS_EVT_FILTERED_EID` 37
EVS Add Filter Command Duplicate Registration Event ID.
- `#define CFE_EVS_LOGMODE_EID` 38
EVS Set Log Mode Command Success Event ID.
- `#define CFE_EVS_ERR_LOGMODE_EID` 39
EVS Set Log Mode Command Invalid Mode Event ID.
- `#define CFE_EVS_ERR_INVALID_BITMASK_EID` 40
EVS Port Or Event Type Bitmask Invalid Event ID.
- `#define CFE_EVS_ERR_UNREGISTERED_EVS_APP` 41
EVS Send Event API App Not Registered With EVS Event ID.
- `#define CFE_EVS_FILTER_MAX_EID` 42
EVS Filter Max Count Reached Event ID.
- `#define CFE_EVS_LEN_ERR_EID` 43
EVS Invalid Command Length Event ID.
- `#define CFE_EVS_SQUELCHED_ERR_EID` 44
EVS Events Squelched Error Event ID.

11.157.1 Detailed Description

cFE Event Services Event IDs

Definition in file [cfe_evs_eventids.h](#).

11.157.2 Macro Definition Documentation

CFE_EVS_ADDFILTER_EID

```
#define CFE_EVS_ADDFILTER_EID 30
```

EVS Add Event Filter Command Success Event ID.

Type: DEBUG

Cause:

[EVS Add Event Filter Command](#) success.

Definition at line 367 of file [cfe_evs_eventids.h](#).

CFE_EVS_DELFILTER_EID

```
#define CFE_EVS_DELFILTER_EID 31
```

EVS Delete Event Filter Command Success Event ID.

Type: DEBUG

Cause:

[EVS Delete Event Filter Command](#) success.

Definition at line 378 of file [cfe_evs_eventids.h](#).

CFE_EVS_DISAPPENTTYPE_EID

```
#define CFE_EVS_DISAPPENTTYPE_EID 24
```

EVS Disable App Event Type Command Success Event ID.

Type: DEBUG

Cause:

[EVS Disable App Event Type Command](#) success.

Definition at line 301 of file [cfe_evs_eventids.h](#).

CFE_EVS_DISAPPEVT_EID

```
#define CFE_EVS_DISAPPEVT_EID 26
```

EVS Disable App Events Command Success Event ID.

Type: DEBUG

Cause:

[EVS Disable App Events Command](#) success.

Definition at line 323 of file [cfe_evs_eventids.h](#).

CFE_EVS_DISEVTTYPE_EID

```
#define CFE_EVS_DISEVTTYPE_EID 21
```

EVS Disable Event Type Command Success Event ID.

Type: DEBUG

Cause:

[EVS Disable Event Type Command](#) success.

Definition at line 268 of file [cfe_evs_eventids.h](#).

CFE_EVS_DISPORT_EID

```
#define CFE_EVS_DISPORT_EID 19
```

EVS Disable Ports Command Success Event ID.

Type: DEBUG

Cause:

[EVS Disable Ports Command](#) success.

Definition at line 246 of file [cfe_evs_eventids.h](#).

CFE_EVS_ENAAPPEVT_EID

```
#define CFE_EVS_ENAAPPEVT_EID 25
```

EVS Enable App Events Command Success Event ID.

Type: DEBUG

Cause:

[EVS Enable App Events Command](#) success.

Definition at line 312 of file [cfe_evs_eventids.h](#).

CFE_EVS_ENAAPPEVTTYPE_EID

```
#define CFE_EVS_ENAAPPEVTTYPE_EID 23
```

EVS Enable App Event Type Command Success Event ID.

Type: DEBUG

Cause:

[EVS Enable App Event Type Command](#) success.

Definition at line 290 of file [cfe_evs_eventids.h](#).

CFE_EVS_ENAEVTTYPE_EID

```
#define CFE_EVS_ENAEVTTYPE_EID 20
```

EVS Enable Event Type Command Success Event ID.

Type: DEBUG

Cause:

[EVS Enable Event Type Command](#) success.

Definition at line 257 of file [cfe_evs_eventids.h](#).

CFE_EVS_ENAPORT_EID

```
#define CFE_EVS_ENAPORT_EID 18
```

EVS Enable Ports Command Success Event ID.

Type: DEBUG

Cause:

[EVS Enable Ports Command](#) success.

Definition at line 235 of file [cfe_evs_eventids.h](#).

CFE_EVS_ERR_APPNOREGS_EID

```
#define CFE_EVS_ERR_APPNOREGS_EID 7
```

EVS Command Application Not Registered With EVS Event ID.

Type: ERROR

Cause:

An EVS command handler failure due to the referenced application not being registered with EVS. OVERLOADED
Definition at line 110 of file [cfe_evs_eventids.h](#).

CFE_EVS_ERR_CC_EID

```
#define CFE_EVS_ERR_CC_EID 15
```

EVS Invalid Command Code Received Event ID.

Type: ERROR

Cause:

Invalid command code for message ID [CFE_EVS_CMD_MID](#) received on the EVS message pipe.
Definition at line 202 of file [cfe_evs_eventids.h](#).

CFE_EVS_ERR_CRDATFILE_EID

```
#define CFE_EVS_ERR_CRDATFILE_EID 13
```

EVS Write Application Data Command Filename Parse or File Create Failed Event ID.

Type: ERROR

Cause:

[Write Application Data Command](#) failed to parse the filename or open/create the file. OVERLOADED
Definition at line 180 of file [cfe_evs_eventids.h](#).

CFE_EVS_ERR_CRLOGFILE_EID

```
#define CFE_EVS_ERR_CRLOGFILE_EID 3
```

EVS Write Event Log Command Filename Parse or File Create Failed Event ID.

Type: ERROR

Cause:

[EVS Write Event Log Command](#) failure parsing the file name or during open/creation of the file. OVERLOADED
Definition at line 77 of file [cfe_evs_eventids.h](#).

CFE_EVS_ERR_EVTIDNOREGS_EID

```
#define CFE_EVS_ERR_EVTIDNOREGS_EID 6
```

EVS Command Event Not Registered For Filtering Event ID.

Type: ERROR

Cause:

An EVS command handler failure due to the event not being registered for filtering. OVERLOADED
Definition at line 99 of file [cfe_evs_eventids.h](#).

CFE_EVS_ERR_ILLAPPIDRANGE_EID

```
#define CFE_EVS_ERR_ILLAPPIDRANGE_EID 8
```

EVS Command Get Application Data Failure Event ID.

Type: ERROR

Cause:

An EVS command handler failure retrieving the application data. OVERLOADED
Definition at line 121 of file [cfe_evs_eventids.h](#).

CFE_EVS_ERR_ILLEGALFMTMOD_EID

```
#define CFE_EVS_ERR_ILLEGALFMTMOD_EID 10
```

EVS Set Event Format Command Invalid Format Event ID.

Type: ERROR

Cause:

[EVS Set Event Format Command](#) failure due to invalid format argument.
Definition at line 144 of file [cfe_evs_eventids.h](#).

CFE_EVS_ERR_INVALID_BITMASK_EID

```
#define CFE_EVS_ERR_INVALID_BITMASK_EID 40
```

EVS Port Or Event Type Bitmask Invalid Event ID.

Type: ERROR

Cause:

Invalid bitmask for EVS port or event type. OVERLOADED
Definition at line 446 of file [cfe_evs_eventids.h](#).

CFE_EVS_ERR_LOGMODE_EID

```
#define CFE_EVS_ERR_LOGMODE_EID 39
```

EVS Set Log Mode Command Invalid Mode Event ID.

Type: ERROR

Cause:

[EVS Set Log Mode Command](#) failure due to invalid log mode.

Definition at line 435 of file [cfe_evs_eventids.h](#).

CFE_EVS_ERR_MAXREGSFILTER_EID

```
#define CFE_EVS_ERR_MAXREGSFILTER_EID 11
```

EVS Add Filter Command Max Filters Exceeded Event ID.

Type: ERROR

Cause:

[EVS Add Filter Command](#) failure due to exceeding the maximum number of filters.

Definition at line 156 of file [cfe_evs_eventids.h](#).

CFE_EVS_ERR_MSGID_EID

```
#define CFE_EVS_ERR_MSGID_EID 5
```

EVS Invalid Message ID Received Event ID.

Type: ERROR

Cause:

Invalid message ID received on the EVS message pipe.

Definition at line 88 of file [cfe_evs_eventids.h](#).

CFE_EVS_ERR_NOAPPIDFOUND_EID

```
#define CFE_EVS_ERR_NOAPPIDFOUND_EID 9
```

EVS Command Get Application ID Failure Event ID.

Type: ERROR

Cause:

An EVS command handler failure retrieving the application ID. OVERLOADED

Definition at line 132 of file [cfe_evs_eventids.h](#).

CFE_EVS_ERR_UNREGISTERED_EVS_APP

#define CFE_EVS_ERR_UNREGISTERED_EVS_APP 41
EVS Send Event API App Not Registered With EVS Event ID.

Type: ERROR

Cause:

An EVS Send Event API called for application not registered with EVS.
Definition at line 457 of file [cfe_evs_eventids.h](#).

CFE_EVS_ERR_WRDATFILE_EID

#define CFE_EVS_ERR_WRDATFILE_EID 12
EVS Write Application Data Command Write Data Failure Event ID.

Type: ERROR

Cause:

[Write Application Data Command](#) failure to write application EVS data.
Definition at line 168 of file [cfe_evs_eventids.h](#).

CFE_EVS_ERR_WRLOGFILE_EID

#define CFE_EVS_ERR_WRLOGFILE_EID 2
EVS Write Event Log Command File Write Entry Failed Event ID.

Type: ERROR

Cause:

[EVS Write Event Log Command](#) failure writing data to the file.
Definition at line 65 of file [cfe_evs_eventids.h](#).

CFE_EVS_EVT_FILTERED_EID

#define CFE_EVS_EVT_FILTERED_EID 37
EVS Add Filter Command Duplicate Registration Event ID.

Type: ERROR

Cause:

[EVS Add Filter Command](#) failure due to event already being registered for filtering.
Definition at line 412 of file [cfe_evs_eventids.h](#).

CFE_EVS_FILTER_MAX_EID

```
#define CFE_EVS_FILTER_MAX_EID 42
```

EVS Filter Max Count Reached Event ID.

Type: INFORMATIONAL

Cause:

Filter count for the event reached CFE_EVS_MAX_FILTER_COUNT and is latched until filter is reset.
Definition at line 468 of file [cfe_evs_eventids.h](#).

CFE_EVS_LEN_ERR_EID

```
#define CFE_EVS_LEN_ERR_EID 43
```

EVS Invalid Command Length Event ID.

Type: ERROR

Cause:

Invalid length for the command code in message ID [CFE_EVS_CMD_MID](#) received on the EVS message pipe.
Definition at line 479 of file [cfe_evs_eventids.h](#).

CFE_EVS_LOGMODE_EID

```
#define CFE_EVS_LOGMODE_EID 38
```

EVS Set Log Mode Command Success Event ID.

Type: DEBUG

Cause:

[EVS Set Log Mode Command](#) success.
Definition at line 423 of file [cfe_evs_eventids.h](#).

CFE_EVS_NOOP_EID

```
#define CFE_EVS_NOOP_EID 0
```

EVS No-op Command Success Event ID.

Type: INFORMATION

Cause:

[EVS NO-OP command](#) success.
Definition at line 42 of file [cfe_evs_eventids.h](#).

CFE_EVS_RSTALLFILTER_EID

```
#define CFE_EVS_RSTALLFILTER_EID 29
```

EVS Reset All Filters Command Success Event ID.

Type: DEBUG

Cause:

[EVS Reset All Filters Command](#) success.
Definition at line 356 of file [cfe_evs_eventids.h](#).

CFE_EVS_RSTCNT_EID

```
#define CFE_EVS_RSTCNT_EID 16
```

EVS Reset Counters Command Success Event ID.

Type: DEBUG

Cause:

[EVS Reset Counters Command](#) success.
Definition at line 213 of file [cfe_evs_eventids.h](#).

CFE_EVS_RSTTEVTCNT_EID

```
#define CFE_EVS_RSTTEVTCNT_EID 27
```

EVS Reset App Event Counter Command Success Event ID.

Type: DEBUG

Cause:

[EVS Reset App Event Counter Command](#) success.
Definition at line 334 of file [cfe_evs_eventids.h](#).

CFE_EVS_RSTFILTER_EID

```
#define CFE_EVS_RSTFILTER_EID 28
```

EVS Reset App Event Filter Command Success Event ID.

Type: DEBUG

Cause:

[EVS Reset App Event Filter Command](#) success.
Definition at line 345 of file [cfe_evs_eventids.h](#).

CFE_EVS_SETEVTFMTMOD_EID

```
#define CFE_EVS_SETEVTFMTMOD_EID 22
```

EVS Set Event Format Mode Command Success Event ID.

Type: DEBUG

Cause:

[EVS Set Event Format Mode Command](#) success.

Definition at line 279 of file [cfe_evs_eventids.h](#).

CFE_EVS_SETFILTERMSK_EID

```
#define CFE_EVS_SETFILTERMSK_EID 17
```

EVS Set Filter Command Success Event ID.

Type: DEBUG

Cause:

[EVS Set Filter Command](#) success.

Definition at line 224 of file [cfe_evs_eventids.h](#).

CFE_EVS_SQUELCHED_ERR_EID

```
#define CFE_EVS_SQUELCHED_ERR_EID 44
```

EVS Events Squelched Error Event ID.

Type: ERROR

Cause:

Events generated in app at a rate in excess of [CFE_PLATFORM_EVS_MAX_APP_EVENT_BURST](#) in one moment or [CFE_PLATFORM_EVS_APP_EVENTS_PER_SEC](#) sustained

Definition at line 492 of file [cfe_evs_eventids.h](#).

CFE_EVS_STARTUP_EID

```
#define CFE_EVS_STARTUP_EID 1
```

EVS Initialization Event ID.

Type: INFORMATION

Cause:

Event Services Task initialization complete.

Definition at line 53 of file [cfe_evs_eventids.h](#).

CFE_EVS_WRDAT_EID

```
#define CFE_EVS_WRDAT_EID 32
```

EVS Write Application Data Command Success Event ID.

Type: DEBUG

Cause:

[EVS Write Application Data Command](#) success.

Definition at line 389 of file [cfe_evs_eventids.h](#).

CFE_EVS_WRITE_HEADER_ERR_EID

```
#define CFE_EVS_WRITE_HEADER_ERR_EID 14
```

EVS Write File Header to Log File Failure Event ID.

Type: ERROR

Cause:

Bytes written during Write File Header to Log File was not equal to the expected header size.

Definition at line 191 of file [cfe_evs_eventids.h](#).

CFE_EVS_WRLOG_EID

```
#define CFE_EVS_WRLOG_EID 33
```

EVS Write Event Log Command Success Event ID.

Type: DEBUG

Cause:

[EVS Write Event Log Command](#) success.

Definition at line 400 of file [cfe_evs_eventids.h](#).

11.158 cfe_evs_eventids.h

[Go to the documentation of this file.](#)

```
00001 /*****
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"
00003  *
00004  * Copyright (c) 2023 United States Government as represented by the
00005  * Administrator of the National Aeronautics and Space Administration.
00006  * All Rights Reserved.
00007  *
00008  * Licensed under the Apache License, Version 2.0 (the "License"); you may
00009  * not use this file except in compliance with the License. You may obtain
00010  * a copy of the License at http://www.apache.org/licenses/LICENSE-2.0
00011  *
00012  * Unless required by applicable law or agreed to in writing, software
```

```
00013  * distributed under the License is distributed on an "AS IS" BASIS,
00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00025 #ifndef CFE_EVS_EVENTS_H
00026 #define CFE_EVS_EVENTS_H
00027
00042 #define CFE_EVS_NOOP_EID 0
00043
00053 #define CFE_EVS_STARTUP_EID 1
00054
00065 #define CFE_EVS_ERR_WRLOGFILE_EID 2
00066
00077 #define CFE_EVS_ERR_CRLOGFILE_EID 3
00078
00088 #define CFE_EVS_ERR_MSGID_EID 5
00089
00099 #define CFE_EVS_ERR_EVTIDNOREGS_EID 6
00100
00110 #define CFE_EVS_ERR_APPNOREGS_EID 7
00111
00121 #define CFE_EVS_ERR_ILLAPPIDRANGE_EID 8
00122
00132 #define CFE_EVS_ERR_NOAPPIDFOUND_EID 9
00133
00144 #define CFE_EVS_ERR_ILLEGALFMTMOD_EID 10
00145
00156 #define CFE_EVS_ERR_MAXREGSFILTER_EID 11
00157
00168 #define CFE_EVS_ERR_WRDATFILE_EID 12
00169
00180 #define CFE_EVS_ERR_CRDATFILE_EID 13
00181
00191 #define CFE_EVS_WRITE_HEADER_ERR_EID 14
00192
00202 #define CFE_EVS_ERR_CC_EID 15
00203
00213 #define CFE_EVS_RSTCNT_EID 16
00214
00224 #define CFE_EVS_SETFILTERMSK_EID 17
00225
00235 #define CFE_EVS_ENAPORT_EID 18
00236
00246 #define CFE_EVS_DISPORT_EID 19
00247
00257 #define CFE_EVS_ENAEVTTYPE_EID 20
00258
00268 #define CFE_EVS_DISEVTTYPE_EID 21
00269
00279 #define CFE_EVS_SETEVTFMTMOD_EID 22
00280
00290 #define CFE_EVS_ENAAPPEVTTYPE_EID 23
00291
00301 #define CFE_EVS_DISAPPENTTYPE_EID 24
00302
00312 #define CFE_EVS_ENAAPPEVT_EID 25
00313
00323 #define CFE_EVS_DISAPPEVT_EID 26
00324
00334 #define CFE_EVS_RSTEVCNT_EID 27
00335
00345 #define CFE_EVS_RSTFILTER_EID 28
00346
00356 #define CFE_EVS_RSTALLFILTER_EID 29
00357
00367 #define CFE_EVS_ADDFILTER_EID 30
00368
00378 #define CFE_EVS_DELFILTER_EID 31
00379
00389 #define CFE_EVS_WRDAT_EID 32
00390
00400 #define CFE_EVS_WRLOG_EID 33
00401
00412 #define CFE_EVS_EVT_FILTERED_EID 37
00413
00423 #define CFE_EVS_LOGMODE_EID 38
00424
00435 #define CFE_EVS_ERR_LOGMODE_EID 39
00436
```

```

00446 #define CFE_EVS_ERR_INVALID_BITMASK_EID 40
00447
00457 #define CFE_EVS_ERR_UNREGISTERED_EVS_APP 41
00458
00468 #define CFE_EVS_FILTER_MAX_EID 42
00469
00479 #define CFE_EVS_LEN_ERR_EID 43
00480
00492 #define CFE_EVS_SQUELCHED_ERR_EID 44
00495 #endif /* CFE_EVS_EVENTS_H */

```

11.159 cfe/modules/evs/fsw/inc/cfe_evs_fcncodes.h File Reference

```
#include "cfe_evs_fcncode_values.h"
```

Macros

Event Services Command Codes

- `#define CFE_EVS_NOOP_CC CFE_EVS_CCVAL(NOOP)`
- `#define CFE_EVS_RESET_COUNTERS_CC CFE_EVS_CCVAL(RESET_COUNTERS)`
- `#define CFE_EVS_ENABLE_EVENT_TYPE_CC CFE_EVS_CCVAL(ENABLE_EVENT_TYPE)`
- `#define CFE_EVS_DISABLE_EVENT_TYPE_CC CFE_EVS_CCVAL(DISABLE_EVENT_TYPE)`
- `#define CFE_EVS_SET_EVENT_FORMAT_MODE_CC CFE_EVS_CCVAL(SET_EVENT_FORMAT_MODE)`
- `#define CFE_EVS_ENABLE_APP_EVENT_TYPE_CC CFE_EVS_CCVAL(ENABLE_APP_EVENT_TYPE)`
- `#define CFE_EVS_DISABLE_APP_EVENT_TYPE_CC CFE_EVS_CCVAL(DISABLE_APP_EVENT_TYPE)`
- `#define CFE_EVS_ENABLE_APP_EVENTS_CC CFE_EVS_CCVAL(ENABLE_APP_EVENTS)`
- `#define CFE_EVS_DISABLE_APP_EVENTS_CC CFE_EVS_CCVAL(DISABLE_APP_EVENTS)`
- `#define CFE_EVS_RESET_APP_COUNTER_CC CFE_EVS_CCVAL(RESET_APP_COUNTER)`
- `#define CFE_EVS_SET_FILTER_CC CFE_EVS_CCVAL(SET_FILTER)`
- `#define CFE_EVS_ENABLE_PORTS_CC CFE_EVS_CCVAL(ENABLE_PORTS)`
- `#define CFE_EVS_DISABLE_PORTS_CC CFE_EVS_CCVAL(DISABLE_PORTS)`
- `#define CFE_EVS_RESET_FILTER_CC CFE_EVS_CCVAL(RESET_FILTER)`
- `#define CFE_EVS_RESET_ALL_FILTERS_CC CFE_EVS_CCVAL(RESET_ALL_FILTERS)`
- `#define CFE_EVS_ADD_EVENT_FILTER_CC CFE_EVS_CCVAL(ADD_EVENT_FILTER)`
- `#define CFE_EVS_DELETE_EVENT_FILTER_CC CFE_EVS_CCVAL(DELETE_EVENT_FILTER)`
- `#define CFE_EVS_WRITE_APP_DATA_FILE_CC CFE_EVS_CCVAL(WRITE_APP_DATA_FILE)`
- `#define CFE_EVS_WRITE_LOG_DATA_FILE_CC CFE_EVS_CCVAL(WRITE_LOG_DATA_FILE)`
- `#define CFE_EVS_SET_LOG_MODE_CC CFE_EVS_CCVAL(SET_LOG_MODE)`
- `#define CFE_EVS_CLEAR_LOG_CC CFE_EVS_CCVAL(CLEAR_LOG)`

11.159.1 Detailed Description

Specification for the CFE Event Services (CFE_EVS) command function codes

Note

This file should be strictly limited to the command/function code (CC) macro definitions. Other definitions such as enums, typedefs, or other macros should be placed in the msgdefs.h or msg.h files.

Definition in file [cfe_evs_fcncodes.h](#).

11.159.2 Macro Definition Documentation

CFE_EVS_ADD_EVENT_FILTER_CC

```
#define CFE_EVS_ADD_EVENT_FILTER_CC CFE_EVS_CCVAL(ADD_EVENT_FILTER)
```

Name Add Application Event Filter

Description

This command adds the given filter for the given application identifier and event identifier. Note: In order for this command to take effect, applications must be registered for Event Service.

Command Mnemonic(s) `$sc_$cpu_EVS_AddEvtFiltr`

Command Structure

`CFE_EVS_AddEventFilterCmd_t`

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_EVS_CMDPC` - command execution counter will increment
- The generation of `CFE_EVS_ADDFILTER_EID` debug event message

Error Conditions

This command may fail for the following reason(s):

- Application name is not valid or not registered with event services
- Specified event ID is already added to the application event filter
- Maximum number of event IDs already added to filter

Evidence of failure may be found in the following telemetry:

- `$sc_$cpu_EVS_CMDEC` - command error counter will increment
- An Error specific event message

Criticality

None.

See also

`CFE_EVS_SET_FILTER_CC`, `CFE_EVS_RESET_FILTER_CC`, `CFE_EVS_RESET_ALL_FILTERS_CC`,
`CFE_EVS_DELETE_EVENT_FILTER_CC`

Definition at line 695 of file `cfe_evs_fcncodes.h`.

CFE_EVS_CLEAR_LOG_CC

```
#define CFE_EVS_CLEAR_LOG_CC CFE_EVS_CCVAL(CLEAR_LOG)
```

Name Clear Event Log

Description

This command clears the contents of the local event log.

Command Mnemonic(s) `$sc_$cpu_EVS_ClrLog`

Command Structure

[CFE_EVS_ClearLogCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_EVS_CMDPC` - command execution counter will increment
- `$sc_$cpu_EVS_LOGFULL` - The LogFullFlag in the Housekeeping telemetry will be cleared
- `$sc_$cpu_EVS_LOGOVERFLOWC` - The LogOverflowCounter in the Housekeeping telemetry will be reset to 0

Error Conditions

There are no error conditions for this command. If the Event Services receives the command, the log is cleared.

Criticality

Clearing the local event log is not particularly hazardous, as the result may be making available space to record valuable event data. However, inappropriately clearing the local event log could result in a loss of critical information. Note: the event log is a back-up log to the on-board recorder.

See also

[CFE_EVS_WRITE_LOG_DATA_FILE_CC](#), [CFE_EVS_SET_LOG_MODE_CC](#)

Definition at line 875 of file [cfe_evs_fcncodes.h](#).

CFE_EVS_DELETE_EVENT_FILTER_CC

```
#define CFE_EVS_DELETE_EVENT_FILTER_CC CFE\_EVS\_CCVAL(DELETE_EVENT_FILTER)
```

Name Delete Application Event Filter

Description

This command removes the given filter for the given application identifier and event identifier. Note: In order for this command to take effect, applications must be registered for Event Service.

Command Mnemonic(s) `$sc_$cpu_EVS_DelEvtFiltr`

Command Structure

[CFE_EVS_DeleteEventFilterCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_EVS_CMDPC` - command execution counter will increment
- The generation of [CFE_EVS_DELFILTER_EID](#) debug event message

Error Conditions

This command may fail for the following reason(s):

- Application name is not valid or not registered with event services
- Specified event ID is not found in the application event filter

Evidence of failure may be found in the following telemetry:

- `$sc_$cpu_EVS_CMDEC` - command error counter will increment
- An Error specific event message

Criticality

None.

See also

[CFE_EVS_SET_FILTER_CC](#), [CFE_EVS_RESET_FILTER_CC](#), [CFE_EVS_RESET_ALL_FILTERS_CC](#),
[CFE_EVS_ADD_EVENT_FILTER_CC](#)

Definition at line 730 of file [cfe_evs_fcncodes.h](#).

CFE_EVS_DISABLE_APP_EVENT_TYPE_CC

```
#define CFE_EVS_DISABLE_APP_EVENT_TYPE_CC CFE_EVS_CCVAL(DISABLE_APP_EVENT_TYPE)
```

Name Disable Application Event Type

Description

This command disables the command specified event type for the command specified application, preventing the application from sending event messages of the command specified event type through Event Service. An Event Type is defined to be a classification of an Event Message such as debug, informational, critical, and error. Note: In order for this command to take effect, applications must be registered for Event Service.

Command Mnemonic(s) `$sc_$cpu_EVS_DisAppEvtType`, `$sc_$cpu_EVS_DisAppEvtTypeMask`

Command Structure

[CFE_EVS_DisableAppEventTypeCmd_t](#) The following bit positions apply to structure member named 'BitMask'. Bit 0 - Debug Bit 1 - Informational Bit 2 - Error Bit 3 - Critical A one in a bit position means the event type will be disabled (or filtered) for the specified application. A zero in a bit position means the filtering state is unchanged for the specified application.

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_EVS_CMDPC` - command execution counter will increment
- The generation of [CFE_EVS_DISAPPENTTYPE_EID](#) debug event message
- The clearing of the Event Type Active Flag in EVS App Data File

Error Conditions

This command may fail for the following reason(s):

- BitMask field invalid - mask cannot be zero, and only bits 0-3 may be set
- Application name is not valid or not registered with event services

Evidence of failure may be found in the following telemetry:

- `$sc_$cpu_EVS_CMDEC` - command error counter will increment
- An Error specific event message

Criticality

Disabling an application's event type is not particularly hazardous, as the result may be shutting off unnecessary event messages and possible event flooding of the system. However, inappropriately disabling an application's event type could result in a loss of critical information and missed behavior for the ground system.

See also

[CFE_EVS_ENABLE_EVENT_TYPE_CC](#), [CFE_EVS_DISABLE_EVENT_TYPE_CC](#), [CFE_EVS_ENABLE_APP_EVENT_TYPE_CC](#),
[CFE_EVS_ENABLE_APP_EVENTS_CC](#), [CFE_EVS_DISABLE_APP_EVENTS_CC](#)

Definition at line 355 of file [cfe_evs_fcncodes.h](#).

CFE_EVS_DISABLE_APP_EVENTS_CC

```
#define CFE_EVS_DISABLE_APP_EVENTS_CC CFE_EVS_CCVAL(DISABLE_APP_EVENTS)
```

Name Disable Event Services for an Application

Description

This command disables the command specified application from sending events through Event Service. Note: In order for this command to take effect, applications must be registered for Event Service.

Command Mnemonic(s) `$sc_$cpu_EVS_DisAppEvGen`

Command Structure

[CFE_EVS_DisableAppEventsCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_EVS_CMDPC` - command execution counter will increment
- The generation of [CFE_EVS_DISAPPEVT_EID](#) debug event message

Error Conditions

This command may fail for the following reason(s):

- Application name is not valid or not registered with event services

Evidence of failure may be found in the following telemetry:

- `$sc_$cpu_EVS_CMDEC` - command error counter will increment
- An Error specific event message

Criticality

Disabling an application's events is not particularly hazardous, as the result may be shutting off unnecessary event messages and possible event flooding of the system. However, inappropriately disabling an application's events could result in a loss of critical information and missed behavior for the ground system.

See also

[CFE_EVS_ENABLE_EVENT_TYPE_CC](#), [CFE_EVS_DISABLE_EVENT_TYPE_CC](#), [CFE_EVS_ENABLE_APP_EVENT_TYPE_CC](#), [CFE_EVS_DISABLE_APP_EVENT_TYPE_CC](#), [CFE_EVS_ENABLE_APP_EVENTS_CC](#)

Definition at line 433 of file [cfe_evs_fcncodes.h](#).

CFE_EVS_DISABLE_EVENT_TYPE_CC

```
#define CFE_EVS_DISABLE_EVENT_TYPE_CC CFE_EVS_CCVAL(DISABLE_EVENT_TYPE)
```

Name Disable Event Type

Description

This command disables the command specified Event Type preventing event messages of this type to be sent through Event Service. An Event Type is defined to be a classification of an Event Message such as debug, informational, error and critical. This command is a global disable of a particular event type, it applies to all applications.

Command Mnemonic(s) `$sc_$cpu_EVS_DisEventType`, `$sc_$cpu_EVS_DisEventTypeMask`

Command Structure

[CFE_EVS_DisableEventTypeCmd_t](#) The following bit positions apply to structure member named 'BitMask'. Bit 0 - Debug Bit 1 - Informational Bit 2 - Error Bit 3 - Critical A one in a bit position means the event type will be disabled (or filtered). A zero in a bit position means the filtering state is unchanged.

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_EVS_CMDPC` - command execution counter will increment
- The generation of [CFE_EVS_DISEVTTYPE_EID](#) debug message

Error Conditions

This command may fail for the following reason(s):

- BitMask field invalid - mask cannot be zero, and only bits 0-3 may be set

Evidence of failure may be found in the following telemetry:

- `$sc_$cpu_EVS_CMDEC` - command error counter will increment
- An Error specific event message

Criticality

Disabling an event type is not particularly hazardous, as the result may be shutting off unnecessary event messages and possible event flooding of the system. However, inappropriately disabling an event type could result in a loss of critical information and missed behavior for the ground system.

See also

[CFE_EVS_ENABLE_EVENT_TYPE_CC](#), [CFE_EVS_ENABLE_APP_EVENT_TYPE_CC](#), [CFE_EVS_DISABLE_APP_EVENT_TYPE_CC](#), [CFE_EVS_ENABLE_APP_EVENTS_CC](#), [CFE_EVS_DISABLE_APP_EVENTS_CC](#)

Definition at line 203 of file [cfe_evs_fcncodes.h](#).

CFE_EVS_DISABLE_PORTS_CC

```
#define CFE_EVS_DISABLE_PORTS_CC CFE_EVS_CCVAL(DISABLE_PORTS)
```

Name Disable Event Services Output Ports

Description

This command disables the specified port from outputting event messages.

Command Mnemonic(s) \$sc_\$cpu_EVS_DisPort, \$sc_\$cpu_EVS_DisPortMask

Command Structure

[CFE_EVS_DisablePortsCmd_t](#) The following bit positions apply to structure member named 'BitMask'. Bit 0 - Port 1 Bit 1 - Port 2 Bit 2 - Port 3 Bit 3 - Port 4 A one in a bit position means the port will be disabled. A zero in a bit position means the port state is unchanged.

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **\$sc_\$cpu_EVS_CMDPC** - command execution counter will increment
- The generation of [CFE_EVS_DISPORT_EID](#) debug event message

Error Conditions

This command may fail for the following reason(s):

- BitMask field invalid - mask cannot be zero, and only bits 0-3 may be set

Evidence of failure may be found in the following telemetry:

- **\$sc_\$cpu_EVS_CMDEC** - command error counter will increment
- An Error specific event message

Criticality

None.

See also

[CFE_EVS_ENABLE_PORTS_CC](#)

Definition at line 589 of file [cfe_evs_fcncodes.h](#).

CFE_EVS_ENABLE_APP_EVENT_TYPE_CC

```
#define CFE_EVS_ENABLE_APP_EVENT_TYPE_CC CFE_EVS_CCVAL(ENABLE_APP_EVENT_TYPE)
```

Name Enable Application Event Type

Description

This command enables the command specified event type for the command specified application, allowing the application to send event messages of the command specified event type through Event Service. An Event Type is defined to be a classification of an Event Message such as debug, informational, critical, and error. Note: In order for this command to take effect, applications must be registered for Event Service.

Command Mnemonic(s) `$sc_$cpu_EVS_EnaAppEvtType`, `$sc_$cpu_EVS_EnaAppEvtTypeMask`

Command Structure

[CFE_EVS_EnableAppEventTypeCmd_t](#) The following bit positions apply to structure member named 'BitMask'. Bit 0 - Debug Bit 1 - Informational Bit 2 - Error Bit 3 - Critical A one in a bit position means the event type will be enabled (or unfiltered) for the specified application. A zero in a bit position means the filtering state is unchanged for the specified application.

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_EVS_CMDPC` - command execution counter will increment
- The generation of [CFE_EVS_ENAAPPEVTTYPE_EID](#) debug event message

Error Conditions

This command may fail for the following reason(s):

- BitMask field invalid - mask cannot be zero, and only bits 0-3 may be set
- Application name is not valid or not registered with event services

Evidence of failure may be found in the following telemetry:

- `$sc_$cpu_EVS_CMDEC` - command error counter will increment
- An Error specific event message

Criticality

Enabling an application event type is not particularly hazardous, as the result may be turning on necessary event messages and communication to the ground system. However, inappropriately enabling an application's event type could result in flooding of the ground system.

See also

[CFE_EVS_ENABLE_EVENT_TYPE_CC](#), [CFE_EVS_DISABLE_EVENT_TYPE_CC](#), [CFE_EVS_DISABLE_APP_EVENT_TYPE_CC](#), [CFE_EVS_ENABLE_APP_EVENTS_CC](#), [CFE_EVS_DISABLE_APP_EVENTS_CC](#)

Definition at line 302 of file [cfe_evs_fcncodes.h](#).

CFE_EVS_ENABLE_APP_EVENTS_CC

```
#define CFE_EVS_ENABLE_APP_EVENTS_CC CFE_EVS_CCVAL(ENABLE_APP_EVENTS)
```

Name Enable Event Services for an Application

Description

This command enables the command specified application to send events through the Event Service. Note: In order for this command to take effect, applications must be registered for Event Service.

Command Mnemonic(s) `$sc_$cpu_EVS_EnaAppEvGen`

Command Structure

`CFE_EVS_EnableAppEventsCmd_t`

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_EVS_CMDPC` - command execution counter will increment
- The generation of `CFE_EVS_ENAAPPEVT_EID` debug event message
- The setting of the Active Flag in EVS App Data File

Error Conditions

This command may fail for the following reason(s):

- Application name is not valid or not registered with event services

Evidence of failure may be found in the following telemetry:

- `$sc_$cpu_EVS_CMDEC` - command error counter will increment
- An Error specific event message

Criticality

Enabling an application events is not particularly hazardous, as the result may be turning on necessary event messages and communication to the ground system. However, inappropriately enabling an application's events could result in flooding of the ground system.

See also

`CFE_EVS_ENABLE_EVENT_TYPE_CC`, `CFE_EVS_DISABLE_EVENT_TYPE_CC`, `CFE_EVS_ENABLE_APP_EVENT_TYPE_CC`,
`CFE_EVS_DISABLE_APP_EVENT_TYPE_CC`, `CFE_EVS_DISABLE_APP_EVENTS_CC`

Definition at line 394 of file `cf_evs_fcncodes.h`.

`CFE_EVS_ENABLE_EVENT_TYPE_CC`

```
#define CFE_EVS_ENABLE_EVENT_TYPE_CC CFE_EVS_CCVAL(ENABLE_EVENT_TYPE)
```

Name Enable Event Type

Description

This command enables the command specified Event Type allowing event messages of this type to be sent through Event Service. An Event Type is defined to be a classification of an Event Message such as debug, informational, error and critical. This command is a global enable of a particular event type, it applies to all applications.

Command Mnemonic(s) `$sc_$cpu_EVS_EnaEventType`, `$sc_$cpu_EVS_EnaEventTypeMask`

Command Structure

[CFE_EVS_EnableEventTypeCmd_t](#) The following bit positions apply to structure member named 'BitMask'. Bit 0 - Debug Bit 1 - Informational Bit 2 - Error Bit 3 - Critical A one in a bit position means the event type will be enabled (or unfiltered). A zero in a bit position means the filtering state is unchanged.

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_EVS_CMDPC` - command execution counter will increment
- The generation of [CFE_EVS_ENAEVTTYPE_EID](#) debug message

Error Conditions

This command may fail for the following reason(s):

- BitMask field invalid - mask cannot be zero, and only bits 0-3 may be set

Evidence of failure may be found in the following telemetry:

- `$sc_$cpu_EVS_CMDEC` - command error counter will increment
- An Error specific event message

Criticality

Enabling an event type is not particularly hazardous, as the result may be turning on necessary event messages and communication to the ground system. However, inappropriately enabling an event type could result in flooding of the system.

See also

[CFE_EVS_DISABLE_EVENT_TYPE_CC](#), [CFE_EVS_ENABLE_APP_EVENT_TYPE_CC](#), [CFE_EVS_DISABLE_APP_EVENT_TYPE_CC](#), [CFE_EVS_ENABLE_APP_EVENTS_CC](#), [CFE_EVS_DISABLE_APP_EVENTS_CC](#)

Definition at line 154 of file [cfe_evs_fcncodes.h](#).

CFE_EVS_ENABLE_PORTS_CC

```
#define CFE_EVS_ENABLE_PORTS_CC CFE_EVS_CCVAL(ENABLE_PORTS)
```

Name Enable Event Services Output Ports

Description

This command enables the command specified port to output event messages

Command Mnemonic(s) `$sc_$cpu_EVS_EnaPort`, `$sc_$cpu_EVS_EnaPortMask`

Command Structure

[CFE_EVS_EnablePortsCmd_t](#) The following bit positions apply to structure member named 'BitMask'. Bit 0 - Port 1 Bit 1 - Port 2 Bit 2 - Port 3 Bit 3 - Port 4 A one in a bit position means the port will be enabled. A zero in a bit position means the port state is unchanged.

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_EVS_CMDPC` - command execution counter will increment
- The generation of `CFE_EVS_ENAPORT_EID` debug event message

Error Conditions

This command may fail for the following reason(s):

- BitMask field invalid - mask cannot be zero, and only bits 0-3 may be set

Evidence of failure may be found in the following telemetry:

- `$sc_$cpu_EVS_CMDEC` - command error counter will increment
- An Error specific event message

Criticality

None.

See also

[CFE_EVS_DISABLE_PORTS_CC](#)

Definition at line 550 of file [cfe_evs_fcncodes.h](#).

CFE_EVS_NOOP_CC

```
#define CFE_EVS_NOOP_CC CFE_EVS_CCVAL(NOOP)
```

Name Event Services No-Op

Description

This command performs no other function than to increment the command execution counter. The command may be used to verify general aliveness of the Event Services task.

Command Mnemonic(s) `$sc_$cpu_EVS_NOOP`

Command Structure

[CFE_EVS_NoopCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_EVS_CMDPC` - command execution counter will increment
- The `CFE_EVS_NOOP_EID` informational event message will be generated

Error Conditions

There are no error conditions for this command. If the Event Services receives the command, the event is sent (although it may be filtered by EVS itself) and the counter is incremented unconditionally.

Criticality

None

See also

Definition at line 67 of file [cfe_evs_fcncodes.h](#).

CFE_EVS_RESET_ALL_FILTERS_CC

```
#define CFE_EVS_RESET_ALL_FILTERS_CC CFE_EVS_CCVAL(RESET_ALL_FILTERS)
```

Name Reset All Event Filters for an Application

Description

This command resets all of the command specified applications event filters. Note: In order for this command to take effect, applications must be registered for Event Service.

Command Mnemonic(s) \$sc_\$cpu_EVS_RstAllFitr

Command Structure

[CFE_EVS_ResetAllFiltersCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- [\\$sc_\\$cpu_EVS_CMDPC](#) - command execution counter will increment
- The generation of [CFE_EVS_RSTALLFILTER_EID](#) debug event message

Error Conditions

This command may fail for the following reason(s):

- Application name is not valid or not registered with event services

Evidence of failure may be found in the following telemetry:

- [\\$sc_\\$cpu_EVS_CMDEC](#) - command error counter will increment
- An Error specific event message

Criticality

None.

See also

[CFE_EVS_SET_FILTER_CC](#), [CFE_EVS_RESET_FILTER_CC](#), [CFE_EVS_ADD_EVENT_FILTER_CC](#), [CFE_EVS_DELETE_EVENT_FILTER_CC](#)

Definition at line 659 of file [cfe_evs_fcncodes.h](#).

CFE_EVS_RESET_APP_COUNTER_CC

```
#define CFE_EVS_RESET_APP_COUNTER_CC CFE_EVS_CCVAL(RESET_APP_COUNTER)
```

Name Reset Application Event Counters

Description

This command sets the command specified application's event counter to zero. Note: In order for this command to take effect, applications must be registered for Event Service.

Command Mnemonic(s) \$sc_\$cpu_EVS_RstAppCtrs

Command Structure

[CFE_EVS_ResetAppCounterCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_EVS_CMDPC` - command execution counter will increment
- The generation of [CFE_EVS_RSTVTCNT_EID](#) debug event message

Error Conditions

This command may fail for the following reason(s):

- Application name is not valid or not registered with event services

Evidence of failure may be found in the following telemetry:

- `$sc_$cpu_EVS_CMDEC` - command error counter will increment
- An Error specific event message

Criticality

This command is not inherently dangerous. However, it is possible for ground systems and on-board safing procedures to be designed such that they react to changes in the counter value that is reset by this command.

See also

[CFE_EVS_RESET_COUNTERS_CC](#)

Definition at line 469 of file [cfe_evs_fcncodes.h](#).

CFE_EVS_RESET_COUNTERS_CC

```
#define CFE_EVS_RESET_COUNTERS_CC CFE\_EVS\_CCVAL(RESET_COUNTERS)
```

Name Event Services Reset Counters

Description

This command resets the following counters within the Event Services housekeeping telemetry:

- Command Execution Counter (`$sc_$cpu_EVS_CMDPC`)
- Command Error Counter (`$sc_$cpu_EVS_CMDEC`)

Command Mnemonic(s) `$sc_$cpu_EVS_ResetCtrs`

Command Structure

[CFE_EVS_ResetCountersCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_EVS_CMDPC` - command execution counter will be reset to 0
- `$sc_$cpu_EVS_CMDEC` - command error counter will be reset to 0
- The [CFE_EVS_RSTCNT_EID](#) debug event message will be generated

Error Conditions

There are no error conditions for this command. If the Event Services receives the command, the event is sent (although it may be filtered by EVS) and the counter is incremented unconditionally.

Criticality

This command is not inherently dangerous. However, it is possible for ground systems and on-board safing procedures to be designed such that they react to changes in the counter values that are reset by this command.

See also

[CFE_EVS_RESET_APP_COUNTER_CC](#)

Definition at line 106 of file [cfe_evs_fcncodes.h](#).

CFE_EVS_RESET_FILTER_CC

```
#define CFE_EVS_RESET_FILTER_CC CFE_EVS_CCVAL(RESET_FILTER)
```

Name Reset an Event Filter for an Application

Description

This command resets the command specified application's event filter for the command specified event ID. Note: In order for this command to take effect, applications must be registered for Event Service.

Command Mnemonic(s) \$sc_\$cpu_EVS_RstBinFltrCtr

Command Structure

[CFE_EVS_ResetFilterCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **\$sc_\$cpu_EVS_CMDPC** - command execution counter will increment
- The generation of [CFE_EVS_RSTFILTER_EID](#) debug event message

Error Conditions

This command may fail for the following reason(s):

- Application name is not valid or not registered with event services
- Specified event ID is not found in the application event filter

Evidence of failure may be found in the following telemetry:

- **\$sc_\$cpu_EVS_CMDEC** - command error counter will increment
- An Error specific event message

Criticality

None.

See also

[CFE_EVS_SET_FILTER_CC](#), [CFE_EVS_RESET_ALL_FILTERS_CC](#), [CFE_EVS_ADD_EVENT_FILTER_CC](#), [CFE_EVS_DELETE_EVENT_FILTER_CC](#)

Definition at line 625 of file [cfe_evs_fcncodes.h](#).

CFE_EVS_SET_EVENT_FORMAT_MODE_CC

```
#define CFE_EVS_SET_EVENT_FORMAT_MODE_CC CFE_EVS_CCVAL(SET_EVENT_FORMAT_MODE)
```

Name Set Event Format Mode

Description

This command sets the event format mode to the command specified value. The event format mode may be either short or long. A short event format detaches the Event Data from the event message and only includes the following information in the event packet: Processor ID, Application ID, Event ID, and Event Type. Refer to section 5.3.3.4 for a description of the Event Service event packet contents. Event Data is defined to be data describing an Event that is supplied to the cFE Event Service. ASCII text strings are used as the primary format for Event Data because heritage ground systems use string compares as the basis for their automated alert systems. Two systems, ANSR and SERS were looked at for interface definitions. The short event format is used to accommodate experiences with limited telemetry bandwidth. The long event format includes all event information included within the short format along with the Event Data.

Command Mnemonic(s) \$sc_\$cpu_EVS_SetEvtFmt

Command Structure

[CFE_EVS_SetEventFormatModeCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **\$sc_\$cpu_EVS_CMDPC** - command execution counter will increment
- The generation of [CFE_EVS_SETEVTFMTMOD_EID](#) debug message

Error Conditions

This command may fail for the following reason(s):

- Invalid MsgFormat mode selection

Evidence of failure may be found in the following telemetry:

- **\$sc_\$cpu_EVS_CMDEC** - command error counter will increment
- An Error specific event message

Criticality

Setting the event format mode is not particularly hazardous, as the result may be saving necessary bandwidth. However, inappropriately setting the event format mode could result in a loss of information and missed behavior for the ground system

See also

Definition at line 250 of file [cfe_evs_fcncodes.h](#).

CFE_EVS_SET_FILTER_CC

```
#define CFE_EVS_SET_FILTER_CC CFE_EVS_CCVAL(SET_FILTER)
```

Name Set Application Event Filter

Description

This command sets the command specified application's event filter mask to the command specified value for the command specified event. Note: In order for this command to take effect, applications must be registered for Event Service.

Command Mnemonic(s) \$sc_\$cpu_EVS_SetBinFiltrMask

Command Structure

[CFE_EVS_SetFilterCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- [\\$sc_\\$cpu_EVS_CMDPC](#) - command execution counter will increment
- The generation of [CFE_EVS_SETFILTERMSK_EID](#) debug event message

Error Conditions

This command may fail for the following reason(s):

- Application name is not valid or not registered with event services
- Specified event ID is not found in the application event filter

Evidence of failure may be found in the following telemetry:

- [\\$sc_\\$cpu_EVS_CMDEC](#) - command error counter will increment
- An Error specific event message

Criticality

Setting an application event filter mask is not particularly hazardous, as the result may be shutting off unnecessary event messages and possible event flooding of the system. However, inappropriately setting an application's event filter mask could result in a loss of critical information and missed behavior for the ground system or flooding of the ground system.

See also

[CFE_EVS_RESET_FILTER_CC](#), [CFE_EVS_RESET_ALL_FILTERS_CC](#), [CFE_EVS_ADD_EVENT_FILTER_CC](#), [CFE_EVS_DELETE_EVENT_FILTER_CC](#)

Definition at line 511 of file [cfe_evs_fcncodes.h](#).

CFE_EVS_SET_LOG_MODE_CC

```
#define CFE_EVS_SET_LOG_MODE_CC CFE_EVS_CCVAL (SET_LOG_MODE)
```

Name Set Logging Mode

Description

This command sets the logging mode to the command specified value.

Command Mnemonic(s) \$sc_\$cpu_EVS_SetLogMode

Command Structure

[CFE_EVS_SetLogModeCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **\$sc_\$cpu_EVS_CMDPC** - command execution counter will increment
- The generation of [CFE_EVS_LOGMODE_EID](#) debug event message

Error Conditions

This command may fail for the following reason(s):

- Invalid LogMode selected - must be either [CFE_EVS_LogMode_OVERWRITE](#) or [CFE_EVS_LogMode_DISCARD](#)

Evidence of failure may be found in the following telemetry:

- **\$sc_\$cpu_EVS_CMDEC** - command error counter will increment
- An Error specific event message

Criticality

Setting the event logging mode is not particularly hazardous, as the result may be saving valuable event data. However, inappropriately setting the log mode could result in a loss of critical information. Note: the event log is a back-up log to the on-board recorder.

See also

[CFE_EVS_WRITE_LOG_DATA_FILE_CC](#), [CFE_EVS_CLEAR_LOG_CC](#)

Definition at line 840 of file [cfe_evs_fcncodes.h](#).

CFE_EVS_WRITE_APP_DATA_FILE_CC

```
#define CFE_EVS_WRITE_APP_DATA_FILE_CC CFE_EVS_CCVAL (WRITE_APP_DATA_FILE)
```

Name Write Event Services Application Information to File

Description

This command writes all application data to a file for all applications that have registered with the EVS. The application data includes the Application ID, Active Flag, Event Count, Event Types Active Flag, and Filter Data.

Command Mnemonic(s) \$sc_\$cpu_EVS_WriteAppData2File

Command Structure

[CFE_EVS_WriteAppDataFileCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_EVS_CMDPC` - command execution counter will increment
- The generation of [CFE_EVS_WRDAT_EID](#) debug event message
- The file specified in the command (or the default specified by the [CFE_PLATFORM_EVS_DEFAULT_APP_DATA_FILE](#) configuration parameter) will be updated with the latest information.

Error Conditions

This command may fail for the following reason(s):

- The specified FileName cannot be parsed
- An Error occurs while trying to write to the file

Evidence of failure may be found in the following telemetry:

- `$sc_$cpu_EVS_CMDEC` - command error counter will increment
- An Error specific event message

Criticality

Writing a file is not particularly hazardous, but if proper file management is not taken, then the file system can fill up if this command is used repeatedly.

See also

[CFE_EVS_WRITE_LOG_DATA_FILE_CC](#), [CFE_EVS_SET_LOG_MODE_CC](#)

Definition at line 769 of file [cfe_evs_fcncodes.h](#).

CFE_EVS_WRITE_LOG_DATA_FILE_CC

```
#define CFE_EVS_WRITE_LOG_DATA_FILE_CC CFE_EVS_CCVAL(WRITE_LOG_DATA_FILE)
```

Name Write Event Log to File

Description

This command requests the Event Service to generate a file containing the contents of the local event log.

Command Mnemonic(s) `$sc_$cpu_EVS_WriteLog2File`

Command Structure

[CFE_EVS_WriteLogDataFileCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_EVS_CMDPC` - command execution counter will increment
- The generation of `CFE_EVS_WRLOG_EID` debug event message

Error Conditions

This command may fail for the following reason(s):

- The specified FileName cannot be parsed
- An Error occurs while trying to write to the file

Evidence of failure may be found in the following telemetry:

- `$sc_$cpu_EVS_CMDEC` - command error counter will increment
- An Error specific event message

Criticality

Writing a file is not particularly hazardous, but if proper file management is not taken, then the file system can fill up if this command is used repeatedly.

See also

[CFE_EVS_WRITE_APP_DATA_FILE_CC](#), [CFE_EVS_SET_LOG_MODE_CC](#), [CFE_EVS_CLEAR_LOG_CC](#)

Definition at line 804 of file [cfe_evs_fcncodes.h](#).

11.160 cfe_evs_fcncodes.h

[Go to the documentation of this file.](#)

```
00001 /*****
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"
00003  *
00004  * Copyright (c) 2023 United States Government as represented by the
00005  * Administrator of the National Aeronautics and Space Administration.
00006  * All Rights Reserved.
00007  *
00008  * Licensed under the Apache License, Version 2.0 (the "License"); you may
00009  * not use this file except in compliance with the License. You may obtain
00010  * a copy of the License at http://www.apache.org/licenses/LICENSE-2.0
00011  *
00012  * Unless required by applicable law or agreed to in writing, software
00013  * distributed under the License is distributed on an "AS IS" BASIS,
00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00028 #ifndef CFE_EVS_FCNCODES_H
00029 #define CFE_EVS_FCNCODES_H
00030
00031 #include "cfe_evs_fcncode_values.h"
00032
00067 #define CFE_EVS_NOOP_CC CFE_EVS_CCVAL(NOOP)
00068
00106 #define CFE_EVS_RESET_COUNTERS_CC CFE_EVS_CCVAL(RESET_COUNTERS)
00107
00154 #define CFE_EVS_ENABLE_EVENT_TYPE_CC CFE_EVS_CCVAL(ENABLE_EVENT_TYPE)
00155
00203 #define CFE_EVS_DISABLE_EVENT_TYPE_CC CFE_EVS_CCVAL(DISABLE_EVENT_TYPE)
00204
00250 #define CFE_EVS_SET_EVENT_FORMAT_MODE_CC CFE_EVS_CCVAL(SET_EVENT_FORMAT_MODE)
00251
00302 #define CFE_EVS_ENABLE_APP_EVENT_TYPE_CC CFE_EVS_CCVAL(ENABLE_APP_EVENT_TYPE)
```

```

00303
00355 #define CFE_EVS_DISABLE_APP_EVENT_TYPE_CC CFE_EVS_CCVAL(DISABLE_APP_EVENT_TYPE)
00356
00394 #define CFE_EVS_ENABLE_APP_EVENTS_CC CFE_EVS_CCVAL(ENABLE_APP_EVENTS)
00395
00433 #define CFE_EVS_DISABLE_APP_EVENTS_CC CFE_EVS_CCVAL(DISABLE_APP_EVENTS)
00434
00469 #define CFE_EVS_RESET_APP_COUNTER_CC CFE_EVS_CCVAL(RESET_APP_COUNTER)
00470
00511 #define CFE_EVS_SET_FILTER_CC CFE_EVS_CCVAL(SET_FILTER)
00512
00550 #define CFE_EVS_ENABLE_PORTS_CC CFE_EVS_CCVAL(ENABLE_PORTS)
00551
00589 #define CFE_EVS_DISABLE_PORTS_CC CFE_EVS_CCVAL(DISABLE_PORTS)
00590
00625 #define CFE_EVS_RESET_FILTER_CC CFE_EVS_CCVAL(RESET_FILTER)
00626
00659 #define CFE_EVS_RESET_ALL_FILTERS_CC CFE_EVS_CCVAL(RESET_ALL_FILTERS)
00660
00695 #define CFE_EVS_ADD_EVENT_FILTER_CC CFE_EVS_CCVAL(ADD_EVENT_FILTER)
00696
00730 #define CFE_EVS_DELETE_EVENT_FILTER_CC CFE_EVS_CCVAL(DELETE_EVENT_FILTER)
00731
00769 #define CFE_EVS_WRITE_APP_DATA_FILE_CC CFE_EVS_CCVAL(WRITE_APP_DATA_FILE)
00770
00804 #define CFE_EVS_WRITE_LOG_DATA_FILE_CC CFE_EVS_CCVAL(WRITE_LOG_DATA_FILE)
00805
00840 #define CFE_EVS_SET_LOG_MODE_CC CFE_EVS_CCVAL(SET_LOG_MODE)
00841
00875 #define CFE_EVS_CLEAR_LOG_CC CFE_EVS_CCVAL(CLEAR_LOG)
00878 #endif

```

11.161 cfe/modules/evs/fsw/inc/cfe_evs_interface_cfg.h File Reference

```
#include "cfe_evs_interface_cfg_values.h"
```

Macros

- #define [CFE_MISSION_EVS_MAX_MESSAGE_LENGTH](#) [CFE_MISSION_EVS_CFGVAL](#)(MAX_MESSAGE_LENGTH)
- #define [DEFAULT_CFE_MISSION_EVS_MAX_MESSAGE_LENGTH](#) 122

11.161.1 Detailed Description

CFE Event Services (CFE_EVS) Interface Configuration Header File
Definition in file [cfe_evs_interface_cfg.h](#).

11.161.2 Macro Definition Documentation

CFE_MISSION_EVS_MAX_MESSAGE_LENGTH

```
#define CFE_MISSION_EVS_MAX_MESSAGE_LENGTH CFE\_MISSION\_EVS\_CFGVAL(MAX_MESSAGE_LENGTH)
```

Purpose Maximum Event Message Length

Description:

Indicates the maximum length (in characters) of the formatted text string portion of an event message

This length does not need to include an extra character for NULL termination.

Limits

Not Applicable

Definition at line 41 of file [cfe_evs_interface_cfg.h](#).

DEFAULT_CFE_MISSION_EVS_MAX_MESSAGE_LENGTH

```
#define DEFAULT_CFE_MISSION_EVS_MAX_MESSAGE_LENGTH 122
```

Definition at line 42 of file [cfe_evs_interface_cfg.h](#).

11.162 cfe_evs_interface_cfg.h

[Go to the documentation of this file.](#)

```
00001 /*****
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"
00003  *
00004  * Copyright (c) 2023 United States Government as represented by the
00005  * Administrator of the National Aeronautics and Space Administration.
00006  * All Rights Reserved.
00007  *
00008  * Licensed under the Apache License, Version 2.0 (the "License"); you may
00009  * not use this file except in compliance with the License. You may obtain
00010  * a copy of the License at http://www.apache.org/licenses/LICENSE-2.0
00011  *
00012  * Unless required by applicable law or agreed to in writing, software
00013  * distributed under the License is distributed on an "AS IS" BASIS,
00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00024 #ifndef CFE_EVS_INTERFACE_CFG_H
00025 #define CFE_EVS_INTERFACE_CFG_H
00026
00027 #include "cfe_evs_interface_cfg_values.h"
00028
00041 #define CFE_MISSION_EVS_MAX_MESSAGE_LENGTH CFE_MISSION_EVS_CFGVAL(MAX_MESSAGE_LENGTH)
00042 #define DEFAULT_CFE_MISSION_EVS_MAX_MESSAGE_LENGTH 122
00043
00044 #endif
```

11.163 cfe/modules/evs/fsw/inc/cfe_evs_internal_cfg.h File Reference

```
#include "cfe_evs_mission_cfg.h"
#include "cfe_evs_internal_cfg_values.h"
```

Macros

- #define [CFE_PLATFORM_EVS_START_TASK_PRIORITY](#) CFE_PLATFORM_EVS_CFGVAL(START_TASK_↵
PRIORITY)
- #define [DEFAULT_CFE_PLATFORM_EVS_START_TASK_PRIORITY](#) 61
- #define [CFE_PLATFORM_EVS_START_TASK_STACK_SIZE](#) CFE_PLATFORM_EVS_CFGVAL(START_↵
TASK_STACK_SIZE)
- #define [DEFAULT_CFE_PLATFORM_EVS_START_TASK_STACK_SIZE](#) CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
- #define [CFE_PLATFORM_EVS_MAX_EVENT_FILTERS](#) CFE_PLATFORM_EVS_CFGVAL(MAX_EVENT_↵
FILTERS)
- #define [DEFAULT_CFE_PLATFORM_EVS_MAX_EVENT_FILTERS](#) 8
- #define [CFE_PLATFORM_EVS_MAX_APP_EVENT_BURST](#) CFE_PLATFORM_EVS_CFGVAL(MAX_APP_↵
EVENT_BURST)
- #define [DEFAULT_CFE_PLATFORM_EVS_MAX_APP_EVENT_BURST](#) 32
- #define [CFE_PLATFORM_EVS_APP_EVENTS_PER_SEC](#) CFE_PLATFORM_EVS_CFGVAL(APP_EVENTS_↵
_PER_SEC)
- #define [DEFAULT_CFE_PLATFORM_EVS_APP_EVENTS_PER_SEC](#) 15
- #define [CFE_PLATFORM_EVS_DEFAULT_LOG_FILE](#) CFE_PLATFORM_EVS_CFGVAL(DEFAULT_LOG_FILE)
- #define [DEFAULT_CFE_PLATFORM_EVS_DEFAULT_LOG_FILE](#) "/ram/cfe_evs.log"
- #define [CFE_PLATFORM_EVS_LOG_MAX](#) CFE_PLATFORM_EVS_CFGVAL(LOG_MAX)
- #define [DEFAULT_CFE_PLATFORM_EVS_LOG_MAX](#) 20

- `#define CFE_PLATFORM_EVS_DEFAULT_APP_DATA_FILE CFE_PLATFORM_EVS_CFGVAL(DEFAULT_APP_DATA_FILE)`
- `#define DEFAULT_CFE_PLATFORM_EVS_DEFAULT_APP_DATA_FILE "/ram/cfe_evs_app.dat"`
- `#define CFE_PLATFORM_EVS_PORT_DEFAULT CFE_PLATFORM_EVS_CFGVAL(PORT_DEFAULT)`
- `#define DEFAULT_CFE_PLATFORM_EVS_PORT_DEFAULT 0x0001`
- `#define CFE_PLATFORM_EVS_DEFAULT_TYPE_FLAG CFE_PLATFORM_EVS_CFGVAL(DEFAULT_TYPE_FLAG)`
- `#define DEFAULT_CFE_PLATFORM_EVS_DEFAULT_TYPE_FLAG 0xE`
- `#define CFE_PLATFORM_EVS_DEFAULT_LOG_MODE CFE_PLATFORM_EVS_CFGVAL(DEFAULT_LOG_MODE)`
- `#define DEFAULT_CFE_PLATFORM_EVS_DEFAULT_LOG_MODE 1`
- `#define CFE_PLATFORM_EVS_DEFAULT_MSG_FORMAT_MODE CFE_PLATFORM_EVS_CFGVAL(DEFAULT_MSG_FORMAT_MODE)`
- `#define DEFAULT_CFE_PLATFORM_EVS_DEFAULT_MSG_FORMAT_MODE CFE_EVS_MsgFormat_LONG`

11.163.1 Detailed Description

CFE Event Services (CFE_EVS) Application Private Config Definitions

This provides default values for configurable items that are internal to this module and do NOT affect the interface(s) of this module. Changes to items in this file only affect the local module and will be transparent to external entities that are using the public interface(s).

Definition in file [cfe_evs_internal_cfg.h](#).

11.163.2 Macro Definition Documentation

CFE_PLATFORM_EVS_APP_EVENTS_PER_SEC

```
#define CFE_PLATFORM_EVS_APP_EVENTS_PER_SEC CFE_PLATFORM_EVS_CFGVAL(APP_EVENTS_PER_SEC)
```

Purpose Sustained number of event messages per second per app before squelching

Description:

Sustained number of events that may be emitted per app per second.

Limits

This number must be less than or equal to [CFE_PLATFORM_EVS_MAX_APP_EVENT_BURST](#). Values lower than 8 may cause functional and unit test failures.

Definition at line 99 of file [cfe_evs_internal_cfg.h](#).

CFE_PLATFORM_EVS_DEFAULT_APP_DATA_FILE

```
#define CFE_PLATFORM_EVS_DEFAULT_APP_DATA_FILE CFE_PLATFORM_EVS_CFGVAL(DEFAULT_APP_DATA_FILE)
```

Purpose Default EVS Application Data Filename

Description:

The value of this constant defines the filename used to store the EVS Application Data(event counts/filtering information). This filename is used only when no filename is specified in the command to dump the event log.

Limits

The length of each string, including the NULL terminator cannot exceed the [OS_MAX_PATH_LEN](#) value.

Definition at line 143 of file [cfe_evs_internal_cfg.h](#).

CFE_PLATFORM_EVS_DEFAULT_LOG_FILE

```
#define CFE_PLATFORM_EVS_DEFAULT_LOG_FILE CFE_PLATFORM_EVS_CFGVAL(DEFAULT_LOG_FILE)
```

Purpose Default Event Log Filename

Description:

The value of this constant defines the filename used to store the Event Services local event log. This filename is used only when no filename is specified in the command to dump the event log.

Limits

The length of each string, including the NULL terminator cannot exceed the [OS_MAX_PATH_LEN](#) value.

Definition at line 114 of file [cfe_evs_internal_cfg.h](#).

CFE_PLATFORM_EVS_DEFAULT_LOG_MODE

```
#define CFE_PLATFORM_EVS_DEFAULT_LOG_MODE CFE_PLATFORM_EVS_CFGVAL(DEFAULT_LOG_MODE)
```

Purpose Default EVS Local Event Log Mode

Description:

Defines a state of overwrite(0) or discard(1) for the operation of the EVS local event log. The log may operate in either Overwrite mode = 0, where once the log becomes full the oldest event in the log will be overwritten, or Discard mode = 1, where once the log becomes full the contents of the log are preserved and the new event is discarded. Overwrite Mode = 0, Discard Mode = 1.

Limits

The valid settings are 0 or 1

Definition at line 193 of file [cfe_evs_internal_cfg.h](#).

CFE_PLATFORM_EVS_DEFAULT_MSG_FORMAT_MODE

```
#define CFE_PLATFORM_EVS_DEFAULT_MSG_FORMAT_MODE CFE_PLATFORM_EVS_CFGVAL(DEFAULT_MSG_FORMAT_MODE)
```

Purpose Default EVS Message Format Mode

Description:

Defines the default message format (long or short) for event messages being sent to the ground. Choose between [CFE_EVS_MsgFormat_LONG](#) or [CFE_EVS_MsgFormat_SHORT](#).

Limits

The valid settings are [CFE_EVS_MsgFormat_LONG](#) or [CFE_EVS_MsgFormat_SHORT](#)

Definition at line 207 of file [cfe_evs_internal_cfg.h](#).

CFE_PLATFORM_EVS_DEFAULT_TYPE_FLAG

```
#define CFE_PLATFORM_EVS_DEFAULT_TYPE_FLAG CFE_PLATFORM_EVS_CFGVAL(DEFAULT_TYPE_FLAG)
```

Purpose Default EVS Event Type Filter Mask

Description:

Defines a state of on or off for all four event types. The term event 'type' refers to the criticality level and may be Debug, Informational, Error or Critical. Each event type has a bit position. (bit 0 = Debug, bit 1 = Info, bit 2 = Error, bit 3 = Critical). This is a global setting, meaning it applies to all applications. To filter an event type, set its bit to zero. For example, 0xE means Debug = OFF, Info = ON, Error = ON, Critical = ON

Limits

The valid settings are 0x0 to 0xF.

Definition at line 176 of file [cfe_evs_internal_cfg.h](#).

CFE_PLATFORM_EVS_LOG_MAX

```
#define CFE_PLATFORM_EVS_LOG_MAX CFE_PLATFORM_EVS_CFGVAL(LOG_MAX)
```

Purpose Maximum Number of Events in EVS Local Event Log

Description:

Dictates the EVS local event log capacity. Units are the number of events.

Limits

There are no restrictions on the lower and upper limits however, the maximum log size is system dependent and should be verified.

Definition at line 127 of file [cfe_evs_internal_cfg.h](#).

CFE_PLATFORM_EVS_MAX_APP_EVENT_BURST

```
#define CFE_PLATFORM_EVS_MAX_APP_EVENT_BURST CFE_PLATFORM_EVS_CFGVAL(MAX_APP_EVENT_BURST)
```

Purpose Maximum number of event before squelching

Description:

Maximum number of events that may be emitted per app per second. Setting this to 0 will cause events to be unrestricted.

Limits

This number must be less than or equal to INT_MAX/1000

Definition at line 86 of file [cfe_evs_internal_cfg.h](#).

CFE_PLATFORM_EVS_MAX_EVENT_FILTERS

```
#define CFE_PLATFORM_EVS_MAX_EVENT_FILTERS CFE_PLATFORM_EVS_CFGVAL(MAX_EVENT_FILTERS)
```

Purpose Define Maximum Number of Event Filters per Application

Description:

Maximum number of events that may be filtered per application.

Limits

There are no restrictions on the lower and upper limits however, the maximum number of event filters is system dependent and should be verified.

Definition at line 73 of file [cfe_evs_internal_cfg.h](#).

CFE_PLATFORM_EVS_PORT_DEFAULT

```
#define CFE_PLATFORM_EVS_PORT_DEFAULT CFE_PLATFORM_EVS_CFGVAL(PORT_DEFAULT)
```

Purpose Default EVS Output Port State

Description:

Defines the default port state (enabled or disabled) for the four output ports defined within the Event Service. Port 1 is usually the uart output terminal. To enable a port, set the proper bit to a 1. Bit 0 is port 1, bit 1 is port2 etc.

Limits

The valid settings are 0x0 to 0xF.

Definition at line 158 of file [cfe_evs_internal_cfg.h](#).

CFE_PLATFORM_EVS_START_TASK_PRIORITY

```
#define CFE_PLATFORM_EVS_START_TASK_PRIORITY CFE_PLATFORM_EVS_CFGVAL(START_TASK_PRIORITY)
```

Purpose Define EVS Task Priority

Description:

Defines the cFE_EVS Task priority.

Limits

Not Applicable

Definition at line 43 of file [cfe_evs_internal_cfg.h](#).

CFE_PLATFORM_EVS_START_TASK_STACK_SIZE

```
#define CFE_PLATFORM_EVS_START_TASK_STACK_SIZE CFE_PLATFORM_EVS_CFGVAL(START_TASK_STACK_SIZE)
```

Purpose Define EVS Task Stack Size

Description:

Defines the cFE_EVS Task Stack Size

Limits

There is a lower limit of 2048 on this configuration parameter. There are no restrictions on the upper limit however, the maximum stack size is system dependent and should be verified. Most operating systems provide tools for measuring the amount of stack used by a task during operation. It is always a good idea to verify that no more than 1/2 of the stack is used.

Definition at line 59 of file [cfe_evs_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_EVS_APP_EVENTS_PER_SEC

```
#define DEFAULT_CFE_PLATFORM_EVS_APP_EVENTS_PER_SEC 15
```

Definition at line 100 of file [cfe_evs_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_EVS_DEFAULT_APP_DATA_FILE

```
#define DEFAULT_CFE_PLATFORM_EVS_DEFAULT_APP_DATA_FILE "/ram/cfe_evs_app.dat"
```

Definition at line 144 of file [cfe_evs_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_EVS_DEFAULT_LOG_FILE

```
#define DEFAULT_CFE_PLATFORM_EVS_DEFAULT_LOG_FILE "/ram/cfe_evs.log"
```

Definition at line 115 of file [cfe_evs_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_EVS_DEFAULT_LOG_MODE

```
#define DEFAULT_CFE_PLATFORM_EVS_DEFAULT_LOG_MODE 1
```

Definition at line 194 of file [cfe_evs_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_EVS_DEFAULT_MSG_FORMAT_MODE

```
#define DEFAULT_CFE_PLATFORM_EVS_DEFAULT_MSG_FORMAT_MODE CFE_EVS_MsgFormat_LONG
```

Definition at line 208 of file [cfe_evs_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_EVS_DEFAULT_TYPE_FLAG

```
#define DEFAULT_CFE_PLATFORM_EVS_DEFAULT_TYPE_FLAG 0xE
```

Definition at line 177 of file [cfe_evs_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_EVS_LOG_MAX

```
#define DEFAULT_CFE_PLATFORM_EVS_LOG_MAX 20
```

Definition at line 128 of file [cfe_evs_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_EVS_MAX_APP_EVENT_BURST

```
#define DEFAULT_CFE_PLATFORM_EVS_MAX_APP_EVENT_BURST 32
```

Definition at line 87 of file [cfe_evs_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_EVS_MAX_EVENT_FILTERS

```
#define DEFAULT_CFE_PLATFORM_EVS_MAX_EVENT_FILTERS 8
```

Definition at line 74 of file [cfe_evs_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_EVS_PORT_DEFAULT

```
#define DEFAULT_CFE_PLATFORM_EVS_PORT_DEFAULT 0x0001
```

Definition at line 159 of file [cfe_evs_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_EVS_START_TASK_PRIORITY

```
#define DEFAULT_CFE_PLATFORM_EVS_START_TASK_PRIORITY 61
```

Definition at line 44 of file [cfe_evs_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_EVS_START_TASK_STACK_SIZE

```
#define DEFAULT_CFE_PLATFORM_EVS_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
```

Definition at line 60 of file [cfe_evs_internal_cfg.h](#).

11.164 cfe_evs_internal_cfg.h

[Go to the documentation of this file.](#)

```
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00017  *****/
00018
00028 #ifndef CFE_EVS_INTERNAL_CFG_H
00029 #define CFE_EVS_INTERNAL_CFG_H
00030
00031 #include "cfe_evs_mission_cfg.h"
00032 #include "cfe_evs_internal_cfg_values.h"
00033
00043 #define CFE_PLATFORM_EVS_START_TASK_PRIORITY CFE_PLATFORM_EVS_CFGVAL(START_TASK_PRIORITY)
00044 #define DEFAULT_CFE_PLATFORM_EVS_START_TASK_PRIORITY 61
00045
00059 #define CFE_PLATFORM_EVS_START_TASK_STACK_SIZE CFE_PLATFORM_EVS_CFGVAL(START_TASK_STACK_SIZE)
00060 #define DEFAULT_CFE_PLATFORM_EVS_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
00061
00073 #define CFE_PLATFORM_EVS_MAX_EVENT_FILTERS CFE_PLATFORM_EVS_CFGVAL(MAX_EVENT_FILTERS)
00074 #define DEFAULT_CFE_PLATFORM_EVS_MAX_EVENT_FILTERS 8
00075
00086 #define CFE_PLATFORM_EVS_MAX_APP_EVENT_BURST CFE_PLATFORM_EVS_CFGVAL(MAX_APP_EVENT_BURST)
00087 #define DEFAULT_CFE_PLATFORM_EVS_MAX_APP_EVENT_BURST 32
00088
00099 #define CFE_PLATFORM_EVS_APP_EVENTS_PER_SEC CFE_PLATFORM_EVS_CFGVAL(APP_EVENTS_PER_SEC)
00100 #define DEFAULT_CFE_PLATFORM_EVS_APP_EVENTS_PER_SEC 15
00101
00114 #define CFE_PLATFORM_EVS_DEFAULT_LOG_FILE CFE_PLATFORM_EVS_CFGVAL(DEFAULT_LOG_FILE)
00115 #define DEFAULT_CFE_PLATFORM_EVS_DEFAULT_LOG_FILE "/ram/cfe_evs.log"
00116
00127 #define CFE_PLATFORM_EVS_LOG_MAX CFE_PLATFORM_EVS_CFGVAL(LOG_MAX)
00128 #define DEFAULT_CFE_PLATFORM_EVS_LOG_MAX 20
00129
00143 #define CFE_PLATFORM_EVS_DEFAULT_APP_DATA_FILE CFE_PLATFORM_EVS_CFGVAL(DEFAULT_APP_DATA_FILE)
00144 #define DEFAULT_CFE_PLATFORM_EVS_DEFAULT_APP_DATA_FILE "/ram/cfe_evs_app.dat"
00145
00158 #define CFE_PLATFORM_EVS_PORT_DEFAULT CFE_PLATFORM_EVS_CFGVAL(PORT_DEFAULT)
00159 #define DEFAULT_CFE_PLATFORM_EVS_PORT_DEFAULT 0x0001
00160
00176 #define CFE_PLATFORM_EVS_DEFAULT_TYPE_FLAG CFE_PLATFORM_EVS_CFGVAL(DEFAULT_TYPE_FLAG)
00177 #define DEFAULT_CFE_PLATFORM_EVS_DEFAULT_TYPE_FLAG 0xE
00178
00193 #define CFE_PLATFORM_EVS_DEFAULT_LOG_MODE CFE_PLATFORM_EVS_CFGVAL(DEFAULT_LOG_MODE)
```

```

00194 #define DEFAULT_CFE_PLATFORM_EVS_DEFAULT_LOG_MODE 1
00195
00207 #define CFE_PLATFORM_EVS_DEFAULT_MSG_FORMAT_MODE          CFE_PLATFORM_EVS_CFGVAL(DEFAULT_MSG_FORMAT_MODE)
00208 #define DEFAULT_CFE_PLATFORM_EVS_DEFAULT_MSG_FORMAT_MODE CFE_EVS_MsgFormat_LONG
00209
00210 #endif

```

11.165 cfe/modules/evs/fsw/inc/cfe_evs_topicsids.h File Reference

```
#include "cfe_evs_topicid_values.h"
```

Macros

- #define [CFE_MISSION_EVS_CMD_TOPICID](#) [CFE_MISSION_EVS_TIDVAL](#)(CMD)
- #define [DEFAULT_CFE_MISSION_EVS_CMD_TOPICID](#) 1
- #define [CFE_MISSION_EVS_SEND_HK_TOPICID](#) [CFE_MISSION_EVS_TIDVAL](#)(SEND_HK)
- #define [DEFAULT_CFE_MISSION_EVS_SEND_HK_TOPICID](#) 9
- #define [CFE_MISSION_EVS_HK_TLM_TOPICID](#) [CFE_MISSION_EVS_TIDVAL](#)(HK_TLM)
- #define [DEFAULT_CFE_MISSION_EVS_HK_TLM_TOPICID](#) 1
- #define [CFE_MISSION_EVS_LONG_EVENT_MSG_TOPICID](#) [CFE_MISSION_EVS_TIDVAL](#)(LONG_EVENT_↵
MSG)
- #define [DEFAULT_CFE_MISSION_EVS_LONG_EVENT_MSG_TOPICID](#) 8
- #define [CFE_MISSION_EVS_SHORT_EVENT_MSG_TOPICID](#) [CFE_MISSION_EVS_TIDVAL](#)(SHORT_↵
EVENT_MSG)
- #define [DEFAULT_CFE_MISSION_EVS_SHORT_EVENT_MSG_TOPICID](#) 9

11.165.1 Detailed Description

CFE Event Services (CFE_EVS) Application Topic IDs
Definition in file [cfe_evs_topicsids.h](#).

11.165.2 Macro Definition Documentation

CFE_MISSION_EVS_CMD_TOPICID

```
#define CFE_MISSION_EVS_CMD_TOPICID CFE\_MISSION\_EVS\_TIDVAL(CMD)
```

Purpose cFE Portable Message Numbers for Commands

Description:

Portable message numbers for the cFE EVS command messages

Limits

Not Applicable

Definition at line 37 of file [cfe_evs_topicsids.h](#).

CFE_MISSION_EVS_HK_TLM_TOPICID

```
#define CFE_MISSION_EVS_HK_TLM_TOPICID CFE\_MISSION\_EVS\_TIDVAL(HK_TLM)
```

Purpose cFE Portable Message Numbers for Telemetry

Description:

Portable message numbers for the cFE EVS telemetry messages

Limits

Not Applicable

Definition at line 51 of file [cfe_evs_topicids.h](#).

CFE_MISSION_EVS_LONG_EVENT_MSG_TOPICID

```
#define CFE_MISSION_EVS_LONG_EVENT_MSG_TOPICID CFE_MISSION_EVS_TIDVAL (LONG_EVENT_MSG)
```

Definition at line 53 of file [cfe_evs_topicids.h](#).

CFE_MISSION_EVS_SEND_HK_TOPICID

```
#define CFE_MISSION_EVS_SEND_HK_TOPICID CFE_MISSION_EVS_TIDVAL (SEND_HK)
```

Definition at line 39 of file [cfe_evs_topicids.h](#).

CFE_MISSION_EVS_SHORT_EVENT_MSG_TOPICID

```
#define CFE_MISSION_EVS_SHORT_EVENT_MSG_TOPICID CFE_MISSION_EVS_TIDVAL (SHORT_EVENT_MSG)
```

Definition at line 55 of file [cfe_evs_topicids.h](#).

DEFAULT_CFE_MISSION_EVS_CMD_TOPICID

```
#define DEFAULT_CFE_MISSION_EVS_CMD_TOPICID 1
```

Definition at line 38 of file [cfe_evs_topicids.h](#).

DEFAULT_CFE_MISSION_EVS_HK_TLM_TOPICID

```
#define DEFAULT_CFE_MISSION_EVS_HK_TLM_TOPICID 1
```

Definition at line 52 of file [cfe_evs_topicids.h](#).

DEFAULT_CFE_MISSION_EVS_LONG_EVENT_MSG_TOPICID

```
#define DEFAULT_CFE_MISSION_EVS_LONG_EVENT_MSG_TOPICID 8
```

Definition at line 54 of file [cfe_evs_topicids.h](#).

DEFAULT_CFE_MISSION_EVS_SEND_HK_TOPICID

```
#define DEFAULT_CFE_MISSION_EVS_SEND_HK_TOPICID 9
```

Definition at line 40 of file [cfe_evs_topicids.h](#).

DEFAULT_CFE_MISSION_EVS_SHORT_EVENT_MSG_TOPICID

```
#define DEFAULT_CFE_MISSION_EVS_SHORT_EVENT_MSG_TOPICID 9
```

Definition at line 56 of file [cfe_evs_topicids.h](#).

11.166 cfe_evs_topicids.h

[Go to the documentation of this file.](#)

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00017  *****/
00018
00023 #ifndef CFE_EVS_TOPICIDS_H
00024 #define CFE_EVS_TOPICIDS_H
00025
00026 #include "cfe-evs-topicid-values.h"
00027
00037 #define CFE_MISSION_EVS_CMD_TOPICID          CFE_MISSION_EVS_TIDVAL(CMD)
00038 #define DEFAULT_CFE_MISSION_EVS_CMD_TOPICID  1
00039 #define CFE_MISSION_EVS_SEND_HK_TOPICID      CFE_MISSION_EVS_TIDVAL(SEND_HK)
00040 #define DEFAULT_CFE_MISSION_EVS_SEND_HK_TOPICID 9
00041
00051 #define CFE_MISSION_EVS_HK_TLM_TOPICID        CFE_MISSION_EVS_TIDVAL(HK_TLM)
00052 #define DEFAULT_CFE_MISSION_EVS_HK_TLM_TOPICID 1
00053 #define CFE_MISSION_EVS_LONG_EVENT_MSG_TOPICID CFE_MISSION_EVS_TIDVAL(LONG_EVENT_MSG)
00054 #define DEFAULT_CFE_MISSION_EVS_LONG_EVENT_MSG_TOPICID 8
00055 #define CFE_MISSION_EVS_SHORT_EVENT_MSG_TOPICID CFE_MISSION_EVS_TIDVAL(SHORT_EVENT_MSG)
00056 #define DEFAULT_CFE_MISSION_EVS_SHORT_EVENT_MSG_TOPICID 9
00057
00058 #endif

```

11.167 cfe/modules/fs/config/default_cfe_fs_extern_typedefs.h File Reference

```

#include "cfe_mission_cfg.h"
#include "cfe_fs_filedef.h"

```

11.167.1 Detailed Description

Public type definitions for the CFE FS module

Definition in file [default_cfe_fs_extern_typedefs.h](#).

11.168 default_cfe_fs_extern_typedefs.h

[Go to the documentation of this file.](#)

```

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00017  *****/
00018
00025 #ifndef DEFAULT_CFE_FS_EXTERN_TYPEDEFS_H
00026 #define DEFAULT_CFE_FS_EXTERN_TYPEDEFS_H
00027
00028 #include "cfe_mission_cfg.h"
00029 #include "cfe_fs_filedef.h"
00030
00031 #endif /* CFE_FS_EXTERN_TYPEDEFS_H */

```


11.169 cfe/modules/fs/config/default_cfe_fs_filedef.h File Reference

```
#include "common_types.h"
#include "cfe_mission_cfg.h"
```

Data Structures

- struct [CFE_FS_Header](#)
Standard cFE File header structure definition.

Typedefs

- typedef [uint32 CFE_FS_SubType_Enum_t](#)
Content descriptor for File Headers.
- typedef struct [CFE_FS_Header](#) [CFE_FS_Header_t](#)
Standard cFE File header structure definition.

Enumerations

- enum [CFE_FS_SubType](#) {
 [CFE_FS_SubType_ES_ERLOG](#) = 1 , [CFE_FS_SubType_ES_SYSLOG](#) = 2 , [CFE_FS_SubType_ES_QUERYALL](#)
 = 3 , [CFE_FS_SubType_ES_PERFDATA](#) = 4 ,
 [CFE_FS_SubType_ES_CDS_REG](#) = 6 , [CFE_FS_SubType_TBL_REG](#) = 9 , [CFE_FS_SubType_TBL_IMG](#) = 8 ,
 [CFE_FS_SubType_EVS_APPDATA](#) = 15 ,
 [CFE_FS_SubType_EVS_EVENTLOG](#) = 16 , [CFE_FS_SubType_SB_PIPEDATA](#) = 20 , [CFE_FS_SubType_SB_ROUTEDATA](#)
 = 21 , [CFE_FS_SubType_SB_MAPDATA](#) = 22 ,
 [CFE_FS_SubType_ES_QUERYALLTASKS](#) = 23 }
File subtypes used within cFE.

11.169.1 Detailed Description

Public type definitions for the CFE FS module
Definition in file [default_cfe_fs_filedef.h](#).

11.169.2 Typedef Documentation

[CFE_FS_Header_t](#)

typedef struct [CFE_FS_Header](#) [CFE_FS_Header_t](#)
Standard cFE File header structure definition.

[CFE_FS_SubType_Enum_t](#)

typedef [uint32](#) [CFE_FS_SubType_Enum_t](#)
Content descriptor for File Headers.

See also

enum [CFE_FS_SubType](#)

Definition at line 176 of file [default_cfe_fs_filedef.h](#).

11.169.3 Enumeration Type Documentation

CFE_FS_SubType

enum [CFE_FS_SubType](#)

File subtypes used within cFE.

This defines all the file subtypes used by cFE. Note apps can extend as needed but need to avoid conflicts (app context not currently included in the file header).

Enumerator

CFE_FS_SubType_ES_ERLOG	Executive Services Exception/Reset Log Type. Executive Services Exception/Reset Log File which is generated in response to a \$sc_\$cpu_ES_WriteERLog2File command.
CFE_FS_SubType_ES_SYSLOG	Executive Services System Log Type. Executive Services System Log File which is generated in response to a \$sc_\$cpu_ES_WriteSysLog2File command.
CFE_FS_SubType_ES_QUERYALL	Executive Services Information on All Applications File. Executive Services Information on All Applications File which is generated in response to a \$sc_\$cpu_ES_WriteAppInfo2File command.
CFE_FS_SubType_ES_PERFDATA	Executive Services Performance Data File. Executive Services Performance Analyzer Data File which is generated in response to a \$sc_\$cpu_ES_StopLADData command.
CFE_FS_SubType_ES_CDS_REG	Executive Services Critical Data Store Registry Dump File. Executive Services Critical Data Store Registry Dump File which is generated in response to a \$sc_\$cpu_ES_WriteCDS2File command.
CFE_FS_SubType_TBL_REG	Table Services Registry Dump File. Table Services Registry Dump File which is generated in response to a \$sc_\$cpu_TBL_WriteReg2File command.
CFE_FS_SubType_TBL_IMG	Table Services Table Image File. Table Services Table Image File which is generated either on the ground or in response to a \$sc_\$cpu_TBL_DUMP command.
CFE_FS_SubType_EVS_APPDATA	Event Services Application Data Dump File. Event Services Application Data Dump File which is generated in response to a \$sc_\$cpu_EVS_WriteAppData2File command.
CFE_FS_SubType_EVS_EVENTLOG	Event Services Local Event Log Dump File. Event Services Local Event Log Dump File which is generated in response to a \$sc_\$cpu_EVS_WriteLog2File command.
CFE_FS_SubType_SB_PIPEDATA	Software Bus Pipe Data Dump File. Software Bus Pipe Data Dump File which is generated in response to a \$sc_\$cpu_SB_WritePipe2File command.
CFE_FS_SubType_SB_ROUTEDATA	Software Bus Message Routing Data Dump File. Software Bus Message Routing Data Dump File which is generated in response to a \$sc_\$cpu_SB_WriteRouting2File command.
CFE_FS_SubType_SB_MAPDATA	Software Bus Message Mapping Data Dump File. Software Bus Message Mapping Data Dump File which is generated in response to a \$sc_\$cpu_SB_WriteMap2File command.
CFE_FS_SubType_ES_QUERYALLTASKS	Executive Services Query All Tasks Data File. Executive Services Query All Tasks Data File which is generated in response to a \$sc_\$cpu_ES_WriteTaskInfo2File command.

Definition at line 39 of file [default_cfe_fs_filedef.h](#).

11.170 default_cfe_fs_filedef.h

[Go to the documentation of this file.](#)

```

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00017  *****/
00018
00025 #ifndef DEFAULT_CFE_FS_FILEHDR_H
00026 #define DEFAULT_CFE_FS_FILEHDR_H
00027
00028 #include "common_types.h"
00029 #include "cfe_mission_cfg.h" /* to define CFE_FS_HDR_DESC_MAX_LEN */
00030
00039 enum CFE_FS_SubType
00040 {
00049     CFE_FS_SubType_ES_ERLOG = 1,
00050
00059     CFE_FS_SubType_ES_SYSLOG = 2,
00060
00069     CFE_FS_SubType_ES_QUERYALL = 3,
00070
00079     CFE_FS_SubType_ES_PERFDATA = 4,
00080
00089     CFE_FS_SubType_ES_CDS_REG = 6,
00090
00099     CFE_FS_SubType_TBL_REG = 9,
00100
00108     CFE_FS_SubType_TBL_IMG = 8,
00109
00118     CFE_FS_SubType_EVS_APPDATA = 15,
00119
00128     CFE_FS_SubType_EVS_EVENTLOG = 16,
00129
00138     CFE_FS_SubType_SB_PIPEDATA = 20,
00139
00148     CFE_FS_SubType_SB_ROUTEDATA = 21,
00149
00158     CFE_FS_SubType_SB_MAPDATA = 22,
00159
00168     CFE_FS_SubType_ES_QUERYALLTASKS = 23
00169 };
00170
00176 typedef uint32 CFE_FS_SubType_Enum_t;
00177
00181 typedef struct CFE_FS_Header
00182 {
00183     uint32 ContentType;
00184     uint32 SubType;
00187     uint32 Length;
00188     uint32 SpacecraftID;
00189     uint32 ProcessorID;
00190     uint32 ApplicationID;
00192     uint32 TimeSeconds;
00193     uint32 TimeSubSeconds;
00195     char Description[CFE_FS_HDR_DESC_MAX_LEN];
00196 } CFE_FS_Header_t;
00197
00198 #endif /* DEFAULT_CFE_FS_FILEHDR_H */

```

11.171 cfe/modules/fs/config/default_cfe_fs_interface_cfg_values.h File Reference

Macros

- #define [CFE_MISSION_FS_CFGVAL\(x\)](#) DEFAULT_CFE_FS_##x

11.171.1 Detailed Description

CFE File Services (CFE_FS) Module Public Definitions

This provides default values for configurable items that affect the interface(s) of this module.

Note

This file may be overridden/superceded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [default_cfe_fs_interface_cfg_values.h](#).

11.171.2 Macro Definition Documentation

CFE_MISSION_FS_CFGVAL

```
#define CFE_MISSION_FS_CFGVAL(  
    x ) DEFAULT_CFE_FS_##x
```

Definition at line 34 of file [default_cfe_fs_interface_cfg_values.h](#).

11.172 default_cfe_fs_interface_cfg_values.h

[Go to the documentation of this file.](#)

```
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00017  *****/  
00018  
00030 #ifndef DEFAULT_CFE_FS_INTERFACE_CFG_VALUES_H  
00031 #define DEFAULT_CFE_FS_INTERFACE_CFG_VALUES_H  
00032  
00033 /* Use the default configuration value for all */  
00034 #define CFE_MISSION_FS_CFGVAL(x) DEFAULT_CFE_FS_##x  
00035  
00036 #endif
```

11.173 cfe/modules/fs/config/default_cfe_fs_mission_cfg.h File Reference

```
#include "cfe_fs_interface_cfg.h"
```

11.173.1 Detailed Description

CFE File Services (CFE_FS) Application Mission Configuration Header File

This is a compatibility header for the "mission_cfg.h" file that has traditionally provided public config definitions for each CFS app.

Note

This file may be overridden/superceded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [default_cfe_fs_mission_cfg.h](#).

11.174 default_cfe_fs_mission_cfg.h

[Go to the documentation of this file.](#)

```

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00016  * limitations under the License.
00017  *****/
00018
00031 #ifndef DEFAULT_CFE_FS_MISSION_CFG_H
00032 #define DEFAULT_CFE_FS_MISSION_CFG_H
00033
00034 #include "cfe_fs_interface_cfg.h"
00035
00036 #endif

```

11.175 cfe/modules/fs/fsw/inc/cfe_fs_interface_cfg.h File Reference

```
#include "cfe_fs_interface_cfg_values.h"
```

Macros

- #define [CFE_FS_HDR_DESC_MAX_LEN](#) [CFE_MISSION_FS_CFGVAL](#)(HDR_DESC_MAX_LEN)
Max length of description field in a standard cFE File Header.
- #define [DEFAULT_CFE_FS_HDR_DESC_MAX_LEN](#) 32
- #define [CFE_FS_FILE_CONTENT_ID](#) [CFE_MISSION_FS_CFGVAL](#)(FILE_CONTENT_ID)
Magic Number for cFE compliant files (= 'cFE1')
- #define [DEFAULT_CFE_FS_FILE_CONTENT_ID](#) 0x63464531

11.175.1 Detailed Description

CFE File Services (CFE_FS) Module Public Definitions

This provides default values for configurable items that affect the interface(s) of this module.

Note

This file may be overridden/superseded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [cfe_fs_interface_cfg.h](#).

11.175.2 Macro Definition Documentation

CFE_FS_FILE_CONTENT_ID

```
#define CFE_FS_FILE_CONTENT_ID CFE_MISSION_FS_CFGVAL(FILE_CONTENT_ID)
```

Magic Number for cFE compliant files (= 'cFE1')

Definition at line 52 of file [cfe_fs_interface_cfg.h](#).

CFE_FS_HDR_DESC_MAX_LEN

```
#define CFE_FS_HDR_DESC_MAX_LEN CFE_MISSION_FS_CFGVAL(HDR_DESC_MAX_LEN)
```

Max length of description field in a standard cFE File Header.

Note

the value of CFE_FS_HDR_DESC_MAX_LEN, if modified, should be constrained to multiples of 4, as it is used within a structure that also contains uint32 types. This ensures that the entire structure remains 32-bit aligned without the need for implicit padding bytes.

Definition at line 46 of file [cfe_fs_interface_cfg.h](#).

DEFAULT_CFE_FS_FILE_CONTENT_ID

```
#define DEFAULT_CFE_FS_FILE_CONTENT_ID 0x63464531
```

Definition at line 53 of file [cfe_fs_interface_cfg.h](#).

DEFAULT_CFE_FS_HDR_DESC_MAX_LEN

```
#define DEFAULT_CFE_FS_HDR_DESC_MAX_LEN 32
```

Definition at line 47 of file [cfe_fs_interface_cfg.h](#).

11.176 cfe_fs_interface_cfg.h

[Go to the documentation of this file.](#)

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00017  *****/
00018
00031 #ifndef CFE_FS_INTERFACE_CFG_H
00032 #define CFE_FS_INTERFACE_CFG_H
00033
00034 #include "cfe_fs_interface_cfg_values.h"
00035
00036 /***** Macro Definitions *****/
00037
00046 #define CFE_FS_HDR_DESC_MAX_LEN CFE_MISSION_FS_CFGVAL(HDR_DESC_MAX_LEN)
00047 #define DEFAULT_CFE_FS_HDR_DESC_MAX_LEN 32
00048
00052 #define CFE_FS_FILE_CONTENT_ID CFE_MISSION_FS_CFGVAL(FILE_CONTENT_ID)
00053 #define DEFAULT_CFE_FS_FILE_CONTENT_ID 0x63464531
00054
00055 #endif /* CFE_FS_INTERFACE_CFG_H */
```

11.177 cfe/modules/msg/fsw/inc/ccsds_hdr.h File Reference

```
#include "common_types.h"
```

Data Structures

- struct [CCSDS_PrimaryHeader](#)

CCSDS packet primary header.

- struct [CCSDS_ExtendedHeader](#)

CCSDS packet extended header.

Typedefs

- typedef struct [CCSDS_PrimaryHeader](#) [CCSDS_PrimaryHeader_t](#)
CCSDS packet primary header.
- typedef struct [CCSDS_ExtendedHeader](#) [CCSDS_ExtendedHeader_t](#)
CCSDS packet extended header.

11.177.1 Detailed Description

Define CCSDS packet header types

- Avoid direct access for portability, use APIs
- Used to construct message structures

Definition in file [ccsds_hdr.h](#).

11.177.2 Typedef Documentation

CCSDS_ExtendedHeader_t

typedef struct [CCSDS_ExtendedHeader](#) [CCSDS_ExtendedHeader_t](#)
CCSDS packet extended header.

CCSDS_PrimaryHeader_t

typedef struct [CCSDS_PrimaryHeader](#) [CCSDS_PrimaryHeader_t](#)
CCSDS packet primary header.

11.178 ccsds_hdr.h

[Go to the documentation of this file.](#)

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00016  * limitations under the License.
00017  *****/
00018
00027 #ifndef CCSDS_HDR_H
00028 #define CCSDS_HDR_H
00029
00030 /*
00031  * Include Files
00032  */
00033
00034 #include "common_types.h"
00035
00036 /*
00037  * Type Definitions
```

```

00038 */
00039
00040 /*****
00041  * Structure definitions for CCSDS headers.
00042  *
00043  * CCSDS headers must always be in network byte order per the standard.
00044  * MSB at the lowest address which is commonly referred to as "BIG Endian"
00045  *
00046  */
00047
00051 typedef struct CCSDS_PrimaryHeader
00052 {
00053     uint8 StreamId[2];
00054     /* bits shift ----- description ----- */
00055     /* 0x07FF 0 : application ID */
00056     /* 0x0800 11 : secondary header: 0 = absent, 1 = present */
00057     /* 0x1000 12 : packet type: 0 = TLM, 1 = CMD */
00058     /* 0xE000 13 : CCSDS version: 0 = ver 1, 1 = ver 2 */
00059
00060     uint8 Sequence[2];
00061     /* bits shift ----- description ----- */
00062     /* 0x3FFF 0 : sequence count */
00063     /* 0xC000 14 : segmentation flags: 3 = complete packet */
00064
00065     uint8 Length[2];
00066     /* bits shift ----- description ----- */
00067     /* 0xFFFF 0 : (total packet length) - 7 */
00068 } CCSDS_PrimaryHeader_t;
00069
00073 typedef struct CCSDS_ExtendedHeader
00074 {
00075     uint8 Subsystem[2];
00076     /* bits shift ----- description ----- */
00077     /* 0x01FF 0 : Subsystem Id mission defined */
00078     /* 0x0200 9 : Playback flag 0 = original, 1 = playback */
00079     /* 0x0400 10 : Endian: Big = 0, Little (Intel) = 1 */
00080     /* 0xF800 11 : EDS Version for packet definition used */
00081
00082     uint8 SystemId[2];
00083     /* 0xFFFF 0 : System Id mission defined */
00084 } CCSDS_ExtendedHeader_t;
00085
00086 #endif /* CCSDS_HDR_H */

```

11.179 cfe/modules/resourceid/fsw/inc/cfe_core_resourceid_basevalues.h File Reference

```
#include "cfe_resourceid_basevalue.h"
```

Enumerations

- enum {
[CFE_RESOURCEID_ES_TASKID_BASE_OFFSET](#) = OS_OBJECT_TYPE_OS_TASK, [CFE_RESOURCEID_ES_APPID_BASE_OFFSET](#) = OS_OBJECT_TYPE_USER + 1, [CFE_RESOURCEID_ES_LIBID_BASE_OFFSET](#) = OS_OBJECT_TYPE_USER + 2, [CFE_RESOURCEID_ES_COUNTID_BASE_OFFSET](#) = OS_OBJECT_TYPE_USER + 3, [CFE_RESOURCEID_ES_POOLID_BASE_OFFSET](#) = OS_OBJECT_TYPE_USER + 4, [CFE_RESOURCEID_ES_CDSBLOCKID_BASE_OFFSET](#) = OS_OBJECT_TYPE_USER + 5, [CFE_RESOURCEID_SB_PIPEID_RESOURCE_BASE_OFFSET](#) = OS_OBJECT_TYPE_USER + 6, [CFE_RESOURCEID_CONFIGID_BASE_OFFSET](#) = OS_OBJECT_TYPE_USER + 7,
[CFE_RESOURCEID_TBL_VALRESULTID_BASE_OFFSET](#) = OS_OBJECT_TYPE_USER + 8, [CFE_RESOURCEID_TBL_DUMPID_BASE_OFFSET](#) = OS_OBJECT_TYPE_USER + 9, [CFE_RESOURCEID_TBL_LOADBUFFID_BASE_OFFSET](#) = OS_OBJECT_TYPE_USER + 10, [CFE_RESOURCEID_TBL_ACCESSID_BASE_OFFSET](#) = OS_OBJECT_TYPE_USER + 11,
[CFE_RESOURCEID_TBL_REGID_BASE_OFFSET](#) = OS_OBJECT_TYPE_USER + 12 }
- enum {
[CFE_ES_TASKID_BASE](#) = CFE_RESOURCEID_MAKE_BASE([CFE_RESOURCEID_ES_TASKID_BASE_OFFSET](#)), [CFE_ES_APPID_BASE](#) = CFE_RESOURCEID_MAKE_BASE([CFE_RESOURCEID_ES_APPID_BASE_OFFSET](#)), [CFE_ES_LIBID_BASE](#) = CFE_RESOURCEID_MAKE_BASE([CFE_RESOURCEID_ES_LIBID_BASE_OFFSET](#))


```

_ES_LIBID_BASE_OFFSET) , CFE_ES_COUNTID_BASE = CFE_RESOURCEID_MAKE_BASE(CFE_↵
RESOURCEID_ES_COUNTID_BASE_OFFSET) ,
CFE_ES_POOLID_BASE = CFE_RESOURCEID_MAKE_BASE(CFE_RESOURCEID_ES_POOLID_BASE_↵
_OFFSET) , CFE_ES_CDSBLOCKID_BASE = CFE_RESOURCEID_MAKE_BASE(CFE_RESOURCEID_↵
ES_CDSBLOCKID_BASE_OFFSET) , CFE_SB_PIPEID_BASE = CFE_RESOURCEID_MAKE_BASE(CFE_↵
RESOURCEID_SB_PIPEID_RESOURCE_BASE_OFFSET) , CFE_CONFIGID_BASE = CFE_RESOURCEID_↵
_MAKE_BASE(CFE_RESOURCEID_CONFIGID_BASE_OFFSET) ,
CFE_TBL_VALRESULTID_BASE = CFE_RESOURCEID_MAKE_BASE(CFE_RESOURCEID_TBL_VALRESULTID_↵
_BASE_OFFSET) , CFE_TBL_DUMPCTRLID_BASE = CFE_RESOURCEID_MAKE_BASE(CFE_RESOURCEID_↵
_TBL_DUMPCTRLID_BASE_OFFSET) , CFE_TBL_LOADBUFFID_BASE = CFE_RESOURCEID_MAKE_↵
_BASE(CFE_RESOURCEID_TBL_LOADBUFFID_BASE_OFFSET) , CFE_TBL_HANDLE_BASE = CFE_↵
RESOURCEID_MAKE_BASE(CFE_RESOURCEID_TBL_ACCESSID_BASE_OFFSET) ,
CFE_TBL_REGID_BASE = CFE_RESOURCEID_MAKE_BASE(CFE_RESOURCEID_TBL_REGID_BASE_↵
OFFSET) }

```

11.179.1 Detailed Description

Contains CFE internal prototypes and definitions related to resource management and related CFE resource IDs.

A CFE ES Resource ID is a common way to identify CFE-managed resources such as apps, tasks, counters, memory pools, CDS blocks, and other entities.

Definition in file [cfe_core_resourceid_basevalues.h](#).

11.180 cfe_core_resourceid_basevalues.h

[Go to the documentation of this file.](#)

```

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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
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00016  * limitations under the License.
00017  *****/
00018
00029 #ifndef CFE_CORE_RESOURCEID_BASEVALUES_H
00030 #define CFE_CORE_RESOURCEID_BASEVALUES_H
00031
00032 /*
00033 ** Include Files
00034 */
00035 #include "cfe_resourceid_basevalue.h"
00036
00041 /*
00042  * Assign unique offsets per resource types used in CFE core apps.
00043  *
00044  * Applications should not use these values directly, but rather
00045  * in conjunction with the CFE_RESOURCEID_MAKE_BASE macro provided
00046  * by the Resource ID module. (see below)
00047  */
00048 enum
00049 {
00050     /*
00051      * Note for Task ID base value --
00052      * This currently shares the same offset as OSAL tasks, such that
00053      * when "simple" (non-enforcing/backward-compatible) IDs are selected,
00054      * the CFE task IDs and the OSAL task IDs end up as the same value.
00055      *
00056      * The "CFE_RESOURCEID_MARK" bit still differentiates the value when
00057      * in strict mode, so there is no overlap in that case.
00058      */

```

```

00059     CFE_RESOURCEID_ES_TASKID_BASE_OFFSET = OS_OBJECT_TYPE_OS_TASK,
00060
00061     /* Other ES managed resources */
00062     CFE_RESOURCEID_ES_APPID_BASE_OFFSET   = OS_OBJECT_TYPE_USER + 1,
00063     CFE_RESOURCEID_ES_LIBID_BASE_OFFSET   = OS_OBJECT_TYPE_USER + 2,
00064     CFE_RESOURCEID_ES_COUNTID_BASE_OFFSET = OS_OBJECT_TYPE_USER + 3,
00065     CFE_RESOURCEID_ES_POOLID_BASE_OFFSET  = OS_OBJECT_TYPE_USER + 4,
00066     CFE_RESOURCEID_ES_CDSBLOCKID_BASE_OFFSET = OS_OBJECT_TYPE_USER + 5,
00067
00068     /* SB managed resources */
00069     CFE_RESOURCEID_SB_PIPEID_RESOURCE_BASE_OFFSET = OS_OBJECT_TYPE_USER + 6,
00070
00071     /* configuration registry */
00072     CFE_RESOURCEID_CONFIGID_BASE_OFFSET = OS_OBJECT_TYPE_USER + 7,
00073
00074     /* TBL managed resources */
00075     CFE_RESOURCEID_TBL_VALRESULTID_BASE_OFFSET = OS_OBJECT_TYPE_USER + 8,
00076     CFE_RESOURCEID_TBL_DUMPCTRLID_BASE_OFFSET = OS_OBJECT_TYPE_USER + 9,
00077     CFE_RESOURCEID_TBL_LOADBUFFID_BASE_OFFSET = OS_OBJECT_TYPE_USER + 10,
00078     CFE_RESOURCEID_TBL_ACCESSID_BASE_OFFSET   = OS_OBJECT_TYPE_USER + 11,
00079     CFE_RESOURCEID_TBL_REGID_BASE_OFFSET      = OS_OBJECT_TYPE_USER + 12,
00080 };
00081
00082 /*
00083  * Assign actual base values from the offsets above
00084  *
00085  * Using "enum" ensures these are resolved as integers now, as opposed to at the point of use like macros.
00086  */
00087 enum
00088 {
00089     /* ES managed resources */
00090     CFE_ES_TASKID_BASE   = CFE_RESOURCEID_MAKE_BASE(CFE_RESOURCEID_ES_TASKID_BASE_OFFSET),
00091     CFE_ES_APPID_BASE    = CFE_RESOURCEID_MAKE_BASE(CFE_RESOURCEID_ES_APPID_BASE_OFFSET),
00092     CFE_ES_LIBID_BASE     = CFE_RESOURCEID_MAKE_BASE(CFE_RESOURCEID_ES_LIBID_BASE_OFFSET),
00093     CFE_ES_COUNTID_BASE  = CFE_RESOURCEID_MAKE_BASE(CFE_RESOURCEID_ES_COUNTID_BASE_OFFSET),
00094     CFE_ES_POOLID_BASE   = CFE_RESOURCEID_MAKE_BASE(CFE_RESOURCEID_ES_POOLID_BASE_OFFSET),
00095     CFE_ES_CDSBLOCKID_BASE = CFE_RESOURCEID_MAKE_BASE(CFE_RESOURCEID_ES_CDSBLOCKID_BASE_OFFSET),
00096
00097     /* SB managed resources */
00098     CFE_SB_PIPEID_BASE = CFE_RESOURCEID_MAKE_BASE(CFE_RESOURCEID_SB_PIPEID_RESOURCE_BASE_OFFSET),
00099
00100     /* configuration registry */
00101     CFE_CONFIGID_BASE = CFE_RESOURCEID_MAKE_BASE(CFE_RESOURCEID_CONFIGID_BASE_OFFSET),
00102
00103     /* TBL managed resources */
00104     CFE_TBL_VALRESULTID_BASE = CFE_RESOURCEID_MAKE_BASE(CFE_RESOURCEID_TBL_VALRESULTID_BASE_OFFSET),
00105     CFE_TBL_DUMPCTRLID_BASE  = CFE_RESOURCEID_MAKE_BASE(CFE_RESOURCEID_TBL_DUMPCTRLID_BASE_OFFSET),
00106     CFE_TBL_LOADBUFFID_BASE  = CFE_RESOURCEID_MAKE_BASE(CFE_RESOURCEID_TBL_LOADBUFFID_BASE_OFFSET),
00107     CFE_TBL_HANDLE_BASE      = CFE_RESOURCEID_MAKE_BASE(CFE_RESOURCEID_TBL_ACCESSID_BASE_OFFSET),
00108     CFE_TBL_REGID_BASE       = CFE_RESOURCEID_MAKE_BASE(CFE_RESOURCEID_TBL_REGID_BASE_OFFSET),
00109 };
00110
00113 #endif /* CFE_CORE_RESOURCEID_BASEVALUES_H */

```

11.181 cfe/modules/resourceid/fsw/inc/cfe_resourceid_basevalue.h File Reference

```

#include "cfe_resourceid_typedef.h"
#include "osapi-idmap.h"

```

Macros

- `#define CFE_RESOURCEID_SHIFT OS_OBJECT_TYPE_SHIFT`
- `#define CFE_RESOURCEID_MAX OS_OBJECT_INDEX_MASK`
- `#define CFE_RESOURCEID_MAKE_BASE(offset) (CFE_RESOURCEID_MARK | ((offset) << CFE_RESOURCEID_SHIFT))`

A macro to generate a CFE resource ID base value from an offset.

11.181.1 Detailed Description

An implementation of CFE resource ID base values/limits that will be compatible with OSAL IDs. This is intended as a transitional tool to provide runtime value uniqueness, particularly when the "simple" (compatible) resource ID

implementation is used. In this mode, compiler type checking is disabled, and so OSAL IDs can be silently interchanged with CFE IDs.

However, by ensuring uniqueness in the runtime values, any ID handling errors may at least be detectable at runtime.

This still works fine with the "strict" resource ID option, but is less important as the compiler type checking should prevent this type of error before the code even runs.

The downside to this implementation is that it has a dependency on the OSAL ID structure.

Definition in file [cfe_resourceid_basevalue.h](#).

11.181.2 Macro Definition Documentation

CFE_RESOURCEID_MAKE_BASE

```
#define CFE_RESOURCEID_MAKE_BASE(  
    offset ) (CFE_RESOURCEID_MARK | ((offset) << CFE_RESOURCEID_SHIFT))
```

A macro to generate a CFE resource ID base value from an offset.

Each CFE ID range is effectively an extension of OSAL ID ranges by starting at OS_OBJECT_TYPE_USER.

Definition at line 73 of file [cfe_resourceid_basevalue.h](#).

CFE_RESOURCEID_MAX

```
#define CFE_RESOURCEID_MAX OS_OBJECT_INDEX_MASK
```

Definition at line 65 of file [cfe_resourceid_basevalue.h](#).

CFE_RESOURCEID_SHIFT

```
#define CFE_RESOURCEID_SHIFT OS_OBJECT_TYPE_SHIFT
```

Definition at line 64 of file [cfe_resourceid_basevalue.h](#).

11.182 cfe_resourceid_basevalue.h

[Go to the documentation of this file.](#)

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00017  *****/  
00018  
00039 #ifndef CFE_RESOURCEID_BASEVALUE_H  
00040 #define CFE_RESOURCEID_BASEVALUE_H  
00041  
00042 /*  
00043 ** Include Files  
00044 */  
00045 #include "cfe_resourceid_typedef.h"  
00046  
00047 /*  
00048  * In this configuration, CFE resource IDs are tailored to not  
00049  * conflict/overlap with OSAL IDs, and are structured in a similar manner.  
00050  */  
00051 #include "osapi-idmap.h"  
00052  
00053 /*  
00054  * Limits/definitions related to CFE_ResourceId_t values.  
00055  *  
00056  * Defining based on OSAL ID values makes this object a superset of
```

```

00057  * the OSAL ID type, such that OSAL IDs can be represented as resource IDs
00058  * and not conflict with/alias each other.
00059  *
00060  * NOTE: This reflects a bit of "inside knowledge" about how OSAL IDs are
00061  * constructed. The overlap between OSAL IDs and ES IDs may not always be
00062  * consistent, and they can diverge in a future version.
00063  */
00064 #define CFE_RESOURCEID_SHIFT OS_OBJECT_TYPE_SHIFT
00065 #define CFE_RESOURCEID_MAX   OS_OBJECT_INDEX_MASK
00066
00073 #define CFE_RESOURCEID_MAKE_BASE(offset) (CFE_RESOURCEID_MARK | ((offset) << CFE_RESOURCEID_SHIFT))
00074
00075 #endif /* CFE_RESOURCEID_BASEVALUE_H */

```

11.183 cfe/modules/sb/config/default_cfe_sb_extern_typedefs.h File Reference

```

#include "common_types.h"
#include "cfe_mission_cfg.h"
#include "cfe_resourceid_typedef.h"

```

Data Structures

- struct [CFE_SB_MsgId_t](#)
CFE_SB_MsgId_t type definition.
- struct [CFE_SB_Qos_t](#)
Quality Of Service Type Definition.

Typedefs

- typedef [uint8 CFE_SB_QosPriority_Enum_t](#)
Selects the priority level for message routing.
- typedef [uint8 CFE_SB_QosReliability_Enum_t](#)
Selects the reliability level for message routing.
- typedef [uint16 CFE_SB_RouteId_Atom_t](#)
An integer type that should be used for indexing into the Routing Table.
- typedef [uint32 CFE_SB_MsgId_Atom_t](#)
CFE_SB_MsgId_Atom_t primitive type definition.
- typedef CFE_RESOURCEID_BASE_TYPE [CFE_SB_PipId_t](#)
CFE_SB_PipId_t to primitive type definition.

Enumerations

- enum [CFE_SB_QosPriority](#) { [CFE_SB_QosPriority_LOW](#) = 0 , [CFE_SB_QosPriority_HIGH](#) = 1 }
 - enum [CFE_SB_QosReliability](#) { [CFE_SB_QosReliability_LOW](#) = 0 , [CFE_SB_QosReliability_HIGH](#) = 1 }
- Label definitions associated with CFE_SB_QosPriority_Enum_t.*
- Label definitions associated with CFE_SB_QosReliability_Enum_t.*

11.183.1 Detailed Description

Declarations and prototypes for cfe_sb_extern_typedefs module
Definition in file [default_cfe_sb_extern_typedefs.h](#).

11.183.2 Typedef Documentation

CFE_SB_MsgId_Atom_t

`typedef uint32 CFE_SB_MsgId_Atom_t`

CFE_SB_MsgId_Atom_t primitive type definition.

This is an integer type capable of holding any Message ID value Note: This value is limited via [CFE_PLATFORM_SB_HIGHEST_VALID_M](#)

Definition at line 89 of file [default_cfe_sb_extern_typedefs.h](#).

CFE_SB_Pipeld_t

`typedef CFE_RESOURCEID_BASE_TYPE CFE_SB_PipeId_t`

CFE_SB_Pipeld_t to primitive type definition.

Software Bus pipe identifier used in many SB APIs, as well as SB Telemetry messages and data files.

Definition at line 112 of file [default_cfe_sb_extern_typedefs.h](#).

CFE_SB_QosPriority_Enum_t

`typedef uint8 CFE_SB_QosPriority_Enum_t`

Selects the priority level for message routing.

See also

enum [CFE_SB_QosPriority](#)

Definition at line 53 of file [default_cfe_sb_extern_typedefs.h](#).

CFE_SB_QosReliability_Enum_t

`typedef uint8 CFE_SB_QosReliability_Enum_t`

Selects the reliability level for message routing.

See also

enum [CFE_SB_QosReliability](#)

Definition at line 76 of file [default_cfe_sb_extern_typedefs.h](#).

CFE_SB_Routeld_Atom_t

`typedef uint16 CFE_SB_RouteId_Atom_t`

An integer type that should be used for indexing into the Routing Table.

Definition at line 81 of file [default_cfe_sb_extern_typedefs.h](#).

11.183.3 Enumeration Type Documentation

CFE_SB_QosPriority

enum [CFE_SB_QosPriority](#)

Label definitions associated with CFE_SB_QosPriority_Enum_t.

Enumerator

CFE_SB_QosPriority_LOW	Normal priority level.
CFE_SB_QosPriority_HIGH	High priority.

Definition at line 35 of file [default_cfe_sb_extern_typedefs.h](#).

CFE_SB_QosReliability

enum [CFE_SB_QosReliability](#)

Label definitions associated with [CFE_SB_QosReliability_Enum_t](#).

Enumerator

CFE_SB_QosReliability_LOW	Normal (best-effort) reliability.
CFE_SB_QosReliability_HIGH	High reliability.

Definition at line 58 of file [default_cfe_sb_extern_typedefs.h](#).

11.184 default_cfe_sb_extern_typedefs.h

[Go to the documentation of this file.](#)

```

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00017  *****/
00018
00025 #ifndef DEFAULT_CFE_SB_EXTERN_TYPEDEFS_H
00026 #define DEFAULT_CFE_SB_EXTERN_TYPEDEFS_H
00027
00028 #include "common_types.h"
00029 #include "cfe_mission_cfg.h"
00030 #include "cfe_resourceid_typedef.h"
00031
00035 enum CFE_SB_QosPriority
00036 {
00040     CFE_SB_QosPriority_LOW = 0,
00041
00045     CFE_SB_QosPriority_HIGH = 1
00046 };
00047
00053 typedef uint8 CFE_SB_QosPriority_Enum_t;
00054
00058 enum CFE_SB_QosReliability
00059 {
00063     CFE_SB_QosReliability_LOW = 0,
00064
00068     CFE_SB_QosReliability_HIGH = 1
00069 };
00070
00076 typedef uint8 CFE_SB_QosReliability_Enum_t;
00077
00081 typedef uint16 CFE_SB_RouteId_Atom_t;
00082
00089 typedef uint32 CFE_SB_MsgId_Atom_t;
00090
00102 typedef struct
00103 {
00104     CFE_SB_MsgId_Atom_t Value;
00105 } CFE_SB_MsgId_t;
00106
00112 typedef CFE_RESOURCEID_BASE_TYPE CFE_SB_PipeId_t;
00113
00119 typedef struct
00120 {
00121     uint8 Priority;
00122     uint8 Reliability;
00124 } CFE_SB_Qos_t;

```

```
00125
00126 #endif /* CFE_SB_EXTERN_TYPEDEFS_H */
```

11.185 cfe/modules/sb/config/default_cfe_sb_fcncode_values.h File Reference

Macros

- #define [CFE_SB_CCVAL\(x\) CFE_SB_FunctionCode_##x](#)

Enumerations

- enum [CFE_SB_FunctionCode_](#) {
[CFE_SB_FunctionCode_NOOP](#) = 0, [CFE_SB_FunctionCode_RESET_COUNTERS](#) = 1, [CFE_SB_FunctionCode_SEND_SB_STATS](#)
= 2, [CFE_SB_FunctionCode_WRITE_ROUTING_INFO](#) = 3,
[CFE_SB_FunctionCode_ENABLE_ROUTE](#) = 4, [CFE_SB_FunctionCode_DISABLE_ROUTE](#) = 5, [CFE_SB_FunctionCode_WRITE_PIPE_INFO](#)
= 7, [CFE_SB_FunctionCode_WRITE_MAP_INFO](#) = 8,
[CFE_SB_FunctionCode_ENABLE_SUB_REPORTING](#) = 9, [CFE_SB_FunctionCode_DISABLE_SUB_REPORTING](#)
= 10, [CFE_SB_FunctionCode_SEND_PREV_SUBS](#) = 11 }

11.185.1 Detailed Description

Specification for the CFE Event Services (CFE_SB) command function codes

Note

This file should be strictly limited to the command/function code (CC) macro definitions. Other definitions such as enums, typedefs, or other macros should be placed in the msgdefs.h or msg.h files.

Definition in file [default_cfe_sb_fcncode_values.h](#).

11.185.2 Macro Definition Documentation

CFE_SB_CCVAL

```
#define CFE_SB_CCVAL(  
    x ) CFE_SB_FunctionCode_##x
```

Definition at line 35 of file [default_cfe_sb_fcncode_values.h](#).

11.185.3 Enumeration Type Documentation

CFE_SB_FunctionCode_

```
enum CFE_SB_FunctionCode_
```

Enumerator

CFE_SB_FunctionCode_NOOP	
CFE_SB_FunctionCode_RESET_COUNTERS	
CFE_SB_FunctionCode_SEND_SB_STATS	
CFE_SB_FunctionCode_WRITE_ROUTING_INFO	
CFE_SB_FunctionCode_ENABLE_ROUTE	
CFE_SB_FunctionCode_DISABLE_ROUTE	
CFE_SB_FunctionCode_WRITE_PIPE_INFO	
CFE_SB_FunctionCode_WRITE_MAP_INFO	
CFE_SB_FunctionCode_ENABLE_SUB_REPORTING	
CFE_SB_FunctionCode_DISABLE_SUB_REPORTING	
CFE_SB_FunctionCode_SEND_PREV_SUBS	

Definition at line 37 of file [default_cfe_sb_fcncode_values.h](#).

11.186 default_cfe_sb_fcncode_values.h

[Go to the documentation of this file.](#)

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00016  * limitations under the License.
00017  *****/
00018
00028 #ifndef DEFAULT_CFE_SB_FCNCODE_VALUES_H
00029 #define DEFAULT_CFE_SB_FCNCODE_VALUES_H
00030
00031 /*****
00032  * Macro Definitions
00033  *****/
00034
00035 #define CFE_SB_CCVAL(x) CFE_SB_FunctionCode_##x
00036
00037 enum CFE_SB_FunctionCode_
00038 {
00039     CFE_SB_FunctionCode_NOOP                = 0,
00040     CFE_SB_FunctionCode_RESET_COUNTERS      = 1,
00041     CFE_SB_FunctionCode_SEND_SB_STATS       = 2,
00042     CFE_SB_FunctionCode_WRITE_ROUTING_INFO = 3,
00043     CFE_SB_FunctionCode_ENABLE_ROUTE        = 4,
00044     CFE_SB_FunctionCode_DISABLE_ROUTE       = 5,
00045     CFE_SB_FunctionCode_WRITE_PIPE_INFO     = 7,
00046     CFE_SB_FunctionCode_WRITE_MAP_INFO      = 8,
00047     CFE_SB_FunctionCode_ENABLE_SUB_REPORTING = 9,
00048     CFE_SB_FunctionCode_DISABLE_SUB_REPORTING = 10,
00049     CFE_SB_FunctionCode_SEND_PREV_SUBS      = 11,
00050 };
00051
00052 #endif
```

11.187 cfe/modules/sb/config/default_cfe_sb_interface_cfg_values.h File Reference

Macros

- #define [CFE_MISSION_SB_CFGVAL\(x\)](#) DEFAULT_CFE_MISSION_SB_##x

11.187.1 Detailed Description

CFE Software Bus (CFE_SB) Application Public Definitions

This provides default values for configurable items that affect the interface(s) of this module. This includes the CMD/TLM message interface, tables definitions, and any other data products that serve to exchange information with other entities.

Note

This file may be overridden/superseded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [default_cfe_sb_interface_cfg_values.h](#).

11.187.2 Macro Definition Documentation

CFE_MISSION_SB_CFGVAL

```
#define CFE_MISSION_SB_CFGVAL(
    x ) DEFAULT_CFE_MISSION_SB_##x
```

Definition at line 36 of file [default_cfe_sb_interface_cfg_values.h](#).

11.188 default_cfe_sb_interface_cfg_values.h

[Go to the documentation of this file.](#)

```
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00017  *****/
00018
00032 #ifndef DEFAULT_CFE_SB_INTERFACE_CFG_VALUES_H
00033 #define DEFAULT_CFE_SB_INTERFACE_CFG_VALUES_H
00034
00035 /* Use the default configuration value for all */
00036 #define CFE_MISSION_SB_CFGVAL(x) DEFAULT_CFE_MISSION_SB_##x
00037
00038 #endif
```

11.189 cfe/modules/sb/config/default_cfe_sb_internal_cfg_values.h File Reference

Macros

- #define [CFE_PLATFORM_SB_CFGVAL](#)(x) DEFAULT_CFE_PLATFORM_SB_##x

11.189.1 Detailed Description

CFE Software Bus (CFE_SB) Application Private Config Definitions

This provides default values for configurable items that are internal to this module and do NOT affect the interface(s) of this module. Changes to items in this file only affect the local module and will be transparent to external entities that are using the public interface(s).

Note

This file may be overridden/superseded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [default_cfe_sb_internal_cfg_values.h](#).

11.189.2 Macro Definition Documentation

CFE_PLATFORM_SB_CFGVAL

```
#define CFE_PLATFORM_SB_CFGVAL(
    x ) DEFAULT_CFE_PLATFORM_SB_##x
```

Definition at line 36 of file [default_cfe_sb_internal_cfg_values.h](#).

11.190 default_cfe_sb_internal_cfg_values.h

[Go to the documentation of this file.](#)

```

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00017  *****/
00018
00032 #ifndef DEFAULT_CFE_SB_INTERNAL_CFG_VALUES_H
00033 #define DEFAULT_CFE_SB_INTERNAL_CFG_VALUES_H
00034
00035 /* Use the default configuration value for all */
00036 #define CFE_PLATFORM_SB_CFGVAL(x) DEFAULT_CFE_PLATFORM_SB_##x
00037
00038 #endif

```

11.191 cfe/modules/sb/config/default_cfe_sb_mission_cfg.h File Reference

```
#include "cfe_sb_interface_cfg.h"
```

11.191.1 Detailed Description

CFE Event Services (CFE_SB) Application Mission Configuration Header File

This is a compatibility header for the "mission_cfg.h" file that has traditionally provided public config definitions for each CFS app.

Note

This file may be overridden/superseded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [default_cfe_sb_mission_cfg.h](#).

11.192 default_cfe_sb_mission_cfg.h

[Go to the documentation of this file.](#)

```

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00017  *****/
00018
00031 #ifndef DEFAULT_CFE_SB_MISSION_CFG_H
00032 #define DEFAULT_CFE_SB_MISSION_CFG_H
00033
00034 #include "cfe_sb_interface_cfg.h"

```

```
00035
00036 #endif
```

11.193 cfe/modules/sb/config/default_cfe_sb_msg.h File Reference

```
#include "cfe_mission_cfg.h"
#include "cfe_sb_fcncodes.h"
#include "cfe_sb_msgdefs.h"
#include "cfe_sb_msgstruct.h"
```

11.193.1 Detailed Description

Specification for the CFE Event Services (CFE_SB) command and telemetry message data types.

This is a compatibility header for the "cfe_sb_msg.h" file that has traditionally provided the message definitions for cFS apps.

Note

This file may be overridden/superceded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [default_cfe_sb_msg.h](#).

11.194 default_cfe_sb_msg.h

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```
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00017  *****/
00018
00031 #ifndef DEFAULT_CFE_SB_MSG_H
00032 #define DEFAULT_CFE_SB_MSG_H
00033
00034 #include "cfe_mission_cfg.h"
00035 #include "cfe_sb_fcncodes.h"
00036 #include "cfe_sb_msgdefs.h"
00037 #include "cfe_sb_msgstruct.h"
00038
00039 #endif
```

11.195 cfe/modules/sb/config/default_cfe_sb_msgdefs.h File Reference

```
#include "common_types.h"
#include "cfe_mission_cfg.h"
#include "cfe_es_extern_typedefs.h"
#include "cfe_sb_extern_typedefs.h"
#include "cfe_sb_fcncodes.h"
```

Data Structures

- struct [CFE_SB_WriteFileInfoCmd_Payload](#)
Write File Info Command Payload.
- struct [CFE_SB_RouteCmd_Payload](#)
Enable/Disable Route Command Payload.
- struct [CFE_SB_HousekeepingTlm_Payload](#)
- struct [CFE_SB_PipeDepthStats](#)
SB Pipe Depth Statistics.
- struct [CFE_SB_PipeInfoEntry](#)
SB Pipe Information File Entry.
- struct [CFE_SB_StatsTlm_Payload](#)
- struct [CFE_SB_RoutingFileEntry](#)
SB Routing File Entry.
- struct [CFE_SB_MsgMapFileEntry](#)
SB Map File Entry.
- struct [CFE_SB_SingleSubscriptionTlm_Payload](#)
- struct [CFE_SB_SubEntries](#)
SB Previous Subscriptions Entry.
- struct [CFE_SB_AllSubscriptionsTlm_Payload](#)

Typedefs

- typedef struct [CFE_SB_WriteFileInfoCmd_Payload](#) [CFE_SB_WriteFileInfoCmd_Payload_t](#)
Write File Info Command Payload.
- typedef struct [CFE_SB_RouteCmd_Payload](#) [CFE_SB_RouteCmd_Payload_t](#)
Enable/Disable Route Command Payload.
- typedef struct [CFE_SB_HousekeepingTlm_Payload](#) [CFE_SB_HousekeepingTlm_Payload_t](#)
- typedef struct [CFE_SB_PipeDepthStats](#) [CFE_SB_PipeDepthStats_t](#)
SB Pipe Depth Statistics.
- typedef struct [CFE_SB_PipeInfoEntry](#) [CFE_SB_PipeInfoEntry_t](#)
SB Pipe Information File Entry.
- typedef struct [CFE_SB_StatsTlm_Payload](#) [CFE_SB_StatsTlm_Payload_t](#)
- typedef struct [CFE_SB_RoutingFileEntry](#) [CFE_SB_RoutingFileEntry_t](#)
SB Routing File Entry.
- typedef struct [CFE_SB_MsgMapFileEntry](#) [CFE_SB_MsgMapFileEntry_t](#)
SB Map File Entry.
- typedef struct [CFE_SB_SingleSubscriptionTlm_Payload](#) [CFE_SB_SingleSubscriptionTlm_Payload_t](#)
- typedef struct [CFE_SB_SubEntries](#) [CFE_SB_SubEntries_t](#)
SB Previous Subscriptions Entry.
- typedef struct [CFE_SB_AllSubscriptionsTlm_Payload](#) [CFE_SB_AllSubscriptionsTlm_Payload_t](#)

11.195.1 Detailed Description

Specification for the CFE Event Services (CFE_SB) command and telemetry message constant definitions. For CFE_SB this is only the function/command code definitions
Definition in file [default_cfe_sb_msgdefs.h](#).

11.195.2 Typedef Documentation

CFE_SB_AllSubscriptionsTlm_Payload_t

```
typedef struct CFE_SB_AllSubscriptionsTlm_Payload CFE_SB_AllSubscriptionsTlm_Payload_t
```

Name SB Previous Subscriptions Packet

This structure defines the pkt(s) sent by SB that contains a list of all current subscriptions. This pkt is generated on cmd and intended to be used primarily by the Software Bus Networking Application (SBN). Typically, when the cmd is received there are more subscriptions than can fit in one pkt. The complete list of subscriptions is sent via a series of segmented pkts.

CFE_SB_HousekeepingTlm_Payload_t

```
typedef struct CFE_SB_HousekeepingTlm_Payload CFE_SB_HousekeepingTlm_Payload_t
```

Name Software Bus task housekeeping Packet

CFE_SB_MsgMapFileEntry_t

```
typedef struct CFE_SB_MsgMapFileEntry CFE_SB_MsgMapFileEntry_t
```

SB Map File Entry.

Structure of one element of the map information in response to [CFE_SB_WRITE_MAP_INFO_CC](#)

CFE_SB_PipeDepthStats_t

```
typedef struct CFE_SB_PipeDepthStats CFE_SB_PipeDepthStats_t
```

SB Pipe Depth Statistics.

Used in SB Statistics Telemetry Packet [CFE_SB_StatsTlm_t](#)

CFE_SB_PipeInfoEntry_t

```
typedef struct CFE_SB_PipeInfoEntry CFE_SB_PipeInfoEntry_t
```

SB Pipe Information File Entry.

This statistics structure is output as part of the CFE SB "Send Pipe Info" command ([CFE_SB_SEND_PIPE_INFO_CC](#)). Previous versions of CFE simply wrote the internal [CFE_SB_PipeD_t](#) object to the file, but this also contains information such as pointers which are not relevant outside the running CFE process.

By defining the pipe info structure separately, it also provides some independence, such that the internal [CFE_SB_PipeD_t](#) definition can evolve without changing the binary format of the information file.

CFE_SB_RouteCmd_Payload_t

```
typedef struct CFE_SB_RouteCmd_Payload CFE_SB_RouteCmd_Payload_t
```

Enable/Disable Route Command Payload.

This structure contains a definition used by two SB commands, 'Enable Route' [CFE_SB_ENABLE_ROUTE_CC](#) and 'Disable Route' [CFE_SB_DISABLE_ROUTE_CC](#). A route is the destination pipe for a particular message and is therefore defined as a MsgId and PipeId combination.

CFE_SB_RoutingFileEntry_t

```
typedef struct CFE_SB_RoutingFileEntry CFE_SB_RoutingFileEntry_t
```

SB Routing File Entry.

Structure of one element of the routing information in response to [CFE_SB_WRITE_ROUTING_INFO_CC](#)

CFE_SB_SingleSubscriptionTlm_Payload_t

```
typedef struct CFE_SB_SingleSubscriptionTlm_Payload CFE_SB_SingleSubscriptionTlm_Payload_t
```

Name SB Subscription Report Packet

This structure defines the pkt sent by SB when a subscription or a request to unsubscribe is received while subscription reporting is enabled. By default subscription reporting is disabled. This feature is intended to be used primarily by Software Bus Networking Application (SBN)

See also

[CFE_SB_ENABLE_SUB_REPORTING_CC](#), [CFE_SB_DISABLE_SUB_REPORTING_CC](#)

CFE_SB_StatsTlm_Payload_t

```
typedef struct CFE_SB_StatsTlm_Payload CFE_SB_StatsTlm_Payload_t
```

Name SB Statistics Telemetry Packet

SB Statistics packet sent in response to [CFE_SB_SEND_SB_STATS_CC](#)

CFE_SB_SubEntries_t

```
typedef struct CFE_SB_SubEntries CFE_SB_SubEntries_t
```

SB Previous Subscriptions Entry.

This structure defines an entry used in the [CFE_SB_PrevSubsPkt_t](#) Intended to be used primarily by Software Bus Networking Application (SBN)

Used in structure definition [CFE_SB_AllSubscriptionsTlm_t](#)

CFE_SB_WriteFileInfoCmd_Payload_t

```
typedef struct CFE_SB_WriteFileInfoCmd_Payload CFE_SB_WriteFileInfoCmd_Payload_t
```

Write File Info Command Payload.

This structure contains a generic definition used by SB commands that write to a file

11.196 default_cfe_sb_msgdefs.h

[Go to the documentation of this file.](#)

```
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00016  * limitations under the License.
00017  *****/
00018
00026 #ifndef DEFAULT_CFE_SB_MSGDEFS_H
00027 #define DEFAULT_CFE_SB_MSGDEFS_H
00028
00029 #include "common_types.h"
00030 #include "cfe_mission_cfg.h"
00031 #include "cfe_es_extern_typedefs.h"
00032 #include "cfe_sb_extern_typedefs.h"
00033 #include "cfe_sb_fcncodes.h"
```

```
00034
00040 typedef struct CFE_SB_WriteFileInfoCmd_Payload
00041 {
00042     char Filename[CFE_MISSION_MAX_PATH_LEN];
00043 } CFE_SB_WriteFileInfoCmd_Payload_t;
00044
00053 typedef struct CFE_SB_RouteCmd_Payload
00054 {
00055     CFE_SB_MsgId_t  MsgId;
00056     CFE_SB_PipeId_t Pipe;
00057     uint8           Spare;
00058 } CFE_SB_RouteCmd_Payload_t;
00059
00060 /*****
00061 **  SB Telemetry Formats  **
00062 *****/
00063
00067 typedef struct CFE_SB_HousekeepingTlm_Payload
00068 {
00069     uint8 CommandCounter;
00071     uint8 CommandErrorCounter;
00073     uint8 NoSubscribersCounter;
00075     uint8 MsgSendErrorCounter;
00078     uint8 MsgReceiveErrorCounter;
00080     uint8 InternalErrorCounter;
00082     uint8 CreatePipeErrorCounter;
00084     uint8 SubscribeErrorCounter;
00086     uint8 PipeOptsErrorCounter;
00088     uint8 DuplicateSubscriptionsCounter;
00090     uint8 GetPipeIdByNameErrorCounter;
00092     uint8 Padding;
00094     uint16 PipeOverflowErrorCounter;
00096     uint16 MsgLimitErrorCounter;
00099     CFE_ES_MemHandle_t MemPoolHandle;
00102     uint32 MemInUse;
00105     uint32 UnmarkedMem;
00107 } CFE_SB_HousekeepingTlm_Payload_t;
00108
00114 typedef struct CFE_SB_PipeDepthStats
00115 {
00116     CFE_SB_PipeId_t PipeId;
00118     uint16           MaxQueueDepth;
00120     uint16           CurrentQueueDepth;
00122     uint16           PeakQueueDepth;
00124     uint16           Spare;
00126 } CFE_SB_PipeDepthStats_t;
00127
00143 typedef struct CFE_SB_PipeInfoEntry
00144 {
00145     CFE_SB_PipeId_t PipeId;
00146     CFE_ES_AppId_t  AppId;
00147     char            PipeName[CFE_MISSION_MAX_API_LEN];
00148     char            AppName[CFE_MISSION_MAX_API_LEN];
00149     uint16           MaxQueueDepth;
00150     uint16           CurrentQueueDepth;
00151     uint16           PeakQueueDepth;
00152     uint16           SendErrors;
00153     uint8            Opts;
00154     uint8            Spare[3];
00155 } CFE_SB_PipeInfoEntry_t;
00156
00162 typedef struct CFE_SB_StatsTlm_Payload
00163 {
00164     uint32 MsgIdsInUse;
00166     uint32 PeakMsgIdsInUse;
00168     uint32 MaxMsgIdsAllowed;
00171     uint32 PipesInUse;
00173     uint32 PeakPipesInUse;
00175     uint32 MaxPipesAllowed;
00178     uint32 MemInUse;
00180     uint32 PeakMemInUse;
00182     uint32 MaxMemAllowed;
00185     uint32 SubscriptionsInUse;
00187     uint32 PeakSubscriptionsInUse;
00189     uint32 MaxSubscriptionsAllowed;
00193     uint32 SBBuffersInUse;
00195     uint32 PeakSBBuffersInUse;
00198     uint32 MaxPipeDepthAllowed;
00200     CFE_SB_PipeDepthStats_t
00201         PipeDepthStats[CFE_MISSION_SB_MAX_PIPES];
00203 } CFE_SB_StatsTlm_Payload_t;
```

```

00204
00210 typedef struct CFE_SB_RoutingFileEntry
00211 {
00212     CFE_SB_MsgId_t   MsgId;
00213     CFE_SB_PipeId_t  PipeId;
00214     uint8            State;
00215     uint16           MsgCnt;
00216     char             AppName[CFE_MISSION_MAX_API_LEN];
00217     char             PipeName[CFE_MISSION_MAX_API_LEN];
00218 } CFE_SB_RoutingFileEntry_t;
00219
00225 typedef struct CFE_SB_MsgMapFileEntry
00226 {
00227     CFE_SB_MsgId_t   MsgId;
00228     CFE_SB_RouteId_Atom_t Index;
00229 } CFE_SB_MsgMapFileEntry_t;
00230
00241 typedef struct CFE_SB_SingleSubscriptionTlm_Payload
00242 {
00243     uint8            SubType;
00244     CFE_SB_MsgId_t   MsgId;
00245     CFE_SB_Qos_t     Qos;
00246     CFE_SB_PipeId_t  Pipe;
00247 } CFE_SB_SingleSubscriptionTlm_Payload_t;
00248
00257 typedef struct CFE_SB_SubEntries
00258 {
00259     CFE_SB_MsgId_t   MsgId;
00260     CFE_SB_Qos_t     Qos;
00261     CFE_SB_PipeId_t  Pipe;
00262 } CFE_SB_SubEntries_t;
00263
00273 typedef struct CFE_SB_AllSubscriptionsTlm_Payload
00274 {
00275     uint32           PktSegment;
00276     uint32           TotalSegments;
00277     uint32           Entries;
00278     CFE_SB_SubEntries_t Entry[CFE_MISSION_SB_SUB_ENTRIES_PER_PKT];
00279 } CFE_SB_AllSubscriptionsTlm_Payload_t;
00280
00281 #endif

```

11.197 cfe/modules/sb/config/default_cfe_sb_msgid_values.h File Reference

```

#include "cfe_core_api_base_msgids.h"
#include "cfe_sb_topicids.h"

```

Macros

- #define `CFE_PLATFORM_SB_CMD_MIDVAL(x)` `CFE_PLATFORM_CMD_TOPICID_TO_MIDV(CFE_MISSION_SB_##x##_TOPICID)`
- #define `CFE_PLATFORM_SB_TLM_MIDVAL(x)` `CFE_PLATFORM_TLM_TOPICID_TO_MIDV(CFE_MISSION_SB_##x##_TOPICID)`

11.197.1 Detailed Description

CFE Event Services (CFE_SB) Application Message IDs
Definition in file [default_cfe_sb_msgid_values.h](#).

11.197.2 Macro Definition Documentation

CFE_PLATFORM_SB_CMD_MIDVAL

```

#define CFE_PLATFORM_SB_CMD_MIDVAL(
    x ) CFE_PLATFORM_CMD_TOPICID_TO_MIDV(CFE_MISSION_SB_##x##_TOPICID)

```

Definition at line 29 of file [default_cfe_sb_msgid_values.h](#).

CFE_PLATFORM_SB_TLM_MIDVAL

```
#define CFE_PLATFORM_SB_TLM_MIDVAL(
    x ) CFE_PLATFORM_TLM_TOPICID_TO_MIDV(CFE_MISSION_SB_##x##_TOPICID)
```

Definition at line 30 of file [default_cfe_sb_msgid_values.h](#).

11.198 default_cfe_sb_msgid_values.h

[Go to the documentation of this file.](#)

```
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00016  * limitations under the License.
00017  *****/
00018
00023 #ifndef DEFAULT_CFE_SB_MSGID_VALUES_H
00024 #define DEFAULT_CFE_SB_MSGID_VALUES_H
00025
00026 #include "cfe_core_api_base_msgids.h"
00027 #include "cfe_sb_topicids.h"
00028
00029 #define CFE_PLATFORM_SB_CMD_MIDVAL(x) CFE_PLATFORM_CMD_TOPICID_TO_MIDV(CFE_MISSION_SB_##x##_TOPICID)
00030 #define CFE_PLATFORM_SB_TLM_MIDVAL(x) CFE_PLATFORM_TLM_TOPICID_TO_MIDV(CFE_MISSION_SB_##x##_TOPICID)
00031
00032 #endif
```

11.199 cfe/modules/sb/config/default_cfe_sb_msgids.h File Reference

```
#include "cfe_sb_msgid_values.h"
```

Macros

- [#define CFE_SB_CMD_MID CFE_PLATFORM_SB_CMD_MIDVAL\(CMD\) /* Default=0x1803 */](#)
- [#define CFE_SB_SEND_HK_MID CFE_PLATFORM_SB_CMD_MIDVAL\(SEND_HK\) /* Default=0x180B */](#)
- [#define CFE_SB_SUB_RPT_CTRL_MID CFE_PLATFORM_SB_CMD_MIDVAL\(SUB_RPT_CTRL\) /* Default=0x180E */](#)
- [#define CFE_SB_HK_TLM_MID CFE_PLATFORM_SB_TLM_MIDVAL\(HK_TLM\) /* Default=0x0803 */](#)
- [#define CFE_SB_STATS_TLM_MID CFE_PLATFORM_SB_TLM_MIDVAL\(STATS_TLM\) /* Default=0x080A */](#)
- [#define CFE_SB_ALLSUBS_TLM_MID CFE_PLATFORM_SB_TLM_MIDVAL\(ALLSUBS_TLM\) /* Default=0x080D */](#)
- [#define CFE_SB_ONESUB_TLM_MID CFE_PLATFORM_SB_TLM_MIDVAL\(ONESUB_TLM\) /* Default=0x080E */](#)

11.199.1 Detailed Description

CFE Event Services (CFE_SB) Application Message IDs

Definition in file [default_cfe_sb_msgids.h](#).

11.199.2 Macro Definition Documentation

CFE_SB_ALLSUBS_TLM_MID

```
#define CFE_SB_ALLSUBS_TLM_MID CFE_PLATFORM_SB_TLM_MIDVAL (ALLSUBS_TLM) /* Default=0x080D */
```

Definition at line 40 of file [default_cfe_sb_msgids.h](#).

CFE_SB_CMD_MID

```
#define CFE_SB_CMD_MID CFE_PLATFORM_SB_CMD_MIDVAL (CMD) /* Default=0x1803 */
```

Definition at line 31 of file [default_cfe_sb_msgids.h](#).

CFE_SB_HK_TLM_MID

```
#define CFE_SB_HK_TLM_MID CFE_PLATFORM_SB_TLM_MIDVAL (HK_TLM) /* Default=0x0803 */
```

Definition at line 38 of file [default_cfe_sb_msgids.h](#).

CFE_SB_ONESUB_TLM_MID

```
#define CFE_SB_ONESUB_TLM_MID CFE_PLATFORM_SB_TLM_MIDVAL (ONESUB_TLM) /* Default=0x080E */
```

Definition at line 41 of file [default_cfe_sb_msgids.h](#).

CFE_SB_SEND_HK_MID

```
#define CFE_SB_SEND_HK_MID CFE_PLATFORM_SB_CMD_MIDVAL (SEND_HK) /* Default=0x180B */
```

Definition at line 32 of file [default_cfe_sb_msgids.h](#).

CFE_SB_STATS_TLM_MID

```
#define CFE_SB_STATS_TLM_MID CFE_PLATFORM_SB_TLM_MIDVAL (STATS_TLM) /* Default=0x080A */
```

Definition at line 39 of file [default_cfe_sb_msgids.h](#).

CFE_SB_SUB_RPT_CTRL_MID

```
#define CFE_SB_SUB_RPT_CTRL_MID CFE_PLATFORM_SB_CMD_MIDVAL (SUB_RPT_CTRL) /* Default=0x180E */
```

Definition at line 33 of file [default_cfe_sb_msgids.h](#).

11.200 default_cfe_sb_msgids.h

[Go to the documentation of this file.](#)

```
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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00023 #ifndef DEFAULT_CFE_SB_MSGIDS_H
00024 #define DEFAULT_CFE_SB_MSGIDS_H
00025
00026 #include "cfe_sb_msgid_values.h"
00027
00028 /*
```

```
00029 ** cFE Command Message Id's
00030 */
00031 #define CFE_SB_CMD_MID          CFE_PLATFORM_SB_CMD_MIDVAL(CMD)          /* Default=0x1803 */
00032 #define CFE_SB_SEND_HK_MID      CFE_PLATFORM_SB_CMD_MIDVAL(SEND_HK)      /* Default=0x180B */
00033 #define CFE_SB_SUB_RPT_CTRL_MID CFE_PLATFORM_SB_CMD_MIDVAL(SUB_RPT_CTRL) /* Default=0x180E */
00034
00035 /*
00036 ** CFE Telemetry Message Id's
00037 */
00038 #define CFE_SB_HK_TLM_MID       CFE_PLATFORM_SB_TLM_MIDVAL(HK_TLM)       /* Default=0x0803 */
00039 #define CFE_SB_STATS_TLM_MID    CFE_PLATFORM_SB_TLM_MIDVAL(STATS_TLM)    /* Default=0x080A */
00040 #define CFE_SB_ALLSUBS_TLM_MID CFE_PLATFORM_SB_TLM_MIDVAL(ALLSUBS_TLM)   /* Default=0x080D */
00041 #define CFE_SB_ONESUB_TLM_MID   CFE_PLATFORM_SB_TLM_MIDVAL(ONESUB_TLM)   /* Default=0x080E */
00042
00043 #endif
```

11.201 cfe/modules/sb/config/default_cfe_sb_msgstruct.h File Reference

```
#include "cfe_sb_msgdefs.h"
#include "cfe_msg_hdr.h"
```

Data Structures

- struct [CFE_SB_NoopCmd](#)
- struct [CFE_SB_ResetCountersCmd](#)
- struct [CFE_SB_EnableSubReportingCmd](#)
- struct [CFE_SB_DisableSubReportingCmd](#)
- struct [CFE_SB_SendSbStatsCmd](#)
- struct [CFE_SB_SendPrevSubsCmd](#)
- struct [CFE_SB_SendHkCmd](#)
- struct [CFE_SB_WriteRoutingInfoCmd](#)
- struct [CFE_SB_WritePipeInfoCmd](#)
- struct [CFE_SB_WriteMapInfoCmd](#)
- struct [CFE_SB_EnableRouteCmd](#)
- struct [CFE_SB_DisableRouteCmd](#)
- struct [CFE_SB_HousekeepingTlm](#)
- struct [CFE_SB_StatsTlm](#)
- struct [CFE_SB_SingleSubscriptionTlm](#)
- struct [CFE_SB_AllSubscriptionsTlm](#)

Typedefs

- typedef struct [CFE_SB_NoopCmd](#) [CFE_SB_NoopCmd_t](#)
- typedef struct [CFE_SB_ResetCountersCmd](#) [CFE_SB_ResetCountersCmd_t](#)
- typedef struct [CFE_SB_EnableSubReportingCmd](#) [CFE_SB_EnableSubReportingCmd_t](#)
- typedef struct [CFE_SB_DisableSubReportingCmd](#) [CFE_SB_DisableSubReportingCmd_t](#)
- typedef struct [CFE_SB_SendSbStatsCmd](#) [CFE_SB_SendSbStatsCmd_t](#)
- typedef struct [CFE_SB_SendPrevSubsCmd](#) [CFE_SB_SendPrevSubsCmd_t](#)
- typedef struct [CFE_SB_SendHkCmd](#) [CFE_SB_SendHkCmd_t](#)
- typedef struct [CFE_SB_WriteRoutingInfoCmd](#) [CFE_SB_WriteRoutingInfoCmd_t](#)
- typedef struct [CFE_SB_WritePipeInfoCmd](#) [CFE_SB_WritePipeInfoCmd_t](#)
- typedef struct [CFE_SB_WriteMapInfoCmd](#) [CFE_SB_WriteMapInfoCmd_t](#)
- typedef struct [CFE_SB_EnableRouteCmd](#) [CFE_SB_EnableRouteCmd_t](#)
- typedef struct [CFE_SB_DisableRouteCmd](#) [CFE_SB_DisableRouteCmd_t](#)
- typedef struct [CFE_SB_HousekeepingTlm](#) [CFE_SB_HousekeepingTlm_t](#)
- typedef struct [CFE_SB_StatsTlm](#) [CFE_SB_StatsTlm_t](#)
- typedef struct [CFE_SB_SingleSubscriptionTlm](#) [CFE_SB_SingleSubscriptionTlm_t](#)
- typedef struct [CFE_SB_AllSubscriptionsTlm](#) [CFE_SB_AllSubscriptionsTlm_t](#)

11.201.1 Detailed Description

Purpose: cFE Executive Services (SB) Command and Telemetry packet definition file.
Definition in file [default_cfe_sb_msgstruct.h](#).

11.201.2 Typedef Documentation

CFE_SB_AllSubscriptionsTlm_t

```
typedef struct CFE_SB_AllSubscriptionsTlm CFE_SB_AllSubscriptionsTlm_t
```

CFE_SB_DisableRouteCmd_t

```
typedef struct CFE_SB_DisableRouteCmd CFE_SB_DisableRouteCmd_t
```

CFE_SB_DisableSubReportingCmd_t

```
typedef struct CFE_SB_DisableSubReportingCmd CFE_SB_DisableSubReportingCmd_t
```

CFE_SB_EnableRouteCmd_t

```
typedef struct CFE_SB_EnableRouteCmd CFE_SB_EnableRouteCmd_t
```

CFE_SB_EnableSubReportingCmd_t

```
typedef struct CFE_SB_EnableSubReportingCmd CFE_SB_EnableSubReportingCmd_t
```

CFE_SB_HousekeepingTlm_t

```
typedef struct CFE_SB_HousekeepingTlm CFE_SB_HousekeepingTlm_t
```

CFE_SB_NoopCmd_t

```
typedef struct CFE_SB_NoopCmd CFE_SB_NoopCmd_t
```

CFE_SB_ResetCountersCmd_t

```
typedef struct CFE_SB_ResetCountersCmd CFE_SB_ResetCountersCmd_t
```

CFE_SB_SendHkCmd_t

```
typedef struct CFE_SB_SendHkCmd CFE_SB_SendHkCmd_t
```

CFE_SB_SendPrevSubsCmd_t

```
typedef struct CFE_SB_SendPrevSubsCmd CFE_SB_SendPrevSubsCmd_t
```

CFE_SB_SendSbStatsCmd_t

```
typedef struct CFE_SB_SendSbStatsCmd CFE_SB_SendSbStatsCmd_t
```

CFE_SB_SingleSubscriptionTlm_t

```
typedef struct CFE_SB_SingleSubscriptionTlm CFE_SB_SingleSubscriptionTlm_t
```

CFE_SB_StatsTlm_t

```
typedef struct CFE_SB_StatsTlm CFE_SB_StatsTlm_t
```

CFE_SB_WriteMapInfoCmd_t

```
typedef struct CFE_SB_WriteMapInfoCmd CFE_SB_WriteMapInfoCmd_t
```

CFE_SB_WritePipeInfoCmd_t

```
typedef struct CFE_SB_WritePipeInfoCmd CFE_SB_WritePipeInfoCmd_t
```

CFE_SB_WriteRoutingInfoCmd_t

```
typedef struct CFE_SB_WriteRoutingInfoCmd CFE_SB_WriteRoutingInfoCmd_t
```

11.202 default_cfe_sb_msgstruct.h

[Go to the documentation of this file.](#)

```
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00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00025 #ifndef DEFAULT_CFE_SB_MSGSTRUCT_H
00026 #define DEFAULT_CFE_SB_MSGSTRUCT_H
00027
00028 /*****
00029  * Includes
00030  *****/
00031 #include "cfe_sb_msgdefs.h"
00032 #include "cfe_msg_hdr.h"
00033
00034 /*****
00035  ** SB Command Formats **
00036 *****/
00037
00038 /*
00039  * SB Messages which have no payload are each
00040  * given unique typedefs to follow the command handler convention
00041  *
00042  * For the SB application these are mapped to the CFE_MSG_CommandHeader_t type,
00043  * as they contain only a primary + command header.
00044  */
00045 typedef struct CFE_SB_NoopCmd
00046 {
00047     CFE_MSG_CommandHeader_t CommandHeader;
00048 } CFE_SB_NoopCmd_t;
00049
00050 typedef struct CFE_SB_ResetCountersCmd
00051 {
00052     CFE_MSG_CommandHeader_t CommandHeader;
00053 } CFE_SB_ResetCountersCmd_t;
00054
00055 typedef struct CFE_SB_EnableSubReportingCmd
00056 {
00057     CFE_MSG_CommandHeader_t CommandHeader;
00058 } CFE_SB_EnableSubReportingCmd_t;
00059
00060 typedef struct CFE_SB_DisableSubReportingCmd
```

```

00061 {
00062     CFE_MSG_CommandHeader_t CommandHeader;
00063 } CFE_SB_DisableSubReportingCmd_t;
00064
00065 typedef struct CFE_SB_SendSbStatsCmd
00066 {
00067     CFE_MSG_CommandHeader_t CommandHeader;
00068 } CFE_SB_SendSbStatsCmd_t;
00069
00070 typedef struct CFE_SB_SendPrevSubsCmd
00071 {
00072     CFE_MSG_CommandHeader_t CommandHeader;
00073 } CFE_SB_SendPrevSubsCmd_t;
00074
00075 typedef struct CFE_SB_SendHkCmd
00076 {
00077     CFE_MSG_CommandHeader_t CommandHeader;
00078 } CFE_SB_SendHkCmd_t;
00079
00080 /*
00081  * Create a unique typedef for each of the commands that share this format.
00082  */
00083 typedef struct CFE_SB_WriteRoutingInfoCmd
00084 {
00085     CFE_MSG_CommandHeader_t CommandHeader;
00086     CFE_SB_WriteFileInfoCmd_Payload_t Payload;
00087 } CFE_SB_WriteRoutingInfoCmd_t;
00088
00089 typedef struct CFE_SB_WritePipeInfoCmd
00090 {
00091     CFE_MSG_CommandHeader_t CommandHeader;
00092     CFE_SB_WriteFileInfoCmd_Payload_t Payload;
00093 } CFE_SB_WritePipeInfoCmd_t;
00094
00095 typedef struct CFE_SB_WriteMapInfoCmd
00096 {
00097     CFE_MSG_CommandHeader_t CommandHeader;
00098     CFE_SB_WriteFileInfoCmd_Payload_t Payload;
00099 } CFE_SB_WriteMapInfoCmd_t;
00100
00101 /*
00102  * Create a unique typedef for each of the commands that share this format.
00103  */
00104 typedef struct CFE_SB_EnableRouteCmd
00105 {
00106     CFE_MSG_CommandHeader_t CommandHeader;
00107     CFE_SB_RouteCmd_Payload_t Payload;
00108 } CFE_SB_EnableRouteCmd_t;
00109
00110 typedef struct CFE_SB_DisableRouteCmd
00111 {
00112     CFE_MSG_CommandHeader_t CommandHeader;
00113     CFE_SB_RouteCmd_Payload_t Payload;
00114 } CFE_SB_DisableRouteCmd_t;
00115
00116 /*****
00117 ** SB Telemetry Formats **
00118 *****/
00119
00120 typedef struct CFE_SB_HousekeepingTlm
00121 {
00122     CFE_MSG_TelemetryHeader_t TelemetryHeader;
00123     CFE_SB_HousekeepingTlm_Payload_t Payload;
00124 } CFE_SB_HousekeepingTlm_t;
00125
00126 typedef struct CFE_SB_StatsTlm
00127 {
00128     CFE_MSG_TelemetryHeader_t TelemetryHeader;
00129     CFE_SB_StatsTlm_Payload_t Payload;
00130 } CFE_SB_StatsTlm_t;
00131
00132 typedef struct CFE_SB_SingleSubscriptionTlm
00133 {
00134     CFE_MSG_TelemetryHeader_t TelemetryHeader;
00135     CFE_SB_SingleSubscriptionTlm_Payload_t Payload;
00136 } CFE_SB_SingleSubscriptionTlm_t;
00137
00138 typedef struct CFE_SB_AllSubscriptionsTlm
00139 {
00140     CFE_MSG_TelemetryHeader_t TelemetryHeader;
00141     CFE_SB_AllSubscriptionsTlm_Payload_t Payload;

```

```
00142 } CFE_SB_AllSubscriptionsTlm_t;
00143
00144 #endif
```

11.203 cfe/modules/sb/config/default_cfe_sb_platform_cfg.h File Reference

```
#include "cfe_sb_mission_cfg.h"
#include "cfe_sb_internal_cfg.h"
```

11.203.1 Detailed Description

CFE Software Bus (CFE_SB) Application Platform Configuration Header File

This is a compatibility header for the "platform_cfg.h" file that has traditionally provided both public and private config definitions for each CFS app.

These definitions are now provided in two separate files, one for the public/mission scope and one for internal scope.

Note

This file may be overridden/superseded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [default_cfe_sb_platform_cfg.h](#).

11.204 default_cfe_sb_platform_cfg.h

[Go to the documentation of this file.](#)

```
00001 /*****
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"
00003  *
00004  * Copyright (c) 2023 United States Government as represented by the
00005  * Administrator of the National Aeronautics and Space Administration.
00006  * All Rights Reserved.
00007  *
00008  * Licensed under the Apache License, Version 2.0 (the "License"); you may
00009  * not use this file except in compliance with the License. You may obtain
00010  * a copy of the License at http://www.apache.org/licenses/LICENSE-2.0
00011  *
00012  * Unless required by applicable law or agreed to in writing, software
00013  * distributed under the License is distributed on an "AS IS" BASIS,
00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00035 #ifndef DEFAULT_CFE_SB_PLATFORM_CFG_H
00036 #define DEFAULT_CFE_SB_PLATFORM_CFG_H
00037
00038 #include "cfe_sb_mission_cfg.h"
00039 #include "cfe_sb_internal_cfg.h"
00040
00041 #endif
```

11.205 cfe/modules/sb/config/default_cfe_sb_topicid_values.h File Reference

Macros

- #define [CFE_MISSION_SB_TIDVAL\(x\)](#) DEFAULT_CFE_MISSION_SB_###x##_TOPICID

11.205.1 Detailed Description

CFE Software Bus (CFE_SB) Application Topic IDs

Definition in file [default_cfe_sb_topicid_values.h](#).

11.205.2 Macro Definition Documentation

CFE_MISSION_SB_TIDVAL

```
#define CFE_MISSION_SB_TIDVAL(
    x ) DEFAULT_CFE_MISSION_SB_##x##_TOPICID
```

Definition at line 26 of file [default_cfe_sb_topicid_values.h](#).

11.206 default_cfe_sb_topicid_values.h

[Go to the documentation of this file.](#)

```
00001 /*****
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"
00003  *
00004  * Copyright (c) 2023 United States Government as represented by the
00005  * Administrator of the National Aeronautics and Space Administration.
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00007  *
00008  * Licensed under the Apache License, Version 2.0 (the "License"); you may
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00011  *
00012  * Unless required by applicable law or agreed to in writing, software
00013  * distributed under the License is distributed on an "AS IS" BASIS,
00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00023 #ifndef DEFAULT_CFE_SB_TOPICID_VALUES_H
00024 #define DEFAULT_CFE_SB_TOPICID_VALUES_H
00025
00026 #define CFE_MISSION_SB_TIDVAL(x) DEFAULT_CFE_MISSION_SB_##x##_TOPICID
00027
00028 #endif
```

11.207 cfe/modules/sb/fsw/inc/cfe_sb_eventids.h File Reference

Macros

SB event IDs

- [#define CFE_SB_INIT_EID 1](#)
SB Initialization Event ID.
- [#define CFE_SB_CR_PIPE_BAD_ARG_EID 2](#)
SB Create Pipe API Bad Argument Event ID.
- [#define CFE_SB_MAX_PIPES_MET_EID 3](#)
SB Create Pipe API Max Pipes Exceeded Event ID.
- [#define CFE_SB_CR_PIPE_ERR_EID 4](#)
SB Create Pipe API Queue Create Failure Event ID.
- [#define CFE_SB_PIPE_ADDED_EID 5](#)
SB Create Pipe API Success Event ID.
- [#define CFE_SB_SUB_ARG_ERR_EID 6](#)
SB Subscribe API Bad Argument Event ID.
- [#define CFE_SB_DUP_SUBSCRIP_EID 7](#)
SB Subscribe API Duplicate MsgId Subscription Event ID.
- [#define CFE_SB_MAX_MSGS_MET_EID 8](#)
SB Subscribe API Max Subscriptions Exceeded Event ID.
- [#define CFE_SB_MAX_DESTS_MET_EID 9](#)
SB Subscribe API Max Destinations Exceeded Event ID.
- [#define CFE_SB_SUBSCRIPTION_RCVD_EID 10](#)
SB Subscribe API Success Event ID.
- [#define CFE_SB_UNSUB_ARG_ERR_EID 11](#)
SB Unsubscribe API Bad Argument Event ID.

- #define CFE_SB_UNSUB_NO_SUBS_EID 12
SB Unsubscribe API No MsgId Subscription Event ID.
- #define CFE_SB_SEND_BAD_ARG_EID 13
SB Transmit API Bad Argument Event ID.
- #define CFE_SB_SEND_NO_SUBS_EID 14
SB Transmit API No MsgId Subscribers Event ID.
- #define CFE_SB_MSG_TOO_BIG_EID 15
SB Transmit API Message Size Limit Exceeded Event ID.
- #define CFE_SB_GET_BUF_ERR_EID 16
SB Transmit API Buffer Request Failure Event ID.
- #define CFE_SB_MSGID_LIM_ERR_EID 17
SB Transmit API MsgId Pipe Limit Exceeded Event ID.
- #define CFE_SB_RCV_BAD_ARG_EID 18
SB Receive Buffer API Bad Argument Event ID.
- #define CFE_SB_BAD_PIPEID_EID 19
SB Receive Buffer API Invalid Pipe Event ID.
- #define CFE_SB_DEST_BLK_ERR_EID 20
SB Subscribe API Get Destination Block Failure Event ID.
- #define CFE_SB_SEND_INV_MSGID_EID 21
SB Transmit API Invalid MsgId Event ID.
- #define CFE_SB_SUBSCRIPTION_RPT_EID 22
SB Subscription Report Sent Event ID.
- #define CFE_SB_HASHCOLLISION_EID 23
SB Subscribe API Message Table Hash Collision Event ID.
- #define CFE_SB_Q_FULL_ERR_EID 25
SB Transmit API Pipe Overflow Event ID.
- #define CFE_SB_Q_WR_ERR_EID 26
SB Transmit API Queue Write Failure Event ID.
- #define CFE_SB_Q_RD_ERR_EID 27
SB Transmit API Queue Read Failure Event ID.
- #define CFE_SB_CMD0_RCVD_EID 28
SB No-op Command Success Event ID.
- #define CFE_SB_CMD1_RCVD_EID 29
SB Reset Counters Command Success Event ID.
- #define CFE_SB_SND_STATS_EID 32
SB Send Statistics Command Success Event ID.
- #define CFE_SB_ENBL_RTE1_EID 33
SB Enable Route Command Invalid MsgId/PipeId Pair Event ID.
- #define CFE_SB_ENBL_RTE2_EID 34
SB Enable Route Command Success Event ID.
- #define CFE_SB_ENBL_RTE3_EID 35
SB Enable Route Command Invalid MsgId or Pipe Event ID.
- #define CFE_SB_DSDL_RTE1_EID 36
SB Disable Route Command Invalid MsgId/PipeId Pair Event ID.
- #define CFE_SB_DSDL_RTE2_EID 37
SB Disable Route Command Success Event ID.
- #define CFE_SB_DSDL_RTE3_EID 38
SB Disable Route Command Invalid MsgId or Pipe Event ID.
- #define CFE_SB_SND_RTG_EID 39
SB File Write Success Event ID.
- #define CFE_SB_SND_RTG_ERR1_EID 40
SB File Write Create File Failure Event ID.
- #define CFE_SB_BAD_CMD_CODE_EID 42
SB Invalid Command Code Received Event ID.
- #define CFE_SB_BAD_MSGID_EID 43

- SB Invalid Message ID Received Event ID.*
- #define [CFE_SB_FULL_SUB_PKT_EID](#) 44
- SB Send Previous Subscriptions Command Full Packet Sent Event ID.*
- #define [CFE_SB_PART_SUB_PKT_EID](#) 45
- SB Send Previous Subscriptions Command Partial Packet Sent Event ID.*
- #define [CFE_SB_DEL_PIPE_ERR1_EID](#) 46
- SB Pipe Delete API Bad Argument Event ID.*
- #define [CFE_SB_PIPE_DELETED_EID](#) 47
- SB Pipe Delete API Success Event ID.*
- #define [CFE_SB_SUBSCRIPTION_REMOVED_EID](#) 48
- SB Unsubscribe API Success Event ID.*
- #define [CFE_SB_FILEWRITE_ERR_EID](#) 49
- SB File Write Failed Event ID.*
- #define [CFE_SB_SUB_INV_PIPE_EID](#) 50
- SB Subscribe API Invalid Pipe Event ID.*
- #define [CFE_SB_SUB_INV_CALLER_EID](#) 51
- SB Subscribe API Not Owner Event ID.*
- #define [CFE_SB_UNSUB_INV_PIPE_EID](#) 52
- SB Unsubscribe API Invalid Pipe Event ID.*
- #define [CFE_SB_UNSUB_INV_CALLER_EID](#) 53
- SB Unsubscribe API Not Owner Event ID.*
- #define [CFE_SB_DEL_PIPE_ERR2_EID](#) 54
- SB Delete Pipe API Not Owner Event ID.*
- #define [CFE_SB_SETPIPEOPTS_ID_ERR_EID](#) 55
- SB Set Pipe Opts API Invalid Pipe Event ID.*
- #define [CFE_SB_SETPIPEOPTS_OWNER_ERR_EID](#) 56
- SB Set Pipe Opts API Not Owner Event ID.*
- #define [CFE_SB_SETPIPEOPTS_EID](#) 57
- SB Set Pipe Opts API Success Event ID.*
- #define [CFE_SB_GETPIPEOPTS_ID_ERR_EID](#) 58
- SB Get Pipe Opts API Invalid Pipe Event ID.*
- #define [CFE_SB_GETPIPEOPTS_PTR_ERR_EID](#) 59
- SB Get Pipe Opts API Invalid Pointer Event ID.*
- #define [CFE_SB_GETPIPEOPTS_EID](#) 60
- SB Get Pipe Opts API Success Event ID.*
- #define [CFE_SB_GETPIPENAME_EID](#) 62
- SB Get Pipe Name API Success Event ID.*
- #define [CFE_SB_GETPIPENAME_NULL_PTR_EID](#) 63
- SB Get Pipe Name API Invalid Pointer Event ID.*
- #define [CFE_SB_GETPIPENAME_ID_ERR_EID](#) 64
- SB Get Pipe Name API Invalid Pipe or Resource Event ID.*
- #define [CFE_SB_GETPIPEIDBYNAME_EID](#) 65
- SB Get Pipe ID By Name API Success Event ID.*
- #define [CFE_SB_GETPIPEIDBYNAME_NULL_ERR_EID](#) 66
- SB Get Pipe ID By Name API Invalid Pointer Event ID.*
- #define [CFE_SB_GETPIPEIDBYNAME_NAME_ERR_EID](#) 67
- SB Get Pipe ID By Name API Name Not Found Or ID Not Matched Event ID.*
- #define [CFE_SB_LEN_ERR_EID](#) 68
- SB Invalid Command Length Event ID.*
- #define [CFE_SB_CR_PIPE_NAME_TAKEN_EID](#) 69
- SB Create Pipe API Name Taken Event ID.*
- #define [CFE_SB_CR_PIPE_NO_FREE_EID](#) 70
- SB Create Pipe API Queues Exhausted Event ID.*
- #define [CFE_SB_SEND_MESSAGE_INTEGRITY_FAIL_EID](#) 71
- SB integrity actions on transmit message failure event.*
- #define [CFE_SB_RCV_MESSAGE_INTEGRITY_FAIL_EID](#) 72
- SB validation of received message failure event.*

11.207.1 Detailed Description

cFE Software Bus Services Event IDs
Definition in file [cfe_sb_eventids.h](#).

11.207.2 Macro Definition Documentation

CFE_SB_BAD_CMD_CODE_EID

```
#define CFE_SB_BAD_CMD_CODE_EID 42
```

SB Invalid Command Code Received Event ID.

Type: ERROR

Cause:

Invalid command code for message ID [CFE_SB_CMD_MID](#) or [CFE_SB_SUB_RPT_CTRL_MID](#) received on the SB message pipe. OVERLOADED
Definition at line 461 of file [cfe_sb_eventids.h](#).

CFE_SB_BAD_MSGID_EID

```
#define CFE_SB_BAD_MSGID_EID 43
```

SB Invalid Message ID Received Event ID.

Type: ERROR

Cause:

Invalid message ID received on the SB message pipe.
Definition at line 472 of file [cfe_sb_eventids.h](#).

CFE_SB_BAD_PIPEID_EID

```
#define CFE_SB_BAD_PIPEID_EID 19
```

SB Receive Buffer API Invalid Pipe Event ID.

Type: ERROR

Cause:

[CFE_SB_ReceiveBuffer](#) API failure due to an invalid Pipe ID.
Definition at line 244 of file [cfe_sb_eventids.h](#).

CFE_SB_CMD0_RCVD_EID

#define CFE_SB_CMD0_RCVD_EID 28
SB No-op Command Success Event ID.

Type: INFORMATION

Cause:

[SB NO-OP Command](#) success.
Definition at line 335 of file [cfe_sb_eventids.h](#).

CFE_SB_CMD1_RCVD_EID

#define CFE_SB_CMD1_RCVD_EID 29
SB Reset Counters Command Success Event ID.

Type: DEBUG

Cause:

[SB Reset Counters Command](#) success.
Definition at line 346 of file [cfe_sb_eventids.h](#).

CFE_SB_CR_PIPE_BAD_ARG_EID

#define CFE_SB_CR_PIPE_BAD_ARG_EID 2
SB Create Pipe API Bad Argument Event ID.

Type: ERROR

Cause:

[CFE_SB_CreatePipe](#) API failure due to a bad input argument.
Definition at line 53 of file [cfe_sb_eventids.h](#).

CFE_SB_CR_PIPE_ERR_EID

#define CFE_SB_CR_PIPE_ERR_EID 4
SB Create Pipe API Queue Create Failure Event ID.

Type: ERROR

Cause:

[CFE_SB_CreatePipe](#) API failure creating the queue.
Definition at line 75 of file [cfe_sb_eventids.h](#).

CFE_SB_CR_PIPE_NAME_TAKEN_EID

#define CFE_SB_CR_PIPE_NAME_TAKEN_EID 69
SB Create Pipe API Name Taken Event ID.

Type: ERROR

Cause:

[CFE_SB_CreatePipe](#) API failure due to pipe name taken.
Definition at line 750 of file [cfe_sb_eventids.h](#).

CFE_SB_CR_PIPE_NO_FREE_EID

#define CFE_SB_CR_PIPE_NO_FREE_EID 70
SB Create Pipe API Queues Exhausted Event ID.

Type: ERROR

Cause:

[CFE_SB_CreatePipe](#) API failure due to no free queues.
Definition at line 761 of file [cfe_sb_eventids.h](#).

CFE_SB_DEL_PIPE_ERR1_EID

#define CFE_SB_DEL_PIPE_ERR1_EID 46
SB Pipe Delete API Bad Argument Event ID.

Type: ERROR

Cause:

An SB Delete Pipe API failed due to an invalid input argument.
Definition at line 507 of file [cfe_sb_eventids.h](#).

CFE_SB_DEL_PIPE_ERR2_EID

#define CFE_SB_DEL_PIPE_ERR2_EID 54
SB Delete Pipe API Not Owner Event ID.

Type: ERROR

Cause:

An SB Delete Pipe API failed due to not being the pipe owner.
Definition at line 595 of file [cfe_sb_eventids.h](#).

CFE_SB_DEST_BLK_ERR_EID

```
#define CFE_SB_DEST_BLK_ERR_EID 20
```

SB Subscribe API Get Destination Block Failure Event ID.

Type: ERROR

Cause:

An SB Subscribe API call failed to get a destination block.

Definition at line 255 of file [cfe_sb_eventids.h](#).

CFE_SB_DSBL_RTE1_EID

```
#define CFE_SB_DSBL_RTE1_EID 36
```

SB Disable Route Command Invalid MsgId/PipeId Pair Event ID.

Type: ERROR

Cause:

[SB Disable Route Command](#) failure due to the Message ID not being subscribed to the pipe.

Definition at line 404 of file [cfe_sb_eventids.h](#).

CFE_SB_DSBL_RTE2_EID

```
#define CFE_SB_DSBL_RTE2_EID 37
```

SB Disable Route Command Success Event ID.

Type: DEBUG

Cause:

[SB Disable Route Command](#) success.

Definition at line 415 of file [cfe_sb_eventids.h](#).

CFE_SB_DSBL_RTE3_EID

```
#define CFE_SB_DSBL_RTE3_EID 38
```

SB Disable Route Command Invalid MsgId or Pipe Event ID.

Type: ERROR

Cause:

[SB Disable Route Command](#) failure due to an invalid MsgId or Pipe.

Definition at line 427 of file [cfe_sb_eventids.h](#).

CFE_SB_DUP_SUBSCRIP_EID

```
#define CFE_SB_DUP_SUBSCRIP_EID 7
```

SB Subscribe API Duplicate MsgId Subscription Event ID.

Type: INFORMATION

Cause:

An SB Subscribe API was called with a Message ID that was already subscribed on the pipe on the pipe.
Definition at line 109 of file [cfe_sb_eventids.h](#).

CFE_SB_ENBL_RTE1_EID

```
#define CFE_SB_ENBL_RTE1_EID 33
```

SB Enable Route Command Invalid MsgId/PipeID Pair Event ID.

Type: ERROR

Cause:

[SB Enable Route Command](#) failure due to the Message ID not being subscribed to the pipe.
Definition at line 369 of file [cfe_sb_eventids.h](#).

CFE_SB_ENBL_RTE2_EID

```
#define CFE_SB_ENBL_RTE2_EID 34
```

SB Enable Route Command Success Event ID.

Type: DEBUG

Cause:

[SB Enable Route Command](#) success.
Definition at line 380 of file [cfe_sb_eventids.h](#).

CFE_SB_ENBL_RTE3_EID

```
#define CFE_SB_ENBL_RTE3_EID 35
```

SB Enable Route Command Invalid MsgId or Pipe Event ID.

Type: ERROR

Cause:

[SB Enable Route Command](#) failure due to an invalid MsgId or Pipe.
Definition at line 392 of file [cfe_sb_eventids.h](#).

CFE_SB_FILEWRITE_ERR_EID

#define CFE_SB_FILEWRITE_ERR_EID 49
SB File Write Failed Event ID.

Type: ERROR

Cause:

An SB file write failure encountered when writing to the file.
Definition at line 540 of file [cfe_sb_eventids.h](#).

CFE_SB_FULL_SUB_PKT_EID

#define CFE_SB_FULL_SUB_PKT_EID 44
SB Send Previous Subscriptions Command Full Packet Sent Event ID.

Type: DEBUG

Cause:

[SB Send Previous Subscriptions Command](#) processing sent a full subscription packet.
Definition at line 484 of file [cfe_sb_eventids.h](#).

CFE_SB_GET_BUF_ERR_EID

#define CFE_SB_GET_BUF_ERR_EID 16
SB Transmit API Buffer Request Failure Event ID.

Type: ERROR

Cause:

An SB Transmit API call buffer request failed.
Definition at line 210 of file [cfe_sb_eventids.h](#).

CFE_SB_GETPIPEIDBYNAME_EID

#define CFE_SB_GETPIPEIDBYNAME_EID 65
SB Get Pipe ID By Name API Success Event ID.

Type: DEBUG

Cause:

[CFE_SB_GetPipeIdByName](#) success.
Definition at line 705 of file [cfe_sb_eventids.h](#).

CFE_SB_GETPIPEIDBYNAME_NAME_ERR_EID

```
#define CFE_SB_GETPIPEIDBYNAME_NAME_ERR_EID 67
```

SB Get Pipe ID By Name API Name Not Found Or ID Not Matched Event ID.

Type: ERROR

Cause:

[CFE_SB_GetPipeIdByName](#) failure due to name not found or ID mismatch. OVERLOADED

Definition at line 727 of file [cfe_sb_eventids.h](#).

CFE_SB_GETPIPEIDBYNAME_NULL_ERR_EID

```
#define CFE_SB_GETPIPEIDBYNAME_NULL_ERR_EID 66
```

SB Get Pipe ID By Name API Invalid Pointer Event ID.

Type: ERROR

Cause:

[CFE_SB_GetPipeIdByName](#) failure due to invalid pointer.

Definition at line 716 of file [cfe_sb_eventids.h](#).

CFE_SB_GETPIPENAME_EID

```
#define CFE_SB_GETPIPENAME_EID 62
```

SB Get Pipe Name API Success Event ID.

Type: DEBUG

Cause:

[CFE_SB_GetPipeName](#) success.

Definition at line 672 of file [cfe_sb_eventids.h](#).

CFE_SB_GETPIPENAME_ID_ERR_EID

```
#define CFE_SB_GETPIPENAME_ID_ERR_EID 64
```

SB Get Pipe Name API Invalid Pipe or Resource Event ID.

Type: ERROR

Cause:

[CFE_SB_GetPipeName](#) failure due to invalid pipe ID or failure in retrieving resource name. OVERLOADED

Definition at line 694 of file [cfe_sb_eventids.h](#).

CFE_SB_GETPIPENAME_NULL_PTR_EID

#define CFE_SB_GETPIPENAME_NULL_PTR_EID 63
SB Get Pipe Name API Invalid Pointer Event ID.

Type: ERROR

Cause:

[CFE_SB_GetPipeName](#) failure due to invalid pointer.
Definition at line 683 of file [cfe_sb_eventids.h](#).

CFE_SB_GETPIPEOPTS_EID

#define CFE_SB_GETPIPEOPTS_EID 60
SB Get Pipe Opts API Success Event ID.

Type: DEBUG

Cause:

[CFE_SB_GetPipeOpts](#) success.
Definition at line 661 of file [cfe_sb_eventids.h](#).

CFE_SB_GETPIPEOPTS_ID_ERR_EID

#define CFE_SB_GETPIPEOPTS_ID_ERR_EID 58
SB Get Pipe Opts API Invalid Pipe Event ID.

Type: ERROR

Cause:

[CFE_SB_GetPipeOpts](#) failure due to invalid pipe ID.
Definition at line 639 of file [cfe_sb_eventids.h](#).

CFE_SB_GETPIPEOPTS_PTR_ERR_EID

#define CFE_SB_GETPIPEOPTS_PTR_ERR_EID 59
SB Get Pipe Opts API Invalid Pointer Event ID.

Type: ERROR

Cause:

[CFE_SB_GetPipeOpts](#) failure due to invalid pointer.
Definition at line 650 of file [cfe_sb_eventids.h](#).

CFE_SB_HASHCOLLISION_EID

```
#define CFE_SB_HASHCOLLISION_EID 23
```

SB Subscribe API Message Table Hash Collision Event ID.

Type: **DEBUG**

Cause:

An SB Subscribe API call caused a message table hash collision, which will impact message transmission performance. This can be resolved by deconflicting MsgId values or increasing [CFE_PLATFORM_SB_MAX_MSG_IDS](#).

Definition at line [290](#) of file [cfe_sb_eventids.h](#).

CFE_SB_INIT_EID

```
#define CFE_SB_INIT_EID 1
```

SB Initialization Event ID.

Type: **INFORMATION**

Cause:

Software Bus Services Task initialization complete.

Definition at line [42](#) of file [cfe_sb_eventids.h](#).

CFE_SB_LEN_ERR_EID

```
#define CFE_SB_LEN_ERR_EID 68
```

SB Invalid Command Length Event ID.

Type: **ERROR**

Cause:

Invalid length for the command code in message ID [CFE_SB_CMD_MID](#) or [CFE_SB_SUB_RPT_CTRL_MID](#) received on the SB message pipe.

Definition at line [739](#) of file [cfe_sb_eventids.h](#).

CFE_SB_MAX_DESTS_MET_EID

```
#define CFE_SB_MAX_DESTS_MET_EID 9
```

SB Subscribe API Max Destinations Exceeded Event ID.

Type: **ERROR**

Cause:

An SB Subscribe API was called with a message id that already has the maximum allowed number of destinations.

Definition at line [133](#) of file [cfe_sb_eventids.h](#).

CFE_SB_MAX_MSGS_MET_EID

```
#define CFE_SB_MAX_MSGS_MET_EID 8
```

SB Subscribe API Max Subscriptions Exceeded Event ID.

Type: ERROR

Cause:

An SB Subscribe API was called on a pipe that already has the maximum allowed number of subscriptions.
Definition at line 121 of file [cfe_sb_eventids.h](#).

CFE_SB_MAX_PIPES_MET_EID

```
#define CFE_SB_MAX_PIPES_MET_EID 3
```

SB Create Pipe API Max Pipes Exceeded Event ID.

Type: ERROR

Cause:

[CFE_SB_CreatePipe](#) API failure to do maximum number of pipes being exceeded.
Definition at line 64 of file [cfe_sb_eventids.h](#).

CFE_SB_MSG_TOO_BIG_EID

```
#define CFE_SB_MSG_TOO_BIG_EID 15
```

SB Transmit API Message Size Limit Exceeded Event ID.

Type: ERROR

Cause:

An SB Transmit API was called with a message that is too big.
Definition at line 199 of file [cfe_sb_eventids.h](#).

CFE_SB_MSGID_LIM_ERR_EID

```
#define CFE_SB_MSGID_LIM_ERR_EID 17
```

SB Transmit API MsgId Pipe Limit Exceeded Event ID.

Type: ERROR

Cause:

An SB Transmit API call failed to deliver the MsgId to a pipe due to the limit for the number of messages with that MsgId for that pipe being exceeded.
Definition at line 222 of file [cfe_sb_eventids.h](#).

CFE_SB_PART_SUB_PKT_EID

```
#define CFE_SB_PART_SUB_PKT_EID 45
```

SB Send Previous Subscriptions Command Partial Packet Sent Event ID.

Type: DEBUG

Cause:

[SB Send Previous Subscriptions Command](#) processing sent a partial subscription packet.

Definition at line [496](#) of file [cfe_sb_eventids.h](#).

CFE_SB_PIPE_ADDED_EID

```
#define CFE_SB_PIPE_ADDED_EID 5
```

SB Create Pipe API Success Event ID.

Type: DEBUG

Cause:

[CFE_SB_CreatePipe](#) API successfully completed.

Definition at line [86](#) of file [cfe_sb_eventids.h](#).

CFE_SB_PIPE_DELETED_EID

```
#define CFE_SB_PIPE_DELETED_EID 47
```

SB Pipe Delete API Success Event ID.

Type: DEBUG

Cause:

An SB Delete Pipe API successfully completed.

Definition at line [518](#) of file [cfe_sb_eventids.h](#).

CFE_SB_Q_FULL_ERR_EID

```
#define CFE_SB_Q_FULL_ERR_EID 25
```

SB Transmit API Pipe Overflow Event ID.

Type: ERROR

Cause:

An SB Transmit API call failed to deliver the Message ID to a pipe due to the pipe queue being full.

Definition at line [302](#) of file [cfe_sb_eventids.h](#).

CFE_SB_Q_RD_ERR_EID

```
#define CFE_SB_Q_RD_ERR_EID 27
```

SB Transmit API Queue Read Failure Event ID.

Type: ERROR

Cause:

An SB Transmit API called failed due to a pipe queue read failure.

Definition at line 324 of file [cfe_sb_eventids.h](#).

CFE_SB_Q_WR_ERR_EID

```
#define CFE_SB_Q_WR_ERR_EID 26
```

SB Transmit API Queue Write Failure Event ID.

Type: ERROR

Cause:

An SB Transmit API call failed due to a pipe queue write failure.

Definition at line 313 of file [cfe_sb_eventids.h](#).

CFE_SB_RCV_BAD_ARG_EID

```
#define CFE_SB_RCV_BAD_ARG_EID 18
```

SB Receive Buffer API Bad Argument Event ID.

Type: ERROR

Cause:

[CFE_SB_ReceiveBuffer](#) API failure due to a bad input argument.

Definition at line 233 of file [cfe_sb_eventids.h](#).

CFE_SB_RCV_MESSAGE_INTEGRITY_FAIL_EID

```
#define CFE_SB_RCV_MESSAGE_INTEGRITY_FAIL_EID 72
```

SB validation of received message failure event.

Type: ERROR

Cause:

A CFE SB receive transaction has rejected a message due to failure of the associated message integrity action(s).

Definition at line 785 of file [cfe_sb_eventids.h](#).

CFE_SB_SEND_BAD_ARG_EID

```
#define CFE_SB_SEND_BAD_ARG_EID 13
```

SB Transmit API Bad Argument Event ID.

Type: ERROR

Cause:

An SB Transmit API failed due to an invalid input argument.
Definition at line 177 of file [cfe_sb_eventids.h](#).

CFE_SB_SEND_INV_MSGID_EID

```
#define CFE_SB_SEND_INV_MSGID_EID 21
```

SB Transmit API Invalid MsgId Event ID.

Type: ERROR

Cause:

An SB Transmit API was called with an invalid message ID.
Definition at line 266 of file [cfe_sb_eventids.h](#).

CFE_SB_SEND_MESSAGE_INTEGRITY_FAIL_EID

```
#define CFE_SB_SEND_MESSAGE_INTEGRITY_FAIL_EID 71
```

SB integrity actions on transmit message failure event.

Type: ERROR

Cause:

A CFE SB transmit transaction has rejected a message due to failure of the associated message integrity action(s).
Definition at line 773 of file [cfe_sb_eventids.h](#).

CFE_SB_SEND_NO_SUBS_EID

```
#define CFE_SB_SEND_NO_SUBS_EID 14
```

SB Transmit API No MsgId Subscribers Event ID.

Type: INFORMATION

Cause:

An SB Transmit API was called with a Message ID with no subscriptions.
Definition at line 188 of file [cfe_sb_eventids.h](#).

CFE_SB_SETPIPEOPTS_EID

```
#define CFE_SB_SETPIPEOPTS_EID 57
```

SB Set Pipe Opts API Success Event ID.

Type: DEBUG

Cause:

[CFE_SB_SetPipeOpts](#) success.
Definition at line 628 of file [cfe_sb_eventids.h](#).

CFE_SB_SETPIPEOPTS_ID_ERR_EID

```
#define CFE_SB_SETPIPEOPTS_ID_ERR_EID 55
```

SB Set Pipe Opts API Invalid Pipe Event ID.

Type: ERROR

Cause:

[CFE_SB_SetPipeOpts](#) API failure due to an invalid pipe ID
Definition at line 606 of file [cfe_sb_eventids.h](#).

CFE_SB_SETPIPEOPTS_OWNER_ERR_EID

```
#define CFE_SB_SETPIPEOPTS_OWNER_ERR_EID 56
```

SB Set Pipe Opts API Not Owner Event ID.

Type: ERROR

Cause:

[CFE_SB_SetPipeOpts](#) API failure due to not being the pipe owner.
Definition at line 617 of file [cfe_sb_eventids.h](#).

CFE_SB_SND_RTG_EID

```
#define CFE_SB_SND_RTG_EID 39
```

SB File Write Success Event ID.

Type: DEBUG

Cause:

An SB file write successfully completed. OVERLOADED
Definition at line 438 of file [cfe_sb_eventids.h](#).

CFE_SB_SND_RTG_ERR1_EID

#define CFE_SB_SND_RTG_ERR1_EID 40
SB File Write Create File Failure Event ID.

Type: ERROR

Cause:

An SB file write failure due to file creation error. OVERLOADED
Definition at line 449 of file [cfe_sb_eventids.h](#).

CFE_SB_SND_STATS_EID

#define CFE_SB_SND_STATS_EID 32
SB Send Statistics Command Success Event ID.

Type: DEBUG

Cause:

[SB Send Statistics Command](#) success.
Definition at line 357 of file [cfe_sb_eventids.h](#).

CFE_SB_SUB_ARG_ERR_EID

#define CFE_SB_SUB_ARG_ERR_EID 6
SB Subscribe API Bad Argument Event ID.

Type: ERROR

Cause:

An SB Subscribe API failed due to an invalid input argument.
Definition at line 97 of file [cfe_sb_eventids.h](#).

CFE_SB_SUB_INV_CALLER_EID

#define CFE_SB_SUB_INV_CALLER_EID 51
SB Subscribe API Not Owner Event ID.

Type: ERROR

Cause:

An SB Subscribe API failed due to not being the pipe owner.
Definition at line 562 of file [cfe_sb_eventids.h](#).

CFE_SB_SUB_INV_PIPE_EID

```
#define CFE_SB_SUB_INV_PIPE_EID 50
```

SB Subscribe API Invalid Pipe Event ID.

Type: ERROR

Cause:

An SB Subscribe API failed due to an invalid pipe ID.
Definition at line 551 of file [cfe_sb_eventids.h](#).

CFE_SB_SUBSCRIPTION_RCVD_EID

```
#define CFE_SB_SUBSCRIPTION_RCVD_EID 10
```

SB Subscribe API Success Event ID.

Type: DEBUG

Cause:

An SB Subscribe API completed successfully.
Definition at line 144 of file [cfe_sb_eventids.h](#).

CFE_SB_SUBSCRIPTION_REMOVED_EID

```
#define CFE_SB_SUBSCRIPTION_REMOVED_EID 48
```

SB Unsubscribe API Success Event ID.

Type: DEBUG

Cause:

An SB Unsubscribe API successfully completed.
Definition at line 529 of file [cfe_sb_eventids.h](#).

CFE_SB_SUBSCRIPTION_RPT_EID

```
#define CFE_SB_SUBSCRIPTION_RPT_EID 22
```

SB Subscription Report Sent Event ID.

Type: DEBUG

Cause:

SB Subscription Report sent in response to a successful subscription.
Definition at line 277 of file [cfe_sb_eventids.h](#).

CFE_SB_UNSUB_ARG_ERR_EID

#define CFE_SB_UNSUB_ARG_ERR_EID 11
SB Unsubscribe API Bad Argument Event ID.

Type: ERROR

Cause:

An SB Unsubscribe API failed due to an invalid input argument.
Definition at line 155 of file [cfe_sb_eventids.h](#).

CFE_SB_UNSUB_INV_CALLER_EID

#define CFE_SB_UNSUB_INV_CALLER_EID 53
SB Unsubscribe API Not Owner Event ID.

Type: ERROR

Cause:

An SB Unsubscribe API failed due to not being the pipe owner.
Definition at line 584 of file [cfe_sb_eventids.h](#).

CFE_SB_UNSUB_INV_PIPE_EID

#define CFE_SB_UNSUB_INV_PIPE_EID 52
SB Unsubscribe API Invalid Pipe Event ID.

Type: ERROR

Cause:

An SB Unsubscribe API failed due to an invalid pipe ID.
Definition at line 573 of file [cfe_sb_eventids.h](#).

CFE_SB_UNSUB_NO_SUBS_EID

#define CFE_SB_UNSUB_NO_SUBS_EID 12
SB Unsubscribe API No MsgId Subscription Event ID.

Type: INFORMATION

Cause:

An SB Unsubscribe API was called with a Message ID that wasn't subscribed on the pipe
Definition at line 166 of file [cfe_sb_eventids.h](#).

11.208 cfe_sb_eventids.h

[Go to the documentation of this file.](#)

```
00001 /*****
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"
00003  *
00004  * Copyright (c) 2023 United States Government as represented by the
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00012  * Unless required by applicable law or agreed to in writing, software
00013  * distributed under the License is distributed on an "AS IS" BASIS,
00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00025 #ifndef CFE_SB_EVENTS_H
00026 #define CFE_SB_EVENTS_H
00027
00042 #define CFE_SB_INIT_EID 1
00043
00053 #define CFE_SB_CR_PIPE_BAD_ARG_EID 2
00054
00064 #define CFE_SB_MAX_PIPES_MET_EID 3
00065
00075 #define CFE_SB_CR_PIPE_ERR_EID 4
00076
00086 #define CFE_SB_PIPE_ADDED_EID 5
00087
00097 #define CFE_SB_SUB_ARG_ERR_EID 6
00098
00109 #define CFE_SB_DUP_SUBSCRIP_EID 7
00110
00121 #define CFE_SB_MAX_MSGS_MET_EID 8
00122
00133 #define CFE_SB_MAX_DESTS_MET_EID 9
00134
00144 #define CFE_SB_SUBSCRIPTION_RCVD_EID 10
00145
00155 #define CFE_SB_UNSUB_ARG_ERR_EID 11
00156
00166 #define CFE_SB_UNSUB_NO_SUBS_EID 12
00167
00177 #define CFE_SB_SEND_BAD_ARG_EID 13
00178
00188 #define CFE_SB_SEND_NO_SUBS_EID 14
00189
00199 #define CFE_SB_MSG_TOO_BIG_EID 15
00200
00210 #define CFE_SB_GET_BUF_ERR_EID 16
00211
00222 #define CFE_SB_MSGID_LIM_ERR_EID 17
00223
00233 #define CFE_SB_RCV_BAD_ARG_EID 18
00234
00244 #define CFE_SB_BAD_PIPEID_EID 19
00245
00255 #define CFE_SB_DEST_BLK_ERR_EID 20
00256
00266 #define CFE_SB_SEND_INV_MSGID_EID 21
00267
00277 #define CFE_SB_SUBSCRIPTION_RPT_EID 22
00278
00290 #define CFE_SB_HASHCOLLISION_EID 23
00291
00302 #define CFE_SB_Q_FULL_ERR_EID 25
00303
00313 #define CFE_SB_Q_WR_ERR_EID 26
00314
00324 #define CFE_SB_Q_RD_ERR_EID 27
00325
00335 #define CFE_SB_CMD0_RCVD_EID 28
00336
00346 #define CFE_SB_CMD1_RCVD_EID 29
00347
```

```
00357 #define CFE_SB_SND_STATS_EID 32
00358
00369 #define CFE_SB_ENBL_RTE1_EID 33
00370
00380 #define CFE_SB_ENBL_RTE2_EID 34
00381
00392 #define CFE_SB_ENBL_RTE3_EID 35
00393
00404 #define CFE_SB_DSBL_RTE1_EID 36
00405
00415 #define CFE_SB_DSBL_RTE2_EID 37
00416
00427 #define CFE_SB_DSBL_RTE3_EID 38
00428
00438 #define CFE_SB_SND_RTG_EID 39
00439
00449 #define CFE_SB_SND_RTG_ERR1_EID 40
00450
00461 #define CFE_SB_BAD_CMD_CODE_EID 42
00462
00472 #define CFE_SB_BAD_MSGID_EID 43
00473
00484 #define CFE_SB_FULL_SUB_PKT_EID 44
00485
00496 #define CFE_SB_PART_SUB_PKT_EID 45
00497
00507 #define CFE_SB_DEL_PIPE_ERR1_EID 46
00508
00518 #define CFE_SB_PIPE_DELETED_EID 47
00519
00529 #define CFE_SB_SUBSCRIPTION_REMOVED_EID 48
00530
00540 #define CFE_SB_FILEWRITE_ERR_EID 49
00541
00551 #define CFE_SB_SUB_INV_PIPE_EID 50
00552
00562 #define CFE_SB_SUB_INV_CALLER_EID 51
00563
00573 #define CFE_SB_UNSUB_INV_PIPE_EID 52
00574
00584 #define CFE_SB_UNSUB_INV_CALLER_EID 53
00585
00595 #define CFE_SB_DEL_PIPE_ERR2_EID 54
00596
00606 #define CFE_SB_SETPIPEOPTS_ID_ERR_EID 55
00607
00617 #define CFE_SB_SETPIPEOPTS_OWNER_ERR_EID 56
00618
00628 #define CFE_SB_SETPIPEOPTS_EID 57
00629
00639 #define CFE_SB_GETPIPEOPTS_ID_ERR_EID 58
00640
00650 #define CFE_SB_GETPIPEOPTS_PTR_ERR_EID 59
00651
00661 #define CFE_SB_GETPIPEOPTS_EID 60
00662
00672 #define CFE_SB_GETPIPENAME_EID 62
00673
00683 #define CFE_SB_GETPIPENAME_NULL_PTR_EID 63
00684
00694 #define CFE_SB_GETPIPENAME_ID_ERR_EID 64
00695
00705 #define CFE_SB_GETPIPEIDBYNAME_EID 65
00706
00716 #define CFE_SB_GETPIPEIDBYNAME_NULL_ERR_EID 66
00717
00727 #define CFE_SB_GETPIPEIDBYNAME_NAME_ERR_EID 67
00728
00739 #define CFE_SB_LEN_ERR_EID 68
00740
00750 #define CFE_SB_CR_PIPE_NAME_TAKEN_EID 69
00751
00761 #define CFE_SB_CR_PIPE_NO_FREE_EID 70
00762
00773 #define CFE_SB_SEND_MESSAGE_INTEGRITY_FAIL_EID 71
00774
00785 #define CFE_SB_RCV_MESSAGE_INTEGRITY_FAIL_EID 72
00786
00789 #endif /* CFE_SB_EVENTS_H */
```

11.209 cfe/modules/sb/fsw/inc/cfe_sb_fcncodes.h File Reference

```
#include "cfe_sb_fcncode_values.h"
```

Macros

- `#define CFE_SB_NOOP_CC CFE_SB_CCVAL(NOOP)`
- `#define CFE_SB_RESET_COUNTERS_CC CFE_SB_CCVAL(RESET_COUNTERS)`
- `#define CFE_SB_SEND_SB_STATS_CC CFE_SB_CCVAL(SEND_SB_STATS)`
- `#define CFE_SB_WRITE_ROUTING_INFO_CC CFE_SB_CCVAL(WRITE_ROUTING_INFO)`
- `#define CFE_SB_ENABLE_ROUTE_CC CFE_SB_CCVAL(ENABLE_ROUTE)`
- `#define CFE_SB_DISABLE_ROUTE_CC CFE_SB_CCVAL(DISABLE_ROUTE)`
- `#define CFE_SB_WRITE_PIPE_INFO_CC CFE_SB_CCVAL(WRITE_PIPE_INFO)`
- `#define CFE_SB_WRITE_MAP_INFO_CC CFE_SB_CCVAL(WRITE_MAP_INFO)`
- `#define CFE_SB_ENABLE_SUB_REPORTING_CC CFE_SB_CCVAL(ENABLE_SUB_REPORTING)`
- `#define CFE_SB_DISABLE_SUB_REPORTING_CC CFE_SB_CCVAL(DISABLE_SUB_REPORTING)`
- `#define CFE_SB_SEND_PREV_SUBS_CC CFE_SB_CCVAL(SEND_PREV_SUBS)`

11.209.1 Detailed Description

Specification for the CFE Event Services (CFE_SB) command function codes

Note

This file should be strictly limited to the command/function code (CC) macro definitions. Other definitions such as enums, typedefs, or other macros should be placed in the msgdefs.h or msg.h files.

Definition in file [cfe_sb_fcncodes.h](#).

11.209.2 Macro Definition Documentation

CFE_SB_DISABLE_ROUTE_CC

```
#define CFE_SB_DISABLE_ROUTE_CC CFE_SB_CCVAL(DISABLE_ROUTE)
```

Name Disable Software Bus Route

Description

This command will disable a particular destination. The destination is specified in terms of MsgID and PipeID. The MsgID and PipeID are parameters in the command. All destinations are enabled by default.

Command Mnemonic(s) \$sc_\$cpu_SB_DisRoute

Command Structure

[CFE_SB_DisableRouteCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_SB_CMDPC` - command execution counter will increment
- View routing information [CFE_SB_WRITE_ROUTING_INFO_CC](#) to verify enable/disable state change
- The [CFE_SB_DSBL RTE2 EID](#) debug event message will be generated
- Destination will stop receiving messages

Error Conditions

This command may fail for the following reason(s):

- the MsgId or PipeId parameters do not pass validation
- the destination does not exist.

Evidence of failure may be found in the following telemetry:

- `$sc_$cpu_SB_CMDEC` - command error counter will increment
- A command specific error event message is issued for all error cases. See [CFE_SB_DSBL_RTE1_EID](#) or [CFE_SB_DSBL_RTE3_EID](#)

Criticality

This command is not intended to be used in nominal conditions. It is possible to get into a state where a destination cannot be re-enabled without resetting the processor. For instance, sending this command with [CFE_SB_CMD_MID](#) and the SB_Cmd_Pipe would inhibit any ground commanding to the software bus until the processor was reset. There are similar problems that may occur when using this command.

Definition at line 273 of file [cfe_sb_fcncodes.h](#).

CFE_SB_DISABLE_SUB_REPORTING_CC

```
#define CFE_SB_DISABLE_SUB_REPORTING_CC CFE_SB_CCVAL(DISABLE_SUB_REPORTING)
```

Name Disable Subscription Reporting Command

Description

This command will disable subscription reporting and is intended to be used only by the CFS SBN (Software Bus Networking) Application. It is not intended to be sent from the ground or used by operations. When subscription reporting is enabled, SB will generate and send a software bus packet for each subscription received. The software bus packet that is sent contains the information received in the subscription API. This subscription report is needed by SBN if offboard routing is required.

Command Mnemonic(s) `$sc_$cpu_SB_DisSubRptg`

Command Structure

[CFE_SB_DisableSubReportingCmd_t](#)

Command Verification

Successful execution of this command will result in the suppression of packets (with the [CFE_SB_ONESUB_TLM_MID](#) MsgId) for each subscription received by SB through the subscription APIs.

Error Conditions

None

Criticality

None

See also

[CFE_SB_SingleSubscriptionTlm_t](#), [CFE_SB_ENABLE_SUB_REPORTING_CC](#), [CFE_SB_SEND_PREV_SUBS_CC](#)

Definition at line 430 of file [cfe_sb_fcncodes.h](#).

CFE_SB_ENABLE_ROUTE_CC

```
#define CFE_SB_ENABLE_ROUTE_CC CFE_SB_CCVAL(ENABLE_ROUTE)
```

Name Enable Software Bus Route

Description

This command will enable a particular destination. The destination is specified in terms of MsgID and PipeID. The MsgID and PipeID are parameters in the command. All destinations are enabled by default. This command is needed only after a [CFE_SB_DISABLE_ROUTE_CC](#) command is used.

Command Mnemonic(s) \$sc_\$cpu_SB_EnaRoute

Command Structure

[CFE_SB_EnableRouteCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- [\\$sc_\\$cpu_SB_CMDPC](#) - command execution counter will increment
- View routing information [CFE_SB_WRITE_ROUTING_INFO_CC](#) to verify enable/disable state change
- The [CFE_SB_ENBL_RTE2_EID](#) debug event message will be generated
- Destination will begin receiving messages

Error Conditions

This command may fail for the following reason(s):

- the MsgID or PipeID parameters do not pass validation
- the destination does not exist.

Evidence of failure may be found in the following telemetry:

- [\\$sc_\\$cpu_SB_CMDEC](#) - command error counter will increment
- A command specific error event message is issued for all error cases. See [CFE_SB_ENBL_RTE1_EID](#) or [CFE_SB_ENBL_RTE3_EID](#)

Criticality

This command is not inherently dangerous.

Definition at line [232](#) of file [cfe_sb_fcncodes.h](#).

CFE_SB_ENABLE_SUB_REPORTING_CC

```
#define CFE_SB_ENABLE_SUB_REPORTING_CC CFE_SB_CCVAL(ENABLE_SUB_REPORTING)
```

Name Enable Subscription Reporting Command

Description

This command will enable subscription reporting and is intended to be used only by the CFS SBN (Software Bus Networking) Application. It is not intended to be sent from the ground or used by operations. When subscription reporting is enabled, SB will generate and send a software bus packet for each subscription received. The software bus packet that is sent contains the information received in the subscription API. This subscription report is needed by SBN if offboard routing is required.

Command Mnemonic(s) `$sc_$cpu_SB_EnaSubRptg`

Command Structure

`CFE_SB_EnableSubReportingCmd_t`

Command Verification

Successful execution of this command will result in the sending of a packet (with the `CFE_SB_ONESUB_TLM_MID` MsgId) for each subscription received by SB through the subscription APIs.

Error Conditions

None

Criticality

None

See also

`CFE_SB_SingleSubscriptionTlm_t`, `CFE_SB_DISABLE_SUB_REPORTING_CC`, `CFE_SB_SEND_PREV_SUBS_CC`

Definition at line 397 of file `cfe_sb_fcncodes.h`.

CFE_SB_NOOP_CC

```
#define CFE_SB_NOOP_CC CFE_SB_CCVAL(NOOP)
```

Name Software Bus No-Op

Description

This command performs no other function than to increment the command execution counter. The command may be used to verify general aliveness of the Software Bus task.

Command Mnemonic(s) `$sc_$cpu_SB_NOOP`

Command Structure

`CFE_SB_NoopCmd_t`

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_SB_CMDPC` - command execution counter will increment
- The `CFE_SB_CMD0_RCVD_EID` informational event message will be generated

Error Conditions

There are no error conditions for this command. If the Software Bus receives the command, the event is sent (although it may be filtered by EVS) and the counter is incremented unconditionally.

Criticality

None

See also

Definition at line 68 of file [cfe_sb_fcncodes.h](#).

CFE_SB_RESET_COUNTERS_CC

```
#define CFE_SB_RESET_COUNTERS_CC CFE_SB_CCVAL(RESET_COUNTERS)
```

Name Software Bus Reset Counters

Description

This command resets the following counters within the Software Bus housekeeping telemetry:

- Command Execution Counter (\$sc_\$cpu_SB_CMDPC)
- Command Error Counter (\$sc_\$cpu_SB_CMDEC)
- No Subscribers Counter (\$sc_\$cpu_SB_NoSubEC)
- Duplicate Subscriptions Counter (\$sc_\$cpu_SB_DupSubCnt)
- Msg Send Error Counter (\$sc_\$cpu_SB_MsgSndEC)
- Msg Receive Error Counter (\$sc_\$cpu_SB_MsgRecEC)
- Internal Error Counter (\$sc_\$cpu_SB_InternalEC)
- Create Pipe Error Counter (\$sc_\$cpu_SB_NewPipeEC)
- Subscribe Error Counter (\$sc_\$cpu_SB_SubscrEC)
- Pipe Overflow Error Counter (\$sc_\$cpu_SB_PipeOvrEC)
- Msg Limit Error Counter (\$sc_\$cpu_SB_MsgLimEC)

Command Mnemonic(s) \$sc_\$cpu_SB_ResetCtrs

Command Structure

[CFE_SB_ResetCountersCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **\$sc_\$cpu_SB_CMDPC** - command execution counter will be reset to 0
- All other counters listed in description will be reset to 0
- The [CFE_SB_CMD1_RCVD_EID](#) informational event message will be generated

Error Conditions

There are no error conditions for this command. If the Software Bus receives the command, the event is sent (although it may be filtered by EVS) and the counter is incremented unconditionally.

Criticality

This command is not inherently dangerous. However, it is possible for ground systems and on-board safing procedures to be designed such that they react to changes in the counter values that are reset by this command.

See also

Definition at line 115 of file [cfe_sb_fcncodes.h](#).

CFE_SB_SEND_PREV_SUBS_CC

```
#define CFE_SB_SEND_PREV_SUBS_CC CFE_SB_CCVAL(SEND_PREV_SUBS)
```

Name Send Previous Subscriptions Command

This command generates a series of packets that contain information

regarding all subscriptions previously received by SB. This command is intended to be used only by the CFS SBN(Software Bus Networking) Application. It is not intended to be sent from the ground or used by operations. When this command is received the software bus will generate and send a series of packets containing information about all subscription previously received.

Command Mnemonic(s) \$sc_\$cpu_SB_SendPrevSubs

Command Structure

[CFE_SB_SendPrevSubsCmd_t](#)

Command Verification

Successful execution of this command will result in a series of packets (with the [CFE_SB_ALLSUBS_TLM_MID](#) MsgId) being sent on the software bus.

Error Conditions

None

Criticality

None

See also

[CFE_SB_AllSubscriptionsTlm_t](#), [CFE_SB_ENABLE_SUB_REPORTING_CC](#), [CFE_SB_DISABLE_SUB_REPORTING_CC](#)

Definition at line 462 of file [cfe_sb_fcncodes.h](#).

CFE_SB_SEND_SB_STATS_CC

```
#define CFE_SB_SEND_SB_STATS_CC CFE_SB_CCVAL(SEND_SB_STATS)
```

Name Send Software Bus Statistics

Description

This command will cause the SB task to send a statistics packet containing current utilization figures and high water marks which may be useful for checking the margin of the SB platform configuration settings.

Command Mnemonic(s) \$sc_\$cpu_SB_DumpStats

Command Structure

[CFE_SB_SendSbStatsCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- [\\$sc_\\$cpu_SB_CMDPC](#) - command execution counter will increment
- Receipt of statistics packet with MsgId [CFE_SB_STATS_TLM_MID](#)
- The [CFE_SB_SND_STATS_EID](#) debug event message will be generated

Error Conditions

There are no error conditions for this command. If the Software Bus receives the command, the debug event is sent and the counter is incremented unconditionally.

Criticality

This command is not inherently dangerous. It will create and send a message on the software bus. If performed repeatedly, it is possible that receiver pipes may overflow.

See also

Definition at line 149 of file [cfe_sb_fcncodes.h](#).

CFE_SB_WRITE_MAP_INFO_CC

```
#define CFE_SB_WRITE_MAP_INFO_CC CFE_SB_CCVAL(WRITE_MAP_INFO)
```

Name Write Map Info to a File

This command will create a file containing the software bus message

map information. The message map is a lookup table (an array of uint16s) that allows fast access to the correct routing table element during a software bus send operation. This is diagnostic information that may be needed due to the dynamic nature of the cFE software bus. An absolute path and filename may be specified in the command. If this command field contains an empty string (NULL terminator as the first character) the default file path and name is used. The default file path and name is defined in the platform configuration file as [CFE_PLATFORM_SB_DEFAULT_MAP_FILENAME](#).

Command Mnemonic(s) \$sc_\$cpu_SB_WriteMap2File

Command Structure

[CFE_SB_WriteMapInfoCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **\$sc_\$cpu_SB_CMDPC** - command execution counter will increment. NOTE: the command counter is incremented when the request is accepted, before writing the file, which is performed as a background task.
- The file specified in the command (or the default specified by the [CFE_PLATFORM_SB_DEFAULT_MAP_FILENAME](#) configuration parameter) will be updated with the latest information.
- The [CFE_SB_SND_RTG_EID](#) debug event message will be generated

Error Conditions

This command may fail for the following reason(s):

- A previous request to write a software bus information file has not yet completed
- The specified FileName cannot be parsed

Evidence of failure may be found in the following telemetry:

- **\$sc_\$cpu_SB_CMDEC** - command error counter will increment
- A command specific error event message is issued for all error cases. See [CFE_SB_SND_RTG_ERR1_EID](#) and [CFE_SB_FILEWRITE_ERR_EID](#)

Criticality

This command is not inherently dangerous. It will create a new file in the file system and could, if performed repeatedly without sufficient file management by the operator, fill the file system.

Definition at line 364 of file [cfe_sb_fcncodes.h](#).

CFE_SB_WRITE_PIPE_INFO_CC

```
#define CFE_SB_WRITE_PIPE_INFO_CC CFE\_SB\_CCVAL(WRITE_PIPE_INFO)
```

Name Write Pipe Info to a File

Description

This command will create a file containing the software bus pipe information. The pipe information contains information about every pipe that has been created through the [CFE_SB_CreatePipe](#) API. An absolute path and filename may be specified in the command. If this command field contains an empty string (NULL terminator as the first character) the default file path and name is used. The default file path and name is defined in the platform configuration file as [CFE_PLATFORM_SB_DEFAULT_PIPE_FILENAME](#).

Command Mnemonic(s) \$sc_\$cpu_SB_WritePipe2File

Command Structure

[CFE_SB_WritePipeInfoCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **\$sc_\$cpu_SB_CMDPC** - command execution counter will increment. NOTE: the command counter is incremented when the request is accepted, before writing the file, which is performed as a background task.
- The file specified in the command (or the default specified by the [CFE_PLATFORM_SB_DEFAULT_PIPE_FILENAME](#) configuration parameter) will be updated with the latest information.
- The [CFE_SB_SND_RTG_EID](#) debug event message will be generated

Error Conditions

This command may fail for the following reason(s):

- A previous request to write a software bus information file has not yet completed
- The specified FileName cannot be parsed

Evidence of failure may be found in the following telemetry:

- **\$sc_\$cpu_SB_CMDEC** - command error counter will increment
- A command specific error event message is issued for all error cases. See [CFE_SB_SND_RTG_ERR1_EID](#) and [CFE_SB_FILEWRITE_ERR_EID](#)

Criticality

This command is not inherently dangerous. It will create a new file in the file system and could, if performed repeatedly without sufficient file management by the operator, fill the file system.

Definition at line 318 of file [cfe_sb_fcncodes.h](#).

CFE_SB_WRITE_ROUTING_INFO_CC

```
#define CFE_SB_WRITE_ROUTING_INFO_CC CFE_SB_CCVAL(WRITE_ROUTING_INFO)
```

Name Write Software Bus Routing Info to a File

Description

This command will create a file containing the software bus routing information. The routing information contains information about every subscription that has been received through the SB subscription APIs. An absolute path and filename may be specified in the command. If this command field contains an empty string (NULL terminator as the first character) the default file path and name is used. The default file path and name is defined in the platform configuration file as [CFE_PLATFORM_SB_DEFAULT_ROUTING_FILENAME](#).

Command Mnemonic(s) \$sc_\$cpu_SB_WriteRouting2File

Command Structure

[CFE_SB_WriteRoutingInfoCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **\$sc_\$cpu_SB_CMDPC** - command execution counter will increment. NOTE: the command counter is incremented when the request is accepted, before writing the file, which is performed as a background task.
- The file specified in the command (or the default specified by the [CFE_PLATFORM_SB_DEFAULT_ROUTING_FILENAME](#) configuration parameter) will be updated with the latest information.
- The [CFE_SB_SND_RTG_EID](#) debug event message will be generated

Error Conditions

This command may fail for the following reason(s):

- A previous request to write a software bus information file has not yet completed
- The specified FileName cannot be parsed

Evidence of failure may be found in the following telemetry:

- `$sc_$cpu_SB_CMDEC` - command error counter will increment
- A command specific error event message is issued for all error cases. See [CFE_SB_SND_RTG_ERR1_EID](#) and [CFE_SB_FILEWRITE_ERR_EID](#)

Criticality

This command is not inherently dangerous. It will create a new file in the file system and could, if performed repeatedly without sufficient file management by the operator, fill the file system.

Definition at line 194 of file [cfe_sb_fcncodes.h](#).

11.210 cfe_sb_fcncodes.h

[Go to the documentation of this file.](#)

```

00001 /*****
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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00028 #ifndef CFE_SB_FCNCODES_H
00029 #define CFE_SB_FCNCODES_H
00030
00031 #include "cfe_sb_fcncode_values.h"
00032
00033 /*****
00034  ** SB task command packet command codes
00035  *****/
00036
00068 #define CFE_SB_NOOP_CC CFE_SB_CCVAL(NOOP)
00069
00115 #define CFE_SB_RESET_COUNTERS_CC CFE_SB_CCVAL(RESET_COUNTERS)
00116
00149 #define CFE_SB_SEND_SB_STATS_CC CFE_SB_CCVAL(SEND_SB_STATS)
00150
00194 #define CFE_SB_WRITE_ROUTING_INFO_CC CFE_SB_CCVAL(WRITE_ROUTING_INFO)
00195
00232 #define CFE_SB_ENABLE_ROUTE_CC CFE_SB_CCVAL(ENABLE_ROUTE)
00233
00273 #define CFE_SB_DISABLE_ROUTE_CC CFE_SB_CCVAL(DISABLE_ROUTE)
00274
00318 #define CFE_SB_WRITE_PIPE_INFO_CC CFE_SB_CCVAL(WRITE_PIPE_INFO)
00319
00364 #define CFE_SB_WRITE_MAP_INFO_CC CFE_SB_CCVAL(WRITE_MAP_INFO)
00365
00397 #define CFE_SB_ENABLE_SUB_REPORTING_CC CFE_SB_CCVAL(ENABLE_SUB_REPORTING)
00398
00430 #define CFE_SB_DISABLE_SUB_REPORTING_CC CFE_SB_CCVAL(DISABLE_SUB_REPORTING)
00431
00462 #define CFE_SB_SEND_PREV_SUBS_CC CFE_SB_CCVAL(SEND_PREV_SUBS)
00463
00464 #endif

```

11.211 cfe/modules/sb/fsw/inc/cfe_sb_interface_cfg.h File Reference

```
#include "cfe_sb_interface_cfg_values.h"
```

Macros

- #define [CFE_MISSION_SB_MAX_SB_MSG_SIZE](#) [CFE_MISSION_SB_CFGVAL](#)(MAX_SB_MSG_SIZE)
- #define [DEFAULT_CFE_MISSION_SB_MAX_SB_MSG_SIZE](#) 32768
- #define [CFE_MISSION_SB_MAX_PIPES](#) [CFE_MISSION_SB_CFGVAL](#)(MAX_PIPES)
- #define [DEFAULT_CFE_MISSION_SB_MAX_PIPES](#) 32
- #define [CFE_MISSION_SB_SUB_ENTRIES_PER_PKT](#) [CFE_MISSION_SB_CFGVAL](#)(SUB_ENTRIES_PER_←
PKT)
- #define [DEFAULT_CFE_MISSION_SB_SUB_ENTRIES_PER_PKT](#) 20

11.211.1 Detailed Description

CFE Software Bus (CFE_SB) Application Public Definitions

This provides default values for configurable items that affect the interface(s) of this module. This includes the CMD/TLM message interface, tables definitions, and any other data products that serve to exchange information with other entities.

Note

This file may be overridden/superseded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [cfe_sb_interface_cfg.h](#).

11.211.2 Macro Definition Documentation

CFE_MISSION_SB_MAX_PIPES

```
#define CFE_MISSION_SB_MAX_PIPES CFE\_MISSION\_SB\_CFGVAL(MAX_PIPES)
```

Purpose Maximum Number of pipes that SB command/telemetry messages may hold

Description:

Dictates the maximum number of unique Pipes the SB message definitions will hold.

This affects the layout of command/telemetry messages but does not affect run time behavior or internal allocation.

Limits

All CPUs within the same SB domain (mission) must share the same definition Note this affects the size of messages, so it must not cause any message to exceed the max length.

Definition at line 70 of file [cfe_sb_interface_cfg.h](#).

CFE_MISSION_SB_MAX_SB_MSG_SIZE

```
#define CFE_MISSION_SB_MAX_SB_MSG_SIZE CFE\_MISSION\_SB\_CFGVAL(MAX_SB_MSG_SIZE)
```

Purpose Maximum SB Message Size

Description:

The following definition dictates the maximum message size allowed on the software bus. SB checks the pkt length field in the header of all messages sent. If the pkt length field indicates the message is larger than this define, SB sends an event and rejects the send.

Limits

This parameter has a lower limit of 6 (CCSDS primary header size). There are no restrictions on the upper limit however, the maximum message size is system dependent and should be verified. Total message size values that are checked against this configuration are defined by a 16 bit data word.

Definition at line 52 of file [cfe_sb_interface_cfg.h](#).

CFE_MISSION_SB_SUB_ENTRIES_PER_PKT

```
#define CFE_MISSION_SB_SUB_ENTRIES_PER_PKT CFE_MISSION_SB_CFGVAL(SUB_ENTRIES_PER_PKT)
```

Purpose Maximum Number of subscription entries per subscription report packet

Description:

The subscription report will group up to this number of entries into each TLM packet

Limits

Must not cause the size of the subscription report telemetry packet to exceed mission limits

Definition at line 82 of file [cfe_sb_interface_cfg.h](#).

DEFAULT_CFE_MISSION_SB_MAX_PIPES

```
#define DEFAULT_CFE_MISSION_SB_MAX_PIPES 32
```

Definition at line 71 of file [cfe_sb_interface_cfg.h](#).

DEFAULT_CFE_MISSION_SB_MAX_SB_MSG_SIZE

```
#define DEFAULT_CFE_MISSION_SB_MAX_SB_MSG_SIZE 32768
```

Definition at line 53 of file [cfe_sb_interface_cfg.h](#).

DEFAULT_CFE_MISSION_SB_SUB_ENTRIES_PER_PKT

```
#define DEFAULT_CFE_MISSION_SB_SUB_ENTRIES_PER_PKT 20
```

Definition at line 83 of file [cfe_sb_interface_cfg.h](#).

11.212 cfe_sb_interface_cfg.h

[Go to the documentation of this file.](#)

```
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```

```

00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00032 #ifndef CFE_SB_INTERFACE_CFG_H
00033 #define CFE_SB_INTERFACE_CFG_H
00034
00035 #include "cfe_sb_interface_cfg_values.h"
00036
00052 #define CFE_MISSION_SB_MAX_SB_MSG_SIZE CFE_MISSION_SB_CFGVAL(MAX_SB_MSG_SIZE)
00053 #define DEFAULT_CFE_MISSION_SB_MAX_SB_MSG_SIZE 32768
00054
00070 #define CFE_MISSION_SB_MAX_PIPES CFE_MISSION_SB_CFGVAL(MAX_PIPES)
00071 #define DEFAULT_CFE_MISSION_SB_MAX_PIPES 32
00072
00082 #define CFE_MISSION_SB_SUB_ENTRIES_PER_PKT CFE_MISSION_SB_CFGVAL(SUB_ENTRIES_PER_PKT)
00083 #define DEFAULT_CFE_MISSION_SB_SUB_ENTRIES_PER_PKT 20
00084
00085 #endif

```

11.213 cfe/modules/sb/fsw/inc/cfe_sb_internal_cfg.h File Reference

```
#include "cfe_sb_internal_cfg_values.h"
```

Macros

- #define CFE_PLATFORM_SB_MAX_MSG_IDS CFE_PLATFORM_SB_CFGVAL(MAX_MSG_IDS)
- #define DEFAULT_CFE_PLATFORM_SB_MAX_MSG_IDS 256
- #define CFE_PLATFORM_SB_MAX_PIPES CFE_PLATFORM_SB_CFGVAL(MAX_PIPES)
- #define DEFAULT_CFE_PLATFORM_SB_MAX_PIPES 64
- #define CFE_PLATFORM_SB_MAX_DEST_PER_PKT CFE_PLATFORM_SB_CFGVAL(MAX_DEST_PER_P←
PKT)
- #define DEFAULT_CFE_PLATFORM_SB_MAX_DEST_PER_PKT 16
- #define CFE_PLATFORM_SB_DEFAULT_MSG_LIMIT CFE_PLATFORM_SB_CFGVAL(DEFAULT_MSG_LIMIT)
- #define DEFAULT_CFE_PLATFORM_SB_DEFAULT_MSG_LIMIT 4
- #define CFE_PLATFORM_SB_BUF_MEMORY_BYTES CFE_PLATFORM_SB_CFGVAL(BUF_MEMORY_←
BYTES)
- #define DEFAULT_CFE_PLATFORM_SB_BUF_MEMORY_BYTES 524288
- #define CFE_PLATFORM_SB_HIGHEST_VALID_MSGID CFE_PLATFORM_SB_CFGVAL(HIGHEST_VALID_←
MSGID)
- #define DEFAULT_CFE_PLATFORM_SB_HIGHEST_VALID_MSGID 0x1FFF
- #define CFE_PLATFORM_SB_DEFAULT_ROUTING_FILENAME CFE_PLATFORM_SB_CFGVAL(DEFAULT_←
ROUTING_FILENAME)
- #define DEFAULT_CFE_PLATFORM_SB_DEFAULT_ROUTING_FILENAME "/ram/cfe_sb_route.dat"
- #define CFE_PLATFORM_SB_DEFAULT_PIPE_FILENAME CFE_PLATFORM_SB_CFGVAL(DEFAULT_PIPE_←
_FILENAME)
- #define DEFAULT_CFE_PLATFORM_SB_DEFAULT_PIPE_FILENAME "/ram/cfe_sb_pipe.dat"
- #define CFE_PLATFORM_SB_DEFAULT_MAP_FILENAME CFE_PLATFORM_SB_CFGVAL(DEFAULT_MAP_←
_FILENAME)
- #define DEFAULT_CFE_PLATFORM_SB_DEFAULT_MAP_FILENAME "/ram/cfe_sb_msgmap.dat"
- #define CFE_PLATFORM_SB_FILTERED_EVENT1 CFE_PLATFORM_SB_CFGVAL(FILTERED_EVENT1)
- #define DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT1 CFE_SB_SEND_NO_SUBS_EID
- #define CFE_PLATFORM_SB_FILTER_MASK1 CFE_PLATFORM_SB_CFGVAL(FILTER_MASK1)
- #define DEFAULT_CFE_PLATFORM_SB_FILTER_MASK1 CFE_EVS_FIRST_4_STOP
- #define CFE_PLATFORM_SB_FILTERED_EVENT2 CFE_PLATFORM_SB_CFGVAL(FILTERED_EVENT2)
- #define DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT2 CFE_SB_DUP_SUBSCRIP_EID
- #define CFE_PLATFORM_SB_FILTER_MASK2 CFE_PLATFORM_SB_CFGVAL(FILTER_MASK2)

```

• #define DEFAULT_CFE_PLATFORM_SB_FILTER_MASK2 CFE_EVS_FIRST_4_STOP
• #define CFE_PLATFORM_SB_FILTERED_EVENT3 CFE_PLATFORM_SB_CFGVAL(FILTERED_EVENT3)
• #define DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT3 CFE_SB_MSGID_LIM_ERR_EID
• #define CFE_PLATFORM_SB_FILTER_MASK3 CFE_PLATFORM_SB_CFGVAL(FILTER_MASK3)
• #define DEFAULT_CFE_PLATFORM_SB_FILTER_MASK3 CFE_EVS_FIRST_16_STOP
• #define CFE_PLATFORM_SB_FILTERED_EVENT4 CFE_PLATFORM_SB_CFGVAL(FILTERED_EVENT4)
• #define DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT4 CFE_SB_Q_FULL_ERR_EID
• #define CFE_PLATFORM_SB_FILTER_MASK4 CFE_PLATFORM_SB_CFGVAL(FILTER_MASK4)
• #define DEFAULT_CFE_PLATFORM_SB_FILTER_MASK4 CFE_EVS_FIRST_16_STOP
• #define CFE_PLATFORM_SB_FILTERED_EVENT5 CFE_PLATFORM_SB_CFGVAL(FILTERED_EVENT5)
• #define DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT5 0
• #define CFE_PLATFORM_SB_FILTER_MASK5 CFE_PLATFORM_SB_CFGVAL(FILTER_MASK5)
• #define DEFAULT_CFE_PLATFORM_SB_FILTER_MASK5 CFE_EVS_NO_FILTER
• #define CFE_PLATFORM_SB_FILTERED_EVENT6 CFE_PLATFORM_SB_CFGVAL(FILTERED_EVENT6)
• #define DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT6 0
• #define CFE_PLATFORM_SB_FILTER_MASK6 CFE_PLATFORM_SB_CFGVAL(FILTER_MASK6)
• #define DEFAULT_CFE_PLATFORM_SB_FILTER_MASK6 CFE_EVS_NO_FILTER
• #define CFE_PLATFORM_SB_FILTERED_EVENT7 CFE_PLATFORM_SB_CFGVAL(FILTERED_EVENT7)
• #define DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT7 0
• #define CFE_PLATFORM_SB_FILTER_MASK7 CFE_PLATFORM_SB_CFGVAL(FILTER_MASK7)
• #define DEFAULT_CFE_PLATFORM_SB_FILTER_MASK7 CFE_EVS_NO_FILTER
• #define CFE_PLATFORM_SB_FILTERED_EVENT8 CFE_PLATFORM_SB_CFGVAL(FILTERED_EVENT8)
• #define DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT8 0
• #define CFE_PLATFORM_SB_FILTER_MASK8 CFE_PLATFORM_SB_CFGVAL(FILTER_MASK8)
• #define DEFAULT_CFE_PLATFORM_SB_FILTER_MASK8 CFE_EVS_NO_FILTER
• #define CFE_PLATFORM_SB_POOL_MAX_BUCKETS CFE_PLATFORM_SB_CFGVAL(POOL_MAX_↵
  BUCKETS)
• #define DEFAULT_CFE_PLATFORM_SB_POOL_MAX_BUCKETS 17
• #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_01 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_↵
  _01)
• #define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_01 8
• #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_02 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_↵
  _02)
• #define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_02 16
• #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_03 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_↵
  _03)
• #define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_03 20
• #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_04 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_↵
  _04)
• #define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_04 36
• #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_05 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_↵
  _05)
• #define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_05 64
• #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_06 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_↵
  _06)
• #define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_06 96
• #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_07 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_↵
  _07)
• #define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_07 128
• #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_08 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_↵
  _08)

```

- #define [DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_08](#) 160
- #define [CFE_PLATFORM_SB_MEM_BLOCK_SIZE_09](#) [CFE_PLATFORM_SB_CFGVAL](#)(MEM_BLOCK_SIZE↔_09)
- #define [DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_09](#) 256
- #define [CFE_PLATFORM_SB_MEM_BLOCK_SIZE_10](#) [CFE_PLATFORM_SB_CFGVAL](#)(MEM_BLOCK_SIZE↔_10)
- #define [DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_10](#) 512
- #define [CFE_PLATFORM_SB_MEM_BLOCK_SIZE_11](#) [CFE_PLATFORM_SB_CFGVAL](#)(MEM_BLOCK_SIZE↔_11)
- #define [DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_11](#) 1024
- #define [CFE_PLATFORM_SB_MEM_BLOCK_SIZE_12](#) [CFE_PLATFORM_SB_CFGVAL](#)(MEM_BLOCK_SIZE↔_12)
- #define [DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_12](#) 2048
- #define [CFE_PLATFORM_SB_MEM_BLOCK_SIZE_13](#) [CFE_PLATFORM_SB_CFGVAL](#)(MEM_BLOCK_SIZE↔_13)
- #define [DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_13](#) 4096
- #define [CFE_PLATFORM_SB_MEM_BLOCK_SIZE_14](#) [CFE_PLATFORM_SB_CFGVAL](#)(MEM_BLOCK_SIZE↔_14)
- #define [DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_14](#) 8192
- #define [CFE_PLATFORM_SB_MEM_BLOCK_SIZE_15](#) [CFE_PLATFORM_SB_CFGVAL](#)(MEM_BLOCK_SIZE↔_15)
- #define [DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_15](#) 16384
- #define [CFE_PLATFORM_SB_MEM_BLOCK_SIZE_16](#) [CFE_PLATFORM_SB_CFGVAL](#)(MEM_BLOCK_SIZE↔_16)
- #define [DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_16](#) 32768
- #define [CFE_PLATFORM_SB_MAX_BLOCK_SIZE](#) [CFE_PLATFORM_SB_CFGVAL](#)(MAX_BLOCK_SIZE)
- #define [DEFAULT_CFE_PLATFORM_SB_MAX_BLOCK_SIZE](#) ([CFE_MISSION_SB_MAX_SB_MSG_SIZE](#) + 128)
- #define [CFE_PLATFORM_SB_START_TASK_PRIORITY](#) [CFE_PLATFORM_SB_CFGVAL](#)(START_TASK↔_PRIORITY)
- #define [DEFAULT_CFE_PLATFORM_SB_START_TASK_PRIORITY](#) 64
- #define [CFE_PLATFORM_SB_START_TASK_STACK_SIZE](#) [CFE_PLATFORM_SB_CFGVAL](#)(START_TASK↔_STACK_SIZE)
- #define [DEFAULT_CFE_PLATFORM_SB_START_TASK_STACK_SIZE](#) [CFE_PLATFORM_ES_DEFAULT_STACK_SIZE](#)

11.213.1 Detailed Description

CFE Software Bus (CFE_SB) Application Private Config Definitions

This provides default values for configurable items that are internal to this module and do NOT affect the interface(s) of this module. Changes to items in this file only affect the local module and will be transparent to external entities that are using the public interface(s).

Note

This file may be overridden/superseded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [cfe_sb_internal_cfg.h](#).

11.213.2 Macro Definition Documentation

CFE_PLATFORM_SB_BUF_MEMORY_BYTES

```
#define CFE_PLATFORM_SB_BUF_MEMORY_BYTES CFE_PLATFORM_SB_CFGVAL (BUF_MEMORY_BYTES)
```

Purpose Size of the SB buffer memory pool

Description:

Dictates the size of the SB memory pool. For each message the SB sends, the SB dynamically allocates from this memory pool, the memory needed to process the message. The memory needed to process each message is msg size + msg descriptor(CFE_SB_BufferD_t). This memory pool is also used to allocate destination descriptors (CFE_SB_DestinationD_t) during the subscription process. To see the run-time, high-water mark and the current utilization figures regarding this parameter, send an SB command to 'Send Statistics Pkt'. Some memory statistics have been added to the SB housekeeping packet. NOTE: It is important to monitor these statistics to ensure the desired memory margin is met.

Limits

This parameter has a lower limit of 512 and an upper limit of UINT_MAX (4 Gigabytes).

Definition at line 129 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_DEFAULT_MAP_FILENAME

```
#define CFE_PLATFORM_SB_DEFAULT_MAP_FILENAME CFE_PLATFORM_SB_CFGVAL (DEFAULT_MAP_FILENAME)
```

Purpose Default Message Map Filename

Description:

The value of this constant defines the filename used to store the software bus message map information. This filename is used only when no filename is specified in the command. The message map is a lookup table (array of 16bit words) that has an element for each possible MsgId value and holds the routing table index for that MsgId. The Msg Map provides fast access to the destinations of a message.

Limits

The length of each string, including the NULL terminator cannot exceed the [OS_MAX_PATH_LEN](#) value.

Definition at line 203 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_DEFAULT_MSG_LIMIT

```
#define CFE_PLATFORM_SB_DEFAULT_MSG_LIMIT CFE_PLATFORM_SB_CFGVAL (DEFAULT_MSG_LIMIT)
```

Purpose Default Subscription Message Limit

Description:

Dictates the default Message Limit when using the [CFE_SB_Subscribe](#) API. This will limit the number of messages with a specific message ID that can be received through a subscription. This only changes the default; other message limits can be set on a per subscription basis using [CFE_SB_SubscribeEx](#) .

Limits

This parameter has a lower limit of 4 and an upper limit of 65535.

Definition at line 106 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_DEFAULT_PIPE_FILENAME

```
#define CFE_PLATFORM_SB_DEFAULT_PIPE_FILENAME CFE_PLATFORM_SB_CFGVAL(DEFAULT_PIPE_FILENAME)
```

Purpose Default Pipe Information Filename

Description:

The value of this constant defines the filename used to store the software bus pipe information. This filename is used only when no filename is specified in the command.

Limits

The length of each string, including the NULL terminator cannot exceed the [OS_MAX_PATH_LEN](#) value.

Definition at line 185 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_DEFAULT_ROUTING_FILENAME

```
#define CFE_PLATFORM_SB_DEFAULT_ROUTING_FILENAME CFE_PLATFORM_SB_CFGVAL(DEFAULT_ROUTING_FILENAME)
```

Purpose Default Routing Information Filename

Description:

The value of this constant defines the filename used to store the software bus routing information. This filename is used only when no filename is specified in the command.

Limits

The length of each string, including the NULL terminator cannot exceed the [OS_MAX_PATH_LEN](#) value.

Definition at line 170 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_FILTER_MASK1

```
#define CFE_PLATFORM_SB_FILTER_MASK1 CFE_PLATFORM_SB_CFGVAL(FILTER_MASK1)
```

Definition at line 224 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_FILTER_MASK2

```
#define CFE_PLATFORM_SB_FILTER_MASK2 CFE_PLATFORM_SB_CFGVAL(FILTER_MASK2)
```

Definition at line 230 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_FILTER_MASK3

```
#define CFE_PLATFORM_SB_FILTER_MASK3 CFE_PLATFORM_SB_CFGVAL(FILTER_MASK3)
```

Definition at line 236 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_FILTER_MASK4

```
#define CFE_PLATFORM_SB_FILTER_MASK4 CFE_PLATFORM_SB_CFGVAL(FILTER_MASK4)
```

Definition at line 242 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_FILTER_MASK5

```
#define CFE_PLATFORM_SB_FILTER_MASK5 CFE_PLATFORM_SB_CFGVAL(FILTER_MASK5)
```

Definition at line 248 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_FILTER_MASK6

```
#define CFE_PLATFORM_SB_FILTER_MASK6 CFE_PLATFORM_SB_CFGVAL (FILTER_MASK6)
```

Definition at line 254 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_FILTER_MASK7

```
#define CFE_PLATFORM_SB_FILTER_MASK7 CFE_PLATFORM_SB_CFGVAL (FILTER_MASK7)
```

Definition at line 260 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_FILTER_MASK8

```
#define CFE_PLATFORM_SB_FILTER_MASK8 CFE_PLATFORM_SB_CFGVAL (FILTER_MASK8)
```

Definition at line 266 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_FILTERED_EVENT1

```
#define CFE_PLATFORM_SB_FILTERED_EVENT1 CFE_PLATFORM_SB_CFGVAL (FILTERED_EVENT1)
```

Purpose SB Event Filtering**Description:**

This group of configuration parameters dictates what SB events will be filtered through SB. The filtering will begin after the SB task initializes and stay in effect until a cmd to SB changes it. This allows the operator to set limits on the number of event messages that are sent during system initialization. NOTE: Set all unused event values and mask values to zero

Limits

This filtering applies only to SB events. These parameters have a lower limit of 0 and an upper limit of 65535.

Definition at line 221 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_FILTERED_EVENT2

```
#define CFE_PLATFORM_SB_FILTERED_EVENT2 CFE_PLATFORM_SB_CFGVAL (FILTERED_EVENT2)
```

Definition at line 227 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_FILTERED_EVENT3

```
#define CFE_PLATFORM_SB_FILTERED_EVENT3 CFE_PLATFORM_SB_CFGVAL (FILTERED_EVENT3)
```

Definition at line 233 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_FILTERED_EVENT4

```
#define CFE_PLATFORM_SB_FILTERED_EVENT4 CFE_PLATFORM_SB_CFGVAL (FILTERED_EVENT4)
```

Definition at line 239 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_FILTERED_EVENT5

```
#define CFE_PLATFORM_SB_FILTERED_EVENT5 CFE_PLATFORM_SB_CFGVAL (FILTERED_EVENT5)
```

Definition at line 245 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_FILTERED_EVENT6

```
#define CFE_PLATFORM_SB_FILTERED_EVENT6 CFE_PLATFORM_SB_CFGVAL (FILTERED_EVENT6)
```

Definition at line 251 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_FILTERED_EVENT7

```
#define CFE_PLATFORM_SB_FILTERED_EVENT7 CFE_PLATFORM_SB_CFGVAL (FILTERED_EVENT7)
```

Definition at line 257 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_FILTERED_EVENT8

```
#define CFE_PLATFORM_SB_FILTERED_EVENT8 CFE_PLATFORM_SB_CFGVAL (FILTERED_EVENT8)
```

Definition at line 263 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_HIGHEST_VALID_MSGID

```
#define CFE_PLATFORM_SB_HIGHEST_VALID_MSGID CFE_PLATFORM_SB_CFGVAL (HIGHEST_VALID_MSGID)
```

Purpose Highest Valid Message Id

Description:

The value of this constant dictates the range of valid message ID's, from 0 to CFE_PLATFORM_SB_HIGHEST_VALID_MSGID (inclusive).

Although this can be defined differently across platforms, each platform can only publish/subscribe to message ids within their allowable range. Typically this value is set the same across all mission platforms to avoid this complexity.

Limits

This parameter has a lower limit is 1, and an upper limit of 0xFFFFFFFF.

When using the direct message map implementation for software bus routing, this value is used to size the map where a value of 0x1FFF results in a 16 KBytes map and 0xFFFF is 128 KBytes.

When using the hash implementation for software bus routing, a multiple of the CFE_PLATFORM_SB_MAX_MSG_IDS is used to size the message map. In that case the range selected here does not impact message map memory use, so it's reasonable to use up to the full range supported by the message ID implementation.

Definition at line 155 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_MAX_BLOCK_SIZE

```
#define CFE_PLATFORM_SB_MAX_BLOCK_SIZE CFE_PLATFORM_SB_CFGVAL (MAX_BLOCK_SIZE)
```

Definition at line 327 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_MAX_DEST_PER_PKT

```
#define CFE_PLATFORM_SB_MAX_DEST_PER_PKT CFE_PLATFORM_SB_CFGVAL (MAX_DEST_PER_PKT)
```

Purpose Maximum Number of unique local destinations a single MsgId can have

Description:

Dictates the maximum number of unique local destinations a single MsgId can have.

Limits

This parameter has a lower limit of 1. There are no restrictions on the upper limit however, the maximum number of destinations per packet is system dependent and should be verified. Destination number values that are checked against this configuration are defined by a 16 bit data word.

Definition at line 90 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_MAX_MSG_IDS

```
#define CFE_PLATFORM_SB_MAX_MSG_IDS CFE_PLATFORM_SB_CFGVAL(MAX_MSG_IDS)
```

Purpose Maximum Number of Unique Message IDs SB Routing Table can hold

Description:

Dictates the maximum number of unique MsgIds the SB routing table will hold. This constant has a direct effect on the size of SB's tables and arrays. Keeping this count as low as possible will save memory. To see the run-time, high-water mark and the current utilization figures regarding this parameter, send an SB command to 'Send Statistics Pkt'.

Limits

This must be a power of two if software bus message routing hash implementation is being used. Lower than 64 will cause unit test failures, and telemetry reporting is impacted below 32. There is no hard upper limit, but impacts memory footprint. For software bus message routing search implementation the number of msg ids subscribed to impacts performance.

Definition at line 55 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_MAX_PIPES

```
#define CFE_PLATFORM_SB_MAX_PIPES CFE_PLATFORM_SB_CFGVAL(MAX_PIPES)
```

Purpose Maximum Number of Unique Pipes SB Routing Table can hold

Description:

Dictates the maximum number of unique Pipes the SB routing table will hold. This constant has a direct effect on the size of SB's tables and arrays. Keeping this count as low as possible will save memory. To see the run-time, high-water mark and the current utilization figures regarding this parameter, send an SB command to 'Send Statistics Pkt'.

Limits

This parameter has a lower limit of 1. This parameter must also be less than or equal to OS_MAX_QUEUES.

Definition at line 73 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_01

```
#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_01 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_01)
```

Purpose Define SB Memory Pool Block Sizes

Description:

Software Bus Memory Pool Block Sizes

Limits

These sizes MUST be increasing and MUST be an integral multiple of 4. The number of block sizes defined should match [CFE_PLATFORM_SB_POOL_MAX_BUCKETS](#)

Definition at line 295 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_02

#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_02 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_02)
Definition at line 297 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_03

#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_03 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_03)
Definition at line 299 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_04

#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_04 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_04)
Definition at line 301 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_05

#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_05 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_05)
Definition at line 303 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_06

#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_06 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_06)
Definition at line 305 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_07

#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_07 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_07)
Definition at line 307 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_08

#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_08 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_08)
Definition at line 309 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_09

#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_09 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_09)
Definition at line 311 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_10

#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_10 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_10)
Definition at line 313 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_11

#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_11 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_11)
Definition at line 315 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_12

#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_12 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_12)
Definition at line 317 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_13

#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_13 CFE_PLATFORM_SB_CFGVAL (MEM_BLOCK_SIZE_13)
Definition at line 319 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_14

#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_14 CFE_PLATFORM_SB_CFGVAL (MEM_BLOCK_SIZE_14)
Definition at line 321 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_15

#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_15 CFE_PLATFORM_SB_CFGVAL (MEM_BLOCK_SIZE_15)
Definition at line 323 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_MEM_BLOCK_SIZE_16

#define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_16 CFE_PLATFORM_SB_CFGVAL (MEM_BLOCK_SIZE_16)
Definition at line 325 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_POOL_MAX_BUCKETS

#define CFE_PLATFORM_SB_POOL_MAX_BUCKETS CFE_PLATFORM_SB_CFGVAL (POOL_MAX_BUCKETS)

Purpose Number of block sizes in SB memory pool structure

Description:

The number of block sizes for the software bus memory pool

Limits:

Must be at least one. No specific upper limit, but the number is anticipated to be reasonably small (i.e. tens, not hundreds). Large values have not been tested.

The SB block size list must correlate with this value

Definition at line 282 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_START_TASK_PRIORITY

#define CFE_PLATFORM_SB_START_TASK_PRIORITY CFE_PLATFORM_SB_CFGVAL (START_TASK_PRIORITY)

Purpose Define SB Task Priority

Description:

Defines the cFE_SB Task priority.

Limits

Not Applicable

Definition at line 339 of file [cfe_sb_internal_cfg.h](#).

CFE_PLATFORM_SB_START_TASK_STACK_SIZE

```
#define CFE_PLATFORM_SB_START_TASK_STACK_SIZE CFE_PLATFORM_SB_CFGVAL (START_TASK_STACK_SIZE)
```

Purpose Define SB Task Stack Size

Description:

Defines the cFE_SB Task Stack Size

Limits

There is a lower limit of 2048 on this configuration parameter. There are no restrictions on the upper limit however, the maximum stack size is system dependent and should be verified. Most operating systems provide tools for measuring the amount of stack used by a task during operation. It is always a good idea to verify that no more than 1/2 of the stack is used.

Definition at line 355 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_BUF_MEMORY_BYTES

```
#define DEFAULT_CFE_PLATFORM_SB_BUF_MEMORY_BYTES 524288
```

Definition at line 130 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_DEFAULT_MAP_FILENAME

```
#define DEFAULT_CFE_PLATFORM_SB_DEFAULT_MAP_FILENAME "/ram/cfe_sb_msgmap.dat"
```

Definition at line 204 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_DEFAULT_MSG_LIMIT

```
#define DEFAULT_CFE_PLATFORM_SB_DEFAULT_MSG_LIMIT 4
```

Definition at line 107 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_DEFAULT_PIPE_FILENAME

```
#define DEFAULT_CFE_PLATFORM_SB_DEFAULT_PIPE_FILENAME "/ram/cfe_sb_pipe.dat"
```

Definition at line 186 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_DEFAULT_ROUTING_FILENAME

```
#define DEFAULT_CFE_PLATFORM_SB_DEFAULT_ROUTING_FILENAME "/ram/cfe_sb_route.dat"
```

Definition at line 171 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_FILTER_MASK1

```
#define DEFAULT_CFE_PLATFORM_SB_FILTER_MASK1 CFE_EVS_FIRST_4_STOP
```

Definition at line 225 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_FILTER_MASK2

```
#define DEFAULT_CFE_PLATFORM_SB_FILTER_MASK2 CFE_EVS_FIRST_4_STOP
```

Definition at line 231 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_FILTER_MASK3

```
#define DEFAULT_CFE_PLATFORM_SB_FILTER_MASK3 CFE_EVS_FIRST_16_STOP
```

Definition at line 237 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_FILTER_MASK4

#define DEFAULT_CFE_PLATFORM_SB_FILTER_MASK4 CFE_EVS_FIRST_16_STOP
Definition at line 243 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_FILTER_MASK5

#define DEFAULT_CFE_PLATFORM_SB_FILTER_MASK5 CFE_EVS_NO_FILTER
Definition at line 249 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_FILTER_MASK6

#define DEFAULT_CFE_PLATFORM_SB_FILTER_MASK6 CFE_EVS_NO_FILTER
Definition at line 255 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_FILTER_MASK7

#define DEFAULT_CFE_PLATFORM_SB_FILTER_MASK7 CFE_EVS_NO_FILTER
Definition at line 261 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_FILTER_MASK8

#define DEFAULT_CFE_PLATFORM_SB_FILTER_MASK8 CFE_EVS_NO_FILTER
Definition at line 267 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT1

#define DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT1 CFE_SB_SEND_NO_SUBS_EID
Definition at line 222 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT2

#define DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT2 CFE_SB_DUP_SUBSCRIP_EID
Definition at line 228 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT3

#define DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT3 CFE_SB_MSGID_LIM_ERR_EID
Definition at line 234 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT4

#define DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT4 CFE_SB_Q_FULL_ERR_EID
Definition at line 240 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT5

#define DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT5 0
Definition at line 246 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT6

#define DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT6 0
Definition at line 252 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT7

#define DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT7 0
Definition at line 258 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT8

#define DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT8 0
Definition at line 264 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_HIGHEST_VALID_MSGID

#define DEFAULT_CFE_PLATFORM_SB_HIGHEST_VALID_MSGID 0x1FFF
Definition at line 156 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_MAX_BLOCK_SIZE

#define DEFAULT_CFE_PLATFORM_SB_MAX_BLOCK_SIZE (CFE_MISSION_SB_MAX_SB_MSG_SIZE + 128)
Definition at line 328 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_MAX_DEST_PER_PKT

#define DEFAULT_CFE_PLATFORM_SB_MAX_DEST_PER_PKT 16
Definition at line 91 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_MAX_MSG_IDS

#define DEFAULT_CFE_PLATFORM_SB_MAX_MSG_IDS 256
Definition at line 56 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_MAX_PIPES

#define DEFAULT_CFE_PLATFORM_SB_MAX_PIPES 64
Definition at line 74 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_01

#define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_01 8
Definition at line 296 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_02

#define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_02 16
Definition at line 298 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_03

#define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_03 20
Definition at line 300 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_04

#define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_04 36
Definition at line 302 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_05

#define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_05 64
Definition at line 304 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_06

#define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_06 96
Definition at line 306 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_07

#define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_07 128
Definition at line 308 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_08

#define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_08 160
Definition at line 310 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_09

#define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_09 256
Definition at line 312 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_10

#define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_10 512
Definition at line 314 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_11

#define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_11 1024
Definition at line 316 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_12

#define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_12 2048
Definition at line 318 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_13

#define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_13 4096
Definition at line 320 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_14

#define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_14 8192
Definition at line 322 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_15

#define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_15 16384
Definition at line 324 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_16

```
#define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_16 32768
```

Definition at line 326 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_POOL_MAX_BUCKETS

```
#define DEFAULT_CFE_PLATFORM_SB_POOL_MAX_BUCKETS 17
```

Definition at line 283 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_START_TASK_PRIORITY

```
#define DEFAULT_CFE_PLATFORM_SB_START_TASK_PRIORITY 64
```

Definition at line 340 of file [cfe_sb_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_SB_START_TASK_STACK_SIZE

```
#define DEFAULT_CFE_PLATFORM_SB_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
```

Definition at line 356 of file [cfe_sb_internal_cfg.h](#).

11.214 cfe_sb_internal_cfg.h

[Go to the documentation of this file.](#)

```
00001 /*****
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"
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00017  *****/
00018
00032 #ifndef CFE_SB_INTERNAL_CFG_H
00033 #define CFE_SB_INTERNAL_CFG_H
00034
00035 #include "cfe_sb_internal_cfg_values.h"
00036
00055 #define CFE_PLATFORM_SB_MAX_MSG_IDS CFE_PLATFORM_SB_CFGVAL(MAX_MSG_IDS)
00056 #define DEFAULT_CFE_PLATFORM_SB_MAX_MSG_IDS 256
00057
00073 #define CFE_PLATFORM_SB_MAX_PIPES CFE_PLATFORM_SB_CFGVAL(MAX_PIPES)
00074 #define DEFAULT_CFE_PLATFORM_SB_MAX_PIPES 64
00075
00090 #define CFE_PLATFORM_SB_MAX_DEST_PER_PKT CFE_PLATFORM_SB_CFGVAL(MAX_DEST_PER_PKT)
00091 #define DEFAULT_CFE_PLATFORM_SB_MAX_DEST_PER_PKT 16
00092
00106 #define CFE_PLATFORM_SB_DEFAULT_MSG_LIMIT CFE_PLATFORM_SB_CFGVAL(DEFAULT_MSG_LIMIT)
00107 #define DEFAULT_CFE_PLATFORM_SB_DEFAULT_MSG_LIMIT 4
00108
00129 #define CFE_PLATFORM_SB_BUF_MEMORY_BYTES CFE_PLATFORM_SB_CFGVAL(BUF_MEMORY_BYTES)
00130 #define DEFAULT_CFE_PLATFORM_SB_BUF_MEMORY_BYTES 524288
00131
00155 #define CFE_PLATFORM_SB_HIGHEST_VALID_MSGID CFE_PLATFORM_SB_CFGVAL(HIGHEST_VALID_MSGID)
00156 #define DEFAULT_CFE_PLATFORM_SB_HIGHEST_VALID_MSGID 0xFFFF
00157
00170 #define CFE_PLATFORM_SB_DEFAULT_ROUTING_FILENAME CFE_PLATFORM_SB_CFGVAL(DEFAULT_ROUTING_FILENAME)
00171 #define DEFAULT_CFE_PLATFORM_SB_DEFAULT_ROUTING_FILENAME "/ram/cfe_sb_route.dat"
00172
00185 #define CFE_PLATFORM_SB_DEFAULT_PIPE_FILENAME CFE_PLATFORM_SB_CFGVAL(DEFAULT_PIPE_FILENAME)
00186 #define DEFAULT_CFE_PLATFORM_SB_DEFAULT_PIPE_FILENAME "/ram/cfe_sb_pipe.dat"
00187
00203 #define CFE_PLATFORM_SB_DEFAULT_MAP_FILENAME CFE_PLATFORM_SB_CFGVAL(DEFAULT_MAP_FILENAME)
```



```
00204 #define DEFAULT_CFE_PLATFORM_SB_DEFAULT_MAP_FILENAME "/ram/cfe_sb_msgmap.dat"
00205
00221 #define CFE_PLATFORM_SB_FILTERED_EVENT1 CFE_PLATFORM_SB_CFGVAL(FILTERED_EVENT1)
00222 #define DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT1 CFE_SB_SEND_NO_SUBS_EID
00223
00224 #define CFE_PLATFORM_SB_FILTER_MASK1 CFE_PLATFORM_SB_CFGVAL(FILTER_MASK1)
00225 #define DEFAULT_CFE_PLATFORM_SB_FILTER_MASK1 CFE_EVS_FIRST_4_STOP
00226
00227 #define CFE_PLATFORM_SB_FILTERED_EVENT2 CFE_PLATFORM_SB_CFGVAL(FILTERED_EVENT2)
00228 #define DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT2 CFE_SB_DUP_SUBSCRIP_EID
00229
00230 #define CFE_PLATFORM_SB_FILTER_MASK2 CFE_PLATFORM_SB_CFGVAL(FILTER_MASK2)
00231 #define DEFAULT_CFE_PLATFORM_SB_FILTER_MASK2 CFE_EVS_FIRST_4_STOP
00232
00233 #define CFE_PLATFORM_SB_FILTERED_EVENT3 CFE_PLATFORM_SB_CFGVAL(FILTERED_EVENT3)
00234 #define DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT3 CFE_SB_MSGID_LIM_ERR_EID
00235
00236 #define CFE_PLATFORM_SB_FILTER_MASK3 CFE_PLATFORM_SB_CFGVAL(FILTER_MASK3)
00237 #define DEFAULT_CFE_PLATFORM_SB_FILTER_MASK3 CFE_EVS_FIRST_16_STOP
00238
00239 #define CFE_PLATFORM_SB_FILTERED_EVENT4 CFE_PLATFORM_SB_CFGVAL(FILTERED_EVENT4)
00240 #define DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT4 CFE_SB_Q_FULL_ERR_EID
00241
00242 #define CFE_PLATFORM_SB_FILTER_MASK4 CFE_PLATFORM_SB_CFGVAL(FILTER_MASK4)
00243 #define DEFAULT_CFE_PLATFORM_SB_FILTER_MASK4 CFE_EVS_FIRST_16_STOP
00244
00245 #define CFE_PLATFORM_SB_FILTERED_EVENT5 CFE_PLATFORM_SB_CFGVAL(FILTERED_EVENT5)
00246 #define DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT5 0
00247
00248 #define CFE_PLATFORM_SB_FILTER_MASK5 CFE_PLATFORM_SB_CFGVAL(FILTER_MASK5)
00249 #define DEFAULT_CFE_PLATFORM_SB_FILTER_MASK5 CFE_EVS_NO_FILTER
00250
00251 #define CFE_PLATFORM_SB_FILTERED_EVENT6 CFE_PLATFORM_SB_CFGVAL(FILTERED_EVENT6)
00252 #define DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT6 0
00253
00254 #define CFE_PLATFORM_SB_FILTER_MASK6 CFE_PLATFORM_SB_CFGVAL(FILTER_MASK6)
00255 #define DEFAULT_CFE_PLATFORM_SB_FILTER_MASK6 CFE_EVS_NO_FILTER
00256
00257 #define CFE_PLATFORM_SB_FILTERED_EVENT7 CFE_PLATFORM_SB_CFGVAL(FILTERED_EVENT7)
00258 #define DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT7 0
00259
00260 #define CFE_PLATFORM_SB_FILTER_MASK7 CFE_PLATFORM_SB_CFGVAL(FILTER_MASK7)
00261 #define DEFAULT_CFE_PLATFORM_SB_FILTER_MASK7 CFE_EVS_NO_FILTER
00262
00263 #define CFE_PLATFORM_SB_FILTERED_EVENT8 CFE_PLATFORM_SB_CFGVAL(FILTERED_EVENT8)
00264 #define DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT8 0
00265
00266 #define CFE_PLATFORM_SB_FILTER_MASK8 CFE_PLATFORM_SB_CFGVAL(FILTER_MASK8)
00267 #define DEFAULT_CFE_PLATFORM_SB_FILTER_MASK8 CFE_EVS_NO_FILTER
00268
00282 #define CFE_PLATFORM_SB_POOL_MAX_BUCKETS CFE_PLATFORM_SB_CFGVAL(POOL_MAX_BUCKETS)
00283 #define DEFAULT_CFE_PLATFORM_SB_POOL_MAX_BUCKETS 17
00284
00295 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_01 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_01)
00296 #define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_01 8
00297 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_02 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_02)
00298 #define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_02 16
00299 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_03 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_03)
00300 #define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_03 20
00301 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_04 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_04)
00302 #define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_04 36
00303 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_05 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_05)
00304 #define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_05 64
00305 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_06 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_06)
00306 #define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_06 96
00307 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_07 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_07)
00308 #define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_07 128
00309 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_08 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_08)
00310 #define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_08 160
00311 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_09 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_09)
00312 #define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_09 256
00313 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_10 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_10)
00314 #define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_10 512
00315 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_11 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_11)
00316 #define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_11 1024
00317 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_12 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_12)
00318 #define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_12 2048
00319 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_13 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_13)
00320 #define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_13 4096
00321 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_14 CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_14)
00322 #define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_14 8192
```

```

00323 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_15      CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_15)
00324 #define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_15 16384
00325 #define CFE_PLATFORM_SB_MEM_BLOCK_SIZE_16      CFE_PLATFORM_SB_CFGVAL(MEM_BLOCK_SIZE_16)
00326 #define DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_16 32768
00327 #define CFE_PLATFORM_SB_MAX_BLOCK_SIZE         CFE_PLATFORM_SB_CFGVAL(MAX_BLOCK_SIZE)
00328 #define DEFAULT_CFE_PLATFORM_SB_MAX_BLOCK_SIZE (CFE_MISSION_SB_MAX_SB_MSG_SIZE + 128)
00329
00339 #define CFE_PLATFORM_SB_START_TASK_PRIORITY      CFE_PLATFORM_SB_CFGVAL(START_TASK_PRIORITY)
00340 #define DEFAULT_CFE_PLATFORM_SB_START_TASK_PRIORITY 64
00341
00355 #define CFE_PLATFORM_SB_START_TASK_STACK_SIZE   CFE_PLATFORM_SB_CFGVAL(START_TASK_STACK_SIZE)
00356 #define DEFAULT_CFE_PLATFORM_SB_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
00357
00358 #endif

```

11.215 cfe/modules/sb/fsw/inc/cfe_sb_topicids.h File Reference

```
#include "cfe_sb_topicid_values.h"
```

Macros

- #define CFE_MISSION_SB_CMD_TOPICID CFE_MISSION_SB_TIDVAL(CMD)
- #define DEFAULT_CFE_MISSION_SB_CMD_TOPICID 3
- #define CFE_MISSION_SB_SEND_HK_TOPICID CFE_MISSION_SB_TIDVAL(SEND_HK)
- #define DEFAULT_CFE_MISSION_SB_SEND_HK_TOPICID 11
- #define CFE_MISSION_SB_SUB_RPT_CTRL_TOPICID CFE_MISSION_SB_TIDVAL(SUB_RPT_CTRL)
- #define DEFAULT_CFE_MISSION_SB_SUB_RPT_CTRL_TOPICID 14
- #define CFE_MISSION_SB_HK_TLM_TOPICID CFE_MISSION_SB_TIDVAL(HK_TLM)
- #define DEFAULT_CFE_MISSION_SB_HK_TLM_TOPICID 3
- #define CFE_MISSION_SB_STATS_TLM_TOPICID CFE_MISSION_SB_TIDVAL(STATS_TLM)
- #define DEFAULT_CFE_MISSION_SB_STATS_TLM_TOPICID 10
- #define CFE_MISSION_SB_ALLSUBS_TLM_TOPICID CFE_MISSION_SB_TIDVAL(ALLSUBS_TLM)
- #define DEFAULT_CFE_MISSION_SB_ALLSUBS_TLM_TOPICID 13
- #define CFE_MISSION_SB_ONESUB_TLM_TOPICID CFE_MISSION_SB_TIDVAL(ONESUB_TLM)
- #define DEFAULT_CFE_MISSION_SB_ONESUB_TLM_TOPICID 14

11.215.1 Detailed Description

CFE Software Bus (CFE_SB) Application Topic IDs
Definition in file [cfe_sb_topicids.h](#).

11.215.2 Macro Definition Documentation

CFE_MISSION_SB_ALLSUBS_TLM_TOPICID

```
#define CFE_MISSION_SB_ALLSUBS_TLM_TOPICID CFE_MISSION_SB_TIDVAL(ALLSUBS_TLM)
```

Definition at line 57 of file [cfe_sb_topicids.h](#).

CFE_MISSION_SB_CMD_TOPICID

```
#define CFE_MISSION_SB_CMD_TOPICID CFE_MISSION_SB_TIDVAL(CMD)
```

Purpose cFE Portable Message Numbers for Commands

Description:

Portable message numbers for the cFE command messages

Limits

Not Applicable

Definition at line 37 of file [cfe_sb_topicsids.h](#).

CFE_MISSION_SB_HK_TLM_TOPICID

```
#define CFE_MISSION_SB_HK_TLM_TOPICID CFE_MISSION_SB_TIDVAL (HK_TLM)
```

Purpose cFE Portable Message Numbers for Telemetry

Description:

Portable message numbers for the cFE telemetry messages

Limits

Not Applicable

Definition at line 53 of file [cfe_sb_topicsids.h](#).

CFE_MISSION_SB_ONESUB_TLM_TOPICID

```
#define CFE_MISSION_SB_ONESUB_TLM_TOPICID CFE_MISSION_SB_TIDVAL (ONESUB_TLM)
```

Definition at line 59 of file [cfe_sb_topicsids.h](#).

CFE_MISSION_SB_SEND_HK_TOPICID

```
#define CFE_MISSION_SB_SEND_HK_TOPICID CFE_MISSION_SB_TIDVAL (SEND_HK)
```

Definition at line 39 of file [cfe_sb_topicsids.h](#).

CFE_MISSION_SB_STATS_TLM_TOPICID

```
#define CFE_MISSION_SB_STATS_TLM_TOPICID CFE_MISSION_SB_TIDVAL (STATS_TLM)
```

Definition at line 55 of file [cfe_sb_topicsids.h](#).

CFE_MISSION_SB_SUB_RPT_CTRL_TOPICID

```
#define CFE_MISSION_SB_SUB_RPT_CTRL_TOPICID CFE_MISSION_SB_TIDVAL (SUB_RPT_CTRL)
```

Definition at line 41 of file [cfe_sb_topicsids.h](#).

DEFAULT_CFE_MISSION_SB_ALLSUBS_TLM_TOPICID

```
#define DEFAULT_CFE_MISSION_SB_ALLSUBS_TLM_TOPICID 13
```

Definition at line 58 of file [cfe_sb_topicsids.h](#).

DEFAULT_CFE_MISSION_SB_CMD_TOPICID

```
#define DEFAULT_CFE_MISSION_SB_CMD_TOPICID 3
```

Definition at line 38 of file [cfe_sb_topicsids.h](#).

DEFAULT_CFE_MISSION_SB_HK_TLM_TOPICID

```
#define DEFAULT_CFE_MISSION_SB_HK_TLM_TOPICID 3
```

Definition at line 54 of file [cfe_sb_topicsids.h](#).

DEFAULT_CFE_MISSION_SB_ONESUB_TLM_TOPICID

```
#define DEFAULT_CFE_MISSION_SB_ONESUB_TLM_TOPICID 14
```

Definition at line 60 of file [cfe_sb_topicsids.h](#).

DEFAULT_CFE_MISSION_SB_SEND_HK_TOPICID

```
#define DEFAULT_CFE_MISSION_SB_SEND_HK_TOPICID 11
```

Definition at line 40 of file [cfe_sb_topicsids.h](#).

DEFAULT_CFE_MISSION_SB_STATS_TLM_TOPICID

```
#define DEFAULT_CFE_MISSION_SB_STATS_TLM_TOPICID 10
```

Definition at line 56 of file [cfe_sb_topicsids.h](#).

DEFAULT_CFE_MISSION_SB_SUB_RPT_CTRL_TOPICID

```
#define DEFAULT_CFE_MISSION_SB_SUB_RPT_CTRL_TOPICID 14
```

Definition at line 42 of file [cfe_sb_topicsids.h](#).

11.216 cfe_sb_topicsids.h

[Go to the documentation of this file.](#)

```
00001 /*****
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"
00003  *
00004  * Copyright (c) 2023 United States Government as represented by the
00005  * Administrator of the National Aeronautics and Space Administration.
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00007  *
00008  * Licensed under the Apache License, Version 2.0 (the "License"); you may
00009  * not use this file except in compliance with the License. You may obtain
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00011  *
00012  * Unless required by applicable law or agreed to in writing, software
00013  * distributed under the License is distributed on an "AS IS" BASIS,
00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00023 #ifndef CFE_SB_TOPICSIDS_H
00024 #define CFE_SB_TOPICSIDS_H
00025
00026 #include "cfe_sb_topicid_values.h"
00027
00037 #define CFE_MISSION_SB_CMD_TOPICID          CFE_MISSION_SB_TIDVAL(CMD)
00038 #define DEFAULT_CFE_MISSION_SB_CMD_TOPICID 3
00039 #define CFE_MISSION_SB_SEND_HK_TOPICID      CFE_MISSION_SB_TIDVAL(SEND_HK)
00040 #define DEFAULT_CFE_MISSION_SB_SEND_HK_TOPICID 11
00041 #define CFE_MISSION_SB_SUB_RPT_CTRL_TOPICID CFE_MISSION_SB_TIDVAL(SUB_RPT_CTRL)
00042 #define DEFAULT_CFE_MISSION_SB_SUB_RPT_CTRL_TOPICID 14
00043
00053 #define CFE_MISSION_SB_HK_TLM_TOPICID      CFE_MISSION_SB_TIDVAL(HK_TLM)
00054 #define DEFAULT_CFE_MISSION_SB_HK_TLM_TOPICID 3
00055 #define CFE_MISSION_SB_STATS_TLM_TOPICID    CFE_MISSION_SB_TIDVAL(STATS_TLM)
00056 #define DEFAULT_CFE_MISSION_SB_STATS_TLM_TOPICID 10
00057 #define CFE_MISSION_SB_ALLSUBS_TLM_TOPICID  CFE_MISSION_SB_TIDVAL(ALLSUBS_TLM)
00058 #define DEFAULT_CFE_MISSION_SB_ALLSUBS_TLM_TOPICID 13
00059 #define CFE_MISSION_SB_ONESUB_TLM_TOPICID   CFE_MISSION_SB_TIDVAL(ONESUB_TLM)
00060 #define DEFAULT_CFE_MISSION_SB_ONESUB_TLM_TOPICID 14
00061
00062 #endif
```

11.217 cfe/modules/tbl/config/default_cfe_tbl_extern_typedefs.h File Reference

```
#include "common_types.h"
#include "cfe_es_extern_typedefs.h"
```

```
#include "cfe_fs_extern_typedefs.h"
#include "cfe_mission_cfg.h"
```

Data Structures

- struct [CFE_TBL_File_Hdr](#)
The definition of the header fields that are included in CFE Table Data files.
- struct [CFE_TBL_CombinedFileHdr](#)
Complete header for CFE table files.

Typedefs

- typedef [uint16](#) [CFE_TBL_BufferSelect_Enum_t](#)
Selects the buffer to operate on for validate or dump commands.
- typedef struct [CFE_TBL_File_Hdr](#) [CFE_TBL_File_Hdr_t](#)
The definition of the header fields that are included in CFE Table Data files.
- typedef struct [CFE_TBL_CombinedFileHdr](#) [CFE_TBL_CombinedFileHdr_t](#)
Complete header for CFE table files.
- typedef [CFE_RESOURCEID_BASE_TYPE](#) [CFE_TBL_RegId_t](#)
A type for Table Registry IDs.
- typedef [CFE_RESOURCEID_BASE_TYPE](#) [CFE_TBL_HandleId_t](#)
A type for Table Handle IDs.

Enumerations

- enum [CFE_TBL_BufferSelect](#) { [CFE_TBL_BufferSelect_INACTIVE](#) = 0 , [CFE_TBL_BufferSelect_ACTIVE](#) = 1 }
- Label definitions associated with CFE_TBL_BufferSelect_Enum_t.*

11.217.1 Detailed Description

Declarations and prototypes for `cfe_tbl_extern_typedefs` module
Definition in file [default_cfe_tbl_extern_typedefs.h](#).

11.217.2 Typedef Documentation

CFE_TBL_BufferSelect_Enum_t

typedef [uint16](#) [CFE_TBL_BufferSelect_Enum_t](#)
Selects the buffer to operate on for validate or dump commands.

See also

enum [CFE_TBL_BufferSelect](#)

Definition at line 54 of file [default_cfe_tbl_extern_typedefs.h](#).

CFE_TBL_CombinedFileHdr_t

typedef struct [CFE_TBL_CombinedFileHdr](#) [CFE_TBL_CombinedFileHdr_t](#)
Complete header for CFE table files.

Table files always have a combination of the standard file header and the table-specific file header. This struct just combines the two and makes for an easier item to pass around, simplifying APIs

CFE_TBL_File_Hdr_t

```
typedef struct CFE_TBL_File_Hdr CFE_TBL_File_Hdr_t
```

The definition of the header fields that are included in CFE Table Data files.

This header follows the CFE_FS header and precedes the actual table data.

Note

The Offset and NumBytes fields in the table header are to 32 bits for backward compatibility with existing CFE versions. This means that even on 64-bit CPUs, individual table files will be limited to 4GiB in size.

CFE_TBL_HandleId_t

```
typedef CFE_RESOURCEID_BASE_TYPE CFE_TBL_HandleId_t
```

A type for Table Handle IDs.

This is the type that is used for any API accepting or returning a table ID

Definition at line 99 of file [default_cfe_tbl_extern_typedefs.h](#).

CFE_TBL_RegId_t

```
typedef CFE_RESOURCEID_BASE_TYPE CFE_TBL_RegId_t
```

A type for Table Registry IDs.

This is the type that is used for any API accepting or returning an Registry ID

Definition at line 92 of file [default_cfe_tbl_extern_typedefs.h](#).

11.217.3 Enumeration Type Documentation**CFE_TBL_BufferSelect**

```
enum CFE_TBL_BufferSelect
```

Label definitions associated with CFE_TBL_BufferSelect_Enum_t.

Enumerator

CFE_TBL_BufferSelect_INACTIVE	Select the Inactive buffer for validate or dump.
CFE_TBL_BufferSelect_ACTIVE	Select the Active buffer for validate or dump.

Definition at line 36 of file [default_cfe_tbl_extern_typedefs.h](#).

11.218 default_cfe_tbl_extern_typedefs.h

[Go to the documentation of this file.](#)

```
00001 /*****
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"
00003  *
00004  * Copyright (c) 2023 United States Government as represented by the
00005  * Administrator of the National Aeronautics and Space Administration.
00006  * All Rights Reserved.
00007  *
00008  * Licensed under the Apache License, Version 2.0 (the "License"); you may
00009  * not use this file except in compliance with the License. You may obtain
00010  * a copy of the License at http://www.apache.org/licenses/LICENSE-2.0
00011  *
00012  * Unless required by applicable law or agreed to in writing, software
00013  * distributed under the License is distributed on an "AS IS" BASIS,
00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00025 #ifndef DEFAULT_CFE_TBL_EXTERN_TYPEDEFS_H
00026 #define DEFAULT_CFE_TBL_EXTERN_TYPEDEFS_H
```

```

00027
00028 #include "common_types.h"
00029 #include "cfe_es_extern_typedefs.h"
00030 #include "cfe_fs_extern_typedefs.h"
00031 #include "cfe_mission_cfg.h" /* for CFE_MISSION_TBL_MAX_FULL_NAME_LEN */
00032
00036 enum CFE_TBL_BufferSelect
00037 {
00041     CFE_TBL_BufferSelect_INACTIVE = 0,
00042
00046     CFE_TBL_BufferSelect_ACTIVE = 1
00047 };
00048
00054 typedef uint16 CFE_TBL_BufferSelect_Enum_t;
00055
00065 typedef struct CFE_TBL_File_Hdr
00066 {
00067     uint32 Reserved;
00068     uint32 Offset;
00069     uint32 NumBytes;
00070     char    TableName[CFE_MISSION_TBL_MAX_FULL_NAME_LEN];
00071 } CFE_TBL_File_Hdr_t;
00072
00080 typedef struct CFE_TBL_CombinedFileHdr
00081 {
00082     CFE_FS_Header_t Std;
00083     CFE_TBL_File_Hdr_t Tbl;
00084
00085 } CFE_TBL_CombinedFileHdr_t;
00086
00092 typedef CFE_RESOURCEID_BASE_TYPE CFE_TBL_RegId_t;
00093
00099 typedef CFE_RESOURCEID_BASE_TYPE CFE_TBL_HandleId_t;
00100
00101 #endif /* CFE_TBL_EXTERN_TTYPEDEFS_H */

```

11.219 cfe/modules/tbl/config/default_cfe_tbl_fcncode_values.h File Reference

Macros

- #define [CFE_TBL_CCVAL\(x\) CFE_TBL_FunctionCode_##x](#)

Enumerations

- enum [CFE_TBL_FunctionCode_](#) {
[CFE_TBL_FunctionCode_NOOP](#) = 0 , [CFE_TBL_FunctionCode_RESET_COUNTERS](#) = 1 , [CFE_TBL_FunctionCode_LOAD](#)
= 2 , [CFE_TBL_FunctionCode_DUMP](#) = 3 ,
[CFE_TBL_FunctionCode_VALIDATE](#) = 4 , [CFE_TBL_FunctionCode_ACTIVATE](#) = 5 , [CFE_TBL_FunctionCode_DUMP_REGISTRY](#)
= 6 , [CFE_TBL_FunctionCode_SEND_REGISTRY](#) = 7 ,
[CFE_TBL_FunctionCode_DELETE_CDS](#) = 8 , [CFE_TBL_FunctionCode_ABORT_LOAD](#) = 9 }

11.219.1 Detailed Description

Specification for the CFE Event Services (CFE_TBL) command function codes

Note

This file should be strictly limited to the command/function code (CC) macro definitions. Other definitions such as enums, typedefs, or other macros should be placed in the msgdefs.h or msg.h files.

Definition in file [default_cfe_tbl_fcncode_values.h](#).

11.219.2 Macro Definition Documentation

CFE_TBL_CCVAL

```

#define CFE_TBL_CCVAL(
    x ) CFE_TBL_FunctionCode_##x

```

Definition at line 35 of file [default_cfe_tbl_fcncode_values.h](#).

11.219.3 Enumeration Type Documentation

CFE_TBL_FunctionCode_

enum CFE_TBL_FunctionCode_

Enumerator

CFE_TBL_FunctionCode_NOOP	
CFE_TBL_FunctionCode_RESET_COUNTERS	
CFE_TBL_FunctionCode_LOAD	
CFE_TBL_FunctionCode_DUMP	
CFE_TBL_FunctionCode_VALIDATE	
CFE_TBL_FunctionCode_ACTIVATE	
CFE_TBL_FunctionCode_DUMP_REGISTRY	
CFE_TBL_FunctionCode_SEND_REGISTRY	
CFE_TBL_FunctionCode_DELETE_CDS	
CFE_TBL_FunctionCode_ABORT_LOAD	

Definition at line 37 of file [default_cfe_tbl_fcncode_values.h](#).

11.220 default_cfe_tbl_fcncode_values.h

[Go to the documentation of this file.](#)

```

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00017  *****/
00018
00028 #ifndef DEFAULT_CFE_TBL_FCNCODE_VALUES_H
00029 #define DEFAULT_CFE_TBL_FCNCODE_VALUES_H
00030
00031 /*****
00032  * Macro Definitions
00033  *****/
00034
00035 #define CFE_TBL_CCVAL(x) CFE_TBL_FunctionCode_##x
00036
00037 enum CFE_TBL_FunctionCode_
00038 {
00039     CFE_TBL_FunctionCode_NOOP          = 0,
00040     CFE_TBL_FunctionCode_RESET_COUNTERS = 1,
00041     CFE_TBL_FunctionCode_LOAD          = 2,
00042     CFE_TBL_FunctionCode_DUMP         = 3,
00043     CFE_TBL_FunctionCode_VALIDATE     = 4,
00044     CFE_TBL_FunctionCode_ACTIVATE     = 5,
00045     CFE_TBL_FunctionCode_DUMP_REGISTRY = 6,
00046     CFE_TBL_FunctionCode_SEND_REGISTRY = 7,
00047     CFE_TBL_FunctionCode_DELETE_CDS   = 8,
00048     CFE_TBL_FunctionCode_ABORT_LOAD   = 9,
00049 };
00050
00051 #endif

```


11.221 cfe/modules/tbl/config/default_cfe_tbl_interface_cfg_values.h File Reference

Macros

- #define [CFE_MISSION_TBL_CFGVAL](#)(x) DEFAULT_CFE_MISSION_TBL_##x

11.221.1 Detailed Description

CFE Executive Services (CFE_ES) Application Public Definitions

This provides default values for configurable items that affect the interface(s) of this module. This includes the CMD/TLM message interface, tables definitions, and any other data products that serve to exchange information with other entities.

Note

This file may be overridden/superceded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [default_cfe_tbl_interface_cfg_values.h](#).

11.221.2 Macro Definition Documentation

CFE_MISSION_TBL_CFGVAL

```
#define CFE_MISSION_TBL_CFGVAL(  
    x ) DEFAULT_CFE_MISSION_TBL_##x
```

Definition at line 36 of file [default_cfe_tbl_interface_cfg_values.h](#).

11.222 default_cfe_tbl_interface_cfg_values.h

[Go to the documentation of this file.](#)

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00016  * limitations under the License.  
00017  *****/  
00018  
00032 #ifndef DEFAULT_CFE_TBL_INTERFACE_CFG_VALUES_H  
00033 #define DEFAULT_CFE_TBL_INTERFACE_CFG_VALUES_H  
00034  
00035 /* Use the default configuration value for all */  
00036 #define CFE_MISSION_TBL_CFGVAL(x) DEFAULT_CFE_MISSION_TBL_##x  
00037  
00038 #endif
```

11.223 cfe/modules/tbl/config/default_cfe_tbl_internal_cfg_values.h File Reference

Macros

- #define [CFE_PLATFORM_TBL_CFGVAL](#)(x) DEFAULT_CFE_PLATFORM_TBL_##x

11.223.1 Detailed Description

CFE Executive Services (CFE_ES) Application Private Config Definitions

This provides default values for configurable items that are internal to this module and do NOT affect the interface(s) of this module. Changes to items in this file only affect the local module and will be transparent to external entities that are using the public interface(s).

Note

This file may be overridden/superseded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [default_cfe_tbl_internal_cfg_values.h](#).

11.223.2 Macro Definition Documentation

CFE_PLATFORM_TBL_CFGVAL

```
#define CFE_PLATFORM_TBL_CFGVAL(  
    x ) DEFAULT_CFE_PLATFORM_TBL_##x
```

Definition at line 36 of file [default_cfe_tbl_internal_cfg_values.h](#).

11.224 default_cfe_tbl_internal_cfg_values.h

[Go to the documentation of this file.](#)

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00017  *****/  
00018  
00032 #ifndef DEFAULT_CFE_TBL_INTERNAL_CFG_VALUES_H  
00033 #define DEFAULT_CFE_TBL_INTERNAL_CFG_VALUES_H  
00034  
00035 /* Use the default configuration value for all */  
00036 #define CFE_PLATFORM_TBL_CFGVAL(x) DEFAULT_CFE_PLATFORM_TBL_##x  
00037  
00038 #endif
```

11.225 cfe/modules/tbl/config/default_cfe_tbl_mission_cfg.h File Reference

```
#include "cfe_tbl_interface_cfg_values.h"
```

Macros

- #define [CFE_MISSION_TBL_MAX_NAME_LENGTH](#) [CFE_MISSION_TBL_CFGVAL](#)(MAX_NAME_LENGTH)
- #define [DEFAULT_CFE_MISSION_TBL_MAX_NAME_LENGTH](#) 16
- #define [CFE_MISSION_TBL_MAX_FULL_NAME_LEN](#) [CFE_MISSION_TBL_CFGVAL](#)(MAX_FULL_NAME_LEN)
- #define [DEFAULT_CFE_MISSION_TBL_MAX_FULL_NAME_LEN](#) 40

11.225.1 Detailed Description

CFE Event Services (CFE_TBL) Application Mission Configuration Header File

This is a compatibility header for the "mission_cfg.h" file that has traditionally provided public config definitions for each CFS app.

Note

This file may be overridden/superseded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [default_cfe_tbl_mission_cfg.h](#).

11.225.2 Macro Definition Documentation**CFE_MISSION_TBL_MAX_FULL_NAME_LEN**

```
#define CFE_MISSION_TBL_MAX_FULL_NAME_LEN CFE_MISSION_TBL_CFGVAL (MAX_FULL_NAME_LEN)
```

Purpose Maximum Length of Full Table Name in messages

Description:

Indicates the maximum length (in characters) of the entire table name within software bus messages, in "App←Name.TableName" notation.

This affects the layout of command/telemetry messages but does not affect run time behavior or internal allocation.

Limits

All CPUs within the same SB domain (mission) must share the same definition Note this affects the size of messages, so it must not cause any message to exceed the max length.

This value should be kept as a multiple of 4, to maintain alignment of any possible neighboring fields without implicit padding.

Definition at line 71 of file [default_cfe_tbl_mission_cfg.h](#).

CFE_MISSION_TBL_MAX_NAME_LENGTH

```
#define CFE_MISSION_TBL_MAX_NAME_LENGTH CFE_MISSION_TBL_CFGVAL (MAX_NAME_LENGTH)
```

Purpose Maximum Table Name Length

Description:

Indicates the maximum length (in characters) of the table name ('TblName') portion of a Full Table Name of the following form: "ApplicationName.TblName"

This length does not need to include an extra character for NULL termination.

Limits

This value should be kept as a multiple of 4, to maintain alignment of any possible neighboring fields without implicit padding.

Definition at line 50 of file [default_cfe_tbl_mission_cfg.h](#).

DEFAULT_CFE_MISSION_TBL_MAX_FULL_NAME_LEN

```
#define DEFAULT_CFE_MISSION_TBL_MAX_FULL_NAME_LEN 40
```

Definition at line 74 of file [default_cfe_tbl_mission_cfg.h](#).

DEFAULT_CFE_MISSION_TBL_MAX_NAME_LENGTH

#define DEFAULT_CFE_MISSION_TBL_MAX_NAME_LENGTH 16
 Definition at line 51 of file [default_cfe_tbl_mission_cfg.h](#).

11.226 default_cfe_tbl_mission_cfg.h

[Go to the documentation of this file.](#)

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00017  *****/
00018
00031 #ifndef DEFAULT_CFE_TBL_MISSION_CFG_H
00032 #define DEFAULT_CFE_TBL_MISSION_CFG_H
00033
00034 #include "cfe_tbl_interface_cfg_values.h"
00035
00050 #define CFE_MISSION_TBL_MAX_NAME_LENGTH CFE_MISSION_TBL_CFGVAL(MAX_NAME_LENGTH)
00051 #define DEFAULT_CFE_MISSION_TBL_MAX_NAME_LENGTH 16
00052
00071 #define CFE_MISSION_TBL_MAX_FULL_NAME_LEN CFE_MISSION_TBL_CFGVAL(MAX_FULL_NAME_LEN)
00072
00073 /* Default value calculated as: (CFE_MISSION_TBL_MAX_NAME_LENGTH + CFE_MISSION_MAX_API_LEN + 4) */
00074 #define DEFAULT_CFE_MISSION_TBL_MAX_FULL_NAME_LEN 40
00075
00076 #endif

```

11.227 cfe/modules/tbl/config/default_cfe_tbl_msg.h File Reference

```

#include "cfe_mission_cfg.h"
#include "cfe_tbl_fcncodes.h"
#include "cfe_tbl_msgdefs.h"
#include "cfe_tbl_msgstruct.h"

```

11.227.1 Detailed Description

Specification for the CFE Event Services (CFE_TBL) command and telemetry message data types.

This is a compatibility header for the "cfe_tbl_msg.h" file that has traditionally provided the message definitions for cFS apps.

Note

This file may be overridden/superseded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [default_cfe_tbl_msg.h](#).

11.228 default_cfe_tbl_msg.h

[Go to the documentation of this file.](#)

```

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00017  *****/
00018
00031 #ifndef DEFAULT_CFE_TBL_MSG_H
00032 #define DEFAULT_CFE_TBL_MSG_H
00033
00034 #include "cfe_mission_cfg.h"
00035 #include "cfe_tbl_fcncodes.h"
00036 #include "cfe_tbl_msgdefs.h"
00037 #include "cfe_tbl_msgstruct.h"
00038
00039 #endif

```

11.229 cfe/modules/tbl/config/default_cfe_tbl_msgdefs.h File Reference

```

#include "common_types.h"
#include "cfe_mission_cfg.h"
#include "cfe_es_extern_typedefs.h"
#include "cfe_time_extern_typedefs.h"
#include "cfe_tbl_extern_typedefs.h"
#include "cfe_tbl_fcncodes.h"

```

Data Structures

- struct [CFE_TBL_LoadCmd_Payload](#)
Load Table Command Payload.
- struct [CFE_TBL_DumpCmd_Payload](#)
Dump Table Command Payload.
- struct [CFE_TBL_ValidateCmd_Payload](#)
Validate Table Command Payload.
- struct [CFE_TBL_ActivateCmd_Payload](#)
Activate Table Command Payload.
- struct [CFE_TBL_DumpRegistryCmd_Payload](#)
Dump Registry Command Payload.
- struct [CFE_TBL_SendRegistryCmd_Payload](#)
Send Table Registry Command Payload.
- struct [CFE_TBL_DeICDSCmd_Payload](#)
Delete Critical Table CDS Command Payload.
- struct [CFE_TBL_AbortLoadCmd_Payload](#)
Abort Load Command Payload.
- struct [CFE_TBL_NotifyCmd_Payload](#)
Table Management Notification Command Payload.
- struct [CFE_TBL_HousekeepingTlm_Payload](#)
- struct [CFE_TBL_TblRegPacket_Payload](#)

Typedefs

- typedef struct [CFE_TBL_LoadCmd_Payload](#) [CFE_TBL_LoadCmd_Payload_t](#)
Load Table Command Payload.
- typedef struct [CFE_TBL_DumpCmd_Payload](#) [CFE_TBL_DumpCmd_Payload_t](#)
Dump Table Command Payload.
- typedef struct [CFE_TBL_ValidateCmd_Payload](#) [CFE_TBL_ValidateCmd_Payload_t](#)
Validate Table Command Payload.
- typedef struct [CFE_TBL_ActivateCmd_Payload](#) [CFE_TBL_ActivateCmd_Payload_t](#)
Activate Table Command Payload.
- typedef struct [CFE_TBL_DumpRegistryCmd_Payload](#) [CFE_TBL_DumpRegistryCmd_Payload_t](#)
Dump Registry Command Payload.
- typedef struct [CFE_TBL_SendRegistryCmd_Payload](#) [CFE_TBL_SendRegistryCmd_Payload_t](#)
Send Table Registry Command Payload.
- typedef struct [CFE_TBL_DeICDSCmd_Payload](#) [CFE_TBL_DeICDSCmd_Payload_t](#)
Delete Critical Table CDS Command Payload.
- typedef struct [CFE_TBL_AbortLoadCmd_Payload](#) [CFE_TBL_AbortLoadCmd_Payload_t](#)
Abort Load Command Payload.
- typedef struct [CFE_TBL_NotifyCmd_Payload](#) [CFE_TBL_NotifyCmd_Payload_t](#)
Table Management Notification Command Payload.
- typedef struct [CFE_TBL_HousekeepingTlm_Payload](#) [CFE_TBL_HousekeepingTlm_Payload_t](#)
- typedef struct [CFE_TBL_TblRegPacket_Payload](#) [CFE_TBL_TblRegPacket_Payload_t](#)

11.229.1 Detailed Description

Specification for the CFE Event Services (CFE_TBL) command and telemetry message payload and constant definitions.

Definition in file [default_cfe_tbl_msgdefs.h](#).

11.229.2 Typedef Documentation

CFE_TBL_AbortLoadCmd_Payload_t

```
typedef struct CFE\_TBL\_AbortLoadCmd\_Payload CFE\_TBL\_AbortLoadCmd\_Payload\_t
Abort Load Command Payload.
```

For command details, see [CFE_TBL_ABORT_LOAD_CC](#)

CFE_TBL_ActivateCmd_Payload_t

```
typedef struct CFE\_TBL\_ActivateCmd\_Payload CFE\_TBL\_ActivateCmd\_Payload\_t
Activate Table Command Payload.
```

For command details, see [CFE_TBL_ACTIVATE_CC](#)

CFE_TBL_DeICDSCmd_Payload_t

```
typedef struct CFE\_TBL\_DeICDSCmd\_Payload CFE\_TBL\_DeICDSCmd\_Payload\_t
Delete Critical Table CDS Command Payload.
```

For command details, see [CFE_TBL_DELETE_CDS_CC](#)

CFE_TBL_DumpCmd_Payload_t

```
typedef struct CFE\_TBL\_DumpCmd\_Payload CFE\_TBL\_DumpCmd\_Payload\_t
Dump Table Command Payload.
```

For command details, see [CFE_TBL_DUMP_CC](#)

CFE_TBL_DumpRegistryCmd_Payload_t

typedef struct CFE_TBL_DumpRegistryCmd_Payload CFE_TBL_DumpRegistryCmd_Payload_t
 Dump Registry Command Payload.

For command details, see [CFE_TBL_DUMP_REGISTRY_CC](#)

CFE_TBL_HousekeepingTlm_Payload_t

typedef struct CFE_TBL_HousekeepingTlm_Payload CFE_TBL_HousekeepingTlm_Payload_t

Name Table Services Housekeeping Packet

CFE_TBL_LoadCmd_Payload_t

typedef struct CFE_TBL_LoadCmd_Payload CFE_TBL_LoadCmd_Payload_t
 Load Table Command Payload.

For command details, see [CFE_TBL_LOAD_CC](#)

CFE_TBL_NotifyCmd_Payload_t

typedef struct CFE_TBL_NotifyCmd_Payload CFE_TBL_NotifyCmd_Payload_t
 Table Management Notification Command Payload.

Description

Whenever an application that owns a table calls the [CFE_TBL_NotifyByMessage](#) API following the table registration, Table services will generate the following command message with the application specified message ID, command code and parameter whenever the table requires management (e.g. - loads and validations).

CFE_TBL_SendRegistryCmd_Payload_t

typedef struct CFE_TBL_SendRegistryCmd_Payload CFE_TBL_SendRegistryCmd_Payload_t
 Send Table Registry Command Payload.

For command details, see [CFE_TBL_SEND_REGISTRY_CC](#)

CFE_TBL_TblRegPacket_Payload_t

typedef struct CFE_TBL_TblRegPacket_Payload CFE_TBL_TblRegPacket_Payload_t

Name Table Registry Info Packet

CFE_TBL_ValidateCmd_Payload_t

typedef struct CFE_TBL_ValidateCmd_Payload CFE_TBL_ValidateCmd_Payload_t
 Validate Table Command Payload.

For command details, see [CFE_TBL_VALIDATE_CC](#)

11.230 default_cfe_tbl_msgdefs.h

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00016  * limitations under the License.
00017  *****/
00018
00024 #ifndef DEFAULT_CFE_TBL_MSGDEFS_H
00025 #define DEFAULT_CFE_TBL_MSGDEFS_H
00026
00027 #include "common_types.h"
00028 #include "cfe_mission_cfg.h"
00029 #include "cfe_es_extern_typedefs.h"
00030 #include "cfe_time_extern_typedefs.h"
00031 #include "cfe_tbl_extern_typedefs.h"
00032 #include "cfe_tbl_fcncodes.h"
00033
00034 /*****/
00035 /* Command Message Data Formats */
00036 /*****/
00037
00044 typedef struct CFE_TBL_LoadCmd_Payload
00045 {
00046     char LoadFilename[CFE_MISSION_MAX_PATH_LEN];
00047 } CFE_TBL_LoadCmd_Payload_t;
00048
00054 typedef struct CFE_TBL_DumpCmd_Payload
00055 {
00056     CFE_TBL_BufferSelect_Enum_t ActiveTableFlag;
00062     char TableName[CFE_MISSION_TBL_MAX_FULL_NAME_LEN];
00065     char DumpFilename[CFE_MISSION_MAX_PATH_LEN];
00068 } CFE_TBL_DumpCmd_Payload_t;
00069
00075 typedef struct CFE_TBL_ValidateCmd_Payload
00076 {
00077     CFE_TBL_BufferSelect_Enum_t ActiveTableFlag;
00083     char TableName[CFE_MISSION_TBL_MAX_FULL_NAME_LEN];
00086 } CFE_TBL_ValidateCmd_Payload_t;
00087
00093 typedef struct CFE_TBL_ActivateCmd_Payload
00094 {
00095     char TableName[CFE_MISSION_TBL_MAX_FULL_NAME_LEN];
00098 } CFE_TBL_ActivateCmd_Payload_t;
00099
00105 typedef struct CFE_TBL_DumpRegistryCmd_Payload
00106 {
00107     char DumpFilename[CFE_MISSION_MAX_PATH_LEN];
00111 } CFE_TBL_DumpRegistryCmd_Payload_t;
00112
00118 typedef struct CFE_TBL_SendRegistryCmd_Payload
00119 {
00120     char TableName[CFE_MISSION_TBL_MAX_FULL_NAME_LEN];
00125 } CFE_TBL_SendRegistryCmd_Payload_t;
00126
00132 typedef struct CFE_TBL_DelCDSCmd_Payload
00133 {
00134     char TableName[CFE_MISSION_TBL_MAX_FULL_NAME_LEN];
00139 } CFE_TBL_DelCDSCmd_Payload_t;
00140
00146 typedef struct CFE_TBL_AbortLoadCmd_Payload
00147 {
00148     char TableName[CFE_MISSION_TBL_MAX_FULL_NAME_LEN];
00151 } CFE_TBL_AbortLoadCmd_Payload_t;
00152
00153 /*****/
00154
00164 typedef struct CFE_TBL_NotifyCmd_Payload
00165 {
00166     uint32 Parameter;
00167 } CFE_TBL_NotifyCmd_Payload_t;
00168
00169 /*****/
00170
00171 /*****/
00172 /* Telemetry Message Data Formats */
00173 /*****/
00177 typedef struct CFE_TBL_HousekeepingTlm_Payload

```



```

00178 {
00179     /*
00180     ** Task command interface counters...
00181     */
00182     uint8 CommandCounter;
00184     uint8 CommandErrorCounter;
00187     /*
00188     ** Table Registry Statistics
00189     */
00190     uint16 NumTables;
00192     uint16 NumLoadPending;
00195     /*
00196     ** Last Table Validation Results
00197     */
00198     uint16 ValidationCounter;
00200     uint32 LastValCrc;
00202     int32 LastValStatus;
00204     bool ActiveBuffer;
00206     char LastValTableName[CFE_MISSION_TBL_MAX_FULL_NAME_LEN];
00208     uint8 SuccessValCounter;
00210     uint8 FailedValCounter;
00212     uint8 NumValRequests;
00215     /*
00216     ** Ground system interface information
00217     */
00218     uint8 NumFreeSharedBufs;
00220     uint8 ByteAlignPad1;
00222     CFE_ES_MemHandle_t MemPoolHandle;
00224     CFE_TIME_SysTime_t LastUpdateTime;
00226     char LastUpdatedTable[CFE_MISSION_TBL_MAX_FULL_NAME_LEN];
00228     char LastFileLoaded[CFE_MISSION_MAX_PATH_LEN];
00230     char LastFileDumped[CFE_MISSION_MAX_PATH_LEN];
00232     char LastTableLoaded[CFE_MISSION_TBL_MAX_FULL_NAME_LEN];
00234 } CFE_TBL_HousekeepingTlm_Payload_t;
00235
00239 typedef struct CFE_TBL_TblRegPacket_Payload
00240 {
00241     CFE_ES_MemOffset_t Size;
00243     uint32 Crc;
00245     uint8 Padding1[4];
00246     CFE_ES_MemAddress_t ActiveBufferAddr;
00248     CFE_ES_MemAddress_t InactiveBufferAddr;
00250     CFE_ES_MemAddress_t ValidationFuncPtr;
00252     CFE_TIME_SysTime_t TimeOfLastUpdate;
00254     CFE_TIME_SysTime_t FileTime;
00256     bool TableLoadedOnce;
00258     bool LoadPending;
00260     bool DumpOnly;
00262     bool DoubleBuffered;
00264     char Name[CFE_MISSION_TBL_MAX_FULL_NAME_LEN];
00266     char LastFileLoaded[CFE_MISSION_MAX_PATH_LEN];
00268     char OwnerAppName[CFE_MISSION_MAX_API_LEN];
00270     bool Critical;
00272     uint8 Padding2[7];
00273 } CFE_TBL_TblRegPacket_Payload_t;
00274
00275 #endif

```

11.231 cfe/modules/tbl/config/default_cfe_tbl_msgid_values.h File Reference

```

#include "cfe_core_api_base_msgids.h"
#include "cfe_tbl_topicids.h"

```

Macros

- #define [CFE_PLATFORM_TBL_CMD_MIDVAL\(x\)](#) [CFE_PLATFORM_CMD_TOPICID_TO_MIDV\(CFE_↵MISSION_TBL_###_TOPICID\)](#)
- #define [CFE_PLATFORM_TBL_TLM_MIDVAL\(x\)](#) [CFE_PLATFORM_TLM_TOPICID_TO_MIDV\(CFE_MISSION_↵_TBL_###_TOPICID\)](#)

11.231.1 Detailed Description

CFE Event Services (CFE_TBL) Application Message IDs
Definition in file [default_cfe_tbl_msgid_values.h](#).

11.231.2 Macro Definition Documentation

CFE_PLATFORM_TBL_CMD_MIDVAL

```
#define CFE_PLATFORM_TBL_CMD_MIDVAL(  
    x ) CFE_PLATFORM_CMD_TOPICID_TO_MIDV(CFE_MISSION_TBL_##x##_TOPICID)
```

Definition at line 29 of file [default_cfe_tbl_msgid_values.h](#).

CFE_PLATFORM_TBL_TLM_MIDVAL

```
#define CFE_PLATFORM_TBL_TLM_MIDVAL(  
    x ) CFE_PLATFORM_TLM_TOPICID_TO_MIDV(CFE_MISSION_TBL_##x##_TOPICID)
```

Definition at line 30 of file [default_cfe_tbl_msgid_values.h](#).

11.232 default_cfe_tbl_msgid_values.h

[Go to the documentation of this file.](#)

```
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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.  
00015  * See the License for the specific language governing permissions and  
00016  * limitations under the License.  
00017  *****/  
00018  
00023 #ifndef DEFAULT_CFE_TBL_MSGID_VALUES_H  
00024 #define DEFAULT_CFE_TBL_MSGID_VALUES_H  
00025  
00026 #include "cfe_core_api_base_msgids.h"  
00027 #include "cfe_tbl_topicsids.h"  
00028  
00029 #define CFE_PLATFORM_TBL_CMD_MIDVAL(x) CFE_PLATFORM_CMD_TOPICID_TO_MIDV(CFE_MISSION_TBL_##x##_TOPICID)  
00030 #define CFE_PLATFORM_TBL_TLM_MIDVAL(x) CFE_PLATFORM_TLM_TOPICID_TO_MIDV(CFE_MISSION_TBL_##x##_TOPICID)  
00031  
00032 #endif
```

11.233 cfe/modules/tbl/config/default_cfe_tbl_msgids.h File Reference

```
#include "cfe_tbl_msgid_values.h"
```

Macros

- #define [CFE_TBL_CMD_MID](#) [CFE_PLATFORM_TBL_CMD_MIDVAL](#)(CMD)
- #define [CFE_TBL_SEND_HK_MID](#) [CFE_PLATFORM_TBL_CMD_MIDVAL](#)(SEND_HK)
- #define [CFE_TBL_HK_TLM_MID](#) [CFE_PLATFORM_TBL_TLM_MIDVAL](#)(HK_TLM)
- #define [CFE_TBL_REG_TLM_MID](#) [CFE_PLATFORM_TBL_TLM_MIDVAL](#)(REG_TLM)

11.233.1 Detailed Description

CFE Event Services (CFE_TBL) Application Message IDs
Definition in file [default_cfe_tbl_msgids.h](#).

11.233.2 Macro Definition Documentation

CFE_TBL_CMD_MID

```
#define CFE_TBL_CMD_MID CFE_PLATFORM_TBL_CMD_MIDVAL (CMD)
```

Definition at line 31 of file [default_cfe_tbl_msgids.h](#).

CFE_TBL_HK_TLM_MID

```
#define CFE_TBL_HK_TLM_MID CFE_PLATFORM_TBL_TLM_MIDVAL (HK_TLM)
```

Definition at line 37 of file [default_cfe_tbl_msgids.h](#).

CFE_TBL_REG_TLM_MID

```
#define CFE_TBL_REG_TLM_MID CFE_PLATFORM_TBL_TLM_MIDVAL (REG_TLM)
```

Definition at line 38 of file [default_cfe_tbl_msgids.h](#).

CFE_TBL_SEND_HK_MID

```
#define CFE_TBL_SEND_HK_MID CFE_PLATFORM_TBL_CMD_MIDVAL (SEND_HK)
```

Definition at line 32 of file [default_cfe_tbl_msgids.h](#).

11.234 default_cfe_tbl_msgids.h

[Go to the documentation of this file.](#)

```
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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00023 #ifndef DEFAULT_CFE_TBL_MSGIDS_H
00024 #define DEFAULT_CFE_TBL_MSGIDS_H
00025
00026 #include "cfe_tbl_msgid_values.h"
00027
00028 /*
00029 ** cFE Command Message Id's
00030 */
00031 #define CFE_TBL_CMD_MID CFE_PLATFORM_TBL_CMD_MIDVAL (CMD)
00032 #define CFE_TBL_SEND_HK_MID CFE_PLATFORM_TBL_CMD_MIDVAL (SEND_HK)
00033
00034 /*
00035 ** CFE Telemetry Message Id's
00036 */
00037 #define CFE_TBL_HK_TLM_MID CFE_PLATFORM_TBL_TLM_MIDVAL (HK_TLM)
00038 #define CFE_TBL_REG_TLM_MID CFE_PLATFORM_TBL_TLM_MIDVAL (REG_TLM)
00039
00040 #endif
```

11.235 cfe/modules/tbl/config/default_cfe_tbl_msgstruct.h File Reference

```
#include "cfe_tbl_msgdefs.h"
#include "cfe_msg_hdr.h"
```

Data Structures

- struct [CFE_TBL_NoopCmd](#)
- struct [CFE_TBL_ResetCountersCmd](#)
- struct [CFE_TBL_SendHkCmd](#)
- struct [CFE_TBL_LoadCmd](#)
Load Table Command.
- struct [CFE_TBL_DumpCmd](#)
- struct [CFE_TBL_ValidateCmd](#)
Validate Table Command.
- struct [CFE_TBL_ActivateCmd](#)
Activate Table Command.
- struct [CFE_TBL_DumpRegistryCmd](#)
Dump Registry Command.
- struct [CFE_TBL_SendRegistryCmd](#)
Send Table Registry Command.
- struct [CFE_TBL_DeleteCDSCmd](#)
Delete Critical Table CDS Command.
- struct [CFE_TBL_AbortLoadCmd](#)
Abort Load Command.
- struct [CFE_TBL_NotifyCmd](#)
- struct [CFE_TBL_HousekeepingTlm](#)
- struct [CFE_TBL_TableRegistryTlm](#)

Typedefs

- typedef struct [CFE_TBL_NoopCmd](#) [CFE_TBL_NoopCmd_t](#)
- typedef struct [CFE_TBL_ResetCountersCmd](#) [CFE_TBL_ResetCountersCmd_t](#)
- typedef struct [CFE_TBL_SendHkCmd](#) [CFE_TBL_SendHkCmd_t](#)
- typedef struct [CFE_TBL_LoadCmd](#) [CFE_TBL_LoadCmd_t](#)
Load Table Command.
- typedef struct [CFE_TBL_DumpCmd](#) [CFE_TBL_DumpCmd_t](#)
- typedef struct [CFE_TBL_ValidateCmd](#) [CFE_TBL_ValidateCmd_t](#)
Validate Table Command.
- typedef struct [CFE_TBL_ActivateCmd](#) [CFE_TBL_ActivateCmd_t](#)
Activate Table Command.
- typedef struct [CFE_TBL_DumpRegistryCmd](#) [CFE_TBL_DumpRegistryCmd_t](#)
Dump Registry Command.
- typedef struct [CFE_TBL_SendRegistryCmd](#) [CFE_TBL_SendRegistryCmd_t](#)
Send Table Registry Command.
- typedef struct [CFE_TBL_DeleteCDSCmd](#) [CFE_TBL_DeleteCDSCmd_t](#)
Delete Critical Table CDS Command.
- typedef struct [CFE_TBL_AbortLoadCmd](#) [CFE_TBL_AbortLoadCmd_t](#)
Abort Load Command.
- typedef struct [CFE_TBL_NotifyCmd](#) [CFE_TBL_NotifyCmd_t](#)
- typedef struct [CFE_TBL_HousekeepingTlm](#) [CFE_TBL_HousekeepingTlm_t](#)
- typedef struct [CFE_TBL_TableRegistryTlm](#) [CFE_TBL_TableRegistryTlm_t](#)

11.235.1 Detailed Description

Purpose: cFE Executive Services (TBL) Command and Telemetry packet definition file.
Definition in file [default_cfe_tbl_msgstruct.h](#).

11.235.2 Typedef Documentation

CFE_TBL_AbortLoadCmd_t

```
typedef struct CFE_TBL_AbortLoadCmd CFE_TBL_AbortLoadCmd_t
```

Abort Load Command.

CFE_TBL_ActivateCmd_t

```
typedef struct CFE_TBL_ActivateCmd CFE_TBL_ActivateCmd_t
```

Activate Table Command.

CFE_TBL_DeleteCDSCmd_t

```
typedef struct CFE_TBL_DeleteCDSCmd CFE_TBL_DeleteCDSCmd_t
```

Delete Critical Table CDS Command.

CFE_TBL_DumpCmd_t

```
typedef struct CFE_TBL_DumpCmd CFE_TBL_DumpCmd_t
```

/brief Dump Table Command

CFE_TBL_DumpRegistryCmd_t

```
typedef struct CFE_TBL_DumpRegistryCmd CFE_TBL_DumpRegistryCmd_t
```

Dump Registry Command.

CFE_TBL_HousekeepingTlm_t

```
typedef struct CFE_TBL_HousekeepingTlm CFE_TBL_HousekeepingTlm_t
```

CFE_TBL_LoadCmd_t

```
typedef struct CFE_TBL_LoadCmd CFE_TBL_LoadCmd_t
```

Load Table Command.

CFE_TBL_NoopCmd_t

```
typedef struct CFE_TBL_NoopCmd CFE_TBL_NoopCmd_t
```

CFE_TBL_NotifyCmd_t

```
typedef struct CFE_TBL_NotifyCmd CFE_TBL_NotifyCmd_t
```

/brief Table Management Notification Command

CFE_TBL_ResetCountersCmd_t

```
typedef struct CFE_TBL_ResetCountersCmd CFE_TBL_ResetCountersCmd_t
```

CFE_TBL_SendHkCmd_t

```
typedef struct CFE_TBL_SendHkCmd CFE_TBL_SendHkCmd_t
```

CFE_TBL_SendRegistryCmd_t

typedef struct CFE_TBL_SendRegistryCmd CFE_TBL_SendRegistryCmd_t
Send Table Registry Command.

CFE_TBL_TableRegistryTlm_t

typedef struct CFE_TBL_TableRegistryTlm CFE_TBL_TableRegistryTlm_t

CFE_TBL_ValidateCmd_t

typedef struct CFE_TBL_ValidateCmd CFE_TBL_ValidateCmd_t
Validate Table Command.

11.236 default_cfe_tbl_msgstruct.h

[Go to the documentation of this file.](#)

```

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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00025 #ifndef DEFAULT_CFE_TBL_MSGSTRUCT_H
00026 #define DEFAULT_CFE_TBL_MSGSTRUCT_H
00027
00028 /*****
00029  * Includes
00030  *****/
00031 #include "cfe_tbl_msgdefs.h"
00032 #include "cfe_msg_hdr.h"
00033
00034 /*****/
00035 /* Command Message Data Formats */
00036 /*****/
00037
00038 /*
00039  * Typedefs for each of the messages that share the command format
00040  * Allows each command to evolve independently, while following the command
00041  * handler prototype convention
00042  */
00043 typedef struct CFE_TBL_NoopCmd
00044 {
00045     CFE_MSG_CommandHeader_t CommandHeader;
00046 } CFE_TBL_NoopCmd_t;
00047
00048 typedef struct CFE_TBL_ResetCountersCmd
00049 {
00050     CFE_MSG_CommandHeader_t CommandHeader;
00051 } CFE_TBL_ResetCountersCmd_t;
00052
00053 typedef struct CFE_TBL_SendHkCmd
00054 {
00055     CFE_MSG_CommandHeader_t CommandHeader;
00056 } CFE_TBL_SendHkCmd_t;
00057
00061 typedef struct CFE_TBL_LoadCmd
00062 {
00063     CFE_MSG_CommandHeader_t CommandHeader;
00064     CFE_TBL_LoadCmd_Payload_t Payload;
00065 } CFE_TBL_LoadCmd_t;
00066

```

```

00070 typedef struct CFE_TBL_DumpCmd
00071 {
00072     CFE_MSG_CommandHeader_t    CommandHeader;
00073     CFE_TBL_DumpCmd_Payload_t Payload;
00074 } CFE_TBL_DumpCmd_t;
00075
00079 typedef struct CFE_TBL_ValidateCmd
00080 {
00081     CFE_MSG_CommandHeader_t    CommandHeader;
00082     CFE_TBL_ValidateCmd_Payload_t Payload;
00083 } CFE_TBL_ValidateCmd_t;
00084
00088 typedef struct CFE_TBL_ActivateCmd
00089 {
00090     CFE_MSG_CommandHeader_t    CommandHeader;
00091     CFE_TBL_ActivateCmd_Payload_t Payload;
00092 } CFE_TBL_ActivateCmd_t;
00093
00097 typedef struct CFE_TBL_DumpRegistryCmd
00098 {
00099     CFE_MSG_CommandHeader_t    CommandHeader;
00100     CFE_TBL_DumpRegistryCmd_Payload_t Payload;
00101 } CFE_TBL_DumpRegistryCmd_t;
00102
00106 typedef struct CFE_TBL_SendRegistryCmd
00107 {
00108     CFE_MSG_CommandHeader_t    CommandHeader;
00109     CFE_TBL_SendRegistryCmd_Payload_t Payload;
00110 } CFE_TBL_SendRegistryCmd_t;
00111
00115 typedef struct CFE_TBL_DeleteCDSCmd
00116 {
00117     CFE_MSG_CommandHeader_t    CommandHeader;
00118     CFE_TBL_DelCDSCmd_Payload_t Payload;
00119 } CFE_TBL_DeleteCDSCmd_t;
00120
00124 typedef struct CFE_TBL_AbortLoadCmd
00125 {
00126     CFE_MSG_CommandHeader_t    CommandHeader;
00127     CFE_TBL_AbortLoadCmd_Payload_t Payload;
00128 } CFE_TBL_AbortLoadCmd_t;
00129
00130 /*****
00131
00135 typedef struct CFE_TBL_NotifyCmd
00136 {
00137     CFE_MSG_CommandHeader_t    CommandHeader;
00138     CFE_TBL_NotifyCmd_Payload_t Payload;
00139 } CFE_TBL_NotifyCmd_t;
00140
00141 /*****
00142
00143 /*****
00144 /* Telemetry Message Data Formats */
00145 /*****
00146
00147 typedef struct CFE_TBL_HousekeepingTlm
00148 {
00149     CFE_MSG_TelemetryHeader_t    TelemetryHeader;
00150     CFE_TBL_HousekeepingTlm_Payload_t Payload;
00151 } CFE_TBL_HousekeepingTlm_t;
00152
00153 typedef struct CFE_TBL_TableRegistryTlm
00154 {
00155     CFE_MSG_TelemetryHeader_t    TelemetryHeader;
00156     CFE_TBL_TblRegPacket_Payload_t Payload;
00157 } CFE_TBL_TableRegistryTlm_t;
00158
00159 #endif

```

11.237 cfe/modules/tbl/config/default_cfe_tbl_platform_cfg.h File Reference

```

#include "cfe_tbl_mission_cfg.h"
#include "cfe_tbl_internal_cfg_values.h"

```

Macros

- #define CFE_PLATFORM_TBL_START_TASK_PRIORITY CFE_PLATFORM_TBL_CFGVAL(START_TASK_PRIORITY)
- #define DEFAULT_CFE_PLATFORM_TBL_START_TASK_PRIORITY 70
- #define CFE_PLATFORM_TBL_START_TASK_STACK_SIZE CFE_PLATFORM_TBL_CFGVAL(START_TASK_STACK_SIZE)
- #define DEFAULT_CFE_PLATFORM_TBL_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
- #define CFE_PLATFORM_TBL_BUF_MEMORY_BYTES CFE_PLATFORM_TBL_CFGVAL(BUF_MEMORY_BYTES)
- #define DEFAULT_CFE_PLATFORM_TBL_BUF_MEMORY_BYTES 524288
- #define CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE CFE_PLATFORM_TBL_CFGVAL(MAX_DBL_TABLE_SIZE)
- #define DEFAULT_CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE 16384
- #define CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE CFE_PLATFORM_TBL_CFGVAL(MAX_SNGL_TABLE_SIZE)
- #define DEFAULT_CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE 16384
- #define CFE_PLATFORM_TBL_MAX_NUM_TABLES CFE_PLATFORM_TBL_CFGVAL(MAX_NUM_TABLES)
- #define DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_TABLES 128
- #define CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES CFE_PLATFORM_TBL_CFGVAL(MAX_CRITICAL_TABLES)
- #define DEFAULT_CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES 32
- #define CFE_PLATFORM_TBL_MAX_NUM_HANDLES CFE_PLATFORM_TBL_CFGVAL(MAX_NUM_HANDLES)
- #define DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_HANDLES 256
- #define CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS CFE_PLATFORM_TBL_CFGVAL(MAX_SIMULTANEOUS_LOADS)
- #define DEFAULT_CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS 4
- #define CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS CFE_PLATFORM_TBL_CFGVAL(MAX_NUM_VALIDATIONS)
- #define DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS 10
- #define CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE CFE_PLATFORM_TBL_CFGVAL(DEFAULT_REG_DUMP_FILE)
- #define DEFAULT_CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE "/ram/cfe_tbl_reg.log"
- #define CFE_PLATFORM_TBL_VALID_SCID_COUNT CFE_PLATFORM_TBL_CFGVAL(VALID_SCID_COUNT)
- #define DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_COUNT 0
- #define CFE_PLATFORM_TBL_U32FROM4CHARS(_C1, _C2, _C3, _C4) ((uint32)(_C1) << 24 | (uint32)(_C2) << 16 | (uint32)(_C3) << 8 | (uint32)(_C4))
- #define CFE_PLATFORM_TBL_VALID_SCID_1 CFE_PLATFORM_TBL_CFGVAL(VALID_SCID_1)
- #define DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_1 (0x42)
- #define CFE_PLATFORM_TBL_VALID_SCID_2 CFE_PLATFORM_TBL_CFGVAL(VALID_SCID_2)
- #define DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_2 (CFE_PLATFORM_TBL_U32FROM4CHARS('a', 'b', 'c', 'd'))
- #define CFE_PLATFORM_TBL_VALID_PRID_COUNT CFE_PLATFORM_TBL_CFGVAL(VALID_PRID_COUNT)
- #define DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_COUNT 0
- #define CFE_PLATFORM_TBL_VALID_PRID_1 CFE_PLATFORM_TBL_CFGVAL(VALID_PRID_1)
- #define DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_1 (1)
- #define CFE_PLATFORM_TBL_VALID_PRID_2 CFE_PLATFORM_TBL_CFGVAL(VALID_PRID_2)
- #define DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_2 (CFE_PLATFORM_TBL_U32FROM4CHARS('a', 'b', 'c', 'd'))

- `#define CFE_PLATFORM_TBL_VALID_PRID_3 CFE_PLATFORM_TBL_CFGVAL(VALID_PRID_3)`
- `#define DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_3 0`
- `#define CFE_PLATFORM_TBL_VALID_PRID_4 CFE_PLATFORM_TBL_CFGVAL(VALID_PRID_4)`
- `#define DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_4 0`

11.237.1 Detailed Description

CFE Table Services (CFE_TBL) Application Platform Configuration Header File

This is a compatibility header for the "platform_cfg.h" file that has traditionally provided both public and private config definitions for each CFS app.

These definitions are now provided in two separate files, one for the public/mission scope and one for internal scope.

Note

This file may be overridden/superseded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [default_cfe_tbl_platform_cfg.h](#).

11.237.2 Macro Definition Documentation

CFE_PLATFORM_TBL_BUF_MEMORY_BYTES

```
#define CFE_PLATFORM_TBL_BUF_MEMORY_BYTES CFE_PLATFORM_TBL_CFGVAL(BUF_MEMORY_BYTES)
```

Purpose Size of Table Services Table Memory Pool

Description:

Defines the TOTAL size of the memory pool that cFE Table Services allocates from the system. The size must be large enough to provide memory for each registered table, the inactive buffers for double buffered tables and for the shared inactive buffers for single buffered tables.

Limits

The cFE does not place a limit on the size of this parameter.

Definition at line 83 of file [default_cfe_tbl_platform_cfg.h](#).

CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE

```
#define CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE CFE_PLATFORM_TBL_CFGVAL(DEFAULT_REG_DUMP_FILE)
```

Purpose Default Filename for a Table Registry Dump

Description:

Defines the file name used to store the table registry when no filename is specified in the dump registry command.

Limits

The length of each string, including the NULL terminator cannot exceed the [OS_MAX_PATH_LEN](#) value.

Definition at line 205 of file [default_cfe_tbl_platform_cfg.h](#).

CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES

```
#define CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES CFE_PLATFORM_TBL_CFGVAL (MAX_CRITICAL_TABLES)
```

Purpose Maximum Number of Critical Tables that can be Registered

Description:

Defines the maximum number of critical tables supported by this processor's Table Services.

Limits

This number must be less than 32767. It should be recognized that this parameter determines the size of the Critical Table Registry which is maintained in the Critical Data Store. An excessively high number will waste Critical Data Store memory. Therefore, this number must not exceed the value defined in CFE_PLATFORM_ES↔_CDS_MAX_NUM_ENTRIES.

Definition at line 142 of file [default_cfe_tbl_platform_cfg.h](#).

CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE

```
#define CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE CFE_PLATFORM_TBL_CFGVAL (MAX_DBL_TABLE_SIZE)
```

Purpose Maximum Size Allowed for a Double Buffered Table

Description:

Defines the maximum allowed size (in bytes) of a double buffered table.

Limits

The cFE does not place a limit on the size of this parameter but it must be less than half of [CFE_PLATFORM_TBL_BUF_MEMORY_B](#)

Definition at line 96 of file [default_cfe_tbl_platform_cfg.h](#).

CFE_PLATFORM_TBL_MAX_NUM_HANDLES

```
#define CFE_PLATFORM_TBL_MAX_NUM_HANDLES CFE_PLATFORM_TBL_CFGVAL (MAX_NUM_HANDLES)
```

Purpose Maximum Number of Table Handles

Description:

Defines the maximum number of Table Handles.

Limits

This number must be less than 32767. This number must be at least as big as the number of tables ([CFE_PLATFORM_TBL_MAX_NUM_TABLES](#)) and should be set higher if tables are shared between applications.

Definition at line 156 of file [default_cfe_tbl_platform_cfg.h](#).

CFE_PLATFORM_TBL_MAX_NUM_TABLES

```
#define CFE_PLATFORM_TBL_MAX_NUM_TABLES CFE_PLATFORM_TBL_CFGVAL (MAX_NUM_TABLES)
```

Purpose Maximum Number of Tables Allowed to be Registered

Description:

Defines the maximum number of tables supported by this processor's Table Services.

Limits

This number must be less than 32767. It should be recognized that this parameter determines the size of the Table Registry. An excessively high number will waste memory.

Definition at line 127 of file [default_cfe_tbl_platform_cfg.h](#).

CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS

```
#define CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS CFE_PLATFORM_TBL_CFGVAL (MAX_NUM_VALIDATIONS)
```

Purpose Maximum Number of Simultaneous Table Validations

Description:

Defines the maximum number of pending validations that the Table Services can handle at any one time. When a table has a validation function, a validation request is made of the application to perform that validation. This number determines how many of those requests can be outstanding at any one time.

Limits

This number must be less than 32767. An excessively high number will degrade system performance and waste memory. A number less than 20 is suggested but not required.

Definition at line 191 of file [default_cfe_tbl_platform_cfg.h](#).

CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS

```
#define CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS CFE_PLATFORM_TBL_CFGVAL (MAX_SIMULTANEOUS_LOADS)
```

Purpose Maximum Number of Simultaneous Loads to Support

Description:

Defines the maximum number of single buffered tables that can be loaded simultaneously. This number is used to determine the number of shared buffers to allocate.

Limits

This number must be less than 32767. An excessively high number will degrade system performance and waste memory. A number less than 5 is suggested but not required.

Definition at line 172 of file [default_cfe_tbl_platform_cfg.h](#).

CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE

```
#define CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE CFE_PLATFORM_TBL_CFGVAL (MAX_SNGL_TABLE_SIZE)
```

Purpose Maximum Size Allowed for a Single Buffered Table

Description:

Defines the maximum allowed size (in bytes) of a single buffered table. **NOTE:** This size determines the size of all shared table buffers. Therefore, this size will be multiplied by [CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS](#) below when allocating memory for shared tables.

Limits

The cFE does not place a limit on the size of this parameter but it must be small enough to allow for [CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS](#) number of tables to fit into [CFE_PLATFORM_TBL_BUF_MEMORY_BYTES](#).

Definition at line 113 of file [default_cfe_tbl_platform_cfg.h](#).

CFE_PLATFORM_TBL_START_TASK_PRIORITY

```
#define CFE_PLATFORM_TBL_START_TASK_PRIORITY CFE_PLATFORM_TBL_CFGVAL (START_TASK_PRIORITY)
```

Purpose Define TBL Task Priority

Description:

Defines the cFE_TBL Task priority.

Limits

Not Applicable

Definition at line 50 of file [default_cfe_tbl_platform_cfg.h](#).

CFE_PLATFORM_TBL_START_TASK_STACK_SIZE

```
#define CFE_PLATFORM_TBL_START_TASK_STACK_SIZE CFE_PLATFORM_TBL_CFGVAL (START_TASK_STACK_SIZE)
```

Purpose Define TBL Task Stack Size

Description:

Defines the cFE_TBL Task Stack Size

Limits

There is a lower limit of 2048 on this configuration parameter. There are no restrictions on the upper limit however, the maximum stack size is system dependent and should be verified. Most operating systems provide tools for measuring the amount of stack used by a task during operation. It is always a good idea to verify that no more than 1/2 of the stack is used.

Definition at line 66 of file [default_cfe_tbl_platform_cfg.h](#).

CFE_PLATFORM_TBL_U32FROM4CHARS

```
#define CFE_PLATFORM_TBL_U32FROM4CHARS(  
    _C1,  
    _C2,  
    _C3,  
    _C4 )  ((uint32) (_C1) << 24 | (uint32) (_C2) << 16 | (uint32) (_C3) << 8 | (uint32) (_C4))
```

Definition at line 229 of file [default_cfe_tbl_platform_cfg.h](#).

CFE_PLATFORM_TBL_VALID_PRID_1

```
#define CFE_PLATFORM_TBL_VALID_PRID_1 CFE_PLATFORM_TBL_CFGVAL(VALID_PRID_1)
```

Purpose Processor ID values used for table load validation

Description:

Defines the processor ID values used for validating the processor ID field in the table file header. To be valid, the spacecraft ID specified in the table file header must match one of the values defined here.

Limits

This value can be any 32 bit unsigned integer.

Definition at line 282 of file [default_cfe_tbl_platform_cfg.h](#).

CFE_PLATFORM_TBL_VALID_PRID_2

```
#define CFE_PLATFORM_TBL_VALID_PRID_2 CFE_PLATFORM_TBL_CFGVAL(VALID_PRID_2)
```

Definition at line 285 of file [default_cfe_tbl_platform_cfg.h](#).

CFE_PLATFORM_TBL_VALID_PRID_3

```
#define CFE_PLATFORM_TBL_VALID_PRID_3 CFE_PLATFORM_TBL_CFGVAL(VALID_PRID_3)
```

Definition at line 288 of file [default_cfe_tbl_platform_cfg.h](#).

CFE_PLATFORM_TBL_VALID_PRID_4

```
#define CFE_PLATFORM_TBL_VALID_PRID_4 CFE_PLATFORM_TBL_CFGVAL(VALID_PRID_4)
```

Definition at line 291 of file [default_cfe_tbl_platform_cfg.h](#).

CFE_PLATFORM_TBL_VALID_PRID_COUNT

```
#define CFE_PLATFORM_TBL_VALID_PRID_COUNT CFE_PLATFORM_TBL_CFGVAL(VALID_PRID_COUNT)
```

Purpose Number of Processor ID's specified for validation

Description:

Defines the number of specified processor ID values that are verified during table loads. If the number is zero then no validation of the processor ID field in the table file header is performed when tables are loaded. Non-zero values indicate how many values from the list of processor ID's defined below are compared to the processor ID field in the table file header. The ELF2CFETBL tool may be used to create table files with specified processor ID values.

Limits

This number must be greater than or equal to zero and less than or equal to 4.

Definition at line 267 of file [default_cfe_tbl_platform_cfg.h](#).

CFE_PLATFORM_TBL_VALID_SCID_1

```
#define CFE_PLATFORM_TBL_VALID_SCID_1 CFE_PLATFORM_TBL_CFGVAL (VALID_SCID_1)
```

Purpose Spacecraft ID values used for table load validation

Description:

Defines the spacecraft ID values used for validating the spacecraft ID field in the table file header. To be valid, the spacecraft ID specified in the table file header must match one of the values defined here.

Limits

This value can be any 32 bit unsigned integer.

Definition at line 244 of file [default_cfe_tbl_platform_cfg.h](#).

CFE_PLATFORM_TBL_VALID_SCID_2

```
#define CFE_PLATFORM_TBL_VALID_SCID_2 CFE_PLATFORM_TBL_CFGVAL (VALID_SCID_2)
```

Definition at line 247 of file [default_cfe_tbl_platform_cfg.h](#).

CFE_PLATFORM_TBL_VALID_SCID_COUNT

```
#define CFE_PLATFORM_TBL_VALID_SCID_COUNT CFE_PLATFORM_TBL_CFGVAL (VALID_SCID_COUNT)
```

Purpose Number of Spacecraft ID's specified for validation

Description:

Defines the number of specified spacecraft ID values that are verified during table loads. If the number is zero then no validation of the spacecraft ID field in the table file header is performed when tables are loaded. Non-zero values indicate how many values from the list of spacecraft ID's defined below are compared to the spacecraft ID field in the table file header. The ELF2CFETBL tool may be used to create table files with specified spacecraft ID values.

Limits

This number must be greater than or equal to zero and less than or equal to 2.

Definition at line 225 of file [default_cfe_tbl_platform_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_BUF_MEMORY_BYTES

```
#define DEFAULT_CFE_PLATFORM_TBL_BUF_MEMORY_BYTES 524288
```

Definition at line 84 of file [default_cfe_tbl_platform_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE

```
#define DEFAULT_CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE "/ram/cfe_tbl_reg.log"
```

Definition at line 206 of file [default_cfe_tbl_platform_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES

```
#define DEFAULT_CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES 32
```

Definition at line 143 of file [default_cfe_tbl_platform_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE

#define DEFAULT_CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE 16384
Definition at line 97 of file [default_cfe_tbl_platform_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_HANDLES

#define DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_HANDLES 256
Definition at line 157 of file [default_cfe_tbl_platform_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_TABLES

#define DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_TABLES 128
Definition at line 128 of file [default_cfe_tbl_platform_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS

#define DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS 10
Definition at line 192 of file [default_cfe_tbl_platform_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS

#define DEFAULT_CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS 4
Definition at line 173 of file [default_cfe_tbl_platform_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE

#define DEFAULT_CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE 16384
Definition at line 114 of file [default_cfe_tbl_platform_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_START_TASK_PRIORITY

#define DEFAULT_CFE_PLATFORM_TBL_START_TASK_PRIORITY 70
Definition at line 51 of file [default_cfe_tbl_platform_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_START_TASK_STACK_SIZE

#define DEFAULT_CFE_PLATFORM_TBL_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
Definition at line 67 of file [default_cfe_tbl_platform_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_1

#define DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_1 (1)
Definition at line 283 of file [default_cfe_tbl_platform_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_2

#define DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_2 (CFE_PLATFORM_TBL_U32FROM4CHARS('a', 'b', 'c', 'd'))
Definition at line 286 of file [default_cfe_tbl_platform_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_3

#define DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_3 0
Definition at line 289 of file [default_cfe_tbl_platform_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_4

#define DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_4 0
 Definition at line 292 of file [default_cfe_tbl_platform_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_COUNT

#define DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_COUNT 0
 Definition at line 268 of file [default_cfe_tbl_platform_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_1

#define DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_1 (0x42)
 Definition at line 245 of file [default_cfe_tbl_platform_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_2

#define DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_2 (CFE_PLATFORM_TBL_U32FROM4CHARS('a', 'b', 'c', 'd'))
 Definition at line 248 of file [default_cfe_tbl_platform_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_COUNT

#define DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_COUNT 0
 Definition at line 226 of file [default_cfe_tbl_platform_cfg.h](#).

11.238 default_cfe_tbl_platform_cfg.h

[Go to the documentation of this file.](#)

```

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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
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00016  * limitations under the License.
00017  *****/
00018
00035 #ifndef DEFAULT_CFE_TBL_PLATFORM_CFG_H
00036 #define DEFAULT_CFE_TBL_PLATFORM_CFG_H
00037
00038 #include "cfe_tbl_mission_cfg.h"
00039 #include "cfe_tbl_internal_cfg_values.h"
00040
00050 #define CFE_PLATFORM_TBL_START_TASK_PRIORITY CFE_PLATFORM_TBL_CFGVAL(START_TASK_PRIORITY)
00051 #define DEFAULT_CFE_PLATFORM_TBL_START_TASK_PRIORITY 70
00052
00066 #define CFE_PLATFORM_TBL_START_TASK_STACK_SIZE CFE_PLATFORM_TBL_CFGVAL(START_TASK_STACK_SIZE)
00067 #define DEFAULT_CFE_PLATFORM_TBL_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
00068
00069 /* Platform Configuration Parameters for Table Service (TBL) */
00070
00083 #define CFE_PLATFORM_TBL_BUF_MEMORY_BYTES CFE_PLATFORM_TBL_CFGVAL(BUF_MEMORY_BYTES)
00084 #define DEFAULT_CFE_PLATFORM_TBL_BUF_MEMORY_BYTES 524288
00085
00096 #define CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE CFE_PLATFORM_TBL_CFGVAL(MAX_DBL_TABLE_SIZE)
00097 #define DEFAULT_CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE 16384
00098
00113 #define CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE CFE_PLATFORM_TBL_CFGVAL(MAX_SNGL_TABLE_SIZE)
00114 #define DEFAULT_CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE 16384
00115

```



```

00127 #define CFE_PLATFORM_TBL_MAX_NUM_TABLES          CFE_PLATFORM_TBL_CFGVAL(MAX_NUM_TABLES)
00128 #define DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_TABLES 128
00129
00142 #define CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES        CFE_PLATFORM_TBL_CFGVAL(MAX_CRITICAL_TABLES)
00143 #define DEFAULT_CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES 32
00144
00156 #define CFE_PLATFORM_TBL_MAX_NUM_HANDLES            CFE_PLATFORM_TBL_CFGVAL(MAX_NUM_HANDLES)
00157 #define DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_HANDLES 256
00158
00172 #define CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS     CFE_PLATFORM_TBL_CFGVAL(MAX_SIMULTANEOUS_LOADS)
00173 #define DEFAULT_CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS 4
00174
00191 #define CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS         CFE_PLATFORM_TBL_CFGVAL(MAX_NUM_VALIDATIONS)
00192 #define DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS 10
00193
00205 #define CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE       CFE_PLATFORM_TBL_CFGVAL(DEFAULT_REG_DUMP_FILE)
00206 #define DEFAULT_CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE "/ram/cfe_tbl_reg.log"
00207
00225 #define CFE_PLATFORM_TBL_VALID_SCID_COUNT            CFE_PLATFORM_TBL_CFGVAL(VALID_SCID_COUNT)
00226 #define DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_COUNT 0
00227
00228 /* macro to construct 32 bit value from 4 chars */
00229 #define CFE_PLATFORM_TBL_U32FROM4CHARS(_C1, _C2, _C3, _C4) \
00230     ((uint32)(_C1) << 24 | (uint32)(_C2) << 16 | (uint32)(_C3) << 8 | (uint32)(_C4))
00231
00244 #define CFE_PLATFORM_TBL_VALID_SCID_1                CFE_PLATFORM_TBL_CFGVAL(VALID_SCID_1)
00245 #define DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_1 (0x42)
00246
00247 #define CFE_PLATFORM_TBL_VALID_SCID_2                CFE_PLATFORM_TBL_CFGVAL(VALID_SCID_2)
00248 #define DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_2 (CFE_PLATFORM_TBL_U32FROM4CHARS('a', 'b', 'c', 'd'))
00249
00267 #define CFE_PLATFORM_TBL_VALID_PRID_COUNT            CFE_PLATFORM_TBL_CFGVAL(VALID_PRID_COUNT)
00268 #define DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_COUNT 0
00269
00282 #define CFE_PLATFORM_TBL_VALID_PRID_1                CFE_PLATFORM_TBL_CFGVAL(VALID_PRID_1)
00283 #define DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_1 (1)
00284
00285 #define CFE_PLATFORM_TBL_VALID_PRID_2                CFE_PLATFORM_TBL_CFGVAL(VALID_PRID_2)
00286 #define DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_2 (CFE_PLATFORM_TBL_U32FROM4CHARS('a', 'b', 'c', 'd'))
00287
00288 #define CFE_PLATFORM_TBL_VALID_PRID_3                CFE_PLATFORM_TBL_CFGVAL(VALID_PRID_3)
00289 #define DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_3 0
00290
00291 #define CFE_PLATFORM_TBL_VALID_PRID_4                CFE_PLATFORM_TBL_CFGVAL(VALID_PRID_4)
00292 #define DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_4 0
00293
00294 #endif

```

11.239 cfe/modules/tbl/config/default_cfe_tbl_topicid_values.h File Reference

Macros

- #define [CFE_MISSION_TBL_TIDVAL](#)(x) DEFAULT_CFE_MISSION_TBL_###x##_TOPICID

11.239.1 Detailed Description

CFE Table Services (CFE_TBL) Application Topic IDs
Definition in file [default_cfe_tbl_topicid_values.h](#).

11.239.2 Macro Definition Documentation

CFE_MISSION_TBL_TIDVAL

```
#define CFE_MISSION_TBL_TIDVAL(
    x ) DEFAULT_CFE_MISSION_TBL_###x##_TOPICID
```

Definition at line 26 of file [default_cfe_tbl_topicid_values.h](#).

11.240 default_cfe_tbl_topicid_values.h

[Go to the documentation of this file.](#)

```
00001 /*****
```

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00016  * limitations under the License.
00017  *****/
00018
00023 #ifndef DEFAULT_CFE_TBL_TOPICID_VALUES_H
00024 #define DEFAULT_CFE_TBL_TOPICID_VALUES_H
00025
00026 #define CFE_MISSION_TBL_TIDVAL(x) DEFAULT_CFE_MISSION_TBL_##x##_TOPICID
00027
00028 #endif

```

11.241 cfe/modules/tbl/fsw/inc/cfe_tbl_eventids.h File Reference

Macros

TBL event IDs

- #define [CFE_TBL_INIT_INF_EID](#) 1
TBL Initialization Event ID.
- #define [CFE_TBL_NOOP_INF_EID](#) 10
TBL No-op Command Success Event ID.
- #define [CFE_TBL_RESET_INF_EID](#) 11
TBL Reset Counters Command Success Event ID.
- #define [CFE_TBL_FILE_LOADED_INF_EID](#) 12
TBL Load Table Command Success Event ID.
- #define [CFE_TBL_OVERWRITE_DUMP_INF_EID](#) 13
TBL Write Table To Existing File Success Event ID.
- #define [CFE_TBL_WRITE_DUMP_INF_EID](#) 14
TBL Write Table To New File Success Event ID.
- #define [CFE_TBL_OVERWRITE_REG_DUMP_INF_EID](#) 15
TBL Write Table Registry To Existing File Success Event ID.
- #define [CFE_TBL_VAL_REQ_MADE_INF_EID](#) 16
TBL Validate Table Request Success Event ID.
- #define [CFE_TBL_LOAD_PEND_REQ_INF_EID](#) 17
TBL Load Table Pending Notification Success Event ID.
- #define [CFE_TBL_TLM_REG_CMD_INF_EID](#) 18
TBL Telemeter Table Registry Entry Command Success Event ID.
- #define [CFE_TBL_LOAD_ABORT_INF_EID](#) 21
TBL Abort Table Load Success Event ID.
- #define [CFE_TBL_WRITE_REG_DUMP_INF_EID](#) 22
TBL Write Table Registry To New File Success Event ID.
- #define [CFE_TBL_ASSUMED_VALID_INF_EID](#) 23
TBL Validate Table Valid Due To No Validation Function Event ID.
- #define [CFE_TBL_LOAD_SUCCESS_INF_EID](#) 35
TBL Load Table API Success Event ID.
- #define [CFE_TBL_VALIDATION_INF_EID](#) 36
TBL Validate Table Success Event ID.
- #define [CFE_TBL_UPDATE_SUCCESS_INF_EID](#) 37
TBL Update Table Success Event ID.

- #define [CFE_TBL_CDS_DELETED_INFO_EID](#) 38
TBL Delete Table CDS Command Success Event ID.
- #define [CFE_TBL_MID_ERR_EID](#) 50
TBL Invalid Message ID Received Event ID.
- #define [CFE_TBL_CC1_ERR_EID](#) 51
TBL Invalid Command Code Received Event ID.
- #define [CFE_TBL_LEN_ERR_EID](#) 52
TBL Invalid Command Length Event ID.
- #define [CFE_TBL_FILE_ACCESS_ERR_EID](#) 53
TBL Load Table File Open Failure Event ID.
- #define [CFE_TBL_FILE_STD_HDR_ERR_EID](#) 54
TBL Load Table File Read Standard Header Failure Event ID.
- #define [CFE_TBL_FILE_TBL_HDR_ERR_EID](#) 55
TBL Load Table File Read Table Header Failure Event ID.
- #define [CFE_TBL_FAIL_HK_SEND_ERR_EID](#) 56
TBL Send Housekeeping Command Transmit Failure Event ID.
- #define [CFE_TBL_NO_SUCH_TABLE_ERR_EID](#) 57
TBL Table Name Not Found Event ID.
- #define [CFE_TBL_FILE_TYPE_ERR_EID](#) 58
TBL Load Table Invalid File Content ID Event ID.
- #define [CFE_TBL_FILE_SUBTYPE_ERR_EID](#) 59
TBL Load Table Invalid File Subtype Event ID.
- #define [CFE_TBL_NO_WORK_BUFFERS_ERR_EID](#) 60
TBL Load Or Dump Table No Working Buffers Available Event ID.
- #define [CFE_TBL_CODEC_ERROR_ERR_EID](#) 61
TBL Encoding/Decoding error Event ID.
- #define [CFE_TBL_CREATING_DUMP_FILE_ERR_EID](#) 62
TBL Write File Creation Failure Event ID.
- #define [CFE_TBL_WRITE_CFE_HDR_ERR_EID](#) 63
TBL Write Standard File Header Failure Event ID.
- #define [CFE_TBL_WRITE_TBL_HDR_ERR_EID](#) 64
TBL Write Table File Header Failure Event ID.
- #define [CFE_TBL_WRITE_TBL_IMG_ERR_EID](#) 65
TBL Write Table File Data Failure Event ID.
- #define [CFE_TBL_TOO_MANY_VALIDATIONS_ERR_EID](#) 67
TBL Validate Table Command Result Storage Exceeded Event ID.
- #define [CFE_TBL_WRITE_TBL_REG_ERR_EID](#) 68
TBL Write Table Registry File Data Failure Event ID.
- #define [CFE_TBL_LOAD_ABORT_ERR_EID](#) 69
TBL Abort Table Load No Load Started Event ID.
- #define [CFE_TBL_ACTIVATE_ERR_EID](#) 70
TBL Activate Table Command No Inactive Buffer Event ID.
- #define [CFE_TBL_FILE_INCOMPLETE_ERR_EID](#) 71
TBL Load Table Incomplete Load Event ID.
- #define [CFE_TBL_LOAD_EXCEEDS_SIZE_ERR_EID](#) 72
TBL Load Table File Exceeds Table Size Event ID.
- #define [CFE_TBL_ZERO_LENGTH_LOAD_ERR_EID](#) 73
TBL Load Table File Zero Length Event ID.
- #define [CFE_TBL_PARTIAL_LOAD_ERR_EID](#) 74
TBL Load Table Uninitialized Partial Load Event ID.
- #define [CFE_TBL_FILE_TOO_BIG_ERR_EID](#) 75
TBL Load Table File Excess Data Event ID.
- #define [CFE_TBL_TOO_MANY_DUMPS_ERR_EID](#) 76
TBL Write Table Command Dump Only Control Blocks Exceeded Event ID.
- #define [CFE_TBL_DUMP_PENDING_ERR_EID](#) 77

- TBL Write Table Command Already In Progress Event ID.*
- #define [CFE_TBL_ACTIVATE_DUMP_ONLY_ERR_EID](#) 78
- TBL Activate Table Command For Dump Only Table Event ID.*
- #define [CFE_TBL_LOADING_A_DUMP_ONLY_ERR_EID](#) 79
- TBL Load Table For Dump Only Table Event ID.*
- #define [CFE_TBL_ILLEGAL_BUFF_PARAM_ERR_EID](#) 80
- TBL Validate Or Write Table Command Invalid Buffer Event ID.*
- #define [CFE_TBL_UNVALIDATED_ERR_EID](#) 81
- TBL Activate Table Command Inactive Image Not Validated Event ID.*
- #define [CFE_TBL_IN_REGISTRY_ERR_EID](#) 82
- TBL Delete Table CDS Command For Registered Table Event ID.*
- #define [CFE_TBL_NOT_CRITICAL_TBL_ERR_EID](#) 83
- TBL Delete Table CDS Command Invalid CDS Type Event ID.*
- #define [CFE_TBL_NOT_IN_CRIT_REG_ERR_EID](#) 84
- TBL Delete Table CDS Command Not In Critical Table Registry Event ID.*
- #define [CFE_TBL_CDS_NOT_FOUND_ERR_EID](#) 85
- TBL Delete Table CDS Command Not In CDS Registry Event ID.*
- #define [CFE_TBL_CDS_DELETE_ERR_EID](#) 86
- TBL Delete Table CDS Command Internal Error Event ID.*
- #define [CFE_TBL_CDS_OWNER_ACTIVE_ERR_EID](#) 87
- TBL Delete Table CDS Command App Active Event ID.*
- #define [CFE_TBL_FAIL_NOTIFY_SEND_ERR_EID](#) 89
- TBL Send Notification Transmit Failed Event ID.*
- #define [CFE_TBL_REGISTER_ERR_EID](#) 90
- TBL Register Table Failed Event ID.*
- #define [CFE_TBL_SHARE_ERR_EID](#) 91
- TBL Share Table Failed Event ID.*
- #define [CFE_TBL_UNREGISTER_ERR_EID](#) 92
- TBL Unregister Table Failed Event ID.*
- #define [CFE_TBL_LOAD_VAL_ERR_EID](#) 93
- TBL Validation Function Invalid Return Code Event ID.*
- #define [CFE_TBL_LOAD_TYPE_ERR_EID](#) 94
- TBL Load Table API Invalid Source Type Event ID.*
- #define [CFE_TBL_UPDATE_ERR_EID](#) 95
- TBL Update Table Failed Event ID.*
- #define [CFE_TBL_VALIDATION_ERR_EID](#) 96
- TBL Validate Table Validation Failed Event ID.*
- #define [CFE_TBL_SPACECRAFT_ID_ERR_EID](#) 97
- TBL Read Header Invalid Spacecraft ID Event ID.*
- #define [CFE_TBL_PROCESSOR_ID_ERR_EID](#) 98
- TBL Read Header Invalid Processor ID Event ID.*
- #define [CFE_TBL_LOAD_IN_PROGRESS_ERR_EID](#) 100
- TBL Load Table API Load Already In Progress Event ID.*
- #define [CFE_TBL_LOAD_FILENAME_LONG_ERR_EID](#) 101
- TBL Load Table Filename Too Long Event ID.*
- #define [CFE_TBL_LOAD_TBLNAME_MISMATCH_ERR_EID](#) 102
- TBL Load Table Name Mismatch Event ID.*
- #define [CFE_TBL_HANDLE_ACCESS_ERR_EID](#) 103
- TBL Load Table API Access Violation Event ID.*
- #define [CFE_TBL_VALIDATION_IN_PROGRESS_ERR_EID](#) 104
- TBL Validation Already In Progress Event ID.*

11.241.1 Detailed Description

cFE Table Services Event IDs

Definition in file [cfe_tbl_eventids.h](#).

11.241.2 Macro Definition Documentation

CFE_TBL_ACTIVATE_DUMP_ONLY_ERR_EID

```
#define CFE_TBL_ACTIVATE_DUMP_ONLY_ERR_EID 78
```

TBL Activate Table Command For Dump Only Table Event ID.

Type: ERROR

Cause:

[TBL Activate Table Command](#) failure due to table being dump only.

Definition at line 553 of file [cfe_tbl_eventids.h](#).

CFE_TBL_ACTIVATE_ERR_EID

```
#define CFE_TBL_ACTIVATE_ERR_EID 70
```

TBL Activate Table Command No Inactive Buffer Event ID.

Type: ERROR

Cause:

[TBL Activate Table Command](#) failure due to no associated inactive buffer.

Definition at line 459 of file [cfe_tbl_eventids.h](#).

CFE_TBL_ASSUMED_VALID_INF_EID

```
#define CFE_TBL_ASSUMED_VALID_INF_EID 23
```

TBL Validate Table Valid Due To No Validation Function Event ID.

Type: INFORMATION

Cause:

[TBL Validate Table Command](#) marking table as valid due to no validation function being registered.

Definition at line 180 of file [cfe_tbl_eventids.h](#).

CFE_TBL_CC1_ERR_EID

```
#define CFE_TBL_CC1_ERR_EID 51
```

TBL Invalid Command Code Received Event ID.

Type: ERROR

Cause:

Invalid command code for message ID [CFE_TBL_CMD_MID](#) received on the TBL message pipe.

Definition at line 246 of file [cfe_tbl_eventids.h](#).

CFE_TBL_CDS_DELETE_ERR_EID

```
#define CFE_TBL_CDS_DELETE_ERR_EID 86
```

TBL Delete Table CDS Command Internal Error Event ID.

Type: **ERROR**

Cause:

[TBL Delete Table CDS Command](#) failure due to an internal error. See the system log for more information.
Definition at line 649 of file [cfe_tbl_eventids.h](#).

CFE_TBL_CDS_DELETED_INFO_EID

```
#define CFE_TBL_CDS_DELETED_INFO_EID 38
```

TBL Delete Table CDS Command Success Event ID.

Type: **INFORMATION**

Cause:

[TBL Delete Table CDS Command](#) success.
Definition at line 224 of file [cfe_tbl_eventids.h](#).

CFE_TBL_CDS_NOT_FOUND_ERR_EID

```
#define CFE_TBL_CDS_NOT_FOUND_ERR_EID 85
```

TBL Delete Table CDS Command Not In CDS Registry Event ID.

Type: **ERROR**

Cause:

[TBL Delete Table CDS Command](#) failure due to the table name not found in the CDS registry.
Definition at line 637 of file [cfe_tbl_eventids.h](#).

CFE_TBL_CDS_OWNER_ACTIVE_ERR_EID

```
#define CFE_TBL_CDS_OWNER_ACTIVE_ERR_EID 87
```

TBL Delete Table CDS Command App Active Event ID.

Type: **ERROR**

Cause:

[TBL Delete Table CDS Command](#) failure due to the owning application being active.
Definition at line 661 of file [cfe_tbl_eventids.h](#).

CFE_TBL_CODEC_ERROR_ERR_EID

#define CFE_TBL_CODEC_ERROR_ERR_EID 61
TBL Encoding/Decoding error Event ID.

Type: ERROR

Cause: Encoding/Decoding of the table binary data from the file content failed

This is possibly due to a corrupt or incompatible file.
Definition at line 358 of file [cfe_tbl_eventids.h](#).

CFE_TBL_CREATING_DUMP_FILE_ERR_EID

#define CFE_TBL_CREATING_DUMP_FILE_ERR_EID 62
TBL Write File Creation Failure Event ID.

Type: ERROR

Cause:

TBL Write Table or Table Registry File failed to create file. OVERLOADED
Definition at line 369 of file [cfe_tbl_eventids.h](#).

CFE_TBL_DUMP_PENDING_ERR_EID

#define CFE_TBL_DUMP_PENDING_ERR_EID 77
TBL Write Table Command Already In Progress Event ID.

Type: ERROR

Cause:

[TBL Write Table Command](#) failure due to a dump already in progress for the same table.
Definition at line 541 of file [cfe_tbl_eventids.h](#).

CFE_TBL_FAIL_HK_SEND_ERR_EID

#define CFE_TBL_FAIL_HK_SEND_ERR_EID 56
TBL Send Housekeeping Command Transmit Failure Event ID.

Type: ERROR

Cause:

[TBL Send Housekeeping Command](#) failure transmitting the housekeeping message.
Definition at line 302 of file [cfe_tbl_eventids.h](#).

CFE_TBL_FAIL_NOTIFY_SEND_ERR_EID

#define CFE_TBL_FAIL_NOTIFY_SEND_ERR_EID 89
TBL Send Notification Transmit Failed Event ID.

Type: ERROR

Cause:

TBL send notification transmit message failure.
Definition at line 672 of file [cfe_tbl_eventids.h](#).

CFE_TBL_FILE_ACCESS_ERR_EID

#define CFE_TBL_FILE_ACCESS_ERR_EID 53
TBL Load Table File Open Failure Event ID.

Type: ERROR

Cause:

Load Table failure opening the file. OVERLOADED
Definition at line 268 of file [cfe_tbl_eventids.h](#).

CFE_TBL_FILE_INCOMPLETE_ERR_EID

#define CFE_TBL_FILE_INCOMPLETE_ERR_EID 71
TBL Load Table Incomplete Load Event ID.

Type: ERROR

Cause:

TBL Load Table failure due to inability to read the size of data specified in the table header from file. OVERLOADED
Definition at line 471 of file [cfe_tbl_eventids.h](#).

CFE_TBL_FILE_LOADED_INF_EID

#define CFE_TBL_FILE_LOADED_INF_EID 12
TBL Load Table Command Success Event ID.

Type: INFORMATION

Cause:

[TBL Load Table Command](#) successfully loaded the new table data to the working buffer.
Definition at line 76 of file [cfe_tbl_eventids.h](#).

CFE_TBL_FILE_STD_HDR_ERR_EID

#define CFE_TBL_FILE_STD_HDR_ERR_EID 54
TBL Load Table File Read Standard Header Failure Event ID.
Type: ERROR

Cause:

Load Table failure reading the file standard header.
Definition at line 279 of file [cfe_tbl_eventids.h](#).

CFE_TBL_FILE_SUBTYPE_ERR_EID

#define CFE_TBL_FILE_SUBTYPE_ERR_EID 59
TBL Load Table Invalid File Subtype Event ID.
Type: ERROR

Cause:

TBL Load Table Failure due to invalid file subtype.
Definition at line 335 of file [cfe_tbl_eventids.h](#).

CFE_TBL_FILE_TBL_HDR_ERR_EID

#define CFE_TBL_FILE_TBL_HDR_ERR_EID 55
TBL Load Table File Read Table Header Failure Event ID.
Type: ERROR

Cause:

Load Table failure reading the file table header.
Definition at line 290 of file [cfe_tbl_eventids.h](#).

CFE_TBL_FILE_TOO_BIG_ERR_EID

#define CFE_TBL_FILE_TOO_BIG_ERR_EID 75
TBL Load Table File Excess Data Event ID.
Type: ERROR

Cause:

TBL Load Table failure due to the file header specified size of data being smaller than the actual data contained in the file. OVERLOADED
Definition at line 517 of file [cfe_tbl_eventids.h](#).

CFE_TBL_FILE_TYPE_ERR_EID

#define CFE_TBL_FILE_TYPE_ERR_EID 58
TBL Load Table Invalid File Content ID Event ID.
Type: ERROR

Cause:

TBL Load Table failure due to invalid file content ID.
Definition at line 324 of file [cfe_tbl_eventids.h](#).

CFE_TBL_HANDLE_ACCESS_ERR_EID

#define CFE_TBL_HANDLE_ACCESS_ERR_EID 103
TBL Load Table API Access Violation Event ID.
Type: ERROR

Cause:

[CFE_TBL_Load](#) API failure due to the application not owning the table.
Definition at line 815 of file [cfe_tbl_eventids.h](#).

CFE_TBL_ILLEGAL_BUFF_PARAM_ERR_EID

#define CFE_TBL_ILLEGAL_BUFF_PARAM_ERR_EID 80
TBL Validate Or Write Table Command Invalid Buffer Event ID.
Type: ERROR

Cause:

[TBL Validate Table Command](#) or [TBL Write Table Command](#) failure due to an invalid buffer selection. OVERLOADED
Definition at line 577 of file [cfe_tbl_eventids.h](#).

CFE_TBL_IN_REGISTRY_ERR_EID

#define CFE_TBL_IN_REGISTRY_ERR_EID 82
TBL Delete Table CDS Command For Registered Table Event ID.
Type: ERROR

Cause:

[TBL Delete Table CDS Command](#) failure due to the table being currently registered.
Definition at line 601 of file [cfe_tbl_eventids.h](#).

CFE_TBL_INIT_INF_EID

#define CFE_TBL_INIT_INF_EID 1
TB Initialization Event ID.

Type: INFORMATION

Cause:

Table Services Task initialization complete.
Definition at line 42 of file [cfe_tbl_eventids.h](#).

CFE_TBL_LEN_ERR_EID

#define CFE_TBL_LEN_ERR_EID 52
TBL Invalid Command Length Event ID.

Type: ERROR

Cause:

Invalid length for the message ID and command code received on the TBL message pipe.
Definition at line 257 of file [cfe_tbl_eventids.h](#).

CFE_TBL_LOAD_ABORT_ERR_EID

#define CFE_TBL_LOAD_ABORT_ERR_EID 69
TBL Abort Table Load No Load Started Event ID.

Type: ERROR

Cause:

[TBL Abort Table Load Command](#) failure due to no load in progress.
Definition at line 447 of file [cfe_tbl_eventids.h](#).

CFE_TBL_LOAD_ABORT_INF_EID

#define CFE_TBL_LOAD_ABORT_INF_EID 21
TBL Abort Table Load Success Event ID.

Type: INFORMATION

Cause:

[TBL Abort Table Load Command](#) success.
Definition at line 157 of file [cfe_tbl_eventids.h](#).

CFE_TBL_LOAD_EXCEEDS_SIZE_ERR_EID

#define CFE_TBL_LOAD_EXCEEDS_SIZE_ERR_EID 72
TBL Load Table File Exceeds Table Size Event ID.

Type: ERROR

Cause:

TBL Load Table failure due to the file header specified offset and/or size of data exceeding the table size. OVERLOADED
Definition at line 483 of file [cfe_tbl_eventids.h](#).

CFE_TBL_LOAD_FILENAME_LONG_ERR_EID

#define CFE_TBL_LOAD_FILENAME_LONG_ERR_EID 101
TBL Load Table Filename Too Long Event ID.

Type: ERROR

Cause:

Load table filename too long.
Definition at line 793 of file [cfe_tbl_eventids.h](#).

CFE_TBL_LOAD_IN_PROGRESS_ERR_EID

#define CFE_TBL_LOAD_IN_PROGRESS_ERR_EID 100
TBL Load Table API Load Already In Progress Event ID.

Type: ERROR

Cause:

[CFE_TBL_Load](#) API failure due to load already in progress.
Definition at line 782 of file [cfe_tbl_eventids.h](#).

CFE_TBL_LOAD_PEND_REQ_INF_EID

#define CFE_TBL_LOAD_PEND_REQ_INF_EID 17
TBL Load Table Pending Notification Success Event ID.

Type: DEBUG

Cause:

TBL load table pending notification successfully sent.
Definition at line 134 of file [cfe_tbl_eventids.h](#).

CFE_TBL_LOAD_SUCCESS_INF_EID

#define CFE_TBL_LOAD_SUCCESS_INF_EID 35
TBL Load Table API Success Event ID.

Type: DEBUG (the first time) and INFORMATION (normally)

Cause:

[CFE_TBL_Load](#) API success for dump only or normal table. OVERLOADED
Definition at line 191 of file [cfe_tbl_eventids.h](#).

CFE_TBL_LOAD_TBLNAME_MISMATCH_ERR_EID

#define CFE_TBL_LOAD_TBLNAME_MISMATCH_ERR_EID 102
TBL Load Table Name Mismatch Event ID.

Type: ERROR

Cause:

Load table name in the table file header does not match the specified table name.
Definition at line 804 of file [cfe_tbl_eventids.h](#).

CFE_TBL_LOAD_TYPE_ERR_EID

#define CFE_TBL_LOAD_TYPE_ERR_EID 94
TBL Load Table API Invalid Source Type Event ID.

Type: ERROR

Cause:

[CFE_TBL_Load](#) API valid due to invalid source type.
Definition at line 727 of file [cfe_tbl_eventids.h](#).

CFE_TBL_LOAD_VAL_ERR_EID

#define CFE_TBL_LOAD_VAL_ERR_EID 93
TBL Validation Function Invalid Return Code Event ID.

Type: ERROR

Cause:

Invalid table validation function return code.
Definition at line 716 of file [cfe_tbl_eventids.h](#).

CFE_TBL_LOADING_A_DUMP_ONLY_ERR_EID

#define CFE_TBL_LOADING_A_DUMP_ONLY_ERR_EID 79
TBL Load Table For Dump Only Table Event ID.

Type: ERROR

Cause:

TBL Load Table failure due to table being dump only. OVERLOADED
Definition at line 564 of file [cfe_tbl_eventids.h](#).

CFE_TBL_MID_ERR_EID

#define CFE_TBL_MID_ERR_EID 50
TBL Invalid Message ID Received Event ID.

Type: ERROR

Cause:

Invalid message ID received on the TBL message pipe.
Definition at line 235 of file [cfe_tbl_eventids.h](#).

CFE_TBL_NO_SUCH_TABLE_ERR_EID

#define CFE_TBL_NO_SUCH_TABLE_ERR_EID 57
TBL Table Name Not Found Event ID.

Type: ERROR

Cause:

TBL command handler unable to find table name. OVERLOADED
Definition at line 313 of file [cfe_tbl_eventids.h](#).

CFE_TBL_NO_WORK_BUFFERS_ERR_EID

#define CFE_TBL_NO_WORK_BUFFERS_ERR_EID 60
TBL Load Or Dump Table No Working Buffers Available Event ID.

Type: ERROR

Cause:

TBL Load or Dump failure due to no working buffers available or internal error. OVERLOADED
Definition at line 346 of file [cfe_tbl_eventids.h](#).

CFE_TBL_NOOP_INF_EID

#define CFE_TBL_NOOP_INF_EID 10
TBL No-op Command Success Event ID.

Type: INFORMATION

Cause:

[NO-OP TBL No-op Command](#) success.
Definition at line 53 of file [cfe_tbl_eventids.h](#).

CFE_TBL_NOT_CRITICAL_TBL_ERR_EID

#define CFE_TBL_NOT_CRITICAL_TBL_ERR_EID 83
TBL Delete Table CDS Command Invalid CDS Type Event ID.

Type: ERROR

Cause:

[TBL Delete Table CDS Command](#) failure due to CDS being in the table registry but not registered as a table within ES.
Definition at line 613 of file [cfe_tbl_eventids.h](#).

CFE_TBL_NOT_IN_CRIT_REG_ERR_EID

#define CFE_TBL_NOT_IN_CRIT_REG_ERR_EID 84
TBL Delete Table CDS Command Not In Critical Table Registry Event ID.

Type: ERROR

Cause:

[TBL Delete Table CDS Command](#) failure due to the table not being in the critical table registry.
Definition at line 625 of file [cfe_tbl_eventids.h](#).

CFE_TBL_OVERWRITE_DUMP_INF_EID

#define CFE_TBL_OVERWRITE_DUMP_INF_EID 13
TBL Write Table To Existing File Success Event ID.

Type: INFORMATION

Cause:

TBL write table to an existing file success.
Definition at line 87 of file [cfe_tbl_eventids.h](#).

CFE_TBL_OVERWRITE_REG_DUMP_INF_EID

#define CFE_TBL_OVERWRITE_REG_DUMP_INF_EID 15
TBL Write Table Registry To Existing File Success Event ID.

Type: DEBUG

Cause:

TBL Write Table Registry to an existing file completed successfully.
Definition at line 109 of file [cfe_tbl_eventids.h](#).

CFE_TBL_PARTIAL_LOAD_ERR_EID

#define CFE_TBL_PARTIAL_LOAD_ERR_EID 74
TBL Load Table Uninitialized Partial Load Event ID.

Type: ERROR

Cause:

TBL Load Table failure due to attempting a partial load to an uninitialized table. OVERLOADED
Definition at line 505 of file [cfe_tbl_eventids.h](#).

CFE_TBL_PROCESSOR_ID_ERR_EID

#define CFE_TBL_PROCESSOR_ID_ERR_EID 98
TBL Read Header Invalid Processor ID Event ID.

Type: ERROR

Cause:

Invalid processor ID in table file header.
Definition at line 771 of file [cfe_tbl_eventids.h](#).

CFE_TBL_REGISTER_ERR_EID

#define CFE_TBL_REGISTER_ERR_EID 90
TBL Register Table Failed Event ID.

Type: ERROR

Cause:

TBL table registration failure. See system log for more information.
Definition at line 683 of file [cfe_tbl_eventids.h](#).

CFE_TBL_RESET_INF_EID

```
#define CFE_TBL_RESET_INF_EID 11
```

TBL Reset Counters Command Success Event ID.

Type: DEBUG

Cause:

[TBL Reset Counters Command](#) success.

Definition at line 64 of file [cfe_tbl_eventids.h](#).

CFE_TBL_SHARE_ERR_EID

```
#define CFE_TBL_SHARE_ERR_EID 91
```

TBL Share Table Failed Event ID.

Type: ERROR

Cause:

TBL share table failure. See system log for more information.

Definition at line 694 of file [cfe_tbl_eventids.h](#).

CFE_TBL_SPACECRAFT_ID_ERR_EID

```
#define CFE_TBL_SPACECRAFT_ID_ERR_EID 97
```

TBL Read Header Invalid Spacecraft ID Event ID.

Type: ERROR

Cause:

Invalid spacecraft ID in table file header.

Definition at line 760 of file [cfe_tbl_eventids.h](#).

CFE_TBL_TLM_REG_CMD_INF_EID

```
#define CFE_TBL_TLM_REG_CMD_INF_EID 18
```

TBL Telemeter Table Registry Entry Command Success Event ID.

Type: DEBUG

Cause:

[TBL Telemeter Table Registry Entry command](#) successfully set the table registry index to telemeter in the next house-keeping packet.

Definition at line 146 of file [cfe_tbl_eventids.h](#).

CFE_TBL_TOO_MANY_DUMPS_ERR_EID

```
#define CFE_TBL_TOO_MANY_DUMPS_ERR_EID 76
```

TBL Write Table Command Dump Only Control Blocks Exceeded Event ID.

Type: ERROR

Cause:

[TBL Write Table Command](#) failure due to exceeding the allocated number of control blocks available to write a dump only table.

Definition at line 529 of file [cfe_tbl_eventids.h](#).

CFE_TBL_TOO_MANY_VALIDATIONS_ERR_EID

```
#define CFE_TBL_TOO_MANY_VALIDATIONS_ERR_EID 67
```

TBL Validate Table Command Result Storage Exceeded Event ID.

Type: ERROR

Cause:

[TBL Validate Table Command](#) failure due to exceeding result storage.

Definition at line 424 of file [cfe_tbl_eventids.h](#).

CFE_TBL_UNREGISTER_ERR_EID

```
#define CFE_TBL_UNREGISTER_ERR_EID 92
```

TBL Unregister Table Failed Event ID.

Type: ERROR

Cause:

TBL unregister table failure. See system log for more information.

Definition at line 705 of file [cfe_tbl_eventids.h](#).

CFE_TBL_UNVALIDATED_ERR_EID

```
#define CFE_TBL_UNVALIDATED_ERR_EID 81
```

TBL Activate Table Command Inactive Image Not Validated Event ID.

Type: ERROR

Cause:

[TBL Activate Table Command](#) failure due to the inactive image not being validated.

Definition at line 589 of file [cfe_tbl_eventids.h](#).

CFE_TBL_UPDATE_ERR_EID

#define CFE_TBL_UPDATE_ERR_EID 95
TBL Update Table Failed Event ID.

Type: ERROR

Cause:

TBL update table failure due to an internal error. OVERLOADED
Definition at line 738 of file [cfe_tbl_eventids.h](#).

CFE_TBL_UPDATE_SUCCESS_INF_EID

#define CFE_TBL_UPDATE_SUCCESS_INF_EID 37
TBL Update Table Success Event ID.

Type: INFORMATION

Cause:

Table update successfully completed.
Definition at line 213 of file [cfe_tbl_eventids.h](#).

CFE_TBL_VAL_REQ_MADE_INF_EID

#define CFE_TBL_VAL_REQ_MADE_INF_EID 16
TBL Validate Table Request Success Event ID.

Type: DEBUG

Cause:

[TBL Validate Table Command](#) success. Note this event signifies the request to validate the table has been successfully submitted. Completion will generate a [CFE_TBL_VALIDATION_INF_EID](#) or [CFE_TBL_VALIDATION_ERR_EID](#) event messages.

Definition at line 123 of file [cfe_tbl_eventids.h](#).

CFE_TBL_VALIDATION_ERR_EID

#define CFE_TBL_VALIDATION_ERR_EID 96
TBL Validate Table Validation Failed Event ID.

Type: ERROR

Cause:

TBL validate table function indicates validation failed. OVERLOADED
Definition at line 749 of file [cfe_tbl_eventids.h](#).

CFE_TBL_VALIDATION_IN_PROGRESS_ERR_EID

#define CFE_TBL_VALIDATION_IN_PROGRESS_ERR_EID 104
TBL Validation Already In Progress Event ID.

Type: ERROR

Cause:

Validation failure due to existing request already pending.
Definition at line 826 of file [cfe_tbl_eventids.h](#).

CFE_TBL_VALIDATION_INF_EID

#define CFE_TBL_VALIDATION_INF_EID 36
TBL Validate Table Success Event ID.

Type: INFORMATION

Cause:

Table active or inactive image successfully validated by the registered validation function. OVERLOADED
Definition at line 202 of file [cfe_tbl_eventids.h](#).

CFE_TBL_WRITE_CFE_HDR_ERR_EID

#define CFE_TBL_WRITE_CFE_HDR_ERR_EID 63
TBL Write Standard File Header Failure Event ID.

Type: ERROR

Cause:

TBL Write Table or Table Registry File failure writing the standard file header. OVERLOADED
Definition at line 380 of file [cfe_tbl_eventids.h](#).

CFE_TBL_WRITE_DUMP_INF_EID

#define CFE_TBL_WRITE_DUMP_INF_EID 14
TBL Write Table To New File Success Event ID.

Type: INFORMATION

Cause:

TBL write table to a new file success.
Definition at line 98 of file [cfe_tbl_eventids.h](#).

CFE_TBL_WRITE_REG_DUMP_INF_EID

#define CFE_TBL_WRITE_REG_DUMP_INF_EID 22
TBL Write Table Registry To New File Success Event ID.

Type: DEBUG

Cause:

TBL Write Table Registry to a new file completed successfully.
Definition at line 168 of file [cfe_tbl_eventids.h](#).

CFE_TBL_WRITE_TBL_HDR_ERR_EID

#define CFE_TBL_WRITE_TBL_HDR_ERR_EID 64
TBL Write Table File Header Failure Event ID.

Type: ERROR

Cause:

TBL Write Table failure writing the table image file header.
Definition at line 391 of file [cfe_tbl_eventids.h](#).

CFE_TBL_WRITE_TBL_IMG_ERR_EID

#define CFE_TBL_WRITE_TBL_IMG_ERR_EID 65
TBL Write Table File Data Failure Event ID.

Type: ERROR

Cause:

TBL Write Table failure writing the table data.
Definition at line 402 of file [cfe_tbl_eventids.h](#).

CFE_TBL_WRITE_TBL_REG_ERR_EID

#define CFE_TBL_WRITE_TBL_REG_ERR_EID 68
TBL Write Table Registry File Data Failure Event ID.

Type: ERROR

Cause:

TB Write Table Registry failure writing file data.
Definition at line 435 of file [cfe_tbl_eventids.h](#).

CFE_TBL_ZERO_LENGTH_LOAD_ERR_EID

```
#define CFE_TBL_ZERO_LENGTH_LOAD_ERR_EID 73
TBL Load Table File Zero Length Event ID.
```

Type: ERROR

Cause:

TBL Load Table failure due to the file header specified size of data being zero.

Definition at line 494 of file [cfe_tbl_eventids.h](#).

11.242 cfe_tbl_eventids.h

[Go to the documentation of this file.](#)

```
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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00025 #ifndef CFE_TBL_EVENTS_H
00026 #define CFE_TBL_EVENTS_H
00027
00042 #define CFE_TBL_INIT_INF_EID 1
00043
00053 #define CFE_TBL_NOOP_INF_EID 10
00054
00064 #define CFE_TBL_RESET_INF_EID 11
00065
00076 #define CFE_TBL_FILE_LOADED_INF_EID 12
00077
00087 #define CFE_TBL_OVERWRITE_DUMP_INF_EID 13
00088
00098 #define CFE_TBL_WRITE_DUMP_INF_EID 14
00099
00109 #define CFE_TBL_OVERWRITE_REG_DUMP_INF_EID 15
00110
00123 #define CFE_TBL_VAL_REQ_MADE_INF_EID 16
00124
00134 #define CFE_TBL_LOAD_PEND_REQ_INF_EID 17
00135
00146 #define CFE_TBL_TLM_REG_CMD_INF_EID 18
00147
00157 #define CFE_TBL_LOAD_ABORT_INF_EID 21
00158
00168 #define CFE_TBL_WRITE_REG_DUMP_INF_EID 22
00169
00180 #define CFE_TBL_ASSUMED_VALID_INF_EID 23
00181
00191 #define CFE_TBL_LOAD_SUCCESS_INF_EID 35
00192
00202 #define CFE_TBL_VALIDATION_INF_EID 36
00203
00213 #define CFE_TBL_UPDATE_SUCCESS_INF_EID 37
00214
00224 #define CFE_TBL_CDS_DELETED_INFO_EID 38
00225
00235 #define CFE_TBL_MID_ERR_EID 50
```

```
00236
00246 #define CFE_TBL_CCI_ERR_EID 51
00247
00257 #define CFE_TBL_LEN_ERR_EID 52
00258
00268 #define CFE_TBL_FILE_ACCESS_ERR_EID 53
00269
00279 #define CFE_TBL_FILE_STD_HDR_ERR_EID 54
00280
00290 #define CFE_TBL_FILE_TBL_HDR_ERR_EID 55
00291
00302 #define CFE_TBL_FAIL_HK_SEND_ERR_EID 56
00303
00313 #define CFE_TBL_NO_SUCH_TABLE_ERR_EID 57
00314
00324 #define CFE_TBL_FILE_TYPE_ERR_EID 58
00325
00335 #define CFE_TBL_FILE_SUBTYPE_ERR_EID 59
00336
00346 #define CFE_TBL_NO_WORK_BUFFERS_ERR_EID 60
00347
00358 #define CFE_TBL_CODEC_ERROR_ERR_EID 61
00359
00369 #define CFE_TBL_CREATING_DUMP_FILE_ERR_EID 62
00370
00380 #define CFE_TBL_WRITE_CFE_HDR_ERR_EID 63
00381
00391 #define CFE_TBL_WRITE_TBL_HDR_ERR_EID 64
00392
00402 #define CFE_TBL_WRITE_TBL_IMG_ERR_EID 65
00403
00404 /*
00405  * NOTE:
00406  * Event ID CFE_TBL_NO_INACTIVE_BUFFER_ERR_EID (66) has been removed, as it
00407  * is superceded by CFE_TBL_ILLEGAL_BUFF_PARAM_ERR_EID (80).
00408  *
00409  * When requesting a dump or validation on a buffer that does not exist the
00410  * FSW will generate the CFE_TBL_ILLEGAL_BUFF_PARAM_ERR_EID regardless of whether
00411  * it was referred to as active or inactive.
00412  */
00413
00424 #define CFE_TBL_TOO_MANY_VALIDATIONS_ERR_EID 67
00425
00435 #define CFE_TBL_WRITE_TBL_REG_ERR_EID 68
00436
00447 #define CFE_TBL_LOAD_ABORT_ERR_EID 69
00448
00459 #define CFE_TBL_ACTIVATE_ERR_EID 70
00460
00471 #define CFE_TBL_FILE_INCOMPLETE_ERR_EID 71
00472
00483 #define CFE_TBL_LOAD_EXCEEDS_SIZE_ERR_EID 72
00484
00494 #define CFE_TBL_ZERO_LENGTH_LOAD_ERR_EID 73
00495
00505 #define CFE_TBL_PARTIAL_LOAD_ERR_EID 74
00506
00517 #define CFE_TBL_FILE_TOO_BIG_ERR_EID 75
00518
00529 #define CFE_TBL_TOO_MANY_DUMPS_ERR_EID 76
00530
00541 #define CFE_TBL_DUMP_PENDING_ERR_EID 77
00542
00553 #define CFE_TBL_ACTIVATE_DUMP_ONLY_ERR_EID 78
00554
00564 #define CFE_TBL_LOADING_A_DUMP_ONLY_ERR_EID 79
00565
00577 #define CFE_TBL_ILLEGAL_BUFF_PARAM_ERR_EID 80
00578
00589 #define CFE_TBL_UNVALIDATED_ERR_EID 81
00590
00601 #define CFE_TBL_IN_REGISTRY_ERR_EID 82
00602
00613 #define CFE_TBL_NOT_CRITICAL_TBL_ERR_EID 83
00614
00625 #define CFE_TBL_NOT_IN_CRIT_REG_ERR_EID 84
00626
00637 #define CFE_TBL_CDS_NOT_FOUND_ERR_EID 85
00638
00649 #define CFE_TBL_CDS_DELETE_ERR_EID 86
00650
```

```

00661 #define CFE_TBL_CDS_OWNER_ACTIVE_ERR_EID 87
00662
00672 #define CFE_TBL_FAIL_NOTIFY_SEND_ERR_EID 89
00673
00683 #define CFE_TBL_REGISTER_ERR_EID 90
00684
00694 #define CFE_TBL_SHARE_ERR_EID 91
00695
00705 #define CFE_TBL_UNREGISTER_ERR_EID 92
00706
00716 #define CFE_TBL_LOAD_VAL_ERR_EID 93
00717
00727 #define CFE_TBL_LOAD_TYPE_ERR_EID 94
00728
00738 #define CFE_TBL_UPDATE_ERR_EID 95
00739
00749 #define CFE_TBL_VALIDATION_ERR_EID 96
00750
00760 #define CFE_TBL_SPACECRAFT_ID_ERR_EID 97
00761
00771 #define CFE_TBL_PROCESSOR_ID_ERR_EID 98
00772
00782 #define CFE_TBL_LOAD_IN_PROGRESS_ERR_EID 100
00783
00793 #define CFE_TBL_LOAD_FILENAME_LONG_ERR_EID 101
00794
00804 #define CFE_TBL_LOAD_TBLNAME_MISMATCH_ERR_EID 102
00805
00815 #define CFE_TBL_HANDLE_ACCESS_ERR_EID 103
00816
00826 #define CFE_TBL_VALIDATION_IN_PROGRESS_ERR_EID 104
00827
00830 #endif /* CFE_TBL_EVENTS_H */

```

11.243 cfe/modules/tbl/fsw/inc/cfe_tbl_fcncodes.h File Reference

```
#include "cfe_tbl_fcncode_values.h"
```

Macros

Table Services Command Codes

- `#define CFE_TBL_NOOP_CC CFE_TBL_CCVAL(NOOP)`
- `#define CFE_TBL_RESET_COUNTERS_CC CFE_TBL_CCVAL(RESET_COUNTERS)`
- `#define CFE_TBL_LOAD_CC CFE_TBL_CCVAL(LOAD)`
- `#define CFE_TBL_DUMP_CC CFE_TBL_CCVAL(DUMP)`
- `#define CFE_TBL_VALIDATE_CC CFE_TBL_CCVAL(VALIDATE)`
- `#define CFE_TBL_ACTIVATE_CC CFE_TBL_CCVAL(ACTIVATE)`
- `#define CFE_TBL_DUMP_REGISTRY_CC CFE_TBL_CCVAL(DUMP_REGISTRY)`
- `#define CFE_TBL_SEND_REGISTRY_CC CFE_TBL_CCVAL(SEND_REGISTRY)`
- `#define CFE_TBL_DELETE_CDS_CC CFE_TBL_CCVAL(DELETE_CDS)`
- `#define CFE_TBL_ABORT_LOAD_CC CFE_TBL_CCVAL(ABORT_LOAD)`

11.243.1 Detailed Description

Specification for the CFE Event Services (CFE_TBL) command function codes

Note

This file should be strictly limited to the command/function code (CC) macro definitions. Other definitions such as enums, typedefs, or other macros should be placed in the msgdefs.h or msg.h files.

Definition in file [cfe_tbl_fcncodes.h](#).

11.243.2 Macro Definition Documentation

CFE_TBL_ABORT_LOAD_CC

```
#define CFE_TBL_ABORT_LOAD_CC CFE_TBL_CCVAL(ABORT_LOAD)
```

Name Abort Table Load

Description

This command will cause Table Services to discard the contents of a table buffer that was previously loaded with the data in a file as specified by a Table Load command. For single buffered tables, the allocated shared working buffer is freed and becomes available for other Table Load commands.

Command Mnemonic(s) \$sc_\$cpu_TBL_LOADABORT

Command Structure

[CFE_TBL_AbortLoadCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **\$sc_\$cpu_TBL_CMDPC** - command execution counter will increment
- The [CFE_TBL_LOAD_ABORT_INF_EID](#) informational event message is generated
- If the load was aborted for a single buffered table, the **\$sc_\$cpu_TBL_NumFreeShrBuf** telemetry point should increment

Error Conditions

This command may fail for the following reason(s):

- The specified table name was not found in the table registry.
- The specified table did not have a load in progress to be aborted.

Evidence of failure may be found in the following telemetry:

- **\$sc_\$cpu_TBL_CMDEC** - command error counter will increment
- Error specific event message

Criticality

This command will cause the loss of data put into an inactive table buffer.

See also

[CFE_TBL_LOAD_CC](#), [CFE_TBL_DUMP_CC](#), [CFE_TBL_VALIDATE_CC](#), [CFE_TBL_ACTIVATE_CC](#)

Definition at line [463](#) of file [cfe_tbl_fcncodes.h](#).

CFE_TBL_ACTIVATE_CC

```
#define CFE_TBL_ACTIVATE_CC CFE_TBL_CCVAL(ACTIVATE)
```

Name Activate Table

Description

This command will cause Table Services to notify a table's owner that an update is pending. The owning application will then update the contents of the active table buffer with the contents of the associated inactive table buffer at a time of their convenience.

Command Mnemonic(s) \$sc_\$cpu_TBL_ACTIVATE

Command Structure

CFE_TBL_ActivateCmd_t

Command Verification

Successful execution of this command may be verified with the following telemetry:

- \$sc_\$cpu_TBL_CMDPC - command execution counter will increment
- The CFE_TBL_UPDATE_SUCCESS_INF_EID informational event message will be generated

Error Conditions

This command may fail for the following reason(s):

- The specified table name was not found in the table registry.
- The table was registered as a "dump only" type and thus cannot be activated
- The table buffer has not been validated.

Evidence of failure may be found in the following telemetry:

- \$sc_\$cpu_TBL_CMDEC - command error counter will increment
- Command specific error event message are issued for all error cases

Criticality

This command will cause the contents of the specified table to be updated with the contents in the inactive table buffer.

See also

CFE_TBL_LOAD_CC, CFE_TBL_DUMP_CC, CFE_TBL_VALIDATE_CC, CFE_TBL_ABORT_LOAD_CC

Definition at line 301 of file [cfe_tbl_fcncodes.h](#).

CFE_TBL_DELETE_CDS_CC

```
#define CFE_TBL_DELETE_CDS_CC CFE_TBL_CCVAL(DELETE_CDS)
```

Name Delete Critical Table from Critical Data Store

Description

This command will delete the Critical Data Store (CDS) associated with the specified Critical Table. Note that any table still present in the Table Registry is unable to be deleted from the Critical Data Store. All Applications that are accessing the critical table must release and unregister their access before the CDS can be deleted.

Command Mnemonic(s) `$sc_$cpu_TBL_DeleteCDS`

Command Structure

`CFE_TBL_DeleteCDSCmd_t`

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_TBL_CMDPC` - command execution counter will increment
- The `CFE_TBL_CDS_DELETED_INFO_EID` informational event message will be generated

Error Conditions

This command may fail for the following reason(s):

- The specified table name was not found in the critical data store registry
- The specified table name WAS found in the table registry (all registrations/sharing of the table must be unregistered before the table's CDS can be deleted)
- The table's owning application is still active

Evidence of failure may be found in the following telemetry:

- `$sc_$cpu_TBL_CMDEC` - command error counter will increment
- Error specific event message

Criticality

This command will cause the loss of the specified table's contents before the owning Application was terminated.

See also

`CFE_ES_DUMP_CDS_REGISTRY_CC`, `CFE_ES_DELETE_CDS_CC`

Definition at line 424 of file `cfe_tbl_fcncodes.h`.

CFE_TBL_DUMP_CC

```
#define CFE_TBL_DUMP_CC CFE_TBL_CCVAL(DUMP)
```

Name Dump Table

Description

This command will cause the Table Services to put the contents of the specified table buffer into the command specified file.

Command Mnemonic(s) `$sc_$cpu_TBL_DUMP`

Command Structure

[CFE_TBL_DumpCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_TBL_CMDPC` - command execution counter will increment
- Either the [CFE_TBL_OVERWRITE_DUMP_INF_EID](#) OR the [CFE_TBL_WRITE_DUMP_INF_EID](#) informational event message will be generated

Error Conditions

This command may fail for the following reason(s):

- A single buffered table's inactive buffer was requested to be dumped and no such buffer is currently allocated.
- Error occurred during write operation to file. Possible causes might be insufficient space in the file system or the filename or file path is improperly specified.
- The specified table name was not found in the table registry.

Evidence of failure may be found in the following telemetry:

- `$sc_$cpu_TBL_CMDEC` - command error counter will increment
- A command specific error event message is issued for all error cases

Criticality

This command is not inherently dangerous. It will create a new file in the file system and could, if performed repeatedly without sufficient file management by the operator, fill the file system.

See also

[CFE_TBL_LOAD_CC](#), [CFE_TBL_VALIDATE_CC](#), [CFE_TBL_ACTIVATE_CC](#), [CFE_TBL_ABORT_LOAD_CC](#)

Definition at line 204 of file [cfe_tbl_fcncodes.h](#).

CFE_TBL_DUMP_REGISTRY_CC

```
#define CFE_TBL_DUMP_REGISTRY_CC CFE\_TBL\_CCVAL(DUMP_REGISTRY)
```

Name Dump Table Registry

Description

This command will cause Table Services to write some of the contents of the Table Registry to the command specified file. This allows the operator to see the current state and configuration of all tables that have been registered with the cFE.

Command Mnemonic(s) `$sc_$cpu_TBL_WriteReg2File`

Command Structure

[CFE_TBL_DumpRegistryCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_TBL_CMDPC` - command execution counter will increment
- The generation of either `CFE_TBL_OVERWRITE_REG_DUMP_INF_EID` or `CFE_TBL_WRITE_REG_DUMP_INF_EID` debug event messages
- The specified file should appear (or be updated) at the specified location in the file system

Error Conditions

This command may fail for the following reason(s):

- A table registry dump is already in progress, not yet completed
- The specified DumpFilename could not be parsed
- Error occurred during write operation to file. Possible causes might be insufficient space in the file system or the filename or file path is improperly specified.

Evidence of failure may be found in the following telemetry:

- `$sc_$cpu_TBL_CMDEC` - command error counter will increment
- An Error specific event message

Criticality

This command is not inherently dangerous. It will create a new file in the file system and could, if performed repeatedly without sufficient file management by the operator, fill the file system.

See also

[CFE_TBL_SEND_REGISTRY_CC](#)

Definition at line 345 of file [cfe_tbl_fcncodes.h](#).

CFE_TBL_LOAD_CC

```
#define CFE_TBL_LOAD_CC CFE_TBL_CCVAL(LOAD)
```

Name Load Table

Description

This command loads the contents of the specified file into an inactive buffer for the table specified within the file.

Command Mnemonic(s) `$sc_$cpu_TBL_Load`

Command Structure

[CFE_TBL_LoadCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_TBL_CMDPC` - command execution counter will increment
- The `CFE_TBL_FILE_LOADED_INF_EID` informational event message will be generated

Error Conditions

This command can fail for the following reasons:

- Table name found in table image file's table header is not found in table registry (ie - The table associated with the table image in the file has not been registered by an application).
- The table image file has an invalid or incorrect size. The size of the image file must match the size field within the header, and must also match the expected size of the table indicated in the registry.
- No working buffers are available for the load. This would indicate that too many single-buffered table loads are in progress at the same time.
- An attempt is being made to load an uninitialized table with a file containing only a partial table image.
- The table image file was unable to be opened. Either the file does not exist at the specified location, the filename is in error, or the file system has been corrupted.

Evidence of failure may be found in the following telemetry:

- `$sc_$cpu_TBL_CMDEC` - command error counter will increment
- Command specific error event messages are issued for all error cases

Criticality

This command is not inherently dangerous. It is performing the first step of loading a table and can be aborted (using the Abort Table Load command described below) without affecting the contents of the active table image.

See also

[CFE_TBL_DUMP_CC](#), [CFE_TBL_VALIDATE_CC](#), [CFE_TBL_ACTIVATE_CC](#), [CFE_TBL_ABORT_LOAD_CC](#)

Definition at line 161 of file [cfe_tbl_fcncodes.h](#).

CFE_TBL_NOOP_CC

```
#define CFE_TBL_NOOP_CC CFE_TBL_CCVAL(NOOP)
```

Name Table No-Op

Description

This command performs no other function than to increment the command execution counter. The command may be used to verify general aliveness of the Table Services task.

Command Mnemonic(s) `$sc_$cpu_TBL_NOOP`

Command Structure

[CFE_TBL_NoopCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_TBL_CMDPC` - command execution counter will increment
- The [CFE_TBL_NOOP_INF_EID](#) informational event message will be generated

Error Conditions

There are no error conditions for this command. If the Table Services receives the command, the event is sent (although it may be filtered by EVS) and the counter is incremented unconditionally.

Criticality

None

See also

Definition at line 70 of file [cfe_tbl_fcncodes.h](#).

CFE_TBL_RESET_COUNTERS_CC

```
#define CFE_TBL_RESET_COUNTERS_CC CFE_TBL_CCVAL(RESET_COUNTERS)
```

Name Table Reset Counters

Description

This command resets the following counters within the Table Services housekeeping telemetry:

- Command Execution Counter (\$sc_\$cpu_TBL_CMDPC)
- Command Error Counter (\$sc_\$cpu_TBL_CMDEC)
- Successful Table Validations Counter (\$sc_\$cpu_TBL_ValSuccessCtr)
- Failed Table Validations Counter (\$sc_\$cpu_TBL_ValFailedCtr)
- Number of Table Validations Requested (\$sc_\$cpu_TBL_ValReqCtr)
- Number of completed table validations (\$sc_\$cpu_TBL_ValCompltdCtr)

Command Mnemonic(s) \$sc_\$cpu_TBL_ResetCtrs

Command Structure

[CFE_TBL_ResetCountersCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **\$sc_\$cpu_TBL_CMDPC** - command execution counter will be reset to 0
- The [CFE_TBL_RESET_INF_EID](#) debug event message will be generated

Error Conditions

There are no error conditions for this command. If the Table Services receives the command, the event is sent (although it may be filtered by EVS) and the counter is incremented unconditionally.

Criticality

This command is not inherently dangerous. However, it is possible for ground systems and on-board safing procedures to be designed such that they react to changes in the counter values that are reset by this command.

See also

Definition at line 111 of file [cfe_tbl_fcncodes.h](#).

CFE_TBL_SEND_REGISTRY_CC

```
#define CFE_TBL_SEND_REGISTRY_CC CFE_TBL_CCVAL(SEND_REGISTRY)
```

Name Telemeter One Table Registry Entry

Description

This command will cause Table Services to telemeter the contents of the Table Registry for the command specified table.

Command Mnemonic(s) \$sc_\$cpu_TBL_TLMReg

Command Structure

[CFE_TBL_SendRegistryCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- [\\$sc_\\$cpu_TBL_CMDPC](#) - command execution counter will increment
- Receipt of a Table Registry Info Packet (see [CFE_TBL_TableRegistryTlm_t](#))
- The [CFE_TBL_TLM_REG_CMD_INF_EID](#) debug event message will be generated

Error Conditions

This command may fail for the following reason(s):

- The specified table name was not found in the table registry.

Evidence of failure may be found in the following telemetry:

- [\\$sc_\\$cpu_TBL_CMDEC](#) - command error counter will increment
- Error specific event message

Criticality

This command is not inherently dangerous. It will generate additional telemetry.

See also

[CFE_TBL_DUMP_REGISTRY_CC](#)

Definition at line 380 of file [cfe_tbl_fcncodes.h](#).

CFE_TBL_VALIDATE_CC

```
#define CFE_TBL_VALIDATE_CC CFE_TBL_CCVAL(VALIDATE)
```

Name Validate Table

Description

This command will cause Table Services to calculate the Data Integrity Value for the specified table and to notify the owning application that the table's validation function should be executed. The results of both the Data Integrity Value computation and the validation function are reported in Table Services Housekeeping Telemetry.

Command Mnemonic(s) `$sc_$cpu_TBL_VALIDATE`

Command Structure

`CFE_TBL_ValidateCmd_t`

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_TBL_CMDPC` - command execution counter will increment
- `$sc_$cpu_TBL_ValReqCtr` - table validation request counter will increment
- `$sc_$cpu_TBL_LastValCRC` - calculated data integrity value will be updated
- The `CFE_TBL_VAL_REQ_MADE_INF_EID` debug event message (indicating the application is being notified of a validation request)

If the specified table has an associated validation function, then the following telemetry will also change:

- Either `$sc_$cpu_TBL_ValSuccessCtr` OR `$sc_$cpu_TBL_ValFailedCtr` will increment
- `$sc_$cpu_TBL_ValCompltdCtr` - table validations performed counter will increment
- `$sc_$cpu_TBL_LastValS` - table validation function return status will update
- The `CFE_TBL_VALIDATION_INF_EID` informational event message (indicating the validation function return status) will be generated

Error Conditions

This command may fail for the following reason(s):

- A single buffered table's inactive buffer was requested to be validated and no such buffer is currently allocated.
- Too many validations have been requested simultaneously. The operator must wait for one or more applications to perform their table validation functions before trying again.
- The specified table name was not found in the table registry.

Evidence of failure may be found in the following telemetry:

- `$sc_$cpu_TBL_CMDEC` - command error counter will increment
- Command specific error event message are issued for all error cases

Criticality

The success or failure of a table validation does not have any immediate impact on table contents. The results are sent to the operator in telemetry and the operator must determine whether the results are acceptable and send a command to activate the validated table image.

See also

`CFE_TBL_LOAD_CC`, `CFE_TBL_DUMP_CC`, `CFE_TBL_ACTIVATE_CC`, `CFE_TBL_ABORT_LOAD_CC`

Definition at line 261 of file `cf_tbl_fcncodes.h`.

11.244 cfe_tbl_fcncodes.h

[Go to the documentation of this file.](#)

```

00001 /*****
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"
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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00028 #ifndef CFE_TBL_FCNCODES_H
00029 #define CFE_TBL_FCNCODES_H
00030
00031 #include "cfe_tbl_fcncode_values.h"
00032
00033 /*
00034  ** Table task command packet command codes
00035  */
00070 #define CFE_TBL_NOOP_CC CFE_TBL_CCVAL(NOOP)
00071
00111 #define CFE_TBL_RESET_COUNTERS_CC CFE_TBL_CCVAL(RESET_COUNTERS)
00112
00161 #define CFE_TBL_LOAD_CC CFE_TBL_CCVAL(LOAD)
00162
00204 #define CFE_TBL_DUMP_CC CFE_TBL_CCVAL(DUMP)
00205
00261 #define CFE_TBL_VALIDATE_CC CFE_TBL_CCVAL(VVALIDATE)
00262
00301 #define CFE_TBL_ACTIVATE_CC CFE_TBL_CCVAL(ACTIVATE)
00302
00345 #define CFE_TBL_DUMP_REGISTRY_CC CFE_TBL_CCVAL(DUMP_REGISTRY)
00346
00380 #define CFE_TBL_SEND_REGISTRY_CC CFE_TBL_CCVAL(SEND_REGISTRY)
00381
00424 #define CFE_TBL_DELETE_CDS_CC CFE_TBL_CCVAL(DELETE_CDS)
00425
00463 #define CFE_TBL_ABORT_LOAD_CC CFE_TBL_CCVAL(ABORT_LOAD)
00466 #endif

```

11.245 cfe/modules/tbl/fsw/inc/cfe_tbl_interface_cfg.h File Reference

```
#include "cfe_tbl_interface_cfg_values.h"
```

Macros

- #define CFE_MISSION_TBL_MAX_NAME_LENGTH CFE_MISSION_TBL_CFGVAL(MAX_NAME_LENGTH)
- #define DEFAULT_CFE_MISSION_TBL_MAX_NAME_LENGTH 16
- #define CFE_MISSION_TBL_MAX_FULL_NAME_LEN CFE_MISSION_TBL_CFGVAL(MAX_FULL_NAME_LEN)
- #define DEFAULT_CFE_MISSION_TBL_MAX_FULL_NAME_LEN 40

11.245.1 Detailed Description

CFE Event Services (CFE_TBL) Application Mission Configuration Header File

This is a compatibility header for the "mission_cfg.h" file that has traditionally provided public config definitions for each CFS app.

Note

This file may be overridden/superseded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [cfe_tbl_interface_cfg.h](#).

11.245.2 Macro Definition Documentation**CFE_MISSION_TBL_MAX_FULL_NAME_LEN**

```
#define CFE_MISSION_TBL_MAX_FULL_NAME_LEN CFE_MISSION_TBL_CFGVAL (MAX_FULL_NAME_LEN)
```

Purpose Maximum Length of Full Table Name in messages

Description:

Indicates the maximum length (in characters) of the entire table name within software bus messages, in "App↵Name.TableName" notation.

This affects the layout of command/telemetry messages but does not affect run time behavior or internal allocation.

Limits

All CPUs within the same SB domain (mission) must share the same definition Note this affects the size of messages, so it must not cause any message to exceed the max length.

This value should be kept as a multiple of 4, to maintain alignment of any possible neighboring fields without implicit padding.

Definition at line 71 of file [cfe_tbl_interface_cfg.h](#).

CFE_MISSION_TBL_MAX_NAME_LENGTH

```
#define CFE_MISSION_TBL_MAX_NAME_LENGTH CFE_MISSION_TBL_CFGVAL (MAX_NAME_LENGTH)
```

Purpose Maximum Table Name Length

Description:

Indicates the maximum length (in characters) of the table name ('TblName') portion of a Full Table Name of the following form: "ApplicationName.TblName"

This length does not need to include an extra character for NULL termination.

Limits

This value should be kept as a multiple of 4, to maintain alignment of any possible neighboring fields without implicit padding.

Definition at line 50 of file [cfe_tbl_interface_cfg.h](#).

DEFAULT_CFE_MISSION_TBL_MAX_FULL_NAME_LEN

```
#define DEFAULT_CFE_MISSION_TBL_MAX_FULL_NAME_LEN 40
```

Definition at line 74 of file [cfe_tbl_interface_cfg.h](#).

DEFAULT_CFE_MISSION_TBL_MAX_NAME_LENGTH

```
#define DEFAULT_CFE_MISSION_TBL_MAX_NAME_LENGTH 16
```

Definition at line 51 of file [cfe_tbl_interface_cfg.h](#).

11.246 cfe_tbl_interface_cfg.h

[Go to the documentation of this file.](#)

```

00001 /*****
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"
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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00031 #ifndef CFE_TBL_MISSION_CFG_H
00032 #define CFE_TBL_MISSION_CFG_H
00033
00034 #include "cfe_tbl_interface_cfg_values.h"
00035
00050 #define CFE_MISSION_TBL_MAX_NAME_LENGTH CFE_MISSION_TBL_CFGVAL(MAX_NAME_LENGTH)
00051 #define DEFAULT_CFE_MISSION_TBL_MAX_NAME_LENGTH 16
00052
00071 #define CFE_MISSION_TBL_MAX_FULL_NAME_LEN CFE_MISSION_TBL_CFGVAL(MAX_FULL_NAME_LEN)
00072
00073 /* Default value calculated as: (CFE_MISSION_TBL_MAX_NAME_LENGTH + CFE_MISSION_MAX_API_LEN + 4) */
00074 #define DEFAULT_CFE_MISSION_TBL_MAX_FULL_NAME_LEN 40
00075
00076 #endif

```

11.247 cfe/modules/tbl/fsw/inc/cfe_tbl_internal_cfg.h File Reference

```

#include "cfe_tbl_mission_cfg.h"
#include "cfe_tbl_internal_cfg_values.h"

```

Macros

- #define CFE_PLATFORM_TBL_START_TASK_PRIORITY CFE_PLATFORM_TBL_CFGVAL(START_TASK_PRIORITY)
- #define DEFAULT_CFE_PLATFORM_TBL_START_TASK_PRIORITY 70
- #define CFE_PLATFORM_TBL_START_TASK_STACK_SIZE CFE_PLATFORM_TBL_CFGVAL(START_TASK_STACK_SIZE)
- #define DEFAULT_CFE_PLATFORM_TBL_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
- #define CFE_PLATFORM_TBL_BUF_MEMORY_BYTES CFE_PLATFORM_TBL_CFGVAL(BUF_MEMORY_BYTES)
- #define DEFAULT_CFE_PLATFORM_TBL_BUF_MEMORY_BYTES 524288
- #define CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE CFE_PLATFORM_TBL_CFGVAL(MAX_DBL_TABLE_SIZE)
- #define DEFAULT_CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE 16384
- #define CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE CFE_PLATFORM_TBL_CFGVAL(MAX_SNGL_TABLE_SIZE)
- #define DEFAULT_CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE 16384
- #define CFE_PLATFORM_TBL_MAX_NUM_TABLES CFE_PLATFORM_TBL_CFGVAL(MAX_NUM_TABLES)
- #define DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_TABLES 128
- #define CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES CFE_PLATFORM_TBL_CFGVAL(MAX_CRITICAL_TABLES)
- #define DEFAULT_CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES 32

- `#define CFE_PLATFORM_TBL_MAX_NUM_HANDLES CFE_PLATFORM_TBL_CFGVAL(MAX_NUM_HANDLES)`
- `#define DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_HANDLES 256`
- `#define CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS CFE_PLATFORM_TBL_CFGVAL(MAX_SIMULTANEOUS_LOADS)`
- `#define DEFAULT_CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS 4`
- `#define CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS CFE_PLATFORM_TBL_CFGVAL(MAX_NUM_VALIDATIONS)`
- `#define DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS 10`
- `#define CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE CFE_PLATFORM_TBL_CFGVAL(DEFAULT_REG_DUMP_FILE)`
- `#define DEFAULT_CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE "/ram/cfe_tbl_reg.log"`
- `#define CFE_PLATFORM_TBL_VALID_SCID_COUNT CFE_PLATFORM_TBL_CFGVAL(VALID_SCID_COUNT)`
- `#define DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_COUNT 0`
- `#define CFE_PLATFORM_TBL_U32FROM4CHARS(_C1, _C2, _C3, _C4) (((uint32)(_C1) << 24 | (uint32)(_C2) << 16 | (uint32)(_C3) << 8 | (uint32)(_C4)))`
- `#define CFE_PLATFORM_TBL_VALID_SCID_1 CFE_PLATFORM_TBL_CFGVAL(VALID_SCID_1)`
- `#define DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_1 (0x42)`
- `#define CFE_PLATFORM_TBL_VALID_SCID_2 CFE_PLATFORM_TBL_CFGVAL(VALID_SCID_2)`
- `#define DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_2 (CFE_PLATFORM_TBL_U32FROM4CHARS('a', 'b', 'c', 'd'))`
- `#define CFE_PLATFORM_TBL_VALID_PRID_COUNT CFE_PLATFORM_TBL_CFGVAL(VALID_PRID_COUNT)`
- `#define DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_COUNT 0`
- `#define CFE_PLATFORM_TBL_VALID_PRID_1 CFE_PLATFORM_TBL_CFGVAL(VALID_PRID_1)`
- `#define DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_1 (1)`
- `#define CFE_PLATFORM_TBL_VALID_PRID_2 CFE_PLATFORM_TBL_CFGVAL(VALID_PRID_2)`
- `#define DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_2 (CFE_PLATFORM_TBL_U32FROM4CHARS('a', 'b', 'c', 'd'))`
- `#define CFE_PLATFORM_TBL_VALID_PRID_3 CFE_PLATFORM_TBL_CFGVAL(VALID_PRID_3)`
- `#define DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_3 0`
- `#define CFE_PLATFORM_TBL_VALID_PRID_4 CFE_PLATFORM_TBL_CFGVAL(VALID_PRID_4)`
- `#define DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_4 0`

11.247.1 Detailed Description

CFE Table Services (CFE_TBL) Application Platform Configuration Header File

This is a compatibility header for the "platform_cfg.h" file that has traditionally provided both public and private config definitions for each CFS app.

These definitions are now provided in two separate files, one for the public/mission scope and one for internal scope.

Note

This file may be overridden/superseded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file `cfe_tbl_internal_cfg.h`.

11.247.2 Macro Definition Documentation

CFE_PLATFORM_TBL_BUF_MEMORY_BYTES

```
#define CFE_PLATFORM_TBL_BUF_MEMORY_BYTES CFE_PLATFORM_TBL_CFGVAL (BUF_MEMORY_BYTES)
```

Purpose Size of Table Services Table Memory Pool

Description:

Defines the TOTAL size of the memory pool that cFE Table Services allocates from the system. The size must be large enough to provide memory for each registered table, the inactive buffers for double buffered tables and for the shared inactive buffers for single buffered tables.

Limits

The cFE does not place a limit on the size of this parameter.

Definition at line 83 of file [cfe_tbl_internal_cfg.h](#).

CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE

```
#define CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE CFE_PLATFORM_TBL_CFGVAL (DEFAULT_REG_DUMP_FILE)
```

Purpose Default Filename for a Table Registry Dump

Description:

Defines the file name used to store the table registry when no filename is specified in the dump registry command.

Limits

The length of each string, including the NULL terminator cannot exceed the [OS_MAX_PATH_LEN](#) value.

Definition at line 205 of file [cfe_tbl_internal_cfg.h](#).

CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES

```
#define CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES CFE_PLATFORM_TBL_CFGVAL (MAX_CRITICAL_TABLES)
```

Purpose Maximum Number of Critical Tables that can be Registered

Description:

Defines the maximum number of critical tables supported by this processor's Table Services.

Limits

This number must be less than 32767. It should be recognized that this parameter determines the size of the Critical Table Registry which is maintained in the Critical Data Store. An excessively high number will waste Critical Data Store memory. Therefore, this number must not exceed the value defined in [CFE_PLATFORM_ES_CDS_MAX_NUM_ENTRIES](#).

Definition at line 142 of file [cfe_tbl_internal_cfg.h](#).

CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE

```
#define CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE CFE_PLATFORM_TBL_CFGVAL (MAX_DBL_TABLE_SIZE)
```

Purpose Maximum Size Allowed for a Double Buffered Table

Description:

Defines the maximum allowed size (in bytes) of a double buffered table.

Limits

The cFE does not place a limit on the size of this parameter but it must be less than half of [CFE_PLATFORM_TBL_BUF_MEMORY_B](#)

Definition at line 96 of file [cfe_tbl_internal_cfg.h](#).

CFE_PLATFORM_TBL_MAX_NUM_HANDLES

```
#define CFE_PLATFORM_TBL_MAX_NUM_HANDLES CFE_PLATFORM_TBL_CFGVAL (MAX_NUM_HANDLES)
```

Purpose Maximum Number of Table Handles

Description:

Defines the maximum number of Table Handles.

Limits

This number must be less than 32767. This number must be at least as big as the number of tables ([CFE_PLATFORM_TBL_MAX_NUM_TABLES](#)) and should be set higher if tables are shared between applications.

Definition at line 156 of file [cfe_tbl_internal_cfg.h](#).

CFE_PLATFORM_TBL_MAX_NUM_TABLES

```
#define CFE_PLATFORM_TBL_MAX_NUM_TABLES CFE_PLATFORM_TBL_CFGVAL (MAX_NUM_TABLES)
```

Purpose Maximum Number of Tables Allowed to be Registered

Description:

Defines the maximum number of tables supported by this processor's Table Services.

Limits

This number must be less than 32767. It should be recognized that this parameter determines the size of the Table Registry. An excessively high number will waste memory.

Definition at line 127 of file [cfe_tbl_internal_cfg.h](#).

CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS

```
#define CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS CFE_PLATFORM_TBL_CFGVAL (MAX_NUM_VALIDATIONS)
```

Purpose Maximum Number of Simultaneous Table Validations

Description:

Defines the maximum number of pending validations that the Table Services can handle at any one time. When a table has a validation function, a validation request is made of the application to perform that validation. This number determines how many of those requests can be outstanding at any one time.

Limits

This number must be less than 32767. An excessively high number will degrade system performance and waste memory. A number less than 20 is suggested but not required.

Definition at line 191 of file [cfe_tbl_internal_cfg.h](#).

CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS

```
#define CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS CFE_PLATFORM_TBL_CFGVAL (MAX_SIMULTANEOUS_LOADS)
```

Purpose Maximum Number of Simultaneous Loads to Support

Description:

Defines the maximum number of single buffered tables that can be loaded simultaneously. This number is used to determine the number of shared buffers to allocate.

Limits

This number must be less than 32767. An excessively high number will degrade system performance and waste memory. A number less than 5 is suggested but not required.

Definition at line 172 of file [cfe_tbl_internal_cfg.h](#).

CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE

```
#define CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE CFE_PLATFORM_TBL_CFGVAL (MAX_SNGL_TABLE_SIZE)
```

Purpose Maximum Size Allowed for a Single Buffered Table

Description:

Defines the maximum allowed size (in bytes) of a single buffered table. **NOTE:** This size determines the size of all shared table buffers. Therefore, this size will be multiplied by [CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS](#) below when allocating memory for shared tables.

Limits

The cFE does not place a limit on the size of this parameter but it must be small enough to allow for [CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS](#) number of tables to fit into [CFE_PLATFORM_TBL_BUF_MEMORY_BYTES](#).

Definition at line 113 of file [cfe_tbl_internal_cfg.h](#).

CFE_PLATFORM_TBL_START_TASK_PRIORITY

```
#define CFE_PLATFORM_TBL_START_TASK_PRIORITY CFE_PLATFORM_TBL_CFGVAL (START_TASK_PRIORITY)
```

Purpose Define TBL Task Priority

Description:

Defines the cFE_TBL Task priority.

Limits

Not Applicable

Definition at line 50 of file [cfe_tbl_internal_cfg.h](#).

CFE_PLATFORM_TBL_START_TASK_STACK_SIZE

```
#define CFE_PLATFORM_TBL_START_TASK_STACK_SIZE CFE_PLATFORM_TBL_CFGVAL (START_TASK_STACK_SIZE)
```

Purpose Define TBL Task Stack Size

Description:

Defines the cFE_TBL Task Stack Size

Limits

There is a lower limit of 2048 on this configuration parameter. There are no restrictions on the upper limit however, the maximum stack size is system dependent and should be verified. Most operating systems provide tools for measuring the amount of stack used by a task during operation. It is always a good idea to verify that no more than 1/2 of the stack is used.

Definition at line 66 of file [cfe_tbl_internal_cfg.h](#).

CFE_PLATFORM_TBL_U32FROM4CHARS

```
#define CFE_PLATFORM_TBL_U32FROM4CHARS(  
    _C1,  
    _C2,  
    _C3,  
    _C4 )  ((uint32) (_C1) << 24 | (uint32) (_C2) << 16 | (uint32) (_C3) << 8 | (uint32) (_C4))
```

Definition at line 229 of file [cfe_tbl_internal_cfg.h](#).

CFE_PLATFORM_TBL_VALID_PRID_1

```
#define CFE_PLATFORM_TBL_VALID_PRID_1 CFE_PLATFORM_TBL_CFGVAL (VALID_PRID_1)
```

Purpose Processor ID values used for table load validation

Description:

Defines the processor ID values used for validating the processor ID field in the table file header. To be valid, the spacecraft ID specified in the table file header must match one of the values defined here.

Limits

This value can be any 32 bit unsigned integer.

Definition at line 282 of file [cfe_tbl_internal_cfg.h](#).

CFE_PLATFORM_TBL_VALID_PRID_2

```
#define CFE_PLATFORM_TBL_VALID_PRID_2 CFE_PLATFORM_TBL_CFGVAL (VALID_PRID_2)
```

Definition at line 285 of file [cfe_tbl_internal_cfg.h](#).

CFE_PLATFORM_TBL_VALID_PRID_3

```
#define CFE_PLATFORM_TBL_VALID_PRID_3 CFE_PLATFORM_TBL_CFGVAL (VALID_PRID_3)
```

Definition at line 288 of file [cfe_tbl_internal_cfg.h](#).

CFE_PLATFORM_TBL_VALID_PRID_4

```
#define CFE_PLATFORM_TBL_VALID_PRID_4 CFE_PLATFORM_TBL_CFGVAL (VALID_PRID_4)
```

Definition at line 291 of file [cfe_tbl_internal_cfg.h](#).

CFE_PLATFORM_TBL_VALID_PRID_COUNT

```
#define CFE_PLATFORM_TBL_VALID_PRID_COUNT CFE_PLATFORM_TBL_CFGVAL (VALID_PRID_COUNT)
```

Purpose Number of Processor ID's specified for validation

Description:

Defines the number of specified processor ID values that are verified during table loads. If the number is zero then no validation of the processor ID field in the table file header is performed when tables are loaded. Non-zero values indicate how many values from the list of processor ID's defined below are compared to the processor ID field in the table file header. The ELF2CFETBL tool may be used to create table files with specified processor ID values.

Limits

This number must be greater than or equal to zero and less than or equal to 4.

Definition at line 267 of file [cfe_tbl_internal_cfg.h](#).

CFE_PLATFORM_TBL_VALID_SCID_1

```
#define CFE_PLATFORM_TBL_VALID_SCID_1 CFE_PLATFORM_TBL_CFGVAL (VALID_SCID_1)
```

Purpose Spacecraft ID values used for table load validation

Description:

Defines the spacecraft ID values used for validating the spacecraft ID field in the table file header. To be valid, the spacecraft ID specified in the table file header must match one of the values defined here.

Limits

This value can be any 32 bit unsigned integer.

Definition at line 244 of file [cfe_tbl_internal_cfg.h](#).

CFE_PLATFORM_TBL_VALID_SCID_2

```
#define CFE_PLATFORM_TBL_VALID_SCID_2 CFE_PLATFORM_TBL_CFGVAL (VALID_SCID_2)
```

Definition at line 247 of file [cfe_tbl_internal_cfg.h](#).

CFE_PLATFORM_TBL_VALID_SCID_COUNT

```
#define CFE_PLATFORM_TBL_VALID_SCID_COUNT CFE_PLATFORM_TBL_CFGVAL (VALID_SCID_COUNT)
```

Purpose Number of Spacecraft ID's specified for validation

Description:

Defines the number of specified spacecraft ID values that are verified during table loads. If the number is zero then no validation of the spacecraft ID field in the table file header is performed when tables are loaded. Non-zero values indicate how many values from the list of spacecraft ID's defined below are compared to the spacecraft ID field in the table file header. The ELF2CFETBL tool may be used to create table files with specified spacecraft ID values.

Limits

This number must be greater than or equal to zero and less than or equal to 2.

Definition at line 225 of file [cfe_tbl_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_BUF_MEMORY_BYTES

```
#define DEFAULT_CFE_PLATFORM_TBL_BUF_MEMORY_BYTES 524288
```

Definition at line 84 of file [cfe_tbl_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE

```
#define DEFAULT_CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE "/ram/cfe_tbl_reg.log"
```

Definition at line 206 of file [cfe_tbl_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES

```
#define DEFAULT_CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES 32
```

Definition at line 143 of file [cfe_tbl_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE

```
#define DEFAULT_CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE 16384
```

Definition at line 97 of file [cfe_tbl_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_HANDLES

```
#define DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_HANDLES 256
```

Definition at line 157 of file [cfe_tbl_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_TABLES

```
#define DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_TABLES 128
```

Definition at line 128 of file [cfe_tbl_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS

```
#define DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS 10
```

Definition at line 192 of file [cfe_tbl_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS

```
#define DEFAULT_CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS 4
```

Definition at line 173 of file [cfe_tbl_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE

```
#define DEFAULT_CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE 16384
```

Definition at line 114 of file [cfe_tbl_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_START_TASK_PRIORITY

```
#define DEFAULT_CFE_PLATFORM_TBL_START_TASK_PRIORITY 70
```

Definition at line 51 of file [cfe_tbl_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_START_TASK_STACK_SIZE

```
#define DEFAULT_CFE_PLATFORM_TBL_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
```

Definition at line 67 of file [cfe_tbl_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_1

```
#define DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_1 (1)
```

Definition at line 283 of file [cfe_tbl_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_2

```
#define DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_2 (CFE_PLATFORM_TBL_U32FROM4CHARS('a', 'b', 'c', 'd'))
```

Definition at line 286 of file [cfe_tbl_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_3

```
#define DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_3 0
```

Definition at line 289 of file [cfe_tbl_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_4

```
#define DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_4 0
```

Definition at line 292 of file [cfe_tbl_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_COUNT

```
#define DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_COUNT 0
```

Definition at line 268 of file [cfe_tbl_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_1

```
#define DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_1 (0x42)
```

Definition at line 245 of file [cfe_tbl_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_2

```
#define DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_2 (CFE_PLATFORM_TBL_U32FROM4CHARS('a', 'b', 'c', 'd'))
```

Definition at line 248 of file [cfe_tbl_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_COUNT

```
#define DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_COUNT 0
```

Definition at line 226 of file [cfe_tbl_internal_cfg.h](#).

11.248 cfe_tbl_internal_cfg.h

[Go to the documentation of this file.](#)

```
00001 /*****
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"
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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00035 #ifndef CFE_TBL_PLATFORM_CFG_H
00036 #define CFE_TBL_PLATFORM_CFG_H
00037
00038 #include "cfe_tbl_mission_cfg.h"
00039 #include "cfe_tbl_internal_cfg_values.h"
00040
00050 #define CFE_PLATFORM_TBL_START_TASK_PRIORITY CFE_PLATFORM_TBL_CFGVAL(START_TASK_PRIORITY)
00051 #define DEFAULT_CFE_PLATFORM_TBL_START_TASK_PRIORITY 70
00052
00066 #define CFE_PLATFORM_TBL_START_TASK_STACK_SIZE CFE_PLATFORM_TBL_CFGVAL(START_TASK_STACK_SIZE)
00067 #define DEFAULT_CFE_PLATFORM_TBL_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
00068
00069 /* Platform Configuration Parameters for Table Service (TBL) */
00070
00083 #define CFE_PLATFORM_TBL_BUF_MEMORY_BYTES CFE_PLATFORM_TBL_CFGVAL(BUF_MEMORY_BYTES)
00084 #define DEFAULT_CFE_PLATFORM_TBL_BUF_MEMORY_BYTES 524288
00085
00096 #define CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE CFE_PLATFORM_TBL_CFGVAL(MAX_DBL_TABLE_SIZE)
00097 #define DEFAULT_CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE 16384
00098
00113 #define CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE CFE_PLATFORM_TBL_CFGVAL(MAX_SNGL_TABLE_SIZE)
00114 #define DEFAULT_CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE 16384
00115
00127 #define CFE_PLATFORM_TBL_MAX_NUM_TABLES CFE_PLATFORM_TBL_CFGVAL(MAX_NUM_TABLES)
00128 #define DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_TABLES 128
00129
00142 #define CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES CFE_PLATFORM_TBL_CFGVAL(MAX_CRITICAL_TABLES)
00143 #define DEFAULT_CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES 32
00144
00156 #define CFE_PLATFORM_TBL_MAX_NUM_HANDLES CFE_PLATFORM_TBL_CFGVAL(MAX_NUM_HANDLES)
00157 #define DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_HANDLES 256
00158
00172 #define CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS CFE_PLATFORM_TBL_CFGVAL(MAX_SIMULTANEOUS_LOADS)
00173 #define DEFAULT_CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS 4
00174
00191 #define CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS CFE_PLATFORM_TBL_CFGVAL(MAX_NUM_VALIDATIONS)
00192 #define DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS 10
00193
00205 #define CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE CFE_PLATFORM_TBL_CFGVAL(DEFAULT_REG_DUMP_FILE)
00206 #define DEFAULT_CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE "/ram/cfe_tbl_reg.log"
00207
00225 #define CFE_PLATFORM_TBL_VALID_SCID_COUNT CFE_PLATFORM_TBL_CFGVAL(VALID_SCID_COUNT)
00226 #define DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_COUNT 0
00227
00228 /* macro to construct 32 bit value from 4 chars */
00229 #define CFE_PLATFORM_TBL_U32FROM4CHARS(_C1, _C2, _C3, _C4) \
00230     ((uint32)(_C1) << 24 | (uint32)(_C2) << 16 | (uint32)(_C3) << 8 | (uint32)(_C4))
00231
00244 #define CFE_PLATFORM_TBL_VALID_SCID_1 CFE_PLATFORM_TBL_CFGVAL(VALID_SCID_1)
00245 #define DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_1 (0x42)
00246
00247 #define CFE_PLATFORM_TBL_VALID_SCID_2 CFE_PLATFORM_TBL_CFGVAL(VALID_SCID_2)
```

```

00248 #define DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_2 (CFE_PLATFORM_TBL_U32FROM4CHARS('a', 'b', 'c', 'd'))
00249
00267 #define CFE_PLATFORM_TBL_VALID_PRID_COUNT CFE_PLATFORM_TBL_CFGVAL(VVALID_PRID_COUNT)
00268 #define DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_COUNT 0
00269
00282 #define CFE_PLATFORM_TBL_VALID_PRID_1 CFE_PLATFORM_TBL_CFGVAL(VVALID_PRID_1)
00283 #define DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_1 (1)
00284
00285 #define CFE_PLATFORM_TBL_VALID_PRID_2 CFE_PLATFORM_TBL_CFGVAL(VVALID_PRID_2)
00286 #define DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_2 (CFE_PLATFORM_TBL_U32FROM4CHARS('a', 'b', 'c', 'd'))
00287
00288 #define CFE_PLATFORM_TBL_VALID_PRID_3 CFE_PLATFORM_TBL_CFGVAL(VVALID_PRID_3)
00289 #define DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_3 0
00290
00291 #define CFE_PLATFORM_TBL_VALID_PRID_4 CFE_PLATFORM_TBL_CFGVAL(VVALID_PRID_4)
00292 #define DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_4 0
00293
00294 #endif

```

11.249 cfe/modules/tbl/fsw/inc/cfe_tbl_topicids.h File Reference

```
#include "cfe_tbl_topicid_values.h"
```

Macros

- `#define CFE_MISSION_TBL_CMD_TOPICID CFE_MISSION_TBL_TIDVAL(CMD)`
- `#define DEFAULT_CFE_MISSION_TBL_CMD_TOPICID 4`
- `#define CFE_MISSION_TBL_SEND_HK_TOPICID CFE_MISSION_TBL_TIDVAL(SEND_HK)`
- `#define DEFAULT_CFE_MISSION_TBL_SEND_HK_TOPICID 12`
- `#define CFE_MISSION_TBL_HK_TLM_TOPICID CFE_MISSION_TBL_TIDVAL(HK_TLM)`
- `#define DEFAULT_CFE_MISSION_TBL_HK_TLM_TOPICID 4`
- `#define CFE_MISSION_TBL_REG_TLM_TOPICID CFE_MISSION_TBL_TIDVAL(REG_TLM)`
- `#define DEFAULT_CFE_MISSION_TBL_REG_TLM_TOPICID 12`

11.249.1 Detailed Description

CFE Table Services (CFE_TBL) Application Topic IDs

Definition in file [cfe_tbl_topicids.h](#).

11.249.2 Macro Definition Documentation

CFE_MISSION_TBL_CMD_TOPICID

```
#define CFE_MISSION_TBL_CMD_TOPICID CFE_MISSION_TBL_TIDVAL(CMD)
```

Purpose cFE Portable Message Numbers for Commands

Description:

Portable message numbers for the cFE command messages

Limits

Not Applicable

Definition at line 37 of file [cfe_tbl_topicids.h](#).

CFE_MISSION_TBL_HK_TLM_TOPICID

```
#define CFE_MISSION_TBL_HK_TLM_TOPICID CFE_MISSION_TBL_TIDVAL (HK_TLM)
```

Purpose cFE Portable Message Numbers for Telemetry

Description:

Portable message numbers for the cFE telemetry messages

Limits

Not Applicable

Definition at line 52 of file [cfe_tbl_topicsids.h](#).

CFE_MISSION_TBL_REG_TLM_TOPICID

```
#define CFE_MISSION_TBL_REG_TLM_TOPICID CFE_MISSION_TBL_TIDVAL (REG_TLM)
```

Definition at line 55 of file [cfe_tbl_topicsids.h](#).

CFE_MISSION_TBL_SEND_HK_TOPICID

```
#define CFE_MISSION_TBL_SEND_HK_TOPICID CFE_MISSION_TBL_TIDVAL (SEND_HK)
```

Definition at line 40 of file [cfe_tbl_topicsids.h](#).

DEFAULT_CFE_MISSION_TBL_CMD_TOPICID

```
#define DEFAULT_CFE_MISSION_TBL_CMD_TOPICID 4
```

Definition at line 38 of file [cfe_tbl_topicsids.h](#).

DEFAULT_CFE_MISSION_TBL_HK_TLM_TOPICID

```
#define DEFAULT_CFE_MISSION_TBL_HK_TLM_TOPICID 4
```

Definition at line 53 of file [cfe_tbl_topicsids.h](#).

DEFAULT_CFE_MISSION_TBL_REG_TLM_TOPICID

```
#define DEFAULT_CFE_MISSION_TBL_REG_TLM_TOPICID 12
```

Definition at line 56 of file [cfe_tbl_topicsids.h](#).

DEFAULT_CFE_MISSION_TBL_SEND_HK_TOPICID

```
#define DEFAULT_CFE_MISSION_TBL_SEND_HK_TOPICID 12
```

Definition at line 41 of file [cfe_tbl_topicsids.h](#).

11.250 cfe_tbl_topicsids.h

[Go to the documentation of this file.](#)

```
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```

```

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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
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00016  * limitations under the License.
00017  *****/
00018
00023 #ifndef CFE_TBL_TOPICIDS_H
00024 #define CFE_TBL_TOPICIDS_H
00025
00026 #include "cfe_tbl_topicid_values.h"
00027
00037 #define CFE_MISSION_TBL_CMD_TOPICID          CFE_MISSION_TBL_TIDVAL(CMD)
00038 #define DEFAULT_CFE_MISSION_TBL_CMD_TOPICID 4
00039
00040 #define CFE_MISSION_TBL_SEND_HK_TOPICID      CFE_MISSION_TBL_TIDVAL(SEND_HK)
00041 #define DEFAULT_CFE_MISSION_TBL_SEND_HK_TOPICID 12
00042
00052 #define CFE_MISSION_TBL_HK_TLM_TOPICID      CFE_MISSION_TBL_TIDVAL(HK_TLM)
00053 #define DEFAULT_CFE_MISSION_TBL_HK_TLM_TOPICID 4
00054
00055 #define CFE_MISSION_TBL_REG_TLM_TOPICID     CFE_MISSION_TBL_TIDVAL(REG_TLM)
00056 #define DEFAULT_CFE_MISSION_TBL_REG_TLM_TOPICID 12
00057
00058 #endif

```

11.251 cfe/modules/time/config/default_cfe_time_extern_typedefs.h File Reference

```
#include "common_types.h"
```

Data Structures

- struct [CFE_TIME_SysTime](#)
Data structure used to hold system time values.

Typedefs

- typedef struct [CFE_TIME_SysTime](#) [CFE_TIME_SysTime_t](#)
Data structure used to hold system time values.
- typedef [uint8](#) [CFE_TIME_FlagBit_Enum_t](#)
Bit positions of the various clock state flags.
- typedef [int16](#) [CFE_TIME_ClockState_Enum_t](#)
Enumerated types identifying the quality of the current time.
- typedef [uint8](#) [CFE_TIME_SourceSelect_Enum_t](#)
Clock Source Selection Parameters.
- typedef [uint8](#) [CFE_TIME_ToneSignalSelect_Enum_t](#)
Tone Signal Selection Parameters.
- typedef [uint8](#) [CFE_TIME_AdjustDirection_Enum_t](#)
STCF adjustment direction (for both one-time and 1Hz adjustments)
- typedef [uint8](#) [CFE_TIME_FlywheelState_Enum_t](#)
Fly-wheel status values.
- typedef [uint8](#) [CFE_TIME_SetState_Enum_t](#)
Clock status values (has the clock been set to correct time)

Enumerations

- enum [CFE_TIME_FlagBit](#) {
[CFE_TIME_FlagBit_CLKSET](#) = 0 , [CFE_TIME_FlagBit_FLYING](#) = 1 , [CFE_TIME_FlagBit_SRCINT](#) = 2 ,
[CFE_TIME_FlagBit_SIGPRI](#) = 3 ,


```
CFE_TIME_FlagBit_SRVFLY = 4 , CFE_TIME_FlagBit_CMDFLY = 5 , CFE_TIME_FlagBit_ADDADJ = 6 ,
CFE_TIME_FlagBit_ADD1HZ = 7 ,
CFE_TIME_FlagBit_ADDTCL = 8 , CFE_TIME_FlagBit_SERVER = 9 , CFE_TIME_FlagBit_GDTONE = 10 }
```

Label definitions associated with CFE_TIME_FlagBit_Enum_t.

- enum `CFE_TIME_ClockState` { `CFE_TIME_ClockState_INVALID` = -1 , `CFE_TIME_ClockState_VALID` = 0 , `CFE_TIME_ClockState_FLYWHEEL` = 1 }

Label definitions associated with CFE_TIME_ClockState_Enum_t.

- enum `CFE_TIME_SourceSelect` { `CFE_TIME_SourceSelect_INTERNAL` = 1 , `CFE_TIME_SourceSelect_EXTERNAL` = 2 }

Label definitions associated with CFE_TIME_SourceSelect_Enum_t.

- enum `CFE_TIME_ToneSignalSelect` { `CFE_TIME_ToneSignalSelect_PRIMARY` = 1 , `CFE_TIME_ToneSignalSelect_REDUNDANT` = 2 }

Label definitions associated with CFE_TIME_ToneSignalSelect_Enum_t.

- enum `CFE_TIME_AdjustDirection` { `CFE_TIME_AdjustDirection_ADD` = 1 , `CFE_TIME_AdjustDirection_SUBTRACT` = 2 }

Label definitions associated with CFE_TIME_AdjustDirection_Enum_t.

- enum `CFE_TIME_FlywheelState` { `CFE_TIME_FlywheelState_NO_FLY` = 0 , `CFE_TIME_FlywheelState_IS_FLY` = 1 }

Label definitions associated with CFE_TIME_FlywheelState_Enum_t.

- enum `CFE_TIME_SetState` { `CFE_TIME_SetState_NOT_SET` = 0 , `CFE_TIME_SetState_WAS_SET` = 1 }

Label definitions associated with CFE_TIME_SetState_Enum_t.

11.251.1 Detailed Description

Declarations and prototypes for `cfe_time_extern_typedefs` module

Definition in file [default_cfe_time_extern_typedefs.h](#).

11.251.2 Typedef Documentation

CFE_TIME_AdjustDirection_Enum_t

```
typedef uint8 CFE_TIME_AdjustDirection_Enum_t
```

STCF adjustment direction (for both one-time and 1Hz adjustments)

See also

enum [CFE_TIME_AdjustDirection](#)

Definition at line 234 of file [default_cfe_time_extern_typedefs.h](#).

CFE_TIME_ClockState_Enum_t

```
typedef int16 CFE_TIME_ClockState_Enum_t
```

Enumerated types identifying the quality of the current time.

Description

The [CFE_TIME_ClockState_Enum_t](#) enumerations identify the three recognized states of the current time. If the clock has never been successfully synchronized with the primary onboard clock source, the time is considered to be [CFE_TIME_ClockState_INVALID](#). If the time is currently synchronized (i.e. - the primary synchronization mechanism has not been dropped for any significant amount of time), then the current time is considered to be [CFE_TIME_ClockState_VALID](#). If the time had, at some point in the past, been synchronized, but the synchronization with the primary onboard clock has since been lost, then the time is considered to be [CFE_TIME_ClockState_FLYWHEEL](#). Since different clocks drift at different rates from one another, the accuracy of the time while in [CFE_TIME_ClockState_FLYWHEEL](#) is dependent upon the time spent in that state.

See also

enum [CFE_TIME_ClockState](#)

Definition at line 165 of file [default_cfe_time_extern_typedefs.h](#).

CFE_TIME_FlagBit_Enum_t

```
typedef uint8 CFE_TIME_FlagBit_Enum_t
```

Bit positions of the various clock state flags.

See also

enum [CFE_TIME_FlagBit](#)

Definition at line 113 of file [default_cfe_time_extern_typedefs.h](#).

CFE_TIME_FlywheelState_Enum_t

```
typedef uint8 CFE_TIME_FlywheelState_Enum_t
```

Fly-wheel status values.

See also

enum [CFE_TIME_FlywheelState](#)

Definition at line 257 of file [default_cfe_time_extern_typedefs.h](#).

CFE_TIME_SetState_Enum_t

```
typedef uint8 CFE_TIME_SetState_Enum_t
```

Clock status values (has the clock been set to correct time)

See also

enum [CFE_TIME_SetState](#)

Definition at line 280 of file [default_cfe_time_extern_typedefs.h](#).

CFE_TIME_SourceSelect_Enum_t

```
typedef uint8 CFE_TIME_SourceSelect_Enum_t
```

Clock Source Selection Parameters.

See also

enum [CFE_TIME_SourceSelect](#)

Definition at line 188 of file [default_cfe_time_extern_typedefs.h](#).

CFE_TIME_SysTime_t

```
typedef struct CFE_TIME_SysTime CFE_TIME_SysTime_t
```

Data structure used to hold system time values.

Description

The [CFE_TIME_SysTime_t](#) data structure is used to hold time values. Time is referred to as the elapsed time (in seconds and subseconds) since a specified epoch time. The subseconds field contains the number of $2^{(-32)}$ second intervals that have elapsed since the epoch.

CFE_TIME_ToneSignalSelect_Enum_t

typedef uint8 CFE_TIME_ToneSignalSelect_Enum_t
Tone Signal Selection Parameters.

See also

enum [CFE_TIME_ToneSignalSelect](#)

Definition at line 211 of file [default_cfe_time_extern_typedefs.h](#).

11.251.3 Enumeration Type Documentation**CFE_TIME_AdjustDirection**

enum [CFE_TIME_AdjustDirection](#)

Label definitions associated with CFE_TIME_AdjustDirection_Enum_t.

Enumerator

CFE_TIME_AdjustDirection_ADD	Add time adjustment.
CFE_TIME_AdjustDirection_SUBTRACT	Subtract time adjustment.

Definition at line 216 of file [default_cfe_time_extern_typedefs.h](#).

CFE_TIME_ClockState

enum [CFE_TIME_ClockState](#)

Label definitions associated with CFE_TIME_ClockState_Enum_t.

Enumerator

CFE_TIME_ClockState_INVALID	The spacecraft time has not been set since the last clock reset. Times returned by clock routines have no relationship to any ground-based time reference.
CFE_TIME_ClockState_VALID	The spacecraft time has been set at least once since the last clock reset, and it is synchronized with the primary on-board time base. Times returned by clock routines can be trusted.
CFE_TIME_ClockState_FLYWHEEL	The spacecraft time has been set at least once since the last clock reset, but it is not currently synchronized with the primary on-board time base. Times returned by clock routines are a "best guess" based on a non-optimal oscillator.

Definition at line 118 of file [default_cfe_time_extern_typedefs.h](#).

CFE_TIME_FlagBit

enum [CFE_TIME_FlagBit](#)

Label definitions associated with CFE_TIME_FlagBit_Enum_t.

Enumerator

CFE_TIME_FlagBit_CLKSET	The spacecraft time has been set.
CFE_TIME_FlagBit_FLYING	This instance of Time Services is flywheeling.
CFE_TIME_FlagBit_SRCINT	The clock source is set to internal.

Enumerator

CFE_TIME_FlagBit_SIGPRI	The clock signal is set to primary.
CFE_TIME_FlagBit_SRVFLY	The Time Server is in flywheel mode.
CFE_TIME_FlagBit_CMDFLY	This instance of Time Services was commanded into flywheel mode.
CFE_TIME_FlagBit_ADDADJ	One time STCF Adjustment is to be done in positive direction.
CFE_TIME_FlagBit_ADD1HZ	1 Hz STCF Adjustment is to be done in a positive direction
CFE_TIME_FlagBit_ADDTCL	Time Client Latency is applied in a positive direction.
CFE_TIME_FlagBit_SERVER	This instance of Time Services is a Time Server.
CFE_TIME_FlagBit_GDTONE	The tone received is good compared to the last tone received.

Definition at line 50 of file [default_cfe_time_extern_typedefs.h](#).

CFE_TIME_FlywheelState

enum [CFE_TIME_FlywheelState](#)

Label definitions associated with CFE_TIME_FlywheelState_Enum_t.

Enumerator

CFE_TIME_FlywheelState_NO_FLY	Not in flywheel state.
CFE_TIME_FlywheelState_IS_FLY	In flywheel state.

Definition at line 239 of file [default_cfe_time_extern_typedefs.h](#).

CFE_TIME_SetState

enum [CFE_TIME_SetState](#)

Label definitions associated with CFE_TIME_SetState_Enum_t.

Enumerator

CFE_TIME_SetState_NOT_SET	Spacecraft time has not been set.
CFE_TIME_SetState_WAS_SET	Spacecraft time has been set.

Definition at line 262 of file [default_cfe_time_extern_typedefs.h](#).

CFE_TIME_SourceSelect

enum [CFE_TIME_SourceSelect](#)

Label definitions associated with CFE_TIME_SourceSelect_Enum_t.

Enumerator

CFE_TIME_SourceSelect_INTERNAL	Use Internal Source.
CFE_TIME_SourceSelect_EXTERNAL	Use External Source.

Definition at line 170 of file [default_cfe_time_extern_typedefs.h](#).

CFE_TIME_ToneSignalSelect

enum [CFE_TIME_ToneSignalSelect](#)

Label definitions associated with CFE_TIME_ToneSignalSelect_Enum_t.

Enumerator

CFE_TIME_ToneSignalSelect_PRIMARY	Primary Source.
CFE_TIME_ToneSignalSelect_REDUNDANT	Redundant Source.

Definition at line 193 of file [default_cfe_time_extern_typedefs.h](#).

11.252 default_cfe_time_extern_typedefs.h

[Go to the documentation of this file.](#)

```

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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
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00016  * limitations under the License.
00017  *****/
00018
00025 #ifndef DEFAULT_CFE_TIME_EXTERN_TYPEDEFS_H
00026 #define DEFAULT_CFE_TIME_EXTERN_TYPEDEFS_H
00027
00028 #include "common_types.h"
00029
00041 typedef struct CFE_TIME_SysTime
00042 {
00043     uint32 Seconds;
00044     uint32 Subseconds;
00045 } CFE_TIME_SysTime_t;
00046
00050 enum CFE_TIME_FlagBit
00051 {
00055     CFE_TIME_FlagBit_CLKSET = 0,
00056
00060     CFE_TIME_FlagBit_FLYING = 1,
00061
00065     CFE_TIME_FlagBit_SRCINT = 2,
00066
00070     CFE_TIME_FlagBit_SIGPRI = 3,
00071
00075     CFE_TIME_FlagBit_SRVFLY = 4,
00076
00080     CFE_TIME_FlagBit_CMDFLY = 5,
00081
00085     CFE_TIME_FlagBit_ADDADJ = 6,
00086
00090     CFE_TIME_FlagBit_ADD1HZ = 7,
00091
00095     CFE_TIME_FlagBit_ADDTCL = 8,
00096
00100     CFE_TIME_FlagBit_SERVER = 9,
00101
00105     CFE_TIME_FlagBit_GDTONE = 10
00106 };
00107
00113 typedef uint8 CFE_TIME_FlagBit_Enum_t;
00114
00118 enum CFE_TIME_ClockState
00119 {
00127     CFE_TIME_ClockState_INVALID = -1,
00128
00136     CFE_TIME_ClockState_VALID = 0,
00137
00146     CFE_TIME_ClockState_FLYWHEEL = 1

```

```

00147 };
00148
00165 typedef int16 CFE_TIME_ClockState_Enum_t;
00166
00170 enum CFE_TIME_SourceSelect
00171 {
00175     CFE_TIME_SourceSelect_INTERNAL = 1,
00176
00180     CFE_TIME_SourceSelect_EXTERNAL = 2
00181 };
00182
00188 typedef uint8 CFE_TIME_SourceSelect_Enum_t;
00189
00193 enum CFE_TIME_ToneSignalSelect
00194 {
00198     CFE_TIME_ToneSignalSelect_PRIMARY = 1,
00199
00203     CFE_TIME_ToneSignalSelect_REDUNDANT = 2
00204 };
00205
00211 typedef uint8 CFE_TIME_ToneSignalSelect_Enum_t;
00212
00216 enum CFE_TIME_AdjustDirection
00217 {
00221     CFE_TIME_AdjustDirection_ADD = 1,
00222
00226     CFE_TIME_AdjustDirection_SUBTRACT = 2
00227 };
00228
00234 typedef uint8 CFE_TIME_AdjustDirection_Enum_t;
00235
00239 enum CFE_TIME_FlywheelState
00240 {
00244     CFE_TIME_FlywheelState_NO_FLY = 0,
00245
00249     CFE_TIME_FlywheelState_IS_FLY = 1
00250 };
00251
00257 typedef uint8 CFE_TIME_FlywheelState_Enum_t;
00258
00262 enum CFE_TIME_SetState
00263 {
00267     CFE_TIME_SetState_NOT_SET = 0,
00268
00272     CFE_TIME_SetState_WAS_SET = 1
00273 };
00274
00280 typedef uint8 CFE_TIME_SetState_Enum_t;
00281
00282 #endif /* CFE_TIME_EXTERN_TYPEDEFS_H */

```

11.253 cfe/modules/time/config/default_cfe_time_fcncode_values.h File Reference

Macros

- #define CFE_TIME_CCVAL(x) CFE_TIME_FunctionCode_##x

Enumerations

- enum CFE_TIME_FunctionCode_ {
CFE_TIME_FunctionCode_NOOP = 0 , CFE_TIME_FunctionCode_RESET_COUNTERS = 1 , CFE_TIME_FunctionCode_SEND_D
= 2 , CFE_TIME_FunctionCode_SET_SOURCE = 3 ,
CFE_TIME_FunctionCode_SET_STATE = 4 , CFE_TIME_FunctionCode_ADD_DELAY = 5 , CFE_TIME_FunctionCode_SUB_DELA
= 6 , CFE_TIME_FunctionCode_SET_TIME = 7 ,
CFE_TIME_FunctionCode_SET_MET = 8 , CFE_TIME_FunctionCode_SET_STCF = 9 , CFE_TIME_FunctionCode_SET_LEAP_SI
= 10 , CFE_TIME_FunctionCode_ADD_ADJUST = 11 ,
CFE_TIME_FunctionCode_SUB_ADJUST = 12 , CFE_TIME_FunctionCode_ADD_ONE_HZ_ADJUSTMENT =
13 , CFE_TIME_FunctionCode_SUB_ONE_HZ_ADJUSTMENT = 14 , CFE_TIME_FunctionCode_SET_SIGNAL
= 15 }

11.253.1 Detailed Description

Specification for the CFE Time Services (CFE_TIME) command function codes

Note

This file should be strictly limited to the command/function code (CC) macro definitions. Other definitions such as enums, typedefs, or other macros should be placed in the msgdefs.h or msg.h files.

Definition in file [default_cfe_time_fcncode_values.h](#).

11.253.2 Macro Definition Documentation

CFE_TIME_CCVAL

```
#define CFE_TIME_CCVAL(  
    x ) CFE_TIME_FunctionCode_##x
```

Definition at line 35 of file [default_cfe_time_fcncode_values.h](#).

11.253.3 Enumeration Type Documentation

CFE_TIME_FunctionCode_

```
enum CFE_TIME_FunctionCode_
```

Enumerator

CFE_TIME_FunctionCode_NOOP	
CFE_TIME_FunctionCode_RESET_COUNTERS	
CFE_TIME_FunctionCode_SEND_DIAGNOSTIC	
CFE_TIME_FunctionCode_SET_SOURCE	
CFE_TIME_FunctionCode_SET_STATE	
CFE_TIME_FunctionCode_ADD_DELAY	
CFE_TIME_FunctionCode_SUB_DELAY	
CFE_TIME_FunctionCode_SET_TIME	
CFE_TIME_FunctionCode_SET_MET	
CFE_TIME_FunctionCode_SET_STCF	
CFE_TIME_FunctionCode_SET_LEAP_SECONDS	
CFE_TIME_FunctionCode_ADD_ADJUST	
CFE_TIME_FunctionCode_SUB_ADJUST	
CFE_TIME_FunctionCode_ADD_ONE_HZ_ADJUSTMENT	
CFE_TIME_FunctionCode_SUB_ONE_HZ_ADJUSTMENT	
CFE_TIME_FunctionCode_SET_SIGNAL	

Definition at line 37 of file [default_cfe_time_fcncode_values.h](#).

11.254 default_cfe_time_fcncode_values.h

[Go to the documentation of this file.](#)

```
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00017  *****/
00018
00028 #ifndef DEFAULT_CFE_TIME_FCNCODE_VALUES_H
00029 #define DEFAULT_CFE_TIME_FCNCODE_VALUES_H
00030
00031 /*****
00032  * Macro Definitions
00033  *****/
00034
00035 #define CFE_TIME_CCVAL(x) CFE_TIME_FunctionCode_##x
00036
00037 enum CFE_TIME_FunctionCode_
00038 {
00039     CFE_TIME_FunctionCode_NOOP                = 0,
00040     CFE_TIME_FunctionCode_RESET_COUNTERS      = 1,
00041     CFE_TIME_FunctionCode_SEND_DIAGNOSTIC     = 2,
00042     CFE_TIME_FunctionCode_SET_SOURCE          = 3,
00043     CFE_TIME_FunctionCode_SET_STATE           = 4,
00044     CFE_TIME_FunctionCode_ADD_DELAY           = 5,
00045     CFE_TIME_FunctionCode_SUB_DELAY           = 6,
00046     CFE_TIME_FunctionCode_SET_TIME            = 7,
00047     CFE_TIME_FunctionCode_SET_MET             = 8,
00048     CFE_TIME_FunctionCode_SET_STCF            = 9,
00049     CFE_TIME_FunctionCode_SET_LEAP_SECONDS    = 10,
00050     CFE_TIME_FunctionCode_ADD_ADJUST          = 11,
00051     CFE_TIME_FunctionCode_SUB_ADJUST          = 12,
00052     CFE_TIME_FunctionCode_ADD_ONE_HZ_ADJUSTMENT = 13,
00053     CFE_TIME_FunctionCode_SUB_ONE_HZ_ADJUSTMENT = 14,
00054     CFE_TIME_FunctionCode_SET_SIGNAL          = 15,
00055 };
00056
00057 #endif

```

11.255 cfe/modules/time/config/default_cfe_time_interface_cfg_values.h File Reference

Macros

- #define [CFE_MISSION_TIME_CFGVAL\(x\)](#) DEFAULT_CFE_MISSION_TIME_##x

11.255.1 Detailed Description

CFE Executive Services (CFE_ES) Application Public Definitions

This provides default values for configurable items that affect the interface(s) of this module. This includes the CMD/TLM message interface, tables definitions, and any other data products that serve to exchange information with other entities.

Note

This file may be overridden/superseded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [default_cfe_time_interface_cfg_values.h](#).

11.255.2 Macro Definition Documentation

CFE_MISSION_TIME_CFGVAL

```
#define CFE_MISSION_TIME_CFGVAL(
    x ) DEFAULT_CFE_MISSION_TIME_##x
```

Definition at line 36 of file [default_cfe_time_interface_cfg_values.h](#).

11.256 default_cfe_time_interface_cfg_values.h

[Go to the documentation of this file.](#)

```

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00017  *****/
00018
00032 #ifndef DEFAULT_CFE_TIME_INTERFACE_CFG_H
00033 #define DEFAULT_CFE_TIME_INTERFACE_CFG_H
00034
00035 /* Use the default configuration value for all */
00036 #define CFE_MISSION_TIME_CFGVAL(x) DEFAULT_CFE_MISSION_TIME_##x
00037
00038 #endif

```

11.257 cfe/modules/time/config/default_cfe_time_internal_cfg_values.h File Reference

Macros

- #define [CFE_PLATFORM_TIME_CFGVAL\(x\)](#) DEFAULT_CFE_PLATFORM_TIME_##x

11.257.1 Detailed Description

CFE Executive Services (CFE_ES) Application Private Config Definitions

This provides default values for configurable items that are internal to this module and do NOT affect the interface(s) of this module. Changes to items in this file only affect the local module and will be transparent to external entities that are using the public interface(s).

Note

This file may be overridden/superseded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [default_cfe_time_internal_cfg_values.h](#).

11.257.2 Macro Definition Documentation

CFE_PLATFORM_TIME_CFGVAL

```

#define CFE_PLATFORM_TIME_CFGVAL(
    x ) DEFAULT_CFE_PLATFORM_TIME_##x

```

Definition at line 36 of file [default_cfe_time_internal_cfg_values.h](#).

11.258 default_cfe_time_internal_cfg_values.h

[Go to the documentation of this file.](#)

```

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00017  *****/
00018
00032 #ifndef DEFAULT_CFE_TIME_INTERNAL_CFG_H
00033 #define DEFAULT_CFE_TIME_INTERNAL_CFG_H
00034
00035 /* Use the default configuration value for all */
00036 #define CFE_PLATFORM_TIME_CFGVAL(x) DEFAULT_CFE_PLATFORM_TIME_##x
00037
00038 /* This define does not really belong here, but it is the only place for it
00039  * for now until the refs can be updated or it can be removed entirely */
00040
00051 #undef CFE_PLATFORM_TIME_CFG_BIGENDIAN
00052
00053 #endif

```

11.259 cfe/modules/time/config/default_cfe_time_mission_cfg.h File Reference

```
#include "cfe_time_interface_cfg.h"
```

11.259.1 Detailed Description

CFE Time Services (CFE_TIME) Application Mission Configuration Header File

This is a compatibility header for the "mission_cfg.h" file that has traditionally provided public config definitions for each CFS app.

Note

This file may be overridden/superseded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [default_cfe_time_mission_cfg.h](#).

11.260 default_cfe_time_mission_cfg.h

[Go to the documentation of this file.](#)

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00017  *****/
00018
00031 #ifndef DEFAULT_CFE_TIME_MISSION_CFG_H
00032 #define DEFAULT_CFE_TIME_MISSION_CFG_H
00033
00034 #include "cfe_time_interface_cfg.h"
00035
00036 #endif

```

11.261 cfe/modules/time/config/default_cfe_time_msg.h File Reference

```
#include "cfe_mission_cfg.h"
#include "cfe_time_fcncodes.h"
#include "cfe_time_msgdefs.h"
#include "cfe_time_msgstruct.h"
```

11.261.1 Detailed Description

Specification for the CFE Time Services (CFE_TIME) command and telemetry message data types.

This is a compatibility header for the "cfe_time_msg.h" file that has traditionally provided the message definitions for cFS apps.

Note

This file may be overridden/supersceded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [default_cfe_time_msg.h](#).

11.262 default_cfe_time_msg.h

[Go to the documentation of this file.](#)

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00017  *****/
00018
00031 #ifndef DEFAULT_CFE_TIME_MSG_H
00032 #define DEFAULT_CFE_TIME_MSG_H
00033
00034 #include "cfe_mission_cfg.h"
00035 #include "cfe_time_fcncodes.h"
00036 #include "cfe_time_msgdefs.h"
00037 #include "cfe_time_msgstruct.h"
00038
00039 #endif
```

11.263 cfe/modules/time/config/default_cfe_time_msgdefs.h File Reference

```
#include "common_types.h"
#include "cfe_mission_cfg.h"
#include "cfe_time_extern_typedefs.h"
#include "cfe_time_fcncodes.h"
```

Data Structures

- struct [CFE_TIME_LeapsCmd_Payload](#)
Set leap seconds command payload.
- struct [CFE_TIME_StateCmd_Payload](#)

- *Set clock state command payload.*
- struct [CFE_TIME_SourceCmd_Payload](#)
- *Set time data source command payload.*
- struct [CFE_TIME_SignalCmd_Payload](#)
- *Set tone signal source command payload.*
- struct [CFE_TIME_TimeCmd_Payload](#)
- *Generic seconds, microseconds command payload.*
- struct [CFE_TIME_OneHzAdjustmentCmd_Payload](#)
- *Generic seconds, subseconds command payload.*
- struct [CFE_TIME_ToneDataCmd_Payload](#)
- *Time at tone data command payload.*
- struct [CFE_TIME_HousekeepingTlm_Payload](#)
- struct [CFE_TIME_DiagnosticTlm_Payload](#)

Macros

- #define [CFE_TIME_FLAG_CLKSET](#) 0x8000
The spacecraft time has been set.
- #define [CFE_TIME_FLAG_FLYING](#) 0x4000
This instance of Time Services is flywheeling.
- #define [CFE_TIME_FLAG_SRCINT](#) 0x2000
The clock source is set to "internal".
- #define [CFE_TIME_FLAG_SIGPRI](#) 0x1000
The clock signal is set to "primary".
- #define [CFE_TIME_FLAG_SRVFLY](#) 0x0800
The Time Server is in flywheel mode.
- #define [CFE_TIME_FLAG_CMDFLY](#) 0x0400
This instance of Time Services was commanded into flywheel mode.
- #define [CFE_TIME_FLAG_ADDADJ](#) 0x0200
One time STCF Adjustment is to be done in positive direction.
- #define [CFE_TIME_FLAG_ADD1HZ](#) 0x0100
1 Hz STCF Adjustment is to be done in a positive direction
- #define [CFE_TIME_FLAG_ADDTCL](#) 0x0080
Time Client Latency is applied in a positive direction.
- #define [CFE_TIME_FLAG_SERVER](#) 0x0040
This instance of Time Services is a Time Server.
- #define [CFE_TIME_FLAG_GDTONE](#) 0x0020
The tone received is good compared to the last tone received.
- #define [CFE_TIME_FLAG_REFERR](#) 0x0010
GetReference read error, will be set if unable to get a consistent ref value.
- #define [CFE_TIME_FLAG_UNUSED](#) 0x000F
Reserved flags - should be zero.

Typedefs

- typedef struct [CFE_TIME_LeapsCmd_Payload](#) [CFE_TIME_LeapsCmd_Payload_t](#)
Set leap seconds command payload.
- typedef struct [CFE_TIME_StateCmd_Payload](#) [CFE_TIME_StateCmd_Payload_t](#)
Set clock state command payload.
- typedef struct [CFE_TIME_SourceCmd_Payload](#) [CFE_TIME_SourceCmd_Payload_t](#)
Set time data source command payload.
- typedef struct [CFE_TIME_SignalCmd_Payload](#) [CFE_TIME_SignalCmd_Payload_t](#)
Set tone signal source command payload.
- typedef struct [CFE_TIME_TimeCmd_Payload](#) [CFE_TIME_TimeCmd_Payload_t](#)
Generic seconds, microseconds command payload.
- typedef struct [CFE_TIME_OneHzAdjustmentCmd_Payload](#) [CFE_TIME_OneHzAdjustmentCmd_Payload_t](#)
Generic seconds, subseconds command payload.
- typedef struct [CFE_TIME_ToneDataCmd_Payload](#) [CFE_TIME_ToneDataCmd_Payload_t](#)
Time at tone data command payload.
- typedef struct [CFE_TIME_HousekeepingTlm_Payload](#) [CFE_TIME_HousekeepingTlm_Payload_t](#)
- typedef struct [CFE_TIME_DiagnosticTlm_Payload](#) [CFE_TIME_DiagnosticTlm_Payload_t](#)

11.263.1 Detailed Description

Specification for the CFE Time Services (CFE_TIME) command and telemetry message payloads and constant definitions.

Definition in file [default_cfe_time_msgdefs.h](#).

11.263.2 Typedef Documentation

CFE_TIME_DiagnosticTlm_Payload_t

```
typedef struct CFE\_TIME\_DiagnosticTlm\_Payload CFE\_TIME\_DiagnosticTlm\_Payload\_t
```

Name Time Services Diagnostics Packet

CFE_TIME_HousekeepingTlm_Payload_t

```
typedef struct CFE\_TIME\_HousekeepingTlm\_Payload CFE\_TIME\_HousekeepingTlm\_Payload\_t
```

Name Time Services Housekeeping Packet

CFE_TIME_LeapsCmd_Payload_t

```
typedef struct CFE\_TIME\_LeapsCmd\_Payload CFE\_TIME\_LeapsCmd\_Payload\_t
```

Set leap seconds command payload.

CFE_TIME_OneHzAdjustmentCmd_Payload_t

```
typedef struct CFE\_TIME\_OneHzAdjustmentCmd\_Payload CFE\_TIME\_OneHzAdjustmentCmd\_Payload\_t
```

Generic seconds, subseconds command payload.

CFE_TIME_SignalCmd_Payload_t

```
typedef struct CFE\_TIME\_SignalCmd\_Payload CFE\_TIME\_SignalCmd\_Payload\_t
```

Set tone signal source command payload.

CFE_TIME_SourceCmd_Payload_t

typedef struct CFE_TIME_SourceCmd_Payload CFE_TIME_SourceCmd_Payload_t
Set time data source command payload.

CFE_TIME_StateCmd_Payload_t

typedef struct CFE_TIME_StateCmd_Payload CFE_TIME_StateCmd_Payload_t
Set clock state command payload.

CFE_TIME_TimeCmd_Payload_t

typedef struct CFE_TIME_TimeCmd_Payload CFE_TIME_TimeCmd_Payload_t
Generic seconds, microseconds command payload.

CFE_TIME_ToneDataCmd_Payload_t

typedef struct CFE_TIME_ToneDataCmd_Payload CFE_TIME_ToneDataCmd_Payload_t
Time at tone data command payload.

11.264 default_cfe_time_msgdefs.h

[Go to the documentation of this file.](#)

```

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00017  *****/
00018
00024 #ifndef DEFAULT_CFE_TIME_MSGDEFS_H
00025 #define DEFAULT_CFE_TIME_MSGDEFS_H
00026
00027 #include "common_types.h"
00028 #include "cfe_mission_cfg.h"
00029 #include "cfe_time_extern_typedefs.h"
00030 #include "cfe_time_fcncodes.h"
00031
00035 #define CFE_TIME_FLAG_CLKSET 0x8000
00036 #define CFE_TIME_FLAG_FLYING 0x4000
00037 #define CFE_TIME_FLAG_SRCINT 0x2000
00038 #define CFE_TIME_FLAG_SIGPRI 0x1000
00039 #define CFE_TIME_FLAG_SRVFLY 0x0800
00040 #define CFE_TIME_FLAG_CMDFLY 0x0400
00041 #define CFE_TIME_FLAG_ADDADJ 0x0200
00042 #define CFE_TIME_FLAG_ADD1HZ 0x0100
00043 #define CFE_TIME_FLAG_ADDTCL 0x0080
00044 #define CFE_TIME_FLAG_SERVER 0x0040
00045 #define CFE_TIME_FLAG_GDTONE 0x0020
00046 #define CFE_TIME_FLAG_REFERR \
00047     0x0010
00048 #define CFE_TIME_FLAG_UNUSED 0x000F
00054 typedef struct CFE_TIME_LeapsCmd_Payload
00055 {
00056     int16 LeapSeconds;
00057 } CFE_TIME_LeapsCmd_Payload_t;
00058
00062 typedef struct CFE_TIME_StateCmd_Payload
00063 {
00064     CFE_TIME_ClockState_Enum_t ClockState;

```

```

00069 } CFE_TIME_StateCmd_Payload_t;
00070
00074 typedef struct CFE_TIME_SourceCmd_Payload
00075 {
00076     int16 TimeSource;
00079 } CFE_TIME_SourceCmd_Payload_t;
00080
00084 typedef struct CFE_TIME_SignalCmd_Payload
00085 {
00086     int16 ToneSource;
00089 } CFE_TIME_SignalCmd_Payload_t;
00090
00094 typedef struct CFE_TIME_TimeCmd_Payload
00095 {
00096     uint32 Seconds;
00097     uint32 MicroSeconds;
00098 } CFE_TIME_TimeCmd_Payload_t;
00099
00103 typedef struct CFE_TIME_OneHzAdjustmentCmd_Payload
00104 {
00105     uint32 Seconds;
00106     uint32 Subseconds;
00107 } CFE_TIME_OneHzAdjustmentCmd_Payload_t;
00108
00112 typedef struct CFE_TIME_ToneDataCmd_Payload
00113 {
00114     CFE_TIME_SysTime_t      AtToneMET;
00115     CFE_TIME_SysTime_t      AtToneSTCF;
00116     int16                  AtToneLeapSeconds;
00117     CFE_TIME_ClockState_Enum_t AtToneState;
00118 } CFE_TIME_ToneDataCmd_Payload_t;
00119
00120 /*****
00121
00125 typedef struct CFE_TIME_HousekeepingTlm_Payload
00126 {
00127     /*
00128     ** Task command interface counters...
00129     */
00130     uint8 CommandCounter;
00132     uint8 CommandErrorCounter;
00135     /*
00136     ** Clock state flags and "as calculated" clock state...
00137     */
00138     uint16                  ClockStateFlags;
00140     CFE_TIME_ClockState_Enum_t ClockStateAPI;
00143     /*
00144     ** Leap Seconds...
00145     */
00146     int16 LeapSeconds;
00149     /*
00150     ** Current MET and STCF time values...
00151     */
00152     CFE_TIME_SysTime_t MET;
00153     CFE_TIME_SysTime_t STCF;
00166     CFE_TIME_SysTime_t AdjustmentFactor;
00167
00168 } CFE_TIME_HousekeepingTlm_Payload_t;
00169
00170 /*****
00171
00175 typedef struct CFE_TIME_DiagnosticTlm_Payload
00176 {
00177     /*
00178     ** Data values used to compute time (in reference to "tone")...
00179     */
00180     CFE_TIME_SysTime_t AtToneMET;
00182     CFE_TIME_SysTime_t AtToneSTCF;
00184     CFE_TIME_SysTime_t AtToneDelay;
00186     CFE_TIME_SysTime_t AtToneLatch;
00189     int16                  AtToneLeapSeconds;
00191     CFE_TIME_ClockState_Enum_t ClockStateAPI;
00194     /*
00195     ** Data values that reflect the time (right now)...
00196     */
00197     CFE_TIME_SysTime_t TimeSinceTone;
00199     CFE_TIME_SysTime_t CurrentLatch;
00201     CFE_TIME_SysTime_t CurrentMET;
00203     CFE_TIME_SysTime_t CurrentTAI;
00205     CFE_TIME_SysTime_t CurrentUTC;
00208     /*

```

```

00209     ** Data values used to define the current clock state...
00210     */
00211     int16 ClockSetState;
00213     int16 ClockFlyState;
00215     int16 ClockSource;
00217     int16 ClockSignal;
00219     int16 ServerFlyState;
00221     int16 Forced2Fly;
00224     /*
00225     ** Clock state flags...
00226     */
00227     uint16 ClockStateFlags;
00230     /*
00231     ** STCF adjustment direction values...
00232     */
00233     int16 OneTimeDirection;
00235     int16 OneHzDirection;
00237     int16 DelayDirection;
00240     /*
00241     ** STCF adjustment values...
00242     */
00243     CFE_TIME_SysTime_t OneTimeAdjust;
00245     CFE_TIME_SysTime_t OneHzAdjust;
00248     /*
00249     ** Most recent local clock latch values...
00250     */
00251     CFE_TIME_SysTime_t ToneSignalLatch;
00253     CFE_TIME_SysTime_t ToneDataLatch;
00256     /*
00257     ** Miscellaneous counters (subject to reset command)...
00258     */
00259     uint32 ToneMatchCounter;
00261     uint32 ToneMatchErrorCounter;
00263     uint32 ToneSignalCounter;
00265     uint32 ToneDataCounter;
00267     uint32 ToneIntCounter;
00269     uint32 ToneIntErrorCounter;
00271     uint32 ToneTaskCounter;
00273     uint32 VersionCounter;
00275     uint32 LocalIntCounter;
00277     uint32 LocalTaskCounter;
00280     /*
00281     ** Miscellaneous counters (not subject to reset command)...
00282     */
00283     uint32 VirtualMET;
00286     /*
00287     ** Time window verification values (converted from micro-secs)...
00288     **
00289     ** Regardless of whether the tone follows the time packet, or vice
00290     ** versa, these values define the acceptable window of time for
00291     ** the second event to follow the first. The minimum value may
00292     ** be as little as zero, and the maximum must be something less
00293     ** than a second.
00294     */
00295     uint32 MinElapsed;
00297     uint32 MaxElapsed;
00300     /*
00301     ** Maximum local clock value (before roll-over)...
00302     */
00303     CFE_TIME_SysTime_t MaxLocalClock;
00306     /*
00307     ** Tone signal tolerance limits...
00308     */
00309     uint32 ToneOverLimit;
00311     uint32 ToneUnderLimit;
00314     /*
00315     ** Reset Area...
00316     */
00317     uint32 DataStoreStatus;
00319 } CFE_TIME_DiagnosticTlm_Payload_t;
00320
00321 #endif

```

11.265 cfe/modules/time/config/default_cfe_time_msgid_values.h File Reference

```

#include "cfe_core_api_base_msgids.h"
#include "cfe_time_topicsids.h"

```


Macros

- `#define CFE_PLATFORM_TIME_CMD_MIDVAL(x) CFE_PLATFORM_CMD_TOPICID_TO_MIDV(CFE_MISSION_TIME_###_TOPICID)`
- `#define CFE_PLATFORM_TIME_TLM_MIDVAL(x) CFE_PLATFORM_TLM_TOPICID_TO_MIDV(CFE_MISSION_TIME_###_TOPICID)`
- `#define CFE_PLATFORM_TIME_GLBCMD_MIDVAL(x) CFE_GLOBAL_CMD_TOPICID_TO_MIDV(CFE_MISSION_TIME_###_TOPICID)`

11.265.1 Detailed Description

CFE Time Services (CFE_TIME) Application Message IDs

Definition in file [default_cfe_time_msgid_values.h](#).

11.265.2 Macro Definition Documentation

CFE_PLATFORM_TIME_CMD_MIDVAL

```
#define CFE_PLATFORM_TIME_CMD_MIDVAL(
    x ) CFE_PLATFORM_CMD_TOPICID_TO_MIDV(CFE_MISSION_TIME_###_TOPICID)
```

Definition at line 29 of file [default_cfe_time_msgid_values.h](#).

CFE_PLATFORM_TIME_GLBCMD_MIDVAL

```
#define CFE_PLATFORM_TIME_GLBCMD_MIDVAL(
    x ) CFE_GLOBAL_CMD_TOPICID_TO_MIDV(CFE_MISSION_TIME_###_TOPICID)
```

Definition at line 33 of file [default_cfe_time_msgid_values.h](#).

CFE_PLATFORM_TIME_TLM_MIDVAL

```
#define CFE_PLATFORM_TIME_TLM_MIDVAL(
    x ) CFE_PLATFORM_TLM_TOPICID_TO_MIDV(CFE_MISSION_TIME_###_TOPICID)
```

Definition at line 30 of file [default_cfe_time_msgid_values.h](#).

11.266 default_cfe_time_msgid_values.h

[Go to the documentation of this file.](#)

```
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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00023 #ifndef DEFAULT_CFE_TIME_MSGID_VALUES_H
00024 #define DEFAULT_CFE_TIME_MSGID_VALUES_H
00025
00026 #include "cfe_core_api_base_msgids.h"
00027 #include "cfe_time_topicids.h"
00028
00029 #define CFE_PLATFORM_TIME_CMD_MIDVAL(x) CFE_PLATFORM_CMD_TOPICID_TO_MIDV(CFE_MISSION_TIME_###_TOPICID)
00030 #define CFE_PLATFORM_TIME_TLM_MIDVAL(x) CFE_PLATFORM_TLM_TOPICID_TO_MIDV(CFE_MISSION_TIME_###_TOPICID)
00031
00032 /* time is unique in that it uses "global" cmd msgids - these are the same on all instances */
```

```
00033 #define CFE_PLATFORM_TIME_GLBCMD_MIDVAL(x) CFE_GLOBAL_CMD_TOPICID_TO_MIDV(CFE_MISSION_TIME_##x##_TOPICID)
00034
00035 #endif
```

11.267 cfe/modules/time/config/default_cfe_time_msgids.h File Reference

```
#include "cfe_time_msgid_values.h"
```

Macros

- `#define CFE_TIME_CMD_MID CFE_PLATFORM_TIME_CMD_MIDVAL(CMD)`
- `#define CFE_TIME_SEND_HK_MID CFE_PLATFORM_TIME_CMD_MIDVAL(SEND_HK)`
- `#define CFE_TIME_TONE_CMD_MID CFE_PLATFORM_TIME_CMD_MIDVAL(TONE_CMD)`
- `#define CFE_TIME_ONEHZ_CMD_MID CFE_PLATFORM_TIME_CMD_MIDVAL(ONEHZ_CMD)`
- `#define CFE_TIME_DATA_CMD_MID CFE_PLATFORM_TIME_GLBCMD_MIDVAL(DATA_CMD)`
- `#define CFE_TIME_SEND_CMD_MID CFE_PLATFORM_TIME_GLBCMD_MIDVAL(SEND_CMD)`
- `#define CFE_TIME_HK_TLM_MID CFE_PLATFORM_TIME_TLM_MIDVAL(HK_TLM)`
- `#define CFE_TIME_DIAG_TLM_MID CFE_PLATFORM_TIME_TLM_MIDVAL(DIAG_TLM)`
- `#define CFE_TIME_1HZ_CMD_MID CFE_TIME_ONEHZ_CMD_MID`

11.267.1 Detailed Description

CFE Time Services (CFE_TIME) Application Message IDs
Definition in file [default_cfe_time_msgids.h](#).

11.267.2 Macro Definition Documentation

CFE_TIME_1HZ_CMD_MID

```
#define CFE_TIME_1HZ_CMD_MID CFE_TIME_ONEHZ_CMD_MID
```

Definition at line 54 of file [default_cfe_time_msgids.h](#).

CFE_TIME_CMD_MID

```
#define CFE_TIME_CMD_MID CFE_PLATFORM_TIME_CMD_MIDVAL(CMD)
```

Definition at line 31 of file [default_cfe_time_msgids.h](#).

CFE_TIME_DATA_CMD_MID

```
#define CFE_TIME_DATA_CMD_MID CFE_PLATFORM_TIME_GLBCMD_MIDVAL(DATA_CMD)
```

Definition at line 39 of file [default_cfe_time_msgids.h](#).

CFE_TIME_DIAG_TLM_MID

```
#define CFE_TIME_DIAG_TLM_MID CFE_PLATFORM_TIME_TLM_MIDVAL(DIAG_TLM)
```

Definition at line 46 of file [default_cfe_time_msgids.h](#).

CFE_TIME_HK_TLM_MID

```
#define CFE_TIME_HK_TLM_MID CFE_PLATFORM_TIME_TLM_MIDVAL(HK_TLM)
```

Definition at line 45 of file [default_cfe_time_msgids.h](#).

CFE_TIME_ONEHZ_CMD_MID

```
#define CFE_TIME_ONEHZ_CMD_MID CFE_PLATFORM_TIME_CMD_MIDVAL(ONEHZ_CMD)
```

Definition at line 34 of file [default_cfe_time_msgids.h](#).

CFE_TIME_SEND_CMD_MID

#define CFE_TIME_SEND_CMD_MID CFE_PLATFORM_TIME_GLB_CMD_MIDVAL (SEND_CMD)
 Definition at line 40 of file [default_cfe_time_msgids.h](#).

CFE_TIME_SEND_HK_MID

#define CFE_TIME_SEND_HK_MID CFE_PLATFORM_TIME_CMD_MIDVAL (SEND_HK)
 Definition at line 32 of file [default_cfe_time_msgids.h](#).

CFE_TIME_TONE_CMD_MID

#define CFE_TIME_TONE_CMD_MID CFE_PLATFORM_TIME_CMD_MIDVAL (TONE_CMD)
 Definition at line 33 of file [default_cfe_time_msgids.h](#).

11.268 default_cfe_time_msgids.h

[Go to the documentation of this file.](#)

```

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00016  * limitations under the License.
00017  *****/
00018
00023 #ifndef DEFAULT_CFE_TIME_MSGIDS_H
00024 #define DEFAULT_CFE_TIME_MSGIDS_H
00025
00026 #include "cfe_time_msgid_values.h"
00027
00028 /*
00029 ** cFE Command Message Id's
00030 */
00031 #define CFE_TIME_CMD_MID CFE_PLATFORM_TIME_CMD_MIDVAL (CMD)
00032 #define CFE_TIME_SEND_HK_MID CFE_PLATFORM_TIME_CMD_MIDVAL (SEND_HK)
00033 #define CFE_TIME_TONE_CMD_MID CFE_PLATFORM_TIME_CMD_MIDVAL (TONE_CMD)
00034 #define CFE_TIME_ONEHZ_CMD_MID CFE_PLATFORM_TIME_CMD_MIDVAL (ONEHZ_CMD)
00035
00036 /*
00037 ** cFE Global Command Message Id's
00038 */
00039 #define CFE_TIME_DATA_CMD_MID CFE_PLATFORM_TIME_GLB_CMD_MIDVAL (DATA_CMD)
00040 #define CFE_TIME_SEND_CMD_MID CFE_PLATFORM_TIME_GLB_CMD_MIDVAL (SEND_CMD)
00041
00042 /*
00043 ** CFE Telemetry Message Id's
00044 */
00045 #define CFE_TIME_HK_TLM_MID CFE_PLATFORM_TIME_TLM_MIDVAL (HK_TLM)
00046 #define CFE_TIME_DIAG_TLM_MID CFE_PLATFORM_TIME_TLM_MIDVAL (DIAG_TLM)
00047
00048 #ifndef CFE_OMIT_DEPRECATED_6_8
00049 /*
00050  * This name was changed/deprecated because technically it constitutes an
00051  * identifier that starts with a digit (1) when the CFE_TIME_ namespace
00052  * qualifier is removed. This matters in some circumstances.
00053  */
00054 #define CFE_TIME_1HZ_CMD_MID CFE_TIME_ONEHZ_CMD_MID
00055 #endif
00056
00057 #endif

```

11.269 cfe/modules/time/config/default_cfe_time_msgstruct.h File Reference

```
#include "cfe_time_msgdefs.h"
#include "cfe_msg_hdr.h"
```

Data Structures

- struct [CFE_TIME_NoopCmd](#)
- struct [CFE_TIME_ResetCountersCmd](#)
- struct [CFE_TIME_SendDiagnosticCmd](#)
- struct [CFE_TIME_OneHzCmd](#)
- struct [CFE_TIME_ToneSignalCmd](#)
- struct [CFE_TIME_FakeToneCmd](#)
- struct [CFE_TIME_SendHkCmd](#)
- struct [CFE_TIME_SetLeapSecondsCmd](#)
Set leap seconds command.
- struct [CFE_TIME_SetStateCmd](#)
Set clock state command.
- struct [CFE_TIME_SetSourceCmd](#)
Set time data source command.
- struct [CFE_TIME_SetSignalCmd](#)
Set tone signal source command.
- struct [CFE_TIME_AddDelayCmd](#)
- struct [CFE_TIME_SubDelayCmd](#)
- struct [CFE_TIME_SetMETCmd](#)
- struct [CFE_TIME_SetSTCFCmd](#)
- struct [CFE_TIME_AddAdjustCmd](#)
- struct [CFE_TIME_SubAdjustCmd](#)
- struct [CFE_TIME_SetTimeCmd](#)
- struct [CFE_TIME_AddOneHzAdjustmentCmd](#)
- struct [CFE_TIME_SubOneHzAdjustmentCmd](#)
- struct [CFE_TIME_ToneDataCmd](#)
Time at tone data command.
- struct [CFE_TIME_HousekeepingTlm](#)
- struct [CFE_TIME_DiagnosticTlm](#)

Typedefs

- typedef struct [CFE_TIME_NoopCmd](#) [CFE_TIME_NoopCmd_t](#)
- typedef struct [CFE_TIME_ResetCountersCmd](#) [CFE_TIME_ResetCountersCmd_t](#)
- typedef struct [CFE_TIME_SendDiagnosticCmd](#) [CFE_TIME_SendDiagnosticCmd_t](#)
- typedef struct [CFE_TIME_OneHzCmd](#) [CFE_TIME_OneHzCmd_t](#)
- typedef struct [CFE_TIME_ToneSignalCmd](#) [CFE_TIME_ToneSignalCmd_t](#)
- typedef struct [CFE_TIME_FakeToneCmd](#) [CFE_TIME_FakeToneCmd_t](#)
- typedef struct [CFE_TIME_SendHkCmd](#) [CFE_TIME_SendHkCmd_t](#)
- typedef struct [CFE_TIME_SetLeapSecondsCmd](#) [CFE_TIME_SetLeapSecondsCmd_t](#)
Set leap seconds command.
- typedef struct [CFE_TIME_SetStateCmd](#) [CFE_TIME_SetStateCmd_t](#)
Set clock state command.
- typedef struct [CFE_TIME_SetSourceCmd](#) [CFE_TIME_SetSourceCmd_t](#)

Set time data source command.

- typedef struct [CFE_TIME_SetSignalCmd](#) [CFE_TIME_SetSignalCmd_t](#)

Set tone signal source command.

- typedef struct [CFE_TIME_AddDelayCmd](#) [CFE_TIME_AddDelayCmd_t](#)
- typedef struct [CFE_TIME_SubDelayCmd](#) [CFE_TIME_SubDelayCmd_t](#)
- typedef struct [CFE_TIME_SetMETCmd](#) [CFE_TIME_SetMETCmd_t](#)
- typedef struct [CFE_TIME_SetSTCFCmd](#) [CFE_TIME_SetSTCFCmd_t](#)
- typedef struct [CFE_TIME_AddAdjustCmd](#) [CFE_TIME_AddAdjustCmd_t](#)
- typedef struct [CFE_TIME_SubAdjustCmd](#) [CFE_TIME_SubAdjustCmd_t](#)
- typedef struct [CFE_TIME_SetTimeCmd](#) [CFE_TIME_SetTimeCmd_t](#)
- typedef struct [CFE_TIME_AddOneHzAdjustmentCmd](#) [CFE_TIME_AddOneHzAdjustmentCmd_t](#)
- typedef struct [CFE_TIME_SubOneHzAdjustmentCmd](#) [CFE_TIME_SubOneHzAdjustmentCmd_t](#)
- typedef struct [CFE_TIME_ToneDataCmd](#) [CFE_TIME_ToneDataCmd_t](#)

Time at tone data command.

- typedef struct [CFE_TIME_HousekeepingTlm](#) [CFE_TIME_HousekeepingTlm_t](#)
- typedef struct [CFE_TIME_DiagnosticTlm](#) [CFE_TIME_DiagnosticTlm_t](#)

11.269.1 Detailed Description

cFE Executive Services (TIME) Command and Telemetry packet definition file.
Definition in file [default_cfe_time_msgstruct.h](#).

11.269.2 Typedef Documentation

CFE_TIME_AddAdjustCmd_t

```
typedef struct CFE\_TIME\_AddAdjustCmd CFE\_TIME\_AddAdjustCmd\_t
```

CFE_TIME_AddDelayCmd_t

```
typedef struct CFE\_TIME\_AddDelayCmd CFE\_TIME\_AddDelayCmd\_t
```

CFE_TIME_AddOneHzAdjustmentCmd_t

```
typedef struct CFE\_TIME\_AddOneHzAdjustmentCmd CFE\_TIME\_AddOneHzAdjustmentCmd\_t
```

CFE_TIME_DiagnosticTlm_t

```
typedef struct CFE\_TIME\_DiagnosticTlm CFE\_TIME\_DiagnosticTlm\_t
```

CFE_TIME_FakeToneCmd_t

```
typedef struct CFE\_TIME\_FakeToneCmd CFE\_TIME\_FakeToneCmd\_t
```

CFE_TIME_HousekeepingTlm_t

```
typedef struct CFE\_TIME\_HousekeepingTlm CFE\_TIME\_HousekeepingTlm\_t
```

CFE_TIME_NoopCmd_t

```
typedef struct CFE\_TIME\_NoopCmd CFE\_TIME\_NoopCmd\_t
```

CFE_TIME_OneHzCmd_t

```
typedef struct CFE\_TIME\_OneHzCmd CFE\_TIME\_OneHzCmd\_t
```

CFE_TIME_ResetCountersCmd_t

```
typedef struct CFE_TIME_ResetCountersCmd CFE_TIME_ResetCountersCmd_t
```

CFE_TIME_SendDiagnosticCmd_t

```
typedef struct CFE_TIME_SendDiagnosticCmd CFE_TIME_SendDiagnosticCmd_t
```

CFE_TIME_SendHkCmd_t

```
typedef struct CFE_TIME_SendHkCmd CFE_TIME_SendHkCmd_t
```

CFE_TIME_SetLeapSecondsCmd_t

```
typedef struct CFE_TIME_SetLeapSecondsCmd CFE_TIME_SetLeapSecondsCmd_t
```

Set leap seconds command.

CFE_TIME_SetMETCmd_t

```
typedef struct CFE_TIME_SetMETCmd CFE_TIME_SetMETCmd_t
```

CFE_TIME_SetSignalCmd_t

```
typedef struct CFE_TIME_SetSignalCmd CFE_TIME_SetSignalCmd_t
```

Set tone signal source command.

CFE_TIME_SetSourceCmd_t

```
typedef struct CFE_TIME_SetSourceCmd CFE_TIME_SetSourceCmd_t
```

Set time data source command.

CFE_TIME_SetStateCmd_t

```
typedef struct CFE_TIME_SetStateCmd CFE_TIME_SetStateCmd_t
```

Set clock state command.

CFE_TIME_SetSTCFCmd_t

```
typedef struct CFE_TIME_SetSTCFCmd CFE_TIME_SetSTCFCmd_t
```

CFE_TIME_SetTimeCmd_t

```
typedef struct CFE_TIME_SetTimeCmd CFE_TIME_SetTimeCmd_t
```

CFE_TIME_SubAdjustCmd_t

```
typedef struct CFE_TIME_SubAdjustCmd CFE_TIME_SubAdjustCmd_t
```

CFE_TIME_SubDelayCmd_t

```
typedef struct CFE_TIME_SubDelayCmd CFE_TIME_SubDelayCmd_t
```

CFE_TIME_SubOneHzAdjustmentCmd_t

```
typedef struct CFE_TIME_SubOneHzAdjustmentCmd CFE_TIME_SubOneHzAdjustmentCmd_t
```

CFE_TIME_ToneDataCmd_t

typedef struct CFE_TIME_ToneDataCmd CFE_TIME_ToneDataCmd_t
Time at tone data command.

CFE_TIME_ToneSignalCmd_t

typedef struct CFE_TIME_ToneSignalCmd CFE_TIME_ToneSignalCmd_t

11.270 default_cfe_time_msgstruct.h

[Go to the documentation of this file.](#)

```

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00016  * limitations under the License.
00017  *****/
00018
00023 #ifndef DEFAULT_CFE_TIME_MSGSTRUCT_H
00024 #define DEFAULT_CFE_TIME_MSGSTRUCT_H
00025
00026 /*****
00027  * Includes
00028  *****/
00029 #include "cfe_time_msgdefs.h"
00030 #include "cfe_msg_hdr.h"
00031
00032 /*
00033  * A separate typedef for each of the commands that share this definition
00034  * This follows the convention for command handler prototypes and allows
00035  * each one to independently evolve as necessary.
00036  */
00037 typedef struct CFE_TIME_NoopCmd
00038 {
00039     CFE_MSG_CommandHeader_t CommandHeader;
00040 } CFE_TIME_NoopCmd_t;
00041
00042 typedef struct CFE_TIME_ResetCountersCmd
00043 {
00044     CFE_MSG_CommandHeader_t CommandHeader;
00045 } CFE_TIME_ResetCountersCmd_t;
00046
00047 typedef struct CFE_TIME_SendDiagnosticCmd
00048 {
00049     CFE_MSG_CommandHeader_t CommandHeader;
00050 } CFE_TIME_SendDiagnosticCmd_t;
00051
00052 typedef struct CFE_TIME_OneHzCmd
00053 {
00054     CFE_MSG_CommandHeader_t CommandHeader;
00055 } CFE_TIME_OneHzCmd_t;
00056
00057 typedef struct CFE_TIME_ToneSignalCmd
00058 {
00059     CFE_MSG_CommandHeader_t CommandHeader;
00060 } CFE_TIME_ToneSignalCmd_t;
00061
00062 typedef struct CFE_TIME_FakeToneCmd
00063 {
00064     CFE_MSG_CommandHeader_t CommandHeader;
00065 } CFE_TIME_FakeToneCmd_t;
00066
00067 typedef struct CFE_TIME_SendHkCmd
00068 {

```

```
00069     CFE_MSG_CommandHeader_t CommandHeader;
00070 } CFE_TIME_SendHkCmd_t;
00071
00075 typedef struct CFE_TIME_SetLeapSecondsCmd
00076 {
00077     CFE_MSG_CommandHeader_t CommandHeader;
00078     CFE_TIME_LeapsCmd_Payload_t Payload;
00079 } CFE_TIME_SetLeapSecondsCmd_t;
00080
00084 typedef struct CFE_TIME_SetStateCmd
00085 {
00086     CFE_MSG_CommandHeader_t CommandHeader;
00087     CFE_TIME_StateCmd_Payload_t Payload;
00088 } CFE_TIME_SetStateCmd_t;
00089
00093 typedef struct CFE_TIME_SetSourceCmd
00094 {
00095     CFE_MSG_CommandHeader_t CommandHeader;
00096     CFE_TIME_SourceCmd_Payload_t Payload;
00097 } CFE_TIME_SetSourceCmd_t;
00098
00102 typedef struct CFE_TIME_SetSignalCmd
00103 {
00104     CFE_MSG_CommandHeader_t CommandHeader;
00105     CFE_TIME_SignalCmd_Payload_t Payload;
00106 } CFE_TIME_SetSignalCmd_t;
00107
00108 /*
00109  * A separate typedef for each of the commands that share this definition
00110  * This follows the convention for command handler prototypes and allows
00111  * each one to independently evolve as necessary.
00112  */
00113 typedef struct CFE_TIME_AddDelayCmd
00114 {
00115     CFE_MSG_CommandHeader_t CommandHeader;
00116     CFE_TIME_TimeCmd_Payload_t Payload;
00117 } CFE_TIME_AddDelayCmd_t;
00118
00119 typedef struct CFE_TIME_SubDelayCmd
00120 {
00121     CFE_MSG_CommandHeader_t CommandHeader;
00122     CFE_TIME_TimeCmd_Payload_t Payload;
00123 } CFE_TIME_SubDelayCmd_t;
00124
00125 typedef struct CFE_TIME_SetMETCmd
00126 {
00127     CFE_MSG_CommandHeader_t CommandHeader;
00128     CFE_TIME_TimeCmd_Payload_t Payload;
00129 } CFE_TIME_SetMETCmd_t;
00130
00131 typedef struct CFE_TIME_SetSTCFCmd
00132 {
00133     CFE_MSG_CommandHeader_t CommandHeader;
00134     CFE_TIME_TimeCmd_Payload_t Payload;
00135 } CFE_TIME_SetSTCFCmd_t;
00136
00137 typedef struct CFE_TIME_AddAdjustCmd
00138 {
00139     CFE_MSG_CommandHeader_t CommandHeader;
00140     CFE_TIME_TimeCmd_Payload_t Payload;
00141 } CFE_TIME_AddAdjustCmd_t;
00142
00143 typedef struct CFE_TIME_SubAdjustCmd
00144 {
00145     CFE_MSG_CommandHeader_t CommandHeader;
00146     CFE_TIME_TimeCmd_Payload_t Payload;
00147 } CFE_TIME_SubAdjustCmd_t;
00148
00149 typedef struct CFE_TIME_SetTimeCmd
00150 {
00151     CFE_MSG_CommandHeader_t CommandHeader;
00152     CFE_TIME_TimeCmd_Payload_t Payload;
00153 } CFE_TIME_SetTimeCmd_t;
00154
00155 /*
00156  * A separate typedef for each of the commands that share this definition
00157  * This follows the convention for command handler prototypes and allows
00158  * each one to independently evolve as necessary.
00159  */
00160 typedef struct CFE_TIME_AddOneHzAdjustmentCmd
00161 {
```



```

00162     CFE_MSG_CommandHeader_t      CommandHeader;
00163     CFE_TIME_OneHzAdjustmentCmd_Payload_t Payload;
00164 } CFE_TIME_AddOneHzAdjustmentCmd_t;
00165
00166 typedef struct CFE_TIME_SubOneHzAdjustmentCmd
00167 {
00168     CFE_MSG_CommandHeader_t      CommandHeader;
00169     CFE_TIME_OneHzAdjustmentCmd_Payload_t Payload;
00170 } CFE_TIME_SubOneHzAdjustmentCmd_t;
00171
00175 typedef struct CFE_TIME_ToneDataCmd
00176 {
00177     CFE_MSG_CommandHeader_t      CommandHeader;
00178     CFE_TIME_ToneDataCmd_Payload_t Payload;
00179 } CFE_TIME_ToneDataCmd_t;
00180
00181 /*****
00182
00183 typedef struct CFE_TIME_HousekeepingTlm
00184 {
00185     CFE_MSG_TelemetryHeader_t      TelemetryHeader;
00186     CFE_TIME_HousekeepingTlm_Payload_t Payload;
00187 } CFE_TIME_HousekeepingTlm_t;
00188
00189 typedef struct CFE_TIME_DiagnosticTlm
00190 {
00191     CFE_MSG_TelemetryHeader_t      TelemetryHeader;
00192     CFE_TIME_DiagnosticTlm_Payload_t Payload;
00193 } CFE_TIME_DiagnosticTlm_t;
00194
00195 #endif

```

11.271 cfe/modules/time/config/default_cfe_time_platform_cfg.h File Reference

```

#include "cfe_time_mission_cfg.h"
#include "cfe_time_internal_cfg.h"

```

11.271.1 Detailed Description

CFE Time Services (CFE_TIME) Application Platform Configuration Header File

This is a compatibility header for the "platform_cfg.h" file that has traditionally provided both public and private config definitions for each CFS app.

These definitions are now provided in two separate files, one for the public/mission scope and one for internal scope.

Note

This file may be overridden/superseded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [default_cfe_time_platform_cfg.h](#).

11.272 default_cfe_time_platform_cfg.h

[Go to the documentation of this file.](#)

```

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```

```

00017  *****/
00018
00035 #ifndef DEFAULT_CFE_TIME_PLATFORM_CFG_H
00036 #define DEFAULT_CFE_TIME_PLATFORM_CFG_H
00037
00038 #include "cfe_time_mission_cfg.h"
00039 #include "cfe_time_internal_cfg.h"
00040
00041 #endif

```

11.273 cfe/modules/time/config/default_cfe_time_topicid_values.h File Reference

Macros

- #define [CFE_MISSION_TIME_TIDVAL](#)(x) DEFAULT_CFE_MISSION_TIME_##x##_TOPICID

11.273.1 Detailed Description

CFE Time Services (CFE_TIME) Application Topic IDs
 Definition in file [default_cfe_time_topicid_values.h](#).

11.273.2 Macro Definition Documentation

CFE_MISSION_TIME_TIDVAL

```
#define CFE_MISSION_TIME_TIDVAL(
    x ) DEFAULT_CFE_MISSION_TIME_##x##_TOPICID
```

Definition at line 26 of file [default_cfe_time_topicid_values.h](#).

11.274 default_cfe_time_topicid_values.h

[Go to the documentation of this file.](#)

```

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00017  *****/
00018
00023 #ifndef DEFAULT_CFE_TIME_TOPICID_VALUES_H
00024 #define DEFAULT_CFE_TIME_TOPICID_VALUES_H
00025
00026 #define CFE_MISSION_TIME_TIDVAL(x) DEFAULT_CFE_MISSION_TIME_##x##_TOPICID
00027
00028 #endif

```

11.275 cfe/modules/time/fsw/inc/cfe_time_eventids.h File Reference

Macros

TIME event IDs

- #define [CFE_TIME_INIT_EID](#) 1
TIME Initialization Event ID.
- #define [CFE_TIME_NOOP_EID](#) 4

- *TIME No-op Command Success Event ID.*
• #define CFE_TIME_RESET_EID 5
- *TIME Reset Counters Command Success Event ID.*
• #define CFE_TIME_DIAG_EID 6
- *TIME Request Diagnostics Command Success Event ID.*
• #define CFE_TIME_STATE_EID 7
- *TIME Set Time State Command Success Event ID.*
• #define CFE_TIME_SOURCE_EID 8
- *TIME Set Time Source Command Success Event ID.*
• #define CFE_TIME_SIGNAL_EID 9
- *TIME Set Tone Source Command Success Event ID.*
• #define CFE_TIME_DELAY_EID 11
- *TIME Add or Subtract Delay Command Success Event ID.*
• #define CFE_TIME_TIME_EID 12
- *TIME Set Time Command Success Event ID.*
• #define CFE_TIME_MET_EID 13
- *TIME Set Mission Elapsed Time Command Success Event ID.*
• #define CFE_TIME_STCF_EID 14
- *TIME Set Spacecraft Time Correlation Factor Command Success Event ID.*
• #define CFE_TIME_DELTA_EID 15
- *TIME Add or Subtract Single STCF Adjustment Command Success Event ID.*
• #define CFE_TIME_ONEHZ_EID 16
- *TIME Add or Subtract STCF Adjustment Each Second Command Success Event ID.*
• #define CFE_TIME_LEAPS_EID 17
- *TIME Set Leap Seconds Command Success Event ID.*
• #define CFE_TIME_FLY_ON_EID 20
- *TIME Entered FLYWHEEL Mode Event ID.*
• #define CFE_TIME_FLY_OFF_EID 21
- *TIME Exited FLYWHEEL Mode Event ID.*
• #define CFE_TIME_ID_ERR_EID 26
- *TIME Invalid Message ID Received Event ID.*
• #define CFE_TIME_CC_ERR_EID 27
- *TIME Invalid Command Code Received Event ID.*
• #define CFE_TIME_STATE_ERR_EID 30
- *TIME Set Clock State Command Invalid State Event ID.*
• #define CFE_TIME_SOURCE_ERR_EID 31
- *TIME Set Clock Source Command Invalid Source Event ID.*
• #define CFE_TIME_SIGNAL_ERR_EID 32
- *TIME Set Clock Tone Source Command Invalid Source Event ID.*
• #define CFE_TIME_DELAY_ERR_EID 33
- *TIME Add or Subtract Tone Delay Command Invalid Time Value Event ID.*
• #define CFE_TIME_TIME_ERR_EID 34
- *TIME Set Spacecraft Time Command Invalid Time Value Event ID.*
• #define CFE_TIME_MET_ERR_EID 35
- *TIME Set Mission Elapsed Time Command Invalid Time Value Event ID.*
• #define CFE_TIME_STCF_ERR_EID 36
- *TIME Set Spacecraft Time Correlation Factor Command Invalid Time Value Event ID.*
• #define CFE_TIME_DELTA_ERR_EID 37
- *TIME Add or Subtract Single STCF Adjustment Command Invalid Time Value Event ID.*
• #define CFE_TIME_SOURCE_CFG_EID 40
- *TIME Set Clock Source Command Incompatible Mode Event ID.*
• #define CFE_TIME_SIGNAL_CFG_EID 41
- *TIME Set Clock Signal Command Incompatible Mode Event ID.*
• #define CFE_TIME_DELAY_CFG_EID 42
- *TIME Add or Subtract Tone Delay Command Incompatible Mode Event ID.*

- `#define CFE_TIME_TIME_CFG_EID 43`
TIME Set Spacecraft Time Command Incompatible Mode Event ID.
- `#define CFE_TIME_MET_CFG_EID 44`
TIME Set Mission Elapsed Time Command Incompatible Mode Event ID.
- `#define CFE_TIME_STCF_CFG_EID 45`
TIME Set Spacecraft Time Correlation Factor Command Incompatible Mode Event ID.
- `#define CFE_TIME_LEAPS_CFG_EID 46`
TIME Set Leap Seconds Command Incompatible Mode Event ID.
- `#define CFE_TIME_DELTA_CFG_EID 47`
TIME Add or Subtract Single STCF Adjustment Command Incompatible Mode Event ID.
- `#define CFE_TIME_ONEHZ_CFG_EID 48`
TIME Add or Subtract STCF Adjustment Each Second Command Incompatible Mode Event ID.
- `#define CFE_TIME_LEN_ERR_EID 49`
TIME Invalid Command Length Event ID.

11.275.1 Detailed Description

cFE Time Services Event IDs

Definition in file [cfe_time_eventids.h](#).

11.275.2 Macro Definition Documentation

CFE_TIME_CC_ERR_EID

```
#define CFE_TIME_CC_ERR_EID 27
```

TIME Invalid Command Code Received Event ID.

Type: ERROR

Cause:

Invalid command code for message ID [CFE_TIME_CMD_MID](#) received on the TIME message pipe.

Definition at line [232](#) of file [cfe_time_eventids.h](#).

CFE_TIME_DELAY_CFG_EID

```
#define CFE_TIME_DELAY_CFG_EID 42
```

TIME Add or Subtract Tone Delay Command Incompatible Mode Event ID.

Type: ERROR

Cause:

[TIME Add Tone Delay Command](#) OR [TIME Subtract Tone Delay Command](#) failure due to being in an incompatible mode.

Definition at line [364](#) of file [cfe_time_eventids.h](#).

CFE_TIME_DELAY_EID

```
#define CFE_TIME_DELAY_EID 11
```

TIME Add or Subtract Delay Command Success Event ID.

Type: INFORMATION

Cause:

[TIME Add Time Delay Command](#) OR a [Subtract Time Delay Command](#) success.

Definition at line 120 of file [cfe_time_eventids.h](#).

CFE_TIME_DELAY_ERR_EID

```
#define CFE_TIME_DELAY_ERR_EID 33
```

TIME Add or Subtract Tone Delay Command Invalid Time Value Event ID.

Type: ERROR

Cause:

[TIME Add Tone Delay Command](#) OR [TIME Subtract Tone Delay Command](#) failure due to an invalid time value.

Definition at line 278 of file [cfe_time_eventids.h](#).

CFE_TIME_DELTA_CFG_EID

```
#define CFE_TIME_DELTA_CFG_EID 47
```

TIME Add or Subtract Single STCF Adjustment Command Incompatible Mode Event ID.

Type: ERROR

Cause:

[TIME Add Single STCF Adjustment Command](#) OR [TIME Subtract Single STCF Adjustment Command](#) failure due to being in an incompatible mode.

Definition at line 425 of file [cfe_time_eventids.h](#).

CFE_TIME_DELTA_EID

```
#define CFE_TIME_DELTA_EID 15
```

TIME Add or Subtract Single STCF Adjustment Command Success Event ID.

Type: INFORMATION

Cause:

[TIME Add Single STCF Adjustment Command](#) OR [TIME Subtract Single STCF Adjustment Command](#) success.

Definition at line 165 of file [cfe_time_eventids.h](#).

CFE_TIME_DELTA_ERR_EID

```
#define CFE_TIME_DELTA_ERR_EID 37
```

TIME Add or Subtract Single STCF Adjustment Command Invalid Time Value Event ID.

Type: ERROR

Cause:

[TIME Add Single STCF Adjustment Command](#) OR [TIME Subtract Single STCF Adjustment Command](#) failure due to an invalid time value.

Definition at line 327 of file [cfe_time_eventids.h](#).

CFE_TIME_DIAG_EID

```
#define CFE_TIME_DIAG_EID 6
```

TIME Request Diagnostics Command Success Event ID.

Type: DEBUG

Cause:

[TIME Request Diagnostics Command](#) success.

Definition at line 75 of file [cfe_time_eventids.h](#).

CFE_TIME_FLY_OFF_EID

```
#define CFE_TIME_FLY_OFF_EID 21
```

TIME Exited FLYWHEEL Mode Event ID.

Type: INFORMATION

Cause:

TIME Exited FLYWHEEL Mode.

Definition at line 210 of file [cfe_time_eventids.h](#).

CFE_TIME_FLY_ON_EID

```
#define CFE_TIME_FLY_ON_EID 20
```

TIME Entered FLYWHEEL Mode Event ID.

Type: INFORMATION

Cause:

TIME Entered FLYWHEEL Mode.

Definition at line 199 of file [cfe_time_eventids.h](#).

CFE_TIME_ID_ERR_EID

#define CFE_TIME_ID_ERR_EID 26
TIME Invalid Message ID Received Event ID.

Type: ERROR

Cause:

Invalid message ID received on the TIME message pipe.
Definition at line 221 of file [cfe_time_eventids.h](#).

CFE_TIME_INIT_EID

#define CFE_TIME_INIT_EID 1
TIME Initialization Event ID.

Type: INFORMATION

Cause:

Time Services Task Initialization complete.
Definition at line 42 of file [cfe_time_eventids.h](#).

CFE_TIME_LEAPS_CFG_EID

#define CFE_TIME_LEAPS_CFG_EID 46
TIME Set Leap Seconds Command Incompatible Mode Event ID.

Type: ERROR

Cause:

[TIME Set Leap Seconds Command](#) failure due to being in an incompatible mode.
Definition at line 412 of file [cfe_time_eventids.h](#).

CFE_TIME_LEAPS_EID

#define CFE_TIME_LEAPS_EID 17
TIME Set Leap Seconds Command Success Event ID.

Type: INFORMATION

Cause:

[TIME Set Leap Seconds Command](#) success.
Definition at line 188 of file [cfe_time_eventids.h](#).

CFE_TIME_LEN_ERR_EID

```
#define CFE_TIME_LEN_ERR_EID 49
```

TIME Invalid Command Length Event ID.

Type: ERROR

Cause:

Invalid length for the command code in message ID [CFE_TIME_CMD_MID](#) received on the TIME message pipe.
Definition at line 450 of file [cfe_time_eventids.h](#).

CFE_TIME_MET_CFG_EID

```
#define CFE_TIME_MET_CFG_EID 44
```

TIME Set Mission Elapsed Time Command Incompatible Mode Event ID.

Type: ERROR

Cause:

[TIME Set Mission Elapsed Time Command](#) failure due to being in an incompatible mode.
Definition at line 388 of file [cfe_time_eventids.h](#).

CFE_TIME_MET_EID

```
#define CFE_TIME_MET_EID 13
```

TIME Set Mission Elapsed Time Command Success Event ID.

Type: INFORMATION

Cause:

[TIME Set Mission Elapsed Time Command](#) success.
Definition at line 142 of file [cfe_time_eventids.h](#).

CFE_TIME_MET_ERR_EID

```
#define CFE_TIME_MET_ERR_EID 35
```

TIME Set Mission Elapsed Time Command Invalid Time Value Event ID.

Type: ERROR

Cause:

[TIME Set Mission Elapsed Time Command](#) failure due to an invalid time value.
Definition at line 302 of file [cfe_time_eventids.h](#).

CFE_TIME_NOOP_EID

#define CFE_TIME_NOOP_EID 4
TIME No-op Command Success Event ID.
Type: INFORMATION

Cause:

[TIME NO-OP Command](#) success.
Definition at line 53 of file [cfe_time_eventids.h](#).

CFE_TIME_ONEHZ_CFG_EID

#define CFE_TIME_ONEHZ_CFG_EID 48
TIME Add or Subtract STCF Adjustment Each Second Command Incompatible Mode Event ID.
Type: ERROR

Cause:

[TIME Add STCF Adjustment Each Second Command](#) OR [TIME Subtract STCF Adjustment Each Second Command](#) failure due to being in an incompatible mode.
Definition at line 438 of file [cfe_time_eventids.h](#).

CFE_TIME_ONEHZ_EID

#define CFE_TIME_ONEHZ_EID 16
TIME Add or Subtract STCF Adjustment Each Second Command Success Event ID.
Type: INFORMATION

Cause:

[TIME Add STCF Adjustment Each Second Command](#) OR [TIME Subtract STCF Adjustment Each Second Command](#) success.
Definition at line 177 of file [cfe_time_eventids.h](#).

CFE_TIME_RESET_EID

#define CFE_TIME_RESET_EID 5
TIME Reset Counters Command Success Event ID.
Type: DEBUG

Cause:

[TIME Reset Counters Command](#) success.
Definition at line 64 of file [cfe_time_eventids.h](#).

CFE_TIME_SIGNAL_CFG_EID

```
#define CFE_TIME_SIGNAL_CFG_EID 41
```

TIME Set Clock Signal Command Incompatible Mode Event ID.

Type: ERROR

Cause:

[TIME Set Clock Signal Command](#) failure due to being in an incompatible mode.
Definition at line 351 of file [cfe_time_eventids.h](#).

CFE_TIME_SIGNAL_EID

```
#define CFE_TIME_SIGNAL_EID 9
```

TIME Set Tone Source Command Success Event ID.

Type: INFORMATION

Cause:

[TIME Set Clock Tone Source Command](#) success.
Definition at line 108 of file [cfe_time_eventids.h](#).

CFE_TIME_SIGNAL_ERR_EID

```
#define CFE_TIME_SIGNAL_ERR_EID 32
```

TIME Set Clock Tone Source Command Invalid Source Event ID.

Type: ERROR

Cause:

[Set Clock Tone Source Command](#) failed due to invalid source requested.
Definition at line 265 of file [cfe_time_eventids.h](#).

CFE_TIME_SOURCE_CFG_EID

```
#define CFE_TIME_SOURCE_CFG_EID 40
```

TIME Set Clock Source Command Incompatible Mode Event ID.

Type: ERROR

Cause:

[TIME Set Clock Source Command](#) failure due to being in an incompatible mode.
Definition at line 339 of file [cfe_time_eventids.h](#).

CFE_TIME_SOURCE_EID

```
#define CFE_TIME_SOURCE_EID 8
```

TIME Set Time Source Command Success Event ID.

Type: INFORMATION

Cause:

[TIME Set Time Source Command](#) success.

Definition at line 97 of file [cfe_time_eventids.h](#).

CFE_TIME_SOURCE_ERR_EID

```
#define CFE_TIME_SOURCE_ERR_EID 31
```

TIME Set Clock Source Command Invalid Source Event ID.

Type: ERROR

Cause:

[TIME Set Clock Source Command](#) failed due to invalid source requested.

Definition at line 254 of file [cfe_time_eventids.h](#).

CFE_TIME_STATE_EID

```
#define CFE_TIME_STATE_EID 7
```

TIME Set Time State Command Success Event ID.

Type: INFORMATION

Cause:

[TIME Set Time State Command](#) success.

Definition at line 86 of file [cfe_time_eventids.h](#).

CFE_TIME_STATE_ERR_EID

```
#define CFE_TIME_STATE_ERR_EID 30
```

TIME Set Clock State Command Invalid State Event ID.

Type: ERROR

Cause:

[TIME Set Clock State Command](#) failed due to invalid state requested.

Definition at line 243 of file [cfe_time_eventids.h](#).

CFE_TIME_STCF_CFG_EID

```
#define CFE_TIME_STCF_CFG_EID 45
```

TIME Set Spacecraft Time Correlation Factor Command Incompatible Mode Event ID.

Type: ERROR

Cause:

[TIME Set Spacecraft Time Correlation Factor Command](#) failure due to being in an incompatible mode.
Definition at line 400 of file [cfe_time_eventids.h](#).

CFE_TIME_STCF_EID

```
#define CFE_TIME_STCF_EID 14
```

TIME Set Spacecraft Time Correlation Factor Command Success Event ID.

Type: INFORMATION

Cause:

[TIME Set Spacecraft Time Correlation Factor Command](#) success.
Definition at line 153 of file [cfe_time_eventids.h](#).

CFE_TIME_STCF_ERR_EID

```
#define CFE_TIME_STCF_ERR_EID 36
```

TIME Set Spacecraft Time Correlation Factor Command Invalid Time Value Event ID.

Type: ERROR

Cause:

[TIME Set Spacecraft Time Correlation Factor Command](#) failure due to an invalid time value.
Definition at line 314 of file [cfe_time_eventids.h](#).

CFE_TIME_TIME_CFG_EID

```
#define CFE_TIME_TIME_CFG_EID 43
```

TIME Set Spacecraft Time Command Incompatible Mode Event ID.

Type: ERROR

Cause:

[TIME Set Spacecraft Time Command](#) failure due to being in an incompatible mode.
Definition at line 376 of file [cfe_time_eventids.h](#).

CFE_TIME_TIME_EID

```
#define CFE_TIME_TIME_EID 12
```

TIME Set Time Command Success Event ID.

Type: INFORMATION

Cause:

[TIME Set Time Command](#) success.

Definition at line 131 of file [cfe_time_eventids.h](#).

CFE_TIME_TIME_ERR_EID

```
#define CFE_TIME_TIME_ERR_EID 34
```

TIME Set Spacecraft Time Command Invalid Time Value Event ID.

Type: ERROR

Cause:

[TIME Set Spacecraft Time Command](#) failure due to an invalid time value.

Definition at line 290 of file [cfe_time_eventids.h](#).

11.276 cfe_time_eventids.h

[Go to the documentation of this file.](#)

```
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00016  * limitations under the License.
00017  *****/
00018
00025 #ifndef CFE_TIME_EVENTS_H
00026 #define CFE_TIME_EVENTS_H
00027
00042 #define CFE_TIME_INIT_EID 1
00043
00053 #define CFE_TIME_NOOP_EID 4
00054
00064 #define CFE_TIME_RESET_EID 5
00065
00075 #define CFE_TIME_DIAG_EID 6
00076
00086 #define CFE_TIME_STATE_EID 7
00087
00097 #define CFE_TIME_SOURCE_EID 8
00098
00108 #define CFE_TIME_SIGNAL_EID 9
```

```

00109
00120 #define CFE_TIME_DELAY_EID 11
00121
00131 #define CFE_TIME_TIME_EID 12
00132
00142 #define CFE_TIME_MET_EID 13
00143
00153 #define CFE_TIME_STCF_EID 14
00154
00165 #define CFE_TIME_DELTA_EID 15
00166
00177 #define CFE_TIME_ONEHZ_EID 16
00178
00188 #define CFE_TIME_LEAPS_EID 17
00189
00199 #define CFE_TIME_FLY_ON_EID 20
00200
00210 #define CFE_TIME_FLY_OFF_EID 21
00211
00221 #define CFE_TIME_ID_ERR_EID 26
00222
00232 #define CFE_TIME_CC_ERR_EID 27
00233
00243 #define CFE_TIME_STATE_ERR_EID 30
00244
00254 #define CFE_TIME_SOURCE_ERR_EID 31
00255
00265 #define CFE_TIME_SIGNAL_ERR_EID 32
00266
00278 #define CFE_TIME_DELAY_ERR_EID 33
00279
00290 #define CFE_TIME_TIME_ERR_EID 34
00291
00302 #define CFE_TIME_MET_ERR_EID 35
00303
00314 #define CFE_TIME_STCF_ERR_EID 36
00315
00327 #define CFE_TIME_DELTA_ERR_EID 37
00328
00339 #define CFE_TIME_SOURCE_CFG_EID 40
00340
00351 #define CFE_TIME_SIGNAL_CFG_EID 41
00352
00364 #define CFE_TIME_DELAY_CFG_EID 42
00365
00376 #define CFE_TIME_TIME_CFG_EID 43
00377
00388 #define CFE_TIME_MET_CFG_EID 44
00389
00400 #define CFE_TIME_STCF_CFG_EID 45
00401
00412 #define CFE_TIME_LEAPS_CFG_EID 46
00413
00425 #define CFE_TIME_DELTA_CFG_EID 47
00426
00438 #define CFE_TIME_ONEHZ_CFG_EID 48
00439
00450 #define CFE_TIME_LEN_ERR_EID 49
00453 #endif /* CFE_TIME_EVENTS_H */

```

11.277 cfe/modules/time/fsw/inc/cfe_time_fcncodes.h File Reference

```
#include "cfe_time_fcncode_values.h"
```

Macros

Time Services Command Codes

- `#define CFE_TIME_NOOP_CC CFE_TIME_CCVAL(NOOP)`
- `#define CFE_TIME_RESET_COUNTERS_CC CFE_TIME_CCVAL(RESET_COUNTERS)`
- `#define CFE_TIME_SEND_DIAGNOSTIC_CC CFE_TIME_CCVAL(SEND_DIAGNOSTIC)`
- `#define CFE_TIME_SET_SOURCE_CC CFE_TIME_CCVAL(SET_SOURCE)`
- `#define CFE_TIME_SET_STATE_CC CFE_TIME_CCVAL(SET_STATE)`

- `#define CFE_TIME_ADD_DELAY_CC CFE_TIME_CCVAL(ADD_DELAY)`
- `#define CFE_TIME_SUB_DELAY_CC CFE_TIME_CCVAL(SUB_DELAY)`
- `#define CFE_TIME_SET_TIME_CC CFE_TIME_CCVAL(SET_TIME)`
- `#define CFE_TIME_SET_MET_CC CFE_TIME_CCVAL(SET_MET)`
- `#define CFE_TIME_SET_STCF_CC CFE_TIME_CCVAL(SET_STCF)`
- `#define CFE_TIME_SET_LEAP_SECONDS_CC CFE_TIME_CCVAL(SET_LEAP_SECONDS)`
- `#define CFE_TIME_ADD_ADJUST_CC CFE_TIME_CCVAL(ADD_ADJUST)`
- `#define CFE_TIME_SUB_ADJUST_CC CFE_TIME_CCVAL(SUB_ADJUST)`
- `#define CFE_TIME_ADD_ONE_HZ_ADJUSTMENT_CC CFE_TIME_CCVAL(ADD_ONE_HZ_ADJUSTMENT)`
- `#define CFE_TIME_SUB_ONE_HZ_ADJUSTMENT_CC CFE_TIME_CCVAL(SUB_ONE_HZ_ADJUSTMENT)`
- `#define CFE_TIME_SET_SIGNAL_CC CFE_TIME_CCVAL(SET_SIGNAL)`

11.277.1 Detailed Description

Specification for the CFE Time Services (CFE_TIME) command function codes

Note

This file should be strictly limited to the command/function code (CC) macro definitions. Other definitions such as enums, typedefs, or other macros should be placed in the msgdefs.h or msg.h files.

Definition in file [cfe_time_fcncodes.h](#).

11.277.2 Macro Definition Documentation

CFE_TIME_ADD_ADJUST_CC

```
#define CFE_TIME_ADD_ADJUST_CC CFE_TIME_CCVAL(ADD_ADJUST)
```

Name Add Delta to Spacecraft Time Correlation Factor

Description

This command adjusts the Spacecraft Time Correlation Factor (STCF) by adding the specified value. The new STCF takes effect immediately upon execution of this command.

Command Mnemonic(s) `$sc_$cpu_TIME_AddSTCFAdj`

Command Structure

[CFE_TIME_AddAdjustCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_TIME_CMDPC` - command execution counter will increment
- `$sc_$cpu_TIME_STCF` - Housekeeping Telemetry point indicating new STCF value
- The [CFE_TIME_DELTA_EID](#) informational event message will be generated

Error Conditions

- An invalid number of microseconds was specified (must be less than 1 million)
- Platform receiving the command is not a Time Server

Evidence of Failure may be found in the following telemetry:

- `$sc_$cpu_TIME_CMDEC` - command error counter will increment
- Error specific event messages will be issued ([CFE_TIME_DELTA_ERR_EID](#) or [CFE_TIME_DELTA_CFG_EID](#))

Criticality

Inappropriately setting the clock may result in other sub-systems performing incorrect time based calculations. The specific risk is dependent upon the behavior of those sub-systems.

See also

[CFE_TIME_ADD_ADJUST_CC](#), [CFE_TIME_SUB_ADJUST_CC](#), [CFE_TIME_ADD_ONE_HZ_ADJUSTMENT_CC](#), [CFE_TIME_SUB_ONE_HZ_ADJUSTMENT_CC](#)

Definition at line 519 of file [cfe_time_fcncodes.h](#).

CFE_TIME_ADD_DELAY_CC

```
#define CFE_TIME_ADD_DELAY_CC CFE_TIME_CCVAL(ADD_DELAY)
```

Name Add Time to Tone Time Delay

Description

This command is used to factor out a known, predictable latency between the Time Server and a particular Time Client. The correction is applied (added) to the current time calculation for Time Clients, so this command has no meaning for Time Servers. Each Time Client can have a unique latency setting. The latency value is a positive number of seconds and microseconds that represent the deviation from the time maintained by the Time Server.

Command Mnemonic(s) \$sc_\$cpu_TIME_AddClockLat

Command Structure

[CFE_TIME_AddDelayCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **\$sc_\$cpu_TIME_CMDPC** - command execution counter will increment
- **\$sc_\$cpu_TIME_DLatentS**, **\$sc_\$cpu_TIME_DLatentSs** - Housekeeping Telemetry point indicating command specified values
- **\$sc_\$cpu_TIME_DLatentDir** - Diagnostic Telemetry point indicating commanded latency direction
- The [CFE_TIME_DELAY_EID](#) informational event message will be generated

Error Conditions

- An invalid number of microseconds was specified (must be less than 1 million)
- Platform receiving the command is not a Time Client

Evidence of Failure may be found in the following telemetry:

- **\$sc_\$cpu_TIME_CMDEC** - command error counter will increment
- Error specific event messages will be issued ([CFE_TIME_DELAY_CFG_EID](#) or [CFE_TIME_DELAY_ERR_EID](#))

Criticality

Inappropriately setting the clock may result in other sub-systems performing incorrect time based calculations. The specific risk is dependent upon the behavior of those sub-systems.

See also

[CFE_TIME_SUB_DELAY_CC](#)

Definition at line 292 of file [cfe_time_fcncodes.h](#).

CFE_TIME_ADD_ONE_HZ_ADJUSTMENT_CC

```
#define CFE_TIME_ADD_ONE_HZ_ADJUSTMENT_CC CFE_TIME_CCVAL(ADD_ONE_HZ_ADJUSTMENT)
```

Name Add Delta to Spacecraft Time Correlation Factor each 1Hz

Description

This command has been updated to take actual sub-seconds ($1/2^{32}$ seconds) rather than micro-seconds as an input argument. This change occurred after the determination was made that one micro-second is too large an increment for a constant 1Hz adjustment.

This command continuously adjusts the Spacecraft Time Correlation Factor (STCF) every second, by adding the specified value. The adjustment to the STCF is applied in the Time Service local 1Hz interrupt handler. As the local 1Hz interrupt is not synchronized to the tone signal, one cannot say when the adjustment will occur, other than once a second, at about the same time relative to the tone.

There was some debate about whether the maximum 1Hz clock drift correction factor would ever need to exceed some small fraction of a second. But, the decision was made to provide the capability to make 1Hz adjustments greater than one second and leave it to the ground system to provide mission specific limits.

Command Mnemonic(s) \$sc_\$cpu_TIME_Add1HzSTCF

Command Structure

```
CFE_TIME_AddOneHzAdjustmentCmd_t
```

Command Verification

Successful execution of this command may be verified with the following telemetry:

- \$sc_\$cpu_TIME_CMDPC - command execution counter will increment
- \$sc_\$cpu_TIME_STCF - Housekeeping Telemetry point indicating new STCF value
- The CFE_TIME_ONEHZ_EID informational event message will be generated

Error Conditions

- Platform receiving the command is not a Time Server

Evidence of Failure may be found in the following telemetry:

- \$sc_\$cpu_TIME_CMDEC - command error counter will increment
- Error specific event message will be issued (CFE_TIME_ONEHZ_CFG_EID)

Criticality

Inappropriately setting the clock may result in other sub-systems performing incorrect time based calculations. The specific risk is dependent upon the behavior of those sub-systems.

See also

CFE_TIME_ADD_ADJUST_CC, CFE_TIME_SUB_ADJUST_CC, CFE_TIME_SUB_ONE_HZ_ADJUSTMENT_CC

Definition at line 597 of file cfe_time_fcncodes.h.

CFE_TIME_NOOP_CC

```
#define CFE_TIME_NOOP_CC CFE_TIME_CCVAL(NOOP)
```

Name Time No-Op

Description

This command performs no other function than to increment the command execution counter. The command may be used to verify general aliveness of the Time Services task.

Command Mnemonic(s) \$sc_\$cpu_TIME_NOOP

Command Structure

[CFE_TIME_NoopCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- [\\$sc_\\$cpu_TIME_CMDPC](#) - command execution counter will increment
- The [CFE_TIME_NOOP_EID](#) informational event message will be generated

Error Conditions

There are no error conditions for this command. If the Time Services receives the command, the event is sent (although it may be filtered by EVS) and the counter is incremented unconditionally.

Criticality

None

See also

Definition at line 68 of file [cfe_time_fcncodes.h](#).

CFE_TIME_RESET_COUNTERS_CC

```
#define CFE_TIME_RESET_COUNTERS_CC CFE_TIME_CCVAL(RESET_COUNTERS)
```

Name Time Reset Counters

Description

This command resets the following counters within the Time Services [Housekeeping Telemetry](#) :

- Command Execution Counter ([\\$sc_\\$cpu_TIME_CMDPC](#))
- Command Error Counter ([\\$sc_\\$cpu_TIME_CMDEC](#)) This command also resets the following counters within the Time Services [Diagnostic Telemetry](#) :
- Tone Signal Detected Software Bus Message Counter ([\\$sc_\\$cpu_TIME_DTSDetCNT](#))
- Time at the Tone Data Software Bus Message Counter ([\\$sc_\\$cpu_TIME_DTatTCNT](#))
- Tone Signal/Data Verify Counter ([\\$sc_\\$cpu_TIME_DVerifyCNT](#))
- Tone Signal/Data Error Counter ([\\$sc_\\$cpu_TIME_DVerifyER](#))

- Tone Signal Interrupt Counter (\$sc_\$cpu_TIME_DTsISRCNT)
- Tone Signal Interrupt Error Counter (\$sc_\$cpu_TIME_DTsISRERR)
- Tone Signal Task Counter (\$sc_\$cpu_TIME_DTsTaskCNT)
- Local 1 Hz Interrupt Counter (\$sc_\$cpu_TIME_D1HzISRCNT)
- Local 1 Hz Task Counter (\$sc_\$cpu_TIME_D1HzTaskCNT)
- Reference Time Version Counter (\$sc_\$cpu_TIME_DVersionCNT)

Command Mnemonic(s) \$sc_\$cpu_TIME_ResetCtrs

Command Structure

[CFE_TIME_ResetCountersCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **\$sc_\$cpu_TIME_CMDPC** - command execution counter will reset to 0
- **\$sc_\$cpu_TIME_CMDEC** - command error counter will reset to 0
- The [CFE_TIME_RESET_EID](#) informational event message will be generated

Error Conditions

There are no error conditions for this command. If the Time Services receives the command, the event is sent (although it may be filtered by EVS) and the counter is reset unconditionally.

Criticality

None

See also

Definition at line 113 of file [cfe_time_fcncodes.h](#).

CFE_TIME_SEND_DIAGNOSTIC_CC

```
#define CFE_TIME_SEND_DIAGNOSTIC_CC CFE_TIME_CCVAL(SEND_DIAGNOSTIC)
```

Name Request TIME Diagnostic Telemetry

Description

This command requests that the Time Service generate a message containing various data values not included in the normal Time Service housekeeping message. The command requests only a single copy of the diagnostic message. Refer to [CFE_TIME_DiagnosticTlm_t](#) for a description of the Time Service diagnostic message contents.

Command Mnemonic(s) \$sc_\$cpu_TIME_RequestDiag

Command Structure

[CFE_TIME_SendDiagnosticCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_TIME_CMDPC` - command execution counter will increment
- Sequence Counter for `CFE_TIME_DiagnosticTlm_t` will increment
- The `CFE_TIME_DIAG_EID` debug event message will be generated

Error Conditions

There are no error conditions for this command. If the Time Services receives the command, the event and telemetry is sent (although one or both may be filtered by EVS and TO) and the counter is incremented unconditionally.

Criticality

None

See also

Definition at line 147 of file `cfe_time_fcncodes.h`.

CFE_TIME_SET_LEAP_SECONDS_CC

```
#define CFE_TIME_SET_LEAP_SECONDS_CC CFE_TIME_CCVAL(SET_LEAP_SECONDS)
```

Name Set Leap Seconds

Description

This command sets the spacecraft Leap Seconds to the specified value. Leap Seconds may be positive or negative, and there is no limit to the value except, of course, the limit imposed by the 16 bit signed integer data type. The new Leap Seconds value takes effect immediately upon execution of this command.

Command Mnemonic(s) `$sc_$cpu_TIME_SetClockLeap`

Command Structure

`CFE_TIME_SetLeapSecondsCmd_t`

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_TIME_CMDPC` - command execution counter will increment
- `$sc_$cpu_TIME_LeapSecs` - Housekeeping Telemetry point indicating new Leap seconds value
- The `CFE_TIME_LEAPS_EID` informational event message will be generated

Error Conditions

- Platform receiving the command is not a Time Server

Evidence of Failure may be found in the following telemetry:

- `$sc_$cpu_TIME_CMDEC` - command error counter will increment
- Error specific event messages will be issued (`CFE_TIME_LEAPS_CFG_EID`)

Criticality

Inappropriately setting the clock may result in other sub-systems performing incorrect time based calculations. The specific risk is dependent upon the behavior of those sub-systems.

See also

[CFE_TIME_SET_TIME_CC](#), [CFE_TIME_SET_MET_CC](#), [CFE_TIME_SET_STCF_CC](#)

Definition at line 484 of file [cfe_time_fcncodes.h](#).

CFE_TIME_SET_MET_CC

```
#define CFE_TIME_SET_MET_CC CFE\_TIME\_CCVAL(SET_MET)
```

Name Set Mission Elapsed Time

Description

This command sets the Mission Elapsed Timer (MET) to the specified value.

Note that the MET (as implemented for cFE Time Service) is a logical representation and not a physical timer. Thus, setting the MET is not dependent on whether the hardware supports a MET register that can be written to.

Note also that Time Service "assumes" that during normal operation, the MET is synchronized to the tone signal. Therefore, unless operating in FLYWHEEL mode, the sub-seconds portion of the MET will be set to zero at the next tone signal interrupt.

The new MET takes effect immediately upon execution of this command.

Command Mnemonic(s) `$sc_$cpu_TIME_SetClockMET`

Command Structure

[CFE_TIME_SetMETCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_TIME_CMDPC` - command execution counter will increment
- `$sc_$cpu_TIME_MET` - Housekeeping Telemetry point indicating new MET value
- The [CFE_TIME_MET_EID](#) informational event message will be generated

Error Conditions

- An invalid number of microseconds was specified (must be less than 1 million)
- Platform receiving the command is not a Time Server

Evidence of Failure may be found in the following telemetry:

- `$sc_$cpu_TIME_CMDEC` - command error counter will increment
- Error specific event messages will be issued ([CFE_TIME_MET_CFG_EID](#) or [CFE_TIME_MET_ERR_EID](#))

Criticality

Inappropriately setting the clock may result in other sub-systems performing incorrect time based calculations. The specific risk is dependent upon the behavior of those sub-systems.

See also

[CFE_TIME_SET_TIME_CC](#), [CFE_TIME_SET_STCF_CC](#), [CFE_TIME_SET_LEAP_SECONDS_CC](#)

Definition at line 413 of file [cfe_time_fcncodes.h](#).

CFE_TIME_SET_SIGNAL_CC

```
#define CFE_TIME_SET_SIGNAL_CC CFE_TIME_CCVAL(SET_SIGNAL)
```

Name Set Tone Signal Source

Description

This command selects the Time Service tone signal source. Although the list of potential tone signal sources is mission specific, a common choice is the selection of primary or redundant tone signal. The selection may be available to both the Time Server and Time Clients, depending on hardware configuration.

Notes:

- This command is only valid when the [CFE_PLATFORM_TIME_CFG_SIGNAL](#) configuration parameter in the [cfe_platform_cfg.h](#) file has been set to true.

Command Mnemonic(s) \$sc_\$cpu_TIME_SetSignal

Command Structure

[CFE_TIME_SetSignalCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- [\\$sc_\\$cpu_TIME_CMDPC](#) - command execution counter will increment
- [\\$sc_\\$cpu_TIME_DSignal](#) - Diagnostic Telemetry point will indicate the command specified value
- The [CFE_TIME_SIGNAL_EID](#) informational event message will be generated

Error Conditions

- Invalid Signal selection (a value other than [CFE_TIME_ToneSignalSelect_PRIMARY](#) or [CFE_TIME_ToneSignalSelect_REDUNDANT](#) was specified)
- Multiple Tone Signal Sources not available on this platform

Evidence of failure may be found in the following telemetry:

- [\\$sc_\\$cpu_TIME_CMDEC](#) - Command Error counter will increment
- Error specific event message (either [CFE_TIME_SIGNAL_CFG_EID](#) or [CFE_TIME_SIGNAL_ERR_EID](#))

Criticality

Although tone signal source selection is important, this command is not critical

See also

[CFE_TIME_SET_STATE_CC](#), [CFE_TIME_SET_SOURCE_CC](#)

Definition at line 686 of file [cfe_time_fcncodes.h](#).

CFE_TIME_SET_SOURCE_CC

```
#define CFE_TIME_SET_SOURCE_CC CFE_TIME_CCVAL(SET_SOURCE)
```

Name Set Time Source

Description

This command selects the Time Service clock source. Although the list of potential clock sources is mission specific and defined via configuration parameters, this command provides a common method for switching between the local processor clock and an external source for time data.

When commanded to accept external time data (GPS, MET, spacecraft time, etc.), the Time Server will enable input via an API function specific to the configuration definitions for the particular source. When commanded to use internal time data, the Time Server will ignore the external data. However, the Time Server will continue to use the API function as the trigger to generate a "time at the tone" command packet regardless of the internal/external command selection.

Notes:

- Operating in FLYWHEEL mode is not considered a choice related to clock source, but rather an element of the clock state. See below for a description of the [CFE_TIME_SET_STATE_CC](#) command.
- This command is only valid when the [CFE_PLATFORM_TIME_CFG_SOURCE](#) configuration parameter in the `cfe_platform_cfg.h` file has been set to true.

Command Mnemonic(s) `$sc_$cpu_TIME_SetSource`

Command Structure

[CFE_TIME_SetSourceCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_TIME_CMDPC` - command execution counter will increment
- `$sc_$cpu_TIME_DSOURCE` - Diagnostic Telemetry point will indicate the command specified value
- The [CFE_TIME_SOURCE_EID](#) informational event message will be generated

Error Conditions

- Invalid Source selection (a value other than [CFE_TIME_SourceSelect_INTERNAL](#) or [CFE_TIME_SourceSelect_EXTERNAL](#) was specified)
- Time source selection not allowed on this platform

Evidence of failure may be found in the following telemetry:

- `$sc_$cpu_TIME_CMDEC` - Command Error counter will increment
- Error specific event message (either [CFE_TIME_SOURCE_CFG_EID](#) or [CFE_TIME_SOURCE_ERR_EID](#))

Criticality

Although clock source selection is important, this command is not critical.

See also

[CFE_TIME_SET_STATE_CC](#), [CFE_TIME_SET_SIGNAL_CC](#)

Definition at line 197 of file [cfe_time_fcncodes.h](#).

CFE_TIME_SET_STATE_CC

```
#define CFE_TIME_SET_STATE_CC CFE_TIME_CCVAL(SET_STATE)
```

Name Set Time State

Description

This command indirectly affects the Time Service on-board determination of clock state. Clock state is a combination of factors, most significantly whether the spacecraft time has been accurately set, and whether Time Service is operating in FLYWHEEL mode.

This command may be used to notify the Time Server that spacecraft time is now correct, or that time is no longer correct. This information will be distributed to Time Clients, and in turn, to any interested sub-systems.

Also, this command may be used to force a Time Server or Time Client into FLYWHEEL mode. Use of FLYWHEEL mode is mainly for debug purposes although in extreme circumstances, it may be of value to force Time Service not to rely on normal time updates. Note that when commanded into FLYWHEEL mode, the Time Service will remain so until receipt of another "set state" command setting the state into a mode other than FLYWHEEL.

Note also that setting the clock state to VALID or INVALID on a Time Client that is currently getting time updates from the Time Server will have very limited effect. As soon as the Time Client receives the next time update, the VALID/INVALID selection will be set to that of the Time Server. However, setting a Time Client to FLYWHEEL cannot be overridden by the Time Server since the Time Client will ignore time updates from the Time Server while in FLYWHEEL mode.

Command Mnemonic(s) \$sc_\$cpu_TIME_SetState

Command Structure

[CFE_TIME_SetStateCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **\$sc_\$cpu_TIME_CMDPC** - command execution counter will increment
- **\$sc_\$cpu_TIME_StateFlg**, \$sc_\$cpu_TIME_FlagSet, \$sc_\$cpu_TIME_FlagFly, \$sc_\$cpu_TIME_↵
FlagSrc, \$sc_\$cpu_TIME_FlagPri, \$sc_\$cpu_TIME_FlagSfly, \$sc_\$cpu_TIME_FlagCfly, \$sc_\$cpu_TIME_↵
FlagAdj, \$sc\$cpu_TIME_Flag1Hzd, \$sc_\$cpu_TIME_FlagClat, \$sc_\$cpu_TIME_FlagSorC, \$sc_\$cpu_↵
_TIME_FlagNIU - Housekeeping Telemetry point "may" indicate the command specified value (see above)
- The [CFE_TIME_STATE_EID](#) informational event message will be generated

Error Conditions

- Invalid State selection (a value other than [CFE_TIME_ClockState_INVALID](#), [CFE_TIME_ClockState_VALID](#) or [CFE_TIME_ClockState_FLYWHEEL](#) was specified)
- Time source selection not allowed on this platform

Evidence of failure may be found in the following telemetry:

- **\$sc_\$cpu_TIME_CMDEC** - Command Error counter will increment
- Error specific event message ([CFE_TIME_STATE_ERR_EID](#))

Criticality

Setting Time Service into FLYWHEEL mode is not particularly hazardous, as the result may be that the calculation of spacecraft time is done using a less than optimal timer. However, inappropriately setting the clock state to VALID (indicating that spacecraft time is accurate) may result in other sub-systems performing incorrect time based calculations. The specific risk is dependent upon the behavior of those sub-systems.

See also

[CFE_TIME_SET_SOURCE_CC](#), [CFE_TIME_SET_SIGNAL_CC](#)

Definition at line 254 of file [cfe_time_fcncodes.h](#).

CFE_TIME_SET_STCF_CC

```
#define CFE_TIME_SET_STCF_CC CFE_TIME_CCVAL(SET_STCF)
```

Name Set Spacecraft Time Correlation Factor

Description

This command sets the Spacecraft Time Correlation Factor (STCF) to the specified value. This command differs from the previously described SET CLOCK in the nature of the command argument. This command sets the STCF value directly, rather than extracting the STCF from a value representing the total of MET, STCF and optionally, Leap Seconds. The new STCF takes effect immediately upon execution of this command.

Command Mnemonic(s) \$sc_\$cpu_TIME_SetClockSTCF

Command Structure

[CFE_TIME_SetSTCFCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- **\$sc_\$cpu_TIME_CMDPC** - command execution counter will increment
- **\$sc_\$cpu_TIME_STCF** - Housekeeping Telemetry point indicating new STCF value
- The [CFE_TIME_STCF_EID](#) informational event message will be generated

Error Conditions

- An invalid number of microseconds was specified (must be less than 1 million)
- Platform receiving the command is not a Time Server

Evidence of Failure may be found in the following telemetry:

- **\$sc_\$cpu_TIME_CMDEC** - command error counter will increment
- Error specific event messages will be issued ([CFE_TIME_STCF_CFG_EID](#) or [CFE_TIME_STCF_ERR_EID](#))

Criticality

Inappropriately setting the clock may result in other sub-systems performing incorrect time based calculations. The specific risk is dependent upon the behavior of those sub-systems.

See also

[CFE_TIME_SET_TIME_CC](#), [CFE_TIME_SET_MET_CC](#), [CFE_TIME_SET_LEAP_SECONDS_CC](#)

Definition at line 449 of file [cfe_time_fcncodes.h](#).

CFE_TIME_SET_TIME_CC

```
#define CFE_TIME_SET_TIME_CC CFE_TIME_CCVAL(SET_TIME)
```

Name Set Spacecraft Time

Description

This command sets the spacecraft clock to a new value, regardless of the current setting (time jam). The new time value represents the desired offset from the mission-defined time epoch and takes effect immediately upon execution of this command. Time Service will calculate a new STCF value based on the current MET and the desired new time using one of the following:

If Time Service is configured to compute current time as TAI

- $STCF = (new\ time) - (current\ MET)$
- $(current\ time) = (current\ MET) + STCF$

If Time Service is configured to compute current time as UTC

- $STCF = ((new\ time) - (current\ MET)) + (Leap\ Seconds)$
- $(current\ time) = ((current\ MET) + STCF) - (Leap\ Seconds)$

Command Mnemonic(s) \$sc_\$cpu_TIME_SetClock

Command Structure

[CFE_TIME_SetTimeCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- \$sc_\$cpu_TIME_CMDPC - command execution counter will increment
- \$sc_\$cpu_TIME_STCF - Housekeeping Telemetry point indicating newly calculated STCF value
- The [CFE_TIME_TIME_EID](#) informational event message will be generated

Error Conditions

- An invalid number of microseconds was specified (must be less than 1 million)
- Platform receiving the command is not a Time Server

Evidence of Failure may be found in the following telemetry:

- \$sc_\$cpu_TIME_CMDEC - command error counter will increment
- Error specific event messages will be issued ([CFE_TIME_TIME_CFG_EID](#) or [CFE_TIME_TIME_ERR_EID](#))

Criticality

Inappropriately setting the clock may result in other sub-systems performing incorrect time based calculations. The specific risk is dependent upon the behavior of those sub-systems.

See also

[CFE_TIME_SET_MET_CC](#), [CFE_TIME_SET_STCF_CC](#), [CFE_TIME_SET_LEAP_SECONDS_CC](#)

Definition at line 374 of file [cfe_time_fcncodes.h](#).

CFE_TIME_SUB_ADJUST_CC

```
#define CFE_TIME_SUB_ADJUST_CC CFE_TIME_CCVAL(SUB_ADJUST)
```

Name Subtract Delta from Spacecraft Time Correlation Factor

Description

This command adjusts the Spacecraft Time Correlation Factor (STCF) by subtracting the specified value. The new STCF takes effect immediately upon execution of this command.

Command Mnemonic(s) \$sc_\$cpu_TIME_SubSTCFAdj

Command Structure

[CFE_TIME_SubAdjustCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry:

- [\\$sc_\\$cpu_TIME_CMDPC](#) - command execution counter will increment
- [\\$sc_\\$cpu_TIME_STCF](#) - Housekeeping Telemetry point indicating new STCF value
- The [CFE_TIME_DELTA_EID](#) informational event message will be generated

Error Conditions

- An invalid number of microseconds was specified (must be less than 1 million)
- Platform receiving the command is not a Time Server

Evidence of Failure may be found in the following telemetry:

- [\\$sc_\\$cpu_TIME_CMDEC](#) - command error counter will increment
- Error specific event messages will be issued ([CFE_TIME_DELTA_ERR_EID](#) or [CFE_TIME_DELTA_CFG_EID](#))

Criticality

Inappropriately setting the clock may result in other sub-systems performing incorrect time based calculations. The specific risk is dependent upon the behavior of those sub-systems.

See also

[CFE_TIME_ADD_ADJUST_CC](#), [CFE_TIME_ADD_ONE_HZ_ADJUSTMENT_CC](#), [CFE_TIME_SUB_ONE_HZ_ADJUSTMENT_CC](#)

Definition at line 552 of file [cfe_time_fcncodes.h](#).

CFE_TIME_SUB_DELAY_CC

```
#define CFE_TIME_SUB_DELAY_CC CFE_TIME_CCVAL(SUB_DELAY)
```

Name Subtract Time from Tone Time Delay

Description

This command is used to factor out a known, predictable latency between the Time Server and a particular Time Client. The correction is applied (subtracted) to the current time calculation for Time Clients, so this command has no meaning for Time Servers. Each Time Client can have a unique latency setting. The latency value is a positive number of seconds and microseconds that represent the deviation from the time maintained by the Time Server.

Note that it is unimaginable that the seconds value will ever be anything but zero.

Command Mnemonic(s) `$sc_$cpu_TIME_SubClockLat`

Command Structure

`CFE_TIME_SubDelayCmd_t`

Command Verification

Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_TIME_CMDPC` - command execution counter will increment
- `$sc_$cpu_TIME_DLatentS`, `$sc_$cpu_TIME_DLatentSs` - Housekeeping Telemetry point indicating command specified values
- `$sc_$cpu_TIME_DLatentDir` - Diagnostic Telemetry point indicating commanded latency direction
- The `CFE_TIME_DELAY_EID` informational event message will be generated

Error Conditions

- An invalid number of microseconds was specified (must be less than 1 million)
- Platform receiving the command is not a Time Client

Evidence of Failure may be found in the following telemetry:

- `$sc_$cpu_TIME_CMDEC` - command error counter will increment
- Error specific event messages will be issued (`CFE_TIME_DELAY_CFG_EID` or `CFE_TIME_DELAY_ERR_EID`)

Criticality

Inappropriately setting the clock may result in other sub-systems performing incorrect time based calculations. The specific risk is dependent upon the behavior of those sub-systems.

See also

`CFE_TIME_ADD_DELAY_CC`

Definition at line 330 of file `cfe_time_fcncodes.h`.

CFE_TIME_SUB_ONE_HZ_ADJUSTMENT_CC

```
#define CFE_TIME_SUB_ONE_HZ_ADJUSTMENT_CC CFE_TIME_CCVAL(SUB_ONE_HZ_ADJUSTMENT)
```

Name Subtract Delta from Spacecraft Time Correlation Factor each 1Hz

Description

This command has been updated to take actual sub-seconds ($1/2^{32}$ seconds) rather than micro-seconds as an input argument. This change occurred after the determination was made that one micro-second is too large an increment for a constant 1Hz adjustment.

This command continuously adjusts the Spacecraft Time Correlation Factor (STCF) every second, by subtracting the specified value. The adjustment to the STCF is applied in the Time Service local 1Hz interrupt handler. As the local 1Hz interrupt is not synchronized to the tone signal, one cannot say when the adjustment will occur, other than once a second, at about the same time relative to the tone.

There was some debate about whether the maximum 1Hz clock drift correction factor would ever need to exceed some small fraction of a second. But, the decision was made to provide the capability to make 1Hz adjustments greater than one second and leave it to the ground system to provide mission specific limits.

Command Mnemonic(s) `$sc_$cpu_TIME_Sub1HzSTCF`

Command Structure

[CFE_TIME_SubOneHzAdjustmentCmd_t](#)

Command Verification

Successful execution of this command may be verified with the following telemetry: Successful execution of this command may be verified with the following telemetry:

- `$sc_$cpu_TIME_CMDPC` - command execution counter will increment
- `$sc_$cpu_TIME_STCF` - Housekeeping Telemetry point indicating new STCF value
- The [CFE_TIME_ONEHZ_EID](#) informational event message will be generated

Error Conditions

- Platform receiving the command is not a Time Server

Evidence of Failure may be found in the following telemetry:

- `$sc_$cpu_TIME_CMDEC` - command error counter will increment
- Error specific event message will be issued ([CFE_TIME_ONEHZ_CFG_EID](#))

Criticality

Inappropriately setting the clock may result in other sub-systems performing incorrect time based calculations. The specific risk is dependent upon the behavior of those sub-systems.

See also

[CFE_TIME_ADD_ADJUST_CC](#), [CFE_TIME_SUB_ADJUST_CC](#), [CFE_TIME_ADD_ONE_HZ_ADJUSTMENT_CC](#)

Definition at line 644 of file [cfe_time_fcncodes.h](#).

11.278 cfe_time_fcncodes.h

[Go to the documentation of this file.](#)

```
00001 /*****
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"
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```

```

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00016  * limitations under the License.
00017  *****/
00018
00028 #ifndef CFE_TIME_FCNCODES_H
00029 #define CFE_TIME_FCNCODES_H
00030
00031 #include "cfe_time_fcncode_values.h"
00032
00033 /*
00034 ** Time task command packet command codes...
00035 */
00068 #define CFE_TIME_NOOP_CC CFE_TIME_CCVAL(NOOP)
00069
00113 #define CFE_TIME_RESET_COUNTERS_CC CFE_TIME_CCVAL(RESET_COUNTERS)
00114
00147 #define CFE_TIME_SEND_DIAGNOSTIC_CC CFE_TIME_CCVAL(SEND_DIAGNOSTIC)
00148
00197 #define CFE_TIME_SET_SOURCE_CC CFE_TIME_CCVAL(SET_SOURCE)
00198
00254 #define CFE_TIME_SET_STATE_CC CFE_TIME_CCVAL(SET_STATE)
00255
00292 #define CFE_TIME_ADD_DELAY_CC CFE_TIME_CCVAL(ADD_DELAY)
00293
00330 #define CFE_TIME_SUB_DELAY_CC CFE_TIME_CCVAL(SUB_DELAY)
00331
00374 #define CFE_TIME_SET_TIME_CC CFE_TIME_CCVAL(SET_TIME)
00375
00413 #define CFE_TIME_SET_MET_CC CFE_TIME_CCVAL(SET_MET)
00414
00449 #define CFE_TIME_SET_STCF_CC CFE_TIME_CCVAL(SET_STCF)
00450
00484 #define CFE_TIME_SET_LEAP_SECONDS_CC CFE_TIME_CCVAL(SET_LEAP_SECONDS)
00485
00519 #define CFE_TIME_ADD_ADJUST_CC CFE_TIME_CCVAL(ADD_ADJUST)
00520
00552 #define CFE_TIME_SUB_ADJUST_CC CFE_TIME_CCVAL(SUB_ADJUST)
00553
00597 #define CFE_TIME_ADD_ONE_HZ_ADJUSTMENT_CC CFE_TIME_CCVAL(ADD_ONE_HZ_ADJUSTMENT)
00598
00644 #define CFE_TIME_SUB_ONE_HZ_ADJUSTMENT_CC CFE_TIME_CCVAL(SUB_ONE_HZ_ADJUSTMENT)
00645
00686 #define CFE_TIME_SET_SIGNAL_CC CFE_TIME_CCVAL(SET_SIGNAL)
00689 #endif

```

11.279 cfe/modules/time/fsw/inc/cfe_time_interface_cfg.h File Reference

```
#include "cfe_time_interface_cfg_values.h"
```

Macros

- #define CFE_MISSION_TIME_CFG_DEFAULT_TAI CFE_MISSION_TIME_CFGVAL(CFG_DEFAULT_TAI)
- #define DEFAULT_CFE_MISSION_TIME_CFG_DEFAULT_TAI true
- #define CFE_MISSION_TIME_CFG_DEFAULT_UTC CFE_MISSION_TIME_CFGVAL(CFG_DEFAULT_UTC)
- #define DEFAULT_CFE_MISSION_TIME_CFG_DEFAULT_UTC false
- #define CFE_MISSION_TIME_CFG_FAKE_TONE CFE_MISSION_TIME_CFGVAL(CFG_FAKE_TONE)
- #define DEFAULT_CFE_MISSION_TIME_CFG_FAKE_TONE true
- #define CFE_MISSION_TIME_AT_TONE_WAS CFE_MISSION_TIME_CFGVAL(AT_TONE_WAS)
- #define DEFAULT_CFE_MISSION_TIME_AT_TONE_WAS true
- #define CFE_MISSION_TIME_AT_TONE_WILL_BE CFE_MISSION_TIME_CFGVAL(AT_TONE_WILL_BE)
- #define DEFAULT_CFE_MISSION_TIME_AT_TONE_WILL_BE false

- `#define CFE_MISSION_TIME_MIN_ELAPSED CFE_MISSION_TIME_CFGVAL(MIN_ELAPSED)`
- `#define DEFAULT_CFE_MISSION_TIME_MIN_ELAPSED 0`
- `#define CFE_MISSION_TIME_MAX_ELAPSED CFE_MISSION_TIME_CFGVAL(MAX_ELAPSED)`
- `#define DEFAULT_CFE_MISSION_TIME_MAX_ELAPSED 200000`
- `#define CFE_MISSION_TIME_DEF_MET_SECS CFE_MISSION_TIME_CFGVAL(DEF_MET_SECS)`
- `#define DEFAULT_CFE_MISSION_TIME_DEF_MET_SECS 1000`
- `#define CFE_MISSION_TIME_DEF_MET_SUBS CFE_MISSION_TIME_CFGVAL(DEF_MET_SUBS)`
- `#define DEFAULT_CFE_MISSION_TIME_DEF_MET_SUBS 0`
- `#define CFE_MISSION_TIME_DEF_STCF_SECS CFE_MISSION_TIME_CFGVAL(DEF_STCF_SECS)`
- `#define DEFAULT_CFE_MISSION_TIME_DEF_STCF_SECS 1000000`
- `#define CFE_MISSION_TIME_DEF_STCF_SUBS CFE_MISSION_TIME_CFGVAL(DEF_STCF_SUBS)`
- `#define DEFAULT_CFE_MISSION_TIME_DEF_STCF_SUBS 0`
- `#define CFE_MISSION_TIME_DEF_LEAPS CFE_MISSION_TIME_CFGVAL(DEF_LEAPS)`
- `#define DEFAULT_CFE_MISSION_TIME_DEF_LEAPS 37`
- `#define CFE_MISSION_TIME_DEF_DELAY_SECS CFE_MISSION_TIME_CFGVAL(DEF_DELAY_SECS)`
- `#define DEFAULT_CFE_MISSION_TIME_DEF_DELAY_SECS 0`
- `#define CFE_MISSION_TIME_DEF_DELAY_SUBS CFE_MISSION_TIME_CFGVAL(DEF_DELAY_SUBS)`
- `#define DEFAULT_CFE_MISSION_TIME_DEF_DELAY_SUBS 1000`
- `#define CFE_MISSION_TIME_EPOCH_YEAR CFE_MISSION_TIME_CFGVAL(EPOCH_YEAR)`
- `#define DEFAULT_CFE_MISSION_TIME_EPOCH_YEAR 1980`
- `#define CFE_MISSION_TIME_EPOCH_DAY CFE_MISSION_TIME_CFGVAL(EPOCH_DAY)`
- `#define DEFAULT_CFE_MISSION_TIME_EPOCH_DAY 1`
- `#define CFE_MISSION_TIME_EPOCH_HOUR CFE_MISSION_TIME_CFGVAL(EPOCH_HOUR)`
- `#define DEFAULT_CFE_MISSION_TIME_EPOCH_HOUR 0`
- `#define CFE_MISSION_TIME_EPOCH_MINUTE CFE_MISSION_TIME_CFGVAL(EPOCH_MINUTE)`
- `#define DEFAULT_CFE_MISSION_TIME_EPOCH_MINUTE 0`
- `#define CFE_MISSION_TIME_EPOCH_SECOND CFE_MISSION_TIME_CFGVAL(EPOCH_SECOND)`
- `#define DEFAULT_CFE_MISSION_TIME_EPOCH_SECOND 0`
- `#define CFE_MISSION_TIME_EPOCH_MICROS CFE_MISSION_TIME_CFGVAL(EPOCH_MICROS)`
- `#define DEFAULT_CFE_MISSION_TIME_EPOCH_MICROS 0`
- `#define CFE_MISSION_TIME_EPOCH_SECONDS CFE_MISSION_TIME_CFGVAL(EPOCH_SECONDS)`
- `#define DEFAULT_CFE_MISSION_TIME_EPOCH_SECONDS 315532800`
- `#define CFE_MISSION_TIME_FS_FACTOR CFE_MISSION_TIME_CFGVAL(FS_FACTOR)`
- `#define DEFAULT_CFE_MISSION_TIME_FS_FACTOR 789004800`

11.279.1 Detailed Description

CFE Time Services (CFE_TIME) Application Mission Configuration Header File

This is a compatibility header for the "mission_cfg.h" file that has traditionally provided public config definitions for each CFS app.

Note

This file may be overridden/superseded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [cfe_time_interface_cfg.h](#).

11.279.2 Macro Definition Documentation

CFE_MISSION_TIME_AT_TONE_WAS

```
#define CFE_MISSION_TIME_AT_TONE_WAS CFE_MISSION_TIME_CFGVAL(AT_TONE_WAS)
```

Purpose Default Time and Tone Order

Description:

Time Services may be configured to expect the time at the tone data packet to either precede or follow the tone signal. If the time at the tone data packet follows the tone signal, then the data within the packet describes what the time "was" at the tone. If the time at the tone data packet precedes the tone signal, then the data within the packet describes what the time "will be" at the tone. One, and only one, of the following symbols must be set to true:

- CFE_MISSION_TIME_AT_TONE_WAS
- CFE_MISSION_TIME_AT_TONE_WILL_BE Note: If Time Services is defined as using a simulated tone signal (see [CFE_MISSION_TIME_CFG_FAKE_TONE](#) above), then the tone data packet must follow the tone signal.

Limits

Either CFE_MISSION_TIME_AT_TONE_WAS or CFE_MISSION_TIME_AT_TONE_WILL_BE must be set to true. They may not both be true and they may not both be false.

Definition at line 93 of file [cfe_time_interface_cfg.h](#).

CFE_MISSION_TIME_AT_TONE_WILL_BE

```
#define CFE_MISSION_TIME_AT_TONE_WILL_BE CFE_MISSION_TIME_CFGVAL(AT_TONE_WILL_BE)
```

Definition at line 96 of file [cfe_time_interface_cfg.h](#).

CFE_MISSION_TIME_CFG_DEFAULT_TAI

```
#define CFE_MISSION_TIME_CFG_DEFAULT_TAI CFE_MISSION_TIME_CFGVAL(CFG_DEFAULT_TAI)
```

Purpose Default Time Format

Description:

The following definitions select either UTC or TAI as the default (mission specific) time format. Although it is possible for an application to request time in a specific format, most callers should use [CFE_TIME_GetTime\(\)](#), which returns time in the default format. This avoids having to modify each individual caller when the default choice is changed.

Limits

if CFE_MISSION_TIME_CFG_DEFAULT_TAI is defined as true then CFE_MISSION_TIME_CFG_DEFAULT_UTC must be defined as false. if CFE_MISSION_TIME_CFG_DEFAULT_TAI is defined as false then CFE_MISSION_←
_TIME_CFG_DEFAULT_UTC must be defined as true.

Definition at line 53 of file [cfe_time_interface_cfg.h](#).

CFE_MISSION_TIME_CFG_DEFAULT_UTC

```
#define CFE_MISSION_TIME_CFG_DEFAULT_UTC CFE_MISSION_TIME_CFGVAL(CFG_DEFAULT_UTC)
```

Definition at line 56 of file [cfe_time_interface_cfg.h](#).

CFE_MISSION_TIME_CFG_FAKE_TONE

```
#define CFE_MISSION_TIME_CFG_FAKE_TONE CFE_MISSION_TIME_CFGVAL(CFG_FAKE_TONE)
```

Purpose Default Time Format

Description:

The following definition enables the use of a simulated time at the tone signal using a software bus message.

Limits

Not Applicable

Definition at line 69 of file [cfe_time_interface_cfg.h](#).

CFE_MISSION_TIME_DEF_DELAY_SECS

```
#define CFE_MISSION_TIME_DEF_DELAY_SECS CFE_MISSION_TIME_CFGVAL(DEF_DELAY_SECS)
```

Definition at line 165 of file [cfe_time_interface_cfg.h](#).

CFE_MISSION_TIME_DEF_DELAY_SUBS

```
#define CFE_MISSION_TIME_DEF_DELAY_SUBS CFE_MISSION_TIME_CFGVAL(DEF_DELAY_SUBS)
```

Definition at line 168 of file [cfe_time_interface_cfg.h](#).

CFE_MISSION_TIME_DEF_LEAPS

```
#define CFE_MISSION_TIME_DEF_LEAPS CFE_MISSION_TIME_CFGVAL(DEF_LEAPS)
```

Definition at line 162 of file [cfe_time_interface_cfg.h](#).

CFE_MISSION_TIME_DEF_MET_SECS

```
#define CFE_MISSION_TIME_DEF_MET_SECS CFE_MISSION_TIME_CFGVAL(DEF_MET_SECS)
```

Purpose Default Time Values

Description:

Default time values are provided to avoid problems due to time calculations performed after startup but before commands can be processed. For example, if the default time format is UTC then it is important that the sum of MET and STCF always exceed the value of Leap Seconds to prevent the UTC time calculation ($\text{time} = \text{MET} + \text{STCF} - \text{Leap Seconds}$) from resulting in a negative (very large) number.

Some past missions have also created known (albeit wrong) default timestamps. For example, assume the epoch is defined as Jan 1, 1970 and further assume the default time values are set to create a timestamp of Jan 1, 2000. Even though the year 2000 timestamps are wrong, it may be of value to keep the time within some sort of bounds acceptable to the software.

Note: Sub-second units are in micro-seconds (0 to 999,999) and all values must be defined

Limits

Not Applicable

Definition at line 150 of file [cfe_time_interface_cfg.h](#).

CFE_MISSION_TIME_DEF_MET_SUBS

#define CFE_MISSION_TIME_DEF_MET_SUBS CFE_MISSION_TIME_CFGVAL (DEF_MET_SUBS)
Definition at line 153 of file [cfe_time_interface_cfg.h](#).

CFE_MISSION_TIME_DEF_STCF_SECS

#define CFE_MISSION_TIME_DEF_STCF_SECS CFE_MISSION_TIME_CFGVAL (DEF_STCF_SECS)
Definition at line 156 of file [cfe_time_interface_cfg.h](#).

CFE_MISSION_TIME_DEF_STCF_SUBS

#define CFE_MISSION_TIME_DEF_STCF_SUBS CFE_MISSION_TIME_CFGVAL (DEF_STCF_SUBS)
Definition at line 159 of file [cfe_time_interface_cfg.h](#).

CFE_MISSION_TIME_EPOCH_DAY

#define CFE_MISSION_TIME_EPOCH_DAY CFE_MISSION_TIME_CFGVAL (EPOCH_DAY)
Definition at line 189 of file [cfe_time_interface_cfg.h](#).

CFE_MISSION_TIME_EPOCH_HOUR

#define CFE_MISSION_TIME_EPOCH_HOUR CFE_MISSION_TIME_CFGVAL (EPOCH_HOUR)
Definition at line 192 of file [cfe_time_interface_cfg.h](#).

CFE_MISSION_TIME_EPOCH_MICROS

#define CFE_MISSION_TIME_EPOCH_MICROS CFE_MISSION_TIME_CFGVAL (EPOCH_MICROS)
Definition at line 201 of file [cfe_time_interface_cfg.h](#).

CFE_MISSION_TIME_EPOCH_MINUTE

#define CFE_MISSION_TIME_EPOCH_MINUTE CFE_MISSION_TIME_CFGVAL (EPOCH_MINUTE)
Definition at line 195 of file [cfe_time_interface_cfg.h](#).

CFE_MISSION_TIME_EPOCH_SECOND

#define CFE_MISSION_TIME_EPOCH_SECOND CFE_MISSION_TIME_CFGVAL (EPOCH_SECOND)
Definition at line 198 of file [cfe_time_interface_cfg.h](#).

CFE_MISSION_TIME_EPOCH_SECONDS

#define CFE_MISSION_TIME_EPOCH_SECONDS CFE_MISSION_TIME_CFGVAL (EPOCH_SECONDS)
Definition at line 205 of file [cfe_time_interface_cfg.h](#).

CFE_MISSION_TIME_EPOCH_YEAR

#define CFE_MISSION_TIME_EPOCH_YEAR CFE_MISSION_TIME_CFGVAL (EPOCH_YEAR)

Purpose Default EPOCH Values

Description:

Default ground time epoch values Note: these values are used only by the [CFE_TIME_Print\(\)](#) API function

Limits

Year - must be within 136 years Day - Jan 1 = 1, Feb 1 = 32, etc. Hour - 0 to 23 Minute - 0 to 59 Second - 0 to 59
Micros - 0 to 999999

Definition at line 186 of file [cfe_time_interface_cfg.h](#).

CFE_MISSION_TIME_FS_FACTOR

```
#define CFE_MISSION_TIME_FS_FACTOR CFE_MISSION_TIME_CFGVAL (FS_FACTOR)
```

Purpose Time File System Factor

Description:

Define the s/c vs file system time conversion constant...

Note: this value is intended for use only by CFE TIME API functions to convert time values based on the ground system epoch (s/c time) to and from time values based on the file system epoch (fs time).

FS time = S/C time + factor S/C time = FS time - factor

Worksheet:

S/C epoch = Jan 1, 2005 (LRO ground system epoch) FS epoch = Jan 1, 1980 (vxWorks DOS file system epoch)

Delta = 25 years, 0 days, 0 hours, 0 minutes, 0 seconds

Leap years = 1980, 1984, 1988, 1992, 1996, 2000, 2004 (divisible by 4 – except if by 100 – unless also by 400)

1 year = 31,536,000 seconds 1 day = 86,400 seconds 1 hour = 3,600 seconds 1 minute = 60 seconds

25 years = 788,400,000 seconds 7 extra leap days = 604,800 seconds

total delta = 789,004,800 seconds

Limits

Not Applicable

Definition at line 244 of file [cfe_time_interface_cfg.h](#).

CFE_MISSION_TIME_MAX_ELAPSED

```
#define CFE_MISSION_TIME_MAX_ELAPSED CFE_MISSION_TIME_CFGVAL (MAX_ELAPSED)
```

Definition at line 124 of file [cfe_time_interface_cfg.h](#).

CFE_MISSION_TIME_MIN_ELAPSED

```
#define CFE_MISSION_TIME_MIN_ELAPSED CFE_MISSION_TIME_CFGVAL (MIN_ELAPSED)
```

Purpose Min and Max Time Elapsed

Description:

Based on the definition of Time and Tone Order (CFE_MISSION_TIME_AT_TONE_WAS/WILL_BE) either the "time at the tone" signal or data packet will follow the other. This definition sets the valid window of time for the second of the pair to lag behind the first. Time Services will invalidate both the tone and packet if the second does not arrive within this window following the first.

For example, if the data packet follows the tone, it might be valid for the data packet to arrive between zero and 100,000 micro-seconds after the tone. But, if the tone follows the packet, it might be valid only if the packet arrived between 200,000 and 700,000 micro-seconds before the tone.

Note: units are in micro-seconds

Limits

0 to 999,999 decimal

Definition at line 121 of file [cfe_time_interface_cfg.h](#).

DEFAULT_CFE_MISSION_TIME_AT_TONE_WAS

```
#define DEFAULT_CFE_MISSION_TIME_AT_TONE_WAS true
```

Definition at line 94 of file [cfe_time_interface_cfg.h](#).

DEFAULT_CFE_MISSION_TIME_AT_TONE_WILL_BE

```
#define DEFAULT_CFE_MISSION_TIME_AT_TONE_WILL_BE false
```

Definition at line 97 of file [cfe_time_interface_cfg.h](#).

DEFAULT_CFE_MISSION_TIME_CFG_DEFAULT_TAI

```
#define DEFAULT_CFE_MISSION_TIME_CFG_DEFAULT_TAI true
```

Definition at line 54 of file [cfe_time_interface_cfg.h](#).

DEFAULT_CFE_MISSION_TIME_CFG_DEFAULT_UTC

```
#define DEFAULT_CFE_MISSION_TIME_CFG_DEFAULT_UTC false
```

Definition at line 57 of file [cfe_time_interface_cfg.h](#).

DEFAULT_CFE_MISSION_TIME_CFG_FAKE_TONE

```
#define DEFAULT_CFE_MISSION_TIME_CFG_FAKE_TONE true
```

Definition at line 70 of file [cfe_time_interface_cfg.h](#).

DEFAULT_CFE_MISSION_TIME_DEF_DELAY_SECS

```
#define DEFAULT_CFE_MISSION_TIME_DEF_DELAY_SECS 0
```

Definition at line 166 of file [cfe_time_interface_cfg.h](#).

DEFAULT_CFE_MISSION_TIME_DEF_DELAY_SUBS

```
#define DEFAULT_CFE_MISSION_TIME_DEF_DELAY_SUBS 1000
```

Definition at line 169 of file [cfe_time_interface_cfg.h](#).

DEFAULT_CFE_MISSION_TIME_DEF_LEAPS

```
#define DEFAULT_CFE_MISSION_TIME_DEF_LEAPS 37
```

Definition at line 163 of file [cfe_time_interface_cfg.h](#).

DEFAULT_CFE_MISSION_TIME_DEF_MET_SECS

```
#define DEFAULT_CFE_MISSION_TIME_DEF_MET_SECS 1000
```

Definition at line 151 of file [cfe_time_interface_cfg.h](#).

DEFAULT_CFE_MISSION_TIME_DEF_MET_SUBS

```
#define DEFAULT_CFE_MISSION_TIME_DEF_MET_SUBS 0
```

Definition at line 154 of file [cfe_time_interface_cfg.h](#).

DEFAULT_CFE_MISSION_TIME_DEF_STCF_SECS

```
#define DEFAULT_CFE_MISSION_TIME_DEF_STCF_SECS 1000000
```

Definition at line 157 of file [cfe_time_interface_cfg.h](#).

DEFAULT_CFE_MISSION_TIME_DEF_STCF_SUBS

```
#define DEFAULT_CFE_MISSION_TIME_DEF_STCF_SUBS 0
```

Definition at line 160 of file [cfe_time_interface_cfg.h](#).

DEFAULT_CFE_MISSION_TIME_EPOCH_DAY

```
#define DEFAULT_CFE_MISSION_TIME_EPOCH_DAY 1
```

Definition at line 190 of file [cfe_time_interface_cfg.h](#).

DEFAULT_CFE_MISSION_TIME_EPOCH_HOUR

```
#define DEFAULT_CFE_MISSION_TIME_EPOCH_HOUR 0
```

Definition at line 193 of file [cfe_time_interface_cfg.h](#).

DEFAULT_CFE_MISSION_TIME_EPOCH_MICROS

```
#define DEFAULT_CFE_MISSION_TIME_EPOCH_MICROS 0
```

Definition at line 202 of file [cfe_time_interface_cfg.h](#).

DEFAULT_CFE_MISSION_TIME_EPOCH_MINUTE

```
#define DEFAULT_CFE_MISSION_TIME_EPOCH_MINUTE 0
```

Definition at line 196 of file [cfe_time_interface_cfg.h](#).

DEFAULT_CFE_MISSION_TIME_EPOCH_SECOND

```
#define DEFAULT_CFE_MISSION_TIME_EPOCH_SECOND 0
```

Definition at line 199 of file [cfe_time_interface_cfg.h](#).

DEFAULT_CFE_MISSION_TIME_EPOCH_SECONDS

```
#define DEFAULT_CFE_MISSION_TIME_EPOCH_SECONDS 315532800
```

Definition at line 206 of file [cfe_time_interface_cfg.h](#).

DEFAULT_CFE_MISSION_TIME_EPOCH_YEAR

```
#define DEFAULT_CFE_MISSION_TIME_EPOCH_YEAR 1980
```

Definition at line 187 of file [cfe_time_interface_cfg.h](#).

DEFAULT_CFE_MISSION_TIME_FS_FACTOR

```
#define DEFAULT_CFE_MISSION_TIME_FS_FACTOR 789004800
```

Definition at line 245 of file [cfe_time_interface_cfg.h](#).

DEFAULT_CFE_MISSION_TIME_MAX_ELAPSED

```
#define DEFAULT_CFE_MISSION_TIME_MAX_ELAPSED 200000
```

Definition at line 125 of file [cfe_time_interface_cfg.h](#).

DEFAULT_CFE_MISSION_TIME_MIN_ELAPSED

```
#define DEFAULT_CFE_MISSION_TIME_MIN_ELAPSED 0
```

Definition at line 122 of file [cfe_time_interface_cfg.h](#).

11.280 cfe_time_interface_cfg.h

[Go to the documentation of this file.](#)

```

00001 /*****
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"
00003  *
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00016  * limitations under the License.
00017  *****/
00018
00031 #ifndef CFE_TIME_INTERFACE_CFG_H
00032 #define CFE_TIME_INTERFACE_CFG_H
00033
00034 #include "cfe_time_interface_cfg_values.h"
00035
00053 #define CFE_MISSION_TIME_CFG_DEFAULT_TAI          CFE_MISSION_TIME_CFGVAL(CFG_DEFAULT_TAI)
00054 #define DEFAULT_CFE_MISSION_TIME_CFG_DEFAULT_TAI true
00055
00056 #define CFE_MISSION_TIME_CFG_DEFAULT_UTC          CFE_MISSION_TIME_CFGVAL(CFG_DEFAULT_UTC)
00057 #define DEFAULT_CFE_MISSION_TIME_CFG_DEFAULT_UTC false
00058
00069 #define CFE_MISSION_TIME_CFG_FAKE_TONE           CFE_MISSION_TIME_CFGVAL(CFG_FAKE_TONE)
00070 #define DEFAULT_CFE_MISSION_TIME_CFG_FAKE_TONE true
00071
00093 #define CFE_MISSION_TIME_AT_TONE_WAS             CFE_MISSION_TIME_CFGVAL(AT_TONE_WAS)
00094 #define DEFAULT_CFE_MISSION_TIME_AT_TONE_WAS true
00095
00096 #define CFE_MISSION_TIME_AT_TONE_WILL_BE         CFE_MISSION_TIME_CFGVAL(AT_TONE_WILL_BE)
00097 #define DEFAULT_CFE_MISSION_TIME_AT_TONE_WILL_BE false
00098
00121 #define CFE_MISSION_TIME_MIN_ELAPSED             CFE_MISSION_TIME_CFGVAL(MIN_ELAPSED)
00122 #define DEFAULT_CFE_MISSION_TIME_MIN_ELAPSED 0
00123
00124 #define CFE_MISSION_TIME_MAX_ELAPSED             CFE_MISSION_TIME_CFGVAL(MAX_ELAPSED)
00125 #define DEFAULT_CFE_MISSION_TIME_MAX_ELAPSED 200000
00126
00150 #define CFE_MISSION_TIME_DEF_MET_SECS           CFE_MISSION_TIME_CFGVAL(DEF_MET_SECS)
00151 #define DEFAULT_CFE_MISSION_TIME_DEF_MET_SECS 1000
00152
00153 #define CFE_MISSION_TIME_DEF_MET_SUBS           CFE_MISSION_TIME_CFGVAL(DEF_MET_SUBS)
00154 #define DEFAULT_CFE_MISSION_TIME_DEF_MET_SUBS 0
00155
00156 #define CFE_MISSION_TIME_DEF_STCF_SECS          CFE_MISSION_TIME_CFGVAL(DEF_STCF_SECS)
00157 #define DEFAULT_CFE_MISSION_TIME_DEF_STCF_SECS 1000000
00158
00159 #define CFE_MISSION_TIME_DEF_STCF_SUBS          CFE_MISSION_TIME_CFGVAL(DEF_STCF_SUBS)
00160 #define DEFAULT_CFE_MISSION_TIME_DEF_STCF_SUBS 0
00161
00162 #define CFE_MISSION_TIME_DEF_LEAPS              CFE_MISSION_TIME_CFGVAL(DEF_LEAPS)
00163 #define DEFAULT_CFE_MISSION_TIME_DEF_LEAPS 37
00164
00165 #define CFE_MISSION_TIME_DEF_DELAY_SECS         CFE_MISSION_TIME_CFGVAL(DEF_DELAY_SECS)
00166 #define DEFAULT_CFE_MISSION_TIME_DEF_DELAY_SECS 0
00167
00168 #define CFE_MISSION_TIME_DEF_DELAY_SUBS         CFE_MISSION_TIME_CFGVAL(DEF_DELAY_SUBS)
00169 #define DEFAULT_CFE_MISSION_TIME_DEF_DELAY_SUBS 1000
00170
00186 #define CFE_MISSION_TIME_EPOCH_YEAR             CFE_MISSION_TIME_CFGVAL(EPOCH_YEAR)
00187 #define DEFAULT_CFE_MISSION_TIME_EPOCH_YEAR 1980
00188
00189 #define CFE_MISSION_TIME_EPOCH_DAY              CFE_MISSION_TIME_CFGVAL(EPOCH_DAY)
00190 #define DEFAULT_CFE_MISSION_TIME_EPOCH_DAY 1
00191
00192 #define CFE_MISSION_TIME_EPOCH_HOUR             CFE_MISSION_TIME_CFGVAL(EPOCH_HOUR)
00193 #define DEFAULT_CFE_MISSION_TIME_EPOCH_HOUR 0
00194
00195 #define CFE_MISSION_TIME_EPOCH_MINUTE           CFE_MISSION_TIME_CFGVAL(EPOCH_MINUTE)
00196 #define DEFAULT_CFE_MISSION_TIME_EPOCH_MINUTE 0
00197

```

```

00198 #define CFE_MISSION_TIME_EPOCH_SECOND      CFE_MISSION_TIME_CFGVAL(EPOCH_SECOND)
00199 #define DEFAULT_CFE_MISSION_TIME_EPOCH_SECOND 0
00200
00201 #define CFE_MISSION_TIME_EPOCH_MICROS      CFE_MISSION_TIME_CFGVAL(EPOCH_MICROS)
00202 #define DEFAULT_CFE_MISSION_TIME_EPOCH_MICROS 0
00203
00204 /* new value added for CFE_TIME_Print() */
00205 #define CFE_MISSION_TIME_EPOCH_SECONDS      CFE_MISSION_TIME_CFGVAL(EPOCH_SECONDS)
00206 #define DEFAULT_CFE_MISSION_TIME_EPOCH_SECONDS 315532800
00207
00244 #define CFE_MISSION_TIME_FS_FACTOR          CFE_MISSION_TIME_CFGVAL(FS_FACTOR)
00245 #define DEFAULT_CFE_MISSION_TIME_FS_FACTOR 789004800
00246
00247 #endif

```

11.281 cfe/modules/time/fsw/inc/cfe_time_internal_cfg.h File Reference

```
#include "cfe_time_internal_cfg_values.h"
```

Macros

- #define CFE_PLATFORM_TIME_CFG_SERVER CFE_PLATFORM_TIME_CFGVAL(CFG_SERVER)
- #define DEFAULT_CFE_PLATFORM_TIME_CFG_SERVER true
- #define CFE_PLATFORM_TIME_CFG_CLIENT CFE_PLATFORM_TIME_CFGVAL(CFG_CLIENT)
- #define DEFAULT_CFE_PLATFORM_TIME_CFG_CLIENT false
- #define CFE_PLATFORM_TIME_CFG_VIRTUAL CFE_PLATFORM_TIME_CFGVAL(CFG_VIRTUAL)
- #define DEFAULT_CFE_PLATFORM_TIME_CFG_VIRTUAL true
- #define CFE_PLATFORM_TIME_CFG_SIGNAL CFE_PLATFORM_TIME_CFGVAL(CFG_SIGNAL)
- #define DEFAULT_CFE_PLATFORM_TIME_CFG_SIGNAL false
- #define CFE_PLATFORM_TIME_CFG_SOURCE CFE_PLATFORM_TIME_CFGVAL(CFG_SOURCE)
- #define DEFAULT_CFE_PLATFORM_TIME_CFG_SOURCE false
- #define CFE_PLATFORM_TIME_CFG_SRC_MET CFE_PLATFORM_TIME_CFGVAL(CFG_SRC_MET)
- #define DEFAULT_CFE_PLATFORM_TIME_CFG_SRC_MET false
- #define CFE_PLATFORM_TIME_CFG_SRC_GPS CFE_PLATFORM_TIME_CFGVAL(CFG_SRC_GPS)
- #define DEFAULT_CFE_PLATFORM_TIME_CFG_SRC_GPS false
- #define CFE_PLATFORM_TIME_CFG_SRC_TIME CFE_PLATFORM_TIME_CFGVAL(CFG_SRC_TIME)
- #define DEFAULT_CFE_PLATFORM_TIME_CFG_SRC_TIME false
- #define CFE_PLATFORM_TIME_MAX_DELTA_SECS CFE_PLATFORM_TIME_CFGVAL(MAX_DELTA_SECS)
- #define DEFAULT_CFE_PLATFORM_TIME_MAX_DELTA_SECS 0
- #define CFE_PLATFORM_TIME_MAX_DELTA_SUBS CFE_PLATFORM_TIME_CFGVAL(MAX_DELTA_SUBS)
- #define DEFAULT_CFE_PLATFORM_TIME_MAX_DELTA_SUBS 500000
- #define CFE_PLATFORM_TIME_MAX_LOCAL_SECS CFE_PLATFORM_TIME_CFGVAL(MAX_LOCAL_SECS)
- #define DEFAULT_CFE_PLATFORM_TIME_MAX_LOCAL_SECS 27
- #define CFE_PLATFORM_TIME_MAX_LOCAL_SUBS CFE_PLATFORM_TIME_CFGVAL(MAX_LOCAL_SUBS)
- #define DEFAULT_CFE_PLATFORM_TIME_MAX_LOCAL_SUBS 0
- #define CFE_PLATFORM_TIME_CFG_TONE_LIMIT CFE_PLATFORM_TIME_CFGVAL(CFG_TONE_LIMIT)
- #define DEFAULT_CFE_PLATFORM_TIME_CFG_TONE_LIMIT 20000
- #define CFE_PLATFORM_TIME_CFG_START_FLY CFE_PLATFORM_TIME_CFGVAL(CFG_START_FLY)
- #define DEFAULT_CFE_PLATFORM_TIME_CFG_START_FLY 2
- #define CFE_PLATFORM_TIME_CFG_LATCH_FLY CFE_PLATFORM_TIME_CFGVAL(CFG_LATCH_FLY)
- #define DEFAULT_CFE_PLATFORM_TIME_CFG_LATCH_FLY 8
- #define CFE_PLATFORM_TIME_START_TASK_PRIORITY CFE_PLATFORM_TIME_CFGVAL(START_TASK←
_PRIORITY)
- #define DEFAULT_CFE_PLATFORM_TIME_START_TASK_PRIORITY 60
- #define CFE_PLATFORM_TIME_TONE_TASK_PRIORITY CFE_PLATFORM_TIME_CFGVAL(TONE_TASK←
PRIORITY)

- `#define DEFAULT_CFE_PLATFORM_TIME_TONE_TASK_PRIORITY 25`
- `#define CFE_PLATFORM_TIME_ONEHZ_TASK_PRIORITY CFE_PLATFORM_TIME_CFGVAL(ONEHZ_↵
TASK_PRIORITY)`
- `#define DEFAULT_CFE_PLATFORM_TIME_ONEHZ_TASK_PRIORITY 25`
- `#define CFE_PLATFORM_TIME_START_TASK_STACK_SIZE CFE_PLATFORM_TIME_CFGVAL(START_↵
TASK_STACK_SIZE)`
- `#define DEFAULT_CFE_PLATFORM_TIME_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE`
- `#define CFE_PLATFORM_TIME_TONE_TASK_STACK_SIZE CFE_PLATFORM_TIME_CFGVAL(TONE_↵
TASK_STACK_SIZE)`
- `#define DEFAULT_CFE_PLATFORM_TIME_TONE_TASK_STACK_SIZE 4096`
- `#define CFE_PLATFORM_TIME_ONEHZ_TASK_STACK_SIZE CFE_PLATFORM_TIME_CFGVAL(ONEHZ_↵
TASK_STACK_SIZE)`
- `#define DEFAULT_CFE_PLATFORM_TIME_ONEHZ_TASK_STACK_SIZE 8192`

11.281.1 Detailed Description

CFE Time Services (CFE_TIME) Application Platform Configuration Header File

This is a compatibility header for the "platform_cfg.h" file that has traditionally provided both public and private config definitions for each CFS app.

These definitions are now provided in two separate files, one for the public/mission scope and one for internal scope.

Note

This file may be overridden/superseded by mission-provided definitions either by overriding this header or by generating definitions from a command/data dictionary tool.

Definition in file [cfe_time_internal_cfg.h](#).

11.281.2 Macro Definition Documentation

CFE_PLATFORM_TIME_CFG_CLIENT

```
#define CFE_PLATFORM_TIME_CFG_CLIENT CFE_PLATFORM_TIME_CFGVAL(CFG_CLIENT)
```

Definition at line 55 of file [cfe_time_internal_cfg.h](#).

CFE_PLATFORM_TIME_CFG_LATCH_FLY

```
#define CFE_PLATFORM_TIME_CFG_LATCH_FLY CFE_PLATFORM_TIME_CFGVAL(CFG_LATCH_FLY)
```

Purpose Define Periodic Time to Update Local Clock Tone Latch

Description:

Define Periodic Time to Update Local Clock Tone Latch. Applies only when in flywheel mode. This define dictates the period at which the simulated 'last tone' time is updated. Units are seconds.

Limits

Not Applicable

Definition at line 217 of file [cfe_time_internal_cfg.h](#).

CFE_PLATFORM_TIME_CFG_SERVER

```
#define CFE_PLATFORM_TIME_CFG_SERVER CFE_PLATFORM_TIME_CFGVAL(CFG_SERVER)
```

Purpose Time Server or Time Client Selection

Description:

This configuration parameter selects whether the Time task functions as a time "server" or "client". A time server generates the "time at the tone" packet which is received by time clients.

Limits

Enable one, and only one by defining either CFE_PLATFORM_TIME_CFG_SERVER or CFE_PLATFORM_TIME_CFG_CLIENT AS true. The other must be defined as false.

Definition at line 52 of file [cfe_time_internal_cfg.h](#).

CFE_PLATFORM_TIME_CFG_SIGNAL

```
#define CFE_PLATFORM_TIME_CFG_SIGNAL CFE_PLATFORM_TIME_CFGVAL(CFG_SIGNAL)
```

Purpose Include or Exclude the Primary/Redundant Tone Selection Cmd

Description:

Depending on the specific hardware system configuration, it may be possible to switch between a primary and redundant tone signal. If supported by hardware, this definition will enable command interfaces to select the active tone signal. Both Time Clients and Time Servers support this feature. Note: Set the CFE_PLATFORM_TIME_CFG_SIGNAL define to true to enable tone signal commands.

Limits

Not Applicable

Definition at line 92 of file [cfe_time_internal_cfg.h](#).

CFE_PLATFORM_TIME_CFG_SOURCE

```
#define CFE_PLATFORM_TIME_CFG_SOURCE CFE_PLATFORM_TIME_CFGVAL(CFG_SOURCE)
```

Purpose Include or Exclude the Internal/External Time Source Selection Cmd

Description:

By default, Time Servers maintain time using an internal MET which may be a h/w register or software counter, depending on available hardware. The following definition enables command interfaces to switch between an internal MET, or external time data received from one of several supported external time sources. Only a Time Server may be configured to use external time data. Note: Set the CFE_PLATFORM_TIME_CFG_SOURCE define to true to include the Time Source Selection Command (command allows selection between the internal or external time source). Then choose the external source with the CFE_TIME_CFG_SRC_??? define.

Limits

Only applies if [CFE_PLATFORM_TIME_CFG_SERVER](#) is set to true.

Definition at line 113 of file [cfe_time_internal_cfg.h](#).

CFE_PLATFORM_TIME_CFG_SRC_GPS

```
#define CFE_PLATFORM_TIME_CFG_SRC_GPS CFE_PLATFORM_TIME_CFGVAL(CFG_SRC_GPS)
```

Definition at line 133 of file [cfe_time_internal_cfg.h](#).

CFE_PLATFORM_TIME_CFG_SRC_MET

```
#define CFE_PLATFORM_TIME_CFG_SRC_MET CFE_PLATFORM_TIME_CFGVAL(CFG_SRC_MET)
```

Purpose Choose the External Time Source for Server only

Description:

If [CFE_PLATFORM_TIME_CFG_SOURCE](#) is set to true, then one of the following external time source types must also be set to true. Do not set any of the external time source types to true unless [CFE_PLATFORM_TIME_CFG_SOURCE](#) is set to true.

Limits

1. If [CFE_PLATFORM_TIME_CFG_SOURCE](#) is set to true then one and only one of the following three external time sources can and must be set true: [CFE_PLATFORM_TIME_CFG_SRC_MET](#), [CFE_PLATFORM_TIME_CFG_SRC_GPS](#), [CFE_PLATFORM_TIME_CFG_SRC_TIME](#)
2. Only applies if [CFE_PLATFORM_TIME_CFG_SERVER](#) is set to true.

Definition at line 130 of file [cfe_time_internal_cfg.h](#).

CFE_PLATFORM_TIME_CFG_SRC_TIME

```
#define CFE_PLATFORM_TIME_CFG_SRC_TIME CFE_PLATFORM_TIME_CFGVAL(CFG_SRC_TIME)
```

Definition at line 136 of file [cfe_time_internal_cfg.h](#).

CFE_PLATFORM_TIME_CFG_START_FLY

```
#define CFE_PLATFORM_TIME_CFG_START_FLY CFE_PLATFORM_TIME_CFGVAL(CFG_START_FLY)
```

Purpose Define Time to Start Flywheel Since Last Tone

Description:

Define time to enter flywheel mode (in seconds since last tone data update) Units are microseconds as measured with the local clock.

Limits

Not Applicable

Definition at line 203 of file [cfe_time_internal_cfg.h](#).

CFE_PLATFORM_TIME_CFG_TONE_LIMIT

```
#define CFE_PLATFORM_TIME_CFG_TONE_LIMIT CFE_PLATFORM_TIME_CFGVAL(CFG_TONE_LIMIT)
```

Purpose Define Timing Limits From One Tone To The Next

Description:

Defines limits to the timing of the 1Hz tone signal. A tone signal is valid only if it arrives within one second (plus or minus the tone limit) from the previous tone signal. Units are microseconds as measured with the local clock.

Limits

Not Applicable

Definition at line 190 of file [cfe_time_internal_cfg.h](#).

CFE_PLATFORM_TIME_CFG_VIRTUAL

```
#define CFE_PLATFORM_TIME_CFG_VIRTUAL CFE_PLATFORM_TIME_CFGVAL(CFG_VIRTUAL)
```

Purpose Local MET or Virtual MET Selection for Time Servers

Description:

Depending on the specific hardware system configuration, it may be possible for Time Servers to read the "local" MET from a h/w register rather than having to track the MET as the count of tone signal interrupts (virtual MET)

Time Clients must be defined as using a virtual MET. Also, a Time Server cannot be defined as having both a h/w MET and an external time source (they both cannot synchronize to the same tone).

Note: "disable" this define (set to false) only for Time Servers with local hardware that supports a h/w MET that is synchronized to the tone signal !!!

Limits

Only applies if [CFE_PLATFORM_TIME_CFG_SERVER](#) is set to true.

Definition at line 76 of file [cfe_time_internal_cfg.h](#).

CFE_PLATFORM_TIME_MAX_DELTA_SECS

```
#define CFE_PLATFORM_TIME_MAX_DELTA_SECS CFE_PLATFORM_TIME_CFGVAL(MAX_DELTA_SECS)
```

Purpose Define the Max Delta Limits for Time Servers using an Ext Time Source

Description:

If [CFE_PLATFORM_TIME_CFG_SOURCE](#) is set to true and one of the external time sources is also set to true, then the delta time limits for range checking is used.

When a new time value is received from an external source, the value is compared against the "expected" time value. If the delta exceeds the following defined amount, then the new time data will be ignored. This range checking is only performed after the clock state has been commanded to "valid". Until then, external time data is accepted unconditionally.

Limits

Applies only if both [CFE_PLATFORM_TIME_CFG_SERVER](#) and [CFE_PLATFORM_TIME_CFG_SOURCE](#) are set to true.

Definition at line 156 of file [cfe_time_internal_cfg.h](#).

CFE_PLATFORM_TIME_MAX_DELTA_SUBS

```
#define CFE_PLATFORM_TIME_MAX_DELTA_SUBS CFE_PLATFORM_TIME_CFGVAL(MAX_DELTA_SUBS)
```

Definition at line 159 of file [cfe_time_internal_cfg.h](#).

CFE_PLATFORM_TIME_MAX_LOCAL_SECS

```
#define CFE_PLATFORM_TIME_MAX_LOCAL_SECS CFE_PLATFORM_TIME_CFGVAL(MAX_LOCAL_SECS)
```

Purpose Define the Local Clock Rollover Value in seconds and subseconds

Description:

Specifies the capability of the local clock. Indicates the time at which the local clock rolls over.

Limits

Not Applicable

Definition at line 172 of file [cfe_time_internal_cfg.h](#).

CFE_PLATFORM_TIME_MAX_LOCAL_SUBS

```
#define CFE_PLATFORM_TIME_MAX_LOCAL_SUBS CFE_PLATFORM_TIME_CFGVAL (MAX_LOCAL_SUBS)
```

Definition at line 175 of file [cfe_time_internal_cfg.h](#).

CFE_PLATFORM_TIME_ONEHZ_TASK_PRIORITY

```
#define CFE_PLATFORM_TIME_ONEHZ_TASK_PRIORITY CFE_PLATFORM_TIME_CFGVAL (ONEHZ_TASK_PRIORITY)
```

Definition at line 239 of file [cfe_time_internal_cfg.h](#).

CFE_PLATFORM_TIME_ONEHZ_TASK_STACK_SIZE

```
#define CFE_PLATFORM_TIME_ONEHZ_TASK_STACK_SIZE CFE_PLATFORM_TIME_CFGVAL (ONEHZ_TASK_STACK_SIZE)
```

Definition at line 263 of file [cfe_time_internal_cfg.h](#).

CFE_PLATFORM_TIME_START_TASK_PRIORITY

```
#define CFE_PLATFORM_TIME_START_TASK_PRIORITY CFE_PLATFORM_TIME_CFGVAL (START_TASK_PRIORITY)
```

Purpose Define TIME Task Priorities

Description:

Defines the cFE_TIME Task priority. Defines the cFE_TIME Tone Task priority. Defines the cFE_TIME 1HZ Task priority.

Limits

There is a lower limit of zero and an upper limit of 255 on these configuration parameters. Remember that the meaning of each task priority is inverted – a "lower" number has a "higher" priority.

Definition at line 233 of file [cfe_time_internal_cfg.h](#).

CFE_PLATFORM_TIME_START_TASK_STACK_SIZE

```
#define CFE_PLATFORM_TIME_START_TASK_STACK_SIZE CFE_PLATFORM_TIME_CFGVAL (START_TASK_STACK_SIZE)
```

Purpose Define TIME Task Stack Sizes

Description:

Defines the cFE_TIME Main Task Stack Size Defines the cFE_TIME Tone Task Stack Size Defines the cFE_TIME 1HZ Task Stack Size

Limits

There is a lower limit of 2048 on these configuration parameters. There are no restrictions on the upper limit however, the maximum stack size is system dependent and should be verified. Most operating systems provide tools for measuring the amount of stack used by a task during operation. It is always a good idea to verify that no more than 1/2 of the stack is used.

Definition at line 257 of file [cfe_time_internal_cfg.h](#).

CFE_PLATFORM_TIME_TONE_TASK_PRIORITY

```
#define CFE_PLATFORM_TIME_TONE_TASK_PRIORITY CFE_PLATFORM_TIME_CFGVAL (TONE_TASK_PRIORITY)
```

Definition at line 236 of file [cfe_time_internal_cfg.h](#).

CFE_PLATFORM_TIME_TONE_TASK_STACK_SIZE

#define CFE_PLATFORM_TIME_TONE_TASK_STACK_SIZE CFE_PLATFORM_TIME_CFGVAL(TONE_TASK_STACK_SIZE)
Definition at line 260 of file [cfe_time_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TIME_CFG_CLIENT

#define DEFAULT_CFE_PLATFORM_TIME_CFG_CLIENT false
Definition at line 56 of file [cfe_time_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TIME_CFG_LATCH_FLY

#define DEFAULT_CFE_PLATFORM_TIME_CFG_LATCH_FLY 8
Definition at line 218 of file [cfe_time_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TIME_CFG_SERVER

#define DEFAULT_CFE_PLATFORM_TIME_CFG_SERVER true
Definition at line 53 of file [cfe_time_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TIME_CFG_SIGNAL

#define DEFAULT_CFE_PLATFORM_TIME_CFG_SIGNAL false
Definition at line 93 of file [cfe_time_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TIME_CFG_SOURCE

#define DEFAULT_CFE_PLATFORM_TIME_CFG_SOURCE false
Definition at line 114 of file [cfe_time_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TIME_CFG_SRC_GPS

#define DEFAULT_CFE_PLATFORM_TIME_CFG_SRC_GPS false
Definition at line 134 of file [cfe_time_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TIME_CFG_SRC_MET

#define DEFAULT_CFE_PLATFORM_TIME_CFG_SRC_MET false
Definition at line 131 of file [cfe_time_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TIME_CFG_SRC_TIME

#define DEFAULT_CFE_PLATFORM_TIME_CFG_SRC_TIME false
Definition at line 137 of file [cfe_time_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TIME_CFG_START_FLY

#define DEFAULT_CFE_PLATFORM_TIME_CFG_START_FLY 2
Definition at line 204 of file [cfe_time_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TIME_CFG_TONE_LIMIT

#define DEFAULT_CFE_PLATFORM_TIME_CFG_TONE_LIMIT 20000
Definition at line 191 of file [cfe_time_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TIME_CFG_VIRTUAL

#define DEFAULT_CFE_PLATFORM_TIME_CFG_VIRTUAL true
Definition at line 77 of file [cfe_time_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TIME_MAX_DELTA_SECS

#define DEFAULT_CFE_PLATFORM_TIME_MAX_DELTA_SECS 0
Definition at line 157 of file [cfe_time_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TIME_MAX_DELTA_SUBS

#define DEFAULT_CFE_PLATFORM_TIME_MAX_DELTA_SUBS 500000
Definition at line 160 of file [cfe_time_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TIME_MAX_LOCAL_SECS

#define DEFAULT_CFE_PLATFORM_TIME_MAX_LOCAL_SECS 27
Definition at line 173 of file [cfe_time_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TIME_MAX_LOCAL_SUBS

#define DEFAULT_CFE_PLATFORM_TIME_MAX_LOCAL_SUBS 0
Definition at line 176 of file [cfe_time_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TIME_ONEHZ_TASK_PRIORITY

#define DEFAULT_CFE_PLATFORM_TIME_ONEHZ_TASK_PRIORITY 25
Definition at line 240 of file [cfe_time_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TIME_ONEHZ_TASK_STACK_SIZE

#define DEFAULT_CFE_PLATFORM_TIME_ONEHZ_TASK_STACK_SIZE 8192
Definition at line 264 of file [cfe_time_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TIME_START_TASK_PRIORITY

#define DEFAULT_CFE_PLATFORM_TIME_START_TASK_PRIORITY 60
Definition at line 234 of file [cfe_time_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TIME_START_TASK_STACK_SIZE

#define DEFAULT_CFE_PLATFORM_TIME_START_TASK_STACK_SIZE [CFE_PLATFORM_ES_DEFAULT_STACK_SIZE](#)
Definition at line 258 of file [cfe_time_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TIME_TONE_TASK_PRIORITY

#define DEFAULT_CFE_PLATFORM_TIME_TONE_TASK_PRIORITY 25
Definition at line 237 of file [cfe_time_internal_cfg.h](#).

DEFAULT_CFE_PLATFORM_TIME_TONE_TASK_STACK_SIZE

#define DEFAULT_CFE_PLATFORM_TIME_TONE_TASK_STACK_SIZE 4096
Definition at line 261 of file [cfe_time_internal_cfg.h](#).

11.282 cfe_time_internal_cfg.h

[Go to the documentation of this file.](#)

```

00001 /*****
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"
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00017  *****/
00018
00035 #ifndef CFE_TIME_INTERNAL_CFG_H
00036 #define CFE_TIME_INTERNAL_CFG_H
00037
00038 #include "cfe_time_internal_cfg_values.h"
00039
00052 #define CFE_PLATFORM_TIME_CFG_SERVER          CFE_PLATFORM_TIME_CFGVAL(CFG_SERVER)
00053 #define DEFAULT_CFE_PLATFORM_TIME_CFG_SERVER true
00054
00055 #define CFE_PLATFORM_TIME_CFG_CLIENT          CFE_PLATFORM_TIME_CFGVAL(CFG_CLIENT)
00056 #define DEFAULT_CFE_PLATFORM_TIME_CFG_CLIENT false
00057
00076 #define CFE_PLATFORM_TIME_CFG_VIRTUAL          CFE_PLATFORM_TIME_CFGVAL(CFG_VIRTUAL)
00077 #define DEFAULT_CFE_PLATFORM_TIME_CFG_VIRTUAL true
00078
00092 #define CFE_PLATFORM_TIME_CFG_SIGNAL          CFE_PLATFORM_TIME_CFGVAL(CFG_SIGNAL)
00093 #define DEFAULT_CFE_PLATFORM_TIME_CFG_SIGNAL false
00094
00113 #define CFE_PLATFORM_TIME_CFG_SOURCE          CFE_PLATFORM_TIME_CFGVAL(CFG_SOURCE)
00114 #define DEFAULT_CFE_PLATFORM_TIME_CFG_SOURCE false
00115
00130 #define CFE_PLATFORM_TIME_CFG_SRC_MET          CFE_PLATFORM_TIME_CFGVAL(CFG_SRC_MET)
00131 #define DEFAULT_CFE_PLATFORM_TIME_CFG_SRC_MET false
00132
00133 #define CFE_PLATFORM_TIME_CFG_SRC_GPS          CFE_PLATFORM_TIME_CFGVAL(CFG_SRC_GPS)
00134 #define DEFAULT_CFE_PLATFORM_TIME_CFG_SRC_GPS false
00135
00136 #define CFE_PLATFORM_TIME_CFG_SRC_TIME          CFE_PLATFORM_TIME_CFGVAL(CFG_SRC_TIME)
00137 #define DEFAULT_CFE_PLATFORM_TIME_CFG_SRC_TIME false
00138
00156 #define CFE_PLATFORM_TIME_MAX_DELTA_SECS          CFE_PLATFORM_TIME_CFGVAL(MAX_DELTA_SECS)
00157 #define DEFAULT_CFE_PLATFORM_TIME_MAX_DELTA_SECS 0
00158
00159 #define CFE_PLATFORM_TIME_MAX_DELTA_SUBS          CFE_PLATFORM_TIME_CFGVAL(MAX_DELTA_SUBS)
00160 #define DEFAULT_CFE_PLATFORM_TIME_MAX_DELTA_SUBS 500000
00161
00172 #define CFE_PLATFORM_TIME_MAX_LOCAL_SECS          CFE_PLATFORM_TIME_CFGVAL(MAX_LOCAL_SECS)
00173 #define DEFAULT_CFE_PLATFORM_TIME_MAX_LOCAL_SECS 27
00174
00175 #define CFE_PLATFORM_TIME_MAX_LOCAL_SUBS          CFE_PLATFORM_TIME_CFGVAL(MAX_LOCAL_SUBS)
00176 #define DEFAULT_CFE_PLATFORM_TIME_MAX_LOCAL_SUBS 0
00177
00190 #define CFE_PLATFORM_TIME_CFG_TONE_LIMIT          CFE_PLATFORM_TIME_CFGVAL(CFG_TONE_LIMIT)
00191 #define DEFAULT_CFE_PLATFORM_TIME_CFG_TONE_LIMIT 20000
00192
00203 #define CFE_PLATFORM_TIME_CFG_START_FLY          CFE_PLATFORM_TIME_CFGVAL(CFG_START_FLY)
00204 #define DEFAULT_CFE_PLATFORM_TIME_CFG_START_FLY 2
00205
00217 #define CFE_PLATFORM_TIME_CFG_LATCH_FLY          CFE_PLATFORM_TIME_CFGVAL(CFG_LATCH_FLY)
00218 #define DEFAULT_CFE_PLATFORM_TIME_CFG_LATCH_FLY 8
00219
00233 #define CFE_PLATFORM_TIME_START_TASK_PRIORITY          CFE_PLATFORM_TIME_CFGVAL(START_TASK_PRIORITY)
00234 #define DEFAULT_CFE_PLATFORM_TIME_START_TASK_PRIORITY 60
00235
00236 #define CFE_PLATFORM_TIME_TONE_TASK_PRIORITY          CFE_PLATFORM_TIME_CFGVAL(TONE_TASK_PRIORITY)
00237 #define DEFAULT_CFE_PLATFORM_TIME_TONE_TASK_PRIORITY 25
00238
00239 #define CFE_PLATFORM_TIME_ONEHZ_TASK_PRIORITY          CFE_PLATFORM_TIME_CFGVAL(ONEHZ_TASK_PRIORITY)
00240 #define DEFAULT_CFE_PLATFORM_TIME_ONEHZ_TASK_PRIORITY 25
00241

```

```

00257 #define CFE_PLATFORM_TIME_START_TASK_STACK_SIZE      CFE_PLATFORM_TIME_CFGVAL(START_TASK_STACK_SIZE)
00258 #define DEFAULT_CFE_PLATFORM_TIME_START_TASK_STACK_SIZE CFE_PLATFORM_ES_DEFAULT_STACK_SIZE
00259
00260 #define CFE_PLATFORM_TIME_TONE_TASK_STACK_SIZE      CFE_PLATFORM_TIME_CFGVAL(TONE_TASK_STACK_SIZE)
00261 #define DEFAULT_CFE_PLATFORM_TIME_TONE_TASK_STACK_SIZE 4096
00262
00263 #define CFE_PLATFORM_TIME_ONEHZ_TASK_STACK_SIZE      CFE_PLATFORM_TIME_CFGVAL(ONEHZ_TASK_STACK_SIZE)
00264 #define DEFAULT_CFE_PLATFORM_TIME_ONEHZ_TASK_STACK_SIZE 8192
00265
00266 #endif

```

11.283 cfe/modules/time/fsw/inc/cfe_time_topicids.h File Reference

```
#include "cfe_time_topicid_values.h"
```

Macros

- `#define CFE_MISSION_TIME_CMD_TOPICID CFE_MISSION_TIME_TIDVAL(CMD)`
- `#define DEFAULT_CFE_MISSION_TIME_CMD_TOPICID 5`
- `#define CFE_MISSION_TIME_SEND_HK_TOPICID CFE_MISSION_TIME_TIDVAL(SEND_HK)`
- `#define DEFAULT_CFE_MISSION_TIME_SEND_HK_TOPICID 13`
- `#define CFE_MISSION_TIME_TONE_CMD_TOPICID CFE_MISSION_TIME_TIDVAL(TONE_CMD)`
- `#define DEFAULT_CFE_MISSION_TIME_TONE_CMD_TOPICID 16`
- `#define CFE_MISSION_TIME_ONEHZ_CMD_TOPICID CFE_MISSION_TIME_TIDVAL(ONEHZ_CMD)`
- `#define DEFAULT_CFE_MISSION_TIME_ONEHZ_CMD_TOPICID 17`
- `#define CFE_MISSION_TIME_DATA_CMD_TOPICID CFE_MISSION_TIME_TIDVAL(DATA_CMD)`
- `#define DEFAULT_CFE_MISSION_TIME_DATA_CMD_TOPICID 0`
- `#define CFE_MISSION_TIME_SEND_CMD_TOPICID CFE_MISSION_TIME_TIDVAL(SEND_CMD)`
- `#define DEFAULT_CFE_MISSION_TIME_SEND_CMD_TOPICID 2`
- `#define CFE_MISSION_TIME_HK_TLM_TOPICID CFE_MISSION_TIME_TIDVAL(HK_TLM)`
- `#define DEFAULT_CFE_MISSION_TIME_HK_TLM_TOPICID 5`
- `#define CFE_MISSION_TIME_DIAG_TLM_TOPICID CFE_MISSION_TIME_TIDVAL(DIAG_TLM)`
- `#define DEFAULT_CFE_MISSION_TIME_DIAG_TLM_TOPICID 6`

11.283.1 Detailed Description

CFE Time Services (CFE_TIME) Application Topic IDs
Definition in file [cfe_time_topicids.h](#).

11.283.2 Macro Definition Documentation

CFE_MISSION_TIME_CMD_TOPICID

```
#define CFE_MISSION_TIME_CMD_TOPICID CFE_MISSION_TIME_TIDVAL(CMD)
```

Purpose cFE Portable Message Numbers for Commands

Description:

Portable message numbers for the cFE command messages

Limits

Not Applicable

Definition at line 37 of file [cfe_time_topicids.h](#).

CFE_MISSION_TIME_DATA_CMD_TOPICID

```
#define CFE_MISSION_TIME_DATA_CMD_TOPICID CFE_MISSION_TIME_TIDVAL (DATA_CMD)
```

Purpose cFE Portable Message Numbers for Global Messages

Description:

Portable message numbers for the cFE global messages

Limits

Not Applicable

Definition at line 55 of file [cfe_time_topicsids.h](#).

CFE_MISSION_TIME_DIAG_TLM_TOPICID

```
#define CFE_MISSION_TIME_DIAG_TLM_TOPICID CFE_MISSION_TIME_TIDVAL (DIAG_TLM)
```

Definition at line 71 of file [cfe_time_topicsids.h](#).

CFE_MISSION_TIME_HK_TLM_TOPICID

```
#define CFE_MISSION_TIME_HK_TLM_TOPICID CFE_MISSION_TIME_TIDVAL (HK_TLM)
```

Purpose cFE Portable Message Numbers for Telemetry

Description:

Portable message numbers for the cFE telemetry messages

Limits

Not Applicable

Definition at line 69 of file [cfe_time_topicsids.h](#).

CFE_MISSION_TIME_ONEHZ_CMD_TOPICID

```
#define CFE_MISSION_TIME_ONEHZ_CMD_TOPICID CFE_MISSION_TIME_TIDVAL (ONEHZ_CMD)
```

Definition at line 43 of file [cfe_time_topicsids.h](#).

CFE_MISSION_TIME_SEND_CMD_TOPICID

```
#define CFE_MISSION_TIME_SEND_CMD_TOPICID CFE_MISSION_TIME_TIDVAL (SEND_CMD)
```

Definition at line 57 of file [cfe_time_topicsids.h](#).

CFE_MISSION_TIME_SEND_HK_TOPICID

```
#define CFE_MISSION_TIME_SEND_HK_TOPICID CFE_MISSION_TIME_TIDVAL (SEND_HK)
```

Definition at line 39 of file [cfe_time_topicsids.h](#).

CFE_MISSION_TIME_TONE_CMD_TOPICID

```
#define CFE_MISSION_TIME_TONE_CMD_TOPICID CFE_MISSION_TIME_TIDVAL (TONE_CMD)
```

Definition at line 41 of file [cfe_time_topicsids.h](#).

DEFAULT_CFE_MISSION_TIME_CMD_TOPICID

#define DEFAULT_CFE_MISSION_TIME_CMD_TOPICID 5
 Definition at line 38 of file [cfe_time_topics.h](#).

DEFAULT_CFE_MISSION_TIME_DATA_CMD_TOPICID

#define DEFAULT_CFE_MISSION_TIME_DATA_CMD_TOPICID 0
 Definition at line 56 of file [cfe_time_topics.h](#).

DEFAULT_CFE_MISSION_TIME_DIAG_TLM_TOPICID

#define DEFAULT_CFE_MISSION_TIME_DIAG_TLM_TOPICID 6
 Definition at line 72 of file [cfe_time_topics.h](#).

DEFAULT_CFE_MISSION_TIME_HK_TLM_TOPICID

#define DEFAULT_CFE_MISSION_TIME_HK_TLM_TOPICID 5
 Definition at line 70 of file [cfe_time_topics.h](#).

DEFAULT_CFE_MISSION_TIME_ONEHZ_CMD_TOPICID

#define DEFAULT_CFE_MISSION_TIME_ONEHZ_CMD_TOPICID 17
 Definition at line 44 of file [cfe_time_topics.h](#).

DEFAULT_CFE_MISSION_TIME_SEND_CMD_TOPICID

#define DEFAULT_CFE_MISSION_TIME_SEND_CMD_TOPICID 2
 Definition at line 58 of file [cfe_time_topics.h](#).

DEFAULT_CFE_MISSION_TIME_SEND_HK_TOPICID

#define DEFAULT_CFE_MISSION_TIME_SEND_HK_TOPICID 13
 Definition at line 40 of file [cfe_time_topics.h](#).

DEFAULT_CFE_MISSION_TIME_TONE_CMD_TOPICID

#define DEFAULT_CFE_MISSION_TIME_TONE_CMD_TOPICID 16
 Definition at line 42 of file [cfe_time_topics.h](#).

11.284 cfe_time_topics.h

[Go to the documentation of this file.](#)

```

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00017  *****/
00018
00023 #ifndef CFE_TIME_TOPICS_H

```

```
00024 #define CFE_TIME_TOPICSIDS_H
00025
00026 #include "cfe_time_topicid_values.h"
00027
00037 #define CFE_MISSION_TIME_CMD_TOPICID          CFE_MISSION_TIME_TIDVAL(CMD)
00038 #define DEFAULT_CFE_MISSION_TIME_CMD_TOPICID  5
00039 #define CFE_MISSION_TIME_SEND_HK_TOPICID      CFE_MISSION_TIME_TIDVAL(SEND_HK)
00040 #define DEFAULT_CFE_MISSION_TIME_SEND_HK_TOPICID 13
00041 #define CFE_MISSION_TIME_TONE_CMD_TOPICID      CFE_MISSION_TIME_TIDVAL(TONE_CMD)
00042 #define DEFAULT_CFE_MISSION_TIME_TONE_CMD_TOPICID 16
00043 #define CFE_MISSION_TIME_ONEHZ_CMD_TOPICID     CFE_MISSION_TIME_TIDVAL(ONEHZ_CMD)
00044 #define DEFAULT_CFE_MISSION_TIME_ONEHZ_CMD_TOPICID 17
00045
00055 #define CFE_MISSION_TIME_DATA_CMD_TOPICID      CFE_MISSION_TIME_TIDVAL(DATA_CMD)
00056 #define DEFAULT_CFE_MISSION_TIME_DATA_CMD_TOPICID 0
00057 #define CFE_MISSION_TIME_SEND_CMD_TOPICID      CFE_MISSION_TIME_TIDVAL(SEND_CMD)
00058 #define DEFAULT_CFE_MISSION_TIME_SEND_CMD_TOPICID 2
00059
00069 #define CFE_MISSION_TIME_HK_TLM_TOPICID        CFE_MISSION_TIME_TIDVAL(HK_TLM)
00070 #define DEFAULT_CFE_MISSION_TIME_HK_TLM_TOPICID 5
00071 #define CFE_MISSION_TIME_DIAG_TLM_TOPICID      CFE_MISSION_TIME_TIDVAL(DIAG_TLM)
00072 #define DEFAULT_CFE_MISSION_TIME_DIAG_TLM_TOPICID 6
00073
00074 #endif
```

11.285 osal/docs/src/osal_frontpage.dox File Reference

11.286 osal/docs/src/osal_fs.dox File Reference

11.287 osal/docs/src/osal_timer.dox File Reference

11.288 osal/src/os/inc/common_types.h File Reference

```
#include <stdint.h>
#include <stddef.h>
#include <stdbool.h>
```

Macros

- `#define CompileTimeAssert(Condition, Message) typedef char Message[(Condition) ? 1 : -1]`
- `#define \_EXTENSION\_`
- `#define OS\_USED`
- `#define OS\_PRINTF(n, m)`
- `#define OSAL\_SIZE\_C(X) ((size_t)(X))`
- `#define OSAL\_BLOCKCOUNT\_C(X) ((osal\_blockcount\_t)(X))`
- `#define OSAL\_INDEX\_C(X) ((osal\_index\_t)(X))`
- `#define OSAL\_OBJTYPE\_C(X) ((osal\_objtype\_t)(X))`
- `#define OSAL\_STATUS\_C(X) ((osal\_status\_t)(X))`

Typedefs

- `typedef int8_t int8`
- `typedef int16_t int16`
- `typedef int32_t int32`
- `typedef int64_t int64`
- `typedef uint8_t uint8`
- `typedef uint16_t uint16`
- `typedef uint32_t uint32`
- `typedef uint64_t uint64`
- `typedef intptr_t intptr`
- `typedef uintptr_t cpuaddr`

- typedef size_t [cpusize](#)
- typedef ptrdiff_t [cpudiff](#)
- typedef uint32 [osal_id_t](#)
- typedef size_t [osal_blockcount_t](#)
- typedef long [osal_offset_t](#)
- typedef uint32 [osal_index_t](#)
- typedef uint32 [osal_obtype_t](#)
- typedef int32 [osal_status_t](#)
- typedef void(* [OS_ArgCallback_t](#)) ([osal_id_t](#) object_id, void *arg)

General purpose OSAL callback function.

Functions

- [CompileTimeAssert](#) (sizeof([uint8](#))==1, TypeUint8WrongSize)
- [CompileTimeAssert](#) (sizeof([uint16](#))==2, TypeUint16WrongSize)
- [CompileTimeAssert](#) (sizeof([uint32](#))==4, TypeUint32WrongSize)
- [CompileTimeAssert](#) (sizeof([uint64](#))==8, TypeUint64WrongSize)
- [CompileTimeAssert](#) (sizeof([int8](#))==1, Typeint8WrongSize)
- [CompileTimeAssert](#) (sizeof([int16](#))==2, Typeint16WrongSize)
- [CompileTimeAssert](#) (sizeof([int32](#))==4, Typeint32WrongSize)
- [CompileTimeAssert](#) (sizeof([int64](#))==8, Typeint64WrongSize)
- [CompileTimeAssert](#) (sizeof([cpuaddr](#)) >=sizeof(void *), TypePtrWrongSize)

11.288.1 Detailed Description

Purpose: Unit specification for common types.

Design Notes: Assumes make file has defined processor family

Definition in file [common_types.h](#).

11.288.2 Macro Definition Documentation

__EXTENSION__

```
#define __EXTENSION__
```

Definition at line 65 of file [common_types.h](#).

CompileTimeAssert

```
#define CompileTimeAssert(  
    Condition,  
    Message ) typedef char Message[(Condition) ? 1 : -1]
```

Definition at line 48 of file [common_types.h](#).

OS_PRINTF

```
#define OS_PRINTF(  
    n,  
    m )
```

Definition at line 67 of file [common_types.h](#).

OS_USED

```
#define OS_USED
```

Definition at line 66 of file [common_types.h](#).

OSAL_BLOCKCOUNT_C

```
#define OSAL_BLOCKCOUNT_C(  
    X ) ((osal_blockcount_t) (X))
```

Definition at line 179 of file [common_types.h](#).

OSAL_INDEX_C

```
#define OSAL_INDEX_C(  
    X ) ((osal_index_t) (X))
```

Definition at line 180 of file [common_types.h](#).

OSAL_OBJTYPE_C

```
#define OSAL_OBJTYPE_C(  
    X ) ((osal_objtype_t) (X))
```

Definition at line 181 of file [common_types.h](#).

OSAL_SIZE_C

```
#define OSAL_SIZE_C(  
    X ) ((size_t) (X))
```

Definition at line 178 of file [common_types.h](#).

OSAL_STATUS_C

```
#define OSAL_STATUS_C(  
    X ) ((osal_status_t) (X))
```

Definition at line 182 of file [common_types.h](#).

11.288.3 Typedef Documentation**cpuaddr**

```
typedef uintptr_t cpuaddr
```

Definition at line 88 of file [common_types.h](#).

cpudiff

```
typedef ptrdiff_t cpudiff
```

Definition at line 90 of file [common_types.h](#).

cpusize

```
typedef size_t cpusize
```

Definition at line 89 of file [common_types.h](#).

int16

```
typedef int16_t int16
```

Definition at line 80 of file [common_types.h](#).

int32

```
typedef int32_t int32
```

Definition at line 81 of file [common_types.h](#).

int64

```
typedef int64_t int64
```

Definition at line 82 of file [common_types.h](#).

int8

```
typedef int8_t int8
```

Definition at line 79 of file [common_types.h](#).

intptr

```
typedef intptr_t intptr
```

Definition at line 87 of file [common_types.h](#).

OS_ArgCallback_t

```
typedef void(* OS_ArgCallback_t) (osal_id_t object_id, void *arg)
```

General purpose OSAL callback function.

This may be used by multiple APIS

Definition at line 150 of file [common_types.h](#).

osal_blockcount_t

```
typedef size_t osal_blockcount_t
```

A type used to represent a number of blocks or buffers

This is used with file system and queue implementations.

Definition at line 116 of file [common_types.h](#).

osal_id_t

```
typedef uint32 osal_id_t
```

A type to be used for OSAL resource identifiers. This typedef is backward compatible with the IDs from older versions of OSAL

Definition at line 108 of file [common_types.h](#).

osal_index_t

```
typedef uint32 osal_index_t
```

A type used to represent an index into a table structure

This is used when referring directly to a table index as opposed to an object ID. It is primarily intended for internal use, but is also output from public APIs such as [OS_ObjectIdToArrayIndex\(\)](#).

Definition at line 133 of file [common_types.h](#).

osal_objtype_t

```
typedef uint32 osal_objtype_t
```

A type used to represent the runtime type or category of an OSAL object

Definition at line 138 of file [common_types.h](#).

osal_offset_t

```
typedef long osal_offset_t
```

A type used to represent an offset into a file

This is used with file system implementation.

Definition at line 123 of file [common_types.h](#).

osal_status_t

```
typedef int32_t osal_status_t
```

The preferred type to represent OSAL status codes defined in [osapi-error.h](#)

Definition at line 143 of file [common_types.h](#).

uint16

```
typedef uint16_t uint16
```

Definition at line 84 of file [common_types.h](#).

uint32

```
typedef uint32_t uint32
```

Definition at line 85 of file [common_types.h](#).

uint64

```
typedef uint64_t uint64
```

Definition at line 86 of file [common_types.h](#).

uint8

```
typedef uint8_t uint8
```

Definition at line 83 of file [common_types.h](#).

11.288.4 Function Documentation**CompileTimeAssert()** [1/9]

```
CompileTimeAssert (
    sizeof(cpuaddr) >=sizeof(void *) ,
    TypePtrWrongSize )
```

CompileTimeAssert() [2/9]

```
CompileTimeAssert (
    sizeof(int16)  ==2,
    Typeint16WrongSize )
```

CompileTimeAssert() [3/9]

```
CompileTimeAssert (
    sizeof(int32)  ==4,
    Typeint32WrongSize )
```

CompileTimeAssert() [4/9]

```
CompileTimeAssert (
    sizeof(int64)  ==8,
    Typeint64WrongSize )
```

CompileTimeAssert() [5/9]

```
CompileTimeAssert (
    sizeof(int8)   ==1,
    Typeint8WrongSize )
```

CompileTimeAssert() [6/9]

```
CompileTimeAssert (
    sizeof(uint16)  = =2,
    TypeUint16WrongSize )
```

CompileTimeAssert() [7/9]

```
CompileTimeAssert (
    sizeof(uint32)  = =4,
    TypeUint32WrongSize )
```

CompileTimeAssert() [8/9]

```
CompileTimeAssert (
    sizeof(uint64)  = =8,
    TypeUint64WrongSize )
```

CompileTimeAssert() [9/9]

```
CompileTimeAssert (
    sizeof(uint8)   = =1,
    TypeUint8WrongSize )
```

11.289 common_types.h

[Go to the documentation of this file.](#)

```
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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00029 #ifndef COMMON_TYPES_H
00030 #define COMMON_TYPES_H
00031
00032 #ifdef __cplusplus
00033 extern "C"
00034 {
00035 #endif
00036
00037 /*
00038 ** Includes
00039 */
00040
00041 /*
00042 ** Macro Definitions
00043 */
00044
00045 /*
00046 ** Condition = TRUE is ok, Condition = FALSE is error
00047 */
00048 #define CompileTimeAssert(Condition, Message) typedef char Message[(Condition) ? 1 : -1]
00049
00050 /*
00051 ** Define compiler specific macros
00052 ** The __extension__ compiler pragma is required
```



```

00053 ** for the uint64 type using GCC with the ANSI C90 standard.
00054 ** Other macros can go in here as needed.
00055 **
00056 ** NOTE: The white-box (coverage) unit testing may need to disable
00057 ** these extra attributes. These test builds define the OSAPI_NO_SPECIAL_ATTRIBS
00058 ** macro to disable this.
00059 */
00060 #if defined(__GNUC__) && !defined(OSAPI_NO_SPECIAL_ATTRIBS)
00061 #define _EXTENSION_      __extension__
00062 #define OS_USED          __attribute__((used))
00063 #define OS_PRINTF(n, m)  __attribute__((format(printf, n, m)))
00064 #else
00065 #define _EXTENSION_
00066 #define OS_USED
00067 #define OS_PRINTF(n, m)
00068 #endif
00069
00070 #include <stdint.h>
00071 #include <stddef.h>
00072 #include <stdbool.h>
00073
00074 /*
00075  * NOTE - NOT DEFINING STRUCT_LOW_BIT_FIRST or STRUCT_HIGH_BIT_FIRST
00076  * We should not make assumptions about the bit order here
00077  */
00078
00079 typedef int8_t      int8;
00080 typedef int16_t     int16;
00081 typedef int32_t     int32;
00082 typedef int64_t     int64;
00083 typedef uint8_t     uint8;
00084 typedef uint16_t    uint16;
00085 typedef uint32_t    uint32;
00086 typedef uint64_t    uint64;
00087 typedef intptr_t    intptr;
00088 typedef uintptr_t   cpuaddr;
00089 typedef size_t      cpusize;
00090 typedef ptrdiff_t   cpudiff;
00091
00092 #ifdef OSAL_OMIT_DEPRECATED
00093     typedef struct
00094     {
00095         uint32_t v;
00096     } osal_id_t;
00097 #else
00098     typedef uint32_t osal_id_t;
00099 #endif
00100
00101 typedef size_t osal_blockcount_t;
00102
00103 typedef long osal_offset_t;
00104
00105 typedef uint32_t osal_index_t;
00106
00107 typedef uint32_t osal_objtype_t;
00108
00109 typedef int32_t osal_status_t;
00110
00111 typedef void (*OS_ArgCallback_t)(osal_id_t object_id, void *arg);
00112
00113 /*
00114  ** Check Sizes
00115  */
00116 CompileTimeAssert(sizeof(uint8) == 1, TypeUint8WrongSize);
00117 CompileTimeAssert(sizeof(uint16) == 2, TypeUint16WrongSize);
00118 CompileTimeAssert(sizeof(uint32) == 4, TypeUint32WrongSize);
00119 CompileTimeAssert(sizeof(uint64) == 8, TypeUint64WrongSize);
00120 CompileTimeAssert(sizeof(int8) == 1, Typeint8WrongSize);
00121 CompileTimeAssert(sizeof(int16) == 2, Typeint16WrongSize);
00122 CompileTimeAssert(sizeof(int32) == 4, Typeint32WrongSize);
00123 CompileTimeAssert(sizeof(int64) == 8, Typeint64WrongSize);
00124 CompileTimeAssert(sizeof(cpuaddr) >= sizeof(void *), TypePtrWrongSize);
00125
00126 #ifdef __cplusplus
00127 }
00128 #endif
00129
00130 /*
00131  * Type macros for literals
00132  */

```

```

00172 * These macros enforce that a literal or other value is
00173 * interpreted as the intended type. Although implicit
00174 * conversions between these types are often possible, using
00175 * this makes it explicit in the code where a type conversion
00176 * is expected.
00177 */
00178 #define OSAL_SIZE_C(X) ((size_t)(X))
00179 #define OSAL_BLOCKCOUNT_C(X) ((osal_blockcount_t)(X))
00180 #define OSAL_INDEX_C(X) ((osal_index_t)(X))
00181 #define OSAL_OBJTYPE_C(X) ((osal_objtype_t)(X))
00182 #define OSAL_STATUS_C(X) ((osal_status_t)(X))
00183
00184 #endif /* COMMON_TYPES_H */

```

11.290 osal/src/os/inc/osapi-binsem.h File Reference

```

#include "osconfig.h"
#include "common_types.h"

```

Data Structures

- struct [OS_bin_sem_prop_t](#)
OSAL binary semaphore properties.

Macros

- #define [OS_SEM_FULL](#) 1
Semaphore full state.
- #define [OS_SEM_EMPTY](#) 0
Semaphore empty state.

Functions

- [int32 OS_BinSemCreate](#) ([osal_id_t](#) *sem_id, const char *sem_name, [uint32](#) sem_initial_value, [uint32](#) options)
Creates a binary semaphore.
- [int32 OS_BinSemFlush](#) ([osal_id_t](#) sem_id)
Unblock all tasks pending on the specified semaphore.
- [int32 OS_BinSemGive](#) ([osal_id_t](#) sem_id)
Increment the semaphore value.
- [int32 OS_BinSemTake](#) ([osal_id_t](#) sem_id)
Decrement the semaphore value.
- [int32 OS_BinSemTimedWait](#) ([osal_id_t](#) sem_id, [uint32](#) msecs)
Decrement the semaphore value with a timeout.
- [int32 OS_BinSemDelete](#) ([osal_id_t](#) sem_id)
Deletes the specified Binary Semaphore.
- [int32 OS_BinSemGetIdByName](#) ([osal_id_t](#) *sem_id, const char *sem_name)
Find an existing semaphore ID by name.
- [int32 OS_BinSemGetInfo](#) ([osal_id_t](#) sem_id, [OS_bin_sem_prop_t](#) *bin_prop)
Fill a property object buffer with details regarding the resource.

11.290.1 Detailed Description

Declarations and prototypes for binary semaphores
Definition in file [osapi-binsem.h](#).

11.291 osapi-binsem.h

[Go to the documentation of this file.](#)

```

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00016  * limitations under the License.
00017  *****/
00018
00025 #ifndef OSAPI_BINSEM_H
00026 #define OSAPI_BINSEM_H
00027
00028 #include "osconfig.h"
00029 #include "common_types.h"
00030
00034 #define OS_SEM_FULL 1
00035 #define OS_SEM_EMPTY 0
00039 typedef struct
00040 {
00041     char      name[OS_MAX_API_NAME];
00042     osal_id_t creator;
00043     int32     value;
00044 } OS_bin_sem_prop_t;
00045
00050 /*-----*/
00071 int32 OS_BinSemCreate(osal_id_t *sem_id, const char *sem_name, uint32 sem_initial_value, uint32 options);
00072
00073 /*-----*/
00087 int32 OS_BinSemFlush(osal_id_t sem_id);
00088
00089 /*-----*/
00106 int32 OS_BinSemGive(osal_id_t sem_id);
00107
00108 /*-----*/
00125 int32 OS_BinSemTake(osal_id_t sem_id);
00126
00127 /*-----*/
00145 int32 OS_BinSemTimedWait(osal_id_t sem_id, uint32 msecs);
00146
00147 /*-----*/
00161 int32 OS_BinSemDelete(osal_id_t sem_id);
00162
00163 /*-----*/
00179 int32 OS_BinSemGetIdByName(osal_id_t *sem_id, const char *sem_name);
00180
00181 /*-----*/
00198 int32 OS_BinSemGetInfo(osal_id_t sem_id, OS_bin_sem_prop_t *bin_prop);
00199
00202 #endif /* OSAPI_BINSEM_H */

```

11.292 osal/src/os/inc/osapi-bsp.h File Reference

```

#include "osconfig.h"
#include "common_types.h"

```

Functions

- void [OS_BSP_SetResourceTypeConfig](#) (uint32 ResourceType, uint32 ConfigOptionValue)
- uint32 [OS_BSP_GetResourceTypeConfig](#) (uint32 ResourceType)
- uint32 [OS_BSP_GetArgC](#) (void)
- char *const * [OS_BSP_GetArgV](#) (void)

- void `OS_BSP_SetExitCode` (int32 code)

11.292.1 Detailed Description

Declarations and prototypes for OSAL BSP

Definition in file `osapi-bsp.h`.

11.293 osapi-bsp.h

[Go to the documentation of this file.](#)

```

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00016  * limitations under the License.
00017  *****/
00018
00025 #ifndef OSAPI_BSP_H
00026 #define OSAPI_BSP_H
00027
00028 #include "osconfig.h"
00029 #include "common_types.h"
00030
00031 /*****
00032  *
00033  *      BSP LOW-LEVEL IMPLEMENTATION FUNCTIONS
00034  *
00035  *****/
00045 /*-----*/
00046
00047     Purpose: Sets BSP/platform-specific flags for the given resource type
00048             Flags and bit meanings are all platform defined.
00049     -----*/
00050 void OS_BSP_SetResourceTypeConfig(uint32 ResourceType, uint32 ConfigOptionValue);
00051
00052 /*-----*/
00053
00054     Purpose: Gets BSP/platform-specific flags for the given resource type
00055             Flags and bit meanings are all platform defined.
00056     -----*/
00057 uint32 OS_BSP_GetResourceTypeConfig(uint32 ResourceType);
00058
00059 /*-----*/
00060
00061     Purpose: Obtain the number of boot arguments passed from the bootloader
00062             or shell if supported by the platform
00063
00064     Returns: The number of boot arguments, or 0 if no arguments were passed
00065             or not supported by the BSP.
00066     -----*/
00067 uint32 OS_BSP_GetArgC(void);
00068
00069 /*-----*/
00070
00071     Purpose: Obtain an array of boot argument strings passed from the bootloader
00072             or shell if supported by the platform
00073
00074     Returns: Pointer to char* array containing the argument strings, or NULL if
00075             no arguments are available or not supported by the BSP.
00076
00077             The array is sized according to OS_BSP_GetArgC()
00078     -----*/
00079 char *const *OS_BSP_GetArgV(void);
00080
00081 /*-----*/
00082
00083     Purpose: Sets the status to be returned to the shell or bootloader

```

```

00084         if supported by the platform. The value is an integer with
00085         platform and application-defined meaning, but BSP's should
00086         attempt to provide consistent meaning for the following values
00087
00088         OS_SUCCESS: normal status (default)
00089         OS_ERROR: any abnormal status
00090
00091         Other more specific status values may be passed, with
00092         implementation-defined behavior. Depending on the system
00093         capabilities, the BSP implementation may either pass the
00094         value through as-is, translate it to defined value, or
00095         ignore it.
00096
00097         Note this does NOT cause the application to exit, it only
00098         sets the state that will be returned if/when the application
00099         exits itself at a future time.
00100
00101         -----*/
00102 void OS_BSP_SetExitCode(int32 code);
00103
00106 #endif /* OSAPI_BSP_H */

```

11.294 osal/src/os/inc/osapi-clock.h File Reference

```

#include "osconfig.h"
#include "common_types.h"

```

Data Structures

- struct [OS_time_t](#)
OSAL time interval structure.

Macros

- #define [OS_TIME_MAX](#) (([OS_time_t](#)) { INT64_MAX })
The maximum value for [OS_time_t](#).
- #define [OS_TIME_ZERO](#) (([OS_time_t](#)) { 0 })
The zero value for [OS_time_t](#).
- #define [OS_TIME_MIN](#) (([OS_time_t](#)) { INT64_MIN })
The minimum value for [OS_time_t](#).

Enumerations

- enum { [OS_TIME_TICK_RESOLUTION_NS](#) = 100 , [OS_TIME_TICKS_PER_SECOND](#) = 1000000000 / [OS_TIME_TICK_RESOLUTION_NS](#) , [OS_TIME_TICKS_PER_MSEC](#) = 1000000 / [OS_TIME_TICK_RESOLUTION_NS](#) , [OS_TIME_TICKS_PER_USEC](#) = 1000 / [OS_TIME_TICK_RESOLUTION_NS](#) }
Multipliers/divisors to convert ticks into standardized units.

Functions

- [int32 OS_GetMonotonicTime](#) ([OS_time_t](#) *time_struct)
Get the monotonic time.
- [int32 OS_GetLocalTime](#) ([OS_time_t](#) *time_struct)
Get the local time.
- [int32 OS_SetLocalTime](#) (const [OS_time_t](#) *time_struct)
Set the local time.
- [OS_time_t OS_TimeFromRelativeMilliseconds](#) ([int32](#) relative_msec)
Gets an absolute time value relative to the current time.

- [int32 OS_TimeToRelativeMilliseconds](#) ([OS_time_t](#) time)
Gets a relative time value from an absolute time.
- static [int64 OS_TimeGetTotalSeconds](#) ([OS_time_t](#) tm)
Get interval from an [OS_time_t](#) object normalized to whole number of seconds.
- static [OS_time_t OS_TimeFromTotalSeconds](#) ([int64](#) tm)
Get an [OS_time_t](#) interval object from an integer number of seconds.
- static [int64 OS_TimeGetTotalMilliseconds](#) ([OS_time_t](#) tm)
Get interval from an [OS_time_t](#) object normalized to millisecond units.
- static [OS_time_t OS_TimeFromTotalMilliseconds](#) ([int64](#) tm)
Get an [OS_time_t](#) interval object from a integer number of milliseconds.
- static [int64 OS_TimeGetTotalMicroseconds](#) ([OS_time_t](#) tm)
Get interval from an [OS_time_t](#) object normalized to microsecond units.
- static [OS_time_t OS_TimeFromTotalMicroseconds](#) ([int64](#) tm)
Get an [OS_time_t](#) interval object from a integer number of microseconds.
- static [int64 OS_TimeGetTotalNanoseconds](#) ([OS_time_t](#) tm)
Get interval from an [OS_time_t](#) object normalized to nanosecond units.
- static [OS_time_t OS_TimeFromTotalNanoseconds](#) ([int64](#) tm)
Get an [OS_time_t](#) interval object from a integer number of nanoseconds.
- static [int64 OS_TimeGetFractionalPart](#) ([OS_time_t](#) tm)
Get subseconds portion (fractional part only) from an [OS_time_t](#) object.
- static [uint32 OS_TimeGetSubsecondsPart](#) ([OS_time_t](#) tm)
Get 32-bit normalized subseconds (fractional part only) from an [OS_time_t](#) object.
- static [uint32 OS_TimeGetMillisecondsPart](#) ([OS_time_t](#) tm)
Get milliseconds portion (fractional part only) from an [OS_time_t](#) object.
- static [uint32 OS_TimeGetMicrosecondsPart](#) ([OS_time_t](#) tm)
Get microseconds portion (fractional part only) from an [OS_time_t](#) object.
- static [uint32 OS_TimeGetNanosecondsPart](#) ([OS_time_t](#) tm)
Get nanoseconds portion (fractional part only) from an [OS_time_t](#) object.
- static [OS_time_t OS_TimeAssembleFromNanoseconds](#) ([int64](#) seconds, [uint32](#) nanoseconds)
Assemble/Convert a number of seconds + nanoseconds into an [OS_time_t](#) interval.
- static [OS_time_t OS_TimeAssembleFromMicroseconds](#) ([int64](#) seconds, [uint32](#) microseconds)
Assemble/Convert a number of seconds + microseconds into an [OS_time_t](#) interval.
- static [OS_time_t OS_TimeAssembleFromMilliseconds](#) ([int64](#) seconds, [uint32](#) milliseconds)
Assemble/Convert a number of seconds + milliseconds into an [OS_time_t](#) interval.
- static [OS_time_t OS_TimeAssembleFromSubseconds](#) ([int64](#) seconds, [uint32](#) subseconds)
Assemble/Convert a number of seconds + subseconds into an [OS_time_t](#) interval.
- static [OS_time_t OS_TimeAdd](#) ([OS_time_t](#) time1, [OS_time_t](#) time2)
Computes the sum of two time intervals.
- static [OS_time_t OS_TimeSubtract](#) ([OS_time_t](#) time1, [OS_time_t](#) time2)
Computes the difference between two time intervals.
- static [bool OS_TimeEqual](#) ([OS_time_t](#) time1, [OS_time_t](#) time2)
Checks if two time values are equal.
- static [int8_t OS_TimeGetSign](#) ([OS_time_t](#) time)
Checks the sign of the time value.
- static [int8_t OS_TimeCompare](#) ([OS_time_t](#) time1, [OS_time_t](#) time2)
Compares two time values.

11.294.1 Detailed Description

Declarations and prototypes for osapi-clock module
Definition in file [osapi-clock.h](#).

11.294.2 Macro Definition Documentation

OS_TIME_MAX

```
#define OS_TIME_MAX ((OS_time_t) { INT64_MAX })
```

The maximum value for [OS_time_t](#).

This is the largest positive (future) time that is representable in an [OS_time_t](#) value.

Definition at line 56 of file [osapi-clock.h](#).

OS_TIME_MIN

```
#define OS_TIME_MIN ((OS_time_t) { INT64_MIN })
```

The minimum value for [OS_time_t](#).

This is the largest negative (past) time that is representable in an [OS_time_t](#) value.

Definition at line 71 of file [osapi-clock.h](#).

OS_TIME_ZERO

```
#define OS_TIME_ZERO ((OS_time_t) { 0 })
```

The zero value for [OS_time_t](#).

This is a reasonable initializer/placeholder value for an [OS_time_t](#)

Definition at line 63 of file [osapi-clock.h](#).

11.294.3 Enumeration Type Documentation

anonymous enum

```
anonymous enum
```

Multipliers/divisors to convert ticks into standardized units.

Various fixed conversion factor constants used by the conversion routines

A 100ns tick time allows max intervals of about +/- 14000 years in a 64-bit signed integer value.

Note

Applications should not directly use these values, but rather use conversion routines below to obtain standardized units (seconds/microseconds/etc).

Enumerator

OS_TIME_TICK_RESOLUTION_NS	
OS_TIME_TICKS_PER_SECOND	
OS_TIME_TICKS_PER_MSEC	
OS_TIME_TICKS_PER_USEC	

Definition at line 84 of file [osapi-clock.h](#).

11.295 osapi-clock.h

[Go to the documentation of this file.](#)

```
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```

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00016  * limitations under the License.
00017  *****/
00018
00025 #ifndef OSAPI_CLOCK_H
00026 #define OSAPI_CLOCK_H
00027
00028 #include "osconfig.h"
00029 #include "common_types.h"
00030
00045 typedef struct
00046 {
00047     int64 ticks;
00048 } OS_time_t;
00049
00056 #define OS_TIME_MAX ((OS_time_t) { INT64_MAX })
00057
00063 #define OS_TIME_ZERO ((OS_time_t) { 0 })
00064
00071 #define OS_TIME_MIN ((OS_time_t) { INT64_MIN })
00072
00084 enum
00085 {
00086     OS_TIME_TICK_RESOLUTION_NS = 100,
00087     OS_TIME_TICKS_PER_SECOND = 1000000000 / OS_TIME_TICK_RESOLUTION_NS,
00088     OS_TIME_TICKS_PER_MSEC = 1000000 / OS_TIME_TICK_RESOLUTION_NS,
00089     OS_TIME_TICKS_PER_USEC = 1000 / OS_TIME_TICK_RESOLUTION_NS
00090 };
00091
00096 /*-----*/
00126 int32 OS_GetMonotonicTime(OS_time_t *time_struct);
00127
00128 /*-----*/
00142 int32 OS_GetLocalTime(OS_time_t *time_struct);
00143
00144 /*-----*/
00158 int32 OS_SetLocalTime(const OS_time_t *time_struct);
00159
00160 /*-----*/
00175 OS_time_t OS_TimeFromRelativeMilliseconds(int32 relative_msec);
00176
00177 /*-----*/
00197 int32 OS_TimeToRelativeMilliseconds(OS_time_t time);
00198
00199 /*-----*/
00200 /*
00201  * Accessor / Unit Conversion routines for OS_time_t
00202  *
00203  * These routines allow the user to simply interpret OS_time_t intervals into
00204  * in normalized units of whole seconds, milliseconds, microseconds, or nanoseconds.
00205  */
00206 /*-----*/
00207
00208 /*-----*/
00225 static inline int64 OS_TimeGetTotalSeconds(OS_time_t tm)
00226 {
00227     return (tm.ticks / OS_TIME_TICKS_PER_SECOND);
00228 }
00229
00230 /*-----*/
00242 static inline OS_time_t OS_TimeFromTotalSeconds(int64 tm)
00243 {
00244     OS_time_t ostm = { tm * OS_TIME_TICKS_PER_SECOND };
00245     return ostm;
00246 }
00247
00248 /*-----*/
00259 static inline int64 OS_TimeGetTotalMilliseconds(OS_time_t tm)
00260 {

```



```

00261     return (tm.ticks / OS_TIME_TICKS_PER_MSEC);
00262 }
00263
00264 /*-----*/
00276 static inline OS_time_t OS_TimeFromTotalMilliseconds(int64 tm)
00277 {
00278     OS_time_t ostm = { tm * OS_TIME_TICKS_PER_MSEC };
00279     return ostm;
00280 }
00281
00282 /*-----*/
00293 static inline int64 OS_TimeGetTotalMicroseconds(OS_time_t tm)
00294 {
00295     return (tm.ticks / OS_TIME_TICKS_PER_USEC);
00296 }
00297
00298 /*-----*/
00310 static inline OS_time_t OS_TimeFromTotalMicroseconds(int64 tm)
00311 {
00312     /* SAD: Overflow is not considered a concern because tm would need to be over 29,227 years in
00313        microseconds */
00314     OS_time_t ostm = { tm * OS_TIME_TICKS_PER_USEC };
00315     return ostm;
00316 }
00317 /*-----*/
00332 static inline int64 OS_TimeGetTotalNanoseconds(OS_time_t tm)
00333 {
00334     return (tm.ticks * OS_TIME_TICK_RESOLUTION_NS);
00335 }
00336
00337 /*-----*/
00349 static inline OS_time_t OS_TimeFromTotalNanoseconds(int64 tm)
00350 {
00351     OS_time_t ostm = { tm / OS_TIME_TICK_RESOLUTION_NS };
00352     return ostm;
00353 }
00354
00355 /*-----*/
00365 static inline int64 OS_TimeGetFractionalPart(OS_time_t tm)
00366 {
00367     return (tm.ticks % OS_TIME_TICKS_PER_SECOND);
00368 }
00369
00370 /*-----*/
00384 static inline uint32 OS_TimeGetSubsecondsPart(OS_time_t tm)
00385 {
00386     /*
00387      * This computation avoids a 32-bit left shift which may not be implemented.
00388      *
00389      * It also must round up, otherwise this may result in a value one
00390      * less than the original when converted back to usec again.
00391      */
00392     int64 frac = (OS_TimeGetFractionalPart(tm) << 30) + (OS_TIME_TICKS_PER_SECOND >> 2);
00393     return (uint32)((frac - 1) / (OS_TIME_TICKS_PER_SECOND >> 2));
00394 }
00395
00396 /*-----*/
00408 static inline uint32 OS_TimeGetMillisecondsPart(OS_time_t tm)
00409 {
00410     return (uint32)OS_TimeGetFractionalPart(tm) / OS_TIME_TICKS_PER_MSEC;
00411 }
00412
00413 /*-----*/
00433 static inline uint32 OS_TimeGetMicrosecondsPart(OS_time_t tm)
00434 {
00435     return (uint32)OS_TimeGetFractionalPart(tm) / OS_TIME_TICKS_PER_USEC;
00436 }
00437
00438 /*-----*/
00452 static inline uint32 OS_TimeGetNanosecondsPart(OS_time_t tm)
00453 {
00454     return (uint32)OS_TimeGetFractionalPart(tm) * OS_TIME_TICK_RESOLUTION_NS;
00455 }
00456
00457 /*-----*/
00473 static inline OS_time_t OS_TimeAssembleFromNanoseconds(int64 seconds, uint32 nanoseconds)
00474 {
00475     OS_time_t result;
00476     result.ticks = seconds * OS_TIME_TICKS_PER_SECOND;
00477     result.ticks += nanoseconds / OS_TIME_TICK_RESOLUTION_NS;

```

```

00478     return result;
00479 }
00480
00481 /*-----*/
00497 static inline OS_time_t OS_TimeAssembleFromMicroseconds(int64 seconds, uint32 microseconds)
00498 {
00499     OS_time_t result;
00500     result.ticks = seconds * OS_TIME_TICKS_PER_SECOND;
00501     result.ticks += microseconds * OS_TIME_TICKS_PER_USEC;
00502     return result;
00503 }
00504
00505 /*-----*/
00521 static inline OS_time_t OS_TimeAssembleFromMilliseconds(int64 seconds, uint32 milliseconds)
00522 {
00523     OS_time_t result;
00524     result.ticks = seconds * OS_TIME_TICKS_PER_SECOND;
00525     result.ticks += milliseconds * OS_TIME_TICKS_PER_MSEC;
00526     return result;
00527 }
00528
00529 /*-----*/
00544 static inline OS_time_t OS_TimeAssembleFromSubseconds(int64 seconds, uint32 subseconds)
00545 {
00546     OS_time_t result;
00547     result.ticks = seconds * OS_TIME_TICKS_PER_SECOND;
00548     /* this should not round in any way, as the 32-bit input value has higher precision */
00549     result.ticks += ((int64)subseconds * (OS_TIME_TICKS_PER_SECOND >> 2)) >> 30;
00550     return result;
00551 }
00552
00553 /*-----*/
00562 static inline OS_time_t OS_TimeAdd(OS_time_t time1, OS_time_t time2)
00563 {
00564     OS_time_t ostm = { time1.ticks + time2.ticks };
00565     return ostm;
00566 }
00567
00568 /*-----*/
00577 static inline OS_time_t OS_TimeSubtract(OS_time_t time1, OS_time_t time2)
00578 {
00579     OS_time_t ostm = { time1.ticks - time2.ticks };
00580     return ostm;
00581 }
00582
00583 /*-----*/
00593 static inline bool OS_TimeEqual(OS_time_t time1, OS_time_t time2)
00594 {
00595     return (time1.ticks == time2.ticks);
00596 }
00597
00598 /*-----*/
00608 static inline int8_t OS_TimeGetSign(OS_time_t time)
00609 {
00610     return (time.ticks > 0) - (time.ticks < 0);
00611 }
00612
00613 /*-----*/
00624 static inline int8_t OS_TimeCompare(OS_time_t time1, OS_time_t time2)
00625 {
00626     return OS_TimeGetSign(OS_TimeSubtract(time1, time2));
00627 }
00628
00631 #endif /* OSAPI_CLOCK_H */

```

11.296 osal/src/os/inc/osapi-common.h File Reference

```

#include "osconfig.h"
#include "common_types.h"

```

Typedefs

- typedef int32(* OS_EventHandler_t) (OS_Event_t event, osal_id_t object_id, void *data)
A callback routine for event handling.

Enumerations

- enum [OS_Event_t](#) {
[OS_EVENT_RESERVED](#) = 0 , [OS_EVENT_RESOURCE_ALLOCATED](#) , [OS_EVENT_RESOURCE_CREATED](#)
[OS_EVENT_RESOURCE_DELETED](#) ,
[OS_EVENT_TASK_STARTUP](#) , [OS_EVENT_MAX](#) }

A set of events that can be used with BSP event callback routines.

Functions

- void [OS_Application_Startup](#) (void)
Application startup.
- void [OS_Application_Run](#) (void)
Application run.
- int32 [OS_API_Init](#) (void)
Initialization of API.
- void [OS_API_Teardown](#) (void)
Teardown/de-initialization of OSAL API.
- void [OS_IdleLoop](#) (void)
Background thread implementation - waits forever for events to occur.
- void [OS_DeleteAllObjects](#) (void)
delete all resources created in OSAL.
- void [OS_ApplicationShutdown](#) (uint8 flag)
Initiate orderly shutdown.
- void [OS_ApplicationExit](#) (int32 Status)
Exit/Abort the application.
- int32 [OS_RegisterEventHandler](#) ([OS_EventHandler_t](#) handler)
Callback routine registration.
- size_t [OS_strlen](#) (const char *s, size_t maxlen)
get string length

11.296.1 Detailed Description

Declarations and prototypes for general OSAL functions that are not part of a subsystem
Definition in file [osapi-common.h](#).

11.296.2 Typedef Documentation

[OS_EventHandler_t](#)

typedef int32(* [OS_EventHandler_t](#)) ([OS_Event_t](#) event, [osal_id_t](#) object_id, void *data)

A callback routine for event handling.

Parameters

in	<i>event</i>	The event that occurred
in	<i>object↔ _id</i>	The associated object_id, or 0 if not associated with an object
in, out	<i>data</i>	An abstract data/context object associated with the event, or NULL.

Returns

status Execution status, see [OSAL Return Code Defines](#).

Definition at line 98 of file [osapi-common.h](#).

11.296.3 Enumeration Type Documentation**OS_Event_t**

enum [OS_Event_t](#)

A set of events that can be used with BSP event callback routines.

Enumerator

OS_EVENT_RESERVED	no-op/reserved event id value
OS_EVENT_RESOURCE_ALLOCATED	resource/id has been newly allocated but not yet created. This event is invoked from WITHIN the locked region, in the context of the task which is allocating the resource. If the handler returns non-success, the error will be returned to the caller and the creation process is aborted.
OS_EVENT_RESOURCE_CREATED	resource/id has been fully created/finalized. Invoked outside locked region, in the context of the task which created the resource. Data object is not used, passed as NULL. Return value is ignored - this is for information purposes only.
OS_EVENT_RESOURCE_DELETED	resource/id has been deleted. Invoked outside locked region, in the context of the task which deleted the resource. Data object is not used, passed as NULL. Return value is ignored - this is for information purposes only.
OS_EVENT_TASK_STARTUP	New task is starting. Invoked outside locked region, in the context of the task which is currently starting, before the entry point is called. Data object is not used, passed as NULL. If the handler returns non-success, task startup is aborted and the entry point is not called.
OS_EVENT_MAX	placeholder for end of enum, not used

Definition at line 34 of file [osapi-common.h](#).

11.297 osapi-common.h

[Go to the documentation of this file.](#)

```

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```

```

00017  *****/
00018
00025 #ifndef OSAPI_COMMON_H
00026 #define OSAPI_COMMON_H
00027
00028 #include "osconfig.h"
00029 #include "common_types.h"
00030
00034 typedef enum
00035 {
00036     OS_EVENT_RESERVED = 0,
00047     OS_EVENT_RESOURCE_ALLOCATED,
00048
00059     OS_EVENT_RESOURCE_CREATED,
00060
00071     OS_EVENT_RESOURCE_DELETED,
00072
00085     OS_EVENT_TASK_STARTUP,
00086
00087     OS_EVENT_MAX
00088 } OS_Event_t;
00089
00098 typedef int32 (*OS_EventHandler_t)(OS_Event_t event,osal_id_t object_id, void *data);
00099
00109 /*-----*/
00115 void OS_Application_Startup(void);
00116
00117 /*-----*/
00123 void OS_Application_Run(void);
00124
00125 /*-----*/
00139 int32 OS_API_Init(void);
00140
00141 /*-----*/
00164 void OS_API_Teardown(void);
00165
00166 /*-----*/
00175 void OS_IdleLoop(void);
00176
00177 /*-----*/
00185 void OS_DeleteAllObjects(void);
00186
00187 /*-----*/
00202 void OS_ApplicationShutdown(uint8 flag);
00203
00204 /*-----*/
00216 void OS_ApplicationExit(int32 Status);
00217
00218 /*-----*/
00234 int32 OS_RegisterEventHandler(OS_EventHandler_t handler);
00235
00236 /*-----*/
00249 size_t OS_strnlen(const char *s, size_t maxlen);
00250
00253 #endif /* OSAPI_COMMON_H */

```

11.298 osal/src/os/inc/osapi-condvar.h File Reference

```

#include "osconfig.h"
#include "common_types.h"
#include "osapi-clock.h"

```

Data Structures

- struct [OS_condvar_prop_t](#)
OSAL condition variable properties.

Functions

- [int32 OS_CondVarCreate](#) ([osal_id_t](#) *var_id, const char *var_name, [uint32](#) options)
Creates a condition variable resource.

- `int32 OS_CondVarLock (osal_id_t var_id)`
Locks/Acquires the underlying mutex associated with a condition variable.
- `int32 OS_CondVarUnlock (osal_id_t var_id)`
Unlocks/Releases the underlying mutex associated with a condition variable.
- `int32 OS_CondVarSignal (osal_id_t var_id)`
Signals the condition variable resource referenced by var_id.
- `int32 OS_CondVarBroadcast (osal_id_t var_id)`
Broadcasts the condition variable resource referenced by var_id.
- `int32 OS_CondVarWait (osal_id_t var_id)`
Waits on the condition variable object referenced by var_id.
- `int32 OS_CondVarTimedWait (osal_id_t var_id, const OS_time_t *abs_wakeup_time)`
Time-limited wait on the condition variable object referenced by var_id.
- `int32 OS_CondVarDelete (osal_id_t var_id)`
Deletes the specified condition variable.
- `int32 OS_CondVarGetIdByName (osal_id_t *var_id, const char *var_name)`
Find an existing condition variable ID by name.
- `int32 OS_CondVarGetInfo (osal_id_t var_id, OS_condvar_prop_t *condvar_prop)`
Fill a property object buffer with details regarding the resource.

11.298.1 Detailed Description

Declarations and prototypes for condition variables

Definition in file `osapi-condvar.h`.

11.299 osapi-condvar.h

[Go to the documentation of this file.](#)

```

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00017  *****/
00018
00025 #ifndef OSAPI_CONDVAR_H
00026 #define OSAPI_CONDVAR_H
00027
00028 #include "osconfig.h"
00029 #include "common_types.h"
00030
00031 #include "osapi-clock.h" /* required for OS_time_t definition */
00032
00034 typedef struct
00035 {
00036     char      name[OS_MAX_API_NAME];
00037     osal_id_t creator;
00038 } OS_condvar_prop_t;
00039
00044 /*-----*/
00083 int32 OS_CondVarCreate(osal_id_t *var_id, const char *var_name, uint32 options);
00084
00085 /*-----*/
00101 int32 OS_CondVarLock(osal_id_t var_id);

```

```

00102
00103 /*-----*/
00116 int32 OS_CondVarUnlock(osal_id_t var_id);
00117
00118 /*-----*/
00135 int32 OS_CondVarSignal(osal_id_t var_id);
00136
00137 /*-----*/
00158 int32 OS_CondVarBroadcast(osal_id_t var_id);
00159
00160 /*-----*/
00186 int32 OS_CondVarWait(osal_id_t var_id);
00187
00188 /*-----*/
00206 int32 OS_CondVarTimedWait(osal_id_t var_id, const OS_time_t *abs_wakeup_time);
00207
00208 /*-----*/
00220 int32 OS_CondVarDelete(osal_id_t var_id);
00221
00222 /*-----*/
00238 int32 OS_CondVarGetIdByName(osal_id_t *var_id, const char *var_name);
00239
00240 /*-----*/
00255 int32 OS_CondVarGetInfo(osal_id_t var_id, OS_condvar_prop_t *condvar_prop);
00258 #endif /* OSAPI_CONDVAR_H */

```

11.300 osal/src/os/inc/osapi-constants.h File Reference

```

#include "osconfig.h"
#include "common_types.h"

```

Macros

- `#define OS_PEND (-1)`
- `#define OS_CHECK (0)`
- `#define OS_OBJECT_ID_UNDEFINED ((osal_id_t) { 0 })`
Initializer for the osal_id_t type which will not match any valid value.
- `#define OS_OBJECT_CREATOR_ANY OS_OBJECT_ID_UNDEFINED`
Constant that may be passed to [OS_ForEachObject\(\)](#)/[OS_ForEachObjectOfType\(\)](#) to match any creator (i.e. get all objects)
- `#define OS_MAX_LOCAL_PATH_LEN (OS_MAX_PATH_LEN + OS_FS_PHYS_NAME_LEN)`
Maximum length of a local/native path name string.

11.300.1 Detailed Description

General constants for OSAL that are shared across subsystems

Definition in file [osapi-constants.h](#).

11.300.2 Macro Definition Documentation

OS_CHECK

```
#define OS_CHECK (0)
```

Definition at line 35 of file [osapi-constants.h](#).

OS_MAX_LOCAL_PATH_LEN

```
#define OS_MAX_LOCAL_PATH_LEN (OS_MAX_PATH_LEN + OS_FS_PHYS_NAME_LEN)
```

Maximum length of a local/native path name string.

This is a concatenation of the OSAL virtual path with the system mount point or device name

Definition at line 54 of file [osapi-constants.h](#).

OS_OBJECT_CREATOR_ANY

```
#define OS_OBJECT_CREATOR_ANY OS_OBJECT_ID_UNDEFINED
```

Constant that may be passed to [OS_ForEachObject\(\)](#)/[OS_ForEachObjectOfType\(\)](#) to match any creator (i.e. get all objects)

Definition at line 46 of file [osapi-constants.h](#).

OS_OBJECT_ID_UNDEFINED

```
#define OS_OBJECT_ID_UNDEFINED ((osal_id_t) { 0 })
```

Initializer for the `osal_id_t` type which will not match any valid value.

Definition at line 40 of file [osapi-constants.h](#).

OS_PEND

```
#define OS_PEND (-1)
```

Definition at line 34 of file [osapi-constants.h](#).

11.301 osapi-constants.h

[Go to the documentation of this file.](#)

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00017  *****/
00018
00025 #ifndef OSAPI_CONSTANTS_H
00026 #define OSAPI_CONSTANTS_H
00027
00028 #include "osconfig.h"
00029 #include "common_types.h"
00030
00031 /*
00032 ** Defines for Queue Timeout parameters
00033 */
00034 #define OS_PEND (-1)
00035 #define OS_CHECK (0)
00036
00040 #define OS_OBJECT_ID_UNDEFINED ((osal_id_t) { 0 })
00041
00046 #define OS_OBJECT_CREATOR_ANY OS_OBJECT_ID_UNDEFINED
00047
00054 #define OS_MAX_LOCAL_PATH_LEN (OS_MAX_PATH_LEN + OS_FS_PHYS_NAME_LEN)
00055
00056 #endif /* OSAPI_CONSTANTS_H */
```

11.302 osal/src/os/inc/osapi-countsem.h File Reference

```
#include "osconfig.h"
#include "common_types.h"
```

Data Structures

- struct [OS_count_sem_prop_t](#)

OSAL counting semaphore properties.

Functions

- `int32 OS_CountSemCreate (osal_id_t *sem_id, const char *sem_name, uint32 sem_initial_value, uint32 options)`
Creates a counting semaphore.
- `int32 OS_CountSemGive (osal_id_t sem_id)`
Increment the semaphore value.
- `int32 OS_CountSemTake (osal_id_t sem_id)`
Decrement the semaphore value.
- `int32 OS_CountSemTimedWait (osal_id_t sem_id, uint32 msecs)`
Decrement the semaphore value with timeout.
- `int32 OS_CountSemDelete (osal_id_t sem_id)`
Deletes the specified counting Semaphore.
- `int32 OS_CountSemGetIdByName (osal_id_t *sem_id, const char *sem_name)`
Find an existing semaphore ID by name.
- `int32 OS_CountSemGetInfo (osal_id_t sem_id, OS_count_sem_prop_t *count_prop)`
Fill a property object buffer with details regarding the resource.

11.302.1 Detailed Description

Declarations and prototypes for counting semaphores

Definition in file [osapi-countsem.h](#).

11.303 osapi-countsem.h

[Go to the documentation of this file.](#)

```

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00017  *****/
00018
00025 #ifndef OSAPI_COUNTSEM_H
00026 #define OSAPI_COUNTSEM_H
00027
00028 #include "osconfig.h"
00029 #include "common_types.h"
00030
00032 typedef struct
00033 {
00034     char      name[OS_MAX_API_NAME];
00035     osal_id_t creator;
00036     int32     value;
00037 } OS_count_sem_prop_t;
00038
00043 /*-----*/
00073 int32 OS_CountSemCreate(osal_id_t *sem_id, const char *sem_name, uint32 sem_initial_value, uint32
options);
00074
00075 /*-----*/
00092 int32 OS_CountSemGive(osal_id_t sem_id);
00093

```

```

00094 /*-----*/
00111 int32 OS_CountSemTake(osal_id_t sem_id);
00112
00113 /*-----*/
00131 int32 OS_CountSemTimedWait(osal_id_t sem_id, uint32 msec);
00132
00133 /*-----*/
00144 int32 OS_CountSemDelete(osal_id_t sem_id);
00145
00146 /*-----*/
00162 int32 OS_CountSemGetIdByName(osal_id_t *sem_id, const char *sem_name);
00163
00164 /*-----*/
00181 int32 OS_CountSemGetInfo(osal_id_t sem_id, OS_count_sem_prop_t *count_prop);
00182
00185 #endif /* OSAPI_COUNTSEM_H */

```

11.304 osal/src/os/inc/osapi-dir.h File Reference

```

#include "osconfig.h"
#include "common_types.h"

```

Data Structures

- struct [os_dirent_t](#)
Directory entry.

Macros

- #define [OS_DIRENTRY_NAME\(x\)](#) ((x).FileName)
Access filename part of the dirent structure.

Functions

- [int32 OS_DirectoryOpen](#) (osal_id_t *dir_id, const char *path)
Opens a directory.
- [int32 OS_DirectoryClose](#) (osal_id_t dir_id)
Closes an open directory.
- [int32 OS_DirectoryRewind](#) (osal_id_t dir_id)
Rewinds an open directory.
- [int32 OS_DirectoryRead](#) (osal_id_t dir_id, [os_dirent_t](#) *dirent)
Reads the next name in the directory.
- [int32 OS_mkdir](#) (const char *path, [uint32](#) access)
Makes a new directory.
- [int32 OS_rmdir](#) (const char *path)
Removes a directory from the file system.

11.304.1 Detailed Description

Declarations and prototypes for directories
Definition in file [osapi-dir.h](#).

11.304.2 Macro Definition Documentation

OS_DIRENTRY_NAME

```
#define OS_DIRENTRY_NAME(  
    x ) ((x).FileName)
```

Access filename part of the dirent structure.

Definition at line 38 of file [osapi-dir.h](#).

11.305 osapi-dir.h

[Go to the documentation of this file.](#)

```
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00017  *****/  
00018  
00025 #ifndef OSAPI_DIR_H  
00026 #define OSAPI_DIR_H  
00027  
00028 #include "osconfig.h"  
00029 #include "common_types.h"  
00030  
00032 typedef struct  
00033 {  
00034     char FileName[OS_MAX_FILE_NAME];  
00035 } os_dirent_t;  
00036  
00038 #define OS_DIRENTRY_NAME(x) ((x).FileName)  
00039  
00040 /*  
00041  * Exported Functions  
00042  */  
00043  
00048 /*-----*/  
00064 int32 OS_DirectoryOpen(osal_id_t *dir_id, const char *path);  
00065  
00066 /*-----*/  
00078 int32 OS_DirectoryClose(osal_id_t dir_id);  
00079  
00080 /*-----*/  
00092 int32 OS_DirectoryRewind(osal_id_t dir_id);  
00093  
00094 /*-----*/  
00109 int32 OS_DirectoryRead(osal_id_t dir_id, os_dirent_t *dirent);  
00110  
00111 /*-----*/  
00131 int32 OS_mkdir(const char *path, uint32 access);  
00132  
00133 /*-----*/  
00149 int32 OS_rmdir(const char *path);  
00152 #endif /* OSAPI_DIR_H */
```

11.306 osal/src/os/inc/osapi-error.h File Reference

```
#include "common_types.h"
```

Macros

- #define `OS_ERROR_NAME_LENGTH` 35
Error string name length.
- #define `OS_STATUS_STRING_LENGTH` 12
Status converted to string length limit.
- #define `OS_SUCCESS` (0)
Successful execution.
- #define `OS_ERROR` (-1)
Failed execution.
- #define `OS_INVALID_POINTER` (-2)
Invalid pointer.
- #define `OS_ERROR_ADDRESS_MISALIGNED` (-3)
Address misalignment.
- #define `OS_ERROR_TIMEOUT` (-4)
Error timeout.
- #define `OS_INVALID_INT_NUM` (-5)
Invalid Interrupt number.
- #define `OS_SEM_FAILURE` (-6)
Semaphore failure.
- #define `OS_SEM_TIMEOUT` (-7)
Semaphore timeout.
- #define `OS_QUEUE_EMPTY` (-8)
Queue empty.
- #define `OS_QUEUE_FULL` (-9)
Queue full.
- #define `OS_QUEUE_TIMEOUT` (-10)
Queue timeout.
- #define `OS_QUEUE_INVALID_SIZE` (-11)
Queue invalid size.
- #define `OS_QUEUE_ID_ERROR` (-12)
Queue ID error.
- #define `OS_ERR_NAME_TOO_LONG` (-13)
name length including null terminator greater than `OS_MAX_API_NAME`
- #define `OS_ERR_NO_FREE_IDS` (-14)
No free IDs.
- #define `OS_ERR_NAME_TAKEN` (-15)
Name taken.
- #define `OS_ERR_INVALID_ID` (-16)
Invalid ID.
- #define `OS_ERR_NAME_NOT_FOUND` (-17)
Name not found.
- #define `OS_ERR_SEM_NOT_FULL` (-18)
Semaphore not full.
- #define `OS_ERR_INVALID_PRIORITY` (-19)
Invalid priority.
- #define `OS_INVALID_SEM_VALUE` (-20)
Invalid semaphore value.

- `#define OS_ERR_FILE` (-27)
File error.
- `#define OS_ERR_NOT_IMPLEMENTED` (-28)
Not implemented.
- `#define OS_TIMER_ERR_INVALID_ARGS` (-29)
Timer invalid arguments.
- `#define OS_TIMER_ERR_TIMER_ID` (-30)
Timer ID error.
- `#define OS_TIMER_ERR_UNAVAILABLE` (-31)
Timer unavailable.
- `#define OS_TIMER_ERR_INTERNAL` (-32)
Timer internal error.
- `#define OS_ERR_OBJECT_IN_USE` (-33)
Object in use.
- `#define OS_ERR_BAD_ADDRESS` (-34)
Bad address.
- `#define OS_ERR_INCORRECT_OBJ_STATE` (-35)
Incorrect object state.
- `#define OS_ERR_INCORRECT_OBJ_TYPE` (-36)
Incorrect object type.
- `#define OS_ERR_STREAM_DISCONNECTED` (-37)
Stream disconnected.
- `#define OS_ERR_OPERATION_NOT_SUPPORTED` (-38)
Requested operation not support on supplied object(s)
- `#define OS_ERR_INVALID_SIZE` (-40)
Invalid Size.
- `#define OS_ERR_OUTPUT_TOO_LARGE` (-41)
Size of output exceeds limit
- `#define OS_ERR_INVALID_ARGUMENT` (-42)
Invalid argument value (other than ID or size)
- `#define OS_ERR_TRY_AGAIN` (-43)
Failure is temporary in nature, call may be repeated.
- `#define OS_ERR_EMPTY_SET` (-44)
Address or name lookup returned no results.
- `#define OS_FS_ERR_PATH_TOO_LONG` (-103)
FS path too long.
- `#define OS_FS_ERR_NAME_TOO_LONG` (-104)
FS name too long.
- `#define OS_FS_ERR_DRIVE_NOT_CREATED` (-106)
FS drive not created.
- `#define OS_FS_ERR_DEVICE_NOT_FREE` (-107)
FS device not free.
- `#define OS_FS_ERR_PATH_INVALID` (-108)
FS path invalid.

Typedefs

- typedef char [os_err_name_t](#)[[OS_ERROR_NAME_LENGTH](#)]
For the [OS_GetErrorName\(\)](#) function, to ensure everyone is making an array of the same length.
- typedef char [os_status_string_t](#)[[OS_STATUS_STRING_LENGTH](#)]
For the [OS_StatusToString\(\)](#) function, to ensure everyone is making an array of the same length.

Functions

- static long [OS_StatusToInteger](#) ([osal_status_t](#) Status)
Convert a status code to a native "long" type.
- [int32 OS_GetErrorName](#) ([int32](#) error_num, [os_err_name_t](#) *err_name)
Convert an error number to a string.
- char * [OS_StatusToString](#) ([osal_status_t](#) status, [os_status_string_t](#) *status_string)
Convert status to a string.

11.306.1 Detailed Description

OSAL error code definitions
Definition in file [osapi-error.h](#).

11.306.2 Macro Definition Documentation

OS_ERROR_NAME_LENGTH

```
#define OS_ERROR_NAME_LENGTH 35
```

Error string name length.

The sizes of strings in OSAL functions are built with this limit in mind. Always check the uses of [os_err_name_t](#) when changing this value.

Definition at line 35 of file [osapi-error.h](#).

OS_STATUS_STRING_LENGTH

```
#define OS_STATUS_STRING_LENGTH 12
```

Status converted to string length limit.

Used for sizing [os_status_string_t](#) intended for use in printing [osal_status_t](#) values Sized to fit LONG_MIN including NULL termination

Definition at line 55 of file [osapi-error.h](#).

11.306.3 Typedef Documentation

os_err_name_t

```
typedef char os_err_name_t[OS_ERROR_NAME_LENGTH]
```

For the [OS_GetErrorName\(\)](#) function, to ensure everyone is making an array of the same length.

Implementation note for developers:

The sizes of strings in OSAL functions are built with this [OS_ERROR_NAME_LENGTH](#) limit in mind. Always check the uses of [os_err_name_t](#) when changing this value.

Definition at line 47 of file [osapi-error.h](#).

os_status_string_t

```
typedef char os_status_string_t[OS_STATUS_STRING_LENGTH]
```

For the [OS_StatusToString\(\)](#) function, to ensure everyone is making an array of the same length.

Definition at line 61 of file [osapi-error.h](#).

11.307 osapi-error.h

[Go to the documentation of this file.](#)

```

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00017  *****/
00018
00025 #ifndef OSAPI_ERROR_H
00026 #define OSAPI_ERROR_H
00027
00028 #include "common_types.h"
00029
00035 #define OS_ERROR_NAME_LENGTH 35
00036
00047 typedef char os_err_name_t[OS_ERROR_NAME_LENGTH];
00048
00055 #define OS_STATUS_STRING_LENGTH 12
00056
00061 typedef char os_status_string_t[OS_STATUS_STRING_LENGTH];
00062
00096 #define OS_SUCCESS (0)
00097 #define OS_ERROR (-1)
00098 #define OS_INVALID_POINTER (-2)
00099 #define OS_ERROR_ADDRESS_MISALIGNED (-3)
00100 #define OS_ERROR_TIMEOUT (-4)
00101 #define OS_INVALID_INT_NUM (-5)
00102 #define OS_SEM_FAILURE (-6)
00103 #define OS_SEM_TIMEOUT (-7)
00104 #define OS_QUEUE_EMPTY (-8)
00105 #define OS_QUEUE_FULL (-9)
00106 #define OS_QUEUE_TIMEOUT (-10)
00107 #define OS_QUEUE_INVALID_SIZE (-11)
00108 #define OS_QUEUE_ID_ERROR (-12)
00109 #define OS_ERR_NAME_TOO_LONG (-13)
00110 #define OS_ERR_NO_FREE_IDS (-14)
00111 #define OS_ERR_NAME_TAKEN (-15)
00112 #define OS_ERR_INVALID_ID (-16)
00113 #define OS_ERR_NAME_NOT_FOUND (-17)
00114 #define OS_ERR_SEM_NOT_FULL (-18)
00115 #define OS_ERR_INVALID_PRIORITY (-19)
00116 #define OS_INVALID_SEM_VALUE (-20)
00117 #define OS_ERR_FILE (-27)
00118 #define OS_ERR_NOT_IMPLEMENTED (-28)
00119 #define OS_TIMER_ERR_INVALID_ARGS (-29)
00120 #define OS_TIMER_ERR_TIMER_ID (-30)
00121 #define OS_TIMER_ERR_UNAVAILABLE (-31)
00122 #define OS_TIMER_ERR_INTERNAL (-32)
00123 #define OS_ERR_OBJECT_IN_USE (-33)
00124 #define OS_ERR_BAD_ADDRESS (-34)
00125 #define OS_ERR_INCORRECT_OBJ_STATE (-35)
00126 #define OS_ERR_INCORRECT_OBJ_TYPE (-36)
00127 #define OS_ERR_STREAM_DISCONNECTED (-37)
00128 #define OS_ERR_OPERATION_NOT_SUPPORTED (-38)
00129 #define OS_ERR_INVALID_SIZE (-40)
00130 #define OS_ERR_OUTPUT_TOO_LARGE (-41)
00131 #define OS_ERR_INVALID_ARGUMENT (-42)
00132 #define OS_ERR_TRY_AGAIN (-43)
00133 #define OS_ERR_EMPTY_SET (-44)
00135 /*
00136 ** Defines for File System Calls
00137 */
00138 /*
00139  * NOTE - these values used to overlap with the
00140  * other OSAPI error codes. They now start at -100
00141  * to avoid this overlap.
00142  */

```

```

00143 #define OS_FS_ERR_PATH_TOO_LONG      (-103)
00144 #define OS_FS_ERR_NAME_TOO_LONG      (-104)
00145 #define OS_FS_ERR_DRIVE_NOT_CREATED  (-106)
00146 #define OS_FS_ERR_DEVICE_NOT_FREE    (-107)
00147 #define OS_FS_ERR_PATH_INVALID       (-108)
00155 /*-----*/
00166 static inline long OS_StatusToInteger(osal_status_t Status)
00167 {
00168     return (long)Status;
00169 }
00170
00171 /*-----*/
00183 int32 OS_GetErrorName(int32 error_num, os_err_name_t *err_name);
00184
00185 /*-----*/
00194 char *OS_StatusToString(osal_status_t status, os_status_string_t *status_string);
00197 #endif /* OSAPI_ERROR_H */

```

11.308 osal/src/os/inc/osapi-file.h File Reference

```

#include "osconfig.h"
#include "common_types.h"
#include "osapi-clock.h"

```

Data Structures

- struct [OS_file_prop_t](#)
OSAL file properties.
- struct [os_fstat_t](#)
File system status.

Macros

- #define [OS_READ_ONLY](#) 0
- #define [OS_WRITE_ONLY](#) 1
- #define [OS_READ_WRITE](#) 2
- #define [OS_SEEK_SET](#) 0
- #define [OS_SEEK_CUR](#) 1
- #define [OS_SEEK_END](#) 2
- #define [OS_FILESTAT_MODE](#)(x) ((x).FileModeBits)
Access file stat mode bits.
- #define [OS_FILESTAT_ISDIR](#)(x) ((x).FileModeBits & [OS_FILESTAT_MODE_DIR](#))
File stat is directory logical.
- #define [OS_FILESTAT_EXEC](#)(x) ((x).FileModeBits & [OS_FILESTAT_MODE_EXEC](#))
File stat is executable logical.
- #define [OS_FILESTAT_WRITE](#)(x) ((x).FileModeBits & [OS_FILESTAT_MODE_WRITE](#))
File stat is write enabled logical.
- #define [OS_FILESTAT_READ](#)(x) ((x).FileModeBits & [OS_FILESTAT_MODE_READ](#))
File stat is read enabled logical.
- #define [OS_FILESTAT_SIZE](#)(x) ((x).FileSize)
Access file stat size field.
- #define [OS_FILESTAT_TIME](#)(x) ([OS_TimeGetTotalSeconds](#)((x).FileTime))
Access file stat time field as a whole number of seconds.

Enumerations

- enum { [OS_FILESTAT_MODE_EXEC](#) = 0x00001 , [OS_FILESTAT_MODE_WRITE](#) = 0x00002 , [OS_FILESTAT_MODE_READ](#) = 0x00004 , [OS_FILESTAT_MODE_DIR](#) = 0x10000 }

File stat mode bits.

- enum [OS_file_flag_t](#) { [OS_FILE_FLAG_NONE](#) = 0x00 , [OS_FILE_FLAG_CREATE](#) = 0x01 , [OS_FILE_FLAG_TRUNCATE](#) = 0x02 }

Flags that can be used with opening of a file (bitmask)

Functions

- [int32 OS_OpenCreate](#) ([osal_id_t](#) *filedes, const char *path, [int32](#) flags, [int32](#) access_mode)
Open or create a file.
- [int32 OS_close](#) ([osal_id_t](#) filedes)
Closes an open file handle.
- [int32 OS_read](#) ([osal_id_t](#) filedes, void *buffer, [size_t](#) nbytes)
Read from a file handle.
- [int32 OS_write](#) ([osal_id_t](#) filedes, const void *buffer, [size_t](#) nbytes)
Write to a file handle.
- [int32 OS_TimedReadAbs](#) ([osal_id_t](#) filedes, void *buffer, [size_t](#) nbytes, [OS_time_t](#) abstime)
File/Stream input read with a timeout.
- [int32 OS_TimedRead](#) ([osal_id_t](#) filedes, void *buffer, [size_t](#) nbytes, [int32](#) timeout)
File/Stream input read with a timeout.
- [int32 OS_TimedWriteAbs](#) ([osal_id_t](#) filedes, const void *buffer, [size_t](#) nbytes, [OS_time_t](#) abstime)
File/Stream output write with a timeout.
- [int32 OS_TimedWrite](#) ([osal_id_t](#) filedes, const void *buffer, [size_t](#) nbytes, [int32](#) timeout)
File/Stream output write with a timeout.
- [int32 OS_FileAllocate](#) ([osal_id_t](#) filedes, [osal_offset_t](#) offset, [osal_offset_t](#) len)
Pre-allocates space at the given file location.
- [int32 OS_FileTruncate](#) ([osal_id_t](#) filedes, [osal_offset_t](#) len)
Changes the size of the file.
- [int32 OS_chmod](#) (const char *path, [uint32](#) access_mode)
Changes the permissions of a file.
- [int32 OS_stat](#) (const char *path, [os_fstat_t](#) *filestats)
Obtain information about a file or directory.
- [int32 OS_lseek](#) ([osal_id_t](#) filedes, [osal_offset_t](#) offset, [uint32](#) whence)
Seeks to the specified position of an open file.
- [int32 OS_remove](#) (const char *path)
Removes a file from the file system.
- [int32 OS_rename](#) (const char *old_filename, const char *new_filename)
Renames a file.
- [int32 OS_cp](#) (const char *src, const char *dest)
Copies a single file from src to dest.
- [int32 OS_mv](#) (const char *src, const char *dest)
Move a single file from src to dest.
- [int32 OS_FDGetInfo](#) ([osal_id_t](#) filedes, [OS_file_prop_t](#) *fd_prop)
Obtain information about an open file.
- [int32 OS_FileOpenCheck](#) (const char *Filename)

Checks to see if a file is open.

- [int32 OS_CloseAllFiles](#) (void)

Close all open files.

- [int32 OS_CloseFileByName](#) (const char *Filename)

Close a file by filename.

11.308.1 Detailed Description

Declarations and prototypes for file objects

Definition in file [osapi-file.h](#).

11.308.2 Macro Definition Documentation

OS_FILESTAT_EXEC

```
#define OS_FILESTAT_EXEC(  
    x ) ((x).FileModeBits & OS_FILESTAT_MODE_EXEC)
```

File stat is executable logical.

Definition at line 92 of file [osapi-file.h](#).

OS_FILESTAT_ISDIR

```
#define OS_FILESTAT_ISDIR(  
    x ) ((x).FileModeBits & OS_FILESTAT_MODE_DIR)
```

File stat is directory logical.

Definition at line 90 of file [osapi-file.h](#).

OS_FILESTAT_MODE

```
#define OS_FILESTAT_MODE(  
    x ) ((x).FileModeBits)
```

Access file stat mode bits.

Definition at line 88 of file [osapi-file.h](#).

OS_FILESTAT_READ

```
#define OS_FILESTAT_READ(  
    x ) ((x).FileModeBits & OS_FILESTAT_MODE_READ)
```

File stat is read enabled logical.

Definition at line 96 of file [osapi-file.h](#).

OS_FILESTAT_SIZE

```
#define OS_FILESTAT_SIZE(  
    x ) ((x).FileSize)
```

Access file stat size field.

Definition at line 98 of file [osapi-file.h](#).

OS_FILESTAT_TIME

```
#define OS_FILESTAT_TIME(  
    x ) (OS_TimeGetTotalSeconds((x).FileTime))
```

Access file stat time field as a whole number of seconds.

Definition at line 100 of file [osapi-file.h](#).

OS_FILESTAT_WRITE

```
#define OS_FILESTAT_WRITE(
    x ) ((x).FileModeBits & OS_FILESTAT_MODE_WRITE)
```

File stat is write enabled logical.

Definition at line 94 of file [osapi-file.h](#).

11.308.3 Enumeration Type Documentation

anonymous enum

```
anonymous enum
```

File stat mode bits.

We must also define replacements for the stat structure's mode bits. This is currently just a small subset since the OSAL just presents a very simplified view of the filesystem to the upper layers. And since not all OS'es are POSIX, the more POSIX-specific bits are not relevant anyway.

Enumerator

OS_FILESTAT_MODE_EXEC	
OS_FILESTAT_MODE_WRITE	
OS_FILESTAT_MODE_READ	
OS_FILESTAT_MODE_DIR	

Definition at line 79 of file [osapi-file.h](#).

OS_file_flag_t

```
enum OS_file_flag_t
```

Flags that can be used with opening of a file (bitmask)

Enumerator

OS_FILE_FLAG_NONE	
OS_FILE_FLAG_CREATE	
OS_FILE_FLAG_TRUNCATE	

Definition at line 105 of file [osapi-file.h](#).

11.309 osapi-file.h

[Go to the documentation of this file.](#)

```
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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
```

```

00025 #ifndef OSAPI_FILE_H
00026 #define OSAPI_FILE_H
00027
00028 #include "osconfig.h"
00029 #include "common_types.h"
00030 #include "osapi-clock.h"
00031
00035 #define OS_READ_ONLY 0
00036 #define OS_WRITE_ONLY 1
00037 #define OS_READ_WRITE 2
00043 #define OS_SEEK_SET 0
00044 #define OS_SEEK_CUR 1
00045 #define OS_SEEK_END 2
00049 typedef struct
00050 {
00051     char      Path[OS_MAX_PATH_LEN];
00052     osal_id_t User;
00053     uint8      IsValid; /* For backward compatibility -- always true if OS_FDGetInfo returned true */
00054 } OS_file_prop_t;
00055
00064 typedef struct
00065 {
00066     uint32      FileModeBits;
00067     OS_time_t   FileTime;
00068     size_t      FileSize;
00069 } os_fstat_t;
00070
00079 enum
00080 {
00081     OS_FILESTAT_MODE_EXEC = 0x00001,
00082     OS_FILESTAT_MODE_WRITE = 0x00002,
00083     OS_FILESTAT_MODE_READ = 0x00004,
00084     OS_FILESTAT_MODE_DIR = 0x10000
00085 };
00086
00088 #define OS_FILESTAT_MODE(x) ((x).FileModeBits)
00090 #define OS_FILESTAT_ISDIR(x) ((x).FileModeBits & OS_FILESTAT_MODE_DIR)
00092 #define OS_FILESTAT_EXEC(x) ((x).FileModeBits & OS_FILESTAT_MODE_EXEC)
00094 #define OS_FILESTAT_WRITE(x) ((x).FileModeBits & OS_FILESTAT_MODE_WRITE)
00096 #define OS_FILESTAT_READ(x) ((x).FileModeBits & OS_FILESTAT_MODE_READ)
00098 #define OS_FILESTAT_SIZE(x) ((x).FileSize)
00100 #define OS_FILESTAT_TIME(x) (OS_TimeGetTotalSeconds((x).FileTime))
00101
00105 typedef enum
00106 {
00107     OS_FILE_FLAG_NONE = 0x00,
00108     OS_FILE_FLAG_CREATE = 0x01,
00109     OS_FILE_FLAG_TRUNCATE = 0x02
00110 } OS_file_flag_t;
00111
00112 /*
00113  * Exported Functions
00114  */
00115
00120 /*-----*/
00143 int32 OS_OpenCreate(osal_id_t *filedes, const char *path, int32 flags, int32 access_mode);
00144
00145 /*-----*/
00159 int32 OS_close(osal_id_t filedes);
00160
00161 /*-----*/
00184 int32 OS_read(osal_id_t filedes, void *buffer, size_t nbytes);
00185
00186 /*-----*/
00207 int32 OS_write(osal_id_t filedes, const void *buffer, size_t nbytes);
00208
00209 /*-----*/
00249 int32 OS_TimedReadAbs(osal_id_t filedes, void *buffer, size_t nbytes, OS_time_t abstime);
00250
00251 /*-----*/
00291 int32 OS_TimedRead(osal_id_t filedes, void *buffer, size_t nbytes, int32 timeout);
00292
00293 /*-----*/
00326 int32 OS_TimedWriteAbs(osal_id_t filedes, const void *buffer, size_t nbytes, OS_time_t abstime);
00327
00328 /*-----*/
00361 int32 OS_TimedWrite(osal_id_t filedes, const void *buffer, size_t nbytes, int32 timeout);
00362
00363 /*-----*/
00382 int32 OS_FileAllocate(osal_id_t filedes, osal_offset_t offset, osal_offset_t len);
00383

```

```

00384 /*-----*/
00403 int32 OS_FileTruncate(osal_id_t filedes, osal_offset_t len);
00404
00405 /*-----*/
00420 int32 OS_chmod(const char *path, uint32 access_mode);
00421
00422 /*-----*/
00439 int32 OS_stat(const char *path, os_fstat_t *filestats);
00440
00441 /*-----*/
00456 int32 OS_lseek(osal_id_t filedes, osal_offset_t offset, uint32 whence);
00457
00458 /*-----*/
00479 int32 OS_remove(const char *path);
00480
00481 /*-----*/
00504 int32 OS_rename(const char *old_filename, const char *new_filename);
00505
00506 /*-----*/
00526 int32 OS_cp(const char *src, const char *dest);
00527
00528 /*-----*/
00554 int32 OS_mv(const char *src, const char *dest);
00555
00556 /*-----*/
00570 int32 OS_FDGetInfo(osal_id_t filedes, OS_file_prop_t *fd_prop);
00571
00572 /*-----*/
00586 int32 OS_FileOpenCheck(const char *Filename);
00587
00588 /*-----*/
00598 int32 OS_CloseAllFiles(void);
00599
00600 /*-----*/
00616 int32 OS_CloseFileByName(const char *Filename);
00619 #endif /* OSAPI_FILE_H */

```

11.310 osal/src/os/inc/osapi-filesys.h File Reference

```

#include "osconfig.h"
#include "common_types.h"

```

Data Structures

- struct [os_fsinfo_t](#)
OSAL file system info.
- struct [OS_statvfs_t](#)

Macros

- #define [OS_CHK_ONLY](#) 0
- #define [OS_REPAIR](#) 1

Functions

- [int32 OS_FileSysAddFixedMap](#) ([osal_id_t](#) *filesys_id, const char *phys_path, const char *virt_path)
Create a fixed mapping between an existing directory and a virtual OSAL mount point.
- [int32 OS_mkfs](#) (char *address, const char *devname, const char *volname, size_t blocksize, [osal_blockcount_t](#) numblocks)
Makes a file system on the target.
- [int32 OS_mount](#) (const char *devname, const char *mountpoint)
Mounts a file system.
- [int32 OS_initfs](#) (char *address, const char *devname, const char *volname, size_t blocksize, [osal_blockcount_t](#) numblocks)

Initializes an existing file system.

- [int32 OS_rmfs](#) (const char *devname)

Removes a file system.

- [int32 OS_unmount](#) (const char *mountpoint)

Unmounts a mounted file system.

- [int32 OS_FileSysStatVolume](#) (const char *name, [OS_statvfs_t](#) *statbuf)

Obtains information about size and free space in a volume.

- [int32 OS_chkfs](#) (const char *name, bool repair)

Checks the health of a file system and repairs it if necessary.

- [int32 OS_FS_GetPhysDriveName](#) (char *PhysDriveName, const char *MountPoint)

Obtains the physical drive name associated with a mount point.

- [int32 OS_TranslatePath](#) (const char *VirtualPath, char *LocalPath)

Translates an OSAL Virtual file system path to a host Local path.

- [int32 OS_GetFsInfo](#) ([os_fsinfo_t](#) *filesys_info)

Returns information about the file system.

11.310.1 Detailed Description

Declarations and prototypes for file systems

Definition in file [osapi-filesys.h](#).

11.310.2 Macro Definition Documentation

OS_CHK_ONLY

```
#define OS_CHK_ONLY 0
```

Unused, API takes bool

Definition at line 31 of file [osapi-filesys.h](#).

OS_REPAIR

```
#define OS_REPAIR 1
```

Unused, API takes bool

Definition at line 32 of file [osapi-filesys.h](#).

11.311 osapi-filesys.h

[Go to the documentation of this file.](#)

```
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00016  * limitations under the License.
00017  *****/
00018
00025 #ifndef OSAPI_FILESYS_H
00026 #define OSAPI_FILESYS_H
00027
00028 #include "osconfig.h"
```

```

00029 #include "common_types.h"
00030
00031 #define OS_CHK_ONLY 0
00032 #define OS_REPAIR 1
00033 typedef struct
00034 {
00035     uint32 MaxFds;
00036     uint32 FreeFds;
00037     uint32 MaxVolumes;
00038     uint32 FreeVolumes;
00039 } os_fsinfo_t;
00040
00041 /*
00042  * @brief The data type filled in by the OS_FileSysStatVolume() call.
00043  *
00044  * Encapsulates detail information about the size and available space
00045  * in a mounted file system volume.
00046  */
00047 typedef struct
00048 {
00049     size_t          block_size;
00050     osal_blockcount_t total_blocks;
00051     osal_blockcount_t blocks_free;
00052 } OS_statvfs_t;
00053
00054 /*
00055  * Exported Functions
00056  */
00057
00058 /*-----*/
00059 int32 OS_FileSysAddFixedMap(osal_id_t *filesys_id, const char *phys_path, const char *virt_path);
00060
00061 /*-----*/
00062 int32 OS_mkfs(char *address, const char *devname, const char *volname, size_t blocksize, osal_blockcount_t
numblocks);
00063
00064 /*-----*/
00065 int32 OS_mount(const char *devname, const char *mountpoint);
00066
00067 /*-----*/
00068 int32 OS_initfs(char *address, const char *devname, const char *volname, size_t blocksize,
osal_blockcount_t numblocks);
00069
00070 /*-----*/
00071 int32 OS_rmfs(const char *devname);
00072
00073 /*-----*/
00074 int32 OS_unmount(const char *mountpoint);
00075
00076 /*-----*/
00077 int32 OS_FileSysStatVolume(const char *name, OS_statvfs_t *statbuf);
00078
00079 /*-----*/
00080 int32 OS_chkfs(const char *name, bool repair);
00081
00082 /*-----*/
00083 int32 OS_FS_GetPhysDriveName(char *PhysDriveName, const char *MountPoint);
00084
00085 /*-----*/
00086 int32 OS_TranslatePath(const char *VirtualPath, char *LocalPath);
00087
00088 /*-----*/
00089 int32 OS_GetFsInfo(os_fsinfo_t *filesys_info);
00090 #endif /* OSAPI_FILESYS_H */

```

11.312 osal/src/os/inc/osapi-heap.h File Reference

```

#include "osconfig.h"
#include "common_types.h"

```

Data Structures

- struct [OS_heap_prop_t](#)
OSAL heap properties.

Functions

- `int32 OS_HeapGetInfo (OS_heap_prop_t *heap_prop)`

Return current info on the heap.

11.312.1 Detailed Description

Declarations and prototypes for heap functions

Definition in file `osapi-heap.h`.

11.313 osapi-heap.h

[Go to the documentation of this file.](#)

```

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00016  * limitations under the License.
00017  *****/
00018
00025 #ifndef OSAPI_HEAP_H
00026 #define OSAPI_HEAP_H
00027
00028 #include "osconfig.h"
00029 #include "common_types.h"
00030
00036 typedef struct
00037 {
00038     size_t      free_bytes;
00039     osal_blockcount_t free_blocks;
00040     size_t      largest_free_block;
00041 } OS_heap_prop_t;
00042
00047 /*-----*/
00057 int32 OS_HeapGetInfo(OS_heap_prop_t *heap_prop);
00060 #endif /* OSAPI_HEAP_H */

```

11.314 osal/src/os/inc/osapi-idmap.h File Reference

```

#include "osconfig.h"
#include "common_types.h"

```

Macros

- `#define OS_OBJECT_INDEX_MASK 0xFFFF`
Object index mask.
- `#define OS_OBJECT_TYPE_SHIFT 16`
Object type shift.
- `#define OS_OBJECT_TYPE_UNDEFINED 0x00`
Object type undefined.
- `#define OS_OBJECT_TYPE_OS_TASK 0x01`
Object task type.
- `#define OS_OBJECT_TYPE_OS_QUEUE 0x02`

- *Object queue type.*
- #define `OS_OBJECT_TYPE_OS_COUNTSEM` 0x03
- *Object counting semaphore type.*
- #define `OS_OBJECT_TYPE_OS_BINSEM` 0x04
- *Object binary semaphore type.*
- #define `OS_OBJECT_TYPE_OS_MUTEX` 0x05
- *Object mutex type.*
- #define `OS_OBJECT_TYPE_OS_STREAM` 0x06
- *Object stream type.*
- #define `OS_OBJECT_TYPE_OS_DIR` 0x07
- *Object directory type.*
- #define `OS_OBJECT_TYPE_OS_TIMEBASE` 0x08
- *Object timebase type.*
- #define `OS_OBJECT_TYPE_OS_TIMECB` 0x09
- *Object timer callback type.*
- #define `OS_OBJECT_TYPE_OS_MODULE` 0x0A
- *Object module type.*
- #define `OS_OBJECT_TYPE_OS_FILESYS` 0x0B
- *Object file system type.*
- #define `OS_OBJECT_TYPE_OS_CONSOLE` 0x0C
- *Object console type.*
- #define `OS_OBJECT_TYPE_OS_CONDVAR` 0x0D
- *Object condition variable type.*
- #define `OS_OBJECT_TYPE_OS_RWLOCK` 0x0E
- *Object readers-writer lock type.*
- #define `OS_OBJECT_TYPE_USER` 0x10
- *Object user type.*

Functions

- static unsigned long `OS_ObjectIdToInteger` (`osal_id_t` object_id)
Obtain an integer value corresponding to an object ID.
- static `osal_id_t` `OS_ObjectIdFromInteger` (unsigned long value)
Obtain an osal ID corresponding to an integer value.
- static bool `OS_ObjectIdEqual` (`osal_id_t` object_id1, `osal_id_t` object_id2)
Check two OSAL object ID values for equality.
- static bool `OS_ObjectIdDefined` (`osal_id_t` object_id)
Check if an object ID is defined.
- `int32` `OS_GetResourceName` (`osal_id_t` object_id, char *buffer, `size_t` buffer_size)
Obtain the name of an object given an arbitrary object ID.
- `osal_objtype_t` `OS_IdentifyObject` (`osal_id_t` object_id)
Obtain the type of an object given an arbitrary object ID.
- `int32` `OS_ConvertToArrayIndex` (`osal_id_t` object_id, `osal_index_t` *ArrayIndex)
Converts an abstract ID into a number suitable for use as an array index.
- `int32` `OS_ObjectIdToArrayIndex` (`osal_objtype_t` idtype, `osal_id_t` object_id, `osal_index_t` *ArrayIndex)
Converts an abstract ID into a number suitable for use as an array index.
- void `OS_ForEachObject` (`osal_id_t` creator_id, `OS_ArgCallback_t` callback_ptr, void *callback_arg)
call the supplied callback function for all valid object IDs

- void [OS_ForEachObjectOfType](#) (osal_objtype_t objtype, osal_id_t creator_id, OS_ArgCallback_t callback_ptr, void *callback_arg)

call the supplied callback function for valid object IDs of a specific type

11.314.1 Detailed Description

Declarations and prototypes for object IDs

Definition in file [osapi-idmap.h](#).

11.314.2 Macro Definition Documentation

OS_OBJECT_INDEX_MASK

```
#define OS_OBJECT_INDEX_MASK 0xFFFF
```

Object index mask.

Definition at line 32 of file [osapi-idmap.h](#).

OS_OBJECT_TYPE_SHIFT

```
#define OS_OBJECT_TYPE_SHIFT 16
```

Object type shift.

Definition at line 33 of file [osapi-idmap.h](#).

11.315 osapi-idmap.h

[Go to the documentation of this file.](#)

```
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00017  *****/
00018
00025 #ifndef OSAPI_IDMAP_H
00026 #define OSAPI_IDMAP_H
00027
00028 #include "osconfig.h"
00029 #include "common_types.h"
00030
00031 /* Defines constants for making object ID's unique */
00032 #define OS_OBJECT_INDEX_MASK 0xFFFF
00033 #define OS_OBJECT_TYPE_SHIFT 16
00038 #define OS_OBJECT_TYPE_UNDEFINED 0x00
00039 #define OS_OBJECT_TYPE_OS_TASK 0x01
00040 #define OS_OBJECT_TYPE_OS_QUEUE 0x02
00041 #define OS_OBJECT_TYPE_OS_COUNTSEM 0x03
00042 #define OS_OBJECT_TYPE_OS_BINSEM 0x04
00043 #define OS_OBJECT_TYPE_OS_MUTEX 0x05
00044 #define OS_OBJECT_TYPE_OS_STREAM 0x06
00045 #define OS_OBJECT_TYPE_OS_DIR 0x07
00046 #define OS_OBJECT_TYPE_OS_TIMEBASE 0x08
00047 #define OS_OBJECT_TYPE_OS_TIMECB 0x09
00048 #define OS_OBJECT_TYPE_OS_MODULE 0x0A
00049 #define OS_OBJECT_TYPE_OS_FILESYS 0x0B
00050 #define OS_OBJECT_TYPE_OS_CONSOLE 0x0C
00051 #define OS_OBJECT_TYPE_OS_CONDVAR 0x0D
00052 #define OS_OBJECT_TYPE_OS_RWLOCK 0x0E
00053 #define OS_OBJECT_TYPE_USER 0x10
```

```

00060 /*-----*/
00081 static inline unsigned long OS_ObjectIdToInteger(osal_id_t object_id)
00082 {
00083     #ifdef OSAL_OMIT_DEPRECATED
00084         return object_id.v;
00085     #else
00086         return object_id;
00087     #endif
00088 }
00089
00090 /*-----*/
00103 static inline osal_id_t OS_ObjectIdFromInteger(unsigned long value)
00104 {
00105     #ifdef OSAL_OMIT_DEPRECATED
00106         osal_id_t idv = { (uint32)value };
00107     #else
00108         osal_id_t idv = (osal_id_t)value;
00109     #endif
00110
00111     return idv;
00112 }
00113
00114 /*-----*/
00130 static inline bool OS_ObjectIdEqual(osal_id_t object_id1, osal_id_t object_id2)
00131 {
00132     return (OS_ObjectIdToInteger(object_id1) == OS_ObjectIdToInteger(object_id2));
00133 }
00134
00135 /*-----*/
00151 static inline bool OS_ObjectIdDefined(osal_id_t object_id)
00152 {
00153     return (OS_ObjectIdToInteger(object_id) != 0);
00154 }
00155
00156 /*-----*/
00173 int32 OS_GetResourceName(osal_id_t object_id, char *buffer, size_t buffer_size);
00174
00175 /*-----*/
00186 osal_objtype_t OS_IdentifyObject(osal_id_t object_id);
00187
00188 /*-----*/
00213 int32 OS_ConvertToArrayIndex(osal_id_t object_id, osal_index_t *ArrayIndex);
00214
00215 /*-----*/
00245 int32 OS_ObjectIdToArrayIndex(osal_objtype_t idtype, osal_id_t object_id, osal_index_t *ArrayIndex);
00246
00247 /*-----*/
00259 void OS_ForEachObject(osal_id_t creator_id, OS_ArgCallback_t callback_ptr, void *callback_arg);
00260
00261 /*-----*/
00274 void OS_ForEachObjectOfType(osal_objtype_t objtype,
00275                             osal_id_t creator_id,
00276                             OS_ArgCallback_t callback_ptr,
00277                             void *callback_arg);
00278
00281 #endif /* OSAPI_IDMAP_H */

```

11.316 osal/src/os/inc/osapi-macros.h File Reference

```

#include <stdio.h>
#include <string.h>
#include "osconfig.h"
#include "common_types.h"
#include "osapi-printf.h"

```

Macros

- #define **BUGREPORT**(...) **OS_printf**(__VA_ARGS__)
- #define **BUGCHECK**(cond, errcode)
Basic Bug-Checking macro.
- #define **ARGCHECK**(cond, errcode)

Generic argument checking macro for non-critical values.

- #define [LENGTHCHECK](#)(str, len, errcode) [ARGCHECK](#)(memchr(str, '\0', len), errcode)

String length limit check macro.

- #define [BUGCHECK_VOID](#)(cond) [BUGCHECK](#)(cond,)

Bug-Check macro for void functions.

11.316.1 Detailed Description

Macro definitions that are used across all OSAL subsystems

Definition in file [osapi-macros.h](#).

11.316.2 Macro Definition Documentation

ARGCHECK

```
#define ARGCHECK(
    cond,
    errcode )
```

Value:

```
if (! (cond) )
{
    return errcode;
}
```

Generic argument checking macro for non-critical values.

This macro checks a conditional that is expected to be true, and return a value if it evaluates false.

ARGCHECK can be used to check for out of range or other invalid argument conditions which may (validly) occur at runtime and do not necessarily indicate bugs in the application.

These argument checks are NOT considered fatal errors. The application continues to run normally. This does not report the error on the console.

As such, ARGCHECK actions are always compiled in - not selectable at compile-time.

See also

[BUGCHECK](#) for checking critical values that indicate bugs

Definition at line 130 of file [osapi-macros.h](#).

BUGCHECK

```
#define BUGCHECK(
    cond,
    errcode )
```

Value:

```
if (! (cond) )
{
    BUGREPORT("\n**BUG** %s():%d:check \'%s\' FAILED --> %s\n\n", __func__, __LINE__, #cond, #errcode);
    return errcode;
}
```

Basic Bug-Checking macro.

This macro checks a conditional, and if it is FALSE, then it generates a report - which may in turn contain additional actions.

BUGCHECK should only be used for conditions which are critical and must always be true. If such a condition is ever false then it indicates a bug in the application which must be resolved. It may or may not be possible to continue operation if a bugcheck fails.

See also

[ARGCHECK](#) for checking non-critical values

Definition at line 104 of file [osapi-macros.h](#).

BUGCHECK_VOID

```
#define BUGCHECK_VOID(
    cond ) BUGCHECK(cond, )
```

Bug-Check macro for void functions.

The basic BUGCHECK macro returns a value, which needs to be empty for functions that do not have a return value. In this case the second argument (errcode) is intentionally left blank.

Definition at line 154 of file [osapi-macros.h](#).

BUGREPORT

```
#define BUGREPORT(
    ... ) OS_printf(__VA_ARGS__)
```

Definition at line 87 of file [osapi-macros.h](#).

LENGTHCHECK

```
#define LENGTHCHECK(
    str,
    len,
    errcode ) ARGCHECK(memchr(str, '\0', len), errcode)
```

String length limit check macro.

This macro is a specialized version of ARGCHECK that confirms a string will fit into a buffer of the specified length, and return an error code if it will not.

Note

this uses ARGCHECK, thus treating a string too long as a normal runtime (i.e. non-bug) error condition with a typical error return to the caller.

Definition at line 145 of file [osapi-macros.h](#).

11.317 osapi-macros.h

[Go to the documentation of this file.](#)

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00016  * limitations under the License.
00017  *****/
00018
00025 #ifndef OSAPI_MACROS_H
00026 #define OSAPI_MACROS_H
00027
00028 #include <stdio.h>
00029 #include <string.h>
00030
00031 #include "osconfig.h"
00032 #include "common_types.h"
00033
00034 /*
00035  * C++ does not support variadic macros until C++11
00036  * These macros should only be used from C code, not headers
00037  * or inline functions. This ifdef prevents the C++ compiler
00038  * from throwing an error about these definitions - as a result
```

```

00039  * these macros are NOT available in C++ source files.
00040  */
00041  #ifndef __cplusplus
00042
00043  #ifdef OSAL_CONFIG_BUGCHECK_DISABLE
00044
00051  #define BUGCHECK(...)
00052
00059  #define BUGREPORT(...)
00060
00061  #else /* Bug checking enabled */
00062
00063  #ifdef OSAL_CONFIG_BUGCHECK_STRICT
00064
00065  /*
00066   * This BUGREPORT implementation aborts the application so that the application
00067   * can be debugged immediately. This prints the message direct to stderr, which is
00068   * typically not buffered, so it should appear on the console before the abort occurs,
00069   * but may appear out of order with respect to calls to OS_printf().
00070   */
00071  #define BUGREPORT(...) \
00072      do \
00073      { \
00074          fprintf(stderr, __VA_ARGS__); \
00075          abort(); \
00076      } while (false)
00077
00078  #else
00079
00080  #include "osapi-printf.h"
00081
00082  /*
00083   * This BUGREPORT simply prints the message using OS_printf, which is buffered. This
00084   * has a minimal realtime impact as it only copies to the buffer, and the print will
00085   * appear in order with respect to other calls to OS_printf() by the application.
00086   */
00087  #define BUGREPORT(...) OS_printf(__VA_ARGS__)
00088
00089  #endif /* OSAL_CONFIG_BUGCHECK_STRICT */
00090
00104  #define BUGCHECK(cond, errcode) \
00105      \
00106      if (!(cond)) \
00107      { \
00108          BUGREPORT("\n**BUG** %s():%d:check '%s' FAILED --> %s\n\n", __func__, __LINE__, #cond, \
00109          #errcode); \
00110          return errcode; \
00111      }
00112
00111  #endif /* OSAL_CONFIG_BUGCHECK_DISABLE */
00112
00130  #define ARGCHECK(cond, errcode) \
00131      if (!(cond)) \
00132      { \
00133          return errcode; \
00134      }
00135
00145  #define LENGTHCHECK(str, len, errcode) ARGCHECK(memchr(str, '\\0', len), errcode)
00146
00154  #define BUGCHECK_VOID(cond) BUGCHECK(cond, )
00155
00156  #endif /* __cplusplus */
00157
00158  #endif /* OSAPI_MACROS_H */

```

11.318 osal/src/os/inc/osapi-module.h File Reference

```

#include "osconfig.h"
#include "common_types.h"

```

Data Structures

- struct [OS_module_address_t](#)
OSAL module address properties.
- struct [OS_module_prop_t](#)
OSAL module properties.
- struct [OS_static_symbol_record_t](#)
Associates a single symbol name with a memory address.

Macros

- `#define OS_MODULE_FLAG_GLOBAL_SYMBOLS 0x00`
Requests [OS_ModuleLoad\(\)](#) to add the symbols to the global symbol table.
- `#define OS_MODULE_FLAG_LOCAL_SYMBOLS 0x01`
Requests [OS_ModuleLoad\(\)](#) to keep the symbols local/private to this module.

Functions

- [int32 OS_SymbolLookup](#) ([cpuaddr](#) *symbol_address, const char *symbol_name)
Find the Address of a Symbol.
- [int32 OS_ModuleSymbolLookup](#) ([osal_id_t](#) module_id, [cpuaddr](#) *symbol_address, const char *symbol_name)
Find the Address of a Symbol within a module.
- [int32 OS_SymbolTableDump](#) (const char *filename, [size_t](#) size_limit)
Dumps the system symbol table to a file.
- [int32 OS_ModuleLoad](#) ([osal_id_t](#) *module_id, const char *module_name, const char *filename, [uint32](#) flags)
Loads an object file.
- [int32 OS_ModuleUnload](#) ([osal_id_t](#) module_id)
Unloads the module file.
- [int32 OS_ModuleInfo](#) ([osal_id_t](#) module_id, [OS_module_prop_t](#) *module_info)
Obtain information about a module.

11.318.1 Detailed Description

Declarations and prototypes for module subsystem
Definition in file [osapi-module.h](#).

11.318.2 Macro Definition Documentation

OS_MODULE_FLAG_GLOBAL_SYMBOLS

```
#define OS_MODULE_FLAG_GLOBAL_SYMBOLS 0x00
```

Requests [OS_ModuleLoad\(\)](#) to add the symbols to the global symbol table.

When supplied as the "flags" argument to [OS_ModuleLoad\(\)](#), this indicates that the symbols in the loaded module should be added to the global symbol table. This will make symbols in this library available for use when resolving symbols in future module loads.

This is the default mode of operation for [OS_ModuleLoad\(\)](#).

Note

On some operating systems, use of this option may make it difficult to unload the module in the future, if the symbols are in use by other entities.

Definition at line 49 of file [osapi-module.h](#).

OS_MODULE_FLAG_LOCAL_SYMBOLS

```
#define OS_MODULE_FLAG_LOCAL_SYMBOLS 0x01
```

Requests [OS_ModuleLoad\(\)](#) to keep the symbols local/private to this module.

When supplied as the "flags" argument to [OS_ModuleLoad\(\)](#), this indicates that the symbols in the loaded module should NOT be added to the global symbol table. This means the symbols in the loaded library will not be available for use by other modules.

Use this option is recommended for cases where no other entities will need to reference symbols within this module. This helps ensure that the module can be more safely unloaded in the future, by preventing other modules from binding to it. It also helps reduce the likelihood of symbol name conflicts among modules.

Note

To look up symbols within a module loaded with this flag, use [OS_SymbolLookupInModule\(\)](#) instead of [OS_SymbolLookup\(\)](#). Also note that references obtained using this method are not tracked by the OS; the application must ensure that all references obtained in this manner have been cleaned up/released before unloading the module.

Definition at line 71 of file [osapi-module.h](#).

11.319 osapi-module.h

[Go to the documentation of this file.](#)

```
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00017  *****/
00018
00025 #ifndef OSAPI_MODULE_H
00026 #define OSAPI_MODULE_H
00027
00028 #include "osconfig.h"
00029 #include "common_types.h"
00030
00031 /*
00032 ** Defines
00033 */
00034
00049 #define OS_MODULE_FLAG_GLOBAL_SYMBOLS 0x00
00050
00071 #define OS_MODULE_FLAG_LOCAL_SYMBOLS 0x01
00072
00073 /*
00074 ** Typedefs
00075 */
00076
00078 typedef struct
00079 {
00080     uint32  valid;
00081     uint32  flags;
00082     cpuaddr code_address;
00083     cpuaddr code_size;
00084     cpuaddr data_address;
00085     cpuaddr data_size;
00086     cpuaddr bss_address;
00087     cpuaddr bss_size;
00088 } OS_module_address_t;
00089
00091 typedef struct
```



```

00092 {
00093     cpuaddr      entry_point;
00094     cpuaddr      host_module_id;
00095     char          filename[OS_MAX_PATH_LEN];
00096     char          name[OS_MAX_API_NAME];
00097     OS_module_address_t addr;
00098 } OS_module_prop_t;
00099
00113 typedef const struct
00114 {
00115     const char *Name;
00116     void (*Address)(void);
00117     const char *Module;
00118 } OS_static_symbol_record_t;
00119
00124 /*-----*/
00142 int32 OS_SymbolLookup(cpuaddr *symbol_address, const char *symbol_name);
00143
00144 /*-----*/
00161 int32 OS_ModuleSymbolLookup(osal_id_t module_id, cpuaddr *symbol_address, const char *symbol_name);
00162
00163 /*-----*/
00184 int32 OS_SymbolTableDump(const char *filename, size_t size_limit);
00185
00186 /*-----*/
00210 int32 OS_ModuleLoad(osal_id_t *module_id, const char *module_name, const char *filename, uint32 flags);
00211
00212 /*-----*/
00225 int32 OS_ModuleUnload(osal_id_t module_id);
00226
00227 /*-----*/
00242 int32 OS_ModuleInfo(osal_id_t module_id, OS_module_prop_t *module_info);
00245 #endif /* OSAPI_MODULE_H */

```

11.320 osal/src/os/inc/osapi-mutex.h File Reference

```

#include "osconfig.h"
#include "common_types.h"

```

Data Structures

- struct [OS_mut_sem_prop_t](#)
OSAL mutex properties.

Functions

- [int32 OS_MutSemCreate](#) ([osal_id_t](#) *sem_id, const char *sem_name, [uint32](#) options)
Creates a mutex semaphore.
- [int32 OS_MutSemGive](#) ([osal_id_t](#) sem_id)
Releases the mutex object referenced by sem_id.
- [int32 OS_MutSemTake](#) ([osal_id_t](#) sem_id)
Acquire the mutex object referenced by sem_id.
- [int32 OS_MutSemDelete](#) ([osal_id_t](#) sem_id)
Deletes the specified Mutex Semaphore.
- [int32 OS_MutSemGetIdByName](#) ([osal_id_t](#) *sem_id, const char *sem_name)
Find an existing mutex ID by name.
- [int32 OS_MutSemGetInfo](#) ([osal_id_t](#) sem_id, [OS_mut_sem_prop_t](#) *mut_prop)
Fill a property object buffer with details regarding the resource.

11.320.1 Detailed Description

Declarations and prototypes for mutexes
Definition in file [osapi-mutex.h](#).

11.321 osapi-mutex.h

[Go to the documentation of this file.](#)

```

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00017  *****/
00018
00025 #ifndef OSAPI_MUTEX_H
00026 #define OSAPI_MUTEX_H
00027
00028 #include "osconfig.h"
00029 #include "common_types.h"
00030
00032 typedef struct
00033 {
00034     char      name[OS_MAX_API_NAME];
00035     osal_id_t creator;
00036 } OS_mut_sem_prop_t;
00037
00042 /*-----*/
00060 int32 OS_MutSemCreate(osal_id_t *sem_id, const char *sem_name, uint32 options);
00061
00062 /*-----*/
00078 int32 OS_MutSemGive(osal_id_t sem_id);
00079
00080 /*-----*/
00096 int32 OS_MutSemTake(osal_id_t sem_id);
00097
00098 /*-----*/
00112 int32 OS_MutSemDelete(osal_id_t sem_id);
00113
00114 /*-----*/
00130 int32 OS_MutSemGetIdByName(osal_id_t *sem_id, const char *sem_name);
00131
00132 /*-----*/
00148 int32 OS_MutSemGetInfo(osal_id_t sem_id, OS_mut_sem_prop_t *mut_prop);
00151 #endif /* OSAPI_MUTEX_H */

```

11.322 osal/src/os/inc/osapi-network.h File Reference

```

#include "osconfig.h"
#include "common_types.h"

```

Functions

- [int32 OS_NetworkGetID](#) (void)
Gets the network ID of the local machine.
- [int32 OS_NetworkGetHostName](#) (char *host_name, size_t name_len)
Gets the local machine network host name.

11.322.1 Detailed Description

Declarations and prototypes for network subsystem
Definition in file [osapi-network.h](#).

11.323 osapi-network.h

[Go to the documentation of this file.](#)

```

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00017  *****/
00018
00025 #ifndef OSAPI_NETWORK_H
00026 #define OSAPI_NETWORK_H
00027
00028 #include "osconfig.h"
00029 #include "common_types.h"
00030
00039 /*-----*/
00053 int32 OS_NetworkGetID(void);
00054
00055 /*-----*/
00070 int32 OS_NetworkGetHostName(char *host_name, size_t name_len);
00071
00074 #endif /* OSAPI_NETWORK_H */

```

11.324 osal/src/os/inc/osapi-printf.h File Reference

```

#include "osconfig.h"
#include "common_types.h"

```

Functions

- void [OS_printf](#) (const char *string,...) [OS_PRINTF](#)(1)
Abstraction for the system printf() call.
- void [OS_printf_disable](#) (void)
This function disables the output from OS_printf.
- void [OS_printf_enable](#) (void)
This function enables the output from OS_printf.

11.324.1 Detailed Description

Declarations and prototypes for printf/console output
Definition in file [osapi-printf.h](#).

11.325 osapi-printf.h

[Go to the documentation of this file.](#)

```

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00016  * limitations under the License.
00017  *****/
00018
00025 #ifndef OSAPI_PRINTF_H
00026 #define OSAPI_PRINTF_H
00027
00028 #include "osconfig.h"
00029 #include "common_types.h"
00030
00035 /*-----*/
00055 void OS_printf(const char *string, ...) OS_PRINTF(1, 2);
00056
00057 /*-----*/
00061 void OS_printf_disable(void);
00062
00063 /*-----*/
00068 void OS_printf_enable(void);
00071 #endif /* OSAPI_PRINTF_H */

```

11.326 osal/src/os/inc/osapi-queue.h File Reference

```

#include "osconfig.h"
#include "common_types.h"

```

Data Structures

- struct [OS_queue_prop_t](#)
OSAL queue properties.

Functions

- [int32 OS_QueueCreate](#) ([osal_id_t](#) *queue_id, const char *queue_name, [osal_blockcount_t](#) queue_depth, [size_t](#) data_size, [uint32](#) flags)
Create a message queue.
- [int32 OS_QueueDelete](#) ([osal_id_t](#) queue_id)
Deletes the specified message queue.
- [int32 OS_QueueGet](#) ([osal_id_t](#) queue_id, void *data, [size_t](#) size, [size_t](#) *size_copied, [int32](#) timeout)
Receive a message on a message queue.
- [int32 OS_QueuePut](#) ([osal_id_t](#) queue_id, const void *data, [size_t](#) size, [uint32](#) flags)
Put a message on a message queue.
- [int32 OS_QueueGetIdByName](#) ([osal_id_t](#) *queue_id, const char *queue_name)
Find an existing queue ID by name.
- [int32 OS_QueueGetInfo](#) ([osal_id_t](#) queue_id, [OS_queue_prop_t](#) *queue_prop)
Fill a property object buffer with details regarding the resource.

11.326.1 Detailed Description

Declarations and prototypes for queue subsystem
Definition in file [osal-queue.h](#).

11.327 osapi-queue.h

[Go to the documentation of this file.](#)

```

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00016  * limitations under the License.
00017  *****/
00018
00025 #ifndef OSAPI_QUEUE_H
00026 #define OSAPI_QUEUE_H
00027
00028 #include "osconfig.h"
00029 #include "common_types.h"
00030
00032 typedef struct
00033 {
00034     char        name[OS_MAX_API_NAME];
00035     osal_id_t creator;
00036 } OS_queue_prop_t;
00037
00067 int32 OS_QueueCreate(osal_id_t      *queue_id,
00068                     const char     *queue_name,
00069                     osal_blockcount_t queue_depth,
00070                     size_t         data_size,
00071                     uint32         flags);
00072
00073 /*-----*/
00090 int32 OS_QueueDelete(osal_id_t queue_id);
00091
00092 /*-----*/
00114 int32 OS_QueueGet(osal_id_t queue_id, void *data, size_t size, size_t *size_copied, int32 timeout);
00115
00116 /*-----*/
00133 int32 OS_QueuePut(osal_id_t queue_id, const void *data, size_t size, uint32 flags);
00134
00135 /*-----*/
00151 int32 OS_QueueGetIdByName(osal_id_t *queue_id, const char *queue_name);
00152
00153 /*-----*/
00168 int32 OS_QueueGetInfo(osal_id_t queue_id, OS_queue_prop_t *queue_prop);
00171 #endif /* OSAPI_QUEUE_H */

```

11.328 osal/src/os/inc/osapi-rwlock.h File Reference

```

#include "osconfig.h"
#include "common_types.h"

```

Data Structures

- struct [OS_rwlock_prop_t](#)
OSAL *rwlock* properties.

Functions

- [int32 OS_RwLockCreate](#) ([osal_id_t](#) *rw_id, const char *rw_name, [uint32](#) options)
Creates an readers-writer lock (*rwlock*)
- [int32 OS_RwLockReadGive](#) ([osal_id_t](#) rw_id)

Releases the reader lock on the rwlock object referenced by rw_id.

- [int32 OS_RwLockWriteGive](#) ([osal_id_t](#) rw_id)

Releases the rwlock object referenced by rw_id.

- [int32 OS_RwLockReadTake](#) ([osal_id_t](#) rw_id)

Acquire the rwlock object as a read lock as referenced by rw_id.

- [int32 OS_RwLockWriteTake](#) ([osal_id_t](#) rw_id)

Acquire the rwlock object as a write lock as referenced by rw_id.

- [int32 OS_RwLockDelete](#) ([osal_id_t](#) rw_id)

Deletes the specified RwLock.

- [int32 OS_RwLockGetIdByName](#) ([osal_id_t](#) *rw_id, const char *rw_name)

Find an existing rwlock ID by name.

- [int32 OS_RwLockGetInfo](#) ([osal_id_t](#) rw_id, [OS_rwlock_prop_t](#) *rw_prop)

Fill a property object buffer with details regarding the resource.

11.328.1 Detailed Description

Declarations and prototypes for rwlocks

Definition in file [osapi-rwlock.h](#).

11.329 osapi-rwlock.h

[Go to the documentation of this file.](#)

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00017  *****/
00018
00025 #ifndef OSAPI_RWLOCK_H
00026 #define OSAPI_RWLOCK_H
00027
00028 #include "osconfig.h"
00029 #include "common_types.h"
00030
00032 typedef struct
00033 {
00034     char      name[OS_MAX_API_NAME];
00035     osal_id_t creator;
00036 } OS_rwlock_prop_t;
00037
00042 /*-----*/
00060 int32 OS_RwLockCreate(osal_id_t *rw_id, const char *rw_name, uint32 options);
00061
00062 /*-----*/
00086 int32 OS_RwLockReadGive(osal_id_t rw_id);
00087
00088 /*-----*/
00110 int32 OS_RwLockWriteGive(osal_id_t rw_id);
00111
00112 /*-----*/
00137 int32 OS_RwLockReadTake(osal_id_t rw_id);
00138
00139 /*-----*/
00154 int32 OS_RwLockWriteTake(osal_id_t rw_id);
00155
00156 /*-----*/
```

```

00170 int32 OS_RwLockDelete(osal_id_t rw_id);
00171
00172 /*-----*/
00188 int32 OS_RwLockGetIdByName(osal_id_t *rw_id, const char *rw_name);
00189
00190 /*-----*/
00205 int32 OS_RwLockGetInfo(osal_id_t rw_id, OS_rwlock_prop_t *rw_prop);
00208 #endif /* OSAPI_RWLOCK_H */

```

11.330 osal/src/os/inc/osapi-select.h File Reference

```

#include "osconfig.h"
#include "common_types.h"
#include "osapi-clock.h"

```

Data Structures

- struct [OS_FdSet](#)
An abstract structure capable of holding several OSAL IDs.

Enumerations

- enum [OS_StreamState_t](#) {
[OS_STREAM_STATE_BOUND](#) = 0x01 , [OS_STREAM_STATE_CONNECTED](#) = 0x02 , [OS_STREAM_STATE_READABLE](#)
= 0x04 , [OS_STREAM_STATE_WRITABLE](#) = 0x08 ,
[OS_STREAM_STATE_LISTENING](#) = 0x10 }

For the [OS_SelectSingle\(\)](#) function's in/out *StateFlags* parameter, the state(s) of the stream and the result of the select is a combination of one or more of these states.

Functions

- [int32 OS_SelectMultipleAbs](#) ([OS_FdSet](#) *ReadSet, [OS_FdSet](#) *WriteSet, [OS_time_t](#) abs_timeout)
Wait for events across multiple file handles.
- [int32 OS_SelectMultiple](#) ([OS_FdSet](#) *ReadSet, [OS_FdSet](#) *WriteSet, [int32](#) msec)
Wait for events across multiple file handles.
- [int32 OS_SelectSingleAbs](#) ([osal_id_t](#) objid, [uint32](#) *StateFlags, [OS_time_t](#) abs_timeout)
Wait for events on a single file handle.
- [int32 OS_SelectSingle](#) ([osal_id_t](#) objid, [uint32](#) *StateFlags, [int32](#) msec)
Wait for events on a single file handle.
- [int32 OS_SelectFdZero](#) ([OS_FdSet](#) *Set)
Clear a FdSet structure.
- [int32 OS_SelectFdAdd](#) ([OS_FdSet](#) *Set, [osal_id_t](#) objid)
Add an ID to an FdSet structure.
- [int32 OS_SelectFdClear](#) ([OS_FdSet](#) *Set, [osal_id_t](#) objid)
Clear an ID from an FdSet structure.
- [bool OS_SelectFdsSet](#) (const [OS_FdSet](#) *Set, [osal_id_t](#) objid)
Check if an FdSet structure contains a given ID.

11.330.1 Detailed Description

Declarations and prototypes for select abstraction
Definition in file [osapi-select.h](#).

11.330.2 Enumeration Type Documentation

OS_StreamState_t

enum [OS_StreamState_t](#)

For the [OS_SelectSingle\(\)](#) function's in/out StateFlags parameter, the state(s) of the stream and the result of the select is a combination of one or more of these states.

See also

[OS_SelectSingle\(\)](#)

Enumerator

OS_STREAM_STATE_BOUND	whether the stream is bound
OS_STREAM_STATE_CONNECTED	whether the stream is connected
OS_STREAM_STATE_READABLE	whether the stream is readable
OS_STREAM_STATE_WRITABLE	whether the stream is writable
OS_STREAM_STATE_LISTENING	whether the stream is listening

Definition at line 56 of file [osapi-select.h](#).

11.331 osapi-select.h

[Go to the documentation of this file.](#)

```

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00017  *****/
00018
00025 #ifndef OSAPI_SELECT_H
00026 #define OSAPI_SELECT_H
00027
00028 #include "osconfig.h"
00029 #include "common_types.h"
00030 #include "osapi-clock.h"
00031
00044 typedef struct
00045 {
00046     uint8 object_ids[(OS_MAX_NUM_OPEN_FILES + 7) / 8];
00047 } OS_FdSet;
00048
00056 typedef enum
00057 {
00058     OS_STREAM_STATE_BOUND      = 0x01,
00059     OS_STREAM_STATE_CONNECTED = 0x02,
00060     OS_STREAM_STATE_READABLE  = 0x04,
00061     OS_STREAM_STATE_WRITABLE  = 0x08,
00062     OS_STREAM_STATE_LISTENING = 0x10
00063 } OS_StreamState_t;
00064
00069 /*-----*/

```



```

00110 int32 OS_SelectMultipleAbs(OS_FdSet *ReadSet, OS_FdSet *WriteSet, OS_time_t abs_timeout);
00111
00112 /*-----*/
00152 int32 OS_SelectMultiple(OS_FdSet *ReadSet, OS_FdSet *WriteSet, int32 msecs);
00153
00154 /*-----*/
00192 int32 OS_SelectSingleAbs(osal_id_t objid, uint32 *StateFlags, OS_time_t abs_timeout);
00193
00194 /*-----*/
00231 int32 OS_SelectSingle(osal_id_t objid, uint32 *StateFlags, int32 msecs);
00232
00233 /*-----*/
00245 int32 OS_SelectFdZero(OS_FdSet *Set);
00246
00247 /*-----*/
00261 int32 OS_SelectFdAdd(OS_FdSet *Set, osal_id_t objid);
00262
00263 /*-----*/
00277 int32 OS_SelectFdClear(OS_FdSet *Set, osal_id_t objid);
00278
00279 /*-----*/
00290 bool OS_SelectFdIsSet(const OS_FdSet *Set, osal_id_t objid);
00293 #endif /* OSAPI_SELECT_H */

```

11.332 osal/src/os/inc/osapi-shell.h File Reference

```

#include "osconfig.h"
#include "common_types.h"

```

Functions

- [int32 OS_ShellOutputToFile](#) (const char *Cmd, osal_id_t filedес)

Executes the command and sends output to a file.

11.332.1 Detailed Description

Declarations and prototypes for shell abstraction

Definition in file [osapi-shell.h](#).

11.333 osapi-shell.h

[Go to the documentation of this file.](#)

```

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00017  *****/
00018
00025 #ifndef OSAPI_SHELL_H
00026 #define OSAPI_SHELL_H
00027
00028 #include "osconfig.h"
00029 #include "common_types.h"
00030
00035 /*-----*/
00051 int32 OS_ShellOutputToFile(const char *Cmd, osal_id_t filedес);
00052
00055 #endif /* OSAPI_SHELL_H */

```

11.334 osal/src/os/inc/osapi-sockets.h File Reference

```
#include "osconfig.h"
#include "common_types.h"
#include "osapi-clock.h"
```

Data Structures

- union [OS_SockAddrData_t](#)
Storage buffer for generic network address.
- struct [OS_SockAddr_t](#)
Encapsulates a generic network address.
- struct [OS_socket_prop_t](#)
Encapsulates socket properties.
- union [OS_socket_optval](#)
Socket option value.

Macros

- #define [OS_SOCKADDR_MAX_LEN](#) 28

Typedefs

- typedef enum [OS_socket_option](#) [OS_socket_option_t](#)
Socket option identifier.
- typedef union [OS_socket_optval](#) [OS_socket_optval_t](#)
Socket option value.

Enumerations

- enum [OS_SocketDomain_t](#) { [OS_SocketDomain_INVALID](#) , [OS_SocketDomain_INET](#) , [OS_SocketDomain_INET6](#) , [OS_SocketDomain_MAX](#) }
Socket domain.
- enum [OS_SocketType_t](#) { [OS_SocketType_INVALID](#) , [OS_SocketType_DATAGRAM](#) , [OS_SocketType_STREAM](#) , [OS_SocketType_MAX](#) }
Socket type.
- enum [OS_SocketShutdownMode_t](#) { [OS_SocketShutdownMode_NONE](#) = 0 , [OS_SocketShutdownMode_SHUT_READ](#) = 1 , [OS_SocketShutdownMode_SHUT_WRITE](#) = 2 , [OS_SocketShutdownMode_SHUT_READWRITE](#) = 3 }
Shutdown Mode.
- enum [OS_socket_option](#) { [OS_socket_option_UNDEFINED](#) , [OS_socket_option_IP_DSCP](#) , [OS_socket_option_MAX](#) }
Socket option identifier.

Functions

- [int32 OS_SocketAddrInit](#) ([OS_SockAddr_t](#) *Addr, [OS_SocketDomain_t](#) Domain)
Initialize a socket address structure to hold an address of the given family.
- [int32 OS_SocketAddrToString](#) (char *buffer, [size_t](#) buflen, const [OS_SockAddr_t](#) *Addr)
Get a string representation of a network host address.
- [int32 OS_SocketAddrFromString](#) ([OS_SockAddr_t](#) *Addr, const char *string)
Set a network host address from a string representation.

- `int32 OS_SocketAddrGetPort (uint16 *PortNum, const OS_SockAddr_t *Addr)`
Get the port number of a network address.
- `int32 OS_SocketAddrSetPort (OS_SockAddr_t *Addr, uint16 PortNum)`
Set the port number of a network address.
- `int32 OS_SocketOpen (osal_id_t *sock_id, OS_SocketDomain_t Domain, OS_SocketType_t Type)`
Opens a socket.
- `int32 OS_SocketBind (osal_id_t sock_id, const OS_SockAddr_t *Addr)`
Binds a socket to a given local address and enter listening (server) mode.
- `int32 OS_SocketListen (osal_id_t sock_id)`
Places the specified socket into a listening state.
- `int32 OS_SocketBindAddress (osal_id_t sock_id, const OS_SockAddr_t *Addr)`
Binds a socket to a given local address.
- `int32 OS_SocketConnectAbs (osal_id_t sock_id, const OS_SockAddr_t *Addr, OS_time_t abs_timeout)`
Connects a socket to a given remote address.
- `int32 OS_SocketConnect (osal_id_t sock_id, const OS_SockAddr_t *Addr, int32 timeout)`
Connects a socket to a given remote address.
- `int32 OS_SocketShutdown (osal_id_t sock_id, OS_SocketShutdownMode_t Mode)`
Implement graceful shutdown of a stream socket.
- `int32 OS_SocketAcceptAbs (osal_id_t sock_id, osal_id_t *connsock_id, OS_SockAddr_t *Addr, OS_time_t abs_timeout)`
Waits for and accept the next incoming connection on the given socket.
- `int32 OS_SocketAccept (osal_id_t sock_id, osal_id_t *connsock_id, OS_SockAddr_t *Addr, int32 timeout)`
Waits for and accept the next incoming connection on the given socket.
- `int32 OS_SocketRecvFromAbs (osal_id_t sock_id, void *buffer, size_t buflen, OS_SockAddr_t *RemoteAddr, OS_time_t abs_timeout)`
Reads data from a message-oriented (datagram) socket.
- `int32 OS_SocketRecvFrom (osal_id_t sock_id, void *buffer, size_t buflen, OS_SockAddr_t *RemoteAddr, int32 timeout)`
Reads data from a message-oriented (datagram) socket.
- `int32 OS_SocketSendTo (osal_id_t sock_id, const void *buffer, size_t buflen, const OS_SockAddr_t *RemoteAddr)`
Sends data to a message-oriented (datagram) socket.
- `int32 OS_SocketGetIdByName (osal_id_t *sock_id, const char *sock_name)`
Gets an OSAL ID from a given name.
- `int32 OS_SocketGetInfo (osal_id_t sock_id, OS_socket_prop_t *sock_prop)`
Gets information about an OSAL Socket ID.
- `int32 OS_SocketGetOption (osal_id_t sock_id, OS_socket_option_t opt_id, OS_socket_optval_t *optval)`
Gets the value of a socket option.
- `int32 OS_SocketSetOption (osal_id_t sock_id, OS_socket_option_t opt_id, const OS_socket_optval_t *optval)`
Sets the value of a socket option.

11.334.1 Detailed Description

Declarations and prototypes for sockets abstraction
 Definition in file `osapi-sockets.h`.

11.334.2 Macro Definition Documentation

OS_SOCKADDR_MAX_LEN

```
#define OS_SOCKADDR_MAX_LEN 28
```

Definition at line 46 of file [osapi-sockets.h](#).

11.334.3 Typedef Documentation

OS_socket_option_t

```
typedef enum OS_socket_option OS_socket_option_t
```

Socket option identifier.

This is used with [OS_SocketGetOption\(\)](#) and [OS_SocketSetOption\(\)](#) to specify which option to get or set, respectively.

OS_socket_optval_t

```
typedef union OS_socket_optval OS_socket_optval_t
```

Socket option value.

This is used with [OS_SocketGetOption\(\)](#) and [OS_SocketSetOption\(\)](#) to store the option value that is get or set, respectively. Currently only integers values are relevant but defining as a union will allow other types to be transparently added in the future if needed.

11.334.4 Enumeration Type Documentation

OS_socket_option

```
enum OS_socket_option
```

Socket option identifier.

This is used with [OS_SocketGetOption\(\)](#) and [OS_SocketSetOption\(\)](#) to specify which option to get or set, respectively.

Enumerator

OS_socket_option_UNDEFINED	Placeholder, no-op if set, always reads 0.
OS_socket_option_IP_DSCP	Get/Set the value for the IP DSCP/Differentiated Services field
OS_socket_option_MAX	Placeholder, marks 1+ the highest valid value

Definition at line 135 of file [osapi-sockets.h](#).

OS_SocketDomain_t

```
enum OS_SocketDomain_t
```

Socket domain.

Enumerator

OS_SocketDomain_INVALID	Invalid.
OS_SocketDomain_INET	IPv4 address family, most commonly used)
OS_SocketDomain_INET6	IPv6 address family, depends on OS/network stack support.
OS_SocketDomain_MAX	Maximum.

Definition at line 61 of file [osapi-sockets.h](#).

OS_SocketShutdownMode_t

enum [OS_SocketShutdownMode_t](#)
Shutdown Mode.

Enumerator

OS_SocketShutdownMode_NONE	Reserved value, no effect.
OS_SocketShutdownMode_SHUT_READ	Disable future reading.
OS_SocketShutdownMode_SHUT_WRITE	Disable future writing.
OS_SocketShutdownMode_SHUT_READWRITE	Disable future reading or writing.

Definition at line 80 of file [osapi-sockets.h](#).

OS_SocketType_t

enum [OS_SocketType_t](#)
Socket type.

Enumerator

OS_SocketType_INVALID	Invalid.
OS_SocketType_DATAGRAM	A connectionless, message-oriented socket.
OS_SocketType_STREAM	A stream-oriented socket with the concept of a connection.
OS_SocketType_MAX	Maximum.

Definition at line 70 of file [osapi-sockets.h](#).

11.335 osapi-sockets.h

[Go to the documentation of this file.](#)

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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00025 #ifndef OSAPI_SOCKETS_H
00026 #define OSAPI_SOCKETS_H
00027
00028 /* NOTE - osconfig.h may optionally specify the value for OS_SOCKADDR_MAX_LEN */
00029 #include "osconfig.h"
00030 #include "common_types.h"
00031 #include "osapi-clock.h"
00032
00033 /*
00034  * The absolute maximum size of a network address
00035  *
00036  * The actual size varies by address type.
00037  *
00038  * This definition should be large enough to capture the
00039  * largest address supported on the system. The default

```

```

00040  * value here should be suitable for many use-cases while
00041  * still keeping the address structure size within reason.
00042  *
00043  * The user may also provide a tuned value through osconfig.h
00044  */
00045 #ifndef OS_SOCKADDR_MAX_LEN
00046 #define OS_SOCKADDR_MAX_LEN 28
00047 #endif
00048
00049 /*
00050  * -----
00051  * Sockets API Data Types
00052  *
00053  * These data types are defined regardless of whether or not networking is compiled-in
00054  *
00055  * They are local to OSAL, they do not directly map to the library-provided types, and
00056  * they do not occupy any space in the executable
00057  * -----
00058  */
00059
00061 typedef enum
00062 {
00063     OS_SocketDomain_INVALID,
00064     OS_SocketDomain_INET,
00065     OS_SocketDomain_INET6,
00066     OS_SocketDomain_MAX
00067 } OS_SocketDomain_t;
00068
00070 typedef enum
00071 {
00072     OS_SocketType_INVALID,
00073     OS_SocketType_DATAGRAM,
00074     OS_SocketType_STREAM,
00075     OS_SocketType_MAX
00076 } OS_SocketType_t;
00077
00078 /* NOTE: The shutdown mode enums are also a bitmask, so the specific values are important here */
00080 typedef enum
00081 {
00082     OS_SocketShutdownMode_NONE          = 0,
00083     OS_SocketShutdownMode_SHUT_READ     = 1,
00084     OS_SocketShutdownMode_SHUT_WRITE    = 2,
00085     OS_SocketShutdownMode_SHUT_READWRITE = 3
00086 } OS_SocketShutdownMode_t;
00087
00096 typedef union
00097 {
00098     uint8  Buffer[OS_SOCKADDR_MAX_LEN];
00099     uint32 AlignU32;
00100     void   *AlignPtr;
00101 } OS_SockAddrData_t;
00102
00110 typedef struct
00111 {
00112     size_t      ActualLength;
00113     OS_SockAddrData_t AddrData;
00114 } OS_SockAddr_t;
00115
00123 typedef struct
00124 {
00125     char      name[OS_MAX_API_NAME];
00126     osal_id_t creator;
00127 } OS_socket_prop_t;
00128
00135 typedef enum OS_socket_option
00136 {
00137     OS_socket_option_UNDEFINED,
00138     OS_socket_option_IP_DSCP,
00139     OS_socket_option_MAX
00140 } OS_socket_option_t;
00141
00150 typedef union OS_socket_optval
00151 {
00152     int32 IntVal;
00153 } OS_socket_optval_t;
00154
00167 /*-----*/
00181 int32 OS_SocketAddrInit(OS_SockAddr_t *Addr, OS_SocketDomain_t Domain);
00182
00183 /*-----*/
00205 int32 OS_SocketAddrToString(char *buffer, size_t buflen, const OS_SockAddr_t *Addr);

```

```

00206
00207 /*-----*/
00230 int32 OS_SocketAddrFromString(OS_SockAddr_t *Addr, const char *string);
00231
00232 /*-----*/
00248 int32 OS_SocketAddrGetPort(uint16 *PortNum, const OS_SockAddr_t *Addr);
00249
00250 /*-----*/
00266 int32 OS_SocketAddrSetPort(OS_SockAddr_t *Addr, uint16 PortNum);
00300 int32 OS_SocketOpen(osal_id_t *sock_id, OS_SocketDomain_t Domain, OS_SocketType_t Type);
00301
00302 /*-----*/
00322 int32 OS_SocketBind(osal_id_t sock_id, const OS_SockAddr_t *Addr);
00323
00324 /*-----*/
00340 int32 OS_SocketListen(osal_id_t sock_id);
00341
00342 /*-----*/
00363 int32 OS_SocketBindAddress(osal_id_t sock_id, const OS_SockAddr_t *Addr);
00364
00365 /*-----*/
00391 int32 OS_SocketConnectAbs(osal_id_t sock_id, const OS_SockAddr_t *Addr, OS_time_t abs_timeout);
00392
00393 /*-----*/
00417 int32 OS_SocketConnect(osal_id_t sock_id, const OS_SockAddr_t *Addr, int32 timeout);
00418
00419 /*-----*/
00436 int32 OS_SocketShutdown(osal_id_t sock_id, OS_SocketShutdownMode_t Mode);
00437
00438 /*-----*/
00469 int32 OS_SocketAcceptAbs(osal_id_t sock_id, osal_id_t *connsock_id, OS_SockAddr_t *Addr, OS_time_t
abs_timeout);
00470
00471 /*-----*/
00500 int32 OS_SocketAccept(osal_id_t sock_id, osal_id_t *connsock_id, OS_SockAddr_t *Addr, int32 timeout);
00501
00502 /*-----*/
00526 int32 OS_SocketRecvFromAbs(osal_id_t sock_id,
00527 void *buffer,
00528 size_t buflen,
00529 OS_SockAddr_t *RemoteAddr,
00530 OS_time_t abs_timeout);
00531
00532 /*-----*/
00556 int32 OS_SocketRecvFrom(osal_id_t sock_id, void *buffer, size_t buflen, OS_SockAddr_t *RemoteAddr, int32
timeout);
00557
00558 /*-----*/
00577 int32 OS_SocketSendTo(osal_id_t sock_id, const void *buffer, size_t buflen, const OS_SockAddr_t
*RemoteAddr);
00578
00579 /*-----*/
00596 int32 OS_SocketGetIdByName(osal_id_t *sock_id, const char *sock_name);
00597
00598 /*-----*/
00613 int32 OS_SocketGetInfo(osal_id_t sock_id, OS_socket_prop_t *sock_prop);
00614
00615 /*-----*/
00630 int32 OS_SocketGetOption(osal_id_t sock_id, OS_socket_option_t opt_id, OS_socket_optval_t *optval);
00631
00632 /*-----*/
00647 int32 OS_SocketSetOption(osal_id_t sock_id, OS_socket_option_t opt_id, const OS_socket_optval_t *optval);
00648
00651 #endif /* OSAPI_SOCKETS_H */

```

11.336 osal/src/os/inc/osapi-task.h File Reference

```

#include "osconfig.h"
#include "common_types.h"

```

Data Structures

- struct [OS_task_prop_t](#)
OSAL task properties.

Macros

- `#define OS_MAX_TASK_PRIORITY 255`
Upper limit for OSAL task priorities.
- `#define OS_FP_ENABLED 1`
Floating point enabled state for a task.
- `#define OSAL_PRIORITY_C(X) ((osal_priority_t) { X })`
- `#define OSAL_STACKPTR_C(X) ((osal_stackptr_t) { X })`
- `#define OSAL_TASK_STACK_ALLOCATE OSAL_STACKPTR_C(NULL)`

Typedefs

- `typedef uint8_t osal_priority_t`
Type to be used for OSAL task priorities.
- `typedef void * osal_stackptr_t`
Type to be used for OSAL stack pointer.
- `typedef void osal_task`
For task entry point.

Functions

- `typedef osal_task ((*osal_task_entry)(void))`
For task entry point.
- `int32 OS_TaskCreate (osal_id_t *task_id, const char *task_name, osal_task_entry function_pointer, osal_stackptr_t stack_pointer, size_t stack_size, osal_priority_t priority, uint32 flags)`
Creates a task and starts running it.
- `int32 OS_TaskDelete (osal_id_t task_id)`
Deletes the specified Task.
- `void OS_TaskExit (void)`
Exits the calling task.
- `int32 OS_TaskInstallDeleteHandler (osal_task_entry function_pointer)`
Installs a handler for when the task is deleted.
- `int32 OS_TaskDelay (uint32 millisecond)`
Delay a task for specified amount of milliseconds.
- `int32 OS_TaskSetPriority (osal_id_t task_id, osal_priority_t new_priority)`
Sets the given task to a new priority.
- `osal_id_t OS_TaskGetId (void)`
Obtain the task id of the calling task.
- `int32 OS_TaskGetIdByName (osal_id_t *task_id, const char *task_name)`
Find an existing task ID by name.
- `int32 OS_TaskGetInfo (osal_id_t task_id, OS_task_prop_t *task_prop)`
Fill a property object buffer with details regarding the resource.
- `int32 OS_TaskFindIdBySystemData (osal_id_t *task_id, const void *sysdata, size_t sysdata_size)`
Reverse-lookup the OSAL task ID from an operating system ID.

11.336.1 Detailed Description

Declarations and prototypes for task abstraction

Definition in file `osapi-task.h`.

11.336.2 Macro Definition Documentation

OS_FP_ENABLED

```
#define OS_FP_ENABLED 1
```

Floating point enabled state for a task.
Definition at line 35 of file [osapi-task.h](#).

OS_MAX_TASK_PRIORITY

```
#define OS_MAX_TASK_PRIORITY 255
```

Upper limit for OSAL task priorities.
Definition at line 32 of file [osapi-task.h](#).

OSAL_PRIORITY_C

```
#define OSAL_PRIORITY_C(  
    X ) ((osal_priority_t) { X })
```

Definition at line 46 of file [osapi-task.h](#).

OSAL_STACKPTR_C

```
#define OSAL_STACKPTR_C(  
    X ) ((osal_stackptr_t) { X })
```

Definition at line 53 of file [osapi-task.h](#).

OSAL_TASK_STACK_ALLOCATE

```
#define OSAL_TASK_STACK_ALLOCATE OSAL_STACKPTR_C(NULL)
```

Definition at line 54 of file [osapi-task.h](#).

11.336.3 Typedef Documentation

osal_priority_t

```
typedef uint8_t osal_priority_t
```

Type to be used for OSAL task priorities.
OSAL priorities are in reverse order, and range from 0 (highest; will preempt all other tasks) to 255 (lowest; will not preempt any other task).
Definition at line 44 of file [osapi-task.h](#).

osal_stackptr_t

```
typedef void* osal_stackptr_t
```

Type to be used for OSAL stack pointer.
Definition at line 51 of file [osapi-task.h](#).

osal_task

```
typedef void osal_task
```

For task entry point.
Definition at line 68 of file [osapi-task.h](#).

11.336.4 Function Documentation

osal_task()

```
typedef osal_task (
    (*) (void) osal_task_entry )
```

For task entry point.

11.337 osapi-task.h

[Go to the documentation of this file.](#)

```
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00016  * limitations under the License.
00017  *****/
00018
00025 #ifndef OSAPI_TASK_H
00026 #define OSAPI_TASK_H
00027
00028 #include "osconfig.h"
00029 #include "common_types.h"
00030
00032 #define OS_MAX_TASK_PRIORITY 255
00033
00035 #define OS_FP_ENABLED 1
00036
00044 typedef uint8_t osal_priority_t;
00045
00046 #define OSAL_PRIORITY_C(X) ((osal_priority_t) { X })
00047
00051 typedef void *osal_stackptr_t;
00052
00053 #define OSAL_STACKPTR_C(X) ((osal_stackptr_t) { X })
00054 #define OSAL_TASK_STACK_ALLOCATE OSAL_STACKPTR_C(NULL)
00055
00057 typedef struct
00058 {
00059     char          name[OS_MAX_API_NAME];
00060     osal_id_t     creator;
00061     size_t        stack_size;
00062     osal_priority_t priority;
00063 } OS_task_prop_t;
00064
00065 /*
00066 ** These typedefs are for the task entry point
00067 */
00068 typedef void osal_task;
00069 typedef osal_task((*osal_task_entry)(void));
00075 /*-----*/
00111 int32 OS_TaskCreate(osal_id_t      *task_id,
00112                    const char     *task_name,
00113                    osal_task_entry function_pointer,
00114                    osal_stackptr_t stack_pointer,
00115                    size_t          stack_size,
00116                    osal_priority_t priority,
00117                    uint32           flags);
00118
00119 /*-----*/
00133 int32 OS_TaskDelete(osal_id_t task_id);
00134
00135 /*-----*/
00141 void OS_TaskExit(void);
00142
00143 /*-----*/
```

```

00157 int32 OS_TaskInstallDeleteHandler(osal_task_entry function_pointer);
00158
00159 /*-----*/
00172 int32 OS_TaskDelay(uint32 millisecond);
00173
00174 /*-----*/
00187 int32 OS_TaskSetPriority(osal_id_t task_id, osal_priority_t new_priority);
00188
00189 /*-----*/
00197 osal_id_t OS_TaskGetId(void);
00198
00199 /*-----*/
00214 int32 OS_TaskGetIdByName(osal_id_t *task_id, const char *task_name);
00215
00216 /*-----*/
00232 int32 OS_TaskGetInfo(osal_id_t task_id, OS_task_prop_t *task_prop);
00233
00234 /*-----*/
00253 int32 OS_TaskFindIdBySystemData(osal_id_t *task_id, const void *sysdata, size_t sysdata_size);
00254
00257 #endif /* OSAPI_TASK_H */

```

11.338 osal/src/os/inc/osapi-timebase.h File Reference

```

#include "osconfig.h"
#include "common_types.h"

```

Data Structures

- struct [OS_timebase_prop_t](#)
Time base properties.

Typedefs

- typedef [uint32](#)(* [OS_TimerSync_t](#)) ([osal_id_t](#) timer_id)
Timer sync.

Functions

- [int32 OS_TimeBaseCreate](#) ([osal_id_t](#) *timebase_id, const char *timebase_name, [OS_TimerSync_t](#) external_sync)
Create an abstract Time Base resource.
- [int32 OS_TimeBaseSet](#) ([osal_id_t](#) timebase_id, [uint32](#) start_time, [uint32](#) interval_time)
Sets the tick period for simulated time base objects.
- [int32 OS_TimeBaseDelete](#) ([osal_id_t](#) timebase_id)
Deletes a time base object.
- [int32 OS_TimeBaseGetIdByName](#) ([osal_id_t](#) *timebase_id, const char *timebase_name)
Find the ID of an existing time base resource.
- [int32 OS_TimeBaseGetInfo](#) ([osal_id_t](#) timebase_id, [OS_timebase_prop_t](#) *timebase_prop)
Obtain information about a timebase resource.
- [int32 OS_TimeBaseGetFreeRun](#) ([osal_id_t](#) timebase_id, [uint32](#) *freerun_val)
Read the value of the timebase free run counter.

11.338.1 Detailed Description

Declarations and prototypes for timebase abstraction
Definition in file [osapi-timebase.h](#).

11.338.2 Typedef Documentation

OS_TimerSync_t

typedef uint32 (* OS_TimerSync_t) (osal_id_t timer_id)

Timer sync.

Definition at line 34 of file [osapi-timebase.h](#).

11.339 osapi-timebase.h

[Go to the documentation of this file.](#)

```

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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00025 #ifndef OSAPI_TIMEBASE_H
00026 #define OSAPI_TIMEBASE_H
00027
00028 #include "osconfig.h"
00029 #include "common_types.h"
00030
00031 /*
00032 ** Typedefs
00033 */
00034 typedef uint32 (*OS_TimerSync_t) (osal_id_t timer_id);
00037 typedef struct
00038 {
00039     char        name[OS_MAX_API_NAME];
00040     osal_id_t   creator;
00041     uint32      nominal_interval_time;
00042     uint32      freerun_time;
00043     uint32      accuracy;
00044 } OS_timebase_prop_t;
00045
00050 /*-----*/
00089 int32 OS_TimeBaseCreate(osal_id_t *timebase_id, const char *timebase_name, OS_TimerSync_t external_sync);
00090
00091 /*-----*/
00118 int32 OS_TimeBaseSet(osal_id_t timebase_id, uint32 start_time, uint32 interval_time);
00119
00120 /*-----*/
00137 int32 OS_TimeBaseDelete(osal_id_t timebase_id);
00138
00139 /*-----*/
00158 int32 OS_TimeBaseGetIdByName(osal_id_t *timebase_id, const char *timebase_name);
00159
00160 /*-----*/
00182 int32 OS_TimeBaseGetInfo(osal_id_t timebase_id, OS_timebase_prop_t *timebase_prop);
00183
00184 /*-----*/
00215 int32 OS_TimeBaseGetFreeRun(osal_id_t timebase_id, uint32 *freerun_val);
00216
00219 #endif /* OSAPI_TIMEBASE_H */

```

11.340 osal/src/os/inc/osapi-timer.h File Reference

```

#include "osconfig.h"
#include "common_types.h"

```

Data Structures

- struct [OS_timer_prop_t](#)
Timer properties.

Typedefs

- typedef void(* [OS_TimerCallback_t](#)) ([osal_id_t](#) timer_id)
Timer callback.

Functions

- [int32 OS_TimerCreate](#) ([osal_id_t](#) *timer_id, const char *timer_name, [uint32](#) *clock_accuracy, [OS_TimerCallback_t](#) callback_ptr)
Create a timer object.
- [int32 OS_TimerAdd](#) ([osal_id_t](#) *timer_id, const char *timer_name, [osal_id_t](#) timebase_id, [OS_ArgCallback_t](#) callback_ptr, void *callback_arg)
Add a timer object based on an existing TimeBase resource.
- [int32 OS_TimerSet](#) ([osal_id_t](#) timer_id, [uint32](#) start_time, [uint32](#) interval_time)
Configures a periodic or one shot timer.
- [int32 OS_TimerDelete](#) ([osal_id_t](#) timer_id)
Deletes a timer resource.
- [int32 OS_TimerGetIdByName](#) ([osal_id_t](#) *timer_id, const char *timer_name)
Locate an existing timer resource by name.
- [int32 OS_TimerGetInfo](#) ([osal_id_t](#) timer_id, [OS_timer_prop_t](#) *timer_prop)
Gets information about an existing timer.

11.340.1 Detailed Description

Declarations and prototypes for timer abstraction (app callbacks)

Definition in file [osapi-timer.h](#).

11.340.2 Typedef Documentation

OS_TimerCallback_t

```
typedef void(* OS_TimerCallback_t) (osal\_id\_t timer_id)
```

Timer callback.

Definition at line 34 of file [osapi-timer.h](#).

11.341 osapi-timer.h

[Go to the documentation of this file.](#)

```
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```

```

00017  *****/
00018
00025 #ifndef OSAPI_TIMER_H
00026 #define OSAPI_TIMER_H
00027
00028 #include "osconfig.h"
00029 #include "common_types.h"
00030
00031 /*
00032 ** Typedefs
00033 */
00034 typedef void (*OS_TimerCallback_t) (osal_id_t timer_id);
00037 typedef struct
00038 {
00039     char        name[OS_MAX_API_NAME];
00040     osal_id_t   creator;
00041     uint32      start_time;
00042     uint32      interval_time;
00043     uint32      accuracy;
00044 } OS_timer_prop_t;
00045
00050 /*-----*/
00088 int32 OS_TimerCreate(osal_id_t      *timer_id,
00089                     const char     *timer_name,
00090                     uint32         *clock_accuracy,
00091                     OS_TimerCallback_t callback_ptr);
00092
00093 /*-----*/
00135 int32 OS_TimerAdd(osal_id_t      *timer_id,
00136                  const char     *timer_name,
00137                  osal_id_t      timebase_id,
00138                  OS_ArgCallback_t callback_ptr,
00139                  void           *callback_arg);
00140
00141 /*-----*/
00176 int32 OS_TimerSet(osal_id_t timer_id, uint32 start_time, uint32 interval_time);
00177
00178 /*-----*/
00196 int32 OS_TimerDelete(osal_id_t timer_id);
00197
00198 /*-----*/
00217 int32 OS_TimerGetIdByName(osal_id_t *timer_id, const char *timer_name);
00218
00219 /*-----*/
00243 int32 OS_TimerGetInfo(osal_id_t timer_id, OS_timer_prop_t *timer_prop);
00246 #endif /* OSAPI_TIMER_H */

```

11.342 osal/src/os/inc/osapi-version.h File Reference

```
#include "common_types.h"
```

Macros

- #define OS_BUILD_NUMBER 0
- #define OS_BUILD_BASELINE "v7.0.1"
- #define OS_BUILD_DEV_CYCLE "v7.0.1"
Development: Release name for current development cycle.
- #define OS_BUILD_CODENAME "Draco"
: Development: Code name for the current build
- #define OS_MAJOR_VERSION 7
Major version number.
- #define OS_MINOR_VERSION 0
Minor version number.
- #define OS_REVISION 1
Revision version number. Value of 99 indicates a development version.
- #define OS_LAST_OFFICIAL "v7.0.1"

- *Last official release.*
• #define `OS_MISSION_REV` 0x0
- *Mission revision.*
• #define `OS_STR_HELPER(x)` #x
- *Helper function to concatenate strings from integer.*
• #define `OS_STR(x)` `OS_STR_HELPER(x)`
- *Helper function to concatenate strings from integer.*
• #define `OS_VERSION` `OS_BUILD_BASELINE` "+dev" `OS_STR(OS_BUILD_NUMBER)`
- *Development Build Version Number.*
• #define `OSAL_API_VERSION` ((`OS_MAJOR_VERSION` * 10000) + (`OS_MINOR_VERSION` * 100) + `OS_REVISION`)
- *Combines the revision components into a single value.*
• #define `OS_CFG_MAX_VERSION_STR_LEN` 256
- *Max Version String length.*

Functions

- const char * `OS_GetVersionString` (void)
- const char * `OS_GetVersionCodeName` (void)
- void `OS_GetVersionNumber` (uint8 VersionNumbers[4])
Obtain the OSAL numeric version number.
- uint32 `OS_GetBuildNumber` (void)
Obtain the OSAL library numeric build number.

11.342.1 Detailed Description

Provide version identifiers for Operating System Abstraction Layer

Note

OSAL follows the same version semantics as cFS, which in turn is based on the Semantic Versioning 2.0 Specification. For more information, see the documentation provided with cFE.

Definition in file [osapi-version.h](#).

11.342.2 Macro Definition Documentation

`OS_BUILD_BASELINE`

```
#define OS_BUILD_BASELINE "v7.0.1"
```

Definition at line 38 of file [osapi-version.h](#).

`OS_BUILD_CODENAME`

```
#define OS_BUILD_CODENAME "Draco"
```

: Development: Code name for the current build

Definition at line 40 of file [osapi-version.h](#).

`OS_BUILD_DEV_CYCLE`

```
#define OS_BUILD_DEV_CYCLE "v7.0.1"
```

Development: Release name for current development cycle.

Definition at line 39 of file [osapi-version.h](#).

OS_BUILD_NUMBER

```
#define OS_BUILD_NUMBER 0
```

Definition at line 37 of file [osapi-version.h](#).

OS_CFG_MAX_VERSION_STR_LEN

```
#define OS_CFG_MAX_VERSION_STR_LEN 256
```

Max Version String length.
Maximum length that an OSAL version string can be.
Definition at line 154 of file [osapi-version.h](#).

OS_LAST_OFFICIAL

```
#define OS_LAST_OFFICIAL "v7.0.1"
```

Last official release.
Definition at line 52 of file [osapi-version.h](#).

OS_MAJOR_VERSION

```
#define OS_MAJOR_VERSION 7
```

Major version number.
Definition at line 45 of file [osapi-version.h](#).

OS_MINOR_VERSION

```
#define OS_MINOR_VERSION 0
```

Minor version number.
Definition at line 46 of file [osapi-version.h](#).

OS_MISSION_REV

```
#define OS_MISSION_REV 0x0
```

Mission revision.
Reserved for mission use to denote patches/customizations as needed. Values 1-254 are reserved for mission use to denote patches/customizations as needed. NOTE: Reserving 0 and 0xFF for cFS open-source development use (pending resolution of nasa/cFS#440)
Definition at line 61 of file [osapi-version.h](#).

OS_REVISION

```
#define OS_REVISION 1
```

Revision version number. Value of 99 indicates a development version.
Definition at line 47 of file [osapi-version.h](#).

OS_STR

```
#define OS_STR(  
    x ) OS_STR_HELPER(x)
```

Helper function to concatenate strings from integer.
Definition at line 67 of file [osapi-version.h](#).

OS_STR_HELPER

```
#define OS_STR_HELPER(  
    x ) #x
```


Helper function to concatenate strings from integer.

Definition at line 66 of file [osapi-version.h](#).

OS_VERSION

```
#define OS_VERSION OS_BUILD_BASELINE "+dev" OS_STR(OS_BUILD_NUMBER)
```

Development Build Version Number.

Baseline git tag + Number of commits since baseline.

Definition at line 72 of file [osapi-version.h](#).

OSAL_API_VERSION

```
#define OSAL_API_VERSION ((OS_MAJOR_VERSION * 10000) + (OS_MINOR_VERSION * 100) + OS_REVISION)
```

Combines the revision components into a single value.

Applications can check against this number

e.g. `"#if OSAL_API_VERSION >= 40100"` would check if some feature added in OSAL 4.1 is present.

Definition at line 79 of file [osapi-version.h](#).

11.342.3 Function Documentation

OS_GetBuildNumber()

```
uint32 OS_GetBuildNumber (
    void )
```

Obtain the OSAL library numeric build number.

The build number is a monotonically increasing number that (coarsely) reflects the number of commits/changes that have been merged since the epoch release. During development cycles this number should increase after each subsequent merge/modification.

Like other version information, this is a fixed number assigned at compile time.

Returns

The OSAL library build number

OS_GetVersionCodeName()

```
const char * OS_GetVersionCodeName (
    void )
```

Gets the OSAL version code name

All NASA CFE/CFS components (including CFE framework, OSAL and PSP) that work together will share the same code name.

Returns

OSAL code name. This is a fixed value string and is never NULL.

OS_GetVersionNumber()

```
void OS_GetVersionNumber (
    uint8 VersionNumbers[4] )
```

Obtain the OSAL numeric version number.

This retrieves the numeric OSAL version identifier as an array of 4 uint8 values.

The array of numeric values is in order of precedence: [0] = Major Number [1] = Minor Number [2] = Revision Number [3] = Mission Revision

The "Mission Revision" (last output) also indicates whether this is an official release, a patched release, or a development version. 0 indicates an official release 1-254 local patch level (reserved for mission use) 255 indicates a development build

Parameters

out	VersionNumbers	A fixed-size array to be filled with the version numbers
-----	----------------	--

OS_GetVersionString()

```
const char * OS_GetVersionString (
    void )
```

Gets the OSAL version/baseline ID as a string

This returns the content of the `OS_VERSION` macro defined above, and is specifically just the baseline and development build ID (if applicable), without any extra info.

Returns

Basic version identifier. This is a fixed value string and is never NULL.

11.343 osapi-version.h

[Go to the documentation of this file.](#)

```
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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00029 #ifndef OSAPI_VERSION_H
00030 #define OSAPI_VERSION_H
00031
00032 #include "common_types.h"
00033
00034 /*
00035  * Development Build Macro Definitions
00036  */
00037 #define OS_BUILD_NUMBER 0
00038 #define OS_BUILD_BASELINE "v7.0.1"
00039 #define OS_BUILD_DEV_CYCLE "v7.0.1"
00040 #define OS_BUILD_CODENAME "Draco"
00042 /*
00043  * Version Macros, see \ref cfsversions for definitions.
00044  */
00045 #define OS_MAJOR_VERSION 7
00046 #define OS_MINOR_VERSION 0
00047 #define OS_REVISION 1
00052 #define OS_LAST_OFFICIAL "v7.0.1"
00053
00061 #define OS_MISSION_REV 0x0
00062
00063 /*
00064  * Tools to construct version string
00065  */
00066 #define OS_STR_HELPER(x) #x
00067 #define OS_STR(x) OS_STR_HELPER(x)
00072 #define OS_VERSION OS_BUILD_BASELINE "+dev" OS_STR(OS_BUILD_NUMBER)
00073
```

```
00079 #define OSAL_API_VERSION ((OS_MAJOR_VERSION * 10000) + (OS_MINOR_VERSION * 100) + OS_REVISION)
00080
00081 /*
00082  * Functions to get OSAL version info
00083  *
00084  * It is preferable to use the functions below to retrieve the OSAL version
00085  * at runtime, because if applications reference the macros above directly, the
00086  * macro will only get evaluated when the _application_ is built.
00087  *
00088  * When using the functions below, the version gets evaluated when the OSAL library
00089  * is built, and therefore if the OSAL library is re-linked without rebuilding the
00090  * application itself, the version will still be correct.
00091  */
00092
00101 const char *OS_GetVersionString(void);
00102
00111 const char *OS_GetVersionCodeName(void);
00112
00132 void OS_GetVersionNumber(uint8 VersionNumbers[4]);
00133
00146 uint32 OS_GetBuildNumber(void);
00147
00154 #define OS_CFG_MAX_VERSION_STR_LEN 256
00155
00156 #endif /* OSAPI_VERSION_H */
```

11.344 osal/src/os/inc/osapi.h File Reference

```
#include <stdio.h>
#include <stdlib.h>
#include <stdarg.h>
#include "common_types.h"
#include "osapi-version.h"
#include "osconfig.h"
#include "osapi-binsem.h"
#include "osapi-clock.h"
#include "osapi-common.h"
#include "osapi-condvar.h"
#include "osapi-constants.h"
#include "osapi-countsem.h"
#include "osapi-dir.h"
#include "osapi-error.h"
#include "osapi-file.h"
#include "osapi-filesys.h"
#include "osapi-heap.h"
#include "osapi-macros.h"
#include "osapi-idmap.h"
#include "osapi-module.h"
#include "osapi-mutex.h"
#include "osapi-network.h"
#include "osapi-printf.h"
#include "osapi-queue.h"
#include "osapi-rwlock.h"
#include "osapi-select.h"
#include "osapi-shell.h"
#include "osapi-sockets.h"
#include "osapi-task.h"
#include "osapi-timebase.h"
#include "osapi-timer.h"
#include "osapi-bsp.h"
```

11.344.1 Detailed Description

Purpose: Contains functions prototype definitions and variables declarations for the OS Abstraction Layer, Core OS module

Definition in file [osapi.h](#).

11.345 osapi.h

[Go to the documentation of this file.](#)

```

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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00026 #ifndef OSAPI_H
00027 #define OSAPI_H
00028
00029 #ifdef __cplusplus
00030 extern "C"
00031 {
00032 #endif
00033
00034 /*
00035  * Note - the "osapi-os-filesys.h" file previously included these system headers
00036  * plus a couple others. Some existing code used stdio/stdlib functions but did
00037  * not #include them, relying on the implicit inclusion that osapi-os-filesys
00038  * provided.
00039  *
00040  * osapi-os-filesys.h now does a more complete abstraction and does not base its
00041  * API on the system API anymore. However, in order to not break other code that
00042  * relied on the implicit inclusion of these system files, these are added here.
00043  *
00044  * (The reasoning is that stdio/stdlib are widely used and well standardized
00045  * across all C implementations, even embedded/lightweight C libraries should
00046  * have some usable version of these includes)
00047  */
00048 #include <stdio.h>
00049 #include <stdlib.h>
00050 #include <stdarg.h>
00051
00052 #include "common_types.h"
00053
00054 #include "osapi-version.h"
00055
00056 /*
00057  ** Include the configuration file
00058  */
00059 #include "osconfig.h"
00060
00061 /*
00062  ** Include the OS API modules
00063  */
00064 #include "osapi-binsem.h"
00065 #include "osapi-clock.h"
00066 #include "osapi-common.h"
00067 #include "osapi-condvar.h"
00068 #include "osapi-constants.h"
00069 #include "osapi-countsem.h"
00070 #include "osapi-dir.h"
00071 #include "osapi-error.h"
00072 #include "osapi-file.h"
00073 #include "osapi-filesys.h"
00074 #include "osapi-heap.h"
00075 #include "osapi-macros.h"
00076 #include "osapi-idmap.h"

```

```

00077 #include "osapi-module.h"
00078 #include "osapi-mutex.h"
00079 #include "osapi-network.h"
00080 #include "osapi-printf.h"
00081 #include "osapi-queue.h"
00082 #include "osapi-rwlock.h"
00083 #include "osapi-select.h"
00084 #include "osapi-shell.h"
00085 #include "osapi-sockets.h"
00086 #include "osapi-task.h"
00087 #include "osapi-timebase.h"
00088 #include "osapi-timer.h"
00089
00090 /*
00091  *****
00092  * Items below here are internal OSAL-use definitions and are not part of the
00093  * OSAL API
00094  *****
00095  */
00096
00097 #include "osapi-bsp.h"
00098
00099 #ifdef __cplusplus
00100 }
00101 #endif
00102
00103 #endif /* OSAPI_H */

```

11.346 psp/fsw/inc/cfe_psp.h File Reference

```

#include "common_types.h"
#include "osapi.h"
#include "pspconfig.h"
#include "cfe_psp_cache_api.h"
#include "cfe_psp_cds_api.h"
#include "cfe_psp_eepromaccess_api.h"
#include "cfe_psp_error.h"
#include "cfe_psp_exception_api.h"
#include "cfe_psp_id_api.h"
#include "cfe_psp_memaccess_api.h"
#include "cfe_psp_memrange_api.h"
#include "cfe_psp_port_api.h"
#include "cfe_psp_ssr_api.h"
#include "cfe_psp_timertick_api.h"
#include "cfe_psp_version_api.h"
#include "cfe_psp_watchdog_api.h"

```

Macros

- `#define PSP_DEBUG_LEV(l, ...)`
- `#define PSP_DEBUG(...)`

Functions

- void `CFE_PSP_Main` (void)
PSP Entry Point to initialize the OSAL and start up the cFE.

11.346.1 Macro Definition Documentation

PSP_DEBUG

```

#define PSP_DEBUG(
    ...
)

```

Definition at line 78 of file [cfe_psp.h](#).

PSP_DEBUG_LEV

```
#define PSP_DEBUG_LEV(
    1,
    ... )
```

Definition at line 77 of file [cfe_psp.h](#).

11.346.2 Function Documentation

CFE_PSP_Main()

```
void CFE_PSP_Main (
    void )
```

PSP Entry Point to initialize the OSAL and start up the cFE.

This is the entry point that the real-time OS calls to start our software. This routine will do any BSP/OS-specific setup, then call the entry point of the flight software (i.e. the cFE main entry point).

Note

The flight software (i.e. cFE) should not call this routine.

11.347 cfe_psp.h

[Go to the documentation of this file.](#)

```
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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00019 /*
00020 ** Author: A. Cudmore
00021 **
00022 ** Purpose: This file contains the cFE Platform Support Package(PSP)
00023 **           prototypes.
00024 **           The PSP routines serve as the "glue" between the RTOS and
00025 **           the cFE Flight Software. The routines fill gaps that are not
00026 **           really considered part of the OS Abstraction, but are required
00027 **           for the cFE flight software implementation. It is possible that
00028 **           some of these routines could migrate into the OS AL.
00029 **
00030 */
00031
00032 #ifndef CFE_PSP_H
00033 #define CFE_PSP_H
00034
00035 /*****
00036 INCLUDE FILES
00037 *****/
00038
00039 #include "common_types.h"
00040 #include "osapi.h"
00041 #include "pspcconfig.h"
00042
00043 /*
00044  * All PSP API sub-components
00045  */
```

```

00046  * Historically these were all declared as part of a monolithic cfe_psp.h header file
00047  * Breaking them up aids in unit testability, allowing each sub-component to be tested
00048  * individually - particularly important for items where the implementation was moved to
00049  * a module. The whole API is included here for backward compatibility.
00050  */
00051 #include "cfe_psp_cache_api.h"
00052 #include "cfe_psp_cds_api.h"
00053 #include "cfe_psp_eepromaccess_api.h"
00054 #include "cfe_psp_error.h"
00055 #include "cfe_psp_exception_api.h"
00056 #include "cfe_psp_id_api.h"
00057 #include "cfe_psp_memaccess_api.h"
00058 #include "cfe_psp_memrange_api.h"
00059 #include "cfe_psp_port_api.h"
00060 #include "cfe_psp_ssr_api.h"
00061 #include "cfe_psp_timertick_api.h"
00062 #include "cfe_psp_version_api.h"
00063 #include "cfe_psp_watchdog_api.h"
00064
00065 /*
00066  * The "PSP_DEBUG" is a no-op unless PSP_CONFIG_DEBUG_PRINTF is enabled.
00067  * When enabled, it is a macro that includes function/line number info.
00068  */
00069 #if defined(PSP_CONFIG_DEBUG_PRINTF)
00070 extern void PSP_DebugPrintf(uint32 Level, const char *Func, uint32 Line, const char *Format, ...);
00071 /* Debug printf's are compiled in, but also can be disabled by a run-time flag.
00072  * Note that the ##__VA_ARGS__ syntax works on GCC but might need tweaks for other compilers... */
00073 #define PSP_DEBUG_LEV(l, ...) PSP_DebugPrintf(l, __func__, __LINE__, __VA_ARGS__);
00074 #define PSP_DEBUG(...)      PSP_DEBUG_LEV(1, __VA_ARGS__)
00075 #else
00076 /* Debug printf's are not compiled in at all */
00077 #define PSP_DEBUG_LEV(l, ...)
00078 #define PSP_DEBUG(...)
00079 #endif
00080
00081 /*****
00082  FUNCTION PROTOTYPES
00083  *****/
00084
00085 /*
00086  ** PSP entry point
00087  */
00088 /*-----*/
00098 void CFE_PSP_Main(void);
00099
00100 #endif

```

11.348 psp/fsw/inc/cfe_psp_cache_api.h File Reference

```

#include "common_types.h"
#include "osapi.h"
#include "cfe_psp_error.h"

```

Functions

- void [CFE_PSP_FlushCaches](#) (uint32 type, void *address, uint32 size)

This is a BSP-specific cache flush routine.

11.348.1 Function Documentation

CFE_PSP_FlushCaches()

```

void CFE_PSP_FlushCaches (
    uint32 type,
    void * address,
    uint32 size )

```

This is a BSP-specific cache flush routine.

Provides a common interface to flush the processor caches. This routine is in the BSP because it is sometimes implemented in hardware and sometimes taken care of by the RTOS.

Parameters

in	type	
in	address	
in	size	

11.349 cfe_psp_cache_api.h

[Go to the documentation of this file.](#)

```

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00016  * limitations under the License.
00017  *****/
00018
00019 /*
00020 ** Author: A. Cudmore
00021 **
00022 ** Purpose: This file contains the cFE Platform Support Package (PSP)
00023 **          prototypes.
00024 **          The PSP routines serve as the "glue" between the RTOS and
00025 **          the cFE Flight Software. The routines fill gaps that are not
00026 **          really considered part of the OS Abstraction, but are required
00027 **          for the cFE flight software implementation. It is possible that
00028 **          some of these routines could migrate into the OS AL.
00029 **
00030 */
00031
00032 #ifndef CFE_PSP_CACHE_API_H
00033 #define CFE_PSP_CACHE_API_H
00034
00035 /*****
00036  INCLUDE FILES
00037  *****/
00038
00039 #include "common_types.h"
00040 #include "osapi.h"
00041
00042 #include "cfe_psp_error.h"
00043
00044 /*****
00045  FUNCTION PROTOTYPES
00046  *****/
00047
00048 /*-----*/
00060 void CFE_PSP_FlushCaches(uint32 type, void *address, uint32 size);
00061
00062 #endif

```

11.350 psp/fsw/inc/cfe_psp_cds_api.h File Reference

```

#include "common_types.h"
#include "osapi.h"
#include "cfe_psp_error.h"

```


Functions

- [int32 CFE_PSP_GetCDSSize](#) (uint32 *SizeOfCDS)
Fetches the size of the OS Critical Data Store area.
- [int32 CFE_PSP_WriteToCDS](#) (const void *PtrToDataToWrite, [uint32](#) CDSOffset, [uint32](#) NumBytes)
Writes to the CDS Block.
- [int32 CFE_PSP_ReadFromCDS](#) (void *PtrToDataFromRead, [uint32](#) CDSOffset, [uint32](#) NumBytes)
Reads from the CDS Block.

11.350.1 Function Documentation

CFE_PSP_GetCDSSize()

```
int32 CFE_PSP_GetCDSSize (
    uint32 * SizeOfCDS )
```

Fetches the size of the OS Critical Data Store area.

Parameters

out	<i>SizeOfCDS</i>	Pointer to the variable that will store the size of the CDS
-----	------------------	---

Returns

0 (OS_SUCCESS or CFE_PSP_SUCCESS) on success, -1 (OS_ERROR or CFE_PSP_ERROR) on error

CFE_PSP_ReadFromCDS()

```
int32 CFE_PSP_ReadFromCDS (
    void * PtrToDataFromRead,
    uint32 CDSOffset,
    uint32 NumBytes )
```

Reads from the CDS Block.

Parameters

out	<i>PtrToDataFromRead</i>	Pointer to the location that will store the data to be read from the CDS
in	<i>CDSOffset</i>	CDS offset
in	<i>NumBytes</i>	Number of bytes to read

Returns

0 (OS_SUCCESS or CFE_PSP_SUCCESS) on success, -1 (OS_ERROR or CFE_PSP_ERROR) on error

CFE_PSP_WriteToCDS()

```
int32 CFE_PSP_WriteToCDS (
    const void * PtrToDataToWrite,
    uint32 CDSOffset,
    uint32 NumBytes )
```

Writes to the CDS Block.

Parameters

in	<i>PtrToDataToWrite</i>	Pointer to the data that will be written to the CDS
in	<i>CDSOffset</i>	CDS offset
in	<i>NumBytes</i>	Number of bytes to write

Returns

0 (OS_SUCCESS or CFE_PSP_SUCCESS) on success, -1 (OS_ERROR or CFE_PSP_ERROR) on error

11.351 cfe_psp_cds_api.h

[Go to the documentation of this file.](#)

```

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00016  * limitations under the License.
00017  *****/
00018
00019 /*
00020 ** Author: A. Cudmore
00021 **
00022 ** Purpose: This file contains the cFE Platform Support Package (PSP)
00023 **          prototypes.
00024 **          The PSP routines serve as the "glue" between the RTOS and
00025 **          the cFE Flight Software. The routines fill gaps that are not
00026 **          really considered part of the OS Abstraction, but are required
00027 **          for the cFE flight software implementation. It is possible that
00028 **          some of these routines could migrate into the OS AL.
00029 **
00030 */
00031
00032 #ifndef CFE_PSP_CDS_API_H
00033 #define CFE_PSP_CDS_API_H
00034
00035 /*****
00036  INCLUDE FILES
00037  *****/
00038
00039 #include "common_types.h"
00040 #include "osapi.h"
00041
00042 #include "cfe_psp_error.h"
00043
00044 /*****
00045  FUNCTION PROTOTYPES
00046  *****/
00047
00048 /*-----*/
00056 int32 CFE_PSP_GetCDSSize(uint32 *SizeOfCDS);
00057
00058 /*-----*/
00068 int32 CFE_PSP_WriteToCDS(const void *PtrToDataToWrite, uint32 CDSOffset, uint32 NumBytes);
00069
00070 /*-----*/
00080 int32 CFE_PSP_ReadFromCDS(void *PtrToDataFromRead, uint32 CDSOffset, uint32 NumBytes);
00081
00082 #endif

```

11.352 psp/fsw/inc/cfe_psp_eepromaccess_api.h File Reference

```
#include "common_types.h"
#include "osapi.h"
#include "cfe_psp_error.h"
```

Functions

- [int32 CFE_PSP_EepromWrite8](#) ([cpuaddr](#) MemoryAddress, [uint8](#) ByteValue)
Write one byte (ByteValue) to EEPROM address MemoryAddress.
- [int32 CFE_PSP_EepromWrite16](#) ([cpuaddr](#) MemoryAddress, [uint16](#) uint16Value)
Write two bytes (uint16Value) to EEPROM address MemoryAddress.
- [int32 CFE_PSP_EepromWrite32](#) ([cpuaddr](#) MemoryAddress, [uint32](#) uint32Value)
Write four bytes (uint32Value) to EEPROM address MemoryAddress.
- [int32 CFE_PSP_EepromWriteEnable](#) ([uint32](#) Bank)
Enable the EEPROM for write operation.
- [int32 CFE_PSP_EepromWriteDisable](#) ([uint32](#) Bank)
Disable the EEPROM from write operation.
- [int32 CFE_PSP_EepromPowerUp](#) ([uint32](#) Bank)
Power up the EEPROM.
- [int32 CFE_PSP_EepromPowerDown](#) ([uint32](#) Bank)
Power down the EEPROM.

11.352.1 Function Documentation

CFE_PSP_EepromPowerDown()

```
int32 CFE_PSP_EepromPowerDown (
    uint32 Bank )
```

Power down the EEPROM.

Parameters

in	Bank	The bank of EEPROM to power down
----	------	----------------------------------

Return values

CFE_PSP_SUCCESS	on success
CFE_PSP_ERROR_NOT_IMPLEMENTED	if not implemented

CFE_PSP_EepromPowerUp()

```
int32 CFE_PSP_EepromPowerUp (
    uint32 Bank )
```

Power up the EEPROM.

Parameters

in	Bank	The bank of EEPROM to power up
----	------	--------------------------------

Return values

<i>CFE_PSP_SUCCESS</i>	on success
<i>CFE_PSP_ERROR_NOT_IMPLEMENTED</i>	if not implemented

CFE_PSP_EepromWrite16()

```
int32 CFE_PSP_EepromWrite16 (
    cpuaddr MemoryAddress,
    uint16 uint16Value )
```

Write two bytes (uint16Value) to EEPROM address MemoryAddress.

Parameters

out	<i>MemoryAddress</i>	Memory address to write to
in	<i>uint16Value</i>	Value to write to memory

Return values

<i>CFE_PSP_SUCCESS</i>	on success
<i>CFE_PSP_ERROR_TIMEOUT</i>	write operation did not go through after a specific timeout.
<i>CFE_PSP_ERROR_ADDRESS_MISALIGNED</i>	if the address is not aligned to a 16-bit addressing scheme.
<i>CFE_PSP_ERROR_NOT_IMPLEMENTED</i>	if not implemented

CFE_PSP_EepromWrite32()

```
int32 CFE_PSP_EepromWrite32 (
    cpuaddr MemoryAddress,
    uint32 uint32Value )
```

Write four bytes (uint32Value) to EEPROM address MemoryAddress.

Parameters

out	<i>MemoryAddress</i>	Memory address to write to
in	<i>uint32Value</i>	Value to write to memory

Return values

<i>CFE_PSP_SUCCESS</i>	on success
<i>CFE_PSP_ERROR_TIMEOUT</i>	write operation did not go through after a specific timeout.
<i>CFE_PSP_ERROR_ADDRESS_MISALIGNED</i>	if the address is not aligned to a 16-bit addressing scheme.
<i>CFE_PSP_ERROR_NOT_IMPLEMENTED</i>	if not implemented

CFE_PSP_EepromWrite8()

```
int32 CFE_PSP_EepromWrite8 (
    cpuaddr MemoryAddress,
```

```
uint8 ByteValue )
```

Write one byte (ByteValue) to EEPROM address MemoryAddress.

Parameters

out	<i>MemoryAddress</i>	Memory address to write to
in	<i>ByteValue</i>	Value to write to memory

Return values

<i>CFE_PSP_SUCCESS</i>	on success
<i>CFE_PSP_ERROR_TIMEOUT</i>	write operation did not go through after a specific timeout.
<i>CFE_PSP_ERROR_ADDRESS_MISALIGNED</i>	if the address is not aligned to a 16-bit addressing scheme.
<i>CFE_PSP_ERROR_NOT_IMPLEMENTED</i>	if not implemented

CFE_PSP_EepromWriteDisable()

```
int32 CFE_PSP_EepromWriteDisable (
    uint32 Bank )
```

Disable the EEPROM from write operation.

Parameters

in	<i>Bank</i>	The bank of EEPROM to disable
----	-------------	-------------------------------

Return values

<i>CFE_PSP_SUCCESS</i>	on success
<i>CFE_PSP_ERROR_NOT_IMPLEMENTED</i>	if not implemented

CFE_PSP_EepromWriteEnable()

```
int32 CFE_PSP_EepromWriteEnable (
    uint32 Bank )
```

Enable the EEPROM for write operation.

Parameters

in	<i>Bank</i>	The bank of EEPROM to enable
----	-------------	------------------------------

Return values

<i>CFE_PSP_SUCCESS</i>	on success
<i>CFE_PSP_ERROR_NOT_IMPLEMENTED</i>	if not implemented

11.353 cfe_psp_eeepromaccess_api.h

[Go to the documentation of this file.](#)

```

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00016  * limitations under the License.
00017  *****/
00018
00019 /*
00020 ** Author: A. Cudmore
00021 **
00022 ** Purpose: This file contains the cFE Platform Support Package (PSP)
00023 **          prototypes.
00024 **          The PSP routines serve as the "glue" between the RTOS and
00025 **          the cFE Flight Software. The routines fill gaps that are not
00026 **          really considered part of the OS Abstraction, but are required
00027 **          for the cFE flight software implementation. It is possible that
00028 **          some of these routines could migrate into the OS AL.
00029 **
00030 */
00031
00032 #ifndef CFE_PSP_EEPROMACCESS_API_H
00033 #define CFE_PSP_EEPROMACCESS_API_H
00034
00035 /*****
00036  INCLUDE FILES
00037  *****/
00038
00039 #include "common_types.h"
00040 #include "osapi.h"
00041
00042 #include "cfe_psp_error.h"
00043
00044 /*-----*/
00056 int32 CFE_PSP_EepromWrite8(cpuaddr MemoryAddress, uint8 ByteValue);
00057
00058 /*-----*/
00070 int32 CFE_PSP_EepromWrite16(cpuaddr MemoryAddress, uint16 uint16Value);
00071
00072 /*-----*/
00084 int32 CFE_PSP_EepromWrite32(cpuaddr MemoryAddress, uint32 uint32Value);
00085
00086 /*-----*/
00095 int32 CFE_PSP_EepromWriteEnable(uint32 Bank);
00096
00097 /*-----*/
00106 int32 CFE_PSP_EepromWriteDisable(uint32 Bank);
00107
00108 /*-----*/
00117 int32 CFE_PSP_EepromPowerUp(uint32 Bank);
00118
00119 /*-----*/
00128 int32 CFE_PSP_EepromPowerDown(uint32 Bank);
00129
00130 #endif

```

11.354 psp/fsw/inc/cfe_psp_endian.h File Reference

```
#include "common_types.h"
```

Functions

- [uint16 CFE_PSP_HtoBE16](#) ([uint16](#) host_16bits)
- [uint16 CFE_PSP_HtoLE16](#) ([uint16](#) host_16bits)
- [uint16 CFE_PSP_BE16toH](#) ([uint16](#) big_endian_16bits)
- [uint16 CFE_PSP_LE16toH](#) ([uint16](#) little_endian_16bits)
- [uint32 CFE_PSP_HtoBE32](#) ([uint32](#) host_32bits)
- [uint32 CFE_PSP_HtoLE32](#) ([uint32](#) host_32bits)
- [uint32 CFE_PSP_BE32toH](#) ([uint32](#) big_endian_32bits)
- [uint32 CFE_PSP_LE32toH](#) ([uint32](#) little_endian_32bits)
- [uint64 CFE_PSP_HtoBE64](#) ([uint64](#) host_64bits)
- [uint64 CFE_PSP_HtoLE64](#) ([uint64](#) host_64bits)
- [uint64 CFE_PSP_BE64toH](#) ([uint64](#) big_endian_64bits)
- [uint64 CFE_PSP_LE64toH](#) ([uint64](#) little_endian_64bits)
- [bool CFE_PSP_IsBigEndian](#) (void)
- [bool CFE_PSP_IsLittleEndian](#) (void)

11.354.1 Detailed Description

Map the PSP endian conversion routines to the system-provided endian.h file This is done using static inline functions, such that the result should be optimized as much as possible.

Definition in file [cfe_psp_endian.h](#).

11.354.2 Function Documentation

CFE_PSP_BE16toH()

```
uint16 CFE_PSP_BE16toH (  
    uint16 big_endian_16bits )
```

CFE_PSP_BE32toH()

```
uint32 CFE_PSP_BE32toH (  
    uint32 big_endian_32bits )
```

CFE_PSP_BE64toH()

```
uint64 CFE_PSP_BE64toH (  
    uint64 big_endian_64bits )
```

CFE_PSP_HtoBE16()

```
uint16 CFE_PSP_HtoBE16 (  
    uint16 host_16bits )
```

CFE_PSP_HtoBE32()

```
uint32 CFE_PSP_HtoBE32 (  
    uint32 host_32bits )
```

CFE_PSP_HtoBE64()

```
uint64 CFE_PSP_HtoBE64 (  
    uint64 host_64bits )
```

CFE_PSP_HtoLE16()

```
uint16 CFE_PSP_HtoLE16 (
    uint16 host_16bits )
```

CFE_PSP_HtoLE32()

```
uint32 CFE_PSP_HtoLE32 (
    uint32 host_32bits )
```

CFE_PSP_HtoLE64()

```
uint64 CFE_PSP_HtoLE64 (
    uint64 host_64bits )
```

CFE_PSP_IsBigEndian()

```
bool CFE_PSP_IsBigEndian (
    void )
```

CFE_PSP_IsLittleEndian()

```
bool CFE_PSP_IsLittleEndian (
    void )
```

CFE_PSP_LE16toH()

```
uint16 CFE_PSP_LE16toH (
    uint16 little_endian_16bits )
```

CFE_PSP_LE32toH()

```
uint32 CFE_PSP_LE32toH (
    uint32 little_endian_32bits )
```

CFE_PSP_LE64toH()

```
uint64 CFE_PSP_LE64toH (
    uint64 little_endian_64bits )
```

11.355 cfe_psp_endian.h

[Go to the documentation of this file.](#)

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00016  * limitations under the License.
00017  *****/
00018
00027 #ifndef CFE_PSP_ENDIAN_H
```



```

00028 #define CFE_PSP_ENDIAN_H
00029
00030 /*
00031 ** Include Files
00032 */
00033 #include "common_types.h"
00034
00035 uint16 CFE_PSP_HtoBE16(uint16 host_16bits);
00036 uint16 CFE_PSP_HtoLE16(uint16 host_16bits);
00037 uint16 CFE_PSP_BE16toH(uint16 big_endian_16bits);
00038 uint16 CFE_PSP_LE16toH(uint16 little_endian_16bits);
00039
00040 uint32 CFE_PSP_HtoBE32(uint32 host_32bits);
00041 uint32 CFE_PSP_HtoLE32(uint32 host_32bits);
00042 uint32 CFE_PSP_BE32toH(uint32 big_endian_32bits);
00043 uint32 CFE_PSP_LE32toH(uint32 little_endian_32bits);
00044
00045 uint64 CFE_PSP_HtoBE64(uint64 host_64bits);
00046 uint64 CFE_PSP_HtoLE64(uint64 host_64bits);
00047 uint64 CFE_PSP_BE64toH(uint64 big_endian_64bits);
00048 uint64 CFE_PSP_LE64toH(uint64 little_endian_64bits);
00049
00050 bool CFE_PSP_IsBigEndian(void);
00051 bool CFE_PSP_IsLittleEndian(void);
00052
00053 #endif /* CFE_PSP_ENDIAN_H */

```

11.356 psp/fsw/inc/cfe_psp_error.h File Reference

cFE PSP Error header

```
#include "common_types.h"
```

Macros

- #define [CFE_PSP_STATUS_C\(X\)](#) ((CFE_PSP_Status_t)(X))
PSP Status macro for literal.
- #define [CFE_PSP_STATUS_STRING_LENGTH](#) 12
PSP Status converted to string length limit.
- #define [CFE_PSP_SUCCESS](#) (CFE_PSP_STATUS_C(0))
- #define [CFE_PSP_ERROR](#) (CFE_PSP_STATUS_C(-1))
- #define [CFE_PSP_INVALID_POINTER](#) (CFE_PSP_STATUS_C(-2))
- #define [CFE_PSP_ERROR_ADDRESS_MISALIGNED](#) (CFE_PSP_STATUS_C(-3))
- #define [CFE_PSP_ERROR_TIMEOUT](#) (CFE_PSP_STATUS_C(-4))
- #define [CFE_PSP_INVALID_INT_NUM](#) (CFE_PSP_STATUS_C(-5))
- #define [CFE_PSP_INVALID_MEM_ADDR](#) (CFE_PSP_STATUS_C(-21))
- #define [CFE_PSP_INVALID_MEM_TYPE](#) (CFE_PSP_STATUS_C(-22))
- #define [CFE_PSP_INVALID_MEM_RANGE](#) (CFE_PSP_STATUS_C(-23))
- #define [CFE_PSP_INVALID_MEM_WORDSIZE](#) (CFE_PSP_STATUS_C(-24))
- #define [CFE_PSP_INVALID_MEM_SIZE](#) (CFE_PSP_STATUS_C(-25))
- #define [CFE_PSP_INVALID_MEM_ATTR](#) (CFE_PSP_STATUS_C(-26))
- #define [CFE_PSP_ERROR_NOT_IMPLEMENTED](#) (CFE_PSP_STATUS_C(-27))
- #define [CFE_PSP_INVALID_MODULE_NAME](#) (CFE_PSP_STATUS_C(-28))
- #define [CFE_PSP_INVALID_MODULE_ID](#) (CFE_PSP_STATUS_C(-29))
- #define [CFE_PSP_NO_EXCEPTION_DATA](#) (CFE_PSP_STATUS_C(-30))

Typedefs

- typedef [int32 CFE_PSP_Status_t](#)
PSP Status type for readability and potentially type safety.
- typedef char [CFE_PSP_StatusString_t](#)[[CFE_PSP_STATUS_STRING_LENGTH](#)]
For the [CFE_PSP_StatusToString\(\)](#) function, to ensure everyone is making an array of the same length.

Functions

- char * [CFE_PSP_StatusToString](#) (CFE_PSP_Status_t status, CFE_PSP_StatusString_t *status_string)
Convert status to a string.

11.356.1 Detailed Description

cFE PSP Error header

Definition in file [cfe_psp_error.h](#).

11.356.2 Macro Definition Documentation

CFE_PSP_ERROR

```
#define CFE_PSP_ERROR (CFE_PSP_STATUS_C(-1))
```

Definition at line 67 of file [cfe_psp_error.h](#).

CFE_PSP_ERROR_ADDRESS_MISALIGNED

```
#define CFE_PSP_ERROR_ADDRESS_MISALIGNED (CFE_PSP_STATUS_C(-3))
```

Definition at line 69 of file [cfe_psp_error.h](#).

CFE_PSP_ERROR_NOT_IMPLEMENTED

```
#define CFE_PSP_ERROR_NOT_IMPLEMENTED (CFE_PSP_STATUS_C(-27))
```

Definition at line 78 of file [cfe_psp_error.h](#).

CFE_PSP_ERROR_TIMEOUT

```
#define CFE_PSP_ERROR_TIMEOUT (CFE_PSP_STATUS_C(-4))
```

Definition at line 70 of file [cfe_psp_error.h](#).

CFE_PSP_INVALID_INT_NUM

```
#define CFE_PSP_INVALID_INT_NUM (CFE_PSP_STATUS_C(-5))
```

Definition at line 71 of file [cfe_psp_error.h](#).

CFE_PSP_INVALID_MEM_ADDR

```
#define CFE_PSP_INVALID_MEM_ADDR (CFE_PSP_STATUS_C(-21))
```

Definition at line 72 of file [cfe_psp_error.h](#).

CFE_PSP_INVALID_MEM_ATTR

```
#define CFE_PSP_INVALID_MEM_ATTR (CFE_PSP_STATUS_C(-26))
```

Definition at line 77 of file [cfe_psp_error.h](#).

CFE_PSP_INVALID_MEM_RANGE

```
#define CFE_PSP_INVALID_MEM_RANGE (CFE_PSP_STATUS_C(-23))
```

Definition at line 74 of file [cfe_psp_error.h](#).

CFE_PSP_INVALID_MEM_SIZE

```
#define CFE_PSP_INVALID_MEM_SIZE (CFE_PSP_STATUS_C(-25))
```

Definition at line 76 of file [cfe_psp_error.h](#).

CFE_PSP_INVALID_MEM_TYPE

```
#define CFE_PSP_INVALID_MEM_TYPE (CFE_PSP_STATUS_C(-22))
```

Definition at line 73 of file [cfe_psp_error.h](#).

CFE_PSP_INVALID_MEM_WORDSIZE

```
#define CFE_PSP_INVALID_MEM_WORDSIZE (CFE_PSP_STATUS_C(-24))
```

Definition at line 75 of file [cfe_psp_error.h](#).

CFE_PSP_INVALID_MODULE_ID

```
#define CFE_PSP_INVALID_MODULE_ID (CFE_PSP_STATUS_C(-29))
```

Definition at line 80 of file [cfe_psp_error.h](#).

CFE_PSP_INVALID_MODULE_NAME

```
#define CFE_PSP_INVALID_MODULE_NAME (CFE_PSP_STATUS_C(-28))
```

Definition at line 79 of file [cfe_psp_error.h](#).

CFE_PSP_INVALID_POINTER

```
#define CFE_PSP_INVALID_POINTER (CFE_PSP_STATUS_C(-2))
```

Definition at line 68 of file [cfe_psp_error.h](#).

CFE_PSP_NO_EXCEPTION_DATA

```
#define CFE_PSP_NO_EXCEPTION_DATA (CFE_PSP_STATUS_C(-30))
```

Definition at line 81 of file [cfe_psp_error.h](#).

CFE_PSP_STATUS_C

```
#define CFE_PSP_STATUS_C(  
    X ) ((CFE_PSP_Status_t) (X))
```

PSP Status macro for literal.

Definition at line 37 of file [cfe_psp_error.h](#).

CFE_PSP_STATUS_STRING_LENGTH

```
#define CFE_PSP_STATUS_STRING_LENGTH 12
```

PSP Status converted to string length limit.

Used for sizing CFE_PSP_StatusString_t intended for use in printing CFE_PSP_Status_t values Sized for `ld (LONG↵_MIN)` including NULL

Definition at line 45 of file [cfe_psp_error.h](#).

CFE_PSP_SUCCESS

```
#define CFE_PSP_SUCCESS (CFE_PSP_STATUS_C(0))
```

Definition at line 66 of file [cfe_psp_error.h](#).

11.356.3 Typedef Documentation**CFE_PSP_Status_t**

```
typedef int32 CFE_PSP_Status_t
```

PSP Status type for readability and potentially type safety.

Definition at line 32 of file [cfe_psp_error.h](#).

CFE_PSP_StatusString_t

```
typedef char CFE_PSP_StatusString_t[CFE_PSP_STATUS_STRING_LENGTH]
```

For the [CFE_PSP_StatusToString\(\)](#) function, to ensure everyone is making an array of the same length.

Definition at line 51 of file [cfe_psp_error.h](#).

11.356.4 Function Documentation**CFE_PSP_StatusToString()**

```
char * CFE_PSP_StatusToString (
    CFE_PSP_Status_t status,
    CFE_PSP_StatusString_t * status_string )
```

Convert status to a string.

Parameters

in	<i>status</i>	Status value to convert
out	<i>status_string</i>	Buffer to store status converted to string

Returns

Passed in string pointer

11.357 cfe_psp_error.h

[Go to the documentation of this file.](#)

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00016  * limitations under the License.
00017  *****/
00018
00019 #ifndef CFE_PSP_ERROR_H
00020 #define CFE_PSP_ERROR_H
00021
00022 #include "common_types.h"
00023
00032 typedef int32 CFE_PSP_Status_t;
00033
00037 #define CFE_PSP_STATUS_C(X) ((CFE_PSP_Status_t)(X))
00038
00045 #define CFE_PSP_STATUS_STRING_LENGTH 12
00046
00051 typedef char CFE_PSP_StatusString_t[CFE_PSP_STATUS_STRING_LENGTH];
00052
00061 char *CFE_PSP_StatusToString(CFE_PSP_Status_t status, CFE_PSP_StatusString_t *status_string);
00062
00063 /*
00064  * Error and return codes
00065  */
00066 #define CFE_PSP_SUCCESS (CFE_PSP_STATUS_C(0))
00067 #define CFE_PSP_ERROR (CFE_PSP_STATUS_C(-1))
00068 #define CFE_PSP_INVALID_POINTER (CFE_PSP_STATUS_C(-2))
00069 #define CFE_PSP_ERROR_ADDRESS_MISALIGNED (CFE_PSP_STATUS_C(-3))
00070 #define CFE_PSP_ERROR_TIMEOUT (CFE_PSP_STATUS_C(-4))
00071 #define CFE_PSP_INVALID_INT_NUM (CFE_PSP_STATUS_C(-5))
```

```

00072 #define CFE_PSP_INVALID_MEM_ADDR      (CFE_PSP_STATUS_C(-21))
00073 #define CFE_PSP_INVALID_MEM_TYPE      (CFE_PSP_STATUS_C(-22))
00074 #define CFE_PSP_INVALID_MEM_RANGE      (CFE_PSP_STATUS_C(-23))
00075 #define CFE_PSP_INVALID_MEM_WORDSIZE   (CFE_PSP_STATUS_C(-24))
00076 #define CFE_PSP_INVALID_MEM_SIZE      (CFE_PSP_STATUS_C(-25))
00077 #define CFE_PSP_INVALID_MEM_ATTR      (CFE_PSP_STATUS_C(-26))
00078 #define CFE_PSP_ERROR_NOT_IMPLEMENTED   (CFE_PSP_STATUS_C(-27))
00079 #define CFE_PSP_INVALID_MODULE_NAME     (CFE_PSP_STATUS_C(-28))
00080 #define CFE_PSP_INVALID_MODULE_ID      (CFE_PSP_STATUS_C(-29))
00081 #define CFE_PSP_NO_EXCEPTION_DATA      (CFE_PSP_STATUS_C(-30))
00082
00083 #endif /* CFE_PSP_ERROR_H */

```

11.358 psp/fsw/inc/cfe_psp_exception_api.h File Reference

```

#include "common_types.h"
#include "osapi.h"
#include "cfe_psp_error.h"

```

Functions

- void [CFE_PSP_AttachExceptions](#) (void)
Sets up the exception environment for the chosen platform.
- void [CFE_PSP_SetDefaultExceptionEnvironment](#) (void)
Defines the CPU and FPU exceptions that are enabled for each cFE Task/App.
- [uint32 CFE_PSP_Exception_GetCount](#) (void)
Returns the unread exception count.
- [int32 CFE_PSP_Exception_GetSummary](#) ([uint32](#) *ContextLogId, [osal_id_t](#) *TaskId, char *ReasonBuf, [uint32](#) ReasonSize)
Retrieves a summary of an exception log entry.
- [int32 CFE_PSP_Exception_CopyContext](#) ([uint32](#) ContextLogId, void *ContextBuf, [uint32](#) ContextSize)
Retrieves exception log entry context information.

11.358.1 Function Documentation

CFE_PSP_AttachExceptions()

```

void CFE_PSP_AttachExceptions (
    void )

```

Sets up the exception environment for the chosen platform.

On a board, this can be configured to look at a debug flag or switch in order to keep the standard OS exception handlers, rather than restarting the system.

CFE_PSP_Exception_CopyContext()

```

int32 CFE_PSP_Exception_CopyContext (
    uint32 ContextLogId,
    void * ContextBuf,
    uint32 ContextSize )

```

Retrieves exception log entry context information.

Parameters

in	<i>ContextLogId</i>	ID of the exception log entry to copy
out	<i>ContextBuf</i>	Pointer to the buffer where the context information is to be copied to
in	<i>ContextSize</i>	Maximum size of context information data to copy

Returns

Size of the copied data

Return values

<code>CFE_PSP_NO_EXCEPTION_DATA</code>	if data has expired from the memory log
--	---

CFE_PSP_Exception_GetCount()

```
uint32 CFE_PSP_Exception_GetCount (
    void )
```

Returns the unread exception count.

Returns

The unread exception count

CFE_PSP_Exception_GetSummary()

```
int32 CFE_PSP_Exception_GetSummary (
    uint32 * ContextLogId,
    osal_id_t * TaskId,
    char * ReasonBuf,
    uint32 ReasonSize )
```

Retrieves a summary of an exception log entry.

Note

This function returns `CFE_PSP_SUCCESS` to indicate that an entry was popped from the queue. This doesn't necessarily mean that the output fields have valid data, but it does mean they are initialized to something.

Parameters

in	<i>ContextLogId</i>	ID of the exception log entry to get a summary for
in, out	<i>TaskId</i>	Pointer to the TaskID buffer
out	<i>ReasonBuf</i>	Pointer to the buffer that will store the exception summary string
in	<i>ReasonSize</i>	Maximum size of the summary string to retrieve

Return values

<code>CFE_PSP_SUCCESS</code>	on success (see note above)
<code>CFE_PSP_NO_EXCEPTION_DATA</code>	if no context available for reading

CFE_PSP_SetDefaultExceptionEnvironment()

```
void CFE_PSP_SetDefaultExceptionEnvironment (
    void )
```

Defines the CPU and FPU exceptions that are enabled for each cFE Task/App.

This function sets a default exception environment that can be used

Note

The exception environment is local to each task. Therefore, this must be Called for each task that wants to do floating point and catch exceptions.

11.359 cfe_psp_exception_api.h

[Go to the documentation of this file.](#)

```

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00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00019 /*
00020 ** Author: A. Cudmore
00021 **
00022 ** Purpose: This file contains the cFE Platform Support Package (PSP)
00023 **          prototypes.
00024 **          The PSP routines serve as the "glue" between the RTOS and
00025 **          the cFE Flight Software. The routines fill gaps that are not
00026 **          really considered part of the OS Abstraction, but are required
00027 **          for the cFE flight software implementation. It is possible that
00028 **          some of these routines could migrate into the OS AL.
00029 **
00030 */
00031
00032 #ifndef CFE_PSP_EXCEPTION_API_H
00033 #define CFE_PSP_EXCEPTION_API_H
00034
00035 /*****
00036  INCLUDE FILES
00037  *****/
00038
00039 #include "common_types.h"
00040 #include "osapi.h"
00041
00042 #include "cfe_psp_error.h"
00043
00044 /*-----*/
00051 void CFE_PSP_AttachExceptions(void);
00052
00053 /*-----*/
00062 void CFE_PSP_SetDefaultExceptionEnvironment(void);
00063
00064 /*-----*/
00070 uint32 CFE_PSP_Exception_GetCount(void);
00071
00072 /*-----*/
00088 int32 CFE_PSP_Exception_GetSummary(uint32 *ContextLogId, osal_id_t *TaskId, char *ReasonBuf, uint32
ReasonSize);
00089
00090 /*-----*/
00101 int32 CFE_PSP_Exception_CopyContext(uint32 ContextLogId, void *ContextBuf, uint32 ContextSize);
00102
00103 #endif

```

11.360 psp/fsw/inc/cfe_psp_id_api.h File Reference

```

#include "common_types.h"
#include "osapi.h"

```

```
#include "cfe_psp_error.h"
```

Functions

- [uint32 CFE_PSP_GetProcessorId](#) (void)
Returns the CPU ID as defined by the specific board and BSP.
- [uint32 CFE_PSP_GetSpacecraftId](#) (void)
Returns the Spacecraft ID (if any)
- `const char *` [CFE_PSP_GetProcessorName](#) (void)
Returns the processor name.

11.360.1 Function Documentation

CFE_PSP_GetProcessorId()

```
uint32 CFE_PSP_GetProcessorId (
    void )
```

Returns the CPU ID as defined by the specific board and BSP.

Returns

The processor ID

CFE_PSP_GetProcessorName()

```
const char * CFE_PSP_GetProcessorName (
    void )
```

Returns the processor name.

Returns

The processor name

CFE_PSP_GetSpacecraftId()

```
uint32 CFE_PSP_GetSpacecraftId (
    void )
```

Returns the Spacecraft ID (if any)

Returns

The Spacecraft ID

11.361 cfe_psp_id_api.h

[Go to the documentation of this file.](#)

```
00001 /*****
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"
00003  *
00004  * Copyright (c) 2023 United States Government as represented by the
00005  * Administrator of the National Aeronautics and Space Administration.
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```



```

00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00019  /*
00020  ** Author: A. Cudmore
00021  **
00022  ** Purpose: This file contains the cFE Platform Support Package(PSP)
00023  **           prototypes.
00024  **           The PSP routines serve as the "glue" between the RTOS and
00025  **           the cFE Flight Software. The routines fill gaps that are not
00026  **           really considered part of the OS Abstraction, but are required
00027  **           for the cFE flight software implementation. It is possible that
00028  **           some of these routines could migrate into the OS AL.
00029  **
00030  */
00031
00032 #ifndef CFE_PSP_ID_API_H
00033 #define CFE_PSP_ID_API_H
00034
00035 /*****
00036  INCLUDE FILES
00037  *****/
00038
00039 #include "common_types.h"
00040 #include "osapi.h"
00041
00042 #include "cfe_psp_error.h"
00043
00044 /*****
00045  FUNCTION PROTOTYPES
00046  *****/
00047
00048 /*-----*/
00054 uint32 CFE_PSP_GetProcessorId(void);
00055
00056 /*-----*/
00062 uint32 CFE_PSP_GetSpacecraftId(void);
00063
00064 /*-----*/
00070 const char *CFE_PSP_GetProcessorName(void);
00071
00072 #endif

```

11.362 psp/fsw/inc/cfe_psp_memaccess_api.h File Reference

```

#include "common_types.h"
#include "osapi.h"
#include "cfe_psp_error.h"

```

Functions

- [int32 CFE_PSP_MemRead8](#) ([cpuaddr](#) MemoryAddress, [uint8](#) *ByteValue)
Read one byte of memory.
- [int32 CFE_PSP_MemWrite8](#) ([cpuaddr](#) MemoryAddress, [uint8](#) ByteValue)
Write one byte of memory.
- [int32 CFE_PSP_MemRead16](#) ([cpuaddr](#) MemoryAddress, [uint16](#) *uint16Value)
Read 2 bytes of memory.
- [int32 CFE_PSP_MemWrite16](#) ([cpuaddr](#) MemoryAddress, [uint16](#) uint16Value)
Write 2 bytes of memory.
- [int32 CFE_PSP_MemRead32](#) ([cpuaddr](#) MemoryAddress, [uint32](#) *uint32Value)
Read 4 bytes of memory.
- [int32 CFE_PSP_MemWrite32](#) ([cpuaddr](#) MemoryAddress, [uint32](#) uint32Value)
Write 4 bytes of memory.
- [int32 CFE_PSP_MemCpy](#) (void *dest, const void *src, [uint32](#) n)

Copy 'n' bytes from 'src' to 'dest'.

- `int32 CFE_PSP_MemSet (void *dest, uint8 value, uint32 n)`

Copy 'n' bytes of value 'value' to 'dest'.

11.362.1 Function Documentation

CFE_PSP_MemCpy()

```
int32 CFE_PSP_MemCpy (
    void * dest,
    const void * src,
    uint32 n )
```

Copy 'n' bytes from 'src' to 'dest'.

Copies 'n' bytes from memory address pointed to by 'src' to memory address pointed to by 'dest'.

Note

For now we are using the standard C library call 'memcpy' but if we find we need to make it more efficient then we'll implement it in assembly.

Parameters

out	<i>dest</i>	Pointer to the destination address to copy to
in	<i>src</i>	Pointer to the address to copy from
in	<i>n</i>	Number of bytes to copy

Returns

Always returns CFE_PSP_SUCCESS

CFE_PSP_MemRead16()

```
int32 CFE_PSP_MemRead16 (
    cpuaddr MemoryAddress,
    uint16 * uint16Value )
```

Read 2 bytes of memory.

Parameters

in	<i>MemoryAddress</i>	Address to be read
out	<i>uint16Value</i>	The address content will be copied to the location pointed to by this argument

Return values

<i>CFE_PSP_SUCCESS</i>	on success
<i>CFE_PSP_ERROR_ADDRESS_MISALIGNED</i>	if the address is not aligned to a 16-bit addressing scheme.
<i>CFE_PSP_ERROR_NOT_IMPLEMENTED</i>	if not implemented

CFE_PSP_MemRead32()

```
int32 CFE_PSP_MemRead32 (
    cpuaddr MemoryAddress,
    uint32 * uint32Value )
```

Read 4 bytes of memory.

Parameters

in	<i>MemoryAddress</i>	Address to be read
out	<i>uint32Value</i>	The address content will be copied to the location pointed to by this argument

Return values

<i>CFE_PSP_SUCCESS</i>	on success
<i>CFE_PSP_ERROR_ADDRESS_MISALIGNED</i>	if the address is not aligned to a 16-bit addressing scheme.
<i>CFE_PSP_ERROR_NOT_IMPLEMENTED</i>	if not implemented

CFE_PSP_MemRead8()

```
int32 CFE_PSP_MemRead8 (
    cpuaddr MemoryAddress,
    uint8 * ByteValue )
```

Read one byte of memory.

Parameters

in	<i>MemoryAddress</i>	Address to be read
out	<i>ByteValue</i>	The address content will be copied to the location pointed to by this argument

Returns

Always returns CFE_PSP_SUCCESS (if implemented)

Return values

<i>CFE_PSP_ERROR_NOT_IMPLEMENTED</i>	if not implemented
--------------------------------------	--------------------

CFE_PSP_MemSet()

```
int32 CFE_PSP_MemSet (
    void * dest,
    uint8 value,
    uint32 n )
```

Copy 'n' bytes of value 'value' to 'dest'.

Copies 'n' number of bytes of value 'value' to memory address pointed to by 'dest'.

Note

For now we are using the standard C library call 'memset' but if we find we need to make it more efficient then we'll implement it in assembly.

Parameters

out	<i>dest</i>	Pointer to the destination address to copy to
in	<i>value</i>	Value to set
in	<i>n</i>	Number of bytes to copy

Returns

Always returns CFE_PSP_SUCCESS

CFE_PSP_MemWrite16()

```
int32 CFE_PSP_MemWrite16 (
    cpuaddr MemoryAddress,
    uint16 uint16Value )
```

Write 2 bytes of memory.

Parameters

out	<i>MemoryAddress</i>	Address to be written to
in	<i>uint16Value</i>	The content pointed to by this argument will be copied to the address

Return values

<i>CFE_PSP_SUCCESS</i>	on success
<i>CFE_PSP_ERROR_ADDRESS_MISALIGNED</i>	if the address is not aligned to a 16-bit addressing scheme.
<i>CFE_PSP_ERROR_NOT_IMPLEMENTED</i>	if not implemented

CFE_PSP_MemWrite32()

```
int32 CFE_PSP_MemWrite32 (
    cpuaddr MemoryAddress,
    uint32 uint32Value )
```

Write 4 bytes of memory.

Parameters

out	<i>MemoryAddress</i>	Address to be written to
in	<i>uint32Value</i>	The content pointed to by this argument will be copied to the address

Return values

<i>CFE_PSP_SUCCESS</i>	on success
<i>CFE_PSP_ERROR_ADDRESS_MISALIGNED</i>	if the address is not aligned to a 16-bit addressing scheme.

Return values

<code>CFE_PSP_ERROR_NOT_IMPLEMENTED</code>	if not implemented
--	--------------------

CFE_PSP_MemWrite8()

```
int32 CFE_PSP_MemWrite8 (
    cpuaddr MemoryAddress,
    uint8 ByteValue )
```

Write one byte of memory.

Parameters

out	<i>MemoryAddress</i>	Address to be written to
in	<i>ByteValue</i>	The content pointed to by this argument will be copied to the address

Returns

Always returns `CFE_PSP_SUCCESS` (if implemented)

Return values

<code>CFE_PSP_ERROR_NOT_IMPLEMENTED</code>	if not implemented
--	--------------------

11.363 cfe_psp_memaccess_api.h

[Go to the documentation of this file.](#)

```
00001 /*****
00002  * NASA Docket No. GSC-19,200-1, and identified as "cFS Draco"
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00008  * Licensed under the Apache License, Version 2.0 (the "License"); you may
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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00019 /*
00020 ** Author: A. Cudmore
00021 **
00022 ** Purpose: This file contains the cFE Platform Support Package (PSP)
00023 **           prototypes.
00024 **           The PSP routines serve as the "glue" between the RTOS and
00025 **           the cFE Flight Software. The routines fill gaps that are not
00026 **           really considered part of the OS Abstraction, but are required
00027 **           for the cFE flight software implementation. It is possible that
00028 **           some of these routines could migrate into the OS AL.
00029 **
00030 */
00031
00032 #ifndef CFE_PSP_MEMACCESS_API_H
00033 #define CFE_PSP_MEMACCESS_API_H
00034
00035 /*****
```

```

00036  INCLUDE FILES
00037  *****/
00038
00039  #include "common_types.h"
00040  #include "osapi.h"
00041
00042  #include "cfe_psp_error.h"
00043
00044  /*
00045  ** Memory API
00046  */
00047  /*-----*/
00057  int32 CFE_PSP_MemRead8(cpuaddr MemoryAddress, uint8 *ByteValue);
00058
00059  /*-----*/
00069  int32 CFE_PSP_MemWrite8(cpuaddr MemoryAddress, uint8 ByteValue);
00070
00071  /*-----*/
00082  int32 CFE_PSP_MemRead16(cpuaddr MemoryAddress, uint16 *uint16Value);
00083
00084  /*-----*/
00095  int32 CFE_PSP_MemWrite16(cpuaddr MemoryAddress, uint16 uint16Value);
00096
00097  /*-----*/
00108  int32 CFE_PSP_MemRead32(cpuaddr MemoryAddress, uint32 *uint32Value);
00109
00110  /*-----*/
00121  int32 CFE_PSP_MemWrite32(cpuaddr MemoryAddress, uint32 uint32Value);
00122
00123  /*-----*/
00139  int32 CFE_PSP_MemCpy(void *dest, const void *src, uint32 n);
00140
00141  /*-----*/
00157  int32 CFE_PSP_MemSet(void *dest, uint8 value, uint32 n);
00158
00159  #endif

```

11.364 psp/fsw/inc/cfe_psp_memrange_api.h File Reference

```

#include "common_types.h"
#include "osapi.h"
#include "cfe_psp_error.h"

```

Macros

- #define CFE_PSP_MEM_RAM 1
- #define CFE_PSP_MEM_EEPROM 2
- #define CFE_PSP_MEM_ANY 3
- #define CFE_PSP_MEM_INVALID 4
- #define CFE_PSP_MEM_ATTR_WRITE 0x01
- #define CFE_PSP_MEM_ATTR_READ 0x02
- #define CFE_PSP_MEM_ATTR_READWRITE 0x03
- #define CFE_PSP_MEM_SIZE_BYTE 0x01
- #define CFE_PSP_MEM_SIZE_WORD 0x02
- #define CFE_PSP_MEM_SIZE_DWORD 0x04

Functions

- [int32 CFE_PSP_GetResetArea](#) (cpuaddr *PtrToResetArea, uint32 *SizeOfResetArea)
Returns the location and size of the ES Reset information area.
- [int32 CFE_PSP_GetUserReservedArea](#) (cpuaddr *PtrToUserArea, uint32 *SizeOfUserArea)
Returns the location and size of the memory used for the cFE user-reserved area.
- [int32 CFE_PSP_GetVolatileDiskMem](#) (cpuaddr *PtrToVolDisk, uint32 *SizeOfVolDisk)
Returns the location and size of the memory used for the cFE volatile disk.

- `int32 CFE_PSP_GetKernelTextSegmentInfo (cpuaddr *PtrToKernelSegment, uint32 *SizeOfKernelSegment)`
Returns the location and size of the kernel memory.
- `int32 CFE_PSP_GetCFETextSegmentInfo (cpuaddr *PtrToCFESegment, uint32 *SizeOfCFESegment)`
Returns the location and size of the kernel memory.
- `int32 CFE_PSP_MemValidateRange (cpuaddr Address, size_t Size, uint32 MemoryType)`
Validates the memory range and type using the global CFE_PSP_MemoryTable.
- `uint32 CFE_PSP_MemRanges (void)`
Returns the number of memory ranges in the CFE_PSP_MemoryTable.
- `int32 CFE_PSP_MemRangeSet (uint32 RangeNum, uint32 MemoryType, cpuaddr StartAddr, size_t Size, size_t WordSize, uint32 Attributes)`
Populates one of the records in the CFE_PSP_MemoryTable.
- `int32 CFE_PSP_MemRangeGet (uint32 RangeNum, uint32 *MemoryType, cpuaddr *StartAddr, size_t *Size, size_t *WordSize, uint32 *Attributes)`
Retrieves one of the records in the CFE_PSP_MemoryTable.

11.364.1 Macro Definition Documentation

CFE_PSP_MEM_ANY

```
#define CFE_PSP_MEM_ANY 3
```

Definition at line 53 of file [cfe_psp_memrange_api.h](#).

CFE_PSP_MEM_ATTR_READ

```
#define CFE_PSP_MEM_ATTR_READ 0x02
```

Definition at line 60 of file [cfe_psp_memrange_api.h](#).

CFE_PSP_MEM_ATTR_READWRITE

```
#define CFE_PSP_MEM_ATTR_READWRITE 0x03
```

Definition at line 61 of file [cfe_psp_memrange_api.h](#).

CFE_PSP_MEM_ATTR_WRITE

```
#define CFE_PSP_MEM_ATTR_WRITE 0x01
```

Definition at line 59 of file [cfe_psp_memrange_api.h](#).

CFE_PSP_MEM_EEPROM

```
#define CFE_PSP_MEM_EEPROM 2
```

Definition at line 52 of file [cfe_psp_memrange_api.h](#).

CFE_PSP_MEM_INVALID

```
#define CFE_PSP_MEM_INVALID 4
```

Definition at line 54 of file [cfe_psp_memrange_api.h](#).

CFE_PSP_MEM_RAM

```
#define CFE_PSP_MEM_RAM 1
```

Definition at line 51 of file [cfe_psp_memrange_api.h](#).

CFE_PSP_MEM_SIZE_BYTE

```
#define CFE_PSP_MEM_SIZE_BYTE 0x01
```

Definition at line 66 of file [cfe_psp_memrange_api.h](#).

CFE_PSP_MEM_SIZE_DWORD

```
#define CFE_PSP_MEM_SIZE_DWORD 0x04
```

Definition at line 68 of file [cfe_psp_memrange_api.h](#).

CFE_PSP_MEM_SIZE_WORD

```
#define CFE_PSP_MEM_SIZE_WORD 0x02
```

Definition at line 67 of file [cfe_psp_memrange_api.h](#).

11.364.2 Function Documentation**CFE_PSP_GetCFETextSegmentInfo()**

```
int32 CFE_PSP_GetCFETextSegmentInfo (
    cpuaddr * PtrToCFESegment,
    uint32 * SizeOfCFESegment )
```

Returns the location and size of the kernel memory.

This function returns the start and end address of the CFE text segment. It may not be implemented on all architectures.

Parameters

out	<i>PtrToCFESegment</i>	Pointer to the variable that will store the location of the cFE text segment
out	<i>SizeOfCFESegment</i>	Pointer to the variable that will store the size of the cFE text segment

Returns

0 (OS_SUCCESS or CFE_PSP_SUCCESS) on success, -1 (OS_ERROR or CFE_PSP_ERROR) on error

CFE_PSP_GetKernelTextSegmentInfo()

```
int32 CFE_PSP_GetKernelTextSegmentInfo (
    cpuaddr * PtrToKernelSegment,
    uint32 * SizeOfKernelSegment )
```

Returns the location and size of the kernel memory.

This function returns the start and end address of the kernel text segment. It may not be implemented on all architectures.

Parameters

out	<i>PtrToKernelSegment</i>	Pointer to the variable that will store the location of the kernel text segment
out	<i>SizeOfKernelSegment</i>	Pointer to the variable that will store the size of the kernel text segment

Returns

0 (OS_SUCCESS or CFE_PSP_SUCCESS) on success, -1 (OS_ERROR or CFE_PSP_ERROR) on error or CFE_PSP_ERROR_NOT_IMPLEMENTED if not implemented

CFE_PSP_GetResetArea()

```
int32 CFE_PSP_GetResetArea (
    cpuaddr * PtrToResetArea,
    uint32 * SizeOfResetArea )
```

Returns the location and size of the ES Reset information area.

This area is preserved during a processor reset and is used to store the ER Log, System Log and reset-related variables

Parameters

out	<i>PtrToResetArea</i>	Pointer to the variable that will store the location of the reset area
out	<i>SizeOfResetArea</i>	Pointer to the variable that will store the reset area size

Returns

0 (OS_SUCCESS or CFE_PSP_SUCCESS) on success, -1 (OS_ERROR or CFE_PSP_ERROR) on error

CFE_PSP_GetUserReservedArea()

```
int32 CFE_PSP_GetUserReservedArea (
    cpuaddr * PtrToUserArea,
    uint32 * SizeOfUserArea )
```

Returns the location and size of the memory used for the cFE user-reserved area.

Parameters

out	<i>PtrToUserArea</i>	Pointer to the variable that will store the location of the user-reserved area
out	<i>SizeOfUserArea</i>	Pointer to the variable that will store the size of the user-reserved area

Returns

0 (OS_SUCCESS or CFE_PSP_SUCCESS) on success, -1 (OS_ERROR or CFE_PSP_ERROR) on error

CFE_PSP_GetVolatileDiskMem()

```
int32 CFE_PSP_GetVolatileDiskMem (
    cpuaddr * PtrToVolDisk,
    uint32 * SizeOfVolDisk )
```

Returns the location and size of the memory used for the cFE volatile disk.

Parameters

out	<i>PtrToVolDisk</i>	Pointer to the variable that will store the location of the cFE volatile disk
out	<i>SizeOfVolDisk</i>	Pointer to the variable that will store the size of the cFE volatile disk

Returns

0 (OS_SUCCESS or CFE_PSP_SUCCESS) on success, -1 (OS_ERROR or CFE_PSP_ERROR) on error

CFE_PSP_MemRangeGet()

```
int32 CFE_PSP_MemRangeGet (
    uint32 RangeNum,
```

```

uint32 * MemoryType,
cpuaddr * StartAddr,
size_t * Size,
size_t * WordSize,
uint32 * Attributes )

```

Retrieves one of the records in the CFE_PSP_MemoryTable.

Note

Because the table is fixed size, the entries are accessed by using the integer index.

Parameters

in	<i>RangeNum</i>	A 32-bit integer (starting with 0) specifying the MemoryTable entry.
out	<i>MemoryType</i>	A pointer to the 32-bit integer where the Memory Type is stored. Any defined CFE_PSP_MEM_* enumeration can be specified
out	<i>StartAddr</i>	Pointer to the 32-bit integer where the 32-bit starting address of the memory range is stored.
out	<i>Size</i>	A pointer to the 32-bit integer where the 32-bit size of the memory range is stored.
out	<i>WordSize</i>	A pointer to the 32-bit integer where the minimum addressable size of the range: (CFE_PSP_MEM_SIZE_BYTE, CFE_PSP_MEM_SIZE_WORD, CFE_PSP_MEM_SIZE_DWORD)
out	<i>Attributes</i>	The attributes of the Memory Range: (CFE_PSP_MEM_ATTR_WRITE, CFE_PSP_MEM_ATTR_READ, CFE_PSP_MEM_ATTR_READWRITE)

Return values

<i>CFE_PSP_SUCCESS</i>	Memory range returned successfully.
<i>CFE_PSP_INVALID_POINTER</i>	Parameter error
<i>CFE_PSP_INVALID_MEM_RANGE</i>	The index into the table is invalid

CFE_PSP_MemRanges()

```

uint32 CFE_PSP_MemRanges (
    void )

```

Returns the number of memory ranges in the CFE_PSP_MemoryTable.

Returns

Positive integer number of entries in the memory range table

CFE_PSP_MemRangeSet()

```

int32 CFE_PSP_MemRangeSet (
    uint32 RangeNum,
    uint32 MemoryType,
    cpuaddr StartAddr,
    size_t Size,
    size_t WordSize,
    uint32 Attributes )

```

Populates one of the records in the CFE_PSP_MemoryTable.

Note

Because the table is fixed size, the entries are set by using the integer index. No validation is done with the address or size.

Parameters

in	<i>RangeNum</i>	A 32-bit integer (starting with 0) specifying the MemoryTable entry.
in	<i>MemoryType</i>	The memory type to validate, including but not limited to: CFE_PSP_MEM_RAM, CFE_PSP_MEM_EEPROM, or CFE_PSP_MEM_ANY Any defined CFE_PSP_MEM_* enumeration can be specified
in	<i>StartAddr</i>	A 32-bit starting address of the memory range
in	<i>Size</i>	A 32-bit size of the memory range (Address + Size = End Address)
in	<i>WordSize</i>	The minimum addressable size of the range: (CFE_PSP_MEM_SIZE_BYTE, CFE_PSP_MEM_SIZE_WORD, CFE_PSP_MEM_SIZE_DWORD)
in	<i>Attributes</i>	The attributes of the Memory Range: (CFE_PSP_MEM_ATTR_WRITE, CFE_PSP_MEM_ATTR_READ, CFE_PSP_MEM_ATTR_READWRITE)

Return values

<i>CFE_PSP_SUCCESS</i>	Memory range set successfully.
<i>CFE_PSP_INVALID_MEM_RANGE</i>	The index into the table is invalid
<i>CFE_PSP_INVALID_MEM_ADDR</i>	Starting address is not valid
<i>CFE_PSP_INVALID_MEM_TYPE</i>	Memory type associated with the range does not match the passed-in type.
<i>CFE_PSP_INVALID_MEM_WORDSIZE</i>	The WordSize parameter is not one of the predefined types.
<i>CFE_PSP_INVALID_MEM_ATTR</i>	The Attributes parameter is not one of the predefined types.
<i>OP_INVALID_MEM_SIZE</i>	The Memory range associated with the address is not large enough to contain Address + Size.

CFE_PSP_MemValidateRange()

```
int32 CFE_PSP_MemValidateRange (
    cpuaddr Address,
    size_t Size,
    uint32 MemoryType )
```

Validates the memory range and type using the global CFE_PSP_MemoryTable.

Parameters

in	<i>Address</i>	A 32-bit starting address of the memory range
in	<i>Size</i>	A 32-bit size of the memory range (Address + Size = End Address)
in	<i>MemoryType</i>	The memory type to validate, including but not limited to: CFE_PSP_MEM_RAM, CFE_PSP_MEM_EEPROM, or CFE_PSP_MEM_ANY Any defined CFE_PSP_MEM_* enumeration can be specified

Return values

<i>CFE_PSP_SUCCESS</i>	Memory range and type information is valid and can be used.
------------------------	---

Return values

<i>CFE_PSP_INVALID_MEM_ADDR</i>	Starting address is not valid
<i>CFE_PSP_INVALID_MEM_TYPE</i>	Memory type associated with the range does not match the passed-in type.
<i>CFE_PSP_INVALID_MEM_RANGE</i>	The Memory range associated with the address is not large enough to contain Address + Size.

11.365 cfe_psp_memrange_api.h

[Go to the documentation of this file.](#)

```

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00017  *****/
00018
00019 /*
00020 ** Author: A. Cudmore
00021 **
00022 ** Purpose: This file contains the cFE Platform Support Package(PSP)
00023 **           prototypes.
00024 **           The PSP routines serve as the "glue" between the RTOS and
00025 **           the cFE Flight Software. The routines fill gaps that are not
00026 **           really considered part of the OS Abstraction, but are required
00027 **           for the cFE flight software implementation. It is possible that
00028 **           some of these routines could migrate into the OS AL.
00029 **
00030 */
00031
00032 #ifndef CFE_PSP_MEMRANGE_API_H
00033 #define CFE_PSP_MEMRANGE_API_H
00034
00035 /*****
00036  INCLUDE FILES
00037  *****/
00038
00039 #include "common_types.h"
00040 #include "osapi.h"
00041
00042 #include "cfe_psp_error.h"
00043
00044 /*****
00045  MACRO DEFINITIONS
00046  *****/
00047
00048 /*
00049 ** Define memory types
00050 */
00051 #define CFE_PSP_MEM_RAM 1
00052 #define CFE_PSP_MEM_EEPROM 2
00053 #define CFE_PSP_MEM_ANY 3
00054 #define CFE_PSP_MEM_INVALID 4
00055
00056 /*
00057 ** Define Memory Read/Write Attributes
00058 */
00059 #define CFE_PSP_MEM_ATTR_WRITE 0x01
00060 #define CFE_PSP_MEM_ATTR_READ 0x02
00061 #define CFE_PSP_MEM_ATTR_READWRITE 0x03
00062
00063 /*
00064 ** Define the Memory Word Sizes

```

```

00065 */
00066 #define CFE_PSP_MEM_SIZE_BYTE 0x01
00067 #define CFE_PSP_MEM_SIZE_WORD 0x02
00068 #define CFE_PSP_MEM_SIZE_DWORD 0x04
00069
00070 /*-----*/
00082 int32 CFE_PSP_GetResetArea(cpuaddr *PtrToResetArea, uint32 *SizeOfResetArea);
00083
00084 /*-----*/
00093 int32 CFE_PSP_GetUserReservedArea(cpuaddr *PtrToUserArea, uint32 *SizeOfUserArea);
00094
00095 /*-----*/
00104 int32 CFE_PSP_GetVolatileDiskMem(cpuaddr *PtrToVolDisk, uint32 *SizeOfVolDisk);
00105
00106 /*-----*/
00119 int32 CFE_PSP_GetKernelTextSegmentInfo(cpuaddr *PtrToKernelSegment, uint32 *SizeOfKernelSegment);
00120
00121 /*-----*/
00133 int32 CFE_PSP_GetCFETextSegmentInfo(cpuaddr *PtrToCFESegment, uint32 *SizeOfCFESegment);
00134
00135 /*-----*/
00151 int32 CFE_PSP_MemValidateRange(cpuaddr Address, size_t Size, uint32 MemoryType);
00152
00153 /*-----*/
00159 uint32 CFE_PSP_MemRanges(void);
00160
00161 /*-----*/
00188 int32 CFE_PSP_MemRangeSet(uint32 RangeNum,
00189                          uint32 MemoryType,
00190                          cpuaddr StartAddr,
00191                          size_t Size,
00192                          size_t WordSize,
00193                          uint32 Attributes);
00194
00195 /*-----*/
00215 int32 CFE_PSP_MemRangeGet(uint32 RangeNum,
00216                          uint32 *MemoryType,
00217                          cpuaddr *StartAddr,
00218                          size_t *Size,
00219                          size_t *WordSize,
00220                          uint32 *Attributes);
00221
00222 #endif

```

11.366 psp/fsw/inc/cfe_psp_port_api.h File Reference

```

#include "common_types.h"
#include "osapi.h"
#include "cfe_psp_error.h"

```

Functions

- `int32 CFE_PSP_PortRead8(cpuaddr PortAddress, uint8 *ByteValue)`
Read one byte of memory.
- `int32 CFE_PSP_PortWrite8(cpuaddr PortAddress, uint8 ByteValue)`
Write one byte of memory.
- `int32 CFE_PSP_PortRead16(cpuaddr PortAddress, uint16 *uint16Value)`
Read 2 bytes of memory.
- `int32 CFE_PSP_PortWrite16(cpuaddr PortAddress, uint16 uint16Value)`
Write 2 bytes of memory.
- `int32 CFE_PSP_PortRead32(cpuaddr PortAddress, uint32 *uint32Value)`
Read 4 bytes of memory.
- `int32 CFE_PSP_PortWrite32(cpuaddr PortAddress, uint32 uint32Value)`
Write 4 bytes of memory.

11.366.1 Function Documentation

CFE_PSP_PortRead16()

```
int32 CFE_PSP_PortRead16 (
    cpuaddr PortAddress,
    uint16 * uint16Value )
```

Read 2 bytes of memory.

Parameters

in	<i>PortAddress</i>	Address to be read
out	<i>uint16Value</i>	The address content will be copied to the location pointed to by this argument

Return values

<i>CFE_PSP_SUCCESS</i>	on success
<i>CFE_PSP_ERROR_ADDRESS_MISALIGNED</i>	if the address is not aligned to a 16-bit addressing scheme.
<i>CFE_PSP_ERROR_NOT_IMPLEMENTED</i>	if not implemented

CFE_PSP_PortRead32()

```
int32 CFE_PSP_PortRead32 (
    cpuaddr PortAddress,
    uint32 * uint32Value )
```

Read 4 bytes of memory.

Parameters

in	<i>PortAddress</i>	Address to be read
out	<i>uint32Value</i>	The address content will be copied to the location pointed to by this argument

Return values

<i>CFE_PSP_SUCCESS</i>	on success
<i>CFE_PSP_ERROR_ADDRESS_MISALIGNED</i>	if the address is not aligned to a 16-bit addressing scheme.
<i>CFE_PSP_ERROR_NOT_IMPLEMENTED</i>	if not implemented

CFE_PSP_PortRead8()

```
int32 CFE_PSP_PortRead8 (
    cpuaddr PortAddress,
    uint8 * ByteValue )
```

Read one byte of memory.

Parameters

in	<i>PortAddress</i>	Address to be read
out	<i>ByteValue</i>	The address content will be copied to the location pointed to by this argument

Returns

Always returns CFE_PSP_SUCCESS (if implemented)

Return values

<i>CFE_PSP_ERROR_NOT_IMPLEMENTED</i>	if not implemented
--------------------------------------	--------------------

CFE_PSP_PortWrite16()

```
int32 CFE_PSP_PortWrite16 (
    cpuaddr PortAddress,
    uint16 uint16Value )
```

Write 2 bytes of memory.

Parameters

out	<i>PortAddress</i>	Address to be written to
in	<i>uint16Value</i>	the content pointed to by this argument will be copied to the address

Return values

<i>CFE_PSP_SUCCESS</i>	on success
<i>CFE_PSP_ERROR_ADDRESS_MISALIGNED</i>	if the address is not aligned to a 16-bit addressing scheme.
<i>CFE_PSP_ERROR_NOT_IMPLEMENTED</i>	if not implemented

CFE_PSP_PortWrite32()

```
int32 CFE_PSP_PortWrite32 (
    cpuaddr PortAddress,
    uint32 uint32Value )
```

Write 4 bytes of memory.

Parameters

out	<i>PortAddress</i>	Address to be written to
in	<i>uint32Value</i>	The content pointed to by this argument will be copied to the address

Return values

<i>CFE_PSP_SUCCESS</i>	on success
<i>CFE_PSP_ERROR_ADDRESS_MISALIGNED</i>	if the address is not aligned to a 16-bit addressing scheme.
<i>CFE_PSP_ERROR_NOT_IMPLEMENTED</i>	if not implemented

CFE_PSP_PortWrite8()

```
int32 CFE_PSP_PortWrite8 (
```

```

    cpuaddr PortAddress,
    uint8 ByteValue )

```

Write one byte of memory.

Parameters

out	<i>PortAddress</i>	Address to be written to
in	<i>ByteValue</i>	The content pointed to by this argument will be copied to the address

Returns

Always returns CFE_PSP_SUCCESS (if implemented)

Return values

CFE_PSP_ERROR_NOT_IMPLEMENTED	if not implemented
-------------------------------	--------------------

11.367 cfe_psp_port_api.h

[Go to the documentation of this file.](#)

```

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00017  *****/
00018
00019 /*
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00022 ** Purpose: This file contains the cFE Platform Support Package (PSP)
00023 **           prototypes.
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00026 **           really considered part of the OS Abstraction, but are required
00027 **           for the cFE flight software implementation. It is possible that
00028 **           some of these routines could migrate into the OS AL.
00029 **
00030 */
00031
00032 #ifndef CFE_PSP_PORT_API_H
00033 #define CFE_PSP_PORT_API_H
00034
00035 /*****
00036 INCLUDE FILES
00037 *****/
00038
00039 #include "common_types.h"
00040 #include "osapi.h"
00041
00042 #include "cfe_psp_error.h"
00043
00044 /*****
00045 FUNCTION PROTOTYPES
00046 *****/
00047
00048 */

```



```

00049 ** I/O Port API
00050 */
00051 /*-----*/
00061 int32 CFE_PSP_PortRead8(cpuaddr PortAddress, uint8 *ByteValue);
00062
00063 /*-----*/
00073 int32 CFE_PSP_PortWrite8(cpuaddr PortAddress, uint8 ByteValue);
00074
00075 /*-----*/
00086 int32 CFE_PSP_PortRead16(cpuaddr PortAddress, uint16 *uint16Value);
00087
00088 /*-----*/
00099 int32 CFE_PSP_PortWrite16(cpuaddr PortAddress, uint16 uint16Value);
00100
00101 /*-----*/
00112 int32 CFE_PSP_PortRead32(cpuaddr PortAddress, uint32 *uint32Value);
00113
00114 /*-----*/
00125 int32 CFE_PSP_PortWrite32(cpuaddr PortAddress, uint32 uint32Value);
00126
00127 #endif

```

11.368 psp/fsw/inc/cfe_psp_ssr_api.h File Reference

```

#include "common_types.h"
#include "osapi.h"
#include "cfe_psp_error.h"

```

Functions

- [int32 CFE_PSP_InitSSR](#) (uint32 bus, uint32 device, char *DeviceName)

Initializes the Solid State recorder memory for a particular platform.

11.368.1 Function Documentation

CFE_PSP_InitSSR()

```

int32 CFE_PSP_InitSSR (
    uint32 bus,
    uint32 device,
    char * DeviceName )

```

Initializes the Solid State recorder memory for a particular platform.

Note

For the MCP750, this simply initializes the Hard Disk device.

Parameters

in	<i>bus</i>	
in	<i>device</i>	
in	<i>DeviceName</i>	

Return values

<i>CFE_PSP_SUCCESS</i>	on success
<i>CFE_PSP_ERROR</i>	on error

11.369 cfe_psp_ssr_api.h

[Go to the documentation of this file.](#)

```

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00017  *****/
00018
00019 /*
00020 ** Author: A. Cudmore
00021 **
00022 ** Purpose: This file contains the cFE Platform Support Package (PSP)
00023 **           prototypes.
00024 **           The PSP routines serve as the "glue" between the RTOS and
00025 **           the cFE Flight Software. The routines fill gaps that are not
00026 **           really considered part of the OS Abstraction, but are required
00027 **           for the cFE flight software implementation. It is possible that
00028 **           some of these routines could migrate into the OS AL.
00029 **
00030 */
00031
00032 #ifndef CFE_PSP_SSR_H
00033 #define CFE_PSP_SSR_H
00034
00035 /*****
00036  INCLUDE FILES
00037  *****/
00038
00039 #include "common_types.h"
00040 #include "osapi.h"
00041
00042 #include "cfe_psp_error.h"
00043
00044 /*****
00045  FUNCTION PROTOTYPES
00046  *****/
00047
00048 /*-----*/
00061 int32 CFE_PSP_InitSSR(uint32 bus, uint32 device, char *DeviceName);
00062
00063 #endif

```

11.370 psp/fsw/inc/cfe_psp_timertick_api.h File Reference

```

#include "common_types.h"
#include "osapi.h"
#include "cfe_psp_error.h"

```

Macros

- `#define CFE_PSP_SOFT_TIMEBASE_NAME "cFS-Master"`
The name of the software/RTOS timebase for general system timers.

Functions

- `void CFE_PSP_GetTime (OS_time_t *LocalTime)`
Sample/Read a monotonic platform clock with normalization.

- [uint32 CFE_PSP_GetTimerTicksPerSecond](#) (void)
- [uint32 CFE_PSP_GetTimerLow32Rollover](#) (void)
- void [CFE_PSP_Get_Timebase](#) (uint32 *Tbu, uint32 *Tbl)

Sample/Read a monotonic platform clock without normalization.

11.370.1 Macro Definition Documentation

CFE_PSP_SOFT_TIMEBASE_NAME

```
#define CFE_PSP_SOFT_TIMEBASE_NAME "cFS-Master"
```

The name of the software/RTOS timebase for general system timers.

This name may be referred to by CFE TIME and/or SCH when setting up its own timers.

Definition at line 53 of file [cfe_psp_timertick_api.h](#).

11.370.2 Function Documentation

CFE_PSP_Get_Timebase()

```
void CFE_PSP_Get_Timebase (
    uint32 * Tbu,
    uint32 * Tbl )
```

Sample/Read a monotonic platform clock without normalization.

Provides a common interface to system timebase. This routine is in the BSP because it is sometimes implemented in hardware and sometimes taken care of by the RTOS.

This is defined as a free-running, monotonically-increasing tick counter. The epoch is not defined, but typically is the system boot time, and the value increases indefinitely as the system runs. The tick period/rate is also not defined.

Rollover events - where the range of representable values is exceeded - are theoretically possible, but would take many years of continuous uptime to occur (typically hundreds of years, if not thousands). System designers should ensure that the actual tick rate and resulting timebase range is sufficiently large to ensure that rollover is not a concern.

Note

This is a "raw" value from the underlying platform with minimal/no conversions or normalization applied. Neither the epoch nor the resolution of this tick counter is specified, and it may vary from platform to platform. Use the [CFE_PSP_GetTime\(\)](#) function to sample the timebase and also convert the units into a normalized/more consistent form.

See also

[CFE_PSP_GetTime\(\)](#)

Parameters

out	<i>Tbu</i>	Buffer to hold the upper 32 bits of a 64-bit tick counter
out	<i>Tbl</i>	Buffer to hold the lower 32 bits of a 64-bit tick counter

CFE_PSP_GetTime()

```
void CFE_PSP_GetTime (
    OS_time_t * LocalTime )
```

Sample/Read a monotonic platform clock with normalization.

Outputs an [OS_time_t](#) value indicating the time elapsed since an epoch. The epoch is not defined, but typically represents the system boot time. The value increases continuously over time and cannot be reset by software.

This is similar to the [CFE_PSP_Get_Timebase\(\)](#), but additionally it normalizes the output value to an [OS_time_t](#), thereby

providing consistent units to the calling application. Any OSAL-provided routine that accepts [OS_time_t](#) inputs may be used to convert this value into other standardized time units.

Note

This should refer to the same time domain as [CFE_PSP_Get_Timebase\(\)](#), the primary difference being the format and units of the output value.

See also

[CFE_PSP_Get_Timebase\(\)](#)

Parameters

out	<i>LocalTime</i>	Value of PSP tick counter as OS_time_t
-----	------------------	--

CFE_PSP_GetTimerLow32Rollover()

```
uint32 CFE_PSP_GetTimerLow32Rollover (
    void )
```

Provides the number that the least significant 32 bits of the 64-bit time stamp returned by [CFE_PSP_Get_Timebase](#) rolls over. If the lower 32 bits rolls at 1 second, then the [CFE_PSP_TIMER_LOW32_ROLLOVER](#) will be 1000000. If the lower 32 bits rolls at its maximum value (2^{32}) then [CFE_PSP_TIMER_LOW32_ROLLOVER](#) will be 0.

Returns

The number that the least significant 32 bits of the 64-bit time stamp returned by [CFE_PSP_Get_Timebase](#) rolls over.

CFE_PSP_GetTimerTicksPerSecond()

```
uint32 CFE_PSP_GetTimerTicksPerSecond (
    void )
```

Provides the resolution of the least significant 32 bits of the 64-bit time stamp returned by [CFE_PSP_Get_Timebase](#) in timer ticks per second. The timer resolution for accuracy should not be any slower than 1000000 ticks per second or 1 us (microsecond) per tick

Returns

The number of timer ticks per second of the time stamp returned by [CFE_PSP_Get_Timebase](#)

11.371 cfe_psp_timertick_api.h

[Go to the documentation of this file.](#)

```
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```

```

00017  *****/
00018
00019  /*
00020  ** Author: A. Cudmore
00021  **
00022  ** Purpose: This file contains the cFE Platform Support Package(PSP)
00023  **           prototypes.
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00026  **           really considered part of the OS Abstraction, but are required
00027  **           for the cFE flight software implementation. It is possible that
00028  **           some of these routines could migrate into the OS AL.
00029  **
00030  */
00031
00032 #ifndef CFE_PSP_TIMERTICK_API_H
00033 #define CFE_PSP_TIMERTICK_API_H
00034
00035 /*****
00036  INCLUDE FILES
00037  *****/
00038
00039 #include "common_types.h"
00040 #include "osapi.h"
00041
00042 #include "cfe_psp_error.h"
00043
00044 /*****
00045  MACRO DEFINITIONS
00046  *****/
00047
00053 #define CFE_PSP_SOFT_TIMEBASE_NAME "cFS-Master"
00054
00055 /*****
00056  FUNCTION PROTOTYPES
00057  *****/
00058
00059 /*-----*/
00079 void CFE_PSP_GetTime(OS_time_t *LocalTime);
00080
00081 /*-----*/
00091 uint32 CFE_PSP_GetTimerTicksPerSecond(void);
00092
00093 /*-----*/
00104 uint32 CFE_PSP_GetTimerLow32Rollover(void);
00105
00135 void CFE_PSP_Get_Timebase(uint32 *Tbu, uint32 *Tbl);
00136
00137 #endif

```

11.372 psp/fsw/inc/cfe_psp_version_api.h File Reference

```

#include "common_types.h"
#include "osapi.h"
#include "cfe_psp_error.h"

```

Functions

- const char * [CFE_PSP_GetVersionString](#) (void)
Obtain the PSP version/baseline identifier string.
- const char * [CFE_PSP_GetVersionCodeName](#) (void)
Obtain the version code name.
- void [CFE_PSP_GetVersionNumber](#) (uint8 VersionNumbers[4])
Retrieves the numeric PSP version identifier as an array of uint8 values.
- uint32 [CFE_PSP_GetBuildNumber](#) (void)
Obtain the PSP library numeric build number.

11.372.1 Function Documentation

CFE_PSP_GetBuildNumber()

```
uint32 CFE_PSP_GetBuildNumber (
    void )
```

Obtain the PSP library numeric build number.

The build number is a monotonically increasing number that (coarsely) reflects the number of commits/changes that have been merged since the epoch release. During development cycles this number should increase after each subsequent merge/modification.

Like other version information, this is a fixed number assigned at compile time.

Returns

The OSAL library build number

CFE_PSP_GetVersionCodeName()

```
const char * CFE_PSP_GetVersionCodeName (
    void )
```

Obtain the version code name.

This retrieves the PSP code name. This is a compatibility indicator for the overall NASA cFS ecosystem. All modular components which are intended to interoperate should report the same code name.

Returns

Code name. This is a fixed string and cannot be NULL.

CFE_PSP_GetVersionNumber()

```
void CFE_PSP_GetVersionNumber (
    uint8 VersionNumbers[4] )
```

Retrieves the numeric PSP version identifier as an array of uint8 values.

The array of numeric values is in order of precedence: [0] = Major Number [1] = Minor Number [2] = Revision Number [3] = Mission Revision

The "Mission Revision" (last output) also indicates whether this is an official release, a patched release, or a development version. 0 indicates an official release 1-254 local patch level (reserved for mission use) 255 indicates a development build

Parameters

out	<i>VersionNumbers</i>	A fixed-size array to be filled with the version numbers
-----	-----------------------	--

CFE_PSP_GetVersionString()

```
const char * CFE_PSP_GetVersionString (
    void )
```

Obtain the PSP version/baseline identifier string.

This retrieves the PSP version identifier string without extra info

Returns

Version string. This is a fixed string and cannot be NULL.

11.373 cfe_psp_version_api.h

[Go to the documentation of this file.](#)

```

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00016  * limitations under the License.
00017  *****/
00018
00019 /*
00020 ** Author: A. Cudmore
00021 **
00022 ** Purpose: This file contains the cFE Platform Support Package(PSP)
00023 **          prototypes.
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00026 **          really considered part of the OS Abstraction, but are required
00027 **          for the cFE flight software implementation. It is possible that
00028 **          some of these routines could migrate into the OS AL.
00029 **
00030 */
00031
00032 #ifndef CFE_PSP_VERSION_API_H
00033 #define CFE_PSP_VERSION_API_H
00034
00035 /*****
00036  INCLUDE FILES
00037  *****/
00038
00039 #include "common_types.h"
00040 #include "osapi.h"
00041
00042 #include "cfe_psp_error.h"
00043
00044 /*****
00045  FUNCTION PROTOTYPES
00046  *****/
00047
00048 /*-----*/
00056 const char *CFE_PSP_GetVersionString(void);
00057
00058 /*-----*/
00068 const char *CFE_PSP_GetVersionCodeName(void);
00069
00070 /*-----*/
00088 void CFE_PSP_GetVersionNumber(uint8 VersionNumbers[4]);
00089
00090 /*-----*/
00104 uint32 CFE_PSP_GetBuildNumber(void);
00105
00106 #endif

```

11.374 psp/fsw/inc/cfe_psp_watchdog_api.h File Reference

```

#include "common_types.h"
#include "osapi.h"
#include "cfe_psp_error.h"

```

Macros

Definitions for PSP PANIC types

- #define [CFE_PSP_PANIC_STARTUP](#) 1
- #define [CFE_PSP_PANIC_VOLATILE_DISK](#) 2
- #define [CFE_PSP_PANIC_MEMORY_ALLOC](#) 3
- #define [CFE_PSP_PANIC_NONVOL_DISK](#) 4
- #define [CFE_PSP_PANIC_STARTUP_SEM](#) 5
- #define [CFE_PSP_PANIC_CORE_APP](#) 6
- #define [CFE_PSP_PANIC_GENERAL_FAILURE](#) 7

Reset Types

- #define [CFE_PSP_RST_TYPE_PROCESSOR](#) 1
- #define [CFE_PSP_RST_TYPE_POWERON](#) 2
- #define [CFE_PSP_RST_TYPE_MAX](#) 3

Reset Sub-Types

- #define [CFE_PSP_RST_SUBTYPE_POWER_CYCLE](#) 1
Reset caused by power having been removed and restored.
- #define [CFE_PSP_RST_SUBTYPE_PUSH_BUTTON](#) 2
Reset caused by reset button on the board.
- #define [CFE_PSP_RST_SUBTYPE_HW_SPECIAL_COMMAND](#) 3
Reset was caused by a reset line having been stimulated by a hardware special command.
- #define [CFE_PSP_RST_SUBTYPE_HW_WATCHDOG](#) 4
Reset was caused by a watchdog timer expiring.
- #define [CFE_PSP_RST_SUBTYPE_RESET_COMMAND](#) 5
Reset was caused by cFE ES processing a [Reset Command](#).
- #define [CFE_PSP_RST_SUBTYPE_EXCEPTION](#) 6
Reset was caused by a Processor Exception.
- #define [CFE_PSP_RST_SUBTYPE_UNDEFINED_RESET](#) 7
Reset was caused in an unknown manner.
- #define [CFE_PSP_RST_SUBTYPE_HWDEBUG_RESET](#) 8
Reset was caused by a JTAG or BDM connection.
- #define [CFE_PSP_RST_SUBTYPE_BANKSWITCH_RESET](#) 9
Reset reverted to a cFE POWERON due to a boot bank switch.
- #define [CFE_PSP_RST_SUBTYPE_MAX](#) 10
Placeholder to indicate 1+ the maximum value that the PSP will ever use.

Functions

- void [CFE_PSP_Restart](#) (uint32 resetType)
Entry point back to the BSP to restart the processor.
- uint32 [CFE_PSP_GetRestartType](#) (uint32 *restartSubType)
Returns the last reset type.
- void [CFE_PSP_WatchdogInit](#) (void)
Configures the watchdog timer.
- void [CFE_PSP_WatchdogEnable](#) (void)
Enables the watchdog timer.
- void [CFE_PSP_WatchdogDisable](#) (void)
Disables the watchdog timer.
- void [CFE_PSP_WatchdogService](#) (void)
Serves the watchdog timer according to the value set in WatchDogSet.

- [uint32 CFE_PSP_WatchdogGet](#) (void)
Gets the watchdog time in milliseconds.
- void [CFE_PSP_WatchdogSet](#) (uint32 WatchdogValue)
Sets the watchdog time in milliseconds.
- void [CFE_PSP_Panic](#) (int32 ErrorCode)
Aborts the cFE startup.

11.374.1 Macro Definition Documentation

CFE_PSP_PANIC_CORE_APP

#define CFE_PSP_PANIC_CORE_APP 6
Definition at line 59 of file [cfe_psp_watchdog_api.h](#).

CFE_PSP_PANIC_GENERAL_FAILURE

#define CFE_PSP_PANIC_GENERAL_FAILURE 7
Definition at line 60 of file [cfe_psp_watchdog_api.h](#).

CFE_PSP_PANIC_MEMORY_ALLOC

#define CFE_PSP_PANIC_MEMORY_ALLOC 3
Definition at line 56 of file [cfe_psp_watchdog_api.h](#).

CFE_PSP_PANIC_NONVOL_DISK

#define CFE_PSP_PANIC_NONVOL_DISK 4
Definition at line 57 of file [cfe_psp_watchdog_api.h](#).

CFE_PSP_PANIC_STARTUP

#define CFE_PSP_PANIC_STARTUP 1
Definition at line 54 of file [cfe_psp_watchdog_api.h](#).

CFE_PSP_PANIC_STARTUP_SEM

#define CFE_PSP_PANIC_STARTUP_SEM 5
Definition at line 58 of file [cfe_psp_watchdog_api.h](#).

CFE_PSP_PANIC_VOLATILE_DISK

#define CFE_PSP_PANIC_VOLATILE_DISK 2
Definition at line 55 of file [cfe_psp_watchdog_api.h](#).

CFE_PSP_RST_SUBTYPE_BANKSWITCH_RESET

#define CFE_PSP_RST_SUBTYPE_BANKSWITCH_RESET 9
Reset reverted to a cFE POWERON due to a boot bank switch.
Definition at line 96 of file [cfe_psp_watchdog_api.h](#).

CFE_PSP_RST_SUBTYPE_EXCEPTION

#define CFE_PSP_RST_SUBTYPE_EXCEPTION 6
Reset was caused by a Processor Exception.
Definition at line 90 of file [cfe_psp_watchdog_api.h](#).

CFE_PSP_RST_SUBTYPE_HW_SPECIAL_COMMAND

```
#define CFE_PSP_RST_SUBTYPE_HW_SPECIAL_COMMAND 3
```

Reset was caused by a reset line having been stimulated by a hardware special command.

Definition at line 84 of file [cfe_psp_watchdog_api.h](#).

CFE_PSP_RST_SUBTYPE_HW_WATCHDOG

```
#define CFE_PSP_RST_SUBTYPE_HW_WATCHDOG 4
```

Reset was caused by a watchdog timer expiring.

Definition at line 86 of file [cfe_psp_watchdog_api.h](#).

CFE_PSP_RST_SUBTYPE_HWDEBUG_RESET

```
#define CFE_PSP_RST_SUBTYPE_HWDEBUG_RESET 8
```

Reset was caused by a JTAG or BDM connection.

Definition at line 94 of file [cfe_psp_watchdog_api.h](#).

CFE_PSP_RST_SUBTYPE_MAX

```
#define CFE_PSP_RST_SUBTYPE_MAX 10
```

Placeholder to indicate 1+ the maximum value that the PSP will ever use.

Definition at line 98 of file [cfe_psp_watchdog_api.h](#).

CFE_PSP_RST_SUBTYPE_POWER_CYCLE

```
#define CFE_PSP_RST_SUBTYPE_POWER_CYCLE 1
```

Reset caused by power having been removed and restored.

Definition at line 80 of file [cfe_psp_watchdog_api.h](#).

CFE_PSP_RST_SUBTYPE_PUSH_BUTTON

```
#define CFE_PSP_RST_SUBTYPE_PUSH_BUTTON 2
```

Reset caused by reset button on the board.

Definition at line 82 of file [cfe_psp_watchdog_api.h](#).

CFE_PSP_RST_SUBTYPE_RESET_COMMAND

```
#define CFE_PSP_RST_SUBTYPE_RESET_COMMAND 5
```

Reset was caused by cFE ES processing a [Reset Command](#).

Definition at line 88 of file [cfe_psp_watchdog_api.h](#).

CFE_PSP_RST_SUBTYPE_UNDEFINED_RESET

```
#define CFE_PSP_RST_SUBTYPE_UNDEFINED_RESET 7
```

Reset was caused in an unknown manner.

Definition at line 92 of file [cfe_psp_watchdog_api.h](#).

CFE_PSP_RST_TYPE_MAX

```
#define CFE_PSP_RST_TYPE_MAX 3
```

Placeholder to indicate 1+ the maximum value that the PSP will ever use.

Definition at line 70 of file [cfe_psp_watchdog_api.h](#).

CFE_PSP_RST_TYPE_POWERON

```
#define CFE_PSP_RST_TYPE_POWERON 2
```

All memory has been cleared

Definition at line 69 of file [cfe_psp_watchdog_api.h](#).

CFE_PSP_RST_TYPE_PROCESSOR

```
#define CFE_PSP_RST_TYPE_PROCESSOR 1
```

Volatile disk, CDS and User Reserved memory may be valid

Definition at line 68 of file [cfe_psp_watchdog_api.h](#).

11.374.2 Function Documentation

CFE_PSP_GetRestartType()

```
uint32 CFE_PSP_GetRestartType (
    uint32 * restartSubType )
```

Returns the last reset type.

Note

If a pointer to a valid memory space is passed in, it returns the reset sub-type in that memory. Right now the reset types are application-specific. For the cFE they are defined in the [cfe_es.h](#) file.

Parameters

<i>restartSubType</i>	
-----------------------	--

CFE_PSP_Panic()

```
void CFE_PSP_Panic (
    int32 ErrorCode )
```

Aborts the cFE startup.

Provides a common interface to abort the cFE startup process and return back to the OS.

Note

This is called by the cFE Core startup code when it needs to abort the cFE startup. This should not be called by applications.

Parameters

in	<i>ErrorCode</i>	Reason for exiting
----	------------------	--------------------

CFE_PSP_Restart()

```
void CFE_PSP_Restart (
    uint32 resetType )
```

Entry point back to the BSP to restart the processor.

The flight software calls this routine to restart the processor.

Parameters

in	<i>resetType</i>	Type of reset
----	------------------	---------------

CFE_PSP_WatchdogDisable()

```
void CFE_PSP_WatchdogDisable (
    void )
```

Disables the watchdog timer.

CFE_PSP_WatchdogEnable()

```
void CFE_PSP_WatchdogEnable (
    void )
```

Enables the watchdog timer.

CFE_PSP_WatchdogGet()

```
uint32 CFE_PSP_WatchdogGet (
    void )
```

Gets the watchdog time in milliseconds.

Returns

The current watchdog value

CFE_PSP_WatchdogInit()

```
void CFE_PSP_WatchdogInit (
    void )
```

Configures the watchdog timer.

To set up the timer resolution and/or other settings custom to this platform.

CFE_PSP_WatchdogService()

```
void CFE_PSP_WatchdogService (
    void )
```

Services the watchdog timer according to the value set in WatchDogSet.

Load the watchdog timer with a count that corresponds to the millisecond time given in the parameter.

Note

Currently an ExpireTime value of zero will result in the minimum reset time of 4.5 seconds. All other ExpireTime values will result in a reset time of 5.5 seconds.

CFE_PSP_WatchdogSet()

```
void CFE_PSP_WatchdogSet (
    uint32 WatchdogValue )
```

Sets the watchdog time in milliseconds.

Parameters

in	<i>WatchdogValue</i>	New watchdog value to set
----	----------------------	---------------------------

11.375 cfe_psp_watchdog_api.h

[Go to the documentation of this file.](#)

```

00001 /*****
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00014  * WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00015  * See the License for the specific language governing permissions and
00016  * limitations under the License.
00017  *****/
00018
00019 /*
00020 ** Author: A. Cudmore
00021 **
00022 ** Purpose: This file contains the cFE Platform Support Package (PSP)
00023 **           prototypes.
00024 **           The PSP routines serve as the "glue" between the RTOS and
00025 **           the cFE Flight Software. The routines fill gaps that are not
00026 **           really considered part of the OS Abstraction, but are required
00027 **           for the cFE flight software implementation. It is possible that
00028 **           some of these routines could migrate into the OS AL.
00029 **
00030 */
00031
00032 #ifndef CFE_PSP_WATCHDOG_API_H
00033 #define CFE_PSP_WATCHDOG_API_H
00034
00035 /*****
00036  INCLUDE FILES
00037  *****/
00038
00039 #include "common_types.h"
00040 #include "osapi.h"
00041
00042 #include "cfe_psp_error.h"
00043
00044 /*****
00045  MACRO DEFINITIONS
00046  *****/
00047
00048 /*
00049 ** Definitions for PSP PANIC types
00050 */
00051
00052 #define CFE_PSP_PANIC_STARTUP 1
00053 #define CFE_PSP_PANIC_VOLATILE_DISK 2
00054 #define CFE_PSP_PANIC_MEMORY_ALLOC 3
00055 #define CFE_PSP_PANIC_NONVOL_DISK 4
00056 #define CFE_PSP_PANIC_STARTUP_SEM 5
00057 #define CFE_PSP_PANIC_CORE_APP 6
00058 #define CFE_PSP_PANIC_GENERAL_FAILURE 7
00059
00060 /*
00061 ** Common definition for reset types at the PSP layer
00062 */
00063
00064 #define CFE_PSP_RST_TYPE_PROCESSOR 1
00065 #define CFE_PSP_RST_TYPE_POWERON 2
00066 #define CFE_PSP_RST_TYPE_MAX 3
00067
00068 /*
00069 ** Reset Sub-Types
00070 */
00071
00072 #define CFE_PSP_RST_SUBTYPE_POWER_CYCLE 1
00073 #define CFE_PSP_RST_SUBTYPE_PUSH_BUTTON 2
00074 #define CFE_PSP_RST_SUBTYPE_HW_SPECIAL_COMMAND 3
00075 #define CFE_PSP_RST_SUBTYPE_HW_WATCHDOG 4
00076 #define CFE_PSP_RST_SUBTYPE_RESET_COMMAND 5
00077 #define CFE_PSP_RST_SUBTYPE_EXCEPTION 6
00078 #define CFE_PSP_RST_SUBTYPE_UNDEFINED_RESET 7
00079 #define CFE_PSP_RST_SUBTYPE_HWDEBUG_RESET 8
00080 #define CFE_PSP_RST_SUBTYPE_BANKSWITCH_RESET 9
00081 #define CFE_PSP_RST_SUBTYPE_MAX 10

```

```
00101 /*****
00102 FUNCTION PROTOTYPES
00103 *****/
00104
00105 /*-----*/
00113 void CFE_PSP_Restart(uint32 resetType);
00114
00115 /*-----*/
00125 uint32 CFE_PSP_GetRestartType(uint32 *restartSubType);
00126
00127 /*-----*/
00133 void CFE_PSP_WatchdogInit(void);
00134
00135 /*-----*/
00139 void CFE_PSP_WatchdogEnable(void);
00140
00141 /*-----*/
00145 void CFE_PSP_WatchdogDisable(void);
00146
00147 /*-----*/
00158 void CFE_PSP_WatchdogService(void);
00159
00160 /*-----*/
00166 uint32 CFE_PSP_WatchdogGet(void);
00167
00168 /*-----*/
00174 void CFE_PSP_WatchdogSet(uint32 WatchdogValue);
00175
00176 /*-----*/
00188 void CFE_PSP_Panic(int32 ErrorCode);
00189
00190 #endif
```


Index

- [_EXTENSION_](#)
 - [common_types.h, 1215](#)
- [accuracy](#)
 - [OS_timebase_prop_t, 595](#)
 - [OS_timer_prop_t, 596](#)
- [Acronyms, 5](#)
- [ActiveBuffer](#)
 - [CFE_TBL_HousekeepingTlm_Payload, 538](#)
- [ActiveBufferAddr](#)
 - [CFE_TBL_TblRegPacket_Payload, 549](#)
- [ActiveTableFlag](#)
 - [CFE_TBL_DumpCmd_Payload, 532](#)
 - [CFE_TBL_ValidateCmd_Payload, 552](#)
- [ActualLength](#)
 - [OS_SockAddr_t, 590](#)
- [ActualType](#)
 - [CFE_Config_ValueEntry, 425](#)
- [addr](#)
 - [OS_module_prop_t, 588](#)
- [AddrData](#)
 - [OS_SockAddr_t, 590](#)
- [Address](#)
 - [OS_static_symbol_record_t, 593](#)
- [AddressesAreValid](#)
 - [CFE_ES_AppInfo, 427](#)
- [Adjusting Time, 71–73](#)
- [AdjustmentFactor](#)
 - [CFE_TIME_HousekeepingTlm_Payload, 565](#)
- [AlignPtr](#)
 - [OS_SockAddrData_t, 591](#)
- [AlignU32](#)
 - [OS_SockAddrData_t, 591](#)
- [All Resets, 39](#)
- [AppData](#)
 - [CFE_EVS_HousekeepingTlm_Payload, 485](#)
- [AppDataFilename](#)
 - [CFE_EVS_AppDataCmd_Payload, 473](#)
- [AppEnableStatus](#)
 - [CFE_EVS_AppTlmData, 476](#)
- [AppEntryPoint](#)
 - [CFE_ES_StartAppCmd_Payload, 465](#)
- [AppFileName](#)
 - [CFE_ES_AppReloadCmd_Payload, 431](#)
 - [CFE_ES_StartAppCmd_Payload, 465](#)
- [AppID](#)
 - [CFE_EVS_AppTlmData, 476](#)
- [AppId](#)
 - [CFE_ES_TaskInfo, 470](#)
 - [CFE_SB_PipeInfoEntry, 513](#)
- [AppInfo](#)
 - [CFE_ES_OneAppTlm_Payload, 451](#)
- [Applicable Documents, 3](#)
- [Application](#)
 - [CFE_ES_AppNameCmd_Payload, 431](#)
 - [CFE_ES_AppReloadCmd_Payload, 431](#)
 - [CFE_ES_SendMemPoolStatsCmd_Payload, 460](#)
 - [CFE_ES_StartAppCmd_Payload, 465](#)
- [Application and Child Task Management, 11](#)
- [ApplicationID](#)
 - [CFE_FS_Header, 500](#)
- [AppMessageSentCounter](#)
 - [CFE_EVS_AppTlmData, 476](#)
- [AppMessageSquelchedCounter](#)
 - [CFE_EVS_AppTlmData, 476](#)
- [AppName](#)
 - [CFE_ES_TaskInfo, 470](#)
 - [CFE_EVS_AppNameBitMaskCmd_Payload, 473](#)
 - [CFE_EVS_AppNameCmd_Payload, 474](#)
 - [CFE_EVS_AppNameEventIDCmd_Payload, 474](#)
 - [CFE_EVS_AppNameEventIDMaskCmd_Payload, 475](#)
 - [CFE_EVS_PacketID, 490](#)
 - [CFE_SB_PipeInfoEntry, 513](#)
 - [CFE_SB_RoutingFileEntry, 517](#)
- [ARGCHECK](#)
 - [osapi-macros.h, 1255](#)
- [AsInteger](#)
 - [CFE_Config_ValueBuffer, 425](#)
- [AsPointer](#)
 - [CFE_Config_ValueBuffer, 425](#)
- [AtToneDelay](#)
 - [CFE_TIME_DiagnosticTlm_Payload, 557](#)
- [AtToneLatch](#)
 - [CFE_TIME_DiagnosticTlm_Payload, 557](#)
- [AtToneLeapSeconds](#)
 - [CFE_TIME_DiagnosticTlm_Payload, 557](#)
 - [CFE_TIME_ToneDataCmd_Payload, 579](#)
- [AtToneMET](#)
 - [CFE_TIME_DiagnosticTlm_Payload, 557](#)
 - [CFE_TIME_ToneDataCmd_Payload, 579](#)
- [AtToneState](#)
 - [CFE_TIME_ToneDataCmd_Payload, 579](#)
- [AtToneSTCF](#)
 - [CFE_TIME_DiagnosticTlm_Payload, 557](#)
 - [CFE_TIME_ToneDataCmd_Payload, 580](#)
- [Autonomous Actions, 38](#)
- [Background, 2](#)
- [Basic File System, 13](#)
- [BitMask](#)
 - [CFE_EVS_AppNameBitMaskCmd_Payload, 473](#)

- CFE_EVS_BitMaskCmd_Payload, 478
- bits
 - CFE_ES_MemAddress_t, 447
 - CFE_ES_MemOffset_t, 447
- block_size
 - OS_statvfs_t, 593
- blocks_free
 - OS_statvfs_t, 593
- BlockSize
 - CFE_ES_BlockStats, 432
- BlockStats
 - CFE_ES_MemPoolStats, 448
- BootSource
 - CFE_ES_HousekeepingTlm_Payload, 440
- bss_address
 - OS_module_address_t, 587
- bss_size
 - OS_module_address_t, 587
- BSSAddress
 - CFE_ES_AppInfo, 427
- BSSSize
 - CFE_ES_AppInfo, 427
- Buffer
 - OS_SockAddrData_t, 591
- BUGCHECK
 - osapi-macros.h, 1255
- BUGCHECK_VOID
 - osapi-macros.h, 1255
- BUGREPORT
 - osapi-macros.h, 1256
- build/osal_public_api/inc/osconfig.h, 597, 604
- ByteAlignPad1
 - CFE_TBL_HousekeepingTlm_Payload, 538
- ByteAlignSpare
 - CFE_ES_CDSRegDumpRec, 433
- CCSDS_ExtendedHeader, 422
 - Subsystem, 423
 - SystemId, 423
- CCSDS_ExtendedHeader_t
 - ccsds_hdr.h, 986
- ccsds_hdr.h
 - CCSDS_ExtendedHeader_t, 986
 - CCSDS_PrimaryHeader_t, 986
- CCSDS_PrimaryHeader, 423
 - Length, 423
 - Sequence, 423
 - StreamId, 424
- CCSDS_PrimaryHeader_t
 - ccsds_hdr.h, 986
- CdsName
 - CFE_ES_DeleteCDSCmd_Payload, 435
- cFE Access Table Content APIs, 262
 - CFE_TBL_GetAddress, 263
- CFE_TBL_GetAddresses, 264
- CFE_TBL_ReleaseAddress, 265
- CFE_TBL_ReleaseAddresses, 266
- cFE Application Behavior APIs, 151
 - CFE_ES_ExitApp, 152
 - CFE_ES_IncrementTaskCounter, 152
 - CFE_ES_RunLoop, 153
 - CFE_ES_WaitForStartupSync, 154
 - CFE_ES_WaitForSystemState, 154
- cFE Application Control APIs, 149
 - CFE_ES_DeleteApp, 149
 - CFE_ES_ReloadApp, 150
 - CFE_ES_RestartApp, 151
- cFE Application Programmer's Interface (API) Reference, 92
- cFE Child Task APIs, 164
 - CFE_ES_CreateChildTask, 164
 - CFE_ES_DeleteChildTask, 165
 - CFE_ES_ExitChildTask, 166
 - CFE_ES_GetTaskIDByName, 166
 - CFE_ES_GetTaskName, 167
- cFE Clock State Flag Defines, 289
 - CFE_TIME_FLAG_ADD1HZ, 289
 - CFE_TIME_FLAG_ADDADJ, 289
 - CFE_TIME_FLAG_ADDTCL, 289
 - CFE_TIME_FLAG_CLKSET, 289
 - CFE_TIME_FLAG_CMDFLY, 290
 - CFE_TIME_FLAG_FLYING, 290
 - CFE_TIME_FLAG_GDTONE, 290
 - CFE_TIME_FLAG_REFERR, 290
 - CFE_TIME_FLAG_SERVER, 290
 - CFE_TIME_FLAG_SIGPRI, 290
 - CFE_TIME_FLAG_SRCINT, 290
 - CFE_TIME_FLAG_SRVFLY, 290
 - CFE_TIME_FLAG_UNUSED, 290
- cFE Command Mnemonic Cross Reference, 77
- cFE Critical Data Store APIs, 170
 - CFE_ES_CopyToCDS, 170
 - CFE_ES_GetCDSBlockIDByName, 171
 - CFE_ES_GetCDSBlockName, 172
 - CFE_ES_RegisterCDS, 172
 - CFE_ES_RestoreFromCDS, 173
- cFE Entry/Exit APIs, 147
 - CFE_ES_Main, 148
 - CFE_ES_ResetCFE, 148
- cFE Event Message Cross Reference, 76
- cFE Event Services Commands, 33
- cFE Event Services Configuration Parameters, 35
- cFE Event Services Overview, 25
- cFE Event Services Telemetry, 34
- cFE Executive Services Commands, 18
- cFE Executive Services Configuration Parameters, 20
- cFE Executive Services Overview, 7
- cFE Executive Services Telemetry, 20

- cFE External Time Source APIs, 281
 - CFE_TIME_ExternalGPS, 282
 - CFE_TIME_ExternalMET, 282
 - CFE_TIME_ExternalTime, 283
 - CFE_TIME_ExternalTone, 284
 - CFE_TIME_RegisterSynchCallback, 284
 - CFE_TIME_UnregisterSynchCallback, 285
- cFE File Header Management APIs, 196
 - CFE_FS_InitHeader, 196
 - CFE_FS_ReadHeader, 197
 - CFE_FS_SetTimestamp, 198
 - CFE_FS_WriteHeader, 198
- cFE File Utility APIs, 200
 - CFE_FS_BackgroundFileDumpsPending, 200
 - CFE_FS_BackgroundFileDumpRequest, 201
 - CFE_FS_ExtractFilenameFromPath, 201
 - CFE_FS_GetDefaultExtension, 202
 - CFE_FS_GetDefaultMountPoint, 202
 - CFE_FS_ParseInputFileName, 203
 - CFE_FS_ParseInputFileNameEx, 203
- cFE Generic Counter APIs, 184
 - CFE_ES_DeleteGenCounter, 185
 - CFE_ES_GetGenCount, 185
 - CFE_ES_GetGenCounterIDByName, 186
 - CFE_ES_GetGenCounterName, 187
 - CFE_ES_IncrementGenCounter, 188
 - CFE_ES_RegisterGenCounter, 188
 - CFE_ES_SetGenCount, 189
- cFE Generic Message APIs, 204
 - CFE_MSG_Init, 205
- cFE Get Current Time APIs, 272
 - CFE_TIME_GetMET, 272
 - CFE_TIME_GetMETseconds, 273
 - CFE_TIME_GetMETsubsecs, 273
 - CFE_TIME_GetTAI, 273
 - CFE_TIME_GetTime, 274
 - CFE_TIME_GetUTC, 274
- cFE Get Table Information APIs, 267
 - CFE_TBL_GetInfo, 267
 - CFE_TBL_GetStatus, 268
 - CFE_TBL_NotifyByMessage, 269
- cFE Get Time Information APIs, 275
 - CFE_TIME_GetClockInfo, 275
 - CFE_TIME_GetClockState, 276
 - CFE_TIME_GetLeapSeconds, 276
 - CFE_TIME_GetSTCF, 276
- cFE Information APIs, 155
 - CFE_ES_GetAppID, 156
 - CFE_ES_GetAppIDByName, 156
 - CFE_ES_GetAppInfo, 157
 - CFE_ES_GetAppName, 158
 - CFE_ES_GetLibIDByName, 158
 - CFE_ES_GetLibInfo, 159
 - CFE_ES_GetLibName, 160
 - CFE_ES_GetModuleInfo, 161
 - CFE_ES_GetResetType, 161
 - CFE_ES_GetTaskID, 162
 - CFE_ES_GetTaskInfo, 163
- cFE Manage Table Content APIs, 257
 - CFE_TBL_DumpToBuffer, 257
 - CFE_TBL_Load, 258
 - CFE_TBL_Manage, 259
 - CFE_TBL_Modified, 260
 - CFE_TBL_Update, 261
 - CFE_TBL_Validate, 262
- cFE Memory Manager APIs, 174
 - CFE_ES_GetMemPoolStats, 175
 - CFE_ES_GetPoolBuf, 175
 - CFE_ES_GetPoolBufInfo, 176
 - CFE_ES_PoolCreate, 177
 - CFE_ES_PoolCreateEx, 178
 - CFE_ES_PoolCreateEx_WithAlignment, 179
 - CFE_ES_PoolCreateNoSem, 180
 - CFE_ES_PoolDelete, 181
 - CFE_ES_PutPoolBuf, 181
- cFE Message Characteristics APIs, 243
 - CFE_SB_GetUserData, 244
 - CFE_SB_GetUserDataLength, 244
 - CFE_SB_MessageStringGet, 245
 - CFE_SB_MessageStringSet, 246
 - CFE_SB_SetUserDataLength, 246
 - CFE_SB_TimeStampMsg, 247
- cFE Message Extended Header APIs, 214
 - CFE_MSG_GetEDSVersion, 214
 - CFE_MSG_GetEndian, 215
 - CFE_MSG_GetPlaybackFlag, 215
 - CFE_MSG_GetSubsystem, 216
 - CFE_MSG_GetSystem, 216
 - CFE_MSG_SetEDSVersion, 217
 - CFE_MSG_SetEndian, 217
 - CFE_MSG_SetPlaybackFlag, 218
 - CFE_MSG_SetSubsystem, 218
 - CFE_MSG_SetSystem, 219
- cFE Message ID APIs, 247
 - CFE_SB_CmdTopicIdToMsgId, 248
 - CFE_SB_GlobalCmdTopicIdToMsgId, 248
 - CFE_SB_GlobalTlmTopicIdToMsgId, 249
 - CFE_SB_IsValidMsgId, 249
 - CFE_SB_LocalCmdTopicIdToMsgId, 249
 - CFE_SB_LocalTlmTopicIdToMsgId, 250
 - CFE_SB_MsgId_Equal, 250
 - CFE_SB_MsgIdToValue, 251
 - CFE_SB_TlmTopicIdToMsgId, 251
 - CFE_SB_ValueToMsgId, 252
- cFE Message Id APIs, 225
 - CFE_MSG_GetMsgId, 225
 - CFE_MSG_GetTypeFromMsgId, 225
 - CFE_MSG_SetMsgId, 226

- cFE Message Integrity APIs, 227
 - CFE_MSG_OriginationAction, 227
 - CFE_MSG_VerificationAction, 228
- cFE Message Primary Header APIs, 205
 - CFE_MSG_GetApId, 206
 - CFE_MSG_GetHasSecondaryHeader, 207
 - CFE_MSG_GetHeaderVersion, 207
 - CFE_MSG_GetNextSequenceCount, 207
 - CFE_MSG_GetSegmentationFlag, 208
 - CFE_MSG_GetSequenceCount, 208
 - CFE_MSG_GetSize, 209
 - CFE_MSG_GetType, 209
 - CFE_MSG_SetApId, 210
 - CFE_MSG_SetHasSecondaryHeader, 210
 - CFE_MSG_SetHeaderVersion, 211
 - CFE_MSG_SetSegmentationFlag, 211
 - CFE_MSG_SetSequenceCount, 212
 - CFE_MSG_SetSize, 212
 - CFE_MSG_SetType, 213
- cFE Message Secondary Header APIs, 219
 - CFE_MSG_GenerateChecksum, 220
 - CFE_MSG_GetFcnCode, 220
 - CFE_MSG_GetMsgTime, 222
 - CFE_MSG_SetFcnCode, 222
 - CFE_MSG_SetMsgTime, 223
 - CFE_MSG_ValidateChecksum, 224
- cFE Message Subscription Control APIs, 233
 - CFE_SB_Subscribe, 234
 - CFE_SB_SubscribeEx, 235
 - CFE_SB_SubscribeLocal, 235
 - CFE_SB_Unsubscribe, 236
 - CFE_SB_UnsubscribeLocal, 238
- cFE Miscellaneous APIs, 168
 - CFE_ES_BackgroundWakeup, 168
 - CFE_ES_CalculateCRC, 168
 - CFE_ES_ProcessAsyncEvent, 169
 - CFE_ES_WriteToSysLog, 169
- cFE Miscellaneous Time APIs, 285
 - CFE_TIME_Local1HzISR, 286
 - CFE_TIME_Print, 286
- cFE Mission Configuration Parameters, 101
- cFE Performance Monitor APIs, 182
 - CFE_ES_PerfLogAdd, 184
 - CFE_ES_PerfLogEntry, 183
 - CFE_ES_PerfLogExit, 183
- cFE Pipe Management APIs, 228
 - CFE_SB_CreatePipe, 229
 - CFE_SB_DeletePipe, 230
 - CFE_SB_GetPipeIdByName, 230
 - CFE_SB_GetPipeName, 231
 - CFE_SB_GetPipeOpts, 231
 - CFE_SB_PipeId_ToIndex, 232
 - CFE_SB_SetPipeOpts, 233
- cFE Registration APIs, 190, 253
 - CFE_EVS_Register, 190
 - CFE_TBL_Register, 253
 - CFE_TBL_Share, 255
 - CFE_TBL_Unregister, 256
- cFE Reset Event Filter APIs, 195
 - CFE_EVS_ResetAllFilters, 195
 - CFE_EVS_ResetFilter, 195
- cFE Resource ID APIs, 145
 - CFE_ES_AppID_ToIndex, 145
 - CFE_ES_CounterID_ToIndex, 145
 - CFE_ES_LibID_ToIndex, 146
 - CFE_ES_TaskID_ToIndex, 147
- cFE Resource ID base values, 287
 - CFE_CONFIGID_BASE, 288
 - CFE_ES_APPID_BASE, 288
 - CFE_ES_CDSBLOCKID_BASE, 288
 - CFE_ES_COUNTID_BASE, 288
 - CFE_ES_LIBID_BASE, 288
 - CFE_ES_POOLID_BASE, 288
 - CFE_ES_TASKID_BASE, 288
 - CFE_RESOURCEID_CONFIGID_BASE_OFFSET, 288
 - CFE_RESOURCEID_ES_APPID_BASE_OFFSET, 288
 - CFE_RESOURCEID_ES_CDSBLOCKID_BASE_OFFSET, 288
 - CFE_RESOURCEID_ES_COUNTID_BASE_OFFSET, 288
 - CFE_RESOURCEID_ES_LIBID_BASE_OFFSET, 288
 - CFE_RESOURCEID_ES_POOLID_BASE_OFFSET, 288
 - CFE_RESOURCEID_ES_TASKID_BASE_OFFSET, 288
 - CFE_RESOURCEID_SB_PIPEID_RESOURCE_BASE_OFFSET, 288
 - CFE_RESOURCEID_TBL_ACCESSID_BASE_OFFSET, 288
 - CFE_RESOURCEID_TBL_DUMPCTRLID_BASE_OFFSET, 288
 - CFE_RESOURCEID_TBL_LOADBUFFID_BASE_OFFSET, 288
 - CFE_RESOURCEID_TBL_REGID_BASE_OFFSET, 288
 - CFE_RESOURCEID_TBL_VALRESULTID_BASE_OFFSET, 288
 - CFE_SB_PIPEID_BASE, 288
 - CFE_TBL_DUMPCTRLID_BASE, 288
 - CFE_TBL_HANDLE_BASE, 288
 - CFE_TBL_LOADBUFFID_BASE, 288
 - CFE_TBL_REGID_BASE, 288
 - CFE_TBL_VALRESULTID_BASE, 288
- cFE Return Code Defines, 121
 - CFE_ES_APP_CLEANUP_ERR, 126

CFE_ES_BAD_ARGUMENT, 126
CFE_ES_BIN_SEM_DELETE_ERR, 127
CFE_ES_BUFFER_NOT_IN_POOL, 127
CFE_ES_CDS_ACCESS_ERROR, 127
CFE_ES_CDS_ALREADY_EXISTS, 127
CFE_ES_CDS_BLOCK_CRC_ERR, 127
CFE_ES_CDS_INSUFFICIENT_MEMORY, 127
CFE_ES_CDS_INVALID, 127
CFE_ES_CDS_INVALID_NAME, 127
CFE_ES_CDS_INVALID_SIZE, 128
CFE_ES_CDS_OWNER_ACTIVE_ERR, 128
CFE_ES_CDS_WRONG_TYPE_ERR, 128
CFE_ES_COUNT_SEM_DELETE_ERR, 128
CFE_ES_ERR_APP_CREATE, 128
CFE_ES_ERR_APP_REGISTER, 128
CFE_ES_ERR_CHILD_TASK_CREATE, 128
CFE_ES_ERR_CHILD_TASK_DELETE, 129
CFE_ES_ERR_CHILD_TASK_DELETE_MAIN_TASK,
129
CFE_ES_ERR_CHILD_TASK_REGISTER, 129
CFE_ES_ERR_DUPLICATE_NAME, 129
CFE_ES_ERR_LOAD_LIB, 129
CFE_ES_ERR_MEM_BLOCK_SIZE, 129
CFE_ES_ERR_NAME_NOT_FOUND, 129
CFE_ES_ERR_RESOURCEID_NOT_VALID, 129
CFE_ES_ERR_SYS_LOG_FULL, 130
CFE_ES_ERR_SYS_LOG_TRUNCATED, 130
CFE_ES_FILE_CLOSE_ERR, 130
CFE_ES_FILE_IO_ERR, 130
CFE_ES_LIB_ALREADY_LOADED, 130
CFE_ES_MUT_SEM_DELETE_ERR, 130
CFE_ES_NO_RESOURCE_IDS_AVAILABLE, 130
CFE_ES_NOT_IMPLEMENTED, 130
CFE_ES_OPERATION_TIMED_OUT, 131
CFE_ES_POOL_BLOCK_INVALID, 131
CFE_ES_QUEUE_DELETE_ERR, 131
CFE_ES_RST_ACCESS_ERR, 131
CFE_ES_TASK_DELETE_ERR, 131
CFE_ES_TIMER_DELETE_ERR, 131
CFE_EVS_APP_FILTER_OVERLOAD, 131
CFE_EVS_APP_ILLEGAL_APP_ID, 132
CFE_EVS_APP_NOT_REGISTERED, 132
CFE_EVS_APP_SQUELCHED, 132
CFE_EVS_EVT_NOT_REGISTERED, 132
CFE_EVS_FILE_WRITE_ERROR, 132
CFE_EVS_INVALID_PARAMETER, 132
CFE_EVS_NOT_IMPLEMENTED, 132
CFE_EVS_RESET_AREA_POINTER, 132
CFE_EVS_UNKNOWN_FILTER, 133
CFE_FS_BAD_ARGUMENT, 133
CFE_FS_FNAME_TOO_LONG, 133
CFE_FS_INVALID_PATH, 133
CFE_FS_NOT_IMPLEMENTED, 133
CFE_SB_BAD_ARGUMENT, 133
CFE_SB_BUF_ALLOC_ERR, 133
CFE_SB_BUFFER_INVALID, 134
CFE_SB_INTERNAL_ERR, 134
CFE_SB_MAX_DESTS_MET, 134
CFE_SB_MAX_MSGS_MET, 134
CFE_SB_MAX_PIPES_MET, 134
CFE_SB_MSG_TOO_BIG, 134
CFE_SB_NO_MESSAGE, 135
CFE_SB_NOT_IMPLEMENTED, 135
CFE_SB_PIPE_CR_ERR, 135
CFE_SB_PIPE_RD_ERR, 135
CFE_SB_TIME_OUT, 135
CFE_SB_WRONG_MSG_TYPE, 135
CFE_STATUS_BAD_COMMAND_CODE, 135
CFE_STATUS_EXTERNAL_RESOURCE_FAIL, 136
CFE_STATUS_INCORRECT_STATE, 136
CFE_STATUS_NO_COUNTER_INCREMENT, 136
CFE_STATUS_NOT_IMPLEMENTED, 136
CFE_STATUS_RANGE_ERROR, 136
CFE_STATUS_REQUEST_ALREADY_PENDING,
136
CFE_STATUS_UNKNOWN_MSG_ID, 136
CFE_STATUS_VALIDATION_FAILURE, 137
CFE_STATUS_WRONG_MSG_LENGTH, 137
CFE_SUCCESS, 137
CFE_TBL_BAD_ARGUMENT, 137
CFE_TBL_ERR_ACCESS, 137
CFE_TBL_ERR_BAD_CONTENT_ID, 137
CFE_TBL_ERR_BAD_PROCESSOR_ID, 137
CFE_TBL_ERR_BAD_SPACECRAFT_ID, 138
CFE_TBL_ERR_BAD_SUBTYPE_ID, 138
CFE_TBL_ERR_DUMP_ONLY, 138
CFE_TBL_ERR_DUPLICATE_DIFF_SIZE, 138
CFE_TBL_ERR_DUPLICATE_NOT_OWNED, 138
CFE_TBL_ERR_FILE_FOR_WRONG_TABLE, 138
CFE_TBL_ERR_FILE_SIZE_INCONSISTENT, 138
CFE_TBL_ERR_FILE_TOO_LARGE, 138
CFE_TBL_ERR_FILENAME_TOO_LONG, 139
CFE_TBL_ERR_HANDLES_FULL, 139
CFE_TBL_ERR_ILLEGAL_SRC_TYPE, 139
CFE_TBL_ERR_INVALID_HANDLE, 139
CFE_TBL_ERR_INVALID_NAME, 139
CFE_TBL_ERR_INVALID_OPTIONS, 139
CFE_TBL_ERR_INVALID_SIZE, 140
CFE_TBL_ERR_LOAD_IN_PROGRESS, 140
CFE_TBL_ERR_LOAD_INCOMPLETE, 140
CFE_TBL_ERR_NEVER_LOADED, 140
CFE_TBL_ERR_NO_ACCESS, 140
CFE_TBL_ERR_NO_BUFFER_AVAIL, 140
CFE_TBL_ERR_NO_STD_HEADER, 141
CFE_TBL_ERR_NO_TBL_HEADER, 141
CFE_TBL_ERR_PARTIAL_LOAD, 141
CFE_TBL_ERR_REGISTRY_FULL, 141
CFE_TBL_ERR_SHORT_FILE, 141

- CFE_TBL_ERR_UNREGISTERED, [141](#)
- CFE_TBL_INFO_DUMP_PENDING, [141](#)
- CFE_TBL_INFO_NO_DUMP_PENDING, [142](#)
- CFE_TBL_INFO_NO_UPDATE_PENDING, [142](#)
- CFE_TBL_INFO_NO_VALIDATION_PENDING, [142](#)
- CFE_TBL_INFO_RECOVERED_TBL, [142](#)
- CFE_TBL_INFO_TABLE_LOCKED, [142](#)
- CFE_TBL_INFO_UPDATE_PENDING, [142](#)
- CFE_TBL_INFO_UPDATED, [142](#)
- CFE_TBL_INFO_VALIDATION_PENDING, [142](#)
- CFE_TBL_MESSAGE_ERROR, [143](#)
- CFE_TBL_NOT_IMPLEMENTED, [143](#)
- CFE_TBL_WARN_DUPLICATE, [143](#)
- CFE_TBL_WARN_NOT_CRITICAL, [143](#)
- CFE_TBL_WARN_PARTIAL_LOAD, [143](#)
- CFE_TBL_WARN_SHORT_FILE, [143](#)
- CFE_TIME_BAD_ARGUMENT, [143](#)
- CFE_TIME_CALLBACK_NOT_REGISTERED, [144](#)
- CFE_TIME_INTERNAL_ONLY, [144](#)
- CFE_TIME_NOT_IMPLEMENTED, [144](#)
- CFE_TIME_OUT_OF_RANGE, [144](#)
- CFE_TIME_TOO_MANY_SYNCH_CALLBACKS, [144](#)
- cFE SB Pipe options, [252](#)
 - CFE_SB_PIPEOPTS_IGNOREMINE, [252](#)
- cFE Send Event APIs, [191](#)
 - CFE_EVS_SendEvent, [191](#)
 - CFE_EVS_SendEventWithApplID, [192](#)
 - CFE_EVS_SendTimedEvent, [193](#)
- cFE Send/Receive Message APIs, [239](#)
 - CFE_SB_ReceiveBuffer, [239](#)
 - CFE_SB_TransmitMsg, [240](#)
- cFE Software Bus Commands, [44](#)
- cFE Software Bus Configuration Parameters, [45](#)
- cFE Software Bus Overview, [36](#)
- cFE Software Bus Telemetry, [45](#)
- cFE Table Services Commands, [53](#)
- cFE Table Services Configuration Parameters, [54](#)
- cFE Table Services Overview, [47](#)
- cFE Table Services Telemetry, [54](#)
- cFE Table Type Defines, [270](#)
 - CFE_TBL_OPT_BUFFER_MSK, [270](#)
 - CFE_TBL_OPT_CRITICAL, [270](#)
 - CFE_TBL_OPT_CRITICAL_MSK, [270](#)
 - CFE_TBL_OPT_DBL_BUFFER, [271](#)
 - CFE_TBL_OPT_DEFAULT, [271](#)
 - CFE_TBL_OPT_DUMP_ONLY, [271](#)
 - CFE_TBL_OPT_LD_DMP_MSK, [271](#)
 - CFE_TBL_OPT_LOAD_DUMP, [271](#)
 - CFE_TBL_OPT_NOT_CRITICAL, [271](#)
 - CFE_TBL_OPT_NOT_USR_DEF, [271](#)
 - CFE_TBL_OPT_SNGL_BUFFER, [271](#)
 - CFE_TBL_OPT_USR_DEF_ADDR, [271](#)
 - CFE_TBL_OPT_USR_DEF_MSK, [272](#)
- cFE Table Types and Table Options, [48](#)
- cFE Telemetry Mnemonic Cross Reference, [81](#)
- cFE Time Arithmetic APIs, [277](#)
 - CFE_TIME_Add, [277](#)
 - CFE_TIME_Compare, [278](#)
 - CFE_TIME_Subtract, [279](#)
- cFE Time Conversion APIs, [279](#)
 - CFE_TIME_MET2SCTime, [280](#)
 - CFE_TIME_Micro2SubSecs, [280](#)
 - CFE_TIME_Sub2MicroSecs, [281](#)
- cFE Time Services Commands, [74](#)
- cFE Time Services Configuration Parameters, [75](#)
- cFE Time Services Overview, [56](#)
- cFE Time Services Telemetry, [75](#)
- cFE Zero Copy APIs, [241](#)
 - CFE_SB_AllocateMessageBuffer, [241](#)
 - CFE_SB_ReleaseMessageBuffer, [242](#)
 - CFE_SB_TransmitBuffer, [242](#)
- cfe/docs/src/cfe_api.dox, [676](#)
- cfe/docs/src/cfe_es.dox, [676](#)
- cfe/docs/src/cfe_evs.dox, [676](#)
- cfe/docs/src/cfe_frontpage.dox, [676](#)
- cfe/docs/src/cfe_glossary.dox, [676](#)
- cfe/docs/src/cfe_sb.dox, [676](#)
- cfe/docs/src/cfe_tbl.dox, [676](#)
- cfe/docs/src/cfe_time.dox, [676](#)
- cfe/docs/src/cfe_xref.dox, [676](#)
- cfe/docs/src/cfs_versions.dox, [676](#)
- cfe/modules/config/fsw/inc/cfe_config_external.h, [676](#), [677](#)
- cfe/modules/config/fsw/inc/cfe_config_init.h, [677](#), [678](#)
- cfe/modules/config/fsw/inc/cfe_config_lookup.h, [678](#)
- cfe/modules/config/fsw/inc/cfe_config_nametable.h, [679](#), [680](#)
- cfe/modules/config/fsw/inc/cfe_config_set.h, [680](#), [681](#)
- cfe/modules/config/fsw/inc/cfe_config_table.h, [681](#), [682](#)
- cfe/modules/core_api/config/default_cfe_core_api_base_msgid_values.h, [683](#), [684](#)
- cfe/modules/core_api/config/default_cfe_core_api_interface_cfg_values.h, [684](#)
- cfe/modules/core_api/config/default_cfe_core_api_msgid_mapping.h, [685](#), [687](#)
- cfe/modules/core_api/config/default_cfe_mission_cfg.h, [687](#)
- cfe/modules/core_api/config/default_cfe_msgids.h, [688](#)
- cfe/modules/core_api/fsw/inc/cfe.h, [689](#)
- cfe/modules/core_api/fsw/inc/cfe_config.h, [690](#), [694](#)
- cfe/modules/core_api/fsw/inc/cfe_config_api_typedefs.h, [694](#), [695](#)
- cfe/modules/core_api/fsw/inc/cfe_core_api_base_msgids.h, [696](#)
- cfe/modules/core_api/fsw/inc/cfe_core_api_interface_cfg.h, [697](#), [699](#)
- cfe/modules/core_api/fsw/inc/cfe_endian.h, [700](#)

- cfe/modules/core_api/fsw/inc/cfe_error.h, 701, 710
cfe/modules/core_api/fsw/inc/cfe_es.h, 715, 718
cfe/modules/core_api/fsw/inc/cfe_es_api_typedefs.h, 721, 726
cfe/modules/core_api/fsw/inc/cfe_evs.h, 728, 729
cfe/modules/core_api/fsw/inc/cfe_evs_api_typedefs.h, 730, 732
cfe/modules/core_api/fsw/inc/cfe_fs.h, 733, 734
cfe/modules/core_api/fsw/inc/cfe_fs_api_typedefs.h, 735, 737
cfe/modules/core_api/fsw/inc/cfe_msg.h, 739, 741
cfe/modules/core_api/fsw/inc/cfe_msg_api_typedefs.h, 743, 747
cfe/modules/core_api/fsw/inc/cfe_resourceid.h, 748, 755
cfe/modules/core_api/fsw/inc/cfe_resourceid_api_typedefs.h, 756, 757
cfe/modules/core_api/fsw/inc/cfe_sb.h, 758, 760
cfe/modules/core_api/fsw/inc/cfe_sb_api_typedefs.h, 762, 765
cfe/modules/core_api/fsw/inc/cfe_tbl.h, 766, 768
cfe/modules/core_api/fsw/inc/cfe_tbl_api_typedefs.h, 770, 773
cfe/modules/core_api/fsw/inc/cfe_tbl_filedef.h, 775, 776
cfe/modules/core_api/fsw/inc/cfe_time.h, 777, 779
cfe/modules/core_api/fsw/inc/cfe_time_api_typedefs.h, 780, 782
cfe/modules/core_api/fsw/inc/cfe_version.h, 783, 785
cfe/modules/es/config/default_cfe_es_extern_typedefs.h, 786, 793
cfe/modules/es/config/default_cfe_es_fcncode_values.h, 796, 797
cfe/modules/es/config/default_cfe_es_interface_cfg_values.h, 798
cfe/modules/es/config/default_cfe_es_internal_cfg_values.h, 799
cfe/modules/es/config/default_cfe_es_memaddress.h, 800, 801
cfe/modules/es/config/default_cfe_es_mission_cfg.h, 802
cfe/modules/es/config/default_cfe_es_msg.h, 802, 803
cfe/modules/es/config/default_cfe_es_msgdefs.h, 803, 807
cfe/modules/es/config/default_cfe_es_msgid_values.h, 810
cfe/modules/es/config/default_cfe_es_msgids.h, 811, 812
cfe/modules/es/config/default_cfe_es_msgstruct.h, 812, 817
cfe/modules/es/config/default_cfe_es_platform_cfg.h, 819, 820
cfe/modules/es/config/default_cfe_es_topicid_values.h, 820
cfe/modules/es/fsw/inc/cfe_es_eventids.h, 821, 846
cfe/modules/es/fsw/inc/cfe_es_fcncodes.h, 849, 871
cfe/modules/es/fsw/inc/cfe_es_interface_cfg.h, 872, 875
cfe/modules/es/fsw/inc/cfe_es_internal_cfg.h, 876, 906
cfe/modules/es/fsw/inc/cfe_es_topicids.h, 909, 911
cfe/modules/evs/config/default_cfe_evs_extern_typedefs.h, 911, 914
cfe/modules/evs/config/default_cfe_evs_fcncode_values.h, 915, 917
cfe/modules/evs/config/default_cfe_evs_interface_cfg_values.h, 917, 918
cfe/modules/evs/config/default_cfe_evs_internal_cfg_values.h, 918, 919
cfe/modules/evs/config/default_cfe_evs_mission_cfg.h, 919
cfe/modules/evs/config/default_cfe_evs_msg.h, 920
cfe/modules/evs/config/default_cfe_evs_msgdefs.h, 921, 925
cfe/modules/evs/config/default_cfe_evs_msgid_values.h, 926, 927
cfe/modules/evs/config/default_cfe_evs_msgids.h, 927, 928
cfe/modules/evs/config/default_cfe_evs_msgstruct.h, 929, 932
cfe/modules/evs/config/default_cfe_evs_platform_cfg.h, 935
cfe/modules/evs/config/default_cfe_evs_topicid_values.h, 936
cfe/modules/evs/fsw/inc/cfe_evs_eventids.h, 936, 948
cfe/modules/evs/fsw/inc/cfe_evs_fcncodes.h, 950, 968
cfe/modules/evs/fsw/inc/cfe_evs_interface_cfg.h, 969, 970
cfe/modules/evs/fsw/inc/cfe_evs_internal_cfg.h, 970, 976
cfe/modules/evs/fsw/inc/cfe_evs_topicids.h, 977, 978
cfe/modules/fs/config/default_cfe_fs_extern_typedefs.h, 979
cfe/modules/fs/config/default_cfe_fs_filedef.h, 980, 982
cfe/modules/fs/config/default_cfe_fs_interface_cfg_values.h, 982, 983
cfe/modules/fs/config/default_cfe_fs_mission_cfg.h, 983, 984
cfe/modules/fs/fsw/inc/cfe_fs_interface_cfg.h, 984, 985
cfe/modules/msg/fsw/inc/ccsds_hdr.h, 985, 986
cfe/modules/resourceid/fsw/inc/cfe_core_resourceid_basevalues.h, 987, 988
cfe/modules/resourceid/fsw/inc/cfe_resourceid_basevalue.h, 989, 990
cfe/modules/sb/config/default_cfe_sb_extern_typedefs.h, 991, 993
cfe/modules/sb/config/default_cfe_sb_fcncode_values.h, 994, 995
cfe/modules/sb/config/default_cfe_sb_interface_cfg_values.h, 995, 996
cfe/modules/sb/config/default_cfe_sb_internal_cfg_values.h, 996, 997
cfe/modules/sb/config/default_cfe_sb_mission_cfg.h, 997
cfe/modules/sb/config/default_cfe_sb_msg.h, 998
cfe/modules/sb/config/default_cfe_sb_msgdefs.h, 998, 1001

- cfe/modules/sb/config/default_cfe_sb_msgid_values.h, 1003, 1004
- cfe/modules/sb/config/default_cfe_sb_msgids.h, 1004, 1005
- cfe/modules/sb/config/default_cfe_sb_msgstruct.h, 1006, 1008
- cfe/modules/sb/config/default_cfe_sb_platform_cfg.h, 1010
- cfe/modules/sb/config/default_cfe_sb_topicid_values.h, 1010, 1011
- cfe/modules/sb/fsw/inc/cfe_sb_eventids.h, 1011, 1031
- cfe/modules/sb/fsw/inc/cfe_sb_fcncodes.h, 1033, 1042
- cfe/modules/sb/fsw/inc/cfe_sb_interface_cfg.h, 1043, 1044
- cfe/modules/sb/fsw/inc/cfe_sb_internal_cfg.h, 1045, 1059
- cfe/modules/sb/fsw/inc/cfe_sb_topicids.h, 1061, 1063
- cfe/modules/tbl/config/default_cfe_tbl_extern_typedefs.h, 1063, 1065
- cfe/modules/tbl/config/default_cfe_tbl_fcncode_values.h, 1066, 1067
- cfe/modules/tbl/config/default_cfe_tbl_interface_cfg_values.h, 1068
- cfe/modules/tbl/config/default_cfe_tbl_internal_cfg_values.h, 1068, 1069
- cfe/modules/tbl/config/default_cfe_tbl_mission_cfg.h, 1069, 1071
- cfe/modules/tbl/config/default_cfe_tbl_msg.h, 1071
- cfe/modules/tbl/config/default_cfe_tbl_msgdefs.h, 1072, 1074
- cfe/modules/tbl/config/default_cfe_tbl_msgid_values.h, 1076, 1077
- cfe/modules/tbl/config/default_cfe_tbl_msgids.h, 1077, 1078
- cfe/modules/tbl/config/default_cfe_tbl_msgstruct.h, 1079, 1081
- cfe/modules/tbl/config/default_cfe_tbl_platform_cfg.h, 1082, 1091
- cfe/modules/tbl/config/default_cfe_tbl_topicid_values.h, 1092
- cfe/modules/tbl/fsw/inc/cfe_tbl_eventids.h, 1093, 1113
- cfe/modules/tbl/fsw/inc/cfe_tbl_fcncodes.h, 1115, 1125
- cfe/modules/tbl/fsw/inc/cfe_tbl_interface_cfg.h, 1125, 1127
- cfe/modules/tbl/fsw/inc/cfe_tbl_internal_cfg.h, 1127, 1136
- cfe/modules/tbl/fsw/inc/cfe_tbl_topicids.h, 1137, 1138
- cfe/modules/time/config/default_cfe_time_extern_typedefs.h, 1139, 1144
- cfe/modules/time/config/default_cfe_time_fcncode_values.h, 1145, 1146
- cfe/modules/time/config/default_cfe_time_interface_cfg_values.h, 1147, 1148
- cfe/modules/time/config/default_cfe_time_internal_cfg_values.h, 1148
- cfe/modules/time/config/default_cfe_time_mission_cfg.h, 1149
- cfe/modules/time/config/default_cfe_time_msg.h, 1150
- cfe/modules/time/config/default_cfe_time_msgdefs.h, 1150, 1153
- cfe/modules/time/config/default_cfe_time_msgid_values.h, 1155, 1156
- cfe/modules/time/config/default_cfe_time_msgids.h, 1157, 1158
- cfe/modules/time/config/default_cfe_time_msgstruct.h, 1159, 1162
- cfe/modules/time/config/default_cfe_time_platform_cfg.h, 1164
- cfe/modules/time/config/default_cfe_time_topicid_values.h, 1165
- cfe/modules/time/fsw/inc/cfe_time_eventids.h, 1165, 1176
- cfe/modules/time/fsw/inc/cfe_time_fcncodes.h, 1177, 1192
- cfe/modules/time/fsw/inc/cfe_time_interface_cfg.h, 1193, 1201
- cfe/modules/time/fsw/inc/cfe_time_internal_cfg.h, 1202, 1210
- cfe/modules/time/fsw/inc/cfe_time_topicids.h, 1211, 1213
- CFE_BIT
 - cfe_sb.h, 760
- CFE_BUILD_BASELINE
 - cfe_version.h, 783
- CFE_BUILD_CODENAME
 - cfe_version.h, 783
- CFE_BUILD_DEV_CYCLE
 - cfe_version.h, 784
- CFE_BUILD_NUMBER
 - cfe_version.h, 784
- CFE_CFG_MAX_VERSION_STR_LEN
 - cfe_version.h, 784
- CFE_CLR
 - cfe_sb.h, 760
- cfe_config.h
 - CFE_Config_GetArrayValue, 690
 - CFE_Config_GetIdByName, 691
 - CFE_Config_GetName, 691
 - CFE_Config_GetObjPointer, 692
 - CFE_Config_GetString, 692
 - CFE_Config_GetValue, 692
 - CFE_Config_GetVersionString, 693
 - CFE_Config_IterateAll, 693
 - cfe_config_api_typedefs.h
 - CFE_Config_ArrayValue_t, 695
 - CFE_Config_Callback_t, 695
 - CFE_CONFIGID_C, 695
 - CFE_ConfigId_t, 695
 - CFE_CONFIGID_UNDEFINED, 695
 - CFE_Config_ArrayValue, 424
 - ElementPtr, 424
 - NumElements, 424

CFE_Config_ArrayValue_t
 cfe_config_api_typedefs.h, 695
CFE_Config_Callback_t
 cfe_config_api_typedefs.h, 695
cfe_config_external.h
 CFE_Config_SetupPlatformConfigInfo, 677
CFE_Config_GetArrayValue
 cfe_config.h, 690
CFE_Config_GetIdByName
 cfe_config.h, 691
CFE_Config_GetName
 cfe_config.h, 691
CFE_Config_GetObjPointer
 cfe_config.h, 692
CFE_Config_GetString
 cfe_config.h, 692
CFE_Config_GetValue
 cfe_config.h, 692
CFE_Config_GetVersionString
 cfe_config.h, 693
CFE_Config_IdNameEntry, 424
 Name, 425
CFE_Config_IdNameEntry_t
 cfe_config_nametable.h, 679
CFE_Config_Init
 cfe_config_init.h, 677
cfe_config_init.h
 CFE_Config_Init, 677
 CFE_Config_SetupBasicBuildInfo, 677
CFE_Config_IterateAll
 cfe_config.h, 693
CFE_Config_LocateConfigRecordById
 cfe_config_lookup.h, 678
cfe_config_lookup.h
 CFE_Config_LocateConfigRecordById, 678
cfe_config_nametable.h
 CFE_Config_IdNameEntry_t, 679
 CFE_CONFIGID_NAMETABLE, 679
cfe_config_set.h
 CFE_Config_SetArrayValue, 680
 CFE_Config_SetObjPointer, 680
 CFE_Config_SetString, 681
 CFE_Config_SetValue, 681
CFE_Config_SetArrayValue
 cfe_config_set.h, 680
CFE_Config_SetObjPointer
 cfe_config_set.h, 680
CFE_Config_SetString
 cfe_config_set.h, 681
CFE_Config_SetupBasicBuildInfo
 cfe_config_init.h, 677
CFE_Config_SetupPlatformConfigInfo
 cfe_config_external.h, 677
CFE_Config_SetValue
 cfe_config_set.h, 681
cfe_config_table.h
 CFE_Config_ValueBuffer_t, 682
 CFE_Config_ValueEntry_t, 682
 CFE_ConfigType, 682
 CFE_ConfigType_ARRAY, 682
 CFE_ConfigType_POINTER, 682
 CFE_ConfigType_STRING, 682
 CFE_ConfigType_t, 682
 CFE_ConfigType_UNDEFINED, 682
 CFE_ConfigType_VALUE, 682
CFE_Config_ValueBuffer, 425
 AsInteger, 425
 AsPointer, 425
CFE_Config_ValueBuffer_t
 cfe_config_table.h, 682
CFE_Config_ValueEntry, 425
 ActualType, 425
 Datum, 425
CFE_Config_ValueEntry_t
 cfe_config_table.h, 682
CFE_CONFIGID_BASE
 cFE Resource ID base values, 288
CFE_CONFIGID_C
 cfe_config_api_typedefs.h, 695
CFE_CONFIGID_NAMETABLE
 cfe_config_nametable.h, 679
CFE_ConfigId_t
 cfe_config_api_typedefs.h, 695
CFE_CONFIGID_UNDEFINED
 cfe_config_api_typedefs.h, 695
CFE_ConfigType
 cfe_config_table.h, 682
CFE_ConfigType_ARRAY
 cfe_config_table.h, 682
CFE_ConfigType_POINTER
 cfe_config_table.h, 682
CFE_ConfigType_STRING
 cfe_config_table.h, 682
CFE_ConfigType_t
 cfe_config_table.h, 682
CFE_ConfigType_UNDEFINED
 cfe_config_table.h, 682
CFE_ConfigType_VALUE
 cfe_config_table.h, 682
cfe_core_api_interface_cfg.h
 CFE_MISSION_MAX_API_LEN, 697
 CFE_MISSION_MAX_FILE_LEN, 697
 CFE_MISSION_MAX_NUM_FILES, 698
 CFE_MISSION_MAX_PATH_LEN, 698
 DEFAULT_CFE_MISSION_CORE_API_MAX_API_LEN,
 699
 DEFAULT_CFE_MISSION_CORE_API_MAX_FILE_LEN,
 699

DEFAULT_CFE_MISSION_CORE_API_MAX_NUM_FILES, CFE_ES_PoolAlign_t, 725
 699
 DEFAULT_CFE_MISSION_CORE_API_MAX_PATH_LEN, CFE_ES_STATIC_POOL_TYPE, 724
 699
 cfe_endian.h
 CFE_MAKE_BIG16, 700
 CFE_MAKE_BIG32, 700
 cfe_error.h
 CFE_ES_StatusToString, 710
 CFE_EVENTS_SERVICE, 708
 CFE_EXECUTIVE_SERVICE, 708
 CFE_FILE_SERVICE, 708
 CFE_GENERIC_SERVICE, 708
 CFE_SERVICE_BITMASK, 708
 CFE_SEVERITY_BITMASK, 708
 CFE_SEVERITY_ERROR, 708
 CFE_SEVERITY_INFO, 708
 CFE_SEVERITY_SUCCESS, 709
 CFE_SOFTWARE_BUS_SERVICE, 709
 CFE_STATUS_C, 709
 CFE_STATUS_STRING_LENGTH, 709
 CFE_Status_t, 709
 CFE_StatusString_t, 709
 CFE_TABLE_SERVICE, 709
 CFE_TIME_SERVICE, 709
 cfe_es.h
 OS_PRINTF, 718
 CFE_ES_ALL_APPS_EID
 cfe_es_eventids.h, 824
 cfe_es_api_typedefs.h
 CFE_ES_APP_RESTART, 722
 CFE_ES_APPID_C, 722
 CFE_ES_APPID_UNDEFINED, 722
 CFE_ES_CDS_BAD_HANDLE, 722
 CFE_ES_CDSHANDLE_C, 723
 CFE_ES_ChildTaskMainFuncPtr_t, 724
 CFE_ES_COUNTERID_C, 723
 CFE_ES_COUNTERID_UNDEFINED, 723
 CFE_ES_CrcType_16_ARC, 726
 CFE_ES_CrcType_CRC_16, 726
 CFE_ES_CrcType_CRC_32, 726
 CFE_ES_CrcType_CRC_8, 726
 CFE_ES_CrcType_Enum, 725
 CFE_ES_CrcType_Enum_t, 724
 CFE_ES_CrcType_MAX, 726
 CFE_ES_CrcType_NONE, 726
 CFE_ES_LIBID_C, 723
 CFE_ES_LIBID_UNDEFINED, 723
 CFE_ES_LibraryEntryFuncPtr_t, 725
 CFE_ES_MEMHANDLE_C, 723
 CFE_ES_MEMHANDLE_UNDEFINED, 723
 CFE_ES_MEMPOOLBUF_C, 723
 CFE_ES_MemPoolBuf_t, 725
 CFE_ES_NO_MUTEX, 723
 CFE_ES_PoolAlign_t, 725
 CFE_ES_StackPointer_t, 725
 CFE_ES_STATIC_POOL_TYPE, 724
 CFE_ES_TASK_STACK_ALLOCATE, 724
 CFE_ES_TaskEntryFuncPtr_t, 725
 CFE_ES_TASKID_C, 724
 CFE_ES_TASKID_UNDEFINED, 724
 CFE_ES_USE_MUTEX, 724
 CFE_ES_APP_CLEANUP_ERR
 cFE Return Code Defines, 126
 CFE_ES_APP_RESTART
 cfe_es_api_typedefs.h, 722
 CFE_ES_APP_TLM_MID
 default_cfe_es_msgids.h, 811
 CFE_ES_APPID_BASE
 cFE Resource ID base values, 288
 CFE_ES_APPID_C
 cfe_es_api_typedefs.h, 722
 CFE_ES_AppId_t
 default_cfe_es_extern_typedefs.h, 788
 CFE_ES_AppID_ToIndex
 cFE Resource ID APIs, 145
 CFE_ES_APPID_UNDEFINED
 cfe_es_api_typedefs.h, 722
 CFE_ES_AppInfo, 426
 AddressesAreValid, 427
 BSSAddress, 427
 BSSSize, 427
 CodeAddress, 427
 CodeSize, 427
 DataAddress, 428
 DataSize, 428
 EntryPoint, 428
 ExceptionAction, 428
 ExecutionCounter, 428
 FileName, 428
 MainTaskId, 428
 MainTaskName, 429
 Name, 429
 NumOfChildTasks, 429
 Padding1, 429
 Padding2, 429
 Padding3, 429
 Priority, 429
 ResourceId, 429
 StackSize, 430
 StartAddress, 430
 Type, 430
 CFE_ES_AppInfo_t
 default_cfe_es_extern_typedefs.h, 788
 CFE_ES_AppNameCmd_Payload, 430
 Application, 431
 CFE_ES_AppNameCmd_Payload_t
 default_cfe_es_msgdefs.h, 805

- CFE_ES_AppReloadCmd_Payload, 431
 - AppFileName, 431
 - Application, 431
- CFE_ES_AppReloadCmd_Payload_t
 - default_cfe_es_msgdefs.h, 805
- CFE_ES_AppState
 - default_cfe_es_extern_typedefs.h, 791
- CFE_ES_AppState_EARLY_INIT
 - default_cfe_es_extern_typedefs.h, 791
- CFE_ES_AppState_Enum_t
 - default_cfe_es_extern_typedefs.h, 788
- CFE_ES_AppState_LATE_INIT
 - default_cfe_es_extern_typedefs.h, 791
- CFE_ES_AppState_MAX
 - default_cfe_es_extern_typedefs.h, 791
- CFE_ES_AppState_RUNNING
 - default_cfe_es_extern_typedefs.h, 791
- CFE_ES_AppState_STOPPED
 - default_cfe_es_extern_typedefs.h, 791
- CFE_ES_AppState_UNDEFINED
 - default_cfe_es_extern_typedefs.h, 791
- CFE_ES_AppState_WAITING
 - default_cfe_es_extern_typedefs.h, 791
- CFE_ES_AppType
 - default_cfe_es_extern_typedefs.h, 791
- CFE_ES_AppType_CORE
 - default_cfe_es_extern_typedefs.h, 791
- CFE_ES_AppType_Enum_t
 - default_cfe_es_extern_typedefs.h, 788
- CFE_ES_AppType_EXTERNAL
 - default_cfe_es_extern_typedefs.h, 791
- CFE_ES_AppType_LIBRARY
 - default_cfe_es_extern_typedefs.h, 791
- CFE_ES_BackgroundWakeup
 - cFE Miscellaneous APIs, 168
- CFE_ES_BAD_ARGUMENT
 - cFE Return Code Defines, 126
- CFE_ES_BIN_SEM_DELETE_ERR
 - cFE Return Code Defines, 127
- CFE_ES_BlockStats, 431
 - BlockSize, 432
 - NumCreated, 432
 - NumFree, 432
- CFE_ES_BlockStats_t
 - default_cfe_es_extern_typedefs.h, 788
- CFE_ES_BOOT_ERR_EID
 - cfe_es_eventids.h, 824
- CFE_ES_BUFFER_NOT_IN_POOL
 - cFE Return Code Defines, 127
- CFE_ES_BUILD_INF_EID
 - cfe_es_eventids.h, 825
- CFE_ES_CalculateCRC
 - cFE Miscellaneous APIs, 168
- CFE_ES_CC1_ERR_EID
 - cfe_es_eventids.h, 825
- CFE_ES_CCVAL
 - default_cfe_es_fcncode_values.h, 796
- CFE_ES_CDS_ACCESS_ERROR
 - cFE Return Code Defines, 127
- CFE_ES_CDS_ALREADY_EXISTS
 - cFE Return Code Defines, 127
- CFE_ES_CDS_BAD_HANDLE
 - cfe_es_api_typedefs.h, 722
- CFE_ES_CDS_BLOCK_CRC_ERR
 - cFE Return Code Defines, 127
- CFE_ES_CDS_DELETE_ERR_EID
 - cfe_es_eventids.h, 825
- CFE_ES_CDS_DELETE_TBL_ERR_EID
 - cfe_es_eventids.h, 826
- CFE_ES_CDS_DELETED_INFO_EID
 - cfe_es_eventids.h, 826
- CFE_ES_CDS_DUMP_ERR_EID
 - cfe_es_eventids.h, 826
- CFE_ES_CDS_INSUFFICIENT_MEMORY
 - cFE Return Code Defines, 127
- CFE_ES_CDS_INVALID
 - cFE Return Code Defines, 127
- CFE_ES_CDS_INVALID_NAME
 - cFE Return Code Defines, 127
- CFE_ES_CDS_INVALID_SIZE
 - cFE Return Code Defines, 128
- CFE_ES_CDS_NAME_ERR_EID
 - cfe_es_eventids.h, 826
- CFE_ES_CDS_OWNER_ACTIVE_EID
 - cfe_es_eventids.h, 827
- CFE_ES_CDS_OWNER_ACTIVE_ERR
 - cFE Return Code Defines, 128
- CFE_ES_CDS_REG_DUMP_INF_EID
 - cfe_es_eventids.h, 827
- CFE_ES_CDS_REGISTER_ERR_EID
 - cfe_es_eventids.h, 827
- CFE_ES_CDS_WRONG_TYPE_ERR
 - cFE Return Code Defines, 128
- CFE_ES_CDSBLOCKID_BASE
 - cFE Resource ID base values, 288
- CFE_ES_CDSHANDLE_C
 - cfe_es_api_typedefs.h, 723
- CFE_ES_CDSHandle_t
 - default_cfe_es_extern_typedefs.h, 788
- CFE_ES_CDSRegDumpRec, 432
 - ByteAlignSpare, 433
 - Handle, 433
 - Name, 433
 - Size, 433
 - Table, 433
- CFE_ES_CDSRegDumpRec_t
 - default_cfe_es_extern_typedefs.h, 788
- CFE_ES_ChildTaskMainFuncPtr_t

- cfe_es_api_typedefs.h, [724](#)
- CFE_ES_CLEAR_ER_LOG_CC
 - cfe_es_fcncodes.h, [850](#)
- CFE_ES_CLEAR_SYS_LOG_CC
 - cfe_es_fcncodes.h, [850](#)
- CFE_ES_ClearERLogCmd, [433](#)
 - CommandHeader, [434](#)
- CFE_ES_ClearERLogCmd_t
 - default_cfe_es_msgstruct.h, [814](#)
- CFE_ES_ClearSysLogCmd, [434](#)
 - CommandHeader, [434](#)
- CFE_ES_ClearSysLogCmd_t
 - default_cfe_es_msgstruct.h, [814](#)
- CFE_ES_CMD_MID
 - default_cfe_es_msgids.h, [811](#)
- CFE_ES_CopyToCDS
 - cFE Critical Data Store APIs, [170](#)
- CFE_ES_COUNT_SEM_DELETE_ERR
 - cFE Return Code Defines, [128](#)
- CFE_ES_COUNTERID_C
 - cfe_es_api_typedefs.h, [723](#)
- CFE_ES_CounterId_t
 - default_cfe_es_extern_typedefs.h, [789](#)
- CFE_ES_CounterID_ToIndex
 - cFE Resource ID APIs, [145](#)
- CFE_ES_COUNTERID_UNDEFINED
 - cfe_es_api_typedefs.h, [723](#)
- CFE_ES_COUNTID_BASE
 - cFE Resource ID base values, [288](#)
- CFE_ES_CrcType_16_ARC
 - cfe_es_api_typedefs.h, [726](#)
- CFE_ES_CrcType_CRC_16
 - cfe_es_api_typedefs.h, [726](#)
- CFE_ES_CrcType_CRC_32
 - cfe_es_api_typedefs.h, [726](#)
- CFE_ES_CrcType_CRC_8
 - cfe_es_api_typedefs.h, [726](#)
- CFE_ES_CrcType_Enum
 - cfe_es_api_typedefs.h, [725](#)
- CFE_ES_CrcType_Enum_t
 - cfe_es_api_typedefs.h, [724](#)
- CFE_ES_CrcType_MAX
 - cfe_es_api_typedefs.h, [726](#)
- CFE_ES_CrcType_NONE
 - cfe_es_api_typedefs.h, [726](#)
- CFE_ES_CreateChildTask
 - cFE Child Task APIs, [164](#)
- CFE_ES_CREATING_CDS_DUMP_ERR_EID
 - cfe_es_eventids.h, [827](#)
- CFE_ES_DELETE_CDS_CC
 - cfe_es_fcncodes.h, [851](#)
- CFE_ES_DeleteApp
 - cFE Application Control APIs, [149](#)
- CFE_ES_DeleteCDSCmd, [434](#)
 - CommandHeader, [435](#)
 - Payload, [435](#)
- CFE_ES_DeleteCDSCmd_Payload, [435](#)
 - CdsName, [435](#)
- CFE_ES_DeleteCDSCmd_Payload_t
 - default_cfe_es_msgdefs.h, [805](#)
- CFE_ES_DeleteCDSCmd_t
 - default_cfe_es_msgstruct.h, [814](#)
- CFE_ES_DeleteChildTask
 - cFE Child Task APIs, [165](#)
- CFE_ES_DeleteGenCounter
 - cFE Generic Counter APIs, [185](#)
- CFE_ES_DUMP_CDS_REGISTRY_CC
 - cfe_es_fcncodes.h, [852](#)
- CFE_ES_DumpCDSRegistryCmd, [435](#)
 - CommandHeader, [436](#)
 - Payload, [436](#)
- CFE_ES_DumpCDSRegistryCmd_Payload, [436](#)
 - DumpFilename, [436](#)
- CFE_ES_DumpCDSRegistryCmd_Payload_t
 - default_cfe_es_msgdefs.h, [805](#)
- CFE_ES_DumpCDSRegistryCmd_t
 - default_cfe_es_msgstruct.h, [814](#)
- CFE_ES_ERLOG1_INF_EID
 - cfe_es_eventids.h, [828](#)
- CFE_ES_ERLOG2_EID
 - cfe_es_eventids.h, [828](#)
- CFE_ES_ERLOG2_ERR_EID
 - cfe_es_eventids.h, [828](#)
- CFE_ES_ERLOG_PENDING_ERR_EID
 - cfe_es_eventids.h, [828](#)
- CFE_ES_ERR_APP_CREATE
 - cFE Return Code Defines, [128](#)
- CFE_ES_ERR_APP_REGISTER
 - cFE Return Code Defines, [128](#)
- CFE_ES_ERR_CHILD_TASK_CREATE
 - cFE Return Code Defines, [128](#)
- CFE_ES_ERR_CHILD_TASK_DELETE
 - cFE Return Code Defines, [129](#)
- CFE_ES_ERR_CHILD_TASK_DELETE_MAIN_TASK
 - cFE Return Code Defines, [129](#)
- CFE_ES_ERR_CHILD_TASK_REGISTER
 - cFE Return Code Defines, [129](#)
- CFE_ES_ERR_DUPLICATE_NAME
 - cFE Return Code Defines, [129](#)
- CFE_ES_ERR_LOAD_LIB
 - cFE Return Code Defines, [129](#)
- CFE_ES_ERR_MEM_BLOCK_SIZE
 - cFE Return Code Defines, [129](#)
- CFE_ES_ERR_NAME_NOT_FOUND
 - cFE Return Code Defines, [129](#)
- CFE_ES_ERR_RESOURCEID_NOT_VALID
 - cFE Return Code Defines, [129](#)
- CFE_ES_ERR_SYS_LOG_FULL

- cFE Return Code Defines, [130](#)
- CFE_ES_ERR_SYS_LOG_TRUNCATED
 - cFE Return Code Defines, [130](#)
- CFE_ES_ERR_SYSLOGMODE_EID
 - cfe_es_eventids.h, [829](#)
- CFE_ES_ERREXIT_APP_ERR_EID
 - cfe_es_eventids.h, [829](#)
- CFE_ES_ERREXIT_APP_INF_EID
 - cfe_es_eventids.h, [829](#)
- cfe_es_eventids.h
 - CFE_ES_ALL_APPS_EID, [824](#)
 - CFE_ES_BOOT_ERR_EID, [824](#)
 - CFE_ES_BUILD_INF_EID, [825](#)
 - CFE_ES_CC1_ERR_EID, [825](#)
 - CFE_ES_CDS_DELETE_ERR_EID, [825](#)
 - CFE_ES_CDS_DELETE_TBL_ERR_EID, [826](#)
 - CFE_ES_CDS_DELETED_INFO_EID, [826](#)
 - CFE_ES_CDS_DUMP_ERR_EID, [826](#)
 - CFE_ES_CDS_NAME_ERR_EID, [826](#)
 - CFE_ES_CDS_OWNER_ACTIVE_EID, [827](#)
 - CFE_ES_CDS_REG_DUMP_INF_EID, [827](#)
 - CFE_ES_CDS_REGISTER_ERR_EID, [827](#)
 - CFE_ES_CREATING_CDS_DUMP_ERR_EID, [827](#)
 - CFE_ES_ERLOG1_INF_EID, [828](#)
 - CFE_ES_ERLOG2_EID, [828](#)
 - CFE_ES_ERLOG2_ERR_EID, [828](#)
 - CFE_ES_ERLOG_PENDING_ERR_EID, [828](#)
 - CFE_ES_ERR_SYSLOGMODE_EID, [829](#)
 - CFE_ES_ERREXIT_APP_ERR_EID, [829](#)
 - CFE_ES_ERREXIT_APP_INF_EID, [829](#)
 - CFE_ES_EXIT_APP_ERR_EID, [829](#)
 - CFE_ES_EXIT_APP_INF_EID, [830](#)
 - CFE_ES_FILEWRITE_ERR_EID, [830](#)
 - CFE_ES_INIT_INF_EID, [830](#)
 - CFE_ES_INITSTATS_INF_EID, [830](#)
 - CFE_ES_INVALID_POOL_HANDLE_ERR_EID, [831](#)
 - CFE_ES_LEN_ERR_EID, [831](#)
 - CFE_ES_MID_ERR_EID, [831](#)
 - CFE_ES_NOOP_INF_EID, [831](#)
 - CFE_ES_ONE_APP_EID, [832](#)
 - CFE_ES_ONE_APPID_ERR_EID, [832](#)
 - CFE_ES_ONE_ERR_EID, [832](#)
 - CFE_ES_OSCREATE_ERR_EID, [832](#)
 - CFE_ES_PCR_ERR1_EID, [833](#)
 - CFE_ES_PCR_ERR2_EID, [833](#)
 - CFE_ES_PERF_DATAWRITTEN_EID, [833](#)
 - CFE_ES_PERF_FILTMSKCMD_EID, [833](#)
 - CFE_ES_PERF_FILTMSKERR_EID, [834](#)
 - CFE_ES_PERF_LOG_ERR_EID, [834](#)
 - CFE_ES_PERF_STARTCMD_EID, [834](#)
 - CFE_ES_PERF_STARTCMD_ERR_EID, [834](#)
 - CFE_ES_PERF_STARTCMD_TRIG_ERR_EID, [835](#)
 - CFE_ES_PERF_STOPCMD_EID, [835](#)
 - CFE_ES_PERF_STOPCMD_ERR2_EID, [835](#)
 - CFE_ES_PERF_TRIGMSKCMD_EID, [835](#)
 - CFE_ES_PERF_TRIGMSKERR_EID, [836](#)
 - CFE_ES_RELOAD_APP_DBG_EID, [836](#)
 - CFE_ES_RELOAD_APP_ERR1_EID, [836](#)
 - CFE_ES_RELOAD_APP_ERR2_EID, [836](#)
 - CFE_ES_RELOAD_APP_ERR3_EID, [837](#)
 - CFE_ES_RELOAD_APP_ERR4_EID, [837](#)
 - CFE_ES_RELOAD_APP_INF_EID, [837](#)
 - CFE_ES_RESET_INF_EID, [837](#)
 - CFE_ES_RESET_PR_COUNT_EID, [838](#)
 - CFE_ES_RESTART_APP_DBG_EID, [838](#)
 - CFE_ES_RESTART_APP_ERR1_EID, [838](#)
 - CFE_ES_RESTART_APP_ERR2_EID, [838](#)
 - CFE_ES_RESTART_APP_ERR3_EID, [839](#)
 - CFE_ES_RESTART_APP_ERR4_EID, [839](#)
 - CFE_ES_RESTART_APP_INF_EID, [839](#)
 - CFE_ES_SET_MAX_PR_COUNT_EID, [839](#)
 - CFE_ES_START_ERR_EID, [840](#)
 - CFE_ES_START_EXC_ACTION_ERR_EID, [840](#)
 - CFE_ES_START_INF_EID, [840](#)
 - CFE_ES_START_INVALID_ENTRY_POINT_ERR_EID, [840](#)
 - CFE_ES_START_INVALID_FILENAME_ERR_EID, [841](#)
 - CFE_ES_START_NULL_APP_NAME_ERR_EID, [841](#)
 - CFE_ES_START_PRIORITY_ERR_EID, [841](#)
 - CFE_ES_STOP_DBG_EID, [841](#)
 - CFE_ES_STOP_ERR1_EID, [842](#)
 - CFE_ES_STOP_ERR2_EID, [842](#)
 - CFE_ES_STOP_ERR3_EID, [842](#)
 - CFE_ES_STOP_INF_EID, [842](#)
 - CFE_ES_SYSLOG1_INF_EID, [843](#)
 - CFE_ES_SYSLOG2_EID, [843](#)
 - CFE_ES_SYSLOG2_ERR_EID, [843](#)
 - CFE_ES_SYSLOGMODE_EID, [843](#)
 - CFE_ES_TASKINFO_EID, [844](#)
 - CFE_ES_TASKINFO_OSCREATE_ERR_EID, [844](#)
 - CFE_ES_TASKINFO_WR_ERR_EID, [844](#)
 - CFE_ES_TASKINFO_WRHDR_ERR_EID, [844](#)
 - CFE_ES_TASKWR_ERR_EID, [845](#)
 - CFE_ES_TLM_POOL_STATS_INFO_EID, [845](#)
 - CFE_ES_VERSION_INF_EID, [845](#)
 - CFE_ES_WRHDR_ERR_EID, [846](#)
 - CFE_ES_WRITE_CFE_HDR_ERR_EID, [846](#)
- CFE_ES_ExceptionAction
 - default_cfe_es_extern_typedefs.h, [791](#)
- CFE_ES_ExceptionAction_Enum_t
 - default_cfe_es_extern_typedefs.h, [789](#)
- CFE_ES_ExceptionAction_PROC_RESTART
 - default_cfe_es_extern_typedefs.h, [792](#)
- CFE_ES_ExceptionAction_RESTART_APP
 - default_cfe_es_extern_typedefs.h, [792](#)
- CFE_ES_EXIT_APP_ERR_EID

- cfe_es_eventids.h, [829](#)
- CFE_ES_EXIT_APP_INF_EID
 - cfe_es_eventids.h, [830](#)
- CFE_ES_ExitApp
 - cFE Application Behavior APIs, [152](#)
- CFE_ES_ExitChildTask
 - cFE Child Task APIs, [166](#)
- cfe_es_fcncodes.h
 - CFE_ES_CLEAR_ER_LOG_CC, [850](#)
 - CFE_ES_CLEAR_SYS_LOG_CC, [850](#)
 - CFE_ES_DELETE_CDS_CC, [851](#)
 - CFE_ES_DUMP_CDS_REGISTRY_CC, [852](#)
 - CFE_ES_NOOP_CC, [853](#)
 - CFE_ES_OVER_WRITE_SYS_LOG_CC, [854](#)
 - CFE_ES_QUERY_ALL_CC, [854](#)
 - CFE_ES_QUERY_ALL_TASKS_CC, [855](#)
 - CFE_ES_QUERY_ONE_CC, [856](#)
 - CFE_ES_RELOAD_APP_CC, [857](#)
 - CFE_ES_RESET_COUNTERS_CC, [858](#)
 - CFE_ES_RESET_PR_COUNT_CC, [859](#)
 - CFE_ES_RESTART_APP_CC, [859](#)
 - CFE_ES_RESTART_CC, [860](#)
 - CFE_ES_SEND_MEM_POOL_STATS_CC, [861](#)
 - CFE_ES_SET_MAX_PR_COUNT_CC, [862](#)
 - CFE_ES_SET_PERF_FILTER_MASK_CC, [863](#)
 - CFE_ES_SET_PERF_TRIGGER_MASK_CC, [864](#)
 - CFE_ES_START_APP_CC, [864](#)
 - CFE_ES_START_PERF_DATA_CC, [865](#)
 - CFE_ES_STOP_APP_CC, [866](#)
 - CFE_ES_STOP_PERF_DATA_CC, [867](#)
 - CFE_ES_WRITE_ER_LOG_CC, [868](#)
 - CFE_ES_WRITE_SYS_LOG_CC, [869](#)
- CFE_ES_FILE_CLOSE_ERR
 - cFE Return Code Defines, [130](#)
- CFE_ES_FILE_IO_ERR
 - cFE Return Code Defines, [130](#)
- CFE_ES_FileNameCmd, [436](#)
 - CommandHeader, [437](#)
 - Payload, [437](#)
- CFE_ES_FileNameCmd_Payload, [437](#)
 - FileName, [437](#)
- CFE_ES_FileNameCmd_Payload_t
 - default_cfe_es_msgdefs.h, [805](#)
- CFE_ES_FileNameCmd_t
 - default_cfe_es_msgstruct.h, [814](#)
- CFE_ES_FILEWRITE_ERR_EID
 - cfe_es_eventids.h, [830](#)
- CFE_ES_FunctionCode_
 - default_cfe_es_fcncode_values.h, [796](#)
- CFE_ES_FunctionCode_CLEAR_ER_LOG
 - default_cfe_es_fcncode_values.h, [797](#)
- CFE_ES_FunctionCode_CLEAR_SYS_LOG
 - default_cfe_es_fcncode_values.h, [797](#)
- CFE_ES_FunctionCode_DELETE_CDS
 - default_cfe_es_fcncode_values.h, [797](#)
- CFE_ES_FunctionCode_DUMP_CDS_REGISTRY
 - default_cfe_es_fcncode_values.h, [797](#)
- CFE_ES_FunctionCode_NOOP
 - default_cfe_es_fcncode_values.h, [796](#)
- CFE_ES_FunctionCode_OVER_WRITE_SYS_LOG
 - default_cfe_es_fcncode_values.h, [797](#)
- CFE_ES_FunctionCode_QUERY_ALL
 - default_cfe_es_fcncode_values.h, [797](#)
- CFE_ES_FunctionCode_QUERY_ALL_TASKS
 - default_cfe_es_fcncode_values.h, [797](#)
- CFE_ES_FunctionCode_QUERY_ONE
 - default_cfe_es_fcncode_values.h, [797](#)
- CFE_ES_FunctionCode_RELOAD_APP
 - default_cfe_es_fcncode_values.h, [797](#)
- CFE_ES_FunctionCode_RESET_COUNTERS
 - default_cfe_es_fcncode_values.h, [797](#)
- CFE_ES_FunctionCode_RESET_PR_COUNT
 - default_cfe_es_fcncode_values.h, [797](#)
- CFE_ES_FunctionCode_RESTART
 - default_cfe_es_fcncode_values.h, [797](#)
- CFE_ES_FunctionCode_RESTART_APP
 - default_cfe_es_fcncode_values.h, [797](#)
- CFE_ES_FunctionCode_SEND_MEM_POOL_STATS
 - default_cfe_es_fcncode_values.h, [797](#)
- CFE_ES_FunctionCode_SET_MAX_PR_COUNT
 - default_cfe_es_fcncode_values.h, [797](#)
- CFE_ES_FunctionCode_SET_PERF_FILTER_MASK
 - default_cfe_es_fcncode_values.h, [797](#)
- CFE_ES_FunctionCode_SET_PERF_TRIGGER_MASK
 - default_cfe_es_fcncode_values.h, [797](#)
- CFE_ES_FunctionCode_START_APP
 - default_cfe_es_fcncode_values.h, [797](#)
- CFE_ES_FunctionCode_START_PERF_DATA
 - default_cfe_es_fcncode_values.h, [797](#)
- CFE_ES_FunctionCode_STOP_APP
 - default_cfe_es_fcncode_values.h, [797](#)
- CFE_ES_FunctionCode_STOP_PERF_DATA
 - default_cfe_es_fcncode_values.h, [797](#)
- CFE_ES_FunctionCode_WRITE_ER_LOG
 - default_cfe_es_fcncode_values.h, [797](#)
- CFE_ES_FunctionCode_WRITE_SYS_LOG
 - default_cfe_es_fcncode_values.h, [797](#)
- CFE_ES_GetAppID
 - cFE Information APIs, [156](#)
- CFE_ES_GetAppIDByName
 - cFE Information APIs, [156](#)
- CFE_ES_GetAppInfo
 - cFE Information APIs, [157](#)
- CFE_ES_GetAppName
 - cFE Information APIs, [158](#)
- CFE_ES_GetCDSBlockIDByName
 - cFE Critical Data Store APIs, [171](#)
- CFE_ES_GetCDSBlockName

- cFE Critical Data Store APIs, [172](#)
- CFE_ES_GetGenCount
 - cFE Generic Counter APIs, [185](#)
- CFE_ES_GetGenCounterIDByName
 - cFE Generic Counter APIs, [186](#)
- CFE_ES_GetGenCounterName
 - cFE Generic Counter APIs, [187](#)
- CFE_ES_GetLibIDByName
 - cFE Information APIs, [158](#)
- CFE_ES_GetLibInfo
 - cFE Information APIs, [159](#)
- CFE_ES_GetLibName
 - cFE Information APIs, [160](#)
- CFE_ES_GetMemPoolStats
 - cFE Memory Manager APIs, [175](#)
- CFE_ES_GetModuleInfo
 - cFE Information APIs, [161](#)
- CFE_ES_GetPoolBuf
 - cFE Memory Manager APIs, [175](#)
- CFE_ES_GetPoolBufInfo
 - cFE Memory Manager APIs, [176](#)
- CFE_ES_GetResetType
 - cFE Information APIs, [161](#)
- CFE_ES_GetTaskID
 - cFE Information APIs, [162](#)
- CFE_ES_GetTaskIDByName
 - cFE Child Task APIs, [166](#)
- CFE_ES_GetTaskInfo
 - cFE Information APIs, [163](#)
- CFE_ES_GetTaskName
 - cFE Child Task APIs, [167](#)
- CFE_ES_HK_TLM_MID
 - default_cfe_es_msgids.h, [811](#)
- CFE_ES_HousekeepingTlm, [438](#)
 - Payload, [438](#)
 - TelemetryHeader, [438](#)
- CFE_ES_HousekeepingTlm_Payload, [438](#)
 - BootSource, [440](#)
 - CFECoreChecksum, [440](#)
 - CFEMajorVersion, [440](#)
 - CFEMinorVersion, [441](#)
 - CFEMissionRevision, [441](#)
 - CFERevision, [441](#)
 - CommandCounter, [441](#)
 - CommandErrorCounter, [441](#)
 - ERLogEntries, [441](#)
 - ERLogIndex, [441](#)
 - HeapBlocksFree, [442](#)
 - HeapBytesFree, [442](#)
 - HeapMaxBlockSize, [442](#)
 - MaxProcessorResets, [442](#)
 - OSALMajorVersion, [442](#)
 - OSALMinorVersion, [442](#)
 - OSALMissionRevision, [442](#)
 - OSALRevision, [443](#)
 - PerfDataCount, [443](#)
 - PerfDataEnd, [443](#)
 - PerfDataStart, [443](#)
 - PerfDataToWrite, [443](#)
 - PerfFilterMask, [443](#)
 - PerfMode, [443](#)
 - PerfState, [444](#)
 - PerfTriggerCount, [444](#)
 - PerfTriggerMask, [444](#)
 - ProcessorResets, [444](#)
 - PSPMajorVersion, [444](#)
 - PSPMinorVersion, [444](#)
 - PSPMissionRevision, [444](#)
 - PSPRevision, [445](#)
 - RegisteredCoreApps, [445](#)
 - RegisteredExternalApps, [445](#)
 - RegisteredLibs, [445](#)
 - RegisteredTasks, [445](#)
 - ResetSubtype, [445](#)
 - ResetType, [445](#)
 - SysLogBytesUsed, [446](#)
 - SysLogEntries, [446](#)
 - SysLogMode, [446](#)
 - SysLogSize, [446](#)
- CFE_ES_HousekeepingTlm_Payload_t
 - default_cfe_es_msgdefs.h, [805](#)
- CFE_ES_HousekeepingTlm_t
 - default_cfe_es_msgstruct.h, [814](#)
- CFE_ES_IncrementGenCounter
 - cFE Generic Counter APIs, [188](#)
- CFE_ES_IncrementTaskCounter
 - cFE Application Behavior APIs, [152](#)
- CFE_ES_INIT_INF_EID
 - cfe_es_eventids.h, [830](#)
- CFE_ES_INITSTATS_INF_EID
 - cfe_es_eventids.h, [830](#)
- cfe_es_interface_cfg.h
 - CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN, [872](#)
 - CFE_MISSION_ES_CDS_MAX_NAME_LENGTH, [873](#)
 - CFE_MISSION_ES_CRC_16, [873](#)
 - CFE_MISSION_ES_CRC_32, [873](#)
 - CFE_MISSION_ES_CRC_8, [873](#)
 - CFE_MISSION_ES_DEFAULT_CRC, [873](#)
 - CFE_MISSION_ES_MAX_APPLICATIONS, [873](#)
 - CFE_MISSION_ES_PERF_MAX_IDS, [874](#)
 - CFE_MISSION_ES_POOL_MAX_BUCKETS, [874](#)
 - DEFAULT_CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN, [874](#)
 - DEFAULT_CFE_MISSION_ES_CDS_MAX_NAME_LENGTH, [875](#)
 - DEFAULT_CFE_MISSION_ES_CRC_16, [875](#)

- DEFAULT_CFE_MISSION_ES_CRC_32, [875](#)
- DEFAULT_CFE_MISSION_ES_CRC_8, [875](#)
- DEFAULT_CFE_MISSION_ES_DEFAULT_CRC, [875](#)
- DEFAULT_CFE_MISSION_ES_MAX_APPLICATIONS, [875](#)
- DEFAULT_CFE_MISSION_ES_PERF_MAX_IDS, [875](#)
- DEFAULT_CFE_MISSION_ES_POOL_MAX_BUCKETS, [875](#)
- cfe_es_internal_cfg.h
 - CFE_PLATFORM_ES_APP_KILL_TIMEOUT, [881](#)
 - CFE_PLATFORM_ES_APP_SCAN_RATE, [881](#)
 - CFE_PLATFORM_ES_CDS_MAX_BLOCK_SIZE, [882](#)
 - CFE_PLATFORM_ES_CDS_MAX_NUM_ENTRIES, [882](#)
 - CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_01, [882](#)
 - CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_02, [882](#)
 - CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_03, [883](#)
 - CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_04, [883](#)
 - CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_05, [883](#)
 - CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_06, [883](#)
 - CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_07, [883](#)
 - CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_08, [883](#)
 - CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_09, [883](#)
 - CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_10, [883](#)
 - CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_11, [883](#)
 - CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_12, [883](#)
 - CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_13, [883](#)
 - CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_14, [884](#)
 - CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_15, [884](#)
 - CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_16, [884](#)
 - CFE_PLATFORM_ES_CDS_SIZE, [884](#)
 - CFE_PLATFORM_ES_DEFAULT_APP_LOG_FILE, [884](#)
 - CFE_PLATFORM_ES_DEFAULT_CDS_REG_DUMP_FILE, [884](#)
 - CFE_PLATFORM_ES_DEFAULT_ER_LOG_FILE, [885](#)
 - CFE_PLATFORM_ES_DEFAULT_PERF_DUMP_FILENAME, [885](#)
 - CFE_PLATFORM_ES_DEFAULT_POR_SYSLOG_MODE, [885](#)
 - CFE_PLATFORM_ES_DEFAULT_PR_SYSLOG_MODE, [886](#)
 - CFE_PLATFORM_ES_DEFAULT_STACK_SIZE, [886](#)
 - CFE_PLATFORM_ES_DEFAULT_SYSLOG_FILE, [886](#)
 - CFE_PLATFORM_ES_DEFAULT_TASK_LOG_FILE, [887](#)
 - CFE_PLATFORM_ES_ER_LOG_ENTRIES, [887](#)
 - CFE_PLATFORM_ES_ER_LOG_MAX_CONTEXT_SIZE, [887](#)
 - CFE_PLATFORM_ES_MAX_APPLICATIONS, [888](#)
 - CFE_PLATFORM_ES_MAX_BLOCK_SIZE, [888](#)
 - CFE_PLATFORM_ES_MAX_GEN_COUNTERS, [888](#)
 - CFE_PLATFORM_ES_MAX_LIBRARIES, [888](#)
 - CFE_PLATFORM_ES_MAX_MEMORY_POOLS, [889](#)
 - CFE_PLATFORM_ES_MAX_PROCESSOR_RESETS, [889](#)
 - CFE_PLATFORM_ES_MEM_BLOCK_SIZE_01, [889](#)
 - CFE_PLATFORM_ES_MEM_BLOCK_SIZE_02, [890](#)
 - CFE_PLATFORM_ES_MEM_BLOCK_SIZE_03, [890](#)
 - CFE_PLATFORM_ES_MEM_BLOCK_SIZE_04, [890](#)
 - CFE_PLATFORM_ES_MEM_BLOCK_SIZE_05, [890](#)
 - CFE_PLATFORM_ES_MEM_BLOCK_SIZE_06, [890](#)
 - CFE_PLATFORM_ES_MEM_BLOCK_SIZE_07, [890](#)
 - CFE_PLATFORM_ES_MEM_BLOCK_SIZE_08, [890](#)
 - CFE_PLATFORM_ES_MEM_BLOCK_SIZE_09, [890](#)
 - CFE_PLATFORM_ES_MEM_BLOCK_SIZE_10, [891](#)
 - CFE_PLATFORM_ES_MEM_BLOCK_SIZE_11, [891](#)
 - CFE_PLATFORM_ES_MEM_BLOCK_SIZE_12, [891](#)
 - CFE_PLATFORM_ES_MEM_BLOCK_SIZE_13, [891](#)
 - CFE_PLATFORM_ES_MEM_BLOCK_SIZE_14, [891](#)
 - CFE_PLATFORM_ES_MEM_BLOCK_SIZE_15, [891](#)
 - CFE_PLATFORM_ES_MEM_BLOCK_SIZE_16, [891](#)
 - CFE_PLATFORM_ES_MEMPOOL_ALIGN_SIZE_MIN, [891](#)
 - CFE_PLATFORM_ES_NONVOL_DISK_MOUNT_STRING, [891](#)
 - CFE_PLATFORM_ES_NONVOL_STARTUP_FILE, [892](#)
 - CFE_PLATFORM_ES_PERF_CHILD_MS_DELAY, [892](#)
 - CFE_PLATFORM_ES_PERF_CHILD_PRIORITY, [892](#)
 - CFE_PLATFORM_ES_PERF_CHILD_STACK_SIZE, [893](#)
 - CFE_PLATFORM_ES_PERF_DATA_BUFFER_SIZE, [893](#)

- CFE_PLATFORM_ES_PERF_ENTRIES_BTWN_DLYS, [893](#)
- CFE_PLATFORM_ES_PERF_FILTMASK_ALL, [893](#)
- CFE_PLATFORM_ES_PERF_FILTMASK_INIT, [894](#)
- CFE_PLATFORM_ES_PERF_FILTMASK_NONE, [894](#)
- CFE_PLATFORM_ES_PERF_TRIGMASK_ALL, [894](#)
- CFE_PLATFORM_ES_PERF_TRIGMASK_INIT, [894](#)
- CFE_PLATFORM_ES_PERF_TRIGMASK_NONE, [894](#)
- CFE_PLATFORM_ES_POOL_MAX_BUCKETS, [895](#)
- CFE_PLATFORM_ES_RAM_DISK_MOUNT_STRING, [895](#)
- CFE_PLATFORM_ES_RAM_DISK_NUM_SECTORS, [895](#)
- CFE_PLATFORM_ES_RAM_DISK_PERCENT_RESERVED, [896](#)
- CFE_PLATFORM_ES_RAM_DISK_SECTOR_SIZE, [896](#)
- CFE_PLATFORM_ES_START_TASK_PRIORITY, [896](#)
- CFE_PLATFORM_ES_START_TASK_STACK_SIZE, [897](#)
- CFE_PLATFORM_ES_STARTUP_SCRIPT_TIMEOUT_MSEC, [897](#)
- CFE_PLATFORM_ES_STARTUP_SYNC_POLL_MSEC, [897](#)
- CFE_PLATFORM_ES_SYSTEM_LOG_SIZE, [898](#)
- CFE_PLATFORM_ES_USER_RESERVED_SIZE, [898](#)
- CFE_PLATFORM_ES_VOLATILE_STARTUP_FILE, [898](#)
- DEFAULT_CFE_PLATFORM_ES_APP_KILL_TIMEOUT, [899](#)
- DEFAULT_CFE_PLATFORM_ES_APP_SCAN_RATE, [899](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MAX_BLOCK_SIZE, [899](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MAX_NUM_ENTRIES, [899](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_01, [899](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_02, [899](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_03, [899](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_04, [899](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_05, [899](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_06, [900](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_07, [900](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_08, [900](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_09, [900](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_10, [900](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_11, [900](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_12, [900](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_13, [900](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_14, [900](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_15, [900](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_16, [900](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_SIZE, [901](#)
- DEFAULT_CFE_PLATFORM_ES_DEFAULT_APP_LOG_FILE, [901](#)
- DEFAULT_CFE_PLATFORM_ES_DEFAULT_CDS_REG_DUMP_FILE, [901](#)
- DEFAULT_CFE_PLATFORM_ES_DEFAULT_ER_LOG_FILE, [901](#)
- DEFAULT_CFE_PLATFORM_ES_DEFAULT_PERF_DUMP_FILENAME, [901](#)
- DEFAULT_CFE_PLATFORM_ES_DEFAULT_POR_SYSLOG_MODE, [901](#)
- DEFAULT_CFE_PLATFORM_ES_DEFAULT_PR_SYSLOG_MODE, [901](#)
- DEFAULT_CFE_PLATFORM_ES_DEFAULT_STACK_SIZE, [901](#)
- DEFAULT_CFE_PLATFORM_ES_DEFAULT_SYSLOG_FILE, [901](#)
- DEFAULT_CFE_PLATFORM_ES_DEFAULT_TASK_LOG_FILE, [901](#)
- DEFAULT_CFE_PLATFORM_ES_ER_LOG_ENTRIES, [901](#)
- DEFAULT_CFE_PLATFORM_ES_ER_LOG_MAX_CONTEXT_SIZE, [901](#)
- DEFAULT_CFE_PLATFORM_ES_MAX_APPLICATIONS, [901](#)
- DEFAULT_CFE_PLATFORM_ES_MAX_BLOCK_SIZE, [901](#)
- DEFAULT_CFE_PLATFORM_ES_MAX_GEN_COUNTERS, [901](#)
- DEFAULT_CFE_PLATFORM_ES_MAX_LIBRARIES, [901](#)
- DEFAULT_CFE_PLATFORM_ES_MAX_MEMORY_POOLS, [901](#)
- DEFAULT_CFE_PLATFORM_ES_MAX_PROCESSOR_RESETS, [901](#)
- DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_01, [901](#)

902	905
DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_02,	DEFAULT_CFE_PLATFORM_ES_PERF_TRIGMASK_INIT,
902	905
DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_03,	DEFAULT_CFE_PLATFORM_ES_PERF_TRIGMASK_NONE,
902	905
DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_04,	DEFAULT_CFE_PLATFORM_ES_POOL_MAX_BUCKETS,
902	905
DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_05,	DEFAULT_CFE_PLATFORM_ES_RAM_DISK_MOUNT_STRING,
903	905
DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_06,	DEFAULT_CFE_PLATFORM_ES_RAM_DISK_NUM_SECTORS,
903	905
DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_07,	DEFAULT_CFE_PLATFORM_ES_RAM_DISK_PERCENT_RESERVED,
903	905
DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_08,	DEFAULT_CFE_PLATFORM_ES_RAM_DISK_SECTOR_SIZE,
903	905
DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_09,	DEFAULT_CFE_PLATFORM_ES_START_TASK_PRIORITY,
903	905
DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_10,	DEFAULT_CFE_PLATFORM_ES_START_TASK_STACK_SIZE,
903	905
DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_11,	DEFAULT_CFE_PLATFORM_ES_STARTUP_SCRIPT_TIMEOUT_MSEC,
903	906
DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_12,	DEFAULT_CFE_PLATFORM_ES_STARTUP_SYNC_POLL_MSEC,
903	906
DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_13,	DEFAULT_CFE_PLATFORM_ES_SYSTEM_LOG_SIZE,
903	906
DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_14,	DEFAULT_CFE_PLATFORM_ES_USER_RESERVED_SIZE,
903	906
DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_15,	DEFAULT_CFE_PLATFORM_ES_VOLATILE_STARTUP_FILE,
903	906
DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_16,	DEFAULT_CFE_PLATFORM_ES_INVALID_POOL_HANDLE_ERR_EID
904	cfe_es_eventids.h, 831
DEFAULT_CFE_PLATFORM_ES_MEMPOOL_ALIGN_SIZE,	DEFAULT_CFE_PLATFORM_ES_NONLEN_ERR_EID
904	cfe_es_eventids.h, 831
DEFAULT_CFE_PLATFORM_ES_NONVOL_DISK_MOUNT_STRING,	ONE_STRING_ALREADY_LOADED
904	cFE Return Code Defines, 130
DEFAULT_CFE_PLATFORM_ES_NONVOL_STARTUP_FILE,	CFEES_LIBID_BASE
904	cFE Resource ID base values, 288
DEFAULT_CFE_PLATFORM_ES_PERF_CHILD_MS_DELAY,	CFEES_LIBID_C
904	cfe_es_api_typedefs.h, 723
DEFAULT_CFE_PLATFORM_ES_PERF_CHILD_PRIORITY,	CFEES_LibId_t
904	default_cfe_es_extern_typedefs.h, 789
DEFAULT_CFE_PLATFORM_ES_PERF_CHILD_STACK_SIZE,	CFEES_LibID_ToIndex
904	cFE Resource ID APIs, 146
DEFAULT_CFE_PLATFORM_ES_PERF_DATA_BUFFER_SIZE,	CFEES_LIBID_UNDEFINED
904	cfe_es_api_typedefs.h, 723
DEFAULT_CFE_PLATFORM_ES_PERF_ENTRIES_BTREE,	CFEES_LibraryEntryFuncPtr_t
904	cfe_es_api_typedefs.h, 725
DEFAULT_CFE_PLATFORM_ES_PERF_FILTMASK_ALIGN,	CFEES_LogEntryType
904	default_cfe_es_extern_typedefs.h, 792
DEFAULT_CFE_PLATFORM_ES_PERF_FILTMASK_INCREMENT,	CFEES_LogEntryType_APPLICATION
904	default_cfe_es_extern_typedefs.h, 792
DEFAULT_CFE_PLATFORM_ES_PERF_FILTMASK_NONE,	CFEES_LogEntryType_CORE
905	default_cfe_es_extern_typedefs.h, 792
DEFAULT_CFE_PLATFORM_ES_PERF_TRIGMASK_ALIGN,	CFEES_LogEntryType_Enum_t

- default_cfe_es_extern_typedefs.h, 789
- CFE_ES_LogMode
 - default_cfe_es_extern_typedefs.h, 792
- CFE_ES_LogMode_DISCARD
 - default_cfe_es_extern_typedefs.h, 792
- CFE_ES_LogMode_Enum_t
 - default_cfe_es_extern_typedefs.h, 789
- CFE_ES_LogMode_OVERWRITE
 - default_cfe_es_extern_typedefs.h, 792
- CFE_ES_Main
 - cFE Entry/Exit APIs, 148
- CFE_ES_MEMADDRESS_C
 - default_cfe_es_memaddress.h, 800
 - example_2x32bit_cfe_es_memaddress.h, 608
 - example_32bit_cfe_es_memaddress.h, 611
 - example_64bit_cfe_es_memaddress.h, 614
- CFE_ES_MemAddress_FromNative
 - example_2x32bit_cfe_es_memaddress.h, 609
- CFE_ES_MemAddress_t, 446
 - bits, 447
 - default_cfe_es_memaddress.h, 801
 - example_32bit_cfe_es_memaddress.h, 612
 - example_64bit_cfe_es_memaddress.h, 614
- CFE_ES_MEMADDRESS_TO_PTR
 - default_cfe_es_memaddress.h, 800
 - example_2x32bit_cfe_es_memaddress.h, 608
 - example_32bit_cfe_es_memaddress.h, 611
 - example_64bit_cfe_es_memaddress.h, 614
- CFE_ES_MemAddress_ToNative
 - example_2x32bit_cfe_es_memaddress.h, 609
- CFE_ES_MEMHANDLE_C
 - cfe_es_api_typedefs.h, 723
- CFE_ES_MemHandle_t
 - default_cfe_es_extern_typedefs.h, 789
- CFE_ES_MEMHANDLE_UNDEFINED
 - cfe_es_api_typedefs.h, 723
- CFE_ES_MEMOFFSET_C
 - default_cfe_es_memaddress.h, 800
 - example_2x32bit_cfe_es_memaddress.h, 609
 - example_32bit_cfe_es_memaddress.h, 611
 - example_64bit_cfe_es_memaddress.h, 614
- CFE_ES_MemOffset_FromNative
 - example_2x32bit_cfe_es_memaddress.h, 609
- CFE_ES_MemOffset_t, 447
 - bits, 447
 - default_cfe_es_memaddress.h, 801
 - example_32bit_cfe_es_memaddress.h, 612
 - example_64bit_cfe_es_memaddress.h, 614
- CFE_ES_MEMOFFSET_TO_SIZE_T
 - default_cfe_es_memaddress.h, 801
 - example_2x32bit_cfe_es_memaddress.h, 609
 - example_32bit_cfe_es_memaddress.h, 612
 - example_64bit_cfe_es_memaddress.h, 614
- CFE_ES_MemOffset_ToNative
 - example_2x32bit_cfe_es_memaddress.h, 609
- CFE_ES_MEMPOOLBUF_C
 - cfe_es_api_typedefs.h, 723
- CFE_ES_MemPoolBuf_t
 - cfe_es_api_typedefs.h, 725
- CFE_ES_MemPoolStats, 448
 - BlockStats, 448
 - CheckErrCtr, 448
 - NumBlocksRequested, 448
 - NumFreeBytes, 448
 - PoolSize, 449
- CFE_ES_MemPoolStats_t
 - default_cfe_es_extern_typedefs.h, 790
- CFE_ES_MEMSTATS_TLM_MID
 - default_cfe_es_msgids.h, 811
- CFE_ES_MemStatsTlm, 449
 - Payload, 449
 - TelemetryHeader, 449
- CFE_ES_MemStatsTlm_t
 - default_cfe_es_msgstruct.h, 815
- CFE_ES_MID_ERR_EID
 - cfe_es_eventids.h, 831
- CFE_ES_MUT_SEM_DELETE_ERR
 - cFE Return Code Defines, 130
- CFE_ES_NO_MUTEX
 - cfe_es_api_typedefs.h, 723
- CFE_ES_NO_RESOURCE_IDS_AVAILABLE
 - cFE Return Code Defines, 130
- CFE_ES_NOOP_CC
 - cfe_es_fcncodes.h, 853
- CFE_ES_NOOP_INF_EID
 - cfe_es_eventids.h, 831
- CFE_ES_NoopCmd, 449
 - CommandHeader, 450
- CFE_ES_NoopCmd_t
 - default_cfe_es_msgstruct.h, 815
- CFE_ES_NOT_IMPLEMENTED
 - cFE Return Code Defines, 130
- CFE_ES_ONE_APP_EID
 - cfe_es_eventids.h, 832
- CFE_ES_ONE_APPID_ERR_EID
 - cfe_es_eventids.h, 832
- CFE_ES_ONE_ERR_EID
 - cfe_es_eventids.h, 832
- CFE_ES_OneAppTlm, 450
 - Payload, 450
 - TelemetryHeader, 450
- CFE_ES_OneAppTlm_Payload, 450
 - AppInfo, 451
- CFE_ES_OneAppTlm_Payload_t
 - default_cfe_es_msgdefs.h, 806
- CFE_ES_OneAppTlm_t
 - default_cfe_es_msgstruct.h, 815
- CFE_ES_OPERATION_TIMED_OUT

- cFE Return Code Defines, [131](#)
- CFE_ES_OSCREATE_ERR_EID
 - cfe_es_eventids.h, [832](#)
- CFE_ES_OVER_WRITE_SYS_LOG_CC
 - cfe_es_fcncodes.h, [854](#)
- CFE_ES_OverWriteSysLogCmd, [451](#)
 - CommandHeader, [451](#)
 - Payload, [451](#)
- CFE_ES_OverWriteSysLogCmd_Payload, [452](#)
 - Mode, [452](#)
- CFE_ES_OverWriteSysLogCmd_Payload_t
 - default_cfe_es_msgdefs.h, [806](#)
- CFE_ES_OverWriteSysLogCmd_t
 - default_cfe_es_msgstruct.h, [815](#)
- CFE_ES_PCR_ERR1_EID
 - cfe_es_eventids.h, [833](#)
- CFE_ES_PCR_ERR2_EID
 - cfe_es_eventids.h, [833](#)
- CFE_ES_PERF_DATAWRITTEN_EID
 - cfe_es_eventids.h, [833](#)
- CFE_ES_PERF_FILTMSKCMD_EID
 - cfe_es_eventids.h, [833](#)
- CFE_ES_PERF_FILTMSKERR_EID
 - cfe_es_eventids.h, [834](#)
- CFE_ES_PERF_LOG_ERR_EID
 - cfe_es_eventids.h, [834](#)
- CFE_ES_PERF_STARTCMD_EID
 - cfe_es_eventids.h, [834](#)
- CFE_ES_PERF_STARTCMD_ERR_EID
 - cfe_es_eventids.h, [834](#)
- CFE_ES_PERF_STARTCMD_TRIG_ERR_EID
 - cfe_es_eventids.h, [835](#)
- CFE_ES_PERF_STOPCMD_EID
 - cfe_es_eventids.h, [835](#)
- CFE_ES_PERF_STOPCMD_ERR2_EID
 - cfe_es_eventids.h, [835](#)
- CFE_ES_PERF_TRIGMSKCMD_EID
 - cfe_es_eventids.h, [835](#)
- CFE_ES_PERF_TRIGMSKERR_EID
 - cfe_es_eventids.h, [836](#)
- CFE_ES_PerfLogAdd
 - cFE Performance Monitor APIs, [184](#)
- CFE_ES_PerfLogEntry
 - cFE Performance Monitor APIs, [183](#)
- CFE_ES_PerfLogExit
 - cFE Performance Monitor APIs, [183](#)
- CFE_ES_PerfMode
 - default_cfe_es_msgdefs.h, [807](#)
- CFE_ES_PerfMode_Enum_t
 - default_cfe_es_msgdefs.h, [806](#)
- CFE_ES_PerfTrigger_CENTER
 - default_cfe_es_msgdefs.h, [807](#)
- CFE_ES_PerfTrigger_END
 - default_cfe_es_msgdefs.h, [807](#)
- CFE_ES_PerfTrigger_START
 - default_cfe_es_msgdefs.h, [807](#)
- CFE_ES_POOL_BLOCK_INVALID
 - cFE Return Code Defines, [131](#)
- CFE_ES_PoolAlign, [452](#)
 - LongDouble, [453](#)
 - LongInt, [453](#)
 - Ptr, [453](#)
- CFE_ES_PoolAlign_t
 - cfe_es_api_typedefs.h, [725](#)
- CFE_ES_PoolCreate
 - cFE Memory Manager APIs, [177](#)
- CFE_ES_PoolCreateEx
 - cFE Memory Manager APIs, [178](#)
- CFE_ES_PoolCreateEx_WithAlignment
 - cFE Memory Manager APIs, [179](#)
- CFE_ES_PoolCreateNoSem
 - cFE Memory Manager APIs, [180](#)
- CFE_ES_PoolDelete
 - cFE Memory Manager APIs, [181](#)
- CFE_ES_POOLID_BASE
 - cFE Resource ID base values, [288](#)
- CFE_ES_PoolStatsTIm_Payload, [453](#)
 - Padding, [453](#)
 - PoolHandle, [453](#)
 - PoolStats, [453](#)
- CFE_ES_PoolStatsTIm_Payload_t
 - default_cfe_es_msgdefs.h, [806](#)
- CFE_ES_ProcessAsyncEvent
 - cFE Miscellaneous APIs, [169](#)
- CFE_ES_PutPoolBuf
 - cFE Memory Manager APIs, [181](#)
- CFE_ES_QUERY_ALL_CC
 - cfe_es_fcncodes.h, [854](#)
- CFE_ES_QUERY_ALL_TASKS_CC
 - cfe_es_fcncodes.h, [855](#)
- CFE_ES_QUERY_ONE_CC
 - cfe_es_fcncodes.h, [856](#)
- CFE_ES_QueryAllCmd, [454](#)
 - CommandHeader, [454](#)
 - Payload, [454](#)
- CFE_ES_QueryAllCmd_t
 - default_cfe_es_msgstruct.h, [815](#)
- CFE_ES_QueryAllTasksCmd, [454](#)
 - CommandHeader, [455](#)
 - Payload, [455](#)
- CFE_ES_QueryAllTasksCmd_t
 - default_cfe_es_msgstruct.h, [815](#)
- CFE_ES_QueryOneCmd, [455](#)
 - CommandHeader, [455](#)
 - Payload, [455](#)
- CFE_ES_QueryOneCmd_t
 - default_cfe_es_msgstruct.h, [815](#)
- CFE_ES_QUEUE_DELETE_ERR

- cFE Return Code Defines, [131](#)
- CFE_ES_RegisterCDS
 - cFE Critical Data Store APIs, [172](#)
- CFE_ES_RegisterGenCounter
 - cFE Generic Counter APIs, [188](#)
- CFE_ES_RELOAD_APP_CC
 - cfe_es_fcncodes.h, [857](#)
- CFE_ES_RELOAD_APP_DBG_EID
 - cfe_es_eventids.h, [836](#)
- CFE_ES_RELOAD_APP_ERR1_EID
 - cfe_es_eventids.h, [836](#)
- CFE_ES_RELOAD_APP_ERR2_EID
 - cfe_es_eventids.h, [836](#)
- CFE_ES_RELOAD_APP_ERR3_EID
 - cfe_es_eventids.h, [837](#)
- CFE_ES_RELOAD_APP_ERR4_EID
 - cfe_es_eventids.h, [837](#)
- CFE_ES_RELOAD_APP_INF_EID
 - cfe_es_eventids.h, [837](#)
- CFE_ES_ReloadApp
 - cFE Application Control APIs, [150](#)
- CFE_ES_ReloadAppCmd, [455](#)
 - CommandHeader, [456](#)
 - Payload, [456](#)
- CFE_ES_ReloadAppCmd_t
 - default_cfe_es_msgstruct.h, [815](#)
- CFE_ES_RESET_COUNTERS_CC
 - cfe_es_fcncodes.h, [858](#)
- CFE_ES_RESET_INF_EID
 - cfe_es_eventids.h, [837](#)
- CFE_ES_RESET_PR_COUNT_CC
 - cfe_es_fcncodes.h, [859](#)
- CFE_ES_RESET_PR_COUNT_EID
 - cfe_es_eventids.h, [838](#)
- CFE_ES_ResetCFE
 - cFE Entry/Exit APIs, [148](#)
- CFE_ES_ResetCountersCmd, [456](#)
 - CommandHeader, [456](#)
- CFE_ES_ResetCountersCmd_t
 - default_cfe_es_msgstruct.h, [815](#)
- CFE_ES_ResetPRCountCmd, [456](#)
 - CommandHeader, [457](#)
- CFE_ES_ResetPRCountCmd_t
 - default_cfe_es_msgstruct.h, [815](#)
- CFE_ES_RESTART_APP_CC
 - cfe_es_fcncodes.h, [859](#)
- CFE_ES_RESTART_APP_DBG_EID
 - cfe_es_eventids.h, [838](#)
- CFE_ES_RESTART_APP_ERR1_EID
 - cfe_es_eventids.h, [838](#)
- CFE_ES_RESTART_APP_ERR2_EID
 - cfe_es_eventids.h, [838](#)
- CFE_ES_RESTART_APP_ERR3_EID
 - cfe_es_eventids.h, [839](#)
- CFE_ES_RESTART_APP_ERR4_EID
 - cfe_es_eventids.h, [839](#)
- CFE_ES_RESTART_APP_INF_EID
 - cfe_es_eventids.h, [839](#)
- CFE_ES_RESTART_CC
 - cfe_es_fcncodes.h, [860](#)
- CFE_ES_RestartApp
 - cFE Application Control APIs, [151](#)
- CFE_ES_RestartAppCmd, [457](#)
 - CommandHeader, [457](#)
 - Payload, [457](#)
- CFE_ES_RestartAppCmd_t
 - default_cfe_es_msgstruct.h, [815](#)
- CFE_ES_RestartCmd, [457](#)
 - CommandHeader, [458](#)
 - Payload, [458](#)
- CFE_ES_RestartCmd_Payload, [458](#)
 - RestartType, [458](#)
- CFE_ES_RestartCmd_Payload_t
 - default_cfe_es_msgdefs.h, [806](#)
- CFE_ES_RestartCmd_t
 - default_cfe_es_msgstruct.h, [815](#)
- CFE_ES_RestoreFromCDS
 - cFE Critical Data Store APIs, [173](#)
- CFE_ES_RST_ACCESS_ERR
 - cFE Return Code Defines, [131](#)
- CFE_ES_RunLoop
 - cFE Application Behavior APIs, [153](#)
- CFE_ES_RunStatus
 - default_cfe_es_extern_typedefs.h, [792](#)
- CFE_ES_RunStatus_APP_ERROR
 - default_cfe_es_extern_typedefs.h, [792](#)
- CFE_ES_RunStatus_APP_EXIT
 - default_cfe_es_extern_typedefs.h, [792](#)
- CFE_ES_RunStatus_APP_RUN
 - default_cfe_es_extern_typedefs.h, [792](#)
- CFE_ES_RunStatus_CORE_APP_INIT_ERROR
 - default_cfe_es_extern_typedefs.h, [792](#)
- CFE_ES_RunStatus_CORE_APP_RUNTIME_ERROR
 - default_cfe_es_extern_typedefs.h, [792](#)
- CFE_ES_RunStatus_Enum_t
 - default_cfe_es_extern_typedefs.h, [790](#)
- CFE_ES_RunStatus_MAX
 - default_cfe_es_extern_typedefs.h, [792](#)
- CFE_ES_RunStatus_SYS_DELETE
 - default_cfe_es_extern_typedefs.h, [792](#)
- CFE_ES_RunStatus_SYS_EXCEPTION
 - default_cfe_es_extern_typedefs.h, [792](#)
- CFE_ES_RunStatus_SYS_RELOAD
 - default_cfe_es_extern_typedefs.h, [792](#)
- CFE_ES_RunStatus_SYS_RESTART
 - default_cfe_es_extern_typedefs.h, [792](#)
- CFE_ES_RunStatus_UNDEFINED
 - default_cfe_es_extern_typedefs.h, [792](#)

- CFE_ES_SEND_HK_MID
 - default_cfe_es_msgids.h, [811](#)
- CFE_ES_SEND_MEM_POOL_STATS_CC
 - cfe_es_fcncodes.h, [861](#)
- CFE_ES_SendHkCmd, [459](#)
 - CommandHeader, [459](#)
- CFE_ES_SendHkCmd_t
 - default_cfe_es_msgstruct.h, [816](#)
- CFE_ES_SendMemPoolStatsCmd, [459](#)
 - CommandHeader, [459](#)
 - Payload, [459](#)
- CFE_ES_SendMemPoolStatsCmd_Payload, [460](#)
 - Application, [460](#)
 - PoolHandle, [460](#)
- CFE_ES_SendMemPoolStatsCmd_Payload_t
 - default_cfe_es_msgdefs.h, [806](#)
- CFE_ES_SendMemPoolStatsCmd_t
 - default_cfe_es_msgstruct.h, [816](#)
- CFE_ES_SET_MAX_PR_COUNT_CC
 - cfe_es_fcncodes.h, [862](#)
- CFE_ES_SET_MAX_PR_COUNT_EID
 - cfe_es_eventids.h, [839](#)
- CFE_ES_SET_PERF_FILTER_MASK_CC
 - cfe_es_fcncodes.h, [863](#)
- CFE_ES_SET_PERF_TRIGGER_MASK_CC
 - cfe_es_fcncodes.h, [864](#)
- CFE_ES_SetGenCount
 - cFE Generic Counter APIs, [189](#)
- CFE_ES_SetMaxPRCountCmd, [460](#)
 - CommandHeader, [461](#)
 - Payload, [461](#)
- CFE_ES_SetMaxPRCountCmd_Payload, [461](#)
 - MaxPRCount, [461](#)
- CFE_ES_SetMaxPRCountCmd_Payload_t
 - default_cfe_es_msgdefs.h, [806](#)
- CFE_ES_SetMaxPRCountCmd_t
 - default_cfe_es_msgstruct.h, [816](#)
- CFE_ES_SetPerfFilterMaskCmd, [461](#)
 - CommandHeader, [462](#)
 - Payload, [462](#)
- CFE_ES_SetPerfFilterMaskCmd_Payload, [462](#)
 - FilterMask, [462](#)
 - FilterMaskNum, [462](#)
- CFE_ES_SetPerfFilterMaskCmd_Payload_t
 - default_cfe_es_msgdefs.h, [806](#)
- CFE_ES_SetPerfFilterMaskCmd_t
 - default_cfe_es_msgstruct.h, [816](#)
- CFE_ES_SetPerfTriggerMaskCmd, [463](#)
 - CommandHeader, [463](#)
 - Payload, [463](#)
- CFE_ES_SetPerfTriggerMaskCmd_t
 - default_cfe_es_msgstruct.h, [816](#)
- CFE_ES_SetPerfTrigMaskCmd_Payload, [463](#)
 - TriggerMask, [464](#)
- TriggerMaskNum, [464](#)
- CFE_ES_SetPerfTrigMaskCmd_Payload_t
 - default_cfe_es_msgdefs.h, [806](#)
- CFE_ES_StackPointer_t
 - cfe_es_api_typedefs.h, [725](#)
- CFE_ES_START_APP_CC
 - cfe_es_fcncodes.h, [864](#)
- CFE_ES_START_ERR_EID
 - cfe_es_eventids.h, [840](#)
- CFE_ES_START_EXC_ACTION_ERR_EID
 - cfe_es_eventids.h, [840](#)
- CFE_ES_START_INF_EID
 - cfe_es_eventids.h, [840](#)
- CFE_ES_START_INVALID_ENTRY_POINT_ERR_EID
 - cfe_es_eventids.h, [840](#)
- CFE_ES_START_INVALID_FILENAME_ERR_EID
 - cfe_es_eventids.h, [841](#)
- CFE_ES_START_NULL_APP_NAME_ERR_EID
 - cfe_es_eventids.h, [841](#)
- CFE_ES_START_PERF_DATA_CC
 - cfe_es_fcncodes.h, [865](#)
- CFE_ES_START_PRIORITY_ERR_EID
 - cfe_es_eventids.h, [841](#)
- CFE_ES_StartApp, [464](#)
 - CommandHeader, [464](#)
 - Payload, [464](#)
- CFE_ES_StartAppCmd_Payload, [464](#)
 - AppEntryPoint, [465](#)
 - AppFileName, [465](#)
 - Application, [465](#)
 - ExceptionAction, [465](#)
 - Padding1, [465](#)
 - Padding2, [465](#)
 - Priority, [466](#)
 - StackSize, [466](#)
- CFE_ES_StartAppCmd_Payload_t
 - default_cfe_es_msgdefs.h, [807](#)
- CFE_ES_StartAppCmd_t
 - default_cfe_es_msgstruct.h, [816](#)
- CFE_ES_StartPerfCmd_Payload, [466](#)
 - TriggerMode, [466](#)
- CFE_ES_StartPerfCmd_Payload_t
 - default_cfe_es_msgdefs.h, [807](#)
- CFE_ES_StartPerfDataCmd, [466](#)
 - CommandHeader, [467](#)
 - Payload, [467](#)
- CFE_ES_StartPerfDataCmd_t
 - default_cfe_es_msgstruct.h, [816](#)
- CFE_ES_STATIC_POOL_TYPE
 - cfe_es_api_typedefs.h, [724](#)
- CFE_ES_StatusToString
 - cfe_error.h, [710](#)
- CFE_ES_STOP_APP_CC
 - cfe_es_fcncodes.h, [866](#)

- CFE_ES_STOP_DBG_EID
 - cfe_es_eventids.h, [841](#)
- CFE_ES_STOP_ERR1_EID
 - cfe_es_eventids.h, [842](#)
- CFE_ES_STOP_ERR2_EID
 - cfe_es_eventids.h, [842](#)
- CFE_ES_STOP_ERR3_EID
 - cfe_es_eventids.h, [842](#)
- CFE_ES_STOP_INF_EID
 - cfe_es_eventids.h, [842](#)
- CFE_ES_STOP_PERF_DATA_CC
 - cfe_es_fcncodes.h, [867](#)
- CFE_ES_StopAppCmd, [467](#)
 - CommandHeader, [467](#)
 - Payload, [467](#)
- CFE_ES_StopAppCmd_t
 - default_cfe_es_msgstruct.h, [816](#)
- CFE_ES_StopPerfCmd_Payload, [468](#)
 - DataFileName, [468](#)
- CFE_ES_StopPerfCmd_Payload_t
 - default_cfe_es_msgdefs.h, [807](#)
- CFE_ES_StopPerfDataCmd, [468](#)
 - CommandHeader, [469](#)
 - Payload, [469](#)
- CFE_ES_StopPerfDataCmd_t
 - default_cfe_es_msgstruct.h, [816](#)
- CFE_ES_SYSLOG1_INF_EID
 - cfe_es_eventids.h, [843](#)
- CFE_ES_SYSLOG2_EID
 - cfe_es_eventids.h, [843](#)
- CFE_ES_SYSLOG2_ERR_EID
 - cfe_es_eventids.h, [843](#)
- CFE_ES_SYSLOGMODE_EID
 - cfe_es_eventids.h, [843](#)
- CFE_ES_SystemState
 - default_cfe_es_extern_typedefs.h, [792](#)
- CFE_ES_SystemState_APPS_INIT
 - default_cfe_es_extern_typedefs.h, [793](#)
- CFE_ES_SystemState_CORE_READY
 - default_cfe_es_extern_typedefs.h, [793](#)
- CFE_ES_SystemState_CORE_STARTUP
 - default_cfe_es_extern_typedefs.h, [793](#)
- CFE_ES_SystemState_EARLY_INIT
 - default_cfe_es_extern_typedefs.h, [793](#)
- CFE_ES_SystemState_Enum_t
 - default_cfe_es_extern_typedefs.h, [790](#)
- CFE_ES_SystemState_MAX
 - default_cfe_es_extern_typedefs.h, [793](#)
- CFE_ES_SystemState_OPERATIONAL
 - default_cfe_es_extern_typedefs.h, [793](#)
- CFE_ES_SystemState_SHUTDOWN
 - default_cfe_es_extern_typedefs.h, [793](#)
- CFE_ES_SystemState_UNDEFINED
 - default_cfe_es_extern_typedefs.h, [793](#)
- CFE_ES_TASK_DELETE_ERR
 - cFE Return Code Defines, [131](#)
- CFE_ES_TASK_STACK_ALLOCATE
 - cfe_es_api_typedefs.h, [724](#)
- CFE_ES_TaskEntryFuncPtr_t
 - cfe_es_api_typedefs.h, [725](#)
- CFE_ES_TASKID_BASE
 - cFE Resource ID base values, [288](#)
- CFE_ES_TASKID_C
 - cfe_es_api_typedefs.h, [724](#)
- CFE_ES_TaskId_t
 - default_cfe_es_extern_typedefs.h, [790](#)
- CFE_ES_TaskID_ToIndex
 - cFE Resource ID APIs, [147](#)
- CFE_ES_TASKID_UNDEFINED
 - cfe_es_api_typedefs.h, [724](#)
- CFE_ES_TaskInfo, [469](#)
 - Appld, [470](#)
 - AppName, [470](#)
 - ExecutionCounter, [470](#)
 - Priority, [470](#)
 - Spare, [470](#)
 - StackSize, [470](#)
 - TaskId, [470](#)
 - TaskName, [470](#)
- CFE_ES_TASKINFO_EID
 - cfe_es_eventids.h, [844](#)
- CFE_ES_TASKINFO_OSCREATE_ERR_EID
 - cfe_es_eventids.h, [844](#)
- CFE_ES_TaskInfo_t
 - default_cfe_es_extern_typedefs.h, [790](#)
- CFE_ES_TASKINFO_WR_ERR_EID
 - cfe_es_eventids.h, [844](#)
- CFE_ES_TASKINFO_WRHDR_ERR_EID
 - cfe_es_eventids.h, [844](#)
- CFE_ES_TaskPriority_Atom_t
 - default_cfe_es_extern_typedefs.h, [791](#)
- CFE_ES_TASKWR_ERR_EID
 - cfe_es_eventids.h, [845](#)
- CFE_ES_TIMER_DELETE_ERR
 - cFE Return Code Defines, [131](#)
- CFE_ES_TLM_POOL_STATS_INFO_EID
 - cfe_es_eventids.h, [845](#)
- cfe_es_topicids.h
 - CFE_MISSION_ES_APP_TLM_TOPICID, [910](#)
 - CFE_MISSION_ES_CMD_TOPICID, [910](#)
 - CFE_MISSION_ES_HK_TLM_TOPICID, [910](#)
 - CFE_MISSION_ES_MEMSTATS_TLM_TOPICID, [910](#)
 - CFE_MISSION_ES_SEND_HK_TOPICID, [910](#)
 - DEFAULT_CFE_MISSION_ES_APP_TLM_TOPICID, [910](#)
 - DEFAULT_CFE_MISSION_ES_CMD_TOPICID, [910](#)

- DEFAULT_CFE_MISSION_ES_HK_TLM_TOPICID, 910
- DEFAULT_CFE_MISSION_ES_MEMSTATS_TLM_TOPICID, 911
- DEFAULT_CFE_MISSION_ES_SEND_HK_TOPICID, 911
- CFE_ES_USE_MUTEX
 - cfe_es_api_typedefs.h, 724
- CFE_ES_VERSION_INF_EID
 - cfe_es_eventids.h, 845
- CFE_ES_WaitForStartupSync
 - cFE Application Behavior APIs, 154
- CFE_ES_WaitForSystemState
 - cFE Application Behavior APIs, 154
- CFE_ES_WRHDR_ERR_EID
 - cfe_es_eventids.h, 846
- CFE_ES_WRITE_CFE_HDR_ERR_EID
 - cfe_es_eventids.h, 846
- CFE_ES_WRITE_ER_LOG_CC
 - cfe_es_fcncodes.h, 868
- CFE_ES_WRITE_SYS_LOG_CC
 - cfe_es_fcncodes.h, 869
- CFE_ES_WriteERLogCmd, 470
 - CommandHeader, 471
 - Payload, 471
- CFE_ES_WriteERLogCmd_t
 - default_cfe_es_msgstruct.h, 816
- CFE_ES_WriteSysLogCmd, 471
 - CommandHeader, 471
 - Payload, 471
- CFE_ES_WriteSysLogCmd_t
 - default_cfe_es_msgstruct.h, 816
- CFE_ES_WriteToSysLog
 - cFE Miscellaneous APIs, 169
- CFE_EVENTS_SERVICE
 - cfe_error.h, 708
- cfe_evs.h
 - CFE_EVS_Send, 729
 - CFE_EVS_SendCrit, 729
 - CFE_EVS_SendDbg, 729
 - CFE_EVS_SendErr, 729
 - CFE_EVS_SendInfo, 729
- CFE_EVS_ADD_EVENT_FILTER_CC
 - cfe_evs_fcncodes.h, 950
- CFE_EVS_AddEventFilterCmd, 472
 - CommandHeader, 472
 - Payload, 472
- CFE_EVS_AddEventFilterCmd_t
 - default_cfe_evs_msgstruct.h, 930
- CFE_EVS_ADDFILTER_EID
 - cfe_evs_eventids.h, 938
- cfe_evs_api_typedefs.h
 - CFE_EVS_BinFilter_t, 732
 - CFE_EVS_EVERY_FOURTH_ONE, 731
 - CFE_EVS_EVERY_OTHER_ONE, 731
 - CFE_EVS_EVERY_OTHER_TWO, 731
 - CFE_EVS_FIRST_16_STOP, 731
 - CFE_EVS_FIRST_32_STOP, 731
 - CFE_EVS_FIRST_4_STOP, 731
 - CFE_EVS_FIRST_64_STOP, 732
 - CFE_EVS_FIRST_8_STOP, 732
 - CFE_EVS_FIRST_ONE_STOP, 732
 - CFE_EVS_FIRST_TWO_STOP, 732
 - CFE_EVS_NO_FILTER, 732
- CFE_EVS_APP_FILTER_OVERLOAD
 - cFE Return Code Defines, 131
- CFE_EVS_APP_ILLEGAL_APP_ID
 - cFE Return Code Defines, 132
- CFE_EVS_APP_NOT_REGISTERED
 - cFE Return Code Defines, 132
- CFE_EVS_APP_SQUELCHED
 - cFE Return Code Defines, 132
- CFE_EVS_AppDataCmd_Payload, 472
 - AppDataFilename, 473
- CFE_EVS_AppDataCmd_Payload_t
 - default_cfe_evs_msgdefs.h, 923
- CFE_EVS_AppNameBitMaskCmd_Payload, 473
 - AppName, 473
 - BitMask, 473
 - Spare, 473
- CFE_EVS_AppNameBitMaskCmd_Payload_t
 - default_cfe_evs_msgdefs.h, 923
- CFE_EVS_AppNameCmd_Payload, 474
 - AppName, 474
- CFE_EVS_AppNameCmd_Payload_t
 - default_cfe_evs_msgdefs.h, 923
- CFE_EVS_AppNameEventIDCmd_Payload, 474
 - AppName, 474
 - EventID, 474
- CFE_EVS_AppNameEventIDCmd_Payload_t
 - default_cfe_evs_msgdefs.h, 923
- CFE_EVS_AppNameEventIDMaskCmd_Payload, 475
 - AppName, 475
 - EventID, 475
 - Mask, 475
- CFE_EVS_AppNameEventIDMaskCmd_Payload_t
 - default_cfe_evs_msgdefs.h, 923
- CFE_EVS_AppTlmData, 475
 - AppEnableStatus, 476
 - AppID, 476
 - AppMessageSentCounter, 476
 - AppMessageSquelchedCounter, 476
- CFE_EVS_AppTlmData_t
 - default_cfe_evs_msgdefs.h, 923
- CFE_EVS_BinFilter, 477
 - EventID, 477
 - Mask, 477
- CFE_EVS_BinFilter_t

- cfe_evs_api_typedefs.h, [732](#)
- CFE_EVS_BitMaskCmd_Payload, [477](#)
 - BitMask, [478](#)
 - Spare, [478](#)
- CFE_EVS_BitMaskCmd_Payload_t
 - default_cfe_evs_msgdefs.h, [924](#)
- CFE_EVS_CCVAL
 - default_cfe_evs_fcncode_values.h, [916](#)
- CFE_EVS_CLEAR_LOG_CC
 - cfe_evs_fcncodes.h, [951](#)
- CFE_EVS_ClearLogCmd, [478](#)
 - CommandHeader, [478](#)
- CFE_EVS_ClearLogCmd_t
 - default_cfe_evs_msgstruct.h, [930](#)
- CFE_EVS_CMD_MID
 - default_cfe_evs_msgids.h, [928](#)
- CFE_EVS_CRITICAL_BIT
 - default_cfe_evs_msgdefs.h, [922](#)
- CFE_EVS_DEBUG_BIT
 - default_cfe_evs_msgdefs.h, [922](#)
- CFE_EVS_DELETE_EVENT_FILTER_CC
 - cfe_evs_fcncodes.h, [952](#)
- CFE_EVS_DeleteEventFilterCmd, [478](#)
 - CommandHeader, [479](#)
 - Payload, [479](#)
- CFE_EVS_DeleteEventFilterCmd_t
 - default_cfe_evs_msgstruct.h, [930](#)
- CFE_EVS_DELFILTER_EID
 - cfe_evs_eventids.h, [938](#)
- CFE_EVS_DISABLE_APP_EVENT_TYPE_CC
 - cfe_evs_fcncodes.h, [953](#)
- CFE_EVS_DISABLE_APP_EVENTS_CC
 - cfe_evs_fcncodes.h, [954](#)
- CFE_EVS_DISABLE_EVENT_TYPE_CC
 - cfe_evs_fcncodes.h, [955](#)
- CFE_EVS_DISABLE_PORTS_CC
 - cfe_evs_fcncodes.h, [955](#)
- CFE_EVS_DisableAppEventsCmd, [479](#)
 - CommandHeader, [479](#)
 - Payload, [479](#)
- CFE_EVS_DisableAppEventsCmd_t
 - default_cfe_evs_msgstruct.h, [930](#)
- CFE_EVS_DisableAppEventTypeCmd, [479](#)
 - CommandHeader, [480](#)
 - Payload, [480](#)
- CFE_EVS_DisableAppEventTypeCmd_t
 - default_cfe_evs_msgstruct.h, [930](#)
- CFE_EVS_DisableEventTypeCmd, [480](#)
 - CommandHeader, [480](#)
 - Payload, [480](#)
- CFE_EVS_DisableEventTypeCmd_t
 - default_cfe_evs_msgstruct.h, [931](#)
- CFE_EVS_DisablePortsCmd, [481](#)
 - CommandHeader, [481](#)
 - Payload, [481](#)
- CFE_EVS_DisablePortsCmd_t
 - default_cfe_evs_msgstruct.h, [931](#)
- CFE_EVS_DISAPPENTTYPE_EID
 - cfe_evs_eventids.h, [938](#)
- CFE_EVS_DISAPPEVT_EID
 - cfe_evs_eventids.h, [939](#)
- CFE_EVS_DISEVTTYPE_EID
 - cfe_evs_eventids.h, [939](#)
- CFE_EVS_DISPORT_EID
 - cfe_evs_eventids.h, [939](#)
- CFE_EVS_ENAAPPEVT_EID
 - cfe_evs_eventids.h, [939](#)
- CFE_EVS_ENAAPPEVTTYPE_EID
 - cfe_evs_eventids.h, [940](#)
- CFE_EVS_ENABLE_APP_EVENT_TYPE_CC
 - cfe_evs_fcncodes.h, [956](#)
- CFE_EVS_ENABLE_APP_EVENTS_CC
 - cfe_evs_fcncodes.h, [957](#)
- CFE_EVS_ENABLE_EVENT_TYPE_CC
 - cfe_evs_fcncodes.h, [958](#)
- CFE_EVS_ENABLE_PORTS_CC
 - cfe_evs_fcncodes.h, [959](#)
- CFE_EVS_EnableAppEventsCmd, [481](#)
 - CommandHeader, [482](#)
 - Payload, [482](#)
- CFE_EVS_EnableAppEventsCmd_t
 - default_cfe_evs_msgstruct.h, [931](#)
- CFE_EVS_EnableAppEventTypeCmd, [482](#)
 - CommandHeader, [482](#)
 - Payload, [482](#)
- CFE_EVS_EnableAppEventTypeCmd_t
 - default_cfe_evs_msgstruct.h, [931](#)
- CFE_EVS_EnableEventTypeCmd, [482](#)
 - CommandHeader, [483](#)
 - Payload, [483](#)
- CFE_EVS_EnableEventTypeCmd_t
 - default_cfe_evs_msgstruct.h, [931](#)
- CFE_EVS_EnablePortsCmd, [483](#)
 - CommandHeader, [483](#)
 - Payload, [483](#)
- CFE_EVS_EnablePortsCmd_t
 - default_cfe_evs_msgstruct.h, [931](#)
- CFE_EVS_ENAEVTTYPE_EID
 - cfe_evs_eventids.h, [940](#)
- CFE_EVS_ENAPORT_EID
 - cfe_evs_eventids.h, [940](#)
- CFE_EVS_ERR_APPNOREGS_EID
 - cfe_evs_eventids.h, [940](#)
- CFE_EVS_ERR_CC_EID
 - cfe_evs_eventids.h, [941](#)
- CFE_EVS_ERR_CRDATFILE_EID
 - cfe_evs_eventids.h, [941](#)
- CFE_EVS_ERR_CRLOGFILE_EID

cfe_evs_eventids.h, 941
 CFE_EVS_ERR_EVTIDNOREGS_EID
 cfe_evs_eventids.h, 941
 CFE_EVS_ERR_ILLAPPIDRANGE_EID
 cfe_evs_eventids.h, 942
 CFE_EVS_ERR_ILLEGALFMTMOD_EID
 cfe_evs_eventids.h, 942
 CFE_EVS_ERR_INVALID_BITMASK_EID
 cfe_evs_eventids.h, 942
 CFE_EVS_ERR_LOGMODE_EID
 cfe_evs_eventids.h, 942
 CFE_EVS_ERR_MAXREGSFILTER_EID
 cfe_evs_eventids.h, 943
 CFE_EVS_ERR_MSGID_EID
 cfe_evs_eventids.h, 943
 CFE_EVS_ERR_NOAPPIDFOUND_EID
 cfe_evs_eventids.h, 943
 CFE_EVS_ERR_UNREGISTERED_EVS_APP
 cfe_evs_eventids.h, 943
 CFE_EVS_ERR_WRDATFILE_EID
 cfe_evs_eventids.h, 944
 CFE_EVS_ERR_WRLOGFILE_EID
 cfe_evs_eventids.h, 944
 CFE_EVS_ERROR_BIT
 default_cfe_evs_msgdefs.h, 922
 CFE_EVS_EventFilter
 default_cfe_evs_extern_typedefs.h, 913
 CFE_EVS_EventFilter_BINARY
 default_cfe_evs_extern_typedefs.h, 913
 CFE_EVS_EventFilter_Enum_t
 default_cfe_evs_extern_typedefs.h, 912
 cfe_evs_eventids.h
 CFE_EVS_ADDFILTER_EID, 938
 CFE_EVS_DELFILTER_EID, 938
 CFE_EVS_DISAPPENTTYPE_EID, 938
 CFE_EVS_DISAPPEVT_EID, 939
 CFE_EVS_DISEVTTYPE_EID, 939
 CFE_EVS_DISPORT_EID, 939
 CFE_EVS_ENAAPPEVT_EID, 939
 CFE_EVS_ENAAPPEVTTYPE_EID, 940
 CFE_EVS_ENAEVTTYPE_EID, 940
 CFE_EVS_ENAPORT_EID, 940
 CFE_EVS_ERR_APPNOREGS_EID, 940
 CFE_EVS_ERR_CC_EID, 941
 CFE_EVS_ERR_CRDATFILE_EID, 941
 CFE_EVS_ERR_CRLOGFILE_EID, 941
 CFE_EVS_ERR_EVTIDNOREGS_EID, 941
 CFE_EVS_ERR_ILLAPPIDRANGE_EID, 942
 CFE_EVS_ERR_ILLEGALFMTMOD_EID, 942
 CFE_EVS_ERR_INVALID_BITMASK_EID, 942
 CFE_EVS_ERR_LOGMODE_EID, 942
 CFE_EVS_ERR_MAXREGSFILTER_EID, 943
 CFE_EVS_ERR_MSGID_EID, 943
 CFE_EVS_ERR_NOAPPIDFOUND_EID, 943
 CFE_EVS_ERR_UNREGISTERED_EVS_APP, 943
 CFE_EVS_ERR_WRDATFILE_EID, 944
 CFE_EVS_ERR_WRLOGFILE_EID, 944
 CFE_EVS_EVT_FILTERED_EID, 944
 CFE_EVS_FILTER_MAX_EID, 944
 CFE_EVS_LEN_ERR_EID, 945
 CFE_EVS_LOGMODE_EID, 945
 CFE_EVS_NOOP_EID, 945
 CFE_EVS_RSTALLFILTER_EID, 945
 CFE_EVS_RSTCNT_EID, 946
 CFE_EVS_RSTEVCNT_EID, 946
 CFE_EVS_RSTFILTER_EID, 946
 CFE_EVS_SETEVTFMTMOD_EID, 946
 CFE_EVS_SETFILTERMSK_EID, 947
 CFE_EVS_SQUELCHED_ERR_EID, 947
 CFE_EVS_STARTUP_EID, 947
 CFE_EVS_WRDAT_EID, 947
 CFE_EVS_WRITE_HEADER_ERR_EID, 948
 CFE_EVS_WRLOG_EID, 948
 CFE_EVS_EventOutput
 default_cfe_evs_extern_typedefs.h, 913
 CFE_EVS_EventOutput_Enum_t
 default_cfe_evs_extern_typedefs.h, 912
 CFE_EVS_EventOutput_PORT1
 default_cfe_evs_extern_typedefs.h, 913
 CFE_EVS_EventOutput_PORT2
 default_cfe_evs_extern_typedefs.h, 913
 CFE_EVS_EventOutput_PORT3
 default_cfe_evs_extern_typedefs.h, 913
 CFE_EVS_EventOutput_PORT4
 default_cfe_evs_extern_typedefs.h, 913
 CFE_EVS_EventType
 default_cfe_evs_extern_typedefs.h, 914
 CFE_EVS_EventType_CRITICAL
 default_cfe_evs_extern_typedefs.h, 914
 CFE_EVS_EventType_DEBUG
 default_cfe_evs_extern_typedefs.h, 914
 CFE_EVS_EventType_Enum_t
 default_cfe_evs_extern_typedefs.h, 912
 CFE_EVS_EventType_ERROR
 default_cfe_evs_extern_typedefs.h, 914
 CFE_EVS_EventType_INFORMATION
 default_cfe_evs_extern_typedefs.h, 914
 CFE_EVS_EVERY_FOURTH_ONE
 cfe_evs_api_typedefs.h, 731
 CFE_EVS_EVERY_OTHER_ONE
 cfe_evs_api_typedefs.h, 731
 CFE_EVS_EVERY_OTHER_TWO
 cfe_evs_api_typedefs.h, 731
 CFE_EVS_EVT_FILTERED_EID
 cfe_evs_eventids.h, 944
 CFE_EVS_EVT_NOT_REGISTERED
 cFE Return Code Defines, 132
 cfe_evs_fcncodes.h

- CFE_EVS_ADD_EVENT_FILTER_CC, [950](#)
- CFE_EVS_CLEAR_LOG_CC, [951](#)
- CFE_EVS_DELETE_EVENT_FILTER_CC, [952](#)
- CFE_EVS_DISABLE_APP_EVENT_TYPE_CC, [953](#)
- CFE_EVS_DISABLE_APP_EVENTS_CC, [954](#)
- CFE_EVS_DISABLE_EVENT_TYPE_CC, [955](#)
- CFE_EVS_DISABLE_PORTS_CC, [955](#)
- CFE_EVS_ENABLE_APP_EVENT_TYPE_CC, [956](#)
- CFE_EVS_ENABLE_APP_EVENTS_CC, [957](#)
- CFE_EVS_ENABLE_EVENT_TYPE_CC, [958](#)
- CFE_EVS_ENABLE_PORTS_CC, [959](#)
- CFE_EVS_NOOP_CC, [960](#)
- CFE_EVS_RESET_ALL_FILTERS_CC, [960](#)
- CFE_EVS_RESET_APP_COUNTER_CC, [961](#)
- CFE_EVS_RESET_COUNTERS_CC, [962](#)
- CFE_EVS_RESET_FILTER_CC, [963](#)
- CFE_EVS_SET_EVENT_FORMAT_MODE_CC, [963](#)
- CFE_EVS_SET_FILTER_CC, [964](#)
- CFE_EVS_SET_LOG_MODE_CC, [965](#)
- CFE_EVS_WRITE_APP_DATA_FILE_CC, [966](#)
- CFE_EVS_WRITE_LOG_DATA_FILE_CC, [967](#)
- CFE_EVS_FILE_WRITE_ERROR
 - cFE Return Code Defines, [132](#)
- CFE_EVS_FILTER_MAX_EID
 - cfe_evs_eventids.h, [944](#)
- CFE_EVS_FIRST_16_STOP
 - cfe_evs_api_typedefs.h, [731](#)
- CFE_EVS_FIRST_32_STOP
 - cfe_evs_api_typedefs.h, [731](#)
- CFE_EVS_FIRST_4_STOP
 - cfe_evs_api_typedefs.h, [731](#)
- CFE_EVS_FIRST_64_STOP
 - cfe_evs_api_typedefs.h, [732](#)
- CFE_EVS_FIRST_8_STOP
 - cfe_evs_api_typedefs.h, [732](#)
- CFE_EVS_FIRST_ONE_STOP
 - cfe_evs_api_typedefs.h, [732](#)
- CFE_EVS_FIRST_TWO_STOP
 - cfe_evs_api_typedefs.h, [732](#)
- CFE_EVS_FunctionCode_
 - default_cfe_evs_fcncode_values.h, [916](#)
- CFE_EVS_FunctionCode_ADD_EVENT_FILTER
 - default_cfe_evs_fcncode_values.h, [916](#)
- CFE_EVS_FunctionCode_CLEAR_LOG
 - default_cfe_evs_fcncode_values.h, [917](#)
- CFE_EVS_FunctionCode_DELETE_EVENT_FILTER
 - default_cfe_evs_fcncode_values.h, [916](#)
- CFE_EVS_FunctionCode_DISABLE_APP_EVENT_TYPE
 - default_cfe_evs_fcncode_values.h, [916](#)
- CFE_EVS_FunctionCode_DISABLE_APP_EVENTS
 - default_cfe_evs_fcncode_values.h, [916](#)
- CFE_EVS_FunctionCode_DISABLE_EVENT_TYPE
 - default_cfe_evs_fcncode_values.h, [916](#)
- CFE_EVS_FunctionCode_DISABLE_PORTS
 - default_cfe_evs_fcncode_values.h, [916](#)
- CFE_EVS_FunctionCode_ENABLE_APP_EVENT_TYPE
 - default_cfe_evs_fcncode_values.h, [916](#)
- CFE_EVS_FunctionCode_ENABLE_APP_EVENTS
 - default_cfe_evs_fcncode_values.h, [916](#)
- CFE_EVS_FunctionCode_ENABLE_EVENT_TYPE
 - default_cfe_evs_fcncode_values.h, [916](#)
- CFE_EVS_FunctionCode_ENABLE_PORTS
 - default_cfe_evs_fcncode_values.h, [916](#)
- CFE_EVS_FunctionCode_NOOP
 - default_cfe_evs_fcncode_values.h, [916](#)
- CFE_EVS_FunctionCode_RESET_ALL_FILTERS
 - default_cfe_evs_fcncode_values.h, [916](#)
- CFE_EVS_FunctionCode_RESET_APP_COUNTER
 - default_cfe_evs_fcncode_values.h, [916](#)
- CFE_EVS_FunctionCode_RESET_COUNTERS
 - default_cfe_evs_fcncode_values.h, [916](#)
- CFE_EVS_FunctionCode_RESET_FILTER
 - default_cfe_evs_fcncode_values.h, [916](#)
- CFE_EVS_FunctionCode_SET_EVENT_FORMAT_MODE
 - default_cfe_evs_fcncode_values.h, [916](#)
- CFE_EVS_FunctionCode_SET_FILTER
 - default_cfe_evs_fcncode_values.h, [916](#)
- CFE_EVS_FunctionCode_SET_LOG_MODE
 - default_cfe_evs_fcncode_values.h, [917](#)
- CFE_EVS_FunctionCode_WRITE_APP_DATA_FILE
 - default_cfe_evs_fcncode_values.h, [917](#)
- CFE_EVS_FunctionCode_WRITE_LOG_DATA_FILE
 - default_cfe_evs_fcncode_values.h, [917](#)
- CFE_EVS_HK_TLM_MID
 - default_cfe_evs_msgids.h, [928](#)
- CFE_EVS_HousekeepingTlm, [484](#)
 - Payload, [484](#)
 - TelemetryHeader, [484](#)
- CFE_EVS_HousekeepingTlm_Payload, [484](#)
 - AppData, [485](#)
 - CommandCounter, [485](#)
 - CommandErrorCounter, [485](#)
 - LogEnabled, [485](#)
 - LogFullFlag, [486](#)
 - LogMode, [486](#)
 - LogOverflowCounter, [486](#)
 - MessageFormatMode, [486](#)
 - MessageSendCounter, [486](#)
 - MessageTruncCounter, [486](#)
 - OutputPort, [486](#)
 - Padding, [487](#)
 - UnregisteredAppCounter, [487](#)
- CFE_EVS_HousekeepingTlm_Payload_t
 - default_cfe_evs_msgdefs.h, [924](#)
- CFE_EVS_HousekeepingTlm_t
 - default_cfe_evs_msgstruct.h, [931](#)
- CFE_EVS_INFORMATION_BIT
 - default_cfe_evs_msgdefs.h, [922](#)

cfe_evs_interface_cfg.h
 CFE_MISSION_EVS_MAX_MESSAGE_LENGTH,
 969
 DEFAULT_CFE_MISSION_EVS_MAX_MESSAGE_LENGTH,
 969
 cfe_evs_internal_cfg.h
 CFE_PLATFORM_EVS_APP_EVENTS_PER_SEC,
 971
 CFE_PLATFORM_EVS_DEFAULT_APP_DATA_FILE,
 971
 CFE_PLATFORM_EVS_DEFAULT_LOG_FILE, 971
 CFE_PLATFORM_EVS_DEFAULT_LOG_MODE,
 972
 CFE_PLATFORM_EVS_DEFAULT_MSG_FORMAT_MODE,
 972
 CFE_PLATFORM_EVS_DEFAULT_TYPE_FLAG,
 972
 CFE_PLATFORM_EVS_LOG_MAX, 973
 CFE_PLATFORM_EVS_MAX_APP_EVENT_BURST,
 973
 CFE_PLATFORM_EVS_MAX_EVENT_FILTERS,
 973
 CFE_PLATFORM_EVS_PORT_DEFAULT, 974
 CFE_PLATFORM_EVS_START_TASK_PRIORITY,
 974
 CFE_PLATFORM_EVS_START_TASK_STACK_SIZE,
 974
 DEFAULT_CFE_PLATFORM_EVS_APP_EVENTS_PER_SEC,
 975
 DEFAULT_CFE_PLATFORM_EVS_DEFAULT_APP_DATA_FILE,
 975
 DEFAULT_CFE_PLATFORM_EVS_DEFAULT_LOG_FILE,
 975
 DEFAULT_CFE_PLATFORM_EVS_DEFAULT_LOG_MODE,
 975
 DEFAULT_CFE_PLATFORM_EVS_DEFAULT_MSG_FORMAT_MODE,
 975
 DEFAULT_CFE_PLATFORM_EVS_DEFAULT_TYPE_FLAG,
 975
 DEFAULT_CFE_PLATFORM_EVS_LOG_MAX, 975
 DEFAULT_CFE_PLATFORM_EVS_MAX_APP_EVENT_BURST,
 975
 DEFAULT_CFE_PLATFORM_EVS_MAX_EVENT_FILTERS,
 975
 DEFAULT_CFE_PLATFORM_EVS_PORT_DEFAULT,
 975
 DEFAULT_CFE_PLATFORM_EVS_START_TASK_PRIORITY,
 976
 DEFAULT_CFE_PLATFORM_EVS_START_TASK_STACK_SIZE,
 976
 CFE_EVS_INVALID_PARAMETER
 cFE Return Code Defines, 132
 CFE_EVS_LEN_ERR_EID
 cfe_evs_eventids.h, 945
 CFE_EVS_LogFileCmd_Payload, 487
 LogFilename, 487
 CFE_EVS_LogFileCmd_Payload_t
 default_cfe_evs_msgdefs.h, 924
 CFE_EVS_LogMode
 default_cfe_evs_extern_typedefs.h, 914
 CFE_EVS_LogMode_DISCARD
 default_cfe_evs_extern_typedefs.h, 914
 CFE_EVS_LOGMODE_EID
 cfe_evs_eventids.h, 945
 CFE_EVS_LogMode_Enum_t
 default_cfe_evs_extern_typedefs.h, 913
 CFE_EVS_LogMode_OVERWRITE
 default_cfe_evs_extern_typedefs.h, 914
 CFE_EVS_LONG_EVENT_MSG_MID
 default_cfe_evs_msgids.h, 928
 CFE_EVS_LongEventTlm, 487
 Payload, 488
 TelemetryHeader, 488
 CFE_EVS_LongEventTlm_Payload, 488
 Message, 488
 PacketID, 488
 Padding, 489
 CFE_EVS_LongEventTlm_Payload_t
 default_cfe_evs_msgdefs.h, 924
 CFE_EVS_LongEventTlm_t
 default_cfe_evs_msgstruct.h, 931
 CFE_EVS_MsgFormat
 default_cfe_evs_extern_typedefs.h, 914
 CFE_EVS_MsgFormat_Enum_t
 default_cfe_evs_extern_typedefs.h, 913
 CFE_EVS_MsgFormat_LONG
 default_cfe_evs_extern_typedefs.h, 914
 CFE_EVS_MsgFormat_SHORT
 default_cfe_evs_extern_typedefs.h, 914
 CFE_EVS_MSG_FORMAT_FILTER
 cfe_evs_api_typedefs.h, 732
 CFE_EVS_NOOP_CC
 cfe_evs_fcncodes.h, 960
 CFE_EVS_NOOP_EID
 cfe_evs_eventids.h, 945
 CFE_EVS_NoopCmd, 489
 CommandHeader, 489
 CFE_EVS_NoopCmd_t
 default_cfe_evs_msgstruct.h, 931
 CFE_EVS_NOT_IMPLEMENTED
 cFE Return Code Defines, 132
 CFE_EVS_PacketID, 489
 AppName, 490
 EventID, 490
 EventType, 490
 ProcessorID, 490
 SpacecraftID, 490
 CFE_EVS_PacketID_t

default_cfe_evs_msgdefs.h, [924](#)
CFE_EVS_PORT1_BIT
 default_cfe_evs_msgdefs.h, [922](#)
CFE_EVS_PORT2_BIT
 default_cfe_evs_msgdefs.h, [923](#)
CFE_EVS_PORT3_BIT
 default_cfe_evs_msgdefs.h, [923](#)
CFE_EVS_PORT4_BIT
 default_cfe_evs_msgdefs.h, [923](#)
CFE_EVS_Register
 cFE Registration APIs, [190](#)
CFE_EVS_RESET_ALL_FILTERS_CC
 cfe_evs_fcncodes.h, [960](#)
CFE_EVS_RESET_APP_COUNTER_CC
 cfe_evs_fcncodes.h, [961](#)
CFE_EVS_RESET_AREA_POINTER
 cFE Return Code Defines, [132](#)
CFE_EVS_RESET_COUNTERS_CC
 cfe_evs_fcncodes.h, [962](#)
CFE_EVS_RESET_FILTER_CC
 cfe_evs_fcncodes.h, [963](#)
CFE_EVS_ResetAllFilters
 cFE Reset Event Filter APIs, [195](#)
CFE_EVS_ResetAllFiltersCmd, [490](#)
 CommandHeader, [491](#)
 Payload, [491](#)
CFE_EVS_ResetAllFiltersCmd_t
 default_cfe_evs_msgstruct.h, [931](#)
CFE_EVS_ResetAppCounterCmd, [491](#)
 CommandHeader, [491](#)
 Payload, [491](#)
CFE_EVS_ResetAppCounterCmd_t
 default_cfe_evs_msgstruct.h, [931](#)
CFE_EVS_ResetCountersCmd, [492](#)
 CommandHeader, [492](#)
CFE_EVS_ResetCountersCmd_t
 default_cfe_evs_msgstruct.h, [931](#)
CFE_EVS_ResetFilter
 cFE Reset Event Filter APIs, [195](#)
CFE_EVS_ResetFilterCmd, [492](#)
 CommandHeader, [492](#)
 Payload, [492](#)
CFE_EVS_ResetFilterCmd_t
 default_cfe_evs_msgstruct.h, [931](#)
CFE_EVS_RSTALLFILTER_EID
 cfe_evs_eventids.h, [945](#)
CFE_EVS_RSTCNT_EID
 cfe_evs_eventids.h, [946](#)
CFE_EVS_RSTEVCNT_EID
 cfe_evs_eventids.h, [946](#)
CFE_EVS_RSTFILTER_EID
 cfe_evs_eventids.h, [946](#)
CFE_EVS_Send
 cfe_evs.h, [729](#)
CFE_EVS_SEND_HK_MID
 default_cfe_evs_msgids.h, [928](#)
CFE_EVS_SendCrit
 cfe_evs.h, [729](#)
CFE_EVS_SendDbg
 cfe_evs.h, [729](#)
CFE_EVS_SendErr
 cfe_evs.h, [729](#)
CFE_EVS_SendEvent
 cFE Send Event APIs, [191](#)
CFE_EVS_SendEventWithAppID
 cFE Send Event APIs, [192](#)
CFE_EVS_SendHkCmd, [493](#)
 CommandHeader, [493](#)
CFE_EVS_SendHkCmd_t
 default_cfe_evs_msgstruct.h, [931](#)
CFE_EVS_SendInfo
 cfe_evs.h, [729](#)
CFE_EVS_SendTimedEvent
 cFE Send Event APIs, [193](#)
CFE_EVS_SET_EVENT_FORMAT_MODE_CC
 cfe_evs_fcncodes.h, [963](#)
CFE_EVS_SET_FILTER_CC
 cfe_evs_fcncodes.h, [964](#)
CFE_EVS_SET_LOG_MODE_CC
 cfe_evs_fcncodes.h, [965](#)
CFE_EVS_SetEventFormatCode_Payload, [493](#)
 MsgFormat, [493](#)
 Spare, [493](#)
CFE_EVS_SetEventFormatMode_Payload_t
 default_cfe_evs_msgdefs.h, [924](#)
CFE_EVS_SetEventFormatModeCmd, [494](#)
 CommandHeader, [494](#)
 Payload, [494](#)
CFE_EVS_SetEventFormatModeCmd_t
 default_cfe_evs_msgstruct.h, [932](#)
CFE_EVS_SETEVTFMOD_EID
 cfe_evs_eventids.h, [946](#)
CFE_EVS_SetFilterCmd, [494](#)
 CommandHeader, [495](#)
 Payload, [495](#)
CFE_EVS_SetFilterCmd_t
 default_cfe_evs_msgstruct.h, [932](#)
CFE_EVS_SETFILTERMSK_EID
 cfe_evs_eventids.h, [947](#)
CFE_EVS_SetLogMode_Payload, [495](#)
 LogMode, [495](#)
 Spare, [495](#)
CFE_EVS_SetLogMode_Payload_t
 default_cfe_evs_msgdefs.h, [924](#)
CFE_EVS_SetLogModeCmd, [496](#)
 CommandHeader, [496](#)
 Payload, [496](#)
CFE_EVS_SetLogModeCmd_t

- default_cfe_evs_msgstruct.h, 932
- CFE_EVS_SHORT_EVENT_MSG_MID
 - default_cfe_evs_msgids.h, 928
- CFE_EVS_ShortEventTlm, 496
 - Payload, 496
 - TelemetryHeader, 496
- CFE_EVS_ShortEventTlm_Payload, 497
 - PacketID, 497
- CFE_EVS_ShortEventTlm_Payload_t
 - default_cfe_evs_msgdefs.h, 924
- CFE_EVS_ShortEventTlm_t
 - default_cfe_evs_msgstruct.h, 932
- CFE_EVS_SQUELCHED_ERR_EID
 - cfe_evs_eventids.h, 947
- CFE_EVS_STARTUP_EID
 - cfe_evs_eventids.h, 947
- cfe_evs_topicids.h
 - CFE_MISSION_EVS_CMD_TOPICID, 977
 - CFE_MISSION_EVS_HK_TLM_TOPICID, 977
 - CFE_MISSION_EVS_LONG_EVENT_MSG_TOPICID, 978
 - CFE_MISSION_EVS_SEND_HK_TOPICID, 978
 - CFE_MISSION_EVS_SHORT_EVENT_MSG_TOPICID, 978
 - DEFAULT_CFE_MISSION_EVS_CMD_TOPICID, 978
 - DEFAULT_CFE_MISSION_EVS_HK_TLM_TOPICID, 978
 - DEFAULT_CFE_MISSION_EVS_LONG_EVENT_MSG_TOPICID, 978
 - DEFAULT_CFE_MISSION_EVS_SEND_HK_TOPICID, 978
 - DEFAULT_CFE_MISSION_EVS_SHORT_EVENT_MSG_TOPICID, 978
- CFE_EVS_UNKNOWN_FILTER
 - cFE Return Code Defines, 133
- CFE_EVS_WRDAT_EID
 - cfe_evs_eventids.h, 947
- CFE_EVS_WRITE_APP_DATA_FILE_CC
 - cfe_evs_fcncodes.h, 966
- CFE_EVS_WRITE_HEADER_ERR_EID
 - cfe_evs_eventids.h, 948
- CFE_EVS_WRITE_LOG_DATA_FILE_CC
 - cfe_evs_fcncodes.h, 967
- CFE_EVS_WriteAppDataFileCmd, 497
 - CommandHeader, 498
 - Payload, 498
- CFE_EVS_WriteAppDataFileCmd_t
 - default_cfe_evs_msgstruct.h, 932
- CFE_EVS_WriteLogDataFileCmd, 498
 - CommandHeader, 498
 - Payload, 498
- CFE_EVS_WriteLogDataFileCmd_t
 - default_cfe_evs_msgstruct.h, 932
- CFE_EVS_WRLOG_EID
 - cfe_evs_eventids.h, 948
- CFE_EXECUTIVE_SERVICE
 - cfe_error.h, 708
- CFE_FILE_SERVICE
 - cfe_error.h, 708
- cfe_fs_api_typedefs.h
 - CFE_FS_FileCategory_BINARY_DATA_DUMP, 737
 - CFE_FS_FileCategory_DYNAMIC_MODULE, 737
 - CFE_FS_FileCategory_MAX, 737
 - CFE_FS_FileCategory_SCRIPT, 737
 - CFE_FS_FileCategory_t, 737
 - CFE_FS_FileCategory_TEMP, 737
 - CFE_FS_FileCategory_TEXT_LOG, 737
 - CFE_FS_FileCategory_UNKNOWN, 737
 - CFE_FS_FileWriteEvent_COMPLETE, 737
 - CFE_FS_FileWriteEvent_CREATE_ERROR, 737
 - CFE_FS_FileWriteEvent_HEADER_WRITE_ERROR, 737
 - CFE_FS_FileWriteEvent_MAX, 737
 - CFE_FS_FileWriteEvent_RECORD_WRITE_ERROR, 737
 - CFE_FS_FileWriteEvent_t, 737
 - CFE_FS_FileWriteEvent_UNDEFINED, 737
 - CFE_FS_FileWriteGetData_t, 736
 - CFE_FS_FileWriteMetaData_t, 736
 - CFE_FS_FileWriteOnEvent_t, 736
- CFE_FS_BackgroundFileDumpsPending
 - cFE File Utility APIs, 200
- CFE_FS_BackgroundFileDumpRequest
 - cFE File Utility APIs, 201
- CFE_FS_BAD_ARGUMENT
 - cFE Return Code Defines, 133
- CFE_FS_ExtractFilenameFromPath
 - cFE File Utility APIs, 201
- CFE_FS_FILE_CONTENT_ID
 - cfe_fs_interface_cfg.h, 984
 - example_mission_cfg.h, 616
- CFE_FS_FileCategory_BINARY_DATA_DUMP
 - cfe_fs_api_typedefs.h, 737
- CFE_FS_FileCategory_DYNAMIC_MODULE
 - cfe_fs_api_typedefs.h, 737
- CFE_FS_FileCategory_MAX
 - cfe_fs_api_typedefs.h, 737
- CFE_FS_FileCategory_SCRIPT
 - cfe_fs_api_typedefs.h, 737
- CFE_FS_FileCategory_t
 - cfe_fs_api_typedefs.h, 737
- CFE_FS_FileCategory_TEMP
 - cfe_fs_api_typedefs.h, 737
- CFE_FS_FileCategory_TEXT_LOG
 - cfe_fs_api_typedefs.h, 737
- CFE_FS_FileCategory_UNKNOWN
 - cfe_fs_api_typedefs.h, 737

- CFE_FS_FileWriteEvent_COMPLETE
 - cfe_fs_api_typedefs.h, [737](#)
- CFE_FS_FileWriteEvent_CREATE_ERROR
 - cfe_fs_api_typedefs.h, [737](#)
- CFE_FS_FileWriteEvent_HEADER_WRITE_ERROR
 - cfe_fs_api_typedefs.h, [737](#)
- CFE_FS_FileWriteEvent_MAX
 - cfe_fs_api_typedefs.h, [737](#)
- CFE_FS_FileWriteEvent_RECORD_WRITE_ERROR
 - cfe_fs_api_typedefs.h, [737](#)
- CFE_FS_FileWriteEvent_t
 - cfe_fs_api_typedefs.h, [737](#)
- CFE_FS_FileWriteEvent_UNDEFINED
 - cfe_fs_api_typedefs.h, [737](#)
- CFE_FS_FileWriteGetData_t
 - cfe_fs_api_typedefs.h, [736](#)
- CFE_FS_FileWriteMetaData, [498](#)
 - Description, [499](#)
 - FileName, [499](#)
 - FileSubType, [499](#)
 - GetData, [499](#)
 - IsPending, [499](#)
 - OnEvent, [499](#)
- CFE_FS_FileWriteMetaData_t
 - cfe_fs_api_typedefs.h, [736](#)
- CFE_FS_FileWriteOnEvent_t
 - cfe_fs_api_typedefs.h, [736](#)
- CFE_FS_FNAME_TOO_LONG
 - cFE Return Code Defines, [133](#)
- CFE_FS_GetDefaultExtension
 - cFE File Utility APIs, [202](#)
- CFE_FS_GetDefaultMountPoint
 - cFE File Utility APIs, [202](#)
- CFE_FS_HDR_DESC_MAX_LEN
 - cfe_fs_interface_cfg.h, [984](#)
 - example_mission_cfg.h, [616](#)
- CFE_FS_Header, [500](#)
 - ApplicationID, [500](#)
 - ContentType, [500](#)
 - Description, [500](#)
 - Length, [500](#)
 - ProcessorID, [501](#)
 - SpacecraftID, [501](#)
 - SubType, [501](#)
 - TimeSeconds, [501](#)
 - TimeSubSeconds, [501](#)
- CFE_FS_Header_t
 - default_cfe_fs_filedef.h, [980](#)
- CFE_FS_InitHeader
 - cFE File Header Management APIs, [196](#)
- cfe_fs_interface_cfg.h
 - CFE_FS_FILE_CONTENT_ID, [984](#)
 - CFE_FS_HDR_DESC_MAX_LEN, [984](#)
 - DEFAULT_CFE_FS_FILE_CONTENT_ID, [985](#)
 - DEFAULT_CFE_FS_HDR_DESC_MAX_LEN, [985](#)
 - CFE_FS_INVALID_PATH
 - cFE Return Code Defines, [133](#)
 - CFE_FS_NOT_IMPLEMENTED
 - cFE Return Code Defines, [133](#)
 - CFE_FS_ParseInputFileName
 - cFE File Utility APIs, [203](#)
 - CFE_FS_ParseInputFileNameEx
 - cFE File Utility APIs, [203](#)
 - CFE_FS_ReadHeader
 - cFE File Header Management APIs, [197](#)
 - CFE_FS_SetTimestamp
 - cFE File Header Management APIs, [198](#)
 - CFE_FS_SubType
 - default_cfe_fs_filedef.h, [981](#)
 - CFE_FS_SubType_Enum_t
 - default_cfe_fs_filedef.h, [980](#)
 - CFE_FS_SubType_ES_CDS_REG
 - default_cfe_fs_filedef.h, [981](#)
 - CFE_FS_SubType_ES_ERLOG
 - default_cfe_fs_filedef.h, [981](#)
 - CFE_FS_SubType_ES_PERFDATA
 - default_cfe_fs_filedef.h, [981](#)
 - CFE_FS_SubType_ES_QUERYALL
 - default_cfe_fs_filedef.h, [981](#)
 - CFE_FS_SubType_ES_QUERYALLTASKS
 - default_cfe_fs_filedef.h, [981](#)
 - CFE_FS_SubType_ES_SYSLOG
 - default_cfe_fs_filedef.h, [981](#)
 - CFE_FS_SubType_EVS_APPDATA
 - default_cfe_fs_filedef.h, [981](#)
 - CFE_FS_SubType_EVS_EVENTLOG
 - default_cfe_fs_filedef.h, [981](#)
 - CFE_FS_SubType_SB_MAPDATA
 - default_cfe_fs_filedef.h, [981](#)
 - CFE_FS_SubType_SB_PIPEDATA
 - default_cfe_fs_filedef.h, [981](#)
 - CFE_FS_SubType_SB_ROUTEDATA
 - default_cfe_fs_filedef.h, [981](#)
 - CFE_FS_SubType_TBL_IMG
 - default_cfe_fs_filedef.h, [981](#)
 - CFE_FS_SubType_TBL_REG
 - default_cfe_fs_filedef.h, [981](#)
 - CFE_FS_WriteHeader
 - cFE File Header Management APIs, [198](#)
- CFE_GENERIC_SERVICE
 - cfe_error.h, [708](#)
- CFE_GLOBAL_BASE_MIDVAL
 - default_cfe_core_api_base_msgid_values.h, [683](#)
- CFE_GLOBAL_CMD_TOPICID_TO_MIDV
 - default_cfe_core_api_msgid_mapping.h, [685](#)
- CFE_GLOBAL_TLM_TOPICID_TO_MIDV
 - default_cfe_core_api_msgid_mapping.h, [686](#)
- CFE_LAST_OFFICIAL

- cfe_version.h, [784](#)
- CFE_MAJOR_VERSION
 - cfe_version.h, [784](#)
- CFE_MAKE_BIG16
 - cfe_endian.h, [700](#)
- CFE_MAKE_BIG32
 - cfe_endian.h, [700](#)
- CFE_MINOR_VERSION
 - cfe_version.h, [784](#)
- CFE_MISSION_CORE_API_CFGVAL
 - default_cfe_core_api_interface_cfg_values.h, [684](#)
- CFE_MISSION_ES_APP_TLM_TOPICID
 - cfe_es_topicids.h, [910](#)
- CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN
 - cfe_es_interface_cfg.h, [872](#)
 - example_mission_cfg.h, [616](#)
- CFE_MISSION_ES_CDS_MAX_NAME_LENGTH
 - cfe_es_interface_cfg.h, [873](#)
 - example_mission_cfg.h, [617](#)
- CFE_MISSION_ES_CFGVAL
 - default_cfe_es_interface_cfg_values.h, [798](#)
- CFE_MISSION_ES_CMD_TOPICID
 - cfe_es_topicids.h, [910](#)
- CFE_MISSION_ES_CRC_16
 - cfe_es_interface_cfg.h, [873](#)
 - example_mission_cfg.h, [617](#)
- CFE_MISSION_ES_CRC_32
 - cfe_es_interface_cfg.h, [873](#)
 - example_mission_cfg.h, [617](#)
- CFE_MISSION_ES_CRC_8
 - cfe_es_interface_cfg.h, [873](#)
 - example_mission_cfg.h, [617](#)
- CFE_MISSION_ES_DEFAULT_CRC
 - cfe_es_interface_cfg.h, [873](#)
 - example_mission_cfg.h, [617](#)
- CFE_MISSION_ES_HK_TLM_TOPICID
 - cfe_es_topicids.h, [910](#)
- CFE_MISSION_ES_MAIN_PERF_ID
 - cfe_perfids.h, [606](#)
- CFE_MISSION_ES_MAX_APPLICATIONS
 - cfe_es_interface_cfg.h, [873](#)
 - example_mission_cfg.h, [618](#)
- CFE_MISSION_ES_MEMSTATS_TLM_TOPICID
 - cfe_es_topicids.h, [910](#)
- CFE_MISSION_ES_PERF_EXIT_BIT
 - cfe_perfids.h, [606](#)
- CFE_MISSION_ES_PERF_MAX_IDS
 - cfe_es_interface_cfg.h, [874](#)
 - example_mission_cfg.h, [618](#)
- CFE_MISSION_ES_POOL_MAX_BUCKETS
 - cfe_es_interface_cfg.h, [874](#)
 - example_mission_cfg.h, [618](#)
- CFE_MISSION_ES_SEND_HK_TOPICID
 - cfe_es_topicids.h, [910](#)
- CFE_MISSION_ES_TIDVAL
 - default_cfe_es_topicid_values.h, [820](#)
- CFE_MISSION_EVS_CFGVAL
 - default_cfe_evs_interface_cfg_values.h, [918](#)
- CFE_MISSION_EVS_CMD_TOPICID
 - cfe_evs_topicids.h, [977](#)
- CFE_MISSION_EVS_HK_TLM_TOPICID
 - cfe_evs_topicids.h, [977](#)
- CFE_MISSION_EVS_LONG_EVENT_MSG_TOPICID
 - cfe_evs_topicids.h, [978](#)
- CFE_MISSION_EVS_MAIN_PERF_ID
 - cfe_perfids.h, [606](#)
- CFE_MISSION_EVS_MAX_MESSAGE_LENGTH
 - cfe_evs_interface_cfg.h, [969](#)
 - example_mission_cfg.h, [619](#)
- CFE_MISSION_EVS_SEND_HK_TOPICID
 - cfe_evs_topicids.h, [978](#)
- CFE_MISSION_EVS_SHORT_EVENT_MSG_TOPICID
 - cfe_evs_topicids.h, [978](#)
- CFE_MISSION_EVS_TIDVAL
 - default_cfe_evs_topicid_values.h, [936](#)
- CFE_MISSION_FS_CFGVAL
 - default_cfe_fs_interface_cfg_values.h, [983](#)
- CFE_MISSION_MAX_API_LEN
 - cfe_core_api_interface_cfg.h, [697](#)
 - example_mission_cfg.h, [619](#)
- CFE_MISSION_MAX_FILE_LEN
 - cfe_core_api_interface_cfg.h, [697](#)
 - example_mission_cfg.h, [619](#)
- CFE_MISSION_MAX_NUM_FILES
 - cfe_core_api_interface_cfg.h, [698](#)
 - example_mission_cfg.h, [620](#)
- CFE_MISSION_MAX_PATH_LEN
 - cfe_core_api_interface_cfg.h, [698](#)
 - example_mission_cfg.h, [620](#)
- CFE_MISSION_REV
 - cfe_version.h, [784](#)
- CFE_MISSION_SB_ALLSUBS_TLM_TOPICID
 - cfe_sb_topicids.h, [1061](#)
- CFE_MISSION_SB_CFGVAL
 - default_cfe_sb_interface_cfg_values.h, [996](#)
- CFE_MISSION_SB_CMD_TOPICID
 - cfe_sb_topicids.h, [1061](#)
- CFE_MISSION_SB_HK_TLM_TOPICID
 - cfe_sb_topicids.h, [1062](#)
- CFE_MISSION_SB_MAIN_PERF_ID
 - cfe_perfids.h, [606](#)
- CFE_MISSION_SB_MAX_PIPES
 - cfe_sb_interface_cfg.h, [1043](#)
 - example_mission_cfg.h, [621](#)
- CFE_MISSION_SB_MAX_SB_MSG_SIZE
 - cfe_sb_interface_cfg.h, [1043](#)
 - example_mission_cfg.h, [621](#)
- CFE_MISSION_SB_MSG_LIM_PERF_ID

- cfe_perfids.h, [606](#)
- CFE_MISSION_SB_ONESUB_TLM_TOPICID
 - cfe_sb_topicsids.h, [1062](#)
- CFE_MISSION_SB_PIPE_OFLOW_PERF_ID
 - cfe_perfids.h, [606](#)
- CFE_MISSION_SB_SEND_HK_TOPICID
 - cfe_sb_topicsids.h, [1062](#)
- CFE_MISSION_SB_STATS_TLM_TOPICID
 - cfe_sb_topicsids.h, [1062](#)
- CFE_MISSION_SB_SUB_ENTRIES_PER_PKT
 - cfe_sb_interface_cfg.h, [1044](#)
- CFE_MISSION_SB_SUB_RPT_CTRL_TOPICID
 - cfe_sb_topicsids.h, [1062](#)
- CFE_MISSION_SB_TIDVAL
 - default_cfe_sb_topicid_values.h, [1011](#)
- CFE_MISSION_TBL_CFGVAL
 - default_cfe_tbl_interface_cfg_values.h, [1068](#)
- CFE_MISSION_TBL_CMD_TOPICID
 - cfe_tbl_topicsids.h, [1137](#)
- CFE_MISSION_TBL_HK_TLM_TOPICID
 - cfe_tbl_topicsids.h, [1137](#)
- CFE_MISSION_TBL_MAIN_PERF_ID
 - cfe_perfids.h, [606](#)
- CFE_MISSION_TBL_MAX_FULL_NAME_LEN
 - cfe_tbl_interface_cfg.h, [1126](#)
 - default_cfe_tbl_mission_cfg.h, [1070](#)
 - example_mission_cfg.h, [621](#)
- CFE_MISSION_TBL_MAX_NAME_LENGTH
 - cfe_tbl_interface_cfg.h, [1126](#)
 - default_cfe_tbl_mission_cfg.h, [1070](#)
 - example_mission_cfg.h, [622](#)
- CFE_MISSION_TBL_REG_TLM_TOPICID
 - cfe_tbl_topicsids.h, [1138](#)
- CFE_MISSION_TBL_SEND_HK_TOPICID
 - cfe_tbl_topicsids.h, [1138](#)
- CFE_MISSION_TBL_TIDVAL
 - default_cfe_tbl_topicid_values.h, [1092](#)
- CFE_MISSION_TIME_AT_TONE_WAS
 - cfe_time_interface_cfg.h, [1195](#)
 - example_mission_cfg.h, [622](#)
- CFE_MISSION_TIME_AT_TONE_WILL_BE
 - cfe_time_interface_cfg.h, [1195](#)
 - example_mission_cfg.h, [623](#)
- CFE_MISSION_TIME_CFG_DEFAULT_TAI
 - cfe_time_interface_cfg.h, [1195](#)
 - example_mission_cfg.h, [623](#)
- CFE_MISSION_TIME_CFG_DEFAULT_UTC
 - cfe_time_interface_cfg.h, [1195](#)
 - example_mission_cfg.h, [623](#)
- CFE_MISSION_TIME_CFG_FAKE_TONE
 - cfe_time_interface_cfg.h, [1195](#)
 - example_mission_cfg.h, [623](#)
- CFE_MISSION_TIME_CFGVAL
 - default_cfe_time_interface_cfg_values.h, [1147](#)
- CFE_MISSION_TIME_CMD_TOPICID
 - cfe_time_topicsids.h, [1211](#)
- CFE_MISSION_TIME_DATA_CMD_TOPICID
 - cfe_time_topicsids.h, [1211](#)
- CFE_MISSION_TIME_DEF_DELAY_SECS
 - cfe_time_interface_cfg.h, [1196](#)
 - example_mission_cfg.h, [623](#)
- CFE_MISSION_TIME_DEF_DELAY_SUBS
 - cfe_time_interface_cfg.h, [1196](#)
 - example_mission_cfg.h, [623](#)
- CFE_MISSION_TIME_DEF_LEAPS
 - cfe_time_interface_cfg.h, [1196](#)
 - example_mission_cfg.h, [624](#)
- CFE_MISSION_TIME_DEF_MET_SECS
 - cfe_time_interface_cfg.h, [1196](#)
 - example_mission_cfg.h, [624](#)
- CFE_MISSION_TIME_DEF_MET_SUBS
 - cfe_time_interface_cfg.h, [1196](#)
 - example_mission_cfg.h, [624](#)
- CFE_MISSION_TIME_DEF_STCF_SECS
 - cfe_time_interface_cfg.h, [1197](#)
 - example_mission_cfg.h, [624](#)
- CFE_MISSION_TIME_DEF_STCF_SUBS
 - cfe_time_interface_cfg.h, [1197](#)
 - example_mission_cfg.h, [624](#)
- CFE_MISSION_TIME_DIAG_TLM_TOPICID
 - cfe_time_topicsids.h, [1212](#)
- CFE_MISSION_TIME_EPOCH_DAY
 - cfe_time_interface_cfg.h, [1197](#)
 - example_mission_cfg.h, [624](#)
- CFE_MISSION_TIME_EPOCH_HOUR
 - cfe_time_interface_cfg.h, [1197](#)
 - example_mission_cfg.h, [624](#)
- CFE_MISSION_TIME_EPOCH_MICROS
 - cfe_time_interface_cfg.h, [1197](#)
 - example_mission_cfg.h, [625](#)
- CFE_MISSION_TIME_EPOCH_MINUTE
 - cfe_time_interface_cfg.h, [1197](#)
 - example_mission_cfg.h, [625](#)
- CFE_MISSION_TIME_EPOCH_SECOND
 - cfe_time_interface_cfg.h, [1197](#)
 - example_mission_cfg.h, [625](#)
- CFE_MISSION_TIME_EPOCH_SECONDS
 - cfe_time_interface_cfg.h, [1197](#)
- CFE_MISSION_TIME_EPOCH_YEAR
 - cfe_time_interface_cfg.h, [1197](#)
 - example_mission_cfg.h, [625](#)
- CFE_MISSION_TIME_FS_FACTOR
 - cfe_time_interface_cfg.h, [1198](#)
 - example_mission_cfg.h, [625](#)
- CFE_MISSION_TIME_HK_TLM_TOPICID
 - cfe_time_topicsids.h, [1212](#)
- CFE_MISSION_TIME_LOCAL1HZISR_PERF_ID
 - cfe_perfids.h, [606](#)

CFE_MISSION_TIME_LOCAL1HZTASK_PERF_ID
 cfe_perfids.h, [606](#)
 CFE_MISSION_TIME_MAIN_PERF_ID
 cfe_perfids.h, [607](#)
 CFE_MISSION_TIME_MAX_ELAPSED
 cfe_time_interface_cfg.h, [1198](#)
 example_mission_cfg.h, [626](#)
 CFE_MISSION_TIME_MIN_ELAPSED
 cfe_time_interface_cfg.h, [1198](#)
 example_mission_cfg.h, [626](#)
 CFE_MISSION_TIME_ONEHZ_CMD_TOPICID
 cfe_time_topics.h, [1212](#)
 CFE_MISSION_TIME_SEND_CMD_TOPICID
 cfe_time_topics.h, [1212](#)
 CFE_MISSION_TIME_SEND_HK_TOPICID
 cfe_time_topics.h, [1212](#)
 CFE_MISSION_TIME_SENDMET_PERF_ID
 cfe_perfids.h, [607](#)
 CFE_MISSION_TIME_TIDVAL
 default_cfe_time_topicid_values.h, [1165](#)
 CFE_MISSION_TIME_TONE1HZISR_PERF_ID
 cfe_perfids.h, [607](#)
 CFE_MISSION_TIME_TONE1HZTASK_PERF_ID
 cfe_perfids.h, [607](#)
 CFE_MISSION_TIME_TONE_CMD_TOPICID
 cfe_time_topics.h, [1212](#)
 cfe_msg_api_typedefs.h
 CFE_MSG_Apld_t, [744](#)
 CFE_MSG_BAD_ARGUMENT, [744](#)
 CFE_MSG_Checksum_t, [744](#)
 CFE_MSG_CommandHeader_t, [745](#)
 CFE_MSG_EDSVersion_t, [745](#)
 CFE_MSG_Endian, [746](#)
 CFE_MSG_Endian_Big, [746](#)
 CFE_MSG_Endian_Invalid, [746](#)
 CFE_MSG_Endian_Little, [746](#)
 CFE_MSG_Endian_t, [745](#)
 CFE_MSG_FcnCode_t, [745](#)
 CFE_MSG_HeaderVersion_t, [745](#)
 CFE_MSG_Message_t, [745](#)
 CFE_MSG_NOT_IMPLEMENTED, [744](#)
 CFE_MSG_PlaybackFlag, [746](#)
 CFE_MSG_PlaybackFlag_t, [745](#)
 CFE_MSG_PlayFlag_Invalid, [746](#)
 CFE_MSG_PlayFlag_Original, [746](#)
 CFE_MSG_PlayFlag_Playback, [746](#)
 CFE_MSG_SegFlag_Continue, [747](#)
 CFE_MSG_SegFlag_First, [747](#)
 CFE_MSG_SegFlag_Invalid, [747](#)
 CFE_MSG_SegFlag_Last, [747](#)
 CFE_MSG_SegFlag_Unsegmented, [747](#)
 CFE_MSG_SegmentationFlag, [747](#)
 CFE_MSG_SegmentationFlag_t, [745](#)
 CFE_MSG_SequenceCount_t, [745](#)
 CFE_MSG_Size_t, [745](#)
 CFE_MSG_Subsystem_t, [746](#)
 CFE_MSG_System_t, [746](#)
 CFE_MSG_TelemetryHeader_t, [746](#)
 CFE_MSG_Type, [747](#)
 CFE_MSG_Type_Cmd, [747](#)
 CFE_MSG_Type_Invalid, [747](#)
 CFE_MSG_Type_t, [746](#)
 CFE_MSG_Type_Tlm, [747](#)
 CFE_MSG_WRONG_MSG_TYPE, [744](#)
 CFE_MSG_Apld_t
 cfe_msg_api_typedefs.h, [744](#)
 CFE_MSG_BAD_ARGUMENT
 cfe_msg_api_typedefs.h, [744](#)
 CFE_MSG_Checksum_t
 cfe_msg_api_typedefs.h, [744](#)
 CFE_MSG_CommandHeader_t
 cfe_msg_api_typedefs.h, [745](#)
 CFE_MSG_EDSVersion_t
 cfe_msg_api_typedefs.h, [745](#)
 CFE_MSG_Endian
 cfe_msg_api_typedefs.h, [746](#)
 CFE_MSG_Endian_Big
 cfe_msg_api_typedefs.h, [746](#)
 CFE_MSG_Endian_Invalid
 cfe_msg_api_typedefs.h, [746](#)
 CFE_MSG_Endian_Little
 cfe_msg_api_typedefs.h, [746](#)
 CFE_MSG_Endian_t
 cfe_msg_api_typedefs.h, [745](#)
 CFE_MSG_FcnCode_t
 cfe_msg_api_typedefs.h, [745](#)
 CFE_MSG_GenerateChecksum
 cFE Message Secondary Header APIs, [220](#)
 CFE_MSG_GetApld
 cFE Message Primary Header APIs, [206](#)
 CFE_MSG_GetEDSVersion
 cFE Message Extended Header APIs, [214](#)
 CFE_MSG_GetEndian
 cFE Message Extended Header APIs, [215](#)
 CFE_MSG_GetFcnCode
 cFE Message Secondary Header APIs, [220](#)
 CFE_MSG_GetHasSecondaryHeader
 cFE Message Primary Header APIs, [207](#)
 CFE_MSG_GetHeaderVersion
 cFE Message Primary Header APIs, [207](#)
 CFE_MSG_GetMsgId
 cFE Message Id APIs, [225](#)
 CFE_MSG_GetMsgTime
 cFE Message Secondary Header APIs, [222](#)
 CFE_MSG_GetNextSequenceCount
 cFE Message Primary Header APIs, [207](#)
 CFE_MSG_GetPlaybackFlag
 cFE Message Extended Header APIs, [215](#)

- CFE_MSG_GetSegmentationFlag
 - cFE Message Primary Header APIs, [208](#)
- CFE_MSG_GetSequenceCount
 - cFE Message Primary Header APIs, [208](#)
- CFE_MSG_GetSize
 - cFE Message Primary Header APIs, [209](#)
- CFE_MSG_GetSubsystem
 - cFE Message Extended Header APIs, [216](#)
- CFE_MSG_GetSystem
 - cFE Message Extended Header APIs, [216](#)
- CFE_MSG_GetType
 - cFE Message Primary Header APIs, [209](#)
- CFE_MSG_GetTypeFromMsgId
 - cFE Message Id APIs, [225](#)
- CFE_MSG_HeaderVersion_t
 - cfe_msg_api_typedefs.h, [745](#)
- CFE_MSG_Init
 - cFE Generic Message APIs, [205](#)
- CFE_MSG_Message_t
 - cfe_msg_api_typedefs.h, [745](#)
- CFE_MSG_NOT_IMPLEMENTED
 - cfe_msg_api_typedefs.h, [744](#)
- CFE_MSG_OriginationAction
 - cFE Message Integrity APIs, [227](#)
- CFE_MSG_PlaybackFlag
 - cfe_msg_api_typedefs.h, [746](#)
- CFE_MSG_PlaybackFlag_t
 - cfe_msg_api_typedefs.h, [745](#)
- CFE_MSG_PlayFlag_Invalid
 - cfe_msg_api_typedefs.h, [746](#)
- CFE_MSG_PlayFlag_Original
 - cfe_msg_api_typedefs.h, [746](#)
- CFE_MSG_PlayFlag_Playback
 - cfe_msg_api_typedefs.h, [746](#)
- CFE_MSG_SegFlag_Continue
 - cfe_msg_api_typedefs.h, [747](#)
- CFE_MSG_SegFlag_First
 - cfe_msg_api_typedefs.h, [747](#)
- CFE_MSG_SegFlag_Invalid
 - cfe_msg_api_typedefs.h, [747](#)
- CFE_MSG_SegFlag_Last
 - cfe_msg_api_typedefs.h, [747](#)
- CFE_MSG_SegFlag_Unsegmented
 - cfe_msg_api_typedefs.h, [747](#)
- CFE_MSG_SegmentationFlag
 - cfe_msg_api_typedefs.h, [747](#)
- CFE_MSG_SegmentationFlag_t
 - cfe_msg_api_typedefs.h, [745](#)
- CFE_MSG_SequenceCount_t
 - cfe_msg_api_typedefs.h, [745](#)
- CFE_MSG_SetApld
 - cFE Message Primary Header APIs, [210](#)
- CFE_MSG_SetEDSVersion
 - cFE Message Extended Header APIs, [217](#)
- CFE_MSG_SetEndian
 - cFE Message Extended Header APIs, [217](#)
- CFE_MSG_SetFcnCode
 - cFE Message Secondary Header APIs, [222](#)
- CFE_MSG_SetHasSecondaryHeader
 - cFE Message Primary Header APIs, [210](#)
- CFE_MSG_SetHeaderVersion
 - cFE Message Primary Header APIs, [211](#)
- CFE_MSG_SetMsgId
 - cFE Message Id APIs, [226](#)
- CFE_MSG_SetMsgTime
 - cFE Message Secondary Header APIs, [223](#)
- CFE_MSG_SetPlaybackFlag
 - cFE Message Extended Header APIs, [218](#)
- CFE_MSG_SetSegmentationFlag
 - cFE Message Primary Header APIs, [211](#)
- CFE_MSG_SetSequenceCount
 - cFE Message Primary Header APIs, [212](#)
- CFE_MSG_SetSize
 - cFE Message Primary Header APIs, [212](#)
- CFE_MSG_SetSubsystem
 - cFE Message Extended Header APIs, [218](#)
- CFE_MSG_SetSystem
 - cFE Message Extended Header APIs, [219](#)
- CFE_MSG_SetType
 - cFE Message Primary Header APIs, [213](#)
- CFE_MSG_Size_t
 - cfe_msg_api_typedefs.h, [745](#)
- CFE_MSG_Subsystem_t
 - cfe_msg_api_typedefs.h, [746](#)
- CFE_MSG_System_t
 - cfe_msg_api_typedefs.h, [746](#)
- CFE_MSG_TelemetryHeader_t
 - cfe_msg_api_typedefs.h, [746](#)
- CFE_MSG_Type
 - cfe_msg_api_typedefs.h, [747](#)
- CFE_MSG_Type_Cmd
 - cfe_msg_api_typedefs.h, [747](#)
- CFE_MSG_Type_Invalid
 - cfe_msg_api_typedefs.h, [747](#)
- CFE_MSG_Type_t
 - cfe_msg_api_typedefs.h, [746](#)
- CFE_MSG_Type_TIm
 - cfe_msg_api_typedefs.h, [747](#)
- CFE_MSG_ValidateChecksum
 - cFE Message Secondary Header APIs, [224](#)
- CFE_MSG_VerificationAction
 - cFE Message Integrity APIs, [228](#)
- CFE_MSG_WRONG_MSG_TYPE
 - cfe_msg_api_typedefs.h, [744](#)
- cfe_perfid.h, [605](#)
 - CFE_MISSION_ES_MAIN_PERF_ID, [606](#)
 - CFE_MISSION_ES_PERF_EXIT_BIT, [606](#)
 - CFE_MISSION_EVS_MAIN_PERF_ID, [606](#)

- CFE_MISSION_SB_MAIN_PERF_ID, [606](#)
- CFE_MISSION_SB_MSG_LIM_PERF_ID, [606](#)
- CFE_MISSION_SB_PIPE_OFLOW_PERF_ID, [606](#)
- CFE_MISSION_TBL_MAIN_PERF_ID, [606](#)
- CFE_MISSION_TIME_LOCAL1HZISR_PERF_ID, [606](#)
- CFE_MISSION_TIME_LOCAL1HZTASK_PERF_ID, [606](#)
- CFE_MISSION_TIME_MAIN_PERF_ID, [607](#)
- CFE_MISSION_TIME_SENDMET_PERF_ID, [607](#)
- CFE_MISSION_TIME_TONE1HZISR_PERF_ID, [607](#)
- CFE_MISSION_TIME_TONE1HZTASK_PERF_ID, [607](#)
- CFE_PLATFORM_BASE_MIDVAL
 - default_cfe_core_api_base_msgid_values.h, [684](#)
- CFE_PLATFORM_CMD_TOPICID_TO_MIDV
 - default_cfe_core_api_msgid_mapping.h, [686](#)
- CFE_PLATFORM_CORE_MAX_STARTUP_MSEC
 - example_platform_cfg.h, [632](#)
- CFE_PLATFORM_ENDIAN
 - example_platform_cfg.h, [632](#)
- CFE_PLATFORM_ES_APP_KILL_TIMEOUT
 - cfe_es_internal_cfg.h, [881](#)
 - example_platform_cfg.h, [633](#)
- CFE_PLATFORM_ES_APP_SCAN_RATE
 - cfe_es_internal_cfg.h, [881](#)
 - example_platform_cfg.h, [633](#)
- CFE_PLATFORM_ES_CDS_MAX_BLOCK_SIZE
 - cfe_es_internal_cfg.h, [882](#)
 - example_platform_cfg.h, [634](#)
- CFE_PLATFORM_ES_CDS_MAX_NUM_ENTRIES
 - cfe_es_internal_cfg.h, [882](#)
 - example_platform_cfg.h, [634](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_01
 - cfe_es_internal_cfg.h, [882](#)
 - example_platform_cfg.h, [634](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_02
 - cfe_es_internal_cfg.h, [882](#)
 - example_platform_cfg.h, [634](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_03
 - cfe_es_internal_cfg.h, [883](#)
 - example_platform_cfg.h, [634](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_04
 - cfe_es_internal_cfg.h, [883](#)
 - example_platform_cfg.h, [635](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_05
 - cfe_es_internal_cfg.h, [883](#)
 - example_platform_cfg.h, [635](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_06
 - cfe_es_internal_cfg.h, [883](#)
 - example_platform_cfg.h, [635](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_07
 - cfe_es_internal_cfg.h, [883](#)
- example_platform_cfg.h, [635](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_08
 - cfe_es_internal_cfg.h, [883](#)
 - example_platform_cfg.h, [635](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_09
 - cfe_es_internal_cfg.h, [883](#)
 - example_platform_cfg.h, [635](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_10
 - cfe_es_internal_cfg.h, [883](#)
 - example_platform_cfg.h, [635](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_11
 - cfe_es_internal_cfg.h, [883](#)
 - example_platform_cfg.h, [635](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_12
 - cfe_es_internal_cfg.h, [883](#)
 - example_platform_cfg.h, [635](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_13
 - cfe_es_internal_cfg.h, [883](#)
 - example_platform_cfg.h, [635](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_14
 - cfe_es_internal_cfg.h, [884](#)
 - example_platform_cfg.h, [635](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_15
 - cfe_es_internal_cfg.h, [884](#)
 - example_platform_cfg.h, [636](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_16
 - cfe_es_internal_cfg.h, [884](#)
 - example_platform_cfg.h, [636](#)
- CFE_PLATFORM_ES_CDS_SIZE
 - cfe_es_internal_cfg.h, [884](#)
 - example_platform_cfg.h, [636](#)
- CFE_PLATFORM_ES_CFGVAL
 - default_cfe_es_internal_cfg_values.h, [799](#)
- CFE_PLATFORM_ES_CMD_MIDVAL
 - default_cfe_es_msgid_values.h, [810](#)
- CFE_PLATFORM_ES_DEFAULT_APP_LOG_FILE
 - cfe_es_internal_cfg.h, [884](#)
 - example_platform_cfg.h, [636](#)
- CFE_PLATFORM_ES_DEFAULT_CDS_REG_DUMP_FILE
 - cfe_es_internal_cfg.h, [884](#)
 - example_platform_cfg.h, [636](#)
- CFE_PLATFORM_ES_DEFAULT_ER_LOG_FILE
 - cfe_es_internal_cfg.h, [885](#)
 - example_platform_cfg.h, [637](#)
- CFE_PLATFORM_ES_DEFAULT_PERF_DUMP_FILENAME
 - cfe_es_internal_cfg.h, [885](#)
 - example_platform_cfg.h, [637](#)
- CFE_PLATFORM_ES_DEFAULT_POR_SYSLOG_MODE
 - cfe_es_internal_cfg.h, [885](#)
 - example_platform_cfg.h, [637](#)
- CFE_PLATFORM_ES_DEFAULT_PR_SYSLOG_MODE
 - cfe_es_internal_cfg.h, [886](#)
 - example_platform_cfg.h, [638](#)
- CFE_PLATFORM_ES_DEFAULT_STACK_SIZE

- [cfe_es_internal_cfg.h](#), 886
 - [example_platform_cfg.h](#), 638
- CFE_PLATFORM_ES_DEFAULT_SYSLOG_FILE
 - [cfe_es_internal_cfg.h](#), 886
 - [example_platform_cfg.h](#), 638
- CFE_PLATFORM_ES_DEFAULT_TASK_LOG_FILE
 - [cfe_es_internal_cfg.h](#), 887
 - [example_platform_cfg.h](#), 639
- CFE_PLATFORM_ES_ER_LOG_ENTRIES
 - [cfe_es_internal_cfg.h](#), 887
 - [example_platform_cfg.h](#), 639
- CFE_PLATFORM_ES_ER_LOG_MAX_CONTEXT_SIZE
 - [cfe_es_internal_cfg.h](#), 887
 - [example_platform_cfg.h](#), 639
- CFE_PLATFORM_ES_MAX_APPLICATIONS
 - [cfe_es_internal_cfg.h](#), 888
 - [example_platform_cfg.h](#), 640
- CFE_PLATFORM_ES_MAX_BLOCK_SIZE
 - [cfe_es_internal_cfg.h](#), 888
 - [example_platform_cfg.h](#), 640
- CFE_PLATFORM_ES_MAX_GEN_COUNTERS
 - [cfe_es_internal_cfg.h](#), 888
 - [example_platform_cfg.h](#), 640
- CFE_PLATFORM_ES_MAX_LIBRARIES
 - [cfe_es_internal_cfg.h](#), 888
 - [example_platform_cfg.h](#), 640
- CFE_PLATFORM_ES_MAX_MEMORY_POOLS
 - [cfe_es_internal_cfg.h](#), 889
 - [example_platform_cfg.h](#), 641
- CFE_PLATFORM_ES_MAX_PROCESSOR_RESETS
 - [cfe_es_internal_cfg.h](#), 889
 - [example_platform_cfg.h](#), 641
- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_01
 - [cfe_es_internal_cfg.h](#), 889
 - [example_platform_cfg.h](#), 641
- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_02
 - [cfe_es_internal_cfg.h](#), 890
 - [example_platform_cfg.h](#), 642
- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_03
 - [cfe_es_internal_cfg.h](#), 890
 - [example_platform_cfg.h](#), 642
- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_04
 - [cfe_es_internal_cfg.h](#), 890
 - [example_platform_cfg.h](#), 642
- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_05
 - [cfe_es_internal_cfg.h](#), 890
 - [example_platform_cfg.h](#), 642
- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_06
 - [cfe_es_internal_cfg.h](#), 890
 - [example_platform_cfg.h](#), 642
- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_07
 - [cfe_es_internal_cfg.h](#), 890
 - [example_platform_cfg.h](#), 642
- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_08
 - [cfe_es_internal_cfg.h](#), 890
 - [example_platform_cfg.h](#), 642
- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_09
 - [cfe_es_internal_cfg.h](#), 890
 - [example_platform_cfg.h](#), 642
- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_10
 - [cfe_es_internal_cfg.h](#), 891
 - [example_platform_cfg.h](#), 643
- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_11
 - [cfe_es_internal_cfg.h](#), 891
 - [example_platform_cfg.h](#), 643
- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_12
 - [cfe_es_internal_cfg.h](#), 891
 - [example_platform_cfg.h](#), 643
- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_13
 - [cfe_es_internal_cfg.h](#), 891
 - [example_platform_cfg.h](#), 643
- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_14
 - [cfe_es_internal_cfg.h](#), 891
 - [example_platform_cfg.h](#), 643
- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_15
 - [cfe_es_internal_cfg.h](#), 891
 - [example_platform_cfg.h](#), 643
- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_16
 - [cfe_es_internal_cfg.h](#), 891
 - [example_platform_cfg.h](#), 643
- CFE_PLATFORM_ES_MEMPOOL_ALIGN_SIZE_MIN
 - [cfe_es_internal_cfg.h](#), 891
 - [example_platform_cfg.h](#), 643
- CFE_PLATFORM_ES_NONVOL_DISK_MOUNT_STRING
 - [cfe_es_internal_cfg.h](#), 891
 - [example_platform_cfg.h](#), 643
- CFE_PLATFORM_ES_NONVOL_STARTUP_FILE
 - [cfe_es_internal_cfg.h](#), 892
 - [example_platform_cfg.h](#), 644
- CFE_PLATFORM_ES_PERF_CHILD_MS_DELAY
 - [cfe_es_internal_cfg.h](#), 892
 - [example_platform_cfg.h](#), 644
- CFE_PLATFORM_ES_PERF_CHILD_PRIORITY
 - [cfe_es_internal_cfg.h](#), 892
 - [example_platform_cfg.h](#), 644
- CFE_PLATFORM_ES_PERF_CHILD_STACK_SIZE
 - [cfe_es_internal_cfg.h](#), 893
 - [example_platform_cfg.h](#), 645
- CFE_PLATFORM_ES_PERF_DATA_BUFFER_SIZE
 - [cfe_es_internal_cfg.h](#), 893
 - [example_platform_cfg.h](#), 645
- CFE_PLATFORM_ES_PERF_ENTRIES_BTWN_DLYS
 - [cfe_es_internal_cfg.h](#), 893
 - [example_platform_cfg.h](#), 645
- CFE_PLATFORM_ES_PERF_FILTERMASK_ALL
 - [cfe_es_internal_cfg.h](#), 893
 - [example_platform_cfg.h](#), 645
- CFE_PLATFORM_ES_PERF_FILTERMASK_INIT

- cfe_es_internal_cfg.h, [894](#)
- example_platform_cfg.h, [646](#)
- CFE_PLATFORM_ES_PERF_FILTERMASK_NONE
 - cfe_es_internal_cfg.h, [894](#)
 - example_platform_cfg.h, [646](#)
- CFE_PLATFORM_ES_PERF_TRIGMASK_ALL
 - cfe_es_internal_cfg.h, [894](#)
 - example_platform_cfg.h, [646](#)
- CFE_PLATFORM_ES_PERF_TRIGMASK_INIT
 - cfe_es_internal_cfg.h, [894](#)
 - example_platform_cfg.h, [646](#)
- CFE_PLATFORM_ES_PERF_TRIGMASK_NONE
 - cfe_es_internal_cfg.h, [894](#)
 - example_platform_cfg.h, [646](#)
- CFE_PLATFORM_ES_POOL_MAX_BUCKETS
 - cfe_es_internal_cfg.h, [895](#)
 - example_platform_cfg.h, [647](#)
- CFE_PLATFORM_ES_RAM_DISK_MOUNT_STRING
 - cfe_es_internal_cfg.h, [895](#)
 - example_platform_cfg.h, [647](#)
- CFE_PLATFORM_ES_RAM_DISK_NUM_SECTORS
 - cfe_es_internal_cfg.h, [895](#)
 - example_platform_cfg.h, [647](#)
- CFE_PLATFORM_ES_RAM_DISK_PERCENT_RESERVED
 - cfe_es_internal_cfg.h, [896](#)
 - example_platform_cfg.h, [648](#)
- CFE_PLATFORM_ES_RAM_DISK_SECTOR_SIZE
 - cfe_es_internal_cfg.h, [896](#)
 - example_platform_cfg.h, [648](#)
- CFE_PLATFORM_ES_START_TASK_PRIORITY
 - cfe_es_internal_cfg.h, [896](#)
 - example_platform_cfg.h, [648](#)
- CFE_PLATFORM_ES_START_TASK_STACK_SIZE
 - cfe_es_internal_cfg.h, [897](#)
 - example_platform_cfg.h, [649](#)
- CFE_PLATFORM_ES_STARTUP_SCRIPT_TIMEOUT_MSEC
 - cfe_es_internal_cfg.h, [897](#)
 - example_platform_cfg.h, [649](#)
- CFE_PLATFORM_ES_STARTUP_SYNC_POLL_MSEC
 - cfe_es_internal_cfg.h, [897](#)
 - example_platform_cfg.h, [649](#)
- CFE_PLATFORM_ES_SYSTEM_LOG_SIZE
 - cfe_es_internal_cfg.h, [898](#)
 - example_platform_cfg.h, [650](#)
- CFE_PLATFORM_ES_TLM_MIDVAL
 - default_cfe_es_msgid_values.h, [810](#)
- CFE_PLATFORM_ES_USER_RESERVED_SIZE
 - cfe_es_internal_cfg.h, [898](#)
 - example_platform_cfg.h, [650](#)
- CFE_PLATFORM_ES_VOLATILE_STARTUP_FILE
 - cfe_es_internal_cfg.h, [898](#)
 - example_platform_cfg.h, [650](#)
- CFE_PLATFORM_EVS_APP_EVENTS_PER_SEC
 - cfe_evs_internal_cfg.h, [971](#)
- example_platform_cfg.h, [651](#)
- CFE_PLATFORM_EVS_CFGVAL
 - default_cfe_evs_internal_cfg_values.h, [919](#)
- CFE_PLATFORM_EVS_CMD_MIDVAL
 - default_cfe_evs_msgid_values.h, [927](#)
- CFE_PLATFORM_EVS_DEFAULT_APP_DATA_FILE
 - cfe_evs_internal_cfg.h, [971](#)
 - example_platform_cfg.h, [651](#)
- CFE_PLATFORM_EVS_DEFAULT_LOG_FILE
 - cfe_evs_internal_cfg.h, [971](#)
 - example_platform_cfg.h, [651](#)
- CFE_PLATFORM_EVS_DEFAULT_LOG_MODE
 - cfe_evs_internal_cfg.h, [972](#)
 - example_platform_cfg.h, [652](#)
- CFE_PLATFORM_EVS_DEFAULT_MSG_FORMAT_MODE
 - cfe_evs_internal_cfg.h, [972](#)
 - example_platform_cfg.h, [652](#)
- CFE_PLATFORM_EVS_DEFAULT_TYPE_FLAG
 - cfe_evs_internal_cfg.h, [972](#)
 - example_platform_cfg.h, [652](#)
- CFE_PLATFORM_EVS_LOG_MAX
 - cfe_evs_internal_cfg.h, [973](#)
 - example_platform_cfg.h, [653](#)
- CFE_PLATFORM_EVS_MAX_APP_EVENT_BURST
 - cfe_evs_internal_cfg.h, [973](#)
 - example_platform_cfg.h, [653](#)
- CFE_PLATFORM_EVS_MAX_EVENT_FILTERS
 - cfe_evs_internal_cfg.h, [973](#)
 - example_platform_cfg.h, [653](#)
- CFE_PLATFORM_EVS_PORT_DEFAULT
 - cfe_evs_internal_cfg.h, [974](#)
 - example_platform_cfg.h, [654](#)
- CFE_PLATFORM_EVS_START_TASK_PRIORITY
 - cfe_evs_internal_cfg.h, [974](#)
 - example_platform_cfg.h, [654](#)
- CFE_PLATFORM_EVS_START_TASK_STACK_SIZE
 - cfe_evs_internal_cfg.h, [974](#)
 - example_platform_cfg.h, [654](#)
- CFE_PLATFORM_EVS_TLM_MIDVAL
 - default_cfe_evs_msgid_values.h, [927](#)
- CFE_PLATFORM_SB_BUF_MEMORY_BYTES
 - cfe_sb_internal_cfg.h, [1048](#)
 - example_platform_cfg.h, [655](#)
- CFE_PLATFORM_SB_CFGVAL
 - default_cfe_sb_internal_cfg_values.h, [996](#)
- CFE_PLATFORM_SB_CMD_MIDVAL
 - default_cfe_sb_msgid_values.h, [1003](#)
- CFE_PLATFORM_SB_DEFAULT_MAP_FILENAME
 - cfe_sb_internal_cfg.h, [1048](#)
 - example_platform_cfg.h, [655](#)
- CFE_PLATFORM_SB_DEFAULT_MSG_LIMIT
 - cfe_sb_internal_cfg.h, [1048](#)
 - example_platform_cfg.h, [655](#)
- CFE_PLATFORM_SB_DEFAULT_PIPE_FILENAME

- cfe_sb_internal_cfg.h, [1048](#)
- example_platform_cfg.h, [656](#)
- CFE_PLATFORM_SB_DEFAULT_ROUTING_FILENAME
 - cfe_sb_internal_cfg.h, [1049](#)
 - example_platform_cfg.h, [656](#)
- CFE_PLATFORM_SB_FILTER_MASK1
 - cfe_sb_internal_cfg.h, [1049](#)
 - example_platform_cfg.h, [656](#)
- CFE_PLATFORM_SB_FILTER_MASK2
 - cfe_sb_internal_cfg.h, [1049](#)
 - example_platform_cfg.h, [656](#)
- CFE_PLATFORM_SB_FILTER_MASK3
 - cfe_sb_internal_cfg.h, [1049](#)
 - example_platform_cfg.h, [657](#)
- CFE_PLATFORM_SB_FILTER_MASK4
 - cfe_sb_internal_cfg.h, [1049](#)
 - example_platform_cfg.h, [657](#)
- CFE_PLATFORM_SB_FILTER_MASK5
 - cfe_sb_internal_cfg.h, [1049](#)
 - example_platform_cfg.h, [657](#)
- CFE_PLATFORM_SB_FILTER_MASK6
 - cfe_sb_internal_cfg.h, [1049](#)
 - example_platform_cfg.h, [657](#)
- CFE_PLATFORM_SB_FILTER_MASK7
 - cfe_sb_internal_cfg.h, [1050](#)
 - example_platform_cfg.h, [657](#)
- CFE_PLATFORM_SB_FILTER_MASK8
 - cfe_sb_internal_cfg.h, [1050](#)
 - example_platform_cfg.h, [657](#)
- CFE_PLATFORM_SB_FILTERED_EVENT1
 - cfe_sb_internal_cfg.h, [1050](#)
 - example_platform_cfg.h, [657](#)
- CFE_PLATFORM_SB_FILTERED_EVENT2
 - cfe_sb_internal_cfg.h, [1050](#)
 - example_platform_cfg.h, [657](#)
- CFE_PLATFORM_SB_FILTERED_EVENT3
 - cfe_sb_internal_cfg.h, [1050](#)
 - example_platform_cfg.h, [657](#)
- CFE_PLATFORM_SB_FILTERED_EVENT4
 - cfe_sb_internal_cfg.h, [1050](#)
 - example_platform_cfg.h, [658](#)
- CFE_PLATFORM_SB_FILTERED_EVENT5
 - cfe_sb_internal_cfg.h, [1050](#)
 - example_platform_cfg.h, [658](#)
- CFE_PLATFORM_SB_FILTERED_EVENT6
 - cfe_sb_internal_cfg.h, [1050](#)
 - example_platform_cfg.h, [658](#)
- CFE_PLATFORM_SB_FILTERED_EVENT7
 - cfe_sb_internal_cfg.h, [1050](#)
 - example_platform_cfg.h, [658](#)
- CFE_PLATFORM_SB_FILTERED_EVENT8
 - cfe_sb_internal_cfg.h, [1051](#)
 - example_platform_cfg.h, [658](#)
- CFE_PLATFORM_SB_HIGHEST_VALID_MSGID
 - cfe_sb_internal_cfg.h, [1051](#)
 - example_platform_cfg.h, [658](#)
- CFE_PLATFORM_SB_MAX_BLOCK_SIZE
 - cfe_sb_internal_cfg.h, [1051](#)
 - example_platform_cfg.h, [658](#)
- CFE_PLATFORM_SB_MAX_DEST_PER_PKT
 - cfe_sb_internal_cfg.h, [1051](#)
 - example_platform_cfg.h, [658](#)
- CFE_PLATFORM_SB_MAX_MSG_IDS
 - cfe_sb_internal_cfg.h, [1051](#)
 - example_platform_cfg.h, [659](#)
- CFE_PLATFORM_SB_MAX_PIPES
 - cfe_sb_internal_cfg.h, [1052](#)
 - example_platform_cfg.h, [659](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_01
 - cfe_sb_internal_cfg.h, [1052](#)
 - example_platform_cfg.h, [659](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_02
 - cfe_sb_internal_cfg.h, [1052](#)
 - example_platform_cfg.h, [660](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_03
 - cfe_sb_internal_cfg.h, [1053](#)
 - example_platform_cfg.h, [660](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_04
 - cfe_sb_internal_cfg.h, [1053](#)
 - example_platform_cfg.h, [660](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_05
 - cfe_sb_internal_cfg.h, [1053](#)
 - example_platform_cfg.h, [660](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_06
 - cfe_sb_internal_cfg.h, [1053](#)
 - example_platform_cfg.h, [660](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_07
 - cfe_sb_internal_cfg.h, [1053](#)
 - example_platform_cfg.h, [660](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_08
 - cfe_sb_internal_cfg.h, [1053](#)
 - example_platform_cfg.h, [660](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_09
 - cfe_sb_internal_cfg.h, [1053](#)
 - example_platform_cfg.h, [660](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_10
 - cfe_sb_internal_cfg.h, [1053](#)
 - example_platform_cfg.h, [660](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_11
 - cfe_sb_internal_cfg.h, [1053](#)
 - example_platform_cfg.h, [661](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_12
 - cfe_sb_internal_cfg.h, [1053](#)
 - example_platform_cfg.h, [661](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_13
 - cfe_sb_internal_cfg.h, [1053](#)
 - example_platform_cfg.h, [661](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_14

- cfe_sb_internal_cfg.h, [1054](#)
- example_platform_cfg.h, [661](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_15
 - cfe_sb_internal_cfg.h, [1054](#)
 - example_platform_cfg.h, [661](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_16
 - cfe_sb_internal_cfg.h, [1054](#)
 - example_platform_cfg.h, [661](#)
- CFE_PLATFORM_SB_POOL_MAX_BUCKETS
 - cfe_sb_internal_cfg.h, [1054](#)
- CFE_PLATFORM_SB_START_TASK_PRIORITY
 - cfe_sb_internal_cfg.h, [1054](#)
 - example_platform_cfg.h, [661](#)
- CFE_PLATFORM_SB_START_TASK_STACK_SIZE
 - cfe_sb_internal_cfg.h, [1054](#)
 - example_platform_cfg.h, [661](#)
- CFE_PLATFORM_SB_TLM_MIDVAL
 - default_cfe_sb_msgid_values.h, [1003](#)
- CFE_PLATFORM_TBL_BUF_MEMORY_BYTES
 - cfe_tbl_internal_cfg.h, [1129](#)
 - default_cfe_tbl_platform_cfg.h, [1084](#)
 - example_platform_cfg.h, [662](#)
- CFE_PLATFORM_TBL_CFGVAL
 - default_cfe_tbl_internal_cfg_values.h, [1069](#)
- CFE_PLATFORM_TBL_CMD_MIDVAL
 - default_cfe_tbl_msgid_values.h, [1077](#)
- CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE
 - cfe_tbl_internal_cfg.h, [1129](#)
 - default_cfe_tbl_platform_cfg.h, [1084](#)
 - example_platform_cfg.h, [662](#)
- CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES
 - cfe_tbl_internal_cfg.h, [1129](#)
 - default_cfe_tbl_platform_cfg.h, [1084](#)
 - example_platform_cfg.h, [662](#)
- CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE
 - cfe_tbl_internal_cfg.h, [1129](#)
 - default_cfe_tbl_platform_cfg.h, [1085](#)
 - example_platform_cfg.h, [662](#)
- CFE_PLATFORM_TBL_MAX_NUM_HANDLES
 - cfe_tbl_internal_cfg.h, [1130](#)
 - default_cfe_tbl_platform_cfg.h, [1085](#)
 - example_platform_cfg.h, [663](#)
- CFE_PLATFORM_TBL_MAX_NUM_TABLES
 - cfe_tbl_internal_cfg.h, [1130](#)
 - default_cfe_tbl_platform_cfg.h, [1085](#)
 - example_platform_cfg.h, [663](#)
- CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS
 - cfe_tbl_internal_cfg.h, [1130](#)
 - default_cfe_tbl_platform_cfg.h, [1086](#)
 - example_platform_cfg.h, [663](#)
- CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS
 - cfe_tbl_internal_cfg.h, [1131](#)
 - default_cfe_tbl_platform_cfg.h, [1086](#)
 - example_platform_cfg.h, [664](#)
- CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE
 - cfe_tbl_internal_cfg.h, [1131](#)
 - default_cfe_tbl_platform_cfg.h, [1086](#)
 - example_platform_cfg.h, [664](#)
- CFE_PLATFORM_TBL_START_TASK_PRIORITY
 - cfe_tbl_internal_cfg.h, [1131](#)
 - default_cfe_tbl_platform_cfg.h, [1087](#)
 - example_platform_cfg.h, [664](#)
- CFE_PLATFORM_TBL_START_TASK_STACK_SIZE
 - cfe_tbl_internal_cfg.h, [1132](#)
 - default_cfe_tbl_platform_cfg.h, [1087](#)
 - example_platform_cfg.h, [665](#)
- CFE_PLATFORM_TBL_TLM_MIDVAL
 - default_cfe_tbl_msgid_values.h, [1077](#)
- CFE_PLATFORM_TBL_U32FROM4CHARS
 - cfe_tbl_internal_cfg.h, [1132](#)
 - default_cfe_tbl_platform_cfg.h, [1087](#)
 - example_platform_cfg.h, [665](#)
- CFE_PLATFORM_TBL_VALID_PRID_1
 - cfe_tbl_internal_cfg.h, [1132](#)
 - default_cfe_tbl_platform_cfg.h, [1088](#)
 - example_platform_cfg.h, [665](#)
- CFE_PLATFORM_TBL_VALID_PRID_2
 - cfe_tbl_internal_cfg.h, [1132](#)
 - default_cfe_tbl_platform_cfg.h, [1088](#)
 - example_platform_cfg.h, [665](#)
- CFE_PLATFORM_TBL_VALID_PRID_3
 - cfe_tbl_internal_cfg.h, [1133](#)
 - default_cfe_tbl_platform_cfg.h, [1088](#)
 - example_platform_cfg.h, [666](#)
- CFE_PLATFORM_TBL_VALID_PRID_4
 - cfe_tbl_internal_cfg.h, [1133](#)
 - default_cfe_tbl_platform_cfg.h, [1088](#)
 - example_platform_cfg.h, [666](#)
- CFE_PLATFORM_TBL_VALID_PRID_COUNT
 - cfe_tbl_internal_cfg.h, [1133](#)
 - default_cfe_tbl_platform_cfg.h, [1088](#)
 - example_platform_cfg.h, [666](#)
- CFE_PLATFORM_TBL_VALID_SCID_1
 - cfe_tbl_internal_cfg.h, [1133](#)
 - default_cfe_tbl_platform_cfg.h, [1088](#)
 - example_platform_cfg.h, [666](#)
- CFE_PLATFORM_TBL_VALID_SCID_2
 - cfe_tbl_internal_cfg.h, [1133](#)
 - default_cfe_tbl_platform_cfg.h, [1089](#)
 - example_platform_cfg.h, [666](#)
- CFE_PLATFORM_TBL_VALID_SCID_COUNT
 - cfe_tbl_internal_cfg.h, [1133](#)
 - default_cfe_tbl_platform_cfg.h, [1089](#)
 - example_platform_cfg.h, [666](#)
- CFE_PLATFORM_TIME_CFG_CLIENT
 - cfe_time_internal_cfg.h, [1203](#)
 - example_platform_cfg.h, [667](#)
- CFE_PLATFORM_TIME_CFG_LATCH_FLY

- [cfe_time_internal_cfg.h](#), [1203](#)
 - [example_platform_cfg.h](#), [667](#)
- CFE_PLATFORM_TIME_CFG_SERVER
 - [cfe_time_internal_cfg.h](#), [1203](#)
 - [example_platform_cfg.h](#), [667](#)
- CFE_PLATFORM_TIME_CFG_SIGNAL
 - [cfe_time_internal_cfg.h](#), [1204](#)
 - [example_platform_cfg.h](#), [667](#)
- CFE_PLATFORM_TIME_CFG_SOURCE
 - [cfe_time_internal_cfg.h](#), [1204](#)
 - [example_platform_cfg.h](#), [668](#)
- CFE_PLATFORM_TIME_CFG_SRC_GPS
 - [cfe_time_internal_cfg.h](#), [1204](#)
 - [example_platform_cfg.h](#), [668](#)
- CFE_PLATFORM_TIME_CFG_SRC_MET
 - [cfe_time_internal_cfg.h](#), [1205](#)
 - [example_platform_cfg.h](#), [668](#)
- CFE_PLATFORM_TIME_CFG_SRC_TIME
 - [cfe_time_internal_cfg.h](#), [1205](#)
 - [example_platform_cfg.h](#), [669](#)
- CFE_PLATFORM_TIME_CFG_START_FLY
 - [cfe_time_internal_cfg.h](#), [1205](#)
 - [example_platform_cfg.h](#), [669](#)
- CFE_PLATFORM_TIME_CFG_TONE_LIMIT
 - [cfe_time_internal_cfg.h](#), [1205](#)
 - [example_platform_cfg.h](#), [669](#)
- CFE_PLATFORM_TIME_CFG_VIRTUAL
 - [cfe_time_internal_cfg.h](#), [1205](#)
 - [example_platform_cfg.h](#), [669](#)
- CFE_PLATFORM_TIME_CFGVAL
 - [default_cfe_time_internal_cfg_values.h](#), [1148](#)
- CFE_PLATFORM_TIME_CMD_MIDVAL
 - [default_cfe_time_msgid_values.h](#), [1156](#)
- CFE_PLATFORM_TIME_GLBCMD_MIDVAL
 - [default_cfe_time_msgid_values.h](#), [1156](#)
- CFE_PLATFORM_TIME_MAX_DELTA_SECS
 - [cfe_time_internal_cfg.h](#), [1206](#)
 - [example_platform_cfg.h](#), [670](#)
- CFE_PLATFORM_TIME_MAX_DELTA_SUBS
 - [cfe_time_internal_cfg.h](#), [1206](#)
 - [example_platform_cfg.h](#), [670](#)
- CFE_PLATFORM_TIME_MAX_LOCAL_SECS
 - [cfe_time_internal_cfg.h](#), [1206](#)
 - [example_platform_cfg.h](#), [670](#)
- CFE_PLATFORM_TIME_MAX_LOCAL_SUBS
 - [cfe_time_internal_cfg.h](#), [1206](#)
 - [example_platform_cfg.h](#), [670](#)
- CFE_PLATFORM_TIME_ONEHZ_TASK_PRIORITY
 - [cfe_time_internal_cfg.h](#), [1207](#)
 - [example_platform_cfg.h](#), [671](#)
- CFE_PLATFORM_TIME_ONEHZ_TASK_STACK_SIZE
 - [cfe_time_internal_cfg.h](#), [1207](#)
 - [example_platform_cfg.h](#), [671](#)
- CFE_PLATFORM_TIME_START_TASK_PRIORITY
 - [cfe_time_internal_cfg.h](#), [1207](#)
 - [example_platform_cfg.h](#), [671](#)
- CFE_PLATFORM_TIME_START_TASK_STACK_SIZE
 - [cfe_time_internal_cfg.h](#), [1207](#)
 - [example_platform_cfg.h](#), [671](#)
- CFE_PLATFORM_TIME_TLM_MIDVAL
 - [default_cfe_time_msgid_values.h](#), [1156](#)
- CFE_PLATFORM_TIME_TONE_TASK_PRIORITY
 - [cfe_time_internal_cfg.h](#), [1207](#)
 - [example_platform_cfg.h](#), [671](#)
- CFE_PLATFORM_TIME_TONE_TASK_STACK_SIZE
 - [cfe_time_internal_cfg.h](#), [1207](#)
 - [example_platform_cfg.h](#), [671](#)
- CFE_PLATFORM_TLM_TOPICID_TO_MIDV
 - [default_cfe_core_api_msgid_mapping.h](#), [686](#)
- [cfe_psp.h](#)
 - [CFE_PSP_Main](#), [1289](#)
 - [PSP_DEBUG](#), [1288](#)
 - [PSP_DEBUG_LEV](#), [1289](#)
- [CFE_PSP_AttachExceptions](#)
 - [cfe_psp_exception_api.h](#), [1304](#)
- [CFE_PSP_BE16toH](#)
 - [cfe_psp_endian.h](#), [1298](#)
- [CFE_PSP_BE32toH](#)
 - [cfe_psp_endian.h](#), [1298](#)
- [CFE_PSP_BE64toH](#)
 - [cfe_psp_endian.h](#), [1298](#)
- [cfe_psp_cache_api.h](#)
 - [CFE_PSP_FlushCaches](#), [1290](#)
- [cfe_psp_cds_api.h](#)
 - [CFE_PSP_GetCDSSize](#), [1292](#)
 - [CFE_PSP_ReadFromCDS](#), [1292](#)
 - [CFE_PSP_WriteToCDS](#), [1292](#)
- [cfe_psp_eepromaccess_api.h](#)
 - [CFE_PSP_EepromPowerDown](#), [1294](#)
 - [CFE_PSP_EepromPowerUp](#), [1294](#)
 - [CFE_PSP_EepromWrite16](#), [1295](#)
 - [CFE_PSP_EepromWrite32](#), [1295](#)
 - [CFE_PSP_EepromWrite8](#), [1295](#)
 - [CFE_PSP_EepromWriteDisable](#), [1296](#)
 - [CFE_PSP_EepromWriteEnable](#), [1296](#)
- [CFE_PSP_EepromPowerDown](#)
 - [cfe_psp_eepromaccess_api.h](#), [1294](#)
- [CFE_PSP_EepromPowerUp](#)
 - [cfe_psp_eepromaccess_api.h](#), [1294](#)
- [CFE_PSP_EepromWrite16](#)
 - [cfe_psp_eepromaccess_api.h](#), [1295](#)
- [CFE_PSP_EepromWrite32](#)
 - [cfe_psp_eepromaccess_api.h](#), [1295](#)
- [CFE_PSP_EepromWrite8](#)
 - [cfe_psp_eepromaccess_api.h](#), [1295](#)
- [CFE_PSP_EepromWriteDisable](#)
 - [cfe_psp_eepromaccess_api.h](#), [1296](#)
- [CFE_PSP_EepromWriteEnable](#)

- cfe_psp_eepromaccess_api.h, [1296](#)
- cfe_psp_endian.h
 - CFE_PSP_BE16toH, [1298](#)
 - CFE_PSP_BE32toH, [1298](#)
 - CFE_PSP_BE64toH, [1298](#)
 - CFE_PSP_HtoBE16, [1298](#)
 - CFE_PSP_HtoBE32, [1298](#)
 - CFE_PSP_HtoBE64, [1298](#)
 - CFE_PSP_HtoLE16, [1298](#)
 - CFE_PSP_HtoLE32, [1299](#)
 - CFE_PSP_HtoLE64, [1299](#)
 - CFE_PSP_IsBigEndian, [1299](#)
 - CFE_PSP_IsLittleEndian, [1299](#)
 - CFE_PSP_LE16toH, [1299](#)
 - CFE_PSP_LE32toH, [1299](#)
 - CFE_PSP_LE64toH, [1299](#)
- CFE_PSP_ERROR
 - cfe_psp_error.h, [1301](#)
- cfe_psp_error.h
 - CFE_PSP_ERROR, [1301](#)
 - CFE_PSP_ERROR_ADDRESS_MISALIGNED, [1301](#)
 - CFE_PSP_ERROR_NOT_IMPLEMENTED, [1301](#)
 - CFE_PSP_ERROR_TIMEOUT, [1301](#)
 - CFE_PSP_INVALID_INT_NUM, [1301](#)
 - CFE_PSP_INVALID_MEM_ADDR, [1301](#)
 - CFE_PSP_INVALID_MEM_ATTR, [1301](#)
 - CFE_PSP_INVALID_MEM_RANGE, [1301](#)
 - CFE_PSP_INVALID_MEM_SIZE, [1301](#)
 - CFE_PSP_INVALID_MEM_TYPE, [1301](#)
 - CFE_PSP_INVALID_MEM_WORDSIZE, [1302](#)
 - CFE_PSP_INVALID_MODULE_ID, [1302](#)
 - CFE_PSP_INVALID_MODULE_NAME, [1302](#)
 - CFE_PSP_INVALID_POINTER, [1302](#)
 - CFE_PSP_NO_EXCEPTION_DATA, [1302](#)
 - CFE_PSP_STATUS_C, [1302](#)
 - CFE_PSP_STATUS_STRING_LENGTH, [1302](#)
 - CFE_PSP_Status_t, [1302](#)
 - CFE_PSP_StatusString_t, [1302](#)
 - CFE_PSP_StatusToString, [1303](#)
 - CFE_PSP_SUCCESS, [1302](#)
- CFE_PSP_ERROR_ADDRESS_MISALIGNED
 - cfe_psp_error.h, [1301](#)
- CFE_PSP_ERROR_NOT_IMPLEMENTED
 - cfe_psp_error.h, [1301](#)
- CFE_PSP_ERROR_TIMEOUT
 - cfe_psp_error.h, [1301](#)
- cfe_psp_exception_api.h
 - CFE_PSP_AttachExceptions, [1304](#)
 - CFE_PSP_Exception_CopyContext, [1304](#)
 - CFE_PSP_Exception_GetCount, [1305](#)
 - CFE_PSP_Exception_GetSummary, [1305](#)
 - CFE_PSP_SetDefaultExceptionEnvironment, [1305](#)
- CFE_PSP_Exception_CopyContext
 - cfe_psp_exception_api.h, [1304](#)
- CFE_PSP_Exception_GetCount
 - cfe_psp_exception_api.h, [1305](#)
- CFE_PSP_Exception_GetSummary
 - cfe_psp_exception_api.h, [1305](#)
- CFE_PSP_FlushCaches
 - cfe_psp_cache_api.h, [1290](#)
- CFE_PSP_Get_Timebase
 - cfe_psp_timertick_api.h, [1326](#)
- CFE_PSP_GetBuildNumber
 - cfe_psp_version_api.h, [1329](#)
- CFE_PSP_GetCDSSize
 - cfe_psp_cds_api.h, [1292](#)
- CFE_PSP_GetCFTextSegmentInfo
 - cfe_psp_memrange_api.h, [1315](#)
- CFE_PSP_GetKernelTextSegmentInfo
 - cfe_psp_memrange_api.h, [1315](#)
- CFE_PSP_GetProcessorId
 - cfe_psp_id_api.h, [1307](#)
- CFE_PSP_GetProcessorName
 - cfe_psp_id_api.h, [1307](#)
- CFE_PSP_GetResetArea
 - cfe_psp_memrange_api.h, [1315](#)
- CFE_PSP_GetRestartType
 - cfe_psp_watchdog_api.h, [1334](#)
- CFE_PSP_GetSpacecraftId
 - cfe_psp_id_api.h, [1307](#)
- CFE_PSP_GetTime
 - cfe_psp_timertick_api.h, [1326](#)
- CFE_PSP_GetTimerLow32Rollover
 - cfe_psp_timertick_api.h, [1327](#)
- CFE_PSP_GetTimerTicksPerSecond
 - cfe_psp_timertick_api.h, [1327](#)
- CFE_PSP_GetUserReservedArea
 - cfe_psp_memrange_api.h, [1316](#)
- CFE_PSP_GetVersionCodeName
 - cfe_psp_version_api.h, [1329](#)
- CFE_PSP_GetVersionNumber
 - cfe_psp_version_api.h, [1329](#)
- CFE_PSP_GetVersionString
 - cfe_psp_version_api.h, [1329](#)
- CFE_PSP_GetVolatileDiskMem
 - cfe_psp_memrange_api.h, [1316](#)
- CFE_PSP_HtoBE16
 - cfe_psp_endian.h, [1298](#)
- CFE_PSP_HtoBE32
 - cfe_psp_endian.h, [1298](#)
- CFE_PSP_HtoBE64
 - cfe_psp_endian.h, [1298](#)
- CFE_PSP_HtoLE16
 - cfe_psp_endian.h, [1298](#)
- CFE_PSP_HtoLE32
 - cfe_psp_endian.h, [1299](#)
- CFE_PSP_HtoLE64

`cfe_psp_endian.h`, 1299
`cfe_psp_id_api.h`
 `CFE_PSP_GetProcessorId`, 1307
 `CFE_PSP_GetProcessorName`, 1307
 `CFE_PSP_GetSpacecraftId`, 1307
`CFE_PSP_InitSSR`
 `cfe_psp_ssr_api.h`, 1324
`CFE_PSP_INVALID_INT_NUM`
 `cfe_psp_error.h`, 1301
`CFE_PSP_INVALID_MEM_ADDR`
 `cfe_psp_error.h`, 1301
`CFE_PSP_INVALID_MEM_ATTR`
 `cfe_psp_error.h`, 1301
`CFE_PSP_INVALID_MEM_RANGE`
 `cfe_psp_error.h`, 1301
`CFE_PSP_INVALID_MEM_SIZE`
 `cfe_psp_error.h`, 1301
`CFE_PSP_INVALID_MEM_TYPE`
 `cfe_psp_error.h`, 1301
`CFE_PSP_INVALID_MEM_WORDSIZE`
 `cfe_psp_error.h`, 1302
`CFE_PSP_INVALID_MODULE_ID`
 `cfe_psp_error.h`, 1302
`CFE_PSP_INVALID_MODULE_NAME`
 `cfe_psp_error.h`, 1302
`CFE_PSP_INVALID_POINTER`
 `cfe_psp_error.h`, 1302
`CFE_PSP_IsBigEndian`
 `cfe_psp_endian.h`, 1299
`CFE_PSP_IsLittleEndian`
 `cfe_psp_endian.h`, 1299
`CFE_PSP_LE16toH`
 `cfe_psp_endian.h`, 1299
`CFE_PSP_LE32toH`
 `cfe_psp_endian.h`, 1299
`CFE_PSP_LE64toH`
 `cfe_psp_endian.h`, 1299
`CFE_PSP_Main`
 `cfe_psp.h`, 1289
`CFE_PSP_MEM_ANY`
 `cfe_psp_memrange_api.h`, 1314
`CFE_PSP_MEM_ATTR_READ`
 `cfe_psp_memrange_api.h`, 1314
`CFE_PSP_MEM_ATTR_READWRITE`
 `cfe_psp_memrange_api.h`, 1314
`CFE_PSP_MEM_ATTR_WRITE`
 `cfe_psp_memrange_api.h`, 1314
`CFE_PSP_MEM_EEPROM`
 `cfe_psp_memrange_api.h`, 1314
`CFE_PSP_MEM_INVALID`
 `cfe_psp_memrange_api.h`, 1314
`CFE_PSP_MEM_RAM`
 `cfe_psp_memrange_api.h`, 1314
`CFE_PSP_MEM_SIZE_BYTE`
 `cfe_psp_memrange_api.h`, 1314
 `cfe_psp_memrange_api.h`, 1314
`CFE_PSP_MEM_SIZE_DWORD`
 `cfe_psp_memrange_api.h`, 1315
`CFE_PSP_MEM_SIZE_WORD`
 `cfe_psp_memrange_api.h`, 1315
`cfe_psp_memaccess_api.h`
 `CFE_PSP_MemCpy`, 1309
 `CFE_PSP_MemRead16`, 1309
 `CFE_PSP_MemRead32`, 1309
 `CFE_PSP_MemRead8`, 1310
 `CFE_PSP_MemSet`, 1310
 `CFE_PSP_MemWrite16`, 1311
 `CFE_PSP_MemWrite32`, 1311
 `CFE_PSP_MemWrite8`, 1312
`CFE_PSP_MemCpy`
 `cfe_psp_memaccess_api.h`, 1309
`cfe_psp_memrange_api.h`
 `CFE_PSP_GetCFETextSegmentInfo`, 1315
 `CFE_PSP_GetKernelTextSegmentInfo`, 1315
 `CFE_PSP_GetResetArea`, 1315
 `CFE_PSP_GetUserReservedArea`, 1316
 `CFE_PSP_GetVolatileDiskMem`, 1316
 `CFE_PSP_MEM_ANY`, 1314
 `CFE_PSP_MEM_ATTR_READ`, 1314
 `CFE_PSP_MEM_ATTR_READWRITE`, 1314
 `CFE_PSP_MEM_ATTR_WRITE`, 1314
 `CFE_PSP_MEM_EEPROM`, 1314
 `CFE_PSP_MEM_INVALID`, 1314
 `CFE_PSP_MEM_RAM`, 1314
 `CFE_PSP_MEM_SIZE_BYTE`, 1314
 `CFE_PSP_MEM_SIZE_DWORD`, 1315
 `CFE_PSP_MEM_SIZE_WORD`, 1315
 `CFE_PSP_MemRangeGet`, 1316
 `CFE_PSP_MemRanges`, 1317
 `CFE_PSP_MemRangeSet`, 1317
 `CFE_PSP_MemValidateRange`, 1318
`CFE_PSP_MemRangeGet`
 `cfe_psp_memrange_api.h`, 1316
`CFE_PSP_MemRanges`
 `cfe_psp_memrange_api.h`, 1317
`CFE_PSP_MemRangeSet`
 `cfe_psp_memrange_api.h`, 1317
`CFE_PSP_MemRead16`
 `cfe_psp_memaccess_api.h`, 1309
`CFE_PSP_MemRead32`
 `cfe_psp_memaccess_api.h`, 1309
`CFE_PSP_MemRead8`
 `cfe_psp_memaccess_api.h`, 1310
`CFE_PSP_MemSet`
 `cfe_psp_memaccess_api.h`, 1310
`CFE_PSP_MemValidateRange`
 `cfe_psp_memrange_api.h`, 1318
`CFE_PSP_MemWrite16`
 `cfe_psp_memaccess_api.h`, 1311

CFE_PSP_MemWrite32
 cfe_psp_memaccess_api.h, [1311](#)
CFE_PSP_MemWrite8
 cfe_psp_memaccess_api.h, [1312](#)
CFE_PSP_NO_EXCEPTION_DATA
 cfe_psp_error.h, [1302](#)
CFE_PSP_Panic
 cfe_psp_watchdog_api.h, [1334](#)
CFE_PSP_PANIC_CORE_APP
 cfe_psp_watchdog_api.h, [1332](#)
CFE_PSP_PANIC_GENERAL_FAILURE
 cfe_psp_watchdog_api.h, [1332](#)
CFE_PSP_PANIC_MEMORY_ALLOC
 cfe_psp_watchdog_api.h, [1332](#)
CFE_PSP_PANIC_NONVOL_DISK
 cfe_psp_watchdog_api.h, [1332](#)
CFE_PSP_PANIC_STARTUP
 cfe_psp_watchdog_api.h, [1332](#)
CFE_PSP_PANIC_STARTUP_SEM
 cfe_psp_watchdog_api.h, [1332](#)
CFE_PSP_PANIC_VOLATILE_DISK
 cfe_psp_watchdog_api.h, [1332](#)
cfe_psp_port_api.h
 CFE_PSP_PortRead16, [1321](#)
 CFE_PSP_PortRead32, [1321](#)
 CFE_PSP_PortRead8, [1321](#)
 CFE_PSP_PortWrite16, [1322](#)
 CFE_PSP_PortWrite32, [1322](#)
 CFE_PSP_PortWrite8, [1322](#)
CFE_PSP_PortRead16
 cfe_psp_port_api.h, [1321](#)
CFE_PSP_PortRead32
 cfe_psp_port_api.h, [1321](#)
CFE_PSP_PortRead8
 cfe_psp_port_api.h, [1321](#)
CFE_PSP_PortWrite16
 cfe_psp_port_api.h, [1322](#)
CFE_PSP_PortWrite32
 cfe_psp_port_api.h, [1322](#)
CFE_PSP_PortWrite8
 cfe_psp_port_api.h, [1322](#)
CFE_PSP_ReadFromCDS
 cfe_psp_cds_api.h, [1292](#)
CFE_PSP_Restart
 cfe_psp_watchdog_api.h, [1334](#)
CFE_PSP_RST_SUBTYPE_BANKSWITCH_RESET
 cfe_psp_watchdog_api.h, [1332](#)
CFE_PSP_RST_SUBTYPE_EXCEPTION
 cfe_psp_watchdog_api.h, [1332](#)
CFE_PSP_RST_SUBTYPE_HW_SPECIAL_COMMAND
 cfe_psp_watchdog_api.h, [1332](#)
CFE_PSP_RST_SUBTYPE_HW_WATCHDOG
 cfe_psp_watchdog_api.h, [1333](#)
CFE_PSP_RST_SUBTYPE_HWDEBUG_RESET
 cfe_psp_watchdog_api.h, [1333](#)
CFE_PSP_RST_SUBTYPE_MAX
 cfe_psp_watchdog_api.h, [1333](#)
CFE_PSP_RST_SUBTYPE_POWER_CYCLE
 cfe_psp_watchdog_api.h, [1333](#)
CFE_PSP_RST_SUBTYPE_PUSH_BUTTON
 cfe_psp_watchdog_api.h, [1333](#)
CFE_PSP_RST_SUBTYPE_RESET_COMMAND
 cfe_psp_watchdog_api.h, [1333](#)
CFE_PSP_RST_SUBTYPE_UNDEFINED_RESET
 cfe_psp_watchdog_api.h, [1333](#)
CFE_PSP_RST_TYPE_MAX
 cfe_psp_watchdog_api.h, [1333](#)
CFE_PSP_RST_TYPE_POWERON
 cfe_psp_watchdog_api.h, [1333](#)
CFE_PSP_RST_TYPE_PROCESSOR
 cfe_psp_watchdog_api.h, [1334](#)
CFE_PSP_SetDefaultExceptionEnvironment
 cfe_psp_exception_api.h, [1305](#)
CFE_PSP_SOFT_TIMEBASE_NAME
 cfe_psp_timertick_api.h, [1326](#)
cfe_psp_ssr_api.h
 CFE_PSP_InitSSR, [1324](#)
CFE_PSP_STATUS_C
 cfe_psp_error.h, [1302](#)
CFE_PSP_STATUS_STRING_LENGTH
 cfe_psp_error.h, [1302](#)
CFE_PSP_Status_t
 cfe_psp_error.h, [1302](#)
CFE_PSP_StatusString_t
 cfe_psp_error.h, [1302](#)
CFE_PSP_StatusToString
 cfe_psp_error.h, [1303](#)
CFE_PSP_SUCCESS
 cfe_psp_error.h, [1302](#)
cfe_psp_timertick_api.h
 CFE_PSP_Get_Timebase, [1326](#)
 CFE_PSP_GetTime, [1326](#)
 CFE_PSP_GetTimerLow32Rollover, [1327](#)
 CFE_PSP_GetTimerTicksPerSecond, [1327](#)
 CFE_PSP_SOFT_TIMEBASE_NAME, [1326](#)
cfe_psp_version_api.h
 CFE_PSP_GetBuildNumber, [1329](#)
 CFE_PSP_GetVersionCodeName, [1329](#)
 CFE_PSP_GetVersionNumber, [1329](#)
 CFE_PSP_GetVersionString, [1329](#)
cfe_psp_watchdog_api.h
 CFE_PSP_GetRestartType, [1334](#)
 CFE_PSP_Panic, [1334](#)
 CFE_PSP_PANIC_CORE_APP, [1332](#)
 CFE_PSP_PANIC_GENERAL_FAILURE, [1332](#)
 CFE_PSP_PANIC_MEMORY_ALLOC, [1332](#)
 CFE_PSP_PANIC_NONVOL_DISK, [1332](#)
 CFE_PSP_PANIC_STARTUP, [1332](#)

- CFE_PSP_PANIC_STARTUP_SEM, [1332](#)
- CFE_PSP_PANIC_VOLATILE_DISK, [1332](#)
- CFE_PSP_Restart, [1334](#)
- CFE_PSP_RST_SUBTYPE_BANKSWITCH_RESET, [1332](#)
- CFE_PSP_RST_SUBTYPE_EXCEPTION, [1332](#)
- CFE_PSP_RST_SUBTYPE_HW_SPECIAL_COMMAND, [1332](#)
- CFE_PSP_RST_SUBTYPE_HW_WATCHDOG, [1333](#)
- CFE_PSP_RST_SUBTYPE_HWDEBUG_RESET, [1333](#)
- CFE_PSP_RST_SUBTYPE_MAX, [1333](#)
- CFE_PSP_RST_SUBTYPE_POWER_CYCLE, [1333](#)
- CFE_PSP_RST_SUBTYPE_PUSH_BUTTON, [1333](#)
- CFE_PSP_RST_SUBTYPE_RESET_COMMAND, [1333](#)
- CFE_PSP_RST_SUBTYPE_UNDEFINED_RESET, [1333](#)
- CFE_PSP_RST_TYPE_MAX, [1333](#)
- CFE_PSP_RST_TYPE_POWERON, [1333](#)
- CFE_PSP_RST_TYPE_PROCESSOR, [1334](#)
- CFE_PSP_WatchdogDisable, [1335](#)
- CFE_PSP_WatchdogEnable, [1335](#)
- CFE_PSP_WatchdogGet, [1335](#)
- CFE_PSP_WatchdogInit, [1335](#)
- CFE_PSP_WatchdogService, [1335](#)
- CFE_PSP_WatchdogSet, [1335](#)
- CFE_PSP_WatchdogDisable
 - cfe_psp_watchdog_api.h, [1335](#)
- CFE_PSP_WatchdogEnable
 - cfe_psp_watchdog_api.h, [1335](#)
- CFE_PSP_WatchdogGet
 - cfe_psp_watchdog_api.h, [1335](#)
- CFE_PSP_WatchdogInit
 - cfe_psp_watchdog_api.h, [1335](#)
- CFE_PSP_WatchdogService
 - cfe_psp_watchdog_api.h, [1335](#)
- CFE_PSP_WatchdogSet
 - cfe_psp_watchdog_api.h, [1335](#)
- CFE_PSP_WriteToCDS
 - cfe_psp_cds_api.h, [1292](#)
- cfe_resourceid.h
 - CFE_ResourceId_CheckFunc_t, [750](#)
 - CFE_ResourceId_Equal, [751](#)
 - CFE_ResourceId_FindNext, [751](#)
 - CFE_ResourceId_FindNextEx, [752](#)
 - CFE_ResourceId_FromInteger, [752](#)
 - CFE_ResourceId_GetBase, [753](#)
 - CFE_ResourceId_GetSerial, [753](#)
 - CFE_ResourceId_IncrementFunc_t, [750](#)
 - CFE_ResourceId_IsDefined, [753](#)
 - CFE_RESOURCEID_TEST_DEFINED, [750](#)
 - CFE_RESOURCEID_TEST_EQUAL, [750](#)
 - CFE_RESOURCEID_TO_ULONG, [750](#)
 - CFE_ResourceId_ToIndex, [754](#)
 - CFE_ResourceId_ToInteger, [755](#)
 - cfe_resourceid_api_typedefs.h
 - CFE_RESOURCEID_RESERVED, [757](#)
 - CFE_RESOURCEID_UNDEFINED, [757](#)
 - cfe_resourceid_basevalue.h
 - CFE_RESOURCEID_MAKE_BASE, [990](#)
 - CFE_RESOURCEID_MAX, [990](#)
 - CFE_RESOURCEID_SHIFT, [990](#)
 - CFE_ResourceId_CheckFunc_t
 - cfe_resourceid.h, [750](#)
 - CFE_RESOURCEID_CONFIGID_BASE_OFFSET
 - cFE Resource ID base values, [288](#)
 - CFE_ResourceId_Equal
 - cfe_resourceid.h, [751](#)
 - CFE_RESOURCEID_ES_APPID_BASE_OFFSET
 - cFE Resource ID base values, [288](#)
 - CFE_RESOURCEID_ES_CDSBLOCKID_BASE_OFFSET
 - cFE Resource ID base values, [288](#)
 - CFE_RESOURCEID_ES_COUNTID_BASE_OFFSET
 - cFE Resource ID base values, [288](#)
 - CFE_RESOURCEID_ES_LIBID_BASE_OFFSET
 - cFE Resource ID base values, [288](#)
 - CFE_RESOURCEID_ES_POOLID_BASE_OFFSET
 - cFE Resource ID base values, [288](#)
 - CFE_RESOURCEID_ES_TASKID_BASE_OFFSET
 - cFE Resource ID base values, [288](#)
 - CFE_ResourceId_FindNext
 - cfe_resourceid.h, [751](#)
 - CFE_ResourceId_FindNextEx
 - cfe_resourceid.h, [752](#)
 - CFE_ResourceId_FromInteger
 - cfe_resourceid.h, [752](#)
 - CFE_ResourceId_GetBase
 - cfe_resourceid.h, [753](#)
 - CFE_ResourceId_GetSerial
 - cfe_resourceid.h, [753](#)
 - CFE_ResourceId_IncrementFunc_t
 - cfe_resourceid.h, [750](#)
 - CFE_ResourceId_IsDefined
 - cfe_resourceid.h, [753](#)
 - CFE_RESOURCEID_MAKE_BASE
 - cfe_resourceid_basevalue.h, [990](#)
 - CFE_RESOURCEID_MAX
 - cfe_resourceid_basevalue.h, [990](#)
 - CFE_RESOURCEID_RESERVED
 - cfe_resourceid_api_typedefs.h, [757](#)
 - CFE_RESOURCEID_SB_PIPEID_RESOURCE_BASE_OFFSET
 - cFE Resource ID base values, [288](#)
 - CFE_RESOURCEID_SHIFT
 - cfe_resourceid_basevalue.h, [990](#)
 - CFE_RESOURCEID_TBL_ACCESSID_BASE_OFFSET
 - cFE Resource ID base values, [288](#)

- CFE_RESOURCEID_TBL_DUMPCTRLID_BASE_OFFSET
 - cFE Resource ID base values, [288](#)
- CFE_RESOURCEID_TBL_LOADBUFFID_BASE_OFFSET
 - cFE Resource ID base values, [288](#)
- CFE_RESOURCEID_TBL_REGID_BASE_OFFSET
 - cFE Resource ID base values, [288](#)
- CFE_RESOURCEID_TBL_VALRESULTID_BASE_OFFSET
 - cFE Resource ID base values, [288](#)
- CFE_RESOURCEID_TEST_DEFINED
 - cfe_resourceid.h, [750](#)
- CFE_RESOURCEID_TEST_EQUAL
 - cfe_resourceid.h, [750](#)
- CFE_RESOURCEID_TO_ULONG
 - cfe_resourceid.h, [750](#)
- CFE_ResourceId_ToIndex
 - cfe_resourceid.h, [754](#)
- CFE_ResourceId_ToInteger
 - cfe_resourceid.h, [755](#)
- CFE_RESOURCEID_UNDEFINED
 - cfe_resourceid_api_typedefs.h, [757](#)
- CFE_REVISION
 - cfe_version.h, [784](#)
- cfe_sb.h
 - CFE_BIT, [760](#)
 - CFE_CLR, [760](#)
 - CFE_SET, [760](#)
 - CFE_TST, [760](#)
- CFE_SB_AllocateMessageBuffer
 - cFE Zero Copy APIs, [241](#)
- CFE_SB_ALLSUBS_TLM_MID
 - default_cfe_sb_msgids.h, [1005](#)
- CFE_SB_AllSubscriptionsTlm, [501](#)
 - Payload, [502](#)
 - TelemetryHeader, [502](#)
- CFE_SB_AllSubscriptionsTlm_Payload, [502](#)
 - Entries, [502](#)
 - Entry, [502](#)
 - PktSegment, [502](#)
 - TotalSegments, [503](#)
- CFE_SB_AllSubscriptionsTlm_Payload_t
 - default_cfe_sb_msgdefs.h, [1000](#)
- CFE_SB_AllSubscriptionsTlm_t
 - default_cfe_sb_msgstruct.h, [1007](#)
- cfe_sb_api_typedefs.h
 - CFE_SB_Buffer_t, [765](#)
 - CFE_SB_DEFAULT_QOS, [763](#)
 - CFE_SB_INVALID_MSG_ID, [763](#)
 - CFE_SB_INVALID_PIPE, [764](#)
 - CFE_SB_MSGID_C, [764](#)
 - CFE_SB_MSGID_RESERVED, [764](#)
 - CFE_SB_MSGID_UNWRAP_VALUE, [764](#)
 - CFE_SB_MSGID_WRAP_VALUE, [764](#)
 - CFE_SB_PEND_FOREVER, [765](#)
 - CFE_SB_PIPEID_C, [765](#)
 - CFE_SB_POLL, [765](#)
 - CFE_SB_SUBSCRIPTION, [765](#)
 - CFE_SB_UNSUBSCRIPTION, [765](#)
- CFE_SB_BAD_ARGUMENT
 - cFE Return Code Defines, [133](#)
- CFE_SB_BAD_CMD_CODE_EID
 - cfe_sb_eventids.h, [1014](#)
- CFE_SB_BAD_MSGID_EID
 - cfe_sb_eventids.h, [1014](#)
- CFE_SB_BAD_PIPEID_EID
 - cfe_sb_eventids.h, [1014](#)
- CFE_SB_BUF_ALLOC_ERR
 - cFE Return Code Defines, [133](#)
- CFE_SB_BUFFER_INVALID
 - cFE Return Code Defines, [134](#)
- CFE_SB_Buffer_t
 - cfe_sb_api_typedefs.h, [765](#)
- CFE_SB_CCVAL
 - default_cfe_sb_fcncode_values.h, [994](#)
- CFE_SB_CMD0_RCVD_EID
 - cfe_sb_eventids.h, [1014](#)
- CFE_SB_CMD1_RCVD_EID
 - cfe_sb_eventids.h, [1015](#)
- CFE_SB_CMD_MID
 - default_cfe_sb_msgids.h, [1005](#)
- CFE_SB_CmdTopicIdToMsgId
 - cFE Message ID APIs, [248](#)
- CFE_SB_CR_PIPE_BAD_ARG_EID
 - cfe_sb_eventids.h, [1015](#)
- CFE_SB_CR_PIPE_ERR_EID
 - cfe_sb_eventids.h, [1015](#)
- CFE_SB_CR_PIPE_NAME_TAKEN_EID
 - cfe_sb_eventids.h, [1015](#)
- CFE_SB_CR_PIPE_NO_FREE_EID
 - cfe_sb_eventids.h, [1016](#)
- CFE_SB_CreatePipe
 - cFE Pipe Management APIs, [229](#)
- CFE_SB_DEFAULT_QOS
 - cfe_sb_api_typedefs.h, [763](#)
- CFE_SB_DEL_PIPE_ERR1_EID
 - cfe_sb_eventids.h, [1016](#)
- CFE_SB_DEL_PIPE_ERR2_EID
 - cfe_sb_eventids.h, [1016](#)
- CFE_SB_DeletePipe
 - cFE Pipe Management APIs, [230](#)
- CFE_SB_DEST_BLK_ERR_EID
 - cfe_sb_eventids.h, [1016](#)
- CFE_SB_DISABLE_ROUTE_CC
 - cfe_sb_fcncodes.h, [1033](#)
- CFE_SB_DISABLE_SUB_REPORTING_CC
 - cfe_sb_fcncodes.h, [1034](#)
- CFE_SB_DisableRouteCmd, [503](#)
 - CommandHeader, [503](#)
 - Payload, [503](#)

CFE_SB_DisableRouteCmd_t
 default_cfe_sb_msgstruct.h, 1007

CFE_SB_DisableSubReportingCmd, 503
 CommandHeader, 504

CFE_SB_DisableSubReportingCmd_t
 default_cfe_sb_msgstruct.h, 1007

CFE_SB_DSBL_RTE1_EID
 cfe_sb_eventids.h, 1017

CFE_SB_DSBL_RTE2_EID
 cfe_sb_eventids.h, 1017

CFE_SB_DSBL_RTE3_EID
 cfe_sb_eventids.h, 1017

CFE_SB_DUP_SUBSCRIP_EID
 cfe_sb_eventids.h, 1017

CFE_SB_ENABLE_ROUTE_CC
 cfe_sb_fcncodes.h, 1034

CFE_SB_ENABLE_SUB_REPORTING_CC
 cfe_sb_fcncodes.h, 1035

CFE_SB_EnableRouteCmd, 504
 CommandHeader, 504
 Payload, 504

CFE_SB_EnableRouteCmd_t
 default_cfe_sb_msgstruct.h, 1007

CFE_SB_EnableSubReportingCmd, 504
 CommandHeader, 505

CFE_SB_EnableSubReportingCmd_t
 default_cfe_sb_msgstruct.h, 1007

CFE_SB_ENBL_RTE1_EID
 cfe_sb_eventids.h, 1018

CFE_SB_ENBL_RTE2_EID
 cfe_sb_eventids.h, 1018

CFE_SB_ENBL_RTE3_EID
 cfe_sb_eventids.h, 1018

cfe_sb_eventids.h
 CFE_SB_BAD_CMD_CODE_EID, 1014
 CFE_SB_BAD_MSGID_EID, 1014
 CFE_SB_BAD_PIPEID_EID, 1014
 CFE_SB_CMD0_RCVD_EID, 1014
 CFE_SB_CMD1_RCVD_EID, 1015
 CFE_SB_CR_PIPE_BAD_ARG_EID, 1015
 CFE_SB_CR_PIPE_ERR_EID, 1015
 CFE_SB_CR_PIPE_NAME_TAKEN_EID, 1015
 CFE_SB_CR_PIPE_NO_FREE_EID, 1016
 CFE_SB_DEL_PIPE_ERR1_EID, 1016
 CFE_SB_DEL_PIPE_ERR2_EID, 1016
 CFE_SB_DEST_BLK_ERR_EID, 1016
 CFE_SB_DSBL_RTE1_EID, 1017
 CFE_SB_DSBL_RTE2_EID, 1017
 CFE_SB_DSBL_RTE3_EID, 1017
 CFE_SB_DUP_SUBSCRIP_EID, 1017
 CFE_SB_ENBL_RTE1_EID, 1018
 CFE_SB_ENBL_RTE2_EID, 1018
 CFE_SB_ENBL_RTE3_EID, 1018
 CFE_SB_FILEWRITE_ERR_EID, 1018
 CFE_SB_FULL_SUB_PKT_EID, 1019
 CFE_SB_GET_BUF_ERR_EID, 1019
 CFE_SB_GETPIPEIDBYNAME_EID, 1019
 CFE_SB_GETPIPEIDBYNAME_NAME_ERR_EID, 1019
 CFE_SB_GETPIPEIDBYNAME_NULL_ERR_EID, 1020
 CFE_SB_GETPIPEIDBYNAME_NULL_PTR_ERR_EID, 1020
 CFE_SB_GETPIPEIDBYNAME_ID_ERR_EID, 1020
 CFE_SB_GETPIPEIDBYNAME_NULL_PTR_ERR_EID, 1020
 CFE_SB_GETPIPEIDBYNAME_ID_ERR_EID, 1021
 CFE_SB_GETPIPEIDBYNAME_PTR_ERR_EID, 1021
 CFE_SB_HASHCOLLISION_EID, 1021
 CFE_SB_INIT_EID, 1022
 CFE_SB_LEN_ERR_EID, 1022
 CFE_SB_MAX_DESTS_MET_EID, 1022
 CFE_SB_MAX_MSGS_MET_EID, 1022
 CFE_SB_MAX_PIPES_MET_EID, 1023
 CFE_SB_MSG_TOO_BIG_EID, 1023
 CFE_SB_MSGID_LIM_ERR_EID, 1023
 CFE_SB_PART_SUB_PKT_EID, 1023
 CFE_SB_PIPE_ADDED_EID, 1024
 CFE_SB_PIPE_DELETED_EID, 1024
 CFE_SB_Q_FULL_ERR_EID, 1024
 CFE_SB_Q_RD_ERR_EID, 1024
 CFE_SB_Q_WR_ERR_EID, 1025
 CFE_SB_RCV_BAD_ARG_EID, 1025
 CFE_SB_RCV_MESSAGE_INTEGRITY_FAIL_EID, 1025
 CFE_SB_SEND_BAD_ARG_EID, 1025
 CFE_SB_SEND_INV_MSGID_EID, 1026
 CFE_SB_SEND_MESSAGE_INTEGRITY_FAIL_EID, 1026
 CFE_SB_SEND_NO_SUBS_EID, 1026
 CFE_SB_SETPIPEIDBYNAME_EID, 1026
 CFE_SB_SETPIPEIDBYNAME_ID_ERR_EID, 1027
 CFE_SB_SETPIPEIDBYNAME_OWNER_ERR_EID, 1027
 CFE_SB_SND_RTG_EID, 1027
 CFE_SB_SND_RTG_ERR1_EID, 1027
 CFE_SB_SND_STATS_EID, 1028
 CFE_SB_SUB_ARG_ERR_EID, 1028
 CFE_SB_SUB_INV_CALLER_EID, 1028
 CFE_SB_SUB_INV_PIPE_EID, 1028
 CFE_SB_SUBSCRIPTION_RCVD_EID, 1029
 CFE_SB_SUBSCRIPTION_REMOVED_EID, 1029
 CFE_SB_SUBSCRIPTION_RPT_EID, 1029
 CFE_SB_UNSUB_ARG_ERR_EID, 1029
 CFE_SB_UNSUB_INV_CALLER_EID, 1030
 CFE_SB_UNSUB_INV_PIPE_EID, 1030
 CFE_SB_UNSUB_NO_SUBS_EID, 1030

cfe_sb_fcncodes.h
 CFE_SB_DISABLE_ROUTE_CC, 1033
 CFE_SB_DISABLE_SUB_REPORTING_CC, 1034

- CFE_SB_ENABLE_ROUTE_CC, [1034](#)
- CFE_SB_ENABLE_SUB_REPORTING_CC, [1035](#)
- CFE_SB_NOOP_CC, [1036](#)
- CFE_SB_RESET_COUNTERS_CC, [1037](#)
- CFE_SB_SEND_PREV_SUBS_CC, [1038](#)
- CFE_SB_SEND_SB_STATS_CC, [1038](#)
- CFE_SB_WRITE_MAP_INFO_CC, [1039](#)
- CFE_SB_WRITE_PIPE_INFO_CC, [1040](#)
- CFE_SB_WRITE_ROUTING_INFO_CC, [1041](#)
- CFE_SB_FILEWRITE_ERR_EID
 - cfe_sb_eventids.h, [1018](#)
- CFE_SB_FULL_SUB_PKT_EID
 - cfe_sb_eventids.h, [1019](#)
- CFE_SB_FunctionCode_
 - default_cfe_sb_fcncode_values.h, [994](#)
- CFE_SB_FunctionCode_DISABLE_ROUTE
 - default_cfe_sb_fcncode_values.h, [994](#)
- CFE_SB_FunctionCode_DISABLE_SUB_REPORTING
 - default_cfe_sb_fcncode_values.h, [994](#)
- CFE_SB_FunctionCode_ENABLE_ROUTE
 - default_cfe_sb_fcncode_values.h, [994](#)
- CFE_SB_FunctionCode_ENABLE_SUB_REPORTING
 - default_cfe_sb_fcncode_values.h, [994](#)
- CFE_SB_FunctionCode_NOOP
 - default_cfe_sb_fcncode_values.h, [994](#)
- CFE_SB_FunctionCode_RESET_COUNTERS
 - default_cfe_sb_fcncode_values.h, [994](#)
- CFE_SB_FunctionCode_SEND_PREV_SUBS
 - default_cfe_sb_fcncode_values.h, [994](#)
- CFE_SB_FunctionCode_SEND_SB_STATS
 - default_cfe_sb_fcncode_values.h, [994](#)
- CFE_SB_FunctionCode_WRITE_MAP_INFO
 - default_cfe_sb_fcncode_values.h, [994](#)
- CFE_SB_FunctionCode_WRITE_PIPE_INFO
 - default_cfe_sb_fcncode_values.h, [994](#)
- CFE_SB_FunctionCode_WRITE_ROUTING_INFO
 - default_cfe_sb_fcncode_values.h, [994](#)
- CFE_SB_GET_BUF_ERR_EID
 - cfe_sb_eventids.h, [1019](#)
- CFE_SB_GetPipeByName
 - cFE Pipe Management APIs, [230](#)
- CFE_SB_GETPIPEIDBYNAME_EID
 - cfe_sb_eventids.h, [1019](#)
- CFE_SB_GETPIPEIDBYNAME_NAME_ERR_EID
 - cfe_sb_eventids.h, [1019](#)
- CFE_SB_GETPIPEIDBYNAME_NULL_ERR_EID
 - cfe_sb_eventids.h, [1020](#)
- CFE_SB_GetPipeName
 - cFE Pipe Management APIs, [231](#)
- CFE_SB_GETPIPIENAME_EID
 - cfe_sb_eventids.h, [1020](#)
- CFE_SB_GETPIPIENAME_ID_ERR_EID
 - cfe_sb_eventids.h, [1020](#)
- CFE_SB_GETPIPIENAME_NULL_PTR_EID
 - cfe_sb_eventids.h, [1020](#)
- CFE_SB_GetPipeOpts
 - cFE Pipe Management APIs, [231](#)
- CFE_SB_GETPIPEOPTS_EID
 - cfe_sb_eventids.h, [1021](#)
- CFE_SB_GETPIPEOPTS_ID_ERR_EID
 - cfe_sb_eventids.h, [1021](#)
- CFE_SB_GETPIPEOPTS_PTR_ERR_EID
 - cfe_sb_eventids.h, [1021](#)
- CFE_SB_GetUserData
 - cFE Message Characteristics APIs, [244](#)
- CFE_SB_GetUserDataLength
 - cFE Message Characteristics APIs, [244](#)
- CFE_SB_GlobalCmdTopicIdToMsgId
 - cFE Message ID APIs, [248](#)
- CFE_SB_GlobalTlmTopicIdToMsgId
 - cFE Message ID APIs, [249](#)
- CFE_SB_HASHCOLLISION_EID
 - cfe_sb_eventids.h, [1021](#)
- CFE_SB_HK_TLM_MID
 - default_cfe_sb_msgids.h, [1005](#)
- CFE_SB_HousekeepingTlm, [505](#)
 - Payload, [505](#)
 - TelemetryHeader, [505](#)
- CFE_SB_HousekeepingTlm_Payload, [505](#)
 - CommandCounter, [506](#)
 - CommandErrorCounter, [506](#)
 - CreatePipeErrorCounter, [506](#)
 - DuplicateSubscriptionsCounter, [507](#)
 - GetPipeIdByNameErrorCounter, [507](#)
 - InternalErrorCounter, [507](#)
 - MemInUse, [507](#)
 - MemPoolHandle, [507](#)
 - MsgLimitErrorCounter, [507](#)
 - MsgReceiveErrorCounter, [507](#)
 - MsgSendErrorCounter, [508](#)
 - NoSubscribersCounter, [508](#)
 - Padding, [508](#)
 - PipeOptsErrorCounter, [508](#)
 - PipeOverflowErrorCounter, [508](#)
 - SubscribeErrorCounter, [508](#)
 - UnmarkedMem, [508](#)
- CFE_SB_HousekeepingTlm_Payload_t
 - default_cfe_sb_msgdefs.h, [1000](#)
- CFE_SB_HousekeepingTlm_t
 - default_cfe_sb_msgstruct.h, [1007](#)
- CFE_SB_INIT_EID
 - cfe_sb_eventids.h, [1022](#)
- cfe_sb_interface_cfg.h
 - CFE_MISSION_SB_MAX_PIPES, [1043](#)
 - CFE_MISSION_SB_MAX_SB_MSG_SIZE, [1043](#)
 - CFE_MISSION_SB_SUB_ENTRIES_PER_PKT, [1044](#)
 - DEFAULT_CFE_MISSION_SB_MAX_PIPES, [1044](#)

- DEFAULT_CFE_MISSION_SB_MAX_SB_MSG_SIZE, [1044](#)
- DEFAULT_CFE_MISSION_SB_SUB_ENTRIES_PER_PKT, [1044](#)
- cfe_sb_internal_cfg.h
 - CFE_PLATFORM_SB_BUF_MEMORY_BYTES, [1048](#)
 - CFE_PLATFORM_SB_DEFAULT_MAP_FILENAME, [1048](#)
 - CFE_PLATFORM_SB_DEFAULT_MSG_LIMIT, [1048](#)
 - CFE_PLATFORM_SB_DEFAULT_PIPE_FILENAME, [1048](#)
 - CFE_PLATFORM_SB_DEFAULT_ROUTING_FILENAME, [1049](#)
 - CFE_PLATFORM_SB_FILTER_MASK1, [1049](#)
 - CFE_PLATFORM_SB_FILTER_MASK2, [1049](#)
 - CFE_PLATFORM_SB_FILTER_MASK3, [1049](#)
 - CFE_PLATFORM_SB_FILTER_MASK4, [1049](#)
 - CFE_PLATFORM_SB_FILTER_MASK5, [1049](#)
 - CFE_PLATFORM_SB_FILTER_MASK6, [1049](#)
 - CFE_PLATFORM_SB_FILTER_MASK7, [1050](#)
 - CFE_PLATFORM_SB_FILTER_MASK8, [1050](#)
 - CFE_PLATFORM_SB_FILTERED_EVENT1, [1050](#)
 - CFE_PLATFORM_SB_FILTERED_EVENT2, [1050](#)
 - CFE_PLATFORM_SB_FILTERED_EVENT3, [1050](#)
 - CFE_PLATFORM_SB_FILTERED_EVENT4, [1050](#)
 - CFE_PLATFORM_SB_FILTERED_EVENT5, [1050](#)
 - CFE_PLATFORM_SB_FILTERED_EVENT6, [1050](#)
 - CFE_PLATFORM_SB_FILTERED_EVENT7, [1050](#)
 - CFE_PLATFORM_SB_FILTERED_EVENT8, [1051](#)
 - CFE_PLATFORM_SB_HIGHEST_VALID_MSGID, [1051](#)
 - CFE_PLATFORM_SB_MAX_BLOCK_SIZE, [1051](#)
 - CFE_PLATFORM_SB_MAX_DEST_PER_PKT, [1051](#)
 - CFE_PLATFORM_SB_MAX_MSG_IDS, [1051](#)
 - CFE_PLATFORM_SB_MAX_PIPES, [1052](#)
 - CFE_PLATFORM_SB_MEM_BLOCK_SIZE_01, [1052](#)
 - CFE_PLATFORM_SB_MEM_BLOCK_SIZE_02, [1052](#)
 - CFE_PLATFORM_SB_MEM_BLOCK_SIZE_03, [1053](#)
 - CFE_PLATFORM_SB_MEM_BLOCK_SIZE_04, [1053](#)
 - CFE_PLATFORM_SB_MEM_BLOCK_SIZE_05, [1053](#)
 - CFE_PLATFORM_SB_MEM_BLOCK_SIZE_06, [1053](#)
 - CFE_PLATFORM_SB_MEM_BLOCK_SIZE_07, [1053](#)
 - CFE_PLATFORM_SB_MEM_BLOCK_SIZE_08, [1053](#)
 - CFE_PLATFORM_SB_MEM_BLOCK_SIZE_09, [1053](#)
 - CFE_PLATFORM_SB_MEM_BLOCK_SIZE_10, [1053](#)
 - CFE_PLATFORM_SB_MEM_BLOCK_SIZE_11, [1053](#)
 - CFE_PLATFORM_SB_MEM_BLOCK_SIZE_12, [1053](#)
 - CFE_PLATFORM_SB_MEM_BLOCK_SIZE_13, [1053](#)
 - CFE_PLATFORM_SB_MEM_BLOCK_SIZE_14, [1054](#)
 - CFE_PLATFORM_SB_MEM_BLOCK_SIZE_15, [1054](#)
 - CFE_PLATFORM_SB_MEM_BLOCK_SIZE_16, [1054](#)
 - CFE_PLATFORM_SB_POOL_MAX_BUCKETS, [1054](#)
 - CFE_PLATFORM_SB_START_TASK_PRIORITY, [1054](#)
 - CFE_PLATFORM_SB_START_TASK_STACK_SIZE, [1054](#)
 - DEFAULT_CFE_PLATFORM_SB_BUF_MEMORY_BYTES, [1055](#)
 - DEFAULT_CFE_PLATFORM_SB_DEFAULT_MAP_FILENAME, [1055](#)
 - DEFAULT_CFE_PLATFORM_SB_DEFAULT_MSG_LIMIT, [1055](#)
 - DEFAULT_CFE_PLATFORM_SB_DEFAULT_PIPE_FILENAME, [1055](#)
 - DEFAULT_CFE_PLATFORM_SB_DEFAULT_ROUTING_FILENAME, [1055](#)
 - DEFAULT_CFE_PLATFORM_SB_FILTER_MASK1, [1055](#)
 - DEFAULT_CFE_PLATFORM_SB_FILTER_MASK2, [1055](#)
 - DEFAULT_CFE_PLATFORM_SB_FILTER_MASK3, [1055](#)
 - DEFAULT_CFE_PLATFORM_SB_FILTER_MASK4, [1055](#)
 - DEFAULT_CFE_PLATFORM_SB_FILTER_MASK5, [1056](#)
 - DEFAULT_CFE_PLATFORM_SB_FILTER_MASK6, [1056](#)
 - DEFAULT_CFE_PLATFORM_SB_FILTER_MASK7, [1056](#)
 - DEFAULT_CFE_PLATFORM_SB_FILTER_MASK8, [1056](#)
 - DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT1, [1056](#)
 - DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT2, [1056](#)
 - DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT3, [1056](#)
 - DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT4, [1056](#)

- 1056
 DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT5, 1059
 1056
 DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT6, CFE_SB_INTERNAL_ERR
 1056 cFE Return Code Defines, 134
 DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT7, CFE_SB_INVALID_MSG_ID
 1056 cfe_sb_api_typedefs.h, 763
 DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT8, CFE_SB_INVALID_PIPE
 1057 cfe_sb_api_typedefs.h, 764
 DEFAULT_CFE_PLATFORM_SB_HIGHEST_VALID_MSG_ID, CFE_SB_IsValidMsgId
 1057 cFE Message ID APIs, 249
 DEFAULT_CFE_PLATFORM_SB_MAX_BLOCK_SIZE, CFE_SB_LEN_ERR_EID
 1057 cfe_sb_eventids.h, 1022
 DEFAULT_CFE_PLATFORM_SB_MAX_DEST_PER_PIPE, CFE_SB_LocalCmdTopicIdToMsgId
 1057 cFE Message ID APIs, 249
 DEFAULT_CFE_PLATFORM_SB_MAX_MSG_IDS, CFE_SB_LocalTlmTopicIdToMsgId
 1057 cFE Message ID APIs, 250
 DEFAULT_CFE_PLATFORM_SB_MAX_PIPES, CFE_SB_MAX_DESTS_MET
 1057 cFE Return Code Defines, 134
 DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_01, CFE_SB_MAX_DESTS_MET_EID
 1057 cfe_sb_eventids.h, 1022
 DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_02, CFE_SB_MAX_MSGS_MET
 1057 cFE Return Code Defines, 134
 DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_03, CFE_SB_MAX_MSGS_MET_EID
 1057 cfe_sb_eventids.h, 1022
 DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_04, CFE_SB_MAX_PIPES_MET
 1057 cFE Return Code Defines, 134
 DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_05, CFE_SB_MAX_PIPES_MET_EID
 1057 cfe_sb_eventids.h, 1023
 DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_06, CFE_SB_MessageStringGet
 1058 cFE Message Characteristics APIs, 245
 DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_07, CFE_SB_MessageStringSet
 1058 cFE Message Characteristics APIs, 246
 DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_08, CFE_SB_Msg, 509
 1058 LongDouble, 509
 DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_09, LongInt, 509
 1058 Msg, 509
 DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_10, CFE_SB_MSG_TOO_BIG
 1058 cFE Return Code Defines, 134
 DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_11, CFE_SB_MSG_TOO_BIG_EID
 1058 cfe_sb_eventids.h, 1023
 DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_12, CFE_SB_MsgId_Atom_t
 1058 default_cfe_sb_extern_typedefs.h, 992
 DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_13, CFE_SB_MSGID_C
 1058 cfe_sb_api_typedefs.h, 764
 DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_14, CFE_SB_MsgId_Equal
 1058 cFE Message ID APIs, 250
 DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_15, CFE_SB_MSGID_LIM_ERR_EID
 1058 cfe_sb_eventids.h, 1023
 DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_16, CFE_SB_MSGID_RESERVED
 1058 cfe_sb_api_typedefs.h, 764
 DEFAULT_CFE_PLATFORM_SB_POOL_MAX_BUCKET, CFE_SB_MsgId_t, 510
 1059 Value, 510
 DEFAULT_CFE_PLATFORM_SB_START_TASK_PRIORITY, CFE_SB_MSGID_UNWRAP_VALUE

- cfe_sb_api_typedefs.h, [764](#)
- CFE_SB_MSGID_WRAP_VALUE
 - cfe_sb_api_typedefs.h, [764](#)
- CFE_SB_MsgIdToValue
 - cFE Message ID APIs, [251](#)
- CFE_SB_MsgMapFileEntry, [510](#)
 - Index, [510](#)
 - MsgId, [510](#)
- CFE_SB_MsgMapFileEntry_t
 - default_cfe_sb_msgdefs.h, [1000](#)
- CFE_SB_NO_MESSAGE
 - cFE Return Code Defines, [135](#)
- CFE_SB_NOOP_CC
 - cfe_sb_fcncodes.h, [1036](#)
- CFE_SB_NoopCmd, [511](#)
 - CommandHeader, [511](#)
- CFE_SB_NoopCmd_t
 - default_cfe_sb_msgstruct.h, [1007](#)
- CFE_SB_NOT_IMPLEMENTED
 - cFE Return Code Defines, [135](#)
- CFE_SB_ONESUB_TLM_MID
 - default_cfe_sb_msgids.h, [1005](#)
- CFE_SB_PART_SUB_PKT_EID
 - cfe_sb_eventids.h, [1023](#)
- CFE_SB_PEND_FOREVER
 - cfe_sb_api_typedefs.h, [765](#)
- CFE_SB_PIPE_ADDED_EID
 - cfe_sb_eventids.h, [1024](#)
- CFE_SB_PIPE_CR_ERR
 - cFE Return Code Defines, [135](#)
- CFE_SB_PIPE_DELETED_EID
 - cfe_sb_eventids.h, [1024](#)
- CFE_SB_PIPE_RD_ERR
 - cFE Return Code Defines, [135](#)
- CFE_SB_PipeDepthStats, [511](#)
 - CurrentQueueDepth, [512](#)
 - MaxQueueDepth, [512](#)
 - PeakQueueDepth, [512](#)
 - PipeId, [512](#)
 - Spare, [512](#)
- CFE_SB_PipeDepthStats_t
 - default_cfe_sb_msgdefs.h, [1000](#)
- CFE_SB_PIPEID_BASE
 - cFE Resource ID base values, [288](#)
- CFE_SB_PIPEID_C
 - cfe_sb_api_typedefs.h, [765](#)
- CFE_SB_PipeId_t
 - default_cfe_sb_extern_typedefs.h, [992](#)
- CFE_SB_PipeId_ToIndex
 - cFE Pipe Management APIs, [232](#)
- CFE_SB_PipeInfoEntry, [512](#)
 - ApplId, [513](#)
 - AppName, [513](#)
 - CurrentQueueDepth, [513](#)
 - MaxQueueDepth, [513](#)
 - Opts, [513](#)
 - PeakQueueDepth, [513](#)
 - PipeId, [514](#)
 - PipeName, [514](#)
 - SendErrors, [514](#)
 - Spare, [514](#)
- CFE_SB_PipeInfoEntry_t
 - default_cfe_sb_msgdefs.h, [1000](#)
- CFE_SB_PIPEOPTS_IGNOREMINE
 - cFE SB Pipe options, [252](#)
- CFE_SB_POLL
 - cfe_sb_api_typedefs.h, [765](#)
- CFE_SB_Q_FULL_ERR_EID
 - cfe_sb_eventids.h, [1024](#)
- CFE_SB_Q_RD_ERR_EID
 - cfe_sb_eventids.h, [1024](#)
- CFE_SB_Q_WR_ERR_EID
 - cfe_sb_eventids.h, [1025](#)
- CFE_SB_Qos_t, [514](#)
 - Priority, [515](#)
 - Reliability, [515](#)
- CFE_SB_QosPriority
 - default_cfe_sb_extern_typedefs.h, [992](#)
- CFE_SB_QosPriority_Enum_t
 - default_cfe_sb_extern_typedefs.h, [992](#)
- CFE_SB_QosPriority_HIGH
 - default_cfe_sb_extern_typedefs.h, [992](#)
- CFE_SB_QosPriority_LOW
 - default_cfe_sb_extern_typedefs.h, [992](#)
- CFE_SB_QosReliability
 - default_cfe_sb_extern_typedefs.h, [992](#)
- CFE_SB_QosReliability_Enum_t
 - default_cfe_sb_extern_typedefs.h, [992](#)
- CFE_SB_QosReliability_HIGH
 - default_cfe_sb_extern_typedefs.h, [993](#)
- CFE_SB_QosReliability_LOW
 - default_cfe_sb_extern_typedefs.h, [993](#)
- CFE_SB_RCV_BAD_ARG_EID
 - cfe_sb_eventids.h, [1025](#)
- CFE_SB_RCV_MESSAGE_INTEGRITY_FAIL_EID
 - cfe_sb_eventids.h, [1025](#)
- CFE_SB_ReceiveBuffer
 - cFE Send/Receive Message APIs, [239](#)
- CFE_SB_ReleaseMessageBuffer
 - cFE Zero Copy APIs, [242](#)
- CFE_SB_RESET_COUNTERS_CC
 - cfe_sb_fcncodes.h, [1037](#)
- CFE_SB_ResetCountersCmd, [515](#)
 - CommandHeader, [515](#)
- CFE_SB_ResetCountersCmd_t
 - default_cfe_sb_msgstruct.h, [1007](#)
- CFE_SB_RouteCmd_Payload, [515](#)
 - MsgId, [516](#)

- Pipe, [516](#)
- Spare, [516](#)
- CFE_SB_RouteCmd_Payload_t
 - default_cfe_sb_msgdefs.h, [1000](#)
- CFE_SB_Routeld_Atomic_t
 - default_cfe_sb_extern_typedefs.h, [992](#)
- CFE_SB_RoutingFileEntry, [516](#)
 - AppName, [517](#)
 - MsgCnt, [517](#)
 - MsgId, [517](#)
 - Pipeld, [517](#)
 - PipeName, [517](#)
 - State, [517](#)
- CFE_SB_RoutingFileEntry_t
 - default_cfe_sb_msgdefs.h, [1000](#)
- CFE_SB_SEND_BAD_ARG_EID
 - cfe_sb_eventids.h, [1025](#)
- CFE_SB_SEND_HK_MID
 - default_cfe_sb_msgids.h, [1005](#)
- CFE_SB_SEND_INV_MSGID_EID
 - cfe_sb_eventids.h, [1026](#)
- CFE_SB_SEND_MESSAGE_INTEGRITY_FAIL_EID
 - cfe_sb_eventids.h, [1026](#)
- CFE_SB_SEND_NO_SUBS_EID
 - cfe_sb_eventids.h, [1026](#)
- CFE_SB_SEND_PREV_SUBS_CC
 - cfe_sb_fcncodes.h, [1038](#)
- CFE_SB_SEND_SB_STATS_CC
 - cfe_sb_fcncodes.h, [1038](#)
- CFE_SB_SendHkCmd, [517](#)
 - CommandHeader, [518](#)
- CFE_SB_SendHkCmd_t
 - default_cfe_sb_msgstruct.h, [1007](#)
- CFE_SB_SendPrevSubsCmd, [518](#)
 - CommandHeader, [518](#)
- CFE_SB_SendPrevSubsCmd_t
 - default_cfe_sb_msgstruct.h, [1007](#)
- CFE_SB_SendSbStatsCmd, [518](#)
 - CommandHeader, [518](#)
- CFE_SB_SendSbStatsCmd_t
 - default_cfe_sb_msgstruct.h, [1007](#)
- CFE_SB_SetPipeOpts
 - cFE Pipe Management APIs, [233](#)
- CFE_SB_SETPIPEOPTS_EID
 - cfe_sb_eventids.h, [1026](#)
- CFE_SB_SETPIPEOPTS_ID_ERR_EID
 - cfe_sb_eventids.h, [1027](#)
- CFE_SB_SETPIPEOPTS_OWNER_ERR_EID
 - cfe_sb_eventids.h, [1027](#)
- CFE_SB_SetUserDataLength
 - cFE Message Characteristics APIs, [246](#)
- CFE_SB_SingleSubscriptionTlm, [518](#)
 - Payload, [519](#)
 - TelemetryHeader, [519](#)
- CFE_SB_SingleSubscriptionTlm_Payload, [519](#)
 - MsgId, [520](#)
 - Pipe, [520](#)
 - Qos, [520](#)
 - SubType, [520](#)
- CFE_SB_SingleSubscriptionTlm_Payload_t
 - default_cfe_sb_msgdefs.h, [1000](#)
- CFE_SB_SingleSubscriptionTlm_t
 - default_cfe_sb_msgstruct.h, [1007](#)
- CFE_SB_SND_RTG_EID
 - cfe_sb_eventids.h, [1027](#)
- CFE_SB_SND_RTG_ERR1_EID
 - cfe_sb_eventids.h, [1027](#)
- CFE_SB_SND_STATS_EID
 - cfe_sb_eventids.h, [1028](#)
- CFE_SB_STATS_TLM_MID
 - default_cfe_sb_msgids.h, [1005](#)
- CFE_SB_StatsTlm, [520](#)
 - Payload, [520](#)
 - TelemetryHeader, [520](#)
- CFE_SB_StatsTlm_Payload, [521](#)
 - MaxMemAllowed, [522](#)
 - MaxMsgIdsAllowed, [522](#)
 - MaxPipeDepthAllowed, [522](#)
 - MaxPipesAllowed, [522](#)
 - MaxSubscriptionsAllowed, [522](#)
 - MemInUse, [522](#)
 - MsgIdsInUse, [522](#)
 - PeakMemInUse, [523](#)
 - PeakMsgIdsInUse, [523](#)
 - PeakPipesInUse, [523](#)
 - PeakSBBuffersInUse, [523](#)
 - PeakSubscriptionsInUse, [523](#)
 - PipeDepthStats, [523](#)
 - PipesInUse, [523](#)
 - SBBuffersInUse, [524](#)
 - SubscriptionsInUse, [524](#)
- CFE_SB_StatsTlm_Payload_t
 - default_cfe_sb_msgdefs.h, [1001](#)
- CFE_SB_StatsTlm_t
 - default_cfe_sb_msgstruct.h, [1007](#)
- CFE_SB_SUB_ARG_ERR_EID
 - cfe_sb_eventids.h, [1028](#)
- CFE_SB_SUB_INV_CALLER_EID
 - cfe_sb_eventids.h, [1028](#)
- CFE_SB_SUB_INV_PIPE_EID
 - cfe_sb_eventids.h, [1028](#)
- CFE_SB_SUB_RPT_CTRL_MID
 - default_cfe_sb_msgids.h, [1005](#)
- CFE_SB_SubEntries, [524](#)
 - MsgId, [524](#)
 - Pipe, [524](#)
 - Qos, [525](#)
- CFE_SB_SubEntries_t

- default_cfe_sb_msgdefs.h, [1001](#)
- CFE_SB_Subscribe
 - cFE Message Subscription Control APIs, [234](#)
- CFE_SB_SubscribeEx
 - cFE Message Subscription Control APIs, [235](#)
- CFE_SB_SubscribeLocal
 - cFE Message Subscription Control APIs, [235](#)
- CFE_SB_SUBSCRIPTION
 - cfe_sb_api_typedefs.h, [765](#)
- CFE_SB_SUBSCRIPTION_RCVD_EID
 - cfe_sb_eventids.h, [1029](#)
- CFE_SB_SUBSCRIPTION_REMOVED_EID
 - cfe_sb_eventids.h, [1029](#)
- CFE_SB_SUBSCRIPTION_RPT_EID
 - cfe_sb_eventids.h, [1029](#)
- CFE_SB_TIME_OUT
 - cFE Return Code Defines, [135](#)
- CFE_SB_TimeStampMsg
 - cFE Message Characteristics APIs, [247](#)
- CFE_SB_TlmTopicIdToMsgId
 - cFE Message ID APIs, [251](#)
- cfe_sb_topicids.h
 - CFE_MISSION_SB_ALLSUBS_TLM_TOPICID, [1061](#)
 - CFE_MISSION_SB_CMD_TOPICID, [1061](#)
 - CFE_MISSION_SB_HK_TLM_TOPICID, [1062](#)
 - CFE_MISSION_SB_ONESUB_TLM_TOPICID, [1062](#)
 - CFE_MISSION_SB_SEND_HK_TOPICID, [1062](#)
 - CFE_MISSION_SB_STATS_TLM_TOPICID, [1062](#)
 - CFE_MISSION_SB_SUB_RPT_CTRL_TOPICID, [1062](#)
 - DEFAULT_CFE_MISSION_SB_ALLSUBS_TLM_TOPICID, [1062](#)
 - DEFAULT_CFE_MISSION_SB_CMD_TOPICID, [1062](#)
 - DEFAULT_CFE_MISSION_SB_HK_TLM_TOPICID, [1062](#)
 - DEFAULT_CFE_MISSION_SB_ONESUB_TLM_TOPICID, [1062](#)
 - DEFAULT_CFE_MISSION_SB_SEND_HK_TOPICID, [1063](#)
 - DEFAULT_CFE_MISSION_SB_STATS_TLM_TOPICID, [1063](#)
 - DEFAULT_CFE_MISSION_SB_SUB_RPT_CTRL_TOPICID, [1063](#)
- CFE_SB_TransmitBuffer
 - cFE Zero Copy APIs, [242](#)
- CFE_SB_TransmitMsg
 - cFE Send/Receive Message APIs, [240](#)
- CFE_SB_UNSUB_ARG_ERR_EID
 - cfe_sb_eventids.h, [1029](#)
- CFE_SB_UNSUB_INV_CALLER_EID
 - cfe_sb_eventids.h, [1030](#)
- CFE_SB_UNSUB_INV_PIPE_EID
 - cfe_sb_eventids.h, [1030](#)
- CFE_SB_UNSUB_NO_SUBS_EID
 - cfe_sb_eventids.h, [1030](#)
- CFE_SB_Unsubscribe
 - cFE Message Subscription Control APIs, [236](#)
- CFE_SB_UnsubscribeLocal
 - cFE Message Subscription Control APIs, [238](#)
- CFE_SB_UNSUBSCRIPTION
 - cfe_sb_api_typedefs.h, [765](#)
- CFE_SB_ValueToMsgId
 - cFE Message ID APIs, [252](#)
- CFE_SB_WRITE_MAP_INFO_CC
 - cfe_sb_fcncodes.h, [1039](#)
- CFE_SB_WRITE_PIPE_INFO_CC
 - cfe_sb_fcncodes.h, [1040](#)
- CFE_SB_WRITE_ROUTING_INFO_CC
 - cfe_sb_fcncodes.h, [1041](#)
- CFE_SB_WriteFileInfoCmd_Payload, [525](#)
 - Filename, [525](#)
- CFE_SB_WriteFileInfoCmd_Payload_t
 - default_cfe_sb_msgdefs.h, [1001](#)
- CFE_SB_WriteMapInfoCmd, [525](#)
 - CommandHeader, [526](#)
 - Payload, [526](#)
- CFE_SB_WriteMapInfoCmd_t
 - default_cfe_sb_msgstruct.h, [1008](#)
- CFE_SB_WritePipeInfoCmd, [526](#)
 - CommandHeader, [526](#)
 - Payload, [526](#)
- CFE_SB_WritePipeInfoCmd_t
 - default_cfe_sb_msgstruct.h, [1008](#)
- CFE_SB_WriteRoutingInfoCmd, [526](#)
 - CommandHeader, [527](#)
 - Payload, [527](#)
- CFE_SB_WriteRoutingInfoCmd_t
 - default_cfe_sb_msgstruct.h, [1008](#)
- CFE_SB_WRONG_MSG_TYPE
 - cFE Return Code Defines, [135](#)
- CFE_SERVICE_BITMASK
 - cfe_error.h, [708](#)
- CFE_SET
 - cfe_sb.h, [760](#)
- CFE_SEVERITY_BITMASK
 - cfe_error.h, [708](#)
- CFE_SEVERITY_ERROR
 - cfe_error.h, [708](#)
- CFE_SEVERITY_INFO
 - cfe_error.h, [708](#)
- CFE_SEVERITY_SUCCESS
 - cfe_error.h, [709](#)
- CFE_SOFTWARE_BUS_SERVICE
 - cfe_error.h, [709](#)
- CFE_SRC_VERSION
 - cfe_version.h, [784](#)

CFE_STATUS_BAD_COMMAND_CODE
 cFE Return Code Defines, [135](#)
 CFE_STATUS_C
 cfe_error.h, [709](#)
 CFE_STATUS_EXTERNAL_RESOURCE_FAIL
 cFE Return Code Defines, [136](#)
 CFE_STATUS_INCORRECT_STATE
 cFE Return Code Defines, [136](#)
 CFE_STATUS_NO_COUNTER_INCREMENT
 cFE Return Code Defines, [136](#)
 CFE_STATUS_NOT_IMPLEMENTED
 cFE Return Code Defines, [136](#)
 CFE_STATUS_RANGE_ERROR
 cFE Return Code Defines, [136](#)
 CFE_STATUS_REQUEST_ALREADY_PENDING
 cFE Return Code Defines, [136](#)
 CFE_STATUS_STRING_LENGTH
 cfe_error.h, [709](#)
 CFE_Status_t
 cfe_error.h, [709](#)
 CFE_STATUS_UNKNOWN_MSG_ID
 cFE Return Code Defines, [136](#)
 CFE_STATUS_VALIDATION_FAILURE
 cFE Return Code Defines, [137](#)
 CFE_STATUS_WRONG_MSG_LENGTH
 cFE Return Code Defines, [137](#)
 CFE_StatusString_t
 cfe_error.h, [709](#)
 CFE_STR
 cfe_version.h, [785](#)
 CFE_STR_HELPER
 cfe_version.h, [785](#)
 CFE_SUCCESS
 cFE Return Code Defines, [137](#)
 CFE_TABLE_SERVICE
 cfe_error.h, [709](#)
 cfe_tbl.h
 CFE_TBL_HandleFromID, [768](#)
 CFE_TBL_HandleID_AsInt, [768](#)
 CFE_TBL_HandleID_IsDefined, [768](#)
 CFE_TBL_HandleID_IsEqual, [768](#)
 CFE_TBL_HandleToID, [768](#)
 CFE_TBL_ABORT_LOAD_CC
 cfe_tbl_fcncodes.h, [1116](#)
 CFE_TBL_AbortLoadCmd, [527](#)
 CommandHeader, [527](#)
 Payload, [527](#)
 CFE_TBL_AbortLoadCmd_Payload, [528](#)
 TableName, [528](#)
 CFE_TBL_AbortLoadCmd_Payload_t
 default_cfe_tbl_msgdefs.h, [1073](#)
 CFE_TBL_AbortLoadCmd_t
 default_cfe_tbl_msgstruct.h, [1080](#)
 CFE_TBL_ACTIVATE_CC
 cfe_tbl_fcncodes.h, [1116](#)
 CFE_TBL_ACTIVATE_DUMP_ONLY_ERR_EID
 cfe_tbl_eventids.h, [1096](#)
 CFE_TBL_ACTIVATE_ERR_EID
 cfe_tbl_eventids.h, [1096](#)
 CFE_TBL_ActivateCmd, [528](#)
 CommandHeader, [529](#)
 Payload, [529](#)
 CFE_TBL_ActivateCmd_Payload, [529](#)
 TableName, [529](#)
 CFE_TBL_ActivateCmd_Payload_t
 default_cfe_tbl_msgdefs.h, [1073](#)
 CFE_TBL_ActivateCmd_t
 default_cfe_tbl_msgstruct.h, [1080](#)
 cfe_tbl_api_typedefs.h
 CFE_TBL_BAD_TABLE_HANDLE, [771](#)
 CFE_TBL_CallbackFuncPtr_t, [772](#)
 CFE_TBL_HANDLE_EQ, [771](#)
 CFE_TBL_HANDLE_INT, [772](#)
 CFE_TBL_HANDLE_IS_VALID, [772](#)
 CFE_TBL_Handle_t, [772](#)
 CFE_TBL_HANDLEID_C, [772](#)
 CFE_TBL_HANDLEID_UNDEFINED, [772](#)
 CFE_TBL_Info_t, [773](#)
 CFE_TBL_MAX_FULL_NAME_LEN, [772](#)
 CFE_TBL_REGID_C, [772](#)
 CFE_TBL_REGID_UNDEFINED, [772](#)
 CFE_TBL_SRC_ADDRESS, [773](#)
 CFE_TBL_SRC_FILE, [773](#)
 CFE_TBL_SrcEnum, [773](#)
 CFE_TBL_SrcEnum_t, [773](#)
 CFE_TBL_ASSUMED_VALID_INF_EID
 cfe_tbl_eventids.h, [1096](#)
 CFE_TBL_BAD_ARGUMENT
 cFE Return Code Defines, [137](#)
 CFE_TBL_BAD_TABLE_HANDLE
 cfe_tbl_api_typedefs.h, [771](#)
 CFE_TBL_BufferSelect
 default_cfe_tbl_extern_typedefs.h, [1065](#)
 CFE_TBL_BufferSelect_ACTIVE
 default_cfe_tbl_extern_typedefs.h, [1065](#)
 CFE_TBL_BufferSelect_Enum_t
 default_cfe_tbl_extern_typedefs.h, [1064](#)
 CFE_TBL_BufferSelect_INACTIVE
 default_cfe_tbl_extern_typedefs.h, [1065](#)
 CFE_TBL_CallbackFuncPtr_t
 cfe_tbl_api_typedefs.h, [772](#)
 CFE_TBL_CC1_ERR_EID
 cfe_tbl_eventids.h, [1096](#)
 CFE_TBL_CCVAL
 default_cfe_tbl_fcncode_values.h, [1066](#)
 CFE_TBL_CDS_DELETE_ERR_EID
 cfe_tbl_eventids.h, [1096](#)
 CFE_TBL_CDS_DELETED_INFO_EID

- cfe_tbl_eventids.h, [1097](#)
- CFE_TBL_CDS_NOT_FOUND_ERR_EID
 - cfe_tbl_eventids.h, [1097](#)
- CFE_TBL_CDS_OWNER_ACTIVE_ERR_EID
 - cfe_tbl_eventids.h, [1097](#)
- CFE_TBL_CMD_MID
 - default_cfe_tbl_msgids.h, [1078](#)
- CFE_TBL_CODEC_ERROR_ERR_EID
 - cfe_tbl_eventids.h, [1097](#)
- CFE_TBL_CombinedFileHdr, [529](#)
 - Std, [530](#)
 - Tbl, [530](#)
- CFE_TBL_CombinedFileHdr_t
 - default_cfe_tbl_extern_typedefs.h, [1064](#)
- CFE_TBL_CREATING_DUMP_FILE_ERR_EID
 - cfe_tbl_eventids.h, [1098](#)
- CFE_TBL_DeICDSCmd_Payload, [530](#)
 - TableName, [530](#)
- CFE_TBL_DeICDSCmd_Payload_t
 - default_cfe_tbl_msgdefs.h, [1073](#)
- CFE_TBL_DELETE_CDS_CC
 - cfe_tbl_fcncodes.h, [1117](#)
- CFE_TBL_DeleteCDSCmd, [530](#)
 - CommandHeader, [531](#)
 - Payload, [531](#)
- CFE_TBL_DeleteCDSCmd_t
 - default_cfe_tbl_msgstruct.h, [1080](#)
- CFE_TBL_DUMP_CC
 - cfe_tbl_fcncodes.h, [1118](#)
- CFE_TBL_DUMP_PENDING_ERR_EID
 - cfe_tbl_eventids.h, [1098](#)
- CFE_TBL_DUMP_REGISTRY_CC
 - cfe_tbl_fcncodes.h, [1119](#)
- CFE_TBL_DumpCmd, [531](#)
 - CommandHeader, [531](#)
 - Payload, [531](#)
- CFE_TBL_DumpCmd_Payload, [532](#)
 - ActiveTableFlag, [532](#)
 - DumpFilename, [532](#)
 - TableName, [532](#)
- CFE_TBL_DumpCmd_Payload_t
 - default_cfe_tbl_msgdefs.h, [1073](#)
- CFE_TBL_DumpCmd_t
 - default_cfe_tbl_msgstruct.h, [1080](#)
- CFE_TBL_DUMPCTRLID_BASE
 - cFE Resource ID base values, [288](#)
- CFE_TBL_DumpRegistryCmd, [533](#)
 - CommandHeader, [533](#)
 - Payload, [533](#)
- CFE_TBL_DumpRegistryCmd_Payload, [533](#)
 - DumpFilename, [533](#)
- CFE_TBL_DumpRegistryCmd_Payload_t
 - default_cfe_tbl_msgdefs.h, [1073](#)
- CFE_TBL_DumpRegistryCmd_t
 - default_cfe_tbl_msgstruct.h, [1080](#)
- CFE_TBL_ERR_ACCESS
 - cFE Return Code Defines, [137](#)
- CFE_TBL_ERR_BAD_CONTENT_ID
 - cFE Return Code Defines, [137](#)
- CFE_TBL_ERR_BAD_PROCESSOR_ID
 - cFE Return Code Defines, [137](#)
- CFE_TBL_ERR_BAD_SPACECRAFT_ID
 - cFE Return Code Defines, [138](#)
- CFE_TBL_ERR_BAD_SUBTYPE_ID
 - cFE Return Code Defines, [138](#)
- CFE_TBL_ERR_DUMP_ONLY
 - cFE Return Code Defines, [138](#)
- CFE_TBL_ERR_DUPLICATE_DIFF_SIZE
 - cFE Return Code Defines, [138](#)
- CFE_TBL_ERR_DUPLICATE_NOT_OWNED
 - cFE Return Code Defines, [138](#)
- CFE_TBL_ERR_FILE_FOR_WRONG_TABLE
 - cFE Return Code Defines, [138](#)
- CFE_TBL_ERR_FILE_SIZE_INCONSISTENT
 - cFE Return Code Defines, [138](#)
- CFE_TBL_ERR_FILE_TOO_LARGE
 - cFE Return Code Defines, [138](#)
- CFE_TBL_ERR_FILENAME_TOO_LONG
 - cFE Return Code Defines, [139](#)
- CFE_TBL_ERR_HANDLES_FULL
 - cFE Return Code Defines, [139](#)
- CFE_TBL_ERR_ILLEGAL_SRC_TYPE
 - cFE Return Code Defines, [139](#)
- CFE_TBL_ERR_INVALID_HANDLE
 - cFE Return Code Defines, [139](#)
- CFE_TBL_ERR_INVALID_NAME
 - cFE Return Code Defines, [139](#)
- CFE_TBL_ERR_INVALID_OPTIONS
 - cFE Return Code Defines, [139](#)
- CFE_TBL_ERR_INVALID_SIZE
 - cFE Return Code Defines, [140](#)
- CFE_TBL_ERR_LOAD_IN_PROGRESS
 - cFE Return Code Defines, [140](#)
- CFE_TBL_ERR_LOAD_INCOMPLETE
 - cFE Return Code Defines, [140](#)
- CFE_TBL_ERR_NEVER_LOADED
 - cFE Return Code Defines, [140](#)
- CFE_TBL_ERR_NO_ACCESS
 - cFE Return Code Defines, [140](#)
- CFE_TBL_ERR_NO_BUFFER_AVAIL
 - cFE Return Code Defines, [140](#)
- CFE_TBL_ERR_NO_STD_HEADER
 - cFE Return Code Defines, [141](#)
- CFE_TBL_ERR_NO_TBL_HEADER
 - cFE Return Code Defines, [141](#)
- CFE_TBL_ERR_PARTIAL_LOAD
 - cFE Return Code Defines, [141](#)

- cFE Return Code Defines, [141](#)
- CFE_TBL_ERR_REGISTRY_FULL
 - cFE Return Code Defines, [141](#)
- CFE_TBL_ERR_SHORT_FILE
 - cFE Return Code Defines, [141](#)
- CFE_TBL_ERR_UNREGISTERED
 - cFE Return Code Defines, [141](#)
- cfe_tbl_eventids.h
 - CFE_TBL_ACTIVATE_DUMP_ONLY_ERR_EID, [1096](#)
 - CFE_TBL_ACTIVATE_ERR_EID, [1096](#)
 - CFE_TBL_ASSUMED_VALID_INF_EID, [1096](#)
 - CFE_TBL_CC1_ERR_EID, [1096](#)
 - CFE_TBL_CDS_DELETE_ERR_EID, [1096](#)
 - CFE_TBL_CDS_DELETED_INFO_EID, [1097](#)
 - CFE_TBL_CDS_NOT_FOUND_ERR_EID, [1097](#)
 - CFE_TBL_CDS_OWNER_ACTIVE_ERR_EID, [1097](#)
 - CFE_TBL_CODEC_ERROR_ERR_EID, [1097](#)
 - CFE_TBL_CREATING_DUMP_FILE_ERR_EID, [1098](#)
 - CFE_TBL_DUMP_PENDING_ERR_EID, [1098](#)
 - CFE_TBL_FAIL_HK_SEND_ERR_EID, [1098](#)
 - CFE_TBL_FAIL_NOTIFY_SEND_ERR_EID, [1098](#)
 - CFE_TBL_FILE_ACCESS_ERR_EID, [1099](#)
 - CFE_TBL_FILE_INCOMPLETE_ERR_EID, [1099](#)
 - CFE_TBL_FILE_LOADED_INF_EID, [1099](#)
 - CFE_TBL_FILE_STD_HDR_ERR_EID, [1099](#)
 - CFE_TBL_FILE_SUBTYPE_ERR_EID, [1100](#)
 - CFE_TBL_FILE_TBL_HDR_ERR_EID, [1100](#)
 - CFE_TBL_FILE_TOO_BIG_ERR_EID, [1100](#)
 - CFE_TBL_FILE_TYPE_ERR_EID, [1100](#)
 - CFE_TBL_HANDLE_ACCESS_ERR_EID, [1101](#)
 - CFE_TBL_ILLEGAL_BUFF_PARAM_ERR_EID, [1101](#)
 - CFE_TBL_IN_REGISTRY_ERR_EID, [1101](#)
 - CFE_TBL_INIT_INF_EID, [1101](#)
 - CFE_TBL_LEN_ERR_EID, [1102](#)
 - CFE_TBL_LOAD_ABORT_ERR_EID, [1102](#)
 - CFE_TBL_LOAD_ABORT_INF_EID, [1102](#)
 - CFE_TBL_LOAD_EXCEEDS_SIZE_ERR_EID, [1102](#)
 - CFE_TBL_LOAD_FILENAME_LONG_ERR_EID, [1103](#)
 - CFE_TBL_LOAD_IN_PROGRESS_ERR_EID, [1103](#)
 - CFE_TBL_LOAD_PEND_REQ_INF_EID, [1103](#)
 - CFE_TBL_LOAD_SUCCESS_INF_EID, [1103](#)
 - CFE_TBL_LOAD_TBLNAME_MISMATCH_ERR_EID, [1104](#)
 - CFE_TBL_LOAD_TYPE_ERR_EID, [1104](#)
 - CFE_TBL_LOAD_VAL_ERR_EID, [1104](#)
 - CFE_TBL_LOADING_A_DUMP_ONLY_ERR_EID, [1104](#)
 - CFE_TBL_LOAD_ERR_EID, [1105](#)
 - CFE_TBL_NO_SUCH_TABLE_ERR_EID, [1105](#)
 - CFE_TBL_NO_WORK_BUFFERS_ERR_EID, [1105](#)
 - CFE_TBL_NOOP_INF_EID, [1105](#)
 - CFE_TBL_NOT_CRITICAL_TBL_ERR_EID, [1106](#)
 - CFE_TBL_NOT_IN_CRIT_REG_ERR_EID, [1106](#)
 - CFE_TBL_OVERWRITE_DUMP_INF_EID, [1106](#)
 - CFE_TBL_OVERWRITE_REG_DUMP_INF_EID, [1106](#)
 - CFE_TBL_PARTIAL_LOAD_ERR_EID, [1107](#)
 - CFE_TBL_PROCESSOR_ID_ERR_EID, [1107](#)
 - CFE_TBL_REGISTER_ERR_EID, [1107](#)
 - CFE_TBL_RESET_INF_EID, [1107](#)
 - CFE_TBL_SHARE_ERR_EID, [1108](#)
 - CFE_TBL_SPACECRAFT_ID_ERR_EID, [1108](#)
 - CFE_TBL_TLM_REG_CMD_INF_EID, [1108](#)
 - CFE_TBL_TOO_MANY_DUMPS_ERR_EID, [1108](#)
 - CFE_TBL_TOO_MANY_VALIDATIONS_ERR_EID, [1109](#)
 - CFE_TBL_UNREGISTER_ERR_EID, [1109](#)
 - CFE_TBL_UNVALIDATED_ERR_EID, [1109](#)
 - CFE_TBL_UPDATE_ERR_EID, [1109](#)
 - CFE_TBL_UPDATE_SUCCESS_INF_EID, [1110](#)
 - CFE_TBL_VAL_REQ_MADE_INF_EID, [1110](#)
 - CFE_TBL_VALIDATION_ERR_EID, [1110](#)
 - CFE_TBL_VALIDATION_IN_PROGRESS_ERR_EID, [1110](#)
 - CFE_TBL_VALIDATION_INF_EID, [1111](#)
 - CFE_TBL_WRITE_CFE_HDR_ERR_EID, [1111](#)
 - CFE_TBL_WRITE_DUMP_INF_EID, [1111](#)
 - CFE_TBL_WRITE_REG_DUMP_INF_EID, [1111](#)
 - CFE_TBL_WRITE_TBL_HDR_ERR_EID, [1112](#)
 - CFE_TBL_WRITE_TBL_IMG_ERR_EID, [1112](#)
 - CFE_TBL_WRITE_TBL_REG_ERR_EID, [1112](#)
 - CFE_TBL_ZERO_LENGTH_LOAD_ERR_EID, [1112](#)
 - CFE_TBL_FAIL_HK_SEND_ERR_EID
 - cfe_tbl_eventids.h, [1098](#)
 - CFE_TBL_FAIL_NOTIFY_SEND_ERR_EID
 - cfe_tbl_eventids.h, [1098](#)
 - cfe_tbl_fcncodes.h
 - CFE_TBL_ABORT_LOAD_CC, [1116](#)
 - CFE_TBL_ACTIVATE_CC, [1116](#)
 - CFE_TBL_DELETE_CDS_CC, [1117](#)
 - CFE_TBL_DUMP_CC, [1118](#)
 - CFE_TBL_DUMP_REGISTRY_CC, [1119](#)
 - CFE_TBL_LOAD_CC, [1120](#)
 - CFE_TBL_NOOP_CC, [1121](#)
 - CFE_TBL_RESET_COUNTERS_CC, [1122](#)
 - CFE_TBL_SEND_REGISTRY_CC, [1122](#)
 - CFE_TBL_VALIDATE_CC, [1123](#)
 - CFE_TBL_FILE_ACCESS_ERR_EID
 - cfe_tbl_eventids.h, [1099](#)
 - CFE_TBL_File_Hdr, [534](#)
 - NumBytes, [534](#)
 - Offset, [534](#)
 - Reserved, [534](#)
 - TableName, [534](#)

CFE_TBL_File_Hdr_t
 default_cfe_tbl_extern_typedefs.h, 1064
CFE_TBL_FILE_INCOMPLETE_ERR_EID
 cfe_tbl_eventids.h, 1099
CFE_TBL_FILE_LOADED_INF_EID
 cfe_tbl_eventids.h, 1099
CFE_TBL_FILE_STD_HDR_ERR_EID
 cfe_tbl_eventids.h, 1099
CFE_TBL_FILE_SUBTYPE_ERR_EID
 cfe_tbl_eventids.h, 1100
CFE_TBL_FILE_TBL_HDR_ERR_EID
 cfe_tbl_eventids.h, 1100
CFE_TBL_FILE_TOO_BIG_ERR_EID
 cfe_tbl_eventids.h, 1100
CFE_TBL_FILE_TYPE_ERR_EID
 cfe_tbl_eventids.h, 1100
CFE_TBL_FILEDEF
 cfe_tbl_filedef.h, 775
CFE_TBL_FileDef, 535
 Description, 535
 ObjectName, 535
 ObjectSize, 535
 TableName, 536
 TgtFilename, 536
cfe_tbl_filedef.h
 CFE_TBL_FILEDEF, 775
 CFE_TBL_FileDef_t, 776
CFE_TBL_FileDef_t
 cfe_tbl_filedef.h, 776
CFE_TBL_FunctionCode_
 default_cfe_tbl_fcncode_values.h, 1067
CFE_TBL_FunctionCode_ABORT_LOAD
 default_cfe_tbl_fcncode_values.h, 1067
CFE_TBL_FunctionCode_ACTIVATE
 default_cfe_tbl_fcncode_values.h, 1067
CFE_TBL_FunctionCode_DELETE_CDS
 default_cfe_tbl_fcncode_values.h, 1067
CFE_TBL_FunctionCode_DUMP
 default_cfe_tbl_fcncode_values.h, 1067
CFE_TBL_FunctionCode_DUMP_REGISTRY
 default_cfe_tbl_fcncode_values.h, 1067
CFE_TBL_FunctionCode_LOAD
 default_cfe_tbl_fcncode_values.h, 1067
CFE_TBL_FunctionCode_NOOP
 default_cfe_tbl_fcncode_values.h, 1067
CFE_TBL_FunctionCode_RESET_COUNTERS
 default_cfe_tbl_fcncode_values.h, 1067
CFE_TBL_FunctionCode_SEND_REGISTRY
 default_cfe_tbl_fcncode_values.h, 1067
CFE_TBL_FunctionCode_VALIDATE
 default_cfe_tbl_fcncode_values.h, 1067
CFE_TBL_GetAddress
 cFE Access Table Content APIs, 263
CFE_TBL_GetAddresses
 cFE Access Table Content APIs, 264
CFE_TBL_GetInfo
 cFE Get Table Information APIs, 267
CFE_TBL_GetStatus
 cFE Get Table Information APIs, 268
CFE_TBL_HANDLE_ACCESS_ERR_EID
 cfe_tbl_eventids.h, 1101
CFE_TBL_HANDLE_BASE
 cFE Resource ID base values, 288
CFE_TBL_HANDLE_EQ
 cfe_tbl_api_typedefs.h, 771
CFE_TBL_HANDLE_INT
 cfe_tbl_api_typedefs.h, 772
CFE_TBL_HANDLE_IS_VALID
 cfe_tbl_api_typedefs.h, 772
CFE_TBL_Handle_t
 cfe_tbl_api_typedefs.h, 772
CFE_TBL_HandleFromID
 cfe_tbl.h, 768
CFE_TBL_HandleID_AsInt
 cfe_tbl.h, 768
CFE_TBL_HANDLEID_C
 cfe_tbl_api_typedefs.h, 772
CFE_TBL_HandleID_IsDefined
 cfe_tbl.h, 768
CFE_TBL_HandleID_IsEqual
 cfe_tbl.h, 768
CFE_TBL_HandleId_t
 default_cfe_tbl_extern_typedefs.h, 1065
CFE_TBL_HANDLEID_UNDEFINED
 cfe_tbl_api_typedefs.h, 772
CFE_TBL_HandleToID
 cfe_tbl.h, 768
CFE_TBL_HK_TLM_MID
 default_cfe_tbl_msgids.h, 1078
CFE_TBL_HousekeepingTlm, 536
 Payload, 536
 TelemetryHeader, 536
CFE_TBL_HousekeepingTlm_Payload, 537
 ActiveBuffer, 538
 ByteAlignPad1, 538
 CommandCounter, 538
 CommandErrorCounter, 538
 FailedValCounter, 538
 LastFileDumped, 538
 LastFileLoaded, 539
 LastTableLoaded, 539
 LastUpdatedTable, 539
 LastUpdateTime, 539
 LastValCrc, 539
 LastValStatus, 539
 LastValTableName, 539
 MemPoolHandle, 540
 NumFreeSharedBufs, 540

- NumLoadPending, [540](#)
- NumTables, [540](#)
- NumValRequests, [540](#)
- SuccessValCounter, [540](#)
- ValidationCounter, [540](#)
- CFE_TBL_HousekeepingTlm_Payload_t
 - default_cfe_tbl_msgdefs.h, [1074](#)
- CFE_TBL_HousekeepingTlm_t
 - default_cfe_tbl_msgstruct.h, [1080](#)
- CFE_TBL_ILLEGAL_BUFF_PARAM_ERR_EID
 - cfe_tbl_eventids.h, [1101](#)
- CFE_TBL_IN_REGISTRY_ERR_EID
 - cfe_tbl_eventids.h, [1101](#)
- CFE_TBL_Info, [541](#)
 - Crc, [541](#)
 - Critical, [541](#)
 - DoubleBuffered, [542](#)
 - DumpOnly, [542](#)
 - FileTime, [542](#)
 - LastFileLoaded, [542](#)
 - NumUsers, [542](#)
 - Size, [542](#)
 - TableLoadedOnce, [542](#)
 - TimeOfLastUpdate, [542](#)
 - UserDefAddr, [542](#)
- CFE_TBL_INFO_DUMP_PENDING
 - cFE Return Code Defines, [141](#)
- CFE_TBL_INFO_NO_DUMP_PENDING
 - cFE Return Code Defines, [142](#)
- CFE_TBL_INFO_NO_UPDATE_PENDING
 - cFE Return Code Defines, [142](#)
- CFE_TBL_INFO_NO_VALIDATION_PENDING
 - cFE Return Code Defines, [142](#)
- CFE_TBL_INFO_RECOVERED_TBL
 - cFE Return Code Defines, [142](#)
- CFE_TBL_Info_t
 - cfe_tbl_api_typedefs.h, [773](#)
- CFE_TBL_INFO_TABLE_LOCKED
 - cFE Return Code Defines, [142](#)
- CFE_TBL_INFO_UPDATE_PENDING
 - cFE Return Code Defines, [142](#)
- CFE_TBL_INFO_UPDATED
 - cFE Return Code Defines, [142](#)
- CFE_TBL_INFO_VALIDATION_PENDING
 - cFE Return Code Defines, [142](#)
- CFE_TBL_INIT_INF_EID
 - cfe_tbl_eventids.h, [1101](#)
- cfe_tbl_interface_cfg.h
 - CFE_MISSION_TBL_MAX_FULL_NAME_LEN, [1126](#)
 - CFE_MISSION_TBL_MAX_NAME_LENGTH, [1126](#)
 - DEFAULT_CFE_MISSION_TBL_MAX_FULL_NAME_LEN, [1126](#)
 - DEFAULT_CFE_MISSION_TBL_MAX_NAME_LENGTH, [1126](#)
- DEFAULT_CFE_MISSION_TBL_MAX_NAME_LENGTH, [1126](#)
- cfe_tbl_internal_cfg.h
 - CFE_PLATFORM_TBL_BUF_MEMORY_BYTES, [1129](#)
 - CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE, [1129](#)
 - CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES, [1129](#)
 - CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE, [1129](#)
 - CFE_PLATFORM_TBL_MAX_NUM_HANDLES, [1130](#)
 - CFE_PLATFORM_TBL_MAX_NUM_TABLES, [1130](#)
 - CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS, [1130](#)
 - CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS, [1131](#)
 - CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE, [1131](#)
 - CFE_PLATFORM_TBL_START_TASK_PRIORITY, [1131](#)
 - CFE_PLATFORM_TBL_START_TASK_STACK_SIZE, [1132](#)
 - CFE_PLATFORM_TBL_U32FROM4CHARS, [1132](#)
 - CFE_PLATFORM_TBL_VALID_PRID_1, [1132](#)
 - CFE_PLATFORM_TBL_VALID_PRID_2, [1132](#)
 - CFE_PLATFORM_TBL_VALID_PRID_3, [1133](#)
 - CFE_PLATFORM_TBL_VALID_PRID_4, [1133](#)
 - CFE_PLATFORM_TBL_VALID_PRID_COUNT, [1133](#)
 - CFE_PLATFORM_TBL_VALID_SCID_1, [1133](#)
 - CFE_PLATFORM_TBL_VALID_SCID_2, [1133](#)
 - CFE_PLATFORM_TBL_VALID_SCID_COUNT, [1133](#)
 - DEFAULT_CFE_PLATFORM_TBL_BUF_MEMORY_BYTES, [1134](#)
 - DEFAULT_CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE, [1134](#)
 - DEFAULT_CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES, [1134](#)
 - DEFAULT_CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE, [1134](#)
 - DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_HANDLES, [1134](#)
 - DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_TABLES, [1134](#)
 - DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS, [1134](#)
 - DEFAULT_CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS, [1134](#)
 - DEFAULT_CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE, [1135](#)
 - DEFAULT_CFE_PLATFORM_TBL_START_TASK_PRIORITY, [1135](#)
 - DEFAULT_CFE_PLATFORM_TBL_START_TASK_STACK_SIZE, [1135](#)

- 1135
- DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_1, 1135
- DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_2, 1135
- DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_3, 1135
- DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_4, 1135
- DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_COUNT, 1135
- DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_1, 1135
- DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_2, 1135
- DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_COUNT, 1135
- CFE_TBL_LEN_ERR_EID
cfe_tbl_eventids.h, 1102
- CFE_TBL_Load
cFE Manage Table Content APIs, 258
- CFE_TBL_LOAD_ABORT_ERR_EID
cfe_tbl_eventids.h, 1102
- CFE_TBL_LOAD_ABORT_INF_EID
cfe_tbl_eventids.h, 1102
- CFE_TBL_LOAD_CC
cfe_tbl_fcncodes.h, 1120
- CFE_TBL_LOAD_EXCEEDS_SIZE_ERR_EID
cfe_tbl_eventids.h, 1102
- CFE_TBL_LOAD_FILENAME_LONG_ERR_EID
cfe_tbl_eventids.h, 1103
- CFE_TBL_LOAD_IN_PROGRESS_ERR_EID
cfe_tbl_eventids.h, 1103
- CFE_TBL_LOAD_PEND_REQ_INF_EID
cfe_tbl_eventids.h, 1103
- CFE_TBL_LOAD_SUCCESS_INF_EID
cfe_tbl_eventids.h, 1103
- CFE_TBL_LOAD_TBLNAME_MISMATCH_ERR_EID
cfe_tbl_eventids.h, 1104
- CFE_TBL_LOAD_TYPE_ERR_EID
cfe_tbl_eventids.h, 1104
- CFE_TBL_LOAD_VAL_ERR_EID
cfe_tbl_eventids.h, 1104
- CFE_TBL_LOADBUFFID_BASE
cFE Resource ID base values, 288
- CFE_TBL_LoadCmd, 543
CommandHeader, 543
Payload, 543
- CFE_TBL_LoadCmd_Payload, 543
LoadFilename, 544
- CFE_TBL_LoadCmd_Payload_t
default_cfe_tbl_msgdefs.h, 1074
- CFE_TBL_LoadCmd_t
default_cfe_tbl_msgstruct.h, 1080
- CFE_TBL_LOADING_A_DUMP_ONLY_ERR_EID
cfe_tbl_eventids.h, 1104
- CFE_TBL_Manage
cFE Manage Table Content APIs, 259
- CFE_TBL_MAX_FULL_NAME_LEN
cfe_tbl_api_typedefs.h, 772
- CFE_TBL_MESSAGE_ERROR
cFE Return Code Defines, 143
- CFE_TBL_MID_ERR_EID
cfe_tbl_eventids.h, 1105
- CFE_TBL_Modified
cFE Manage Table Content APIs, 260
- CFE_TBL_NO_SUCH_TABLE_ERR_EID
cfe_tbl_eventids.h, 1105
- CFE_TBL_NO_WORK_BUFFERS_ERR_EID
- CFE_TBL_NOOP_CC
cfe_tbl_fcncodes.h, 1121
- CFE_TBL_NOOP_INF_EID
cfe_tbl_eventids.h, 1105
- CFE_TBL_NoopCmd, 544
CommandHeader, 544
- CFE_TBL_NoopCmd_t
default_cfe_tbl_msgstruct.h, 1080
- CFE_TBL_NOT_CRITICAL_TBL_ERR_EID
cfe_tbl_eventids.h, 1106
- CFE_TBL_NOT_IMPLEMENTED
cFE Return Code Defines, 143
- CFE_TBL_NOT_IN_CRIT_REG_ERR_EID
cfe_tbl_eventids.h, 1106
- CFE_TBL_NotifyByMessage
cFE Get Table Information APIs, 269
- CFE_TBL_NotifyCmd, 544
CommandHeader, 545
Payload, 545
- CFE_TBL_NotifyCmd_Payload, 545
Parameter, 545
- CFE_TBL_NotifyCmd_Payload_t
default_cfe_tbl_msgdefs.h, 1074
- CFE_TBL_NotifyCmd_t
default_cfe_tbl_msgstruct.h, 1080
- CFE_TBL_OPT_BUFFER_MSK
cFE Table Type Defines, 270
- CFE_TBL_OPT_CRITICAL
cFE Table Type Defines, 270
- CFE_TBL_OPT_CRITICAL_MSK
cFE Table Type Defines, 270
- CFE_TBL_OPT_DBL_BUFFER
cFE Table Type Defines, 271
- CFE_TBL_OPT_DEFAULT
cFE Table Type Defines, 271
- CFE_TBL_OPT_DUMP_ONLY
cFE Table Type Defines, 271
- CFE_TBL_OPT_LD_DMP_MSK

- cFE Table Type Defines, [271](#)
- CFE_TBL_OPT_LOAD_DUMP
 - cFE Table Type Defines, [271](#)
- CFE_TBL_OPT_NOT_CRITICAL
 - cFE Table Type Defines, [271](#)
- CFE_TBL_OPT_NOT_USR_DEF
 - cFE Table Type Defines, [271](#)
- CFE_TBL_OPT_SNGL_BUFFER
 - cFE Table Type Defines, [271](#)
- CFE_TBL_OPT_USR_DEF_ADDR
 - cFE Table Type Defines, [271](#)
- CFE_TBL_OPT_USR_DEF_MSK
 - cFE Table Type Defines, [272](#)
- CFE_TBL_OVERWRITE_DUMP_INF_EID
 - cfe_tbl_eventids.h, [1106](#)
- CFE_TBL_OVERWRITE_REG_DUMP_INF_EID
 - cfe_tbl_eventids.h, [1106](#)
- CFE_TBL_PARTIAL_LOAD_ERR_EID
 - cfe_tbl_eventids.h, [1107](#)
- CFE_TBL_PROCESSOR_ID_ERR_EID
 - cfe_tbl_eventids.h, [1107](#)
- CFE_TBL_REG_TLM_MID
 - default_cfe_tbl_msgids.h, [1078](#)
- CFE_TBL_REGID_BASE
 - cFE Resource ID base values, [288](#)
- CFE_TBL_REGID_C
 - cfe_tbl_api_typedefs.h, [772](#)
- CFE_TBL_RegId_t
 - default_cfe_tbl_extern_typedefs.h, [1065](#)
- CFE_TBL_REGID_UNDEFINED
 - cfe_tbl_api_typedefs.h, [772](#)
- CFE_TBL_Register
 - cFE Registration APIs, [253](#)
- CFE_TBL_REGISTER_ERR_EID
 - cfe_tbl_eventids.h, [1107](#)
- CFE_TBL_ReleaseAddress
 - cFE Access Table Content APIs, [265](#)
- CFE_TBL_ReleaseAddresses
 - cFE Access Table Content APIs, [266](#)
- CFE_TBL_RESET_COUNTERS_CC
 - cfe_tbl_fcncodes.h, [1122](#)
- CFE_TBL_RESET_INF_EID
 - cfe_tbl_eventids.h, [1107](#)
- CFE_TBL_ResetCountersCmd, [545](#)
 - CommandHeader, [546](#)
- CFE_TBL_ResetCountersCmd_t
 - default_cfe_tbl_msgstruct.h, [1080](#)
- CFE_TBL_SEND_HK_MID
 - default_cfe_tbl_msgids.h, [1078](#)
- CFE_TBL_SEND_REGISTRY_CC
 - cfe_tbl_fcncodes.h, [1122](#)
- CFE_TBL_SendHkCmd, [546](#)
 - CommandHeader, [546](#)
- CFE_TBL_SendHkCmd_t
 - default_cfe_tbl_msgstruct.h, [1080](#)
- CFE_TBL_SendRegistryCmd, [546](#)
 - CommandHeader, [547](#)
 - Payload, [547](#)
- CFE_TBL_SendRegistryCmd_Payload, [547](#)
 - TableName, [547](#)
- CFE_TBL_SendRegistryCmd_Payload_t
 - default_cfe_tbl_msgdefs.h, [1074](#)
- CFE_TBL_SendRegistryCmd_t
 - default_cfe_tbl_msgstruct.h, [1080](#)
- CFE_TBL_Share
 - cFE Registration APIs, [255](#)
- CFE_TBL_SHARE_ERR_EID
 - cfe_tbl_eventids.h, [1108](#)
- CFE_TBL_SPACECRAFT_ID_ERR_EID
 - cfe_tbl_eventids.h, [1108](#)
- CFE_TBL_SRC_ADDRESS
 - cfe_tbl_api_typedefs.h, [773](#)
- CFE_TBL_SRC_FILE
 - cfe_tbl_api_typedefs.h, [773](#)
- CFE_TBL_SrcEnum
 - cfe_tbl_api_typedefs.h, [773](#)
- CFE_TBL_SrcEnum_t
 - cfe_tbl_api_typedefs.h, [773](#)
- CFE_TBL_TableRegistryTlm, [547](#)
 - Payload, [548](#)
 - TelemetryHeader, [548](#)
- CFE_TBL_TableRegistryTlm_t
 - default_cfe_tbl_msgstruct.h, [1081](#)
- CFE_TBL_TblRegPacket_Payload, [548](#)
 - ActiveBufferAddr, [549](#)
 - Crc, [549](#)
 - Critical, [549](#)
 - DoubleBuffered, [549](#)
 - DumpOnly, [549](#)
 - FileTime, [549](#)
 - InactiveBufferAddr, [550](#)
 - LastFileLoaded, [550](#)
 - LoadPending, [550](#)
 - Name, [550](#)
 - OwnerAppName, [550](#)
 - Padding1, [550](#)
 - Padding2, [550](#)
 - Size, [551](#)
 - TableLoadedOnce, [551](#)
 - TimeOfLastUpdate, [551](#)
 - ValidationFuncPtr, [551](#)
- CFE_TBL_TblRegPacket_Payload_t
 - default_cfe_tbl_msgdefs.h, [1074](#)
- CFE_TBL_TLM_REG_CMD_INF_EID
 - cfe_tbl_eventids.h, [1108](#)
- CFE_TBL_TOO_MANY_DUMPS_ERR_EID
 - cfe_tbl_eventids.h, [1108](#)
- CFE_TBL_TOO_MANY_VALIDATIONS_ERR_EID

- cfe_tbl_eventids.h, [1109](#)
- cfe_tbl_topicsids.h
 - CFE_MISSION_TBL_CMD_TOPICID, [1137](#)
 - CFE_MISSION_TBL_HK_TLM_TOPICID, [1137](#)
 - CFE_MISSION_TBL_REG_TLM_TOPICID, [1138](#)
 - CFE_MISSION_TBL_SEND_HK_TOPICID, [1138](#)
 - DEFAULT_CFE_MISSION_TBL_CMD_TOPICID, [1138](#)
 - DEFAULT_CFE_MISSION_TBL_HK_TLM_TOPICID, [1138](#)
 - DEFAULT_CFE_MISSION_TBL_REG_TLM_TOPICID, [1138](#)
 - DEFAULT_CFE_MISSION_TBL_SEND_HK_TOPICID, [1138](#)
- CFE_TBL_Unregister
 - cFE Registration APIs, [256](#)
- CFE_TBL_UNREGISTER_ERR_EID
 - cfe_tbl_eventids.h, [1109](#)
- CFE_TBL_UNVALIDATED_ERR_EID
 - cfe_tbl_eventids.h, [1109](#)
- CFE_TBL_Update
 - cFE Manage Table Content APIs, [261](#)
- CFE_TBL_UPDATE_ERR_EID
 - cfe_tbl_eventids.h, [1109](#)
- CFE_TBL_UPDATE_SUCCESS_INF_EID
 - cfe_tbl_eventids.h, [1110](#)
- CFE_TBL_VAL_REQ_MADE_INF_EID
 - cfe_tbl_eventids.h, [1110](#)
- CFE_TBL_Validate
 - cFE Manage Table Content APIs, [262](#)
- CFE_TBL_VALIDATE_CC
 - cfe_tbl_fcncodes.h, [1123](#)
- CFE_TBL_ValidateCmd, [551](#)
 - CommandHeader, [552](#)
 - Payload, [552](#)
- CFE_TBL_ValidateCmd_Payload, [552](#)
 - ActiveTableFlag, [552](#)
 - TableName, [552](#)
- CFE_TBL_ValidateCmd_Payload_t
 - default_cfe_tbl_msgdefs.h, [1074](#)
- CFE_TBL_ValidateCmd_t
 - default_cfe_tbl_msgstruct.h, [1081](#)
- CFE_TBL_VALIDATION_ERR_EID
 - cfe_tbl_eventids.h, [1110](#)
- CFE_TBL_VALIDATION_IN_PROGRESS_ERR_EID
 - cfe_tbl_eventids.h, [1110](#)
- CFE_TBL_VALIDATION_INF_EID
 - cfe_tbl_eventids.h, [1111](#)
- CFE_TBL_VALRESULTID_BASE
 - cFE Resource ID base values, [288](#)
- CFE_TBL_WARN_DUPLICATE
 - cFE Return Code Defines, [143](#)
- CFE_TBL_WARN_NOT_CRITICAL
 - cFE Return Code Defines, [143](#)
- CFE_TBL_WARN_PARTIAL_LOAD
 - cFE Return Code Defines, [143](#)
- CFE_TBL_WARN_SHORT_FILE
 - cFE Return Code Defines, [143](#)
- CFE_TBL_WRITE_CFE_HDR_ERR_EID
 - cfe_tbl_eventids.h, [1111](#)
- CFE_TBL_WRITE_DUMP_INF_EID
 - cfe_tbl_eventids.h, [1111](#)
- CFE_TBL_WRITE_REG_DUMP_INF_EID
 - cfe_tbl_eventids.h, [1111](#)
- CFE_TBL_WRITE_TBL_HDR_ERR_EID
 - cfe_tbl_eventids.h, [1112](#)
- CFE_TBL_WRITE_TBL_IMG_ERR_EID
 - cfe_tbl_eventids.h, [1112](#)
- CFE_TBL_WRITE_TBL_REG_ERR_EID
 - cfe_tbl_eventids.h, [1112](#)
- CFE_TBL_ZERO_LENGTH_LOAD_ERR_EID
 - cfe_tbl_eventids.h, [1112](#)
- cfe_time.h
 - CFE_TIME_Copy, [778](#)
- CFE_TIME_1HZ_CMD_MID
 - default_cfe_time_msgids.h, [1157](#)
- CFE_TIME_A_GT_B
 - cfe_time_api_typedefs.h, [782](#)
- CFE_TIME_A_LT_B
 - cfe_time_api_typedefs.h, [782](#)
- CFE_TIME_Add
 - cFE Time Arithmetic APIs, [277](#)
- CFE_TIME_ADD_ADJUST_CC
 - cfe_time_fcncodes.h, [1178](#)
- CFE_TIME_ADD_DELAY_CC
 - cfe_time_fcncodes.h, [1179](#)
- CFE_TIME_ADD_ONE_HZ_ADJUSTMENT_CC
 - cfe_time_fcncodes.h, [1179](#)
- CFE_TIME_AddAdjustCmd, [553](#)
 - CommandHeader, [553](#)
 - Payload, [553](#)
- CFE_TIME_AddAdjustCmd_t
 - default_cfe_time_msgstruct.h, [1160](#)
- CFE_TIME_AddDelayCmd, [553](#)
 - CommandHeader, [553](#)
 - Payload, [553](#)
- CFE_TIME_AddDelayCmd_t
 - default_cfe_time_msgstruct.h, [1160](#)
- CFE_TIME_AddOneHzAdjustmentCmd, [554](#)
 - CommandHeader, [554](#)
 - Payload, [554](#)
- CFE_TIME_AddOneHzAdjustmentCmd_t
 - default_cfe_time_msgstruct.h, [1160](#)
- CFE_TIME_AdjustDirection
 - default_cfe_time_extern_typedefs.h, [1142](#)
- CFE_TIME_AdjustDirection_ADD
 - default_cfe_time_extern_typedefs.h, [1142](#)
- CFE_TIME_AdjustDirection_Enum_t

- default_cfe_time_extern_typedefs.h, 1140
- CFE_TIME_AdjustDirection_SUBTRACT
 - default_cfe_time_extern_typedefs.h, 1142
- cfe_time_api_typedefs.h
 - CFE_TIME_A_GT_B, 782
 - CFE_TIME_A_LT_B, 782
 - CFE_TIME_Compare, 782
 - CFE_TIME_Compare_t, 781
 - CFE_TIME_EQUAL, 782
 - CFE_TIME_PRINTED_STRING_SIZE, 781
 - CFE_TIME_SynchCallbackPtr_t, 781
 - CFE_TIME_ZERO_VALUE, 781
- CFE_TIME_BAD_ARGUMENT
 - cFE Return Code Defines, 143
- CFE_TIME_CALLBACK_NOT_REGISTERED
 - cFE Return Code Defines, 144
- CFE_TIME_CC_ERR_EID
 - cfe_time_eventids.h, 1167
- CFE_TIME_CCVAL
 - default_cfe_time_fcncode_values.h, 1146
- CFE_TIME_ClockState
 - default_cfe_time_extern_typedefs.h, 1142
- CFE_TIME_ClockState_Enum_t
 - default_cfe_time_extern_typedefs.h, 1140
- CFE_TIME_ClockState_FLYWHEEL
 - default_cfe_time_extern_typedefs.h, 1142
- CFE_TIME_ClockState_INVALID
 - default_cfe_time_extern_typedefs.h, 1142
- CFE_TIME_ClockState_VALID
 - default_cfe_time_extern_typedefs.h, 1142
- CFE_TIME_CMD_MID
 - default_cfe_time_msgids.h, 1157
- CFE_TIME_Compare
 - cFE Time Arithmetic APIs, 278
 - cfe_time_api_typedefs.h, 782
- CFE_TIME_Compare_t
 - cfe_time_api_typedefs.h, 781
- CFE_TIME_Copy
 - cfe_time.h, 778
- CFE_TIME_DATA_CMD_MID
 - default_cfe_time_msgids.h, 1157
- CFE_TIME_DELAY_CFG_EID
 - cfe_time_eventids.h, 1167
- CFE_TIME_DELAY_EID
 - cfe_time_eventids.h, 1167
- CFE_TIME_DELAY_ERR_EID
 - cfe_time_eventids.h, 1168
- CFE_TIME_DELTA_CFG_EID
 - cfe_time_eventids.h, 1168
- CFE_TIME_DELTA_EID
 - cfe_time_eventids.h, 1168
- CFE_TIME_DELTA_ERR_EID
 - cfe_time_eventids.h, 1168
- CFE_TIME_DIAG_EID
 - cfe_time_eventids.h, 1169
- CFE_TIME_DIAG_TLM_MID
 - default_cfe_time_msgids.h, 1157
- CFE_TIME_DiagnosticTlm, 554
 - Payload, 555
 - TelemetryHeader, 555
- CFE_TIME_DiagnosticTlm_Payload, 555
 - AtToneDelay, 557
 - AtToneLatch, 557
 - AtToneLeapSeconds, 557
 - AtToneMET, 557
 - AtToneSTCF, 557
 - ClockFlyState, 558
 - ClockSetState, 558
 - ClockSignal, 558
 - ClockSource, 558
 - ClockStateAPI, 558
 - ClockStateFlags, 558
 - CurrentLatch, 559
 - CurrentMET, 559
 - CurrentTAI, 559
 - CurrentUTC, 559
 - DataStoreStatus, 559
 - DelayDirection, 559
 - Forced2Fly, 560
 - LocalIntCounter, 560
 - LocalTaskCounter, 560
 - MaxElapsed, 560
 - MaxLocalClock, 560
 - MinElapsed, 560
 - OneHzAdjust, 560
 - OneHzDirection, 561
 - OneTimeAdjust, 561
 - OneTimeDirection, 561
 - ServerFlyState, 561
 - TimeSinceTone, 561
 - ToneDataCounter, 561
 - ToneDataLatch, 561
 - ToneIntCounter, 562
 - ToneIntErrorCounter, 562
 - ToneMatchCounter, 562
 - ToneMatchErrorCounter, 562
 - ToneOverLimit, 562
 - ToneSignalCounter, 562
 - ToneSignalLatch, 562
 - ToneTaskCounter, 563
 - ToneUnderLimit, 563
 - VersionCounter, 563
 - VirtualMET, 563
- CFE_TIME_DiagnosticTlm_Payload_t
 - default_cfe_time_msgdefs.h, 1152
- CFE_TIME_DiagnosticTlm_t
 - default_cfe_time_msgstruct.h, 1160
- CFE_TIME_EQUAL

- cfe_time_api_typedefs.h, [782](#)
- cfe_time_eventids.h
 - CFE_TIME_CC_ERR_EID, [1167](#)
 - CFE_TIME_DELAY_CFG_EID, [1167](#)
 - CFE_TIME_DELAY_EID, [1167](#)
 - CFE_TIME_DELAY_ERR_EID, [1168](#)
 - CFE_TIME_DELTA_CFG_EID, [1168](#)
 - CFE_TIME_DELTA_EID, [1168](#)
 - CFE_TIME_DELTA_ERR_EID, [1168](#)
 - CFE_TIME_DIAG_EID, [1169](#)
 - CFE_TIME_FLY_OFF_EID, [1169](#)
 - CFE_TIME_FLY_ON_EID, [1169](#)
 - CFE_TIME_ID_ERR_EID, [1169](#)
 - CFE_TIME_INIT_EID, [1170](#)
 - CFE_TIME_LEAPS_CFG_EID, [1170](#)
 - CFE_TIME_LEAPS_EID, [1170](#)
 - CFE_TIME_LEN_ERR_EID, [1170](#)
 - CFE_TIME_MET_CFG_EID, [1171](#)
 - CFE_TIME_MET_EID, [1171](#)
 - CFE_TIME_MET_ERR_EID, [1171](#)
 - CFE_TIME_NOOP_EID, [1171](#)
 - CFE_TIME_ONEHZ_CFG_EID, [1172](#)
 - CFE_TIME_ONEHZ_EID, [1172](#)
 - CFE_TIME_RESET_EID, [1172](#)
 - CFE_TIME_SIGNAL_CFG_EID, [1172](#)
 - CFE_TIME_SIGNAL_EID, [1173](#)
 - CFE_TIME_SIGNAL_ERR_EID, [1173](#)
 - CFE_TIME_SOURCE_CFG_EID, [1173](#)
 - CFE_TIME_SOURCE_EID, [1173](#)
 - CFE_TIME_SOURCE_ERR_EID, [1174](#)
 - CFE_TIME_STATE_EID, [1174](#)
 - CFE_TIME_STATE_ERR_EID, [1174](#)
 - CFE_TIME_STCF_CFG_EID, [1174](#)
 - CFE_TIME_STCF_EID, [1175](#)
 - CFE_TIME_STCF_ERR_EID, [1175](#)
 - CFE_TIME_TIME_CFG_EID, [1175](#)
 - CFE_TIME_TIME_EID, [1175](#)
 - CFE_TIME_TIME_ERR_EID, [1176](#)
- CFE_TIME_ExternalGPS
 - cFE External Time Source APIs, [282](#)
- CFE_TIME_ExternalMET
 - cFE External Time Source APIs, [282](#)
- CFE_TIME_ExternalTime
 - cFE External Time Source APIs, [283](#)
- CFE_TIME_ExternalTone
 - cFE External Time Source APIs, [284](#)
- CFE_TIME_FakeToneCmd, [563](#)
 - CommandHeader, [564](#)
- CFE_TIME_FakeToneCmd_t
 - default_cfe_time_msgstruct.h, [1160](#)
- cfe_time_fnccodes.h
 - CFE_TIME_ADD_ADJUST_CC, [1178](#)
 - CFE_TIME_ADD_DELAY_CC, [1179](#)
 - CFE_TIME_ADD_ONE_HZ_ADJUSTMENT_CC, [1179](#)
 - CFE_TIME_NOOP_CC, [1180](#)
 - CFE_TIME_RESET_COUNTERS_CC, [1181](#)
 - CFE_TIME_SEND_DIAGNOSTIC_CC, [1182](#)
 - CFE_TIME_SET_LEAP_SECONDS_CC, [1183](#)
 - CFE_TIME_SET_MET_CC, [1184](#)
 - CFE_TIME_SET_SIGNAL_CC, [1185](#)
 - CFE_TIME_SET_SOURCE_CC, [1185](#)
 - CFE_TIME_SET_STATE_CC, [1186](#)
 - CFE_TIME_SET_STCF_CC, [1188](#)
 - CFE_TIME_SET_TIME_CC, [1188](#)
 - CFE_TIME_SUB_ADJUST_CC, [1189](#)
 - CFE_TIME_SUB_DELAY_CC, [1190](#)
 - CFE_TIME_SUB_ONE_HZ_ADJUSTMENT_CC, [1191](#)
- CFE_TIME_FLAG_ADD1HZ
 - cFE Clock State Flag Defines, [289](#)
- CFE_TIME_FLAG_ADDADJ
 - cFE Clock State Flag Defines, [289](#)
- CFE_TIME_FLAG_ADDTCL
 - cFE Clock State Flag Defines, [289](#)
- CFE_TIME_FLAG_CLKSET
 - cFE Clock State Flag Defines, [289](#)
- CFE_TIME_FLAG_CMDFLY
 - cFE Clock State Flag Defines, [290](#)
- CFE_TIME_FLAG_FLYING
 - cFE Clock State Flag Defines, [290](#)
- CFE_TIME_FLAG_GDTONE
 - cFE Clock State Flag Defines, [290](#)
- CFE_TIME_FLAG_REFERR
 - cFE Clock State Flag Defines, [290](#)
- CFE_TIME_FLAG_SERVER
 - cFE Clock State Flag Defines, [290](#)
- CFE_TIME_FLAG_SIGPRI
 - cFE Clock State Flag Defines, [290](#)
- CFE_TIME_FLAG_SRCINT
 - cFE Clock State Flag Defines, [290](#)
- CFE_TIME_FLAG_SRVFLY
 - cFE Clock State Flag Defines, [290](#)
- CFE_TIME_FLAG_UNUSED
 - cFE Clock State Flag Defines, [290](#)
- CFE_TIME_FlagBit
 - default_cfe_time_extern_typedefs.h, [1142](#)
- CFE_TIME_FlagBit_ADD1HZ
 - default_cfe_time_extern_typedefs.h, [1143](#)
- CFE_TIME_FlagBit_ADDADJ
 - default_cfe_time_extern_typedefs.h, [1143](#)
- CFE_TIME_FlagBit_ADDTCL
 - default_cfe_time_extern_typedefs.h, [1143](#)
- CFE_TIME_FlagBit_CLKSET
 - default_cfe_time_extern_typedefs.h, [1142](#)
- CFE_TIME_FlagBit_CMDFLY
 - default_cfe_time_extern_typedefs.h, [1143](#)

- CFE_TIME_FlagBit_Enum_t
 - default_cfe_time_extern_typedefs.h, [1141](#)
- CFE_TIME_FlagBit_FLYING
 - default_cfe_time_extern_typedefs.h, [1142](#)
- CFE_TIME_FlagBit_GDTONE
 - default_cfe_time_extern_typedefs.h, [1143](#)
- CFE_TIME_FlagBit_SERVER
 - default_cfe_time_extern_typedefs.h, [1143](#)
- CFE_TIME_FlagBit_SIGPRI
 - default_cfe_time_extern_typedefs.h, [1143](#)
- CFE_TIME_FlagBit_SRCINT
 - default_cfe_time_extern_typedefs.h, [1142](#)
- CFE_TIME_FlagBit_SRVFLY
 - default_cfe_time_extern_typedefs.h, [1143](#)
- CFE_TIME_FLY_OFF_EID
 - cfe_time_eventids.h, [1169](#)
- CFE_TIME_FLY_ON_EID
 - cfe_time_eventids.h, [1169](#)
- CFE_TIME_FlywheelState
 - default_cfe_time_extern_typedefs.h, [1143](#)
- CFE_TIME_FlywheelState_Enum_t
 - default_cfe_time_extern_typedefs.h, [1141](#)
- CFE_TIME_FlywheelState_IS_FLY
 - default_cfe_time_extern_typedefs.h, [1143](#)
- CFE_TIME_FlywheelState_NO_FLY
 - default_cfe_time_extern_typedefs.h, [1143](#)
- CFE_TIME_FunctionCode_
 - default_cfe_time_fcncode_values.h, [1146](#)
- CFE_TIME_FunctionCode_ADD_ADJUST
 - default_cfe_time_fcncode_values.h, [1146](#)
- CFE_TIME_FunctionCode_ADD_DELAY
 - default_cfe_time_fcncode_values.h, [1146](#)
- CFE_TIME_FunctionCode_ADD_ONE_HZ_ADJUSTMENT
 - default_cfe_time_fcncode_values.h, [1146](#)
- CFE_TIME_FunctionCode_NOOP
 - default_cfe_time_fcncode_values.h, [1146](#)
- CFE_TIME_FunctionCode_RESET_COUNTERS
 - default_cfe_time_fcncode_values.h, [1146](#)
- CFE_TIME_FunctionCode_SEND_DIAGNOSTIC
 - default_cfe_time_fcncode_values.h, [1146](#)
- CFE_TIME_FunctionCode_SET_LEAP_SECONDS
 - default_cfe_time_fcncode_values.h, [1146](#)
- CFE_TIME_FunctionCode_SET_MET
 - default_cfe_time_fcncode_values.h, [1146](#)
- CFE_TIME_FunctionCode_SET_SIGNAL
 - default_cfe_time_fcncode_values.h, [1146](#)
- CFE_TIME_FunctionCode_SET_SOURCE
 - default_cfe_time_fcncode_values.h, [1146](#)
- CFE_TIME_FunctionCode_SET_STATE
 - default_cfe_time_fcncode_values.h, [1146](#)
- CFE_TIME_FunctionCode_SET_STCF
 - default_cfe_time_fcncode_values.h, [1146](#)
- CFE_TIME_FunctionCode_SET_TIME
 - default_cfe_time_fcncode_values.h, [1146](#)
- CFE_TIME_FunctionCode_SUB_ADJUST
 - default_cfe_time_fcncode_values.h, [1146](#)
- CFE_TIME_FunctionCode_SUB_DELAY
 - default_cfe_time_fcncode_values.h, [1146](#)
- CFE_TIME_FunctionCode_SUB_ONE_HZ_ADJUSTMENT
 - default_cfe_time_fcncode_values.h, [1146](#)
- CFE_TIME_GetClockInfo
 - cFE Get Time Information APIs, [275](#)
- CFE_TIME_GetClockState
 - cFE Get Time Information APIs, [276](#)
- CFE_TIME_GetLeapSeconds
 - cFE Get Time Information APIs, [276](#)
- CFE_TIME_GetMET
 - cFE Get Current Time APIs, [272](#)
- CFE_TIME_GetMETseconds
 - cFE Get Current Time APIs, [273](#)
- CFE_TIME_GetMETsubsecs
 - cFE Get Current Time APIs, [273](#)
- CFE_TIME_GetSTCF
 - cFE Get Time Information APIs, [276](#)
- CFE_TIME_GetTAI
 - cFE Get Current Time APIs, [273](#)
- CFE_TIME_GetTime
 - cFE Get Current Time APIs, [274](#)
- CFE_TIME_GetUTC
 - cFE Get Current Time APIs, [274](#)
- CFE_TIME_HK_TLM_MID
 - default_cfe_time_msgids.h, [1157](#)
- CFE_TIME_HousekeepingTlm, [564](#)
 - Payload, [564](#)
 - TelemetryHeader, [564](#)
- CFE_TIME_HousekeepingTlm_Payload, [564](#)
 - AdjustmentFactor, [565](#)
 - ClockStateAPI, [565](#)
 - ClockStateFlags, [565](#)
 - CommandCounter, [565](#)
 - CommandErrorCounter, [566](#)
 - LeapSeconds, [566](#)
 - MET, [566](#)
 - STCF, [566](#)
- CFE_TIME_HousekeepingTlm_Payload_t
 - default_cfe_time_msgdefs.h, [1152](#)
- CFE_TIME_HousekeepingTlm_t
 - default_cfe_time_msgstruct.h, [1160](#)
- CFE_TIME_ID_ERR_EID
 - cfe_time_eventids.h, [1169](#)
- CFE_TIME_INIT_EID
 - cfe_time_eventids.h, [1170](#)
- cfe_time_interface_cfg.h
 - CFE_MISSION_TIME_AT_TONE_WAS, [1195](#)
 - CFE_MISSION_TIME_AT_TONE_WILL_BE, [1195](#)
 - CFE_MISSION_TIME_CFG_DEFAULT_TAI, [1195](#)
 - CFE_MISSION_TIME_CFG_DEFAULT_UTC, [1195](#)
 - CFE_MISSION_TIME_CFG_FAKE_TONE, [1195](#)

- CFE_MISSION_TIME_DEF_DELAY_SECS, [1196](#)
- CFE_MISSION_TIME_DEF_DELAY_SUBS, [1196](#)
- CFE_MISSION_TIME_DEF_LEAPS, [1196](#)
- CFE_MISSION_TIME_DEF_MET_SECS, [1196](#)
- CFE_MISSION_TIME_DEF_MET_SUBS, [1196](#)
- CFE_MISSION_TIME_DEF_STCF_SECS, [1197](#)
- CFE_MISSION_TIME_DEF_STCF_SUBS, [1197](#)
- CFE_MISSION_TIME_EPOCH_DAY, [1197](#)
- CFE_MISSION_TIME_EPOCH_HOUR, [1197](#)
- CFE_MISSION_TIME_EPOCH_MICROS, [1197](#)
- CFE_MISSION_TIME_EPOCH_MINUTE, [1197](#)
- CFE_MISSION_TIME_EPOCH_SECOND, [1197](#)
- CFE_MISSION_TIME_EPOCH_SECONDS, [1197](#)
- CFE_MISSION_TIME_EPOCH_YEAR, [1197](#)
- CFE_MISSION_TIME_FS_FACTOR, [1198](#)
- CFE_MISSION_TIME_MAX_ELAPSED, [1198](#)
- CFE_MISSION_TIME_MIN_ELAPSED, [1198](#)
- DEFAULT_CFE_MISSION_TIME_AT_TONE_WAS, [1198](#)
- DEFAULT_CFE_MISSION_TIME_AT_TONE_WILL_BE, [1199](#)
- DEFAULT_CFE_MISSION_TIME_CFG_DEFAULT_TAI, [1199](#)
- DEFAULT_CFE_MISSION_TIME_CFG_DEFAULT_UTC, [1199](#)
- DEFAULT_CFE_MISSION_TIME_CFG_FAKE_TONE, [1199](#)
- DEFAULT_CFE_MISSION_TIME_DEF_DELAY_SECS, [1199](#)
- DEFAULT_CFE_MISSION_TIME_DEF_DELAY_SUBS, [1199](#)
- DEFAULT_CFE_MISSION_TIME_DEF_LEAPS, [1199](#)
- DEFAULT_CFE_MISSION_TIME_DEF_MET_SECS, [1199](#)
- DEFAULT_CFE_MISSION_TIME_DEF_MET_SUBS, [1199](#)
- DEFAULT_CFE_MISSION_TIME_DEF_STCF_SECS, [1199](#)
- DEFAULT_CFE_MISSION_TIME_DEF_STCF_SUBS, [1199](#)
- DEFAULT_CFE_MISSION_TIME_EPOCH_DAY, [1200](#)
- DEFAULT_CFE_MISSION_TIME_EPOCH_HOUR, [1200](#)
- DEFAULT_CFE_MISSION_TIME_EPOCH_MICROS, [1200](#)
- DEFAULT_CFE_MISSION_TIME_EPOCH_MINUTE, [1200](#)
- DEFAULT_CFE_MISSION_TIME_EPOCH_SECOND, [1200](#)
- DEFAULT_CFE_MISSION_TIME_EPOCH_SECONDS, [1200](#)
- DEFAULT_CFE_MISSION_TIME_EPOCH_YEAR, [1200](#)
- DEFAULT_CFE_MISSION_TIME_FS_FACTOR, [1200](#)
- DEFAULT_CFE_MISSION_TIME_MAX_ELAPSED, [1200](#)
- DEFAULT_CFE_MISSION_TIME_MIN_ELAPSED, [1200](#)
- cfe_time_internal_cfg.h
 - CFE_PLATFORM_TIME_CFG_CLIENT, [1203](#)
 - CFE_PLATFORM_TIME_CFG_LATCH_FLY, [1203](#)
 - CFE_PLATFORM_TIME_CFG_SERVER, [1203](#)
 - CFE_PLATFORM_TIME_CFG_SIGNAL, [1204](#)
 - CFE_PLATFORM_TIME_CFG_SOURCE, [1204](#)
 - CFE_PLATFORM_TIME_CFG_SRC_GPS, [1204](#)
 - CFE_PLATFORM_TIME_CFG_SRC_MET, [1205](#)
 - CFE_PLATFORM_TIME_CFG_SRC_TIME, [1205](#)
 - CFE_PLATFORM_TIME_CFG_START_FLY, [1205](#)
 - CFE_PLATFORM_TIME_CFG_TONE_LIMIT, [1205](#)
 - CFE_PLATFORM_TIME_CFG_VIRTUAL, [1205](#)
 - CFE_PLATFORM_TIME_MAX_DELTA_SECS, [1206](#)
 - CFE_PLATFORM_TIME_MAX_DELTA_SUBS, [1206](#)
 - CFE_PLATFORM_TIME_MAX_LOCAL_SECS, [1206](#)
 - CFE_PLATFORM_TIME_MAX_LOCAL_SUBS, [1206](#)
 - CFE_PLATFORM_TIME_ONEHZ_TASK_PRIORITY, [1207](#)
 - CFE_PLATFORM_TIME_ONEHZ_TASK_STACK_SIZE, [1207](#)
 - CFE_PLATFORM_TIME_START_TASK_PRIORITY, [1207](#)
 - CFE_PLATFORM_TIME_START_TASK_STACK_SIZE, [1207](#)
 - CFE_PLATFORM_TIME_TONE_TASK_PRIORITY, [1207](#)
 - CFE_PLATFORM_TIME_TONE_TASK_STACK_SIZE, [1207](#)
 - DEFAULT_CFE_PLATFORM_TIME_CFG_CLIENT, [1208](#)
 - DEFAULT_CFE_PLATFORM_TIME_CFG_LATCH_FLY, [1208](#)
 - DEFAULT_CFE_PLATFORM_TIME_CFG_SERVER, [1208](#)
 - DEFAULT_CFE_PLATFORM_TIME_CFG_SIGNAL, [1208](#)
 - DEFAULT_CFE_PLATFORM_TIME_CFG_SOURCE, [1208](#)
 - DEFAULT_CFE_PLATFORM_TIME_CFG_SRC_GPS, [1208](#)
 - DEFAULT_CFE_PLATFORM_TIME_CFG_SRC_MET, [1208](#)
 - DEFAULT_CFE_PLATFORM_TIME_CFG_SRC_TIME, [1208](#)
 - DEFAULT_CFE_PLATFORM_TIME_CFG_START_FLY, [1208](#)
 - DEFAULT_CFE_PLATFORM_TIME_CFG_TONE_LIMIT, [1208](#)

- 1208
- DEFAULT_CFE_PLATFORM_TIME_CFG_VIRTUAL, 1208
- DEFAULT_CFE_PLATFORM_TIME_MAX_DELTA_SECS, 1209
- DEFAULT_CFE_PLATFORM_TIME_MAX_DELTA_SUBSECS, 1209
- DEFAULT_CFE_PLATFORM_TIME_MAX_LOCAL_SECONDS, 1209
- DEFAULT_CFE_PLATFORM_TIME_MAX_LOCAL_SUBSECS, 1209
- DEFAULT_CFE_PLATFORM_TIME_ONEHZ_TASK_PRIORITY, 1209
- DEFAULT_CFE_PLATFORM_TIME_ONEHZ_TASK_STACK_SIZE, 1209
- DEFAULT_CFE_PLATFORM_TIME_START_TASK_PRIORITY, 1209
- DEFAULT_CFE_PLATFORM_TIME_START_TASK_STACK_SIZE, 1209
- DEFAULT_CFE_PLATFORM_TIME_TONE_TASK_PRIORITY, 1209
- DEFAULT_CFE_PLATFORM_TIME_TONE_TASK_STACK_SIZE, 1209
- CFE_TIME_INTERNAL_ONLY
 - cFE Return Code Defines, 144
- CFE_TIME_LEAPS_CFG_EID
 - cfe_time_eventids.h, 1170
- CFE_TIME_LEAPS_EID
 - cfe_time_eventids.h, 1170
- CFE_TIME_LeapsCmd_Payload, 566
 - LeapSeconds, 567
- CFE_TIME_LeapsCmd_Payload_t
 - default_cfe_time_msgdefs.h, 1152
- CFE_TIME_LEN_ERR_EID
 - cfe_time_eventids.h, 1170
- CFE_TIME_Local1HzISR
 - cFE Miscellaneous Time APIs, 286
- CFE_TIME_MET2SCTime
 - cFE Time Conversion APIs, 280
- CFE_TIME_MET_CFG_EID
 - cfe_time_eventids.h, 1171
- CFE_TIME_MET_EID
 - cfe_time_eventids.h, 1171
- CFE_TIME_MET_ERR_EID
 - cfe_time_eventids.h, 1171
- CFE_TIME_Micro2SubSecs
 - cFE Time Conversion APIs, 280
- CFE_TIME_NOOP_CC
 - cfe_time_fcncodes.h, 1180
- CFE_TIME_NOOP_EID
 - cfe_time_eventids.h, 1171
- CFE_TIME_NoopCmd, 567
 - CommandHeader, 567
- CFE_TIME_NoopCmd_t
 - default_cfe_time_msgstruct.h, 1160
- CFE_RETURN_CODE_DEFINES, 144
 - cFE Return Code Defines, 144
- CFE_TIME_ONEHZ_CFG_EID
 - cfe_time_eventids.h, 1172
- CFE_TIME_ONEHZ_CMD_MID
 - default_cfe_time_msgids.h, 1157
- CFE_TIME_ONEHZ_EID
 - cfe_time_eventids.h, 1172
- CFE_TIME_OneHzAdjustmentCmd_Payload, 567
 - Seconds, 568
 - Subseconds, 568
- CFE_TIME_OneHzAdjustmentCmd_Payload_t
 - default_cfe_time_msgdefs.h, 1152
- CFE_TIME_OneHzCmd, 568
 - CommandHeader, 568
- CFE_TIME_OneHzCmd_t
 - default_cfe_time_msgstruct.h, 1160
- CFE_TIME_OUT_OF_RANGE
 - cFE Return Code Defines, 144
- CFE_TIME_Print
 - Miscellaneous Time APIs, 286
- CFE_TIME_PRINTED_STRING_SIZE
 - cfe_time_api_typedefs.h, 781
- CFE_TIME_RegisterSynchCallback
 - cFE External Time Source APIs, 284
- CFE_TIME_RESET_COUNTERS_CC
 - cfe_time_fcncodes.h, 1181
- CFE_TIME_RESET_EID
 - cfe_time_eventids.h, 1172
- CFE_TIME_ResetCountersCmd, 568
 - CommandHeader, 569
- CFE_TIME_ResetCountersCmd_t
 - default_cfe_time_msgstruct.h, 1160
- CFE_TIME_SEND_CMD_MID
 - default_cfe_time_msgids.h, 1157
- CFE_TIME_SEND_DIAGNOSTIC_CC
 - cfe_time_fcncodes.h, 1182
- CFE_TIME_SEND_HK_MID
 - default_cfe_time_msgids.h, 1158
- CFE_TIME_SendDiagnosticCmd, 569
 - CommandHeader, 569
- CFE_TIME_SendDiagnosticCmd_t
 - default_cfe_time_msgstruct.h, 1161
- CFE_TIME_SendHkCmd, 569
 - CommandHeader, 569
- CFE_TIME_SendHkCmd_t
 - default_cfe_time_msgstruct.h, 1161
- CFE_TIME_SERVICE
 - cfe_error.h, 709
- CFE_TIME_SET_LEAP_SECONDS_CC
 - cfe_time_fcncodes.h, 1183
- CFE_TIME_SET_MET_CC
 - cfe_time_fcncodes.h, 1184

CFE_TIME_SET_SIGNAL_CC
 cfe_time_fcncodes.h, 1185

CFE_TIME_SET_SOURCE_CC
 cfe_time_fcncodes.h, 1185

CFE_TIME_SET_STATE_CC
 cfe_time_fcncodes.h, 1186

CFE_TIME_SET_STCF_CC
 cfe_time_fcncodes.h, 1188

CFE_TIME_SET_TIME_CC
 cfe_time_fcncodes.h, 1188

CFE_TIME_SetLeapSecondsCmd, 570
 CommandHeader, 570
 Payload, 570

CFE_TIME_SetLeapSecondsCmd_t
 default_cfe_time_msgstruct.h, 1161

CFE_TIME_SetMETCmd, 570
 CommandHeader, 570
 Payload, 570

CFE_TIME_SetMETCmd_t
 default_cfe_time_msgstruct.h, 1161

CFE_TIME_SetSignalCmd, 571
 CommandHeader, 571
 Payload, 571

CFE_TIME_SetSignalCmd_t
 default_cfe_time_msgstruct.h, 1161

CFE_TIME_SetSourceCmd, 571
 CommandHeader, 572
 Payload, 572

CFE_TIME_SetSourceCmd_t
 default_cfe_time_msgstruct.h, 1161

CFE_TIME_SetState
 default_cfe_time_extern_typedefs.h, 1143

CFE_TIME_SetState_Enum_t
 default_cfe_time_extern_typedefs.h, 1141

CFE_TIME_SetState_NOT_SET
 default_cfe_time_extern_typedefs.h, 1143

CFE_TIME_SetState_WAS_SET
 default_cfe_time_extern_typedefs.h, 1143

CFE_TIME_SetStateCmd, 572
 CommandHeader, 572
 Payload, 572

CFE_TIME_SetStateCmd_t
 default_cfe_time_msgstruct.h, 1161

CFE_TIME_SetSTCFCmd, 573
 CommandHeader, 573
 Payload, 573

CFE_TIME_SetSTCFCmd_t
 default_cfe_time_msgstruct.h, 1161

CFE_TIME_SetTimeCmd, 573
 CommandHeader, 573
 Payload, 573

CFE_TIME_SetTimeCmd_t
 default_cfe_time_msgstruct.h, 1161

CFE_TIME_SIGNAL_CFG_EID
 cfe_time_eventids.h, 1172

CFE_TIME_SIGNAL_EID
 cfe_time_eventids.h, 1173

CFE_TIME_SIGNAL_ERR_EID
 cfe_time_eventids.h, 1173

CFE_TIME_SignalCmd_Payload, 574
 ToneSource, 574

CFE_TIME_SignalCmd_Payload_t
 default_cfe_time_msgdefs.h, 1152

CFE_TIME_SOURCE_CFG_EID
 cfe_time_eventids.h, 1173

CFE_TIME_SOURCE_EID
 cfe_time_eventids.h, 1173

CFE_TIME_SOURCE_ERR_EID
 cfe_time_eventids.h, 1174

CFE_TIME_SourceCmd_Payload, 574
 TimeSource, 575

CFE_TIME_SourceCmd_Payload_t
 default_cfe_time_msgdefs.h, 1152

CFE_TIME_SourceSelect
 default_cfe_time_extern_typedefs.h, 1143

CFE_TIME_SourceSelect_Enum_t
 default_cfe_time_extern_typedefs.h, 1141

CFE_TIME_SourceSelect_EXTERNAL
 default_cfe_time_extern_typedefs.h, 1143

CFE_TIME_SourceSelect_INTERNAL
 default_cfe_time_extern_typedefs.h, 1143

CFE_TIME_STATE_EID
 cfe_time_eventids.h, 1174

CFE_TIME_STATE_ERR_EID
 cfe_time_eventids.h, 1174

CFE_TIME_StateCmd_Payload, 575
 ClockState, 575

CFE_TIME_StateCmd_Payload_t
 default_cfe_time_msgdefs.h, 1153

CFE_TIME_STCF_CFG_EID
 cfe_time_eventids.h, 1174

CFE_TIME_STCF_EID
 cfe_time_eventids.h, 1175

CFE_TIME_STCF_ERR_EID
 cfe_time_eventids.h, 1175

CFE_TIME_Sub2MicroSecs
 cFE Time Conversion APIs, 281

CFE_TIME_SUB_ADJUST_CC
 cfe_time_fcncodes.h, 1189

CFE_TIME_SUB_DELAY_CC
 cfe_time_fcncodes.h, 1190

CFE_TIME_SUB_ONE_HZ_ADJUSTMENT_CC
 cfe_time_fcncodes.h, 1191

CFE_TIME_SubAdjustCmd, 575
 CommandHeader, 576
 Payload, 576

CFE_TIME_SubAdjustCmd_t
 default_cfe_time_msgstruct.h, 1161

- CFE_TIME_SubDelayCmd, 576
 - CommandHeader, 576
 - Payload, 576
- CFE_TIME_SubDelayCmd_t
 - default_cfe_time_msgstruct.h, 1161
- CFE_TIME_SubOneHzAdjustmentCmd, 576
 - CommandHeader, 577
 - Payload, 577
- CFE_TIME_SubOneHzAdjustmentCmd_t
 - default_cfe_time_msgstruct.h, 1161
- CFE_TIME_Subtract
 - cFE Time Arithmetic APIs, 279
- CFE_TIME_SynchCallbackPtr_t
 - cfe_time_api_typedefs.h, 781
- CFE_TIME_SysTime, 577
 - Seconds, 578
 - Subseconds, 578
- CFE_TIME_SysTime_t
 - default_cfe_time_extern_typedefs.h, 1141
- CFE_TIME_TIME_CFG_EID
 - cfe_time_eventids.h, 1175
- CFE_TIME_TIME_EID
 - cfe_time_eventids.h, 1175
- CFE_TIME_TIME_ERR_EID
 - cfe_time_eventids.h, 1176
- CFE_TIME_TimeCmd_Payload, 578
 - MicroSeconds, 578
 - Seconds, 578
- CFE_TIME_TimeCmd_Payload_t
 - default_cfe_time_msgdefs.h, 1153
- CFE_TIME_TONE_CMD_MID
 - default_cfe_time_msgids.h, 1158
- CFE_TIME_ToneDataCmd, 578
 - CommandHeader, 579
 - Payload, 579
- CFE_TIME_ToneDataCmd_Payload, 579
 - AtToneLeapSeconds, 579
 - AtToneMET, 579
 - AtToneState, 579
 - AtToneSTCF, 580
- CFE_TIME_ToneDataCmd_Payload_t
 - default_cfe_time_msgdefs.h, 1153
- CFE_TIME_ToneDataCmd_t
 - default_cfe_time_msgstruct.h, 1161
- CFE_TIME_ToneSignalCmd, 580
 - CommandHeader, 580
- CFE_TIME_ToneSignalCmd_t
 - default_cfe_time_msgstruct.h, 1162
- CFE_TIME_ToneSignalSelect
 - default_cfe_time_extern_typedefs.h, 1143
- CFE_TIME_ToneSignalSelect_Enum_t
 - default_cfe_time_extern_typedefs.h, 1141
- CFE_TIME_ToneSignalSelect_PRIMARY
 - default_cfe_time_extern_typedefs.h, 1144
- CFE_TIME_ToneSignalSelect_REDUNDANT
 - default_cfe_time_extern_typedefs.h, 1144
- CFE_TIME_TOO_MANY_SYNCH_CALLBACKS
 - cFE Return Code Defines, 144
- cfe_time_topicids.h
 - CFE_MISSION_TIME_CMD_TOPICID, 1211
 - CFE_MISSION_TIME_DATA_CMD_TOPICID, 1211
 - CFE_MISSION_TIME_DIAG_TLM_TOPICID, 1212
 - CFE_MISSION_TIME_HK_TLM_TOPICID, 1212
 - CFE_MISSION_TIME_ONEHZ_CMD_TOPICID, 1212
 - CFE_MISSION_TIME_SEND_CMD_TOPICID, 1212
 - CFE_MISSION_TIME_SEND_HK_TOPICID, 1212
 - CFE_MISSION_TIME_TONE_CMD_TOPICID, 1212
 - DEFAULT_CFE_MISSION_TIME_CMD_TOPICID, 1212
 - DEFAULT_CFE_MISSION_TIME_DATA_CMD_TOPICID, 1213
 - DEFAULT_CFE_MISSION_TIME_DIAG_TLM_TOPICID, 1213
 - DEFAULT_CFE_MISSION_TIME_HK_TLM_TOPICID, 1213
 - DEFAULT_CFE_MISSION_TIME_ONEHZ_CMD_TOPICID, 1213
 - DEFAULT_CFE_MISSION_TIME_SEND_CMD_TOPICID, 1213
 - DEFAULT_CFE_MISSION_TIME_SEND_HK_TOPICID, 1213
 - DEFAULT_CFE_MISSION_TIME_TONE_CMD_TOPICID, 1213
- CFE_TIME_UnregisterSynchCallback
 - cFE External Time Source APIs, 285
- CFE_TIME_ZERO_VALUE
 - cfe_time_api_typedefs.h, 781
- CFE_TST
 - cfe_sb.h, 760
- cfe_version.h
 - CFE_BUILD_BASELINE, 783
 - CFE_BUILD_CODENAME, 783
 - CFE_BUILD_DEV_CYCLE, 784
 - CFE_BUILD_NUMBER, 784
 - CFE_CFG_MAX_VERSION_STR_LEN, 784
 - CFE_LAST_OFFICIAL, 784
 - CFE_MAJOR_VERSION, 784
 - CFE_MINOR_VERSION, 784
 - CFE_MISSION_REV, 784
 - CFE_REVISION, 784
 - CFE_SRC_VERSION, 784
 - CFE_STR, 785
 - CFE_STR_HELPER, 785
- CFECoreChecksum
 - CFE_ES_HousekeepingTlm_Payload, 440
- CFEMajorVersion
 - CFE_ES_HousekeepingTlm_Payload, 440

- CFEMinorVersion
 - CFE_ES_HousekeepingTlm_Payload, [441](#)
- CFEMissionRevision
 - CFE_ES_HousekeepingTlm_Payload, [441](#)
- CFERevision
 - CFE_ES_HousekeepingTlm_Payload, [441](#)
- CheckErrCtr
 - CFE_ES_MemPoolStats, [448](#)
- Client, [70](#), [71](#)
- ClockFlyState
 - CFE_TIME_DiagnosticTlm_Payload, [558](#)
- ClockSetState
 - CFE_TIME_DiagnosticTlm_Payload, [558](#)
- ClockSignal
 - CFE_TIME_DiagnosticTlm_Payload, [558](#)
- ClockSource
 - CFE_TIME_DiagnosticTlm_Payload, [558](#)
- ClockState
 - CFE_TIME_StateCmd_Payload, [575](#)
- ClockStateAPI
 - CFE_TIME_DiagnosticTlm_Payload, [558](#)
 - CFE_TIME_HousekeepingTlm_Payload, [565](#)
- ClockStateFlags
 - CFE_TIME_DiagnosticTlm_Payload, [558](#)
 - CFE_TIME_HousekeepingTlm_Payload, [565](#)
- code_address
 - OS_module_address_t, [587](#)
- code_size
 - OS_module_address_t, [587](#)
- CodeAddress
 - CFE_ES_AppInfo, [427](#)
- CodeSize
 - CFE_ES_AppInfo, [427](#)
- CommandCounter
 - CFE_ES_HousekeepingTlm_Payload, [441](#)
 - CFE_EVS_HousekeepingTlm_Payload, [485](#)
 - CFE_SB_HousekeepingTlm_Payload, [506](#)
 - CFE_TBL_HousekeepingTlm_Payload, [538](#)
 - CFE_TIME_HousekeepingTlm_Payload, [565](#)
- CommandErrorCounter
 - CFE_ES_HousekeepingTlm_Payload, [441](#)
 - CFE_EVS_HousekeepingTlm_Payload, [485](#)
 - CFE_SB_HousekeepingTlm_Payload, [506](#)
 - CFE_TBL_HousekeepingTlm_Payload, [538](#)
 - CFE_TIME_HousekeepingTlm_Payload, [566](#)
- CommandHeader
 - CFE_ES_ClearERLogCmd, [434](#)
 - CFE_ES_ClearSysLogCmd, [434](#)
 - CFE_ES_DeleteCDSCmd, [435](#)
 - CFE_ES_DumpCDSRegistryCmd, [436](#)
 - CFE_ES_FileNameCmd, [437](#)
 - CFE_ES_NoopCmd, [450](#)
 - CFE_ES_OverWriteSysLogCmd, [451](#)
 - CFE_ES_QueryAllCmd, [454](#)
 - CFE_ES_QueryAllTasksCmd, [455](#)
 - CFE_ES_QueryOneCmd, [455](#)
 - CFE_ES_ReloadAppCmd, [456](#)
 - CFE_ES_ResetCountersCmd, [456](#)
 - CFE_ES_ResetPRCountCmd, [457](#)
 - CFE_ES_RestartAppCmd, [457](#)
 - CFE_ES_RestartCmd, [458](#)
 - CFE_ES_SendHkCmd, [459](#)
 - CFE_ES_SendMemPoolStatsCmd, [459](#)
 - CFE_ES_SetMaxPRCountCmd, [461](#)
 - CFE_ES_SetPerfFilterMaskCmd, [462](#)
 - CFE_ES_SetPerfTriggerMaskCmd, [463](#)
 - CFE_ES_StartApp, [464](#)
 - CFE_ES_StartPerfDataCmd, [467](#)
 - CFE_ES_StopAppCmd, [467](#)
 - CFE_ES_StopPerfDataCmd, [469](#)
 - CFE_ES_WriteERLogCmd, [471](#)
 - CFE_ES_WriteSysLogCmd, [471](#)
 - CFE_EVS_AddEventFilterCmd, [472](#)
 - CFE_EVS_ClearLogCmd, [478](#)
 - CFE_EVS_DeleteEventFilterCmd, [479](#)
 - CFE_EVS_DisableAppEventsCmd, [479](#)
 - CFE_EVS_DisableAppEventTypeCmd, [480](#)
 - CFE_EVS_DisableEventTypeCmd, [480](#)
 - CFE_EVS_DisablePortsCmd, [481](#)
 - CFE_EVS_EnableAppEventsCmd, [482](#)
 - CFE_EVS_EnableAppEventTypeCmd, [482](#)
 - CFE_EVS_EnableEventTypeCmd, [483](#)
 - CFE_EVS_EnablePortsCmd, [483](#)
 - CFE_EVS_NoopCmd, [489](#)
 - CFE_EVS_ResetAllFiltersCmd, [491](#)
 - CFE_EVS_ResetAppCounterCmd, [491](#)
 - CFE_EVS_ResetCountersCmd, [492](#)
 - CFE_EVS_ResetFilterCmd, [492](#)
 - CFE_EVS_SendHkCmd, [493](#)
 - CFE_EVS_SetEventFormatModeCmd, [494](#)
 - CFE_EVS_SetFilterCmd, [495](#)
 - CFE_EVS_SetLogModeCmd, [496](#)
 - CFE_EVS_WriteAppDataFileCmd, [498](#)
 - CFE_EVS_WriteLogDataFileCmd, [498](#)
 - CFE_SB_DisableRouteCmd, [503](#)
 - CFE_SB_DisableSubReportingCmd, [504](#)
 - CFE_SB_EnableRouteCmd, [504](#)
 - CFE_SB_EnableSubReportingCmd, [505](#)
 - CFE_SB_NoopCmd, [511](#)
 - CFE_SB_ResetCountersCmd, [515](#)
 - CFE_SB_SendHkCmd, [518](#)
 - CFE_SB_SendPrevSubsCmd, [518](#)
 - CFE_SB_SendSbStatsCmd, [518](#)
 - CFE_SB_WriteMapInfoCmd, [526](#)
 - CFE_SB_WritePipeInfoCmd, [526](#)
 - CFE_SB_WriteRoutingInfoCmd, [527](#)
 - CFE_TBL_AbortLoadCmd, [527](#)
 - CFE_TBL_ActivateCmd, [529](#)

- CFE_TBL_DeleteCDSCmd, 531
- CFE_TBL_DumpCmd, 531
- CFE_TBL_DumpRegistryCmd, 533
- CFE_TBL_LoadCmd, 543
- CFE_TBL_NoopCmd, 544
- CFE_TBL_NotifyCmd, 545
- CFE_TBL_ResetCountersCmd, 546
- CFE_TBL_SendHkCmd, 546
- CFE_TBL_SendRegistryCmd, 547
- CFE_TBL_ValidateCmd, 552
- CFE_TIME_AddAdjustCmd, 553
- CFE_TIME_AddDelayCmd, 553
- CFE_TIME_AddOneHzAdjustmentCmd, 554
- CFE_TIME_FakeToneCmd, 564
- CFE_TIME_NoopCmd, 567
- CFE_TIME_OneHzCmd, 568
- CFE_TIME_ResetCountersCmd, 569
- CFE_TIME_SendDiagnosticCmd, 569
- CFE_TIME_SendHkCmd, 569
- CFE_TIME_SetLeapSecondsCmd, 570
- CFE_TIME_SetMETCmd, 570
- CFE_TIME_SetSignalCmd, 571
- CFE_TIME_SetSourceCmd, 572
- CFE_TIME_SetStateCmd, 572
- CFE_TIME_SetSTCFCmd, 573
- CFE_TIME_SetTimeCmd, 573
- CFE_TIME_SubAdjustCmd, 576
- CFE_TIME_SubDelayCmd, 576
- CFE_TIME_SubOneHzAdjustmentCmd, 577
- CFE_TIME_ToneDataCmd, 579
- CFE_TIME_ToneSignalCmd, 580
- common_types.h
 - _EXTENSION_, 1215
 - CompileTimeAssert, 1215, 1218, 1219
 - cpuaddr, 1216
 - cpudiff, 1216
 - cpusize, 1216
 - int16, 1216
 - int32, 1216
 - int64, 1216
 - int8, 1217
 - intptr, 1217
 - OS_ArgCallback_t, 1217
 - OS_PRINTF, 1215
 - OS_USED, 1215
 - OSAL_BLOCKCOUNT_C, 1215
 - osal_blockcount_t, 1217
 - osal_id_t, 1217
 - OSAL_INDEX_C, 1216
 - osal_index_t, 1217
 - OSAL_OBJTYPE_C, 1216
 - osal_objtype_t, 1217
 - osal_offset_t, 1217
 - OSAL_SIZE_C, 1216
 - OSAL_STATUS_C, 1216
 - osal_status_t, 1217
 - uint16, 1218
 - uint32, 1218
 - uint64, 1218
 - uint8, 1218
- CompileTimeAssert
 - common_types.h, 1215, 1218, 1219
- ContentType
 - CFE_FS_Header, 500
- Control of Packet Routing, 42
- Core Flight Executive Documentation, 1
- cpuaddr
 - common_types.h, 1216
- cpudiff
 - common_types.h, 1216
- cpusize
 - common_types.h, 1216
- Crc
 - CFE_TBL_Info, 541
 - CFE_TBL_TblRegPacket_Payload, 549
- CreatePipeErrorCounter
 - CFE_SB_HousekeepingTlm_Payload, 506
- creator
 - OS_bin_sem_prop_t, 581
 - OS_condvar_prop_t, 581
 - OS_count_sem_prop_t, 582
 - OS_mut_sem_prop_t, 588
 - OS_queue_prop_t, 589
 - OS_rwlock_prop_t, 590
 - OS_socket_prop_t, 592
 - OS_task_prop_t, 594
 - OS_timebase_prop_t, 595
 - OS_timer_prop_t, 596
- Critical
 - CFE_TBL_Info, 541
 - CFE_TBL_TblRegPacket_Payload, 549
- Critical Data Store, 15
- Critical Tables, 50
- CurrentLatch
 - CFE_TIME_DiagnosticTlm_Payload, 559
- CurrentMET
 - CFE_TIME_DiagnosticTlm_Payload, 559
- CurrentQueueDepth
 - CFE_SB_PipeDepthStats, 512
 - CFE_SB_PipeInfoEntry, 513
- CurrentTAI
 - CFE_TIME_DiagnosticTlm_Payload, 559
- CurrentUTC
 - CFE_TIME_DiagnosticTlm_Payload, 559
- data_address
 - OS_module_address_t, 587
- data_size

- OS_module_address_t, [587](#)
- DataAddress
 - CFE_ES_AppInfo, [428](#)
- DataFileName
 - CFE_ES_StopPerfCmd_Payload, [468](#)
- DataSize
 - CFE_ES_AppInfo, [428](#)
- DataStoreStatus
 - CFE_TIME_DiagnosticTlm_Payload, [559](#)
- Datum
 - CFE_Config_ValueEntry, [425](#)
- default_cfe_core_api_base_msgid_values.h
 - CFE_GLOBAL_BASE_MIDVAL, [683](#)
 - CFE_PLATFORM_BASE_MIDVAL, [684](#)
- default_cfe_core_api_interface_cfg_values.h
 - CFE_MISSION_CORE_API_CFGVAL, [684](#)
- default_cfe_core_api_msgid_mapping.h
 - CFE_GLOBAL_CMD_TOPICID_TO_MIDV, [685](#)
 - CFE_GLOBAL_TLM_TOPICID_TO_MIDV, [686](#)
 - CFE_PLATFORM_CMD_TOPICID_TO_MIDV, [686](#)
 - CFE_PLATFORM_TLM_TOPICID_TO_MIDV, [686](#)
 - DEFAULT_CFE_PLATFORM_CMD_MID_BASE, [686](#)
 - DEFAULT_CFE_PLATFORM_TLM_MID_BASE, [686](#)
 - DEFAULT_GLOBAL_CMD_MID_BASE, [686](#)
 - DEFAULT_GLOBAL_TLM_MID_BASE, [686](#)
- default_cfe_es_extern_typedefs.h
 - CFE_ES_AppId_t, [788](#)
 - CFE_ES_AppInfo_t, [788](#)
 - CFE_ES_AppState, [791](#)
 - CFE_ES_AppState_EARLY_INIT, [791](#)
 - CFE_ES_AppState_Enum_t, [788](#)
 - CFE_ES_AppState_LATE_INIT, [791](#)
 - CFE_ES_AppState_MAX, [791](#)
 - CFE_ES_AppState_RUNNING, [791](#)
 - CFE_ES_AppState_STOPPED, [791](#)
 - CFE_ES_AppState_UNDEFINED, [791](#)
 - CFE_ES_AppState_WAITING, [791](#)
 - CFE_ES_AppType, [791](#)
 - CFE_ES_AppType_CORE, [791](#)
 - CFE_ES_AppType_Enum_t, [788](#)
 - CFE_ES_AppType_EXTERNAL, [791](#)
 - CFE_ES_AppType_LIBRARY, [791](#)
 - CFE_ES_BlockStats_t, [788](#)
 - CFE_ES_CDSHandle_t, [788](#)
 - CFE_ES_CDSRegDumpRec_t, [788](#)
 - CFE_ES_CounterId_t, [789](#)
 - CFE_ES_ExceptionAction, [791](#)
 - CFE_ES_ExceptionAction_Enum_t, [789](#)
 - CFE_ES_ExceptionAction_PROC_RESTART, [792](#)
 - CFE_ES_ExceptionAction_RESTART_APP, [792](#)
 - CFE_ES_LibId_t, [789](#)
 - CFE_ES_LogEntryType, [792](#)
 - CFE_ES_LogEntryType_APPLICATION, [792](#)
 - CFE_ES_LogEntryType_CORE, [792](#)
 - CFE_ES_LogEntryType_Enum_t, [789](#)
 - CFE_ES_LogMode, [792](#)
 - CFE_ES_LogMode_DISCARD, [792](#)
 - CFE_ES_LogMode_Enum_t, [789](#)
 - CFE_ES_LogMode_OVERWRITE, [792](#)
 - CFE_ES_MemHandle_t, [789](#)
 - CFE_ES_MemPoolStats_t, [790](#)
 - CFE_ES_RunStatus, [792](#)
 - CFE_ES_RunStatus_APP_ERROR, [792](#)
 - CFE_ES_RunStatus_APP_EXIT, [792](#)
 - CFE_ES_RunStatus_APP_RUN, [792](#)
 - CFE_ES_RunStatus_CORE_APP_INIT_ERROR, [792](#)
 - CFE_ES_RunStatus_CORE_APP_RUNTIME_ERROR, [792](#)
 - CFE_ES_RunStatus_Enum_t, [790](#)
 - CFE_ES_RunStatus_MAX, [792](#)
 - CFE_ES_RunStatus_SYS_DELETE, [792](#)
 - CFE_ES_RunStatus_SYS_EXCEPTION, [792](#)
 - CFE_ES_RunStatus_SYS_RELOAD, [792](#)
 - CFE_ES_RunStatus_SYS_RESTART, [792](#)
 - CFE_ES_RunStatus_UNDEFINED, [792](#)
 - CFE_ES_SystemState, [792](#)
 - CFE_ES_SystemState_APPS_INIT, [793](#)
 - CFE_ES_SystemState_CORE_READY, [793](#)
 - CFE_ES_SystemState_CORE_STARTUP, [793](#)
 - CFE_ES_SystemState_EARLY_INIT, [793](#)
 - CFE_ES_SystemState_Enum_t, [790](#)
 - CFE_ES_SystemState_MAX, [793](#)
 - CFE_ES_SystemState_OPERATIONAL, [793](#)
 - CFE_ES_SystemState_SHUTDOWN, [793](#)
 - CFE_ES_SystemState_UNDEFINED, [793](#)
 - CFE_ES_TaskId_t, [790](#)
 - CFE_ES_TaskInfo_t, [790](#)
 - CFE_ES_TaskPriority_Atom_t, [791](#)
- default_cfe_es_fcncode_values.h
 - CFE_ES_CCVAL, [796](#)
 - CFE_ES_FunctionCode, [796](#)
 - CFE_ES_FunctionCode_CLEAR_ER_LOG, [797](#)
 - CFE_ES_FunctionCode_CLEAR_SYS_LOG, [797](#)
 - CFE_ES_FunctionCode_DELETE_CDS, [797](#)
 - CFE_ES_FunctionCode_DUMP_CDS_REGISTRY, [797](#)
 - CFE_ES_FunctionCode_NOOP, [796](#)
 - CFE_ES_FunctionCode_OVER_WRITE_SYS_LOG, [797](#)
 - CFE_ES_FunctionCode_QUERY_ALL, [797](#)
 - CFE_ES_FunctionCode_QUERY_ALL_TASKS, [797](#)
 - CFE_ES_FunctionCode_QUERY_ONE, [797](#)
 - CFE_ES_FunctionCode_RELOAD_APP, [797](#)
 - CFE_ES_FunctionCode_RESET_COUNTERS, [797](#)
 - CFE_ES_FunctionCode_RESET_PR_COUNT, [797](#)
 - CFE_ES_FunctionCode_RESTART, [797](#)
 - CFE_ES_FunctionCode_RESTART_APP, [797](#)

- CFE_ES_FunctionCode_SEND_MEM_POOL_STATS, 797
- CFE_ES_FunctionCode_SET_MAX_PR_COUNT, 797
- CFE_ES_FunctionCode_SET_PERF_FILTER_MASK, 797
- CFE_ES_FunctionCode_SET_PERF_TRIGGER_MASK, 797
- CFE_ES_FunctionCode_START_APP, 797
- CFE_ES_FunctionCode_START_PERF_DATA, 797
- CFE_ES_FunctionCode_STOP_APP, 797
- CFE_ES_FunctionCode_STOP_PERF_DATA, 797
- CFE_ES_FunctionCode_WRITE_ER_LOG, 797
- CFE_ES_FunctionCode_WRITE_SYS_LOG, 797
- default_cfe_es_interface_cfg_values.h
 - CFE_MISSION_ES_CFGVAL, 798
- default_cfe_es_internal_cfg_values.h
 - CFE_PLATFORM_ES_CFGVAL, 799
- default_cfe_es_memaddress.h
 - CFE_ES_MEMADDRESS_C, 800
 - CFE_ES_MemAddress_t, 801
 - CFE_ES_MEMADDRESS_TO_PTR, 800
 - CFE_ES_MEMOFFSET_C, 800
 - CFE_ES_MemOffset_t, 801
 - CFE_ES_MEMOFFSET_TO_SIZET, 801
- default_cfe_es_msgdefs.h
 - CFE_ES_AppNameCmd_Payload_t, 805
 - CFE_ES_AppReloadCmd_Payload_t, 805
 - CFE_ES_DeleteCDSCmd_Payload_t, 805
 - CFE_ES_DumpCDSRegistryCmd_Payload_t, 805
 - CFE_ES_FileNameCmd_Payload_t, 805
 - CFE_ES_HousekeepingTlm_Payload_t, 805
 - CFE_ES_OneAppTlm_Payload_t, 806
 - CFE_ES_OverWriteSysLogCmd_Payload_t, 806
 - CFE_ES_PerfMode, 807
 - CFE_ES_PerfMode_Enum_t, 806
 - CFE_ES_PerfTrigger_CENTER, 807
 - CFE_ES_PerfTrigger_END, 807
 - CFE_ES_PerfTrigger_START, 807
 - CFE_ES_PoolStatsTlm_Payload_t, 806
 - CFE_ES_RestartCmd_Payload_t, 806
 - CFE_ES_SendMemPoolStatsCmd_Payload_t, 806
 - CFE_ES_SetMaxPRCountCmd_Payload_t, 806
 - CFE_ES_SetPerfFilterMaskCmd_Payload_t, 806
 - CFE_ES_SetPerfTrigMaskCmd_Payload_t, 806
 - CFE_ES_StartAppCmd_Payload_t, 807
 - CFE_ES_StartPerfCmd_Payload_t, 807
 - CFE_ES_StopPerfCmd_Payload_t, 807
- default_cfe_es_msgid_values.h
 - CFE_PLATFORM_ES_CMD_MIDVAL, 810
 - CFE_PLATFORM_ES_TLM_MIDVAL, 810
- default_cfe_es_msgids.h
 - CFE_ES_APP_TLM_MID, 811
 - CFE_ES_CMD_MID, 811
 - CFE_ES_HK_TLM_MID, 811
 - CFE_ES_MEMSTATS_TLM_MID, 811
 - CFE_ES_SEND_HK_MID, 811
- default_cfe_es_msgstruct.h
 - CFE_ES_ClearERLogCmd_t, 814
 - CFE_ES_ClearSysLogCmd_t, 814
 - CFE_ES_DeleteCDSCmd_t, 814
 - CFE_ES_DumpCDSRegistryCmd_t, 814
 - CFE_ES_FileNameCmd_t, 814
 - CFE_ES_HousekeepingTlm_t, 814
 - CFE_ES_MemStatsTlm_t, 815
 - CFE_ES_NoopCmd_t, 815
 - CFE_ES_OneAppTlm_t, 815
 - CFE_ES_OverWriteSysLogCmd_t, 815
 - CFE_ES_QueryAllCmd_t, 815
 - CFE_ES_QueryAllTasksCmd_t, 815
 - CFE_ES_QueryOneCmd_t, 815
 - CFE_ES_ReloadAppCmd_t, 815
 - CFE_ES_ResetCountersCmd_t, 815
 - CFE_ES_ResetPRCountCmd_t, 815
 - CFE_ES_RestartAppCmd_t, 815
 - CFE_ES_RestartCmd_t, 815
 - CFE_ES_SendHkCmd_t, 816
 - CFE_ES_SendMemPoolStatsCmd_t, 816
 - CFE_ES_SetMaxPRCountCmd_t, 816
 - CFE_ES_SetPerfFilterMaskCmd_t, 816
 - CFE_ES_SetPerfTriggerMaskCmd_t, 816
 - CFE_ES_StartAppCmd_t, 816
 - CFE_ES_StartPerfDataCmd_t, 816
 - CFE_ES_StopAppCmd_t, 816
 - CFE_ES_StopPerfDataCmd_t, 816
 - CFE_ES_WriteERLogCmd_t, 816
 - CFE_ES_WriteSysLogCmd_t, 816
- default_cfe_es_topicid_values.h
 - CFE_MISSION_ES_TIDVAL, 820
- default_cfe_evs_extern_typedefs.h
 - CFE_EVS_EventFilter, 913
 - CFE_EVS_EventFilter_BINARY, 913
 - CFE_EVS_EventFilter_Enum_t, 912
 - CFE_EVS_EventOutput, 913
 - CFE_EVS_EventOutput_Enum_t, 912
 - CFE_EVS_EventOutput_PORT1, 913
 - CFE_EVS_EventOutput_PORT2, 913
 - CFE_EVS_EventOutput_PORT3, 913
 - CFE_EVS_EventOutput_PORT4, 913
 - CFE_EVS_EventType, 914
 - CFE_EVS_EventType_CRITICAL, 914
 - CFE_EVS_EventType_DEBUG, 914
 - CFE_EVS_EventType_Enum_t, 912
 - CFE_EVS_EventType_ERROR, 914
 - CFE_EVS_EventType_INFORMATION, 914
 - CFE_EVS_LogMode, 914
 - CFE_EVS_LogMode_DISCARD, 914
 - CFE_EVS_LogMode_Enum_t, 913

- CFE_EVS_LogMode_OVERWRITE, 914
- CFE_EVS_MsgFormat, 914
- CFE_EVS_MsgFormat_Enum_t, 913
- CFE_EVS_MsgFormat_LONG, 914
- CFE_EVS_MsgFormat_SHORT, 914
- default_cfe_evs_fcncode_values.h
 - CFE_EVS_CCVAL, 916
 - CFE_EVS_FunctionCode_, 916
 - CFE_EVS_FunctionCode_ADD_EVENT_FILTER, 916
 - CFE_EVS_FunctionCode_CLEAR_LOG, 917
 - CFE_EVS_FunctionCode_DELETE_EVENT_FILTER, 916
 - CFE_EVS_FunctionCode_DISABLE_APP_EVENT_TYPE, 916
 - CFE_EVS_FunctionCode_DISABLE_APP_EVENTS, 916
 - CFE_EVS_FunctionCode_DISABLE_EVENT_TYPE, 916
 - CFE_EVS_FunctionCode_DISABLE_PORTS, 916
 - CFE_EVS_FunctionCode_ENABLE_APP_EVENT_TYPE, 916
 - CFE_EVS_FunctionCode_ENABLE_APP_EVENTS, 916
 - CFE_EVS_FunctionCode_ENABLE_EVENT_TYPE, 916
 - CFE_EVS_FunctionCode_ENABLE_PORTS, 916
 - CFE_EVS_FunctionCode_NOOP, 916
 - CFE_EVS_FunctionCode_RESET_ALL_FILTERS, 916
 - CFE_EVS_FunctionCode_RESET_APP_COUNTER, 916
 - CFE_EVS_FunctionCode_RESET_COUNTERS, 916
 - CFE_EVS_FunctionCode_RESET_FILTER, 916
 - CFE_EVS_FunctionCode_SET_EVENT_FORMAT_MODE, 916
 - CFE_EVS_FunctionCode_SET_FILTER, 916
 - CFE_EVS_FunctionCode_SET_LOG_MODE, 917
 - CFE_EVS_FunctionCode_WRITE_APP_DATA_FILE, 917
 - CFE_EVS_FunctionCode_WRITE_LOG_DATA_FILE, 917
- default_cfe_evs_interface_cfg_values.h
 - CFE_MISSION_EVS_CFGVAL, 918
- default_cfe_evs_internal_cfg_values.h
 - CFE_PLATFORM_EVS_CFGVAL, 919
- default_cfe_evs_msgdefs.h
 - CFE_EVS_AppDataCmd_Payload_t, 923
 - CFE_EVS_AppNameBitMaskCmd_Payload_t, 923
 - CFE_EVS_AppNameCmd_Payload_t, 923
 - CFE_EVS_AppNameEventIDCmd_Payload_t, 923
 - CFE_EVS_AppNameEventIDMaskCmd_Payload_t, 923
 - CFE_EVS_AppTlmData_t, 923
 - CFE_EVS_BitMaskCmd_Payload_t, 924
 - CFE_EVS_CRITICAL_BIT, 922
 - CFE_EVS_DEBUG_BIT, 922
 - CFE_EVS_ERROR_BIT, 922
 - CFE_EVS_HousekeepingTlm_Payload_t, 924
 - CFE_EVS_INFORMATION_BIT, 922
 - CFE_EVS_LogFileCmd_Payload_t, 924
 - CFE_EVS_LongEventTlm_Payload_t, 924
 - CFE_EVS_PacketID_t, 924
 - CFE_EVS_PORT1_BIT, 922
 - CFE_EVS_PORT2_BIT, 923
 - CFE_EVS_PORT3_BIT, 923
 - CFE_EVS_PORT4_BIT, 923
 - CFE_EVS_SetEventFormatMode_Payload_t, 924
 - CFE_EVS_SetLogMode_Payload_t, 924
 - CFE_EVS_ShortEventTlm_Payload_t, 924
- default_cfe_evs_msgid_values.h
 - CFE_PLATFORM_EVS_CMD_MIDVAL, 927
 - CFE_PLATFORM_EVS_TLM_MIDVAL, 927
- default_cfe_evs_msgids.h
 - CFE_EVS_CMD_MID, 928
 - CFE_EVS_HK_TLM_MID, 928
 - CFE_EVS_LONG_EVENT_MSG_MID, 928
 - CFE_EVS_SEND_HK_MID, 928
 - CFE_EVS_SHORT_EVENT_MSG_MID, 928
- default_cfe_evs_msgstruct.h
 - CFE_EVS_AddEventFilterCmd_t, 930
 - CFE_EVS_ClearLogCmd_t, 930
 - CFE_EVS_DeleteEventFilterCmd_t, 930
 - CFE_EVS_DisableAppEventsCmd_t, 930
 - CFE_EVS_DisableAppEventTypeCmd_t, 930
 - CFE_EVS_DisableEventTypeCmd_t, 931
 - CFE_EVS_DisablePortsCmd_t, 931
 - CFE_EVS_EnableAppEventsCmd_t, 931
 - CFE_EVS_EnableAppEventTypeCmd_t, 931
 - CFE_EVS_EnableEventTypeCmd_t, 931
 - CFE_EVS_EnablePortsCmd_t, 931
 - CFE_EVS_HousekeepingTlm_t, 931
 - CFE_EVS_LongEventTlm_t, 931
 - CFE_EVS_NoopCmd_t, 931
 - CFE_EVS_ResetAllFiltersCmd_t, 931
 - CFE_EVS_ResetAppCounterCmd_t, 931
 - CFE_EVS_ResetCountersCmd_t, 931
 - CFE_EVS_ResetFilterCmd_t, 931
 - CFE_EVS_SendHkCmd_t, 931
 - CFE_EVS_SetEventFormatModeCmd_t, 932
 - CFE_EVS_SetFilterCmd_t, 932
 - CFE_EVS_SetLogModeCmd_t, 932
 - CFE_EVS_ShortEventTlm_t, 932
 - CFE_EVS_WriteAppDataFileCmd_t, 932
 - CFE_EVS_WriteLogDataFileCmd_t, 932
- default_cfe_evs_topicid_values.h
 - CFE_MISSION_EVS_TIDVAL, 936

DEFAULT_CFE_FS_FILE_CONTENT_ID
 cfe_fs_interface_cfg.h, 985
 default_cfe_fs_filedef.h
 CFE_FS_Header_t, 980
 CFE_FS_SubType, 981
 CFE_FS_SubType_Enum_t, 980
 CFE_FS_SubType_ES_CDS_REG, 981
 CFE_FS_SubType_ES_ERLOG, 981
 CFE_FS_SubType_ES_PERFDATA, 981
 CFE_FS_SubType_ES_QUERYALL, 981
 CFE_FS_SubType_ES_QUERYALLTASKS, 981
 CFE_FS_SubType_ES_SYSLOG, 981
 CFE_FS_SubType_EVS_APPDATA, 981
 CFE_FS_SubType_EVS_EVENTLOG, 981
 CFE_FS_SubType_SB_MAPDATA, 981
 CFE_FS_SubType_SB_PIPEDATA, 981
 CFE_FS_SubType_SB_ROUTEDATA, 981
 CFE_FS_SubType_TBL_IMG, 981
 CFE_FS_SubType_TBL_REG, 981
 DEFAULT_CFE_FS_HDR_DESC_MAX_LEN
 cfe_fs_interface_cfg.h, 985
 default_cfe_fs_interface_cfg_values.h
 CFE_MISSION_FS_CFGVAL, 983
 DEFAULT_CFE_MISSION_CORE_API_MAX_API_LEN
 cfe_core_api_interface_cfg.h, 699
 DEFAULT_CFE_MISSION_CORE_API_MAX_FILE_LEN
 cfe_core_api_interface_cfg.h, 699
 DEFAULT_CFE_MISSION_CORE_API_MAX_NUM_FILES
 cfe_core_api_interface_cfg.h, 699
 DEFAULT_CFE_MISSION_CORE_API_MAX_PATH_LEN
 cfe_core_api_interface_cfg.h, 699
 DEFAULT_CFE_MISSION_ES_APP_TLM_TOPICID
 cfe_es_topicids.h, 910
 DEFAULT_CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN
 cfe_es_interface_cfg.h, 874
 DEFAULT_CFE_MISSION_ES_CDS_MAX_NAME_LENGTH
 cfe_es_interface_cfg.h, 875
 DEFAULT_CFE_MISSION_ES_CMD_TOPICID
 cfe_es_topicids.h, 910
 DEFAULT_CFE_MISSION_ES_CRC_16
 cfe_es_interface_cfg.h, 875
 DEFAULT_CFE_MISSION_ES_CRC_32
 cfe_es_interface_cfg.h, 875
 DEFAULT_CFE_MISSION_ES_CRC_8
 cfe_es_interface_cfg.h, 875
 DEFAULT_CFE_MISSION_ES_DEFAULT_CRC
 cfe_es_interface_cfg.h, 875
 DEFAULT_CFE_MISSION_ES_HK_TLM_TOPICID
 cfe_es_topicids.h, 910
 DEFAULT_CFE_MISSION_ES_MAX_APPLICATIONS
 cfe_es_interface_cfg.h, 875
 DEFAULT_CFE_MISSION_ES_MEMSTATS_TLM_TOPICID
 cfe_es_topicids.h, 911
 DEFAULT_CFE_MISSION_ES_PERF_MAX_IDS
 cfe_es_interface_cfg.h, 875
 DEFAULT_CFE_MISSION_ES_POOL_MAX_BUCKETS
 cfe_es_interface_cfg.h, 875
 DEFAULT_CFE_MISSION_ES_SEND_HK_TOPICID
 cfe_es_topicids.h, 911
 DEFAULT_CFE_MISSION_EVS_CMD_TOPICID
 cfe_es_topicids.h, 978
 DEFAULT_CFE_MISSION_EVS_HK_TLM_TOPICID
 cfe_es_topicids.h, 978
 DEFAULT_CFE_MISSION_EVS_LONG_EVENT_MSG_TOPICID
 cfe_es_topicids.h, 978
 DEFAULT_CFE_MISSION_EVS_MAX_MESSAGE_LENGTH
 cfe_es_interface_cfg.h, 969
 DEFAULT_CFE_MISSION_EVS_SEND_HK_TOPICID
 cfe_es_topicids.h, 978
 DEFAULT_CFE_MISSION_EVS_SHORT_EVENT_MSG_TOPICID
 cfe_es_topicids.h, 978
 DEFAULT_CFE_MISSION_SB_ALLSUBS_TLM_TOPICID
 cfe_sb_topicids.h, 1062
 DEFAULT_CFE_MISSION_SB_CMD_TOPICID
 cfe_sb_topicids.h, 1062
 DEFAULT_CFE_MISSION_SB_HK_TLM_TOPICID
 cfe_sb_topicids.h, 1062
 DEFAULT_CFE_MISSION_SB_MAX_PIPES
 cfe_sb_interface_cfg.h, 1044
 DEFAULT_CFE_MISSION_SB_MAX_SB_MSG_SIZE
 cfe_sb_interface_cfg.h, 1044
 DEFAULT_CFE_MISSION_SB_ONESUB_TLM_TOPICID
 cfe_sb_topicids.h, 1062
 DEFAULT_CFE_MISSION_SB_SEND_HK_TOPICID
 cfe_sb_topicids.h, 1063
 DEFAULT_CFE_MISSION_SB_STATS_TLM_TOPICID
 cfe_sb_topicids.h, 1063
 DEFAULT_CFE_MISSION_SB_SUB_ENTRIES_PER_PKT
 cfe_sb_interface_cfg.h, 1044
 DEFAULT_CFE_MISSION_SB_SUB_RPT_CTRL_TOPICID
 cfe_sb_topicids.h, 1063
 DEFAULT_CFE_MISSION_TBL_CMD_TOPICID
 cfe_tbl_topicids.h, 1138
 DEFAULT_CFE_MISSION_TBL_HK_TLM_TOPICID
 cfe_tbl_topicids.h, 1138
 DEFAULT_CFE_MISSION_TBL_MAX_FULL_NAME_LEN
 cfe_tbl_interface_cfg.h, 1126
 default_cfe_tbl_mission_cfg.h, 1070
 DEFAULT_CFE_MISSION_TBL_MAX_NAME_LENGTH
 cfe_tbl_interface_cfg.h, 1126
 default_cfe_tbl_mission_cfg.h, 1070
 DEFAULT_CFE_MISSION_TBL_REG_TLM_TOPICID
 cfe_tbl_topicids.h, 1138
 DEFAULT_CFE_MISSION_TBL_SEND_HK_TOPICID
 cfe_tbl_topicids.h, 1138
 DEFAULT_CFE_MISSION_TIME_AT_TONE_WAS
 cfe_time_interface_cfg.h, 1198
 DEFAULT_CFE_MISSION_TIME_AT_TONE_WILL_BE

- cfe_time_interface_cfg.h, [1199](#)
- DEFAULT_CFE_MISSION_TIME_CFG_DEFAULT_TAI
 - cfe_time_interface_cfg.h, [1199](#)
- DEFAULT_CFE_MISSION_TIME_CFG_DEFAULT_UTC
 - cfe_time_interface_cfg.h, [1199](#)
- DEFAULT_CFE_MISSION_TIME_CFG_FAKE_TONE
 - cfe_time_interface_cfg.h, [1199](#)
- DEFAULT_CFE_MISSION_TIME_CMD_TOPICID
 - cfe_time_topics.h, [1212](#)
- DEFAULT_CFE_MISSION_TIME_DATA_CMD_TOPICID
 - cfe_time_topics.h, [1213](#)
- DEFAULT_CFE_MISSION_TIME_DEF_DELAY_SECS
 - cfe_time_interface_cfg.h, [1199](#)
- DEFAULT_CFE_MISSION_TIME_DEF_DELAY_SUBS
 - cfe_time_interface_cfg.h, [1199](#)
- DEFAULT_CFE_MISSION_TIME_DEF_LEAPS
 - cfe_time_interface_cfg.h, [1199](#)
- DEFAULT_CFE_MISSION_TIME_DEF_MET_SECS
 - cfe_time_interface_cfg.h, [1199](#)
- DEFAULT_CFE_MISSION_TIME_DEF_MET_SUBS
 - cfe_time_interface_cfg.h, [1199](#)
- DEFAULT_CFE_MISSION_TIME_DEF_STCF_SECS
 - cfe_time_interface_cfg.h, [1199](#)
- DEFAULT_CFE_MISSION_TIME_DEF_STCF_SUBS
 - cfe_time_interface_cfg.h, [1199](#)
- DEFAULT_CFE_MISSION_TIME_DIAG_TLM_TOPICID
 - cfe_time_topics.h, [1213](#)
- DEFAULT_CFE_MISSION_TIME_EPOCH_DAY
 - cfe_time_interface_cfg.h, [1200](#)
- DEFAULT_CFE_MISSION_TIME_EPOCH_HOUR
 - cfe_time_interface_cfg.h, [1200](#)
- DEFAULT_CFE_MISSION_TIME_EPOCH_MICROS
 - cfe_time_interface_cfg.h, [1200](#)
- DEFAULT_CFE_MISSION_TIME_EPOCH_MINUTE
 - cfe_time_interface_cfg.h, [1200](#)
- DEFAULT_CFE_MISSION_TIME_EPOCH_SECOND
 - cfe_time_interface_cfg.h, [1200](#)
- DEFAULT_CFE_MISSION_TIME_EPOCH_SECONDS
 - cfe_time_interface_cfg.h, [1200](#)
- DEFAULT_CFE_MISSION_TIME_EPOCH_YEAR
 - cfe_time_interface_cfg.h, [1200](#)
- DEFAULT_CFE_MISSION_TIME_FS_FACTOR
 - cfe_time_interface_cfg.h, [1200](#)
- DEFAULT_CFE_MISSION_TIME_HK_TLM_TOPICID
 - cfe_time_topics.h, [1213](#)
- DEFAULT_CFE_MISSION_TIME_MAX_ELAPSED
 - cfe_time_interface_cfg.h, [1200](#)
- DEFAULT_CFE_MISSION_TIME_MIN_ELAPSED
 - cfe_time_interface_cfg.h, [1200](#)
- DEFAULT_CFE_MISSION_TIME_ONEHZ_CMD_TOPICID
 - cfe_time_topics.h, [1213](#)
- DEFAULT_CFE_MISSION_TIME_SEND_CMD_TOPICID
 - cfe_time_topics.h, [1213](#)
- DEFAULT_CFE_MISSION_TIME_SEND_HK_TOPICID
 - cfe_time_topics.h, [1213](#)
- cfe_time_topics.h, [1213](#)
- DEFAULT_CFE_MISSION_TIME_TONE_CMD_TOPICID
 - cfe_time_topics.h, [1213](#)
- DEFAULT_CFE_PLATFORM_CMD_MID_BASE
 - default_cfe_core_api_msgid_mapping.h, [686](#)
- DEFAULT_CFE_PLATFORM_ES_APP_KILL_TIMEOUT
 - cfe_es_internal_cfg.h, [899](#)
- DEFAULT_CFE_PLATFORM_ES_APP_SCAN_RATE
 - cfe_es_internal_cfg.h, [899](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MAX_BLOCK_SIZE
 - cfe_es_internal_cfg.h, [899](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MAX_NUM_ENTRIES
 - cfe_es_internal_cfg.h, [899](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_01
 - cfe_es_internal_cfg.h, [899](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_02
 - cfe_es_internal_cfg.h, [899](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_03
 - cfe_es_internal_cfg.h, [899](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_04
 - cfe_es_internal_cfg.h, [899](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_05
 - cfe_es_internal_cfg.h, [899](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_06
 - cfe_es_internal_cfg.h, [900](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_07
 - cfe_es_internal_cfg.h, [900](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_08
 - cfe_es_internal_cfg.h, [900](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_09
 - cfe_es_internal_cfg.h, [900](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_10
 - cfe_es_internal_cfg.h, [900](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_11
 - cfe_es_internal_cfg.h, [900](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_12
 - cfe_es_internal_cfg.h, [900](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_13
 - cfe_es_internal_cfg.h, [900](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_14
 - cfe_es_internal_cfg.h, [900](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_15
 - cfe_es_internal_cfg.h, [900](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_16
 - cfe_es_internal_cfg.h, [900](#)
- DEFAULT_CFE_PLATFORM_ES_CDS_SIZE
 - cfe_es_internal_cfg.h, [901](#)
- DEFAULT_CFE_PLATFORM_ES_DEFAULT_APP_LOG_FILE
 - cfe_es_internal_cfg.h, [901](#)
- DEFAULT_CFE_PLATFORM_ES_DEFAULT_CDS_REG_DUMP_FILE
 - cfe_es_internal_cfg.h, [901](#)
- DEFAULT_CFE_PLATFORM_ES_DEFAULT_ER_LOG_FILE
 - cfe_es_internal_cfg.h, [901](#)
- DEFAULT_CFE_PLATFORM_ES_DEFAULT_PERF_DUMP_FILENAME

cfe_es_internal_cfg.h, 901	cfe_es_internal_cfg.h, 903
DEFAULT_CFE_PLATFORM_ES_DEFAULT_POR_SYSLOG_MODE	DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_15
cfe_es_internal_cfg.h, 901	cfe_es_internal_cfg.h, 903
DEFAULT_CFE_PLATFORM_ES_DEFAULT_PR_SYSLOG_MODE	DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_16
cfe_es_internal_cfg.h, 901	cfe_es_internal_cfg.h, 904
DEFAULT_CFE_PLATFORM_ES_DEFAULT_STACK_SIZE	DEFAULT_CFE_PLATFORM_ES_MEMPOOL_ALIGN_SIZE_MIN
cfe_es_internal_cfg.h, 901	cfe_es_internal_cfg.h, 904
DEFAULT_CFE_PLATFORM_ES_DEFAULT_SYSLOG_FILE	DEFAULT_CFE_PLATFORM_ES_NONVOL_DISK_MOUNT_STRING
cfe_es_internal_cfg.h, 901	cfe_es_internal_cfg.h, 904
DEFAULT_CFE_PLATFORM_ES_DEFAULT_TASK_LOG_FILE	DEFAULT_CFE_PLATFORM_ES_NONVOL_STARTUP_FILE
cfe_es_internal_cfg.h, 901	cfe_es_internal_cfg.h, 904
DEFAULT_CFE_PLATFORM_ES_ER_LOG_ENTRIES	DEFAULT_CFE_PLATFORM_ES_PERF_CHILD_MS_DELAY
cfe_es_internal_cfg.h, 901	cfe_es_internal_cfg.h, 904
DEFAULT_CFE_PLATFORM_ES_ER_LOG_MAX_CONTEXT_SIZE	DEFAULT_CFE_PLATFORM_ES_PERF_CHILD_PRIORITY
cfe_es_internal_cfg.h, 902	cfe_es_internal_cfg.h, 904
DEFAULT_CFE_PLATFORM_ES_MAX_APPLICATIONS	DEFAULT_CFE_PLATFORM_ES_PERF_CHILD_STACK_SIZE
cfe_es_internal_cfg.h, 902	cfe_es_internal_cfg.h, 904
DEFAULT_CFE_PLATFORM_ES_MAX_BLOCK_SIZE	DEFAULT_CFE_PLATFORM_ES_PERF_DATA_BUFFER_SIZE
cfe_es_internal_cfg.h, 902	cfe_es_internal_cfg.h, 904
DEFAULT_CFE_PLATFORM_ES_MAX_GEN_COUNTERS	DEFAULT_CFE_PLATFORM_ES_PERF_ENTRIES_BTWN_DLYS
cfe_es_internal_cfg.h, 902	cfe_es_internal_cfg.h, 904
DEFAULT_CFE_PLATFORM_ES_MAX_LIBRARIES	DEFAULT_CFE_PLATFORM_ES_PERF_FILTERMASK_ALL
cfe_es_internal_cfg.h, 902	cfe_es_internal_cfg.h, 904
DEFAULT_CFE_PLATFORM_ES_MAX_MEMORY_POOLS	DEFAULT_CFE_PLATFORM_ES_PERF_FILTERMASK_INIT
cfe_es_internal_cfg.h, 902	cfe_es_internal_cfg.h, 904
DEFAULT_CFE_PLATFORM_ES_MAX_PROCESSOR_RESOURCES	DEFAULT_CFE_PLATFORM_ES_PERF_FILTERMASK_NONE
cfe_es_internal_cfg.h, 902	cfe_es_internal_cfg.h, 905
DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_01	DEFAULT_CFE_PLATFORM_ES_PERF_TRIGMASK_ALL
cfe_es_internal_cfg.h, 902	cfe_es_internal_cfg.h, 905
DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_02	DEFAULT_CFE_PLATFORM_ES_PERF_TRIGMASK_INIT
cfe_es_internal_cfg.h, 902	cfe_es_internal_cfg.h, 905
DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_03	DEFAULT_CFE_PLATFORM_ES_PERF_TRIGMASK_NONE
cfe_es_internal_cfg.h, 902	cfe_es_internal_cfg.h, 905
DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_04	DEFAULT_CFE_PLATFORM_ES_POOL_MAX_BUCKETS
cfe_es_internal_cfg.h, 902	cfe_es_internal_cfg.h, 905
DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_05	DEFAULT_CFE_PLATFORM_ES_RAM_DISK_MOUNT_STRING
cfe_es_internal_cfg.h, 903	cfe_es_internal_cfg.h, 905
DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_06	DEFAULT_CFE_PLATFORM_ES_RAM_DISK_NUM_SECTORS
cfe_es_internal_cfg.h, 903	cfe_es_internal_cfg.h, 905
DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_07	DEFAULT_CFE_PLATFORM_ES_RAM_DISK_PERCENT_RESERVED
cfe_es_internal_cfg.h, 903	cfe_es_internal_cfg.h, 905
DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_08	DEFAULT_CFE_PLATFORM_ES_RAM_DISK_SECTOR_SIZE
cfe_es_internal_cfg.h, 903	cfe_es_internal_cfg.h, 905
DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_09	DEFAULT_CFE_PLATFORM_ES_START_TASK_PRIORITY
cfe_es_internal_cfg.h, 903	cfe_es_internal_cfg.h, 905
DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_10	DEFAULT_CFE_PLATFORM_ES_START_TASK_STACK_SIZE
cfe_es_internal_cfg.h, 903	cfe_es_internal_cfg.h, 905
DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_11	DEFAULT_CFE_PLATFORM_ES_STARTUP_SCRIPT_TIMEOUT_MSEC
cfe_es_internal_cfg.h, 903	cfe_es_internal_cfg.h, 906
DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_12	DEFAULT_CFE_PLATFORM_ES_STARTUP_SYNC_POLL_MSEC
cfe_es_internal_cfg.h, 903	cfe_es_internal_cfg.h, 906
DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_13	DEFAULT_CFE_PLATFORM_ES_SYSTEM_LOG_SIZE
cfe_es_internal_cfg.h, 903	cfe_es_internal_cfg.h, 906
DEFAULT_CFE_PLATFORM_ES_MEM_BLOCK_SIZE_14	DEFAULT_CFE_PLATFORM_ES_USER_RESERVED_SIZE

[cfe_es_internal_cfg.h](#), 906
[DEFAULT_CFE_PLATFORM_ES_VOLATILE_STARTUP_FILE](#)
[cfe_es_internal_cfg.h](#), 906
[DEFAULT_CFE_PLATFORM_EVS_APP_EVENTS_PER_SECOND](#)
[cfe_evs_internal_cfg.h](#), 975
[DEFAULT_CFE_PLATFORM_EVS_DEFAULT_APP_DATA_FILE](#)
[cfe_evs_internal_cfg.h](#), 975
[DEFAULT_CFE_PLATFORM_EVS_DEFAULT_LOG_FILE](#)
[cfe_evs_internal_cfg.h](#), 975
[DEFAULT_CFE_PLATFORM_EVS_DEFAULT_LOG_MODE](#)
[cfe_evs_internal_cfg.h](#), 975
[DEFAULT_CFE_PLATFORM_EVS_DEFAULT_MSG_FORMAT_MODE](#)
[cfe_evs_internal_cfg.h](#), 975
[DEFAULT_CFE_PLATFORM_EVS_DEFAULT_TYPE_FLAG](#)
[cfe_evs_internal_cfg.h](#), 975
[DEFAULT_CFE_PLATFORM_EVS_LOG_MAX](#)
[cfe_evs_internal_cfg.h](#), 975
[DEFAULT_CFE_PLATFORM_EVS_MAX_APP_EVENT_BUFFER_SIZE](#)
[cfe_evs_internal_cfg.h](#), 975
[DEFAULT_CFE_PLATFORM_EVS_MAX_EVENT_FILTERS](#)
[cfe_evs_internal_cfg.h](#), 975
[DEFAULT_CFE_PLATFORM_EVS_PORT_DEFAULT](#)
[cfe_evs_internal_cfg.h](#), 975
[DEFAULT_CFE_PLATFORM_EVS_START_TASK_PRIORITY](#)
[cfe_evs_internal_cfg.h](#), 976
[DEFAULT_CFE_PLATFORM_EVS_START_TASK_STACK_SIZE](#)
[cfe_evs_internal_cfg.h](#), 976
[DEFAULT_CFE_PLATFORM_SB_BUF_MEMORY_BYTES](#)
[cfe_sb_internal_cfg.h](#), 1055
[DEFAULT_CFE_PLATFORM_SB_DEFAULT_MAP_FILENAME](#)
[cfe_sb_internal_cfg.h](#), 1055
[DEFAULT_CFE_PLATFORM_SB_DEFAULT_MSG_LIMIT](#)
[cfe_sb_internal_cfg.h](#), 1055
[DEFAULT_CFE_PLATFORM_SB_DEFAULT_PIPE_FILENAME](#)
[cfe_sb_internal_cfg.h](#), 1055
[DEFAULT_CFE_PLATFORM_SB_DEFAULT_ROUTING_FILE](#)
[cfe_sb_internal_cfg.h](#), 1055
[DEFAULT_CFE_PLATFORM_SB_FILTER_MASK1](#)
[cfe_sb_internal_cfg.h](#), 1055
[DEFAULT_CFE_PLATFORM_SB_FILTER_MASK2](#)
[cfe_sb_internal_cfg.h](#), 1055
[DEFAULT_CFE_PLATFORM_SB_FILTER_MASK3](#)
[cfe_sb_internal_cfg.h](#), 1055
[DEFAULT_CFE_PLATFORM_SB_FILTER_MASK4](#)
[cfe_sb_internal_cfg.h](#), 1055
[DEFAULT_CFE_PLATFORM_SB_FILTER_MASK5](#)
[cfe_sb_internal_cfg.h](#), 1056
[DEFAULT_CFE_PLATFORM_SB_FILTER_MASK6](#)
[cfe_sb_internal_cfg.h](#), 1056
[DEFAULT_CFE_PLATFORM_SB_FILTER_MASK7](#)
[cfe_sb_internal_cfg.h](#), 1056
[DEFAULT_CFE_PLATFORM_SB_FILTER_MASK8](#)
[cfe_sb_internal_cfg.h](#), 1056
[DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT1](#)
[cfe_sb_internal_cfg.h](#), 1056
[DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT2](#)
[cfe_sb_internal_cfg.h](#), 1056
[DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT3](#)
[cfe_sb_internal_cfg.h](#), 1056
[DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT4](#)
[cfe_sb_internal_cfg.h](#), 1056
[DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT5](#)
[cfe_sb_internal_cfg.h](#), 1056
[DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT6](#)
[cfe_sb_internal_cfg.h](#), 1056
[DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT7](#)
[cfe_sb_internal_cfg.h](#), 1056
[DEFAULT_CFE_PLATFORM_SB_FILTERED_EVENT8](#)
[cfe_sb_internal_cfg.h](#), 1057
[DEFAULT_CFE_PLATFORM_SB_HIGHEST_VALID_MSGID](#)
[cfe_sb_internal_cfg.h](#), 1057
[DEFAULT_CFE_PLATFORM_SB_MAX_BLOCK_SIZE](#)
[cfe_sb_internal_cfg.h](#), 1057
[DEFAULT_CFE_PLATFORM_SB_MAX_DEST_PER_PKT](#)
[cfe_sb_internal_cfg.h](#), 1057
[DEFAULT_CFE_PLATFORM_SB_MAX_MSG_IDS](#)
[cfe_sb_internal_cfg.h](#), 1057
[DEFAULT_CFE_PLATFORM_SB_MAX_PIPES](#)
[cfe_sb_internal_cfg.h](#), 1057
[DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_01](#)
[cfe_sb_internal_cfg.h](#), 1057
[DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_02](#)
[cfe_sb_internal_cfg.h](#), 1057
[DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_03](#)
[cfe_sb_internal_cfg.h](#), 1057
[DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_04](#)
[cfe_sb_internal_cfg.h](#), 1057
[DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_05](#)
[cfe_sb_internal_cfg.h](#), 1057
[DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_06](#)
[cfe_sb_internal_cfg.h](#), 1058
[DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_07](#)
[cfe_sb_internal_cfg.h](#), 1058
[DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_08](#)
[cfe_sb_internal_cfg.h](#), 1058
[DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_09](#)
[cfe_sb_internal_cfg.h](#), 1058
[DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_10](#)
[cfe_sb_internal_cfg.h](#), 1058
[DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_11](#)
[cfe_sb_internal_cfg.h](#), 1058
[DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_12](#)
[cfe_sb_internal_cfg.h](#), 1058
[DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_13](#)
[cfe_sb_internal_cfg.h](#), 1058
[DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_14](#)
[cfe_sb_internal_cfg.h](#), 1058
[DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_15](#)

- cfe_sb_internal_cfg.h, [1058](#)
- DEFAULT_CFE_PLATFORM_SB_MEM_BLOCK_SIZE_16
 - cfe_sb_internal_cfg.h, [1058](#)
- DEFAULT_CFE_PLATFORM_SB_POOL_MAX_BUCKETS
 - cfe_sb_internal_cfg.h, [1059](#)
- DEFAULT_CFE_PLATFORM_SB_START_TASK_PRIORITY
 - cfe_sb_internal_cfg.h, [1059](#)
- DEFAULT_CFE_PLATFORM_SB_START_TASK_STACK_SIZE
 - cfe_sb_internal_cfg.h, [1059](#)
- DEFAULT_CFE_PLATFORM_TBL_BUF_MEMORY_BYTES
 - cfe_tbl_internal_cfg.h, [1134](#)
 - default_cfe_tbl_platform_cfg.h, [1089](#)
- DEFAULT_CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_DEPTH
 - cfe_tbl_internal_cfg.h, [1134](#)
 - default_cfe_tbl_platform_cfg.h, [1089](#)
- DEFAULT_CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES
 - cfe_tbl_internal_cfg.h, [1134](#)
 - default_cfe_tbl_platform_cfg.h, [1089](#)
- DEFAULT_CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE
 - cfe_tbl_internal_cfg.h, [1134](#)
 - default_cfe_tbl_platform_cfg.h, [1089](#)
- DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_HANDLES
 - cfe_tbl_internal_cfg.h, [1134](#)
 - default_cfe_tbl_platform_cfg.h, [1090](#)
- DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_TABLES
 - cfe_tbl_internal_cfg.h, [1134](#)
 - default_cfe_tbl_platform_cfg.h, [1090](#)
- DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS
 - cfe_tbl_internal_cfg.h, [1134](#)
 - default_cfe_tbl_platform_cfg.h, [1090](#)
- DEFAULT_CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_READS
 - cfe_tbl_internal_cfg.h, [1134](#)
 - default_cfe_tbl_platform_cfg.h, [1090](#)
- DEFAULT_CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE
 - cfe_tbl_internal_cfg.h, [1135](#)
 - default_cfe_tbl_platform_cfg.h, [1090](#)
- DEFAULT_CFE_PLATFORM_TBL_START_TASK_PRIORITY
 - cfe_tbl_internal_cfg.h, [1135](#)
 - default_cfe_tbl_platform_cfg.h, [1090](#)
- DEFAULT_CFE_PLATFORM_TBL_START_TASK_STACK_SIZE
 - cfe_tbl_internal_cfg.h, [1135](#)
 - default_cfe_tbl_platform_cfg.h, [1090](#)
- DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_1
 - cfe_tbl_internal_cfg.h, [1135](#)
 - default_cfe_tbl_platform_cfg.h, [1090](#)
- DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_2
 - cfe_tbl_internal_cfg.h, [1135](#)
 - default_cfe_tbl_platform_cfg.h, [1090](#)
- DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_3
 - cfe_tbl_internal_cfg.h, [1135](#)
 - default_cfe_tbl_platform_cfg.h, [1090](#)
- DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_4
 - cfe_tbl_internal_cfg.h, [1135](#)
 - default_cfe_tbl_platform_cfg.h, [1090](#)
- DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_COUNT
 - cfe_tbl_internal_cfg.h, [1135](#)
 - default_cfe_tbl_platform_cfg.h, [1091](#)
- DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_1
 - cfe_tbl_internal_cfg.h, [1135](#)
 - default_cfe_tbl_platform_cfg.h, [1091](#)
- DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_2
 - cfe_tbl_internal_cfg.h, [1135](#)
 - default_cfe_tbl_platform_cfg.h, [1091](#)
- DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_COUNT
 - cfe_tbl_internal_cfg.h, [1135](#)
 - default_cfe_tbl_platform_cfg.h, [1091](#)
- DEFAULT_CFE_PLATFORM_TIME_CFG_CLIENT
 - cfe_time_internal_cfg.h, [1208](#)
- DEFAULT_CFE_PLATFORM_TIME_CFG_LATCH_FLY
 - cfe_time_internal_cfg.h, [1208](#)
- DEFAULT_CFE_PLATFORM_TIME_CFG_SERVER
 - cfe_time_internal_cfg.h, [1208](#)
- DEFAULT_CFE_PLATFORM_TIME_CFG_SIGNAL
 - cfe_time_internal_cfg.h, [1208](#)
- DEFAULT_CFE_PLATFORM_TIME_CFG_SOURCE
 - cfe_time_internal_cfg.h, [1208](#)
- DEFAULT_CFE_PLATFORM_TIME_CFG_SRC_GPS
 - cfe_time_internal_cfg.h, [1208](#)
- DEFAULT_CFE_PLATFORM_TIME_CFG_SRC_MET
 - cfe_time_internal_cfg.h, [1208](#)
- DEFAULT_CFE_PLATFORM_TIME_CFG_SRC_TIME
 - cfe_time_internal_cfg.h, [1208](#)
- DEFAULT_CFE_PLATFORM_TIME_CFG_START_FLY
 - cfe_time_internal_cfg.h, [1208](#)
- DEFAULT_CFE_PLATFORM_TIME_CFG_TONE_LIMIT
 - cfe_time_internal_cfg.h, [1208](#)
- DEFAULT_CFE_PLATFORM_TIME_CFG_VIRTUAL
 - cfe_time_internal_cfg.h, [1208](#)
- DEFAULT_CFE_PLATFORM_TIME_MAX_DELTA_SECS
 - cfe_time_internal_cfg.h, [1209](#)
- DEFAULT_CFE_PLATFORM_TIME_MAX_DELTA_SUBS
 - cfe_time_internal_cfg.h, [1209](#)
- DEFAULT_CFE_PLATFORM_TIME_MAX_LOCAL_SECS
 - cfe_time_internal_cfg.h, [1209](#)
- DEFAULT_CFE_PLATFORM_TIME_MAX_LOCAL_SUBS
 - cfe_time_internal_cfg.h, [1209](#)
- DEFAULT_CFE_PLATFORM_TIME_ONEHZ_TASK_PRIORITY
 - cfe_time_internal_cfg.h, [1209](#)
- DEFAULT_CFE_PLATFORM_TIME_ONEHZ_TASK_STACK_SIZE
 - cfe_time_internal_cfg.h, [1209](#)
- DEFAULT_CFE_PLATFORM_TIME_START_TASK_PRIORITY
 - cfe_time_internal_cfg.h, [1209](#)
- DEFAULT_CFE_PLATFORM_TIME_START_TASK_STACK_SIZE
 - cfe_time_internal_cfg.h, [1209](#)
- DEFAULT_CFE_PLATFORM_TIME_TONE_TASK_PRIORITY
 - cfe_time_internal_cfg.h, [1209](#)
- DEFAULT_CFE_PLATFORM_TIME_TONE_TASK_STACK_SIZE
 - cfe_time_internal_cfg.h, [1209](#)

- DEFAULT_CFE_PLATFORM_TLM_MID_BASE
 - default_cfe_core_api_msgid_mapping.h, [686](#)
- default_cfe_sb_extern_typedefs.h
 - CFE_SB_MsgId_Atomic_t, [992](#)
 - CFE_SB_PipeId_t, [992](#)
 - CFE_SB_QosPriority, [992](#)
 - CFE_SB_QosPriority_Enum_t, [992](#)
 - CFE_SB_QosPriority_HIGH, [992](#)
 - CFE_SB_QosPriority_LOW, [992](#)
 - CFE_SB_QosReliability, [992](#)
 - CFE_SB_QosReliability_Enum_t, [992](#)
 - CFE_SB_QosReliability_HIGH, [993](#)
 - CFE_SB_QosReliability_LOW, [993](#)
 - CFE_SB_RouteId_Atomic_t, [992](#)
- default_cfe_sb_fcncode_values.h
 - CFE_SB_CCVAL, [994](#)
 - CFE_SB_FunctionCode_, [994](#)
 - CFE_SB_FunctionCode_DISABLE_ROUTE, [994](#)
 - CFE_SB_FunctionCode_DISABLE_SUB_REPORTING, [994](#)
 - CFE_SB_FunctionCode_ENABLE_ROUTE, [994](#)
 - CFE_SB_FunctionCode_ENABLE_SUB_REPORTING, [994](#)
 - CFE_SB_FunctionCode_NOOP, [994](#)
 - CFE_SB_FunctionCode_RESET_COUNTERS, [994](#)
 - CFE_SB_FunctionCode_SEND_PREV_SUBS, [994](#)
 - CFE_SB_FunctionCode_SEND_SB_STATS, [994](#)
 - CFE_SB_FunctionCode_WRITE_MAP_INFO, [994](#)
 - CFE_SB_FunctionCode_WRITE_PIPE_INFO, [994](#)
 - CFE_SB_FunctionCode_WRITE_ROUTING_INFO, [994](#)
- default_cfe_sb_interface_cfg_values.h
 - CFE_MISSION_SB_CFGVAL, [996](#)
- default_cfe_sb_internal_cfg_values.h
 - CFE_PLATFORM_SB_CFGVAL, [996](#)
- default_cfe_sb_msgdefs.h
 - CFE_SB_AllSubscriptionsTlm_Payload_t, [1000](#)
 - CFE_SB_HousekeepingTlm_Payload_t, [1000](#)
 - CFE_SB_MsgMapFileEntry_t, [1000](#)
 - CFE_SB_PipeDepthStats_t, [1000](#)
 - CFE_SB_PipeInfoEntry_t, [1000](#)
 - CFE_SB_RouteCmd_Payload_t, [1000](#)
 - CFE_SB_RoutingFileEntry_t, [1000](#)
 - CFE_SB_SingleSubscriptionTlm_Payload_t, [1000](#)
 - CFE_SB_StatsTlm_Payload_t, [1001](#)
 - CFE_SB_SubEntries_t, [1001](#)
 - CFE_SB_WriteFileInfoCmd_Payload_t, [1001](#)
- default_cfe_sb_msgid_values.h
 - CFE_PLATFORM_SB_CMD_MIDVAL, [1003](#)
 - CFE_PLATFORM_SB_TLM_MIDVAL, [1003](#)
- default_cfe_sb_msgids.h
 - CFE_SB_ALLSUBS_TLM_MID, [1005](#)
 - CFE_SB_CMD_MID, [1005](#)
 - CFE_SB_HK_TLM_MID, [1005](#)
 - CFE_SB_ONESUB_TLM_MID, [1005](#)
 - CFE_SB_SEND_HK_MID, [1005](#)
 - CFE_SB_STATS_TLM_MID, [1005](#)
 - CFE_SB_SUB_RPT_CTRL_MID, [1005](#)
- default_cfe_sb_msgstruct.h
 - CFE_SB_AllSubscriptionsTlm_t, [1007](#)
 - CFE_SB_DisableRouteCmd_t, [1007](#)
 - CFE_SB_DisableSubReportingCmd_t, [1007](#)
 - CFE_SB_EnableRouteCmd_t, [1007](#)
 - CFE_SB_EnableSubReportingCmd_t, [1007](#)
 - CFE_SB_HousekeepingTlm_t, [1007](#)
 - CFE_SB_NoopCmd_t, [1007](#)
 - CFE_SB_ResetCountersCmd_t, [1007](#)
 - CFE_SB_SendHkCmd_t, [1007](#)
 - CFE_SB_SendPrevSubsCmd_t, [1007](#)
 - CFE_SB_SendSbStatsCmd_t, [1007](#)
 - CFE_SB_SingleSubscriptionTlm_t, [1007](#)
 - CFE_SB_StatsTlm_t, [1007](#)
 - CFE_SB_WriteMapInfoCmd_t, [1008](#)
 - CFE_SB_WritePipeInfoCmd_t, [1008](#)
 - CFE_SB_WriteRoutingInfoCmd_t, [1008](#)
- default_cfe_sb_topicid_values.h
 - CFE_MISSION_SB_TIDVAL, [1011](#)
- default_cfe_tbl_extern_typedefs.h
 - CFE_TBL_BufferSelect, [1065](#)
 - CFE_TBL_BufferSelect_ACTIVE, [1065](#)
 - CFE_TBL_BufferSelect_Enum_t, [1064](#)
 - CFE_TBL_BufferSelect_INACTIVE, [1065](#)
 - CFE_TBL_CombinedFileHdr_t, [1064](#)
 - CFE_TBL_File_Hdr_t, [1064](#)
 - CFE_TBL_HandleId_t, [1065](#)
 - CFE_TBL_RegId_t, [1065](#)
- default_cfe_tbl_fcncode_values.h
 - CFE_TBL_CCVAL, [1066](#)
 - CFE_TBL_FunctionCode_, [1067](#)
 - CFE_TBL_FunctionCode_ABORT_LOAD, [1067](#)
 - CFE_TBL_FunctionCode_ACTIVATE, [1067](#)
 - CFE_TBL_FunctionCode_DELETE_CDS, [1067](#)
 - CFE_TBL_FunctionCode_DUMP, [1067](#)
 - CFE_TBL_FunctionCode_DUMP_REGISTRY, [1067](#)
 - CFE_TBL_FunctionCode_LOAD, [1067](#)
 - CFE_TBL_FunctionCode_NOOP, [1067](#)
 - CFE_TBL_FunctionCode_RESET_COUNTERS, [1067](#)
 - CFE_TBL_FunctionCode_SEND_REGISTRY, [1067](#)
 - CFE_TBL_FunctionCode_VALIDATE, [1067](#)
- default_cfe_tbl_interface_cfg_values.h
 - CFE_MISSION_TBL_CFGVAL, [1068](#)
- default_cfe_tbl_internal_cfg_values.h
 - CFE_PLATFORM_TBL_CFGVAL, [1069](#)
- default_cfe_tbl_mission_cfg.h
 - CFE_MISSION_TBL_MAX_FULL_NAME_LEN, [1070](#)
 - CFE_MISSION_TBL_MAX_NAME_LENGTH, [1070](#)

- DEFAULT_CFE_MISSION_TBL_MAX_FULL_NAME_LEN, [1070](#)
- DEFAULT_CFE_MISSION_TBL_MAX_NAME_LENGTH, [1070](#)
- default_cfe_tbl_msgdefs.h
 - CFE_TBL_AbortLoadCmd_Payload_t, [1073](#)
 - CFE_TBL_ActivateCmd_Payload_t, [1073](#)
 - CFE_TBL_DelCDSCmd_Payload_t, [1073](#)
 - CFE_TBL_DumpCmd_Payload_t, [1073](#)
 - CFE_TBL_DumpRegistryCmd_Payload_t, [1073](#)
 - CFE_TBL_HousekeepingTlm_Payload_t, [1074](#)
 - CFE_TBL_LoadCmd_Payload_t, [1074](#)
 - CFE_TBL_NotifyCmd_Payload_t, [1074](#)
 - CFE_TBL_SendRegistryCmd_Payload_t, [1074](#)
 - CFE_TBL_TblRegPacket_Payload_t, [1074](#)
 - CFE_TBL_ValidateCmd_Payload_t, [1074](#)
- default_cfe_tbl_msgid_values.h
 - CFE_PLATFORM_TBL_CMD_MIDVAL, [1077](#)
 - CFE_PLATFORM_TBL_TLM_MIDVAL, [1077](#)
- default_cfe_tbl_msgids.h
 - CFE_TBL_CMD_MID, [1078](#)
 - CFE_TBL_HK_TLM_MID, [1078](#)
 - CFE_TBL_REG_TLM_MID, [1078](#)
 - CFE_TBL_SEND_HK_MID, [1078](#)
- default_cfe_tbl_msgstruct.h
 - CFE_TBL_AbortLoadCmd_t, [1080](#)
 - CFE_TBL_ActivateCmd_t, [1080](#)
 - CFE_TBL_DeleteCDSCmd_t, [1080](#)
 - CFE_TBL_DumpCmd_t, [1080](#)
 - CFE_TBL_DumpRegistryCmd_t, [1080](#)
 - CFE_TBL_HousekeepingTlm_t, [1080](#)
 - CFE_TBL_LoadCmd_t, [1080](#)
 - CFE_TBL_NoopCmd_t, [1080](#)
 - CFE_TBL_NotifyCmd_t, [1080](#)
 - CFE_TBL_ResetCountersCmd_t, [1080](#)
 - CFE_TBL_SendHkCmd_t, [1080](#)
 - CFE_TBL_SendRegistryCmd_t, [1080](#)
 - CFE_TBL_TableRegistryTlm_t, [1081](#)
 - CFE_TBL_ValidateCmd_t, [1081](#)
- default_cfe_tbl_platform_cfg.h
 - CFE_PLATFORM_TBL_BUF_MEMORY_BYTES, [1084](#)
 - CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE, [1084](#)
 - CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES, [1084](#)
 - CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE, [1085](#)
 - CFE_PLATFORM_TBL_MAX_NUM_HANDLES, [1085](#)
 - CFE_PLATFORM_TBL_MAX_NUM_TABLES, [1085](#)
 - CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS, [1086](#)
 - CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS, [1086](#)
 - CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE, [1086](#)
 - CFE_PLATFORM_TBL_START_TASK_PRIORITY, [1087](#)
 - CFE_PLATFORM_TBL_START_TASK_STACK_SIZE, [1087](#)
 - CFE_PLATFORM_TBL_U32FROM4CHARS, [1087](#)
 - CFE_PLATFORM_TBL_VALID_PRID_1, [1088](#)
 - CFE_PLATFORM_TBL_VALID_PRID_2, [1088](#)
 - CFE_PLATFORM_TBL_VALID_PRID_3, [1088](#)
 - CFE_PLATFORM_TBL_VALID_PRID_4, [1088](#)
 - CFE_PLATFORM_TBL_VALID_PRID_COUNT, [1088](#)
 - CFE_PLATFORM_TBL_VALID_SCID_1, [1088](#)
 - CFE_PLATFORM_TBL_VALID_SCID_2, [1089](#)
 - CFE_PLATFORM_TBL_VALID_SCID_COUNT, [1089](#)
 - DEFAULT_CFE_PLATFORM_TBL_BUF_MEMORY_BYTES, [1089](#)
 - DEFAULT_CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE, [1089](#)
 - DEFAULT_CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES, [1089](#)
 - DEFAULT_CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE, [1089](#)
 - DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_HANDLES, [1090](#)
 - DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_TABLES, [1090](#)
 - DEFAULT_CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS, [1090](#)
 - DEFAULT_CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS, [1090](#)
 - DEFAULT_CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE, [1090](#)
 - DEFAULT_CFE_PLATFORM_TBL_START_TASK_PRIORITY, [1090](#)
 - DEFAULT_CFE_PLATFORM_TBL_START_TASK_STACK_SIZE, [1090](#)
 - DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_1, [1090](#)
 - DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_2, [1090](#)
 - DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_3, [1090](#)
 - DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_4, [1090](#)
 - DEFAULT_CFE_PLATFORM_TBL_VALID_PRID_COUNT, [1091](#)
 - DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_1, [1091](#)
 - DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_2, [1091](#)
 - DEFAULT_CFE_PLATFORM_TBL_VALID_SCID_COUNT, [1091](#)

- 1091
- default_cfe_tbl_topcid_values.h
 - CFE_MISSION_TBL_TIDVAL, 1092
- default_cfe_time_extern_typedefs.h
 - CFE_TIME_AdjustDirection, 1142
 - CFE_TIME_AdjustDirection_ADD, 1142
 - CFE_TIME_AdjustDirection_Enum_t, 1140
 - CFE_TIME_AdjustDirection_SUBTRACT, 1142
 - CFE_TIME_ClockState, 1142
 - CFE_TIME_ClockState_Enum_t, 1140
 - CFE_TIME_ClockState_FLYWHEEL, 1142
 - CFE_TIME_ClockState_INVALID, 1142
 - CFE_TIME_ClockState_VALID, 1142
 - CFE_TIME_FlagBit, 1142
 - CFE_TIME_FlagBit_ADD1HZ, 1143
 - CFE_TIME_FlagBit_ADDADJ, 1143
 - CFE_TIME_FlagBit_ADDTCL, 1143
 - CFE_TIME_FlagBit_CLKSET, 1142
 - CFE_TIME_FlagBit_CMDFLY, 1143
 - CFE_TIME_FlagBit_Enum_t, 1141
 - CFE_TIME_FlagBit_FLYING, 1142
 - CFE_TIME_FlagBit_GDTONE, 1143
 - CFE_TIME_FlagBit_SERVER, 1143
 - CFE_TIME_FlagBit_SIGPRI, 1143
 - CFE_TIME_FlagBit_SRCINT, 1142
 - CFE_TIME_FlagBit_SRVFLY, 1143
 - CFE_TIME_FlywheelState, 1143
 - CFE_TIME_FlywheelState_Enum_t, 1141
 - CFE_TIME_FlywheelState_IS_FLY, 1143
 - CFE_TIME_FlywheelState_NO_FLY, 1143
 - CFE_TIME_SetState, 1143
 - CFE_TIME_SetState_Enum_t, 1141
 - CFE_TIME_SetState_NOT_SET, 1143
 - CFE_TIME_SetState_WAS_SET, 1143
 - CFE_TIME_SourceSelect, 1143
 - CFE_TIME_SourceSelect_Enum_t, 1141
 - CFE_TIME_SourceSelect_EXTERNAL, 1143
 - CFE_TIME_SourceSelect_INTERNAL, 1143
 - CFE_TIME_SysTime_t, 1141
 - CFE_TIME_ToneSignalSelect, 1143
 - CFE_TIME_ToneSignalSelect_Enum_t, 1141
 - CFE_TIME_ToneSignalSelect_PRIMARY, 1144
 - CFE_TIME_ToneSignalSelect_REDUNDANT, 1144
- default_cfe_time_fcncode_values.h
 - CFE_TIME_CCVAL, 1146
 - CFE_TIME_FunctionCode_, 1146
 - CFE_TIME_FunctionCode_ADD_ADJUST, 1146
 - CFE_TIME_FunctionCode_ADD_DELAY, 1146
 - CFE_TIME_FunctionCode_ADD_ONE_HZ_ADJUSTMENT, 1146
 - CFE_TIME_FunctionCode_NOOP, 1146
 - CFE_TIME_FunctionCode_RESET_COUNTERS, 1146
 - CFE_TIME_FunctionCode_SEND_DIAGNOSTIC, 1146
 - CFE_TIME_FunctionCode_SET_LEAP_SECONDS, 1146
 - CFE_TIME_FunctionCode_SET_MET, 1146
 - CFE_TIME_FunctionCode_SET_SIGNAL, 1146
 - CFE_TIME_FunctionCode_SET_SOURCE, 1146
 - CFE_TIME_FunctionCode_SET_STATE, 1146
 - CFE_TIME_FunctionCode_SET_STCF, 1146
 - CFE_TIME_FunctionCode_SET_TIME, 1146
 - CFE_TIME_FunctionCode_SUB_ADJUST, 1146
 - CFE_TIME_FunctionCode_SUB_DELAY, 1146
 - CFE_TIME_FunctionCode_SUB_ONE_HZ_ADJUSTMENT, 1146
- default_cfe_time_interface_cfg_values.h
 - CFE_MISSION_TIME_CFGVAL, 1147
- default_cfe_time_internal_cfg_values.h
 - CFE_PLATFORM_TIME_CFGVAL, 1148
- default_cfe_time_msgdefs.h
 - CFE_TIME_DiagnosticTlm_Payload_t, 1152
 - CFE_TIME_HousekeepingTlm_Payload_t, 1152
 - CFE_TIME_LeapsCmd_Payload_t, 1152
 - CFE_TIME_OneHzAdjustmentCmd_Payload_t, 1152
 - CFE_TIME_SignalCmd_Payload_t, 1152
 - CFE_TIME_SourceCmd_Payload_t, 1152
 - CFE_TIME_StateCmd_Payload_t, 1153
 - CFE_TIME_TimeCmd_Payload_t, 1153
 - CFE_TIME_ToneDataCmd_Payload_t, 1153
- default_cfe_time_msgid_values.h
 - CFE_PLATFORM_TIME_CMD_MIDVAL, 1156
 - CFE_PLATFORM_TIME_GLB_CMD_MIDVAL, 1156
 - CFE_PLATFORM_TIME_TLM_MIDVAL, 1156
- default_cfe_time_msgids.h
 - CFE_TIME_1HZ_CMD_MID, 1157
 - CFE_TIME_CMD_MID, 1157
 - CFE_TIME_DATA_CMD_MID, 1157
 - CFE_TIME_DIAG_TLM_MID, 1157
 - CFE_TIME_HK_TLM_MID, 1157
 - CFE_TIME_ONEHZ_CMD_MID, 1157
 - CFE_TIME_SEND_CMD_MID, 1157
 - CFE_TIME_SEND_HK_MID, 1158
 - CFE_TIME_TONE_CMD_MID, 1158
- default_cfe_time_msgstruct.h
 - CFE_TIME_AddAdjustCmd_t, 1160
 - CFE_TIME_AddDelayCmd_t, 1160
 - CFE_TIME_AddOneHzAdjustmentCmd_t, 1160
 - CFE_TIME_DiagnosticTlm_t, 1160
 - CFE_TIME_FakeToneCmd_t, 1160
 - CFE_TIME_HousekeepingTlm_t, 1160
 - CFE_TIME_NoopCmd_t, 1160
 - CFE_TIME_OneHzCmd_t, 1160
 - CFE_TIME_ResetCountersCmd_t, 1160
 - CFE_TIME_SendDiagnosticCmd_t, 1161
 - CFE_TIME_SendHkCmd_t, 1161

- CFE_TIME_SetLeapSecondsCmd_t, 1161
- CFE_TIME_SetMETCmd_t, 1161
- CFE_TIME_SetSignalCmd_t, 1161
- CFE_TIME_SetSourceCmd_t, 1161
- CFE_TIME_SetStateCmd_t, 1161
- CFE_TIME_SetSTCFCmd_t, 1161
- CFE_TIME_SetTimeCmd_t, 1161
- CFE_TIME_SubAdjustCmd_t, 1161
- CFE_TIME_SubDelayCmd_t, 1161
- CFE_TIME_SubOneHzAdjustmentCmd_t, 1161
- CFE_TIME_ToneDataCmd_t, 1161
- CFE_TIME_ToneSignalCmd_t, 1162
- default_cfe_time_topcid_values.h
 - CFE_MISSION_TIME_TIDVAL, 1165
- DEFAULT_GLOBAL_CMD_MID_BASE
 - default_cfe_core_api_msgid_mapping.h, 686
- DEFAULT_GLOBAL_TLM_MID_BASE
 - default_cfe_core_api_msgid_mapping.h, 686
- DelayDirection
 - CFE_TIME_DiagnosticTlm_Payload, 559
- Dependencies, 5
- Description
 - CFE_FS_FileWriteMetaData, 499
 - CFE_FS_Header, 500
 - CFE_TBL_FileDef, 535
- Diagnostic Data, 42
- Double Buffered Tables, 49
- DoubleBuffered
 - CFE_TBL_Info, 542
 - CFE_TBL_TblRegPacket_Payload, 549
- Dump Only Tables, 50
- DumpFilename
 - CFE_ES_DumpCDSRegistryCmd_Payload, 436
 - CFE_TBL_DumpCmd_Payload, 532
 - CFE_TBL_DumpRegistryCmd_Payload, 533
- DumpOnly
 - CFE_TBL_Info, 542
 - CFE_TBL_TblRegPacket_Payload, 549
- DuplicateSubscriptionsCounter
 - CFE_SB_HousekeepingTlm_Payload, 507
- Effects of a Processor Reset on EVS, 31
- Effects of Processor Reset on Tables, 51
- ElementPtr
 - CFE_Config_ArrayValue, 424
- Enabling Fake Tone Signal, 60, 63
- Entries
 - CFE_SB_AllSubscriptionsTlm_Payload, 502
- Entry
 - CFE_SB_AllSubscriptionsTlm_Payload, 502
- entry_point
 - OS_module_prop_t, 588
- EntryPoint
 - CFE_ES_AppInfo, 428
- ERLogEntries
 - CFE_ES_HousekeepingTlm_Payload, 441
- ERLogIndex
 - CFE_ES_HousekeepingTlm_Payload, 441
- Event Message Control, 27
- Event Message Filtering, 29
- Event Message Format, 26
- EventID
 - CFE_EVS_AppNameEventIDCmd_Payload, 474
 - CFE_EVS_AppNameEventIDMaskCmd_Payload, 475
 - CFE_EVS_BinFilter, 477
 - CFE_EVS_PacketID, 490
- EventType
 - CFE_EVS_PacketID, 490
- EVS Counters, 30
- EVS Registry, 29
- EVS squelching of misbehaving apps, 31
- example_2x32bit_cfe_es_memaddress.h, 608
 - CFE_ES_MEMADDRESS_C, 608
 - CFE_ES_MemAddress_FromNative, 609
 - CFE_ES_MEMADDRESS_TO_PTR, 608
 - CFE_ES_MemAddress_ToNative, 609
 - CFE_ES_MEMOFFSET_C, 609
 - CFE_ES_MemOffset_FromNative, 609
 - CFE_ES_MEMOFFSET_TO_SIZET, 609
 - CFE_ES_MemOffset_ToNative, 609
- example_32bit_cfe_es_memaddress.h, 611
 - CFE_ES_MEMADDRESS_C, 611
 - CFE_ES_MemAddress_t, 612
 - CFE_ES_MEMADDRESS_TO_PTR, 611
 - CFE_ES_MEMOFFSET_C, 611
 - CFE_ES_MemOffset_t, 612
 - CFE_ES_MEMOFFSET_TO_SIZET, 612
- example_64bit_cfe_es_memaddress.h, 613
 - CFE_ES_MEMADDRESS_C, 614
 - CFE_ES_MemAddress_t, 614
 - CFE_ES_MEMADDRESS_TO_PTR, 614
 - CFE_ES_MEMOFFSET_C, 614
 - CFE_ES_MemOffset_t, 614
 - CFE_ES_MEMOFFSET_TO_SIZET, 614
- example_mission_cfg.h, 615
 - CFE_FS_FILE_CONTENT_ID, 616
 - CFE_FS_HDR_DESC_MAX_LEN, 616
 - CFE_MISSION_ES_CDS_MAX_FULL_NAME_LEN, 616
 - CFE_MISSION_ES_CDS_MAX_NAME_LENGTH, 617
 - CFE_MISSION_ES_CRC_16, 617
 - CFE_MISSION_ES_CRC_32, 617
 - CFE_MISSION_ES_CRC_8, 617
 - CFE_MISSION_ES_DEFAULT_CRC, 617
 - CFE_MISSION_ES_MAX_APPLICATIONS, 618
 - CFE_MISSION_ES_PERF_MAX_IDS, 618

- CFE_MISSION_ES_POOL_MAX_BUCKETS, [618](#)
- CFE_MISSION_EVS_MAX_MESSAGE_LENGTH, [619](#)
- CFE_MISSION_MAX_API_LEN, [619](#)
- CFE_MISSION_MAX_FILE_LEN, [619](#)
- CFE_MISSION_MAX_NUM_FILES, [620](#)
- CFE_MISSION_MAX_PATH_LEN, [620](#)
- CFE_MISSION_SB_MAX_PIPES, [621](#)
- CFE_MISSION_SB_MAX_SB_MSG_SIZE, [621](#)
- CFE_MISSION_TBL_MAX_FULL_NAME_LEN, [621](#)
- CFE_MISSION_TBL_MAX_NAME_LENGTH, [622](#)
- CFE_MISSION_TIME_AT_TONE_WAS, [622](#)
- CFE_MISSION_TIME_AT_TONE_WILL_BE, [623](#)
- CFE_MISSION_TIME_CFG_DEFAULT_TAI, [623](#)
- CFE_MISSION_TIME_CFG_DEFAULT_UTC, [623](#)
- CFE_MISSION_TIME_CFG_FAKE_TONE, [623](#)
- CFE_MISSION_TIME_DEF_DELAY_SECS, [623](#)
- CFE_MISSION_TIME_DEF_DELAY_SUBS, [623](#)
- CFE_MISSION_TIME_DEF_LEAPS, [624](#)
- CFE_MISSION_TIME_DEF_MET_SECS, [624](#)
- CFE_MISSION_TIME_DEF_MET_SUBS, [624](#)
- CFE_MISSION_TIME_DEF_STCF_SECS, [624](#)
- CFE_MISSION_TIME_DEF_STCF_SUBS, [624](#)
- CFE_MISSION_TIME_EPOCH_DAY, [624](#)
- CFE_MISSION_TIME_EPOCH_HOUR, [624](#)
- CFE_MISSION_TIME_EPOCH_MICROS, [625](#)
- CFE_MISSION_TIME_EPOCH_MINUTE, [625](#)
- CFE_MISSION_TIME_EPOCH_SECOND, [625](#)
- CFE_MISSION_TIME_EPOCH_YEAR, [625](#)
- CFE_MISSION_TIME_FS_FACTOR, [625](#)
- CFE_MISSION_TIME_MAX_ELAPSED, [626](#)
- CFE_MISSION_TIME_MIN_ELAPSED, [626](#)
- example_platform_cfg.h, [628](#)
- CFE_PLATFORM_CORE_MAX_STARTUP_MSEC, [632](#)
- CFE_PLATFORM_ENDIAN, [632](#)
- CFE_PLATFORM_ES_APP_KILL_TIMEOUT, [633](#)
- CFE_PLATFORM_ES_APP_SCAN_RATE, [633](#)
- CFE_PLATFORM_ES_CDS_MAX_BLOCK_SIZE, [634](#)
- CFE_PLATFORM_ES_CDS_MAX_NUM_ENTRIES, [634](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_01, [634](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_02, [634](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_03, [634](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_04, [635](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_05, [635](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_06, [635](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_07, [635](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_08, [635](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_09, [635](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_10, [635](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_11, [635](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_12, [635](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_13, [635](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_14, [635](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_15, [636](#)
- CFE_PLATFORM_ES_CDS_MEM_BLOCK_SIZE_16, [636](#)
- CFE_PLATFORM_ES_CDS_SIZE, [636](#)
- CFE_PLATFORM_ES_DEFAULT_APP_LOG_FILE, [636](#)
- CFE_PLATFORM_ES_DEFAULT_CDS_REG_DUMP_FILE, [636](#)
- CFE_PLATFORM_ES_DEFAULT_ER_LOG_FILE, [637](#)
- CFE_PLATFORM_ES_DEFAULT_PERF_DUMP_FILENAME, [637](#)
- CFE_PLATFORM_ES_DEFAULT_POR_SYSLOG_MODE, [637](#)
- CFE_PLATFORM_ES_DEFAULT_PR_SYSLOG_MODE, [638](#)
- CFE_PLATFORM_ES_DEFAULT_STACK_SIZE, [638](#)
- CFE_PLATFORM_ES_DEFAULT_SYSLOG_FILE, [638](#)
- CFE_PLATFORM_ES_DEFAULT_TASK_LOG_FILE, [639](#)
- CFE_PLATFORM_ES_ER_LOG_ENTRIES, [639](#)
- CFE_PLATFORM_ES_ER_LOG_MAX_CONTEXT_SIZE, [639](#)
- CFE_PLATFORM_ES_MAX_APPLICATIONS, [640](#)
- CFE_PLATFORM_ES_MAX_BLOCK_SIZE, [640](#)
- CFE_PLATFORM_ES_MAX_GEN_COUNTERS, [640](#)
- CFE_PLATFORM_ES_MAX_LIBRARIES, [640](#)
- CFE_PLATFORM_ES_MAX_MEMORY_POOLS, [641](#)
- CFE_PLATFORM_ES_MAX_PROCESSOR_RESETS, [641](#)
- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_01, [641](#)
- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_02, [642](#)
- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_03, [642](#)
- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_04, [642](#)

- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_05, [642](#)
- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_06, [642](#)
- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_07, [642](#)
- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_08, [642](#)
- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_09, [642](#)
- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_10, [643](#)
- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_11, [643](#)
- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_12, [643](#)
- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_13, [643](#)
- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_14, [643](#)
- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_15, [643](#)
- CFE_PLATFORM_ES_MEM_BLOCK_SIZE_16, [643](#)
- CFE_PLATFORM_ES_MEMPOOL_ALIGN_SIZE_MIN, [643](#)
- CFE_PLATFORM_ES_NONVOL_DISK_MOUNT_STRING, [643](#)
- CFE_PLATFORM_ES_NONVOL_STARTUP_FILE, [644](#)
- CFE_PLATFORM_ES_PERF_CHILD_MS_DELAY, [644](#)
- CFE_PLATFORM_ES_PERF_CHILD_PRIORITY, [644](#)
- CFE_PLATFORM_ES_PERF_CHILD_STACK_SIZE, [645](#)
- CFE_PLATFORM_ES_PERF_DATA_BUFFER_SIZE, [645](#)
- CFE_PLATFORM_ES_PERF_ENTRIES_BTWN_DLVS, [645](#)
- CFE_PLATFORM_ES_PERF_FILTMASK_ALL, [645](#)
- CFE_PLATFORM_ES_PERF_FILTMASK_INIT, [646](#)
- CFE_PLATFORM_ES_PERF_FILTMASK_NONE, [646](#)
- CFE_PLATFORM_ES_PERF_TRIGMASK_ALL, [646](#)
- CFE_PLATFORM_ES_PERF_TRIGMASK_INIT, [646](#)
- CFE_PLATFORM_ES_PERF_TRIGMASK_NONE, [646](#)
- CFE_PLATFORM_ES_POOL_MAX_BUCKETS, [647](#)
- CFE_PLATFORM_ES_RAM_DISK_MOUNT_STRING, [647](#)
- CFE_PLATFORM_ES_RAM_DISK_NUM_SECTORS, [647](#)
- CFE_PLATFORM_ES_RAM_DISK_PERCENT_RESERVED, [648](#)
- CFE_PLATFORM_ES_RAM_DISK_SECTOR_SIZE, [648](#)
- CFE_PLATFORM_ES_START_TASK_PRIORITY, [648](#)
- CFE_PLATFORM_ES_START_TASK_STACK_SIZE, [649](#)
- CFE_PLATFORM_ES_STARTUP_SCRIPT_TIMEOUT_MSEC, [649](#)
- CFE_PLATFORM_ES_STARTUP_SYNC_POLL_MSEC, [649](#)
- CFE_PLATFORM_ES_SYSTEM_LOG_SIZE, [650](#)
- CFE_PLATFORM_ES_USER_RESERVED_SIZE, [650](#)
- CFE_PLATFORM_ES_VOLATILE_STARTUP_FILE, [650](#)
- CFE_PLATFORM_EVS_APP_EVENTS_PER_SEC, [651](#)
- CFE_PLATFORM_EVS_DEFAULT_APP_DATA_FILE, [651](#)
- CFE_PLATFORM_EVS_DEFAULT_LOG_FILE, [651](#)
- CFE_PLATFORM_EVS_DEFAULT_LOG_MODE, [652](#)
- CFE_PLATFORM_EVS_DEFAULT_MSG_FORMAT_MODE, [652](#)
- CFE_PLATFORM_EVS_DEFAULT_TYPE_FLAG, [652](#)
- CFE_PLATFORM_EVS_LOG_MAX, [653](#)
- CFE_PLATFORM_EVS_MAX_APP_EVENT_BURST, [653](#)
- CFE_PLATFORM_EVS_MAX_EVENT_FILTERS, [653](#)
- CFE_PLATFORM_EVS_PORT_DEFAULT, [654](#)
- CFE_PLATFORM_EVS_START_TASK_PRIORITY, [654](#)
- CFE_PLATFORM_EVS_START_TASK_STACK_SIZE, [654](#)
- CFE_PLATFORM_SB_BUF_MEMORY_BYTES, [655](#)
- CFE_PLATFORM_SB_DEFAULT_MAP_FILENAME, [655](#)
- CFE_PLATFORM_SB_DEFAULT_MSG_LIMIT, [655](#)
- CFE_PLATFORM_SB_DEFAULT_PIPE_FILENAME, [656](#)
- CFE_PLATFORM_SB_DEFAULT_ROUTING_FILENAME, [656](#)
- CFE_PLATFORM_SB_FILTER_MASK1, [656](#)
- CFE_PLATFORM_SB_FILTER_MASK2, [656](#)
- CFE_PLATFORM_SB_FILTER_MASK3, [657](#)
- CFE_PLATFORM_SB_FILTER_MASK4, [657](#)
- CFE_PLATFORM_SB_FILTER_MASK5, [657](#)
- CFE_PLATFORM_SB_FILTER_MASK6, [657](#)
- CFE_PLATFORM_SB_FILTER_MASK7, [657](#)
- CFE_PLATFORM_SB_FILTER_MASK8, [657](#)
- CFE_PLATFORM_SB_FILTERED_EVENT1, [657](#)
- CFE_PLATFORM_SB_FILTERED_EVENT2, [657](#)
- CFE_PLATFORM_SB_FILTERED_EVENT3, [657](#)
- CFE_PLATFORM_SB_FILTERED_EVENT4, [658](#)
- CFE_PLATFORM_SB_FILTERED_EVENT5, [658](#)
- CFE_PLATFORM_SB_FILTERED_EVENT6, [658](#)
- CFE_PLATFORM_SB_FILTERED_EVENT7, [658](#)
- CFE_PLATFORM_SB_FILTERED_EVENT8, [658](#)
- CFE_PLATFORM_SB_HIGHEST_VALID_MSGID, [658](#)
- CFE_PLATFORM_SB_MAX_BLOCK_SIZE, [658](#)
- CFE_PLATFORM_SB_MAX_DEST_PER_PKT, [658](#)
- CFE_PLATFORM_SB_MAX_MSG_IDS, [659](#)

- CFE_PLATFORM_SB_MAX_PIPES, [659](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_01, [659](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_02, [660](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_03, [660](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_04, [660](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_05, [660](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_06, [660](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_07, [660](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_08, [660](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_09, [660](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_10, [660](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_11, [661](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_12, [661](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_13, [661](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_14, [661](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_15, [661](#)
- CFE_PLATFORM_SB_MEM_BLOCK_SIZE_16, [661](#)
- CFE_PLATFORM_SB_START_TASK_PRIORITY, [661](#)
- CFE_PLATFORM_SB_START_TASK_STACK_SIZE, [661](#)
- CFE_PLATFORM_TBL_BUF_MEMORY_BYTES, [662](#)
- CFE_PLATFORM_TBL_DEFAULT_REG_DUMP_FILE, [662](#)
- CFE_PLATFORM_TBL_MAX_CRITICAL_TABLES, [662](#)
- CFE_PLATFORM_TBL_MAX_DBL_TABLE_SIZE, [662](#)
- CFE_PLATFORM_TBL_MAX_NUM_HANDLES, [663](#)
- CFE_PLATFORM_TBL_MAX_NUM_TABLES, [663](#)
- CFE_PLATFORM_TBL_MAX_NUM_VALIDATIONS, [663](#)
- CFE_PLATFORM_TBL_MAX_SIMULTANEOUS_LOADS, [664](#)
- CFE_PLATFORM_TBL_MAX_SNGL_TABLE_SIZE, [664](#)
- CFE_PLATFORM_TBL_START_TASK_PRIORITY, [664](#)
- CFE_PLATFORM_TBL_START_TASK_STACK_SIZE, [665](#)
- CFE_PLATFORM_TBL_U32FROM4CHARS, [665](#)
- CFE_PLATFORM_TBL_VALID_PRID_1, [665](#)
- CFE_PLATFORM_TBL_VALID_PRID_2, [665](#)
- CFE_PLATFORM_TBL_VALID_PRID_3, [666](#)
- CFE_PLATFORM_TBL_VALID_PRID_4, [666](#)
- CFE_PLATFORM_TBL_VALID_PRID_COUNT, [666](#)
- CFE_PLATFORM_TBL_VALID_SCID_1, [666](#)
- CFE_PLATFORM_TBL_VALID_SCID_2, [666](#)
- CFE_PLATFORM_TBL_VALID_SCID_COUNT, [666](#)
- CFE_PLATFORM_TIME_CFG_CLIENT, [667](#)
- CFE_PLATFORM_TIME_CFG_LATCH_FLY, [667](#)
- CFE_PLATFORM_TIME_CFG_SERVER, [667](#)
- CFE_PLATFORM_TIME_CFG_SIGNAL, [667](#)
- CFE_PLATFORM_TIME_CFG_SOURCE, [668](#)
- CFE_PLATFORM_TIME_CFG_SRC_GPS, [668](#)
- CFE_PLATFORM_TIME_CFG_SRC_MET, [668](#)
- CFE_PLATFORM_TIME_CFG_SRC_TIME, [669](#)
- CFE_PLATFORM_TIME_CFG_START_FLY, [669](#)
- CFE_PLATFORM_TIME_CFG_TONE_LIMIT, [669](#)
- CFE_PLATFORM_TIME_CFG_VIRTUAL, [669](#)
- CFE_PLATFORM_TIME_MAX_DELTA_SECS, [670](#)
- CFE_PLATFORM_TIME_MAX_DELTA_SUBS, [670](#)
- CFE_PLATFORM_TIME_MAX_LOCAL_SECS, [670](#)
- CFE_PLATFORM_TIME_MAX_LOCAL_SUBS, [670](#)
- CFE_PLATFORM_TIME_ONEHZ_TASK_PRIORITY, [671](#)
- CFE_PLATFORM_TIME_ONEHZ_TASK_STACK_SIZE, [671](#)
- CFE_PLATFORM_TIME_START_TASK_PRIORITY, [671](#)
- CFE_PLATFORM_TIME_START_TASK_STACK_SIZE, [671](#)
- CFE_PLATFORM_TIME_TONE_TASK_PRIORITY, [671](#)
- CFE_PLATFORM_TIME_TONE_TASK_STACK_SIZE, [671](#)
- Exception and Reset (ER) Log, [10](#)
- ExceptionAction
 - CFE_ES_AppInfo, [428](#)
 - CFE_ES_StartAppCmd_Payload, [465](#)
- ExecutionCounter
 - CFE_ES_AppInfo, [428](#)
 - CFE_ES_TaskInfo, [470](#)
- FailedValCounter
 - CFE_TBL_HousekeepingTlm_Payload, [538](#)
- File Descriptors In Osal, [101](#)
- File System Overview, [100](#)
- FileModeBits
 - os_fstat_t, [585](#)
- FileName
 - CFE_ES_AppInfo, [428](#)
 - CFE_ES_FileNameCmd_Payload, [437](#)
 - CFE_FS_FileWriteMetaData, [499](#)
 - os_dirent_t, [582](#)
- Filename
 - CFE_SB_WriteFileInfoCmd_Payload, [525](#)
- filename
 - OS_module_prop_t, [588](#)
- FileSize
 - os_fstat_t, [585](#)
- FileSubType
 - CFE_FS_FileWriteMetaData, [499](#)
- FileTime
 - CFE_TBL_Info, [542](#)
 - CFE_TBL_TblRegPacket_Payload, [549](#)
 - os_fstat_t, [585](#)

- FilterMask
 - CFE_ES_SetPerfFilterMaskCmd_Payload, [462](#)
- FilterMaskNum
 - CFE_ES_SetPerfFilterMaskCmd_Payload, [462](#)
- flags
 - OS_module_address_t, [587](#)
- Flywheeling, [67](#)
- Forced2Fly
 - CFE_TIME_DiagnosticTlm_Payload, [560](#)
- free_blocks
 - OS_heap_prop_t, [586](#)
- free_bytes
 - OS_heap_prop_t, [586](#)
- FreeFds
 - os_fsinfo_t, [584](#)
- freerun_time
 - OS_timebase_prop_t, [596](#)
- FreeVolumes
 - os_fsinfo_t, [584](#)
- Frequently Asked Questions about Event Services, [32](#)
- Frequently Asked Questions about Executive Services, [18](#)
- Frequently Asked Questions about Software Bus, [43](#)
- Frequently Asked Questions about Table Services, [52](#)
- Frequently Asked Questions about Time Services, [74](#)
- GetData
 - CFE_FS_FileWriteMetaData, [499](#)
- GetPipeIdByNameErrorCounter
 - CFE_SB_HousekeepingTlm_Payload, [507](#)
- Glossary of Terms, [91](#)
- Handle
 - CFE_ES_CDSRegDumpRec, [433](#)
- HeapBlocksFree
 - CFE_ES_HousekeepingTlm_Payload, [442](#)
- HeapBytesFree
 - CFE_ES_HousekeepingTlm_Payload, [442](#)
- HeapMaxBlockSize
 - CFE_ES_HousekeepingTlm_Payload, [442](#)
- host_module_id
 - OS_module_prop_t, [588](#)
- InactiveBufferAddr
 - CFE_TBL_TblRegPacket_Payload, [550](#)
- Index
 - CFE_SB_MsgMapFileEntry, [510](#)
- Initialization, [39](#), [68](#)
- int16
 - common_types.h, [1216](#)
- int32
 - common_types.h, [1216](#)
- int64
 - common_types.h, [1216](#)
- int8
 - common_types.h, [1217](#)
- InternalErrorCounter
 - CFE_SB_HousekeepingTlm_Payload, [507](#)
- interval_time
 - OS_timer_prop_t, [596](#)
- intptr
 - common_types.h, [1217](#)
- IntVal
 - OS_socket_optval, [592](#)
- IsPending
 - CFE_FS_FileWriteMetaData, [499](#)
- IsValid
 - OS_file_prop_t, [583](#)
- Known Problem, [43](#)
- largest_free_block
 - OS_heap_prop_t, [586](#)
- LastFileDumped
 - CFE_TBL_HousekeepingTlm_Payload, [538](#)
- LastFileLoaded
 - CFE_TBL_HousekeepingTlm_Payload, [539](#)
 - CFE_TBL_Info, [542](#)
 - CFE_TBL_TblRegPacket_Payload, [550](#)
- LastTableLoaded
 - CFE_TBL_HousekeepingTlm_Payload, [539](#)
- LastUpdatedTable
 - CFE_TBL_HousekeepingTlm_Payload, [539](#)
- LastUpdateTime
 - CFE_TBL_HousekeepingTlm_Payload, [539](#)
- LastValCrc
 - CFE_TBL_HousekeepingTlm_Payload, [539](#)
- LastValStatus
 - CFE_TBL_HousekeepingTlm_Payload, [539](#)
- LastValTableName
 - CFE_TBL_HousekeepingTlm_Payload, [539](#)
- LeapSeconds
 - CFE_TIME_HousekeepingTlm_Payload, [566](#)
 - CFE_TIME_LeapsCmd_Payload, [567](#)
- Length
 - CCSDS_PrimaryHeader, [423](#)
 - CFE_FS_Header, [500](#)
- LENGTHCHECK
 - osapi-macros.h, [1256](#)
- Listing Current Applications, [12](#)
- Listing Current Tasks, [13](#)
- LoadFilename
 - CFE_TBL_LoadCmd_Payload, [544](#)
- Loading Common Libraries, [13](#)
- LoadPending
 - CFE_TBL_TblRegPacket_Payload, [550](#)
- Local Event Log, [27](#)
- LocalIntCounter
 - CFE_TIME_DiagnosticTlm_Payload, [560](#)
- LocalTaskCounter
 - CFE_TIME_DiagnosticTlm_Payload, [560](#)

- LogEnabled
 - CFE_EVS_HousekeepingTlm_Payload, [485](#)
- LogFilename
 - CFE_EVS_LogFileCmd_Payload, [487](#)
- LogFullFlag
 - CFE_EVS_HousekeepingTlm_Payload, [486](#)
- LogMode
 - CFE_EVS_HousekeepingTlm_Payload, [486](#)
 - CFE_EVS_SetLogMode_Payload, [495](#)
- LogOverflowCounter
 - CFE_EVS_HousekeepingTlm_Payload, [486](#)
- LongDouble
 - CFE_ES_PoolAlign, [453](#)
 - CFE_SB_Msg, [509](#)
- LongInt
 - CFE_ES_PoolAlign, [453](#)
 - CFE_SB_Msg, [509](#)
- MainTaskId
 - CFE_ES_AppInfo, [428](#)
- MainTaskName
 - CFE_ES_AppInfo, [429](#)
- Managing Tables, [47](#)
- Mask
 - CFE_EVS_AppNameEventIDMaskCmd_Payload, [475](#)
 - CFE_EVS_BinFilter, [477](#)
- MaxElapsed
 - CFE_TIME_DiagnosticTlm_Payload, [560](#)
- MaxFds
 - os_fsinfo_t, [584](#)
- MaxLocalClock
 - CFE_TIME_DiagnosticTlm_Payload, [560](#)
- MaxMemAllowed
 - CFE_SB_StatsTlm_Payload, [522](#)
- MaxMsgIdsAllowed
 - CFE_SB_StatsTlm_Payload, [522](#)
- MaxPipeDepthAllowed
 - CFE_SB_StatsTlm_Payload, [522](#)
- MaxPipesAllowed
 - CFE_SB_StatsTlm_Payload, [522](#)
- MaxPRCount
 - CFE_ES_SetMaxPRCountCmd_Payload, [461](#)
- MaxProcessorResets
 - CFE_ES_HousekeepingTlm_Payload, [442](#)
- MaxQueueDepth
 - CFE_SB_PipeDepthStats, [512](#)
 - CFE_SB_PipeInfoEntry, [513](#)
- MaxSubscriptionsAllowed
 - CFE_SB_StatsTlm_Payload, [522](#)
- MaxVolumes
 - os_fsinfo_t, [584](#)
- MemInUse
 - CFE_SB_HousekeepingTlm_Payload, [507](#)
 - CFE_SB_StatsTlm_Payload, [522](#)
- Memory, [38](#)
- Memory Pool, [16](#)
- MemPoolHandle
 - CFE_SB_HousekeepingTlm_Payload, [507](#)
 - CFE_TBL_HousekeepingTlm_Payload, [540](#)
- Message
 - CFE_EVS_LongEventTlm_Payload, [488](#)
- Message Limit Error, [41](#)
- Message Routing, [40](#)
- MessageFormatMode
 - CFE_EVS_HousekeepingTlm_Payload, [486](#)
- Messages, [37](#)
- MessageSendCounter
 - CFE_EVS_HousekeepingTlm_Payload, [486](#)
- MessageTruncCounter
 - CFE_EVS_HousekeepingTlm_Payload, [486](#)
- MET
 - CFE_TIME_HousekeepingTlm_Payload, [566](#)
- MicroSeconds
 - CFE_TIME_TimeCmd_Payload, [578](#)
- MinElapsed
 - CFE_TIME_DiagnosticTlm_Payload, [560](#)
- Mode
 - CFE_ES_OverWriteSysLogCmd_Payload, [452](#)
- Module
 - OS_static_symbol_record_t, [593](#)
- Msg
 - CFE_SB_Msg, [509](#)
- MsgCnt
 - CFE_SB_RoutingFileEntry, [517](#)
- MsgFormat
 - CFE_EVS_SetEventFormatCode_Payload, [493](#)
- MsgId
 - CFE_SB_MsgMapFileEntry, [510](#)
 - CFE_SB_RouteCmd_Payload, [516](#)
 - CFE_SB_RoutingFileEntry, [517](#)
 - CFE_SB_SingleSubscriptionTlm_Payload, [520](#)
 - CFE_SB_SubEntries, [524](#)
- MsgIdsInUse
 - CFE_SB_StatsTlm_Payload, [522](#)
- MsgLimitErrorCounter
 - CFE_SB_HousekeepingTlm_Payload, [507](#)
- MsgReceiveErrorCounter
 - CFE_SB_HousekeepingTlm_Payload, [507](#)
- MsgSendErrorCounter
 - CFE_SB_HousekeepingTlm_Payload, [508](#)
- Name
 - CFE_Config_IdNameEntry, [425](#)
 - CFE_ES_AppInfo, [429](#)
 - CFE_ES_CDSRegDumpRec, [433](#)
 - CFE_TBL_TblRegPacket_Payload, [550](#)
 - OS_static_symbol_record_t, [593](#)

- name
 - OS_bin_sem_prop_t, [581](#)
 - OS_condvar_prop_t, [581](#)
 - OS_count_sem_prop_t, [582](#)
 - OS_module_prop_t, [588](#)
 - OS_mut_sem_prop_t, [588](#)
 - OS_queue_prop_t, [589](#)
 - OS_rwlock_prop_t, [590](#)
 - OS_socket_prop_t, [592](#)
 - OS_task_prop_t, [594](#)
 - OS_timebase_prop_t, [596](#)
 - OS_timer_prop_t, [596](#)
- nominal_interval_time
 - OS_timebase_prop_t, [596](#)
- Normal Operation, [69](#)
- NoSubscribersCounter
 - CFE_SB_HousekeepingTlm_Payload, [508](#)
- NumBlocksRequested
 - CFE_ES_MemPoolStats, [448](#)
- NumBytes
 - CFE_TBL_File_Hdr, [534](#)
- NumCreated
 - CFE_ES_BlockStats, [432](#)
- NumElements
 - CFE_Config_ArrayValue, [424](#)
- NumFree
 - CFE_ES_BlockStats, [432](#)
- NumFreeBytes
 - CFE_ES_MemPoolStats, [448](#)
- NumFreeSharedBufs
 - CFE_TBL_HousekeepingTlm_Payload, [540](#)
- NumLoadPending
 - CFE_TBL_HousekeepingTlm_Payload, [540](#)
- NumOfChildTasks
 - CFE_ES_AppInfo, [429](#)
- NumTables
 - CFE_TBL_HousekeepingTlm_Payload, [540](#)
- NumUsers
 - CFE_TBL_Info, [542](#)
- NumValRequests
 - CFE_TBL_HousekeepingTlm_Payload, [540](#)
- object_ids
 - OS_FdSet, [583](#)
- ObjectName
 - CFE_TBL_FileDef, [535](#)
- ObjectSize
 - CFE_TBL_FileDef, [535](#)
- Offset
 - CFE_TBL_File_Hdr, [534](#)
- OneHzAdjust
 - CFE_TIME_DiagnosticTlm_Payload, [560](#)
- OneHzDirection
 - CFE_TIME_DiagnosticTlm_Payload, [561](#)
- OneTimeAdjust
 - CFE_TIME_DiagnosticTlm_Payload, [561](#)
- OneTimeDirection
 - CFE_TIME_DiagnosticTlm_Payload, [561](#)
- OnEvent
 - CFE_FS_FileWriteMetaData, [499](#)
- Operation of the SB Software, [39](#)
- Opts
 - CFE_SB_PipeInfoEntry, [513](#)
- OS_ADD_TASK_FLAGS
 - osconfig.h, [598](#)
- OS_API_Init
 - OSAL Core Operation APIs, [313](#)
- OS_API_Teardown
 - OSAL Core Operation APIs, [313](#)
- OS_Application_Run
 - OSAL Core Operation APIs, [313](#)
- OS_Application_Startup
 - OSAL Core Operation APIs, [313](#)
- OS_ApplicationExit
 - OSAL Core Operation APIs, [314](#)
- OS_ApplicationShutdown
 - OSAL Core Operation APIs, [314](#)
- OS_ArgCallback_t
 - common_types.h, [1217](#)
- OS_bin_sem_prop_t, [580](#)
 - creator, [581](#)
 - name, [581](#)
 - value, [581](#)
- OS_BinSemCreate
 - OSAL Binary Semaphore APIs, [292](#)
- OS_BinSemDelete
 - OSAL Binary Semaphore APIs, [292](#)
- OS_BinSemFlush
 - OSAL Binary Semaphore APIs, [293](#)
- OS_BinSemGetIdByName
 - OSAL Binary Semaphore APIs, [293](#)
- OS_BinSemGetInfo
 - OSAL Binary Semaphore APIs, [294](#)
- OS_BinSemGive
 - OSAL Binary Semaphore APIs, [294](#)
- OS_BinSemTake
 - OSAL Binary Semaphore APIs, [295](#)
- OS_BinSemTimedWait
 - OSAL Binary Semaphore APIs, [295](#)
- OS_BSP_GetArgC
 - OSAL BSP low level access APIs, [296](#)
- OS_BSP_GetArgV
 - OSAL BSP low level access APIs, [296](#)
- OS_BSP_GetResourceTypeConfig
 - OSAL BSP low level access APIs, [296](#)
- OS_BSP_SetExitCode
 - OSAL BSP low level access APIs, [296](#)
- OS_BSP_SetResourceTypeConfig

- OSAL BSP low level access APIs, 296
- OS_BUFFER_MSG_DEPTH
 - osconfig.h, 598
- OS_BUFFER_SIZE
 - osconfig.h, 599
- OS_BUILD_BASELINE
 - osapi-version.h, 1282
- OS_BUILD_CODENAME
 - osapi-version.h, 1282
- OS_BUILD_DEV_CYCLE
 - osapi-version.h, 1282
- OS_BUILD_NUMBER
 - osapi-version.h, 1282
- OS_CFG_MAX_VERSION_STR_LEN
 - osapi-version.h, 1283
- OS_CHECK
 - osapi-constants.h, 1234
- OS_CHK_ONLY
 - osapi-filesys.h, 1249
- OS_chkfs
 - OSAL File System Level APIs, 354
- OS_chmod
 - OSAL Standard File APIs, 341
- OS_close
 - OSAL Standard File APIs, 341
- OS_CloseAllFiles
 - OSAL Standard File APIs, 342
- OS_CloseFileByName
 - OSAL Standard File APIs, 342
- OS_condvar_prop_t, 581
 - creator, 581
 - name, 581
- OS_CondVarBroadcast
 - OSAL Condition Variable APIs, 316
- OS_CondVarCreate
 - OSAL Condition Variable APIs, 316
- OS_CondVarDelete
 - OSAL Condition Variable APIs, 317
- OS_CondVarGetIdByName
 - OSAL Condition Variable APIs, 318
- OS_CondVarGetInfo
 - OSAL Condition Variable APIs, 318
- OS_CondVarLock
 - OSAL Condition Variable APIs, 319
- OS_CondVarSignal
 - OSAL Condition Variable APIs, 319
- OS_CondVarTimedWait
 - OSAL Condition Variable APIs, 320
- OS_CondVarUnlock
 - OSAL Condition Variable APIs, 320
- OS_CondVarWait
 - OSAL Condition Variable APIs, 321
- OS_ConvertToArrayIndex
 - OSAL Object ID Utility APIs, 364
- OS_count_sem_prop_t, 581
 - creator, 582
 - name, 582
 - value, 582
- OS_CountSemCreate
 - OSAL Counting Semaphore APIs, 322
- OS_CountSemDelete
 - OSAL Counting Semaphore APIs, 323
- OS_CountSemGetIdByName
 - OSAL Counting Semaphore APIs, 323
- OS_CountSemGetInfo
 - OSAL Counting Semaphore APIs, 324
- OS_CountSemGive
 - OSAL Counting Semaphore APIs, 324
- OS_CountSemTake
 - OSAL Counting Semaphore APIs, 325
- OS_CountSemTimedWait
 - OSAL Counting Semaphore APIs, 325
- OS_cp
 - OSAL Standard File APIs, 342
- OS_DeleteAllObjects
 - OSAL Core Operation APIs, 314
- OS_DirectoryClose
 - OSAL Directory APIs, 326
- OS_DirectoryOpen
 - OSAL Directory APIs, 327
- OS_DirectoryRead
 - OSAL Directory APIs, 327
- OS_DirectoryRewind
 - OSAL Directory APIs, 328
- os_dirent_t, 582
 - FileName, 582
- OS_DIRENTRY_NAME
 - osapi-dir.h, 1238
- OS_ERR_BAD_ADDRESS
 - OSAL Return Code Defines, 331
- OS_ERR_EMPTY_SET
 - OSAL Return Code Defines, 331
- OS_ERR_FILE
 - OSAL Return Code Defines, 332
- OS_ERR_INCORRECT_OBJ_STATE
 - OSAL Return Code Defines, 332
- OS_ERR_INCORRECT_OBJ_TYPE
 - OSAL Return Code Defines, 332
- OS_ERR_INVALID_ARGUMENT
 - OSAL Return Code Defines, 332
- OS_ERR_INVALID_ID
 - OSAL Return Code Defines, 332
- OS_ERR_INVALID_PRIORITY
 - OSAL Return Code Defines, 332
- OS_ERR_INVALID_SIZE
 - OSAL Return Code Defines, 332
- OS_ERR_NAME_NOT_FOUND
 - OSAL Return Code Defines, 332

- os_err_name_t
 - osapi-error.h, [1241](#)
- OS_ERR_NAME_TAKEN
 - OSAL Return Code Defines, [332](#)
- OS_ERR_NAME_TOO_LONG
 - OSAL Return Code Defines, [333](#)
- OS_ERR_NO_FREE_IDS
 - OSAL Return Code Defines, [333](#)
- OS_ERR_NOT_IMPLEMENTED
 - OSAL Return Code Defines, [333](#)
- OS_ERR_OBJECT_IN_USE
 - OSAL Return Code Defines, [333](#)
- OS_ERR_OPERATION_NOT_SUPPORTED
 - OSAL Return Code Defines, [333](#)
- OS_ERR_OUTPUT_TOO_LARGE
 - OSAL Return Code Defines, [333](#)
- OS_ERR_SEM_NOT_FULL
 - OSAL Return Code Defines, [333](#)
- OS_ERR_STREAM_DISCONNECTED
 - OSAL Return Code Defines, [333](#)
- OS_ERR_TRY_AGAIN
 - OSAL Return Code Defines, [333](#)
- OS_ERROR
 - OSAL Return Code Defines, [334](#)
- OS_ERROR_ADDRESS_MISALIGNED
 - OSAL Return Code Defines, [334](#)
- OS_ERROR_NAME_LENGTH
 - osapi-error.h, [1241](#)
- OS_ERROR_TIMEOUT
 - OSAL Return Code Defines, [334](#)
- OS_EVENT_MAX
 - osapi-common.h, [1231](#)
- OS_EVENT_RESERVED
 - osapi-common.h, [1231](#)
- OS_EVENT_RESOURCE_ALLOCATED
 - osapi-common.h, [1231](#)
- OS_EVENT_RESOURCE_CREATED
 - osapi-common.h, [1231](#)
- OS_EVENT_RESOURCE_DELETED
 - osapi-common.h, [1231](#)
- OS_Event_t
 - osapi-common.h, [1231](#)
- OS_EVENT_TASK_STARTUP
 - osapi-common.h, [1231](#)
- OS_EventHandler_t
 - osapi-common.h, [1230](#)
- OS_FDGetInfo
 - OSAL Standard File APIs, [343](#)
- OS_FdSet, [583](#)
 - object_ids, [583](#)
- OS_FILE_FLAG_CREATE
 - osapi-file.h, [1246](#)
- OS_FILE_FLAG_NONE
 - osapi-file.h, [1246](#)
- OS_file_flag_t
 - osapi-file.h, [1246](#)
- OS_FILE_FLAG_TRUNCATE
 - osapi-file.h, [1246](#)
- OS_file_prop_t, [583](#)
 - IsValid, [583](#)
 - Path, [583](#)
 - User, [584](#)
- OS_FileAllocate
 - OSAL Standard File APIs, [343](#)
- OS_FileOpenCheck
 - OSAL Standard File APIs, [344](#)
- OS_FILESTAT_EXEC
 - osapi-file.h, [1245](#)
- OS_FILESTAT_ISDIR
 - osapi-file.h, [1245](#)
- OS_FILESTAT_MODE
 - osapi-file.h, [1245](#)
- OS_FILESTAT_MODE_DIR
 - osapi-file.h, [1246](#)
- OS_FILESTAT_MODE_EXEC
 - osapi-file.h, [1246](#)
- OS_FILESTAT_MODE_READ
 - osapi-file.h, [1246](#)
- OS_FILESTAT_MODE_WRITE
 - osapi-file.h, [1246](#)
- OS_FILESTAT_READ
 - osapi-file.h, [1245](#)
- OS_FILESTAT_SIZE
 - osapi-file.h, [1245](#)
- OS_FILESTAT_TIME
 - osapi-file.h, [1245](#)
- OS_FILESTAT_WRITE
 - osapi-file.h, [1245](#)
- OS_FileSysAddFixedMap
 - OSAL File System Level APIs, [354](#)
- OS_FileSysStatVolume
 - OSAL File System Level APIs, [355](#)
- OS_FileTruncate
 - OSAL Standard File APIs, [345](#)
- OS_ForEachObject
 - OSAL Object ID Utility APIs, [365](#)
- OS_ForEachObjectOfType
 - OSAL Object ID Utility APIs, [365](#)
- OS_FP_ENABLED
 - osapi-task.h, [1276](#)
- OS_FS_DEV_NAME_LEN
 - osconfig.h, [599](#)
- OS_FS_ERR_DEVICE_NOT_FREE
 - OSAL Return Code Defines, [334](#)
- OS_FS_ERR_DRIVE_NOT_CREATED
 - OSAL Return Code Defines, [334](#)
- OS_FS_ERR_NAME_TOO_LONG
 - OSAL Return Code Defines, [334](#)

- OS_FS_ERR_PATH_INVALID
 - OSAL Return Code Defines, [334](#)
- OS_FS_ERR_PATH_TOO_LONG
 - OSAL Return Code Defines, [334](#)
- OS_FS_GetPhysDriveName
 - OSAL File System Level APIs, [356](#)
- OS_FS_PHYS_NAME_LEN
 - osconfig.h, [599](#)
- OS_FS_VOL_NAME_LEN
 - osconfig.h, [599](#)
- os_fsinfo_t, [584](#)
 - FreeFds, [584](#)
 - FreeVolumes, [584](#)
 - MaxFds, [584](#)
 - MaxVolumes, [584](#)
- os_fstat_t, [585](#)
 - FileModeBits, [585](#)
 - FileSize, [585](#)
 - FileTime, [585](#)
- OS_GetBuildNumber
 - osapi-version.h, [1284](#)
- OS_GetErrorName
 - OSAL Error Info APIs, [336](#)
- OS_GetFsInfo
 - OSAL File System Level APIs, [356](#)
- OS_GetLocalTime
 - OSAL Real Time Clock APIs, [298](#)
- OS_GetMonotonicTime
 - OSAL Real Time Clock APIs, [298](#)
- OS_GetResourceName
 - OSAL Object ID Utility APIs, [365](#)
- OS_GetVersionCodeName
 - osapi-version.h, [1284](#)
- OS_GetVersionNumber
 - osapi-version.h, [1284](#)
- OS_GetVersionString
 - osapi-version.h, [1285](#)
- OS_heap_prop_t, [585](#)
 - free_blocks, [586](#)
 - free_bytes, [586](#)
 - largest_free_block, [586](#)
- OS_HeapGetInfo
 - OSAL Heap APIs, [361](#)
- OS_IdentifyObject
 - OSAL Object ID Utility APIs, [366](#)
- OS_IdleLoop
 - OSAL Core Operation APIs, [314](#)
- OS_initfs
 - OSAL File System Level APIs, [357](#)
- OS_INVALID_INT_NUM
 - OSAL Return Code Defines, [334](#)
- OS_INVALID_POINTER
 - OSAL Return Code Defines, [335](#)
- OS_INVALID_SEM_VALUE
 - OSAL Return Code Defines, [335](#)
- OS_LAST_OFFICIAL
 - osapi-version.h, [1283](#)
- OS_lseek
 - OSAL Standard File APIs, [345](#)
- OS_MAJOR_VERSION
 - osapi-version.h, [1283](#)
- OS_MAX_API_NAME
 - osconfig.h, [599](#)
- OS_MAX_BIN_SEMAPHORES
 - osconfig.h, [599](#)
- OS_MAX_CMD_LEN
 - osconfig.h, [599](#)
- OS_MAX_CONDVARS
 - osconfig.h, [600](#)
- OS_MAX_CONSOLES
 - osconfig.h, [600](#)
- OS_MAX_COUNT_SEMAPHORES
 - osconfig.h, [600](#)
- OS_MAX_FILE_NAME
 - osconfig.h, [600](#)
- OS_MAX_FILE_SYSTEMS
 - osconfig.h, [600](#)
- OS_MAX_LOCAL_PATH_LEN
 - osapi-constants.h, [1234](#)
- OS_MAX_MODULES
 - osconfig.h, [600](#)
- OS_MAX_MUTEXES
 - osconfig.h, [600](#)
- OS_MAX_NUM_OPEN_DIRS
 - osconfig.h, [601](#)
- OS_MAX_NUM_OPEN_FILES
 - osconfig.h, [601](#)
- OS_MAX_PATH_LEN
 - osconfig.h, [601](#)
- OS_MAX_QUEUES
 - osconfig.h, [601](#)
- OS_MAX_RWLOCKS
 - osconfig.h, [601](#)
- OS_MAX_SYM_LEN
 - osconfig.h, [601](#)
- OS_MAX_TASK_PRIORITY
 - osapi-task.h, [1276](#)
- OS_MAX_TASKS
 - osconfig.h, [601](#)
- OS_MAX_TIMEBASES
 - osconfig.h, [602](#)
- OS_MAX_TIMERS
 - osconfig.h, [602](#)
- OS_MINOR_VERSION
 - osapi-version.h, [1283](#)
- OS_MISSION_REV
 - osapi-version.h, [1283](#)
- OS_mkdir

- OSAL Directory APIs, 328
- OS_mkfs
 - OSAL File System Level APIs, 357
- OS_module_address_t, 586
 - bss_address, 587
 - bss_size, 587
 - code_address, 587
 - code_size, 587
 - data_address, 587
 - data_size, 587
 - flags, 587
 - valid, 587
- OS_MODULE_FILE_EXTENSION
 - osconfig.h, 602
- OS_MODULE_FLAG_GLOBAL_SYMBOLS
 - osapi-module.h, 1258
- OS_MODULE_FLAG_LOCAL_SYMBOLS
 - osapi-module.h, 1258
- OS_module_prop_t, 587
 - addr, 588
 - entry_point, 588
 - filename, 588
 - host_module_id, 588
 - name, 588
- OS_ModuleInfo
 - OSAL Dynamic Loader and Symbol APIs, 369
- OS_ModuleLoad
 - OSAL Dynamic Loader and Symbol APIs, 369
- OS_ModuleSymbolLookup
 - OSAL Dynamic Loader and Symbol APIs, 370
- OS_ModuleUnload
 - OSAL Dynamic Loader and Symbol APIs, 371
- OS_mount
 - OSAL File System Level APIs, 358
- OS_mut_sem_prop_t, 588
 - creator, 588
 - name, 588
- OS_MutSemCreate
 - OSAL Mutex APIs, 373
- OS_MutSemDelete
 - OSAL Mutex APIs, 373
- OS_MutSemGetIdByName
 - OSAL Mutex APIs, 374
- OS_MutSemGetInfo
 - OSAL Mutex APIs, 374
- OS_MutSemGive
 - OSAL Mutex APIs, 375
- OS_MutSemTake
 - OSAL Mutex APIs, 375
- OS_mv
 - OSAL Standard File APIs, 346
- OS_NetworkGetHostName
 - OSAL Network ID APIs, 376
- OS_NetworkGetID
 - OSAL Network ID APIs, 377
- OS_OBJECT_CREATOR_ANY
 - osapi-constants.h, 1234
- OS_OBJECT_ID_UNDEFINED
 - osapi-constants.h, 1235
- OS_OBJECT_INDEX_MASK
 - osapi-idmap.h, 1253
- OS_OBJECT_TYPE_OS_BINSEM
 - OSAL Object Type Defines, 362
- OS_OBJECT_TYPE_OS_CONDVAR
 - OSAL Object Type Defines, 362
- OS_OBJECT_TYPE_OS_CONSOLE
 - OSAL Object Type Defines, 362
- OS_OBJECT_TYPE_OS_COUNTSEM
 - OSAL Object Type Defines, 362
- OS_OBJECT_TYPE_OS_DIR
 - OSAL Object Type Defines, 362
- OS_OBJECT_TYPE_OS_FILESYS
 - OSAL Object Type Defines, 362
- OS_OBJECT_TYPE_OS_MODULE
 - OSAL Object Type Defines, 362
- OS_OBJECT_TYPE_OS_MUTEX
 - OSAL Object Type Defines, 362
- OS_OBJECT_TYPE_OS_QUEUE
 - OSAL Object Type Defines, 363
- OS_OBJECT_TYPE_OS_RWLOCK
 - OSAL Object Type Defines, 363
- OS_OBJECT_TYPE_OS_STREAM
 - OSAL Object Type Defines, 363
- OS_OBJECT_TYPE_OS_TASK
 - OSAL Object Type Defines, 363
- OS_OBJECT_TYPE_OS_TIMEBASE
 - OSAL Object Type Defines, 363
- OS_OBJECT_TYPE_OS_TIMECB
 - OSAL Object Type Defines, 363
- OS_OBJECT_TYPE_SHIFT
 - osapi-idmap.h, 1253
- OS_OBJECT_TYPE_UNDEFINED
 - OSAL Object Type Defines, 363
- OS_OBJECT_TYPE_USER
 - OSAL Object Type Defines, 363
- OS_ObjectIdDefined
 - OSAL Object ID Utility APIs, 366
- OS_ObjectIdEqual
 - OSAL Object ID Utility APIs, 367
- OS_ObjectIdFromInteger
 - OSAL Object ID Utility APIs, 367
- OS_ObjectIdToArrayIndex
 - OSAL Object ID Utility APIs, 367
- OS_ObjectIdToInteger
 - OSAL Object ID Utility APIs, 368
- OS_OpenCreate
 - OSAL Standard File APIs, 346
- OS_PEND

- osapi-constants.h, [1235](#)
- OS_PRINTF
 - cfe_es.h, [718](#)
 - common_types.h, [1215](#)
- OS_printf
 - OSAL Printf APIs, [377](#)
- OS_PRINTF_CONSOLE_NAME
 - osconfig.h, [602](#)
- OS_printf_disable
 - OSAL Printf APIs, [378](#)
- OS_printf_enable
 - OSAL Printf APIs, [378](#)
- OS_QUEUE_EMPTY
 - OSAL Return Code Defines, [335](#)
- OS_QUEUE_FULL
 - OSAL Return Code Defines, [335](#)
- OS_QUEUE_ID_ERROR
 - OSAL Return Code Defines, [335](#)
- OS_QUEUE_INVALID_SIZE
 - OSAL Return Code Defines, [335](#)
- OS_QUEUE_MAX_DEPTH
 - osconfig.h, [602](#)
- OS_queue_prop_t, [589](#)
 - creator, [589](#)
 - name, [589](#)
- OS_QUEUE_TIMEOUT
 - OSAL Return Code Defines, [335](#)
- OS_QueueCreate
 - OSAL Message Queue APIs, [378](#)
- OS_QueueDelete
 - OSAL Message Queue APIs, [379](#)
- OS_QueueGet
 - OSAL Message Queue APIs, [379](#)
- OS_QueueGetIdByName
 - OSAL Message Queue APIs, [380](#)
- OS_QueueGetInfo
 - OSAL Message Queue APIs, [381](#)
- OS_QueuePut
 - OSAL Message Queue APIs, [381](#)
- OS_read
 - OSAL Standard File APIs, [347](#)
- OS_READ_ONLY
 - OSAL File Access Option Defines, [339](#)
- OS_READ_WRITE
 - OSAL File Access Option Defines, [339](#)
- OS_RegisterEventHandler
 - OSAL Core Operation APIs, [314](#)
- OS_remove
 - OSAL Standard File APIs, [347](#)
- OS_rename
 - OSAL Standard File APIs, [348](#)
- OS_REPAIR
 - osapi-filesys.h, [1249](#)
- OS_REVISION
 - osapi-version.h, [1283](#)
- OS_rmdir
 - OSAL Directory APIs, [329](#)
- OS_rmfs
 - OSAL File System Level APIs, [359](#)
- OS_rwlock_prop_t, [589](#)
 - creator, [590](#)
 - name, [590](#)
- OS_RwLockCreate
 - OSAL RwLock APIs, [382](#)
- OS_RwLockDelete
 - OSAL RwLock APIs, [383](#)
- OS_RwLockGetIdByName
 - OSAL RwLock APIs, [383](#)
- OS_RwLockGetInfo
 - OSAL RwLock APIs, [384](#)
- OS_RwLockReadGive
 - OSAL RwLock APIs, [384](#)
- OS_RwLockReadTake
 - OSAL RwLock APIs, [385](#)
- OS_RwLockWriteGive
 - OSAL RwLock APIs, [385](#)
- OS_RwLockWriteTake
 - OSAL RwLock APIs, [386](#)
- OS_SEEK_CUR
 - OSAL Reference Point For Seek Offset Defines, [339](#)
- OS_SEEK_END
 - OSAL Reference Point For Seek Offset Defines, [339](#)
- OS_SEEK_SET
 - OSAL Reference Point For Seek Offset Defines, [339](#)
- OS_SelectFdAdd
 - OSAL Select APIs, [387](#)
- OS_SelectFdClear
 - OSAL Select APIs, [387](#)
- OS_SelectFdsSet
 - OSAL Select APIs, [388](#)
- OS_SelectFdZero
 - OSAL Select APIs, [388](#)
- OS_SelectMultiple
 - OSAL Select APIs, [389](#)
- OS_SelectMultipleAbs
 - OSAL Select APIs, [390](#)
- OS_SelectSingle
 - OSAL Select APIs, [391](#)
- OS_SelectSingleAbs
 - OSAL Select APIs, [391](#)
- OS_SEM_EMPTY
 - OSAL Semaphore State Defines, [291](#)
- OS_SEM_FAILURE
 - OSAL Return Code Defines, [335](#)
- OS_SEM_FULL
 - OSAL Semaphore State Defines, [291](#)
- OS_SEM_TIMEOUT
 - OSAL Return Code Defines, [335](#)

- OS_SetLocalTime
 - OSAL Real Time Clock APIs, [299](#)
- OS_SHELL_CMD_INPUT_FILE_NAME
 - osconfig.h, [602](#)
- OS_ShellOutputToFile
 - OSAL Shell APIs, [393](#)
- OS_SOCKADDR_MAX_LEN
 - osapi-sockets.h, [1271](#)
 - osconfig.h, [602](#)
- OS_SockAddr_t, [590](#)
 - ActualLength, [590](#)
 - AddrData, [590](#)
- OS_SockAddrData_t, [590](#)
 - AlignPtr, [591](#)
 - AlignU32, [591](#)
 - Buffer, [591](#)
- OS_socket_option
 - osapi-sockets.h, [1271](#)
- OS_socket_option_IP_DSCP
 - osapi-sockets.h, [1271](#)
- OS_socket_option_MAX
 - osapi-sockets.h, [1271](#)
- OS_socket_option_t
 - osapi-sockets.h, [1271](#)
- OS_socket_option_UNDEFINED
 - osapi-sockets.h, [1271](#)
- OS_socket_optval, [591](#)
 - IntVal, [592](#)
- OS_socket_optval_t
 - osapi-sockets.h, [1271](#)
- OS_socket_prop_t, [592](#)
 - creator, [592](#)
 - name, [592](#)
- OS_SocketAccept
 - OSAL Socket Management APIs, [397](#)
- OS_SocketAcceptAbs
 - OSAL Socket Management APIs, [398](#)
- OS_SocketAddrFromString
 - OSAL Socket Address APIs, [394](#)
- OS_SocketAddrGetPort
 - OSAL Socket Address APIs, [394](#)
- OS_SocketAddrInit
 - OSAL Socket Address APIs, [395](#)
- OS_SocketAddrSetPort
 - OSAL Socket Address APIs, [395](#)
- OS_SocketAddrToString
 - OSAL Socket Address APIs, [396](#)
- OS_SocketBind
 - OSAL Socket Management APIs, [399](#)
- OS_SocketBindAddress
 - OSAL Socket Management APIs, [399](#)
- OS_SocketConnect
 - OSAL Socket Management APIs, [400](#)
- OS_SocketConnectAbs
 - OSAL Socket Management APIs, [401](#)
- OS_SocketDomain_INET
 - osapi-sockets.h, [1271](#)
- OS_SocketDomain_INET6
 - osapi-sockets.h, [1271](#)
- OS_SocketDomain_INVALID
 - osapi-sockets.h, [1271](#)
- OS_SocketDomain_MAX
 - osapi-sockets.h, [1271](#)
- OS_SocketDomain_t
 - osapi-sockets.h, [1271](#)
- OS_SocketGetIdByName
 - OSAL Socket Management APIs, [402](#)
- OS_SocketGetInfo
 - OSAL Socket Management APIs, [402](#)
- OS_SocketGetOption
 - OSAL Socket Management APIs, [403](#)
- OS_SocketListen
 - OSAL Socket Management APIs, [403](#)
- OS_SocketOpen
 - OSAL Socket Management APIs, [404](#)
- OS_SocketRecvFrom
 - OSAL Socket Management APIs, [404](#)
- OS_SocketRecvFromAbs
 - OSAL Socket Management APIs, [405](#)
- OS_SocketSendTo
 - OSAL Socket Management APIs, [406](#)
- OS_SocketSetOption
 - OSAL Socket Management APIs, [406](#)
- OS_SocketShutdown
 - OSAL Socket Management APIs, [407](#)
- OS_SocketShutdownMode_NONE
 - osapi-sockets.h, [1272](#)
- OS_SocketShutdownMode_SHUT_READ
 - osapi-sockets.h, [1272](#)
- OS_SocketShutdownMode_SHUT_READWRITE
 - osapi-sockets.h, [1272](#)
- OS_SocketShutdownMode_SHUT_WRITE
 - osapi-sockets.h, [1272](#)
- OS_SocketShutdownMode_t
 - osapi-sockets.h, [1271](#)
- OS_SocketType_DATAGRAM
 - osapi-sockets.h, [1272](#)
- OS_SocketType_INVALID
 - osapi-sockets.h, [1272](#)
- OS_SocketType_MAX
 - osapi-sockets.h, [1272](#)
- OS_SocketType_STREAM
 - osapi-sockets.h, [1272](#)
- OS_SocketType_t
 - osapi-sockets.h, [1272](#)
- OS_stat
 - OSAL Standard File APIs, [349](#)
- OS_static_symbol_record_t, [592](#)

- Address, [593](#)
- Module, [593](#)
- Name, [593](#)
- OS_STATUS_STRING_LENGTH
 - osapi-error.h, [1241](#)
- os_status_string_t
 - osapi-error.h, [1241](#)
- OS_StatusToInteger
 - OSAL Error Info APIs, [338](#)
- OS_StatusToString
 - OSAL Error Info APIs, [338](#)
- OS_statvfs_t, [593](#)
 - block_size, [593](#)
 - blocks_free, [593](#)
 - total_blocks, [593](#)
- OS_STR
 - osapi-version.h, [1283](#)
- OS_STR_HELPER
 - osapi-version.h, [1283](#)
- OS_STREAM_STATE_BOUND
 - osapi-select.h, [1267](#)
- OS_STREAM_STATE_CONNECTED
 - osapi-select.h, [1267](#)
- OS_STREAM_STATE_LISTENING
 - osapi-select.h, [1267](#)
- OS_STREAM_STATE_READABLE
 - osapi-select.h, [1267](#)
- OS_STREAM_STATE_WRITABLE
 - osapi-select.h, [1267](#)
- OS_StreamState_t
 - osapi-select.h, [1267](#)
- OS_strlen
 - OSAL Core Operation APIs, [315](#)
- OS_SUCCESS
 - OSAL Return Code Defines, [336](#)
- OS_SymbolLookup
 - OSAL Dynamic Loader and Symbol APIs, [371](#)
- OS_SymbolTableDump
 - OSAL Dynamic Loader and Symbol APIs, [372](#)
- OS_task_prop_t, [594](#)
 - creator, [594](#)
 - name, [594](#)
 - priority, [594](#)
 - stack_size, [594](#)
- OS_TaskCreate
 - OSAL Task APIs, [408](#)
- OS_TaskDelay
 - OSAL Task APIs, [409](#)
- OS_TaskDelete
 - OSAL Task APIs, [409](#)
- OS_TaskExit
 - OSAL Task APIs, [410](#)
- OS_TaskFindIdBySystemData
 - OSAL Task APIs, [410](#)
- OS_TaskGetId
 - OSAL Task APIs, [410](#)
- OS_TaskGetIdByName
 - OSAL Task APIs, [411](#)
- OS_TaskGetInfo
 - OSAL Task APIs, [411](#)
- OS_TaskInstallDeleteHandler
 - OSAL Task APIs, [412](#)
- OS_TaskSetPriority
 - OSAL Task APIs, [412](#)
- OS_TIME_MAX
 - osapi-clock.h, [1226](#)
- OS_TIME_MIN
 - osapi-clock.h, [1226](#)
- OS_time_t, [594](#)
 - ticks, [595](#)
- OS_TIME_TICK_RESOLUTION_NS
 - osapi-clock.h, [1226](#)
- OS_TIME_TICKS_PER_MSEC
 - osapi-clock.h, [1226](#)
- OS_TIME_TICKS_PER_SECOND
 - osapi-clock.h, [1226](#)
- OS_TIME_TICKS_PER_USEC
 - osapi-clock.h, [1226](#)
- OS_TIME_ZERO
 - osapi-clock.h, [1226](#)
- OS_TimeAdd
 - OSAL Real Time Clock APIs, [299](#)
- OS_TimeAssembleFromMicroseconds
 - OSAL Real Time Clock APIs, [300](#)
- OS_TimeAssembleFromMilliseconds
 - OSAL Real Time Clock APIs, [300](#)
- OS_TimeAssembleFromNanoseconds
 - OSAL Real Time Clock APIs, [301](#)
- OS_TimeAssembleFromSubseconds
 - OSAL Real Time Clock APIs, [301](#)
- OS_timebase_prop_t, [595](#)
 - accuracy, [595](#)
 - creator, [595](#)
 - freerun_time, [596](#)
 - name, [596](#)
 - nominal_interval_time, [596](#)
- OS_TimeBaseCreate
 - OSAL Time Base APIs, [413](#)
- OS_TimeBaseDelete
 - OSAL Time Base APIs, [414](#)
- OS_TimeBaseGetFreeRun
 - OSAL Time Base APIs, [414](#)
- OS_TimeBaseGetIdByName
 - OSAL Time Base APIs, [415](#)
- OS_TimeBaseGetInfo
 - OSAL Time Base APIs, [416](#)
- OS_TimeBaseSet
 - OSAL Time Base APIs, [416](#)

- OS_TimeCompare
 - OSAL Real Time Clock APIs, [302](#)
- OS_TimedRead
 - OSAL Standard File APIs, [349](#)
- OS_TimedReadAbs
 - OSAL Standard File APIs, [350](#)
- OS_TimedWrite
 - OSAL Standard File APIs, [351](#)
- OS_TimedWriteAbs
 - OSAL Standard File APIs, [352](#)
- OS_TimeEqual
 - OSAL Real Time Clock APIs, [302](#)
- OS_TimeFromRelativeMilliseconds
 - OSAL Real Time Clock APIs, [303](#)
- OS_TimeFromTotalMicroseconds
 - OSAL Real Time Clock APIs, [303](#)
- OS_TimeFromTotalMilliseconds
 - OSAL Real Time Clock APIs, [304](#)
- OS_TimeFromTotalNanoseconds
 - OSAL Real Time Clock APIs, [304](#)
- OS_TimeFromTotalSeconds
 - OSAL Real Time Clock APIs, [305](#)
- OS_TimeGetFractionalPart
 - OSAL Real Time Clock APIs, [305](#)
- OS_TimeGetMicrosecondsPart
 - OSAL Real Time Clock APIs, [305](#)
- OS_TimeGetMillisecondsPart
 - OSAL Real Time Clock APIs, [306](#)
- OS_TimeGetNanosecondsPart
 - OSAL Real Time Clock APIs, [307](#)
- OS_TimeGetSign
 - OSAL Real Time Clock APIs, [307](#)
- OS_TimeGetSubsecondsPart
 - OSAL Real Time Clock APIs, [309](#)
- OS_TimeGetTotalMicroseconds
 - OSAL Real Time Clock APIs, [309](#)
- OS_TimeGetTotalMilliseconds
 - OSAL Real Time Clock APIs, [310](#)
- OS_TimeGetTotalNanoseconds
 - OSAL Real Time Clock APIs, [310](#)
- OS_TimeGetTotalSeconds
 - OSAL Real Time Clock APIs, [311](#)
- OS_TIMER_ERR_INTERNAL
 - OSAL Return Code Defines, [336](#)
- OS_TIMER_ERR_INVALID_ARGS
 - OSAL Return Code Defines, [336](#)
- OS_TIMER_ERR_TIMER_ID
 - OSAL Return Code Defines, [336](#)
- OS_TIMER_ERR_UNAVAILABLE
 - OSAL Return Code Defines, [336](#)
- OS_timer_prop_t, [596](#)
 - accuracy, [596](#)
 - creator, [596](#)
 - interval_time, [596](#)
 - name, [596](#)
 - start_time, [597](#)
- OS_TimerAdd
 - OSAL Timer APIs, [417](#)
- OS_TimerCallback_t
 - osapi-timer.h, [1280](#)
- OS_TimerCreate
 - OSAL Timer APIs, [418](#)
- OS_TimerDelete
 - OSAL Timer APIs, [419](#)
- OS_TimerGetIdByName
 - OSAL Timer APIs, [420](#)
- OS_TimerGetInfo
 - OSAL Timer APIs, [421](#)
- OS_TimerSet
 - OSAL Timer APIs, [421](#)
- OS_TimerSync_t
 - osapi-timebase.h, [1279](#)
- OS_TimeSubtract
 - OSAL Real Time Clock APIs, [311](#)
- OS_TimeToRelativeMilliseconds
 - OSAL Real Time Clock APIs, [311](#)
- OS_TranslatePath
 - OSAL File System Level APIs, [359](#)
- OS_unmount
 - OSAL File System Level APIs, [360](#)
- OS_USED
 - common_types.h, [1215](#)
- OS_UTILITYTASK_PRIORITY
 - osconfig.h, [603](#)
- OS_UTILITYTASK_STACK_SIZE
 - osconfig.h, [603](#)
- OS_VERSION
 - osapi-version.h, [1284](#)
- OS_write
 - OSAL Standard File APIs, [352](#)
- OS_WRITE_ONLY
 - OSAL File Access Option Defines, [339](#)
- Osapi API Documentation, [98](#)
- OSAL Binary Semaphore APIs, [291](#)
 - OS_BinSemCreate, [292](#)
 - OS_BinSemDelete, [292](#)
 - OS_BinSemFlush, [293](#)
 - OS_BinSemGetIdByName, [293](#)
 - OS_BinSemGetInfo, [294](#)
 - OS_BinSemGive, [294](#)
 - OS_BinSemTake, [295](#)
 - OS_BinSemTimedWait, [295](#)
- OSAL BSP low level access APIs, [296](#)
 - OS_BSP_GetArgC, [296](#)
 - OS_BSP_GetArgV, [296](#)
 - OS_BSP_GetResourceTypeConfig, [296](#)
 - OS_BSP_SetExitCode, [296](#)
 - OS_BSP_SetResourceTypeConfig, [296](#)

- OSAL Condition Variable APIs, [315](#)
 - OS_CondVarBroadcast, [316](#)
 - OS_CondVarCreate, [316](#)
 - OS_CondVarDelete, [317](#)
 - OS_CondVarGetIdByName, [318](#)
 - OS_CondVarGetInfo, [318](#)
 - OS_CondVarLock, [319](#)
 - OS_CondVarSignal, [319](#)
 - OS_CondVarTimedWait, [320](#)
 - OS_CondVarUnlock, [320](#)
 - OS_CondVarWait, [321](#)
- OSAL Core Operation APIs, [312](#)
 - OS_API_Init, [313](#)
 - OS_API_Teardown, [313](#)
 - OS_Application_Run, [313](#)
 - OS_Application_Startup, [313](#)
 - OS_ApplicationExit, [314](#)
 - OS_ApplicationShutdown, [314](#)
 - OS_DeleteAllObjects, [314](#)
 - OS_IdleLoop, [314](#)
 - OS_RegisterEventHandler, [314](#)
 - OS_strnlen, [315](#)
- OSAL Counting Semaphore APIs, [322](#)
 - OS_CountSemCreate, [322](#)
 - OS_CountSemDelete, [323](#)
 - OS_CountSemGetIdByName, [323](#)
 - OS_CountSemGetInfo, [324](#)
 - OS_CountSemGive, [324](#)
 - OS_CountSemTake, [325](#)
 - OS_CountSemTimedWait, [325](#)
- OSAL Directory APIs, [326](#)
 - OS_DirectoryClose, [326](#)
 - OS_DirectoryOpen, [327](#)
 - OS_DirectoryRead, [327](#)
 - OS_DirectoryRewind, [328](#)
 - OS_mkdir, [328](#)
 - OS_rmdir, [329](#)
- OSAL Dynamic Loader and Symbol APIs, [369](#)
 - OS_ModuleInfo, [369](#)
 - OS_ModuleLoad, [369](#)
 - OS_ModuleSymbolLookup, [370](#)
 - OS_ModuleUnload, [371](#)
 - OS_SymbolLookup, [371](#)
 - OS_SymbolTableDump, [372](#)
- OSAL Error Info APIs, [336](#)
 - OS_GetErrorName, [336](#)
 - OS_StatusToInteger, [338](#)
 - OS_StatusToString, [338](#)
- OSAL File Access Option Defines, [339](#)
 - OS_READ_ONLY, [339](#)
 - OS_READ_WRITE, [339](#)
 - OS_WRITE_ONLY, [339](#)
- OSAL File System Level APIs, [353](#)
 - OS_chkfs, [354](#)
 - OS_FileSysAddFixedMap, [354](#)
 - OS_FileSysStatVolume, [355](#)
 - OS_FS_GetPhysDriveName, [356](#)
 - OS_GetFsInfo, [356](#)
 - OS_initfs, [357](#)
 - OS_mkfs, [357](#)
 - OS_mount, [358](#)
 - OS_rmfs, [359](#)
 - OS_TranslatePath, [359](#)
 - OS_unmount, [360](#)
- OSAL Heap APIs, [360](#)
 - OS_HeapGetInfo, [361](#)
- OSAL Introduction, [100](#)
- OSAL Message Queue APIs, [378](#)
 - OS_QueueCreate, [378](#)
 - OS_QueueDelete, [379](#)
 - OS_QueueGet, [379](#)
 - OS_QueueGetIdByName, [380](#)
 - OS_QueueGetInfo, [381](#)
 - OS_QueuePut, [381](#)
- OSAL Mutex APIs, [372](#)
 - OS_MutSemCreate, [373](#)
 - OS_MutSemDelete, [373](#)
 - OS_MutSemGetIdByName, [374](#)
 - OS_MutSemGetInfo, [374](#)
 - OS_MutSemGive, [375](#)
 - OS_MutSemTake, [375](#)
- OSAL Network ID APIs, [376](#)
 - OS_NetworkGetHostName, [376](#)
 - OS_NetworkGetID, [377](#)
- OSAL Object ID Utility APIs, [364](#)
 - OS_ConvertToArrayIndex, [364](#)
 - OS_ForEachObject, [365](#)
 - OS_ForEachObjectOfType, [365](#)
 - OS_GetResourceName, [365](#)
 - OS_IdentifyObject, [366](#)
 - OS_ObjectIdDefined, [366](#)
 - OS_ObjectIdEqual, [367](#)
 - OS_ObjectIdFromInteger, [367](#)
 - OS_ObjectIdToArrayIndex, [367](#)
 - OS_ObjectIdToInteger, [368](#)
- OSAL Object Type Defines, [361](#)
 - OS_OBJECT_TYPE_OS_BINSEM, [362](#)
 - OS_OBJECT_TYPE_OS_CONDVAR, [362](#)
 - OS_OBJECT_TYPE_OS_CONSOLE, [362](#)
 - OS_OBJECT_TYPE_OS_COUNTSEM, [362](#)
 - OS_OBJECT_TYPE_OS_DIR, [362](#)
 - OS_OBJECT_TYPE_OS_FILESYS, [362](#)
 - OS_OBJECT_TYPE_OS_MODULE, [362](#)
 - OS_OBJECT_TYPE_OS_MUTEX, [362](#)
 - OS_OBJECT_TYPE_OS_QUEUE, [363](#)
 - OS_OBJECT_TYPE_OS_RWLOCK, [363](#)
 - OS_OBJECT_TYPE_OS_STREAM, [363](#)
 - OS_OBJECT_TYPE_OS_TASK, [363](#)

- OS_OBJECT_TYPE_OS_TIMEBASE, 363
- OS_OBJECT_TYPE_OS_TIMECB, 363
- OS_OBJECT_TYPE_UNDEFINED, 363
- OS_OBJECT_TYPE_USER, 363
- OSAL Printf APIs, 377
 - OS_printf, 377
 - OS_printf_disable, 378
 - OS_printf_enable, 378
- OSAL Real Time Clock APIs, 297
 - OS_GetLocalTime, 298
 - OS_GetMonotonicTime, 298
 - OS_SetLocalTime, 299
 - OS_TimeAdd, 299
 - OS_TimeAssembleFromMicroseconds, 300
 - OS_TimeAssembleFromMilliseconds, 300
 - OS_TimeAssembleFromNanoseconds, 301
 - OS_TimeAssembleFromSubseconds, 301
 - OS_TimeCompare, 302
 - OS_TimeEqual, 302
 - OS_TimeFromRelativeMilliseconds, 303
 - OS_TimeFromTotalMicroseconds, 303
 - OS_TimeFromTotalMilliseconds, 304
 - OS_TimeFromTotalNanoseconds, 304
 - OS_TimeFromTotalSeconds, 305
 - OS_TimeGetFractionalPart, 305
 - OS_TimeGetMicrosecondsPart, 305
 - OS_TimeGetMillisecondsPart, 306
 - OS_TimeGetNanosecondsPart, 307
 - OS_TimeGetSign, 307
 - OS_TimeGetSubsecondsPart, 309
 - OS_TimeGetTotalMicroseconds, 309
 - OS_TimeGetTotalMilliseconds, 310
 - OS_TimeGetTotalNanoseconds, 310
 - OS_TimeGetTotalSeconds, 311
 - OS_TimeSubtract, 311
 - OS_TimeToRelativeMilliseconds, 311
- OSAL Reference Point For Seek Offset Defines, 339
 - OS_SEEK_CUR, 339
 - OS_SEEK_END, 339
 - OS_SEEK_SET, 339
- OSAL Return Code Defines, 329
 - OS_ERR_BAD_ADDRESS, 331
 - OS_ERR_EMPTY_SET, 331
 - OS_ERR_FILE, 332
 - OS_ERR_INCORRECT_OBJ_STATE, 332
 - OS_ERR_INCORRECT_OBJ_TYPE, 332
 - OS_ERR_INVALID_ARGUMENT, 332
 - OS_ERR_INVALID_ID, 332
 - OS_ERR_INVALID_PRIORITY, 332
 - OS_ERR_INVALID_SIZE, 332
 - OS_ERR_NAME_NOT_FOUND, 332
 - OS_ERR_NAME_TAKEN, 332
 - OS_ERR_NAME_TOO_LONG, 333
 - OS_ERR_NO_FREE_IDS, 333
 - OS_ERR_NOT_IMPLEMENTED, 333
 - OS_ERR_OBJECT_IN_USE, 333
 - OS_ERR_OPERATION_NOT_SUPPORTED, 333
 - OS_ERR_OUTPUT_TOO_LARGE, 333
 - OS_ERR_SEM_NOT_FULL, 333
 - OS_ERR_STREAM_DISCONNECTED, 333
 - OS_ERR_TRY_AGAIN, 333
 - OS_ERROR, 334
 - OS_ERROR_ADDRESS_MISALIGNED, 334
 - OS_ERROR_TIMEOUT, 334
 - OS_FS_ERR_DEVICE_NOT_FREE, 334
 - OS_FS_ERR_DRIVE_NOT_CREATED, 334
 - OS_FS_ERR_NAME_TOO_LONG, 334
 - OS_FS_ERR_PATH_INVALID, 334
 - OS_FS_ERR_PATH_TOO_LONG, 334
 - OS_INVALID_INT_NUM, 334
 - OS_INVALID_POINTER, 335
 - OS_INVALID_SEM_VALUE, 335
 - OS_QUEUE_EMPTY, 335
 - OS_QUEUE_FULL, 335
 - OS_QUEUE_ID_ERROR, 335
 - OS_QUEUE_INVALID_SIZE, 335
 - OS_QUEUE_TIMEOUT, 335
 - OS_SEM_FAILURE, 335
 - OS_SEM_TIMEOUT, 335
 - OS_SUCCESS, 336
 - OS_TIMER_ERR_INTERNAL, 336
 - OS_TIMER_ERR_INVALID_ARGS, 336
 - OS_TIMER_ERR_TIMER_ID, 336
 - OS_TIMER_ERR_UNAVAILABLE, 336
- OSAL RwLock APIs, 382
 - OS_RwLockCreate, 382
 - OS_RwLockDelete, 383
 - OS_RwLockGetIdByName, 383
 - OS_RwLockGetInfo, 384
 - OS_RwLockReadGive, 384
 - OS_RwLockReadTake, 385
 - OS_RwLockWriteGive, 385
 - OS_RwLockWriteTake, 386
- OSAL Select APIs, 386
 - OS_SelectFdAdd, 387
 - OS_SelectFdClear, 387
 - OS_SelectFdsSet, 388
 - OS_SelectFdZero, 388
 - OS_SelectMultiple, 389
 - OS_SelectMultipleAbs, 390
 - OS_SelectSingle, 391
 - OS_SelectSingleAbs, 391
- OSAL Semaphore State Defines, 291
 - OS_SEM_EMPTY, 291
 - OS_SEM_FULL, 291
- OSAL Shell APIs, 392
 - OS_ShellOutputToFile, 393
- OSAL Socket Address APIs, 393

- OS_SocketAddrFromString, 394
- OS_SocketAddrGetPort, 394
- OS_SocketAddrInit, 395
- OS_SocketAddrSetPort, 395
- OS_SocketAddrToString, 396
- OSAL Socket Management APIs, 396
 - OS_SocketAccept, 397
 - OS_SocketAcceptAbs, 398
 - OS_SocketBind, 399
 - OS_SocketBindAddress, 399
 - OS_SocketConnect, 400
 - OS_SocketConnectAbs, 401
 - OS_SocketGetIdByName, 402
 - OS_SocketGetInfo, 402
 - OS_SocketGetOption, 403
 - OS_SocketListen, 403
 - OS_SocketOpen, 404
 - OS_SocketRecvFrom, 404
 - OS_SocketRecvFromAbs, 405
 - OS_SocketSendTo, 406
 - OS_SocketSetOption, 406
 - OS_SocketShutdown, 407
- OSAL Standard File APIs, 340
 - OS_chmod, 341
 - OS_close, 341
 - OS_CloseAllFiles, 342
 - OS_CloseFileByName, 342
 - OS_cp, 342
 - OS_FDGetInfo, 343
 - OS_FileAllocate, 343
 - OS_FileOpenCheck, 344
 - OS_FileTruncate, 345
 - OS_lseek, 345
 - OS_mv, 346
 - OS_OpenCreate, 346
 - OS_read, 347
 - OS_remove, 347
 - OS_rename, 348
 - OS_stat, 349
 - OS_TimedRead, 349
 - OS_TimedReadAbs, 350
 - OS_TimedWrite, 351
 - OS_TimedWriteAbs, 352
 - OS_write, 352
- OSAL Task APIs, 407
 - OS_TaskCreate, 408
 - OS_TaskDelay, 409
 - OS_TaskDelete, 409
 - OS_TaskExit, 410
 - OS_TaskFindIdBySystemData, 410
 - OS_TaskGetId, 410
 - OS_TaskGetIdByName, 411
 - OS_TaskGetInfo, 411
 - OS_TaskInstallDeleteHandler, 412
 - OS_TaskSetPriority, 412
- OSAL Time Base APIs, 413
 - OS_TimeBaseCreate, 413
 - OS_TimeBaseDelete, 414
 - OS_TimeBaseGetFreeRun, 414
 - OS_TimeBaseGetIdByName, 415
 - OS_TimeBaseGetInfo, 416
 - OS_TimeBaseSet, 416
- OSAL Timer APIs, 417
 - OS_TimerAdd, 417
 - OS_TimerCreate, 418
 - OS_TimerDelete, 419
 - OS_TimerGetIdByName, 420
 - OS_TimerGetInfo, 421
 - OS_TimerSet, 421
- osal/docs/src/osal_frontpage.dox, 1214
- osal/docs/src/osal_fs.dox, 1214
- osal/docs/src/osal_timer.dox, 1214
- osal/src/os/inc/common_types.h, 1214, 1219
- osal/src/os/inc/osapi-binsem.h, 1221, 1222
- osal/src/os/inc/osapi-bsp.h, 1222, 1223
- osal/src/os/inc/osapi-clock.h, 1224, 1226
- osal/src/os/inc/osapi-common.h, 1229, 1231
- osal/src/os/inc/osapi-condvar.h, 1232, 1233
- osal/src/os/inc/osapi-constants.h, 1234, 1235
- osal/src/os/inc/osapi-countsem.h, 1235, 1236
- osal/src/os/inc/osapi-dir.h, 1237, 1238
- osal/src/os/inc/osapi-error.h, 1238, 1242
- osal/src/os/inc/osapi-file.h, 1243, 1246
- osal/src/os/inc/osapi-filesys.h, 1248, 1249
- osal/src/os/inc/osapi-heap.h, 1250, 1251
- osal/src/os/inc/osapi-idmap.h, 1251, 1253
- osal/src/os/inc/osapi-macros.h, 1254, 1256
- osal/src/os/inc/osapi-module.h, 1257, 1259
- osal/src/os/inc/osapi-mutex.h, 1260, 1261
- osal/src/os/inc/osapi-network.h, 1261, 1262
- osal/src/os/inc/osapi-printf.h, 1262
- osal/src/os/inc/osapi-queue.h, 1263, 1264
- osal/src/os/inc/osapi-rwlock.h, 1264, 1265
- osal/src/os/inc/osapi-select.h, 1266, 1267
- osal/src/os/inc/osapi-shell.h, 1268
- osal/src/os/inc/osapi-sockets.h, 1269, 1272
- osal/src/os/inc/osapi-task.h, 1274, 1277
- osal/src/os/inc/osapi-timebase.h, 1278, 1279
- osal/src/os/inc/osapi-timer.h, 1279, 1280
- osal/src/os/inc/osapi-version.h, 1281, 1285
- osal/src/os/inc/osapi.h, 1286, 1287
- OSAL_API_VERSION
 - osapi-version.h, 1284
- OSAL_BLOCKCOUNT_C
 - common_types.h, 1215
- osal_blockcount_t
 - common_types.h, 1217
- OSAL_CONFIG_CONSOLE_ASYNC

- osconfig.h, [603](#)
- OSAL_CONFIG_INCLUDE_DYNAMIC_LOADER
 - osconfig.h, [603](#)
- OSAL_CONFIG_INCLUDE_NETWORK
 - osconfig.h, [603](#)
- OSAL_CONFIG_INCLUDE_STATIC_LOADER
 - osconfig.h, [603](#)
- osal_id_t
 - common_types.h, [1217](#)
- OSAL_INDEX_C
 - common_types.h, [1216](#)
- osal_index_t
 - common_types.h, [1217](#)
- OSAL_OBJTYPE_C
 - common_types.h, [1216](#)
- osal_objtype_t
 - common_types.h, [1217](#)
- osal_offset_t
 - common_types.h, [1217](#)
- OSAL_PRIORITY_C
 - osapi-task.h, [1276](#)
- osal_priority_t
 - osapi-task.h, [1276](#)
- OSAL_SIZE_C
 - common_types.h, [1216](#)
- OSAL_STACKPTR_C
 - osapi-task.h, [1276](#)
- osal_stackptr_t
 - osapi-task.h, [1276](#)
- OSAL_STATUS_C
 - common_types.h, [1216](#)
- osal_status_t
 - common_types.h, [1217](#)
- osal_task
 - osapi-task.h, [1276](#), [1277](#)
- OSAL_TASK_STACK_ALLOCATE
 - osapi-task.h, [1276](#)
- OSALMajorVersion
 - CFE_ES_HousekeepingTlm_Payload, [442](#)
- OSALMinorVersion
 - CFE_ES_HousekeepingTlm_Payload, [442](#)
- OSALMissionRevision
 - CFE_ES_HousekeepingTlm_Payload, [442](#)
- OSALRevision
 - CFE_ES_HousekeepingTlm_Payload, [443](#)
- osapi-clock.h
 - OS_TIME_MAX, [1226](#)
 - OS_TIME_MIN, [1226](#)
 - OS_TIME_TICK_RESOLUTION_NS, [1226](#)
 - OS_TIME_TICKS_PER_MSEC, [1226](#)
 - OS_TIME_TICKS_PER_SECOND, [1226](#)
 - OS_TIME_TICKS_PER_USEC, [1226](#)
 - OS_TIME_ZERO, [1226](#)
- osapi-common.h
 - OS_EVENT_MAX, [1231](#)
 - OS_EVENT_RESERVED, [1231](#)
 - OS_EVENT_RESOURCE_ALLOCATED, [1231](#)
 - OS_EVENT_RESOURCE_CREATED, [1231](#)
 - OS_EVENT_RESOURCE_DELETED, [1231](#)
 - OS_Event_t, [1231](#)
 - OS_EVENT_TASK_STARTUP, [1231](#)
 - OS_EventHandler_t, [1230](#)
- osapi-constants.h
 - OS_CHECK, [1234](#)
 - OS_MAX_LOCAL_PATH_LEN, [1234](#)
 - OS_OBJECT_CREATOR_ANY, [1234](#)
 - OS_OBJECT_ID_UNDEFINED, [1235](#)
 - OS_PEND, [1235](#)
- osapi-dir.h
 - OS_DIRENTRY_NAME, [1238](#)
- osapi-error.h
 - os_err_name_t, [1241](#)
 - OS_ERROR_NAME_LENGTH, [1241](#)
 - OS_STATUS_STRING_LENGTH, [1241](#)
 - os_status_string_t, [1241](#)
- osapi-file.h
 - OS_FILE_FLAG_CREATE, [1246](#)
 - OS_FILE_FLAG_NONE, [1246](#)
 - OS_file_flag_t, [1246](#)
 - OS_FILE_FLAG_TRUNCATE, [1246](#)
 - OS_FILESTAT_EXEC, [1245](#)
 - OS_FILESTAT_ISDIR, [1245](#)
 - OS_FILESTAT_MODE, [1245](#)
 - OS_FILESTAT_MODE_DIR, [1246](#)
 - OS_FILESTAT_MODE_EXEC, [1246](#)
 - OS_FILESTAT_MODE_READ, [1246](#)
 - OS_FILESTAT_MODE_WRITE, [1246](#)
 - OS_FILESTAT_READ, [1245](#)
 - OS_FILESTAT_SIZE, [1245](#)
 - OS_FILESTAT_TIME, [1245](#)
 - OS_FILESTAT_WRITE, [1245](#)
- osapi-filesys.h
 - OS_CHK_ONLY, [1249](#)
 - OS_REPAIR, [1249](#)
- osapi-idmap.h
 - OS_OBJECT_INDEX_MASK, [1253](#)
 - OS_OBJECT_TYPE_SHIFT, [1253](#)
- osapi-macros.h
 - ARGCHECK, [1255](#)
 - BUGCHECK, [1255](#)
 - BUGCHECK_VOID, [1255](#)
 - BUGREPORT, [1256](#)
 - LENGTHCHECK, [1256](#)
- osapi-module.h
 - OS_MODULE_FLAG_GLOBAL_SYMBOLS, [1258](#)
 - OS_MODULE_FLAG_LOCAL_SYMBOLS, [1258](#)
- osapi-select.h
 - OS_STREAM_STATE_BOUND, [1267](#)

- OS_STREAM_STATE_CONNECTED, [1267](#)
- OS_STREAM_STATE_LISTENING, [1267](#)
- OS_STREAM_STATE_READABLE, [1267](#)
- OS_STREAM_STATE_WRITABLE, [1267](#)
- OS_StreamState_t, [1267](#)
- osapi-sockets.h
 - OS_SOCKADDR_MAX_LEN, [1271](#)
 - OS_socket_option, [1271](#)
 - OS_socket_option_IP_DSCP, [1271](#)
 - OS_socket_option_MAX, [1271](#)
 - OS_socket_option_t, [1271](#)
 - OS_socket_option_UNDEFINED, [1271](#)
 - OS_socket_optval_t, [1271](#)
 - OS_SocketDomain_INET, [1271](#)
 - OS_SocketDomain_INET6, [1271](#)
 - OS_SocketDomain_INVALID, [1271](#)
 - OS_SocketDomain_MAX, [1271](#)
 - OS_SocketDomain_t, [1271](#)
 - OS_SocketShutdownMode_NONE, [1272](#)
 - OS_SocketShutdownMode_SHUT_READ, [1272](#)
 - OS_SocketShutdownMode_SHUT_READWRITE, [1272](#)
 - OS_SocketShutdownMode_SHUT_WRITE, [1272](#)
 - OS_SocketShutdownMode_t, [1271](#)
 - OS_SocketType_DATAGRAM, [1272](#)
 - OS_SocketType_INVALID, [1272](#)
 - OS_SocketType_MAX, [1272](#)
 - OS_SocketType_STREAM, [1272](#)
 - OS_SocketType_t, [1272](#)
- osapi-task.h
 - OS_FP_ENABLED, [1276](#)
 - OS_MAX_TASK_PRIORITY, [1276](#)
 - OSAL_PRIORITY_C, [1276](#)
 - osal_priority_t, [1276](#)
 - OSAL_STACKPTR_C, [1276](#)
 - osal_stackptr_t, [1276](#)
 - osal_task, [1276](#), [1277](#)
 - OSAL_TASK_STACK_ALLOCATE, [1276](#)
- osapi-timebase.h
 - OS_TimerSync_t, [1279](#)
- osapi-timer.h
 - OS_TimerCallback_t, [1280](#)
- osapi-version.h
 - OS_BUILD_BASELINE, [1282](#)
 - OS_BUILD_CODENAME, [1282](#)
 - OS_BUILD_DEV_CYCLE, [1282](#)
 - OS_BUILD_NUMBER, [1282](#)
 - OS_CFG_MAX_VERSION_STR_LEN, [1283](#)
 - OS_GetBuildNumber, [1284](#)
 - OS_GetVersionCodeName, [1284](#)
 - OS_GetVersionNumber, [1284](#)
 - OS_GetVersionString, [1285](#)
 - OS_LAST_OFFICIAL, [1283](#)
 - OS_MAJOR_VERSION, [1283](#)
 - OS_MINOR_VERSION, [1283](#)
 - OS_MISSION_REV, [1283](#)
 - OS_REVISION, [1283](#)
 - OS_STR, [1283](#)
 - OS_STR_HELPER, [1283](#)
 - OS_VERSION, [1284](#)
 - OSAL_API_VERSION, [1284](#)
- osconfig.h
 - OS_ADD_TASK_FLAGS, [598](#)
 - OS_BUFFER_MSG_DEPTH, [598](#)
 - OS_BUFFER_SIZE, [599](#)
 - OS_FS_DEV_NAME_LEN, [599](#)
 - OS_FS_PHYS_NAME_LEN, [599](#)
 - OS_FS_VOL_NAME_LEN, [599](#)
 - OS_MAX_API_NAME, [599](#)
 - OS_MAX_BIN_SEMAPHORES, [599](#)
 - OS_MAX_CMD_LEN, [599](#)
 - OS_MAX_CONDVARS, [600](#)
 - OS_MAX_CONSOLES, [600](#)
 - OS_MAX_COUNT_SEMAPHORES, [600](#)
 - OS_MAX_FILE_NAME, [600](#)
 - OS_MAX_FILE_SYSTEMS, [600](#)
 - OS_MAX_MODULES, [600](#)
 - OS_MAX_Mutexes, [600](#)
 - OS_MAX_NUM_OPEN_DIRS, [601](#)
 - OS_MAX_NUM_OPEN_FILES, [601](#)
 - OS_MAX_PATH_LEN, [601](#)
 - OS_MAX_QUEUES, [601](#)
 - OS_MAX_RWLOCKS, [601](#)
 - OS_MAX_SYM_LEN, [601](#)
 - OS_MAX_TASKS, [601](#)
 - OS_MAX_TIMEBASES, [602](#)
 - OS_MAX_TIMERS, [602](#)
 - OS_MODULE_FILE_EXTENSION, [602](#)
 - OS_PRINTF_CONSOLE_NAME, [602](#)
 - OS_QUEUE_MAX_DEPTH, [602](#)
 - OS_SHELL_CMD_INPUT_FILE_NAME, [602](#)
 - OS_SOCKADDR_MAX_LEN, [602](#)
 - OS_UTILITYTASK_PRIORITY, [603](#)
 - OS_UTILITYTASK_STACK_SIZE, [603](#)
 - OSAL_CONFIG_CONSOLE_ASYNC, [603](#)
 - OSAL_CONFIG_INCLUDE_DYNAMIC_LOADER, [603](#)
 - OSAL_CONFIG_INCLUDE_NETWORK, [603](#)
 - OSAL_CONFIG_INCLUDE_STATIC_LOADER, [603](#)
- OutputPort
 - CFE_EVS_HousekeepingTlm_Payload, [486](#)
- OwnerAppName
 - CFE_TBL_TblRegPacket_Payload, [550](#)
- Packet Sequence Values, [40](#)
- PacketID
 - CFE_EVS_LongEventTlm_Payload, [488](#)
 - CFE_EVS_ShortEventTlm_Payload, [497](#)

Padding

CFE_ES_PoolStatsTlm_Payload, [453](#)
 CFE_EVS_HousekeepingTlm_Payload, [487](#)
 CFE_EVS_LongEventTlm_Payload, [489](#)
 CFE_SB_HousekeepingTlm_Payload, [508](#)

Padding1

CFE_ES_AppInfo, [429](#)
 CFE_ES_StartAppCmd_Payload, [465](#)
 CFE_TBL_TblRegPacket_Payload, [550](#)

Padding2

CFE_ES_AppInfo, [429](#)
 CFE_ES_StartAppCmd_Payload, [465](#)
 CFE_TBL_TblRegPacket_Payload, [550](#)

Padding3

CFE_ES_AppInfo, [429](#)

Parameter

CFE_TBL_NotifyCmd_Payload, [545](#)

Path

OS_file_prop_t, [583](#)

Payload

CFE_ES_DeleteCDSCmd, [435](#)
 CFE_ES_DumpCDSRegistryCmd, [436](#)
 CFE_ES_FileNameCmd, [437](#)
 CFE_ES_HousekeepingTlm, [438](#)
 CFE_ES_MemStatsTlm, [449](#)
 CFE_ES_OneAppTlm, [450](#)
 CFE_ES_OverWriteSysLogCmd, [451](#)
 CFE_ES_QueryAllCmd, [454](#)
 CFE_ES_QueryAllTasksCmd, [455](#)
 CFE_ES_QueryOneCmd, [455](#)
 CFE_ES_ReloadAppCmd, [456](#)
 CFE_ES_RestartAppCmd, [457](#)
 CFE_ES_RestartCmd, [458](#)
 CFE_ES_SendMemPoolStatsCmd, [459](#)
 CFE_ES_SetMaxPRCountCmd, [461](#)
 CFE_ES_SetPerfFilterMaskCmd, [462](#)
 CFE_ES_SetPerfTriggerMaskCmd, [463](#)
 CFE_ES_StartApp, [464](#)
 CFE_ES_StartPerfDataCmd, [467](#)
 CFE_ES_StopAppCmd, [467](#)
 CFE_ES_StopPerfDataCmd, [469](#)
 CFE_ES_WriteERLogCmd, [471](#)
 CFE_ES_WriteSysLogCmd, [471](#)
 CFE_EVS_AddEventFilterCmd, [472](#)
 CFE_EVS_DeleteEventFilterCmd, [479](#)
 CFE_EVS_DisableAppEventsCmd, [479](#)
 CFE_EVS_DisableAppEventTypeCmd, [480](#)
 CFE_EVS_DisableEventCmd, [480](#)
 CFE_EVS_DisablePortsCmd, [481](#)
 CFE_EVS_EnableAppEventsCmd, [482](#)
 CFE_EVS_EnableAppEventTypeCmd, [482](#)
 CFE_EVS_EnableEventCmd, [483](#)
 CFE_EVS_EnableEventCmd, [483](#)
 CFE_EVS_EnablePortsCmd, [483](#)
 CFE_EVS_HousekeepingTlm, [484](#)

CFE_EVS_LongEventTlm, [488](#)
 CFE_EVS_ResetAllFiltersCmd, [491](#)
 CFE_EVS_ResetAppCounterCmd, [491](#)
 CFE_EVS_ResetFilterCmd, [492](#)
 CFE_EVS_SetEventFormatModeCmd, [494](#)
 CFE_EVS_SetFilterCmd, [495](#)
 CFE_EVS_SetLogModeCmd, [496](#)
 CFE_EVS_ShortEventTlm, [496](#)
 CFE_EVS_WriteAppDataFileCmd, [498](#)
 CFE_EVS_WriteLogDataFileCmd, [498](#)
 CFE_SB_AllSubscriptionsTlm, [502](#)
 CFE_SB_DisableRouteCmd, [503](#)
 CFE_SB_EnableRouteCmd, [504](#)
 CFE_SB_HousekeepingTlm, [505](#)
 CFE_SB_SingleSubscriptionTlm, [519](#)
 CFE_SB_StatsTlm, [520](#)
 CFE_SB_WriteMapInfoCmd, [526](#)
 CFE_SB_WritePipeInfoCmd, [526](#)
 CFE_SB_WriteRoutingInfoCmd, [527](#)
 CFE_TBL_AbortLoadCmd, [527](#)
 CFE_TBL_ActivateCmd, [529](#)
 CFE_TBL_DeleteCDSCmd, [531](#)
 CFE_TBL_DumpCmd, [531](#)
 CFE_TBL_DumpRegistryCmd, [533](#)
 CFE_TBL_HousekeepingTlm, [536](#)
 CFE_TBL_LoadCmd, [543](#)
 CFE_TBL_NotifyCmd, [545](#)
 CFE_TBL_SendRegistryCmd, [547](#)
 CFE_TBL_TableRegistryTlm, [548](#)
 CFE_TBL_ValidateCmd, [552](#)
 CFE_TIME_AddAdjustCmd, [553](#)
 CFE_TIME_AddDelayCmd, [553](#)
 CFE_TIME_AddOneHzAdjustmentCmd, [554](#)
 CFE_TIME_DiagnosticTlm, [555](#)
 CFE_TIME_HousekeepingTlm, [564](#)
 CFE_TIME_SetLeapSecondsCmd, [570](#)
 CFE_TIME_SetMETCmd, [570](#)
 CFE_TIME_SetSignalCmd, [571](#)
 CFE_TIME_SetSourceCmd, [572](#)
 CFE_TIME_SetStateCmd, [572](#)
 CFE_TIME_SetSTCFCmd, [573](#)
 CFE_TIME_SetTimeCmd, [573](#)
 CFE_TIME_SubAdjustCmd, [576](#)
 CFE_TIME_SubDelayCmd, [576](#)
 CFE_TIME_SubOneHzAdjustmentCmd, [577](#)
 CFE_TIME_ToneDataCmd, [579](#)

PeakMemInUse

CFE_SB_StatsTlm_Payload, [523](#)

PeakMsgIdsInUse

CFE_SB_StatsTlm_Payload, [523](#)

PeakPipesInUse

CFE_SB_StatsTlm_Payload, [523](#)

PeakQueueDepth

CFE_SB_PipeDepthStats, [512](#)

- CFE_SB_PipeInfoEntry, 513
- PeakSBBuffersInUse
 - CFE_SB_StatsTlm_Payload, 523
- PeakSubscriptionsInUse
 - CFE_SB_StatsTlm_Payload, 523
- PerfDataCount
 - CFE_ES_HousekeepingTlm_Payload, 443
- PerfDataEnd
 - CFE_ES_HousekeepingTlm_Payload, 443
- PerfDataStart
 - CFE_ES_HousekeepingTlm_Payload, 443
- PerfDataToWrite
 - CFE_ES_HousekeepingTlm_Payload, 443
- PerfFilterMask
 - CFE_ES_HousekeepingTlm_Payload, 443
- PerfMode
 - CFE_ES_HousekeepingTlm_Payload, 443
- Performance Data Collection, 14
- Performance Data Collection Trigger Masks, 14
- PerfState
 - CFE_ES_HousekeepingTlm_Payload, 444
- PerfTriggerCount
 - CFE_ES_HousekeepingTlm_Payload, 444
- PerfTriggerMask
 - CFE_ES_HousekeepingTlm_Payload, 444
- Pipe
 - CFE_SB_RouteCmd_Payload, 516
 - CFE_SB_SingleSubscriptionTlm_Payload, 520
 - CFE_SB_SubEntries, 524
- Pipe Overflow Error, 41
- PipeDepthStats
 - CFE_SB_StatsTlm_Payload, 523
- Pipeld
 - CFE_SB_PipeDepthStats, 512
 - CFE_SB_PipeInfoEntry, 514
 - CFE_SB_RoutingFileEntry, 517
- PipeName
 - CFE_SB_PipeInfoEntry, 514
 - CFE_SB_RoutingFileEntry, 517
- PipeOptsErrorCounter
 - CFE_SB_HousekeepingTlm_Payload, 508
- PipeOverflowErrorCounter
 - CFE_SB_HousekeepingTlm_Payload, 508
- Pipes, 37
- PipesInUse
 - CFE_SB_StatsTlm_Payload, 523
- PktSegment
 - CFE_SB_AllSubscriptionsTlm_Payload, 502
- PoolHandle
 - CFE_ES_PoolStatsTlm_Payload, 453
 - CFE_ES_SendMemPoolStatsCmd_Payload, 460
- PoolSize
 - CFE_ES_MemPoolStats, 449
- PoolStats
 - CFE_ES_PoolStatsTlm_Payload, 453
- Power-On Reset, 68, 69
- Priority
 - CFE_ES_ApplInfo, 429
 - CFE_ES_StartAppCmd_Payload, 466
 - CFE_ES_TaskInfo, 470
 - CFE_SB_Qos_t, 515
- priority
 - OS_task_prop_t, 594
- Processor Reset, 68, 69
- ProcessorID
 - CFE_EVS_PacketID, 490
 - CFE_FS_Header, 501
- ProcessorResets
 - CFE_ES_HousekeepingTlm_Payload, 444
- psp/fsw/inc/cfe_psp.h, 1288, 1289
- psp/fsw/inc/cfe_psp_cache_api.h, 1290, 1291
- psp/fsw/inc/cfe_psp_cds_api.h, 1291, 1293
- psp/fsw/inc/cfe_psp_eepromaccess_api.h, 1294, 1297
- psp/fsw/inc/cfe_psp_endian.h, 1297, 1299
- psp/fsw/inc/cfe_psp_error.h, 1300, 1303
- psp/fsw/inc/cfe_psp_exception_api.h, 1304, 1306
- psp/fsw/inc/cfe_psp_id_api.h, 1306, 1307
- psp/fsw/inc/cfe_psp_memaccess_api.h, 1308, 1312
- psp/fsw/inc/cfe_psp_memrange_api.h, 1313, 1319
- psp/fsw/inc/cfe_psp_port_api.h, 1320, 1323
- psp/fsw/inc/cfe_psp_ssr_api.h, 1324, 1325
- psp/fsw/inc/cfe_psp_timertick_api.h, 1325, 1327
- psp/fsw/inc/cfe_psp_version_api.h, 1328, 1330
- psp/fsw/inc/cfe_psp_watchdog_api.h, 1330, 1336
- PSP_DEBUG
 - cfe_psp.h, 1288
- PSP_DEBUG_LEV
 - cfe_psp.h, 1289
- PSPMajorVersion
 - CFE_ES_HousekeepingTlm_Payload, 444
- PSPMinorVersion
 - CFE_ES_HousekeepingTlm_Payload, 444
- PSPMissionRevision
 - CFE_ES_HousekeepingTlm_Payload, 444
- PSPRevision
 - CFE_ES_HousekeepingTlm_Payload, 445
- Ptr
 - CFE_ES_PoolAlign, 453
- Qos
 - CFE_SB_SingleSubscriptionTlm_Payload, 520
 - CFE_SB_SubEntries, 525
- Quality of Service, 42
- RegisteredCoreApps
 - CFE_ES_HousekeepingTlm_Payload, 445
- RegisteredExternalApps
 - CFE_ES_HousekeepingTlm_Payload, 445
- RegisteredLibs

- CFE_ES_HousekeepingTlm_Payload, 445
- RegisteredTasks
 - CFE_ES_HousekeepingTlm_Payload, 445
- Reliability
 - CFE_SB_Qos_t, 515
- Reloading an Application, 12
- Reserved
 - CFE_TBL_File_Hdr, 534
- Reset Types and Subtypes, 10
- ResetSubtype
 - CFE_ES_HousekeepingTlm_Payload, 445
- Resetting EVS Counters, 30
- ResetType
 - CFE_ES_HousekeepingTlm_Payload, 445
- ResourceId
 - CFE_ES_AppInfo, 429
- Restarting an Application, 12
- RestartType
 - CFE_ES_RestartCmd_Payload, 458
- SB Event Filtering, 41
- SBBuffersInUse
 - CFE_SB_StatsTlm_Payload, 524
- Seconds
 - CFE_TIME_OneHzAdjustmentCmd_Payload, 568
 - CFE_TIME_SysTime, 578
 - CFE_TIME_TimeCmd_Payload, 578
- Selecting Tone and Data Ordering, 60, 64
- SendErrors
 - CFE_SB_PipeInfoEntry, 514
- Sequence
 - CCSDS_PrimaryHeader, 423
- Server, 70, 71
- ServerFlyState
 - CFE_TIME_DiagnosticTlm_Payload, 561
- Setting MET, 71–73
- Setting Time, 70, 72, 73
- Single Buffered Tables, 49
- Size
 - CFE_ES_CDSRegDumpRec, 433
 - CFE_TBL_Info, 542
 - CFE_TBL_TblRegPacket_Payload, 551
- Software Bus Terminology, 36
- Software Reset, 10
- SpacecraftID
 - CFE_EVS_PacketID, 490
 - CFE_FS_Header, 501
- Spare
 - CFE_ES_TaskInfo, 470
 - CFE_EVS_AppNameBitMaskCmd_Payload, 473
 - CFE_EVS_BitMaskCmd_Payload, 478
 - CFE_EVS_SetEventFormatCode_Payload, 493
 - CFE_EVS_SetLogMode_Payload, 495
 - CFE_SB_PipeDepthStats, 512
 - CFE_SB_PipeInfoEntry, 514
 - CFE_SB_RouteCmd_Payload, 516
- Specifying Time Server/Client, 61, 64
- Specifying Time Signal, 63, 66
- Specifying Time Source, 62, 65
- Specifying Time Tone Byte Order, 61, 65
- Specifying Tone and Data Window, 61, 64
- stack_size
 - OS_task_prop_t, 594
- StackSize
 - CFE_ES_AppInfo, 430
 - CFE_ES_StartAppCmd_Payload, 466
 - CFE_ES_TaskInfo, 470
- start_time
 - OS_timer_prop_t, 597
- StartAddress
 - CFE_ES_AppInfo, 430
- Starting an Application, 11
- Starting to Collect Performance Data, 14
- State
 - CFE_SB_RoutingFileEntry, 517
- STCF
 - CFE_TIME_HousekeepingTlm_Payload, 566
- Std
 - CFE_TBL_CombinedFileHdr, 530
- Stopping an Application, 11
- Stopping the Collection of Performance Data, 14
- StreamId
 - CCSDS_PrimaryHeader, 424
- SubscribeErrorCounter
 - CFE_SB_HousekeepingTlm_Payload, 508
- Subscriptions, 37
- SubscriptionsInUse
 - CFE_SB_StatsTlm_Payload, 524
- Subseconds
 - CFE_TIME_OneHzAdjustmentCmd_Payload, 568
 - CFE_TIME_SysTime, 578
- Subsystem
 - CCSDS_ExtendedHeader, 423
- SubType
 - CFE_FS_Header, 501
 - CFE_SB_SingleSubscriptionTlm_Payload, 520
- SuccessValCounter
 - CFE_TBL_HousekeepingTlm_Payload, 540
- SysLogBytesUsed
 - CFE_ES_HousekeepingTlm_Payload, 446
- SysLogEntries
 - CFE_ES_HousekeepingTlm_Payload, 446
- SysLogMode
 - CFE_ES_HousekeepingTlm_Payload, 446
- SysLogSize
 - CFE_ES_HousekeepingTlm_Payload, 446
- System Log, 18
- SystemId

- CCSDS_ExtendedHeader, [423](#)
- Table
 - CFE_ES_CDSRegDumpRec, [433](#)
- Table Registry, [50](#)
- Table Services Telemetry, [51](#)
- TableLoadedOnce
 - CFE_TBL_Info, [542](#)
 - CFE_TBL_TblRegPacket_Payload, [551](#)
- TableName
 - CFE_TBL_AbortLoadCmd_Payload, [528](#)
 - CFE_TBL_ActivateCmd_Payload, [529](#)
 - CFE_TBL_DeICDSCmd_Payload, [530](#)
 - CFE_TBL_DumpCmd_Payload, [532](#)
 - CFE_TBL_File_Hdr, [534](#)
 - CFE_TBL_FileDef, [536](#)
 - CFE_TBL_SendRegistryCmd_Payload, [547](#)
 - CFE_TBL_ValidateCmd_Payload, [552](#)
- Tables with Validation Functions, [49](#)
- TaskId
 - CFE_ES_TaskInfo, [470](#)
- TaskName
 - CFE_ES_TaskInfo, [470](#)
- Tbl
 - CFE_TBL_CombinedFileHdr, [530](#)
- TelemetryHeader
 - CFE_ES_HousekeepingTlm, [438](#)
 - CFE_ES_MemStatsTlm, [449](#)
 - CFE_ES_OneAppTlm, [450](#)
 - CFE_EVS_HousekeepingTlm, [484](#)
 - CFE_EVS_LongEventTlm, [488](#)
 - CFE_EVS_ShortEventTlm, [496](#)
 - CFE_SB_AllSubscriptionsTlm, [502](#)
 - CFE_SB_HousekeepingTlm, [505](#)
 - CFE_SB_SingleSubscriptionTlm, [519](#)
 - CFE_SB_StatsTlm, [520](#)
 - CFE_TBL_HousekeepingTlm, [536](#)
 - CFE_TBL_TableRegistryTlm, [548](#)
 - CFE_TIME_DiagnosticTlm, [555](#)
 - CFE_TIME_HousekeepingTlm, [564](#)
- Terminology, [8](#)
- TgtFilename
 - CFE_TBL_FileDef, [536](#)
- ticks
 - OS_time_t, [595](#)
- Time Components, [57](#)
- Time Configuration, [59](#)
- Time Format Selection, [60](#), [63](#)
- Time Formats, [58](#)
- Time Services Paradigm(s), [66](#)
- Time State, [68](#)
- Time Structure, [58](#)
- TimeOfLastUpdate
 - CFE_TBL_Info, [542](#)
 - CFE_TBL_TblRegPacket_Payload, [551](#)
- Timer Overview, [101](#)
- TimeSeconds
 - CFE_FS_Header, [501](#)
- TimeSinceTone
 - CFE_TIME_DiagnosticTlm_Payload, [561](#)
- TimeSource
 - CFE_TIME_SourceCmd_Payload, [575](#)
- TimeSubSeconds
 - CFE_FS_Header, [501](#)
- ToneDataCounter
 - CFE_TIME_DiagnosticTlm_Payload, [561](#)
- ToneDataLatch
 - CFE_TIME_DiagnosticTlm_Payload, [561](#)
- ToneIntCounter
 - CFE_TIME_DiagnosticTlm_Payload, [562](#)
- ToneIntErrorCounter
 - CFE_TIME_DiagnosticTlm_Payload, [562](#)
- ToneMatchCounter
 - CFE_TIME_DiagnosticTlm_Payload, [562](#)
- ToneMatchErrorCounter
 - CFE_TIME_DiagnosticTlm_Payload, [562](#)
- ToneOverLimit
 - CFE_TIME_DiagnosticTlm_Payload, [562](#)
- ToneSignalCounter
 - CFE_TIME_DiagnosticTlm_Payload, [562](#)
- ToneSignalLatch
 - CFE_TIME_DiagnosticTlm_Payload, [562](#)
- ToneSource
 - CFE_TIME_SignalCmd_Payload, [574](#)
- ToneTaskCounter
 - CFE_TIME_DiagnosticTlm_Payload, [563](#)
- ToneUnderLimit
 - CFE_TIME_DiagnosticTlm_Payload, [563](#)
- total_blocks
 - OS_statvfs_t, [593](#)
- TotalSegments
 - CFE_SB_AllSubscriptionsTlm_Payload, [503](#)
- TriggerMask
 - CFE_ES_SetPerfTrigMaskCmd_Payload, [464](#)
- TriggerMaskNum
 - CFE_ES_SetPerfTrigMaskCmd_Payload, [464](#)
- TriggerMode
 - CFE_ES_StartPerfCmd_Payload, [466](#)
- Type
 - CFE_ES_ApplInfo, [430](#)
- uint16
 - common_types.h, [1218](#)
- uint32
 - common_types.h, [1218](#)
- uint64
 - common_types.h, [1218](#)
- uint8

- [common_types.h](#), [1218](#)
- UnmarkedMem
 - [CFE_SB_HousekeepingTlm_Payload](#), [508](#)
- UnregisteredAppCounter
 - [CFE_EVS_HousekeepingTlm_Payload](#), [487](#)
- User
 - [OS_file_prop_t](#), [584](#)
- User Defined Address Tables, [50](#)
- UserDefAddr
 - [CFE_TBL_Info](#), [542](#)
- valid
 - [OS_module_address_t](#), [587](#)
- ValidationCounter
 - [CFE_TBL_HousekeepingTlm_Payload](#), [540](#)
- ValidationFuncPtr
 - [CFE_TBL_TblRegPacket_Payload](#), [551](#)
- Value
 - [CFE_SB_MsgId_t](#), [510](#)
- value
 - [OS_bin_sem_prop_t](#), [581](#)
 - [OS_count_sem_prop_t](#), [582](#)
- Version Identification, [18](#)
- Version Numbers, [4](#)
- VersionCounter
 - [CFE_TIME_DiagnosticTlm_Payload](#), [563](#)
- Viewing the Collection of Performance Data, [15](#)
- Virtual MET, [62](#), [65](#)
- VirtualMET
 - [CFE_TIME_DiagnosticTlm_Payload](#), [563](#)